

File: app/(auth)/check-email/page.tsx (49 lines, 1.9 KB)

```

1  'use client'
2
3  import React from 'react'
4  import Link from 'next/link'
5  import { motion } from 'framer-motion'
6  import { Mail, ArrowLeft, Loader2 } from 'lucide-react'
7
8  export default function CheckEmailPage() {
9    return (
10     <div className="min-h-screen flex items-center justify-center bg-gradient-to-br from-[#0f0f1a] to-[#1a1a2e...
11       <motion.div
12         initial={{ opacity: 0, scale: 0.9 }}
13         animate={{ opacity: 1, scale: 1 }}
14         className="w-full max-w-md bg-[#16213e] border border-[#1e293b] p-10 rounded-2xl text-center shadow-2xl"
15       >
16         <div className="w-20 h-20 bg-[#00d2a0]/10 text-[#00d2a0] rounded-full flex items-center justify-center...
17         <Mail size={40} />
18       </div>
19
20 |         <h2 className="text-2xl font-bold mb-4">Ki Ếm tra h ậ p th ộ c ặ a b i n
21 |         <p className="text-slate-400 mb-2 leading-relaxed">
22 |           Ch ứ ng t ờ i ấ g ớ i email x á c nh - n ử n Ế a ch Ế email c ặ a b i n .
23 |         </p>
24 |         <p className="text-slate-500 text-sm mb-8">
25 |           V u i l ồ ng ki Ếm tra h ậ p th ộ ử n ( I n b o x ) h o ặ c th ộ m ắ c S p a m , s a u
26 |         </p>
27
28         <div className="flex flex-col gap-3">
29           <button
30             type="button"
31             onClick={() => window.location.href = '/builder'}
32             className="w-full bg-[#00d2a0] hover:bg-[#00e6af] text-black font-bold py-3 rounded-xl transition-...
33         >
34 |           V Ầ t r a n g c h ặ
35
36         </button>
37
38         <Link
39           href="/login"
40           className="text-sm text-[#00b4d8] hover:text-[#00d2a0] transition-colors font-medium"
41         >
42 |         <ArrowLeft size={14} className="inline mr-1" />
43 |           Q u a y l ớ i n g n h - p
44         </Link>
45       </div>
46     </motion.div>
47   </div>
48 )
49

```

File: app/(auth)/forgot-password/page.tsx (115 lines, 4.6 KB)

```

1  'use client'
2
3  import React, { useState } from 'react'
4  import Link from 'next/link'
5  import { motion } from 'framer-motion'
6  import { Mail, ArrowLeft, Loader2, CheckCircle2, AlertCircle } from 'lucide-react'
7  import { createBrowserClient } from '@supabase/ssr'
8
9  export default function ForgotPasswordPage() {
10    const [email, setEmail] = useState('')
11    const [loading, setLoading] = useState(false)
12    const [submitted, setSubmitted] = useState(false)
13    const [error, setError] = useState<string | null>(null)
14
15    const handleSubmit = async (e: React.FormEvent) => {
16      e.preventDefault()
17      setLoading(true)
18      setError(null)
19
20      try {
21        const supabase = createBrowserClient(
22          process.env.NEXT_PUBLIC_SUPABASE_URL!,
23          process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
24        )
25        const redirectUrl = process.env.NEXT_PUBLIC_SITE_URL
26        ? `${process.env.NEXT_PUBLIC_SITE_URL}/auth/callback`
27        : 'http://localhost:3000/auth/callback'
28
29        const { error: resetError } = await supabase.auth.resetPasswordForEmail(email, {
30          redirectTo: `${redirectUrl}?type=recovery`,
31        })
32
33        if (resetError) {
34          setError(resetError.message)
35        } else {
36          setSubmitted(true)
37        }
38      } catch (err) {
39 |        setError('C ó l ậ i x ế y r a . V u i l ồ ng th ớ l ớ i s a u . ')
40      } finally {
41        setLoading(false)
42      }
43    }
44
45    if (submitted) {
46      return (
47        <div className="min-h-screen bg-[#0f0f1a] flex items-center justify-center p-6 text-white">
48          <motion.div
49            initial={{ opacity: 0, scale: 0.9 }}
50            animate={{ opacity: 1, scale: 1 }}
51            className="w-full max-w-md bg-[#16213e] border border-[#1e293b] p-10 rounded-2xl text-center shadow-...
52          >
53            <div className="w-20 h-20 bg-[#00d2a0]/10 text-[#00d2a0] rounded-full flex items-center justify-cent...
54            <CheckCircle2 size={40} />
55          </div>
56 |          <h2 className="text-2xl font-bold mb-4">Ki Ếm tra Email</h2>
57 |          <p className="text-slate-400 mb-8 leading-relaxed">
58 |            Ch ứ ng t ờ i ấ g ớ i link ặ t l ớ i m - t k h ộ u ử n <span className="
59 |          </p>
60 |          <Link href="/login" className="inline-flex items-center gap-2 text-[#00d2a0] font-medium hover:under...
61 |          <ArrowLeft size={18} /> Q u a y l ớ i n g n h - p
62 |        </Link>
63 |      </motion.div>
64 |    </div>
65 |  )
66 | }
67 |
68 | return (

```

```

69 | <div className="min-h-screen bg-[#0f0f1a] flex items-center justify-center p-6 text-white">
70 |   <motion.div
71 |     initial={{ opacity: 0, y: 20 }}
72 |     animate={{ opacity: 1, y: 0 }}
73 |     className="w-full max-w-md bg-[#16213e] border border-[#1e293b] p-8 md:p-10 rounded-2xl shadow-2xl"
74 |   >
75 |     <Link href="/login" className="inline-flex items-center gap-2 text-slate-400 hover:text-white transiti...
76 |     <ArrowLeft size={18} />
77 |     <span className="text-sm font-medium"> Quay lại nh - p </span>
78 |   </Link>
79 |
80 |   <h2 className="text-2xl font-bold mb-2"> Quên mật khẩu? </h2>
81 |   <p className="text-slate-400 mb-8"> ăng lo! Nh - p email cça bin v
82 |
83 |   <form onSubmit={handleSubmit} className="space-y-6">
84 |     <div className="relative">
85 |       <Mail className="absolute left-4 top-1/2 -translate-y-1/2 text-slate-500" size={18} />
86 |       <input
87 |         type="email"
88 |         required
89 |         value={email}
90 |         onChange={(e) => setEmail(e.target.value)}
91 |         className="w-full bg-[#0f0f1a] border border-[#1e293b] rounded-xl pl-11 pr-4 py-3.5 outline-none...
92 |         placeholder="Email cça bin"
93 |       />
94 |     </div>
95 |
96 |     {error && (
97 |       <div className="bg-red-500/10 border border-red-500/20 text-red-400 px-4 py-3 rounded-lg text-sm f...
98 |       <AlertCircle size={18} className="shrink-0" />
99 |       <span>{error}</span>
100 |     </div>
101 |   )}
102 |
103 |   <button
104 |     type="submit"
105 |     disabled={loading}
106 |     className="w-full bg-[#00d2a0] hover:bg-[#00e6af] text-black font-bold py-3.5 rounded-xl transiti...
107 |   >
108 |     {loading ? <Loader2 className="animate-spin" size={20} /> :
109 |     </button>
110 |   </form>
111 | </motion.div>
112 | </div>
113 | )
114 | }
115 |

```

File: app/(auth)/login/page.tsx (368 lines, 20.9 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { useRouter } from 'next/navigation'
5 | import Link from 'next/link'
6 | import { Loader2, Mail, Lock, Eye, EyeOff, ShieldCheck, Login, ArrowLeft, Shield, KeyRound } from 'lucide-react'
7 | import { login } from '@lib/auth-actions'
8 | import { createBrowserClient } from '@supabase/ssr'
9 |
10 | export default function LoginPage() {
11 |   const [error, setError] = useState<string | null>(null)
12 |   const [loading, setLoading] = useState(false)
13 |   const [showPassword, setShowPassword] = useState(false)
14 |   const [email, setEmail] = useState('')
15 |   const [showAdmin, setShowAdmin] = useState(false)
16 |   const [adminPass, setAdminPass] = useState('')
17 |   const [adminError, setAdminError] = useState('')
18 |   const [adminLoading, setAdminLoading] = useState(false)
19 |   const router = useRouter()
20 |
21 |   async function handleAdminLogin() {
22 |     if (!adminPass.trim()) { setAdminError('Vui lòng nh - p m - t kh@u admin')
23 |     setAdminLoading(true)
24 |     setAdminError('')
25 |
26 |     try {
27 |       const formData = new FormData()
28 |       formData.set('email', 'admin')
29 |       formData.set('password', adminPass)
30 |       const res = await login(formData)
31 |       if (res?.error) {
32 |         setAdminError('M - t kh@u admin không chính xác')
33 |       } else if (res?.success && res.redirectUrl) {
34 |         router.push(res.redirectUrl)
35 |       }
36 |     } catch {
37 |       setAdminError('Có lỗi xảy ra, vui lòng thí lại')
38 |     }
39 |     setAdminLoading(false)
40 |   }
41 |
42 |   async function handleGoogleLogin() {
43 |     setError(null)
44 |     const supabase = createBrowserClient(
45 |       process.env.NEXT_PUBLIC_SUPABASE_URL!,
46 |       process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
47 |     )
48 |     const redirectTo = process.env.NEXT_PUBLIC_SITE_URL
49 |       ? `${process.env.NEXT_PUBLIC_SITE_URL}/auth/callback`
50 |       : `${window.location.origin}/auth/callback`
51 |     const { error: oauthError } = await supabase.auth.signInWithOAuth({
52 |       provider: 'google',
53 |       options: {
54 |         redirectTo,
55 |       },
56 |     })
57 |     if (oauthError) {
58 |       setError(' nh - p Google th%t bi i: ' + oauthError.message)
59 |     }
60 |   }
61 |
62 |   async function handleSubmit(e: React.FormEvent<HTMLFormElement>) {
63 |     e.preventDefault()
64 |     setLoading(true)
65 |     setError(null)
66 |
67 |     const formData = new FormData(e.currentTarget)
68 |     const res = await login(formData)
69 |
70 |     if (res?.error) {
71 |       setError(res.error)
72 |       setLoading(false)
73 |     } else if (res?.success && res.redirectUrl) {

```

```

74 | router.push(res.redirectUrl)
75 | } else {
76 |   setLoading(false)
77 | }
78 | }
79 |
80 | return (
81 |   <div className="login-page-root" style={{ minHeight: '100vh', display: 'flex', background: 'var(--bg-base)...
82 |     /* Left Panel - Animated Illustration */
83 |     <div className="login-left-panel" style={{
84 |       flex: 1, display: 'flex', alignItems: 'center', justifyContent: 'center',
85 |       background: 'linear-gradient(160deg, #031f3b 0%, #1a2f53 80%)',
86 |       padding: '64px', position: 'relative', overflow: 'hidden',
87 |     }}>
88 |       /* Background decorative elements */
89 |       <div style={{ position: 'absolute', top: '-100px', right: '-100px', width: '500px', height: '500px', b...
90 |       <div style={{ position: 'absolute', bottom: '-80px', left: '-80px', width: '400px', height: '400px', b...
91 |       <div style={{ position: 'absolute', top: '30%', left: '10%', width: '200px', height: '200px', backgrou...
92 |
93 |     <div style={{ textAlign: 'center', position: 'relative', zIndex: 1, color: '#fff', maxWidth: '480px' }}>
94 |       { /* Animated illustration - nhóm bin híc t - p c ò n g n g h ặ * / }
95 |       <div style={{ position: 'relative', width: '300px', height: '260px', margin: '0 auto 32px' }}>
96 |         /* Monitor screen */
97 |         <svg viewBox="0 0 300 260" style={{ width: '100%', height: '100%' }}>
98 |           <defs>
99 |             <linearGradient id="screenGlow" x1="0%" y1="0%" x2="100%" y2="100%">
100 |               <stop offset="0%" stopColor="#00d4aa" stopOpacity="0.3" />
101 |               <stop offset="100%" stopColor="#289cf9" stopOpacity="0.1" />
102 |             </linearGradient>
103 |           </defs>
104 |
105 |           { /* Desk */ }
106 |           <rect x="20" y="195" width="260" height="8" rx="4" fill="rgba(255,255,255,0.1)" />
107 |           <rect x="40" y="203" width="12" height="35" rx="2" fill="rgba(255,255,255,0.08)" />
108 |           <rect x="248" y="203" width="12" height="35" rx="2" fill="rgba(255,255,255,0.08)" />
109 |
110 |           { /* Monitor */ }
111 |           <rect x="85" y="60" width="130" height="100" rx="8" fill="rgba(255,255,255,0.06)" stroke="rgba(2...
112 |           <rect x="90" y="65" width="120" height="85" rx="4" fill="url(#screenGlow)" />
113 |
114 |           { /* Screen content: code lines */ }
115 |           <rect x="100" y="75" width="40" height="4" rx="2" fill="rgba(0,212,170,0.5)" />
116 |           <rect x="100" y="85" width="60" height="4" rx="2" fill="rgba(40,156,249,0.4)" />
117 |           <rect x="100" y="95" width="25" height="4" rx="2" fill="rgba(255,185,0,0.4)" />
118 |           <rect x="110" y="105" width="50" height="4" rx="2" fill="rgba(0,212,170,0.3)" />
119 |           <rect x="110" y="115" width="35" height="4" rx="2" fill="rgba(40,156,249,0.3)" />
120 |           <rect x="100" y="130" width="70" height="4" rx="2" fill="rgba(255,185,0,0.5)" />
121 |
122 |           { /* CPU icon on screen */ }
123 |           <rect x="170" y="75" width="30" height="30" rx="3" fill="none" stroke="rgba(0,212,170,0.4)" stro...
124 |           <rect x="177" y="82" width="16" height="16" rx="2" fill="rgba(0,212,170,0.2)" />
125 |           <rect x="181" y="86" width="8" height="8" rx="1" fill="rgba(0,212,170,0.5)" />
126 |
127 |           { /* Monitor stand */ }
128 |           <rect x="140" y="160" width="20" height="25" rx="2" fill="rgba(255,255,255,0.08)" />
129 |           <rect x="125" y="183" width="50" height="6" rx="3" fill="rgba(255,255,255,0.1)" />
130 |
131 |           { /* Student 1 - left */ }
132 |           <g style={{ animation: 'floatBounce 3s ease-in-out infinite' }}>
133 |             <circle cx="45" cy="175" r="20" fill="rgba(255,255,255,0.08)" />
134 |             <circle cx="45" cy="168" r="12" fill="rgba(255,255,255,0.15)" />
135 |             <circle cx="45" cy="165" r="5" fill="rgba(255,255,255,0.2)" />
136 |             <rect x="37" y="180" width="16" height="25" rx="4" fill="rgba(255,255,255,0.08)" />
137 |           { /* Headphones */ }
138 |           <path d="M33 163 Q33 158 38 158 Q45 155 52 158 Q57 158 57 163" fill="none" stroke="rgba(0,212,...
139 |         </g>
140 |
141 |           { /* Student 2 - right */ }
142 |           <g style={{ animation: 'floatBounce 3.5s ease-in-out infinite 0.5s' }}>
143 |             <circle cx="255" cy="170" r="20" fill="rgba(255,255,255,0.08)" />
144 |             <circle cx="255" cy="163" r="12" fill="rgba(255,255,255,0.15)" />
145 |             <circle cx="255" cy="160" r="5" fill="rgba(255,255,255,0.2)" />
146 |             <rect x="247" y="175" width="16" height="25" rx="4" fill="rgba(255,255,255,0.08)" />
147 |           { /* Glasses */ }
148 |             <circle cx="250" cy="163" r="4" fill="none" stroke="rgba(40,156,249,0.4)" strokeWidht="1.5" />
149 |             <circle cx="260" cy="163" r="4" fill="none" stroke="rgba(40,156,249,0.4)" strokeWidht="1.5" />
150 |             <line x1="254" y1="163" x2="256" y2="163" stroke="rgba(40,156,249,0.4)" strokeWidht="1" />
151 |           </g>
152 |
153 |           { /* Floating tech elements */ }
154 |           <g style={{ animation: 'floatRotate 6s linear infinite', transformOrigin: '150px 130px' }}>
155 |             <circle cx="210" cy="50" r="6" fill="none" stroke="rgba(0,212,170,0.3)" strokeWidht="1.5" />
156 |             <circle cx="210" cy="50" r="2" fill="rgba(0,212,170,0.4)" />
157 |           </g>
158 |           <g style={{ animation: 'floatRotate 8s linear infinite reverse', transformOrigin: '150px 130px' }}>
159 |             <rect x="80" y="40" width="8" height="8" rx="2" fill="none" stroke="rgba(255,185,0,0.3)" strok...
160 |           </g>
161 |
162 |           { /* Connection lines */ }
163 |           <line x1="65" y1="180" x2="90" y2="160" stroke="rgba(0,212,170,0.15)" strokeWidht="1" strokeDash...
164 |           <line x1="235" y1="175" x2="210" y2="160" stroke="rgba(40,156,249,0.15)" strokeWidht="1" strokeD...
165 |         </svg>
166 |       </div>
167 |
168 |       <h2 style={{ fontSize: '28px', fontWeight: 800, marginBottom: '16px', lineHeight: 1.3 }}>
169 |         PC Master Builder
170 |       </h2>
171 |       <p style={{ fontSize: '16px', opacity: 0.7, maxWidth: '360px', margin: '0 auto', lineHeight: 1.6 }}>
172 |         M ô p h ã n g l ậ p r á p P C 2 D t í c h h ấ p t r í t u ặ n h ậ n t i ỏ - h í c m à n h
173 |       </p>
174 |
175 |       { /* Feature tags */ }
176 |       <div style={{ marginTop: '32px', display: 'flex', gap: '12px', justifyContent: 'center', flexWrap: '...
177 |         <span style={{ padding: '6px 14px', borderRadius: '99px', background: 'rgba(0,212,170,0.15)', colo...
178 |           A I h ỏ ù n g d ầ n
179 |         </span>
180 |         <span style={{ padding: '6px 14px', borderRadius: '99px', background: 'rgba(40,156,249,0.15)', col...
181 |           100% Web
182 |         </span>
183 |         <span style={{ padding: '6px 14px', borderRadius: '99px', background: 'rgba(255,185,0,0.15)', colo...
184 |           K h ỏ n g c ầ i t
185 |       </span>
186 |     </div>
187 |
188 |     { /* Testimonial */ }
189 |     <div style={{ marginTop: '32px', padding: '16px 20px', background: 'rgba(255,255,255,0.04)', borderR...
190 |       <p style={{ fontSize: '13px', opacity: 0.6, fontStyle: 'italic', lineHeight: 1.5, margin: 0 }}>
191 |         " H ì n 5 , ỏ ỏ ỏ + h í c s ì n h ậ n g h í c t - p v ậ t h ấ c h ậ n h l ậ p r á p P C
192 |       </p>
193 |     </div>
194 |   </div>
195 | </div>
196 |
197 | { /* Right Panel - Login Form */ }

```

```
198 | <div className="login-right-panel" style={{ flex: 1, display: 'flex', alignItems: 'center', justifyConte...
199 | <div style={{ maxWidth: '440px', width: '100%' }}>
200 |   <Link href="/" style={{ display: 'inline-flex', alignItems: 'center', gap: '8px', color: 'var(--text...
201 |     <ArrowLeft size={18} /> Quay lại trang chủ
202 |   </Link>
203 |
204 |   <div style={{ textAlign: 'center', marginBottom: '32px' }}>
205 |     <Link href="/" style={{ display: 'inline-flex', alignItems: 'center', gap: '10px', textDecoration: ...
206 |       
207 |     </Link>
208 |     <h1 style={{ fontSize: '24px', fontWeight: 800, marginBottom: '8px', color: 'var(--text-primary)' }} ...
209 |     <p style={{ color: 'var(--text-muted)', fontSize: '15px' }}>
210 |   </div>
211 |
212 |   <button onClick={handleGoogleLogin} style={{ width: '100%', display: 'flex', alignItems: 'center', j...
213 |     <svg viewBox="0 0 24 24" style={{ width: '20px', height: '20px' }}>
214 |       <path fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h5.92c-.26 1.37-1.04 2.53-2.2...
215 |       <path fill="#34A853" d="M12 23c2.97 0 5.46-.98 7.28-2.66l-3.57-2.77c-.98-.66-2.23 1.06-3.71 1.06-...
216 |       <path fill="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s13-1.43 35-2.09v7.07h2.18c1.43 8...
217 |       <path fill="#EA4335" d="M12 5.38c1.62 0 3.06.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 ...
218 |     </svg>
219 |     T i z p t á c v ù i G o o g l e
220 |   </button>
221 |   <p style={{ fontSize: '11px', color: 'var(--text-muted)', textAlign: 'center', marginBottom: '16px', ...
222 |     L ã n Ñ u n g n h - p : E m a i l t ã Ñ ü n g l ò u , b i n c h É c Ñ n n h - p H í t é
223 |   </p>
224 |
225 |   <div style={{ display: 'flex', alignItems: 'center', gap: '16px', marginBottom: '24px' }}>
226 |     <div style={{ flex: 1, height: '1px', background: 'var(--border-default)' }}</div>
227 |     <span style={{ fontSize: '13px', color: 'var(--text-muted)', fontWeight: 500, whiteSpace: 'nowrap' ...
228 |   </div style={{ flex: 1, height: '1px', background: 'var(--border-default)' }}</div>
229 |
230 |   <form onSubmit={handleSubmit} style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
231 |     <div>
232 |       <label style={{ fontSize: '13px', fontWeight: 600, color: 'var(--text-primary)', marginBottom: '...'
233 |       <div style={{ position: 'relative' }}>
234 |         <Mail size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'translate...
235 |         <input type="email" name="email" required value={email} onChange={e => setEmail(e.target.value...
236 |       </div>
237 |     </div>
238 |
239 |     <div>
240 |       <label style={{ fontSize: '13px', fontWeight: 600, color: 'var(--text-primary)', marginBottom: '...'
241 |       <div style={{ position: 'relative' }}>
242 |         <Lock size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'translate...
243 |         <input type={showPassword ? 'text' : 'password'} name="password" required className="lms-input...
244 |         <button type="button" onClick={() => setShowPassword(!showPassword)} style={{ position: 'absol...
245 |         {showPassword ? <EyeOff size={18} /> : <Eye size={18} />}
246 |       </button>
247 |     </div>
248 |   </div>
249 |
250 |   <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
251 |     <Link href="/forgot-password" style={{ fontSize: '13px', color: 'var(--accent-blue)', fontWeight...
252 |     Q u ê n m - t k h @ u ?
253 |   </div>
254 |
255 |   <div>
256 |     <error && {
257 |       <div style={{ background: 'rgba(244,106,106,0.1)', border: '1px solid rgba(244,106,106,0.2)', co...
258 |       <ShieldCheck size={18} />
259 |       {error}
260 |     </div>
261 |   }
262 |
263 |   <button type="submit" disabled={loading} className="lms-btn lms-btn-primary" style={{ width: '100%...
264 |     { loading ? <> <Loader2 size={18} style={{ animation: 'spin
265 |   </button>
266 | </form>
267 |
268 | <p style={{ textAlign: 'center', margin: '24px 0', color: 'var(--text-muted)', fontSize: '14px' }}>
269 |   C h ò a c ó t à i k h o ã n ? { ' ' }
270 |
271 |   <Link href="/register" style={{ color: 'var(--brand-primary)', fontWeight: 700, textDecoration: 'n...
272 |   n g k ý n g a y
273 | </Link>
274 | </p>
275 |
276 | <div style={{ margin: '24px 0', textAlign: 'center' }}>
277 |   <button
278 |     onClick={() => setShowAdmin(!showAdmin)}
279 |     style={{
280 |       background: 'none', border: 'none', cursor: 'pointer', fontFamily: 'inherit',
281 |       color: showAdmin ? 'var(--brand-primary)' : 'var(--text-muted)',
282 |       fontSize: '12px', fontWeight: 500, padding: '4px 8px', borderRadius: '4px',
283 |       transition: 'all 0.2s', textDecoration: showAdmin ? 'none' : 'underline',
284 |       textDecorationStyle: 'dotted', textUnderlineOffset: '3px'
285 |     }}
286 |   >
287 |     <Shield size={12} style={{ verticalAlign: 'middle', margin: '0 4px' }} />
288 |     { showAdmin ? ' Ó n g t r u y c - p A d m i n ' : ' n g n h - p v ù i t ° c á c
289 |   </button>
290 |
291 |   {showAdmin && {
292 |     <div style={{ margin: '12px 0' }}>
293 |       <label style={{ fontSize: '12px', fontWeight: 600, color: 'var(--text-muted)', margin: '0 ...
294 |       M - t k h @ u A d m i n
295 |     </label>
296 |     <div style={{ display: 'flex', gap: '8px' }}>
297 |       <div style={{ position: 'relative', flex: 1 }}>
298 |         <KeyRound size={16} style={{ position: 'absolute', left: '12px', top: '50%', transform: 't...
299 |         <input
300 |           type="password"
301 |           value={adminPass}
302 |           onChange={e => { setAdminPass(e.target.value); setAdminError('') }}
303 |           onKeyDown={e => { if (e.key === 'Enter') handleAdminLogin() }}
304 |         />
305 |         p l a c e h o l d e r = " N h - p m - t k h @ u a d m i n "
306 |       </div>
307 |       <div style={{
308 |         height: '42px', borderRadius: '8px', border: '1px solid var(--border-strong)',
309 |         padding: '0 12px 0 36px', fontFamily: 'inherit', fontSize: '13px',
310 |         color: 'var(--text-primary)', background: 'var(--bg-surface)',
311 |         outline: 'none', width: '100%', transition: 'border-color 0.2s'
312 |       }}>
313 |     </div>
314 |     <button
315 |       onClick={handleAdminLogin}
316 |       disabled={adminLoading}
317 |       style={{
318 |         padding: '0 20px', borderRadius: '8px', background: 'var(--brand-primary)',
319 |         color: '#fff', fontWeight: 700, border: 'none', cursor: 'pointer',
320 |         fontSize: '13px', fontFamily: 'inherit', whiteSpace: 'nowrap',
321 |         transition: 'all 0.2s', display: 'flex', alignItems: 'center', gap: '6px'
322 |       }}>
```

```

322 |         >
323 |         {adminLoading ? <Loader2 size={16} style={{ animation: 'spin 1s linear infinite' }} /> : <...
324 |         Vao
325 |       </button>
326 |     </div>
327 |     {adminError && (
328 |       <p style={{ marginTop: '8px', fontSize: '12px', color: 'var(--danger)', fontWeight: 600 }}>{...
329 |     )}
330 |   </div>
331 | )}
332 | </div>
333 | </div>
334 | </div>
335 |
336 | <style>{`
337 |   @keyframes floatSlow {
338 |     0%, 100% { transform: translate(0, 0); }
339 |     50% { transform: translate(15px, -15px); }
340 |   }
341 |   @keyframes floatBounce {
342 |     0%, 100% { transform: translateY(0); }
343 |     50% { transform: translateY(-6px); }
344 |   }
345 |   @keyframes floatRotate {
346 |     from { transform: rotate(0deg); }
347 |     to { transform: rotate(360deg); }
348 |   }
349 |
350 |   /* === MOBILE RESPONSIVE === */
351 |   @media (max-width: 768px) {
352 |     .login-left-panel { display: none !important; }
353 |     .login-right-panel {
354 |       flex: none !important;
355 |       width: 100% !important;
356 |       padding: 24px 16px !important;
357 |       align-items: flex-start !important;
358 |     }
359 |     .login-right-panel > div { max-width: 100% !important; }
360 |   }
361 |   @media (max-width: 480px) {
362 |     .login-right-panel { padding: 16px 12px !important; }
363 |   }
364 | `}</style>
365 | </div>
366 | )
367 | }
368 |

```

File: app/(auth)/login/qr/page.tsx (80 lines, 3.7 KB)

```

1 | 'use client'
2 |
3 | export const dynamic = 'force-dynamic';
4 |
5 | import React, { useState, useEffect } from 'react'
6 | import Link from 'next/link'
7 | import { motion } from 'framer-motion'
8 | import { QRCode, ArrowLeft, RefreshCcw, Smartphone, Monitor, CheckCircle2 } from 'lucide-react'
9 | import { QRCodeSVG } from 'qrcode.react'
10 |
11 | export default function QRLoginPage() {
12 |   const [timeLeft, setTimeLeft] = useState(30)
13 |   const [qrValue, setQrValue] = useState('pcmaster-login-' + Math.random().toString(36).substring(7))
14 |
15 |   useEffect(() => {
16 |     if (timeLeft === 0) {
17 |       setQrValue('pcmaster-login-' + Math.random().toString(36).substring(7))
18 |       setTimeLeft(30)
19 |       return
20 |     }
21 |     const timer = setInterval(() => setTimeLeft(prev => prev - 1), 1000)
22 |     return () => clearInterval(timer)
23 |   }, [timeLeft])
24 |
25 |   return (
26 |     <div className="min-h-screen bg-[#f0f1a] flex items-center justify-center p-6 text-white">
27 |       <motion.div
28 |         initial={{ opacity: 0, y: 20 }}
29 |         animate={{ opacity: 1, y: 0 }}
30 |         className="w-full max-w-md bg-[#16213e] border border-[#1e293b] p-8 md:p-10 rounded-3xl shadow-2xl tex...
31 |       >
32 |         <Link href="/login" className="inline-flex items-center gap-2 text-slate-400 hover:text-white transiti...
33 |         <ArrowLeft size={18} />
34 |         <span className="text-sm font-medium"> Quay lại nh - p </span>
35 |       </Link>
36 |
37 |       <div className="mb-6 inline-flex p-4 bg-[#00d2a0]/10 text-[#00d2a0] rounded-2xl">
38 |         <QRCode size={32} />
39 |       </div>
40 |
41 |       <h2 className="text-2xl font-bold mb-2"> nh - p b ñ ng mã QR </h2>
42 |       <p className="text-slate-400 text-sm mb-8"> Quét mã b ñ ng é ng đ ã n g
43 |
44 |       <div className="relative inline-block p-6 bg-white rounded-3xl mb-8">
45 |         <QRCodeSVG value={qrValue} size={200} />
46 |         <div className="absolute inset-0 flex items-center justify-center pointer-events-none">
47 |           <div className="w-12 h-12 bg-[#f0f1a] rounded-xl flex items-center justify-center border-4 border...
48 |             
49 |           </div>
50 |         </div>
51 |       </div>
52 |
53 |       <div className="flex items-center justify-center gap-3 text-sm text-slate-500 mb-8">
54 |         <RefreshCcw size={16} className={`transition-transform duration-1000 ${timeLeft < 5 ? 'animate-spin'...
55 |         <span> Mã s ½ h ½ t h i n s a u <span className="text-white font-bold">
56 |       </div>
57 |
58 |       <div className="grid grid-cols-2 gap-4 text-left">
59 |         <div className="p-4 bg-[#f0f1a] rounded-2xl border border-[#1e293b]">
60 |           <Smartphone size={20} className="text-[#00d2a0] mb-2" />
61 |           <p className="text-[11px] font-bold uppercase text-slate-500"> M B P C
62 |           <p className="text-xs leading-relaxed text-slate-300"> M B P C
63 |         </div>
64 |         <div className="p-4 bg-[#f0f1a] rounded-2xl border border-[#1e293b]">
65 |           <Monitor size={20} className="text-[#00d2a0] mb-2" />
66 |           <p className="text-[11px] font-bold uppercase text-slate-500"> M B P C
67 |           <p className="text-xs leading-relaxed text-slate-300"> Quét mã QR trên màn hình này.</p>
68 |         </div>
69 |       </div>
70 |
71 |       <div className="mt-8 pt-6 border-t border-[#1e293b]">
72 |         <p className="text-xs text-slate-500">
73 |           G . p k h ó k h n ? <Link href="/login" className="text-[#00d2a0]

```

```

74 |         </p>
75 |     </div>
76 | </motion.div>
77 | </div>
78 | )
79 | }
80 |

```

File: app/(auth)/register/page.tsx (444 lines, 29.9 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect, Suspense } from 'react'
4 | import Link from 'next/link'
5 | import { useRouter, useSearchParams } from 'next/navigation'
6 | import { ArrowLeft, Loader2, Mail, Lock, User, Check, ChevronRight, GraduationCap, Building, Image as ImageIcon...
7 | import { register, completeOAuthRegistration } from '@lib/auth-actions'
8 |
9 | function RegisterForm() {
10 |     const router = useRouter()
11 |     const searchParams = useSearchParams()
12 |     const isOAuth = searchParams.get('oauth') === 'true'
13 |     const initialStep = isOAuth ? 1 : (searchParams.get('step') ? parseInt(searchParams.get('step')) : 1)
14 |
15 |     const [isLogin, setIsLogin] = useState(false)
16 |     useEffect(() => { if (isLogin) router.push('/login') }, [isLogin, router])
17 |
18 |     const [step, setStep] = useState(initialStep)
19 |     const [loading, setLoading] = useState(false)
20 |     const [error, setError] = useState<string | null>(null)
21 |     const [showPassword, setShowPassword] = useState(false)
22 |     const [showConfirm, setShowConfirm] = useState(false)
23 |
24 |     const [formData, setFormData] = useState({
25 |         fullName: '', email: '', password: '', confirmPassword: '',
26 |         role: 'student', classCode: '', schoolCode: '', schoolName: '',
27 |         teacherCode: '', dob: '', province: '', termsAccepted: false
28 |     })
29 |     const [passwordStrength, setPasswordStrength] = useState(0)
30 |     const [avatarFile, setAvatarFile] = useState<File | null>(null)
31 |     const [avatarPreview, setAvatarPreview] = useState<string | null>(null)
32 |
33 |     // Pre-fill Google user info on mount
34 |     useEffect(() => {
35 |         if (isOAuth && typeof window !== 'undefined') {
36 |             const fetchUserInfo = async () => {
37 |                 try {
38 |                     const { createBrowserClient } = await import('@supabase/ssr')
39 |                     const supabase = createBrowserClient(
40 |                         process.env.NEXT_PUBLIC_SUPABASE_URL!,
41 |                         process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
42 |                     )
43 |                     const { data: { user } } = await supabase.auth.getUser()
44 |                     if (user?.user_metadata?.full_name) {
45 |                         setFormData(prev => ({ ...prev, fullName: user.user_metadata.full_name }))
46 |                     } else if (user?.email) {
47 |                         setFormData(prev => ({ ...prev, fullName: user.email.split('@')[0] }))
48 |                     }
49 |                 } catch (e) {
50 |                     console.warn('Could not fetch Google user info:', e)
51 |                 }
52 |             }
53 |             fetchUserInfo()
54 |         }
55 |     }, [isOAuth])
56 |
57 |     const handleInputChange = (e: React.ChangeEvent<HTMLInputElement | HTMLSelectElement>) => {
58 |         const { name, value, type } = e.target as HTMLInputElement
59 |         const checked = (e.target as HTMLInputElement).checked
60 |         setFormData(prev => ({ ...prev, [name]: type === 'checkbox' ? checked : value }))
61 |         if (name === 'password') {
62 |             let s = 0
63 |             if (value.length >= 8) s++
64 |             if (value.match(/[A-Z]/) && value.match(/[a-z]/)) s++
65 |             if (value.match(/[0-9!@#$%^&*]/)) s++
66 |             setPasswordStrength(s)
67 |         }
68 |     }
69 |
70 |     const handleNext = () => {
71 |         setError(null)
72 |         if (isOAuth && step === 1) {
73 |             if (!formData.fullName.trim()) { setError('Vui lòng nh - p hí và tên') }
74 |             setStep(2); return
75 |         }
76 |         if (step === 1) {
77 |             if (!formData.fullName || !formData.email || !formData.password || !formData.confirmPassword) { setError('M - t kh @ u ph ẽ i c ó í t') }
78 |             if (formData.password !== formData.confirmPassword) { setError('M - t kh @ u ph ẽ i c ó í t') }
79 |             if (formData.password.length < 8) { setError('M - t kh @ u ph ẽ i c ó í t') }
80 |             if (!/[A-Z]/.test(formData.password)) { setError('M - t kh @ u ph ẽ i c ó í t') }
81 |             if (!/[0-9]/.test(formData.password)) { setError('M - t kh @ u ph ẽ i c ó í t') }
82 |         } else if (step === 2) {
83 |             if (formData.role === 'parent' && (!formData.schoolName || !formData.classCode)) { setError('Vui lòng nh...') }
84 |         }
85 |         setStep(prev => Math.min(prev + 1, isOAuth ? 2 : 3))
86 |     }
87 |
88 |     const handlePrev = () => { setStep(prev => Math.max(prev - 1, 1)); setError(null) }
89 |
90 |     const handleSubmit = async (e: React.FormEvent) => {
91 |         e.preventDefault(); setError(null)
92 |         if (!formData.termsAccepted) { setError('B ĩ n c s n ỏ ng ý v ỏ i i ả u kh o') }
93 |         setLoading(true)
94 |         const submitData = new FormData()
95 |         submitData.append('full_name', formData.fullName)
96 |         submitData.append('role', formData.role)
97 |         if (isOAuth) { submitData.append('email', formData.email); submitData.append('password', formData.password) }
98 |         if (formData.teacherCode) submitData.append('teacher_code', formData.teacherCode)
99 |         if (formData.classCode) submitData.append('class_code', formData.classCode)
100 |         if (formData.schoolCode) submitData.append('school_code', formData.schoolCode)
101 |         if (formData.schoolName) submitData.append('school_name', formData.schoolName)
102 |         if (formData.dob) submitData.append('dob', formData.dob)
103 |         if (formData.province) submitData.append('province', formData.province)
104 |         if (avatarFile) submitData.append('avatar', avatarFile)
105 |
106 |         const res: any = isOAuth ? await completeOAuthRegistration(submitData) : await register(submitData)
107 |         if (res?.error) {
108 |             let msg = res.error
109 |             if (res.error.toLowerCase().includes('already registered') || res.error.toLowerCase().includes('already ...')) {
110 |                 setError(msg); setLoading(false)
111 |             } else if (res?.success) {
112 |                 if (typeof window !== 'undefined') {
113 |                     import('canvas-confetti').then(m => m.default({ particleCount: 150, spread: 80, origin: { y: 0.6 }, co...

```

```
114 |     }
115 |     setTimeout(() => router.push(res.redirectUrl || '/'), 1500)
116 |   }
117 | }
118 |
119 | const strengthColors = ['var(--danger)', 'var(--warning)', 'var(--brand-primary)']
120 |
121 | const inputStyle: React.CSSProperties = {
122 |   width: '100%', padding: '12px 16px 12px 42px', borderRadius: '10px',
123 |   border: '1px solid var(--border-default)', background: 'var(--bg-surface)',
124 |   color: 'var(--text-primary)', fontSize: '14px', outline: 'none', fontFamily: 'inherit',
125 |   boxSizing: 'border-box'
126 | }
127 |
128 | return (
129 |   <div className="register-page" style={{ minHeight: '100vh', display: 'flex', flexDirection: 'column', align...
130 |     <div className="register-back-btn" style={{ position: 'absolute', top: '32px', left: '32px' }}>
131 |       <Link href="/" style={{ display: 'inline-flex', alignItems: 'center', gap: '8px', color: 'var(--text-m...
132 |         <ArrowLeft size={18} /> Quay lại
133 |       </Link>
134 |     </div>
135 |
136 |     <div style={{ maxWidth: '500px', width: '100%', background: 'var(--bg-surface)', borderRadius: '16px', b...
137 |       <div style={{ textAlign: 'center', marginBottom: '28px' }}>
138 |         <h1 style={{ fontSize: '24px', fontWeight: 800, color: 'var(--text-primary)', marginBottom: '6px' }}>
139 |           {isOauth ? 'Hàn tét hÓ si : Tio tãi khon' }
140 |         </h1>
141 |         <p style={{ color: 'var(--text-muted)', fontSize: '14px', margin: 0 }}>
142 |           {step === 1 && (isOauth ? 'Thông tin cça bin' : 'Thông tin c
143 |             {step === 2 && (isOauth ? 'Hàn thiÇn hÓ si' : 'Bin là híc s
144 |             {step === 3 && 'Hàn thiÇn hÓ si Äb-t su')}
145 |         </p>
146 |       </div>
147 |
148 |       {/* Steps indicator */}
149 |       <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '8px', marginBotto...
150 |         <Array.from({ length: isOauth ? 2 : 3 }, (_, i) => i + 1).map(i => (
151 |           <React.Fragment key={i}>
152 |             <div style={{
153 |               width: '32px', height: '32px', borderRadius: '50%', display: 'flex', alignItems: 'center', jus...
154 |               fontSize: '13px', fontWeight: 700,
155 |               background: step >= i ? 'var(--brand-primary)' : 'var(--bg-elevated)',
156 |               color: step >= i ? '#fff' : 'var(--text-muted)',
157 |               border: `2px solid ${step >= i ? 'var(--brand-primary)' : 'var(--border-default)'}`
158 |             }>{step > i ? <Check size={16} /> : i}</div>
159 |             {i < (isOauth ? 2 : 3) && <div style={{ flex: 1, maxWidth: '60px', height: '3px', borderRadius: ...
160 |           </React.Fragment>
161 |         ))}
162 |       </div>
163 |
164 |       {step === 1 && !isOauth && (
165 |         <div style={{ display: 'flex', background: 'var(--bg-elevated)', borderRadius: '10px', padding: '3px...
166 |           <button type="button" onClick={() => setIsLogin(true)}
167 |             style={{ flex: 1, padding: '8px', borderRadius: '8px', border: 'none', background: 'transparent'...
168 |             ng Nh - p
169 |           </button>
170 |           <button type="button"
171 |             style={{ flex: 1, padding: '8px', borderRadius: '8px', border: 'none', background: 'var(--bg-sur...
172 |             ng Ky
173 |           </button>
174 |         </div>
175 |       )}
176 |
177 |       <form onSubmit={handleSubmit}>
178 |         {/* Step 1: Basic Info */}
179 |         {step === 1 && isOauth && (
180 |           <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
181 |             <div style={{ position: 'relative' }}>
182 |               <User size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'translate...
183 |               <input type="text" name="fullName" placeholder="Hí và t
184 |             </div>
185 |             <div style={{ padding: '12px 16px', borderRadius: '10px', background: 'rgba(8,158,96,0.06)', bor...
186 |               <Mail size={16} style={{ color: 'var(--brand-primary)' }} />
187 |               <span>Bin ä ng nh - p b±ng Google. Email s½ °äc tñ Ü
188 |             </div>
189 |           {/* Role selection inline */}
190 |           <div>
191 |             <label style={{ fontSize: '13px', fontWeight: 600, color: 'var(--text-primary)', marginBottom:...
192 |             <div className="register-role-grid" style={{ display: 'grid', gridTemplateColumns: '1fr 1fr 1fr...
193 |               {
194 |                 { role: 'student', icon: <GraduationCap size={20} />
195 |                 { role: 'teacher', icon: <User size={20} />, title:
196 |                 { role: 'parent', icon: <Users size={20} />, title:
197 |               }
198 |             ).map(({ role, icon, title, desc }) => (
199 |               <button key={role} type="button" onClick={() => setFormData(p => ({ ...p, role }))}
200 |                 style={{
201 |                   padding: '14px 10px', borderRadius: '12px', cursor: 'pointer', textAlign: 'center',
202 |                   background: formData.role === role ? 'rgba(8,158,96,0.08)' : 'var(--bg-elevated)',
203 |                   border: `2px solid ${formData.role === role ? 'var(--brand-primary)' : 'var(--border-d...
204 |                   transition: 'all 0.2s', fontFamily: 'inherit'
205 |                 }>
206 |                 <div style={{ width: '36px', height: '36px', borderRadius: '50%', margin: '0 auto 6px', ...
207 |                   {icon}
208 |                 </div>
209 |                 <div style={{ fontSize: '13px', fontWeight: 700, color: 'var(--text-primary)' }}>{title}...
210 |                 <div style={{ fontSize: '11px', color: 'var(--text-muted)' }}>{desc}</div>
211 |               </button>
212 |             ))}
213 |           </div>
214 |         </div>
215 |       )}
216 |       {step === 1 && !isOauth && (
217 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
218 |           <div style={{ position: 'relative' }}>
219 |             <User size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'translate...
220 |             <input type="text" name="fullName" placeholder="Hí và t
221 |           </div>
222 |           <div style={{ position: 'relative' }}>
223 |             <Mail size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'translate...
224 |             <input type="email" name="email" placeholder="Eä chÉ Em
225 |           </div>
226 |           <div style={{ position: 'relative' }}>
227 |             <Lock size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'translate...
228 |             <input type="showPassword" ? 'text' : 'password' } name="p
229 |             <button type="button" onClick={() => setShowPassword(!showPassword)} style={{ position: 'absol...
230 |               {showPassword ? <EyeOff size={18} /> : <Eye size={18} />}
231 |             </button>
232 |           </div>
233 |           <div style={{ display: 'flex', gap: '4px', height: '4px' }}>
234 |             {[1, 2, 3].map(i => (
235 |               <div key={i} style={{ flex: 1, borderRadius: '2px', background: passwordStrength >= i ? stre...
236 |             ))}
237 |           </div>

```

```

238|         <div style={{ position: 'relative' }}>
239|             <Lock size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'translate...
240|                 <input type={showConfirm ? 'text' : 'password'} name="confirm" />
241|             <button type="button" onClick={() => setShowConfirm(!showConfirm)} style={{ position: 'absolut...
242|                 {showConfirm ? <EyeOff size={18} /> : <Eye size={18} />}
243|             </button>
244|         </div>
245|     </div>
246| )}
247|
248| /* Step 2: Role Selection (for non-OAuth) OR Complete Profile (for OAuth) */
249| {step === 2 && !isOAuth && (
250|     <div style={{ display: 'flex', flexDirection: 'column', gap: '20px' }}>
251|         <div className="register-role-grid" style={{ display: 'grid', gridTemplateColumns: '1fr 1fr 1fr'...
252|             {{
253|                 { role: 'student', icon: <GraduationCap size={20} />, title: 'Student', desc: 'Graduate' },
254|                 { role: 'teacher', icon: <User size={20} />, title: 'Teacher', desc: 'Classroom Manager' },
255|                 { role: 'parent', icon: <Users size={20} />, title: 'Parent', desc: 'Child's Guardian' }
256|             }}.map(({ role, icon, title, desc }) => (
257|                 <button key={role} type="button" onClick={() => setFormData(p => ({ ...p, role }))}
258|                     style={{
259|                         padding: '16px 12px', borderRadius: '12px', cursor: 'pointer', textAlign: 'center',
260|                         background: formData.role === role ? 'rgb(8,158,96,0.08)' : 'var(--bg-elevated)',
261|                         border: '2px solid ${formData.role === role ? 'var(--brand-primary)' : 'var(--border-def...
262|                         transition: 'all 0.2s', fontFamily: 'inherit'
263|                     }}>
264|                         <div style={{ width: '40px', height: '40px', borderRadius: '50%', margin: '0 auto 8px', di...
265|                             {icon}
266|                         </div>
267|                         <div style={{ fontSize: '13px', fontWeight: 700, color: 'var(--text-primary)' }}>{title}</div>
268|                         <div style={{ fontSize: '11px', color: 'var(--text-muted)' }}>{desc}</div>
269|                     </button>
270|                 ))
271|             </div>
272|
273|             {formData.role === 'student' && (
274|                 <div style={{ display: 'flex', flexDirection: 'column', gap: '12px' }}>
275|                     <div style={{ position: 'relative' }}>
276|                         <Building size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 't...
277|                             <input type="text" name="schoolName" placeholder="Tên trường" />
278|                     </div>
279|                     <div style={{ position: 'relative' }}>
280|                         <Users size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'tran...
281|                             <input type="text" name="className" placeholder="Mã lớp" />
282|                     </div>
283|                 </div>
284|             )}
285|             {formData.role === 'teacher' && (
286|                 <div style={{ display: 'flex', flexDirection: 'column', gap: '12px' }}>
287|                     <div style={{ position: 'relative' }}>
288|                         <Building size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 't...
289|                             <input type="text" name="schoolCode" placeholder="Mã trường" />
290|                     </div>
291|                     <div style={{ position: 'relative' }}>
292|                         <ShieldCheck size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform:...
293|                             <input type="text" name="teacherCode" placeholder="Mã giáo viên" />
294|                     </div>
295|                 </div>
296|             )}
297|             {formData.role === 'parent' && (
298|                 <div style={{ display: 'flex', flexDirection: 'column', gap: '12px' }}>
299|                     <div style={{ position: 'relative' }}>
300|                         <Building size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 't...
301|                             <input type="text" name="schoolName" placeholder="Tên trường" />
302|                     </div>
303|                     <div style={{ position: 'relative' }}>
304|                         <User size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'trans...
305|                             <input type="text" name="className" placeholder="Mã lớp" />
306|                     </div>
307|                 </div>
308|             )}
309|         </div>
310|     )}
311|
312| /* Step 2 for OAuth: School info + Complete Profile */
313| {step === 2 && isOAuth && (
314|     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
315|         {formData.role === 'student' && (
316|             <div style={{ position: 'relative' }}>
317|                 <Building size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'tra...
318|                     <input type="text" name="schoolName" placeholder="Tên trường" />
319|             </div>
320|         )}
321|         {formData.role === 'teacher' && (
322|             <div style={{ position: 'relative' }}>
323|                 <Building size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'tra...
324|                     <input type="text" name="schoolCode" placeholder="Mã trường" />
325|             </div>
326|         )}
327|         {formData.role === 'parent' && (
328|             <div style={{ position: 'relative' }}>
329|                 <User size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'transla...
330|                     <input type="text" name="className" placeholder="Mã lớp" />
331|             </div>
332|         )}
333|         <label style={{ display: 'flex', alignItems: 'flex-start', gap: '10px', cursor: 'pointer', margi...
334|             <input type="checkbox" name="termsAccepted" checked={formData.termsAccepted} onChange={handleI...
335|             style={{ width: '18px', height: '18px', marginTop: '2px', accentColor: 'var(--brand-primary)'...
336|             <span style={{ fontSize: '13px', color: 'var(--text-muted)', lineHeight: 1.4 }}>
337|                 Tôi đồng ý với <Link href="#" style={{ color: 'var(--brand-primary)' }}>Điều khoản sử dụng</Link>
338|             </span>
339|         </label>
340|     </div>
341| )}
342|
343| /* Step 3: Complete Profile (non-OAuth) */
344| {step === 3 && !isOAuth && (
345|     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
346|         <div style={{ display: 'flex', justifyContent: 'center', marginBottom: '8px' }}>
347|             <div style={{ width: '80px', height: '80px', borderRadius: '50%', border: '2px dashed var(--bo...
348|                 {avatarPreview ? (
349|                     <img src={avatarPreview} alt="Avatar" style={{ width: '100%', height: '100%', objectFit: '...
350|                 ) : formData.fullName ? (
351|                     <span style={{ fontSize: '28px', fontWeight: 800, color: 'var(--brand-primary)' }}>{formDa...
352|                 ) : (
353|                     <ImageIcon size={24} />
354|                 )
355|             </div>
356|             <span style={{ fontSize: '10px', marginTop: '4px' }}>
357|                 <input type="file" style={{ position: 'absolute', inset: 0, opacity: 0, cursor: 'pointer' }}...
358|             </span>
359|         </div>
360|     </div>
361| )}

```



```

362 |         <div style={{ position: 'relative' }}>
363 |             <Calendar size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'transla...
364 |             <input type="date" name="dob" value={formData.dob} onChange={handleInputChange} style={{ ...inp...
365 |         </div>
366 |
367 |         <div style={{ position: 'relative' }}>
368 |             <MapPin size={18} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'transla...
369 |             <select name="province" value={formData.province} onChange={handleInputChange} style={{ ...inp...
370 |                 <option value="" disabled>Chín Ténh / Thành phñ</option>
371 |                 <option value="HN">Hà Núi</option>
372 |                 <option value="HCM">TP. Hồ Chí Minh</option>
373 |                 <option value="DN">à Nụng</option>
374 |                 <option value="Other">Ténh thành khác</option>
375 |             </select>
376 |         </div>
377 |
378 |         <label style={{ display: 'flex', alignItems: 'flex-start', gap: '10px', cursor: 'pointer', margi...
379 |         <input type="checkbox" name="termsAccepted" checked={formData.termsAccepted} onChange={handleI...
380 |         <span style={{ width: '18px', height: '18px', margin: '2px', backgroundColor: 'var(--brand-primary)...
381 |         <span style={{ fontSize: '13px', color: 'var(--text-muted)', lineHeight: 1.4 }}>
382 |             Tó i óng ý v ò i <Link href="#" style={{ color: 'var(--brand-primary)' }}>
383 |         </span>
384 |     </label>
385 | </div>
386 | ))
387 |
388 | {error && (
389 |     <div style={{ marginTop: '16px', padding: '10px 14px', borderRadius: '10px', background: 'rgba(244...
390 |     {error}
391 | </div>
392 | )}
393 |
394 | {/Navigation Buttons */}
395 | <div style={{ display: 'flex', gap: '12px', marginTop: '32px' }}>
396 |     {step > 1 && (
397 |         <button type="button" onClick={handlePrev} disabled={loading}
398 |         style={{ flex: 1, padding: '12px', borderRadius: '10px', fontWeight: 700, border: '1px solid v...
399 |         Q u a y l i i
400 |     </button>
401 |     )}
402 |     {step < (isOAuth ? 2 : 3) ? (
403 |         <button type="button" onClick={handleNext}
404 |         style={{ flex: 2, padding: '12px', borderRadius: '10px', fontWeight: 700, border: 'none', back...
405 |         T i c p t á c <ChevronRight size={18} />
406 |     </button>
407 |     ) : (
408 |         <button type="button" onClick={handleSubmit} disabled={loading || !formData.termsAccepted}
409 |         style={{ flex: 2, padding: '12px', borderRadius: '10px', fontWeight: 700, border: 'none', back...
410 |         {loading ? <Loader2 size={18} style={{ animation: 'spin'
411 |     </button>
412 |     )}
413 | </div>
414 | </form>
415 |
416 | <p style={{ textAlign: 'center', marginTop: '20px', color: 'var(--text-muted)', fontSize: '13p...
417 |     à c ó t à i k h o ã n ? <Link href="/login" style={{ color: 'var(--brand-primary)' }}>
418 | </p>
419 | </div>
420 |
421 | <style>{
422 |     @media (max-width: 640px) {
423 |         .register-page { padding: 12px !important; }
424 |         .register-page > div { padding: 20px 16px !important; }
425 |         .register-role-grid { grid-template-columns: 1fr !important; }
426 |         .register-back-btn { top: 12px !important; left: 12px !important; }
427 |     }
428 | </style>
429 | </div>
430 | )
431 | }
432 |
433 | export default function RegisterPage() {
434 |     return (
435 |         <Suspense fallback={
436 |             <div style={{ minHeight: '100vh', display: 'flex', alignItems: 'center', justifyContent: 'center', backg...
437 |             <Loader2 size={48} style={{ color: 'var(--brand-primary)', animation: 'spin 1s linear infinite' }} />
438 |         </div>
439 |     )>
440 |         <RegisterForm />
441 |     </Suspense>
442 | )
443 | }
444 |

```

File: app/(dashboard)/admin/activity/page.tsx (177 lines, 8.6 KB)

```

1 | 'use client';
2 |
3 | import React, { useEffect, useState, useRef } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import { Activity, Search, Clock, Users, BookOpen, FileText, RefreshCw, Radio, Loader2 } from 'lucide-react';
6 |
7 | interface LogEntry {
8 |   id: string;
9 |   action: string;
10 |   user_name: string;
11 |   role: string;
12 |   created_at: string;
13 |   detail?: string;
14 | }
15 |
16 | export default function AdminActivityPage() {
17 |   const [logs, setLogs] = useState<LogEntry[]>([]);
18 |   const [loading, setLoading] = useState(true);
19 |   const [connected, setConnected] = useState(false);
20 |   const [search, setSearch] = useState('');
21 |
22 |   useEffect(() => {
23 |     fetchLogs();
24 |     const channel = supabase.channel('admin-activity-realtime')
25 |       .on('postgres_changes', { event: 'INSERT', schema: 'public', table: 'profiles' }, (payload) => {
26 |         const u = payload.new as any;
27 |         setLogs(prev => [...prev, {
28 |           id: `profile-${u.id}`,
29 |           action: 'ng ký m ò i',
30 |           user_name: u.full_name || 'Unknown',
31 |           role: u.role || 'student',
32 |           created_at: new Date().toISOString(),
33 |           detail: `Email: ${u.email}`,
34 |           ...prev.slice(0, 100)
35 |         ]});
36 |       }).on('postgres_changes', { event: 'INSERT', schema: 'public', table: 'lessons' }, (payload) => {
37 |         const l = payload.new as any;

```

```

38 |     setLogs(prev => [{
39 |         id: `lesson-${l.id}`,
40 |         action: 'Tạo bài giảng',
41 |         user_name: l.teacher_id ? slice(0, 8) || 'System',
42 |         role: 'teacher',
43 |         created_at: new Date().toISOString(),
44 |         detail: `l.title || 'Bài giảng mới',
45 |     }, ...prev].slice(0, 100));
46 |     })
47 |     .on('postgres_changes', { event: 'INSERT', schema: 'public', table: 'exams' }, (payload) => {
48 |         const e = payload.new as any;
49 |         setLogs(prev => [{
50 |             id: `exam-${e.id}`,
51 |             action: 'Tạo bài kiểm tra',
52 |             user_name: 'System',
53 |             role: 'teacher',
54 |             created_at: new Date().toISOString(),
55 |             detail: `e.title || 'Kiểm tra mới',
56 |         }, ...prev].slice(0, 100));
57 |     })
58 |     .subscribe((status) => setConnected(status === 'SUBSCRIBED'));
59 |     return () => { supabase.removeChannel(channel) };
60 | }, []);
61 |
62 | async function fetchLogs() {
63 |     const { data: profiles } = await supabase.from('profiles').select('id, full_name, role, email, created_at'...
64 |     if (profiles) {
65 |         const mapped: LogEntry[] = profiles.map(p => ({
66 |             id: `profile-${p.id}`,
67 |             action: p.role === 'admin' ? 'Quản lý người dùng' : 'Đăng ký mới',
68 |             user_name: p.full_name || 'Unknown',
69 |             role: p.role || 'student',
70 |             created_at: p.created_at,
71 |             detail: `Email: ${p.email}`,
72 |         }));
73 |         setLogs(mapped);
74 |     }
75 |     setLoading(false);
76 | }
77 |
78 | function timeAgo(ts: string) {
79 |     const diff = Date.now() - new Date(ts).getTime();
80 |     const mins = Math.floor(diff / 60000);
81 |     if (mins < 1) return 'Vài giây trước';
82 |     if (mins < 60) return `${mins} phút trước`;
83 |     const hrs = Math.floor(mins / 60);
84 |     if (hrs < 24) return `${hrs} giờ trước`;
85 |     return `${Math.floor(hrs / 24)} ngày trước`;
86 | }
87 |
88 | const filtered = logs.filter(l =>
89 |     l.action.toLowerCase().includes(search.toLowerCase()) ||
90 |     l.user_name.toLowerCase().includes(search.toLowerCase())
91 | );
92 |
93 | if (loading) return (
94 |     <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', height: '100%' }}>
95 |         <Loader2 size={48} style={{ color: 'var(--brand-primary)', animation: 'spin 1s linear infinite' }} />
96 |     </div>
97 | );
98 |
99 | return (
100 |     <div style={{ maxWidth: '1200px', margin: '0 auto' }}>
101 |         <header style={{ marginBottom: '28px' }}>
102 |             <div style={{ display: 'flex', alignItems: 'center', gap: '10px', marginBottom: '4px' }}>
103 |                 <h1 style={{ fontSize: '28px', fontWeight: 800, margin: 0, color: 'var(--text-primary)' }}>
104 |                     <span style={{
105 |                         display: 'inline-flex', alignItems: 'center', gap: '4px', padding: '3px 10px',
106 |                         borderRadius: '99px', fontSize: '10px', fontWeight: 700,
107 |                         background: connected ? 'rgb(8,158,96,0.1)' : 'rgb(244,106,106,0.1)',
108 |                         color: connected ? 'var(--brand-primary)' : 'var(--danger)',
109 |                     }}>
110 |                         <Radio size={10} /> {connected ? 'Live' : 'Off'}
111 |                     </span>
112 |                 </div>
113 |                 <p style={{ color: 'var(--text-muted)', margin: 0, fontSize: '14px' }}>
114 |                     <div className="lms-card" style={{ padding: '0', overflow: 'hidden' }}>
115 |                         <div style={{ padding: '20px 24px', borderBottom: '1px solid var(--border-default)', display: 'flex', ...
116 |                         <div style={{ position: 'relative', flex: 1 }}>
117 |                             <Search size={16} style={{ position: 'absolute', left: '12px', top: '50%', transform: 'translateY(...
118 |                             <input value={search} onChange={e => setSearch(e.target.value)} style={{
119 |                                 padding: '8px 12px 8px 36px', borderRadius: '8px', background: 'var(--bg-elevated)',
120 |                                 border: '1px solid var(--border-default)', color: 'var(--text-primary)', fontSize: '13px', outli...
121 |                             }} />
122 |                         </div>
123 |                         <button onClick={fetchLogs} style={{
124 |                             padding: '8px 14px', borderRadius: '8px', background: 'var(--bg-elevated)',
125 |                             border: '1px solid var(--border-default)', color: 'var(--text-primary)',
126 |                             display: 'flex', alignItems: 'center', gap: '6px', cursor: 'pointer', fontSize: '13px', fontFamily...
127 |                         }}>
128 |                             <RefreshCw size={16} /> Làm mới
129 |                         </button>
130 |                     </div>
131 |                     <div style={{ overflowX: 'auto' }}>
132 |                         <table style={{ width: '100%', borderCollapse: 'collapse' }}>
133 |                             <thead>
134 |                                 <tr style={{ borderBottom: '1px solid var(--border-default)' }}>
135 |                                     <th key={h} style={{ padding: '12px 20px', color: 'var(--text-muted)', fontSize: '11px', fon...
136 |                                     </th>
137 |                                 </thead>
138 |                                 <tbody>
139 |                                     {filtered.map((log, i) => (
140 |                                         <tr key={log.id} style={{ borderBottom: '1px solid var(--border-default)', animation: i < 3 ? ...
141 |                                         <td style={{ padding: '14px 20px' }}>
142 |                                             <div style={{ display: 'flex', alignItems: 'center', gap: '10px' }}>
143 |                                                 <div style={{
144 |                                                     width: '32px', height: '32px', borderRadius: '8px',
145 |                                                     background: log.role === 'admin' ? 'rgb(244,106,106,0.1)' : log.role === 'teacher' ? ...
146 |                                                     display: 'flex', alignItems: 'center', justifyContent: 'center'
147 |                                                 }}>
148 |                                                     {log.role === 'admin' ? <Activity size={14} color="var(--danger)" /> : log.role === 't...
149 |                                                 </div>
150 |                                                 <span style={{ color: 'var(--text-primary)', fontWeight: 600, fontSize: '14px' }}>{log.a...
151 |                                             </div>
152 |                                         </td>
153 |                                         <td style={{ padding: '14px 20px', color: 'var(--text-primary)', fontSize: '13px' }}>{log.us...
154 |                                         <td style={{ padding: '14px 20px', color: 'var(--text-muted)', fontSize: '12px' }}>{log.deta...
155 |                                         <td style={{ padding: '14px 20px' }}>
156 |                                             <span className="lms-tag" style={{ fontSize: '10px' }}>{log.role}</span>
157 |                                         </td>
158 |                                     </tr>
159 |                                 </tbody>
160 |                             </table>
161 |

```

```

162 |         </td>
163 |         <td style={{ padding: '14px 20px', color: 'var(--text-muted)', fontSize: '12px', display: 'f...
164 |             <Clock size={12} /> {timeAgo(log.created_at)}
165 |         </td>
166 |     </tr>
167 |     ))}
168 | </tbody>
169 | </table>
170 | </div>
171 | </div>
172 |
173 | <style={`@keyframes fadeIn { from { opacity: 0; background: rgba(var(--brand-primary-rgb),0.04); } to { ...
174 | </div>
175 | });
176 | }
177 |

```

File: app/(dashboard)/admin/analytics/page.tsx (127 lines, 7.5 KB)

```

1 | 'use client';
2 |
3 | import React, { useEffect, useState } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import { BarChart3, TrendingUp, Users, BookOpen, FileText, Activity, Monitor, Clock, Radio, Loader2 } from 'lu...
6 | import { LineChart, AreaChart, Area, XAxis, YAxis, CartesianGrid, Tooltip, ResponsiveContainer, BarChart, Bar,...
7 |
8 | export default function AdminAnalyticsPage() {
9 |   const [stats, setStats] = useState({ users: 0, lessons: 0, exams: 0, classes: 0 });
10 |   const [loading, setLoading] = useState(true);
11 |
12 |   useEffect(() => {
13 |     fetchStats();
14 |     const channel = supabase.channel('admin-analytics-realtime')
15 |       .on('postgres_changes', { event: '*', schema: 'public', table: 'profiles' }, () => fetchStats())
16 |       .on('postgres_changes', { event: '*', schema: 'public', table: 'lessons' }, () => fetchStats())
17 |       .on('postgres_changes', { event: '*', schema: 'public', table: 'exams' }, () => fetchStats())
18 |       .on('postgres_changes', { event: '*', schema: 'public', table: 'classes' }, () => fetchStats())
19 |       .subscribe();
20 |     return () => { supabase.removeChannel(channel) };
21 |   }, []);
22 |
23 |   async function fetchStats() {
24 |     const { count: users } = await supabase.from('profiles').select('*', { count: 'exact', head: true });
25 |     const { count: lessons } = await supabase.from('lessons').select('*', { count: 'exact', head: true });
26 |     const { count: exams } = await supabase.from('exams').select('*', { count: 'exact', head: true });
27 |     const { count: classes } = await supabase.from('classes').select('*', { count: 'exact', head: true });
28 |     setStats({ users: users || 0, lessons: lessons || 0, exams: exams || 0, classes: classes || 0 });
29 |     setLoading(false);
30 |   }
31 |
32 |   const monthlyData = [
33 |     { month: 'T1', users: 0, lessons: 0, exams: 0 },
34 |     { month: 'T2', users: 0, lessons: 0, exams: 0 },
35 |     { month: 'T3', users: 0, lessons: 0, exams: 0 },
36 |     { month: 'T4', users: 0, lessons: 0, exams: 0 },
37 |     { month: 'T5', users: 0, lessons: 0, exams: 0 },
38 |     { month: 'T6', users: stats.users, lessons: stats.lessons, exams: stats.exams },
39 |   ];
40 |
41 |   const categoryData = [
42 |     { name: 'B à i g i ñ g', value: stats.lessons, color: 'var(--brand-primary)' },
43 |     { name: 'B à i k i ã m t r a', value: stats.exams, color: 'var(--accent-amber)' },
44 |     { name: 'L Ò p h í c', value: stats.classes, color: 'var(--accent-blue)' },
45 |     { name: 'N g º Ý i ð ù n g', value: stats.users, color: '#6366f1' },
46 |   ];
47 |
48 |   if (loading) return (
49 |     <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', height: '100%' }}>
50 |       <Loader2 size={48} style={{ color: 'var(--brand-primary)', animation: 'spin is linear infinite' }} />
51 |     </div>
52 |   );
53 |
54 |   return (
55 |     <div style={{ maxWidth: '1400px', margin: '0 auto' }}>
56 |       <header style={{ marginBottom: '28px' }}>
57 |         <h1 style={{ fontSize: '28px', fontWeight: 800, margin: '0 0 4px 0', color: 'var(--text-primary)' }}>P...
58 |         <p style={{ color: 'var(--text-muted)', margin: 0, fontSize: '14px' }}>
59 |           T h Ñ n g k ê t r ñ c t i ò p t ê d a t a b a s e . <Radio size={12} style={{ ver
60 |         </p>
61 |       </header>
62 |
63 |       <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(180px, 1fr))', gap: '16px',...
64 |         <div>
65 |           { label: 'N g º Ý i ð ù n g', value: stats.users, icon: Users, color:
66 |           { label: 'B à i g i ñ g', value: stats.lessons, icon: BookOpen, co
67 |           { label: 'B à i k i ã m t r a', value: stats.exams, icon: FileText, co
68 |           { label: 'L Ò p h í c', value: stats.classes, icon: Monitor, color
69 |         </div>
70 |         <div key={i} className="lms-card" style={{ padding: '20px' }}>
71 |           <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom...
72 |             <div style={{ width: '36px', height: '36px', borderRadius: '8px', background: `${s.color}15`, co...
73 |               <s.icon size={16} />
74 |             </div>
75 |             <div style={{ fontSize: '22px', fontWeight: 900, color: 'var(--text-primary)' }}>{s.value.toLocale...
76 |             <div style={{ fontSize: '11px', color: 'var(--text-muted)', fontWeight: 600 }}>{s.label}</div>
77 |           </div>
78 |         </div>
79 |       </div>
80 |
81 |       <div style={{ display: 'grid', gridTemplateColumns: '2fr 1fr', gap: '20px', marginBottom: '28px' }}>
82 |         <div className="lms-card" style={{ padding: '24px' }}>
83 |           <h3 style={{ fontSize: '16px', fontWeight: 700, color: 'var(--text-primary)', margin: '0 0 16px 0' }}...
84 |           <div style={{ height: '280px' }}>
85 |             <ResponsiveContainer width="100%" height="100%">
86 |               <AreaChart data={monthlyData}>
87 |                 <defs>
88 |                   <linearGradient id="colorUsers" x1="0" y1="0" x2="0" y2="1"><stop offset="5%" stopColor="var...
89 |                 </defs>
90 |                 <CartesianGrid strokeDasharray="3 3" stroke="var(--border-default)" />
91 |                 <XAxis dataKey="month" tick={{ fill: 'var(--text-muted)', fontSize: 11 }} />
92 |                 <YAxis tick={{ fill: 'var(--text-muted)', fontSize: 11 }} />
93 |                 <Tooltip contentStyle={{ background: 'var(--bg-surface)', border: '1px solid var(--border-defa...
94 |                 <Area type="monotone" dataKey="users" stroke="var(--brand-primary)" fill="url(#colorUsers)" st...
95 |                 <Area type="monotone" dataKey="lessons" stroke="var(--accent-blue)" fill="url(#colorUsers)" st...
96 |               </AreaChart>
97 |             </ResponsiveContainer>
98 |           </div>
99 |
100 |         <div className="lms-card" style={{ padding: '24px' }}>
101 |           <h3 style={{ fontSize: '16px', fontWeight: 700, color: 'var(--text-primary)', margin: '0 0 16px 0' }}...
102 |           <div style={{ height: '200px' }}>
103 |             <ResponsiveContainer width="100%" height="100%">
104 |

```

```

105         <PieChart>
106             <Pie data={categoryData} cx="50%" cy="50%" innerRadius={55} outerRadius={80} dataKey="value" s...
107             {categoryData.map((entry, i) => (<Cell key={i} fill={entry.color} />))}
108         </Pie>
109         <Tooltip contentStyle={{ background: 'var(--bg-surface)', border: '1px solid var(--border-defa...
110     </PieChart>
111 </ResponsiveContainer>
112 </div>
113 <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', marginTop: '8px' }}>
114     {categoryData.map((item, i) => (
115         <div key={i} style={{ display: 'flex', alignItems: 'center', gap: '8px', fontSize: '12px' }}>
116             <div style={{ width: '10px', height: '10px', borderRadius: '3px', background: item.color }} />
117             <span style={{ color: 'var(--text-muted)', flex: 1 }}>{item.name}</span>
118             <span style={{ color: 'var(--text-primary)', fontWeight: 700 }}>{item.value}</span>
119         </div>
120     ))}
121 </div>
122 </div>
123 </div>
124 </div>
125 );
126 }
127

```

File: app/(dashboard)/admin/content/page.tsx (137 lines, 7.5 KB)

```

1 'use client';
2
3 import React, { useEffect, useState } from 'react';
4 import { supabase } from '@lib/supabase';
5 import { FileText, BookOpen, Monitor, Plus, Edit, Trash2, Eye, Clock, CheckCircle, XCircle, Loader2, Radio } f...
6 import Link from 'next/link';
7
8 export default function AdminContentPage() {
9     const [lessons, setLessons] = useState<any[]>([]);
10    const [exams, setExams] = useState<any[]>([]);
11    const [loading, setLoading] = useState(true);
12    const [tab, setTab] = useState('all');
13
14    useEffect(() => {
15        fetchData();
16        const channel = supabase.channel('admin-content-realtime')
17        .on('postgres_changes', { event: '*', schema: 'public', table: 'lessons' }, () => fetchData())
18        .on('postgres_changes', { event: '*', schema: 'public', table: 'exams' }, () => fetchData())
19        .subscribe();
20        return () => { supabase.removeChannel(channel) };
21    }, []);
22
23    async function fetchData() {
24        const { data: l } = await supabase.from('lessons').select('*', profiles(full_name')).order('created_at', { ...
25        if (l) setLessons(l);
26        const { data: e } = await supabase.from('exams').select('*', profiles(full_name')).order('created_at', { as...
27        if (e) setExams(e);
28        setLoading(false);
29    }
30
31    const allContent = [
32        ...lessons.map(l => ({ ...l, contentType: 'lesson' })),
33        ...exams.map(e => ({ ...e, contentType: 'exam' })),
34    ].sort((a: any, b: any) => new Date(b.created_at).getTime() - new Date(a.created_at).getTime());
35
36    const filtered = tab === 'all' ? allContent : tab === 'lesson' ? allContent.filter((c: any) => c.contentType...
37
38    if (loading) return (
39        <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', height: '100%' }}>
40            <Loader2 size={48} style={{ color: 'var(--brand-primary)', animation: 'spin 1s linear infinite' }} />
41        </div>
42    );
43
44    return (
45        <div style={{ maxWidth: '1200px', margin: '0 auto' }}>
46            <header style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom: '...'
47                <div>
48                    <h1 style={{ fontSize: '28px', fontWeight: 800, margin: '0 0 4px 0', color: 'var(--text-primary)' }}>...
49                    <p style={{ color: 'var(--text-muted)', margin: 0, fontSize: '14px' }}>
50                        Q u ê n l ý b à i g i ê n g v à b à i k i Ā m t r a . <Radio size={12} style=
51                </p>
52            </div>
53            </header>
54
55            <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(180px, 1fr))', gap: '16px',...
56                {[
57                    { label: 'B à i g i ê n g ', value: lessons.length, icon: BookOpen, c
58                    { label: 'B à i k i Ā m t r a ', value: exams.length, icon: FileText, c
59                ]}.map((s, i) => (
60                    <div key={i} className="lms-card" style={{ padding: '20px', display: 'flex', alignItems: 'center', g...
61                    <div style={{ width: '40px', height: '40px', borderRadius: '10px', background: `${s.color}15`, col...
62                    <s.icon size={18} />
63                    </div>
64                    <div>
65                        <div style={{ fontSize: '22px', fontWeight: 900, color: 'var(--text-primary)' }}>{s.value}</div>
66                        <div style={{ fontSize: '12px', color: 'var(--text-muted)', fontWeight: 600 }}>{s.label}</div>
67                    </div>
68                </div>
69            ))}
70            </div>
71
72            <div className="lms-card" style={{ overflow: 'hidden', padding: '0' }}>
73                <div style={{ padding: '20px 24px', borderBottom: '1px solid var(--border-default)', display: 'flex', ...
74                    {[
75                        { key: 'a l l ', label: 'T ấ t c ế ' },
76                        { key: 'l e s s o n ', label: 'B à i g i ê n g ' },
77                        { key: 'e x a m ', label: 'B à i k i Ā m t r a ' },
78                    ]}.map(t => (
79                        <button key={t.key} onClick={() => setTab(t.key)} style={{
80                            padding: '6px 16px', borderRadius: '8px', border: '1px solid var(--border-default)',
81                            background: tab === t.key ? 'var(--brand-primary)' : 'transparent',
82                            color: tab === t.key ? '#fff' : 'var(--text-muted)',
83                            fontWeight: 600, cursor: 'pointer', fontSize: '12px', fontFamily: 'inherit'
84                        }}>{t.label}</button>
85                    ))}
86                </div>
87
88                <div style={{ overflow: 'auto' }}>
89                    <table style={{ width: '100%', borderCollapse: 'collapse' }}>
90                        <thead>
91                            <tr style={{ borderBottom: '1px solid var(--border-default)' }}>
92                                { [ 'T i ê u Á ', 'L o i i ', 'T á c g i ê ', 'T r i n g t h á i ', 'C - p n h - t '
93                                <th key={h} style={{ padding: '12px 20px', color: 'var(--text-muted)', fontSize: '11px', fon...
94                            </tr>
95                        </thead>
96                        <tbody>
97

```

```

98 |         {filtered.map((item: any) => (
99 |             <tr key={`_${item.contentType}-${item.id}`} style={{ borderBottom: '1px solid var(--border-defa...
100 |             <td style={{ padding: '14px 20px' }}>
101 |                 <div style={{ display: 'flex', alignItems: 'center', gap: '10px' }}>
102 |                     <div style={{
103 |                         width: '32px', height: '32px', borderRadius: '8px',
104 |                         background: item.contentType === 'lesson' ? 'rgba(8,158,96,0.1)' : 'rgba(255,163,0,0.1)',
105 |                         display: 'flex', alignItems: 'center', justifyContent: 'center'
106 |                     }}>
107 |                         {item.contentType === 'lesson' ? <BookOpen size={14} color="var(--brand-primary)" /> : ...
108 |                     </div>
109 |                     <div>
110 |                         <div style={{ color: 'var(--text-primary)', fontWeight: 600, fontSize: '14px' }}>{item...
111 |                         <div style={{ color: 'var(--text-muted)', fontSize: '11px' }}>ID: {item.id?.slice(0, 8...
112 |                     </div>
113 |                 </td>
114 |             </tr>
115 |             <td style={{ padding: '14px 20px' }}>
116 |                 <span className="lms-tag" style={{ fontSize: '10px' }}>{item.contentType === 'lesson' ? 'B...
117 |             </td>
118 |             <td style={{ padding: '14px 20px', color: 'var(--text-muted)', fontSize: '13px' }}>{item.pro...
119 |             <td style={{ padding: '14px 20px' }}>
120 |                 <div style={{ display: 'flex', alignItems: 'center', gap: '6px', color: item.is_published ...
121 |                 {item.is_published ? <CheckCircle size={12} /> : <XCircle size={12} />}
122 |                 {item.is_published ? 'ã x u ¤ t b £ n' : 'B £ n n h á p'}
123 |             </div>
124 |             </td>
125 |             <td style={{ padding: '14px 20px', color: 'var(--text-muted)', fontSize: '12px', display: 'f...
126 |                 <Clock size={12} /> {new Date(item.created_at).toLocaleDateString('vi-VN')}
127 |             </td>
128 |         </tr>
129 |     </tbody>
130 | </table>
131 | </div>
132 | </div>
133 | </div>
134 | </div>
135 | });
136 | }
137 |

```

File: app/(dashboard)/admin/error.tsx (26 lines, 0.9 KB)

```

1 | // Path: app/(dashboard)/admin/error.tsx
2 | 'use client'
3 | import { useEffect } from 'react'
4 |
5 | export default function Error({
6 |   error, reset
7 | }: { error: Error & { digest?: string }; reset: () => void }) {
8 |   useEffect(() => { console.error(error) }, [error])
9 |   return (
10 |     <div className="flex flex-col items-center justify-center h-full gap-6 p-8">
11 |       <div className="text-center">
12 |         <div className="text-center">
13 |           <h2 className="text-lg font-semibold text-[#d8dbe8] mb-2">C ó l x
14 |           <p className="text-sm text-[#5a5d72] mb-4">{error.message} | 'V
15 |         <button
16 |           onClick={reset}
17 |           className="px-4 py-2 bg-[#00d4aa]/15 border border-[#00d4aa]/30
18 |           text-[#00d4aa] rounded-lg text-sm font-medium hover:bg-[#00d4aa]/25 transition-colors"
19 |         >
20 |           T h í l i
21 |         </button>
22 |       </div>
23 |     </div>
24 |   )
25 | }
26 |

```

File: app/(dashboard)/admin/layout.tsx (130 lines, 6.7 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect, useCallback, useRef } from 'react'
4 | import Link from 'next/link'
5 | import { usePathname, useRouter } from 'next/navigation'
6 | import { LayoutDashboard, Users, Settings, Logout, Book, ShieldAlert, GraduationCap, Sun, Moon, BarChart3, Act...
7 | import PageTransition from '@components/PageTransition'
8 |
9 | function ThemeToggle() {
10 |   const [theme, setThemeState] = useState<'light' | 'dark'>('dark')
11 |   useEffect(() => {
12 |     const saved = (localStorage.getItem('theme') as 'light' | 'dark') || 'dark'
13 |     setThemeState(saved)
14 |     document.documentElement.setAttribute('data-theme', saved)
15 |   }, [])
16 |   const toggle = useCallback(() => {
17 |     const next = theme === 'light' ? 'dark' : 'light'
18 |     setThemeState(next)
19 |     document.documentElement.setAttribute('data-theme', next)
20 |     localStorage.setItem('theme', next)
21 |   }, [theme])
22 |   return (
23 |     <button onClick={toggle} style={{ display: 'flex', alignItems: 'center', gap: '12px', padding: '12px 16px'...
24 |       {theme === 'light' ? <Moon size={16} /> : <Sun size={16} />}
25 |       {theme === 'light' ? 'G i a o d i ç n t Ñ i' : 'G i a o d i ç n s á n g'}
26 |     </button>
27 |   )
28 | }
29 |
30 | export default function AdminLayout({ children }: { children: React.ReactNode }) {
31 |   const pathname = usePathname()
32 |   const router = useRouter()
33 |   const [connected, setConnected] = useState(true)
34 |   const [onlineCount, setOnlineCount] = useState(0)
35 |   const [currentTime, setCurrentTime] = useState('')
36 |
37 |   useEffect(() => {
38 |     const auth = localStorage.getItem('admin_auth')
39 |     if (auth !== 'true') { router.push('/login') }
40 |
41 |     const timeInt = setInterval(() => {
42 |       setCurrentTime(new Date().toLocaleTimeString('vi-VN', { hour: '2-digit', minute: '2-digit', second: '2-d...
43 |     }, 1000)
44 |
45 |     const onlineInt = setInterval(() => {
46 |       setOnlineCount(Math.floor(Math.random() * 12 + 3))
47 |     }, 5000)
48 |
49 |     const connInt = setInterval(() => {

```

```

50 |     setConnected(navigator.onLine)
51 | }, 3000)
52 |
53 | return () => { clearInterval(timeInt); clearInterval(onlineInt); clearInterval(connInt) }
54 | }, [])
55 |
56 | function handleLogout() {
57 |     localStorage.removeItem('admin_auth')
58 |     localStorage.removeItem('admin_login_time')
59 |     router.push('/login')
60 | }
61 |
62 | const isActive = (href: string) => pathname === href || pathname?.startsWith(href + '/')
63 |
64 | const NavItem = ({ href, icon: Icon, label }: { href: string; icon: any; label: string }) => {
65 |     const active = isActive(href)
66 |     return (
67 |         <Link href={href} style={{
68 |             display: 'flex', alignItems: 'center', gap: '10px', padding: '10px 14px', borderRadius: '8px', textDec...
69 |             background: active ? 'rgba(8,158,96,0.15)' : 'transparent',
70 |             color: active ? '#fff' : 'rgba(255,255,255,0.55)',
71 |             fontWeight: active ? 700 : 500,
72 |             transition: 'all 0.2s'
73 |         }}>
74 |             <Icon size={18} /> {label}
75 |         </Link>
76 |     )
77 | }
78 |
79 | return (
80 |     <div style={{ display: 'flex', minHeight: '100vh', background: 'var(--bg-base)' }}>
81 |         <aside style={{ width: '260px', background: 'linear-gradient(180deg, #031f3b 0%, #1a2f53 100%)', display: ...
82 |             <div style={{ padding: '20px 20px 16px', display: 'flex', alignItems: 'center', gap: '10px', borderBot...
83 |                 <div style={{ width: '36px', height: '36px', borderRadius: '8px', background: 'var(--accent-blue)', ...
84 |                     <ShieldAlert size={18} color="#fff" />
85 |                 </div>
86 |                 <div style={{
87 |                     <div style={{ fontWeight: 800, fontSize: '16px', color: '#fff', letterSpacing: '-0.3px' }}>PC Mast...
88 |                     <div style={{
89 |                         </div>
90 |                     </div>
91 |
92 |                     <div style={{ padding: '12px 16px', borderBottom: '1px solid rgba(255,255,255,0.06)', display: 'flex',...
93 |                         <div style={{ display: 'flex', alignItems: 'center', gap: '6px' }}>
94 |                             <span style={{ width: '6px', height: '6px', borderRadius: '50%', background: connected ? '#30c080'...
95 |                             <span style={{ fontSize: '10px', color: 'rgba(255,255,255,0.45)', fontWeight: 500 }}>Realtime</span>
96 |                         </div>
97 |                         <div style={{ fontSize: '10px', color: 'rgba(255,255,255,0.35)' }}>
98 |                             <Wifi size={10} style={{ verticalAlign: 'middle', marginRight: '3px' }} />
99 |                             {onlineCount} online
100 |                         </div>
101 |                     </div>
102 |
103 |                     <nav style={{ padding: '16px 12px', display: 'flex', flexDirection: 'column', gap: '2px', flex: 1 }}>
104 |                         <NavItem href="/admin" icon={LayoutDashboard} label="Tổng quan" />
105 |                         <NavItem href="/admin/users" icon={Users} label="Quản lý người dùng" />
106 |                         <NavItem href="/admin/schools" icon={Book} label="Quản lý trường" />
107 |                         <NavItem href="/admin/settings" icon={Settings} label="Cài đặt" />
108 |                         <div style={{ height: '1px', background: 'rgba(255,255,255,0.06)', margin: '6px 0' }} />
109 |                         <NavItem href="/admin/analytics" icon={BarChart3} label="Phân tích" />
110 |                         <NavItem href="/admin/activities" icon={Activity} label="Nhật ký" />
111 |                         <NavItem href="/admin/notifications-admin" icon={Bell} label="Thông báo" />
112 |                         <NavItem href="/admin/content" icon={FileText} label="Quản lý nội dung" />
113 |                     </nav>
114 |
115 |                     <div style={{ padding: '12px 16px', borderTop: '1px solid rgba(255,255,255,0.06)', display: 'flex', fl...
116 |                         <div style={{ fontSize: '10px', color: 'rgba(255,255,255,0.3)', textAlign: 'center', marginBottom: '...
117 |                         <ThemeToggle />
118 |                         <button onClick={handleLogout} style={{ display: 'flex', alignItems: 'center', gap: '10px', padding:...
119 |                             <LogOut size={16} />
120 |                             Ngủ
121 |                         </div>
122 |                     </aside>
123 |
124 |                     <main style={{ flex: 1, padding: '28px 32px', overflowY: 'auto', minWidth: 0 }}>
125 |                         <PageTransition>{children}</PageTransition>
126 |                     </main>
127 |                 </div>
128 |             )
129 |         }
130 |     )

```

File: app/(dashboard)/admin/loading.tsx (15 lines, 0.5 KB)

```

1 | // Path: app/(dashboard)/admin/loading.tsx
2 | export default function Loading() {
3 |     return (
4 |         <div className="p-6 space-y-4 animate-pulse">
5 |             <div className="h-8 bg-[#1d1f2a] rounded-lg w-48" />
6 |             <div className="grid grid-cols-1 md:grid-cols-4 gap-4">
7 |                 {[...Array(4)].map((_, i) => (
8 |                     <div key={i} className="h-24 bg-[#1d1f2a] rounded-xl" />
9 |                 ))}
10 |             </div>
11 |             <div className="h-64 bg-[#1d1f2a] rounded-xl" />
12 |         </div>
13 |     )
14 | }
15 |

```

File: app/(dashboard)/admin/notifications-admin/page.tsx (90 lines, 6.0 KB)

```

1 | 'use client';
2 |
3 | import React, { useState } from 'react';
4 | import { Bell, Send, Users, BookOpen, Megaphone, Clock, CheckCircle, AlertTriangle, Info } from 'lucide-react';
5 |
6 | export default function AdminNotificationsPage() {
7 |     const [title, setTitle] = useState('');
8 |     const [content, setContent] = useState('');
9 |     const [target, setTarget] = useState('all');
10 |
11 |     return (
12 |         <div style={{ maxWidth: '1200px', margin: '0 auto' }}>
13 |             <header style={{ marginBottom: '28px' }}>
14 |                 <h1 style={{ fontSize: '28px', fontWeight: 800, margin: '0 0 4px 0', color: 'var(--text-primary)' }}>T...
15 |                 <p style={{ color: 'var(--text-muted)', margin: 0, fontSize: '14px' }}>
16 |             </header>
17 |
18 |             <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '20px' }}>
19 |                 <div className="lms-card" style={{ padding: '24px' }}>

```

```

20| <h3 style={{ fontSize: '16px', fontWeight: 700, color: 'var(--text-primary)', margin: '0 0 20px 0', ...
21| 2 1 | < M e g a p h o n e   s i z e = { 1 8 }   c o l o r = " v a r ( - - b r a n d - p r i m a r y ) "   / >   T i o   t h
22| </h3>
23|
24| <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
25|   <div>
26|     <label style={{ fontSize: '13px', fontWeight: 600, color: 'var(--text-primary)', marginBottom: '...'
27|     <input value={title} onChange={e => setTitle(e.target.value)} className="lms-input" placeholder=...
28|   </div>
29|   <div>
30|     <label style={{ fontSize: '13px', fontWeight: 600, color: 'var(--text-primary)', marginBottom: '...'
31|     <textarea value={content} onChange={e => setContent(e.target.value)} rows={4} className="lms-inp...
32|   </div>
33|   <div>
34|     <label style={{ fontSize: '13px', fontWeight: 600, color: 'var(--text-primary)', marginBottom: '...'
35|     <div style={{ display: 'flex', gap: '8px', flexWrap: 'wrap' }}>
36|       {[
37|         { value: 'all', label: 'Tất cả', icon: Users },
38|         { value: 'students', label: 'Học sinh', icon: Users },
39|         { value: 'teachers', label: 'Giáo viên', icon: BookOpen },
40|       ].map(opt => (
41|         <button key={opt.value} onClick={() => setTarget(opt.value)} style={{
42|           padding: '8px 16px', borderRadius: '8px', border: '2px solid ${target === opt.value ? 'var...'
43|           background: target === opt.value ? 'var(--brand-subtle)' : 'transparent',
44|           color: target === opt.value ? 'var(--brand-primary)' : 'var(--text-muted)',
45|           fontWeight: 600, cursor: 'pointer', fontSize: '12px', fontFamily: 'inherit',
46|           display: 'flex', alignItems: 'center', gap: '6px', transition: 'all 0.2s'
47|         }}>
48|           <opt.icon size={14} /> {opt.label}
49|         </button>
50|       ))}
51|     </div>
52|   </div>
53|   <button style={{ padding: '12px 24px', borderRadius: '10px', background: 'var(--brand-primary)', c...
54|     < S e n d   s i z e = { 1 8 }   / >   G i i   t h ô n g   b á o
55|   </button>
56| </div>
57| </div>
58|
59| <div>
60| <h3 style={{ fontSize: '16px', fontWeight: 700, color: 'var(--text-primary)', margin: '0 0 16px 0', ...
61| 6 1 | < B e l l   s i z e = { 1 8 }   c o l o r = " v a r ( - - a c c e n t - a m b e r ) "   / >   T h ô n g   b á o   ä
62| </h3>
63| <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
64|   {[
65|     { title: 'Lêch thi gia kó', time: '2 gi Ý tr ° Û c', status:
66|     { title: 'Bêo tri hÇ th Ñng', time: '1 ng ày tr ° Û c', status:
67|     { title: 'B à i   g i Ë n g   m Ò i', time: '2   ng ày tr ° Û c', status:
68|     { title: 'K ÷ t   q u Ë   t h i   h í c   k ó', time: '3   ng ày tr ° Û c', statu
69|   ].map((notif, i) => (
70|     <div key={i} className="lms-card" style={{ padding: '16px 20px', display: 'flex', alignItems: 'c...
71|     <div style={{ width: '36px', height: '36px', borderRadius: '8px', background: notif.type === '...'
72|     {notif.type === 'success' ? <CheckCircle size={16} color="var(--brand-primary)" /> : notif.t...
73|   </div>
74|   <div style={{ flex: 1 }}>
75|     <div style={{ fontSize: '14px', fontWeight: 700, color: 'var(--text-primary)' }}>{notif.titl...
76|     <div style={{ fontSize: '11px', color: 'var(--text-muted)', display: 'flex', gap: '12px', ma...
77|       <span><Clock size={10} /> {notif.time}</span>
78|       <span><notif.target></span>
79|     </div>
80|   </div>
81|   <span className="lms-tag lms-tag-green" style={{ fontSize: '10px' }}>{notif.status}</span>
82| </div>
83|   ))}
84| </div>
85| </div>
86| </div>
87| </div>
88| );
89| }
90|

```

File: app/(dashboard)/admin/page.tsx (502 lines, 27.0 KB)

```

1| 'use client'
2|
3| import React, { useEffect, useState, useCallback, useRef } from 'react'
4| import { useRouter } from 'next/navigation'
5| import { supabase } from '@lib/supabase'
6| import {
7|   Users, Shield, Activity, Database, Settings, Search, Filter,
8|   Loader2, AlertCircle, BarChart3, TrendingUp, Award, Clock,
9|   BookOpen, GraduationCap, Monitor, Bell, FileText, MessageSquare,
10|   Cpu, Wifi, HardDrive, Globe, ChevronRight, Plus, RefreshCw,
11|   Zap, Download, Upload, CheckCircle, XCircle, Radio, HeartPulse, UserPlus
12| } from 'lucide-react'
13| import { LineChart, Line, AreaChart, Area, XAxis, YAxis, CartesianGrid, Tooltip, ResponsiveContainer, BarChart...
14|
15| const SYSTEM_START = Date.now()
16|
17| function formatUptime(ms: number) {
18|   const d = Math.floor(ms / 86400000)
19|   const h = Math.floor((ms % 86400000) / 3600000)
20|   const m = Math.floor((ms % 3600000) / 60000)
21|   return d > 0 ? `${d}d ${h}h ${m}m` : `${h}h ${m}m`
22| }
23|
24| function formatTime(ts: string | number) {
25|   const d = new Date(ts)
26|   return d.toLocaleTimeString('vi-VN', { hour: '2-digit', minute: '2-digit', second: '2-digit' })
27| }
28|
29| function timeAgo(ts: string | number) {
30|   const diff = Date.now() - new Date(ts).getTime()
31|   const mins = Math.floor(diff / 60000)
32|   if (mins < 1) return 'v à i   g i â y   tr ° Û c'
33|   if (mins < 60) return ` $ { mins }   p h ú t   tr ° Û c `
34|   const hrs = Math.floor(mins / 60)
35|   if (hrs < 24) return ` $ { hrs }   g i Ý   tr ° Û c `
36|   return ` $ { Math . f l o o r ( hrs / 24 ) }   n g à y   tr ° Û c `
37| }
38|
39| type RealtimeEvent = {
40|   id: string
41|   type: 'user_created' | 'lesson_created' | 'exam_completed' | 'submission' | 'system'
42|   label: string
43|   detail: string
44|   time: string
45| }
46|
47| export default function AdminDashboard() {
48|   const router = useRouter()
49|   const [users, setUsers] = useState<any[]>([])

```

```

50 | const [stats, setStats] = useState({ totalUsers: 0, teachers: 0, students: 0, activeClasses: 0, totalLessons...
51 | const [loading, setLoading] = useState(true)
52 | const [realtimeEvents, setRealtimeEvents] = useState<RealtimeEvent[]>([])
53 | const [connected, setConnected] = useState(false)
54 | const [latency, setLatency] = useState<number | null>(null)
55 | const [uptime, setUptime] = useState('')
56 | const [autoRefresh, setAutoRefresh] = useState(true)
57 | const [weeklyData, setWeeklyData] = useState([
58 |   { day: 'T2', users: 0, lessons: 0, exams: 0 },
59 |   { day: 'T3', users: 0, lessons: 0, exams: 0 },
60 |   { day: 'T4', users: 0, lessons: 0, exams: 0 },
61 |   { day: 'T5', users: 0, lessons: 0, exams: 0 },
62 |   { day: 'T6', users: 0, lessons: 0, exams: 0 },
63 |   { day: 'T7', users: 0, lessons: 0, exams: 0 },
64 |   { day: 'CN', users: 0, lessons: 0, exams: 0 },
65 | ])
66 | const pollRef = useRef<NodeJS.Timeout | null>(null)
67 | const latencyRef = useRef<number[]>([])
68 | const eventsRef = useRef<HTMLDivElement>(null)
69 |
70 | useEffect(() => {
71 |   const auth = localStorage.getItem('admin_auth')
72 |   if (auth !== 'true') { router.push('/login'); return }
73 |   fetchAdminData()
74 |
75 |   const uptimeInt = setInterval(() => {
76 |     setUptime(formatUptime(Date.now() - SYSTEM_START))
77 |   }, 1000)
78 |
79 |   return () => {
80 |     clearInterval(uptimeInt)
81 |     if (pollRef.current) clearInterval(pollRef.current)
82 |   }
83 | }, [])
84 |
85 | useEffect(() => {
86 |   if (!autoRefresh) return
87 |   pollRef.current = setInterval(fetchAdminData, 30000)
88 |   return () => { if (pollRef.current) clearInterval(pollRef.current) }
89 | }, [autoRefresh])
90 |
91 | useEffect(() => {
92 |   const channel = supabase.channel('admin-dashboard-realtime', {
93 |     config: { broadcast: { self: true } },
94 |   })
95 |
96 |   channel
97 |     .on('postgres_changes', {
98 |       event: 'INSERT',
99 |       schema: 'public',
100 |      table: 'profiles',
101 |    }, (payload) => {
102 |      const newUser = payload.new as any
103 |      setStats(prev => ({ ...prev, totalUsers: prev.totalUsers + 1 }))
104 |      const evt: RealtimeEvent = {
105 |        id: `user-${payload.commit_timestamp}`,
106 |        type: 'user_created',
107 |        label: 'Thêm người dùng mới',
108 |        detail: `${newUser.full_name || 'Unknown'} (${newUser.role || 'student'})`,
109 |        time: formatTime(payload.commit_timestamp || Date.now()),
110 |      }
111 |      setRealtimeEvents(prev => [evt, ...prev].slice(0, 50))
112 |      setUsers(prev => [newUser, ...prev])
113 |    })
114 |     .on('postgres_changes', {
115 |       event: 'INSERT',
116 |       schema: 'public',
117 |       table: 'lessons',
118 |     }, (payload) => {
119 |       const newLesson = payload.new as any
120 |       setStats(prev => ({ ...prev, totalLessons: prev.totalLessons + 1 }))
121 |       const evt: RealtimeEvent = {
122 |         id: `lesson-${payload.commit_timestamp}`,
123 |         type: 'lesson_created',
124 |         label: 'Thêm bài giảng mới',
125 |         detail: newLesson.title || 'Bài giảng không tiêu đề',
126 |         time: formatTime(payload.commit_timestamp || Date.now()),
127 |       }
128 |       setRealtimeEvents(prev => [evt, ...prev].slice(0, 50))
129 |     })
130 |     .on('postgres_changes', {
131 |       event: 'INSERT',
132 |       schema: 'public',
133 |       table: 'exams',
134 |     }, (payload) => {
135 |       const newExam = payload.new as any
136 |       setStats(prev => ({ ...prev, totalExams: prev.totalExams + 1 }))
137 |       const evt: RealtimeEvent = {
138 |         id: `exam-${payload.commit_timestamp}`,
139 |         type: 'exam_completed',
140 |         label: 'Bài kiểm tra mới',
141 |         detail: (payload.new as any).title || 'Kiểm tra không tiêu đề',
142 |         time: formatTime(payload.commit_timestamp || Date.now()),
143 |       }
144 |       setRealtimeEvents(prev => [evt, ...prev].slice(0, 50))
145 |     })
146 |     .subscribe((status) => {
147 |       setConnected(status === 'SUBSCRIBED')
148 |     })
149 |
150 |   const latInt = setInterval(async () => {
151 |     const start = Date.now()
152 |     try {
153 |       const { error } = await supabase.from('profiles').select('id', { count: 'exact', head: true })
154 |       if (!error) {
155 |         const ms = Date.now() - start
156 |         latencyRef.current.push(ms)
157 |         if (latencyRef.current.length > 10) latencyRef.current.shift()
158 |         setLatency(Math.round(latencyRef.current.reduce((a, b) => a + b, 0) / latencyRef.current.length))
159 |       } catch { }
160 |     }, 5000)
161 |
162 |     return () => {
163 |       supabase.removeChannel(channel)
164 |       clearInterval(latInt)
165 |     }
166 |   }, [])
167 |
168 |   async function fetchAdminData() {
169 |     try {
170 |       const { data: profiles } = await supabase.from('profiles').select('*').order('created_at', { ascending: ...
171 |       if (profiles) setUsers(profiles)
172 |       const teachers = profiles?.filter(u => u.role === 'teacher').length || 0
173 |       const students = profiles?.filter(u => u.role === 'student').length || 0

```



```

174 |     const { count: classesCount } = await supabase.from('classes').select('*', { count: 'exact', head: true })
175 |     const { count: lessonsCount } = await supabase.from('lessons').select('*', { count: 'exact', head: true })
176 |     const { count: examsCount } = await supabase.from('exams').select('*', { count: 'exact', head: true })
177 |     setStats({ totalUsers: profiles?.length || 0, teachers, students, activeClasses: classesCount || 0, tota...
178 |
179 |     const sevenDaysAgo = new Date(Date.now() - 7 * 86400000).toISOString()
180 |     const dayNames = ['CN', 'T2', 'T3', 'T4', 'T5', 'T6', 'T7']
181 |     const weeklyMap: Record<string, { users: number; lessons: number; exams: number }> = {}
182 |     for (let i = 6; i >= 0; i--) {
183 |         const d = new Date(Date.now()) - i * 86400000
184 |         const key = dayNames[d.getDay()]
185 |         weeklyMap[key] = { users: 0, lessons: 0, exams: 0 }
186 |     }
187 |     const { data: recentProfiles } = await supabase.from('profiles').select('created_at').gte('created_at', ...
188 |     recentProfiles?.forEach(p => {
189 |         const key = dayNames[new Date(p.created_at).getDay()]
190 |         if (weeklyMap[key]) weeklyMap[key].users++
191 |     })
192 |     const { data: recentLessons } = await supabase.from('lessons').select('created_at').gte('created_at', se...
193 |     recentLessons?.forEach(l => {
194 |         const key = dayNames[new Date(l.created_at).getDay()]
195 |         if (weeklyMap[key]) weeklyMap[key].lessons++
196 |     })
197 |     const { data: recentExams } = await supabase.from('exams').select('created_at').gte('created_at', sevenD...
198 |     recentExams?.forEach(e => {
199 |         const key = dayNames[new Date(e.created_at).getDay()]
200 |         if (weeklyMap[key]) weeklyMap[key].exams++
201 |     })
202 |     setWeeklyData(dayNames.map(day => ({ day, ...weeklyMap[day] || { users: 0, lessons: 0, exams: 0 } })))
203 | } catch (err) { console.error('fetchAdminData error:', err) }
204 | finally { setLoading(false) }
205 | }
206 |
207 | const addSystemEvent = useCallback((label: string, detail: string) => {
208 |     const evt: RealtimeEvent = {
209 |         id: `sys-${Date.now()}-${Math.random()}`,
210 |         type: 'system', label, detail,
211 |         time: formatTime(Date.now()),
212 |     }
213 |     setRealtimeEvents(prev => [evt, ...prev].slice(0, 50))
214 | }, [])
215 |
216 | useEffect(() => {
217 |     if (!eventsRef.current) return
218 |     eventsRef.current.scrollTop = 0
219 | }, [realtimeEvents.length])
220 |
221 | if (loading) return (
222 |     <div style={{ display: 'flex', flexDirection: 'column', alignItems: 'center', justifyContent: 'center', he...
223 |     <Loader2 size={48} style={{ color: 'var(--brand-primary)', animation: 'spin 1s linear infinite' }} />
224 |     <p style={{ color: 'var(--text-muted)', fontSize: '14px' }}> ang t
225 | </div>
226 | )
227 |
228 | const cardStyle: React.CSSProperties = {
229 |     background: 'var(--bg-surface)', border: '1px solid var(--border-default)',
230 |     borderRadius: '12px', padding: '20px',
231 | }
232 |
233 | return (
234 |     <div style={{ maxWidth: '1400px', margin: '0 auto' }}>
235 |         <header style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom: '...'
236 |         <div>
237 |             <div style={{ display: 'flex', alignItems: 'center', gap: '10px', marginBottom: '4px' }}>
238 |                 <h1 style={{ fontSize: '26px', fontWeight: 800, margin: 0, c
239 |                 <span style={{
240 |                     display: 'inline-flex', alignItems: 'center', gap: '4px', padding: '3px 10px',
241 |                     borderRadius: '99px', fontSize: '10px', fontWeight: 700,
242 |                     background: connected ? 'rgba(8,158,96,0.1)' : 'rgba(244,106,106,0.1)',
243 |                     color: connected ? 'var(--brand-primary)' : 'var(--danger)',
244 |                     textTransform: 'uppercase', letterSpacing: '0.3px',
245 |                 }}>
246 |                     <Radio size={10} />
247 |                     {connected ? 'Real-time' : 'Disconnected'}
248 |                 </span>
249 |             </div>
250 |             <p style={{ color: 'var(--text-muted)', margin: 0, fontSize: '13px', display: 'flex', alignItems: 'c...
251 |                 D a s h b o a r d   q u e n   t r e   .   P C   M a s t e r   L M S   v 2 . 4 . 1
252 |                 {latency !== null && {
253 |                     <span style={{
254 |                         display: 'inline-flex', alignItems: 'center', gap: '3px', fontSize: '11px',
255 |                         color: latency < 100 ? 'var(--brand-primary)' : latency < 300 ? 'var(--accent-amber)' : 'var(-...
256 |                     }}>
257 |                         <HeartPulse size={11} /> {latency}ms
258 |                     </span>
259 |                 )}
260 |                 <span style={{ fontSize: '11px', color: 'var(--text-muted)' }}> Uptime: {uptime}</span>
261 |             </p>
262 |         </div>
263 |         <div style={{ display: 'flex', gap: '8px', alignItems: 'center' }}>
264 |             <label style={{ display: 'flex', alignItems: 'center', gap: '6px', fontSize: '12px', color: 'var(--t...
265 |                 <input type="checkbox" checked={autoRefresh} onChange={e => setAutoRefresh(e.target.checked)} styl...
266 |                 T ã ù n g
267 |             </label>
268 |             <button onClick={fetchAdminData} style={{
269 |                 padding: '8px 14px', borderRadius: '8px', background: 'var(--bg-elevated)',
270 |                 border: '1px solid var(--border-default)', color: 'var(--text-primary)',
271 |                 display: 'flex', alignItems: 'center', gap: '6px', cursor: 'pointer',
272 |                 fontSize: '12px', fontWeight: 600, fontFamily: 'inherit',
273 |             }}>
274 |                 < R e f r e s h C w   s i z e = { 1 4 }   / >   L à m   m ù i
275 |             </button>
276 |         </div>
277 |     </header>
278 |
279 |     <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(180px, 1fr))', gap: '12px',...
280 |     {
281 |         { label: 'Giáo viên', value: stats.teachers, icon: Shield, color: 'var(--accent-blue)', change: '+2' },
282 |         { label: 'Hìc sinh', value: stats.students, icon: GraduationCap, color: 'var(--text-primary)', change: '+1' },
283 |         { label: 'L ù p h ì c', value: stats.activeClasses, icon: Database, color: 'var(--text-primary)', change: '+1' },
284 |         { label: 'B à i g i ẽ n g', value: stats.totalLessons, icon: BookOpen, color: 'var(--text-primary)', change: '+1' },
285 |         { label: 'B à i k i ả m t r a', value: stats.totalExams, icon: FileText, color: 'var(--text-primary)', change: '+1' }
286 |     ]
287 |     .map((s, i) => {
288 |         <div key={i} style={{
289 |             ...cardStyle, padding: '16px 18px', transition: 'all 0.2s',
290 |             borderLeft: '3px solid ${s.color}',
291 |         }}>
292 |             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'flex-start', marginBo...
293 |             <div style={{
294 |                 width: '34px', height: '34px', borderRadius: '8px',
295 |                 background: `${s.color}15`, color: s.color,
296 |                 display: 'flex', alignItems: 'center', justifyContent: 'center',
297 |             }}>

```

```
298 |         <s.icon size={16} />
299 |       </div>
300 |       <span style={{ fontSize: '10px', fontWeight: 700, color: 'var(--brand-primary)', background: 'va...
301 |     </div>
302 |     <div style={{ fontSize: '24px', fontWeight: 900, color: 'var(--text-primary)' }}>{s.value.toLocale...
303 |     <div style={{ fontSize: '11px', color: 'var(--text-muted)', fontWeight: 600, marginTop: '1px' }}>{...
304 |   </div>
305 | ))}
306 | </div>
307 |
308 | <div style={{ display: 'grid', gridTemplateColumns: '1.5fr 1fr', gap: '20px', marginBottom: '24px' }}>
309 |   <div style={cardStyle}>
310 |     <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom: ...
311 |     <h3 style={{ fontSize: '15px', fontWeight: 700, color: 'var(--text-primary)', margin: 0, display: ...
312 |       < T r e n d i n g U p   s i z e = { 1 6 }   c o l o r = " v a r ( - - b r a n d - p r i m a r y ) "   / >   H o i t
313 |     </h3>
314 |     <div style={{ display: 'flex', gap: '12px', fontSize: '10px', color: 'var(--text-muted)' }}>
315 |       <span style={{ display: 'flex', alignItems: 'center', gap: '3px' }}><div style={{ width: '7px', ...
316 |       <span style={{ display: 'flex', alignItems: 'center', gap: '3px' }}><div style={{ width: '7px', ...
317 |     </div>
318 |   </div>
319 |   <div style={{ height: '220px' }}>
320 |     <ResponsiveContainer width="100%" height="100%">
321 |       <BarChart data={weeklyData}>
322 |         <CartesianGrid strokeDasharray="3 3" stroke="var(--border-default)" />
323 |         <XAxis dataKey="day" tick={{ fill: 'var(--text-muted)', fontSize: 10 }} />
324 |         <YAxis tick={{ fill: 'var(--text-muted)', fontSize: 10 }} />
325 |         <Tooltip contentStyle={{ background: 'var(--bg-surface)', border: '1px solid var(--border-defa...
326 |         <Bar dataKey="users" fill="var(--brand-primary)" radius={[3, 3, 0, 0]} />
327 |         <Bar dataKey="lessons" fill="var(--accent-blue)" radius={[3, 3, 0, 0]} />
328 |       </BarChart>
329 |     </ResponsiveContainer>
330 |   </div>
331 | </div>
332 |
333 |   <div style={cardStyle}>
334 |     <h3 style={{ fontSize: '15px', fontWeight: 700, color: 'var(--text-primary)', margin: '0 0 14px 0', ...
335 |       < C p u   s i z e = { 1 6 }   c o l o r = " v a r ( - - b r a n d - p r i m a r y ) "   / >   H ẹ   t h ấ   n   g
336 |     </h3>
337 |     <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '8px' }}>
338 |       {
339 |         { icon: Cpu, label: 'CPU Server', value: 'N/A', status: 'ok' as const },
340 |         { icon: HardDrive, label: 'RAM', value: 'N/A', status: 'ok' as const },
341 |         { icon: Database, label: 'Database', value: latency ? ` ${latency}ms` : 'N/A' },
342 |         { icon: Globe, label: 'Uptime', value: uptime ? ` ${uptime}mins` : 'N/A' },
343 |         { icon: Wifi, label: 'API', value: latency ? ` ${latency}ms` : 'N/A' },
344 |         { icon: Activity, label: 'Realtime', value: connected ? 'N/A' : 'N/A' }
345 |       ].map((item, i) => {
346 |         const statusColor = item.status === 'ok' ? 'var(--brand-primary)' : item.status === 'warn' ? 'va...
347 |         const statusBg = item.status === 'ok' ? 'rgba(8,158,96,0.1)' : item.status === 'warn' ? 'rgba(25...
348 |         return (
349 |           <div key={i} style={{
350 |             padding: '12px 14px', borderRadius: '8px', background: 'var(--bg-elevated)',
351 |             border: '1px solid var(--border-default)', display: 'flex', alignItems: 'center', gap: '10px',
352 |           }}>
353 |             <div style={{ width: '32px', height: '32px', borderRadius: '8px', background: statusBg, disp...
354 |             <item.icon size={15} />
355 |             </div>
356 |             <div style={{ flex: 1, minWidth: 0 }}>
357 |               <div style={{ fontSize: '10px', color: 'var(--text-muted)', fontWeight: 600 }}>{item.label...
358 |               <div style={{ fontSize: '13px', fontWeight: 700, color: 'var(--text-primary)' }}>{item.val...
359 |             </div>
360 |             <div style={{ width: '6px', height: '6px', borderRadius: '50%', background: statusColor, box...
361 |             </div>
362 |           )
363 |         )
364 |       }
365 |     </div>
366 |   </div>
367 |
368 |   <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '20px', marginBottom: '24px' }}>
369 |     <div style={cardStyle}>
370 |       <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom: ...
371 |       <h3 style={{ fontSize: '15px', fontWeight: 700, color: 'var(--text-primary)', margin: 0, display: ...
372 |         < A c t i v i t y   s i z e = { 1 6 }   c o l o r = " v a r ( - - b r a n d - p r i m a r y ) "   / >   S ấ   k i
373 |         {realtimeEvents.length > 0 && (
374 |           <span style={{
375 |             fontSize: '10px', fontWeight: 700, padding: '1px 6px', borderRadius: '99px',
376 |             background: 'var(--brand-subtle)', color: 'var(--brand-primary)', marginLeft: '4px',
377 |           }}>
378 |             {realtimeEvents.length}
379 |           </span>
380 |         )
381 |       </h3>
382 |       {realtimeEvents.length > 0 && (
383 |         <button onClick={() => setRealtimeEvents([])}
384 |           style={{ background: 'none', border: 'none', color: 'var(--text-muted)', cursor: 'pointer', fo...
385 |           X ó a t ấ t c ấ
386 |         </button>
387 |       )
388 |     </div>
389 |     <div ref={eventsRef} style={{
390 |       maxHeight: '320px', overflowY: 'auto', display: 'flex', flexDirection: 'column', gap: '2px',
391 |     }}>
392 |       {realtimeEvents.length === 0 ? (
393 |         <div style={{ padding: '24px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '13px'...
394 |         <Radio size={28} style={{ opacity: 0.15, marginBottom: '8px' }} />
395 |         < p >   a n g   c h ấ   s ấ   k i c n   r e a l t i m e . . .   < / p >
396 |         < p   s t y l e = { {   f o n t S i z e :   ' 1 1 p x ' ,   o p a c i t y :   0 . 6   } } >   C á c   t h a y
397 |       </div>
398 |     ) : (
399 |       realtimeEvents.map((evt, i) => (
400 |         <div key={evt.id} style={{
401 |           display: 'flex', alignItems: 'center', gap: '10px', padding: '8px 10px',
402 |           borderRadius: '6px', transition: 'background 0.15s',
403 |           background: i === 0 ? 'rgba(8,158,96,0.04)' : 'transparent',
404 |           animation: i === 0 ? 'fadeIn 0.3s ease' : undefined,
405 |         }}>
406 |           <span style={{ fontSize: '16px', flexShrink: 0 }}>{evt.label.split(' ')[0]}</span>
407 |           <div style={{ flex: 1, minWidth: 0 }}>
408 |             <div style={{ fontSize: '12px', fontWeight: 700, color: 'var(--text-primary)' }}>{evt.labe...
409 |             <div style={{ fontSize: '11px', color: 'var(--text-muted)' }}>{evt.detail}</div>
410 |           </div>
411 |           <span style={{ fontSize: '10px', color: 'var(--text-muted)', whiteSpace: 'nowrap', flexShrin...
412 |           </div>
413 |         )
414 |       )
415 |     </div>
416 |   </div>
417 |
418 |   <div style={cardStyle}>
419 |     <h3 style={{ fontSize: '15px', fontWeight: 700, color: 'var(--text-primary)', margin: '0 0 14px 0', ...
420 |     <Zap size={16} color="var(--brand-primary)" /> Thao tác nhanh
421 |   </h3>
```

File: app/(dashboard)/admin/schools/page.tsx (80 lines, 3.9 KB)

3 9 | `< p style={ { color: 'var(--text-muted)', margin: 0, fontSize:`

```

40 |     </div>
41 | </header>
42 |
43 | <div style={{ marginBottom: '20px' }}>
44 |   <div style={{ position: 'relative', maxWidth: '400px' }}>
45 |     <Search size={16} style={{ position: 'absolute', left: '12px', top: '50%', transform: 'translateY(-50%...'
46 |       <input value={search} onChange={e => setSearch(e.target.value)}
47 |         width: '100%', padding: '10px 12px 10px 36px', borderRadius: '10px', background: 'var(--bg-surface)',
48 |         border: '1px solid var(--border-default)', color: 'var(--text-primary)', fontSize: '13px', outline...
49 |       </input>
50 |     </div>
51 |   </div>
52 |
53 |   <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill, minmax(300px, 1fr))', gap: '16px'...
54 |     {filtered.map((school, i) => {
55 |       <div key={i} className="lms-card" style={{ padding: '20px' }}>
56 |         <div style={{ display: 'flex', alignItems: 'center', gap: '12px', marginBottom: '12px' }}>
57 |           <div style={{ width: '40px', height: '40px', borderRadius: '10px', background: 'rgba(var(--brand...
58 |             <Book size={20} />
59 |           </div>
60 |           <div>
61 |             <h3 style={{ margin: 0, fontSize: '15px', fontWeight: 700, color: 'var(--text-primary)' }}>{sc...
62 |           </div>
63 |           <div>
64 |             <div style={{ display: 'flex', alignItems: 'center', gap: '6px', fontSize: '13px', color: 'var(--t...
65 |               <Users size={14} /> {school.count} híc sinh
66 |             </div>
67 |           </div>
68 |         </div>
69 |       </div>
70 |
71 |       {filtered.length === 0 && (
72 |         <div style={{ textAlign: 'center', padding: '60px 20px', color: 'var(--text-muted)' }}>
73 |           <Book size={48} style={{ opacity: 0.2, marginBottom: '12px' }} />
74 |           <p>Chưa có trường híc nào.</p>
75 |         </div>
76 |       )}
77 |     </div>
78 |   );
79 | }
80 |

```

File: app/(dashboard)/admin/settings/page.tsx (120 lines, 6.9 KB)

```

1 | 'use client';
2 |
3 | import React, { useEffect, useState } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import { Settings, Save, Globe, Bell, Shield, Database, RefreshCw, CheckCircle, AlertCircle, Loader2 } from 'l...
6 |
7 | export default function AdminSettingsPage() {
8 |   const [saving, setSaving] = useState(false);
9 |   const [toast, setToast] = useState<{ message: string; type: 'success' | 'error' } | null>(null);
10 |   const [settings, setSettings] = useState({
11 |     siteName: 'PC Master Builder LMS',
12 |     description: 'NĂn tēng híc t-p và thíc hành l-p ráp máy tính tríc t...
13 |     maintenanceMode: false,
14 |     allowRegistration: true,
15 |     defaultRole: 'student',
16 |   });
17 |
18 |   const showToast = (message: string, type: 'success' | 'error' = 'success') => {
19 |     setToast({ message, type });
20 |     setTimeout(() => setToast(null), 3000);
21 |   };
22 |
23 |   const handleSave = async () => {
24 |     setSaving(true);
25 |     await new Promise(r => setTimeout(r, 800));
26 |     showToast('Đã lưu cýu hình hç thñng!');
27 |     setSaving(false);
28 |   };
29 |
30 |   return (
31 |     <div style={{ maxWidth: '900px', margin: '0 auto' }}>
32 |       <header style={{ marginBottom: '28px' }}>
33 |         <h1 style={{ fontSize: '28px', fontWeight: 800, margin: '0 0 4px 0', color: 'var(--text-primary)' }}>C...
34 |         <p style={{ color: 'var(--text-muted)', margin: 0, fontSize: '14'
35 |       </header>
36 |
37 |       {toast && (
38 |         <div style={{
39 |           padding: '12px 20px', borderRadius: '10px', marginBottom: '20px',
40 |           background: toast.type === 'success' ? 'rgba(8,158,96,0.1)' : 'rgba(244,106,106,0.1)',
41 |           border: '1px solid ${toast.type === 'success' ? 'var(--brand-primary)' : 'var(--danger)'}',
42 |           color: toast.type === 'success' ? 'var(--brand-primary)' : 'var(--danger)',
43 |           display: 'flex', alignItems: 'center', gap: '8px', fontSize: '14px', fontWeight: 600
44 |         }}>
45 |           {toast.type === 'success' ? <CheckCircle size={18} /> : <AlertCircle size={18} />}
46 |           {toast.message}
47 |         </div>
48 |       )}
49 |
50 |       <div className="lms-card" style={{ padding: '28px', marginBottom: '20px' }}>
51 |         <h2 style={{ fontSize: '16px', fontWeight: 700, margin: '0 0 20px 0', color: 'var(--text-primary)', di...
52 |         <Globe size={18} color="var(--brand-primary)" /> Thông tin chung
53 |       </h2>
54 |       <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
55 |         <div>
56 |           <label style={{ display: 'block', fontSize: '12px', fontWeight: 700, color: 'var(--text-muted)', m...
57 |           <input value={settings.siteName} onChange={e => setSettings({ ...settings, siteName: e.target.valu...
58 |             width: '100%', padding: '10px 14px', borderRadius: '10px', background: 'var(--bg-elevated)',
59 |             border: '1px solid var(--border-default)', color: 'var(--text-primary)', fontSize: '14px', outli...
60 |           </input>
61 |         </div>
62 |         <div>
63 |           <label style={{ display: 'block', fontSize: '12px', fontWeight: 700, color: 'var(--text-muted)', m...
64 |           <textarea value={settings.description} onChange={e => setSettings({ ...settings, description: e.ta...
65 |             width: '100%', padding: '10px 14px', borderRadius: '10px', background: 'var(--bg-elevated)',
66 |             border: '1px solid var(--border-default)', color: 'var(--text-primary)', fontSize: '14px', outli...
67 |           </textarea>
68 |         </div>
69 |       </div>
70 |     </div>
71 |
72 |     <div className="lms-card" style={{ padding: '28px', marginBottom: '20px' }}>
73 |       <h2 style={{ fontSize: '16px', fontWeight: 700, margin: '0 0 20px 0', color: 'var(--text-primary)', di...
74 |       <Shield size={18} color="var(--brand-primary)" /> Béo m-t & Qu
75 |     </h2>
76 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
77 |       {
78 |         { key: 'allowRegistration', label: 'Cho phép ng ký tài kho
79 |         { key: 'maintenanceMode', label: 'Chú ù béo tríc', desc: 'Ch

```

```
80 |         ].map((item: any) => (  
81 |             <label key={item.key} style={{ display: 'flex', alignItems: 'center', gap: '12px', cursor: 'pointe...  
82 |             <input type="checkbox" checked={settings as any}[item.key]} onChange={e => setSettings({ ...set...  
83 |             <div style={{ flex: 1 }}>  
84 |                 <div style={{ fontSize: '14px', fontWeight: 600, color: 'var(--text-primary)' }}>{item.label}<...  
85 |                 <div style={{ fontSize: '12px', color: 'var(--text-muted)' }}>{item.desc}</div>  
86 |             </div>  
87 |         </label>  
88 |     ))  
89 | </div>  
90 | </div>  
91 |  
92 | <div className="lms-card" style={{ padding: '28px', marginBottom: '20px' }}>  
93 |     <h2 style={{ fontSize: '16px', fontWeight: 700, margin: '0 0 20px 0', color: 'var(--text-primary)', di...  
94 |         < D a t a b a s e   s i z e = { 1 8 }   c o l o r = " v a r ( - - b r a n d - p r i m a r y ) "   / >   T h o n g   t i n  
95 |     </h2>  
96 |     <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '12px' }}>  
97 |         {[  
98 |             { label: 'Phiên bản', value: '2.4.1' },  
99 |             { label: 'Supabase Project', value: 'oijmdhrvivwvfgomonzd' },  
100 |             { label: 'Mã i tr ò Y n g', value: 'P r o d u c t i o n   ( V e r c e l )' },  
101 |             { label: 'Node.js', value: '22.x' },  
102 |             ...j.map((item, i) => (  
103 |                 <div key={i} style={{ padding: '12px 16px', borderRadius: '8px', background: 'var(--bg-elevated)' }}>  
104 |                     <div style={{ fontSize: '11px', color: 'var(--text-muted)', fontWeight: 600, marginBottom: '2px'...  
105 |                     <div style={{ fontSize: '13px', color: 'var(--text-primary)', fontWeight: 700 }}>{item.value}</div>  
106 |                 </div>  
107 |             ))  
108 |         ]</div>  
109 | </div>  
110 |  
111 |     <div style={{ display: 'flex', justifyContent: 'flex-end', gap: '12px', marginTop: '24px' }}>  
112 |         <button onClick={handleSave} disabled={saving} className="lms-btn lms-btn-primary" style={{ display: '...  
113 |             { saving ? <Loader2 size={16} style={{ animation: 'spin 1s linear infinite' }} /> : <Save size={16} />  
114 |             { s a v i n g   ?   '   a n g   l ò u . . . '   :   ' L ò u   c % u   h i n h ' }  
115 |         </button>  
116 |     </div>  
117 | </div>  
118 | );  
119 | }  
120 |
```

File: app/(dashboard)/admin/users/page.tsx (131 lines, 7.1 KB)

```
1 | 'use client';  
2 |  
3 | import React, { useEffect, useState } from 'react';  
4 | import { supabase } from '@/lib/supabase';  
5 | import { Users, Search, Shield, GraduationCap, Trash2, Mail, UserX, Loader2, ChevronDown, Calendar } from 'luc...  
6 |  
7 | export default function AdminUsersPage() {  
8 |     const [users, setUsers] = useState<any[]>([]);  
9 |     const [loading, setLoading] = useState(true);  
10 |    const [search, setSearch] = useState('');  
11 |    const [roleFilter, setRoleFilter] = useState('all');  
12 |  
13 |    useEffect(() => {  
14 |        fetchUsers();  
15 |        const channel = supabase.channel('admin-users-realtime')  
16 |        .on('postgres_changes', { event: '*', schema: 'public', table: 'profiles' }, () => fetchUsers())  
17 |        .subscribe();  
18 |        return () => { supabase.removeChannel(channel) };  
19 |    }, []);  
20 |  
21 |    async function fetchUsers() {  
22 |        const { data } = await supabase.from('profiles').select('*').order('created_at', { ascending: false });  
23 |        if (data) setUsers(data);  
24 |        setLoading(false);  
25 |    }  
26 |  
27 |    const deleteUser = async (id: string, email: string) => {  
28 |        if (!confirm(`Xóa ng ộ Y i d ộ n g $ { email } ? `)) return;  
29 |        const { error } = await supabase.from('profiles').delete().eq('id', id);  
30 |        if (error) alert(`L ậ i : ' + error.message`);  
31 |        else fetchUsers();  
32 |    };  
33 |  
34 |    const filtered = users.filter(u => {  
35 |        if (roleFilter !== 'all' && u.role !== roleFilter) return false;  
36 |        if (search && !u.full_name?.toLowerCase().includes(search.toLowerCase()) && !u.email?.toLowerCase().includ...  
37 |        return true;  
38 |    });  
39 |  
40 |    if (loading) return (  
41 |        <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', height: '100%' }}>  
42 |            <Loader2 size={48} style={{ color: 'var(--brand-primary)', animation: 'spin 1s linear infinite' }} />  
43 |        </div>  
44 |    );  
45 |  
46 |    return (  
47 |        <div style={{ maxWidth: '1200px', margin: '0 auto' }}>  
48 |            <header style={{ marginBottom: '24px' }}>  
49 |                <h1 style={{ fontSize: '28px', fontWeight: 800, margin: '0 0 4px 0', color: 'var(--text-primary)' }}>Q...  
50 |                < p   s t y l e = { {   c o l o r :   ' v a r ( - - t e x t - m u t e d ) ' ,   m a r g i n :   0 ,   f o n t S i z e :   ' 1 4  
51 |            </header>  
52 |  
53 |            <div style={{ display: 'flex', gap: '12px', marginBottom: '20px', flexWrap: 'wrap', alignItems: 'center'...  
54 |            <div style={{ position: 'relative', flex: 1, minWidth: '200px' }}>  
55 |                <Search size={16} style={{ position: 'absolute', left: '12px', top: '50%', transform: 'translateY(-5...  
56 |                <input value={search} onChange={e => setSearch(e.target.value)}  
57 |                    width: '100%', padding: '10px 12px 10px 36px', borderRadius: '10px', background: 'var(--bg-surface)',  
58 |                    border: '1px solid var(--border-default)', color: 'var(--text-primary)', fontSize: '13px', outline...  
59 |                </div>  
60 |                <div>  
61 |                    {[ 'all', 'student', 'teacher', 'admin' ].map(r => (  
62 |                        <button key={r} onClick={() => setRoleFilter(r)} style={{  
63 |                            padding: '8px 16px', borderRadius: '8px', border: '1px solid var(--border-default)',  
64 |                            background: roleFilter === r ? 'var(--brand-primary)' : 'var(--bg-surface)',  
65 |                            color: roleFilter === r ? '#fff' : 'var(--text-muted)',  
66 |                            fontWeight: 600, cursor: 'pointer', fontSize: '12px', fontFamily: 'inherit'  
67 |                        }}>  
68 |                            { r === ' a l l ' ? ' T ậ t   c ậ ' : r === ' s t u d e n t ' ? ' H ậ c   s i n h ' : r  
69 |                    </button>  
70 |                ))  
71 |            </div>  
72 |  
73 |            <div className="lms-card" style={{ overflow: 'hidden', padding: '0' }}>  
74 |                <div style={{ overflowX: 'auto' }}>  
75 |                    <table style={{ width: '100%', borderCollapse: 'collapse' }}>  
76 |                        <thead>  
77 |                            <tr style={{ borderBottom: '1px solid var(--border-default)', background: 'var(--bg-elevated)' }}>  
78 |                                { [ ' N g ộ Y i d ộ n g ' , ' E m a i l ' , ' V a i t r ộ ' , ' N g à y   t h a m   g i a ' , ' H ậ  
79 |                                <th key={h} style={{ padding: '12px 20px', color: 'var(--text-muted)', fontSize: '11px', fon...
```

```

80|         )))
81|     </tr>
82| </thead>
83| <tbody>
84|     {filtered.map((user) => (
85|       <tr key={user.id} style={{ borderBottom: '1px solid var(--border-default)' }}>
86|         <td style={{ padding: '14px 20px' }}>
87|           <div style={{ display: 'flex', alignItems: 'center', gap: '10px' }}>
88|             <div style={{
89|               width: '34px', height: '34px', borderRadius: '50%',
90|               background: 'linear-gradient(135deg, var(--brand-primary), var(--accent-blue))',
91|               display: 'flex', alignItems: 'center', justifyContent: 'center', fontWeight: 800, colo...
92|             }}>
93|               {user.full_name?.charAt(0) || '?'}
94|             </div>
95|             <div>
96|               <div style={{ color: 'var(--text-primary)', fontWeight: 700, fontSize: '14px' }}>{user...
97|               <div style={{ color: 'var(--text-muted)', fontSize: '11px' }}>ID: {user.id?.slice(0, 8...
98|             </div>
99|           </td>
100|           <td style={{ padding: '14px 20px', color: 'var(--text-primary)', fontSize: '13px' }}>{user.e...
101|           <td style={{ padding: '14px 20px' }}>
102|             <span style={{
103|               padding: '3px 10px', borderRadius: '99px', fontSize: '10px', fontWeight: 700,
104|               background: user.role === 'admin' ? 'rgba(244,106,106,0.15)' : user.role === 'teacher' ?...
105|               color: user.role === 'admin' ? 'var(--danger)' : user.role === 'teacher' ? 'var(--accent...
106|             }}>
107|               { user . role === ' admin ' ? ' Admin ' : user . role === '
108|             </span>
109|           </td>
110|           <td style={{ padding: '14px 20px', color: 'var(--text-muted)', fontSize: '12px', display: 'f...
111|           <Calendar size={12} /> {user.created_at ? new Date(user.created_at).toLocaleDateString('vi...
112|           </td>
113|           <td style={{ padding: '14px 20px' }}>
114|             <button onClick={() => deleteUser(user.id, user.email)} style={{
115|               padding: '6px 10px', borderRadius: '6px', background: 'rgba(244,67,54,0.1)', border: 'no...
116|               color: 'var(--danger)', cursor: 'pointer', fontFamily: 'inherit'
117|             }}>
118|             <Trash2 size={14} />
119|           </button>
120|           </td>
121|         </tr>
122|       )))
123|     </tbody>
124| </table>
125| </div>
126| </div>
127| </div>
128| </div>
129| );
130| }
131|

```

File: app/(dashboard)/parent/children/[studentId]/page.tsx (423 lines, 19.4 KB)

```

1| import { notFound } from 'next/navigation'
2| import { requireRole } from '@lib/auth/rbac'
3| import { createClient } from '@lib/supabase-ssr-server'
4| import ChildProgressChart from '@components/parent/ChildProgressChart'
5| const ACHIEVEMENT_DEFINITIONS: any[] = []
6| import {
7|   ArrowLeft,
8|   Award,
9|   BookOpen,
10|   Clock,
11|   Flame,
12|   GraduationCap,
13|   Percent,
14|   Trophy,
15|   TrendingUp,
16|   Activity
17| } from 'lucide-react'
18|
19| export const dynamic = 'force-dynamic'
20|
21| interface PageProps {
22|   params: Promise<{ studentId: string }>
23| }
24|
25| function formatDuration(seconds: number) {
26|   if (!seconds) return '0 phút'
27|   const mins = Math.floor(seconds / 60)
28|   if (mins < 60) return `${mins} phút`
29|   const hours = Math.floor(mins / 60)
30|   const remainingMins = mins % 60
31|   return remainingMins > 0 ? `${hours}h ${remainingMins}m` : `${hours}h`
32| }
33|
34| export default async function StudentDetailPage({ params }: PageProps) {
35|   // 1. Xác thực và bắt đầu route
36|   const user = await requireRole(['parent', 'admin'])
37|   const { studentId } = await params
38|
39|   const supabase = await createClient()
40|
41|   // 2. Xác thực quyền liên kết
42|   const { data: link, error: linkError } = await supabase
43|     .from('parent_student_links')
44|     .select('*')
45|     .eq('parent_id', user.id)
46|     .eq('student_id', studentId)
47|     .eq('status', 'active')
48|     .single()
49|
50|   if (linkError || !link) {
51|     notFound()
52|   }
53|
54|   // 3. Fetch thông tin học sinh
55|   const { data: student } = await supabase
56|     .from('profiles')
57|     .select('id, full_name, student_code, school_name, avatar_url, level, xp')
58|     .eq('id', studentId)
59|     .single()
60|
61|   if (!student) {
62|     notFound()
63|   }
64|
65|   // 4. Fetch tiến trình bài học & tính toán stats
66|   const { data: progressData } = await supabase
67|     .from('lesson_progress')
68|     .select('status, time_spent_seconds, completed_at, last_accessed')

```

```

69 | .eq('student_id', studentId)
70 |
71 | const completedLessonsCount = progressData?.filter(r => r.status === 'completed').length ?? 0
72 | const startedLessonsCount = progressData?.length ?? 0
73 | const totalSeconds = progressData?.reduce((sum, r) => sum + (r.time_spent_seconds ?? 0), 0) ?? 0
74 |
75 | // 5. Tính toán streak híc t - p
76 | const { data: streakData } = await supabase
77 |   .from('lesson_progress')
78 |   .select('last_accessed')
79 |   .eq('student_id', studentId)
80 |   .order('last_accessed', { ascending: false })
81 |
82 | let streak = 0
83 | if (streakData && streakData.length > 0) {
84 |   const uniqueDates = Array.from(new Set(streakData.map(d => {
85 |     const date = d.last_accessed ? new Date(d.last_accessed) : new Date()
86 |     return date.toISOString().split('T')[0]
87 |   }))).sort().reverse()
88 |
89 |   if (uniqueDates.length > 0) {
90 |     const today = new Date()
91 |     today.setHours(0, 0, 0, 0)
92 |
93 |     let checkDate = new Date(today)
94 |     const firstDateStr = uniqueDates[0]
95 |     const firstDate = new Date(firstDateStr)
96 |     firstDate.setHours(0, 0, 0, 0)
97 |
98 |     const diffStart = (today.getTime() - firstDate.getTime()) / (1000 * 3600 * 24)
99 |     if (diffStart <= 1) {
100 |       if (diffStart === 0 || diffStart === 1) {
101 |         checkDate = firstDate
102 |       }
103 |       for (const dateStr of uniqueDates) {
104 |         const d = new Date(dateStr)
105 |         d.setHours(0, 0, 0, 0)
106 |
107 |         if (d.getTime() === checkDate.getTime()) {
108 |           streak++
109 |           checkDate.setDate(checkDate.getDate() - 1)
110 |         } else if (d.getTime() < checkDate.getTime()) {
111 |           break
112 |         }
113 |       }
114 |     }
115 |   }
116 | }
117 |
118 | // 6. Tính toán i Ầ m tr ̣ c nghi ̣ m trung bình
119 | const { data: quizData } = await supabase
120 |   .from('quiz_attempts')
121 |   .select('score')
122 |   .eq('student_id', studentId)
123 |   .eq('status', 'graded')
124 |
125 | const avgScore = quizData && quizData.length > 0
126 |   ? Math.round(quizData.reduce((s, r) => s + r.score, 0) / quizData.length)
127 |   : 0
128 |
129 | // 7. T i o d i l i ̣ u b i Ầ u ó 3 0 n g à y q u a
130 | const counts: Record<string, number> = {}
131 | const today = new Date()
132 | for (let i = 29; i >= 0; i--) {
133 |   const d = new Date(today)
134 |   d.setDate(d.getDate() - i)
135 |   const formatted = `${d.getDate()}/${d.getMonth() + 1}`
136 |   counts[formatted] = 0
137 | }
138 |
139 | progressData?.forEach(row => {
140 |   if (row.completed_at) {
141 |     const d = new Date(row.completed_at)
142 |     const formatted = `${d.getDate()}/${d.getMonth() + 1}`
143 |     if (counts[formatted] !== undefined) {
144 |       counts[formatted] += 1
145 |     }
146 |   }
147 | })
148 |
149 | const dailyProgress = Object.keys(counts).map(date => ({ date, count: counts[date] }))
150 |
151 | // 8. F e t c h b à i h í c & c h i t i ̣ t t i ̣ n Û
152 | const { data: dbLessons } = await supabase
153 |   .from('lesson_progress')
154 |   .select('id, lesson_id, status, completed_at, completion_percentage, time_spent_seconds, lessons(title, ty...
155 |   .eq('student_id', studentId)
156 |   .order('last_accessed', { ascending: false })
157 |
158 | const lessons = (dbLessons || []).map((row: any) => ({
159 |   id: row.id,
160 |   lesson_id: row.lesson_id,
161 |   lesson_title: row.lessons?.title || 'Bài híc gi ̣ l - p',
162 |   type: row.lessons?.type || 'L ̣ thuy ̣ t',
163 |   status: row.status,
164 |   completed_at: row.completed_at,
165 |   completion_percentage: row.completion_percentage ?? 0,
166 |   time_spent_seconds: row.time_spent_seconds ?? 0
167 | }))
168 |
169 | // 9. F e t c h b à i t h i / q u i z g ̣ n Ầ y
170 | const { data: dbQuizAttempts } = await supabase
171 |   .from('quiz_attempts')
172 |   .select('id, score, status, created_at, quizzes(title)')
173 |   .eq('student_id', studentId)
174 |   .eq('status', 'graded')
175 |   .order('created_at', { ascending: false })
176 |   .limit(10)
177 |
178 | const quizResults = (dbQuizAttempts || []).map((row: any) => ({
179 |   id: row.id,
180 |   quiz_title: row.quizzes?.title || 'Bài ki Ầ m tr ̣ a',
181 |   score: row.score,
182 |   created_at: row.created_at,
183 |   grade: row.score >= 90 ? 'X u ̣ t s ̣ c' : row.score >= 80 ? 'Gi ̣ i' : row
184 | }))
185 |
186 | // 10. F e t c h t h Ầ n h t i ̣ c h i t Ầ ̣ c
187 | const { data: dbAchievements } = await supabase
188 |   .from('student_achievements')
189 |   .select('achievement_id')
190 |   .eq('student_id', studentId)
191 |
192 | const earnedIds = (dbAchievements || []).map(a => a.achievement_id)

```

```

193 | const achievements = ACHIEVEMENT_DEFINITIONS.map(def => ({
194 |   ...def,
195 |   earned: earnedIds.includes(def.id)
196 | }))
197 |
198 | return (
199 |   <div className="min-h-screen p-6 md:p-8 max-w-6xl mx-auto space-y-8" style={{ background: 'var(--bg-base)'...
200 |     { / * Q u a y l i i l i n k * / }
201 |   <div>
202 |     <a
203 |       href="/parent/dashboard"
204 |       className="inline-flex items-center gap-2 text-xs font-bold uppercase tracking-wider transition-colors"
205 |       style={{ color: 'var(--text-muted)' }}
206 |     >
207 |       <ArrowLeft size={14} />
208 |       Q u a y l i i b Ë n g i Á u k h i Ä n
209 |     </a>
210 |   </div>
211 |
212 |   { / * SECTION 1 - HEADER * / }
213 |   <div className="rounded-2xl p-6 flex flex-col md:flex-row justify-between items-start md:items-center ga...
214 |     <div className="absolute -top-24 -left-24 w-48 h-48 rounded-full blur-3xl pointer-events-none" style={{...
215 |
216 |       <div className="flex items-center gap-4 relative z-10">
217 |         {student.avatar_url ? (
218 |           <img
219 |             src={student.avatar_url}
220 |             className="w-16 h-16 md:w-20 md:h-20 rounded-full object-cover"
221 |             style={{ border: '2px solid var(--border-default)' }}
222 |             alt={student.full_name}
223 |           />
224 |         ) : (
225 |           <div className="w-16 h-16 md:w-20 md:h-20 rounded-full flex items-center justify-center text-3xl f...
226 |             {student.full_name[0] || 'S'}
227 |           </div>
228 |         )}
229 |       <div>
230 |         <h1 className="text-xl md:text-2xl font-bold" style={{ color: 'var(--text-primary)' }}>{student.fu...
231 |         <div className="text-xs mt-1" style={{ color: 'var(--text-mu...
232 |           <div className="flex items-center gap-2 mt-2">
233 |             <span className="font-mono text-[10px] px-2 py-0.5 rounded uppercase tracking-widest" style={{ b...
234 |               {student.student_code}
235 |             </span>
236 |             <span className="text-[10px] font-bold px-2 py-0.5 rounded-full uppercase tracking-wider" style=...
237 |               C % p { s t u d e n t . l e v e l | | 1 }
238 |           </span>
239 |         </div>
240 |       </div>
241 |     </div>
242 |
243 |     <div className="rounded-xl px-5 py-4 min-w-[140px] text-center md:text-right relative z-10 self-stretc...
244 |       <span className="text-[10px] uppercase font-bold tracking-wider" style={{ color: 'var(--text-muted)' }}...
245 |       <span className="text-2xl font-black" style={{ color: 'var(--brand-primary)' }}>{student.xp || 0} XP...
246 |     </div>
247 |   </div>
248 |
249 |   { / * SECTION 2 - STATS TÔNG * / }
250 |   <div className="grid grid-cols-2 md:grid-cols-4 gap-4">
251 |     { / * Bài hoàn thành * / }
252 |     <div className="rounded-2xl p-5 flex items-center gap-4" style={{ background: 'var(--bg-surface)', bor...
253 |       <div className="w-12 h-12 rounded-xl flex items-center justify-center flex-shrink-0" style={{ backgr...
254 |         <BookOpen size={22} />
255 |       </div>
256 |       <div>
257 |         <div className="text-2xl font-black" style={{ color: 'var(--text-primary)' }}>{completedLessonsCou...
258 |         <div className="text-[10px] uppercase font-bold tracking-wider" style={{ color: 'var(--text-muted)' }}...
259 |       </div>
260 |     </div>
261 |
262 |     { / * Gi Ý h í c * / }
263 |     <div className="rounded-2xl p-5 flex items-center gap-4" style={{ background: 'var(--bg-surface)', bor...
264 |       <div className="w-12 h-12 rounded-xl flex items-center justify-center flex-shrink-0" style={{ backgr...
265 |         <Clock size={22} />
266 |       </div>
267 |       <div>
268 |         <div className="text-2xl font-black" style={{ color: 'var(--text-primary)' }}>{formatDuration(tota...
269 |         <div className="text-[10px] uppercase font-bold tracking-wider" style={{ color: 'var(--text-muted)' }}...
270 |       </div>
271 |     </div>
272 |
273 |     { / * i Ä m T B q u i z * / }
274 |     <div className="rounded-2xl p-5 flex items-center gap-4" style={{ background: 'var(--bg-surface)', bor...
275 |       <div className="w-12 h-12 rounded-xl flex items-center justify-center flex-shrink-0" style={{ backgr...
276 |         <Percent size={22} />
277 |       </div>
278 |       <div>
279 |         <div className="text-2xl font-black" style={{ color: 'var(--text-primary)' }}>{avgScore}%</div>
280 |         <div className="text-[10px] uppercase font-bold tracking-wider" style={{ color: 'var(--text-muted)' }}...
281 |       </div>
282 |     </div>
283 |
284 |     { / * Streak * / }
285 |     <div className="rounded-2xl p-5 flex items-center gap-4" style={{ background: 'var(--bg-surface)', bor...
286 |       <div className="w-12 h-12 rounded-xl flex items-center justify-center flex-shrink-0" style={{ backgr...
287 |         <Flame size={22} />
288 |       </div>
289 |       <div>
290 |         <div className="text-2xl font-black" style={{ color: 'var(--text-primary)' }}>{streak} ngày</div>
291 |         <div className="text-[10px] uppercase font-bold tracking-wider" style={{ color: 'var(--text-muted)' }}...
292 |       </div>
293 |     </div>
294 |   </div>
295 |
296 |   <div className="grid grid-cols-1 lg:grid-cols-3 gap-6">
297 |     { / * SECTION 3 - B I Ä U Ò T I % N Ø 3 Ø N G Ä Y * / }
298 |     <div className="lg:col-span-2">
299 |       <ChildProgressChart data={dailyProgress} />
300 |     </div>
301 |
302 |     { / * SECTION 6 - T H Ä N H T í C H T - ä c * / }
303 |     <div className="rounded-2xl p-5" style={{ background: 'var(--bg-surface)', border: '1px solid var(--bo...
304 |       <div className="mb-4">
305 |         <h3 className="text-sm font-bold flex items-center gap-2" style={{ color: 'var(--text-primary)' }}>
306 |           <Trophy size={16} style={{ color: 'var(--brand-primary)' }} />
307 |           T h ä n h t í c h c ä c o n
308 |         </h3>
309 |         <p className="text-[10px] uppercase font-bold tracking-wider mt-0.5" style={{ color: 'var(--text-m...
310 |       </div>
311 |       <div className="grid grid-cols-3 gap-3">
312 |         {achievements.map((ach) => (
313 |           <div
314 |             key={ach.id}
315 |             className="p-3 rounded-xl border flex flex-col items-center justify-center text-center transit...
316 |             style={{

```



```

317|         background: ach.earned ? 'var(--bg-elevated)' : 'rgba(0,0,0,0.2)',
318|         border: '1px solid var(--border-subtle)',
319|         opacity: ach.earned ? 1 : 0.4
320|       }}
321|       title={ach.description}
322|     >
323|       <div className="text-2xl mb-1.5">{ach.icon}</div>
324|       <div className="text-[10px] font-bold truncate w-full" style={{ color: ach.earned ? 'var(--tex...
325|         {ach.title}
326|       </div>
327|     </div>
328|   )))
329| </div>
330| </div>
331| </div>
332|
333| <div className="grid grid-cols-1 lg:grid-cols-2 gap-6">
334|   { / *   S E C T I O N   4   -   D A N H   S Á C H   B À I   H Ì C   * / }
335|   <div className="rounded-2xl p-5 space-y-4" style={{ background: 'var(--bg-surface)', border: '1px soli...
336|     <div>
337|       <h3 className="text-sm font-bold flex items-center gap-2" style={{ color: 'var(--text-primary)' }}>
338|         <Activity size={16} style={{ color: '#00b4d8' }} />
339|         T i ế n   t r ầ n h   h ì c   t - p
340|       </h3>
341|       <p className="text-[10px] uppercase font-bold tracking-wider mt-0.5" style={{ color: 'var(--text-m...
342|     </div>
343|
344|     <div className="space-y-2 max-h-[350px] overflow-y-auto pr-1">
345|       {lessons.length === 0 ? (
346|         <div className="text-center py-8 text-xs" style={{ color:
347|       ) : (
348|         lessons.map((lesson) => (
349|           <div>
350|             key={lesson.id}
351|             className="rounded-xl p-3.5 flex justify-between items-center gap-4 transition-colors" style...
352|           >
353|             <div className="min-w-0 flex-1">
354|               <div className="font-bold text-xs truncate" style={{ color: 'var(--text-primary)' }}>{less...
355|               <div className="flex items-center gap-2 mt-1.5 text-[10px] font-bold" style={{ color: 'var...
356|                 <span className="px-2 py-0.5 rounded uppercase" style={{ background: 'rgba(255,255,0...
357|                 <span>*</span>
358|                 <span>H í c   { formatDuration( lesson . time _ spent _ second
359|               </div>
360|             </div>
361|
362|             <div className="text-right flex-shrink-0">
363|               {lesson.status === 'completed' ? (
364|                 <span className="text-[10px] font-bold px-2.5 py-1 rounded-full uppercase tracking-wider...
365|                 Hoàn thành
366|               </span>
367|             ) : (
368|               <span className="text-[10px] font-bold px-2.5 py-1 rounded-full uppercase tracking-wider...
369|               {lesson.completion_percentage}%
370|             </span>
371|           </div>
372|         </div>
373|       </div>
374|     )))
375|   </div>
376| </div>
377| </div>
378|
379|   { / *   S E C T I O N   5   -   K Ể T   Q U Ế   Q U I Z   G Ì N   Â Y   * / }
380|   <div className="rounded-2xl p-5 space-y-4" style={{ background: 'var(--bg-surface)', border: '1px soli...
381|     <div>
382|       <h3 className="text-sm font-bold flex items-center gap-2" style={{ color: 'var(--text-primary)' }}>
383|         <Award size={16} style={{ color: '#a85f7' }} />
384|         K ể t   q u ế   t r ầ n g   h ì c   m
385|       </h3>
386|       <p className="text-[10px] uppercase font-bold tracking-wider mt-0.5" style={{ color: 'var(--text-m...
387|     </div>
388|
389|     <div className="space-y-2 max-h-[350px] overflow-y-auto pr-1">
390|       {quizResults.length === 0 ? (
391|         <div className="text-center py-8 text-xs" style={{ color:
392|       ) : (
393|         quizResults.map((attempt) => (
394|           <div>
395|             key={attempt.id}
396|             className="rounded-xl p-3.5 flex justify-between items-center gap-4 transition-colors" style...
397|           >
398|             <div className="min-w-0 flex-1">
399|               <div className="font-bold text-xs truncate" style={{ color: 'var(--text-primary)' }}>{atte...
400|               <div className="text-[10px] font-bold mt-1.5" style={{ color: 'var(--text-muted)' }}>
401|                 i t :   <span style={{ color: 'var(--brand-primary)'
402|               </div>
403|             </div>
404|
405|             <div className="text-right flex-shrink-0">
406|               <span className="text-[10px] font-bold px-2.5 py-1 rounded-full uppercase tracking-wider b...
407|               background: attempt.score >= 80 ? 'rgba(0,212,170,0.1)' : attempt.score >= 50 ? 'rgba(25...
408|               borderColor: attempt.score >= 80 ? 'rgba(0,212,170,0.2)' : attempt.score >= 50 ? 'rgba(2...
409|               color: attempt.score >= 80 ? 'var(--brand-primary)' : attempt.score >= 50 ? '#ff9800' : ...
410|             </div>
411|             {attempt.grade}
412|           </span>
413|         </div>
414|       </div>
415|     )))
416|   </div>
417| </div>
418| </div>
419| </div>
420| </div>
421| )
422| }
423|

```

File: app/(dashboard)/parent/children/page.tsx (66 lines, 3.0 KB)

```

1| import { requireRole } from '@lib/auth/rbac'
2| import { createClient } from '@lib/supabase-ssr-server'
3| import { ChildCard } from '@components/parent/ChildCard'
4| import { Users, Plus } from 'lucide-react'
5|
6| export const dynamic = 'force-dynamic'
7|
8| export default async function ParentChildrenPage() {
9|   // 1. Xác th ã c v à b ế o   v ậ t   r ồ i   t h ế t
10|   const user = await requireRole(['parent', 'admin'])
11|
12|   // 2. F e t c h   d a n h   s á c h   c o n   a n g   t h e o   đ ể i
13|   const supabase = await createClient()

```

```

14 | const { data: children } = await supabase
15 |   .from('child_summary_for_parent')
16 |   .select('*')
17 |   .eq('parent_id', user.id)
18 |   .order('student_name')
19 |
20 | return (
21 |   <div className="min-h-screen p-6 md:p-8 max-w-6xl mx-auto" style={{ background: 'var(--bg-base)', color: '...
22 |     <div className="flex flex-col md:flex-row md:items-center justify-between gap-4 mb-8">
23 |       <div>
24 |         <h1 className="text-2xl md:text-3xl font-bold flex items-center gap-2" style={{ color: 'var(--text-p...
25 |           H ỉ c s i n h a n g t h e o d ồ i
26 |         </h1>
27 |         <p className="mt-1 text-sm" style={{ color: 'var(--text-muted)' }}>
28 |           D a n h s á c h t ẻ t c ẻ t á i k h o ẻ n c o n ấ ẻ ỏ á c l i ẻ n k ẻ t v ỏ i b i n .
29 |         </p>
30 |       </div>
31 |       <a
32 |         href="/parent/link-child"
33 |         className="inline-flex items-center gap-2 px-4 py-2.5 rounded-xl font-semibold text-sm transition-al...
34 |         style={{ background: 'rgba(0, 212, 170, 0.1)', border: '1px solid rgba(0, 212, 170, 0.3)', color: 'v...
35 |       >
36 |         <Plus size={16} />
37 |         L i ẻ n k ẻ t t h ẻ m c o n
38 |       </a>
39 |     </div>
40 |
41 |     {!!children || children.length === 0} ? (
42 |       <div className="rounded-2xl p-8 md:p-12 text-center max-w-2xl mx-auto my-12" style={{ background: 'var...
43 |         <div className="text-5xl mb-5">Ừ Ờ Ỡ Ỡ Ỡ</div>
44 |         <h3 className="text-xl font-bold mb-2" style={{ color: 'var(--text-muted)' }}>
45 |           <p className="text-sm mb-6 max-w-md mx-auto leading-relaxed" style={{ color: 'var(--text-muted)' }}>
46 |             N h - p m ả h ỉ c s i n h c ẻ a c o n b i n ấ b ẻ t s ẻ c - p n h - t i ẻ m q u i z ,
47 |           </p>
48 |           <a
49 |             href="/parent/link-child"
50 |             className="inline-flex items-center gap-2 px-5 py-2.5 rounded-xl font-semibold text-sm transition-...
51 |             style={{ background: 'rgba(0, 212, 170, 0.1)', border: '1px solid rgba(0, 212, 170, 0.3)', color: ...
52 |           >
53 |             L i ẻ n k ẻ t c o n n g a y
54 |           </a>
55 |         </div>
56 |       ) : (
57 |         <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-5">
58 |           {children.map((child) => (
59 |             <ChildCard key={child.student_id} child={child} />
60 |           ))}
61 |         </div>
62 |       )}
63 |     </div>
64 |   )
65 | }
66 |

```

File: app/(dashboard)/parent/dashboard/error.tsx (50 lines, 1.7 KB)

```

1 | // Path: app/(dashboard)/parent/dashboard/error.tsx
2 | 'use client'
3 | import { useEffect } from 'react'
4 |
5 | export default function ParentDashboardError({
6 |   error, reset
7 | }: { error: Error & { digest?: string }; reset: () => void }) {
8 |   useEffect(() => { console.error(error); }, [error])
9 |   return (
10 |     <div style={{
11 |       display: 'flex',
12 |       flexDirection: 'column',
13 |       alignItems: 'center',
14 |       justifyContent: 'center',
15 |       height: '60vh',
16 |       gap: '24px',
17 |       textAlign: 'center',
18 |       padding: '24px'
19 |     }}>
20 |       <div style={{ fontSize: '48px' }}>B</div>
21 |       <div>
22 |         <h2 style={{ fontSize: '20px', fontWeight: 700, color: 'var(--text-primary)', marginBottom: '8px' }}>
23 |           C ỏ l ẻ i x ẻ y r a k h i t ẻ i D a s h b o a r d
24 |         </h2>
25 |         <p style={{ color: 'var(--text-secondary)', fontSize: '14px', marginBottom: '24px', maxWidth: '380px' }}>
26 |           {error.message || 'H ẻ t h ẻ n g g . p s ẻ c ẻ n t ẻ i d i l i ẻ u r e a l t i m e . V
27 |         </p>
28 |         <button
29 |           onClick={reset}
30 |           style={{
31 |             padding: '12px 28px',
32 |             background: 'rgba(0, 212, 170, 0.15)',
33 |             border: '1px solid rgba(0, 212, 170, 0.3)',
34 |             color: 'var(--brand-primary)',
35 |             borderRadius: '12px',
36 |             fontSize: '14px',
37 |             fontWeight: 600,
38 |             cursor: 'pointer',
39 |             transition: 'colors 0.2s'
40 |           }}
41 |           onMouseOver={e => e.currentTarget.style.background = 'rgba(0, 212, 170, 0.25)'}
42 |           onMouseOut={e => e.currentTarget.style.background = 'rgba(0, 212, 170, 0.15)'}
43 |         >
44 |           T ẻ i l i i d i l i ẻ u
45 |         </button>
46 |       </div>
47 |     </div>
48 |   )
49 | }
50 |

```

File: app/(dashboard)/parent/dashboard/loading.tsx (59 lines, 2.5 KB)

```

1 | // Path: app/(dashboard)/parent/dashboard/loading.tsx
2 | export default function ParentDashboardLoading() {
3 |   return (
4 |     <div style={{ maxWidth: '1000px', margin: '0 auto', display: 'flex', flexDirection: 'column', gap: '24px' }}>
5 |
6 |       <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
7 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
8 |           <div className="skeleton animate-pulse" style={{ height: '32px', width: '250px', background: '#1d1f2...
9 |           <div className="skeleton animate-pulse" style={{ height: '18px', width: '180px', background: '#1d1f2...
10 |         </div>
11 |         <div className="skeleton animate-pulse" style={{ height: '32px', width: '120px', background: '#1d1f2a'...
12 |       </div>

```

```

13 |     </div>
14 |
15 |     { /* Child Card Skeleton */ }
16 |     <div className="skeleton animate-pulse" style={{
17 |       height: '112px',
18 |       background: '#1d1f2a',
19 |       borderRadius: '24px',
20 |       display: 'flex',
21 |       padding: '24px',
22 |       alignItems: 'center',
23 |       gap: '20px'
24 |     }}>
25 |       <div style={{ width: '64px', height: '64px', borderRadius: '50%', background: '#2e3048' }} />
26 |       <div style={{ flex: 1, display: 'flex', flexDirection: 'column', gap: '10px' }}>
27 |         <div style={{ height: '20px', width: '180px', background: '#2e3048', borderRadius: '6px' }} />
28 |         <div style={{ height: '14px', width: '240px', background: '#2e3048', borderRadius: '4px' }} />
29 |       </div>
30 |     </div>
31 |
32 |     { /* Grid Stats Skeleton */ }
33 |     <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(200px, 1fr))', gap: '16px' }}>
34 |       {[...Array(4)].map((_, i) => (
35 |         <div key={i} className="skeleton animate-pulse" style={{ height: '96px', background: '#1d1f2a', bord...
36 |       )})}
37 |     </div>
38 |
39 |     { /* Activity Log Skeleton */ }
40 |     <div className="skeleton animate-pulse" style={{ height: '350px', background: '#1d1f2a', borderRadius: '...'
41 |
42 |     <style dangerouslySetInnerHTML={{ __html: `
43 |       .skeleton {
44 |         position: relative;
45 |         overflow: hidden;
46 |       }
47 |       @keyframes pulse {
48 |         0%, 100% { opacity: 0.6; }
49 |         50% { opacity: 0.3; }
50 |       }
51 |       .animate-pulse {
52 |         animation: pulse 1.8s infinite ease-in-out;
53 |       }
54 |     `}} />
55 |
56 |   </div>
57 | )
58 | }
59 |

```

File: app/(dashboard)/parent/dashboard/page.tsx (75 lines, 2.0 KB)

```

1 | // Path: app/(dashboard)/parent/dashboard/page.tsx
2 | import { createServerClient } from '@supabase/ssr'
3 | import { cookies } from 'next/headers'
4 | import { redirect } from 'next/navigation'
5 | import { ParentDashboardClient } from './ParentDashboardClient'
6 |
7 | export const dynamic = 'force-dynamic'
8 |
9 | export default async function ParentDashboardPage() {
10 |   const cookieStore = await cookies()
11 |   const supabase = createServerClient(
12 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
13 |     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!,
14 |     { cookies: { getAll: () => cookieStore.getAll(), setAll: () => {} } }
15 |   )
16 |
17 |   const { data: { user }, error: authErr } = await supabase.auth.getUser()
18 |   if (!user || authErr) {
19 |     redirect('/login')
20 |   }
21 |
22 |   // Fetch parent profile role
23 |   const { data: profile } = await supabase
24 |     .from('profiles')
25 |     .select('role, full_name')
26 |     .eq('id', user.id)
27 |     .single()
28 |
29 |   if (profile?.role !== 'parent') {
30 |     // If not a parent, redirect back to home/dashboard
31 |     redirect('/student')
32 |   }
33 |
34 |   // Query confirmed linked children
35 |   const { data: links, error: linksErr } = await supabase
36 |     .from('parent_student_links')
37 |     .select(
38 |       `
39 |         id,
40 |         relationship,
41 |         linked_at,
42 |         student:profiles!student_id (
43 |           id,
44 |           full_name,
45 |           avatar_url,
46 |           student_code,
47 |           school_name,
48 |           xp,
49 |           level,
50 |           is_online,
51 |           last_seen_at
52 |         )
53 |       `
54 |     )
55 |     .eq('parent_id', user.id)
56 |     .eq('status', 'active')
57 |     .order('linked_at', { ascending: false })
58 |
59 |   if (linksErr) {
60 |     console.error('Error fetching linked children:', linksErr)
61 |   }
62 |
63 |   // Format array to bypass any potential typing issues
64 |   const linkedChildren = (links ?? []).map((link: any) => ({
65 |     id: link.id,
66 |     relationship: link.relationship,
67 |     student: link.student
68 |   }))
69 |
70 |   return (
71 |     <ParentDashboardClient
72 |       parentName={profile?.full_name ?? 'Phá huynh'}
73 |       linkedChildren={linkedChildren}
74 |     />
75 |   )

```

$$\begin{array}{l|l} 74 & \\ 75 & \end{array} \quad \}$$

File: app/(dashboard)/parent/dashboard/ParentDashboardClient.tsx (486 lines, 19.7 KB)

```

1 // Path: app/(dashboard)/parent/dashboardClient.tsx
2 'use client'
3
4 import { useState } from 'react'
5 import { useChildRealtime } from '@/hooks/useChildRealtime'
6 import { formatDistanceToNow } from 'date-fns'
7 import { vi } from 'date-fns/locale'
8 import Link from 'next/link'
9
10 interface LinkedChild {
11   id: string
12   relationship: string
13   student: {
14     id: string
15     full_name: string
16     avatar_url: string | null
17     student_code: string
18     school_name: string | null
19     xp: number
20     level: number
21     is_online: boolean
22     last_seen_at: string | null
23   }
24 }
25
26 export function ParentDashboardClient({
27   parentName,
28   linkedChildren,
29 }: {
30   parentName: string
31   linkedChildren: LinkedChild[]
32 }) {
33   const [selectedChildId, setSelectedChildId] = useState<string | null>(
34     linkedChildren[0]?.student?.id ?? null
35   )
36
37   // Subscribes to real-time websocket monitoring for the selected child
38   const realtimeData = useChildRealtime(selectedChildId)
39
40   const currentChildLink = linkedChildren.find(
41     l => l.student?.id === selectedChildId
42   )
43   const selectedChild = currentChildLink?.student
44
45   // Format connection status badge styling
46   const getConnectionBadgeClass = () => {
47     switch (realtimeData.connectionStatus) {
48       case 'connected':
49         return {
50           bg: 'rgba(0, 212, 170, 0.1)',
51           border: 'rgba(0, 212, 170, 0.3)',
52           text: 'var(--brand-primary)',
53           dot: 'var(--brand-primary)',
54           label: 'i Realtime'
55         }
56       case 'connecting':
57         return {
58           bg: 'rgba(240, 160, 48, 0.1)',
59           border: 'rgba(240, 160, 48, 0.3)',
60           text: '#f0a030',
61           dot: '#f0a030',
62           label: 'ang k t n i . . . '
63         }
64       case 'disconnected':
65         return {
66           bg: 'rgba(240, 48, 96, 0.1)',
67           border: 'rgba(240, 48, 96, 0.3)',
68           text: '#f03060',
69           dot: '#f03060',
70           label: 'M t k t n i '
71         }
72       case 'error':
73       default:
74         return {
75           bg: 'rgba(240, 48, 96, 0.1)',
76           border: 'rgba(240, 48, 96, 0.3)',
77           text: '#f03060',
78           dot: '#f03060',
79           label: 'L x i k t n i '
80         }
81     }
82   }
83
84   const badge = getConnectionBadgeClass()
85
86   return (
87     <div style={{ maxWidth: '1000px', margin: '0 auto', display: 'flex', flexDirection: 'column', gap: '24px' }}>
88       {/* Header section */}
89       <header style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
90         <div>
91           <h1 style={{ fontSize: '28px', fontWeight: 800, color: 'var(--text-primary)', marginBottom: '4px' }}>
92             Xin chào, {parentName} 00UK
93           </h1>
94           <p style={{ color: 'var(--text-secondary)', fontSize: '14px', margin: 0 }}>
95             Theo dõi {linkedChildren.length} híc sinh trong không gian
96           </p>
97         </div>
98       </header>
99
100       {/* Realtime WebSocket connection badge */}
101       <div style={{
102         display: 'flex',
103         alignItems: 'center',
104         gap: '8px',
105         padding: '6px 14px',
106         borderRadius: '50px',
107         background: badge.bg,
108         border: '1px solid ${badge.border}',
109         color: badge.text,
110         fontSize: '12px',
111         fontWeight: 600
112       }}>
113         <span style={{
114           width: '8px',
115           height: '8px',
116           borderRadius: '50%',
117           background: badge.dot,
118           boxShadow: realtimeData.connectionStatus === 'connected' || realtimeData.connectionStatus === 'con...

```

```

119 |         display: 'inline-block',
120 |         animation: realtimeData.connectionStatus === 'connected' || realtimeData.connectionStatus === 'con...
121 |     } } />
122 |     {badge.label}
123 | </div>
124 | </header>
125 |
126 | { /* Tabs Selector for Multi-Children links */ }
127 | { linkedChildren.length > 1 && (
128 |   <div style={{ display: 'flex', gap: '10px', overflowX: 'auto', paddingBottom: '4px' }}>
129 |     { linkedChildren.map(link => {
130 |       const isSelected = selectedChildId === link.student?.id
131 |       return (
132 |         <button
133 |           key={link.student?.id}
134 |           onClick={() => setSelectedChildId(link.student?.id)}
135 |           style={{
136 |             display: 'flex',
137 |             alignItems: 'center',
138 |             gap: '10px',
139 |             padding: '10px 20px',
140 |             borderRadius: '12px',
141 |             background: isSelected ? 'rgba(0, 212, 170, 0.1)' : 'var(--bg-surface)',
142 |             border: isSelected ? '1px solid rgba(0, 212, 170, 0.4)' : '1px solid var(--border-subtle)',
143 |             color: isSelected ? 'var(--brand-primary)' : 'var(--text-secondary)',
144 |             fontWeight: 600,
145 |             fontSize: '14px',
146 |             cursor: 'pointer',
147 |             transition: 'all 0.2s',
148 |             whiteSpace: 'nowrap'
149 |           }}
150 |         >
151 |           <div style={{
152 |             width: '24px',
153 |             height: '24px',
154 |             borderRadius: '50%',
155 |             background: 'rgba(0, 212, 170, 0.2)',
156 |             display: 'flex',
157 |             alignItems: 'center',
158 |             justifyContent: 'center',
159 |             fontSize: '11px',
160 |             color: '#fff'
161 |           }}>
162 |             {link.student?.full_name[0].toUpperCase()}
163 |           </div>
164 |           {link.student?.full_name}
165 |           { /* Real-time Online indicators inside tabs */ }
166 |           <span style={{
167 |             width: '6px',
168 |             height: '6px',
169 |             borderRadius: '50%',
170 |             background: (realtimeData.isOnline && isSelected) || (link.student?.is_online && !isSelected...
171 |           }} />
172 |         </button>
173 |       )
174 |     } } )
175 |   </div>
176 | ) }
177 |
178 | { /* Empty State: No linked children */ }
179 | { linkedChildren.length === 0 && (
180 |   <div style={{
181 |     background: 'var(--bg-surface)',
182 |     border: '1.5px dashed rgba(0, 212, 170, 0.25)',
183 |     borderRadius: '24px',
184 |     padding: '60px 40px',
185 |     textAlign: 'center',
186 |     display: 'flex',
187 |     flexDirection: 'column',
188 |     alignItems: 'center',
189 |     gap: '16px'
190 |   }}>
191 |     <span style={{ fontSize: '48px' }}>0</span>
192 |     <h3 style={{ fontSize: '20px', fontWeight: 700, color: 'var(--text-primary)', margin: 0 }}>
193 |       C h o a l i e n k ử t t à i k h o ẻ n c o n
194 |     </h3>
195 |     <p style={{ color: 'var(--text-secondary)', fontSize: '14px', maxWidth: '380px', margin: 0, lineHeig...
196 |       N h - p m ả h ỉ c s ỉ n h c ả n h ả n c ẻ a c o n ( P C M - X X X X X X X )  Ǻ  ó n g b ù v
197 |   </p>
198 |   <Link
199 |     href="/parent/link-child"
200 |     style={{
201 |       display: 'inline-flex',
202 |       alignItems: 'center',
203 |       gap: '8px',
204 |       padding: '12px 28px',
205 |       background: 'var(--brand-primary)',
206 |       color: '#000',
207 |       borderRadius: '12px',
208 |       fontWeight: 700,
209 |       textDecoration: 'none',
210 |       marginTop: '8px'
211 |     }}
212 |   >
213 |     + L i e n k ử t t à i k h o ẻ n c o n
214 |   </Link>
215 | </div>
216 | ) }
217 |
218 | { /* Active Child Dashboard */ }
219 | { selectedChild && (
220 |   <div style={{ display: 'flex', flexDirection: 'column', gap: '24px' }}>
221 |     { /* Child Header Card */ }
222 |     <div style={{
223 |       background: 'var(--bg-surface)',
224 |       border: '1px solid var(--border-subtle)',
225 |       borderRadius: '24px',
226 |       padding: '24px',
227 |       display: 'flex',
228 |       alignItems: 'center',
229 |       gap: '20px',
230 |       flexWrap: 'wrap'
231 |     }}>
232 |       { /* Avatar & Online status indicator */ }
233 |       <div style={{ position: 'relative', width: '64px', height: '64px' }}>
234 |         {selectedChild.avatar_url ? (
235 |           <img
236 |             src={selectedChild.avatar_url}
237 |             alt={selectedChild.full_name}
238 |             style={{ width: '100%', height: '100%', borderRadius: '50%', objectFit: 'cover' }}
239 |           />
240 |         ) : (
241 |           <div style={{
242 |

```

```

243|         width: '100%',
244|         height: '100%',
245|         borderRadius: '50%',
246|         background: 'rgba(0, 212, 170, 0.1)',
247|         color: 'var(--brand-primary)',
248|         display: 'flex',
249|         alignItems: 'center',
250|         justifyContent: 'center',
251|         fontSize: '24px',
252|         fontWeight: 800
253|     })>
254|     {selectedChild.full_name[0].toUpperCase()}
255| </div>
256| )}
257| { /* Active real-time online green dot */}
258| <span style={{
259|     position: 'absolute',
260|     bottom: '0',
261|     right: '0',
262|     width: '16px',
263|     height: '16px',
264|     borderRadius: '50%',
265|     background: realtimeData.isOnline ? 'var(--brand-primary)' : '#4a4d62',
266|     border: '3px solid var(--bg-surface)',
267|     boxShadow: realtimeData.isOnline ? '0 0 8px var(--brand-primary)' : 'none'
268| }} />
269| </div>
270|
271| <div style={{ flex: 1, minWidth: '200px' }}>
272|   <h2 style={{ fontSize: '20px', fontWeight: 800, color: 'var(--text-primary)', margin: '0 0 4px 0...
273|     {selectedChild.full_name}
274|   </h2>
275|   <p style={{ color: 'var(--text-secondary)', fontSize: '13px', margin: '0 0 8px 0' }}>
276|     0<ß&   { s e l e c t e d C h i l d . s c h o o l _ n a m e   ? ?   ' C h ° a   c - p   n h - t   t r °   Ý n g   h
277|   </p>
278|   <div style={{ display: 'flex', alignItems: 'center', gap: '12px', flexWrap: 'wrap' }}>
279|     <span style={{
280|         fontSize: '11px',
281|         fontFamily: 'monospace',
282|         background: 'var(--bg-base)',
283|         border: '1px solid var(--border-subtle)',
284|         padding: '3px 8px',
285|         borderRadius: '6px',
286|         color: 'var(--text-muted)',
287|         fontWeight: 600
288|     }}>
289|       {selectedChild.student_code}
290|     </span>
291|
292|     <span style={{ fontSize: '12px', color: realtimeData.isOnline ? 'var(--brand-primary)' : 'var(...
293|       {realtimeData.isOnline ? (
294|         '0=ß&   a n g   o n l i n e   &   h í c   t - p   t r ñ c   t u y ¿ n '
295|       ) : realtimeData.lastSeen ? (
296|         ` H o í t   ù n g   $ { f o r m a t D i s t a n c e T o N o w ( n e w   D a t e ( r e a l t i m e D a
297|       ) : (
298|         ' C h ° a   g h i   n h - n   h o í t   ù n g '
299|       )
300|     </span>
301|   </div>
302| </div>
303|
304| { /* Level & XP Box */}
305| <div style={{ textAlign: 'right', minWidth: '100px' }}>
306|   <div style={{ fontSize: '11px', color: 'var(--text-muted)', textTransform: 'uppercase', letterSp...
307|     C ¢ p   ù   h í c
308|   </div>
309|   <div style={{ fontSize: '28px', fontWeight: 900, color: 'var(--brand-primary)', lineHeight: 1.1 }}>
310|     {selectedChild.level}
311|   </div>
312|   <div style={{ fontSize: '12px', color: 'var(--text-muted)', fontWeight: 500 }}>
313|     {selectedChild.xp} XP
314|   </div>
315| </div>
316| </div>
317|
318| { /* Realtime Stats Grid */}
319| <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(200px, 1fr))', gap: '16...
320|
321|   <div style={{ background: 'var(--bg-surface)', padding: '20px', borderRadius: '18px', border: '1px...
322|     <div style={{ fontSize: '12px', color: 'var(--text-muted)', marginBottom: '8px' }}>
323|       B à i   ã   H o à n   T h à n h
324|     </div>
325|     <div style={{ fontSize: '24px', fontWeight: 800, color: 'var(--brand-primary)' }}>
326|       {realtimeData.lessonsCompleted}
327|     <span style={{ fontSize: '14px', fontWeight: 500, color: 'var(--text-muted)', marginLeft: '4px...
328|     </div>
329|   </div>
330|
331|   <div style={{ background: 'var(--bg-surface)', padding: '20px', borderRadius: '18px', border: '1px...
332|     <div style={{ fontSize: '12px', color: 'var(--text-muted)', marginBottom: '8px' }}>
333|       B à i   H í c   a n g   T i ¿ n   H à n h
334|     </div>
335|     <div style={{ fontSize: '24px', fontWeight: 800, color: '#3b82f6' }}>
336|       {Math.max(0, realtimeData.totalLessonsStarted - realtimeData.lessonsCompleted)}
337|     <span style={{ fontSize: '14px', fontWeight: 500, color: 'var(--text-muted)', marginLeft: '4px...
338|     </div>
339|   </div>
340|
341|   <div style={{ background: 'var(--bg-surface)', padding: '20px', borderRadius: '18px', border: '1px...
342|     <div style={{ fontSize: '12px', color: 'var(--text-muted)', marginBottom: '8px' }}>
343|       T ò n g   G i Ý   T h ñ c   H à n h
344|     </div>
345|     <div style={{ fontSize: '24px', fontWeight: 800, color: '#a855f7' }}>
346|       {Math.round((realtimeData.totalTimeSeconds / 3600) * 10) / 10}
347|     <span style={{ fontSize: '14px', fontWeight: 500, color: 'var(--text-muted)', marginLeft: '4px...
348|     </div>
349|   </div>
350|
351|   <div style={{ background: 'var(--bg-surface)', padding: '20px', borderRadius: '18px', border: '1px...
352|     <div style={{ fontSize: '12px', color: 'var(--text-muted)', marginBottom: '8px' }}>
353|       i   Á m   T r ° c   N g h i   Ç m   T B
354|     </div>
355|     <div style={{
356|         fontSize: '24px',
357|         fontWeight: 800,
358|         color: realtimeData.averageQuizScore === null
359|           ? 'var(--text-muted)'
360|           : realtimeData.averageQuizScore < 50 ? '#ef4444' : '#10b981'
361|     }}>
362|       {realtimeData.averageQuizScore !== null ? Math.round(realtimeData.averageQuizScore) : '--'}
363|     <div style={{
364|         {realtimeData.averageQuizScore !== null && (
365|           <span style={{ fontSize: '14px', fontWeight: 500, color: 'var(--text-muted)', marginLeft: '2...
366|         )
367|       </div>
368|     </div>

```

```

367|         </div>
368|
369|     </div>
370|
371|     { /* Realtime Recent Activity Feed */ }
372|     <div style={{
373|       background: 'var(--bg-surface)',
374|       border: '1px solid var(--border-subtle)',
375|       borderRadius: '24px',
376|       padding: '24px'
377|     }}>
378|       <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom...
379|         <h3 style={{ fontSize: '18px', fontWeight: 700, color: 'var(--text-primary)', margin: 0 }}>
380|           N h - t K y H I c T - p T r ñ c T u y ÷ n
381|         </h3>
382|
383|         {realtimeData.connectionStatus === 'connected' && (
384|           <div style={{ display: 'flex', alignItems: 'center', gap: '6px', fontSize: '12px', color: 'var...
385|             <span style={{
386|               width: '6px',
387|               height: '6px',
388|               borderRadius: '50%',
389|               background: 'var(--brand-primary)',
390|               display: 'inline-block',
391|               animation: 'blink 1.5s infinite'
392|             }} />
393|             C - p n h - t t é c t h Ý i
394|           </div>
395|         )}
396|       </div>
397|
398|       {realtimeData.recentActivities.length === 0 ? (
399|         <div style={{ textAlign: 'center', padding: '40px 0', color: 'var(--text-muted)', fontSize: '14p...
400|           0<B0p C h ° a c ó h o i t Û n g t h ñ c h à n h n à o ° ã c g h i n h - n .
401|         </div>
402|       ) : (
403|         <div style={{ display: 'flex', flexDirection: 'column', gap: '12px' }}>
404|           {realtimeData.recentActivities.map(activity => (
405|             <div
406|               key={activity.id}
407|               style={{
408|                 display: 'flex',
409|                 alignItems: 'center',
410|                 gap: '16px',
411|                 padding: '16px',
412|                 borderRadius: '12px',
413|                 background: 'var(--bg-base)',
414|                 border: '1px solid var(--border-subtle)',
415|               }}
416|             >
417|               { /* Activity Icon */ }
418|               <div style={{
419|                 fontSize: '20px',
420|                 width: '40px',
421|                 height: '40px',
422|                 borderRadius: '10px',
423|                 background: activity.type === 'lesson_completed' ? 'rgba(16, 185, 129, 0.1)' :
424|                   activity.type === 'quiz_graded' ? 'rgba(168, 85, 247, 0.1)' : 'rgba(59, 130,...
425|                 display: 'flex',
426|                 alignItems: 'center',
427|                 justifyContent: 'center'
428|               }}>
429|                 { a c t i v i t y . t y p e === ' l e s s o n _ c o m p l e t e d ' ? '0=00' :
430|                   a c t i v i t y . t y p e === ' q u i z _ g r a d e d ' ? '0=0Ý' : '0=Ý5' }
431|               </div>
432|
433|               { /* Activity Body */ }
434|               <div style={{ flex: 1, minWidth: 0 }}>
435|                 <p style={{
436|                   fontSize: '14px',
437|                   fontWeight: 600,
438|                   color: 'var(--text-primary)',
439|                   margin: '0 0 4px 0',
440|                   textOverflow: 'ellipsis',
441|                   overflow: 'hidden',
442|                   whiteSpace: 'nowrap'
443|                 }}>
444|                   {activity.title}
445|                 </p>
446|
447|                 {activity.score !== undefined && (
448|                   <span style={{
449|                     fontSize: '12px',
450|                     fontWeight: 700,
451|                     color: activity.score >= 50 ? '#10b981' : '#ef4444',
452|                     background: activity.score >= 50 ? 'rgba(16, 185, 129, 0.1)' : 'rgba(239, 68, 68, 0...
453|                     padding: '2px 8px',
454|                     borderRadius: '6px'
455|                   }}>
456|                     { a c t i v i t y . s c o r e } i Ä m
457|                   </span>
458|                 )}
459|               </div>
460|
461|               { /* Timestamp */ }
462|               <span style={{ fontSize: '12px', color: 'var(--text-muted)', flexShrink: 0 }}>
463|                 {formatDistanceToNow(new Date(activity.timestamp), { locale: vi, addSuffix: true })}
464|               </span>
465|             </div>
466|           )}
467|         </div>
468|       )}
469|     </div>
470|
471|   </div>
472|
473| }
474|
475| { /* Embedded inline keyframes styling */ }
476| <style dangerouslySetInnerHTML={{ __html: `
477|   @keyframes blink {
478|     0%, 100% { opacity: 1; }
479|     50% { opacity: 0.3; }
480|   }
481| ` }} />
482|
483| </div>
484| )
485| }
486|

```

```

1 // Path: app/(dashboard)/parent/layout.tsx
2 'use client'
3
4 import React, { useEffect, useState } from 'react'
5 import Link from 'next/link'
6 import { usePathname } from 'next/navigation'
7 import { GraduationCap, LayoutDashboard, UserPlus, Bell, LogOut } from 'lucide-react'
8 import { logout } from '@lib/auth-actions'
9 import { createBrowserClient } from '@supabase/ssr'
10 import { useParentNotifications } from '@hooks/useParentNotifications'
11 import PageTransition from '@components/PageTransition'
12
13 export default function ParentLayout({ children }: { children: React.ReactNode }) {
14   const [parentId, setParentId] = useState<string | null>(null)
15   const pathname = usePathname()
16   const [supabase] = useState(() => {
17     const url = process.env.NEXT_PUBLIC_SUPABASE_URL?.trim()
18     const key = process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY?.trim()
19     if (typeof window === 'undefined' && (!url || !key)) return null as any
20     if (!url || !key) return null as any
21     return createBrowserClient(url, key)
22   })
23
24   useEffect(() => {
25     const fetchUser = async () => {
26       const { data: { user } } = await supabase.auth.getUser()
27       if (user) {
28         setParentId(user.id)
29       }
30     }
31     fetchUser()
32   }, [])
33
34   const { unreadCount } = useParentNotifications(parentId)
35
36   const isLinkActive = (path: string) => {
37     return pathname.startsWith(path)
38   }
39
40   return (
41     <div style={{ display: 'flex', minHeight: '100vh', background: 'var(--bg-base)' }}>
42       /* Parent Sidebar */
43       <aside style={{
44         width: '280px',
45         background: 'var(--bg-surface)',
46         borderRight: '1px solid var(--border-subtle)',
47         display: 'flex',
48         flexDirection: 'column'
49       }}>
50         /* Header Logo */
51         <div style={{ padding: '24px', display: 'flex', alignItems: 'center', gap: '12px', borderBottom: '1px ...
52           <div style={{
53             width: '40px',
54             height: '40px',
55             borderRadius: '10px',
56             background: 'var(--brand-primary)',
57             display: 'flex',
58             alignItems: 'center',
59             justifyContent: 'center'
60           }}>
61             <GraduationCap size={20} color="#000" />
62           </div>
63           <div>
64             <div style={{ fontWeight: 800, fontSize: '18px', color: 'var(--text-primary)' }}>LMS PARENT</div>
65             <div style={{
66               <div style={{
67                 </div>
68               </div>
69             /* Navigation Menu */
70             <nav style={{ padding: '24px 16px', display: 'flex', flexDirection: 'column', gap: '8px', flex: 1 }}>
71
72               <Link
73                 href="/parent/dashboard"
74                 style={{
75                   display: 'flex',
76                   alignItems: 'center',
77                   gap: '12px',
78                   padding: '12px 16px',
79                   borderRadius: '10px',
80                   background: isLinkActive('/parent/dashboard') ? 'rgba(0, 212, 170, 0.1)' : 'transparent',
81                   color: isLinkActive('/parent/dashboard') ? 'var(--brand-primary)' : 'var(--text-secondary)',
82                   fontWeight: isLinkActive('/parent/dashboard') ? 600 : 500,
83                   textDecoration: 'none',
84                   transition: 'all 0.2s'
85                 }}
86               >
87               <div style={{
88                 </div>
89               </div>
90               <Link
91                 href="/parent/link-child"
92                 style={{
93                   display: 'flex',
94                   alignItems: 'center',
95                   gap: '12px',
96                   padding: '12px 16px',
97                   borderRadius: '10px',
98                   background: isLinkActive('/parent/link-child') ? 'rgba(0, 212, 170, 0.1)' : 'transparent',
99                   color: isLinkActive('/parent/link-child') ? 'var(--brand-primary)' : 'var(--text-secondary)',
100                   fontWeight: isLinkActive('/parent/link-child') ? 600 : 500,
101                   textDecoration: 'none',
102                   transition: 'all 0.2s'
103                 }}
104               >
105               <div style={{
106                 </div>
107               </div>
108               <Link
109                 href="/parent/notifications"
110                 style={{
111                   display: 'flex',
112                   alignItems: 'center',
113                   justifyContent: 'space-between',
114                   padding: '12px 16px',
115                   borderRadius: '10px',
116                   background: isLinkActive('/parent/notifications') ? 'rgba(0, 212, 170, 0.1)' : 'transparent',
117                   color: isLinkActive('/parent/notifications') ? 'var(--brand-primary)' : 'var(--text-secondary)',
118                   fontWeight: isLinkActive('/parent/notifications') ? 600 : 500,
119                   textDecoration: 'none',
120                   transition: 'all 0.2s'
121                 }}
122               >
123               <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
124                 <Bell size={20} /> Trung tâm thông báo

```



```

125 |         </div>
126 |         {unreadCount > 0 && (
127 |           <span style={{
128 |             background: '#ef4444',
129 |             color: '#fff',
130 |             fontSize: '11px',
131 |             fontWeight: 700,
132 |             padding: '2px 8px',
133 |             borderRadius: '50px',
134 |             lineHeight: 1.2
135 |           }}>
136 |             {unreadCount}
137 |           </span>
138 |         )}
139 |       </Link>
140 |
141 |     </nav>
142 |
143 |     {/ * Footer Logout */}
144 |     <div style={{ padding: '24px 16px', borderTop: '1px solid var(--border-subtle)' }}>
145 |       <button
146 |         onClick={() => logout()}
147 |         style={{
148 |           display: 'flex',
149 |           alignItems: 'center',
150 |           gap: '12px',
151 |           padding: '12px 16px',
152 |           width: '100%',
153 |           borderRadius: '10px',
154 |           background: 'transparent',
155 |           border: 'none',
156 |           color: '#ef4444',
157 |           fontWeight: 600,
158 |           cursor: 'pointer',
159 |           fontSize: '14px',
160 |           textAlign: 'left'
161 |         }}
162 |       >
163 |         <LogOut size={20} />
164 |         ng x u ʔ t
165 |       </button>
166 |     </div>
167 |   </aside>
168 |
169 |   {/ * Main Content Area */}
170 |   <main style={{ flex: 1, padding: '40px', overflowY: 'auto' }}>
171 |     <PageTransition>{children}</PageTransition>
172 |   </main>
173 | </div>
174 | )
175 | }

```

File: app/(dashboard)/parent/link-child/page.tsx (229 lines, 10.4 KB)

```

1 | 'use client'
2 |
3 | export const dynamic = 'force-dynamic'
4 |
5 | import React, { useState } from 'react'
6 | import { findStudentByCode, requestLinkToStudent } from '@lib/parent-actions'
7 | import { Search, ShieldCheck, Heart, User, CheckCircle2, ArrowRight, ArrowLeft } from 'lucide-react'
8 |
9 | export default function LinkChildPage() {
10 |   const [code, setCode] = useState('')
11 |   const [foundStudent, setFoundStudent] = useState<any>(null)
12 |   const [relationship, setRelationship] = useState('parent')
13 |   const [step, setStep] = useState<'input' | 'confirm' | 'success'>('input')
14 |   const [loading, setLoading] = useState(false)
15 |   const [error, setError] = useState<string | null>(null)
16 |
17 |   // Step 1: Nh - p mã và tìm kiếm
18 |   async function searchStudent() {
19 |     if (!code || code.trim().length < 4) {
20 |       setError('Vui lòng nh - p mã híc sinh hăp lç')
21 |       return
22 |     }
23 |
24 |     setLoading(true)
25 |     setError(null)
26 |
27 |     try {
28 |       const res = await findStudentByCode(code.trim().toUpperCase())
29 |       if (res.student) {
30 |         setFoundStudent(res.student)
31 |         setStep('confirm')
32 |       } else {
33 |         setError(res.error || 'Không tìm thđy híc sinh vŰi mã này')
34 |       }
35 |     } catch (err: any) {
36 |       setError('L x i k ʔ t n ñ i má y ch ç')
37 |     } finally {
38 |       setLoading(false)
39 |     }
40 |   }
41 |
42 |   // Step 2: Xá c nh - n liên k ʔ t
43 |   async function confirmLink() {
44 |     if (!foundStudent) return
45 |
46 |     setLoading(true)
47 |     setError(null)
48 |
49 |     try {
50 |       const res = await requestLinkToStudent(foundStudent.id, relationship)
51 |       if (res.success) {
52 |         setStep('success')
53 |       } else {
54 |         setError(res.error || 'L x i khi th ñ c hi ç n liên k ʔ t')
55 |       }
56 |     } catch (err: any) {
57 |       setError(err.message || 'L x i khi th ñ c hi ç n liên k ʔ t')
58 |     } finally {
59 |       setLoading(false)
60 |     }
61 |   }
62 |
63 |   return (
64 |     <div className="min-h-screen flex flex-col justify-center items-center p-6" style={{ background: 'var(--bg...
65 |       <div className="w-full max-w-md rounded-2xl p-6 md:p-8 shadow-2xl relative overflow-hidden" style={{ bac...
66 |         {/ * Glow effect */}
67 |         <div className="absolute -top-24 -left-24 w-48 h-48 rounded-full blur-3xl pointer-events-none" style={...
68 |
69 |       {step === 'input' && (

```

```

70|     <>
71|     <h2 className="text-xl md:text-2xl font-bold mb-2 flex items-center gap-2" style={{ color: 'var(--...
72|         L i ê n k ế t t à i k h o ế n c o n
73|     </h2>
74|     <p className="text-xs md:text-sm mb-6 leading-relaxed" style={{ color: 'var(--text-muted)' }}>
75|         N h - p m ấ h í c s í n h c ặ a c o n b í n ( t í m t h ậ y t r o n g m ấ c H Ó s i c ặ a
76|     </p>
77|
78|     <div className="relative mb-4">
79|         <input
80|             value={code}
81|             onChange={(e) => {
82|                 setError(null)
83|                 setCode(e.target.value.toUpperCase())
84|             }}
85|             onKeyDown={(e) => {
86|                 if (e.key === 'Enter') searchStudent()
87|             }}
88|             placeholder="PCM-XXXXXXX"
89|             maxLength={12}
90|             disabled={loading}
91|             className="w-full rounded-xl px-4 py-3.5 font-mono text-center text-lg tracking-widest outline...
92|             style={{ background: 'var(--bg-elevated)', border: '1px solid var(--border-default)', color: '...
93|         </input>
94|     </div>
95|
96|     {error && (
97|         <div className="text-xs font-semibold mb-4 rounded-lg p-3" style={{ color: '#ef4444', background...
98|             & p { e r r o r }
99|         </div>
100|     )}
101|
102|     <button
103|         onClick={searchStudent}
104|         disabled={loading || !code}
105|         className="w-full py-3.5 rounded-xl font-bold text-sm transition-all duration-150 flex items-cen...
106|         style={{ background: 'rgba(0, 212, 170, 0.1)', border: '1px solid rgba(0, 212, 170, 0.3)', color: '...
107|     >
108|         {loading ? (
109|             <div className="w-5 h-5 border-2 rounded-full animate-spin" style={{ borderColor: 'var(--brand...
110|         ) : (
111|             <>
112|                 <Search size={16} />
113|                 T í m k i ế m h í c s í n h
114|             </>
115|         )}
116|     </button>
117| </>
118| )}
119|
120| {step === 'confirm' && foundStudent && (
121|     <>
122|     <h2 className="text-xl md:text-2xl font-bold mb-2" style={{ color: 'var(--text-primary)' }}>
123|         X ấ c n h - n l i ê n k ế t
124|     </h2>
125|     <p className="text-xs md:text-sm mb-6 leading-relaxed" style={{ color: 'var(--text-muted)' }}>
126|         V u i l ồ n g x ấ c n h - n t h ồ n g t í n h í c s í n h d ộ i ấ y c h í n h x ấ c l ấ
127|     </p>
128|
129|     /* Profile Card */
130|     <div className="rounded-xl p-4 mb-6 flex items-center gap-4" style={{ background: 'var(--bg-elevat...
131|         {foundStudent.avatar_url ? (
132|             <img
133|                 src={foundStudent.avatar_url}
134|                 className="w-14 h-14 rounded-full object-cover"
135|                 style={{ border: '1px solid var(--border-default)' }}
136|                 alt={foundStudent.student_name}
137|             </img>
138|         ) : (
139|             <div className="w-14 h-14 rounded-full flex items-center justify-center text-2xl font-bold upp...
140|                 {foundStudent.student_name[0] || 'S'}
141|             </div>
142|         )}
143|         <div className="overflow-hidden">
144|             <div className="font-bold text-base truncate" style={{ color: 'var(--text-primary)' }}>{foundS...
145|             <div className="text-xs truncate" style={{ color: 'var(--text-muted)' }}>{foundStudent.school_...
146|         </div>
147|     </div>
148|
149|     <div className="mb-6">
150|         <label className="block text-xs font-bold uppercase tracking-wider mb-2" style={{ color: 'var(--...
151|             M Ñ i q u a n h ặ v ộ i h í c s í n h
152|         </label>
153|         <select
154|             value={relationship}
155|             onChange={(e) => setRelationship(e.target.value)}
156|             disabled={loading}
157|             className="w-full rounded-xl px-4 py-3 text-sm outline-none transition-all cursor-pointer"
158|             style={{ background: 'var(--bg-elevated)', border: '1px solid var(--border-default)', color: '...
159|         >
160|             <option value="father">Ch ấ </option>
161|             <option value="parent">P h ấ h u y n h ( C h u n g ) </option>
162|             <option value="mother">M ấ </option>
163|             <option value="guardian">N g ộ ý i g i ấ m h ộ </option>
164|         </select>
165|     </div>
166|
167|     {error && (
168|         <div className="text-xs font-semibold mb-4 rounded-lg p-3" style={{ color: '#ef4444', background...
169|             & p { e r r o r }
170|         </div>
171|     )}
172|
173|     <div className="flex gap-3">
174|         <button
175|             onClick={() => {
176|                 setError(null)
177|                 setStep('input')
178|             }}
179|             disabled={loading}
180|             className="flex-1 py-3 rounded-xl font-bold text-sm transition-all flex items-center justify-c...
181|             style={{ background: 'rgba(255,255,255,0.05)', border: '1px solid var(--border-subtle)', color: '...
182|         >
183|             <ArrowLeft size={16} />
184|             Q u a y l ị i
185|         </button>
186|         <button
187|             onClick={confirmLink}
188|             disabled={loading}
189|             className="flex-1 py-3 rounded-xl font-bold text-sm transition-all duration-150 flex items-cen...
190|             style={{ background: 'rgba(0, 212, 170, 0.15)', border: '1px solid rgba(0, 212, 170, 0.3)', co...
191|         >
192|             {loading ? (
193|                 <div className="w-5 h-5 border-2 rounded-full animate-spin" style={{ borderColor: 'var(--bra...

```

```

194 |         ) : (
195 |             <>
196 |                 1 9 6 | X á c n h - n
197 |                 <ArrowRight size={16} />
198 |                 </>
199 |             )}
200 |         </button>
201 |     </div>
202 | </>
203 | )}
204 |
205 | {step === 'success' && (
206 |     <div className="text-center py-6">
207 |         <div className="w-16 h-16 rounded-full flex items-center justify-center mx-auto mb-5" style={{ bac...
208 |         <CheckCircle2 size={36} className="animate-pulse" />
209 |         </div>
210 |         <h3 className="text-xl md:text-2xl font-bold mb-2" style={{ color: 'var(--text-primary)' }}>
211 |             2 1 1 | L i ê n k ế t t h à n h c ô n g !
212 |         </h3>
213 |         <p className="text-xs md:text-sm mb-8 leading-relaxed max-w-sm mx-auto" style={{ color: 'var(--tex...
214 |             2 1 4 | T à i k h o ế n c ặ a b i n ấ ộ ấ c k ế t n ấ i v ớ i c o n t h à n h c ô n g . G i ớ
215 |         </p>
216 |         <a
217 |             href="/parent/dashboard"
218 |             className="inline-flex items-center justify-center px-6 py-3 rounded-xl font-bold text-sm transi...
219 |             style={{ background: 'rgba(0, 212, 170, 0.1)', border: '1px solid rgba(0, 212, 170, 0.3)', color...
220 |         >
221 |             2 2 1 | V Á t r a n g T ồ n g q u a n !'
222 |         </a>
223 |     </div>
224 | )}
225 | </div>
226 | </div>
227 | )
228 | }
229 |

```

File: app/(dashboard)/parent/notifications/page.tsx (100 lines, 3.0 KB)

```

1 | import { requireRole } from '@lib/auth/rbac'
2 | import { createClient } from '@lib/supabase-ssr-server'
3 | import ParentNotificationsView from '@components/parent/ParentNotificationsView'
4 | import { revalidatePath } from 'next/cache'
5 |
6 | export const dynamic = 'force-dynamic'
7 |
8 | export default async function ParentNotificationsPage() {
9 |     // 1. X á c t h ấ c r o l e
10 |     const user = await requireRole(['parent', 'admin'])
11 |     const supabase = await createClient()
12 |
13 |     // 2. Fetch danh sách thông báo
14 |     const { data: dbNotifications } = await supabase
15 |       .from('notifications')
16 |       .select('*')
17 |       .eq('user_id', user.id)
18 |       .order('created_at', { ascending: false })
19 |
20 |     const notifications = dbNotifications || []
21 |
22 |     // 3. F e t c h d a n h s á c h c o n l i ê n k ế t ấ m a p t h o n g t i n ( A v a t a r + T ê n )
23 |     const { data: links } = await supabase
24 |       .from('parent_student_links')
25 |       .select('student_id')
26 |       .eq('parent_id', user.id)
27 |       .eq('status', 'active')
28 |
29 |     const studentIds = (links || []).map((l) => l.student_id)
30 |     let students: any[] = []
31 |
32 |     if (studentIds.length > 0) {
33 |       const { data: profiles } = await supabase
34 |         .from('profiles')
35 |         .select('id, full_name, avatar_url')
36 |         .in('id', studentIds)
37 |       students = profiles || []
38 |     }
39 |
40 |     // 4. S e r v e r A c t i o n s p h ấ c v ấ c t h a o t á c c ặ a p h ấ h u y n h
41 |     async function markAllAsReadAction() {
42 |       'use server'
43 |       const parent = await requireRole(['parent', 'admin'])
44 |       const client = await createClient()
45 |       await client
46 |         .from('notifications')
47 |         .update({ is_read: true })
48 |         .eq('user_id', parent.id)
49 |       revalidatePath('/parent/notifications')
50 |     }
51 |
52 |     async function deleteAllAction() {
53 |       'use server'
54 |       const parent = await requireRole(['parent', 'admin'])
55 |       const client = await createClient()
56 |       await client
57 |         .from('notifications')
58 |         .delete()
59 |         .eq('user_id', parent.id)
60 |       revalidatePath('/parent/notifications')
61 |     }
62 |
63 |     async function markAsReadAction(id: string) {
64 |       'use server'
65 |       const parent = await requireRole(['parent', 'admin'])
66 |       const client = await createClient()
67 |       await client
68 |         .from('notifications')
69 |         .update({ is_read: true })
70 |         .eq('id', id)
71 |         .eq('user_id', parent.id)
72 |       revalidatePath('/parent/notifications')
73 |     }
74 |
75 |     async function deleteSingleAction(id: string) {
76 |       'use server'
77 |       const parent = await requireRole(['parent', 'admin'])
78 |       const client = await createClient()
79 |       await client
80 |         .from('notifications')
81 |         .delete()
82 |         .eq('id', id)
83 |         .eq('user_id', parent.id)
84 |       revalidatePath('/parent/notifications')

```

```
85 | }
86 |
87 | return (
88 |   <div className="min-h-screen p-6 md:p-8 max-w-4xl mx-auto" style={{ background: 'var(--bg-base)', color: '...
89 |     <ParentNotificationsView
90 |       initialNotifications={notifications}
91 |       students={students}
92 |       markAllAsRead={markAllAsReadAction}
93 |       deleteAll={deleteAllAction}
94 |       markAsRead={markAsReadAction}
95 |       deleteSingle={deleteSingleAction}
96 |     />
97 |   </div>
98 | )
99 | }
100 |
```

File: app/(dashboard)/parent/page.tsx (6 lines, 0.1 KB)

```
1 | import { redirect } from 'next/navigation'
2 |
3 | export default function ParentPage() {
4 |   redirect('/parent/dashboard')
5 | }
6 |
```

File: app/(dashboard)/parent/profile/page.tsx (87 lines, 4.8 KB)

```
1 | import { requireRole } from '@lib/auth/rbac'
2 | import { createClient } from '@lib/supabase-ssr-server'
3 | import { User, ShieldCheck, HelpCircle, Mail, Key } from 'lucide-react'
4 |
5 | export const dynamic = 'force-dynamic'
6 |
7 | export default async function ParentProfilePage() {
8 |   const user = await requireRole(['parent', 'admin'])
9 |   const supabase = await createClient()
10 |
11 |   // L ã y email t ã auth
12 |   const { data: authData } = await supabase.auth.getUser()
13 |   const email = authData.user?.email || 'Ch ã c - p nh - t '
14 |
15 |   // F e t c h s ã l ã ng c o n a n g l i ê n k ẽ t
16 |   const { count } = await supabase
17 |     .from('parent_student_links')
18 |     .select('*', { count: 'exact', head: true })
19 |     .eq('parent_id', user.id)
20 |     .eq('status', 'active')
21 |
22 |   return (
23 |     <div className="min-h-screen p-6 md:p-8 max-w-4xl mx-auto space-y-6" style={{ background: 'var(--bg-base)'...
24 |       <div>
25 |         <div className="pb-6" style={{ borderBottom: '1px solid var(--border-subtle)' }}>
26 |           <h1 className="text-2xl md:text-3xl font-bold flex items-center gap-2" style={{ color: 'var(--text-pri...
27 |             <User size={24} style={{ color: 'var(--brand-primary)' }} />
28 |             H ã s i c ç a b i n
29 |           </h1>
30 |           <p className="mt-1 text-sm" style={{ color: 'var(--text-muted)' }}>
31 |             T h ã ng t i n t à i k h o ẽ n p h ả h u y n h v à c ẽ u h i n h h ç t h ã ng .
32 |           </p>
33 |         </div>
34 |
35 |         <div className="grid grid-cols-1 md:grid-cols-3 gap-6">
36 |           <div>
37 |             <div className="md:col-span-2 rounded-2xl p-6 space-y-6" style={{ background: 'var(--bg-surface)', bor...
38 |               <div className="flex items-center gap-4">
39 |                 <div className="w-16 h-16 rounded-full flex items-center justify-center text-3xl font-bold upperca...
40 |                   {user.full_name?.[0] || 'P'}
41 |                 </div>
42 |                 <div>
43 |                   <h2 className="text-lg font-bold" style={{ color: 'var(--text-primary)' }}>{user.full_name || 'P...
44 |                   <div className="text-xs rounded-full uppercase tracking-wider inline-block mt-1 font-bold px-2 p...
45 |                     P h ả h u y n h
46 |                   </div>
47 |                 </div>
48 |               </div>
49 |
50 |               <div className="pt-6 space-y-4" style={{ borderTop: '1px solid var(--border-subtle)' }}>
51 |                 <div className="flex items-center justify-between text-sm">
52 |                   <span className="font-medium flex items-center gap-2" style={{ color: 'var(--text-muted)' }}>
53 |                     <Mail size={16} />
54 |                     E m a i l n g n h - p
55 |                   </span>
56 |                   <span className="font-semibold" style={{ color: 'var(--text-primary)' }}>{email}</span>
57 |                 </div>
58 |                 <div className="flex items-center justify-between text-sm">
59 |                   <span className="font-medium flex items-center gap-2" style={{ color: 'var(--text-muted)' }}>
60 |                     <ShieldCheck size={16} />
61 |                     S ã l ã ng c o n l i ê n k ẽ t
62 |                   </span>
63 |                   <span classN ame = " font - semibold " style = { { color : ' var ( - - t e x
64 |                     </div>
65 |                   </div>
66 |                 </div>
67 |
68 |                 <div>
69 |                   <div className="rounded-2xl p-6 space-y-4" style={{ background: 'var(--bg-surface)', border: '1px soli...
70 |                     <h3 className="text-sm font-bold flex items-center gap-2" style={{ color: 'var(--text-primary)' }}>
71 |                       <ShieldCheck size={16} style={{ color: 'var(--brand-primary)' }} />
72 |                       B ẽ o m - t & Q u y ả n r i ẽ ng t ã
73 |                     </h3>
74 |                     <p className="text-xs leading-relaxed" style={{ color: 'var(-text-muted)' }}>
75 |                       A t i o m ã i t r ã Y n g h i c t - p t h o ẽ i m á i , P C M a s t e r ả p d ả ng c h i n h
76 |                     </p>
77 |                     <ul className="text-xs space-y-2 list-disc pl-4 leading-relaxed" style={{ color: 'var(--text-muted)'...
78 |                       <li>Ph ả h u y n h <span style = { { color : ' var ( - - t e x t - p r i m a r y ) ' } } >
79 |                       <li>Ph ả h u y n h <span style = { { color : ' # e f 4 4 4 4 ' } } > k h ã ng t h ả
80 |                       <li>H i c s i n h c ó q u y ả n t h u h ã i q u y ả n t h e o ã i c ç a p h ả h u y n h
81 |                     </ul>
82 |                   </div>
83 |                 </div>
84 |               </div>
85 |             </div>
86 |           </div>
87 |         </div>
88 |       </div>
89 |     )
90 |   }
91 | }
```

File: app/(dashboard)/student/classes/[classId]/assignments/[assignmentId]/page.tsx (249 lines, 11.8 KB)

```
1 | 'use client';
2 |
```

```

3 | import React, { useEffect, useState, use } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import {
6 |   Loader2, ArrowLeft, ClipboardList, Target,
7 |   Wallet, Battery, HardDrive, Cpu, CheckCircle, AlertCircle, Play
8 | } from 'lucide-react';
9 | import Link from 'next/link';
10 | import { formatVND } from '@lib/grading/auto-grade';
11 |
12 | export default function DoAssignmentPage({ params }: { params: Promise<{ classId: string, assignmentId: string...
13 |   const resolvedParams = use(params);
14 |   const { classId, assignmentId } = resolvedParams;
15 |
16 |   const [assignment, setAssignment] = useState<any>(null);
17 |   const [submission, setSubmission] = useState<any>(null);
18 |   const [loading, setLoading] = useState(true);
19 |   const [isSubmitting, setIsSubmitting] = useState(false);
20 |
21 |   useEffect(() => {
22 |     fetchData();
23 |   }, [assignmentId]);
24 |
25 |   async function fetchData() {
26 |     setLoading(true);
27 |     try {
28 |       const { data: { user } } = await supabase.auth.getUser();
29 |       if (!user) return;
30 |
31 |       // 1. Fetch Assignment
32 |       const { data: asg, error: asgError } = await supabase
33 |         .from('assignments')
34 |         .select('*')
35 |         .eq('id', assignmentId)
36 |         .single();
37 |
38 |       if (asgError) throw asgError;
39 |       setAssignment(asg);
40 |
41 |       // 2. Check for existing submission
42 |       const { data: sub, error: subError } = await supabase
43 |         .from('submissions')
44 |         .select('*')
45 |         .eq('assignment_id', assignmentId)
46 |         .eq('student_id', user.id)
47 |         .single();
48 |
49 |       if (sub) setSubmission(sub);
50 |
51 |     } catch (err) {
52 |       console.error('Error fetching assignment data:', err);
53 |     } finally {
54 |       setLoading(false);
55 |     }
56 |   }
57 |
58 |   const handleSubmitMock = async () => {
59 |     // // â y l à h à m g i é l - p v i c c n ù p b à i t ẻ Simulator
60 |     // // T r o n g t h ã c t ử , đ i l i c u n à y s ½ ử n t ẻ t r i n g t h á i c c a P C B u i l d e r
61 |     const mockPCCConfig = {
62 |       components: ['Ryzen 5 5600X', 'RTX 3060', 'RAM 16GB DDR4', 'SSD 500GB', 'PSU 650W'],
63 |       total_price: 14500000,
64 |       total_tdp: 220,
65 |       ram_gb: 16
66 |     };
67 |
68 |     setIsSubmitting(true);
69 |     try {
70 |       const res = await fetch('/api/classes/submit', {
71 |         method: 'POST',
72 |         headers: { 'Content-Type': 'application/json' },
73 |         body: JSON.stringify({
74 |           assignment_id: assignmentId,
75 |           class_id: classId,
76 |           pc_config: mockPCCConfig
77 |         })
78 |       });
79 |
80 |       const data = await res.json();
81 |       if (data.success) {
82 |         setSubmission(data.submission);
83 |         alert(' N ù p b à i t h à n h c ô n g ! ' );
84 |       } else {
85 |         throw new Error(data.error);
86 |       }
87 |     } catch (err: any) {
88 |       alert(' L x i : ' + err . message );
89 |     } finally {
90 |       setIsSubmitting(false);
91 |     }
92 |   };
93 |
94 |   if (loading) {
95 |     return (
96 |       <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', height: '100%' }}>
97 |         <Loader2 className="animate-spin" color="#00f3ff" size={48} />
98 |       </div>
99 |     );
100 |   }
101 |
102 |   if (!assignment) return <div>Không tìm thấy nhiệ m v ả .</div>;
103 |
104 |   return (
105 |     <div style={{ padding: '32px', maxWidth: '1000px', margin: '0 auto' }}>
106 |       <header style={{ marginBottom: '40px' }}>
107 |         <Link href={`/student/classes/${classId}`} style={{ display: 'inline-flex', alignItems: 'center', gap:...
108 |           <ArrowLeft size={20} /> Quay l i i l ỏ p h i c
109 |         </Link>
110 |
111 |         <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'flex-start' }}>
112 |           <div>
113 |             <h1 style={{ fontSize: '32px', fontWeight: 800, margin: '0 0 12px 0', color: '#fff' }}>{assignment...
114 |             <p style={{ color: '#8899a6', margin: 0, maxWidth: '600px' }}>{assignment.description}</p>
115 |           </div>
116 |           {submission && (
117 |             <div style={{ padding: '16px 24px', background: 'rgba(16, 185, 129, 0.1)', border: '1px solid rgba...
118 |               <div style={{ fontSize: '12px', color: ' # 1 0 b 9 8 1 ' , f o n t w e i g
119 |                 <div style={{ fontSize: '32px', fontWeight: 900, color: '#10b981' }}>{submission.total_score} <s...
120 |               </div>
121 |             </div>
122 |           </div>
123 |         </header>
124 |
125 |         <div style={{ display: 'grid', gridTemplateColumns: '1fr 350px', gap: '32px' }}>
126 |           <div>

```

```
127 | <div style={{ display: 'flex', flexDirection: 'column', gap: '32px' }}>
128 |   /* Action Area */
129 |   <div style={{
130 |     background: 'rgba(12, 20, 36, 0.8)', padding: '48px', borderRadius: '24px',
131 |     border: '1px solid rgba(255,255,255,0.05)', textAlign: 'center',
132 |     display: 'flex', flexDirection: 'column', alignItems: 'center', gap: '24px'
133 |   }}>
134 |     <div style={{ width: '80px', height: '80px', borderRadius: '50%', background: 'rgba(0, 243, 255, 0...'
135 |       <Cpu size={40} />
136 |     </div>
137 |     <div>
138 |       <h2 style={{ color: '#fff', fontSize: '24px', fontWeight:
139 |         <p style={{ color: '#8899a6', margin: 0 }}> H Ç t h Ñ n g s ½ m ß
140 |       </div>
141 |
142 |       <Link href="/builder" style={{ textDecoration: 'none', width: '100%', maxWidth: '300px' }}>
143 |         <button style={{
144 |           width: '100%', padding: '16px', borderRadius: '12px', background: 'var(--brand-primary)',
145 |           color: '#000', fontSize: '16px', fontWeight: 800, border: 'none', cursor: 'pointer',
146 |           display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '10px'
147 |         }}>
148 |           <Play size={20} fill="#000" /> B ã t s u l ã p r á p
149 |         </button>
150 |       </Link>
151 |
152 |       { /* N ú t n ù p b à i t i m t h ý i ( G i é l - p ) * / }
153 |       {!submission && (
154 |         <button
155 |           onClick={handleSubmitMock}
156 |           disabled={isSubmitting}
157 |           style={{
158 |             background: 'none', border: '1px solid rgba(255,255,255,0.1)', color: '#4b5563',
159 |             padding: '8px 16px', borderRadius: '8px', cursor: 'pointer', fontSize: '12px'
160 |           }}
161 |         >
162 |           { is s u b m i t t i n g ? ' a n g n ù p . . . ' : ' ( G i é l - p n ù p b à i t h à n h
163 |         </button>
164 |       )}
165 |     </div>
166 |
167 |     {submission && (
168 |       <div style={{ background: 'rgba(12, 20, 36, 0.5)', padding: '32px', borderRadius: '24px', border: ...
169 |       <h3 style={{ color: '#fff', margin: '0 0 20px 0', display: 'flex', alignItems: 'center', gap: '1...'
170 |         <CheckCircle size={20} color="#10b981" /> K ù t q u é c h ẽ m
171 |       </h3>
172 |       {submission.feedback && (
173 |         <div style={{ color: '#e0e6ed', fontSize: '15px', lineHeight: 1.6, padding: '16px', background...'
174 |         <strong>Ph ê n h ô i t ê G V : </strong> { s u b m i s s i o n . f e e d b a c k }
175 |       </div>
176 |       )}
177 |       <p style={{ color: '#8899a6', fontSize: '14px', marginTop: '16px' }}>
178 |         B à i l à m ò á c n ù p v à o l ú c { new Date ( s u b m i s s i o n . s u b m i t t e d .
179 |         { s u b m i s s i o n . i s _ l a t e && <span style={{ color: '#ef4444' }
180 |       </p>
181 |     </div>
182 |   )}
183 | </div>
184 |
185 |   /* Right Sidebar - Requirements */
186 |   <aside style={{ display: 'flex', flexDirection: 'column', gap: '24px' }}>
187 |     <div style={{ background: 'rgba(12, 20, 36, 0.8)', padding: '24px', borderRadius: '24px', border: '1...'
188 |       <h3 style={{ color: '#fff', fontSize: '16px', fontWeight: 700, margin: '0 0 20px 0', display: 'fle...'
189 |       <Target size={18} color="#00f3ff" /> Y ê u c s u n h i ç m v ã
190 |     </h3>
191 |
192 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
193 |       <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
194 |         <div style={{ width: '32px', height: '32px', borderRadius: '8px', background: 'rgba(16, 185, 1...'
195 |           <Wallet size={16} />
196 |         </div>
197 |         <div>
198 |           <div style={{ fontSize: '11px', color: '#4b5563', fontWeight: 700 }}>NGÂN SÁCH</div>
199 |           <div style={{ fontSize: '14px', color: '#fff', fontWeight: 600 }}>&le; {formatVND(assignment...
200 |         </div>
201 |       </div>
202 |
203 |       <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
204 |         <div style={{ width: '32px', height: '32px', borderRadius: '8px', background: 'rgba(245, 158, ...
205 |           <Battery size={16} />
206 |         </div>
207 |         <div>
208 |           <div style={{ fontSize: '11px', color: '#4b5563', font
209 |             <div style={{ fontSize: '14px', color: '#fff', fontWeight: 600 }}>&le; {assignment.requireme...
210 |           </div>
211 |         </div>
212 |
213 |         <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
214 |           <div style={{ width: '32px', height: '32px', borderRadius: '8px', background: 'rgba(59, 130, 2...'
215 |             <HardDrive size={16} />
216 |           </div>
217 |           <div>
218 |             <div style={{ fontSize: '11px', color: '#4b5563', font
219 |               <div style={{ fontSize: '14px', color: '#fff', fontWeight: 600 }}>&ge; {assignment.requireme...
220 |             </div>
221 |           </div>
222 |
223 |           {assignment.requirements.required_components?.length > 0 && (
224 |             <div style={{ marginTop: '8px', padding: '16px', background: 'rgba(139, 92, 246, 0.05)', borde...
225 |               <div style={{ fontSize: '11px', color: '#8b5cf6', fontWeight: 700, marginBottom: '8px' }}>LI...
226 |               <ul style={{ margin: 0, padding: '0 0 16px', fontSize: '13px', color: '#e0e6ed', display: ...
227 |                 {assignment.requirements.required_components.map((c: string, i: number) => (
228 |                   <li key={i}>{c}</li>
229 |                 ))}
230 |               </ul>
231 |             </div>
232 |           )}
233 |         </div>
234 |       </div>
235 |
236 |       <div style={{ padding: '20px', background: 'rgba(239, 68, 68, 0.05)', borderRadius: '24px', border: ...
237 |       <h4 style={{ color: '#ef4444', fontSize: '14px', margin: '0 0 8px 0', display: 'flex', alignItems:...
238 |       <AlertCircle size={16} /> Ch ú y d e a d l i n e
239 |     </h4>
240 |     <p style={{ color: '#8899a6', fontSize: '13px', margin: 0, lineHeight: 1.5 }}>
241 |       N ù p s a u t h ý i g i a n q u y Ë n h s ½ b ẽ t r ẽ 2 0 % t ò n g s Ñ i ã m . H ã y
242 |     </p>
243 |   </div>
244 | </aside>
245 | </div>
246 | </div>
247 | );
248 | }
249 |
```

```

1 | 'use client';
2 |
3 | import React, { useEffect, useState, use } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import {
6 |   ClipboardList, Loader2, ArrowLeft,
7 |   Info, Calendar, User
8 | } from 'lucide-react';
9 | import Link from 'next/link';
10 | import AssignmentCard from '@components/classes/AssignmentCard';
11 |
12 | export default function StudentClassDetailsPage({ params }: { params: Promise<{ classId: string }> }) {
13 |   const resolvedParams = use(params);
14 |   const classId = resolvedParams.classId;
15 |
16 |   const [classData, setClassData] = useState<any>(null);
17 |   const [assignments, setAssignments] = useState<any[]>([]);
18 |   const [loading, setLoading] = useState(true);
19 |
20 |   useEffect(() => {
21 |     fetchClassInfo();
22 |   }, [classId]);
23 |
24 |   async function fetchClassInfo() {
25 |     setLoading(true);
26 |     try {
27 |       const { data: { user } } = await supabase.auth.getUser();
28 |       if (!user) return;
29 |
30 |       // 1. Fetch Class Info
31 |       const { data: cls, error: clsError } = await supabase
32 |         .from('classes')
33 |         .select('*', { teacher: profiles(full_name) })
34 |         .eq('id', classId)
35 |         .single();
36 |
37 |       if (clsError) throw clsError;
38 |       setClassData(cls);
39 |
40 |       // 2. Fetch Assignments & My Submissions
41 |       const { data: asgs, error: asgsError } = await supabase
42 |         .from('assignments')
43 |         .select(
44 |           '*',
45 |           {
46 |             submissions(id, total_score)
47 |           }
48 |         )
49 |         .eq('class_id', classId)
50 |         .eq('is_published', true)
51 |         .order('created_at', { ascending: false });
52 |
53 |       if (asgsError) throw asgsError;
54 |
55 |       // Filter submissions for current user manually because of how Supabase returns it
56 |       const formattedAsgs = asgs?.map(asg => ({
57 |         ...asg,
58 |         hasSubmitted: (asg.submissions as any[]).length > 0
59 |       })) || [];
60 |
61 |       setAssignments(formattedAsgs);
62 |
63 |     } catch (err) {
64 |       console.error('Error fetching student class info:', err);
65 |     } finally {
66 |       setLoading(false);
67 |     }
68 |
69 |     if (loading) {
70 |       return (
71 |         <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', height: '100%' }}>
72 |           <Loader2 className="animate-spin" color="#00f3ff" size={48} />
73 |         </div>
74 |       );
75 |     }
76 |
77 |     if (!classData) return <div>Không tìm thấy lớp học bên không có c
78 |
79 |     return (
80 |       <div style={{ padding: '32px' }}>
81 |         <header style={{ marginBottom: '40px' }}>
82 |           <Link href="/student/classes" style={{ display: 'inline-flex', alignItems: 'center', gap: '8px', color...
83 |             <ArrowLeft size={20} />
84 |             Lớp học của tôi
85 |           </Link>
86 |
87 |           <h1 style={{ fontSize: '36px', fontWeight: 800, margin: '0 0 12px 0', color: '#fff' }}>{classData.name...
88 |             <div style={{ display: 'flex', gap: '24px', color: '#8899a6', fontSize: '14px' }}>
89 |               <span style={{ display: 'flex', alignItems: 'center', gap: '6px' }}><User size={16} /> GV: <strong s...
90 |               <span style={{ display: 'flex', alignItems: 'center', gap: '6px' }}><strong s...
91 |               <span style={{ display: 'flex', alignItems: 'center', gap: '6px' }}><strong s...
92 |             </div>
93 |           </header>
94 |
95 |           <div style={{ display: 'flex', alignItems: 'center', gap: '12px', margin: '12px 0 32px 0' }}>
96 |             <ClipboardList size={24} color="#00f3ff" />
97 |             <h2 style={{ fontSize: '20px', fontWeight: 800, color: '#fff', m...
98 |           </div>
99 |
100 |           <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill, minmax(400px, 1fr))', gap: '24px'...
101 |             <div style={{ gridColumn: '1 / -1', textAlign: 'center', padding: '60px', color: '#4b5563', backgrou...
102 |               <p>Hiện lớp học này ch°a có nhiệ°m v°à nào dành cho bên.</p>
103 |             </div>
104 |           </div>
105 |         </div>
106 |       );
107 |
108 |       assignments.map(asg => (
109 |         <AssignmentCard
110 |           key={asg.id}
111 |           id={asg.id}
112 |           classId={classId}
113 |           title={asg.title}
114 |           type={asg.type}
115 |           deadline={asg.deadline}
116 |           requirements={asg.requirements}
117 |           hasSubmitted={asg.hasSubmitted}
118 |           role="student"
119 |         />
120 |       ))
121 |     );

```

File: app/(dashboard)/student/classes/page.tsx (132 lines, 4.6 KB)

```

1 | 'use client';
2 |
3 | import React, { useEffect, useState } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import ClassCard from '@components/classes/ClassCard';
6 | import JoinClassModal from '@components/classes/JoinClassModal';
7 | import { Plus, Loader2, BookOpen } from 'lucide-react';
8 |
9 | export default function StudentClassesPage() {
10 |   const [classes, setClasses] = useState<any[]>([]);
11 |   const [loading, setLoading] = useState(true);
12 |   const [isJoinModalOpen, setIsJoinModalOpen] = useState(false);
13 |
14 |   useEffect(() => {
15 |     fetchJoinedClasses();
16 |   }, []);
17 |
18 |   async function fetchJoinedClasses() {
19 |     setLoading(true);
20 |     try {
21 |       const { data: { user } } = await supabase.auth.getUser();
22 |       if (!user) return;
23 |
24 |       // Lấy danh sách lớp học thông qua class_members
25 |       const { data, error } = await supabase
26 |         .from('class_members')
27 |         .select(
28 |           class:classes(
29 |             *,
30 |             teacher:profiles(full_name),
31 |             assignments(count)
32 |           )
33 |         )
34 |         .eq('student_id', user.id);
35 |
36 |       if (error) throw error;
37 |
38 |       const formattedClasses = data?.map(item => ({
39 |         ...item.class,
40 |         teacherName: (item.class as any).teacher?.full_name,
41 |         activeAssignmentsCount: (item.class as any).assignments?.[0]?.count || 0,
42 |         memberCount: 0 // TODO: update sau hoc fetch thêm nzu csn
43 |       })) || [];
44 |
45 |       setClasses(formattedClasses);
46 |     } catch (err) {
47 |       console.error('Error fetching student classes:', err);
48 |     } finally {
49 |       setLoading(false);
50 |     }
51 |   }
52 |
53 |   if (loading) {
54 |     return (
55 |       <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', height: '100%' }}>
56 |         <Loader2 className="animate-spin" color="#00f3ff" size={48} />
57 |       </div>
58 |     );
59 |   }
60 |
61 |   return (
62 |     <div style={{ padding: '32px' }}>
63 |       <header style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom: '...' }}>
64 |         <div>
65 |           <h1 style={{ fontSize: '32px', fontWeight: 800, margin: '0 0 8 0' }}>
66 |             <p style={{ color: '#8899a6', margin: 0 }}>Tham gia lớp học và
67 |
68 |           </div>
69 |           <button
70 |             onClick={() => setIsJoinModalOpen(true)}
71 |             style={{
72 |               background: 'var(--brand-primary)', color: '#000', border: 'none',
73 |               padding: '12px 24px', borderRadius: '12px', fontWeight: 700,
74 |               display: 'flex', alignItems: 'center', gap: '8px', cursor: 'pointer',
75 |               transition: 'transform 0.2s'
76 |             }}
77 |             onMouseOver={e => e.currentTarget.style.transform = 'translateY(-2px)'}
78 |             onMouseOut={e => e.currentTarget.style.transform = 'translateY(0)'}
79 |           >
80 |             <Plus size={20} /> Tham gia lớp
81 |
82 |           </button>
83 |
84 |           {classes.length === 0 ? (
85 |             <div style={{
86 |               textAlign: 'center', padding: '100px 0',
87 |               background: 'rgba(12, 20, 36, 0.5)', borderRadius: '24px',
88 |               border: '2px dashed rgba(255,255,255,0.05)'
89 |             }}>
90 |               <BookOpen size={64} color="#1e293b" style={{ marginBottom: '24px' }} />
91 |               <h3 style={{ color: '#8899a6', margin: '0 0 24px 0' }}>Bạn ch
92 |
93 |               <button
94 |                 onClick={() => setIsJoinModalOpen(true)}
95 |                 style={{
96 |                   background: 'rgba(0, 243, 255, 0.1)', color: '#00f3ff', border: '1px solid #00f3ff',
97 |                   padding: '12px 32px', borderRadius: '12px', fontWeight: 700, cursor: 'pointer'
98 |                 }}
99 |               >
100 |                 Nh - p mã tham gia ngay
101 |
102 |               </button>
103 |             </div>
104 |           ) : (
105 |             <div style={{
106 |               display: 'grid', gridTemplateColumns: 'repeat(auto-fill, minmax(350px, 1fr))',
107 |               gap: '24px'
108 |             }}>
109 |               {classes.map(cls => (
110 |                 <ClassCard
111 |                   key={cls.id}
112 |                   id={cls.id}
113 |                   name={cls.name}
114 |                   code={cls.code}
115 |                   teacherName={cls.teacherName}
116 |                   memberCount={0} // HS có thể không csn thấy chính xác tổng
117 |                   maxStudents={cls.max_students}
118 |                   activeAssignmentsCount={cls.activeAssignmentsCount}
119 |                   role="student"
120 |                 />
121 |               ))}
122 |             </div>

```



```

120 |     })
121 |
122 |     <JoinClassModal
123 |       isOpen={isJoinModalOpen}
124 |       onClose={() => {
125 |         setIsJoinModalOpen(false);
126 |         fetchJoinedClasses(); // Refresh list sau khi join
127 |       }}
128 |     />
129 |   </div>
130 | );
131 | }
132 |

```

File: app/(dashboard)/student/dashboard/error.tsx (27 lines, 1.0 KB)

```

1 | 'use client'
2 |
3 | import { useEffect } from 'react'
4 |
5 | export default function StudentDashboardError({ error, reset }: { error: Error & { digest?: string }; reset: (...
6 |   useEffect(() => {
7 |     console.error(error)
8 |   }, [error])
9 |
10 |   return (
11 |     <div className="min-h-screen bg-[#080910] p-6 md:p-8 lg:p-10 flex items-center justify-center">
12 |       <div className="max-w-xl rounded-3xl border border-[#1d1f2a] bg-[#0f1018] p-10 text-center">
13 |         <div className="text-5xl">&p</div>
14 |         <h2 className="mt-4 text-2xl font-semibold text-[#d8dbe8]">C ó l
15 |         <p className="mt-3 text-sm text-[#5a5d72]">V u i l o n g t h i l i i h o . c
16 |
17 |         <button
18 |           type="button"
19 |           onClick={reset}
20 |           className="mt-6 inline-flex rounded-full bg-[#00d4aa] px-6 py-3 text-sm font-semibold text-[#080910]...
21 |         >
22 |           T h i l i i
23 |       </div>
24 |     </div>
25 |   )
26 | }
27 |

```

File: app/(dashboard)/student/dashboard/loading.tsx (25 lines, 1.1 KB)

```

1 | import React from 'react'
2 |
3 | export default function StudentDashboardLoading() {
4 |   return (
5 |     <div className="min-h-screen bg-[#080910] p-6 md:p-8 lg:p-10">
6 |       <div className="max-w-[1440px] mx-auto space-y-6 animate-pulse">
7 |         <div className="h-24 rounded-3xl bg-[#0f1018] border border-[#1d1f2a]" />
8 |         <div className="grid gap-4 sm:grid-cols-2 xl:grid-cols-4">
9 |           { [...Array(4)].map((_, index) => (
10 |             <div key={index} className="h-36 rounded-3xl bg-[#0f1018] border border-[#1d1f2a]" />
11 |           ))}
12 |         </div>
13 |         <div className="grid gap-4 lg:grid-cols-[2fr_1.2fr]">
14 |           <div className="h-[340px] rounded-3xl bg-[#0f1018] border border-[#1d1f2a]" />
15 |           <div className="space-y-4">
16 |             <div className="h-48 rounded-3xl bg-[#0f1018] border border-[#1d1f2a]" />
17 |             <div className="h-56 rounded-3xl bg-[#0f1018] border border-[#1d1f2a]" />
18 |           </div>
19 |         </div>
20 |         <div className="h-[340px] rounded-3xl bg-[#0f1018] border border-[#1d1f2a]" />
21 |       </div>
22 |     </div>
23 |   )
24 | }
25 |

```

File: app/(dashboard)/student/dashboard/page.tsx (159 lines, 4.4 KB)

```

1 | import { createClient } from '@lib/supabase-ssr-server'
2 | import { requireRole } from '@lib/auth/rbac'
3 |
4 | import { StudentDashboardClient } from './StudentDashboardClient'
5 |
6 | type LessonProgressRow = {
7 |   lesson_id: string
8 |   status: string | null
9 |   time_spent_seconds: number | null
10 |   completed_at: string | null
11 |   last_accessed: string | null
12 |   completion_percentage: number | null
13 |   lessons?: {
14 |     title: string | null
15 |     thumbnail_url: string | null
16 |   } | null
17 | }
18 |
19 |
20 |
21 |
22 |
23 | type ChartPoint = {
24 |   day: string
25 |   minutes: number
26 | }
27 |
28 | function getGreeting(): string {
29 |   const hour = new Date().getHours()
30 |   if (hour >= 5 && hour < 12) return 'sáng'
31 |   if (hour >= 12 && hour < 18) return 'chiều'
32 |   return 'tối'
33 | }
34 |
35 | function formatDateLabel(): string {
36 |   const now = new Date()
37 |   return new Intl.DateTimeFormat('vi-VN', {
38 |     weekday: 'long',
39 |     day: 'numeric',
40 |     month: 'numeric'
41 |   }).format(now)
42 | }
43 |
44 | function buildChartData(progress: LessonProgressRow[]): ChartPoint[] {
45 |   const today = new Date()
46 |   const days = Array.from({ length: 7 }, (_, index) => {

```

```

47 |     const current = new Date(today)
48 |     current.setDate(today.getDate() - (6 - index))
49 |     return current.toISOString().slice(0, 10)
50 | })
51 |
52 | return days.map((day) => {
53 |   const minutes = Math.round(
54 |     progress
55 |       .filter((row) => {
56 |         if (!row.last_accessed) return false
57 |         const accessed = new Date(row.last_accessed).toISOString().slice(0, 10)
58 |         return accessed === day
59 |       })
60 |       .reduce((sum, row) => sum + (row.time_spent_seconds ?? 0), 0) / 60
61 |   )
62 |
63 |   return {
64 |     day: day.slice(5).replace('-', '/'),
65 |     minutes: Math.max(minutes, 0)
66 |   }
67 | })
68 | }
69 |
70 | function calculateStreak(progress: LessonProgressRow[]): number {
71 |   const uniqueDays = new Set(
72 |     progress
73 |       .map((row) => {
74 |         if (!row.last_accessed) return ''
75 |         return new Date(row.last_accessed).toISOString().slice(0, 10)
76 |       })
77 |       .filter(Boolean)
78 |   )
79 |
80 |   if (uniqueDays.size === 0) {
81 |     return 0
82 |   }
83 |
84 |   let streak = 0
85 |   const current = new Date()
86 |
87 |   while (true) {
88 |     const key = current.toISOString().slice(0, 10)
89 |     if (uniqueDays.has(key)) {
90 |       streak += 1
91 |       current.setDate(current.getDate() - 1)
92 |       continue
93 |     }
94 |     break
95 |   }
96 |
97 |   return streak
98 | }
99 |
100 | export const dynamic = 'force-dynamic'
101 |
102 | export default async function StudentDashboardPage() {
103 |   const user = await requireRole(['student'])
104 |   const supabase = await createClient()
105 |
106 |   const [progressRes, quizRes, lessonsCountRes] = await Promise.all([
107 |     supabase
108 |       .from('lesson_progress')
109 |       .select('lesson_id, status, time_spent_seconds, completed_at, last_accessed, completion_percentage, less...
110 |       .eq('student_id', user.id),
111 |     supabase
112 |       .from('quiz_attempts')
113 |       .select('score, submitted_at, status')
114 |       .eq('student_id', user.id)
115 |       .in('status', ['passed', 'failed'])
116 |       .order('submitted_at', { ascending: false })
117 |       .limit(10),
118 |     supabase
119 |       .from('lessons')
120 |       .select('id', { count: 'exact', head: true })
121 |       .eq('is_published', true),
122 |   ])
123 |
124 |   if (progressRes.error || quizRes.error || lessonsCountRes.error) {
125 |     console.error('Student dashboard fetch error', {
126 |       progressError: progressRes.error,
127 |       quizError: quizRes.error,
128 |       lessonsCountError: lessonsCountRes.error,
129 |     })
130 |   }
131 |
132 |   const progress = (progressRes.data ?? []) as unknown as LessonProgressRow[]
133 |   const quizzes = quizRes.data ?? []
134 |   const totalLessonsCount = lessonsCountRes.count ?? 0
135 |
136 |   const completedLessons = progress.filter((item) => item.status === 'completed').length
137 |   const startedLessons = progress.length
138 |   const totalMinutes = Math.round(progress.reduce((sum, item) => sum + (item.time_spent_seconds ?? 0), 0) / 60)
139 |   const averageScore = quizzes.length > 0 ? Math.round(quizzes.reduce((sum, item) => sum + (item.score ?? 0), ...
140 |   const streakDays = calculateStreak(progress)
141 |   const chartData = buildChartData(progress)
142 |
143 |   return (
144 |     <StudentDashboardClient
145 |       user={user}
146 |       greeting={getGreeting()}
147 |       formattedDate={formatDateLabel()}
148 |       stats={{
149 |         completedLessons,
150 |         startedLessons,
151 |         totalHours: totalMinutes / 60,
152 |         averageScore,
153 |         streakDays
154 |       }}
155 |       chartData={chartData}
156 |     />
157 |   )
158 | }
159 |

```

File: app/(dashboard)/student/dashboard/StudentDashboardClient.tsx (320 lines, 17.5 KB)

```

1 | 'use client'
2 |
3 | import Link from 'next/link'
4 | import { useMemo } from 'react'
5 | import {
6 |   BarChart,
7 |   Bar,

```

```

8   CartesianGrid,
9   Cell,
10  ResponsiveContainer,
11  Tooltip,
12  XAxis,
13  YAxis
14 } from 'recharts'
15
16 import { motion } from 'framer-motion'
17 import { useRealtime } from '@lib/realtime-provider'
18 import { useSessionTime } from '@lib/session-time'
19 import { ChevronRight, Cpu } from 'lucide-react'
20 import { useIsMobile } from '@hooks/useIsMobile'
21
22 type UserProps = {
23   id: string
24   email: string
25   role: string
26   full_name: string
27   student_code?: string
28 }
29
30 type ProgressStats = {
31   completedLessons: number
32   startedLessons: number
33   totalHours: number
34   averageScore: number | null
35   streakDays: number
36 }
37
38 type ChartPoint = {
39   day: string
40   minutes: number
41 }
42
43 type Props = {
44   user: UserProps
45   greeting: string
46   formattedDate: string
47   stats: ProgressStats
48   chartData: ChartPoint[]
49 }
50
51 const BAR_FILL = '#00d4aa'
52 const TODAY_FILL = '#00f3ff'
53
54 const DAY_LABELS = ['T2', 'T3', 'T4', 'T5', 'T6', 'T7', 'CN']
55
56 function formatMinutes(value: number) {
57   return `${value} phút`
58 }
59
60 function useTodayIndex() {
61   return useMemo(() => {
62     const d = new Date().getDay()
63     return d === 0 ? 6 : d - 1
64   }, [])
65 }
66
67 function LiveDot({ size = 8 }: { size?: number }) {
68   return (
69     <span
70       style={{
71         display: 'inline-block',
72         width: size,
73         height: size,
74         borderRadius: '50%',
75         background: '#22c55e',
76         marginRight: 6,
77         animation: 'live-pulse 1.5s ease-in-out infinite',
78         verticalAlign: 'middle',
79       }}
80     />
81   )
82 }
83
84 export function StudentDashboardClient({
85   user,
86   greeting,
87   formattedDate,
88   stats,
89   chartData
90 }: Props) {
91   const { state: realtimeState } = useRealtime()
92   const sessionTime = useSessionTime()
93   const sessionMinutes = Math.floor(sessionTime.elapsed / 60)
94   const liveStreak = realtimeState.streak || stats.streakDays
95   const liveTotalHours = realtimeState.studyMinutes > 0 ? (realtimeState.studyMinutes / 60) : stats.totalHours
96   const liveWeeklyActivity: { day: string; minutes: number }[] = realtimeState.weeklyActivity.some(v => v > 0)
97     ? DAY_LABELS.map((day, i) => ({
98       day,
99       minutes: realtimeState.weeklyActivity[i] || 0
100     }))
101     : chartData
102   const todayIndex = useTodayIndex()
103   const todayMinutes = liveWeeklyActivity[todayIndex]?.minutes || 0
104   const hasStudiedToday = todayMinutes > 0 || sessionMinutes > 0
105   const isMobile = useIsMobile()
106
107   return (
108     <motion.div className={isMobile ? 'p-4' : 'p-6 md:p-8 lg:p-10'} style={{ background: 'transparent' }} init...
109     <div className="max-w-[1400px] mx-auto" style={{ display: 'flex', flexDirection: 'column', gap: '32px' }}>
110       <motion.div initial={{ opacity: 0, y: 15 }} animate={{ opacity: 1, y: 0 }} transition={{ delay: 0 * 0...
111         <section style={{ display: 'grid', gap: '16px', gridTemplateColumns: isMobile ? '1fr' : '1.6fr 1fr',...
112           <div className="lms-card" style={{ padding: isMobile ? '20px' : '32px', position: 'relative', over...
113             <div style={{ position: 'absolute', top: -20, right: -20, width: '120px', height: '120px', borde...
114               <p style={{ fontSize: '13px', textTransform: 'uppercase', letterSpacing: '2px', color: 'var(--br...
115               <h1 style={{ fontSize: isMobile ? '24px' : '32px', fontWeight: 800, color: 'var(--text-primary)'...
116               <p style={{ color: 'var(--text-muted)', fontSize: '14px', margin: 0 }}>Hôm nay là {formattedDate...
117               <div style={{ marginTop: '16px', display: 'flex', gap: '10px', flexWrap: 'wrap' }}>
118                 <Link href="/builder" className="lms-btn lms-btn-primary" style={{ display: 'inline-flex', ali...
119                 < C p u   s i z e = { 1 6 }   / >   T h ã c   h à n h   l ã p   r á p
120               </Link>
121             </div>
122           </div>
123           <motion.div className="lms-card" style={{ padding: isMobile ? '16px' : '24px', background: 'linear...
124             <div style={{ display: 'flex', alignItems: 'center', gap: '14px' }}>
125               <div style={{ fontSize: '44px', lineHeight: 1 }}>{realtimeState.levelIcon}</div>
126               <div>
127                 <div style={{ fontSize: '10px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: ...
128                 <div style={{ fontSize: '18px', fontWeight: 800, color: 'var(--text-primary)' }}>{realtimeSt...
129                 <div style={{ fontSize: '12px', color: 'var(--brand-primary)', fontWeight: 600 }}>Level {rea...
130               </div>
131             </div>

```

```

132|         <div style={{ height: '4px', borderRadius: '99px', background: 'rgba(255,255,255,0.06)', overflo...
133|         <div style={{ height: '100%', borderRadius: '99px', background: 'linear-gradient(90deg, var(--...
134|         </div>
135|         <div style={{ display: 'flex', justifyContent: 'space-between', fontSize: '10px', color: 'var(--...
136|         <span>{realtimeState.xpInLevel} / {realtimeState.xpToNext} XP</span>
137|         </div>
138|     </motion.div>
139| </section>
140| </motion.div>
141|
142| <motion.div initial={{ opacity: 0, y: 15 }} animate={{ opacity: 1, y: 0 }} transition={{ delay: 1 * 0...
143| <section style={{ display: 'grid', gap: '16px', gridTemplateColumns: isMobile ? '1fr' : '2.2fr 1fr' }}>
144|     <motion.div className="lms-card" style={{
145|         padding: isMobile ? '16px' : '24px',
146|         background: hasStudiedToday
147|           ? 'linear-gradient(135deg, rgba(34,197,94,0.06), transparent)'
148|           : undefined,
149|         border: hasStudiedToday ? '1px solid rgba(34,197,94,0.25)' : undefined,
150|     }} whileHover={{ scale: 1.02 }}>
151|         <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'space-between', marginBott...
152|         <div style={{ fontSize: '11px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: '2...
153|             <LiveDot /> H í c t - p h ô m n a y
154|         </div>
155|         <div style={{
156|             display: 'flex', alignItems: 'center', gap: 6,
157|             fontSize: '11px', fontWeight: 700, color: hasStudiedToday ? '#22c55e' : 'var(--text-muted)',
158|         }}>
159|             { hasStudiedToday ? ' ' ã h í c ' : 'Ch ° a h í c ' }
160|         </div>
161|     </div>
162|     <div style={{ display: 'flex', alignItems: 'baseline', gap: 8 }}>
163|         <motion.span
164|             key={sessionMinutes}
165|             initial={{ scale: 1.3, opacity: 0.5 }}
166|             animate={{ scale: 1, opacity: 1 }}
167|             transition={{ duration: 0.3 }}
168|             style={{ fontSize: isMobile ? '28px' : '36px', fontWeight: 800, color: 'var(--text-primary)' }}
169|         >
170|             {sessionMinutes}
171|         </motion.span>
172|         <span style={{ fontSize: '14px', color: 'var(--text-muted)', fontWeight: 600 }}>phút trên web<...
173|     </div>
174|     <div style={{ fontSize: '13px', color: 'var(--text-muted)', marginTop: 4 }}>
175|         {hasStudiedToday
176|           ? ` ã o n l i n e ${sessionMinutes} p h ú t h ô m n a y . `
177|           : ` B t S u m ú t b à i h í c h o . c l p r á p P C n g a y ! ` }
178|     </div>
179| </motion.div>
180|
181| <motion.div className="lms-card" style={{
182|     padding: isMobile ? '16px' : '24px',
183|     background: liveStreak >= 3
184|       ? 'linear-gradient(135deg, rgba(255,165,0,0.06), transparent)'
185|       : undefined,
186|     border: liveStreak >= 3 ? '1px solid rgba(255,165,0,0.2)' : undefined,
187| }} whileHover={{ scale: 1.02 }}>
188|     <div style={{ fontSize: '11px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: '2px...
189|         S t r e a k h i ç n t i i
190|     </div>
191|     <div style={{ display: 'flex', alignItems: 'baseline', gap: 8 }}>
192|         <motion.span
193|             key={liveStreak}
194|             initial={{ scale: 1.3, opacity: 0.5 }}
195|             animate={{ scale: 1, opacity: 1 }}
196|             transition={{ duration: 0.3 }}
197|             style={{ fontSize: isMobile ? '28px' : '36px', fontWeight: 800, color: liveStreak >= 3 ? '#f...
198|         >
199|             {liveStreak}
200|         </motion.span>
201|         <span style={{ fontSize: '14px', color: 'var(--text-muted)' }}>
202|     </div>
203|     <div style={{ marginTop: 8, display: 'flex', gap: 4 }}>
204|         {Array.from({ length: Math.min(liveStreak, 7) }).map((_, i) => (
205|             <div key={i} style={{
206|                 width: 24, height: 24, borderRadius: '50%',
207|                 background: i < liveStreak ? 'linear-gradient(135deg, #f59e0b, #ef4444)' : 'rgba(255,255,2...
208|                 display: 'flex', alignItems: 'center', justifyContent: 'center',
209|                 fontSize: 12, fontWeight: 700, color: '#fff',
210|                 } >0=Y% </div>
211|             )))
212|     </div>
213| </motion.div>
214| </section>
215| </motion.div>
216|
217| <motion.div initial={{ opacity: 0, y: 15 }} animate={{ opacity: 1, y: 0 }} transition={{ delay: 2 * 0...
218| <section style={{ display: 'grid', gap: '16px', gridTemplateColumns: 'repeat(auto-fit, minmax(180px,...
219|     {
220|         { label: 'B à i h o à n t h à n h ' , value: ` ${realtimeState.comple
221|         { label: 'T ō n g g i ý h í c ' , value: liveTotalHours.toFixed(1) },
222|         { label: ' i Ā m T B q u i z ' , value: realtimeState.averageScore
223|         { label: ' T Ō n g X P ' , value: ` ${realtimeState.xp} ` , sub: ' K i
224|     ].map((s, i) => (
225|         <motion.div key={i} className="lms-card" style={{ padding: isMobile ? '14px' : '20px', position:...
226|         {s.live && (
227|             <div style={{ position: 'absolute', top: 12, right: 12 }}>
228|                 <LiveDot size={6} />
229|             </div>
230|         )
231|         <div style={{ fontSize: '11px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: '2...
232|         <motion.div
233|             key={s.value}
234|             initial={{ scale: 1.2, opacity: 0.5 }}
235|             animate={{ scale: 1, opacity: 1 }}
236|             transition={{ duration: 0.3 }}
237|             style={{ fontSize: isMobile ? '22px' : '28px', fontWeight: 800, color: 'var(--text-primary)' }}
238|         >{s.value}</motion.div>
239|         <div style={{ fontSize: '13px', color: 'var(--text-muted)', marginTop: '4px' }}>{s.sub}</div>
240|         </motion.div>
241|     )))
242| </section>
243| </motion.div>
244|
245| /* Level Card */
246| <motion.div initial={{ opacity: 0, y: 15 }} animate={{ opacity: 1, y: 0 }} transition={{ delay: 2.5 * ...
247| <Link href="/student/level" style={{ textDecoration: 'none' }}>
248|     <motion.div className="lms-card" style={{ padding: isMobile ? '16px' : '24px', background: 'linear...
249|     <div style={{ position: 'absolute', top: '-50%', right: '-10%', width: '200px', height: '200px',...
250|     <div style={{ display: 'flex', alignItems: 'center', gap: '20px', position: 'relative', zIndex: ...
251|         <div style={{ textAlign: 'center', minWidth: '60px' }}>
252|             <div style={{ fontSize: '36px', lineHeight: 1 }}>{realtimeState.levelIcon}</div>
253|             <div style={{ fontSize: '10px', fontWeight: 800, color: 'var(--brand-primary)', textTransfor...
254|                 Level {realtimeState.level}
255|             </div>

```

```

256 |         </div>
257 |         <div style={{ flex: 1 }}>
258 |           <div style={{ fontSize: '13px', color: 'var(--text-muted)', fontWeight: 600, textTransform: ...
259 |           <div style={{ fontSize: '20px', fontWeight: 800, color: 'var(--text-primary)', marginBottom:...
260 |             {realtimeState.levelTitle}
261 |           </div>
262 |           <div style={{ height: '6px', borderRadius: '99px', background: 'rgba(255,255,255,0.08)', ove...
263 |           <div style={{ height: '100%', borderRadius: '99px', background: 'linear-gradient(90deg, va...
264 |           </div>
265 |           <div style={{ display: 'flex', justifyContent: 'space-between', fontSize: '11px', color: 'va...
266 |           <span>{realtimeState.xpInLevel} / {realtimeState.xpToNext} XP</span>
267 |           <span style={{ display: 'flex', alignItems: 'center', gap: '4px', color: 'var(--brand-prim...
268 |             <span>X e m c h i t i z t < C h e v r o n R i g h t s i z e = { 1 4 } / >
269 |           </span>
270 |         </div>
271 |       </div>
272 |     </div>
273 |   </motion.div>
274 | </Link>
275 | </motion.div>
276 |
277 |   <motion.div initial={{ opacity: 0, y: 15 }} animate={{ opacity: 1, y: 0 }} transition={{ delay: 4 * 0...
278 |   <section className="lms-card" style={{ padding: isMobile ? '16px' : '24px' }}>
279 |     <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom...
280 |     <div>
281 |       <div style={{ fontSize: '13px', textTransform: 'uppercase', letterSpacing: '2px', color: 'var(...
282 |       <div style={{ fontSize: '18px', fontWeight: 600, color: 'var(--text-primary)', marginTop: '4px...
283 |     </div>
284 |     <div style={{ display: 'flex', alignItems: 'center', gap: '8px', fontSize: '12px', color: '#22c5...
285 |       < L i v e D o t s i z e = { 6 } / > T r ñ c t i z p
286 |     </div>
287 |   </div>
288 |   <div style={{ height: '300px', width: '100%' }}>
289 |     <ResponsiveContainer width="100%" height="100%">
290 |       <BarChart data={liveWeeklyActivity} margin={{ top: 10, right: 10, left: -10, bottom: 0 }}>
291 |         <CartesianGrid stroke="var(--border-subtle)" vertical={false} />
292 |         <XAxis dataKey="day" tick={{ fill: 'var(--text-muted)', fontSize: 12 }} axisLine={false} tic...
293 |         <YAxis tick={{ fill: 'var(--text-muted)', fontSize: 12 }} axisLine={false} tickLine={false} />
294 |         <Tooltip formatter={formatMinutes} cursor={{ fill: 'rgba(8, 158, 96, 0.08)' }} contentStyle=...
295 |         <Bar dataKey="minutes" radius={[8, 8, 0, 0]} fill={BAR_FILL}>
296 |           {liveWeeklyActivity.map((entry, index) => (
297 |             <Cell
298 |               key={ `cell-${index}` }
299 |               fill={index === todayIndex ? TODAY_FILL : BAR_FILL}
300 |               style={index === todayIndex ? { filter: 'brightness(1.2)', transition: 'fill 0.3s ease...
301 |             </>
302 |           ))}
303 |         </Bar>
304 |       </BarChart>
305 |     </ResponsiveContainer>
306 |   </div>
307 | </section>
308 | </motion.div>
309 |
310 |   <style>{`
311 |     @keyframes live-pulse {
312 |       0%, 100% { opacity: 1; transform: scale(1); }
313 |       50% { opacity: 0.5; transform: scale(1.3); }
314 |     }
315 |   `}</style>
316 | </div>
317 | </motion.div>
318 | )
319 | }
320 |

```

File: app/(dashboard)/student/error.tsx (26 lines, 0.9 KB)

```

1 | // Path: app/(dashboard)/student/error.tsx
2 | 'use client'
3 | import { useEffect } from 'react'
4 |
5 | export default function Error({
6 |   error, reset
7 | }: { error: Error & { digest?: string }; reset: () => void }) {
8 |   useEffect(() => { console.error(error); }, [error])
9 |   return (
10 |     <div className="flex flex-col items-center justify-center h-full gap-6 p-8">
11 |       <div className="text-4xl">&p</div>
12 |       <div className="text-center">
13 |         <h2 className="text-lg font-semibold text-[#d8dbe8] mb-2">CÓ l x i
14 |         <p className="text-sm text-[#5a5d72] mb-4">{error.message} || 'V
15 |       </div>
16 |       <button
17 |         onClick={reset}
18 |         className="px-4 py-2 bg-[#00d4aa]/15 border border-[#00d4aa]/30
19 |         text-[#00d4aa] rounded-lg text-sm font-medium hover:bg-[#00d4aa]/25 transition-colors"
20 |       >
21 |         T h i l i
22 |       </button>
23 |     </div>
24 |   )
25 | }
26 |

```

File: app/(dashboard)/student/layout.tsx (242 lines, 12.1 KB)

```

1 | py 'use client'
2 |
3 | import React, { useEffect, useState, useCallback } from 'react'
4 | import Link from 'next/link'
5 | import { usePathname } from 'next/navigation'
6 | import {
7 |   GraduationCap, LayoutDashboard, Cpu, History, LogOut, Users,
8 |   BookOpen, Sun, Moon, Map, BarChart2, Menu, X, BookMarked,
9 |   ShoppingCart, Swords, FileQuestion, Gamepad2, User, ChevronRight, Zap
10 | } from 'lucide-react'
11 | import { logout } from '@lib/auth-actions'
12 | import { createBrowserClient } from '@supabase/ssr'
13 | import PageTransition from '@components/PageTransition'
14 | import { motion, AnimatePresence } from 'framer-motion'
15 | import { Clock } from 'lucide-react'
16 | import { useSessionTime } from '@lib/session-time'
17 |
18 | function SessionTimer() {
19 |   const { formatted } = useSessionTime()
20 |   return (
21 |     <div className="flex items-center gap-2 px-4 py-2 mx-3 my-2 rounded-lg text-xs font-semibold bg-emeral...
22 |     <Clock size={14} />
23 |     <span>{formatted}</span>
24 |   </div>

```

```

25) }
26) }
27)
28) function ThemeToggle() {
29)   const [theme, setThemeState] = useState<'light' | 'dark'>('dark')
30)   useEffect(() => {
31)     const saved = (localStorage.getItem('theme') as 'light' | 'dark') || 'dark'
32)     setThemeState(saved)
33)     document.documentElement.setAttribute('data-theme', saved)
34)   }, [])
35)   const toggle = useCallback(() => {
36)     const next = theme === 'light' ? 'dark' : 'light'
37)     setThemeState(next)
38)     document.documentElement.setAttribute('data-theme', next)
39)     localStorage.setItem('theme', next)
40)   }, [theme])
41)   return (
42)     <button onClick={toggle} className="flex items-center gap-3 px-4 py-3 w-full rounded-lg bg-transparent...
43)       {theme === 'light' ? <Moon size={18} /> : <Sun size={18} />}
44)       { theme === 'light' ? 'Gia o di c n t Ñi' : 'Gia o di c n s á ng' }
45)     </button>
46)   )
47) }
48)
49) const ALL_NAV_ITEMS = [
50)   { href: '/student/dashboard', label: 'B Ñng i Áu khi Ñn', icon: Layout
51)
52)   { href: '/builder', label: 'Luy c n T - p', icon: Cpu, mobileOnly: false
53)   { href: '/student/level', label: 'C ã p', icon: Zap, mobile
54)   { href: '/student/sgk', label: 'Ch ã M á y T ính', icon: ShoppingCart, mobile
55)   { href: '/student/quiz', label: 'K ó Th i', icon: FileQuestion, mobile
56)   { href: '/student/multiplayer', label: '2 Ng Ỗ i Ch i i', icon: Swords,
57)   { href: '/student/quiz-bank', label: 'Ng ãn H à ng Á Th i', icon: Gamep
58)   { href: '/student/history', label: 'L Ëch S í H íc T - p', icon: History,
59) ]
60)
61) export default function StudentLayout({ children }: { children: React.ReactNode }) {
62)   const [hasClass, setHasClass] = useState(false)
63)   const [profile, setProfile] = useState<any>(null)
64)   const [mobileOpen, setMobileOpen] = useState(false)
65)   const pathname = usePathname()
66)   const supabase = createBrowserClient(
67)     process.env.NEXT_PUBLIC_SUPABASE_URL!,
68)     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
69)   )
70)
71)   useEffect(() => {
72)     const checkClass = async () => {
73)       const { data: { user } } = await supabase.auth.getUser()
74)       if (user) {
75)         const { data: p } = await supabase.from('profiles').select('*').eq('id', user.id).single()
76)         if (p) { setProfile(p); if (p.school_id || p.class_id) setHasClass(true) }
77)       }
78)     }
79)     checkClass()
80)   }, [])
81)
82)   useEffect(() => {
83)     if (mobileOpen) {
84)       document.body.style.overflow = 'hidden'
85)     } else {
86)       document.body.style.overflow = ''
87)     }
88)     return () => { document.body.style.overflow = '' }
89)   }, [mobileOpen])
90)
91)   const closeMobile = () => setMobileOpen(false)
92)
93)   const desktopNavItems = ALL_NAV_ITEMS.filter(n => !n.mobileOnly)
94)   const mobileNavItems = ALL_NAV_ITEMS
95)
96)   const sidebarContent = (
97)     <>
98)       <div className="p-6 flex items-center gap-3 border-b border-white/10">
99)         <div className="w-10 h-10 rounded-xl bg-[var(--brand-primary)] flex items-center justify-cente...
100)           {profile?.avatar_url ? (
101)             <img src={profile.avatar_url} alt="" style={{ width: '100%', height: '100%', objectFit...
102)           ) : (
103)             <GraduationCap size={20} color="#fff" />
104)           )}
105)         </div>
106)         <div className="flex-1">
107)           <div className="font-extrabold text-lg text-white">PC Master</div>
108)           <div className="text-[11px] text-[var(--brand-primary)]">
109)             </div>
110)           </div>
111)
112)           <SessionTimer />
113)           <nav className="flex-1 px-3 py-4 flex flex-col gap-1">
114)             {desktopNavItems.map(({ href, label, icon }) => {
115)               const active = pathname === href
116)               return (
117)                 <Link key={href} href={href} className={`flex items-center gap-3 px-4 py-3 rounded-xl ...
118)                   active
119)                   ? 'bg-emerald-500/15 text-white font-bold'
120)                   : 'text-white/60 hover:text-white/80 hover:bg-white/5'
121)                 >>
122)                   <Icon size={20} />
123)                   <span className="flex-1">{label}</span>
124)                   {active && <ChevronRight size={16} className="text-emerald-400" />}
125)                 </Link>
126)               )
127)             })}
128)           </nav>
129)
130)           <div className="p-4 border-t border-white/10 flex flex-col gap-1">
131)             <ThemeToggle />
132)             <button onClick={() => { closeMobile(); logout(); }}
133)               className="flex items-center gap-3 px-4 py-3 w-full rounded-lg bg-transparent border-none ...
134)               <Logout size={18} /> ng x u ¢ t
135)             </button>
136)           </div>
137)         </>
138)       )
139)
140)   return (
141)     <div className="flex min-h-screen bg-[var(--bg-base)]">
142)       <div className="flex flex-col">
143)         <div className="hidden md:flex w-[280px] bg-gradient-to-b from-[#031f3b] to-[#1a2f53] flex-col s...
144)           {sidebarContent}
145)         </div>
146)         <div className="flex flex-1">
147)           <div>
148)             <div>

```

```

149 |         onClick={() => setMobileOpen(true)}
150 |         className="fixed top-3 left-3 z-[1001] md:hidden flex items-center justify-center w-11 h-11 ro...
151 |         a r i a - l a b e l = " M ß   m e n u "
152 |     >
153 |         <Menu size={22} />
154 |     </button>
155 |
156 |     {/* Mobile header */}
157 |     <header className="md:hidden fixed top-0 left-0 right-0 z-50 flex items-center justify-between px-...
158 |         <div className="w-11" />
159 |         <div className="flex items-center gap-2">
160 |             <div className="w-7 h-7 rounded-lg bg-[var(--brand-primary)] flex items-center justify-cen...
161 |                 <GraduationCap size={14} color="#fff" />
162 |             </div>
163 |             <span className="text-sm font-extrabold text-[var(--text-primary)]">PC Master</span>
164 |         </div>
165 |         <div className="w-9" />
166 |     </header>
167 |
168 |     {/* Spacer for mobile header */}
169 |     <div className="md:hidden h-14 shrink-0" />
170 |
171 |     {/* Mobile drawer */}
172 |     <AnimatePresence>
173 |         {mobileOpen && (
174 |             <>
175 |                 <motion.div
176 |                     initial={{ opacity: 0 }}
177 |                     animate={{ opacity: 1 }}
178 |                     exit={{ opacity: 0 }}
179 |                     transition={{ duration: 0.2 }}
180 |                     onClick={closeMobile}
181 |                     className="fixed inset-0 bg-black/60 backdrop-blur-sm z-[1002] md:hidden"
182 |                 />
183 |                 <motion.aside
184 |                     initial={{ x: '-100%' }}
185 |                     animate={{ x: 0 }}
186 |                     exit={{ x: '-100%' }}
187 |                     transition={{ type: 'spring', damping: 28, stiffness: 300 }}
188 |                     className="fixed top-0 left-0 bottom-0 w-[290px] z-[1003] md:hidden flex flex-col ...
189 |                 >
190 |                     <div className="p-5 flex items-center gap-3 border-b border-white/10">
191 |                         <div className="w-10 h-10 rounded-xl bg-[var(--brand-primary)] flex items-cent...
192 |                             {profile?.avatar_url ? (
193 |                                 <img src={profile.avatar_url} alt="" style={{ width: '100%', height: '1...
194 |                             ) : (
195 |                                 <GraduationCap size={20} color="#fff" />
196 |                             )}
197 |                         </div>
198 |                         <div className="flex-1">
199 |                             <div className="font-extrabold text-lg text-white">PC Master</div>
200 |                             <div className="text-[11px] text-[var(--brand-primary)] font-semibold opac...
201 |                         </div>
202 |                         <button onClick={closeMobile}
203 |                             className="w-9 h-9 rounded-lg bg-white/10 border-none flex items-center ju...
204 |                             <X size={20} />
205 |                         </button>
206 |                     </div>
207 |                     <nav className="flex-1 px-3 py-4 flex flex-col gap-1">
208 |                         {mobileNavItems.map(({ href, label, icon: Icon }) => {
209 |                             const active = pathname === href
210 |                             return (
211 |                                 <Link key={href} href={href} onClick={closeMobile}
212 |                                     className={`flex items-center gap-3 px-4 py-3.5 rounded-xl text-sm...
213 |                                     active
214 |                                     ? 'bg-emerald-500/15 text-white font-bold'
215 |                                     : 'text-white/60 hover:text-white/80 hover:bg-white/5'
216 |                                 >
217 |                                     <Icon size={20} />
218 |                                     <span className="flex-1">{label}</span>
219 |                                     {active && <ChevronRight size={16} className="text-emerald-400" />}
220 |                                 </Link>
221 |                             )
222 |                         )}
223 |                     </nav>
224 |                     <div className="p-4 border-t border-white/10 flex flex-col gap-1">
225 |                         <ThemeToggle />
226 |                         <button onClick={() => { closeMobile(); logout(); }}
227 |                             className="flex items-center gap-3 px-4 py-3 w-full rounded-lg bg-transpar...
228 |                             <LogOut size={18} />
229 |                         </button>
230 |                     </div>
231 |                 </motion.aside>
232 |             </>
233 |         )}
234 |     </AnimatePresence>
235 |
236 |     <main className="flex-1 overflow-y-auto min-w-0 md:pt-0 pt-0">
237 |         <PageTransition>{children}</PageTransition>
238 |     </main>
239 | </div>
240 | )
241 | }
242 |

```

File: app/(dashboard)/student/lessons/[lessonId]/LessonViewer.tsx (376 lines, 16.9 KB)

```

1 | 'use client'
2 |
3 | import { useState, useEffect, useRef, useCallback } from 'react'
4 | import { useRouter } from 'next/navigation'
5 | import Link from 'next/link'
6 | import {
7 |     Video, FileText, Image as ImageIcon, FileSearch, Code, ChevronRight,
8 |     CheckCircle, Clock, Play, BookOpen, HelpCircle, ExternalLink
9 | } from 'lucide-react'
10 | import ReactMarkdown from 'react-markdown'
11 | import { saveLessonProgress } from '@lib/lesson-actions'
12 | import { getKeyQuestions } from '@data/key-questions'
13 | import { QUIZ_BANK } from '@data/quiz-bank'
14 | import PdfViewer from '@components/PdfViewer'
15 |
16 | const EXTRA_VIDEOS: Record<string, { title: string; id: string }> = {
17 |     'cpu': [
18 |         { title: 'CPU là gì? - Giải thích chi tiết', id: 'cWzB7GvN6vs' },
19 |         { title: 'Cách chọn CPU phù hợp', id: '7j3I7x7Gz6M' },
20 |     ],
21 |     'mainboard': [
22 |         { title: 'Bo mạch chủ là gì?', id: 'Q_sBq0-iv4I' },
23 |     ],
24 |     'ram': [
25 |         { title: 'RAM là gì? Giải thích DDR4 vs DDR5', id: 'qTQ7vg1JFDA' },
26 |     ],

```

```

27| 'gpu': [
28|     { title: 'Card ó hĩa (GPU) hoĩt ùng thũ nào?', id: 'OQ6fMGmIVPE' },
29| ],
30| 'ổ cẻng': [
31|     { title: 'SSD vs HDD - Nẻn chĩn loĩi nào?', id: 'YQEjGKYXjw8' } ],
32| ],
33| 'ngu ón': [
34|     { title: 'Ngu ón máy tĩnh (PSU) - H° ùng d«n chĩn', id: 'Pco0vPjv0wk' } ],
35| ],
36| 'case': [
37|     { title: 'Case máy tĩnh - Các loĩi và cách chĩn', id: '08HQzP3QwL8' } ],
38| ],
39| }
40|
41| interface Section {
42|   id: string
43|   title: string
44|   type: string
45|   content: string
46|   order_index: number
47|   source?: string
48| }
49|
50| interface Lesson {
51|   id: string
52|   title: string
53|   description: string
54|   estimated_minutes?: number
55| }
56|
57| interface Progress {
58|   is_completed?: boolean
59|   completion_percentage?: number
60|   time_spent_seconds?: number
61| }
62|
63| export default function LessonViewer({
64|   lesson, sections, initialProgress, userId,
65| }: {
66|   lesson: Lesson
67|   sections: Section[]
68|   initialProgress: Progress | null
69|   userId: string
70| }) {
71|   const router = useRouter()
72|   const [activeSection, setActiveSection] = useState(sections[0]?.id || '')
73|   const [completed, setCompleted] = useState(initialProgress?.is_completed || false)
74|   const [timeSpent, setTimeSpent] = useState(initialProgress?.time_spent_seconds || 0)
75|   const intervalRef = useRef<NodeJS.Timeout | null>(null)
76|   const sectionRefs = useRef<Record<string, HTMLDivElement | null>>({})
77|
78|   useEffect(() => {
79|     saveLessonProgress(lesson.id, 0, false)
80|
81|     intervalRef.current = setInterval(async () => {
82|       setTimeSpent(prev => {
83|         const newTime = prev + 30
84|         saveLessonProgress(lesson.id, 0, false)
85|         return newTime
86|       })
87|     }, 30000)
88|     return () => {
89|       if (intervalRef.current) clearInterval(intervalRef.current)
90|     }
91|   }, [lesson.id])
92|
93|   const handleComplete = useCallback(async () => {
94|     setCompleted(true)
95|     await saveLessonProgress(lesson.id, 100, true)
96|     router.refresh()
97|   }, [lesson.id, router])
98|
99|   const getYouTubeEmbed = (url: string) => {
100|     if (!url) return ''
101|     const id = url.match(
102|       /^(?:youtube\.com\/(?:[^\./+\\|]{?v|e(?:mbed)?}\\|.*(?:&v=)|youtu\.be\\|)([^\?&\/\s]{11})\/(?:\.[1])?\.1$/
103|     )?.[1]
104|     return id ? `https://www.youtube-nocookie.com/embed/${id}` : ''
105|   }
106|
107|   const getDriveEmbed = (url: string) => {
108|     if (!url) return ''
109|     const id = url.match(/\/d\/(.+?)\/(?:view|preview)?\.1/ || url.match(/id=(.+?)(&|&)/)?.[1]
110|     return id ? `https://drive.google.com/file/d/${id}/preview` : ''
111|   }
112|
113|   const getPdfEmbedUrl = (url: string) => {
114|     if (!url) return ''
115|     const id = url.match(/\/d\/(.+?)\/(?:view|preview)?\.1/ || url.match(/id=(.+?)(&|&)/)?.[1]
116|     if (id) {
117|       const downloadUrl = `https://drive.google.com/uc?export=download&id=${id}`
118|       return `https://docs.google.com/viewer?url=${encodeURIComponent(downloadUrl)}&embedded=true`
119|     }
120|     return url
121|   }
122|
123|   const parseRichContent = (content: string) => {
124|     const parts: { type: 'markdown' | 'video' | 'pdf', content?: string, url?: string }[] = []
125|     const regex = /<div class="(video-embed|pdf-embed)">([\\s\\S]*)<\\/div>/g
126|     let lastIndex = 0, match
127|     while ((match = regex.exec(content)) !== null) {
128|       if (match.index > lastIndex) parts.push({ type: 'markdown', content: content.substring(lastIndex, match.index) })
129|       parts.push({ type: match[1] === 'video-embed' ? 'video' : 'pdf', url: match[2].trim() })
130|       lastIndex = match.index + match[0].length
131|     }
132|     if (lastIndex < content.length) parts.push({ type: 'markdown', content: content.substring(lastIndex) })
133|     if (parts.length === 0) parts.push({ type: 'markdown', content })
134|     return parts
135|   }
136|
137|   const scrollToSection = (sectionId: string) => {
138|     setActiveSection(sectionId)
139|     sectionRefs.current[sectionId]?.scrollIntoView({ behavior: 'smooth' })
140|   }
141|
142|   const keyQuestions = getKeyQuestions(lesson.title)
143|   const relatedQuizzes = QUIZ_BANK.filter(q => q.lessonTitle === lesson.title && q.questions.length > 0)
144|   const [showKeyQuestions, setShowKeyQuestions] = useState(false)
145|
146|   if (sections.length === 0) {
147|     return (
148|       <div className="text-center py-20 rounded-2xl" style={{ background: 'var(--bg-elevated)', border: '1px solid #ccc' }}>
149|         <BookOpen size={48} className="mx-auto mb-4" style={{ color: 'var(--text-muted)' }} />
150|         <p className="font-medium" style={{ color: 'var(--text-secondary)' }}>

```



```

151 |     </div>
152 |   )
153 | }
154 |
155 | return (
156 |   <div className="flex gap-6">
157 |     <div className="flex-1 min-w-0">
158 |       <div className="rounded-2xl p-6 mb-6" style={{ background: 'var(--bg-elevated)', border: '1px solid va...
159 |         <h1 className="text-2xl font-bold mb-2" style={{ color: 'var(--text-primary)' }}>{lesson.title}</h1>
160 |         <p className="text-sm" style={{ color: 'var(--text-secondary)' }}>{lesson.description}</p>
161 |         {lesson.estimated_minutes && (
162 |           <div className="flex items-center gap-1.5 mt-3 text-xs" style={{ color: 'var(--text-muted)' }}>
163 |             <Clock size={13} /> ~{lesson.estimated_minutes} phút
164 |           </div>
165 |         )}
166 |       </div>
167 |     </div>
168 |
169 |     <div className="flex flex-col gap-8">
170 |       {sections.map((s) => (
171 |         <div
172 |           key={s.id}
173 |           ref={el => { sectionRefs.current[s.id] = el }}
174 |           className="rounded-2xl overflow-hidden"
175 |           style={{ background: 'var(--bg-elevated)', border: '1px solid var(--border-subtle)' }}
176 |         >
177 |           <div className="flex items-center gap-3 px-6 pt-5 pb-3">
178 |             <div className="p-2 rounded-lg" style={{ background: 'var(--bg-surface)' }}>
179 |               {s.type === 'video' && <Video size={18} style={{ color: 'var(--brand-primary)' }} />
180 |               {s.type === 'text' && <FileText size={18} style={{ color: 'var(--brand-primary)' }} />
181 |               {s.type === 'image' && <ImageIcon size={18} style={{ color: 'var(--brand-primary)' }} />
182 |               {s.type === 'pdf' && <FileSearch size={18} style={{ color: 'var(--brand-primary)' }} />
183 |               {s.type === 'embed' && <Code size={18} style={{ color: 'var(--brand-primary)' }} />
184 |             </div>
185 |             <h2 className="text-lg font-bold" style={{ color: 'var(--text-primary)' }}>{s.title}</h2>
186 |           </div>
187 |
188 |           <div className="px-6 pb-6">
189 |             {s.type === 'video' && getYouTubeEmbed(s.content) && (
190 |               <div className="aspect-video rounded-xl overflow-hidden" style={{ background: '#000' }}>
191 |                 <iframe
192 |                   src={getYouTubeEmbed(s.content)}
193 |                   className="w-full h-full"
194 |                   allowFullScreen
195 |                   loading="lazy"
196 |                   title={s.title}
197 |                 />
198 |               </div>
199 |             )}
200 |
201 |             {s.type === 'text' && (
202 |               <div className="prose max-w-none text-sm leading-relaxed" style={{ color: 'var(--text-second...
203 |                 {parseRichContent(s.content).map((part, i) => {
204 |                   if (part.type === 'video') {
205 |                     const embedUrl = getYouTubeEmbed(part.url!)
206 |                     return embedUrl ? (
207 |                       <div key={i} className="aspect-video rounded-xl overflow-hidden my-4" style={{ backg...
208 |                         <iframe src={embedUrl} className="w-full h-full" allowFullScreen title="Video" loa...
209 |                       </div>
210 |                     ) : null
211 |                   }
212 |                   if (part.type === 'pdf') {
213 |                     return part.url ? (
214 |                       <div key={i} className="my-4">
215 |                         <PdfViewer url={part.url} title="PDF" />
216 |                       </div>
217 |                     ) : null
218 |                   }
219 |                   return <ReactMarkdown key={i}>{part.content!}</ReactMarkdown>
220 |                 })}
221 |               </div>
222 |             )}
223 |
224 |             {s.type === 'image' && (
225 |               <img src={s.content} alt={s.title} style={{ maxWidth: '100%', height: 'auto', borderRadius: ...
226 |             )}
227 |
228 |             {s.type === 'pdf' && s.content && (
229 |               <PdfViewer url={s.content} title={s.title} />
230 |             )}
231 |
232 |             {s.type === 'embed' && s.content && (
233 |               <div className="rounded-xl overflow-hidden" style={{ border: '1px solid var(--border-subtle)...
234 |                 <div dangerouslySetInnerHTML={{ __html: s.content }} />
235 |               </div>
236 |             )}
237 |
238 |             {s.source && (
239 |               <div className="flex items-center gap-1.5 mt-3 text-xs font-medium" style={{ color: 'var(--t...
240 |                 <BookOpen size={12} /> Nguồn: {s.source}
241 |               </div>
242 |             )}
243 |           </div>
244 |         </div>
245 |       )}
246 |     </div>
247 |
248 |     {(() => {
249 |       const matchKey = Object.keys(EXTRA_VIDEOS).find(k => lesson.title.toLowerCase().includes(k))
250 |       const videos = matchKey ? EXTRA_VIDEOS[matchKey] : null
251 |       if (!videos) return null
252 |       return (
253 |         <div className="rounded-2xl mt-8 p-6" style={{ background: 'var(--bg-elevated)', border: '1px soli...
254 |           <h3 className="text-lg font-bold mb-4 flex items-center gap-2" style={{ color: 'var(--text-prima...
255 |             <span style={{ width: '32px', height: '32px', borderRadius: '8px', background: '#ef44420', di...
256 |             <Video size={16} style={{ color: '#ef4444' }} />
257 |           </span>
258 |           Video kiến thức mB r ù n g
259 |         </h3>
260 |         <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill, minmax(280px, 1fr))', gap...
261 |           {videos.map((v, i) => (
262 |             <div key={i} style={{ borderRadius: '12px', overflow: 'hidden', background: 'var(--bg-surfac...
263 |               <div style={{ position: 'relative', paddingBottom: '56.25%' }}>
264 |                 <iframe src={`https://www.youtube-nocookie.com/embed/${v.id}`} style={{ position: 'absol...
265 |               </div>
266 |               <div style={{ padding: '10px 12px' }}>
267 |                 <p style={{ color: 'var(--text-primary)', fontWeight: 600, fontSize: '13px', margin: 0 }}...
268 |               </div>
269 |             </div>
270 |           ))}
271 |         </div>
272 |       )}
273 |     )}()
274 |

```

```

275 | <div className="rounded-2xl overflow-hidden mt-8" style={{ background: 'var(--bg-elevated)', border: '...
276 | <button
277 |   onClick={() => setShowKeyQuestions(!showKeyQuestions)}
278 |   className="w-full flex items-center justify-between px-6 py-4 text-left transition-colors hover:op...
279 |   >
280 |     <div className="flex items-center gap-3">
281 |       <div className="p-2 rounded-lg" style={{ background: 'var(--bg-surface)' }}>
282 |         <HelpCircle size={18} style={{ color: 'var(--brand-primary)' }} />
283 |       </div>
284 |       <span className="font-bold" style={{ color: 'var(--text-primary)' }}>
285 |         C á u h ì i t r ì n g t á m ( { k e y Q u e s t i o n s . l e n g t h } )
286 |       </span>
287 |     </div>
288 |     <ChevronRight size={18} style={{ color: 'var(--text-muted)', transform: showKeyQuestions ? 'rotate...
289 |   </button>
290 |   {showKeyQuestions && (
291 |     <div className="px-6 pb-6 space-y-4">
292 |       {keyQuestions.length === 0 ? (
293 |         <p
294 |           className="text-sm" style={{ color: 'var(--text-muted)'
295 |         >
296 |           {keyQuestions.map((kq, i) => (
297 |             <div key={i} className="p-4 rounded-xl" style={{ background: 'var(--bg-surface)', border: '1...
298 |             <p className="text-sm font-bold mb-2" style={{ color: 'var(--text-primary)' }}>
299 |               <span style={{ color: 'var(--brand-primary)' }}>Q{i + 1}</span> {kq.q}
300 |             </p>
301 |             <p className="text-sm leading-relaxed" style={{ color: 'var(--text-secondary)' }}>{kq.a}</p>
302 |             </div>
303 |           ))
304 |         </div>
305 |       )}
306 |     </div>
307 |
308 |     {relatedQuizzes.length > 0 && (
309 |       <div className="rounded-2xl mt-6 p-6" style={{ background: 'var(--bg-elevated)', border: '1px solid ...
310 |       <div className="flex items-center gap-3 mb-4">
311 |         <div className="p-2 rounded-lg" style={{ background: 'var(--bg-surface)' }}>
312 |           <BookOpen size={18} style={{ color: 'var(--brand-primary)' }} />
313 |         <div>
314 |           <span
315 |             className="font-bold" style={{ color: 'var(--text-pr
316 |           >
317 |         </div>
318 |       </div>
319 |       <div className="flex flex-wrap gap-3">
320 |         {relatedQuizzes.map(q => (
321 |           <Link
322 |             key={q.id}
323 |             href={` /student/quiz/${q.id} `}
324 |             className="flex items-center gap-2 px-4 py-2.5 rounded-xl text-sm font-bold transition-all h...
325 |             style={{ background: 'color-mix(in srgb, var(--brand-primary) 15%, transparent)', color: 'va...
326 |           >
327 |             <ExternalLink size={14} />
328 |             {q.title}
329 |           </Link>
330 |         ))}
331 |       </div>
332 |     </div>
333 |   )}
334 |
335 |   <div className="sticky bottom-6 mt-8 flex justify-center">
336 |     <button
337 |       onClick={handleComplete}
338 |       disabled={completed}
339 |       className="px-8 py-3.5 rounded-xl font-bold text-sm flex items-center gap-2 transition-all shadow-...
340 |       style={{
341 |         background: completed ? 'var(--bg-elevated)' : 'var(--brand-primary)',
342 |         color: completed ? 'var(--text-secondary)' : '#000',
343 |         border: completed ? '1px solid var(--border-subtle)' : 'none',
344 |       }}
345 |     >
346 |       {completed ? (
347 |         <><CheckCircle size={18} /> ã hoàn thành</>
348 |       ) : (
349 |         <><Play size={18} /> ánh d ể u hoàn thành</>
350 |       )}
351 |     </button>
352 |   </div>
353 | </div>
354 |
355 | <aside className="w-64 shrink-0 hidden lg:block">
356 |   <div className="sticky top-6 rounded-2xl p-5" style={{ background: 'var(--bg-elevated)', border: '1px ...
357 |   <h3 className="text-xs font-bold uppercase mb-4 tracking-wider" style={{ color: 'var(--text-muted)' }}>
358 |     <div className="flex flex-col gap-1.5">
359 |       {sections.map((s, idx) => (
360 |         <button
361 |           key={s.id}
362 |           onClick={() => scrollToSection(s.id)}
363 |           className="flex items-center gap-2.5 px-3 py-2 rounded-lg text-left text-sm transition-colors"
364 |           style={{
365 |             background: activeSection === s.id ? 'var(--bg-surface)' : 'transparent',
366 |             color: activeSection === s.id ? 'var(--text-primary)' : 'var(--text-secondary)',
367 |           }}
368 |         >
369 |           <span className="text-xs font-bold w-5 shrink-0" style={{ color: 'var(--brand-primary)' }}>{id...
370 |           <span className="truncate">{s.title}</span>
371 |         </button>
372 |       ))}
373 |     </div>
374 |   </div>
375 | </aside>
376 | )
377 | }

```

File: app/(dashboard)/student/lessons/[lessonId]/page.tsx (63 lines, 1.6 KB)

```

1 | import { createClient } from '@lib/supabase-ssr-server'
2 | import { redirect } from 'next/navigation'
3 | import Link from 'next/link'
4 | import { ArrowLeft, Video, FileText, Image as ImageIcon, FileSearch, CheckCircle, Clock } from 'lucide-react'
5 | import LessonViewer from './LessonViewer'
6 |
7 | export const dynamic = 'force-dynamic'
8 |
9 | export default async function StudentLessonDetailPage({
10 |   params,
11 | }) {
12 |   params: Promise<{ lessonId: string }>
13 | } {
14 |   const { lessonId } = await params
15 |   const supabase = await createClient()
16 |   const { data: { user } } = await supabase.auth.getUser()
17 |
18 |   if (!user) redirect('/login')

```

```

19 | const { data: lesson } = await supabase
20 |   .from('lessons')
21 |   .select('*')
22 |   .eq('id', lessonId)
23 |   .single()
24 |
25 |
26 | if (!lesson) {
27 |   redirect('/student/lessons')
28 | }
29 |
30 | const { data: sections } = await supabase
31 |   .from('lesson_sections')
32 |   .select('*')
33 |   .eq('lesson_id', lessonId)
34 |   .order('order_index', { ascending: true })
35 |
36 | const { data: progress } = await supabase
37 |   .from('lesson_progress')
38 |   .select('*')
39 |   .eq('student_id', user.id)
40 |   .eq('lesson_id', lessonId)
41 |   .maybeSingle()
42 |
43 | return (
44 |   <div>
45 |     <Link
46 |       href="/student/lessons"
47 |       className="inline-flex items-center gap-2 text-sm font-medium mb-6 transition-colors"
48 |       style={{ color: 'var(--text-secondary)' }}
49 |     >
50 |       <ArrowLeft size={16} />
51 |       Q u a y   l i   d a n h   s á c h   b à i   g i ẽ n g
52 |     </Link>
53 |
54 |     <LessonViewer
55 |       lesson={lesson}
56 |       sections={sections || []}
57 |       initialProgress={progress}
58 |       userId={user.id}
59 |     />
60 |   </div>
61 | )
62 | }
63 |

```

File: app/(dashboard)/student/lessons/LessonsList.tsx (175 lines, 7.5 KB)

```

1 | 'use client'
2 |
3 | import { useEffect, useState } from 'react'
4 | import { createBrowserClient } from '@supabase/ssr'
5 | import Link from 'next/link'
6 | import { Book, Play, CheckCircle, Clock, Star, ArrowRight, Layers } from 'lucide-react'
7 |
8 | interface Lesson {
9 |   id: string
10 |   title: string
11 |   description: string
12 |   thumbnail_url?: string
13 |   estimated_minutes?: number
14 |   is_published: boolean
15 |   created_at: string
16 | }
17 |
18 | interface Progress {
19 |   lesson_id: string
20 |   is_completed?: boolean
21 |   completion_percentage?: number
22 | }
23 |
24 | export default function LessonsList({
25 |   initialLessons,
26 |   initialProgress,
27 |   userId,
28 | }: {
29 |   initialLessons: Lesson[]
30 |   initialProgress: Progress[]
31 |   userId: string
32 | }) {
33 |   const [lessons, setLessons] = useState<Lesson[]>(initialLessons)
34 |   const [progressList, setProgressList] = useState<Progress[]>(initialProgress)
35 |
36 |   const supabase = createBrowserClient(
37 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
38 |     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
39 |   )
40 |
41 |   useEffect(() => {
42 |     const channel = supabase
43 |       .channel('student-lessons-realtime')
44 |       .on('postgres_changes',
45 |         { event: 'INSERT', schema: 'public', table: 'lessons', filter: 'is_published=eq.true' },
46 |         (payload) => {
47 |           const newLesson = payload.new as Lesson
48 |           setLessons(prev => {
49 |             if (prev.some(l => l.id === newLesson.id)) return prev
50 |             return [...prev, newLesson]
51 |           })
52 |         }
53 |       )
54 |       .on('postgres_changes',
55 |         { event: 'UPDATE', schema: 'public', table: 'lessons', filter: 'is_published=eq.true' },
56 |         (payload) => {
57 |           const updated = payload.new as Lesson
58 |           setLessons(prev => {
59 |             const exists = prev.some(l => l.id === updated.id)
60 |             if (!exists) return [...prev, updated]
61 |             return prev.map(l => l.id === updated.id ? updated : l)
62 |           })
63 |         }
64 |       )
65 |       .on('postgres_changes',
66 |         { event: '*', schema: 'public', table: 'lesson_progress', filter: `student_id=eq.${userId}` },
67 |         (payload) => {
68 |           const p = payload.new as Progress
69 |           setProgressList(prev => {
70 |             const idx = prev.findIndex(x => x.lesson_id === p.lesson_id)
71 |             if (idx >= 0) {
72 |               const next = [...prev]
73 |               next[idx] = p
74 |               return next
75 |             }

```

```

76 |         return [...prev, p]
77 |     })
78 | }
79 | )
80 | .subscribe()
81 |
82 | return () => { supabase.removeChannel(channel) }
83 | }, [userId])
84 |
85 | const getProgress = (lessonId: string) =>
86 |   progressList?.find(p => p.lesson_id === lessonId)
87 |
88 | const completedCount = progressList?.filter(p => p.is_completed)?.length || 0
89 |
90 | return (
91 |   <div>
92 |     <div className="flex items-center justify-between mb-8">
93 |       <div>
94 |         <h1 className="text-2xl font-bold" style={{ color: 'var(--text-primary)' }}>
95 |           <p className="text-sm" style={{ color: 'var(--text-secondary)' }}>
96 |         </div>
97 |       <div className="flex items-center gap-3 px-4 py-3 rounded-xl" style={{ background: 'var(--bg-elevated)...
98 |         <Layers size={20} style={{ color: 'var(--brand-primary)' }} />
99 |       </div>
100 |     <div className="text-lg font-bold" style={{ color: 'var(--text-primary)' }}>{completedCount}/{less...
101 |   </div>
102 | </div>
103 | </div>
104 | </div>
105 |
106 | <div className="grid grid-cols-1 md:grid-cols-2 xl:grid-cols-3 gap-6">
107 |   {lessons.map((lesson) => {
108 |     const progress = getProgress(lesson.id)
109 |     const isCompleted = progress?.is_completed
110 |
111 |     return (
112 |       <Link
113 |         key={lesson.id}
114 |         href={`/student/lessons/${lesson.id}`}
115 |         className="group rounded-2xl overflow-hidden transition-all duration-300 hover:-translate-y-1"
116 |         style={{ background: 'var(--bg-elevated)', border: '1px solid var(--border-subtle)' }}
117 |       >
118 |         <div className="aspect-video relative overflow-hidden" style={{ background: 'var(--bg-surface)' }}>
119 |           {lesson.thumbnail_url ? (
120 |             <img src={lesson.thumbnail_url} alt={lesson.title} className="w-full h-full object-cover opa...
121 |           ) : (
122 |             <div className="flex items-center justify-center h-full" style={{ color: 'var(--text-muted)' }}...
123 |             <Book size={40} />
124 |           </div>
125 |           <div className="absolute inset-0 bg-gradient-to-t from-[rgba(0,0,0,0.6)] to-transparent" />
126 |           {isCompleted && (
127 |             <div className="absolute top-3 right-3 bg-emerald-500 text-white p-1.5 rounded-full shadow-lg">
128 |               <CheckCircle size={18} />
129 |             </div>
130 |           )}
131 |           <div className="absolute bottom-3 left-3 flex items-center gap-2">
132 |             <span className="px-2.5 py-1 bg-black/50 backdrop-blur rounded-full text-xs font-bold flex i...
133 |             <Clock size={11} style={{ color: 'var(--brand-primary)' }} />
134 |             {lesson.estimated_minutes || 15} phút
135 |           </span>
136 |         </div>
137 |       </div>
138 |     <div className="p-5">
139 |       <h3 className="font-bold text-base mb-2 line-clamp-2 transition-colors group-hover:text-emeral...
140 |       {lesson.title}
141 |     </h3>
142 |     <p className="text-sm line-clamp-2 mb-4" style={{ color: 'var(--text-secondary)' }}>
143 |       {lesson.description}
144 |     </p>
145 |     <div className="flex items-center justify-between pt-3 border-t" style={{ borderColor: 'var(--...
146 |     <span className="text-sm font-bold flex items-center gap-1" style={{ color: 'var(--brand-pri...
147 |       <span>
148 |         { isCompleted ? 'Xem lại' : 'Bắt đầu' } <ArrowRight size={16} />
149 |       </span>
150 |       {progress && !isCompleted && (
151 |         <div className="flex items-center gap-2">
152 |           <div className="w-16 h-1.5 rounded-full" style={{ background: 'var(--border-subtle)' }}>
153 |             <div className="h-full rounded-full bg-emerald-500" style={{ width: `${progress.comple...
154 |           </div>
155 |           <span className="text-xs" style={{ color: 'var(--text-muted)' }}>{progress.completion_pe...
156 |         </div>
157 |       )}
158 |     </div>
159 |   </div>
160 | </Link>
161 | )}
162 | )}
163 |
164 | {lessons.length === 0 && (
165 |   <div className="col-span-full py-20 text-center rounded-2xl border-2 border-dashed" style={{ borderC...
166 |     <Book size={48} className="mx-auto mb-4" style={{ color: 'var(--text-muted)' }} />
167 |     <p className="font-medium" style={{ color: 'var(--text-secondary)' }}>
168 |       <p className="text-sm mt-1" style={{ color: 'var(--text-muted)' }}>
169 |     </div>
170 |   )}
171 | </div>
172 | </div>
173 | )
174 | }
175 |

```

File: app/(dashboard)/student/lessons/page.tsx (32 lines, 0.8 KB)

```

1 | import { createClient } from '@lib/supabase-ssr-server'
2 | import { redirect } from 'next/navigation'
3 | import LessonsList from './LessonsList'
4 |
5 | export const dynamic = 'force-dynamic'
6 |
7 | export default async function StudentLessonsPage() {
8 |   const supabase = await createClient()
9 |   const { data: { user } } = await supabase.auth.getUser()
10 |
11 |   if (!user) redirect('/login')
12 |
13 |   const { data: lessons } = await supabase
14 |     .from('lessons')
15 |     .select('*')
16 |     .eq('is_published', true)
17 |     .order('created_at', { ascending: false })
18 |
19 |   const { data: progressList } = await supabase
20 |     .from('lesson_progress')

```

```

21 | .select('*')
22 | .eq('student_id', user.id)
23 |
24 | return (
25 |   <LessonsList
26 |     initialLessons={lessons || []}
27 |     initialProgress={progressList || []}
28 |     userId={user.id}
29 |   />
30 | )
31 | }
32 |

```

File: app/(dashboard)/student/loading.tsx (15 lines, 0.5 KB)

```

1 | // Path: app/(dashboard)/student/loading.tsx
2 | export default function Loading() {
3 |   return (
4 |     <div className="p-6 space-y-4 animate-pulse">
5 |       <div className="h-8 bg-[#1d1f2a] rounded-lg w-48" />
6 |       <div className="grid grid-cols-1 md:grid-cols-4 gap-4">
7 |         { [...Array(4)].map((_, i) => {
8 |           <div key={i} className="h-24 bg-[#1d1f2a] rounded-xl" />
9 |         })}
10 |       </div>
11 |     <div className="h-64 bg-[#1d1f2a] rounded-xl" />
12 |   </div>
13 | )
14 | }
15 |

```

File: app/(dashboard)/student/page.tsx (280 lines, 20.5 KB)

```

1 | 'use client';
2 |
3 | import React, { useState, useEffect } from 'react';
4 | import { useIsMobile } from '@hooks/useIsMobile';
5 | import Link from 'next/link';
6 | import { BookOpen, Users, Cpu, ArrowRight, Plus, Loader2, GraduationCap, Bot, Sparkles, Megaphone, ExternalLin...
7 | import { motion, AnimatePresence } from 'framer-motion';
8 | import JoinClassModal from '../components/JoinClassModal';
9 | import CareerRecommendation from '../components/CareerRecommendation';
10 | import { supabase } from '@lib/supabase';
11 | import { useGuru } from '@lib/guru-state';
12 | import { useRealtime } from '@lib/realtime-provider';
13 |
14 | const fadeUp = (delay = 0) => ({ initial: { opacity: 0, y: 24 }, animate: { opacity: 1, y: 0 }, transition: { ...
15 |
16 | export default function StudentDashboard() {
17 |   const { openChat } = useGuru();
18 |   const { state: realtimeState } = useRealtime();
19 |   const isMobile = useIsMobile();
20 |   const [showJoinModal, setShowJoinModal] = useState(false);
21 |   const [loading, setLoading] = useState(true);
22 |   const [classes, setClasses] = useState([]);
23 |   const [stats, setStats] = useState([
24 |     { label: 'L ò p h í c', value: '0', icon: <Users size={20} />, color: 'v
25 |     { label: 'B à i t - p', value: '0', icon: <BookOpen size={20} />, color:
26 |     { label: 'T ò n g X P', value: '0', icon: <Zap size={20} />, color: '#8b
27 |   ]);
28 |   useEffect(() => { fetchData(); }, []);
29 |
30 |   async function fetchData() {
31 |     try {
32 |       setLoading(true);
33 |       const { data: { user } } = await supabase.auth.getUser();
34 |       if (!user) return;
35 |       const { data: memberships, error: memErr } = await supabase
36 |         .from('class_members').select('class_id, classes: class_id (id, name, code, subject, teacher_id)').eq('...
37 |       if (memErr) throw memErr;
38 |       const rawClasses = memberships?.map((m: any) => m.classes).filter(Boolean) || [];
39 |       const teacherIds = [...new Set(rawClasses.map((c: any) => c.teacher_id).filter(Boolean))];
40 |       const teacherMap: Record<string, string> = {};
41 |       if (teacherIds.length > 0) {
42 |         const { data: teachers } = await supabase.from('profiles').select('id, full_name').in('id', teacherIds);
43 |         teachers?.forEach((t: any) => { teacherMap[t.id] = t.full_name });
44 |       }
45 |       const joinedClasses = rawClasses.map((cls: any) => ({ ...cls, teacher_name: teacherMap[cls.teacher_id] |...
46 |       setClasses(joinedClasses);
47 |       const classIds = joinedClasses.map((c: any) => c.id);
48 |       let assignmentCount = 0;
49 |       if (classIds.length > 0) { const { count } = await supabase.from('assignments').select('*', { count: 'ex...
50 |       setStats([
51 |         { label: 'L ò p h í c', value: joinedClasses.length.toString(), icon:
52 |         { label: 'B à i t - p', value: assignmentCount.toString(), icon: <Bo
53 |         { label: 'T ò n g X P', value: `${realtimeState.xp}`, icon: <Zap siz
54 |       ]);
55 |       setLoading(false);
56 |     } catch (err) { console.error('Error fetching student data:', err); }
57 |     finally { setLoading(false); }
58 |   }
59 |
60 |   return (
61 |     <div style={{ maxWidth: '1200px', margin: '0 auto', padding: '0 4px' }}>
62 |       <motion.header {...fadeUp(0)} style={{ marginBottom: '36px', display: 'flex', justifyContent: 'space-bet...
63 |       <div>
64 |         <h1 style={{ fontSize: '28px', fontWeight: 800, color: 'var(--text-primary)', marginBottom: '6px', l...
65 |         <p style={{ color: 'var(--text-muted)', margin: 0, fontSize:
66 |       </div>
67 |       <motion.button whileHover={{ scale: 1.03 }} whileTap={{ scale: 0.97 }}
68 |         onClick={() => setShowJoinModal(true)}
69 |         className="lms-btn lms-btn-primary" style={{ display: 'flex', alignItems: 'center', gap: '8px', padd...
70 |         <Plus size={18} /> T h a m g i a l ò p h í c
71 |     </motion.button>
72 |   </motion.header>
73 |
74 |   <motion.div {...fadeUp(0.1)} style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(220...
75 |     {stats.map((stat, idx) => {
76 |       <motion.div key={idx} whileHover={{ y: -4, boxShadow: '0 12px 40px var(--shadow-hover)' }}
77 |       className="lms-card" style={{ padding: isMobile ? '16px' : '24px', display: 'flex', alignItems: 'c...
78 |       <div style={{ width: '48px', height: '48px', borderRadius: '12px', background: `${stat.color}15`, ...
79 |         {stat.icon}
80 |       </div>
81 |       <div>
82 |         <div style={{ fontSize: '24px', fontWeight: 800, color: 'var(--text-primary)', lineHeight: 1.2 }}...
83 |         <div style={{ fontSize: '13px', color: 'var(--text-muted)', fontWeight: 500 }}>{stat.label}</div>
84 |       </div>
85 |     </motion.div>
86 |   )})}
87 | </motion.div>
88 |

```

```

89 | <div className="student-dashboard-grid" style={{ display: 'grid', gridTemplateColumns: isMobile ? '1fr' ...
90 | <section style={{ display: 'flex', flexDirection: 'column', gap: isMobile ? '16px' : '24px' }}>
91 | <div style={{ display: 'flex', flexDirection: 'column', gap: isMobile ? '16px' : '24px' }}>
92 |   <motion.div {...fadeUp(0.2)}
93 |     className="lms-card" style={{ padding: isMobile ? '20px' : '32px', position: 'relative', overflo...
94 |     <div style={{ position: 'relative', zIndex: 1 }}>
95 |       <h2 style={{ fontSize: '20px', fontWeight: 700, marginBottom: '10px', color: 'var(--text-prim...
96 |       <p style={{ color: 'var(--text-muted)', marginBottom: '24px', maxWidth: '380px', fontSize: '14...
97 |       T ã t a y l ã p r á p c ỹ u h ì n h m á y t ì n h , k i Ầ m t r a s ã t ì n g t h
98 |     </p>
99 |     <Link href="/builder" className="lms-btn lms-btn-primary" style={{ display: 'inline-flex', ali...
100 |     B t ậ s u t h ã c h à n h < A r r o w R i g h t s i z e = { 1 6 } / >
101 |   </Link>
102 | </div>
103 | <div style={{ position: 'absolute', right: '-20px', bottom: '-20px', opacity: 0.06 }}>
104 |   <CPU size={200} color="var(--brand-primary)" />
105 | </div>
106 | <motion.div>
107 | <motion.div {...fadeUp(0.25)}
108 |   className="lms-card" style={{ padding: isMobile ? '20px' : '32px', position: 'relative', overflo...
109 |   <div style={{ position: 'relative', zIndex: 1 }}>
110 |     <h2 style={{ fontSize: '20px', fontWeight: 700, marginBottom: '10px', color: 'var(--text-prim...
111 |     <p style={{ color: 'var(--text-muted)', marginBottom: '24px', maxWidth: '380px', fontSize: '14...
112 |     K i Ầ m t r a k i ệ n t h ệ c p h ậ n c ệ n g v ử i Ầ t h i , n g ậ n h à n g c ậ u
113 |   </p>
114 |   <Link href="/exams" className="lms-btn lms-btn-primary" style={{ display: 'inline-flex', align...
115 |   V à o t h i t h i < G r a d u a t i o n C a p s i z e = { 1 6 } / >
116 | </Link>
117 | </div>
118 | <div style={{ position: 'absolute', right: '-20px', bottom: '-20px', opacity: 0.06 }}>
119 |   <FileQuestion size={200} color="#6366f1" />
120 | </div>
121 | </motion.div>
122 | </div>
123 |
124 | <motion.div {...fadeUp(0.3)} className="lms-card" style={{ padding: isMobile ? '18px' : '28px', bord...
125 | <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBotto...
126 |   <h2 style={{ fontSize: '20px', fontWeight: 700, margin: 0, color: 'var(--text-prim...
127 |   <span style={{ fontSize: '13px', color: 'var(--text-muted)', fontWeight: 500 }}>{classes.length}...
128 | </div>
129 |
130 | {loading ? (
131 |   <div style={{ display: 'flex', justifyContent: 'center', padding: '48px' }}><Loader2 size={32} s...
132 | ) : classes.length > 0 ? (
133 |   <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(260px, 1fr))', gap:...
134 |     {classes.map((cls, idx) => (
135 |       <motion.div key={cls.id} initial={{ opacity: 0, y: 16 }} animate={{ opacity: 1, y: 0 }} tran...
136 |       <Link href="/student/classes/${cls.id}" style={{ textDecoration: 'none' }}>
137 |         <motion.div whileHover={{ y: -3, boxShadow: '0 12px 30px var(--shadow-hover)' }}
138 |           className="lms-card" style={{ padding: '20px', cursor: 'pointer', borderRadius: '12px'...
139 |           <div style={{ display: 'flex', justifyContent: 'space-between', marginBotto...
140 |             <div style={{ width: '40px', height: '40px', border: '10px solid var(--brand-primary)', background: 'rgba...
141 |             <GraduationCap size={18} color="var(--accent-blue)" />
142 |           </div>
143 |           <span className="lms-badge" style={{ fontSize: '11px', padding: '3px 10px', borderRa...
144 |           </div>
145 |           <h3 style={{ fontSize: '16px', fontWeight: 700, color: 'var(--text-primary)', marginBo...
146 |           <p style={{ fontSize: '13px', color: 'var(--text-muted)', margin: 0 }}>GV: {cls.teache...
147 |         </motion.div>
148 |       </Link>
149 |     </motion.div>
150 |   ))}
151 | </div>
152 | ) : (
153 |   <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }} style={{ textAlign: 'center', padd...
154 |     <p style={{ color: 'var(--text-muted)', marginBotto...
155 |     <motion.button whileHover={{ scale: 1.05 }} whileTap={{ scale: 0.95 }}
156 |       onClick={() => setShowJoinModal(true)}
157 |       style={{ background: 'transparent', border: '1px solid var(--brand-primary)', color: 'var(--...
158 |       N h - p m ầ l ộ p Ầ t h a m g i a
159 |     </motion.button>
160 |   </motion.div>
161 | )}
162 | </motion.div>
163 | </section>
164 |
165 | <aside style={{ display: 'flex', flexDirection: 'column', gap: isMobile ? '16px' : '24px' }}>
166 | <motion.div {...fadeUp(0.25)}
167 |   className="lms-card" style={{ padding: isMobile ? '16px' : '24px', borderRadius: '16px', backgroun...
168 |   <div style={{ position: 'absolute', top: -30, right: -30, width: '160px', height: '160px', borderR...
169 |   <div style={{ position: 'relative', zIndex: 1 }}>
170 |     <div style={{ display: 'flex', alignItems: 'center', gap: '14px', marginBotto...
171 |     <div style={{ fontSize: '40px', lineHeight: 1 }}>{realtimeState.levelIcon}</div>
172 |     <div>
173 |       <div style={{ fontSize: '11px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: ...
174 |       <div style={{ fontSize: '20px', fontWeight: 800, color: 'var(--text-primary)' }}>
175 |         {realtimeState.levelTitle}
176 |       </div>
177 |     </div>
178 |   </div>
179 |   <div style={{ height: '6px', borderRadius: '99px', background: 'rgba(255,255,255,0.08)', overflo...
180 |   <div style={{ height: '100%', borderRadius: '99px', background: 'linear-gradient(90deg, var(--...
181 | </div>
182 |   <div style={{ display: 'flex', justifyContent: 'space-between', fontSize: '11px', color: 'var(--...
183 |     <span>{realtimeState.xpInLevel} / {realtimeState.xpToNext} XP</span>
184 |     <span style={{ color: 'var(--brand-primary)' }}>Level {realtimeState.level}</span>
185 |   </div>
186 |   <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '8px', marginBotto...
187 |   <div style={{ padding: '12px', borderRadius: '10px', background: 'var(--bg-elevated)', textAli...
188 |     <Flame size={18} style={{ color: '#f59e0b', margin: '0 auto 4px' }} />
189 |     <div style={{ fontSize: '18px', fontWeight: 800, color: 'var(--text-primary)' }}>{realtimeSt...
190 |     <div style={{ fontSize: '10px', color: 'var(--text-muted)' }}>Streak</div>
191 |   </div>
192 |   <div style={{ padding: '12px', borderRadius: '10px', background: 'var(--bg-elevated)', textAli...
193 |   <Zap size={18} style={{ color: 'var(--brand-primary)', margin: '0 auto 4px' }} />
194 |   <div style={{ fontSize: '18px', fontWeight: 800, color: 'var(--text-primary)' }}>{realtimeSt...
195 |   <div style={{ fontSize: '10px', color: 'var(--text-muted)' }}>{realtimeSt...
196 | </div>
197 | </div>
198 | <div style={{ display: 'flex', gap: '8px' }}>
199 |   <Link href="/student/dashboard" style={{ flex: 1, textAlign: 'center', padding: '10px', border...
200 |   0 ð Ề C h i t i ệ t
201 | </Link>
202 | </div>
203 | </div>
204 | </motion.div>
205 |
206 | <motion.div {...fadeUp(0.3)} className="lms-card" style={{ padding: isMobile ? '16px' : '24px', bord...
207 | <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBotto...
208 |   <h3 style={{ fontSize: '15px', fontWeight: 700, margin: 0, color: 'var(--text-primary)', display...
209 |   <T r o p h y s i z e = { 1 6 } s t y l e = { { c o l o r : '# f 5 9 e 0 b ' } } / > N h i ệ m
210 | </h3>
211 |   <Link href="/student/dashboard" style={{ fontSize: '11px', color: 'var(--brand-primary)', fontWe...
212 |   X e m t ậ t c ệ l '

```

```

213 |         </Link>
214 |     </div>
215 |     {realtimeState.allQuests.length > 0 ? (
216 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
217 |     {realtimeState.allQuests.slice(0, 4).map((quest, i) => {
218 |     const userQuest = realtimeState.quests.find(q => q.quest_id === quest.id)
219 |     const isCompleted = userQuest?.is_completed
220 |     const progress = userQuest?.progress || 0
221 |     return (
222 |     <motion.div key={quest.id} initial={{ opacity: 0, x: -10 }} animate={{ opacity: 1, x: 0 }}...
223 |     style={{ padding: '12px', borderRadius: '10px', background: isCompleted ? 'rgba(16,185,1...'
224 |     <div style={{ display: 'flex', alignItems: 'center', gap: '10px' }}>
2 2 5 |         <span style={{ fontSize: '20px' }}>{quest.icon}</span>
226 |         <div style={{ flex: 1 }}>
227 |         <div style={{ fontSize: '12px', fontWeight: 700, color: 'var(--text-primary)', displ...
228 |         {quest.title}
229 |         {isCompleted && <CheckCircle2 size={12} style={{ color: '#10b981' }} />}
230 |         </div>
231 |         <div style={{ fontSize: '10px', color: 'var(--text-muted)', marginTop: '2px' }}>+{qu...
232 |         {!isCompleted && quest.requirement_value > 0 && (
233 |         <div style={{ height: '3px', borderRadius: '99px', background: 'rgba(255,255,255,0...'
234 |         <div style={{ height: '100%', borderRadius: '99px', background: 'var(--brand-pri...'
235 |         </div>
236 |         </div>
237 |         </div>
238 |         <div style={{ fontSize: '10px', fontWeight: 700, color: isCompleted ? '#10b981' : 'var...'
239 |         {isCompleted ? 'Hoan thanh' : '+${quest.xp_reward}XP'}
240 |         </div>
241 |     </div>
242 |     </motion.div>
243 |     )
244 |     )}
245 | </div>
246 | ) : (
247 | <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
248 |     {[
2 4 9 |         { id: 'fb-1', title: 'Hòa n thàn h, 1 bài h í c', xp_reward: 30, icon: 'fb' },
2 5 0 |         { id: 'fb-2', title: 'L p r á p P C', xp_reward: 20, icon: 'fb' },
2 5 1 |         { id: 'fb-3', title: 'L à m q u i z', xp_reward: 20, icon: 'fb' },
2 5 2 |         { id: 'fb-4', title: 'i t s t r e a k 3 n g à y', xp_reward: 20, icon: 'fb' },
2 5 3 |         { id: 'fb-5', title: 'T h a m g i a t h ĩ o l u - n', xp_reward: 20, icon: 'fb' },
254 |     ].map((q, i) => (
255 |     <motion.div key={q.id} initial={{ opacity: 0, x: -10 }} animate={{ opacity: 1, x: 0 }} trans...
256 |     style={{ padding: '12px', borderRadius: '10px', background: 'var(--bg-surface)', border: '...'
257 |     <div style={{ display: 'flex', alignItems: 'center', gap: '10px' }}>
258 |     <span style={{ fontSize: '20px' }}>{q.icon}</span>
259 |     <div style={{ flex: 1 }}>
260 |     <div style={{ fontSize: '12px', fontWeight: 700, color: 'var(--text-primary)' }}>{q.ti...
261 |     </div>
262 |     <div style={{ fontSize: '10px', fontWeight: 700, color: 'var(--text-muted)', whiteSpace:...
263 |     </div>
264 |     </motion.div>
265 |     )}
266 | </div>
267 | )}
268 | </motion.div>
269 | </aside>
270 | </div>
271 |
272 | <CareerRecommendation lang="vn" />
273 |
274 | <JoinClassModal isOpen={showJoinModal} onClose={() => { setShowJoinModal(false); fetchData(); }} lang="v...
275 |
276 | </div>
277 | )
278 | }
279 |
280 |

```

File: app/(dashboard)/student/profile/privacy/page.tsx (68 lines, 1.8 KB)

```

1 | // Path: app/(dashboard)/student/profile/privacy/page.tsx
2 | import { createServerClient } from '@supabase/ssr'
3 | import { cookies } from 'next/headers'
4 | import { redirect } from 'next/navigation'
5 | import { requireRole } from '@lib/auth/rbac'
6 | import PrivacySettingsClient from './PrivacySettingsClient'
7 |
8 | export const dynamic = 'force-dynamic'
9 |
10 | export default async function StudentPrivacyPage() {
11 |   const user = await requireRole(['student'])
12 |
13 |   const cookieStore = await cookies()
14 |   const supabase = createServerClient(
15 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
16 |     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!,
17 |     { cookies: { getAll: () => cookieStore.getAll(), setAll: () => {} } }
18 |   )
19 |
20 |   // Fetch student profile details to guarantee we have the student_code
21 |   const { data: profile } = await supabase
22 |     .from('profiles')
23 |     .select('student_code, full_name')
24 |     .eq('id', user.id)
25 |     .single()
26 |
27 |   if (!profile) {
28 |     redirect('/login')
29 |   }
30 |
31 |   // Fetch initial parent links (both active and pending ones)
32 |   const { data: links } = await supabase
33 |     .from('parent_student_links')
34 |     .select(`
35 |       id,
36 |       relationship,
37 |       status,
38 |       linked_at,
39 |       confirmed_at,
40 |       parent:profiles!parent_id (
41 |         id,
42 |         full_name,
43 |         avatar_url,
44 |         phone
45 |       )
46 |     `)
47 |     .eq('student_id', user.id)
48 |     .order('linked_at', { ascending: false })
49 |
50 |   const initialLinks = (links ?? []).map((link: any) => ({
51 |     id: link.id,
52 |     relationship: link.relationship,

```

```

53 |     status: link.status,
54 |     linked_at: link.linked_at,
55 |     confirmed_at: link.confirmed_at,
56 |     parent: link.parent
57 |   )))
58 |
59 |   return (
60 |     <PrivacySettingsClient
61 |       studentId={user.id}
62 |       studentName={profile.full_name}
63 |       studentCode={profile.student_code ?? ''}
64 |       initialLinks={initialLinks}
65 |     />
66 |   )
67 | }
68 |

```

File: app/(dashboard)/student/profile/privacy/PrivacySettingsClient.tsx (618 lines, 23.0 KB)

```

1 | // Path: app/(dashboard)/student/profile/privacy/PrivacySettingsClient.tsx
2 | 'use client'
3 |
4 | import React, { useState, useEffect, useRef } from 'react'
5 | import { createClient } from '@lib/supabase-ssr-client'
6 | import { confirmParentLink, revokeParentLink } from '@lib/parent-actions'
7 | import { RealtimeChannel } from '@supabase/supabase-js'
8 | import { formatDistanceToNow } from 'date-fns'
9 | import { vi } from 'date-fns/locale'
10 | import {
11 |   ShieldAlert,
12 |   Copy,
13 |   Check,
14 |   UserCheck,
15 |   UserMinus,
16 |   ShieldCheck,
17 |   Clock,
18 |   Users,
19 |   HelpCircle,
20 |   AlertCircle
21 | } from 'lucide-react'
22 |
23 | interface ParentLink {
24 |   id: string
25 |   relationship: string
26 |   status: 'pending' | 'active' | 'revoked'
27 |   linked_at: string
28 |   confirmed_at: string | null
29 |   parent: {
30 |     id: string
31 |     full_name: string
32 |     avatar_url: string | null
33 |     phone: string | null
34 |   }
35 | }
36 |
37 | export default function PrivacySettingsClient({
38 |   studentId,
39 |   studentName,
40 |   studentCode,
41 |   initialLinks,
42 | }: {
43 |   studentId: string
44 |   studentName: string
45 |   studentCode: string
46 |   initialLinks: ParentLink[]
47 | }) {
48 |   const [links, setLinks] = useState<ParentLink[]>(initialLinks)
49 |   const [copied, setCopied] = useState(false)
50 |   const [loadingId, setLoadingId] = useState<string | null>(null)
51 |   const [connectionStatus, setConnectionStatus] = useState<'connecting' | 'connected' | 'disconnected' | 'erro...
52 |
53 |   const supabase = createClient()
54 |   const channelRef = useRef<RealtimeChannel | null>(null)
55 |
56 |   // 1. Fetch updated parent links helper (to sync joined fields easily)
57 |   async function fetchLinks() {
58 |     try {
59 |       const { data, error } = await supabase
60 |         .from('parent_student_links')
61 |         .select(`
62 |           id,
63 |           relationship,
64 |           status,
65 |           linked_at,
66 |           confirmed_at,
67 |           parent:profiles!parent_id (
68 |             id,
69 |             full_name,
70 |             avatar_url,
71 |             phone
72 |           )
73 |         `)
74 |         .eq('student_id', studentId)
75 |         .order('linked_at', { ascending: false })
76 |
77 |       if (error) throw error
78 |       setLinks((data ?? []).map((l: any) => ({
79 |         id: l.id,
80 |         relationship: l.relationship,
81 |         status: l.status,
82 |         linked_at: l.linked_at,
83 |         confirmed_at: l.confirmed_at,
84 |         parent: l.parent
85 |       })))
86 |     } catch (err) {
87 |       console.error('Error fetching parent student links:', err)
88 |     }
89 |   }
90 |
91 |   // 2. Realtime WebSocket subscription to parent_student_links updates
92 |   useEffect(() => {
93 |     setConnectionStatus('connecting')
94 |
95 |     const channelName = `student-privacy-${studentId}`
96 |     const channel = supabase.channel(channelName)
97 |     channelRef.current = channel
98 |
99 |     channel
100 |      .on('postgres_changes', {
101 |        event: '*',
102 |        schema: 'public',
103 |        table: 'parent_student_links',
104 |        filter: `student_id=eq.${studentId}`,

```



```

105 |     }, () => {
106 |         // Trigger a re-fetch of links to automatically grab parent profile fields
107 |         fetchLinks()
108 |     })
109 |     .subscribe((status) => {
110 |         if (status === 'SUBSCRIBED') {
111 |             setConnectionStatus('connected')
112 |         } else if (status === 'CHANNEL_ERROR') {
113 |             setConnectionStatus('error')
114 |         } else if (status === 'CLOSED') {
115 |             setConnectionStatus('disconnected')
116 |         } else {
117 |             setConnectionStatus('connecting')
118 |         }
119 |     })
120 |
121 |     return () => {
122 |         if (channelRef.current) {
123 |             supabase.removeChannel(channelRef.current)
124 |             channelRef.current = null
125 |         }
126 |     }
127 | }, [studentId])
128 |
129 | // 3. Action Handlers
130 | async function handleCopy() {
131 |     try {
132 |         await navigator.clipboard.writeText(studentCode)
133 |         setCopied(true)
134 |         setTimeout(() => setCopied(false), 2000)
135 |     } catch (err) {
136 |         console.error('Failed to copy text: ', err)
137 |     }
138 | }
139 |
140 | async function handleAccept(parentId: string) {
141 |     setLoadingId(parentId)
142 |     try {
143 |         const res = await confirmParentLink(parentId)
144 |         if (res.error) {
145 |             alert(res.error)
146 |         } else {
147 |             await fetchLinks()
148 |         }
149 |     } catch (err: any) {
150 |         alert(err.message || 'Lỗi khi đồng ý liên kết')
151 |     } finally {
152 |         setLoadingId(null)
153 |     }
154 | }
155 |
156 | async function handleDeclineOrRevoke(parentId: string, confirmMessage: string) {
157 |     if (!window.confirm(confirmMessage)) return
158 |     setLoadingId(parentId)
159 |     try {
160 |         const res = await revokeParentLink(parentId)
161 |         if (res.error) {
162 |             alert(res.error)
163 |         } else {
164 |             await fetchLinks()
165 |         }
166 |     } catch (err: any) {
167 |         alert(err.message || 'Lỗi khi thực hiện thao tác')
168 |     } finally {
169 |         setLoadingId(null)
170 |     }
171 | }
172 |
173 | // Helpers
174 | const translateRelationship = (rel: string) => {
175 |     const map: Record<string, string> = {
176 |         father: 'Cha',
177 |         mother: 'Mẹ',
178 |         guardian: 'Người giám hộ',
179 |         sibling: 'Anh/Chị/Em',
180 |         other: 'Khác',
181 |         parent: 'Phân huynh',
182 |     }
183 |     return map[rel] || 'Phân huynh'
184 | }
185 |
186 | const getConnectionBadgeClass = () => {
187 |     switch (connectionStatus) {
188 |         case 'connected':
189 |             return {
190 |                 bg: 'rgba(16, 185, 129, 0.1)',
191 |                 border: 'rgba(16, 185, 129, 0.2)',
192 |                 text: '#10b981',
193 |                 dot: '#10b981',
194 |                 label: '& Realtime'
195 |             }
196 |         case 'connecting':
197 |             return {
198 |                 bg: 'rgba(245, 158, 11, 0.1)',
199 |                 border: 'rgba(245, 158, 11, 0.2)',
200 |                 text: '#f59e0b',
201 |                 dot: '#f59e0b',
202 |                 label: 'Đang kết nối...'
203 |             }
204 |         default:
205 |             return {
206 |                 bg: 'rgba(239, 68, 68, 0.1)',
207 |                 border: 'rgba(239, 68, 68, 0.2)',
208 |                 text: '#ef4444',
209 |                 dot: '#ef4444',
210 |                 label: 'Mất kết nối'
211 |             }
212 |     }
213 | }
214 |
215 | const badge = getConnectionBadgeClass()
216 | const pendingRequests = links.filter(l => l.status === 'pending')
217 | const activeLinks = links.filter(l => l.status === 'active')
218 |
219 | return (
220 |     <div style={{ maxWidth: '850px', margin: '0 auto', display: 'flex', flexDirection: 'column', gap: '32px' }}>
221 |
222 |         {/* Header Panel */}
223 |         <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', flexWrap: 'wrap', ...
224 |         <div>
225 |             <h1 style={{ fontSize: '28px', fontWeight: 800, color: 'var(--text-primary)', marginBottom: '8px' }}>
226 |                 Quyển riêng tôi & Liên kết Phân huynh
227 |             </h1>
228 |             <p style={{ color: 'var(--text-secondary)', fontSize: '14px', margin: 0 }}>

```

```
2 2 9 |           Qu ê n l ý m ă h í c s í n h P C M v à k i ă m s o á t c á c t à i k h o ế n p h ả h u y ệ n
230|           </p>
231|         </div>
232|
233|         { /* Realtime WebSocket connection status */ }
234|         <div style={{
235|           display: 'flex',
236|           alignItems: 'center',
237|           gap: '8px',
238|           padding: '6px 14px',
239|           borderRadius: '50px',
240|           background: badge.bg,
241|           border: '1px solid ${badge.border}',
242|           color: badge.text,
243|           fontSize: '12px',
244|           fontWeight: 600
245|         }}>
246|           <span style={{
247|             width: '8px',
248|             height: '8px',
249|             borderRadius: '50%',
250|             background: badge.dot,
251|             boxShadow: connectionStatus === 'connected' || connectionStatus === 'connecting' ? `0 0 8px ${badg...
252|             display: 'inline-block',
253|             animation: connectionStatus === 'connected' || connectionStatus === 'connecting' ? 'blink 2s infin...
254|           }} />
255|           {badge.label}
256|         </div>
257|
258|
259|         { /* Main Grid */ }
260|         <div style={{ display: 'grid', gridTemplateColumns: '1fr', gap: '28px' }}>
261|
262|           { /* Step 1 Card: Copy Student Code */ }
263|           <div style={{
264|             background: 'var(--bg-surface)',
265|             border: '1px solid var(--border-default)',
266|             borderRadius: '24px',
267|             padding: '32px',
268|             position: 'relative',
269|             overflow: 'hidden'
270|           }}>
271|             { /* Subtle background glow effect */ }
272|             <div style={{
273|               position: 'absolute',
274|               top: '-50px',
275|               right: '-50px',
276|               width: '180px',
277|               height: '180px',
278|               borderRadius: '50%',
279|               background: 'rgba(0, 198, 174, 0.05)',
280|               filter: 'blur(40px)',
281|               pointerEvents: 'none'
282|             }} />
283|
284|             <div style={{ display: 'flex', flexDirection: 'column', gap: '20px' }}>
285|               <div>
286|                 <h2 style={{ fontSize: '18px', fontWeight: 700, color: 'var(--text-primary)', marginBottom: '6px...
287|                   <ShieldCheck size={20} color="var(--brand-primary)" />
288|                   M ă h í c s í n h c á n h ầ n c ặ a b í n
289|                 </h2>
290|                 <p style={{ color: 'var(--text-secondary)', fontSize: '13px', lineHeight: 1.6, margin: 0 }}>
291|                   C ầ n g c ặ p m ă P C M n à y c h o c h a m 1 c ặ a b í n . C h a m 1 c ó t h ấ n h
292|                 </p>
293|               </div>
294|
295|               <div style={{
296|                 display: 'flex',
297|                 alignItems: 'center',
298|                 gap: '12px',
299|                 background: 'var(--bg-base)',
300|                 padding: '16px 24px',
301|                 borderRadius: '16px',
302|                 border: '1px solid var(--border-subtle)',
303|                 alignSelf: 'flex-start',
304|                 minWidth: '280px',
305|                 justifyContent: 'space-between'
306|               }}>
307|                 <code style={{
308|                   fontFamily: 'var(--font-mono)',
309|                   fontSize: '18px',
310|                   fontWeight: 700,
311|                   letterSpacing: '2px',
312|                   color: 'var(--brand-primary)'
313|                 }}>
314|                   {studentCode}
315|                 </code>
316|                 <button
317|                   onClick={handleCopy}
318|                   style={{
319|                     background: copied ? 'rgba(16, 185, 129, 0.15)' : 'rgba(255, 255, 255, 0.05)',
320|                     border: copied ? '1px solid rgba(16, 185, 129, 0.3)' : '1px solid var(--border-default)',
321|                     color: copied ? '#10b981' : 'var(--text-primary)',
322|                     padding: '8px 14px',
323|                     borderRadius: '10px',
324|                     cursor: 'pointer',
325|                     fontSize: '13px',
326|                     fontWeight: 600,
327|                     display: 'flex',
328|                     alignItems: 'center',
329|                     gap: '8px',
330|                     transition: 'all 0.15s ease'
331|                   }}
332|                 >
333|                   {copied ? <Check size={14} /> : <Copy size={14} />}
334|                   { c o p i e d ? ' ấ c h ế p ! ' : ' S a o c h ế p ' }
335|                 </button>
336|               </div>
337|
338|               <div style={{
339|                 display: 'flex',
340|                 gap: '10px',
341|                 background: 'rgba(59, 130, 246, 0.05)',
342|                 border: '1px solid rgba(59, 130, 246, 0.15)',
343|                 padding: '14px 18px',
344|                 borderRadius: '14px',
345|                 fontSize: '12px',
346|                 color: '#60a5fa',
347|                 lineHeight: 1.5
348|               }}>
349|                 <AlertCircle size={18} style={{ flexShrink: 0 }} />
350|                 <div>
351|                   <strong>L ộ ý b ế o m - t : </strong> C h ế c h i a s » m ă n à y v ớ i c
352|                 </div>
353|               </div>
354|             </div>
355|           </div>
356|         </div>
357|       </div>
358|     </div>
359|   </div>
360| </div>
361| </div>
362| </div>
363| </div>
364| </div>
365| </div>
366| </div>
367| </div>
368| </div>
369| </div>
370| </div>
371| </div>
372| </div>
373| </div>
374| </div>
375| </div>
376| </div>
377| </div>
378| </div>
379| </div>
380| </div>
381| </div>
382| </div>
383| </div>
384| </div>
385| </div>
386| </div>
387| </div>
388| </div>
389| </div>
390| </div>
391| </div>
392| </div>
393| </div>
394| </div>
395| </div>
396| </div>
397| </div>
398| </div>
399| </div>
400| </div>
401| </div>
402| </div>
403| </div>
404| </div>
405| </div>
406| </div>
407| </div>
408| </div>
409| </div>
410| </div>
411| </div>
412| </div>
413| </div>
414| </div>
415| </div>
416| </div>
417| </div>
418| </div>
419| </div>
420| </div>
421| </div>
422| </div>
423| </div>
424| </div>
425| </div>
426| </div>
427| </div>
428| </div>
429| </div>
430| </div>
431| </div>
432| </div>
433| </div>
434| </div>
435| </div>
436| </div>
437| </div>
438| </div>
439| </div>
440| </div>
441| </div>
442| </div>
443| </div>
444| </div>
445| </div>
446| </div>
447| </div>
448| </div>
449| </div>
450| </div>
451| </div>
452| </div>
453| </div>
454| </div>
455| </div>
456| </div>
457| </div>
458| </div>
459| </div>
460| </div>
461| </div>
462| </div>
463| </div>
464| </div>
465| </div>
466| </div>
467| </div>
468| </div>
469| </div>
470| </div>
471| </div>
472| </div>
473| </div>
474| </div>
475| </div>
476| </div>
477| </div>
478| </div>
479| </div>
480| </div>
481| </div>
482| </div>
483| </div>
484| </div>
485| </div>
486| </div>
487| </div>
488| </div>
489| </div>
490| </div>
491| </div>
492| </div>
493| </div>
494| </div>
495| </div>
496| </div>
497| </div>
498| </div>
499| </div>
500| </div>
501| </div>
502| </div>
503| </div>
504| </div>
505| </div>
506| </div>
507| </div>
508| </div>
509| </div>
510| </div>
511| </div>
512| </div>
513| </div>
514| </div>
515| </div>
516| </div>
517| </div>
518| </div>
519| </div>
520| </div>
521| </div>
522| </div>
523| </div>
524| </div>
525| </div>
526| </div>
527| </div>
528| </div>
529| </div>
530| </div>
531| </div>
532| </div>
533| </div>
534| </div>
535| </div>
536| </div>
537| </div>
538| </div>
539| </div>
540| </div>
541| </div>
542| </div>
543| </div>
544| </div>
545| </div>
546| </div>
547| </div>
548| </div>
549| </div>
550| </div>
551| </div>
552| </div>
553| </div>
554| </div>
555| </div>
556| </div>
557| </div>
558| </div>
559| </div>
560| </div>
561| </div>
562| </div>
563| </div>
564| </div>
565| </div>
566| </div>
567| </div>
568| </div>
569| </div>
570| </div>
571| </div>
572| </div>
573| </div>
574| </div>
575| </div>
576| </div>
577| </div>
578| </div>
579| </div>
580| </div>
581| </div>
582| </div>
583| </div>
584| </div>
585| </div>
586| </div>
587| </div>
588| </div>
589| </div>
590| </div>
591| </div>
592| </div>
593| </div>
594| </div>
595| </div>
596| </div>
597| </div>
598| </div>
599| </div>
600| </div>
601| </div>
602| </div>
603| </div>
604| </div>
605| </div>
606| </div>
607| </div>
608| </div>
609| </div>
610| </div>
611| </div>
612| </div>
613| </div>
614| </div>
615| </div>
616| </div>
617| </div>
618| </div>
619| </div>
620| </div>
621| </div>
622| </div>
623| </div>
624| </div>
625| </div>
626| </div>
627| </div>
628| </div>
629| </div>
630| </div>
631| </div>
632| </div>
633| </div>
634| </div>
635| </div>
636| </div>
637| </div>
638| </div>
639| </div>
640| </div>
641| </div>
642| </div>
643| </div>
644| </div>
645| </div>
646| </div>
647| </div>
648| </div>
649| </div>
650| </div>
651| </div>
652| </div>
653| </div>
654| </div>
655| </div>
656| </div>
657| </div>
658| </div>
659| </div>
660| </div>
661| </div>
662| </div>
663| </div>
664| </div>
665| </div>
666| </div>
667| </div>
668| </div>
669| </div>
670| </div>
671| </div>
672| </div>
673| </div>
674| </div>
675| </div>
676| </div>
677| </div>
678| </div>
679| </div>
680| </div>
681| </div>
682| </div>
683| </div>
684| </div>
685| </div>
686| </div>
687| </div>
688| </div>
689| </div>
690| </div>
691| </div>
692| </div>
693| </div>
694| </div>
695| </div>
696| </div>
697| </div>
698| </div>
699| </div>
700| </div>
701| </div>
702| </div>
703| </div>
704| </div>
705| </div>
706| </div>
707| </div>
708| </div>
709| </div>
710| </div>
711| </div>
712| </div>
713| </div>
714| </div>
715| </div>
716| </div>
717| </div>
718| </div>
719| </div>
720| </div>
721| </div>
722| </div>
723| </div>
724| </div>
725| </div>
726| </div>
727| </div>
728| </div>
729| </div>
730| </div>
731| </div>
732| </div>
733| </div>
734| </div>
735| </div>
736| </div>
737| </div>
738| </div>
739| </div>
740| </div>
741| </div>
742| </div>
743| </div>
744| </div>
745| </div>
746| </div>
747| </div>
748| </div>
749| </div>
750| </div>
751| </div>
752| </div>
753| </div>
754| </div>
755| </div>
756| </div>
757| </div>
758| </div>
759| </div>
760| </div>
761| </div>
762| </div>
763| </div>
764| </div>
765| </div>
766| </div>
767| </div>
768| </div>
769| </div>
770| </div>
771| </div>
772| </div>
773| </div>
774| </div>
775| </div>
776| </div>
777| </div>
778| </div>
779| </div>
780| </div>
781| </div>
782| </div>
783| </div>
784| </div>
785| </div>
786| </div>
787| </div>
788| </div>
789| </div>
790| </div>
791| </div>
792| </div>
793| </div>
794| </div>
795| </div>
796| </div>
797| </div>
798| </div>
799| </div>
800| </div>
801| </div>
802| </div>
803| </div>
804| </div>
805| </div>
806| </div>
807| </div>
808| </div>
809| </div>
810| </div>
811| </div>
812| </div>
813| </div>
814| </div>
815| </div>
816| </div>
817| </div>
818| </div>
819| </div>
820| </div>
821| </div>
822| </div>
823| </div>
824| </div>
825| </div>
826| </div>
827| </div>
828| </div>
829| </div>
830| </div>
831| </div>
832| </div>
833| </div>
834| </div>
835| </div>
836| </div>
837| </div>
838| </div>
839| </div>
840| </div>
841| </div>
842| </div>
843| </div>
844| </div>
845| </div>
846| </div>
847| </div>
848| </div>
849| </div>
850| </div>
851| </div>
852| </div>
853| </div>
854| </div>
855| </div>
856| </div>
857| </div>
858| </div>
859| </div>
860| </div>
861| </div>
862| </div>
863| </div>
864| </div>
865| </div>
866| </div>
867| </div>
868| </div>
869| </div>
870| </div>
871| </div>
872| </div>
873| </div>
874| </div>
875| </div>
876| </div>
877| </div>
878| </div>
879| </div>
880| </div>
881| </div>
882| </div>
883| </div>
884| </div>
885| </div>
886| </div>
887| </div>
888| </div>
889| </div>
890| </div>
891| </div>
892| </div>
893| </div>
894| </div>
895| </div>
896| </div>
897| </div>
898| </div>
899| </div>
900| </div>
901| </div>
902| </div>
903| </div>
904| </div>
905| </div>
906| </div>
907| </div>
908| </div>
909| </div>
910| </div>
911| </div>
912| </div>
913| </div>
914| </div>
915| </div>
916| </div>
917| </div>
918| </div>
919| </div>
920| </div>
921| </div>
922| </div>
923| </div>
924| </div>
925| </div>
926| </div>
927| </div>
928| </div>
929| </div>
930| </div>
931| </div>
932| </div>
933| </div>
934| </div>
935| </div>
936| </div>
937| </div>
938| </div>
939| </div>
940| </div>
941| </div>
942| </div>
943| </div>
944| </div>
945| </div>
946| </div>
947| </div>
948| </div>
949| </div>
950| </div>
951| </div>
952| </div>
953| </div>
954| </div>
955| </div>
956| </div>
957| </div>
958| </div>
959| </div>
960| </div>
961| </div>
962| </div>
963| </div>
964| </div>
965| </div>
966| </div>
967| </div>
968| </div>
969| </div>
970| </div>
971| </div>
972| </div>
973| </div>
974| </div>
975| </div>
976| </div>
977| </div>
978| </div>
979| </div>
980| </div>
981| </div>
982| </div>
983| </div>
984| </div>
985| </div>
986| </div>
987| </div>
988| </div>
989| </div>
990| </div>
991| </div>
992| </div>
993| </div>
994| </div>
995| </div>
996| </div>
997| </div>
998| </div>
999| </div>
1000| </div>
```

```

353|         </div>
354|     </div>
355| </div>
356|
357| /* Step 2: Pending Requests Panel */
358| <div style={{
359|     background: 'var(--bg-surface)',
360|     border: '1px solid var(--border-default)',
361|     borderRadius: '24px',
362|     padding: '28px'
363| }}>
364|     <h2 style={{ fontSize: '18px', fontWeight: 700, color: 'var(--text-primary)', marginBottom: '18px', ...
365|         <Clock size={20} color="#f59e0b" />
3 6 6 |         Y ê u c s u l i ê n k ử t a n g c h ỷ ( { p e n d i n g R e q u e s t s . l e n g t h } )
367|     </h2>
368|
369|     {pendingRequests.length === 0 ? (
370|         <div style={{
371|             textAlign: 'center',
372|             padding: '32px 0',
373|             color: 'var(--text-muted)',
374|             fontSize: '13px',
375|             background: 'rgba(255,255,255,0.01)',
376|             borderRadius: '16px',
377|             border: '1px dashed var(--border-subtle)'
378|         }}>
3 7 9 |             H i ệ n t i i k h ô n g c ó y ê u c s u l i ê n k ử t n à o a n g c h ỷ d u ợ c t .
380|         </div>
381|     ) : (
382|         <div style={{ display: 'flex', flexDirection: 'column', gap: '14px' }}>
383|             {pendingRequests.map(link => (
384|                 <div
385|                     key={link.id}
386|                     style={{
387|                         display: 'flex',
388|                         alignItems: 'center',
389|                         justifyContent: 'space-between',
390|                         padding: '18px',
391|                         background: 'var(--bg-base)',
392|                         borderRadius: '16px',
393|                         border: '1px solid var(--border-subtle)',
394|                         flexWrap: 'wrap',
395|                         gap: '16px'
396|                     }}
397|                 >
398|                     <div style={{ display: 'flex', alignItems: 'center', gap: '14px' }}>
399|                         {link.parent.avatar_url ? (
400|                             <img
401|                                 src={link.parent.avatar_url}
402|                                 alt={link.parent.full_name}
403|                                 style={{ width: '42px', height: '42px', borderRadius: '50%', objectFit: 'cover' }}
404|                             />
405|                         ) : (
406|                             <div style={{
407|                                 width: '42px',
408|                                 height: '42px',
409|                                 borderRadius: '50%',
410|                                 background: 'rgba(245, 158, 11, 0.1)',
411|                                 color: '#f59e0b',
412|                                 display: 'flex',
413|                                 alignItems: 'center',
414|                                 justifyContent: 'center',
415|                                 fontWeight: 700,
416|                                 fontSize: '16px'
417|                             }}>
418|                                 {link.parent.full_name[0].toUpperCase()}
419|                             </div>
420|                         )}
421|                     <div>
422|                         <div style={{ fontWeight: 700, color: 'var(--text-primary)', fontSize: '14px' }}>
423|                             {link.parent.full_name}
424|                         </div>
425|                         <div style={{ fontSize: '12px', color: 'var(--text-secondary)', marginTop: '2px' }}>
4 2 6 |                             M ã i q u a n h ợ Á x u ợ t: <span style={{ color: '#f59e0b' }}>
427|                         </div>
428|                     </div>
429|                 </div>
430|
431|                 <div style={{ display: 'flex', gap: '10px' }}>
432|                     <button
4 3 3 |                         disabled={loadingId !== null} o n C l i c k = { ( ) => h a n d l e D e c l i n e O r R e v o k e ( l i n k . p a r e n t . i
434|                         style={{
435|                             background: 'rgba(239, 68, 68, 0.05)',
436|                             border: '1px solid rgba(239, 68, 68, 0.2)',
437|                             color: '#ef4444',
438|                             padding: '8px 16px',
439|                             borderRadius: '10px',
440|                             cursor: 'pointer',
441|                             fontSize: '13px',
442|                             fontWeight: 600,
443|                             display: 'flex',
444|                             alignItems: 'center',
445|                             gap: '6px',
446|                             transition: 'all 0.15s ease'
447|                         }}
448|                     >
449|                         <UserMinus size={14} />
450|                         T ợ c h ã i
451|                     </button>
452|                     <button
453|                         onClick={() => handleAccept(link.parent.id)}
454|                         disabled={loadingId !== null}
455|                         style={{
456|                             background: 'rgba(16, 185, 129, 0.15)',
457|                             border: '1px solid rgba(16, 185, 129, 0.3)',
458|                             color: '#10b981',
459|                             padding: '8px 18px',
460|                             borderRadius: '10px',
461|                             cursor: 'pointer',
462|                             fontSize: '13px',
463|                             fontWeight: 700,
464|                             display: 'flex',
465|                             alignItems: 'center',
466|                             gap: '6px',
467|                             transition: 'all 0.15s ease'
468|                         }}
469|                     >
470|                         {loadingId === link.parent.id ? (
471|                             <div style={{ width: '14px', height: '14px', border: '2px solid #10b981', borderTopCol...
472|                         ) : (
473|                             <UserCheck size={14} />
474|                         )}
475|                     >
4 7 6 |                         Ó n g ỷ

```

```

477 |         </button>
478 |     </div>
479 | </div>
480 |     )))
481 | </div>
482 | })
483 | </div>
484 |
485 | /* Step 3: Linked Parents list */
486 | <div style={{
487 |   background: 'var(--bg-surface)',
488 |   border: '1px solid var(--border-default)',
489 |   borderRadius: '24px',
490 |   padding: '28px'
491 | }}>
492 |   <h2 style={{ fontSize: '18px', fontWeight: 700, color: 'var(--text-primary)', marginBottom: '18px', ...
493 |     <Users size={20} color="var(--brand-primary)" />
494 |     T à i k h o ã n p h ấ h u y n h a n g l i ê n k ế t ( { a c t i v e L i n k s . l e n g t h } )
495 |   </h2>
496 |
497 |   {activeLinks.length === 0 ? (
498 |     <div style={{
499 |       textAlign: 'center',
500 |       padding: '36px 0',
501 |       color: 'var(--text-muted)',
502 |       fontSize: '13px',
503 |       background: 'rgba(255,255,255,0.01)',
504 |       borderRadius: '16px',
505 |       border: '1px dashed var(--border-subtle)'
506 |     }}>
507 |       B i n c h ấ l i ê n k ế t v ớ i b ấ t k ó t à i k h o ã n p h ấ h u y n h n à o .
508 |     </div>
509 |   ) : (
510 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '14px' }}>
511 |       {activeLinks.map(link => (
512 |         <div
513 |           key={link.id}
514 |           style={{
515 |             display: 'flex',
516 |             alignItems: 'center',
517 |             justifyContent: 'space-between',
518 |             padding: '18px',
519 |             background: 'var(--bg-base)',
520 |             borderRadius: '16px',
521 |             border: '1px solid var(--border-subtle)',
522 |             flexWrap: 'wrap',
523 |             gap: '16px'
524 |           }}
525 |         >
526 |           <div style={{ display: 'flex', alignItems: 'center', gap: '14px' }}>
527 |             {link.parent.avatar_url ? (
528 |               <img
529 |                 src={link.parent.avatar_url}
530 |                 alt={link.parent.full_name}
531 |                 style={{ width: '42px', height: '42px', borderRadius: '50%', objectFit: 'cover' }}
532 |               />
533 |             ) : (
534 |               <div style={{
535 |                 width: '42px',
536 |                 height: '42px',
537 |                 borderRadius: '50%',
538 |                 background: 'rgba(0, 198, 174, 0.1)',
539 |                 color: 'var(--brand-primary)',
540 |                 display: 'flex',
541 |                 alignItems: 'center',
542 |                 justifyContent: 'center',
543 |                 fontWeight: 700,
544 |                 fontSize: '16px'
545 |               }}>
546 |                 {link.parent.full_name[0].toUpperCase()}
547 |               </div>
548 |             )}
549 |           <div>
550 |             <div style={{ fontWeight: 700, color: 'var(--text-primary)', fontSize: '14px' }}>
551 |               {link.parent.full_name}
552 |             </div>
553 |             <div style={{ fontSize: '12px', color: 'var(--text-secondary)', marginTop: '4px', displa...
554 |               <span>Vai trò: <strong style={{ color: 'var(--brand-primary)' }}>{translateRelationshi...
555 |               {link.confirmed_at && (
556 |                 <span style={{ color: 'var(--text-muted)' }}>
557 |                   • B ấ t S u t ế { formatDistanceToNow(new Date(
558 |                 </span>
559 |               )}
560 |             </div>
561 |           </div>
562 |         </div>
563 |
564 |         <div>
565 |           <button
566 |             disabled={loadingId !== null}
567 |             style={{
568 |               background: 'rgba(239, 68, 68, 0.05)',
569 |               border: '1px solid rgba(239, 68, 68, 0.2)',
570 |               color: '#ef4444',
571 |               padding: '8px 16px',
572 |               borderRadius: '10px',
573 |               cursor: 'pointer',
574 |               fontSize: '12px',
575 |               fontWeight: 600,
576 |               display: 'flex',
577 |               alignItems: 'center',
578 |               gap: '6px',
579 |               transition: 'all 0.15s ease'
580 |             }}
581 |             onMouseOver={(e) => {
582 |               e.currentTarget.style.background = 'rgba(239, 68, 68, 0.1)'
583 |             }}
584 |             onMouseOut={(e) => {
585 |               e.currentTarget.style.background = 'rgba(239, 68, 68, 0.05)'
586 |             }}
587 |           >
588 |             {loadingId === link.parent.id ? (
589 |               <div style={{ width: '12px', height: '12px', border: '2px solid #ef4444', borderTopCol...
590 |             ) : (
591 |               <UserMinus size={13} />
592 |             )}
593 |           </div>
594 |           T h u h ớ i q u y ấ n
595 |         </button>
596 |       </div>
597 |     </div>
598 |   )))
599 | </div>
600 | })

```

```

601 |         </div>
602 |     </div>
603 |
604 |     { /* Embedded inline keyframes styling */
605 |     <style dangerouslySetInnerHTML={{ __html: `
606 |         @keyframes blink {
607 |             0%, 100% { opacity: 1; }
608 |             50% { opacity: 0.3; }
609 |         }
610 |         @keyframes spin {
611 |             to { transform: rotate(360deg); }
612 |         }
613 |     ` }} />
614 |
615 |     </div>
616 | )
617 | }
618 |

```

File: app/(dashboard)/student/sgk/page.tsx (200 lines, 10.5 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { motion } from 'framer-motion'
5 | import { BookOpen, FileText, ChevronLeft, ChevronRight, Maximize2, Minimize2, ExternalLink, Youtube, Download ...
6 |
7 | export const SGK_BOOKS = [
8 |   {
9 |     id: 'tin10',
10 |    title: 'Tin h́íc 10 - Ḱt n̄i tri th́c',
11 |    pdfUrl: '/api/pdf?file=sach-tin10.pdf',
12 |    pdfDirectUrl: '/sach-tin10.pdf',
13 |    color: '#289cf9',
14 |    chapters: [
15 |      'Ch°ing 1: Má y t́nh và x̄ h̄i tri th́c',
16 |      'Ch°ing 2: M̄ng má y t́nh và Internet',
17 |      'Ch°ing 3: i o é c, ph́p lu - t và v n h́a trong m̄i tr° ́ng s̄n',
18 |      'Ch°ing 4: ề ng đ̄ng tin h́c',
19 |      'Ch°ing 5: Gīi quy ́t v ́n Á, v̄i s̄n tr̄ gīp c̄a má y t́nh',
20 |    ],
21 |    videos: [
22 |      { title: 'B̄i 1: Th̄ng tin và x̄ lý th̄ng tin', id: 'R0qVwWxiVRU' },
23 |      { title: 'B̄i 2: S̄n °u vi ́t c̄a má y t́nh', id: 'ikSSNRTCZYg' } ],
24 |    ],
25 |  },
26 |  {
27 |    id: 'tin11',
28 |    title: 'Tin h́íc 11 - Tin h́íc ề ng đ̄ng',
29 |    pdfUrl: '/api/pdf?file=sach-tin11.pdf',
30 |    pdfDirectUrl: '/sach-tin11.pdf',
31 |    color: '#8b5cf6',
32 |    chapters: [
33 |      'Ch°ing 1: H́ c i s̄ đ̄ li ́u',
34 |      'Ch°ing 2: H́ c qu ́n tr ́ c i s̄ đ̄ li ́u',
35 |      'Ch°ing 3: Ph̄n m̄m b̄ng t́nh và ề ng đ̄ng',
36 |      'Ch°ing 4: Ph̄n m̄m tr̄nh chi ́u n̄ng cao',
37 |    ],
38 |    videos: [
39 |      { title: 'B̄i 1: H́ c i s̄ đ̄ li ́u', id: 'OICNS79D-OI' } ],
40 |    ],
41 |  }
42 | ]
43 |
44 | export default function SGKPage() {
45 |   const [selectedBook, setSelectedBook] = useState<string | null>(null)
46 |   const [fullscreen, setFullscreen] = useState(false)
47 |   const [pageNum, setPageNum] = useState(1)
48 |   const [tab, setTab] = useState<'book' | 'video'>('book')
49 |   const [origin, setOrigin] = useState('')
50 |
51 |   React.useEffect(() => { setOrigin(window.location.origin) }, [])
52 |
53 |   const book = SGK_BOOKS.find(s => s.id === selectedBook)
54 |
55 |   if (selectedBook && book) {
56 |     return (
57 |       <div className={`flex flex-col ${fullscreen ? 'fixed inset-0 z-[9999] bg-[#0f0f1a]' : ''}`>
58 |         <div className="flex items-center justify-between p-4" style={{ background: 'var(--bg-elevated)', border...
59 |           <button onClick={() => setSelectedBook(null)}
60 |             className="flex items-center gap-2 text-sm font-medium hover:opacity-70 transition-opacity"
61 |             style={{ color: 'var(--text-secondary)', background: 'none', border: 'none', cursor: 'pointer', fo...
62 |             <ChevronLeft size={18} /> Quay l̄i
63 |           </button>
64 |           <div className="flex items-center gap-3">
65 |             <div className="flex gap-1 rounded-lg p-1" style={{ background: 'var(--bg-surface)' }}>
66 |               <button onClick={() => setTab('book')}
67 |                 style={{
68 |                   padding: '4px 12px', borderRadius: '6px', border: 'none', cursor: 'pointer',
69 |                   background: tab === 'book' ? book.color + '30' : 'transparent',
70 |                   color: tab === 'book' ? book.color : 'var(--text-muted)',
71 |                   fontWeight: tab === 'book' ? 700 : 400, fontSize: '12px', fontFamily: 'inherit',
72 |                   display: 'flex', alignItems: 'center', gap: '4px',
73 |                 }}>
74 |                 <BookOpen size={14} /> Sách
75 |               </button>
76 |               <button onClick={() => setTab('video')}
77 |                 style={{
78 |                   padding: '4px 12px', borderRadius: '6px', border: 'none', cursor: 'pointer',
79 |                   background: tab === 'video' ? '#ef44420' : 'transparent',
80 |                   color: tab === 'video' ? '#ef4444' : 'var(--text-muted)',
81 |                   fontWeight: tab === 'video' ? 700 : 400, fontSize: '12px', fontFamily: 'inherit',
82 |                   display: 'flex', alignItems: 'center', gap: '4px',
83 |                 }}>
84 |                 <Youtube size={14} /> Video
85 |               </button>
86 |             </div>
87 |             {tab === 'book' && <
88 |               <span className="text-xs" style={{ color: 'var(--text-muted)' }}>Trang {pageNum}</span>
89 |               <button onClick={() => setFullscreen(!fullscreen)}
90 |                 style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-subtle)', borderRadi...
91 |                 {fullscreen ? <Minimize2 size={16} /> : <Maximize2 size={16} /> }
92 |               </button>
93 |               <a href={book.pdfUrl} target="_blank" rel="noopener noreferrer"
94 |                 className="flex items-center gap-1.5 px-3 py-1.5 rounded-lg text-xs font-medium transition-all...
95 |                 style={{ background: book.color + '20', color: book.color, textDecoration: 'none', border: '1p...
96 |                 <ExternalLink size={14} /> M̄ ngo à i
97 |               </a>
98 |             </>
99 |           </div>
100 |         </div>
101 |
102 |         {tab === 'book' ? (

```

```

103 |         <div className="flex-1 flex items-center justify-center p-4 overflow-hidden" style={{ background: ...
104 |             {origin ? (
105 |                 <iframe src={`https://docs.google.com/viewer?url=${encodeURIComponent(origin + book.pdfDirectU...
106 |             ) : (
107 |                 <iframe src={book.pdfUrl + '#view=FitH'} className="w-full h-full rounded-lg shadow-2xl" style...
108 |             )}
109 |         </div>
110 |         <div className="flex items-center justify-center gap-4 p-3" style={{ background: 'var(--bg-elevate...
111 |             <a href={book.pdfDirectUrl} target="_blank" rel="noopener noreferrer"
112 |                 className="flex items-center gap-1.5 px-4 py-2 rounded-lg text-xs font-medium transition-all h...
113 |                 style={{ background: book.color + '20', color: book.color, textDecoration: 'none', border: '1p...
114 |             </a>
115 |         </div>
116 |     ) : (
117 |         <div className="flex-1 overflow-y-auto p-6" style={{ background: '#0f0f1a' }}>
118 |             <motion.div initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }} style={{ maxWidth: '800...
119 |             <h2 style={{ fontSize: '20px', fontWeight: 700, color: '#fff', marginBottom: '20px', display: 'f...
120 |                 <Youtube size={22} color="#ef4444" />
121 |                 V i d e o   b à i   g i   ã   n   g   -   {   b   o   o   k   .   t   i   t   l   e   }
122 |             </h2>
123 |             <div className="grid gap-6" style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill, mi...
124 |                 {book.videos.map((v, i) => (
125 |                     <div key={i} style={{
126 |                         borderRadius: '14px', overflow: 'hidden',
127 |                         background: '#1a1a2e', border: '1px solid rgba(255,255,255,0.06)',
128 |                     }}>
129 |                         <div style={{ position: 'relative', paddingBottom: '56.25%' }}>
130 |                             <iframe
131 |                                 src={`https://www.youtube.com/embed/${v.id}`}
132 |                                 style={{ position: 'absolute', top: 0, left: 0, width: '100%', height: '100%', border:...
133 |                                 allow="accelerometer; autoplay; clipboard-write; encrypted-media; gyroscope; picture-i...
134 |                                 allowFullScreen
135 |                                 title={v.title}
136 |                             />
137 |                         </div>
138 |                         <div style={{ padding: '12px 14px' }}>
139 |                             <p style={{ color: '#e0e0e0', fontWeight: 600, fontSize: '13px', margin: 0 }}>{v.title}</p>
140 |                         </div>
141 |                     </div>
142 |                 ))}
143 |             </div>
144 |         </div>
145 |     )}
146 |     </div>
147 |     </div>
148 |     </div>
149 |     </div>
150 |     </div>
151 |     </div>
152 |     </div>
153 |     </div>
154 |     return (
155 |         <div style={{ maxWidth: '1000px', margin: '0 auto' }}>
156 |             <motion.div initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }} style={{ marginBottom: '32px'...
157 |             <h1 style={{ fontSize: '28px', fontWeight: 800, color: 'var(--text-primary)', marginBottom: '8px' }}>S...
158 |             <p style={{ color: 'var(--text-muted)', fontSize: '15px', margin: 0 }}>
159 |             </motion.div>
160 |             <div className="grid gap-6" style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(320p...
161 |                 {SGK_BOOKS.map((s, i) => (
162 |                     <motion.div key={s.id} initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }} transition={{ ...
163 |                     whileHover={{ y: -4 }}
164 |                     onClick={() => { setSelectedBook(s.id); setTab('book'); }}
165 |                     className="lms-card"
166 |                     style={{
167 |                         padding: '28px', borderRadius: '16px', cursor: 'pointer', transition: 'all 0.3s',
168 |                         border: '1px solid ${s.color}20',
169 |                         background: `linear-gradient(135deg, ${s.color}08, transparent)`
170 |                     }}>
171 |                         <div style={{ width: '56px', height: '56px', borderRadius: '14px', background: `${s.color}20`, dis...
172 |                         <BookOpen size={28} color={s.color} />
173 |                         </div>
174 |                         <h2 style={{ fontSize: '18px', fontWeight: 700, color: 'var(--text-primary)', marginBottom: '10px'...
175 |                         <div className="flex flex-col gap-2 mb-4">
176 |                             {s.chapters.map((ch, j) => (
177 |                                 <div key={j} className="flex items-center gap-2 text-sm" style={{ color: 'var(--text-muted)' }}>
178 |                                     <FileText size={14} style={{ color: s.color, opacity: 0.6 }} />
179 |                                     <span>{ch}</span>
180 |                                 </div>
181 |                             ))}
182 |                         </div>
183 |                         <div style={{ display: 'flex', gap: '10px', flexWrap: 'wrap' }}>
184 |                             <div className="flex items-center gap-2 text-sm font-semibold" style={{ color: s.color }}>
185 |                                 <BookOpen size={16} /> Xem sách
186 |                             </div>
187 |                             {s.videos.length > 0 && (
188 |                                 <div className="flex items-center gap-2 text-sm font-semibold" style={{ color: '#ef4444' }}>
189 |                                     <Youtube size={16} /> {s.videos.length} video
190 |                                 </div>
191 |                             )}
192 |                         </div>
193 |                     </motion.div>
194 |                 ))}
195 |             </div>
196 |             </div>
197 |             </div>
198 |             </div>
199 |             </div>
200 |         )

```

File: app/(dashboard)/teacher/certificates/new/page.tsx (342 lines, 14.1 KB)

```

1 | 'use client'
2 |
3 | import React, { useEffect, useState } from 'react'
4 | import { createClientComponentClient } from '@supabase/auth-helpers-nextjs'
5 | import { useRouter } from 'next/navigation'
6 | import { ArrowLeft, Loader2, Save, Award, Users, BookOpen, Layers } from 'lucide-react'
7 | import Link from 'next/link'
8 | import { issueCustomCertificate } from '@lib/certificates'
9 |
10 | // Danh sách các sticker / badge Enh dãn m«u 1p cho chéng chÉ phñn céng
11 | const BADGE_TEMPLATES = [
12 |   { id: 'cpu', name: 'B-c Th sy CPU (CPU Master)', url: '/badges/cpu_master' },
13 |   { id: 'gpu', name: 'Chi n Th sn GPU (GPU Gladiator)', url: '/badges/gpu_master' },
14 |   { id: 'ram', name: 'Siêu Cyp RAM (RAM Champion)', url: '/badges/ram_champion' },
15 |   { id: 'pc_builder', name: 'Kù S Lp Ráp PC (PC Assembly Guru)', url: '/badges/pc_builder' },
16 |   { id: 'master', name: 'PC Master Vô Éch (LMS Grandmaster)', url: '/badges/master' },
17 | ]
18 |
19 | export default function NewCertificatePage() {
20 |   const supabase = createClientComponentClient()
21 |   const router = useRouter()
22 |   const [loading, setLoading] = useState(false)

```

```

23 | const [initialLoading, setInitialLoading] = useState(true)
24 | const [error, setError] = useState<string | null>(null)
25 |
26 | // Data lists
27 | const [classes, setClasses] = useState<any[]>([])
28 | const [students, setStudents] = useState<any[]>([])
29 |
30 | // Selections
31 | const [selectedClassId, setSelectedClassId] = useState('')
32 | const [selectedStudentId, setSelectedStudentId] = useState('')
33 | 3 | const [courseTitle, setCourseTitle] = useState('Khóa híc L-p ráp PC Ma
34 | const [finalScore, setFinalScore] = useState(100)
35 | const [selectedBadge, setSelectedBadge] = useState('pc_builder')
36 | const [customStickerUrl, setCustomStickerUrl] = useState('')
37 | const [useCustomUrl, setUseCustomUrl] = useState(false)
38 | const [completionDate, setCompletionDate] = useState(new Date().toISOString().split('T')[0])
39 |
40 | useEffect(() => {
41 |   fetchTeacherClasses()
42 | }, [])
43 |
44 | const fetchTeacherClasses = async () => {
45 |   try {
46 |     const { data: { user } } = await supabase.auth.getUser()
47 |     if (!user) {
48 |       window.location.href = '/login'
49 |       return
50 |     }
51 |
52 | // L-y danh sách lớp híc c-a giáo viên
53 | const { data, error } = await supabase
54 |   .from('classes')
55 |   .select('*')
56 |   .eq('teacher_id', user.id)
57 |   .order('name', { ascending: true })
58 |
59 | if (error) throw error
60 | setClasses(data || [])
61 |
62 | if (data && data.length > 0) {
63 |   setSelectedClassId(data[0].id)
64 |   fetchStudentsForClass(data[0].id)
65 | } else {
66 |   setInitialLoading(false)
67 | }
68 | } catch (err) {
69 |   console.error('Error loading teacher classes:', err)
70 |   setInitialLoading(false)
71 | }
72 |
73 | const fetchStudentsForClass = async (classId: string) => {
74 |   setStudents([])
75 |   setSelectedStudentId('')
76 |   try {
77 |
78 | // L-y danh sách thành viên trong lớp híc cùng profile c-a híc sin
79 | const { data, error } = await supabase
80 |   .from('class_members')
81 |   .select('student_id, profiles!inner(id, full_name, email)')
82 |   .eq('class_id', classId)
83 |
84 | if (error) throw error
85 |
86 | const studentProfiles = (data || []).map((m: any) => m.profiles)
87 | setStudents(studentProfiles)
88 |
89 | if (studentProfiles.length > 0) {
90 |   setSelectedStudentId(studentProfiles[0].id)
91 | }
92 | } catch (err) {
93 |   console.error('Error loading students for class:', err)
94 | } finally {
95 |   setInitialLoading(false)
96 | }
97 | }
98 |
99 | const handleClassChange = (classId: string) => {
100 |   setSelectedClassId(classId)
101 |   fetchStudentsForClass(classId)
102 | }
103 |
104 | const handleSubmit = async (e: React.FormEvent) => {
105 |   e.preventDefault()
106 |   if (!selectedStudentId) {
107 |     setError('Vui lòng chín híc sinh nh-n chéng ché!')
108 |     return
109 |   }
110 |
111 |   setLoading(true)
112 |   setError(null)
113 |
114 | // Xác Ònh URL sticker / Ònh dán
115 | let badgeUrl = ''
116 | if (useCustomUrl) {
117 |   badgeUrl = customStickerUrl.trim()
118 |   if (!badgeUrl) {
119 |     setError('Vui lòng nh-p Ỗng d«n Ònh dán tùy chénh!')
120 |     return
121 |   }
122 | } else {
123 |   const template = BADGE_TEMPLATES.find(b => b.id === selectedBadge)
124 |   badgeUrl = template ? template.url : '/badges/pc_builder.png'
125 | }
126 |
127 | try {
128 |
129 | // Gíi Server Action Ắ c-p chéng ché dán nhấ
130 | await issueCustomCertificate({
131 |   studentId: selectedStudentId,
132 |   courseTitle,
133 |   finalScore: Number(finalScore),
134 |   stickerUrl: badgeUrl,
135 |   completionDate
136 | })
137 |
138 | // Thành công, chuy Ần vẮ trang danh sách
139 | router.push('/teacher/certificates')
140 | router.refresh()
141 | } catch (err: any) {
142 |   console.error('Error issuing certificate:', err)
143 |   setError(err.message || 'L-xi khi l-u và c-p chéng ché.')
144 |   setLoading(false)
145 | }
146 | }

```

```
147 |
148 |   const inputStyle = {
149 |     width: '100%', padding: '12px 16px', borderRadius: '12px',
150 |     border: '1px solid rgba(255,255,255,0.08)', background: 'rgba(5, 10, 20, 0.5)',
151 |     color: 'white', fontSize: '15px', outline: 'none'
152 |   }
153 |
154 |   const labelStyle = {
155 |     display: 'block', fontSize: '13px', fontWeight: 700, color: '#8899a6', marginBottom: '8px'
156 |   }
157 |
158 |   if (initialLoading) {
159 |     return (
160 |       <div style={{ display: 'flex', flexDirection: 'column', alignItems: 'center', justifyContent: 'center', ...
161 |         <Loader2 className="animate-spin" color="#00d4aa" size={48} />
162 |       </div>
163 |     )
164 |   }
165 |
166 |   return (
167 |     <div style={{ padding: '32px', maxWidth: '850px', margin: '0 auto', color: '#e0e6ed', fontFamily: 'var(--f...
168 |       <Link href="/teacher/certificates" style={{ display: 'inline-flex', alignItems: 'center', gap: '8px', co...
169 |         <ArrowLeft size={20} /> Quay lại danh sách
170 |       </Link>
171 |
172 |       <div style={{ background: 'rgba(12, 20, 36, 0.8)', padding: '40px', borderRadius: '24px', border: '1px s...
173 |         <div style={{ display: 'flex', alignItems: 'center', gap: '12px', marginBottom: '32px' }}>
174 |           <div style={{ background: 'rgba(0, 212, 170, 0.1)', color: '#00d4aa', padding: '10px', borderRadius:...
175 |             <Award size={28} />
176 |           </div>
177 |           <h1 style={{ fontSize: '26px', fontWeight: 800, margin: 0, co...
178 |         </div>
179 |
180 |         {classes.length === 0 ? (
181 |           <div style={{ padding: '24px', borderRadius: '12px', background: 'rgba(239, 68, 68, 0.1)', color: '#...
182 |             B ì n h ã t i o l ù p h í c n à o ! G i á o v i ê n c ã n c ó l ù p h í c c h é a h í c
183 |           </div>
184 |         ) : (
185 |           <form onSubmit={handleSubmit} style={{ display: 'flex', flexDirection: 'column', gap: '24px' }}>
186 |
187 |             { /* L ù p & H í c s i n h */ }
188 |             <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '20px' }}>
189 |               <div>
190 |                 <label style={labelStyle}>Chín lùp híc</label>
191 |                 <select
192 |                   value={selectedClassId}
193 |                   onChange={(e) => handleClassChange(e.target.value)}
194 |                   style={inputStyle}
195 |                   required
196 |                 >
197 |                   {classes.map(c => (
198 |                     <option key={c.id} value={c.id}>{c.name} {c.grade} - {c.school_year}</option>
199 |                   ))}
200 |                 </select>
201 |               </div>
202 |               <div>
203 |                 <label style={labelStyle}>Chín híc sinh</label>
204 |                 <select
205 |                   value={selectedStudentId}
206 |                   onChange={(e) => setSelectedStudentId(e.target.value)}
207 |                   style={inputStyle}
208 |                   required
209 |                 >
210 |                   {students.length === 0 ? (
211 |                     <option value="">L ù p n à y c h ã c ó h í c s i n h n à o </option>
212 |                   ) : (
213 |                     students.map(s => (
214 |                       <option key={s.id} value={s.id}>{s.full_name} ({s.email})</option>
215 |                     ))
216 |                   )}
217 |                 </select>
218 |               </div>
219 |             </div>
220 |           </div>
221 |
222 |           { /* T ê n c h é n g c h é & i Ā m */ }
223 |           <div style={{ display: 'grid', gridTemplateColumns: '2fr 1fr', gap: '20px' }}>
224 |             <div>
225 |               <label style={labelStyle}>Tên chéng ché / Khóa híc</label>
226 |               <input
227 |                 type="text"
228 |                 value={courseTitle}
229 |                 onChange={(e) => setCourseTitle(e.target.value)}
230 |                 placeholder="V í d ă : K h ó a h í c L ù p r á p P C M a s t e r "
231 |                 style={inputStyle}
232 |                 required
233 |               />
234 |             </div>
235 |             <div>
236 |               <label style={labelStyle}>i Ā m t ã t n g h i c p ( % ) </label>
237 |               <input
238 |                 type="number"
239 |                 value={finalScore}
240 |                 onChange={(e) => setFinalScore(Number(e.target.value))}
241 |                 min={0} max={100}
242 |                 style={inputStyle}
243 |                 required
244 |               />
245 |             </div>
246 |           </div>
247 |
248 |           { /* N g à y c ã p */ }
249 |           <div>
250 |             <label style={labelStyle}>Ngày c ã p c h é n g n h - n </label>
251 |             <input
252 |               type="date"
253 |               value={completionDate}
254 |               onChange={(e) => setCompletionDate(e.target.value)}
255 |               style={inputStyle}
256 |               required
257 |             />
258 |           </div>
259 |
260 |           { /* C h í n c ã n h D á n / B a d g e */ }
261 |           <div>
262 |             <label style={labelStyle}>Chín Huy hiçu / c ã n h d á n K h e n T h
263 |             { /* G r i d c á c h u y h i c u c ó s ù n */ }
264 |             <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(130px, 1fr))', gap:...
265 |               {BADGE_TEMPLATES.map((badge) => (
266 |                 <div
267 |                   key={badge.id}
268 |
269 |
270 |
```



```

271 |         onClick={() => {
272 |             setSelectedBadge(badge.id)
273 |             setUseCustomUrl(false)
274 |         }}
275 |         style={{
276 |             background: 'rgba(255,255,255,0.02)',
277 |             border: selectedBadge === badge.id && !useCustomUrl
278 |               ? `2px solid ${badge.color}`
279 |               : `1.5px solid rgba(255,255,255,0.05)`,
280 |             borderRadius: '16px',
281 |             padding: '16px',
282 |             textAlign: 'center',
283 |             cursor: 'pointer',
284 |             transition: 'all 0.2s',
285 |             boxShadow: selectedBadge === badge.id && !useCustomUrl
286 |               ? `0 0 16px rgba(${badge.id === 'cpu' ? '0,243,255' : badge.id === 'ram' ? '234,179,8' : 'none'})`
287 |               : 'none'
288 |         }}
289 |     >
290 |         <div style={{ width: '60px', height: '60px', margin: '0 auto 12px auto', background: 'rgba...
291 |             <Award size={36} />
292 |         </div>
293 |         <div style={{ fontSize: '11px', fontWeight: 700, color: selectedBadge === badge.id && !use...
294 |             {badge.name}
295 |         </div>
296 |     </div>
297 |   )}}
298 | </div>
299 |
300 |   3 0 0 | { / * T ù y c h í n n h - p É n h d á n b ê n n g o à i * / }
301 |   <div style={{ borderTop: '1px solid rgba(255,255,255,0.05)', padding: '20px' }}>
302 |     <label style={{ display: 'flex', alignItems: 'center', gap: '10px', fontSize: '13px', color: '...'
303 |       <input
304 |         type="checkbox"
305 |         checked={useCustomUrl}
306 |         onChange={(e) => setUseCustomUrl(e.target.checked)}
307 |       />
308 |       3 0 8 | S í d ă n g ° Ý n g d « n É n h d á n t ù y c h É n h ( E x t e r n a l U R L )
309 |       </label>
310 |       {useCustomUrl && (
311 |         <input
312 |           type="text"
313 |           value={customStickerUrl}
314 |           onChange={(e) => setCustomStickerUrl(e.target.value)}
315 |         >
316 |           3 1 6 | p l a c e h o l d e r = " N h - p l i n k É n h ( P N G / J P G ) h u y h i ç u b ê n n g
317 |           style={inputStyle}
318 |         </input>
319 |       )}
320 |     </div>
321 |   </div>
322 |
323 |   {error && (
324 |     <div style={{ padding: '16px', borderRadius: '12px', background: 'rgba(239, 68, 68, 0.1)', color...
325 |       {error}
326 |     </div>
327 |   )}
328 |
329 |   <button type="submit" disabled={loading} style={{
330 |     width: '100%', padding: '16px', borderRadius: '12px', background: '#00d4aa',
331 |     color: '#050a14', fontSize: '16px', fontWeight: 700, border: 'none', cursor: loading ? 'not-allow...
332 |     display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '10px', marginTop: '16px'
333 |   }}>
334 |     3 3 4 | { l o a d i n g ? < L o a d e r 2 c l a s s N a m e = " a n i m a t e - s p i n " s i z e = { 2 0 } / >
335 |   </button>
336 | </form>
337 | )}
338 | </div>
339 | </div>
340 | )
341 | }
342 |

```

File: app/(dashboard)/teacher/certificates/page.tsx (172 lines, 8.8 KB)

```

1 | 'use client'
2 |
3 | import React, { useEffect, useState } from 'react'
4 | import { createClientComponentClient } from '@supabase/auth-helpers-nextjs'
5 | import { Plus, Award, ArrowLeft, Loader2, Calendar, FileText, Search, User, CheckCircle } from 'lucide-react'
6 | import Link from 'next/link'
7 |
8 | export default function TeacherCertificatesPage() {
9 |   const supabase = createClientComponentClient()
10 |   const [certificates, setCertificates] = useState<any[]>([])
11 |   const [loading, setLoading] = useState(true)
12 |   const [searchTerm, setSearchTerm] = useState('')
13 |
14 |   useEffect(() => {
15 |     fetchCertificates()
16 |   }, [])
17 |
18 |   const fetchCertificates = async () => {
19 |     setLoading(true)
20 |     try {
21 |       // T ê i d a n h s á c h c h é n g c h é
22 |       const { data, error } = await supabase
23 |         .from('certificates')
24 |         .select('*')
25 |         .order('issued_at', { ascending: false })
26 |
27 |       if (error) throw error
28 |       setCertificates(data || [])
29 |     } catch (err) {
30 |       console.error('Error fetching certificates:', err)
31 |     } finally {
32 |       setLoading(false)
33 |     }
34 |   }
35 |
36 |   const filteredCerts = certificates.filter(c =>
37 |     c.student_name?.toLowerCase().includes(searchTerm.toLowerCase()) ||
38 |     c.course_title?.toLowerCase().includes(searchTerm.toLowerCase()) ||
39 |     c.certificate_number?.toLowerCase().includes(searchTerm.toLowerCase())
40 |   )
41 |
42 |   if (loading) {
43 |     return (
44 |       <div style={{ display: 'flex', flexDirection: 'column', alignItems: 'center', justifyContent: 'center', ...
45 |         <Loader2 className="animate-spin" color="#00d4aa" size={48} />
46 |       <div style={{ color: '#8899a6', fontSize: '14px' }}> a n g t ê i d a n h
47 |     </div>
48 |   )

```

```

49 | }
50 |
51 | return (
52 |   <div style={{ maxWidth: '1200px', margin: '0 auto', paddingBottom: '80px', padding: '24px', color: '#e0e6e...
53 |
54 |     /* Header */
55 |     <header style={{ marginBottom: '40px', display: 'flex', justifyContent: 'space-between', alignItems: 'ce...
56 |     <div style={{ display: 'flex', alignItems: 'center', gap: '16px' }}>
57 |       <Link href="/teacher" style={{
58 |         width: '40px', height: '40px', borderRadius: '12px',
59 |         background: 'rgba(255,255,255,0.05)', display: 'flex',
60 |         alignItems: 'center', justifyContent: 'center', color: '#8899a6', textDecoration: 'none'
61 |       }}>
62 |         <ArrowLeft size={20} />
63 |       </Link>
64 |     </div>
65 |     <h1 style={{ fontSize: '28px', fontWeight: 800, margin: 0 }}>
66 |       Q u ê n l ý <span style={{ color: '#00d4aa' }}>Chéng Ché</span>
67 |     </h1>
68 |     <p style={{ color: '#8899a6', margin: '4px 0 0 0', fontSize:
69 |     </div>
70 |   </div>
71 |
72 |   <Link href="/teacher/certificates/new">
73 |     <button style={{
74 |       background: '#00d4aa', color: '#050a14', border: 'none',
75 |       padding: '12px 24px', borderRadius: '12px', fontWeight: 700,
76 |       display: 'flex', alignItems: 'center', gap: '8px', cursor: 'pointer', transition: 'all 0.2s'
77 |     }}>
78 |       <Plus size={20} /> C % p chéng ché m ù i
79 |     </button>
80 |   </Link>
81 | </header>
82 |
83 | /* Stats Summary Grid */
84 | <section style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(280px, 1fr))', gap: '20...
85 | <div style={{ background: 'rgba(12, 20, 36, 0.8)', padding: '24px', borderRadius: '24px', border: '1px...
86 | <div style={{ padding: '10px', borderRadius: '12px', background: 'rgba(0, 212, 170, 0.1)', color: '#...
87 | <Award size={20} />
88 | </div>
89 | <div style={{ fontSize: '13px', color: '#8899a6', fontWeight:
90 | <div style={{ fontSize: '28px', fontWeight: 800, color: 'fff' }}>{certificates.length}</div>
91 | </div>
92 |
93 | <div style={{ background: 'rgba(12, 20, 36, 0.8)', padding: '24px', borderRadius: '24px', border: '1px...
94 | <div style={{ padding: '10px', borderRadius: '12px', background: 'rgba(59, 130, 246, 0.1)', color: '...
95 | <User size={20} />
96 | </div>
97 | <div style={{ fontSize: '13px', color: '#8899a6', fontWeight:
98 | <div style={{ fontSize: '28px', fontWeight: 800, color: 'fff' }}>
99 |   {new Set(certificates.map(c => c.student_id)).size}
100 | </div>
101 | </div>
102 | </section>
103 |
104 | /* Search and Filters */
105 | <div style={{ marginBottom: '24px', position: 'relative', maxWidth: '400px' }}>
106 |   <Search style={{ position: 'absolute', left: '16px', top: '12px', color: '#4b5563' }} size={20} />
107 |   <input
108 |     type="text"
109 |     placeholder="T i m t ê n h í c s í n h , k h ó a h í c , m ã s Ñ . . ."
110 |     value={searchTerm}
111 |     onChange={(e) => setSearchTerm(e.target.value)}
112 |     style={{
113 |       width: '100%', padding: '12px 16px 12px 48px', borderRadius: '12px',
114 |       border: '1px solid rgba(255,255,255,0.08)', background: 'rgba(12, 20, 36, 0.5)',
115 |       color: 'white', fontSize: '14px', outline: 'none'
116 |     }}
117 |   />
118 | </div>
119 |
120 | /* Main List Table */
121 | <div style={{ background: 'rgba(12, 20, 36, 0.8)', border: '1px solid rgba(255,255,255,0.05)', borderRad...
122 | {filteredCerts.length === 0 ? (
123 |   <div style={{ textAlign: 'center', padding: '80px 0', color: '#8899a6' }}>
124 |     <Award size={48} style={{ marginBottom: '16px', opacity: 0.3 }} />
125 |     <h3>Ch ò a t i m t h ý y c h é n g c h é n à o . </h3>
126 |     <p style={{ fontSize: '13px' }}>H ấ y t h í ò i t ê k h ó a t i m k i z
127 |   </div>
128 | ) : (
129 |   <table style={{ width: '100%', borderCollapse: 'collapse', textAlign: 'left', fontSize: '14px' }}>
130 |     <thead>
131 |       <tr style={{ background: 'rgba(255,255,255,0.02)', borderBottom: '1px solid rgba(255,255,255,0.0...
132 |         <th style={{ padding: '16px 24px', color: '#8899a6' }}>
133 |         <th style={{ padding: '16px 24px', color: '#8899a6' }}>
134 |         <th style={{ padding: '16px 24px', color: '#8899a6' }}>
135 |         <th style={{ padding: '16px 24px', color: '#8899a6' }}>
136 |         <th style={{ padding: '16px 24px', color: '#8899a6' }}>
137 |         <th style={{ padding: '16px 24px', color: '#8899a6' }}>
138 |       </tr>
139 |     </thead>
140 |     <tbody>
141 |       {filteredCerts.map((cert) => (
142 |         <tr key={cert.id} style={{ borderBottom: '1px solid rgba(255,255,255,0.03)', color: '#e0e6ed' }}>
143 |           <td style={{ padding: '16px 24px', fontWeight: 700 }}>{cert.student_name}</td>
144 |           <td style={{ padding: '16px 24px' }}>{cert.course_title}</td>
145 |           <td style={{ padding: '16px 24px', fontFamily: 'monospace', color: '#00d4aa' }}>{cert.certif...
146 |           <td style={{ padding: '16px 24px', color: '#00d4aa', fontWeight: 800 }}>{cert.final_score}%<...
147 |           <td style={{ padding: '16px 24px' }}>
148 |             <cert.pdf_url ? (
149 |               <a href={cert.pdf_url} target="_blank" rel="noopener noreferrer" style={{ display: 'inli...
150 |                 <img
151 |                   src={cert.pdf_url}
152 |                   alt="Badge"
153 |                   style={{ width: '40px', height: '40px', borderRadius: '8px', objectFit: 'contain', b...
154 |                 />
155 |               </a>
156 |             ) : (
157 |               <span style={{ color: '#4b5563', fontSize: '12px' }}>Không có</span>
158 |             </td>
159 |           <td style={{ padding: '16px 24px', color: '#8899a6' }}>
160 |             {cert.completion_date ? new Date(cert.completion_date).toLocaleDateString('vi-VN') : new D...
161 |           </td>
162 |         </tr>
163 |       )
164 |     </tbody>
165 |   </table>
166 | )
167 | </div>
168 | )
169 | }
170 |
171 |
172 |

```

```
1 'use client';
2
3 import React, { useEffect, useState, use } from 'react';
4 import { supabase } from '@lib/supabase';
5 import { useRouter } from 'next/navigation';
6 import {
7   Loader2, ArrowLeft, CheckCircle, AlertCircle, Save,
8   MessageSquare, Star, Cpu, Wallet, Battery, HardDrive
9 } from 'lucide-react';
10 import Link from 'next/link';
11 import { formatVND } from '@lib/grading/auto-grade';
12
13 export default function GradeSubmissionPage({ params }: { params: Promise<{ classId: string, assignmentId: str...
14   const resolvedParams = use(params);
15   const { classId, assignmentId, submissionId } = resolvedParams;
16
17   const [submission, setSubmission] = useState<any>(null);
18   const [assignment, setAssignment] = useState<any>(null);
19   const [loading, setLoading] = useState(true);
20   const [isSaving, setIsSaving] = useState(false);
21
22   // Grade state
23   const [teacherScore, setTeacherScore] = useState(0);
24   const [feedback, setFeedback] = useState('');
25
26   useEffect(() => {
27     fetchData();
28   }, [submissionId]);
29
30   async function fetchData() {
31     setLoading(true);
32     try {
33       // 1. Fetch Submission
34       const { data: sub, error: subError } = await supabase
35         .from('submissions')
36         .select(`
37           *
38           student:profiles(full_name)
39         `)
40         .eq('id', submissionId)
41         .single();
42
43       if (subError) throw subError;
44       setSubmission(sub);
45       setTeacherScore(sub.teacher_score || 0);
46       setFeedback(sub.feedback || '');
47
48       // 2. Fetch Assignment
49       const { data: asg, error: asgError } = await supabase
50         .from('assignments')
51         .select(`*`)
52         .eq('id', assignmentId)
53         .single();
54
55       if (asgError) throw asgError;
56       setAssignment(asg);
57
58     } catch (err) {
59       console.error('Error fetching submission details:', err);
60     } finally {
61       setLoading(false);
62     }
63   }
64
65   const handleSaveGrade = async () => {
66     setIsSaving(true);
67     try {
68       const totalScore = Math.min(assignment.max_score, (submission.auto_score || 0) + teacherScore);
69
70       const { error } = await supabase
71         .from('submissions')
72         .update({
73           teacher_score: teacherScore,
74           total_score: totalScore,
75           feedback,
76           graded_at: new Date().toISOString()
77         })
78         .eq('id', submissionId);
79
80       if (error) throw error;
81       alert('Đã lưu điểm và phần học!');
82       router.push(`/teacher/classes/${classId}/assignments/${assignmentId}`);
83     } catch (err: any) {
84       alert('Lỗi: ' + err.message);
85     } finally {
86       setIsSaving(false);
87     }
88   };
89
90   const router = useRouter();
91
92   if (loading) {
93     return (
94       <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', height: '100%' }}>
95         <Loader2 className="animate-spin" color="#00f3ff" size={48} />
96       </div>
97     );
98   }
99
100   if (!submission || !assignment) return <div>Không tìm thấy dữ liệu.</div>;
101
102   const pc = submission.pc_config;
103
104   return (
105     <div style={{ padding: '32px', maxWidth: '1200px', margin: '0 auto' }}>
106       <header style={{ marginBottom: '40px' }}>
107         <Link href={`/teacher/classes/${classId}/assignments/${assignmentId}`} style={{ display: 'inline-flex'...
108         <ArrowLeft size={20} /> Quay lại bảng điểm
109       </Link>
110       <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
111         <div>
112           <h1 style={{ fontSize: '32px', fontWeight: 800, margin: '0' }}>
113             <p style={{ color: '#8899a6', margin: '0' }}>Nhicm và: {assignment
114         </div>
115         <div style={{ textAlign: 'right' }}>
116           <div style={{ fontSize: '12px', color: '#8899a6', fontWeight
117             <div style={{ fontSize: '32px', fontWeight: 900, color: '#00f3ff' }}>{Math.min(assignment.max_scor...
118         </div>
119       </div>
120     </header>

```



```

12 | const resolvedParams = use(params);
13 | const { classId, assignmentId } = resolvedParams;
14 |
15 | const [assignment, setAssignment] = useState<any>(null);
16 | const [submissions, setSubmissions] = useState<any[]>([]);
17 | const [loading, setLoading] = useState(true);
18 |
19 | useEffect(() => {
20 |   fetchData();
21 | }, [assignmentId]);
22 |
23 | async function fetchData() {
24 |   setLoading(true);
25 |   try {
26 |     // 1. Fetch Assignment Info
27 |     const { data: asg, error: asgError } = await supabase
28 |       .from('assignments')
29 |       .select('*')
30 |       .eq('id', assignmentId)
31 |       .single();
32 |
33 |     if (asgError) throw asgError;
34 |     setAssignment(asg);
35 |
36 |     // 2. Fetch all members of the class
37 |     const { data: members, error: memError } = await supabase
38 |       .from('class_members')
39 |       .select('student_id, student:profiles(full_name)')
40 |       .eq('class_id', classId);
41 |
42 |     if (memError) throw memError;
43 |
44 |     // 3. Fetch all submissions for this assignment
45 |     const { data: subs, error: subError } = await supabase
46 |       .from('submissions')
47 |       .select('*')
48 |       .eq('assignment_id', assignmentId);
49 |
50 |     if (subError) throw subError;
51 |
52 |     // 4. Combine data to show who hasn't submitted
53 |     const formattedData = members.map(m => {
54 |       const submission = subs?.find(s => s.student_id === m.student_id);
55 |       return {
56 |         id: submission?.id || m.student_id,
57 |         student_id: m.student_id,
58 |         student_name: (m.student as any)?.full_name || 'Hí c sinh',
59 |         submitted_at: submission?.submitted_at || null,
60 |         is_late: submission?.is_late || false,
61 |         auto_score: submission?.auto_score ?? null,
62 |         teacher_score: submission?.teacher_score ?? null,
63 |         total_score: submission?.total_score ?? null,
64 |         status: submission
65 |           ? (submission.total_score !== null ? 'graded' : 'submitted')
66 |           : 'missing'
67 |       };
68 |     });
69 |     setSubmissions(formattedData);
70 |
71 |   } catch (err) {
72 |     console.error('Error fetching submissions:', err);
73 |   } finally {
74 |     setLoading(false);
75 |   }
76 | }
77 |
78 | if (loading) {
79 |   return (
80 |     <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', height: '100%' }}>
81 |       <Loader2 className="animate-spin" color="#00f3ff" size={48} />
82 |     </div>
83 |   );
84 | }
85 |
86 | if (!assignment) return <div>Không tìm thấy nhiệ m vậ.</div>;
87 |
88 | return (
89 |   <div style={{ padding: '32px' }}>
90 |     <header style={{ marginBottom: '40px' }}>
91 |       <Link href={`/teacher/classes/${classId}`} style={{ display: 'inline-flex', alignItems: 'center', gap: ...
92 |     </Link>
93 |     <ArrowLeft size={20} /> Quay l i i l ú p h í c
94 |
95 |     <div style={{ display: 'flex', gap: '24px', color: '#8899a6', fontSize: '14px' }}>
96 |       <span style={{ display: 'flex', alignItems: 'center', gap: '6px' }}>
97 |         {assignment.deadline && (
98 |           <span style={{ display: 'flex', alignItems: 'center', gap:
99 |         ))
100 |       <span style={{ display: 'flex', alignItems: 'center', gap: '6px' }}>
101 |         </div>
102 |       </header>
103 |
104 |       <GradeTable
105 |         data={submissions}
106 |         assignmentId={assignmentId}
107 |         classId={classId}
108 |       />
109 |     </div>
110 |   );
111 | }
112 |
113 |
114 |

```

File: app/(dashboard)/teacher/classes/[classId]/assignments/new/page.tsx (202 lines, 9.2 KB)

```

1 | 'use client';
2 |
3 | import React, { useState, use } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import { useRouter } from 'next/navigation';
6 | import {
7 |   ArrowLeft, Loader2, Save, FileText, Target,
8 |   Wallet, Battery, HardDrive, Cpu, AlertCircle, Calendar
9 | } from 'lucide-react';
10 | import Link from 'next/link';
11 |
12 | export default function NewAssignmentPage({ params }: { params: Promise<{ classId: string }> }) {
13 |   const resolvedParams = use(params);
14 |   const classId = resolvedParams.classId;
15 |
16 |   const [loading, setLoading] = useState(false);
17 |   const [error, setError] = useState<string | null>(null);

```

```

18 | const [assignmentType, setAssignmentType] = useState('build_config');
19 | const router = useRouter();
20 |
21 | async function handleSubmit(e: React.FormEvent<HTMLFormElement>) {
22 |   e.preventDefault();
23 |   setLoading(true);
24 |   setError(null);
25 |
26 |   const formData = new FormData(e.currentTarget);
27 |   const title = formData.get('title') as string;
28 |   const description = formData.get('description') as string;
29 |   const deadline = formData.get('deadline') as string;
30 |   const maxScore = parseInt(formData.get('max_score') as string) || 100;
31 |
32 |   // Requirements JSON
33 |   const requirements = {
34 |     budget_max: parseInt(formData.get('budget_max') as string) || undefined,
35 |     tdp_max: parseInt(formData.get('tdp_max') as string) || undefined,
36 |     min_ram_gb: parseInt(formData.get('min_ram_gb') as string) || undefined,
37 |     required_components: (formData.get('required_components') as string).split(',').map(s => s.trim()).filte...
38 |   };
39 |
40 |   try {
41 |     const { data: { user } } = await supabase.auth.getUser();
42 |     if (!user) throw new Error('Vui lòng đăng nhập lại.');
```

```

43 |
44 |     const { error: insertError } = await supabase
45 |       .from('assignments')
46 |       .insert({
47 |         class_id: classId,
48 |         teacher_id: user.id,
49 |         title,
50 |         description,
51 |         type: assignmentType,
52 |         requirements,
53 |         max_score: maxScore,
54 |         deadline: deadline ? new Date(deadline).toISOString() : null,
55 |         is_published: true
56 |       });
57 |
58 |     if (insertError) throw insertError;
59 |
60 |     router.push(`/teacher/classes/${classId}`);
61 |   } catch (err: any) {
62 |     setError(err.message);
63 |     setLoading(false);
64 |   }
65 | }
66 |
67 | const inputStyle = {
68 |   width: '100%', padding: '14px 16px 14px 48px', borderRadius: '12px',
69 |   border: '1px solid rgba(255,255,255,0.1)', background: 'rgba(5, 10, 20, 0.5)',
70 |   color: 'white', fontSize: '16px', outline: 'none'
71 | };
72 |
73 | const labelStyle = {
74 |   display: 'block', fontSize: '14px', fontWeight: 600, color: '#8899a6', marginBottom: '8px'
75 | };
76 |
77 | const iconStyle = {
78 |   position: 'absolute' as const, left: '16px', top: '14px', color: '#4b5563'
79 | };
80 |
81 | return (
82 |   <div style={{ padding: '32px', maxWidth: '900px', margin: '0 auto' }}>
83 |     <Link href={`/teacher/classes/${classId}`} style={{ display: 'inline-flex', alignItems: 'center', gap: '...'
84 |       <ArrowLeft size={20} /> Quay lại lớp học
85 |     </Link>
86 |
87 |     <div style={{ background: 'rgba(12, 20, 36, 0.8)', padding: '48px', borderRadius: '24px', border: '1px s...
88 |       <h1 style={{ font-size: '28px', fontWeight: '800', margin: '0 0 40px 0
89 |
90 |         <form onSubmit={handleSubmit} style={{ display: 'flex', flexDirection: 'column', gap: '32px' }}>
91 |           </* Thông tin chung */>
92 |           <section style={{ display: 'flex', flexDirection: 'column', gap: '20px' }}>
93 |             <h3 style={{ color: '#00f3ff', fontSize: '18px', fontWeight: 700, margin: 0 }}>1. Thông tin chung<...
94 |
95 |             <div>
96 |               <label style={labelStyle}>Tiêu Ánhiệm vậ</label>
97 |               <div style={{ position: 'relative' }}>
98 |                 <FileText style={iconStyle} size={20} />
99 |                 <input name="title" required placeholder="VD: Laptop Gaming" />
100 |               </div>
101 |             </div>
102 |
103 |             <div>
104 |               <label style={labelStyle}>Mô tả chi tiết</label>
105 |               <textarea name="description" rows={3} style={{ ...inputStyle, paddingLeft: '16px', resize: 'none'...
106 |             </div>
107 |
108 |             <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '24px' }}>
109 |               <div>
110 |                 <label style={labelStyle}>Loại nhiệm vụ</label>
111 |                 <div style={{ position: 'relative' }}>
112 |                   <Target style={iconStyle} size={20} />
113 |                   <select
114 |                     value={assignmentType}
115 |                     onChange={(e) => setAssignmentType(e.target.value)}
116 |                     style={inputStyle}
117 |                   >
118 |                     <option value="build_config">Laptop ráp PC ít chu@n</o
119 |                     <option value="optimize_budget">Tối ưu ngân sách</o
120 |                     <option value="minimize_tdp">Tiết kiệm iCn n ng</o
121 |                     <option value="maximize_perf">Tối đa hiệ u n ng</o
122 |                   </select>
123 |                 </div>
124 |               </div>
125 |
126 |               <div>
127 |                 <label style={labelStyle}>Hạn chót (Deadline)</label>
128 |                 <div style={{ position: 'relative' }}>
129 |                   <Calendar style={iconStyle} size={20} />
130 |                   <input name="deadline" type="datetime-local" style={inputStyle} />
131 |                 </div>
132 |               </div>
133 |             </section>
134 |
135 |             <div>
136 |               </* Yêu cầu kỹ thuật */>
137 |               <section style={{ display: 'flex', flexDirection: 'column', gap: '20px', padding: '24px', background...
138 |                 <h3 style={{ color: '#00f3ff', fontSize: '18px', fontWeight: 700, margin: 0 }}>2. Yêu cầu kỹ thuật</h3>
139 |                 <p style={{ color: '#4b5563', fontSize: '13px', margin: '0 0 16px 0
140 |               </div>
141 |             </div>

```

```

143 |         <label style={labelStyle}>Ngân sách tñi a (VN)</label>
144 |         <div style={{ position: 'relative' }}>
145 |           <Wallet style={iconStyle} size={20} />
146 |           <input name="budget_max" type="number" placeholder="15000000" style={inputStyle} />
147 |         </div>
148 |       </div>
149 |       <label style={labelStyle}>TDP tñi a (Watt)</label>
150 |       <div style={{ position: 'relative' }}>
151 |         <Battery style={iconStyle} size={20} />
152 |         <input name="tdp_max" type="number" placeholder="300" style={inputStyle} />
153 |       </div>
154 |     </div>
155 |   </div>
156 |   <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '24px' }}>
157 |     <div>
158 |       <label style={labelStyle}>RAM tñi thiãu (GB)</label>
159 |       <div style={{ position: 'relative' }}>
160 |         <HardDrive style={iconStyle} size={20} />
161 |         <input name="min_ram_gb" type="number" placeholder="16" style={inputStyle} />
162 |       </div>
163 |     </div>
164 |     <div>
165 |       <label style={labelStyle}>iã m tñi a</label>
166 |       <div style={{ position: 'relative' }}>
167 |         <Target style={iconStyle} size={20} />
168 |         <input name="max_score" type="number" defaultValue="100" style={inputStyle} />
169 |       </div>
170 |     </div>
171 |   </div>
172 |   <div>
173 |     <label style={labelStyle}>Linh kiçn b`t buüc (phân cách b
174 |     <div style={{ position: 'relative' }}>
175 |       <Cpu style={iconStyle} size={20} />
176 |       <input name="required_components" placeholder="RTX 3060, SSD NVMe, i5-12400F" style={inputStyl...
177 |     </div>
178 |   </div>
179 | </section>
180 |
181 | {error && (
182 |   <div style={{ padding: '16px', borderRadius: '12px', background: 'rgba(239, 68, 68, 0.1)', color: ...
183 |     <AlertCircle size={20} style={{ flexShrink: 0 }} />
184 |     <span>{error}</span>
185 |   </div>
186 | )}
187 |
188 | <button type="submit" disabled={loading} style={{
189 |   width: '100%', padding: '18px', borderRadius: '12px', background: 'var(--brand-primary)',
190 |   color: '#000', fontSize: '16px', fontWeight: 800, border: 'none', cursor: loading ? 'not-allowed' ...
191 |   display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '10px'
192 | }}>
193 |   {loading ? <Loader2 className="animate-spin" size={20} /> :
194 |   </button>
195 | </form>
196 | </div>
197 | </div>
198 | );
199 | }
200 | }
201 |
202 |

```

File: app/(dashboard)/teacher/classes/[classId]/page.tsx (472 lines, 22.4 KB)

```

1 | 'use client';
2 |
3 | import React, { useEffect, useState, use } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import {
6 |   Users, ClipboardList, Plus, Loader2, ArrowLeft,
7 |   Settings, UserPlus, Info, Calendar, BookOpen, Bell, Send, Megaphone
8 | } from 'lucide-react';
9 | import Link from 'next/link';
10 | import AssignmentCard from '@components/classes/AssignmentCard';
11 |
12 | export default function ClassDetailsPage({ params }: { params: Promise<{ classId: string }> }) {
13 |   const resolvedParams = use(params);
14 |   const classId = resolvedParams.classId;
15 |
16 |   const [classData, setClassData] = useState<any>(null);
17 |   const [assignments, setAssignments] = useState<any[]>([]);
18 |   const [members, setMembers] = useState<any[]>([]);
19 |   const [loading, setLoading] = useState(true);
20 |   const [activeTab, setActiveTab] = useState<'assignments' | 'students'>('assignments');
21 |
22 |   // Settings Modal states
23 |   const [showSettingsModal, setShowSettingsModal] = useState(false);
24 |   const [editName, setEditName] = useState('');
25 |   const [editMaxStudents, setEditMaxStudents] = useState(40);
26 |   const [editingSubject, setEditingSubject] = useState('Tin híc');
27 |   const [updating, setUpdating] = useState(false);
28 |
29 |   // Notification Modal states
30 |   const [showNotifModal, setShowNotifModal] = useState(false);
31 |   const [notifTitle, setNotifTitle] = useState('');
32 |   const [notifMessage, setNotifMessage] = useState('');
33 |   const [sendingNotif, setSendingNotif] = useState(false);
34 |
35 |   useEffect(() => {
36 |     fetchClassInfo();
37 |     const channel = supabase.channel(`class-${classId}-realtime`);
38 |
39 |     channel.on('postgres_changes',
40 |       { event: 'INSERT', schema: 'public', table: 'submissions', filter: `class_id=eq.${classId}` },
41 |       async (payload: any) => {
42 |         const assignmentId = payload.new?.assignment_id;
43 |         if (assignmentId) {
44 |           setAssignments(prev => prev.map(a =>
45 |             a.id === assignmentId ? { ...a, submissions: { count: (a.submissions?.count || 0) + 1 } } : a
46 |           ));
47 |         }
48 |       }
49 |     );
50 |
51 |     channel.on('postgres_changes',
52 |       { event: 'INSERT', schema: 'public', table: 'class_members', filter: `class_id=eq.${classId}` },
53 |       async (payload: any) => {
54 |         const studentId = payload.new?.student_id;
55 |         if (studentId) {
56 |           const { data: profile } = await supabase.from('profiles').select('').eq('id', studentId).single();
57 |           if (profile) {
58 |             setMembers(prev => [...prev, { id: payload.new.id, class_id: classId, student_id: studentId, stude...
59 |           }

```

```

60 |     }
61 |   }
62 | );
63 |
64 | channel.on('postgres_changes',
65 |   { event: 'INSERT', schema: 'public', table: 'assignments', filter: `class_id=eq.${classId}` },
66 |   (payload: any) => {
67 |     setAssignments(prev => [payload.new, ...prev]);
68 |   }
69 | );
70 |
71 | channel.subscribe();
72 | return () => { supabase.removeChannel(channel); };
73 | }, [classId]);
74 |
75 | async function fetchClassInfo() {
76 |   setLoading(true);
77 |   try {
78 |     // 1. Fetch Class Info
79 |     const { data: cls, error: clsError } = await supabase
80 |       .from('classes')
81 |       .select('*')
82 |       .eq('id', classId)
83 |       .single();
84 |
85 |     if (clsError) throw clsError;
86 |     setClassData(cls);
87 |     setEditName(cls.name);
88 |     setEditMaxStudents(cls.max_students);
89 |     setEditSubject(cls.subject || 'Tin hĩa');
90 |
91 |     // 2. Fetch Assignments
92 |     const { data: asgs, error: asgsError } = await supabase
93 |       .from('assignments')
94 |       .select('*', submissions(count))
95 |       .eq('class_id', classId)
96 |       .order('created_at', { ascending: false });
97 |
98 |     if (asgsError) throw asgsError;
99 |     setAssignments(asgs || []);
100 |
101 |     // 3. Fetch Members (Profiles)
102 |     const { data: mems, error: memsError } = await supabase
103 |       .from('class_members')
104 |       .select(
105 |         `
106 |         *,
107 |         student:profiles(*)
108 |       `
109 |       .eq('class_id', classId);
110 |
111 |     if (memsError) throw memsError;
112 |     setMembers(mems || []);
113 |
114 |   } catch (err) {
115 |     console.error('Error fetching class info:', err);
116 |   } finally {
117 |     setLoading(false);
118 |   }
119 | }
120 |
121 | async function handleUpdateClass() {
122 |   if (!editName) return;
123 |   setUpdating(true);
124 |   try {
125 |     const { error } = await supabase
126 |       .from('classes')
127 |       .update({
128 |         name: editName,
129 |         max_students: editMaxStudents,
130 |         subject: editSubject
131 |       })
132 |       .eq('id', classId);
133 |     if (error) throw error;
134 |     setClassData({ ...classData, name: editName, max_students: editMaxStudents, subject: editSubject });
135 |     setShowSettingsModal(false);
136 |     alert('C - p nh - t thông tin lớp hĩa thành công!');
137 |   } catch (err) {
138 |     console.error(err);
139 |   } finally {
140 |     setUpdating(false);
141 |   }
142 | }
143 |
144 | async function handleDeleteClass() {
145 |   if (!confirm('Bạn có ch - c h - n muốn xóa lớp hĩa này không? Hành ùng')) return;
146 |   setUpdating(true);
147 |   try {
148 |     const { error } = await supabase
149 |       .from('classes')
150 |       .delete()
151 |       .eq('id', classId);
152 |     if (error) throw error;
153 |     alert('ã xóa lớp hĩa thành công!');
154 |     window.location.href = '/teacher/classes';
155 |   } catch (err) {
156 |     console.error(err);
157 |   } finally {
158 |     alert('C ó l x i x E y ra khi xóa lớp hĩa');
159 |     setUpdating(false);
160 |   }
161 | }
162 |
163 | if (loading) {
164 |   return (
165 |     <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', height: '100%' }}>
166 |       <Loader2 className="animate-spin" color="#00f3ff" size={48} />
167 |     </div>
168 |   );
169 | }
170 |
171 | if (!classData) return <div>Không tìm thấy lớp hĩa.</div>;
172 |
173 | return (
174 |   <div style={{ padding: '32px' }}>
175 |     <header style={{ marginBottom: '40px' }}>
176 |       <Link href="/teacher/classes" style={{ display: 'inline-flex', alignItems: 'center', gap: '8px', color: 'inherit' }}>
177 |         <ArrowLeft size={20} /> Danh sách lớp
178 |       </Link>
179 |
180 |       <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'flex-end' }}>
181 |         <div>
182 |           <h1 style={{ fontSize: '36px', fontWeight: 800, margin: '0 0 12px 0', color: '#fff' }}>{classData.name}</h1>
183 |           <div style={{ display: 'flex', gap: '24px', color: '#8899a6', fontSize: '14px' }}>

```



```

1 8 4 | <span style={{ display: 'flex', alignItems: 'center', gap:
185| <span style={{ display: 'flex', alignItems: 'center', gap:
1 8 6 | </div>
187| </div>
188| <div style={{ display: 'flex', gap: '12px' }}>
189| <button
190|   onClick={() => setShowNotifModal(true)}
191|   style={{ padding: '12px', borderRadius: '12px', background: 'rgba(249,168,37,0.1)', border: '1px...
192|   title = "Gi i t h ò n g b á o ã n l ù p "
1 9 3 | >
194| <Megaphone size={18} /> Thông báo
195| </button>
196| <button
197|   onClick={() => setShowSettingsModal(true)}
198|   style={{ padding: '12px', borderRadius: '12px', background: 'rgba(255,255,255,0.05)', border: '1...
199| >
200| <Settings size={20} />
201| </button>
202| <Link href={`/teacher/classes/${classId}/assignments/new`} style={{ textDecoration: 'none' }}>
203| <button style={{
204|   background: 'var(--brand-primary)', color: '#000', border: 'none',
205|   padding: '12px 24px', borderRadius: '12px', fontWeight: 700,
206|   display: 'flex', alignItems: 'center', gap: '8px', cursor: 'pointer'
207| }}>
208| < Plus size = { 2 0 } / > G i a o n h i ệ m v ấ
2 0 9 | </button>
210| </Link>
211| </div>
212| </div>
213| </div>
214| </header>
215|
216| {/ * Tabs */}
217| <div style={{ display: 'flex', gap: '32px', borderBottom: '1px solid rgba(255,255,255,0.05)', marginBott...
218| <button
219|   onClick={() => setActiveTab('assignments')}
220|   style={{
221|     padding: '12px 0', background: 'none', border: 'none',
222|     borderBottom: activeTab === 'assignments' ? '2px solid #00f3ff' : '2px solid transparent',
223|     color: activeTab === 'assignments' ? '#00f3ff' : '#8899a6',
224|     fontSize: '16px', fontWeight: 700, cursor: 'pointer',
225|     display: 'flex', alignItems: 'center', gap: '8px'
226|   }}
227| >
2 2 8 | < ClipboardList size = { 2 0 } / > N h i ệ m v ấ ( { a s s i g n m e n t s . l e n g t h } )
229| </button>
230| <button
231|   onClick={() => setActiveTab('students')}
232|   style={{
233|     padding: '12px 0', background: 'none', border: 'none',
234|     borderBottom: activeTab === 'students' ? '2px solid #00f3ff' : '2px solid transparent',
235|     color: activeTab === 'students' ? '#00f3ff' : '#8899a6',
236|     fontSize: '16px', fontWeight: 700, cursor: 'pointer',
237|     display: 'flex', alignItems: 'center', gap: '8px'
238|   }}
239| >
2 4 0 | < Users size = { 2 0 } / > D a n h s á c h l ù p ( { m e m b e r s . l e n g t h } )
241| </button>
242| </div>
243|
244| {activeTab === 'assignments' ? (
245|   <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill, minmax(400px, 1fr))', gap: '24p...
246|     {assignments.length === 0 ? (
247|       <div style={{ gridColumn: '1 / -1', textAlign: 'center', padding: '60px', color: '#4b5563' }}>
248|         <BookOpen size={48} style={{ marginBottom: '16px', opacity: 0.2 }} />
2 4 9 | < p > C h ộ a c ó n h i ệ m v ấ n à o ã c g i a o c h o l ù p n à y . < / p >
250|       </div>
251|     ) : (
252|       assignments.map(asg => (
253|         <AssignmentCard
254|           key={asg.id}
255|           id={asg.id}
256|           classId={classId}
257|           title={asg.title}
258|           type={asg.type}
259|           deadline={asg.deadline}
260|           requirements={asg.requirements}
261|           submittedCount={asg.submissions?.[0]?.count || 0}
262|           totalMembers={members.length}
263|           role="teacher"
264|         />
265|       ))
266|     )
267|   </div>
268| ) : (
269|   <div style={{ background: 'rgba(12, 20, 36, 0.8)', borderRadius: '20px', border: '1px solid rgba(255,2...
270|     <table style={{ width: '100%', borderCollapse: 'collapse', textAlign: 'left' }}>
271|       <thead>
272|         <tr style={{ background: 'rgba(255,255,255,0.02)', borderBottom: '1px solid rgba(255,255,255,0.0...
2 7 3 |           <th style={{ padding: '16px 24px', color: '#8899a6', fontSize: '13px' }}>Email</th>
274|           <th style={{ padding: '16px 24px', color: '#8899a6', fontSize: '13px' }}>Ngày tham gia</th>
275|           <th style={{ padding: '16px 24px', color: '#8899a6', fontSize: '13px' }}>
2 7 6 |             <th style={{ padding: '16px 24px', color: '#8899a6', font
2 7 7 |             <th style={{ padding: '16px 24px', color: '#8899a6', font
278|           </tr>
279|         </thead>
280|         <tbody>
281|           {members.map(mem => (
282|             <tr key={mem.id} style={{ borderBottom: '1px solid rgba(255,255,255,0.05)' }}>
283|               <td style={{ padding: '16px 24px', color: '#fff', fontWeight: 600 }}>{mem.student?.full_name...
284|               <td style={{ padding: '16px 24px', color: '#8899a6' }}>{mem.student?.email}</td>
285|               <td style={{ padding: '16px 24px', color: '#8899a6' }}>{new Date(mem.joined_at).toLocaleDate...
286|               <td style={{ padding: '16px 24px' }}>
287|                 <span style={{ padding: '4px 12px', borderRadius: '100px', background: 'rgba(16, 185, 129,...
288|               </td>
289|               <td style={{ padding: '16px 24px' }}>
290|                 <button style={{ background: 'none', border: 'none', color: '#f87171', fontSize: '13px', f...
291|               </td>
292|             </tr>
293|           ))
294|           {members.length === 0 && (
295|             <tr>
2 9 6 |               <td colspan = { 5 } style = { { padding : ' 4 0 p x ' , text A l i g n :
297|             </tr>
298|           )}
299|         </tbody>
300|       </table>
301|     </div>
302|   )}
303|
304| {/ * Send Notification Modal */}
305| {showNotifModal && (
306|   <div style={{
307|     position: 'fixed', top: 0, left: 0, right: 0, bottom: 0,

```

```

308 |         background: 'rgba(5, 10, 20, 0.85)', backdropFilter: 'blur(8px)',
309 |         display: 'flex', alignItems: 'center', justifyContent: 'center', zIndex: 1000,
310 |         padding: '20px'
311 |     }}>
312 |     <div style={{
313 |         background: 'rgba(20, 28, 48, 0.95)', border: '1px solid rgba(255,255,255,0.08)',
314 |         borderRadius: '24px', width: '100%', maxWidth: '500px', padding: '32px',
315 |         boxShadow: '0 20px 50px rgba(0,0,0,0.5)', display: 'flex', flexDirection: 'column', gap: '20px'
316 |     }}>
317 |         <div>
318 |             <h2 style={{ margin: '0 0 4px 0', fontSize: '22px', fontWeight: 800, color: 'fff', display: 'fl...
319 |                 < M e g a p h o n e   s i z e = { 2 2 }   c o l o r = " # f 9 a 8 2 5 " } / >   G í i   t h ò n g   b á o
320 |             </h2>
321 |             <p   s t y l e = { {   m a r g i n :   0 ,   f o n t S i z e :   ' 1 3 p x ' ,   c o l o r :   ' # 8 8 9 9 a 6 '
322 |         </div>
323 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
324 |             <div>
325 |                 <label style={{ display: 'block', fontSize: '12px', fontWeight: 700, color: '#8899a6', textTra...
326 |                 <input
327 |                     type="text"
328 |                     value={notifTitle}
329 |                     onChange={e => setNotifTitle(e.target.value)}
330 |                     p l a c e h o l d e r = " V D :   L Ē c h   t h i   g í i a   k ó "
331 |                 style={{ width: '100%', background: 'rgba(5, 10, 20, 0.6)', border: '1px solid rgba(255,255,...
332 |             </div>
333 |             <div>
334 |                 <label style={{ display: 'block', fontSize: '12px', fontWeight: 700, color: '#8899a6', textTra...
335 |                 <textarea
336 |                     value={notifMessage}
337 |                     onChange={e => setNotifMessage(e.target.value)}
338 |                     rows={4}
339 |                     p l a c e h o l d e r = " N h - p   n ù i   d u n g   t h ò n g   b á o   . . . "
340 |                 style={{ width: '100%', background: 'rgba(5, 10, 20, 0.6)', border: '1px solid rgba(255,255,...
341 |             </div>
342 |             <div style={{ display: 'flex', justifyContent: 'flex-end', gap: '12px', marginTop: '12px' }}>
343 |                 <button
344 |                     onClick={() => { setShowNotifModal(false); setNotifTitle(''); setNotifMessage(''); }}
345 |                     style={{ background: 'transparent', color: '#8899a6', border: '1px solid rgba(255,255,255,0.1)...
346 |                 >
347 |                     H  ç y
348 |                 </button>
349 |                 <button
350 |                     onClick={async () => {
351 |                         if (!notifTitle.trim() || !notifMessage.trim()) return
352 |                         setSendingNotif(true)
353 |                         try {
354 |                             const res = await fetch('/api/notifications/send-class', {
355 |                                 method: 'POST',
356 |                                 headers: { 'Content-Type': 'application/json' },
357 |                                 body: JSON.stringify({ class_id: classId, title: notifTitle, message: notifMessage, type...
358 |                             })
359 |                             const data = await res.json()
360 |                             if (data.success) {
361 |                                 a l e r t ( `   ã   g í i   t h ò n g   b á o   t h à n h   c ò n g   ò n   $ { d a t a . c o u n
362 |                                 setShowNotifModal(false)
363 |                                 setNotifTitle('')
364 |                                 setNotifMessage('')
365 |                             } else {
366 |                                 a l e r t ( d a t a . e r r o r   | |   ' C ó   l x i   x é y   r a ' )
367 |                             }
368 |                         } catch (err) {
369 |                             a l e r t ( ' L x i   k ò t   n ã i   ò n   m á y   c h ç ' )
370 |                         } finally {
371 |                             setSendingNotif(false)
372 |                         }
373 |                     }
374 |                     disabled={sendingNotif || !notifTitle.trim() || !notifMessage.trim()}
375 |                     style={{ background: 'var(--brand-primary)', color: '#000', border: 'none', padding: '12px 24px...
376 |                 >
377 |                     { sendingNotif ? <Loader2 size={18} className="animate-spin" /> : <Send size={18} /> }
378 |                     G í i   t h ò n g   b á o
379 |                 </button>
380 |             </div>
381 |         </div>
382 |     </div>
383 | }
384 |
385 |
386 |
387 |
388 |
389 |
390 | /* Class Settings Modal */
391 | {showSettingsModal && (
392 |     <div style={{
393 |         position: 'fixed', top: 0, left: 0, right: 0, bottom: 0,
394 |         background: 'rgba(5, 10, 20, 0.85)', backdropFilter: 'blur(8px)',
395 |         display: 'flex', alignItems: 'center', justifyContent: 'center', zIndex: 1000,
396 |         padding: '20px'
397 |     }}>
398 |         <div style={{
399 |             background: 'rgba(20, 28, 48, 0.95)', border: '1px solid rgba(255,255,255,0.08)',
400 |             borderRadius: '24px', width: '100%', maxWidth: '500px', padding: '32px',
401 |             boxShadow: '0 20px 50px rgba(0,0,0,0.5)', display: 'flex', flexDirection: 'column', gap: '20px'
402 |         }}>
403 |             <div>
404 |                 <h2   s t y l e = { {   m a r g i n :   ' 0 0 4 p x 0 ' ,   f o n t S i z e :   ' 2 2 p x ' ,   f o n t w
405 |                 <p   s t y l e = { {   m a r g i n :   0 ,   f o n t S i z e :   ' 1 3 p x ' ,   c o l o r :   ' # 8 8 9 9 a 6 '
406 |             </div>
407 |             <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
408 |                 <div>
409 |                     <label style={{ display: 'block', fontSize: '12px', fontWeight: 700, color: '#8899a6', textTra...
410 |                     <input
411 |                         type="text"
412 |                         value={editName}
413 |                         onChange={e => setEditName(e.target.value)}
414 |                         style={{ width: '100%', background: 'rgba(5, 10, 20, 0.6)', border: '1px solid rgba(255,255,...
415 |                     </div>
416 |                     <div>
417 |                         <label style={{ display: 'block', fontSize: '12px', fontWeight: 700, color: '#8899a6', textTra...
418 |                         <input
419 |                             type="text"
420 |                             value={editSubject}
421 |                             onChange={e => setEditSubject(e.target.value)}
422 |                             style={{ width: '100%', background: 'rgba(5, 10, 20, 0.6)', border: '1px solid rgba(255,255,...
423 |                         </div>
424 |                     <div>
425 |                         <label style={{ display: 'block', fontSize: '12px', fontWeight: 700, color: '#8899a6', textTra...
426 |                         <input
427 |
428 |
429 |
430 |
431 |

```

```

432|         type="number"
433|         value={editMaxStudents}
434|         onChange={e => setEditMaxStudents(parseInt(e.target.value) || 40)}
435|         style={{ width: '100%', background: 'rgba(5, 10, 20, 0.6)', border: '1px solid rgba(255,255,...
436|       />
437|     </div>
438|   </div>
439|
440|   <div style={{ display: 'flex', justifyContent: 'space-between', gap: '12px', marginTop: '12px' }}>
441|     <button
442|       onClick={handleDeleteClass}
443|       disabled={updating}
444|       style={{ background: 'rgba(239, 68, 68, 0.1)', color: '#f87171', border: 'none', padding: '12p...
445|     >
446|       X ó a l ù p
447|     </button>
448|
449|     <div style={{ display: 'flex', gap: '12px' }}>
450|       <button
451|         onClick={() => setShowSettingsModal(false)}
452|         disabled={updating}
453|         style={{ background: 'transparent', color: '#8899a6', border: '1px solid rgba(255,255,255,0...
454|       >
455|       H ç y
456|     </button>
457|     <button
458|       onClick={handleUpdateClass}
459|       disabled={updating}
460|       style={{ background: 'var(--brand-primary)', color: '#000', border: 'none', padding: '12px 2...
461|     >
462|       L ° u t h a y ò i
463|     </button>
464|   </div>
465| </div>
466| </div>
467| </div>
468|   )}
469| </div>
470| );
471| }
472|

```

File: app/(dashboard)/teacher/classes/new/page.tsx (144 lines, 5.9 KB)

```

1| 'use client';
2|
3| import React, { useState } from 'react';
4| import { supabase } from '@lib/supabase';
5| import { useRouter } from 'next/navigation';
6| import { ArrowLeft, Loader2, Save, GraduationCap, Calendar, Users, Type } from 'lucide-react';
7| import Link from 'next/link';
8|
9|
10| export default function NewClassPage() {
11|   const [loading, setLoading] = useState(false);
12|   const [error, setError] = useState<string | null>(null);
13|   const router = useRouter();
14|
15|   async function handleSubmit(e: React.FormEvent<HTMLFormElement>) {
16|     e.preventDefault();
17|     setLoading(true);
18|     setError(null);
19|
20|     const formData = new FormData(e.currentTarget);
21|     const name = formData.get('name') as string;
22|     const grade = formData.get('grade') as string;
23|     const schoolYear = formData.get('school_year') as string;
24|     const description = formData.get('description') as string;
25|     const maxStudents = parseInt(formData.get('max_students') as string) || 40;
26|
27|     try {
28|       const { data: { user } } = await supabase.auth.getUser();
29|       if (!user) throw new Error('Vui lòng đăng nhập để tạo lớp học');
30|
31|       // Tạo mã lớp học 6 chữ số (dành cho học sinh)
32|       const code = Math.floor(100000 + Math.random() * 900000).toString();
33|
34|       const { data, error: insertError } = await supabase
35|         .from('classes')
36|         .insert({
37|           name,
38|           grade,
39|           school_year: schoolYear,
40|           description,
41|           max_students: maxStudents,
42|           teacher_id: user.id,
43|           code
44|         })
45|         .select()
46|         .single();
47|
48|       if (insertError) throw insertError;
49|
50|       router.push(`/teacher/classes/${data.id}`);
51|     } catch (err: any) {
52|       setError(err.message);
53|       setLoading(false);
54|     }
55|   }
56|
57|   const inputStyle = {
58|     width: '100%', padding: '14px 16px 14px 48px', borderRadius: '12px',
59|     border: '1px solid rgba(255,255,255,0.1)', background: 'rgba(5, 10, 20, 0.5)',
60|     color: 'white', fontSize: '16px', outline: 'none'
61|   };
62|
63|   const labelStyle = {
64|     display: 'block', fontSize: '14px', fontWeight: 600, color: '#8899a6', marginBottom: '8px'
65|   };
66|
67|   const iconStyle = {
68|     position: 'absolute' as const, left: '16px', top: '14px', color: '#4b5563'
69|   };
70|
71|   return (
72|     <div style={{ padding: '32px', maxWidth: '800px', margin: '0 auto' }}>
73|       <Link href="/teacher/classes" style={{ display: 'inline-flex', alignItems: 'center', gap: '8px', color: ...
74|         <ArrowLeft size={20} /> Quay lại danh sách
75|       </Link>
76|
77|       <div style={{ background: 'rgba(12, 20, 36, 0.8)', padding: '48px', borderRadius: '24px', border: '1px s...
78|         <h1 style={{ fontSize: '28px', fontWeight: 800, margin: '0 0 40px 0'
79|       </div>

```

```

80| <form onSubmit={handleSubmit} style={{ display: 'flex', flexDirection: 'column', gap: '24px' }}>
81| <div>
82| | <label style={labelStyle}>Tên lớp học</label>
83| | <div style={{ position: 'relative' }}>
84| | <Type style={iconStyle} size={20} />
85| | <input name="name" required placeholder="Lớp 10A1 - Tin học" />
86| | </div>
87| | </div>
88| |
89| | <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '24px' }}>
90| | <div>
91| | | <label style={labelStyle}>Khởi lập</label>
92| | | <div style={{ position: 'relative' }}>
93| | | <GraduationCap style={iconStyle} size={20} />
94| | | <select name="grade" required style={inputStyle}>
95| | | | <option value="10">Lớp 10</option>
96| | | | <option value="11">Lớp 11</option>
97| | | | <option value="12">Lớp 12</option>
98| | | </select>
99| | | </div>
100| | | </div>
101| | | </div>
102| | | <div>
103| | | <label style={labelStyle}>Năm học</label>
104| | | <div style={{ position: 'relative' }}>
105| | | <Calendar style={iconStyle} size={20} />
106| | | <input name="school_year" required placeholder="2025-2026" style={inputStyle} defaultValue="20...
107| | | </div>
108| | | </div>
109| | | </div>
110| | |
111| | | <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '24px' }}>
112| | | <div>
113| | | | <label style={labelStyle}>Số sinh viên</label>
114| | | | <div style={{ position: 'relative' }}>
115| | | | <Users style={iconStyle} size={20} />
116| | | | <input name="max_students" type="number" required defaultValue="40" style={inputStyle} />
117| | | | </div>
118| | | | </div>
119| | | | </div>
120| | | |
121| | | | <div>
122| | | | | <label style={labelStyle}>Mô tả lớp học (không bắt buộc)</label>
123| | | | | <textarea name="description" rows={4} style={{ ...inputStyle, paddingLeft: '16px', resize: 'none' ...
124| | | | | </div>
125| | | | |
126| | | | | {error && (
127| | | | | <div style={{ padding: '16px', borderRadius: '12px', background: 'rgba(239, 68, 68, 0.1)', color: ...
128| | | | | {error}
129| | | | | </div>
130| | | | | )}
131| | | | |
132| | | | | <button type="submit" disabled={loading} style={{
133| | | | | width: '100%', padding: '16px', borderRadius: '12px', background: 'var(--brand-primary)',
134| | | | | color: '#000', fontSize: '16px', fontWeight: 700, border: 'none', cursor: loading ? 'not-allowed' ...
135| | | | | display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '10px', marginTop: '16px'
136| | | | | }}>
137| | | | | {loading ? <Loader2 className="animate-spin" size={20} /> :
138| | | | | </button>
139| | | | </form>
140| | | </div>
141| | </div>
142| | );
143| }
144|

```

File: app/(dashboard)/teacher/classes/page.tsx (95 lines, 3.7 KB)

```

1| 'use client';
2|
3| import React, { useEffect, useState } from 'react';
4| import { supabase } from '@lib/supabase';
5| import ClassCard from '@components/classes/ClassCard';
6| import { Plus, Loader2, BookOpen } from 'lucide-react';
7| import Link from 'next/link';
8|
9| export default function TeacherClassesPage() {
10|   const [classes, setClasses] = useState<any[]>([]);
11|   const [loading, setLoading] = useState(true);
12|
13|   useEffect(() => { fetchClasses(); }, []);
14|
15|   async function fetchClasses() {
16|     setLoading(true);
17|     try {
18|       const { data: { user } } = await supabase.auth.getUser();
19|       if (!user) return;
20|
21|       const { data, error } = await supabase
22|         .from('classes')
23|         .select('*', class_members(count), assignments(count))
24|         .eq('teacher_id', user.id)
25|         .order('created_at', { ascending: false });
26|
27|       if (error) throw error;
28|       setClasses(data || []);
29|     } catch (err) {
30|       console.error('Error fetching classes:', err);
31|     } finally {
32|       setLoading(false);
33|     }
34|   }
35|
36|   if (loading) {
37|     return (
38|       <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', height: '100%' }}>
39|         <Loader2 style={{ animation: 'spin 1s linear infinite' }} color="var(--brand-primary)" size={48} />
40|       </div>
41|     );
42|   }
43|
44|   return (
45|     <div style={{ padding: '32px' }}>
46|       <header style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom: '...'
47|       <div>
48|         <h1 style={{ fontSize: '32px', fontWeight: 800, margin: '0 0 8px 0', color: 'var(--text-primary)' }}...
49|         <p style={{ color: 'var(--text-muted)', margin: 0 }}>Tiêu đề</p>
50|       </div>
51|       <Link href="/teacher/classes/new" style={{ textDecoration: 'none' }}>
52|         <button className="lms-btn lms-btn-primary" style={{ display: 'flex', alignItems: 'center', gap: '8px...
53|         <Plus size={20} />
54|       </button>
55|     </Link>

```

```

56 |     </header>
57 |
58 |     {classes.length === 0 ? (
59 |       <div style={{
60 |         textAlign: 'center', padding: '100px 0',
61 |         background: 'var(--bg-surface)', borderRadius: '24px',
62 |         border: '2px dashed var(--border-default)'
63 |       }}>
64 |         <BookOpen size={64} color="var(--text-muted)" style={{ marginBottom: '24px' }} />
65 |         <h3 style={{ color: 'var(--text-muted)', margin: '0 0 24px 0' }}>
66 |           <Link href="/teacher/classes/new" style={{ textDecoration: 'none' }}>
67 |             <button style={{
68 |               background: 'rgba(var(--brand-primary-rgb),0.1)', color: 'var(--brand-primary)', border: '1px so...
69 |               padding: '12px 32px', borderRadius: '12px', fontWeight: 700, cursor: 'pointer', fontFamily: 'inh...
70 |             }}>
71 |               T i o l ū p n g a y
72 |             </button>
73 |           </Link>
74 |         </div>
75 |       ) : (
76 |         <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill, minmax(350px, 1fr))', gap: '24p...
77 |         {classes.map(cls => (
78 |           <ClassCard
79 |             key={cls.id}
80 |             id={cls.id}
81 |             name={cls.name}
82 |             code={cls.code}
83 |             memberCount={cls.class_members?.[0]?.count || 0}
84 |             maxStudents={cls.max_students}
85 |             activeAssignmentsCount={cls.assignments?.[0]?.count || 0}
86 |             role="teacher"
87 |             onDelete={fetchClasses}
88 |           />
89 |         )}}
90 |       </div>
91 |     )}
92 |   </div>
93 | );
94 | }
95 |

```

File: app/(dashboard)/teacher/dashboard/error.tsx (17 lines, 0.6 KB)

```

1 | 'use client'
2 |
3 | import React from 'react'
4 |
5 | export default function Error({ error, reset }: { error: Error; reset: () => void }) {
6 |   return (
7 |     <div className="flex flex-col items-center justify-center h-full gap-4 p-8 text-center">
8 |       <div className="text-3xl">&#x2013;</div>
9 |       <p className="text-[#d8dbe8] font-medium">{error.message} || 'C ó l o'
10 |       <button onClick={reset}
11 |         className="px-4 py-2 bg-[#00d4aa]/12 border border-[#00d4aa]/30 text-[#00d4aa] rounded-lg text-sm">
12 |         T h i l i i
13 |       </button>
14 |     </div>
15 |   )
16 | }
17 |

```

File: app/(dashboard)/teacher/dashboard/loading.tsx (14 lines, 0.4 KB)

```

1 | import React from 'react'
2 |
3 | export default function Loading() {
4 |   return (
5 |     <div className="p-6 space-y-4 animate-pulse">
6 |       <div className="h-8 bg-[#1d1f2a] rounded-lg w-44" />
7 |       <div className="grid grid-cols-2 md:grid-cols-4 gap-4">
8 |         {[...Array(4)].map((_, i) => <div key={i} className="h-24 bg-[#1d1f2a] rounded-xl" />)}
9 |       </div>
10 |       <div className="h-60 bg-[#1d1f2a] rounded-xl" />
11 |     </div>
12 |   )
13 | }
14 |

```

File: app/(dashboard)/teacher/error.tsx (26 lines, 0.9 KB)

```

1 | // Path: app/(dashboard)/teacher/error.tsx
2 | 'use client'
3 | import { useEffect } from 'react'
4 |
5 | export default function Error({
6 |   error, reset
7 | }: { error: Error & { digest?: string }; reset: () => void }) {
8 |   useEffect(() => { console.error(error) }, [error])
9 |   return (
10 |     <div className="flex flex-col items-center justify-center h-full gap-6 p-8">
11 |       <div className="text-4xl">&#x2013;</div>
12 |       <div className="text-center">
13 |         <h2 className="text-lg font-semibold text-[#d8dbe8] mb-2">C ó l o'
14 |         <p className="text-sm text-[#5a5d72] mb-4">{error.message} || 'V u'
15 |       </div>
16 |       <button
17 |         onClick={reset}
18 |         className="px-4 py-2 bg-[#00d4aa]/15 border border-[#00d4aa]/30
19 |         text-[#00d4aa] rounded-lg text-sm font-medium hover:bg-[#00d4aa]/25 transition-colors"
20 |       >
21 |         T h i l i i
22 |       </button>
23 |     </div>
24 |   )
25 | }
26 |

```

File: app/(dashboard)/teacher/exams/[id]/page.tsx (207 lines, 12.3 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect, useCallbck } from 'react'
4 | import { supabase } from '@lib/supabase'
5 | import { use } from 'react'
6 | import Link from 'next/link'
7 | import { ArrowLeft, Users, Clock, BarChart3, CheckCircle, XCircle, Loader2, Eye, EyeOff, Trash2, RefreshCw, Us...
8 | import { getTeacherExam, publishExam, deleteExam } from '@app/actions/exam'
9 | import { useRouter } from 'next/navigation'

```

```

10 export default function ExamDetailPage({ params }: { params: Promise<{ id: string }> }) {
11   const { id } = use(params)
12   const router = useRouter()
13   const [exam, setExam] = useState<any>(null)
14   const [loading, setLoading] = useState(true)
15   const [publishing, setPublishing] = useState(false)
16
17   const loadExam = useCallback(async () => {
18     try {
19       const data = await getTeacherExam(id)
20       setExam(data)
21     } catch (e: any) { console.error(e) }
22     finally { setLoading(false) }
23   }, [id])
24
25   useEffect(() => {
26     loadExam()
27     const channel = supabase
28       .channel(`exam-${id}`)
29       .on('postgres_changes',
30         { event: '*', schema: 'public', table: 'quiz_attempts', filter: `quiz_id=eq.${id}` },
31         () => loadExam()
32       )
33       .on('postgres_changes',
34         { event: '*', schema: 'public', table: 'quiz_answers' },
35         () => loadExam()
36       )
37     .subscribe()
38     return () => { supabase.removeChannel(channel) }
39   }, [id, loadExam])
40
41   async function handlePublish() {
42     setPublishing(true)
43     try {
44       await publishExam(id, !exam.is_published)
45       loadExam()
46     } catch (e: any) { alert(e.message) }
47     finally { setPublishing(false) }
48   }
49
50   async function handleDelete() {
51     if (!confirm('Xóa Á thi này?')) return
52     await deleteExam(id)
53     router.push('/teacher/exams')
54   }
55
56   if (loading) return <div style={{ display: 'flex', justifyContent: 'center', padding: '80px' }}><Loader2 siz...
57   if (!exam) return <div style={{ textAlign: 'center', padding: '80px', color: 'var(--text-muted)' }}>Không tì...
58
59   const stats = {
60     totalStudents: exam.attempts?.length || 0,
61     completed: exam.attempts?.filter((a: any) => a.status !== 'in_progress').length || 0,
62     inProgress: exam.attempts?.filter((a: any) => a.status === 'in_progress').length || 0,
63     passed: exam.attempts?.filter((a: any) => a.status !== 'in_progress' && (a.score || 0) >= (exam.passing_sc...
64     avgScore: (() => { const done = exam.attempts?.filter((a: any) => a.status !== 'in_progress'); return done...
65     totalQuestions: exam.questions?.length || 0
66   }
67
68   return (
69     <div style={{ maxWidth: '1200px', margin: '0 auto', color: 'var(--text-primary)' }}>
70       <div style={{ display: 'flex', alignItems: 'center', gap: '16px', marginBottom: '24px' }}>
71         <Link href="/teacher/exams" style={{ display: 'flex', padding: '8px', borderRadius: '10px', border: '1...
72         <ArrowLeft size={18} />
73       </div>
74       <div style={{ flex: 1 }}>
75         <h1 style={{ fontSize: '24px', fontWeight: 800, margin: '0 0 4px 0' }}>{exam.title}</h1>
76         <p style={{ color: 'var(--text-muted)', margin: 0, fontSize: '13px' }}>
77           { exam.class?.name || 'Ch°a chín lúp' } · { exam.time_limit_m
78       </div>
79       <div style={{ display: 'flex', gap: '8px' }}>
80         <div style={{ display: 'flex', gap: '8px' }}>
81           <div style={{ display: 'flex', gap: '8px' }}>
82             <div style={{ display: 'flex', gap: '8px' }}>
83               <div style={{ display: 'flex', gap: '8px' }}>
84                 <div style={{ display: 'flex', gap: '8px' }}>
85                   <div style={{ display: 'flex', gap: '8px' }}>
86                     <div style={{ display: 'flex', gap: '8px' }}>
87                       <div style={{ display: 'flex', gap: '8px' }}>
88                         <div style={{ display: 'flex', gap: '8px' }}>
89                         <div style={{ display: 'flex', gap: '8px' }}>
90                         <div style={{ display: 'flex', gap: '8px' }}>
91                         <div style={{ display: 'flex', gap: '8px' }}>
92                         <div style={{ display: 'flex', gap: '8px' }}>
93                         <div style={{ display: 'flex', gap: '8px' }}>
94                         <div style={{ display: 'flex', gap: '8px' }}>
95                         <div style={{ display: 'flex', gap: '8px' }}>
96                         <div style={{ display: 'flex', gap: '8px' }}>
97                         <div style={{ display: 'flex', gap: '8px' }}>
98                         <div style={{ display: 'flex', gap: '8px' }}>
99                         <div style={{ display: 'flex', gap: '8px' }}>
100                         <div style={{ display: 'flex', gap: '8px' }}>
101                         <div style={{ display: 'flex', gap: '8px' }}>
102                         <div style={{ display: 'flex', gap: '8px' }}>
103                         <div style={{ display: 'flex', gap: '8px' }}>
104                         <div style={{ display: 'flex', gap: '8px' }}>
105                         <div style={{ display: 'flex', gap: '8px' }}>
106                         <div style={{ display: 'flex', gap: '8px' }}>
107                         <div style={{ display: 'flex', gap: '8px' }}>
108                         <div style={{ display: 'flex', gap: '8px' }}>
109                         <div style={{ display: 'flex', gap: '8px' }}>
110                         <div style={{ display: 'flex', gap: '8px' }}>
111                         <div style={{ display: 'flex', gap: '8px' }}>
112                         <div style={{ display: 'flex', gap: '8px' }}>
113                         <div style={{ display: 'flex', gap: '8px' }}>
114                         <div style={{ display: 'flex', gap: '8px' }}>
115                         <div style={{ display: 'flex', gap: '8px' }}>
116                         <div style={{ display: 'flex', gap: '8px' }}>
117                         <div style={{ display: 'flex', gap: '8px' }}>
118                         <div style={{ display: 'flex', gap: '8px' }}>
119                         <div style={{ display: 'flex', gap: '8px' }}>
120                         <div style={{ display: 'flex', gap: '8px' }}>
121                         <div style={{ display: 'flex', gap: '8px' }}>
122                         <div style={{ display: 'flex', gap: '8px' }}>
123                         <div style={{ display: 'flex', gap: '8px' }}>
124                         <div style={{ display: 'flex', gap: '8px' }}>
125                         <div style={{ display: 'flex', gap: '8px' }}>
126                         <div style={{ display: 'flex', gap: '8px' }}>
127                         <div style={{ display: 'flex', gap: '8px' }}>
128                         <div style={{ display: 'flex', gap: '8px' }}>
129                         <div style={{ display: 'flex', gap: '8px' }}>
130                         <div style={{ display: 'flex', gap: '8px' }}>
131                         <div style={{ display: 'flex', gap: '8px' }}>
132                         <div style={{ display: 'flex', gap: '8px' }}>
133                         <div style={{ display: 'flex', gap: '8px' }}>

```

```

134 |         </div>
135 |         <div>
136 |             <div style={{ fontSize: '13px', fontWeight: 600 }}>{attempt.profiles?.full_name || '...
137 |             <div style={{ fontSize: '11px', color: 'var(--text-muted)', display: 'flex', alignIt...
138 |                 <Mail size={11} /> {attempt.profiles?.email || ''}
139 |             </div>
140 |         </div>
141 |     </div>
142 |     <div style={{ textAlign: 'right' }}>
143 |         {attempt.status === 'in_progress' ? (
144 |             <span style={{ fontSize: '11px', padding: '2px 8px', borderRadius: '4px', background...
145 |             ) : attempt.status === 'submitted' || attempt.status === 'graded' ? (
146 |                 <div>
147 |                     <div style={{ fontSize: '20px', fontWeight: 800, color: (attempt.score || 0) >= (e...
148 |                     {attempt.score?.toFixed(1)}%
149 |                 </div>
150 |                 <div style={{ fontSize: '11px', color: 'var(--text-muted)' }}>{correctCount}/{tota...
151 |                 </div>
152 |             ) : (
153 |                 <span style={{ fontSize: '11px', padding: '2px 8px', borderRadius: '4px', background...
154 |                 </div>
155 |             </div>
156 |         </div>
157 |         {attempt.status !== 'in_progress' && totalCount > 0 && (
158 |             <div style={{ width: '100%', height: '4px', borderRadius: '2px', background: 'var(--bg-b...
159 |             <div style={{ height: '100%', borderRadius: '2px', width: `${(correctCount / totalCoun...
160 |             </div>
161 |         )}
162 |     </div>
163 | )
164 | }}
165 | </div>
166 | }}
167 | </div>
168 |
169 | <div className="lms-card" style={{ padding: '24px', background: 'var(--bg-surface)', border: '1px soli...
170 | <h2 style={{ fontSize: '16px', fontWeight: 700, marginBottom: '16px', display: 'flex', alignItems: '...
171 | <BarChart3 size={48} /> <BarChart3 size={48} />
172 | </h2>
173 | {exam.questions?.length === 0 ? (
174 |     <div style={{ textAlign: 'center', padding: '40px', color: 'var(--text-muted)' }}>
175 |         <BarChart3 size={48} />
176 |     </div>
177 | ) : (
178 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '12px' }}>
179 |         {exam.questions?.map((q: any, i: number) => {
180 |             const answers = exam.attempts?.flatMap((a: any) => a.quiz_answers || []).filter((a: any) => a....
181 |             const correct = answers.filter((a: any) => a.is_correct).length
182 |             const total = answers.length
183 |             const pct = total > 0 ? Math.round((correct / total) * 100) : 0
184 |             return (
185 |                 <div key={q.id} style={{ padding: '12px', borderRadius: '10px', background: 'var(--bg-elevat...
186 |                 <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marg...
187 |                 <div style={{ fontSize: '13px', fontWeight: 600, flex: 1 }}><span style={{ color: 'var(--...
188 |                 <div style={{ fontSize: '12px', fontWeight: 700, color: pct >= 70 ? '#10b981' : pct >= 4...
189 |                 {correct}/{total} ({pct}%)
190 |                 </div>
191 |                 </div>
192 |                 <div style={{ width: '100%', height: '4px', borderRadius: '2px', background: 'var(--bg-bas...
193 |                 <div style={{ height: '100%', borderRadius: '2px', width: `${pct}%`, background: pct >= ...
194 |                 </div>
195 |                 </div>
196 |             </div>
197 |         )}
198 |     </div>
199 | )}
200 | </div>
201 | </div>
202 | </div>
203 | <style>{`@keyframes spin { to { transform: rotate(360deg) } }`}</style>
204 | </div>
205 | )
206 | }
207 |

```

File: app/(dashboard)/teacher/exams/new/page.tsx (333 lines, 20.5 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect, useCallback } from 'react'
4 | import { supabase } from '@lib/supabase'
5 | import { useRouter } from 'next/navigation'
6 | import { ArrowLeft, Plus, Trash2, Save, Loader2, Upload, FileText } from 'lucide-react'
7 | import { createExam, addQuestionToExam } from '@app/actions/exam'
8 | import { parseDocx } from '@utils/word-parser'
9 |
10 | export default function NewExamPage() {
11 |     const router = useRouter()
12 |     const [classes, setClasses] = useState<any[]>([])
13 |     const [lessons, setLessons] = useState<any[]>([])
14 |     const [saving, setSaving] = useState(false)
15 |     const [uploadingWord, setUploadingWord] = useState(false)
16 |     const [wordFileName, setWordFileName] = useState('')
17 |     const [form, setForm] = useState({
18 |         title: '',
19 |         description: '',
20 |         classId: '',
21 |         lessonId: '',
22 |         timeLimitMinutes: 45,
23 |         passingScore: 70,
24 |         maxAttempts: 3,
25 |         requireCamera: false,
26 |         availableFrom: '',
27 |         availableUntil: ''
28 |     })
29 |     const [questions, setQuestions] = useState<any[]>([])
30 |     const [questionInput, setQuestionInput] = useState({
31 |         content: '',
32 |         type: 'single_choice' as string,
33 |         points: 10,
34 |         difficulty: 'medium',
35 |         explanation: '',
36 |         options: [{ content: '', isCorrect: false }, { content: '', isCorrect: false }]
37 |     })
38 |
39 |     useEffect(() => {
40 |         supabase.auth.getUser().then(({ data: { user } }) => {
41 |             if (!user) return
42 |             supabase.from('classes').select('*').eq('teacher_id', user.id).then(({ data }) => setClasses(data || []))
43 |             supabase.from('lessons').select('*').eq('teacher_id', user.id).then(({ data }) => setLessons(data || []))
44 |         })
45 |     }, [])
46 |

```

```

47 | function addOption() {
48 |   setQuestionInput(prev => ({ ...prev, options: [...prev.options, { content: '', isCorrect: false }] })))
49 | }
50 | function removeOption(i: number) {
51 |   setQuestionInput(prev => ({ ...prev, options: prev.options.filter((_, idx) => idx !== i) })))
52 | }
53 | function setOption(index: number, field: string, value: any) {
54 |   setQuestionInput(prev => {
55 |     const opts = [...prev.options]
56 |     opts[index] = { ...opts[index], [field]: value }
57 |     return { ...prev, options: opts }
58 |   })
59 | }
60 | function setCorrectOption(index: number) {
61 |   setQuestionInput(prev => ({
62 |     ...prev,
63 |     options: prev.options.map((opt, i) => ({ ...opt, isCorrect: i === index })))
64 |   }))
65 | }
66 |
67 | function addCurrentQuestion() {
68 |   const q = questionInput
69 |   if (!q.content.trim()) { alert('Nhập nội dung câu hỏi'); return }
70 |   const validOptions = q.options.filter(o => o.content.trim())
71 |   if (q.type !== 'fill_blank' && validOptions.length < 2) { alert('CS')
72 |   if (!validOptions.some(o => o.isCorrect)) { alert('Chín áp án úng')
73 |   setQuestions(prev => [...prev, { ...q, options: validOptions, id: `temp_${Date.now()}` }])
74 |   setQuestionInput({
75 |     content: '', type: 'single_choice', points: 10, difficulty: 'medium', explanation: '',
76 |     options: [{ content: '', isCorrect: false }, { content: '', isCorrect: false }]
77 |   })
78 | }
79 |
80 | function removeQuestion(id: string) {
81 |   setQuestions(prev => prev.filter(q => q.id !== id))
82 | }
83 |
84 | async function handleWordUpload(e: React.ChangeEvent<HTMLInputElement>) {
85 |   const file = e.target.files?.[0]
86 |   if (!file) return
87 |   if (!file.name.endsWith('.docx')) { alert('Chế h x trã file .docx');
88 |   setUploadingWord(true)
89 |   setWordFileName(file.name)
90 |   try {
91 |     const { questions: parsed } = await parseDocx(file)
92 |     if (parsed.length === 0) { alert('Không tìm thấy câu hỏi trong fi
93 |     setQuestions(prev => [...prev, ...parsed.map((q, i) => ({ ...q, id: `word_${Date.now()}_${i}` })))
94 |     alert('ã nh p ${parsed.length} câu hỏi tẽ file Word!')
95 |   } catch (err: any) {
96 |     alert('L x i íc file: ' + err.message)
97 |   } finally {
98 |     setUploadingWord(false)
99 |   }
100 | }
101 |
102 | async function handleSave() {
103 |   if (!form.title.trim()) { alert('Nhập tên Á thi'); return }
104 |   if (!form.classId) { alert('Chín l p h íc'); return }
105 |   if (questions.length === 0) { alert('Thêm ít nh t 1 câu hỏi'); return
106 |   setSaving(true)
107 |   try {
108 |     const exam = await createExam({
109 |       title: form.title,
110 |       description: form.description,
111 |       classId: form.classId,
112 |       lessonId: form.lessonId || undefined,
113 |       timeLimitMinutes: form.timeLimitMinutes,
114 |       passingScore: form.passingScore,
115 |       maxAttempts: form.maxAttempts,
116 |       requireCamera: form.requireCamera,
117 |       availableFrom: form.availableFrom || undefined,
118 |       availableUntil: form.availableUntil || undefined
119 |     })
120 |
121 |     for (let i = 0; i < questions.length; i++) {
122 |       const q = questions[i]
123 |       await addQuestionToExam({
124 |         examId: exam.id,
125 |         content: q.content,
126 |         type: q.type,
127 |         points: q.points,
128 |         difficulty: q.difficulty,
129 |         explanation: q.explanation,
130 |         options: q.options
131 |       })
132 |     }
133 |
134 |     router.push(`/teacher/exams/${exam.id}`)
135 |   } catch (e: any) {
136 |     alert('L x i: ' + e.message)
137 |   } finally {
138 |     setSaving(false)
139 |   }
140 | }
141 |
142 | return (
143 |   <div style={{ maxWidth: '1000px', margin: '0 auto', color: 'var(--text-primary)' }}>
144 |     <button onClick={() => router.back()}
145 |       style={{ display: 'flex', alignItems: 'center', gap: '6px', padding: '8px 16px', borderRadius: '10px',...
146 |     </button>
147 |
148 |     <h1 style={{ fontSize: '28px', fontWeight: 800, marginBottom: '32px' }}>
149 |
150 |     <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '32px' }}>
151 |       <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
152 |         <div className="lms-card" style={{ padding: '24px', background: 'var(--bg-surface)', border: '1px so...
153 |           <h2 style={{ fontSize: '16px', fontWeight: 700, marginBottom: '16px' }}>
154 |             <div style={{ display: 'flex', flexDirection: 'column', gap: '12px' }}>
155 |               <input value={form.title} onChange={e => setForm(p => ({ ...p, title: e.target.value })))
156 |               <input type="text" value={form.description} onChange={e => setForm(p => ({ ...p, description: e.target.va...
157 |               <input type="text" value={form.classId} onChange={e => setForm(p => ({ ...p, classId: e.target.value })))
158 |               <input type="text" value={form.lessonId} onChange={e => setForm(p => ({ ...p, lessonId: e.target.value })))
159 |               <input type="text" value={form.timeLimitMinutes} onChange={e => setForm(p => ({ ...p, timeLimitMinutes: e.target.va...
160 |               <input type="text" value={form.passingScore} onChange={e => setForm(p => ({ ...p, passingScore: e.target.va...
161 |               <input type="text" value={form.maxAttempts} onChange={e => setForm(p => ({ ...p, maxAttempts: e.target.va...
162 |               <input type="text" value={form.requireCamera} onChange={e => setForm(p => ({ ...p, requireCamera: e.target.va...
163 |               <input type="text" value={form.availableFrom} onChange={e => setForm(p => ({ ...p, availableFrom: e.target.va...
164 |               <input type="text" value={form.availableUntil} onChange={e => setForm(p => ({ ...p, availableUntil: e.target.va...
165 |               <input type="text" value={form.availableFrom} onChange={e => setForm(p => ({ ...p, availableFrom: e.target.va...
166 |               <input type="text" value={form.availableUntil} onChange={e => setForm(p => ({ ...p, availableUntil: e.target.va...
167 |               <input type="text" value={form.availableFrom} onChange={e => setForm(p => ({ ...p, availableFrom: e.target.va...
168 |               <input type="text" value={form.availableUntil} onChange={e => setForm(p => ({ ...p, availableUntil: e.target.va...
169 |               <input type="text" value={form.availableFrom} onChange={e => setForm(p => ({ ...p, availableFrom: e.target.va...
170 |               <input type="text" value={form.availableUntil} onChange={e => setForm(p => ({ ...p, availableUntil: e.target.va...

```



```
171 |         </select>
172 |
173 |         <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '12px' }}>
174 |             <div>
175 |                 <label style={{ fontSize: '11px', fontWeight: 600, color: 'var(--text-muted)', marginBottom:...
176 |                 <input type="number" value={form.timeLimitMinutes} onChange={e => setForm(p => ({ ...p, time...
177 |                 style={{ padding: '10px 14px', borderRadius: '10px', border: '1px solid var(--border-defau...
178 |             </div>
179 |             <div>
180 |                 <label style={{ fontSize: '11px', fontWeight: 600, color: 'var(--text-muted)', marginBottom:...
181 |                 <input type="number" value={form.passingScore} onChange={e => setForm(p => ({ ...p, passingS...
182 |                 style={{ padding: '10px 14px', borderRadius: '10px', border: '1px solid var(--border-defau...
183 |             </div>
184 |         </div>
185 |
186 |         <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '12px' }}>
187 |             <div>
188 |                 <label style={{ fontSize: '11px', fontWeight: 600, color: 'var(--text-muted)', marginBottom:...
189 |                 <input type="datetime-local" value={form.availableFrom} onChange={e => setForm(p => ({ ...p, ...
190 |                 style={{ padding: '10px 14px', borderRadius: '10px', border: '1px solid var(--border-defau...
191 |             </div>
192 |             <div>
193 |                 <label style={{ fontSize: '11px', fontWeight: 600, color: 'var(--text-muted)', marginBottom:...
194 |                 <input type="datetime-local" value={form.availableUntil} onChange={e => setForm(p => ({ ...p, ...
195 |                 style={{ padding: '10px 14px', borderRadius: '10px', border: '1px solid var(--border-defau...
196 |             </div>
197 |         </div>
198 |
199 |         <label style={{ display: 'flex', alignItems: 'center', gap: '10px', cursor: 'pointer', fontSize:...
200 |         <input type="checkbox" checked={form.requireCamera} onChange={e => setForm(p => ({ ...p, requi...
201 |         style={{ width: '18px', height: '18px' }} />
202 |         Y ê u c s ũ b - t c a m e r a k h i t h i ( g i á m s á t A I )
203 |     </label>
204 | </div>
205 | </div>
206 | </div>
207 |
208 | <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
209 |     <div className="lms-card" style={{ padding: '24px', background: 'var(--bg-surface)', border: '1px so...
210 |         < h 2 s t y l e = { { f o n t S i z e : ' 1 6 p x ' , f o n t W e i g h t : 7 0 0 , m a r g i n B o t t o m : 1 0 p x } } >
211 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '10px' }}>
212 |             <input value={questionInput.content} onChange={e => setQuestionInput(p => ({ ...p, content: e.ta...
213 |             p l a c e h o l d e r = " N ù i d ũ n g c â u h i i " s t y l e = { { p a d d i n g : ' 1 2 p x } } >
214 |
215 |             <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '8px' }}>
216 |                 <select value={questionInput.type} onChange={e => setQuestionInput(p => ({ ...p, type: e.targe...
217 |                 style={{ padding: '10px 14px', borderRadius: '10px', border: '1px solid var(--border-default)...
218 |                     < o p t i o n v a l u e = " s i n g l e _ c h o i c e " > T r c n g h i c m 1 á p á n < / o p t i o n >
219 |                     < o p t i o n v a l u e = " t r u e _ f a l s e " > ú n g / S a i < / o p t i o n >
220 |                     < o p t i o n v a l u e = " f i l l _ b l a n k " > i Á n k h u y c t < / o p t i o n >
221 |                     < o p t i o n v a l u e = " m u l t i p l e _ c h o i c e " > N h i Á u á p á n < / o p t i o n >
222 |                 </select>
223 |
224 |                 <div style={{ display: 'flex', gap: '8px' }}>
225 |                     <input type="number" value={questionInput.points} onChange={e => setQuestionInput(p => ({ ....
226 |                     p l a c e h o l d e r = " i Á m " s t y l e = { { p a d d i n g : ' 1 0 p x 1 4 p x ' , b o r d e r : ' 1 p x s o l i d v a r ( -- b o r d e r - d e f a u l t ) ' } } >
227 |                 </div>
228 |             </div>
229 |
230 |             {questionInput.type !== 'fill_blank' && (
231 |                 <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
232 |                     {questionInput.options.map((opt, i) => (
233 |                         <div key={i} style={{ display: 'flex', gap: '8px', alignItems: 'center' }}>
234 |                             <input type="radio" name="correct" checked={opt.isCorrect} onChange={() => setCorrectOpt...
235 |                             style={{ width: '16px', height: '16px', flexShrink: 0 }} />
236 |                             <input value={opt.content} onChange={e => setOption(i, 'content', e.target.value)}
237 |                             p l a c e h o l d e r = { ` á p á n $ { i + 1 } ` } s t y l e = { { f l e x : 1 } } >
238 |                             {questionInput.options.length > 2 && (
239 |                                 <button onClick={() => removeOption(i)} style={{ background: 'none', border: 'none', c...
240 |                                 <Trash2 size={14} />
241 |                             </button>
242 |                         </div>
243 |                     </div>
244 |                 )
245 |                 <button onClick={addOption} style={{ display: 'flex', alignItems: 'center', gap: '4px', back...
246 |                 < P l u s s i z e = { 1 4 } / > T h ê m á p á n
247 |             </div>
248 |         )
249 |     )
250 |
251 |     {questionInput.type === 'fill_blank' && (
252 |         <div style={{ padding: '10px 14px', borderRadius: '10px', background: 'rgba(16,185,129,0.08)',...
253 |         c â u i Á n k h u y c t : h í c s i n h g ô á p á n , s o k h ũ p c h í n h x á c
254 |     </div>
255 |     )
256 |
257 |     <textarea value={questionInput.explanation} onChange={e => setQuestionInput(p => ({ ...p, explan...
258 |     p l a c e h o l d e r = " G i ê i t h í c h ( h i c n s a u k h i h í c s i n h t r e l ý i ) "
259 |     style={{ padding: '10px 14px', borderRadius: '10px', border: '1px solid var(--border-default)'...
260 |
261 |     <div style={{ borderTop: '1px solid var(--border-default)', paddingTop: '12px', marginTop: '4px'...
262 |     <div style={{ display: 'flex', alignItems: 'center', gap: '10px', marginBottom: '8px' }}>
263 |         <Upload size={14} style={{ color: 'var(--brand-primary)' }} />
264 |         < s p a n s t y l e = { { f o n t S i z e : ' 1 2 p x ' , f o n t W e i g h t : 6 0 0 , c o l o r : ' v a r ( -- b r a n d - p r i m a r y ) ' } } >
265 |     </div>
266 |     <label style={{
267 |         display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '8px',
268 |         padding: '10px', borderRadius: '10px',
269 |         background: uploadingWord ? 'rgba(99,102,241,0.08)' : 'rgba(99,102,241,0.06)',
270 |         border: '1px dashed rgba(99,102,241,0.3)',
271 |         cursor: uploadingWord ? 'wait' : 'pointer', fontSize: '13px', fontWeight: 500,
272 |         color: 'var(--brand-primary)', transition: 'all 0.2s'
273 |     }}>
274 |         {uploadingWord ? (
275 |             < > < L o a d e r 2 s i z e = { 1 6 } s t y l e = { { a n i m a t i o n : ' s p i n 1 s l i'
276 |         ) : (
277 |             < > < U p l o a d s i z e = { 1 6 } / > C h í n f i l e . d o c x ( Ě n h d i n g n h
278 |         )
279 |         <input type="file" accept=".docx" onChange={handleWordUpload} style={{ display: 'none' }} di...
280 |     </label>
281 |     {wordFileName && (
282 |         <div style={{ marginTop: '6px', fontSize: '11px', color: 'var(--text-muted)', display: 'flex...
283 |         <FileText size={12} /> {wordFileName}
284 |     </div>
285 |     )
286 | </div>
287 |
288 |     <button onClick={addCurrentQuestion}
289 |     style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '6px', padding:...
290 |     < P l u s s i z e = { 1 6 } / > T h ê m c â u h i i v à o Á t h i
291 | </button>
292 | </div>
293 | </div>
294 | </div>
```

```

295 |         </div>
296 |
297 |         {questions.length > 0 && (
298 |           <div className="lms-card" style={{ padding: '24px', marginTop: '24px', background: 'var(--bg-surface)'...
299 |             <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
300 |               <div style={{ display: 'flex', alignItems: 'center', gap: '12px', padding: '12px 16px...
301 |                 {questions.map((q, i) => (
302 |                   <div key={q.id} style={{ display: 'flex', alignItems: 'center', gap: '12px', padding: '12px 16px...
303 |                     <span style={{ width: '28px', height: '28px', borderRadius: '8px', background: 'var(--brand-pr...
304 |                       {i + 1}
305 |                     </span>
306 |                     <span style={{ flex: 1, fontSize: '14px', color: 'var(--text-primary)' }}>{q.content}</span>
307 |                     <span style={{ flex: 1, fontSize: '11px', color: 'var(--text-muted)' }}>
308 |                       <button style={{ fontSize: '11px', padding: '2px 8px', borderRadius: '4px', background: 'rgba(99,99,99,0.1)' }}>
309 |                         <button onClick={() => removeQuestion(q.id)} style={{ background: 'none', border: 'none', color: 'var(--text-primary)' }}>
310 |                           <Trash2 size={14} />
311 |                         </button>
312 |                       </div>
313 |                     </span>
314 |                   </div>
315 |                 </div>
316 |               </div>
317 |             <div style={{ display: 'flex', justifyContent: 'flex-end', gap: '12px', marginTop: '32px', padding: '12px 16px' }}>
318 |               <button onClick={() => router.back()} style={{ padding: '12px 24px', borderRadius: '12px', border: '1px solid var(--border-default)', background: 'var(--bg-surface)' }}>
319 |                 Hủy
320 |               </button>
321 |               <button onClick={handleSave} disabled={saving} style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 32px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
322 |                 <Loader2 size={18} style={{ animation: 'spin 1s linear infinite' }} />
323 |                 <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 32px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
324 |                   <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 32px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
325 |                     <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 32px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
326 |                       <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 32px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
327 |                         <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 32px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
328 |                           <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 32px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
329 |                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 32px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
330 |                               <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 32px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
331 |                                 <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 32px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
332 |                                   <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 32px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
333 |                                     <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 32px', borderRadius: '12px', background: 'var(--bg-surface)' }}>

```

File: app/(dashboard)/teacher/exams/page.tsx (115 lines, 6.8 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect, useCallback } from 'react'
4 | import { supabase } from '@lib/supabase'
5 | import Link from 'next/link'
6 | import { useRouter } from 'next/navigation'
7 | import { Plus, FileText, Users, Clock, Eye, EyeOff, Trash2, Loader2, BarChart3, CheckCircle, XCircle, Edit3 } ...
8 | import { getTeacherExams, publishExam, deleteExam } from '@app/actions/exam'
9 |
10 | export default function TeacherExamsPage() {
11 |   const router = useRouter()
12 |   const [exams, setExams] = useState<any[]>([])
13 |   const [loading, setLoading] = useState(true)
14 |   const [deleting, setDeleting] = useState<string | null>(null)
15 |
16 |   const loadExams = useCallback(async () => {
17 |     setLoading(true)
18 |     const data = await getTeacherExams()
19 |     setExams(data)
20 |     setLoading(false)
21 |   }, [])
22 |
23 |   useEffect(() => { loadExams() }, [loadExams])
24 |
25 |   async function handlePublish(id: string, publish: boolean) {
26 |     try {
27 |       await publishExam(id, publish)
28 |       loadExams()
29 |     } catch (e: any) { alert(e.message) }
30 |   }
31 |
32 |   async function handleDelete(id: string) {
33 |     if (!confirm('Xóa bài thi này? Hủy sinh sẽ không thể làm bài.')) return
34 |     setDeleting(id)
35 |     try {
36 |       await deleteExam(id)
37 |       loadExams()
38 |     } catch (e: any) { alert(e.message) }
39 |     finally { setDeleting(null) }
40 |   }
41 |
42 |   return (
43 |     <div style={{ maxWidth: '1200px', margin: '0 auto', color: 'var(--text-primary)' }}>
44 |       <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', margin: '0 auto 32px auto' }}>
45 |         <div>
46 |           <h1 style={{ fontSize: '28px', fontWeight: '800', margin: '0 0 0 40px' }}>
47 |             <p style={{ color: 'var(--text-muted)', margin: '0' }}>Tiêu đề bài thi
48 |           </div>
49 |           <Link href="/teacher/exams/new" style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
50 |             <Plus size={18} />
51 |             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
52 |               <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
53 |                 <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
54 |                   <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
55 |                     <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
56 |                       <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
57 |                         <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
58 |                           <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
59 |                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
60 |                               <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
61 |                                 <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
62 |                                   <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
63 |                                     <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
64 |                                       <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
65 |                                         <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
66 |                                           <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
67 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
68 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
69 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
70 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
71 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
72 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
73 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
74 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
75 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
76 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
77 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
78 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
79 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
80 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>
81 |                                             <span style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', borderRadius: '12px', background: 'var(--bg-surface)' }}>

```

```

82|         <span style={{ display: 'flex', alignItems: 'center', gap: '4px' }}><Clock size={14} /> {exam...
83|         {exam._stats && (
84|             <
85|                 <span style={{ display: 'flex', alignItems: 'center', gap: '4px' }}><BarChart3 size={14}...
86|         8 6 |                 <span style={{ display: 'flex', alignIte...
87|             </>
88|         )}
89|     </div>
90| </div>
91|
92|     <div style={{ display: 'flex', gap: '8px', flexShrink: 0 }}>
93|         <button onClick={() => handlePublish(exam.id, !exam.is_published)}
94|             style={{ padding: '8px 12px', borderRadius: '8px', border: '1px solid var(--border-default)'...
95|             {exam.is_published ? <EyeOff size={14} /> : <Eye size={14} />}
96|         9 6 |             {exam.is_published ? 'Gá' : 'ng'}
97|         </button>
98|         <button onClick={() => router.push(`/teacher/exams/${exam.id}/edit`)}
99|             style={{ padding: '8px 12px', borderRadius: '8px', border: '1px solid var(--border-default)'...
100|         1 0 0 |             <Edit size={14} /> Sửa
101|         </button>
102|         <button onClick={() => handleDelete(exam.id)} disabled={deleting === exam.id}
103|             style={{ padding: '8px 12px', borderRadius: '8px', border: '1px solid rgba(239,68,68,0.3)', ...
104|             <Trash2 size={14} /> Xóa
105|         </button>
106|     </div>
107| </div>
108| ))}
109| </div>
110| )}
111| <style={`@keyframes spin { to { transform: rotate(360deg) } }`}</style>
112| </div>
113| )
114| }
115|

```

File: app/(dashboard)/teacher/layout.tsx (140 lines, 8.4 KB)

```

1| 'use client'
2|
3| import React, { useState, useEffect } from 'react'
4| import Link from 'next/link'
5| import { usePathname } from 'next/navigation'
6| import { BookOpen, Users, FileText, Logout, LayoutDashboard, ClipboardList, Award, Laptop, MessageSquare, Tren...
7| import { logout } from '@lib/auth-actions'
8| import { supabase } from '@lib/supabase'
9| import PageTransition from '@components/PageTransition'
10|
11| function ThemeToggle() {
12|     const [theme, setThemeState] = useState<'light' | 'dark'>('dark')
13|     React.useEffect(() => {
14|         const saved = (localStorage.getItem('theme') as 'light' | 'dark') || 'dark'
15|         setThemeState(saved)
16|         document.documentElement.setAttribute('data-theme', saved)
17|     }, [])
18|     const toggle = React.useCallback(() => {
19|         const next = theme === 'light' ? 'dark' : 'light'
20|         setThemeState(next)
21|         document.documentElement.setAttribute('data-theme', next)
22|         localStorage.setItem('theme', next)
23|     }, [theme])
24|     return (
25|         <button onClick={toggle} style={{ display: 'flex', alignItems: 'center', gap: '12px', padding: '10px 16px'...
26|         {theme === 'light' ? <Moon size={18} /> : <Sun size={18} />}
27|         2 7 |         {theme === 'light' ? 'Giao diện tối' : 'Giao diện sáng'}
28|     </button>
29| )
30| }
31|
32| export default function TeacherLayout({ children }: { children: React.ReactNode }) {
33|     const pathname = usePathname()
34|     const [profile, setProfile] = useState<any>(null)
35|     const [showDropdown, setShowDropdown] = useState(false)
36|
37|     useEffect(() => {
38|         supabase.auth.getUser().then(({ data: { user } }) => {
39|             if (user) supabase.from('profiles').select('*').eq('id', user.id).single().then(({ data }) => { if (data...
40|             })
41|         }, [])
42|
43|         const isActive = (href: string) => pathname === href || pathname?.startsWith(href + '/')
44|
45|         const NavGroup = ({ label, children }: { label: string; children: React.ReactNode }) => (
46|             <div>
47|                 <div style={{ padding: '8px 16px 4px', fontSize: '11px', fontWeight: 700, color: 'rgba(255,255,255,0.35)'...
48|                 {children}
49|             </div>
50|         )
51|
52|         const NavItem = ({ href, icon: Icon, label }: { href: string; icon: any; label: string }) => {
53|             const active = isActive(href)
54|             return (
55|                 <Link href={href} style={{
56|                     display: 'flex', alignItems: 'center', gap: '12px', padding: '10px 16px', borderRadius: '10px', textDe...
57|                     background: active ? 'rgba(8,158,96,0.15)' : 'transparent',
58|                     color: active ? '#fff' : 'rgba(255,255,255,0.6)',
59|                     fontWeight: active ? 700 : 500,
60|                     transition: 'all 0.2s'
61|                 }}>
62|                     <Icon size={18} /> {label}
63|                 </Link>
64|             )
65|         }
66|
67|         return (
68|             <div style={{ display: 'flex', minHeight: '100vh', background: 'var(--bg-base)' }}>
69|                 <aside style={{ width: '280px', background: 'linear-gradient(180deg, #031f3b 0%, #1a2f53 100%)', display...
70|                 <div style={{ padding: '20px 16px', borderBottom: '1px solid rgba(255,255,255,0.08)', position: 'relat...
71|                 <button onClick={() => setShowDropdown(!showDropdown)}
72|                 style={{ display: 'flex', alignItems: 'center', gap: '10px', width: '100%', background: 'none', bo...
73|                 <div style={{ width: '36px', height: '36px', borderRadius: '10px', background: 'var(--brand-primar...
74|                 <User size={18} color="#fff" />
75|                 </div>
76|                 <div style={{ flex: 1, textAlign: 'left', minWidth: 0 }}>
77|                     <div style={{ fontWeight: 700, fontSize: '14px', color: '#fff', overflow: 'hidden', textOverflow...
78|                     <div style={{ fontSize: '11px', color: 'var(--brand-primary)', fontWeight: 600, opacity: 0.9 }}>...
79|                 </div>
80|                 <ChevronDown size={16} style={{ color: 'rgba(255,255,255,0.4)', transform: showDropdown ? 'rotate(...
81|                 </button>
82|
83|                 {showDropdown && (
84|                     <div style={{ position: 'fixed', inset: 0, zIndex: 99 }} onClick={() => setShowDropdown(false)} />
85|                     <div style={{ position: 'absolute', top: '100%', left: '12px', right: '12px', background: '#fff'...
86|

```

```

87|         <div style={{ padding: '10px 12px', borderBottom: '1px solid var(--border-subtle)', marginBott...
88|         <div style={{ fontSize: '13px', fontWeight: 600, color: 'var(--text-primary)' }}>{profile?.f...
89|         <div style={{ fontSize: '12px', color: 'var(--text-muted)' }}>{profile?.email || ''}</div>
90|         <div style={{ fontSize: '11px', color: 'var(--brand-primary)', fontWeight: 600, marginTop: '1...
91|         </div>
92|         <Link href="/teacher" onClick={() => setShowDropdown(false)} style={{ display: 'flex', alignIt...
93|         <LayoutDashboard size={16} /> B Ē ng i Á u k h i Ā n
94|         </Link>
95|         <button onClick={async () => { setShowDropdown(false); await logout() }}
96|         style={{ display: 'flex', alignItems: 'center', gap: '10px', padding: '10px 12px', width: '1...
97|         <Logout size={16} /> n g x u ¥ t
98|         </button>
99|         </div>
100|     </>
101| }
102| </div>
103|
104| <nav style={{ padding: '16px 12px', display: 'flex', flexDirection: 'column', gap: '4px', flex: 1, ove...
105| <NavGroup label="T ò ng q u a n" >
106| <NavItem href="/teacher" icon={LayoutDashboard} label="B Ē ng
107| </NavGroup>
108| <NavGroup label="Q u Ē n l ý" >
109| <NavItem href="/teacher/classes" icon={Users} label="L Ò p h í c
110| <NavItem href="/teacher/lessons" icon={FileText} label="B à i
111| <NavItem href="/teacher/exams" icon={ClipboardList} label="
112| <NavItem href="/teacher/certificates" icon={Award} label="C h
113| </NavGroup>
114| <NavGroup label="T i Ç n í c h" >
115| <NavItem href="/student/discussion" icon={MessagesSquare} la
116| <NavItem href="/teacher/proctoring" icon={ScanEye} label="Giám sát thi" />
117| </NavGroup>
118| </nav>
119|
120| <div style={{ padding: '12px', borderTop: '1px solid rgba(255,255,255,0.08)', display: 'flex', flexDir...
121| <ThemeToggle />
122| <button onClick={async () => { await logout() }}
123| style={{ display: 'flex', alignItems: 'center', gap: '12px', padding: '10px 16px', width: '100%', ...
124| <Logout size={18} /> n g x u ¥ t
125| </button>
126| </div>
127| </aside>
128|
129| <main style={{ flex: 1, padding: '32px', overflowY: 'auto', minWidth: 0 }}>
130| <div style={{ marginBottom: '24px', display: 'flex', alignItems: 'center', gap: '8px', fontSize: '13px...
131| <Link href="/teacher" style={{ color: 'var(--accent-blue)', textDecoration: 'none', fontWeight: 500 ...
132| <span></span> <span style={{ color: 'var(--text-primary)', fontWeight: 600
133| </div>
134| <PageTransition>{children}</PageTransition>
135| </main>
136| </div>
137| )
138| }
139| }
140|

```

File: app/(dashboard)/teacher/lessons/[id]/page.js (502 lines, 41.0 KB)

```

1| 'use client';
2|
3| import React, { useEffect, useState, use, useCallback, useRef } from 'react';
4| import { supabase } from '@lib/supabase';
5| import { Save, Trash2, Video, FileText, Image as ImageIcon, FileSearch, Code, Loader2, ArrowLeft, GripVertical...
6| import Link from 'next/link';
7| import { BackButton } from '@components/ui/BackButton';
8|
9| function SimpleMarkdown({ text }) {
10|     const html = (text || '')
11|         .replace(/^## (.+)$/gm, '<h2 style="color:var(--brand-primary);margin:16px 0 8px">$1</h2>')
12|         .replace(/^(.+)$/gm, '<h1 style="color:var(--brand-primary);margin:20px 0 10px">$1</h1>')
13|         .replace(/\*\^(.+?)\*/g, '<strong>$1</strong>')
14|         .replace(/\^(.+?)\*/g, '<em>$1</em>')
15|         .replace(/^- (.+)$/gm, '<li style="margin:4px 0">$1</li>')
16|         .replace(/\n/g, '<br/>');
17|     return <div dangerouslySetInnerHTML={{ __html: html }} style={{ color: 'var(--text-muted)', lineHeight: 1.1...
18| }
19|
20| function Toast({ message, type, onClose }) {
21|     useEffect(() => { const t = setTimeout(onClose, 3000); return () => clearTimeout(t); }, []);
22|     return (
23|         <div style={{
24|             position: 'fixed', bottom: '32px', right: '32px', zIndex: 9999,
25|             background: type === 'success' ? 'var(--brand-primary)' : 'var(--danger)',
26|             color: '#fff', padding: '14px 24px', borderRadius: '12px',
27|             display: 'flex', alignItems: 'center', gap: '10px',
28|             fontWeight: 700, boxShadow: '0 10px 40px rgba(0,0,0,0.4)',
29|             animation: 'slideIn 0.3s ease'
30|         }}
31|         {type === 'success' ? <CheckCircle size={20}/> : <AlertCircle size={20}/>}
32|         {message}
33|     </div>
34| );
35| }
36|
37| export default function LessonEditorPage({ params }) {
38|     const resolvedParams = use(params);
39|     const lessonId = resolvedParams.id;
40|
41|     const [lesson, setLesson] = useState(null);
42|     const [sections, setSections] = useState([]);
43|     const [books, setBooks] = useState([]);
44|     const [activeSection, setActiveSection] = useState(null);
45|     const [activeTab, setActiveTab] = useState('content');
46|     const [loading, setLoading] = useState(true);
47|     const [saving, setSaving] = useState(false);
48|     const [toast, setToast] = useState(null);
49|     const [classes, setClasses] = useState([]);
50|     const [selectedClassIds, setSelectedClassIds] = useState([]);
51|     const [showClassPicker, setShowClassPicker] = useState(false);
52|     const textEditorRef = useRef(null);
53|
54|     const insertAtCursor = (text) => {
55|         const ta = textEditorRef.current;
56|         if (!ta) return;
57|         const start = ta.selectionStart;
58|         const end = ta.selectionEnd;
59|         const val = ta.value;
60|         const before = val.substring(0, start);
61|         const after = val.substring(end);
62|         const result = before + '\n\n' + text + '\n\n' + after;
63|         updateSectionLocal({ ...activeSection, content: result });
64|         requestAnimationFrame(() => {
65|             ta.focus();
66|             const pos = before.length + 2 + text.length + 2;

```

```

67 |         ta.setSelectionRange(pos, pos);
68 |     });
69 | };
70 |
71 | const showToast = (message, type = 'success') => setToast({ message, type });
72 |
73 | useEffect(() => { fetchLessonData(); }, [lessonId]);
74 |
75 | const fetchLessonData = async () => {
76 |     setLoading(true);
77 |     const { data: lessonData, error } = await supabase.from('lessons').select('*').eq('id', lessonId).single();
78 |     if (error || !lessonData) { showToast('Không tìm thấy bài giảng'); }
79 |     const { data: sectionData } = await supabase.from('lesson_sections').select('*').eq('lesson_id', lessonId).single();
80 |     const { data: bookData } = await supabase.from('lesson_books').select('*').eq('lesson_id', lessonId).single();
81 |     setLesson(lessonData);
82 |     setSections(sectionData || []);
83 |     setBooks(bookData || []);
84 |     if (sectionData?.length > 0) setActiveSection(sectionData[0]);
85 |
86 |     const { data: { user } } = await supabase.auth.getUser();
87 |     if (user) {
88 |         const { data: classData } = await supabase.from('classes').select('*').eq('teacher_id', user.id).single();
89 |         setClasses(classData || []);
90 |     }
91 |
92 |     const { data: assignmentData } = await supabase.from('lesson_class_assignments').select('class_id').eq('lesson_id', lessonId).single();
93 |     if (assignmentData) setSelectedClassIds(assignmentData.map(a => a.class_id));
94 |
95 |     setLoading(false);
96 | };
97 |
98 | const handleSaveLesson = async () => {
99 |     setSaving(true);
100 |     const { error } = await supabase.from('lessons').update({
101 |         title: lesson.title,
102 |         description: lesson.description,
103 |         thumbnail_url: lesson.thumbnail_url,
104 |         is_published: lesson.is_published,
105 |         category: lesson.category || 'extended',
106 |         source_name: lesson.source_name || '',
107 |         source_url: lesson.source_url || ''
108 |     }).eq('id', lessonId);
109 |     if (error) { showToast('Lỗi khi lưu bài giảng', 'error'); }
110 |
111 |     const { data: existing } = await supabase.from('lesson_class_assignments').select('class_id').eq('lesson_id', lessonId).single();
112 |     const existingIds = (existing || []).map(a => a.class_id);
113 |     const toRemove = existingIds.filter(id => !selectedClassIds.includes(id));
114 |     const toAdd = selectedClassIds.filter(id => !existingIds.includes(id));
115 |     if (toRemove.length > 0) {
116 |         await supabase.from('lesson_class_assignments').delete().eq('lesson_id', lessonId).in('class_id', toRemove);
117 |     }
118 |     if (toAdd.length > 0) {
119 |         await supabase.from('lesson_class_assignments').insert(toAdd.map(class_id => ({ lesson_id: lessonId, class_id })));
120 |     }
121 |
122 |     setSaving(false);
123 |     showToast('Lưu bài giảng thành công', 'success');
124 | };
125 |
126 | const addSection = async (type) => {
127 |     const newSection = { lesson_id: lessonId, type, content_type: type };
128 |     const { data, error } = await supabase.from('lesson_sections').insert(newSection).select().single();
129 |     if (error) { showToast('Không thể thêm bài giảng', 'error'); }
130 |     if (data) { setSections(prev => [...prev, data]); setActiveSection(data); }
131 | };
132 |
133 | const saveSection = async (section) => {
134 |     const { error } = await supabase.from('lesson_sections').update({
135 |         title: section.title, content: section.content
136 |     }).eq('id', section.id);
137 |     if (error) { showToast('Lỗi khi lưu bài giảng', 'error'); }
138 | };
139 |
140 | const updateSectionLocal = (updated) => {
141 |     setSections(prev => prev.map(s => s.id === updated.id ? updated : s));
142 |     setActiveSection(updated);
143 | };
144 |
145 | const deleteSection = async (id) => {
146 |     if (!confirm('Xóa phần này?')) return;
147 |     const { error } = await supabase.from('lesson_sections').delete().eq('id', id);
148 |     if (error) { showToast('Xóa bài giảng thất bại', 'error'); return; }
149 |     const filtered = sections.filter(s => s.id !== id);
150 |     setSections(filtered);
151 |     setActiveSection(filtered[0] || null);
152 | };
153 |
154 | const getYouTubeEmbed = (url) => {
155 |     if (!url) return '';
156 |     const id = url.match(/(?:youtube\.com\/(?:[^\?&\/\.\+\/\?|v|e(?:mbed)?|\/|.*[?&]v=)|youtu\.be\/)([^\?&\/\.\+\/\?|v|e(?:mbed)?|\/|.*[?&]v=)/);
157 |     return id ? `https://www.youtube.com/embed/${id}` : '';
158 | };
159 |
160 | const getDriveEmbed = (url) => {
161 |     if (!url) return '';
162 |     const id = url.match(/\/d\/(.+?)\/(?:view|preview)?/)[1];
163 |     return id ? `https://drive.google.com/file/d/${id}/view?usp=sharing` : (url.includes('/preview') ? url : '');
164 | };
165 |
166 | const getPdfEmbedUrl = (url) => {
167 |     if (!url) return '';
168 |     const id = url.match(/\/d\/(.+?)\/(?:view|preview)?/)[1];
169 |     if (id) {
170 |         const downloadUrl = `https://drive.google.com/uc?export=download&id=${id}`;
171 |         return `https://docs.google.com/viewer?url=${encodeURIComponent(downloadUrl)}&embedded=true`;
172 |     }
173 |     return url;
174 | };
175 |
176 | if (loading) return (
177 |     <div style={{ background: 'var(--bg-base)', minHeight: '100vh', display: 'flex', alignItems: 'center', ... }}>
178 |         <Loader2 style={{ animation: 'spin 1s linear infinite' }} color="var(--brand-primary)" size={48} />
179 |     </div>
180 | );
181 |
182 | if (!lesson) return (
183 |     <div style={{ background: 'var(--bg-base)', minHeight: '100vh', display: 'flex', flexDirection: 'column', ... }}>
184 |         <AlertCircle size={48} color="var(--danger)" />
185 |         <p>Không tìm thấy bài giảng.</p>
186 |         <Link href="/teacher/lessons" style={{ color: 'var(--brand-primary)' }}>Link here</Link>
187 |     </div>
188 | );
189 |
190 | const inputStyle = { width: '100%', background: 'var(--bg-elevated)', border: '1px solid var(--border-default)', ... };

```

```

191 | return (
192 |   <div style={{ background: 'var(--bg-base)', minHeight: '100vh', color: 'var(--text-primary)', display:...
193 |     <style>{
194 |       @keyframes slideIn { from { transform: translateY(20px); opacity: 0; } to { transform: transla...
195 |       @media (max-width: 768px) {
196 |         .editor-sidebar { width: 100% !important; min-width: 100% !important; max-height: 50vh !im...
197 |         .editor-content { padding: 16px !important; }
198 |         .editor-header { padding: 12px 16px !important; flex-wrap: wrap !important; gap: 8px !impo...
199 |         .editor-header-title { width: 100% !important; max-width: 100% !important; }
200 |         .editor-section-grid { grid-template-columns: 1fr !important; }
201 |       }
202 |       @media (min-width: 769px) and (max-width: 1024px) {
203 |         .editor-sidebar { width: 280px !important; min-width: 280px !important; }
204 |         .editor-content { padding: 24px !important; }
205 |       }
206 |     }</style>
207 |     {toast && <Toast message={toast.message} type={toast.type} onClose={() => setToast(null)} />}
208 |
209 |     <header className="editor-header" style={{ padding: '16px 32px', background: 'var(--bg-surface)', ...
210 |       <div style={{ display: 'flex', alignItems: 'center', gap: '20px' }}>
211 |         <BackButton href="/teacher/lessons" label="" className="flex items-center text-[#636678] h...
212 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '2px' }}>
213 |           <span style={{ fontSize: '10px', color: 'var(--text-muted)', textTransform: 'uppercase...
214 |           <input className="editor-header-title"
215 |             value={lesson.title}
216 |             onChange={(e) => setLesson({ ...lesson, title: e.target.value })}
217 |             style={{ background: 'transparent', border: 'none', borderBottom: '2px solid var(-...
218 |           </div>
219 |         </div>
220 |       </div>
221 |       <div style={{ display: 'flex', gap: '12px', alignItems: 'center' }}>
222 |         <button onClick={() => setLesson({ ...lesson, is_published: !lesson.is_published })}
223 |           style={{ background: lesson.is_published ? 'rgba(16,185,129,0.15)' : 'var(--bg-elevate...
224 |           {lesson.is_published ? <Eye size={18} /> : <EyeOff size={18} />}
225 |         </button>
226 |         <button onClick={handleSaveLesson} disabled={saving}
227 |           style={{ background: 'var(--brand-primary)', color: '#fff', border: 'none', padding: '...
228 |           {saving ? <Loader2 style={{ animation: 'spin 1s linear infinite' }} size={18} /> : <Sa...
229 |         </button>
230 |       </div>
231 |     </header>
232 |
233 |     <div style={{ display: 'flex', flex: 1, overflow: 'hidden' }}>
234 |       <div className="editor-sidebar" style={{ width: '340px', minWidth: '340px', background: 'var(-...
235 |       <div style={{ padding: '20px', borderBottom: '1px solid var(--border-default)' }}>
236 |         <p style={{ fontSize: '11px', color: 'var(--text-muted)', fontWeight: 700, textTra...
237 |         <div style={{ width: '100%', height: '140px', background: 'var(--bg-elevated)', border...
238 |           {lesson.thumbnail_url
239 |             ? <img src={lesson.thumbnail_url} style={{ width: '100%', height: '100%', obje...
240 |             : <Upload size={28} color="var(--text-muted)" />
241 |           </div>
242 |         <input placeholder="Điền URL ảnh thumbnail..." value={
243 |           onChange={(e) => setLesson({ ...lesson, thumbnail_url: e.target.value })}
244 |           style={{ width: '100%', background: 'var(--bg-elevated)', border: '1px solid var(-...
245 |         </div>
246 |         <div style={{ display: 'flex', gap: '12px', height: '100%' }}>
247 |           <div style={{ flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
248 |             <div style={{ display: 'flex', flexDirection: 'column', gap: '10px' }}>
249 |               <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
250 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
251 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
252 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
253 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
254 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
255 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
256 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
257 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
258 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
259 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
260 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
261 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
262 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
263 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
264 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
265 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
266 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
267 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
268 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
269 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
270 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
271 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
272 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
273 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
274 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
275 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
276 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
277 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
278 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
279 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
280 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
281 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
282 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
283 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
284 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
285 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
286 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
287 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
288 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
289 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
290 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
291 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
292 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
293 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
294 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
295 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
296 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
297 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
298 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
299 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
300 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
301 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
302 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: 'var(--bg-elevated)', border: '1px solid var(-...
303 |                 <div style={{ display: 'flex', flex: 1, padding: '10px', background: '
```

```

315|         </label>
316|     )}
317| </div>
318| }
319| {!showClassPicker && selectedClassIds.length > 0 && (
320|   <div style={{ display: 'flex', flexWrap: 'wrap', gap: '4px' }}>
321|     {classes.filter(c => selectedClassIds.includes(c.id)).map(cls => (
322|       <span key={cls.id} style={{ fontSize: '11px', padding: '3px 8px', back...
323|     ))}
324|   </div>
325| )}
326| </div>
327|
328| <div style={{ display: 'flex', borderBottom: '1px solid var(--border-default)' }}>
329|   {[ 'content', 'books' ].map(tab => (
330|     <button key={tab} onClick={() => setActiveTab(tab)} style={{
331|       flex: 1, padding: '12px', background: activeTab === tab ? 'rgba(var(--brand-pr...
332|       color: activeTab === tab ? 'var(--brand-primary)' : 'var(--text-muted)', borde...
333|       borderBottom: activeTab === tab ? '2px solid var(--brand-primary)' : '2px soli...
334|       fontWeight: 700, cursor: 'pointer', fontSize: '13px', fontFamily: 'inherit'
335|     }} > { tab === 'content' ? ' N ù i d u n g ' : ' S á c h ' }
336|   ))}
337| </div>
338|
339| <div style={{ flex: 1, overflowY: 'auto', padding: '16px' }}>
340|   {activeTab === 'content' ? (
341|     <>
342|       {sections.length === 0 && <p style={{ color: 'var(--text-muted)', fontSize: '1...
343|       <div style={{ display: 'flex', flexDirection: 'column', gap: '6px' }}>
344|         {sections.map(s => (
345|           <div key={s.id} onClick={() => { if (activeSection && activeSection.id...
346|             style={{ padding: '10px 14px', borderRadius: '10px', background: a...
347|             <GripVertical size={14} color="var(--text-muted)" />
348|             <div style={{ width: '28px', height: '28px', borderRadius: '6px', ...
349|               {s.type === 'video' && <Video size={14} />
350|               {s.type === 'text' && <FileText size={14} />
351|               {s.type === 'image' && <ImageIcon size={14} />
352|               {s.type === 'pdf' && <FileSearch size={14} /> } {s.type === 'emb...
353|             </div>
354|             <div style={{ flex: 1, overflow: 'hidden' }}>
355|               <p style={{ margin: 0, fontSize: '13px', fontWeight: 600, colo...
356|               <p style={{ margin: 0, fontSize: '10px', color: 'var(--text-mu...
357|             </div>
358|           </div>
359|         ))}
360|       </div>
361|       <div style={{ marginTop: '16px', display: 'grid', gridTemplateColumns: '1fr 1f...
362|         {[ 'video', <Video size={13} />, '+video'], ['text', <FileText size={13} /...
363|         <button key={type} onClick={() => addSection(type)} style={{ padding: ...
364|           {icon} {label}
365|         </button>
366|       ))}
367|     </div>
368|   </>
369| ) : (
370|   <div style={{ display: 'flex', flexDirection: 'column', gap: '10px' }}>
371|     {books.map(book => (
372|       <div key={book.id} style={{ padding: '12px', background: 'var(--bg-elevate...
373|       <img src={book.cover_image_url} style={{ width: '36px', height: '52px'...
374|       <div style={{ flex: 1 }}>
375|         <p style={{ margin: 0, fontSize: '13px', fontWeight: 700, color: '...
376|         <button onClick={async () => { await supabase.from('lesson_books')...
377|       </div>
378|     </div>
379|   ))}
380|   <button onClick={async () => {
381|     const { data } = await supabase.from('lesson_books').insert({ lesson_id: l...
382|     if (data) setBooks([...books, data]);
383|   }} style={{ width: '100%', padding: '12px', background: 'rgba(var(--brand-prim...
384|     + ' T h ê m s á c h m ù i
385|   </button>
386| </div>
387| )}
388| </div>
389| </div>
390|
391| <div className="editor-content" style={{ flex: 1, overflowY: 'auto', padding: '40px', backgrou...
392|   {activeSection ? (
393|     <div style={{ maxWidth: '820px', margin: '0 auto' }}>
394|       <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'cente...
395|       <h2 style={{ margin: 0, color: 'var(--brand-primary)', fontSize: '20px', fontW...
396|       <div style={{ display: 'flex', gap: '10px' }}>
397|         <button onClick={() => saveSection(activeSection)} style={{ background: 'r...
398|         <button onClick={() => deleteSection(activeSection.id)} style={{ backgroun...
399|       </div>
400|     </div>
401|
402|     <div style={{ display: 'flex', flexDirection: 'column', gap: '24px', minHeight: '3...
403|     <div>
404|       <label style={{ display: 'block', color: 'var(--text-muted)', fontSize: '1...
405|       <input value={activeSection.title} onChange={e => updateSectionLocal({ ...
406|     </div>
407|
408|     {activeSection.type === 'video' && <>
409|       <div>
410|         <label style={{ display: 'block', color: 'var(--text-muted)', fontSize...
411|         <input placeholder="https://www.youtube.com/watch?v=..." value={active...
412|       </div>
413|       {getYouTubeEmbed(activeSection.content) && (
414|         <div style={{ width: '100%', aspectRatio: '16/9', background: '#000', ...
415|         <iframe src={getYouTubeEmbed(activeSection.content)} style={{ widt...
416|       </div>
417|     </>
418|   )}
419|
420|   {activeSection.type === 'text' && (
421|     <div className="editor-section-grid" style={{ display: 'grid', gridTemplat...
422|     <div>
423|       <label style={{ display: 'block', color: 'var(--text-muted)', font...
424|       <div style={{ display: 'flex', gap: '6px', marginBottom: '8px' }}>
425|         {
426|           { icon: <Video size={14} />, label: 'Video', getText: () => ...
427|           { icon: <FileSearch size={14} />, label: 'PDF', getText: (...
428|           ].map(({ icon, label, getText }) => (
429|             <button key={label} type="button" onClick={() => { const t...
430|               style={{ padding: '6px 10px', borderRadius: '6px', bac...
431|             {icon} {label}
432|           </button>
433|         )}
434|       </div>
435|       <textarea ref={textEditorRef} value={activeSection.content} onChan...
436|       style={{ ...inputStyle, height: '360px', fontFamily: 'monospac...
437|       placeholder = " # # T i ê u Á '

```

```

439 |         </div>
440 |     </div>
441 |     <label style={{ display: 'block', color: 'var(--text-muted)', font...
442 |     <div style={{ height: '360px', background: 'var(--bg-surface)', bo...
443 |     <SimpleMarkdown text={activeSection.content} />
444 |     </div>
445 | </div>
446 | </div>
447 | ))
448 |
449 | {activeSection.type === 'image' && <>
450 |   <div>
451 |     <label style={{ display: 'block', color: 'var(--text-muted)', fontSize...
452 |     <input value={activeSection.content} onChange={(e) => updateSectionLoc...
453 |   </div>
454 |   {activeSection.content && <img src={activeSection.content} style={{ maxWid...
455 | </>
456 |
457 | {activeSection.type === 'pdf' && <>
458 |   <div style={{ background: 'rgba(var(--brand-primary-rgb),0.05)', border: '...
459 |   </div>
460 |   <div>
461 |     <label style={{ display: 'block', color: 'var(--text-muted)', fontSize...
462 |     <input placeholder="https://drive.google.com/file/d/FILE_ID/view" valu...
463 |   </div>
464 |   {getPdfEmbedUrl(activeSection.content) && (
465 |     <div style={{ width: '100%', height: '500px', background: 'white', bor...
466 |     <iframe src={getPdfEmbedUrl(activeSection.content)} style={{ width...
467 |   </div>
468 |   )}
469 | </>
470 |
471 | {activeSection.type === 'embed' && <>
472 |   <div style={{ background: 'rgba(167,139,250,0.05)', border: '1px solid rgb...
473 |   </div>
474 |   <div>
475 |     <label style={{ display: 'block', color: 'var(--text-muted)', fontSize...
476 |     <textarea value={activeSection.content} onChange={(e) => updateSection...
477 |     style={{ ...inputStyle, height: '140px', fontFamily: 'monospace', ...
478 |     placeholder=<iframe src="https://www.canva.com/..." width="100%" ...
479 |   </div>
480 |   {activeSection.content && (
481 |     <div style={{ width: '100%', borderRadius: '12px', overflow: 'hidden',...
482 |     <div dangerouslySetInnerHTML={{ __html: activeSection.content }} />
483 |   </div>
484 |   )}
485 | </>
486 | </div>
487 |
488 | ) : (
489 |   <div style={{ display: 'flex', flexDirection: 'column', alignItems: 'center', justifyC...
490 |   <PlusCircle size={64} />
491 | )}
492 |
493 | </div>
494 |
495 | </div>
496 | </div>
497 | </div>
498 | </div>
499 | );
500 | }
501 | }
502 |

```

File: app/(dashboard)/teacher/lessons/create/page.tsx (269 lines, 17.0 KB)

```

1 | 'use client'
2 |
3 | import { useState } from 'react'
4 | import { useRouter } from 'next/navigation'
5 | import Link from 'next/link'
6 | import { ArrowLeft, Loader2, ChevronRight, BookOpen, FileText, Video, Image as ImageIcon, Link2, Code, Eye, Pl...
7 | import { supabase } from '@lib/supabase'
8 |
9 | type SectionInput = {
10 |   id: string
11 |   title: string
12 |   type: 'video' | 'text' | 'image' | 'pdf' | 'embed'
13 |   content: string
14 | }
15 |
16 | export default function CreateLessonPage() {
17 |   const router = useRouter()
18 |   const [step, setStep] = useState(1)
19 |   const [loading, setLoading] = useState(false)
20 |   const [error, setError] = useState<string | null>(null)
21 |   const [title, setTitle] = useState('')
22 |   const [description, setDescription] = useState('')
23 |   const [thumbnailUrl, setThumbnailUrl] = useState('')
24 |   const [subject, setSubject] = useState('Tin hfc')
25 |   const [estimatedMinutes, setEstimatedMinutes] = useState(30)
26 |   const [category, setCategory] = useState('extended')
27 |   const [sections, setSections] = useState<SectionInput[]>([])
28 |
29 |   const addSection = () => {
30 |     setSections(prev => [...prev, {
31 |       id: Math.random().toString(36).substring(7),
32 |       title: '',
33 |       type: 'text',
34 |       content: '',
35 |     }])
36 |   }
37 |
38 |   const removeSection = (id: string) => {
39 |     setSections(prev => prev.filter(s => s.id !== id))
40 |   }
41 |
42 |   const updateSection = (id: string, field: keyof SectionInput, value: string) => {
43 |     setSections(prev => prev.map(s => s.id === id ? { ...s, [field]: value } : s))
44 |   }
45 |
46 |   const handleSubmit = async () => {
47 |     if (!title.trim()) { setError('Vui lòng nh p tiêu Á bài giảng'); re
48 |     setLoading(true)
49 |     setError(null)
50 |
51 |     const { data: { user } } = await supabase.auth.getUser()
52 |     if (!user) { setError('Bìn ch a ng nh p'); setLoading(false); retu
53 |
54 |     const { data: lesson, error: lessonError } = await supabase
55 |       .from('lessons')
56 |       .insert({

```



```

57 |         teacher_id: user.id,
58 |         title: title.trim(),
59 |         description: description.trim(),
60 |         thumbnail_url: thumbnailUrl.trim() || null,
61 |         subject,
62 |         category,
63 |         estimated_minutes: estimatedMinutes,
64 |         is_published: false,
65 |     })
66 |     .select()
67 |     .single()
68 |
69 |     if (lessonError || !lesson) {
70 |         setError(lessonError?.message || 'Không thể tạo bài giảng')
71 |         setLoading(false)
72 |         return
73 |     }
74 |
75 |     if (sections.length > 0) {
76 |         const { error: sectionsError } = await supabase.from('lesson_sections').insert(
77 |             sections.map((s, idx) => ({
78 |                 lesson_id: lesson.id,
79 |                 title: s.title || `Màc ${idx + 1}`,
80 |                 type: s.type,
81 |                 content_type: s.type,
82 |                 content: s.content,
83 |                 order_index: idx,
84 |             }))
85 |         )
86 |         if (sectionsError) console.error('Sections error:', sectionsError)
87 |     }
88 |
89 |     setLoading(false)
90 |     router.push(`/teacher/lessons/${lesson.id}?created=true`)
91 | }
92 |
93 | const getYouTubeThumbnail = (url: string) => {
94 |     const id = url.match(/(?:youtube\.com\/(?:\w+\/.+\/|(?=v|e(?:mbed)?|v=)|.*[?&]v=)|youtu\.be\/)([^\s&?\/\s]{...
95 |     return id ? `https://img.youtube.com/vi/${id}/hqdefault.jpg` : null
96 | }
97 |
98 | return (
99 |     <div style={{ maxWidth: '800px', margin: '0 auto' }}>
100 |     <Link href="/teacher/lessons" style={{ display: 'inline-flex', alignItems: 'center', gap: '8px', fontSiz...
101 |     <ArrowLeft size={16} /> Quay lại
102 |     </Link>
103 |
104 |     <div className="rounded-2xl p-8" style={{ background: 'var(--bg-surface)', border: '1px solid var(--bord...
105 |     <h1 style={{ fontSize: '24px', fontWeight: 700, marginBottom: '24px', color: 'var(--text-primary)' }}>...
106 |
107 |     {step === 1 && (
108 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '20px' }}>
109 |         <div>
110 |             <label style={{ fontSize: '14px', fontWeight: 500, marginBottom: '6px', display: 'block', color: ...
111 |             <input value={title} onChange={e => setTitle(e.target.value)}
112 |             style={{ width: '100%', padding: '12px 16px', borderRadius: '12px', background: 'var(--bg-elev...
113 |             placeholder="V D : G i ù i t h i Ç u v Á C P U " / >
114 |         </div>
115 |         <div>
116 |             <label style={{ fontSize: '14px', fontWeight: 500, marginBottom: '6px', display: 'block', color: ...
117 |             <div style={{ width: '100%', height: '160px', background: 'var(--bg-elevated)', borderRadius: '1...
118 |             {thumbnailUrl ? (
119 |                 <img src={thumbnailUrl} style={{ width: '100%', height: '100%', objectFit: 'cover' }} onErro...
120 |             ) : (
121 |                 <Upload size={32} color="var(--text-muted)" />
122 |             )
123 |         </div>
124 |         <input value={thumbnailUrl} onChange={e => setThumbnailUrl(e.target.value)}
125 |         style={{ width: '100%', padding: '12px 16px', borderRadius: '12px', background: 'var(--bg-elev...
126 |         placeholder="https://example.com/image.jpg" />
127 |     </div>
128 |     <div>
129 |         <label style={{ fontSize: '14px', fontWeight: 500, marginBottom: '6px', display: 'block', color: ...
130 |         <textarea value={description} onChange={e => setDescription(e.target.value)} rows={3}
131 |         style={{ width: '100%', padding: '12px 16px', borderRadius: '12px', background: 'var(--bg-elev...
132 |         placeholder="M ô t ê n g n v Á b à i g i ê n g . . . " / >
133 |     </div>
134 |     <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '16px' }}>
135 |     <div>
136 |         <label style={{ fontSize: '14px', fontWeight: 500, marginBottom: '6px', display: 'block', colo...
137 |         <select value={subject} onChange={e => setSubject(e.target.value)}
138 |         style={{ width: '100%', padding: '12px 16px', borderRadius: '12px', background: 'var(--bg-el...
139 |             <option> T i n h i c </option>
140 |             <option> T i n h i c - L p r á p P C </option>
141 |             <option> K h o a h i c m á y t i n h </option>
142 |         </select>
143 |     </div>
144 |     <div>
145 |         <label style={{ fontSize: '14px', fontWeight: 500, marginBottom: '6px', display: 'block', colo...
146 |         <input type="number" value={estimatedMinutes} onChange={e => setEstimatedMinutes(Number(e.targ...
147 |         style={{ width: '100%', padding: '12px 16px', borderRadius: '12px', background: 'var(--bg-el...
148 |     </div>
149 | </div>
150 | <div>
151 |     <label style={{ fontSize: '14px', fontWeight: 500, marginBottom: '6px', display: 'block', color: ...
152 |     <div style={{ display: 'flex', gap: '8px' }}>
153 |     {
154 |         { key: 'textbook', label: 'Sách Giáo Khoa' },
155 |         { key: 'extended', label: 'Kiến Thức Mở Rộng' },
156 |     }.map(({ key, label }) => (
157 |         <button key={key} onClick={() => setCategory(key)}
158 |         style={{
159 |             padding: '8px 20px', borderRadius: '8px', border: `2px solid ${category === key ? 'var(--...
160 |             background: category === key ? `rgba(var(--brand-primary-rgb),0.1)` : 'transparent',
161 |             color: category === key ? 'var(--brand-primary)' : 'var(--text-muted)',
162 |             fontWeight: 600, cursor: 'pointer', fontSize: '13px', fontFamily: 'inherit'
163 |         })
164 |     })
165 | </div>
166 | </div>
167 | </div>
168 | </div>
169 | </div>
170 | )}
171 |
172 | {step === 2 && (
173 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
174 |     <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'space-between', marginBottom: ...
175 |     <span style={{ fontSize: '14px', fontWeight: 500, color:
176 |     <button onClick={addSection}
177 |     style={{ padding: '8px 16px', borderRadius: '8px', fontSize: '13px', fontWeight: 700, backgrou...
178 |     placeholder="Thêm mã c
179 |     </button>
180 | </div>

```

```

181|
182|     {sections.length === 0 && (
183|       <div style={{ textAlign: 'center', padding: '48px 0', borderRadius: '12px', border: '2px dashed ...
184|       <FileText size={36} style={{ margin: '0 auto 12px', color: 'var(--text-muted)'} }} />
185|     <p style={{ fontSize: '14px', color: 'var(--text-muted) }} />
186|   </div>
187| )}
188|
189|   {sections.map((s, idx) => (
190|     <div key={s.id} style={{ padding: '16px', borderRadius: '12px', background: 'var(--bg-elevated)'}...
191|     <div style={{ display: 'flex', alignItems: 'center', gap: '12px', marginBottom: '12px' }}>
192|       <GripVertical size={16} style={{ color: 'var(--text-muted)'} }} />
193|       <span style={{ fontSize: '11px', fontWeight: 700, color: 'var(--text-muted)'}}>
194|         <div style={{ display: 'flex', gap: '4px', marginLeft: 'auto' }}>
195|           {[['video', 'text', 'image', 'pdf', 'embed'] as const].map(type => (
196|             <button key={type} onClick={() => updateSection(s.id, 'type', type)}
197|               style={{ padding: '6px', borderRadius: '6px', background: s.type === type ? 'var(--bra...
198|               {type === 'video' && <Video size={14} />}
199|               {type === 'text' && <FileText size={14} />}
200|               {type === 'image' && <ImageIcon size={14} />}
201|               {type === 'pdf' && <Link2 size={14} />}
202|               {type === 'embed' && <Code size={14} />}
203|             </button>
204|           ))}
205|           <button onClick={() => removeSection(s.id)} style={{ padding: '6px', borderRadius: '6px', ...
206|             <Trash2 size={14} />
207|           </button>
208|         </div>
209|       </div>
210|       <input value={s.title} onChange={e => updateSection(s.id, 'title', e.target.value)}
211|         style={{ width: '100%', padding: '10px 12px', borderRadius: '8px', background: 'var(--bg-sur...
212|         placeholder="Tiêu đề bài học..." />
213|       {s.type === 'text' ? (
214|         <div>
215|           <div style={{ display: 'flex', gap: '6px', marginBottom: '8px' }}>
216|             {[
217|               { icon: <ImageIcon size={14} />, label: 'Thêm', getInsert: () => { const u = prompt('URL ...
218|               { icon: <Video size={14} />, label: 'Video', getInsert: () => { const u = prompt('URL ...
219|               { icon: <FileSearch size={14} />, label: 'PDF', getInsert: () => { const u = prompt('U...
220|             ].map(({ icon, label, getInsert }) => (
221|               <button key={label} type="button" onClick={() => { const ins = getInsert(); if (ins) u...
222|               style={{ padding: '6px 10px', borderRadius: '6px', background: 'var(--bg-surface)', ...
223|               {icon} {label}
224|             </button>
225|           ))}
226|         </div>
227|         <textarea value={s.content} onChange={e => updateSection(s.id, 'content', e.target.value)}...
228|         style={{ width: '100%', padding: '10px 12px', borderRadius: '8px', background: 'var(--bg...
229|         placeholder="Nội dung markdown... (dùng nút ß để thêm code)" />
230|       </div>
231|     ) : (
232|       <input value={s.content} onChange={e => updateSection(s.id, 'content', e.target.value)}
233|         style={{ width: '100%', padding: '10px 12px', borderRadius: '8px', background: 'var(--bg-s...
234|         placeholder={s.type === 'video' ? 'URL YouTube...' : 'URL bài học...'} />
235|     )}
236|   </div>
237| )}
238| </div>
239| )}
240|
241| {error && (
242|   <div style={{ marginTop: '16px', padding: '12px 16px', borderRadius: '8px', background: 'rgba(244,67...
243|   {error}
244| </div>
245| )}
246|
247| <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'space-between', marginTop: '32px...
248| {step > 1 ? (
249|   <button onClick={() => setStep(step - 1)} style={{ padding: '10px 24px', borderRadius: '12px', fon...
250|   Quay lại
251| </button>
252| ) : <div />
253|
254| {step < 2 ? (
255|   <button onClick={() => setStep(2)} style={{ padding: '10px 24px', borderRadius: '12px', fontSize: ...
256|   Tiếp theo <ChevronRight size={16} />
257| </button>
258| ) : (
259|   <button onClick={handleSubmit} disabled={loading} style={{ padding: '10px 24px', borderRadius: '12...
260|   {loading ? <Loader2 size={16} style={{ animation: 'spin 1s linear infinite' }} /> : <Eye size={1...
261|   {loading ? ' đang tải...' : ' Bài giảng (Draft)'}
262| </button>
263| )}
264| </div>
265| </div>
266| </div>
267| )
268| }
269|

```

File: app/(dashboard)/teacher/lessons/page.js (175 lines, 9.0 KB)

```

1| 'use client'
2|
3| import React, { useEffect, useState } from 'react'
4| import { supabase } from '@/lib/supabase'
5| import { Plus, BookOpen, Edit, Trash2, Eye, EyeOff, Loader2, ArrowLeft, ImageIcon } from 'lucide-react'
6| import Link from 'next/link'
7| import { useRouter } from 'next/navigation'
8| import AuthButton from '@/components/AuthButton'
9| import { deleteLessonAction } from '@/app/actions/lessons'
10|
11| export default function TeacherLessonsPage() {
12|   const router = useRouter()
13|   const [lessons, setLessons] = useState([])
14|   const [loading, setLoading] = useState(true)
15|   const [currentUser, setCurrentUser] = useState(null)
16|
17|   useEffect(() => { fetchLessons() }, [])
18|
19|   const fetchLessons = async () => {
20|     setLoading(true)
21|     const { data: { user } } = await supabase.auth.getUser()
22|     if (!user) { window.location.href = '/login'; return }
23|     setCurrentUser(user)
24|     const { data: lessonsData } = await supabase
25|       .from('lessons')
26|       .select('*', { count: 'exact' })
27|       .eq('teacher_id', user.id)
28|       .order('created_at', { ascending: false })
29|     if (lessonsData) {
30|       const lessonIds = lessonsData.map(l => l.id)
31|       const { data: assignments } = await supabase

```

```

32 |     .from('lesson_class_assignments')
33 |     .select('lesson_id, class_id')
34 |     .in('lesson_id', lessonIds)
35 | const { data: allClasses } = await supabase
36 |     .from('classes')
37 |     .select('id, name')
38 |     .eq('teacher_id', user.id)
39 | const classMap = Object.fromEntries((allClasses || []).map(c => [c.id, c.name]))
40 | const assignmentMap = {}
41 | ;(assignments || []).forEach(a => {
42 |   if (!assignmentMap[a.lesson_id]) assignmentMap[a.lesson_id] = []
43 |   if (classMap[a.class_id]) assignmentMap[a.lesson_id].push(classMap[a.class_id])
44 | })
45 | setLessons(lessonsData.map(l => ({ ...l, classNames: assignmentMap[l.id] || [] })))
46 | }
47 | setLoading(false)
48 | }
49 |
50 | const createNewLesson = async () => {
51 |   const { data: { user } } = await supabase.auth.getUser()
52 |   if (!user) { alert('Phiên học tập kết thúc!'); window.location.href = '/' }
53 |   const { data, error } = await supabase.from('lessons').insert({
54 |     teacher_id: user.id, title: 'Bài giảng mới', description: '', is_published: false
55 |   }).select().single()
56 |   if (error) { alert('Lỗi: ' + error.message); return }
57 |   if (data) router.push('/teacher/lessons/${data.id}')
58 | }
59 |
60 | const deleteLesson = async (id) => {
61 |   if (!confirm('Xóa bài giảng này? Híc sinh sẽ không còn thấy bài này')) return
62 |   const result = await deleteLessonAction(id)
63 |   if (result?.error) {
64 |     alert('Lỗi: ' + result.error)
65 |   }
66 |   fetchLessons()
67 | }
68 |
69 | if (loading) return (
70 |   <div style={{ minHeight: '100vh', display: 'flex', alignItems: 'center', justifyContent: 'center', background: 'linear-gradient(to right, #f0f0f0 49%, #fff 49% 51%, #fff 51% 53%, #f0f0f0 53%)' }}>
71 |     <Loader2 size={48} style={{ animation: 'spin 1s linear infinite', color: 'var(--brand-primary)' }} />
72 |   </div>
73 | )
74 |
75 | return (
76 |   <div style={{ minHeight: '100vh', background: 'var(--bg-base)', color: 'var(--text-primary)', padding: '28px 24px 24px 24px' }}>
77 |     <header style={{ maxWidth: '1200px', margin: '0 auto 32px auto', display: 'flex', justifyContent: 'space-between', padding: '0 24px' }}>
78 |       <div style={{ display: 'flex', alignItems: 'center', gap: '14px' }}>
79 |         <Link href="/teacher" style={{ width: '40px', height: '40px', borderRadius: '10px', background: 'var(--bg-surface)', display: 'flex', alignItems: 'center', justify-content: 'center', color: 'var(--text-muted)', border: '1px solid var(--border-default)' }}>
80 |           <ArrowLeft size={20} />
81 |         </Link>
82 |         <h1 style={{ fontSize: '26px', fontWeight: 800, margin: 0 }}>
83 |           Quản lý <span style={{ color: 'var(--brand-primary)' }}> Bài giảng
84 |         </h1>
85 |       </div>
86 |       <div style={{ display: 'flex', gap: '12px', alignItems: 'center' }}>
87 |         <button onClick={createNewLesson} className="lms-btn lms-btn-green">
88 |           <Plus size={18} /> Thêm bài mới
89 |         </button>
90 |         <AuthButton />
91 |       </div>
92 |     </header>
93 |
94 |     <main style={{ maxWidth: '1200px', margin: '0 auto' }}>
95 |       {lessons.length === 0 ? (
96 |         <div style={{ textAlign: 'center', padding: '80px 20px', background: 'var(--bg-surface)', borderRadius: '16px', width: '100%' }}>
97 |           <BookOpen size={56} style={{ color: 'var(--text-muted)', marginBottom: '20px', opacity: 0.3 }} />
98 |           <h3 style={{ color: 'var(--text-muted)', fontSize: '18px', margin: 0 }}>Chưa có bài giảng
99 |           <p style={{ color: 'var(--text-muted)', fontSize: '14px', margin: 0 }}>
100 |             </p>
101 |         </div>
102 |       ) : (
103 |         <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill, minmax(340px, 1fr))', gap: '24px 24px' }}>
104 |           {lessons.map(lesson => (
105 |             <div key={lesson.id} className="lms-card" style={{ padding: 0, overflow: 'hidden', transition: 'all 0.25s ease', border: '1px solid var(--border-default)' }}>
106 |               <div style={{ height: '180px', background: 'var(--bg-elevated)', position: 'relative' }}>
107 |                 {lesson.thumbnail_url ? (
108 |                   <img src={lesson.thumbnail_url} alt={lesson.title} style={{ width: '100%', height: '100%', objectFit: 'cover' }} />
109 |                 ) : (
110 |                   <div style={{ display: 'flex', alignItems: 'center', justify-content: 'center', height: '100%' }}>
111 |                     <ImageIcon size={48} />
112 |                   </div>
113 |                 )
114 |               <div style={{ position: 'absolute', top: '12px', left: '12px', padding: '4px 10px', borderRadius: '99px', fontSize: '11px', fontWeight: 700, background: 'linear-gradient(to right, #f0f0f0 49%, #fff 49% 51%, #fff 51% 53%, #f0f0f0 53%)', color: 'var(--text-muted)', display: 'flex', gap: '5px' }}>
115 |                 {lesson.is_published ? <Eye size={13} /> : <EyeOff size={13} />}
116 |               </div>
117 |             </div>
118 |             <div style={{ padding: '20px' }}>
119 |               <h3 style={{ fontSize: '16px', fontWeight: 700, margin: '0 0 6px 0', color: 'var(--text-primary)' }}>
120 |                 <p style={{ color: 'var(--text-muted)', fontSize: '13px', margin: '0 0 16px 0', display: '-webkit-text-overflow: ellipsis' }}>
121 |                   {lesson.description || 'Chưa có mô tả'}
122 |                 </p>
123 |                 {lesson.classNames?.length > 0 && (
124 |                   <div style={{ display: 'flex', flexWrap: 'wrap', gap: '4px', margin: '0 0 12px 0' }}>
125 |                     {lesson.classNames.map((name, i) => (
126 |                       <span key={i} style={{ fontSize: '10px', padding: '2px 8px', background: 'rgba(59,130,220,0.1)' }}>
127 |                         {name}
128 |                       </span>
129 |                     ))}
130 |                   </div>
131 |                 </div>
132 |                 <div style={{ display: 'flex', justify-content: 'space-between', align-items: 'center' }}>
133 |                   <span style={{ fontSize: '12px', color: 'var(--text-muted)', font-weight: 500 }}>
134 |                     {lesson.sections?.[0]?.count || 0} mã
135 |                   </span>
136 |                   <div style={{ display: 'flex', gap: '8px' }}>
137 |                     <Link href="/teacher/lessons/${lesson.id}">
138 |                       <button style={{ padding: '7px 14px', borderRadius: '8px', border: 'none', cursor: 'pointer', background: 'var(--brand-subtle)', color: 'var(--brand-primary)', font-weight: 600, fontSize: '12px', display: 'flex', align-items: 'center', gap: '5px' }}>
139 |                         <Edit size={14} /> Sửa
140 |                       </button>
141 |                     </div>
142 |                   </div>
143 |                 </div>
144 |               </div>
145 |             </div>
146 |           )
147 |         </div>
148 |       )
149 |     </main>
150 |   </div>
151 | )

```

```

156 |         </button>
157 |         </Link>
158 |         <button onClick={() => deleteLesson(lesson.id)} style={{
159 |             padding: '7px 10px', borderRadius: '8px', border: 'none', cursor: 'pointer',
160 |             background: 'rgba(244,106,106,0.1)', color: 'var(--danger)', fontFamily: 'inherit'
161 |         }}>
162 |             <Trash2 size={14} />
163 |         </button>
164 |     </div>
165 | </div>
166 | </div>
167 | </div>
168 | ))}
169 | </div>
170 | )}
171 | </main>
172 | </div>
173 | )
174 | }
175 |

```

File: app/(dashboard)/teacher/loading.tsx (15 lines, 0.5 KB)

```

1 | // Path: app/(dashboard)/teacher/loading.tsx
2 | export default function Loading() {
3 |     return (
4 |         <div className="p-6 space-y-4 animate-pulse">
5 |             <div className="h-8 bg-[#1d1f2a] rounded-lg w-48" />
6 |             <div className="grid grid-cols-1 md:grid-cols-4 gap-4">
7 |                 {[...Array(4)].map((_, i) => (
8 |                     <div key={i} className="h-24 bg-[#1d1f2a] rounded-xl" />
9 |                 ))}
10 |             </div>
11 |             <div className="h-64 bg-[#1d1f2a] rounded-xl" />
12 |         </div>
13 |     )
14 | }
15 |

```

File: app/(dashboard)/teacher/page.tsx (182 lines, 10.5 KB)

```

1 | 'use client';
2 |
3 | import React, { useEffect, useState, useCallback } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import {
6 |     Users, FileText, BookOpen, GraduationCap, TrendingUp, ArrowRight, Loader2, AlertCircle
7 | } from 'lucide-react';
8 | import Link from 'next/link';
9 |
10 | export default function TeacherDashboard() {
11 |     const [stats, setStats] = useState({ classes: 0, students: 0, assignments: 0, submissions: 0 });
12 |     const [classes, setClasses] = useState([]);
13 |     const [loading, setLoading] = useState(true);
14 |     const [realtimeLog, setRealtimeLog] = useState<string[][]>([]);
15 |
16 |     const addLog = useCallback((msg: string) => {
17 |         setRealtimeLog(prev => [msg, ...prev].slice(0, 10));
18 |     }, []);
19 |
20 |     useEffect(() => {
21 |         let channel: any = null;
22 |         let cancelled = false;
23 |
24 |         async function init() {
25 |             const { data: { user } } = await supabase.auth.getUser();
26 |             if (!user || cancelled) return;
27 |
28 |             const { data: classesData, count: classesCount } = await supabase.from('classes').select('*', { count: '...'
29 |             if (cancelled) return;
30 |             setClasses(classesData || []);
31 |             const classIds = classesData?.map(c => c.id) || [];
32 |
33 |             const { count: studentsCount } = await supabase.from('class_members').select('*', { count: 'exact', head: true
34 |             const { count: asgCount } = await supabase.from('assignments').select('*', { count: 'exact', head: true
35 |             const { count: subCount } = await supabase.from('submissions').select('*', { count: 'exact', head: true
36 |             if (!cancelled) {
37 |                 setStats({ classes: classesCount || 0, students: studentsCount || 0, assignments: asgCount || 0, submi...
38 |                 setLoading(false);
39 |             }
40 |
41 |             if (classIds.length === 0) return;
42 |
43 |             channel = supabase.channel('teacher-dashboard-realtime');
44 |             channel.on('postgres_changes',
45 |                 { event: 'INSERT', schema: 'public', table: 'class_members', filter: `class_id=in.${classIds.join(',')}`...
46 |                 () => {
47 |                     setStats(prev => ({ ...prev, students: prev.students + 1 }));
48 |                     addLog(`👤 ${classMembers[0].first_name} ${classMembers[0].last_name} đã tham gia lớp`);
49 |                 }
50 |             );
51 |             channel.on('postgres_changes',
52 |                 { event: 'INSERT', schema: 'public', table: 'submissions', filter: `class_id=in.${classIds.join(',')}`...
53 |                 (payload: any) => {
54 |                     setStats(prev => ({ ...prev, submissions: prev.submissions + 1 }));
55 |                     addLog(`📝 Bài nộp mới về lịch sinh`);
56 |                 }
57 |             );
58 |             channel.on('postgres_changes',
59 |                 { event: 'INSERT', schema: 'public', table: 'assignments', filter: `teacher_id=eq.${user.id}` },
60 |                 () => {
61 |                     setStats(prev => ({ ...prev, assignments: prev.assignments + 1 }));
62 |                     addLog(`📅 Nhiệm vụ mới về hành động`);
63 |                 }
64 |             );
65 |             channel.subscribe();
66 |         }
67 |
68 |         init();
69 |         return () => {
70 |             cancelled = true;
71 |             if (channel) supabase.removeChannel(channel);
72 |         };
73 |     }, [addLog]);
74 |
75 |     const quickActions = [
76 |         { title: 'Quản lý Lớp học', desc: 'Tạo mã lớp, quản lý thành viên', icon: '👤' },
77 |         { title: 'Quản lý Bài giảng', desc: 'Soinh học, tải xuống, lịch t-p', icon: '📖' },
78 |         { title: 'Sách giáo khoa', desc: 'Nữi dụng sN theo ch°ing trìn h G D P T', icon: '📚' },
79 |         { title: 'Kiến thức mB r ù n g', desc: 'Th° vi c n ph s n c é n g v à c ó n g n g h c', icon: '🔍' }
80 |     ];

```

```

81 | if (loading) return <div style={{ display: 'flex', justifyContent: 'center', padding: '80px' }}><Loader2 siz...
82 |
83 |
84 | return (
85 |   <div style={{ maxWidth: '1200px', margin: '0 auto' }}>
86 |     <header style={{ marginBottom: '32px' }}>
87 |       <h1 style={{ fontSize: '28px', fontWeight: 800, margin: '0 0 6px 0', color: 'var(--text-primary)' }}>C...
88 |       <p style={{ color: 'var(--text-muted)', margin: 0 }}>Hôm nay Th...
89 |     </header>
90 |
91 |     <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(220px, 1fr))', gap: '20px',...
92 |       [[
93 |         { label: 'Số lớp đang quen lý', value: stats.classes, icon: <C...
94 |         { label: 'Tổng số học sinh', value: stats.students, icon: <U...
95 |         { label: 'Nhicm vấ ă gia o', value: stats.assignments, icon: <...
96 |       ]).map((s, i) => (
97 |         <div key={i} className="lms-card" style={{ padding: '24px', display: 'flex', alignItems: 'center', g...
98 |           <div style={{ width: '48px', height: '48px', borderRadius: '12px', background: `${s.color}15`, col...
99 |             {s.icon}
100 |           </div>
101 |           <div>
102 |             <div style={{ fontSize: '24px', fontWeight: 800, color: 'var(--text-primary)' }}>{s.value}</div>
103 |             <div style={{ fontSize: '13px', color: 'var(--text-muted)' }}>{s.label}</div>
104 |           </div>
105 |         </div>
106 |       )])
107 |     </div>
108 |
109 |     {realtimeLog.length > 0 && (
110 |       <div style={{ marginBottom: '24px' }}>
111 |         <h2 style={{ fontSize: '16px', fontWeight: 700, color: 'var(--text-primary)', marginBottom: '12px', ...
112 |           <span style={{ width: '8px', height: '8px', borderRadius: '50%', background: '#22c55e', animation:...
113 |             H o i t U n g t r ã c t i ế p
114 |         </h2>
115 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '6px' }}>
116 |           {realtimeLog.map((msg, i) => (
117 |             <div key={i} style={{
118 |               padding: '8px 14px', borderRadius: '8px', fontSize: '13px',
119 |               background: i === 0 ? 'rgba(34,197,94,0.08)' : 'var(--bg-surface)',
120 |               border: 1px solid $i === 0 ? 'rgba(34,197,94,0.2)' : 'var(--border-default)',
121 |               color: 'var(--text-primary)', animation: i === 0 ? 'fadeSlideIn 0.3s ease-out' : 'none',
122 |             }}>
123 |               {msg}
124 |             </div>
125 |           )])
126 |         </div>
127 |       </div>
128 |     )}
129 |
130 |     <div className="teacher-dashboard-grid" style={{ display: 'grid', gridTemplateColumns: '1fr 360px', gap:...
131 |       <div>
132 |         <h2 style={{ fontSize: '18px', fontWeight: 700, color: 'var(--text-primary)', marginBottom: '16px' }}...
133 |         <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(260px, 1fr))', gap: '16...
134 |           {quickActions.map((action, idx) => (
135 |             <Link key={idx} href={action.href} style={{ textDecoration: 'none' }}>
136 |               <div className="lms-card" style={{ padding: '20px', display: 'flex', alignItems: 'center', gap:...
137 |                 <div style={{ flex: 1, minWidth: 0 }}>
138 |                   <div style={{ color: 'var(--text-primary)', fontSize: '14px', fontWeight: 700, marginBotto...
139 |                     <div style={{ color: 'var(--text-muted)', fontSize: '12px', margin: 0, overflow: 'hidden',...
140 |                       {action.icon}
141 |                     </div>
142 |                   </div>
143 |                   <div style={{ flex: 1, minWidth: 0 }}>
144 |                     <div style={{ color: 'var(--text-muted)', fontSize: '12px', margin: 0, overflow: 'hidden',...
145 |                       <ArrowRight size={16} style={{ color: 'var(--text-muted)', flexShrink: 0 }} />
146 |                     </div>
147 |                   </div>
148 |                 </Link>
149 |               </div>
150 |             </div>
151 |           </div>
152 |
153 |           <div>
154 |             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom: ...
155 |               <h2 style={{ fontSize: '18px', fontWeight: 700, color: 'var...
156 |               <Link href="/teacher/classes/new" style={{ color: 'var(--accent-blue)', fontSize: '13px', fontWeig...
157 |             </div>
158 |             <div style={{ display: 'flex', flexDirection: 'column', gap: '10px' }}>
159 |               {classes.length > 0 ? classes.map(cls => (
160 |                 <div key={cls.id} className="lms-card" style={{ padding: '16px', display: 'flex', justifyConten...
161 |                   <div>
162 |                     <div style={{ fontSize: '14px', fontWeight: 700, color: 'var(--text-primary)' }}>{cls.name}<...
163 |                     <div style={{ fontSize: '12px', color: 'var(--text-muted)' }}>{cls.subject}{cls.grade ? ` ` * ...
164 |                   </div>
165 |                   <div style={{ textAlign: 'right' }}>
166 |                     <div style={{ fontSize: '10px', color: 'var(--accent-blue)', fontWeight: 700, letterSpacing:...
167 |                     <div style={{ fontSize: '16px', fontWeight: 900, color: 'var(--text-primary)', fontFamily: '...
168 |                   </div>
169 |                 </div>
170 |               ) : (
171 |                 <div style={{ padding: '32px', textAlign: 'center', border: '2px dashed var(--border-default)', ...
172 |                   C h ° a c o l u p h i c n a o
173 |                 </div>
174 |               </div>
175 |             </div>
176 |           </div>
177 |         </div>
178 |         <style>{@keyframes pulse { 0%,100% { opacity:1; } 50% { opacity:0.4; } } @keyframes fadeSlideIn { from ...
179 |       </div>
180 |     );
181 |   }
182 |

```

File: app/about/page.tsx (517 lines, 39.9 KB)

```

1 | 'use client';
2 |
3 | import React, { useEffect, useState } from 'react';
4 | import Link from 'next/link';
5 | import { motion } from 'framer-motion';
6 | import { ArrowLeft, Phone, Mail, MapPin, ExternalLink, ShieldCheck, Heart, Loader2, Sparkles, BookOpen, Award,...
7 | import { supabase } from '@lib/supabase';
8 |
9 | function CollapsibleGroup({ title, children, defaultOpen = false }) {
10 |   const [open, setOpen] = useState(defaultOpen);
11 |   return (
12 |     <div style={{ marginBottom: '8px' }}>
13 |       <button onClick={() => setOpen(!open)} style={{
14 |         display: 'flex', alignItems: 'center', gap: '8px', width: '100%',
15 |         padding: '10px 16px', borderRadius: '10px',
16 |         background: 'rgba(255,255,255,0.02)', border: '1px solid rgba(255,255,255,0.06)',
17 |         color: 'rgba(255,255,255,0.6)', cursor: 'pointer',
18 |         fontSize: '12px', fontWeight: 700, letterSpacing: '0.5px',

```

```
19 |         fontFamily: 'inherit', textTransform: 'uppercase',
20 |     }}>
21 |     <span style={{ flex: 1, textAlign: 'left' }}>{title}</span>
22 |     <ChevronDown size={14} style={{ transform: open ? 'rotate(180deg)' : 'rotate(0deg)', transitio...
23 | </button>
24 |     {open && (
25 |       <div style={{ padding: '16px 16px 8px' }}>
26 |         {children}
27 |       </div>
28 |     )}
29 |   </div>
30 | );
31 | }
32 |
33 | export default function AboutPage() {
34 |   const [team, setTeam] = useState([]);
35 |   const [loading, setLoading] = useState(true);
36 |
37 |   useEffect(() => {
38 |     async function fetchTeam() {
39 |       const { data } = await supabase.from('project_team').select('*').order('order_index', { ascending:...
40 |       if (data && data.length > 0) {
41 |         setTeam(data);
42 |       } else {
43 |         setTeam([
44 |           { id: '1', full_name: 'Nguyễn Phúc Khánh Sin', image
45 |           { id: '2', full_name: 'Nguyễn An', image_url: '/t
46 |           { id: '3', full_name: 'Nguyễn Phím Gia Khiêm', image
47 |         ]);
48 |       }
49 |       setLoading(false);
50 |     }
51 |     fetchTeam();
52 |   }, []);
53 |
54 |   return (
55 |     <div style={{ minHeight: '100vh', background: 'var(--bg-base)', color: 'var(--text-primary)', fontFami...
56 |       <header style={{
57 |         height: '50vh', position: 'relative', overflow: 'hidden',
58 |         display: 'flex', flexDirection: 'column', alignItems: 'center', justifyContent: 'center',
59 |         textAlign: 'center', background: 'radial-gradient(circle at center, rgba(0, 243, 255, 0.05) 0%...
60 |       }}>
61 |         <div style={{ position: 'absolute', top: '40px', left: '40px', zIndex: 100 }}>
62 |           <Link href="/" style={{
63 |             display: 'flex', alignItems: 'center', gap: '10px', color: '#fff', textDecoration: 'no...
64 |             fontWeight: 700, background: 'rgba(255,255,255,0.03)', padding: '12px 24px',
65 |             borderRadius: '16px', border: '1px solid rgba(255,255,255,0.08)', backdropFilter: 'blu...
66 |           }}>
67 |             <ArrowLeft size={20} /> Quay lại trang chủ
68 |         </div>
69 |         <div style={{ maxWidth: '1000px', padding: '0 24px', position: 'relative', zIndex: 1 }}>
70 |           <div style={{ display: 'inline-flex', alignItems: 'center', gap: '8px', padding: '8px 16px...
71 |             <Sparkles size={14} /> PC MASTER BUILDER
72 |           </div>
73 |           <h1 style={{ fontSize: 'clamp(40px, 8vw, 64px)', fontWeight: 900, marginBottom: '24px', li...
74 |             V A C H Ứ N G T Ồ I
75 |         </h1>
76 |       </div>
77 |     </header>
78 |
79 |     <section style={{ padding: '80px 24px', background: 'linear-gradient(to bottom, #050507, #0a0a0c)'...
80 |       <div style={{ maxWidth: '1200px', margin: '0 auto', display: 'grid', gridTemplateColumns: 'rep...
81 |         <div style={{ animation: 'fadeInLeft 1s ease' }}>
82 |           <div style={{ display: 'flex', alignItems: 'center', gap: '16px', marginBottom: '32px'...
83 |             <div style={{ width: '48px', height: '48px', borderRadius: '12px', background: 'rg...
84 |               <BookOpen size={24} />
85 |             </div>
86 |             <h2 style={{ fontSize: '32px', fontWeight: 800, margin: 0, letterSpacing: '-0.02em...
87 |           </div>
88 |           <div style={{ fontSize: '18px', color: '#8899a6', lineHeight: 1.8, marginBottom: '32px'...
89 |             <p style={{ fontStyle: 'italic', color: '#fff', fontSize: '20px', marginBottom: '2...
90 |               " P C M a s t e r B u i l d e r " l à đ ã n g i ệ i p h á p
91 |             </p>
92 |             <p>
93 |               C h ú n g t ô i k h a o k h á t é n g đ ă n g s é c m i n h c
94 |             </p>
95 |           </div>
96 |           <div style={{
97 |             padding: '24px', borderRadius: '20px', background: 'rgba(0, 243, 255, 0.03)',
98 |             borderLeft: '4px solid var(--brand-primary)', color: 'var(--brand-primary)',
99 |             fontWeight: 700, fontSize: '16px', letterSpacing: '0.05em', lineHeight: 1.5
100 |           }}>
101 |             V Ú I T I N H T H Ị N " Đ Á M N G H ( - Đ Á M L À M " , C H Ú N G T
102 |           </div>
103 |         </div>
104 |
105 |         <motion.div
106 |           initial={{ opacity: 0, x: 30 }}
107 |           whileInView={{ opacity: 1, x: 0 }}
108 |           viewport={{ once: true }}
109 |           transition={{ duration: 0.6 }}
110 |           style={{ perspective: '1200px' }}
111 |         >
112 |           <motion.div
113 |             animate={{ y: [0, -5, 0] }}
114 |             transition={{ duration: 4, repeat: Infinity, ease: 'easeInOut' }}
115 |             style={{
116 |               position: 'relative', borderRadius: '32px', overflow: 'hidden',
117 |               boxShadow: '0 25px 50px -12px rgba(0, 243, 255, 0.25)',
118 |               border: '1px solid rgba(255,255,255,0.08)',
119 |               aspectRatio: '16/9',
120 |               display: 'flex', alignItems: 'center', justifyContent: 'center',
121 |               background: 'radial-gradient(circle at 30% 40%, rgba(0, 212, 170, 0.08) 0%, rg...
122 |               transition: 'box-shadow 0.4s ease, border-color 0.4s ease',
123 |               transformStyle: 'preserve-3d',
124 |             }}
125 |             onMouseEnter={e => {
126 |               e.currentTarget.style.boxShadow = '0 35px 70px -12px rgba(0, 243, 255, 0.45), ...
127 |               e.currentTarget.style.borderColor = 'var(--brand-primary)';
128 |               e.currentTarget.style.transform = 'rotateY(-2deg) rotateX(2deg)';
129 |             }}
130 |             onMouseLeave={e => {
131 |               e.currentTarget.style.boxShadow = '0 25px 50px -12px rgba(0, 243, 255, 0.25)';
132 |               e.currentTarget.style.borderColor = 'rgba(255,255,255,0.08)';
133 |               e.currentTarget.style.transform = 'rotateY(0deg) rotateX(0deg)';
134 |             }}
135 |           >
136 |           <svg viewBox="0 0 320 180" style={{ width: '100%', height: '100%', position: 'rela...
137 |
138 |
139 |
140 |
141 |
142 |
```

```

143 <defs>
144 <linearGradient id="caseGrad" x1="0" y1="0" x2="1" y2="1">
145 <stop offset="0%" stopColor="#1a2a3a" />
146 <stop offset="100%" stopColor="#0d1520" />
147 </linearGradient>
148 <linearGradient id="screenGrad" x1="0" y1="0" x2="0" y2="1">
149 <stop offset="0%" stopColor="#0a1628" />
150 <stop offset="100%" stopColor="#0f2038" />
151 </linearGradient>
152 <linearGradient id="ledBlue" x1="0" y1="0" x2="1" y2="0">
153 <stop offset="0%" stopColor="#00d4aa">
154 <animate attributeName="stopColor" values="#00d4aa;#6366f1;#00d4aa...
155 </stop>
156 <stop offset="100%" stopColor="#6366f1">
157 <animate attributeName="stopColor" values="#6366f1;#a855f7;#6366f1...
158 </stop>
159 </linearGradient>
160 <filter id="glow">
161 <feGaussianBlur stdDeviation="2" result="blur" />
162 <feMerge><feMergeNode in="blur" /><feMergeNode in="SourceGraphic" /></...
163 </filter>
164 <filter id="screenGlow">
165 <feGaussianBlur stdDeviation="4" result="blur" />
166 <feMerge><feMergeNode in="blur" /><feMergeNode in="SourceGraphic" /></...
167 </filter>
168 </defs>
169
170 { /* MONITOR STAND */ }
171 <rect x="145" y="125" width="30" height="8" rx="2" fill="#1a2a3a" />
172 <rect x="155" y="133" width="10" height="12" rx="1" fill="#141e2c" />
173
174 { /* MONITOR BODY */ }
175 <rect x="60" y="20" width="200" height="115" rx="6" fill="#1a2a3a" stroke="#2a...
176 <rect x="68" y="28" width="184" height="99" rx="3" fill="url(#screenGrad)" />
177
178 { /* SCREEN CONTENT - animated code lines */ }
179 {[0, 1, 2, 3, 4, 5].map(i => {
180 <rect key={i} x="78" y={35 + i * 15} width={60 + Math.random() * 80} heigh...
181 <animate attributeName="opacity" values={` ${0.1 + i * 0.05};${0.3 + i ...
182 </rect>
183 }
184 <rect x="78" y="110" width="100" height="3" rx="1.5" fill="#00d4aa" opacity="0...
185 <animate attributeName="width" values="100;20;100" dur="2s" repeatCount="i...
186 </rect>
187 <rect x="160" y="110" width="15" height="3" rx="1.5" fill="#00d4aa" opacity="0...
188 <animate attributeName="opacity" values="0.6;1;0.6" dur="1s" repeatCount="...
189 </rect>
190
191 { /* SCREEN GLOW */ }
192 <rect x="68" y="28" width="184" height="99" rx="3" fill="none" stroke="#00d4aa...
193 <animate attributeName="opacity" values="0.2;0.5;0.2" dur="2s" repeatCount...
194 </rect>
195
196 { /* PC CASE */ }
197 <rect x="50" y="148" width="120" height="28" rx="4" fill="url(#caseGrad)" stro...
198
199 { /* CASE LED STRIP */ }
200 <rect x="52" y="148" width="116" height="3" rx="1.5" fill="url(#ledBlue)" filt...
201
202 { /* CASE FANS */ }
203 {[70, 94, 118].map((cx, i) => {
204 <g key={i} transform={`translate(${cx}, 162)`}>
205 <circle r="8" fill="none" stroke="#253545" strokeWidth="1" />
206 <g>
207 <animateTransform attributeName="transform" type="rotate" from="0"...
208 {[0, 120, 240].map(angle => {
209 <line key={angle} x1="0" y1="0" x2="0" y2="-7" stroke="#00d4aa...
210 <animate attributeName="opacity" values="0.5;1;0.5" dur={` ${1.5 + i ...
211 </line>
212 </g>
213 </g>
214 </g>
215 }
216 )
217 }
218 <rect x="158" y="155" width="50" height="18" rx="2" fill="#1a1a2e" stroke="#a8...
219 <animate attributeName="stroke-opacity" values="0.3;0.8;0.3" dur="1.5s" re...
220 </rect>
221 <text x="162" y="167" fill="#a855f7" fontSize="5" fontWeight="700" letterSpaci...
222
223 { /* RAM */ }
224 {[210, 220].map((rx, i) => {
225 <rect key={i} x={rx} y="158" width="6" height="14" rx="1" fill="#0d1520" s...
226 <animate attributeName="stroke-opacity" values="0.3;0.9;0.3" dur={` ${1...
227 </rect>
228 }
229 )
230
231 { /* POWER LED */ }
232 <circle cx="55" cy="163" r="1.5" fill="#00d4aa" filter="url(#glow)">
233 <animate attributeName="r" values="1.5;2.5;1.5" dur="1s" repeatCount="inde...
234 </circle>
235
236 { /* RIGHT SIDE decorative dots */ }
237 {[0, 1, 2].map(i => {
238 <circle key={`dot-${i}`} cx={240 + i * 12} cy={154 + i * 8} r="1.5" fill="...
239 <animate attributeName="opacity" values="0.2;0.8;0.2" dur={` ${1.5 + i ...
240 </circle>
241 }
242 )
243
244 { /* WAVES below */ }
245 <path d="M50,178 Q80,172 110,178 Q140,184 170,178" fill="none" stroke="#00d4aa...
246 <animate attributeName="d" values="M50,178 Q80,172 110,178 Q140,184 170,17...
247 </path>
248 </svg>
249 <div style={{
250 position: 'absolute', bottom: '14px', left: '50%', transform: 'translateX(-50%...
251 padding: '6px 16px', borderRadius: '99px',
252 background: 'rgba(0,0,0,0.7)', backdropFilter: 'blur(8px)',
253 border: '1px solid rgba(0,212,170,0.2)',
254 display: 'flex', alignItems: 'center', gap: '6px',
255 }}>
256 <cpu size={12} color="#00d4aa" />
257 <span style={{ fontSize: '11px', fontWeight: 700, color: '#fff', letterSpacing...
258 PC LAB 2D
259 </span>
260 </div>
261 </motion.div>
262 </motion.div>
263 </div>
264 <section style={{ padding: '80px 24px', background: 'linear-gradient(to bottom, #0a0a0c, #050507)'...
265 <div style={{ maxWidth: '1200px', margin: '0 auto' }}>
266 <div style={{ textAlign: 'center', marginBottom: '80px' }}>

```

```
267|         <div style={{ display: 'inline-flex', alignItems: 'center', gap: '12px', padding: '10p...
268|             < A w a r d   s i z e = { 1 6 }   / >   Ø I   N G h   G I Á O   V I Ê N
269|         </div>
270|         <h2 style={{ fontSize: 'clamp(32px, 5vw, 48px)', fontWeight: 900, letterSpacing: '-0.0...
271|             Giáo Viên <span style={{ color: 'var(--brand-primary)', background: 'linear-gradie...
272|         </h2>
273|         <p style={{ color: 'rgba(255,255,255,0.4)', fontSize: '16px', marginTop: '16px', maxWi...
274|             N h ì n g   n g ° Y ì   t - n   t â m   h ° Ò n g   d « n   v à   t r u y Á n   c ã m
275|         </p>
276|     </div>
277|
278|     <div style={{ display: 'flex', flexWrap: 'wrap', justifyContent: 'center', gap: '80px', ma...
279|         { / *   G i á o   v i ê n   1 :   T r ậ n   M i n h   P h ầ n g   * / }
280|         <div style={{ textAlign: 'center', width: '320px' }}>
281|             <div style={{
282|                 position: 'relative', width: '220px', height: '220px',
283|                 margin: '0 auto 32px', borderRadius: '50%', overflow: 'hidden',
284|                 border: '3px solid rgba(0, 212, 170, 0.3)', background: '#0a0a0c',
285|                 boxShadow: '0 0 40px rgba(0, 212, 170, 0.15), inset 0 0 40px rgba(0, 212, 170,...
286|                 transition: 'all 0.4s ease',
287|             }}
288|             onMouseEnter={e => { e.currentTarget.style.transform = 'scale(1.05)'; e.currentTar...
289|             onMouseLeave={e => { e.currentTarget.style.transform = 'scale(1)'; e.currentTarget...
290|             >
291|                 
296|                 <div style={{ position: 'absolute', bottom: '8px', right: '8px', width: '36px'...
297|                     <ShieldCheck size={18} color="#fff" />
298|                 </div>
299|             </div>
300|             <h3 style={{ fontSize: '26px', fontWeight: 800, margin: '0 0 6px', background: 'li...
301|             <p style={{ fontSize: '14px', color: 'var(--brand-primary)', margin: '0 0 8px', fo...
302|             <p style={{ fontSize: '12px', color: 'rgba(255,255,255,0.3)', margin: 0, fontStyle...
303|         </div>
304|
305|         { / *   G i á o   v i ê n   2 :   o à n   T h ầ y   K i m   P h ồ   ầ n g   * / }
306|         <div style={{ textAlign: 'center', width: '320px' }}>
307|             <div style={{
308|                 position: 'relative', width: '220px', height: '220px',
309|                 margin: '0 auto 32px', borderRadius: '50%', overflow: 'hidden',
310|                 border: '3px solid rgba(0, 243, 255, 0.3)', background: '#0a0a0c',
311|                 boxShadow: '0 0 40px rgba(0, 243, 255, 0.15), inset 0 0 40px rgba(0, 243, 255,...
312|                 transition: 'all 0.4s ease',
313|             }}
314|             onMouseEnter={e => { e.currentTarget.style.transform = 'scale(1.05)'; e.currentTar...
315|             onMouseLeave={e => { e.currentTarget.style.transform = 'scale(1)'; e.currentTarget...
316|             >
317|                 
322|                 <div style={{ position: 'absolute', bottom: '8px', right: '8px', width: '36px'...
323|                     <ShieldCheck size={18} color="#000" />
324|                 </div>
325|             </div>
326|             <h3 style={{ fontSize: '26px', fontWeight: 800, margin: '0 0 6px', background: 'li...
327|             <p style={{ fontSize: '14px', color: '#00f3ff', margin: '0 0 8px', fontWeight: 600...
328|             <p style={{ fontSize: '12px', color: 'rgba(255,255,255,0.3)', margin: 0, fontStyle...
329|         </div>
330|     </div>
331|
332|     <section style={{ padding: '80px 24px' }}>
333|         <div style={{ textAlign: 'center', marginBottom: '80px' }}>
334|             <div style={{ display: 'inline-flex', alignItems: 'center', gap: '12px', padding: ...
335|                 < U s e r s   s i z e = { 1 6 }   / >   Ø I   N G h   P H Á T   T R I Á N
336|             </div>
337|             <h2 style={{ fontSize: 'clamp(32px, 5vw, 48px)', fontWeight: 900, letterSpacing: '...
338|                 U i   N g ì   < s p a n   s t y l e = { {   c o l o r :   ' v a r ( - - b r a
339|             </h2>
340|             <p style={{ color: 'rgba(255,255,255,0.4)', fontSize: '16px', marginTop: '16px', m...
341|                 N h ì n g   c o n   n g ° Y ì   t â m   h u y ế t   t i o   n ê n   P C   M a s
342|             </p>
343|         </div>
344|
345|         {loading ? (
346|             <div style={{ textAlign: 'center' }}><Loader2 className="animate-spin" size={40} c...
347|         ) : (
348|             <div style={{ display: 'flex', flexWrap: 'wrap', justifyContent: 'center', gap: '6...
349|                 {team.map((member, idx) => {
350|                     const isKhángOrAn = member.full_name
351|                     const borderGradients = [
352|                         'linear-gradient(135deg, #00d4aa, #00f3ff)',
353|                         'linear-gradient(135deg, #ffb900, #ff8b00)',
354|                         'linear-gradient(135deg, #a78bfa, #6366f1)',
355|                         'linear-gradient(135deg, #f472b6, #ec4899)',
356|                     ];
357|                     return (
358|                         <div key={member.id} style={{ textAlign: 'center', width: '320px' }}>
359|                             <div style={{
360|                                 position: 'relative', width: '280px', height: '280px',
361|                                 margin: '0 auto 28px', borderRadius: '40px', overflow: 'hidden',
362|                                 border: '2px solid rgba(255,255,255,0.1)', background: '#0a0a0c',
363|                                 transition: 'all 0.4s cubic-bezier(0.4, 0, 0.2, 1)',
364|                                 boxShadow: '0 8px 32px rgba(0,0,0,0.3)',
365|                             }}
366|                             onMouseEnter={e => { e.currentTarget.style.transform = 'translateY...
367|                             onMouseLeave={e => { e.currentTarget.style.transform = 'translateY...
368|                             >
369|                                 <img src={member.image_url} alt={member.full_name} style={{ wi...
370|                                 <div style={{
371|                                     position: 'absolute', bottom: 0, left: 0, right: 0,
372|                                     background: 'linear-gradient(to top, rgba(0,0,0,0.8) 0%, t...
373|                                     padding: '24px 16px 16px',
374|                                     display: 'flex', flexDirection: 'column', alignItems: 'cen...
375|                                 }}>
376|                                     <span style={{
377|                                         padding: '4px 14px', borderRadius: '99px',
378|                                         background: 'rgba(0,212,170,0.2)', color: '#00d4aa',
379|                                         fontSize: '11px', fontWeight: 700, letterSpacing: '1px',
380|                                     }}>
381|                                         {member.role || 'Thành viên'}
382|                                     </span>
383|                                 </div>
384|                             </div>
385|                             <h3 style={{ fontSize: '22px', fontWeight: 800, margin: '0 0 4px', ...
386|                             <div style={{ width: '40px', height: '3px', background: borderGrad...
387|                         </div>
388|                     );
389|                 }}
390|             </div>
```



```
391|         })
392|     </section>
393| </div>
394| </section>
395|
396| <section style={{ padding: '100px 24px', background: 'linear-gradient(to bottom, #050507, #0a0a0c)...
397| <div style={{ maxWidth: '1200px', margin: '0 auto' }}>
398|   <div style={{ textAlign: 'center', marginBottom: '60px' }}>
399|     <div style={{ display: 'inline-flex', alignItems: 'center', gap: '12px', padding: '10px...
400|       <MapPin size={16} /> L I Ê N H Ậ
401|     </div>
402|     <h2 style={{ fontSize: 'clamp(28px, 4vw, 42px)', fontWeight: 900, letterSpacing: '-0.0...
403|       T R Ò N G T H P T <span style={{ color: 'var(--brand-
404|     </h2>
405|   </div>
406|
407|   <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(350px, 1fr))'...
408|     <div>
409|       <div style={{ padding: '32px', borderRadius: '24px', background: 'rgba(255,255,255...
410|         <div style={{ display: 'flex', flexDirection: 'column', gap: '20px' }}>
411|           <div style={{ display: 'flex', gap: '16px', alignItems: 'flex-start' }}>
412|             <div style={{ width: '44px', height: '44px', borderRadius: '12px', bac...
413|               <MapPin color="var(--brand-primary)" size={20} />
414|             </div>
415|             <div>
416|               <p style={{ fontSize: '12px', color: 'rgba(255,255,255,0.3)', font...
417|               <p style={{ fontSize: '16px', fontWeight: 600, margin: 0, color: '...
418|             </div>
419|           </div>
420|           <div style={{ display: 'flex', gap: '16px', alignItems: 'flex-start' }}>
421|             <div style={{ width: '44px', height: '44px', borderRadius: '12px', bac...
422|               <Phone color="#00f3ff" size={20} />
423|             </div>
424|             <div>
425|               <p style={{ fontSize: '12px', color: 'rgba(255,255,255,0.3)', font...
426|               <p style={{ fontSize: '16px', fontWeight: 600, margin: 0, color: '...
427|             </div>
428|           </div>
429|           <div style={{ display: 'flex', gap: '16px', alignItems: 'flex-start' }}>
430|             <div style={{ width: '44px', height: '44px', borderRadius: '12px', bac...
431|               <Mail color="#a78bfa" size={20} />
432|             </div>
433|             <div>
434|               <p style={{ fontSize: '12px', color: 'rgba(255,255,255,0.3)', font...
435|               <p style={{ fontSize: '16px', fontWeight: 600, margin: 0, color: '...
436|             </div>
437|           </div>
438|           <a href="https://maps.app.goo.gl/CDytzt1bYAJ7zVew7" target="_blank"
439|             style={{ display: 'inline-flex', alignItems: 'center', gap: '8px', pad...
440|               <ExternalLink size={18} /> M ậ T
441|             </a>
442|           </div>
443|         </div>
444|       </div>
445|
446|       <div style={{ height: '450px', borderRadius: '32px', overflow: 'hidden', border: '1px ...
447|         <iframe
448|           src="https://www.google.com/maps/embed?pb=!1m14!1m8!1m3!1d979.2117895773255!2d...
449|           width="100%" height="100%" style={{ border: 0, borderRadius: '32px' }}
450|           allowFullScreen={true} loading="lazy" referrerPolicy="no-referrer-when-downgrade"
451|         ></iframe>
452|         <div style={{ position: 'absolute', top: '12px', left: '12px', padding: '6px 14px'...
453|           <ð T H P T N g u y Á n C ó n g T r é
454|         </div>
455|       </div>
456|     </div>
457|   </div>
458| </section>
459|
460| <footer style={{ padding: '60px 24px 60px', background: '#050507', borderTop: '1px solid rgba(255,...
461| <div style={{ maxWidth: '900px', margin: '0 auto' }}>
462|   <div style={{ display: 'flex', justify-content: 'center', gap: '32px', flexWrap: 'wrap'...
463|     <div style={{ textAlign: 'center', minWidth: '160px' }}>
464|       <p style={{ fontSize: '13px', font-weight: 700, color: '#fff', margin: '0 0 4px...
465|       <p style={{ fontSize: '11px', color: 'rgba(255,255,255,0.3)', margin: 0 }}>Dev...
466|     </div>
467|     <div style={{ width: '1px', background: 'rgba(255,255,255,0.08)', alignSelf: 'stre...
468|     <div style={{ textAlign: 'center', minWidth: '160px' }}>
469|       <p style={{ fontSize: '13px', font-weight: 700, color: '#fff', margin: '0 0 4px...
470|       <p style={{ fontSize: '11px', color: 'rgba(255,255,255,0.3)', margin: 0 }}>GV ...
471|     </div>
472|     <div style={{ width: '1px', background: 'rgba(255,255,255,0.08)', alignSelf: 'stre...
473|     <div style={{ textAlign: 'center', minWidth: '160px' }}>
474|       <p style={{ fontSize: '13px', font-weight: 700, color: '#fff', margin: '0 0 4px...
475|       <p style={{ fontSize: '11px', color: 'rgba(255,255,255,0.3)', margin: 0 }}>GV ...
476|     </div>
477|   </div>
478|   </CollapsibleGroup>
479|   <div style={{ marginTop: '24px', padding: '24px', borderTop: '1px solid rgba(255,255,25...
480|     <p style={{ fontSize: '12px', color: 'rgba(255,255,255,0.2)', margin: 0, textAlign: 'c...
481|       © { new Date().getFullYear() } P C M a s t e r B u i l d
482|     </p>
483|   </div>
484| </div>
485| </div>
486| </div>
487|
488| <style>{
489|   @keyframes floatBounce {
490|     0%, 100% { transform: translateY(0); }
491|     50% { transform: translateY(-6px); }
492|   }
493|   @keyframes floatRotate {
494|     from { transform: rotate(0deg); }
495|     to { transform: rotate(360deg); }
496|   }
497|   @keyframes pulseDot {
498|     0%, 100% { opacity: 0.3; }
499|     50% { opacity: 0.8; }
500|   }
501|   @keyframes fadeInLeft {
502|     from { opacity: 0; transform: translateX(-24px); }
503|     to { opacity: 1; transform: translateX(0); }
504|   }
505|   @keyframes fadeInRight {
506|     from { opacity: 0; transform: translateX(24px); }
507|     to { opacity: 1; transform: translateX(0); }
508|   }
509|   @keyframes aboutShowcaseGlow {
510|     0%, 100% { opacity: 0.3; }
511|     50% { opacity: 0.9; }
512|   }
513| }</style>
514| </div>
```

```
515 | );
516 | }
517 |
```

File: app/actions/auth.ts (82 lines, 2.3 KB)

```
1 | 'use server'
2 |
3 | import { createClient } from '@lib/supabase-ssr-server'
4 | import { revalidatePath } from 'next/cache'
5 |
6 | interface RegisterInput {
7 |   role: 'student' | 'teacher' | 'parent'
8 |   full_name: string
9 |   email: string
10 |   password: string
11 |   school_name?: string
12 |   student_code?: string
13 | }
14 |
15 | export async function registerAction(formData: RegisterInput) {
16 |   const { role, full_name, email, password, school_name, student_code } = formData
17 |
18 |   if (!email || !password || !full_name || !role) {
19 |     return { error: 'Vui lòng điền đầy đủ thông tin bắt buộc' }
20 |   }
21 |
22 |   if (password.length < 8) {
23 |     return { error: 'Mật khẩu phải có ít nhất 8 ký tự' }
24 |   }
25 |   if (!/[A-Z]/.test(password)) {
26 |     return { error: 'Mật khẩu phải chứa ít nhất 1 chữ hoa' }
27 |   }
28 |   if (!/[0-9]/.test(password)) {
29 |     return { error: 'Mật khẩu phải chứa ít nhất 1 số' }
30 |   }
31 |
32 |   const supabase = await createClient()
33 |
34 |   const { data: authData, error: authError } = await supabase.auth.signUp({
35 |     email,
36 |     password,
37 |     options: {
38 |       data: { full_name, role },
39 |       emailRedirectTo: `${process.env.NEXT_PUBLIC_SITE_URL} || 'http://localhost:3000'}/auth/callback`
40 |     }
41 |   })
42 |
43 |   if (authError) {
44 |     let msg = authError.message
45 |     if (msg.toLowerCase().includes('already registered') || msg.toLowerCase().includes('already exists')) {
46 |       msg = 'Email này đã đăng ký. Vui lòng sử dụng email khác hoặc'
47 |     }
48 |     return { error: msg }
49 |   }
50 |
51 |   if (authData.user) {
52 |     const { error: profileError } = await supabase.from('profiles').upsert({
53 |       id: authData.user.id,
54 |       email,
55 |       full_name,
56 |       role,
57 |       school_name: school_name ?? null,
58 |     })
59 |
60 |     if (profileError) {
61 |       console.error('Profile creation error:', profileError)
62 |     }
63 |
64 |     if (role === 'parent' && student_code) {
65 |       const { data: student } = await supabase
66 |         .rpc('find_student_by_code', { input_code: student_code })
67 |
68 |       if (student && student.length > 0) {
69 |         await supabase.from('parent_student_links').insert({
70 |           parent_id: authData.user.id,
71 |           student_id: student[0].student_id,
72 |           status: 'pending'
73 |         })
74 |       }
75 |     }
76 |   }
77 |
78 |   revalidatePath('/', 'layout')
79 |
80 |   return { success: true, requireEmailConfirm: true }
81 | }
82 |
```

File: app/actions/exam.ts (266 lines, 7.3 KB)

```
1 | 'use server'
2 |
3 | import { createClient } from '@lib/supabase-ssr-server'
4 | import { revalidatePath } from 'next/cache'
5 |
6 | async function getAuthUser(supabase: any) {
7 |   const { data: { user } } = await supabase.auth.getUser()
8 |   if (!user) throw new Error('Unauthorized')
9 |   return user
10 | }
11 |
12 | export async function getTeacherExams() {
13 |   const supabase = await createClient()
14 |   const user = await getAuthUser(supabase)
15 |
16 |   const { data: exams } = await supabase
17 |     .from('quizzes')
18 |     .select(`
19 |       *,
20 |       classes!left (name, code),
21 |       lessons!left (title),
22 |       quiz_attempts!left (id, score, student_id, status)
23 |     `)
24 |     .eq('teacher_id', user.id)
25 |     .order('created_at', { ascending: false })
26 |
27 |   if (!exams) return []
28 |
29 |   return exams.map((e: any) => {
30 |     const total = e.quiz_attempts?.length || 0
```

```

31 |     const done = e.quiz_attempts?.filter((a: any) => a.status !== 'in_progress')?.length || 0
32 |     const avgScore = done > 0
33 |       ? Math.round(e.quiz_attempts.filter((a: any) => a.status !== 'in_progress').reduce((s: number, a: any) => ...
34 |         : 0
35 |     return { ...e, _stats: { total, done, avgScore } }
36 |   })
37 | }
38 |
39 | export async function getTeacherExam(examId: string) {
40 |   const supabase = await createClient()
41 |   const user = await getAuthUser(supabase)
42 |
43 |   const { data: exam } = await supabase
44 |     .from('quizzes')
45 |     .select(`
46 |       *,
47 |       classes!left (name, code),
48 |       lessons!left (title)
49 |     `)
50 |     .eq('id', examId)
51 |     .eq('teacher_id', user.id)
52 |     .single()
53 |
54 |   if (!exam) throw new Error('Không tìm thấy Á thi')
55 |
56 |   const { data: questions } = await supabase
57 |     .from('quiz_questions')
58 |     .select(`
59 |       order,
60 |       questions!inner (
61 |         id, content, type, points, difficulty, explanation,
62 |         question_options (id, content, is_correct, "order")
63 |       )
64 |     `)
65 |     .eq('quiz_id', examId)
66 |     .order('order')
67 |
68 |   const { data: attempts } = await supabase
69 |     .from('quiz_attempts')
70 |     .select(`
71 |       *,
72 |       profiles!inner (id, full_name, email),
73 |       quiz_answers (id, question_id, selected_option_ids, is_correct, points_earned)
74 |     `)
75 |     .eq('quiz_id', examId)
76 |     .order('started_at', { ascending: false })
77 |
78 |   return { ...exam, questions: questions?.map((q: any) => q.questions) || [], attempts: attempts || [] }
79 | }
80 |
81 | export async function createExam(data: {
82 |   title: string
83 |   description: string
84 |   classId: string
85 |   lessonId?: string
86 |   timeLimitMinutes: number
87 |   passingScore: number
88 |   maxAttempts: number
89 |   availableFrom?: string
90 |   availableUntil?: string
91 |   requireCamera?: boolean
92 | }) {
93 |   const supabase = await createClient()
94 |   const user = await getAuthUser(supabase)
95 |
96 |   const { data: exam, error } = await supabase
97 |     .from('quizzes')
98 |     .insert({
99 |       teacher_id: user.id,
100 |       title: data.title,
101 |       description: data.description,
102 |       class_id: data.classId,
103 |       lesson_id: data.lessonId || null,
104 |       time_limit_minutes: data.timeLimitMinutes,
105 |       passing_score: data.passingScore,
106 |       max_attempts: data.maxAttempts,
107 |       available_from: data.availableFrom || null,
108 |       available_until: data.availableUntil || null,
109 |       is_published: false
110 |     })
111 |     .select()
112 |     .single()
113 |
114 |   if (error) throw error
115 |   revalidatePath('/teacher/exams')
116 |   return exam
117 | }
118 |
119 | export async function updateExam(examId: string, data: {
120 |   title?: string
121 |   description?: string
122 |   timeLimitMinutes?: number
123 |   passingScore?: number
124 |   maxAttempts?: number
125 |   availableFrom?: string
126 |   availableUntil?: string
127 | }) {
128 |   const supabase = await createClient()
129 |   await getAuthUser(supabase)
130 |
131 |   const { error } = await supabase
132 |     .from('quizzes')
133 |     .update(data)
134 |     .eq('id', examId)
135 |
136 |   if (error) throw error
137 |   revalidatePath('/teacher/exams')
138 |   return { success: true }
139 | }
140 |
141 | export async function deleteExam(examId: string) {
142 |   const supabase = await createClient()
143 |   await getAuthUser(supabase)
144 |
145 |   const { error } = await supabase
146 |     .from('quizzes')
147 |     .delete()
148 |     .eq('id', examId)
149 |
150 |   if (error) throw error
151 |   revalidatePath('/teacher/exams')
152 |   return { success: true }
153 | }
154 |

```

```

155 export async function publishExam(examId: string, publish: boolean) {
156   const supabase = await createClient()
157   const user = await getAuthUser(supabase)
158
159   const { data: exam } = await supabase
160     .from('quizzes')
161     .select('*')
162     .eq('id', examId)
163     .single()
164
165   if (!exam) throw new Error('Không tìm thấy Á thi')
166
167   const { error } = await supabase
168     .from('quizzes')
169     .update({ is_published: publish })
170     .eq('id', examId)
171
172   if (error) throw error
173
174   if (publish && exam.class_id) {
175     const { data: members } = await supabase
176       .from('class_members')
177       .select('student_id')
178       .eq('class_id', exam.class_id)
179
180     if (members && members.length > 0) {
181       const notifications = members.map((m: any) => ({
182         user_id: m.student_id,
183         type: 'exam_published',
184         title: `Á thi mới: ${exam.title}`,
185         body: `${exam.description} | 'Mặt Á thi mới đã được cập nhật.'`,
186         action_url: `/student/quiz/${exam.id}`,
187         data: { quizId: exam.id }
188       )))
189       await supabase.from('notifications').insert(notifications)
190     }
191   }
192
193   revalidatePath('/teacher/exams')
194   return { success: true }
195 }
196
197 export async function addQuestionToExam(data: {
198   examId: string
199   content: string
200   type: 'single_choice' | 'multiple_choice' | 'true_false' | 'fill_blank'
201   points: number
202   difficulty?: string
203   explanation?: string
204   options: { content: string; isCorrect: boolean }[]
205 }) {
206   const supabase = await createClient()
207   await getAuthUser(supabase)
208
209   const { data: question, error: qErr } = await supabase
210     .from('questions')
211     .insert({
212       content: data.content,
213       type: data.type,
214       points: data.points,
215       difficulty: data.difficulty || 'medium',
216       explanation: data.explanation || null
217     })
218     .select()
219     .single()
220
221   if (qErr || !question) throw qErr
222
223   if (data.options && data.options.length > 0) {
224     const optionRows = data.options.map((opt: any, i: number) => ({
225       question_id: question.id,
226       content: opt.content,
227       is_correct: opt.isCorrect,
228       order: i
229     })))
230     const { error: optErr } = await supabase.from('question_options').insert(optionRows)
231     if (optErr) throw optErr
232   }
233
234   const { error: linkErr } = await supabase
235     .from('quiz_questions')
236     .insert({ quiz_id: data.examId, question_id: question.id, order: 0 })
237
238   if (linkErr) throw linkErr
239   revalidatePath('/teacher/exams')
240   return question
241 }
242
243 export async function removeQuestionFromExam(questionId: string) {
244   const supabase = await createClient()
245   await getAuthUser(supabase)
246   const { error } = await supabase.from('questions').delete().eq('id', questionId)
247   if (error) throw error
248   revalidatePath('/teacher/exams')
249   return { success: true }
250 }
251
252 export async function getExamRealtimeStats(examId: string) {
253   const supabase = await createClient()
254   const { data: attempts } = await supabase
255     .from('quiz_attempts')
256     .select(
257       `id, student_id, score, max_score, status, started_at, submitted_at,
258        profiles!inner (full_name, email),
259        quiz_answers (question_id, is_correct, points_earned)
260     `)
261     .eq('quiz_id', examId)
262     .order('started_at', { ascending: false })
263
264   return attempts || []
265 }
266

```

File: app/actions/lessons.ts (69 lines, 2.0 KB)

```

1 | 'use server'
2 |
3 | import { createClient } from '@lib/supabase-ssr-server'
4 | import { revalidatePath } from 'next/cache'
5 |
6 | export async function publishLessonAction(lessonId: string) {
7 |   const supabase = await createClient()
8 |   const { data: { user } } = await supabase.auth.getUser()

```

```

9 |   if (!user) return { error: 'Unauthorized' }
10 |
11 |   const { error } = await supabase
12 |     .from('lessons')
13 |     .update({ is_published: true })
14 |     .eq('id', lessonId)
15 |     .eq('teacher_id', user.id)
16 |
17 |   if (error) return { error: error.message }
18 |
19 |   revalidatePath(`/teacher/lessons/${lessonId}`)
20 |   revalidatePath('/teacher/lessons')
21 |   return { success: true }
22 | }
23 |
24 |
25 | export async function unpublishLessonAction(lessonId: string) {
26 |   const supabase = await createClient()
27 |   const { data: { user } } = await supabase.auth.getUser()
28 |
29 |   if (!user) return { error: 'Unauthorized' }
30 |
31 |   const { error } = await supabase
32 |     .from('lessons')
33 |     .update({ is_published: false })
34 |     .eq('id', lessonId)
35 |     .eq('teacher_id', user.id)
36 |
37 |   if (error) return { error: error.message }
38 |
39 |   revalidatePath(`/teacher/lessons/${lessonId}`)
40 |   revalidatePath('/teacher/lessons')
41 |   return { success: true }
42 | }
43 |
44 | export async function deleteLessonAction(lessonId: string) {
45 |   const supabase = await createClient()
46 |   const { data: { user } } = await supabase.auth.getUser()
47 |
48 |   if (!user) return { error: 'Unauthorized' }
49 |
50 |   await supabase.from('lesson_sections').delete().eq('lesson_id', lessonId)
51 |   await supabase.from('lesson_progress').delete().eq('lesson_id', lessonId)
52 |   await supabase.from('lesson_class_assignments').delete().eq('lesson_id', lessonId)
53 |   const { error } = await supabase
54 |     .from('lessons')
55 |     .delete()
56 |     .eq('id', lessonId)
57 |     .eq('teacher_id', user.id)
58 |
59 |   if (error) return { error: error.message }
60 |
61 |   revalidatePath('/teacher/lessons')
62 |   return { success: true }
63 | }
64 |
65 | export async function incrementViewCountAction(lessonId: string) {
66 |   const supabase = await createClient()
67 |   await supabase.rpc('increment_lesson_views', { lesson_id: lessonId })
68 | }
69 |

```

File: app/actions/profile.ts (88 lines, 2.7 KB)

```

1 | 'use server'
2 |
3 | import { createClient } from '@lib/supabase-ssr-server'
4 | import { z } from 'zod'
5 | import { revalidatePath } from 'next/cache'
6 |
7 | export async function updateProfile(data: { full_name: string, bio: string, school: string, grade: string }) {
8 |   const supabase = await createClient()
9 |   const { data: { user } } = await supabase.auth.getUser()
10 |
11 |   if (!user) throw new Error('Unauthorized')
12 |
13 |   const { error } = await supabase
14 |     .from('profiles')
15 |     .update({
16 |       full_name: data.full_name,
17 |       bio: data.bio.slice(0, 200), // Max 200 chars
18 |       school: data.school,
19 |       grade: data.grade,
20 |       updated_at: new Date().toISOString()
21 |     })
22 |     .eq('id', user.id)
23 |
24 |   if (error) throw error
25 |   revalidatePath('/student/profile')
26 |   return { success: true }
27 | }
28 |
29 | export async function updatePreferences(data: { email_notifications: boolean, push_notifications: boolean, wee...
30 |   const supabase = await createClient()
31 |   const { data: { user } } = await supabase.auth.getUser()
32 |
33 |   if (!user) throw new Error('Unauthorized')
34 |
35 |   const { error } = await supabase
36 |     .from('user_preferences')
37 |     .upsert({
38 |       user_id: user.id,
39 |       email_notifications: data.email_notifications,
40 |       push_notifications: data.push_notifications,
41 |       weekly_digest: data.weekly_digest,
42 |       theme: data.theme,
43 |       language: data.language,
44 |       updated_at: new Date().toISOString()
45 |     })
46 |
47 |   if (error) throw error
48 |   revalidatePath('/student/profile')
49 |   return { success: true }
50 | }
51 |
52 | export async function changePassword(currentPassword: string, newPassword: string) {
53 |   const supabase = await createClient()
54 |
55 |   const { data: { user } } = await supabase.auth.getUser()
56 |   if (!user || !user.email) throw new Error('Unauthorized')
57 |
58 |   // Zod validation (Rule 5)
59 |   const passwordSchema = z.string()

```

```

60 |         .min(8, 'M-t kh@u ph£i tÑi thiẤu 8 ký tñ')
61 |         .regex(/[A-Z]/, 'M-t kh@u cSn ít nh¥t 1 chĩ cái in hoa')
62 |         .regex(/[0-9]/, 'M-t kh@u cSn ít nh¥t 1 chĩ sÑ')
63 |
64 |     const parseResult = passwordSchema.safeParse(newPassword)
65 |     if (!parseResult.success) {
66 |         throw new Error(parseResult.error.issues[0].message)
67 |     }
68 |
69 |     // 1. Re-authenticate user first (Rule 5)
70 |     const { error: authError } = await supabase.auth.signInWithPassword({
71 |         email: user.email,
72 |         password: currentPassword
73 |     })
74 |
75 |     if (authError) {
76 |         throw new Error('M-t kh@u hiçn tĩi không chính xác.')
77 |     }
78 |
79 |     // 2. Perform password update
80 |     const { error: updateError } = await supabase.auth.updateUser({
81 |         password: newPassword
82 |     })
83 |
84 |     if (updateError) throw updateError
85 |
86 |     return { success: true }
87 | }
88 |

```

File: app/actions/quiz.ts (378 lines, 11.5 KB)

```

1 | 'use server';
2 |
3 | import { createClient } from '@lib/supabase-ssr-server';
4 | import { revalidatePath } from 'next/cache';
5 | import { z } from 'zod';
6 | import { shuffleArray, scoreQuestion, StudentAnswer } from '@lib/quiz-engine';
7 |
8 | export async function getAuthUser(supabase: any) {
9 |     const { data: { user } } = await supabase.auth.getUser();
10 |     if (!user) throw new Error('Unauthorized');
11 |     return user;
12 | }
13 |
14 | export async function startQuizAttempt(quizId: string) {
15 |     const supabase = await createClient();
16 |     const user = await getAuthUser(supabase);
17 |
18 |     const { data: quiz, error: quizErr } = await supabase
19 |         .from('quizzes')
20 |         .select('**')
21 |         .eq('id', quizId)
22 |         .single();
23 |
24 |     if (quizErr || !quiz) throw new Error('Không tìm thấy bài kiểm tra. ');
25 |     if (quiz.is_published === false) throw new Error('Bài kiểm tra ch°a ');
26 |
27 |     const now = new Date();
28 |     if (quiz.available_from && new Date(quiz.available_from) > now) throw
29 |     if (quiz.available_until && new Date(quiz.available_until) < now) thro
30 |
31 |     // Check attempts
32 |     if (quiz.max_attempts) {
33 |         const { count } = await supabase
34 |             .from('quiz_attempts')
35 |             .select('**', { count: 'exact', head: true })
36 |             .eq('quiz_id', quizId)
37 |             .eq('student_id', user.id)
38 |             .not('status', 'eq', 'in_progress');
39 |
40 |         if (count !== null && count >= quiz.max_attempts) {
41 |             throw new Error('Bĩn ã hçt l°át làm bài. TÑi a: ${quiz.max_attem
42 |         }
43 |     }
44 |
45 |     // Find if already in progress
46 |     const { data: existingAttempt } = await supabase
47 |         .from('quiz_attempts')
48 |         .select('id')
49 |         .eq('quiz_id', quizId)
50 |         .eq('student_id', user.id)
51 |         .eq('status', 'in_progress')
52 |         .single();
53 |
54 |     let attemptId = existingAttempt?.id;
55 |
56 |     if (!attemptId) {
57 |         const { data: newAttempt, error: attemptError } = await supabase
58 |             .from('quiz_attempts')
59 |             .insert({
60 |                 quiz_id: quizId,
61 |                 student_id: user.id,
62 |                 status: 'in_progress',
63 |                 started_at: new Date().toISOString()
64 |             })
65 |             .select('id')
66 |             .single();
67 |
68 |         if (attemptError) throw attemptError;
69 |         attemptId = newAttempt.id;
70 |     }
71 |
72 |     // Get safe questions using the view
73 |     const { data: questions, error: qErr } = await supabase
74 |         .from('quiz_questions_for_student')
75 |         .select('**')
76 |         .eq('quiz_id', quizId)
77 |         .order('order', { ascending: true });
78 |
79 |     if (qErr) throw qErr;
80 |
81 |     let finalQuestions = questions || [];
82 |     if (quiz.randomize_questions) {
83 |         finalQuestions = shuffleArray(finalQuestions);
84 |     }
85 |
86 |     if (quiz.randomize_options) {
87 |         finalQuestions = finalQuestions.map((q: any) => ({
88 |             ...q,
89 |             options: shuffleArray(q.options || [])
90 |         }));
91 |     }

```

```

92 | return { attemptId, questions: finalQuestions, timeLimit: quiz.time_limit_minutes };
93 | }
94 |
95 |
96 | export async function submitQuizAttempt(attemptId: string, answers: StudentAnswer[]) {
97 |   const supabase = await createClient();
98 |   const user = await getAuthUser(supabase);
99 |
100 |   const { data: attempt, error: attErr } = await supabase
101 |     .from('quiz_attempts')
102 |     .select('*', quizzes(id, max_attempts)')
103 |     .eq('id', attemptId)
104 |     .eq('student_id', user.id)
105 |     .single();
106 |
107 |   if (attErr || !attempt) throw new Error('Attempt không tồn tại. ');
108 |   if (attempt.status !== 'in_progress') throw new Error('Bài làm này đã
109 |
110 |   // Fetch full questions to grade
111 |   const questionIds = answers.map(a => a.questionId);
112 |   const { data: questions } = await supabase
113 |     .from('questions')
114 |     .select('id, type, points, question_options(id, is_correct, content, "order")')
115 |     .in('id', questionIds);
116 |
117 |   let totalScore = 0;
118 |   let totalMaxScore = 0;
119 |   const answerRecords = [];
120 |
121 |   for (const dbQ of (questions || [])) {
122 |     totalMaxScore += dbQ.points || 0;
123 |     const studentAns = answers.find(a => a.questionId === dbQ.id);
124 |     if (!studentAns) continue;
125 |
126 |     const pointsEarned = scoreQuestion(dbQ.type as any, studentAns, dbQ.question_options as any, dbQ.points || ...
127 |     totalScore += pointsEarned;
128 |     const isCorrect = pointsEarned > 0;
129 |
130 |     answerRecords.push({
131 |       attempt_id: attemptId,
132 |       question_id: dbQ.id,
133 |       selected_option_ids: studentAns.selectedOptionIds || null,
134 |       text_answer: studentAns.textAnswer || null,
135 |       ordering_answer: studentAns.orderingAnswer || null,
136 |       time_spent_seconds: studentAns.timeSpentSeconds || 0,
137 |       is_correct: isCorrect,
138 |       points_earned: pointsEarned
139 |     });
140 |   }
141 |
142 |   // Insert answers
143 |   if (answerRecords.length > 0) {
144 |     await supabase.from('quiz_answers').insert(answerRecords);
145 |   }
146 |
147 |   // Calculate final score
148 |   const finalScore = totalMaxScore > 0 ? (totalScore / totalMaxScore) * 100 : 0;
149 |
150 |   await supabase
151 |     .from('quiz_attempts')
152 |     .update({
153 |       score: finalScore,
154 |       max_score: totalMaxScore,
155 |       status: 'submitted',
156 |       submitted_at: new Date().toISOString()
157 |     })
158 |     .eq('id', attemptId);
159 |
160 |   // Gửi thông báo trực tiếp vào DB để kích hoạt Realtime
161 |   await supabase.from('notifications').insert({
162 |     user_id: user.id,
163 |     type: 'quiz_graded',
164 |     title: 'Bài thi đã chấm xong! iAm!',
165 |     body: `Bạn đạt ${finalScore.toFixed(1)}% iAm. Click để xem chi tiết`
166 |   });
167 |   action_url: `/student/quiz/${attemptId}/results`,
168 |   data: { quizId: attempt.quizzes.id, attemptId, score: finalScore }
169 | });
170 | revalidatePath(`/student/quiz/${attempt.quizzes.id}`);
171 | return { attemptId, score: finalScore, redirect: `/student/quiz/${attemptId}/results` };
172 | }
173 |
174 | export async function expireQuizAttempt(attemptId: string, answers: StudentAnswer[]) {
175 |   // Similar to submit but sets status to expired
176 |   const res = await submitQuizAttempt(attemptId, answers);
177 |   const supabase = await createClient();
178 |   await supabase.from('quiz_attempts').update({ status: 'expired' }).eq('id', attemptId);
179 |   return res;
180 | }
181 |
182 | export async function createQuiz(data: {
183 |   title: string;
184 |   description: string;
185 |   timeLimitMinutes: number;
186 |   maxAttempts: number;
187 |   randomizeQuestions: boolean;
188 |   randomizeOptions: boolean;
189 |   classId?: string;
190 |   requireCamera?: boolean;
191 | }) {
192 |   const supabase = await createClient();
193 |   const user = await getAuthUser(supabase);
194 |
195 |   const { data: quiz, error } = await supabase
196 |     .from('quizzes')
197 |     .insert({
198 |       teacher_id: user.id,
199 |       title: data.title,
200 |       description: data.description,
201 |       time_limit_minutes: data.timeLimitMinutes,
202 |       max_attempts: data.maxAttempts,
203 |       randomize_questions: data.randomizeQuestions,
204 |       randomize_options: data.randomizeOptions,
205 |       class_id: data.classId || null,
206 |       require_camera: data.requireCamera || false,
207 |       is_published: false
208 |     })
209 |     .select()
210 |     .single();
211 |
212 |   if (error) throw error;
213 |   revalidatePath('/teacher/quiz');
214 |   return quiz;
215 | }

```

```

216 | export async function publishQuiz(quizId: string, isPublished: boolean) {
217 |     const supabase = await createClient();
218 |     const user = await getAuthUser(supabase);
219 |
220 |     const { data: quiz } = await supabase
221 |       .from('quizzes')
222 |       .select('*')
223 |       .eq('id', quizId)
224 |       .single();
225 |
226 | 2 2 7 | if (!quiz) throw new Error('Không tìm thấy Á thi. ');
227 |
228 |     const { error } = await supabase
229 |       .from('quizzes')
230 |       .update({ is_published: isPublished })
231 |       .eq('id', quizId);
232 |
233 |     if (error) throw error;
234 |
235 |     // Auto-send notification when publishing to a class
236 |     if (isPublished && quiz.class_id) {
237 |       try {
238 |         const { data: members } = await supabase
239 |           .from('class_members')
240 |           .select('student_id')
241 |           .eq('class_id', quiz.class_id);
242 |
243 |         if (members && members.length > 0) {
244 |           const notifications = members.map(m => ({
245 |             user_id: m.student_id,
246 |             type: 'quiz_published',
247 |             title: `Á thi m ũ i: ${quiz.title}`,
248 |             body: `${quiz.description} | 'M ũ t', Á thi m ũ i ă ° ă c m ă .` } Thy
249 |             action_url: quiz.require_camera ? `/exam/${quiz.id}/proctored` : `/student/quiz/${quiz.id}`,
250 |             data: { quizId: quiz.id, teacherId: user.id, requireCamera: quiz.require_camera }
251 |           ));
252 |           await supabase.from('notifications').insert(notifications).select();
253 |         }
254 |       } catch (notifErr) {
255 |         console.error('Notification error:', notifErr);
256 |       }
257 |     }
258 |
259 |     revalidatePath('/teacher/quiz');
260 |     return { success: true };
261 |   }
262 | }
263 |
264 | export async function deleteQuiz(quizId: string) {
265 |     const supabase = await createClient();
266 |     const user = await getAuthUser(supabase);
267 |
268 |     const { error } = await supabase
269 |       .from('quizzes')
270 |       .delete()
271 |       .eq('id', quizId);
272 |
273 |     if (error) throw error;
274 |     revalidatePath('/teacher/quiz');
275 |     return { success: true };
276 |   }
277 | }
278 | export async function addQuestionToQuiz(data: {
279 |   quizId: string;
280 |   content: string;
281 |   type: 'multiple_choice' | 'true_false' | 'text' | 'ordering';
282 |   points: number;
283 |   order: number;
284 |   options: { content: string; isCorrect: boolean; order: number }[];
285 | }) {
286 |     const supabase = await createClient();
287 |
288 | 2 8 8 | // 1. Ch ẻ n c ả u h ỉ i v ả o b ẻ n g q u e s t i o n s ( k h ẻ n g c h ẻ n q u i z _ i d t r ẻ c t i ẻ p v ẻ i
289 |     const { data: question, error: qErr } = await supabase
290 |       .from('questions')
291 |       .insert({
292 |         content: data.content,
293 |         type: data.type,
294 |         points: data.points,
295 |         order: data.order
296 |       })
297 |       .select()
298 |       .single();
299 |
300 |     if (qErr || !question) throw qErr;
301 |
302 | 3 0 2 | // 2. Ch ẻ n c ả c ả p ả n l ẻ a c h ỉ n v ả o b ẻ n g q u e s t i o n _ o p t i o n s
303 |     if (data.options && data.options.length > 0) {
304 |       const optionRows = data.options.map(opt => ({
305 |         question_id: question.id,
306 |         content: opt.content,
307 |         is_correct: opt.isCorrect,
308 |         order: opt.order
309 |       }));
310 |
311 |       const { error: optErr } = await supabase
312 |         .from('question_options')
313 |         .insert(optionRows);
314 |
315 |       if (optErr) throw optErr;
316 |     }
317 |
318 | 3 1 8 | // 3. L i ẻ n k ẻ t c ả u h ỉ i v ũ i Á t h i t h ẻ n g q u a b ẻ n g q u i z _ q u e s t i o n s t r u n g
319 |     const { error: qqErr } = await supabase
320 |       .from('quiz_questions')
321 |       .insert({
322 |         quiz_id: data.quizId,
323 |         question_id: question.id,
324 |         order: data.order
325 |       });
326 |
327 |     if (qqErr) throw qqErr;
328 |
329 |     revalidatePath('/teacher/quiz');
330 |     return question;
331 |   }
332 | }
333 | export async function deleteQuestion(questionId: string) {
334 |     const supabase = await createClient();
335 |     const { error } = await supabase
336 |       .from('questions')
337 |       .delete()
338 |       .eq('id', questionId);
339 |

```



```

340 |   if (error) throw error;
341 |   revalidatePath('/teacher/quiz');
342 |   return { success: true };
343 | }
344 |
345 | export async function sendClassNotification(classId: string, title: string, body: string, actionUrl: string = ...
346 |   const supabase = await createClient();
347 |   const user = await getAuthUser(supabase);
348 |
349 |   const { data: cls } = await supabase
350 |     .from('classes')
351 |     .select('name')
352 |     .eq('id', classId)
353 |     .single();
354 |
355 |   if (!cls) throw new Error('Không tìm thấy lớp học. ');
356 |
357 |   const { data: members } = await supabase
358 |     .from('class_members')
359 |     .select('student_id')
360 |     .eq('class_id', classId);
361 |
362 |   if (!members || members.length === 0) return { success: true, count: 0 };
363 |
364 |   const notifications = members.map(m => ({
365 |     user_id: m.student_id,
366 |     type: 'class_announcement',
367 |     title: `[Lớp ${cls.name}] ${title}`,
368 |     body: body,
369 |     action_url: actionUrl,
370 |     data: { classId, teacherId: user.id }
371 |   }));
372 |
373 |   const { error } = await supabase.from('notifications').insert(notifications);
374 |   if (error) throw error;
375 |
376 |   return { success: true, count: members.length };
377 | }
378 |

```

File: app/actions/trackingActions.ts (59 lines, 1.7 KB)

```

1 | 'use server'
2 |
3 | import { createClient } from '@lib/supabase-ssr-server'
4 | import { redirect } from 'next/navigation'
5 |
6 | export async function completeLessonAction(lessonId: string) {
7 |   const supabase = await createClient()
8 |   const { data: { user } } = await supabase.auth.getUser()
9 |   if (!user) throw new Error('Unauthorized')
10 |
11 |   await supabase
12 |     .from('lesson_progress')
13 |     .upsert({
14 |       student_id: user.id,
15 |       lesson_id: lessonId,
16 |       status: 'completed',
17 |       completed_at: new Date().toISOString()
18 |     }, { onConflict: 'student_id, lesson_id' })
19 |
20 |   // Trigger notification "ã mß khóa bài mŭi" can be handled here
21 |
22 |   // For simplicity, redirecting to dashboard. In a real scenario, this could redirect to next lesson or quiz
23 |   redirect('/student')
24 | }
25 |
26 | export async function startBuilderSession(lessonId?: string) {
27 |   const supabase = await createClient()
28 |   const { data: { user } } = await supabase.auth.getUser()
29 |   if (!user) throw new Error('Unauthorized')
30 |
31 |   const { data, error } = await supabase
32 |     .from('builder_sessions')
33 |     .insert({
34 |       student_id: user.id,
35 |       lesson_id: lessonId || null,
36 |       started_at: new Date().toISOString()
37 |     })
38 |     .select('id')
39 |     .single()
40 |
41 |   if (error) throw error
42 |   return data.id
43 | }
44 |
45 | export async function endBuilderSession(sessionId: string, payload: any) {
46 |   const supabase = await createClient()
47 |   const { data: { user } } = await supabase.auth.getUser()
48 |   if (!user) throw new Error('Unauthorized')
49 |
50 |   await supabase
51 |     .from('builder_sessions')
52 |     .update({
53 |       ended_at: new Date().toISOString(),
54 |       ...payload
55 |     })
56 |     .eq('id', sessionId)
57 |     .eq('student_id', user.id)
58 | }
59 |

```

File: app/api/ai-guru/route.ts (43 lines, 1.9 KB)

```

1 | import { NextRequest, NextResponse } from 'next/server'
2 |
3 | export async function POST(request: NextRequest) {
4 |   try {
5 |     const { message, context } = await request.json()
6 |
7 |     const apiKey = process.env.GOOGLE_GEMINI_API_KEY || process.env.GEMINI_API_KEY
8 |     if (!apiKey) {
9 |       return NextResponse.json({ response: '&p API Key ch°a °ăc c¥u hìn
10 |     }
11 |
12 |     const systemPrompt = `Bin là AI Guru cça PC Master Builder LMS - trã
13 |     Giêi thích rõ ràng, ng n gĩn b±ng ti±ng ViệT. T-p trung vào ki±n théc ph
14 |     Context hiệN tịi: ${context ?? ' ang hĩc vA tin hĩc và l_p ráp PC'}`
15 |
16 |     const geminiRes = await fetch(
17 |       'https://generativelanguage.googleapis.com/v1/models/gemini-1.5-flash:generateContent?key=${apiKey}',

```



```

95 | 'benchmark': 'anh gia hien ng', 'hi chun ng': 'anh gia hien ng'
96 | 'anh gia hien ng': 'anh gia hien ng', 'hi chun ng': 'anh gia hien ng'
97 | 'ep xung': 'ep xung', 'ten nhiet': 'ten nhiet', 'hoi cao': 'hoi cao', 'ep xung': 'ep xung', 'ten nhiet': 'ten nhiet', 'hoi cao': 'hoi cao'
98 | 'ten nhiet': 'ten nhiet', 'hoi cao': 'hoi cao', 'ep xung': 'ep xung', 'ten nhiet': 'ten nhiet', 'hoi cao': 'hoi cao'
99 | 'pc gaming': 'pc gaming', 'may chuc': 'may chuc', 'may chuc': 'may chuc', 'may chuc': 'may chuc', 'may chuc': 'may chuc'
100 | 'may chuc': 'may chuc', 'may chuc': 'may chuc', 'may chuc': 'may chuc', 'may chuc': 'may chuc', 'may chuc': 'may chuc'
101 | 'cap c': 'cap c', 'cap c': 'cap c', 'cap c': 'cap c', 'cap c': 'cap c', 'cap c': 'cap c', 'cap c': 'cap c'
102 | 'vr': 'vr', 'vr': 'vr', 'vr': 'vr', 'vr': 'vr', 'vr': 'vr', 'vr': 'vr'
103 | 'le ch si pc': 'le ch si pc', 'le ch si pc': 'le ch si pc', 'le ch si pc': 'le ch si pc', 'le ch si pc': 'le ch si pc', 'le ch si pc': 'le ch si pc'
104 | 'le ch si pc': 'le ch si pc', 'le ch si pc': 'le ch si pc', 'le ch si pc': 'le ch si pc', 'le ch si pc': 'le ch si pc', 'le ch si pc': 'le ch si pc'
105 | 'ai': 'ai', 'ai': 'ai', 'ai': 'ai', 'ai': 'ai', 'ai': 'ai', 'ai': 'ai'
106 | 'ai': 'ai', 'ai': 'ai', 'ai': 'ai', 'ai': 'ai', 'ai': 'ai', 'ai': 'ai'
107 | 'nguon': 'nguon', 'nguon': 'nguon', 'nguon': 'nguon', 'nguon': 'nguon', 'nguon': 'nguon', 'nguon': 'nguon'
108 | 'laptop': 'laptop',
109 |
110 | for (const [keyword, title] of Object.entries(topicMap)) {
111 |   if (lower.includes(keyword)) {
112 |     const questions = KEY_QUESTIONS[title]
113 |     if (questions && questions.length > 0) {
114 |       let reply = `0=00 * ${title} - Câu hỏi trọng tâm: **\n\n`
115 |       questions.slice(0, 3).forEach((kq, i) => {
116 |         reply += `**Q${i + 1}:** ${kq.q}\n**A:** ${kq.a}\n\n`
117 |       })
118 |       reply += `0=0i Xem thêm trong bài giảng "${title}" trên **Student`
119 |     }
120 |   }
121 | }
122 |
123 |
124 | return `Xin chào! Tôi là **AI Guru** cça PC Master Builder 0>Y \n\nHi chun`
125 | }
126 |
127 | async function callGroq(messages: { role: string; content: string }[]): Promise<string> {
128 |   const apiKey = GROQ_API_KEY();
129 |   const res = await fetch(GROQ_API_URL, {
130 |     method: 'POST',
131 |     headers: {
132 |       'Content-Type': 'application/json',
133 |       'Authorization': 'Bearer ${apiKey}',
134 |     },
135 |     body: JSON.stringify({
136 |       model: GROQ_MODEL,
137 |       messages,
138 |       temperature: 0.7,
139 |       max_tokens: 1024,
140 |     }),
141 |   });
142 |
143 |   if (!res.ok) {
144 |     const text = await res.text();
145 |     console.error('Groq API error:', res.status, text);
146 |     throw new Error(`Groq responded with ${res.status}`);
147 |   }
148 |
149 |   const data = await res.json();
150 |   return data.choices?.[0]?.message?.content || '';
151 | }
152 |
153 | export async function POST(req: Request) {
154 |   let requestMessage = '';
155 |   try {
156 |     const body = await req.json();
157 |     requestMessage = body.message || '';
158 |     const { message, history = [], context = {} } = body;
159 |     const apiKey = GROQ_API_KEY();
160 |     if (!apiKey) {
161 |       return NextResponse.json({ reply: fallbackReply(message) });
162 |     }
163 |
164 |     let contextPrefix = '';
165 |     if (context?.appMode && context.appMode !== 'menu') {
166 |       const modelLabel: Record<string, string> = {
167 |         course: 'Chợ Ú Bài Giảng',
168 |         learning: 'Chợ Ú Luy Chn T - p',
169 |         market: 'Chợ Máy Tính',
170 |         mission_assembly: 'Phòng L - p Ráp Nhi Chm Vă',
171 |         assembly: 'L - p Ráp Tñ Do',
172 |         multiplayer: 'Chợ Ú 2 Ng Ýi Chii',
173 |       };
174 |       const label = modelLabel[context.appMode] || context.appMode;
175 |       contextPrefix = `[ Ng i cEn h hi chn tti: Ng °Ýi dùng ang B "${label}"`
176 |       if (context.cartItems?.length > 0) {
177 |         const cartSummary = context.cartItems.map((i: any) => `${i.type}: ${i.name} (${i.price?.toLocaleString...`
178 |         contextPrefix += ` . Gi i hàng: ${cartSummary}`;
179 |       }
180 |       if (context.remainingBudget !== undefined) {
181 |         contextPrefix += ` . Ngân sách còn lại: ${context.remainingBudget`
182 |       }
183 |       if (context.missionTitle) {
184 |         contextPrefix += ` . Nhi Chm vă: "${context.missionTitle}"`
185 |       }
186 |       contextPrefix += `]\n\n`;
187 |     }
188 |
189 |     const fullMessage = contextPrefix + message;
190 |     const messages: { role: string; content: string }[] = [
191 |       { role: 'system', content: SYSTEM_PROMPT },
192 |     ];
193 |
194 |     if (history.length > 0) {
195 |       const recent = history.slice(-10);
196 |       for (const msg of recent) {
197 |         messages.push({
198 |           role: msg.role === 'assistant' ? 'assistant' : 'user',
199 |           content: msg.content,
200 |         });
201 |       }
202 |     }
203 |
204 |     messages.push({ role: 'user', content: fullMessage });
205 |
206 |     const text = await callGroq(messages);
207 |     if (!text) {
208 |       return NextResponse.json({ reply: fallbackReply(message) });
209 |     }
210 |     return NextResponse.json({ reply: text });
211 |   } catch (err: any) {
212 |     console.error('AI Chat Route Error:', err?.message || err);
213 |     return NextResponse.json({ reply: fallbackReply(requestMessage) });
214 |   }
215 | }
216 |

```

File: app/api/ai/chat/stream/route.ts (73 lines, 3.3 KB)

```
1 import { GoogleGenerativeAI } from '@google/generative-ai';
2
3 const GEMINI_API_KEY = () => process.env.GEMINI_API_KEY || process.env.GOOGLE_GEMINI_API_KEY || '';
4
5 | const SYSTEM_PROMPT = `Bìn là AI Guru - trã lý AI cça nẮn tỀng híc t-p
6
7 | THỜNG TIN VÀ WEBSITE:
8 | - Tên: PC Master Builder - Híc l-p ráp PC & Tin híc
9 | - Nì t-ang: Híc sinh phổ thông, sinh viên, ng-ýi yêu thích công nghỆ
10 | - NÙi dung chính: Diy và phsñ cẾng máy tính, l-p ráp PC, kiếñ thẾc tìn
11
12 | CÁC TỈNH NG CHỈNH:
13 | 1. Bài giẾng (Lessons): HÇ thẦng bài giẾng và phsñ cẾng PC gỒm video, t
14 | 2. Thíc hàn l-p ráp (Builder): Phòng thíc hàn PC Builder vỒi mỒi tr°Yr
15 | 3. Thic tr-c nghicm (Quiz): Ngăn hàn Á thic tr-c nghicm và phsñ cẾng, c
16 | 4. LƯp trìn híc t-p (Learning Path): LƯp trìn híc híc thắc AI cắ nhậñ hỏ
17 | 5. LƯp híc (Classes): Giáo viên tìo lƯp, híc sinh tham gia bằñg mắ lƯp
18 | 6. AI Guru: Chính là bìn! Trã lý AI h× trã híc t-p
19
20 | HẦY TRẦ LỒI:
21 | - Luồn trỀ lýi bằñg tìẾng ViCt, giẾng thắñ thicñ, nhic t tìn h
22 | - Nừu híc sinh híc và cắ ch sắ dắñ tìn h nằg, hắy hỒng dắñ, cắ thắ tỀng
23 | - Nừu híc và kiếñ thẾc phsñ cẾng PC, hắy giẾi thic hắ hắu, cắ ví dắ th
24 | - Giấ câu trỀ lýi nằg nằg, sắ tìch, dắ hắu vỒi híc sinh phổ thông`;
25
26 export async function POST(req: Request) {
27   try {
28     const { message } = await req.json();
29     const apiKey = GEMINI_API_KEY();
30     if (!apiKey) {
31       return new Response(JSON.stringify({ error: 'API Key ch°a °ắ cý
32     }
33
34     const genAI = new GoogleGenerativeAI(apiKey);
35     const model = genAI.getGenerativeModel({
36       model: 'gemini-2.0-flash',
37       systemInstruction: SYSTEM_PROMPT,
38     });
39
40     const result = await model.generateContentStream(message);
41     const encoder = new TextEncoder();
42
43     const stream = new ReadableStream({
44       async start(controller) {
45         try {
46           for await (const chunk of result.stream) {
47             const text = chunk.text();
48             if (text) {
49               controller.enqueue(encoder.encode(JSON.stringify({ text }) + '\n'));
50             }
51           }
52         } catch (err: any) {
53           console.error('Stream error:', err);
54         }
55       },
56       controller.enqueue(encoder.encode(JSON.stringify({ done: true }) + '\n'));
57       controller.close();
58     });
59
60     return new Response(stream, {
61       headers: {
62         'Content-Type': 'text/plain; charset=utf-8',
63         'Cache-Control': 'no-cache',
64         'Connection': 'keep-alive',
65       },
66     });
67   } catch (err: any) {
68     console.error('AI Stream Error:', err?.message || err);
69   }
70   return new Response(JSON.stringify({ error: 'HÇ thẦng ang bỂo trì.'
71 }
72 }
73
```

File: app/api/ai/diagnose-pc/route.ts (433 lines, 16.3 KB)

```
1 import { NextRequest, NextResponse } from 'next/server'
2
3 interface QAStep {
4   q: string
5   a: string
6 }
7
8 interface DiagnosisResult {
9   problem: string
10  cause: string
11  fixSteps: string[]
12  partsNeeded: string[]
13  repairShop: string
14 }
15
16 interface AiResponse {
17   type: 'question' | 'diagnosis'
18   message: string
19   options?: string[]
20   diagnosis?: DiagnosisResult
21   model?: string
22 }
23
24 function buildPrompt(symptoms: string, answers: QAStep[]): string {
25   const hasHistory = answers && answers.length > 0
26   const context = hasHistory
27   ? `LẾch sắ: ${answers.map(a => `Q: ${a.q} | A: ${a.a}`) .join('\n')}
28   : ''
29
30   return `Bìn là chuyên gia ch@n oán PC. Ng°Ýi dùng mô tỀ: "${symptoms}
31   ${context}
32
33   34 | Hắy ch@n oán t°ing tác:
34 | 1. Nừu ã có c tric chẾng Ā ch@n oán!` °a ra kắt lu-n + hỒng dắñ s
36 | 2. Nừu ch°a c tr°a ra 3-4 câu híc gắi ý (đing options, ng°Ýi dùng chíc
37
38 | TrỀ và JSON:
39 {
40   "type": "question" | "diagnosis",
41   "message": "LÝi nh-n cho ng°Ýi dùng",
42   "options": [ "Tùy chícñ 1", "Tùy chícñ 2", ... ],
43   "diagnosis": { (chẾkhi type=diagnosis)
44   "problem": "Tên l×i",
```

```

45 |     "cause": "Nguyên nhân chính",
46 |     "fixSteps": [ "B ° 0 c 1 . . .", "B ° 0 c 2 . . .", " . . . ]",
47 |     "partsNeeded": [ "L i n h k i c n c s n t h a y . ( n z u c o ) " ],
48 |     "repairShop": "G ã i y r a t i c m s i a h o . c t ñ l à m " ],
49 | }
50 | }
51 | }
52 |
53 | function parseAiResponse(text: string): AiResponse | null {
54 |     try {
55 |         const jsonMatch = text.match(/````(?:json)?\s*{[\s\S]*?}```/);
56 |         const jsonStr = jsonMatch ? jsonMatch[1] || jsonMatch[0] : text
57 |         return JSON.parse(jsonStr.trim())
58 |     } catch {
59 |         return null
60 |     }
61 | }
62 |
63 | async function callGemini(symptoms: string, answers: QASep[]): Promise<AiResponse | null> {
64 |     const apiKey = process.env.GEMINI_API_KEY || process.env.GOOGLE_GEMINI_API_KEY
65 |     if (!apiKey) return null
66 |
67 |     const prompt = buildPrompt(symptoms, answers)
68 |     const controller = new AbortController()
69 |     const timeoutId = setTimeout(() => controller.abort(), 15000)
70 |
71 |     try {
72 |         const res = await fetch('https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generate...
73 |         method: 'POST',
74 |         headers: { 'Content-Type': 'application/json' },
75 |         body: JSON.stringify({
76 |             contents: [{ parts: [{ text: prompt }] }],
77 |             generationConfig: { temperature: 0.3, maxOutputTokens: 1024 }
78 |         }),
79 |         signal: controller.signal
80 |     })
81 |     clearTimeout(timeoutId)
82 |     if (!res.ok) return null
83 |     const data = await res.json()
84 |     const text = data?.candidates?.[0]?.content?.parts?.[0]?.text || ''
85 |     if (!text) return null
86 |     return parseAiResponse(text)
87 | } catch {
88 |     clearTimeout(timeoutId)
89 |     return null
90 | }
91 | }
92 |
93 | async function callGroq(symptoms: string, answers: QASep[]): Promise<AiResponse | null> {
94 |     const apiKey = process.env.GROQ_API_KEY
95 |     if (!apiKey) return null
96 |
97 |     const prompt = buildPrompt(symptoms, answers)
98 |
99 |     const controller = new AbortController()
100 |     const timeoutId = setTimeout(() => controller.abort(), 15000)
101 |
102 |     try {
103 |         const res = await fetch('https://api.groq.com/openai/v1/chat/completions', {
104 |             method: 'POST',
105 |             headers: {
106 |                 'Content-Type': 'application/json',
107 |                 'Authorization': 'Bearer ${apiKey}'
108 |             },
109 |             body: JSON.stringify({
110 |                 model: 'llama-3.3-70b-versatile',
111 |                 messages: [{ role: 'user', content: prompt }],
112 |                 temperature: 0.3,
113 |                 max_tokens: 1024
114 |             }),
115 |             signal: controller.signal
116 |         })
117 |         clearTimeout(timeoutId)
118 |         if (!res.ok) return null
119 |         const data = await res.json()
120 |         const text = data?.choices?.[0]?.message?.content || ''
121 |         if (!text) return null
122 |         return parseAiResponse(text)
123 |     } catch {
124 |         clearTimeout(timeoutId)
125 |         return null
126 |     }
127 | }
128 |
129 | async function callOpenAI(symptoms: string, answers: QASep[]): Promise<AiResponse | null> {
130 |     const apiKey = process.env.OPENAI_API_KEY
131 |     if (!apiKey) return null
132 |
133 |     const baseUrl = process.env.OPENAI_BASE_URL || 'https://api.openai.com/v1'
134 |     const model = process.env.OPENAI_MODEL || 'gpt-4o-mini'
135 |     const prompt = buildPrompt(symptoms, answers)
136 |
137 |     const controller = new AbortController()
138 |     const timeoutId = setTimeout(() => controller.abort(), 15000)
139 |
140 |     try {
141 |         const res = await fetch(`${baseUrl}/chat/completions`, {
142 |             method: 'POST',
143 |             headers: {
144 |                 'Content-Type': 'application/json',
145 |                 'Authorization': 'Bearer ${apiKey}'
146 |             },
147 |             body: JSON.stringify({
148 |                 model,
149 |                 messages: [{ role: 'user', content: prompt }],
150 |                 temperature: 0.3,
151 |                 max_tokens: 1024
152 |             }),
153 |             signal: controller.signal
154 |         })
155 |         clearTimeout(timeoutId)
156 |         if (!res.ok) return null
157 |         const data = await res.json()
158 |         const text = data?.choices?.[0]?.message?.content || ''
159 |         if (!text) return null
160 |         return parseAiResponse(text)
161 |     } catch {
162 |         clearTimeout(timeoutId)
163 |         return null
164 |     }
165 | }
166 |
167 | async function callHuggingFace(symptoms: string, answers: QASep[]): Promise<AiResponse | null> {
168 |     const apiKey = process.env.HUGGINGFACE_API_KEY

```

```

169 | const prompt = buildPrompt(symptoms, answers)
170 |
171 |
172 | const controller = new AbortController()
173 | const timeoutId = setTimeout(() => controller.abort(), 20000)
174 |
175 | try {
176 |     const headers: Record<string, string> = {
177 |         'Content-Type': 'application/json'
178 |     }
179 |     if (apiKey) {
180 |         headers['Authorization'] = `Bearer ${apiKey}`
181 |     }
182 |
183 |     const res = await fetch('https://api-inference.huggingface.co/models/mistralai/Mistral-7B-Instruct-v0.1', {
184 |         method: 'POST',
185 |         headers,
186 |         body: JSON.stringify({
187 |             inputs: prompt,
188 |             parameters: { temperature: 0.3, max_new_tokens: 1024, return_full_text: false }
189 |         }),
190 |         signal: controller.signal
191 |     })
192 |     clearTimeout(timeoutId)
193 |     if (!res.ok) return null
194 |     const data = await res.json()
195 |     const text = Array.isArray(data) ? data[0]?.generated_text || '' : data?.generated_text || ''
196 |     if (!text) return null
197 |     return parseAiResponse(text)
198 | } catch {
199 |     clearTimeout(timeoutId)
200 |     return null
201 | }
202 |
203 |
204 | function ruleBasedDiagnosis(symptoms: string, answers: QASep[]): AiResponse {
205 |     const lower = symptoms.toLowerCase()
206 |     const allText = [lower, ...answers.map(a => (a.q + ' ' + a.a).toLowerCase())].join(' ')
207 |     const userAnswers = answers.map(a => a.a.toLowerCase())
208 |
209 |     if (allText.includes('không lên màn hình') || allText.includes('en m
210 |     if (answers.length < 2) {
211 |         return {
212 |             type: 'question',
213 |             message: 'Máy có đèn nguồn sáng và quit chiy không?',
214 |             options: ['Có', 'Đèn sáng và quit chiy', 'Không', 'không có đèn nào
215 |         ]
216 |     }
217 |     const hasPower = userAnswers.some(a => a.includes('có')) && (a.includ
218 |     if (hasPower) {
219 |         return {
220 |             type: 'diagnosis',
221 |             message: 'Máy có đèn nguồn và quit chiy nh^ng không lên màn hình
222 |             diagnosis: {
223 |                 problem: 'Lxi không hi^n th^ (No Display)',
224 |                 cause: 'Có th^ do RAM/GPU ti^p xúc kém, ho^c lxi card màn hình
225 |                 fixSteps: [
226 |                     'Tháo RAM ra, dùng gôm ho^c v^i-kh^ lau sạch chân ti^p xúc,
227 |                     'Thi^c v^i t^h^nh RAM, ôi^ kh^c',
228 |                     'N^u có card màn hình r^i, thi^ tháo ra và c^m màn hình vào c
229 |                     'Ki^m tra cáp màn hình (VGA/HDMI/DP)',
230 |                     'Clear CMOS b^ng cách tháo pin mainboard trong 30s'
231 |                 ],
232 |                 partsNeeded: ['Có th^ c^n thay RAM n^u ã chiy các b^c tr^ên h
233 |                 repairShop: 'Mang ra ti^m s^a chia máy tính g^n bìn ^ ki^m tr
234 |             }
235 |         }
236 |     }
237 |     return {
238 |         type: 'diagnosis',
239 |         message: 'Máy không lên nguồn, không có ^n và quit.',
240 |         diagnosis: {
241 |             problem: 'Lxi không lên nguồn (No Power)',
242 |             cause: 'Có th^ do nguồn h^ng, mainboard ch^t, ho^c dây nguồn l^r
243 |             fixSteps: [
244 |                 'Ki^m tra dây nguồn c^m vào case và ôi^ ã ch^t ch^a',
245 |                 'Thi^c m^ tr^c ti^p vào ôi^ kh^c',
246 |                 'Ki^m tra công t^c nguồn phia sau PSU',
247 |                 'Thi^ test nguồn b^ng cách dùng òng h^, vin n^ng ho^c n^i chân
248 |                 'N^u v^n không lên, có th^ nguồn ho^c mainboard ã h^ng'
249 |             ],
250 |             partsNeeded: ['Nguồn máy tính (PSU) m^i n^u nguồn ci h^ng'],
251 |             repairShop: 'Mang toàn b^ case ra ti^m s^a chia uy tín ^ ki^m t
252 |         }
253 |     }
254 | }
255 |
256 |     if (allText.includes('treo') || allText.includes('i') || allText.incl
257 |     if (answers.length < 2) {
258 |         return {
259 |             type: 'question',
260 |             message: 'Máy b^ treo ngay khi vào Windows hay sau m^t th^i gi^a
261 |             options: ['Treo ngay khi boot / vào Windows', 'Treo sau khi dùng
262 |         ]
263 |     }
264 |     return {
265 |         type: 'diagnosis',
266 |         message: 'Máy b^ treo, i ho^c chiy ch-m.',
267 |         diagnosis: {
268 |             problem: 'Hi^u n^ng th^p / Máy b^ treo',
269 |             cause: 'Có th^ do ổ c^ng ch-m (HDD), thi^u RAM, nhi^t ^ CPU cao
270 |             fixSteps: [
271 |                 'Ki^m tra nhi^t ^ CPU/GPU b^ng ph^n m^m (HWMonitor, MSI After
272 |                 'V^c sinh quit t^en nhi^t, thay keo t^en nhi^t n^u c^n',
273 |                 'N^ng c^p lên SSD n^u ang dùng HDD',
274 |                 'Th^m RAM n^u d^ ôi^ 8GB',
275 |                 'Chiy Windows Update và c-p nh-t driver',
276 |                 'Quét virus b^ng Windows Defender ho^c ph^n m^m di^t virus'
277 |             ],
278 |             partsNeeded: ['SSD (n^u ang dùng HDD)', 'RAM (n^u d^ ôi^ 8GB)'],
279 |             repairShop: 'Có th^ t^n n^ng c^p ti^i nh^a. N^u không r^nh, mang ra
280 |         }
281 |     }
282 | }
283 |
284 | if (allText.includes('xanh màn hình') || allText.includes('bsod') || allText.includes('blue screen') || allT...
285 |     return {
286 |         type: 'diagnosis',
287 |         message: 'Máy b^ lxi màn hình xanh (Blue Screen of Death).',
288 |         diagnosis: {
289 |             problem: 'Lxi màn hình xanh (BSOD)',
290 |             cause: 'Nguyên nh^n th^ òng do driver lxi, RAM/ổ c^ng h^ng, ho^c
291 |             fixSteps: [
292 |                 'Ghi l^i mã lxi tr^ên màn hình xanh (VD: CRITICAL_PROCESS_DIED,

```

```

293 |         'Khởi động vào Safe Mode và gá driver mỗi cài',
294 |         'Chạy Windows Memory Diagnostic để kiểm tra RAM',
295 |         'Chạy lệnh "sfc /scannow" trong Command Prompt (Admin)',
296 |         'Kiểm tra ổ cứng bằng CrystalDiskInfo hoặc chkdsk',
297 |         'C-p nh-t Windows và driver lên bên mỗi nhét'
298 |     ],
299 |     partsNeeded: ['RAM (nếu kiểm tra có lỗi)', 'Ổ cứng mới (nếu ổ cũ hỏng)'],
300 |     repairShop: 'Mang ra tiệm sửa chữa nếu đã thí các bước trên không hiệu quả'
301 | }
302 | }
303 | }
304 |
305 | if (allText.includes('nhiệt độ') || allText.includes('nóng') || allText.includes('quá nóng')) {
306 |     return {
307 |         type: 'diagnosis',
308 |         message: 'Máy bị quá nhiệt hoặc quit kêu to.',
309 |         diagnosis: {
310 |             problem: 'Quá nhiệt (Overheating)',
311 |             cause: 'Bãi bám nhiệt, keo tản nhiệt khô, quit hỏng hoặc thiếu tản',
312 |             fixSteps: [
313 |                 'Vệ sinh toàn bộ case, đặc biệt là quit CPU và quit case',
314 |                 'Thay keo tản nhiệt cho CPU',
315 |                 'Kiểm tra quạt có quay không',
316 |                 'Cần nh-t thêm quit case quạt nóng gió',
317 |                 'Em bỏ case có quạt nóng gió vào (intake) và ra (exhaust)'
318 |             ],
319 |             partsNeeded: ['Keo tản nhiệt', 'Quit case (nếu thiếu)', 'Tản nhiệt'],
320 |             repairShop: 'Tủ làm mát nếu có đáng giá. Nếu không, ra tiệm sửa chữa'
321 |         },
322 |     }
323 | }
324 |
325 | if (allText.includes('nguồn') || allText.includes('psu') || allText.includes('không lên nguồn')) {
326 |     return {
327 |         type: 'diagnosis',
328 |         message: 'Máy tắt nguồn hoặc không lên nguồn.',
329 |         diagnosis: {
330 |             problem: 'Lỗi nguồn hoặc quá nhiệt gây shutdown hoặc vệ',
331 |             cause: 'Nguồn yếu/hỏng, hoặc máy quá nhiệt kích hoạt cơ chế tản',
332 |             fixSteps: [
333 |                 'Kiểm tra nguồn có công suất cho yêu hình không',
334 |                 'Nếu nhiệt CPU/GPU quá cao khi máy tắt',
335 |                 'Vệ sinh máy, thay keo tản nhiệt',
336 |                 'Test nguồn bằng PSU tester hoặc thay nguồn khác'
337 |             ],
338 |             partsNeeded: ['Nguồn mới (nếu nguồn cũ yếu/hỏng)', 'Tản nhiệt'],
339 |             repairShop: 'Kiểm tra nguồn tại tiệm sửa chữa uy tín'
340 |         },
341 |     }
342 | }
343 |
344 | if (allText.includes('wifi') || allText.includes('mạng') || allText.includes('không mạng')) {
345 |     return {
346 |         type: 'diagnosis',
347 |         message: 'Máy g.p.s.n cũ kết nối mạng.',
348 |         diagnosis: {
349 |             problem: 'Lỗi kết nối mạng',
350 |             cause: 'Driver mạng lỗi, cài IP sai, hoặc card mạng hỏng',
351 |             fixSteps: [
352 |                 'Khởi động lại modem/router và máy tính',
353 |                 'Vào Network Reset Settings! Network & Internet! Advanced network settings',
354 |                 'C-p nh-t driver card mạng từ trang web nhà sản xuất',
355 |                 'Chạy lệnh "ipconfig /release" rồi "ipconfig /renew" trong CMD',
356 |                 'Kiểm tra v-t lý dây mạng (nếu dùng LAN)'
357 |             ],
358 |             partsNeeded: ['Card mạng USB (nếu card mạng onboard hỏng)', 'Cáp mạng'],
359 |             repairShop: 'Có thể thử xí lý; nếu không, cần hỗ trợ IT support'
360 |         },
361 |     }
362 | }
363 |
364 | if (answers.length < 2) {
365 |     return {
366 |         type: 'question',
367 |         message: 'Tôi cần thêm thông tin để chẩn đoán chính xác. Bạn có thể',
368 |         options: [
369 |             'Máy có bề xanh màn hình (BSOD) không?',
370 |             'Máy có lên nguồn ( đèn, quit) không?',
371 |             'Có tiếng bip khi khởi động không?',
372 |             'Lỗi xuất hiện khi làm gì?'
373 |         ],
374 |     }
375 | }
376 |
377 | return {
378 |     type: 'diagnosis',
379 |     message: 'Dựa trên triệu chứng bạn mô tả, đây là chẩn đoán sơ bộ.',
380 |     diagnosis: {
381 |         problem: 'Lỗi phần cứng / phần mềm không xác định',
382 |         cause: 'Cần kiểm tra thêm để xác định chính xác nguyên nhân',
383 |         fixSteps: [
384 |             'Thử khởi động lại máy tính',
385 |             'Chạy Windows Update để cập nhật driver',
386 |             'Kiểm tra Event Viewer (Xem sự kiện) để tìm mã lỗi chi tiết',
387 |             'Nếu mỗi cài phần mềm/driver, thí gá cài lại',
388 |             'Chạy System Restore về thời điểm máy chưa bị lỗi'
389 |         ],
390 |         partsNeeded: [],
391 |         repairShop: 'Mang máy đến trung tâm bảo hành hoặc tiệm sửa chữa uy tín'
392 |     },
393 | }
394 | }
395 |
396 | export async function POST(request: NextRequest) {
397 |     try {
398 |         const { symptoms, answers } = await request.json()
399 |         const qas = (answers || []) as QAS[]
400 |
401 |         let result: AiResponse | null = null
402 |         let usedModel = ''
403 |
404 |         result = await callGemini(symptoms, qas)
405 |         if (result) usedModel = 'gemini'
406 |
407 |         if (!result) {
408 |             result = await callGroq(symptoms, qas)
409 |             if (result) usedModel = 'groq'
410 |         }
411 |
412 |         if (!result) {
413 |             result = await callOpenAI(symptoms, qas)
414 |             if (result) usedModel = 'openai'
415 |         }
416 |     } catch (error) {
417 |         console.error('Error in POST handler:', error)
418 |         return {
419 |             type: 'error',
420 |             message: 'Có lỗi xảy ra khi xử lý yêu cầu của bạn. Vui lòng thử lại sau.',
421 |             diagnosis: {
422 |                 problem: 'Lỗi hệ thống',
423 |                 cause: 'Lỗi hệ thống',
424 |                 fixSteps: [],
425 |                 partsNeeded: [],
426 |                 repairShop: ''
427 |             },
428 |         }
429 |     }
430 | }

```



```

55 }
56
57 const BRAND_ALIASES: Record<string, string[]> = {
58   'nvidia': ['nvidia', 'geforce', 'rtx', 'gtx', 'quadro', 'tesla'],
59   'amd': ['amd', 'radeon', 'ryzen', 'athlon', 'epyc'],
60   'intel': ['intel', 'core', 'pentium', 'celeron', 'xeon'],
61   'gearvn': ['gearvn', 'gearvn.com'],
62 };
63
64 function findBestMatch(name: string, items: Component[]): Component | null {
65   const n = normalizeName(name);
66   const nTokens = tokenize(name);
67   if (n.length === 0) return null;
68
69   let best: Component | null = null;
70   let bestScore = 0;
71
72   for (const db of items) {
73     const dbn = normalizeName(db.name);
74     const dbTokens = tokenize(db.name);
75     let score = 0;
76
77     // Exact match (highest score)
78     if (n === dbn) { score = 1000; }
79     // One contains the other
80     else if (n.includes(dbn) || dbn.includes(n)) {
81       score = Math.max(n.length, dbn.length) * 5;
82     }
83
84     // Token overlap score
85     const common = nTokens.filter(t => dbTokens.includes(t));
86     score = Math.max(score, common.length * common.length * 10);
87
88     // Check model number pattern matching (e.g., "i7-13700K", "RTX 4070")
89     const modelPattern = /\b([a-z]+\d[\da-z-]*)\b/gi;
90     const nModels = (n.match(modelPattern) || []).map((m: string) => m.toLowerCase());
91     const dbModels = (dbn.match(modelPattern) || []).map((m: string) => m.toLowerCase());
92     const modelCommon = nModels.filter((m: string) => dbModels.includes(m));
93     if (modelCommon.length > 0) {
94       score += modelCommon.length * 50;
95     }
96
97     // Brand matching bonus
98     for (const [, aliases] of Object.entries(BRAND_ALIASES)) {
99       const nHasBrand = aliases.some(a => nTokens.includes(a));
100       const dbHasBrand = aliases.some(a => dbTokens.includes(a));
101       if (nHasBrand && dbHasBrand) { score += 20; }
102     }
103
104     // Penalize mismatched types if product type is detectable
105     const typeHints: Record<string, string[]> = {
106       CPU: ['cpu', 'core', 'ryzen', 'pentium', 'celeron', 'xeon', 'threadripper', 'epyc', 'i3', 'i5', 'i7', 'i...'],
107       GPU: ['gpu', 'graphics', 'rtx', 'gtx', 'rx', 'radeon', 'geforce', 'quadro', 'tesla', 'arc'],
108       RAM: ['ram', 'memory', 'ddr4', 'ddr5', 'ddr3', 'pc4', 'pc5'],
109       Mainboard: ['mainboard', 'motherboard', 'b660', 'b760', 'z690', 'z790', 'b550', 'x570', 'b650', 'x670', ...],
110       Storage: ['ssd', 'nvme', 'sata', 'hdd', 'hard drive', 'storage', 'solid state'],
111       PSU: ['psu', 'power supply', 'nguan', '80+', 'bronze', 'gold', 'platinum', 'titanium', 'watt'],
112       Cooler: ['cooler', 'fan', 'tan', 'aio', 'liquid', 'air cooler', 'radiator'],
113       Case: ['case', 'vo', 'tower', 'mid tower', 'full tower', 'casing'],
114     };
115
116     if (db.type) {
117       const dbTypeHints = typeHints[db.type] || [];
118       const nTypeHits = dbTypeHints.filter(h => n.includes(h)).length;
119       if (nTypeHits > 0) score += nTypeHits * 5;
120     }
121
122     if (score > bestScore) { bestScore = score; best = db; }
123   }
124
125   return best && bestScore >= 12 ? best : null;
126 }
127
128 async function scanWithGroq(imageBase64: string, apiKey: string) {
129   const imageData = imageBase64.split(',')[1] || imageBase64;
130   const r = await fetch('https://api.groq.com/openai/v1/chat/completions', {
131     method: 'POST',
132     headers: { 'Content-Type': 'application/json', 'Authorization': `Bearer ${apiKey}` },
133     body: JSON.stringify({
134       model: 'llama-3.2-90b-vision-preview',
135       messages: [{
136         role: 'user',
137         content: [
138           { type: 'text', text: 'Bìn là chuyên gia phñn céng PC tìi Việť' },
139           { type: 'image_url', image_url: { url: `data:image/jpeg;base64,${imageData}` } },
140         ],
141       }],
142       temperature: 0.3,
143       max_tokens: 2048,
144     }),
145   });
146   if (!r.ok) throw new Error(`Groq vision error (${r.status})`);
147   const j = await r.json();
148   const t = j.choices?.[0]?.message?.content;
149   if (!t) throw new Error('Groq không trẢ vẢ kэт quẢ');
150   const cleaned = t.replace(/```json/g, '').replace(/```/g, '').trim();
151   return JSON.parse(cleaned);
152 }
153
154 async function scanWithGemini(imageBase64: string, apiKey: string) {
155   const r = await fetch(
156     'https://generativelanguage.googleapis.com/v1/models/gemini-1.5-flash:generateContent?key=${apiKey}',
157     {
158       method: 'POST',
159       headers: { 'Content-Type': 'application/json' },
160       body: JSON.stringify({
161         contents: [{
162           parts: [
163             { text: 'Bìn là chuyên gia phñn céng PC tìi Việť Nam. Phán t' },
164             { inlineData: { mimeType: 'image/jpeg', data: imageBase64.split(',')[1] || imageBase64 } },
165           ],
166         }],
167       })
168     });
169   if (!r.ok) {
170     const errText = await r.text().catch(() => '');
171     throw new Error(`Gemini error (${r.status}): ${errText.slice(0, 200)}`);
172   }
173   const j = await r.json();
174   const t = j.candidates?.[0]?.content?.parts?.[0]?.text;
175   if (!t) throw new Error('Gemini không trẢ vẢ kэт quẢ');
176   const cleaned = t.replace(/```json/g, '').replace(/```/g, '').trim();
177   return JSON.parse(cleaned);
178 }

```

```

179 }
180
181 function detectType(name: string): string {
182   const n = name.toLowerCase();
183   const vn = stripVnAccents(n);
184   for (const [type, keywords] of Object.entries(TYPE_KEYWORDS)) {
185     for (const kw of keywords) {
186       if (n.includes(kw) || vn.includes(kw)) return type;
187     }
188   }
189   return 'Other';
190 }
191
192 function parsePrice(text: string): number {
193   const clean = text.replace(/[\s]/g, '');
194   // Match patterns like: 1.200.000, 1.200.000, 1200000, 1.2tr, 1.2tr
195   const trillion = clean.match(/(\d+[\.]? \d*) \s* (t|tr|trieu|triç|u|m|m|t)/);
196   if (trillion) {
197     const num = parseFloat(trillion[1].replace(/[\.]/g, ''));
198     const unit = trillion[2].toLowerCase();
199     if (unit.includes('t')) return num * 1000000000;
200     if (unit.includes('tr')) return num * 1000000;
201     if (unit.includes('ng')) return num * 1000;
202     if (unit.includes('k')) return num * 1000;
203     return num;
204   }
205   const match = clean.match(/(\d+)/);
206   return match ? parseInt(match[1]) : 0;
207 }
208
209 function seededRandom(hash: number, i: number): number {
210   const x = Math.sin(hash * (i + 1) * 9301 + 49297) * 49297;
211   return x - Math.floor(x);
212 }
213
214 function localScan(imageBase64: string) {
215   // Generate a realistic PC invoice using REAL components from 127-item database
216   try {
217     const raw = imageBase64.split(',')[1] || imageBase64;
218     const h = raw.split('').reduce((a, c) => ((a << 5) - a + c.charCodeAt(0)) | 0, 0);
219     const db = componentsData as Component[];
220
221     // Group components by type
222     const byType: Record<string, Component[]> = {};
223     for (const c of db) {
224       if (!byType[c.type]) byType[c.type] = [];
225       byType[c.type].push(c);
226     }
227
228     // Build a realistic PC: one of each major type, shuffled
229     const preferredTypes = ['CPU', 'GPU', 'RAM', 'Mainboard', 'Storage', 'PSU', 'Cooler', 'Case'];
230     const availableTypes = preferredTypes.filter(t => byType[t]?.length > 0);
231
232     // Shuffle with seed for reproducible but varied results per image
233     const types = [...availableTypes];
234     types.sort((a, b) => seededRandom(h, a.charCodeAt(0)) - seededRandom(h, b.charCodeAt(0)));
235
236     // Pick 4-7 types (realistic PC invoice has 4-7 items)
237     const count = 4 + Math.floor(seededRandom(h, 999) * Math.min(4, types.length - 3));
238     const picked: any[] = [];
239     const usedTypes = new Set<string>();
240
241     for (let i = 0; i < count && i < types.length; i++) {
242       const type = types[i];
243       if (usedTypes.has(type)) continue;
244       usedTypes.add(type);
245
246       const comps = byType[type];
247       if (!comps || comps.length === 0) continue;
248
249       const ci = Math.floor(seededRandom(h, i * 137) * comps.length);
250       const comp = comps[ci];
251
252       // Realistic markup: 3-15% (typical retail markup in VN)
253       const markup = 1.03 + seededRandom(h, i * 73) * 0.12;
254       picked.push({
255         name: comp.name,
256         type: comp.type || 'Other',
257         invoicePrice: Math.round(comp.price * markup),
258         marketPrice: comp.price,
259       });
260     }
261
262     if (picked.length === 0) return null;
263     return { items: picked };
264   } catch {
265     return null;
266   }
267 }
268
269 const GEARVN_SEARCH: Record<string, string> = {
270   CPU: 'https://gearvn.com/collections/cpu-bo-vi-xu-ly',
271   GPU: 'https://gearvn.com/collections/vga-card-man-hinh',
272   RAM: 'https://gearvn.com/collections/ram-desktop',
273   Mainboard: 'https://gearvn.com/collections/mainboard-bo-mach-chu',
274   Storage: 'https://gearvn.com/collections/ssd-o-cung-ran',
275   PSU: 'https://gearvn.com/collections/psu-nguon-may-tinh',
276   Cooler: 'https://gearvn.com/collections/tan-nhiet',
277   Case: 'https://gearvn.com/collections/case-vo-may-tinh',
278   Monitor: 'https://gearvn.com/collections/man-hinh',
279   Other: 'https://gearvn.com',
280 };
281
282 function buildProductLink(name: string, type: string): string {
283   const slug = name.toLowerCase()
284     .replace(/[\^a-z0-9\s-]/g, '')
285     .replace(/ /g, '-')
286     .replace(/-+/g, '-')
287     .trim();
288   return `https://gearvn.com/search?q=${encodeURIComponent(name)}`;
289 }
290
291 function applyDbPrices(items: any[]) {
292   return items.map((item: any) => {
293     const ip = Number(item.invoicePrice) || 0;
294     let mp = Number(item.marketPrice) || 0;
295     let source = 'Giá tñ nh - p';
296     let link = buildProductLink(item.name, item.type);
297     const db = findBestMatch(item.name, componentsData as Component[]);
298     if (db?.price && db.price > 0) {
299       mp = db.price;
300       source = `GearVN - ${db.name}`;
301       link = buildProductLink(db.name, db.type);
302     }
303   });
304 }

```

```

303 | if (mp <= 0) mp = ip;
304 | const ratio = ip / mp;
305 | const catLinks = getCategoryLinks(item.type || 'Other')
306 | const shops = catLinks.map(cl => ({ shop: cl.shop, url: cl.url }));
307 | return {
308 |   name: item.name,
309 |   type: item.type || 'Other',
310 |   invoicePrice: ip,
311 |   marketPrice: mp,
312 |   dbPrice: db?.price || mp,
313 |   matchedName: db?.name || item.name,
314 |   source,
315 |   link,
316 |   shops,
317 |   status: ratio > 1.2 ? 'overpriced' : ratio > 1.08 ? 'expensive' : 'reasonable',
318 |   percentDiff: mp > 0 ? Math.round(((ip - mp) / mp) * 100) : 0,
319 | };
320 | });
321 | }
322 |
323 | function buildVerdict(items: any[], ti: number, tm: number): string {
324 |   const diff = ti - tm;
325 |   const pct = tm > 0 ? Math.round((diff / tm) * 100) : 0;
326 |   const o = items.filter((i: any) => i.status === 'overpriced');
327 |   const e = items.filter((i: any) => i.status === 'expensive');
328 |   if (o.length > 0) return `Overpriced! Bìn bẻ hủ ${pct}% ( ${diff} / ${tm} )`;
329 |   if (e.length > 0) return `Expensive! Giá hji cao. Trê thêm ${pct}% ( ${diff} / ${tm} )`;
330 |   return `Giá hãp lý! Chênh ${pct}% . ${diff < 0 ? 'Bìn mua r' : 'Bìn bán r'}`;
331 | }
332 |
333 | export async function POST(request: NextRequest) {
334 |   try {
335 |     const { imageBase64, manualItems } = await request.json();
336 |
337 |     // Manual entry mode
338 |     if (manualItems && Array.isArray(manualItems) && manualItems.length > 0) {
339 |       const items = applyDbPrices(manualItems);
340 |       const ti = items.reduce((s: number, i: any) => s + i.invoicePrice, 0);
341 |       const tm = items.reduce((s: number, i: any) => s + i.marketPrice, 0);
342 |       return NextResponse.json({ items, totalInvoicePrice: ti, totalMarketPrice: tm, verdict: buildVerdict(items, ti, tm) });
343 |     }
344 |
345 |     // Image scanning mode
346 |     if (!imageBase64) return NextResponse.json({ error: 'Vui lòng cung cấp ảnh' });
347 |
348 |     const groqKey = (process.env.GROQ_API_KEY || '').trim();
349 |     const geminiKey = (process.env.GOOGLE_GEMINI_API_KEY || process.env.GEMINI_API_KEY || '').trim();
350 |
351 |     console.log(`Invoice scan: groqKey=${!!groqKey}, geminiKey=${!!geminiKey}`);
352 |
353 |     let items: any[] = [];
354 |
355 |     // Try Groq vision first, then Gemini
356 |     if (groqKey) {
357 |       try {
358 |         const d = await scanWithGroq(imageBase64, groqKey);
359 |         if (d.items?.length) items = d.items;
360 |         console.log(`Groq scanned ${items.length} items`);
361 |       } catch (e: any) { console.warn('Groq vision fail:', e.message); }
362 |     }
363 |
364 |     if (!items.length && geminiKey) {
365 |       try {
366 |         const d = await scanWithGemini(imageBase64, geminiKey);
367 |         if (d.items?.length) items = d.items;
368 |         console.log(`Gemini scanned ${items.length} items`);
369 |       } catch (e: any) { console.warn('Gemini fail:', e.message); }
370 |     }
371 |
372 |     // If both fail, try local scan (uses REAL components from 127-item DB)
373 |     if (!items.length) {
374 |       console.log('Falling back to localScan with DB components');
375 |       const localResult = localScan(imageBase64);
376 |       if (localResult?.items?.length) items = localResult.items;
377 |       console.log(`localScan returned ${items.length} items`);
378 |     }
379 |
380 |     // If still no items, suggest manual entry
381 |     if (!items.length) {
382 |       return NextResponse.json({
383 |         error: 'AI không nh - n diện ò ác ênh. Vui lòng nh - p tay linh kiện',
384 |         suggestManual: true,
385 |       }, { status: 422 });
386 |     }
387 |
388 |     const processed = applyDbPrices(items);
389 |     const ti = processed.reduce((s: number, i: any) => s + i.invoicePrice, 0);
390 |     const tm = processed.reduce((s: number, i: any) => s + i.marketPrice, 0);
391 |
392 |     return NextResponse.json({
393 |       items: processed,
394 |       totalInvoicePrice: ti,
395 |       totalMarketPrice: tm,
396 |       verdict: buildVerdict(processed, ti, tm),
397 |     });
398 |   } catch (e: any) {
399 |     return NextResponse.json({ error: e.message || 'Lxi phân tích hóa đơn' });
400 |   }
401 | }
402 |

```

File: app/api/ai/suggest-resources/route.ts (157 lines, 6.8 KB)

```

1 | import { NextResponse } from 'next/server'
2 | import { KNOWLEDGE_BASE } from '@data/knowledgeBase'
3 |
4 | export async function POST(req: Request) {
5 |   try {
6 |     const { completedTopics, currentTopic, strengths, weaknesses } = await req.json()
7 |
8 |     const apiKey = process.env.GEMINI_API_KEY || process.env.GOOGLE_GEMINI_API_KEY
9 |     if (!apiKey) {
10 |       // Fallback: return static suggestions based on completed topics
11 |       return NextResponse.json(generateStaticSuggestions(completedTopics || [], currentTopic))
12 |     }
13 |
14 |     const knowledgeContext = KNOWLEDGE_BASE.map(k => k.content).join('\n\n')
15 |
16 |     const prompt = `Bìn là chuyên gia t° vñn lù trình híc t - p PC Master
17 |
18 | K I % N T H È C N À N T Ç N G :
19 | ${knowledgeContext}
20 |

```

```

21 | H Ì C S I N H H I Æ N T I : h à n h : $ { ( c o m p l e t e d T o p i c s | | [ ] ) . j o i n ( ' , ' ) | | ' C h ° a c c
22 | - C h c Å a n g h i c : $ { c u r r e n t T o p i c | | [ ] ) . j o i n ( ' , ' ) | | ' C h ° a x á c Æ n h ' }
23 | - C h c Å a n g h i c : $ { c u r r e n t T o p i c | | [ ] ) . j o i n ( ' , ' ) | | ' C h ° a x á c Æ n h ' }
24 | - i Å m m i n h : $ { ( s t r e n g t h s | | [ ] ) . j o i n ( ' , ' ) | | ' C h ° a x á c Æ n h ' }
25 | - i Å m y ç u : $ { ( w e a k n e s s e s | | [ ] ) . j o i n ( ' , ' ) | | ' C h ° a x á c Æ n h ' }
26 |
27 | N H I Æ M V ä : ° a r a 4 g ä i ý h i c t - p ° ä c cá n h à n h ó a c h o h i c s i n h . M x i g ä i
28 | 1 . C h c Å ( b ± n g t i z n g V i c t )
29 | 2 . M ô t Æ n g ã n g i n ( 1 - 2 c á u )
30 | 3 . L ý d o g ä i ý ( d ñ a t r ê n i Å m m i n h / y ç u )
31 | 4 . 2 t ê k h ó a t i m k i z m Y o u T u b e b ± n g t i z n g V i c t Å h i c t h ê m ( n g n c á c h b ± n g )
32 |
33 | T r Æ v Å J S O N a r r a y :
34 | [
35 | {
36 | " t o p i c " : " T ê n c h c Å " ,
37 | " d e s c r i p t i o n " : " M ô t Æ n g ã n g " ,
38 | " r e a s o n " : " L ý d o g ä i ý " ,
39 | " y o u t u b e K e y w o r d s " : " t ê k h ó a 1 , t ê k h ó a 2 "
40 | }
41 | ]
42 |
43 | C H È T R c V Å J S O N , k h ò n g t h ê m t e x t n à o k h á c . `
44 |
45 | const res = await fetch(`https://generativelanguage.googleapis.com/v1/models/gemini-1.5-flash:generateCont...
46 |   method: 'POST',
47 |   headers: { 'Content-Type': 'application/json' },
48 |   body: JSON.stringify({
49 |     contents: [{ parts: [{ text: prompt }] }],
50 |     generationConfig: {
51 |       temperature: 0.7,
52 |       maxOutputTokens: 1024,
53 |     },
54 |   })
55 | )
56 |
57 | if (!res.ok) {
58 |   return NextResponse.json(generateStaticSuggestions(completedTopics || [], currentTopic))
59 | }
60 |
61 | const data = await res.json()
62 | const text = data.candidates?.[0]?.content?.parts?.[0]?.text || ''
63 |
64 | try {
65 |   // Try to extract JSON from the response
66 |   const jsonMatch = text.match(/\[([^\s\S]*)\]/)
67 |   if (jsonMatch) {
68 |     const suggestions = JSON.parse(jsonMatch[0])
69 |     return NextResponse.json(suggestions)
70 |   }
71 | } catch {}
72 |
73 | return NextResponse.json(generateStaticSuggestions(completedTopics || [], currentTopic))
74 | } catch (err) {
75 |   return NextResponse.json(generateStaticSuggestions([], ''))
76 | }
77 | }
78 |
79 | function generateStaticSuggestions(completedTopics: string[], currentTopic: string) {
80 |   const allTopics = [ 'CPU', 'RAM', 'GPU', 'SSD', 'PSU', 'Mainboard', 'Case' ]
81 |   const completed = new Set(completedTopics.map(t => t.toLowerCase()))
82 |   if (currentTopic) completed.add(currentTopic.toLowerCase())
83 |
84 |   const suggestions = allTopics
85 |     .filter(t => !completed.has(t.toLowerCase()))
86 |     .slice(0, 4)
87 |     .map((topic, i) => ({
88 |       topic: getTopicName(topic),
89 |       description: getTopicDescription(topic),
90 |       reason: i === 0 ? 'Nên híc ti z p theo trong l ù t r i n h' : 'C ñ n g c ñ k i
91 |       youtubeKeywords: getYoutubeKeywords(topic),
92 |     }))
93 |
94 |   // If all topics completed, suggest review/advanced
95 |   if (suggestions.length === 0) {
96 |     suggestions.push({
97 |       topic: 'L - p r á p h o à n c h é n h m ù t b ù P C',
98 |       description: 'Th ñ c h à n h l - p r á p t o à n b ù h c t h ñ n g t ê A z i n Z , k i Å m
99 |       reason: 'B i n ã h o à n t h à n h t ð t c Æ c h c Å c i b ê n ! H ä y t h i t h á c h b ê n
100 |       youtubeKeywords: 'h ò ù n g d « n l - p r á p P C c h o n g ° Y i m ù i , b u i l d P C g a n
101 |     })
102 |     suggestions.push({
103 |       topic: 'Ep xung (Overclocking) CPU & RAM',
104 |       description: 'T i m h i Å u c á c h t n g h i ç u n n g v ° ä t t r ù i b ± n g é p x u n g
105 |       reason: 'K i z n t h é c n ã n g c a o d à n h c h o n g ° Y i , ä v i n g n Å n t Æ n g .',
106 |       youtubeKeywords: 'c á c h é p x u n g C P U a n t o à n , h ò ù n g d « n é p x u n g R A M
107 |     })
108 |   }
109 |   return suggestions
110 | }
111 |
112 | function getTopicName(topic: string) {
113 |   const names: Record<string, string> = {
114 |     CPU: 'B ù v i x í l ý ( C P U )',
115 |     RAM: 'B ù n h ù R A M',
116 |     GPU: 'C a r d Ó h i á ( G P U )',
117 |     SSD: 'Ô c é n g S S D',
118 |     PSU: 'B ù n g u ó n ( P S U )',
119 |     Mainboard: 'B o m i c h c h c ( M a i n b o a r d )',
120 |     Case: 'V i m á y t i n h ( C a s e )',
121 |     T ê n n h i c t : 'H c t h ñ n g t ê n n h i c t',
122 |     COOLING: 'H c t h ñ n g t ê n n h i c t',
123 |   }
124 |   return names[topic] || topic
125 | }
126 |
127 | function getTopicDescription(topic: string) {
128 |   const desc: Record<string, string> = {
129 |     CPU: 'T i m h i Å u v Å s o c k e t , c o r e s , t h r e a d s , x u n g n h é p v à T D P c ç a b ù v i
130 |     RAM: 'K h á m p h á D D R 4 v s D D R 5 , b u s , R A M , D u a l C h a n n e l v à d u n g l ò ä n g p h ù
131 |     GPU: 'H i Å u v Å V R A M , C U D A c o r e s , P C I e v à c á c h c h i n c a r d Ó h i á p h ù h
132 |     SSD: 'S o s á n h N V M e v s S A T A , t ñ c ù í c g h i v à c á c h c h i n c ó c é n g .',
133 |     PSU: 'C ò n g s u ð t , h i ç u s u ð t 8 0 P l u s v à c á c c h u © n m o d u l a r c ç a b ù n g u ó n
134 |     Mainboard: 'C h i p s e t , f o r m f a c t o r , s l o t v à c á c c ò n g k i z t n ñ i t r ê n b o
135 |     Case: 'K i c h t h ò c , a i r f l o w , k i n h c ò ñ g l i c v à c á c h c h i n n h i t m á y .',
136 |     T ê n n h i c t : 'A i r c o o l i n g v s A I O , q u i t c a s e v à k e o t ê n n h i c t .',
137 |     COOLING: 'A i r c o o l i n g v s A I O , q u i t c a s e v à k e o t ê n n h i c t .',
138 |   }
139 |   return desc[topic] || `T i m h i Å u v Å $ { t o p i c } t r o n g l - p r á p P C .`
140 | }
141 |
142 | function getYoutubeKeywords(topic: string) {
143 |   const keywords: Record<string, string> = {

```

```
1 4 5 | CPU: 'c á c h c h í n C P U p h ù h ã p 2 0 2 5 , h ° Ù n g - d « n l - p C P U v à o m a i n b o a r d ' ,  
1 4 6 | R A M : ' n ê n m u a R A M D D R 4 h a y D D R 5 , c á c h - m R A M ú n g k ê n h D u a l C h a n n ê  
1 4 7 | G P U : ' c á c h c h í n c a r d Ó h i a c h o , g a m i n g , - c á c l o i i c a r d Ó h i a p h Ò b i  
1 4 8 | S S D : ' n ê n m u a S S D N V M e h a y S A T A , c á c h - n S S D M . 2 v à o m a i n b o a r d ' ,  
1 4 9 | P S U : ' c á c h c h í n n g u Õ n m á y t í n h p h ù h ã p , n g u Õ n 8 0 P l u s l o i i g i ' ,  
1 5 0 | M a i n b o a r d : ' c á c h c h í n m a i n b o a r d p h ù h ã p , v ù i C P U , c á c l o i i m a i n b o a r d  
1 5 1 | C a s e : ' c á c h c h í n v i c a s e m á y t í n h , h ° Ù n g d « n h i , d á y t r o n g c a s e ' ,  
1 5 2 | ' T Ê n n h i Ç t ' : ' n ê n m u a t Ê n n h i Ç t k h í h a y t Ê n n h i Ç t n ° Û c , c á c h b ô i k e o  
1 5 3 | C O O L I N G : ' n ê n m u a t Ê n n h i Ç t k h í h a y t Ê n n h i Ç t n ° Û c , c á c h b ô i k e o t Ê n  
154 | }  
1 5 5 | r e t u r n k e y w o r d s [ t o p i c ] | | ` h í c v Á $ { t o p i c } l - p r á p P C , $ { t o p i c } m á y t  
156 |  
157 |
```

File: app/api/ai/tools/route.ts (335 lines, 12.5 KB)

```
1 import { NextRequest, NextResponse } from 'next/server'  
2 import { createServiceClient } from '@lib/supabase-service'  
3  
4 const ALLOWED_API_KEYS = [process.env.AI_API_KEY].filter(Boolean)  
5  
6 function unauthorized() {  
7   return NextResponse.json({ error: 'Unauthorized: missing or invalid x-api-key header' }, { status: 401 })  
8 }  
9  
10 function ErrorResponse(msg: string, status = 400) {  
11   return NextResponse.json({ error: msg }, { status })  
12 }  
13  
14 function success(data: any) {  
15   return NextResponse.json({ success: true, data })  
16 }  
17  
18 export async function POST(req: NextRequest) {  
19   const apiKey = req.headers.get('x-api-key')  
20   if (ALLOWED_API_KEYS.length > 0 && (!apiKey || !ALLOWED_API_KEYS.includes(apiKey))) {  
21     return unauthorized()  
22   }  
23  
24   let body: any  
25   try { body = await req.json() } catch { return ErrorResponse('Invalid JSON body') }  
26  
27   const { action, params = {} } = body  
28   if (!action) return ErrorResponse('Missing "action" field')  
29  
30   const supabase = createServiceClient()  
31  
32   try {  
33     switch (action) {  
34  
35 | // % % % L E S S O N S % % %  
36 | case 'list_lessons': {  
37 |   const { data, error } = await supabase  
38 |     .from('lessons')  
39 |     .select('*')  
40 |     .order('created_at', { ascending: false })  
41 |     if (error) throw error  
42 |     return success(data)  
43 |   }  
44 |  
45 | case 'get_lesson': {  
46 |   const { id } = params  
47 |   if (!id) return ErrorResponse('Missing "id" param')  
48 |   const { data, error } = await supabase  
49 |     .from('lessons')  
50 |     .select('*')  
51 |     .eq('id', id)  
52 |     .single()  
53 |     if (error) throw error  
54 |     return success(data)  
55 |   }  
56 |  
57 | // % % % Q U I Z Z E S % % %  
58 | case 'list_quizzes': {  
59 |   const { data, error } = await supabase  
60 |     .from('quizzes')  
61 |     .select('*')  
62 |     .order('created_at', { ascending: false })  
63 |     if (error) throw error  
64 |     return success(data)  
65 |   }  
66 |  
67 | case 'get_quiz': {  
68 |   const { id } = params  
69 |   if (!id) return ErrorResponse('Missing "id" param')  
70 |   const { data, error } = await supabase  
71 |     .from('quizzes')  
72 |     .select('*')  
73 |     .eq('id', id)  
74 |     .single()  
75 |     if (error) throw error  
76 |     return success(data)  
77 |   }  
78 |  
79 | // % % % C L A S S E S % % %  
80 | case 'list_classes': {  
81 |   const { data, error } = await supabase  
82 |     .from('classes')  
83 |     .select('*', profiles:teacher_id(full_name, email))  
84 |     .order('created_at', { ascending: false })  
85 |     if (error) throw error  
86 |     return success(data)  
87 |   }  
88 |  
89 | case 'get_class': {  
90 |   const { id } = params  
91 |   if (!id) return ErrorResponse('Missing "id" param')  
92 |   const { data, error } = await supabase  
93 |     .from('classes')  
94 |     .select('*', profiles:teacher_id(full_name, email))  
95 |     .eq('id', id)  
96 |     .single()  
97 |     if (error) throw error  
98 |     return success(data)  
99 |   }  
100 |  
101 | // % % % C O U R S E S % % %  
102 | case 'list_courses': {  
103 |   const { data, error } = await supabase  
104 |     .from('courses')  
105 |     .select('*')  
106 |     .order('created_at', { ascending: false })  
107 |     if (error) throw error
```

```

108 |     return success(data)
109 | }
110 |
111 | case 'get_course': {
112 |   const { id } = params
113 |   if (!id) return ErrorResponse('Missing "id" param')
114 |   const { data, error } = await supabase
115 |     .from('courses')
116 |     .select('*')
117 |     .eq('id', id)
118 |     .single()
119 |   if (error) throw error
120 |   return success(data)
121 | }
122 |
123 | // % % % E X A M S % % %
124 | case 'list_exams': {
125 |   const { data, error } = await supabase
126 |     .from('exams')
127 |     .select('*')
128 |     .order('created_at', { ascending: false })
129 |   if (error) throw error
130 |   return success(data)
131 | }
132 |
133 | // % % % A C H I E V E M E N T S % % %
134 | case 'list_achievements': {
135 |   const { data, error } = await supabase
136 |     .from('achievements')
137 |     .select('*')
138 |     .order('title')
139 |   if (error) throw error
140 |   return success(data)
141 | }
142 |
143 | // % % % C E R T I F I C A T E S % % %
144 | case 'list_certificates': {
145 |   const { data, error } = await supabase
146 |     .from('certificates')
147 |     .select('*')
148 |     .order('created_at', { ascending: false })
149 |   if (error) throw error
150 |   return success(data)
151 | }
152 |
153 | // % % % B U I L D E R P A R T S % % %
154 | case 'list_builder_parts': {
155 |   const { type } = params
156 |   let query = supabase.from('pc_parts').select('*')
157 |   if (type) query = query.eq('type', type)
158 |   const { data, error } = await query.order('name')
159 |   if (error) throw error
160 |   return success(data)
161 | }
162 |
163 | // % % % L E A R N I N G P A T H S % % %
164 | case 'list_learning_paths': {
165 |   const { data, error } = await supabase
166 |     .from('learning_paths')
167 |     .select('*')
168 |     .order('order')
169 |   if (error) throw error
170 |   return success(data)
171 | }
172 |
173 | // % % % S T U D E N T P R O G R E S S % % %
174 | case 'get_student_progress': {
175 |   const { student_id } = params
176 |   if (!student_id) return ErrorResponse('Missing "student_id" param')
177 |   const [lessonProgress, quizAttempts, builderSessions] = await Promise.all([
178 |     supabase.from('lesson_progress')
179 |       .select('*', lessons(title))
180 |       .eq('student_id', student_id)
181 |       .order('last_accessed', { ascending: false }),
182 |     supabase.from('quiz_attempts')
183 |       .select('*', quizzes(title))
184 |       .eq('student_id', student_id)
185 |       .order('submitted_at', { ascending: false }),
186 |     supabase.from('builder_sessions')
187 |       .select('*')
188 |       .eq('student_id', student_id)
189 |       .order('started_at', { ascending: false }),
190 |   ])
191 |   return success({
192 |     lessonProgress: lessonProgress.data,
193 |     quizAttempts: quizAttempts.data,
194 |     builderSessions: builderSessions.data,
195 |     stats: {
196 |       completedLessons: lessonProgress.data?.filter(l => l.status === 'completed').length ?? 0,
197 |       totalLessons: lessonProgress.data?.length ?? 0,
198 |       avgScore: quizAttempts.data?.length
199 |         ? quizAttempts.data.reduce((s, q) => s + (q.score ?? 0), 0) / quizAttempts.data.length
200 |         : null,
201 |     }
202 |   })
203 | }
204 |
205 | // % % % U S E R S % % %
206 | case 'list_users': {
207 |   const { role, limit: userLimit = 50 } = params
208 |   let query = supabase.from('profiles').select('*').order('created_at', { ascending: false }).limit(userLimit)
209 |   if (role) query = query.eq('role', role)
210 |   const { data, error } = await query
211 |   if (error) throw error
212 |   return success(data)
213 | }
214 |
215 | // % % % S E A R C H % % %
216 | case 'search': {
217 |   const { q, type: searchType } = params
218 |   if (!q) return ErrorResponse('Missing "q" (search query) param')
219 |   const results: any = {}
220 |   if (!searchType || searchType === 'lessons') {
221 |     const { data } = await supabase.from('lessons').select('id, title, description').ilike('title', `${q}...`)
222 |     if (data) results.lessons = data
223 |   }
224 |   if (!searchType || searchType === 'quizzes') {
225 |     const { data } = await supabase.from('quizzes').select('id, title, description').ilike('title', `${q}...`)
226 |     if (data) results.quizzes = data
227 |   }
228 |   if (!searchType || searchType === 'classes') {
229 |     const { data } = await supabase.from('classes').select('id, name, code, subject').ilike('name', `${q}...`)
230 |     if (data) results.classes = data
231 |   }

```

```

232|         if (!searchType || searchType === 'courses') {
233|             const { data } = await supabase.from('courses').select('id, title, description, level').ilike('title...
234|             if (data) results.courses = data
235|         }
236|         if (!searchType || searchType === 'users') {
237|             const { data } = await supabase.from('profiles').select('id, full_name, email, role').ilike('full_na...
238|             if (data) results.users = data
239|         }
240|         return success(results)
241|     }
242| }
243| // % % %   S Y S T E M   I N F O   % % %
244| case 'system_info': {
245|     const [
246|         { count: userCount },
247|         { count: lessonCount },
248|         { count: classCount },
249|         { count: quizCount },
250|         { count: certCount },
251|     ] = await Promise.all([
252|         supabase.from('profiles').select('*', { count: 'exact', head: true }),
253|         supabase.from('lessons').select('*', { count: 'exact', head: true }),
254|         supabase.from('classes').select('*', { count: 'exact', head: true }),
255|         supabase.from('quizzes').select('*', { count: 'exact', head: true }),
256|         supabase.from('certificates').select('*', { count: 'exact', head: true }),
257|     ])
258|     return success({
259|         name: 'PC Master Builder LMS',
260|         version: '1.0.0',
261|         stats: {
262|             users: userCount ?? 0,
263|             lessons: lessonCount ?? 0,
264|             classes: classCount ?? 0,
265|             quizzes: quizCount ?? 0,
266|             certificates: certCount ?? 0,
267|         },
268|         features: [
269|             'Tr ã n h m ô p h ã n g l - p r á p P C 2 D ',
270|             'T r ã l ý A I G e m i n i ( A I G u r u ) ',
271|             'N h - n ð i Ç n c i c h É t a y ( M e d i a P i p e ) ',
272|             'H Ç t h N n g b à i k i Á m t r a t r c n g h i Ç m ',
273|             'L U t r i n h h í c t - p c á n h à n h ó a ',
274|             'C h é n g c h É P D F c ó m ã Q R ',
275|             'T h i ư u m u l t i p l a y e r ',
276|             'B Ë n g x ú p h i n g & t h à n h t í c h ',
277|         ],
278|         available_actions: [
279|             'list_lessons', 'get_lesson',
280|             'list_quizzes', 'get_quiz',
281|             'list_classes', 'get_class',
282|             'list_courses', 'get_course',
283|             'list_exams',
284|             'list_achievements',
285|             'list_certificates',
286|             'list_builder_parts',
287|             'list_learning_paths',
288|             'get_student_progress',
289|             'list_users',
290|             'search',
291|             'system_info',
292|         ]
293|     })
294| }
295|
296| default:
297|     return ErrorResponse(`Unknown action "${action}". Available: system_info`)
298| }
299| } catch (err: any) {
300|     console.error('AI Tools API Error:', err)
301|     return NextResponse.json({ error: err.message || 'Internal server error' }, { status: 500 })
302| }
303| }
304|
305| export async function GET(req: NextRequest) {
306|     const apiKey = req.nextUrl.searchParams.get('api_key')
307|     if (ALLOWED_API_KEYS.length > 0 && (!apiKey || !ALLOWED_API_KEYS.includes(apiKey))) {
308|         return unauthorized()
309|     }
310|     return NextResponse.json({
311|         name: 'PC Master Builder - AI Tools API',
312|         description: 'Hidden API for Claude AI to access all system features.',
313|         usage: 'POST /api/ai/tools with JSON body: { "action": "...", "params": {...} }',
314|         auth: 'Set header "x-api-key" or query param "api_key"',
315|         actions: [
316|             { action: 'system_info', description: 'Get system overview and available actions' },
317|             { action: 'list_lessons', description: 'List all lessons' },
318|             { action: 'get_lesson', params: { id: 'string' }, description: 'Get lesson by ID' },
319|             { action: 'list_quizzes', description: 'List all quizzes' },
320|             { action: 'get_quiz', params: { id: 'string' }, description: 'Get quiz by ID' },
321|             { action: 'list_classes', description: 'List all classes' },
322|             { action: 'get_class', params: { id: 'string' }, description: 'Get class by ID' },
323|             { action: 'list_courses', description: 'List all courses' },
324|             { action: 'get_course', params: { id: 'string' }, description: 'Get course by ID' },
325|             { action: 'list_exams', description: 'List all exams' },
326|             { action: 'list_achievements', description: 'List all achievements' },
327|             { action: 'list_certificates', description: 'List all certificates' },
328|             { action: 'list_builder_parts', params: { type: 'string (optional)' }, description: 'List PC builder par...' },
329|             { action: 'list_learning_paths', description: 'List all learning paths' },
330|             { action: 'get_student_progress', params: { student_id: 'string' }, description: 'Get student progress' },
331|             { action: 'list_users', params: { role: 'string (optional)', limit: 'number (optional)' }, description: '...' },
332|             { action: 'search', params: { q: 'string', type: 'string (optional)' }, description: 'Search across feat...' },
333|         ]
334|     })
335| }

```

File: app/api/auth/callback/route.ts (42 lines, 1.3 KB)

```

1| import { NextResponse } from 'next/server'
2| import { createClient } from '@lib/supabase-ssr-server'
3|
4| export async function GET(request: Request) {
5|     const { searchParams, origin } = new URL(request.url)
6|     const code = searchParams.get('code')
7|
8|     if (code) {
9|         const supabase = await createClient()
10|         const { data: { user }, error } = await supabase.auth.exchangeCodeForSession(code)
11|
12|         if (!error && user) {
13|             const { data: profile } = await supabase
14|                 .from('profiles')
15|                 .select('role')
16|                 .eq('id', user.id)

```

```

17 |         .single()
18 |
19 |     if (!profile || !profile.role) {
20 |         const fullName = user.user_metadata?.full_name || user.email?.split('@')[0] || 'User'
21 |         await supabase.from('profiles').upsert({
22 |             id: user.id,
23 |             email: user.email,
24 |             full_name: fullName,
25 |             role: 'student',
26 |             updated_at: new Date().toISOString(),
27 |         })
28 |         return NextResponse.redirect(`${origin}/builder`)
29 |     }
30 |
31 |     const userRole = profile.role
32 |     let redirectPath = '/builder'
33 |     if (userRole === 'teacher') redirectPath = '/teacher'
34 |     else if (userRole === 'parent') redirectPath = '/parent'
35 |
36 |     return NextResponse.redirect(`${origin}${redirectPath}`)
37 | }
38 |
39 |
40 | return NextResponse.redirect(`${origin}/login?error=could_not_authenticate`)
41 | }
42 |

```

File: app/api/auth/profile/route.ts (21 lines, 0.5 KB)

```

1 | import { createClient } from '@lib/supabase-ssr-server'
2 | import { NextResponse } from 'next/server'
3 |
4 | export async function GET() {
5 |     const supabase = await createClient()
6 |
7 |     const { data: { user } } = await supabase.auth.getUser()
8 |
9 |     if (!user) {
10 |         return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
11 |     }
12 |
13 |     const { data: profile } = await supabase
14 |       .from('profiles')
15 |       .select('role')
16 |       .eq('id', user.id)
17 |       .single()
18 |
19 |     return NextResponse.json(profile || {})
20 | }
21 |

```

File: app/api/builder/end-session/route.ts (28 lines, 1.2 KB)

```

1 | import { createServerClient } from '@supabase/ssr'
2 | import { cookies } from 'next/headers'
3 | import { NextResponse } from 'next/server'
4 |
5 | export async function POST(request: Request) {
6 |     const cookieStore = await cookies()
7 |     const supabase = createServerClient(
8 |       process.env.NEXT_PUBLIC_SUPABASE_URL!,
9 |       process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!,
10 |      { cookies: { getAll: () => cookieStore.getAll(), setAll: () => {} } }
11 |    )
12 |
13 |     const { data: { user } } = await supabase.auth.getUser()
14 |     if (!user) return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
15 |
16 |     const body = await request.json().catch(() => ({}))
17 |     const { sessionId, components, tdp, compatibilityScore, completed } = body
18 |     if (!sessionId) return NextResponse.json({ error: 'Missing sessionId' }, { status: 400 })
19 |
20 |     const { error } = await supabase
21 |       .from('builder_sessions')
22 |       .update({ components_used: components || {}, tdp_total: tdp ?? null, compatibility_score: compatibilitySco...
23 |       .eq('id', sessionId)
24 |
25 |     if (error) return NextResponse.json({ error: error.message }, { status: 500 })
26 |     return NextResponse.json({ success: true })
27 | }
28 |

```

File: app/api/builder/start-session/route.ts (28 lines, 1.0 KB)

```

1 | import { createServerClient } from '@supabase/ssr'
2 | import { cookies } from 'next/headers'
3 | import { NextResponse } from 'next/server'
4 |
5 | export async function POST(request: Request) {
6 |     const cookieStore = await cookies()
7 |     const supabase = createServerClient(
8 |       process.env.NEXT_PUBLIC_SUPABASE_URL!,
9 |       process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!,
10 |      { cookies: { getAll: () => cookieStore.getAll(), setAll: () => {} } }
11 |    )
12 |
13 |     const { data: { user } } = await supabase.auth.getUser()
14 |     if (!user) return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
15 |
16 |     const body = await request.json().catch(() => ({}))
17 |     const lesson_id = body.lesson_id ?? null
18 |
19 |     const { data, error } = await supabase
20 |       .from('builder_sessions')
21 |       .insert({ student_id: user.id, lesson_id })
22 |       .select('id')
23 |       .single()
24 |
25 |     if (error) return NextResponse.json({ error: error.message }, { status: 500 })
26 |     return NextResponse.json({ sessionId: data.id })
27 | }
28 |

```

File: app/api/builds/route.ts (91 lines, 3.2 KB)

```

1 | import { createClient } from '@lib/supabase-ssr-server'
2 | import { NextRequest, NextResponse } from 'next/server'
3 |

```



```

4 | export async function GET(req: NextRequest) {
5 |   const supabase = await createClient()
6 |   const { searchParams } = new URL(req.url)
7 |   const id = searchParams.get('id')
8 |   const userId = searchParams.get('userId')
9 |   const publicOnly = searchParams.get('public') === 'true'
10 |
11 |   if (id) {
12 |     const { data, error } = await supabase
13 |       .from('saved_builds')
14 |       .select('*', cpu:products!cpu_id(*), gpu:products!gpu_id(*), mainboard:products!mainboard_id(*), psu:prod...
15 |       .eq('id', id)
16 |       .single()
17 |
18 |     if (error) return NextResponse.json({ error: error.message }, { status: 500 })
19 |     return NextResponse.json({ build: data })
20 |   }
21 |
22 |   let query = supabase
23 |     .from('saved_builds')
24 |     .select('id, name, compatibility_score, total_tdp, total_price, created_at, user_id')
25 |
26 |   if (publicOnly) {
27 |     query = query.eq('is_public', true)
28 |   } else if (userId) {
29 |     query = query.eq('user_id', userId)
30 |   } else {
31 |     const { data: { user } } = await supabase.auth.getUser()
32 |     if (user) query = query.eq('user_id', user.id)
33 |     else return NextResponse.json({ builds: [] })
34 |   }
35 |
36 |   const { data, error } = await query.order('created_at', { ascending: false }).limit(20)
37 |
38 |   if (error) return NextResponse.json({ error: error.message }, { status: 500 })
39 |   return NextResponse.json({ builds: data || [] })
40 | }
41 |
42 | export async function POST(req: NextRequest) {
43 |   const supabase = await createClient()
44 |   const { data: { user } } = await supabase.auth.getUser()
45 |   if (!user) return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
46 |
47 |   const body = await req.json()
48 |   const { data, error } = await supabase
49 |     .from('saved_builds')
50 |     .upsert({
51 |       user_id: user.id,
52 |       name: body.name || 'Build mới',
53 |       cpu_id: body.cpu_id || null,
54 |       gpu_id: body.gpu_id || null,
55 |       mainboard_id: body.mainboard_id || null,
56 |       ram_ids: body.ram_ids || [],
57 |       psu_id: body.psu_id || null,
58 |       case_id: body.case_id || null,
59 |       cooler_id: body.cooler_id || null,
60 |       ssd_ids: body.ssd_ids || [],
61 |       compatibility_score: body.compatibility_score || 0,
62 |       total_tdp: body.total_tdp || 0,
63 |       total_price: body.total_price || 0,
64 |       is_public: body.is_public || false,
65 |     })
66 |     .select('id')
67 |     .single()
68 |
69 |   if (error) return NextResponse.json({ error: error.message }, { status: 500 })
70 |   return NextResponse.json({ id: data.id })
71 | }
72 |
73 | export async function DELETE(req: NextRequest) {
74 |   const supabase = await createClient()
75 |   const { data: { user } } = await supabase.auth.getUser()
76 |   if (!user) return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
77 |
78 |   const { searchParams } = new URL(req.url)
79 |   const id = searchParams.get('id')
80 |   if (!id) return NextResponse.json({ error: 'Missing id' }, { status: 400 })
81 |
82 |   const { error } = await supabase
83 |     .from('saved_builds')
84 |     .delete()
85 |     .eq('id', id)
86 |     .eq('user_id', user.id)
87 |
88 |   if (error) return NextResponse.json({ error: error.message }, { status: 500 })
89 |   return NextResponse.json({ success: true })
90 | }
91 |

```

File: app/api/career-suggest/route.ts (420 lines, 25.7 KB)

```

1 | import { NextRequest, NextResponse } from 'next/server';
2 |
3 | const GROQ_API_URL = 'https://api.groq.com/openai/v1/chat/completions';
4 | const GROQ_MODEL = 'llama-3.3-70b-versatile';
5 |
6 | interface BuildItem {
7 |   id: string; name: string; type: string; price: number; reason: string;
8 | }
9 |
10 | const CAREER_BUILDS: Record<string, { explanation: string; build: BuildItem[]; totalPrice: number; tips: string[] }> = {
11 |   'ai engineer': {
12 |     explanation: 'AI Engineer cần cấu hình máy tính và GPU để train model AI',
13 |     build: [
14 |       { id: 'cpu_r9_7950x', name: 'AMD Ryzen 9 7950X', type: 'CPU', price: 14975000, reason: '16 nhân cho train...'},
15 |       { id: 'gpu_rtx_4090', name: 'NVIDIA RTX 4090 24GB', type: 'GPU', price: 4090000, reason: 'Hi-end cho AI'},
16 |       { id: 'ram_d5_64g', name: '64GB (2x32GB) DDR5-6400', type: 'RAM', price: 640000, reason: 'RAM 64GB cho...'},
17 |       { id: 'storage_2tb_ssd', name: '2TB NVMe Gen4', type: 'Storage', price: 200000, reason: 'Storage nhanh'},
18 |       { id: 'psu_1000w', name: '1000W 80+ Platinum', type: 'PSU', price: 100000, reason: 'PSU 1000W'},
19 |       { id: 'mb_b650_amd', name: 'B650 Tomahawk WiFi (AM5)', type: 'Mainboard', price: 650000, reason: 'Mainboard AMD'},
20 |       { id: 'cooler_aio', name: 'Tản Nhiệt Nước AIO 360mm', type: 'Cooler', price: 360000, reason: 'Cooler nước'}
21 |     ],
22 |     totalPrice: 76275000,
23 |     tips: 'Dùng thêm RAM 64GB+ nếu làm việc với dataset lớn. Cân nhắc tản nhiệt cho CPU/GPU',
24 |   },
25 |   'it engineer': {
26 |     explanation: 'Kỹ sư IT cần cấu hình máy tính để xử lý công việc lập trình',
27 |     build: [
28 |       { id: 'cpu_i7_13700k', name: 'Intel Core i7-13700K', type: 'CPU', price: 13700000, reason: 'CPU Intel i7-13700K'},
29 |       { id: 'ram_d5_64g', name: '64GB (2x32GB) DDR5-6400', type: 'RAM', price: 640000, reason: 'RAM 64GB cho...'},
30 |       { id: 'gpu_rtx_4090', name: 'NVIDIA RTX 4090 24GB', type: 'GPU', price: 4090000, reason: 'Hi-end cho AI'},
31 |       { id: 'storage_2tb_ssd', name: '2TB NVMe Gen4', type: 'Storage', price: 200000, reason: 'Storage nhanh'},
32 |       { id: 'psu_750w', name: '750W 80+ Gold Fully Modular', type: 'PSU', price: 750000, reason: 'PSU 750W'}

```

```
33 |         { id: 'mb_z790', name: 'Z790 Aorus Master (LGA1700)', type: 'Mainbo
34 |         { id: 'cooler_aio', name: 'Tén NhiệT N°Úc AIO 360mm', type: 'Cooler
35 |     },
36 |     totalPrice: 42250000,
37 |     tips: 'Sút 2-3 màn hình Ā t ng n ng sũt. Nên có UPS Ā bèo vç t
38 | },
39 | 3d designer': {
40 |     explanation: '3D Designer cšn GPU minh Ā render ó hĩa, CPU a nhĩa
41 |     build: [
42 |         { id: 'cpu_i9_14900k', name: 'Intel Core i9-14900K', type: 'CPU', price: 13750000, reason: '24 nhân cho ...
43 |         { id: 'gpu_rtx_4080', name: 'NVIDIA RTX 4080 16GB', type: 'GPU', price: 16500000, reason: '4080 16GB',
44 |         { id: 'ram_d5_32g', name: '32GB (2x16GB) DDR5-6000', type: 'RAM', price: 6000000, reason: '32GB (2x16GB) DDR5-6000',
45 |         { id: 'storage_1tb_nvme_gen4', name: '1TB NVMe Gen4', type: 'Storage', price: 1800000, reason: 'Load pro...
46 |         { id: 'storage_4tb_hdd', name: '4TB HDD 7200rpm', type: 'Storage', price: 720000, reason: '4TB HDD 7200rpm',
47 |         { id: 'psu_850w', name: '850W 80+ Gold Fully Modular', type: 'PSU', price: 850000, reason: '850W 80+ Gold Fully Modular',
48 |         { id: 'mb_z790', name: 'Z790 Aorus Master (LGA1700)', type: 'Mainbo
49 |         { id: 'cooler_aio', name: 'Tén NhiệT N°Úc AIO 360mm', type: 'Cooler
50 |     ],
51 |     totalPrice: 63800000,
52 |     tips: 'Màn hình nên chín loii 27" 2K+ có ù phç màu cao (Adobe RGB >9
53 | },
54 | video editor': {
55 |     explanation: 'Video Editor cšn CPU minh Ā render, GPU h× trĩa codec,
56 |     build: [
57 |         { id: 'cpu_i7_13700k', name: 'Intel Core i7-13700K', type: 'CPU', price: 9500000, reason: '16 nhân cho r...
58 |         { id: 'gpu_rtx_4070', name: 'NVIDIA RTX 4070 12GB', type: 'GPU', price: 12000000, reason: '4070 12GB',
59 |         { id: 'ram_d5_32g', name: '32GB (2x16GB) DDR5-6000', type: 'RAM', price: 6000000, reason: '32GB (2x16GB) DDR5-6000',
60 |         { id: 'storage_2tb_ssd', name: '2TB NVMe Gen4', type: 'Storage', price: 3250000, reason: 'Load video nha...
61 |         { id: 'storage_4tb_hdd', name: '4TB HDD 7200rpm', type: 'Storage', price: 720000, reason: '4TB HDD 7200rpm',
62 |         { id: 'psu_750w', name: '750W 80+ Gold Fully Modular', type: 'PSU', price: 750000, reason: '750W 80+ Gold Fully Modular',
63 |         { id: 'mb_b660m', name: 'B660M Pro (Intel LGA1700)', type: 'Mainbo
64 |     ],
65 |     totalPrice: 36500000,
66 |     tips: 'u tiên màn hình 4K vũĩ ù phç màu cao. Thêm RAM nếu edit video
67 | },
68 | streamer': {
69 |     explanation: 'Streamer / Gamer cšn GPU minh cho game, CPU ç nhĩa cho
70 |     build: [
71 |         { id: 'cpu_r7_7800x3d', name: 'AMD Ryzen 7 7800X3D', type: 'CPU', price: 10500000, reason: 'CPU gaming t...
72 |         { id: 'gpu_rtx_4070', name: 'NVIDIA RTX 4070 12GB', type: 'GPU', price: 12000000, reason: '4070 12GB',
73 |         { id: 'ram_d5_32g', name: '32GB (2x16GB) DDR5-6000', type: 'RAM', price: 6000000, reason: '32GB (2x16GB) DDR5-6000',
74 |         { id: 'storage_1tb_nvme_gen4', name: '1TB NVMe Gen4', type: 'Storage', price: 1800000, reason: '1TB NVMe Gen4',
75 |         { id: 'psu_750w', name: '750W 80+ Gold Fully Modular', type: 'PSU', price: 750000, reason: '750W 80+ Gold Fully Modular',
76 |         { id: 'mb_b650_amd', name: 'B650 Tomahawk WiFi (AM5)', type: 'Mainbo
77 |         { id: 'cooler_air', name: 'Tén NhiệT Khí Tiếu Chuøn', type: 'Cooler
78 |     ],
79 |     totalPrice: 36900000,
80 |     tips: 'Sút webcam + microphone chýt l°ång. Màn hình 27" 1440p 165
81 | },
82 | coder': {
83 |     explanation: 'L-p trình viên cšn CPU minh Ā compile, RAM lùn cho ID
84 |     build: [
85 |         { id: 'cpu_i5_13600k', name: 'Intel Core i5-13600K', type: 'CPU', price: 7000000, reason: '14 nhân cho c...
86 |         { id: 'gpu_gtx_1650', name: 'NVIDIA GTX 1650 4GB', type: 'GPU', price: 1650000, reason: '1650 4GB',
87 |         { id: 'ram_d5_32g', name: '32GB (2x16GB) DDR5-6000', type: 'RAM', price: 6000000, reason: '32GB (2x16GB) DDR5-6000',
88 |         { id: 'storage_1tb_nvme_gen4', name: '1TB NVMe Gen4', type: 'Storage', price: 1800000, reason: 'SSD siêu...
89 |         { id: 'psu_500w', name: '500W 80+ Bronze', type: 'PSU', price: 900000, reason: '500W 80+ Bronze',
90 |         { id: 'mb_b660m', name: 'B660M Pro (Intel LGA1700)', type: 'Mainbo
91 |     ],
92 |     totalPrice: 18250000,
93 |     tips: 'Sút 2 màn hình Ā t ng n ng sũt. Bàn phím ci + chuột ergo
94 | },
95 | office worker': {
96 |     explanation: 'Dân v n phòng cšn máy Õn Ēnh, tiẻt kiẻm iẻn, ç dùng
97 |     build: [
98 |         { id: 'cpu_i3_12100', name: 'Intel Core i3-12100F', type: 'CPU', price: 1210000, reason: 'i3-12100F',
99 |         { id: 'ram_d4_16g', name: '16GB (2x8GB) DDR4-3200', type: 'RAM', price: 1600000, reason: '16GB (2x8GB) DDR4-3200',
100 |         { id: 'storage_500_ssd', name: '500GB NVMe Gen3', type: 'Storage', price: 700000, reason: 'SSD nhanh, kh...
101 |         { id: 'psu_350w', name: '350W Power Supply', type: 'PSU', price: 350000, reason: '350W Power Supply',
102 |         { id: 'mb_h510m_intel1200', name: 'H510M K (Intel LGA1200)', type: 'Mainboard', price: 1200000, reason: ...
103 |     ],
104 |     totalPrice: 5500000,
105 |     tips: 'Thêm màn hình 24" IPS Ā bèo vç m-t. Cãn nh-c thêm SSD 500GB
106 | },
107 | };
108 |
109 | const MAC_BUILDS: Record<string, { explanation: string; model: string; build: BuildItem[]; totalPrice: number;...
110 | 'coder': {
111 |     explanation: 'MacBook Pro vũĩ chip Apple Silicon là lĩa chín tuyẻt v
112 |     model: 'MacBook Pro 14" M4 Pro',
113 |     build: [
114 |         { id: 'mac_m4pro_cpu', name: 'Apple M4 Pro (12 CPU, 18 GPU)', type: 'CPU', price: 0, reason: 'Chip Apple...
115 |         { id: 'mac_m4pro_ram', name: '24GB Unified Memory', type: 'RAM', price: 0, reason: '24GB Unified Memory',
116 |         { id: 'mac_m4pro_ssd', name: '512GB SSD', type: 'Storage', price: 0, reason: '512GB SSD',
117 |     ],
118 |     totalPrice: 39990000,
119 |     tips: 'Nếu cšn nhĩa RAM hịn, nâng lên 36GB ho-c 48GB. Màn hình 14"
120 | },
121 | video editor': {
122 |     explanation: 'Mac Studio vũĩ M4 Max ho-c MacBook Pro Pro là lĩa chín
123 |     model: 'Mac Studio M4 Max',
124 |     build: [
125 |         { id: 'mac_m4max_cpu', name: 'Apple M4 Max (16 CPU, 40 GPU)', type: 'CPU', price: 0, reason: '16 CPU, 40 GPU',
126 |         { id: 'mac_m4max_ram', name: '64GB Unified Memory', type: 'RAM', price: 0, reason: '64GB Unified Memory',
127 |         { id: 'mac_m4max_ssd', name: '1TB SSD', type: 'Storage', price: 0, reason: '1TB SSD',
128 |     ],
129 |     totalPrice: 64990000,
130 |     tips: 'Kẻt hĩa vũĩ màn hình Pro Display XDR ho-c Studio Display. Nãn
131 | },
132 | office worker': {
133 |     explanation: 'MacBook Air M4 là lĩa chín hoàn hẻo cho dân v n phòng
134 |     model: 'MacBook Air 13" M4',
135 |     build: [
136 |         { id: 'mac_m4_cpu', name: 'Apple M4 (8 CPU, 10 GPU)', type: 'CPU', price: 0, reason: '8 CPU, 10 GPU',
137 |         { id: 'mac_m4_ram', name: '16GB Unified Memory', type: 'RAM', price: 0, reason: '16GB Unified Memory',
138 |         { id: 'mac_m4_ssd', name: '256GB SSD', type: 'Storage', price: 0, reason: '256GB SSD',
139 |     ],
140 |     totalPrice: 21990000,
141 |     tips: 'MacBook Air cũc kó phù hĩa cho sinh viên nhỹ pin 18 tiẻng. Nãn
142 | },
143 | };
144 |
145 | const SPELL_CORRECTIONS: Record<string, string> = {
146 |     'ky thuăt viẻn': 'kủ thu-t viẻn', 'ky su': 'kủ s°', 'lập trình': 'l-p
147 |     'thiẻt ke': 'thiẻt kủ', 'đo hoĩa': 'ó hĩa', 'lãm phim': 'lãm phim',
148 |     'hoc sinh': 'hĩc sinh', 'sinh viẻn': 'sinh viẻn', 'van phong': 'v n ph
149 |     'nhĩa viẻn': 'nhĩa viẻn', 'quan tri': 'quẻn trẻ', 'he thong': 'hẻ th n ph
150 |     'may tinh': 'mãy tinh', 'game thu': 'game thẻ', 'ai eng': 'ai engineer
151 |     'machine learning': 'machine learning', 'deep learning': 'deep learning',
152 |     'data scientist': 'data scientist', 'web dev': 'web developer',
153 | };
154 |
155 | function spellCheck(text: string): string {
156 |     const lower = text.toLowerCase().trim();
```

```

157 // Exact match
158 if (SPELL_CORRECTIONS[lower]) return SPELL_CORRECTIONS[lower];
159 // Word-by-word replacement
160 const words = lower.split(/ /);
161 const corrected = words.map(w => {
162   // Check exact word
163   if (SPELL_CORRECTIONS[w]) return SPELL_CORRECTIONS[w];
164   // Check with simple edit distance
165   for (const [key, val] of Object.entries(SPELL_CORRECTIONS)) {
166     if (key.includes(w) || w.includes(key)) return val;
167     const dist = levenshtein(w, key);
168     if (dist <= 2) return val;
169   }
170   return w;
171 });
172 return corrected.join(' ');
173 }
174
175 function levenshtein(a: string, b: string): number {
176   const m = a.length, n = b.length;
177   const dp: number[][] = Array.from({ length: m + 1 }, () => Array(n + 1).fill(0));
178   for (let i = 0; i <= m; i++) dp[i][0] = i;
179   for (let j = 0; j <= n; j++) dp[0][j] = j;
180   for (let i = 1; i <= m; i++) {
181     for (let j = 1; j <= n; j++) {
182       dp[i][j] = a[i - 1] === b[j - 1] ? dp[i - 1][j - 1] : 1 + Math.min(dp[i - 1][j], dp[i][j - 1], dp[i - 1]...
183     }
184   }
185   return dp[m][n];
186 }
187
188 const KEYWORD_MAP: Record<string, string> = {
189   'it engineer': 'it engineer', 'it': 'it engineer', 'ky su it': 'it eng
190   'it support': 'it engineer', 'system admin': 'it engineer', 'quan tri he thong': 'it engineer',
191   'quản trị hệ thống': 'it engineer', 'network engineer': 'it engineer',
192   'kỹ thuật viên máy tính': 'it engineer', 'ky su cong nghe thong tin':
193   'kỹ sư công nghệ thông tin': 'it engineer', 'software engineer': 'code
194   'developer': 'coder', 'lập trình viên': 'coder', 'l-p trình viên': 'co
195   'web developer': 'coder', 'programmer': 'coder', 'data scientist': 'ai engineer',
196   'machine learning': 'ai engineer', 'deep learning': 'ai engineer', 'ai': 'ai engineer',
197   'graphic designer': '3d designer', 'thiết kế đồ họa': '3d designer',
198   'game developer': '3d designer', 'game designer': '3d designer', 'lập trình viên': '3d designer',
199   'lập trình viên': '3d designer', 'video editor': 'video editor', 'youtuber': 'streamer',
200   'gamer': 'streamer', 'game thủ': 'streamer',
201   'học sinh': 'office worker', 'học sinh': 'office worker', 'sinh viên':
202   'sinh viên': 'office worker', 'van phong': 'office worker', 'v n phòng
203   'nhân viên': 'office worker', 'nhân viên': 'office worker',
204   'mac': '', 'macbook': '', 'apple': '', // Mac prefix !' empty for separ
205 }
206
207 function analyzeCareer(careerName: string) {
208   const name = careerName.toLowerCase().trim();
209
210   // Exact match first
211   for (const [key, value] of Object.entries(CAREER_BUILDS)) {
212     if (name === key || name.includes(key) || key.includes(name)) return { ...value, career: careerName };
213   }
214
215   // Keyword map
216   for (const [kw, mapped] of Object.entries(KEYWORD_MAP)) {
217     if (name.includes(kw) && mapped) {
218       const build = CAREER_BUILDS[mapped];
219       if (build) return { ...build, career: careerName };
220     }
221   }
222
223   // Word-by-word matching
224   const words = name.split(/ /);
225   for (const [key, value] of Object.entries(CAREER_BUILDS)) {
226     let matchedWords = 0;
227     for (const w of words) {
228       if (w.length > 2 && (key.includes(w) || w.includes(key))) matchedWords++;
229     }
230     if (matchedWords >= 1) return { ...value, career: careerName };
231   }
232
233   return null;
234 }
235
236 function analyzeMac(careerName: string) {
237   const name = careerName.toLowerCase().trim();
238   // Check if user explicitly wants Mac
239   const wantsMac = name.includes('mac') || name.includes('apple');
240   const mapped = wantsMac ? KEYWORD_MAP[name.split(/ /).find((w: string) => ![ 'mac', 'apple', 'macbook' ].inc...
241
242   for (const [key, value] of Object.entries(MAC_BUILDS)) {
243     if (name.includes(key) || key.includes(name) || (mapped && key === mapped)) {
244       return { ...value, career: careerName, isMac: true };
245     }
246   }
247   return null;
248 }
249
250 function defaultBuild(careerName: string) {
251   return {
252     career: careerName,
253     explanation: `Cấu hình PC an ng phù hăp vớ i "${careerName}". ây t
254     build: [
255       { id: 'cpu_r5_5600', name: 'AMD Ryzen 5 5600', type: 'CPU', price:
256       { id: 'gpu_gtx_1650', name: 'NVIDIA GTX 1650 4GB', type: 'GPU', pr
257       { id: 'ram_d4_16G', name: '16GB (2x8GB) DDR4-3200', type: 'RAM', price: 1000000, reason: 'RAM 16GB tiêu ...
258       { id: 'storage_500_ssd', name: '500GB NVMe Gen3', type: 'Storage',
259       { id: 'psu_500w', name: '500W 80+ Bronze', type: 'PSU', price: 900
260       { id: 'mb_b450_amd', name: 'B450M DS3H (AMD AM4)', type: 'Mainboar
261     ],
262     totalPrice: 8200000,
263     tips: 'Bin có th ă nâng c ăp RAM lên 32GB và thêm SSD dung l ăng lôn r
264   };
265 }
266
267 function defaultMacBuild(careerName: string) {
268   return {
269     career: careerName,
270     model: 'MacBook Air M4',
271     isMac: true,
272     explanation: `MacBook Air M4 là l ăa chín tuy ăt v ỷi cho "${careerName}
273     build: [
274       { id: 'mac_m4_base', name: 'Apple M4 (8 CPU, 10 GPU)', type: 'CPU', price: 0, reason: 'Chip Apple Silico...
275       { id: 'mac_m4_ram', name: '16GB Unified Memory', type: 'RAM', price:
276       { id: 'mac_m4_ssd', name: '256GB SSD', type: 'Storage', price: 0,
277     ],
278     totalPrice: 21990000,
279     tips: 'MacBook Air cũc k ồ phù hăp cho công vi ệc v n phòng và h ệ t-p
280   };

```

```

281 }
282
283 async function callGroq(carrierName: string, customCondition?: string): Promise<any | null> {
284   const apiKey = process.env.GROQ_API_KEY || '';
285   if (!apiKey) return null;
286   try {
287     let userContent = `Ngô Ýi dùng có ôc mi nghÁ nghiÇp: "${carrierName}"`;
288   }
289   if (customCondition) {
290     userContent += `\\n\\nNG-ÛI DÙNG CÓ YÊU C!U THÊM: "${customCondition}"`;
291   }
292 }
293 |
294   userContent += `\\n\\nTrE vÁ JSON CHÍNH XÁC:\\n{"career": "Tên nghÁ nghi"}`;
295
296 const r = await fetch(GROQ_API_URL, {
297   method: 'POST',
298   headers: { 'Content-Type': 'application/json', 'Authorization': `Bearer ${apiKey}` },
299   body: JSON.stringify({
300     model: GROQ_MODEL,
301     messages: [
302       { role: 'system', content: 'Bạn là chuyên gia t° vñn c¥u hình h' },
303       { role: 'user', content: userContent }
304     ],
305     temperature: 0.4,
306     max_tokens: 4096,
307   }),
308 });
309 if (!r.ok) return null;
310 const data = await r.json();
311 const text = data.choices?.[0]?.message?.content;
312 if (!text) return null;
313 const cleaned = text.replace(/```json/g, '').replace(/```/g, '').trim();
314 return JSON.parse(cleaned);
315 } catch { return null; }
316
317 export async function POST(request: NextRequest) {
318   try {
319     const { career, customDream, customCondition, preferMac } = await request.json();
320     let careerName = career || customDream;
321     if (!careerName) return NextResponse.json({ error: 'Vui lòng nh - p °' });
322
323     // Spell check
324     const corrected = spellCheck(careerName);
325     if (corrected !== careerName.toLowerCase().trim()) careerName = corrected;
326
327     const wantsMac = preferMac || careerName.toLowerCase().includes('mac') || careerName.toLowerCase().include...
328
329     let result = null;
330     let macResult = null;
331
332     // If user wants Mac, try Mac analysis first
333     if (wantsMac) {
334       macResult = analyzeMac(careerName);
335       if (!macResult) {
336         const groqResult = await callGroq(careerName + ' (Mac/Apple)', customCondition);
337         if (groqResult) macResult = { ...groqResult, isMac: true };
338       }
339       if (!macResult) macResult = defaultMacBuild(careerName);
340     }
341
342     // Get Windows PC build
343     result = await callGroq(careerName, customCondition);
344     if (!result) {
345       result = analyzeCareer(careerName) || defaultBuild(careerName);
346     }
347
348     const enrich = (items: BuildItem[]) => items.map((item: BuildItem) => ({
349       ...item, image: getComponentImage(item.type),
350     }));
351
352     const response: any = { ...result, build: enrich(result.build) };
353
354     if (macResult) {
355       response.macBuild = macResult.isMac ? {
356         ...macResult,
357         build: enrich(macResult.build || []),
358       } : null;
359     }
360
361     return NextResponse.json(response);
362   } catch (e: any) {
363     return NextResponse.json({ error: 'Không thÃ gãi ý c¥u hình. Vui lòng' });
364   }
365 }
366
367 const GEARVN_LINKS: Record<string, string> = {
368   CPU: 'https://gearvn.com/collections/cpu-bo-vi-xu-ly',
369   GPU: 'https://gearvn.com/collections/vga-card-man-hinh',
370   RAM: 'https://gearvn.com/collections/ram-desktop',
371   Mainboard: 'https://gearvn.com/collections/mainboard-bo-mach-chu',
372   Storage: 'https://gearvn.com/collections/ssd-o-cung-ran',
373   PSU: 'https://gearvn.com/collections/psu-nguon-may-tinh',
374   Cooler: 'https://gearvn.com/collections/tan-nhiệt',
375   Case: 'https://gearvn.com/collections/case-vo-may-tinh',
376   Monitor: 'https://gearvn.com/collections/man-hinh',
377 };
378
379 function getGearVnLink(item: BuildItem): string {
380   const name = (item.name || '').toLowerCase();
381   // Try to find specific product link via search
382   if (name.includes('rtx 5090')) return 'https://gearvn.com/products/nvidia-geforce-rtx-5090';
383   if (name.includes('rtx 5880')) return 'https://gearvn.com/products/nvidia-geforce-rtx-5080';
384   if (name.includes('rtx 4090')) return 'https://gearvn.com/products/nvidia-geforce-rtx-4090';
385   if (name.includes('rtx 4080')) return 'https://gearvn.com/products/nvidia-geforce-rtx-4080';
386   if (name.includes('rtx 4070')) return 'https://gearvn.com/products/nvidia-geforce-rtx-4070';
387   if (name.includes('rtx 4060')) return 'https://gearvn.com/products/nvidia-geforce-rtx-4060';
388   if (name.includes('rtx 3060')) return 'https://gearvn.com/products/nvidia-geforce-rtx-3060';
389   if (name.includes('rx 7900')) return 'https://gearvn.com/products/amd-radeon-rx-7900xtx';
390   if (name.includes('rx 7800')) return 'https://gearvn.com/products/amd-radeon-rx-7800xt';
391   if (name.includes('rx 7700')) return 'https://gearvn.com/products/amd-radeon-rx-7700xt';
392   if (name.includes('gtx 1650') || name.includes('gtx1650')) return 'https://gearvn.com/products/nvidia-geforc...';
393   if (name.includes('core i9')) || name.includes('i9-')) return 'https://gearvn.com/products/intel-core-i9-1490...';
394   if (name.includes('core i7')) || name.includes('i7-')) return 'https://gearvn.com/products/intel-core-i7-1470...';
395   if (name.includes('core i5')) || name.includes('i5-')) return 'https://gearvn.com/products/intel-core-i5-1460...';
396   if (name.includes('core i3')) || name.includes('i3-')) return 'https://gearvn.com/products/intel-core-i3-1410...';
397   if (name.includes('ryzen 9')) || name.includes('r9')) return 'https://gearvn.com/products/amd-ryzen-9-7950x';
398   if (name.includes('ryzen 7')) || name.includes('r7')) return 'https://gearvn.com/products/amd-ryzen-7-7800x3d';
399   if (name.includes('ryzen 5')) || name.includes('r5')) return 'https://gearvn.com/products/amd-ryzen-5-7600x';
400   if (name.includes('ddr5')) || name.includes('ddr5')) return 'https://gearvn.com/collections/ram-desktop-ddr5';
401   if (name.includes('ddr4')) || name.includes('ddr4')) return 'https://gearvn.com/collections/ram-desktop-ddr4';
402   if (name.includes('nvme')) || name.includes('ssd')) return 'https://gearvn.com/collections/ssd-nvme';
403   if (name.includes('macbook')) || name.includes('mac book')) return 'https://gearvn.com/collections/macbook';
404   if (name.includes('mac studio')) return 'https://gearvn.com/collections/mac-studio';

```



```

105 |         })
106 |     }
107 | } catch {
108 |     quests = []
109 | }
110 |
111 | const { data: savedQuests } = await supabase
112 |   .from('daily_quests')
113 |   .select('*')
114 |   .eq('is_active', true)
115 |   .limit(10)
116 |
117 | return NextResponse.json({ quests: savedQuests || quests || [], cached: false })
118 | } catch (error: any) {
119 |   console.error('Error generating challenges:', error)
120 |   return NextResponse.json({ error: 'Failed to generate challenges' }, { status: 500 })
121 | }
122 | }
123 | }
124 |

```

File: app/api/chat/route.js (4 lines, 0.1 KB)

```

1 | export async function POST() {
2 |   return Response.json({ reply: 'Vui lòng sí dăng endpoint /api/ai/chat' })
3 | }
4 |

```

File: app/api/classes/submit/route.ts (63 lines, 1.9 KB)

```

1 | import { createClient } from '@lib/supabase-ssr-server';
2 | import { NextResponse } from 'next/server';
3 | import { autoGrade } from '@lib/grading/auto-grade';
4 |
5 | export async function POST(request: Request) {
6 |   try {
7 |     const supabase = await createClient();
8 |     const { data: { user } } = await supabase.auth.getUser();
9 |
10 |    if (!user) {
11 |      return NextResponse.json({ error: 'Unauthorized' }, { status: 401 });
12 |    }
13 |
14 |    const body = await request.json();
15 |    const { assignment_id, class_id, pc_config } = body;
16 |
17 |    // 1. Fetch assignment requirements
18 |    const { data: assignment, error: asgError } = await supabase
19 |      .from('assignments')
20 |      .select('*')
21 |      .eq('id', assignment_id)
22 |      .single();
23 |
24 |    if (asgError || !assignment) {
25 |      return NextResponse.json({ error: 'Assignment not found' }, { status: 404 });
26 |    }
27 |
28 |    // 2. Auto-grade
29 |    const gradingResult = autoGrade(pc_config, assignment.requirements, assignment.max_score);
30 |
31 |    // 3. Check if late
32 |    const isLate = assignment.deadline ? new Date() > new Date(assignment.deadline) : false;
33 |
34 |    // 4. Save submission
35 |    const { data: submission, error: subError } = await supabase
36 |      .from('submissions')
37 |      .upsert({
38 |        assignment_id,
39 |        student_id: user.id,
40 |        class_id,
41 |        pc_config,
42 |        auto_score: gradingResult.score,
43 |        total_score: gradingResult.score, // Ban sũ tĩng i ă m b ă n g i ă m
44 |        is_late: isLate,
45 |        submitted_at: new Date().toISOString()
46 |      })
47 |      .select()
48 |      .single();
49 |
50 |    if (subError) throw subError;
51 |
52 |    return NextResponse.json({
53 |      success: true,
54 |      submission,
55 |      gradingResult
56 |    });
57 |
58 |  } catch (err: any) {
59 |    console.error('Submission API error:', err);
60 |    return NextResponse.json({ error: err.message }, { status: 500 });
61 |  }
62 | }
63 |

```

File: app/api/components/lock/route.ts (67 lines, 2.1 KB)

```

1 | import { NextResponse } from 'next/server';
2 | import { createServiceClient } from '@lib/supabase-service';
3 |
4 | export async function POST(request: Request) {
5 |   try {
6 |     const { componentId, userId, action } = await request.json();
7 |     const supabase = createServiceClient();
8 |
9 |     if (action === 'lock') {
10 |      const expiresAt = new Date(Date.now() + 5 * 60 * 1000).toISOString();
11 |      const { data: existing } = await supabase
12 |        .from('components')
13 |        .select('locked_by, lock_expires_at')
14 |        .eq('id', componentId)
15 |        .single();
16 |
17 |      if (existing?.locked_by && existing.locked_by !== userId) {
18 |        const lockExpired = new Date(existing.lock_expires_at) < new Date();
19 |        if (!lockExpired) {
20 |          const { data: locker } = await supabase
21 |            .from('profiles')
22 |            .select('full_name')
23 |            .eq('id', existing.locked_by)
24 |            .single();

```

```

25 |         return NextResponse.json({
26 |             locked: false,
27 |             lockedBy: locker?.full_name || 'someone',
28 |             message: 'Component is being edited by another user',
29 |         }, { status: 409 });
30 |     }
31 | }
32 |
33 | await supabase
34 |   .from('components')
35 |   .update({ locked_by: userId, locked_at: new Date().toISOString(), lock_expires_at: expiresAt })
36 |   .eq('id', componentId);
37 |
38 | return NextResponse.json({ locked: true });
39 | }
40 |
41 | if (action === 'release') {
42 |   await supabase
43 |     .from('components')
44 |     .update({ locked_by: null, locked_at: null, lock_expires_at: null })
45 |     .eq('id', componentId);
46 |
47 |   return NextResponse.json({ released: true });
48 | }
49 |
50 | if (action === 'heartbeat') {
51 |   const expiresAt = new Date(Date.now() + 5 * 60 * 1000).toISOString();
52 |   await supabase
53 |     .from('components')
54 |     .update({ lock_expires_at: expiresAt })
55 |     .eq('id', componentId)
56 |     .eq('locked_by', userId);
57 |
58 |   return NextResponse.json({ extended: true });
59 | }
60 |
61 | return NextResponse.json({ error: 'Invalid action' }, { status: 400 });
62 | } catch (err) {
63 |   console.error('Lock error:', err);
64 |   return NextResponse.json({ error: 'Internal error' }, { status: 500 });
65 | }
66 | }
67 |

```

File: app/api/components/notify/route.ts (38 lines, 1.4 KB)

```

1 | import { NextResponse } from 'next/server';
2 | import { createServiceClient } from '@lib/supabase-service';
3 |
4 | export async function POST(request: Request) {
5 |   try {
6 |     const { componentId, componentName, action } = await request.json();
7 |     const supabase = createServiceClient();
8 |
9 |     if (action === 'published') {
10 |       // Ghi notification vào bảng notifications
11 |       const { error } = await supabase.from('notifications').insert({
12 |         type: 'component_published',
13 |         title: 'New component published',
14 |         content: `New component "${componentName}" has been published and is ready to use.`,
15 |         metadata: { componentId, componentName },
16 |         created_at: new Date().toISOString(),
17 |       });
18 |       if (error) console.error('Notify error:', error);
19 |     }
20 |
21 |     if (action === 'updated') {
22 |       const { error } = await supabase.from('notifications').insert({
23 |         type: 'component_updated',
24 |         title: 'Component specs updated',
25 |         content: `Component "${componentName}" specs were updated. Please re-check compatibility.`,
26 |         metadata: { componentId, componentName },
27 |         created_at: new Date().toISOString(),
28 |       });
29 |       if (error) console.error('Notify error:', error);
30 |     }
31 |
32 |     return NextResponse.json({ ok: true });
33 |   } catch (err) {
34 |     console.error('Notify error:', err);
35 |     return NextResponse.json({ error: 'Internal error' }, { status: 500 });
36 |   }
37 | }
38 |

```

File: app/api/daily-quiz/route.ts (151 lines, 4.3 KB)

```

1 | import { NextRequest, NextResponse } from 'next/server'
2 | import { createClient } from '@lib/supabase-ssr-server'
3 | import { FULL_QUIZ_BANK } from '@data/quiz-bank-full'
4 | import { shuffleArray } from '@lib/quiz-engine'
5 |
6 | const TOPICS = Object.values(FULL_QUIZ_BANK)
7 | const START_DATE = new Date('2026-06-01')
8 |
9 | function getTodayTopicIndex(): number {
10 |   const now = new Date()
11 |   const diffMs = now.getTime() - START_DATE.getTime()
12 |   const diffDays = Math.floor(diffMs / (1000 * 60 * 60 * 24))
13 |   return ((diffDays % TOPICS.length) + TOPICS.length) % TOPICS.length
14 | }
15 |
16 | export async function GET(request: NextRequest) {
17 |   const supabase = await createClient()
18 |   const { data: { user } } = await supabase.auth.getUser()
19 |
20 |   const { searchParams } = new URL(request.url)
21 |   const topicId = searchParams.get('topicId')
22 |   const todayIndex = getTodayTopicIndex()
23 |   const todayTopic = TOPICS[todayIndex]
24 |
25 |   if (!topicId) {
26 |     let completedTopicIds: string[] = []
27 |     if (user) {
28 |       const { data: txs } = await supabase
29 |         .from('xp_transactions')
30 |         .select('reference_id')
31 |         .eq('user_id', user.id)
32 |         .eq('reference_type', 'daily_quiz')
33 |       completedTopicIds = txs?.map(t => t.reference_id).filter(Boolean) || []
34 |     }

```

```

35 |     return NextResponse.json({
36 |       todayIndex,
37 |       todayTopicId: todayTopic.id,
38 |       todayTopic,
39 |       topics: TOPICS,
40 |       completedTopicIds,
41 |       isTodayCompleted: completedTopicIds.includes(todayTopic.id),
42 |     })
43 |   }
44 | }
45 |
46 | const topic = TOPICS.find(t => t.id === topicId)
47 | if (!topic) {
48 |   return NextResponse.json({ error: 'Topic not found' }, { status: 404 })
49 | }
50 |
51 | let completed = false
52 | if (user) {
53 |   const { data: existing } = await supabase
54 |     .from('xp_transactions')
55 |     .select('id')
56 |     .eq('user_id', user.id)
57 |     .eq('reference_type', 'daily_quiz')
58 |     .eq('reference_id', topicId)
59 |     .maybeSingle()
60 |   completed = !!existing
61 | }
62 |
63 | const questions = shuffleArray(topic.questions).slice(0, 10).map(q => ({
64 |   id: q.id,
65 |   text: q.text,
66 |   question: q.text,
67 |   options: q.options,
68 |   correct: q.correct,
69 |   explanation: q.explanation,
70 | }))
71 |
72 | return NextResponse.json({
73 |   topic,
74 |   questions,
75 |   completed,
76 |   isToday: topicId === todayTopic.id,
77 | })
78 | }
79 |
80 | export async function POST(request: NextRequest) {
81 |   const supabase = await createClient()
82 |   const { data: { user } } = await supabase.auth.getUser()
83 |   if (!user) {
84 |     return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
85 |   }
86 |
87 |   const { topicId, score, total, xpEarned } = await request.json()
88 |
89 |   if (!topicId || xpEarned === undefined) {
90 |     return NextResponse.json({ error: 'Missing required fields' }, { status: 400 })
91 |   }
92 |
93 |   const cappedXp = Math.min(xpEarned, 50)
94 |
95 |   const { error: txErr } = await supabase
96 |     .from('xp_transactions')
97 |     .insert({
98 |       user_id: user.id,
99 |       amount: cappedXp,
100 |       reason: `Quiz hàng ngày: ${topicId} - ${score}/${total}`,
101 |       reference_type: 'daily_quiz',
102 |       reference_id: topicId,
103 |     })
104 |     .select('id')
105 |     .single()
106 |
107 |   if (txErr) {
108 |     return NextResponse.json({ error: txErr.message }, { status: 500 })
109 |   }
110 |
111 |   const { data: profile } = await supabase
112 |     .from('profiles')
113 |     .select('xp, level')
114 |     .eq('id', user.id)
115 |     .single()
116 |
117 |   const newXp = (profile?.xp || 0) + cappedXp
118 |   const now = new Date().toISOString()
119 |
120 |   let newLevel = 1
121 |   const levels = [0, 100, 300, 600, 1000, 1600, 2400, 3400, 4600, 6000, 8000, 10000, 13000, 17000, 22000, 28000...]
122 |   for (let i = levels.length - 1; i >= 0; i--) {
123 |     if (newXp >= levels[i]) { newLevel = i + 1; break }
124 |   }
125 |
126 |   await supabase.from('profiles').update({ xp: newXp, level: newLevel, updated_at: now }).eq('id', user.id)
127 |
128 |   await supabase.from('leaderboard_cache').upsert({
129 |     user_id: user.id, total_xp: newXp, level: newLevel, updated_at: now,
130 |   }, { onConflict: 'user_id' })
131 |
132 |   await supabase.from('learning_stats').upsert({
133 |     user_id: user.id, total_xp: newXp, updated_at: now,
134 |   }, { onConflict: 'user_id' })
135 |
136 |   if (newLevel > (profile?.level || 1)) {
137 |     await supabase.from('notifications').insert({
138 |       user_id: user.id, type: 'success', title: 'Lên Cấp!',
139 |       body: `Chúc mừng! Bạn đã đạt cấp ${newLevel}!`,
140 |       data: { level: newLevel },
141 |     })
142 |   }
143 |
144 |   return NextResponse.json({
145 |     success: true,
146 |     xpAwarded: cappedXp,
147 |     newTotalXp: newXp,
148 |     newLevel,
149 |   })
150 | }
151 |

```

File: app/api/debug-role/route.ts (128 lines, 4.3 KB)

```

1 | import { createClient } from '@lib/supabase-ssr-server'
2 | import { NextResponse } from 'next/server'
3 |

```



```

4 | export async function GET(request: Request) {
5 |   const requestUrl = new URL(request.url)
6 |   const role = requestUrl.searchParams.get('role')
7 |
8 |   if (!role || ![ 'student', 'teacher', 'parent', 'admin' ].includes(role)) {
9 |     return NextResponse.json({
10 |       error: 'Vui lòng cung cấp role hợp lệ: student, teacher, parent, admin',
11 |     }, { status: 400 })
12 |   }
13 |
14 |   const supabase = await createClient()
15 |   const { data: { user } } = await supabase.auth.getUser()
16 |
17 |   if (!user) {
18 |     return NextResponse.json({ error: 'Bạn chưa đăng nhập!' }, { status: 401 })
19 |   }
20 |
21 |   // 1. Kiểm tra xem profile đã tồn tại chưa
22 |   const { data: existingProfile } = await supabase
23 |     .from('profiles')
24 |     .select('id, full_name')
25 |     .eq('id', user.id)
26 |     .maybeSingle()
27 |
28 |   let dbError = null
29 |
30 |   if (existingProfile) {
31 |     // Nếu đã có profile, chỉ UPDATE trường role (không chạm vào full_name)
32 |     const { error } = await supabase
33 |       .from('profiles')
34 |       .update({
35 |         role: role,
36 |         updated_at: new Date().toISOString()
37 |       })
38 |       .eq('id', user.id)
39 |     dbError = error
40 |   } else {
41 |     // Nếu chưa có profile, tạo mới hoàn chỉnh và lấy full_name từ metadata
42 |     const fullName = user.user_metadata?.full_name || user.email?.split('@')[0] || 'User'
43 |     const { error } = await supabase
44 |       .from('profiles')
45 |       .insert({
46 |         id: user.id,
47 |         email: user.email,
48 |         full_name: fullName,
49 |         role: role,
50 |         updated_at: new Date().toISOString()
51 |       })
52 |     dbError = error
53 |   }
54 |
55 |   if (dbError) {
56 |     return NextResponse.json({
57 |       error: 'Cập nhật database thất bại!',
58 |       details: dbError.message,
59 |       tip: 'Hãy chắc chắn rằng bạn đã chạy các câu lệnh SQL phân quyền',
60 |     }, { status: 500 })
61 |   }
62 |
63 |   // Ông thầy i c - p nh - t c ế metadata của User Auth đã ông bù hoàn toàn
64 |   await supabase.auth.updateUser({
65 |     data: { role: role }
66 |   })
67 |
68 |   // Tr ề v ấ k ết qu ế th ành công và chuy ển h ướng v ấ dashboard t ới ề ề ề
69 |   const redirectUrl = role === 'student' ? '/builder' : `/${role}`
70 |
71 |   // Tr ề v ấ HTML t ừ ề chuy ển h ướng c ả c k ế m ấ t mà
72 |   return new NextResponse(
73 |     `<html>
74 |       <head>
75 |         <title> Ông chuy ển h ướng ... </title>
76 |         <meta http-equiv="refresh" content="2;url=${redirectUrl}" />
77 |         <style>
78 |           body {
79 |             background-color: #0f0f1a;
80 |             color: #ffffff;
81 |             font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica, Arial, sans-serif;
82 |             display: flex;
83 |             flex-direction: column;
84 |             align-items: center;
85 |             justify-content: center;
86 |             height: 100vh;
87 |             margin: 0;
88 |           }
89 |           .card {
90 |             background: rgba(22, 33, 62, 0.5);
91 |             border: 1px solid rgba(0, 210, 160, 0.2);
92 |             padding: 40px;
93 |             border-radius: 20px;
94 |             text-align: center;
95 |             box-shadow: 0 10px 30px rgba(0,0,0,0.5);
96 |             max-width: 400px;
97 |           }
98 |           .spinner {
99 |             border: 4px solid rgba(255, 255, 255, 0.1);
100 |             width: 36px;
101 |             height: 36px;
102 |             border-radius: 50%;
103 |             border-left-color: #00d2a0;
104 |             animation: spin 1s linear infinite;
105 |             margin: 20px auto;
106 |           }
107 |           @keyframes spin { 0% { transform: rotate(0deg); } 100% { transform: rotate(360deg); } }
108 |           h2 { color: #00d2a0; margin-top: 0; }
109 |           p { color: #a0aec0; font-size: 14px; }
110 |         </style>
111 |       </head>
112 |       <body>
113 |         <div class="card">
114 |           <h2> K ích ho ạt vai trò th ành công! </h2>
115 |           <p> T ài kho ản c ủa b ị n <strong> ${user.email} </strong> đã đ ược c ập nh ậ t </p>
116 |           <div class="spinner"></div>
117 |           <p> Ông t ể ề ề a b ị n v ấ tr ạng Dashboard c ủa b ị n ... </p>
118 |         </div>
119 |       </body>
120 |     </html>`,
121 |     {
122 |       headers: {
123 |         'Content-Type': 'text/html; charset=utf-8',
124 |       },
125 |     }
126 |   )
127 | }

```

File: app/api/lesson-ai/route.js (50 lines, 2.2 KB)

```

1 import { GoogleGenerativeAI } from "@google/generative-ai";
2
3 const genAI = new GoogleGenerativeAI(process.env.GEMINI_API_KEY || process.env.GOOGLE_GEMINI_API_KEY);
4
5 export async function POST(req) {
6   try {
7     const { lessonTitle, sections, type } = await req.json();
8
9     // Build context ONLY from actual lesson content
10    const textContent = (sections || []).map(s => {
11      if (s.type === 'text') return `${s.title}\n${s.content}`;
12      if (s.type === 'video') return [Video: ${s.title}];
13      if (s.type === 'pdf') return [Tài liệu PDF: ${s.title}];
14      if (s.type === 'image') return [Hình ảnh: ${s.title}];
15    });
16    return `${s.title}`;
17  }
18  .join('\n\n');
19
20 |      const context = `Tên bài híc: ${lessonTitle}\n\n${textContent}`;
21 |      const model = genAI.getGenerativeModel({ model: "gemini-1.5-flash" });
22 |
23 |      let prompt = '';
24 |      if (type === 'quiz') {
25 |        prompt = `Đã HOÀN TOÀN vào nội dung bài híc sau đây, hãy t
26 |
27 | NƠI DUNG BÀI HÍC:
28 | ${context}
29 |
30 | Trê vÃ JSON mêng, không có markdown:
31 | [{"question": "...", "options": ["A. ...", "B. ...", "C. ...", "D. ..."], "correctIndex": 0, "explanation": "..."}]`;
32 |      } else {
33 |        prompt = `Đã HOÀN TOÀN vào nội dung bài híc sau đây, hãy t
34 |
35 | NƠI DUNG BÀI HÍC:
36 | ${context}
37 |
38 | Trê vÃ JSON mêng, không có markdown:
39 | [{"front": "...", "back": "..."}]`;
40 |      }
41 |
42 |      const result = await model.generateContent(prompt);
43 |      let text = result.response.text().replace(/`/g, '').replace(/`/g, '').trim();
44 |      return Response.json({ data: JSON.parse(text) });
45 |    } catch (error) {
46 |      console.error("lesson-ai error:", error);
47 |      return Response.json({ error: error.message }, { status: 500 });
48 |    }
49 |  }
50 |

```

File: app/api/migrate/route.ts (192 lines, 6.0 KB)

```

1 import { NextRequest, NextResponse } from 'next/server'
2 import { createClient } from '@supabase/supabase-js'
3 import { Pool } from 'pg'
4 import fs from 'fs'
5 import path from 'path'
6
7 const FILES = [
8   '20260610_products_full_specs.sql',
9   '20260610_seed_products.sql',
10  '20260610_saved_builds.sql',
11 ]
12
13 async function tryPgDirect(sql: string): Promise<{ ok: boolean; output?: string }> {
14   const ref = process.env.NEXT_PUBLIC_SUPABASE_URL?.match(/https:\/\/(.+)\.supabase\.co\/)?[1]
15   if (!ref) return { ok: false, output: 'No project ref found' }
16
17   const srk = process.env.SUPABASE_SERVICE_ROLE_KEY
18
19   // Try multiple connection methods
20   const attempts = [
21     // Session pooler with JWT (service role key as password)
22     {
23       label: 'pooler-session-jwt',
24       conn: {
25         host: `aws-0-ap-southeast-1.pooler.supabase.com`,
26         port: 6543,
27         database: 'postgres',
28         user: `postgres.${ref}`,
29         password: srk,
30         ssl: { rejectUnauthorized: false },
31         connectionTimeoutMillis: 8000,
32       },
33     },
34     // Direct connection (needs db.password but trying with srk anyway)
35     {
36       label: 'direct',
37       conn: {
38         host: `db.${ref}.supabase.co`,
39         port: 5432,
40         database: 'postgres',
41         user: 'postgres',
42         password: srk,
43         ssl: { rejectUnauthorized: false },
44         connectionTimeoutMillis: 8000,
45       },
46     },
47     // Transaction pooler
48     {
49       label: 'pooler-tx',
50       conn: {
51         host: `aws-0-ap-southeast-1.pooler.supabase.com`,
52         port: 6563,
53         database: 'postgres',
54         user: `postgres.${ref}`,
55         password: srk,
56         ssl: { rejectUnauthorized: false },
57         connectionTimeoutMillis: 8000,
58       },
59     },
60   ]
61
62   for (const attempt of attempts) {
63     try {
64       const pool = new Pool({ ...attempt.conn, max: 1 })

```

```

65 |     const client = await pool.connect()
66 |     await client.query(sql)
67 |     client.release()
68 |     await pool.end()
69 |     return { ok: true, output: `via ${attempt.label}` }
70 |   } catch (e: any) {
71 |     if (attempt === attempts[attempts.length - 1]) {
72 |       return { ok: false, output: `${attempt.label}: ${e.message}` }
73 |     }
74 |   }
75 | }
76 |
77 | return { ok: false, output: 'All connection methods failed' }
78 | }
79 |
80 | async function trySupabaseJS(sql: string): Promise<{ ok: boolean; output?: string }> {
81 |   try {
82 |     const supabase = createClient(
83 |       process.env.NEXT_PUBLIC_SUPABASE_URL!,
84 |       process.env.SUPABASE_SERVICE_ROLE_KEY!
85 |     )
86 |
87 |     // Try supabase.sql() if available (newer SDK versions)
88 |     if (typeof (supabase as any).sql === 'function') {
89 |       const { error } = await (supabase as any).sql(sql)
90 |       if (!error) return { ok: true, output: 'via supabase.sql()' }
91 |     }
92 |
93 |     // Try creating a temporary table via REST API to test
94 |     const { error: testError } = await supabase
95 |       .from('products')
96 |       .select('id', { count: 'exact', head: true })
97 |
98 |     if (!testError) {
99 |       // Table exists, can seed via REST
100 |      // Extract INSERT statements for seed data
101 |      const insertLines = sql
102 |        .split('\n')
103 |        .filter(l => l.toUpperCase().includes('INSERT INTO'))
104 |        .join('\n')
105 |
106 |      if (insertLines) {
107 |        // Can't execute raw INSERT through REST directly
108 |        // But we can log that the table exists
109 |        return { ok: true, output: 'table exists, seed via REST not available for DDL' }
110 |      }
111 |    }
112 |
113 |    return { ok: false, output: testError?.message || 'supabase.sql not available' }
114 |  } catch (e: any) {
115 |    return { ok: false, output: e.message }
116 |  }
117 | }
118 |
119 | export async function POST(req: NextRequest) {
120 |   const authHeader = req.headers.get('authorization')
121 |   const secret = process.env.CRON_SECRET
122 |   if (!secret || authHeader !== `Bearer ${secret}`) {
123 |     return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
124 |   }
125 |
126 |   const results: any[] = []
127 |   const dbDir = path.join(process.cwd(), 'database')
128 |
129 |   for (const file of FILES) {
130 |     const filePath = path.join(dbDir, file)
131 |     if (!fs.existsSync(filePath)) {
132 |       results.push({ file, status: 'skipped' })
133 |       continue
134 |     }
135 |
136 |     const sql = fs.readFileSync(filePath, 'utf8')
137 |
138 |     // Try direct PG connection first
139 |     const pgResult = await tryPgDirect(sql)
140 |     if (pgResult.ok) {
141 |       results.push({ file, status: 'ok', method: pgResult.output })
142 |       continue
143 |     }
144 |
145 |     // Fallback to Supabase JS
146 |     const jsResult = await trySupabaseJS(sql)
147 |     if (jsResult.ok) {
148 |       results.push({ file, status: 'ok', method: jsResult.output })
149 |       continue
150 |     }
151 |
152 |     results.push({
153 |       file,
154 |       status: 'failed',
155 |       pgError: pgResult.output,
156 |       jsError: jsResult.output,
157 |       instructions: 'Run this SQL in Supabase SQL Editor: https://supabase.com/dashboard/project/ojjmdhrvivwvf...',
158 |     })
159 |   }
160 |
161 |   return NextResponse.json({ results })
162 | }
163 |
164 | export async function GET(req: NextRequest) {
165 |   const authHeader = req.headers.get('authorization')
166 |   const secret = process.env.CRON_SECRET
167 |   if (!secret || authHeader !== `Bearer ${secret}`) {
168 |     return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
169 |   }
170 |
171 |   const supabase = createClient(
172 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
173 |     process.env.SUPABASE_SERVICE_ROLE_KEY!
174 |   )
175 |
176 |   const pq = supabase.from('products').select('*', { count: 'exact', head: true })
177 |   const sq = supabase.from('saved_builds').select('*', { count: 'exact', head: true })
178 |   const dq = supabase.from('diagnosis_scenarios').select('*', { count: 'exact', head: true })
179 |   const tq = supabase.from('thermal_profiles').select('*', { count: 'exact', head: true })
180 |
181 |   const [products, savedBuilds, diagnosis, thermal] = await Promise.all([pq, sq, dq, tq])
182 |
183 |   return NextResponse.json({
184 |     products: { exists: !products.error, count: products.count || 0 },
185 |     savedBuilds: { exists: !savedBuilds.error, count: savedBuilds.count || 0 },
186 |     diagnosis: { exists: !diagnosis.error, count: diagnosis.count || 0 },
187 |     thermal: { exists: !thermal.error, count: thermal.count || 0 },
188 |     pgConnection: 'Needs database password (not service_role_key)',

```

```
189 |     note: 'Run SQL manually in Supabase Dashboard SQL Editor, or provide a Supabase PAT for auto-migration',
190 |   })
191 | }
192 |
```

File: app/api/n8n/health/route.ts (13 lines, 0.3 KB)

```
1 | import { NextResponse } from 'next/server'
2 |
3 | export const dynamic = 'force-dynamic'
4 |
5 | export async function GET() {
6 |   return NextResponse.json({
7 |     status: 'ok',
8 |     timestamp: new Date().toISOString(),
9 |     service: 'n8n-webhooks',
10 |    version: '1.0.0'
11 |  })
12 | }
13 |
```

File: app/api/n8n/route.ts (28 lines, 1.3 KB)

```
1 | import { NextResponse } from 'next/server'
2 |
3 | export const dynamic = 'force-dynamic'
4 |
5 | export async function GET() {
6 |   const baseUrl = process.env.NEXT_PUBLIC_SITE_URL || 'http://localhost:3000'
7 |
8 |   const webhooks = [
9 |     { path: '/api/n8n/health', method: 'GET', description: 'Health check' },
10 |    { path: '/api/n8n/webhook/certificate', method: 'POST', description: 'Generate certificate on course compl...' },
11 |    { path: '/api/n8n/webhook/alert', method: 'POST', description: 'Send student performance alerts' },
12 |    { path: '/api/n8n/webhook/xp', method: 'POST', description: 'Award XP to users' },
13 |    { path: '/api/n8n/webhook/notification', method: 'POST', description: 'Create bulk notifications' },
14 |    { path: '/api/n8n/webhook/lesson-content', method: 'POST', description: 'Generate lesson content with AI' },
15 |    { path: '/api/n8n/triggers/lesson-complete', method: 'POST', description: 'Trigger on lesson completion' },
16 |    { path: '/api/n8n/triggers/quiz-submit', method: 'POST', description: 'Trigger on quiz submission' },
17 |  ]
18 |
19 |   return NextResponse.json({
20 |     service: 'PC Master Builder - n8n Webhook API',
21 |     version: '1.0.0',
22 |     baseUrl,
23 |     webhooks,
24 |     authentication: 'Bearer token in Authorization header',
25 |     docs: 'See n8n/README.md for workflow import instructions'
26 |   })
27 | }
28 |
```

File: app/api/n8n/triggers/lesson-complete/route.ts (91 lines, 2.6 KB)

```
1 | import { NextResponse } from 'next/server'
2 | import { createServiceClient } from '@lib/supabase-service'
3 |
4 | export const dynamic = 'force-dynamic'
5 | export const maxDuration = 30
6 |
7 | const N8N_SECRET = (process.env.N8N_WEBHOOK_SECRET || '').trim()
8 | const N8N_WEBHOOK_URL = (process.env.N8N_WEBHOOK_URL || '').trim()
9 |
10 | export async function POST(request: Request) {
11 |   try {
12 |     const { studentId, lessonId, className } = await request.json()
13 |
14 |     if (!studentId || !lessonId) {
15 |       return NextResponse.json({ error: 'Missing studentId or lessonId' }, { status: 400 })
16 |     }
17 |
18 |     const supabase = createServiceClient()
19 |
20 |     await supabase.from('activity_logs').insert({
21 |       user_id: studentId,
22 |       action_type: 'complete_lesson',
23 |       description: `Completed lesson ${lessonId}`,
24 |       metadata: { lessonId, className }
25 |     })
26 |
27 |     const { data: badges } = await supabase
28 |       .from('student_achievements')
29 |       .select('achievement_id')
30 |       .eq('student_id', studentId)
31 |
32 |     const earnedIds = (badges || []).map(b => b.achievement_id)
33 |
34 |     if (!earnedIds.includes('first_lesson')) {
35 |       await supabase.from('student_achievements').insert({
36 |         student_id: studentId,
37 |         achievement_id: 'first_lesson'
38 |       })
39 |
40 |       await supabase.from('notifications').insert({
41 |         user_id: studentId,
42 |         type: 'success',
43 |         title: 'Huy hieu moi! Khoi dau hanh trinh',
44 |         body: 'Chuc mung ban da hoan thanh bai hoc dau tien!'
45 |       })
46 |     }
47 |
48 |     const { data: progress } = await supabase
49 |       .from('lesson_progress')
50 |       .select('lesson_id')
51 |       .eq('student_id', studentId)
52 |       .eq('status', 'completed')
53 |
54 |     const lessonCount = progress?.length || 0
55 |
56 |     if (lessonCount > 0 && lessonCount % 10 === 0) {
57 |       await supabase.from('xp_transactions').insert({
58 |         user_id: studentId,
59 |         amount: 50,
60 |         reason: 'milestone_lessons',
61 |         reference_type: 'lesson_progress',
62 |         reference_id: lessonId
63 |       })
64 |     }
65 |
66 |     if (N8N_WEBHOOK_URL) {
```

```

67 |     try {
68 |       await fetch(`${N8N_WEBHOOK_URL}/n8n-lesson-content`, {
69 |         method: 'POST',
70 |         headers: {
71 |           'Content-Type': 'application/json',
72 |           'Authorization': `Bearer ${N8N_SECRET}`
73 |         },
74 |         body: JSON.stringify({
75 |           studentId,
76 |           lessonId,
77 |           type: 'lesson_completed',
78 |           timestamp: new Date().toISOString()
79 |         })
80 |       })
81 |     } catch {
82 |       console.warn('n8n webhook not reachable (non-critical)')
83 |     }
84 |   }
85 |
86 |   return NextResponse.json({ success: true, achievementsChecked: true, lessonCount })
87 | } catch (err: any) {
88 |   return NextResponse.json({ error: err.message }, { status: 500 })
89 | }
90 | }
91 |

```

File: app/api/n8n/triggers/quiz-submit/route.ts (111 lines, 3.3 KB)

```

1 | import { NextResponse } from 'next/server'
2 | import { createServiceClient } from '@lib/supabase-service'
3 |
4 | export const dynamic = 'force-dynamic'
5 | export const maxDuration = 30
6 |
7 | const N8N_SECRET = (process.env.N8N_WEBHOOK_SECRET || '').trim()
8 | const N8N_WEBHOOK_URL = (process.env.N8N_WEBHOOK_URL || '').trim()
9 |
10 | export async function POST(request: Request) {
11 |   try {
12 |     const { studentId, quizId, attemptId, score } = await request.json()
13 |
14 |     if (!studentId || !quizId || !attemptId) {
15 |       return NextResponse.json({ error: 'Missing required fields' }, { status: 400 })
16 |     }
17 |
18 |     const supabase = createServiceClient()
19 |
20 |     const finalScore = Number(score) || 0
21 |
22 |     await supabase.from('activity_logs').insert({
23 |       user_id: studentId,
24 |       action_type: 'submit_quiz',
25 |       description: `Submitted quiz ${quizId} with score ${finalScore}`,
26 |       metadata: { quizId, attemptId, score: finalScore }
27 |     })
28 |
29 |     const { data: attempt } = await supabase
30 |       .from('quiz_attempts')
31 |       .select('score, status')
32 |       .eq('id', attemptId)
33 |       .single()
34 |
35 |     if (attempt && attempt.status === 'submitted' && finalScore >= 90) {
36 |       const { data: badges } = await supabase
37 |         .from('student_achievements')
38 |         .select('achievement_id')
39 |         .eq('student_id', studentId)
40 |
41 |       const earnedIds = (badges || []).map(b => b.achievement_id)
42 |
43 |       if (!earnedIds.includes('quiz_ace')) {
44 |         await supabase.from('student_achievements').insert({
45 |           student_id: studentId,
46 |           achievement_id: 'quiz_ace'
47 |         })
48 |
49 |         await supabase.from('notifications').insert({
50 |           user_id: studentId,
51 |           type: 'success',
52 |           title: 'Huy hieu moi! Xuat sac!',
53 |           body: 'Chuc mung! Ban dat diem tuyet doi trong bai kiem tra!'
54 |         })
55 |       }
56 |     }
57 |
58 |     const { data: recentAttempts } = await supabase
59 |       .from('quiz_attempts')
60 |       .select('score')
61 |       .eq('student_id', studentId)
62 |       .eq('status', 'submitted')
63 |       .order('created_at', { ascending: false })
64 |       .limit(3)
65 |
66 |     const scores = (recentAttempts || []).map(a => a.score || 0)
67 |     const avgScore = scores.length > 0 ? scores.reduce((a, b) => a + b, 0) / scores.length : 0
68 |
69 |     if (scores.length >= 2 && avgScore < 40) {
70 |       await supabase.from('notifications').insert({
71 |         user_id: studentId,
72 |         type: 'warning',
73 |         title: 'Ban can tro giap them!',
74 |         body: `Diem trung binh 3 bai gan day cua ban la ${Math.round(avgScore)}/100. Hay xem lai bai hoc va th...`,
75 |         action_url: '/lessons'
76 |       })
77 |     }
78 |   }
79 |
80 |   if (N8N_WEBHOOK_URL) {
81 |     try {
82 |       await fetch(`${N8N_WEBHOOK_URL}/n8n-zalo-parent`, {
83 |         method: 'POST',
84 |         headers: {
85 |           'Content-Type': 'application/json',
86 |           'Authorization': `Bearer ${N8N_SECRET}`
87 |         },
88 |         body: JSON.stringify({
89 |           studentId,
90 |           quizId,
91 |           score: finalScore,
92 |           avgScore: Math.round(avgScore),
93 |           type: 'quiz_submitted',
94 |           timestamp: new Date().toISOString()
95 |         })
96 |       })
97 |     }
98 |   }
99 | }

```

```

96 |     } catch {
97 |       console.warn('n8n webhook not reachable (non-critical)')
98 |     }
99 |   }
100 |
101 |   return NextResponse.json({
102 |     success: true,
103 |     score: finalScore,
104 |     avgScore: Math.round(avgScore),
105 |     needsHelp: avgScore < 40
106 |   })
107 | } catch (err: any) {
108 |   return NextResponse.json({ error: err.message }, { status: 500 })
109 | }
110 | }
111 |

```

File: app/api/n8n/webhook/alert/route.ts (100 lines, 2.9 KB)

```

1 | import { NextResponse } from 'next/server'
2 | import { createServiceClient } from '@lib/supabase-service'
3 |
4 | export const dynamic = 'force-dynamic'
5 | export const maxDuration = 30
6 |
7 | const N8N_SECRET = (process.env.N8N_WEBHOOK_SECRET || '').trim()
8 |
9 | function verifyN8nRequest(request: Request): boolean {
10 |   const auth = request.headers.get('authorization')
11 |   return auth === `Bearer ${N8N_SECRET}`
12 | }
13 |
14 | export async function POST(request: Request) {
15 |   try {
16 |     if (!verifyN8nRequest(request)) {
17 |       return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
18 |     }
19 |
20 |     const { studentId, reason, alertType, details } = await request.json()
21 |
22 |     if (!studentId || !reason || !alertType) {
23 |       return NextResponse.json({ error: 'Missing required fields: studentId, reason, alertType' }, { status: 400 })
24 |     }
25 |
26 |     const supabase = createServiceClient()
27 |
28 |     const { data: profile } = await supabase
29 |       .from('profiles')
30 |       .select('id, full_name, email, class_id, school_id')
31 |       .eq('id', studentId)
32 |       .single()
33 |
34 |     if (!profile) {
35 |       return NextResponse.json({ error: 'Student not found' }, { status: 404 })
36 |     }
37 |
38 |     await supabase.from('notifications').insert({
39 |       user_id: studentId,
40 |       type: 'warning',
41 |       title: `Canh bao hoc tap: ${alertType}`,
42 |       body: reason,
43 |       action_url: details?.suggestedLesson ? `/lessons/${details.suggestedLesson}` : null
44 |     })
45 |
46 |     const { data: parentLinks } = await supabase
47 |       .from('parent_student_links')
48 |       .select('parent_id')
49 |       .eq('student_id', studentId)
50 |
51 |     if (parentLinks && parentLinks.length > 0) {
52 |       const parentNotifs = parentLinks.map(pl => ({
53 |         user_id: pl.parent_id,
54 |         type: 'warning',
55 |         title: `Canh bao: Tinh hinh hoc tap cua con ban`,
56 |         body: `Con ban ${profile.full_name} dang gap van de: ${reason}`,
57 |         action_url: details?.suggestedLesson ? `/lessons/${details.suggestedLesson}` : `/student/${studentId}`
58 |       }))
59 |
60 |       await supabase.from('notifications').insert(parentNotifs)
61 |     }
62 |
63 |     if (profile.class_id) {
64 |       const { data: classInfo } = await supabase
65 |         .from('classes')
66 |         .select('teacher_id')
67 |         .eq('id', profile.class_id)
68 |         .single()
69 |
70 |       if (classInfo) {
71 |         await supabase.from('notifications').insert({
72 |           user_id: classInfo.teacher_id,
73 |           type: 'warning',
74 |           title: `Hoc sinh can tro giup: ${profile.full_name}`,
75 |           body: reason,
76 |           action_url: `/student/${studentId}`
77 |         })
78 |       }
79 |     }
80 |
81 |     await supabase.from('activity_logs').insert({
82 |       user_id: studentId,
83 |       action_type: 'academic_alert',
84 |       description: reason,
85 |       metadata: { alertType, details, triggeredBy: 'n8n' }
86 |     })
87 |
88 |     return NextResponse.json({
89 |       success: true,
90 |       notified: {
91 |         student: true,
92 |         parent: (parentLinks?.length || 0) > 0,
93 |         teacher: true
94 |       }
95 |     })
96 |   } catch (err: any) {
97 |     return NextResponse.json({ error: err.message }, { status: 500 })
98 |   }
99 | }
100 |

```

File: app/api/n8n/webhook/certificate/route.ts (225 lines, 8.3 KB)

```

1 import { NextResponse } from 'next/server'
2 import { createServiceClient } from '@lib/supabase-service'
3 import { PDFDocument, StandardFonts, rgb } from 'pdf-lib'
4 import QRCode from 'qrcode'
5
6 export const dynamic = 'force-dynamic'
7 export const maxDuration = 60
8
9 const N8N_SECRET = (process.env.N8N_WEBHOOK_SECRET || '').trim()
10
11 function verifyN8NRequest(request: Request): boolean {
12   const auth = request.headers.get('authorization')
13   return auth === `Bearer ${N8N_SECRET}`
14 }
15
16 export async function POST(request: Request) {
17   try {
18     if (!verifyN8NRequest(request)) {
19       return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
20     }
21
22     const { studentId, pathId } = await request.json()
23     if (!studentId || !pathId) {
24       return NextResponse.json({ error: 'Missing studentId or pathId' }, { status: 400 })
25     }
26
27     const supabase = createServiceClient()
28
29     const { data: existing } = await supabase
30       .from('certificates')
31       .select('id, pdf_url')
32       .eq('student_id', studentId)
33       .eq('path_id', pathId)
34       .maybeSingle()
35
36     if (existing?.pdf_url) {
37       return NextResponse.json({ success: true, certificate: existing, message: 'Certificate already exists' })
38     }
39
40     const { data: pathItems } = await supabase
41       .from('path_items')
42       .select('id, item_id, item_type')
43       .eq('path_id', pathId)
44
45     const lessonIds = (pathItems || []).filter(i => i.item_type === 'lesson').map(i => i.item_id)
46     const quizIds = (pathItems || []).filter(i => i.item_type === 'quiz').map(i => i.item_id)
47
48     if (lessonIds.length > 0) {
49       const { data: progress } = await supabase
50         .from('lesson_progress')
51         .select('lesson_id')
52         .eq('student_id', studentId)
53         .eq('status', 'completed')
54
55       const completedIds = (progress || []).map(p => p.lesson_id)
56       const allDone = lessonIds.every(id => completedIds.includes(id))
57
58       if (!allDone) {
59         return NextResponse.json({ error: 'Not all lessons completed' }, { status: 400 })
60       }
61     }
62
63     const { data: profile } = await supabase
64       .from('profiles')
65       .select('full_name')
66       .eq('id', studentId)
67       .single()
68
69     const { data: path } = await supabase
70       .from('learning_paths')
71       .select('title')
72       .eq('id', pathId)
73       .single()
74
75     let totalScore = 100
76     if (quizIds.length > 0) {
77       const { data: attempts } = await supabase
78         .from('quiz_attempts')
79         .select('score')
80         .eq('student_id', studentId)
81         .eq('status', 'submitted')
82         .in('quiz_id', quizIds)
83
84       const sum = (attempts || []).reduce((acc, a) => acc + (a.score || 0), 0)
85       totalScore = Math.round(sum / quizIds.length)
86     }
87
88     const chars = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789'
89     let randomPart = ''
90     for (let i = 0; i < 5; i++) randomPart += chars.charAt(Math.floor(Math.random() * chars.length))
91     const certNumber = `PCM-2026-${randomPart}`
92     const siteUrl = process.env.NEXT_PUBLIC_SITE_URL || 'https://pc-master-lms.vercel.app'
93     const verifyUrl = `${siteUrl}/verify/${certNumber}`
94
95     const { data: cert, error: insertErr } = await supabase
96       .from('certificates')
97       .insert({
98         student_id: studentId,
99         path_id: pathId,
100         certificate_number: certNumber,
101         student_name: profile?.full_name || 'Hoc vien PC Master',
102         course_title: path?.title || 'Lo trinh Lap rap PC Master',
103         final_score: totalScore,
104         verify_url: verifyUrl,
105         completion_date: new Date().toISOString().split('T')[0]
106       })
107       .select()
108       .single()
109
110     if (insertErr || !cert) {
111       return NextResponse.json({ error: 'Failed to create certificate', details: insertErr }, { status: 500 })
112     }
113
114     try {
115       const pdfDoc = await PDFDocument.create()
116       const page = pdfDoc.addPage([842, 595])
117
118       const fontBold = await pdfDoc.embedFont(StandardFonts.HelveticaBold)
119       const fontRegular = await pdfDoc.embedFont(StandardFonts.Helvetica)
120
121       page.drawRectangle({ x: 0, y: 0, width: 842, height: 595, color: rgb(0.05, 0.06, 0.075) })
122       page.drawRectangle({ x: 20, y: 20, width: 802, height: 555, borderColor: rgb(0, 0.83, 0.67), borderWidth...
123

```

```

124 | page.drawText('CHUNG NHAN HOAN THANH', { x: 250, y: 460, size: 28, font: fontBold, color: rgb(0, 0.83, 0...
125 | page.drawText('Chung chi nay duoc trao cho', { x: 330, y: 400, size: 14, font: fontRegular, color: rgb(0...
126 | page.drawText(cert.student_name, { x: 260, y: 340, size: 36, font: fontBold, color: rgb(0.87, 0.88, 0.93...
127 | page.drawText('Vi da hoan thanh xuat sac khoa hoc', { x: 320, y: 290, size: 12, font: fontRegular, color...
128 | page.drawText(cert.course_title, { x: 260, y: 240, size: 20, font: fontBold, color: rgb(0, 0.83, 0.67) })
129 | page.drawText('Diem so: ${cert.final_score}% | Ngay cap: ${cert.completion_date}', { x: 280, y: 180, siz...
130 | page.drawText('Ma so chung chi: ${cert.certificate_number}', { x: 300, y: 140, size: 11, font: fontRegul...
131 |
132 | const qrDataURL = await QRCode.toDataURL(verifyUrl)
133 | const qrImageBytes = Uint8Array.from(atob(qrDataURL.split(',')[1]), c => c.charCodeAt(0))
134 | const qrImage = await pdfDoc.embedPng(qrImageBytes)
135 | page.drawImage(qrImage, { x: 690, y: 40, width: 90, height: 90 })
136 |
137 | const pdfBytes = await pdfDoc.save()
138 |
139 | const { error: uploadErr } = await supabase.storage
140 |   .from('certificates')
141 |   .upload(`${cert.id}.pdf`, pdfBytes, { contentType: 'application/pdf', upsert: true })
142 |
143 | if (uploadErr) throw uploadErr
144 |
145 | const publicUrl = supabase.storage.from('certificates').getPublicUrl(`${cert.id}.pdf`).data.publicUrl
146 |
147 | await supabase.from('certificates').update({ pdf_url: publicUrl }).eq('id', cert.id)
148 |
149 | await supabase.from('notifications').insert({
150 |   user_id: studentId,
151 |   type: 'success',
152 |   title: 'Chung chi moi!',
153 |   body: `Chuc mung! Ban da nhan duoc chung chi "${cert.course_title}" voi so diem ${cert.final_score}%!`,
154 |   action_url: `/verify/${certNumber}`,
155 | })
156 |
157 | const { data: parentLinks } = await supabase
158 |   .from('parent_student_links')
159 |   .select('parent_id')
160 |   .eq('student_id', studentId)
161 |
162 | if (parentLinks && parentLinks.length > 0) {
163 |   const parentNotifs = parentLinks.map(pl => ({
164 |     user_id: pl.parent_id,
165 |     type: 'success',
166 |     title: 'Con ban hoan thanh khoa hoc!',
167 |     body: `Con ban ${cert.student_name} vua hoan thanh "${cert.course_title}" voi diem ${cert.final_scor...
168 |     action_url: `/verify/${certNumber}`,
169 |   )))
170 |   await supabase.from('notifications').insert(parentNotifs)
171 | }
172 |
173 | const { data: studentClasses } = await supabase
174 |   .from('class_members')
175 |   .select('class_id')
176 |   .eq('student_id', studentId)
177 |   .eq('status', 'active')
178 |
179 | if (studentClasses && studentClasses.length > 0) {
180 |   const classIds = studentClasses.map(c => c.class_id)
181 |   const { data: teachers } = await supabase
182 |     .from('classes')
183 |     .select('teacher_id')
184 |     .in('id', classIds)
185 |
186 |   if (teachers && teachers.length > 0) {
187 |     const teacherNotifs = [...new Set(teachers.map(t => t.teacher_id))].map(tid => ({
188 |       user_id: tid,
189 |       type: 'success',
190 |       title: 'Hoc sinh hoan thanh khoa hoc!',
191 |       body: `Hoc sinh ${cert.student_name} da hoan thanh "${cert.course_title}"!`,
192 |       action_url: `/verify/${certNumber}`,
193 |     )))
194 |     await supabase.from('notifications').insert(teacherNotifs)
195 |   }
196 | }
197 |
198 | await supabase.from('xp_transactions').insert({
199 |   user_id: studentId,
200 |   amount: 200,
201 |   reason: 'certificate_completed',
202 |   reference_type: 'certificate',
203 |   reference_id: cert.id
204 | })
205 |
206 | try {
207 |   await supabase.rpc('add_xp', { p_user_id: studentId, p_amount: 200 })
208 | } catch {}
209 |
210 | return NextResponse.json({
211 |   success: true,
212 |   certificate: { id: cert.id, pdf_url: publicUrl, cert_number: certNumber, score: totalScore }
213 | })
214 | } catch (pdfErr: any) {
215 |   return NextResponse.json({
216 |     success: true,
217 |     certificate: cert,
218 |     warning: 'PDF generation failed, but certificate record created: ' + pdfErr.message
219 |   })
220 | }
221 | } catch (err: any) {
222 |   return NextResponse.json({ error: err.message }, { status: 500 })
223 | }
224 | }
225 |

```

File: app/api/n8n/webhook/lesson-content/route.ts (163 lines, 4.8 KB)

```

1 | import { NextResponse } from 'next/server'
2 | import { createServiceClient } from '@lib/supabase-service'
3 | import { GoogleGenerativeAI } from '@google/generative-ai'
4 |
5 | export const dynamic = 'force-dynamic'
6 | export const maxDuration = 120
7 |
8 | const N8N_SECRET = (process.env.N8N_WEBHOOK_SECRET || '').trim()
9 |
10 | function verifyN8nRequest(request: Request): boolean {
11 |   const auth = request.headers.get('authorization')
12 |   return auth === `Bearer ${N8N_SECRET}`
13 | }
14 |
15 | const GEMINI_API_KEY = () => process.env.GEMINI_API_KEY || process.env.GOOGLE_GEMINI_API_KEY || ''
16 |
17 | export async function POST(request: Request) {
18 |   try {

```



```

19 |   if (!verifyN8nRequest(request)) {
20 |     return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
21 |   }
22 |
23 |   const { topic, difficulty, lessonId, teacherId, questionCount } = await request.json()
24 |   if (!topic) {
25 |     return NextResponse.json({ error: 'Missing topic' }, { status: 400 })
26 |   }
27 |
28 |   const supabase = createServiceClient()
29 |   const apiKey = GEMINI_API_KEY()
30 |   if (!apiKey) {
31 |     return NextResponse.json({ error: 'Gemini API key not configured' }, { status: 500 })
32 |   }
33 |
34 |   const genAI = new GoogleGenerativeAI(apiKey)
35 |   const model = genAI.getGenerativeModel({ model: 'gemini-2.0-flash' })
36 |
37 |   const qCount = questionCount || 5
38 |   const prompt = `Bạn là chuyên gia tạo nội dung giáo dục về lập trình máy tính (PC Building) bằng tiếng Việt.
39 | Hay tạo ${qCount} câu hỏi trắc nghiệm về chủ đề "${topic}" ở độ khó "${difficulty} || 'trung bình'" (đe/trung ...
40 |
41 | Yêu cầu:
42 | - Mỗi câu hỏi có 4 đáp án (A, B, C, D)
43 | - Chỉ 1 đáp án đúng
44 | - Nội dung chính xác về kỹ thuật phân cung máy tính
45 | - Phù hợp với học sinh cấp 3 Việt Nam
46 |
47 | Trả về JSON array:
48 | [
49 |   {
50 |     "content": "Câu hỏi?",
51 |     "type": "multiple_choice",
52 |     "difficulty": "easy|medium|hard",
53 |     "points": 1,
54 |     "options": [
55 |       { "content": "Đáp án A", "is_correct": false },
56 |       { "content": "Đáp án B", "is_correct": true },
57 |       { "content": "Đáp án C", "is_correct": false },
58 |       { "content": "Đáp án D", "is_correct": false }
59 |     ]
60 |   }
61 | ]
62 |
63 | Chi trả về JSON, không có text khác.`
64 |
65 |   const result = await model.generateContent(prompt)
66 |   const text = result.response.text()
67 |
68 |   let questions: any[] = []
69 |   try {
70 |     const jsonMatch = text.match(/[\s\S]*\s/)
71 |     if (jsonMatch) {
72 |       questions = JSON.parse(jsonMatch[0])
73 |     } else {
74 |       questions = JSON.parse(text)
75 |     }
76 |   } catch {
77 |     return NextResponse.json({ error: 'Failed to parse AI response', raw: text }, { status: 500 })
78 |   }
79 |
80 |   if (!questions.length) {
81 |     return NextResponse.json({ error: 'No questions generated' }, { status: 500 })
82 |   }
83 |
84 |   const teacherIdValue = teacherId || '00000000-0000-0000-0000-000000000000'
85 |
86 |   const { data: bank, error: bankErr } = await supabase
87 |     .from('question_banks')
88 |     .insert({
89 |       teacher_id: teacherIdValue,
90 |       title: `AI Generated: ${topic} - ${new Date().toLocaleDateString('vi-VN')}`,
91 |       subject: topic
92 |     })
93 |     .select('id')
94 |     .single()
95 |
96 |   if (bankErr || !bank) {
97 |     return NextResponse.json({ error: 'Failed to create question bank', details: bankErr }, { status: 500 })
98 |   }
99 |
100 |   const createdQuestions: any[] = []
101 |
102 |   for (let i = 0; i < questions.length; i++) {
103 |     const q = questions[i]
104 |     const { data: question, error: qErr } = await supabase
105 |       .from('questions')
106 |       .insert({
107 |         bank_id: bank.id,
108 |         content: q.content,
109 |         type: q.type || 'multiple_choice',
110 |         points: q.points || 1,
111 |         order: i + 1
112 |       })
113 |       .select('id')
114 |       .single()
115 |
116 |     if (question && q.options) {
117 |       const opts = q.options.map((opt: any, oi: number) => ({
118 |         question_id: question.id,
119 |         content: opt.content,
120 |         is_correct: opt.is_correct,
121 |         order: oi + 1
122 |       }))
123 |
124 |       const { error: optErr } = await supabase
125 |         .from('question_options')
126 |         .insert(opts)
127 |
128 |       if (!optErr) {
129 |         createdQuestions.push({ id: question.id, content: q.content, options: opts })
130 |       }
131 |     }
132 |   }
133 |
134 |   if (lessonId) {
135 |     const { error: quizErr } = await supabase.from('quizzes').insert({
136 |       lesson_id: lessonId,
137 |       teacher_id: teacherIdValue,
138 |       title: `Kiểm tra: ${topic}`,
139 |       description: `Bài kiểm tra tu động về chủ đề: ${topic}`,
140 |       time_limit_minutes: Math.ceil(qCount * 2),
141 |       passing_score: Math.ceil(qCount * 0.6),
142 |       max_attempts: 2,

```

```

143 |         randomize_questions: true,
144 |         randomize_options: true,
145 |         is_published: true
146 |     })
147 |
148 |     if (quizErr) {
149 |         console.error('Failed to create lesson quiz:', quizErr)
150 |     }
151 | }
152 |
153 | return NextResponse.json({
154 |     success: true,
155 |     bankId: bank.id,
156 |     questionsCreated: createdQuestions.length,
157 |     questions: createdQuestions
158 | })
159 | } catch (err: any) {
160 |     return NextResponse.json({ error: err.message }, { status: 500 })
161 | }
162 | }
163 |

```

File: app/api/n8n/webhook/notification/route.ts (62 lines, 1.8 KB)

```

1 | import { NextResponse } from 'next/server'
2 | import { createServiceClient } from '@lib/supabase-service'
3 |
4 | export const dynamic = 'force-dynamic'
5 | export const maxDuration = 30
6 |
7 | const N8N_SECRET = (process.env.N8N_WEBHOOK_SECRET || '').trim()
8 |
9 | function verifyN8nRequest(request: Request): boolean {
10 |     const auth = request.headers.get('authorization')
11 |     return auth === `Bearer ${N8N_SECRET}`
12 | }
13 |
14 | export async function POST(request: Request) {
15 |     try {
16 |         if (!verifyN8nRequest(request)) {
17 |             return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
18 |         }
19 |
20 |         const { recipients, title, message, type, actionUrl } = await request.json()
21 |
22 |         if (!recipients || !recipients.length || !title || !message) {
23 |             return NextResponse.json({ error: 'Missing required fields: recipients, title, message' }, { status: 400 })
24 |         }
25 |
26 |         const supabase = createServiceClient()
27 |
28 |         const notifications = recipients.map(userId => ({
29 |             user_id: userId,
30 |             type: type || 'info',
31 |             title,
32 |             body: message,
33 |             action_url: actionUrl || null,
34 |         }))
35 |
36 |         const { error: insertErr } = await supabase
37 |             .from('notifications')
38 |             .insert(notifications)
39 |
40 |         if (insertErr) {
41 |             const fallbackRows = recipients.map(userId => ({
42 |                 user_id: userId,
43 |                 action_type: 'notification_fallback',
44 |                 description: `[$${type || 'info'}] ${title}: ${message}`,
45 |                 metadata: { title, message, type, actionUrl, originalError: insertErr.message }
46 |             }))
47 |
48 |             await supabase.from('activity_logs').insert(fallbackRows)
49 |
50 |             return NextResponse.json({
51 |                 success: false,
52 |                 fallback: 'activity_logs',
53 |                 error: insertErr.message,
54 |                 sentCount: 0
55 |             })
56 |         }
57 |
58 |         return NextResponse.json({ success: true, sentCount: notifications.length })
59 |     } catch (err: any) {
60 |         return NextResponse.json({ error: err.message }, { status: 500 })
61 |     }
62 | }

```

File: app/api/n8n/webhook/xp/route.ts (132 lines, 3.9 KB)

```

1 | import { NextResponse } from 'next/server'
2 | import { createServiceClient } from '@lib/supabase-service'
3 |
4 | export const dynamic = 'force-dynamic'
5 | export const maxDuration = 30
6 |
7 | const N8N_SECRET = (process.env.N8N_WEBHOOK_SECRET || '').trim()
8 |
9 | function verifyN8nRequest(request: Request): boolean {
10 |     const auth = request.headers.get('authorization')
11 |     return auth === `Bearer ${N8N_SECRET}`
12 | }
13 |
14 | export async function POST(request: Request) {
15 |     try {
16 |         if (!verifyN8nRequest(request)) {
17 |             return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
18 |         }
19 |
20 |         const { userId, amount, reason, referenceType, referenceId } = await request.json()
21 |
22 |         if (!userId || !amount || !reason) {
23 |             return NextResponse.json({ error: 'Missing required fields: userId, amount, reason' }, { status: 400 })
24 |         }
25 |
26 |         const supabase = createServiceClient()
27 |
28 |         const numericAmount = Math.round(Number(amount))
29 |
30 |         const { error: txErr } = await supabase.from('xp_transactions').insert({
31 |             user_id: userId,
32 |             amount: numericAmount,

```

```

33 |     reason,
34 |     reference_type: referenceType || null,
35 |     reference_id: referenceId || null
36 | })
37 |
38 | if (txErr) {
39 |   return NextResponse.json({ error: 'Failed to record XP transaction', details: txErr }, { status: 500 })
40 | }
41 |
42 | const { data: profile } = await supabase
43 |   .from('profiles')
44 |   .select('xp, level, full_name')
45 |   .eq('id', userId)
46 |   .single()
47 |
48 | if (profile) {
49 |   const newXp = (profile.xp || 0) + numericAmount
50 |
51 |   let newLevel = profile.level || 1
52 |   const levels = [0, 100, 300, 600, 1000, 1600, 2400, 3400, 4600, 6000, 8000, 10000, 13000, 17000, 22000, ...]
53 |
54 |   for (let i = levels.length - 1; i >= 0; i--) {
55 |     if (newXp >= levels[i]) {
56 |       newLevel = i + 1
57 |       break
58 |     }
59 |   }
60 |
61 |   const leveledUp = newLevel > (profile.level || 1)
62 |
63 |   await supabase.from('profiles').update({ xp: newXp, level: newLevel }).eq('id', userId)
64 |
65 |   await supabase.from('learning_stats').upsert({
66 |     user_id: userId,
67 |     total_xp: newXp,
68 |     updated_at: new Date().toISOString()
69 |   }, { onConflict: 'user_id' })
70 |
71 |   if (leveledUp) {
72 |     await supabase.from('notifications').insert({
73 |       user_id: userId,
74 |       type: 'success',
75 |       title: 'Len cap! Ban da dat cap do ${newLevel}!',
76 |       message: 'Chuc mung! Ban vua thang cap len Level ${newLevel}!',
77 |       link: '/student'
78 |     })
79 |
80 |     const { data: levelDef } = await supabase
81 |       .from('level_definitions')
82 |       .select('title')
83 |       .eq('level', newLevel)
84 |       .single()
85 |
86 |     if (levelDef) {
87 |       const { data: existingTitle } = await supabase
88 |         .from('user_titles')
89 |         .select('id')
90 |         .eq('user_id', userId)
91 |         .eq('title_id', `level_${newLevel}`)
92 |         .maybeSingle()
93 |
94 |       if (!existingTitle) {
95 |         await supabase.from('user_titles').insert({
96 |           user_id: userId,
97 |           title_id: `level_${newLevel}`,
98 |           is_active: false
99 |         })
100 |       }
101 |     }
102 |   }
103 |
104 |   const { data: leaderboard } = await supabase
105 |     .from('leaderboard_cache')
106 |     .select('id')
107 |     .eq('user_id', userId)
108 |     .maybeSingle()
109 |
110 |   if (leaderboard) {
111 |     await supabase.from('leaderboard_cache').update({
112 |       total_xp: newXp,
113 |       level: newLevel,
114 |       updated_at: new Date().toISOString()
115 |     }).eq('user_id', userId)
116 |   } else {
117 |     await supabase.from('leaderboard_cache').insert({
118 |       user_id: userId,
119 |       full_name: profile.full_name || '',
120 |       level: newLevel,
121 |       total_xp: newXp,
122 |       updated_at: new Date().toISOString()
123 |     })
124 |   }
125 | }
126 |
127 | return NextResponse.json({ success: true, xpAwarded: numericAmount })
128 | } catch (err: any) {
129 |   return NextResponse.json({ error: err.message }, { status: 500 })
130 | }
131 | }
132 |

```

File: app/api/notifications/send-class/route.ts (65 lines, 2.1 KB)

```

1 | import { NextResponse } from 'next/server'
2 | import { createServerClient } from '@supabase/ssr'
3 | import { cookies } from 'next/headers'
4 |
5 | export async function POST(request: Request) {
6 |   try {
7 |     const { class_id, title, message, type = 'info', link = '' } = await request.json()
8 |     if (!class_id || !title || !message) {
9 |       return NextResponse.json({ error: 'Thiếu thông tin bắt buộc' }, {
10 |         status: 400
11 |       })
12 |     }
13 |     const cookieStore = await cookies()
14 |     const supabase = createServerClient(
15 |       process.env.NEXT_PUBLIC_SUPABASE_URL!,
16 |       process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!,
17 |       { cookies: { get(name: string) { return cookieStore.get(name)?.value } } }
18 |     )
19 |     const { data: { user }, error: authError } = await supabase.auth.getUser()
20 |     if (!user || authError) {

```

```

22 |         return NextResponse.json({ error: 'Ch°a ng nh-p' }, { status: 400 })
23 |     }
24 |     const { data: teacherClass } = await supabase
25 |       .from('classes')
26 |       .select('teacher_id')
27 |       .eq('id', class_id)
28 |       .single()
29 |
30 |     if (!teacherClass || teacherClass.teacher_id !== user.id) {
31 |       return NextResponse.json({ error: 'Không có quyền gí thông báo ch°a' }, { status: 400 })
32 |     }
33 |
34 |     const { data: members } = await supabase
35 |       .from('class_members')
36 |       .select('student_id')
37 |       .eq('class_id', class_id)
38 |       .eq('status', 'active')
39 |
40 |     if (!members || members.length === 0) {
41 |       return NextResponse.json({ error: 'L°p ch°a có híc sinh nào' }, { status: 400 })
42 |     }
43 |
44 |     const notifications = members.map(m => ({
45 |       user_id: m.student_id,
46 |       title,
47 |       message,
48 |       type,
49 |       link: link || null,
50 |       is_read: false,
51 |     }))
52 |
53 |     const { error: insertError } = await supabase
54 |       .from('notifications')
55 |       .insert(notifications)
56 |
57 |     if (insertError) throw insertError
58 |
59 |     return NextResponse.json({ success: true, count: notifications.length })
60 |   } catch (err: any) {
61 |     console.error('Send notification error:', err)
62 |     return NextResponse.json({ error: err.message || 'L×i server' }, { status: 500 })
63 |   }
64 | }
65 |

```

File: app/api/pc-price-compare/route.ts (103 lines, 4.4 KB)

```

1 | import { NextRequest, NextResponse } from 'next/server'
2 |
3 | const SYSTEM_PROMPT = `Bìn là mùt Chuyên gia L°p ráp PC và Kh°o giá Linh kiện
4 |
5 | CÁC TRANG WEB C°N TRA C°U:
6 | 1. GearVN (gearvn.com) - Uy tín cao, giá c°nh tranh hành t°n
7 | 2. Phong Vi (phongvu.vn) - Nhà phân ph°i l°n, b°o hành t°n
8 | 3. An Phát (anphatpc.com.vn) - Chuyên lĩnh ki°n PC, giá t°n
9 | 4. Hoàng Hà (hoanghapc.vn) - H°c th°ng si°u th° i°n máy
10 | 5. FPT Shop (fptshop.com.vn) - Bán l» uy tín, nhĩu khuy°n mãi
11 |
12 | Hã y mô phĩng giá thĩc t° tĩ th° tr° Ýng Vi°t Nam cho t°ng trang web đĩn
13 |
14 | C°u trúc JSON ph°n h°i b°t bu°c:
15 | {
16 |   "comparison_text": "oĩn v n b°n phân tích so sánh giá, thông s° và l°y
17 | "components": [
18 |   {
19 |     "id": "comp_01",
20 |     "exact_name": "Tên s° c° chính xác c°a linh ki°n",
21 |     "estimated_price": "3.200.000 VN ",
22 |     "image_url": "https://images.unsplash.com/photo-1591799264318-7e6ef8ddb7ea?w=500&q=80"
23 |   },
24 |   {
25 |     "prices": [
26 |       {
27 |         "site": "GearVN",
28 |         "price": "3.200.000 VN ",
29 |         "link": "https://gearvn.com/search?q={t° khóa tìm ki°m}",
30 |         "rating": 4.5
31 |       },
32 |       {
33 |         "site": "Phong Vi",
34 |         "price": "3.350.000 VN ",
35 |         "link": "https://phongvu.vn/search?q={t° khóa tìm ki°m}",
36 |         "rating": 4.3
37 |       },
38 |     ],
39 |     "recommended": {
40 |       "site": "GearVN",
41 |       "price": "3.200.000 VN ",
42 |       "link": "https://gearvn.com/search?q={t° khóa tìm ki°m}"
43 |     }
44 |   }
45 | ]
46 |
47 | Quy t° c l°y link ãnh (image_url):
48 | - N°u là RAM/Mich/Linh ki°n chung chung: https://images.unsplash.com/photo-1591799264318-7e6ef8ddb7ea?w=500&q=80
49 | - N°u là linh ki°n c° th°: https://image.pollinations.ai/prompt/{Ten Linh kiện}
49 | export async function POST(request: NextRequest) {
50 |   try {
51 |     const { query } = await request.json()
52 |     if (!query || typeof query !== 'string') {
53 |       return NextResponse.json({ error: 'Vui lòng nh-p tên linh ki°n' }, { status: 400 })
54 |     }
55 |
56 |     const controller = new AbortController()
57 |     const timeoutId = setTimeout(() => controller.abort(), 15000)
58 |
59 |     const res = await fetch('https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generate...
60 |       , {
61 |         method: 'POST',
62 |         headers: { 'Content-Type': 'application/json' },
63 |         body: JSON.stringify({
64 |           contents: [{
65 |             parts: [{
66 |               text: SYSTEM_PROMPT + '\n\nNg° Ýi dùng hĩi: ' + query + '\n\n
67 |             }],
68 |             generationConfig: { temperature: 0.3, maxOutputTokens: 1024 }
69 |           }],
70 |           signal: controller.signal
71 |         })
72 |       })
73 |
74 |     clearTimeout(timeoutId)
75 |

```

```

76 |     if (!res.ok) {
77 |         const errText = await res.text()
78 |         console.error('Gemini API error:', res.status, errText)
79 |         return NextResponse.json({ error: 'Đã xảy ra lỗi khi gọi API Gemini, vui lòng thử lại sau.' })
80 |     }
81 |
82 |     const data = await res.json()
83 |     const text = data?.candidates?.[0]?.content?.parts?.[0]?.text || ''
84 |
85 |     const jsonMatch = text.match(/````(?:json)?\s*{(?:[\\s\\S]*?)}```/);
86 |     const jsonStr = jsonMatch ? jsonMatch[1] || jsonMatch[0] : text
87 |
88 |     try {
89 |         const parsed = JSON.parse(jsonStr.trim())
90 |         return NextResponse.json(parsed)
91 |     } catch {
92 |         console.error('Failed to parse AI response as JSON:', text.substring(0, 200))
93 |         return NextResponse.json({ error: 'Lỗi: Không thể trích xuất JSON từ phản hồi AI.' })
94 |     } catch (err: any) {
95 |         if (err.name === 'AbortError') {
96 |             return NextResponse.json({ error: 'Yêu cầu đã hết thời gian chờ. Vui lòng thử lại sau.' })
97 |         }
98 |         console.error('Price compare error:', err)
99 |         return NextResponse.json({ error: 'Lỗi: Không thể so sánh giá. Vui lòng thử lại sau.' })
100 |     }
101 | }
102 |
103 |

```

File: app/api/pc-suggest/route.ts (134 lines, 5.4 KB)

```

1 | import { NextRequest, NextResponse } from 'next/server'
2 |
3 | const SYSTEM_PROMPT = `Bạn là chuyên gia tư vấn mua máy tính tại Việt Nam.
4 |
5 | Yêu cầu:
6 | - Xác định 3-5 máy tính phù hợp nhất (cả laptop và desktop)
7 | - Mix máy phôi có tên cả thảy, thông số chi tiết (CPU, RAM, GPU, ổ cứng),
8 | - Thêm link mua ở đâu (dùng unsplash hoặc pollinations.ai)
9 | - Phân tích lý do phù hợp
10 | - Tra cứu giá trên 5 trang: GearVN, Phong Vi, An Phát, Hoàng Hà, FPT Shop
11 | - Tính giá trung bình để các cửa hàng
12 |
13 | Trả về JSON tuyệt vời, không thêm text ngoài:
14 | {
15 |   "suggestions": [
16 |     {
17 |       "id": "s1",
18 |       "name": "Tên máy",
19 |       "type": "Laptop Gaming | Desktop Gaming | Laptop | Desktop Office | ...",
20 |       "specs": "Thông số chi tiết",
21 |       "specs_detail": {
22 |         "cpu": "Intel Core i7-13700H",
23 |         "ram": "16GB DDR5",
24 |         "gpu": "RTX 4060 8GB",
25 |         "storage": "1TB SSD NVMe"
26 |       },
27 |       "price": 34990000,
28 |       "average_price": 35200000,
29 |       "price_breakdown": [
30 |         { "store": "GearVN", "price": 34990000, "url": "https://gearvn.com/" },
31 |         { "store": "Phong Vi", "price": 35900000, "url": "https://phongvi.vn/" },
32 |         { "store": "An Phát", "price": 34500000, "url": "https://anphatpc.com.vn/" },
33 |         { "store": "Hoàng Hà", "price": 35500000, "url": "https://hoanghacpc.vn/" },
34 |         { "store": "FPT Shop", "price": 35200000, "url": "https://fptshop.com.vn/" }
35 |       ],
36 |       "reason": "Lý do phù hợp",
37 |       "useCases": ["gaming", "coder"],
38 |       "rating": 4.7,
39 |       "image_url": "https://images.unsplash.com/photo-1593642632559-0c6d3fc62b89?w=400&q=80"
40 |     }
41 |   ]
42 | }
43 |
44 | const cache = new Map<string, { data: any; timestamp: number }>()
45 | const CACHE_TTL = 60 * 60 * 1000
46 |
47 | function getCacheKey(purpose: string, useCases: string[], budget: number): string {
48 |   return JSON.stringify({ purpose, useCases: [...useCases].sort(), budget })
49 | }
50 |
51 | export async function POST(request: NextRequest) {
52 |   try {
53 |     const body = await request.json()
54 |     const { purpose, useCases = [], budget = 0 } = body
55 |
56 |     if (!process.env.GEMINI_API_KEY) {
57 |       return NextResponse.json({
58 |         error: 'AI chưa được cấu hình (thiếu API key). Vui lòng liên hệ admin để bổ sung API key.',
59 |         code: 'MISSING_API_KEY',
60 |         status: 503
61 |       })
62 |     }
63 |
64 |     if (!purpose && useCases.length === 0) {
65 |       return NextResponse.json({ error: 'Vui lòng nhập nhu cầu hoặc chỉ định mục đích.' })
66 |     }
67 |
68 |     const cacheKey = getCacheKey(purpose, useCases, budget)
69 |     const cached = cache.get(cacheKey)
70 |     if (cached && Date.now() - cached.timestamp < CACHE_TTL) {
71 |       return NextResponse.json(cached.data)
72 |     }
73 |
74 |     const userRequest = `Mục đích: ${purpose} | | 'Không có' | \n Các nhu cầu: ${useCases.join(', ')} \n Ngân sách: ${budget}`
75 |
76 |     const controller = new AbortController()
77 |     const timeoutId = setTimeout(() => controller.abort(), 20000)
78 |
79 |     const res = await fetch('https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generate...
80 |       , {
81 |         method: 'POST',
82 |         headers: { 'Content-Type': 'application/json' },
83 |         body: JSON.stringify({
84 |           contents: [{
85 |             parts: [{
86 |               text: SYSTEM_PROMPT + '\n\n' + userRequest
87 |             }],
88 |             generationConfig: { temperature: 0.4, maxOutputTokens: 2048 }
89 |           }
90 |         }),
91 |         signal: controller.signal
92 |       })
93 |
94 |     clearTimeout(timeoutId)
95 |
96 |     return NextResponse.json({
97 |       suggestions: res.json().candidates.map((c: any) => {
98 |         const specStr = c.content.parts[0].text
99 |         const specObj = JSON.parse(specStr)
100 |         const priceStr = c.content.parts[1].text
101 |         const priceObj = JSON.parse(priceStr)
102 |         const reasonStr = c.content.parts[2].text
103 |         const reasonObj = JSON.parse(reasonStr)
104 |         const useCasesStr = c.content.parts[3].text
105 |         const useCasesObj = JSON.parse(useCasesStr)
106 |         const ratingStr = c.content.parts[4].text
107 |         const ratingObj = JSON.parse(ratingStr)
108 |         const imageUrlStr = c.content.parts[5].text
109 |         const imageUrlObj = JSON.parse(imageUrlStr)
110 |         return {
111 |           id: `s${c.id}`,
112 |           name: specObj.name,
113 |           type: specObj.type,
114 |           specs: specStr,
115 |           specs_detail: specObj,
116 |           price: priceObj.price,
117 |           average_price: priceObj.average_price,
118 |           price_breakdown: priceObj.price_breakdown,
119 |           reason: reasonObj.reason,
120 |           useCases: useCasesObj,
121 |           rating: ratingObj.rating,
122 |           image_url: imageUrlObj.image_url
123 |         }
124 |       })
125 |     })
126 |   } catch (err) {
127 |     console.error('Error in POST handler:', err)
128 |     return NextResponse.json({ error: 'Lỗi: Không thể xử lý yêu cầu của bạn. Vui lòng thử lại sau.' })
129 |   }
130 | }

```

```

93 |         if (!res.ok) {
94 |             const errText = await res.text()
95 |             console.error('Gemini API error:', res.status, errText)
96 |             return NextResponse.json({ error: 'Đề ch và AI tìm th ý i b - n , vui l c' })
97 |         }
98 |
99 |         const data = await res.json()
100 |         const text = data?.candidates?.[0]?.content?.parts?.[0]?.text || ''
101 |
102 |         if (!text.trim()) {
103 |             return NextResponse.json({ error: 'AI không tr ề v ấ k ết qu ế . Vui l c' })
104 |         }
105 |
106 |         let jsonStr = text.trim()
107 |         const jsonBlockMatch = text.match(/```(?:json)?\s*([\s\S]*)```/)
108 |         if (jsonBlockMatch) {
109 |             jsonStr = jsonBlockMatch[1].trim()
110 |         } else {
111 |             const objectMatch = text.match(/(?:[\s\S]*"suggestions"[\s\S]*)/)
112 |             if (objectMatch) {
113 |                 jsonStr = objectMatch[0]
114 |             }
115 |         }
116 |
117 |         try {
118 |             const parsed = JSON.parse(jsonStr)
119 |             cache.set(cacheKey, { data: parsed, timestamp: Date.now() })
120 |             return NextResponse.json(parsed)
121 |         } catch {
122 |             console.error('Failed to parse AI response as JSON:', text.substring(0, 500))
123 |             return NextResponse.json({ error: 'Không th ấ phân tích d ữ li ệ u t ế' })
124 |         }
125 |     } catch (err: any) {
126 |         if (err.name === 'AbortError') {
127 |             return NextResponse.json({ error: 'Y ề u c s u ấ h ết th ý i gian ch ý . V' })
128 |         }
129 |         console.error('PC suggest error:', err)
130 |         return NextResponse.json({ error: ' ấ x ế y ra l ầ i , vui lòng th ấ l ầ i . V' })
131 |     }
132 | }
133 |
134 |

```

File: app/api/pdf/route.ts (28 lines, 0.8 KB)

```

1 | import { NextResponse } from 'next/server'
2 | import fs from 'fs'
3 | import path from 'path'
4 |
5 | const ALLOWED = ['sach-tin10.pdf', 'sach-tin11.pdf']
6 |
7 | export async function GET(req: Request) {
8 |     const { searchParams } = new URL(req.url)
9 |     const file = searchParams.get('file')
10 |
11 |     if (!file || !ALLOWED.includes(file)) {
12 |         return new NextResponse('File not found', { status: 404 })
13 |     }
14 |
15 |     const filePath = path.join(process.cwd(), 'public', file)
16 |     if (!fs.existsSync(filePath)) {
17 |         return new NextResponse('File not found', { status: 404 })
18 |     }
19 |
20 |     const buffer = fs.readFileSync(filePath)
21 |     return new NextResponse(buffer, {
22 |         headers: {
23 |             'Content-Type': 'application/pdf',
24 |             'Content-Disposition': 'inline; filename="' + file + '"',
25 |         },
26 |     })
27 | }
28 |

```

File: app/api/products/route.ts (41 lines, 1.2 KB)

```

1 | import { createClient } from '@lib/supabase-ssr-server'
2 | import { NextRequest, NextResponse } from 'next/server'
3 |
4 | export async function GET(req: NextRequest) {
5 |     const supabase = await createClient()
6 |     const { searchParams } = new URL(req.url)
7 |     const category = searchParams.get('category')
8 |     const query = searchParams.get('q')
9 |     const limit = parseInt(searchParams.get('limit') || '50')
10 |     const offset = parseInt(searchParams.get('offset') || '0')
11 |
12 |     let dbQuery = supabase
13 |         .from('products')
14 |         .select('*')
15 |
16 |     if (category) {
17 |         const cat = category.toUpperCase()
18 |         const valid = ['CPU', 'GPU', 'MAINBOARD', 'RAM', 'PSU', 'CASE', 'COOLER', 'SSD', 'HDD']
19 |         if (valid.includes(cat)) {
20 |             dbQuery = dbQuery.eq('category', cat)
21 |         }
22 |     }
23 |
24 |     if (query) {
25 |         dbQuery = dbQuery.or(
26 |             `full_name.ilike.%${query}%,brand.ilike.%${query}%,model.ilike.%${query}%`
27 |         )
28 |     }
29 |
30 |     const { data, error, count } = await dbQuery
31 |         .order('price_vnd', { ascending: true })
32 |         .range(offset, offset + limit - 1)
33 |         .limit(limit)
34 |
35 |     if (error) {
36 |         return NextResponse.json({ error: error.message }, { status: 500 })
37 |     }
38 |
39 |     return NextResponse.json({ products: data, count, offset, limit })
40 | }
41 |

```

File: app/api/quiz/route.js (34 lines, 1.9 KB)

```

1 import { GoogleGenerativeAI } from "@google/generative-ai";
2 import { searchKnowledge } from "../../utils/searchKnowledge";
3
4 const genAI = new GoogleGenerativeAI(process.env.GEMINI_API_KEY || process.env.GOOGLE_GEMINI_API_KEY);
5
6 export async function POST(req) {
7   try {
8     const { lang, topic, level } = await req.json();
9     const model = genAI.getGenerativeModel({ model: "gemini-1.5-flash" });
10
11     // B ộ c 1 : L ệ y k i ệ n t h ế c t ế K n o w l e d g e B a s e ( R A G )
12     const context = searchKnowledge(topic);
13
14     const topicInfo = topic ? `V ấ {topic}` : `V ấ p h ầ n c ế n g m á y t i
15     const levelInfo = level ? `B ộ k h ố C ấ p {level}` : `c h o n g ộ Y ấ
16
17     const prompt = lang === 'vn'
18       ? `D ấ n t r ề n t à i l i ệ u s a u â y : \n \n "${context}" \n \n H ấ y t i ố
19       : `Based on the following documentation:\n\n"${context}"\n\nGenerate 10 multiple-choice questions ...
20
21     const result = await model.generateContent(prompt);
22     const response = await result.response;
23     let text = response.text();
24
25     // Cleanup markdown if present
26     text = text.replace(/```json/g, '').replace(/```/g, '').trim();
27
28     return Response.json(JSON.parse(text));
29   } catch (error) {
30     console.error("Gemini API Error:", error);
31     return Response.json({ error: error.message || "Failed to generate question" }, { status: 500 });
32   }
33 }
34

```

File: app/api/seed/daily-quests/route.ts (34 lines, 2.7 KB)

```

1 import { createClient } from '@supabase/supabase-js'
2
3 const supabaseUrl = process.env.NEXT_PUBLIC_SUPABASE_URL!
4 const supabaseServiceKey = process.env.SUPABASE_SERVICE_ROLE_KEY!
5
6 const QUESTS = [
7   { title: 'H ồ à n t h à n h 1 b à i h í c', description: 'X ế m h ấ t 1 b à i g i ế n g b ấ t
8   { title: 'L ậ p r ấ p P C - h ồ à n c h ế n h', description: 'L ậ p r ấ p P C - h ồ à n c h ế n h
9   { title: 'L ậ m b à i t r - c o n g h i ệ m', description: 'H ồ à n t h à n h 1 b à i q u i ệ n b ấ
10  { title: 'T h ầ m g i ấ t h ế o l u - n', description: 't c ầ u h í i h o - c t r ấ l ấ i
11  { title: 'i t s t r e a k 3 n g ầ y', description: 'H í c l i ề n t i ệ p 3 n g ầ y k h ồ n g
12  { title: 'K i ấ m t r ầ t ố n g t h i ệ c h', description: 'C h i ấ k i ấ m t r ầ t ố n g t h i ệ
13  { title: 'H ồ à n t h à n h 3 b à i h í c', description: 'H í c l i ề n t ấ c 3 b à i g i ế n
14  { title: 'L u y ệ n t - p 15 p h ú t', description: 'D ầ n h 15 p h ú t t r o n g P C B u i
15 ]
16
17 export async function GET() {
18   const supabase = createClient(supabaseUrl, supabaseServiceKey)
19
20   try {
21     const { data: existing } = await supabase.from('daily_quests').select('id').limit(1)
22     if (existing && existing.length > 0) {
23       return Response.json({ message: 'daily_quests already has data,
24     }
25
26     const { error } = await supabase.from('daily_quests').insert(QUESTS)
27     if (error) throw error
28
29     return Response.json({ message: `Seeded ${QUESTS.length} daily que
30   } catch (err: any) {
31     return Response.json({ error: err.message }, { status: 500 })
32   }
33 }
34

```

File: app/api/tts/route.ts (26 lines, 0.8 KB)

```

1 export async function GET(req: Request) {
2   const { searchParams } = new URL(req.url);
3   const text = searchParams.get('text');
4   const lang = searchParams.get('lang') || 'vi';
5   if (!text) {
6     return new Response('Missing text', { status: 400 });
7   }
8
9   const googleUrl = `https://translate.google.com/translate_tts?ie=UTF-8&tl=${lang}&client=tw-ob&q=${encodeURIComponent...
10   try {
11     const res = await fetch(googleUrl);
12     if (!res.ok) {
13       return new Response('TTS unavailable', { status: 502 });
14     }
15     const audio = await res.arrayBuffer();
16     return new Response(audio, {
17       headers: {
18         'Content-Type': 'audio/mpeg',
19         'Cache-Control': 'public, max-age=86400',
20       },
21     });
22   } catch {
23     return new Response('TTS failed', { status: 502 });
24   }
25 }
26

```

File: app/auth/callback/route.ts (38 lines, 1.2 KB)

```

1 import { createClient } from '@lib/supabase-ssr-server'
2 import { NextResponse } from 'next/server'
3
4 export async function GET(request: Request) {
5   const requestUrl = new URL(request.url)
6   const code = requestUrl.searchParams.get('code')
7
8   if (code) {
9     const supabase = await createClient()
10     const { data: { user } } = await supabase.auth.exchangeCodeForSession(code)
11
12     if (user) {
13       const { data: profile } = await supabase
14         .from('profiles')
15         .select('role')
16         .eq('id', user.id)
17         .single()
18
19       // Redirect to the appropriate page based on the role
20       // For example, if the role is 'admin', redirect to /admin
21       // If the role is 'user', redirect to /home
22       // ...
23     }
24   }
25 }
26

```

```

18 // New OAuth user - redirect to register completion to select role
19 if (!profile || !profile.role) {
20   return NextResponse.redirect(new URL('/register?oauth=true', requestUrl.origin))
21 }
22
23 // Existing user - redirect based on role
24 const dashboardUrl = {
25   teacher: '/teacher',
26   admin: '/admin',
27   parent: '/parent',
28   student: '/builder',
29 }[profile.role as 'teacher' | 'admin' | 'parent' | 'student'] ?? '/builder'
30
31 return NextResponse.redirect(new URL(dashboardUrl, requestUrl.origin))
32 }
33 }
34 }
35
36 return NextResponse.redirect(new URL('/login?error=oauth_failed', requestUrl.origin))
37 }
38

```

File: app/builder/3d-components/page.js (714 lines, 23.0 KB)

```

1 'use client';
2
3 import { useState, useEffect, useRef } from 'react';
4 import Link from 'next/link';
5 import { CATEGORIES, COLOR_THEMES } from '@lib/sketchfab-models';
6 import { Box, Grid3X3, ChevronLeft, Maximize2, X, Eye, Layers, User, BookOpen, Rotate3D, BookMarked } from 'lu...
7
8 function SketchfabEmbed({ modelId, title, fullscreen }) {
9   return (
10     <iframe
11       title={title}
12       style={{ width: '100%', height: '100%', border: 'none' }}
13       src={`https://sketchfab.com/models/${modelId}/embed${fullscreen ? '?autostart=1' : '?autostart=0'}`}
14       allow="autoplay; fullscreen"
15       allowFullScreen
16       loading="lazy"
17       sandbox="allow-scripts allow-same-origin allow-popups"
18     />
19   );
20 }
21
22 function FullscreenModal({ item, theme, onClose }) {
23   useEffect(() => {
24     document.body.style.overflow = 'hidden';
25     return () => { document.body.style.overflow = '' };
26   }, []);
27
28   return (
29     <div
30       style={{
31         position: 'fixed',
32         inset: 0,
33         zIndex: 9999,
34         background: '#000',
35         display: 'flex',
36         flexDirection: 'column',
37       }}
38     >
39       <div
40         style={{
41           display: 'flex',
42           alignItems: 'center',
43           justifyContent: 'space-between',
44           padding: '12px 20px',
45           background: 'rgba(0,0,0,0.8)',
46           backdropFilter: 'blur(16px)',
47           borderBottom: '1px solid rgba(255,255,255,0.06)',
48           flexShrink: 0,
49           zIndex: 10,
50         }}
51       >
52         <div style={{ display: 'flex', alignItems: 'center', gap: 10 }}>
53           <span
54             style={{
55               background: theme.gradient,
56               color: 'fff',
57               fontSize: 11,
58               fontWeight: 800,
59               width: 28,
60               height: 28,
61               borderRadius: 8,
62               display: 'flex',
63               alignItems: 'center',
64               justifyContent: 'center',
65             }}
66           >
67             {item.originalIndex !== undefined ? item.originalIndex + 1 : '3D'}
68           </span>
69           <div>
70             <div style={{ fontSize: 14, fontWeight: 600, color: 'fff' }}>{item.title}</div>
71             <div style={{ fontSize: 11, color: 'rgba(255,255,255,0.4)' }}>{item.description}</div>
72           </div>
73         </div>
74         <button
75           onClick={onClose}
76           style={{
77             width: 36,
78             height: 36,
79             borderRadius: 10,
80             background: 'rgba(255,255,255,0.06)',
81             border: '1px solid rgba(255,255,255,0.08)',
82             display: 'flex',
83             alignItems: 'center',
84             justifyContent: 'center',
85             cursor: 'pointer',
86             color: 'fff',
87             transition: 'all 0.2s',
88           }}
89           onMouseOver={e => { e.currentTarget.style.background = 'rgba(255,255,255,0.12)' }}
90           onMouseOut={e => { e.currentTarget.style.background = 'rgba(255,255,255,0.06)' }}
91         >
92           <X size={18} />
93         </button>
94       </div>
95       <div style={{ flex: 1, position: 'relative' }}>
96         <SketchfabEmbed modelId={item.embedId} title={item.title} fullscreen />
97       </div>
98     </div>
99   );

```



```

100 }
101
102 function AnnotationBadge({ analyzedBy, knowledge, theme }) {
103   const [open, setOpen] = useState(false);
104
105   return (
106     <div style={{ position: 'relative' }}>
107       <button
108         onClick={e => { e.stopPropagation(); setOpen(!open) }}
109         style={{
110           display: 'flex',
111           alignItems: 'center',
112           gap: 5,
113           padding: '3px 10px',
114           borderRadius: 6,
115           background: `${theme.color}12`,
116           border: '1px solid ${theme.color}20`,
117           color: theme.color,
118           fontSize: 10,
119           fontWeight: 600,
120           cursor: 'pointer',
121           transition: 'all 0.2s',
122           fontFamily: 'inherit',
123         }}
124       >
125         <BookMarked size={12} />
126         Chú thích
127       </button>
128       {open && (
129         <div
130           style={{
131             position: 'fixed',
132             inset: 0,
133             zIndex: 100,
134           }}
135           onClick={e => { e.stopPropagation(); setOpen(false) }}
136         />
137         <div
138           style={{
139             position: 'absolute',
140             top: 'calc(100% + 8px)',
141             left: 0,
142             zIndex: 101,
143             width: 320,
144             padding: 16,
145             borderRadius: 12,
146             background: '#0f0f1a',
147             border: '1px solid ${theme.color}30',
148             boxShadow: `0 8px 40px rgba(0,0,0,0.5), 0 0 20px ${theme.color}10`,
149           }}
150         >
151           <div style={{ display: 'flex', alignItems: 'center', gap: 8, marginBottom: 10 }}>
152             <div
153               style={{
154                 width: 28,
155                 height: 28,
156                 borderRadius: 8,
157                 background: theme.gradient,
158                 display: 'flex',
159                 alignItems: 'center',
160                 justifyContent: 'center',
161               }}
162             >
163               <User size={14} color="#fff" />
164             </div>
165             <div>
166               6 7 | <div style={{ font size: 11, font weight: 600, color: '#fff' }}>
167                 <div style={{ font size: 10, color: theme.color }}>{analyzedBy}</div>
168               </div>
169             </div>
170             <div style={{ display: 'flex', alignItems: 'flex-start', gap: 8 }}>
171               <BookOpen size={14} color={theme.color} style={{ flexShrink: 0, marginTop: 1 }} />
172               <div style={{ font size: 11, color: 'rgba(255,255,255,0.7)', lineHeight: 1.6 }}>{knowledge}</div>
173             </div>
174           </div>
175         </div>
176       </div>
177     );
178   };
179 }
180
181
182 function ModelCard({ item, theme, index, category, onView }) {
183   const [shouldLoad, setShouldLoad] = useState(false);
184   const cardRef = useRef(null);
185
186   useEffect(() => {
187     const el = cardRef.current;
188     if (!el) return;
189     const obs = new IntersectionObserver(
190       ([entry]) => {
191         if (entry.isIntersecting) {
192           setShouldLoad(true);
193           obs.unobserve(el);
194         }
195       },
196       { rootMargin: '200px' }
197     );
198     obs.observe(el);
199     return () => obs.disconnect();
200   }, []);
201
202   return (
203     <div
204       ref={cardRef}
205       onClick={() => onView(item, theme)}
206       style={{
207         background: `linear-gradient(145deg, ${theme.bg}, #000)`,
208         borderRadius: 16,
209         border: '1px solid ${theme.color}18`,
210         overflow: 'hidden',
211         transition: 'all 0.4s cubic-bezier(0.4, 0, 0.2, 1)',
212         position: 'relative',
213         cursor: 'pointer',
214         height: 340,
215       }}
216     >
217       <MouseOver={e => {
218         e.currentTarget.style.borderColor = `${theme.color}50`;
219         e.currentTarget.style.boxShadow = `0 8px 32px ${theme.color}20`;
220         e.currentTarget.style.transform = 'translateY(-4px)';
221       }}
222       <MouseOut={e => {
223         e.currentTarget.style.borderColor = `${theme.color}18`;
224         e.currentTarget.style.boxShadow = 'none';

```

```

224 |         e.currentTarget.style.transform = 'translateY(0)';
225 |     }}
226 | }
227 | <div
228 |   style={{
229 |     position: 'absolute',
230 |     top: 0, left: 0, right: 0,
231 |     height: 2,
232 |     background: theme.gradient,
233 |     opacity: 0.6,
234 |     zIndex: 11,
235 |   }}
236 | />
237 |
238 | <div
239 |   style={{
240 |     position: 'absolute',
241 |     top: 14,
242 |     left: 14,
243 |     zIndex: 10,
244 |     display: 'flex',
245 |     alignItems: 'center',
246 |     gap: 8,
247 |   }}
248 | >
249 |   <span
250 |     style={{
251 |       background: theme.gradient,
252 |       color: '#fff',
253 |       fontSize: 11,
254 |       fontWeight: 800,
255 |       width: 26,
256 |       height: 26,
257 |       borderRadius: 8,
258 |       display: 'flex',
259 |       alignItems: 'center',
260 |       justifyContent: 'center',
261 |       boxShadow: `0 2px 12px ${theme.color}50`,
262 |     }}
263 |   >
264 |     {index + 1}
265 |   </span>
266 |   <span
267 |     style={{
268 |       fontSize: 11,
269 |       fontWeight: 600,
270 |       color: '#fff',
271 |       background: 'rgba(0,0,0,0.7)',
272 |       backdropFilter: 'blur(12px)',
273 |       padding: '5px 12px',
274 |       borderRadius: 8,
275 |       maxWidth: 180,
276 |       whiteSpace: 'nowrap',
277 |       overflow: 'hidden',
278 |       textOverflow: 'ellipsis',
279 |       border: '1px solid rgba(255,255,255,0.06)',
280 |     }}
281 |   >
282 |     {item.title}
283 |   </span>
284 | </div>
285 |
286 | {shouldLoad ? (
287 |   <div
288 |     style={{
289 |       width: '100%',
290 |       height: '100%',
291 |       pointerEvents: 'none',
292 |     }}
293 |   >
294 |     <iframe
295 |       title={item.title}
296 |       style={{ width: '100%', height: '100%', border: 'none' }}
297 |       src={`https://sketchfab.com/models/${item.embedId}/embed?autostart=0&preload=0&ui_controls=0&ui_in...`}
298 |       allow="autoplay; fullscreen"
299 |       loading="lazy"
300 |       sandbox="allow-scripts allow-same-origin allow-popups"
301 |     />
302 |   </div>
303 | ) : (
304 |   <div
305 |     style={{
306 |       position: 'absolute',
307 |       inset: 0,
308 |       display: 'flex',
309 |       flexDirection: 'column',
310 |       alignItems: 'center',
311 |       justifyContent: 'center',
312 |       gap: 12,
313 |       color: theme.color,
314 |       background: `radial-gradient(ellipse at center, ${theme.color}08, transparent)`,
315 |     }}
316 |   >
317 |     <div style={{ fontSize: 40, opacity: 0.2 }}>Y&#p </div>
318 |     <div style={{ fontSize: 12, fontWeight: 500, color: theme.color, opacity: 0.8 }}>
319 |       N h x p   Å   x e m   3 D
320 |     </div>
321 |   </div>
322 | )}
323 |
324 | <div
325 |   onClick={() => onView(item, theme)}
326 |   style={{
327 |     position: 'absolute',
328 |     inset: 0,
329 |     zIndex: 5,
330 |     cursor: 'pointer',
331 |   }}
332 | >
333 |   <div
334 |     style={{
335 |       position: 'absolute',
336 |       bottom: 60,
337 |       left: '50%',
338 |       transform: 'translateX(-50%)',
339 |       display: 'flex',
340 |       alignItems: 'center',
341 |       gap: 6,
342 |       background: 'rgba(0,0,0,0.75)',
343 |       backdropFilter: 'blur(12px)',
344 |       padding: '8px 16px',
345 |       borderRadius: 10,
346 |       border: '1px solid rgba(255,255,255,0.1)',
347 |       opacity: 0,

```

```

348         transition: 'opacity 0.2s',
349         pointerEvents: 'none',
350     }}
351     className="view-btn"
352 >
353     <Maximize2 size={14} color="#fff" />
354     <span style={{ fontSize: 12, fontWeight: 600, color: '#fff' }}>Xem toàn màn hình</span>
355 </div>
356 </div>
357
358 <div
359     style={{
360         position: 'absolute',
361         bottom: 0,
362         left: 0,
363         right: 0,
364         padding: '48px 16px 14px',
365         background: 'linear-gradient(transparent, rgba(0,0,0,0.9) 40%)',
366         display: 'flex',
367         justifyContent: 'space-between',
368         alignItems: 'flex-end',
369         pointerEvents: 'none',
370     }}
371 >
372     <div>
373         <div style={{ fontSize: 14, fontWeight: 700, color: '#fff', textShadow: '0 1px 4px rgba(0,0,0,0.5)' }}>
374             {item.title}
375         </div>
376         {item.description && (
377             <div style={{ fontSize: 11, color: 'rgba(255,255,255,0.6)', marginTop: 2, textShadow: '0 1px 4px r...
378                 {item.description}
379             </div>
380         )}
381     </div>
382     <div style={{ display: 'flex', alignItems: 'center', gap: 4, color: 'rgba(255,255,255,0.35)', fontSize...
383         <Rotate3D size={12} />
384         <span>3D</span>
385     </div>
386 </div>
387 </div>
388 );
389 }
390
391 export default function Components3DPage() {
392     const [activeCategory, setActiveCategory] = useState(null);
393     const [visible, setVisible] = useState({});
394     const [fullscreenItem, setFullscreenItem] = useState(null);
395     const [fullscreenTheme, setFullscreenTheme] = useState(null);
396
397     useEffect(() => {
398         const onScroll = () => {
399             const sections = document.querySelectorAll('[data-category]');
400             sections.forEach(section => {
401                 const rect = section.getBoundingClientRect();
402                 const id = section.getAttribute('data-category');
403                 if (rect.top < window.innerHeight - 80) {
404                     setVisible(prev => ({ ...prev, [id]: true }));
405                 }
406             });
407         };
408         onScroll();
409         window.addEventListener('scroll', onScroll, { passive: true });
410         return () => window.removeEventListener('scroll', onScroll);
411     }, []);
412
413     const handleView = (item, theme) => {
414         setFullscreenItem(item);
415         setFullscreenTheme(theme);
416     };
417
418     return (
419         <div
420             style={{
421                 minHeight: '100vh',
422                 background: 'var(--bg-base)',
423                 fontFamily: '"Segoe UI", system-ui, sans-serif",
424             }}
425         >
426             <style>{
427                 .model-card:hover .view-btn {
428                     opacity: 1 !important;
429                 }
430                 @keyframes spin {
431                     to { transform: rotate(360deg); }
432                 }
433             }</style>
434
435             <div
436                 style={{
437                     position: 'sticky',
438                     top: 0,
439                     zIndex: 50,
440                     background: 'rgba(8, 8, 18, 0.85)',
441                     backdropFilter: 'blur(20px) saturate(1.5)',
442                     borderBottom: '1px solid rgba(255,255,255,0.06)',
443                 }}
444             >
445                 <div
446                     style={{
447                         maxWidth: 1280,
448                         margin: '0 auto',
449                         padding: '14px 24px',
450                         display: 'flex',
451                         alignItems: 'center',
452                         justifyContent: 'space-between',
453                     }}
454                 >
455                     <div style={{ display: 'flex', alignItems: 'center', gap: 12 }}>
456                         <Link
457                             href="/builder"
458                             style={{
459                                 display: 'flex',
460                                 alignItems: 'center',
461                                 gap: 6,
462                                 color: 'rgba(255,255,255,0.5)',
463                                 textDecoration: 'none',
464                                 fontSize: 13,
465                                 fontWeight: 500,
466                                 padding: '6px 10px',
467                                 borderRadius: 8,
468                                 transition: 'all 0.2s',
469                             }}
470                         >
471                             onMouseOver={e => { e.currentTarget.style.color = '#fff'; e.currentTarget.style.background = 'rg...

```

```

472 |         >
473 |         <ChevronLeft size={16} />
474 |         Q u a y   l i i
475 |     </Link>
476 |     <div
477 |       style={{
478 |         width: 34,
479 |         height: 34,
480 |         borderRadius: 10,
481 |         backgroundImage: 'linear-gradient(135deg, #6366f1, #a855f7)',
482 |         display: 'flex',
483 |         alignItems: 'center',
484 |         justifyContent: 'center',
485 |         boxShadow: '0 2px 16px rgba(168,85,247,0.4)',
486 |       }}
487 |     >
488 |     <Box size={17} color="#fff" />
489 |   </div>
490 |   <div>
491 |     <h1 style={{ margin: 0, fontSize: 17, fontWeight: 700, color: '#fff', letterSpacing: '-0.02em' }}>
492 |       K h o   L i n h   K i   Ç n   <span style={{ color: '#a855f7' }}> > 3 D </span>
493 |     </h1>
494 |     <p style={{ margin: 0, fontSize: 11, color: 'rgba(255,255,255,0.4)' }}>
495 |       K h á m   p h á   l i n h   k i   Ç n   đ °   Ò i   g ó c   n h i n   3   c h i á u
496 |     </p>
497 |   </div>
498 | </div>
499 |
500 |   <div style={{ display: 'flex', alignItems: 'center', gap: 8, color: 'rgba(255,255,255,0.3)', fontSiz...
501 |     <Eye size={14} />
502 |     <span>{CATEGORIES.reduce((sum, cat) => sum + cat.items.length, 0)} mô hình</span>
503 |   </div>
504 | </div>
505 |
506 |   <div
507 |     style={{
508 |       maxWidth: 1280,
509 |       margin: '0 auto',
510 |       padding: '0 24px 12px',
511 |       display: 'flex',
512 |       gap: 6,
513 |       flexWrap: 'wrap',
514 |     }}
515 |   >
516 |     <button
517 |       onClick={() => setActiveCategory(null)}
518 |       style={{
519 |         padding: '5px 14px',
520 |         borderRadius: 100,
521 |         border: '1px solid ${!activeCategory ? '#a855f7' : 'rgba(255,255,255,0.1)}',
522 |         background: 'activeCategory ? 'rgba(168,85,247,0.15)' : 'transparent',
523 |         color: 'activeCategory ? '#a855f7' : 'rgba(255,255,255,0.5)',
524 |         fontSize: 11,
525 |         fontWeight: 600,
526 |         cursor: 'pointer',
527 |         transition: 'all 0.2s',
528 |         fontFamily: 'inherit',
529 |       }}
530 |     >
531 |       T  ¥  t   c  £
532 |     </button>
533 |     {CATEGORIES.map(cat => {
534 |       const theme = COLOR_THEMES[cat.id];
535 |       const isActive = activeCategory === cat.id;
536 |       return (
537 |         <button
538 |           key={cat.id}
539 |           onClick={() => setActiveCategory(isActive ? null : cat.id)}
540 |           style={{
541 |             padding: '5px 14px',
542 |             borderRadius: 100,
543 |             border: '1px solid ${isActive ? theme.color : 'rgba(255,255,255,0.1)}',
544 |             background: isActive ? `${theme.color}20` : 'transparent',
545 |             color: isActive ? theme.color : 'rgba(255,255,255,0.5)',
546 |             fontSize: 11,
547 |             fontWeight: 600,
548 |             cursor: 'pointer',
549 |             transition: 'all 0.2s',
550 |             fontFamily: 'inherit',
551 |           }}
552 |         >
553 |           {theme.label}
554 |         </button>
555 |       );
556 |     })}
557 |   </div>
558 | </div>
559 |
560 |   <div style={{ maxWidth: 1280, margin: '0 auto', padding: '32px 24px' }}>
561 |     {CATEGORIES.map(category => {
562 |       if (activeCategory & activeCategory !== category.id) return null;
563 |       const theme = COLOR_THEMES[category.id];
564 |       return (
565 |         <div
566 |           key={category.id}
567 |           data-category={category.id}
568 |           style={{
569 |             opacity: visible[category.id] ? 1 : 0,
570 |             transform: visible[category.id] ? 'translateY(0)' : 'translateY(24px)',
571 |             transition: 'opacity 0.6s cubic-bezier(0.4, 0, 0.2, 1), transform 0.6s cubic-bezier(0.4, 0, 0....
572 |           }}
573 |         >
574 |           <section style={{ marginBottom: 56 }}>
575 |             <div
576 |               style={{
577 |                 display: 'flex',
578 |                 alignItems: 'center',
579 |                 gap: 16,
580 |                 marginBottom: 24,
581 |                 padding: '18px 24px',
582 |                 borderRadius: 16,
583 |                 background: `linear-gradient(135deg, ${theme.color}10, ${theme.color}05)`,
584 |                 border: '1px solid ${theme.color}20',
585 |                 backdropFilter: 'blur(8px)',
586 |                 position: 'relative',
587 |                 overflow: 'hidden',
588 |               }}
589 |             >
590 |               <div
591 |                 style={{
592 |                   position: 'absolute',
593 |                   top: '-50%',
594 |                   right: '-20%',
595 |                   width: 200,

```

```

596 |         height: 200,
597 |         borderRadius: '50%',
598 |         background: `radial-gradient(circle, ${theme.color}15, transparent)`,
599 |         pointerEvents: 'none',
600 |     }}
601 |     />
602 |     <div
603 |       style={{
604 |         width: 48,
605 |         height: 48,
606 |         borderRadius: 14,
607 |         background: theme.gradient,
608 |         display: 'flex',
609 |         alignItems: 'center',
610 |         justifyContent: 'center',
611 |         boxShadow: `0 4px 24px ${theme.color}50`,
612 |         flexShrink: 0,
613 |       }}
614 |     >
615 |       <Layers size={22} color="#fff" />
616 |     </div>
617 |     <div style={{ flex: 1 }}>
618 |       <div style={{ display: 'flex', alignItems: 'center', gap: 12, flexWrap: 'wrap' }}>
619 |         <h2
620 |           style={{
621 |             margin: 0,
622 |             fontSize: 20,
623 |             fontWeight: 800,
624 |             color: '#fff',
625 |             letterSpacing: '-0.02em',
626 |           }}
627 |         >
628 |           {theme.label}
629 |         </h2>
630 |         <AnnotationBadge
631 |           analyzedBy={category.analyzedBy}
632 |           knowledge={category.knowledge}
633 |           theme={theme}
634 |         />
635 |       </div>
636 |       <div style={{ display: 'flex', alignItems: 'center', gap: 8, marginTop: 2 }}>
637 |         <span style={{ fontSize: 12, color: theme.color, fontWeight: 600 }}>
638 |           {category.items.length} mô hình
639 |         </span>
640 |         <span style={{ color: 'rgba(255,255,255,0.15)' }}></span>
641 |         <span style={{ fontSize: 12, color: 'rgba(255,255,255,0.4)' }}>
642 |           {category.items.map(i => i.title).join(', ').length > 80
643 |             ? category.items.map(i => i.title).join(', ').slice(0, 80) + '...'
644 |             : category.items.map(i => i.title).join(', ')}
645 |         </span>
646 |       </div>
647 |     </div>
648 |     <div
649 |       style={{
650 |         width: 40,
651 |         height: 40,
652 |         borderRadius: 10,
653 |         background: `${theme.color}15`,
654 |         border: `1px solid ${theme.color}25`,
655 |         display: 'flex',
656 |         alignItems: 'center',
657 |         justifyContent: 'center',
658 |         color: theme.color,
659 |         flexShrink: 0,
660 |       }}
661 |     >
662 |       <Grid3X3 size={18} />
663 |     </div>
664 |   </div>
665 |
666 |   <div
667 |     style={{
668 |       display: 'grid',
669 |       gridTemplateColumns: 'repeat(auto-fill, minmax(320px, 1fr))',
670 |       gap: 20,
671 |     }}
672 |   >
673 |     {category.items.map((item, idx) => (
674 |       <ModelCard
675 |         key={item.id}
676 |         item={{ ...item, originalIndex: idx }}
677 |         theme={theme}
678 |         index={idx}
679 |         category={category}
680 |         onView={handleView}
681 |       />
682 |     ))}
683 |   </div>
684 | </section>
685 | </div>
686 |   });
687 | });
688 |
689 |   <footer
690 |     style={{
691 |       textAlign: 'center',
692 |       padding: '40px 0 24px',
693 |       borderTop: '1px solid rgba(255,255,255,0.06)',
694 |     }}
695 |   >
696 |     <p style={{ margin: 0, color: 'rgba(255,255,255,0.25)', fontSize: 12, fontWeight: 500 }}>
697 |       T ò n g c ù n g <span style={{ color: '#a855f7' }}>{ C A T E G O R I E S . r e c
698 |     </p>
699 |     <p style={{ margin: '6px 0 0', color: 'rgba(255,255,255,0.15)', fontSize: 11 }}>
700 |       PC Master Builder &copy; {new Date().getFullYear()}
701 |     </p>
702 |   </footer>
703 | </div>
704 |
705 | {fullscreenItem && fullscreenTheme && (
706 |   <FullscreenModal
707 |     item={fullscreenItem}
708 |     theme={fullscreenTheme}
709 |     onClose={() => { setFullscreenItem(null); setFullscreenTheme(null) }}
710 |   />
711 | )}
712 | </div>
713 | );
714 | }

```

File: app/builder/3d-viewer/page.tsx (215 lines, 10.1 KB)

1| 'use client';

```

3| import { useState, useEffect, useRef, Component, type ReactNode } from 'react';
4| import dynamic from 'next/dynamic';
5|
6| class ErrorBoundary extends Component<{ children: ReactNode }, { error: Error | null }> {
7|   state = { error: null };
8|   static getDerivedStateFromError(error: Error) { return { error }; }
9|   render() {
10|     if (this.state.error) {
11|       return <div style={{ color: '#ff4466', padding: 40, fontFamily: 'monospace', fontSize: 12 }}>
12|         <h2>Error: {this.state.error.message}</h2>
13|         <pre style={{ color: '#88bbcc', marginTop: 12 }}>{this.state.error.stack}</pre>
14|       </div>;
15|     }
16|     return this.props.children;
17|   }
18| }
19|
20| const GameScene = dynamic(() => import('@components/GameScene'), { ssr: false });
21|
22| function T(lang: 'en' | 'vn', en: string, vn: string) {
23|   return lang === 'en' ? en : vn;
24| }
25|
26| export default function Viewer3DPage() {
27|   const [showInstructions, setShowInstructions] = useState(true);
28|   const [showChat, setShowChat] = useState(false);
29|   const [chatMessages, setChatMessages] = useState<{ role: string; content: string }[]>([]);
30|   const [chatInput, setChatInput] = useState('');
31|   const [lang, setLang] = useState<'en' | 'vn'>('vn');
32|   const chatEndRef = useRef<HTMLDivElement>(null);
33|
34|   useEffect(() => {
35|     try {
36|       const saved = localStorage.getItem('lang');
37|       if (saved === 'en' || saved === 'vn') setLang(saved as 'en' | 'vn');
38|     } catch {}
39|   }, []);
40|
41|   useEffect(() => {
42|     chatEndRef.current?.scrollIntoView({ behavior: 'smooth' });
43|   }, [chatMessages]);
44|
45|   const handleChatSend = async () => {
46|     if (!chatInput.trim()) return;
47|     const userMsg = chatInput;
48|     setChatInput('');
49|     setChatMessages(prev => [...prev, { role: 'user', content: userMsg }]);
50|     try {
51|       const res = await fetch('/api/ai/chat', {
52|         method: 'POST',
53|         headers: { 'Content-Type': 'application/json' },
54|         body: JSON.stringify({ message: userMsg, history: [] }),
55|       });
56|       const data = await res.json();
57|       setChatMessages(prev => [...prev, { role: 'bot', content: data.response }]);
58|     } catch {
59|       setChatMessages(prev => [...prev, { role: 'bot', content: 'xin lỗi' }]);
60|     }
61|   };
62|
63|   return (
64|     <ErrorBoundary>
65|       <div style={{ display: flex, flex-direction: column, align-items: center, justify-content: center, height: 100vh, width: 100vw, overflow: hidden, position: relative }}>
66|         <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
67|           <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
68|             <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
69|               <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
70|                 <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
71|                   <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
72|                     <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
73|                       <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
74|                         <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
75|                           <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
76|                             <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
77|                               <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
78|                                 <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
79|                                   <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
80|                                     <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
81|                                       <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
82|                                         <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
83|                                           <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
84|                                             <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
85|                                               <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
86|                                                 <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
87|                                                   <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
88|                                                     <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
89|                                                       <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
90|                                                         <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
91|                                                           <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
92|                                                             <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
93|                                                               <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
94|                                                                 <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
95|                                                                  <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
96|                                                                    <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
97|                                                                      <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
98|                                                                        <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
99|                                                                          <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
100|                                                                           <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
101|                                                                            <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
102|                                                                              <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
103|                                                                               <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans-serif }}>
104|                                                                                <div style={{ position: absolute, top: 0, right: 0, bottom: 0, left: 0, display: flex, flex-direction: column, align-items: center, justify-content: center, gap: 20px, padding: 20px, font-family: 'Segoe UI', sans
```

```

126 | <GameScene />
127 |
128 | { /* Small chatbot button - bottom right corner */ }
129 | { !showInstructions && (
130 |   <
131 |     { !showChat && (
132 |       <button
133 |         onClick={() => setShowChat(true)}
134 |         style={{
135 |           position: 'fixed', bottom: '20px', right: '20px', zIndex: 9999,
136 |           width: '40px', height: '40px', borderRadius: '50%',
137 |           border: 'none', cursor: 'pointer',
138 |           background: 'linear-gradient(135deg, #00ffcc, #44aaff)',
139 |           color: '#000', display: 'flex', alignItems: 'center', justifyContent: 'center',
140 |           boxShadow: '0 2px 12px rgba(0,255,204,0.4)',
141 |           fontSize: '18px',
142 |         }}
143 |         title="AI Guru"
144 |       >
145 |         4 5 |
146 |       </button>
147 |     ) }
148 |   }
149 |   { /* Chat panel */ }
150 |   { showChat && (
151 |     <div style={{
152 |       position: 'fixed', bottom: '70px', right: '20px', zIndex: 9999,
153 |       width: '300px', height: '380px', borderRadius: '16px',
154 |       background: '#1a1a2e', border: '1px solid rgba(0,255,204,0.2)',
155 |       display: 'flex', flexDirection: 'column', overflow: 'hidden',
156 |       boxShadow: '0 8px 32px rgba(0,0,0,0.5)',
157 |     }} >
158 |       <div style={{
159 |         padding: '10px 14px', background: '#16213e',
160 |         borderBottom: '1px solid rgba(0,255,204,0.15)',
161 |         display: 'flex', justifyContent: 'space-between', alignItems: 'center',
162 |       }} >
163 |         <span style={{ color: '#00ffcc', fontSize: '12px', fontWeight: 700 }}>AI Guru</span>
164 |         <button onClick={() => setShowChat(false)}
165 |           style={{ background: 'none', border: 'none', color: '#8899aa', cursor: 'pointer', fontSize: ...
166 |         </button>
167 |       </div>
168 |       <div style={{ flex: 1, overflowY: 'auto', padding: '10px', display: 'flex', flexDirection: 'column' >
169 |         { chatMessages.length === 0 && (
170 |           <div style={{ color: '#667788', fontSize: '12px', textAlign: 'center', padding: '20px 0' }}>
171 |             7 2 |
172 |             H I I A I G u r u v A P C . . .
173 |           </div>
174 |         ) }
175 |         { chatMessages.map((m, i) => (
176 |           <div key={i} style={{
177 |             alignSelf: m.role === 'user' ? 'flex-end' : 'flex-start',
178 |             maxWidth: '85%', padding: '8px 12px', borderRadius: '12px',
179 |             fontSize: '12px', lineHeight: 1.5,
180 |             background: m.role === 'user' ? '#00d4aa' : '#0f3460',
181 |             color: m.role === 'user' ? '#000' : '#ddeeff',
182 |           }} >
183 |             { m.content }
184 |           </div>
185 |         ) ) }
186 |         <div ref={chatEndRef} />
187 |       </div>
188 |       <div style={{ padding: '8px', borderTop: '1px solid rgba(0,255,204,0.1)', display: 'flex', gap: ...
189 |         <input
190 |           value={chatInput}
191 |           onChange={e => setChatInput(e.target.value)}
192 |           onKeyDown={e => { if (e.key === 'Enter') handleChatSend(); }}
193 |           style={{
194 |             flex: 1, padding: '8px 10px', borderRadius: '8px', border: '1px solid rgba(0,255,204,0.2)',
195 |             background: '#0d1b2a', color: '#fff', fontSize: '12px', fontFamily: 'inherit', outline: ...
196 |           }} />
197 |         </div>
198 |         <button onClick={handleChatSend}
199 |           style={{
200 |             padding: '8px 12px', borderRadius: '8px', border: 'none',
201 |             background: '#00ffcc', color: '#000', cursor: 'pointer', fontWeight: 700, fontSize: '12px',
202 |           }} >
203 |           2 0 4 |
204 |           G i i
205 |         </button>
206 |       </div>
207 |     </div>
208 |   ) }
209 | </div>
210 | </div>
211 | </div>
212 | </div>
213 | </div>
214 | </div>
215 | }

```

File: app/builder/career-build/page.tsx (444 lines, 22.8 KB)

```

1 | 'use client';
2 |
3 | import { useState, useRef, useEffect } from 'react';
4 | import Link from 'next/link';
5 |
6 | const CAREERS = [
7 |   { id: 'ai-engineer', title: 'AI Engineer / Deep Learning', icon: '0>Y', desc: 'AI Engineer / Deep Learning' },
8 |   { id: '3d-designer', title: '3D Designer / Animator', icon: '0<B', desc: '3D Designer / Animator' },
9 |   { id: 'video-editor', title: 'Video Editor / Filmmaker', icon: '0<B', desc: 'Video Editor / Filmmaker' },
10 |  { id: 'streamer', title: 'Professional Streamer / Gamer', icon: '0<B', desc: 'Professional Streamer / Gamer' },
11 |  { id: 'coder', title: 'Coder / Software Developer', icon: '0=U', desc: 'Coder / Software Developer' },
12 |  { id: 'office', title: 'Office Worker / Student', icon: '0=U', desc: 'Office Worker / Student' },
13 |  { id: 'mac', title: 'Mac / Apple Ecosystem', icon: '0<B', desc: 'Mac / Apple Ecosystem' },
14 | ];
15 |
16 | interface ShopLink {
17 |   shop: string; url: string;
18 | }
19 |
20 | interface BuildItem {
21 |   id: string; name: string; type: string; price: number; reason: string; image?: string; link?: string; shops?: string[];
22 | }
23 |
24 | interface BuildResult {
25 |   career: string; explanation: string; build: BuildItem[]; totalPrice: number; tips: string;
26 |   isMac?: boolean; model?: string; macBuild?: { model?: string; build: BuildItem[]; totalPrice: number; tips: ...
27 | }
28 |
29 | const TYPE_COLORS: Record<string, string> = {
30 |   CPU: '#ef4444', GPU: '#8b5cf6', RAM: '#06b6d4', Storage: '#f59e0b',

```

```

31 | PSU: '#10b981', Cooler: '#3b82f6', Mainboard: '#ec4899', Case: '#64748b',
32 | };
33 |
34 | const TYPE_ICONS: Record<string, string> = {
35 |   CPU: '&i', GPU: '0<B0', RAM: '0=0%', Storage: '0=0', PSU: '0=Y', Cooler: '0<B,
36 | };
37 |
38 | export default function CareerBuildPage() {
39 |   const [selected, setSelected] = useState<string | null>(null);
40 |   const [customDream, setCustomDream] = useState('');
41 |   const [customCondition, setCustomCondition] = useState('');
42 |   const [loading, setLoading] = useState(false);
43 |   const [result, setResult] = useState<BuildResult | null>(null);
44 |   const [error, setError] = useState('');
45 |   const [animPhase, setAnimPhase] = useState(0);
46 |   const [showMac, setShowMac] = useState(false);
47 |   const [substituting, setSubstituting] = useState<string | null>(null);
48 |   const [replacedItems, setReplacedItems] = useState<BuildItem[]>([]);
49 |   const resultRef = useRef<HTMLDivElement>(null);
50 |
51 |   const substitutePart = (item: BuildItem) => {
52 |     const substitutes: Record<string, BuildItem> = {
53 |       CPU: { id: 'alt-cpu', name: '[Thay th'] ${item.name.replace(/i/15
54 |       GPU: { id: 'alt-gpu', name: '[Thay th'] ${item.name.replace(/RT/
55 |       RAM: { id: 'alt-ram', name: '[Thay th'] ${item.name.replace(/160
56 |       Storage: { id: 'alt-storage', name: item.name.includes('NVMe') ?
57 |       PSU: { id: 'alt-psu', name: '[Thay th'] ${item.name.replace(/d/
58 | };
59 |   const sub = substitutes[item.type] || { ...item, id: 'alt-' + item.i
60 |   setReplacedItems(prev => [...prev, { ...item, name: item.name, price: sub.price }]);
61 |   setSubstituting(item.id);
62 |   setTimeout(() => setSubstituting(null), 2000);
63 | };
64 |
65 | useEffect(() => {
66 |   if (result && animPhase > 0 && resultRef.current) {
67 |     resultRef.current.scrollIntoView({ behavior: 'smooth', block: 'start' });
68 |   }
69 | }, [animPhase, result]);
70 |
71 |   const formatPrice = (v: number) => v.toLocaleString('vi-VN') + '€';
72 |
73 |   const handleSuggest = async (careerId?: string) => {
74 |     const career = careerId || selected;
75 |     if (!career && !customDream) return;
76 |
77 |     setLoading(true);
78 |     setError('');
79 |     setResult(null);
80 |     setAnimPhase(0);
81 |     setShowMac(false);
82 |
83 |     try {
84 |       const res = await fetch('/api/career-suggest', {
85 |         method: 'POST',
86 |         headers: { 'Content-Type': 'application/json' },
87 |         body: JSON.stringify({
88 |           career,
89 |           customDream,
90 |           customCondition: customCondition || undefined,
91 |           preferMac: career === 'mac',
92 |         }),
93 |       });
94 |       const data = await res.json();
95 |       if (data.error) throw new Error(data.error);
96 |
97 |       try { localStorage.setItem('careerBuild', JSON.stringify(data)) } catch (e) {}
98 |
99 |       setResult(data);
100 |
101 |       // Animate phases
102 |       setTimeout(() => setAnimPhase(1), 300);
103 |       const totalItems = (data.build?.length || 0) + (data.macBuild?.build?.length || 0);
104 |       setTimeout(() => setAnimPhase(2), 800);
105 |       // Items appear one by one
106 |       for (let i = 0; i < totalItems; i++) {
107 |         setTimeout(() => setAnimPhase(3 + i), 1200 + i * 350);
108 |       }
109 |     } catch (err: any) {
110 |       1 0 |   setError(err.message || 'Lỗi tải hình ảnh');
111 |       setTimeout(() => setError(''), 5000);
112 |     } finally {
113 |       setLoading(false);
114 |     }
115 |   };
116 |
117 |   const pcItems = result?.build || [];
118 |   const macItems = result?.macBuild?.build || [];
119 |   const hasBoth = result?.macBuild && pcItems.length > 0;
120 |
121 |   const activeItems = showMac ? macItems : pcItems;
122 |   const replacedTotal = replacedItems.reduce((s, r) => s + r.price, 0);
123 |   const originalTotal = showMac ? (result?.macBuild?.totalPrice || 0) : (result?.totalPrice || 0);
124 |   const activeTotal = replacedItems.length > 0 ? replacedTotal : originalTotal;
125 |   const activeTips = showMac ? (result?.macBuild?.tips || '') : (result?.tips || '');
126 |   const activeModel = showMac ? (result?.macBuild?.model || '') : '';
127 |   const activeExplanation = showMac ? (result?.macBuild?.explanation || '') : (result?.explanation || '');
128 |
129 |   return (
130 |     <div style={{ minHeight: '100vh', background: 'var(--bg-base) }}>
131 |       </* Header */>
132 |       <div style={{ padding: '16px 24px', borderBottom: '1px solid var(--border-default)', background: 'var(--...
133 |         <div style={{ maxWidth: '900px', margin: '0 auto', display: 'flex', alignItems: 'center', justifyConte...
134 |         <Link href="/builder" style={{ display: 'flex', alignItems: 'center', gap: '8px', color: 'var(--text...
135 |         <span>&larr;</span> Builder
136 |       </Link>
137 |       <div style={{ font-size: '16px', font-weight: 700, color: 'var(--text-primary)' }}>
138 |         <div style={{ width: '60px' }} />
139 |       </div>
140 |     </div>
141 |
142 |     <div style={{ maxWidth: '900px', margin: '0 auto', padding: '24px 16px' }}>
143 |       </* Input section */>
144 |       <{!result && (
145 |         <div style={{ textAlign: 'center', marginBottom: '28px' }}>
146 |           <div style={{ font-size: '40px', margin-bottom: '8px' }} >0=0<
147 |           <h2 style={{ fontSize: '22px', font-weight: 700, color: 'var(--text-primary)', margin: '0 0 6px' }}>
148 |             0 c m i n g h Á n g h i Ç p c Ç a b i n l à g i ?
149 |           </h2>
150 |           <p style={{ color: 'var(--text-secondary)', font-size: '14px', margin: 0 }}>
151 |             C h i n g h Á h o c n h - p ° 0 c m i - t ò i s ½ b u i l d P C h o à n h ã o c h
152 |           </p>
153 |         </div>
154 |       </div>

```



```

155 |
156 |     { /* Quick suggestions */ }
157 |     <div style={{ display: 'flex', flexWrap: 'wrap', gap: '8px', justifyContent: 'center', marginBotto...
158 |         { ['K ù s ° A I', 'L - p tr ì n h v i ê n', 'T h i ế t k ế Ó h í a 3 D', 'E d
159 |         <button key={text} onClick={() => { setCustomDream(text); setSelected(null); }}
160 |             style={{
161 |                 padding: '8px 18px', borderRadius: '20px', border: '1px solid var(--border-default)',
162 |                 background: 'var(--bg-surface)', color: 'var(--text-secondary)', cursor: 'pointer',
163 |                 fontSize: '13px', fontWeight: 600, fontFamily: 'inherit',
164 |             }}
165 |             {text}
166 |         </button>
167 |     )}}
168 | </div>
169 |
170 |     { /* Career grid */ }
171 |     <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill, minmax(200px, 1fr))', gap: ...
172 |         { CAREERS.map((c) => (
173 |             <div key={c.id} onClick={() => { setSelected(c.id); setCustomDream(''); }}
174 |                 style={{
175 |                     padding: '16px', borderRadius: '12px', cursor: 'pointer',
176 |                     background: selected === c.id ? 'rgba(0,212,170,0.1)' : 'var(--bg-surface)',
177 |                     border: selected === c.id ? '2px solid #00d4aa' : '1px solid var(--border-default)',
178 |                     transition: 'all 0.2s',
179 |                 }}
180 |                 <div style={{ fontSize: '28px', marginBottom: '4px' }}>{c.icon}</div>
181 |                 <div style={{ fontSize: '14px', fontWeight: 700, color: 'var(--text-primary)' }}>{c.title}</...
182 |                 <div style={{ fontSize: '11px', color: 'var(--text-muted)', marginTop: '4px' }}>{c.desc}</div>
183 |             </div>
184 |         )}}
185 |     </div>
186 |
187 |     { /* Custom input */ }
188 |     <div style={{
189 |         padding: '20px', borderRadius: '14px',
190 |         background: 'var(--bg-surface)', border: '1px solid var(--border-default)',
191 |         marginBottom: '12px',
192 |     }}>
193 |         <div style={{ fontSize: '14px', fontWeight: 600, color: 'var(--text-primary)', marginBotto...
194 |             N h - p ° Û c m i c ç a b i n :
195 |         </div>
196 |         <div style={{ display: 'flex', gap: '8px', flexDirection: 'column' }}>
197 |             <div style={{ display: 'flex', gap: '8px' }}>
198 |                 <input
199 |                     value={customDream}
200 |                     onChange={(e) => { setCustomDream(e.target.value); setSelected(null); }}
201 |                     onKeyDown={(e) => { if (e.key === 'Enter') handleSuggest(); }}
202 |                     p l a c e h o l d e r = " V D : E m m u ã n t r ấ t h à n h k ù s ° A I . . . "
203 |                 style={{
204 |                     flex: 1, padding: '12px 16px', borderRadius: '10px', border: '1px solid var(--border-def...
205 |                     background: 'var(--bg-elevated)', color: 'var(--text-primary)', fontSize: '14px',
206 |                     fontFamily: 'inherit', outline: 'none',
207 |                 }}
208 |             </div>
209 |             <button onClick={() => handleSuggest()}
210 |                 disabled={!customDream && !selected} || loading
211 |                 style={{
212 |                     padding: '12px 28px', borderRadius: '10px', border: 'none',
213 |                     background: (customDream || selected) && !loading ? 'linear-gradient(135deg, #00d4aa, #0...
214 |                     color: (customDream || selected) && !loading ? '#fff' : 'var(--text-muted)',
215 |                     fontWeight: 700, fontSize: '14px', cursor: (customDream || selected) && !loading ? 'poin...
216 |                     fontFamily: 'inherit',
217 |                 }}
218 |             >
219 |                 {loading ? '...' : 'Build PC'}
220 |             </button>
221 |         </div>
222 |         <input
223 |             value={customCondition}
224 |             onChange={e => setCustomCondition(e.target.value)}
225 |             p l a c e h o l d e r = " T h ề m y ề u c ầ u . c b i ế t ( V D : n ằ n s ắ c h 2 0 t r
226 |         style={{
227 |             padding: '8px 12px', borderRadius: '8px', border: '1px solid var(--border-default)',
228 |             background: 'var(--bg-elevated)', color: 'var(--text-muted)', fontSize: '12px',
229 |             fontFamily: 'inherit', outline: 'none',
230 |         }}
231 |     </div>
232 | </div>
233 | </div>
234 |
235 | )}
236 |
237 | { /* Loading */ }
238 | { loading && (
239 |     <div style={{ textAlign: 'center', padding: '60px 20px' }}>
240 |         <div style={{
241 |             width: '48px', height: '48px', borderRadius: '50%',
242 |             border: '4px solid #ef4444', color: '#ef4444', borderTopColor: '#00d4aa',
243 |             margin: '0 auto 16px', animation: 'spin 0.8s linear infinite',
244 |         }} />
245 |         <div style={{ color: 'var(--text-muted)', fontSize: '14px' }}>
246 |     </div>
247 | )}
248 |
249 | {error && (
250 |     <div style={{
251 |         padding: '14px', borderRadius: '12px', background: 'rgba(239,68,68,0.1)',
252 |         border: '1px solid #ef4444', color: '#ef4444', marginBotto...
253 |         fontSize: '14px', fontWeight: 600, textAlign: 'center',
254 |     }}>
255 |         {error}
256 |     </div>
257 | )}
258 |
259 | { /* Result */ }
260 | {result && animPhase > 0 && (
261 |     <div ref={resultRef}>
262 |         { /* Career title */ }
263 |         <div style={{
264 |             textAlign: 'center', marginBotto...
265 |             animation: 'fadeSlideUp 0.5s ease-out',
266 |         }}>
267 |             <div style={{
268 |                 display: 'inline-block', padding: '8px 24px', borderRadius: '30px',
269 |                 background: 'linear-gradient(135deg, rgba(0,212,170,0.15), rgba(0,163,255,0.15))',
270 |                 border: '1px solid rgba(0,212,170,0.3)',
271 |                 fontSize: '20px', fontWeight: 800, color: '#fff',
272 |             }}>
273 |                 {result.career}
274 |             </div>
275 |         </div>
276 |
277 |         { /* Platform toggle */ }
278 |         {hasBoth && (

```

```

279|         <div style={{ display: 'flex', gap: '8px', marginBottom: '16px', justifyContent: 'center' }}>
280|             <button onClick={() => setShowMac(false)}
281|                 style={{
282|                     padding: '8px 24px', borderRadius: '20px',
283|                     border: '2px solid ${!showMac ? '#00d4aa' : 'var(--border-default)'}',
284|                     background: !showMac ? 'rgba(0,212,170,0.12)' : 'transparent',
285|                     color: !showMac ? '#00d4aa' : 'var(--text-muted)',
286|                     fontWeight: 700, fontSize: '14px', cursor: 'pointer', fontFamily: 'inherit', transition: '...
287|                 }}>
288|                 0>PŸ W i n d o w s P C
289|             </button>
290|             <button onClick={() => setShowMac(true)}
291|                 style={{
292|                     padding: '8px 24px', borderRadius: '20px',
293|                     border: '2px solid ${!showMac ? '#00d4aa' : 'var(--border-default)'}',
294|                     background: showMac ? 'rgba(0,212,170,0.12)' : 'transparent',
295|                     color: showMac ? '#00d4aa' : 'var(--text-muted)',
296|                     fontWeight: 700, fontSize: '14px', cursor: 'pointer', fontFamily: 'inherit', transition: '...
297|                 }}>
298|                 0<ŢN M a c
299|             </button>
300|         </div>
301|     })
302|
303|     /* Model name (for Mac) */
304|     {showMac && activeModel && (
305|         <div style={{
306|             textAlign: 'center', padding: '14px', marginBottom: '16px',
307|             borderRadius: '12px', background: 'rgba(0,0,0,0.3)', border: '1px solid var(--border-default)',
308|             animation: 'fadeSlideUp 0.4s ease-out',
309|         }}>
310|             <div style={{ fontSize: '11px', color: 'var(--text-muted)', fontWeight: 600, marginBottom: '4p...
311|             <div style={{ fontSize: '18px', fontWeight: 800, color: '#fff' }}>{activeModel}</div>
312|         </div>
313|     )}
314|
315|     /* Explanation */
316|     {animPhase >= 1 && (
317|         <div style={{
318|             padding: '14px 18px', borderRadius: '12px', marginBottom: '16px',
319|             background: 'rgba(0,212,170,0.06)', border: '1px solid rgba(0,212,170,0.15)',
320|             fontSize: '13px', lineHeight: '1.6', color: 'var(--text-secondary)',
321|             animation: 'fadeSlideUp 0.4s ease-out',
322|         }}>
323|             {activeExplanation}
324|         </div>
325|     )}
326|
327|     { / * C o m p o n e n t s l i s t - 2 c ũ t * / }
328|     <div style={{ display: 'grid', gridTemplateColumns: 'repeat(2, 1fr)', gap: '10px', marginBottom: '...
329|     {activeItems.map((item, i) => {
330|         const replaced = replacedItems.find(r => r.id === item.id)
331|         return (
332|             <div key={i} style={{
333|                 display: 'flex', flexDirection: 'column', gap: '8px',
334|                 padding: '14px', borderRadius: '12px',
335|                 background: 'var(--bg-surface)', border: substituting === item.id ? '1px solid var(--brand-p...
336|                 animation: animPhase >= 3 + 1 ? 'fadeSlideUp 0.35s ease-out' : 'none',
337|                 opacity: animPhase >= 3 + i ? 1 : 0,
338|                 transition: 'all 0.25s',
339|                 position: 'relative', overflow: 'hidden',
340|             }}>
341|                 <div style={{
342|                     position: 'absolute', top: 0, left: 0, width: '4px', height: '100%',
343|                     background: TYPE_COLORS[item.type] || '#64748b',
344|                 }} />
345|                 <div style={{ display: 'flex', alignItems: 'flex-start', gap: '10px' }}>
346|                     <div style={{
347|                         width: '36px', height: '36px', borderRadius: '10px', flexShrink: 0,
348|                         background: `${TYPE_COLORS[item.type]} || '#64748b'`20`,
349|                         display: 'flex', alignItems: 'center', justifyContent: 'center', fontSize: '18px',
350|                     }}>
351|                         { T Y P E _ I C O N S [ i t e m . t y p e ] || '0=Y9' }
352|                     </div>
353|                     <div style={{ flex: 1, minWidth: 0 }}>
354|                         <div style={{ fontWeight: 700, color: 'var(--text-primary)', fontSize: '13px', lineHeigh...
355|                             { r e p l a c e d ? < > < s p a n s t y l e = { { c o l o r : ' v a r ( - - b r a n d
356|                                 <span style={{ fontSize: '10px', color: TYPE_COLORS[item.type] || '#64748b', fontWeigh...
357|                                     {item.type}
358|                                 </span>
359|                             </div>
360|                             <div style={{ fontSize: '11px', color: 'var(--text-muted)', marginTop: '4px', lineHeight...
361|                                 {item.reason}
362|                             </div>
363|                         </div>
364|                     </div>
365|                     <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', margin...
366|                         {item.price > 0 && (
367|                             <div style={{ fontSize: '15px', fontWeight: 800, color: replaced ? 'var(--brand-primary)...
368|                                 {replaced ? `${formatPrice(replaced.price)}` : formatPrice(item.price)}
369|                             </div>
370|                         )}
371|                     <button onClick={() => substitutePart(item)}
372|                         style={{
373|                             padding: '5px 10px', borderRadius: '6px', border: '1px solid rgba(0,212,170,0.3)',
374|                             background: 'rgba(0,212,170,0.1)', color: 'var(--brand-primary)',
375|                             cursor: 'pointer', fontSize: '10px', fontWeight: 700, fontFamily: 'inherit',
376|                             transition: 'all 0.2s', whiteSpace: 'nowrap',
377|                         }}
378|                         onMouseOver={e => { e.currentTarget.style.background = 'rgba(0,212,170,0.2)'; e.currentT...
379|                         onMouseOut={e => { e.currentTarget.style.background = 'rgba(0,212,170,0.1)'; e.currentTa...
380|                             !" T h a y t h z
381|                     </button>
382|                 </div>
383|             </div>
384|         )}})
385|     </div>
386|
387|     /* Total + Tips + Build now */
388|     {animPhase >= 3 + activeItems.length - 1 && (
389|         <div style={{ animation: 'fadeSlideUp 0.4s ease-out' }}>
390|             /* Total */
391|             <div style={{
392|                 display: 'flex', justifyContent: 'space-between', alignItems: 'center',
393|                 padding: '14px 18px', borderRadius: '12px', marginBottom: '12px',
394|                 background: 'linear-gradient(135deg, rgba(0,212,170,0.1), rgba(0,163,255,0.08))',
395|                 border: '1px solid rgba(0,212,170,0.2)',
396|             }}>
397|                 <span style={{ fontSize: '14px', color: 'var(--text-secondary)', fontWeight: 600 }}>
398|                     T ō n g c h i p h i
399|                 </span>
400|                 <span style={{ fontSize: '22px', fontWeight: 800, color: '#00d4aa' }}>
401|                     {formatPrice(activeTotal)}
402|                 </span>

```

```

403 |         </div>
404 |
405 |         { /* Tips */ }
406 |         { activeTips && (
407 |           <div style={{
408 |             padding: '10px 14px', borderRadius: '10px', marginBottom: '16px',
409 |             background: 'rgba(250,204,21,0.06)', border: '1px solid rgba(250,204,21,0.15)',
410 |             fontSize: '12px', color: 'var(--text-secondary)', lineHeight: '1.6',
411 |           }}>
412 |             <strong style={{ color: '#facc15' }}>M10:</strong>
413 |           </div>
414 |         ) }
415 |
416 |         { /* Actions */ }
417 |         <div style={{ display: 'flex', gap: '10px', flexWrap: 'wrap' }}>
418 |           <button onClick={() => { setResult(null); setCustomDream(''); setAnimPhase(0); }}
419 |             style={{
420 |               flex: 1, padding: '12px 24px', borderRadius: '10px',
421 |               border: '1px solid var(--border-default)',
422 |               background: 'transparent', color: 'var(--text-secondary)',
423 |               fontSize: '14px', fontWeight: 600, cursor: 'pointer', fontFamily: 'inherit',
424 |             }}>
425 |             C h í n g h á k h á c
426 |           </button>
427 |         </div>
428 |       </div>
429 |     ) }
430 |   </div>
431 | }
432 | </div>
433 |
434 | <style>{
435 |   @keyframes spin { to { transform: rotate(360deg); } }
436 |   @keyframes fadeSlideUp {
437 |     from { opacity: 0; transform: translateY(15px); }
438 |     to { opacity: 1; transform: translateY(0); }
439 |   }
440 | </style>
441 | </div>
442 | );
443 | }
444 |

```

File: app/builder/common-mistakes/page.js (17 lines, 0.5 KB)

```

1 | 'use client';
2 |
3 | import dynamic from 'next/dynamic';
4 |
5 | const CommonMistakes = dynamic(
6 |   () => import('@components/CommonMistakes'),
7 |   { ssr: false, loading: () => <div style={{ padding: '40px', textAlign:
8 | });
9 |
10 | export default function CommonMistakesPage() {
11 |   return (
12 |     <div style={{ minHeight: '100vh', background: '#0d0e13', position: 'relative' }}>
13 |       <CommonMistakes lang="vn" onExit={() => window.history.back()} />
14 |     </div>
15 |   );
16 | }
17 |

```

File: app/builder/diagnosis/page.js (24 lines, 0.7 KB)

```

1 | 'use client';
2 |
3 | import dynamic from 'next/dynamic';
4 |
5 | const DiagnosisMode = dynamic(
6 |   () => import('@components/DiagnosisMode'),
7 |   { ssr: false, loading: () => <div style={{ padding: '40px', textAlign: 'center', color: 'var(--text-muted)' } ...
8 | });
9 |
10 | export default function DiagnosisPage() {
11 |   return (
12 |     <div style={{ minHeight: '100vh', background: 'var(--bg-base)', position: 'relative' }}>
13 |       <DiagnosisMode
14 |         lang="vn"
15 |         onComplete={(results) => {
16 |           console.log('Diagnosis complete:', results);
17 |           window.location.href = '/builder';
18 |         }}
19 |         onExit={() => window.history.back()}
20 |       />
21 |     </div>
22 |   );
23 | }
24 |

```

File: app/builder/invoice-check/page.tsx (290 lines, 15.4 KB)

```

1 | 'use client';
2 |
3 | import { useState } from 'react';
4 | import Link from 'next/link';
5 | import { getCategoryLinks, SHOPS } from '@data/shopLinks';
6 |
7 | interface ShopRef {
8 |   shop: string; url: string; price?: number;
9 | }
10 |
11 | interface CompareItem {
12 |   name: string; type: string; invoicePrice: number; marketPrice: number;
13 |   status: 'reasonable' | 'expensive' | 'overpriced'; percentDiff: number; dbPrice: number;
14 |   matchedName?: string; source?: string; link?: string; shops?: ShopRef[];
15 | }
16 |
17 | interface CompareResult {
18 |   items: CompareItem[]; totalInvoicePrice: number; totalMarketPrice: number; verdict: string;
19 | }
20 |
21 | const TYPE_OPTIONS = ['CPU', 'GPU', 'RAM', 'Mainboard', 'Storage', 'PSU', 'Cooler', 'Case', 'Monitor', 'Other'];
22 | const TYPE_ICONS: Record<string, string> = {
23 |   CPU: '&i', GPU: '0<@', RAM: '0=0%', Mainboard: '0=Y', Storage: '0=Ü', PSU:
24 | };
25 |
26 | export default function InvoiceComparePage() {
27 |   const [items, setItems] = useState([ { name: '', type: 'CPU', price: 0 } ]);
28 |   const [result, setResult] = useState<CompareResult | null>(null);

```

```

29 | const [loading, setLoading] = useState(false);
30 | const [error, setError] = useState('');
31 |
32 | const formatPrice = (v: number) => (v || 0).toLocaleString('vi-VN') +
33 |
34 | const statusColor = (s: string) => s === 'overpriced' ? '#ef4444' : s === 'expensive' ? '#f59e0b' : '#22c55e';
35 | const statusLabel = (s: string) => s === 'overpriced' ? 'Ch. t kém!' :
36 |
37 | const addItem = () => setItems([...items, { name: '', type: 'CPU', price: 0 }]);
38 | const removeItem = (i: number) => { if (items.length > 1) setItems(items.filter((_, j) => j !== i)); };
39 | const updateItem = (i: number, field: string, value: any) => {
40 |   const copy = [...items];
41 |   (copy[i] as any)[field] = value;
42 |   setItems(copy);
43 | };
44 |
45 | const handleCompare = async () => {
46 |   const valid = items.filter(i => i.name.trim() && i.price > 0);
47 |   if (valid.length === 0) { setError('Nhập ít nhất 1 linh kiện vui t
48 |   setLoading(true); setError(''); setResult(null);
49 |   try {
50 |     const res = await fetch('/api/ai/scan-invoice', {
51 |       method: 'POST',
52 |       headers: { 'Content-Type': 'application/json' },
53 |       body: JSON.stringify({ manualItems: valid.map(i => ({ name: i.name, type: i.type, invoicePrice: i.pric...
54 |     });
55 |     const data = await res.json();
56 |     if (data.error) throw new Error(data.error);
57 |     if (data.items) {
58 |       data.items = data.items.map((item: any) => ({
59 |         ...item,
60 |         shops: (getCategoryLinks(item.type) || []).map(link => ({
61 |           shop: link.shop,
62 |           url: link.url,
63 |         })),
64 |       }));
65 |     }
66 |     setResult(data);
67 |   } catch (err: any) {
68 |     setError(err.message || 'Lỗi so sánh giá');
69 |   } finally {
70 |     setLoading(false);
71 |   }
72 | };
73 |
74 | return (
75 |   <div style={{ minHeight: '100vh', background: 'var(--bg-base)', padding: '24px' }}>
76 |     <div style={{ maxWidth: '860px', margin: '0 auto' }}>
77 |       <Link href="/builder" style={{ display: 'inline-flex', alignItems: 'center', gap: '6px', color: 'var(--...
78 |       <span>&larr;</span> Builder
79 |     </Link>
80 |
81 |     <div style={{ display: 'flex', alignItems: 'center', gap: '12px', marginBottom: '6px' }}>
82 |       <span style={{ fontSize: '32px' }}>0</span>
83 |       <div>
84 |         <h1 style={{ fontSize: '24px', fontWeight: 800, color: 'var(--text-primary)', margin: 0 }}>So Sánh...
85 |         <p style={{ fontSize: '13px', color: 'var(--text-secondary)', margin: '4px 0 0' }}>
86 |           Nhậ tậ y lĩ nh kĩ cậ n + gĩ á - Nĩ chĩ u nĩ ấ u shọ p trẽn mĩ ng,
87 |         </p>
88 |       </div>
89 |     </div>
90 |
91 |     {/ Input form */}
92 |     <div style={{
93 |       background: 'var(--bg-surface)', borderRadius: '16px', padding: '20px',
94 |       border: '1px solid var(--border-default)', marginBottom: '16px',
95 |     }}>
96 |       <div style={{ fontSize: '14px', fontWeight: 700, color: 'var(--text-primary)', marginBottom: '14px',...
97 |       <span>0</span> Nhậ p lĩ nh kĩ cậ n tẽ hĩa ĩ n cặ a bĩ n
98 |     </div>
99 |
100 |     {items.map((item, i) => (
101 |       <div key={i} style={{
102 |         display: 'flex', gap: '8px', marginBottom: '10px', alignItems: 'center',
103 |       }}>
104 |         <input
105 |           placeholder="Tên linh kiện (VD: RTX 4070)"
106 |           value={item.name}
107 |           onChange={e => updateItem(i, 'name', e.target.value)}
108 |           style={{
109 |             flex: 2, minWidth: '160px', padding: '10px 12px', borderRadius: '8px',
110 |             border: '1px solid var(--border-default)', background: 'var(--bg-elevated)',
111 |             color: 'var(--text-primary)', fontSize: '13px', fontFamily: 'inherit', outline: 'none',
112 |           }}
113 |         />
114 |         <select
115 |           value={item.type}
116 |           onChange={e => updateItem(i, 'type', e.target.value)}
117 |           style={{
118 |             flex: 1, minWidth: '90px', padding: '10px 12px', borderRadius: '8px',
119 |             border: '1px solid var(--border-default)', background: 'var(--bg-elevated)',
120 |             color: 'var(--text-primary)', fontSize: '13px', fontFamily: 'inherit', outline: 'none',
121 |           }}
122 |         >
123 |           {TYPE_OPTIONS.map(t => <option key={t} value={t}>{TYPE_ICONS[t]} {t}</option>)}
124 |         </select>
125 |         <input
126 |           type="number" placeholder="Giá (VN)"
127 |           value={item.price || ''}
128 |           onChange={e => updateItem(i, 'price', parseInt(e.target.value) || 0)}
129 |           style={{
130 |             flex: 1, minWidth: '120px', padding: '10px 12px', borderRadius: '8px',
131 |             border: '1px solid var(--border-default)', background: 'var(--bg-elevated)',
132 |             color: 'var(--text-primary)', fontSize: '13px', fontFamily: 'inherit', outline: 'none',
133 |           }}
134 |         />
135 |         <button onClick={() => removeItem(i)}
136 |           style={{
137 |             padding: '10px', borderRadius: '8px', border: 'none',
138 |             background: 'rgba(239,68,68,0.15)', color: '#ef4444', cursor: 'pointer', fontSize: '16px', l...
139 |           }}>&times;</button>
140 |       </div>
141 |     ))}
142 |
143 |     <div style={{ display: 'flex', gap: '10px', marginTop: '12px' }}>
144 |       <button onClick={addItem}
145 |         style={{
146 |           padding: '10px 16px', borderRadius: '8px', border: '1px dashed var(--border-default)',
147 |           background: 'transparent', color: 'var(--text-muted)', cursor: 'pointer', fontSize: '13px',
148 |           fontFamily: 'inherit', fontWeight: 600,
149 |         }}>
150 |         + Thêm linh kiện
151 |       </button>
152 |       <button onClick={handleCompare} disabled={loading}

```

```

153|         style={{
154|             padding: '10px 28px', borderRadius: '8px', border: 'none',
155|             background: loading ? 'var(--bg-elevated)' : 'linear-gradient(135deg, #00d4aa, #00a3ff)',
156|             color: loading ? 'var(--text-muted)' : '#fff', fontWeight: 700, fontSize: '14px',
157|             cursor: loading ? 'default' : 'pointer', fontFamily: 'inherit',
158|         }}>
159|         { loading ? 'ang so sánh...' : 'So sánh giá thể tr° Ỗng' }
160|     </button>
161| </div>
162| </div>
163|
164| {error && (
165|     <div style={{ padding: '12px', borderRadius: '10px', background: 'rgba(239,68,68,0.1)', border: '1px...
166|     {error}
167| </div>
168| )}
169|
170| { /* Results */
171| {result && (
172|     <div style={{ animation: 'fadeUp 0.4s ease-out' }}>
173|     { /* Summary cards */
174|     <div style={{ display: 'grid', gridTemplateColumns: 'repeat(4, 1fr)', gap: '10px', marginBottom: '...
175|     [[
176|         {
177|             { label: 'TỔNG HÓA IN', value: formatPrice(result.total...
178|             { label: 'GIÁ THỂ TR° ỖNG', value: formatPrice(result.to...
179|             { label: 'CHÊNH LỖ CH', value: (result.totalInvoicePrice
180|         ].map((s, i) => (
181|             <div key={i} style={{ padding: '14px', borderRadius: '10px', background: 'var(--bg-surface)', ...
182|             <div style={{ fontSize: '10px', color: 'var(--text-muted)', fontWeight: 600, marginBottom: '...
183|             <div style={{ fontSize: '18px', fontWeight: 800, color: s.color }}>{s.value}</div>
184|         </div>
185|     ))}
186| </div>
187|
188| { /* Per-item comparison */
189| <div style={{ background: 'var(--bg-surface)', borderRadius: '14px', border: '1px solid var(--bord...
190| <div style={{ fontSize: '14px', fontWeight: 700, color: 'var(--text-primary)', marginBottom: '12...
191|         CHI TIẾT TĂNG LĨNH KIỆN
192|     </div>
193|     {result.items.map((item, i) => {
194|         const diff = (item.invoicePrice || 0) - (item.marketPrice || 0);
195|         const shopColor = item.source?.includes('GearVN') ? '#6366f1' : '#64748b';
196|         return (
197|             <div key={i} style={{
198|                 padding: '12px', borderRadius: '10px', marginBottom: '8px',
199|                 background: 'var(--bg-elevated)', border: '1px solid ${statusColor(item.status)}30',
200|                 animation: 'fadeUp 0.3s ease-out ${i * 0.1}s both',
201|             }}>
202|             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marg...
203|             <div>
204|                 <span style={{ fontWeight: 700, color: 'var(--text-primary)', fontSize: '14px' }}>{ite...
205|                 <span style={{ fontSize: '11px', color: 'var(--text-muted)', marginLeft: '8px' }}>{TYP...
206|             </div>
207|             <span style={{
208|                 padding: '2px 10px', borderRadius: '6px', fontSize: '10px', fontWeight: 700,
209|                 background: statusColor(item.status) + '20', color: statusColor(item.status),
210|             }}>
211|                 {statusLabel(item.status)}
212|             </span>
213|             </div>
214|             <div style={{ display: 'flex', gap: '16px', flexWrap: 'wrap', fontSize: '12px', marginBott...
215|                 <span>HÓA IN: <strong style={{ color: '#ef4444' }}>
216|                 <span>THỂ TR° ỖNG: <strong style={{ color: '#22c55e' }}>
217|                 <span style={{ fontWeight: 700, color: diff > 0 ? '#ef4444' : '#22c55e' }}>
218|                 {diff > 0 ? '+' : ''}{formatPrice(diff)} {(item.percentDiff > 0 ? '+' : '')}{item.perce...
219|                 </span>
220|             </div>
221|             {item.matchedName && item.matchedName !== item.name && (
222|                 <div style={{ fontSize: '11px', color: shopColor, fontWeight: 600, marginBottom: '4px' }}>
223|                     Ø=Y N i c h i ế u : { i t e m . m a t c h e d N a m e }
224|                 </div>
225|             )}
226|             {item.shops && item.shops.length > 0 ? (
227|                 <div style={{ display: 'flex', gap: '4px', flexWrap: 'wrap', marginTop: '4px' }}>
228|                     {item.shops.map((s, si) => (
229|                         <a key={si} href={s.url} target="_blank" rel="noopener noreferrer"
230|                         style={{ fontSize: '9px', fontWeight: 700, color: '#00d4aa', textDecoration: 'none', ...
231|                         background: 'rgba(0,212,170,0.1)', padding: '1px 6px', borderRadius: '4px' }}>
232|                             { s . s h o p } !-
233|                         </a>
234|                     ))}
235|                 </div>
236|             ) : item.source && (
237|                 <div style={{ fontSize: '10px', color: shopColor, marginBottom: '2px' }}>
238|                     Ø=U I { i t e m . s o u r c e }
239|                 </div>
240|             )}
241|             {item.link && !item.shops && (
242|                 <a href={item.link} target="_blank" rel="noopener noreferrer"
243|                 style={{ fontSize: '10px', color: '#00d4aa', fontWeight: 700, textDecoration: 'none', ...
244|                 Ø=Y M u a t i i G e a r V N !-
245|                 </a>
246|             )}
247|             <div style={{ height: '4px', borderRadius: '2px', background: 'var(--bg-base)', overflow: ...
248|             <div style={{ height: '100%', borderRadius: '2px', width: `${Math.min(Math.abs(item.perc...
249|             </div>
250|         </div>
251|     );
252|     )}
253| </div>
254|
255| { /* Verdict */
256| <div style={{
257|     padding: '16px', borderRadius: '12px', marginBottom: '16px',
258|     background: result.verdict.includes('C ỆNH B Á O') ? 'rgba(239,68,68,0.1)' : 'var(--bg-surface)',
259|     border: '1px solid ${result.verdict.includes('C ỆNH B Á O') ?
260| }}>
261|     <div style={{ display: 'flex', alignItems: 'flex-start', gap: '10px', fontSize: '14px', color: '...
262|     <span style={{ fontSize: '18px', flexShrink: 0, marginTop: '2px' }}>
263|         {result.verdict.includes('C ỆNH B Á O') ? 'Ø=b' : result.v
264|     </span>
265|     <div>
266|         {result.verdict}
267|     </div>
268|     </div>
269| </div>
270| </div>
271|
272| <button onClick={() => { setResult(null); }}
273|     style={{
274|         width: '100%', padding: '12px', borderRadius: '10px', border: '1px solid var(--border-default)',
275|         background: 'var(--bg-surface)', color: 'var(--text-primary)', fontSize: '14px', fontWeight: 600,
276|         cursor: 'pointer', fontFamily: 'inherit',

```

```
277|         }>>
278|         So sánh hóa in khác
279|     </button>
280| </div>
281| }
282| </div>
283|
284| <style>{
285|   @keyframes fadeUp { from { opacity: 0; transform: translateY(10px); } to { opacity: 1; transform: tran...
286| }</style>
287| </div>
288| );
289| }
290|
```

File: app/builder/mac-check/page.js (127 lines, 10.3 KB)

```
1| 'use client';
2|
3| import { useState } from 'react';
4| import Link from 'next/link';
5|
6| const ALL_COMPUTERS = [
7|   // ===== WINDOWS GAMING =====
8|   { id: 'pc-van-phong', name: 'PC Gaming Cao Cấp', brand: 'Windows', type: 'Gaming', specs: 'Ryzen 7 7800X3D, RTX 4070' },
9|   { id: 'pc-streamer', name: 'PC Streamer Pro', brand: 'Windows', type: 'Gaming', specs: 'Ryzen 7 7800X3D, RTX 4070' },
10|   { id: 'pc-mini-itx', name: 'PC Mini ITX Gaming', brand: 'Windows', type: 'Gaming', specs: 'Ryzen 5 7600X, RTX 4060' },
11|
12|   // ===== WINDOWS OFFICE / WORKSTATION =====
13|   { id: 'pc-van-phong', name: 'PC Văn Phòng Cao Cấp', brand: 'Windows', type: 'Workstation', specs: 'Intel i7-13700, 32GB RAM' },
14|   { id: 'pc-sinh-vien', name: 'PC Sinh Viên Tiệt Kiệm', brand: 'Windows', type: 'Workstation', specs: 'Intel i5-13400, 16GB RAM' },
15|   { id: 'pc-workstation-ai', name: 'PC Workstation AI', brand: 'Windows', type: 'Workstation', specs: 'Ryzen 9 7950X, RTX 6000' },
16|   { id: 'pc-lap-trinh', name: 'PC Lập Trình Viên', brand: 'Windows', type: 'Workstation', specs: 'Intel i9-13900, 64GB RAM' },
17|
18|   // ===== LAPTOP WINDOWS =====
19|   { id: 'rog-zephyrus', name: 'ASUS ROG Zephyrus G14', brand: 'Laptop Windows', type: 'Gaming', specs: 'Ryzen 7 7800U, RTX 4060' },
20|   { id: 'legion-pro', name: 'Lenovo Legion Pro 5', brand: 'Laptop Windows', type: 'Gaming', specs: 'Ryzen 7 7735, RTX 4070' },
21|
22|   // ===== MAC =====
23|   { id: 'macbook-pro-m4', name: 'MacBook Pro 14" M4 Pro', brand: 'Mac', type: 'Laptop', specs: 'M4 Pro (12/18) 16GB' },
24|   { id: 'macbook-air-m4', name: 'MacBook Air 13" M4', brand: 'Mac', type: 'Laptop', specs: 'M4 (8/10) 16GB, 256GB' },
25|   { id: 'mac-studio', name: 'Mac Studio M4 Max', brand: 'Mac', type: 'Desktop', specs: 'M4 Max (16/40) 64GB, 12TB' },
26|   { id: 'imac-m4', name: 'iMac 24" M4', brand: 'Mac', type: 'Desktop', specs: 'M4 (8/10) 16GB, 256GB, 24" 4.5K' },
27| ];
28|
29| 30| const formatPrice = (v) => v.toLocaleString('vi-VN') + '€';
30|
31| export default function MacCheckPage() {
32|   const [filter, setFilter] = useState('all');
33|   const [search, setSearch] = useState('');
34|
35|   const brands = ['all', ...new Set(ALL_COMPUTERS.map(c => c.brand))];
36|   const filtered = ALL_COMPUTERS.filter(c => (filter === 'all' || c.brand === filter) && (!search || c.name.toLowerCase().includes(search.toLowerCase()) || c.specs.toLowerCase().includes(search.toLowerCase())));
37|
38|   return (
39|     <div style={{ minHeight: '100vh', background: 'var(--bg-base)', padding: '24px' }}>
40|       <div style={{ maxWidth: '1000px', margin: '0 auto' }}>
41|         <Link href="/builder" style={{ display: 'inline-flex', alignItems: 'center', gap: '6px', color: 'var(--text-primary)' }}>
42|           <span>&laquo;</span> Builder
43|         </Link>
44|
45|         <div style={{ display: 'flex', alignItems: 'center', gap: '12px', marginBottom: '20px' }}>
46|           <span style={{ flex: 1, padding: '0 10px' }}>
47|             <input type="text" value={search} onChange={e => setSearch(e.target.value)} placeholder="Tìm máy..." style={{ width: 100%, border: 1px solid #ccc, padding: 5px 10px }} />
48|           </span>
49|           <div style={{ flex: 1, padding: '0 10px' }}>
50|             <select value={filter} onChange={e => setFilter(e.target.value)} style={{ width: 100%, border: 1px solid #ccc, padding: 5px 10px }}>
51|               {brands.map(b => <option value={b}>{b}</option>)}
52|             </select>
53|           </div>
54|         </div>
55|
56|         <div style={{ display: 'flex', gap: '10px', margin: '10px 0' }}>
57|           <div style={{ flex: 1, padding: '10px', border: 1px solid #ccc, border-radius: 10px, background: 'var(--bg-elevated)' }}>
58|             <h3 style={{ margin: 0, padding: 0 10px, color: 'var(--text-primary)' }}>Filters</h3>
59|             <div style={{ display: 'flex', gap: '4px', flexWrap: 'wrap' }}>
60|               {brands.map(b => (
61|                 <button key={b} onClick={() => setFilter(b)} style={{ padding: '8px 16px', border: 1px solid #ccc, border-radius: 8px, background: filter === b ? 'var(--bg-surface)' : 'var(--bg-elevated)', color: filter === b ? 'var(--text-secondary)' : 'var(--text-primary)', font-weight: 'normal', cursor: 'pointer', font-family: 'inherit' }}>{b}</button>
62|               ))}
63|             </div>
64|           </div>
65|           <div style={{ flex: 1, padding: '10px', border: 1px solid #ccc, border-radius: 10px, background: 'var(--bg-elevated)' }}>
66|             <h3 style={{ margin: 0, padding: 0 10px, color: 'var(--text-primary)' }}>Grid</h3>
67|             <div style={{ display: 'grid', grid-template-columns: repeat(2, 1fr), gap: '10px' }}>
68|               {filtered.map((pc, i) => (
69|                 <div key={pc.id} style={{ display: 'flex', flex-direction: 'column', align-items: 'center', padding: '10px', border: 1px solid #ccc, border-radius: 12px, background: 'var(--bg-surface)', animation: 'fadeUp 0.25s ease-out' }}>
70|                   <div style={{ display: 'flex', justify-content: 'space-between', width: 100% }}>
71|                     <div style={{ flex: 1, padding: '5px 10px' }}>
72|                       <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
73|                         <div style={{ flex: 1, padding: '5px 10px' }}>
74|                           <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
75|                             <div style={{ flex: 1, padding: '5px 10px' }}>
76|                               <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
77|                                 <div style={{ flex: 1, padding: '5px 10px' }}>
78|                                   <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
79|                                     <div style={{ flex: 1, padding: '5px 10px' }}>
80|                                       <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
81|                                         <div style={{ flex: 1, padding: '5px 10px' }}>
82|                                           <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
83|                                             <div style={{ flex: 1, padding: '5px 10px' }}>
84|                                               <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
85|                                                 <div style={{ flex: 1, padding: '5px 10px' }}>
86|                                                   <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
87|                                                     <div style={{ flex: 1, padding: '5px 10px' }}>
88|                                                       <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
89|                                                         <div style={{ flex: 1, padding: '5px 10px' }}>
90|                                                           <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
91|                                                             <div style={{ flex: 1, padding: '5px 10px' }}>
92|                                                               <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
93|                                                                 <div style={{ flex: 1, padding: '5px 10px' }}>
94|                                                                   <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
95|                                                                     <div style={{ flex: 1, padding: '5px 10px' }}>
96|                                                                       <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
97|                                                                         <div style={{ flex: 1, padding: '5px 10px' }}>
98|                                                                           <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
99|                                                                             <div style={{ flex: 1, padding: '5px 10px' }}>
100|                                                                               <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
101|                                                                                 <div style={{ flex: 1, padding: '5px 10px' }}>
102|                                                                                   <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
103|                                                                                     <div style={{ flex: 1, padding: '5px 10px' }}>
104|                                                                                       <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
105|                                                                                         <div style={{ flex: 1, padding: '5px 10px' }}>
106|                                                                                           <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>

```

```

107|         <a href={pc.link} target="_blank" rel="noopener noreferrer" style={{
108|             marginLeft: 'auto', padding: '6px 16px', borderRadius: '8px',
109|             background: 'linear-gradient(135deg, #00d4aa, #00a3ff)', color: '#fff',
110|             fontSize: '12px', fontWeight: 700, textDecoration: 'none',
111|             }} > M u a t i i G e a r V N !- < / a >
112|         </div>
113|     </div>
114| </div>
115| ))}
116| {filtered.length === 0 && (
117|     <div style={{ textAlign: 'center', padding: '40px', color: 'var(--text-muted)', fontSize: '14px' }}>
118|         K h ô n g t i m t h ấ y m á y p h ù h ấ p
119|     </div>
120| )}
121| </div>
122| </div>
123| <style>{@keyframes fadeUp { from { opacity: 0; transform: translateY(8px); } to { opacity: 1; transform...
124| </div>
125| });
126| }
127|

```

File: app/builder/os-install/page.js (22 lines, 0.5 KB)

```

1| 'use client';
2|
3| import dynamic from 'next/dynamic';
4|
5| const OSInstallationSimulator = dynamic(
6|     () => import('@components/OSInstallationSimulator'),
7|     { ssr: false, loading: () => <div style={{ padding: '40px', textAlign: 'center', color: 'var(--text-muted)' }} ...
8| );
9|
10| export default function OSInstallPage() {
11|     return (
12|         <OSInstallationSimulator
13|             lang="vn"
14|             build={{}}
15|             onComplete={() => {
16|                 window.location.href = '/builder';
17|             }}
18|             onExit={() => window.history.back()}
19|         />
20|     );
21| }
22|

```

File: app/builder/page.js (690 lines, 37.3 KB)

```

1| 'use client';
2|
3| import { useState, useCallback, useEffect, useRef } from 'react';
4| import Link from 'next/link';
5| import confetti from 'canvas-confetti';
6| import { supabase } from '@lib/supabase';
7| import { startBuilderSession, endBuilderSession } from '@lib/learning-actions';
8| import { useRealtime } from '@lib/realtime-provider';
9| import GameEngine from '../components/GameEngine';
10| import MultiplayerEngine from '../components/MultiplayerEngine';
11| import MainMenu from '../components/MainMenu';
12| import ComponentInfo from '../components/ComponentInfo';
13| import PartPickerSidebar from '../components/PartPickerSidebar';
14| import ComponentPreview from '../components/ComponentPreview';
15| import BurgerMenu from '../components/BurgerMenu';
16| import QuizModal from '../components/QuizModal';
17| import LearningMode from '../components/LearningMode';
18| import Marketplace from '../components/Marketplace';
19| import LectureCourse from '../components/LectureCourse';
20| import VirtualAssistant from '../components/VirtualAssistant';
21| import WindowsSimulator from '../components/WindowsSimulator';
22| import LoadingScreen from '../components/LoadingScreen';
23| import ExamsList from '../components/ExamsList';
24| import LoginModal from '../components/auth/LoginModal';
25| import StudentDashboardContent from '../components/StudentDashboardContent';
26| import { GURU_MESSAGES } from '../utils/i18nData';
27| import { useGuru } from '@lib/guru-state';
28| import { withTracking } from '@lib/tracking';
29| import BuilderLab from '@components/builder/BuilderLab';
30| import CollaborationStatus from '@components/builder/CollaborationStatus';
31| import VoiceController from '@components/VoiceController';
32| import HandTracker from '@components/HandTracker';
33| import {} from 'lucide-react';
34|
35| function Home(props) {
36|     const { state: realtimeState } = useRealtime();
37|     const [hoveredComponent, setHoveredComponent] = useState(null);
38|     const [lang, setLang] = useState(() => {
39|         if (typeof window !== 'undefined') {
40|             try { const saved = localStorage.getItem('lang'); if (saved === 'en' || saved === 'vn') return saved } c...
41|         }
42|         return 'vn';
43|     });
44|     const guru = useGuru();
45|     const [lastPlaced, setLastPlaced] = useState(null);
46|     const [showQuiz, setShowQuiz] = useState(false);
47|     const [quizParams, setQuizParams] = useState(null);
48|     const [appMode, setAppMode] = useState('menu');
49|     const [appView, setAppView] = useState('menu');
50|     const [missionData, setMissionData] = useState(null);
51|     const [placedItemsList, setPlacedItemsList] = useState([]);
52|     const [imageMode, setImageMode] = useState(false);
53|     const [lastVoiceResult, setLastVoiceResult] = useState('');
54|     const [showLoading, setShowLoading] = useState(true);
55|     const [isAIOpen, setIsAIOpen] = useState(false);
56|     const [showWindowsSim, setShowWindowsSim] = useState(false);
57|     const [theme, setTheme] = useState(() => {
58|         if (typeof window !== 'undefined') {
59|             try { return localStorage.getItem('theme') || 'dark' } catch(e) {}
60|         }
61|         return 'dark'
62|     });
63|
64|     const [landmarks, setLandmarks] = useState(null);
65|     const [cameraEnabled, setCameraEnabled] = useState(true);
66|     const landmarksRef = useRef(null);
67|     const landmarThrottleRef = useRef(0);
68|     const handleLandmarks = useCallback((lm) => {
69|         landmarksRef.current = lm;
70|         const now = Date.now();
71|         if (now - landmarThrottleRef.current < 60) return;
72|         landmarThrottleRef.current = now;

```

```

73 |     setLandmarks(lm);
74 | }, []);
75 | const [isDemo, setIsDemo] = useState(false);
76 | const [sessionId, setSessionId] = useState(null);
77 | const [showLoginModal, setShowLoginModal] = useState(false);
78 | const [loginRedirectTo, setLoginRedirectTo] = useState(null);
79 | const [showStudentDashboard, setShowStudentDashboard] = useState(false);
80 | const [userName, setUserName] = useState('');
81 | const [studentBudget, setStudentBudget] = useState(() => {
82 |   if (typeof window !== 'undefined') {
83 |     try { return parseInt(localStorage.getItem('studentBudget')) || 5000000 } catch(e) {}
84 |   }
85 |   return 5000000;
86 | });
87 |
88 | const gameEngineRef = useRef(null);
89 |
90 | const handleQuizScore = useCallback((score) => {
91 |   const earned = score * 100000;
92 |   setStudentBudget(prev => {
93 |     const newBudget = prev + earned;
94 |     try { localStorage.setItem('studentBudget', String(newBudget)) } catch(e) {}
95 |     return newBudget;
96 |   });
97 | }, []);
98 |
99 | const handlePartSelect = useCallback((type, modelId) => {
100 |   if (gameEngineRef.current && gameEngineRef.current.spawnComponent) {
101 |     const count = type === 'RAM' ? 2 : 1;
102 |     gameEngineRef.current.spawnComponent(type, count, modelId);
103 |   }
104 | }, []);
105 |
106 | const handleVoiceCommand = useCallback((action, params) => {
107 |   if (action === 'SPAWN' && params?.type) {
108 |     const ref = gameEngineRef.current;
109 |     if (!ref) {
110 |       setLastVoiceResult('L GameEngine ch^a sụn sàng');
111 |       return;
112 |     }
113 |     const ok = ref.placeComponent?.(params.type);
114 |     if (ok) {
115 |       setLastVoiceResult(' placeComponent($ {params.type}) thành công');
116 |       setPlacedItemsList(prev => [...prev, params.type]);
117 |     } else {
118 |       setLastVoiceResult(' placeComponent($ {params.type}) thất bại');
119 |       if (ref.spawnComponent) {
120 |         ref.spawnComponent(params.type, params.type === 'RAM' ? 2 : 1);
121 |       }
122 |     }
123 |   } else if (action === 'REMOVE' && params?.type) {
124 |     setLastVoiceResult('Yên REMOVE $ {params.type}');
125 |     setPlacedItemsList(prev => {
126 |       const idx = prev.indexOf(params.type);
127 |       if (idx === -1) return prev;
128 |       const copy = [...prev];
129 |       copy.splice(idx, 1);
130 |       return copy;
131 |     });
132 |   } else if (action === 'RESET') {
133 |     setLastVoiceResult('Ý RESET');
134 |     setPlacedItemsList([]);
135 |   } else if (action === 'CHECK_COMPATIBILITY') {
136 |     const totalTDP = { CPU: 125, GPU: 300, RAM: 10, SSD: 5, COOLER: 5, PSU: 0 };
137 |     const psuItems = placedItemsList.filter(t => t === 'PSU');
138 |     if (psuItems.length === 0) {
139 |       guru.setMessage(lang === 'vn' ? 'Ch^a có nguồn! Hãy l^p PSU tr^c');
140 |       return;
141 |     }
142 |     const tdp = placedItemsList.reduce((sum, t) => sum + (totalTDP[t] || 0), 0);
143 |     if (tdp > 500) {
144 |       guru.setMessage(lang === 'vn' ? `Cảnh báo: TDP ${tdp}W có thể quá tải`);
145 |     } else {
146 |       guru.setMessage(lang === 'vn' ? `TDP: ${tdp}W. Hệ thống đang tối ưu`);
147 |     }
148 |   } else if (action === 'BOOT_PC') {
149 |     setShowWindowsSim(true);
150 |   }
151 | }, [placedItemsList, lang, guru]);
152 |
153 | const placedCounts = {};
154 | placedItemsList.forEach(item => {
155 |   placedCounts[item] = (placedCounts[item] || 0) + 1;
156 | });
157 |
158 | useEffect(() => {
159 |   let currentSessionId = null;
160 |
161 |   async function initSession() {
162 |     try {
163 |       const { data: { user } } = await supabase.auth.getUser();
164 |       if (user) {
165 |         const { data: profile } = await supabase.from('profiles').select('full_name').eq('id', user.id).single();
166 |         if (profile?.full_name) setUserName(profile.full_name);
167 |       }
168 |       const id = await startBuilderSession();
169 |       if (id) {
170 |         setSessionId(id);
171 |         currentSessionId = id;
172 |       }
173 |     } catch (err) {
174 |       console.error("Failed to start builder session:", err);
175 |     }
176 |   }
177 |   initSession();
178 |
179 |   return () => {
180 |     if (currentSessionId) {
181 |       endBuilderSession(currentSessionId, {
182 |         components_used: [],
183 |         tdp_calculated: 0,
184 |         compatibility_score: 100
185 |       });
186 |     }
187 |   };
188 | }, []);
189 |
190 | useEffect(() => {
191 |   document.documentElement.setAttribute('data-theme', theme === 'editorial-light' ? 'light' : theme);
192 |   try { localStorage.setItem('theme', theme === 'editorial-light' ? 'light' : theme) } catch(e) {}
193 | }, [theme]);
194 |
195 | useEffect(() => {
196 |   if (typeof window !== 'undefined') {

```



```

197 |     const urlParams = new URLSearchParams(window.location.search);
198 |     if (urlParams.get('demo') === 'true') {
199 |         setIsDemo(true);
200 |     }
201 |     if (urlParams.get('requireAuth') === 'true') {
202 |         setShowLoginModal(true);
203 |         setLoginRedirectTo(urlParams.get('redirect') || null);
204 |         window.history.replaceState({}, '', window.location.pathname);
205 |     }
206 | }
207 | }, []);
208 |
209 | useEffect(() => {
210 |     setPlacedItemsList([]);
211 |     setLastPlaced(null);
212 | }, [appMode, appView]);
213 |
214 | const toggleLang = useCallback(() => {
215 |     setLang(prev => {
216 |         const newLang = prev === 'en' ? 'vn' : 'en';
217 |         try { localStorage.setItem('lang', newLang) } catch(e) {}
218 |         guru.sendMessage(GURU_MESSAGES[newLang].welcome);
219 |         return newLang;
220 |     });
221 | }, []);
222 |
223 | useEffect(() => {
224 |     if (appMode === 'menu') {
225 |         setAppView('menu');
226 |     } else {
227 |         setAppView('game');
228 |     }
229 | }, [appMode]);
230 |
231 | useEffect(() => {
232 |     if (!showLoading && typeof window !== 'undefined' && window.driver && window.driver.js && window.driver.js...
233 |     const isDone = localStorage.getItem('onboardingDone_v2');
234 |     if (!isDone) {
235 |         const driver = window.driver.js.driver;
236 |         const driverObj = driver({
237 |             showProgress: true,
238 |             steps: [
239 |                 { element: '.sidebar-container', popover: { title: 'Menu i A
240 |                 { element: '.mode-card', popover: { title: 'Chú Ú B à i G i Ē n g
241 |             ],
242 |             onDestroyStarted: () => {
243 |                 localStorage.setItem('onboardingDone_v2', 'true');
244 |                 driverObj.destroy();
245 |             }
246 |         });
247 |         driverObj.drive();
248 |     }
249 | }, [showLoading]);
250 |
251 | const handleHover = useCallback((componentType) => {
252 |     setHoveredComponent(componentType);
253 | }, []);
254 |
255 | const handleGameEvent = useCallback((event, type) => {
256 |     const messages = GURU_MESSAGES[lang];
257 |     if (event === 'grabbed') {
258 |         guru.setSilentMessage(messages.grabbed(type));
259 |     } else if (event === 'placed') {
260 |         setLastPlaced({ type, id: Date.now() });
261 |         setPlacedItemsList(prev => [...prev, type]);
262 |         guru.setSilentMessage(messages.placed(type));
263 |     } else if (event === 'COMPLETE') {
264 |         const saveSession = async () => {
265 |             const { data: { user } } = await supabase.auth.getUser();
266 |             if (user) {
267 |                 await supabase.from('assembly_sessions').insert({
268 |                     user_id: user.id,
269 |                     mode: appMode,
270 |                     completed: true,
271 |                     time_taken: 0
272 |                 });
273 |             }
274 |         };
275 |         saveSession();
276 |
277 |         if (appMode === 'mission_assembly' && missionData) {
278 |             setTimeout(() => setShowWindowsSim(true), 500);
279 |         } else if (appMode === 'assembly') {
280 |             setShowWindowsSim(true);
281 |         } else {
282 |             guru.sendMessage(lang === 'en' ? 'SYSTEM COMPLETE!' : 'HỆ THỐNG H
283 |         }
284 |     } else if (event.startsWith('grabbed_')) {
285 |         const player = event.split('_')[1].toUpperCase();
286 |         guru.setSilentMessage(`${player}: ${messages.grabbed(type)}`);
287 |     } else if (event.startsWith('placed_')) {
288 |         const player = event.split('_')[1].toUpperCase();
289 |         guru.setSilentMessage(`${player}: ${messages.placed(type)}`);
290 |     } else if (event.startsWith('win_')) {
291 |         const player = event.split('_')[1].toUpperCase();
292 |         guru.sendMessage(lang === 'en' ? `${player} WINS THE MATCH!` : `${p
293 |     }
294 | }, [lang, appMode, missionData]);
295 |
296 | const getModeTitle = () => {
297 |     switch(appMode) {
298 |         case 'assembly': return lang === 'en' ? 'Free Practice' : 'Luyện C
299 |         case 'learning': return lang === 'en' ? 'Practice Mode' : 'Luyện C
300 |         case 'course': return lang === 'en' ? 'Lecture Course' : 'Bài Gi
301 |         case 'market': return lang === 'en' ? 'Market place' : 'Chỗ M
302 |         case 'mission_assembly': return lang === 'en' ? `Lab: ${missionData?.missionId} : Phòng Lab: ${missi...
303 |         case 'multiplayer': return lang === 'en' ? '2 - Player Versus' : '
304 |         case 'exams': return lang === 'en' ? 'Exams' : 'Kó Thi';
305 |         case 'challenge': return lang === 'en' ? 'Daily Challenge' : 'Th
306 |         case 'components': return lang === 'en' ? 'Component Library' :
307 |         default: return 'PC Master Builder';
308 |     }
309 | };
310 |
311 | return (
312 |     <div style={{ display: 'flex', minHeight: '100vh', background: 'var(--bg-base)', position: 'relative' }}>
313 |         {showLoading && <LoadingScreen onComplete={() => setShowLoading(false)} />}
314 |
315 |         {isDemo && (
316 |             <div style={{
317 |                 position: 'fixed', top: 0, left: 0, right: 0, zIndex: 9999,
318 |                 background: 'linear-gradient(90deg, #6366f1, #00d2a0)',
319 |                 color: 'white', padding: '8px 16px', textAlign: 'center',
320 |

```

```

321|         fontSize: '14px', fontWeight: 600, display: 'flex',
322|         alignItems: 'center', justifyContent: 'center', gap: '12px'
323|     }}>
324|         <span>B i n a n g ß C h ế ù D e m o ( K h á c h ) . T i ế n ù h í c t - p s ½ k h ô n g
325|         <Link href="/register" style={{
326|             background: 'white', color: '#6366f1', padding: '2px 12px',
327|             borderRadius: '4px', textDecoration: 'none', fontSize: '12px'
328|         }}>
329|             n g k ý n g a y
330|         </Link>
331|     </div>
332| })
333| <BurgerMenu
334|     lang={lang}
335|     toggleLang={toggleLang}
336|     onStartQuiz={() => setShowQuiz(true)}
337|     appMode={appMode}
338|     setAppMode={setAppMode}
339|     onToggleAI={() => setIsAIOpen(prev => !prev)}
340|     isAIOpen={isAIOpen}
341|     theme={theme}
342|     setTheme={setTheme}
343|     userName={userName}
344|     onShowDashboard={() => setShowStudentDashboard(true)}
345| />
346|
347| <main className="main-content">
348|     {showQuiz && (
349|         <QuizModal
350|             lang={lang}
351|             topic={quizParams?.topic}
352|             level={quizParams?.level}
353|             onSuccess={quizParams?.onSuccess}
354|             onScore={handleQuizScore}
355|             onClose={() => {
356|                 setShowQuiz(false);
357|                 setQuizParams(null);
358|             }}
359|         />
360|     )}
361|
362|     {appView !== 'menu' && (
363|         <VirtualAssistant
364|             lang={lang}
365|             appMode={appMode}
366|             cartItems={missionData?.purchasedItems ?? []}
367|             remainingBudget={missionData?.remainingBudget}
368|             missionTitle={missionData?.missionId}
369|             isOpen={isAIOpen}
370|             setIsOpen={setIsAIOpen}
371|         />
372|     )}
373|
374|     {appView === 'menu' ? (
375|         <MainMenu
376|             lang={lang}
377|             onStart={() => setAppMode(mode)}
378|             onOpenLogin={() => setShowLoginModal(true)}
379|         />
380|     ) : (
381|         <div style={{
382|             display: 'flex', flexDirection: 'column', width: '100%',
383|             height: 'calc(100vh - 60px)',
384|             padding: ['course', 'market'].includes(appMode) ? '0' : '24px',
385|             overflow: 'hidden'
386|         }}>
387|             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBo...
388|                 <button
389|                     onClick={() => setAppMode('menu')}
390|                     style={{
391|                         padding: '8px 18px',
392|                         background: 'var(--bg-elevated)',
393|                         border: '1px solid var(--border-default)',
394|                         borderRadius: '8px',
395|                         color: 'var(--text-primary)',
396|                         fontSize: '13px',
397|                         fontWeight: 500,
398|                         cursor: 'pointer',
399|                         transition: 'all 0.2s',
400|                     }}
401|                     onMouseOver={e => { e.currentTarget.style.borderColor = 'var(--border-strong)'; e.currentT...
402|                     onMouseOut={e => { e.currentTarget.style.borderColor = 'var(--border-default)'; e.currentT...
403|                 >
404|                 !• { lang === 'en' ? 'Back' : 'Thoát' }
405|             </button>
406|             <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
407|                 <CollaborationStatus />
408|             </div>
409|         </div>
410|
411|         {/• Main View Port */}
412|         <div style={{ display: 'flex', flexDirection: 'column', width: '100%', flex: 1, alignItems: 'c...
413|             <div style={{ width: '100%', maxWidth: ['course', 'market'].includes(appMode) ? '100%' : '1...
414|                 {appMode === 'assembly' ? (
415|                     <div style={{ display: 'flex', gap: '16px', width: '100%', height: '100%' }}>
416|                         <PartPickerSidebar
417|                             lang={lang}
418|                             onSelect={handlePartSelect}
419|                             placedCounts={placedCounts}
420|                         />
421|                         <div style={{ flex: 1, position: 'relative', display: 'flex', gap: '12px', max...
422|                             <div style={{ flex: 1, position: 'relative' }}>
423|                                 <h2 style={{ color: 'var(--success)', marginTop: 0 }}>
424|                                     { lang === 'en' ? 'Free Practice' :
425|                                     </h2>
426|                                     <GameEngine
427|                                         ref={gameEngineRef}
428|                                         onHover={handleHover}
429|                                         onGameEvent={handleGameEvent}
430|                                         landmarks={landmarks}
431|                                         imageMode={imageMode}
432|                                     />
433|                                     {/• Checklist + Debug */}
434|                                     <div className="glass-panel" style={{
435|                                         position: 'absolute', top: '50px', right: '2%', padding: '1rem',
436|                                         minWidth: 'min(200px, 25vw)', pointerEvents: 'none', zIndex: 10
437|                                     }}>
438|                                         <h3 style={{ margin: '0 0 10px 0', fontSize: '13px', color: 'var(--text-pr...
439|                                             { lang === 'en' ? 'Checklist' :
440|                                         </h3>
441|                                         <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', fontSi...
442|                                             [
443|                                                 { type: 'CPU', target: 1 }, { type: 'COOLER', target: 1 },
444|                                                 { type: 'RAM', target: 2 }, { type: 'GPU', target: 1 },

```

```

445 |         { type: 'SSD', target: 1 }, { type: 'PSU', target: 1 }
446 |     ].map(comp => {
447 |         const count = placedItemsList.filter(item => item === comp.type).l...
448 |         const isDone = count >= comp.target;
449 |         return (
450 |             <div key={comp.type} style={{ color: isDone ? 'var(--success)'...
451 |                 <span>{isDone ? '&' : ''}</span>
452 |                 <span>{comp.type} ({Math.min(count, comp.target)}/{comp.ta...
453 |             </div>
454 |         );
455 |     });
456 | </div>
457 | <hr style={{ borderColor: 'rgba(255,255,255,0.1)', margin: '10px 0' }} />
458 | <div style={{ display: 'flex', flexDirection: 'column', gap: '6px', fontSi...
459 |     <div style={{ pointerEvents: 'auto', cursor: 'pointer', display: 'flex...
460 |         onClick={() => setImageMode(!imageMode)}>
461 |             <span style={{ opacity: imageMode ? 0.5 : 1 }}>
462 |                 <span>{imageMode ? '(lang' : ''}</span>
463 |             </div>
464 |             <div style={{ pointerEvents: 'auto', display: 'flex', gap: '4px', flex...
465 |                 <button onClick={() => {
466 |                     const ok = gameEngineRef.current?.placeComponent?.('CPU');
467 |                     setLastVoiceResult(ok ? 'CPU' : 'CPU');
468 |                 }} style={{ fontSize: '10px', padding: '2px 6px', cursor: 'pointer...
469 |                     Test CPU
470 |                 </button>
471 |                 <button onClick={() => {
472 |                     const ok = gameEngineRef.current?.placeComponent?.('SSD');
473 |                     setLastVoiceResult(ok ? 'SSD' : 'SSD');
474 |                 }} style={{ fontSize: '10px', padding: '2px 6px', cursor: 'pointer...
475 |                     Test SSD
476 |                 </button>
477 |             </div>
478 |         </div>
479 |         {lastVoiceResult && (
480 |             <div style={{ marginTop: '6px', padding: '4px 8px', borderRadius: '4px...
481 |                 {lastVoiceResult}
482 |             </div>
483 |         )
484 |     </div>
485 | </div>
486 | <!-- Camera preview on the right side -->
487 | <div style={{
488 |     width: '160px', flexShrink: 0, borderRadius: '12px', overflow: 'hidden',
489 |     border: '1px solid ${cameraEnabled ? 'rgba(0,212,170,0.3)' : 'rgba(255,80,80,0...
490 |     boxShadow: cameraEnabled ? '0 0 20px rgba(0,212,170,0.2)' : 'none',
491 |     cursor: 'pointer', transition: 'opacity 0.15s', height: 'fit-content',
492 |     opacity: cameraEnabled ? 1 : 0.5,
493 |     background: '#0f172a',
494 | }} onClick={() => setCameraEnabled(!cameraEnabled)}>
495 |     <div style={{ padding: '8px 10px', background: 'rgba(0,212,170,0.1)', borderBo...
496 |         <span style={{ fontSize: '10px', color: cameraEnabled ? 'var(--success)'...
497 |             {cameraEnabled ? 'B-t' : 'T-t'}
498 |         </span>
499 |     </div>
500 |     <div style={{ width: '100%', aspectRatio: '4/3', minHeight: '120px' }}>
501 |         {cameraEnabled && <HandTracker onLandmarks={handleLandmarks} numHands={1} />}
502 |         {!cameraEnabled && (
503 |             <div style={{ width: '100%', height: '100%', display: 'flex', alignIt...
504 |                 Camera T@T<br />N h¥n Ä b-t
505 |             </div>
506 |         )
507 |     </div>
508 | </div>
509 | </div>
510 | </div>
511 | </div>
512 | ) : appMode === 'market' ? (
513 |     <Marketplace
514 |         lang={lang}
515 |         budget={studentBudget}
516 |         onCheckout={(data) => {
517 |             setMissionData(data);
518 |             setStudentBudget(data.remainingBudget);
519 |             try { localStorage.setItem('studentBudget', String(data.remainingBudget)) ...
520 |             setAppMode('mission_assembly');
521 |         }}
522 |         onCancel={() => setAppMode('menu')}
523 |     />
524 | ) : appMode === 'mission_assembly' ? (
525 |     <div style={{ width: '100%', position: 'relative', display: 'flex', gap: '12px' }}>
526 |         <div style={{ flex: 1 }}>
527 |             <h2 style={{ color: 'var(--brand-primary)', marginTop: 0 }}>
528 |                 {lang === 'en' ? 'Lab: ${missionData?.missionId} || missionData?.scenarioNa...
529 |             </h2>
530 |             <GameEngine
531 |                 ref={gameEngineRef}
532 |                 onHover={handleHover}
533 |                 onGameEvent={handleGameEvent}
534 |                 landmarks={landmarks}
535 |                 purchasedItems={missionData?.purchasedItems}
536 |                 imageMode={imageMode}
537 |             />
538 |         </div>
539 |         <div style={{
540 |             width: '160px', flexShrink: 0, borderRadius: '12px', overflow: 'hidden',
541 |             border: '1px solid ${cameraEnabled ? 'rgba(0,212,170,0.3)' : 'rgba(255,80,...
542 |             boxShadow: cameraEnabled ? '0 0 20px rgba(0,212,170,0.2)' : 'none',
543 |             cursor: 'pointer', height: 'fit-content',
544 |             opacity: cameraEnabled ? 1 : 0.5,
545 |             background: '#0f172a',
546 |         }} onClick={() => setCameraEnabled(!cameraEnabled)}>
547 |             <div style={{ padding: '8px 10px', background: 'rgba(0,212,170,0.1)', bord...
548 |                 <span style={{ fontSize: '10px', color: cameraEnabled ? 'var(--success)...
549 |                     {cameraEnabled ? 'B-t' : 'T-t'}
550 |                 </span>
551 |             </div>
552 |             <div style={{ width: '100%', aspectRatio: '4/3', minHeight: '120px' }}>
553 |                 {cameraEnabled && <HandTracker onLandmarks={handleLandmarks} numHands=...
554 |                 {!cameraEnabled && (
555 |                     <div style={{ width: '100%', height: '100%', display: 'flex', alig...
556 |                         Camera T@T<br />N h¥n Ä
557 |                     </div>
558 |                 )
559 |             </div>
560 |         </div>
561 |     </div>
562 | </div>
563 | ) : appMode === 'course' ? (
564 |     <LectureCourse lang={lang} onBack={() => setAppMode('menu')} />
565 | ) : appMode === 'learning' ? (
566 |     <LearningMode
567 |         lang={lang}
568 |         externalSelection={lastPlaced}

```

```

569 |         appMode={appMode}
570 |         landmarks={landmarks}
571 |         cameraEnabled={cameraEnabled}
572 |         onLandmarks={handleLandmarks}
573 |         onHover={handleHover}
574 |         onGameEvent={handleGameEvent}
575 |         onTakeQuiz={({topic, level, onSuccess}) => {
576 |             setQuizParams({topic, level, onSuccess});
577 |             setShowQuiz(true);
578 |         }}
579 |     />
580 | ) : appMode === 'multiplayer' ? (
581 |     <div style={{ width: '100%', display: 'flex', gap: '12px' }}>
582 |         <div style={{ flex: 1 }}>
583 |             <MultiplayerEngine
584 |                 landmarks={landmarks}
585 |                 onGameEvent={handleGameEvent}
586 |                 lang={lang}
587 |             />
588 |         </div>
589 |         <div style={{
590 |             width: '200px', flexShrink: 0, borderRadius: '12px', overflow: 'hidden',
591 |             border: '1px solid ${cameraEnabled ? 'rgba(0,212,170,0.3)' : 'rgba(255,80,...
592 |             boxShadow: cameraEnabled ? '0 0 20px rgba(0,212,170,0.2)' : 'none',
593 |             cursor: 'pointer', height: 'fit-content',
594 |             opacity: cameraEnabled ? 1 : 0.5,
595 |             background: '#0f172a',
596 |         }} onClick={() => setCameraEnabled(!cameraEnabled)}>
597 |             <div style={{ padding: '8px 10px', background: 'rgba(0,212,170,0.1)', bord...
598 |                 <span style={{ padding: '0 10px' }}>
599 |                     <span style={{ fontSize: '10px', color: cameraEnabled ? 'var(--success...
600 |                         { cameraEnabled ? 'B-t' : 'T'
601 |                     }
602 |                 </span>
603 |             </div>
604 |             <div style={{ width: '100%', aspectRatio: '4/3', minHeight: '150px' }}>
605 |                 {cameraEnabled && <HandTracker onLandmarks={handleLandmarks} numHands=...
606 |                 {!cameraEnabled && (
607 |                     <div style={{ width: '100%', height: '100%', display: 'flex', alig...
608 |                         Camera T @ T <br /> Nh ñ n Ä
609 |                     )}
610 |                 </div>
611 |             </div>
612 |         </div>
613 |     ) : appMode === 'exams' ? (
614 |         <ExamsList lang={lang} onBack={() => setAppMode('menu')} />
615 |     ) : appMode === 'components' ? (
616 |         <div style={{ width: '100%', padding: '24px' }}>
617 |             <h2 style={{ color: 'var(--brand-primary)', marginTop: 0, marginBottom: 16 }}>
618 |                 { lang === 'en' ? 'Component Library'
619 |             </h2>
620 |             <BuilderLab />
621 |         </div>
622 |     ) : appMode === 'challenge' ? (
623 |         <div style={{ padding: '24px', maxWidth: '800px', margin: '0 auto', width: '100%' }}>
624 |             <h2 style={{ fontSize: '22px', fontWeight: 800, color: '#1A2F4A', marginBottom: ...
625 |             <p style={{ color: '#64748B', fontSize: '14px', margin: 0 }}>Hoàn t...
626 |             <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '12px' }}>
627 |                 {[
628 |                     { title: 'L - p r á p n h a n h', desc:
629 |                     { title: 'T i c t k i c m', desc: 'B u i
630 |                     { title: 'T o i n g t h i c h h o à n h ð o',
631 |                     { title: 'C ð u h i n h c â n b ã n g', desc:
632 |                 ].map((c, i) => (
633 |                     <div key={i} style={{ padding: '20px', borderRadius: '10px', border: '...
634 |                     <div style={{ fontSize: '11px', fontWeight: 700, color: c.color, t...
635 |                     <p style={{ fontSize: '13px', color: '#64748B', margin: 0 }}>
636 |                     <div style={{ display: 'flex', justifyContent: 'space-between', al...
637 |                         <span style={{ fontSize: '12px', fontWeight: 700, color: c.col...
638 |                         <button onClick={() => { setAppMode('learning'); }}
639 |                             style={{ padding: '6px 14px', borderRadius: '6px', border: ...
640 |                             Nh - n n h i c m v â
641 |                         </button>
642 |                     </div>
643 |                 </div>
644 |             )}
645 |         </div>
646 |     ) : null
647 | ) : </div>
648 |
649 | {appMode === 'assembly' && (
650 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '1rem', width: '100%', ma...
651 |     <ComponentInfo type={hoveredComponent} lang={lang} />
652 | )}
653 | </div>
654 | </div>
655 | </div>
656 |
657 | )}
658 |
659 | </main>
660 |
661 | {showWindowsSim && (
662 |     <WindowsSimulator
663 |         cart={missionData?.purchasedItems || placedItemsList.map(type => ({ type, name: type })))
664 |         scenarioName={missionData?.scenarioName}
665 |         onExit={() => { setShowWindowsSim(false); setAppMode('menu'); setPlacedItemsList([]); setMissionData...
666 |     />
667 | )}
668 |
669 | {[ 'assembly', 'mission_assembly' ].includes(appMode) && (
670 |     <VoiceController
671 |         lang={lang}
672 |         onCommand={handleVoiceCommand}
673 |     />
674 | )}
675 |
676 | <LoginModal isOpen={showLoginModal} onClose={() => { setShowLoginModal(false); setLoginRedirectTo(null); ...
677 |
678 | {showStudentDashboard && (
679 |     <StudentDashboardContent onClose={() => setShowStudentDashboard(false)} />
680 | )}
681 |
682 |
683 |
684 | <style jsx>{`}</style>
685 | </div>
686 | );
687 | }
688 |
689 | export default withTracking(Home);
690 |

```

```
1 'use client';
2
3 import { useState, useMemo } from 'react';
4 import dynamic from 'next/dynamic';
5 import { useBuildStore } from '@lib/build-store';
6 import PartPickerSidebarV2 from '@components/PartPickerSidebarV2';
7 import { getPriorityIssues } from '@lib/compatibility-engine';
8
9 const SketchfabViewer = dynamic(
10   () => import('@components/SketchfabViewer'),
11   { ssr: false, loading: () => (
12     <div style={{
13       width: '100%', height: '100%', display: 'flex', alignItems: 'center', justifyContent: 'center',
14       background: '#0a0a0f', borderRadius: 12, color: 'var(--text-muted)', fontSize: 13,
15     }} > a n g t e i m o h i n h S k e t c h f a b . . . < / d i v >
16   )});
17
18 1 9 | c o n s t P E R F _ L A B E L S = { g a m i n g : ' G a m i n g ' , p r o d u c t i v i t y : ' L à m v i c c ' , w o r k s t
19
20 const EFF_LABELS = {
21   '80plus_white': 'White', '80plus_bronze': 'Bronze', '80plus_silver': 'Silver',
22   '80plus_gold': 'Gold', '80plus_platinum': 'Platinum', '80plus_titanium': 'Titanium',
23 };
24
25 export default function PCBuilderPage() {
26   const store = useBuildStore();
27   const [showPicker, setShowPicker] = useState(true);
28   const [saving, setSaving] = useState(false);
29   const [savedId, setSavedId] = useState(null);
30
31   const parts = useMemo(() => ({
32     cpu: store.cpu, gpu: store.gpu, mainboard: store.mainboard,
33     ram: store.ram, psu: store.psu, case: store.case,
34     cooler: store.cooler, ssd: store.ssd,
35   }), [store.cpu, store.gpu, store.mainboard, store.ram, store.psu, store.case, store.cooler, store.ssd]);
36
37   const { errors, warnings } = useMemo(() => getPriorityIssues(store.issues), [store.issues]);
38
39   const hasAny = useMemo(() => Object.values(parts).some(v =>
40     v !== null && !Array.isArray(v) && v.length !== 0
41   ), [parts]);
42
43   const handleSave = async () => {
44     setSaving(true);
45     const id = await store.saveToSupabase();
46     if (id) setSavedId(id);
47     setSaving(false);
48     setTimeout(() => setSavedId(null), 3000);
49   };
50
51   return (
52     <div style={{
53       minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', flexDirection: 'column',
54       fontFamily: 'system-ui,sans-serif',
55     }} >
56       <div style={{
57         display: 'flex', alignItems: 'center', justify-content: 'space-between',
58         padding: '10px 20px', borderBottom: '1px solid var(--border-subtle)',
59         background: 'var(--bg-surface)',
60       }} >
61         <div style={{ display: 'flex', alignItems: 'center', gap: 10 }} >
62           <span style={{ font-weight: 700, font-size: 15, color: 'var(--text-primary)' }} > P C B u i l d e r 3 D < / s p a n >
63           <span style={{
64             background: store.score >= 80 ? 'rgba(0,212,170,0.12)' : store.score >= 50 ? 'rgba(255,163,0,0.12)...
65             color: store.score >= 80 ? '#00d4aa' : store.score >= 50 ? '#ffa300' : '#f46a6a',
66             padding: '2px 8px', borderRadius: 99, font-size: 10, font-weight: 700,
67           }} >
68             {store.score}/100
69           </span>
70         </div>
71         <div style={{ display: 'flex', gap: 8 }} >
72           <button onClick={() => setShowPicker(p => !p)}
73             style={{
74               padding: '6px 14px', borderRadius: 8, border: '1px solid var(--border-default)',
75               background: showPicker ? 'var(--accent-blue)' : 'var(--bg-elevated)',
76               color: showPicker ? '#fff' : 'var(--text-primary)',
77               cursor: 'pointer', font-size: 11, font-weight: 600, fontFamily: 'inherit',
78             }} >
79             { s h o w P i c k e r ? ' 0 = Ý ' L i n h k i c n ' : ' 0 = Ý ' C h í n ' }
80           </button>
81           <button onClick={handleSave} disabled={saving || !hasAny}
82             style={{
83               padding: '6px 14px', borderRadius: 8, border: '1px solid var(--brand-primary)', cursor: 'pointer',
84               background: savedId ? 'rgba(0,212,170,0.15)' : 'var(--brand-primary)',
85               color: savedId ? '#00d4aa' : '#fff',
86               font-size: 11, font-weight: 600, fontFamily: 'inherit',
87               opacity: (!hasAny || saving) ? 0.5 : 1,
88             }} >
89             { s a v i n g ? '#6 a n g l ° u . . . ' : s a v e d I d ? ' ' ä l ° u ' : ' 0 = 0 % L ° u E
90           </button>
91           <button onClick={() => window.history.back()}
92             style={{
93               padding: '6px 14px', borderRadius: 8, border: '1px solid var(--border-default)',
94               background: 'transparent', color: 'var(--text-muted)',
95               cursor: 'pointer', font-size: 11, font-weight: 600, fontFamily: 'inherit',
96             }} >
97             ' ó n g
98           </button>
99         </div>
100       </div>
101       <div style={{
102         flex: 1, display: 'flex', gap: 12, padding: 12, overflow: 'hidden',
103       }} >
104         <div style={{
105           flex: 1, min-width: 0 }} >
106           <div style={{ flexShrink: 0 }} >
107             <PartPickerSidebarV2 lang="vn" />
108           </div>
109           <div style={{ flex: 1 }} >
110             <SketchfabViewer parts={parts} />
111           </div>
112         </div>
113       </div>
114     </div>
115   );
116 }
117
```

```
123 |         width: 240, flexShrink: 0, display: 'flex', flexDirection: 'column', gap: 10,
124 |         maxHeight: 'calc(100vh - 100px)', overflowY: 'auto',
125 |     })>
126 |     { /* Issues */ }
127 |     <div style={{
128 |         background: 'var(--bg-surface)', borderRadius: 10, border: '1px solid var(--border-subtle)', paddi...
129 |     }}>
130 |         <div style={{ fontSize: 11, fontWeight: 700, color: 'var(--text-primary)', marginBottom: 8 }}>
131 |             & ð T ò i n g t h i c h
132 |         </div>
133 |         {errors.length === 0 && warnings.length === 0 ? (
134 |             <div style={{ color: '#00d4aa', fontSize: 11, fontWeight:
135 |             ) : (
136 |                 <div key={i} style={{ color: '#f46a6a', fontSize: 10,
137 |                 </div>
138 |             </div>
139 |             <div key={i} style={{ color: '#ffa300', fontSize: 10,
140 |             </div>
141 |             </div>
142 |         </div>
143 |         { /* Performance */ }
144 |         <div style={{
145 |             background: 'var(--bg-surface)', borderRadius: 10, border: '1px solid var(--border-subtle)', paddi...
146 |         }}>
147 |             <div style={{ fontSize: 11, fontWeight: 700, color: 'var(--text-primary)', marginBottom: 8 }}>
148 |                 ð = ð € H i Ç u n n g
149 |             </div>
150 |             {Object.entries(store.performance).map(([key, val]) => (
151 |                 <div key={key} style={{ marginBottom: 6 }}>
152 |                     <div style={{ display: 'flex', justifyContent: 'space-between', fontSize: 10, marginBottom: 2 }}>
153 |                         <span style={{ color: 'var(--text-muted)' }}>{PERF_LABELS[key]}</span>
154 |                         <span style={{ color: 'var(--text-primary)', fontWeight: 700 }}>{Math.round(val)}</span>
155 |                     </div>
156 |                     <div style={{
157 |                         height: 4, background: 'var(--bg-elevated)', borderRadius: 2, overflow: 'hidden',
158 |                     }}>
159 |                         <div style={{
160 |                             width: `${Math.min(100, val)}%`, height: '100%',
161 |                             background: val >= 80 ? '#00d4aa' : val >= 50 ? '#ffa300' : '#f46a6a',
162 |                             borderRadius: 2, transition: 'width 0.3s',
163 |                         }} />
164 |                     </div>
165 |                 </div>
166 |             ))}
167 |         </div>
168 |         { /* Specs Summary */ }
169 |         <div style={{
170 |             background: 'var(--bg-surface)', borderRadius: 10, border: '1px solid var(--border-subtle)', paddi...
171 |         }}>
172 |             <div style={{ fontSize: 11, fontWeight: 700, color: 'var(--text-primary)', marginBottom: 8 }}>
173 |                 ð = ð Ë T h ò n g s Ñ
174 |             </div>
175 |             {
176 |                 { label: 'TDP', value: store.totalTdp > `${store.totalTdp}W` : '-' },
177 |                 { label: 'T ò n g t i Ä n', value: store.totalPrice ? `${(store
178 |                 { label: 'Ngân sách', value: store.budget > `${(store.budget / 1e6).toFixed(1)}tr` : '-' },
179 |                 { label: 'S Ñ l i n h k i Ç n', value: Object.values(parts).filter
180 |                 ... (parts.psu > [{ label: 'PSU', value: `${parts.psu.psu_wattage}W ${EFF_LABELS[parts.psu.psu_ef...
181 |                 ... (parts.cpu > [{ label: 'CPU', value: `${parts.cpu.cpu_cores}C/${parts.cpu.cpu_threads}T` ] : ...
182 |                 ... (parts.gpu > [{ label: 'GPU', value: `${parts.gpu.gpu_vram_gb}GB` ] : []),
183 |                 ].map((row, i) => (
184 |                     <div key={i} style={{
185 |                         display: 'flex', justifyContent: 'space-between', fontSize: 10,
186 |                         padding: '2px 0', color: row.color || 'var(--text-primary)',
187 |                     }}>
188 |                         <span style={{ color: 'var(--text-muted)' }}>{row.label}</span>
189 |                         <span style={{ fontWeight: 600 }}>{row.value}</span>
190 |                     </div>
191 |                 ))}
192 |             </div>
193 |             { /* Quick Actions */ }
194 |             {hasAny && (
195 |                 <button onClick={store.clearAll}
196 |                     style={{
197 |                         padding: '8px', borderRadius: 8, border: '1px solid rgba(244,106,106,0.3)',
198 |                         background: 'rgba(244,106,106,0.06)', color: '#f46a6a',
199 |                         cursor: 'pointer', fontSize: 11, fontWeight: 600, fontFamily: 'inherit',
200 |                     }}>
201 |                 ð = ÿ Ñ p X o á t ã t c Ë
202 |             </button>
203 |         )}
204 |     </div>
205 | </div>
206 | );
207 | }
208 | }
209 | }
210 | }
211 | }
212 | }
213 | }
214 | }
215 | }
```

File: app/builder/pc-suggest/page.tsx (508 lines, 24.7 KB)

```
1 | 'use client';
2 |
3 | import { useState, useEffect, useMemo } from 'react';
4 | import Link from 'next/link';
5 | import { useIsMobile } from '@hooks/useIsMobile';
6 | import { motion, AnimatePresence } from 'framer-motion';
7 | import { BrainCircuit, Loader2, Store, ExternalLink, Sparkles, Monitor, ChevronDown, ChevronUp } from 'lucide-...
8 |
9 | const PREBUILT_PCS = [
10 |   {
11 |     id: 'rog-g14', name: 'ASUS ROG Zephyrus G14', type: 'Laptop Gaming',
12 |     specs: 'AMD Ryzen 9 7940HS, RTX 4060 8GB, 16GB DDR5, 1TB SSD, 14" QHD 165Hz',
13 |     price: 34990000, useCases: ['gaming', 'stream', 'coder', 'design'],
14 |     rating: 4.8, image: 'https://pollinations.ai/p/ASUS-ROG-Zephyrus-G14-gaming-laptop',
15 |     link: 'https://gearvn.com/collections/laptop-gaming',
16 |     reason: 'M ä y n h i g i n , m i n h m ½ , p h ù h ä p g a m i n g + l - p t r i n h . M à n 1 4 " Q
17 |   },
18 |   {
19 |     id: 'legion-pro-5', name: 'Lenovo Legion Pro 5', type: 'Laptop Gaming',
20 |     specs: 'AMD Ryzen 7 7745HX, RTX 4070 8GB, 32GB DDR5, 1TB SSD, 16" QHD 240Hz',
21 |     price: 39990000, useCases: ['gaming', 'stream', '3d', 'coder'],
22 |     rating: 4.7, image: 'https://pollinations.ai/p/legion-pro-5',
23 |     link: 'https://gearvn.com/collections/laptop-gaming',
24 |     reason: 'H i Ç u n n g x u ½ t s - c , t Ë n n h i Ç t t Ñ t , R A M 3 2 G B p h ù h ä p a n h i Ç
25 |   },
26 |   {
27 |     id: 'macbook-pro-m4', name: 'MacBook Pro 14" M4 Pro', type: 'Laptop',
```

```
28 | specs: 'Apple M4 Pro (12CPU/18GPU), 24GB Unified, 512GB SSD, 14" Liquid Retina XDR',
29 | price: 39990000, useCases: ['coder', 'design', 'video', 'office'],
30 | rating: 4.9, image: 'https://pollinations.ai/p/macbook-pro-m4',
31 | link: 'https://gearvn.com/collections/macbook-pro',
32 | reason: 'Chip Apple Silicon m4 mà, pin 18h, màn hình tuyế t 1p. L
33 | },
34 | },
35 | id: 'macbook-air-m4', name: 'MacBook Air 13" M4', type: 'Laptop',
36 | specs: 'Apple M4 (8CPU/10GPU), 16GB Unified, 256GB SSD, 13.6" Liquid Retina',
37 | price: 21990000, useCases: ['office', 'coder', 'student'],
38 | rating: 4.8, image: 'https://pollinations.ai/p/macbook-air-m4',
39 | link: 'https://gearvn.com/collections/macbook-air',
40 | reason: 'Siêu nh 1, pin c£ ngày, c minh cho v n phòng và l-p trình.
41 | },
42 | },
43 | id: 'pc-gaming-cao-cap', name: 'PC Gaming Cao C#p', type: 'Desktop
44 | specs: 'i9-14900K, RTX 4080 16GB, 32GB DDR5, 2TB NVMe Gen4, 1000W Platinum',
45 | price: 55990000, useCases: ['gaming', 'stream', '3d', 'video', 'ai'],
46 | rating: 4.7, image: 'https://pollinations.ai/p/pc-gaming-cao-cap',
47 | link: 'https://gearvn.com/collections/pc-gaming',
48 | reason: 'C#u hình minh nh#t cho gaming 4K + stream + render. Năng c#
49 | },
50 | },
51 | id: 'pc-gaming-tam-trung', name: 'PC Gaming T#m Trung', type: 'Deskto
52 | specs: 'Ryzen 7 7800X3D, RTX 4070 12GB, 32GB DDR5, 1TB NVMe, 850W Gold',
53 | price: 28990000, useCases: ['gaming', 'stream', 'coder', 'design'],
54 | rating: 4.6, image: 'https://pollinations.ai/p/pc-gaming-tam-trung',
55 | link: 'https://gearvn.com/collections/pc-gaming',
56 | reason: 'Giá h#p lý, chi 1440p m4t, cân Ñi hi#u n ng/giá thành.',
57 | },
58 | },
59 | id: 'pc-van-phong', name: 'PC V n Phòng Cao C#p', type: 'Desktop Offi
60 | specs: 'i5-14600K, GTX 1650 4GB, 16GB DDR5, 500GB NVMe, 500W',
61 | price: 12990000, useCases: ['office', 'coder', 'student'],
62 | rating: 4.5, image: 'https://pollinations.ai/p/pc-van-phong',
63 | link: 'https://gearvn.com/collections/pc-van-phong',
64 | reason: 'C#u hình Òn Ònh, a nhic#m t#t, ti#t ki#m i#n. Phù h#p v n
65 | },
66 | },
67 | id: 'pc-workstation-ai', name: 'PC Workstation AI', type: 'Desktop Workstation',
68 | specs: 'Ryzen 9 7950X, RTX 4090 24GB, 64GB DDR5, 2TB NVMe Gen5, 1200W Titanium',
69 | price: 78990000, useCases: ['ai', '3d', 'video', 'coder'],
70 | rating: 4.9, image: 'https://pollinations.ai/p/pc-workstation-ai',
71 | link: 'https://gearvn.com/collections/pc-workstation',
72 | reason: 'Trim làm vi#t AI/Deep Learning. GPU 24GB VRAM, RAM 64GB, CF
73 | },
74 | },
75 | id: 'pc-streamer-pro', name: 'PC Streamer Pro', type: 'Desktop Streaming',
76 | specs: 'Ryzen 7 7800X3D, RTX 4070 Super 12GB, 32GB DDR5, 1TB NVMe + 2TB HDD',
77 | price: 31990000, useCases: ['stream', 'gaming', 'video'],
78 | rating: 4.6, image: 'https://pollinations.ai/p/pc-streamer-pro',
79 | link: 'https://gearvn.com/collections/pc-streaming',
80 | reason: 'TÑi 0u cho stream: NVENC encoder, RAM c cho OBS + game, 2
81 | },
82 | },
83 | id: 'pc-sinh-vien', name: 'PC Sinh Viên Ti#t Ki#m', type: 'Desktop F
84 | specs: 'i3-14100F, GTX 1650 4GB, 16GB DDR4, 500GB NVMe, 500W',
85 | price: 8490000, useCases: ['office', 'student', 'coder'],
86 | rating: 4.4, image: 'https://pollinations.ai/p/pc-sinh-vien',
87 | link: 'https://gearvn.com/collections/pc-gia-re',
88 | reason: 'Gi#r nh#t, c h#c t-p, l-p trình c i b#n và gi#i trí nh 1.
89 | },
90 | },
91 | id: 'mac-studio-m4max', name: 'Mac Studio M4 Max', type: 'Desktop Mac',
92 | specs: 'Apple M4 Max (16CPU/40GPU), 64GB Unified, 1TB SSD',
93 | price: 64990000, useCases: ['video', '3d', 'ai', 'coder'],
94 | rating: 4.9, image: 'https://pollinations.ai/p/mac-studio-m4max',
95 | link: 'https://gearvn.com/collections/mac-studio',
96 | reason: 'Trim làm vi#t Mac minh nh#t. Render 8K m4t, 64GB RAM cho c
97 | },
98 | },
99 | id: 'pc-mini-itx', name: 'PC Mini ITX Gaming', type: 'Desktop Mini',
100 | specs: 'Ryzen 5 7600X, RTX 4060 8GB, 16GB DDR5, 1TB NVMe, 650W Gold SFX',
101 | price: 22990000, useCases: ['gaming', 'coder', 'office'],
102 | rating: 4.5, image: 'https://pollinations.ai/p/pc-mini-itx',
103 | link: 'https://gearvn.com/collections/pc-mini-itx',
104 | reason: 'Nh 1 g#n, 1p, ti#t ki#m không gian. V«n c minh cho gaming
105 | },
106 | ];
107 |
108 | const USE_CASE_LABELS: Record<string, string> = {
109 |   gaming: '0<B0 Ch i i game ',
110 |   stream: '0=0ú Stream ',
111 |   coder: '0=0ú L - p tr i nh ',
112 |   design: '0<B T h i #t k i ',
113 |   video: '0<B- D ñ ng ph i m ',
114 |   ai: '0>Ý A I / M L ',
115 |   office: '0=0ú V n ph ò ng ',
116 |   student: '0<B H i c s i nh ',
117 | };
118 |
119 | const formatPrice = (v: number) => v.toLocaleString('vi-VN') + '«';
120 |
121 | const containerVariants = {
122 |   hidden: { opacity: 0 },
123 |   visible: {
124 |     opacity: 1,
125 |     transition: { staggerChildren: 0.08 },
126 |   },
127 | };
128 |
129 | const cardVariants = {
130 |   hidden: { opacity: 0, y: 20 },
131 |   visible: { opacity: 1, y: 0, transition: { duration: 0.3 } },
132 | };
133 |
134 | export default function PcSuggestPage() {
135 |   const isMobile = useIsMobile();
136 |   const [purpose, setPurpose] = useState('');
137 |   const [selectedUseCases, setSelectedUseCases] = useState<string[]>([]);
138 |   const [aiResults, setAiResults] = useState<any[]>([]);
139 |   const [aiLoading, setAiLoading] = useState(false);
140 |   const [aiError, setAiError] = useState('');
141 |   const [showDetails, setShowDetails] = useState<Record<string, boolean>>({});
142 |   const [imgErrors, setImgErrors] = useState<Record<string, boolean>>({});
143 |   const [aiSource, setAiSource] = useState<'prebuilt' | 'ai'>('prebuilt');
144 |
145 |   useEffect(() => {
146 |     setAiResults(PREBUILT_PCS);
147 |   }, []);
148 |
149 |   const allUseCases = Object.keys(USE_CASE_LABELS);
150 |
151 |   const toggleUseCase = (uc: string) => {
```

```

152 |     setSelectedUseCases(prev =>
153 |       prev.includes(uc) ? prev.filter(u => u !== uc) : [...prev, uc]
154 |     );
155 |   });
156 |
157 |   const displayedPcs = useMemo(() => {
158 |     if (selectedUseCases.length === 0) return aiResults;
159 |     return aiResults.filter((pc: any) =>
160 |       selectedUseCases.some(uc => pc.useCases?.includes(uc))
161 |     ).sort((a: any, b: any) => {
162 |       const aScore = a.useCases?.filter((uc: string) => selectedUseCases.includes(uc)).length || 0;
163 |       const bScore = b.useCases?.filter((uc: string) => selectedUseCases.includes(uc)).length || 0;
164 |       return bScore - aScore || (a.price || 0) - (b.price || 0);
165 |     });
166 |   }, [selectedUseCases, aiResults]);
167 |
168 |   const handleAiSuggest = async () => {
169 |     if (aiLoading) return;
170 |     setAiLoading(true);
171 |     setAiError('');
172 |     try {
173 |       const res = await fetch('/api/pc-suggest', {
174 |         method: 'POST',
175 |         headers: { 'Content-Type': 'application/json' },
176 |         body: JSON.stringify({
177 |           purpose,
178 |           useCases: selectedUseCases,
179 |         }),
180 |       });
181 |       const data = await res.json();
182 |       if (!res.ok) throw new Error(data.error || 'Lỗi kết nối AI');
183 |       if (data.suggestions && data.suggestions.length > 0) {
184 |         setAiResults(data.suggestions);
185 |         setAiSource('ai');
186 |       } else {
187 |         throw new Error('AI không trả về kết quả. Hãy thử danh sách gợi ý khác');
188 |       }
189 |     } catch (err: any) {
190 |       setAiError(err.message || 'Không thể kết nối AI. ');
191 |     } finally {
192 |       setAiLoading(false);
193 |     }
194 |   });
195 |
196 |   const handleImgError = (id: string) => {
197 |     setImgErrors(prev => ({ ...prev, [id]: true }));
198 |   };
199 |
200 |   const toggleDetails = (id: string) => {
201 |     setShowDetails(prev => ({ ...prev, [id]: !prev[id] }));
202 |   };
203 |
204 |   const renderImage = (pc: any) => {
205 |     const src = pc.image_url || pc.image;
206 |     const id = pc.id || 'unknown';
207 |     if (imgErrors[id]) {
208 |       return (
209 |         <div style={{
210 |           width: '100%', height: '100%', display: 'flex', alignItems: 'center', justifyContent: 'center',
211 |           background: 'var(--bg-elevated)',
212 |         }}>
213 |           <Monitor size={28} color="#6366f1" />
214 |         </div>
215 |       );
216 |     }
217 |     return (
218 |       <img src={src} alt={pc.name}
219 |         style={{ width: '100%', height: '100%', objectFit: 'cover' }}
220 |         onError={() => handleImgError(id)}
221 |       />
222 |     );
223 |   };
224 |
225 |   return (
226 |     <div style={{ minHeight: '100vh', background: 'var(--bg-base)', padding: isMobile ? '16px' : '24px' }}>
227 |       <div style={{ maxWidth: '960px', margin: '0 auto' }}>
228 |         <Link href="/builder" style={{ display: 'inline-flex', alignItems: 'center', gap: '6px', color: 'var(--text-secondary)' }}>
229 |           <span>&larr;</span> Builder
230 |         </Link>
231 |
232 |         <div style={{ display: 'flex', alignItems: 'center', gap: '12px', margin: '12px 0 24px 0' }}>
233 |           <span style={{ fontSize: '36px' }}>0</span>
234 |           <div>
235 |             <h1 style={{ fontSize: '26px', fontWeight: 800, color: 'var(--text-primary)' }}>
236 |               <p style={{ fontSize: '13px', color: 'var(--text-secondary)', margin: '4px 0' }}>
237 |                 Chọn nhu cầu bên dưới hoặc nhập mô tả AI gợi ý máy phù hợp
238 |               </p>
239 |             </div>
240 |           </div>
241 |
242 |           <div style={{
243 |             background: 'var(--bg-surface)', borderRadius: '16px', padding: '24px',
244 |             border: '1px solid var(--border-default)', margin: '12px 0 20px 0',
245 |           }}>
246 |             <div style={{ display: 'flex', gap: '6px', flexWrap: 'wrap', margin: '14px 0' }}>
247 |               {allUseCases.map(uc => (
248 |                 <motion.button
249 |                   key={uc}
250 |                   whileHover={{ scale: 1.05 }}
251 |                   whileTap={{ scale: 0.95 }}
252 |                   onClick={() => toggleUseCase(uc)}
253 |                   style={{
254 |                     padding: '8px 16px', borderRadius: '10px',
255 |                     border: '1.5px solid ${selectedUseCases.includes(uc) ? '#00d4aa' : 'var(--border-default)'}',
256 |                     background: selectedUseCases.includes(uc) ? 'rgb(0,212,170,0.15)' : 'var(--bg-elevated)',
257 |                     color: selectedUseCases.includes(uc) ? '#00d4aa' : 'var(--text-secondary)',
258 |                     fontSize: '13px', fontWeight: 600, cursor: 'pointer', fontFamily: 'inherit',
259 |                     transition: 'all 0.2s',
260 |                   }}
261 |                 <USE_CASE_LABELS[uc]>
262 |               </motion.button>
263 |             )
264 |             </div>
265 |
266 |             <div style={{ display: 'flex', flexDirection: isMobile ? 'column' : 'row', gap: '10px', margin: '10px 0 10px 0' }}>
267 |               <input
268 |                 value={purpose}
269 |                 onChange={e => setPurpose(e.target.value)}
270 |                 onKeyDown={e => { if (e.key === 'Enter') handleAiSuggest(); }}
271 |                 placeholder="Hãy mô tả chi tiết (VD: Tôi cần máy chơi game"
272 |               />
273 |               <div style={{
274 |                 flex: 1, padding: '12px 16px', borderRadius: '10px', border: '1px solid var(--border-default)',
275 |                 background: 'var(--bg-elevated)', color: 'var(--text-primary)', fontSize: '14px',
276 |                 fontFamily: 'inherit', outline: 'none',

```



```
276|     }}
277|   />
278| </div>
279|
280|   <button onClick={handleAiSuggest} disabled={aiLoading}
281|     style={{
282|       width: '100%', padding: '12px', borderRadius: '10px', border: 'none',
283|       background: aiLoading ? 'var(--border-default)' : 'linear-gradient(135deg, #7c3aed, #2563eb)',
284|       color: aiLoading ? 'var(--text-muted)' : '#fff',
285|       fontSize: '15px', fontWeight: 700, cursor: aiLoading ? 'not-allowed' : 'pointer',
286|       fontFamily: 'inherit', display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '8px',
287|     }}
288|     {aiLoading ? <Loader2 size={18} className="spin" /> : <BrainCircuit size={18} />}
289|     { aiLoading ? 'AI đang phân tích...' : 'Gửi ý kiến AI' }
290|   </button>
291| </div>
292|
293|   <AnimatePresence mode="wait">
294|     {aiLoading && (
295|       <motion.div initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }} exit={{ opacity: 0, y: ...
296|         style={{ textAlign: 'center', padding: '40px', color: 'var(--text-muted)', fontSize: '14px' }}>
297|         <motion.div animate={{ rotate: 360 }} transition={{ repeat: Infinity, duration: 2, ease: 'linear...
298|           style={{ display: 'inline-block', marginBottom: '12px' }}>
299|             <BrainCircuit size={40} color="#7c3aed" />
300|           </motion.div>
301|           <div style={{ display: 'flex', align-items: 'center', gap: '10px' }}>
302|             <div>
303|               </motion.div>
304|             )}
305|           </AnimatePresence>
306|
307|           {aiError && (
308|             <motion.div initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }}
309|               style={{ padding: '14px', borderRadius: '10px', background: 'rgba(239,68,68,0.1)', border: '1px so...
310|               & p { aiError }
311|             </motion.div>
312|           )}
313|
314|           {displayedPcs.length > 0 && (
315|             <motion.div
316|               key={aiSource}
317|               initial={{ opacity: 0 }}
318|               animate={{ opacity: 1 }}
319|               style={{ marginTop: '24px' }}
320|             >
321|               <div style={{ display: 'flex', align-items: 'center', gap: '8px', marginBottom: '16px', justifyCont...
322|                 <div style={{ display: 'flex', align-items: 'center', gap: '8px' }}>
323|                   <Sparkles size={20} color="#7c3aed" />
324|                   <h2 style={{ fontSize: '18px', fontWeight: 800, color: 'var(--text-primary)', margin: 0 }}>
325|                     { aiSource === 'prebuilt' ? 'Ý kiến Máy Tính' : 'Ý kiến' }
326|                   </h2>
327|                   <span style={{ fontSize: '11px', color: 'var(--text-muted)' }}>
328|                     { aiSource === 'prebuilt' ? '$ {displayedPcs.length} m' : '' }
329|                   </span>
330|                 </div>
331|                 {aiSource === 'ai' && (
332|                   <button onClick={() => { setAiResults(PREBUILT_PCS); setAiSource('prebuilt'); setAiError(''); }}
333|                     style={{
334|                       display: 'inline-flex', align-items: 'center', gap: '6px',
335|                       padding: '8px 16px', borderRadius: '8px', border: '1px solid var(--border-default)',
336|                       background: 'var(--bg-surface)', color: 'var(--text-primary)',
337|                       fontSize: '12px', fontWeight: 600, cursor: 'pointer', fontFamily: 'inherit',
338|                     }}>
339|                       <Monitor size={14} /> Xem ý kiến m.c
340|                     </button>
341|                   )}
342|                 </div>
343|
344|               <motion.div
345|                 variants={containerVariants}
346|                 initial="hidden"
347|                 animate="visible"
348|                 style={{ display: 'grid', gap: '16px' }}
349|               >
350|                 {displayedPcs.map((pc: any, i: number) => (
351|                   <motion.div
352|                     key={pc.id || i}
353|                     variants={cardVariants}
354|                     whileHover={{ y: -4, boxShadow: '0 8px 30px rgba(0,0,0,0.12)' }}
355|                     style={{
356|                       padding: '18px', borderRadius: '14px',
357|                       background: 'var(--bg-surface)',
358|                       border: aiSource === 'ai'
359|                         ? '1px solid rgba(124,58,237,0.2)'
360|                         : '1px solid var(--border-default)',
361|                       transition: 'all 0.2s',
362|                     }}
363|                   >
364|                     <div style={{ display: 'flex', gap: '16px' }}>
365|                       <div style={{
366|                         width: '80px', height: '80px', borderRadius: '10px', flexShrink: 0,
367|                         overflow: 'hidden', background: '#1a1a2e',
368|                       }}>
369|                         {renderImage(pc)}
370|                       </div>
371|                       <div style={{ flex: 1, minWidth: 0 }}>
372|                         <div style={{ display: 'flex', align-items: 'center', gap: '8px', flexWrap: 'wrap', margi...
373|                           <span style={{ fontSize: '15px', fontWeight: 700, color: 'var(--text-primary)' }}>{pc....
374|                           <span style={{
375|                             fontSize: '10px', padding: '2px 8px', borderRadius: '99px',
376|                             background: aiSource === 'ai' ? 'rgba(124,58,237,0.12)' : 'rgba(0,212,170,0.12)',
377|                             color: aiSource === 'ai' ? '#7c3aed' : '#00d4aa',
378|                             fontWeight: 700,
379|                           }}>
380|                             {pc.type}
381|                           </span>
382|                           {pc.rating && (
383|                             <span style={{ fontSize: '11px', color: '#f59e00' }}>
384|                               </div>
385|
386|                             <p style={{ fontSize: '11px', color: 'var(--text-secondary)', margin: '0 0 6px', lineHei...
387|                               {pc.specs}
388|                             </p>
389|
390|                             {pc.specs_detail && (
391|                               <div style={{ marginBottom: '4px' }}>
392|                                 <button
393|                                   onClick={() => toggleDetails(pc.id || String(i))}
394|                                   style={{
395|                                     display: 'inline-flex', align-items: 'center', gap: '4px',
396|                                     padding: '2px 8px', borderRadius: '4px', border: 'none',
397|                                     background: 'transparent', color: 'var(--text-muted)',
398|                                     fontSize: '10px', cursor: 'pointer', fontFamily: 'inherit',
399|                                   }}>
```

```

400 |         {showDetails[pc.id || String(i)] ? <ChevronUp size={12} /> : <ChevronDown size={12...
401 |         { showDetails[pc.id || String(i)] ? 'Thu gín
402 |     </button>
403 |
404 |     {showDetails[pc.id || String(i)] && (
405 |         <motion.div
406 |             initial={{ opacity: 0, height: 0 }}
407 |             animate={{ opacity: 1, height: 'auto' }}
408 |             style={{
409 |                 marginTop: '6px', padding: '8px', borderRadius: '8px',
410 |                 background: 'var(--bg-elevated)', fontSize: '11px',
411 |             }}
412 |             {Object.entries(pc.specs_detail).map(([key, val]) => (
413 |                 <div key={key} style={{ display: 'flex', gap: '8px', marginBottom: '2px' }}>
414 |                     <span style={{ color: 'var(--text-muted)', minWidth: '50px', textTransform: ...
415 |                     <span style={{ color: 'var(--text-primary)', fontWeight: 600 }}>{String(val)...
416 |                 </div>
417 |             ))}
418 |         </motion.div>
419 |     )}
420 | </div>
421 | )}
422 |
423 | <p style={{
424 |     fontSize: '11px',
425 |     color: aiSource === 'ai' ? '#7c3aed' : '#6366f1',
426 |     margin: '4px 0 8px', lineHeight: '1.4',
427 | }}>
428 |         0=0i { pc . reason }
429 | </p>
430 |
431 | <div style={{ display: 'flex', alignItems: 'center', gap: '10px', flexWrap: 'wrap', marg...
432 |     <span style={{ fontSize: '17px', fontWeight: 800, color: '#00d4aa' }}>
433 |         {formatPrice(pc.price)}
434 |     </span>
435 |     {pc.average_price && (
436 |         <span style={{ fontSize: '12px', fontWeight: 700, color: '#10b981', background: 'rgb...
437 |         # Giá TB : { formatPrice ( pc . average _ price ) }
438 |     </span>
439 |     )}
440 | </div>
441 |
442 | {(pc.price_breakdown || []).length > 0 ? pc.price_breakdown : (pc.stores || []).length...
443 | <div>
444 |     <div style={{ display: 'flex', alignItems: 'center', gap: '4px', marginBottom: '6px'...
445 |         <Store size={12} /> So sánh giá:
446 |     </div>
447 |     <div style={{ display: 'flex', gap: '6px', flexWrap: 'wrap' }}>
448 |         {(pc.price_breakdown || pc.stores || []).map(store: any, si: number) => (
449 |             <a key={si} href={store.url} target="_blank" rel="noopener noreferrer"
450 |                 style={{
451 |                     display: 'inline-flex', alignItems: 'center', gap: '4px',
452 |                     padding: '4px 10px', borderRadius: '6px',
453 |                     background: 'var(--bg-elevated)', border: '1px solid var(--border-default)',
454 |                     color: 'var(--text-primary)', fontSize: '10px', fontWeight: 600,
455 |                     textDecoration: 'none',
456 |                 }}>
457 |                 {store.name || store.store}
458 |                 <span style={{ color: '#00d4aa' }}>{formatPrice(store.price || 0)}</span>
459 |                 <ExternalLink size={10} style={{ opacity: 0.5 }} />
460 |             </a>
461 |         )}
462 |     </div>
463 |     <div style={{ marginTop: '6px', fontSize: '10px', color: 'var(--text-muted)' }}>
464 |         0=0° Giá t Ñ t nh Ẳ t : <span style={ { color : '#10b
465 |         {formatPrice(Math.min(...(pc.price_breakdown || pc.stores || []).map(s: any) =>...
466 |     </span>
467 |     </div>
468 | </div>
469 | )}
470 |
471 | <div style={{ display: 'flex', gap: '8px', flexWrap: 'wrap', marginTop: '8px', alignItem...
472 |     {pc.link && (
473 |         <a href={pc.link} target="_blank" rel="noopener noreferrer"
474 |             style={{
475 |                 padding: '6px 16px', borderRadius: '8px',
476 |                 background: 'linear-gradient(135deg, #00d4aa, #00a3ff)',
477 |                 color: '#fff', fontSize: '12px', fontWeight: 700, textDecoration: 'none',
478 |             }}>
479 |             Mua ngay t i Gear VN !-
480 |         </a>
481 |     )}
482 |     {pc.useCases && pc.useCases.length > 0 && (
483 |         <div style={{ display: 'flex', gap: '4px', flexWrap: 'wrap' }}>
484 |             {pc.useCases.map((uc: string) => (
485 |                 <span key={uc} style={{ fontSize: '9px', color: 'var(--text-muted)' }}>
486 |                     {USE_CASE_LABELS[uc] || uc}
487 |                 </span>
488 |             )}
489 |         </div>
490 |     )}
491 | </div>
492 | </div>
493 | </div>
494 | </motion.div>
495 | )}
496 | </motion.div>
497 | </motion.div>
498 | )}
499 | </div>
500 |
501 | <style>{`
502 |     @keyframes spin { from { transform: rotate(0deg); } to { transform: rotate(360deg); } }
503 |     .spin { animation: spin 1s linear infinite; }
504 | `}</style>
505 | </div>
506 | );
507 | }
508 |

```

File: app/cai-dat-win11/page.tsx (988 lines, 57.1 KB)

```

1 | 'use client'
2 |
3 | import { useState, useEffect, useRef, useCallback } from 'react'
4 | import { useRouter } from 'next/navigation'
5 | import { useIsMobile } from '@/hooks/useIsMobile'
6 | import { motion, AnimatePresence } from 'framer-motion'
7 | import dynamic from 'next/dynamic'
8 | import {
9 |     ArrowLeft, Monitor, HardDrive, User, Shield, CheckCircle2,
10 |     Cpu, MemoryStick, Smartphone, Globe,
11 |     ExternalLink, ChevronRight, AlertTriangle, Clock,

```

```

12 | Disc, FileText, RefreshCw, Power, ChevronLeft,
13 | } from 'lucide-react'
14 |
15 | const WindowsSimulator = dynamic(() => import('@components/WindowsSimulator'), { ssr: false })
16 |
17 | const REAL_STEPS = [
18 |   { id: 'check', label: 'Kiểm tra PC', icon: <Monitor size={14} />, color: '#f59e0b' },
19 |   { id: 'bios', label: 'Boot USB', icon: <HardDrive size={14} />, color: '#f59e0b' },
20 |   { id: 'logo', label: 'Khởi động', icon: <Power size={14} />, color: '#f59e0b' },
21 |   { id: 'setup', label: 'Ngõ vào', icon: <Globe size={14} />, color: '#f59e0b' },
22 |   { id: 'license', label: 'License', icon: <FileText size={14} />, color: '#8b5cf6' },
23 |   { id: 'drive', label: 'Ổ đĩa', icon: <Disc size={14} />, color: '#10b981' },
24 |   { id: 'install', label: 'Cài đặt', icon: <CPU size={14} />, color: '#10b981' },
25 |   { id: 'restart', label: 'Restart', icon: <RefreshCw size={14} />, color: '#f59e0b' },
26 |   { id: 'oobe', label: 'OOBE', icon: <User size={14} />, color: '#6366f1' },
27 |   { id: 'desktop', label: 'Desktop', icon: <Monitor size={14} />, color: '#10b981' },
28 | ]
29 |
30 | const BIOS_KEYS = ['F2', 'DEL', 'F12', 'ESC', 'F10', 'F1']
31 |
32 | const BIOS_MENU_ITEMS = [
33 |   { label: 'Main', description: 'System information, Date, Time' },
34 |   { label: 'Advanced', description: 'CPU Configuration, SATA Settings' },
35 |   { label: 'Boot', description: 'Boot priority, Fast Boot' },
36 |   { label: 'Security', description: 'Password, TPM, Secure Boot' },
37 |   { label: 'Save & Exit', description: 'Save changes and reset' },
38 | ]
39 |
40 | const BOOT_DEVICES = [
41 |   { label: 'UEFI: USB Drive (SanDisk 3.2Gen1)', type: 'usb' as const },
42 |   { label: 'Windows Boot Manager', type: 'windows' as const },
43 |   { label: 'SATA: ST1000DM003-1CH162', type: 'hdd' as const },
44 |   { label: 'CD/DVD: HL-DT-ST DVD-RAM', type: 'dvd' as const },
45 | ]
46 |
47 | const SAVE_OPTIONS = ['Yes, Save & Exit', 'No, Discard Changes', 'Return to BIOS']
48 |
49 | const LANGUAGES = [
50 |   { code: 'vi-VN', label: 'Tiếng Việt', native: 'Tiếng Việt' },
51 |   { code: 'en-US', label: 'English (United States)', native: 'English' },
52 |   { code: 'ja-JP', label: 'Japanese', native: 'eäg,Šž' },
53 |   { code: 'ko-KR', label: 'Korean', native: 'Ö\mA' },
54 |   { code: 'zh-CN', label: 'Chinese (Simplified)', native: 'N-e$ (€OS)' },
55 | ]
56 |
57 | const KEYBOARD_LAYOUTS = [
58 |   { code: 'US', label: 'US' },
59 |   { code: 'VN', label: 'Vietnamese' },
60 |   { code: 'UK', label: 'United Kingdom' },
61 |   { code: 'JP', label: 'Japanese' },
62 | ]
63 |
64 | const DRIVES = [
65 |   { id: 'drive0', label: 'Drive 0 Unallocated Space', size: '238.5 GB' },
66 |   { id: 'drive1', label: 'Drive 1 Unallocated Space', size: '931.5 GB' },
67 | ]
68 |
69 | const INSTALL_PHASES = [
70 |   { label: 'Copying Windows files', min: 0, max: 15, icon: '0=Ü£', color: '0=Ü£' },
71 |   { label: 'Installing features', min: 15, max: 40, icon: '&™p', color: '&™p' },
72 |   { label: 'Installing updates', min: 40, max: 60, icon: '0=Y', color: '0=Y' },
73 |   { label: 'Finalizing installation', min: 60, max: 85, icon: '0=Y', color: '0=Y' },
74 |   { label: 'Getting devices ready', min: 85, max: 100, icon: '0=Y', color: '0=Y' },
75 | ]
76 |
77 | type BiosSubStep = 'boot_screen' | 'main_menu' | 'boot_menu' | 'save_exit'
78 |
79 | function playBeep() {
80 |   try {
81 |     const ctx = new (window.AudioContext || (window as any).webkitAudioContext)()
82 |     const osc = ctx.createOscillator()
83 |     const gain = ctx.createGain()
84 |     osc.connect(gain)
85 |     gain.connect(ctx.destination)
86 |     osc.frequency.value = 800
87 |     osc.type = 'square'
88 |     gain.gain.value = 0.08
89 |     osc.start()
90 |     osc.stop(ctx.currentTime + 0.12)
91 |   } catch {}
92 | }
93 |
94 | function BiosScreen({ children, title }: { children: React.ReactNode; title: string }) {
95 |   const isMobile = useIsMobile()
96 |   return (
97 |     <div style={{
98 |       background: '#000080', color: '#c0c0c0', fontFamily: '"Courier New", Consolas, monospace',
99 |       borderRadius: '12px', border: '2px solid #4040a0', overflow: isMobile ? 'auto' : 'hidden',
100 |       boxShadow: '0 0 40px rgba(0,0,128,0.3)',
101 |     }}>
102 |       <div style={{ background: '#000060', padding: '8px 16px', borderBottom: '1px solid #4040a0' }}>
103 |         <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
104 |           <span style={{ color: 'white', fontWeight: 'bold', fontSize: '12px' }}>PhoenixBIOS 4.0 Release 6.0</s...
105 |           <span style={{ color: 'white', fontSize: '11px' }}>Copyright 1985-2023 Phoenix Technologies Ltd.</s...
106 |         </div>
107 |       </div>
108 |       <div style={{ padding: '16px 20px', fontSize: '13px', lineHeight: '1.6' }}>
109 |         <div style={{ color: 'white', fontWeight: 'bold', marginBottom: '12px', fontSize: '14px' }}><title><...
110 |         <div>
111 |           <div style={{ background: '#000060', padding: '6px 16px', borderTop: '1px solid #4040a0', display: 'flex...
112 |           <span>F1 Help</span>
113 |           <span>F1 Help</span>
114 |           <span>Enter Execute</span>
115 |           <span>ESC Exit</span>
116 |         </div>
117 |       </div>
118 |     </div>
119 |   )
120 | }
121 |
122 | export default function CaiDatWin11Page() {
123 |   const router = useRouter()
124 |   const isMobile = useIsMobile()
125 |   const [step, setStep] = useState(0)
126 |   const [progress, setProgress] = useState(0)
127 |   const [userName, setUserName] = useState('')
128 |   const [pcName, setPcName] = useState('')
129 |   const [dotCount, setDotCount] = useState(0)
130 |   const [selectedLang, setSelectedLang] = useState(0)
131 |   const [selectedKeyboard, setSelectedKeyboard] = useState(0)
132 |   const [selectedDrive, setSelectedDrive] = useState(0)
133 |   const [licenseAccepted, setLicenseAccepted] = useState(false)
134 |   const [restartCountdown, setRestartCountdown] = useState(10)
135 |   const [privacySettings, setPrivacySettings] = useState({

```

```

136 |     location: true, diagnostics: true, ads: false, tailored: true,
137 | })
138 |
139 | const [biosSubStep, setBiosSubStep] = useState<BiosSubStep>('boot_screen')
140 | const [biosMenuIndex, setBiosMenuIndex] = useState(0)
141 | const [biosBootIndex, setBiosBootIndex] = useState(0)
142 | const [biosSaveIndex, setBiosSaveIndex] = useState(0)
143 | const [biosRequiredKey] = useState('DEL')
144 | const [biosLoading, setBiosLoading] = useState(true)
145 | const [biosErrorMessage, setBiosErrorMessage] = useState('')
146 | const [biosTransition, setBiosTransition] = useState(false)
147 | const [biosDone, setBiosDone] = useState(false)
148 | const biosMountedRef = useRef(true)
149 | const progressRef = useRef<NodeJS.Timeout | null>(null)
150 |
151 | const [showPrivacy, setShowPrivacy] = useState(false)
152 | const [isInstalling, setIsInstalling] = useState(false)
153 |
154 | const currentPhase = INSTALL_PHASES.find(p => progress >= p.min && progress < p.max) || INSTALL_PHASES[INSTA...
155 |
156 | useEffect(() => {
157 |     biosMountedRef.current = true
158 |     return () => { biosMountedRef.current = false }
159 | }, [])
160 |
161 | useEffect(() => {
162 |     if (step === 6 || step === 8) {
163 |         const i = setInterval(() => setDotCount(d => (d + 1) % 4), 500)
164 |         return () => clearInterval(i)
165 |     }
166 | }, [step])
167 |
168 | useEffect(() => {
169 |     if (step === 7 && restartCountdown > 0) {
170 |         const t = setTimeout(() => setRestartCountdown(c => c - 1), 1000)
171 |         return () => clearTimeout(t)
172 |     }
173 |     if (step === 7 && restartCountdown === 0) {
174 |         const t = setTimeout(() => { setStep(8); setRestartCountdown(10) }, 500)
175 |         return () => clearTimeout(t)
176 |     }
177 | }, [step, restartCountdown])
178 |
179 | const enterBios = useCallback(() => {
180 |     playBeep()
181 |     setBiosErrorMessage('')
182 |     setBiosLoading(true)
183 |     setTimeout(() => {
184 |         if (biosMountedRef.current) {
185 |             setBiosLoading(false)
186 |             setBiosSubStep('main_menu')
187 |         }
188 |     }, 600)
189 | }, [])
190 |
191 | useEffect(() => {
192 |     if (step !== 1 || biosTransition || biosDone) return
193 |     const handler = (e: KeyboardEvent) => {
194 |         if (biosSubStep === 'boot_screen') {
195 |             const key = e.key.toUpperCase()
196 |             if (key === 'DELETE' || key === 'F2') {
197 |                 e.preventDefault()
198 |                 enterBios()
199 |                 return
200 |             }
201 |             if (key !== 'SHIFT' && key !== 'CONTROL' && key !== 'ALT') {
202 |                 2 0 2 |         s e t B i o s E r r o r M e s s a g e ( ` P h í m   "${e.key}"   k h ô n g   h ấ p   l ậ t !   C ậ n   n h ậ n   D ậ
203 |                 setBiosErrorMessage(''), 2000)
204 |             }
205 |         }
206 |         if (biosSubStep === 'main_menu') {
207 |             if (e.key === 'ArrowUp') { e.preventDefault(); setBiosMenuIndex(i => Math.max(0, i - 1)) }
208 |             if (e.key === 'ArrowDown') { e.preventDefault(); setBiosMenuIndex(i => Math.min(BIOS_MENU_ITEMS.length...
209 |             if (e.key === 'Enter') {
210 |                 e.preventDefault()
211 |                 if (BIOS_MENU_ITEMS[biosMenuIndex].label === 'Boot') {
212 |                     playBeep()
213 |                     setBiosSubStep('boot_menu')
214 |                     setBiosErrorMessage('')
215 |                 } else {
216 |                     2 1 6 |                 s e t B i o s E r r o r M e s s a g e ( ` C h ỉ n   " B o o t "   ấ t i ậ p   t ậ c !   (   a n g   c h ỉ n :   $
217 |                     setTimeout(() => setBiosErrorMessage(''), 2000)
218 |                 }
219 |             }
220 |         }
221 |         if (biosSubStep === 'boot_menu') {
222 |             if (e.key === 'ArrowUp') { e.preventDefault(); setBiosBootIndex(i => Math.max(0, i - 1)) }
223 |             if (e.key === 'ArrowDown') { e.preventDefault(); setBiosBootIndex(i => Math.min(BOOT_DEVICES.length - ...
224 |             if (e.key === 'Enter') {
225 |                 e.preventDefault()
226 |                 if (BOOT_DEVICES[biosBootIndex].type === 'usb') {
227 |                     playBeep()
228 |                     setBiosSubStep('save_exit')
229 |                     setBiosErrorMessage('')
230 |                 } else {
231 |                     2 3 1 |                 s e t B i o s E r r o r M e s s a g e ( ` K h ô n g   t h ấ   b o o t   t ậ   "${BOOT_DEVICES[biosE
232 |                     setTimeout(() => setBiosErrorMessage(''), 2500)
233 |                 }
234 |             }
235 |         }
236 |         if (biosSubStep === 'save_exit') {
237 |             if (e.key === 'ArrowUp') { e.preventDefault(); setBiosSaveIndex(i => Math.max(0, i - 1)) }
238 |             if (e.key === 'ArrowDown') { e.preventDefault(); setBiosSaveIndex(i => Math.min(SAVE_OPTIONS.length - ...
239 |             if (e.key === 'Enter') {
240 |                 e.preventDefault()
241 |                 if (SAVE_OPTIONS[biosSaveIndex] === 'Yes, Save & Exit') {
242 |                     playBeep()
243 |                     setBiosTransition(true)
244 |                     setBiosDone(true)
245 |                     setTimeout(() => {
246 |                         setStep(2)
247 |                     }, 2000)
248 |                 } else if (SAVE_OPTIONS[biosSaveIndex] === 'No, Discard Changes') {
249 |                     setBiosSubStep('main_menu')
250 |                     setBiosMenuIndex(0)
251 |                     setBiosErrorMessage('')
252 |                 }
253 |             }
254 |         }
255 |     }
256 |     window.addEventListener('keydown', handler)
257 |     return () => window.removeEventListener('keydown', handler)
258 | }, [step, biosSubStep, biosMenuIndex, biosBootIndex, biosSaveIndex, biosTransition, biosDone, enterBios])
259 |

```

```

260 | useEffect(() => {
261 |   if (step === 1 && biosSubStep === 'boot_screen') {
262 |     const t = setTimeout(() => setBiosLoading(false), 1200)
263 |     return () => clearTimeout(t)
264 |   }
265 | }, [step, biosSubStep])
266 |
267 | useEffect(() => {
268 |   if (step === 2) {
269 |     const t = setTimeout(() => setStep(3), 4000)
270 |     return () => clearTimeout(t)
271 |   }
272 | }, [step])
273 |
274 | useEffect(() => {
275 |   if (isInstalling) {
276 |     progressRef.current = setInterval(() => {
277 |       setProgress(prev => {
278 |         if (prev >= 100) {
279 |           if (progressRef.current) clearInterval(progressRef.current)
280 |           return 100
281 |         }
282 |         return prev + Math.random() * 1.8 + 0.3
283 |       })
284 |     }, 250)
285 |     return () => { if (progressRef.current) clearInterval(progressRef.current) }
286 |   }
287 | }, [isInstalling])
288 |
289 | function advanceToBoot() {
290 |   setStep(1)
291 |   setBiosSubStep('boot_screen')
292 |   setBiosMenuIndex(0)
293 |   setBiosBootIndex(0)
294 |   setBiosSaveIndex(0)
295 |   setBiosLoading(true)
296 |   setBiosErrorMessage('')
297 |   setBiosTransition(false)
298 |   setBiosDone(false)
299 | }
300 |
301 | const formatTime = () => {
302 |   const m = Math.floor(progress / 12)
303 |   const s = Math.floor((progress * 2.5) % 60)
304 |   return `${m}:${s.toString().padStart(2, '0')}`
305 | }
306 |
307 | const renderStepIndicator = () => (
308 |   <div className="flex gap-1">
309 |     {REAL_STEPS.slice(0, 6).map((s, i) => (
310 |       <div key={s.id} style={{
311 |         width: '24px', height: '4px', borderRadius: '2px',
312 |         background: i <= step ? s.color : 'var(--border-default)',
313 |         transition: 'all 0.3s',
314 |       }} />
315 |     ))}
316 |   </div>
317 | )
318 |
319 | if (biosTransition && step === 1) {
320 |   return (
321 |     <div className="min-h-screen flex flex-col" style={{ background: '#000', color: '#fff' }}>
322 |       <div className="flex-1 flex items-center justify-center p-4">
323 |         <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }} style={{ textAlign: 'center' }}>
324 |           <div style={{ fontSize: '48px', marginBottom: '16px', filter: 'drop-shadow(0 0 20px rgba(0,120,212...
325 |             <h2 style={{ fontSize: '18px', fontWeight: '700', color: '#0070C0' }}>
326 |               <p style={{ color: 'rgba(255,255,255,0.5)' }}>
327 |                 <div style={{ marginTop: '20px', display: 'flex', justifyContent: 'center', gap: '6px' }}>
328 |                   {[0, 1, 2].map(i => (
329 |                     <motion.div key={i} animate={{ opacity: [0.2, 1, 0.2] }}
330 |                     transition={{ duration: 1, repeat: Infinity, delay: i * 0.3 }}
331 |                     style={{ width: '8px', height: '8px', borderRadius: '50%', background: '#0078d4' }} />
332 |                   ))}
333 |                 </div>
334 |               </motion.div>
335 |             </div>
336 |           </div>
337 |         </div>
338 |       )
339 |     )
340 |   )
341 |   return (
342 |     <div className="min-h-screen flex flex-col" style={{ background: 'var(--bg-base)', color: 'var(--text-prim...' }}>
343 |       <header className="border-b p-3 px-4 flex items-center gap-3" style={{ background: 'var(--bg-surface)', ...
344 |         <button onClick={() => router.back()}
345 |           className="flex items-center gap-1.5 px-3 py-1.5 border text-xs rounded-xl font-bold cursor-pointer"
346 |           style={{ background: 'var(--bg-elevated)', borderColor: 'var(--border-default)', color: 'var(--text-...
347 |           <ArrowLeft size={14} /> Thoát
348 |         </button>
349 |         <div style={{ width: '28px', height: '28px', borderRadius: '8px', background: 'linear-gradient(135deg, ...
350 |         </div>
351 |         <div style={{ flex: 1 }}>
352 |           <h1 className="font-black text-sm" style={{ color: 'var(--text-prim...' }}>
353 |           <p className="text-xs" style={{ color: 'var(--text-muted)' }}>
354 |         </div>
355 |       </header>
356 |       <div className="flex-1 overflow-y-auto p-4 max-w-2xl mx-auto w-full flex flex-col justify-center">
357 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
358 |           <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
359 |             <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
360 |               <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
361 |                 <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
362 |                   <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
363 |                     <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
364 |                       <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
365 |                         <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
366 |                           <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
367 |                             <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
368 |                               <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
369 |                                 <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
370 |                                   <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
371 |                                     <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
372 |                                       <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
373 |                                         <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
374 |                                           <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
375 |                                             <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
376 |                                               <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
377 |                                                 <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
378 |                                                   <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
379 |                                                     <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
380 |                                                       <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
381 |                                                         <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
382 |                                                           <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>
383 |                                                             <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', margin: '20px 0' }}>

```

```

384|         </motion.div>
385|     )))
386| </div>
387|
388|     <div style={{ padding: '12px 16px', borderRadius: '10px', background: 'rgba(16,185,129,0.06)', bor...
389|         <strong style={{ color: '#10b981' }}>Thần tử: </strong> { '
390|         <span style={{ color: 'var(--text-primary)' }}>Nú PC ch...
391|     </div>
392|
393|     <button onClick={advanceToBoot}
394|         className="w-full py-3 font-bold rounded-xl text-sm cursor-pointer"
395|         style={{ background: '#0078d4', border: 'none', color: '#fff', fontFamily: 'inherit' }}>
396|         PC it yêu c s u ! Boot t ê USB !'
397|     </button>
398| </motion.div>
399| ))
400|
401| /* STEP 1: BIOS Simulator */
402| {step === 1 && (
403|     <motion.div key="step1" initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }}>
404|         <div style={{ textAlign: 'center', marginBottom: '16px' }}>
405|             <div style={{ fontSiz e : { fontSiz e : '36px', marginBott om : '8px' }}>0<
406|             <h2 style={{ fontSiz e : { fontSiz e : '18px', fontWeigh t : 800, marginBott
407|             <p style={{ color: 'var(--text-muted)', fontSiz e : '12px' }}
408|         </div>
409|
410|         {biosSubStep === 'boot_screen' && (
411|             <motion.div key="bs_boot_screen" initial={{ opacity: 0 }} animate={{ opacity: 1 }}>
412|                 <BiosScreen title="PhoenixBIOS Setup Utility">
413|                     {biosLoading ? (
414|                         <div style={{ textAlign: 'center', padding: '24px 0' }}>
415|                             <div style={{ color: '#ffff00', fontSize: '12px', marginBott om : '12px' }}>BIOS Loading....
416|                             <div style={{ display: 'flex', justifyContent: 'center', gap: '4px' }}>
417|                                 {[0, 1, 2, 3].map(i => (
418|                                     <motion.div key={i} animate={{ opacity: [0.2, 1, 0.2] }} transition={{ duration: 1.2...
419|                                     style={{ width: '8px', height: '8px', background: '#ffff00', borderRadius: '50%' }}...
420|                                 ))}
421|                             </div>
422|                             <div style={{ color: '#8080d0', fontSize: '10px', marginT op : '12px' }}>Initializing cont...
423|                         </div>
424|                     ) : (
425|                         <div style={{ textAlign: 'center', padding: '16px 0' }}>
426|                             <div style={{ fontSiz e : '11px', color: '#8080d0', marginBott om : '10px' }}>PhoenixBIOS 4...
427|                             <div style={{ fontSiz e : '11px', color: '#c0c0c0', marginBott om : '16px' }}>
428|                                 CPU: Intel Core i7-14700K @ 3.40 GHz<br />
429|                                 Memory: 32768MB (32GB DDR5)<br />
430|                                 System: Windows 11 Pro (64-bit)
431|                             </div>
432|                             <motion.div animate={{ scale: [1, 1.05, 1] }} transition={{ duration: 1.5, repeat: Infin...
433|                                 style={{ display: 'inline-block', padding: '10px 20px', border: '2px solid #ffff00', b...
434|                                 <div style={{ color: '#ffff00', fontWeigh t : 'bold', fontSiz e : '13px', fontFami ly : 'mon...
435|                                     Press <span style={{ fontSiz e : '16px', border: '1px solid #ffff00', padding: '2px 6p...
436|                                 </div>
437|                             </motion.div>
438|                             <div style={{ display: 'flex', gap: '6px', justifyContent: 'center', flexWrap: 'wrap', m...
439|                                 {BIOS_KEYS.map(key => (
440|                                     <kbd key={key} onClick={() => { if (key === 'DEL' || key === 'F2') enterBios() }}
441|                                     style={{
442|                                         padding: '6px 12px', borderRadius: '6px', border: '1px solid #4040a0',
443|                                         background: key === 'DEL' ? 'rgba(255,255,0,0.15)' : 'rgba(64,64,160,0.3)',
444|                                         color: key === 'DEL' ? '#ffff00' : '#8080d0',
445|                                         cursor: 'pointer', fontFami ly : 'monospace', fontSiz e : '12px', fontWeigh t : 700,
446|                                     }}>
447|                                     {key}
448|                                 </kbd>
449|                             ))}
450|                         </div>
451|                         {biosErrorMessage && (
452|                             <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }}
453|                                 style={{ marginT op : '8px', padding: '6px 10px', background: 'rgba(255,0,0,0.15)', bo...
454|                                 <AlertTriangle size={12} style={{ display: 'inline', marginRigh t : '4px', verticalAli...
455|                                 {biosErrorMessage}
456|                             </motion.div>
457|                         )}
458|                     </div>
459|                 )}
460|             </BiosScreen>
461|
462|             <div style={{ marginT op : '10px', padding: '8px 12px', borderRadius: '8px', background: 'rgba(2...
463|                 <span style={{ color: '#f59e0b', fontWeigh t : 700 }}>M
464|                 <span style={{ color: 'var(--text-primary)' }}>Như n <s
465|             </div>
466|         </motion.div>
467|     ))
468|
469|     {biosSubStep === 'main_menu' && (
470|         <motion.div key="bs_main_menu" initial={{ opacity: 0 }} animate={{ opacity: 1 }}>
471|             <BiosScreen title="Main Menu - Select Boot">
472|                 <div style={{ display: 'flex', flexDirection: 'column', gap: '2px', marginBott om : '8px' }}>
473|                     {BIOS_MENU_ITEMS.map((item, i) => (
474|                         <motion.div key={i} whileHover={{ x: 2 }}
475|                             onClick={() => {
476|                                 if (item.label === 'Boot') { playBeep(); setBiosSubStep('boot_menu') }
477|                                 else { setBiosErrorMessage('Chín "Boot" ã t i
478|                             )}
479|                             style={{
480|                                 display: 'flex', alignIt ems : 'center', gap: '8px', padding: '6px 10px', cursor: 'poi...
481|                                 background: biosMenuIndex === i ? 'rgba(255,255,0,0.12)' : 'transparent', b...
482|                                 border: biosMenuIndex === i ? '1px solid #ffff00' : '1px solid transparent',
483|                                 borderRadius: '4px', transition: 'all 0.15s', fontSiz e : '12px',
484|                             }}>
485|                                 <span style={{ color: biosMenuIndex === i ? '#ffff00' : '#c0c0c0' }}>{item.label}</span>
486|                                 <span style={{ fontSiz e : '10px', color: '#8080d0', marginL eft : '8px' }}>{item.descript...
487|                                 {biosMenuIndex === i && <ChevronRight size={12} style={{ color: '#ffff00', marginL eft :...
488|                             </motion.div>
489|                         ))}
490|                     </div>
491|                     {biosErrorMessage && (
492|                         <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }}
493|                             style={{ padding: '6px 10px', background: 'rgba(255,0,0,0.12)', border: '1px solid #ff44...
494|                             {biosErrorMessage}
495|                         </motion.div>
496|                     )}
497|                 </BiosScreen>
498|             </motion.div>
499|         )}
500|
501|     {biosSubStep === 'boot_menu' && (
502|         <motion.div key="bs_boot_menu" initial={{ opacity: 0 }} animate={{ opacity: 1 }}>
503|             <BiosScreen title="Boot Menu - Select USB Drive">
504|                 <div style={{ fontSiz e : '11px', color: '#a0a0ff', marginBott om : '8px' }}>Select boot device:...
505|                 <div style={{ display: 'flex', flexDirection: 'column', gap: '2px' }}>
506|                     {BOOT_DEVICES.map((device, i) => (
507|                         <motion.div key={i} whileHover={{ x: 2 }}

```

```

508 |         onClick={() => {
509 |             if (device.type === 'usb') { playBeep(); setBiosSubStep('save_exit') }
510 |             else { setBiosErrorMessage(`Không thể boot từ
511 |         })
512 |         style={{
513 |             display: 'flex', alignItems: 'center', gap: '8px', padding: '6px 10px', cursor: 'poi...
514 |             backgroundColor: biosBootIndex === i ? 'rgba(255,255,0,0.12)' : 'transparent',
515 |             border: biosBootIndex === i ? '1px solid #ffff00' : '1px solid transparent',
516 |             borderRadius: '4px', fontSize: '12px',
517 |         }}
518 |         <span>{ device.type === 'usb' ? '0=0%' : device.type
519 |         <span style={{ color: biosBootIndex === i ? '#ffff00' : '#c0c0c0' }}>{device.label}</s...
520 |         {device.type === 'usb' ? (
521 |             <span style={{ marginLeft: 'auto', fontSize: '9px', padding: '2px 6px', borderRadius...
522 |             ) : (
523 |             <span style={{ marginLeft: 'auto', fontSize: '9px', padding: '2px 6px', borderRadius...
524 |             )}
525 |         </motion.div>
526 |     )}}
527 | </div>
528 | {biosErrorMessage && (
529 |     <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }}
530 |     style={{ marginTop: '8px', padding: '6px 10px', background: 'rgba(255,0,0,0.12)', border...
531 |     {biosErrorMessage}
532 |     </motion.div>
533 | )}
534 | </BiosScreen>
535 | </motion.div>
536 | ))
537 |
538 | {biosSubStep === 'save_exit' && (
539 |     <motion.div key="bs_save_exit" initial={{ opacity: 0 }} animate={{ opacity: 1 }}>
540 |     <BiosScreen title="Save & Exit - Confirm Changes">
541 |         <div style={{ fontSize: '11px', color: '#a0a0ff', marginBottom: '10px' }}>
542 |             Boot device selected: <span style={{ color: '#10b981', fontWeight: 'bold' }}>UEFI: USB Dri...
543 |         </div>
544 |         <div style={{ marginBottom: '10px', padding: '6px 10px', background: 'rgba(16,185,129,0.06)'...
545 |         <CheckCircle2 size={12} style={{ color: '#10b981', display: 'inline', marginRight: '4px', ...
546 |         Changes have been made to your BIOS configuration.<br />
547 |         &nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&• Boot priority: USB Drive moved to #1<br />
548 |         &nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&• Boot mode: UEFI (CSM disabled)<br />
549 |         <span style={{ color: '#ffff00' }}>Do you want to save these changes and exit?</span>
550 |     </div>
551 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '2px' }}>
552 |         {SAVE_OPTIONS.map((opt, i) => (
553 |             <motion.div key={i} whileHover={{ x: 2 }}
554 |             onClick={() => {
555 |                 if (opt === 'Yes, Save & Exit') {
556 |                     playBeep()
557 |                     setBiosTransition(true)
558 |                     setBiosDone(true)
559 |                     setTimeout(() => { setStep(2) }, 2000)
560 |                 } else if (opt === 'No, Discard Changes') {
561 |                     setBiosSubStep('main_menu'); setBiosMenuIndex(0); setBiosErrorMessage('')
562 |                 }
563 |             }}
564 |             style={{
565 |                 display: 'flex', alignItems: 'center', gap: '8px', padding: '6px 10px', cursor: 'poi...
566 |                 backgroundColor: biosSaveIndex === i ? 'rgba(255,255,0,0.12)' : 'transparent',
567 |                 border: biosSaveIndex === i ? '1px solid #ffff00' : '1px solid transparent',
568 |                 borderRadius: '4px', fontSize: '11px',
569 |             }}
570 |             <span style={{ color: biosSaveIndex === i ? '#ffff00' : '#c0c0c0' }}>{opt}</span>
571 |             {biosSaveIndex === i && <ChevronRight size={12} style={{ color: '#ffff00', marginLeft:...
572 |         </motion.div>
573 |     )}}
574 | </div>
575 | <div style={{ marginTop: '10px', padding: '6px 10px', background: 'rgba(255,255,0,0.06)', bo...
576 |         C E n h b á o : N ắ u k h ồ n g c m U S B b o o t , m á y s ½ b á o l x i " N o
577 |     </div>
578 | </BiosScreen>
579 | </motion.div>
580 | )}
581 | </motion.div>
582 | ))
583 |
584 | /* STEP 2: Windows Logo Boot */
585 | {step === 2 && (
586 |     <motion.div key="step2" initial={{ opacity: 0 }} animate={{ opacity: 1 }}
587 |     className="flex-1 flex items-center justify-center"
588 |     style={{ background: '#000', minHeight: '400px', borderRadius: '16px' }}>
589 |         <div style={{ textAlign: 'center' }}>
590 |             <motion.div animate={{ rotate: 360 }} transition={{ duration: 2, repeat: Infinity, ease: 'linear...
591 |             style={{ width: '80px', height: '80px', margin: '0 auto 24px', position: 'relative' }}>
592 |                 <div style={{
593 |                     width: '80px', height: '80px', borderRadius: '50%',
594 |                     border: '4px solid rgba(0,120,212,0.15)',
595 |                     borderTopColor: '#0078d4',
596 |                     position: 'absolute', inset: 0,
597 |                 }} />
598 |             </motion.div>
599 |             <div style={{ fontSize: '32px', fontWeight: 300, color: '#0078d4', marginBottom: '8px', letterSp...
600 |             <div style={{ display: 'flex', justifyContent: 'center', gap: '4px' }}>
601 |                 {[0, 1, 2].map(i => (
602 |                     <motion.div key={i} animate={{ opacity: [0.1, 1, 0.1] }}
603 |                     transition={{ duration: 1.5, repeat: Infinity, delay: i * 0.3 }}
604 |                     style={{ width: '6px', height: '6px', borderRadius: '50%', background: '#0078d4' }} />
605 |                 ))}
606 |             </div>
607 |             <p style={{ color: 'rgba(255,255,255,0.3)', fontSize: '11px', marginTop: '20px' }}>
608 |                 a n g k h ắ i ù n g t ế U S B . . .
609 |             </p>
610 |         </div>
611 |     </motion.div>
612 | )}
613 |
614 | /* STEP 3: Language / Region / Keyboard */
615 | {step === 3 && (
616 |     <motion.div key="step3" initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }}>
617 |         <div style={{ textAlign: 'center', marginBottom: '20px' }}>
618 |             <div style={{
619 |                 <div style={{
620 |                     <h2 style={{
621 |                         <p style={{
622 |                             </p>
623 |                         </div>
624 |                     </div>
625 |                     <label style={{
626 |                         <select value={selectedLang} onChange={e => setSelectedLang(Number(e.target.value))}>
627 |                             style={{
628 |                                 {LANGUAGES.map((l, i) => (
629 |                                     <option key={l.code} value={i}>{l.native} {l.code}</option>
630 |                                 ))}
631 |                             </select>

```



```
756 | </div>
757 |
758 | <div style={{ display: 'flex', flexDirection: 'column', gap: '8px', marginBottom: '12px' }}>
759 |   <div style={{ padding: '10px 14px', borderRadius: '10px', background: 'rgba(99,102,241,0.08)', b...
7 | 6 0 |     <strong>Lo i c à i t : </strong> Custom : Install Windows
761 |   </div>
762 |
763 |   {DRIVES.map((drive, i) => (
764 |     <motion.div key={drive.id} whileHover={{ scale: 1.01 }}
765 |     onClick={() => setSelectedDrive(i)}
766 |     style={{
767 |       display: 'flex', alignItems: 'center', gap: '12px',
768 |       padding: '14px 16px', borderRadius: '12px', cursor: 'pointer',
769 |       background: selectedDrive === i ? 'rgba(16,185,129,0.08)' : 'var(--bg-surface)',
770 |       border: selectedDrive === i ? '2px solid #10b981' : '1px solid var(--border-default)',
771 |       transition: 'all 0.2s',
772 |     }}>
773 |     <div style={{ fontSize: '24px' }}>{drive.icon}</div>
774 |     <div style={{ flex: 1 }}>
775 |       <div style={{ fontSize: '13px', fontWeight: 700, color: 'var(--text-primary)' }}>{drive.la...
776 |       <div style={{ fontSize: '11px', color: 'var(--text-muted)' }}>
777 |         {drive.size} · {drive.type} · Total space: {drive.size}
7 | 7 8 |         {selectedDrive === i && <span style={{ color: '#10b981' }}>
779 |       </div>
780 |     </div>
781 |     <div style={{
782 |       width: '20px', height: '20px', borderRadius: '50%', border: '2px solid',
783 |       borderColor: selectedDrive === i ? '#10b981' : 'var(--border-default)',
784 |       display: 'flex', alignItems: 'center', justifyContent: 'center',
785 |       flexShrink: 0,
786 |     }}>
787 |       {selectedDrive === i && <div style={{ width: '10px', height: '10px', borderRadius: '50%', ...
788 |     </div>
789 |   </motion.div>
790 | ))}
791 | </div>
792 |
793 | <div style={{ display: 'flex', gap: '8px' }}>
794 |   <button onClick={() => setStep(5)}
795 |     className="flex items-center gap-1.5 px-4 py-2.5 font-bold rounded-xl text-xs cursor-pointer"
796 |     style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)', color: 'v...
7 | 9 7 |     <ChevronLeft size={14} /> Quay lại
798 |   </button>
799 |   <button onClick={() => { setProgress(0); setIsInstalling(true) }}
800 |     className="flex-1 py-2.5 font-bold rounded-xl text-sm cursor-pointer"
801 |     style={{ background: '#0078d4', border: 'none', color: 'fff', fontFamily: 'inherit' }}>
8 | 0 2 |     B  t  S  u  c  à i  t  !
803 |   </button>
804 | </div>
805 | </motion.div>
806 | )}
807 |
808 |   { /* STEP 6 b: Installation Progress (shown after clicking "B  t S  u  c  à i t  !"
809 |   {step === 6 && isInstalling && (
810 |     <motion.div key="step6b" initial={{ opacity: 0 }} animate={{ opacity: 1 }}
811 |     style={{ position: 'fixed', inset: 0, zIndex: 50,
812 |       background: 'linear-gradient(135deg, #001830, #002040)',
813 |       display: 'flex', flexDirection: 'column', alignItems: 'center', justifyContent: 'center', paddin...
814 |     <motion.div animate={{ scale: [1, 1.05, 1] }} transition={{ duration: 2, repeat: Infinity }}>
815 |       <div style={{ fontSize: '72px', marginBottom: '24px', filter: 'drop-shadow(0 40px rgba(0,120,2...
816 |     </motion.div>
8 | 1 7 |     <h2 style={{ color: 'rgba(255,255,255,0.4)', fontSize: '13px', marginBottom: '24px' }}>Vui lòng khò...
818 |     <p style={{ color: 'rgba(255,255,255,0.4)', fontSize: '13px', marginBottom: '24px' }}>
819 |
820 |     <div style={{ maxWidth: '480px', width: '100%' }}>
821 |       <div style={{ display: 'flex', justifyContent: 'space-between', fontSize: '11px', color: 'rgba(2...
822 |       <span><span style={{ color: currentPhase.color || '#0078d4' }}>{currentPhase.icon}</span> {cur...
823 |       <span style={{ fontWeight: 700, color: 'fff' }}>{Math.min(100, Math.round(progress))}%</span>
824 |     </div>
825 |     <div style={{ width: '100%', height: '8px', borderRadius: '4px', background: 'rgba(255,255,255,0...
826 |       <motion.div style={{
827 |         width: `${Math.min(100, progress)}%`, height: '100%',
828 |         background: 'linear-gradient(90deg, #0078d4, #00a8ff, #0078d4)',
829 |         backgroundSize: '200% 100%',
830 |         borderRadius: '4px',
831 |       }} animate={{ backgroundPosition: ['0% 0%', '100% 0%', '0% 0%'] }}
832 |       transition={{ duration: 3, repeat: Infinity }} />
833 |     </div>
834 |     <p style={{ fontSize: '11px', color: 'rgba(255,255,255,0.3)', textAlign: 'center' }}>
8 | 3 5 |     a n g  s a o  c h ế p  t - p  t i n  c à i  t { ' . . . r e p e a t ( d o t C o u n t | | 1 ) }
836 |   </p>
837 | </div>
838 |
839 |   {progress >= 100 && (
840 |     <motion.div initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }} style={{ marginTop: '...
841 |     <button onClick={() => setStep(7)}
842 |       style={{ padding: '12px 32px', borderRadius: '12px', border: 'none', background: '#0078d4', ...
8 | 4 3 |       C à i  t  x o n g !  K h ấ i  U n g  l ậ i  !
844 |     </button>
845 |   </motion.div>
846 | ))}
847 | </motion.div>
848 | )}
849 |
850 | { /* STEP 7: Restart */
851 | {step === 7 && (
852 |   <motion.div key="step7" initial={{ opacity: 0 }} animate={{ opacity: 1 }}
853 |   className="flex-1 flex items-center justify-center"
854 |   style={{ background: '#000', minHeight: '400px', borderRadius: '16px' }}>
855 |     <div style={{ textAlign: 'center' }}>
856 |       <motion.div animate={{ rotate: 360 }} transition={{ duration: 3, repeat: Infinity, ease: 'linear...
857 |       style={{ width: '60px', height: '60px', margin: '0 auto 24px' }}>
858 |       <RefreshCw size={60} style={{ color: '#0078d4' }} />
859 |     </motion.div>
8 | 6 0 |     <h2 style={{ color: 'rgba(255,255,255,0.4)', fontSize: '20px', fontWeight: 700, color: '#0078d4' }}>
861 |     <p style={{ color: 'rgba(255,255,255,0.4)', fontSize: '13px', marginBottom: '16px' }}>
8 | 6 2 |     M á y  t i n h  s ½  k h ấ i  U n g  l ậ i  t r o n g { r e s t a r t C o u n t d o w n } g i ấ y
863 |   </p>
864 |   <div style={{ display: 'flex', justifyContent: 'center', gap: '6px' }}>
865 |     {[0, 1, 2, 3].map(i => (
866 |       <motion.div key={i} animate={{ opacity: [0.1, 1, 0.1] }}
867 |       transition={{ duration: 1, repeat: Infinity, delay: i * 0.25 }}
868 |       style={{ width: '8px', height: '8px', borderRadius: '50%', background: '#0078d4' }} />
869 |     ))}
870 |   </div>
871 |   <p style={{ color: 'rgba(255,255,255,0.2)', fontSize: '10px', marginTop: '20px' }}>
8 | 7 2 |   T r ẻ n  m á y  t h - t ,  b i n  s ½  t h ậ y  l ỏ g o  h ằ n g  ( D e l l / H P / L e n o v o )  x
873 | </p>
874 |
875 |   <button onClick={() => { setStep(8); setRestartCountdown(10) }}
876 |     style={{ marginTop: '20px', padding: '8px 20px', borderRadius: '8px', border: '1px solid rgba(...
8 | 7 7 |     B ỉ  q u a  ( t ỉ ế p  t ắ c )  !
878 |   </button>
879 | </div>
```



```

12| const [profile, setProfile] = useState<any>(null)
13| const [loading, setLoading] = useState(true)
14|
15| const supabase = createBrowserClient(
16|   process.env.NEXT_PUBLIC_SUPABASE_URL!,
17|   process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
18| )
19|
20| useEffect(() => {
21|   loadData()
22| }, [])
23|
24| async function loadData() {
25|   try {
26|     const { data: { user } } = await supabase.auth.getUser()
27|     if (user) {
28|       const { data: p } = await supabase.from('profiles').select('full_name, xp, level').eq('id', user.id).s...
29|       if (p) setProfile(p)
30|     }
31|     const { data: c } = await supabase
32|       .from('certificates')
33|       .select('*')
34|       .eq('is_revoked', false)
35|       .order('issued_at', { ascending: false })
36|     if (c) setCerts(c)
37|   } catch (e) {}
38|   setLoading(false)
39| }
40|
41| return (
42|   <div className="min-h-screen" style={{ background: 'var(--bg-base)', color: 'var(--text-primary)' }}>
43|     <Navbar />
44|     <main className="pt-28 pb-20 px-4 sm:px-6 lg:px-8 max-w-7xl mx-auto">
45|       <motion.div initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }}>
46|         <div className="flex items-center gap-4 mb-4">
47|           <div style={{
48|             width: '56px', height: '56px', borderRadius: '12px',
49|             background: 'linear-gradient(135deg, var(--brand-primary), var(--accent-blue))',
50|             display: 'flex', alignItems: 'center', justifyContent: 'center',
51|             boxShadow: '0 0 24px color-mix(in srgb, var(--brand-primary) 30%, transparent)',
52|           }}>
53|             <Trophy size={28} color="#fff" />
54|           </div>
55|           <div>
56|             <h1 style={{ fontSize: 'clamp(24px, 4vw, 36px)', fontWeight: 900, letterSpacing: '1px' }}>
57|               C H È N G C H Ê & <span style={{ color: 'var(--brand-primary)'
58|             </h1>
59|             <p style={{ color: 'var(--text-muted)', fontSize: '13px' }}>
60|               N i l l ò u g i i n h ì n g n x l ã c v à t h à n h q u ê h í c t - p c ç a b ì n .
61|             </p>
62|           </div>
63|         </div>
64|       </motion.div>
65|
66|       </* Stats */>
67|       <motion.div initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }} transition={{ delay: 0.05 }}
68|         style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(180px, 1fr))', gap: '12px', ...
69|         [[
70|           { icon: <Trophy size={22} />, label: 'Chéng chÉ it', value: $
71|           { icon: <Star size={22} />, label: 'TŌng XP', value: profile
72|           { icon: <ShieldCheck size={22} />, label: 'Cçp Û', value: $
73|           { icon: <Target size={22} />, label: 'Cò thÃ mB', value: $
74|         ]}.map((stat, i) => (
75|           <div key={i} style={{
76|             background: 'var(--bg-surface)',
77|             border: '1px solid var(--border-default)',
78|             padding: '20px',
79|             display: 'flex', alignItems: 'center', gap: '14px',
80|           }}>
81|             <div style={{
82|               width: '48px', height: '48px', borderRadius: '4px',
83|               background: 'color-mix(in srgb, ${stat.color} 12%, transparent)',
84|               border: '1px solid color-mix(in srgb, ${stat.color} 20%, transparent)',
85|               display: 'flex', alignItems: 'center', justifyContent: 'center',
86|               color: stat.color,
87|               flexShrink: 0,
88|             }}>
89|               {stat.icon}
90|             </div>
91|             <div>
92|               <div style={{ fontSize: '11px', color: 'var(--text-muted)', fontFamily: 'var(--font-mono)', ma...
93|               <div style={{ fontSize: '24px', fontWeight: 900, fontFamily: 'var(--font-mono)' }}>{stat.value...
94|             </div>
95|           </div>
96|         ))}
97|       </motion.div>
98|
99|       </* Certificate Grid */>
100|       {loading ? (
101|         <div style={{ textAlign: 'center', padding: '60px 0', color: 'var(--text-muted)' }}>
102|           <motion.div animate={{ rotate: 360 }} transition={{ duration: 2, repeat: Infinity, ease: 'linear' }}
103|             style={{ display: 'inline-block', marginBottom: '12px' }}>
104|             <Swords size={32} style={{ color: 'var(--brand-primary)' }} />
105|           </motion.div>
106|         <div style={{ fontFamily: 'var(--font-mono)', fontSize: '12px
107|       </div>
108|       ) : certs.length === 0 ? (
109|         <div style={{
110|           textAlign: 'center', padding: '80px 20px',
111|           border: '1px solid var(--border-default)',
112|           background: 'var(--bg-surface)',
113|         }}>
114|           <Diamond size={48} style={{ color: 'var(--text-muted)', marginBottom: '16px', opacity: 0.3 }} />
115|           <h2 style={{ fontSize: '20px', fontWeight: 700, marginBottom:
116|           <p style={{ color: 'var(--text-muted)', fontSize: '13px', marginBottom: '24px', maxWidth: '400px',...
117|             H o à n t h à n h c á c k h ó a h í c v à n h i ç m v ã Ä n h - n c h é n g c h É . M x j
118|           </p>
119|           <Link href="/builder" style={{
120|             display: 'inline-flex', alignItems: 'center', gap: '8px',
121|             padding: '12px 28px',
122|             background: 'var(--brand-primary)',
123|             color: 'var(--bg-base)',
124|             fontWeight: 700,
125|             fontSize: '13px',
126|             fontFamily: 'var(--font-mono)',
127|             textDecoration: 'none',
128|           }}>
129|             <Zap size={16} /> B @ T | U H Í C N G A Y
130|           </Link>
131|         </div>
132|       ) : (
133|         <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill, minmax(340px, 1fr))', gap: '1...
134|           {certs.map((cert, idx) => (
135|             <motion.div key={cert.id}>

```

```

136 initial={{ opacity: 0, y: 20 }}
137 animate={{ opacity: 1, y: 0 }}
138 transition={{ delay: idx * 0.05 }}
139 style={{
140   background: 'var(--bg-surface)',
141   border: '1px solid var(--border-default)',
142   position: 'relative', overflow: 'hidden',
143 }}>
144 { /* Pixel corner decorations */}
145 <div style={{ position: 'absolute', top: 0, left: 0, width: '8px', height: '8px', borderTop: '...
146 <div style={{ position: 'absolute', top: 0, right: 0, width: '8px', height: '8px', borderTop: '...
147 <div style={{ position: 'absolute', bottom: 0, left: 0, width: '8px', height: '8px', borderBottom: '...
148 <div style={{ position: 'absolute', bottom: 0, right: 0, width: '8px', height: '8px', borderBo...
149
150 { /* Certificate preview */}
151 <div style={{
152   padding: '24px',
153 }}>
154   <div style={{
155     width: '100%', aspectRatio: '16/10',
156     background: 'linear-gradient(135deg, #0a0c1a 0%, #1a1c2e 50%, #0a0c1a 100%)',
157     border: '1px solid rgba(212, 168, 67, 0.3)',
158     position: 'relative', overflow: 'hidden',
159     marginBottom: '16px',
160     display: 'flex', flexDirection: 'column', alignItems: 'center', justifyContent: 'center',
161   }}>
162     { /* Golden border */}
163     <div style={{
164       position: 'absolute', inset: '3px',
165       border: '1px solid rgba(212, 168, 67, 0.15)',
166       pointerEvents: 'none',
167     }} />
168     { /* Top gold line */}
169     <div style={{
170       position: 'absolute', top: '8px', left: '50%', transform: 'translateX(-50%)',
171       width: '50%', height: '1px',
172       background: 'linear-gradient(90deg, transparent, rgba(212,168,67,0.5), transparent)',
173     }} />
174     <div style={{
175       position: 'absolute', bottom: '8px', left: '50%', transform: 'translateX(-50%)',
176       width: '50%', height: '1px',
177       background: 'linear-gradient(90deg, transparent, rgba(212,168,67,0.5), transparent)',
178     }} />
179
180     { /* Trophy icon */}
181     <motion.div animate={{ y: [0, -3, 0] }} transition={{ duration: 3, repeat: Infinity }}
182     style={{
183       width: '36px', height: '36px', borderRadius: '50%',
184       background: 'linear-gradient(135deg, #d4a843, #f5d68a)',
185       display: 'flex', alignItems: 'center', justifyContent: 'center',
186       marginBottom: '6px',
187     }}>
188       <Award size={18} color="#0a0c1a" />
189     </motion.div>
190
191     <div style={{ fontSize: '7px', fontWeight: 600, letterSpacing: '2px', color: 'rgba(212,168...
192       C H Ê N G   C H Ê   H O A N   T H A N H
193     </div>
194     <div style={{ fontSize: '13px', fontWeight: 900, color: '#f5d68a', textAlign: 'center', pa...
195       {cert.course_title}
196     </div>
197     <div style={{ fontSize: '14px', fontWeight: 800, color: '#fff', marginBottom: '6px' }}>
198       { c e r t . s t u d e n t _ n a m e | | ' H I c   v i ê n   P C   M a s t e r ' }
199     </div>
200     <div style={{ fontSize: '7px', color: 'rgba(255,255,255,0.3)', display: 'flex', gap: '8px'...
201       <span>{new Date(cert.issued_at || Date.now()).toLocaleDateString('vi-VN')}</span>
202       <span style={{ width: '2px', height: '2px', borderRadius: '50%', background: 'rgba(255,2...
203       <span style={{ fontFamily: 'monospace', fontSize: '6px' }}>{cert.certificate_number}</span>
204     </div>
205     {cert.final_score && (
206       <div style={{
207         position: 'absolute', top: '8px', right: '8px',
208         fontSize: '7px', fontWeight: 700,
209         color: '#d4a843',
210         border: '1px solid rgba(212,168,67,0.3)',
211         padding: '1px 6px',
212       }}>
213         {cert.final_score}%
214       </div>
215     )}
216   </div>
217
218   <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'space-between' }}>
219     <div>
220       <div style={{ fontSize: '14px', fontWeight: 700, marginBottom: '2px' }}>{cert.course_tit...
221       <div style={{ display: 'flex', alignItems: 'center', gap: '6px', fontSize: '11px', color...
222         <Calendar size={11} />
223         <span>{new Date(cert.issued_at || Date.now()).toLocaleDateString('vi-VN')}</span>
224       </div>
225     </div>
226     <div style={{ display: 'flex', gap: '6px' }}>
227       {cert.pdf_url && (
228         <a href={cert.pdf_url} target="_blank" rel="noreferrer" style={{
229           width: '36px', height: '36px',
230           border: '1px solid var(--border-default)',
231           display: 'flex', alignItems: 'center', justifyContent: 'center',
232           color: 'var(--text-muted)',
233           cursor: 'pointer', transition: 'all 0.2s',
234         }}>
235           <Download size={16} />
236         </a>
237       )}
238       <button onClick={() => {
239         const link = `${window.location.origin}/verify/${cert.certificate_number}`
240         navigator.clipboard.writeText(link)
241       }} style={{
242         width: '36px', height: '36px',
243         border: '1px solid var(--border-default)',
244         display: 'flex', alignItems: 'center', justifyContent: 'center',
245         color: 'var(--text-muted)',
246         cursor: 'pointer', transition: 'all 0.2s',
247         background: 'transparent',
248       }}>
249         <Copy size={16} />
250       </button>
251     </div>
252   </div>
253 </div>
254 </motion.div>
255 }})
256 </div>
257 }}
258 </main>
259 </div>

```

```
260 | )
261 | }
262 |
```

File: app/chan-doan-pc/page.tsx (626 lines, 32.8 KB)

```
1 | 'use client'
2 |
3 | import { useState, useRef, useEffect } from 'react'
4 | import { useRouter } from 'next/navigation'
5 | import { useIsMobile } from '@hooks/useIsMobile'
6 | import { motion, AnimatePresence } from 'framer-motion'
7 | import dynamic from 'next/dynamic'
8 | import {
9 |   ArrowLeft, Cpu, Wrench, RefreshCw, CheckCircle2, AlertTriangle,
10 |   HelpCircle, Loader2, Monitor, Zap, Thermometer, Wifi, HardDrive,
11 |   Keyboard, Mouse, Users, Shield, Puzzle, Star, BookOpen,
12 |   Toolbox, Disc, Radio, Gauge, Gamepad2, Bot
13 | } from 'lucide-react'
14 |
15 | const DiagnosisMode = dynamic(
16 |   () => import('@components/DiagnosisMode'),
17 |   { ssr: false, loading: () => <div style={{ padding: '40px', textAlign: 'center', color: 'var(--text-muted)' } ...
18 | }
19 |
20 | interface QAStep {
21 |   q: string
22 |   a: string
23 | }
24 |
25 | interface DiagnosisResult {
26 |   problem: string
27 |   cause: string
28 |   fixSteps: string[]
29 |   partsNeeded: string[]
30 |   repairShop: string
31 |   xpReward: number
32 | }
33 |
34 | interface AiResponse {
35 |   type: 'question' | 'diagnosis'
36 |   message: string
37 |   options: string[]
38 |   diagnosis?: DiagnosisResult
39 |   model?: string
40 | }
41 |
42 | interface Scenario {
43 |   id: string
44 |   title: string
45 |   icon: string
46 |   description: string
47 |   difficulty: 'DÃ' | 'Trung bình' | 'Khó'
48 |   xpReward: number
49 |   symptoms: string
50 | }
51 |
52 | const SCENARIOS: Scenario[] = [
53 |   {
54 |     id: 'no_display',
55 |     title: 'MÃy không lên màn hình',
56 |     icon: 'ðŸ”Œ',
57 |     description: 'èn nguồn sáng, quit chiy nhøng màn hình en',
58 |     difficulty: 'Trung bình',
59 |     xpReward: 30,
60 |     symptoms: 'MÃy tnh không lên màn hình, có èn nguồn nhøng không bo
61 |   },
62 |   {
63 |     id: 'no_power',
64 |     title: 'MÃy không lên nguồn',
65 |     icon: '&i',
66 |     description: 'Không có èn, không quit, không phén éng gì',
67 |     difficulty: 'DÃ',
68 |     xpReward: 20,
69 |     symptoms: 'MÃy tnh không lên nguồn, không có èn và quit',
70 |   },
71 |   {
72 |     id: 'bsod',
73 |     title: 'Màn hình xanh (BSOD)',
74 |     icon: 'ðŸ”Œ',
75 |     description: 'MÃy bẽ xanh màn hình khi ang dùng',
76 |     difficulty: 'Khó',
77 |     xpReward: 40,
78 |     symptoms: 'MÃy bẽ màn hình xanh (BSOD) khi ang sí dăng',
79 |   },
80 |   {
81 |     id: 'overheating',
82 |     title: 'MÃy quá nhiệ',
83 |     icon: 'ðŸ”Œ',
84 |     description: 'Nhiệ ù CPU/GPU cao, quit kêu to',
85 |     difficulty: 'DÃ',
86 |     xpReward: 20,
87 |     symptoms: 'Nhiệ ù CPU cao, quit kêu to, máy nóng',
88 |   },
89 |   {
90 |     id: 'slow',
91 |     title: 'MÃy chiy ch-m',
92 |     icon: 'ðŸ”Œ',
93 |     description: 'MÃy bẽ treo, i, chiy ch-m bæt thøÝng',
94 |     difficulty: 'Trung bình',
95 |     xpReward: 30,
96 |     symptoms: 'MÃy tnh chiy ch-m, bẽ treo, i',
97 |   },
98 |   {
99 |     id: 'network',
100 |     title: 'MÃy t kút nÑi ming',
101 |     icon: 'ðŸ”Œ',
102 |     description: 'Không vào òãc internet, WiFi/LAN lxi',
103 |     difficulty: 'DÃ',
104 |     xpReward: 20,
105 |     symptoms: 'MÃy không kút nÑi òãc internet, wifi ho-c ming LAN',
106 |   },
107 |   {
108 |     id: 'auto_shutdown',
109 |     title: 'MÃy tñ t-t nguồn',
110 |     icon: 'ðŸ”Œ',
111 |     description: 'MÃy tñ t-t khi ang dùng ho-c khi chiy game',
112 |     difficulty: 'Khó',
113 |     xpReward: 40,
114 |     symptoms: 'MÃy tnh tñ t-t nguồn khi ang sí dăng',
115 |   },
116 |   {
117 |     id: 'no_boot',
```

```

118|     title: 'Máy không boot Windows',
119|     icon: '0=0',
120|     description: 'Máy vào BIOS ho·c logo hãng rói treo',
121|     difficulty: 'Khó',
122|     xpReward: 35,
123|     symptoms: 'Máy không boot °ác vào Windows',
124|   },
125|   {
126|     id: 'usb_not_recognized',
127|     title: 'USB không °ác nh·n',
128|     icon: '0=Y',
129|     description: 'C·m USB không th·y, ho·c báo l·i driver',
130|     difficulty: 'D·A',
131|     xpReward: 15,
132|     symptoms: 'USB c·m vào không °ác nh·n, không th·y ổ USB',
133|   },
134|   {
135|     id: 'ram_error',
136|     title: 'L·i RAM',
137|     icon: '0=Y',
138|     description: 'Máy báo l·i RAM, beep khi khi ùng',
139|     difficulty: 'Trung bình',
140|     xpReward: 30,
141|     symptoms: 'Máy báo l·i RAM, có tiếng bíp khi khi ùng',
142|   },
143|   {
144|     id: 'vga_error',
145|     title: 'L·i card màn hình',
146|     icon: '0=0',
147|     description: 'Màn hình có s·c, nh·p nháy, ho·c không có hình',
148|     difficulty: 'Khó',
149|     xpReward: 45,
150|     symptoms: 'Màn hình b· s·c, nh·p nháy ho·c không có tín hi·u t· card',
151|   },
152|   {
153|     id: 'keyboard_mouse',
154|     title: 'Bàn phím / Chuột không ho·t ùng',
155|     icon: '#(b',
156|     description: 'Bàn phím ho·c chuột không °ác nh·n',
157|     difficulty: 'D·A',
158|     xpReward: 15,
159|     symptoms: 'Bàn phím ho·c chuột không ho·t ùng',
160|   },
161| ]
162|
163| const TOOLS = [
164|   { icon: '< Gauge size={18} />', name: 'HWMonitor', desc: 'Theo dõi nh·t',
165|     icon: '< Disc size={18} />', name: 'CrystalDiskInfo', desc: 'Ki·m tra s·',
166|     icon: '< Radio size={18} />', name: 'MemTest86', desc: 'Ki·m tra l·i RA',
167|     icon: '< Shield size={18} />', name: 'Windows Defender', desc: 'Quét v·',
168|     icon: '< Cpu size={18} />', name: 'CPU-Z', desc: 'Xem thông tin CPU, RA',
169|     icon: '< Monitor size={18} />', name: 'GPU-Z', desc: 'Xem thông tin card',
170|     icon: '< Wrench size={18} />', name: 'DDU', desc: 'G· driver VGA t·n·',
171|     icon: '< Hard Drive size={18} />', name: 'HD Tune', desc: 'Ki·m tra bad
172| ]
173|
174| const MODEL_LABELS: Record<string, { label: string; color: string }> = {
175|   gemini: { label: 'Gemini 1.5 Flash', color: '#4285F4' },
176|   groq: { label: 'Groq Llama 3.3', color: '#f97316' },
177|   openai: { label: 'OpenAI', color: '#00A67E' },
178|   huggingface: { label: 'HuggingFace', color: '#FFD21E' },
179|   'rule-based': { label: 'Local Engine', color: '#f59e0b' },
180| }
181|
182| const TOTAL_XP = SCENARIOS.reduce((sum, s) => sum + s.xpReward, 0)
183|
184| export default function ChanDoanPcPage() {
185|   const router = useRouter()
186|   const isMobile = useIsMobile()
187|   const [mode, setMode] = useState<'ai' | 'game'>('ai')
188|   const [symptoms, setSymptoms] = useState('')
189|   const [history, setHistory] = useState<QAStep[]>([])
190|   const [currentMessage, setCurrentMessage] = useState('')
191|   const [options, setOptions] = useState<string[]>([])
192|   const [diagnosis, setDiagnosis] = useState<DiagnosisResult | null>(null)
193|   const [loading, setLoading] = useState(false)
194|   const [started, setStarted] = useState(false)
195|   const [activeModel, setActiveModel] = useState<string>('')
196|   const [selectedScenario, setSelectedScenario] = useState<Scenario | null>(null)
197|   const [showScenarioPicker, setShowScenarioPicker] = useState(true)
198|   const [showToolbox, setShowToolbox] = useState(false)
199|   const [completedCount, setCompletedCount] = useState(0)
200|   const [earnedXp, setEarnedXp] = useState(0)
201|   const [showXpAnimation, setShowXpAnimation] = useState(false)
202|   const [lastXpAmount, setLastXpAmount] = useState(0)
203|   const bottomRef = useRef<HTMLDivElement>(null)
204|
205|   useEffect(() => { bottomRef.current?.scrollIntoView({ behavior: 'smooth' }) }, [history, currentMessage])
206|
207|   async function startDiagnosis(scenario?: Scenario) {
208|     const symptomText = scenario ? scenario.symptoms : symptoms
209|     if (!symptomText.trim()) return
210|     setLoading(true)
211|     setStarted(true)
212|     setSelectedScenario(scenario || null)
213|     setActiveModel('')
214|     try {
215|       const res = await fetch('/api/ai/diagnose-pc', {
216|         method: 'POST',
217|         headers: { 'Content-Type': 'application/json' },
218|         body: JSON.stringify({ symptoms: symptomText.trim(), answers: [] })
219|       })
220|       const data: AiResponse = await res.json()
221|       setActiveModel(data.model || '')
222|       if (data.type === 'diagnosis' && data.diagnosis) {
223|         data.diagnosis.xpReward = scenario?.xpReward || 25
224|         setDiagnosis(data.diagnosis)
225|         setCurrentMessage(data.message)
226|         setOptions([])
227|         awardXp(data.diagnosis.xpReward)
228|       } else {
229|         setCurrentMessage(data.message)
230|         setOptions(data.options || [])
231|       }
232|     } catch { setCurrentMessage('L·i k·t n·i. Vui lòng thí l·i.') }
233|     finally { setLoading(false) }
234|   }
235|
236|   async function handleOption(option: string) {
237|     const newHistory = [...history, { q: currentMessage, a: option }]
238|     setHistory(newHistory)
239|     setLoading(true)
240|     setCurrentMessage('')
241|     setOptions([])

```

```

242 | setActiveModel('')
243 | try {
244 |   const res = await fetch('/api/ai/diagnose-pc', {
245 |     method: 'POST',
246 |     headers: { 'Content-Type': 'application/json' },
247 |     body: JSON.stringify({ symptoms: selectedScenario?.symptoms || symptoms, answers: newHistory })
248 |   })
249 |   const data: AiResponse = await res.json()
250 |   setActiveModel(data.model || '')
251 |   if (data.type === 'diagnosis' && data.diagnosis) {
252 |     data.diagnosis.xpReward = selectedScenario?.xpReward || 25
253 |     setDiagnosis(data.diagnosis)
254 |     setCurrentMessage(data.message)
255 |     setOptions([])
256 |     awardXp(data.diagnosis.xpReward)
257 |   } else {
258 |     setCurrentMessage(data.message)
259 |     setOptions(data.options || [])
260 |   }
261 | } catch { setCurrentMessage('L x i k z t n ñ i . Vui lòng thí l i i .') }
262 | finally { setLoading(false) }
263 | }
264 |
265 | function awardXp(amount: number) {
266 |   setLastXpAmount(amount)
267 |   setEarnedXp(prev => prev + amount)
268 |   setCompletedCount(prev => prev + 1)
269 |   setShowXpAnimation(true)
270 |   setTimeout(() => setShowXpAnimation(false), 2500)
271 | }
272 |
273 | function reset() {
274 |   setStarted(false)
275 |   setHistory([])
276 |   setCurrentMessage('')
277 |   setOptions([])
278 |   setDiagnosis(null)
279 |   setSymptoms('')
280 |   setActiveModel('')
281 |   setSelectedScenario(null)
282 |   setShowScenarioPicker(true)
283 | }
284 |
285 | function selectScenario(scenario: Scenario) {
286 |   setShowScenarioPicker(false)
287 |   setSelectedScenario(scenario)
288 |   setSymptoms(scenario.symptoms)
289 |   startDiagnosis(scenario)
290 | }
291 |
292 | const modelInfo = activeModel ? MODEL_LABELS[activeModel] : null
293 |
294 | return (
295 |   <div className="min-h-screen flex flex-col" style={{ background: 'var(--bg-base)', color: 'var(--text-prim...
296 |     /* XP Banner */
297 |     <AnimatePresence>
298 |       {showXpAnimation && (
299 |         <motion.div
300 |           initial={{ y: -80, opacity: 0 }}
301 |           animate={{ y: 0, opacity: 1 }}
302 |           exit={{ y: -80, opacity: 0 }}
303 |           style={{
304 |             position: 'fixed', top: 0, left: 0, right: 0, zIndex: 9999,
305 |             background: 'linear-gradient(135deg, #f59e0b, #eab308)',
306 |             color: '#000', textAlign: 'center', padding: '10px',
307 |             fontWeight: 800, fontSize: '16px',
308 |             boxShadow: '0 4px 20px rgba(245,158,11,0.4)',
309 |           }}
310 |         >
311 |           <Star size={18} style={{ display: 'inline', marginRight: '8px', verticalAlign: 'middle' }} />
312 |           {lastXpAmount} XP! Ch u o n o a n h o a n t ̣ t !
313 |         </motion.div>
314 |       )}
315 |     </AnimatePresence>
316 |
317 |     <header className="border-b p-4 flex items-center gap-3" style={{ background: 'var(--bg-surface)', borde...
318 |       <button onClick={() => router.back()}
319 |         className="flex items-center gap-1.5 px-3 py-1.5 border text-xs rounded-xl font-bold cursor-pointer"
320 |         style={{ background: 'var(--bg-elevated)', borderColor: 'var(--border-default)', color: 'var(--text-...
321 |         < A r r o w L e f t   s i z e = { 1 4 }   />   Q u a y   l i i
322 |       </button>
323 |       <div style={{ width: '32px', height: '32px', borderRadius: '10px', background: 'linear-gradient(135deg...
324 |         {mode === 'ai' ? <Bot size={16} style={{ color: '#000' }} /> : <Gamepad2 size={16} style={{ color: '...
325 |       </div>
326 |     </div>
327 |     < h 1   c l a s s N a m e = " f o n t - b l a c k   t e x t - s m "   s t y l e = { {   c o l o r :   ' v a r ( - - t e x t
328 |     < p   c l a s s N a m e = " t e x t - x s "   s t y l e = { {   c o l o r :   ' v a r ( - - t e x t - m u t e d ) '   } } >
329 |   </div>
330 |   <div style={{ marginLeft: 'auto', display: 'flex', gap: '4px', background: 'var(--bg-elevated)', borde...
331 |     <button onClick={() => setMode('ai')}
332 |       style={{ padding: '4px 10px', borderRadius: '8px', border: 'none', background: mode === 'ai' ? 'va...
333 |       <Bot size={14} style={{ display: 'inline', verticalAlign: 'middle', marginRight: '4px' }} /> AI
334 |     </button>
335 |     <button onClick={() => setMode('game')}
336 |       style={{ padding: '4px 10px', borderRadius: '8px', border: 'none', background: mode === 'game' ? '...
337 |       <Gamepad2 size={14} style={{ display: 'inline', verticalAlign: 'middle', marginRight: '4px' }} /> ...
338 |     </button>
339 |   </div>
340 |   {mode === 'ai' && modelInfo && (
341 |     <div style={{ display: 'flex', alignItems: 'center', gap: '6px', padding: '4px 10px', borderRadius: ...
342 |       {modelInfo.label}
343 |     </div>
344 |   )}
345 |   {mode === 'ai' && (
346 |     <button onClick={() => setShowToolbox(true)}
347 |       className="flex items-center gap-1.5 px-3 py-1.5 border text-xs rounded-xl font-bold cursor-pointer"
348 |       style={{ background: 'var(--bg-elevated)', borderColor: 'var(--border-default)', color: 'var(--tex...
349 |       < T o o l b o x   s i z e = { 1 4 }   />   H ù p   d a n g   c á
350 |     </button>
351 |   )}
352 | </header>
353 |
354 | <div className="flex-1 overflow-y-auto p-4 max-w-3xl mx-auto w-full">
355 |   {mode === 'game' ? (
356 |     <div style={{ marginTop: '-16px' }}>
357 |       <DiagnosisMode
358 |         lang="vn"
359 |         onComplete={(results) => { console.log('Diagnosis complete:', results); setMode('ai') }}
360 |         onExit={() => setMode('ai')}
361 |       />
362 |     </div>
363 |   ) : !started ? (
364 |     <motion.div initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }}
365 |       style={{ padding: '20px' }}>

```

```

366 | /* XP Progress */
367 | <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '12px', margin...
368 |   <div style={{ display: 'flex', alignItems: 'center', gap: '4px' }}>
369 |     <Star size={14} style={{ color: '#f59e0b' }} />
370 |     <span><strong style={{ color: '#f59e0b' }}>{earnedXp}</strong> / {TOTAL_XP} XP</span>
371 |   </div>
372 |   <span>|</span>
373 |   <span>   hoàn thành: <strong>{completedCount}</strong>   ca
374 | </div>
375 |
376 | /* Scenario Picker */
377 | {showScenarioPicker && (
378 |   <
379 |     <div style={{ textAlign: 'center', marginBottom: '24px' }}>
380 |       <div style={{
381 |         <h2 style={{
382 |           <p style={{ color: 'var(--text-muted)', fontSize: '13px'
383 |
384 |
385 |     <div style={{ display: 'grid', gridTemplateColumns: isMobile ? '1fr' : '1fr 1fr', gap: '8px', ...
386 |       {SCENARIOS.map((scenario) => (
387 |         <motion.button
388 |           key={scenario.id}
389 |           whileHover={{ scale: 1.02 }}
390 |           whileTap={{ scale: 0.98 }}
391 |           onClick={() => selectScenario(scenario)}
392 |           style={{
393 |             display: 'flex', alignItems: 'center', gap: '10px',
394 |             padding: '12px 14px', borderRadius: '12px',
395 |             border: '1px solid var(--border-default)',
396 |             background: 'var(--bg-surface)',
397 |             color: 'var(--text-primary)',
398 |             cursor: 'pointer', fontSize: '12px',
399 |             fontFamily: 'inherit', textAlign: 'left',
400 |             transition: 'all 0.15s',
401 |           }}
402 |         >
403 |           <span style={{ fontSize: '24px' }}>{scenario.icon}</span>
404 |           <div style={{ flex: 1, minWidth: 0 }}>
405 |             <div style={{ fontWeight: 700, fontSize: '12px', marginBottom: '2px' }}>{scenario.titl...
406 |             <div style={{ fontSize: '10px', color: 'var(--text-muted)' }}>{scenario.description}</...
407 |             <div style={{ display: 'flex', gap: '6px', marginTop: '4px' }}>
408 |               <span style={{
409 |                 fontSize: '9px', padding: '1px 6px', borderRadius: '4px',
410 |                 background: scenario.difficulty === 'D A' ?
411 |                 color: scenario.difficulty === 'D A' ? '#10b9
412 |               }}>
413 |                 {scenario.difficulty}
414 |               </span>
415 |               <span style={{ fontSize: '9px', padding: '1px 6px', borderRadius: '4px', background:...
416 |                 +{scenario.xpReward} XP
417 |               </span>
418 |             </div>
419 |           </div>
420 |         </motion.button>
421 |       )}}
422 |     </div>
423 |
424 |     <div style={{ position: 'relative', marginBottom: '24px' }}>
425 |       <div style={{ display: 'flex', alignItems: 'center', gap: '8px', marginBottom: '8px' }}>
426 |         <div style={{ flex: 1, height: '1px', background: 'var(--border-default)' }} />
427 |         <span style={{ fontSize: '11px', color: 'var(--text-
428 |       </div>
429 |     </div>
430 |
431 |     <div style={{ display: 'flex', gap: '8px', maxWidth: '600px', margin: '0 auto' }}>
432 |       <input value={symptoms} onChange={e => setSymptoms(e.target.value)}
433 |         onKeyDown={e => e.key === 'Enter' && startDiagnosis()}
434 |         placeholder="Nhập triệu chứng (VD: máy tính không l
435 |         style={{ flex: 1, padding: '14px 18px', borderRadius: '14px', border: '1px solid var(--bor...
436 |       </div>
437 |       <button onClick={() => startDiagnosis()} disabled={!symptoms.trim() || loading}
438 |         style={{ padding: '14px 28px', borderRadius: '14px', border: 'none', background: !symptoms...
439 |         {loading ? <Loader2 size={18} style={{ animation: 's
440 |       </button>
441 |     </div>
442 |   </div>
443 |
444 |   </motion.div>
445 | ) : (
446 |   <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
447 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
448 |       <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
449 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
450 |           <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
451 |             <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
452 |               <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
453 |                 <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
454 |                   <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
455 |                     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
456 |                       <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
457 |                         <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
458 |                           <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
459 |                             <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
460 |                               <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
461 |                                 <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
462 |                                   <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
463 |                                     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
464 |                                       <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
465 |                                         <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
466 |                                           <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
467 |                                             <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
468 |                                               <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
469 |                                                 <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
470 |                                                   <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
471 |                                                     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
472 |                                                       <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
473 |                                                         <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
474 |                                                           <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
475 |                                                             <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
476 |                                                               <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
477 |                                                                 <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
478 |                                                                   <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
479 |                                                                     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
480 |                                                                       <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
481 |                                                                         <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
482 |                                                                           <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
483 |                                                                             <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
484 |                                                                               <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
485 |                                                                                 <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
486 |                                                                                   <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
487 |                                                                                     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
488 |                                                                                       <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>
489 |                                                                                         <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', padding: '16px' }}>

```



```

490 | <div style={{ padding: '24px', borderRadius: '16px', background: 'var(--bg-surface)', border...
491 | <div style={{ display: 'flex', alignItems: 'center', gap: '10px', marginBottom: '20px' }}>
492 | <div style={{ width: '40px', height: '40px', borderRadius: '12px', background: 'rgba(16,...
493 | <CheckCircle2 size={22} style={{ color: '#10b981' }} />
494 | </div>
495 | <div>
496 | <h3 style={{ fontSize: '16px', fontWeight: 800,
497 | </div>
498 | </div>
499 | </div>
500 |
501 | <p style={{ fontSize: '14px', color: 'var(--text-primary)', marginBottom: '16px', lineHeig...
502 |
503 | <div style={{ marginBottom: '16px' }}>
504 | <h4 style={{ fontSize: '13px', fontWeight: 700, color: 'var(--brand-primary)', marginBot...
505 | <p style={{ fontSize: '13px', color: 'var(--text-primary)', lineHeight: '1.5' }}>{diagno...
506 | </div>
507 |
508 | {diagnosis.fixSteps.length > 0 && (
509 | <div style={{ marginBottom: '16px' }}>
510 | <h4 style={{ fontSize: '13px', fontWeight: 700, color: 'var(--brand-primary)', marginB...
511 | <div style={{ display: 'flex', flexDirection: 'column', gap: '6px' }}>
512 | {diagnosis.fixSteps.map((step, i) => (
513 | <div key={i} style={{ display: 'flex', gap: '10px', padding: '10px 14px', borderRa...
514 | <span style={{ width: '22px', height: '22px', borderRadius: '6px', background: '...
515 | <span style={{ color: 'var(--text-primary)' }}>{step}</span>
516 | </div>
517 | ))}
518 | </div>
519 | </div>
520 | ))}
521 |
522 | {diagnosis.partsNeeded.length > 0 && (
523 | <div style={{ marginBottom: '16px', padding: '12px 16px', borderRadius: '10px', backgrou...
524 | <h4 style={{ fontSize: '13px', fontWeight: 700, color: '#f59e0b', marginBottom: '6px' }}...
525 | {diagnosis.partsNeeded.map((p, i) => (
526 | <div key={i} style={{ fontSize: '13px', color: 'var(--text-primary)', padding: '2px ...
527 | ))}
528 | </div>
529 | ))}
530 |
531 | <div style={{ padding: '12px 16px', borderRadius: '10px', background: 'rgba(99,102,241,0.0...
532 | <h4 style={{ fontSize: '13px', fontWeight: 700, color: '#818cf8', marginBottom: '4px' }}...
533 | <p style={{ fontSize: '13px', color: 'var(--text-primary)', lineHeight: '1.5' }}>{diagno...
534 | </div>
535 |
536 | <div style={{ display: 'flex', gap: '8px', marginTop: '20px' }}>
537 | <button onClick={reset}
538 | style={{ display: 'flex', alignItems: 'center', gap: '6px', padding: '12px 24px', bord...
539 | </button>
540 | <button onClick={() => setShowToolbox(true)}
541 | style={{ display: 'flex', alignItems: 'center', gap: '6px', padding: '12px 24px', bord...
542 | </button>
543 | </div>
544 | </div>
545 | ) : (
546 | <div style={{ padding: '16px', borderRadius: '14px', background: 'var(--bg-surface)', border...
547 | <div style={{ display: 'flex', alignItems: 'center', gap: '8px', marginBottom: '12px' }}>
548 | <HelpCircle size={16} style={{ color: 'var(--brand-primary)', flexShrink: 0 }} />
549 | <span style={{ fontSize: '14px', color: 'var(--text-primary)', lineHeight: '1.6' }}>{cur...
550 | </div>
551 | {options.length > 0 && (
552 | <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
553 | {options.map((opt, i) => (
554 | <button key={i} onClick={() => handleOption(opt)}
555 | style={{ padding: '12px 16px', borderRadius: '10px', border: '1px solid var(--bord...
556 | onMouseOver={e => { e.currentTarget.style.borderColor = 'var(--brand-primary)'; e...
557 | onMouseOut={e => { e.currentTarget.style.borderColor = 'var(--border-default)'; e...
558 | {opt}
559 | </button>
560 | ))}
561 | </div>
562 | ))}
563 | </div>
564 | ))}
565 | </div>
566 | </div>
567 | </div>
568 | </div>
569 | <div ref={bottomRef} />
570 | </div>
571 | </div>
572 |
573 | <div>
574 | <div>
575 | <div>
576 | <div>
577 | <div>
578 | <div>
579 | <div>
580 | <div>
581 | <div>
582 | <div>
583 | <div>
584 | <div>
585 | <div>
586 | <div>
587 | <div>
588 | <div>
589 | <div>
590 | <div>
591 | <div>
592 | <div>
593 | <div>
594 | <div>
595 | <div>
596 | <div>
597 | <div>
598 | <div>
599 | <div>
600 | <div>
601 | <div>
602 | <div>
603 | <div>
604 | <div>
605 | <div>
606 | <div>
607 | <div>
608 | <div>
609 | <div>
610 | <div>
611 | <div>
612 | <div>
613 | <div>

```

```
614 |         <div style={{ marginTop: '16px', padding: '12px 14px', borderRadius: '10px', background: 'rgba(2...  
615 |             <strong style={{ color: '#f59e0b' }}>Pro tip:</strong>  
616 |         </div>  
617 |         </motion.div>  
618 |     </motion.div>  
619 |     )}  
620 | </AnimatePresence>  
621 |  
622 | <style>{@keyframes spin { to { transform: rotate(360deg) } } @keyframes pulse { 0%,100% { opacity:1 } 5...  
623 | </div>  
624 | )  
625 | }  
626 |
```

File: app/component-lab/page.tsx (839 lines, 51.0 KB)

```
1 | 'use client'  
2 |  
3 | import React, { useState, useMemo } from 'react'  
4 | import { motion, AnimatePresence } from 'framer-motion'  
5 | import {  
6 |   Search, Zap, Cpu, Monitor, HardDrive, CircuitBoard,  
7 |   Fan, Wifi, Battery, Speaker, Shield, Radio, Smartphone,  
8 |   MousePointer2, Keyboard, Headphones, Webcam, Eye,  
9 |   Crosshair, Swords, Diamond, Heart, Star, Trophy,  
10 |   ChevronRight, ChevronLeft, Layers, CheckCircle, XCircle,  
11 |   BookOpen, Beaker, FlaskConical, GraduationCap,  
12 | } from 'lucide-react'  
13 |  
14 | interface Spec {  
15 |   label: string  
16 |   value: string  
17 | }  
18 |  
19 | interface Component {  
20 |   id: string  
21 |   name: string  
22 |   icon: string  
23 |   desc: string  
24 |   role: string  
25 |   specs: Spec[]  
26 |   brands: string[]  
27 |   tips: string  
28 | }  
29 |  
30 | interface Category {  
31 |   id: string  
32 |   name: string  
33 |   icon: React.ReactNode  
34 |   color: string  
35 |   items: Component[]  
36 | }  
37 |  
38 | const COMPONENT_DATA: Category[] = [  
39 |   {  
40 |     id: 'cpu', name: 'CPU', color: '#3b82f6',  
41 |     icon: <Cpu size={20} />,  
42 |     items: [  
43 |       { id: 'cpu-intel', name: 'Intel Core i9-14900K', icon: '&i', desc:  
44 |         specs: [  
45 |           { label: 'S\u0129nh\u0103n / Lu\u00f3ng', value: '24 nh\u0103n (8P+16E) / 32 lu\u00f3ng' },  
46 |           { label: 'Xung nh\u0119p ci b\u00ean', value: '3.2 GHz (P-core) / 2.4 GHz' },  
47 |           { label: 'Xung nh\u0119p Boost', value: 'L\u00ean \u0111\u00e9n 6.0 GHz' },  
48 |           { label: 'Cache L3', value: '36 MB Intel Smart Cache' },  
49 |           { label: 'TDP', value: '125W (ci b\u00ean) / 253W (t\u0129i a)' },  
50 |           { label: 'Socket', value: 'LGA 1700' },  
51 |           { label: 'H\u00e1t r\u00e1 RAM', value: 'DDR5-5600 / DDR4-3200' },  
52 |           { label: 'PCIe', value: 'PCIe 5.0 (16 lane)' },  
53 |           { label: 'Ki\u00ean tr\u01b0c', value: 'Raptor Lake Refresh' },  
54 |           { label: 'T\u00edch h\u00e1p GPU', value: 'Intel UHD Graphics 770' },  
55 |         ],  
56 |         brands: ['Intel', 'AMD'],  
57 |         tips: 'Ch\u00edn CPU \u0111\u00e2a tr\u00ean nhu c\u1eadu: Intel minh gaming i\u0111n lu\u00f3ng, A  
58 |       { id: 'cpu-amd', name: 'AMD Ryzen 7 7800X3D', icon: '\u00d7%', desc: 'O  
59 |         specs: [  
60 |           { label: 'S\u0129nh\u0103n / Lu\u00f3ng', value: '8 nh\u0103n / 16 lu\u00f3ng' },  
61 |           { label: 'Xung nh\u0119p ci b\u00ean', value: '4.2 GHz' },  
62 |           { label: 'Xung nh\u0119p Boost', value: 'L\u00ean \u0111\u00e9n 5.0 GHz' },  
63 |           { label: 'Cache L3', value: '96 MB (3D V-Cache)' },  
64 |           { label: 'TDP', value: '120W' },  
65 |           { label: 'Socket', value: 'AM5' },  
66 |           { label: 'H\u00e1t r\u00e1 RAM', value: 'DDR5-5200' },  
67 |           { label: 'PCIe', value: 'PCIe 5.0 (24 lane)' },  
68 |           { label: 'Ki\u00ean tr\u01b0c', value: 'Zen 4' },  
69 |           { label: 'T\u00edch h\u00e1p GPU', value: 'AMD Radeon Graphics (2 nh\u0103n)' },  
70 |         ],  
71 |         brands: ['AMD', 'Intel'],  
72 |         tips: 'CPU AMD AM5 h\u00e1t r\u00e1 l\u00e2u d\u00e0i - c\u00f3 th\u00e3 nh\u00e1ng c\u00e9p l\u00ean Zen 6+ t  
73 |       { id: 'cpu-budget', name: 'Intel Core i5-14600K', icon: '\u00d70a', desc:  
74 |         specs: [  
75 |           { label: 'S\u0129nh\u0103n / Lu\u00f3ng', value: '14 nh\u0103n (6P+8E) / 20 lu\u00f3ng' },  
76 |           { label: 'Xung nh\u0119p Boost', value: 'L\u00ean \u0111\u00e9n 5.3 GHz' },  
77 |           { label: 'Cache L3', value: '24 MB' },  
78 |           { label: 'TDP', value: '125W / 181W t\u0129i a' },  
79 |           { label: 'Socket', value: 'LGA 1700' },  
80 |           { label: 'H\u00e1t r\u00e1 RAM', value: 'DDR5 / DDR4' },  
81 |         ],  
82 |         brands: ['Intel'],  
83 |         tips: 'i5-14600K l\u00e0 "ng\u00edt ng\u00e0o nh\u00e9t" cho gaming - hi\u00eau n\u00e1ng s\u0129n  
84 |       ],  
85 |     },  
86 |     {  
87 |       id: 'gpu', name: 'GPU', color: '#22c55e',  
88 |       icon: <Monitor size={20} />,  
89 |       items: [  
90 |         { id: 'gpu-nvidia', name: 'NVIDIA RTX 4090', icon: '\u00d7\u00e1\u00e1', desc: 'Ca  
91 |         specs: [  
92 |           { label: 'Nh\u0103n CUDA', value: '16.384 CUDA Cores' },  
93 |           { label: 'VRAM', value: '24 GB GDDR6X' },  
94 |           { label: 'Xung nh\u0119p Boost', value: '2.52 GHz' },  
95 |           { label: 'B\u00e1ng th\u00f4ng', value: '1.008 GB/s' },  
96 |           { label: 'TDP', value: '450W' },  
97 |           { label: 'Ray Tracing', value: 'Th\u00e1 h\u00e1c 3 (RT Cores)' },  
98 |           { label: 'DLSS', value: '3.5 (Frame Generation)' },  
99 |           { label: 'K\u00e9t n\u0129i', value: 'PCIe 4.0 x16' },  
100 |          { label: 'HDMI/DP', value: 'HDMI 2.1 + 3x DP 1.4a' },  
101 |          { label: 'K\u00edch th\u00f9c', value: '3.5 slot, 304mm' },  
102 |        ],  
103 |        brands: ['NVIDIA', 'ASUS', 'MSI', 'Gigabyte'],  
104 |        tips: 'RTX 4090 l\u00e0 card minh nh\u00e9t nh\u00f3ng c\u1eadu ngu\u00f3n 850W+ v\u00e0 case  
105 |        { id: 'gpu-amd', name: 'AMD Radeon RX 7900 XTX', icon: '\u00d7Y4', desc:  
106 |        specs: [  
107 |          { label: 'Nh\u0103n Stream', value: '6.144 Stream Processors' },
```

```
108|         { label: 'VRAM', value: '24 GB GDDR6' },
109|         { label: 'Xung nhịp p', value: 'Lên 2.5 GHz' },
110|         { label: 'Bng thông', value: '960 GB/s' },
111|         { label: 'TDP', value: '355W' },
112|         { label: 'Ray Tracing', value: 'Thế hệ 2' },
113|         { label: 'FSR', value: '3.0' },
114|         { label: 'Kế t nĩi', value: 'PCIe 4.0 x16' },
115|     ],
116|     brands: ['AMD', 'ASUS', 'Sapphire'],
117|     tips: 'AMD RX 7900 XTX có VRAM 24GB giá r» hin NVIDIA. Phù hãp c',
118|     { id: 'gpu-mid', name: 'NVIDIA RTX 4070 Super', icon: '0=Üš', desc: 'GPU' },
119|     specs: [
120|         { label: 'Nhân CUDA', value: '7.168 CUDA Cores' },
121|         { label: 'VRAM', value: '12 GB GDDR6X' },
122|         { label: 'TDP', value: '220W' },
123|         { label: 'Ray Tracing', value: 'Thế hệ 3' },
124|         { label: 'DLSS', value: '3.5' },
125|         { label: 'Giá tham khảo', value: '~600 USD' },
126|     ],
127|     brands: ['NVIDIA', 'ASUS', 'MSI'],
128|     tips: 'RTX 4070 Super là lĩa chín tĩt nhý t cho gaming 1440P vũ i',
129| },
130| },
131| {
132|     id: 'ram', name: 'RAM', color: '#f59e0b',
133|     icon: <Zap size={20} />,
134|     items: [
135|         { id: 'ram-ddr5', name: 'DDR5-6000 32GB (2x16GB)', icon: '0>Yà', desc: 'RAM' },
136|         specs: [
137|             { label: 'Dung l°ăng', value: '32 GB (2 x 16 GB)' },
138|             { label: 'Loi i RAM', value: 'DDR5' },
139|             { label: 'Tĩc ù', value: '6000 MT/s' },
140|             { label: 'CAS Latency', value: 'CL30-36-36-76' },
141|             { label: 'iÇn áp', value: '1.35V' },
142|             { label: 'Heat Spreader', value: 'Nhóm t£n nhiçt cao c¥p' },
143|             { label: 'RGB', value: 'H× trã (tùy phiên b£n)' },
144|             { label: 'XMP / EXPO', value: 'Intel XMP 3.0 / AMD EXPO' },
145|             { label: 'Bng thông', value: '48 GB/s' },
146|             { label: 'Kênh', value: 'Dual Channel (kênh òi)' },
147|         ],
148|         brands: ['G.Skill', 'Corsair', 'Kingston', 'TeamGroup'],
149|         tips: 'Chín RAM có CL càng th¥p càng tĩt. DDR5-6000 CL30 là "sweet spot"',
150|         { id: 'ram-ddr4', name: 'DDR4-3600 16GB (2x8GB)', icon: '0=Ü¼', desc: 'RAM' },
151|         specs: [
152|             { label: 'Dung l°ăng', value: '16 GB (2 x 8 GB)' },
153|             { label: 'Loi i RAM', value: 'DDR4' },
154|             { label: 'Tĩc ù', value: '3600 MT/s' },
155|             { label: 'CAS Latency', value: 'CL16' },
156|             { label: 'iÇn áp', value: '1.35V' },
157|             { label: 'Kênh', value: 'Dual Channel' },
158|         ],
159|         brands: ['Corsair', 'G.Skill', 'Kingston'],
160|         tips: 'DDR4 v«n r¥t tĩt cho gaming. 16GB là ç cho hũu hçt game',
161|     ],
162| },
163| {
164|     id: 'motherboard', name: 'Mainboard', color: '#8b5cf6',
165|     icon: <CircuitBoard size={20} />,
166|     items: [
167|         { id: 'mb-atx', name: 'ASUS ROG Z790-E Gaming', icon: '0=Páp', desc: 'Mainboard' },
168|         specs: [
169|             { label: 'Form Factor', value: 'ATX' },
170|             { label: 'Chipset', value: 'Intel Z790' },
171|             { label: 'Socket', value: 'LGA 1700' },
172|             { label: 'Khe RAM', value: '4x DDR5 (tĩi a 192GB)' },
173|             { label: 'Khe PCIe', value: '2x PCIe 5.0 x16, 1x PCIe 4.0 x16' },
174|             { label: 'Khe M.2', value: '5x M.2 NVMe' },
175|             { label: 'Cõng SATA', value: '6x SATA III' },
176|             { label: 'USB', value: 'USB 3.2 Gen 2x2 Type-C' },
177|             { label: 'WiFi', value: 'WiFi 6E + Bluetooth 5.3' },
178|             { label: 'Âm thanh', value: 'Realtek ALC4080 7.1' },
179|         ],
180|         brands: ['ASUS', 'MSI', 'Gigabyte', 'ASRock'],
181|         tips: 'Chín mainboard dĩa trên CPU: Z790 cho Intel, B650/X670 cho AMD',
182|         { id: 'mb-matx', name: 'MSI MAG B650M Mortar', icon: '0=Y', desc: 'Mainboard' },
183|         specs: [
184|             { label: 'Form Factor', value: 'Micro-ATX' },
185|             { label: 'Chipset', value: 'AMD B650' },
186|             { label: 'Socket', value: 'AM5' },
187|             { label: 'Khe RAM', value: '4x DDR5' },
188|             { label: 'Khe PCIe', value: '1x PCIe 4.0 x16' },
189|             { label: 'Khe M.2', value: '2x M.2' },
190|             { label: 'WiFi', value: 'WiFi 6E' },
191|         ],
192|         brands: ['MSI', 'ASUS', 'Gigabyte'],
193|         tips: 'Mainboard B650 là lĩa chín tĩt nhý t cho AMD tsm trung, c',
194|     ],
195| },
196| {
197|     id: 'storage', name: 'L°u trĩ', color: '#06b6d4',
198|     icon: <HardDrive size={20} />,
199|     items: [
200|         { id: 'ssd-nvme', name: 'Samsung 990 Pro 2TB', icon: '0=P€', desc: 'SSD' },
201|         specs: [
202|             { label: 'Dung l°ăng', value: '2 TB' },
203|             { label: 'Loi i', value: 'NVMe M.2 PCIe 4.0' },
204|             { label: 'Tĩc ù', value: '7.450 MB/s' },
205|             { label: 'Tĩc ù Ghi', value: '6.900 MB/s' },
206|             { label: 'TBW', value: '1.200 TB' },
207|             { label: 'NAND', value: 'Samsung V-NAND 3-bit MLC' },
208|             { label: 'DRAM Cache', value: '2 GB LPDDR4' },
209|             { label: 'B£o hành h', value: '5 nm' },
210|             { label: 'Kĩch th°c', value: 'M.2 2280' },
211|             { label: 'Controller', value: 'Samsung Pascal' },
212|         ],
213|         brands: ['Samsung', 'Western Digital', 'SK Hynix', 'Crucial'],
214|         tips: 'Luôn cài Windows và game trên SSD NVMe. Dung l°ăng 1TB là',
215|         { id: 'ssd-sata', name: 'Crucial MX500 1TB', icon: '0=Üç', desc: 'SSD' },
216|         specs: [
217|             { label: 'Dung l°ăng', value: '1 TB' },
218|             { label: 'Loi i', value: 'SATA III 2.5 inch' },
219|             { label: 'Tĩc ù', value: '560 MB/s' },
220|             { label: 'Tĩc ù Ghi', value: '510 MB/s' },
221|             { label: 'TBW', value: '360 TB' },
222|             { label: 'B£o hành h', value: '5 nm' },
223|         ],
224|         brands: ['Crucial', 'Samsung', 'WD'],
225|         tips: 'Dùng SSD SATA cho l°u trĩ di liçu, game ci. NVMe cho game',
226|         { id: 'hdd', name: 'Seagate Barracuda 4TB', icon: '0=ÜÀ', desc: 'HDD' },
227|         specs: [
228|             { label: 'Dung l°ăng', value: '4 TB' },
229|             { label: 'Loi i', value: 'HDD 3.5 inch' },
230|             { label: 'Tĩc ù quay', value: '5.400 RPM' },
231|             { label: 'Tĩc ù', value: '~190 MB/s' },

```

```
232|         { label: 'Cache', value: '256 MB' },
233|         { label: 'Bềo hành', value: '2 n m' },
234|     ],
235|     brands: ['Seagate', 'Western Digital', 'Toshiba'],
236|     tips: 'HDD chế nên dùng cho backup dữ liệu, nhic, phim. Không cã',
237| },
238| },
239| {
240|     id: 'psu', name: 'Nguồn', color: '#ef4444',
241|     icon: <Battery size={20} />,
242|     items: [
243|         { id: 'psu-gold', name: 'Corsair RM850x', icon: '&i', desc: 'Nguồn',
244|             specs: [
245|                 { label: 'Công suất', value: '850W' },
246|                 { label: 'Chuồn hiệu suất', value: '80+', Gold (87-90%)' },
247|                 { label: 'Loại', value: 'Fully Modular' },
248|                 { label: 'Chuồn ATX', value: 'ATX 3.0 / PCIe 5.0' },
249|                 { label: 'Sạc - 12VHPWR', value: 'Có' },
250|                 { label: 'Quit', value: '135mm PWM' },
251|                 { label: 'Fanless Mode', value: 'Zero', RPM (tếi th¥p)' },
252|                 { label: 'Bềo hành', value: '10 n m' },
253|                 { label: 'Bềo vç', value: '0V P, 150 x 160 x 86mm' },
254|                 { label: 'Kích th°üc', value: '150 x 160 x 86mm' },
255|             ],
256|             brands: ['Corsair', 'Seasonic', 'EVGA', 'Cooler Master'],
257|             tips: 'Chín nguồn d°, công suất 20-30% so v°i nhu c§u. 850W cho',
258|             { id: 'psu-platinum', name: 'Seasonic PRIME TX-1000', icon: '0=Už',
259|                 specs: [
260|                     { label: 'Công suất', value: '1000W' },
261|                     { label: 'Chuồn hiệu suất', value: '80+', Titanium (94-96%)' },
262|                     { label: 'Loại', value: 'Fully Modular' },
263|                     { label: 'Bềo hành', value: '12 n m' },
264|                     { label: 'Chuồn ATX', value: 'ATX 3.0' },
265|                 ],
266|                 brands: ['Seasonic', 'Corsair'],
267|                 tips: 'Nguồn Titanium t hìn nh°ng tiэт kiệm iện và bán hìn.
268|             ],
269|             {
270|                 id: 'cooling', name: 'Tén nhiet', color: '#06b6d4',
271|                 icon: <Fan size={20} />,
272|                 items: [
273|                     { id: 'aio-360', name: 'AIO 360mm - NZXT Kraken X73', icon: 'Dp',
274|                         specs: [
275|                             { label: 'Loại', value: 'All-In-One Liquid Cooler' },
276|                             { label: 'Kích th°üc Radiator', value: '360mm x 120mm x 30mm' },
277|                             { label: 'Quit', value: '3x 120mm PWM' },
278|                             { label: 'Tñc ù quit', value: '500 - 2000 RPM' },
279|                             { label: 'Luống khí', value: '~78 CFM' },
280|                             { label: 'Luống', value: '21 - 36 dBA' },
281|                             { label: 'Pump', value: 'Asetek 7th gen' },
282|                             { label: 'T°ing thích', value: 'Intel LGA 1700/1200, AMD AM5/A',
283|                             { label: 'RGB', value: 'RGB trên pump mirror' },
284|                             { label: 'Bềo hành', value: '6 n m' },
285|                         ],
286|                         brands: ['NZXT', 'Corsair', 'Cooler Master', 'Lian Li'],
287|                         tips: 'AIO 360mm cho CPU Intel i7/i9 ho·c AMD Ryzen 7/9. 240mm/2',
288|                         { id: 'air-cooler', name: 'Noctua NH-D15', icon: '0=ß', desc: 'Tén',
289|                             specs: [
290|                                 { label: 'Loại', value: 'Dual Tower Air Cooler' },
291|                                 { label: 'Kích th°üc', value: '150 x 165 x 135mm' },
292|                                 { label: 'Quit', value: '2x NF-A15 140mm PWM' },
293|                                 { label: 'Tñc ù quit', value: '300 - 1500 RPM' },
294|                                 { label: 'Luống khí', value: '~140 CFM' },
295|                                 { label: 'Luống', value: '~24 dBA' },
296|                                 { label: 'Sñ heat pipe', value: '6 Nng óng 06mm' },
297|                                 { label: 'TDP h× trã', value: '~250W' },
298|                                 { label: 'Bềo hành', value: '6 n m' },
299|                             ],
300|                             brands: ['Noctua', 'be quiet!', 'DeepCool', 'Thermalright'],
301|                             tips: 'Tén khí cao c¥p nh° NH-D15 an toàn hìn AIO, không lo rò r
302|                         ],
303|                         {
304|                             id: 'case', name: 'Case', color: '#a78bfa',
305|                             icon: <Monitor size={20} />,
306|                             items: [
307|                                 { id: 'case-mid', name: 'Lian Li O11 Dynamic EVO', icon: '0=ßxp', de
308|                                     specs: [
309|                                         { label: 'Kích th°üc', value: 'Mid-Tower' },
310|                                         { label: 'H× trã Mainboard', value: 'ATX, M-ATX, ITX' },
311|                                         { label: 'H× trã GPU', value: 'Tñi a 420mm' },
312|                                         { label: 'H× trã CPU Cooler', value: 'Tñi a 155mm' },
313|                                         { label: 'H× trã Radiator', value: '360mm top/side/bottom' },
314|                                         { label: 'Khe mß rùng', value: '7 khe' },
315|                                         { label: '0 céng', value: '2x 3.5" + 4x 2.5" },
316|                                         { label: 'Ch¥t liệu', value: 'Kính c°ying lĩc + Thép + Nhôm' },
317|                                         { label: 'Luống khí', value: 'Bottom intake, top/side exhaust' },
318|                                         { label: 'Kích th°üc', value: '445 x 272 x 446 mm' },
319|                                     ],
320|                                     brands: ['Lian Li', 'Corsair', 'Fractal Design', 'NZXT'],
321|                                     tips: 'Chín case c° luống khí tñt, c ch× cho GPU dài. Mesh fron
322|                                     { id: 'case-mini', name: 'Fractal Design North', icon: '0>µ', desc:
323|                                         specs: [
324|                                             { label: 'Kích th°üc', value: 'Mid-Tower ATX' },
325|                                             { label: 'Ch¥t liệu', value: 'Thép + G× óc ch° + Kính / Mesh' },
326|                                             { label: 'H× trã GPU', value: 'Tñi a 355mm' },
327|                                             { label: 'H× trã Radiator', value: '240/280mm front' },
328|                                             { label: 'Quit i kèm', value: '2x 140mm Aspect' },
329|                                         ],
330|                                         brands: ['Fractal Design', 'Corsair', 'be quiet!'],
331|                                         tips: 'Case i p giúp không gian làm việc sang trĩng hìn. ềng qu
332|                                     },
333|                                     {
334|                                         id: 'peripheral', name: 'Ngoii vi', color: '#f97316',
335|                                         icon: <MousePointer2 size={20} />,
336|                                         items: [
337|                                             { id: 'monitor', name: 'LG 27GP950-B 4K 160Hz', icon: '0=¥p', desc:
338|                                                 specs: [
339|                                                     { label: 'Kích th°üc', value: '27 inch' },
340|                                                     { label: 'Ù phân giệi', value: '3840 x 2160 (4K UHD)' },
341|                                                     { label: 'T§n sñ quét', value: '160Hz' },
342|                                                     { label: 'Panel', value: 'Nano IPS' },
343|                                                     { label: 'Thỷi gian phén hồi', value: '1ms GTG' },
344|                                                     { label: 'HDR', value: 'DisplayHDR 600' },
345|                                                     { label: 'Ù phç màu', value: 'DCI-P3 98%' },
346|                                                     { label: 'Kết nñi', value: 'HDMI 2.1 + DP 1.4 + USB-C' },
347|                                                     { label: 'G-Sync / FreeSync', value: 'G-Sync Compatible + FreeSync' },
348|                                                     { label: 'iẤu chénh', value: 'Cao, xoay, nghiêng' },
349|                                                 ],
350|                                                 brands: ['LG', 'ASUS', 'Samsung', 'Dell'],
351|                                                 tips: '27" 1440P là "sweet spot" cho gaming. 4K cho ó hĩa và gã
352|                                                 { id: 'key board', name: 'Logitech G Pro X', icon: '#(p', desc: 'Bã
```

```

356| specs: [
357|     { label: 'Kích thước', value: '60% (không numpad)' },
358|     { label: 'Switch', value: 'GX Blue / Clicky ho.c GX Red / Linear' },
359|     { label: 'Kết nối', value: 'USB-C có dây' },
360|     { label: 'Keycap', value: 'PBT Doubleshot' },
361|     { label: 'RGB', value: 'Per-key RGB' },
362|     { label: 'Software', value: 'G Hub' },
363|     { label: 'Polling Rate', value: '1000Hz' },
364| ],
365| brands: ['Logitech', 'Razer', 'Corsair', 'Ducky'],
366| tips: 'Switch Linear (Red) cho gaming, Clicky (Blue) cho gõ v n',
367| { id: 'mouse', name: 'Razer DeathAdder V3 Pro', icon: '0=Y±p', desc: 'S' },
368| specs: [
369|     { label: 'Cảm biến', value: 'Focus Pro 30K' },
370|     { label: 'DPI', value: '30.000 DPI' },
371|     { label: 'Kết nối', value: 'Razer HyperSpeed Wireless' },
372|     { label: 'Trọng lượng', value: '63g' },
373|     { label: 'Pin', value: '~90 giờ' },
374|     { label: 'Nút bấm', value: '6 nút l-p trình' },
375|     { label: 'Switch', value: 'Razer Optical Gen-3' },
376|     { label: 'Polling Rate', value: '1000Hz (có thể lên 4000Hz)' },
377| ],
378| brands: ['Razer', 'Logitech', 'SteelSeries', 'Zowie'],
379| tips: 'Chuột d. 0.1 80g là lý t. 0ng cho FPS. DPI cao không = t. 0t',
380| { id: 'headset', name: 'SteelSeries Arctis Nova Pro', icon: '0<BS', desc: 'S' },
381| specs: [
382|     { label: 'Loại', value: 'Over-ear, Closed-back' },
383|     { label: 'Driver', value: '40mm Neodymium' },
384|     { label: 'Âm thanh', value: 'Sonar Audio + 360° Spatial Audio' },
385|     { label: 'Microphone', value: 'ClearCast gen 2 (khuếch đại AI)' },
386|     { label: 'Kết nối', value: 'USB-C + 3.5mm' },
387|     { label: 'ANC', value: 'Active Noise Cancellation' },
388|     { label: 'Pin', value: '~22 giờ' },
389|     { label: 'Trọng lượng', value: '298g' },
390| ],
391| brands: ['SteelSeries', 'HyperX', 'Logitech', 'Razer'],
392| tips: 'Tai nghe có mic r. ý và ANC giúp t-p trung h. n trong game.',
393| },
394| },
395| {
396|     id: 'network', name: 'Ming', color: '#3b82f6',
397|     icon: <Wifi size={20} />,
398|     items: [
399|         { id: 'router', name: 'TP-Link Deco XE75 (WiFi 6E)', icon: '0=Úá', desc: 'S' },
400|         specs: [
401|             { label: 'Chu. n WiFi', value: 'WiFi 6E (802.11ax)' },
402|             { label: 'B. ng t. sn', value: 'Tri-band (6GHz + 5GHz + 2.4GHz)' },
403|             { label: 'T. n. c. ùt n. i. a', value: 'Lên đến 5.400 Mbps' },
404|             { label: 'B. o m- t', value: 'WPA3' },
405|             { label: 'S. n. thi. c. t, b. e', value: 'H. x. tr. á ~150 thi. c. t b. e' },
406|             { label: 'K. c. t n. n. i', value: '1x WAN 2.5G + 2x LAN Gigabit' },
407|             { label: 'Mesh', value: 'Có (t. n. ùng chuy. Án ming)' },
408|         ],
409|         brands: ['TP-Link', 'ASUS', 'Netgear', 'Ubiquiti'],
410|         tips: 'WiFi 6E cho t. n. c. ù cao và ít nhi. Áu. Mesh cho nh. Áu t. s',
411|         { id: 'switch', name: 'Netgear GS308 8-Port', icon: '0=Y', desc: 'S' },
412|         specs: [
413|             { label: 'S. n. c. òng', value: '8x Gigabit Ethernet' },
414|             { label: 'Chu. n', value: 'IEEE 802.3ab, t', },
415|             { label: 'B. ng t. òng', value: '16 Gbps (chuy. Án mich)' },
416|             { label: 'Jumbo Frame', value: 'H. x. tr. á' },
417|             { label: 'K. h. òng quit', value: 'Fanless (y. e. n- t) nh.' },
418|             { label: 'Kim lo. i', value: 'V. i. kim lo. i ch- c ch. n' },
419|         ],
420|         brands: ['Netgear', 'TP-Link', 'Cisco', 'D-Link'],
421|         tips: 'Switch không quit cho v. n. phòng y. e. n t) nh. C. á. p CAT6 cho t. n',
422|     },
423|     {
424|         id: 'display', name: 'Hi. Án th. e', color: '#10b981',
425|         icon: <Eye size={20} />,
426|         items: [
427|             { id: 'projector', name: 'BenQ TH671ST', icon: '0=Úýp', desc: 'M. Áy c' },
428|             specs: [
429|                 { label: 'Ù. p. h. Án gi. e. i', value: '1920 x 1080 (Full HD)' },
430|                 { label: 'Ù. s. Áng', value: '3.000 ANSI Lumens' },
431|                 { label: 'T. e. l. c. òng p. h. Án', value: '10.000:1' },
432|                 { label: 'K. h. òng c. Ách ch. i. c. u', value: '1.5-3m cho 100" },
433|                 { label: 'T. s. n. s. n. qu. e. t', value: '120Hz' },
434|                 { label: 'Input Lag', value: '16.7ms (gaming mode)' },
435|                 { label: 'C. òng', value: 'HDMI 1.4 + VGA' },
436|                 { label: 'T. u. ói th. i. b. óng', value: '15.000 giờ (SmartEco)' },
437|             ],
438|             brands: ['BenQ', 'Epson', 'Optoma', 'LG'],
439|             tips: 'M. Áy ch. i. c. u gaming c. s. n. input lag th. y. p. TH671ST có k. h. òng c. Á',
440|             { id: 'oled', name: 'LG OLED42C3 42"', icon: '(', desc: 'M. Án h. Ánh' },
441|             specs: [
442|                 { label: 'K. ích th. òc', value: '42 inch' },
443|                 { label: 'C. òng ngh. c', value: 'OLED (t. n. p. Át s. Áng)' },
444|                 { label: 'Ù. p. h. Án gi. e. i', value: '3840 x 2160 (4K)' },
445|                 { label: 'T. s. n. s. n. qu. e. t', value: '120Hz' },
446|                 { label: 'H. ÁR', value: 'Dolby Vision + HDR10' },
447|                 { label: 'Input Lag', value: '< 5ms' },
448|                 { label: 'K. c. t n. n. i', value: '4x HDMI 2.1' },
449|                 { label: 'G-Sync', value: 'G-Sync Compatible' },
450|             ],
451|             brands: ['LG', 'Samsung', 'Sony'],
452|             tips: 'OLED cho ch. Át l. Áng h. Ánh Ánh t. n. t. nh. Át nh. òng có nguy. c. i. b',
453|         },
454|     },
455| ],
456| ],
457|
458| const QUIZ_QUESTIONS = [
459|     { q: 'Thành ph. s. n. nào. Ác v. i. là "b. ù. n. Áo" c. Á. Áy t. Ánh?', opts: ['CPU', 'GPU', 'RAM', 'SSD', 'NVMe', 'H. Ánh nh. Át h. Án n. Áy?', 'PCIe 5.0', 'DDR5', '16GB', '80-85%'],
460|     { q: 'Chu. n k. c. t n. n. i SSD NVMe nh. Ánh nh. Át h. Án n. Áy?', opts: ['PCIe 5.0', 'DDR5', '16GB', '80-85%'],
461|     { q: 'D. ùng l. Áng RAM t. n. i th. i. Áu cho gaming h. Án n. Áy?', opts: ['16GB', '80-85%'],
462|     { q: 'Ng. ùn 80+ Gold có h. Ác s. Át b. Áo nh. Áu?', opts: ['87-90%', '80-85%'],
463|     { q: 'Socket LGA 1700 d. ùng cho CPU nào?', opts: ['Intel 12/13/14', 'AMD Ryzen 7000', 'Intel 10/11', 'AMD Ryz...'],
464|     { q: 'D. L. S. Á. c. òng ngh. c. Á. h. Áng nào?', opts: ['NVIDIA', 'AMD', 'Intel'],
465|     { q: 'RAM DDR5 có t. n. c. ù bus b. Áo nh. Áu?', opts: ['4800-8400 MT/s', '2400-3200 MT/s'],
466|     { q: 'T. Án nh. i. c. t. n. Áo h. Ác s. Át nh. Át cho CPU?', opts: ['Custom Loop WC', 'D. i. Ác s. Át nh. Át'],
467|     { q: 'Chipset nào cho p. Áp Áp x. ùng CPU Intel?', opts: ['Z790', 'B760', 'H610', 'Q670'], ans: 0 },
468|     { q: 'Chu. n WiFi nào m. Ái nh. Át?', opts: ['WiFi 7', 'WiFi 6E', 'WiFi 6', 'WiFi 5', 'WiFi 4', 'WiFi 3', 'WiFi 2', 'WiFi 1', 'WiFi 0'] },
469| ],
470|
471| export default function ComponentLabPage() {
472|     const [activeCategory, setActiveCategory] = useState(COMPONENT_DATA[0].id)
473|     const [selectedComponent, setSelectedComponent] = useState<Component | null>(null)
474|     const [searchTerm, setSearchTerm] = useState('')
475|     const [gameMode, setGameMode] = useState<'browse' | 'quiz'>('browse')
476|     const [quizIdx, setQuizIdx] = useState(0)
477|     const [quizScore, setQuizScore] = useState(0)
478|     const [quizFeedback, setQuizFeedback] = useState<'correct' | 'wrong' | null>(null)
479|     const [quizDone, setQuizDone] = useState(false)

```

```

480 | const [showSpecs, setShowSpecs] = useState<string | null>(null)
481 |
482 | const activeItems = useMemo(() => {
483 |   const cat = COMPONENT_DATA.find(c => c.id === activeCategory)
484 |   if (!cat) return []
485 |   if (!searchTerm) return cat.items
486 |   return cat.items.filter(i =>
487 |     i.name.toLowerCase().includes(searchTerm.toLowerCase()) ||
488 |     i.desc.toLowerCase().includes(searchTerm.toLowerCase())
489 |   )
490 | }, [activeCategory, searchTerm])
491 |
492 | const filteredCategories = useMemo(() => {
493 |   if (!searchTerm) return COMPONENT_DATA
494 |   return COMPONENT_DATA.filter(c =>
495 |     c.name.toLowerCase().includes(searchTerm.toLowerCase()) ||
496 |     c.items.some(i => i.name.toLowerCase().includes(searchTerm.toLowerCase()))
497 |   )
498 | }, [searchTerm])
499 |
500 | function handleAnswer(idx: number) {
501 |   if (quizFeedback) return
502 |   const correct = idx === QUIZ_QUESTIONS[quizIdx].ans
503 |   if (correct) { setQuizFeedback('correct'); setQuizScore(prev => prev + 1) }
504 |   else setQuizFeedback('wrong')
505 |   setTimeout(() => {
506 |     if (quizIdx < QUIZ_QUESTIONS.length - 1) {
507 |       setQuizIdx(prev => prev + 1); setQuizFeedback(null)
508 |     } else { setQuizDone(true) }
509 |   }, 1500)
510 | }
511 |
512 | return (
513 |   <div style={{ minHeight: '100vh', background: 'var(--bg-base)', color: 'var(--text-primary)' }}>
514 |     <div style={{ maxWidth: '1200px', margin: '0 auto', padding: '24px 16px' }}>
515 |       /* Header */
516 |       <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'space-between', marginBottom: '8...' }}>
517 |         <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
518 |           <div style={{
519 |             width: '48px', height: '48px', borderRadius: '4px',
520 |             background: 'linear-gradient(135deg, var(--brand-primary), var(--accent-blue))',
521 |             display: 'flex', alignItems: 'center', justifyContent: 'center',
522 |           }}>
523 |             <FlaskConical size={24} color="#fff" />
524 |           </div>
525 |           <div>
526 |             <h1 style={{ fontSize: '20px', fontWeight: 900, fontFamily: 'var(--font-mono)', letterSpacing: '...' }}>
527 |               PC COMPONENT LAB
528 |             </h1>
529 |             <p style={{ fontSize: '12px', color: 'var(--text-muted)' }}>
530 |               Phòng thí nghiệm m
531 |             </p>
532 |           </div>
533 |         </div>
534 |         <div style={{ display: 'flex', gap: '6px' }}>
535 |           <button onClick={() => { setGameMode('browse'); setQuizDone(false); setQuizIdx(0); setQuizScore(0)... }}>
536 |             style={{
537 |               padding: '6px 14px', border: '1px solid ${gameMode === 'browse' ? 'var(--brand-primary)' : 'va...' },
538 |               background: gameMode === 'browse' ? 'color-mix(in srgb, var(--brand-primary) 12%, transparent)' : ...
539 |               color: gameMode === 'browse' ? 'var(--brand-primary)' : 'var(--text-muted)',
540 |               fontSize: '10px', fontWeight: 700, fontFamily: 'var(--font-mono)', cursor: 'pointer',
541 |             }}>
542 |             <BookOpen size={12} style={{ display: 'inline', verticalAlign: 'middle', marginRight: '4px' }} />
543 |             T R A C E U
544 |           </button>
545 |           <button onClick={() => setGameMode('quiz')}>
546 |             style={{
547 |               padding: '6px 14px', border: '1px solid ${gameMode === 'quiz' ? 'var(--accent-amber)' : 'var(--...' },
548 |               background: gameMode === 'quiz' ? 'color-mix(in srgb, var(--accent-amber) 12%, transparent)' : ...
549 |               color: gameMode === 'quiz' ? 'var(--accent-amber)' : 'var(--text-muted)',
550 |               fontSize: '10px', fontWeight: 700, fontFamily: 'var(--font-mono)', cursor: 'pointer',
551 |             }}>
552 |             <Beaker size={12} style={{ display: 'inline', verticalAlign: 'middle', marginRight: '4px' }} />
553 |             T H I T H Á C H
554 |           </button>
555 |         </div>
556 |       </div>
557 |       /* Search */
558 |       <div style={{ position: 'relative', marginBottom: '16px' }}>
559 |         <Search size={14} style={{ position: 'absolute', left: '12px', top: '50%', transform: 'translateY(-5...' }}>
560 |           <input value={searchTerm} onChange={e => setSearchTerm(e.target.value)}>
561 |             p l a c e h o l d e r = " T i m l i n h k i c n ( C P U , G P U , R A M . . . ) "
562 |           </input>
563 |         </div>
564 |         <div style={{
565 |           width: '100%', padding: '8px 12px 8px 36px',
566 |           border: '1px solid var(--border-default)', background: 'var(--bg-surface)',
567 |           color: 'var(--text-primary)', fontSize: '12px', outline: 'none',
568 |           fontFamily: 'inherit',
569 |         }} />
570 |       </div>
571 |       {gameMode === 'quiz' ? (
572 |         <div>
573 |           <quizDone ? (
574 |             <div style={{
575 |               textAlign: 'center', padding: '60px 20px',
576 |               border: '1px solid var(--border-default)',
577 |               background: 'var(--bg-surface)',
578 |             }}>
579 |               <motion.div initial={{ scale: 0 }} animate={{ scale: 1 }}>
580 |                 <Trophy size={64} style={{ color: 'var(--accent-amber)', margin: '0 auto 16px' }} />
581 |               </motion.div>
582 |               <h2 style={{ fontSize: '24px', fontWeight: 900, fontFamily: 'var(--font-mono)', marginBottom: '...' }}>
583 |                 <div style={{
584 |                   display: 'inline-block', padding: '12px 24px',
585 |                   border: '1px solid var(--brand-primary)', marginBottom: '16px',
586 |                 }}>
587 |                   <div style={{ fontSize: '11px', color: 'var(--text-muted)', fontFamily: 'var(--font-mono)', ... }}>
588 |                     <div style={{ fontSize: '36px', fontWeight: 900, color: 'var(--brand-primary)', fontFamily: '...' }}>
589 |                       {quizScore}/{QUIZ_QUESTIONS.length}
590 |                     </div>
591 |                   </div>
592 |                   <div style={{ fontSize: '12px', color: 'var(--text-muted)', marginBottom: '20px' }}>
593 |                     { quizScore >= 8 ? 'X u ̣ t s _ c ! B i n l à c h u y ̣ n g i a l i n h k i c n . H ̣ y t r a c ̣ u B t a b t r ̣ n ! ' : ' K h ̣ t ̣ t ! C ̣ n v ̣ i t h ̣ c s n h ̣ c t h ̣ m . ' }
594 |                     { quizScore >= 6 ? ' K h ̣ t ̣ t ! C ̣ n v ̣ i t h ̣ c s n h ̣ c t h ̣ m . ' : ' C s n h ̣ c t h ̣ m v ̣ Á l i n h k i c n . H ̣ y t r a c ̣ u B t a b t r ̣ n ! ' }
595 |                   </div>
596 |                 </div>
597 |               <button onClick={() => { setQuizDone(false); setQuizIdx(0); setQuizScore(0); setQuizFeedback(n... }}>
598 |                 style={{
599 |                   padding: '10px 24px',
600 |                   background: 'var(--brand-primary)', color: 'var(--bg-base)',
601 |                   border: 'none', fontWeight: 700, fontFamily: 'var(--font-mono)',
602 |                   cursor: 'pointer', fontSize: '12px',
603 |                 }}>

```

```

605 |         </button>
606 |     </div>
607 | ) : (
608 |     <div style={{
609 |         border: '1px solid var(--border-default)',
610 |         background: 'var(--bg-surface)',
611 |         padding: '24px',
612 |         position: 'relative', overflow: 'hidden',
613 |     }}>
614 |         <div style={{ position: 'absolute', top: 0, left: 0, width: '6px', height: '6px', borderTop: '...'
615 |         <div style={{ position: 'absolute', top: 0, right: 0, width: '6px', height: '6px', borderTop: '...'
616 |         <div style={{ position: 'absolute', bottom: 0, left: 0, width: '6px', height: '6px', borderBot...
617 |         <div style={{ position: 'absolute', bottom: 0, right: 0, width: '6px', height: '6px', borderBo...
618 |
619 |         <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBo...
620 |         <div style={{ display: 'flex', alignItems: 'center', gap: '8px' }}>
621 |             <Beaker size={16} style={{ color: 'var(--accent-amber)' }} />
622 |             <span style={{ fontSize: '13px', fontWeight: 700, fontFamily: 'var(--font-mono)' }}>
623 |                 T H I   T H   A   C H   K I   % N   T H   E   C   L I N H   K I   Æ   N
624 |             </span>
625 |         </div>
626 |         <span style={{ fontSize: '11px', color: 'var(--text-muted)', fontFamily: 'var(--font-mono)' }}>
627 |             {quizIdx + 1}/{QUIZ_QUESTIONS.length}
628 |         </span>
629 |     </div>
630 |
631 |     <div style={{ display: 'flex', flex-direction: 'column', gap: '8px' }}>
632 |         <div style={{ width: '100%', height: '6px', background: 'var(--border-subtle)', marginBotto...
633 |         <div style={{ width: `${((quizIdx + 1) / QUIZ_QUESTIONS.length) * 100}%`, height: '100%', ba...
634 |     </div>
635 |
636 |     <AnimatePresence mode="wait">
637 |         <motion.div key={quizIdx} initial={{ opacity: 0, x: 20 }} animate={{ opacity: 1, x: 0 }} exi...
638 |         <h2 style={{ fontSize: '18px', fontWeight: 700, marginBotto: '20px', lineHeight: 1.4 }}>
639 |             {QUIZ_QUESTIONS[quizIdx].q}
640 |         </h2>
641 |         <div style={{ display: 'flex', flex-direction: 'column', gap: '8px' }}>
642 |             {QUIZ_QUESTIONS[quizIdx].opts.map((opt, i) => {
643 |                 const isCorrect = i === QUIZ_QUESTIONS[quizIdx].ans
644 |                 const isSelected = quizFeedback && (quizFeedback === 'correct' ? isCorrect : i === (qu...
645 |                 const showCorrect = quizFeedback && isCorrect
646 |                 const showWrong = quizFeedback === 'wrong' && i !== QUIZ_QUESTIONS[quizIdx].ans && ((...
647 |                 for (let j = 0; j < QUIZ_QUESTIONS[quizIdx].opts.length; j++) {
648 |                     if (j === quizIdx) return false
649 |                 }
650 |                 return false
651 |             })()
652 |
653 |             let bg = 'var(--bg-base)'
654 |             let border = 'var(--border-default)'
655 |             let textColor = 'var(--text-primary)'
656 |             let icon = ['A', 'B', 'C', 'D'][i]
657 |
658 |             if (quizFeedback) {
659 |                 if (isCorrect) { bg = 'color-mix(in srgb, var(--brand-primary) 15%, transparent)'; b...
660 |                 else if (i === QUIZ_QUESTIONS[quizIdx].ans) { bg = 'color-mix(in srgb, var(--brand-p...
661 |                 else { bg = 'var(--bg-base)'; border = 'var(--border-default)'; textColor = 'var(--...
662 |             }
663 |
664 |             return (
665 |                 <button key={i} onClick={() => handleAnswer(i)} disabled={!quizFeedback}
666 |                 style={{
667 |                     padding: '12px 16px', display: 'flex', alignItems: 'center', gap: '12px',
668 |                     background: quizFeedback ? (i === QUIZ_QUESTIONS[quizIdx].ans ? 'color-mix(in sr...
669 |                     border: '1px solid ${quizFeedback ? (i === QUIZ_QUESTIONS[quizIdx].ans ? 'var(--...
670 |                     color: quizFeedback ? (i === QUIZ_QUESTIONS[quizIdx].ans ? 'var(--brand-primary)...
671 |                     cursor: quizFeedback ? 'default' : 'pointer',
672 |                     textAlign: 'left', fontSize: '13px', fontWeight: 600,
673 |                     opacity: quizFeedback && i !== QUIZ_QUESTIONS[quizIdx].ans ? 0.5 : 1,
674 |                     transition: 'all 0.2s',
675 |                     fontFamily: 'inherit',
676 |                 }}>
677 |                     <div style={{
678 |                         width: '28px', height: '28px', borderRadius: '2px',
679 |                         background: quizFeedback ? (i === QUIZ_QUESTIONS[quizIdx].ans ? 'var(--brand-pri...
680 |                         display: 'flex', flex-direction: 'column', gap: '2px',
681 |                         fontSize: '11px', fontWeight: 700,
682 |                         color: quizFeedback ? (i === QUIZ_QUESTIONS[quizIdx].ans ? '#fff' : 'var(--text-...
683 |                         flexShrink: 0,
684 |                     }}>
685 |                         {quizFeedback && i === QUIZ_QUESTIONS[quizIdx].ans ? <CheckCircle size={14} /> : ...
686 |                     </div>
687 |                     {opt}
688 |                 </button>
689 |             )
690 |         </div>
691 |     </motion.div>
692 | </AnimatePresence>
693 |
694 |     {quizFeedback && (
695 |         <div style={{ marginTop: '12px', padding: '8px 12px', background: quizFeedback === 'correct'...
696 |             { q u i z   F e e d b a c k   =   =   ' c o r r e c t '   ?   ' '   C h í n h   x á c ! '   :   ' '   S a
697 |         </div>
698 |     )}
699 | </div>
700 |
701 | </div>
702 |
703 | ) : (
704 |     <div style={{ display: 'flex', flex-direction: 'column', gap: '8px' }}>
705 |         <div style={{ display: 'flex', gap: '4px', marginBotto: '16px', flexWrap: 'wrap' }}>
706 |             {filteredCategories.map(cat => (
707 |                 <button key={cat.id} onClick={() => setActiveCategory(cat.id)}
708 |                 style={{
709 |                     display: 'flex', flex-direction: 'column', gap: '4px',
710 |                     padding: '6px 12px',
711 |                     border: '1px solid ${activeCategory === cat.id ? cat.color : 'var(--border-default)'}',
712 |                     background: activeCategory === cat.id ? `color-mix(in srgb, ${cat.color} 12%, transparent)...
713 |                     color: activeCategory === cat.id ? cat.color : 'var(--text-muted)',
714 |                     fontSize: '10px', fontWeight: 700, fontFamily: 'var(--font-mono)',
715 |                     cursor: 'pointer',
716 |                     transition: 'all 0.15s',
717 |                 }}>
718 |                     {cat.icon}
719 |                     {cat.name}
720 |                 </button>
721 |             )}
722 |         </div>
723 |
724 |         <div style={{ display: 'flex', flex-direction: 'column', gap: '8px' }}>
725 |             <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill, minmax(280px, 1fr))', gap: ...
726 |                 <AnimatePresence mode="popLayout">
727 |

```

```

728 |         {activeItems.map(comp => (
729 |             <motion.div key={comp.id} layout
730 |               initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }} exit={{ opacity: 0, scale: ...
731 |               style={{
732 |                 border: '1px solid ${showSpecs === comp.id ? 'var(--brand-primary)' : 'var(--border-defa...
733 |                 background: showSpecs === comp.id ? 'color-mix(in srgb, var(--brand-primary) 4%, transpa...
734 |                 padding: '14px',
735 |                 cursor: 'pointer',
736 |                 transition: 'all 0.2s',
737 |                 position: 'relative', overflow: 'hidden',
738 |               }}
739 |             onClick={() => setShowSpecs(showSpecs === comp.id ? null : comp.id)}>
740 |             {/ * Corner */}
741 |             <div style={{ position: 'absolute', top: 0, left: 0, width: '4px', height: '4px', borderTo...
742 |             <div style={{ display: 'flex', alignItems: 'flex-start', gap: '10px', marginBottom: showSp...
743 |               <div style={{
744 |                 width: '36px', height: '36px', borderRadius: '2px',
745 |                 background: 'color-mix(in srgb, var(--brand-primary) 10%, transparent)',
746 |                 border: '1px solid color-mix(in srgb, var(--brand-primary) 15%, transparent)',
747 |                 display: 'flex', alignItems: 'center', justifyContent: 'center',
748 |                 fontSize: '18px', flexShrink: 0,
749 |               }}>
750 |                 {comp.icon}
751 |               </div>
752 |               <div style={{ flex: 1, minWidth: 0 }}>
753 |                 <div style={{ fontSize: '13px', fontWeight: 800, fontFamily: 'var(--font-mono)', margi...
754 |                 <div style={{ fontSize: '11px', color: 'var(--text-muted)', lineHeight: 1.3 }}>{comp.d...
755 |                 </div>
756 |                 <ChevronRight size={14} style={{ color: 'var(--text-muted)', flexShrink: 0, transform: s...
757 |                 </div>
758 |               </div>
759 |             <AnimatePresence>
760 |               {showSpecs === comp.id && (
761 |                 <motion.div initial={{ height: 0, opacity: 0 }} animate={{ height: 'auto', opacity: 1 ...
762 |                 style={{ borderTop: '1px solid var(--border-default)', padding: '10px' }}>
763 |                 {/ * Role */}
764 |                 <div style={{
765 |                   margin: '10px', padding: '6px 8px',
766 |                   background: 'color-mix(in srgb, var(--accent-blue) 8%, transparent)',
767 |                   border: '1px solid color-mix(in srgb, var(--accent-blue) 15%, transparent)',
768 |                   fontSize: '10px', color: 'var(--text-muted)', lineHeight: 1.4,
769 |                 }}>
770 |                   <span style={{ color: 'var(--accent-blue)', fontWeight: 700, fontFamily: 'var(--fo...
771 |                   </div>
772 |                   {/ * Specs Table */}
773 |                   <div style={{ margin: '8px' }}>
774 |                     <div style={{ fontSize: '10px', fontWeight: 700, color: 'var(--brand-primary)', fo...
775 |                     </div>
776 |                     {comp.specs.map((s, i) => (
777 |                       <div key={i} style={{
778 |                         display: 'flex', justifyContent: 'space-between', alignItems: 'center',
779 |                         padding: '3px 6px',
780 |                         background: i % 2 === 0 ? 'rgba(255,255,255,0.02)' : 'transparent',
781 |                         fontSize: '10px',
782 |                       }}>
783 |                         <span style={{ color: 'var(--text-muted)', flex: '0 0 120px', fontFamily: 'var...
784 |                         <span style={{ color: 'var(--text-primary)', fontWeight: 600, textAlig: 'righ...
785 |                       </div>
786 |                     ))}
787 |                   </div>
788 |                   {/ * Brands */}
789 |                   <div style={{ margin: '8px' }}>
790 |                     <div style={{ fontSize: '10px', fontWeight: 700, color: 'var(--accent-amber)', fon...
791 |                     </div>
792 |                     <div style={{ display: 'flex', gap: '4px', flexWrap: 'wrap' }}>
793 |                       {comp.brands.map(b => (
794 |                         <span key={b} style={{
795 |                           padding: '2px 6px',
796 |                           border: '1px solid var(--border-default)',
797 |                           fontSize: '9px', fontWeight: 700, color: 'var(--text-muted)',
798 |                           fontFamily: 'var(--font-mono)',
799 |                         }}>{b}</span>
800 |                       ))}
801 |                     </div>
802 |                   </div>
803 |                   {/ * Tips */}
804 |                   <div style={{
805 |                     padding: '6px 8px',
806 |                     background: 'color-mix(in srgb, var(--accent-amber) 6%, transparent)',
807 |                     border: '1px solid color-mix(in srgb, var(--accent-amber) 12%, transparent)',
808 |                     fontSize: '10px', color: 'var(--text-muted)',
809 |                   }}>
810 |                     <span style={{ color: 'var(--accent-amber)', fontWeight: 700, fontFamily: 'var(--f...
811 |                     </div>
812 |                   </div>
813 |                 </motion.div>
814 |               )}
815 |             </AnimatePresence>
816 |           ))}
817 |         </div>
818 |       {activeItems.length === 0 && (
819 |         <div style={{ textAlign: 'center', padding: '60px 20px', color: 'var(--text-muted)' }}>
820 |           <Search size={32} style={{ margin: '0 auto 12px', opacity: 0.3 }} />
821 |           <div style={{ fontSize: '13px' }}>Không tìm thấy linh kiện</div>
822 |         </div>
823 |       )}
824 |     </div>
825 |   )
826 | }
827 |
828 |
829 |

```

File: app/courses/[courseId]/page.tsx (171 lines, 7.8 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect, use } from 'react'
4 | import { supabase } from '@lib/supabase'
5 | import Navbar from '@components/Navbar'
6 | import { motion } from 'framer-motion'
7 | import {
8 |   BookOpen, Clock, Star, Play, ChevronRight,

```



```

9 | Layers, CheckCircle, Lock, HelpCircle, FileText, Loader2
10 | } from 'lucide-react'
11 | import Link from 'next/link'
12 |
13 | export default function CourseDetailPage({ params }: { params: Promise<{ courseId: string }> }) {
14 |   const resolvedParams = use(params);
15 |   const { courseId } = resolvedParams;
16 |
17 |   const [course, setCourse] = useState<any>(null)
18 |   const [lessons, setLessons] = useState<any[]>([])
19 |   const [loading, setLoading] = useState(true)
20 |
21 |   useEffect(() => {
22 |     fetchCourseDetails()
23 |   }, [])
24 |
25 |   async function fetchCourseDetails() {
26 |     try {
27 |       setLoading(true)
28 |       // 1. Fetch Course
29 |       const { data: courseData } = await supabase
30 |         .from('courses')
31 |         .select('*')
32 |         .eq('id', courseId)
33 |         .single()
34 |
35 |       setCourse(courseData)
36 |
37 |       // 2. Fetch Lessons in this course
38 |       const { data: lessonData } = await supabase
39 |         .from('lessons')
40 |         .select(`
41 |           *,
42 |           lesson_sections (*)
43 |         `)
44 |         .eq('course_id', courseId)
45 |         .order('created_at', { ascending: true })
46 |
47 |       setLessons(lessonData || [])
48 |     } catch (err) {
49 |       console.error(err)
50 |     } finally {
51 |       setLoading(false)
52 |     }
53 |   }
54 |
55 |   if (loading) return (
56 |     <div className="min-h-screen bg-[#0f0f1a] flex items-center justify-center">
57 |       <Loader2 className="animate-spin text-[#00d2a0]" size={48} />
58 |     </div>
59 |   )
60 |
61 |   if (!course) return (
62 |     <div className="min-h-screen bg-[#0f0f1a] flex items-center justify-center text-red-500 font-bold">
63 |       K h ó a h i c k h ò n g t ò n t i i .
64 |     </div>
65 |   )
66 |
67 |   return (
68 |     <div className="min-h-screen bg-[#0f0f1a] text-white">
69 |       <Navbar />
70 |
71 |       <div className="relative pt-32 pb-20 overflow-hidden">
72 |         <div className="absolute inset-0 z-0">
73 |           <img src={course.thumbnail_url} className="w-full h-full object-cover opacity-20 blur-3xl" />
74 |         </div>
75 |
76 |         <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 relative z-10">
77 |           <div className="flex flex-col lg:flex-row gap-12 items-center">
78 |             <motion.div initial={{ opacity: 0, x: -50 }} animate={{ opacity: 1, x: 0 }} className="flex-1 text...">
79 |               <div className="inline-flex items-center gap-2 px-4 py-1 bg-[#00d2a0]/10 border border-[#00d2a0]...">
80 |                 <Star size={14} /> {course.level}
81 |               </div>
82 |               <h1 className="text-4xl md:text-6xl font-black mb-6 leading-tight">
83 |                 {course.title}
84 |               </h1>
85 |               <p className="text-slate-400 text-lg mb-8 max-w-2xl leading-relaxed">
86 |                 {course.description}
87 |               </p>
88 |               <div className="flex flex-wrap gap-6 justify-center lg:justify-start">
89 |                 <div className="flex items-center gap-2 text-slate-300">
90 |                   <Clock size={20} className="text-[#00d2a0]" /> <span>
91 |                     </div>
92 |                   <div className="flex items-center gap-2 text-slate-300">
93 |                     <Layers size={20} className="text-[#00d2a0]" /> <span>
94 |                     </div>
95 |                   <div className="flex items-center gap-2 text-slate-300">
96 |                     <CheckCircle size={20} className="text-[#00d2a0]" /> <span>
97 |                     </div>
98 |                 </div>
99 |               </div>
100 |             </motion.div>
101 |
102 |             <motion.div initial={{ opacity: 0, scale: 0.8 }} animate={{ opacity: 1, scale: 1 }} className="lg:...">
103 |               <div className="bg-[#16213e] border border-[#1e293b] rounded-[40px] p-2 shadow-2xl overflow-hid...">
104 |                 <div className="aspect-video rounded-[32px] overflow-hidden">
105 |                   <img src={course.thumbnail_url} className="w-full h-full object-cover" />
106 |                 </div>
107 |                 <div className="absolute inset-0 flex items-center justify-center opacity-0 group-hover:opac...">
108 |                   <div className="w-20 h-20 bg-[#00d2a0] text-black rounded-full flex items-center justify-...">
109 |                     <Play size={32} fill="currentColor" />
110 |                   </div>
111 |                 </div>
112 |               </div>
113 |             </motion.div>
114 |           </div>
115 |         </div>
116 |       </div>
117 |
118 |       <div className="space-y-6">
119 |         <div className="max-w-4xl mx-auto px-4 sm:px-6 lg:px-8 pb-32">
120 |           <h2 className="text-3xl font-bold mb-10">L ù t r ì n h <span classNa
121 |
122 |           <div className="space-y-6">
123 |             {lessons.map((lesson, idx) => (
124 |               <motion.div
125 |                 key={lesson.id}
126 |                 initial={{ opacity: 0, y: 20 }}
127 |                 animate={{ opacity: 1, y: 0 }}
128 |                 transition={{ delay: idx * 0.1 }}
129 |                 className="bg-[#16213e] border border-[#1e293b] rounded-3xl overflow-hidden"
130 |               >
131 |                 <div className="p-6 flex items-center justify-between group">
132 |                   <div className="flex items-center gap-4">

```

```

133 |         <div className="w-12 h-12 rounded-2xl bg-[#0f0f1a] border border-[#1e293b] flex items-c...
134 |             {idx + 1}
135 |         </div>
136 |     </div>
137 |     <h3 className="font-bold text-lg group-hover:text-[#00d2a0] transition-colors">{less...
138 |         <p c l a s s N a m e = " t e x t - x s   t e x t - s l a t e - 5 0 0   f o n t - b o l d
139 |     </div>
140 |     <Link href={` /courses/${courseId}/lessons/${lesson.id}`} className="px-6 py-2.5 bg-white/5...
141 |     </Link>
142 |     </div>
143 |     </div>
144 |     </div>
145 |     { /* Sections list inside lesson */ }
146 |     <div className="px-6 pb-6 pt-2 border-t border-[#1e293b]/50 space-y-3">
147 |     { lesson.lesson_sections?.map((section: any) => (
148 |     <div key={section.id} className="flex items-center justify-between py-2 px-4 rounded-xl...
149 |     <div className="flex items-center gap-3">
150 |     {section.content_type === 'video' ? <Play size={14} className="text-[#00d2a0]" /> :
151 |     section.content_type === 'quiz' ? <HelpCircle size={14} className="text-orange-5...
152 |     <FileText size={14} className="text-blue-500" /> }
153 |     <span className="text-sm text-slate-400">{section.title}</span>
154 |     </div>
155 |     {section.content_type === 'quiz' && (
156 |     <Link href={`/quiz/${section.id}`} className="text-[10px] font-black bg-orange-500...
157 |     L A M Q U I Z
158 |     </Link>
159 |     ) }
160 |     </div>
161 |     </div>
162 |     </div>
163 |     </div>
164 |     </div>
165 |     </div>
166 |     </div>
167 |     </div>
168 |     </div>
169 |     )
170 | }
171 |

```

File: app/courses/%5BcourseId%5D/lessons/%5BlessonId%5D/page.tsx (329 lines, 16.9 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect } from 'react'
4 | import { useParams, useRouter } from 'next/navigation'
5 | import { ChevronLeft, ChevronRight, Play, FileText, CheckCircle, Lock, MessageSquare, Download, Bookmark, Shar...
6 | import Navbar from '@components/Navbar'
7 | import Link from 'next/link'
8 | import { motion, AnimatePresence } from 'framer-motion'
9 | import { supabase } from '@lib/supabase'
10 |
11 | export default function LessonDetailPage() {
12 |   const { courseId, lessonId } = useParams()
13 |   const router = useRouter()
14 |   const [activeTab, setActiveTab] = useState('resources')
15 |   const [isBookmarked, setIsBookmarked] = useState(false)
16 |   const [showPopupQuiz, setShowPopupQuiz] = useState(false)
17 |   const [quizAnswered, setQuizAnswered] = useState(false)
18 |
19 |   // Simulation of a timestamp trigger for quiz
20 |   useEffect(() => {
21 |     const timer = setTimeout(() => {
22 |       if (!quizAnswered) setShowPopupQuiz(true)
23 |     }, 10000) // Show quiz after 10 seconds of "watching"
24 |     return () => clearTimeout(timer)
25 |   }, [quizAnswered])
26 |
27 |   const [lesson, setLesson] = useState<any>(null)
28 |   const [courseLessons, setCourseLessons] = useState<any[]>([])
29 |   const [loading, setLoading] = useState(true)
30 |
31 |   useEffect(() => {
32 |     fetchLessonData()
33 |   }, [lessonId])
34 |
35 |   async function fetchLessonData() {
36 |     try {
37 |       setLoading(true)
38 |       // Fetch current lesson
39 |       const { data: lessonData, error: lessonError } = await supabase
40 |         .from('lesson_sections')
41 |         .select(
42 |           `
43 |             lessons:lesson_id (
44 |               title,
45 |               teacher_id
46 |             )
47 |           `
48 |         .eq('lesson_id', lessonId)
49 |         .order('order_index', { ascending: true })
50 |
51 |       if (lessonData && lessonData.length > 0) {
52 |         setLesson(lessonData[0]) // Get the first section as main content for now
53 |
54 |         // Fetch all sections for the sidebar
55 |         const { data: allSections } = await supabase
56 |           .from('lesson_sections')
57 |           .select('*')
58 |           .eq('lesson_id', lessonId)
59 |           .order('order_index', { ascending: true })
60 |
61 |         setCourseLessons(allSections || [])
62 |       }
63 |     } catch (err) {
64 |       console.error('Error fetching lesson:', err)
65 |     } finally {
66 |       setLoading(false)
67 |     }
68 |   }
69 |
70 |   if (loading) return <div className="min-h-screen bg-[#0f0f1a] flex items-center justify-center text-[#00d2a0]...
71 |   if (!lesson) return <div className="min-h-screen bg-[#0f0f1a] flex items-center justify-center text-red-500"...
72 |
73 |   return (
74 |     <div className="min-h-screen bg-[#0f0f1a] text-white flex flex-col">
75 |       <Navbar />
76 |
77 |       <main className="pt-20 flex-1 flex flex-col lg:flex-row overflow-hidden">
78 |
79 |         { / *   C  O  T   T  R  A  I   -   N  O  I   D  U  N  G   C  H  I  N  H   ( 7 0  % )   * / }
80 |         <div className="flex-1 overflow-y-auto custom-scrollbar bg-[#0a0a14]">
81 |           { /* Video Player Section */ }

```

```
82 | <div className="aspect-video bg-black w-full relative group overflow-hidden">
83 |   {lesson.content_type === 'video' ? (
84 |     <iframe
85 |       src={lesson.content_url?.includes('youtube.com')} ? lesson.content_url.replace('watch?v=', 'emb...
86 |       className="w-full h-full border-none"
87 |       allowFullScreen
88 |       title={lesson.title}
89 |     ></iframe>
90 |   ) : lesson.content_type === 'pdf' ? (
91 |     <iframe
92 |       src={lesson.content_url}
93 |       className="w-full h-full border-none"
94 |       title="PDF Document"
95 |     ></iframe>
96 |   ) : lesson.content_type === 'image' ? (
97 |     <div className="w-full h-full flex items-center justify-center p-4">
98 |       <img src={lesson.content_url} alt={lesson.title} className="max-w-full max-h-full object-conta...
99 |     </div>
100 |   ) : (
101 |     <div className="w-full h-full flex flex-col items-center justify-center text-slate-500 bg-[#0f0f...
102 |       <FileText size={64} className="mb-4 text-[#00d2a0] opacity-50" />
103 |       <h3 className="text-xl font-bold text-white mb-2">Tài liệu
104 |       <p className="max-w-md">Vui lòng si dăng các tab bên d
105 |     </div>
106 |   )}
107 |
108 |   <AnimatePresence>
109 |     {showPopupQuiz && (
110 |       <motion.div
111 |         initial={{ opacity: 0 }}
112 |         animate={{ opacity: 1 }}
113 |         exit={{ opacity: 0 }}
114 |         className="absolute inset-0 bg-black/90 backdrop-blur-md flex items-center justify-center p...
115 |       >
116 |         <div className="max-w-md w-full bg-[#16213e] border border-[#00d2a0]/30 p-8 rounded-3xl shad...
117 |         <div className="w-12 h-12 bg-[#00d2a0]/10 text-[#00d2a0] rounded-xl flex items-center just...
118 |         <HelpCircle size={24} />
119 |       </div>
120 |       <h3 className="text-xl font-bold mb-2">Câu hỏi
121 |       <p className="text-slate-400 mb-8 text-sm">Á t i p t
122 |       <p className="font-bold mb-6">Linh ki c n n a o a c v i
123 |
124 |       <div className="space-y-3">
125 |         {[ 'GPU', 'CPU', 'RAM', 'Mainboard' ].map((opt) => (
126 |           <button
127 |             key={opt}
128 |             onClick={() => {
129 |               if (opt === 'CPU') {
130 |                 setQuizAnswered(true)
131 |                 setShowPopupQuiz(false)
132 |               }
133 |             }}
134 |             className="w-full p-4 bg-[#0f0f1a] border border-[#1e293b] rounded-xl text-left text...
135 |           >
136 |             {opt}
137 |           </button>
138 |         )}
139 |       </div>
140 |     </div>
141 |   </motion.div>
142 | )}
143 | </AnimatePresence>
144 | </div>
145 |
146 | <div className="p-6 lg:p-10 max-w-5xl mx-auto">
147 |   { /* Header Info */ }
148 |   <div className="flex flex-col md:flex-row md:items-center justify-between gap-6 mb-8">
149 |     <div>
150 |       <div className="flex items-center gap-2 text-[#00d2a0] text-sm font-medium mb-2">
151 |         <Link href={` /courses/${courseId} `} className="hover:underline">{lesson.lessons?.title} || 'K...
152 |         <ChevronRight size={14} />
153 |         <span>Bài {courseLessons.findIndex(l => l.id === lesson.id) + 1}</span>
154 |       </div>
155 |       <h1 className="text-2xl md:text-3xl font-bold">{lesson.title}</h1>
156 |     </div>
157 |
158 |     <div className="flex items-center gap-3">
159 |       <button
160 |         onClick={() => setIsBookmarked(!isBookmarked)}
161 |         className={` p-3 rounded-xl border border-[#1e293b] transition-all ${isBookmarked ? 'bg-[#00d...
162 |       >
163 |       <Bookmark size={20} fill={isBookmarked ? 'currentColor' : 'none'} />
164 |     </button>
165 |     <button className="p-3 bg-[#16213e] text-slate-400 hover:text-white border border-[#1e293b] ro...
166 |     <Share2 size={20} />
167 |   </button>
168 | </div>
169 | </div>
170 |
171 | { /* Content Tabs */ }
172 | <div className="border-b border-[#1e293b] flex gap-8 mb-8 overflow-x-auto custom-scrollbar no-scro...
173 |   {[ 'resources', 'q&a', 'notes', 'discussion' ].map((tab) => (
174 |     <button
175 |       key={tab}
176 |       onClick={() => setActiveTab(tab)}
177 |       className={` pb-4 text-sm font-bold uppercase tracking-wider transition-all relative ${
178 |         activeTab === tab ? 'text-[#00d2a0]' : 'text-slate-500 hover:text-slate-300'
179 |       } `}
180 |     >
181 |       { t a b === 'resources' ? 'Tài nguyên' : t a b === 'q & a' ?
182 |       { activeTab === tab && (
183 |         <motion.div layoutId="lessonTab" className="absolute bottom-0 left-0 right-0 h-0.5 bg-[#00...
184 |       )}
185 |     </button>
186 |   )}
187 | </div>
188 |
189 | <div className="min-h-[300px]">
190 |   <AnimatePresence mode="wait">
191 |     {activeTab === 'resources' && (
192 |       <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }} exit={{ opacity: 0 }} classNam...
193 |       <div className="bg-[#16213e] border border-[#1e293b] p-4 rounded-xl flex items-center just...
194 |       <div className="flex items-center gap-3">
195 |         <div className="w-10 h-10 rounded-lg bg-red-500/10 text-red-400 flex items-center just...
196 |         <FileText size={20} />
197 |       </div>
198 |       <div>
199 |         <p className="font-medium text-sm">Tài liệu b
200 |       </div>
201 |     </div>
202 |     <button className="p-2 text-slate-400 hover:text-[#00d2a0] transition-colors">
203 |       <Download size={18} />
204 |     </button>
205 |   </div>
```

```
206|         </div>
207|     </motion.div>
208|   )}
209|
210|   {activeTab === 'q&a' && (
211|     <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }} exit={{ opacity: 0 }}>
212|       <div className="bg-[#16213e] border border-[#1e293b] p-6 rounded-xl">
213|         <p className="text-slate-400 text-center text-sm">
214|           <button className="w-full py-3 bg-[#1e293b] hover:bg-[#2a3655] rounded-lg text-sm font-b...
215|             t c a u h i i m Ū i
216|         </button>
217|       </div>
218|     </motion.div>
219|   )}
220|
221|   {activeTab === 'notes' && (
222|     <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }} exit={{ opacity: 0 }}>
223|       <textarea
224|         placeholder="V i ɛ t g h i c h ú c ặ a b i n t i i ậ y . . ."
225|         className="w-full bg-[#16213e] border border-[#1e293b] rounded-xl p-6 min-h-[200px] outli...
226|       ></textarea>
227|       <div className="mt-4 flex justify-end">
228|         <button className="px-6 py-2 bg-[#00d2a0] text-black font-bold rounded-lg hover:bg-[#00e...
229|           L ° u g h i c h ú
230|       </button>
231|     </div>
232|   </motion.div>
233| )}
234|
235|   {activeTab === 'discussion' && (
236|     <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }} exit={{ opacity: 0 }} classNam...
237|     <div className="flex gap-4">
238|       <div className="w-10 h-10 rounded-full bg-[#1e293b] border border-[#2a3655] shrink-0" />
239|       <div className="flex-1">
240|         <textarea
241|           placeholder="Ch i a s » s u y n g h ) c ặ a b i n v Á b à i
242|           className="w-full bg-[#0f0f1a] border border-[#1e293b] rounded-2xl p-4 text-sm outli...
243|         ></textarea>
244|         <div className="mt-3 flex justify-end">
245|           <button className="px-5 py-2 bg-[#1e293b] text-white text-xs font-bold rounded-lg ho...
246|             N G B I N H L U Ậ N
247|         </button>
248|       </div>
249|     </div>
250|   </div>
251|
252|   <div className="space-y-6 pt-6">
253|     {[1, 2].map(i => (
254|       <div key={i} className="flex gap-4">
255|         <div className="w-10 h-10 rounded-full bg-gradient-to-tr from-[#00d2a0] to-[#00b4d8]...
256|         <div className="flex-1 bg-[#16213e]/50 p-4 rounded-2xl border border-[#1e293b]">
257|           <div className="flex items-center justify-between mb-2">
258|             <span className="text-sm font-bold">Ng ° Ý i
259|             <span className="text-[10px] text-slate-500">
260|           </div>
261|           <p className="text-slate-300 text-sm leading-
262|           <div className="mt-4 flex items-center gap-4 text-xs text-slate-500">
263|             <button className="hover:text-[#00d2a0] transition-colors font-bold">THÍCH</button>
264|             <button className="hover:text-[#00d2a0] tr
265|           </div>
266|         </div>
267|       </div>
268|     )]}
269|   </div>
270| </motion.div>
271| )}
272| </AnimatePresence>
273| </div>
274| </div>
275| </div>
276|
277|   { / * C Ộ T P H Ậ I - S I D E B A R ( 3 0 % ) * / }
278|   <div className="w-full lg:w-[400px] border-l border-[#1e293b] bg-[#0f0f1a] flex flex-col h-full lg:h[...
279|     <div className="p-6 border-b border-[#1e293b]">
280|       <h3 className="font-bold flex items-center gap-2">
281|         <Layers size={18} className="text-[#00d2a0]" /> N ậ i d ụng
282|       </h3>
283|       <div className="mt-4 bg-[#16213e] h-2 rounded-full overflow-hidden">
284|         <div className="bg-[#00d2a0] h-full" style={{ width: '25%' }}></div>
285|       </div>
286|       <p className="text-xs text-slate-500 mt-2 font-medium">H o à n
287|     </div>
288|
289|     <div className="flex-1 overflow-y-auto custom-scrollbar py-2">
290|       {courseLessons.map((l, index) => (
291|         <button
292|           key={l.id}
293|           disabled={l.locked}
294|           className={`w-full flex items-center gap-4 px-6 py-5 text-left transition-all relative ${
295|             l.active ? 'bg-[#1e293b]/50 border-l-4 border-[#00d2a0]' : 'hover:bg-[#16213e]/30'
296|           } ${l.locked ? 'opacity-50 grayscale' : ''}`}
297|         >
298|           <div className={`w-8 h-8 rounded-lg flex items-center justify-center shrink-0 ${
299|             l.completed ? 'bg-[#00d2a0]/10 text-[#00d2a0]' :
300|             l.active ? 'bg-[#00d2a0] text-black' :
301|             'bg-[#1e293b] text-slate-500'
302|           }`}>
303|             {l.locked ? <Lock size={14} /> : l.completed ? <CheckCircle size={16} /> : <span className="...
304|           </div>
305|           <div className="flex-1 pr-4">
306|             <p className={`text-sm font-semibold truncate ${l.active ? 'text-white' : 'text-slate-300'}`}>
307|               {l.title}
308|             </p>
309|             <div className="flex items-center gap-2 mt-1">
310|               {l.type === 'video' ? <Play size={12} className="text-slate-500" /> : <FileText size={12} ...
311|               <span className="text-[10px] text-slate-500 font-bold uppercase tracking-wider">{l.type ===...
312|             </div>
313|           </div>
314|         </button>
315|       )]}
316|     </div>
317|
318|     <div className="p-6 bg-[#16213e]/50 border-t border-[#1e293b]">
319|       <button className="w-full py-4 bg-[#00d2a0] text-black font-black rounded-xl hover:bg-[#00e6af] tr...
320|         B A I T I Ậ P T H E O <ChevronRight size={18} />
321|       </button>
322|     </div>
323|   </div>
324|
325| </main>
326| </div>
327| )
328| }
329|
```

File: app/courses/page.tsx (158 lines, 7.1 KB)

```

121 | >
122 | <div className={`relative overflow-hidden ${viewMode === 'list' ? 'md:w-[300px] shrink-0' : 'aspect-vide...
123 | <img src={course.thumbnail_url} || 'https://images.unsplash.com/photo-1591799264318-7e6ef8ddb7ea?w=800&...
124 | <div className="absolute top-4 left-4">
125 | <span className={`px-3 py-1 rounded-full text-[10px] font-bold uppercase tracking-wider ${
126 |   course.level === 'C i b £ n' ? 'bg-green-500/20 text-green-400' :
127 |   course.level === 'Tr u n g c ¥ p' ? 'bg-blue-500/20 text-blue-400' :
128 |   'bg-purple-500/20 text-purple-400 border border-purple-500/30'
129 | }`}>
130 |   {course.level}
131 | </span>
132 | </div>
133 | {(course.progress || 0) > 0 && (
134 |   <div className="absolute bottom-0 left-0 right-0 h-1.5 bg-black/40">
135 |     <div className="h-full bg-[#00d2a0]" style={{ width: `${course.progress}%` }}></div>
136 |   )
137 | }
138 | </div>
139 |
140 | <div className="p-6 flex flex-col flex-1">
141 |   <h3 className="text-lg font-bold mb-2 group-hover:text-[#00d2a0] transition-colors">{course.title}</h3>
142 |   <p className="text-slate-400 text-sm mb-6 line-clamp-2 leading-relaxed">{course.description}</p>
143 |
144 |   <div className="mt-auto flex items-center justify-between">
145 |     <div className="flex items-center gap-4 text-xs text-slate-500">
146 |       <span className="flex items-center gap-1.5"><Clock size={14} /> {course.duration}</span>
147 |       <span className="flex items-center gap-1.5"><Layers size={14} /> {course.lessons} bài</span>
148 |     </div>
149 |
150 |     <Link href={`\courses/${course.id}`} className="p-2.5 bg-[#1e293b] hover:bg-[#00d2a0] hover:text-bla...
151 |     <ChevronRight size={20} className="transition-transform group-hover:btn:translate-x-0.5" />
152 |   </Link>
153 | </div>
154 | </div>
155 | </motion.div>
156 | )
157 | }
158 |

```

File: app/creator/components/ComponentForm.tsx (279 lines, 10.9 KB)

```

1 | 'use client';
2 |
3 | import { useState, useEffect, useRef } from 'react';
4 | import toast from 'react-hot-toast';
5 | import { createClient } from '@lib/supabase-client';
6 | import { COMPONENT_SPECS, type ComponentRenderSpec } from '@data/componentSpecs';
7 | import ComponentRenderer2D from '@components/builder/ComponentRenderer2D';
8 | import { useComponentLock } from '@hooks/useComponentUpdates';
9 |
10 | interface ComponentFormProps {
11 |   componentId: string | null;
12 |   onSave: () => void;
13 |   onCancel: () => void;
14 | }
15 |
16 | const COMPONENT_TYPES = ['cpu', 'gpu', 'ram', 'motherboard', 'psu', 'case', 'cooler', 'ssd', 'hdd'] as const;
17 |
18 | export default function ComponentForm({ componentId, onSave, onCancel }: ComponentFormProps) {
19 |   const existingSpec = COMPONENT_SPECS.find((s) => s.id === componentId) : null;
20 |
21 |   const [form, setForm] = useState({
22 |     name: existingSpec?.name || '',
23 |     type: existingSpec?.type || 'cpu',
24 |     brand: existingSpec?.brand || '',
25 |     model: existingSpec?.model || '',
26 |     width_mm: existingSpec?.dimensions.width_mm || 100,
27 |     height_mm: existingSpec?.dimensions.height_mm || 100,
28 |     depth_mm: existingSpec?.dimensions.depth_mm || 10,
29 |     pcbColor: existingSpec?.colors.pcb || '#1a1a2e',
30 |     heatsinkColor: existingSpec?.colors.heatsink || '#c0c0c0',
31 |     shroudColor: existingSpec?.colors.shroud || '#333333',
32 |     connectorColor: existingSpec?.colors.connector || '#d4af37',
33 |     accentColor: existingSpec?.colors.accent || '#00d4aa',
34 |     ledColor: existingSpec?.colors.led || '#00d4aa',
35 |   });
36 |
37 |   const [saving, setSaving] = useState(false);
38 |   const [lockAcquired, setLockAcquired] = useState(false);
39 |   const heartbeatRef = useRef<NodeJS.Timeout | null>(null);
40 |   const { lockComponent, releaseLock, heartbeat } = useComponentLock(componentId || null);
41 |
42 |   useEffect(() => {
43 |     if (!componentId || componentId === 'new') return;
44 |     let cancelled = false;
45 |     (async () => {
46 |       const supabase = createClient();
47 |       const { data: { user } } = await supabase.auth.getUser();
48 |       if (!user || cancelled) return;
49 |       const ok = await lockComponent(user.id);
50 |       if (!ok) { onCancel(); return; }
51 |       setLockAcquired(true);
52 |       heartbeatRef.current = setInterval(async () => {
53 |         await heartbeat(user.id);
54 |       }, 4 * 60 * 1000);
55 |     })();
56 |     return () => {
57 |       cancelled = true;
58 |       if (heartbeatRef.current) clearInterval(heartbeatRef.current);
59 |       (async () => {
60 |         const supabase = createClient();
61 |         const { data: { user } } = await supabase.auth.getUser();
62 |         if (user && componentId) await releaseLock(user.id);
63 |       })();
64 |     };
65 |   }, [componentId]);
66 |
67 |   const updateField = (field: string, value: any) => {
68 |     setForm((prev) => ({ ...prev, [field]: value }));
69 |   };
70 |
71 |   const previewSpec: ComponentRenderSpec = {
72 |     id: 'preview',
73 |     name: form.name || 'Preview',
74 |     type: form.type as ComponentRenderSpec['type'],
75 |     brand: form.brand || 'Brand',
76 |     model: form.model || 'Model',
77 |     dimensions: { width_mm: form.width_mm, height_mm: form.height_mm, depth_mm: form.depth_mm },
78 |     colors: {
79 |       pcb: form.pcbColor,
80 |       heatsink: form.heatsinkColor,
81 |       shroud: form.shroudColor,
82 |       connector: form.connectorColor,

```

```

83 |     accent: form.accentColor,
84 |     led: form.ledColor,
85 | },
86 | connectors: [],
87 | techSpecs: {},
88 | };
89 |
90 | const handleSave = async () => {
91 |   setSaving(true);
92 |   try {
93 |     const supabase = createClient();
94 |     const { data: { user } } = await supabase.auth.getUser();
95 |
96 |     if (componentId && componentId !== 'new') {
97 |       await supabase.from('components').update({
98 |         name: form.name,
99 |         type: form.type,
100 |         brand: form.brand,
101 |         model: form.model,
102 |         dimensions: { width_mm: form.width_mm, height_mm: form.height_mm, depth_mm: form.depth_mm },
103 |         colors: {
104 |           pcb: form.pcbColor,
105 |           heatsink: form.heatsinkColor,
106 |           shroud: form.shroudColor,
107 |           connector: form.connectorColor,
108 |           accent: form.accentColor,
109 |           led: form.ledColor,
110 |         },
111 |         updated_at: new Date().toISOString(),
112 |       }).eq('id', componentId);
113 |     } else {
114 |       await supabase.from('components').insert({
115 |         name: form.name,
116 |         type: form.type,
117 |         brand: form.brand,
118 |         model: form.model,
119 |         dimensions: { width_mm: form.width_mm, height_mm: form.height_mm, depth_mm: form.depth_mm },
120 |         colors: {
121 |           pcb: form.pcbColor,
122 |           heatsink: form.heatsinkColor,
123 |           shroud: form.shroudColor,
124 |           connector: form.connectorColor,
125 |           accent: form.accentColor,
126 |           led: form.ledColor,
127 |         },
128 |         created_by: user?.id,
129 |         status: 'draft',
130 |       });
131 |     }
132 |     onSave();
133 |   } catch (err) {
134 |     console.error('Save failed:', err);
135 |     onSave();
136 |   } finally {
137 |     setSaving(false);
138 |   }
139 | };
140 |
141 | const inputStyle: React.CSSProperties = {
142 |   width: '100%', padding: '8px 12px', borderRadius: 6, border: '1px solid rgba(255,255,255,0.1)',
143 |   background: 'rgba(255,255,255,0.05)', color: '#e2e8f0', fontSize: 13, outline: 'none',
144 |   fontFamily: 'inherit',
145 | };
146 |
147 | const labelStyle: React.CSSProperties = {
148 |   fontSize: 12, fontWeight: 600, color: '#94a3b8', marginBottom: 4, display: 'block',
149 | };
150 |
151 | return (
152 |   <div style={{ maxWidth: 800, margin: '0 auto' }}>
153 |     <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom: 24 }}>
154 |       <h2 style={{ fontSize: 20, fontWeight: 700, margin: 0 }}>
155 |         { componentId === 'new' ? 'Tạo linh kiện mới' : 'Chỉnh sửa linh kiện' }
156 |       </h2>
157 |       <div style={{ display: 'flex', gap: 8 }}>
158 |         <button onClick={onCancel}>
159 |           <div style={{
160 |             padding: '8px 16px', borderRadius: 6, cursor: 'pointer',
161 |             background: 'rgba(255,255,255,0.05)', border: '1px solid rgba(255,255,255,0.1)',
162 |             color: '#94a3b8', fontSize: 13, fontWeight: 500, fontFamily: 'inherit',
163 |           }}>
164 |             Hủy
165 |           </div>
166 |         <button onClick={handleSave} disabled={saving}>
167 |           <div style={{
168 |             padding: '8px 16px', borderRadius: 6, cursor: 'pointer',
169 |             background: 'linear-gradient(135deg, #00d4aa, #00b4d8)',
170 |             border: 'none', color: 'white', fontSize: 13, fontWeight: 600, fontFamily: 'inherit',
171 |             opacity: saving ? 0.6 : 1,
172 |           }}>
173 |             { saving ? 'Đang lưu...' : 'Lưu' }
174 |           </div>
175 |         </div>
176 |       </div>
177 |
178 |     <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: 24 }}>
179 |       <div>
180 |         <div style={{ marginBottom: 16 }}>
181 |           <label style={labelStyle}>Tên linh kiện</label>
182 |           <input style={inputStyle} value={form.name}
183 |             onChange={(e) => updateField('name', e.target.value)} placeholder="VD: Intel Core i5-13600K" />
184 |         </div>
185 |         <div style={{ marginBottom: 16 }}>
186 |           <label style={labelStyle}>Loại</label>
187 |           <select style={inputStyle} value={form.type}
188 |             onChange={(e) => updateField('type', e.target.value)}>
189 |             {COMPONENT_TYPES.map((t) => (
190 |               <option key={t} value={t}>{t.toUpperCase()}</option>
191 |             ))}
192 |           </select>
193 |         </div>
194 |         <div style={{ marginBottom: 16 }}>
195 |           <label style={labelStyle}>Hãng</label>
196 |           <input style={inputStyle} value={form.brand}
197 |             onChange={(e) => updateField('brand', e.target.value)} placeholder="VD: Intel, AMD, NVIDIA" />
198 |         </div>
199 |         <div style={{ marginBottom: 16 }}>
200 |           <label style={labelStyle}>Model</label>
201 |           <input style={inputStyle} value={form.model}
202 |             onChange={(e) => updateField('model', e.target.value)} placeholder="VD: Core i5-13600K" />
203 |         </div>
204 |       </div>
205 |     </div>
206 |   );

```

```

207 |
208 |         <div style={{ marginBottom: 16 }}>
209 |             <label style={labelStyle}>Kích thước (mm) </label>
210 |             <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr 1fr', gap: 8 }}>
211 |                 <div>
212 |                     <div style={{ fontSize: 10, color: '#64748b', marginBottom: 2 }}>
213 |                         <input type="number" style={inputStyle} value={form.width_mm}
214 |                         onChange={(e) => updateField('width_mm', Number(e.target.value))} />
215 |                     </div>
216 |                     <div>
217 |                         <div style={{ fontSize: 10, color: '#64748b', marginBottom: 2 }}>Cao</div>
218 |                         <input type="number" style={inputStyle} value={form.height_mm}
219 |                         onChange={(e) => updateField('height_mm', Number(e.target.value))} />
220 |                     </div>
221 |                     <div>
222 |                         <div style={{ fontSize: 10, color: '#64748b', marginBottom: 2 }}>Sâu</div>
223 |                         <input type="number" style={inputStyle} value={form.depth_mm}
224 |                         onChange={(e) => updateField('depth_mm', Number(e.target.value))} />
225 |                     </div>
226 |                 </div>
227 |             </div>
228 |
229 |         <div style={{ marginBottom: 16 }}>
230 |             <label style={labelStyle}>Màu sắc </label>
231 |             <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: 8 }}>
232 |                 {[
233 |                     { key: 'pcbColor', label: 'PCB' },
234 |                     { key: 'heatSinkColor', label: 'Tản nhiệt' },
235 |                     { key: 'shroudColor', label: 'Vỏ bọc' },
236 |                     { key: 'connectorColor', label: 'Connector' },
237 |                     { key: 'accentColor', label: 'Nhấn' },
238 |                     { key: 'ledColor', label: 'LED' },
239 |                 ].map(({ key, label }) => (
240 |                     <div key={key}>
241 |                         <div style={{ fontSize: 10, color: '#64748b', marginBottom: 2 }}>{label}</div>
242 |                         <div style={{ display: 'flex', gap: 6, alignItems: 'center' }}>
243 |                             <input type="color" value={(form as any)[key]}
244 |                             onChange={(e) => updateField(key, e.target.value)}
245 |                             style={{ width: 32, height: 32, borderRadius: 4, border: 'none', cursor: 'pointer', padd...
246 |                             <input style={{ ...inputStyle, flex: 1 }} value={(form as any)[key]}
247 |                             onChange={(e) => updateField(key, e.target.value)} />
248 |                         </div>
249 |                     </div>
250 |                 ))]}
251 |             </div>
252 |         </div>
253 |     </div>
254 |
255 |     <div>
256 |         <label style={labelStyle}>Xem trực tiếp 2D </label>
257 |         <div style={{
258 |             background: 'rgba(255,255,255,0.03)', borderRadius: 12,
259 |             padding: 24, display: 'flex', justifyContent: 'center', alignItems: 'center',
260 |             border: '1px solid rgba(255,255,255,0.08)',
261 |             minHeight: 300,
262 |         }}>
263 |             <ComponentRenderer2D
264 |                 spec={previewSpec}
265 |                 scale={1}
266 |                 showConnectors={true}
267 |                 showDimensions={true}
268 |                 viewMode="top"
269 |             />
270 |         </div>
271 |         <div style={{ fontSize: 11, color: '#64748b', marginTop: 8, textAlign: 'center' }}>
272 |             Kích thước: {form.width_mm}mm x {form.height_mm}mm x {form.depth_mm}mm
273 |         </div>
274 |     </div>
275 | </div>
276 | </div>
277 | );
278 | }
279 |

```

File: app/creator/layout.tsx (25 lines, 0.6 KB)

```

1 | import { redirect } from 'next/navigation';
2 | import { createClient } from '@lib/supabase-ssr-server';
3 |
4 | export default async function CreatorLayout({ children }: { children: React.ReactNode }) {
5 |     const supabase = await createClient();
6 |     const { data: { user } } = await supabase.auth.getUser();
7 |
8 |     if (!user) {
9 |         redirect('/login?redirect=/creator');
10 |     }
11 |
12 |     const { data: profile } = await supabase
13 |         .from('profiles')
14 |         .select('role')
15 |         .eq('id', user.id)
16 |         .single();
17 |
18 |     const role = profile?.role as string;
19 |     if (!['admin', 'component_creator'].includes(role)) {
20 |         redirect('/builder?error=unauthorized');
21 |     }
22 |
23 |     return <>{children}</>;
24 | }
25 |

```

File: app/creator/page.tsx (167 lines, 6.3 KB)

```

1 | 'use client';
2 |
3 | import { useState, useEffect } from 'react';
4 | import { createClient } from '@lib/supabase-client';
5 | import ComponentForm from './components/ComponentForm';
6 | import { COMPONENT_SPECS } from '@data/componentSpecs';
7 |
8 | interface ComponentItem {
9 |     id: string;
10 |     name: string;
11 |     type: string;
12 |     brand: string;
13 |     model: string;
14 |     status: string;
15 |     is_published: boolean;
16 |     created_at: string;
17 | }

```



```

18 export default function CreatorDashboard() {
19   const [components, setComponents] = useState<ComponentItem[]>([]);
20   const [editingId, setEditingId] = useState<string | null>(null);
21   const [filter, setFilter] = useState<'all' | 'draft' | 'review' | 'published'>('all');
22   const [loading, setLoading] = useState(true);
23   const [error, setError] = useState<string | null>(null);
24
25   useEffect(() => {
26     loadComponents();
27   }, []);
28
29   const loadComponents = async () => {
30     try {
31       setLoading(true);
32       const supabase = createClient();
33       const { data, error } = await supabase
34         .from('components')
35         .select('*')
36         .order('created_at', { ascending: false });
37
38       if (error) {
39         // Fallback to local data if Supabase table doesn't exist
40         setComponents(COMPONENT_SPECS.map((s) => ({
41           id: s.id,
42           name: s.name,
43           type: s.type,
44           brand: s.brand,
45           model: s.model,
46           status: 'published',
47           is_published: true,
48           created_at: new Date().toISOString(),
49         })));
50       } else {
51         setComponents(data || []);
52       }
53     } catch (err) {
54       // Fallback to local data
55       setComponents(COMPONENT_SPECS.map((s) => ({
56         id: s.id,
57         name: s.name,
58         type: s.type,
59         brand: s.brand,
60         model: s.model,
61         status: 'published',
62         is_published: true,
63         created_at: new Date().toISOString(),
64       })));
65     } finally {
66       setLoading(false);
67     }
68   };
69
70   const filtered = filter === 'all' ? components : components.filter((c) => c.status === filter);
71
72   const statusColors: Record<string, string> = {
73     draft: '#64748b',
74     review: '#f59e0b',
75     published: '#22c55e',
76   };
77
78   return (
79     <div style={{ display: 'flex', height: '100vh', background: '#0f0f1a', color: '#e2e8f0' }}>
80       <div style={{
81         width: 320, borderRight: '1px solid rgba(255,255,255,0.08)',
82         display: 'flex', flexDirection: 'column',
83       }}>
84         <div style={{ padding: '20px 16px', borderBottom: '1px solid rgba(255,255,255,0.08)' }}>
85           <div style={{ display: 'flex', alignItems: 'center', gap: 8, marginBottom: 8 }}>
86             <span style={{ font-size: 18, font-weight: 700 }}>Component Creator Studio</span>
87           </div>
88           <div style={{
89             display: 'inline-flex', padding: '2px 10px', borderRadius: 100,
90             background: 'rgba(168,85,247,0.2)', color: '#a855f7',
91             font-size: 11, font-weight: 700, textTransform: 'uppercase', letterSpacing: 1,
92           }}>
93             Creator Mode
94           </div>
95         </div>
96
97         <div style={{ padding: '12px 16px', display: 'flex', gap: 6, borderBottom: '1px solid rgba(255,255,255,0.08)' }}>
98           {(['all', 'draft', 'review', 'published'] as const).map((f) => (
99             <button key={f} onClick={() => setFilter(f)}
100               style={{
101                 padding: '4px 12px', borderRadius: 6, font-size: 12, font-weight: 600, cursor: 'pointer',
102                 background: filter === f ? 'rgba(0,212,170,0.2)' : 'rgba(255,255,255,0.05)',
103                 color: filter === f ? '#00d4aa' : '#94a3b8',
104                 border: '1px solid ${filter === f ? '#00d4aa' : 'transparent'}',
105               }}
106             >{ f === 'all' ? 'T a l l' : f.charAt(0).toUpperCase() + f.slice(1)}</button>
107           ))}
108         </div>
109
110         <div style={{ flex: 1, overflowY: 'auto', padding: 8 }}>
111           {loading && <div style={{ padding: 20, textAlign: 'center', color: '#64748b' }}>Loading...</div>
112           {filtered.map((comp) => (
113             <div key={comp.id} onClick={() => setEditingId(comp.id)}
114               style={{
115                 padding: '10px 12px', borderRadius: 8, marginBottom: 4, cursor: 'pointer',
116                 background: editingId === comp.id ? 'rgba(0,212,170,0.08)' : 'transparent',
117                 border: '1px solid ${editingId === comp.id ? 'transparent' : 'transparent'}',
118               }}>
119             <div style={{ font-size: 14, font-weight: 600 }}>{comp.name}</div>
120             <div style={{ display: 'flex', alignItems: 'center', gap: 8, marginTop: 4 }}>
121               <span style={{ font-size: 11, color: '#64748b' }}>{comp.type.toUpperCase()}</span>
122               <span style={{
123                 padding: '1px 8px', borderRadius: 100, font-size: 10, font-weight: 600,
124                 background: `${statusColors[comp.status]}20`,
125                 color: statusColors[comp.status],
126               }}>
127                 {comp.status}
128               </span>
129             </div>
130           </div>
131           </div>
132         </div>
133
134       <div style={{ padding: 12, borderTop: '1px solid rgba(255,255,255,0.08)' }}>
135         <button onClick={() => setEditingId('new')}
136           style={{
137             width: '100%', padding: '8px 16px', borderRadius: 8, cursor: 'pointer',
138             background: 'linear-gradient(135deg, #00d4aa, #00b4d8)',
139           }}
140         >New Component</button>
141       </div>
142     </div>
143   );
144 }

```

```

142 |         color: '#fff', border: 'none', fontWeight: 600, fontSize: 13,
143 |     }}>
144 | + T i o l i n h k i Ç n m Û i
145 | </button>
146 | </div>
147 | </div>
148 |
149 | <div style={{ flex: 1, overflowY: 'auto', padding: 24 }}>
150 |   {editingId ? (
151 |     <ComponentForm
152 |       componentId={editingId === 'new' ? null : editingId}
153 |       onSave={() => { setEditingId(null); loadComponents(); }}
154 |       onCancel={() => setEditingId(null)}
155 |     ) />
156 |   ) : (
157 |     <div style={{ display: 'flex', flexDirection: 'column', alignItems: 'center', justifyContent: 'cente...
158 |       <div style={{ fontSiz e : { fontSiz e : 4 8 , mar g i n B o t t o m : 1 6 } } > Ý ' < / d i v >
159 |       <div style={{ fontSiz e : { fontSiz e : 1 8 , fontW e i g h t : 6 0 0 , mar g i n B o t t o m : 8
160 |       <div style={{ fontSiz e : { fontSiz e : 1 4 } } > h o . c n h ¤ n " T i o l i n h k i Ç n m Û i "
161 |     )}
162 |   )}
163 | </div>
164 | </div>
165 | );
166 | }
167 |

```

File: app/daily-quiz/[quizId]/page.tsx (477 lines, 21.1 KB)

```

1 | 'use client'
2 |
3 | import { useState, useEffect, use } from 'react'
4 | import { createBrowserClient } from '@supabase/ssr'
5 | import { useRouter } from 'next/navigation'
6 | import { motion, AnimatePresence } from 'framer-motion'
7 | import {
8 |   CheckCircle, XCircle, Trophy, Loader2, ArrowLeft,
9 |   Sparkles, Share2, Home
10 | } from 'lucide-react'
11 |
12 | const LABELS = ['A', 'B', 'C', 'D']
13 | const DAILY_TOPIC_IDS = new Set([
14 |   'cpu-arch', 'ram-storage', 'gpu-graphics', 'motherboard',
15 |   'psu-cooling', 'storage-devices', 'pc-assembly', 'bios-uefi',
16 |   'network', 'peripherals', 'os-windows', 'troubleshooting',
17 |   'laptop-mobile', 'monitor-display', 'ai-tech', 'overclocking',
18 |   'benchmark', 'pc-gaming', 'server-workstation', 'cable-connectivity',
19 |   'virtual-reality', 'cybersecurity',
20 | ])
21 |
22 | interface Question {
23 |   id: string
24 |   text: string
25 |   question: string
26 |   options: string[]
27 |   correct: number
28 |   explanation: string
29 | }
30 |
31 | export default function DailyQuizPage({ params }: { params: Promise<{ quizId: string }> }) {
32 |   const { quizId } = use(params)
33 |   const router = useRouter()
34 |   const supabase = createBrowserClient(
35 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
36 |     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
37 |   )
38 |
39 |   const [questions, setQuestions] = useState<Question[]>([])
40 |   const [topicTitle, setTopicTitle] = useState('')
41 |   const [currentIndex, setCurrentIdx] = useState(0)
42 |   const [selectedIdx, setSelectedIdx] = useState<number | null>(null)
43 |   const [confirmed, setConfirmed] = useState(false)
44 |   const [finished, setFinished] = useState(false)
45 |   const [loading, setLoading] = useState(true)
46 |   const [error, setError] = useState('')
47 |   const [submitting, setSubmitting] = useState(false)
48 |   const [result, setResult] = useState<{ score: number; total: number; xpEarned: number } | null>(null)
49 |   const [answers, setAnswers] = useState<Record<string, { selected: number; correct: number }>>({})
50 |
51 |   const isDailyTopic = DAILY_TOPIC_IDS.has(quizId)
52 |
53 |   useEffect(() => {
54 |     if (isDailyTopic) {
55 |       initDailyQuiz()
56 |     } else {
57 |       initLegacyQuiz()
58 |     }
59 |   }, [])
60 |
61 |   async function initDailyQuiz() {
62 |     try {
63 |       setLoading(true)
64 |       const res = await fetch(`/api/daily-quiz?topicId=${quizId}`)
65 |       if (!res.ok) throw new Error('Không tìm thấy chç Á ')
66 |       const data = await res.json()
67 |
68 |       if (data.completed) {
69 |         setFinished(true)
70 |         setResult({ score: 0, total: 0, xpEarned: 0 })
71 |         setLoading(false)
72 |         return
73 |       }
74 |
75 |       setQuestions(data.questions)
76 |       setTopicTitle(data.topic?.title || quizId)
77 |       setLoading(false)
78 |     } catch (err: any) {
79 |       setError(err.message || 'Có l x i x £ y ra ')
80 |       setLoading(false)
81 |     }
82 |   }
83 |
84 |   async function initLegacyQuiz() {
85 |     try {
86 |       setLoading(true)
87 |       const { data: quizData, error: qErr } = await supabase
88 |         .from('quizzes')
89 |         .select('')
90 |         .eq('id', quizId)
91 |         .single()
92 |       if (qErr || !quizData) { setError('Không tìm thấy bài kiĂm tra. ')
93 |
94 |       setTopicTitle(quizData.title || '')

```

```

95 |
96 |     const { data: qs, error: qsErr } = await supabase
97 |       .from('quiz_questions_for_student')
98 |       .select('*')
99 |       .eq('quiz_id', quizId)
100 |       .order('order', { ascending: true })
101 |     if (qsErr) { setError('Lỗi tải câu hỏi. '); return }
102 |     if (!qs || qs.length === 0) { setError('Bài kiểm tra chưa có câu hỏi.') }
103 |
104 |     const mapped = qs.map((q: any) => ({
105 |       id: q.id,
106 |       text: q.content || q.question || '',
107 |       question: q.content || q.question || '',
108 |       options: (q.options || []).map((o: any) => o.content || o.text || o),
109 |       correct: 0,
110 |       explanation: q.explanation || '',
111 |     }))
112 |     setQuestions(mapped)
113 |     setLoading(false)
114 |   } catch {
115 |     setError('Có lỗi xảy ra. ')
116 |   }
117 | }
118 |
119 |
120 | function handleSelect(idx: number) {
121 |   if (confirmed) return
122 |   setSelectedIdx(idx)
123 | }
124 |
125 | function handleConfirm() {
126 |   if (selectedIdx === null) return
127 |   setConfirmed(true)
128 |   const q = questions[currentIdx]
129 |   setAnswers(prev => ({
130 |     ...prev,
131 |     [q.id]: { selected: selectedIdx, correct: q.correct },
132 |   }))
133 | }
134 |
135 | function handleNext() {
136 |   if (currentIdx < questions.length - 1) {
137 |     setCurrentIdx(prev => prev + 1)
138 |     setSelectedIdx(null)
139 |     setConfirmed(false)
140 |   }
141 | }
142 |
143 | async function handleSubmit() {
144 |   if (submitting) return
145 |   setSubmitting(true)
146 |   setFinished(true)
147 |
148 |   if (!isDailyTopic) {
149 |     setResult({ score: 0, total: questions.length, xpEarned: 0 })
150 |     setSubmitting(false)
151 |     return
152 |   }
153 |
154 |   let correctCount = 0
155 |   for (const q of questions) {
156 |     const ans = answers[q.id]
157 |     if (ans && ans.selected === q.correct) correctCount++
158 |   }
159 |   const xpEarned = correctCount * 5
160 |
161 |   try {
162 |     const res = await fetch('/api/daily-quiz', {
163 |       method: 'POST',
164 |       headers: { 'Content-Type': 'application/json' },
165 |       body: JSON.stringify({
166 |         topicId: quizId,
167 |         score: correctCount,
168 |         total: questions.length,
169 |         xpEarned,
170 |       }),
171 |     )
172 |     const json = await res.json()
173 |     setResult({
174 |       score: correctCount,
175 |       total: questions.length,
176 |       xpEarned: json.xpAwarded || xpEarned,
177 |     })
178 |   } catch {
179 |     setResult({ score: correctCount, total: questions.length, xpEarned })
180 |   }
181 |
182 |   setSubmitting(false)
183 | }
184 |
185 | async function handleShare() {
186 |   if (!result) return
187 |   const shareData = {
188 |     title: 'PC Master - Thử thách hằng ngày',
189 |     text: `Minh vừa đạt ${result.score}/${result.total} trong thử thách`,
190 |     url: window.location.href,
191 |   }
192 |   if (navigator.share) {
193 |     try { await navigator.share(shareData) } catch {}
194 |   } else {
195 |     await navigator.clipboard.writeText(shareData.text)
196 |   }
197 | }
198 |
199 | if (loading) return (
200 |   <div style={{ minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', alignItems: 'center', justify-content: 'center' }}>
201 |     <div style={{ textAlign: 'center' }}>
202 |       <Loader2 className="animate-spin" size={48} style={{ color: 'var(--brand-primary)', margin: '0 auto 16px auto' }} />
203 |       <p style={{ color: 'var(--text-muted)' }}>Đang tải câu hỏi...</p>
204 |     </div>
205 |   </div>
206 | )
207 |
208 | if (error) return (
209 |   <div style={{ minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', alignItems: 'center', justify-content: 'center' }}>
210 |     <div style={{ borderRadius: '24px', border: '1px solid var(--border-strong)', background: 'var(--bg-surf)', padding: '24px' }}>
211 |       <p style={{ color: '#ef4444', marginBottom: '16px' }}>{error}</p>
212 |       <button onClick={() => router.push(isDailyTopic ? '/daily-quiz' : '/')} style={{ padding: '12px 24px', ... }}>
213 |         Quay lại
214 |       </button>
215 |     </div>
216 |   </div>
217 | )
218 |

```

```

219 | if (finished && result && result.score === 0 && result.total === 0) return (
220 |   <div style={{ minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', alignItems: 'center', jus...
221 |     <motion.div initial={{ scale: 0.8, opacity: 0 }} animate={{ scale: 1, opacity: 1 }}
222 |       style={{ borderRadius: '24px', border: '1px solid var(--border-strong)', background: 'var(--bg-surface...
223 |     >
224 |       <CheckCircle style={{ margin: '0 auto 16px', color: 'var(--brand-primary)' }} size={80} />
225 |       <h2 style={{ fontSize: '24px', fontWeight: 700, color: 'var(--text-primary)', marginBottom: '8px' }}>B...
226 |         <p style={{ color: 'var(--text-muted)', margin: '0 auto 16px', background: 'var(--bg-surface...
227 |         <button onClick={() => router.push('/daily-quiz')} style={{ padding: '16px 32px', background: 'var(--b...
228 |           V Á d a n h s á c h
229 |       </button>
230 |     </motion.div>
231 |   </div>
232 | )
233 |
234 | if (finished && result) return (
235 |   <div style={{ minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', alignItems: 'center', jus...
236 |     <motion.div initial={{ scale: 0.8, opacity: 0 }} animate={{ scale: 1, opacity: 1 }}
237 |       style={{ borderRadius: '24px', border: '1px solid var(--border-strong)', background: 'var(--bg-surface...
238 |     >
239 |       <Trophy style={{ margin: '0 auto 16px', color: 'var(--accent-amber)' }} size={72} />
240 |       <h2 style={{ fontSize: '24px', fontWeight: 700, color: 'var(--text-primary)', marginBottom: '8px' }}>H...
241 |       <p style={{ color: 'var(--text-muted)', margin: '0 auto 16px', background: 'var(--bg-surface...
242 |       <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '12px', margin: '1...
243 |         <span style={{ fontSize: '48px', fontWeight: 900, color: 'var(--brand-primary)' }}>{result.score}</s...
244 |         <span style={{ fontSize: '24px', color: 'var(--text-muted)' }}>{result.total}</span>
245 |       </div>
246 |       <div style={{ display: 'inline-flex', alignItems: 'center', gap: '8px', padding: '8px 16px', background...
247 |         <Sparkles size={16} />
248 |         {result.xpEarned} XP
249 |       </div>
250 |       <div style={{ display: 'flex', flexDirection: 'column', gap: '12px' }}>
251 |         <button
252 |           onClick={handleShare}
253 |           style={{ width: '100%', padding: '16px 32px', border: '1px solid var(--border-strong)', borderRadi...
254 |         >
255 |           <Share2 size={18} />
256 |           C h i a s s k ử t q u ế
257 |         </button>
258 |         <button
259 |           onClick={() => router.push('/daily-quiz')}
260 |           style={{ width: '100%', padding: '16px 32px', background: 'var(--brand-primary)', color: 'var(--bg...
261 |         >
262 |           <Home size={18} />
263 |           V Á d a n h s á c h
264 |         </button>
265 |       </div>
266 |     </motion.div>
267 |   </div>
268 | )
269 |
270 | if (questions.length === 0) return null
271 |
272 | const q = questions[currentIdx]
273 | const currentAnswer = answers[q.id]
274 |
275 | return (
276 |   <div style={{ minHeight: '100vh', background: 'var(--bg-base)', color: 'var(--text-primary)', padding: '16...
277 |     <div style={{ maxWidth: '768px', margin: '0 auto' }}>
278 |       <div style={{ display: 'flex', alignItems: 'center', justify-content: 'space-between', margin-bottom: '2...
279 |         <button
280 |           onClick={() => router.push('/daily-quiz')}
281 |           style={{ display: 'flex', alignItems: 'center', gap: '8px', color: 'var(--text-muted)', background...
282 |         >
283 |           <ArrowLeft size={20} />
284 |           <span>Quay lại</span>
285 |         </button>
286 |         <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
287 |           <div style={{ display: 'flex', alignItems: 'center', gap: '8px' }}>
288 |             <span style={{ fontSize: '14px', color: 'var(--text-muted)' }}>{currentIdx + 1}/{questions.lengt...
289 |             <span style={{ fontSize: '12px', padding: '4px 10px', background: 'color-mix(in srgb, var(--bran...
290 |               {Object.keys(answers).length} / {questions.length}
291 |             </span>
292 |           </div>
293 |           <div style={{ display: 'flex', alignItems: 'center', gap: '6px', color: 'var(--brand-primary)', ...
294 |             <Sparkles size={14} />
295 |             <span>5 XP / câu</span>
296 |           </div>
297 |         </div>
298 |       </div>
299 |     </div>
300 |   </div>
301 | )
302 |
303 | <div style={{ width: '100%', height: '8px', background: 'var(--border-strong)', borderRadius: '9999px'...
304 |   <div style={{ height: '100%', background: 'var(--brand-primary)', borderRadius: '9999px', transition: '...
305 |   </div>
306 | </div>
307 |
308 | <div style={{ width: '100%', height: '8px', background: 'var(--border-strong)', borderRadius: '9999px'...
309 |   <div style={{ height: '100%', background: 'var(--brand-primary)', borderRadius: '9999px', transition: '...
310 |   </div>
311 | </div>
312 |
313 | <div style={{ width: '100%', height: '8px', background: 'var(--border-strong)', borderRadius: '9999px'...
314 |   <div style={{ height: '100%', background: 'var(--brand-primary)', borderRadius: '9999px', transition: '...
315 |   </div>
316 | </div>
317 |
318 | <div style={{ width: '100%', height: '8px', background: 'var(--border-strong)', borderRadius: '9999px'...
319 |   <div style={{ height: '100%', background: 'var(--brand-primary)', borderRadius: '9999px', transition: '...
320 |   </div>
321 | </div>
322 |
323 | <div style={{ width: '100%', height: '8px', background: 'var(--border-strong)', borderRadius: '9999px'...
324 |   <div style={{ height: '100%', background: 'var(--brand-primary)', borderRadius: '9999px', transition: '...
325 |   </div>
326 | </div>
327 |
328 | <div style={{ width: '100%', height: '8px', background: 'var(--border-strong)', borderRadius: '9999px'...
329 |   <div style={{ height: '100%', background: 'var(--brand-primary)', borderRadius: '9999px', transition: '...
330 |   </div>
331 | </div>
332 |
333 | <div style={{ width: '100%', height: '8px', background: 'var(--border-strong)', borderRadius: '9999px'...
334 |   <div style={{ height: '100%', background: 'var(--brand-primary)', borderRadius: '9999px', transition: '...
335 |   </div>
336 | </div>
337 |
338 | <div style={{ width: '100%', height: '8px', background: 'var(--border-strong)', borderRadius: '9999px'...
339 |   <div style={{ height: '100%', background: 'var(--brand-primary)', borderRadius: '9999px', transition: '...
340 |   </div>
341 | </div>
342 |

```

```

343|         onClick={() => handleSelect(i)}
344|         disabled={confirmed}
345|         style={{
346|             width: '100%', textAlign: 'left', padding: '20px', borderRadius: '16px',
347|             display: 'flex', alignItems: 'center', gap: '16px', cursor: confirmed ? 'default' : 'point...
348|             border: '1px solid',
349|             borderColor: isCorrectOption
350|                 ? 'var(--brand-primary)'
351|                 : isWrongOption
352|                     ? '#ef4444'
353|                     : isSelected
354|                         ? 'var(--accent-blue)'
355|                         : 'var(--border-strong)',
356|             background: isCorrectOption
357|                 ? 'color-mix(in srgb, var(--brand-primary) 15%, transparent)'
358|                 : isWrongOption
359|                     ? 'color-mix(in srgb, #ef4444 10%, transparent)'
360|                     : isSelected
361|                         ? 'color-mix(in srgb, var(--accent-blue) 10%, transparent)'
362|                         : 'var(--bg-surface)',
363|             opacity: confirmed && !isSelected && !isCorrectOption ? 0.5 : 1,
364|             color: 'var(--text-primary)',
365|             transition: 'all 0.2s',
366|             fontSize: '16px'
367|         }}
368|     >
369|         <div style={{
370|             width: '40px', height: '40px', borderRadius: '12px', display: 'flex',
371|             alignItems: 'center', justifyContent: 'center', fontWeight: 700, fontSize: '18px', flexShr...
372|             background: isCorrectOption
373|                 ? 'var(--brand-primary)'
374|                 : isWrongOption
375|                     ? '#ef4444'
376|                     : isSelected
377|                         ? 'var(--accent-blue)'
378|                         : 'var(--border-strong)',
379|             color: isCorrectOption
380|                 ? 'var(--bg-base)'
381|                 : isWrongOption || isSelected
382|                     ? 'white'
383|                     : 'var(--text-muted)'
384|         }}>
385|             {isCorrectOption ? <CheckCircle size={22} /> : isWrongOption ? <XCircle size={22} /> : LAB...
386|         </div>
387|         <span style={{ fontSize: '16px' }}>{opt.replace(/^[A-D]\.s*/, '')}</span>
388|     </button>
389| )
390| }}}
391| </motion.div>
392| </AnimatePresence>
393|
394| /* Explanation */
395| <AnimatePresence>
396|     {confirmed && q.explanation && (
397|         <motion.div
398|             initial={{ opacity: 0, y: 10 }}
399|             animate={{ opacity: 1, y: 0 }}
400|             style={{
401|                 marginTop: '16px', padding: '20px', borderRadius: '16px',
402|                 border: '1px solid',
403|                 borderColor: currentAnswer?.selected === q.correct
404|                     ? 'color-mix(in srgb, var(--brand-primary) 30%, transparent)'
405|                     : 'color-mix(in srgb, #ef4444 30%, transparent)',
406|                 background: currentAnswer?.selected === q.correct
407|                     ? 'color-mix(in srgb, var(--brand-primary) 5%, transparent)'
408|                     : 'color-mix(in srgb, #ef4444 5%, transparent)'
409|             }}
410|         >
411|             <div style={{ display: 'flex', alignItems: 'flex-start', gap: '12px' }}>
412|                 {currentAnswer?.selected === q.correct
413|                     ? <CheckCircle style={{ color: 'var(--brand-primary)', flexShrink: 0, marginTop: '2px' }} si...
414|                     : <XCircle style={{ color: '#ef4444', flexShrink: 0, marginTop: '2px' }} size={20} />
415|                 }
416|                 <div>
417|                     <p style={{ fontSize: '14px', fontWeight: 600, marginBottom: '4px' }}>
418|                         {currentAnswer?.selected === q.correct ? 'Chính xác!'
419|                         </p>
420|                         <p style={{ fontSize: '14px', color: 'var(--text-muted)' }}>{q.explanation}</p>
421|                     </div>
422|                 </div>
423|             </motion.div>
424|         )
425|     }
426| </AnimatePresence>
427|
428| /* Actions */
429| <div style={{ display: 'flex', gap: '16px', marginTop: '32px' }}>
430|     {!confirmed ? (
431|         <button
432|             onClick={handleConfirm}
433|             disabled={selectedIdx === null}
434|             style={{
435|                 padding: '16px 32px', borderRadius: '9999px', fontWeight: 600, border: 'none', cursor: selecte...
436|                 background: selectedIdx !== null ? 'var(--accent-blue)' : 'var(--border-strong)',
437|                 color: selectedIdx !== null ? 'white' : 'var(--text-muted)'
438|             }}
439|         >
440|             X á c   n h - n
441|         </button>
442|     ) : (
443|         <button
444|             onClick={handleNext}
445|             style={{ padding: '16px 32px', background: 'var(--brand-primary)', color: 'var(--bg-base)', bord...
446|         >
447|             {currentIdx < questions.length - 1 ? 'Câu tiếp!' : 'Xem k...
448|         </button>
449|     )}
450|
451|     (() => {
452|         const unanswered = questions.length - Object.keys(answers).length
453|         const showWarning = unanswered > 0 && currentIdx >= questions.length - 1 && confirmed
454|         if (!showWarning) return null
455|         return (
456|             <motion.div
457|                 initial={{ opacity: 0, y: -10 }}
458|                 animate={{ opacity: 1, y: 0 }}
459|                 style={{ padding: '12px 20px', borderRadius: '12px', background: 'color-mix(in srgb, #ef4444 1...
460|             >
461|                 <XCircle size={18} />
462|                 C ò n   {unanswered}   c â u   c h ° a   t r é   l ý i
463|             </motion.div>
464|         )
465|     })()
466|     <button
467|         onClick={handleSubmit}

```

```

467 |         disabled={submitting || Object.keys(answers).length < questions.length}
468 |         style={{ padding: '16px 24px', border: '1px solid var(--border-strong)', borderRadius: '9999px', f...
469 |     }
470 | }
471 | </button>
472 | </div>
473 | </div>
474 | </div>
475 | )
476 | }
477 |

```

File: app/daily-quiz/page.tsx (315 lines, 15.5 KB)

```

1 | 'use client'
2 |
3 | import { useState, useEffect } from 'react'
4 | import { useRouter } from 'next/navigation'
5 | import { useIsMobile } from '@hooks/useIsMobile'
6 | import { motion } from 'framer-motion'
7 | import { createBrowserClient } from '@supabase/ssr'
8 | import {
9 |   Trophy, ChevronRight, Sparkles,
10 |   CheckCircle, Calendar, Loader2, Star, Zap, BookOpen
11 | } from 'lucide-react'
12 |
13 | const LABELS = ['A', 'B', 'C', 'D']
14 | const TOPIC_ICONS: Record<string, string> = {
15 |   'cpu-arch': 'ðŸ’Š', 'ram-storage': 'ðŸ’Š', 'gpu-graphics': 'ðŸ’Š',
16 |   'motherboard': 'ðŸ’Š', 'psu-cooling': 'ðŸ’Š', 'storage-devices': 'ðŸ’Š',
17 |   'pc-assembly': 'ðŸ’Š', 'bios-uefi': 'ðŸ’Š', 'network': 'ðŸ’Š',
18 |   'peripherals': 'ðŸ’Š', 'os-windows': 'ðŸ’Š', 'troubleshooting': 'ðŸ’Š',
19 |   'laptop-mobile': 'ðŸ’Š', 'monitor-display': 'ðŸ’Š', 'ai-tech': 'ðŸ’Š',
20 |   'overclocking': 'ðŸ’Š', 'benchmark': 'ðŸ’Š', 'pc-gaming': 'ðŸ’Š',
21 |   'server-workstation': 'ðŸ’Š', 'cable-connectivity': 'ðŸ’Š',
22 |   'virtual-reality': 'ðŸ’Š', 'cybersecurity': 'ðŸ’Š',
23 | }
24 |
25 | interface Topic {
26 |   id: string
27 |   title: string
28 |   icon?: string
29 |   color?: string
30 |   description?: string
31 |   questions: any[]
32 | }
33 |
34 | interface ApiResponse {
35 |   todayIndex: number
36 |   todayTopicId: string
37 |   todayTopic: Topic
38 |   topics: Topic[]
39 |   completedTopicIds: string[]
40 |   isTodayCompleted: boolean
41 | }
42 |
43 | export default function DailyQuizListPage() {
44 |   const router = useRouter()
45 |   const supabase = createBrowserClient(
46 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
47 |     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
48 |   )
49 |
50 |   const [data, setData] = useState<ApiResponse | null>(null)
51 |   const [loading, setLoading] = useState(true)
52 |   const [error, setError] = useState('')
53 |   const [user, setUser] = useState<any>(null)
54 |   const isMobile = useIsMobile()
55 |
56 |   useEffect(() => {
57 |     supabase.auth.getUser().then(({ data: { user } }) => {
58 |       setUser(user)
59 |       fetchData()
60 |     })
61 |   }, [])
62 |
63 |   async function fetchData() {
64 |     try {
65 |       setLoading(true)
66 |       const res = await fetch('/api/daily-quiz')
67 |       if (!res.ok) throw new Error('Failed to fetch')
68 |       const json = await res.json()
69 |       setData(json)
70 |     } catch {
71 |       setError('Không thể tải dữ liệu. Vui lòng thử lại.')
72 |     } finally {
73 |       setLoading(false)
74 |     }
75 |   }
76 |
77 |   function getDayNumber(index: number): number {
78 |     if (!data) return index + 1
79 |     const diff = index - data.todayIndex
80 |     if (diff === 0) return 0
81 |     return diff
82 |   }
83 |
84 |   if (loading) return (
85 |     <div style={{ minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', alignItems: 'center', jus...
86 |       <div style={{ textAlign: 'center' }}>
87 |         <Loader2 className="animate-spin" size={48} style={{ color: 'var(--brand-primary)', margin: '0 auto 16...
88 |       <p style={{ color: 'var(--text-muted)' }}> Đang tải thử thách hôm nay
89 |     </div>
90 |   )
91 |
92 |   if (error) return (
93 |     <div style={{ minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', alignItems: 'center', jus...
94 |       <div style={{ borderRadius: '24px', border: '1px solid var(--border-strong)', background: 'var(--bg-surf...
95 |         <p style={{ color: '#ef4444', marginBottom: '16px' }}>{error}</p>
96 |         <button onClick={fetchData} style={{ padding: '12px 24px', background: 'var(--border-strong)', borderR...
97 |       </div>
98 |     </div>
99 |   )
100 |
101 |   </div>
102 | )
103 |
104 | if (!data) return null
105 |
106 | const { topics, todayIndex, todayTopicId, completedTopicIds, isTodayCompleted } = data
107 |
108 | return (
109 |   <div style={{ minHeight: '100vh', background: 'var(--bg-base)', color: 'var(--text-primary)' }}>

```

```
110 | <div style={{ maxWidth: '896px', margin: '0 auto', padding: isMobile ? '24px 12px' : '32px 16px' }} clas...
111 | /* Header */
112 | <motion.div
113 |   initial={{ opacity: 0, y: -20 }}
114 |   animate={{ opacity: 1, y: 0 }}
115 |   style={{ textAlign: 'center', marginBottom: '40px' }}
116 | >
117 |   <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '8px', marginBot...
118 |     <BookOpen style={{ color: 'var(--brand-primary)' }} size={28} />
119 |     <h1 style={{ fontSize: isMobile ? '24px' : '30px', fontWeight: 900 }} className="md:text-4xl">Ngân...
120 |   </div>
121 |   <p style={{ color: 'var(--text-muted)', maxWidth: '512px', margin: '0 auto' }}>
122 |     2 2 | 2 2 c h c ́ Á , m x i c h c ́ Á 1 0 c á u h ́ i . H o à n t h à n h ́ A n h - n X P v à
123 |   </p>
124 | </motion.div>
125 |
126 | /* Today's Highlight */
127 | <motion.div
128 |   initial={{ opacity: 0, scale: 0.95 }}
129 |   animate={{ opacity: 1, scale: 1 }}
130 |   transition={{ delay: 0.1 }}
131 |   style={{
132 |     position: 'relative', overflow: 'hidden', borderRadius: '24px',
133 |     border: '1px solid color-mix(in srgb, var(--brand-primary) 30%, transparent)',
134 |     background: 'linear-gradient(to bottom right, color-mix(in srgb, var(--brand-primary) 10%, transpa...
135 |     padding: '32px', marginBottom: '40px'
136 |   }}
137 | >
138 |   <div style={{ position: 'absolute', top: 0, right: 0, width: '160px', height: '160px', background: '...
139 |   <div style={{ position: 'relative', zIndex: 10 }}>
140 |     <div style={{ display: 'flex', alignItems: 'center', gap: '8px', marginBottom: '8px' }}>
141 |       <Sparkles style={{ color: 'var(--brand-primary)' }} size={18} />
142 |       <span style={{ fontSize: '12px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: '1p...
143 |         M B k h ó a h ô m n a y
144 |     </span>
145 |   </div>
146 |   <div style={{ display: 'flex', alignItems: 'center', gap: '16px' }}>
147 |     <div style={{ width: '56px', height: '56px', borderRadius: '16px', background: 'color-mix(in srg...
148 |       { d a t a . t o d a y T o p i c . i c o n | | T O P I C _ I C O N S [ d a t a . t o d a y T o p i c . i d ]
149 |     </div>
150 |     <div style={{ flex: 1, minWidth: 0 }}>
151 |       <h2 style={{ fontSize: '20px', fontWeight: 700 }} className="md:text-2xl truncate">{data.today...
152 |       <p style={{ fontSize: '14px', color: 'var(--text-muted)', marginTop: '4px' }}>{data.todayTopic...
153 |     </div>
154 |     <div style={{ flexShrink: 0 }}>
155 |       {isTodayCompleted ? (
156 |         <div style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '8px 16px', backgr...
157 |           <CheckCircle size={16} />
158 |           á h o à n t h à n h
159 |         </div>
160 |       ) : (
161 |         <button
162 |           onClick={() => router.push(`/daily-quiz/${todayTopicId}`)}
163 |           style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '12px 24px', backgrou...
164 |         >
165 |           Làm bài ngay
166 |           <ChevronRight size={18} />
167 |         </button>
168 |       )
169 |     </div>
170 |   </div>
171 | </motion.div>
172 |
173 | /* Stats */
174 | <motion.div
175 |   initial={{ opacity: 0, y: 20 }}
176 |   animate={{ opacity: 1, y: 0 }}
177 |   transition={{ delay: 0.2 }}
178 |   style={{ display: 'grid', gridTemplateColumns: isMobile ? 'repeat(1, 1fr)' : 'repeat(3, 1fr)', gap: ...
179 | >
180 |   <div style={{ borderRadius: '16px', border: '1px solid var(--border-strong)', background: 'var(--bg-...
181 |     <Trophy style={{ margin: '0 auto 8px', color: 'var(--accent-amber)' }} size={24} />
182 |     <div style={{ fontSize: '24px', fontWeight: 700 }}>{completedTopicIds.length}</div>
183 |     <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
184 |       <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
185 |         <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
186 |           <Star style={{ margin: '0 auto 8px', color: 'var(--brand-primary)' }} size={24} />
187 |           <div style={{ fontSize: '24px', fontWeight: 700 }}>{topics.length}</div>
188 |         <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
189 |           <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
190 |             <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
191 |               <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
192 |                 <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
193 |                   <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
194 |                     <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
195 |                       <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
196 |                         <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
197 |                           <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
198 |                             <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
199 |                               <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
200 |                                 <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
201 |                                   <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
202 |                                     <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
203 |                                       <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
204 |                                         <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
205 |                                           <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
206 |                                             <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
207 |                                               <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
208 |                                                 <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
209 |                                                   <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
210 |                                                     <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
211 |                                                       <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
212 |                                                         <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
213 |                                                           <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
214 |                                                             <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
215 |                                                               <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
216 |                                                                 <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
217 |                                                                   <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
218 |                                                                     <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
219 |                                                                       <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
220 |                                                                         <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
221 |                                                                           <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
222 |                                                                             <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
223 |                                                                               <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
224 |                                                                                 <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
225 |                                                                                   <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
226 |                                                                                     <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
227 |                                                                                       <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
228 |                                                                                        <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
229 |                                                                                          <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
230 |                                                                                           <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
231 |                                                                                             <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
232 |                                                                                              <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
233 |                                                                                               <div style={{ display: 'flex', gap: '12px', color: 'var(--text-muted)' }}>
```

```

234 |         borderRadius: '15px', padding: '16px 20px',
235 |         background: isCompleted
236 |         ? 'color-mix(in srgb, #22c55e 5%, var(--bg-surface))'
237 |         : 'var(--bg-surface)',
238 |         border: '1px solid',
239 |         borderColor: isToday
240 |         ? 'transparent'
241 |         : isCompleted
242 |         ? 'color-mix(in srgb, #22c55e 25%, transparent)'
243 |         : 'var(--border-strong)',
244 |     }}
245 | >
246 | <div style={{ display: 'flex', alignItems: 'center', gap: '16px' }}>
247 |   <div
248 |     style={{
249 |       width: '44px', height: '44px', borderRadius: '14px', display: 'flex',
250 |       alignItems: 'center', justifyContent: 'center', fontSize: '16px', fontWeight: 700,
251 |       flexShrink: 0,
252 |       background: isCompleted
253 |       ? 'linear-gradient(135deg, #22c55e, #16a34a)'
254 |       : isToday
255 |       ? 'linear-gradient(135deg, var(--brand-primary), #6366f1, #a855f7)'
256 |       : 'var(--border-strong)',
257 |       color: isCompleted || isToday ? 'white' : 'var(--text-muted)',
258 |       boxShadow: isCompleted || isToday ? '0 4px 12px color-mix(in srgb, currentColor 30...'
259 |     }}
260 |     >
261 |     {isCompleted ? <CheckCircle size={22} /> : topicNum}
262 |   </div>
263 |
264 |   <div style={{ flex: 1, minWidth: 0 }}>
265 |     <div style={{ display: 'flex', alignItems: 'center', gap: '8px', marginBottom: '4px'...
266 |       <span style={{ fontSize: '16px', fontWeight: 700, color: 'var(--text-primary)' }}>
267 |         {topic.title}
268 |       </span>
269 |       {isToday && (
270 |         <span style={{ fontSize: '10px', padding: '2px 8px', background: 'linear-gradien...
271 |           Hôm nay
272 |         </span>
273 |       )}
274 |     </div>
275 |     <p style={{ fontSize: '13px', color: 'var(--text-secondary)', margin: 0, overflow: '...
276 |       {topic.description} || `Chợ Á ${topicNum} v
277 |     </p>
278 |   </div>
279 |
280 |   <div style={{ flexShrink: 0 }}>
281 |     {isCompleted ? (
282 |       <div style={{ display: 'flex', alignItems: 'center', gap: '6px', fontSize: '14px',...
283 |         <CheckCircle size={18} />
284 |         ã l à m
285 |       </div>
286 |     ) : (
287 |       <button
288 |         onClick={() => router.push(`/daily-quiz/${topic.id}`)}
289 |         style={{
290 |           display: 'flex', alignItems: 'center', gap: '6px', padding: '10px 20px',
291 |           borderRadius: '9999px', fontSize: '14px', fontWeight: 600, border: 'none', cur...
292 |           background: isToday ? 'linear-gradient(135deg, var(--brand-primary), #6366f1)'...
293 |           color: isToday ? 'white' : 'var(--text-primary)',
294 |           boxShadow: isToday ? '0 4px 16px color-mix(in srgb, var(--brand-primary) 30%, ...
295 |           transition: 'all 0.2s',
296 |         }}
297 |       >
298 |         { isToday ? 'L à m b à i' : 'V à o h í c' }
299 |     <ChevronRight size={16} />
300 |   </button>
301 | )}
302 | </div>
303 | </div>
304 | </div>
305 | </div>
306 | </motion.div>
307 | )
308 | }}
309 | </div>
310 | </motion.div>
311 | </div>
312 | </div>
313 | )
314 | }
315 |

```

File: app/exam/[examId]/ExamPlayer.tsx (271 lines, 12.4 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect } from 'react'
4 | import { motion, AnimatePresence } from 'framer-motion'
5 | import {
6 |   ChevronLeft, ChevronRight, Clock, Send,
7 |   AlertTriangle, CheckCircle2, List, Timer
8 | } from 'lucide-react'
9 | import { useRouter } from 'next/navigation'
10 | import { submitExamAttempt } from '@lib/exam-actions'
11 | import { supabase } from '@lib/supabase'
12 |
13 | type Question = {
14 |   id: string;
15 |   text: string;
16 |   type: 'single' | 'multiple' | 'boolean' | 'fill' | 'essay';
17 |   options?: { id: string; text: string }[];
18 |   correctAnswer?: any;
19 | }
20 |
21 | type ExamPlayerProps = {
22 |   examId: string;
23 |   attemptId: string;
24 |   questions: Question[];
25 |   timeLimit: number;
26 | }
27 |
28 | export default function ExamPlayer({ examId, attemptId, questions, timeLimit }: ExamPlayerProps) {
29 |   const router = useRouter()
30 |   const [currentIdx, setCurrentIdx] = useState(0)
31 |   const [answers, setAnswers] = useState<Record<string, any>>({})
32 |   const [timeLeft, setTimeLeft] = useState(timeLimit * 60)
33 |   const [isSubmitting, setIsSubmitting] = useState(false)
34 |
35 |   useEffect(() => {
36 |     if (timeLeft <= 0) {
37 |       handleAutoSubmit()
38 |       return

```



```

39 |     }
40 |     const timer = setInterval(() => setTimeLeft(prev => prev - 1), 1000)
41 |     return () => clearInterval(timer)
42 | }, [timeLeft])
43 |
44 | useEffect(() => {
45 |     const handleBeforeUnload = (e: BeforeUnloadEvent) => {
46 |         e.preventDefault()
47 |         e.returnValue = 'Bị n c h - c h - n m u ñ n r ý i k h i b à i t h i ? T i z n t r i
48 |     }
49 |     window.addEventListener('beforeunload', handleBeforeUnload)
50 |     return () => { window.removeEventListener('beforeunload', handleBeforeUnload) }
51 | }, [])
52 |
53 | const formatTime = (seconds: number) => {
54 |     const m = Math.floor(seconds / 60)
55 |     const s = seconds % 60
56 |     return `${m}:${s < 10 ? '0' : ''}${s}`
57 | }
58 |
59 | const handleAnswer = (qId: string, val: any) => {
60 |     setAnswers(prev => ({ ...prev, [qId]: val }))
61 | }
62 |
63 |
64 | useEffect(() => {
65 |     const updateProgress = async () => {
66 |         const answeredCount = Object.keys(answers).length
67 |         if (answeredCount > 0) {
68 |             await supabase
69 |                 .from('exam_attempts')
70 |                 .update({ progress: answeredCount })
71 |                 .eq('id', attemptId)
72 |         }
73 |     }
74 |     updateProgress()
75 | }, [answers, attemptId])
76 |
77 | const handleAutoSubmit = () => {
78 |     if (!isSubmitting) handleSubmit()
79 | }
80 |
81 | const handleSubmit = async () => {
82 |     setIsSubmitting(true)
83 |     const score = Math.floor(Math.random() * 50) + 50
84 |     const timeSpent = timeLimit * 60 - timeLeft
85 |
86 |     const res = await submitExamAttempt(attemptId, answers, score, timeSpent)
87 |     if (res.success) {
88 |         router.push(`/exam/${examId}/result/${attemptId}`)
89 |     } else {
90 |         alert("L x i k h i n ù p b à i : " + res.error)
91 |         setIsSubmitting(false)
92 |     }
93 | }
94 |
95 | const activeQ = questions[currentIdx]
96 |
97 | return (
98 |     <div className="min-h-screen flex flex-col" style={{ background: 'var(--bg-base)', color: 'var(--text-prim...
99 |     <header className="h-20 flex items-center justify-between px-6 md:px-12 sticky top-0 z-50" style={{ back...
100 |     <div className="flex items-center gap-4">
101 |     <div className="p-2 rounded-lg" style={{ background: 'color-mix(in srgb, var(--brand-primary) 10%, t...
102 |     <Timer size={24} />
103 | </div>
104 | <div>
105 |     <div className="text-[10px] font-black uppercase tracking-widest" style={{ color: 'var(--text-mute...
106 |     <div className={`text-2xl font-mono font-bold ${timeLeft < 60 ? 'text-red-500 animate-pulse' : ''}}...
107 |     {formatTime(timeLeft)}
108 | </div>
109 | </div>
110 | </div>
111 |
112 |     <div className="hidden md:flex items-center gap-2">
113 |     {questions.map((_, i) => (
114 |         <div
115 |             key={i}
116 |             className="rounded-full transition-all"
117 |             style={{
118 |                 width: i === currentIdx ? '24px' : '8px',
119 |                 height: '8px',
120 |                 background: i === currentIdx ? 'var(--brand-primary)' : (answers[questions[i].id] ? 'var(--acc...
121 |             }}
122 |         />
123 |     ))}
124 | </div>
125 |
126 |     <button
127 |         onClick={() => { if (confirm("Bị n c h - c h - n m u ñ n n ù p b à i ?")) h
128 |         disabled={isSubmitting}
129 |         className="px-6 py-2.5 rounded-xl font-bold flex items-center gap-2 transition-all disabled:opacity-50"
130 |         style={{ background: 'var(--brand-primary)', color: 'var(--bg-base)' }}
131 |     >
132 |         <Send size={18} /> {isSubmitting ? 'ang n ù p . . . ' : 'N ù p b à i ' }
133 |     </button>
134 | </header>
135 |
136 |     <div className="flex-1 flex overflow-hidden">
137 |     <aside className="w-20 md:w-64 overflow-y-auto p-4 hidden md:block" style={{ background: 'color-mix(in...
138 |     <h3 className="text-xs font-black uppercase tracking-widest mb-6 px-2" style={{ color: 'var(--text-m...
139 |     <div className="grid grid-cols-1 gap-2">
140 |     {questions.map((q, i) => (
141 |         <button
142 |             key={q.id}
143 |             onClick={() => setCurrentIdx(i)}
144 |             className="flex items-center gap-3 p-3 rounded-xl border transition-all text-left"
145 |             style={{
146 |                 background: currentIdx === i ? 'color-mix(in srgb, var(--brand-primary) 10%, transparent)' : ...
147 |                 borderColor: currentIdx === i ? 'var(--brand-primary)' : (answers[q.id] ? 'var(--border-defa...
148 |                 color: currentIdx === i ? 'var(--brand-primary)' : (answers[q.id] ? 'var(--accent-blue)' : ...
149 |             }}
150 |         >
151 |         <span className="font-bold text-sm">Câu {i + 1}</span>
152 |         {answers[q.id] && <CheckCircle2 size={14} className="ml-auto" />}
153 |     </button>
154 |     ))}
155 | </div>
156 | </aside>
157 |
158 |     <main className="flex-1 overflow-y-auto p-6 md:p-12 lg:p-20 relative">
159 |     <AnimatePresence mode="wait">
160 |     <motion.div
161 |         key={currentIdx}
162 |         initial={{ opacity: 0, x: 20 }}

```

```

163 |         animate={{ opacity: 1, x: 0 }}
164 |         exit={{ opacity: 0, x: -20 }}
165 |         transition={{ duration: 0.2 }}
166 |         className="max-w-3xl mx-auto"
167 |     >
168 |     <div className="mb-10">
169 |         <span className="inline-block px-3 py-1 rounded-lg text-xs font-bold uppercase tracking-wider ...
170 |             { activeQ.type === 'single' ? 'Tr_c_ngh_iÇm_1_áp_án' :
171 |               activeQ.type === 'multiple' ? 'Tr_c_ngh_iÇm_nhi_Áu_áp
172 |               activeQ.type === 'boolean' ? 'úng / Sai' :
173 |               activeQ.type === 'fill' ? 'i_Án_vào_ch_x_tr_Ñng' : 'Tñ
174 |         </span>
175 |         <h2 className="text-2xl md:text-3xl font-bold leading-tight">
176 |         <span className="mr-4 italic" style={{ color: 'var(--text-muted)' }}>Câu {currentIdx + 1}</...
177 |         {activeQ.text}
178 |         </h2>
179 |     </div>
180 |
181 |     <div className="space-y-4 mb-20">
182 |         {activeQ.type === 'single' && activeQ.options?.map(opt => (
183 |             <button
184 |                 key={opt.id}
185 |                 onClick={() => handleAnswer(activeQ.id, opt.id)}
186 |                 className="w-full p-6 rounded-2xl border-2 text-left transition-all flex items-center gap-4"
187 |                 style={{
188 |                     background: answers[activeQ.id] === opt.id ? 'color-mix(in srgb, var(--brand-primary) 10...
189 |                     borderColor: answers[activeQ.id] === opt.id ? 'var(--brand-primary)' : 'var(--border-def...
190 |                 }}
191 |             >
192 |                 <div className="w-6 h-6 rounded-full border-2 flex items-center justify-center font-bold t...
193 |                     style={{
194 |                         borderColor: answers[activeQ.id] === opt.id ? 'var(--brand-primary)' : 'var(--text-mut...
195 |                         background: answers[activeQ.id] === opt.id ? 'var(--brand-primary)' : 'transparent',
196 |                         color: answers[activeQ.id] === opt.id ? 'var(--bg-base)' : 'var(--text-muted)'
197 |                     }}
198 |                 >{opt.id}</div>
199 |                 <span className="text-lg font-medium">{opt.text}</span>
200 |             </button>
201 |         )]}
202 |
203 |         {activeQ.type === 'boolean' && (
204 |             <div className="flex gap-4">
205 |                 2 0 5 |             { [ ' ú ng ' , ' Sai ' ] . map ( val => (
206 |                     <button
207 |                         key={val}
208 |                         onClick={() => handleAnswer(activeQ.id, val)}
209 |                         className="flex-1 p-8 rounded-3xl border-2 font-bold text-xl transition-all"
210 |                         style={{
211 |                             background: answers[activeQ.id] === val ? 'var(--brand-primary)' : 'var(--bg-surface)',
212 |                             borderColor: answers[activeQ.id] === val ? 'var(--brand-primary)' : 'var(--border-de...
213 |                             color: answers[activeQ.id] === val ? 'var(--bg-base)' : 'var(--text-primary)'
214 |                         }}
215 |                     >
216 |                         {val}
217 |                     </button>
218 |                 ))}
219 |             </div>
220 |         )}
221 |
222 |         {activeQ.type === 'fill' && (
223 |             <input
224 |                 type="text"
225 |                 value={answers[activeQ.id] || ''}
226 |                 onChange={(e) => handleAnswer(activeQ.id, e.target.value)}
227 |                 2 2 7 |                 p l a c e h o l d e r = " N h - p c â u t r é l ý i c ç a b i n . . . "
228 |                 className="w-full p-6 border-2 rounded-2xl focus:outline-none text-xl font-bold transition...
229 |                 style={{ background: 'var(--bg-surface)', borderColor: 'var(--border-default)', color: 'va...
230 |             </input>
231 |         )}
232 |
233 |         {activeQ.type === 'essay' && (
234 |             <textarea
235 |                 rows={8}
236 |                 value={answers[activeQ.id] || ''}
237 |                 onChange={(e) => handleAnswer(activeQ.id, e.target.value)}
238 |                 2 3 8 |                 p l a c e h o l d e r = " V i ế t c â u t r é l ý i t ã l u - n t i i â y . . . "
239 |                 className="w-full p-6 border-2 rounded-2xl focus:outline-none text-lg leading-relaxed tran...
240 |                 style={{ background: 'var(--bg-surface)', borderColor: 'var(--border-default)', color: 'va...
241 |             </textarea>
242 |         )}
243 |     </div>
244 |
245 |     <div className="flex items-center justify-between pt-10" style={{ borderTop: '1px solid var(--bo...
246 |         <button
247 |             disabled={currentIdx === 0}
248 |             onClick={() => setCurrentIdx(prev => prev - 1)}
249 |             className="p-4 rounded-xl disabled:opacity-30 transition-all flex items-center gap-2"
250 |             style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)', color: ...
251 |         >
252 |             2 5 2 |             <ChevronLeft size={20} /> C â u t r ề
253 |         </button>
254 |         <div className="font-bold text-sm" style={{ color: 'var(--text-muted)' }}>Câu {currentIdx + 1}...
255 |         <button
256 |             disabled={currentIdx === questions.length - 1}
257 |             onClick={() => setCurrentIdx(prev => prev + 1)}
258 |             className="p-4 rounded-xl disabled:opacity-30 transition-all flex items-center gap-2"
259 |             style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)', color: ...
260 |         >
261 |             C â u s a u <ChevronRight size={20} />
262 |         </button>
263 |     </div>
264 | </motion.div>
265 | </AnimatePresence>
266 | </main>
267 | </div>
268 | </div>
269 | )
270 | }
271 |

```

File: app/exam/[examId]/lab-exam/page.tsx (159 lines, 6.7 KB)

```

1 | 'use client'
2 |
3 | import { useState, useCallback, useEffect, use } from 'react'
4 | import { useRouter } from 'next/navigation'
5 | import RawGameEngine from '@components/GameEngine'
6 | const GameEngine = RawGameEngine as any;
7 | import VirtualAssistant from '@components/VirtualAssistant'
8 | import { useGuru } from '@lib/guru-state'
9 | import { GURU_MESSAGES } from '@utils/i18nData'
10 | import { Timer, Send, AlertCircle, Info, ChevronLeft } from 'lucide-react'
11 | import { motion } from 'framer-motion'

```

```

12 | import { submitExamAttempt } from '@lib/exam-actions'
13 |
14 | export default function LabExamPage({ params }: { params: Promise<{ examId: string }> }) {
15 |   const resolvedParams = use(params);
16 |   const { examId } = resolvedParams;
17 |   const router = useRouter();
18 |
19 |   // 1. Exam State
20 |   const [timeLeft, setTimeLeft] = useState(1800); // 30 phút
21 |   const [isSubmitting, setIsSubmitting] = useState(false);
22 |   const guru = useGuru();
23 |
24 |   // 2. Mission Data (Mocked from Mega Prompt Scenario)
25 |   const [missionData] = useState({
26 |     missionId: "L - p r á p P C V n P h ò n g",
27 |     scenario: "H à y l - p r á p m ù t b ù m à y", tính v n phòng vũ ngân sách 15 t
28 |     budget: 15000000,
29 |     purchasedItems: [
30 |       { id: 1, name: "Core i3-12100", type: "CPU", price: 3000000, power: 65, socket: "LGA1700" },
31 |       { id: 2, name: "H610M-K", type: "Mainboard", price: 2500000, socket: "LGA1700" },
32 |       { id: 3, name: "RAM 8GB DDR4", type: "RAM", price: 800000, power: 5 },
33 |       { id: 4, name: "RAM 8GB DDR4", type: "RAM", price: 800000, power: 5 },
34 |       { id: 5, name: "SSD 500GB NVMe", type: "SSD", price: 1200000, power: 10 },
35 |       { id: 6, name: "PSU 450W 80 Plus", type: "PSU", price: 1000000, wattage: 450 },
36 |       { id: 7, name: "Stock Cooler", type: "COOLER", price: 0, power: 0 }
37 |     ],
38 |     targets: { minPower: 50 }
39 |   });
40 |
41 |   // 3. Timer Logic
42 |   useEffect(() => {
43 |     const timer = setInterval(() => setTimeLeft(prev => prev - 1), 1000);
44 |     if (timeLeft <= 0) handleSubmit();
45 |     return () => clearInterval(timer);
46 |   }, [timeLeft]);
47 |
48 |   const formatTime = (s: number) => {
49 |     const m = Math.floor(s / 60);
50 |     const sec = s % 60;
51 |     return `${m}:${sec < 10 ? '0' : ''}${sec}`;
52 |   };
53 |
54 |   // 4. Handle Game Events (Tá i sí d à n g l o g i c c h ư m i ă m c ặ a b i n )
55 |   const handleGameEvent = useCallback((event: string, type: string) => {
56 |     if (event === 'grabbed') {
57 |       guru.setMessag e ( ` B i n a n g c ư m : ${type} ` );
58 |     } else if (event === 'placed') {
59 |       guru.setMessag e ( ` ă l - p x o n g : ${type} . T i ếp t ă c n ă o ! ` );
60 |     } else if (event === 'COMPLETE') {
61 |       guru.setMessag e ( " H ặ t h ư ợ n ă ă c l - p r á p h o à n t ă t ! H à y k i ă m t r a
62 |     }, []);
63 |
64 |     const handleSubmit = async () => {
65 |       setIsSubmitting(true);
66 |
67 |       // Logic ch ư m i ă m t ă ù n g ( B ệ t ệ b u i l d e r / p a g e . j s c ặ a b i n )
68 |       let totalTDP = 0, psuWattage = 0;
69 |       missionData.purchasedItems.forEach(item => {
70 |         if (item.power) totalTDP += item.power;
71 |         if (item.wattage) psuWattage += item.wattage;
72 |       });
73 |
74 |       let score = 100;
75 |       if ( psuWattage < totalTDP + 50 ) score -= 30; // N g u ồ n y ế u
76 |       // T ă m c ă c t i ệ u c h i c h ư m i ă m k h ă c t i i ă y . . .
77 |
78 |       // N ặ p b à i ( S i d à n g s e r v e r a c t i o n c h u n g )
79 |       const res = await submitExamAttempt("dummy_attempt_id", {}, score, 1800 - timeLeft);
80 |       if (res.success) {
81 |         router.push(`/exam/${examId}/result/dummy_attempt_id`);
82 |       } else {
83 |         alert ( " L x i n ặ p b à i ! " );
84 |         setIsSubmitting(false);
85 |       }
86 |     };
87 |   });
88 |
89 |   return (
90 |     <div className="min-h-screen bg-[#0f0f1a] text-white flex flex-col overflow-hidden">
91 |       <div className="h-20 bg-[#16213e] border-b border-[#1e293b] flex items-center justify-between px-8 z-...>
92 |         <div className="flex items-center gap-6">
93 |           <button onClick={() => router.back()} className="p-2 hover:bg-white/5 rounded-lg transition-colors">
94 |             <ChevronLeft size={24} />
95 |           </button>
96 |           <div>
97 |             <h1 className="text-xl font-black uppercase tracking-wider">{missionData.missionId}</h1>
98 |             <div className="flex items-center gap-2 text-xs text-[#00d2a0] font-bold">
99 |               <Timer size={14} /> Th ư i g i a n c ò n l i i : {formatTime(timeLeft)}
100 |             </div>
101 |           </div>
102 |         </div>
103 |       </div>
104 |
105 |       <div className="flex items-center gap-4">
106 |         <button
107 |           onClick={handleSubmit}
108 |           disabled={isSubmitting}
109 |           className="px-8 py-2.5 bg-[#00d2a0] text-black font-black rounded-xl hover:shadow-[0_0_20px_rgba(...
110 |         >
111 |         { isSubmitting ? ' a n g n ặ p . . . ' : ' N ặ p b à i t h i ' }
112 |       </button>
113 |     </div>
114 |   </header>
115 |
116 |   <div className="flex-1 flex relative">
117 |     <div className="absolute top-6 left-6 z-10 w-80">
118 |       <div>
119 |         <div>
120 |           initial={{ x: -20, opacity: 0 }}
121 |           animate={{ x: 0, opacity: 1 }}
122 |           className="bg-[#16213e]/80 backdrop-blur-md p-6 rounded-3xl border border-[#1e293b] shadow-2xl"
123 |         >
124 |           <h3 className="text-sm font-black text-[#00d2a0] uppercase tracking-widest mb-3 flex items-center ...>
125 |             <Info size={16} /> Y ế u c ấ u ă b à i
126 |           </h3>
127 |           <p className="text-sm text-slate-300 leading-relaxed italic">
128 |             "{missionData.scenario}"
129 |           </p>
130 |           <div className="mt-4 border-t border-white/5 flex justify-between items-end">
131 |             <div>
132 |               <div className="text-[10px] text-slate-500 font-bold uppercase">Ngân sách</div>
133 |               <div className="text-lg font-black text-white">{missionData}
134 |             </div>
135 |             <div className="text-right">

```

```
1 3 6 | <div className="text-lg font-black text-[#00d2a0]">{missionData.purchasedItems.length}</div>
137 | </div>
138 | </div>
139 | </motion.div>
140 | </div>
141 |
142 |
143 | { /* The 2D Lab Canvas (Reusing your GameEngine) */ }
144 | <main className="flex-1 flex items-center justify-center bg-black/40">
145 |   <div className="relative w-full max-w-[1400px] aspect-[14/8]">
146 |     <GameEngine
147 |       onGameEvent={handleGameEvent}
148 |       purchasedItems={missionData.purchasedItems}
149 |     />
150 |
151 |   </div>
152 | </main>
153 | </div>
154 |
155 | </div>
156 | )
157 | }
158 |
159 |
```

File: app/exam/[examId]/page.tsx (118 lines, 5.1 KB)

```
1 import { createClient } from '@lib/supabase-ssr-server'
2 import { redirect } from 'next/navigation'
3 import ExamPlayer from './ExamPlayer'
4 import { startExamAttempt } from '@lib/exam-actions'
5
6 export const dynamic = 'force-dynamic'
7
8 export default async function ExamPage({ params }: { params: { examId: string } }) {
9   const supabase = await createClient()
10   const { data: { user } } = await supabase.auth.getUser()
11   if (!user) redirect('/login')
12
13   // 1. Khỏi t i o phi ê n t h i ( A t t e m p t )
14   const res = await startExamAttempt(params.examId)
15   if (res.error) {
16     return <div className="min-h-screen flex items-center justify-center">
17   }
18
19   // 2. Ngân hàng câu hỏi t h - t v á k i z n t r ú c m á y t í n h v à p h ã n c é n g P C ( C h o i t r ú c m á y t í n h v à p h ã n c é n g P C )
20   const realQuestions: any = [
21     {
22       id: 'q1',
23       text: "Trong k i z n t r ú c m á y t í n h V o n N e u m a n n , b ù x í l ý t r u n g t â m ( C a c h e )",
24       type: 'single',
25       options: [
26         { id: 'A', text: "B ù i á u k h i ã n ( C U ) v à B ù t í n h t o á n s ã h í c / l o g i c" },
27         { id: 'B', text: "B ù n h ù R A M v à R O M B I O S" },
28         { id: 'C', text: "T h i z t b ê v à o ( I n p u t ) v à T h i z t b ê r a ( O u t p u t )" },
29         { id: 'D', text: "Ô c é n g H D D v à n g u o ã n i c ã n P S U" }
30       ],
31     },
32     {
33       id: 'q2',
34       text: "B ù n h ù Ç m ( C a c h e ) t í c h h á p t r ã c t i z p t r o n g C P U c ó t á c d ä n g",
35       type: 'single',
36       options: [
37         { id: 'A', text: "T n g t ã c ù t r u y x u z t đ i l i Ç u b ä n g c á c h l o u t r i" },
38         { id: 'B', text: "T n g d u n g l o ä n g l o u t r i t o ä n g t h ä c c a h c i á u h" },
39         { id: 'C', text: "N g n c h n v i r u s ù c h i i x ä m n h - p v à o n h ä n x í l" },
40         { id: 'D', text: "T h a y t h z h o à n t o à n b ù n h ù t r o n g R A M k h i s y d u" }
41       ],
42     },
43     {
44       id: 'q3',
45       text: "B ù n h ù R A M ( R a n d o m A c c e s s M e m o r y ) l à b ù n h ù c h é í c , k h o n g",
46       type: 'boolean'
47     },
48     {
49       id: 'q4',
50       text: "T h i z t b ê n à o s a u ä y l à t h i z t b ê l o u t r i t h é c z p ( S e c o n d a r y )",
51       type: 'single',
52       options: [
53         { id: 'A', text: "SSD (Solid State Drive)" },
54         { id: 'B', text: "RAM (Random Access Memory)" },
55         { id: 'C', text: "Cache L2" },
56         { id: 'D', text: "Register (Thanh ghi)" }
57       ],
58     },
59     {
60       id: 'q5',
61       text: "M ä c í c h c h í n h c c a c o n g n g c H y p e r - T h r e a d i n g ( S i e u p h ä n l u c )",
62       type: 'single',
63       options: [
64         { id: 'A', text: "C h o p h é p m ù t n h ä n v - t l ý ( P h y s i c a l C o r e ) x í l ý" },
65         { id: 'B', text: "T ã ù n g t ã n g x u n g n h é p c i b ä n ( B a s e C l o c k ) c c a" },
66         { id: 'C', text: "G i ä m 5 0 % l o ä n g i c ã n n n g t i e u t h ä k h i t ä i n n o" },
67         { id: 'D', text: "H x t r ä C P U k z t n ã i t r ã c t i z p v ù i c a r d ô h í a n" }
68       ],
69     },
70     {
71       id: 'q6',
72       text: "i n v é F L O P S ( F l o a t i n g - p o i n t O p e r a t i o n s P e r S e c o n d ) t h o ä n g",
73       type: 'single',
74       options: [
75         { id: 'A', text: "G P U ( C a r d ó h í a r ý i / B ù x í l ý ó h í a )" },
76         { id: 'B', text: "T ã c ù q u a y c c a Q u i t t ä n n h i c t ( R P M )" },
77         { id: 'C', text: "T ã c ù t r u y ä n đ i l i Ç u c c a c ä p m i n g L a n" },
78         { id: 'D', text: "X u n g n h é p h o i t ù n g c c a R A M" }
79       ],
80     },
81     {
82       id: 'q7',
83       text: "C h u o n g i a o t i z p P C I e 4 . 0 x 1 6 c ó b n g t h o n g l ý t h u y z t t ã i a",
84       type: 'single',
85       options: [
86         { id: 'A', text: "3 1 . 5 G B / s ( k h o ä n g 3 2 G B / s )" },
87         { id: 'B', text: "1 5 . 8 G B / s ( k h o ä n g 1 6 G B / s )" },
88         { id: 'C', text: "6 3 . 0 G B / s ( k h o ä n g 6 4 G B / s )" },
89         { id: 'D', text: "8.0 GB/s" }
90       ],
91     },
92     {
93       id: 'q8',
94       text: "i ä n t ê n v i z t t t c c a b ù p h - n c z p n g u o n , c h u y ä n ô i d o n g",
95       type: 'fill'
96     },
97   ]
```

```

97 | {
98 |   id: 'q9',
99 |   text: "Keo tẽn nhiệ t (Thermal Paste) có tác dẫ ng lặ p sý khoẽ ng t
100 |   type: 'boolean'
101 | },
102 | {
103 |   id: 'q10',
104 |   text: "Hẫ y trìn h bẫ y ng - n gĩ n (tẽ 2 - 3 cầ u) lý do tầ i sả o k hông nẽ n
105 |   type: 'essay'
106 | }
107 | ]
108 |
109 | return (
110 |   <ExamPlayer
111 |     examId={params.examId}
112 |     attemptId={res.attemptId}
113 |     questions={realQuestions}
114 |     timeLimit={30}
115 |   />
116 | )
117 | }
118 |

```

File: app/exam/[examId]/proctored/page.tsx (335 lines, 19.7 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect, useCallback } from 'react'
4 | import { motion, AnimatePresence } from 'framer-motion'
5 | import {
6 |   Camera, ShieldAlert, Monitor, Lock, AlertCircle, CheckCircle2, ScanFace, Zap,
7 |   ArrowRight, Video, EyeOff, UserX, UserCheck, Clock, AlertTriangle, RefreshCw,
8 |   Gauge, Maximize, Wifi
9 | } from 'lucide-react'
10 | import { useRouter } from 'next/navigation'
11 | import { supabase } from '@lib/supabase'
12 | import ExamPlayer from '../ExamPlayer'
13 | import { useProctorEngine } from '@hooks/useProctorEngine'
14 |
15 | const MOCK_EXAM_ID = 'proctored-mock-exam'
16 | const QUESTIONS: any = [
17 |   { id: 'q1', text: "Tròn g kiẽ n trư c mẫ y tĩ n h Von Neumann, bõ xu lý trư n g tẫ m (CPU) gõ m nhũ n g thẫ n g phẫ n cõt l...
18 |   { id: 'q2', text: "Bõ nhõ dem (Cache) tĩ ch hõ p trư c tĩ p trõ n g CPU cõ tậ dũ n g chĩ n h lầ gĩ?", type: 'single'...
19 |   { id: 'q3', text: "Bõ nhõ RAM (Random Access Memory) lầ bõ nhõ chĩ dõ c, k hông bĩ mậ t dũ liệ u k hĩ mậ t dũ liệ u...
20 |   { id: 'q4', text: "Thĩ e t bĩ n ả o sẫ dẫ y lầ thĩ e t bĩ lũ u trũ thu cậ p (Secondary Storage) k hông bĩ mậ t dũ liệ u...
21 |   { id: 'q5', text: "Mũ c dĩ ch chĩ n h cũ a cõ n g n g hĩ Hyper-Threading (Sĩ eũ phẫ n lũ oĩ n g) trẽn CPU Intel lầ gĩ?", ty...
22 |   { id: 'q6', text: "Đõ n vĩ FLOPS (Floating-point Operations Per Second) thườ n g đũ c dũ n g đẽ dõ hĩ eũ n ả n g cũ a ...
23 |   { id: 'q7', text: "Chũ a n g ả o tĩ p PCIe 4.0 x16 cõ b ả n g thõ n g lý thũ e t tõ i dầ lầ bả o nhĩ eũ GB/s?", type: 's...
24 |   { id: 'q8', text: "Đĩ eũ n vĩ eũ t tậ t cũ a bõ phẫ n cậ p n gũ oĩ n , chũ eũ n đõ i dũ n g dĩ eũ n xõ a y chĩ eũ (AC) thẫ n g mõt c...
25 |   { id: 'q9', text: "Keo tẽn nhiệ t (Thermal Paste) cõ tậ dũ n g lậ p dẫ y k hỏ a n g trõ n g k hông k hĩ sĩ eũ nhõ gĩ uầ bẽ...
26 |   { id: 'q10', text: "Hẫ y trìn h bẫ y n ả n g gõ n (tũ 2-3 cầ u) lý do tầ i sả o k hông nẽ n chõ n bõ n gũ oĩ n PSU kẽ m chậ t l...
27 | ]
28 |
29 | export default function ProctoredExamPage({ params }: { params: Promise<{ examId: string }> }) {
30 |   const router = useRouter()
31 |   const [step, setStep] = useState<'check' | 'exam'>('check')
32 |   const [hasPermission, setHasPermission] = useState(false)
33 |   const [showViolationOverlay, setShowViolationOverlay] = useState(false)
34 |   const [examSubmitted, setExamSubmitted] = useState(false)
35 |   const [violationHistory, setViolationHistory] = useState<{ time: string; type: string }[]>([])
36 |   const [realtimeConnected, setRealtimeConnected] = useState(false)
37 |
38 |   const { state, videoRef, canvasRef, startCamera, stopCamera, startFaceDetection, captureIdentity, recordViol...
39 |   examId: MOCK_EXAM_ID,
40 |   attemptId: 'proctored-attempt',
41 |   identityPhoto: null,
42 |   onViolation: (type) => {
43 |     setViolationHistory(prev => [...prev, { time: new Date().toLocaleTimeString(), type }])
44 |     setShowViolationOverlay(true)
45 |   },
46 |   onMaxViolations: () => {
47 |     setTimeout(() => {
48 |       alert('Bẫ n dầ vĩ phẫ m quầ 3 lẫ n. Hẽ thõ n g tũ dũ n g n ỏ p bầ i!')
49 |       setExamSubmitted(true)
50 |     }, 2000)
51 |   },
52 | })
53 |
54 |   useEffect(() => {
55 |     if (step !== 'exam') return
56 |
57 |     const handleVisibilityChange = () => {
58 |       if (document.visibilityState === 'hidden') {
59 |         recordViolation('tab_switch')
60 |       }
61 |     }
62 |     const handleFullscreenChange = () => {
63 |       if (!document.fullscreenElement) {
64 |         recordViolation('fullscreen_exit')
65 |       }
66 |     }
67 |     const disableEvent = (e: any) => e.preventDefault()
68 |
69 |     document.addEventListener('visibilitychange', handleVisibilityChange)
70 |     document.addEventListener('fullscreenchange', handleFullscreenChange)
71 |     document.addEventListener('copy', disableEvent)
72 |     document.addEventListener('paste', disableEvent)
73 |     document.addEventListener('contextmenu', disableEvent)
74 |     document.addEventListener('keydown', (e) => {
75 |       if ((e.ctrlKey || e.metaKey) && ['c', 'v', 'p', 'u'].includes(e.key.toLowerCase())) e.preventDefault()
76 |     })
77 |
78 |     startCamera().then(stream => {
79 |       if (stream) {
80 |         setHasPermission(true)
81 |         startFaceDetection()
82 |       }
83 |     })
84 |
85 |     let realtimeChannel: any = null
86 |     try {
87 |       realtimeChannel = supabase
88 |         .channel(`proctor-${MOCK_EXAM_ID}`)
89 |         .subscribe((status) => {
90 |           if (status === 'SUBSCRIBED') setRealtimeConnected(true)
91 |         })
92 |     } catch (e) {}
93 |
94 |     return () => {
95 |       document.removeEventListener('visibilitychange', handleVisibilityChange)
96 |       document.removeEventListener('fullscreenchange', handleFullscreenChange)
97 |       document.removeEventListener('copy', disableEvent)
98 |       document.removeEventListener('paste', disableEvent)

```

```

99     document.removeEventListener('contextmenu', disableEvent)
100     stopCamera()
101     if (realtimeChannel) supabase.removeChannel(realtimeChannel)
102 }
103 }, [step, startCamera, startFaceDetection, recordViolation, stopCamera])
104
105 const startExam = useCallback(() => {
106     const photo = captureIdentity()
107     if (!photo) {
108         alert('Vui lòng chụp ảnh xác thực danh tính trước!')
109         return
110     }
111     document.documentElement.requestFullscreen().catch(() => {
112         alert('Trình duyệt không hỗ trợ Fullscreen. Vui lòng dùng Chrome/Edge.')
113     })
114     setStep('exam')
115 }, [captureIdentity])
116
117 if (examSubmitted) {
118     router.push(`/exam/${MOCK_EXAM_ID}/result/proctored`)
119     return null
120 }
121
122 if (step === 'check') {
123     return (
124         <div className="min-h-screen flex items-center justify-center p-6 overflow-hidden relative" style={{ bac...
125             <div className="absolute top-0 left-0 w-full h-full opacity-10 pointer-events-none">
126                 <div className="absolute top-1/4 left-1/4 w-96 h-96 blur-[120px]" style={{ background: 'var(--brand-...
127                 <div className="absolute bottom-1/4 right-1/4 w-96 h-96 blur-[120px]" style={{ background: 'var(--ac...
128             </div>
129
130             <motion.div initial={{ opacity: 0, scale: 0.9 }} animate={{ opacity: 1, scale: 1 }}
131                 className="max-w-4xl w-full rounded-[48px] p-10 md:p-16 shadow-2xl relative z-10"
132                 style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)' }}>
133                 <div className="flex flex-col md:flex-row gap-12 items-center">
134                     <div className="w-full md:w-1/2 space-y-6">
135                         <div className="aspect-video rounded-[32px] border-2 overflow-hidden relative group"
136                             style={{ background: 'var(--bg-elevated)', borderColor: state.cameraError ? 'var(--danger)' : ...
137                             {state.cameraError ? (
138                                 <div className="absolute inset-0 flex flex-col items-center justify-center gap-4 p-6">
139                                     <Camera size={48} style={{ color: 'var(--danger)' }} />
140                                     <p className="text-sm font-bold text-center" style={{ color: 'var(--danger)' }}>{state.cam...
141                                     <button onClick={startCamera} className="px-6 py-2 rounded-xl text-sm font-bold transition...
142                                         style={{ background: 'var(--brand-primary)', color: '#fff', border: 'none' }}>
143                                         <RefreshCw size={14} className="inline mr-2" />Thử lại
144                                     </button>
145                                 </div>
146                             ) : !hasPermission ? (
147                                 <div className="absolute inset-0 flex flex-col items-center justify-center gap-4">
148                                     <Camera size={48} style={{ color: 'var(--text-muted)' }} />
149                                     <p className="text-sm text-center px-4" style={{ color: 'var(--text-muted)' }}>
150                                         Can cấp quyền Camera để tham gia kỳ thi có giám sát
151                                     </p>
152                                     <button onClick={startCamera}
153                                         className="px-6 py-2 rounded-xl text-sm font-bold hover:scale-105 transition-all flex it...
154                                         style={{ background: 'var(--bg-elevated)', color: 'var(--text-primary)', border: '1px so...
155                                         <Video size={16} />Kích hoạt Camera
156                                     </button>
157                                 </div>
158                             ) : (
159                                 <div>
160                                     <video ref={videoRef} autoPlay playsInline muted className="w-full h-full object-cover mir...
161                                     {state.identityCapture && (
162                                         <div className="absolute inset-0 border-4 rounded-[32px] z-20 pointer-events-none border...
163                                         <div className="absolute top-4 right-4 bg-green-500 text-black px-3 py-1 rounded-full ...
164                                         <CheckCircle2 size={12} />Đã xác minh
165                                         </div>
166                                     </div>
167                                 )}
168                                 <div className="absolute bottom-3 left-3 flex items-center gap-2 bg-black/60 px-3 py-1.5 r...
169                                 <div className="w-2 h-2 rounded-full ${state.faceDetected ? 'bg-green-500 animate-pulse...
170                                 <span className="text-[10px] font-bold text-white/80">
171                                     {state.faceDetected ? 'Đã phát hiện khuôn mặt' : 'Đang tìm khuôn mặt...'}
172                                 </span>
173                                 </div>
174                             </div>
175                         </div>
176                     </div>
177                     <canvas ref={canvasRef} width="320" height="240" className="hidden" />
178
179                     <button onClick={startExam} disabled={!state.identityCapture && !state.faceDetected}
180                         className="w-full py-4 font-black uppercase tracking-wider rounded-2xl flex items-center justi...
181                         style={{ background: 'var(--brand-primary)', color: '#fff' }}>
182                         <ScanFace size={20} />
183                         {state.identityCapture ? 'Chụp lại ảnh xác thực' : 'Chụp ảnh xác thực & bắt đầu'}
184                     </button>
185                 </div>
186
187                 <div className="w-full md:w-1/2 space-y-8">
188                     <div>
189                         <h1 className="text-3xl font-black mb-2 uppercase tracking-tight" style={{ color: 'var(--text-...
190                             Ky thi <span style={{ color: 'var(--brand-primary)' }}>Giám sát</span>
191                         </h1>
192                         <p style={{ color: 'var(--text-muted)', fontWeight: 500 }}>
193                             He thống AI Proctoring với MediaPipe FaceLandmarker đang bảo vệ tính minh bạch của kỳ thi.
194                         </p>
195                     </div>
196
197                     <div className="space-y-4">
198                         {[
199                             { icon: Monitor, text: "Bắt buộc che đậy toàn màn hình", color: "var(--accent-blue)" },
200                             { icon: ShieldAlert, text: "Phát hiện và cảnh báo khi rời Tab", color: "var(--danger)" },
201                             { icon: Lock, text: "Khoa chức năng Copy, Paste, chuột phải", color: "#8b5cf6" },
202                             { icon: Camera, text: "Phát hiện khuôn mặt bằng AI (MediaPipe)", color: "var(--brand-primary...
203                             { icon: Gauge, text: "Theo dõi hướng nhìn & phát hiện nhiều mặt", color: "#f59e0b" },
204                             { icon: Wifi, text: "Cảnh báo thời gian thực đến giao viên", color: "#06b6d4" },
205                         ].map((item, i) => (
206                             <div key={i} className="flex items-center gap-4 p-4 rounded-2xl"
207                                 style={{ background: 'var(--bg-elevated)', border: '1px solid var(--border-default)' }}>
208                                 <div className="p-2 rounded-lg" style={{ backgroundColor: `${item.color}15`, color: item.c...
209                                     <item.icon size={18} />
210                                 </div>
211                                 <span className="text-sm font-bold" style={{ color: 'var(--text-muted)' }}>{item.text}</span>
212                             </div>
213                         ))}
214                     </div>
215
216                     <div className="p-4 rounded-2xl"
217                         style={{ background: 'rgba(245,158,11,0.08)', border: '1px solid rgba(245,158,11,0.2)' }}>
218                     <p className="text-xs font-bold flex items-center gap-2" style={{ color: '#f59e0b' }}>
219                         <AlertTriangle size={14} />
220                         Nếu bạn rời khỏi màn hình, không có khuôn mặt, hoặc phát hiện nhiều người qua 3 lần, bài thi...
221                     </p>
222                 </div>

```

```

223 |         </div>
224 |     </div>
225 | </motion.div>
226 | </div>
227 | )
228 | }
229 |
230 | return (
231 | <div className="relative min-h-screen" style={{ background: 'var(--bg-base)' }}>
232 |   <ExamPlayer examId={MOCK_EXAM_ID} attemptId="proctored_attempt" questions={QUESTIONS} timeLimit={60} />
233 |
234 |   <AnimatePresence>
235 |     {showViolationOverlay && (
236 |       <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }} exit={{ opacity: 0 }}
237 |         className="fixed inset-0 z-[9999] backdrop-blur-xl flex items-center justify-center p-10 text-cent...
238 |         style={{ background: 'rgba(244,67,54,0.95)' }}>
239 |         <div className="max-w-xl space-y-8">
240 |           <div className="w-32 h-32 bg-white/10 rounded-full flex items-center justify-center mx-auto bord...
241 |             <AlertCircle size={64} />
242 |           </div>
243 |           <h1 className="text-6xl font-black uppercase tracking-tighter italic">Canh bao vi pham!</h1>
244 |           <p className="text-2xl font-bold opacity-80 leading-relaxed">
245 |             Phat hien hanh vi bat thuong trong qua trinh thi.
246 |           </p>
247 |
248 |           <div className="grid grid-cols-2 gap-4 max-w-sm mx-auto">
249 |             <div className="bg-black/20 p-4 rounded-2xl border border-white/10">
250 |               <div className="text-[10px] font-black uppercase tracking-widest opacity-50 mb-1">So lan VP<...
251 |               <div className="text-4xl font-black">{state.violations} / {state.maxViolations}</div>
252 |             </div>
253 |             <div className="bg-black/20 p-4 rounded-2xl border border-white/10">
254 |               <div className="text-[10px] font-black uppercase tracking-widest opacity-50 mb-1">Trang thai...
255 |               <div className="flex items-center justify-center gap-2 mt-2">
256 |                 <div className={w-3 h-3 rounded-full ${state.faceDetected ? 'bg-green-500' : 'bg-red-500'}...
257 |                 <span className="text-lg font-bold">{state.faceDetected ? 'Co mat' : 'Mat mat'}</span>
258 |               </div>
259 |             </div>
260 |           </div>
261 |
262 |           {violationHistory.length > 0 && (
263 |             <div className="bg-black/20 p-4 rounded-2xl border border-white/10 max-w-sm mx-auto">
264 |               <div className="text-[10px] font-black uppercase tracking-widest opacity-50 mb-2">Lich su VP...
265 |               <div className="space-y-1 max-h-24 overflow-y-auto">
266 |                 {violationHistory.slice(-5).map((v, i) => (
267 |                   <div key={i} className="text-xs opacity-80 flex justify-between">
268 |                     <span>{v.time}</span>
269 |                     <span className="font-bold">{v.type.replace(/_/g, ' ')}</span>
270 |                   </div>
271 |                 ))}
272 |               </div>
273 |             </div>
274 |           )}
275 |
276 |           <button onClick={() => { document.documentElement.requestFullscreen(); setShowViolationOverlay(f...
277 |             className="px-12 py-5 bg-white text-red-600 font-black uppercase rounded-2xl shadow-2xl hover:...
278 |             Quay lai bai thi ngay
279 |           </button>
280 |           <p className="text-sm opacity-50 font-bold uppercase italic tracking-widest">
281 |             Hanh dong cua ban da duoc ghi lai va gui den giao vien.
282 |           </p>
283 |         </div>
284 |       </motion.div>
285 |     )}
286 |   </AnimatePresence>
287 |
288 |   <div className="fixed bottom-6 right-6 w-56 rounded-xl overflow-hidden z-[100] shadow-2xl"
289 |     style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)' }}>
290 |     <div className="relative aspect-video">
291 |       <video ref={videoRef} autoPlay playsInline muted className="w-full h-full object-cover mirror opacit...
292 |         {state.multipleFaces && (
293 |           <div className="absolute inset-0 bg-red-500/20 border-2 border-red-500 rounded-lg flex items-cente...
294 |           <UserX size={32} className="text-red-500" />
295 |         </div>
296 |       )}
297 |       <div className="absolute top-2 left-2 flex items-center gap-1.5 bg-black/60 px-2 py-1 rounded-md bac...
298 |         <div className={w-2 h-2 rounded-full ${state.faceDetected ? 'bg-green-500' : 'bg-red-500'} animat...
299 |         <span className="text-[9px] font-black uppercase text-white/80">
300 |           {state.faceDetected ? (state.multipleFaces ? 'NHIEU MAT!' : 'FACE OK') : 'NO FACE'}
301 |         </span>
302 |       </div>
303 |       <div className="absolute top-2 right-2 flex items-center gap-1">
304 |         <div className="w-2 h-2 bg-red-500 rounded-full animate-pulse" />
305 |         <span className="text-[9px] font-black uppercase text-white/70">REC</span>
306 |       </div>
307 |       {realtimeConnected && (
308 |         <div className="absolute top-2 right-12 bg-green-500/80 text-black px-1.5 py-0.5 rounded text-[8px]...
309 |         LIVE
310 |       </div>
311 |     )}
312 |     <div className="absolute bottom-2 left-2 flex items-center gap-1 bg-black/60 px-2 py-0.5 rounded">
313 |       <Clock size={10} className="text-white/70" />
314 |       <span className="text-[8px] font-black uppercase text-white/70">
315 |         VP: {state.violations}/{state.maxViolations}
316 |       </span>
317 |     </div>
318 |     <div className="absolute bottom-2 right-2 bg-black/60 px-1.5 py-0.5 rounded">
319 |       <span className="text-[8px] font-mono text-white/60">
320 |         Y: {state.gazeDirection.yaw.toFixed(1)} P: {state.gazeDirection.pitch.toFixed(1)}
321 |       </span>
322 |     </div>
323 |   </div>
324 |   <div className="flex items-center justify-between px-3 py-1.5"
325 |     style={{ background: 'var(--bg-elevated)', borderTop: '1px solid var(--border-default)' }}>
326 |     <span className="text-[8px] font-bold uppercase" style={{ color: state.lookingAway ? 'var(--danger)' : ...
327 |     {state.lookingAway ? 'Dang nhin di cho khac!' : 'Dang tap trung'}}
328 |   </span>
329 |   <Maximize size={10} style={{ color: 'var(--text-muted)' }} onClick={() => document.documentElement.r...
330 | </div>
331 | </div>
332 | )
333 | }
334 | )
335 |

```

File: app/exam/[examId]/result/[attemptId]/page.tsx (246 lines, 10.8 KB)

```

1 | 'use client';
2 |
3 | import React, { useEffect, useState, use } from 'react';
4 | import Link from 'next/link';
5 | import { motion } from 'framer-motion';
6 | import {
7 |   Trophy, Clock, Target, ArrowRight, RotateCcw,

```

```

8 | Share2, CheckCircle2, XCircle, AlertCircle,
9 | BarChart3, PieChart, Layout
10 | } from 'lucide-react';
11 | import confetti from 'canvas-confetti';
12 |
13 | export default function ExamResultPage({ params }: { params: Promise<{ examId: string, attemptId: string }> }) {
14 |   const resolvedParams = use(params);
15 |   const { examId, attemptId } = resolvedParams;
16 |
17 |   // Mock data matching Mega Prompt examples
18 |   const [result, setResult] = useState({
19 |     score: 85,
20 |     total: 100,
21 |     correctCount: 17,
22 |     wrongCount: 2,
23 |     blankCount: 1,
24 |     timeSpent: "23 phút 15 giây",
25 |     rank: 5,
26 |     passScore: 50,
27 |     topics: [
28 |       { name: "CPU & Mainboard", score: 9, total: 10 },
29 |       { name: "RAM & Storage", score: 6, total: 10 },
30 |       { name: "GPU & PSU", score: 7, total: 10 }
31 |     ]
32 |   });
33 |
34 |   const isPassed = result.score >= result.passScore;
35 |
36 |   const getGrade = (score: number) => {
37 |     if (score >= 90) return { label: "Xuất sắc", color: "#00d2a0" };
38 |     if (score >= 80) return { label: "Giỏi", color: "#00b4d8" };
39 |     if (score >= 65) return { label: "Khá", color: "#f59e0b" };
40 |     if (score >= 50) return { label: "Trung bình", color: "#94a3b8" };
41 |     return { label: "Yếu", color: "#ef4444" };
42 |   };
43 |
44 |   const grade = getGrade(result.score);
45 |
46 |   useEffect(() => {
47 |     if (isPassed) {
48 |       const duration = 3 * 1000;
49 |       const animationEnd = Date.now() + duration;
50 |       const defaults = { startVelocity: 30, spread: 360, ticks: 60, zIndex: 0 };
51 |
52 |       const randomInRange = (min: number, max: number) => Math.random() * (max - min) + min;
53 |
54 |       const interval: any = setInterval(function() {
55 |         const timeLeft = animationEnd - Date.now();
56 |
57 |         if (timeLeft <= 0) {
58 |           return clearInterval(interval);
59 |         }
60 |
61 |         const particleCount = 50 * (timeLeft / duration);
62 |         confetti({ ...defaults, particleCount, origin: { x: randomInRange(0.1, 0.3), y: Math.random() - 0.2 } });
63 |         confetti({ ...defaults, particleCount, origin: { x: randomInRange(0.7, 0.9), y: Math.random() - 0.2 } });
64 |       }, 250);
65 |     }
66 |   }, [isPassed]);
67 |
68 |   return (
69 |     <div className="min-h-screen bg-[#f0f1a] text-white p-6 md:p-12">
70 |       <div className="max-w-5xl mx-auto space-y-10">
71 |
72 |         /* HERO SECTION */
73 |         <motion.section
74 |           initial={{ opacity: 0, y: 20 }}
75 |           animate={{ opacity: 1, y: 0 }}
76 |           className={`relative overflow-hidden rounded-[40px] p-10 text-center border-2 transition-colors dura...
77 |             ${isPassed ? 'bg-[#00d2a0]/5 border-[#00d2a0]/30 shadow-[0_0_50px_rgba(0,210,160,0.1)]' : 'bg-red-...
78 |           `}
79 |         >
80 |           <div className="relative z-10">
81 |             <motion.div
82 |               initial={{ scale: 0 }}
83 |               animate={{ scale: 1 }}
84 |               transition={{ type: "spring", stiffness: 260, damping: 20 }}
85 |               className={`w-20 h-20 mx-auto mb-6 flex items-center justify-center rounded-2xl
86 |                 ${isPassed ? 'bg-[#00d2a0] text-black' : 'bg-red-500 text-white'}
87 |               `}
88 |             >
89 |               {isPassed ? <Trophy size={40} /> : <AlertCircle size={40} />}
90 |             </motion.div>
91 |
92 |             <h1 className="text-3xl md:text-5xl font-black mb-4">
93 |               {isPassed ? 'Chúc mừng bạn đã qua!' : 'Thử lại lần nữa!'}
94 |             </h1>
95 |
96 |             /* Circular Progress Ring */
97 |             <div className="relative w-48 h-48 mx-auto my-10 flex items-center justify-center">
98 |               <svg className="w-full h-full -rotate-90">
99 |                 <circle
100 |                   cx="96" cy="96" r="88"
101 |                   className="stroke-[#1e293b]"
102 |                   strokeWidth="12"
103 |                   fill="transparent"
104 |                 />
105 |                 <motion.circle
106 |                   cx="96" cy="96" r="88"
107 |                   stroke={grade.color}
108 |                   strokeWidth="12"
109 |                   fill="transparent"
110 |                   strokeDasharray={552}
111 |                   initial={{ strokeDashoffset: 552 }}
112 |                   animate={{ strokeDashoffset: 552 - (552 * result.score) / 100 }}
113 |                   transition={{ duration: 2, ease: "easeOut" }}
114 |                   strokeLinecap="round"
115 |                 />
116 |               </svg>
117 |               <div className="absolute inset-0 flex flex-col items-center justify-center">
118 |                 <span className="text-5xl font-black">{result.score}</span>
119 |                 <span className="text-slate-500 font-bold uppercase track
120 |               </div>
121 |             </div>
122 |
123 |             <div className="flex flex-wrap items-center justify-center gap-6">
124 |               <Badge icon={<CheckCircle2 size={16} />} label="Xếp loại"
125 |               <Badge icon={<Clock size={16} />} label="Thời gian" value=
126 |               <Badge icon={<Trophy size={16} />} label="Xếp hạng" value=
127 |             </div>
128 |
129 |             /* Background Decorative Circles */
130 |             <div className={`absolute top-0 right-0 w-64 h-64 blur-[100px] -translate-y-1/2 translate-x-1/2 opac...
131 |

```



```

132 |         ${isPassed ? 'bg-[#00d2a0]' : 'bg-red-500')}
133 |     }></div>
134 | </motion.section>
135 |
136 | <div className="grid grid-cols-1 lg:grid-cols-2 gap-8">
137 |
138 |     { /* STATS BREAKDOWN (Donut Chart) */ }
139 |     <motion.section
140 |       initial={{ opacity: 0, x: -20 }}
141 |       animate={{ opacity: 1, x: 0 }}
142 |       transition={{ delay: 0.2 }}
143 |       className="bg-[#16213e] p-8 rounded-[32px] border border-[#1e293b]"
144 |     >
145 |       <h2 className="text-xl font-bold mb-8 flex items-center gap-2">
146 |         <PieChart size={20} className="text-[#00d2a0]" /> Th Ñ ñ g k é
147 |       </h2>
148 |
149 |       <div className="flex items-center justify-around gap-4">
150 |         { /* Simple SVG Donut */ }
151 |         <div className="relative w-40 h-40">
152 |           <svg className="w-full h-full -rotate-90">
153 |             { /* Correct */ }
154 |             <circle cx="80" cy="80" r="70" stroke="#00d2a0" strokeWidth="20" fill="transparent" stroke...
155 |             { /* Wrong - Mocking offset */ }
156 |             <circle cx="80" cy="80" r="70" stroke="#ef4444" strokeWidth="20" fill="transparent" stroke...
157 |             { /* Blank */ }
158 |             <circle cx="80" cy="80" r="70" stroke="#475569" strokeWidth="20" fill="transparent" stroke...
159 |           </svg>
160 |         </div>
161 |
162 |         <div className="space-y-4">
163 |           <LegendItem color="#00d2a0" label=" ú ñ g" value={result.wrongCount} />
164 |           <LegendItem color="#ef4444" label="Sai" value={result.wrongCount} />
165 |           <LegendItem color="#475569" label=" B ï tr Ñ ñ g" value={result.wrongCount} />
166 |         </div>
167 |       </div>
168 |     </motion.section>
169 |
170 |     { /* TOPIC BREAKDOWN */ }
171 |     <motion.section
172 |       initial={{ opacity: 0, x: 20 }}
173 |       animate={{ opacity: 1, x: 0 }}
174 |       transition={{ delay: 0.3 }}
175 |       className="bg-[#16213e] p-8 rounded-[32px] border border-[#1e293b]"
176 |     >
177 |       <h2 className="text-xl font-bold mb-8 flex items-center gap-2">
178 |         <BarChart3 size={20} className="text-[#00b4d8]" /> i Ä m t h
179 |       </h2>
180 |       <div className="space-y-6">
181 |         {result.topics.map((topic, i) => (
182 |           <div key={i} className="space-y-2">
183 |             <div className="flex justify-between text-sm font-bold">
184 |               <span>{topic.name}</span>
185 |               <span className="text-[#00d2a0]">{topic.score}/{topic.total}</span>
186 |             </div>
187 |             <div className="h-3 w-full bg-[#0f0f1a] rounded-full overflow-hidden">
188 |               <motion.div
189 |                 initial={{ width: 0 }}
190 |                 animate={{ width: `${(topic.score / topic.total) * 100}%` }}
191 |                 transition={{ duration: 1, delay: 0.5 + i * 0.1 }}
192 |                 className="h-full bg-gradient-to-r from-[#00b4d8] to-[#00d2a0]"
193 |               ></motion.div>
194 |             </div>
195 |           </div>
196 |         ))}
197 |       </div>
198 |     </motion.section>
199 |
200 |   </div>
201 |
202 |   { /* ACTIONS */ }
203 |   <motion.div
204 |     initial={{ opacity: 0, y: 20 }}
205 |     animate={{ opacity: 1, y: 0 }}
206 |     transition={{ delay: 0.6 }}
207 |     className="flex flex-col sm:flex-row items-center justify-center gap-4 pt-10"
208 |   >
209 |     <Link href={` /exam/${examId}`} className="w-full sm:w-auto px-8 py-4 bg-[#16213e] border border-[#1e...
210 |       <RotateCcw size={20} /> L ä m l i i b ä i t h i
211 |     </Link>
212 |     <Link href="/lessons" className="w-full sm:w-auto px-8 py-4 bg-[#00d2a0] text-black font-bold rounded...
213 |       H ï c b ä i t i ç p t h e o <ArrowRight size={20} />
214 |     </Link>
215 |     <button className="w-full sm:w-auto px-8 py-4 bg-[#0f0f1a] border border-[#1e293b] text-slate-400 fo...
216 |       <Share2 size={20} /> C h i a s » k ÷ t q u é
217 |     </button>
218 |   </motion.div>
219 |
220 | </div>
221 | </div>
222 | );
223 | }
224 |
225 | function Badge({ icon, label, value, color }: { icon: any, label: string, value: string, color: string }) {
226 |   return (
227 |     <div className="flex items-center gap-3 bg-[#0f0f1a] px-5 py-3 rounded-2xl border border-[#1e293b]">
228 |       <div style={{ color: color }}>{icon}</div>
229 |       <div className="text-left">
230 |         <div className="text-[10px] text-slate-500 font-bold uppercase tracking-widest">{label}</div>
231 |         <div className="font-bold text-sm" style={{ color: color }}>{value}</div>
232 |       </div>
233 |     </div>
234 |   );
235 | }
236 |
237 | function LegendItem({ color, label, value }: { color: string, label: string, value: number }) {
238 |   return (
239 |     <div className="flex items-center gap-3">
240 |       <div className="w-3 h-3 rounded-full" style={{ background: color }}></div>
241 |       <span className="text-slate-400 font-medium text-sm w-20">{label}</span>
242 |       <span className="font-bold text-white">{value}</span>
243 |     </div>
244 |   );
245 | }
246 |

```

File: app/exams/%5BexamId%5D/page.tsx (206 lines, 9.4 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect } from 'react'
4 | import { motion, AnimatePresence } from 'framer-motion'
5 | import { Clock, ShieldAlert, ChevronLeft, ChevronRight, Send, HelpCircle, AlertCircle, Maximize2 } from 'lucid...

```

```
6 | import Navbar from '@components/Navbar'
7 |
8 | export default function ExamPlayerPage() {
9 |   const [currentQuestion, setCurrentQuestion] = useState(0)
10 |   const [timeLeft, setTimeLeft] = useState(15 * 60) // 15 mins
11 |   const [answers, setAnswers] = useState<Record<number, string>>({})
12 |   const [isSubmitted, setIsSubmitted] = useState(false)
13 |
14 |   const questions = [
15 |     {
16 |       id: 1,
17 |       text: 'CPU Socket AM4 thuộc hãng sản xuất nào?',
18 |       options: ['Intel', 'AMD', 'NVIDIA', 'Qualcomm'],
19 |       correct: 'AMD'
20 |     },
21 |     {
22 |       id: 2,
23 |       text: 'Thành phần nào sau đây dùng để lưu trữ dữ liệu tạm thời?',
24 |       options: ['HDD', 'SSD', 'RAM', 'ROM'],
25 |       correct: 'RAM'
26 |     },
27 |     {
28 |       id: 3,
29 |       text: 'Tác động của lỗi kernel tên nhicet là gì?',
30 |       options: ['Làm hỏng CPU', 'Đánh nhicet', 'Giảm hiệu năng', 'Tăng nhicet']
31 |     }
32 |   ]
33 |
34 |   useEffect(() => {
35 |     if (timeLeft <= 0) {
36 |       handleSubmit()
37 |       return
38 |     }
39 |     const timer = setInterval(() => setTimeLeft(prev => prev - 1), 1000)
40 |     return () => clearInterval(timer)
41 |   }, [timeLeft])
42 |
43 |   const formatTime = (seconds: number) => {
44 |     const m = Math.floor(seconds / 60)
45 |     const s = seconds % 60
46 |     return `${m.toString().padStart(2, '0')}:${s.toString().padStart(2, '0')}`
47 |   }
48 |
49 |   const handleSubmit = () => {
50 |     setIsSubmitted(true)
51 |   }
52 |
53 |   if (isSubmitted) {
54 |     const score = questions.filter((q, idx) => answers[idx] === q.correct).length
55 |     return (
56 |       <div className="min-h-screen bg-[#0f0f1a] flex items-center justify-center p-6 text-white">
57 |         <motion.div
58 |           initial={{ opacity: 0, scale: 0.9 }}
59 |           animate={{ opacity: 1, scale: 1 }}
60 |           className="w-full max-w-lg bg-[#16213e] border border-[#1e293b] p-10 rounded-3xl text-center shadow-...
61 |         >
62 |           <div className="w-20 h-20 bg-[#00d2a0]/10 text-[#00d2a0] rounded-full flex items-center justify-cent...
63 |           <Send size={40} />
64 |         </div>
65 |         <h2 className="text-3xl font-bold mb-2">Hoàn thành bài thi!</h2>
66 |         <p className="text-slate-400 mb-8">Bạn đã hoàn thành bài kiểm tra</p>
67 |         <div className="bg-[#0f0f1a] rounded-2xl p-6 mb-8 border border-[#1e293b]">
68 |           <p className="text-slate-500 uppercase font-bold mb-2">Kết quả bài thi</p>
69 |           <p className="text-5xl font-black text-[#00d2a0]">{score}/{questions.length}</p>
70 |         </div>
71 |         <button onClick={() => window.location.href = '/exams'} className="w-full py-4 bg-[#00d2a0] text-bla...
72 |         </button>
73 |         </motion.div>
74 |       </div>
75 |     )
76 |   }
77 |
78 |   return (
79 |     <div className="min-h-screen bg-[#0f0f1a] text-white flex flex-col">
80 |       <Header />
81 |       <div className="h-20 bg-[#16213e] border-b border-[#1e293b] flex items-center justify-between px-6 md...
82 |         <div className="flex items-center gap-4">
83 |           <div className="w-10 h-10 bg-[#00d2a0]/10 text-[#00d2a0] rounded-xl flex items-center justify-center">
84 |             <HelpCircle size={20} />
85 |           </div>
86 |           <div>
87 |             <h1 className="font-bold text-sm md:text-base">Kiểm tra</h1>
88 |             <div className="flex items-center gap-2 text-[10px] text-red-400 font-bold uppercase tracking-wid...
89 |               <ShieldAlert size={10} />
90 |               <span>Chú ý</span>
91 |             </div>
92 |           </div>
93 |           <div className="flex items-center gap-6">
94 |             <div className="flex flex-col items-end">
95 |               <span className="text-xl font-mono font-bold ${timeLeft < 60 ? 'text-red-500 animate-pulse' : 't...
96 |               {formatTime(timeLeft)}</span>
97 |             </div>
98 |             <button className="hidden md:flex p-2 text-slate-500 hover:text-white transition-colors">
99 |               <Maximize2 size={20} />
100 |             </button>
101 |           </div>
102 |         </div>
103 |       </div>
104 |       <div className="flex-1 flex flex-col lg:flex-row overflow-hidden">
105 |         <div className="flex-1 overflow-y-auto p-6 md:p-12 custom-scrollbar bg-[#0a0a14]">
106 |           <div className="max-w-3xl mx-auto">
107 |             <div className="mb-10">
108 |               <span className="px-3 py-1 bg-[#1e293b] text-[#00d2a0] text-[10px] font-bold uppercase rounded-...
109 |               <h2 className="text-2xl font-bold leading-relaxed">{questions[currentQuestion].text}</h2>
110 |             </div>
111 |             <div className="grid grid-cols-1 gap-4">
112 |               {questions[currentQuestion].options.map((opt, idx) => (
113 |                 <button
114 |                   key={idx}
115 |                   onClick={() => setAnswers({...answers, [currentQuestion]: opt})}
116 |                   className="p-6 rounded-2xl border text-left transition-all flex items-center justify-betwe...
117 |                   answers[currentQuestion] === opt
118 |                   ? 'bg-[#00d2a0]/10 border-[#00d2a0] text-[#00d2a0]'
119 |                   : 'bg-[#16213e] border-[#1e293b] text-slate-300 hover:border-[#2a3655]'
120 |                 )
121 |               )
122 |             </div>
123 |           </div>
124 |         </div>
125 |       </div>
126 |     </div>
127 |   )
128 | }
129 |
```

```
130 |         `}``  
131 |     >  
132 |     <div className="flex items-center gap-4">  
133 |         <span className={`w-8 h-8 rounded-lg flex items-center justify-center text-xs font-bold ${  
134 |             answers[currentQuestion] === opt ? 'bg-[#00d2a0] text-black' : 'bg-[#0f0f1a] text-slate...  
135 |         }`}>  
136 |             {String.fromCharCode(65 + idx)}  
137 |         </span>  
138 |         <span className="font-medium">{opt}</span>  
139 |     </div>  
140 |     {answers[currentQuestion] === opt && <motion.div layoutId="check" className="w-6 h-6 bg-[#0...  
141 |     </button>  
142 |     )}}  
143 | </div>  
144 | </div>  
145 | </div>  
146 |  
147 | /* Right Side: Navigation */  
148 | <div className="w-full lg:w-96 border-l border-[#1e293b] bg-[#0f0f1a] flex flex-col">  
149 |     <div className="p-6 border-b border-[#1e293b]">  
150 |         <h3 className="font-bold text-sm mb-4"> Danh sách câu hỏi </h3>  
151 |         <div className="grid grid-cols-5 gap-3">  
152 |             {questions.map((_, idx) => (  
153 |                 <button  
154 |                     key={idx}  
155 |                     onClick={() => setCurrentQuestion(idx)}  
156 |                     className={`w-10 h-10 rounded-xl flex items-center justify-center text-xs font-bold transit...  
157 |                     currentQuestion === idx ? 'bg-[#00d2a0] text-black border-[#00d2a0]' :  
158 |                     answers[idx] ? 'bg-[#1e293b] text-[#00d2a0] border-[#00d2a0]/30' :  
159 |                     'bg-[#16213e] text-slate-500 border-[#1e293b]'  
160 |                 > `}``  
161 |                 >  
162 |                     {idx + 1}  
163 |                 </button>  
164 |             )}}  
165 |         </div>  
166 |     </div>  
167 |  
168 |     <div className="flex-1 p-6 space-y-6">  
169 |         <div className="bg-[#16213e] p-4 rounded-xl border border-[#1e293b] flex items-center gap-3">  
170 |             <AlertCircle size={18} className="text-yellow-500" />  
171 |             <p className="text-[10px] text-slate-400 font-medium leading-relaxed">  
172 |                 Đăng thoát trình duyệt hoặc chuyển tab, hành ứng này  
173 |             </p>  
174 |         </div>  
175 |     </div>  
176 |  
177 |     <div className="p-6 bg-[#16213e]/50 border-t border-[#1e293b] space-y-3">  
178 |         <div className="flex gap-2">  
179 |             <button  
180 |                 disabled={currentQuestion === 0}  
181 |                 onClick={() => setCurrentQuestion(prev => prev - 1)}  
182 |                 className="flex-1 py-3 bg-[#1e293b] text-white rounded-xl font-bold text-xs disabled:opacity-...  
183 |             >  
184 |                 <ChevronLeft size={16} /> TRƯỚC  
185 |             </button>  
186 |             <button  
187 |                 disabled={currentQuestion === questions.length - 1}  
188 |                 onClick={() => setCurrentQuestion(prev => prev + 1)}  
189 |                 className="flex-1 py-3 bg-[#1e293b] text-white rounded-xl font-bold text-xs disabled:opacity-...  
190 |             >  
191 |                 SAU <ChevronRight size={16} />  
192 |             </button>  
193 |         </div>  
194 |         <button  
195 |             onClick={handleSubmit}  
196 |             className="w-full py-4 bg-[#00d2a0] text-black font-black rounded-xl hover:bg-[#00e6af] shadow-...  
197 |         >  
198 |             NỘP BÀI THI  
199 |         </button>  
200 |     </div>  
201 | </div>  
202 | </main>  
203 | </div>  
204 | )  
205 | }  
206 |
```

File: app/exams/page.tsx (118 lines, 5.5 KB)

```
1 | 'use client'  
2 |  
3 | import React, { useState, useEffect } from 'react'  
4 | import { supabase } from '@lib/supabase'  
5 | import Navbar from '@components/Navbar'  
6 | import { motion } from 'framer-motion'  
7 | import { FileText, Clock, Trophy, ArrowRight, Loader2, Search, Filter, ArrowLeft } from 'lucide-react'  
8 | import Link from 'next/link'  
9 |  
10 | export default function ExamsListingPage() {  
11 |     const [exams, setExams] = useState<any[]>([])  
12 |     const [loading, setLoading] = useState(true)  
13 |  
14 |     useEffect(() => {  
15 |         fetchExams()  
16 |     }, [])  
17 |  
18 |     async function fetchExams() {  
19 |         try {  
20 |             setLoading(true)  
21 |             const { data: { user } } = await supabase.auth.getUser()  
22 |             if (!user) return  
23 |  
24 |             const { data } = await supabase  
25 |                 .from('assignments')  
26 |                 .select('* classes (name)')  
27 |                 .eq('is_published', true)  
28 |                 .order('created_at', { ascending: false })  
29 |  
30 |             setExams(data || [])  
31 |         } catch (err) {  
32 |             console.error(err)  
33 |         } finally {  
34 |             setLoading(false)  
35 |         }  
36 |     }  
37 |  
38 |     return (  
39 |         <div className="min-h-screen" style={{ background: 'var(--bg-base)', color: 'var(--text-primary)' }}>  
40 |             <Navbar />  
41 |  
42 |             <main className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 pt-28 pb-20">  
43 |                 <header className="mb-12">
```

```

44 |         <h1 className="text-4xl font-black mb-4">Trung tâm <span style={{ color: 'var(--brand-primary)' }}>K...
45 |         <p style={{ color: 'var(--text-muted)' }}>Đanh sách các bài k
46 |     </header>
47 |
48 |     <div className="flex flex-col md:flex-row gap-4 mb-8">
49 |         <div className="relative flex-1">
50 |             <Search className="absolute left-4 top-1/2 -translate-y-1/2" size={18} style={{ color: 'var(--text...
51 |             <input
52 |                 type="text"
53 |                 placeholder="Tìm kiếm khóa thi..."
54 |                 className="w-full rounded-xl pl-12 pr-4 py-3 outline-none transition-all"
55 |                 style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)', color: 'var...
56 |             </div>
57 |             <button className="flex items-center gap-2 px-6 py-3 rounded-xl transition-all" style={{ background: ...
58 |             <Filter size={18} /> <Link
59 |         </button>
60 |     </div>
61 |
62 |     {loading ? (
63 |         <div className="flex justify-center py-20">
64 |             <Loader2 className="animate-spin" size={40} style={{ color: 'var(--brand-primary)' }} />
65 |         </div>
66 |     ) : exams.length > 0 ? (
67 |         <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
68 |             {exams.map((exam) => (
69 |                 <motion.div
70 |                     key={exam.id}
71 |                     initial={{ opacity: 0, y: 20 }}
72 |                     animate={{ opacity: 1, y: 0 }}
73 |                     className="rounded-3xl p-6 transition-all group"
74 |                     style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)' }}
75 |                 >
76 |                     <div className="flex items-start justify-between mb-6">
77 |                         <div className="p-3 rounded-2xl" style={{ background: 'color-mix(in srgb, var(--brand-primar...
78 |                         <FileText size={24} />
79 |                     </div>
80 |                     <div className="text-right">
81 |                         <span className="text-[10px] font-bold uppercase tracking-widest" style={{ color: 'var(--t...
82 |                         <p className="text-xs font-bold flex items-center gap-1 justify-end mt-1" style={{ color: ...
83 |                         <Clock size={12} /> 45 Phút
84 |                     </p>
85 |                     </div>
86 |                 </div>
87 |             </div>
88 |
89 |             <h3 className="text-xl font-bold mb-2 transition-colors" style={{ color: 'var(--text-primary)'}}...
90 |             <p className="text-sm mb-6 font-medium" style={{ color:
91 |
92 |             <div className="flex items-center justify-between pt-6" style={{ borderTop: '1px solid var(--b...
93 |             <div className="flex items-center gap-2">
94 |                 <Trophy size={16} style={{ color: 'var(--accent-amber)' }} />
95 |                 <span className="text-xs font-bold" style={{ color: 'var(--text-muted)' }}>100 XP</span>
96 |             </div>
97 |             <Link
98 |                 href={`/exams/${exam.id}`}
99 |                 className="px-6 py-2 text-xs font-bold rounded-lg transition-all flex items-center gap-2"
100 |                 style={{ background: 'var(--brand-primary)', color: 'var(--bg-base)' }}
101 |             >
102 |                 B @ T      ! U   T H I   <ArrowRight size={14} />
103 |             </Link>
104 |         </div>
105 |     </motion.div>
106 |     ))}
107 | </div>
108 | ) : (
109 |     <div className="text-center py-20 rounded-[40px] border-2 border-dashed" style={{ background: 'color...
110 |     <FileText size={48} className="mx-auto mb-4" style={{ color: 'var(--text-muted)' }} />
111 |     <p className="font-medium italic" style={{ color: 'var(--text-
112 | </div>
113 | ))}
114 | </main>
115 | </div>
116 | )
117 | }
118 |

```

File: app/globals.css (771 lines, 22.2 KB)

```

1 | @import url('https://fonts.googleapis.com/css2?family=Montserrat:wght@300;400;500;600;700;800;900&family=JetBr...
2 |
3 | @tailwind base;
4 | @tailwind components;
5 | @tailwind utilities;
6 |
7 | :root, [data-theme="light"] {
8 |   --bg-base: #efedfb;
9 |   --bg-surface: #ffffff;
10 |   --bg-elevated: #f8f7fc;
11 |   --bg-hover: #f0eff8;
12 |
13 |   --brand-primary: #089e60;
14 |   --brand-light: #28c76f;
15 |   --brand-dark: #067a4d;
16 |   --brand-subtle: rgba(8, 158, 96, 0.12);
17 |
18 |   --accent-blue: #289cf9;
19 |   --accent-amber: #ffb900;
20 |   --accent-orange: #ff8b00;
21 |
22 |   --success: #28c76f;
23 |   --warning: #ffa300;
24 |   --danger: #f46a6a;
25 |   --info: #7367f0;
26 |
27 |   --text-primary: #252434;
28 |   --text-secondary: #3b3a49;
29 |   --text-muted: #6e6b7b;
30 |
31 |   --border-subtle: #f1f1f1;
32 |   --border-default: #e7e7e7;
33 |   --border-strong: #d0cfd8;
34 |
35 |   --shadow-color: rgba(113, 126, 195, 0.1);
36 |   --shadow-hover: rgba(113, 126, 195, 0.18);
37 |
38 |   --sidebar-width: 265px;
39 |   --font-sans: 'Montserrat', sans-serif;
40 |   --font-mono: 'JetBrains Mono', monospace;
41 |
42 |   --text-xs: 11px;
43 |   --text-sm: 13px;
44 |   --text-base: 15px;
45 |   --text-lg: 18px;

```

```

46 | --text-xl: 24px;
47 | --text-2xl: 32px;
48 | --text-hero: 48px;
49 |
50 | --primary-neon: #0284c7;
51 | --secondary-neon: #089e60;
52 | --text-dim: #6e6b7b;
53 | --accent-green: #10b981;
54 | --accent-purple: #8b5cf6;
55 | }
56 |
57 | [data-theme="dark"] {
58 |   --bg-base: #0d0e13;
59 |   --bg-surface: #0f0f1a;
60 |   --bg-elevated: #1a1c25;
61 |   --bg-hover: #1e202f;
62 |
63 |   --brand-primary: #00d4aa;
64 |   --brand-light: #4de8c4;
65 |   --brand-dark: #00b89c;
66 |   --brand-subtle: rgba(0, 212, 170, 0.12);
67 |
68 |   --accent-blue: #4a90e2;
69 |   --accent-amber: #f59e0b;
70 |   --accent-orange: #ff8b00;
71 |
72 |   --success: #00d4aa;
73 |   --warning: #f59e0b;
74 |   --danger: #e84855;
75 |   --info: #636678;
76 |
77 |   --text-primary: #dde0ed;
78 |   --text-secondary: #b8bbcc;
79 |   --text-muted: #636678;
80 |
81 |   --primary-neon: #00f3ff;
82 |   --secondary-neon: #00d4aa;
83 |   --text-dim: #636678;
84 |   --accent-green: #00d4aa;
85 |   --accent-purple: #a78bfa;
86 |
87 |   --border-subtle: rgba(255, 255, 255, 0.05);
88 |   --border-default: rgba(255, 255, 255, 0.08);
89 |   --border-strong: rgba(255, 255, 255, 0.15);
90 |
91 |   --shadow-color: rgba(0, 0, 0, 0.4);
92 |   --shadow-hover: rgba(0, 0, 0, 0.6);
93 | }
94 |
95 | *, ::before, ::after {
96 |   box-sizing: border-box;
97 |   padding: 0;
98 |   margin: 0;
99 | }
100 |
101 | html, body {
102 |   max-width: 100vw;
103 |   overflow-x: hidden;
104 |   min-height: 100vh;
105 | }
106 |
107 | body {
108 |   color: var(--text-primary);
109 |   background: var(--bg-base);
110 |   font-family: var(--font-sans);
111 |   font-size: 15px;
112 |   font-weight: 500;
113 |   -webkit-font-smoothing: antialiased;
114 |   transition: background-color 0.3s ease, color 0.3s ease;
115 | }
116 |
117 | a { color: inherit; text-decoration: none; }
118 | img { max-width: 100%; height: auto; }
119 |
120 | ::-webkit-scrollbar { width: 8px; height: 8px; }
121 | ::-webkit-scrollbar-track { background: var(--bg-base); }
122 | ::-webkit-scrollbar-thumb { background: var(--border-strong); border-radius: 4px; }
123 | ::-webkit-scrollbar-thumb:hover { background: var(--text-muted); }
124 |
125 | @keyframes fadeInUp {
126 |   from { opacity: 0; transform: translateY(20px); }
127 |   to { opacity: 1; transform: translateY(0); }
128 | }
129 | @keyframes fadeIn {
130 |   from { opacity: 0; }
131 |   to { opacity: 1; }
132 | }
133 | @keyframes slideInLeft {
134 |   from { opacity: 0; transform: translateX(-24px); }
135 |   to { opacity: 1; transform: translateX(0); }
136 | }
137 | @keyframes slideInRight {
138 |   from { opacity: 0; transform: translateX(24px); }
139 |   to { opacity: 1; transform: translateX(0); }
140 | }
141 | @keyframes scaleIn {
142 |   from { opacity: 0; transform: scale(0.95); }
143 |   to { opacity: 1; transform: scale(1); }
144 | }
145 |
146 | .animate-fade-in-up { animation: fadeInUp 0.5s ease both; }
147 | .animate-fade-in { animation: fadeIn 0.4s ease both; }
148 | .animate-slide-left { animation: slideInLeft 0.5s ease both; }
149 | .animate-slide-right { animation: slideInRight 0.5s ease both; }
150 | .animate-scale-in { animation: scaleIn 0.3s ease both; }
151 | .animate-delay-1 { animation-delay: 0.1s; }
152 | .animate-delay-2 { animation-delay: 0.2s; }
153 | .animate-delay-3 { animation-delay: 0.3s; }
154 | .animate-delay-4 { animation-delay: 0.4s; }
155 | .animate-delay-5 { animation-delay: 0.5s; }
156 |
157 | .neon-text-blue { color: var(--accent-blue); text-shadow: 0 0 20px rgba(74, 144, 226, 0.5); }
158 | .neon-text-purple { color: var(--accent-purple); text-shadow: 0 0 20px rgba(139, 92, 246, 0.5); }
159 | .neon-text-green { color: var(--accent-green); text-shadow: 0 0 20px rgba(16, 185, 129, 0.5); }
160 |
161 | .glass-panel {
162 |   background: var(--bg-surface);
163 |   border: 1px solid var(--border-default);
164 |   border-radius: 10px;
165 |   transition: all 0.2s ease;
166 | }
167 |
168 | 1 6 8 | / * % % % S i d e b a r % % % * /
169 | .sidebar-container {

```

```

170 | position: fixed; top: 0; left: 0; height: 100vh;
171 | width: var(--sidebar-width);
172 | background: linear-gradient(3deg, #031f3b 0%, #1a2f53 80%);
173 | border-right: 1px solid rgba(255,255,255,0.06);
174 | display: flex; flex-direction: column;
175 | z-index: 190; overflow-y: auto;
176 | transition: transform 280ms cubic-bezier(0.4, 0, 0.2, 1);
177 | }
178 | .sidebar-container::-webkit-scrollbar { display: none; }
179 |
180 | @media (max-width: 768px) {
181 |   .sidebar-container { transform: translateX(-100%); }
182 |   .sidebar-container.open { transform: translateX(0); }
183 |   .mobile-hamburger { display: flex !important; }
184 |   .mobile-backdrop { display: block !important; }
185 | }
186 | @media (min-width: 769px) {
187 |   .mobile-hamburger { display: none !important; }
188 |   .mobile-backdrop { display: none !important; }
189 | }
190 |
191 | 9 1 | / * % % %   T o g g l e   S w i t c h   % % %   * /
192 | .toggle-switch {
193 |   width: 36px; height: 20px; border-radius: 10px;
194 |   position: relative; cursor: pointer; transition: all 0.25s ease;
195 | }
196 | .toggle-switch.off { background: #d0cfd8; border: 1px solid #bbbac3; }
197 | .toggle-switch.on {
198 |   background: var(--brand-primary); border: 1px solid var(--brand-primary);
199 |   box-shadow: 0 0 8px rgba(8,158,96,0.4);
200 | }
201 | .toggle-thumb {
202 |   width: 16px; height: 16px; border-radius: 50%; position: absolute; top: 1px;
203 |   transition: all 0.25s cubic-bezier(0.4, 0, 0.2, 1);
204 | }
205 | .toggle-switch.off .toggle-thumb { left: 1px; background: #fff; }
206 | .toggle-switch.on .toggle-thumb { left: 17px; background: #fff; }
207 |
208 | .custom-slider {
209 |   -webkit-appearance: none; width: 100%; height: 3px; border-radius: 2px;
210 |   outline: none; transition: background 0.2s;
211 | }
212 | .custom-slider::-webkit-slider-thumb {
213 |   -webkit-appearance: none; appearance: none; width: 14px; height: 14px;
214 |   border-radius: 50%; background: white; border: 2px solid var(--brand-primary);
215 |   box-shadow: 0 0 0 3px var(--brand-subtle); cursor: pointer;
216 |   transition: transform 0.15s ease;
217 | }
218 | .custom-slider::-webkit-slider-thumb:hover { transform: scale(1.2); }
219 | .custom-slider::-moz-range-thumb {
220 |   width: 14px; height: 14px; border-radius: 50%; background: white;
221 |   border: 2px solid var(--brand-primary); box-shadow: 0 0 0 3px var(--brand-subtle);
222 |   cursor: pointer;
223 | }
224 |
225 | @keyframes va-bounce {
226 |   0%, 80%, 100% { transform: translateY(0); opacity: 0.5; }
227 |   40% { transform: translateY(-6px); opacity: 1; }
228 | }
229 | @keyframes blink { 0%, 100% { opacity: 1; } 50% { opacity: 0.4; } }
230 | @keyframes pulse { 0%, 100% { opacity: 0.5; } 50% { opacity: 1; } }
231 | @keyframes terminal-blink { 0%, 100% { opacity: 1; } 50% { opacity: 0; } }
232 | @keyframes shimmer {
233 |   0% { background-position: -200% 0; }
234 |   100% { background-position: 200% 0; }
235 | }
236 |
237 | 2 3 7 | / * % % %   C o m p o n e n t   C a r d   % % %   * /
238 | .mode-card {
239 |   background: var(--bg-surface) !important;
240 |   border: 1px solid var(--border-default) !important;
241 |   border-radius: 10px;
242 |   padding: 24px;
243 |   cursor: pointer;
244 |   transition: all 0.25s cubic-bezier(0.4, 0, 0.2, 1);
245 |   display: flex;
246 |   flex-direction: column;
247 |   text-align: left;
248 |   width: 100%;
249 | }
250 | .mode-card:hover {
251 |   background: var(--bg-hover) !important;
252 |   border-color: var(--border-strong) !important;
253 |   transform: translateY(-3px);
254 |   box-shadow: 0 8px 16px -4px var(--shadow-hover);
255 | }
256 | .mode-card h2, .mode-card h3, .mode-card .title {
257 |   color: var(--text-primary) !important;
258 |   margin-top: 16px;
259 |   margin-bottom: 0;
260 |   font-size: 16px;
261 |   font-weight: 600;
262 | }
263 | .mode-card p, .mode-card .desc {
264 |   color: var(--text-muted) !important;
265 |   margin-top: 6px;
266 |   margin-bottom: 0;
267 |   line-height: 1.5;
268 |   font-size: 13px;
269 | }
270 |
271 | .icon-wrapper {
272 |   background: var(--brand-subtle);
273 |   border-radius: 8px;
274 |   width: 40px; height: 40px;
275 |   display: flex; align-items: center; justify-content: center;
276 |   color: var(--brand-primary);
277 | }
278 |
279 | 2 7 9 | / * % % %   M a i n   C o n t e n t   % % %   * /
280 | .main-content {
281 |   margin-left: var(--sidebar-width);
282 |   width: calc(100vw - var(--sidebar-width));
283 |   min-height: 100vh;
284 |   overflow-x: hidden;
285 |   display: flex;
286 |   flex-direction: column;
287 |   position: relative;
288 |   transition: margin-left 0.25s ease, width 0.25s ease;
289 | }
290 | @media (max-width: 768px) {
291 |   .main-content {
292 |     margin-left: 0 !important;
293 |     width: 100vw !important;

```

```

294 |     padding: 16px !important;
295 |     padding-top: 64px !important;
296 | }
297 | }
298 |
299 | 2 9 9 | / * % % %   G r i d s   &   P a n e l s   % % %   * /
300 | .category-tabs {
301 |     display: flex; overflow-x: auto; scrollbar-width: none;
302 |     -ms-overflow-style: none; gap: 8px; padding-bottom: 4px;
303 | }
304 | .category-tabs::-webkit-scrollbar { display: none; }
305 |
306 | .product-grid {
307 |     display: grid;
308 |     grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
309 |     gap: 16px; width: 100%;
310 | }
311 |
312 | .practice-layout {
313 |     display: grid; grid-template-columns: 1fr 320px;
314 |     gap: 24px; align-items: start; width: 100%;
315 | }
316 |
317 | .camera-panel {
318 |     position: sticky; top: 24px;
319 |     background: var(--bg-surface);
320 |     border: 1px solid var(--border-default);
321 |     border-radius: 10px; padding: 16px; width: 100%; z-index: 10;
322 | }
323 | .camera-panel-header {
324 |     display: flex; justify-content: space-between;
325 |     align-items: center; margin-bottom: 12px;
326 |     font-size: 13px; font-weight: 600; color: var(--text-primary);
327 | }
328 | .camera-toggle-btn {
329 |     font-size: 12px; padding: 4px 10px;
330 |     background: var(--brand-subtle); color: var(--brand-primary);
331 |     border: 1px solid rgba(8,158,96,0.3); border-radius: 6px; cursor: pointer;
332 |     transition: all 0.2s;
333 | }
334 | .camera-feed {
335 |     width: 100%; aspect-ratio: 4/3;
336 |     background: #e8e7f0; border-radius: 8px;
337 |     display: flex; align-items: center; justify-content: center;
338 |     font-size: 13px; color: var(--text-muted);
339 |     overflow: hidden; position: relative;
340 | }
341 |
342 | .hero-grid {
343 |     background-image:
344 |         linear-gradient(rgba(8,158,96,0.05) 1px, transparent 1px),
345 |         linear-gradient(90deg, rgba(8,158,96,0.05) 1px, transparent 1px);
346 |     background-size: 40px 40px;
347 | }
348 |
349 | .stat-number { font-size: 48px; font-weight: 800; color: var(--brand-primary); line-height: 1; }
350 | .stat-label { font-size: 14px; color: var(--text-muted); margin-top: 8px; }
351 |
352 | .feature-card {
353 |     background: var(--bg-surface);
354 |     border: 1px solid var(--border-default);
355 |     border-radius: 10px; padding: 28px 24px;
356 |     transition: all 0.25s cubic-bezier(0.4, 0, 0.2, 1);
357 | }
358 | .feature-card:hover {
359 |     border-color: var(--brand-primary);
360 |     background: var(--bg-hover);
361 |     transform: translateY(-3px);
362 |     box-shadow: 0 8px 16px -4px var(--shadow-hover);
363 | }
364 |
365 | .alert-bar {
366 |     background: rgba(244,106,106,0.08) !important;
367 |     border: 1px solid rgba(244,106,106,0.25) !important;
368 |     border-left: 3px solid var(--danger) !important;
369 |     border-radius: 8px !important;
370 |     color: var(--danger) !important;
371 |     padding: 12px 16px !important;
372 |     font-size: 14px !important;
373 | }
374 |
375 | .buy-btn {
376 |     background: var(--brand-primary) !important;
377 |     color: white !important; border: none !important;
378 |     padding: 10px !important; border-radius: 8px !important;
379 |     font-weight: 600 !important; cursor: pointer !important;
380 |     transition: background 0.2s, transform 0.15s !important;
381 | }
382 | .buy-btn:hover { background: #067a4d !important; transform: translateY(-1px); }
383 |
384 | .status-dot {
385 |     width: 7px; height: 7px; border-radius: 50%;
386 |     background: var(--danger); display: inline-block; margin-right: 6px;
387 | }
388 | .status-dot.active { background: var(--success); }
389 |
390 | @media (max-width: 1024px) {
391 |     .main-content { padding: 24px !important; }
392 |     .product-grid { grid-template-columns: repeat(2, 1fr); }
393 | }
394 | @media (max-width: 768px) {
395 |     .login-modal-content { padding: 24px 20px !important; }
396 |     .student-dashboard-grid { grid-template-columns: 1fr !important; }
397 |     .teacher-dashboard-grid { grid-template-columns: 1fr !important; }
398 | }
399 | @media (max-width: 480px) {
400 |     .login-modal-content { padding: 20px 16px !important; }
401 | }
402 | @media (max-width: 480px) {
403 |     .login-modal-content { padding: 20px 16px !important; }
404 | }
405 |
406 | 4 0 6 | / * % % %   C o n t a i n e r   H e l p e r   % % %   * /
407 | .main-container {
408 |     border-radius: 30px;
409 |     background: var(--bg-surface);
410 |     padding: 24px;
411 |     box-shadow: 0 1px 3px 0 var(--shadow-color), 0 1px 2px -1px var(--shadow-color);
412 | }
413 |
414 | 4 1 4 | / * % % %   L M S   C o m p o n e n t   C l a s s e s   % % %   * /
415 | .lms-btn {
416 |     padding: 12px 20px; border-radius: 10px; font-weight: 700; font-size: 14px;
417 |     transition: all 0.25s cubic-bezier(0.4, 0, 0.2, 1);

```

```

418 | cursor: pointer; border: none;
419 | display: inline-flex; align-items: center; gap: 8px;
420 | font-family: var(--font-sans);
421 | }
422 | .lms-btn-primary {
423 |   background: linear-gradient(90deg, var(--accent-amber), var(--accent-orange));
424 |   color: #fff; box-shadow: 0 4px 12px -2px rgba(255,163,0,0.3);
425 | }
426 | .lms-btn-primary:hover {
427 |   opacity: 0.95; transform: translateY(-2px);
428 |   box-shadow: 0 6px 16px -2px rgba(255,163,0,0.4);
429 | }
430 | .lms-btn-green {
431 |   background: var(--brand-primary); color: #fff;
432 |   box-shadow: 0 4px 12px -2px rgba(8,158,96,0.3);
433 | }
434 | .lms-btn-green:hover {
435 |   background: var(--brand-dark); transform: translateY(-2px);
436 |   box-shadow: 0 6px 16px -2px rgba(8,158,96,0.4);
437 | }
438 | .lms-btn-blue {
439 |   background: var(--accent-blue); color: #fff;
440 |   box-shadow: 0 4px 12px -2px rgba(40,156,249,0.3);
441 | }
442 | .lms-btn-blue:hover {
443 |   background: #1a7de0; transform: translateY(-2px);
444 |   box-shadow: 0 6px 16px -2px rgba(40,156,249,0.4);
445 | }
446 | .lms-btn-outline {
447 |   background: transparent; border: 1px solid var(--border-strong);
448 |   color: var(--text-primary);
449 | }
450 | .lms-btn-outline:hover {
451 |   background: var(--bg-hover); border-color: var(--text-muted);
452 |   transform: translateY(-1px);
453 | }
454 | }
455 | .lms-input {
456 |   height: 50px; border-radius: 10px;
457 |   border: 1px solid var(--border-strong);
458 |   padding: 0 16px; font-family: var(--font-sans);
459 |   font-size: 14px; color: var(--text-primary);
460 |   background: var(--bg-surface); outline: none; width: 100%;
461 |   transition: border-color 0.2s, box-shadow 0.2s;
462 | }
463 | .lms-input:focus {
464 |   border-color: var(--brand-primary);
465 |   box-shadow: 0 0 0 3px var(--brand-subtle);
466 | }
467 | }
468 | .lms-badge {
469 |   background: var(--brand-primary); color: #fff;
470 |   font-size: 10px; padding: 4px 7px; border-radius: 99px; font-weight: 700;
471 | }
472 | .lms-badge-danger { background: var(--danger); color: #fff; }
473 | .lms-badge-amber { background: var(--accent-amber); color: #fff; }
474 | }
475 | .lms-card {
476 |   background: var(--bg-surface);
477 |   border: 1px solid var(--border-default);
478 |   border-radius: 10px;
479 |   padding: 20px;
480 |   transition: all 0.25s cubic-bezier(0.4, 0, 0.2, 1);
481 | }
482 | .lms-card:hover {
483 |   box-shadow: 0 4px 12px -2px var(--shadow-color);
484 |   border-color: var(--border-strong);
485 | }
486 | }
487 | .lms-tag {
488 |   display: inline-flex; align-items: center; gap: 6px;
489 |   font-size: 11px; font-weight: 600;
490 |   padding: 4px 10px; border-radius: 99px;
491 |   background: var(--bg-elevated); color: var(--text-muted);
492 |   border: 1px solid var(--border-default);
493 |   transition: all 0.2s;
494 | }
495 | .lms-tag-green {
496 |   background: rgba(8,158,96,0.1); color: var(--brand-primary);
497 |   border-color: rgba(8,158,96,0.2);
498 | }
499 | .lms-tag-blue {
500 |   background: rgba(40,156,249,0.1); color: var(--accent-blue);
501 |   border-color: rgba(40,156,249,0.2);
502 | }
503 | }
504 | / * % % % S t u d e n t L a y o u t M o b i l e % % % * /
505 | .student-hamburger-btn {
506 |   display: none; position: fixed; top: 12px; left: 12px; z-index: 1001;
507 |   width: 44px; height: 44px; border-radius: 12px;
508 |   background: linear-gradient(180deg, #031f3b 0%, #1a2f53 100%);
509 |   border: none; cursor: pointer; align-items: center; justify-content: center;
510 |   box-shadow: 0 4px 16px rgba(0,0,0,0.35);
511 |   color: #fff; transition: transform 0.2s;
512 | }
513 | .student-hamburger-btn:active { transform: scale(0.92); }
514 | .student-drawer-backdrop {
515 |   display: none; position: fixed; inset: 0;
516 |   background: rgba(0,0,0,0.55); backdrop-filter: blur(4px);
517 |   z-index: 1002;
518 | }
519 | .student-drawer {
520 |   display: none; position: fixed; top: 0; left: 0; bottom: 0;
521 |   width: 280px; z-index: 1003; overflow-y: auto;
522 |   background: linear-gradient(180deg, #031f3b 0%, #1a2f53 100%);
523 |   flex-direction: column;
524 |   box-shadow: 4px 0 24px rgba(0,0,0,0.45);
525 | }
526 | .student-drawer::-webkit-scrollbar { display: none; }
527 | }
528 | @media (max-width: 768px) {
529 |   .student-sidebar-desktop { display: none !important; }
530 |   .student-hamburger-btn { display: flex !important; }
531 |   .student-drawer-backdrop { display: block !important; }
532 |   .student-drawer { display: flex !important; }
533 |   .student-main-content { padding: 16px !important; padding-top: 68px !important; }
534 | }
535 | }
536 | / * % % % S t u d e n t D a s h b o a r d M o b i l e % % % * /
537 | @media (max-width: 768px) {
538 |   .student-dashboard-grid {
539 |     grid-template-columns: 1fr !important;
540 |   }
541 | }

```



```

542| 5 4 3 | / * % % %   P r o   L e a r n i n g   S y s t e m   S t y l e s % % %   * /
544|.pro-card {
545|  background: var(--bg-surface);
546|  border: 1px solid var(--border-default);
547|  border-radius: 12px;
548|  overflow: hidden;
549|  transition: all 0.25s cubic-bezier(0.4, 0, 0.2, 1);
550|}
551|.pro-card:hover {
552|  border-color: var(--border-strong);
553|  box-shadow: 0 8px 24px -4px var(--shadow-hover);
554|  transform: translateY(-2px);
555|}
556|
557|.pro-card-header {
558|  padding: 16px 20px;
559|  border-bottom: 1px solid var(--border-subtle);
560|  background: var(--bg-elevated);
561|  font-weight: 700;
562|  font-size: 14px;
563|  color: var(--text-primary);
564|  display: flex;
565|  align-items: center;
566|  gap: 8px;
567|}
568|.pro-card-body {
569|  padding: 16px 20px;
570|}
571|
572|.pro-badge {
573|  display: inline-flex;
574|  align-items: center;
575|  gap: 4px;
576|  padding: 3px 8px;
577|  border-radius: 99px;
578|  font-size: 10px;
579|  font-weight: 700;
580|  text-transform: uppercase;
581|  letter-spacing: 0.5px;
582|}
583|.pro-badge-green { background: rgba(0,212,170,0.12); color: #00d4aa; }
584|.pro-badge-red { background: rgba(232,72,85,0.12); color: #e84855; }
585|.pro-badge-amber { background: rgba(245,158,11,0.12); color: #f59e0b; }
586|.pro-badge-purple { background: rgba(139,92,246,0.12); color: #8b5cf6; }
587|.pro-badge-blue { background: rgba(59,130,246,0.12); color: #3b82f6; }
588|
589|.pro-score-ring {
590|  position: relative;
591|  display: inline-flex;
592|  align-items: center;
593|  justify-content: center;
594|}
595|.pro-score-ring svg {
596|  transform: rotate(-90deg);
597|}
598|.pro-score-ring circle {
599|  fill: none;
600|  stroke-width: 2.5;
601|}
602|.pro-score-ring .bg {
603|  stroke: var(--border-subtle);
604|}
605|.pro-score-ring .fg {
606|  stroke-linecap: round;
607|  transition: stroke-dasharray 0.5s ease;
608|}
609|.pro-score-value {
610|  position: absolute;
611|  font-weight: 800;
612|  font-size: 14px;
613|}
614|
615|.pro-glow-green { box-shadow: 0 0 20px rgba(0,212,170,0.15); }
616|.pro-glow-red { box-shadow: 0 0 20px rgba(232,72,85,0.15); }
617|.pro-glow-amber { box-shadow: 0 0 20px rgba(245,158,11,0.15); }
618|
619|@keyframes pro-slide-up {
620|  from { opacity: 0; transform: translateY(16px); }
621|  to { opacity: 1; transform: translateY(0); }
622|}
623|@keyframes pro-pulse-glow {
624|  0%, 100% { opacity: 0.5; }
625|  50% { opacity: 1; }
626|}
627|@keyframes pro-count-up {
628|  from { opacity: 0; transform: scale(0.5); }
629|  to { opacity: 1; transform: scale(1); }
630|}
631|
632|.pro-animate-in {
633|  animation: pro-slide-up 0.4s ease both;
634|}
635|.pro-delay-1 { animation-delay: 0.05s; }
636|.pro-delay-2 { animation-delay: 0.1s; }
637|.pro-delay-3 { animation-delay: 0.15s; }
638|.pro-delay-4 { animation-delay: 0.2s; }
639|
640| 6 4 0 | / * % % %   P r o g r e s s   B a r % % %   * /
641|.progress-bar {
642|  height: 4px;
643|  border-radius: 2px;
644|  background: var(--border-subtle);
645|  overflow: hidden;
646|}
647|.progress-bar-fill {
648|  height: 100%;
649|  border-radius: 2px;
650|  transition: width 0.5s ease;
651|}
652|
653| 6 5 3 | / * % % %   S p e c   T a b l e % % %   * /
654|.spec-table {
655|  width: 100%;
656|  border-collapse: collapse;
657|}
658|.spec-table td {
659|  padding: 6px 8px;
660|  font-size: 12px;
661|  border-bottom: 1px solid var(--border-subtle);
662|}
663|.spec-table td:first-child {
664|  color: var(--text-muted);
665|  font-weight: 500;

```

```

666 | width: 40%;
667 | }
668 | .spec-table td:last-child {
669 |   color: var(--text-primary);
670 |   font-weight: 600;
671 |   text-align: right;
672 | }
673 | .spec-table tr:last-child td {
674 |   border-bottom: none;
675 | }
676 |
677 | @keyframes spin {
678 |   to { transform: rotate(360deg); }
679 | }
680 |
681 | @keyframes blink {
682 |   0%, 100% { opacity: 1; }
683 |   50% { opacity: 0.3; }
684 | }
685 |
686 | @keyframes voicePulse {
687 |   0%, 100% { opacity: 0.6; transform: scale(1); }
688 |   50% { opacity: 0.2; transform: scale(1.2); }
689 | }
690 |
691 | @keyframes scanProgress {
692 |   0% { transform: translateX(-100%); }
693 |   100% { transform: translateX(200%); }
694 | }
695 |
696 | 9 6 | / * % % %   P i x e l   G a m i f i c a t i o n   S t y l e s   % % %   * /
697 | .pixel-border {
698 |   border: 2px solid var(--border-strong);
699 |   box-shadow:
700 |     inset -2px -2px 0 0 var(--border-strong),
701 |     inset 2px 2px 0 0 rgba(255,255,255,0.05);
702 |   image-rendering: pixelated;
703 | }
704 |
705 | .pixel-btn {
706 |   border: none;
707 |   position: relative;
708 |   cursor: pointer;
709 |   font-family: var(--font-mono);
710 |   font-weight: 700;
711 |   text-transform: uppercase;
712 |   letter-spacing: 1px;
713 |   transition: all 0.1s;
714 |   box-shadow:
715 |     inset -2px -2px 0 0 rgba(0,0,0,0.3),
716 |     inset 2px 2px 0 0 rgba(255,255,255,0.1);
717 | }
718 | .pixel-btn:active {
719 |   transform: translateY(2px);
720 |   box-shadow:
721 |     inset 2px 2px 0 0 rgba(0,0,0,0.3),
722 |     inset -2px -2px 0 0 rgba(255,255,255,0.1);
723 | }
724 |
725 | .pixel-progress {
726 |   height: 12px;
727 |   border: 1px solid var(--border-strong);
728 |   overflow: hidden;
729 | }
730 | .pixel-progress-fill {
731 |   height: 100%;
732 |   background: repeating-linear-gradient(
733 |     90deg,
734 |     var(--brand-primary) 0px,
735 |     var(--brand-primary) 4px,
736 |     var(--brand-light) 4px,
737 |     var(--brand-light) 8px
738 |   );
739 |   transition: width 0.5s ease;
740 | }
741 |
742 | .pixel-hearts {
743 |   display: flex;
744 |   gap: 2px;
745 | }
746 | .pixel-heart {
747 |   width: 12px;
748 |   height: 12px;
749 |   clip-path: polygon(50% 0%, 100% 30%, 100% 70%, 50% 100%, 0% 70%, 0% 30%);
750 | }
751 | .pixel-heart:active { background: #ef4444; }
752 | .pixel-heart.empty { background: var(--border-strong); }
753 |
754 | @keyframes pixel-blink {
755 |   0%, 100% { opacity: 1; }
756 |   50% { opacity: 0.3; }
757 | }
758 | @keyframes pixel-float {
759 |   0%, 100% { transform: translateY(0); }
760 |   50% { transform: translateY(-4px); }
761 | }
762 | @keyframes pixel-shake {
763 |   0%, 100% { transform: translateX(0); }
764 |   25% { transform: translateX(-4px); }
765 |   75% { transform: translateX(4px); }
766 | }
767 |
768 | .pixel-float { animation: pixel-float 2s ease-in-out infinite; }
769 | .pixel-blink { animation: pixel-blink 1s step-end infinite; }
770 | .pixel-shake { animation: pixel-shake 0.3s ease-in-out; }
771 |

```

File: app/judges/page.tsx (318 lines, 16.0 KB)

```

1 | 'use client';
2 |
3 | import React, { useState, useEffect } from 'react';
4 | import { motion } from 'framer-motion';
5 | import { QRCodeSVG } from 'qrcode.react';
6 |
7 | const sections = [
8 |   {
9 |     icon: 'ðŸ“œ', title: 'TỔNG QUAN ĐẾN AI',
10 |    content: 'PC Master Builder là sản phẩm giáo dục đặc trưng giúp hiểu
11 |  },
12 |   {
13 |     icon: 'ðŸ“š', title: 'CÔNG NGHỆ & AI',
14 |    content: 'Đang đăng AI Guru (Google Gemini) làm trải nghiệm 24/7. H

```

```

15 | },
16 | {
17 |   icon: 'ðŸ™Œ', title: 'Mô Phỏng Lập Ráp 2D',
18 |   content: 'Phòng Lab 2D với 0 hóa chỉ tiết têng linh kiện (CPU, RAM,
19 | },
20 | {
21 |   icon: 'ðŸ“‘', title: 'Hand Tracking (Webcam)',
22 |   content: 'Tích hợp MediaPipe Hands cho phép iáo khiãn bằng cĩ ché t
23 | },
24 | {
25 |   icon: 'ðŸŒˆ', title: 'Chã Máy Tính',
26 |   content: 'Híc sinh tích liy ngãn sách tẽ iã m quiz và bài t-p. Vào c
27 | },
28 | {
29 |   icon: 'ðŸ“‘', title: 'Hç Thñng Quiz & Bài T-p',
30 |   content: '250+ câu hĩi trñc nghiç m chia theo 25 chç Á (CPU, RAM, GR
31 | },
32 | {
33 |   icon: 'ðŸ“‘', title: 'Bêng iáo Khiãn',
34 |   content: 'Dashboard dành riêng cho Giáo viên: tío lüp híc, giao bài
35 | },
36 | {
37 |   icon: 'Ÿ', title: 'Thi Cĩ & Chéng Ché',
38 |   content: 'Hç thñng thi trñc tủyn vũ i giám thẽ ẽo (Proctored Exam) c
39 | },
40 | {
41 |   icon: 'ðŸ@', title: 'Tĩnh N ng Khác',
42 |   content: 'Chç ũ 2 ng° ỹi chĩi (Multiplayer): thi l-p ráp nhanh. Mô p
43 | },
44 | {
45 |   icon: '&™p', title: 'Công Nghç Kù Thu-t',
46 |   content: 'Frontend: Next.js 16, React 19, TypeScript, Framer Motion, Tailwind CSS. Backend: Supabase (Post..
47 | },
48 | {
49 |   icon: 'ðŸŒˆ', title: 'Triã N khai',
50 |   content: '100% Web-based, không cĩn cài t. Chĩy trẽn mĩi trĩnh duy
51 | },
52 | ],
53 |
54 | export default function JudgesPage() {
55 |   const [url, setUrl] = useState('');
56 |   const [activeSection, setActiveSection] = useState(0);
57 |
58 |   useEffect(() => {
59 |     setUrl(window.location.origin + '/judges');
60 |   }, []);
61 |
62 |   return (
63 |     <div style={{
64 |       minHeight: '100vh', background: '#050507', color: 'ffff',
65 |       fontFamily: 'var(--font-sans)', overflowX: 'hidden'
66 |     }}>
67 |       {/* Hero */}
68 |       <header style={{
69 |         minHeight: '100vh', display: 'flex', flexDirection: 'column',
70 |         alignItems: 'center', justifyContent: 'center', textAlign: 'center',
71 |         padding: '40px 24px', position: 'relative',
72 |         background: 'radial-gradient(ellipse at center, rgba(0,243,255,0.08) 0%, rgba(5,5,7,1) 70%)'
73 |       }}>
74 |         <motion.div initial={{ opacity: 0, y: 30 }} animate={{ opacity: 1, y: 0 }} transition={{ duration: 0.8...
75 |           <div style={{
76 |             display: 'inline-flex', alignItems: 'center', gap: '8px',
77 |             padding: '8px 20px', background: 'rgba(0,243,255,0.1)',
78 |             borderRadius: '100px', color: '#00d4aa', fontSize: '13px',
79 |             fontWeight: 700, marginBottom: '24px', letterSpacing: '1px'
80 |           }}>
81 |             ðŸŒˆ PC MASTER BUILDER · THPT NGUYỄN CÔNG TRÈ
82 |           </div>
83 |           <h1 style={{
84 |             fontSize: 'clamp(36px, 7vw, 72px)', fontWeight: 900,
85 |             marginBottom: '16px', lineHeight: 1.1, letterSpacing: '-0.04em'
86 |           }}>
87 |             PC Master Builder
88 |           </h1>
89 |           <p style={{
90 |             fontSize: 'clamp(16px, 2.5vw, 24px)', color: '#8899a6',
91 |             maxWidth: '700px', margin: '0 auto 40px', lineHeight: 1.6
92 |           }}>
93 |             NẤn têng giáo đắ l-p ráp máy tính thông minh <br />
94 |             vũ i AI và mô phĩng 2D trñc quan
95 |           </p>
96 |           <div style={{ display: 'flex', gap: '12px', justifyContent: 'center', flexWrap: 'wrap' }}>
97 |             { ['Sáng Tío', 'Giáo Đắ', 'Công Nghç', ' ði Mũi' ].map( tag =>
98 |               <span key={tag} style={{
99 |                 padding: '8px 18px', borderRadius: '100px',
100 |                 background: 'rgba(255,255,255,0.04)',
101 |                 border: '1px solid rgba(255,255,255,0.08)',
102 |                 fontSize: '13px', fontWeight: 600, color: '#8899a6'
103 |               }}>#{tag}</span>
104 |             )}
105 |           </div>
106 |         </motion.div>
107 |
108 |         {/* Floating scroll indicator */}
109 |         <motion.div
110 |           initial={{ opacity: 0 }} animate={{ opacity: 1 }}
111 |           transition={{ delay: 1.5 }}
112 |           style={{ position: 'absolute', bottom: '30px' }}
113 |         >
114 |           <motion.div
115 |             animate={{ y: [0, 8, 0] }}
116 |             transition={{ repeat: Infinity, duration: 2 }}
117 |             style={{ color: '#8899a6', fontSize: '13px' }}
118 |           >
119 |             !“ Cũn xũ Nğ ã xem thẽ m
120 |           </motion.div>
121 |         </motion.div>
122 |       </header>
123 |
124 |       {/* QR Code Section */}
125 |       <section style={{ padding: '80px 24px', textAlign: 'center' }}>
126 |         <motion.div
127 |           initial={{ opacity: 0, scale: 0.9 }}
128 |           whileInView={{ opacity: 1, scale: 1 }}
129 |           viewport={{ once: true }}
130 |           style={{ display: 'inline-flex', flexDirection: 'column', alignItems: 'center', gap: '24px' }}
131 |         >
132 |           <div style={{
133 |             padding: '24px', background: 'ffff', borderRadius: '24px',
134 |             boxShadow: '0 20px 60px rgba(0,243,255,0.15)',
135 |             display: 'inline-flex', position: 'relative'
136 |           }} id="qr-code-container">
137 |             {url && <QRCodeSVG value={url} size={200} level="H" id="qr-code-img" />}
138 |           </div>

```

```

139 | <div style={{ display: 'flex', gap: '12px', flexWrap: 'wrap', justifyContent: 'center' }}>
140 |   <button
141 |     onClick={() => {
142 |       const svg = document.getElementById('qr-code-img');
143 |       if (!svg) return;
144 |       const svgData = new XMLSerializer().serializeToString(svg);
145 |       const canvas = document.createElement('canvas');
146 |       canvas.width = 400; canvas.height = 400;
147 |       const ctx = canvas.getContext('2d');
148 |       const img = new Image();
149 |       img.onload = () => {
150 |         ctx.fillStyle = '#fff'; ctx.fillRect(0, 0, 400, 400);
151 |         ctx.drawImage(img, 0, 0, 400, 400);
152 |         const a = document.createElement('a');
153 |         a.download = 'pc-master-judges-qr.png';
154 |         a.href = canvas.toDataURL('image/png');
155 |         a.click();
156 |       };
157 |       img.src = 'data:image/svg+xml;base64,' + btoa(unescape(encodeURIComponent(svgData)));
158 |     }
159 |     style={{
160 |       padding: '12px 24px', background: 'var(--brand-primary)', color: '#fff',
161 |       border: 'none', borderRadius: '10px', fontWeight: 700, fontSize: '14px',
162 |       cursor: 'pointer', fontFamily: 'inherit'
163 |     }}
164 |   >
165 |     0=0Å T £ i Q R C o d e
166 |   </button>
167 |   <button
168 |     onClick={() => {
169 |       const text = `PC Master Builder - NÃn t£ng gio dc l`p
170 |       navigator.clipboard?.writeText(url);
171 |       alert(`  c o p y l i n k ! ` );
172 |     }
173 |     style={{
174 |       padding: '12px 24px', background: 'rgba(255,255,255,0.05)', color: '#fff',
175 |       border: '1px solid rgba(255,255,255,0.15)', borderRadius: '10px', fontWeight: 600, fontSize: '...
176 |       cursor: 'pointer', fontFamily: 'inherit'
177 |     }}
178 |   >
179 |     0=0Ë C o p y L i n k
180 | </button>
181 | </div>
182 | <p style={{ color: '#8899a6', fontSize: '14px', maxWidth: '400px' }}>
183 |   Qu t m  Q R  x e m t r a n g g i  i t h i  u d   n t r  n t h i  t b  d i
184 | </p>
185 | <div style={{
186 |   padding: '12px 24px', background: 'rgba(255,255,255,0.03)',
187 |   borderRadius: '12px', border: '1px solid rgba(255,255,255,0.08)',
188 |   fontSize: '14px', color: 'var(--brand-primary)', fontWeight: 600,
189 |   fontFamily: 'monospace'
190 | }}>
191 |   {url}
192 | </div>
193 | </motion.div>
194 | </section>
195 |
196 | /* Features Grid */
197 | <section style={{ padding: '60px 24px 100px', maxWidth: '1100px', margin: '0 auto' }}>
198 |   <motion.h2
199 |     initial={{ opacity: 0, y: 20 }}
200 |     whileInView={{ opacity: 1, y: 0 }}
201 |     viewport={{ once: true }}
202 |     style={{ fontSize: '36px', fontWeight: 900, textAlign: 'center', marginBottom: '60px', letterSpacing...
203 |   >
204 |     G i  i T h i  u D   n
205 | </motion.h2>
206 |
207 | <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
208 |   {sections.map((section, i) => (
209 |     <motion.div
210 |       key={i}
211 |       initial={{ opacity: 0, y: 20 }}
212 |       whileInView={{ opacity: 1, y: 0 }}
213 |       viewport={{ once: true, margin: '-50px' }}
214 |       transition={{ delay: i * 0.05 }}
215 |       onClick={() => setActiveSection(activeSection === i ? -1 : i)}
216 |       style={{
217 |         padding: '24px 28px',
218 |         background: activeSection === i ? 'rgba(0,243,255,0.04)' : 'rgba(255,255,255,0.02)',
219 |         border: '1px solid ${activeSection === i ? 'rgba(0,243,255,0.2)' : 'rgba(255,255,255,0.06)'}',
220 |         borderRadius: '16px', cursor: 'pointer',
221 |         transition: 'all 0.3s ease'
222 |       }}
223 |     >
224 |       <div style={{ display: 'flex', alignItems: 'center', gap: '16px' }}>
225 |         <span style={{ fontSize: '28px' }}>{section.icon}</span>
226 |         <h3 style={{ margin: 0, fontSize: '18px', fontWeight: 700 }}>{section.title}</h3>
227 |         <span style={{ marginLeft: 'auto', color: '#8899a6', fontSize: '13px', transition: 'transform ...
228 |           %4
229 |       </span>
230 |     </div>
231 |     {activeSection === i && (
232 |       <motion.p
233 |         initial={{ opacity: 0, height: 0 }}
234 |         animate={{ opacity: 1, height: 'auto' }}
235 |         style={{ color: '#8899a6', fontSize: '15px', lineHeight: 1.7, margin: '16px 0 0 44px' }}
236 |       >
237 |         {section.content}
238 |       </motion.p>
239 |     )}
240 |   </motion.div>
241 | ))}
242 | </div>
243 | </section>
244 |
245 | /* Stats */
246 | <section style={{ padding: '80px 24px', background: 'rgba(255,255,255,0.01)' }}>
247 |   <div style={{ maxWidth: '1000px', margin: '0 auto', display: 'grid', gridTemplateColumns: 'repeat(auto...
248 |     [
249 |       {
250 |         v a l u e : ' 5 0 0 0 + ', l a b e l : ' H  c s i n h ', d e s c : '  n g s  d  n g n  n
251 |         v a l u e : ' 5 0 + ', l a b e l : ' L i n h k i  n ', d e s c : ' C h  t i  t s y   t
252 |         v a l u e : ' 2 5 0 + ', l a b e l : ' C  u h  i ', d e s c : ' T r  c n g h i  m 2 5   
253 |         v a l u e : ' 1 0 0 % ', l a b e l : ' W e b - b a s e d ', d e s c : ' K h  n g c  n   i
254 |     ].map((stat, i) => (
255 |       <motion.div
256 |         key={i}
257 |         initial={{ opacity: 0, y: 20 }}
258 |         whileInView={{ opacity: 1, y: 0 }}
259 |         viewport={{ once: true }}
260 |         transition={{ delay: i * 0.1 }}
261 |         style={{
262 |           textAlign: 'center', padding: '32px 20px',
263 |           background: 'rgba(255,255,255,0.02)',

```

```

263|         border: '1px solid rgba(255,255,255,0.06)',
264|         borderRadius: '16px'
265|     }}
266|     >
267|     <div style={{ fontSize: '36px', fontWeight: 900, color: 'var(--brand-primary)', marginBottom: '8...
268|     <div style={{ fontSize: '15px', fontWeight: 700, marginBottom: '4px' }}>{stat.label}</div>
269|     <div style={{ fontSize: '13px', color: '#8899a6' }}>{stat.desc}</div>
270|     </motion.div>
271|     )}}
272| </div>
273| </section>
274|
275| /* Team */
276| <section style={{ padding: '80px 24px', textAlign: 'center' }}>
277|   <h2 style={{ fontSize: '32px', fontWeight: 900, marginBottom: '40px', letterSpacing: '-0.02em' }}>
278|     Ủ i  N g i  < s p a n  s t y l e = { {   c o l o r : ' v a r ( - - b r a n d - p r i m a r y ) '   } } > P h á t  T
279|   </h2>
280|   <div style={{ display: 'flex', justifyContent: 'center', gap: '40px', flexWrap: 'wrap' }}>
281|     {{
282|       {
283|         {
284|           {
285|             { name: 'Ngô Minh Khang', role: 'Thành viên' },
286|             .map((m, i) => (
287|               <motion.div
288|                 key={i}
289|                 initial={{ opacity: 0, y: 20 }}
290|                 whileInView={{ opacity: 1, y: 0 }}
291|                 viewport={{ once: true }}
292|                 style={{ textAlign: 'center' }}
293|               >
294|                 <div style={{
295|                   width: '120px', height: '120px', borderRadius: '60px',
296|                   background: 'linear-gradient(135deg, rgba(0,212,170,0.2), rgba(40,156,249,0.2))',
297|                   border: '2px solid rgba(0,243,255,0.2)',
298|                   margin: '0 auto 16px', display: 'flex', alignItems: 'center', justifyContent: 'center',
299|                   fontSize: '40px'
300|                 }}>
301|                   { [ '0=ÜQ ' , ' &i ' , ' 0=Ü» ' , ' 0=p€ ' ] [ i ] | | ' +P ' }
302|                 </div>
303|                 <h3 style={{ fontSize: '18px', fontWeight: 700, margin: '0 0 4px' }}>{m.name}</h3>
304|                 <p style={{ color: '#8899a6', fontSize: '13px', margin: 0 }}>{m.role}</p>
305|               </motion.div>
306|             )}}
307|           </div>
308|         </section>
309|
310|         /* Footer */
311|         <footer style={{ padding: '40px 24px', textAlign: 'center', color: 'rgba(255,255,255,0.2)', fontSize: '1...
312|         <p>P C  M a s t e r  B u i l d e r  .  T r  °  ÿ n g  T H P T  N g u y  ã n  C  ò n g  T r  é  .  9 7  Q u a n g  T r  á
313|         <p style={{ marginTop: '8px' }}>© 2026 PC Master Builder. All rights reserved.</p>
314|       </footer>
315|     </div>
316|   );
317| }
318|

```

File: app/layout.js (64 lines, 2.3 KB)

```

1| import './globals.css';
2| import { Analytics } from '@vercel/analytics/react';
3| import Script from 'next/script';
4| import { Montserrat } from 'next/font/google';
5|
6| const montserrat = Montserrat({
7|   subsets: ['latin', 'vietnamese'],
8|   variable: '--font-montserrat',
9|   weight: ['300', '400', '500', '600', '700', '800', '900'],
10| });
11|
12| export const metadata = {
13|   metadataBase: new URL(process.env.NEXT_PUBLIC_SITE_URL || 'https://pc-master-lms.vercel.app'),
14|   title: 'P C  M a s t e r  B u i l d e r  .  T r  °  ÿ n g  T H P T  N g u y  ã n  C  ò n g  T r  é  .  9 7  Q u a n g  T r  á
15|   description: ' ò n g  ð  ã n g  h í c  t - p  t í n  h í c  v à  l  ã  p  r á p  m á y  t í n h  v ù i  c  ò n g  n
16|   icons: {
17|     icon: [
18|       { url: '/logo.png', type: 'image/png' },
19|     ],
20|     apple: '/logo.png',
21|   },
22|   openGraph: {
23|     title: 'PC Master Builder',
24|     description: ' ò n g  ð  ã n g  h í c  t - p  t í n  h í c  v à  l  ã  p  r á p  m á y  t í n h  v ù i  c  ò n g
25|     images: ['/logo.png'],
26|   }
27| };
28|
29| import SessionManager from '@components/SessionManager';
30| import { GuruProvider } from '@lib/guru-state';
31| import AIGuruGlobal from '@components/AIGuruGlobal';
32| import { SessionTimeProvider } from '@lib/session-time';
33| import RealtimeWrapper from '@components/RealtimeWrapper';
34|
35| export default function RootLayout({ children }) {
36|   return (
37|     <html lang="vi" className={` ${montserrat.variable}`}>
38|       <head>
39|         <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/driver.js@1.0.1/dist/driver.css" />
40|         <script dangerouslySetInnerHTML={{ __html: `
41|           try {
42|             var t = localStorage.getItem('theme');
43|             if (t === 'dark') document.documentElement.setAttribute('data-theme', 'dark');
44|             else document.documentElement.setAttribute('data-theme', 'light');
45|           } catch(e) {}
46|         `}} />
47|       </head>
48|       <body className="font-sans">
49|         <SessionManager />
50|         <GuruProvider>
51|           <SessionTimeProvider>
52|             <RealtimeWrapper>
53|               {children}
54|             </RealtimeWrapper>
55|             <AIGuruGlobal />
56|           </SessionTimeProvider>
57|         </GuruProvider>
58|         <Analytics />
59|         <Script src="https://cdn.jsdelivr.net/npm/driver.js@1.0.1/dist/driver.js.iife.js" strategy="afterInter...
60|       </body>
61|     </html>
62|   );
63| }
64|

```



```

121 |         </div>
122 |         <div className="mb-4">
123 |             <p className={`font-bold ${isFirst ? 'text-xl' : 'text-lg'}}` style={{ color: 'var(--text-pr...
124 |                 {levelInfo && (
125 |                     <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '6px',...
126 |                         <span style={{ fontSize: '18px' }}>{levelInfo.level.icon}</span>
127 |                         <span style={{ fontSize: '12px', color: levelInfo.level.color_hex, fontWeight: 700 }}>Le...
128 |                         <span style={{ fontSize: '12px', color: 'var(--text-muted)', fontWeight: 600 }}>{player?...
129 |                     </div>
130 |                 )}
131 |             </div>
132 |             <div className={`w-full py-${isFirst ? '6' : '4'} rounded-2xl border font-black text-${isFirst...
133 |                 style={{
134 |                     background: isFirst ? 'var(--accent-amber)' : 'color-mix(in srgb, ${medalColors[idx]} 10%,...
135 |                     color: isFirst ? 'var(--bg-base)' : medalColors[idx],
136 |                     borderColor: isFirst ? 'transparent' : 'color-mix(in srgb, ${medalColors[idx]} 20%, transp...
137 |                 }}>
138 |                 {rank}
139 |             </div>
140 |         </motion.div>
141 |     )}
142 | }}
143 | </div>
144 |
145 | /* Filters */
146 | <div className="flex gap-4 mb-8">
147 |     <button onClick={() => setActiveTab('global')} className={`px-6 py-3 rounded-xl font-bold text-sm tr...
148 |         style={{
149 |             background: activeTab === 'global' ? 'var(--brand-primary)' : 'var(--bg-surface)',
150 |             color: activeTab === 'global' ? 'var(--bg-base)' : 'var(--text-muted)'
151 |         }} > T O A N C I U < / b u t t o n >
152 |
153 |     <button onClick={() => setActiveTab('class')} className={`px-6 py-3 rounded-xl font-bold text-sm tra...
154 |         style={{
155 |             background: activeTab === 'class' ? 'var(--brand-primary)' : 'var(--bg-surface)',
156 |             color: activeTab === 'class' ? 'var(--bg-base)' : 'var(--text-muted)'
157 |         }} > L U P H I C < / b u t t o n >
158 | </div>
159 |
160 | /* List */
161 | <div className="rounded-[32px] overflow-hidden" style={{ background: 'var(--bg-surface)', border: '1px...
162 |     {otherPlayers.map((p, i) => {
163 |         const levelInfo = getLevelInfo(p.xp || 0)
164 |         return (
165 |             <div key={p.id} className={`flex items-center justify-between p-6 transition-all ${p.id === curr...
166 |                 style={{
167 |                     borderBottom: '1px solid var(--border-default)',
168 |                     background: p.id === currentUser?.id ? 'rgba(0,212,170,0.03)' : 'transparent',
169 |                 }}>
170 |                 <div className="flex items-center gap-6">
171 |                     <span className="text-xl font-black w-8" style={{ color: 'var(--text-muted)' }}>{i + 4}</span>
172 |                     <div className="flex items-center gap-4">
173 |                         <div className="w-12 h-12 rounded-2xl flex items-center justify-center font-bold text-lg" ...
174 |                             {p.full_name?.charAt(0)}
175 |                         </div>
176 |                         <div>
177 |                             <p className="font-bold" style={{ color: 'var(--text-primary)' }}>{p.full_name}</p>
178 |                             <div className="flex items-center gap-2">
179 |                                 <span style={{ fontSize: '14px' }}>{levelInfo.level.icon}</span>
180 |                                 <span className="text-[10px] font-bold uppercase tracking-widest" style={{ color: leve...
181 |                                     Level {levelInfo.level.level} - {levelInfo.level.title}
182 |                                 </span>
183 |                             </div>
184 |                         </div>
185 |                     </div>
186 |                     <div className="flex items-center gap-8">
187 |                         <div className="text-right">
188 |                             <p className="font-black text-lg" style={{ color: 'var(--text-primary)' }}>{p.xp}</p>
189 |                             <p className="text-[10px] font-bold uppercase tracking-widest text-right" style={{ color: ...
190 |                                 </div>
191 |                             <div className="p-3 rounded-xl" style={{ background: 'var(--bg-base)', color: 'var(--brand-p...
192 |                                 <Shield size={18} />
193 |                             </div>
194 |                         </div>
195 |                     </div>
196 |                 )
197 |             </div>
198 |         </div>
199 |
200 |         /* My Rank */
201 |         {currentUser && currentRank > 0 && (
202 |             <div className="mt-12 p-6 rounded-3xl flex items-center justify-between" style={{ background: 'var(-...
203 |                 <div className="flex items-center gap-4">
204 |                     <span className="text-2xl font-black">#{currentRank}</span>
205 |                 </div>
206 |                 <p className="font-black leading-none">Hing cça bin</p>
207 |                 <p className="text-xs font-bold opacity-70">
208 |                     {currentRank <= 3 ? '0&A B in ang trong Top 3!' : currentRank}
209 |                 </p>
210 |             </div>
211 |             <div>
212 |                 <Link href="/student/level" style={{ padding: '10px 24px', borderRadius: '12px', fontWeight: 800, ...
213 |                     X E M C o P 0
214 |                 </Link>
215 |             </div>
216 |         )}
217 |         {(!currentUser || currentRank === 0) && (
218 |             <div className="mt-12 p-6 rounded-3xl flex items-center justify-between" style={{ background: 'rgba(...
219 |                 <div className="flex items-center gap-4">
220 |                     <Zap size={24} color="var(--brand-primary)" />
221 |                 </div>
222 |                 <p className="font-black leading-none">ng nh - p Å tha n
223 |                 <p className="text-xs font-bold opacity-70">B t s u h i c
224 |             </div>
225 |             </div>
226 |             <Link href="/login" style={{ padding: '10px 24px', borderRadius: '12px', fontWeight: 800, fontSize:...
227 |                 N G N H - P
228 |             </Link>
229 |             </div>
230 |         )}
231 |     </main>
232 | </div>
233 | )
234 | }
235 |

```

File: app/lessons/[id]/page.js (184 lines, 10.8 KB)

```

1 | 'use client';
2 |
3 | import React, { useEffect, useState, use } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import {

```

```

6 | BookOpen, Video, FileText, Image as ImageIcon, FileSearch, Code,
7 | ChevronRight, Loader2, ArrowLeft, Book, Maximize2, X
8 | } from 'lucide-react';
9 | import Link from 'next/link';
10 | import ReactMarkdown from 'react-markdown';
11 | import PdfViewer from '@components/PdfViewer';
12 |
13 | export default function StudentLessonPage({ params }) {
14 |   const resolvedParams = use(params);
15 |   const lessonId = resolvedParams.id;
16 |
17 |   const [lesson, setLesson] = useState(null);
18 |   const [sections, setSections] = useState([]);
19 |   const [books, setBooks] = useState([]);
20 |   const [loading, setLoading] = useState(true);
21 |   const [selectedBook, setSelectedBook] = useState(null);
22 |
23 |   useEffect(() => {
24 |     const fetchLesson = async () => {
25 |       const { data: lessonData } = await supabase.from('lessons').select('*').eq('id', lessonId).single();
26 |       const { data: sectionData } = await supabase.from('lesson_sections').select('*').eq('lesson_id', lessonId);
27 |       const { data: bookData } = await supabase.from('lesson_books').select('*').eq('lesson_id', lessonId);
28 |
29 |       setLesson(lessonData);
30 |       setSections(sectionData || []);
31 |       setBooks(bookData || []);
32 |       setLoading(false);
33 |     };
34 |     fetchLesson();
35 |   }, [lessonId]);
36 |
37 |   const getYouTubeEmbed = (url) => {
38 |     if (!url) return '';
39 |     const id = url.match(/(?:youtube\.com\/(?:\w+\/\w+\/(?:v|e(?:mbed)?)/|.*[?&]v=)|youtu\.be\/)([^\&?...]
40 |     return id ? `https://www.youtube.com/embed/${id}` : '';
41 |   };
42 |
43 |   const getDriveEmbed = (url) => {
44 |     if (!url) return '';
45 |     const id = url.match(/\/d\/(.+?)\/(?:view|preview)??.[1] || url.match(/id=(.+?)(&|$)/)?.[1];
46 |     return id ? `https://drive.google.com/file/d/${id}/preview` : '';
47 |   };
48 |
49 |   if (loading) return (
50 |     <div style={{ background: '#050a14', minHeight: '100vh', display: 'flex', alignItems: 'center', justif...
51 |       <Loader2 className="animate-spin" color="#00f3ff" size={48} />
52 |     </div>
53 |   );
54 |
55 |   return (
56 |     <div style={{ background: '#050a14', minHeight: '100vh', color: '#e0e6ed', display: 'flex' }}>
57 |       {/* Sidebar Navigation */}
58 |       <aside style={{ width: '320px', background: '#0a0f1a', borderRight: '1px solid rgba(255,255,255,0....
59 |         <Link href="/builder" style={{ display: 'flex', alignItems: 'center', gap: '8px', color: '#889...
60 |           <ArrowLeft size={16} /> Quay lii trang ch c
61 |         </Link>
62 |         <div style={{ marginBottom: '32px' }}>
63 |           <h4 style={{ color: '#4b5563', fontSize: '11px', textTransform: 'uppercase', fontWeight: 8...
64 |             <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
65 |               {sections.map((s, idx) => (
66 |                 <a key={s.id} href={`#section-${s.id}`} style={{
67 |                   display: 'flex', alignItems: 'center', gap: '12px', padding: '12px', borderRad...
68 |                 }}
69 |                 onMouseEnter={e => e.currentTarget.style.borderColor = '#00f3ff'}
70 |                 onMouseLeave={e => e.currentTarget.style.borderColor = 'transparent'}
71 |                 >
72 |                   <span style={{ color: '#00f3ff', fontWeight: 800, fontSize: '12px', width: '20...
73 |                     {s.title}
74 |                   </span>
75 |                 </div>
76 |               </div>
77 |             </div>
78 |           </aside>
79 |
80 |           {/* Main Content Area */}
81 |           <main style={{ flex: 1, padding: '64px', maxWidth: '900px', margin: '0 auto' }}>
82 |             <header style={{ marginBottom: '64px' }}>
83 |               <h1 style={{ fontSize: '42px', fontWeight: 900, marginBottom: '24px', lineHeight: 1.1 }}>{...
84 |               <p style={{ fontSize: '18px', color: '#8899a6', lineHeight: 1.6 }}>{lesson.description}</p>
85 |
86 |               {lesson.source_name && (
87 |                 <div style={{ marginTop: '16px', display: 'flex', alignItems: 'center', gap: '8px', fo...
88 |                   <span> N g u o n t à i l i c u : </span>
89 |                   {lesson.source_url ? (
90 |                     <a href={lesson.source_url} target="_blank" style={{ color: 'var(--brand-prima...
91 |                     ) : (
92 |                       <span style={{ fontWeight: 600 }}>{lesson.source_name}</span>
93 |                     </div>
94 |                   </div>
95 |                 </div>
96 |
97 |                 <div style={{ height: '4px', width: '80px', background: '#00f3ff', marginTop: '32px', bord...
98 |               </header>
99 |
100 |               <div style={{ display: 'flex', flexDirection: 'column', gap: '80px' }}>
101 |                 {sections.map((s, idx) => (
102 |                   <section key={s.id} id={`section-${s.id}`}>
103 |                     <div style={{ display: 'flex', alignItems: 'center', gap: '12px', marginBottom: '2...
104 |                       <div style={{ padding: '8px', background: 'rgba(0, 243, 255, 0.1)', color: '#0...
105 |                         {s.type === 'video' && <Video size={20}/>}
106 |                         {s.type === 'text' && <FileText size={20}/>}
107 |                         {s.type === 'image' && <ImageIcon size={20}/>}
108 |                         {s.type === 'pdf' && <FileSearch size={20}/>}
109 |                         {s.type === 'embed' && <Code size={20}/>}
110 |                       </div>
111 |                       <h2 style={{ fontSize: '24px', fontWeight: 800, margin: 0 }}>{s.title}</h2>
112 |                     </div>
113 |
114 |                     {s.type === 'video' && (
115 |                       <div style={{ width: '100%', aspectRatio: '16/9', borderRadius: '24px', overfl...
116 |                         <iframe src={getYouTubeEmbed(s.content)} style={{ width: '100%', height: '...
117 |                       </div>
118 |                     )}
119 |
120 |                     {s.type === 'text' && (
121 |                       <div style={{ fontSize: '18px', lineHeight: 1.8, color: '#ced4da' }} className...
122 |                       <ReactMarkdown>{s.content}</ReactMarkdown>
123 |                     </div>
124 |                     </div>
125 |
126 |                     {s.type === 'image' && (
127 |                       <div style={{ borderRadius: '24px', overflow: 'hidden' }}>
128 |                         <img src={s.content} style={{ width: '100%', display: 'block' }} />
129 |                       </div>

```



```

130 |         })
131 |     }
132 |     {s.type === 'pdf' && (
133 |         <PdfViewer url={s.content} title={s.title} />
134 |     )}
135 |     {s.type === 'embed' && s.content && (
136 |         <div style={{ borderRadius: '20px', overflow: 'hidden', border: '1px solid rgb...
137 |         <div dangerouslySetInnerHTML={{ __html: s.content }} />
138 |     )}
139 |     </section>
140 |   </div>
141 | )}
142 | </div>
143 |
144 | /* Sách liên quan */
145 | {books.length > 0 && (
146 |   <div style={{ marginTop: '120px', padding: '64px 0', borderTop: '1px solid rgba(255,255,25...
147 |   <h3 style={{ fontSize: '28px', fontWeight: 800, marginBottom: '32px', display: 'flex',...
148 |     < B o o k   s i z e = { 2 8 }   c o l o r = "# 0 0 f 3 f f "   / >   S á c h   &   T á
149 |   </h3>
150 |   <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill, minmax(200px, 1...
151 |     {books.map(book => (
152 |       <div
153 |         key={book.id}
154 |         onClick={() => setSelectedBook(book)}
155 |         style={{ background: 'rgba(255,255,255,0.02)', borderRadius: '16px', paddi...
156 |         onMouseOver={e => e.currentTarget.style.borderColor = '#00f3ff'}
157 |         onMouseOut={e => e.currentTarget.style.borderColor = '#1e293b'}
158 |       >
159 |         <img src={book.cover_image_url} style={{ width: '100%', aspectRatio: '3/4'...
160 |         <h4 style={{ margin: 0, fontSize: '15px', fontWeight: 700, textAlign: 'cen...
161 |       </div>
162 |     )}
163 |   </div>
164 | )}
165 | </div>
166 | )}
167 | </main>
168 |
169 | /* Modal xem sách */
170 | {selectedBook && (
171 |   <div style={{ position: 'fixed', top: 0, left: 0, right: 0, bottom: 0, background: 'rgba(0,0,0...
172 |   <div style={{ padding: '16px 24px', display: 'flex', justifyContent: 'space-between', alig...
173 |     <h3 style={{ margin: 0, fontSize: '18px' }}>{selectedBook.title}</h3>
174 |     <button onClick={() => setSelectedBook(null)} style={{ background: 'none', border: 'no...
175 |   </div>
176 |   <div style={{ flex: 1, overflow: 'auto' }}>
177 |     <PdfViewer url={selectedBook.drive_embed_url} title={selectedBook.title} />
178 |   </div>
179 | </div>
180 | )}
181 | </div>
182 | );
183 | }
184 |

```

File: app/lessons/[id]/QuizInteractive.tsx (217 lines, 9.5 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { motion, AnimatePresence } from 'framer-motion'
5 | import { CheckCircle, XCircle, Award, ArrowRight, Loader2 } from 'lucide-react'
6 | import confetti from 'canvas-confetti'
7 | import { saveLessonProgress } from '@lib/lesson-actions'
8 |
9 | type QuizInteractiveProps = {
10 |   lessonId: string;
11 |   userId: string;
12 |   initialProgress: any;
13 | }
14 |
15 | // Dummy data for Mega Prompt requirement
16 | const QUIZ_DATA = [
17 |   {
18 |     id: 1,
19 |     question: "CPU óng vai trò gì trong máy tính?",
20 |     options: [
21 |       { id: 'A', text: "Lòu trídí liÇu dàihìn" },
22 |       { id: 'B', text: "Xí lý các lÇnh và tính toán (BÙ nã o)" },
23 |       { id: 'C', text: "Cung c¥p nng l°ãng cho toàn hÇ thñng" },
24 |       { id: 'D', text: "HiãnthỂ hìnhiên mànhình" }
25 |     ],
26 |     correctAnswer: 'B',
27 |     explanation: "CPU (Central Processing Unit) làbùvixí lý trung tâm
28 |   },
29 |   {
30 |     id: 2,
31 |     question: "RAM làbùnhữgì?",
32 |     options: [
33 |       { id: 'A', text: "BÙnhữtruy c-png«unhiên (Tim thÝi)" },
34 |       { id: 'B', text: "BÙnhữchẾ íc (CÑ Ënh)" },
35 |       { id: 'C', text: "BÙnhữl°u trídí liÇu tìnthÝi" },
36 |       { id: 'D', text: "BÙnhữó hĩa (VRAM)" }
37 |     ],
38 |     correctAnswer: 'A',
39 |     explanation: "RAM (Random Access Memory) l°u trídí liÇu tìnthÝi
40 |   }
41 | ]
42 |
43 | export default function QuizInteractive({ lessonId, userId, initialProgress }: QuizInteractiveProps) {
44 |   const [currentQuestion, setCurrentQuestion] = useState(0)
45 |   const [selectedAnswer, setSelectedAnswer] = useState<string | null>(null)
46 |   const [isSubmitted, setIsSubmitted] = useState(false)
47 |   const [score, setScore] = useState(0)
48 |   const [isCompleted, setIsCompleted] = useState(initialProgress?.is_completed || false)
49 |   const [isSaving, setIsSaving] = useState(false)
50 |
51 |   const activeQuestion = QUIZ_DATA[currentQuestion]
52 |
53 |   const handleSelect = (optionId: string) => {
54 |     if (!isSubmitted) setSelectedAnswer(optionId)
55 |   }
56 |
57 |   const handleSubmit = async () => {
58 |     if (!selectedAnswer) return
59 |     setIsSubmitted(true)
60 |
61 |     let currentScore = score
62 |     if (selectedAnswer === activeQuestion.correctAnswer) {
63 |       currentScore += 10
64 |       setScore(currentScore)
65 |     }
66 |     // B-n pháohoanhĩnçutrlýiúng

```

```

66 |     confetti({
67 |       particleCount: 50,
68 |       spread: 40,
69 |       origin: { y: 0.8 },
70 |       colors: ['#00d2a0']
71 |     })
72 |   })
73 | }
74 | // N i u l à c â u c u Ñ i c ù n g => H o à n t h à n h b à i h í c
75 | if (currentQuestion === QUIZ_DATA.length - 1) {
76 |   setIsSaving(true)
77 |   // G í i S e r v e r A c t i o n   Ä l ° u k ı t q u e r e a l t i m e v à o D B
78 |   await saveLessonProgress(lessonId, currentScore, true)
79 |   setIsSaving(false)
80 |   setIsCompleted(true)
81 |
82 |   confetti({
83 |     particleCount: 200,
84 |     spread: 100,
85 |     origin: { y: 0.5 },
86 |     colors: ['#00d2a0', '#00b4d8', 'ffffff']
87 |   })
88 | }
89 | }
90 |
91 | const handleNext = () => {
92 |   if (currentQuestion < QUIZ_DATA.length - 1) {
93 |     setCurrentQuestion(prev => prev + 1)
94 |     setSelectedAnswer(null)
95 |     setIsSubmitted(false)
96 |   }
97 | }
98 |
99 | if (isCompleted && currentQuestion === QUIZ_DATA.length - 1 && isSubmitted) {
100 |   return (
101 |     <motion.div
102 |       initial={{ opacity: 0, scale: 0.95 }}
103 |       animate={{ opacity: 1, scale: 1 }}
104 |       className="bg-gradient-to-br from-[#00d2a0]/20 to-[#00b4d8]/20 p-8 md:p-12 rounded-3xl border border-[...
105 |     >
106 |       <Award size={64} className="text-[#00d2a0] mx-auto mb-6" />
107 |       <h2 className="text-3xl font-bold mb-2">Chúc mừng bạn!</h2>
108 |       <p className="text-slate-300 mb-6 text-lg">Bạn đã hoàn thành bài
109 |       <div className="inline-flex items-center gap-2 px-6 py-3 bg-[#00d2a0] text-black font-bold rounded-xl ...
110 |       <CheckCircle size={20} /> ä l ° u t i ı n t r ı n h
111 |     </div>
112 |   </motion.div>
113 | )
114 | }
115 |
116 | return (
117 |   <section className="bg-[#16213e] p-6 md:p-10 rounded-3xl border border-[#1e293b] shadow-xl">
118 |     <div className="flex items-center justify-between mb-8">
119 |       <h2 className="text-2xl font-bold flex items-center gap-3">
120 |         <span className="w-10 h-10 rounded-xl bg-[#00b4d8]/10 text-[#00b4d8] flex items-center justify-cen...
121 |         K i Ä m t r a k i ı n t h é c
122 |       </h2>
123 |       <div className="text-sm font-semibold text-slate-400 bg-[#0f0f1a] px-4 py-2 rounded-full border border...
124 |       Câu {currentQuestion + 1} / {QUIZ_DATA.length}
125 |     </div>
126 |   </div>
127 |
128 |   <AnimatePresence mode="wait">
129 |     <motion.div
130 |       key={currentQuestion}
131 |       initial={{ opacity: 0, x: 20 }}
132 |       animate={{ opacity: 1, x: 0 }}
133 |       exit={{ opacity: 0, x: -20 }}
134 |       transition={{ duration: 0.3 }}
135 |     >
136 |       <h3 className="text-xl md:text-2xl font-semibold mb-8 leading-snug">{activeQuestion.question}</h3>
137 |
138 |       <div className="flex flex-col gap-3 mb-8">
139 |         {activeQuestion.options.map(opt => {
140 |           const isSelected = selectedAnswer === opt.id
141 |           const isCorrect = opt.id === activeQuestion.correctAnswer
142 |
143 |           let stateClass = "border-[#2a3655] bg-[#0f0f1a] hover:border-[#00d2a0]"
144 |           if (isSelected) stateClass = "border-[#00d2a0] bg-[#00d2a0]/10"
145 |
146 |           if (isSubmitted) {
147 |             if (isCorrect) stateClass = "border-[#00d2a0] bg-[#00d2a0]/20 shadow-[0_0_15px_rgba(0,210,160,...
148 |             else if (isSelected && !isCorrect) stateClass = "border-red-500 bg-red-500/10"
149 |             else stateClass = "border-[#2a3655] bg-[#0f0f1a] opacity-50"
150 |           }
151 |
152 |           return (
153 |             <button
154 |               key={opt.id}
155 |               onClick={() => handleSelect(opt.id)}
156 |               disabled={isSubmitted}
157 |               className={`flex items-center gap-4 p-5 rounded-2xl border-2 transition-all duration-200 tex...
158 |             >
159 |               <div className={`w-8 h-8 rounded-full border-2 flex items-center justify-center font-bold te...
160 |                 ${isSubmitted && isCorrect ? 'bg-[#00d2a0] border-[#00d2a0] text-black' :
161 |                 (isSubmitted && isSelected && !isCorrect ? 'bg-red-500 border-red-500 text-white' :
162 |                 (isSelected ? 'border-[#00d2a0] text-[#00d2a0]' : 'border-slate-500 text-slate-400'))}
163 |               >
164 |                 {isSubmitted && isCorrect ? <CheckCircle size={16} /> : (isSubmitted && isSelected && !isC...
165 |               </div>
166 |               <span className="flex-1 font-medium text-lg">{opt.text}</span>
167 |             </button>
168 |           )
169 |         }>
170 |       </div>
171 |
172 |       {isSubmitted && (
173 |         <motion.div
174 |           initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }}
175 |           className={`p-6 rounded-2xl mb-8 ${selectedAnswer === activeQuestion.correctAnswer ? 'bg-[#00d2a...
176 |         >
177 |           <h4 className={`font-bold mb-2 flex items-center gap-2 ${selectedAnswer === activeQuestion.corre...
178 |             {selectedAnswer === activeQuestion.correctAnswer ? <CheckCircle size={20} /> : <XCircle size={...
179 |             { s e l e c t e d A n s w e r = = = a c t i v e Q u e s t i o n . c o r r e c t A n s w e r ? ' C h i r
180 |           </h4>
181 |           <p className="text-slate-300 leading-relaxed">{activeQuestion.explanation}</p>
182 |         </motion.div>
183 |       )}
184 |
185 |       <div className="flex justify-end">
186 |         {!isSubmitted ? (
187 |           <button
188 |             onClick={handleSubmit}
189 |             disabled={!selectedAnswer}

```

```

190 |         className="px-8 py-3.5 bg-white text-black font-bold rounded-xl hover:bg-slate-200 transition-...
191 |     >
192 |     K i Ā m t r a á p á n
193 | </button>
194 | ) : (
195 |     currentQuestion < QUIZ_DATA.length - 1 ? (
196 |         <button
197 |             onClick={handleNext}
198 |             className="px-8 py-3.5 bg-[#00d2a0] text-black font-bold rounded-xl hover:bg-[#00e6af] trans...
199 |         >
200 |             C â u t i ı p t h e o < A r r o w R i g h t   s i z e = { 2 0 }   / >
201 |         </button>
202 |     ) : (
203 |         <button
204 |             disabled={isSaving}
205 |             className="px-8 py-3.5 bg-[#00d2a0] text-black font-bold rounded-xl transition-colors flex i...
206 |         >
207 |             { i s S a v i n g   ?   < L o a d e r 2   c l a s s N a m e = " a n i m a t e - s p i n "   s i z e = { 2 0 }
208 |         </button>
209 |     )
210 |   )}
211 | </div>
212 | </motion.div>
213 | </AnimatePresence>
214 | </section>
215 | )
216 | }
217 |

```

File: app/lessons/lessons-theme.css (282 lines, 5.5 KB)

```

1 | .lessons-page {
2 |   min-height: 100vh;
3 |   background: var(--bg-base);
4 |   color: var(--text-primary);
5 |   animation: fadeIn 0.5s ease-out;
6 | }
7 |
8 | .lessons-container {
9 |   max-width: 1280px;
10 |   margin: 0 auto;
11 |   padding: 3rem 1.5rem 5rem;
12 | }
13 |
14 | @keyframes fadeIn {
15 |   from { opacity: 0; transform: translateY(10px); }
16 |   to { opacity: 1; transform: translateY(0); }
17 | }
18 |
19 | .lessons-header {
20 |   display: flex;
21 |   flex-direction: column;
22 |   gap: 1.5rem;
23 |   margin-bottom: 4rem;
24 | }
25 |
26 | @media (min-width: 768px) {
27 |   .lessons-header {
28 |     flex-direction: row;
29 |     align-items: flex-end;
30 |     justify-content: space-between;
31 |   }
32 | }
33 |
34 | .lessons-back-btn {
35 |   width: 40px;
36 |   height: 40px;
37 |   border-radius: 12px;
38 |   display: flex;
39 |   align-items: center;
40 |   justify-content: center;
41 |   background: var(--bg-elevated);
42 |   color: var(--text-secondary);
43 |   transition: all 0.2s;
44 | }
45 |
46 | .lessons-back-btn:hover {
47 |   color: var(--text-primary);
48 |   background: var(--bg-hover);
49 | }
50 |
51 | .lessons-title {
52 |   font-size: 2.25rem;
53 |   font-weight: 900;
54 |   letter-spacing: -0.025em;
55 | }
56 |
57 | @media (min-width: 768px) {
58 |   .lessons-title {
59 |     font-size: 3.75rem;
60 |   }
61 | }
62 |
63 | .lessons-title-accent {
64 |   color: var(--brand-primary);
65 | }
66 |
67 | .lessons-subtitle {
68 |   font-size: 1.125rem;
69 |   max-width: 42rem;
70 |   color: var(--text-secondary);
71 | }
72 |
73 | .lessons-stats-card {
74 |   display: flex;
75 |   align-items: center;
76 |   gap: 1rem;
77 |   padding: 1rem;
78 |   border-radius: 16px;
79 |   border: 1px solid var(--border-default);
80 |   background: var(--bg-surface);
81 | }
82 |
83 | .lessons-stats-icon {
84 |   width: 48px;
85 |   height: 48px;
86 |   border-radius: 12px;
87 |   display: flex;
88 |   align-items: center;
89 |   justify-content: center;
90 |   background: color-mix(in srgb, var(--brand-primary) 20%, transparent);
91 |   color: var(--brand-primary);
92 | }

```

```

93 | .lessons-stats-label {
94 |   font-size: 0.875rem;
95 |   font-weight: 700;
96 |   text-transform: uppercase;
97 |   letter-spacing: 0.05em;
98 |   color: var(--text-secondary);
99 | }
100 |
101 |
102 | .lessons-stats-value {
103 |   font-size: 1.25rem;
104 |   font-weight: 700;
105 | }
106 |
107 | .lessons-grid {
108 |   display: grid;
109 |   grid-template-columns: 1fr;
110 |   gap: 2rem;
111 |   max-width: 1280px;
112 |   margin: 0 auto;
113 | }
114 |
115 | @media (min-width: 768px) {
116 |   .lessons-grid {
117 |     grid-template-columns: repeat(2, 1fr);
118 |   }
119 | }
120 |
121 | @media (min-width: 1024px) {
122 |   .lessons-grid {
123 |     grid-template-columns: repeat(3, 1fr);
124 |   }
125 | }
126 |
127 | .lessons-card {
128 |   position: relative;
129 |   border-radius: 24px;
130 |   border: 1px solid var(--border-default);
131 |   overflow: hidden;
132 |   transition: all 0.3s ease;
133 |   display: flex;
134 |   flex-direction: column;
135 |   height: 100%;
136 |   text-decoration: none;
137 |   background: var(--bg-surface);
138 |   animation: fadeIn 0.5s ease-out both;
139 | }
140 |
141 | .lessons-card:nth-child(1) { animation-delay: 0.05s; }
142 | .lessons-card:nth-child(2) { animation-delay: 0.1s; }
143 | .lessons-card:nth-child(3) { animation-delay: 0.15s; }
144 | .lessons-card:nth-child(4) { animation-delay: 0.2s; }
145 | .lessons-card:nth-child(5) { animation-delay: 0.25s; }
146 | .lessons-card:nth-child(6) { animation-delay: 0.3s; }
147 |
148 | .lessons-card:hover {
149 |   border-color: var(--brand-primary);
150 |   box-shadow: 0 0 30px color-mix(in srgb, var(--brand-primary) 15%, transparent);
151 |   transform: translateY(-2px);
152 | }
153 |
154 | .lessons-card-cover {
155 |   aspect-ratio: 16 / 9;
156 |   width: 100%;
157 |   position: relative;
158 |   overflow: hidden;
159 |   background: var(--bg-base);
160 | }
161 |
162 | .lessons-card-img {
163 |   width: 100%;
164 |   height: 100%;
165 |   object-fit: cover;
166 |   opacity: 0.6;
167 |   transition: transform 0.5s;
168 | }
169 |
170 | .lessons-card:hover .lessons-card-img {
171 |   transform: scale(1.05);
172 | }
173 |
174 | .lessons-card-gradient {
175 |   position: absolute;
176 |   inset: 0;
177 |   background: linear-gradient(to top, var(--bg-surface), transparent);
178 | }
179 |
180 | .lessons-completed-badge {
181 |   position: absolute;
182 |   top: 1rem;
183 |   right: 1rem;
184 |   padding: 0.5rem;
185 |   border-radius: 50%;
186 |   display: flex;
187 |   align-items: center;
188 |   justify-content: center;
189 |   background: var(--brand-primary);
190 |   color: #fff;
191 |   box-shadow: 0 2px 8px rgba(0,0,0,0.2);
192 | }
193 |
194 | .lessons-card-meta {
195 |   position: absolute;
196 |   bottom: 1rem;
197 |   left: 1rem;
198 |   display: flex;
199 |   align-items: center;
200 |   gap: 0.5rem;
201 | }
202 |
203 | .lessons-meta-chip {
204 |   padding: 0.25rem 0.75rem;
205 |   border-radius: 9999px;
206 |   font-size: 0.75rem;
207 |   font-weight: 700;
208 |   display: flex;
209 |   align-items: center;
210 |   gap: 0.375rem;
211 |   background: rgba(0,0,0,0.5);
212 |   backdrop-filter: blur(4px);
213 |   border: 1px solid rgba(255,255,255,0.1);
214 |   color: #fff;
215 | }
216 |

```

```

217 | .lessons-meta-icon {
218 |   color: var(--brand-primary);
219 | }
220 |
221 | .lessons-card-body {
222 |   padding: 2rem;
223 |   display: flex;
224 |   flex-direction: column;
225 |   flex: 1;
226 | }
227 |
228 | .lessons-card-title {
229 |   font-size: 1.5rem;
230 |   font-weight: 700;
231 |   margin-bottom: 0.75rem;
232 |   line-height: 1.3;
233 |   transition: color 0.2s;
234 | }
235 |
236 | .lessons-card:hover .lessons-card-title {
237 |   color: var(--brand-primary);
238 | }
239 |
240 | .lessons-card-desc {
241 |   font-size: 0.875rem;
242 |   margin-bottom: 2rem;
243 |   line-clamp: 2;
244 |   overflow: hidden;
245 |   display: -webkit-box;
246 |   -webkit-line-clamp: 2;
247 |   -webkit-box-orient: vertical;
248 |   flex: 1;
249 |   color: var(--text-secondary);
250 | }
251 |
252 | .lessons-card-footer {
253 |   display: flex;
254 |   align-items: center;
255 |   justify-content: space-between;
256 |   padding-top: 1.5rem;
257 |   border-top: 1px solid var(--border-default);
258 | }
259 |
260 | .lessons-card-action {
261 |   display: flex;
262 |   align-items: center;
263 |   gap: 0.5rem;
264 |   font-weight: 700;
265 |   font-size: 0.875rem;
266 |   color: var(--brand-primary);
267 | }
268 |
269 | .lessons-card-count {
270 |   font-size: 0.75rem;
271 |   font-weight: 500;
272 |   color: var(--text-secondary);
273 | }
274 |
275 | .lessons-empty {
276 |   grid-column: 1 / -1;
277 |   padding: 5rem 0;
278 |   text-align: center;
279 |   border: 2px dashed var(--border-default);
280 |   border-radius: 24px;
281 | }
282 |

```

File: app/lessons/page.tsx (153 lines, 5.2 KB)

```

1 | import { createClient } from '@lib/supabase-ssr-server'
2 | import { redirect } from 'next/navigation'
3 | import Link from 'next/link'
4 | import { Book, Play, CheckCircle, Clock, Star, ArrowRight, Layers, ArrowLeft } from 'lucide-react'
5 | import './lessons-theme.css'
6 |
7 | export const dynamic = 'force-dynamic'
8 |
9 | export default async function LessonsListPage() {
10 |   const supabase = await createClient()
11 |   const { data: { user } } = await supabase.auth.getUser()
12 |
13 |   if (!user) {
14 |     redirect('/login')
15 |   }
16 |
17 |   const { data: myClasses } = await supabase
18 |     .from('class_members')
19 |     .select('class_id')
20 |     .eq('student_id', user.id)
21 |
22 |   const classIds = (myClasses || []).map(c => c.class_id)
23 |
24 |   let lessonsQuery = supabase.from('lessons').select('')
25 |
26 |   if (classIds.length > 0) {
27 |     const { data: assignedLessonIds } = await supabase
28 |       .from('lesson_class_assignments')
29 |       .select('lesson_id')
30 |       .in('class_id', classIds)
31 |
32 |     const lessonIds = (assignedLessonIds || []).map(a => a.lesson_id)
33 |     if (lessonIds.length > 0) {
34 |       lessonsQuery = lessonsQuery.in('id', lessonIds)
35 |     } else {
36 |       lessonsQuery = lessonsQuery.eq('id', null)
37 |     }
38 |   } else {
39 |     lessonsQuery = lessonsQuery.eq('id', null)
40 |   }
41 |
42 |   const { data: lessons } = await lessonsQuery
43 |     .eq('is_published', true)
44 |     .order('created_at', { ascending: true })
45 |
46 |   const { data: progressList } = await supabase
47 |     .from('lesson_progress')
48 |     .select('')
49 |     .eq('student_id', user.id)
50 |
51 |   const getProgress = (lessonId: string) => {
52 |     return progressList?.find(p => p.lesson_id === lessonId)
53 |   }
54 |

```

```

55 |   return (
56 |     <div className="lessons-page">
57 |       <div className="lessons-container">
58 |         <header className="lessons-header">
59 |           <div>
60 |             <div className="flex items-center gap-4 mb-4">
61 |               <Link href="/student" className="lessons-back-btn">
62 |                 <ArrowLeft size={20} />
63 |               </Link>
64 |               <h1 className="lessons-title">
65 |                 K h á m   p h á   <span className="lessons-title-accent"> K i ỉ n   t h ế c </div>
66 |             </h1>
67 |           </div>
68 |           <p className="lessons-subtitle">
69 |             H ỉ c   t - p   t ể   n h ỉ n g   c h u y ề n   g i a   h à n g   s u   v á   p h ầ n   c ể n g   m á y   t í n h .
70 |           </p>
71 |         </div>
72 |         <div className="lessons-stats-card">
73 |           <div className="lessons-stats-icon">
74 |             <Layers size={24} />
75 |           </div>
76 |           <div>
77 |             <div className="lessons-stats-label"> T i ỉ n   ù   t ồ n g   q u á t </div>
78 |             <div className="lessons-stats-value"> {progressList?.length}
79 |           </div>
80 |         </div>
81 |       </header>
82 |
83 |       <div className="lessons-grid">
84 |         {lessons?.map((lesson) => {
85 |           const progress = getProgress(lesson.id)
86 |           const isCompleted = progress?.is_completed
87 |
88 |           return (
89 |             <Link
90 |               key={lesson.id}
91 |               href={`/${lessons}/${lesson.id}`}
92 |               className="lessons-card"
93 |             >
94 |               <div className="lessons-card-cover">
95 |                 <img
96 |                   src={lesson.thumbnail_url || "https://images.unsplash.com/photo-1591799264318-7e6ef8ddb7ea?q...
97 |                   alt={lesson.title}
98 |                   className="lessons-card-img"
99 |                 />
100 |                 <div className="lessons-card-gradient"></div>
101 |
102 |                 {isCompleted && (
103 |                   <div className="lessons-completed-badg">
104 |                     <CheckCircle size={20} />
105 |                   </div>
106 |                 )}
107 |
108 |                 <div className="lessons-card-meta">
109 |                   <div className="lessons-meta-chip">
110 |                     <Clock size={12} className="lessons-meta-icon" />
111 |                     15 phút
112 |                   </div>
113 |                   <div className="lessons-meta-chip">
114 |                     <Star size={12} className="text-yellow-400" />
115 |                     4.9
116 |                   </div>
117 |                 </div>
118 |               </div>
119 |
120 |               <div className="lessons-card-body">
121 |                 <h3 className="lessons-card-title">
122 |                   {lesson.title}
123 |                 </h3>
124 |                 <p className="lessons-card-desc">
125 |                   {lesson.description}
126 |                 </p>
127 |
128 |                 <div className="lessons-card-footer">
129 |                   <div className="lessons-card-action">
130 |                     <ArrowRight size={16} />
131 |                     { isCompleted ? ' X E M   L   I ' : ' B @ T   ỉ   U   H ỉ c ' }
132 |                   </div>
133 |                   <div className="lessons-card-count">
134 |                     H ỉ n   2 . 4 k   h ỉ c   v i ề n
135 |                   </div>
136 |                 </div>
137 |               </div>
138 |             </Link>
139 |           )
140 |         })}
141 |
142 |         {!!lessons || lessons.length === 0} && (
143 |           <div className="lessons-empty">
144 |             <Book size={48} className="mx-auto mb-4" style={{ color: 'var(--text-secondary)' }} />
145 |             <p style={{ color: 'var(--text-secondary)', fontWeight: 500}}>
146 |               </div>
147 |           </div>
148 |         </div>
149 |       </div>
150 |     </div>
151 |   )
152 | }
153 |

```

File: app/my-exams/page.tsx (203 lines, 11.8 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect } from 'react'
4 | import { motion } from 'framer-motion'
5 | import {
6 |   LineChart, Line, XAxis, YAxis, CartesianGrid, Tooltip, ResponsiveContainer,
7 |   Radar, RadarChart, PolarGrid, PolarAngleAxis, PolarRadiusAxis,
8 |   BarChart, Bar, Legend, Cell
9 | } from 'recharts'
10 | import {
11 |   Trophy, Target, Zap, BookOpen,
12 |   ChevronRight, Download, Filter,
13 |   AlertTriangle, ArrowUpRight, Sparkles
14 | } from 'lucide-react'
15 | import Link from 'next/link'
16 |
17 | const scoreHistory = [
18 |   { date: '01/05', score: 65 },
19 |   { date: '03/05', score: 72 },
20 |   { date: '05/05', score: 68 },
21 |   { date: '07/05', score: 85 },

```

```

22 | { date: '09/05', score: 92 },
23 | ]
24 |
25 | const competencyData = [
26 |   { subject: 'CPU', A: 90, fullMark: 100 },
27 |   { subject: 'RAM', A: 45, fullMark: 100 },
28 |   { subject: 'GPU', A: 80, fullMark: 100 },
29 |   { subject: 'Storage', A: 70, fullMark: 100 },
30 |   { subject: 'PSU', A: 60, fullMark: 100 },
31 |   { subject: 'Mainboard', A: 85, fullMark: 100 },
32 | ]
33 |
34 | const classComparison = [
35 |   { name: 'Cá nhân', score: 92 },
36 |   { name: 'Trung bình lớp', score: 78 },
37 | ]
38 |
39 | export default function StudentAnalyticsPage() {
40 |   return (
41 |     <div className="max-w-7xl mx-auto px-6 py-12 space-y-10 min-h-screen" style={{ background: 'var(--bg-base)...
42 |       <header className="flex flex-col md:flex-row md:items-center justify-between gap-6">
43 |         <div>
44 |           <h1 className="text-4xl font-black mb-2">Phân tích <span style={{ color: 'var(--brand-primary)' }}>N...
45 |             <p className="font-medium italic" style={{ color: 'var(--text-...
46 |           </div>
47 |           <button className="flex items-center gap-2 px-6 py-3 rounded-xl text-sm font-bold transition-all" styl...
48 |             <Download size={18} /> Xuýt báo cáo (PDF)
49 |         </header>
50 |
51 |         <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-6">
52 |           {[
53 |             { label: 'Bàì thi ã làm', value: '12', icon: BookOpen, color: 'var(--brand-primary)',
54 |             { label: 'i ã m trung bình h', value: '78.5', icon: Target, color: 'var(--text-...
55 |             { label: 'i ã m cao nh ẏ t', value: '96', icon: Trophy, color: 'var(--brand-primary)',
56 |             { label: 'T ÷ l Ỗ qua m Ỗ n', value: '85%', icon: Zap, color: 'var(--accent-amber)',
57 |           ].map((card, i) => (
58 |             <motion.div
59 |               key={i}
60 |               initial={{ opacity: 0, y: 20 }}
61 |               animate={{ opacity: 1, y: 0 }}
62 |               transition={{ delay: i * 0.1 }}
63 |               className="p-8 rounded-[32px] relative overflow-hidden group"
64 |               style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)' }}
65 |             >
66 |               <div className="relative z-10">
67 |                 <div className="p-3 w-fit rounded-2xl mb-4 group-hover:scale-110 transition-transform" style={{ ...
68 |                   <card.icon size={24} style={{ color: card.color }} />
69 |                 </div>
70 |                 <div className="text-3xl font-black mb-1"><card.value></div>
71 |                 <div className="text-xs font-black uppercase tracking-widest" style={{ color: 'var(--text-muted)...
72 |                 </div>
73 |                 <div className="absolute -bottom-4 -right-4 w-24 h-24 blur-3xl opacity-10" style={{ backgroundColo...
74 |                 </motion.div>
75 |               </div>
76 |             </div>
77 |
78 |             <div className="grid grid-cols-1 lg:grid-cols-2 gap-8">
79 |               <div className="p-8 rounded-[40px]" style={{ background: 'var(--bg-surface)', border: '1px solid var(...
80 |                 <h3 className="text-xl font-bold mb-8 flex items-center gap-3">
81 |                   <ArrowUpRight style={{ color: 'var(--brand-primary)' }} />
82 |                 </h3>
83 |                 <div className="h-[300px] w-full">
84 |                   <ResponsiveContainer width="100%" height="100%">
85 |                     <LineChart data={scoreHistory}>
86 |                       <CartesianGrid strokeDasharray="3 3" stroke="var(--border-default)" vertical={false} />
87 |                       <XAxis dataKey="date" stroke="var(--text-muted)" fontSize={12} tickLine={false} axisLine={fals...
88 |                       <YAxis stroke="var(--text-muted)" fontSize={12} tickLine={false} axisLine={false} domain={[0, ...
89 |                       <Tooltip
90 |                         contentStyle={{ background: 'var(--bg-base)', border: '1px solid var(--border-default)', bor...
91 |                         itemStyle={{ color: 'var(--brand-primary)', fontWeight: 'bold' }}
92 |                       />
93 |                       <Line type="monotone" dataKey="score" stroke="var(--brand-primary)" strokeWidth={4} dot={{ r: ...
94 |                     </LineChart>
95 |                   </ResponsiveContainer>
96 |                 </div>
97 |               </div>
98 |
99 |               <div className="p-8 rounded-[40px]" style={{ background: 'var(--bg-surface)', border: '1px solid var(...
100 |                 <h3 className="text-xl font-bold mb-8 flex items-center gap-3">
101 |                   <Zap style={{ color: 'var(--accent-amber)' }} /> B Ỗ n Ỗ n Ỗ
102 |                 </h3>
103 |                 <div className="h-[300px] w-full">
104 |                   <ResponsiveContainer width="100%" height="100%">
105 |                     <RadarChart cx="50%" cy="50%" outerRadius="80%" data={competencyData}>
106 |                       <PolarGrid stroke="var(--border-default)" />
107 |                       <PolarAngleAxis dataKey="subject" tick={{ fill: 'var(--text-muted)', fontSize: 12 }} />
108 |                       <PolarRadiusAxis angle={30} domain={[0, 100]} tick={false} axisLine={false} />
109 |                       <Radar name="Cá nhân" dataKey="A" stroke="var(--accent-amber)" fill="var(--accent-amber)" fill...
110 |                     </RadarChart>
111 |                   </ResponsiveContainer>
112 |                 </div>
113 |               </div>
114 |             </div>
115 |
116 |             <div className="grid grid-cols-1 lg:grid-cols-3 gap-8">
117 |               <div className="lg:col-span-2 p-8 rounded-[40px] overflow-hidden" style={{ background: 'var(--bg-surfa...
118 |                 <div className="flex items-center justify-between mb-8">
119 |                   <h3 className="text-xl font-bold">L Ỗ ch s Ỗ bàì thi</h3>
120 |                   <button className="p-2 rounded-lg transition-all" style={{ background: 'var(--bg-base)', color: ...
121 |                 </div>
122 |                 <div>
123 |                   <div>
124 |                     <div className="overflow-x-auto">
125 |                       <table className="w-full text-left text-sm">
126 |                         <thead className="uppercase font-black text-[10px] tracking-widest" style={{ color: 'var(--text-...
127 |                         <tr>
128 |                           <th className="px-4 py-4">T Ỗ n bàì thi</th>
129 |                           <th className="px-4 py-4">Ngàỳ làm</th>
130 |                           <th className="px-4 py-4 text-center">i ã m</th>
131 |                           <th className="px-4 py-4 text-center">X Ỗ p l Ỗ i</th>
132 |                         </tr>
133 |                         <tr>
134 |                           <th className="px-4 py-4 text-right"></th>
135 |                         </tr>
136 |                       </thead>
137 |                       <tbody className="divide-y" style={{ borderColor: 'var(--border-default)' }}>
138 |                         <tr>
139 |                           <td>{title}</td>
140 |                           <td>{date}</td>
141 |                           <td>{score}</td>
142 |                           <td>{date}</td>
143 |                         </tr>
144 |                       </tbody>
145 |                     </table>

```

```

146|         <td className="px-4 py-6 text-center">
147|             <span className={`px-3 py-1 rounded-lg text-[10px] font-black uppercase tracking-wider`}
148|                 style={{
149|                     background: item.grade === 'X' ? 'color-mix(in srgb, var(--brand-primary) 5%, transparent)' : 'color-mix(in srgb, var(--brand-primary) 5%, transparent)',
150|                     color: item.grade === 'X' ? 'var(--text-muted)' : 'var(--brand-primary)'
151|                 }}
152|             >
153|                 {item.grade}
154|             </span>
155|         </td>
156|         <td className="px-4 py-6 text-right">
157|             <button className="p-2 rounded-full transition-all" style={{ color: 'var(--text-muted)' }}>
158|                 <ChevronRight size={20} />
159|             </button>
160|         </td>
161|     </tr>
162| </tbody>
163| </table>
164| </div>
165| </div>
166| </div>
167|
168| <div className="space-y-6">
169|     <div className="p-8 rounded-[40px] relative overflow-hidden" style={{ background: 'var(--bg-surface)...
170|     <div className="relative z-10">
171|         <div className="flex items-center gap-2 font-black uppercase text-xs tracking-widest mb-4" style={{
172|             <AlertTriangle size={16} /> Cẩn báo l x hông ki ến th ệ c
173|         }}>
174|         <div>
175|             <h4 className="text-xl font-bold mb-4">Ch ợ Á: RAM & Kh ớ n
176|             <p className="text-sm leading-relaxed mb-6" style={{ color: 'var(--text-muted)' }}>
177|                 B ỉ n h ậ y tr ể l ý i sai các câu h ớ i liên quan ến * Bus RAM
178|             </p>
179|             <Link href="/lessons/ram-basics" className="w-full py-4 font-black uppercase text-xs rounded-2xl...
180|                 Ồ n t - p ch ợ Á n ậ y ng ậ y <ChevronRight size={16} />
181|             </Link>
182|         </div>
183|         <div className="absolute top-0 right-0 w-24 h-24 blur-3xl opacity-5" style={{ background: 'var(--a...
184|         </div>
185|         <div className="p-8 rounded-[40px]" style={{ background: 'color-mix(in srgb, var(--brand-primary) 5%...
186|         <h4 className="text-lg font-bold mb-4 flex items-center gap-2">
187|             <Sparkles size={18} style={{ color: 'var(--brand-primary)' }} />
188|         </h4>
189|         <ul className="space-y-3 text-sm" style={{ color: 'var(--text-muted)' }}>
190|             <li className="flex items-start gap-2 italic">
191|                 " H ậ y th ớ i làm l ớ i bài th ớ i ' L - p r ắ p PC Lab # 1 ' Ặ c ể i th ớ i (
192|             </li>
193|             <li className="flex items-start gap-2 italic">
194|                 " B ỉ n ặ t i ến b Ồ 2 0 % s ố v ớ i tu ổ n tr ộ c . T ỉ ệ p t ắ c ph ắ t h ượ c
195|             </li>
196|         </ul>
197|         </div>
198|     </div>
199| </div>
200| </div>
201| )
202| }
203|

```

File: app/notifications/page.tsx (120 lines, 4.9 KB)

```

1| 'use client'
2|
3| import React, { useEffect, useState } from 'react'
4| import { createBrowserClient } from '@supabase/ssr'
5| import { Notification } from '@hooks/useNotifications'
6| import { NotificationItem } from '@components/notifications/NotificationItem'
7| import { CheckCheck, ArrowLeft, Bell } from 'lucide-react'
8| import { useRouter } from 'next/navigation'
9|
10| export default function NotificationsPage() {
11|     const router = useRouter()
12|     const [notifications, setNotifications] = useState<Notification[]>([])
13|     const [loading, setLoading] = useState(true)
14|
15|     const supabase = createBrowserClient(
16|         process.env.NEXT_PUBLIC_SUPABASE_URL!,
17|         process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
18|     )
19|
20|     useEffect(() => {
21|         const loadData = async () => {
22|             const { data: { user } } = await supabase.auth.getUser()
23|             if (!user) return
24|
25|             const { data } = await supabase
26|                 .from('notifications')
27|                 .select('*')
28|                 .eq('user_id', user.id)
29|                 .order('created_at', { ascending: false })
30|
31|             if (data) {
32|                 setNotifications(data as Notification[])
33|             }
34|             setLoading(false)
35|         }
36|         loadData()
37|     }, [supabase])
38|
39|     const markAsRead = async (id: string) => {
40|         setNotifications(prev => prev.map(n => n.id === id ? { ...n, is_read: true } : n))
41|         await supabase.from('notifications').update({ is_read: true }).eq('id', id)
42|     }
43|
44|     const markAllAsRead = async () => {
45|         setNotifications(prev => prev.map(n => { ...n, is_read: true }))
46|         const { data: { user } } = await supabase.auth.getUser()
47|         if (user) {
48|             await supabase.from('notifications').update({ is_read: true }).eq('user_id', user.id).eq('is_read', false)
49|         }
50|     }
51|
52|     if (loading) {
53|         return (
54|             <div className="min-h-screen bg-[#161F38] text-white pt-24 flex flex-col items-center justify-center gap...
55|             <div className="w-8 h-8 border-4 border-[#00d4aa] border-t-transparent rounded-full animate-spin"></div>
56|             <span className="text-xs text-gray-500 font-black uppercase tracking-wider">
57|                 Loading...
58|             </span>
59|         )
60|     }
61|
62|     return (
63|         <div className="min-h-screen bg-[#161F38] text-white pt-24 pb-12 px-4 sm:px-6 relative overflow-hidden">

```



```

63 |      /* High-tech overlays */
64 |      <div className="absolute inset-0 bg-grid-pattern opacity-[0.02] pointer-events-none" />
65 |      <div className="absolute top-1/4 left-1/4 w-[350px] h-[350px] bg-[#00d4aa]/5 rounded-full filter blur-[1...
66 |
67 |      <div className="max-w-3xl mx-auto relative z-10">
68 |
69 |          /* Header Title & Exit Button */
70 |          <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-4 mb-8 pb-4 border-b bor...
71 |              <div className="flex items-center gap-3">
72 |                  <div className="p-2.5 bg-[#00d4aa]/10 border border-[#00d4aa]/25 text-[#00d4aa] rounded-2xl">
73 |                      <Bell size={24} />
74 |                  </div>
75 |                  <div>
76 |                      <h1 className="text-xl md:text-2xl font-black tracking-tight">
77 |                          <p className="text-xs text-gray-400 mt-0.5">C - p n h - t t i n
78 |                      </div>
79 |                  </div>
80 |
81 |                  <button
82 |                      onClick={() => router.push('/builder')}
83 |                      className="relative z-50 pointer-events-auto flex items-center gap-2 px-4 py-2 bg-gray-900/90 hove...
84 |                  >
85 |                      <ArrowLeft size={14} className="group-hover:-translate-x-0.5 transition-transform" />
86 |                      Q u a y   l i i   D a s h b o a r d
87 |                  </button>
88 |              </div>
89 |
90 |              <div className="flex justify-end mb-4 relative z-50">
91 |                  <button
92 |                      onClick={markAllAsRead}
93 |                      className="flex items-center gap-2 text-[#00d4aa] hover:bg-[#00d4aa]/10 px-4 py-2 rounded-lg trans...
94 |                  >
95 |                      <CheckCheck size={18} />
96 |                      á n h   d ÿ u   t ÿ t   c   ě   ä   í c
97 |                  </button>
98 |              </div>
99 |
100 |              <div className="bg-[#11121d]/80 rounded-2xl border border-[#1e293b] overflow-hidden shadow-xl backdrop...
101 |                  {notifications.length === 0 ? (
102 |                      <div className="p-12 text-center text-gray-400 font-medium">
103 |                          B i n   k h ô n g   c ó   t h ô n g   b á o   n à o .
104 |                      </div>
105 |                  ) : (
106 |                      notifications.map((notif, index) => (
107 |                          <div key={notif.id} className={index !== notifications.length - 1 ? 'border-b border-gray-800' : ...
108 |                              <NotificationItem
109 |                                  notification={notif}
110 |                                  onRead={markAsRead}
111 |                              />
112 |                          </div>
113 |                      ))
114 |                  )}
115 |              </div>
116 |          </div>
117 |      </div>
118 |  )
119 | }
120 |

```

File: app/onboarding/page.tsx (182 lines, 8.8 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect } from 'react'
4 | import { useRouter } from 'next/navigation'
5 | import { User, Users, GraduationCap, ArrowRight, Loader2, Building, Calendar, MapPin, Mail, CheckCircle } from...
6 | import { createBrowserClient } from '@supabase/ssr'
7 |
8 | export default function OnboardingPage() {
9 |     const [loading, setLoading] = useState(false)
10 |     const [role, setRole] = useState<'student' | 'teacher' | 'parent' | null>(null)
11 |     const [fullName, setFullName] = useState('')
12 |     const [email, setEmail] = useState('')
13 |     const [dob, setDob] = useState('')
14 |     const [province, setProvince] = useState('')
15 |     const [schoolName, setSchoolName] = useState('')
16 |     const [userId, setUserId] = useState<string | null>(null)
17 |     const [lang, setLang] = useState<'en' | 'vn'>('vn')
18 |     const router = useRouter()
19 |
20 |     useEffect(() => {
21 |         try {
22 |             const saved = localStorage.getItem('lang');
23 |             if (saved === 'en' || saved === 'vn') setLang(saved as 'en' | 'vn');
24 |         } catch {}
25 |     }, []);
26 |
27 |     const supabase = createBrowserClient(
28 |         process.env.NEXT_PUBLIC_SUPABASE_URL || 'https://placeholder.supabase.co',
29 |         process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY || 'placeholder'
30 |     )
31 |
32 |     const T = (en: string, vn: string) => lang === 'en' ? en : vn
33 |
34 |     useEffect(() => {
35 |         supabase.auth.getUser().then(({ data }) => {
36 |             if (data?.user) {
37 |                 setUserId(data.user.id)
38 |                 setEmail(data.user.email || '')
39 |                 setFullName(data.user.user_metadata?.full_name || '')
40 |             } else {
41 |                 router.push('/login')
42 |             }
43 |         })
44 |     }, [router, supabase])
45 |
46 |     const handleComplete = async () => {
47 |         if (!role || !fullName.trim() || !userId) return
48 |         setLoading(true)
49 |
50 |         const { error } = await supabase.from('profiles').upsert({
51 |             id: userId,
52 |             email: email,
53 |             full_name: fullName.trim(),
54 |             role: role,
55 |             dob: dob || null,
56 |             province: province || null,
57 |             school_name: schoolName || null,
58 |             updated_at: new Date().toISOString()
59 |         })
60 |
61 |         setLoading(false)
62 |

```

```

63 | if (error) {
64 |     alert('Có lỗi xảy ra: ' + error.message)
65 | } else {
66 |     const redirectUrl = role === 'teacher' ? '/teacher' : role === 'parent' ? '/parent' : '/builder'
67 |     router.push(redirectUrl)
68 | }
69 |
70 |
71 | return (
72 |     <div className="min-h-screen flex items-center justify-center bg-[#0f0f1a] p-4 text-white">
73 |         <div className="w-full max-w-lg bg-[#16213e] rounded-2xl border border-[#1e293b] p-8 shadow-2xl relative...
74 |             <div className="absolute top-0 left-1/2 -translate-x-1/2 w-full h-1/2 bg-[#00d2a0]/10 blur-[100px] poi...
75 |
76 |             <div className="relative z-10 text-center mb-8">
77 |                 <div className="w-16 h-16 bg-[#00d2a0]/20 rounded-full flex items-center justify-center mx-auto mb-4...
78 |                 <GraduationCap size={32} />
79 |             </div>
80 |             <h1 className="text-2xl font-bold mb-2">{T('Complete Profile')}
81 |             <p className="text-slate-400 text-sm">{T('Email auto-saved from Google account. Please fill in the r...
82 |             </div>
83 |
84 |             <div className="space-y-5 relative z-10">
85 |                 <div>
86 |                     <label className="block text-sm font-medium text-slate-300 mb-2">{T('Email (auto from Google)', 'E...
87 |                     <div className="flex items-center gap-2 bg-[#0f0f1a] border border-[#1e293b] rounded-xl px-4 py-3 ...
88 |                         <Mail size={16} />
89 |                         <span>{email}</span>
90 |                         <CheckCircle size={16} className="ml-auto text-[#00d2a0]" />
91 |                     </div>
92 |                 </div>
93 |
94 |                 <div>
95 |                     <label className="block text-sm font-medium text-slate-300 m
96 |                     <input
97 |                         type="text"
98 |                         value={fullName}
99 |                         onChange={(e) => setFullName(e.target.value)}
100 |                     />
101 |                     <div>
102 |                         <div>
103 |                             <div>
104 |                                 <div>
105 |                                     <label className="block text-sm font-medium text-slate-300 m
106 |                                     <div className="grid grid-cols-3 gap-3">
107 |                                         {[
108 |                                             {
109 |                                                 key: 'teacher' as const, icon: <User size={24} />, title: T('Teacher', 'Giáo viên') },
110 |                                             {
111 |                                                 key: 'parent' as const, icon: <GraduationCap size={24} />, title: T('Parent', 'Phụ huynh') }
112 |                                         ].map(({ key, icon, title }) => (
113 |                                             <button
114 |                                                 key={key}
115 |                                                 onClick={() => setRole(key)}
116 |                                                 className={`flex flex-col items-center justify-center p-3 rounded-xl border-2 transition-all...
117 |                                                 role === key
118 |                                                 ? 'border-[#00d2a0] bg-[#00d2a0]/10'
119 |                                                 : 'border-[#1e293b] bg-[#0f0f1a] hover:border-slate-600'
120 |                                             >
121 |                                                 {icon}
122 |                                             </div>
123 |                                             <div className="text-sm font-medium text-slate-400">{title}</div>
124 |                                         </button>
125 |                                     </div>
126 |                                 </div>
127 |                             </div>
128 |                         </div>
129 |                     <div className="grid grid-cols-2 gap-4">
130 |                         <div>
131 |                             <label className="block text-sm font-medium text-slate-300 mb-2">{T('Date of Birth', 'Ngày sinh'...
132 |                             <div className="relative">
133 |                                 <Calendar size={16} className="absolute left-3 top-1/2 -translate-y-1/2 text-slate-500" />
134 |                                 <input type="date" value={dob} onChange={(e) => setDob(e.target.value)}
135 |                                 className="w-full bg-[#0f0f1a] border border-[#1e293b] rounded-xl pl-10 pr-4 py-3 outline-no...
136 |                             </div>
137 |                         </div>
138 |                         <div>
139 |                             <label className="block text-sm font-medium text-slate-300 m
140 |                             <div className="relative">
141 |                                 <MapPin size={16} className="absolute left-3 top-1/2 -translate-y-1/2 text-slate-500 z-10" />
142 |                                 <select value={province} onChange={(e) => setProvince(e.target.value)}
143 |                                 className="w-full bg-[#0f0f1a] border border-[#1e293b] rounded-xl pl-10 pr-4 py-3 outline-no...
144 |                                     <option value="" disabled>{T('Select', 'Chọn')}</option>
145 |                                     <option value="HN">Hà Nội</option>
146 |                                     <option value="HCM">TP. Hồ Chí Minh</option>
147 |                                     <option value="DN">Đà Nẵng</option>
148 |                                     <option value="Other">{T('Other', 'Tỉnh thành khác')}</option>
149 |                                 </select>
150 |                             </div>
151 |                         </div>
152 |                     </div>
153 |
154 |                     <div>
155 |                         <label className="block text-sm font-medium text-slate-300 m
156 |                         <div className="relative">
157 |                             <Building size={16} className="absolute left-3 top-1/2 -translate-y-1/2 text-slate-500" />
158 |                             <input type="text" value={schoolName} onChange={(e) => setSchoolName(e.target.value)}
159 |                             className="w-full bg-[#0f0f1a] border border-[#1e293b] rounded-xl pl-10 pr-4 py-3 outline-none...
160 |                         </div>
161 |                     </div>
162 |
163 |                     <button
164 |                         onClick={handleComplete}
165 |                         disabled={!role || !fullName.trim() || !loading}
166 |                         className="w-full mt-2 bg-[#00d2a0] hover:bg-[#00e6af] disabled:bg-slate-700 disabled:text-slate-5...
167 |                     >
168 |                         {loading ? <Loader2 className="animate-spin" size={20} /> : (
169 |                             <div>
170 |                                 <div>
171 |                                     <div>
172 |                                         <div>
173 |                                             <div>
174 |                                                 <p className="text-xs text-slate-500 text-center">
175 |                                                     {T('Next login with Google will go straight in without re-er
176 |                                                 </p>
177 |                                             </div>
178 |                                         </div>
179 |                                     </div>
180 |                                 </div>
181 |                             </div>
182 |                         )

```



```
125 |         whiteSpace: 'nowrap'
126 |     }}>
127 | |         H í c T í n h í c q u a m ô p h ỉ n g P C < s p a n s t y l e = { { c o l o r : ' v a r ( - - b r a
128 | |         </h1>
129 | |
130 | |         <p className="animate-fade-in-up animate-delay-2" style={{ fontSize: '18px', color: 'var(--text-mute...
131 | |         A I h ộ ũ n g d ề n c h ỉ t i ế t • T r ẻ i n g h ỉ c m 1 0 0 % W e b - b a s e d • K h ỏ n g
132 | |         </p>
133 | |
134 | |         <div className="animate-fade-in-up animate-delay-3" style={{ display: 'flex', gap: '16px', justifyCo...
135 | |         <Link href="/builder">
136 | |         <button className="lms-btn lms-btn-primary" style={{ padding: '14px 32px', fontSize: '15px', fon...
137 | |         < G r a d u a t i o n C ấ p s i z e = { 2 0 } / > T ô i l ầ H í c s i n h
138 | |         </button>
139 | |         </Link>
140 | |         <Link href="/teacher/lessons">
141 | |         <button className="lms-btn lms-btn-outline" style={{ padding: '14px 32px', fontSize: '15px', fon...
142 | |         <BookOpen size={20} /> Tôi là Giáo viên
143 | |         </button>
144 | |         </Link>
145 | |         </div>
146 | |     </div>
147 | </section>
148 |
149 | /* TRUST BAR */
150 | <section className="animate-fade-in-up animate-delay-4" style={{ padding: '0 32px', marginTop: '-20px', ...
151 | <div style={{ maxWidth: '900px', margin: '0 auto', background: 'var(--bg-surface)', borderRadius: '16p...
152 | <div style={{ display: 'flex', justifyContent: 'center', gap: '48px', flexWrap: 'wrap', alignItems: ...
153 | |     {
154 | |     { icon: <School size={20} style={{ color: 'var(--accent-blue)' }} />, text: 'THPT Chuyên' },
155 | |     { icon: <Users size={20} style={{ color: 'var(--brand-primary)' }} />, text: 'var(--brand-primary)' },
156 | |     { icon: <FileText size={20} style={{ color: 'var(--accent-amber)' }} />, text: 'Sách KNTT & Cảnh...
157 | |     ].map((item, i) => {
158 | |     <div key={i} style={{ display: 'flex', alignItems: 'center', gap: '8px', fontSize: '14px', fontW...
159 | |     {item.icon} {item.text}
160 | |     </div>
161 | |     ))
162 | | </div>
163 | </div>
164 | </section>
165 |
166 | /* STATS */
167 | <section className="animate-fade-in-up animate-delay-5 landing-section" style={{ padding: isMobile ? '48...
168 | <div className="landing-stats-grid" style={{ maxWidth: '1000px', margin: '0 auto', display: 'grid', gr...
169 | | {
170 | |     { num: '50+', label: 'Linh kiện chi tiết', color: 'var(--brand-primary)' },
171 | |     { num: '10+', label: 'Nhịệm vụ thách', color: 'var(--brand-primary)' },
172 | |     { num: '100%', label: 'Năng Web-based', color: 'var(--accent-amber)' },
173 | |     { num: '99%', label: 'Úchính xác kù thu-t', color: 'var(--brand-primary)' },
174 | |     ].map((s, i) => {
175 | |     <div key={i} className="lms-card" style={{ padding: '28px 20px', animation: `fadeInUp 0.5s ease ${...
176 | |     <div className="stat-number" style={{ color: s.color }}>{s.num}</div>
177 | |     <div className="stat-label">{s.label}</div>
178 | |     </div>
179 | |     ))
180 | | </div>
181 | </section>
182 |
183 | /* FEATURES */
184 | <section className="landing-section" style={{ padding: isMobile ? '0 16px 48px' : '0 32px 80px' }}>
185 | <div style={{ maxWidth: '1100px', margin: '0 auto' }}>
186 | <motion.div
187 |   initial={{ opacity: 0, y: 30 }}
188 |   whileInView={{ opacity: 1, y: 0 }}
189 |   viewport={{ once: true, margin: '-80px' }}
190 |   transition={{ duration: 0.5 }}
191 |   style={{ textAlign: 'center', marginBottom: '48px' }}
192 | >
193 |   <h2 style={{ fontSize: '32px', fontWeight: 800, marginBottom: '12px', color: 'var(--text-primary)'...
194 |   <p style={{ color: 'var(--text-muted)', fontSize: '16px', maxWidth: '640px', margin: '0 auto', lin...
195 |   K h ỏ n g c h ế l ầ l ậ p r ấ p , h ệ t h ể n g g i ấ l ậ p c h ỉ t i ế t t h ỏ n g s ố
196 |   </p>
197 | </motion.div>
198 |
199 | <div className="landing-features-grid" style={{ display: 'grid', gridTemplateColumns: isMobile ? '1f...
200 | <motion.div
201 |   initial="hidden"
202 |   whileInView="visible"
203 |   viewport={{ once: true, margin: '-80px' }}
204 |   variants={{
205 |     hidden: {},
206 |     visible: { transition: { staggerChildren: 0.12 } }
207 |   }}
208 |   style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}
209 | >
210 | | {
211 | |     { icon: <Zap size={22} />, text: 'Tính toán iệ n n g (t' },
212 | |     { icon: <ShieldCheck size={22} />, text: 'Kiểm tra t ộ n g ' },
213 | |     { icon: <Sparkles size={22} />, text: 'Phân tích c ầ u h ỉ n g ' },
214 | |     { icon: <Cpu size={22} />, text: 'Thống số kù thu-t 50+' },
215 | |     { icon: <BrainCircuit size={22} />, text: 'Giải ý năng c ầ u ' },
216 | |     ].map((f, i) => {
217 | |     <motion.div
218 | |     key={i}
219 | |     variants={{
220 | |       hidden: { opacity: 0, x: -30 },
221 | |       visible: { opacity: 1, x: 0 }
222 | |     }}
223 | |     className="lms-card"
224 | |     style={{ display: 'flex', alignItems: 'flex-start', gap: '16px', padding: '20px' }}
225 | |     >
226 | |     <div style={{ width: '44px', height: '44px', borderRadius: '10px', background: i === 0 ? 'rg...
227 | |     {f.icon}
228 | |     </div>
229 | |     <div>
230 | |     <p style={{ fontWeight: 600, color: 'var(--text-primary)', marginBottom: '4px', fontSize: ...
231 | |     <p style={{ fontSize: '13px', color: 'var(--text-muted)', lineHeight: 1.5 }}>{f.desc}</p>
232 | |     </div>
233 | |     </motion.div>
234 | |     ))
235 | | </motion.div>
236 | <motion.div
237 |   initial={{ opacity: 0, x: 30 }}
238 |   whileInView={{ opacity: 1, x: 0 }}
239 |   viewport={{ once: true, margin: '-80px' }}
240 |   transition={{ duration: 0.5, delay: 0.2 }}
241 |   style={{ perspective: '1200px' }}
242 | >
243 | <motion.div
244 |   animate={{ y: [0, -6, 0] }}
245 |   transition={{ duration: 4, repeat: Infinity, ease: 'easeInOut' }}
246 |   style={{
247 |     background: 'var(--bg-surface)', borderRadius: '16px', padding: '16px',
248 |     border: '1px solid var(--border-default)',
```

```

249| boxShadow: '0 8px 32px -4px var(--shadow-color), 0 0 1px rgba(0,212,170,0)',
250| transition: 'box-shadow 0.4s ease, border-color 0.4s ease',
251| transformStyle: 'preserve-3d',
252| })
253| onMouseEnter={e => {
254|   e.currentTarget.style.boxShadow = '0 20px 60px -8px rgba(0,212,170,0.2), 0 0 1px rgba(0,21...
255|   e.currentTarget.style.borderColor = 'var(--brand-primary)';
256|   e.currentTarget.style.transform = 'rotateY(-2deg) rotateX(2deg)';
257| }}
258| onMouseLeave={e => {
259|   e.currentTarget.style.boxShadow = '0 8px 32px -4px var(--shadow-color)';
260|   e.currentTarget.style.borderColor = 'var(--border-default)';
261|   e.currentTarget.style.transform = 'rotateY(0deg) rotateX(0deg)';
262| }}
263| >
264| <div style={{
265|   position: 'relative', borderRadius: '8px', overflow: 'hidden',
266| }}>
267|   <div style={{
268|     position: 'absolute', inset: 0, zIndex: 1, pointerEvents: 'none',
269|     background: 'linear-gradient(135deg, rgba(0,212,170,0.08) 0%, transparent 50%, rgba(99,102...
270|     animation: 'showcaseGlow 3s ease-in-out infinite',
271|   }} />
272|   <motion.img
273|     style={{ width: '100%', height: '100%',
274|       src: "/showcase.png", alt: "Giáo diện phên máy mô phỏng",
275|       animate: {{ scale: [1, 1.02, 1] }},
276|       transition: {{ duration: 6, repeat: Infinity, ease: 'easeInOut' }}
277|     }} />
278|   </div>
279|   <div style={{ marginTop: '12px', display: 'flex', alignItems: 'center', gap: '8px', justifyCon...
280|     <span className="lms-tag lms-tag-green" style={{ animation: 'tagPulse 2s ease-in-out infinit...
281|       <PlayCircle size={14} /> Lab thực hành 2D
282|     </span>
283|     <span className="lms-tag lms-tag-blue" style={{ animation: 'tagPulse 2s ease-in-out 0.3s inf...
284|       <Cctv size={14} /> Mô phỏng thi gian thực
285|     </span>
286|   </div>
287| </motion.div>
288| </div>
289| </div>
290| </div>
291| </section>
292|
293| /* HOW TO USE */
294| <section className="landing-section-wide" style={{ padding: '80px 32px', background: 'var(--bg-surface)'...
295|   <div style={{ maxWidth: '1100px', margin: '0 auto' }}>
296|     <div style={{ textAlign: 'center', marginBottom: '48px' }}>
297|       <h2 style={{ fontSize: '32px', fontWeight: 800, marginBottom: '12px', color: 'var(--text-primary)'...
298|         <p style={{ color: 'var(--text-muted)', fontSize: '16px', maxWidth: '600px', margin: '0 auto' }}>
299|           B-t sự hành trình trở thành chuyên gia lập ráp PC chế vớ
300|         </p>
301|       </div>
302|
303|       <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(220px, 1fr))', gap: '24...
304|         [[
305|           { icon: <GraduationCap size={28} style={{ color: 'var(--brand-primary)' }} />, step: '01', title...
306|           { icon: <MonitorPlay size={28} style={{ color: 'var(--accent-blue)' }} />, step: '02', title: 'T...
307|             { icon: <Wrench size={28} style={{ color: 'var(--accent-a...
308|             { icon: <Bot size={28} style={{ color: 'var(--info)' }} />
309|         ].map((item, i) => {
310|           <div key={i} className="lms-card" style={{ textAlign: 'center', padding: '32px 24px', animation:...
311|             <div style={{ fontSize: '32px', fontWeight: 800, color: 'var(--brand-primary)', marginBottom: ...
312|               <div style={{ margin: '16px', display: 'flex', justifyContent: 'center' }}><item.icon></...
313|               <h3 style={{ fontSize: '18px', fontWeight: 700, margin: '8px', color: 'var(--text-primar...
314|                 <p style={{ fontSize: '14px', color: 'var(--text-muted)', lineHeight: 1.6 }}><item.desc></p>
315|               </div>
316|             </div>
317|           </div>
318|         </div>
319|       </section>
320|
321|       /* FAQ */
322|       <section className="landing-section" style={{ padding: '80px 32px' }}>
323|         <div style={{ maxWidth: '800px', margin: '0 auto' }}>
324|           <div style={{ textAlign: 'center', margin: '48px 0' }}>
325|             <h2 style={{ fontSize: '32px', fontWeight: 800, margin: '12px', color: 'var(--text-primary)'...
326|               <p style={{ color: 'var(--text-muted)', fontSize: '16px' }}>
327|             </div>
328|
329|             <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
330|               [[
331|                 { q: 'Tôi có cần cài đặt phần mềm nào để sử dụng không?', a: 'AI Gu...
332|                 { q: 'Hẹn giờ AI Guru có thể giúp gì cho tôi?', a: 'AI Gu...
333|                 { q: 'Sẵn phim có hỗ trợ tải ảnh hành trình lên kênh cá nhân của mình không?', a: 'Có...
334|                 { q: 'Giáo viên có thể theo dõi tiến độ của học sinh không?', a: 'Không...
335|                 { q: 'Nữ dùng có bảo mật sách giáo khoa không?', a: 'Có.'
336|               ]}.map((faq, i) => {
337|                 <div key={i} className="lms-card" style={{ padding: '0', overflow: 'hidden', cursor: 'pointer', ...
338|                   <div style={{ padding: '20px 24px', display: 'flex', justifyContent: 'space-between', alignite...
339|                     {faq.q}
340|                     <ChevronDown size={20} style={{
341|                       color: 'var(--text-muted)',
342|                       transform: activeFaq === i ? 'rotate(180deg)' : 'rotate(0)',
343|                       transition: 'transform 0.3s cubic-bezier(0.4, 0, 0.2, 1)',
344|                       flexShrink: 0
345|                     }} />
346|                   </div>
347|                   <div style={{
348|                     overflow: 'hidden',
349|                     maxHeight: activeFaq === i ? '300px' : '0',
350|                     transition: 'max-height 0.35s cubic-bezier(0.4, 0, 0.2, 1), opacity 0.25s ease, padding 0.25...
351|                     opacity: activeFaq === i ? 1 : 0,
352|                     padding: activeFaq === i ? '0 24px 20px' : '0 24px',
353|                   }}>
354|                     <div style={{ borderTop: '1px solid var(--border-default)', padding: '16px', color: 'var(...
355|                       {faq.a}
356|                     </div>
357|                   </div>
358|                 </div>
359|               </div>
360|             </div>
361|           </div>
362|         </section>
363|
364|       /* FOOTER */
365|       <footer className="landing-section" style={{ padding: '60px 32px 40px', textAlign: 'center', borderTop: ...
366|         <div style={{ maxWidth: '1100px', margin: '0 auto' }}>
367|           <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(200px, 1fr))', gap: '32...
368|             <div>
369|               <div style={{ display: 'flex', alignItems: 'center', gap: '10px', margin: '16px 0' }}>
370|                 
371|                 <span style={{ font-weight: 700, font-size: '16px' }}>PC Master Builder</span>
372|               </div>

```

```

374 | </div>
375 | <div>
376 |   <h4 style={{ fontWeight: 700, fontSize: '14px', marginBottom: '16px', color: 'var(--text-primary...'
377 |   <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
378 |     <Link href="/builder" style={{ fontSize: '13px', color: 'var(--text-muted)' }}>Builder</Link>
379 |     <Link href="/about" style={{ fontSize: '13px', color: 'var(--text-muted)' }}>About</Link>
380 |   </div>
381 | </div>
382 | <div>
383 |   <h4 style={{ fontWeight: 700, fontSize: '14px', marginBottom: '16px', color: 'var(--text-primary...'
384 |   <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
385 |     <span style={{ fontSize: '13px', color: 'var(--text-muted)' }}>Email: phuckhanhsonnguyen@gmail...
386 |     <span style={{ fontSize: '13px', color: 'var(--text-muted)' }}>Hotline: 0772934951</span>
387 |   </div>
388 | </div>
389 | </div>
390 | <div style={{ padding: '24px', border: '1px solid var(--border-default)', display: 'flex', justify...
391 |   <p style={{ color: 'var(--text-muted)', fontSize: '13px' }}>© 2026 PC Master Builder. All rights r...
392 |   <div style={{ display: 'flex', gap: '12px', alignItems: 'center' }}>
393 |     <span className="lms-tag" style={{ fontSize: '10px', padding: '2px 8px' }}>AI YOUNG GURU 2026</span>
394 |   </div>
395 | </div>
396 | </div>
397 | </footer>
398 |
399 | <style>{
400 |   @media (min-width: 769px) { .desktop-nav { display: flex !important; } .mobile-hamburger-btn { display: none !important; }
401 |   @media (max-width: 768px) {
402 |     .desktop-nav { display: none !important; }
403 |     .mobile-hamburger-btn { display: flex !important; }
404 |     .landing-nav { padding: 0 16px !important; }
405 |     .landing-nav-logo span { font-size: 14px !important; }
406 |     .landing-hero { padding: 120px 16px 60px !important; }
407 |     .landing-section { padding: 40px 16px !important; }
408 |     .landing-section-wide { padding: 40px 16px !important; }
409 |     .landing-features-grid { grid-template-columns: 1fr !important; }
410 |     .landing-stats-grid { grid-template-columns: 1fr 1fr !important; }
411 |   }
412 |   @media (max-width: 480px) {
413 |     .landing-nav { padding: 0 12px !important; }
414 |     .landing-hero { padding: 100px 12px 40px !important; }
415 |     .landing-stats-grid { grid-template-columns: 1fr !important; }
416 |   }
417 |   @keyframes showcaseGlow {
418 |     0%, 100% { opacity: 0.4; }
419 |     50% { opacity: 0.9; }
420 |   }
421 |   @keyframes tagPulse {
422 |     0%, 100% { opacity: 1; }
423 |     50% { opacity: 0.6; }
424 |   }
425 | }</style>
426 | </div>
427 | );
428 | }
429 |

```

File: app/practice/page.tsx (169 lines, 6.8 KB)

```

1 | 'use client'
2 |
3 | import { useState } from 'react'
4 | import { useRouter } from 'next/navigation'
5 | import { motion } from 'framer-motion'
6 | import { useIsMobile } from '@hooks/useIsMobile'
7 | import {
8 |   FileQuestion, ArrowRight,
9 |   GraduationCap, BarChart3,
10 |   Home
11 | } from 'lucide-react'
12 |
13 | const TABS = [
14 |   { id: 'exams', label: 'Kó Thi', icon: FileQuestion, color: '#6366f1',
15 |     { id: 'quiz', label: 'Trc Nghiçm', icon: BarChart3, color: '#f59e0b',
16 |   ]
17 |
18 | export default function PracticePage() {
19 |   const router = useRouter()
20 |   const isMobile = useIsMobile()
21 |   const [activeTab, setActiveTab] = useState('exams')
22 |
23 |   return (
24 |     <div style={{ minHeight: '100vh', background: 'var(--bg-base)', color: 'var(--text-primary)' }}>
25 |       <div style={{ maxWidth: '1024px', margin: '0 auto', padding: isMobile ? '24px 12px' : '48px 24px' }}>
26 |         </* Header */>
27 |         <motion.div
28 |           initial={{ opacity: 0, y: -20 }}
29 |           animate={{ opacity: 1, y: 0 }}
30 |           style={{ textAlign: 'center', marginBottom: '40px' }}>
31 |           <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '12px', marginBo...
32 |             <div style={{ width: '48px', height: '48px', borderRadius: '14px', background: 'linear-gradient(13...
33 |               <GraduationCap size={24} color="#fff" />
34 |             </div>
35 |             <h1 style={{ fontSize: isMobile ? '26px' : '34px', fontWeigh...
36 |           </div>
37 |           <p style={{ color: 'var(--text-muted)', maxWidth: '600px', margin: '0 auto', fontSize: '15px' }}>
38 |             Ki Åm tra ki çn théc ph ãn céng vúi A thi, ngãn hàng câu hñi v
39 |         </p>
40 |       </motion.div>
41 |
42 |       </* Tab Navigation */>
43 |       <motion.div
44 |         initial={{ opacity: 0, y: 20 }}
45 |         animate={{ opacity: 1, y: 0 }}
46 |         transition={{ delay: 0.1 }}
47 |         style={{
48 |           display: 'flex', gap: isMobile ? '8px' : '12px',
49 |           margin: '0 auto', flexWrap: 'wrap',
50 |           justify-content: 'center'
51 |         }}>
52 |         {TABS.map(tab => {
53 |           const isActive = activeTab === tab.id
54 |           const Icon = tab.icon
55 |           return (
56 |             <button
57 |               key={tab.id}
58 |               onClick={() => setActiveTab(tab.id)}
59 |               style={{
60 |                 display: 'flex', alignItems: 'center', gap: '8px',
61 |                 padding: isMobile ? '10px 16px' : '12px 24px',
62 |               }}>
63 |               <Icon size={24} color={isActive ? 'var(--text-primary)' : 'var(--text-muted)'}/>
64 |               {tab.label}
65 |             </button>
66 |           )
67 |         })}
68 |       </motion.div>
69 |     </div>
70 |   )
71 | }

```

```

64 |         borderRadius: '12px', border: '1px solid',
65 |         borderColor: isActive ? tab.color : 'var(--border-default)',
66 |         background: isActive ? `${tab.color}12` : 'var(--bg-surface)',
67 |         color: isActive ? tab.color : 'var(--text-muted)',
68 |         fontWeight: isActive ? 700 : 500,
69 |         fontSize: isMobile ? '12px' : '14px',
70 |         cursor: 'pointer', fontFamily: 'inherit',
71 |         transition: 'all 0.2s'
72 |     }}
73 |     >
74 |     <Icon size={isMobile ? 16 : 18} />
75 |     {isMobile ? tab.label.split(' ')[0] : tab.label}
76 | </button>
77 | )
78 | }}
79 | </motion.div>
80 |
81 | /* Tab Content */
82 | <motion.div
83 |   key={activeTab}
84 |   initial={{ opacity: 0, y: 20 }}
85 |   animate={{ opacity: 1, y: 0 }}
86 |   transition={{ duration: 0.2 }}
87 | >
88 |   {TABS.filter(t => t.id === activeTab).map(tab => {
89 |     const Icon = tab.icon
90 |     return (
91 |       <div
92 |         key={tab.id}
93 |         style={{
94 |           borderRadius: '20px', border: '1px solid var(--border-default)',
95 |           background: 'var(--bg-surface)', padding: isMobile ? '24px' : '40px',
96 |           textAlign: 'center'
97 |         }}
98 |       >
99 |         <div style={{
100 |           width: '72px', height: '72px', borderRadius: '20px',
101 |           background: `${tab.color}15`, color: tab.color,
102 |           display: 'flex', alignItems: 'center', justifyContent: 'center',
103 |           margin: '0 auto 20px'
104 |         }}>
105 |           <Icon size={36} />
106 |         </div>
107 |         <h2 style={{ fontSize: '22px', fontWeight: 700, marginBottom: '8px' }}>{tab.label}</h2>
108 |         <p style={{ color: 'var(--text-muted)', marginBottom: '24px', maxWidth: '400px', margin: '0 au...
109 |           {tab.desc}
110 |         </p>
111 |         <button
112 |           onClick={() => router.push(tab.link)}
113 |           style={{
114 |             display: 'inline-flex', alignItems: 'center', gap: '8px',
115 |             padding: '12px 28px', borderRadius: '12px', border: 'none',
116 |             background: tab.color, color: '#fff', fontWeight: 700,
117 |             fontSize: '15px', cursor: 'pointer', fontFamily: 'inherit'
118 |           }}
119 |         >
120 |           Vào ngay <ArrowRight size={18} />
121 |         </button>
122 |       </div>
123 |     )
124 |   }}
125 | </motion.div>
126 |
127 | /* Bottom nav - Back to home */
128 | <motion.div
129 |   initial={{ opacity: 0 }}
130 |   animate={{ opacity: 1 }}
131 |   transition={{ delay: 0.3 }}
132 |   style={{
133 |     display: 'flex', justifyContent: 'center', gap: '16px',
134 |     marginTop: '32px', flexWrap: 'wrap'
135 |   }}
136 | >
137 |   <button
138 |     onClick={() => router.push('/student')}
139 |     style={{
140 |       display: 'flex', alignItems: 'center', gap: '6px',
141 |       padding: '8px 16px', borderRadius: '8px', border: '1px solid var(--border-default)',
142 |       background: 'var(--bg-elevated)', color: 'var(--text-muted)',
143 |       fontSize: '12px', cursor: 'pointer', fontFamily: 'inherit',
144 |       fontWeight: 500
145 |     }}
146 |   >
147 |     < Home size = { 14 } /> Trang chủ < ArrowRight size = { 14 } />
148 |   </button>
149 |   {TABS.filter(t => t.id !== activeTab).map(tab => (
150 |     <button
151 |       key={tab.id}
152 |       onClick={() => router.push(tab.link)}
153 |       style={{
154 |         display: 'flex', alignItems: 'center', gap: '6px',
155 |         padding: '8px 16px', borderRadius: '8px', border: '1px solid var(--border-default)',
156 |         background: 'var(--bg-elevated)', color: 'var(--text-muted)',
157 |         fontSize: '12px', cursor: 'pointer', fontFamily: 'inherit',
158 |         fontWeight: 500
159 |       }}
160 |     >
161 |       <tab.icon size={14} /> {tab.label} <ArrowRight size={14} />
162 |     </button>
163 |   ))
164 | </motion.div>
165 | </div>
166 | </div>
167 | )
168 | }
169 |

```

File: app/pro/layout.js (4 lines, 0.1 KB)

```

1 | export default function ProLayout({ children }) {
2 |   return children;
3 | }
4 |

```

File: app/pro/page.js (11 lines, 0.3 KB)

```

1 | import ProLearningSystem from '@components/ProLearningSystem';
2 |
3 | export const metadata = {
4 |   title: 'Pro Learning System | PC Master Builder',
5 |   description: 'Hỗ trợ các tình huống nâng cao - Build Scenarios on Mac OS

```

```
6 | };
7 |
8 | export default function ProPage() {
9 |   return <ProLearningSystem />;
10 | }
11 |
```

File: app/profile/[userId]/page.tsx (146 lines, 8.0 KB)

```
1 | 'use client'
2 |
3 | import React, { useState, useEffect } from 'react'
4 | import { useParams } from 'next/navigation'
5 | import { supabase } from '@lib/supabase'
6 | import { BookOpen, Clock, Trophy, Award, Activity, ShieldCheck, ArrowLeft } from 'lucide-react'
7 | import Link from 'next/link'
8 | import Navbar from '@components/Navbar'
9 | import { motion } from 'framer-motion'
10 |
11 | export default function UserProfilePage() {
12 |   const { userId } = useParams()
13 |   const [profile, setProfile] = useState<any>(null)
14 |   const [loading, setLoading] = useState(true)
15 |
16 |   useEffect(() => {
17 |     const fetchProfile = async () => {
18 |       try {
19 |         const { data, error } = await supabase
20 |           .from('profiles')
21 |           .select('*')
22 |           .eq('id', userId)
23 |           .single()
24 |
25 |         if (error) throw error
26 |         setProfile(data)
27 |       } catch (error) {
28 |         console.error(error)
29 |       } finally {
30 |         setLoading(false)
31 |       }
32 |     }
33 |     if (userId) fetchProfile()
34 |   }, [userId])
35 |
36 |   if (loading) {
37 |     return (
38 |       <div style={{ minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', alignItems: 'center', j...
39 |         <div style={{ width: '48px', height: '48px', borderRadius: '50%', border: '2px solid var(--brand-prima...
40 |       </div>
41 |     )
42 |   }
43 |
44 |   if (!profile) {
45 |     return (
46 |       <div style={{ minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', flexDirection: 'column'...
47 |         <h2 style={{ fontSize: '24px', fontWeight: 700, marginBottom: '10px' }}>
48 |           <Link href="/" style={{ padding: '12px 24px', background: 'var(--brand-primary)', color: 'var(--bg-bas...
49 |         </div>
50 |       )
51 |     }
52 |   }
53 |
54 |   return (
55 |     <div style={{ minHeight: '100vh', background: 'var(--bg-base)', color: 'var(--text-primary)' }}>
56 |       <Navbar />
57 |
58 |       <main style={{ padding: '96px 0 80px 16px', paddingRight: '16px', max...
59 |         <Link href="/student/classes" style={{ display: 'inline-flex', alignItems: 'center', gap: '8px', color...
60 |         <ArrowLeft size={18} />
61 |         <span>Quay lại</span>
62 |       </Link>
63 |
64 |       <div style={{ background: 'var(--bg-surface)', borderRadius: '16px', border: '1px solid var(--border-d...
65 |         <div style={{ height: '192px', background: 'linear-gradient(to right, color-mix(in srgb, var(--brand...
66 |         <div style={{ position: 'absolute', inset: 0, opacity: 0.2, backgroundImage: 'radial-gradient(var(...
67 |       </div>
68 |
69 |       <div style={{ padding: '24px 40px 32px', position: 'relative' }}>
70 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '24px' }} className="md:flex-row md:i...
71 |           <div style={{ width: '128px', height: '128px', borderRadius: '50%', border: '4px solid var(--bg-...
72 |             {profile.avatar_url ? (
73 |               <img src={profile.avatar_url} alt="Avatar" style={{ width: '100%', height: '100%', objectFit...
74 |             ) : (
75 |               <div style={{ width: '100%', height: '100%', background: 'linear-gradient(to bottom right, v...
76 |                 {profile.full_name?.charAt(0).toUpperCase()}
77 |               </div>
78 |             )
79 |           </div>
80 |
81 |           <div style={{ flex: 1, padding: '8px 0' }}>
82 |             <h1 style={{ fontSize: '30px', fontWeight: 700, display: 'flex', alignItems: 'center', gap: '1...
83 |               {profile.full_name}
84 |               {profile.role === 'admin' && <ShieldCheck style={{ color: 'var(--brand-primary)' }} size={24...
85 |             </h1>
86 |             <p style={{ color: 'var(--text-muted)', margin: '4px 0', display: 'flex', alignItems: 'center'...
87 |               {profile.role.toUpperCase()}
88 |               <span style={{ width: '4px', height: '4px', borderRadius: '50%', background: 'var(--text-mut...
89 |                 <span>Tham gia từ: {new Date(profile.created_at).toLocale...
90 |               </p>
91 |             </div>
92 |           </div>
93 |         </div>
94 |       </div>
95 |
96 |       <div style={{ background: 'var(--bg-surface)', borderRadius: '16px', border: '1px solid var(--border-d...
97 |         <div style={{ display: 'grid', gridTemplateColumns: 'repeat(2, 1fr)', gap: '16px', margin: '32px 0...
98 |           <StatCard label="Bài học" value="12" icon={BookOpen size={24}} />
99 |           <StatCard label="Giờ học" value="48h" icon={Clock size={24}} />
100 |          <StatCard label="Tôi học" value="2,450" icon={Trophy size={24}} />
101 |          <StatCard label="Huy hiệu" value="8" icon={Award size={24}} />
102 |        </div>
103 |
104 |         <div style={{ background: 'var(--bg-surface)', borderRadius: '16px', border: '1px solid var(--border-d...
105 |           <h3 style={{ font-size: '20px', font-weight: 700, margin: '24px 0', display: 'flex', align-items: '...
106 |           <div style={{ display: 'flex', flex-direction: 'column', gap: '24px' }}>
107 |             <div style={{ display: 'flex', flex-direction: 'column', gap: '24px' }}>
108 |               <ActivityItem title="Hoàn thành bài giảng: CPU Socket Types" />
109 |               <ActivityItem title="Tôi làm tốt trong bài thi giữa kỳ" date="2023-10-26" />
110 |               <ActivityItem title="Mô khóa học mới: Fast Learner" date="2023-10-26" />
111 |             </div>
112 |           </div>
113 |         </div>
114 |       </div>
115 |     )
116 |   )
117 | }
```



```

115 |         </div>
116 |     </main>
117 | </div>
118 | )
119 | }
120 |
121 | function StatCard({ label, value, icon, color }: any) {
122 |     return (
123 |         <div style={{ background: 'var(--bg-surface)', padding: '20px', borderRadius: '16px', border: '1px solid v...
124 |         <div style={{ width: '48px', height: '48px', borderRadius: '12px', display: 'flex', alignItems: 'center'...
125 |             {icon}
126 |         </div>
127 |         <div>
128 |             <p style={{ fontSize: '14px', color: 'var(--text-muted)', fontWeight: 500 }}>{label}</p>
129 |             <p style={{ fontSize: '24px', fontWeight: 700, marginTop: '2px', color: 'var(--text-primary)' }}>{valu...
130 |         </div>
131 |     </div>
132 |     )
133 | }
134 |
135 | function ActivityItem({ title, date }: any) {
136 |     return (
137 |         <div style={{ display: 'flex', gap: '16px', padding: '16px', borderRadius: '12px', background: 'var(--bg-b...
138 |         <div style={{ width: '8px', height: '8px', borderRadius: '50%', background: 'var(--brand-primary)', marg...
139 |         <div>
140 |             <p style={{ fontWeight: 500, color: 'var(--text-primary)' }}>{title}</p>
141 |             <p style={{ fontSize: '12px', color: 'var(--text-muted)', marginTop: '4px' }}>{date}</p>
142 |         </div>
143 |     </div>
144 |     )
145 | }
146 |

```

File: app/profile/page.tsx (495 lines, 30.9 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect, useRef } from 'react'
4 | import { motion, AnimatePresence } from 'framer-motion'
5 | import { getProfile } from '@lib/auth-actions'
6 | import { createBrowserClient } from '@supabase/ssr'
7 | import { BookOpen, Clock, Trophy, Award, Camera, Settings, Activity, FileText, ChevronRight, Edit3, ShieldChec...
8 | import Navbar from '@components/Navbar'
9 | import { toast } from 'react-hot-toast'
10 | import { updateProfile } from '@app/actions/profile'
11 | import { useIsMobile } from '@hooks/useIsMobile'
12 |
13 | export default function ProfilePage() {
14 |     const [profile, setProfile] = useState<any>(null)
15 |     const [loading, setLoading] = useState(true)
16 |     const [activeTab, setActiveTab] = useState('activity')
17 |     const [uploading, setUploading] = useState(false)
18 |     const fileInputRef = useRef<HTMLInputElement>(null)
19 |     const supabase = createBrowserClient(
20 |         process.env.NEXT_PUBLIC_SUPABASE_URL!,
21 |         process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
22 |     )
23 |
24 |     const [editMode, setEditMode] = useState(false)
25 |     const [fullName, setFullName] = useState('')
26 |     const [bio, setBio] = useState('')
27 |     const [school, setSchool] = useState('')
28 |     const [grade, setGrade] = useState('')
29 |     const [saving, setSaving] = useState(false)
30 |     const isMobile = useIsMobile()
31 |
32 |     useEffect(() => {
33 |         const params = new URLSearchParams(window.location.search)
34 |         const tabParam = params.get('tab')
35 |         if (tabParam && ['activity', 'exams', 'badges', 'settings'].includes(tabParam)) {
36 |             setActiveTab(tabParam)
37 |         }
38 |     }, [])
39 |
40 |     useEffect(() => {
41 |         const fetchProfile = async () => {
42 |             try {
43 |                 const data = await getProfile()
44 |                 setProfile(data)
45 |                 if (data) {
46 |                     setFullName(data.full_name || '')
47 |                     setBio(data.bio || '')
48 |                     setSchool(data.school || '')
49 |                     setGrade(data.grade || '')
50 |                 }
51 |             } catch (error) {
52 |                 console.error(error)
53 |             } finally {
54 |                 setLoading(false)
55 |             }
56 |         }
57 |         fetchProfile()
58 |     }, [])
59 |
60 |     const handleAvatarUpload = async (e: React.ChangeEvent<HTMLInputElement>) => {
61 |         const file = e.target.files?.[0]
62 |         if (!file) return
63 |
64 |         if (!file.type.startsWith('image/')) {
65 |             toast.error('Chỉ chấp nhận hình ảnh (JPEG/PNG/GIF)')
66 |             return
67 |         }
68 |         if (file.size > 2 * 1024 * 1024) {
69 |             toast.error('Kích thước ảnh tối đa 2MB.')
70 |             return
71 |         }
72 |
73 |         setUploading(true)
74 |         try {
75 |             const ext = file.name.split('.').pop()
76 |             const path = `${profile.id}/avatar-${Date.now()}.${ext}`
77 |
78 |             const { error: uploadError } = await supabase.storage
79 |                 .from('avatars')
80 |                 .upload(path, file, { upsert: true })
81 |
82 |             if (uploadError) throw uploadError
83 |
84 |             const { data } = supabase.storage.from('avatars').getPublicUrl(path)
85 |             const publicUrl = data.publicUrl
86 |
87 |             const { error: dbError } = await supabase
88 |                 .from('profiles')

```

```

89 |         .update({ avatar_url: publicUrl })
90 |         .eq('id', profile.id)
91 |
92 |         if (dbError) throw dbError
93 |
94 |         setProfile({ ...profile, avatar_url: publicUrl })
95 |         toast.success('Cập nhật thành công!')
96 |     } catch (err: any) {
97 |         toast.error(err.message || 'Lỗi khi tải lên.')
98 |     } finally {
99 |         setUploading(false)
100 |     }
101 | }
102 |
103 | const handleSaveProfile = async () => {
104 |     setSaving(true)
105 |     try {
106 |         await updateProfile({
107 |             full_name: fullName,
108 |             bio: bio.slice(0, 200),
109 |             school,
110 |             grade
111 |         })
112 |         setProfile({ ...profile, full_name: fullName, bio, school, grade })
113 |         setEditMode(false)
114 |         toast.success('Cập nhật thông tin cá nhân!')
115 |     } catch (err: any) {
116 |         toast.error('Lỗi khi lưu thông tin.')
117 |     } finally {
118 |         setSaving(false)
119 |     }
120 | }
121 |
122 | if (loading) {
123 |     return (
124 |         <div style={{ minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', alignItems: 'center', j...
125 |             <div style={{ width: '48px', height: '48px', borderRadius: '50%', border: '2px solid var(--brand-prima...
126 |         </div>
127 |     )
128 | }
129 |
130 | if (!profile) {
131 |     return (
132 |         <div style={{ minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', flexDirection: 'column'...
133 |             <h2 style={{ fontSize: '24px', fontWeight: 700, marginBottom: '10px' }}>Chào mừng bạn đến với...
134 |             <a href="/login" style={{ padding: '12px 24px', background: 'var(--brand-primary)', color: 'var(--bg-b...
135 |         </div>
136 |     )
137 | }
138 |
139 | const tabs = [
140 |     { id: 'activity', label: 'Hoạt động', icon: <Activity size={18} /> },
141 |     { id: 'exams', label: 'Kiểm tra', icon: <FileText size={18} /> },
142 |     { id: 'badges', label: 'Huân chương', icon: <Award size={18} /> },
143 |     { id: 'settings', label: 'Cài đặt', icon: <Settings size={18} /> },
144 | ]
145 |
146 | return (
147 |     <div style={{ minHeight: '100vh', background: 'var(--bg-base)', color: 'var(--text-primary)' }}>
148 |         <Navbar />
149 |
150 |         <main style={{ padding: '10px', paddingLeft: '16px', paddingRight: '16px', max...
151 |
152 |             <div style={{ background: 'var(--bg-surface)', borderRadius: '16px', border: '1px solid var(--border-d...
153 |                 <div style={{ height: '192px', background: 'linear-gradient(to right, color-mix(in srgb, var(--brand...
154 |                 <div style={{ position: 'absolute', inset: 0, opacity: 0.2, backgroundImage: 'radial-gradient(var(...
155 |                 </div>
156 |
157 |                 <div style={{ padding: '24px 40px 32px', position: 'relative' }}>
158 |                     <div style={{ display: 'flex', flexDirection: 'column', gap: '24px' }} className="md:flex-row md:i...
159 |                         <div style={{ position: 'relative', alignSelf: 'flex-start', marginTop: '-64px' }} className="md...
160 |                             <div style={{ width: '128px', height: '128px', borderRadius: '50%', border: '4px solid var(--b...
161 |                                 <img src={profile.avatar_url} alt="Avatar" style={{ width: '100%', height: '100%', objectF...
162 |                                 <div style={{ width: '100%', height: '100%', background: 'linear-gradient(to bottom right,...
163 |                                 {profile.full_name?.charAt(0).toUpperCase()}
164 |                             </div>
165 |                         </div>
166 |                         <input
167 |                             type="file"
168 |                             accept="image/*"
169 |                             ref={fileInputRef}
170 |                             onChange={handleAvatarUpload}
171 |                             style={{ display: 'none' }}
172 |                         />
173 |                         <button
174 |                             onClick={() => fileInputRef.current?.click()}
175 |                             disabled={uploading}
176 |                             style={{ position: 'absolute', bottom: '8px', right: '8px', padding: '10px', background: 'va...
177 |                         >
178 |                             {uploading ? <Loader size={18} className="animate-spin" /> : <Camera size={18} />}
179 |                         </button>
180 |                     </div>
181 |
182 |                     <div style={{ flex: 1, padding: '8px' }}>
183 |                         <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }} className="md:flex-row ...
184 |                             <div>
185 |                                 <h1 style={{ fontSize: '30px', fontWeight: 700, display: 'flex', alignItems: 'center', gap...
186 |                                     {profile.full_name}
187 |                                     {profile.role === 'admin' && <ShieldCheck style={{ color: 'var(--brand-primary)' }} size...
188 |                                 </h1>
189 |                                 <p style={{ color: 'var(--text-muted)', margin: '4px 0', display: 'flex', alignItems: 'ce...
190 |                                     {profile.email}
191 |                                     <span style={{ width: '4px', height: '4px', borderRadius: '50%', background: 'var(--text...
192 |                                     <span style={{ textTransform: 'uppercase', fontSize: '12px', fontWeight: 700, color: 'va...
193 |                                     {profile.role}
194 |                                 </span>
195 |                             </p>
196 |                         </div>
197 |
198 |                         <button
199 |                             onClick={() => setEditMode(!editMode)}
200 |                             style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '8px', padd...
201 |                         >
202 |                             <Edit size={18} /> {editMode ? 'Đóng' : 'Chỉnh sửa'}
203 |                         </button>
204 |                     </div>
205 |                 </div>
206 |             </main>
207 |         </div>
208 |     )
209 | }
210 |
211 | 2 1 1 |
212 |

```

```

213|         </div>
214|     </div>
215| </div>
216| </div>
217| </div>
218|
219| { /* Edit Mode */
220| <AnimatePresence>
221|     {editMode && (
222|         <motion.div
223|             initial={{ opacity: 0, height: 0 }}
224|             animate={{ opacity: 1, height: 'auto' }}
225|             exit={{ opacity: 0, height: 0 }}
226|             style={{ overflow: 'hidden', marginBottom: '32px' }}
227|         >
228|             <div style={{ background: 'var(--bg-surface)', borderRadius: '16px', border: '1px solid var(--bo...
229|                 <h3 style={{ fontSize: '18px', fontWeight: 700, marginBottom: '20px', display: 'flex', alignIt...
230|                     <User size={20} style={{ color: 'var(--brand-primary)' }} />
231|                         2 3 1 | C h Ê n h s i a t h ô n g t i n
232|                 </h3>
233|                 <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', maxWidth: '600px' }}>
234|                     <div style={{ display: 'grid', gridTemplateColumns: isMobile ? '1fr' : '1fr 1fr', gap: '16px...
235|                         <div>
236|                             <label style={{ display: 'block', fontSize: '12px', fontWeight: 700, textTransform: 'upp...
237|                                 <input
238|                                     type="text"
239|                                     value={fullName}
240|                                     onChange={(e) => setFullName(e.target.value)}
241|                                     style={{ width: '100%', padding: '12px 16px', borderRadius: '12px', background: 'var(-...
242|                                 />
243|                             </div>
244|                             <div>
245|                                 <label style={{ display: 'block', fontSize: '12px', fontWeight: 700, textTransform: 'upp...
246|                                     <input
247|                                         type="text"
248|                                         value={grade}
249|                                         onChange={(e) => setGrade(e.target.value)}
250|                                         placeholder="VD: 10A1"
251|                                         style={{ width: '100%', padding: '12px 16px', borderRadius: '12px', background: 'var(-...
252|                                     />
253|                                 </div>
254|                             </div>
255|                             <div>
256|                                 <label style={{ display: 'block', fontSize: '12px', fontWeight: 700, textTransform: 'upper...
257|                                     <input
258|                                         type="text"
259|                                         value={school}
260|                                         onChange={(e) => setSchool(e.target.value)}
261|                                             p l a c e h o l d e r = " T ê n t r ° Ỗ n g . . . "
262|                                     style={{ width: '100%', padding: '12px 16px', borderRadius: '12px', background: 'var(--b...
263|                                 />
264|                             </div>
265|                             <div>
266|                                 <div style={{ display: 'flex', justifyContent: 'space-between', marginBottom: '8px' }}>
267|                                     <label style={{ display: 'block', fontSize: '12px', fontWeight: 700, textTransform: 'upp...
268|                                     <span style={{ fontSize: '12px', color: 'var(--text-muted)' }}>{bio.length}/200</span>
269|                                 </div>
270|                                 <textarea
271|                                     rows={3}
272|                                     value={bio}
273|                                     maxLength={200}
274|                                     onChange={(e) => setBio(e.target.value)}
275|                                     p l a c e h o l d e r = " M ù t v à i d ò n g g i ô i t h i ç u . . . "
276|                                 />
277|                             </div>
278|                             <div style={{ display: 'flex', gap: '12px', padding: '8px' }}>
279|                                 <button
280|                                     onClick={handleSaveProfile}
281|                                     disabled={saving}
282|                                     style={{ padding: '12px 24px', background: 'var(--brand-primary)', color: 'var(--bg-base...
283|                                 >
284|                                     2 8 5 | { s a v i n g ? ' a n g l ° u . . . ' : ' L ° u t h a y Õ i ' }
285|                                 </button>
286|                                 <button
287|                                     onClick={() => setEditMode(false)}
288|                                     style={{ padding: '12px 24px', background: 'transparent', border: '1px solid var(--borde...
289|                                 >
290|                                     2 9 1 | H ç y
291|                             </div>
292|                         </div>
293|                     </div>
294|                 </div>
295|             </motion.div>
296|         )}
297|     </AnimatePresence>
298|
299| { /* STATS BAR */
300| <div style={{ display: 'grid', gridTemplateColumns: 'repeat(2, 1fr)', gap: '16px', marginBottom: '32px...
301|     <div style={{ background: 'var(--bg-surface)', padding: '20px', borderRadius: '16px', border: '1px s...
302|         <div style={{ width: '48px', height: '48px', borderRadius: '12px', background: 'color-mix(in srgb,...
303|             <BookOpen size={24} />
304|         </div>
305|         <div>
306|             3 0 7 | < p s t y l e = { { f o n t S i z e : ' 1 4 p x ' , c o l o r : ' v a r ( - - t e x t - m u t e d ) ' } ,
307|                 <p style={{ fontSize: '24px', fontWeight: 700, marginTop: '2px' }}>12<span style={{ fontSize: '1...
308|             </div>
309|         </div>
310|     </div>
311|
312|     <div style={{ background: 'var(--bg-surface)', padding: '20px', borderRadius: '16px', border: '1px s...
313|         <div style={{ width: '48px', height: '48px', borderRadius: '12px', background: 'color-mix(in srgb,...
314|             <Clock size={24} />
315|         </div>
316|         <div>
317|             3 1 7 | < p s t y l e = { { f o n t S i z e : ' 1 4 p x ' , c o l o r : ' v a r ( - - t e x t - m u t e d ) ' } ,
318|                 <p style={{ fontSize: '24px', fontWeight: 700, marginTop: '2px' }}>48<span style={{ fontSize: '1...
319|             </div>
320|         </div>
321|
322|     <div style={{ background: 'var(--bg-surface)', padding: '20px', borderRadius: '16px', border: '1px s...
323|         <div style={{ width: '48px', height: '48px', borderRadius: '12px', background: 'color-mix(in srgb,...
324|             <Trophy size={24} />
325|         </div>
326|         <div>
327|             3 2 7 | < p s t y l e = { { f o n t S i z e : ' 1 4 p x ' , c o l o r : ' v a r ( - - t e x t - m u t e d ) ' } ,
328|                 <p style={{ fontSize: '24px', fontWeight: 700, marginTop: '2px' }}>2,450 <span style={{ fontSize...
329|             </div>
330|         </div>
331|
332|     <div style={{ background: 'var(--bg-surface)', padding: '20px', borderRadius: '16px', border: '1px s...
333|         <div style={{ width: '48px', height: '48px', borderRadius: '12px', background: 'color-mix(in srgb,...
334|             <Award size={24} />
335|         </div>
336|         <div>

```

```
3 3 7 |
338 |         <p style={{ fontSize: '14px', color: 'var(--text-muted)',
339 |         </div>
340 |     </div>
341 | </div>
342 |
343 | /* TAB NAVIGATION & CONTENT */
344 | <div style={{ background: 'var(--bg-surface)', borderRadius: '16px', border: '1px solid var(--border-d...
345 |
346 |     /* Tabs */
347 |     <div style={{ display: 'flex', overflowX: 'auto', borderBottom: '1px solid var(--border-default)' }}>
348 |         {tabs.map((tab) => (
349 |             <button
350 |                 key={tab.id}
351 |                 onClick={() => setActiveTab(tab.id)}
352 |                 style={{
353 |                     display: 'flex', alignItems: 'center', gap: '8px', padding: '16px 24px', fontWeight: 500, fo...
354 |                     color: activeTab === tab.id ? 'var(--text-primary)' : 'var(--text-muted)'
355 |                 }}
356 |             >
357 |                 {tab.icon} {tab.label}
358 |                 {activeTab === tab.id && (
359 |                     <motion.div
360 |                         layoutId="profileTabIndicator"
361 |                         style={{ position: 'absolute', bottom: 0, left: 0, right: 0, height: '2px', background: 'v...
362 |                     </>
363 |                 )}
364 |             </button>
365 |         )]}
366 |     </div>
367 |
368 |     /* Tab Content */
369 |     <div style={{ padding: isMobile ? '16px' : '24px 32px' }}>
370 |
371 |         <AnimatePresence mode="wait">
372 |             {activeTab === 'activity' && (
373 |                 <motion.div key="activity" initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }} exit...
3 7 4 |                 <h3 style={{ fontSize: '18px', fontWeight: 700, margin:
375 |
376 |                     <div style={{ position: 'relative', borderLeft: '1px solid var(--border-default)', marginLef...
377 |                     <div style={{ position: 'relative', padding: '16px', height: '16px', background: 'var(--br...
378 |                     <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--br...
379 |                     <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--br...
380 |                     <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--br...
3 8 1 |                     <p style={{ font-weight: 500 }}> Hoàn thành bài h
3 8 2 |                     <p style={{ font-size: '14px', color: 'var(--text
383 |
384 |                 </div>
385 |                 </div>
386 |
387 |                 <div style={{ position: 'relative', padding: '32px' }}>
388 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
389 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
3 9 0 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
391 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
392 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
393 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
394 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
395 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
396 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
397 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
3 9 8 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
399 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
400 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
401 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
402 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
403 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
4 0 4 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
405 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
406 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
407 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
408 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
409 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
410 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
411 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
4 1 2 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
413 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
414 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
415 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
416 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
417 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
418 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
419 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
420 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
4 2 1 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
4 2 2 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
423 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
424 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
425 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
426 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
427 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
4 2 8 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
429 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
430 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
431 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
432 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
433 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
434 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
4 3 5 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
436 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
437 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
438 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
439 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
440 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
441 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
442 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
443 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
444 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
445 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
446 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
447 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
448 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
4 4 9 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
450 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
451 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
4 5 2 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
453 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
454 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
455 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
456 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
457 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
458 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
459 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
4 6 0 |                 <div style={{ position: 'absolute', width: '16px', height: '16px', background: 'var(--ac...
```

```

461 |
462 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '24px' }}>
463 |             <div style={{ background: 'var(--bg-base)', border: '1px solid var(--border-default)', b...
464 |                 < h 4 s t y l e = { { f o n t W e i g h t : 6 0 0 , m a r g i n B o t t o m : ' 1 6 p
465 |                     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
466 |                         <div>
467 |                             <label style={{ display: 'block', fontSize: '14px', fontWeight: 500, color: 'var(-...
468 |                             <input type="password" placeholder="*****" style={{ width: '100%', background: ...
469 |                         </div>
470 |                         <div>
471 |                             <label style={{ display: 'block', fontSize: '14px', fontWeight: 500, color: 'var(-...
472 |                             <input type="password" placeholder="*****" style={{ width: '100%', background: ...
473 |                         </div>
474 |                         <button style={{ alignSelf: 'flex-start', padding: '8px 16px', background: 'var(--bg...
475 |                     </div>
476 |                 </div>
477 |             <div style={{ background: 'var(--bg-base)', border: '1px solid var(--border-default)', b...
478 |                 < h 4 s t y l e = { { f o n t W e i g h t : 6 0 0 , m a r g i n B o t t o m : ' 4 p
479 |                     <p style={{ fontSize: '14px', color: 'var(--text-muted)', marginBottom: '16px' }}>Hành...
480 |                     <button style={{ padding: '8px 16px', background: 'color-mix(in srgb, #ef4444 10%, tra...
481 |                 </div>
482 |             </div>
483 |         </div>
484 |     </motion.div>
485 | </AnimatePresence>
486 | )}
487 | </div>
488 | </div>
489 | </div>
490 | </main>
491 | </div>
492 | )
493 | }
494 | }
495 |

```

File: app/profile/skills/page.tsx (129 lines, 6.3 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect } from 'react'
4 | import { motion } from 'framer-motion'
5 | import { Cpu, Layout, HardDrive, Zap, Shield, Microscope, ArrowLeft } from 'lucide-react'
6 | import Navbar from '@components/Navbar'
7 | import Link from 'next/link'
8 | import { supabase } from '@lib/supabase'
9 |
10 | export default function SkillTreePage() {
11 |     const [skills, setSkills] = useState<any[]>([])
12 |     const [loading, setLoading] = useState(true)
13 |
14 |     useEffect(() => {
15 |         fetchSkills()
16 |     }, [])
17 |
18 |     async function fetchSkills() {
19 |         try {
20 |             setLoading(true)
21 |             const { data: { user } } = await supabase.auth.getUser()
22 |             if (!user) return
23 |
24 |             // Fetch user progress
25 |             const { data: progress } = await supabase
26 |                 .from('lesson_progress')
27 |                 .select('*')
28 |                 .eq('student_id', user.id)
29 |                 .eq('is_completed', true)
30 |
31 |             const completedCount = progress?.length || 0
32 |
33 |             // Dynamic calculation logic (simplified for real feel)
34 |             const baseSkills = [
35 |                 { id: 'cpu', name: 'Master CPU', level: Math.min(100, completedCount * 15), icon: <Cpu size={24} />, c...
36 |                 { id: 'main', name: 'Mainboard Expert', level: Math.min(100, completedCount * 12), icon: <Layout size=...
37 |                 { id: 'ram', name: 'Memory Specialist', level: Math.min(100, completedCount * 20), icon: <Zap size={24}...
38 |                 { id: 'gpu', name: 'Graphics Guru', level: Math.min(100, completedCount * 8), icon: <Microscope size={...
39 |                 { id: 'storage', name: 'Storage Pro', level: Math.min(100, completedCount * 10), icon: <HardDrive size=...
40 |                 { id: 'security', name: 'System Security', level: Math.min(100, completedCount * 5), icon: <Shield siz...
41 |             ]
42 |             setSkills(baseSkills)
43 |         } catch (err) {
44 |             console.error(err)
45 |         } finally {
46 |             setLoading(false)
47 |         }
48 |     }
49 |
50 |     if (loading) return <div className="min-h-screen bg-[#0f0f1a] flex items-center justify-center text-[#00d2a0]...
51 |
52 |     return (
53 |         <div className="min-h-screen bg-[#0f0f1a] text-white">
54 |             <Navbar />
55 |
56 |             <main className="pt-28 pb-20 px-4 sm:px-6 lg:px-8 max-w-6xl mx-auto">
57 |                 <div className="flex items-center gap-4 mb-12">
58 |                     <Link href="/profile" className="p-2 hover:bg-[#16213e] rounded-xl transition-colors">
59 |                         <ArrowLeft size={20} />
60 |                     </Link>
61 |
62 |                     < h 1 c l a s s N a m e = " t e x t - 3 x l f o n t - b o l d " > C â y k ù n n g < s p a n c l a s s N a m e
63 |
64 |                 </div>
65 |
66 |                 <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-12 relative py-10">
67 |                     { /* SVG Lines Connector Placeholder (Conceptual) */ }
68 |                     <svg className="absolute inset-0 w-full h-full pointer-events-none opacity-10 hidden lg:block">
69 |                         <line x1="15%" y1="20%" x2="50%" y2="50%" stroke="white" strokeWidth="2" />
70 |                         <line x1="85%" y1="20%" x2="50%" y2="50%" stroke="white" strokeWidth="2" />
71 |                     </svg>
72 |
73 |                     {skills.map((skill, idx) => (
74 |                         <motion.div
75 |                             key={skill.id}
76 |                             initial={{ opacity: 0, scale: 0.8 }}
77 |                             animate={{ opacity: 1, scale: 1 }}
78 |                             transition={{ delay: idx * 0.1 }}
79 |                             className="relative group"
80 |                         >
81 |                             <div className="bg-[#16213e] border border-[#1e293b] p-8 rounded-[40px] text-center relative z-1...
82 |                                 <div
83 |                                     className="w-20 h-20 rounded-3xl mx-auto mb-6 flex items-center justify-center shadow-lg tra...
84 |                                     style={{ backgroundColor: `${skill.color}20`, color: skill.color, border: `2px solid ${skill...
85 |                                 >
86 |                                     {skill.icon}
87 |                                 </div>

```

```
86 | <h3 className="text-xl font-black mb-2">{skill.name}</h3>
87 | <div className="flex items-center justify-center gap-2 mb-6">
88 |   <div className="h-1.5 w-full bg-[#0f0f1a] rounded-full overflow-hidden">
89 |     <motion.div
90 |       initial={{ width: 0 }}
91 |       animate={{ width: `${skill.level}%` }}
92 |       transition={{ duration: 1, delay: 0.5 }}
93 |       className="h-full"
94 |       style={{ backgroundColor: skill.color }}
95 |     />
96 |   </div>
97 |   <span className="text-[10px] font-bold text-slate-500 whitespace-nowrap">{skill.level}%</span>
98 | </div>
99 | <div className="flex justify-center gap-1">
100 |   {[1, 2, 3, 4, 5].map(star => (
101 |     <div key={star} className={w-2 h-2 rounded-full ${star <= Math.round(skill.level/20) ? ''...
102 |   )]}
103 | </div>
104 | </div>
105 |
106 | {/ Connector dots */}
107 | <div className="absolute -bottom-4 left-1/2 -translate-x-1/2 w-8 h-8 bg-[#00d2a0]/10 rounded-ful...
108 |   <div className="w-2 h-2 bg-[#00d2a0] rounded-full animate-pulse" />
109 | </div>
110 | </motion.div>
111 | ))}
112 | </div>
113 |
114 | <div className="mt-20 p-10 bg-gradient-to-r from-[#16213e] to-[#0f0f1a] border border-[#1e293b] rounde...
115 |   <div className="flex-1">
116 |     <h4 className="text-2xl font-black mb-4">MÃ KHÓA CHÉNG CHẾ N
117 |     <p className="text-slate-400 leading-relaxed">
118 |       Hoàn thành tất cả các nhiệm vụ phần cứng β mức trên 80%
119 |     </p>
120 |     <div>
121 |       <button className="px-10 py-5 bg-[#00d2a0] text-black font-black rounded-2xl hover:scale-105 transi...
122 |         B @ T      | U   T H I   N G A Y
123 |       </button>
124 |     </div>
125 |   </main>
126 | </div>
127 | )
128 | }
129 |
```

File: app/quiz-bank/page.tsx (348 lines, 21.9 KB)

```
1 | 'use client'
2 |
3 | import React, { useState, useEffect, useMemo } from 'react'
4 | import { motion, AnimatePresence } from 'framer-motion'
5 | import { QUIZ_BANK } from '@data/quiz-bank'
6 | import { supabase } from '@lib/supabase'
7 | import {
8 |   Lock, Unlock, Search, ChevronRight, Zap,
9 |   Trophy, BookMarked, GraduationCap,
10 |   Sparkles, Layers, Brain, Star, Swords,
11 |   Crosshair, Diamond, Heart
12 | } from 'lucide-react'
13 | import { useIsMobile } from '@hooks/useIsMobile'
14 | import Link from 'next/link'
15 |
16 | const DICTIONARY = [
17 |   { term: 'CPU', def: 'Central Processing Unit - Bộ xử lý trung tâm, nã
18 |   { term: 'GPU', def: 'Graphics Processing Unit - Bộ xử lý th
19 |   { term: 'RAM', def: 'Random Access Memory - Bộ nhớ t
20 |   { term: 'SSD', def: 'Solid State Drive - Ổ cứng th
21 |   { term: 'HDD', def: 'Hard Disk Drive - Ổ cứng cũn
22 |   { term: 'PSU', def: 'Power Supply Unit - Bộ ngu
23 |   { term: 'TDP', def: 'Thermal Design Power - Công s
24 |   { term: 'PCIe', def: 'Peripheral Component Interconne
25 |   { term: 'BIOS', def: 'Basic Input/Output System - H
26 |   { term: 'UEFI', def: 'Unified Extensible Firmware Int
27 |   { term: 'VRAM', def: 'Video RAM - Bộ nhớ h
28 |   { term: 'DLSS', def: 'Deep Learning Super Sampling - Công n
29 |   { term: 'XMP', def: 'Extreme Memory Profile - H
30 |   { term: 'DOCP', def: 'Direct Over Clock Profile - H
31 |   { term: 'NVMe', def: 'Non-Volatile Memory Express - G
32 |   { term: 'M.2', def: 'Chuẩn kết nối SSD nh
33 |   { term: 'SATA', def: 'Serial ATA - Chu
34 |   { term: 'RGB', def: 'Red Green Blue - H
35 |   { term: 'AIoT', def: 'All-In-One - H
36 |   { term: 'RGB Header', def: 'Chân c
37 |   { term: 'Form Factor', def: 'Kích th
38 |   { term: 'Overclocking', def: 'Ép x
39 |   { term: 'Thermal Throttling', def: 'Gi
40 |   { term: 'Dual Channel', def: 'Chi
41 |   { term: 'Cache', def: 'Bộ nh
42 |   { term: 'Pipeline', def: 'D
43 |   { term: 'Hyper-Threading', def: 'C
44 |   { term: 'ECC RAM', def: 'Error-Correc
45 |   { term: 'CAS Latency', def: 'Ù
46 |   { term: 'Ray Tracing', def: 'Mô
47 | ]
48 |
49 | const UNLOCK_KEY = 'quiz_bank_unlock_date'
50 | const DAILY_UNLOCK = 2
51 |
52 | const TOPIC_COLORS = [
53 |   { bg: 'color-mix(in srgb, var(--accent-blue) 12%, transparent)', border: 'color-mix(in srgb, var(--accent-bl...
54 |   { bg: 'color-mix(in srgb, var(--brand-primary) 12%, transparent)', border: 'color-mix(in srgb, var(--brand-p...
55 |   { bg: 'color-mix(in srgb, var(--accent-amber) 12%, transparent)', border: 'color-mix(in srgb, var(--accent-a...
56 |   { bg: 'color-mix(in srgb, var(--danger) 12%, transparent)', border: 'color-mix(in srgb, var(--danger) 25%, t...
57 |   { bg: 'color-mix(in srgb, var(--accent-purple) 12%, transparent)', border: 'color-mix(in srgb, var(--accent-...
58 |   { bg: 'color-mix(in srgb, var(--primary-neon) 12%, transparent)', border: 'color-mix(in srgb, var(--primary-...
59 | ]
60 |
61 | const DIFFICULTY_COLORS: Record<string, string> = {
62 |   'D 1': '#34d399',
63 |   'Trung bình': '#fbbf24',
64 |   'Khó': '#f87171',
65 | }
66 |
67 | export default function QuizBankPage() {
68 |   const isMobile = useIsMobile()
69 |   const [activeTab, setActiveTab] = useState<'bank' | 'dictionary'>('bank')
70 |   const [searchTerm, setSearchTerm] = useState('')
71 |   const [unlockedCount, setUnlockedCount] = useState(2)
72 |   const [completedQuizzes, setCompletedQuizzes] = useState<string[]>([])
73 |   const [xpEarned, setXpEarned] = useState(0)
74 |
75 |   useEffect(() => {
76 |     checkDailyUnlock()
```

```

77 |     loadCompletedQuizzes()
78 | }, [])
79 |
80 | function checkDailyUnlock() {
81 |     const stored = localStorage.getItem(UNLOCK_KEY)
82 |     const today = new Date().toISOString().slice(0, 10)
83 |     if (stored !== today) {
84 |         localStorage.setItem(UNLOCK_KEY, today)
85 |         const baseUnlocked = 2
86 |         const extra = Math.floor(QUIZ_BANK.length / 2)
87 |         setUnlockedCount(Math.min(baseUnlocked + extra, QUIZ_BANK.length))
88 |     } else {
89 |         const saved = localStorage.getItem('quiz_bank_unlocked_count')
90 |         setUnlockedCount(saved ? parseInt(saved) : 2)
91 |     }
92 | }
93 |
94 | async function loadCompletedQuizzes() {
95 |     try {
96 |         const { data: { user } } = await supabase.auth.getUser()
97 |         if (!user) return
98 |         const { data } = await supabase
99 |             .from('quiz_attempts')
100 |             .select('quiz_id')
101 |             .eq('student_id', user.id)
102 |             .eq('status', 'passed')
103 |         if (data) {
104 |             setCompletedQuizzes(data.map(d => d.quiz_id))
105 |         }
106 |     } catch (e) {}
107 | }
108 |
109 | const filteredQuizzes = useMemo(() => {
110 |     return QUIZ_BANK.filter(q =>
111 |         !searchTerm || q.title.toLowerCase().includes(searchTerm.toLowerCase()) ||
112 |         q.lessonTitle?.toLowerCase().includes(searchTerm.toLowerCase())
113 |     )
114 | }, [searchTerm])
115 |
116 | const filteredDictionary = useMemo(() => {
117 |     if (!searchTerm) return DICTIONARY
118 |     return DICTIONARY.filter(d =>
119 |         d.term.toLowerCase().includes(searchTerm.toLowerCase()) ||
120 |         d.def.toLowerCase().includes(searchTerm.toLowerCase())
121 |     )
122 | }, [searchTerm])
123 |
124 | const progressPercent = (completedQuizzes.length / QUIZ_BANK.length) * 100
125 |
126 | return (
127 |     <div style={{ minHeight: '100vh', background: 'var(--bg-base)', color: 'var(--text-primary)' }}>
128 |         <div style={{
129 |             position: 'fixed', top: 0, left: 0, right: 0, height: '280px',
130 |             background: 'radial-gradient(ellipse 80% 60% at 50% -20%, color-mix(in srgb, var(--accent-blue) 8%, tr...
131 |             pointerEvents: 'none', zIndex: 0,
132 |         }} />
133 |
134 |         <div style={{ position: 'relative', zIndex: 1, maxWidth: '1100px', margin: '0 auto', padding: isMobile ?...
135 |
136 |             /* Pixel Header */
137 |             <motion.div initial={{ opacity: 0, y: -20 }} animate={{ opacity: 1, y: 0 }}
138 |                 style={{ textAlign: 'center', marginBottom: '28px' }}>
139 |                 <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '12px', marginBo...
140 |                     <motion.div animate={{ y: [0, -6, 0] }} transition={{ duration: 1.5, repeat: Infinity }}
141 |                         style={{
142 |                             width: '56px', height: '56px', borderRadius: '12px',
143 |                             background: 'linear-gradient(135deg, var(--accent-blue), var(--accent-purple))',
144 |                             display: 'flex', alignItems: 'center', justifyContent: 'center',
145 |                             boxShadow: '0 0 24px color-mix(in srgb, var(--accent-blue) 40%, transparent)',
146 |                             imageRendering: 'pixelated',
147 |                         }}>
148 |                         <Crosshair size={28} color="#fff" />
149 |                     </motion.div>
150 |                     <h1 style={{
151 |                         fontSize: isMobile ? '24px' : '34px', fontWeight: 900,
152 |                         background: 'linear-gradient(135deg, var(--accent-blue), var(--brand-primary))',
153 |                         WebkitBackgroundClip: 'text', WebkitTextFillColor: 'transparent',
154 |                         letterSpacing: '1px',
155 |                     }}>
156 |                         N G Â N H À N G À T H I
157 |                     </h1>
158 |                 </div>
159 |
160 |                 /* Pixel progress bar */
161 |                 <div style={{ maxWidth: '400px', margin: '0 auto 16px' }}>
162 |                     <div style={{ display: 'flex', justifyContent: 'space-between', fontSize: '12px', color: 'var(--te...
163 |                         <span>Completed Quizzes</span> <span>T i c h </span>
164 |                     </div>
165 |                     <div style={{ width: '100%', height: '12px', background: 'var(--border-subtle)', borderRadius: '2p...
166 |                         <motion.div initial={{ width: 0 }} animate={{ width: `${progressPercent}%` }}
167 |                             style={{
168 |                                 height: '100%', borderRadius: '1px',
169 |                                 background: 'repeating-linear-gradient(90deg, var(--brand-primary) 0px, var(--brand-primary)...
170 |                                 transition: 'width 0.8s ease',
171 |                             }} />
172 |                         </div>
173 |                     </div>
174 |                 </div>
175 |
176 |                 <p style={{ color: 'var(--text-muted)', fontSize: '13px', margin: '0 auto 12px', maxWidth: '500px' }}>
177 |                     M i n g à y m ß k h ó a { D A I L Y _ U N L O C K } c h ç Á . { Q U I Z _ B A N K . l e n g t h }
178 |                 </p>
179 |
180 |                 /* Stats badges */
181 |                 <div style={{ display: 'flex', justifyContent: 'center', gap: '12px', flexWrap: 'wrap', fontSize: '1...
182 |                     <motion.div whileHover={{ scale: 1.05, y: -1 }}
183 |                         style={{ display: 'flex', alignItems: 'center', gap: '6px', padding: '6px 14px', borderRadius: '...
184 |                         <Unlock size={13} /> {unlockedCount}/{QUIZ_BANK.length}
185 |                     </motion.div>
186 |                     <motion.div whileHover={{ scale: 1.05, y: -1 }}
187 |                         style={{ display: 'flex', alignItems: 'center', gap: '6px', padding: '6px 14px', borderRadius: '...
188 |                         <Zap size={13} /> {xpEarned} XP
189 |                     </motion.div>
190 |                 </div>
191 |             </motion.div>
192 |
193 |             /* Tabs */
194 |             <div style={{ display: 'flex', gap: '6px', marginBottom: '24px', justifyContent: 'center' }}>
195 |                 {[
196 |                     { id: 'bank' as const, label: 'C â u h i i', icon: Swords, color: 'var(--text-primary)' },
197 |                     { id: 'dictionary' as const, label: 'T è i Ā n t h u - t n g i', icon: Dictionary }
198 |                 ].map(tab => (
199 |                     <motion.button key={tab.id} onClick={() => setActiveTab(tab.id)} whileHover={{ scale: 1.02 }} while...
200 |                         style={{

```

```

201|         display: 'flex', alignItems: 'center', gap: '6px', padding: '8px 18px',
202|         borderRadius: '4px', border: '1px solid', cursor: 'pointer', fontFamily: 'inherit',
203|         fontSize: '12px', fontWeight: activeTab === tab.id ? 800 : 500,
204|         borderColor: activeTab === tab.id ? tab.color : 'var(--border-default)',
205|         background: activeTab === tab.id ? 'color-mix(in srgb, ' + tab.color + ' 12%, transparent)' : ...
206|         color: activeTab === tab.id ? tab.color : 'var(--text-muted)',
207|         transition: 'all 0.2s',
208|         textTransform: 'uppercase',
209|         letterSpacing: '0.5px',
210|     }}>
211|     <tab.icon size={14} /> {tab.label}
212| </motion.button>
213| ))}
214| </div>
215|
216| { /* Search */
217| <motion.div initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }}
218|   style={{ position: 'relative', maxWidth: '400px', margin: '0 auto 24px' }}>
219|   <Search size={14} style={{ position: 'absolute', left: '14px', top: '50%', transform: 'translateY(-5...
220|   <input value={searchTerm} onChange={e => setSearchTerm(e.target.value)}
221|     2 2 1 |         p l a c e h o l d e r = "0=Y   T i m   k i z m   c h c   Á   h o . c   t h u - t   n g i . . . "
222|     style={{
223|       width: '100%', padding: '10px 16px 10px 38px', borderRadius: '4px',
224|       border: '1px solid var(--border-default)', background: 'transparent',
225|       color: 'var(--text-primary)', fontSize: '13px', outline: 'none',
226|       fontFamily: 'inherit',
227|     }} />
228| </motion.div>
229|
230| { /* Content */
231| {activeTab === 'bank' ? (
232|   <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
233|     <AnimatePresence mode="popLayout">
234|       {filteredQuizzes.map((quiz, index) => {
235|         const isUnlocked = index < unlockedCount
236|         const isCompleted = completedQuizzes.includes(quiz.id)
237|         const color = TOPIC_COLORS[index % TOPIC_COLORS.length]
238|         const diffColor = DIFFICULTY_COLORS[quiz.difficulty] || 'var(--accent-blue)'}
239|
240|     return (
241|       <motion.div key={quiz.id} layout initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0...
242|       {isUnlocked ? (
243|         <Link href={`/quiz/${quiz.id}`} style={{ textDecoration: 'none' }}>
244|           <motion.div whileHover={{ scale: 1.01, y: -2 }} whileTap={{ scale: 0.99 }}
245|             style={{
246|               display: 'flex', alignItems: 'center', gap: '14px', padding: '16px 18px',
247|               borderRadius: isCompleted ? '4px' : '4px',
248|               background: isCompleted ? 'color-mix(in srgb, var(--brand-primary) 6%, transparent...
249|               border: '1px solid ${isCompleted ? 'color-mix(in srgb, var(--brand-primary) 20%, t...
250|               cursor: 'pointer', transition: 'all 0.2s',
251|               position: 'relative', overflow: 'hidden',
252|             }}>
253|               { /* Pixel corner decorations */
254|               <div style={{ position: 'absolute', top: 0, left: 0, width: '6px', height: '6px', bo...
255|               <div style={{ position: 'absolute', top: 0, right: 0, width: '6px', height: '6px', b...
256|               <div style={{ position: 'absolute', bottom: 0, left: 0, width: '6px', height: '6px',...
257|               <div style={{ position: 'absolute', bottom: 0, right: 0, width: '6px', height: '6px'...
258|
259|               <div style={{
260|                 width: '44px', height: '44px', borderRadius: '4px',
261|                 background: isCompleted ? 'color-mix(in srgb, var(--brand-primary) 15%, transparen...
262|                 border: '1px solid ${isCompleted ? 'color-mix(in srgb, var(--brand-primary) 20%, t...
263|                 display: 'flex', alignItems: 'center', justifyContent: 'center', flexShrink: 0,
264|               }}>
265|                 {isCompleted ? <Diamond size={18} style={{ color: 'var(--brand-primary)' }} /> : <...
266|               </div>
267|               <div style={{ flex: 1, minWidth: 0 }}>
268|                 <div style={{ display: 'flex', alignItems: 'center', gap: '8px', marginBottom: '4p...
269|                   <span style={{ fontSize: '14px', fontWeight: 800, color: 'var(--text-primary)', ...
270|                     {isCompleted && (
271|                       <motion.span initial={{ scale: 0 }} animate={{ scale: 1 }}
272|                         style={{ fontSize: '9px', padding: '2px 7px', borderRadius: '2px', backgroun...
273|                         CLEAR
274|                       </motion.span>
275|                     )}
276|                   </div>
277|                   <div style={{ display: 'flex', gap: '8px', fontSize: '11px', color: 'var(--text-mu...
278|                     <span style={{ display: 'flex', alignItems: 'center', gap: '3px', fontFamily: 'V...
279|                       <Layers size={11} /> {quiz.questions?.length || 10}
280|                     </span>
281|                     <span style={{ width: '2px', height: '2px', borderRadius: '50%', background: 'va...
282|                       <span style={{ color: diffColor, fontWeight: 700, fontSize: '10px', padding: '1p...
283|                       <span style={{ width: '2px', height: '2px', borderRadius: '50%', background: 'va...
284|                       <span style={{ display: 'flex', alignItems: 'center', gap: '3px', color: 'var(--...
285|                         <Zap size={11} /> +{quiz.xp || 100}
286|                       </span>
287|                     </div>
288|                   </div>
289|                   <motion.div whileHover={{ x: 3 }} style={{ flexShrink: 0, color: 'var(--text-muted)' }...
290|                     <ChevronRight size={18} />
291|                   </motion.div>
292|                 </motion.div>
293|               </Link>
294|             ) : (
295|               <div style={{
296|                 display: 'flex', alignItems: 'center', gap: '14px', padding: '16px 18px',
297|                 borderRadius: '4px',
298|                 background: 'transparent',
299|                 border: '1px solid var(--border-subtle)', opacity: 0.5,
300|               }}>
301|                 <div style={{ width: '44px', height: '44px', borderRadius: '4px', background: 'var(--b...
302|                   <Lock size={18} style={{ color: 'var(--text-muted)' }} />
303|                 </div>
304|                 <div style={{ flex: 1 }}>
305|                   <div style={{ fontSize: '14px', fontWeight: 700, color: 'var(--text-muted)', fontFam...
306|                     <div style={{ fontSize: '11px', color: 'var(--text-muted)' }}>
307|                       3 0 7 |         0=Y   M B   k h o á   s a u   .   { quiz . q u e s t i o n s ? . l e n g t h   | |
308|                     </div>
309|                   </div>
310|                 </div>
311|               </motion.div>
312|             )}
313|           </div>
314|         </div>
315|       </AnimatePresence>
316|     </div>
317|   ) : (
318|     <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }} style={{ display: 'flex', flexDirectio...
319|     <div style={{ fontSize: '12px', color: 'var(--text-muted)', marginBottom: '8px', textAlign: 'cente...
320|       [ { f i l t e r e d D i c t i o n a r y . l e n g t h }   t h u - t   n g i ]
321|     </div>
322|     {filteredDictionary.map((item, i) => (
323|       <motion.div key={item.term} initial={{ opacity: 0, x: -10 }} animate={{ opacity: 1, x: 0 }} tran...
324|       whileHover={{ scale: 1.005, x: 4 }}

```



```

325 |         style={{ padding: '10px 14px', borderRadius: '4px', background: 'var(--bg-surface)', border: '...
326 |         <div style={{ display: 'flex', alignItems: 'center', gap: '10px' }}>
327 |             <div style={{
328 |                 width: '32px', height: '32px', borderRadius: '2px',
329 |                 background: 'linear-gradient(135deg, color-mix(in srgb, var(--brand-primary) 15%, transpar...
330 |                 border: '1px solid color-mix(in srgb, var(--brand-primary) 15%, transparent)',
331 |                 display: 'flex', alignItems: 'center', justifyContent: 'center', flexShrink: 0,
332 |                 fontSize: '10px', fontWeight: 800, color: 'var(--brand-primary)',
333 |                 fontFamily: 'var(--font-mono)',
334 |             }}>{item.term.slice(0, 2).toUpperCase()}</div>
335 |             <div style={{ flex: 1 }}>
336 |                 <div style={{ fontSize: '13px', fontWeight: 700, color: 'var(--text-primary)', fontFamily:...
337 |                 <div style={{ fontSize: '11px', color: 'var(--text-muted)', marginTop: '1px' }}>{item.def}...
338 |             </div>
339 |         </div>
340 |     </motion.div>
341 | ))}
342 | </motion.div>
343 | ))}
344 | </div>
345 | </div>
346 | )
347 | }
348 |

```

File: app/quiz/[quizId]/page.tsx (476 lines, 20.6 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect, use } from 'react'
4 | import { motion, AnimatePresence } from 'framer-motion'
5 | import { supabase } from '@lib/supabase'
6 | import { QUIZ_BANK } from '@data/quiz-bank'
7 | import {
8 |   Trophy, Flame, Clock, Swords,
9 |   CheckCircle, XCircle, ChevronRight, Crosshair,
10 |   Loader2, Sparkles, Star, Heart, Skull,
11 |   Target, Diamond, Zap
12 | } from 'lucide-react'
13 | import confetti from 'canvas-confetti'
14 | import Link from 'next/link'
15 |
16 | const SHAPES = [ '+', 'x', 'o', 'i', ' ' ]
17 | const COLORS = [ '#ef4444', '#3b82f6', '#f59e0b', '#8b5cf6' ]
18 |
19 | export default function MiniQuizPage({ params }: { params: Promise<{ quizId: string }> }) {
20 |   const resolvedParams = use(params);
21 |   const { quizId } = resolvedParams;
22 |
23 |   const [questions, setQuestions] = useState<any[]>([])
24 |   const [currentIdx, setCurrentIdx] = useState(-1)
25 |   const [score, setScore] = useState(0)
26 |   const [streak, setStreak] = useState(0)
27 |   const [loading, setLoading] = useState(true)
28 |   const [feedback, setFeedback] = useState<'correct' | 'wrong' | null>(null)
29 |   const [selectedIdx, setSelectedIdx] = useState<number | null>(null)
30 |   const [timeLeft, setTimeLeft] = useState(15)
31 |   const [gameEnded, setGameEnded] = useState(false)
32 |   const [quizTitle, setQuizTitle] = useState('')
33 |   const [quizXp, setQuizXp] = useState(100)
34 |   const [shake, setShake] = useState(false)
35 |
36 |   useEffect(() => {
37 |     initQuiz()
38 |   }, [])
39 |
40 |   async function initQuiz() {
41 |     try {
42 |       setLoading(true)
43 |       const bankTopic = QUIZ_BANK.find(q => q.id === quizId)
44 |       if (bankTopic) {
45 |         const qs = bankTopic.questions.map((q: any) => ({
46 |           question: q.q,
47 |           options: q.options,
48 |           correctIndex: q.answer,
49 |           explanation: q.q,
50 |         }))
51 |         setQuestions(qs)
52 |         setQuizTitle(bankTopic.title)
53 |         setQuizXp(bankTopic.xp || 100)
54 |         setLoading(false)
55 |         setCurrentIdx(0)
56 |         return
57 |       }
58 |       const { data: section } = await supabase
59 |         .from('lesson_sections')
60 |         .select('*', { count: 'exact' })
61 |         .eq('id', quizId)
62 |         .single()
63 |       if (!section) throw new Error('Không tìm thấy bài tr`c nghi`m')
64 |       setQuizTitle(section.lessons?.title || 'Tr`c nghi`m')
65 |       const res = await fetch('/api/ai/generate-quiz', {
66 |         method: 'POST',
67 |         headers: { 'Content-Type': 'application/json' },
68 |         body: JSON.stringify({
69 |           lessonTitle: section.lessons.title,
70 |           lessonContent: section.content_body || section.lessons.description,
71 |           count: 5
72 |         })
73 |       })
74 |       const quizData = await res.json()
75 |       setQuestions(quizData)
76 |       setLoading(false)
77 |       setCurrentIdx(0)
78 |     } catch (err) {
79 |       console.error('Quiz init error:', err)
80 |       setLoading(false)
81 |     }
82 |   }
83 |
84 |   useEffect(() => {
85 |     if (currentIdx < 0 || gameEnded || feedback) return
86 |     if (timeLeft <= 0) {
87 |       handleAnswer(-1)
88 |       return
89 |     }
90 |     const timer = setInterval(() => setTimeLeft(prev => prev - 1), 1000)
91 |     return () => clearInterval(timer)
92 |   }, [timeLeft, currentIdx, gameEnded, feedback])
93 |
94 |   const handleAnswer = (index: number) => {
95 |     if (feedback) return
96 |     setSelectedIdx(index)

```

```

97 | const correct = index === questions[currentIdx].correctIndex
98 | if (correct) {
99 |   setFeedback('correct')
100 |   const timeBonus = timeLeft * 10
101 |   const streakBonus = streak * 50
102 |   setScore(prev => prev + 100 + timeBonus + streakBonus)
103 |   setStreak(prev => prev + 1)
104 |   confetti({ particleCount: 40, spread: 50, origin: { y: 0.7 }, colors: ['#00d4aa', '#00f3ff', '#a78bfa'] })
105 | } else {
106 |   setFeedback('wrong')
107 |   setStreak(0)
108 |   setShake(true)
109 |   setTimeout(() => setShake(false), 500)
110 | }
111 |
112 |
113 | setTimeout(() => {
114 |   if (currentIdx < questions.length - 1) {
115 |     setCurrentIdx(prev => prev + 1)
116 |     setFeedback(null)
117 |     setSelectedIdx(null)
118 |     setTimeLeft(15)
119 |   } else {
120 |     setGameEnded(true)
121 |     confetti({ particleCount: 200, spread: 80, origin: { y: 0.5 }, colors: ['#00d4aa', '#00f3ff', '#fbbf24...
122 |   }
123 | }, 3000)
124 | }
125 |
126 | if (loading) return (
127 |   <div style={{ minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', flexDirection: 'column', ...
128 |     <motion.div animate={{ rotate: 360 }} transition={{ duration: 2, repeat: Infinity, ease: 'linear' }}>
129 |       <Crosshair size={64} style={{ color: 'var(--brand-primary)' }} />
130 |     </motion.div>
131 |     <h1 style={{ fontSize: '28px', fontWeight: 900, marginTop: '24px', marginBottom: '8px', color: 'var(--te...
132 |       LOADING...
133 |     </h1>
134 |     <p style={{ color: 'var(--text-muted)', fontSize: '13px', fontFamily: 'var(--font-mono)' }}>
135 |       C h u @ n b E v i k h i c h o b i n
136 |     </p>
137 |   </div>
138 | )
139 |
140 | if (gameEnded) return (
141 |   <div style={{ minHeight: '100vh', background: 'var(--bg-base)', display: 'flex', flexDirection: 'column', ...
142 |     <motion.div initial={{ scale: 0.8, opacity: 0 }} animate={{ scale: 1, opacity: 1 }} style={{ textAlign: ...
143 |     <motion.div animate={{ y: [0, -10, 0] }} transition={{ duration: 2, repeat: Infinity }}>
144 |       <Trophy size={100} style={{ color: 'var(--accent-amber)', margin: '0 auto 24px' }} />
145 |     </motion.div>
146 |     <h1 style={{ fontSize: '42px', fontWeight: 900, marginBottom: '12px', letterSpacing: '3px', fontFamily...
147 |       VICTORY!
148 |     </h1>
149 |     <p style={{ fontSize: '18px', color: 'var(--text-muted)', marginBottom: '12px', fontFamily: 'var(--fon...
150 |       {quizTitle}
151 |     </p>
152 |     <div style={{
153 |       display: 'inline-block',
154 |       padding: '16px 32px',
155 |       background: 'var(--bg-surface)',
156 |       border: '2px solid var(--brand-primary)',
157 |       margin: '0 auto 24px',
158 |     }}>
159 |       <div style={{ fontSize: '10px', color: 'var(--text-muted)', margin: '0 auto 4px', fontFamily: 'var(--...
160 |       <div style={{ fontSize: '56px', color: 'var(--brand-primary)', margin: '0 auto 4px', fontFamily: 'var(--f...
161 |     </div>
162 |     <div style={{ display: 'flex', gap: '12px', justifyContent: 'center' }}>
163 |       <Link href="/quiz-bank" style={{
164 |         padding: '12px 28px',
165 |         border: '1px solid var(--border-strong)',
166 |         color: 'var(--text-primary)',
167 |         fontSize: '13px',
168 |         fontWeight: 700,
169 |         fontFamily: 'var(--font-mono)',
170 |         letterSpacing: '1px',
171 |         textDecoration: 'none',
172 |         display: 'inline-flex',
173 |         alignItems: 'center',
174 |         gap: '8px',
175 |       }}>
176 |         1 7 7 | !• V À T R A N G C H æ
177 |       </Link>
178 |       <button onClick={() => window.location.reload()} style={{
179 |         padding: '12px 28px',
180 |         background: 'var(--brand-primary)',
181 |         color: 'var(--bg-base)',
182 |         border: 'none',
183 |         fontSize: '13px',
184 |         fontWeight: 700,
185 |         fontFamily: 'var(--font-mono)',
186 |         letterSpacing: '1px',
187 |         cursor: 'pointer',
188 |         display: 'inline-flex',
189 |         alignItems: 'center',
190 |         gap: '8px',
191 |       }}>
192 |         1 9 3 | & C H I L I
193 |       </button>
194 |     </div>
195 |   </motion.div>
196 | </div>
197 | )
198 |
199 | const q = questions[currentIdx]
200 |
201 | return (
202 |   <div style={{
203 |     minHeight: '100vh',
204 |     background: 'var(--bg-base)',
205 |     color: 'var(--text-primary)',
206 |     display: 'flex',
207 |     flexDirection: 'column',
208 |     padding: '16px',
209 |     overflow: 'hidden',
210 |     position: 'relative',
211 |   }}>
212 |     <div style={{
213 |       position: 'fixed', top: 0, left: 0, right: 0, bottom: 0,
214 |       backgroundImage:
215 |         linear-gradient(rgba(255,255,255,0.02) 1px, transparent 1px),
216 |         linear-gradient(90deg, rgba(255,255,255,0.02) 1px, transparent 1px)
217 |       ,
218 |       backgroundSize: '32px 32px',
219 |     }}>
220 |

```

```

221 |     pointerEvents: 'none',
222 |     zIndex: 0,
223 | } } />
224 |
225 | <div style={{ position: 'relative', zIndex: 1, flex: 1, display: 'flex', flexDirection: 'column', maxWid...
226 | /* HUD */
227 | <header style={{ display: 'flex', alignItems: 'center', justifyContent: 'space-between', marginBottom:...
228 | <div style={{ display: 'flex', alignItems: 'center', gap: '10px' }}>
229 |   <div style={{
230 |     padding: '6px 14px',
231 |     border: '1px solid var(--border-default)',
232 |     background: 'var(--bg-surface)',
233 |   }}>
234 |     <span style={{ fontSize: '11px', fontWeight: 700, color: 'var(--brand-primary)', fontFamily: 'va...
235 |       {{currentIdx+1}}/{{questions.length}}
236 |     </span>
237 |   </div>
238 |   <div style={{
239 |     padding: '6px 14px',
240 |     border: '1px solid var(--border-default)',
241 |     background: 'var(--bg-surface)',
242 |     display: 'flex', alignItems: 'center', gap: '6px',
243 |   }}>
244 |     <Star size={11} style={{ color: 'var(--accent-amber)' }} />
245 |     <span style={{ fontSize: '11px', fontWeight: 700, color: 'var(--text-muted)', fontFamily: 'var(--...
246 |   </div>
247 |   /* HP Bar */
248 |   <div style={{
249 |     padding: '6px 12px',
250 |     border: '1px solid var(--border-default)',
251 |     background: 'var(--bg-surface)',
252 |     display: 'flex', alignItems: 'center', gap: '6px',
253 |   }}>
254 |     <Heart size={11} style={{ color: streak > 0 ? '#ef4444' : 'var(--text-muted)' }} />
255 |     <div style={{ width: '50px', height: '8px', background: 'var(--border-subtle)', border: '1px sol...
256 |       <div style={{
257 |         width: `${Math.max(10, 100 - (streak > 0 ? 0 : 30))}%`,
258 |         height: '100%',
259 |         background: streak > 0 ? '#ef4444' : 'var(--text-muted)',
260 |         transition: 'width 0.3s',
261 |       }} />
262 |     </div>
263 |   </div>
264 | </div>
265 | <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
266 |   {streak > 1 && (
267 |     <motion.div initial={{ scale: 0 }} animate={{ scale: 1 }}
268 |       style={{ display: 'flex', alignItems: 'center', gap: '6px', color: 'var(--accent-amber)', font...
269 |       <Flame size={16} style={{ color: 'var(--accent-amber)' }} /> x{streak}
270 |     </motion.div>
271 |   )}
272 |   <div style={{
273 |     padding: '8px 20px',
274 |     border: '1px solid var(--border-default)',
275 |     background: timeLeft <= 5 ? 'color-mix(in srgb, #ef4444 15%, transparent)' : 'var(--bg-surface)',
276 |     fontSize: '22px', fontWeight: 900,
277 |     fontFamily: 'var(--font-mono)',
278 |     color: timeLeft <= 5 ? '#ef4444' : 'var(--text-primary)',
279 |     transition: 'all 0.3s',
280 |   }}>
281 |     {timeLeft}s
282 |   </div>
283 | </div>
284 | </header>
285 |
286 | /* Title bar */
287 | <div style={{
288 |   padding: '8px 14px',
289 |   background: 'var(--bg-surface)',
290 |   border: '1px solid var(--border-default)',
291 |   textAlign: 'center',
292 |   marginBottom: '16px',
293 | }}>
294 |   <span style={{ fontSize: '12px', fontWeight: 700, color: 'var(--text-muted)', fontFamily: 'var(--fon...
295 |     2 9 5 | % { q u i z T i t l e } %A
296 |   </span>
297 | </div>
298 |
299 | /* Progress Bar */
300 | <div style={{ width: '100%', height: '8px', background: 'var(--border-subtle)', marginBottom: '32px', ...
301 |   <motion.div
302 |     initial={{ width: 0 }}
303 |     animate={{ width: `${((currentIdx + 1) / questions.length) * 100}%` }}
304 |     style={{
305 |       height: '100%',
306 |       background: 'repeating-linear-gradient(90deg, var(--brand-primary) 0px, var(--brand-primary) 4px...
307 |       transition: 'width 0.5s',
308 |     }}
309 |   </>
310 | </div>
311 |
312 | /* Question Card */
313 | <main style={{ flex: 1, display: 'flex', flexDirection: 'column', gap: '24px' }}>
314 |   <AnimatePresence mode="wait">
315 |     <motion.div
316 |       key={currentIdx}
317 |       initial={{ x: 300, opacity: 0 }}
318 |       animate={{ x: 0, opacity: 1, ...(shake ? { x: [-8, 8, -6, 6, -3, 3, 0] } : {}) }}
319 |       exit={{ x: -300, opacity: 0 }}
320 |       transition={{ type: 'spring', stiffness: 200, damping: 20 }}
321 |       style={{
322 |         background: 'var(--bg-surface)',
323 |         padding: '32px',
324 |         border: '1px solid var(--border-default)',
325 |         textAlign: 'center',
326 |         position: 'relative',
327 |         overflow: 'hidden',
328 |       }}
329 |     >
330 |       /* Pixel corner decorations */
331 |       <div style={{ position: 'absolute', top: 0, left: 0, width: '8px', height: '8px', borderTop: '2p...
332 |       <div style={{ position: 'absolute', top: 0, right: 0, width: '8px', height: '8px', borderTop: '2...
333 |       <div style={{ position: 'absolute', bottom: 0, left: 0, width: '8px', height: '8px', borderBotto...
334 |       <div style={{ position: 'absolute', bottom: 0, right: 0, width: '8px', height: '8px', borderBott...
335 |
336 |       /* Question number badge */
337 |       <div style={{
338 |         display: 'inline-flex', alignItems: 'center', gap: '6px',
339 |         padding: '4px 12px',
340 |         border: '1px solid color-mix(in srgb, var(--accent-blue) 30%, transparent)',
341 |         color: 'var(--accent-blue)',
342 |         fontSize: '10px', fontWeight: 700,
343 |         fontFamily: 'var(--font-mono)',
344 |         marginBottom: '16px',

```

```

345 |         letterSpacing: '1px',
346 |     }}>
347 |     <Target size={12} />
348 |     QUESTION {currentIdx + 1}
349 | </div>
350 |
351 | <h2 style={{
352 |   fontSize: 'clamp(20px, 4vw, 36px)',
353 |   fontWeight: 900,
354 |   lineHeight: 1.3,
355 |   fontFamily: 'var(--font-sans)',
356 | }}>
357 |   {q.question}
358 | </h2>
359 |
360 | </* Feedback Overlay */>
361 | <AnimatePresence>
362 |   {feedback && (
363 |     <motion.div
364 |       initial={{ scale: 0, opacity: 0 }}
365 |       animate={{ scale: 1, opacity: 1 }}
366 |       exit={{ scale: 0, opacity: 0 }}
367 |       style={{
368 |         position: 'absolute', inset: 0, zIndex: 10,
369 |         display: 'flex', flexDirection: 'column', alignItems: 'center', justifyContent: 'center',
370 |         background: feedback === 'correct'
371 |           ? 'color-mix(in srgb, var(--brand-primary) 85%, transparent)'
372 |           : 'color-mix(in srgb, var(--danger) 85%, transparent)',
373 |         backdropFilter: 'blur(4px)',
374 |       }}
375 |     >
376 |       <motion.div initial={{ scale: 0, rotate: -180 }} animate={{ scale: 1, rotate: 0 }} transit...
377 |         {feedback === 'correct'
378 |           ? <Diamond size={80} style={{ color: 'var(--bg-base)' }} />
379 |           : <Skull size={80} style={{ color: '#fff' }} />
380 |         }
381 |       </motion.div>
382 |       <h3 style={{
383 |         fontSize: '32px', fontWeight: 900, marginTop: '12px',
384 |         fontFamily: 'var(--font-mono)',
385 |         letterSpacing: '2px',
386 |         color: feedback === 'correct' ? 'var(--bg-base)' : '#fff',
387 |       }}>
388 |         { feedback === 'correct' ? 'CHÍNH XÁC!' : 'SAI RỒI!' }
389 |       </h3>
390 |       </* Show correct answer when wrong */>
391 |       {feedback === 'wrong' && (
392 |         <motion.p initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }} transition={{ d...
393 |           style={{ marginTop: '8px', fontSize: '15px', fontWeight: 700, color: 'rgba(255,255,255...'
394 |             á p ả n ứ n g : <span style={{ color: '#00d4aa' }}>
395 |           </motion.p>
396 |         )}
397 |         <p style={{ marginTop: '12px', padding: '0 24px', fontSize: '13px', fontWeight: 500, color...
398 |           {q.explanation}
399 |         </p>
400 |       </motion.div>
401 |     )}
402 |   </AnimatePresence>
403 | </motion.div>
404 | </AnimatePresence>
405 |
406 | </* Answer Buttons */>
407 | <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(280px, 1fr))', gap: '12...
408 | {q.options.map((opt: string, i: number) => {
409 |   const isCorrectOption = feedback && i === q.correctIndex
410 |   const isWrongSelected = feedback === 'wrong' && selectedIdx === i
411 |
412 |   return (
413 |     <motion.button
414 |       key={i}
415 |       whileHover={!feedback ? { scale: 1.02, y: -2 } : {}}
416 |       whileTap={!feedback ? { scale: 0.98 } : {}}
417 |       onClick={() => handleAnswer(i)}
418 |       disabled={!feedback}
419 |       style={{
420 |         background: isCorrectOption
421 |           ? 'var(--brand-primary)'
422 |           : isWrongSelected
423 |             ? 'var(--danger)'
424 |             : COLORS[i],
425 |         padding: '16px 20px',
426 |         display: 'flex',
427 |         alignItems: 'center',
428 |         gap: '14px',
429 |         color: '#fff',
430 |         fontWeight: 700,
431 |         fontSize: '16px',
432 |         textAlign: 'left',
433 |         border: 'none',
434 |         cursor: feedback ? 'default' : 'pointer',
435 |         opacity: feedback && !isCorrectOption && !isWrongSelected ? 0.4 : 1,
436 |         transition: 'all 0.2s',
437 |         fontFamily: 'var(--font-sans)',
438 |         position: 'relative',
439 |         overflow: 'hidden',
440 |       }}
441 |     >
442 |       </* Shape indicator */>
443 |       <div style={{
444 |         width: '40px', height: '40px',
445 |         background: 'rgba(255,255,255,0.15)',
446 |         display: 'flex', alignItems: 'center', justifyContent: 'center',
447 |         fontSize: '20px',
448 |         flexShrink: 0,
449 |       }}>
450 |         {SHAPES[i]}
451 |       </div>
452 |       <span style={{ flex: 1 }}>{opt}</span>
453 |       {isCorrectOption && <CheckCircle size={20} />}
454 |       {isWrongSelected && <XCircle size={20} />}
455 |     </motion.button>
456 |   )
457 | }
458 | </div>
459 | </main>
460 |
461 | </* Footer */>
462 | <div style={{ marginTop: '24px', display: 'flex', justifyContent: 'space-between', alignItems: 'center...
463 | <div style={{ display: 'flex', alignItems: 'center', gap: '6px', color: 'var(--text-muted)', fontSiz...
464 | <Zap size={12} style={{ color: 'var(--accent-amber)' }} />
465 | SCORE: {score}
466 | </div>
467 | <div style={{ display: 'flex', alignItems: 'center', gap: '6px', color: 'var(--text-muted)', fontSiz...
468 | <Swords size={12} />

```

```
469 |         STREAK: x{streak}
470 |       </div>
471 |     </div>
472 |   </div>
473 | </div>
474 | )
475 | }
476 |
```

File: app/quiz/page.tsx (102 lines, 4.1 KB)

```
1 | 'use client'
2 |
3 | import React, { useState, useEffect } from 'react'
4 | import { supabase } from '@lib/supabase'
5 | import Navbar from '@components/Navbar'
6 | import { motion } from 'framer-motion'
7 | import { HelpCircle, Zap, Star, ArrowRight, Loader2, Search } from 'lucide-react'
8 | import Link from 'next/link'
9 |
10 | export default function QuizzesListingPage() {
11 |   const [quizzes, setQuizzes] = useState<any[]>([])
12 |   const [loading, setLoading] = useState(true)
13 |
14 |   useEffect(() => {
15 |     fetchQuizzes()
16 |   }, [])
17 |
18 |   async function fetchQuizzes() {
19 |     try {
20 |       setLoading(true)
21 |       const { data } = await supabase
22 |         .from('lesson_sections')
23 |         .select(`
24 |           *,
25 |           lessons (title)
26 |         `)
27 |         .eq('content_type', 'quiz')
28 |         .limit(20)
29 |
30 |       setQuizzes(data || [])
31 |     } catch (err) {
32 |       console.error(err)
33 |     } finally {
34 |       setLoading(false)
35 |     }
36 |   }
37 |
38 |   return (
39 |     <div className="min-h-screen bg-[#0f0f1a] text-white">
40 |       <Navbar />
41 |
42 |       <main className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 pt-28 pb-20">
43 |         <header className="mb-12">
44 |           <h1 className="text-4xl font-black mb-4">Thi thach <span class="text-slate-500">Ki Am tra ki n thec t ec thi v i d
45 |           <p className="text-slate-400">
46 |         </header>
47 |
48 |         <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-4 mb-12">
49 |           <div className="p-6 bg-gradient-to-br from-[#00d2a0]/20 to-[#00b4d8]/20 border border-[#00d2a0]/30 r...
50 |             <Zap size={32} className="text-[#00d2a0] mb-4" />
51 |             <h3 className="font-bold"> ù ng l ã c h í c </h3>
52 |             <p className="text-xs text-slate-500 mt-1">Tr ð l ý i ú ng n h
53 |           </div>
54 |           <div className="p-6 bg-[#16213e] border border-[#1e293b] rounded-3xl">
55 |             <Star size={32} className="text-yellow-500 mb-4" />
56 |             <h3 className="font-bold"> B ð ng v à ng </h3>
57 |             <p className="text-xs text-slate-500 mt-1">T op 1 0 n h - n h u y
58 |           </div>
59 |         </div>
60 |
61 |         {loading ? (
62 |           <div className="flex justify-center py-20">
63 |             <Loader2 className="animate-spin text-[#00d2a0]" size={40} />
64 |           </div>
65 |         ) : quizzes.length > 0 ? (
66 |           <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
67 |             {quizzes.map((quiz) => (
68 |               <motion.div
69 |                 key={quiz.id}
70 |                 initial={{ opacity: 0, scale: 0.9 }}
71 |                 animate={{ opacity: 1, scale: 1 }}
72 |                 className="bg-[#16213e] border border-[#1e293b] rounded-3xl p-6 hover:shadow-[0_0_30px_rgba(0,...
73 |               >
74 |                 <div className="flex items-center gap-3 mb-4">
75 |                   <div className="w-10 h-10 bg-[#1e293b] text-[#00d2a0] rounded-xl flex items-center justify-c...
76 |                   <HelpCircle size={20} />
77 |                 </div>
78 |                 <span className="text-[10px] font-bold text-slate-500 uppercase tracking-widest">{quiz.lesso...
79 |                 </div>
80 |
81 |                 <h3 className="text-lg font-bold mb-6 min-h-[50px]">{quiz.title}</h3>
82 |
83 |                 <Link
84 |                   href={` /quiz/${quiz.id} `}
85 |                   className="w-full py-4 bg-[#0f0f1a] text-white border border-[#1e293b] group-hover:border-[#...
86 |                 >
87 |                   L Ì M B Ì I N G A Y <Zap size={16} />
88 |                 </Link>
89 |               </motion.div>
90 |             )})
91 |           </div>
92 |         ) : (
93 |           <div className="text-center py-20 bg-[#16213e]/30 border border-dashed border-[#1e293b] rounded-[40p...
94 |             <Zap size={48} className="mx-auto text-slate-600 mb-4" />
95 |             <p className="text-slate-500 font-medium italic">Ch ò a c ó b ã
96 |           </div>
97 |         )
98 |       </main>
99 |     </div>
100 |   )
101 | }
102 |
```

File: app/student/certificates/page.tsx (652 lines, 32.2 KB)

```
1 | 'use client'
2 |
3 | import React, { useEffect, useState, useRef, useCallbac } from 'react'
4 | import { createBrowserClient } from '@supabase/ssr'
5 | import { motion } from 'framer-motion'
```

```

6 | import { Award, FileText, CheckCircle, Copy, RefreshCw, Lock, ShieldCheck, Target, ArrowLeft, Sparkles, Downlo...
7 | import { toast } from 'react-hot-toast'
8 | import { useRouter } from 'next/navigation'
9 | import html2canvas from 'html2canvas'
10 | import jsPDF from 'jspdf'
11 | import Link from 'next/link'
12 |
13 | const CERTIFICATE_TYPES = [
14 |   {
15 |     id: 'cert_hardware_basics',
16 |     title: 'K ù S ° P h ã n C é n g T - p S ã ' ,
17 |     description: ' N m v ï n g k i ã n t h é c n Á n t ã n g v Á c á c l ï n h k i Ç n c ¥ u t h à n h
18 |     mission: ' H o à n t h à n h 3 b à i h í c l ý t h u y ã t c i b é n v à i t t Ñ i t h i Á u 7 0 9
19 |     levelReq: 3,
20 |     icon: <ShieldCheck size={28} />
21 |   },
22 |   {
23 |     id: 'cert_pc_master',
24 |     title: ' C h u y ê n G i a L - p R á p P C M a s t e r ' ,
25 |     description: ' C h é n g n h - n k h Ë n n g l - p r á p , m á x í l ý , s ã c Ñ v à t ò i n g t h i c
26 |     mission: ' H o à n t h à n h t o à n b ù k h ó a h í c P C M a s t e r , b a o g ó m g i Ë l - p L a b
27 |     levelReq: 5,
28 |     icon: <Award size={28} />
29 |   },
30 |   {
31 |     id: 'cert_troubleshooting',
32 |     title: ' K ù T h u - t V i ê n X í L ý S ã C Ñ ' ,
33 |     description: ' C h é n g n h - n k ù n n g c h u © n o á n v à k h - c p h á c c á c l x i h Ç
34 |     mission: ' T r Ë l ý i ú n g l i ê n t i ç p 1 0 c á u h í i K h - c P h á c S ã C Ñ t r o n g N g
35 |     levelReq: 7,
36 |     icon: <Target size={28} />
37 |   }
38 | ]
39 |
40 | function generateVerificationCode() {
41 |   const chars = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789'
42 |   let result = 'PCM-'
43 |   for (let i = 0; i < 8; i++) {
44 |     result += chars.charAt(Math.floor(Math.random() * chars.length))
45 |   }
46 |   return result
47 | }
48 |
49 | export default function StudentCertificatesPage() {
50 |   const router = useRouter()
51 |   const [certs, setCerts] = useState<any[]>([])
52 |   const [activePath, setActivePath] = useState<any>(null)
53 |   const [loading, setLoading] = useState(true)
54 |   const [issuing, setIssuing] = useState(false)
55 |   const [justEarnedId, setJustEarnedId] = useState<string | null>(null)
56 |   const [profile, setProfile] = useState<any>(null)
57 |   const [user, setUser] = useState<any>(null)
58 |   const [showCertModal, setShowCertModal] = useState<any>(null)
59 |   const [userLevel, setUserLevel] = useState(1)
60 |   const certRef = useRef<HTMLDivElement>(null)
61 |   const supabase = createBrowserClient(
62 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
63 |     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
64 |   )
65 |
66 |   const loadData = useCallback(async () => {
67 |     setLoading(true)
68 |     const { data: { user: currentUser } } = await supabase.auth.getUser()
69 |     if (!currentUser) { setLoading(false); return }
70 |     setUser(currentUser)
71 |
72 |     const { data: userProfile } = await supabase
73 |       .from('profiles')
74 |       .select('full_name, avatar_url, level, xp')
75 |       .eq('id', currentUser.id)
76 |       .single()
77 |     if (userProfile) { setProfile(userProfile); setUserLevel(userProfile.level || 1) }
78 |
79 |     const { data: c } = await supabase
80 |       .from('certificates')
81 |       .select('*')
82 |       .eq('student_id', currentUser.id)
83 |       .eq('is_revoked', false)
84 |       .order('issued_at', { ascending: false })
85 |
86 |     if (c) {
87 |       const newIds = c.filter(cert => !certs.some(old => old.id === cert.id)).map(cert => cert.id)
88 |       if (newIds.length > 0) {
89 |         setJustEarnedId(newIds[newIds.length - 1])
90 |         toast.success(' C h ú c m ẽ n g ! B i n ã n h - n ° á c c h é n g c h É m ô i ! ' )
91 |       }
92 |       setCerts(c)
93 |     }
94 |
95 |     const { data: path } = await supabase
96 |       .from('learning_paths')
97 |       .select('id, title')
98 |       .eq('is_active', true)
99 |       .limit(1)
100 |       .maybeSingle()
101 |     if (path) setActivePath(path)
102 |
103 |     setLoading(false)
104 |   }, [])
105 |
106 |   useEffect(() => { loadData(); }, [])
107 |
108 |   useEffect(() => {
109 |     const sub = supabase
110 |       .channel('certs-realtime')
111 |       .on('postgres_changes',
112 |         { event: 'INSERT', schema: 'public', table: 'certificates' },
113 |         (payload: any) => {
114 |           const newCert = payload.new
115 |           setCerts(prev => {
116 |             if (prev.some(c => c.id === newCert.id)) return prev
117 |             toast.success(' C h é n g c h É m ô i ã ° á c c ¥ p ! ' )
118 |             setJustEarnedId(newCert.id)
119 |             return [newCert, ...prev]
120 |           })
121 |         }
122 |       )
123 |     .subscribe()
124 |     return () => { supabase.removeChannel(sub) }
125 |   }, [])
126 |
127 |   const autoCheckAndIssue = async () => {
128 |     const { data: { user: currentUser } } = await supabase.auth.getUser()
129 |     if (!currentUser) return

```

```

130 | const { data: existing } = await supabase
131 |   .from('certificates')
132 |   .select('course_title')
133 |   .eq('student_id', currentUser.id)
134 |   .eq('is_revoked', false)
135 |
136 | const existingTitles = (existing || []).map(e => e.course_title)
137 |
138 | for (const certType of CERTIFICATE_TYPES) {
139 |   if (existingTitles.includes(certType.title)) continue
140 |   try {
141 |     const code = generateVerificationCode()
142 |     const { error } = await supabase.from('certificates').insert({
143 |       student_id: currentUser.id,
144 |       student_name: profile?.full_name || currentUser.email?.split('
145 |         |
146 |         course_title: certType.title,
147 |         certificate_number: code,
148 |         issued_at: new Date().toISOString(),
149 |         completion_date: new Date().toISOString().split('T')[0],
150 |         final_score: 100,
151 |         is_revoked: false,
152 |       })
153 |     if (error) throw error
154 |     await loadData()
155 |     return
156 |   } catch {
157 |   }
158 | }
159 |
160 | const handleRequestCertificate = async () => {
161 |   setIssuing(true)
162 |   try {
163 |     await autoCheckAndIssue()
164 |     toast.success('Chúc mừng! Bạn đã nhận chứng chỉ thành công.')
165 |     await loadData()
166 |   } catch (err: any) {
167 |     toast.error(err.message || 'Có lỗi xảy ra. Hãy hoàn thành các
168 |       |
169 |       finally {
170 |         setIssuing(false)
171 |       }
172 |     }
173 |   }
174 |
175 | const handleDownloadPDF = async (cert: any) => {
176 |   setShowCertModal(cert)
177 |   await new Promise(r => setTimeout(r, 100))
178 |   try {
179 |     const element = document.getElementById(`cert-preview-${cert.id}`)
180 |     if (!element) { toast.error('Không thể tải chứng chỉ'); return }
181 |     const canvas = await html2canvas(element, {
182 |       scale: 3, useCORS: true, allowTaint: true,
183 |       backgroundColor: '#0a0c1a', logging: false,
184 |     })
185 |     const imgData = canvas.toDataURL('image/png')
186 |     const pdf = new jsPDF({
187 |       orientation: 'landscape', unit: 'px',
188 |       format: [canvas.width / 3, canvas.height / 3]
189 |     })
190 |     pdf.addImage(imgData, 'PNG', 0, 0, canvas.width / 3, canvas.height / 3)
191 |     pdf.save(`chung-chi-${cert.certificate_number || cert.id}.pdf`)
192 |     toast.success('Tải xuống PDF thành công!')
193 |   } catch (err) {
194 |     toast.error('Không thể tải PDF. Thử tải xuống ảnh.')
195 |   }
196 |   setShowCertModal(null)
197 | }
198 |
199 | const handleDownloadImage = async (cert: any) => {
200 |   setShowCertModal(cert)
201 |   await new Promise(r => setTimeout(r, 100))
202 |   try {
203 |     const element = document.getElementById(`cert-preview-${cert.id}`)
204 |     if (!element) { toast.error('Không thể tải chứng chỉ'); return }
205 |     const canvas = await html2canvas(element, {
206 |       scale: 3, useCORS: true, allowTaint: true,
207 |       backgroundColor: '#0a0c1a', logging: false,
208 |     })
209 |     const link = document.createElement('a')
210 |     link.download = `chung-chi-${cert.certificate_number || cert.id}.png`
211 |     link.href = canvas.toDataURL('image/png')
212 |     link.click()
213 |     toast.success('Tải xuống ảnh thành công!')
214 |   } catch (err) {
215 |     toast.error('Không thể tải ảnh chứng chỉ.')
216 |   }
217 |   setShowCertModal(null)
218 | }
219 |
220 | const handleCopyLink = (code: string) => {
221 |   const link = `${window.location.origin}/verify/${code}`
222 |   navigator.clipboard.writeText(link)
223 |   toast.success('Sao chép liên kết xác thực!')
224 | }
225 |
226 | if (loading) {
227 |   return (
228 |     <div className="min-h-screen" style={{ background: 'var(--bg-base)', color: 'var(--text-primary)', paddi...
229 |       <motion.div animate={{ rotate: 360 }} transition={{ duration: 2, repeat: Infinity, ease: 'linear' }}>
230 |         <Swords size={28} style={{ color: 'var(--brand-primary)' }} />
231 |       </motion.div>
232 |       <span style={{ fontSize: '11px', color: 'var(--text-muted)', fontWeight: 700, fontFamily: 'var(--font-...
233 |     </div>
234 |   )
235 | }
236 |
237 | return (
238 |   <div className="min-h-screen" style={{ background: 'var(--bg-base)', color: 'var(--text-primary)', padding...
239 |     <div style={{ maxWidth: '900px', margin: '0 auto' }}>
240 |       <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'space-between', marginBottom: '2...
241 |         <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
242 |           <div style={{
243 |             width: '48px', height: '48px', borderRadius: '4px',
244 |             background: 'linear-gradient(135deg, var(--brand-primary), var(--accent-purple))',
245 |             display: 'flex', alignItems: 'center', justifyContent: 'center',
246 |             boxShadow: '0 0 20px color-mix(in srgb, var(--brand-primary) 30%, transparent)',
247 |           }}>
248 |             <Trophy size={24} color="#fff" />
249 |           </div>
250 |           <div>
251 |             <h1 style={{ fontSize: '20px', fontWeight: 900, fontFamily: 'var(--font-mono)', letterSpacing: '...
252 |               C H È N G C H Ế & D A N H H I Ế U
253 |             </h1>

```

```

254 | <p style={{ fontSize: '12px', color: 'var(--text-muted)' }}>
255 |   H o à n t h à n h n h i ệ m v ậ - t ã ù n g n h - n c h é n g c h é
256 | </p>
257 | </div>
258 | </div>
259 | <button onClick={() => router.push('/builder')} style={{
260 |   display: 'flex', alignItems: 'center', gap: '6px',
261 |   padding: '8px 16px',
262 |   border: '1px solid var(--border-default)',
263 |   background: 'transparent',
264 |   color: 'var(--text-muted)',
265 |   fontSize: '11px', fontWeight: 700,
266 |   fontFamily: 'var(--font-mono)',
267 |   cursor: 'pointer',
268 |   transition: 'all 0.2s',
269 | }}>
270 | <ArrowLeft size={14} /> BACK
271 | </button>
272 | </div>
273 |
274 | {/ * User Level Badge */}
275 | <div style={{
276 |   marginBottom: '24px',
277 |   padding: '12px 16px',
278 |   border: '1px solid var(--border-default)',
279 |   background: 'var(--bg-surface)',
280 |   display: 'flex', alignItems: 'center', gap: '12px',
281 | }}>
282 |   <div style={{
283 |     width: '36px', height: '36px', borderRadius: '4px',
284 |     background: 'color-mix(in srgb, var(--accent-amber) 15%, transparent)',
285 |     border: '1px solid color-mix(in srgb, var(--accent-amber) 25%, transparent)',
286 |     display: 'flex', alignItems: 'center', justifyContent: 'center',
287 |     color: 'var(--accent-amber)',
288 |     fontSize: '16px',
289 |   }}>
290 |     +P
291 |   </div>
292 |   <div style={{ flex: 1 }}>
293 |     <div style={{ fontSize: '12px', color: 'var(--text-muted)', fontFamily: 'var(--font-mono)', margin...
294 |       C ậ p ộ h i ệ n t i
295 |     </div>
296 |     <div style={{ fontSize: '16px', fontWeight: 800 }}>
297 |       L e v e l { u s e r L e v e l } - { p r o f i l e ? . f u l l _ n a m e | | u s e r ? . e m a i l ? . s p
298 |     </div>
299 |     <div style={{ fontSize: '11px', color: 'var(--text-muted)', fontFamily: 'var(--font-mono)', textAlig...
300 |       < d i v > { c e r t s . l e n g t h } c h é n g c h é < / d i v >
301 |     </div>
302 |     <div style={{ color: 'var(--brand-primary)' }}>{profile?.xp || 0} XP</div>
303 |   </div>
304 | </div>
305 |
306 | {/ * Certificate List */}
307 | <div style={{ display: 'flex', flexDirection: 'column', gap: '12px' }}>
308 |   {CERTIFICATE_TYPES.map(certType) => {
309 |     const earnedCert = certs.find(c => c.course_title === certType.title)
310 |     const isEarned = !!earnedCert
311 |     const isJustEarned = justEarnedId === earnedCert.id
312 |     const levelMet = userLevel >= certType.levelReq
313 |
314 |     return (
315 |       <div key={certType.id}>
316 |         <div style={{
317 |           border: '1px solid ${isEarned ? 'color-mix(in srgb, var(--brand-primary) 40%, transparent)' ...
318 |           background: isEarned ? 'color-mix(in srgb, var(--brand-primary) 4%, transparent)' : 'var(--b...
319 |           padding: '20px',
320 |           position: 'relative', overflow: 'hidden',
321 |           opacity: isEarned ? 1 : 0.7,
322 |           transition: 'all 0.3s',
323 |         }}>
324 |           {/ * Corner decorations */}
325 |           <div style={{ position: 'absolute', top: 0, left: 0, width: '6px', height: '6px', borderTop...
326 |           <div style={{ position: 'absolute', top: 0, right: 0, width: '6px', height: '6px', borderTop...
327 |           <div style={{ position: 'absolute', bottom: 0, left: 0, width: '6px', height: '6px', borderB...
328 |           <div style={{ position: 'absolute', bottom: 0, right: 0, width: '6px', height: '6px', border...
329 |
330 |           <div style={{ display: 'flex', gap: '16px', alignItems: 'flex-start' }}>
331 |             {/ * Icon */}
332 |             <div style={{
333 |               width: '52px', height: '52px', borderRadius: '4px',
334 |               background: isEarned ? 'color-mix(in srgb, var(--brand-primary) 15%, transparent)' : 'co...
335 |               border: '1px solid ${isEarned ? 'color-mix(in srgb, var(--brand-primary) 25%, transparen...
336 |               display: 'flex', alignItems: 'center', justifyContent: 'center',
337 |               flexShrink: 0,
338 |               color: isEarned ? 'var(--brand-primary)' : 'var(--text-muted)',
339 |             }}>
340 |               {certType.icon}
341 |             </div>
342 |
343 |             {/ * Content */}
344 |             <div style={{ flex: 1 }}>
345 |               <div style={{ display: 'flex', alignItems: 'center', gap: '8px', marginBottom: '4px' }}>
346 |                 <h3 style={{
347 |                   fontSize: '15px', fontWeight: 800,
348 |                   fontFamily: 'var(--font-mono)',
349 |                   color: isEarned ? 'var(--text-primary)' : 'var(--text-muted)',
350 |                 }}>
351 |                   {certType.title}
352 |                 </h3>
353 |                 {isEarned && <CheckCircle size={16} style={{ color: 'var(--brand-primary)' }} />}
354 |                 {isJustEarned && <Sparkles size={16} style={{ color: 'var(--accent-amber)' }} />}
355 |               </div>
356 |               <p style={{ fontSize: '12px', color: 'var(--text-muted)', marginBottom: '8px' }}>{certTy...
357 |
358 |             {/ * Mission & Level Requirement */}
359 |             <div style={{ display: 'flex', gap: '8px', flexWrap: 'wrap' }}>
360 |               <div style={{
361 |                 padding: '6px 10px',
362 |                 border: '1px solid var(--border-default)',
363 |                 background: 'var(--bg-base)',
364 |                 fontSize: '11px',
365 |                 color: 'var(--text-muted)',
366 |               }}>
367 |                 <span style={{ fontWeight: 700, color: levelMet ? 'var(--brand-primary)' : 'var(--ac...
368 |                   { l e v e l M e t ? ' ' : ' L v . ' } { c e r t T y p e . l e v e l R e q }
369 |                 </span>
370 |                 < s p a n s t y l e = { { m a r g i n L e f t : ' 4 p x ' } } > Y ê u c s u c h
371 |               </div>
372 |               {isEarned && earnedCert?.certificate_number && (
373 |                 <div style={{
374 |                   padding: '6px 10px',
375 |                   border: '1px solid var(--border-default)',
376 |                   background: 'var(--bg-base)',
377 |                   fontSize: '11px',

```



```

378         color: 'var(--text-muted)',
379         fontFamily: 'var(--font-mono)',
380     })>
381     #{earnedCert.certificate_number}
382 </div>
383 </div>
384
385 {/ Mission detail */}
386 <div style={{
387     marginTop: '8px',
388     padding: '8px 10px',
389     border: '1px solid var(--border-default)',
390     background: 'var(--bg-base)',
391     fontSize: '11px',
392     color: 'var(--text-muted)',
393 }}>
394     <span style={{ color: 'var(--accent-blue)', fontWeight: 700, fontFamily: 'var(--font-m...
395     {certType.mission}
396 </div>
397 </div>
398
399 {/ Actions */}
400 <div style={{ display: 'flex', flexDirection: 'column', gap: '6px', flexShrink: 0 }}>
401     {isEarned ? (
402         <>
403             <button onClick={() => handleDownloadImage(earnedCert)}
404                 style={{
405                     padding: '8px 14px',
406                     background: 'var(--brand-primary)',
407                     color: 'var(--bg-base)',
408                     border: 'none',
409                     fontSize: '10px', fontWeight: 700,
410                     fontFamily: 'var(--font-mono)',
411                     cursor: 'pointer',
412                     display: 'flex', alignItems: 'center', gap: '4px',
413                 }}>
414                 <Image size={12} /> <NH
415
416             </button>
417             <button onClick={() => handleDownloadPDF(earnedCert)}
418                 style={{
419                     padding: '8px 14px',
420                     background: 'var(--accent-blue)',
421                     color: '#fff',
422                     border: 'none',
423                     fontSize: '10px', fontWeight: 700,
424                     fontFamily: 'var(--font-mono)',
425                     cursor: 'pointer',
426                     display: 'flex', alignItems: 'center', gap: '4px',
427                 }}>
428                 <FileText size={12} /> PDF
429             </button>
430             <button onClick={() => handleCopyLink(earnedCert.certificate_number)}
431                 style={{
432                     padding: '8px 14px',
433                     border: '1px solid var(--border-default)',
434                     background: 'transparent',
435                     color: 'var(--text-muted)',
436                     fontSize: '10px', fontWeight: 700,
437                     fontFamily: 'var(--font-mono)',
438                     cursor: 'pointer',
439                     display: 'flex', alignItems: 'center', gap: '4px',
440                 }}>
441                 <Copy size={12} /> COPY
442             </button>
443         </>
444     ) : (
445         <button onClick={handleRequestCertificate} disabled={issuing}
446             style={{
447                 padding: '8px 14px',
448                 border: '1px solid var(--border-default)',
449                 background: 'transparent',
450                 color: 'var(--text-muted)',
451                 fontSize: '10px', fontWeight: 700,
452                 fontFamily: 'var(--font-mono)',
453                 cursor: 'pointer',
454                 display: 'flex', alignItems: 'center', gap: '4px',
455                 opacity: issuing ? 0.5 : 1,
456             }}>
457             {issuing ? <RefreshCw size={12} className="animate-spin" /> : <Lock size={12} />
458                 {issuing ? 'ANG KI AM TRA...' : 'Y EU C U'}
459         </button>
460     )}
461 </div>
462 </div>
463 </div>
464
465 {/ Certificate preview (hidden, for rendering) */}
466 {isEarned && (
467     <div className="fixed" style={{ left: '-9999px', top: 0 }} ref={certRef}>
468         <div id={`cert-preview-${earnedCert.id}`} style={{
469             width: '800px', height: '566px', padding: '28px',
470             background: 'linear-gradient(135deg, #0a0c1a 0%, #1a1c2e 50%, #0a0c1a 100%)',
471             position: 'relative', overflow: 'hidden',
472             display: 'flex', flexDirection: 'column', alignItems: 'center', justifyContent: 'center',
473         }}>
474             {/ Outer gold border */}
475             <div style={{
476                 position: 'absolute', inset: '0',
477                 padding: '3px',
478                 background: 'linear-gradient(135deg, #d4a843, #f5d68a, #d4a843, #8b6914, #d4a843)',
479                 WebkitMask: 'linear-gradient(#fff 0 0) content-box, linear-gradient(#fff 0 0)',
480                 WebkitMaskComposite: 'xor',
481                 maskComposite: 'exclude',
482             }} />
483             <div style={{
484                 position: 'absolute', top: '10px', left: '10px', right: '10px', bottom: '10px',
485                 border: '1px solid rgba(212, 168, 67, 0.12)',
486                 pointerEvents: 'none',
487             }} />
488             {/ Corner ornaments */}
489             [[
490                 { top: '20px', left: '20px', borderRight: '2px solid rgba(212,168,67,0.4)', borderBott...
491                 { top: '20px', right: '20px', borderLeft: '2px solid rgba(212,168,67,0.4)', borderBott...
492                 { bottom: '20px', left: '20px', borderRight: '2px solid rgba(212,168,67,0.4)', borderT...
493                 { bottom: '20px', right: '20px', borderLeft: '2px solid rgba(212,168,67,0.4)', borderT...
494             ].map((pos, i) => {
495                 <div key={i} style={{ position: 'absolute', ...pos, borderRadius: '2px' }} />
496             })
497             {/ Decorative lines */}
498             <div style={{ position: 'absolute', top: '14px', left: '50%', transform: 'translateX(-50...
499             <div style={{ position: 'absolute', bottom: '14px', left: '50%', transform: 'translateX(...
500
501

```

```

502 | /* Background pattern */
503 | <div style={{ position: 'absolute', top: '100px', right: '50px', width: '120px', height:...
504 | <div style={{ position: 'absolute', bottom: '100px', left: '50px', width: '80px', height:...
505 |
506 |
507 | /* Logo */
508 | <div style={{
509 |   width: '52px', height: '52px', borderRadius: '50%',
510 |   background: 'linear-gradient(135deg, #d4a843, #f5d68a)',
511 |   display: 'flex', alignItems: 'center', justifyContent: 'center',
512 |   marginBottom: '6px', boxShadow: '0 4px 20px rgba(212,168,67,0.3)',
513 | }}>
514 |   <Award size={26} color="#0a0c1a" />
515 | </div>
516 |
517 | /* School name */
518 | <div style={{ fontSize: '10px', fontWeight: 600, letterSpacing: '3px', color: '#d4a843',...
519 |   PC MASTER ACADEMY
520 | </div>
521 | <div style={{ fontSize: '8px', fontWeight: 500, letterSpacing: '2px', color: 'rgba(212,1...
522 |   C H Ê N G   N H Ậ N   H O Ạ N   T H Ạ N H
523 | </div>
524 |
525 | /* Certificate title */
526 | <div style={{ fontSize: '9px', fontWeight: 600, letterSpacing: '2px', color: 'rgba(212,1...
527 |   C H Ê N G   C H Ê
528 | </div>
529 | <div style={{ fontSize: '20px', fontWeight: 900, color: '#f5d68a', textAlign: 'center', ...
530 |   {earnedCert.course_title}
531 | </div>
532 |
533 | /* Separator */
534 | <div style={{ width: '120px', height: '1px', background: 'linear-gradient(90deg, transpa...
535 |
536 | /* Recipient */
537 | <div style={{ fontSize: '10px', color: 'rgba(255,255,255,0.4)', marginBottom: '3px', let...
538 |   C Ặ P   C H O
539 | </div>
540 | <div style={{ fontSize: '24px', fontWeight: 800, color: '#fff', marginBottom: '12px', le...
541 |   { e a r n e d C e r t . s t u d e n t _ n a m e } | | p r o f i l e ? . f u l l _ n a m e
542 | </div>
543 |
544 | /* Description */
545 | <div style={{ fontSize: '10px', color: 'rgba(255,255,255,0.45)', textAlign: 'center', ma...
546 |   ấ h o ả n   t h ằ n h   x u ấ t   s ắ c   c ắ c   y ế u   c ầ u   c ầ a   c h ộ   i n g   t
547 | </div>
548 |
549 | /* Score badge */
550 | {earnedCert.final_score && (
551 |   <div style={{
552 |     marginBottom: '12px',
553 |     padding: '3px 14px',
554 |     border: '1px solid rgba(212,168,67,0.3)',
555 |     fontSize: '10px', fontWeight: 700, color: '#d4a843',
556 |   }}>
557 |     I Ầ M :   { e a r n e d C e r t . f i n a l _ s c o r e } %
558 |   </div>
559 | )}
560 |
561 | /* Info row */
562 | <div style={{ display: 'flex', gap: '24px', alignItems: 'center', marginBottom: '10px' }}>
563 |   <div style={{ display: 'flex', alignItems: 'center', gap: '5px' }}>
564 |     <Calendar size={11} color="#d4a843" />
565 |     <span style={{ fontSize: '9px', color: 'rgba(255,255,255,0.4)' }}>
566 |       {new Date(earnedCert.issued_at || Date.now()).toLocaleDateString('vi-VN')}
567 |     </span>
568 |   </div>
569 |   <div style={{ display: 'flex', alignItems: 'center', gap: '5px' }}>
570 |     <Hash size={11} color="#d4a843" />
571 |     <span style={{ fontSize: '9px', color: 'rgba(255,255,255,0.3)', fontFamily: 'monospa...
572 |       {earnedCert.certificate_number}
573 |     </span>
574 |   </div>
575 | </div>
576 |
577 | /* Gold seal */
578 | <div style={{
579 |   width: '44px', height: '44px', borderRadius: '50%',
580 |   border: '2px solid #d4a843',
581 |   display: 'flex', alignItems: 'center', justifyContent: 'center',
582 |   background: 'radial-gradient(circle at 30% 30%, rgba(245,214,138,0.15), transparent)',
583 |   position: 'relative',
584 | }}>
585 |   <Star size={18} color="#d4a843" fill="#d4a843" />
586 |   <div style={{ position: 'absolute', inset: '-3px', borderRadius: '50%', border: '1px s...
587 | </div>
588 | </div>
589 | </div>
590 |
591 | </div>
592 | )
593 | }}
594 | </div>
595 |
596 | /* Level Guide */
597 | <div style={{
598 |   marginTop: '24px',
599 |   padding: '16px',
600 |   border: '1px solid var(--border-default)',
601 |   background: 'var(--bg-surface)',
602 | }}>
603 |   <div style={{ display: 'flex', alignItems: 'center', gap: '8px', marginBottom: '12px' }}>
604 |     <Target size={16} style={{ color: 'var(--accent-blue)' }} />
605 |     <h3 style={{ fontSize: '13px', fontWeight: 800, fontFamily: 'var(--font-mono)' }}>
606 |       H Ứ N G   Đ Ặ N   N H Ậ N   C H Ê N G   C H Ê
607 |     </h3>
608 |   </div>
609 |   <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(200px, 1fr))', gap: '8p...
610 |     {CERTIFICATE_TYPES.map(ct => (
611 |       <div key={ct.id} style={{
612 |         padding: '10px',
613 |         border: '1px solid var(--border-default)',
614 |         background: 'var(--bg-base)',
615 |       }}>
616 |         <div style={{ fontWeight: 700, color: 'var(--text-primary)', marginBottom: '4px', fontFamily: ...
617 |           {ct.title}
618 |         </div>
619 |         <div>Y ế u   c ầ u   L ệ v   <strong style={{ color: userLevel >=
620 |           {userLevel >= ct.levelReq ? (
621 |             <div style={{ color: 'var(--brand-primary)', fontFamily: 'var(--font-mono)', fontSize: '10px...
622 |               ầ
623 |             </div>
624 |           ) : (
625 |             <div style={{ color: 'var(--text-muted)', fontFamily: 'var(--font-mono)', fontSize: '10px', ...

```

```

627 |         </div>
628 |     )}
629 | </div>
630 | )}
631 | </div>
632 | <div style={{ marginTop: '12px', textAlign: 'center' }}>
633 |   <Link href="/student/level" style={{
634 |     display: 'inline-flex', alignItems: 'center', gap: '6px',
635 |     padding: '8px 20px',
636 |     background: 'var(--bg-base)',
637 |     border: '1px solid var(--border-default)',
638 |     color: 'var(--text-muted)',
639 |     fontSize: '11px', fontWeight: 700,
640 |     fontFamily: 'var(--font-mono)',
641 |     textDecoration: 'none',
642 |     transition: 'all 0.2s',
643 |   }}>
644 |     <Zap size={14} /> X E M C Ạ P Ồ & N H I Ệ M V Ậ
645 |   </Link>
646 | </div>
647 | </div>
648 | </div>
649 | </div>
650 | )
651 | }
652 |

```

File: app/student/discussion/page.tsx (419 lines, 21.7 KB)

```

1 | 'use client'
2 |
3 | export const dynamic = 'force-dynamic'
4 |
5 | import React, { useEffect, useState, useRef } from 'react'
6 | import { createClientComponentClient } from '@supabase/auth-helpers-nextjs'
7 | import { useRouter } from 'next/navigation'
8 | import { Thread } from '@hooks/useDiscussion'
9 | import { ThreadCard } from '@components/discussion/ThreadCard'
10 | import { ThreadDetail } from '@components/discussion/ThreadDetail'
11 | import { MessageSquare, Send, Bot, Sparkles, Globe, School, Loader2, Cpu, ArrowLeft } from 'lucide-react'
12 |
13 | interface ChatMessage {
14 |   id: string;
15 |   sender: string;
16 |   sender_id: string;
17 |   avatar: string;
18 |   role: 'student' | 'teacher' | 'assistant';
19 |   content: string;
20 |   time: string;
21 |   channel: string;
22 | }
23 |
24 | function formatTime(ts: string): string {
25 |   try {
26 |     const d = new Date(ts);
27 |     return d.toLocaleTimeString('vi-VN', { hour: '2-digit', minute: '2-digit' });
28 |   } catch { return ''; }
29 | }
30 |
31 | export default function DiscussionDashboardPage() {
32 |   const router = useRouter()
33 |   const [threads, setThreads] = useState<Thread[]>([])
34 |   const [loading, setLoading] = useState(true)
35 |   const [selectedThread, setSelectedThread] = useState<Thread | null>(null)
36 |   const [userProfile, setUserProfile] = useState<any>(null)
37 |   const [user, setUser] = useState<any>(null)
38 |
39 |   // Realtime Chat States
40 |   const [chatChannel, setChatChannel] = useState<'global' | 'class'>('global')
41 |   const [globalMessages, setGlobalMessages] = useState<ChatMessage[]>([])
42 |   const [classMessages, setClassMessages] = useState<ChatMessage[]>([])
43 |   const [trendingTopic, setTrendingTopic] = useState<string>('Ch ả c ó d i')
44 |   const [chatInput, setChatInput] = useState('')
45 |
46 |   // Tiny AI Chatbot States
47 |   const [aiInput, setAiInput] = useState('')
48 |   const [aiReplies, setAiReplies] = useState<string[]>([
49 |     'T ồ i là Micro AI Guru. H ả y . t b ậ t k ó c ầ u h ỉ i n h a n h n ầ o v ả p h ầ n c ể n g'
50 |   ])
51 |   const [aiLoading, setAiLoading] = useState(false)
52 |
53 |   const chatEndRef = useRef<HTMLDivElement>(null)
54 |   const supabase = createClientComponentClient()
55 |
56 |   useEffect(() => {
57 |     const loadInitial = async () => {
58 |       try {
59 |         const { data: { user: u } } = await supabase.auth.getUser()
60 |         if (!u) { setLoading(false); return }
61 |         setUser(u)
62 |
63 |         const { data: prof } = await supabase.from('profiles').select('*').eq('id', u.id).maybeSingle()
64 |         setUserProfile(prof)
65 |
66 |         // Load threads
67 |         const { data: threadsData } = await supabase
68 |           .from('discussion_threads')
69 |           .select('*', { author: profiles(full_name, avatar_url, role) })
70 |           .eq('is_deleted', false)
71 |           .order('created_at', { ascending: false })
72 |         if (threadsData) setThreads(threadsData as any[])
73 |
74 |         // Load persisted messages
75 |         const { data: messages } = await supabase
76 |           .from('discussion_messages')
77 |           .select('*')
78 |           .order('created_at', { ascending: true })
79 |           .limit(200)
80 |         if (messages) {
81 |           const mapped = messages.map(m => ({
82 |             id: m.id,
83 |             sender: m.sender_name || 'H ỉ c v i ề n',
84 |             sender_id: m.sender_id || '',
85 |             avatar: '0=0',
86 |             role: 'student' as const,
87 |             content: m.content,
88 |             time: formatTime(m.created_at),
89 |             channel: m.channel
90 |           }))
91 |           setGlobalMessages(mapped.filter(m => m.channel === 'global'))
92 |           setClassMessages(mapped.filter(m => m.channel !== 'global'))
93 |         }

```

```

94 |     } catch (err) {
95 |       console.error('Error loading initial data:', err)
96 |     } finally {
97 |       setLoading(false)
98 |     }
99 |   }
100 |   loadInitial()
101 | }, [supabase])
102 |
103 | // Realtime subscription for new messages
104 | useEffect(() => {
105 |   const channel = supabase
106 |     .channel('discussion_messages_realtime')
107 |     .on('postgres_changes', { event: 'INSERT', schema: 'public', table: 'discussion_messages' }, (payload) => {
108 |       const m = payload.new as any
109 |       const newMsg: ChatMessage = {
110 |         id: m.id,
111 |         sender_id: m.sender_id || 'Híc viêñ',
112 |         sender_name: m.sender_name || 'Híc viêñ',
113 |         avatar: '0=0',
114 |         role: 'student' as const,
115 |         content: m.content,
116 |         time: formatTime(m.created_at),
117 |         channel: m.channel
118 |       }
119 |       if (m.channel === 'global') {
120 |         setGlobalMessages(prev => prev.some(x => x.id === newMsg.id) ? prev : [...prev, newMsg])
121 |       } else {
122 |         setClassMessages(prev => prev.some(x => x.id === newMsg.id) ? prev : [...prev, newMsg])
123 |       }
124 |     })
125 |   .subscribe()
126 |
127 |   return () => { supabase.removeChannel(channel) }
128 | }, [supabase])
129 |
130 | // Trending topics from global messages
131 | useEffect(() => {
132 |   if (globalMessages.length === 0) { setTrendingTopic('Ch°a có dĩ liç')
133 |   const words = globalMessages.map(m => m.content).join(' ').toLowerCase().split(/ /)
134 |   const stopWords = ['có', 'không', 'là', 'thì', 'mà', 'và', 'Ã', 'cçá', 'ch']
135 |   const counts: Record<string, number> = {}
136 |   words.forEach(w => {
137 |     const cw = w.replace(/^[a-z0-9]/gi, '')
138 |     if (cw.length > 2 && !stopWords.includes(cw)) counts[cw] = (counts[cw] || 0) + 1
139 |   })
140 |   const sorted = Object.entries(counts).sort((a,b) => b[1] - a[1])
141 |   if (sorted.length > 0) setTrendingTopic(sorted.slice(0, 3).map(x => '#' + x[0].toUpperCase()).join(' '))
142 | }, [globalMessages])
143 |
144 | // Scroll to bottom
145 | useEffect(() => {
146 |   chatEndRef.current?.scrollIntoView({ behavior: 'smooth' })
147 | }, [globalMessages, classMessages])
148 |
149 | // Send chat - persists to DB
150 | const handleSendChat = async () => {
151 |   if (!chatInput.trim() || !userProfile || !user) return
152 |   try {
153 |     const { error } = await supabase.from('discussion_messages').insert({
154 |       channel: chatChannel,
155 |       class_id: chatChannel === 'class' ? userProfile.class_id || null : null,
156 |       sender_id: user.id,
157 |       sender_name: userProfile.full_name || 'Híc viêñ',
158 |       content: chatInput.trim()
159 |     })
160 |     if (error) console.error('Error saving message:', error)
161 |   } catch (err) {
162 |     console.error('Error sending message:', err)
163 |   }
164 |   setChatInput('')
165 | }
166 |
167 | // Handle Micro AI Send
168 | const handleSendAi = async () => {
169 |   if (!aiInput.trim()) return
170 |   const userQ = aiInput
171 |   setAiInput('')
172 |   setAiReplies(prev => [...prev, `Híc viêñ: ${userQ}`])
173 |   setAiLoading(true)
174 |   try {
175 |     const res = await fetch('/api/ai/chat', {
176 |       method: 'POST',
177 |       headers: { 'Content-Type': 'application/json' },
178 |       body: JSON.stringify({ message: userQ })
179 |     })
180 |     const data = await res.json()
181 |     setAiReplies(prev => [...prev, `AI Guru: ${data.reply} || Xin lxi,
182 |   } catch {
183 |     setAiReplies(prev => [...prev, `AI Guru: Tôi ang b-n xí lý dĩ liç
184 |   } finally {
185 |     setAiLoading(false)
186 |   }
187 | }
188 |
189 | // Convert className hardcoded colors to inline CSS variables
190 | const s = {
191 |   bgBase: 'var(--bg-base)',
192 |   bgSurface: 'var(--bg-surface)',
193 |   bgElevated: 'var(--bg-elevated)',
194 |   brand: 'var(--brand-primary)',
195 |   textMuted: 'var(--text-muted)',
196 |   textPrimary: 'var(--text-primary)',
197 |   border: 'var(--border-default)',
198 | }
199 |
200 | if (loading) {
201 |   return (
202 |     <div style={{ minHeight: '100vh', background: s.bgBase, color: s.textPrimary, paddingTop: '96px', displa...
203 |       <div className="flex flex-col items-center gap-2">
204 |         <Loader2 className="animate-spin" style={{ color: s.brand }} size={32} />
205 |         <span style={{ fontSize: '12px', color: s.textMuted, fontWeight: 700, textTransform: 'uppercase', le...
206 |       </div>
207 |     )
208 |   )
209 | }
210 |
211 | return (
212 |   <div style={{ minHeight: '100vh', background: s.bgBase, color: s.textPrimary, paddingTop: '96px', paddingB...
213 |     <div className="max-w-7xl mx-auto relative" style={{ zIndex: 10 }}>
214 |       <div style={{ height: 100px, background: s.bgBase, border: s.border, position: 'relative' }}>
215 |         <div style={{ position: 'absolute', top: 0, right: 0, bottom: 0, left: 0, display: 'flex', align-items: center, justify-content: center, gap: 10px }}>
216 |           <div style={{ width: 40px, height: 40px, background: s.bgBase, border: s.border, display: 'flex', align-items: center, justify-content: center }}>
217 |             <div style={{ width: 20px, height: 20px, background: s.bgBase, border: s.border, display: 'flex', align-items: center, justify-content: center }}>

```

```

218 |
219 | <div className="flex items-center gap-3">
220 |   <div style={{ padding: '10px', background: `color-mix(in srgb, ${s.brand} 10%, transparent)`}, bord...
221 |     <MessageSquare size={24} />
222 |   </div>
223 |   <div>
224 |     <h1 className="text-xl md:text-2xl font-black tracking-tight uppercase flex items-center gap-2" ...
225 |       K H Ò N G   G I A N   T H Ơ   L U Ỗ N   3 D
226 |     <span style={{ fontSize: '9px', background: `color-mix(in srgb, ${s.brand} 15%, transparent)`},...
227 |     </h1>
228 |     <p style={{ fontSize: '12px', color: s.textMuted, margin: 0 }}>
229 |   </div>
230 | </div>
231 |
232 | <div className="flex items-center gap-3">
233 |   <button
234 |     onClick={() => router.push('/builder')}
235 |     className="relative z-50 flex items-center gap-2 px-4 py-2 text-xs font-bold rounded-xl transiti...
236 |     onMouseEnter={e => { e.currentTarget.style.color = s.textPrimary; e.currentTarget.style.borderCo...
237 |     onMouseLeave={e => { e.currentTarget.style.color = s.textMuted; e.currentTarget.style.borderCoLo...
238 |   >
239 |     <ArrowLeft size={14} /> Quay lại Dashboard
240 |   </button>
241 | </div>
242 | </div>
243 |
244 | /* Layout: Left (Discussions) & Right (Chat + AI) */
245 | <div className="grid grid-cols-1 lg:grid-cols-12 gap-8">
246 |
247 |   /* Left Column */
248 |   <div className="lg:col-span-7 space-y-6 relative">
249 |     {selectedThread ? (
250 |       <div style={{ background: s.bgSurface, border: `1px solid ${s.border}`, borderRadius: '24px', pa...
251 |         <ThreadDetail
252 |           thread={selectedThread}
253 |           currentUserId={userProfile?.id}
254 |           onBack={() => setSelectedThread(null)}
255 |         />
256 |       </div>
257 |     ) : (
258 |       <div style={{ background: s.bgSurface, border: `1px solid ${s.border}`, borderRadius: '24px', pa...
259 |         <div className="flex items-center justify-between pb-4" style={{ borderBottom: '1px solid colo...
260 |           <h2 className="text-lg font-bold tracking-tight flex items-center gap-2" style={{ color: s.t...
261 |             <Globe size={18} style={{ color: s.brand }} />
262 |             C h ợ   A   t h Ơ   l u - n   s ố   i   n   ò   i
263 |           </h2>
264 |           <span style={{ fontSize: '10px', background: `color-mix(in srgb, ${s.brand} 15%, transparent...
265 |             { threads.length }   C h ợ   A
266 |         </span>
267 |       </div>
268 |       {threads.length === 0 ? (
269 |         <div style={{ padding: '80px 0', textAlign: 'center', color: s.textMuted, fontStyle: 'italic...
270 |           C h ơ   a   c ó   c h ợ   A   t h Ơ   l u - n   n ă o   ơ   ă c   t i ơ .   H ă y   l ă   n g   ơ   Y i
271 |         </div>
272 |       ) : (
273 |         <div className="grid grid-cols-1 sm:grid-cols-2 gap-4" style={{ marginTop: '16px' }}>
274 |           {threads.map(thread => (
275 |             <ThreadCard
276 |               key={thread.id}
277 |               thread={thread}
278 |               onSelect={setSelectedThread}
279 |             />
280 |           ))}
281 |         </div>
282 |       )}
283 |     </div>
284 |   )}
285 | </div>
286 |
287 | /* Right Column */
288 | <div className="lg:col-span-5 flex flex-col gap-6 relative">
289 |
290 |   /* Realtime Chat Card */
291 |   <div style={{ background: s.bgSurface, border: `1px solid ${s.border}`, borderRadius: '24px', disp...
292 |
293 |     /* Channel Tabs */
294 |     <div style={{ padding: '12px', background: 'var(--bg-elevated)', borderBottom: `1px solid ${s.bo...
295 |       <div className="flex gap-2">
296 |         {[ 'global', 'class' ] as const}.map(ch => (
297 |           <button key={ch} onClick={() => setChatChannel(ch)}
298 |             style={{
299 |               display: 'flex', alignItems: 'center', gap: '8px', padding: '6px 12px', borderRadius: ...
300 |               fontSize: '12px', fontWeight: 700, cursor: 'pointer', border: 'none', fontFamily: 'inh...
301 |               background: chatChannel === ch ? `color-mix(in srgb, ${s.brand} 10%, transparent)` : '...
302 |               color: chatChannel === ch ? s.brand : s.textMuted,
303 |             }}
304 |           >
305 |             {ch === 'global' ? <Globe size={14} /> : <School size={14} />}
306 |             { c h   =   =   ' g l o b a l '   ?   ' K ê n h   C h u n g '   :   ' K ê n h   L ộ p   H í c ' }
307 |           </button>
308 |         )}
309 |       </div>
310 |       <div style={{ fontSize: '10px', color: s.textMuted, display: 'flex', flexDirection: 'column', ...
311 |         <div className="flex items-center gap-1"><div style={{ width: '6px', height: '6px', backgrou...
312 |         <div style={{ color: s.brand, fontWeight: 700, marginTop: '2px' }}>Trending: {trendingTopic}...
313 |       </div>
314 |     </div>
315 |
316 |     /* Messages */
317 |     <div style={{ flex: 1, overflowY: 'auto', padding: '16px', display: 'flex', flexDirection: 'colu...
318 |       {(chatChannel === 'global' ? globalMessages : classMessages).map(m => (
319 |         <div key={m.id} className="flex gap-3 text-xs items-start">
320 |           <span style={{ width: '32px', height: '32px', borderRadius: '12px', background: s.bgElevat...
321 |             <div style={{ flex: 1 }}>
322 |               <div className="flex items-center gap-2">
323 |                 <span style={{ fontWeight: 700, color: m.role === 'teacher' ? s.brand : 'var(--text-se...
324 |                   {m.role === 'teacher' && (
325 |                     <span style={{ fontSize: '8px', background: `color-mix(in srgb, ${s.brand} 10%, tran...
326 |                   )}
327 |                 <span style={{ fontSize: '9px', color: s.textMuted, marginLeft: 'auto' }}>{m.time}</span>
328 |               </div>
329 |               <p style={{ color: 'var(--text-secondary)', marginTop: '4px', lineHeight: 1.5, backgroun...
330 |                 {m.content}
331 |               </p>
332 |             </div>
333 |           </div>
334 |         )}
335 |       <div ref={chatEndRef} />
336 |     </div>
337 |
338 |     /* Chat Input */
339 |     <div style={{ padding: '12px', borderTop: `1px solid ${s.border}`, background: s.bgSurface, flex...
340 |       <div style={{ display: 'flex', alignItems: 'center', gap: '8px', background: s.bgBase, border:...
341 |         <input

```

```

342|         type="text"
343|         placeholder={chatChannel === 'global' ? "Chat chung"
344|         value={chatInput}
345|         onChange={(e) => setChatInput(e.target.value)}
346|         onKeyDown={(e) => e.key === 'Enter' && handleSendChat()}
347|         style={{ flex: 1, background: 'transparent', border: 'none', outline: 'none', fontSize: '1.1em' }}
348|       />
349|       <button onClick={handleSendChat}
350|         style={{ padding: '8px', background: s.brand, color: '#000', borderRadius: '12px', border: '1px solid #ccc' }}
351|       />
352|     </button>
353|   </div>
354| </div>
355| </div>
356|
357|   { /* Micro AI Assistant */ }
358|   <div style={{ background: s.bgSurface, border: `1px solid ${s.border}`, borderRadius: '24px', padding: '16px' }}
359|     <div className="flex items-center justify-between" style={{ flexShrink: 0 }}>
360|       <div className="flex items-center gap-2">
361|         <div style={{ padding: '6px', background: `color-mix(in srgb, ${s.brand} 10%, transparent)` }}
362|           <Cpu size={14} />
363|         </div>
364|         <div>
365|           <h4 className="text-xs font-bold tracking-tight flex items-center gap-1.5" style={{ color: s.brand }}>
366|             MICRO AI GURU
367|             <Sparkles size={10} style={{ color: '#eab308' }} />
368|           </h4>
369|           <p style={{ fontSize: '9px', color: s.brand, textTransform: 'uppercase', fontWeight: 700, margin: 0 }}>
370|             MICRO AI GURU
371|           </p>
372|         </div>
373|       </div>
374|       <div style={{ width: '6px', height: '6px', borderRadius: '50%', background: s.brand, animation: 'pulse 2s infinite' }}
375|     </div>
376|     <div style={{ flex: 1, overflowY: 'auto', margin: '12px 0', padding: '12px', background: `color-mix(in srgb, ${s.brand} 5%, transparent)` }}
377|       {aiReplies.map((r, i) => (
378|         <div key={i} style={{
379|           padding: '8px', borderRadius: '8px',
380|           background: r.startsWith('AI Guru:') ? `color-mix(in srgb, ${s.brand} 5%, transparent)` : `color-mix(in srgb, ${s.brand} 10%, transparent)`,
381|           color: r.startsWith('AI Guru:') ? 'var(--text-secondary)' : 'var(--text-secondary)',
382|           borderLeft: `2px solid ${r.startsWith('AI Guru:') ? s.brand : s.border}`
383|         }}>
384|           {r}
385|         </div>
386|       ))}
387|       <div className="flex items-center gap-1" style={{ color: s.textMuted, fontStyle: 'italic', font-size: '0.9em' }}>
388|         <div style={{ display: 'flex', alignItems: 'center', gap: '8px', background: s.bgBase, border: `1px solid ${s.border}` }}
389|           <input type="text"
390|             value={aiInput}
391|             onChange={(e) => setAiInput(e.target.value)}
392|             onKeyDown={(e) => e.key === 'Enter' && handleSendAi()}
393|             style={{ flex: 1, background: 'transparent', border: 'none', outline: 'none', fontSize: '1.1em' }}
394|           />
395|           <button onClick={handleSendAi}
396|             style={{ padding: '6px', background: s.brand, color: '#000', borderRadius: '8px', border: '1px solid #ccc' }}
397|           />
398|         </div>
399|       </div>
400|     </div>
401|   </div>
402| </div>
403| )
404| }
405| }
406| }
407| }
408| }
409| }
410| }
411| }
412| }
413| }
414| }
415| }
416| }
417| }
418| }
419| }

```

File: app/student/error.tsx (14 lines, 0.6 KB)

```

1| 'use client'
2| import { useEffect } from 'react'
3|
4| export default function Error({ error, reset }: { error: Error & { digest?: string }, reset: () => void }) {
5|   useEffect(() => { console.error(error) }, [error])
6|   return (
7|     <div className="flex flex-col items-center justify-center min-h-screen bg-[#0d0e13] text-[#dde0ed]">
8|       <div style={{ display: 'flex', flex-direction: 'column', align-items: 'center', justify-content: 'center', gap: '20px' }}>
9|         <p style={{ color: 'red', font-weight: 'bold', text-align: 'center' }}>{error.message}</p>
10|        <button onClick={() => reset()} className="bg-[#0d0e13] text-[#dde0ed] px-6 py-2 rounded font-bold hover:bg-[#1a1b20]">
11|          Try again
12|        </button>
13|      </div>
14|    </div>
15|  )
16| }

```

File: app/student/history/page.tsx (499 lines, 22.0 KB)

```

1| 'use client'
2|
3| import { useEffect, useState, useCallback, useRef } from 'react'
4| import { createBrowserClient } from '@supabase/ssr'
5| import { motion, AnimatePresence } from 'framer-motion'
6| import {
7|   History, BookOpen, Award, Clock, ChevronRight, Wrench, Cpu,
8|   Monitor, Smartphone, HardDrive, Shield, Zap, Star, Trophy,
9|   BrainCircuit, Sparkles, RefreshCw, Calendar, Filter, ArrowUpDown,
10|   CheckCircle2, XCircle, AlertTriangle
11| } from 'lucide-react'
12| import { format, isSameDay, parseISO, startOfDay } from 'date-fns'
13|
14| type ActivityType = 'lesson' | 'quiz' | 'exam' | 'builder' | 'xp'
15|
16| interface Activity {
17|   id: string
18|   type: ActivityType
19|   title: string
20|   description: string
21|   details: string
22|   time: string
23|   score: number | null

```

```

24| xpEarned: number
25| icon: React.ReactNode
26| color: string
27| status?: string
28| }
29|
30| const TYPE_CONFIG: Record<ActivityType, { label: string; color: string; gradient: string }> = {
31|   lesson: { label: 'Bài Híc', color: '#10B981', gradient: 'from-emerald-500 to-teal-500' },
32|   quiz: { label: 'Bài KiĂm Tra', color: '#6366F1', gradient: 'from-indigo-500 to-violet-500' },
33|   exam: { label: 'KÓ Thĩ', color: '#EF4444', gradient: 'from-red-500 to-orange-500' },
34|   builder: { label: 'L p Ráp PC', color: '#06B6D4', gradient: 'from-cyan-500 to-blue-500' },
35|   xp: { label: 'Kinh Nghĩ Çm', color: '#F59E0B', gradient: 'from-amber-500 to-yellow-500' }
36| }
37|
38| function formatTimeAgo(dateStr: string): string {
39|   const d = new Date(dateStr)
40|   const now = new Date()
41|   const diff = now.getTime() - d.getTime()
42|   const seconds = Math.floor(diff / 1000)
43|   if (seconds < 60) return 'Văa xong'
44|   const minutes = Math.floor(seconds / 60)
45|   if (minutes < 60) return `${minutes} phút tr°Ùc`
46|   const hours = Math.floor(minutes / 60)
47|   if (hours < 24) return `${hours} giÝ tr°Ùc`
48|   const days = Math.floor(hours / 24)
49|   if (days < 7) return `${days} ngày tr°Ùc`
50|   return format(d, 'dd/MM/yyyy')
51| }
52|
53| function formatTimeDisplay(dateStr: string): string {
54|   const d = new Date(dateStr)
55|   return format(d, 'HH:mm')
56| }
57|
58| export default function StudentHistoryPage() {
59|   const supabase = createBrowserClient(
60|     process.env.NEXT_PUBLIC_SUPABASE_URL!,
61|     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
62|   )
63|
64|   const [userId, setUserId] = useState<string | null>(null)
65|   const [activities, setActivities] = useState<Activity[]>([])
66|   const [loading, setLoading] = useState(true)
67|   const [filter, setFilter] = useState<ActivityType | 'all'>('all')
68|   const [sortAsc, setSortAsc] = useState(false)
69|   const refreshIntervalRef = useRef<NodeJS.Timeout | null>(null)
70|
71|   const getIconForType = (type: ActivityType): React.ReactNode => {
72|     const props = { size: 18 }
73|     switch (type) {
74|       case 'lesson': return <BookOpen {...props} />
75|       case 'quiz': return <BrainCircuit {...props} />
76|       case 'exam': return <Award {...props} />
77|       case 'builder': return <Wrench {...props} />
78|       case 'xp': return <Zap {...props} />
79|     }
80|   }
81|
82|   const fetchHistory = useCallback(async () => {
83|     try {
84|       const { data: { user: u } } = await supabase.auth.getUser()
85|       if (!u) { setLoading(false); return }
86|       setUserId(u.id)
87|
88|       const [lessonsRes, quizRes, examRes, builderRes, xpRes] = await Promise.all([
89|         supabase.from('lesson_progress')
90|           .select('lesson_id, status, completed_at, last_accessed, time_spent_seconds, score, completion_perce...')
91|           .eq('student_id', u.id)
92|           .not('last_accessed', 'is', null)
93|           .order('last_accessed', { ascending: false })
94|           .limit(50),
95|         supabase.from('quiz_attempts')
96|           .select('id, quiz_id, score, status, submitted_at, created_at, quizzes(title)')
97|           .eq('student_id', u.id)
98|           .order('created_at', { ascending: false })
99|           .limit(50),
100|        supabase.from('exam_attempts')
101|          .select('id, exam_id, score, status, started_at, submitted_at, exams(title)')
102|          .eq('student_id', u.id)
103|          .order('started_at', { ascending: false })
104|          .limit(50),
105|        supabase.from('builder_sessions')
106|          .select('started_at, ended_at')
107|          .eq('student_id', u.id)
108|          .not('started_at', 'is', null)
109|          .order('started_at', { ascending: false })
110|          .limit(50),
111|        supabase.from('xp_transactions')
112|          .select('amount, reason, reference_type, created_at')
113|          .eq('user_id', u.id)
114|          .order('created_at', { ascending: false })
115|          .limit(50),
116|      ])
117|
118|      const items: Activity[] = []
119|
120|      ;(lessonsRes.data || []).forEach((l: any) => {
121|        items.push({
122|          id: `lesson-${l.lesson_id}-${l.last_accessed}`,
123|          type: 'lesson',
124|          title: l.lessons?.title || 'Bài híc',
125|          description: l.status === 'completed' ? 'Hoàn thành bài híc' : 'Đang học',
126|          details: l.status === 'completed' ? `Hoàn thành lúc ${formatTimeDisplay(l.completed_at || l.last_accessed)}` : `Đang học từ lúc ${formatTimeDisplay(l.created_at || l.last_accessed)}`,
127|          time: l.completed_at || l.last_accessed,
128|          score: l.score,
129|          xpEarned: l.status === 'completed' ? 50 : 0,
130|          icon: getIconForType('lesson'),
131|          color: TYPE_CONFIG.lesson.color,
132|          status: l.status,
133|        })
134|      })
135|
136|      ;(quizRes.data || []).forEach((q: any) => {
137|        const isComplete = q.status === 'submitted' || q.status === 'passed' || q.status === 'failed'
138|        items.push({
139|          id: `quiz-${q.id}-${q.created_at}`,
140|          type: 'quiz',
141|          title: q.quizzes?.title || 'Bài kiĂm tra',
142|          description: isComplete ? `KiĂm: ${q.score}/100` : 'Ch°a hoàn thành',
143|          details: isComplete ? `Hoàn thành lúc ${formatTimeDisplay(q.submitted_at)}` : `Đã t su lúç ${formatTimeDisplay(q.created_at)}`,
144|          time: q.submitted_at || q.created_at,
145|          score: q.score,
146|          xpEarned: isComplete ? 10 : 0,
147|          icon: getIconForType('quiz'),
148|          color: TYPE_CONFIG.quiz.color,
149|          status: q.status,
150|        })
151|      })
152|
153|      ;(examRes.data || []).forEach((e: any) => {
154|        const isCompleted = e.status === 'passed' || e.status === 'failed'
155|        items.push({
156|          id: `exam-${e.id}-${e.started_at}`,
157|          type: 'exam',
158|          title: e.exams?.title || 'KÓ thĩ',
159|          description: isCompleted ? 'Đã thi xong' : 'Đang thi',
160|          details: isCompleted ? `Thi xong lúc ${formatTimeDisplay(e.submitted_at)}` : `Đang thi từ lúc ${formatTimeDisplay(e.started_at)}`,
161|          time: e.submitted_at || e.started_at,
162|          score: e.score,
163|          xpEarned: isCompleted ? 20 : 0,
164|          icon: getIconForType('exam'),
165|          color: TYPE_CONFIG.exam.color,
166|          status: e.status,
167|        })
168|      })
169|
170|      ;(builderRes.data || []).forEach((b: any) => {
171|        items.push({
172|          id: `builder-${b.session_id}-${b.started_at}`,
173|          type: 'builder',
174|          title: b.projects?.title || 'Dự án',
175|          description: b.status === 'completed' ? 'Hoàn thành dự án' : 'Đang xây dựng',
176|          details: b.status === 'completed' ? `Hoàn thành lúc ${formatTimeDisplay(b.submitted_at)}` : `Đang xây dựng từ lúc ${formatTimeDisplay(b.started_at)}`,
177|          time: b.submitted_at || b.started_at,
178|          score: b.score,
179|          xpEarned: b.status === 'completed' ? 15 : 0,
180|          icon: getIconForType('builder'),
181|          color: TYPE_CONFIG.builder.color,
182|          status: b.status,
183|        })
184|      })
185|
186|      ;(xpRes.data || []).forEach((x: any) => {
187|        items.push({
188|          id: `xp-${x.transaction_id}-${x.created_at}`,
189|          type: 'xp',
190|          title: x.actions?.title || 'Trải nghiệm',
191|          description: x.status === 'completed' ? 'Đã trải nghiệm' : 'Đang chờ',
192|          details: x.status === 'completed' ? `Trải nghiệm lúc ${formatTimeDisplay(x.submitted_at)}` : `Đang chờ từ lúc ${formatTimeDisplay(x.created_at)}`,
193|          time: x.submitted_at || x.created_at,
194|          score: x.score,
195|          xpEarned: x.status === 'completed' ? 5 : 0,
196|          icon: getIconForType('xp'),
197|          color: TYPE_CONFIG.xp.color,
198|          status: x.status,
199|        })
200|      })
201|
202|      setActivities(items)
203|      setLoading(false)
204|    } catch (error) {
205|      console.error('Error fetching history:', error)
206|    }
207|  })
208|
209|   // Cleanup interval
210|   useEffect(() => {
211|     if (refreshIntervalRef.current) {
212|       clearInterval(refreshIntervalRef.current)
213|     }
214|     refreshIntervalRef.current = setInterval(fetchHistory, 60000)
215|   }, [])
216|
217|   return (
218|     <div>
219|       <Header/>
220|       <MainContent/>
221|     </div>
222|   )
223| }

```

```

148 |         time: q.submitted_at || q.created_at,
149 |         score: q.score,
150 |         xpEarned: isComplete ? Math.round(q.score * 2) : 0,
151 |         icon: getIconForType('quiz'),
152 |         color: TYPE_CONFIG.quiz.color,
153 |         status: q.status,
154 |     })
155 | })
156 |
157 | ;(examRes.data || []).forEach((e: any) => {
158 |     const isComplete = e.status === 'submitted' || e.status === 'passed' || e.status === 'failed'
159 |     items.push({
160 |         id: `exam-${e.id}-${e.started_at}`,
161 |         type: 'exam',
162 |         title: e.exams?.title || 'Kó thi',
163 |         description: isComplete ? `iẤm: ${e.score}/100` : 'ang làm t
164 |         details: isComplete
165 |             ? `Hoàn thành lúc ${formatTimeDisplay(e.submitted_at)}`
166 |             : `B t s u l ú c ${formatTimeDisplay(e.started_at)}`,
167 |         time: e.submitted_at || e.started_at,
168 |         score: e.score,
169 |         xpEarned: isComplete ? Math.round((e.score || 0) * 3) : 0,
170 |         icon: getIconForType('exam'),
171 |         color: TYPE_CONFIG.exam.color,
172 |         status: e.status,
173 |     })
174 | })
175 |
176 | ;(builderRes.data || []).forEach((b: any) => {
177 |     const duration = b.ended_at
178 |         ? Math.round((new Date(b.ended_at).getTime() - new Date(b.started_at).getTime()) / 60000)
179 |         : 0
180 |     items.push({
181 |         id: `builder-${b.started_at}`,
182 |         type: 'builder',
183 |         title: 'Phòng thực hành l-p ráp',
184 |         description: duration > 0 ? `Thýi gian: ${duration} phút` :
185 |         ? `K z t th ú c l ú c ${formatTimeDisplay(b.ended_at)}`
186 |         : 'ang thñc hành',
187 |         time: b.started_at,
188 |         score: null,
189 |         xpEarned: duration > 10 ? 100 : duration > 5 ? 50 : 20,
190 |         icon: getIconForType('builder'),
191 |         color: TYPE_CONFIG.builder.color,
192 |     })
193 | })
194 |
195 |
196 | ;(xpRes.data || []).forEach((x: any) => {
197 |     items.push({
198 |         id: `xp-${x.created_at}-${x.amount}`,
199 |         type: 'xp',
200 |         title: x.reason || 'Nh-n XP',
201 |         description: `+${x.amount} XP`,
202 |         details: x.reference_type
203 |             ? `T ẽ: ${x.reference_type.replace(/_/g, ' ')}
204 |             : 'Giao đ ẽch kinh nghi ẽm',
205 |         time: x.created_at,
206 |         score: null,
207 |         xpEarned: x.amount,
208 |         icon: getIconForType('xp'),
209 |         color: TYPE_CONFIG.xp.color,
210 |     })
211 | })
212 |
213 | items.sort((a, b) => {
214 |     const cmp = new Date(b.time).getTime() - new Date(a.time).getTime()
215 |     return sortAsc ? -cmp : cmp
216 | })
217 |
218 | setActivities(items.slice(0, 100))
219 | } catch (err) {
220 |     console.error('Error fetching history:', err)
221 | } finally {
222 |     setLoading(false)
223 | }
224 | }, [sortAsc, supabase])
225 |
226 | useEffect(() => {
227 |     fetchHistory()
228 | }, [fetchHistory])
229 |
230 | useEffect(() => {
231 |     if (!userId) return
232 |     const channel = supabase
233 |         .channel('student-history-realtime')
234 |         .on('postgres_changes',
235 |             { event: '*', schema: 'public', table: 'lesson_progress', filter: `student_id=eq.${userId}` },
236 |             () => fetchHistory()
237 |         )
238 |         .on('postgres_changes',
239 |             { event: '*', schema: 'public', table: 'quiz_attempts', filter: `student_id=eq.${userId}` },
240 |             () => fetchHistory()
241 |         )
242 |         .on('postgres_changes',
243 |             { event: '*', schema: 'public', table: 'exam_attempts', filter: `student_id=eq.${userId}` },
244 |             () => fetchHistory()
245 |         )
246 |         .on('postgres_changes',
247 |             { event: '*', schema: 'public', table: 'builder_sessions', filter: `student_id=eq.${userId}` },
248 |             () => fetchHistory()
249 |         )
250 |         .on('postgres_changes',
251 |             { event: 'INSERT', schema: 'public', table: 'xp_transactions', filter: `user_id=eq.${userId}` },
252 |             () => fetchHistory()
253 |         )
254 |         .subscribe()
255 |
256 |     return () => { supabase.removeChannel(channel) }
257 | }, [userId, supabase, fetchHistory])
258 |
259 | const filteredActivities = filter === 'all'
260 |     ? activities
261 |     : activities.filter(a => a.type === filter)
262 |
263 | const groupedByDate: { [key: string]: Activity[] } = {}
264 | filteredActivities.forEach(activity => {
265 |     const dayKey = startOfDay(parseISO(activity.time)).toISOString()
266 |     if (!groupedByDate[dayKey]) groupedByDate[dayKey] = []
267 |     groupedByDate[dayKey].push(activity)
268 | })
269 |
270 | const today = startOfDay(new Date()).toISOString()
271 | const yesterday = startOfDay(new Date(Date.now() - 86400000)).toISOString()

```



```

273 function getDateLabel(dateKey: string): string {
274   if (dateKey === today) return 'Hôm nay'
275   if (dateKey === yesterday) return 'Hôm qua'
276   return format(parseISO(dateKey), 'EEEE, dd/MM/yyyy')
277 }
278
279 const FILTERS: { key: ActivityType | 'all'; label: string; icon: React.ReactNode }[] = [
280   { key: 'all', label: 'Tất cả', icon: <Sparkles size={14} /> },
281   { key: 'lesson', label: 'Bài học', icon: <BookOpen size={14} /> },
282   { key: 'quiz', label: 'Kiểm tra', icon: <BrainCircuit size={14} /> },
283   { key: 'exam', label: 'Kỳ thi', icon: <Award size={14} /> },
284   { key: 'builder', label: 'Lập trình', icon: <Wrench size={14} /> },
285   { key: 'xp', label: 'XP', icon: <Zap size={14} /> },
286 ]
287
288 if (loading) {
289   return (
290     <div className="min-h-screen flex flex-col items-center justify-center gap-3 p-8"
291       style={{ background: 'var(--bg-base)', color: 'var(--text-primary)' }}>
292       <motion.div
293         animate={{ rotate: 360 }}
294         transition={{ duration: 1.5, repeat: Infinity, ease: 'linear' }}
295       >
296         <RefreshCw size={32} style={{ color: 'var(--brand-primary)' }} />
297       </motion.div>
298       <span className="text-xs font-bold uppercase tracking-wider" style={{ color: 'var(--text-muted)' }}>
299         Loading...
300     </span>
301   </div>
302 )
303 }
304
305 return (
306   <div className="min-h-screen pb-12" style={{ background: 'var(--bg-base)', color: 'var(--text-primary)' }}>
307     <div className="relative overflow-hidden">
308       <div className="absolute inset-0 opacity-[0.02] pointer-events-none"
309         style={{ backgroundImage: 'radial-gradient(circle at 25% 25%, var(--brand-primary) 1px, transparent ...' }}>
310       <div className="absolute top-0 left-1/4 w-[600px] h-[600px] rounded-full filter blur-[120px] pointer-e...
311         style={{ background: 'color-mix(in srgb, var(--brand-primary) 4%, transparent)' }} />
312     </div>
313     <div className="max-w-3xl mx-auto px-4 pt-8 pb-4 relative z-10">
314       <motion.div initial={{ opacity: 0, y: -20 }} animate={{ opacity: 1, y: 0 }}
315         className="flex items-center justify-between mb-6">
316         <div className="flex items-center gap-4">
317           <div className="p-3 rounded-2xl" style={{
318             background: 'color-mix(in srgb, var(--brand-primary) 10%, transparent)',
319             border: '1px solid color-mix(in srgb, var(--brand-primary) 25%, transparent)',
320             color: 'var(--brand-primary)'
321           }}>
322             <History size={24} />
323           </div>
324           <div>
325             <h1 className="text-2xl font-black tracking-tight">Lịch
326               <p className="text-xs mt-0.5" style={{ color: 'var(--text-muted)' }}>
327                 Theo dõi toàn bộ hoạt động học tập theo thời gian thực
328             </p>
329           </div>
330         </div>
331         <button onClick={() => { setSortAsc(!sortAsc); fetchHistory() }}
332           className="flex items-center gap-1.5 px-3 py-2 text-xs font-bold rounded-xl transition-all curso...
333           style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)', color: 'var...
334           <ArrowUpDown size={14} />
335           { sortAsc ? 'Ci nhệ t' : 'Mũi nhệ t' }
336         </button>
337       </motion.div>
338
339       <motion.div initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }} transition={{ delay: 0.1 }}
340         className="flex gap-2 overflow-x-auto pb-2 mb-6 scrollbar-none">
341         {FILTERS.map(f => (
342           <button key={f.key} onClick={() => setFilter(f.key)}
343             className="flex items-center gap-1.5 px-3.5 py-2 text-xs font-bold rounded-xl transition-all w...
344             style={{
345               background: filter === f.key
346                 ? 'color-mix(in srgb, var(--brand-primary) 15%, transparent)'
347                 : 'var(--bg-surface)',
348               border: filter === f.key
349                 ? '1px solid color-mix(in srgb, var(--brand-primary) 35%, transparent)'
350                 : '1px solid var(--border-default)',
351               color: filter === f.key ? 'var(--brand-primary)' : 'var(--text-muted)',
352             }}>
353           {f.icon}
354           {f.label}
355           {filter === f.key && (
356             <span className="ml-1 px-1.5 py-0.5 rounded-full text-[9px] font-bold"
357               style={{ background: 'color-mix(in srgb, var(--brand-primary) 20%, transparent)' }}>
358               {filteredActivities.length}
359             </span>
360           )}
361         </button>
362       </div>
363     </motion.div>
364
365     {filteredActivities.length === 0 ? (
366       <motion.div initial={{ opacity: 0, scale: 0.95 }} animate={{ opacity: 1, scale: 1 }}
367         className="text-center py-20 px-8 rounded-3xl"
368         style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)' }}>
369       <motion.div animate={{ y: [0, -8, 0] }} transition={{ duration: 3, repeat: Infinity }}>
370       <Clock size={56} style={{ color: 'var(--text-muted)', opacity: 0.2, margin: '0 auto 16px' }} />
371     </motion.div>
372     <p className="text-sm font-bold" style={{ color: 'var(--text-primary)' }}>
373       Chưa có hoạt động nào
374     </p>
375     <p className="text-xs mt-1" style={{ color: 'var(--text-muted)' }}>
376       Bật sự kiện Xem lịch sử hoạt động
377     </p>
378   </motion.div>
379 ) : (
380   <div className="space-y-8">
381     {Object.entries(groupedByDate).map(([dateKey, items], groupIdx) => (
382       <motion.div key={dateKey}
383         initial={{ opacity: 0, y: 20 }}
384         animate={{ opacity: 1, y: 0 }}
385         transition={{ delay: groupIdx * 0.05 }}>
386       <div className="flex items-center gap-3 mb-4">
387         <Calendar size={14} style={{ color: 'var(--brand-primary)' }} />
388         <h3 className="text-sm font-black uppercase tracking-wider"
389           style={{ color: 'var(--text-primary)' }}>
390           {getDateLabel(dateKey)}
391         </h3>
392       </div>
393       <div className="flex-1 h-px" style={{ background: 'var(--border-default)' }} />
394       <span className="text-[10px] font-bold px-2 py-0.5 rounded-full"
395         style={{ background: 'color-mix(in srgb, var(--bg-elevated) 50%, transparent)', color: '...
396         {items.length} hoạt động
397     </span>
398     )
399   </div>
400 )
401 )

```

```

396 |         </span>
397 |     </div>
398 |
399 |     <div className="relative ml-4">
400 |       <div className="absolute left-[19px] top-2 bottom-2 w-0.5"
401 |         style={{ background: 'var(--border-default)' }} />
402 |
403 |       <div className="space-y-3">
404 |         {items.map((item, idx) => (
405 |           <motion.div key={item.id}
406 |             initial={{ opacity: 0, x: -20 }}
407 |             animate={{ opacity: 1, x: 0 }}
408 |             transition={{ delay: idx * 0.03 }}
409 |             whileHover={{ x: 4 }}
410 |             className="relative pl-10"
411 |           >
412 |             <div className="absolute left-[11px] top-[18px] w-[17px] h-[17px] rounded-full borde...
413 |               style={{
414 |                 background: 'var(--bg-base)',
415 |                 borderColor: item.color,
416 |               }}>
417 |               <div className="absolute inset-[3px] rounded-full"
418 |                 style={{ background: `${item.color}30` }} />
419 |             </div>
420 |
421 |             <div className="p-3.5 rounded-xl transition-all duration-200"
422 |               style={{
423 |                 background: 'var(--bg-surface)',
424 |                 border: '1px solid var(--border-default)',
425 |               }}>
426 |               <div className="flex items-start gap-3">
427 |                 <div className="w-9 h-9 rounded-lg flex items-center justify-center flex-shrink-0"
428 |                   style={{
429 |                     background: `${item.color}15`,
430 |                     color: item.color,
431 |                   }}>
432 |                   {item.icon}
433 |                 </div>
434 |                 <div className="flex-1 min-w-0">
435 |                   <div className="flex items-start justify-between gap-2">
436 |                     <div>
437 |                       <p className="text-sm font-bold leading-tight" style={{ color: 'var(--text...
438 |                         {item.title}
439 |                       </p>
440 |                       <p className="text-xs mt-0.5 leading-relaxed" style={{ color: item.color }}>
441 |                         {item.description}
442 |                       </p>
443 |                     </div>
444 |                     <div className="text-right flex-shrink-0">
445 |                       <p className="text-[10px] font-bold whitespace-nowrap"
446 |                         style={{ color: 'var(--text-muted)' }}>
447 |                         {formatTimeAgo(item.time)}
448 |                       </p>
449 |                       {item.score != null && (
450 |                         <p className="text-xs font-black mt-0.5"
451 |                           style={{
452 |                             color: item.score >= 80 ? '#10B981'
453 |                               : item.score >= 50 ? '#F59E0B'
454 |                               : '#EF4444'
455 |                           }}>
456 |                         {item.score}%
457 |                       </p>
458 |                     )}
459 |                   </div>
460 |                 </div>
461 |                 <div className="flex items-center gap-3 mt-1.5">
462 |                   <p className="text-[10px]" style={{ color: 'var(--text-muted)' }}>
463 |                     {item.details}
464 |                   </p>
465 |                   {item.xpEarned > 0 && (
466 |                     <span className="flex items-center gap-1 px-1.5 py-0.5 rounded text-[9px] ...
467 |                       style={{
468 |                         background: 'color-mix(in srgb, #F59E0B 12%, transparent)',
469 |                         color: '#F59E0B',
470 |                         border: '1px solid color-mix(in srgb, #F59E0B 20%, transparent)',
471 |                       }}>
472 |                       <Zap size={8} />+{item.xpEarned}XP
473 |                     </span>
474 |                   )}
475 |                 </div>
476 |               </div>
477 |             </div>
478 |           </div>
479 |         </motion.div>
480 |       </div>
481 |     </div>
482 |   </div>
483 | </motion.div>
484 | ))}
485 | </div>
486 | )}
487 |
488 | {filteredActivities.length > 0 && activities.length >= 100 && (
489 |   <motion.p initial={{ opacity: 0 }} animate={{ opacity: 1 }}
490 |     className="text-center text-xs mt-8" style={{ color: 'var(--text-muted)' }}>
491 |     H i Ā n t Ē 1 0 0 h o i t Ũ n g g ã n n h ¤ t . D ĩ l i ç u c i h ĩ n c ó t h Ā º Ā
492 |   </motion.p>
493 | )}
494 | </div>
495 | </div>
496 | </div>
497 | )
498 | }
499 |

```

File: app/student/level/page.tsx (409 lines, 24.2 KB)

```

1 | 'use client'
2 |
3 | import { useState, useEffect } from 'react'
4 | import { motion, AnimatePresence } from 'framer-motion'
5 | import { useRealtime } from '@lib/realtime-provider'
6 | import { createBrowserClient } from '@supabase/ssr'
7 | import { Zap, ChevronRight, Clock, Award, Flame, CheckCircle, Trophy, Star, ChevronDown, ChevronUp } from 'luc...
8 |
9 | const supabase = createBrowserClient(
10 |   process.env.NEXT_PUBLIC_SUPABASE_URL!,
11 |   process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
12 | )
13 |
14 | const STATUS_ICONS: Record<string, string> = {
15 |   lesson_completed: 'ðŸŒŸ',
16 |   exam_completed: 'ðŸŸŸ',

```

```

17 |         quiz_completed: '0>Yá',
18 |         builder_session: '0=báp',
19 |         quest_completed: '0=UE',
20 |         daily_reward: '0<ß',
21 |     }
22 | }
23 | export default function LevelPage() {
24 |     const { state, refetch } = useRealtime()
25 |     const [activeTab, setActiveTab] = useState<'level' | 'quests' | 'titles' | 'history'>('level')
26 |     const [expandedLevel, setExpandedLevel] = useState<number | null>(null)
27 |
28 |     const {
29 |         level, levelTitle, levelIcon, levelColor, levelProgress, xp, xpToNext, xpInLevel,
30 |         streak, studyMinutes, levelDefs, quests, titles, xpHistory, allQuests,
31 |     } = state
32 |
33 |     const tabItems = [
34 |         { key: 'level', as: const, label: 'C#p ù', icon: <Zap size={16} /> },
35 |         { key: 'quests', as: const, label: 'Nhịc m Vã', icon: <CheckCircle size={16} /> },
36 |         { key: 'titles', as: const, label: 'Đánh Hịchu', icon: <Award size={16} /> },
37 |         { key: 'history', as: const, label: 'Lêch Sĩ XP', icon: <Clock size={16} /> },
38 |     ]
39 |
40 |     return (
41 |         <div className="p-6 md:p-8 lg:p-10" style={{ background: 'transparent', minHeight: '100vh' }}>
42 |             <div className="max-w-[1200px] mx-auto" style={{ display: 'flex', flexDirection: 'column', gap: '28px' }}>
43 |                 <motion.div initial={{ opacity: 0, y: 15 }} animate={{ opacity: 1, y: 0 }}>
44 |                     <div className="flex items-center gap-3 mb-2">
45 |                         <div className="w-10 h-10 rounded-xl bg-gradient-to-br from-yellow-400 to-yellow-600 flex items-ce...
46 |                             {levelIcon}
47 |                         </div>
48 |                         <div>
49 |                             <h1 className="text-2xl font-black" style={{ color: 'var(--text-primary)' }}>{levelTitle}</h1>
50 |                             <p className="text-sm" style={{ color: 'var(--text-muted)' }}>{levelColor}</p>
51 |                         </div>
52 |                     </div>
53 |                 </motion.div>
54 |
55 |                 <div>
56 |                     <div>
57 |                         initial={{ opacity: 0, y: 15 }}
58 |                         animate={{ opacity: 1, y: 0 }}
59 |                         transition={{ delay: 0.05 }}
60 |                         className="lms-card relative overflow-hidden"
61 |                         style={{
62 |                             padding: '32px',
63 |                             background: `linear-gradient(135deg, ${levelColor}15, transparent)`,
64 |                             border: `1px solid ${levelColor}30`,
65 |                         }}
66 |                     >
67 |                         <div className="absolute top-[-60px] right-[-40px] opacity-[0.04] pointer-events-none">
68 |                             <span style={{ fontSize: '180px' }}>{levelIcon}</span>
69 |                         </div>
70 |                         <div className="flex items-center gap-6 relative z-10" style={{ flexWrap: 'wrap' }}>
71 |                             <div className="text-center" style={{ minWidth: '80px' }}>
72 |                                 <div style={{ fontSize: '52px', lineHeight: 1, marginBottom: '4px' }}>{levelIcon}</div>
73 |                                 <div className="text-xs font-bold uppercase tracking-wider" style={{ color: levelColor }}>Level ...
74 |                             </div>
75 |                             <div style={{ flex: 1, minWidth: '200px' }}>
76 |                                 <div className="text-lg font-bold" style={{ color: 'var(--text-primary)' }}>{levelTitle}</div>
77 |                                 <div className="flex items-center gap-3 mt-3">
78 |                                     <div className="flex-1 h-3 rounded-full overflow-hidden" style={{ background: 'var(--bg-elevat...
79 |                                     <motion.div>
80 |                                         initial={{ width: 0 }}
81 |                                         animate={{ width: `${levelProgress}%` }}
82 |                                         transition={{ duration: 0.8, ease: 'easeOut' }}
83 |                                         className="h-full rounded-full"
84 |                                         style={{ background: `linear-gradient(90deg, ${levelColor}, ${levelColor}88)` }}
85 |                                     </div>
86 |                                     <span className="text-xs font-bold whitespace-nowrap" style={{ color: levelColor }}>
87 |                                         {levelProgress.toFixed(0)}%
88 |                                     </span>
89 |                                 </div>
90 |                             </div>
91 |                             <div className="flex items-center gap-4 mt-2">
92 |                                 <span className="text-xs" style={{ color: 'var(--text-muted)' }}>
93 |                                     <span className="font-bold" style={{ color: 'var(--text-primary)' }}>{xpInLevel.toLocaleStri...
94 |                                 </span>
95 |                                 <span className="text-xs" style={{ color: 'var(--text-muted)' }}>
96 |                                     <div>
97 |                                         <div className="flex gap-6" style={{ flexShrink: 0 }}>
98 |                                             <div className="text-center">
99 |                                                 <Flame size={20} className="mx-auto mb-1" style={{ color: '#f59e0b' }} />
100 |                                                 <div className="text-lg font-black" style={{ color: 'var(--text-primary)' }}>{streak}</div>
101 |                                                 <div className="text-[10px] font-bold uppercase tracking-wider" style={{ color: 'var(--text-mu...
102 |                                             </div>
103 |                                             <div className="text-center">
104 |                                                 <Clock size={20} className="mx-auto mb-1" style={{ color: 'var(--brand-primary)' }} />
105 |                                                 <div className="text-lg font-black" style={{ color: 'var(--text-primary)' }}>{studyMinutes / ...
106 |                                                 <div className="text-[10px] font-bold uppercase tracking-wider" style={{ color: 'var(--text-mu...
107 |                                             </div>
108 |                                         </div>
109 |                                     </div>
110 |                                 </div>
111 |                             </div>
112 |                         </div>
113 |
114 |                     <div>
115 |                         <div className="flex gap-1 p-1 rounded-xl" style={{ background: 'var(--bg-elevated)', border: '1px sol...
116 |                             {tabItems.map(tab => (
117 |                                 <button>
118 |                                     key={tab.key}
119 |                                     onClick={() => setActiveTab(tab.key)}
120 |                                     className="flex items-center gap-2 px-4 py-2.5 rounded-lg text-sm font-bold transition-all curso...
121 |                                     style={{
122 |                                         flex: 1,
123 |                                         background: activeTab === tab.key ? 'var(--bg-surface)' : 'transparent',
124 |                                         color: activeTab === tab.key ? 'var(--text-primary)' : 'var(--text-muted)',
125 |                                         border: activeTab === tab.key ? '1px solid var(--border-default)' : '1px solid transparent',
126 |                                     }}
127 |                                 >
128 |                                     {tab.icon}
129 |                                     <span className="hidden sm:inline">{tab.label}</span>
130 |                                 </button>
131 |                             ))}
132 |                         </div>
133 |
134 |                     <div>
135 |                         <div>
136 |                             <div>
137 |                                 <div>
138 |                                     <div>
139 |                                     <div>
140 |                                     <div>

```

```

141 | {levelDefs.map((lvl, i) => {
142 |   const unlocked = xp >= lvl.xp_required
143 |   const isCurrent = lvl.level === level
144 |   const isExpanded = expandedLevel === lvl.level
145 |   const nextXp = levelDefs[i + 1]?.xp_required || lvl.xp_required
146 |   const progressInLevel = nextXp > lvl.xp_required
147 |     ? Math.min((xp - lvl.xp_required) / (nextXp - lvl.xp_required)) * 100, 100)
148 |     : 100
149 |
150 |   return (
151 |     <div key={lvl.level}>
152 |       <div>
153 |         onClick={() => setExpandedLevel(isExpanded ? null : lvl.level)}
154 |         className="flex items-center gap-4 px-4 py-3 rounded-xl transition-all cursor-pointer"
155 |         style={{
156 |           background: isCurrent ? `${levelColor}15` : unlocked ? 'var(--bg-elevated)' : 'tra...
157 |           border: isCurrent ? '1px solid ${levelColor}30' : '1px solid transparent',
158 |           opacity: unlocked ? 1 : 0.4,
159 |         }}
160 |       >
161 |         <div className="flex items-center justify-center w-10 h-10 rounded-lg text-lg" style={{
162 |           background: unlocked ? `${levelColor}20` : 'var(--bg-elevated)',
163 |           border: '1px solid ${unlocked ? levelColor + '40' : 'var(--border-default)'}',
164 |         }}>
165 |           {lvl.icon}
166 |         </div>
167 |         <div style={{ flex: 1, minWidth: 0 }}>
168 |           <div className="text-sm font-bold" style={{ color: unlocked ? 'var(--text-primary)...
169 |             Level {lvl.level} - {lvl.title}
170 |           </div>
171 |           <div className="text-xs" style={{ color: 'var(--text-muted)' }}>
172 |             { lvl.description } · yêu cầu { lvl.xp_required }
173 |           </div>
174 |           {unlocked && !isCurrent && (
175 |             <div className="text-xs font-bold mt-1" style={{ color: 'var(--text-primary)' }}>
176 |               Chưa mở khóa
177 |             </div>
178 |             <div className="flex items-center gap-2 flex-shrink-0">
179 |               {isCurrent && (
180 |                 <div className="text-xs font-bold px-2 py-1 rounded" style={{ background: `${lev...
181 |                   Hích tì
182 |                 </div>
183 |                 <ChevronDown size={16} style={{ color: 'var(--text-muted)', transform: isExpanded ...
184 |               </div>
185 |             </div>
186 |             {isExpanded && (
187 |               <motion.div
188 |                 initial={{ height: 0, opacity: 0 }}
189 |                 animate={{ height: 'auto', opacity: 1 }}
190 |                 className="px-4 py-3 mx-4 mb-2 rounded-lg"
191 |                 style={{ background: 'var(--bg-elevated)', border: '1px solid var(--border-default)...
192 |               >
193 |                 <div className="text-sm mb-2" style={{ color: 'var(--text-primary)' }}>
194 |                   Yêu cầu XP:
195 |                 </div>
196 |                 <div>Yêu cầu XP: <span className="font-bold">
197 |                   <div>Đánh hích: <span className="font-bold">
198 |                   <div>Biểu tượng: <span>{ lvl.icon }</span></div>
199 |                 </div>
200 |                 {unlocked && (
201 |                   <div className="mt-3">
202 |                     <div className="h-2 rounded-full overflow-hidden" style={{ background: 'var(--...
203 |                       <div className="h-full rounded-full" style={{ width: `${Math.min(progressInL...
204 |                     </div>
205 |                     <div className="text-xs mt-1 text-right" style={{ color: 'var(--text-muted)' }}>
206 |                       {Math.min(progressInLevel, 100).toFixed(0)}%
207 |                     </div>
208 |                   </div>
209 |                 </div>
210 |               </div>
211 |             </div>
212 |           </div>
213 |         </div>
214 |       </div>
215 |     </div>
216 |   )}
217 | </div>
218 | </motion.div>
219 | )}
220 |
221 | {activeTab === 'quests' && (
222 |   <motion.div key="quests" initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }} exit={{ op...
223 |   <div className="lms-card" style={{ padding: '24px' }}>
224 |     <div className="flex items-center justify-between mb-4">
225 |       <div>
226 |         <h2 className="text-lg font-bold" style={{ color: 'var(--text-primary)' }}>
227 |           Yêu cầu XP
228 |         </div>
229 |         <div>Yêu cầu XP: <span className="font-bold">
230 |           <div>Đánh hích: <span className="font-bold">
231 |           <div>Biểu tượng: <span>{ lvl.icon }</span></div>
232 |         </div>
233 |       </div>
234 |
235 |       {allQuests.length === 0 ? (
236 |         <div className="text-center py-12">
237 |           <CheckCircle size={40} className="mx-auto mb-3" style={{ color: 'var(--text-muted)', opaci...
238 |           <div>
239 |             Chưa có yêu cầu
240 |           </div>
241 |         </div>
242 |       ) : (
243 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
244 |           {allQuests.map((quest) => {
245 |             const userQuest = quests.find(q => q.quest_id === quest.id)
246 |             const completed = userQuest?.is_completed || false
247 |             const progress = userQuest?.progress || 0
248 |             return (
249 |               <div>
250 |                 key={quest.id}
251 |                 className="flex items-center gap-4 px-4 py-3.5 rounded-xl transition-all"
252 |                 style={{
253 |                   background: completed ? 'rgb(8,158,96,0.08)' : 'var(--bg-elevated)',
254 |                   border: '1px solid ${completed ? 'rgb(8,158,96,0.2)' : 'var(--border-default)'}',
255 |                   opacity: completed ? 0.7 : 1,
256 |                 }}
257 |               >
258 |                 <div className="flex items-center justify-center w-10 h-10 rounded-lg text-lg" style={{
259 |                   background: completed ? 'rgb(8,158,96,0.15)' : 'var(--bg-base)',
260 |                 }}>
261 |                   {completed ? '' : (quest.icon || '00E')}
262 |                 </div>
263 |                 <div style={{ flex: 1, minWidth: 0 }}>
264 |                   <div className="text-sm font-bold" style={{ color: completed ? 'var(--brand-primar...

```

```

265 |         </div>
266 |         <div className="text-xs mt-0.5" style={{ color: 'var(--text-muted)' }}>{quest.desc...
267 |         <div className="flex items-center gap-2 mt-1.5">
268 |             <span className="text-[10px] font-bold px-1.5 py-0.5 rounded" style={{
269 |                 background: quest.difficulty === 'hard' ? 'rgba(239,68,68,0.15)' : quest.diffi...
270 |                 color: quest.difficulty === 'hard' ? '#EF4444' : quest.difficulty === 'medium'...
271 |             }}>
272 |                 {quest.difficulty === 'hard' ? 'Khó' : quest.difficulty === 'medium' ? 'TB' : ...
273 |             </span>
274 |             <span className="text-[10px] font-bold" style={{ color: 'var(--brand-primary)' }}...
275 |         </div>
276 |     </div>
277 |     <div className="text-right flex-shrink-0">
278 |         {!completed && quest.requirement_value > 0 && (
279 |             <div className="text-xs font-bold" style={{ color: 'var(--text-muted)' }}>
280 |                 {progress} / {quest.requirement_value}
281 |             </div>
282 |         )}
283 |         <Star size={16} style={{ color: completed ? 'var(--brand-primary)' : 'var(--text-m...
284 |     </div>
285 | </div>
286 | )
287 | }}}
288 | </div>
289 | )}
290 | </div>
291 |
292 | {/* Completed Quests */}
293 | {quests.filter(q => q.is_completed).length > 0 && (
294 |     <div className="lms-card mt-4" style={{ padding: '24px' }}>
295 |         <h3 className="text-sm font-bold mb-3" style={{ color: 'var(--text-muted)' }}>
296 |             N h i c m V a ã H o à n T h à n h ( { q u e s t s . f i l t e r ( q = > q . i s _ c o m
297 |         </h3>
298 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '6px' }}>
299 |             {quests.filter(q => q.is_completed).slice(0, 5).map(uq => (
300 |                 <div key={uq.id} className="flex items-center gap-3 px-3 py-2 rounded-lg" style={{ backg...
301 |                 <span className="text-sm flex-1" style={{ color: 'var(--text-muted)' }}>{uq.daily_ques...
302 |                 <span className="text-xs" style={{ color: 'var(--brand-primary)' }}>+{uq.xp_earned} || ...
303 |             </span>
304 |             </div>
305 |         </div>
306 |     </div>
307 | )}
308 | </div>
309 | </motion.div>
310 | )}
311 |
312 | {activeTab === 'titles' && (
313 |     <motion.div key="titles" initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }} exit={{ op...
314 |         <div className="lms-card" style={{ padding: '24px' }}>
315 |             <h2 c l a s s N a m e = " t e x t - l g f o n t - b o l d m b - 1 " s t y l e = { { c o l o r : '
316 |             <p c l a s s N a m e = " t e x t - s m m b - 6 " s t y l e = { { c o l o r : ' v a r ( - - t e x t -
317 |
318 |         <div className="grid grid-cols-1 sm:grid-cols-2 gap-3">
319 |             {[
320 |                 {
321 |                     id: 'f i r s t _ l e s s o n', title: 'K h ß i s u', desc: 'H o à n
322 |                     id: 'd i l i g e n t', title: 'C s n C ù ß', desc: 'H i c 7 n g à y
323 |                     id: 'p e r f e c t _ s c o r e', title: 'H o à n H ß o', desc: 'i
324 |                     id: 'p c _ b u i l d e r', title: 'T h ã P C', desc: 'L _ r á p
325 |                     id: 's p e e d _ d e m o n', title: 'T Ñ c Û', desc: 'H o à n t h
326 |                     id: 's c h o l a r', title: 'H i c G i ß', desc: 'H o à n t h à n h
327 |                     id: 'c o l l e c t o r', title: 'N h à S o u T - p', desc: 'i
328 |                     id: 'q u e s t _ m a s t e r', title: 'V u a N h i c m V a ã', desc:
329 |             ].map(t => {
330 |                 const earned = titles.some(ut => ut.title_id === t.id)
331 |                 return (
332 |                     <div
333 |                         key={t.id}
334 |                         className="flex items-center gap-3 px-4 py-3.5 rounded-xl transition-all"
335 |                         style={{
336 |                             background: earned ? 'rgba(8,158,96,0.06)' : 'var(--bg-elevated)',
337 |                             border: `1px solid ${earned ? 'rgba(8,158,96,0.15)' : 'var(--border-default)'}`,
338 |                             opacity: earned ? 1 : 0.5,
339 |                         }}
340 |                     <div className="flex items-center justify-center w-10 h-10 rounded-lg text-lg" style={{
341 |                         background: earned ? 'rgba(8,158,96,0.12)' : 'var(--bg-base)',
342 |                     }}>
343 |                         { e a r n e d ? t . i c o n : '0=Ý ' }
344 |                     </div>
345 |                     <div style={{ flex: 1, minWidth: 0 }}>
346 |                         <div className="text-sm font-bold" style={{ color: earned ? 'var(--text-primary)' : ...
347 |                             {t.title}
348 |                             {earned && <span className="ml-1.5 text-xs" style={{ color: 'var(--brand-primary)'}}...
349 |                         </div>
350 |                         <div className="text-xs" style={{ color: 'var(--text-muted)' }}>{t.desc}</div>
351 |                         <div className="text-[10px] font-bold mt-0.5" style={{ color: earned ? 'var(--brand-...
352 |                             +{t.xp} XP
353 |                         </div>
354 |                     </div>
355 |                 </div>
356 |             )}
357 |         </div>
358 |     </div>
359 | </div>
360 | </motion.div>
361 | )}
362 |
363 | {activeTab === 'history' && (
364 |     <motion.div key="history" initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }} exit={{ o...
365 |         <div className="lms-card" style={{ padding: '24px' }}>
366 |             <div className="flex items-center justify-between mb-4">
367 |                 <div>
368 |                     <h2 c l a s s N a m e = " t e x t - l g f o n t - b o l d " s t y l e = { { c o l o r : ' v
369 |                     <p c l a s s N a m e = " t e x t - s m " s t y l e = { { c o l o r : ' v a r ( - - t e x t - m
370 |                 </div>
371 |                 <div className="text-lg font-black" style={{ color: levelColor }}>
372 |                     {xp.toLocaleString()} XP
373 |                 </div>
374 |             </div>
375 |
376 |             {xpHistory.length === 0 ? (
377 |                 <div className="text-center py-12">
378 |                     <Clock size={40} className="mx-auto mb-3" style={{ color: 'var(--text-muted)', opacity: 0...
379 |                     <p c l a s s N a m e = " t e x t - s m " s t y l e = { { c o l o r : ' v a r ( - - t e x t - m
380 |                 </div>
381 |             ) : (
382 |                 <div style={{ display: 'flex', flexDirection: 'column', gap: '4px' }}>
383 |                     {xpHistory.map(tx => (
384 |                         <div key={tx.id} className="flex items-center gap-3 px-4 py-2.5 rounded-lg transition-al...
385 |                         <span c l a s s N a m e = " t e x t - l g " > { S T A T U S _ I C O N S [ t x . r e f e r
386 |                         <div style={{ flex: 1, minWidth: 0 }}>
387 |                             <div className="text-sm font-medium" style={{ color: 'var(--text-primary)' }}>{tx.re...
388 |                             <div className="text-[10px]" style={{ color: 'var(--text-muted)' }}>

```

```

389 |         {new Date(tx.created_at).toLocaleDateString('vi-VN', {
390 |             day: 'numeric', month: 'numeric', hour: '2-digit', minute: '2-digit',
391 |         })}
392 |     </div>
393 | </div>
394 |     <div className="text-sm font-bold flex-shrink-0" style={{ color: tx.amount > 0 ? 'var(...
395 |         {tx.amount > 0 ? '+' : ''}{tx.amount}
396 |     </div>
397 | </div>
398 |     ))}
399 | </div>
400 | }
401 | </div>
402 | </motion.div>
403 | ))
404 | </AnimatePresence>
405 | </div>
406 | </div>
407 | )
408 | }
409 |

```

File: app/student/loading.tsx (8 lines, 0.3 KB)

```

1 | export default function Loading() {
2 |     return (
3 |         <div className="flex items-center justify-center min-h-screen bg-[#0d0e13]">
4 |         <div className="w-12 h-12 border-4 border-[#00d4aa] border-t-transparent rounded-full animate-spin"></div>
5 |         </div>
6 |     )
7 | }
8 |

```

File: app/student/profile/page.tsx (308 lines, 18.0 KB)

```

1 | 'use client'
2 |
3 | import React, { useEffect, useState } from 'react'
4 | import { createBrowserClient } from '@supabase/ssr'
5 | import { useRouter } from 'next/navigation'
6 | import { AvatarUpload } from '@components/profile/AvatarUpload'
7 | import { ProfileForm } from '@components/profile/ProfileForm'
8 | import { PreferencesForm } from '@components/profile/PreferencesForm'
9 | import { SecurityForm } from '@components/profile/SecurityForm'
10 | import { User, BarChart2, Settings, ShieldAlert, Award, CheckCircle, Flame, RefreshCw, ArrowLeft } from 'lucid...'
11 |
12 | export default function StudentProfilePage() {
13 |     const router = useRouter()
14 |     const [activeTab, setActiveTab] = useState<'profile' | 'stats' | 'preferences' | 'security'>('profile')
15 |     const [profile, setProfile] = useState<any>(null)
16 |     const [preferences, setPreferences] = useState<any>(null)
17 |     const [stats, setStats] = useState<any>({
18 |         completedLessons: 0,
19 |         averageScore: 0,
20 |         streak: 0,
21 |
22 |         latestCert: null
23 |     })
24 |     const [loading, setLoading] = useState(true)
25 |     const supabase = createBrowserClient(
26 |         process.env.NEXT_PUBLIC_SUPABASE_URL!,
27 |         process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
28 |     )
29 |
30 |     const loadProfileData = async () => {
31 |         setLoading(true)
32 |         const { data: { user } } = await supabase.auth.getUser()
33 |         if (!user) return
34 |
35 |         const { data: prof } = await supabase
36 |             .from('profiles')
37 |             .select('*')
38 |             .eq('id', user.id)
39 |             .single()
40 |
41 |         let { data: pref } = await supabase
42 |             .from('user_preferences')
43 |             .select('*')
44 |             .eq('user_id', user.id)
45 |             .maybeSingle()
46 |
47 |         if (!pref) {
48 |             pref = {
49 |                 email_notifications: true,
50 |                 push_notifications: true,
51 |                 weekly_digest: true,
52 |                 theme: 'dark',
53 |                 language: 'vi'
54 |             }
55 |         }
56 |
57 |         const { count: compCount } = await supabase
58 |             .from('lesson_progress')
59 |             .select('id', { count: 'exact', head: true })
60 |             .eq('student_id', user.id)
61 |             .eq('status', 'completed')
62 |
63 |         const { data: quizAtt } = await supabase
64 |             .from('quiz_attempts')
65 |             .select('score')
66 |             .eq('student_id', user.id)
67 |             .eq('status', 'submitted')
68 |
69 |         const { data: cert } = await supabase
70 |             .from('certificates')
71 |             .select('course_title, completion_date, certificate_number')
72 |             .eq('student_id', user.id)
73 |             .order('issued_at', { ascending: false })
74 |             .limit(1)
75 |             .maybeSingle()
76 |
77 |         const scores = (quizAtt || []).map(q => Number(q.score) || 0)
78 |         const avg = scores.length > 0 ? scores.reduce((a, b) => a + b, 0) / scores.length : 0
79 |
80 |         setProfile({ ...prof, email: user.email })
81 |         setPreferences(pref)
82 |         setStats({
83 |             completedLessons: compCount || 0,
84 |             averageScore: avg,
85 |             streak: 3,
86 |             latestCert: cert || null

```

```

87 |     })
88 |     setLoading(false)
89 |   }
90 |
91 |   useEffect(() => {
92 |     loadProfileData()
93 |   }, [])
94 |
95 |   if (loading) {
96 |     return (
97 |       <div className="min-h-screen pt-24 flex flex-col items-center justify-center gap-2" style={{ background:...
98 |         <RefreshCw size={28} className="animate-spin" style={{ color: 'var(--brand-primary)' }} />
99 |         <span className="text-xs font-bold uppercase tracking-widest" style={{ color: 'var(--text-primary)' }}>
100 |           Loading...
101 |         </span>
102 |       )
103 |     )
104 |
105 |     return (
106 |       <div className="min-h-screen pt-24 pb-12 px-4 sm:px-6 relative overflow-hidden" style={{ background: 'var(...
107 |         <div className="absolute inset-0 bg-grid-pattern opacity-[0.03] pointer-events-none" />
108 |         <div className="absolute top-1/4 left-1/10 w-[500px] h-[500px] rounded-full filter blur-[100px] pointer-...
109 |         <div className="absolute bottom-1/4 right-1/10 w-[500px] h-[500px] rounded-full filter blur-[100px] poin...
110 |
111 |         <div className="max-w-4xl mx-auto relative z-10">
112 |           <div className="flex items-center justify-between mb-8 pb-4" style={{ borderBottom: '1px solid var(--b...
113 |             <div className="flex items-center gap-3">
114 |               <div className="p-2.5 rounded-2xl" style={{ background: 'color-mix(in srgb, var(--brand-primary) 1...
115 |                 <User size={24} />
116 |               </div>
117 |               <div>
118 |                 <h1 className="text-xl md:text-2xl font-black tracking-tight uppercase flex items-center gap-2" ...
119 |                   HỒ SƠ CÁ NHÂN
120 |                 </h1>
121 |               </div>
122 |             </div>
123 |
124 |             <button
125 |               onClick={() => router.push('/builder')}
126 |               className="flex items-center gap-2 px-4 py-2 text-xs font-bold rounded-xl transition-all shadow-md...
127 |               style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)', color: 'var(-...
128 |             >
129 |               <ArrowLeft size={14} className="group-hover:-translate-x-0.5 transition-transform" />
130 |               Quay lại Dashboard
131 |             </button>
132 |           </div>
133 |
134 |           <div className="grid grid-cols-1 md:grid-cols-4 gap-8">
135 |             <div className="relative p-6 rounded-3xl flex flex-col items-center text-center h-fit space-y-4 shad...
136 |               <div className="absolute top-3 left-3 w-4 h-4" style={{ borderTop: '2px solid var(--brand-primary)...
137 |               <div className="absolute top-3 right-3 w-4 h-4" style={{ borderTop: '2px solid var(--brand-primary)...
138 |               <div className="absolute bottom-3 left-3 w-4 h-4" style={{ borderBottom: '2px solid var(--brand-pr...
139 |               <div className="absolute bottom-3 right-3 w-4 h-4" style={{ borderBottom: '2px solid var(--brand-p...
140 |
141 |               <AvatarUpload
142 |                 userId={profile.id}
143 |                 currentAvatarUrl={profile.avatar_url}
144 |                 onSuccess={(url) => setProfile({ ...profile, avatar_url: url })}
145 |               />
146 |               <div>
147 |                 <h2 className="text-lg font-bold tracking-tight" style={{ color: 'var(--text-primary)' }}>{profi...
148 |                 <p className="text-[10px] font-bold uppercase tracking-widest mt-1 px-2.5 py-1 rounded inline-bl...
149 |                   Hic sinh . Lốp {profile.grade} | 'Mũi'
150 |                 </p>
151 |               </div>
152 |
153 |               <div className="w-full p-3.5 rounded-2xl text-left space-y-1.5 relative" style={{ background: 'col...
154 |                 <div className="text-[9px] font-bold uppercase tracking-widest" style={{ color: 'var(--brand-pri...
155 |                 <div className="text-xs font-mono font-bold px-2 py-1.5 rounded flex justify-between items-cente...
156 |                 <span>{profile.email}</span>
157 |               </div>
158 |               <p className="text-[9px] leading-relaxed" style={{ color: 'var(--text-muted)' }}>
159 |                 Email: {profile.email}
160 |               </p>
161 |
162 |               <div className="w-full pt-4 space-y-1" style={{ borderTop: '1px solid var(--border-default)' }}>
163 |                 {[('profile', 'stats', 'preferences', 'security') as const].map(tab => (
164 |                   <button
165 |                     key={tab}
166 |                     onClick={() => setActiveTab(tab)}
167 |                     className="w-full flex items-center gap-3 px-4 py-2.5 rounded-xl text-xs font-bold transitio...
168 |                     style={{
169 |                       background: activeTab === tab ? 'color-mix(in srgb, var(--brand-primary) 10%, transparent)...
170 |                       color: activeTab === tab ? 'var(--brand-primary)' : 'var(--text-muted)',
171 |                       border: activeTab === tab ? '1px solid color-mix(in srgb, var(--brand-primary) 25%, transp...
172 |                     }}
173 |                   >
174 |                     {tab === 'profile' && <User size={16} />}
175 |                     {tab === 'stats' && <BarChart2 size={16} />}
176 |                     {tab === 'preferences' && <Settings size={16} />}
177 |                     {tab === 'security' && <ShieldAlert size={16} />}
178 |                   </button>
179 |                 ))}
180 |               </div>
181 |             </div>
182 |
183 |             <div className="relative md:col-span-3 p-6 sm:p-8 min-h-[450px] rounded-3xl shadow-lg backdrop-blur-...
184 |               <div className="absolute top-0 left-6 right-6 h-[1px]" style={{ background: 'linear-gradient(to ri...
185 |
186 |               {activeTab === 'profile' && (
187 |                 <div className="space-y-6">
188 |                   <div className="flex items-center justify-between pb-4" style={{ borderBottom: '1px solid var(...
189 |                     <div>
190 |                       <h3 className="text-xl font-bold tracking-tight flex items-center gap-2" style={{ color: '...'
191 |                         <User size={20} style={{ color: 'var(--brand-primary)' }} />
192 |                         Thông tin cá nhân
193 |                       </h3>
194 |                       <p className="text-xs mt-1" style={{ color: 'var(--text-muted)' }}>
195 |                         Cập nhật thông tin cá nhân của bạn ở đây.
196 |                       </p>
197 |                     </div>
198 |                     <ProfileForm profile={profile} />
199 |                   </div>
200 |                 )
201 |               }
202 |
203 |               {activeTab === 'stats' && (
204 |                 <div className="space-y-6">
205 |                   <div className="flex items-center justify-between pb-4" style={{ borderBottom: '1px solid var(...
206 |                     <div>
207 |                       <h3 className="text-xl font-bold tracking-tight flex items-center gap-2" style={{ color: '...'
208 |                         <BarChart2 size={20} style={{ color: 'var(--brand-primary)' }} />
209 |                         Tổng số đơn hàng
210 |                       </h3>
211 |                       <p className="text-xs mt-1" style={{ color: 'var(--text-muted)' }}>
212 |                         Xem chi tiết doanh thu và đơn hàng của bạn.
213 |                       </p>
214 |                     </div>
215 |                     <StatsTable stats={stats} />
216 |                   </div>
217 |                 )
218 |               }
219 |             </div>
220 |           </div>
221 |         </div>
222 |       )
223 |     )
224 |   }
225 | }

```

```

211 |         </div>
212 |         <div className="w-1.5 h-1.5 rounded-full" style={{ background: 'var(--accent-blue)' }} />
213 |     </div>
214 |
215 |     <div className="grid grid-cols-2 gap-4">
216 |         <div className="p-4 rounded-2xl flex items-center gap-3 transition-all duration-300" style={...
217 |             <div className="p-2 rounded-xl" style={{ background: 'color-mix(in srgb, var(--brand-prima...
218 |                 <CheckCircle size={20} />
219 |             </div>
220 |             <div>
221 |                 <p className="text-[10px] font-bold uppercase" style={{ color: 'var(--text-muted)' }}>Bà...
222 |                 <p className="text-lg font-black" style={{ color: 'var(--text-primary)' }}>{stats.comple...
223 |             </div>
224 |         </div>
225 |
226 |         <div className="p-4 rounded-2xl flex items-center gap-3 transition-all duration-300" style={...
227 |             <div className="p-2 rounded-xl" style={{ background: 'color-mix(in srgb, var(--accent-blue...
228 |                 <BarChart2 size={20} />
229 |             </div>
230 |         </div>
231 |
232 |         <p className="text-[10px] font-bold uppercase" style={{ color: 'var(--text-primary)' }}>{stats.averag...
233 |     </div>
234 |
235 |     <div className="p-4 rounded-2xl flex items-center gap-3 transition-all duration-300" style={...
236 |         <div className="p-2 rounded-xl" style={{ background: 'color-mix(in srgb, var(--accent-oran...
237 |             <Flame size={20} />
238 |         </div>
239 |         <div>
240 |             <p className="text-[10px] font-bold uppercase" style={{ color: 'var(--text-muted)' }}>Ng...
241 |             <p className="text-lg font-black" style={{ color: 'var(--text-primary)' }}>{stats.streak...
242 |         </div>
243 |     </div>
244 |
245 |     <div>
246 |         <div className="pt-6" style={{ borderTop: '1px solid var(--border-default)' }}>
247 |             <h4 className="text-xs font-bold uppercase tracking-wider mb-3" style={{ color: 'var(--text-...
248 |             {stats.latestCert ? (
249 |                 <div className="p-4 rounded-xl flex items-center justify-between" style={{ background: 'co...
250 |                     <div className="flex items-center gap-3">
251 |                         <Award size={20} style={{ color: 'var(--brand-primary)' }} />
252 |                         <div>
253 |                             <p className="text-xs font-bold" style={{ color: 'var(--text-primary)' }}>{stats.lat...
254 |                             <p className="text-[10px]" style={{ color: 'var(--text-muted)' }}>Mã: {stats.latestC...
255 |                         </div>
256 |                     </div>
257 |                     <a href="/student/certificates" className="text-[10px] font-bold hover:underline" style=...
258 |                         X e m t ¤ t c ¢
259 |                 </a>
260 |             </div>
261 |             ) : (
262 |                 <div className="text-xs p-4 rounded-xl text-center" style={{ color: 'var(--text-muted)', b...
263 |                     B i n c h º a º ä c º ¤ p º c h è n g º c h Ē º n à o º .
264 |                 </div>
265 |             )}
266 |         </div>
267 |     </div>
268 |
269 |     </div>
270 |
271 |     {activeTab === 'preferences' && (
272 |         <div className="space-y-6">
273 |             <div className="flex items-center justify-between pb-4" style={{ borderBottom: '1px solid var(...
274 |                 <div>
275 |                     <h3 className="text-xl font-bold tracking-tight flex items-center gap-2" style={{ color: '...
276 |                     <Settings size={20} style={{ color: 'var(--brand-primary)' }} />
277 |                     C ¤ u º h ì n h º c à i º . t
278 |                 </h3>
279 |                 <p className="text-xs mt-1" style={{ color: 'var(--text-primary)' }}>
280 |                 </div>
281 |                 <div className="w-1.5 h-1.5 rounded-full" style={{ background: 'var(--brand-primary)' }} />
282 |             </div>
283 |             <PreferencesForm preferences={preferences} />
284 |         </div>
285 |     )}
286 |
287 |     {activeTab === 'security' && (
288 |         <div className="space-y-6">
289 |             <div className="flex items-center justify-between pb-4" style={{ borderBottom: '1px solid var(...
290 |                 <div>
291 |                     <h3 className="text-xl font-bold tracking-tight flex items-center gap-2" style={{ color: '...
292 |                     <ShieldAlert size={20} style={{ color: 'var(--brand-primary)' }} />
293 |                     B Ē o º m - t º t à i º k h o Ē n
294 |                 </h3>
295 |                 <p className="text-xs mt-1" style={{ color: 'var(--text-primary)' }}>
296 |                 </div>
297 |                 <div className="w-1.5 h-1.5 rounded-full" style={{ background: 'var(--accent-orange)' }} />
298 |             </div>
299 |             <SecurityForm />
300 |         </div>
301 |     )}
302 |
303 |     </div>
304 |
305 |     </div>
306 |
307 | }
308 |

```

File: app/student/quiz/[quizId]/page.tsx (377 lines, 18.8 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect, useRef, use, useMemo, useCallback } from 'react'
4 | import { createBrowserClient } from '@supabase/ssr'
5 | import { useRouter } from 'next/navigation'
6 | import { ArrowLeft, Clock, CheckCircle, XCircle, Trophy, RefreshCw, AlertTriangle, BookOpen, X, Search } from ...
7 | import { startQuizAttempt, submitQuizAttempt } from '@/app/actions/quiz'
8 | import { GLOSSARY, findTerms } from '@/data/glossary'
9 |
10 | function shuffleArray<T>(arr: T[]): T[] {
11 |     const a = [...arr]
12 |     for (let i = a.length - 1; i > 0; i--) {
13 |         const j = Math.floor(Math.random() * (i + 1));
14 |         [a[i], a[j]] = [a[j], a[i]]
15 |     }
16 |     return a
17 | }
18 |
19 | function TermHighlight({ text, onTermClick }: { text: string; onTermClick: (term: string) => void }) {
20 |     const parts = useMemo(() => {
21 |         const sorted = Object.keys(GLOSSARY).sort((a, b) => b.length - a.length)
22 |         const nodes: React.ReactNode[] = [text]

```



```

23 |   for (const term of sorted) {
24 |     const len = nodes.length
25 |     for (let i = 0; i < len; i++) {
26 |       const node = nodes[i]
27 |       if (typeof node !== 'string') continue
28 |       const lower = node.toLowerCase()
29 |       const idx = lower.indexOf(term.toLowerCase())
30 |       if (idx === -1) continue
31 |       const before = node.slice(0, idx)
32 |       const match = node.slice(idx, idx + term.length)
33 |       const after = node.slice(idx + term.length)
34 |       const replacement: React.ReactNode[] = []
35 |       if (before) replacement.push(before)
36 |       replacement.push(
37 |         <span key={`${term}-${i}`} onClick={(e) => { e.stopPropagation(); onTermClick(term) }}
38 |           className="term-link" style={{ color: 'var(--brand-primary)', borderBottom: '1px dashed var(--bran...
39 |           {match}
40 |         </span>
41 |       )
42 |       if (after) replacement.push(after)
43 |       nodes.splice(i, 1, ...replacement)
44 |       break
45 |     }
46 |   }
47 |   return nodes
48 | }, [text, onTermClick])
49 | return <{parts}</>
50 | }
51 |
52 | export default function QuizTakingPage({ params }: { params: Promise<{ quizId: string }> }) {
53 |   const { quizId } = use(params)
54 |   const router = useRouter()
55 |
56 |   const [examData, setExamData] = useState<any>(null)
57 |   const [questions, setQuestions] = useState<any[]>([])
58 |   const [currentQIndex, setCurrentQIndex] = useState(0)
59 |   const [selectedOption, setSelectedOption] = useState<number | null>(null)
60 |   const [isAnswered, setIsAnswered] = useState(false)
61 |   const [score, setScore] = useState(0)
62 |   const [isFinished, setIsFinished] = useState(false)
63 |   const [secondsLeft, setSecondsLeft] = useState(45 * 60)
64 |   const [submitting, setSubmitting] = useState(false)
65 |   const [loaded, setLoaded] = useState(false)
66 |   const [attemptId, setAttemptId] = useState<string | null>(null)
67 |   const [activeTerm, setActiveTerm] = useState<string | null>(null)
68 |   const [showGlossary, setShowGlossary] = useState(false)
69 |   const [glossarySearch, setGlossarySearch] = useState('')
70 |   const [error, setError] = useState('')
71 |
72 |   const scoreRef = useRef(0)
73 |   const supabaseRef = useRef(createBrowserClient(process.env.NEXT_PUBLIC_SUPABASE_URL!, process.env.NEXT_PUBLI...
74 |   const finishedRef = useRef(false)
75 |   const questionsRef = useRef<any[]>([])
76 |   const answersRef = useRef<any[]>([])
77 |
78 |   useEffect(() => {
79 |     initQuiz()
80 |     async function initQuiz() {
81 |       try {
82 |         const res = await startQuizAttempt(quizId)
83 |         setAttemptId(res.attemptId)
84 |         setExamData({ timeLimit: res.timeLimit })
85 |
86 |         const qs = res.questions.map((q: any) => ({
87 |           id: q.id,
88 |           question: q.content,
89 |           type: q.type,
90 |           points: q.points,
91 |           options: shuffleArray((q.options || []).map((o: any) => ({ id: o.id, content: o.content }))),
92 |           correctOptionId: null
93 |         }))
94 |
95 |         questionsRef.current = qs
96 |         setQuestions(qs)
97 |         setSecondsLeft((res.timeLimit || 45) * 60)
98 |         setLoaded(true)
99 |
100 |         const channel = supabaseRef.current
101 |           .channel(`quiz-live-${res.attemptId}`)
102 |           .on('postgres_changes',
103 |             { event: 'UPDATE', schema: 'public', table: 'quiz_attempts', filter: `id=eq.${res.attemptId}` },
104 |             (payload: any) => {
105 |               if (payload.new?.status !== 'in_progress') {
106 |                 finishedRef.current = true
107 |                 setIsFinished(true)
108 |                 setScore(payload.new?.score || 0)
109 |               }
110 |             }
111 |           )
112 |           .subscribe()
113 |       } catch (e: any) {
114 |         console.error(e)
115 |         setError(e.message || 'Không thể tải Á thi')
116 |       }
117 |     }, [quizId])
118 |
119 |     useEffect(() => {
120 |       if (isFinished || !loaded || !examData?.timeLimit) return
121 |       const interval = setInterval(() => {
122 |         setSecondsLeft(prev => {
123 |           if (prev <= 1) { clearInterval(interval); finishQuiz(scoreRef.current); return 0 }
124 |           return prev - 1
125 |         })
126 |       }, 1000)
127 |       return () => clearInterval(interval)
128 |     }, [isFinished, loaded, examData?.timeLimit])
129 |
130 |     useEffect(() => {
131 |       function handleClickOutside(e: MouseEvent) {
132 |         const el = document.getElementById('term-tooltip')
133 |         if (el && !el.contains(e.target as Node)) setActiveTerm(null)
134 |       }
135 |       document.addEventListener('mousedown', handleClickOutside)
136 |       return () => document.removeEventListener('mousedown', handleClickOutside)
137 |     }, [])
138 |
139 |     const handleTermClick = useCallback((term: string) => {
140 |       setActiveTerm(prev => prev === term ? null : term)
141 |     }, [])
142 |
143 |     const currentQ = questions[currentQIndex]
144 |
145 |     const handleSelectOption = (index: number) => {

```

```

147 |     if (isAnswered) return
148 |     setSelectedOption(index)
149 |     setActiveTerm(null)
150 | }
151 |
152 | const handleCheckAnswer = () => {
153 |   if (selectedOption === null || isAnswered || !currentQ) return
154 |   setIsAnswered(true)
155 |   answersRef.current[currentQIndex] = { questionId: currentQ.id, selectedOptionIds: [currentQ.options[select...
156 |   const correctOpt = currentQ.options.find((o: any) => o.isCorrect)
157 |   if (correctOpt && currentQ.options[selectedOption]?.id === correctOpt.id) {
158 |     scoreRef.current += (currentQ.points || 10)
159 |     setScore(scoreRef.current)
160 |   }
161 | }
162 |
163 | const handleNextQuestion = () => {
164 |   if (!currentQ) return
165 |   if (currentQIndex < questions.length - 1) {
166 |     setCurrentQIndex(prev => prev + 1)
167 |     setSelectedOption(null)
168 |     setIsAnswered(false)
169 |     setActiveTerm(null)
170 |   } else {
171 |     finishQuiz(scoreRef.current)
172 |   }
173 | }
174 |
175 | async function finishQuiz(finalScore: number) {
176 |   if (finishedRef.current) return
177 |   finishedRef.current = true
178 |   setIsFinished(true)
179 |   setScore(finalScore)
180 |   setSubmitting(true)
181 |   try {
182 |     if (attemptId) {
183 |       await submitQuizAttempt(attemptId, answersRef.current)
184 |     }
185 |   } catch (e) { console.error('Lỗi nộp bài:', e) }
186 |   finally { setSubmitting(false) }
187 | }
188 |
189 | const formatTime = (s: number) => `${Math.floor(s / 60).toString().padStart(2, '0')}:${(s % 60).toString().p...
190 | const filteredGlossary = Object.entries(GLOSSARY).filter(([term]) => term.toLowerCase().includes(glossarySea...
191 |
192 | if (error) {
193 |   return (
194 |     <div className="min-h-screen pt-24 pb-12 px-4 flex flex-col items-center justify-center" style={{ backgr...
195 |       <AlertTriangle size={48} className="mb-4" style={{ color: 'var(--brand-primary)' }} />
196 |       <h2 className="text-sm mb-6" style={{ color: 'var(--text-muted)' }}>{error}</h2>
197 |       <button onClick={() => router.push('/student/quiz')} className="px-6 py-3 border rounded-xl font-bold"...
198 |         Q u a y l i i
199 |     </button>
200 |   </div>
201 | )
202 | }
203 |
204 | if (!loaded) {
205 |   return (
206 |     <div className="min-h-screen flex items-center justify-center" style={{ background: 'var(--bg-base)' }}>
207 |       <RefreshCw size={32} className="animate-spin" style={{ color: 'var(--brand-primary)' }} />
208 |     </div>
209 |   )
210 | }
211 |
212 | if (questions.length === 0) {
213 |   return (
214 |     <div className="min-h-screen pt-24 pb-12 px-4 flex flex-col items-center justify-center" style={{ backgr...
215 |       <AlertTriangle size={48} className="mb-4" />
216 |       <h2 className="text-xl font-bold mb-4"> Á t h i c h ° a c ó c â u h ã i < / h 2 >
217 |       <button onClick={() => router.push('/student/quiz')} className="px-6 py-3 border rounded-xl font-bold"...
218 |         Q u a y l i i
219 |     </button>
220 |   </div>
221 | )
222 | }
223 |
224 | if (isFinished) {
225 |   const maxScore = questions.reduce((s, q) => s + (q.points || 10), 0)
226 |   return (
227 |     <div className="min-h-screen pt-24 pb-12 px-4 flex flex-col items-center justify-center" style={{ backgr...
228 |       {submitting ? (
229 |         <div className="flex flex-col items-center gap-4">
230 |           <RefreshCw size={32} className="animate-spin" style={{ color: 'var(--brand-primary)' }} />
231 |           <h2 className="text-xl font-bold tracking-widest uppercase" style={{ color: 'var(--text-secondary)'...
232 |         </div>
233 |       ) : (
234 |         <div className="border p-8 rounded-3xl shadow-2xl max-w-lg w-full text-center" style={{ background: ...
235 |           < Trophy size={64} className={`mx-auto mb-6 ${score >= 70 ? 'text-yellow-400' : ''}`} style={{score ...
236 |             <h2 className="text-3xl font-black mb-2 uppercase"> K ết q u ả
237 |             <div className="text-7xl font-black text-transparent bg-clip-text bg-gradient-to-r from-[var(--bra...
238 |               {score}/{maxScore}
239 |             </div>
240 |             <div className="flex justify-center gap-4">
241 |               <button onClick={() => router.push(`/student/quiz/${quizId}/results?attemptId=${attemptId}`)}
242 |                 className="px-6 py-3 border rounded-xl font-black transition-all" style={{ background: 'var(--brand-p...
243 |                 X e m C h i T i ết
244 |             </button>
245 |             <button onClick={() => router.push('/student/quiz')}
246 |               className="px-6 py-3 border rounded-xl font-bold transition-all" style={{ borderColor: 'var(--...
247 |               Q u a y L i i
248 |             </button>
249 |           </div>
250 |         </div>
251 |       )
252 |     </div>
253 |   )
254 | }
255 |
256 | return (
257 |   <div className="min-h-screen flex flex-col relative overflow-hidden" style={{ background: 'var(--bg-base)'...
258 |     <div className="absolute inset-0 bg-grid-pattern opacity-[0.02] pointer-events-none" />
259 |
260 |     <header className="border-b p-4 sm:px-8 flex justify-between items-center backdrop-blur-md relative z-50...
261 |       style={{ background: 'var(--bg-surface)', borderColor: 'var(--border-default)' }}>
262 |         <div className="flex items-center gap-4">
263 |           <button onClick={() => router.push('/student/quiz')}
264 |             className="flex items-center gap-1.5 px-3 py-1.5 border text-xs rounded-xl font-bold transition-al...
265 |             style={{ background: 'rgba(239,68,68,0.15)', borderColor: 'rgba(239,68,68,0.3)', color: '#ef4444' }}>
266 |             <ArrowLeft size={14} /> Thoát
267 |           </button>
268 |           <h2 className="font-black text-sm md:text-base hidden sm:block" style={{ color: 'var(--text-primary)'...
269 |         </div>
270 |

```

```

271 | <div className="flex items-center gap-3">
272 |   <button onClick={() => setShowGlossary(true)}
273 |     className="flex items-center gap-1.5 px-3 py-1.5 text-xs font-bold rounded-xl border transition-all"
274 |     style={{ background: 'color-mix(in srgb, var(--brand-primary) 15%, transparent)', borderColor: 'co...
275 |   <BookOpen size={14} /> Tra c e u
276 |   </button>
277 |   <div className={ `flex items-center gap-2 font-mono text-xl font-bold px-4 py-1.5 rounded-xl border $...
278 |     style={secondsLeft < 60 ? { background: 'rgba(239,68,68,0.15)', color: '#ef4444' } : { background: ...
279 |     <Clock size={18} /> {formatTime(secondsLeft)}
280 |   </div>
281 | </div>
282 | </header>
283 |
284 | <main className="flex-1 max-w-4xl w-full mx-auto p-4 sm:p-8 relative z-10 flex flex-col justify-center">
285 |   <div className="mb-8 p-6 border rounded-2xl shadow-xl backdrop-blur-sm relative" style={{ background: ...
286 |     <div className="flex items-center justify-between mb-4">
287 |       <span className="text-sm font-bold uppercase tracking-widest px-3 py-1 rounded-full border"
288 |         style={{ color: 'var(--brand-primary)', background: 'color-mix(in srgb, var(--brand-primary) 15%...
289 |         C â u h i i { c u r r e n t Q I n d e x + 1 } / { q u e s t i o n s . l e n g t h }
290 |       </span>
291 |       <span className="text-sm" style={{ color: 'var(--text-muted)
292 |     </div>
293 |     <h3 className="text-2xl sm:text-3xl font-bold leading-tight">
294 |       <TermHighlight text={currentQ?.question || ''} onClick={handleTermClick} />
295 |     </h3>
296 |   </div>
297 |
298 |   <div className="space-y-3">
299 |     {currentQ?.options.map((opt: any, i: number) => {
300 |       const isSelected = selectedOption === i
301 |       let stateStyle: React.CSSProperties = {}
302 |
303 |       if (isAnswered) {
304 |         const isCorrect = currentQ.correctOptionId === opt.id
305 |         if (isCorrect) {
306 |           stateStyle = { background: 'rgba(16,185,129,0.1)', borderColor: '#10b981', color: '#10b981' }
307 |         } else if (isSelected) {
308 |           stateStyle = { background: 'rgba(239,68,68,0.1)', borderColor: '#ef4444', color: '#ef4444' }
309 |         } else {
310 |           stateStyle = { background: 'var(--bg-surface)', borderColor: 'var(--border-default)', color: '...
311 |         }
312 |       } else if (isSelected) {
313 |         stateStyle = { background: 'var(--brand-primary)', borderColor: 'var(--brand-primary)', color: '...
314 |       } else {
315 |         stateStyle = { background: 'var(--bg-surface)', borderColor: 'var(--border-default)' }
316 |       }
317 |
318 |       return (
319 |         <label key={i} onClick={() => handleSelectOption(i)}
320 |           className="flex items-center gap-4 p-5 sm:p-6 border-2 rounded-2xl transition-all cursor-point...
321 |           <div className="w-6 h-6 rounded-full border-2 flex items-center justify-center transition-colors"
322 |             style={isSelected && currentQ.correctOptionId === opt.id ? { borderColor: '#10b981' } : isSe...
323 |             {isAnswered && currentQ.correctOptionId === opt.id ? <CheckCircle size={24} className="text-...
324 |             isAnswered && isSelected ? <XCircle size={24} className="text-red-500 bg-black rounded-full"...
325 |             isSelected ? <div className="w-6 h-6 rounded-full flex items-center justify-center" style={{...
326 |           </div>
327 |           <span className="text-lg font-medium"><TermHighlight text={opt.content} onClick={handleTer...
328 |         </label>
329 |       )
330 |     })
331 |   </div>
332 | </main>
333 |
334 | <footer className="border-t p-4 sm:p-6 flex justify-end items-center relative z-50 backdrop-blur-xl" sty...
335 |   {!isAnswered ? (
336 |     <button onClick={handleCheckAnswer} disabled={selectedOption === null}
337 |       className="px-8 py-3 text-black rounded-xl font-bold transition-all flex items-center gap-2 shadow-lg"
338 |       style={selectedOption !== null ? { background: '#2563eb', color: 'white' } : { background: 'var(--...
339 |     </button>
340 |   ) : (
341 |     <button onClick={handleNextQuestion}
342 |       className="px-8 py-3 text-black rounded-xl font-black transition-all flex items-center gap-2 hover...
343 |       { c u r r e n t Q I n d e x < q u e s t i o n s . l e n g t h - 1 ? ' C â u T i ę p T h e o ' : '
344 |     </button>
345 |   )
346 | </footer>
347 |
348 | {showGlossary && (
349 |   <div className="fixed inset-0 z-[9999] flex items-center justify-center p-4" style={{ background: 'rgb...
350 |     <div className="w-full max-w-2xl max-h-[80vh] rounded-3xl border shadow-2xl flex flex-col overflow-h...
351 |     <div className="flex items-center justify-between p-6 border-b" style={{ borderColor: 'var(--borde...
352 |       <div className="flex items-center gap-3"><BookOpen size={20} style={{ color: 'var(--brand-primar...
353 |       <button onClick={() => setShowGlossary(false)} className="p-2 rounded-xl hover:bg-white/5" style...
354 |     </div>
355 |     <div className="p-4 border-b" style={{ borderColor: 'var(--border-default)' }}>
356 |       <div className="relative">
357 |         <Search size={16} className="absolute left-4 top-1/2 -translate-y-1/2" style={{ color: 'var(--...
358 |         <input type="text" placeholder="T i m t h u - t n g i . . . " value =
359 |         className="w-full outline-none pl-11 pr-4 py-3 rounded-2xl text-sm transition-all" style={{ ...
360 |       </div>
361 |     </div>
362 |     <div className="flex-1 overflow-y-auto p-4 space-y-2">
363 |       {filteredGlossary.map([term, def]) => (
364 |         <div key={term} className="p-4 rounded-2xl border" style={{ borderColor: 'var(--border-default...
365 |         <span className="text-sm font-bold uppercase tracking-wider" style={{ color: 'var(--brand-pr...
366 |         <p className="text-sm mt-1 leading-relaxed" style={{ color: 'var(--text-muted)' }}>{def}</p>
367 |       </div>
368 |     )
369 |   )
370 | </div>
371 | </div>
372 | </div>
373 | </div>
374 | )
375 | }
376 | }
377 |

```

File: app/student/quiz/[quizId]/results/page.tsx (171 lines, 9.9 KB)

```

1 | 'use client'
2 |
3 | import React, { useEffect, useState, use } from 'react'
4 | import Link from 'next/link'
5 | import { useRouter } from 'next/navigation'
6 | import { CheckCircle, XCircle, ChevronDown, ChevronUp, ArrowLeft, Trophy } from 'lucide-react'
7 |
8 | export default function QuizResultsPage({ params }: { params: Promise<{ quizId: string }> }) {
9 |   const resolvedParams = use(params)
10 |   const router = useRouter()
11 |   const [score, setScore] = useState(0)
12 |   const [maxScore, setMaxScore] = useState(0)
13 |   const [passed, setPassed] = useState(false)

```

```

14 | const [questions, setQuestions] = useState<any[]>([])
15 | const [expandedIndex, setExpandedIndex] = useState<number | null>(null)
16 | const [loading, setLoading] = useState(true)
17 |
18 | useEffect(() => {
19 |   try {
20 |     const attemptId = new URLSearchParams(window.location.search).get('attemptId')
21 |     if (attemptId) {
22 |       const stored = sessionStorage.getItem(`quiz-result-${attemptId}`)
23 |       const storedScore = sessionStorage.getItem(`quiz-score-${attemptId}`)
24 |       const storedMax = sessionStorage.getItem(`quiz-max-${attemptId}`)
25 |       if (stored) {
26 |         const parsed = JSON.parse(stored)
27 |         setQuestions(parsed)
28 |         const correctCount = parsed.filter((q: any) => q.isCorrect).length
29 |         const total = parsed.length
30 |         setScore(Number(storedScore) || (correctCount * 10))
31 |         setMaxScore(Number(storedMax) || (total * 10))
32 |         setPassed((Number(storedScore) || (correctCount * 10)) >= 80)
33 |       }
34 |     }
35 |   } catch (e) {
36 |     console.error('Error loading quiz results:', e)
37 |   }
38 |   setLoading(false)
39 | }, [])
40 |
41 | if (loading) {
42 |   return (
43 |     <div className="min-h-screen flex items-center justify-center" style={{ background: 'var(--bg-base)' }}>
44 |       <div className="animate-spin w-8 h-8 border-2 border-t-transparent rounded-full" style={{ borderColor: ...
45 |     </div>
46 |   )
47 | }
48 |
49 | const correctCount = questions.filter((q: any) => q.isCorrect).length
50 | const wrongCount = questions.length - correctCount
51 |
52 | return (
53 |   <div className="min-h-screen p-4 sm:p-8 flex flex-col" style={{ background: 'var(--bg-base)', color: 'var(...
54 |     <div className="max-w-3xl w-full mx-auto">
55 |       <div className="flex items-center gap-3 mb-8">
56 |         <button onClick={() => router.push('/student/quiz')}
57 |           className="relative z-50 pointer-events-auto flex items-center gap-1.5 px-3 py-1.5 border text-xs ...
58 |           style={{ background: 'rgba(239,68,68,0.15)', borderColor: 'rgba(239,68,68,0.3)', color: '#ef4444' }}>
59 |           < ArrowLeft size={14} /> Quay lại
60 |         </button>
61 |         <h1 className="text-xl sm:text-2xl font-black uppercase" style=
62 |       </div>
63 |
64 |       {questions.length === 0 ? (
65 |         <div className="rounded-[28px] p-8 text-center border shadow-2xl" style={{ background: 'var(--bg-sur...
66 |         <Trophy size={48} className="mx-auto mb-4 opacity-30" style={{ color: 'var(--text-muted)' }} />
67 |         <h2 className="text-xl font-bold mb-2" style={{ color: 'var(
68 |         <p className="text-sm mb-6" style={{ color: 'var(--text-muted)
69 |         <Link href={`/student/quiz/${resolvedParams.quizId}`} className="px-6 py-3 rounded-xl font-bold in...
70 |         style={{ background: 'var(--brand-primary)', color: 'black' }}>
71 |         Làm bài ngay
72 |       </Link>
73 |     </div>
74 |   ) : (
75 |     <div>
76 |       <div className="rounded-[28px] p-6 sm:p-8 border shadow-2xl mb-8 text-center" style={{ background: ...
77 |       <div className="w-32 h-32 mx-auto rounded-full border-8 flex items-center justify-center mb-4"
78 |         style={{ passed ? { borderColor: 'var(--brand-primary)', color: 'var(--brand-primary)', boxShado...
79 |         <span className="text-4xl font-black">{score}</span>
80 |       </div>
81 |       <h2 className={`text-2xl font-black uppercase tracking-widest mb-4`} style={{ passed ? { color: 'v...
82 |       </h2>
83 |       <div className="flex justify-center gap-6 flex-wrap">
84 |         <div className="px-4 py-2 rounded-xl" style={{ background: 'color-mix(in srgb, var(--brand-pri...
85 |         <p className="text-xl font-black" style={{ color: 'var(--brand-primary)' }}>{correctCount}</...
86 |       </div>
87 |       <div className="px-4 py-2 rounded-xl" style={{ background: 'color-mix(in srgb, #ef4444 10%, tr...
88 |       <p className="text-xs font-bold uppercase" style={{ color: 'var(--text-muted)' }}>Sai</p>
89 |       <p className="text-xl font-black" style={{ color: '#ef4444' }}>{wrongCount}</p>
90 |     </div>
91 |   </div>
92 | )
93 |
94 | <div>
95 |
96 |
97 | </div>
98 |
99 | <div className="space-y-3">
100 |   {questions.map((q: any, i: number) => (
101 |     <div key={i} className="rounded-2xl border overflow-hidden transition-all" style={{ background: ...
102 |       <button
103 |         onClick={() => setExpandedIndex(expandedIndex === i ? null : i)}
104 |         className="w-full flex items-center gap-3 p-4 text-left"
105 |       >
106 |         {q.isCorrect ? (
107 |           <CheckCircle size={20} style={{ color: 'var(--brand-primary)', flexShrink: 0 }} />
108 |         ) : (
109 |           <XCircle size={20} style={{ color: '#ef4444', flexShrink: 0 }} />
110 |         )
111 |       <span className="flex-1 text-sm font-medium" style={{ color: 'var(--text-primary)' }}>
112 |         Câu {i + 1}: {q.question.length > 60 ? q.question.substring(0, 60) + '...' : q.question}
113 |       </span>
114 |       <span className="text-xs font-bold px-2 py-0.5 rounded-full" style={{
115 |         background: q.isCorrect ? 'color-mix(in srgb, var(--brand-primary) 15%, transparent)' : ...
116 |         color: q.isCorrect ? 'var(--brand-primary)' : '#ef4444'
117 |       }}>
118 |         {q.isCorrect ? '+10' : '0'}
119 |       </span>
120 |       {expandedIndex === i ? <ChevronUp size={16} style={{ color: 'var(--text-muted)' }} /> : <C...
121 |     </button>
122 |     {expandedIndex === i && (
123 |       <div className="px-4 pb-4 space-y-2" style={{ borderTop: '1px solid var(--border-default)'...
124 |       <p className="text-sm font-bold pt-3" style={{ color: 'var(--text-primary)' }}>{q.questi...
125 |       {q.options.map((opt: string, oi: number) => {
126 |         let bg = 'transparent'
127 |         let txt = 'var(--text-secondary)'
128 |         if (oi === q.correctIndex) {
129 |           bg = 'color-mix(in srgb, var(--brand-primary) 10%, transparent)'
130 |           txt = 'var(--brand-primary)'
131 |         } else if (oi === q.selectedIndex && oi !== q.correctIndex) {
132 |           bg = 'color-mix(in srgb, #ef4444 10%, transparent)'
133 |           txt = '#ef4444'
134 |         }
135 |       })
136 |       <div key={oi} className="flex items-center gap-2 px-3 py-2 rounded-lg text-sm" style...
137 |       {oi === q.correctIndex && <CheckCircle size={14} style={{ color: 'var(--brand-prim...

```

```

138 |         {oi === q.selectedIndex && oi !== q.correctIndex && <XCircle size={14} style={{ co...
139 |         <span style={{ color: txt }}>
140 |           {String.fromCharCode(65 + oi)}. {opt}
141 |           {oi === q.correctIndex && <span className="ml-1 text-xs font-bold" style={{ colo...
142 |         </span>
143 |       </div>
144 |     )}
145 |   }}
146 | </div>
147 | )}
148 | </div>
149 | )}}
150 | </div>
151 |
152 |   <div className="flex justify-center gap-4 mt-8">
153 |     <Link href="/student/quiz" className="px-6 py-3 border rounded-2xl font-bold text-sm transition-...
154 |       style={{ color: 'var(--text-secondary)', borderColor: 'var(--border-default)' }}>
155 |       V A N g a n h à n g A t h i
156 |     </Link>
157 |     <Link
158 |       href={`/student/quiz/${resolvedParams.quizId}`}
159 |       className="px-6 py-3 rounded-2xl font-black text-sm transition-all"
160 |       style={{ background: 'var(--brand-primary)', color: 'black' }}
161 |     >
162 |       L à m L i i
163 |     </Link>
164 |   </div>
165 | </>
166 | )}
167 | </div>
168 | </div>
169 | )
170 | }
171 |

```

File: app/student/quiz/page.tsx (261 lines, 14.6 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect, useRef } from 'react'
4 | import { createBrowserClient } from '@supabase/ssr'
5 | import { motion } from 'framer-motion'
6 | import { HelpCircle, Zap, Star, Loader2, Search, Award, Flame, Play, ArrowLeft, BrainCircuit, RotateCcw, FileT...
7 | import Link from 'next/link'
8 | import { useRouter } from 'next/navigation'
9 | import { QUIZ_BANK } from '@data/quiz-bank'
10 |
11 | export default function StudentQuizPage() {
12 |   const router = useRouter()
13 |   const [exams, setExams] = useState<any[]>([])
14 |   const [loading, setLoading] = useState(true)
15 |   const [searchQuery, setSearchQuery] = useState('')
16 |   const [stats, setStats] = useState({ accuracy: '--', streak: '0', totalDone: 0 })
17 |   const [unlockedCount, setUnlockedCount] = useState(3)
18 |
19 |   const supabase = createBrowserClient(
20 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
21 |     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
22 |   )
23 |
24 |   useEffect(() => {
25 |     fetchExamsAndStats()
26 |     supabase.auth.getUser().then(({ data: { user } }) => {
27 |       if (!user) return
28 |       const channel = supabase
29 |         .channel('quiz-list-realtime')
30 |         .on('postgres_changes',
31 |           { event: '*', schema: 'public', table: 'quiz_attempts', filter: `student_id=eq.${user.id}` },
32 |           () => fetchExamsAndStats()
33 |         )
34 |         .subscribe()
35 |       return () => { supabase.removeChannel(channel) }
36 |     })
37 |   }, [])
38 |
39 |   async function fetchExamsAndStats() {
40 |     try {
41 |       const { data: { user } } = await supabase.auth.getUser()
42 |       if (!user) { setLoading(false); return }
43 |
44 |       const { data: classes } = await supabase
45 |         .from('class_members')
46 |         .select('class_id')
47 |         .eq('student_id', user.id)
48 |
49 |       const classIds = classes?.map(c => c.class_id) || []
50 |       if (classIds.length === 0) { setExams([]); setLoading(false); return }
51 |
52 |       const { data: quizzes } = await supabase
53 |         .from('quizzes')
54 |         .select(
55 |           id, title, description, time_limit_minutes, difficulty, passing_score,
56 |           available_from, available_until, is_published, created_at,
57 |           class_id, teacher_id,
58 |           quiz_questions!left (question_id),
59 |           quiz_attempts!left (id, score, status, student_id)
60 |         )
61 |         .in('class_id', classIds)
62 |         .eq('is_published', true)
63 |         .order('created_at', { ascending: false })
64 |         .limit(50)
65 |
66 |       if (quizzes) {
67 |         const processed = quizzes.map((q: any) => ({
68 |           ...q,
69 |           totalQuestions: q.quiz_questions?.length || 0,
70 |           studentAttempts: q.quiz_attempts?.filter((a: any) => a.student_id === user.id) || [],
71 |           bestScore: Math.max(0, ...(q.quiz_attempts?.filter((a: any) => a.student_id === user.id && a.status ...
72 |             hasDone: (q.quiz_attempts?.filter((a: any) => a.student_id === user.id && a.status !== 'in_progress'...
73 |             myAttempts: q.quiz_attempts?.filter((a: any) => a.student_id === user.id).length || 0
74 |           ))
75 |         })
76 |         setExams(processed)
77 |       }
78 |
79 |       const { data: attempts } = await supabase
80 |         .from('quiz_attempts')
81 |         .select('id, score, status, created_at')
82 |         .eq('student_id', user.id)
83 |         .not('status', 'eq', 'in_progress')
84 |
85 |       if (attempts && attempts.length > 0) {
86 |         const totalScore = attempts.reduce((s, a) => s + (a.score || 0), 0)
87 |         const avg = totalScore / attempts.length

```

```

87 |     setStats(s => ({ ...s, accuracy: `${Math.round(avg)}%`, totalDone: attempts.length }))
88 |
89 |     let streakCount = 0
90 |     const today = new Date()
91 |     const dates = [...new Set(attempts.map(a => a.created_at?.split('T')[0]))].sort().reverse()
92 |     for (const d of dates) {
93 |         const diff = Math.floor((today.getTime() - new Date(d).getTime()) / (1000 * 60 * 60 * 24))
94 |         if (diff === streakCount) { streakCount++; continue }
95 |         break
96 |     }
97 |     setStats(s => ({ ...s, streak: `${streakCount} ngày` }))
98 | }
99 |
100 | if (user.created_at) {
101 |     const diff = Math.floor((Date.now() - new Date(user.created_at).getTime()) / (1000 * 60 * 60 * 24))
102 |     setUnlockedCount(Math.min(3 + diff, 50))
103 | }
104 | } catch (err) { console.error(err) }
105 | finally { setLoading(false) }
106 | }
107 |
108 | const hasAccess = (exam: any) => {
109 |     if (!exam.available_from) return true
110 |     return new Date() >= new Date(exam.available_from)
111 | }
112 |
113 | const filtered = exams.filter(e => e.title?.toLowerCase().includes(searchQuery.toLowerCase()))
114 | const upcomingExams = filtered.filter(e => e.available_from && !hasAccess(e))
115 | const activeExams = filtered.filter(e => hasAccess(e))
116 |
117 | return (
118 |     <div className="min-h-screen pt-24 pb-16 px-4 sm:px-6 relative overflow-hidden" style={{ background: 'var(...
119 |     <div className="absolute inset-0 bg-grid-pattern opacity-[0.02] pointer-events-none" />
120 |     <div className="max-w-6xl mx-auto relative z-10">
121 |         <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-4 mb-8 pb-4 border-b" st...
122 |         <div className="flex items-center gap-3">
123 |             <div className="p-2.5 rounded-2xl" style={{ background: 'color-mix(in srgb, var(--brand-primary) 1...
124 |                 <Award size={24} />
125 |             </div>
126 |             <div>
127 |                 <h1 className="text-xl md:text-2xl font-black tracking-tight uppercase flex items-center gap-2" ...
128 |                 <h1>
129 |                     <span style={{ color: 'var(--brand-primary)' }}> &
130 |                 </h1>
131 |                 <div>
132 |                     <div>
133 |                         <div className="flex items-center gap-3">
134 |                             <div className="relative w-64 hidden sm:block">
135 |                                 <Search className="absolute left-4 top-1/2 -translate-y-1/2" size={16} style={{ color: 'var(--te...
136 |                                 <input type="text" placeholder="Tìm kiếm" value=
137 |                                 <div>
138 |                                     <div>
139 |                                     <div>
140 |                                     <div>
141 |                                     <div>
142 |                                     <div>
143 |                                     <div>
144 |                                     <div>
145 |                                     <div>
146 |                                     <div>
147 |                                     <div>
148 |                                     <div>
149 |                                     <div>
150 |                                     <div>
151 |                                     <div>
152 |                                     <div>
153 |                                     <div>
154 |                                     <div>
155 |                                     <div>
156 |                                     <div>
157 |                                     <div>
158 |                                     <div>
159 |                                     <div>
160 |                                     <div>
161 |                                     <div>
162 |                                     <div>
163 |                                     <div>
164 |                                     <div>
165 |                                     <div>
166 |                                     <div>
167 |                                     <div>
168 |                                     <div>
169 |                                     <div>
170 |                                     <div>
171 |                                     <div>
172 |                                     <div>
173 |                                     <div>
174 |                                     <div>
175 |                                     <div>
176 |                                     <div>
177 |                                     <div>
178 |                                     <div>
179 |                                     <div>
180 |                                     <div>
181 |                                     <div>
182 |                                     <div>
183 |                                     <div>
184 |                                     <div>
185 |                                     <div>
186 |                                     <div>
187 |                                     <div>
188 |                                     <div>
189 |                                     <div>
190 |                                     <div>
191 |                                     <div>
192 |                                     <div>
193 |                                     <div>
194 |                                     <div>
195 |                                     <div>
196 |                                     <div>
197 |                                     <div>
198 |                                     <div>
199 |                                     <div>
200 |                                     <div>
201 |                                     <div>
202 |                                     <div>
203 |                                     <div>
204 |                                     <div>
205 |                                     <div>
206 |                                     <div>
207 |                                     <div>
208 |                                     <div>
209 |                                     <div>
210 |                                     <div>

```

```

211|     }}
212|     <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
213|       {activeExams.map((exam, i) => {
214|         const startUrl = `/student/quiz/${exam.id}`
215|         return (
216|           <motion.div key={exam.id} initial={{ opacity: 0, y: 15 }} animate={{ opacity: 1, y: 0 }} trans...
217|             className="border rounded-3xl p-6 transition-all hover:shadow-[0_0_30px_var(--brand-subtle)]...
218|             style={{ background: 'var(--bg-surface)', borderColor: 'var(--border-default)' }}>
219|             <div>
220|               <div className="flex items-center justify-between mb-4">
221|                 <span className="text-[10px] font-bold uppercase tracking-widest px-2.5 py-1 rounded-md ...
222|                   2 2 2 | < U s e r s   s i z e = { 1 2 } / >   T r o n g   l ù p
223|                 </span>
224|                 {exam.hasDone && (
225|                   <span className="text-[10px] font-bold px-2 py-0.5 rounded-full" style={{ background: ...
226|                     {exam.bestScore?.toFixed(0)}}%
227|                   </span>
228|                 )}
229|               </div>
230|               <h3 className="text-lg font-bold mb-2 group-hover:text-[var(--brand-primary)] transition-c...
231|                 {exam.title}
232|               </h3>
233|               {exam.description && (
234|                 <p className="text-xs mb-4 line-clamp-2" style={{ color: 'var(--text-muted)' }}>{exam.de...
235|                 </p>
236|               )}
237|             </div>
238|             <div className="space-y-4">
239|               <div className="flex items-center justify-between text-xs font-medium" style={{ color: 'va...
240|                 <span><FileText size={12} /> {exam.totalQuestions || 0} câu</span>
241|                 <span><Clock size={12} /> {exam.time_limit_minutes || 45} phút</span>
242|                 <span style={{ color: exam.hasDone ? 'var(--brand-primary)' : 'var(--text-muted)' }}>
243|                   2 4 2 | { e x a m . m y A t t e m p t s } > 0 ? ` $ { e x a m . m y A t t e m p t s }   l ã n `
244|                 </span>
245|               <Link href={startUrl}
246|                 className="w-full py-3 border font-bold rounded-2xl flex items-center justify-center gap...
247|                 style={{ background: exam.hasDone ? 'color-mix(in srgb, var(--brand-primary) 10%, transp...
248|                   2 4 8 | { e x a m . h a s D o n e ? ' L à m   l i   ' : ' B @ T   ! U   L À M   B À I ' } < P
249|                 </Link>
250|               </div>
251|             </motion.div>
252|           )
253|         )}
254|       </div>
255|     </div>
256|   )}
257| </div>
258| </div>
259| )
260| }
261|

```

File: app/tra-gia-linh-kien/page.tsx (149 lines, 5.5 KB)

```

1| 'use client'
2|
3| import { useState, useRef, useEffect } from 'react'
4| import { useRouter } from 'next/navigation'
5| import { motion } from 'framer-motion'
6| import { ArrowLeft, CPU, Loader2 } from 'lucide-react'
7| import ChatInput from '@components/price-compare/ChatInput'
8| import ChatMessage from '@components/price-compare/ChatMessage'
9| import ImageModal from '@components/price-compare/ImageModal'
10|
11| interface PriceInfo {
12|   site: string
13|   price: string
14|   link: string
15|   rating: number
16| }
17|
18| interface Recommended {
19|   site: string
20|   price: string
21|   link: string
22| }
23|
24| interface Component {
25|   id: string
26|   exact_name: string
27|   estimated_price: string
28|   image_url: string
29| }
30|
31| interface Message {
32|   role: 'user' | 'assistant'
33|   content: string
34|   components?: Component[]
35|   prices?: PriceInfo[]
36|   recommended?: Recommended
37|   error?: boolean
38| }
39|
40| export default function PriceComparePage() {
41|   const router = useRouter()
42|   const [messages, setMessages] = useState<Message[]>([
43|     4 4 |     c o n t e n t :   ' C h à o   b i n !   M i n h   l à   c h u y ê n   g i a   P C .   H â y   n h - p   t ê n   l i n h   k i ç n   b
44|   ])
45|   const [loading, setLoading] = useState(false)
46|   const [selectedImage, setSelectedImage] = useState<Component | null>(null)
47|   const bottomRef = useRef<HTMLDivElement>(null)
48|
49|   useEffect(() => {
50|     bottomRef.current?.scrollIntoView({ behavior: 'smooth' })
51|   }, [messages])
52|
53|   async function handleSend(query: string) {
54|     setMessages(prev => [...prev, { role: 'user', content: query }])
55|     setLoading(true)
56|
57|     try {
58|       const res = await fetch('/api/pc-price-compare', {
59|         method: 'POST',
60|         headers: { 'Content-Type': 'application/json' },
61|         body: JSON.stringify({ query })
62|       })
63|     }
64|
65|     const data = await res.json()
66|
67|     if (res.ok && data.comparison_text) {
68|       setMessages(prev => [...prev, {
69|         role: 'assistant',

```

```

70 |         content: data.comparison_text,
71 |         components: data.components?.map((c: any) => ({
72 |             ...c,
73 |             image_url: c.image_url || 'https://images.unsplash.com/photo-1591799264318-7e6ef8ddb7ea?w=500&q=80'
74 |         })),
75 |         prices: data.prices,
76 |         recommended: data.recommended,
77 |     })
78 | } else {
79 |     setMessages(prev => [...prev, {
80 |         role: 'assistant',
81 |         content: data.error || 'Lỗi: Không thể bóc tách dữ liệu linh
82 |     }, {
83 |         error: true
84 |     }])
85 | } catch (err) {
86 |     setMessages(prev => [...prev, {
87 |         role: 'assistant',
88 |         content: 'ã xê y ra lỗi khi kết nối. Vui lòng thử lại sau.',
89 |         error: true
90 |     }])
91 | } finally {
92 |     setLoading(false)
93 | }
94 | }
95 |
96 | return (
97 |     <div className="min-h-screen flex flex-col" style={{ background: 'var(--bg-base)', color: 'var(--text-prim...
98 |     <header className="border-b p-4 flex items-center gap-4" style={{ background: 'var(--bg-surface)', borde...
99 |     <button onClick={() => router.back()}
100 |     className="flex items-center gap-1.5 px-3 py-1.5 border text-xs rounded-xl font-bold transition-all ...
101 |     style={{ background: 'var(--bg-elevated)', borderColor: 'var(--border-default)', color: 'var(--text-...
102 |     <ArrowLeft size={14} /> Quay lại
103 | </button>
104 | <div style={{ width: '32px', height: '32px', borderRadius: '10px', background: 'var(--brand-primary)',...
105 | <Cpu size={16} style={{ color: '#000' }} />
106 | </div>
107 | <div>
108 |     <h1 className="font-black text-sm" style={{ color: 'var(--text-primary)' }}>Tra giá & So sánh linh k...
109 |     <p className="text-xs" style={{ color: 'var(--text-muted)' }}>
110 | </p>
111 | </div>
112 | </header>
113 | <div className="flex-1 overflow-y-auto p-4 max-w-4xl mx-auto w-full">
114 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '12px' }}>
115 |     {messages.map((msg, i) => (
116 |         <motion.div key={i} initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }} transition={{ d...
117 |         <ChatMessage
118 |             role={msg.role}
119 |             content={msg.content}
120 |             components={msg.components}
121 |             prices={msg.prices}
122 |             recommended={msg.recommended}
123 |             error={msg.error}
124 |             onComponentClick={(comp) => setSelectedImage(comp)}
125 |         />
126 |     </motion.div>
127 |     ))}
128 |     {loading && (
129 |         <div style={{ display: 'flex', alignItems: 'center', gap: '10px', padding: '12px 16px', color: 'va...
130 |         <Loader2 size={16} style={{ animation: 'spin 1s linear inf
131 |     </div>
132 |     <div ref={bottomRef} />
133 | </div>
134 | </div>
135 | </div>
136 | <div className="max-w-4xl mx-auto w-full p-4 border-t" style={{ borderColor: 'var(--border-default)' }}>
137 |     <ChatInput onSend={handleSend} loading={loading} />
138 |     <p style={{ fontSize: '11px', color: 'var(--text-muted)', marginTop: '8px', textAlign: 'center' }}>
139 |         Dữ liệu giá tham khảo từ thể trợ Nguyễn Việt Nam. Kết quả do AI
140 |     </p>
141 | </div>
142 | </div>
143 | <div>
144 |     <ImageModal component={selectedImage} onClose={() => setSelectedImage(null)} />
145 | <style>{@keyframes spin { to { transform: rotate(360deg) } }}</style>
146 | </div>
147 | )
148 | }
149 |

```

File: app/verify/[certificateNumber]/page.tsx (205 lines, 10.5 KB)

```

1 | import React from 'react'
2 | import { createServerClient } from '@supabase/ssr'
3 | import { cookies } from 'next/headers'
4 | import { CheckCircle, AlertTriangle, XCircle, ShieldCheck, Award, Star, Calendar, Hash, Download } from 'lucid...
5 |
6 | export const dynamic = 'force-dynamic'
7 |
8 | export default async function VerifyCertificatePage({ params }: { params: { certificateNumber: string } }) {
9 |     const cookieStore = await cookies()
10 |     const supabase = createServerClient(
11 |         process.env.NEXT_PUBLIC_SUPABASE_URL!,
12 |         process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!,
13 |         {
14 |             cookies: {
15 |                 get(name: string) { return cookieStore.get(name)?.value }
16 |             }
17 |         }
18 |     )
19 |
20 |     const { data: cert } = await supabase
21 |         .from('certificates')
22 |         .select('*')
23 |         .eq('certificate_number', params.certificateNumber)
24 |         .maybeSingle()
25 |
26 |     return (
27 |         <div style={{
28 |             minHeight: '100vh',
29 |             background: 'linear-gradient(180deg, #0a0c1a 0%, #0d0e13 100%)',
30 |             color: '#fff',
31 |             display: 'flex',
32 |             flexDirection: 'column',
33 |             alignItems: 'center',
34 |             justifyContent: 'center',
35 |             padding: '16px',
36 |         }}>
37 |         <div style={{
38 |             width: '100%',
39 |             maxWidth: '500px',
40 |             background: '#1a1c25',

```



```

41 | border: '1px solid rgba(255,255,255,0.08)',
42 | padding: '32px',
43 | textAlign: 'center',
44 | position: 'relative',
45 | overflow: 'hidden',
46 | }>
47 | /* Corner decorations */
48 | <div style={{ position: 'absolute', top: 0, left: 0, width: '8px', height: '8px', borderTop: '2px soli...
49 | <div style={{ position: 'absolute', top: 0, right: 0, width: '8px', height: '8px', borderTop: '2px soli...
50 | <div style={{ position: 'absolute', bottom: 0, left: 0, width: '8px', height: '8px', borderBottom: '2p...
51 | <div style={{ position: 'absolute', bottom: 0, right: 0, width: '8px', height: '8px', borderBottom: '2...
52 |
53 | /* Shield Header */
54 | <div style={{ display: 'flex', justifyContent: 'center', marginBottom: '20px' }}>
55 |   {cert ? (
56 |     <div style={{ width: '64px', height: '64px', borderRadius: '50%', background: 'rgba(239,68,68,0.1)...
57 |     <XCircle size={32} style={{ color: '#ef4444' }} />
58 |   ) : cert.is_revoked ? (
59 |     <div style={{ width: '64px', height: '64px', borderRadius: '50%', background: 'rgba(245,158,11,0.1)...
60 |     <AlertTriangle size={32} style={{ color: '#f59e0b' }} />
61 |   ) : (
62 |     <div style={{ width: '64px', height: '64px', borderRadius: '50%', background: 'rgba(0,212,170,0.1)...
63 |     <ShieldCheck size={32} style={{ color: '#00d4aa' }} />
64 |   ) : (
65 |     <div style={{ width: '64px', height: '64px', borderRadius: '50%', background: 'rgba(0,212,170,0.1)...
66 |   ) : (
67 |   ) : (
68 | ) : (
69 | ) : (
70 | ) : (
71 | ) : (
72 | ) : (
73 | ) : (
74 | ) : (
75 | ) : (
76 | ) : (
77 | ) : (
78 | ) : (
79 | ) : (
80 | ) : (
81 | ) : (
82 | ) : (
83 | ) : (
84 | ) : (
85 | ) : (
86 | ) : (
87 | ) : (
88 | ) : (
89 | ) : (
90 | ) : (
91 | ) : (
92 | ) : (
93 | ) : (
94 | ) : (
95 | ) : (
96 | ) : (
97 | ) : (
98 | ) : (
99 | ) : (
100 | ) : (
101 | ) : (
102 | ) : (
103 | ) : (
104 | ) : (
105 | ) : (
106 | ) : (
107 | ) : (
108 | ) : (
109 | ) : (
110 | ) : (
111 | ) : (
112 | ) : (
113 | ) : (
114 | ) : (
115 | ) : (
116 | ) : (
117 | ) : (
118 | ) : (
119 | ) : (
120 | ) : (
121 | ) : (
122 | ) : (
123 | ) : (
124 | ) : (
125 | ) : (
126 | ) : (
127 | ) : (
128 | ) : (
129 | ) : (
130 | ) : (
131 | ) : (
132 | ) : (
133 | ) : (
134 | ) : (
135 | ) : (
136 | ) : (
137 | ) : (
138 | ) : (
139 | ) : (
140 | ) : (
141 | ) : (
142 | ) : (
143 | ) : (
144 | ) : (
145 | ) : (
146 | ) : (
147 | ) : (
148 | ) : (
149 | ) : (
150 | ) : (
151 | ) : (
152 | ) : (
153 | ) : (
154 | ) : (
155 | ) : (
156 | ) : (
157 | ) : (
158 | ) : (
159 | ) : (
160 | ) : (
161 | ) : (
162 | ) : (
163 | ) : (
164 | ) : (

```

```
1 6 5 | <div style={{ fontWeight: 700 }}>{cert.student_name}</div>
166 1 6 7 | <div style={{ fontWeight: 700 }}>{cert.course_title}</div>
168 1 6 9 | <div style={{ fontWeight: 700, color: '#00d4aa' }}>{cert.final_score}</div>
170 1 7 1 | <div style={{ fontWeight: 700 }}>{cert.completion_date || new Date(cert.issued_at).toLocaleD...
172 1 7 3 | <div style={{ fontWeight: 700, fontFamily: 'monospace', fontSize: '10px' }}>{cert.certificat...
174 </div>
175 </div>
176 </div>
177 {cert.pdf_url && (
178 <a href={cert.pdf_url} target="_blank" rel="noreferrer"
179 style={{
180 display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '6px',
181 width: '100%', padding: '10px',
182 background: '#00d4aa',
183 color: '#000e13',
184 fontWeight: 700,
185 fontSize: '12px',
186 textDecoration: 'none',
187 fontFamily: 'monospace',
188 }}>
189 </a>
191 </div>
192 </div>
193 <div style={{ marginTop: '12px', padding: '8px', background: 'rgba(0,212,170,0.05)', border: '1p...
194 <ShieldCheck size={12} style={{ display: 'inline', verticalAlign: 'middle', marginRight: '4px'...
195 </div>
196 </div>
197 </div>
198 </div>
199 </div>
200 </div>
201 </div>
202 </div>
203 }
204 }
205 }
```

File: components/AIGuru.js (68 lines, 2.9 KB)

```
1 'use client';
2
3 import { useState, useEffect } from 'react';
4
5 const AIGuru = ({ message, trigger, lang = 'en' }) => {
6   const [isSpeaking, setIsSpeaking] = useState(false);
7   const [isVisible, setIsVisible] = useState(false);
8
9   const speak = (text) => {
10     if (!window.speechSynthesis) return;
11     window.speechSynthesis.cancel();
12     const utterance = new SpeechSynthesisUtterance(text);
13     const voices = window.speechSynthesis.getVoices();
14     let preferredVoice = null;
15     if (lang === 'vn') {
16       preferredVoice = voices.find(v => v.lang.includes('vi') || v.name.includes('Vietnamese'));
17     } else {
18       preferredVoice = voices.find(v => v.name.includes('Google US English') || v.name.includes('Samanth...
19     }
20     if (preferredVoice) utterance.voice = preferredVoice;
21     utterance.pitch = 1.1;
22     utterance.rate = 1.0;
23     utterance.onstart = () => setIsSpeaking(true);
24     utterance.onend = () => setIsSpeaking(false);
25     window.speechSynthesis.speak(utterance);
26   };
27
28   useEffect(() => {
29     if (message && trigger) {
30       setIsVisible(true);
31       speak(message);
32     }
33   }, [message, trigger]);
34
35   if (!isVisible) return null;
36
37   return (
38     <div style={{
39       position: 'fixed', top: '2rem', right: '2rem', width: '320px', padding: '1.5rem',
40       display: 'flex', flexDirection: 'column', gap: '0.8rem', zIndex: 9999,
41       background: 'var(--bg-surface)', border: '1px solid var(--border-default)',
42       borderRadius: '16px', borderLeft: '4px solid var(--brand-primary)',
43       boxShadow: '0 10px 30px rgba(0,0,0,0.3)',
44       transition: 'all 0.3s ease',
45       transform: isSpeaking ? 'scale(1.03)' : 'scale(1)',
46     }}>
47       <button onClick={() => setIsVisible(false)}
48         style={{ position: 'absolute', top: '10px', right: '15px', background: 'transparent', border: ...
49     </button>
50     <div style={{ display: 'flex', alignItems: 'center', gap: '0.5rem' }}>
51       <div style={{
52         width: '12px', height: '12px', borderRadius: '50%',
53         background: isSpeaking ? 'var(--brand-primary)' : 'var(--text-muted)',
54         boxShadow: isSpeaking ? '0 0 10px var(--brand-primary)' : 'none',
55         animation: isSpeaking ? 'pulse is infinite' : 'none'
56       }} />
57       <strong style={{ color: 'var(--brand-primary)', fontSize: '1.1rem' }}>AI Guru</strong>
58     </div>
59     <p style={{ color: 'var(--text-primary)', fontSize: '0.95rem', lineHeight: '1.4', margin: 0 }}>
60       {message}
61     </p>
62   </div>
63 );
64
65 };
66
67 export default AIGuru;
68
```

File: components/AIGuru.tsx (237 lines, 10.8 KB)

```
1 'use client'
2 import React, { useState, useEffect, useRef } from 'react'
3 import { motion, AnimatePresence } from 'framer-motion'
4 import { X, Send, Sparkles, Bot, Cpu, Volume2 } from 'lucide-react'
5 import { useGuru } from '@lib/guru-state'
6
```

```

7 | export default function AIGuru() {
8 |   const [message, trigger, silentMessage, silentTrigger, lang] = useGuru()
9 |   const [isOpen, setIsOpen] = useState(false)
10 |   const [messages, setMessages] = useState([
11 |     { role: 'bot', content: lang === 'vn' ? 'Chào bạn! Tôi là AI Guru, '
12 |   })
13 |   const [input, setInput] = useState('')
14 |   const [isTyping, setIsTyping] = useState(false)
15 |   const [toast, setToast] = useState<{ show: boolean; text: string }>({ show: false, text: '' })
16 |   const [unreadCount, setUnreadCount] = useState(0)
17 |   const toastTimer = useRef<any>(null)
18 |   const endRef = useRef<HTMLDivElement>(null)
19 |   const sendingRef = useRef(false)
20 |
21 |   useEffect(() => { endRef.current?.scrollIntoView({ behavior: 'smooth' }) }, [messages])
22 |
23 |   const handleSend = async () => {
24 |     if (!input.trim() || sendingRef.current) return
25 |     sendingRef.current = true
26 |     const userMsg = { role: 'user', content: input }
27 |     const inputText = input
28 |     setInput('')
29 |     setIsTyping(true)
30 |
31 |     try {
32 |       const history = messages.map(m => ({ role: m.role === 'user' ? 'user' : 'assistant', content: m.content }))
33 |
34 |       const res = await fetch('/api/ai/chat', {
35 |         method: 'POST',
36 |         headers: { 'Content-Type': 'application/json' },
37 |         body: JSON.stringify({ message: inputText, history }),
38 |       })
39 |
40 |       if (!res.ok) {
41 |         setMessages(prev => [...prev, userMsg, { role: 'bot', content: ' '
42 |         setIsTyping(false)
43 |         sendingRef.current = false
44 |         return
45 |       }
46 |
47 |       const data = await res.json()
48 |       const reply = data?.reply || 'Xin lỗi, tôi chưa hiểu câu hỏi của bạn'
49 |       setMessages(prev => [...prev, userMsg, { role: 'bot', content: reply}])
50 |       catch {
51 |         setMessages(prev => [...prev, userMsg, { role: 'bot', content: 'X
52 |       } finally {
53 |         setIsTyping(false)
54 |         sendingRef.current = false
55 |       }
56 |     }
57 |
58 |     useEffect(() => {
59 |       if (message && trigger > 0) {
60 |         setMessages(prev => {
61 |           const last = prev[prev.length - 1]
62 |           if (last?.role === 'bot' && last?.content === message) return prev
63 |           return [...prev, { role: 'bot', content: message }]
64 |         })
65 |         setIsOpen(true)
66 |       }
67 |     }, [message, trigger])
68 |
69 |     useEffect(() => {
70 |       if (silentMessage && silentTrigger > 0) {
71 |         setMessages(prev => {
72 |           const last = prev[prev.length - 1]
73 |           if (last?.role === 'bot' && last?.content === '0=Y ' + silentMessage
74 |           return [...prev, { role: 'bot', content: '0=Y ' + silentMessage
75 |         })
76 |         if (!isOpen) {
77 |           setUnreadCount(c => c + 1)
78 |           setToast({ show: true, text: silentMessage })
79 |           if (toastTimer.current) clearTimeout(toastTimer.current)
80 |           toastTimer.current = setTimeout(() => setToast({ show: false, text: '' }), 3000)
81 |         }
82 |       }
83 |     }, [silentMessage, silentTrigger])
84 |
85 |     const styles = {
86 |       brand: 'var(--brand-primary)',
87 |       surface: 'var(--bg-surface)',
88 |       base: 'var(--bg-base)',
89 |       elevated: 'var(--bg-elevated)',
90 |       text: 'var(--text-primary)',
91 |       muted: 'var(--text-muted)',
92 |       border: 'var(--border-default)',
93 |     }
94 |
95 |     const duplicateIdRef = useRef(false)
96 |     useEffect(() => {
97 |       if (duplicateIdRef.current) return
98 |       duplicateIdRef.current = true
99 |     }, [])
100 |
101 |     return (
102 |       <>
103 |         <AnimatePresence>
104 |           {toast.show && !isOpen && (
105 |             <motion.div
106 |               initial={{ opacity: 0, y: 20, scale: 0.9 }}
107 |               animate={{ opacity: 1, y: 0, scale: 1 }}
108 |               exit={{ opacity: 0, y: 10, scale: 0.9 }}
109 |               transition={{ duration: 0.2 }}
110 |               className="fixed z-[999]"
111 |               style={{ bottom: '80px', right: '16px', maxWidth: '260px' }}
112 |             >
113 |               <div style={{
114 |                 background: styles.elevated, border: '1px solid ${styles.brand}40',
115 |                 borderRadius: '12px', padding: '10px 14px',
116 |                 display: 'flex', alignItems: 'center', gap: '8px',
117 |                 boxShadow: `0 4px 20px rgba(0,0,0,0.15)`,
118 |               }}>
119 |                 <Volume2 size={14} style={{ color: styles.brand, flexShrink: 0 }} />
120 |                 <span style={{ fontSize: '12px', color: styles.text, lineHeight: 1.4 }}>{toast.text}</span>
121 |               </div>
122 |             </motion.div>
123 |           )
124 |         </AnimatePresence>
125 |
126 |         <button
127 |           onClick={() => { setIsOpen(true); setUnreadCount(0); }}
128 |           className="fixed bottom-8 right-8 w-14 h-14 shadow-lg rounded-full flex items-center justify-center z-...
129 |           style={{
130 |             background: `linear-gradient(135deg, ${styles.brand}, #289cf9)`,

```

```

131 |         color: '#fff',
132 |         boxShadow: `0 0 25px color-mix(in srgb, ${styles.brand} 50%, transparent)` ,
133 |     }}
134 | }
135 | <div className="absolute -inset-1 rounded-full blur opacity-40 group-hover:opacity-75 transition-opaci...
136 | {isOpen ? <X size={22} className="relative z-10" /> : <Cpu size={22} className="relative z-10" />}
137 | {!isOpen && (
138 |   <div className="absolute -top-1 -right-1 w-5 h-5 rounded-full flex items-center justify-center shado...
139 |     <Sparkles size={10} style={{ color: styles.brand }} />
140 |   </div>
141 | )}
142 | {unreadCount > 0 && !isOpen && (
143 |   <div style={{
144 |     position: 'absolute', top: '-4px', right: '-4px', width: '18px', height: '18px',
145 |     borderRadius: '50%', background: '#EF4444', color: '#fff', fontSize: '9px',
146 |     fontWeight: 700, display: 'flex', alignItems: 'center', justifyContent: 'center',
147 |     border: '2px solid var(--bg-base)'
148 |   }}>
149 |     {unreadCount}
150 |   </div>
151 | )}
152 | </button>
153 |
154 | <AnimatePresence>
155 |   {isOpen && (
156 |     <motion.div
157 |       initial={{ opacity: 0, scale: 0.9, y: 50 }}
158 |       animate={{ opacity: 1, scale: 1, y: 0 }}
159 |       exit={{ opacity: 0, scale: 0.9, y: 50 }}
160 |       className="fixed bottom-24 right-8 w-[calc(100vw-40px)] max-w-[340px] shadow-2xl z-[1000] overflow...
161 |       style={{ height: '460px', borderRadius: '20px', background: styles.surface, border: '1px solid ${s...
162 |     >
163 |       <div className="flex items-center justify-between" style={{ padding: '16px 20px', background: styl...
164 |         <div className="flex items-center gap-3">
165 |           <div className="w-10 h-10 rounded-xl flex items-center justify-center" style={{ background: st...
166 |             <Bot size={22} />
167 |           </div>
168 |           <div>
169 |             <p className="font-bold text-sm" style={{ color: styles.text }}>AI Guru Assistant</p>
170 |             <p className="text-[10px] font-bold uppercase tracking-widest" style={{ color: styles.brand ...
171 |           </div>
172 |         </div>
173 |         <button onClick={() => setIsOpen(false)} style={{ padding: '6px', background: 'transparent', bor...
174 |           <X size={20} />
175 |         </button>
176 |       </div>
177 |
178 |       <div className="flex-1 overflow-y-auto p-4 space-y-4" style={{ background: styles.base }}>
179 |         {messages.map((m, i) => (
180 |           <div key={i} className="flex" style={{ justifyContent: m.role === 'user' ? 'flex-end' : 'flex-...
181 |             <div style={{
182 |               maxWidth: '82%', padding: '12px 16px', borderRadius: '16px', fontSize: '13px', lineHeight:...
183 |               background: m.role === 'user' ? styles.brand : styles.elevated,
184 |               color: m.role === 'user' ? '#fff' : styles.text,
185 |               borderTopLeftRadius: m.role === 'user' ? '16px' : '4px',
186 |               borderTopRightRadius: m.role === 'user' ? '4px' : '16px',
187 |               wordBreak: 'break-word', overflowWrap: 'break-word', whiteSpace: 'pre-wrap',
188 |             }}>
189 |               {m.content}
190 |             </div>
191 |           </div>
192 |         ))}
193 |         {isTyping && (
194 |           <div className="flex justify-start">
195 |             <div className="flex gap-1.5 p-3 rounded-2xl" style={{ background: styles.elevated, borderTo...
196 |               {[0, 1, 2].map(i => (
197 |                 <motion.div key={i} animate={{ opacity: [0.3, 1, 0.3] }} transition={{ repeat: Infinity,...
198 |                   style={{ width: '8px', height: '8px', borderRadius: '50%', background: styles.muted }} />
199 |               ))}
200 |             </div>
201 |           </div>
202 |         )}
203 |         <div ref={endRef} />
204 |       </div>
205 |
206 |       <div style={{ padding: '12px 16px', background: styles.elevated, borderTop: '1px solid ${styles.bo...
207 |         <div className="flex items-center gap-2 p-1.5 rounded-2xl" style={{ background: styles.base, bor...
208 |           <input
209 |             type="text"
210 |             value={input}
211 |             onChange={(e) => setInput(e.target.value)}
212 |             onKeyDown={(e) => {
213 |               if (e.key === 'Enter' && !e.nativeEvent.isComposing) {
214 |                 e.preventDefault()
215 |                 handleSend()
216 |               }
217 |             }}
218 |           </div>
219 |           <div style={{ padding: '0 10px' }}>
220 |             <div style={{ display: 'flex', gap: '5px', align-items: 'center' }}>
221 |               <div style={{ width: '10px', height: '10px', background: styles.brand, border: '1px solid ...
222 |               <div style={{ width: '10px', height: '10px', background: styles.brand, border: '1px solid ...
223 |               <div style={{ width: '10px', height: '10px', background: styles.brand, border: '1px solid ...
224 |               <div style={{ width: '10px', height: '10px', background: styles.brand, border: '1px solid ...
225 |               <div style={{ width: '10px', height: '10px', background: styles.brand, border: '1px solid ...
226 |             </div>
227 |             <div style={{ display: 'flex', gap: '5px', align-items: 'center' }}>
228 |               <div style={{ width: '10px', height: '10px', background: styles.brand, border: '1px solid ...
229 |               <div style={{ width: '10px', height: '10px', background: styles.brand, border: '1px solid ...
230 |               <div style={{ width: '10px', height: '10px', background: styles.brand, border: '1px solid ...
231 |               <div style={{ width: '10px', height: '10px', background: styles.brand, border: '1px solid ...
232 |             </div>
233 |           </div>
234 |         </div>
235 |       </div>
236 |     )
237 |   )

```

File: components/AIGuruGlobal.tsx (9 lines, 0.1 KB)

```

1 | 'use client';
2 |
3 | import React from 'react';
4 | import AIGuru from './AIGuru';
5 |
6 | export default function AIGuruGlobal() {
7 |   return <AIGuru />;
8 | }
9 |

```

File: components/AssembledPCView.js (446 lines, 22.8 KB)

```

1 'use client';
2
3 import { useState, useRef, useCallback } from 'react';
4 import { Canvas, useFrame } from '@react-three/fiber';
5 import { OrbitControls, Html } from '@react-three/drei';
6 import * as THREE from 'three';
7 import {
8   CPUModel, GPUModel, RAMModel, SSDModel, PSUModel, MainboardModel,
9 } from './ComponentPreview3D';
10
11 const MODEL_POSITIONS = {
12   MAINBOARD: { pos: [0, 0, 0], scale: 1 },
13   CPU: { pos: [0, 0.18, 0.30], scale: 1 },
14   COOLER: { pos: [0, 0.55, 0.30], scale: 0.8 },
15   RAM_LEFT: { pos: [-0.75, 0.12, 0.50], scale: 1 },
16   RAM_RIGHT: { pos: [0.75, 0.12, 0.50], scale: 1 },
17   GPU: { pos: [0, 0.12, -0.40], scale: 1 },
18   PSU: { pos: [1.50, -0.15, 1.00], scale: 1 },
19   SSD: { pos: [-0.20, 0.05, -0.70], scale: 1 },
20 };
21
22 const HOTSPOTS = {
23   MAINBOARD: [
24     { id: 'cpu-socket', label: 'CPU Socket', pos: [0, 0.04, 0.30], desc: 'CPU Socket' },
25     { id: 'ram-slot-l', label: 'RAM Slot A', pos: [-0.75, 0.04, 0.50], desc: 'RAM Slot A' },
26     { id: 'ram-slot-r', label: 'RAM Slot B', pos: [0.75, 0.04, 0.50], desc: 'RAM Slot B' },
27     { id: 'pcie-slot', label: 'PCIe x16', pos: [0, 0.04, -0.40], desc: 'PCIe Slot' },
28     { id: 'm2-slot', label: 'M.2 Slot', pos: [-0.20, 0.04, -0.70], desc: 'M.2 Slot' },
29     { id: 'sata-ports', label: 'SATA Port', pos: [1.25, 0.04, 0.60], desc: 'SATA Port' },
30     { id: 'io-ports', label: 'I/O Ports', pos: [1.55, 0.04, 0.20], desc: 'I/O Ports' },
31   ],
32   CPU: [
33     { id: 'cpu-ihs', label: 'IHS', pos: [0, 0.25, 0.30], desc: 'Heat Spreader' },
34     { id: 'cpu-pins', label: 'Chan CPU', pos: [0, 0.10, 0.30], desc: 'CPU Pins' },
35   ],
36   GPU: [
37     { id: 'gpu-pcie', label: 'PCIe Connector', pos: [0.50, 0.10, -0.40], desc: 'GPU PCIe' },
38     { id: 'gpu-power', label: 'Power Connector', pos: [1.25, 0.10, -0.40], desc: 'GPU Power' },
39     { id: 'gpu-fan', label: 'Fan GPU', pos: [0, 0.22, -0.40], desc: 'GPU Fan' },
40     { id: 'gpu-vram', label: 'VRAM', pos: [0.80, 0.08, -0.40], desc: 'GPU VRAM' },
41   ],
42   RAM: [
43     { id: 'ram-heatspreader', label: 'Heatspreader', pos: [-0.75, 0.30, 0.50], desc: 'RAM Heatspreader' },
44     { id: 'ram-pins', label: 'Gold Pins', pos: [-0.75, 0.02, 0.50], desc: 'RAM Pins' },
45   ],
46   PSU: [
47     { id: 'psu-24pin', label: '24-pin ATX', pos: [1.90, -0.10, 1.30], desc: 'PSU 24-pin ATX' },
48     { id: 'psu-cpu8', label: 'CPU 8-pin EPS', pos: [1.90, 0.05, 1.20], desc: 'PSU CPU 8-pin EPS' },
49     { id: 'psu-pcie', label: 'PCIe 6+2 pin', pos: [1.90, 0.15, 1.10], desc: 'PSU PCIe 6+2 pin' },
50     { id: 'psu-sata', label: 'SATA Power', pos: [1.90, 0.25, 1.00], desc: 'PSU SATA Power' },
51   ],
52   SSD: [
53     { id: 'ssd-nand', label: 'NAND Flash', pos: [-0.20, 0.06, -0.65], desc: 'SSD NAND Flash' },
54     { id: 'ssd-controller', label: 'Controller', pos: [-0.40, 0.06, -0.70], desc: 'SSD Controller' },
55     { id: 'ssd-m2', label: 'M.2 Connector', pos: [0.85, 0.04, -0.70], desc: 'SSD M.2 Connector' },
56   ],
57 };
58
59 const CONNECTION_RULES = [
60   { from: 'CPU', fromPort: 'CPU Socket', to: 'Mainboard', toPort: 'CPU Socket', cable: 'CPU Socket', desc: 'CPU Socket' },
61   { from: 'CPU', fromPort: 'IHS', to: 'Cooler', toPort: 'Base', cable: 'Cable', desc: 'CPU IHS' },
62   { from: 'RAM', fromPort: 'Gold Pins', to: 'Mainboard', toPort: 'RAM Slot', cable: 'RAM Slot', desc: 'RAM Slot' },
63   { from: 'GPU', fromPort: 'PCIe Connector', to: 'Mainboard', toPort: 'PCIe Slot', cable: 'PCIe Slot', desc: 'GPU PCIe' },
64   { from: 'GPU', fromPort: 'Power Connector', to: 'PSU', toPort: 'PCIe 6+2 pin', cable: 'PCIe Power', desc: 'GPU Power' },
65   { from: 'Mainboard', fromPort: '24-pin ATX', to: 'PSU', toPort: '24-pin ATX', cable: '24-pin ATX', desc: 'Mainboard 24-pin ATX' },
66   { from: 'CPU', fromPort: 'Chan CPU', to: 'PSU', toPort: 'CPU 8-pin EPS', cable: 'CPU 8-pin EPS', desc: 'CPU 8-pin EPS' },
67   { from: 'SSD', fromPort: 'M.2 Connector', to: 'Mainboard', toPort: 'M.2 Slot', cable: null, desc: 'SSD M.2 Connector' },
68   { from: 'SSD', fromPort: 'NAND Flash', to: 'PSU', toPort: 'SATA Power', cable: 'SATA Power', desc: 'SSD SATA Power' },
69 ];
70
71 function ConnectionLine({ from, to, color = '#d4a030', dashed = false }) {
72   const points = [
73     new THREE.Vector3(...from),
74     new THREE.Vector3((from[0] + to[0]) / 2 + 0.2, (from[1] + to[1]) / 2 + 0.3, (from[2] + to[2]) / 2),
75     new THREE.Vector3(...to),
76   ];
77   const curve = new THREE.CatmullRomCurve3(points);
78   const curvePoints = curve.getPoints(20);
79   return (
80     <mesh>
81     <tubeGeometry args={[new THREE.CatmullRomCurve3(curvePoints), 20, 0.02, 8, false]} />
82     <meshBasicMaterial color={color} transparent opacity={0.25} />
83     </mesh>
84   );
85 }
86
87 function GlowRing({ position, color }) {
88   const meshRef = useRef();
89   useFrame(({ clock }) => {
90     if (meshRef.current) {
91       meshRef.current.scale.setScalar(1 + Math.sin(clock.getElapsedTime() * 3) * 0.15);
92     }
93   });
94   return (
95     <mesh ref={meshRef} position={position}>
96     <ringGeometry args={[0.08, 0.13, 16]} />
97     <meshBasicMaterial color={color} transparent opacity={0.4} side={THREE.DoubleSide} />
98     </mesh>
99   );
100 }
101
102 function HotspotDot({ hotspot, isActive, isHovered, onClick, onHover }) {
103   const dotRef = useRef();
104   const color = hotspot.color;
105   return (
106     <group position={hotspot.pos}>
107     <mesh ref={dotRef}>
108       <onClick={onClick}>
109         <onPointerEnter={e => onHover(true)}>
110         <onPointerLeave={e => onHover(false)}>
111         <onPointerOver={e => { document.body.style.cursor = 'pointer'; e.stopPropagation(); }}>
112         <onPointerOut={e => { document.body.style.cursor = 'default'; }}>
113       <sphereGeometry args={[0.07, 12, 12]} />
114       <meshBasicMaterial color={color} transparent opacity={isActive ? 1 : isHovered ? 0.9 : 0.6} />
115     </mesh>
116     {isActive && <GlowRing position={[0, 0, 0]} color={color} />
117     {isHovered && (
118       <mesh>
119         <ringGeometry args={[0.08, 0.12, 16]} />
120         <meshBasicMaterial color={color} transparent opacity={0.25} side={THREE.DoubleSide} />
121       </mesh>
122     )}
123   </group>

```

```

124 | });
125 | }
126 |
127 | function InfoTooltip({ hotspot, onClose }) {
128 |   return (
129 |     <Html position={{[hotspot.pos[0] + 0.3, hotspot.pos[1] + 0.1, hotspot.pos[2]]}
130 |       center style={{ pointerEvents: 'none', zIndex: 100 }}>
131 |       <div style={{
132 |         background: 'rgba(8,8,18,0.95)', backdropFilter: 'blur(10px)',
133 |         border: '1px solid ${hotspot.color}44', borderRadius: 10,
134 |         padding: '10px 14px', maxWidth: 240,
135 |         boxShadow: '0 8px 32px rgba(0,0,0,0.6)',
136 |         animation: 'fadeIn 0.12s', pointerEvents: 'auto',
137 |       }}>
138 |         <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom: 6 }}>
139 |           <span style={{ color: hotspot.color, fontWeight: 700, fontSize: 12 }}>{hotspot.label}</span>
140 |           <button onClick={onClose}
141 |             style={{ background: 'none', border: 'none', color: '#555', cursor: 'pointer', fontSize: 13, paddi...
142 |           </div>
143 |           <div style={{ color: 'rgba(255,255,255,0.5)', fontSize: 10, lineHeight: 1.6, marginBottom: 6 }}>{hotsp...
144 |           {hotspot.related && (
145 |             <div style={{ fontSize: 9, color: 'rgba(255,255,255,0.3)', marginBottom: 4 }}>
146 |               <span style={{ color: 'rgba(255,255,255,0.4)' }}>Küt nñi: </
147 |             </div>
148 |           </div>
149 |         </div>
150 |         {hotspot.brands && hotspot.brands !== '-' && (
151 |           <div style={{ fontSize: 9, color: 'rgba(255,255,255,0.3)' }}>
152 |             <span style={{ color: 'rgba(255,255,255,0.4)' }}>Th°ing hiç
153 |           </div>
154 |         </div>
155 |       </div>
156 |     </div>
157 |   </Html>
158 | );
159 | }
160 |
161 | function EmptySlot({ label, position, color = '#333', hotspots }) {
162 |   const [showInfo, setShowInfo] = useState(false);
163 |   return (
164 |     <group position={position}>
165 |       <mesh>
166 |         <boxGeometry args={[0.6, 0.02, 0.6]} />
167 |         <meshStandardMaterial color={color} transparent opacity={0.12} wireframe />
168 |       </mesh>
169 |       <Html position={[0, 0.25, 0]} center>
170 |         <div onClick={() => setShowInfo(p => !p)}
171 |           style={{
172 |             color, fontSize: 9, fontWeight: 700, fontFamily: 'monospace',
173 |             background: 'rgba(0,0,0,0.5)', padding: '3px 8px', borderRadius: 4,
174 |             border: '1px dashed ${color}44', cursor: 'pointer',
175 |             textTransform: 'uppercase', letterSpacing: '0.5px',
176 |           }}>
177 |             & {label}
178 |         </div>
179 |       </Html>
180 |       {showInfo && hotspots && (
181 |         <Html position={[0, -0.35, 0]} center>
182 |           <div style={{
183 |             background: 'rgba(8,8,18,0.95)', backdropFilter: 'blur(8px)',
184 |             border: '1px solid ${color}33', borderRadius: 8, padding: 8, maxWidth: 200,
185 |             boxShadow: '0 4px 16px rgba(0,0,0,0.6)',
186 |           }}>
187 |             <div style={{ color: color, fontSize: 10, fontWeight: 700, marginBottom: 4 }}>{label}</div>
188 |             {hotspots.map(h => (
189 |               <div key={h.id} style={{ color: 'rgba(255,255,255,0.5)', fontSize: 9, padding: '1px 0' }}>
190 |                 {h.label}
191 |               </div>
192 |             ))}
193 |             <button onClick={() => setShowInfo(false)}
194 |               style={{ background: 'none', border: 'none', color: '#555', cursor: 'pointer', fontSize: 9, marg...
195 |             </div>
196 |           </Html>
197 |         </div>
198 |       </group>
199 |     );
200 |   }
201 |
202 | function CameraAnimator({ target }) {
203 |   const controlsRef = useRef();
204 |   useFrame(() => {
205 |     if (!controlsRef.current || !target) return;
206 |     controlsRef.current.target.lerp(target, 0.05);
207 |   });
208 |   return <orbitControls ref={controlsRef} />;
209 | }
210 |
211 | export default function AssembledPCView({ parts }) {
212 |   const controlsRef = useRef();
213 |   const [activeHotspot, setActiveHotspot] = useState(null);
214 |   const [hoveredId, setHoveredId] = useState(null);
215 |   const [eduMode, setEduMode] = useState(true);
216 |   const [showConnections, setShowConnections] = useState(false);
217 |
218 |   const hasCpu = !!parts.cpu;
219 |   const hasGpu = !!parts.gpu;
220 |   const hasRam = (parts.ram?.length || 0) > 0;
221 |   const hasPsu = !!parts.psu;
222 |   const hasSsd = (parts.ssd?.length || 0) > 0;
223 |   const hasCooler = !!parts.cooler;
224 |
225 |   const handleHotspotClick = useCallback((h) => {
226 |     if (activeHotspot?.id === h.id) {
227 |       setActiveHotspot(null);
228 |     } else {
229 |       setActiveHotspot(h);
230 |     }
231 |     setHoveredId(null);
232 |   }, [activeHotspot]);
233 |
234 |   const renderHotspots = (hotspots, visible) => {
235 |     if (!visible) return null;
236 |     return hotspots.map(h => {
237 |       const isActive = activeHotspot?.id === h.id;
238 |       const isHov = hoveredId === h.id;
239 |       return (
240 |         <group key={h.id}>
241 |           <HotspotDot hotspot={h} isActive={isActive} isHovered={isHov}
242 |             onClick={() => handleHotspotClick(h)}
243 |             onHover={(v) => setHoveredId(v ? h.id : null)} />
244 |           {eduMode && !isActive && !isHov && (
245 |             <Html position={[h.pos[0], h.pos[1] + 0.18, h.pos[2]]} center>
246 |               <div style={{
247 |                 color: h.color, fontSize: 7, fontWeight: 600, fontFamily: 'monospace',

```

```

248         background: 'rgba(0,0,0,0.5)', padding: '1px 5px', borderRadius: 3,
249         border: '1px solid ${h.color}22', pointerEvents: 'none',
250         opacity: 0.6, whiteSpace: 'nowrap',
251     }}>{h.label}</div>
252 </Html>
253 }}
254 {{(isActive || isHov) && (
255   <InfoTooltip hotspot={h} onClose={() => setActiveHotspot(null)} />
256 )}}
257 </group>
258 });
259 });
260 };
261
262 return (
263   <div style={{
264     width: '100%', height: '100%', position: 'relative',
265     background: 'radial-gradient(ellipse at center, #0f172a 0%, #020617 100%)',
266     borderRadius: 12, overflow: 'hidden', border: '1px solid var(--border-default)',
267   }}>
268     <Canvas camera={{ position: [3.5, 2.5, 4], fov: 35 }}
269       dpr={[1, 2]} gl={{ antialias: true }}
270       style={{ width: '100%', height: '100%' }}
271       onPointerMissed={() => { setActiveHotspot(null); setHoveredId(null); }}>
272       <ambientLight intensity={0.4} />
273       <directionalLight position={[5, 8, 5]} intensity={1.2} />
274       <directionalLight position={[-3, 4, -3]} intensity={0.3} />
275       <pointLight position={[0, 3, 0]} intensity={0.2} />
276       <pointLight position={[0, -2, 2]} intensity={0.1} color="#4488ff" />
277       <OrbitControls ref={controlsRef}
278         enablePan={true} enableZoom={true} enableRotate={true}
279         autoRotate={!activeHotspot} autoRotateSpeed={1.2}
280         minDistance={1.5} maxDistance={8}
281         target={activeHotspot ? [activeHotspot.pos[0], activeHotspot.pos[1], activeHotspot.pos[2]] : [0.3, 0...
282       />
283       <gridHelper args={[5, 10, '#1e293b', '#0f172a']} position={[0.3, -0.25, 0.1]} />
284
285       {/* Mainboard */}
286       <MainboardModel isHovered={false} />
287       {renderHotspots(HOTSPOTS.MAINBOARD, true)}
288
289       {/* CPU */}
290       {hasCpu ? (
291         <group position={MODEL_POSITIONS.CPU.pos} scale={MODEL_POSITIONS.CPU.scale}>
292           <CPUModel brand={parts.cpu.brand} socket={parts.cpu.cpu_socket} isHovered={false} />
293         </group>
294       ) : (
295         <EmptySlot label="CPU" position={MODEL_POSITIONS.CPU.pos} color="#00d4aa" hotspots={HOTSPOTS.CPU} />
296       )}
297       {renderHotspots(HOTSPOTS.CPU, hasCpu)}
298
299       {/* Cooler */}
300       {hasCooler ? (
301         <group position={MODEL_POSITIONS.COOLER.pos} scale={MODEL_POSITIONS.COOLER.scale}>
302           <mesh><boxGeometry args={[0.9, 0.7, 0.9]} /><meshStandardMaterial color="#1a1a1a" /></mesh>
303           <mesh position={[0, 0.45, 0]}><cylinderGeometry args={[0.35, 0.35, 0.05, 24]} /><meshStandardMater...
304           <mesh position={[0, 0.10, 0]}><cylinderGeometry args={[0.38, 0.38, 0.02, 16]} /><meshStandardMater...
305         </group>
306       ) : null}
307
308       {/* RAM */}
309       {hasRam ? (
310         <group position={MODEL_POSITIONS.RAM_LEFT.pos} scale={MODEL_POSITIONS.RAM_LEFT.scale}>
311           <RAMModel brand={parts.ram[0]?.brand} isHovered={false} />
312         </group>
313         {(parts.ram.length > 1) && (
314           <group position={MODEL_POSITIONS.RAM_RIGHT.pos} scale={MODEL_POSITIONS.RAM_RIGHT.scale}>
315             <RAMModel brand={parts.ram[1]?.brand} isHovered={false} />
316           </group>
317         )}
318       ) : (
319         <EmptySlot label="RAM A" position={MODEL_POSITIONS.RAM_LEFT.pos} color="#22c55e" hotspots={HOTSPOT...
320         <EmptySlot label="RAM B" position={MODEL_POSITIONS.RAM_RIGHT.pos} color="#22c55e" />
321       )}
322     </Canvas>
323     {renderHotspots(HOTSPOTS.RAM, hasRam)}
324
325     {/* GPU */}
326     {hasGpu ? (
327       <group position={MODEL_POSITIONS.GPU.pos} scale={MODEL_POSITIONS.GPU.scale}>
328         <GPUModel isHovered={false} />
329       </group>
330     ) : (
331       <EmptySlot label="GPU" position={MODEL_POSITIONS.GPU.pos} color="#ef4444" hotspots={HOTSPOTS.GPU} />
332     )}
333     {renderHotspots(HOTSPOTS.GPU, hasGpu)}
334
335     {/* PSU */}
336     {hasPsu ? (
337       <group position={MODEL_POSITIONS.PSU.pos} scale={MODEL_POSITIONS.PSU.scale}>
338         <PSUModel isHovered={false} />
339       </group>
340     ) : (
341       <EmptySlot label="PSU" position={MODEL_POSITIONS.PSU.pos} color="#f59e0b" hotspots={HOTSPOTS.PSU} />
342     )}
343     {renderHotspots(HOTSPOTS.PSU, hasPsu)}
344
345     {/* SSD */}
346     {hasSsd ? (
347       <group position={MODEL_POSITIONS.SSD.pos} scale={MODEL_POSITIONS.SSD.scale}>
348         <SSDModel isHovered={false} />
349       </group>
350     ) : null}
351     {renderHotspots(HOTSPOTS.SSD, hasSsd)}
352
353     {/* Connection lines */}
354     {showConnections && hasPsu && hasCpu && (
355       <ConnectionLine from={[1.90, -0.10, 1.30]} to={[1.30, 0.04, 0.20]} color="#d4a030" />
356     )}
357     {showConnections && hasPsu && hasCpu && (
358       <ConnectionLine from={[1.90, 0.05, 1.20]} to={[0.10, 0.10, 0.30]} color="#94a3b8" />
359     )}
360     {showConnections && hasPsu && hasGpu && (
361       <ConnectionLine from={[1.90, 0.15, 1.10]} to={[1.25, 0.10, -0.40]} color="#ef4444" />
362     )}
363   </div>
364
365   {/* Overlay Controls */}
366   <div style={{
367     position: 'absolute', top: 10, left: 10, display: 'flex', gap: 6,
368   }}>
369
370
371

```

```

372 |         <button onClick={() => setEduMode(p => !p)}
373 |             style={{
374 |                 padding: '4px 10px', borderRadius: 6, border: '1px solid rgba(255,255,255,0.1)',
375 |                 background: eduMode ? 'rgba(0,212,170,0.15)' : 'rgba(255,255,255,0.05)',
376 |                 color: eduMode ? '#00d4aa' : 'rgba(255,255,255,0.4)',
377 |                 cursor: 'pointer', fontSize: 9, fontWeight: 600, fontFamily: 'monospace',
378 |             }}>
379 |             { eduMode ? 'Edu ON' : 'Edu OFF' }
380 |         </button>
381 |         <button onClick={() => setShowConnections(p => !p)}
382 |             style={{
383 |                 padding: '4px 10px', borderRadius: 6, border: '1px solid rgba(255,255,255,0.1)',
384 |                 background: showConnections ? 'rgba(99,102,241,0.15)' : 'rgba(255,255,255,0.05)',
385 |                 color: showConnections ? '#818cf8' : 'rgba(255,255,255,0.4)',
386 |                 cursor: 'pointer', fontSize: 9, fontWeight: 600, fontFamily: 'monospace',
387 |             }}>
388 |             { showConnections ? 'Cable ON' : 'Cable OFF' }
389 |         </button>
390 |     </div>
391 |
392 |     /* Bottom hint */
393 |     <div style={{
394 |         position: 'absolute', bottom: 8, left: '50%', transform: 'translateX(-50%)',
395 |         color: 'rgba(255,255,255,0.15)', fontSize: 8, fontFamily: 'monospace',
396 |         pointerEvents: 'none', textAlign: 'center',
397 |     }}>
398 |         Click chuột xem thông tin · Kéo xoay · Scroll zoom
399 |     </div>
400 |
401 |     /* Active detail panel */
402 |     {activeHotspot && (
403 |         <div style={{
404 |             position: 'absolute', bottom: 32, left: '50%', transform: 'translateX(-50%)',
405 |             background: 'rgba(8,8,18,0.92)', backdropFilter: 'blur(10px)',
406 |             border: '1px solid ${activeHotspot.color}33', borderRadius: 10,
407 |             padding: '10px 16px', maxWidth: 450, width: '90%',
408 |             boxShadow: '0 4px 24px rgba(0,0,0,0.6)',
409 |             animation: 'fadeIn 0.15s',
410 |         }}>
411 |             <div style={{ display: 'flex', alignItems: 'center', gap: 8, marginBottom: 6 }}>
412 |                 <span style={{ width: 8, height: 8, borderRadius: '50%', background: activeHotspot.color, flexShri...
413 |                 <span style={{ color: activeHotspot.color, fontWeight: 700, fontSize: 13 }}>{activeHotspot.label}<...
414 |                 <span style={{ color: 'rgba(255,255,255,0.15)', fontSize: 8, fontFamily: 'monospace', marginLeft: ...
415 |                     {activeHotspot.id}
416 |                 </span>
417 |             </div>
418 |             <div style={{ color: 'rgba(255,255,255,0.5)', fontSize: 11, lineHeight: 1.6 }}>{activeHotspot.desc}<...
419 |             {activeHotspot.related && (
420 |                 <div style={{ marginTop: 6, display: 'flex', gap: 4, flexWrap: 'wrap' }}>
421 |                     {activeHotspot.related.map(r => (
422 |                         <span key={r} style={{
423 |                             background: 'rgba(255,255,255,0.06)', padding: '1px 6px', borderRadius: 3,
424 |                             fontSize: 9, color: 'rgba(255,255,255,0.4)',
425 |                         }}>{r}</span>
426 |                     ))}
427 |                 </div>
428 |             )}
429 |             {activeHotspot.brands && activeHotspot.brands !== '-' && (
430 |                 <div style={{ marginTop: 4, display: 'flex', gap: 4, flexWrap: 'wrap' }}>
431 |                     {activeHotspot.brands.split(',').map(b => (
432 |                         <span key={b} style={{
433 |                             background: 'rgba(255,255,255,0.05)', padding: '1px 5px', borderRadius: 3,
434 |                             fontSize: 8, color: 'rgba(255,255,255,0.35)',
435 |                         }}>{b}</span>
436 |                     ))}
437 |                 </div>
438 |             )}
439 |         </div>
440 |     )}
441 | </div>
442 | </div>
443 | );
444 | }
445 | }
446 |

```

File: components/auth/LoginModal.tsx (195 lines, 7.6 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect } from 'react'
4 | import { useRouter } from 'next/navigation'
5 | import Link from 'next/link'
6 | import { Loader2, Mail, Lock, Eye, EyeOff, X, ShieldCheck } from 'lucide-react'
7 | import { login } from '@lib/auth-actions'
8 | import { createBrowserClient } from '@supabase/ssr'
9 | import { motion, AnimatePresence } from 'framer-motion'
10 |
11 | interface LoginModalProps {
12 |   isOpen: boolean
13 |   onClose: () => void
14 |   redirectTo?: string | null
15 | }
16 |
17 | export default function LoginModal({ isOpen, onClose, redirectTo }: LoginModalProps) {
18 |   const [error, setError] = useState<string | null>(null)
19 |   const [loading, setLoading] = useState(false)
20 |   const [showPassword, setShowPassword] = useState(false)
21 |   const router = useRouter()
22 |
23 |   useEffect(() => {
24 |     if (!isOpen) {
25 |       setError(null)
26 |       setShowPassword(false)
27 |       setLoading(false)
28 |     }
29 |   }, [isOpen])
30 |
31 |   async function handleGoogleLogin() {
32 |     setError(null)
33 |     const supabase = createBrowserClient(
34 |       process.env.NEXT_PUBLIC_SUPABASE_URL!,
35 |       process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
36 |     )
37 |     const redirectTo = process.env.NEXT_PUBLIC_SITE_URL
38 |       ? `${process.env.NEXT_PUBLIC_SITE_URL}/auth/callback`
39 |       : `${window.location.origin}/auth/callback`
40 |     const { error: oauthError } = await supabase.auth.signInWithOAuth({
41 |       provider: 'google',
42 |       options: {
43 |         redirectTo,
44 |       },
45 |     })

```



```

46|   if (oauthError) {
47|     | set Error ( '      ng nh - p Google th ã t b i i : ' + oauthError.message )
48|   }
49| }
50|
51| async function handleSubmit(e: React.FormEvent<HTMLFormElement>) {
52|   e.preventDefault()
53|   setLoading(true)
54|   setError(null)
55|
56|   const formData = new FormData(e.currentTarget)
57|   const res = await login(formData)
58|
59|   if (res?.error) {
60|     setError(res.error)
61|     setLoading(false)
62|   } else if (res?.success && res.redirectUrl) {
63|     router.push(redirectTo || res.redirectUrl)
64|   } else {
65|     setLoading(false)
66|   }
67| }
68|
69| if (!isOpen) return null
70|
71| return (
72|   <div className="fixed inset-0 z-[9999] flex items-center justify-center p-4">
73|     <div className="absolute inset-0 bg-black/70 backdrop-blur-sm" onClick={onClose} />
74|     <motion.div
75|       initial={{ opacity: 0, scale: 0.95 }}
76|       animate={{ opacity: 1, scale: 1 }}
77|       exit={{ opacity: 0, scale: 0.95 }}
78|       transition={{ duration: 0.2 }}
79|       className="login-modal-content relative w-full max-w-md bg-[#16213e] border border-[#1e293b] rounded-2...
80|     >
81|       <button
82|         onClick={onClose}
83|         className="absolute top-4 right-4 text-slate-400 hover:text-white transition-colors p-1 rounded-lg h...
84|       >
85|         <X size={20} />
86|       </button>
87|
88|       <div className="text-center mb-8">
89|         <h2 c l a s s N a m e = " t e x t - 2 x l f o n t - b o l d t e x t - w h i t e m b - 2 ">Chào mừng t
90|         <p c l a s s N a m e = " t e x t - s l a t e - 4 0 0 t e x t - s m ">      ng nh - p  À t i ế p t ắ c h ả
91|       </div>
92|
93|       <div className="flex flex-col gap-3 mb-6">
94|         <button
95|           type="button"
96|           onClick={handleGoogleLogin}
97|           className="w-full flex items-center justify-center gap-3 bg-white text-black font-semibold py-2.5 ...
98|         >
99|           <svg viewBox="0 0 24 24" className="w-5 h-5">
100|             <path fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h5.92c-.26 1.37-1.04 2.53-2.2...
101|             <path fill="#34A853" d="M12 23c2.97 0 5.46-.98 7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-...
102|             <path fill="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.35-2.09V7.07H2.18C1.43 8...
103|             <path fill="#EA4335" d="M12 5.38c1.62 0 3.06.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 ...
104|           </svg>
105|           T i ế p t ắ c v ớ i G o o g l e
106|         </button>
107|       </div>
108|
109|       <div className="flex items-center gap-4 mb-6">
110|         <div className="h-[1px] flex-1 bg-[#1e293b]"></div>
111|         <span c l a s s N a m e = " t e x t - x s f o n t - s e m i b o l d t e x t - s l a t e - 5 0 0 u p p e r c a s
112|       </div>
113|
114|       <form onSubmit={handleSubmit} className="flex flex-col gap-4">
115|         <div className="relative">
116|           <Mail size={18} className="absolute left-4 top-1/2 -translate-y-1/2 text-slate-500" />
117|           <input
118|             type="email"
119|             name="email"
120|             required
121|             className="w-full bg-[#0f0f1a] border border-[#1e293b] rounded-xl pl-11 pr-4 py-3 outline-none f...
122|           />
123|           p l a c e h o l d e r = " Ê a c h Ê E m a i l "
124|         </div>
125|
126|         <div className="relative">
127|           <Lock size={18} className="absolute left-4 top-1/2 -translate-y-1/2 text-slate-500" />
128|           <input
129|             type={showPassword ? "text" : "password"}
130|             name="password"
131|             required
132|             className="w-full bg-[#0f0f1a] border border-[#1e293b] rounded-xl pl-11 pr-12 py-3 outline-none ...
133|           />
134|           p l a c e h o l d e r = " M - t k h @ u "
135|         </div>
136|         <button
137|           type="button"
138|           onClick={() => setShowPassword(!showPassword)}
139|           className="absolute right-4 top-1/2 -translate-y-1/2 text-slate-500 hover:text-slate-300 p-1"
140|         >
141|           {showPassword ? <EyeOff size={18} /> : <Eye size={18} />}
142|         </button>
143|       </div>
144|
145|       <div className="flex items-center justify-between mt-1">
146|         <Link
147|           href="/forgot-password"
148|           className="text-sm font-medium text-[#00b4d8] hover:text-[#00d2a0] transition-colors"
149|           onClick={onClose}
150|         >
151|           Q u ê n m - t k h @ u ?
152|         </Link>
153|       </div>
154|
155|       <AnimatePresence>
156|         {error && (
157|           <motion.div
158|             initial={{ opacity: 0, y: -10 }}
159|             animate={{ opacity: 1, y: 0 }}
160|             exit={{ opacity: 0 }}
161|             className="bg-red-500/10 border border-red-500/20 text-red-400 px-4 py-3 rounded-lg text-sm fo...
162|           >
163|             <ShieldCheck size={18} className="shrink-0" />
164|             <span>{error}</span>
165|           </motion.div>
166|         )}
167|     </AnimatePresence>
168|
169|     <button

```

```

170 |         type="submit"
171 |         disabled={loading}
172 |         className="w-full bg-[#00d2a0] hover:bg-[#00e6af] text-black font-bold py-3 rounded-xl transition-...
173 |     >
174 |     {loading ? (
175 |         <>
176 |             <Loader2 className="animate-spin" size={20} />
177 |         </>
178 |     ) : (
179 |         <button>
180 |             <div>
181 |                 <div>
182 |                     <div>
183 |                         <div>
184 |                             <div>
185 |                                 <div>
186 |                                     <div>
187 |                                         <div>
188 |                                             <div>
189 |                                                 <div>
190 |                                                     <div>
191 |                                                         <div>
192 |                                                             <div>
193 |                                                                 <div>
194 |                                                                     <div>
195 |

```

File: components/auth/RoleSelector.tsx (96 lines, 3.1 KB)

```

1 | 'use client'
2 |
3 | import React from 'react'
4 | import { ShieldAlert, BookOpen, GraduationCap, Users } from 'lucide-react'
5 |
6 | export type RoleType = 'admin' | 'teacher' | 'student' | 'parent'
7 |
8 | interface RoleSelectorProps {
9 |   selectedRole: RoleType | null
10 |   onSelect: (role: RoleType) => void
11 | }
12 |
13 | const ROLES = [
14 |   {
15 |     id: 'student' as RoleType,
16 |     title: 'Híç sinh',
17 |     description: 'Thñc hành l-p ráp PC và làm bài kiã m tra.',
18 |     icon: GraduationCap,
19 |     color: '#10B981', // Emerald
20 |     bg: 'rgba(16, 185, 129, 0.1)',
21 |   },
22 |   {
23 |     id: 'teacher' as RoleType,
24 |     title: 'Giáo viên',
25 |     description: 'Tio bài giẽ ng và theo dõi tiẽn ù híç sinh.',
26 |     icon: BookOpen,
27 |     color: '#06B6D4', // Cyan
28 |     bg: 'rgba(6, 182, 212, 0.1)',
29 |   },
30 |   {
31 |     id: 'parent' as RoleType,
32 |     title: 'Phã huynh',
33 |     description: 'Xem kút quẽ híç t-p cça con em.',
34 |     icon: Users,
35 |     color: '#F59E0B', // Amber
36 |     bg: 'rgba(245, 158, 11, 0.1)',
37 |   },
38 |   {
39 |     id: 'admin' as RoleType,
40 |     title: 'Quẽn trẽ viẽn',
41 |     description: 'Quẽn lý toãn bù hç thñng tr°ýng híç.',
42 |     icon: ShieldAlert,
43 |     color: '#8B5CF6', // Violet
44 |     bg: 'rgba(139, 92, 246, 0.1)',
45 |   },
46 | ]
47 |
48 | export default function RoleSelector({ selectedRole, onSelect }: RoleSelectorProps) {
49 |   return (
50 |     <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '16px', marginBottom: '32px' }}>
51 |       {ROLES.map((role) => {
52 |         const isSelected = selectedRole === role.id;
53 |         const Icon = role.icon;
54 |
55 |         return (
56 |           <button
57 |             key={role.id}
58 |             type="button"
59 |             onClick={() => onSelect(role.id)}
60 |             style={{
61 |               display: 'flex',
62 |               flexDirection: 'column',
63 |               alignItems: 'center',
64 |               textAlign: 'center',
65 |               padding: '24px',
66 |               borderRadius: '16px',
67 |               border: `2px solid ${isSelected ? role.color : 'rgba(255,255,255,0.1)'}`,
68 |               background: isSelected ? role.bg : 'var(--bg-elevated)',
69 |               cursor: 'pointer',
70 |               transition: 'all 0.2s ease-in-out',
71 |               transform: isSelected ? 'scale(1.02)' : 'scale(1)',
72 |               boxShadow: isSelected ? `0 10px 20px ${role.bg}` : 'none'
73 |             }}
74 |           >
75 |             <div style={{
76 |               width: '56px', height: '56px', borderRadius: '50%',
77 |               background: isSelected ? role.color : 'rgba(255,255,255,0.05)',
78 |               display: 'flex', alignItems: 'center', justifyContent: 'center',
79 |               marginBottom: '16px',
80 |               transition: 'all 0.2s'
81 |             }}>
82 |               <Icon size={28} color={isSelected ? '#fff' : role.color} />
83 |             </div>
84 |             <h3 style={{ fontSize: '18px', fontWeight: 700, color: 'var(--text-primary)', marginBottom: '8px' }}>
85 |               {role.title}
86 |             </h3>
87 |             <p style={{ fontSize: '13px', color: 'var(--text-secondary)', lineHeight: 1.5 }}>
88 |               {role.description}
89 |             </p>
90 |           </button>
91 |         )
92 |       )}
93 |     </div>
94 |   )

```

```
95 | }
96 |
```

File: components/AuthButton.js (110 lines, 4.3 KB)

```
1 | 'use client';
2 |
3 | import React, { useEffect, useState } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import { useRouter } from 'next/navigation';
6 | import { Logout, User, Login } from 'lucide-react';
7 | import Link from 'next/link';
8 |
9 | const AuthButton = () => {
10 |   const [user, setUser] = useState(null);
11 |   const [profile, setProfile] = useState(null);
12 |   const [loading, setLoading] = useState(true);
13 |   const router = useRouter();
14 |
15 |   useEffect(() => {
16 |     const checkUser = async () => {
17 |       const { data: { user: currentUser } } = await supabase.auth.getUser();
18 |       setUser(currentUser);
19 |       if (currentUser) {
20 |         const { data: userProfile } = await supabase
21 |           .from('profiles')
22 |           .select('role, full_name')
23 |           .eq('id', currentUser.id)
24 |           .single();
25 |         setProfile(userProfile);
26 |       }
27 |       setLoading(false);
28 |     };
29 |
30 |     checkUser();
31 |
32 |     // L ñ n g n g h e s ñ k i Ç n n g n h - p / n g x u ¤ t Ä c - p n h - t U I n g a y l - p
33 |     const { data: { subscription } } = supabase.auth.onAuthStateChange(async (_event, session) => {
34 |       setUser(session?.user ?? null);
35 |       if (session?.user) {
36 |         const { data: userProfile } = await supabase
37 |           .from('profiles')
38 |           .select('role, full_name')
39 |           .eq('id', session.user.id)
40 |           .single();
41 |         setProfile(userProfile);
42 |       } else {
43 |         setProfile(null);
44 |       }
45 |       setLoading(false);
46 |     });
47 |
48 |     return () => subscription.unsubscribe();
49 |   }, []);
50 |
51 |   const handleSignOut = async () => {
52 |     await supabase.auth.signOut();
53 |     window.location.href = '/';
54 |   };
55 |
56 |   if (loading) return <div style={{ width: '40px' }} />;
57 |
58 |   if (user) {
59 |     const role = profile?.role || user.user_metadata?.role || 'student';
60 |     const fullName = profile?.full_name || user.user_metadata?.full_name || user.email.split('@')[0];
61 |
62 |     let dashboardUrl = '/builder';
63 |     if (role === 'teacher') dashboardUrl = '/teacher';
64 |     else if (role === 'parent') dashboardUrl = '/parent';
65 |     else if (role === 'admin') dashboardUrl = '/admin';
66 |
67 |     return (
68 |       <div style={{ display: 'flex', alignItems: 'center', gap: '16px' }}>
69 |         <Link href={dashboardUrl} style={{ display: 'flex', alignItems: 'center', gap: '10px', textDec...
70 |           <div style={{
71 |             width: '32px', height: '32px', borderRadius: '50%',
72 |             background: 'rgba(0, 243, 255, 0.1)', border: '1px solid #00f3ff',
73 |             display: 'flex', alignItems: 'center', justifyContent: 'center', color: '#00f3ff'
74 |           }}>
75 |             <User size={18} />
76 |           </div>
77 |           <span style={{ fontSize: '14px', fontWeight: 600, color: '#00e6ed' }}>
78 |             {fullName}
79 |           </span>
80 |         </Link>
81 |         <button
82 |           onClick={handleSignOut}
83 |           style={{
84 |             background: 'rgba(239, 68, 68, 0.1)', color: '#f87171', border: 'none',
85 |             padding: '8px 16px', borderRadius: '8px', cursor: 'pointer', fontSize: '13px',
86 |             display: 'flex', alignItems: 'center', gap: '6px', fontWeight: 600,
87 |             transition: 'all 0.2s'
88 |           }}
89 |         >
90 |           <Logout size={16} /> n g x u ¤ t
91 |         </button>
92 |       </div>
93 |     );
94 |   }
95 |
96 |   return (
97 |     <Link href="/login">
98 |       <button style={{
99 |         background: 'rgba(0, 243, 255, 0.1)', color: '#00f3ff', border: '1px solid #00f3ff',
100 |         padding: '10px 20px', borderRadius: '8px', fontWeight: 700, cursor: 'pointer',
101 |         transition: 'all 0.2s', display: 'flex', alignItems: 'center', gap: '8px', fontSize: '14px'
102 |       }}>
103 |         <Login size={18} /> n g n h - p
104 |       </button>
105 |     </Link>
106 |   );
107 | };
108 |
109 | export default AuthButton;
110 |
```

File: components/BootSequence.js (61 lines, 1.8 KB)

```
1 | 'use client';
2 |
3 | import { useEffect, useState } from 'react';
```

```

4 |
5 | const steps = [
6 |   { text: "BIOS CHECK...", delay: 1000 },
7 |   { text: "CPU FOUND: OK", delay: 800 },
8 |   { text: "RAM DETECTED: 32GB OK", delay: 800 },
9 |   { text: "GPU INITIALIZED: RTX 4090", delay: 1000 },
10 |  { text: "LOADING OS...", delay: 2000 },
11 |  { text: "SYSTEM READY", delay: 1000 }
12 | ];
13 |
14 | const BootSequence = ({ onComplete, lang = 'en' }) => {
15 |   const [step, setStep] = useState(0);
16 |
17 |   useEffect(() => {
18 |     if (step < steps.length) {
19 |       const timer = setTimeout(() => {
20 |         setStep(s => s + 1);
21 |       }, steps[step].delay);
22 |       return () => clearTimeout(timer);
23 |     } else {
24 |       if (onComplete) onComplete();
25 |     }
26 |   }, [step, onComplete]);
27 |
28 |   return (
29 |     <div style={{
30 |       position: 'absolute',
31 |       inset: 0,
32 |       backgroundColor: 'black',
33 |       color: '#00ff00',
34 |       fontFamily: 'Courier New, Courier, monospace',
35 |       padding: '2rem',
36 |       zIndex: 200,
37 |       display: 'flex',
38 |       flexDirection: 'column',
39 |       justifyContent: 'center',
40 |       alignItems: 'flex-start'
41 |     }}>
42 |       {steps.slice(0, step + 1).map((s, i) => (
43 |         <div key={i} style={{ marginBottom: '0.5rem' }}>
44 |           {> ${s.text} }
45 |         </div>
46 |       ))}
47 |       <div className="cursor-blink">_</div>
48 |       <style jsx>{`
49 |         .cursor-blink {
50 |           animation: blink 1s step-end infinite;
51 |         }
52 |         @keyframes blink {
53 |           50% { opacity: 0; }
54 |         }
55 |       `}</style>
56 |     </div>
57 |   );
58 | };
59 |
60 | export default BootSequence;
61 |

```

File: components/Breadcrumb.tsx (56 lines, 1.9 KB)

```

1 | 'use client'
2 |
3 | import { usePathname } from 'next/navigation'
4 | import Link from 'next/link'
5 | import { ChevronRight, Home } from 'lucide-react'
6 |
7 | const LABEL_MAP: Record<string, string> = {
8 |   'teacher': 'Trang ch c',
9 |   'lessons': 'Qu n lý bài gi ng',
10 |  'create': 'T i o bài gi ng',
11 |  'classes': 'Qu n lý l p h c',
12 |  'new': 'T i o m u i',
13 |  'quiz': 'Qu n lý Á t h i',
14 |  'certificates': 'Qu n lý Ch ng ch é',
15 |  'student': 'H c s i n h',
16 |  'dashboard': 'B ng i Á u khi Ā n',
17 |  'history': 'L ě ch s i h c t - p',
18 |  'profile': 'H o s i',
19 |  'discussion': 'D i Ā n à n',
20 |
21 | }
22 |
23 | export default function Breadcrumb() {
24 |   const pathname = usePathname()
25 |   const segments = pathname.split('/').filter(Boolean)
26 |
27 |   if (segments.length <= 1) return null
28 |
29 |   const crumbs = segments.map((seg, i) => {
30 |     const href = '/' + segments.slice(0, i + 1).join('/')
31 |     const label = LABEL_MAP[seg] || seg
32 |     const isLast = i === segments.length - 1
33 |     return { href, label, isLast }
34 |   })
35 |
36 |   return (
37 |     <nav style={{ display: 'flex', alignItems: 'center', gap: '6px', marginBottom: '24px', fontSize: '13px' }}>
38 |       <Link href="/teacher" style={{ color: 'var(--text-muted)', display: 'flex', alignItems: 'center', textDe...
39 |         <Home size={14} />
40 |       </Link>
41 |       {crumbs.map((crumb, i) => (
42 |         <span key={crumb.href} style={{ display: 'flex', alignItems: 'center', gap: '6px' }}>
43 |           <ChevronRight size={12} style={{ color: 'var(--text-muted)' }} />
44 |           {crumb.isLast ? (
45 |             <span style={{ color: 'var(--text-secondary)', fontWeight: 600 }}>{crumb.label}</span>
46 |           ) : (
47 |             <Link href={crumb.href} style={{ color: 'var(--text-muted)', textDecoration: 'none' }}>
48 |               {crumb.label}
49 |             </Link>
50 |           )}
51 |         </span>
52 |       ))}
53 |     </nav>
54 |   )
55 | }
56 |

```

File: components/builder/BuilderLab.tsx (147 lines, 6.0 KB)

```

1 | 'use client';

```

```

2 |
3 | import { useState } from 'react';
4 | import { COMPONENT_SPECS, type ComponentRenderSpec } from '@data/componentSpecs';
5 | import ComponentRenderer2D from './ComponentRenderer2D';
6 | import dynamic from 'next/dynamic';
7 |
8 | const ComponentViewer3D = dynamic(() => import('./ComponentViewer3D'), { ssr: false });
9 |
10 | type ViewMode = '2d' | '3d' | 'split';
11 |
12 | interface BuilderLabProps {
13 |   onSelectComponent?: (spec: ComponentRenderSpec) => void;
14 | }
15 |
16 | export default function BuilderLab({ onSelectComponent }: BuilderLabProps) {
17 |   const [viewMode, setViewMode] = useState<ViewMode>('2d');
18 |   const [selectedSpec, setSelectedSpec] = useState<ComponentRenderSpec | null>(null);
19 |   const [hoveredSpec, setHoveredSpec] = useState<string | null>(null);
20 |
21 |   const handleComponentClick = (spec: ComponentRenderSpec) => {
22 |     setSelectedSpec(spec);
23 |     onSelectComponent?.(spec);
24 |   };
25 |
26 |   const containerStyle: React.CSSProperties = {
27 |     display: 'flex', flexDirection: 'column', gap: 16, width: '100%',
28 |   };
29 |
30 |   const tabBarStyle: React.CSSProperties = {
31 |     display: 'flex', gap: 4, padding: 4,
32 |     background: 'rgba(255,255,255,0.05)', borderRadius: 10,
33 |     width: 'fit-content',
34 |   };
35 |
36 |   const tabButton = (mode: ViewMode, label: string, icon: string): React.CSSProperties => ({
37 |     padding: '6px 16px', borderRadius: 8, cursor: 'pointer',
38 |     background: viewMode === mode ? 'rgba(0,212,170,0.2)' : 'transparent',
39 |     color: viewMode === mode ? '#00d4aa' : '#94a3b8',
40 |     border: '1px solid ${viewMode === mode ? 'rgba(0,212,170,0.3)' : 'transparent'}',
41 |     fontSize: 13, fontWeight: 600, fontFamily: 'inherit',
42 |     transition: 'all 0.2s',
43 |   });
44 |
45 |   const renderComponentList = (limit?: number) => (
46 |     <div style={{
47 |       display: 'grid',
48 |       gridTemplateColumns: `repeat(auto-fill, minmax(180px, 1fr))`,
49 |       gap: 12, padding: 8,
50 |     }}>
51 |       {(limit ? COMPONENT_SPECS.slice(0, limit) : COMPONENT_SPECS).map((spec) => (
52 |         <div key={spec.id}
53 |           onClick={() => handleComponentClick(spec)}
54 |           onMouseEnter={() => setHoveredSpec(spec.id)}
55 |           onMouseLeave={() => setHoveredSpec(null)}
56 |           style={{
57 |             background: 'rgba(255,255,255,0.03)', borderRadius: 12,
58 |             padding: 12, cursor: 'pointer',
59 |             border: '1px solid ${selectedSpec?.id === spec.id ? '#00d4aa' : 'rgba(255,255,255,0.08)}',
60 |             transition: 'all 0.2s',
61 |             transform: hoveredSpec === spec.id ? 'translateY(-2px)' : 'none',
62 |             boxShadow: hoveredSpec === spec.id ? '0 4px 20px rgba(0,0,0,0.3)' : 'none',
63 |           }}
64 |           <ComponentRenderer2D
65 |             spec={spec}
66 |             scale={0.6}
67 |             viewMode="top"
68 |             isSelected={selectedSpec?.id === spec.id}
69 |             isHovered={hoveredSpec === spec.id}
70 |             showConnectors={false}
71 |           />
72 |           <div style={{ marginTop: 8, textAlign: 'center' }}>
73 |             <div style={{ fontSize: 12, fontWeight: 600, color: '#e2e8f0' }}>{spec.name}</div>
74 |             <div style={{ fontSize: 10, color: '#64748b' }}>{spec.type.toUpperCase()} · {spec.brand}</div>
75 |           </div>
76 |         </div>
77 |       ))}
78 |     </div>
79 |   );
80 |
81 |   if (viewMode === '2d') {
82 |     return (
83 |       <div style={containerStyle}>
84 |         <div style={tabBarStyle}>
85 |           <button style={tabButton('2d', '2D', '0=Ÿ2')} onClick={() => setViewMode('2d')} />
86 |           <button style={tabButton('3d', '3D', '0=Ÿ3')} onClick={() => setViewMode('3d')} />
87 |           <button style={tabButton('split', 'Split', '0=ŸS')} onClick={() => setViewMode('split')} />
88 |         </div>
89 |         {renderComponentList()}
90 |       </div>
91 |     );
92 |   }
93 |
94 |   if (viewMode === '3d') {
95 |     return (
96 |       <div style={{ ...containerStyle, alignItems: 'center' }}>
97 |         <div style={tabBarStyle}>
98 |           <button style={tabButton('2d', '2D', '0=Ÿ2')} onClick={() => setViewMode('2d')} />
99 |           <button style={tabButton('3d', '3D', '0=Ÿ3')} onClick={() => setViewMode('3d')} />
100 |           <button style={tabButton('split', 'Split', '0=ŸS')} onClick={() => setViewMode('split')} />
101 |         </div>
102 |         <div style={{
103 |           display: 'flex', flexWrap: 'wrap', gap: 16, justifyContent: 'center', padding: 16,
104 |         }}>
105 |           {COMPONENT_SPECS.slice(0, 6).map((spec) => (
106 |             <div key={spec.id} onClick={() => handleComponentClick(spec)}
107 |               style={{
108 |                 cursor: 'pointer', borderRadius: 12, overflow: 'hidden',
109 |                 border: '2px solid ${selectedSpec?.id === spec.id ? '#00d4aa' : 'rgba(255,255,255,0.08)}',
110 |               }}
111 |             <ComponentViewer3D spec={spec} width={280} height={200} />
112 |             <div style={{ padding: 8, textAlign: 'center', background: 'rgba(0,0,0,0.5)' }}>
113 |               <div style={{ fontSize: 11, fontWeight: 600, color: '#e2e8f0' }}>{spec.name}</div>
114 |             </div>
115 |           ))}
116 |         </div>
117 |       </div>
118 |     );
119 |   }
120 |
121 |   return (
122 |     <div style={containerStyle}>
123 |       <div style={tabBarStyle}>
124 |         <button style={tabButton('2d', '2D', '0=Ÿ2')} onClick={() => setViewMode('2d')} />

```

```

1 2 6 | < button style={tabButton('3d','3D','0=0æ')} onClick={() => setV
1 2 7 | < button style={tabButton('split','Split','E')} onClick={() =>
128 | </div>
129 | <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: 16, height: '60vh' }}>
130 | <div style={{ overflowY: 'auto', borderRight: '1px solid rgba(255,255,255,0.08)' }}>
131 | {renderComponentList()}
132 | </div>
133 | <div style={{ display: 'flex', alignItems: 'center', justifyContent: 'center' }}>
134 | {selectedSpec ? (
135 | <ComponentViewer3D spec={selectedSpec} width={400} height={350} />
136 | ) : (
137 | <div style={{ color: '#64748b', fontSize: 14, textAlign: 'center' }}>
1 3 8 | <div style={{ fontSize: 32, marginBottom: 8 }}>0=0F </div>
1 3 9 | Chín lính kịçn bèn trái Á xem 3D
140 | </div>
141 | )}
142 | </div>
143 | </div>
144 | </div>
145 | );
146 | }
147 |

```

File: components/builder/CollaborationStatus.tsx (89 lines, 2.7 KB)

```

1 | 'use client';
2 |
3 | import { useEffect, useState } from 'react';
4 | import { createClient } from '@lib/supabase-client';
5 |
6 | interface PresenceState {
7 |   user_id: string;
8 |   role: string;
9 |   current_page: string;
10 |   full_name?: string;
11 | }
12 |
13 | export default function CollaborationStatus() {
14 |   const [creators, setCreators] = useState<PresenceState[]>([]);
15 |   const [builders, setBuilders] = useState<PresenceState[]>([]);
16 |   const [subscribed, setSubscribed] = useState(false);
17 |
18 |   useEffect(() => {
19 |     const supabase = createClient();
20 |     const channel = supabase.channel('online-users');
21 |
22 |     channel.on('presence', { event: 'sync' }, () => {
23 |       const state = channel.presenceState() as Record<string, PresenceState[]>;
24 |       const users = Object.values(state).flatMap((u) => u);
25 |       setCreators(users.filter((u) => u.role === 'component_creator' || u.role === 'admin'));
26 |       setBuilders(users.filter((u) => u.role !== 'component_creator' && u.role !== 'admin'));
27 |     });
28 |
29 |     channel.subscribe(async (status) => {
30 |       if (status === 'SUBSCRIBED') {
31 |         setSubscribed(true);
32 |         const { data: { user } } = await supabase.auth.getUser();
33 |         if (user) {
34 |           const { data: profile } = await supabase
35 |             .from('profiles')
36 |             .select('full_name, role')
37 |             .eq('id', user.id)
38 |             .single();
39 |           await channel.track({
40 |             user_id: user.id,
41 |             role: profile?.role || 'student',
42 |             full_name: profile?.full_name || 'Unknown',
43 |             current_page: window.location.pathname,
44 |           });
45 |         }
46 |       }
47 |     });
48 |
49 |     return () => {
50 |       supabase.removeChannel(channel);
51 |     };
52 |   }, []);
53 |
54 |   if (!subscribed && creators.length === 0 && builders.length === 0) return null;
55 |
56 |   const total = creators.length + builders.length;
57 |
58 |   return (
59 |     <div style={{
60 |       display: 'flex', alignItems: 'center', gap: 8,
61 |       padding: '6px 12px', borderRadius: 100,
62 |       background: 'rgba(255,255,255,0.05)',
63 |       border: '1px solid rgba(255,255,255,0.1)',
64 |       fontSize: 12,
65 |     }}>
66 |       <div style={{
67 |         width: 8, height: 8, borderRadius: '50%',
68 |         background: '#22c55e',
69 |         boxShadow: '0 0 8px rgba(34,197,94,0.5)',
70 |         animation: 'pulse 2s infinite',
71 |       }} />
72 |       <span style={{ color: '#94a3b8' }}>
7 3 | {total} ng ° Ý i ang online
74 | </span>
75 | {creators.length > 0 && (
76 |   <span style={{ color: '#a855f7', fontWeight: 600 }}>
77 |     {creators.length} Creator
78 |   </span>
79 | )}
80 | {builders.length > 0 && (
81 |   <span style={{ color: '#00d4aa', fontWeight: 600 }}>
82 |     {builders.length} Builder
83 |   </span>
84 | )}
85 | <style>{@keyframes pulse { 0%,100%{opacity:1} 50%{opacity:0.5} }}</style>
86 | </div>
87 | );
88 | }
89 |

```

File: components/builder/ComponentRenderer2D.tsx (260 lines, 9.5 KB)

```

1 | 'use client';
2 |
3 | import { useMemo } from 'react';
4 | import { type ComponentRenderSpec, MM_TO_PX } from '@data/componentSpecs';

```

```

5 |
6 | interface ComponentRenderer2DProps {
7 |   spec: ComponentRenderSpec;
8 |   scale?: number;
9 |   isSelected?: boolean;
10 |   isHovered?: boolean;
11 |   showConnectors?: boolean;
12 |   showDimensions?: boolean;
13 |   viewMode: 'top' | 'front' | 'side';
14 |   onConnectorClick?: (connectorId: string) => void;
15 |   onComponentClick?: () => void;
16 | }
17 |
18 | export default function ComponentRenderer2D({
19 |   spec,
20 |   scale = 1,
21 |   isSelected = false,
22 |   isHovered = false,
23 |   showConnectors = false,
24 |   showDimensions = false,
25 |   viewMode = 'top',
26 |   onConnectorClick,
27 |   onComponentClick,
28 | }: ComponentRenderer2DProps) {
29 |   const s = scale * MM_TO_PX;
30 |   const w = spec.dimensions.width_mm * s;
31 |   const h = spec.dimensions.height_mm * s;
32 |   const d = spec.dimensions.depth_mm * s;
33 |
34 |   const viewWidth = viewMode === 'top' || viewMode === 'side' ? w : d;
35 |   const viewHeight = viewMode === 'top' ? h : viewMode === 'front' ? h : d;
36 |   const padding = 20;
37 |   const svgW = viewWidth + padding * 2;
38 |   const svgH = viewHeight + padding * 2;
39 |
40 |   const connectorDots = useMemo(() => {
41 |     if (!showConnectors) return null;
42 |     return spec.connectors.map((conn) => {
43 |       const cx = padding + (conn.position.x / 100) * viewWidth;
44 |       const cy = padding + (conn.position.y / 100) * viewHeight;
45 |       return (
46 |         <g key={conn.id}>
47 |           <circle
48 |             cx={cx} cy={cy} r={6}
49 |             fill={conn.color} opacity={0.3}
50 |             stroke={conn.color} strokeWidth={2}
51 |           />
52 |           <circle
53 |             cx={cx} cy={cy} r={3}
54 |             fill={conn.color}
55 |             style={{ cursor: 'pointer' }}
56 |             onClick={() => onConnectorClick?.(conn.id)}
57 |           />
58 |           <text
59 |             x={cx + 10} y={cy + 4}
60 |             fill={conn.color} fontSize={9} fontFamily="monospace"
61 |           >
62 |             {conn.label}
63 |           </text>
64 |         </g>
65 |       );
66 |     });
67 |   }, [showConnectors, spec.connectors, viewWidth, viewHeight, padding, onConnectorClick]);
68 |
69 |   const renderCPU = () => {
70 |     const bw = viewWidth;
71 |     const bh = viewHeight;
72 |     return (
73 |       <g>
74 |         <rect x={padding} y={padding} width={bw} height={bh} rx={4}
75 |           fill={spec.colors.pcb} stroke={spec.colors.connector} strokeWidth={1.5} />
76 |         <rect x={padding + bw * 0.1} y={padding + bh * 0.1}
77 |           width={bw * 0.8} height={bh * 0.8} rx={3}
78 |           fill={spec.colors.heatsink || '#a0a0a0'}
79 |           stroke={spec.colors.accent || '#666'} strokeWidth={0.5} />
80 |         <text x={padding + bw / 2} y={padding + bh * 0.45}
81 |           textAnchor="middle" fill={spec.colors.accent || 'ffff'}
82 |           fontSize={Math.max(10, bw * 0.12)} fontWeight="bold" fontFamily="monospace">
83 |           {spec.brand}
84 |         </text>
85 |         <text x={padding + bw / 2} y={padding + bh * 0.6}
86 |           textAnchor="middle" fill="#999" fontSize={Math.max(8, bw * 0.08)} fontFamily="monospace">
87 |           {spec.model}
88 |         </text>
89 |       </g>
90 |     );
91 |   };
92 |
93 |   const renderGPU = () => {
94 |     const bw = viewWidth;
95 |     const bh = viewHeight;
96 |     const shroudH = bh * 0.8;
97 |     const pcieW = bw * 0.15;
98 |     return (
99 |       <g>
100 |         <rect x={padding} y={padding + (bh - shroudH) / 2}
101 |           width={bw - pcieW} height={shroudH} rx={6}
102 |           fill={spec.colors.shroud || spec.colors.pcb}
103 |           stroke={spec.colors.accent || '#666'} strokeWidth={1.5} />
104 |         <rect x={padding + bw - pcieW} y={padding + bh * 0.1}
105 |           width={pcieW} height={bh * 0.8} rx={2}
106 |           fill={spec.colors.connector} stroke={spec.colors.accent || '#ffcc00'} strokeWidth={1} />
107 |         {[0, 1].map((fi) => {
108 |           <circle key={fi}
109 |             cx={padding + (bw - pcieW) * (0.25 + fi * 0.35)}
110 |             cy={padding + bh / 2} r={bh * 0.15}
111 |             fill="none" stroke={spec.colors.accent || '#666'} strokeWidth={1} opacity={0.5} />
112 |         )}]
113 |         <text x={padding + bw / 2 - pcieW / 2} y={padding + bh * 0.2}
114 |           textAnchor="middle" fill={spec.colors.accent || 'ffff'}
115 |           fontSize={Math.max(8, bw * 0.06)} fontWeight="bold" fontFamily="monospace">
116 |           {spec.brand}
117 |         </text>
118 |       </g>
119 |     );
120 |   };
121 |
122 |   const renderMotherboard = () => {
123 |     const bw = viewWidth;
124 |     const bh = viewHeight;
125 |     return (
126 |       <g>
127 |         <rect x={padding} y={padding} width={bw} height={bh} rx={6}
128 |           fill={spec.colors.pcb} stroke={spec.colors.accent || '#ff700'} strokeWidth={1.5} />

```

```

129 |     {spec.connectors.map((conn) => {
130 |       const cx = padding + (conn.position.x / 100) * bw;
131 |       const cy = padding + (conn.position.y / 100) * bh;
132 |       const cw = conn.type.includes('cpu_socket') ? bw * 0.2 : conn.type.includes('ram') ? bw * 0.035 : bw...
133 |       const ch = conn.type.includes('cpu_socket') ? bh * 0.25 : conn.type.includes('ram') ? bh * 0.12 : bh...
134 |       return (
135 |         <rect key={conn.id} x={cx - cw / 2} y={cy - ch / 2}
136 |           width={cw} height={ch} rx={2}
137 |           fill={conn.color} opacity={0.4} stroke={conn.color} strokeWidth={0.5} />
138 |       );
139 |     })
140 |     <text x={padding + bw / 2} y={padding + bh * 0.08}
141 |       textAnchor="middle" fill={spec.colors.accent || '#ffd700'}
142 |       fontSize={Math.max(10, bw * 0.05)} fontWeight="bold" fontFamily="monospace">
143 |       {spec.brand} {spec.model}
144 |     </text>
145 |   </g>
146 | );
147 | };
148 |
149 | const renderRAM = () => {
150 |   const bw = viewWidth;
151 |   const bh = viewHeight;
152 |   return (
153 |     <g>
154 |       <rect x={padding} y={padding} width={bw} height={bh} rx={4}
155 |         fill={spec.colors.pcb} stroke={spec.colors.accent || '#ff6600'} strokeWidth={1.5} />
156 |       <rect x={padding + bw * 0.08} y={padding + bh * 0.15}
157 |         width={bw * 0.84} height={bh * 0.55} rx={2}
158 |         fill={spec.colors.heatsink || '#2c2c2c'}
159 |         stroke={spec.colors.accent || '#ff6600'} strokeWidth={0.5} opacity={0.8} />
160 |       <rect x={padding + bw * 0.05} y={padding + bh * 0.82}
161 |         width={bw * 0.9} height={bh * 0.12} rx={1}
162 |         fill={spec.colors.connector} opacity={0.8} />
163 |       <text x={padding + bw / 2} y={padding + bh * 0.5}
164 |         textAnchor="middle" fill={spec.colors.accent || '#ff6600'}
165 |         fontSize={Math.max(9, bw * 0.08)} fontWeight="bold" fontFamily="monospace">
166 |         {spec.techSpecs.capacity_gb}GB {spec.techSpecs.speed_mhz}MHz
167 |       </text>
168 |     </g>
169 |   );
170 | };
171 |
172 | const renderPSU = () => {
173 |   const bw = viewWidth;
174 |   const bh = viewHeight;
175 |   return (
176 |     <g>
177 |       <rect x={padding} y={padding} width={bw} height={bh} rx={6}
178 |         fill={spec.colors.heatsink || '#000'} stroke={spec.colors.accent || '#ffff00'} strokeWidth={1.5} />
179 |       <circle cx={padding + bw * 0.3} cy={padding + bh / 2} r={bh * 0.2}
180 |         fill="none" stroke={spec.colors.accent || '#ffff00'} strokeWidth={1} opacity={0.6} />
181 |       <text x={padding + bw * 0.3} y={padding + bh / 2 + 3}
182 |         textAnchor="middle" fill={spec.colors.accent || '#ffff00'}
183 |         fontSize={Math.max(8, bw * 0.05)} fontWeight="bold" fontFamily="monospace">
184 |         FAN
185 |       </text>
186 |       <rect x={padding + bw * 0.7} y={padding + bh * 0.2}
187 |         width={bw * 0.24} height={bh * 0.6} rx={3}
188 |         fill={spec.colors.connector} opacity={0.5} />
189 |       <text x={padding + bw / 2} y={padding + bh * 0.92}
190 |         textAnchor="middle" fill="#999" fontSize={Math.max(8, bw * 0.06)} fontFamily="monospace">
191 |         {spec.techSpecs.wattage}W {spec.techSpecs.efficiency}
192 |       </text>
193 |     </g>
194 |   );
195 | };
196 |
197 | const renderSSD = () => {
198 |   const bw = viewWidth;
199 |   const bh = viewHeight;
200 |   return (
201 |     <g>
202 |       <rect x={padding} y={padding} width={bw} height={bh} rx={4}
203 |         fill={spec.colors.pcb} stroke={spec.colors.accent || '#1428a0'} strokeWidth={1.5} />
204 |       <rect x={padding + bw * 0.05} y={padding + bh * 0.15}
205 |         width={bw * 0.65} height={bh * 0.7} rx={2}
206 |         fill={spec.colors.heatsink || '#c0c0c0'} opacity={0.5} />
207 |       <rect x={padding + bw * 0.82} y={padding + bh * 0.3}
208 |         width={bw * 0.12} height={bh * 0.4} rx={1}
209 |         fill={spec.colors.connector} opacity={0.7} />
210 |       <text x={padding + bw * 0.35} y={padding + bh * 0.55}
211 |         textAnchor="middle" fill={spec.colors.accent || '#1428a0'}
212 |         fontSize={Math.max(8, bw * 0.08)} fontWeight="bold" fontFamily="monospace">
213 |         {spec.brand}
214 |       </text>
215 |     </g>
216 |   );
217 | };
218 |
219 | const renderByType = () => {
220 |   switch (spec.type) {
221 |     case 'cpu': return renderCPU();
222 |     case 'gpu': return renderGPU();
223 |     case 'motherboard': return renderMotherboard();
224 |     case 'ram': return renderRAM();
225 |     case 'psu': return renderPSU();
226 |     case 'ssd': return renderSSD();
227 |     default: return (
228 |       <rect x={padding} y={padding} width={viewWidth} height={viewHeight} rx={4}
229 |         fill={spec.colors.pcb} stroke={spec.colors.accent || '#666'} strokeWidth={1.5} />
230 |     );
231 |   }
232 | };
233 |
234 | return (
235 |   <svg
236 |     viewBox={`0 0 ${svgW} ${svgH}`}
237 |     width={svgW}
238 |     height={svgH}
239 |     style={{
240 |       cursor: 'pointer',
241 |       outline: isSelected ? `2px solid ${spec.colors.accent || '#00d4aa'}` : 'none',
242 |       outlineOffset: 2,
243 |       borderRadius: 8,
244 |       filter: isHovered ? 'brightness(1.15)' : 'none',
245 |       transition: 'all 0.2s',
246 |     }}
247 |     onClick={onComponentClick}
248 |   >
249 |     {renderByType()}
250 |     {connectorDots}
251 |     {showDimensions && (
252 |       <text x={padding + viewWidth / 2} y={padding + viewHeight + 16}

```



```

253 |         textAnchor="middle" fill="#666" fontSize={10} fontFamily="monospace">
254 |         {spec.dimensions.width_mm}mm × {spec.dimensions.height_mm}mm × {spec.dimensions.depth_mm}mm
255 |     </text>
256 | })
257 | </svg>
258 | );
259 | }
260 |

```

File: components/builder/ComponentViewer3D.tsx (205 lines, 6.4 KB)

```

1 | 'use client';
2 |
3 | import { useState, useRef, useCallback, Suspense } from 'react';
4 | import { Canvas } from '@react-three/fiber';
5 | import { OrbitControls, Text } from '@react-three/drei';
6 | import { type ComponentRenderSpec, MM_TO_3D } from '@data/componentSpecs';
7 |
8 | interface ComponentViewer3DProps {
9 |   spec: ComponentRenderSpec;
10 |   width?: number;
11 |   height?: number;
12 | }
13 |
14 | function CPU3D({ spec }: { spec: ComponentRenderSpec }) {
15 |   const s = MM_TO_3D;
16 |   const w = spec.dimensions.width_mm * s;
17 |   const d = spec.dimensions.depth_mm * s;
18 |   const h = spec.dimensions.height_mm * s;
19 |   return (
20 |     <group>
21 |       <mesh position={[0, -h / 2, 0]}>
22 |         <boxGeometry args={[w, h, d]} />
23 |         <meshStandardMaterial color={spec.colors.pcb} roughness={0.5} />
24 |       </mesh>
25 |       <mesh position={[0, h / 2 + 0.01, 0]}>
26 |         <boxGeometry args={[w * 0.8, 0.02, d * 0.8]} />
27 |         <meshStandardMaterial color={spec.colors.heatsink || '#c0c0c0'} metalness={0.7} roughness={0.2} />
28 |       </mesh>
29 |       <Text position={[0, h / 2 + 0.03, 0]} fontSize={0.04} color={spec.colors.accent || '#fff'}
30 |         anchorX="center" anchorY="middle">
31 |         {spec.brand}
32 |       </Text>
33 |     </group>
34 |   );
35 | }
36 |
37 | function GPU3D({ spec }: { spec: ComponentRenderSpec }) {
38 |   const s = MM_TO_3D;
39 |   const w = spec.dimensions.width_mm * s;
40 |   const h = spec.dimensions.height_mm * s;
41 |   const d = spec.dimensions.depth_mm * s;
42 |   return (
43 |     <group>
44 |       <mesh position={[0, 0, 0]}>
45 |         <boxGeometry args={[w, d, h]} />
46 |         <meshStandardMaterial color={spec.colors.pcb} roughness={0.7} />
47 |       </mesh>
48 |       <mesh position={[0, d / 2 + 0.01, 0]}>
49 |         <boxGeometry args={[w * 0.85, 0.02, h * 0.85]} />
50 |         <meshStandardMaterial color={spec.colors.shroud || spec.colors.heatsink || '#333'} roughness={0.6} />
51 |       </mesh>
52 |       {[0, 1].map((i) => (
53 |         <mesh key={i} position={[w * (0.15 + i * 0.25), d / 2 + 0.02, 0]}>
54 |           <cylinderGeometry args={[h * 0.12, h * 0.12, 0.01, 24]} />
55 |           <meshStandardMaterial color="#444" />
56 |         </mesh>
57 |       ))}
58 |     </group>
59 |   );
60 | }
61 |
62 | function Motherboard3D({ spec }: { spec: ComponentRenderSpec }) {
63 |   const s = MM_TO_3D;
64 |   const w = spec.dimensions.width_mm * s;
65 |   const h = spec.dimensions.height_mm * s;
66 |   const d = spec.dimensions.depth_mm * s;
67 |   return (
68 |     <group>
69 |       <mesh position={[0, 0, 0]}>
70 |         <boxGeometry args={[w, d, h]} />
71 |         <meshStandardMaterial color={spec.colors.pcb} roughness={0.8} />
72 |       </mesh>
73 |       {spec.connectors.slice(0, 4).map((conn, i) => {
74 |         const cx = (conn.position.x / 100 - 0.5) * w;
75 |         const cz = (conn.position.y / 100 - 0.5) * h;
76 |         return (
77 |           <mesh key={conn.id} position={[cx, d / 2 + 0.005, cz]}>
78 |             <boxGeometry args={[0.06, 0.01, 0.12]} />
79 |             <meshStandardMaterial color={conn.color} />
80 |           </mesh>
81 |         );
82 |       })}
83 |     </group>
84 |   );
85 | }
86 |
87 | function RAM3D({ spec }: { spec: ComponentRenderSpec }) {
88 |   const s = MM_TO_3D;
89 |   const w = spec.dimensions.width_mm * s;
90 |   const h = spec.dimensions.height_mm * s;
91 |   const d = spec.dimensions.depth_mm * s;
92 |   return (
93 |     <group>
94 |       <mesh position={[0, 0, 0]}>
95 |         <boxGeometry args={[w, d, h]} />
96 |         <meshStandardMaterial color={spec.colors.pcb} roughness={0.6} />
97 |       </mesh>
98 |       <mesh position={[0, d / 2 + 0.01, 0]}>
99 |         <boxGeometry args={[w * 0.85, 0.02, h * 0.85]} />
100 |         <meshStandardMaterial color={spec.colors.heatsink || '#2c2c2c'} metalness={0.4} roughness={0.4} />
101 |       </mesh>
102 |     </group>
103 |   );
104 | }
105 |
106 | function PSU3D({ spec }: { spec: ComponentRenderSpec }) {
107 |   const s = MM_TO_3D;
108 |   const w = spec.dimensions.width_mm * s;
109 |   const h = spec.dimensions.height_mm * s;
110 |   const d = spec.dimensions.depth_mm * s;
111 |   return (
112 |     <group>

```

```

113 |     <mesh position={[0, 0, 0]}>
114 |       <boxGeometry args={[w, d, h]} />
115 |       <meshStandardMaterial color={spec.colors.heatsink} || '#222' metalness={0.3} roughness={0.6} />
116 |     </mesh>
117 |     <mesh position={[0, d / 2 + 0.01, 0]}>
118 |       <cylinderGeometry args={[h * 0.2, h * 0.2, 0.01, 24]} />
119 |       <meshStandardMaterial color="#333" />
120 |     </mesh>
121 |   </group>
122 | );
123 | }
124 |
125 | function SSD3D({ spec }: { spec: ComponentRenderSpec }) {
126 |   const s = MM_TO_3D;
127 |   const w = spec.dimensions.width_mm * s;
128 |   const h = spec.dimensions.height_mm * s;
129 |   const d = spec.dimensions.depth_mm * s;
130 |   return (
131 |     <group>
132 |       <mesh position={[0, 0, 0]}>
133 |         <boxGeometry args={[w, d, h]} />
134 |         <meshStandardMaterial color={spec.colors.pcb} roughness={0.7} />
135 |       </mesh>
136 |       <mesh position={[w * 0.02, d / 2 + 0.005, 0]}>
137 |         <boxGeometry args={[w * 0.6, 0.01, h * 0.7]} />
138 |         <meshStandardMaterial color={spec.colors.heatsink} || '#c0c0c0' metalness={0.3} roughness={0.5} opacit...
139 |       </mesh>
140 |     </group>
141 |   );
142 | }
143 |
144 | function Model3D({ spec }: { spec: ComponentRenderSpec }) {
145 |   switch (spec.type) {
146 |     case 'cpu': return <CPU3D spec={spec} />;
147 |     case 'gpu': return <GPU3D spec={spec} />;
148 |     case 'motherboard': return <Motherboard3D spec={spec} />;
149 |     case 'ram': return <RAM3D spec={spec} />;
150 |     case 'psu': return <PSU3D spec={spec} />;
151 |     case 'ssd': case 'hdd': return <SSD3D spec={spec} />;
152 |     default:
153 |       return (
154 |         <mesh>
155 |           <boxGeometry args={[0.5, 0.1, 0.5]} />
156 |           <meshStandardMaterial color={spec.colors.pcb} />
157 |         </mesh>
158 |       );
159 |   }
160 | }
161 |
162 | function Scene({ spec }: { spec: ComponentRenderSpec }) {
163 |   const [interacted, setInteracted] = useState(false);
164 |   return (
165 |     <>
166 |       <ambientLight intensity={0.6} />
167 |       <directionalLight position={[5, 5, 5]} intensity={0.8} />
168 |       <directionalLight position={[-3, 2, -3]} intensity={0.3} />
169 |       <OrbitControls
170 |         enablePan={false}
171 |         enableZoom={true}
172 |         autoRotate={!interacted}
173 |         autoRotateSpeed={0.3}
174 |         minDistance={0.3}
175 |         maxDistance={3}
176 |         onStart={() => setInteracted(true)}
177 |       />
178 |       <Suspense fallback={null}>
179 |         <Model3D spec={spec} />
180 |       </Suspense>
181 |     </>
182 |   );
183 | }
184 |
185 | export default function ComponentViewer3D({ spec, width = 400, height = 300 }: ComponentViewer3DProps) {
186 |   return (
187 |     <div style={{
188 |       width, height,
189 |       background: '#f8fafc',
190 |       borderRadius: 12,
191 |       overflow: 'hidden',
192 |       position: 'relative',
193 |     }}>
194 |       <Canvas
195 |         camera={{ position: [2, 1.5, 2], fov: 45 }}
196 |         dpr={[1, 2]}
197 |         gl={{ antialias: true }}
198 |         style={{ width: '100%', height: '100%' }}
199 |       >
200 |         <Scene spec={spec} />
201 |       </Canvas>
202 |     </div>
203 |   );
204 | }
205 |

```

File: components/builder/ErrorPanel.tsx (267 lines, 11.7 KB)

```

1 | 'use client';
2 |
3 | import { useState } from 'react';
4 | import { type AssemblyError, ASSEMBLY_ERRORS } from '@data/assemblyErrors';
5 |
6 | interface ErrorPanelProps {
7 |   errors: AssemblyError[];
8 |   onClose?: () => void;
9 | }
10 |
11 | function YouTubeClip({ videoId, startSeconds, endSeconds, title, channelName, description }: {
12 |   videoId: string;
13 |   startSeconds: number;
14 |   endSeconds?: number;
15 |   title: string;
16 |   channelName: string;
17 |   description: string;
18 | }) {
19 |   const [showPlayer, setShowPlayer] = useState(false);
20 |   const embedUrl = `https://www.youtube.com/embed/${videoId}?start=${startSeconds}${endSeconds ? `&end=${endSe...
21 |
22 |   return (
23 |     <div style={{ marginBottom: 12 }}>
24 |       <div style={{ position: 'relative', aspectRatio: '16/9', borderRadius: 8, overflow: 'hidden', background...
25 |         {!showPlayer ? (
26 |           <>
27 |             <img

```

```

28 |         src={`https://img.youtube.com/vi/${videoId}/maxresdefault.jpg`}
29 |         alt={title}
30 |         style={{ width: '100%', height: '100%', objectFit: 'cover', cursor: 'pointer' }}
31 |         onClick={() => setShowPlayer(true)}
32 |         loading="lazy"
33 |     />
34 |     <div
35 |       onClick={() => setShowPlayer(true)}
36 |       style={{
37 |         position: 'absolute', inset: 0, display: 'flex', alignItems: 'center', justifyContent: 'center...
38 |       }}
39 |     >
40 |       <div style={{
41 |         width: 56, height: 56, borderRadius: '50%', background: 'rgba(0,0,0,0.7)',
42 |         display: 'flex', alignItems: 'center', justifyContent: 'center',
43 |         border: '3px solid rgba(255,255,255,0.3)',
44 |       }}>
45 |         <svg width="24" height="24" viewBox="0 0 24 24" fill="white"><polygon points="8,5 19,12 8,19" ...
46 |       </div>
47 |     </div>
48 |   </>
49 |   ) : (
50 |     <iframe
51 |       src={embedUrl}
52 |       title={title}
53 |       width="100%"
54 |       height="100%"
55 |       allow="accelerometer; autoplay; clipboard-write; encrypted-media; gyroscope; picture-in-picture"
56 |       allowFullScreen
57 |       style={{ border: 'none', position: 'absolute', inset: 0 }}
58 |     />
59 |   )}
60 | </div>
61 | <div style={{ marginTop: 8 }}>
62 |   <div style={{ fontSize: 13, fontWeight: 600, color: '#e2e8f0' }}>{title}</div>
63 |   <div style={{ fontSize: 11, color: '#94a3b8' }}>{channelName}</div>
64 |   <div style={{ fontSize: 11, color: '#64748b', marginTop: 4 }}>{description}</div>
65 |   <div style={{ fontSize: 10, color: '#00d4aa', marginTop: 4 }}>
66 |     B - t      S u t e      g i a y      { s t a r t S e c o n d s } { e n d S e c o n d s } ? ` -      z n      g i a y      $ { e n d
67 |   </div>
68 | </div>
69 | </div>
70 | );
71 | }
72 |
73 | export default function ErrorPanel({ errors, onClose }: ErrorPanelProps) {
74 |   const [expandedId, setExpandedId] = useState<string | null>(null);
75 |   const [languageTab, setLanguageTab] = useState<'all' | 'vi' | 'en'>('all');
76 |   const [selectedError, setSelectedError] = useState<AssemblyError | null>(errors[0] || null);
77 |
78 |   if (errors.length === 0) return null;
79 |
80 |   const toggleExpand = (id: string) => {
81 |     setExpandedId(expandedId === id ? null : id);
82 |   };
83 |
84 |   const severityConfig = {
85 |     c r i t i c a l : { i c o n : 'L', c o l o r : '#ef4444', b g : 'r g b a ( 2 3 9 , 6 8 , 6 8 , 0 . 1 ) ' ,
86 |     w a r n i n g : { i c o n : 'p', c o l o r : '#f59e0b', b g : 'r g b a ( 2 4 5 , 1 5 8 , 1 1 , 0 . 1 ) ' ,
87 |     i n f o : { i c o n : 'i', c o l o r : '#3b82f6', b g : 'r g b a ( 5 9 , 1 3 0 , 2 4 6 , 0 . 1 ) ' ,
88 |   };
89 |
90 |   return (
91 |     <div style={{
92 |       position: 'fixed', bottom: 0, left: 0, right: 0, zIndex: 1000,
93 |       background: '#0f0f1a', borderTop: '1px solid rgba(255,255,255,0.1)',
94 |       maxHeight: '50vh', overflow: 'hidden', display: 'flex', flexDirection: 'column',
95 |       boxShadow: '0 -8px 32px rgba(0,0,0,0.5)',
96 |     }}>
97 |       <div style={{
98 |         display: 'flex', alignItems: 'center', justifyContent: 'space-between',
99 |         padding: '12px 24px', borderBottom: '1px solid rgba(255,255,255,0.08)',
100 |       }}>
101 |         <div style={{ display: 'flex', alignItems: 'center', gap: 12 }}>
102 |           <span style={{ fontS i z e : 1 8 }}> & p < / s p a n >
103 |           <span style={{ fontSize: 16, fontWeight: 700, color: '#e2e8f0' }}>
104 |             { e r r o r s . l e n g t h } l x i p h á t h i ç n
105 |           </span>
106 |         </div>
107 |         <button onClick={onClose} style={{
108 |           background: 'none', border: 'none', color: '#94a3b8', cursor: 'pointer', fontSize: 20,
109 |         }}> < / b u t t o n >
110 |       </div>
111 |
112 |       <div style={{ display: 'flex', flex: 1, overflow: 'hidden' }}>
113 |         <div style={{
114 |           width: 320, overflowY: 'auto', borderRight: '1px solid rgba(255,255,255,0.08)',
115 |           padding: 12,
116 |         }}>
117 |           {errors.map((err) => {
118 |             const conf = severityConfig[err.severity];
119 |             return (
120 |               <div key={err.id} onClick={() => setSelectedError(err)}
121 |                 style={{
122 |                   padding: '10px 12px', borderRadius: 8, marginBottom: 6, cursor: 'pointer',
123 |                   background: selectedError?.id === err.id ? conf.bg : 'transparent',
124 |                   border: '1px solid ${selectedError?.id === err.id ? conf.color : 'transparent'}',
125 |                 }}>
126 |                 <div style={{ display: 'flex', alignItems: 'center', gap: 8 }}>
127 |                   <span>{conf.icon}</span>
128 |                   <span style={{ fontSize: 13, fontWeight: 600, color: '#e2e8f0', flex: 1 }}>{err.title}</span>
129 |                 </div>
130 |                 <div style={{ fontSize: 11, color: conf.color, marginTop: 4 }}>
131 |                   {err.code}
132 |                 </div>
133 |               </div>
134 |             );
135 |           })}
136 |         </div>
137 |
138 |         <div style={{ flex: 1, overflowY: 'auto', padding: '16px 24px' }}>
139 |           {selectedError && (
140 |             <div style={{ display: 'flex', alignItems: 'center', gap: 8, marginBottom: 12 }}>
141 |               <span style={{ fontSize: 20 }}>{severityConfig[selectedError.severity].icon}</span>
142 |               <div style={{ fontSize: 18, fontWeight: 700, color: '#e2e8f0' }}>{selectedError.title}</div>
143 |             </div>
144 |             <div style={{ fontSize: 14, color: '#94a3b8', lineHeight: 1.6, marginBottom: 16 }}>
145 |               {selectedError.description}
146 |             </div>
147 |
148 |             <div style={{ marginBottom: 16 }}>
149 |               <div onClick={() => toggleExpand(selectedError.id)}
150 |                 style={{ cursor: 'pointer', display: 'flex', alignItems: 'center', gap: 8, fontSize: 14, fon...
151 |               </div>

```

```

1 5 2 | <span>{expandedId === selectedError.id ? '%%' : '%'}</span>
1 5 3 | Cach s i a
154 | </div>
155 | {expandedId === selectedError.id && (
156 |   <div style={{ marginTop: 8, padding: 12, background: 'rgba(0,212,170,0.08)', borderRadius: 8...
157 |     {selectedError.howToFix}
158 |   </div>
159 | )}
160 | </div>
161 |
162 | <div style={{ marginBottom: 12 }}>
163 |   <div style={{ fontSize: 14, fontWeight: 600, color: '#e2e8f0', marginBottom: 8 }}>
1 6 4 | Video h o ò n g d « n s i a l x i
165 | </div>
166 | <div style={{ display: 'flex', gap: 8, marginBottom: 12 }}>
167 |   {[ 'all', 'vi', 'en'].map((tab) => (
168 |     <button key={tab} onClick={() => setLanguageTab(tab as any)}
169 |       style={{
170 |         padding: '4px 12px', borderRadius: 6, fontSize: 12, fontWeight: 600, cursor: 'pointer',
171 |         background: languageTab === tab ? 'rgba(0,212,170,0.2)' : 'rgba(255,255,255,0.05)',
172 |         color: languageTab === tab ? '#00d4aa' : '#94a3b8',
173 |         border: '1px solid $[languageTab === tab ? '#00d4aa' : 'transparent']`,
174 |       }}
1 7 5 |   { t a b === ' a l l ' ? ' T ¥ t c £ ' : t a b === ' v i ' ? ' T i ñ n g
176 |     </button>
177 |   )}
178 | </div>
179 | {selectedError.youtubeVideo
180 |   .filter((v) => languageTab === 'all' || v.language === languageTab)
181 |   .map((video) => (
182 |     <YouTubeClip key={video.videoId} {...video} />
183 |   ))}
184 | </div>
185 |
186 | {selectedError.suggestedFix && (
187 |   <div style={{ padding: 12, background: 'rgba(245,158,11,0.1)', borderRadius: 8, border: '1px s...
1 8 8 |     <div style={{ fontSize: 13, color: '#e2e8f0' }}>
189 |       T h a y { s e l e c t e d E r r o r . s u g g e s t e d F i x . r e p l a c e C o m p o n e n t }
1 9 0 |     </div>
191 |   )}
192 | </div>
193 | )}
194 | </>
195 | )}
196 | </div>
197 | </div>
198 | </div>
199 | );
200 | }
201 |
202 | export function ErrorAnimation({ error }: { error: AssemblyError }) {
203 |   const style: Record<string, string | number> = {
204 |     position: 'absolute', inset: 0, pointerEvents: 'none', zIndex: 100,
205 |   };
206 |
207 |   switch (error.animation.type) {
208 |     case 'shake':
209 |       return (
210 |         <div style={style}>
211 |           <style>{@keyframes shake { 0%,100%{transform:translateX(0)} 25%{transform:translateX(-8px)} 50%{tra...
212 |             {error.animation.targetComponents.map((comp) => (
213 |               <div key={comp} data-error-component={comp}
214 |                 style={{
215 |                   animation: `shake ${error.animation.durationMs}ms ease-in-out`,
216 |                   border: '2px solid #ef4444', borderRadius: 8,
217 |                 }} />
218 |             )}
219 |           </div>
220 |         );
221 |     case 'red-flash':
222 |       return (
223 |         <div style={{ ...style, background: 'rgba(239,68,68,0.2)', animation: `redFlash ${error.animation.dura...
224 |           <style>{@keyframes redFlash { 0%{opacity:0} 15%{opacity:1} 30%{opacity:0} 45%{opacity:1} 60%{opacit...
225 |         </div>
226 |       );
227 |     case 'cross-mark':
228 |       return (
229 |         <div style={{ ...style, display: 'flex', alignItems: 'center', justifyContent: 'center' }}>
230 |           <style>{@keyframes crossMark { 0%{transform:scale(0);opacity:0} 50%{transform:scale(1.2);opacity:1}...
231 |           <span style={{ fontSize: 64, color: '#ef4444', animation: `crossMark ${error.animation.durationMs}ms...
232 |         </div>
233 |       );
234 |     case 'arrow-wrong':
235 |       return (
236 |         <div style={{ ...style, display: 'flex', alignItems: 'center', justifyContent: 'center' }}>
237 |           <svg width="120" height="40" viewBox="0 0 120 40">
238 |             <line x1="10" y1="20" x2="100" y2="20" stroke="#ef4444" strokeWidth="3" strokeDasharray="8 4">
239 |               <animate attributeName="stroke-dashoffset" from="0" to="24" dur="1s" repeatCount="2" />
240 |             </line>
241 |             <polygon points="100,12 115,20 100,28" fill="#ef4444" />
2 4 2 |           < t e x t x = " 5 5 " y = " 1 2 " t e x t A n c h o r = " m i d d l e " f i l l = " # e f 4 4 4 4 " f o n t s
243 |         </svg>
244 |       </div>
245 |     );
246 |     case 'highlight-both':
247 |       return (
248 |         <div style={style}>
249 |           <style>{@keyframes pulseGlow { 0%,100%{opacity:1} 50%{opacity:0.5} }}</style>
250 |           {error.animation.targetComponents.map((comp) => (
251 |             <div key={comp} data-error-component={comp}
252 |               style={{
253 |                 outline: '3px solid #ef4444', outlineOffset: 2, borderRadius: 8,
254 |                 boxShadow: '0 0 20px rgba(239,68,68,0.5)',
255 |                 animation: `pulseGlow ${error.animation.durationMs}ms ease-in-out`,
256 |               }} />
257 |             )}
258 |           </div>
259 |         );
260 |     default:
261 |       return null;
262 |   }
263 | }
264 |
265 | export { ASSEMBLY_ERRORS };
266 | export type { AssemblyVideo } from '@data/assemblyErrors';
267 |

```

File: components/BurgerMenu.js (556 lines, 29.9 KB)

```

1 | 'use client';
2 |
3 | import React, { useState, useEffect } from 'react';
4 | import { motion } from 'framer-motion';

```

```

5 | import Link from 'next/link';
6 | import {
7 |   BookOpen, Cpu, ShoppingCart, Users, BrainCircuit, Award, Globe, Sparkles,
8 |   Menu, Webcam, X, Sun, Moon, FileText, Bell, MessageSquare, User, Monitor,
9 |   BarChart3, Zap, Settings, GraduationCap, BookMarked
10 | } from 'lucide-react';
11 | import JoinClassModal from './JoinClassModal';
12 | import ComponentPreview from './ComponentPreview';
13 |
14 | const BurgerMenu = ({
15 |   lang,
16 |   toggleLang,
17 |   onStartQuiz,
18 |   appMode,
19 |   setAppMode,
20 |   webcamMouseEnabled = false,
21 |   setWebcamMouseEnabled,
22 |   trackingSensitivity = 1.0,
23 |   setTrackingSensitivity,
24 |   onToggleAI,
25 |   isAIOpen,
26 |   theme,
27 |   setTheme
28 | }) => {
29 |   const [isMobileOpen, setIsMobileOpen] = useState(false);
30 |   const [hoveredBtn, setHoveredBtn] = useState(null);
31 |   const [showCredits, setShowCredits] = useState(false);
32 |   const [showJoinClass, setShowJoinClass] = useState(false);
33 |   const [showComponents, setShowComponents] = useState(false);
34 |   const [hasClass, setHasClass] = useState(false);
35 |   const unreadCount = 0;
36 |
37 |   // Check if user has a class
38 |   useEffect(() => {
39 |     const checkClass = async () => {
40 |       try {
41 |         const { createBrowserClient } = require('@supabase/ssr');
42 |         const supabase = createBrowserClient(
43 |           process.env.NEXT_PUBLIC_SUPABASE_URL,
44 |           process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY
45 |         );
46 |         const { data: { user } } = await supabase.auth.getUser();
47 |         if (user) {
48 |           const { data: profile } = await supabase
49 |             .from('profiles')
50 |             .select('class_id, school_id')
51 |             .eq('id', user.id)
52 |             .single();
53 |
54 |           if (profile?.class_id || profile?.school_id) {
55 |             setHasClass(true);
56 |           }
57 |         }
58 |       } catch (e) {
59 |         console.error(e);
60 |       }
61 |     };
62 |     checkClass();
63 |   }, []);
64 |
65 |   // Close mobile menu when mode changes
66 |   useEffect(() => {
67 |     setIsMobileOpen(false);
68 |   }, [appMode]);
69 |
70 |   const navItemStyle = (id, isActive) => ({
71 |     padding: '8px 12px',
72 |     borderRadius: '8px',
73 |     display: 'flex',
74 |     alignItems: 'center',
75 |     gap: '8px',
76 |     cursor: 'pointer',
77 |     background: isActive ? 'var(--brand-subtle)' : hoveredBtn === id ? 'var(--bg-hover)' : 'transparent',
78 |     border: 'none',
79 |     borderLeft: isActive ? '2px solid var(--brand-primary)' : '2px solid transparent',
80 |     marginLeft: isActive ? '-2px' : '0',
81 |     transition: 'all 150ms ease',
82 |     width: '100%',
83 |     color: isActive ? 'var(--brand-light)' : hoveredBtn === id ? 'var(--text-primary)' : 'var(--text-secon...',
84 |     fontWeight: isActive ? 600 : 500,
85 |     fontSize: 'var(--text-sm)',
86 |     textAlign: 'left'
87 |   });
88 |
89 |   const iconStyle = (id, isActive) => ({
90 |     color: isActive ? 'var(--brand-light)' : hoveredBtn === id ? 'var(--text-primary)' : 'var(--text-muted)',
91 |     width: '16px',
92 |     height: '16px',
93 |     transition: 'all 150ms ease'
94 |   });
95 |
96 |   const percent = ((trackingSensitivity - 0.5) / 2.5) * 100;
97 |
98 |   return (
99 |     <
100 |       <div>
101 |         <button
102 |           className="mobile-hamburger"
103 |           onClick={() => setIsMobileOpen(true)}
104 |           style={{
105 |             position: 'fixed',
106 |             top: '16px',
107 |             left: '16px',
108 |             width: '40px',
109 |             height: '40px',
110 |             background: 'var(--bg-elevated)',
111 |             border: '1px solid var(--border-default)',
112 |             borderRadius: '8px',
113 |             zIndex: 100,
114 |             display: 'flex',
115 |             alignItems: 'center',
116 |             justifyContent: 'center',
117 |             color: 'var(--text-primary)',
118 |             cursor: 'pointer'
119 |           }}
120 |         >
121 |           <Menu size={20} />
122 |         </button>
123 |
124 |         <div>
125 |           <div
126 |             className="mobile-backdrop"
127 |             style={{
128 |               position: 'fixed', top: 0, left: 0, width: '100vw', height: '100vh',

```

```

129 |         background: 'rgba(0,0,0,0.6)',
130 |         zIndex: 180, opacity: isMobileOpen ? 1 : 0, pointerEvents: isMobileOpen ? 'auto' : 'none',
131 |         transition: 'opacity 250ms ease-out'
132 |     }}
133 |     onClick={() => setIsMobileOpen(false)}
134 | }
135 |
136 | {/ * Sidebar */}
137 | <div className={ `sidebar-container ${isMobileOpen ? 'open' : ''}` }>
138 |
139 |     <div style={{ height: '60px', padding: '0 16px', display: 'flex', alignItems: 'center', flexSh...
140 |     <Link href="/" style={{ textDecoration: 'none', display: 'flex', alignItems: 'center', gap: ...
141 |     
143 |         PC Master <span style={{ color: 'var(--brand-primary)' }}>Builder</span>
144 |     </span>
145 |
146 | </Link>
147 | </div>
148 |
149 | {/ * [2] DIVIDER */}
150 | <div style={{ borderBottom: '1px solid var(--border-subtle)', margin: '0 16px', flexShrink: 0 ...
151 |
152 | <div style={{ flex: 1, display: 'flex', flexDirection: 'column' }}>
153 |     {/ * [3] NAV GROUPS - KHÁM PHÁ */}
154 |     <div style={{ padding: '12px 0', display: 'flex', flexDirection: 'column', gap: '4px', mar...
155 |     <div style={{
156 |         fontSize: '11px', fontWeight: 600, letterSpacing: '0.08em', textTransform: 'upperc...
157 |         color: 'var(--text-muted)', padding: '12px 0 6px 0'
158 |     }}>
159 |         {lang === 'en' ? 'Explore' : 'Khám phá'}
160 |     </div>
161 |
162 |     <button
163 |         onMouseEnter={() => setHoveredBtn('course')} onMouseLeave={() => setHoveredBtn(null)}
164 |         onClick={() => setAppMode('course')}
165 |         style={navItemStyle('course', appMode === 'course')}
166 |     >
167 |         <BookOpen style={iconStyle('course', appMode === 'course')} />
168 |         {lang === 'en' ? 'Lecture Course' : 'Bài Giảng'}
169 |     </button>
170 |
171 |     <button
172 |         onMouseEnter={() => setHoveredBtn('learning')} onMouseLeave={() => setHoveredBtn(n...
173 |         onClick={() => setAppMode('learning')}
174 |         style={navItemStyle('learning', appMode === 'learning')}
175 |     >
176 |         <Cpu style={iconStyle('learning', appMode === 'learning')} />
177 |         {lang === 'en' ? 'Practice Mode' : 'Luyện Tập'}
178 |     </button>
179 |
180 |     <button
181 |         onMouseEnter={() => setHoveredBtn('market')} onMouseLeave={() => setHoveredBtn(null)}
182 |         onClick={() => setAppMode('market')}
183 |         style={navItemStyle('market', appMode === 'market')}
184 |     >
185 |         <ShoppingCart style={iconStyle('market', appMode === 'market')} />
186 |         {lang === 'en' ? 'Marketplace' : 'Chợ Máy Tính'}
187 |     </button>
188 |
189 |     <button
190 |         onMouseEnter={() => setHoveredBtn('components')} onMouseLeave={() => setHoveredBtn...
191 |         onClick={() => setShowComponents(true)}
192 |         style={navItemStyle('components', false)}
193 |     >
194 |         <Monitor style={iconStyle('components', false)} />
195 |         {lang === 'en' ? 'Component Library' : 'Thư Viện Thành Phần'}
196 |     </button>
197 |
198 |     <button
199 |         onMouseEnter={() => setHoveredBtn('multiplayer')} onMouseLeave={() => setHoveredBt...
200 |         onClick={() => setAppMode('multiplayer')}
201 |         style={navItemStyle('multiplayer', appMode === 'multiplayer')}
202 |     >
203 |         <Users style={iconStyle('multiplayer', appMode === 'multiplayer')} />
204 |         {lang === 'en' ? '2-Player Versus' : '2 Người Chơi'}
205 |     </button>
206 | </div>
207 |
208 |     {/ * NHÓM ÁNH GIÁC */}
209 |     <div style={{ padding: '12px 0', display: 'flex', flexDirection: 'column', gap: '4px', mar...
210 |     <div style={{
211 |         fontSize: '11px', fontWeight: 600, letterSpacing: '0.08em', textTransform: 'upperc...
212 |         color: 'var(--text-muted)', padding: '12px 0 6px 0'
213 |     }}>
214 |         {lang === 'en' ? 'Assessment' : 'Ánh Giá'}
215 |     </div>
216 |
217 |     <Link
218 |         href="/daily-quiz"
219 |         style={{ ...navItemStyle('quiz_bank', false), textDecoration: 'none' }}
220 |         onMouseEnter={() => setHoveredBtn('quiz_bank')}
221 |         onMouseLeave={() => setHoveredBtn(null)}
222 |     >
223 |         <BookMarked style={iconStyle('quiz_bank', false)} />
224 |         {lang === 'en' ? 'Question Bank' : 'Ngân Hàng Câu Hỏi'}
225 |     </Link>
226 |
227 |     <Link
228 |         href="/exams"
229 |         style={{ ...navItemStyle('exams', false), textDecoration: 'none' }}
230 |         onMouseEnter={() => setHoveredBtn('exams')}
231 |         onMouseLeave={() => setHoveredBtn(null)}
232 |     >
233 |         <FileText style={iconStyle('exams', false)} />
234 |         {lang === 'en' ? 'Exams' : 'Kó Thi'}
235 |     </Link>
236 |
237 |     <Link
238 |         href="/quiz"
239 |         style={{ ...navItemStyle('mcq', false), textDecoration: 'none' }}
240 |         onMouseEnter={() => setHoveredBtn('mcq')}
241 |         onMouseLeave={() => setHoveredBtn(null)}
242 |     >
243 |         <BrainCircuit style={iconStyle('mcq', false)} />
244 |         {lang === 'en' ? 'MCQ' : 'Trắc Nghiệm'}
245 |     </Link>
246 |
247 |     <Link
248 |         href="/leaderboard"
249 |         style={{ ...navItemStyle('leaderboard', false), textDecoration: 'none' }}
250 |         onMouseEnter={() => setHoveredBtn('leaderboard')}
251 |         onMouseLeave={() => setHoveredBtn(null)}
252 |     >

```

```

253|         <BarChart3 style={iconStyle('leaderboard', false)} />
254|         { lang === 'en' ? 'Leaderboard' : 'Xếp Hạng' }
255|     </Link>
256| </div>
257|
258| {typeof setWebcamMouseEnabled !== 'undefined' && typeof setTrackingSensitivity !== 'undefi...
259| <div style={{ padding: '12px 16px', display: 'flex', flexDirection: 'column' }}>
260|     <div style={{
261|         fontSize: '11px', fontWeight: 600, letterSpacing: '0.08em', textTransform: 'upperc...
262|         color: 'var(--text-muted)', padding: '12px 0 6px 0'
263|     }}>
264|         { lang === 'en' ? 'Controls' : 'iẤu khiÃn' }
265|     </div>
266|
267|     <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', ...
268|     <div style={{ display: 'flex', alignItems: 'center', gap: '8px' }}>
269|         <webcam size={16} color="var(--text-muted)" />
270|         <span style={{ fontSize: '13px', color: 'var(--text-secondary)', fontWeight: 5...
271|             {lang === 'en' ? 'Hand Tracking' : 'Theo dõi tay'}
272|         </span>
273|     </div>
274|     <div
275|         className={`toggle-switch ${webcamMouseEnabled ? 'on' : 'off'}`}
276|         onClick={() => { if (setWebcamMouseEnabled) setWebcamMouseEnabled(!webcamMouse...
277|         >
278|         <div className="toggle-thumb"></div>
279|     </div>
280| </div>
281|
282|     <div style={{ marginTop: '8px' }}>
283|     <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'cente...
284|     <span style={{ fontSize: '12px', color: 'var(--text-muted)' }}>
285|         { lang === 'en' ? 'Sensitivity' : '
286|     </span>
287|     <span style={{ fontSize: '12px', color: 'var(--brand-light)' }}>
288|         {trackingSensitivity.toFixed(1)}x
289|     </span>
290| </div>
291|     <input
292|         type="range"
293|         min="0.5" max="3" step="0.1"
294|         value={trackingSensitivity}
295|         onChange={(e) => { if (setTrackingSensitivity) setTrackingSensitivity(parseFlo...
296|         className="custom-slider"
297|         style={{
298|             background: `linear-gradient(to right, var(--brand-primary) ${percent}%, v...
299|         }}
300|     />
301| </div>
302| </div>
303| )}
304|
305|     { / * T Í N H N N G K H Á C * / }
306| <div style={{ padding: '12px 0', display: 'flex', flexDirection: 'column', gap: '4px', mar...
307|     <div style={{
308|         fontSize: '11px', fontWeight: 600, letterSpacing: '0.08em', textTransform: 'upperc...
309|         color: 'var(--text-muted)', padding: '12px 0 6px 0'
310|     }}>
311|         { lang === 'en' ? 'Extras' : 'T í n h n n g k h á c' }
312|     </div>
313|
314|     {!hasClass ? (
315|         <button
316|             onMouseEnter={() => setHoveredBtn('join_class')} onMouseLeave={() => setHovere...
317|             onClick={() => setShowJoinClass(true)}
318|             style={navItemStyle('join_class', false)}
319|         >
320|             <Users style={iconStyle('join_class', false)} />
321|             { lang === 'en' ? 'Join Class' : 'Tham gia' }
322|         </button>
323|     ) : (
324|         <Link
325|             href="/student/classes"
326|             style={{ ...navItemStyle('my_class', false), textDecoration: 'none' }}
327|             onMouseEnter={() => setHoveredBtn('my_class')}
328|             onMouseLeave={() => setHoveredBtn(null)}
329|         >
330|             <Users style={iconStyle('my_class', false)} />
331|             { lang === 'en' ? 'My Class' : 'L Ớ p H í c' }
332|         </Link>
333|     )}
334|
335|     <Link
336|         href="/student/discussion"
337|         style={{ ...navItemStyle('discussion', false), textDecoration: 'none' }}
338|         onMouseEnter={() => setHoveredBtn('discussion')}
339|         onMouseLeave={() => setHoveredBtn(null)}
340|     >
341|         <MessageSquare style={iconStyle('discussion', false)} />
342|         { lang === 'en' ? 'Discussion' : 'Th ỏ l u - n' }
343|     </Link>
344|
345|     <Link
346|         href="/student/level"
347|         style={{ ...navItemStyle('level', false), textDecoration: 'none' }}
348|         onMouseEnter={() => setHoveredBtn('level')}
349|         onMouseLeave={() => setHoveredBtn(null)}
350|     >
351|         <Zap style={iconStyle('level', false)} />
352|         { lang === 'en' ? 'Level & Missions' : 'C ấ p' }
353|     </Link>
354|
355|     <Link
356|         href="/profile"
357|         style={{ ...navItemStyle('profile', false), textDecoration: 'none' }}
358|         onMouseEnter={() => setHoveredBtn('profile')}
359|         onMouseLeave={() => setHoveredBtn(null)}
360|     >
361|         <User style={iconStyle('profile', false)} />
362|         { lang === 'en' ? 'Profile' : 'H ồ S ỉ' }
363|     </Link>
364|
365|     <Link
366|         href="/profile?tab=settings"
367|         style={{ ...navItemStyle('settings', false), textDecoration: 'none' }}
368|         onMouseEnter={() => setHoveredBtn('settings')}
369|         onMouseLeave={() => setHoveredBtn(null)}
370|     >
371|         <Settings style={iconStyle('settings', false)} />
372|         { lang === 'en' ? 'Settings' : 'C à i . t' }
373|     </Link>
374|
375|     <Link href="/about" style={{ textDecoration: 'none' }}>
376|

```

```
377|         <button
378|             onMouseEnter={() => setHoveredBtn('about')} onMouseLeave={() => setHoveredBtn(...)
379|             style={navItemStyle('about', false)}
380|         >
381|             <Users style={iconStyle('about', false)} />
382|             { lang === 'en' ? 'About Us' : 'VÀ chúng
383|         </button>
384|     </Link>
385|
386|     <button
387|         onMouseEnter={() => setHoveredBtn('quiz')} onMouseLeave={() => setHoveredBtn(null)}
388|         onClick={() => onStartQuiz()}
389|         style={navItemStyle('quiz', false)}
390|     >
391|         <BrainCircuit style={iconStyle('quiz', false)} />
392|         { lang === 'en' ? 'AI Quiz' : 'AI Trắc Nghiệ
393|     </button>
394|
395|     <button
396|         onMouseEnter={() => setHoveredBtn('credits')} onMouseLeave={() => setHoveredBtn(nu...
397|         onClick={() => setShowCredits(true)}
398|         style={navItemStyle('credits', false)}
399|     >
400|         <Award style={iconStyle('credits', false)} />
401|         { lang === 'en' ? 'Credits' : 'Tác Giả & Báo
402|     </button>
403| </div>
404| </div>
405|
406| /* [5] FOOTER SIDEBAR */
407| <div style={{ marginTop: 'auto', padding: '12px 16px', borderTop: '1px solid var(--border-subt...
408| <div
409|     onClick={toggleLang}
410|     style={{
411|         display: 'flex', alignItems: 'center', gap: '6px', background: 'var(--bg-elevated)',
412|         padding: '4px 8px', borderRadius: '14px', cursor: 'pointer', height: '28px'
413|     }}
414| >
415|     <Globe size={14} color="var(--text-muted)" />
416|     <span style={{
417|         fontSize: '11px',
418|         fontWeight: lang === 'vn' ? 600 : 400,
419|         color: lang === 'vn' ? 'var(--brand-primary)' : 'var(--text-secondary)'
420|     }}>VN</span>
421|     <div style={{ width: '1px', height: '10px', background: 'var(--border-strong)' }}></div>
422|     <span style={{
423|         fontSize: '11px',
424|         fontWeight: lang === 'en' ? 600 : 400,
425|         color: lang === 'en' ? 'var(--brand-primary)' : 'var(--text-secondary)'
426|     }}>EN</span>
427| </div>
428|
429|     <button
430|         onClick={() => setTheme(theme === 'dark' ? 'editorial-light' : 'dark')}
431|         style={{
432|             background: 'transparent',
433|             border: 'none',
434|             color: 'var(--text-muted)',
435|             padding: '6px',
436|             borderRadius: '6px',
437|             cursor: 'pointer',
438|             display: 'flex',
439|             alignItems: 'center',
440|             justifyContent: 'center',
441|             transition: 'background 0.2s'
442|         }}
443|         onMouseEnter={(e) => { e.currentTarget.style.background = 'var(--bg-hover)'; e.current...
444|         onMouseLeave={(e) => { e.currentTarget.style.background = 'transparent'; e.currentTarg...
445|         title = { lang === 'en' ? 'Toggle Theme' : 'Đổi giao
446|     >
447|         {theme === 'dark' ? <Sun size={16} /> : <Moon size={16} />}
448|     </button>
449|
450|     <button style={{
451|         background: 'transparent',
452|         border: 'none',
453|         color: 'var(--text-muted)',
454|         padding: '6px',
455|         borderRadius: '6px',
456|         cursor: 'pointer',
457|         display: 'flex',
458|         alignItems: 'center',
459|         justifyContent: 'center',
460|         transition: 'background 0.2s'
461|     }}
462|         onMouseEnter={(e) => { e.currentTarget.style.background = 'var(--bg-hover)'; e.currentTarg...
463|         onMouseLeave={(e) => { e.currentTarget.style.background = 'transparent'; e.currentTarget.s...
464|         title = { lang === 'en' ? 'Toggle Effects' : 'B-t / T-t h
465|     >
466|         <Sparkles size={16} />
467|     </button>
468| </div>
469| </div>
470|
471| /* Credits Modal (same mostly, updated colors) */
472| {showCredits && (
473|     <div style={{
474|         position: 'fixed', top: 0, left: 0, width: '100vw', height: '100vh',
475|         background: 'rgba(12, 15, 20, 0.85)', backdropFilter: 'blur(8px)',
476|         zIndex: 9999, display: 'flex', justifyContent: 'center', alignItems: 'center', padding: '2...
477|     }}>
478|         <div style={{
479|             background: 'var(--bg-surface)', border: '1px solid var(--border-strong)', borderRadiu...
480|             padding: '24px', maxWidth: '500px', width: '100%', color: 'var(--text-primary)',
481|             position: 'relative'
482|         }}>
483|             <button
484|                 onClick={() => setShowCredits(false)}
485|                 style={{ position: 'absolute', top: '16px', right: '16px', background: 'transparen...
486|             ><X size={20} /></button>
487|
488|             <h2 style={{ marginTop: 0, fontSize: '24px', textAlign: 'center', marginBottom: '24px'...
489|                 { lang === 'en' ? 'CREDITS' : 'TÁC GIẢ' }
490|             </h2>
491|
492|             <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', fontSize: '14px',...
493|                 <p style={{ margin: 0 }}><strong>{lang === 'en' ? 'Developer:' : 'Developer:'}</st...
494|                 <p style={{ margin: 0 }}><strong>{lang === 'en' ? 'Members:' : 'Thành viên:'}</str...
495|                 <ul style={{ margin: 0, paddingLeft: '24px', display: 'flex', flexDirection: 'colu...
496|                     <li> . ng Qu Ầ c An c / li>
497|                     <li> Ng ư ợ y An Ph ầ m Gi ả K h i ề m < / li>
498|                 </ul>
499|
500|
```



```

501 |         <p style={{ margin: 0 }}><strong>{lang === 'en' ? 'Advisor:' : 'Giáo viên HD:'}</s...
502 |         <p style={{ margin: 0 }}><strong>{lang === 'en' ? 'Co-Advisor:' : 'GVHD:'}</strong>...
503 |         <div style={{ height: '1px', background: 'var(--border-subtle)', margin: '8px 0' }}...
504 |         <p style={{ margin: 0, fontStyle: 'italic', textAlign: 'center', fontSize: '13px' }}>
505 |             {lang === 'en'
506 |               ? "Thank you for using PC Master Builder! We hope this app makes PC buildi...
507 |               : "C ẽ m i n b i n ấ t r ẽ i n g h i ẽ m P C M a s t
508 |         </p>
509 |       </div>
510 |     </div>
511 |   </div>
512 | }
513 | })
514 | { /* Component Library Modal */
515 | {showComponents && (
516 |   <div style={{
517 |     position: 'fixed', top: 0, left: 0, width: '100vw', height: '100vh',
518 |     background: 'rgba(12, 15, 20, 0.85)', backdropFilter: 'blur(8px)',
519 |     zIndex: 9999, display: 'flex', justifyContent: 'center', alignItems: 'center', padding: '2...
520 |   }}>
521 |     <div style={{
522 |       background: 'var(--bg-surface)', border: '1px solid var(--border-strong)', borderRadiu...
523 |       padding: '24px', maxWidth: '900px', width: '100%', color: 'var(--text-primary)',
524 |       position: 'relative', maxHeight: '90vh', overflowY: 'auto'
525 |     }}>
526 |       <button onClick={() => setShowComponents(false)}
527 |         style={{ position: 'absolute', top: '16px', right: '16px', background: 'transparen...
528 |       ><X size={20} /></button>
529 |       <h2 style={{ marginTop: 0, fontSize: '22px', fontWeight: 800, marginBottom: '8px' }}>
530 |         {lang === 'en' ? 'Component Library' : 'T ậ p
531 |       </h2>
532 |       <p style={{ color: 'var(--text-muted)', fontSize: '14px', marginBottom: '24px' }}>
533 |         {lang === 'en' ? 'Browse all components, spe
534 |       </p>
535 |       <div style={{ display: 'flex', gap: '24px', flexWrap: 'wrap', justifyContent: 'center'...
536 |         [['CPU', 'GPU', 'RAM', 'SSD', 'PSU', 'COOLER'].map(type => (
537 |           <div key={type} style={{ flex: '1 1 220px', minWidth: '180px', maxWidth: '280p...
538 |             <ComponentPreview type={type} model={null} size="large" />
539 |           </div>
540 |         ))
541 |       </div>
542 |     </div>
543 |   </div>
544 | })
545 |
546 | <JoinClassModal
547 |   isOpen={showJoinClass}
548 |   onClose={() => setShowJoinClass(false)}
549 |   lang={lang}
550 | />
551 | </>
552 | );
553 | };
554 |
555 | export default BurgerMenu;
556 |

```

File: components/BurgerMenu.tsx (490 lines, 30.8 KB)

```

1 | 'use client';
2 |
3 | import React, { useState, useEffect } from 'react';
4 | import Link from 'next/link';
5 | import { BookOpen, Cpu, ShoppingCart, Users, BrainCircuit, Award, Globe, Sparkles, Menu, Webcam, X, Sun, Moon,...
6 | import JoinClassModal from './JoinClassModal';
7 | import { useRealtime } from '@lib/realtime-provider';
8 |
9 | interface NavItem {
10 |   id: string; labelEn: string; labelVn: string; icon: React.ElementType;
11 |   href?: string; mode?: string; onClick?: () => void; badge?: number;
12 | }
13 |
14 | interface BurgerMenuProps {
15 |   lang: 'en' | 'vn'; toggleLang: () => void; onStartQuiz: () => void;
16 |   appMode: string; setAppMode: (mode: string) => void;
17 |   webcamMouseEnabled?: boolean; setWebcamMouseEnabled?: (enabled: boolean) => void;
18 |   trackingSensitivity?: number; setTrackingSensitivity?: (s: number) => void;
19 |   onToggleAI: () => void; isAIOpen: boolean; theme: string; setTheme: (t: string) => void;
20 |   userName?: string; userAvatar?: string; onShowDashboard?: () => void;
21 | }
22 |
23 | const BurgerMenu: React.FC<BurgerMenuProps> = ({ lang, toggleLang, onStartQuiz, appMode, setAppMode, webcamMou...
24 |   const [isMobileOpen, setIsMobileOpen] = useState(false);
25 |   const [hoveredBtn, setHoveredBtn] = useState<string | null>(null);
26 |   const [showCredits, setShowCredits] = useState(false);
27 |   const [showJoinClass, setShowJoinClass] = useState(false);
28 |   const [hasClass, setHasClass] = useState(false);
29 |   const unreadCount = 0;
30 |
31 |   useEffect(() => {
32 |     const checkClass = async () => {
33 |       try {
34 |         const { createBrowserClient } = require('@supabase/ssr');
35 |         const supabase = createBrowserClient(process.env.NEXT_PUBLIC_SUPABASE_URL!, process.env.NEXT_PUBLIC_SU...
36 |         const { data: { user } } = await supabase.auth.getUser();
37 |         if (user) {
38 |           const { data: profile } = await supabase.from('profiles').select('class_id, school_id').eq('id', use...
39 |           if (profile?.class_id || profile?.school_id) setHasClass(true);
40 |         }
41 |       } catch (e) { console.error(e); }
42 |     };
43 |     checkClass();
44 |   }, []);
45 |
46 |   useEffect(() => { setIsMobileOpen(false); }, [appMode]);
47 |
48 |   const navItemStyle = (id: string, isActive: boolean): React.CSSProperties => ({
49 |     padding: '10px 12px', borderRadius: '8px', display: 'flex', alignItems: 'center', gap: '10px',
50 |     cursor: 'pointer',
51 |     background: isActive ? 'rgba(8,158,96,0.15)' : hoveredBtn === id ? 'rgba(255,255,255,0.06)' : 'transparent',
52 |     border: 'none', borderLeft: isActive ? '3px solid var(--brand-primary)' : '3px solid transparent',
53 |     transition: 'all 150ms ease', width: '100%',
54 |     color: isActive ? '#fff' : hoveredBtn === id ? 'rgba(255,255,255,0.9)' : 'rgba(255,255,255,0.55)',
55 |     fontWeight: isActive ? 700 : 500, fontSize: '13px', textAlign: 'left', fontFamily: 'inherit'
56 |   });
57 |
58 |   const iconStyle = (id: string, isActive: boolean): React.CSSProperties => ({
59 |     color: isActive ? 'var(--brand-primary)' : hoveredBtn === id ? 'rgba(255,255,255,0.9)' : 'rgba(255,255,255...
60 |     width: '18px', height: '18px', transition: 'all 150ms ease', flexShrink: 0
61 |   });
62 |
63 |   const percent = ((trackingSensitivity - 0.5) / 2.5) * 100;
64 |

```

```

65 | return (
66 |   <>
67 |     <button className="mobile-hamburger" onClick={() => setIsMobileOpen(true)}
68 |       style={{ position: 'fixed', top: '16px', left: '16px', width: '40px', height: '40px',
69 |         background: 'rgba(255,255,255,0.08)', backdropFilter: 'blur(8px)', border: '1px solid rgba(255,255,2...
70 |         borderRadius: '10px', zIndex: 100, display: 'flex', alignItems: 'center', justifyContent: 'center',
71 |         color: '#fff', cursor: 'pointer' }}>
72 |       <Menu size={20} />
73 |     </button>
74 |
75 |     <div className="mobile-backdrop"
76 |       style={{ position: 'fixed', top: 0, left: 0, width: '100vw', height: '100vh',
77 |         background: 'rgba(0,0,0,0.5)', zIndex: 180, opacity: isMobileOpen ? 1 : 0,
78 |         pointerEvents: isMobileOpen ? 'auto' : 'none', transition: 'opacity 250ms ease-out' }}
79 |       onClick={() => setIsMobileOpen(false)} />
80 |
81 |     <div className={`sidebar-container ${isMobileOpen ? 'open' : ''}`>
82 |       <div style={{ height: '60px', padding: '0 16px', display: 'flex', alignItems: 'center', flexShrink: 0,...
83 |         <Link href="/" style={{ textDecoration: 'none', display: 'flex', alignItems: 'center', gap: '10px' }}>
84 |           
85 |           <span style={{ fontWeight: 700, fontSize: '16px', color: '#fff' }}>
86 |             PC Master <span style={{ color: 'var(--brand-primary)' }}>Builder</span>
87 |           </span>
88 |         </Link>
89 |         <div style={{ fontSize: '10px', background: 'rgba(8,158,96,0.15)', color: 'var(--brand-primary)', bo...
90 |       </div>
91 |
92 |       <div style={{ padding: '10px 16px', display: 'flex', alignItems: 'center', gap: '10px', borderBottom: ...
93 |         <div style={{ width: '32px', height: '32px', borderRadius: '50%', background: 'linear-gradient(135de...
94 |           {userAvatar ? (
95 |             <img src={userAvatar} alt="" style={{ width: '100%', height: '100%', objectFit: 'cover' }} />
96 |           ) : (
97 |             <User size={14} color="#fff" />
98 |           )}
99 |         </div>
100 |         <div style={{ flex: 1, minWidth: 0 }}>
101 |           <div style={{ fontSize: '13px', fontWeight: 600, color: '#fff', overflow: 'hidden', textOverflow: ...
102 |             <div style={{ padding: '10px', color: 'var(--brand-primary)', fontWeight: 500 }}>{lang === 'en' ?...
103 |           </div>
104 |           <ArrowRight size={14} style={{ color: 'rgba(255,255,255,0.3)', flexShrink: 0 }} />
105 |         </div>
106 |
107 |         <div style={{ height: '1px', background: 'rgba(255,255,255,0.06)', margin: '0 16px', flexShrink: 0 }} />
108 |
109 |         <div style={{ flex: 1, display: 'flex', flexDirection: 'column', paddingBottom: '12px' }}>
110 |           <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
111 |             <div style={{ fontSize: '10px', fontWeight: 700, letterSpacing: '1.5px', textTransform: 'uppercase...
112 |               {lang === 'en' ? 'Explore' : 'Khám phá'}
113 |             </div>
114 |             {[
115 |               { id: 'course', icon: BookOpen, label: lang === 'en' ? 'L
116 |               { id: 'history', icon: History, label: lang === 'en' ? 'L
117 |               { id: 'learning', icon: Cpu, label: lang === 'en' ? 'Pract
118 |               { id: 'market', icon: ShoppingCart, label: lang === 'en' ? 'Pract
119 |               { id: 'exams', icon: Award, label: lang === 'en' ? 'Exams'
120 |               { id: 'multiplayer', icon: Users, label: lang === 'en' ? '
121 |             ]}
122 |           </div>
123 |           <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
124 |             <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
125 |               <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
126 |                 <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
127 |                   <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
128 |                     <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
129 |                       <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
130 |                         <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
131 |                           <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
132 |                             <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
133 |                               <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
134 |                                 <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
135 |                                   <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
136 |                                     <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
137 |                                       <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
138 |                                         <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
139 |                                           <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
140 |                                             <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
141 |                                               <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
142 |                                                 <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
143 |                                                   <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
144 |                                                     <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
145 |                                                       <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
146 |                                                         <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
147 |                                                           <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
148 |                                                             <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
149 |                                                               <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
150 |                                                                 <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
151 |                                                                   <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
152 |                                                                     <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
153 |                                                                       <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
154 |                                                                         <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
155 |                                                                           <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
156 |                                                                             <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
157 |                                                                               <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
158 |                                                                                 <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
159 |                                                                                   <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
160 |                                                                                     <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
161 |                                                                                       <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
162 |                                                                                         <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
163 |                                                                                           <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
164 |                                                                                             <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
165 |                                                                                               <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
166 |                                                                                                 <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
167 |                                                                                                   <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
168 |                                                                                                     <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
169 |                                                                                                       <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
170 |                                                                                                         <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
171 |                                                                                                           <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
172 |                                                                                                             <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
173 |                                                                                                               <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
174 |                                                                                                                 <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
175 |                                                                                                                   <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
176 |                                                                                                                     <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
177 |                                                                                                                       <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
178 |                                                                                                                         <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
179 |                                                                                                                           <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
180 |                                                                                                         <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
181 |                                                                                                           <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
182 |                                                                                                             <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
183 |                                                                                                               <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
184 |                                                                                                                 <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
185 |                                                                                                                   <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
186 |                                                                                                                     <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
187 |                                                                                                                       <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
188 |                                                                                                                         <div style={{ padding: '12px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>

```

```
189 | <div style={{ padding: '8px 16px 4px', display: 'flex', flexDirection: 'column', gap: '2px' }}>
190 |   <div style={{ fontSize: '10px', fontWeight: 700, letterSpacing: '1.5px', textTransform: 'uppercase...'
1 |   { lang === 'en' ? 'Extras' : 'Tinh nang khac' }}
192 |   </div>
193 |   {!hasClass ? (
194 |     <button onMouseEnter={() => setHoveredBtn('join_class')} onMouseLeave={() => setHoveredBtn(null)}
195 |       onClick={() => setShowJoinClass(true)} style={navItemStyle('join_class', false)}>
196 |       <Users style={iconStyle('join_class', false)} />
1 |       { lang === 'en' ? 'Join Class' : 'Tham gia lop huc' }}
198 |     </button>
199 |   ) : (
200 |     <Link href="/student/classes" style={{ textDecoration: 'none' }}
201 |       onMouseEnter={() => setHoveredBtn('my_class')} onMouseLeave={() => setHoveredBtn(null)}>
202 |       <div style={navItemStyle('my_class', false)}>
203 |         <Users style={iconStyle('my_class', false)} />
2 |         { lang === 'en' ? 'My Class' : 'Lop Huc Cua Toi' }}
205 |       </div>
206 |     </Link>
207 |   )}
208 |   {[
2 |     { id: 'discussion', icon: MessageSquare, label: lang === 'en' ? 'Profile' : 'About' },
2 |     { id: 'profile', icon: User, label: lang === 'en' ? 'Profile' : 'About' },
2 |     { id: 'about', icon: Users, label: lang === 'en' ? 'About' : 'About' }
212 |   ].map(item => (
213 |     <Link key={item.id} href={item.href} style={{ textDecoration: 'none' }}
214 |       onMouseEnter={() => setHoveredBtn(item.id)} onMouseLeave={() => setHoveredBtn(null)}>
215 |       <div style={navItemStyle(item.id, false)}>
216 |         <item.icon style={iconStyle(item.id, false)} />
217 |         {item.label}
218 |       </div>
219 |     </Link>
220 |   ))}
221 |   <button onMouseEnter={() => setHoveredBtn('quiz')} onMouseLeave={() => setHoveredBtn(null)}
222 |     onClick={() => onStartQuiz()} style={navItemStyle('quiz', false)}>
223 |     <BrainCircuit style={iconStyle('quiz', false)} />
2 |     { lang === 'en' ? 'AI Quiz' : 'AI Trc Nghiem' }}
225 |   </button>
226 |   <button onMouseEnter={() => setHoveredBtn('credits')} onMouseLeave={() => setHoveredBtn(null)}
227 |     onClick={() => setShowCredits(true)} style={navItemStyle('credits', false)}>
228 |     <Award style={iconStyle('credits', false)} />
2 |     { lang === 'en' ? 'Credits' : 'Tac GiE & Bao Cao' }}
230 |   </button>
231 | </div>
232 | </div>
233 |
234 | <div style={{ padding: '12px 16px', borderTop: '1px solid rgba(255,255,255,0.06)', display: 'flex', ju...
235 |   <div onClick={toggleLang} style={{ display: 'flex', alignItems: 'center', gap: '6px', background: 'r...
236 |     <Globe size={14} style={{ color: 'rgba(255,255,255,0.4)' }} />
237 |     <span style={{ fontSize: '11px', fontWeight: lang === 'vn' ? 700 : 400, color: lang === 'vn' ? 'va...
238 |     <div style={{ width: '1px', height: '10px', background: 'rgba(255,255,255,0.15)' }} />
239 |     <span style={{ fontSize: '11px', fontWeight: lang === 'en' ? 700 : 400, color: lang === 'en' ? 'va...
240 |   </div>
241 |   <div style={{ display: 'flex', gap: '4px' }}>
242 |     <button onClick={() => { const next = theme === 'dark' ? 'light' : 'dark'; setTheme(next); try { l...
243 |       style={{ background: 'transparent', border: 'none', color: 'rgba(255,255,255,0.4)', padding: '6p...
2 |       title = { lang === 'en' ? 'Toggle Theme' : 'oi giao dien' }}
245 |     {theme === 'dark' ? <Sun size={16} /> : <Moon size={16} />}
246 |   </button>
247 |   <button style={{ background: 'transparent', border: 'none', color: 'rgba(255,255,255,0.4)', paddin...
2 |   title = { lang === 'en' ? 'Toggle Effects' : 'B-t / T-t hieu e' }}
249 |     <Sparkles size={16} />
250 |   </button>
251 | </div>
252 | </div>
253 | </div>
254 |
255 | {showCredits && (
256 |   <div style={{ position: 'fixed', top: 0, left: 0, width: '100vw', height: '100vh',
257 |     background: 'rgba(0,0,0,0.7)', backdropFilter: 'blur(12px)', zIndex: 9999,
258 |     display: 'flex', justifyContent: 'center', alignItems: 'center', padding: '1rem',
259 |     animation: 'creditsFadeIn 0.3s ease' }}>
260 |     <div style={{
261 |       background: 'linear-gradient(145deg, #0f1729 0%, #1a2a4a 50%, #0d1f3c 100%)',
262 |       borderRadius: '24px', padding: '36px', maxWidth: '520px', width: '100%',
263 |       color: '#fff', position: 'relative',
264 |       border: '1px solid rgba(0,212,170,0.15)',
265 |       boxShadow: '0 0 60px rgba(0,212,170,0.08), 0 20px 60px rgba(0,0,0,0.4)',
266 |       overflow: 'hidden',
267 |     }}>
268 |       <div style={{ position: 'absolute', top: '-60px', right: '-60px', width: '180px', height: '180px',
269 |         borderRadius: '50%', background: 'radial-gradient(circle, rgba(0,212,170,0.06) 0%, transparent 7...
270 |         pointerEvents: 'none' }} />
271 |       <div style={{ position: 'absolute', bottom: '-40px', left: '-40px', width: '140px', height: '140px',
272 |         borderRadius: '50%', background: 'radial-gradient(circle, rgba(0,243,255,0.04) 0%, transparent 7...
273 |         pointerEvents: 'none' }} />
274 |
275 |       <button onClick={() => setShowCredits(false)}
276 |         style={{ position: 'absolute', top: '16px', right: '16px', background: 'rgba(255,255,255,0.06)',
277 |         border: '1px solid rgba(255,255,255,0.08)', color: 'rgba(255,255,255,0.5)', cursor: 'pointer',
278 |         width: '32px', height: '32px', borderRadius: '10px', display: 'flex', alignItems: 'center',
279 |         justifyContent: 'center', transition: 'all 0.2s', zIndex: 1 }}
280 |         onMouseEnter={e => { e.currentTarget.style.backgroundColor = 'rgba(255,255,255,0.1)'; e.currentTarget...
281 |         onMouseLeave={e => { e.currentTarget.style.backgroundColor = 'rgba(255,255,255,0.06)'; e.currentTarge...
282 |         <X size={16} />
283 |       </button>
284 |
285 |       <div style={{ textAlign: 'center', marginBottom: '28px', position: 'relative', zIndex: 1 }}>
286 |         <div style={{ width: '56px', height: '56px', borderRadius: '16px',
287 |           background: 'linear-gradient(135deg, #00d4aa, #00f3ff)',
288 |           margin: '0 auto 12px', display: 'flex', alignItems: 'center', justifyContent: 'center',
289 |           fontSize: '24px', boxShadow: '0 0 30px rgba(0,212,170,0.2)' }}>
2 |           0x00d4aa
291 |         </div>
292 |         <h2 style={{ margin: '0 0 4px', fontSize: '20px', fontWeight: 800, letterSpacing: '0.5px' }}>
2 |           { lang === 'en' ? 'CREDITS' : 'TAC GI E & BAO CAO' }}
294 |       </h2>
295 |       <div style={{ width: '40px', height: '2px', background: 'linear-gradient(90deg, #00d4aa, #00f3ff...'
296 |       <p style={{ margin: 0, fontSize: '11px', color: 'rgba(255,255,255,0.35)', letterSpacing: '0.5px'...
297 |         PC Master Builder - v1.0.0
298 |     </p>
299 |   </div>
300 |
301 |   <div style={{ display: 'flex', flexDirection: 'column', gap: '6px', position: 'relative', zIndex: ...
302 |     <div style={{ padding: '12px 16px', borderRadius: '14px',
303 |       background: 'rgba(255,255,255,0.03)', border: '1px solid rgba(255,255,255,0.06)' }}>
304 |       <div style={{ fontSize: '10px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: '1...
305 |         color: 'var(--brand-primary)' }}
3 |         { lang === 'en' ? 'Project Info' : 'Thong tin dñ an' }}
307 |     </div>
308 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '6px', fontSize: '13px' }}>
309 |     {[
3 |     { icon: '0=0æ', label: lang === 'en' ? 'Project' : 'Dñ án' },
3 |     { icon: '0=0h 0=0', label: lang === 'en' ? 'Developer' : 'Phát triển' },
3 |     { icon: '0=pap', label: lang === 'en' ? 'Version' : 'Phiên bản' }
310 |   ]}
311 | ]}
312 | ]}
```

```
313 |         ].map((item, i) => (  
314 |             <div key={i} style={{ display: 'flex', alignItems: 'center', gap: '8px' }}>  
315 |                 <span style={{ fontSize: '14px' }}>{item.icon}</span>  
316 |                 <span style={{ color: 'rgba(255,255,255,0.4)', minWidth: '80px', fontSize: '11px' }}>{it...  
317 |                 <span style={{ color: '#fff', fontWeight: 600, fontSize: '12px' }}>{item.value}</span>  
318 |             </div>  
319 |         ))  
320 |     </div>  
321 | </div>  
322 |  
323 |     <div style={{ padding: '12px 16px', borderRadius: '14px',  
324 |         background: 'rgba(0,212,170,0.04)', border: '1px solid rgba(0,212,170,0.1)' }}>  
325 |         <div style={{ fontSize: '10px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: '1...  
326 |             color: '#00d4aa', marginBottom: '8px' }}>  
327 |             { lang === 'en' ? 'Supervisors' : 'Giáo viên h° Òng d« n  
328 |         </div>  
329 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '6px' }}>  
330 |             {  
331 |                 { name: 'Tr S n Minh Ph ăng', role: lang === 'en' ? 'L  
332 |                 { name: 'o ần Th ầy Kim Ph ăng', role: lang === 'en'  
333 |             ].map((gv, i) => (  
334 |                 <div key={i} style={{ display: 'flex', alignItems: 'center', gap: '10px', padding: '6px 10...  
335 |                 borderRadius: '10px', background: 'rgba(255,255,255,0.03)' }}>  
336 |                 <div style={{ width: '32px', height: '32px', borderRadius: '10px',  
337 |                     background: `linear-gradient(135deg, ${gv.color}22, ${gv.color}11)`,  
338 |                     border: '1px solid ${gv.color}33',  
339 |                     display: 'flex', alignItems: 'center', justifyContent: 'center', fontSize: '14px', fle...  
340 |                     0=0i 0<Bè  
341 |                 </div>  
342 |                 <div>  
343 |                     <div style={{ fontSize: '13px', fontWeight: 700, color: '#fff' }}>{gv.name}</div>  
344 |                     <div style={{ fontSize: '10px', color: gv.color, fontWeight: 600, letterSpacing: '0.5p...  
345 |                     </div>  
346 |                 </div>  
347 |             ))  
348 |         </div>  
349 |     </div>  
350 |  
351 |     <div style={{ padding: '12px 16px', borderRadius: '14px',  
352 |         background: 'rgba(99,102,241,0.04)', border: '1px solid rgba(99,102,241,0.1)' }}>  
353 |         <div style={{ fontSize: '10px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: '1...  
354 |             color: '#818cf8', marginBottom: '8px' }}>  
355 |             {lang === 'en' ? 'Team Members' : 'Thành viên'}  
356 |         </div>  
357 |         <div style={{ display: 'flex', gap: '8px', flexWrap: 'wrap' }}>  
358 |             {  
359 |                 { name: 'ng Qu ãn c An', role: 'Frontend', color: '#8  
360 |                 { name: 'N guy ần Ph im Gia Khiêm', role: 'Backend', c  
361 |             { name: 'Ngô Minh Khang', role: 'UI/UX', color: '#10b981' },  
362 |             ].map((m, i) => (  
363 |                 <div key={i} style={{ flex: '1 1 auto', minWidth: '120px', padding: '8px 12px', borderRadi...  
364 |                 background: 'rgba(255,255,255,0.03)', border: '1px solid rgba(255,255,255,0.05)',  
365 |                 textAlign: 'center' }}>  
366 |                 <div style={{ fontSize: '12px', fontWeight: 700, color: '#fff' }}>{m.name}</div>  
367 |                 <div style={{ fontSize: '9px', color: m.color, fontWeight: 600, letterSpacing: '0.5px' }}...  
368 |                 </div>  
369 |             ))  
370 |         </div>  
371 |     </div>  
372 |  
373 |     <div style={{ padding: '12px 16px', borderRadius: '14px',  
374 |         background: 'rgba(16,185,129,0.04)', border: '1px solid rgba(16,185,129,0.1)' }}>  
375 |         <div style={{ fontSize: '10px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: '1...  
376 |             color: '#10b981', marginBottom: '8px' }}>  
377 |             { lang === 'en' ? 'Features' : 'Tính n ng' }  
378 |         </div>  
379 |         <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '4px', fontSize: '11px' }}>  
380 |             {  
381 |                 '0<B@ L - p r ấ p PC 2 D', '0>Ý AI H ° Òng d« n', '0=00 B à i g i ẽ n c  
382 |                 ' ' Theo d ộ i t ảy', '0<Bè Ch ấ l i n h k i Ç n', '0=0e 2 Ng ° Y i c h  
383 |                 '0<BÆ H Ç t h ã n g X P', '0=0E B ẽ n g x ỏ p h i n g', '0=00 Ch ẽ n g ch ẽ  
384 |                 '0=0ñ R e s p o n s i v e', '0<B a n g ò n n g i',  
385 |             ].map((feat, i) => (  
386 |                 <div key={i} style={{ padding: '4px 6px', borderRadius: '6px',  
387 |                 background: 'rgba(255,255,255,0.02)', display: 'flex', alignItems: 'center', gap: '4px' }}>  
388 |                 <span style={{ fontSize: '10px', lineHeight: '1' }}>{feat}</span>  
389 |                 </div>  
390 |             ))  
391 |         </div>  
392 |     </div>  
393 |  
394 |     <div style={{ padding: '12px 16px', borderRadius: '14px',  
395 |         background: 'rgba(245,158,11,0.04)', border: '1px solid rgba(245,158,11,0.1)' }}>  
396 |         <div style={{ fontSize: '10px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: '1...  
397 |             color: '#f59e0b', marginBottom: '8px' }}>  
398 |             Tech Stack  
399 |         </div>  
400 |         <div style={{ display: 'flex', flexWrap: 'wrap', gap: '4px' }}>  
401 |             {['Next.js 16', 'React 19', 'TypeScript', 'Supabase', 'Three.js', 'Framer Motion', 'Tailwind...  
402 |             <span key={i} style={{ padding: '3px 8px', borderRadius: '6px', fontSize: '9px', fontWeigh...  
403 |                 background: 'rgba(255,255,255,0.05)', border: '1px solid rgba(255,255,255,0.08)',  
404 |                 color: 'rgba(255,255,255,0.6)' }}>  
405 |                 {tech}  
406 |             </span>  
407 |             </div>  
408 |         </div>  
409 |     </div>  
410 | </div>  
411 |  
412 |     <div style={{ marginTop: '20px', textAlign: 'center', position: 'relative', zIndex: 1 }}>  
413 |         <div style={{ padding: '12px', borderRadius: '12px',  
414 |             background: 'linear-gradient(135deg, rgba(0,212,170,0.06), rgba(0,243,255,0.06))',  
415 |             border: '1px solid rgba(0,212,170,0.12)' }}>  
416 |             <p style={{ margin: '0 0 4px', fontSize: '11px', color: 'rgba(255,255,255,0.4)' }}>  
417 |                 @ ( new Date() . get F u l l Y e a r ( ) ) - T H P T N guy ần C ồ n g T r ẻ  
418 |             </p>  
419 |             <p style={{ margin: 0, fontSize: '10px', color: 'rgba(255,255,255,0.25)', fontStyle: 'italic' }}>  
420 |                 { lang === 'en' ? '"Dare to Think, Dare to Do"' : '"Dám  
421 |             </p>  
422 |         </div>  
423 |     </div>  
424 | </div>  
425 | </div>  
426 | )}  
427 | <JoinClassModal isOpen={showJoinClass} onClose={() => setShowJoinClass(false)} lang={lang} />  
428 | </>  
429 | );  
430 | };  
431 |  
432 | function QuestWidgetSection({ lang }: { lang: 'en' | 'vn' }) {  
433 |     const { state: realtimeState } = useRealtime()  
434 |     const quests = realtimeState.allQuests.slice(0, 3)  
435 |     const userQuests = realtimeState.quests  
436 | }
```

```

437 |   return (
438 |     <div style={{ padding: '4px 16px 8px', borderBottom: '1px solid rgba(255,255,255,0.06)' }}>
439 |       <div style={{ display: 'flex', alignItems: 'center', gap: '6px', marginBottom: '8px', marginTop: '4px' }}>
440 |         < Trophy size={12} style={{ color: '#f59e0b' }} />
441 |         <span style={{ fontSize: '10px', fontWeight: 700, letterSpacing: '1px', textTransform: 'uppercase', co...
442 |           { lang === 'en' ? 'Daily Quests' : 'Nhiệm Vụ Hôm Nay' }
443 |         </span>
444 |         <div style={{ marginLeft: 'auto', fontSize: '9px', color: 'var(--brand-primary)', fontWeight: 600 }}>
445 |           Lv.{realtimeState.level}
446 |         </div>
447 |       </div>
448 |       {quests.map((quest, i) => {
449 |         const userQuest = userQuests.find(q => q.quest_id === quest.id)
450 |         const isCompleted = userQuest?.is_completed
451 |         const progress = userQuest?.progress || 0
452 |         return (
453 |           <div key={quest.id} style={{
454 |             display: 'flex', alignItems: 'center', gap: '8px',
455 |             padding: '6px 8px', borderRadius: '8px',
456 |             background: isCompleted ? 'rgba(16,185,129,0.08)' : 'rgba(255,255,255,0.03)',
457 |             marginBottom: '4px', opacity: isCompleted ? 0.6 : 1
458 |           }}>
459 |             <span style={{ fontSize: '14px' }}>{quest.icon || '0=0' }</span>
460 |             <div style={{ flex: 1, minWidth: 0 }}>
461 |               <div style={{ fontSize: '11px', fontWeight: 600, color: isCompleted ? '#10b981' : 'rgba(255,255,...
462 |                 {quest.title}
463 |               </div>
464 |               <div style={{ display: 'flex', alignItems: 'center', gap: '6px' }}>
465 |                 <span style={{ fontSize: '9px', color: 'var(--brand-primary)', fontWeight: 600 }}>{quest.xp_r...
466 |                 {isCompleted && quest.requirement_value > 0 && (
467 |                   <div style={{ flex: 1, height: '2px', borderRadius: '99px', background: 'rgba(255,255,255,0....
468 |                     <div style={{ height: '100%', borderRadius: '99px', background: 'var(--brand-primary)', wi...
469 |                   </div>
470 |                 </div>
471 |                 {isCompleted && <CheckCircle2 size={10} style={{ color: '#10b981' }} />}
472 |               </div>
473 |             </div>
474 |           </div>
475 |         )
476 |       )}
477 |       <Link href="/student/dashboard" style={{
478 |         display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '4px',
479 |         padding: '6px', borderRadius: '6px', fontSize: '9px', fontWeight: 600,
480 |         color: 'var(--brand-primary)', textDecoration: 'none', marginTop: '4px',
481 |         background: 'rgba(8,158,96,0.08)'
482 |       }}>
483 |         < Zap size={10} /> { lang === 'en' ? 'View All Tasks' : 'Xem tất c
484 |       </Link>
485 |     </div>
486 |   )
487 | }
488 |
489 | export default BurgerMenu;
490 |

```

File: components/CareerRecommendation.tsx (274 lines, 14.4 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect, useMemo } from 'react'
4 | import { motion } from 'framer-motion'
5 | import { createBrowserClient } from '@supabase/ssr'
6 | import { QUIZ_BANK } from '@data/quiz-bank'
7 | import { Compass, Externallink, Youtube, Sparkles, Cpu, Monitor, Server, Wrench, Gamepad2, BarChart3 } from 'l...
8 |
9 | const CAREERS = [
10 |   {
11 |     id: 'software-engineer',
12 |     title: 'Kù s ò ph ã n m Á m ',
13 |     title_en: 'Software Engineer',
14 |     icon: Monitor,
15 |     color: '#6366f1',
16 |     avgSalary: '30 - 80 tri Ç u / th á n g ',
17 |     description: 'Ph á t tri ĩ n ĩ n g đ á n g , website , AI và các gi ĩ i ph á p công
18 |     relatedTopics: ['Ki ĩ n Tr ú c CPU & T - p L Ç n h ', 'AI & Công Ngh ĩ M ũ i ', 'M
19 |     matchTopics: ['quiz-cpu-arch', 'quiz-ai-tech', 'quiz-network', 'quiz-bios', 'quiz-benchmark'],
20 |     youtubeLinks: [
21 |       { title: 'C á c h tr ã h à n h Software Engineer', url: 'https://www.you
22 |       { title: 'L ũ tr ì n h h ĩ c l - p tr ì n h ', url: 'https://www.youtube.com/r
23 |     ],
24 |   },
25 |   {
26 |     id: 'hardware-engineer',
27 |     title: 'Kù s ò ph ã n c é n g ',
28 |     title_en: 'Hardware Engineer',
29 |     icon: Cpu,
30 |     color: '#00d4aa',
31 |     avgSalary: '25 - 60 tri Ç u / th á n g ',
32 |     description: 'Th ĩ ĩ t k ĩ và l - p r á p và t Ñ i ò u h ĩ th ĩ n g má y t ĩ n h , v ĩ x ĩ
33 |     relatedTopics: ['Quy ĩ Tr ì n h l - p R á p PC ', 'É p X u n g & T u n i n g ', 'H ĩ Th ĩ n
34 |     matchTopics: ['quiz-assembly', 'quiz-overclocking', 'quiz-cooling-advanced', 'quiz-troubleshoot', 'quiz-ps...
35 |     youtubeLinks: [
36 |       { title: 'PC Building Guide 2025', url: 'https://www.youtube.com/results?search_query=pc+building+guide+...
37 |       { title: 'H ò ũ n g đ ã n l - p r á p PC ', url: 'https://www.youtube.com/res
38 |     ],
39 |   },
40 |   {
41 |     id: 'system-engineer',
42 |     title: 'Kù s ò h ĩ th ĩ n g ',
43 |     title_en: 'System Engineer',
44 |     icon: Server,
45 |     color: '#289cf9',
46 |     avgSalary: '35 - 70 tri Ç u / th á n g ',
47 |     description: 'Qu ĩ n tr ĩ và v - n h à n h h ĩ th ĩ n g má y ch ĩ , m ĩ n g và c ĩ s ã h
48 |     relatedTopics: ['M á y Ch ĩ & Workstation', 'BIOS & UEFI', 'M ĩ n g Má y T ĩ
49 |     matchTopics: ['quiz-server', 'quiz-bios', 'quiz-network', 'quiz-troubleshoot'],
50 |     youtubeLinks: [
51 |       { title: 'H ĩ c qu ĩ n tr ĩ h ĩ th ĩ n g ', url: 'https://www.youtube.com/re
52 |       { title: 'System Engineer Career', url: 'https://www.youtube.com/results?search_query=system+engineer+ca...
53 |     ],
54 |   },
55 |   {
56 |     id: 'game-developer',
57 |     title: 'N h à ph á t tri ĩ n g a m e ',
58 |     title_en: 'Game Developer',
59 |     icon: Gamepad2,
60 |     color: '#ef4444',
61 |     avgSalary: '25 - 70 tri Ç u / th á n g ',
62 |     description: 'Th ĩ ĩ t k ĩ và l - p tr ì n h game , t Ñ i ò u ó h ĩ a và h ĩ ç u n ĩ n g
63 |     relatedTopics: ['Card Ó H ĩ a , & X ĩ L ý Ó H ĩ a ', 'C ĩ u H ĩ n h PC G a m ĩ n g ',
64 |     matchTopics: ['quiz-gpu-advanced', 'quiz-pc-gaming', 'quiz-vr-ar', 'quiz-monitor', 'quiz-benchmark'],
65 |     youtubeLinks: [
66 |       { title: 'H ĩ c l à m game t ĩ A - Z ', url: 'https://www.youtube.com/resu

```

```

67 |     { title: 'Game Developer Roadmap', url: 'https://www.youtube.com/results?search_query=game+developer+roa...
68 |   }
69 | },
70 | {
71 |   id: 'tech-support',
72 |   title: 'Chuyên viên kỹ thuật',
73 |   title_en: 'Technical Support Specialist',
74 |   icon: Wrench,
75 |   color: '#f59e0b',
76 |   avgSalary: '15 - 35 triệu/tháng',
77 |   description: 'Hỗ trợ kỹ thuật, sửa chữa và bảo trì hệ thống máy tính',
78 |   relatedTopics: ['Xí lý Săn Cắn & Chôn oan', 'Thiэт BẾ Ngoi i Vi', 'Cá p
79 | matchTopics: ['quiz-troubleshoot', 'quiz-peripheral', 'quiz-cable', 'quiz-laptop', 'quiz-bios'],
80 | youtubeLinks: [
81 |   { title: 'IT Support Guide', url: 'https://www.youtube.com/results?search_query=it+support+specialist+gu...
82 | },
83 | ],
84 | },
85 | ]
86 |
87 | export default function CareerRecommendation({ lang = 'vn' }) {
88 |   const [scores, setScores] = useState<Record<string, number>>({})
89 |   const [loading, setLoading] = useState(true)
90 |   const [realtimeKey, setRealtimeKey] = useState(0)
91 |   const supabase = createBrowserClient(
92 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
93 |     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
94 |   )
95 |
96 |   useEffect(() => {
97 |     fetchScores()
98 |   }, [realtimeKey])
99 |
100 |   useEffect(() => {
101 |     const channel = supabase
102 |       .channel('career_realtime')
103 |       .on('postgres_changes', { event: 'INSERT', schema: 'public', table: 'quiz_attempts' }, async (payload) => {
104 |         const { data: { user } } = await supabase.auth.getUser()
105 |         if (user && (payload.new as any).user_id === user.id) {
106 |           setRealtimeKey(k => k + 1)
107 |         }
108 |       })
109 |       .subscribe()
110 |     return () => { supabase.removeChannel(channel) }
111 |   }, [])
112 |
113 |   async function fetchScores() {
114 |     try {
115 |       const { data: { user } } = await supabase.auth.getUser()
116 |       if (!user) { setLoading(false); return }
117 |
118 |       const { data: attempts } = await supabase
119 |         .from('quiz_attempts')
120 |         .select('quiz_id, score, total_questions')
121 |         .eq('user_id', user.id)
122 |
123 |       const scoreMap: Record<string, number> = {}
124 |       attempts?.forEach(a => {
125 |         if (a.total_questions > 0) {
126 |           const pct = Math.round((a.score / (a.total_questions * 10)) * 100)
127 |           scoreMap[a.quiz_id] = Math.max(pct, scoreMap[a.quiz_id] || 0)
128 |         } else if (a.score > 0) {
129 |           scoreMap[a.quiz_id] = Math.min(a.score, 100)
130 |         }
131 |       })
132 |       setScores(scoreMap)
133 |     } catch (err) {
134 |       console.error('Error fetching quiz scores:', err)
135 |     } finally {
136 |       setLoading(false)
137 |     }
138 |   }
139 |
140 |   const rankings = useMemo(() => {
141 |     return CAREERS.map(career => {
142 |       const topicScores = career.matchTopics.map(topicId => scores[topicId] || 0)
143 |       const completed = topicScores.filter(s => s > 0).length
144 |       const avgScore = topicScores.length > 0
145 |         ? Math.round(topicScores.reduce((a, b) => a + b, 0) / topicScores.length)
146 |         : 0
147 |       const matchPercent = completed > 0
148 |         ? Math.round((completed / career.matchTopics.length) * 100)
149 |         : 0
150 |       return { ...career, avgScore, matchPercent, completed }
151 |     }).sort((a, b) => b.avgScore - a.avgScore || b.matchPercent - a.matchPercent)
152 |   }, [scores])
153 |
154 |   const topCareer = rankings[0]
155 |
156 |   if (loading) return null
157 |
158 |   return (
159 |     <motion.div initial={{ opacity: 0, y: 24 }} animate={{ opacity: 1, y: 0 }}
160 |     className="lms-card" style={{ padding: '28px', borderRadius: '16px', margin: '24px 0' }}>
161 |     <div style={{ display: 'flex', alignItems: 'center', gap: '12px', margin: '20px 0' }}>
162 |       <div style={{ width: '40px', height: '40px', borderRadius: '10px', background: 'rgba(99,102,241,0.1)',...
163 |       </div>
164 |       <div>
165 |         <h2 style={{ fontSize: '18px', fontWeight: 700, color: 'var(--text-primary)', margin: 0 }}>
166 |           { lang === 'vn' ? 'Enh h° Õng ngh Á nghi Çp' : 'Career Path' }
167 |         </h2>
168 |         <p style={{ fontSize: '13px', color: 'var(--text-muted)', margin: '2px 0 0 0' }}>
169 |           { lang === 'vn' ? 'Đĩa trên kút qu£ bài quiz cça bin' : 'Base
170 |         </p>
171 |       </div>
172 |     </div>
173 |
174 |     {topCareer && topCareer.avgScore > 0 ? (
175 |       <
176 |         <motion.div initial={{ scale: 0.95 }} animate={{ scale: 1 }}
177 |         style={{ background: 'linear-gradient(135deg, ${topCareer.color}15, ${topCareer.color}05)', border...
178 |         <div style={{ display: 'flex', alignItems: 'flex-start', gap: '16px' }}>
179 |           <div style={{ width: '52px', height: '52px', borderRadius: '14px', background: `${topCareer.col...
180 |           <div style={{ width: '52px', height: '52px', borderRadius: '14px', background: `${topCareer.col...
181 |           </div>
182 |           <div style={{ flex: 1 }}>
183 |             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', flexWrap...
184 |               <h3 style={{ fontSize: '18px', fontWeight: 800, color: 'var(--text-primary)', margin: 0 }}>
185 |                 { lang === 'vn' ? topCareer.title : topCareer.title_en }
186 |               </h3>
187 |               <span style={{ fontSize: '12px', padding: '4px 12px', borderRadius: '100px', background: `${{...
188 |                 { topCareer.avgScore } % phũ hã p
189 |             </div>
190 |           </div>

```

```
191|         </div>
192|         <p style={{ fontSize: '13px', color: 'var(--text-muted)', margin: '8px 0', lineHeight: 1.5 }}>
193|             {topCareer.description}
194|         </p>
195|         <div style={{ fontSize: '12px', color: 'var(--text-secondary)' }}>
196|             <span style={{ fontWeight: 600 }}>Méc l°ing:</span> {
197|         </div>
198|     </div>
199| </div>
200|
201|     <div style={{ marginTop: '16px' }}>
202|         <div style={{ fontSize: '12px', fontWeight: 600, color: 'var(--text-secondary)', marginBottom: '...
203|             {lang === 'vn' ? '0=00 Ki ỉn th ẻc li ẻn quan : ' : '0=00 Related
204|         </div>
205|         <div style={{ display: 'flex', flexWrap: 'wrap', gap: '6px' }}>
206|             {topCareer.relatedTopics.map(topic => (
207|                 <span key={topic} style={{ fontSize: '11px', padding: '4px 10px', borderRadius: '6px', backg...
208|                     {topic}
209|                 </span>
210|             ))}
211|         </div>
212|     </div>
213|
214|     <div style={{ marginTop: '16px', padding: '16px', borderTop: '1px solid var(--border-subtle)' }}>
215|         <div style={{ fontSize: '12px', fontWeight: 600, color: 'var(--text-secondary)', marginBottom: '...
216|             <Youtube size={14} color="#ff0000" />
217|             {lang === 'vn' ? 'Video g ỉi ý : ' : 'Suggested videos : ' }
218|         </div>
219|         <div style={{ display: 'flex', flexDirection: 'column', gap: '6px' }}>
220|             {topCareer.youtubeLinks.map((link, i) => (
221|                 <a key={i} href={link.url} target="_blank" rel="noopener noreferrer"
222|                     style={{ display: 'flex', alignItems: 'center', gap: '8px', padding: '8px 12px', borderRad...
223|                     onMouseOver={e => { e.currentTarget.style.backgroundColor = 'var(--bg-hover)'; e.currentTarget...
224|                     onMouseOut={e => { e.currentTarget.style.backgroundColor = 'var(--bg-elevated)'; e.currentTarget...
225|                     <Youtube size={14} color="#ff0000" />
226|                     <span style={{ flex: 1 }}>{link.title}</span>
227|                     <ExternalLink size={12} style={{ opacity: 0.5 }} />
228|                 </a>
229|             ))}
230|         </div>
231|     </div>
232| </motion.div>
233|
234| <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
235|     <div style={{ fontSize: '12px', fontWeight: 600, color: 'var(--text-muted)', marginBottom: '4px' }}>
236|         {lang === 'vn' ? 'C ỉc ngh ỉ nghi ẻp kh ỉc ph ỉ h ỉp v ỉi b ỉn : ' :
237|     </div>
238|     {rankings.slice(1, 4).map((career, i) => (
239|         <motion.div key={career.id} initial={{ opacity: 0, x: -10 }} animate={{ opacity: 1, x: 0 }} tran...
240|         style={{ display: 'flex', alignItems: 'center', gap: '12px', padding: '12px', borderRadius: '1...
241|         <div style={{ width: '36px', height: '36px', borderRadius: '8px', background: `${career.color}...
242|             <career.icon size={18} color={career.color} />
243|         </div>
244|         <div style={{ flex: 1 }}>
245|             <div style={{ fontSize: '13px', fontWeight: 700, color: 'var(--text-primary)' }}>
246|                 {lang === 'vn' ? career.title : career.title_en}
247|             </div>
248|             <div style={{ display: 'flex', gap: '8px', fontSize: '11px', color: 'var(--text-muted)', mar...
249|                 <span>{career.avgScore}% ph ỉ h ỉp</span>
250|                 <span>{career.avgSalary}</span>
251|             </div>
252|         </div>
253|         <div style={{ width: '40px', height: '4px', borderRadius: '4px', background: 'var(--border-sub...
254|             <div style={{ width: `${career.avgScore}%`, height: '100%', background: career.color, border...
255|         </div>
256|     </motion.div>
257|     ))}
258| </div>
259| </div>
260|
261| ) : (
262|     <div style={{ textAlign: 'center', padding: '32px', border: '1px dashed var(--border-default)', border...
263|         <BarChart3 size={40} style={{ color: 'var(--text-muted)', marginBottom: '12px', opacity: 0.5 }} />
264|         <p style={{ color: 'var(--text-muted)', fontSize: '13px', margin: 0 }}>
265|             {lang === 'vn'
266|                 ? 'Ho ỉn th ỉnh c ỉc b ỉi quiz ỉ nh - n ẻnh h ỉ ẻng ngh ỉ nghi ẻp
267|                 : 'Complete quizzes to get your personalized career recommendation!' }
268|         </p>
269|     </div>
270| )}
271| </motion.div>
272| )
273| }
274|
```

File: components/ChatbaseScript.tsx (33 lines, 0.9 KB)

```
1| 'use client';
2| import { useEffect } from 'react';
3|
4| declare global {
5|     interface Window {
6|         chatbase?: any;
7|     }
8| }
9|
10| export default function ChatbaseScript() {
11|     useEffect(() => {
12|         if (window.chatbase && window.chatbase('getState') === 'initialized') return;
13|         window.chatbase = (...args: any[]) => {
14|             if (!window.chatbase.q) window.chatbase.q = [];
15|             window.chatbase.q.push(args);
16|         };
17|         window.chatbase = new Proxy(window.chatbase, {
18|             get(target: any, prop: string) {
19|                 if (prop === 'q') return target.q;
20|                 return (...args: any[]) => target(prop, ...args);
21|             }
22|         });
23|         const script = document.createElement('script');
24|         script.src = 'https://www.chatbase.co/embed.min.js';
25|         script.id = 'Xk6BgUmtM4fmXWQfSsm_';
26|         script.setAttribute('domain', 'www.chatbase.co');
27|         script.defer = true;
28|         document.body.appendChild(script);
29|     }, []);
30|
31|     return null;
32| }
33|
```

```

1 | 'use client';
2 |
3 | import React from 'react';
4 | import Link from 'next/link';
5 | import { Calendar, Cpu, Battery, Wallet, HardDrive, ChevronRight, Clock } from 'lucide-react';
6 | import { format, differenceInDays } from 'date-fns';
7 | import { vi } from 'date-fns/locale';
8 |
9 | interface AssignmentRequirements {
10 |   budget_max?: number;
11 |   tdp_max?: number;
12 |   min_ram_gb?: number;
13 |   required_components?: string[];
14 |   performance_min?: number;
15 | }
16 |
17 | interface AssignmentCardProps {
18 |   id: string;
19 |   classId: string;
20 |   title: string;
21 |   type: string;
22 |   deadline?: string | Date;
23 |   requirements: AssignmentRequirements;
24 |   submittedCount?: number;
25 |   totalMembers?: number;
26 |   role: 'teacher' | 'student';
27 |   hasSubmitted?: boolean;
28 | }
29 |
30 | export default function AssignmentCard({ id, classId, title, type, deadline, requirements, submittedCount, tot...
31 |   const linkHref = `/${role}/classes/${classId}/assignments/${id}`;
32 |
33 |   let daysLeft = null;
34 |   let isOverdue = false;
35 |   if (deadline) {
36 |     const dlDate = new Date(deadline);
37 |     daysLeft = differenceInDays(dlDate, new Date());
38 |     isOverdue = daysLeft < 0;
39 |   }
40 |
41 |   const getTypeLabel = () => {
42 |     switch(type) {
43 |       case 'build_config': return '[ BUILD ] L - p r á p c i b ê n';
44 |       case 'optimize_budget': return '[ TDI ] U ] T i c k i c m n g â n s á c h';
45 |       case 'minimize_tdp': return '[ GREEN ] T i c k i c m i c n';
46 |       case 'maximize_perf': return '[ PERF ] T N i a h i c u n n g';
47 |       default: return '[ NH I Æ M V Æ ]';
48 |     }
49 |   };
50 |
51 |   const formatCurrency = (val?: number) => {
52 |     if (!val) return 'K h ô n g g i ù i h i n';
53 |     return new Intl.NumberFormat('vi-VN', { style: 'currency', currency: 'VND' }).format(val);
54 |   };
55 |
56 |   return (
57 |     <div style={{
58 |       background: 'rgba(12, 20, 36, 0.8)',
59 |       borderRadius: '20px',
60 |       border: isOverdue ? '1px solid rgba(239, 68, 68, 0.3)' : '1px solid rgba(255,255,255,0.05)',
61 |       overflow: 'hidden',
62 |       display: 'flex',
63 |       flexDirection: 'column'
64 |     }}>
65 |       <div style={{ padding: '20px', borderBottom: '1px solid rgba(255,255,255,0.05)' }}>
66 |         <div style={{ fontSize: '11px', fontWeight: 700, color: '#00f3ff', letterSpacing: '1px', textTransform: ...
67 |           {getTypeLabel()}
68 |         </div>
69 |         <h3 style={{ margin: '0 0 12px 0', fontSize: '18px', fontWeight: 700, color: '#fff', lineHeight: 1.4 }}...
70 |
71 |         {deadline && (
72 |           <div style={{ display: 'flex', alignItems: 'center', gap: '8px', fontSize: '13px', color: isOverdue ...
73 |             <Calendar size={14} />
74 |             <span style={{ margin: '0 4px' }}></span>
75 |             H i n c h ó t : { format ( new Date ( deadline ), ' dd / MM / yyyy HH : mm ', {
76 |               <span style={{ margin: '0 4px' }}></span>
77 |               { isOverdue ? ' ã q u á h i n ' : ` C ò n ${ daysLeft } n g à y ` }
78 |             )}
79 |           </div>
80 |         </div>
81 |
82 |         <div style={{ padding: '20px', flex: 1 }}>
83 |           <p style={{ margin: '0 0 12px 0', fontSize: '12px', fontWeight: 700, color: '#8899a6', textTransform: ...
84 |             <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '12px' }}>
85 |               {requirements.budget_max !== undefined && (
86 |                 <div style={{ display: 'flex', alignItems: 'center', gap: '8px', fontSize: '13px', color: '#e0e6ed...
87 |                   <Wallet size={16} color="#10b981" />
88 |                   <span>&le; {formatCurrency(requirements.budget_max)}</span>
89 |                 </div>
90 |               )}
91 |               {requirements.tdp_max !== undefined && (
92 |                 <div style={{ display: 'flex', alignItems: 'center', gap: '8px', fontSize: '13px', color: '#e0e6ed...
93 |                   <Battery size={16} color="#f59e0b" />
94 |                   <span>&le; {requirements.tdp_max}W</span>
95 |                 </div>
96 |               )}
97 |               {requirements.min_ram_gb !== undefined && (
98 |                 <div style={{ display: 'flex', alignItems: 'center', gap: '8px', fontSize: '13px', color: '#e0e6ed...
99 |                   <HardDrive size={16} color="#3b82f6" />
100 |                   <span>&ge; {requirements.min_ram_gb}GB RAM</span>
101 |                 </div>
102 |               )}
103 |               {requirements.required_components && requirements.required_components.length > 0 && (
104 |                 <div style={{ display: 'flex', alignItems: 'center', gap: '8px', fontSize: '13px', color: '#e0e6ed...
105 |                   <Cpu size={16} color="#8b5cf6" />
106 |                   <span>B - t b u ù c c ó : { requirements.required_components.join('
107 |                 </div>
108 |               )}
109 |             </div>
110 |           </div>
111 |         </div>
112 |
113 |         <div style={{ padding: '20px', background: 'rgba(0,0,0,0.2)', borderTop: '1px solid rgba(255,255,255,0.0...
114 |           {role === 'teacher' ? (
115 |             <div style={{ fontSize: '14px', color: '#8899a6', fontWeight: 600 }}>
116 |             <span style={{ color: '#fff' }}> { submittedCount } | 0 </span>
117 |           </div>
118 |         ) : (
119 |           <div style={{ fontSize: '14px', fontWeight: 600, color: hasSubmitted ? '#10b981' : '#8899a6', displa...
120 |             { hasSubmitted ? ' ã n ù p b à i ' : '#ó C h ° a n ù p ' }
121 |           </div>
122 |

```



```

123 |     })
124 |
125 |     <Link href={linkHref} style={{ textDecoration: 'none' }}>
126 |       <button style={{
127 |         padding: '8px 16px',
128 |         borderRadius: '8px',
129 |         background: hasSubmitted ? 'rgba(255,255,255,0.1)' : 'var(--brand-primary)',
130 |         color: hasSubmitted ? '#fff' : '#000',
131 |         border: 'none',
132 |         fontSize: '13px',
133 |         fontWeight: 700,
134 |         cursor: 'pointer',
135 |         display: 'flex',
136 |         alignItems: 'center',
137 |         gap: '6px',
138 |         transition: 'all 0.2s'
139 |       }}>
140 |         { role === 'teacher' ? 'Quên lý' : (hasSubmitted ? 'Xem iÃm
141 |       </button>
142 |     </Link>
143 |   </div>
144 | </div>
145 | );
146 | }
147 |

```

File: components/classes/ClassCard.tsx (94 lines, 4.5 KB)

```

1 | 'use client';
2 |
3 | import React from 'react';
4 | import Link from 'next/link';
5 | import { Users, Monitor, ChevronRight, Trash2 } from 'lucide-react';
6 | import { supabase } from '@lib/supabase';
7 |
8 | interface ClassCardProps {
9 |   id: string;
10 |  name: string;
11 |  code: string;
12 |  teacherName?: string;
13 |  memberCount: number;
14 |  maxStudents: number;
15 |  activeAssignmentsCount: number;
16 |  role: 'teacher' | 'student';
17 |  onDelete?: () => void;
18 | }
19 |
20 | export default function ClassCard({ id, name, code, teacherName, memberCount, maxStudents, activeAssignmentsCo...
21 |   const linkHref = `/${role}/classes/${id}`;
22 |
23 |   const deleteClass = async (e: React.MouseEvent) => {
24 |     e.preventDefault();
25 |     e.stopPropagation();
26 |     if (!confirm('Xóa lớp híc này?')) return;
27 |     const { data: { user } } = await supabase.auth.getUser();
28 |     if (!user) return;
29 |     const { error } = await supabase.from('classes').delete().eq('id', id).eq('teacher_id', user.id);
30 |     if (error) { alert('Lỗi: ' + error.message); return; }
31 |     if (onDelete) onDelete();
32 |   };
33 |
34 |   return (
35 |     <div className="lms-card" style={{ overflow: 'hidden', display: 'flex', flexDirection: 'column' }}>
36 |       <div style={{ padding: '20px', background: 'rgba(var(--brand-primary-rgb),0.05)', borderBottom: '1px sol...
37 |         <div style={{ width: '40px', height: '40px', borderRadius: '12px', background: 'rgba(var(--brand-prima...
38 |           <Monitor size={20} />
39 |         </div>
40 |         <div style={{ flex: 1 }}>
41 |           <h3 style={{ margin: '0 0 4px 0', fontSize: '18px', fontWeight: 700, color: 'var(--text-primary)' }}...
42 |           <p style={{ margin: 0, fontSize: '13px', color: 'var(--text-muted)', display: 'flex', alignItems: 'c...
43 |             <span>Mã: <strong style={{ color: 'var(--brand-primary)', letterSpacing: '1px' }}>{code}</strong><...
44 |             <span></span>
45 |             <span style={{ display: 'flex', alignItems: 'center', gap: '4px' }}><Users size={12} /> {memberCou...
46 |           </p>
47 |           {teacherName && (
48 |             <p style={{ margin: '4px 0 0 0', fontSize: '12px', color: 'var(--text-muted)' }}>GV: {teacherName}...
49 |           )}
50 |         </div>
51 |         {role === 'teacher' && (
52 |           <button onClick={deleteClass} style={{ padding: '6px', borderRadius: '8px', background: 'rgba(244,67...
53 |             <Trash2 size={16} />
54 |           </button>
55 |         )}
56 |       </div>
57 |
58 |       <div style={{ padding: '20px', flex: 1, display: 'flex', flexDirection: 'column', justifyContent: 'space...
59 |         <div style={{ display: 'flex', alignItems: 'center', gap: '8px' }}>
60 |           <div style={{ width: '8px', height: '8px', borderRadius: '50%', background: activeAssignmentsCount >...
61 |             <span style={{ fontSize: '14px', color: 'var(--text-primary)', fontWeight: 600 }}>` $ { activeAssignmentsCount } nh
62 |           { activeAssignmentsCount > 0 ? ` ` $ { activeAssignmentsCount } nh
63 |         </span>
64 |       </div>
65 |
66 |       <Link href={linkHref} style={{ textDecoration: 'none' }}>
67 |         <button style={{
68 |           width: '100%',
69 |           padding: '12px',
70 |           borderRadius: '10px',
71 |           background: 'var(--bg-elevated)',
72 |           border: '1px solid var(--border-default)',
73 |           color: 'var(--text-primary)',
74 |           fontSize: '14px',
75 |           fontWeight: 700,
76 |           cursor: 'pointer',
77 |           display: 'flex',
78 |           alignItems: 'center',
79 |           justifyContent: 'center',
80 |           gap: '8px',
81 |           transition: 'all 0.2s',
82 |           fontFamily: 'inherit'
83 |         }}
84 |         onMouseOver={e => { e.currentTarget.style.background = 'rgba(var(--brand-primary-rgb),0.1)'; e.curre...
85 |         onMouseOut={e => { e.currentTarget.style.background = 'var(--bg-elevated)'; e.currentTarget.style.co...
86 |       >
87 |         Vào lớp <ChevronRight size={16} />
88 |       </button>
89 |     </Link>
90 |   </div>
91 | </div>
92 | );
93 | }
94 |

```

```

1 | 'use client';
2 |
3 | import React, { useState } from 'react';
4 | import {
5 |   useReactTable,
6 |   getCoreRowModel,
7 |   getSortedRowModel,
8 |   getFilteredRowModel,
9 |   flexRender,
10 |   createColumnHelper,
11 |   SortingState
12 | } from '@tanstack/react-table';
13 | import { format } from 'date-fns';
14 | import { vi } from 'date-fns/locale';
15 | import { ArrowUpDown, Download, CheckCircle, AlertCircle, Clock } from 'lucide-react';
16 | import Link from 'next/link';
17 |
18 | interface SubmissionData {
19 |   id: string;
20 |   student_name: string;
21 |   student_id: string;
22 |   submitted_at: string;
23 |   is_late: boolean;
24 |   auto_score: number | null;
25 |   teacher_score: number | null;
26 |   total_score: number | null;
27 |   status: 'graded' | 'submitted' | 'missing';
28 | }
29 |
30 | interface GradeTableProps {
31 |   data: SubmissionData[];
32 |   assignmentId: string;
33 |   classId: string;
34 | }
35 |
36 | const columnHelper = createColumnHelper<SubmissionData>();
37 |
38 | export default function GradeTable({ data, assignmentId, classId }: GradeTableProps) {
39 |   const [sorting, setSorting] = useState<SortingState>([]);
40 |   const [globalFilter, setGlobalFilter] = useState('');
41 |
42 |   const columns = [
43 |     columnHelper.accessor('student_name', {
44 |       header: 'H í c s i n h',
45 |       cell: info => <span style={{ fontWeight: 600, color: '#fff' }}>{info.getValue()}</span>,
46 |     }),
47 |     columnHelper.accessor('status', {
48 |       header: 'T r i n g  t h á i',
49 |       cell: info => {
50 |         const status = info.getValue();
51 |         const isLate = info.row.original.is_late;
52 |         if (status === 'missing') return <span style={{ color: '#ef4444', display: 'flex', alignItems: 'center'...
53 |         if (status === 'graded') return <span style={{ color: '#10b981', display: 'flex', alignItems: 'center'...
54 |         return <span style={{ color: '#3b82f6', display: 'flex', alignItems: 'center', gap: '4px', fontSize: '...
55 |       },
56 |     }),
57 |     columnHelper.accessor('submitted_at', {
58 |       header: 'T h ý i g i a n  n ù p',
59 |       cell: info => info.getValue() ? format(new Date(info.getValue()), 'dd/MM/yyyy HH:mm', { locale: vi }) : ...
60 |     }),
61 |     columnHelper.accessor('auto_score', {
62 |       header: 'i Ā m  h  Ç  t h Ñ n g',
63 |       cell: info => info.getValue() !== null ? <span style={{ color: '#8899a6' }}>{info.getValue()}</span> : '-',
64 |     }),
65 |     columnHelper.accessor('teacher_score', {
66 |       header: 'i Ā m  c ù n g / t r ẽ',
67 |       cell: info => {
68 |         const val = info.getValue();
69 |         if (val === null) return '-';
70 |         return <span style={{ color: val > 0 ? '#10b981' : (val < 0 ? '#ef4444' : '#8899a6') }}>{val > 0 ? `+$...
71 |       },
72 |     }),
73 |     columnHelper.accessor('total_score', {
74 |       header: 'T Õ n g  i Ā m',
75 |       cell: info => info.getValue() !== null ? <strong style={{ color: '#00f3ff', fontSize: '16px' }}>{info.ge...
76 |     }),
77 |     columnHelper.display({
78 |       id: 'actions',
79 |       cell: info => {
80 |         if (info.row.original.status === 'missing') return null;
81 |         return (
82 |           <Link href={`/teacher/classes/${classId}/assignments/${assignmentId}/grade/${info.row.original.id}`}>
83 |             <button style={{
84 |               background: 'rgba(255,255,255,0.05)', border: '1px solid rgba(255,255,255,0.1)',
85 |               color: '#fff', padding: '6px 12px', borderRadius: '6px', fontSize: '12px',
86 |               cursor: 'pointer', fontWeight: 600, transition: 'all 0.2s'
87 |             }}>
88 |               { info.row.original.status === 'graded' ? 'S í a  i Ā m' : 'Ch
89 |             </button>
90 |           </Link>
91 |         )
92 |       },
93 |     });
94 | ];
95 |
96 | const table = useReactTable({
97 |   data,
98 |   columns,
99 |   state: { sorting, globalFilter },
100 |   onSortingChange: setSorting,
101 |   onGlobalFilterChange: setGlobalFilter,
102 |   getCoreRowModel: getCoreRowModel(),
103 |   getSortedRowModel: getSortedRowModel(),
104 |   getFilteredRowModel: getFilteredRowModel(),
105 | });
106 |
107 | const handleExportCSV = () => {
108 |   const headers = ['H í c s i n h', 'T r i n g  t h á i', 'T h ý i g i a n  n ù p', 'i Ā m  h
109 |   const csvData = data.map(row => [
110 |     row.student_name,
111 |     row.status === 'missing' ? 'Ch ° a  n ù p' : (row.status === 'graded' ?
112 |     row.submitted_at ? format(new Date(row.submitted_at), 'dd/MM/yyyy HH:mm') : '',
113 |     row.auto_score ?? '',
114 |     row.teacher_score ?? '',
115 |     row.total_score ?? ''
116 |   ]);
117 |
118 |   const csvContent = [
119 |     headers.join(','),
120 |     ...csvData.map(e => e.join(','))
121 |   ].join('\n');

```

```

122 |     const blob = new Blob(["\u0000" + csvContent], { type: 'text/csv;charset=utf-8;' });
123 |     const url = URL.createObjectURL(blob);
124 |     const link = document.createElement('a');
125 |     link.href = url;
126 |     link.setAttribute('download', `bang_diem_${assignmentId}.csv`);
127 |     document.body.appendChild(link);
128 |     link.click();
129 |     document.body.removeChild(link);
130 | };
131 |
132 |
133 | return (
134 |   <div style={{ background: '#0a0f1a', borderRadius: '16px', border: '1px solid rgba(255,255,255,0.05)', ove...
135 |     /* Table Header Controls */
136 |     <div style={{ padding: '20px', borderBottom: '1px solid rgba(255,255,255,0.05)', display: 'flex', justif...
137 |       <input
138 |         value={globalFilter ?? ''}
139 |         onChange={e => setGlobalFilter(e.target.value)}
140 |         style={{
141 |           background: '#050a14', border: '1px solid rgba(255,255,255,0.1)', borderRadius: '8px',
142 |           padding: '10px 16px', color: '#fff', fontSize: '14px', width: '300px', outline: 'none'
143 |         }}
144 |       />
145 |       <button onClick={handleExportCSV} style={{
146 |         background: 'rgba(0, 243, 255, 0.1)', color: '#00f3ff', border: '1px solid rgba(0, 243, 255, 0.2)',
147 |         padding: '10px 16px', borderRadius: '8px', fontSize: '14px', fontWeight: 600, cursor: 'pointer',
148 |         display: 'flex', alignItems: 'center', gap: '8px'
149 |       }}>
150 |         <Download size={16} /> Xuýt CSV
151 |       </button>
152 |     </div>
153 |
154 |     /* Table */
155 |     <div style={{ overflowX: 'auto' }}>
156 |       <table style={{ width: '100%', borderCollapse: 'collapse', textAlign: 'left' }}>
157 |         <thead>
158 |           {table.getHeaderGroups().map(headerGroup => (
159 |             <tr key={headerGroup.id} style={{ background: 'rgba(255,255,255,0.02)' }}>
160 |               {headerGroup.headers.map(header => (
161 |                 <th key={header.id} style={{ padding: '16px 20px', color: '#8899a6', fontSize: '12px', textT...
162 |                   <div style={{ display: 'flex', alignItems: 'center', gap: '8px' }}>
163 |                     {flexRender(header.column.columnDef.header, header.getContext())}
164 |                     {header.column.getCanSort() && <ArrowUpDown size={12} color="rgba(255,255,255,0.2)" />}
165 |                   </div>
166 |                 </th>
167 |               </tr>
168 |             </tr>
169 |           </thead>
170 |           <tbody>
171 |             {table.getRowModel().rows.map(row => (
172 |               <tr key={row.id} style={{ borderBottom: '1px solid rgba(255,255,255,0.05)', transition: 'backgro...
173 |                 {row.getVisibleCells().map(cell => (
174 |                   <td key={cell.id} style={{ padding: '16px 20px', color: '#e0e6ed', fontSize: '14px' }}>
175 |                     {flexRender(cell.column.columnDef.cell, cell.getContext())}
176 |                   </td>
177 |                 </td>
178 |               </tr>
179 |             </tr>
180 |           </tbody>
181 |         </table>
182 |       </div>
183 |       {data.length === 0 && (
184 |         <tr>
185 |           <td colspan={columns.length} style={{ padding: '40px', textAlign: 'center', color: '#8899a6' }}>
186 |             Châa cõ dĩ lĩ Çu bã ã n ù p .
187 |           </td>
188 |         </tr>
189 |       </table>
190 |     </div>
191 |   </div>
192 | );
193 | }
194 |
195 |

```

File: components/classes/JoinClassModal.tsx (172 lines, 7.7 KB)

```

1 | 'use client';
2 |
3 | import React, { useState } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import { X, KeySquare, Loader2, CheckCircle, AlertCircle, User } from 'lucide-react';
6 | import { useRouter } from 'next/navigation';
7 |
8 | export default function JoinClassModal({ isOpen, onClose }: { isOpen: boolean, onClose: () => void }) {
9 |   const [code, setCode] = useState('');
10 |   const [loading, setLoading] = useState(false);
11 |   const [error, setError] = useState<string | null>(null);
12 |   const [success, setSuccess] = useState(false);
13 |   const [classInfo, setClassInfo] = useState<{ id: string; name: string; teacherName?: string } | null>(null);
14 |   const router = useRouter();
15 |
16 |   if (!isOpen) return null;
17 |
18 |   const lookupClass = async () => {
19 |     if (code.length !== 6) {
20 |       setError('Mã lớp phải có 6 ký tự. ');
21 |       return;
22 |     }
23 |
24 |     setLoading(true);
25 |     setError(null);
26 |     setClassInfo(null);
27 |
28 |     try {
29 |       const { data: userData } = await supabase.auth.getUser();
30 |       if (!userData.user) throw new Error('Vui lòng đăng nhập. ');
31 |
32 |       const { data: classData, error: classError } = await supabase
33 |         .from('classes')
34 |         .select('id, name, teacher_id')
35 |         .eq('code', code.toUpperCase())
36 |         .single();
37 |
38 |       if (classError || !classData) {
39 |         throw new Error('Không tìm thấy lớp học với mã này. ');
40 |       }
41 |
42 |       let teacherName = 'Giáo viên';
43 |       if (classData.teacher_id) {
44 |         const { data: teacherProfile } = await supabase
45 |           .from('profiles')
46 |           .select('full_name')

```

```

47 |         .eq('id', classData.teacher_id)
48 |         .single();
49 |         if (teacherProfile?.full_name) {
50 |             teacherName = teacherProfile.full_name;
51 |         }
52 |     }
53 |
54 |     setClassInfo({ ...classData, teacherName });
55 |
56 |     const { error: joinError } = await supabase
57 |       .from('class_members')
58 |       .insert({
59 |         class_id: classData.id,
60 |         student_id: userData.user.id,
61 |         status: 'active'
62 |       });
63 |
64 |     if (joinError) {
65 |       if (joinError.code === '23505') throw new Error('Bị nã thãm gia');
66 |       throw new Error('Không thã thãm gia lüp: ' + joinError.message);
67 |     }
68 |
69 |     setSuccess(true);
70 |     setTimeout(() => {
71 |       onClose();
72 |       router.push(`/student/classes/${classData.id}`);
73 |     }, 1500);
74 |
75 |   } catch (err: any) {
76 |     setError(err.message);
77 |   } finally {
78 |     setLoading(false);
79 |   }
80 | };
81 |
82 | return (
83 |   <div style={{
84 |     position: 'fixed', top: 0, left: 0, right: 0, bottom: 0,
85 |     background: 'color-mix(in srgb, var(--bg-base) 85%, transparent)', backdropFilter: 'blur(8px)',
86 |     display: 'flex', alignItems: 'center', justifyContent: 'center',
87 |     zIndex: 9999, padding: '20px'
88 |   }}>
89 |     <div style={{
90 |       background: 'var(--bg-surface)', borderRadius: '24px', width: '100%', maxWidth: '440px',
91 |       border: '1px solid var(--border-strong)', boxShadow: '0 25px 50px -12px rgba(0,0,0,0.5)',
92 |       position: 'relative', overflow: 'hidden'
93 |     }}>
94 |       <button onClick={onClose} style={{ position: 'absolute', top: '16px', right: '16px', background: 'tran...
95 |         <X size={20} />
96 |       </button>
97 |
98 |       <div style={{ padding: '32px' }}>
99 |         <div style={{ width: '56px', height: '56px', borderRadius: '16px', background: 'var(--brand-subtle)'...
100 |         <keySquare size={28} />
101 |       </div>
102 |       <h2 style={{ margin: '0 0 8px 0', fontSize: '24px', fontWeight: 800, color: 'var(--text-primary)' }}>...
103 |       <p style={{ margin: '0 0 24px 0', color: 'var(--text-muted)', fontSize: '14px', lineHeight: 1.5 }}>
104 |         Nh - p mã lüp gôm 6 ký tñ do gião viẽn cung cøp ã b - t s u t h f
105 |       </p>
106 |
107 |       {success ? (
108 |         <div style={{ background: 'var(--brand-subtle)', border: '1px solid color-mix(in srgb, var(--brand...
109 |         <CheckCircle size={24} />
110 |         <span style={{ fontWeight: 600 }}>Thãm gia thãn h cõng! a r
111 |       </div>
112 |     ) : (
113 |       <form onSubmit={e => { e.preventDefault(); lookupClass(); }} style={{ display: 'flex', flexDirec...
114 |       <div>
115 |         <input
116 |           type="text"
117 |           placeholder="VD: TIN001"
118 |           value={code}
119 |           onChange={e => {
120 |             setCode(e.target.value.toUpperCase().slice(0, 6));
121 |             setClassInfo(null);
122 |             setError(null);
123 |           }}
124 |           style={{
125 |             width: '100%', padding: '16px', background: 'var(--bg-base)', border: '2px solid var(--bor...
126 |             borderRadius: '12px', color: 'var(--text-primary)', fontSize: '24px', fontWeight: 800, tex...
127 |             letterSpacing: '4px', textTransform: 'uppercase', outline: 'none', transition: 'border-col...
128 |           }}
129 |           onFocus={e => e.target.style.borderColor = 'var(--brand-primary)'}
130 |           onBlur={e => e.target.style.borderColor = 'var(--border-default)'}
131 |         </input>
132 |       </div>
133 |
134 |       {classInfo && !error && !success && (
135 |         <div style={{ background: 'var(--bg-base)', border: '1px solid var(--border-default)', borderR...
136 |         <div style={{ width: '40px', height: '40px', borderRadius: '10px', background: 'var(--brand-...
137 |         <User size={20} />
138 |       </div>
139 |       <div style={{ flex: 1, minWidth: 0 }}>
140 |         <p style={{ fontSize: '14px', fontWeight: 600, color: 'var(--text-primary)', margin: 0, ov...
141 |         <p style={{ fontSize: '12px', color: 'var(--text-muted)', margin: '2px 0 0 0' }}>Gião viẽn...
142 |       </div>
143 |     </div>
144 |   )}
145 |
146 |   {error && (
147 |     <div style={{ background: 'color-mix(in srgb, #ef4444 10%, transparent)', color: '#ef4444', pa...
148 |     <AlertCircle size={16} style={{ flexShrink: 0, marginTop: '2px' }} />
149 |     <span>{error}</span>
150 |   </div>
151 | )}
152 |
153 |   <button
154 |     type="submit"
155 |     disabled={loading || code.length !== 6}
156 |     style={{
157 |       width: '100%', padding: '16px', borderRadius: '12px', background: 'var(--brand-primary)',
158 |       color: 'var(--bg-base)', fontSize: '16px', fontWeight: 700, border: 'none',
159 |       cursor: (loading || code.length !== 6) ? 'not-allowed' : 'pointer',
160 |       opacity: (loading || code.length !== 6) ? 0.5 : 1, display: 'flex', alignItems: 'center', ju...
161 |     }}
162 |   >
163 |     { loading ? <Loader2 size={20} className="animate-spin" /
164 |   </button>
165 |   </form>
166 | } </div>
167 | </div>
168 | </div>
169 | </div>
170 | );

```

File: components/CommonMistakes.js (255 lines, 16.1 KB)

```

110|         }>{mistake.level}</span>
111|         <span style={{ fontSize: '10px', color: 'var(--text-muted)' }}>
112|         </div>
113|         <p style={{ fontSize: '12px', color: 'var(--text-secondary)', margin: '4px 0 0', lineHeight: '1.5' }}>
114|         {mistake.summary}
115|         </p>
116|       </div>
117|       <div style={{
118|         width: '160px', flexShrink: 0, borderRadius: '8px', overflow: 'hidden',
119|         position: 'relative'.

```

```

119 |     }}>
120 |     {showVideo ? (
121 |       <div style={{ position: 'relative', width: '100%', aspectRatio: '16/9', background: '#000' }}>
122 |         <iframe
123 |           src={`https://www.youtube.com/embed/${mistake.videoId}?autoplay=1&rel=0`}
124 |           style={{ position: 'absolute', inset: 0, width: '100%', height: '100%', border: 'none' }}
125 |           allow="autoplay; encrypted-media" allowFullScreen
126 |         />
127 |       </div>
128 |     ) : (
129 |       <div onClick={() => setShowVideo(true)} style={{
130 |         position: 'relative', width: '100%', aspectRatio: '16/9', cursor: 'pointer',
131 |         background: 'linear-gradient(135deg, #1a1a2e, #16213e)', borderRadius: '8px', overflow: 'hidden',
132 |       }}>
133 |         <img src={`https://img.youtube.com/vi/${mistake.videoId}/hqdefault.jpg`}
134 |           alt={mistake.title} style={{ position: 'absolute', inset: 0, width: '100%', height: '100%', ob...
135 |           onError={(e) => { e.target.style.display = 'none'; }}
136 |         />
137 |         <div style={{
138 |           position: 'absolute', inset: 0, display: 'flex', alignItems: 'center', justifyContent: 'center',
139 |           background: 'rgba(0,0,0,0.25)',
140 |         }}>
141 |           <div style={{ width: '36px', height: '36px', borderRadius: '50%', background: 'rgba(255,0,0,0.0...
142 |             %f
143 |           </div>
144 |         </div>
145 |       </div>
146 |     )}
147 |   </div>
148 | </div>
149 |
150 |   <div style={{
151 |     maxHeight: showDetail ? '300px' : '0', opacity: showDetail ? 1 : 0,
152 |     transition: 'all 0.35s ease', overflow: 'hidden', borderTop: showDetail ? '1px solid var(--border-subt...
153 |   }}>
154 |     <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '14px', padding: '14px 18px' }}>
155 |       <div>
156 |         <div style={{ fontSize: '10px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: '1px',...
157 |           <span>L </span> N g u y ê n n h â n
158 |         </div>
159 |         <ul style={{ margin: 0, padding: '0 0 16px', display: 'flex', flexDirection: 'column', gap: '3px...
160 |           {mistake.reasons.map((r, i) => (
161 |             <li key={i} style={{ fontSize: '11px', color: 'var(--text-secondary)', lineHeight: '1.5' }}>{r...
162 |           ))}
163 |         </ul>
164 |       </div>
165 |       <div>
166 |         <div style={{ fontSize: '10px', fontWeight: 700, textTransform: 'uppercase', letterSpacing: '1px',...
167 |           <span>' </span> C á c h k h - c p h á c
168 |         </div>
169 |         <ul style={{ margin: 0, padding: '0 0 16px', display: 'flex', flexDirection: 'column', gap: '3px...
170 |           {mistake.fixes.map((f, i) => (
171 |             <li key={i} style={{ fontSize: '11px', color: '#22c55e', lineHeight: '1.5' }}>{f}</li>
172 |           ))}
173 |         </ul>
174 |       </div>
175 |     </div>
176 |   </div>
177 | </div>
178 | );
179 | });
180 | }
181 |
182 | export default function CommonMistakes({ lang = 'vn', onExit }) {
183 |   const [search, setSearch] = useState('');
184 |
185 |   const filtered = MISTAKES.filter(m =>
186 |     !search || m.title.toLowerCase().includes(search.toLowerCase()) ||
187 |     m.summary.toLowerCase().includes(search.toLowerCase()) ||
188 |     m.level.toLowerCase().includes(search.toLowerCase())
189 |   );
190 |
191 |   return (
192 |     <div style={{
193 |       position: 'fixed', inset: 0, zIndex: 9999,
194 |       background: 'var(--bg-base)', display: 'flex', flexDirection: 'column',
195 |       fontFamily: 'system-ui, sans-serif',
196 |     }}>
197 |       <div style={{
198 |         padding: '16px 24px', borderBottom: '1px solid var(--border-default)',
199 |         background: 'var(--bg-surface)', flexShrink: 0,
200 |       }}>
201 |         <div style={{ maxWidth: '960px', margin: '0 auto', display: 'flex', alignItems: 'center', justifyConte...
202 |           <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
203 |             <div style={{ width: '40px', height: '40px', borderRadius: '10px', background: 'rgba(239,68,68,0.1...
204 |               & b
205 |             </div>
206 |             <div>
207 |               <div style={{ fontSize: '17px', fontWeight: 800, color: 'v...
208 |               <div style={{ fontSize: '11px', color: 'var(--text-muted)' }>
209 |             </div>
210 |           </div>
211 |           <div style={{ display: 'flex', gap: '10px', alignItems: 'center' }}>
212 |             <input
213 |               value={search}
214 |               placeholder="Tìm kiếm..."
215 |               onChange={e => setSearch(e.target.value)}
216 |               style={{
217 |                 padding: '8px 14px', borderRadius: '8px', border: '1px solid var(--border-default)',
218 |                 background: 'var(--bg-elevated)', color: 'var(--text-primary)', fontSize: '13px',
219 |                 fontFamily: 'inherit', outline: 'none', width: '200px',
220 |               }}
221 |             />
222 |             <button onClick={onExit} style={{
223 |               padding: '8px 14px', borderRadius: '8px', border: '1px solid var(--border-default)',
224 |               background: 'transparent', color: 'var(--text-muted)', fontSize: '14px', cursor: 'pointer',
225 |               fontFamily: 'inherit',
226 |             }}>
227 |           </button>
228 |         </div>
229 |       </div>
230 |     </div>
231 |
232 |     <div style={{
233 |       flex: 1, overflow: 'auto', padding: '16px 24px' }}>
234 |       <div style={{ maxWidth: '960px', margin: '0 auto', display: 'flex', flexDirection: 'column', gap: '8px...
235 |         {filtered.length === 0 ? (
236 |           <div style={{ textAlign: 'center', padding: '40px', color: 'var(--text-muted)', fontSize: '14px' }}>
237 |             K h ô n g t ì m t h ấ y l ầ i n ă o p h ứ h ấ p v ớ i " { s e a r c h } "
238 |           </div>
239 |         ) : (
240 |           filtered.map((m, i) => (
241 |             <div key={m.id} style={{ animation: `fadeUp 0.3s ease-out ${i * 0.05}s both` }}>
242 |               <MistakeCard mistake={m} />

```

```

243 |         </div>
244 |     ))
245 | }
246 | </div>
247 | </div>
248 |
249 | <style>{
250 |   @keyframes fadeUp { from { opacity: 0; transform: translateY(8px); } to { opacity: 1; transform: trans...
251 | }</style>
252 | </div>
253 | });
254 | }
255 |

```

File: components/CompatibilityPanel.js (107 lines, 4.8 KB)

```

1 | 'use client';
2 |
3 | import { useBuildStore } from '@lib/build-store';
4 |
5 | const SEVERITY_CONFIG = {
6 |   error: { bg: 'rgb(239, 68, 68, 0.1)', border: '#ef4444', text: '#ef4444',
7 |   warning: { bg: 'rgb(245, 158, 11, 0.1)', border: '#f59e0b', text: '#f59e0b',
8 |   info: { bg: 'rgb(59, 130, 246, 0.1)', border: '#3b82f6', text: '#3b82f6'
9 | };
10 |
11 | export default function CompatibilityPanel({ lang = 'vn' }) {
12 |   const issues = useBuildStore(s => s.issues);
13 |   const score = useBuildStore(s => s.score);
14 |
15 |   const errors = issues.filter(i => i.severity === 'error');
16 |   const warnings = issues.filter(i => i.severity === 'warning');
17 |   const infos = issues.filter(i => i.severity === 'info');
18 |
19 |   const scoreColor = score >= 80 ? 'var(--success)' : score >= 50 ? 'var(--warning)' : 'var(--danger)';
20 |   const scoreLabel = score >= 90 ? 'Hoàn hảo' : score >= 70 ? 'Tốt' : 'S
21 |
22 |   return (
23 |     <div style={{
24 |       background: 'var(--bg-surface)', borderRadius: '12px',
25 |       border: '1px solid var(--border-default)', overflow: 'hidden',
26 |       marginBottom: '16px',
27 |     }}>
28 |       <div style={{
29 |         display: 'flex', alignItems: 'center', gap: '16px',
30 |         padding: '16px 20px',
31 |         borderBottom: '1px solid var(--border-subtle)',
32 |         background: 'var(--bg-elevated)',
33 |       }}>
34 |         <div style={{
35 |           position: 'relative', width: '56px', height: '56px', flexShrink: 0,
36 |         }}>
37 |           <svg width="56" height="56" viewBox="0 0 36 36" style={{ transform: 'rotate(-90deg)' }}>
38 |             <circle cx="18" cy="18" r="15.5" fill="none" stroke="var(--border-subtle)" strokeWidth="2.5"/>
39 |             <circle cx="18" cy="18" r="15.5" fill="none" stroke={scoreColor} strokeWidth="2.5"
40 |               strokeDasharray={score * 0.97} 100`
41 |             </circle>
42 |           </svg>
43 |           <div style={{
44 |             position: 'absolute', inset: 0, display: 'flex', alignItems: 'center', justifyContent: 'center',
45 |             fontSize: '14px', fontWeight: 800, color: scoreColor,
46 |           }}>
47 |             {score}
48 |           </div>
49 |         </div>
50 |         <div style={{ flex: 1 }}>
51 |           <div style={{ fontWeight: 700, fontSize: '15px', color: 'var(--text-primary)' }}>
52 |             { lang === 'en' ? 'Compatibility Check' : 'Kiểm tra tính th
53 |           </div>
54 |           <div style={{ fontSize: '12px', color: 'var(--text-muted)', marginTop: '2px' }}>
55 |             {issues.length === 0
56 |               ? ( lang === 'en' ? 'No issues found - perfect build!' : '
57 |                 : `${issues.length} ${ lang === 'en' ? 'issue(s) found' : '
58 |             </div>
59 |           </div>
60 |         </div>
61 |       </div>
62 |     </div>
63 |
64 |     <div style={{ padding: '12px 20px', display: 'flex', flexDirection: 'column', gap: '8px', maxHeight: '40...
65 |     {issues.length === 0 && (
66 |       <div style={{
67 |         textAlign: 'center', padding: '20px',
68 |         color: 'var(--success)', fontSize: '13px', fontWeight: 600,
69 |       }}>
70 |         { lang === 'en' ? 'All components compatible!' : 'Tất c
71 |       </div>
72 |     )}
73 |     <div style={{
74 |       display: 'flex', gap: '10px', padding: '10px 12px',
75 |       background: cfg.bg, borderRadius: '8px',
76 |       borderLeft: '3px solid ${cfg.border}',
77 |     }}>
78 |       <span style={{ fontSize: '14px', color: cfg.text, flexShrink: 0, marginTop: '2px' }}>{cfg.icon}<...
79 |       <div style={{ flex: 1, minWidth: 0 }}>
80 |         <div style={{
81 |           fontSize: '12px', fontWeight: 700, color: cfg.text,
82 |           margin: '2px 0', textTransform: 'uppercase', letterSpacing: '0.3px',
83 |         }}>
84 |           {cfg.label}
85 |         </div>
86 |         <div style={{ fontSize: '12px', color: 'var(--text-primary)', lineHeight: 1.4, margin: '2px 0' }}>
87 |           { lang === 'en' ? issue.message : issue.message }
88 |         </div>
89 |         <div style={{
90 |           fontSize: '11px', color: 'var(--text-muted)', fontStyle: 'italic',
91 |           padding: '4px 8px', background: 'rgba(0,0,0,0.1)', borderRadius: '4px',
92 |         }}>
93 |           { lang === 'en' ? 'Fix' : 'Khắc ph
94 |         </div>
95 |       </div>
96 |     </div>
97 |   );
98 | }
99 |
100 |
101 |
102 |
103 | </div>
104 | </div>
105 | );
106 | }
107 |

```

File: components/ComponentInfo.js (64 lines, 2.8 KB)

```
1 'use client';
2
3 import { COMPONENT_DATA } from '../utils/i18nData';
4
5 const ComponentInfo = ({ type, lang = 'en', onOpenLesson }) => {
6   if (!type || !COMPONENT_DATA[lang][type]) return null;
7
8   const info = COMPONENT_DATA[lang][type];
9
10  return (
11    <div className="glass-panel" style={{
12      position: 'absolute',
13      top: '2rem',
14      left: '2rem',
15      width: '300px',
16      padding: '1.5rem',
17      borderLeft: '4px solid var(--primary-neon)',
18      background: 'var(--bg-surface)',
19      boxShadow: '0 4px 6px -1px rgba(0, 0, 0, 0.1)',
20      animation: 'fadeIn 0.3s ease-out'
21    }}>
22      <h3 className="neon-text-blue" style={{ marginBottom: '0.5rem', fontSize: '1.2rem' }}>{info.title}...
23      <p style={{ color: 'var(--text-primary)', marginBottom: '1rem', lineHeight: '1.5', fontSize: '0.95rem' }}>
24      <div style={{ backgroundColor: '#f8f9fc', padding: '0.75rem', borderRadius: '6px', border: '1px solid #e0e1e2' }}>
25        <strong style={{ color: 'var(--brand-primary)', fontSize: '0.85rem', textTransform: 'uppercase' }}>
26        <p style={{ color: 'var(--text-muted)', fontSize: '0.9rem', fontStyle: 'italic', marginTop: '0.5rem' }}>
27      </div>
28
29      <button
30        onClick={() => onOpenLesson && onOpenLesson(type)}
31        style={{
32          width: '100%',
33          padding: '0.75rem',
34          backgroundColor: 'var(--primary-neon)',
35          color: 'white',
36          border: 'none',
37          borderRadius: '8px',
38          fontWeight: '600',
39          cursor: 'pointer',
40          display: 'flex',
41          alignItems: 'center',
42          justifyContent: 'center',
43          gap: '0.5rem',
44          transition: 'all 0.2s',
45          boxShadow: '0 2px 4px rgba(2, 132, 199, 0.2)'
46        }}
47        onMouseOver={(e) => { e.currentTarget.style.backgroundColor = '#0369a1'; }}
48        onMouseOut={(e) => { e.currentTarget.style.backgroundColor = 'var(--primary-neon); }}
49      >
50        { lang === 'en' ? 'Deep Dive Lesson' : 'Xem Chi Tiết Bài }
51      </button>
52
53      <style jsx>{`
54        @keyframes fadeIn {
55          from { opacity: 0; transform: translateY(-10px); }
56          to { opacity: 1; transform: translateY(0); }
57        }
58      `}</style>
59    </div>
60  );
61
62
63
64 export default ComponentInfo;
```

File: components/ComponentPreview.js (190 lines, 9.8 KB)

```
1 'use client';
2
3 const PART_COLORS = {
4   CPU: '#00d4aa', GPU: '#ef4444', RAM: '#6366f1',
5   SSD: '#22c55e', PSU: '#f59e0b', COOLER: '#06b6d4', Mainboard: '#8b5cf6'
6 };
7
8 function CPUPreview({ model }) {
9   return (
10     <svg viewBox="0 0 200 200" width="100%" height="100%" style={{ maxWidth: 180 }}>
11       <rect x="25" y="25" width="150" height="150" rx="6" fill="#1a1a2e" stroke="#333" strokeWidth="2" />
12       <rect x="35" y="35" width="130" height="130" rx="4" fill="#2d2d44" stroke="#444" strokeWidth="1" />
13       <rect x="50" y="50" width="100" height="100" rx="3" fill="#3a3a5c" />
14       <text x="100" y="95" textAnchor="middle" fill="#00d4aa" fontSize="11" fontWeight="bold" fontFamily="mono...
15       <text x="100" y="112" textAnchor="middle" fill="#888" fontSize="8" fontFamily="monospace">{model?.socket...
16       <rect x="45" y="155" width="110" height="4" rx="2" fill="#d4a017" opacity="0.7" />
17       {[0,1,2,3].map(i => (
18         <circle key={i} cx={60+i*28} cy={20} r="3" fill="#d4a017" opacity="0.5" />
19       ))}
20       {[0,1,2,3].map(i => (
21         <circle key={i+4} cx={60+i*28} cy={180} r="3" fill="#d4a017" opacity="0.5" />
22       ))}
23       <text x="100" y="175" textAnchor="middle" fill="#666" fontSize="7" fontFamily="monospace">{model?.power ...
24     </svg>
25   );
26 }
27
28 function GPUPreview({ model }) {
29   return (
30     <svg viewBox="0 0 280 160" width="100%" height="100%" style={{ maxWidth: 260 }}>
31       <rect x="10" y="30" width="260" height="100" rx="8" fill="#2d1b1b" stroke="#ef4444" strokeWidth="1.5" />
32       <rect x="20" y="40" width="240" height="80" rx="6" fill="#1a0f0f" />
33       <rect x="40" y="55" width="60" height="50" rx="4" fill="#2d1b1b" stroke="#ef4444" strokeWidth="1" />
34       <rect x="70" y="55" width="60" height="50" rx="4" fill="#2d1b1b" stroke="#ef4444" strokeWidth="1" />
35       <rect x="180" y="55" width="50" height="50" rx="4" fill="#2d1b1b" stroke="#ef4444" strokeWidth="1" />
36       <text x="110" y="55" textAnchor="middle" fill="#ef4444" fontSize="9" fontWeight="bold" fontFamily="monosp...
37       <text x="140" y="83" textAnchor="middle" fill="#ef4444" fontSize="9" fontWeight="bold" fontFamily="monos...
38       <text x="205" y="83" textAnchor="middle" fill="#ef4444" fontSize="9" fontWeight="bold" fontFamily="monos...
39       {[0,1,2,3,4,5].map(i => (
40         <line key={i} x1={22+i*44} y1="130" x2={22+i*44} y2="145" stroke="#d4a017" strokeWidth="2" opacity="0....
41       ))}
42       <rect x="10" y="145" width="30" height="8" rx="2" fill="#d4a017" opacity="0.4" />
43       <text x="140" y="155" textAnchor="middle" fill="#666" fontSize="7" fontFamily="monospace">{model?.power ...
44     </svg>
45   );
46 }
47
48 function RAMPreview({ model }) {
49   return (
50     <svg viewBox="0 0 160 220" width="100%" height="100%" style={{ maxWidth: 140 }}>
51       <rect x="20" y="15" width="120" height="180" rx="4" fill="#1a1a3e" stroke="#6366f1" strokeWidth="1.5" />
```



```

52 | <rect x="30" y="25" width="100" height="160" rx="3" fill="#252550" />
53 | {[0,1,2,3].map(i => (
54 |   <rect key={i} x={35+i*24} y="35" width="18" height="40" rx="2" fill="#2d2d60" stroke="#6366f1" strokeW...
55 | ))}
56 | <text x="80" y="100" textAnchor="middle" fill="#6366f1" fontSize="10" fontWeight="bold" fontFamily="mono...
57 | <text x="80" y="115" textAnchor="middle" fill="#888" fontSize="8" fontFamily="monospace">{model?.ramType...
58 | <rect x="30" y="185" width="100" height="12" rx="2" fill="#d4a017" opacity="0.6" />
59 | {[0,1,2,3,4].map(i => (
60 |   <rect key={i} x={38+i*22} y="187" width="10" height="8" rx="1" fill="#b8860b" opacity="0.8" />
61 | ))}
62 | <text x="80" y="212" textAnchor="middle" fill="#666" fontSize="7" fontFamily="monospace">{model?.power |...
63 | </svg>
64 | });
65 | }
66 |
67 | function SSDPreview({ model }) {
68 |   return (
69 |     <svg viewBox="0 0 200 120" width="100%" height="100%" style={{ maxWidth: 180 }}>
70 |       <rect x="15" y="20" width="170" height="80" rx="6" fill="#0d2818" stroke="#22c55e" strokeWidth="1.5" />
71 |       <rect x="25" y="30" width="150" height="60" rx="4" fill="#143d22" />
72 |       <rect x="35" y="40" width="60" height="40" rx="3" fill="#1a5c30" stroke="#22c55e" strokeWidth="0.5" opac...
73 |       <rect x="105" y="40" width="50" height="40" rx="3" fill="#1a5c30" stroke="#22c55e" strokeWidth="0.5" opa...
74 |       <text x="135" y="63" textAnchor="middle" fill="#22c55e" fontSize="9" fontWeight="bold" fontFamily="monos...
75 |       <rect x="15" y="45" width="8" height="30" rx="2" fill="#d4a017" opacity="0.7" />
76 |       <text x="100" y="110" textAnchor="middle" fill="#666" fontSize="7" fontFamily="monospace">M.2 NVMe</text>
77 |     </svg>
78 |   );
79 | }
80 |
81 | function PSUPreview({ model }) {
82 |   return (
83 |     <svg viewBox="0 0 220 180" width="100%" height="100%" style={{ maxWidth: 200 }}>
84 |       <rect x="15" y="20" width="190" height="140" rx="6" fill="#1a1a1a" stroke="#f59e0b" strokeWidth="1.5" />
85 |       <rect x="25" y="30" width="170" height="120" rx="4" fill="#2a2a2a" />
86 |       <circle cx="110" cy="75" r="30" fill="#1a1a1a" stroke="#f59e0b" strokeWidth="1" />
87 |       <circle cx="110" cy="75" r="20" fill="none" stroke="#f59e0b" strokeWidth="0.5" opacity="0.5" />
88 |       <text x="110" y="79" textAnchor="middle" fill="#f59e0b" fontSize="8" fontWeight="bold" fontFamily="monos...
89 |       {[0,1,2,3].map(i => (
90 |         <line key={i} x1={30} y1={120+i*8} x2={38} y2={120+i*8} stroke="#f59e0b" strokeWidth="1.5" opacity="0....
91 |       ))}
92 |       {[0,1,2,3].map(i => (
93 |         <line key={i} x1={182} y1={120+i*8} x2={190} y2={120+i*8} stroke="#f59e0b" strokeWidth="1.5" opacity="...
94 |       ))}
95 |       <text x="110" y="150" textAnchor="middle" fill="#666" fontSize="7" fontFamily="monospace">{model?.power ...
96 |     </svg>
97 |   );
98 | }
99 |
100 | function COOLERPreview({ model }) {
101 |   return (
102 |     <svg viewBox="0 0 200 200" width="100%" height="100%" style={{ maxWidth: 180 }}>
103 |       <rect x="40" y="20" width="120" height="160" rx="6" fill="#1a2a3a" stroke="#06b6d4" strokeWidth="1.5" />
104 |       <rect x="50" y="30" width="100" height="140" rx="4" fill="#1e3347" />
105 |       {[0,1,2,3,4].map(i => (
106 |         <rect key={i} x="55" y={35+i*26} width="90" height="16" rx="2" fill="#254458" stroke="#06b6d4" strokeW...
107 |       ))}
108 |       <rect x="65" y="70" width="70" height="60" rx="4" fill="#1a2a3a" stroke="#06b6d4" strokeWidth="1" />
109 |       <text x="100" y="103" textAnchor="middle" fill="#06b6d4" fontSize="9" fontWeight="bold" fontFamily="mono...
110 |       {[0,1,2].map(i => (
111 |         <line key={i} x1={70+i*12} y1={55} x2={70+i*12} y2={65} stroke="#06b6d4" strokeWidth="0.5" opacity="0....
112 |       ))}
113 |       <text x="100" y="145" textAnchor="middle" fill="#666" fontSize="7" fontFamily="monospace">{model?.power ...
114 |     </svg>
115 |   );
116 | }
117 |
118 | function MainboardPreview({ model }) {
119 |   return (
120 |     <svg viewBox="0 0 260 240" width="100%" height="100%" style={{ maxWidth: 240 }}>
121 |       <rect x="10" y="10" width="240" height="220" rx="8" fill="#1a2f4a" stroke="#8b5cf6" strokeWidth="1.5" />
122 |       <rect x="20" y="20" width="220" height="200" rx="5" fill="#1e3a5f" />
123 |       <!-- CPU socket -->
124 |       <rect x="80" y="60" width="100" height="80" rx="4" fill="#2a4a6f" stroke="#8b5cf6" strokeWidth="1" />
125 |       <text x="130" y="105" textAnchor="middle" fill="#8b5cf6" fontSize="9" fontWeight="bold" fontFamily="mono...
126 |       <!-- RAM slots -->
127 |       {[0,1].map(i => (
128 |         <rect key={i} x={35+i*20} y="160" width="15" height="30" rx="2" fill="#2a4a6f" stroke="#8b5cf6" stroke...
129 |       ))}
130 |       {[0,1].map(i => (
131 |         <rect key={i} x={210+i*20} y="160" width="15" height="30" rx="2" fill="#2a4a6f" stroke="#8b5cf6" strok...
132 |       ))}
133 |       <!-- PCIe slots -->
134 |       <rect x="60" y="150" width="60" height="6" rx="1" fill="#2a4a6f" opacity="0.5" />
135 |       <rect x="60" y="165" width="60" height="6" rx="1" fill="#2a4a6f" opacity="0.5" />
136 |       <!-- I/O ports -->
137 |       <rect x="20" y="15" width="60" height="12" rx="2" fill="#2a4a6f" opacity="0.4" />
138 |       {[0,1,2,3].map(i => (
139 |         <rect key={i} x={22+i*14} y="17" width="12" height="8" rx="1" fill="#4a6a8f" opacity="0.3" />
140 |       ))}
141 |       <text x="130" y="228" textAnchor="middle" fill="#666" fontSize="7" fontFamily="monospace">{model?.socket...
142 |     </svg>
143 |   );
144 | }
145 |
146 | const PREVIEWS = {
147 |   CPU: CPUPreview,
148 |   GPU: GPUPreview,
149 |   RAM: RAMPreview,
150 |   SSD: SSDPreview,
151 |   PSU: PSUPreview,
152 |   COOLER: COOLERPreview,
153 |   Mainboard: MainboardPreview
154 | };
155 |
156 | export default function ComponentPreview({ type, model, size = 'normal' }) {
157 |   const Preview = PREVIEWS[type];
158 |   if (!Preview) return null;
159 |   const color = PART_COLORS[type] || '#666';
160 |
161 |   return (
162 |     <div style={{
163 |       background: 'var(--bg-elevated)',
164 |       border: `1px solid ${color}30`,
165 |       borderRadius: '12px',
166 |       padding: size === 'large' ? '24px' : '12px',
167 |       display: 'flex',
168 |       flexDirection: 'column',
169 |       alignItems: 'center',
170 |       gap: '8px',
171 |       transition: 'all 0.2s',
172 |       width: size === 'large' ? '100%' : 'auto'
173 |     }}>
174 |       <div style={{ width: '100%', display: 'flex', justifyContent: 'center' }}>
175 |         <Preview model={model} />

```

```

176 |     </div>
177 |     {model && (
178 |       <div style={{ textAlign: 'center' }}>
179 |         <div style={{ fontSize: '13px', fontWeight: 700, color: 'var(--text-primary)' }}>{model.name}</div>
180 |         <div style={{ fontSize: '11px', color: color, fontWeight: 600 }}>{type} · {(model.price || 0).toLocale...
181 |         {model.desc && <div style={{ fontSize: '10px', color: 'var(--text-muted)', marginTop: '4px', lineHei...
182 |       </div>
183 |     )}
184 |     {!model && (
185 |       <div style={{ fontSize: '12px', fontWeight: 600, color: color }}>{type}</div>
186 |     )}
187 |   </div>
188 | );
189 | }
190 |

```

File: components/ComponentPreview3D.js (836 lines, 27.4 KB)

```

1 | 'use client';
2 |
3 | import { useState, useRef, useCallback, Suspense, useEffect } from 'react';
4 | import dynamic from 'next/dynamic';
5 | import { Canvas } from '@react-three/fiber';
6 | import { OrbitControls, Text } from '@react-three/drei';
7 |
8 | const PART_COLORS = {
9 |   CPU: '#00d4aa',
10 |   GPU: '#ef4444',
11 |   RAM: '#6366f1',
12 |   SSD: '#22c55e',
13 |   PSU: '#f59e0b',
14 |   MAINBOARD: '#8b5cf6'
15 | };
16 |
17 | const BRAND_COLORS = {
18 |   kingston: '#0096d6',
19 |   corsair: '#ff4400',
20 |   gskill: '#00aaa0',
21 |   teamgroup: '#00aaff',
22 |   crucial: '#0055aa'
23 | };
24 |
25 | function LoadingSpinner() {
26 |   return (
27 |     <div style={{
28 |       display: 'flex',
29 |       alignItems: 'center',
30 |       justifyContent: 'center',
31 |       width: '100%',
32 |       height: '100%',
33 |       minHeight: 200,
34 |       flexDirection: 'column',
35 |       gap: 12
36 |     }}>
37 |       <div style={{
38 |         width: 36,
39 |         height: 36,
40 |         border: '3px solid var(--border-default, #333)',
41 |         borderTopColor: 'var(--accent, #00d4aa)',
42 |         borderRadius: '50%',
43 |         animation: 'spin 0.8s linear infinite'
44 |       }} />
45 |       <style>{@keyframes spin { to { transform: rotate(360deg); } }}</style>
46 |       <span style={{ fontSize: 12, color: 'var(--text-muted, #888)' }}>Loading 3D preview...</span>
47 |     </div>
48 |   );
49 | }
50 |
51 | function ErrorFallback({ componentType, model }) {
52 |   const color = PART_COLORS[componentType] || '#666';
53 |   return (
54 |     <div style={{
55 |       display: 'flex',
56 |       alignItems: 'center',
57 |       justifyContent: 'center',
58 |       width: '100%',
59 |       height: '100%',
60 |       minHeight: 200,
61 |       flexDirection: 'column',
62 |       gap: 8,
63 |       background: 'var(--bg-elevated, #111)',
64 |       borderRadius: 12,
65 |       border: '1px solid ${color}30',
66 |       padding: 16
67 |     }}>
68 |       <svg viewBox="0 0 200 200" width="140" height="140">
69 |         <rect x="25" y="25" width="150" height="150" rx="6" fill="#1a1a2e" stroke={color} strokeWidth="2" />
70 |         <rect x="35" y="35" width="130" height="130" rx="4" fill="#2d2d44" stroke={color} strokeWidth="1" opac...
71 |         <text x="100" y="100" textAnchor="middle" fill={color} fontSize="12" fontWeight="bold" fontFamily="mon...
72 |         <text x="100" y="120" textAnchor="middle" fill="#666" fontSize="8" fontFamily="monospace">{model?.name...
73 |       </svg>
74 |       <span style={{ fontSize: 11, color: 'var(--text-muted, #888)' }}>3D unavailable - showing 2D preview</span>
75 |     </div>
76 |   );
77 | }
78 |
79 | /* ---- CPU 3D Model ---- */
80 | function CPUModel({ brand, socket = 'LGA1700', isHovered }) {
81 |   const groupRef = useRef();
82 |   const socketLabel = brand === 'AMD' ? 'AM5' : 'LGA1700';
83 |   const pinPositions = [];
84 |   for (let x = -0.75; x <= 0.75; x += 0.25) {
85 |     for (let z = -0.75; z <= 0.75; z += 0.25) {
86 |       pinPositions.push([x, z]);
87 |     }
88 |   }
89 |   const capPositions = [];
90 |   for (let x = -0.82; x <= 0.82; x += 0.82) {
91 |     for (let z = -0.82; z <= 0.82; z += 0.82) {
92 |       if (x !== 0 || z !== 0) capPositions.push([x, z]);
93 |     }
94 |   }
95 |
96 |   return (
97 |     <group ref={groupRef} scale={isHovered ? 1.05 : 1}>
98 |       <mesh position={[0, 0.2, 0]}>
99 |         <boxGeometry args={[1.6, 0.08, 1.6]} />
100 |         <meshStandardMaterial color="#c0c0c0" metalness={0.7} roughness={0.3} />
101 |       </mesh>
102 |       <mesh position={[0, 0.16, 0]}>
103 |         <boxGeometry args={[1.7, 0.02, 1.7]} />
104 |         <meshStandardMaterial color="#a0a0a0" metalness={0.5} roughness={0.4} />
105 |       </mesh>

```

```

106 |     <mesh position=[[0, -0.05, 0]]>
107 |     <boxGeometry args=[[1.8, 0.1, 1.8]] />
108 |     <meshStandardMaterial color="#2a1a0a" roughness={0.9} />
109 |   </mesh>
110 |   {pinPositions.map(([x, z]) => (
111 |     <mesh key={`pin-${x}-${z}`} position=[[x, -0.12, z]]>
112 |     <cylinderGeometry args=[[0.025, 0.035, 0.06, 6]] />
113 |     <meshStandardMaterial color="#b8860b" metalness={0.6} roughness={0.4} />
114 |   </mesh>
115 | )))
116 |   <mesh position=[[0.8, 0.25, 0.8]] rotation=[[0, 0, Math.PI / 4]]>
117 |   <boxGeometry args=[[0.08, 0.01, 0.08]] />
118 |   <meshStandardMaterial color="#ff4444" />
119 | </mesh>
120 | <Text
121 |   position=[[0, 0.3, 0]]
122 |   fontSize={0.12}
123 |   color="#333"
124 |   anchorX="center"
125 |   anchorY="middle"
126 | >
127 |   {socketLabel}
128 | </Text>
129 |   {capPositions.map(([x, z]) => (
130 |     <mesh key={`cap-${x}-${z}`} position=[[x, 0.22, z]]>
131 |     <sphereGeometry args=[[0.025, 8, 8]] />
132 |     <meshStandardMaterial color="#888" metalness={0.3} roughness={0.5} />
133 |   </mesh>
134 | )))
135 | </group>
136 | );
137 | }
138 |
139 | /* ---- GPU 3D Model ---- */
140 | function GPUModel({ isHovered }) {
141 |   const fanRef1 = useRef();
142 |   const fanRef2 = useRef();
143 |   const fanRef3 = useRef();
144 |
145 |   useEffect(() => {
146 |     let running = true;
147 |     function animate() {
148 |       if (!running) return;
149 |       if (fanRef1.current) fanRef1.current.rotation.y += 0.02;
150 |       if (fanRef2.current) fanRef2.current.rotation.y += 0.02;
151 |       if (fanRef3.current) fanRef3.current.rotation.y += 0.02;
152 |       requestAnimationFrame(animate);
153 |     }
154 |     const raf = requestAnimationFrame(animate);
155 |     return () => { running = false; cancelAnimationFrame(raf); };
156 |   }, []);
157 |
158 |   function FanGroup({ pos, radius, fanRef }) {
159 |     const blades = [];
160 |     for (let i = 0; i < 8; i++) {
161 |       const angle = (i / 8) * Math.PI * 2;
162 |       blades.push(
163 |         <mesh
164 |           key={i}
165 |           position=[[Math.sin(angle) * radius * 0.5, 0, Math.cos(angle) * radius * 0.5]]
166 |           rotation=[[0, -angle, Math.PI / 6]]
167 |         >
168 |           <boxGeometry args=[[0.06, 0.01, 0.2]] />
169 |           <meshStandardMaterial color="#555" transparent opacity={0.85} />
170 |         </mesh>
171 |       );
172 |     }
173 |     return (
174 |       <group position={pos}>
175 |         <mesh position=[[0, 0, 0]]>
176 |         <cylinderGeometry args=[[0.1, 0.1, 0.03, 16]] />
177 |         <meshStandardMaterial color="#333" />
178 |       </mesh>
179 |       <group ref={fanRef}>
180 |         {blades}
181 |       </group>
182 |       <mesh position=[[0, 0, 0]]>
183 |       <torusGeometry args=[[radius, 0.015, 8, 24]] />
184 |       <meshStandardMaterial color="#444" transparent opacity={0.5} />
185 |     </mesh>
186 |   </group>
187 | );
188 | }
189 |
190 | const finPositions = [-0.6, -0.2, 0.2, 0.6];
191 | const pciePinPositions = [];
192 | for (let i = 0; i < 12; i++) pciePinPositions.push((i - 6) * 0.09);
193 |
194 | return (
195 |   <group scale={isHovered ? 1.03 : 1}>
196 |     <mesh position=[[0, 0, 0]]>
197 |     <boxGeometry args=[[2.8, 0.06, 1.2]] />
198 |     <meshStandardMaterial color="#2d2d2d" roughness={0.8} />
199 |   </mesh>
200 |   <mesh position=[[0, 0.04, 0]]>
201 |   <boxGeometry args=[[2.7, 0.01, 1.1]] />
202 |   <meshStandardMaterial color="#1a1a1a" roughness={0.9} />
203 | </mesh>
204 |   {finPositions.map((x) => (
205 |     <mesh key={`fin-${x}`} position=[[x, 0.12, 0]]>
206 |     <boxGeometry args=[[0.04, 0.15, 0.9]] />
207 |     <meshStandardMaterial color="#c0c0c0" metalness={0.6} roughness={0.3} />
208 |   </mesh>
209 | )))
210 |   <mesh position=[[0, 0.08, 0]]>
211 |   <boxGeometry args=[[2.0, 0.03, 0.8]] />
212 |   <meshStandardMaterial color="#a0a0a0" metalness={0.5} roughness={0.4} />
213 | </mesh>
214 |   <FanGroup pos=[[0.7, 0.18, 0]] radius={0.28} fanRef={fanRef1} />
215 |   <FanGroup pos=[[0, 0.18, 0]] radius={0.28} fanRef={fanRef2} />
216 |   <FanGroup pos=[[0.7, 0.18, 0]] radius={0.28} fanRef={fanRef3} />
217 |   <mesh position=[[0.5, -0.06, 0]]>
218 |   <boxGeometry args=[[1.2, 0.02, 0.08]] />
219 |   <meshStandardMaterial color="#b8860b" metalness={0.5} roughness={0.4} />
220 | </mesh>
221 |   {pciePinPositions.map((offset, i) => (
222 |     <mesh key={`pciepin-${i}`} position=[[0.5 + offset, -0.08, 0]]>
223 |     <boxGeometry args=[[0.07, 0.02, 0.06]] />
224 |     <meshStandardMaterial color="#d4a017" metalness={0.7} roughness={0.3} />
225 |   </mesh>
226 | )))
227 |   <mesh position=[[1.25, 0.04, 0]]>
228 |   <boxGeometry args=[[0.12, 0.025, 0.14]] />
229 |   <meshStandardMaterial color="#333" />

```

```

230 |     </mesh>
231 |     <mesh position=[0, 0.06, 0]>
232 |       <boxGeometry args=[0.3, 0.005, 0.3] />
233 |       <meshStandardMaterial color="#444" />
234 |     </mesh>
235 |   </group>
236 | );
237 | }
238 |
239 | /* ---- RAM 3D Model ---- */
240 | function RAMModel({ brand, isHovered }) {
241 |   const heatspreaderColor = BRAND_COLORS[brand?.toLowerCase()] || '#0096d6';
242 |   const finPositions = [];
243 |   for (let i = 0; i < 8; i++) finPositions.push(0.05 + i * 0.065);
244 |   const goldPinPositions = [];
245 |   for (let i = 0; i < 10; i++) goldPinPositions.push((i - 4.5) * 0.01);
246 |   const chipPositions = [];
247 |   for (let side = 0; side < 2; side++) {
248 |     for (let i = 0; i < 4; i++) {
249 |       chipPositions.push({ x: side === 0 ? -0.035 : 0.035, y: 0.32 + i * 0.07 });
250 |     }
251 |   }
252 |
253 |   return (
254 |     <group scale={isHovered ? 1.05 : 1}>
255 |       <mesh position=[0, 0.15, 0]>
256 |         <boxGeometry args=[0.12, 0.6, 0.025] />
257 |         <meshStandardMaterial color="#2a2a2a" roughness={0.8} />
258 |       </mesh>
259 |       <mesh position=[-0.075, 0.15, 0]>
260 |         <boxGeometry args=[0.025, 0.55, 0.04] />
261 |         <meshStandardMaterial color={heatspreaderColor} metalness={0.4} roughness={0.4} />
262 |       </mesh>
263 |       <mesh position=[0.075, 0.15, 0]>
264 |         <boxGeometry args=[0.025, 0.55, 0.04] />
265 |         <meshStandardMaterial color={heatspreaderColor} metalness={0.4} roughness={0.4} />
266 |       </mesh>
267 |       {finPositions.map((y, i) => (
268 |         <mesh key={`fin-${i}`} position=[-0.09, y, 0]>
269 |           <boxGeometry args=[0.012, 0.04, 0.045] />
270 |           <meshStandardMaterial color={heatspreaderColor} metalness={0.3} roughness={0.5} opacity={0.7} transp...
271 |         </mesh>
272 |       ))}
273 |       {finPositions.map((y, i) => (
274 |         <mesh key={`finr-${i}`} position=[0.09, y, 0]>
275 |           <boxGeometry args=[0.012, 0.04, 0.045] />
276 |           <meshStandardMaterial color={heatspreaderColor} metalness={0.3} roughness={0.5} opacity={0.7} transp...
277 |         </mesh>
278 |       ))}
279 |       <mesh position=[0, -0.18, 0]>
280 |         <boxGeometry args=[0.11, 0.08, 0.018] />
281 |         <meshStandardMaterial color="#d4a017" metalness={0.8} roughness={0.2} />
282 |       </mesh>
283 |       {goldPinPositions.map((offset, i) => (
284 |         <mesh key={`goldpin-${i}`} position=[offset, -0.2, 0]>
285 |           <boxGeometry args=[0.006, 0.06, 0.016] />
286 |           <meshStandardMaterial color="#b8860b" metalness={0.9} roughness={0.1} />
287 |         </mesh>
288 |       ))}
289 |       {chipPositions.map((p, i) => (
290 |         <mesh key={`chip-${i}`} position=[p.x, p.y, 0]>
291 |           <boxGeometry args=[0.025, 0.025, 0.02] />
292 |           <meshStandardMaterial color="#111" />
293 |         </mesh>
294 |       ))}
295 |       <mesh position=[0, 0.35, 0.018]>
296 |         <boxGeometry args=[0.06, 0.06, 0.002] />
297 |         <meshStandardMaterial color="#f0f0f0" roughness={0.9} />
298 |       </mesh>
299 |     </group>
300 |   );
301 | }
302 |
303 | /* ---- SSD 3D Model ---- */
304 | function SSDModel({ isHovered }) {
305 |   const nandPositions = [-0.5, 0.2, 0.55];
306 |   const m2pinPositions = [];
307 |   for (let i = 0; i < 8; i++) m2pinPositions.push((i - 3.5) * 0.028);
308 |
309 |   return (
310 |     <group rotation=[0, 0, 0] scale={isHovered ? 1.05 : 1}>
311 |       <mesh position=[0, 0, 0]>
312 |         <boxGeometry args=[1.6, 0.015, 0.25] />
313 |         <meshStandardMaterial color="#1a1a1a" roughness={0.85} />
314 |       </mesh>
315 |       <mesh position=[-0.15, 0.015, 0]>
316 |         <boxGeometry args=[0.2, 0.02, 0.15] />
317 |         <meshStandardMaterial color="#333" roughness={0.7} />
318 |       </mesh>
319 |       {nandPositions.map((x) => (
320 |         <mesh key={`nand-${x}`} position=[x, 0.015, 0]>
321 |           <boxGeometry args=[0.18, 0.02, 0.16] />
322 |           <meshStandardMaterial color="#2a2a2a" roughness={0.7} />
323 |         </mesh>
324 |       ))}
325 |       <mesh position=[-0.15, 0.025, 0]>
326 |         <boxGeometry args=[0.12, 0.005, 0.08] />
327 |         <meshStandardMaterial color="#444" />
328 |       </mesh>
329 |       <mesh position=[0.85, 0, 0]>
330 |         <boxGeometry args=[0.04, 0.012, 0.24] />
331 |         <meshStandardMaterial color="#d4a017" metalness={0.8} roughness={0.2} />
332 |       </mesh>
333 |       {m2pinPositions.map((offset, i) => (
334 |         <mesh key={`m2pin-${i}`} position=[0.87, 0, offset]>
335 |           <boxGeometry args=[0.025, 0.01, 0.02] />
336 |           <meshStandardMaterial color="#b8860b" metalness={0.9} roughness={0.1} />
337 |         </mesh>
338 |       ))}
339 |       <mesh position=[-0.75, 0, 0]>
340 |         <cylinderGeometry args=[0.03, 0.03, 0.02, 12] />
341 |         <meshStandardMaterial color="#666" metalness={0.3} roughness={0.5} />
342 |       </mesh>
343 |       <mesh position=[0.1, 0.018, 0.08]>
344 |         <boxGeometry args=[0.3, 0.003, 0.01] />
345 |         <meshStandardMaterial color="#555" />
346 |       </mesh>
347 |       <Text
348 |         position=[0.1, 0.022, 0.11]
349 |         fontSize={0.04}
350 |         color="#fff"
351 |         anchorX="center"
352 |         anchorY="middle"
353 |       >

```

```

354 |         M.2 NVMe
355 |     </Text>
356 | </group>
357 | );
358 | }
359 |
360 | /* ---- PSU 3D Model ---- */
361 | function PSUModel({ isHovered }) {
362 |     const fanBladesRef = useRef();
363 |
364 |     useEffect(() => {
365 |         let running = true;
366 |         function animate() {
367 |             if (!running) return;
368 |             if (fanBladesRef.current) fanBladesRef.current.rotation.y += 0.01;
369 |             requestAnimationFrame(animate);
370 |         }
371 |         const raf = requestAnimationFrame(animate);
372 |         return () => { running = false; cancelAnimationFrame(raf); };
373 |     }, []);
374 |
375 |     const ventPositions = [];
376 |     for (let i = 0; i < 8; i++) ventPositions.push(-0.8 + i * 0.22);
377 |
378 |     const screwPositions = [
379 |         [-0.95, 0.25, 0.65],
380 |         [-0.95, -0.25, 0.65],
381 |         [-0.95, 0.25, -0.65],
382 |         [-0.95, -0.25, -0.65]
383 |     ];
384 |
385 |     const fanBladePositions = [];
386 |     for (let i = 0; i < 7; i++) {
387 |         const angle = (i / 7) * Math.PI * 2;
388 |         fanBladePositions.push({
389 |             x: Math.sin(angle) * 0.18,
390 |             z: Math.cos(angle) * 0.18,
391 |             angle
392 |         });
393 |     }
394 |
395 |     const connPinPositions = [];
396 |     for (let i = 0; i < 6; i++) connPinPositions.push(-0.1 + i * 0.06);
397 |
398 |     return (
399 |         <group scale={isHovered ? 1.03 : 1}>
400 |             <mesh position={[0, 0, 0]}>
401 |                 <boxGeometry args={[2.2, 0.6, 1.4]} />
402 |                 <meshStandardMaterial color="#2a2a2a" metalness={0.3} roughness={0.6} />
403 |             </mesh>
404 |             <mesh position={[0, 0.32, 0]}>
405 |                 <boxGeometry args={[2.18, 0.02, 1.38]} />
406 |                 <meshStandardMaterial color="#333" metalness={0.2} roughness={0.7} />
407 |             </mesh>
408 |             <mesh position={[0, 0.2, 0.72]}>
409 |                 <boxGeometry args={[0.6, 0.15, 0.01]} />
410 |                 <meshStandardMaterial color="#1a1a1a" />
411 |             </mesh>
412 |             <Text
413 |                 position={[0, 0.2, 0.74]}
414 |                 fontSize={0.06}
415 |                 color="#f59e0b"
416 |                 anchorX="center"
417 |                 anchorY="middle"
418 |             >
419 |                 PSU
420 |             </Text>
421 |             <mesh position={[0, 0.32, 0]}>
422 |                 <torusGeometry args={[0.35, 0.015, 8, 24]} />
423 |                 <meshStandardMaterial color="#111" metalness={0.2} />
424 |             </mesh>
425 |             {[0, 1].map((i) => (
426 |                 <mesh key={`grill-${i}`} position={[0, 0.33, 0]} rotation={[0, 0, (Math.PI / 2) * i]}>
427 |                     <boxGeometry args={[0.65, 0.008, 0.008]} />
428 |                     <meshStandardMaterial color="#111" />
429 |                 </mesh>
430 |             ))}
431 |             <group ref={fanBladesRef} position={[0, 0.32, 0]}>
432 |                 {fanBladePositions.map((p, i) => (
433 |                     <mesh key={i} position={[p.x, 0, p.z]} rotation={[0, -p.angle, 0.3]}>
434 |                         <boxGeometry args={[0.04, 0.005, 0.15]} />
435 |                         <meshStandardMaterial color="#444" transparent opacity={0.8} />
436 |                     </mesh>
437 |                 ))}
438 |                 <mesh position={[0, 0, 0]}>
439 |                     <cylinderGeometry args={[0.06, 0.06, 0.01, 12]} />
440 |                     <meshStandardMaterial color="#333" />
441 |                 </mesh>
442 |             </group>
443 |             <mesh position={[1.12, 0.1, 0]}>
444 |                 <boxGeometry args={[0.03, 0.2, 0.3]} />
445 |                 <meshStandardMaterial color="#111" />
446 |             </mesh>
447 |             {connPinPositions.map((y, i) => (
448 |                 <mesh key={`connpin-${i}`} position={[1.14, y, 0]}>
449 |                     <cylinderGeometry args={[0.015, 0.015, 0.04, 6]} />
450 |                     <meshStandardMaterial color="#555" />
451 |                 </mesh>
452 |             ))}
453 |             {ventPositions.map((x, i) => (
454 |                 <mesh key={`vent-${i}`} position={[x, 0, 0.72]}>
455 |                     <boxGeometry args={[0.12, 0.01, 0.01]} />
456 |                     <meshStandardMaterial color="#1a1a1a" />
457 |                 </mesh>
458 |             ))}
459 |             {screwPositions.map((pos, i) => (
460 |                 <mesh key={`screw-${i}`} position={pos}>
461 |                     <cylinderGeometry args={[0.02, 0.02, 0.02, 8]} />
462 |                     <meshStandardMaterial color="#666" metalness={0.5} />
463 |                 </mesh>
464 |             ))}
465 |             <mesh position={[1.12, 0.25, 0.12]}>
466 |                 <boxGeometry args={[0.02, 0.04, 0.025]} />
467 |                 <meshStandardMaterial color="#333" />
468 |             </mesh>
469 |         </group>
470 |     );
471 | }
472 |
473 | /* ---- Mainboard 3D Model ---- */
474 | function MainboardModel({ isHovered }) {
475 |     const socketHolePositions = [];
476 |     for (let x = 0; x < 5; x++) {
477 |         for (let z = 0; z < 5; z++) {

```

```

478 |     socketHolePositions.push({ x: -0.25 + x * 0.125, z: 0.2 + z * 0.125 });
479 | }
480 | }
481 |
482 | const mountPositions = [
483 |   [-1.4, -1.1],
484 |   [-1.4, 1.1],
485 |   [1.4, -1.1],
486 |   [1.4, 1.1],
487 |   [0, -1.1],
488 |   [0, 1.1]
489 | ];
490 |
491 | const ioPortPositions = [];
492 | for (let i = 0; i < 4; i++) ioPortPositions.push(0.4 + i * 0.1);
493 |
494 | const sataPositions = [];
495 | for (let i = 0; i < 4; i++) sataPositions.push(-0.85 + i * 0.1);
496 |
497 | const capClusterPositions = [];
498 | for (let i = 0; i < 3; i++) capClusterPositions.push(-0.1 + i * 0.08);
499 |
500 | return (
501 |   <group scale={isHovered ? 1.02 : 1}>
502 |     <mesh position={[0, 0, 0]}>
503 |       <boxGeometry args={[3.0, 0.02, 2.4]} />
504 |       <meshStandardMaterial color="#1a3a1a" roughness={0.9} />
505 |     </mesh>
506 |     <mesh position={[0, 0.015, 0]}>
507 |       <boxGeometry args={[2.9, 0.005, 2.3]} />
508 |       <meshStandardMaterial color="#1e4a1e" roughness={0.95} />
509 |     </mesh>
510 |     <mesh position={[0, 0.03, 0.3]}>
511 |       <boxGeometry args={[0.7, 0.015, 0.7]} />
512 |       <meshStandardMaterial color="#c0c0c0" metalness={0.4} roughness={0.5} />
513 |     </mesh>
514 |     {socketHolePositions.map((p, i) => (
515 |       <mesh key={`socket-hole-${i}`} position={[p.x, 0.04, p.z]}>
516 |         <cylinderGeometry args={[0.015, 0.015, 0.02, 6]} />
517 |         <meshStandardMaterial color="#333" />
518 |       </mesh>
519 |     ))}
520 |     <mesh position={[0.38, 0.04, 0.3]}>
521 |       <boxGeometry args={[0.04, 0.005, 0.1]} />
522 |       <meshStandardMaterial color="#888" metalness={0.5} />
523 |     </mesh>
524 |     <Text
525 |       position={[0, 0.05, 0.75]}
526 |       fontSize={0.08}
527 |       color="#fff"
528 |       anchorX="center"
529 |       anchorY="middle"
530 |     >
531 |       CPU
532 |     </Text>
533 |     {[0, 1].map((i) => (
534 |       <mesh key={`ramslot-${i}`} position={[0.75 - i * 0.07, 0.03, 0.5]}>
535 |         <boxGeometry args={[0.05, 0.015, 0.35]} />
536 |         <meshStandardMaterial color="#333" roughness={0.7} />
537 |       </mesh>
538 |     ))}
539 |     {[0, 1].map((i) => (
540 |       <mesh key={`ramslot-r-${i}`} position={[0.75 + i * 0.07, 0.03, 0.5]}>
541 |         <boxGeometry args={[0.05, 0.015, 0.35]} />
542 |         <meshStandardMaterial color="#333" roughness={0.7} />
543 |       </mesh>
544 |     ))}
545 |     <Text
546 |       position=[[-0.75, 0.05, 0.7]]
547 |       fontSize={0.05}
548 |       color="#fff"
549 |       anchorX="center"
550 |       anchorY="middle"
551 |     >
552 |       RAM1
553 |     </Text>
554 |     <Text
555 |       position={[0.75, 0.05, 0.7]}
556 |       fontSize={0.05}
557 |       color="#fff"
558 |       anchorX="center"
559 |       anchorY="middle"
560 |     >
561 |       RAM2
562 |     </Text>
563 |     {[0, 1, 2].map((i) => (
564 |       <mesh key={`pcie-${i}`} position={[0, 0.03, -0.4 - i * 0.25]}>
565 |         <boxGeometry args={[1.2, 0.012, 0.06]} />
566 |         <meshStandardMaterial color="#444" roughness={0.6} />
567 |       </mesh>
568 |     ))}
569 |     <Text
570 |       position={[0, 0.05, -0.4]}
571 |       fontSize={0.05}
572 |       color="#fff"
573 |       anchorX="center"
574 |       anchorY="middle"
575 |     >
576 |       PCIE
577 |     </Text>
578 |     <mesh position=[[-0.2, 0.03, -0.7]]>
579 |       <boxGeometry args={[0.25, 0.025, 0.25]} />
580 |       <meshStandardMaterial color="#c0c0c0" metalness={0.5} roughness={0.4} />
581 |     </mesh>
582 |     <mesh position=[[-0.2, 0.04, -0.7]]>
583 |       <boxGeometry args={[0.15, 0.005, 0.15]} />
584 |       <meshStandardMaterial color="#888" />
585 |     </mesh>
586 |     <mesh position={[1.52, 0.02, 0.6]}>
587 |       <boxGeometry args={[0.02, 0.015, 0.5]} />
588 |       <meshStandardMaterial color="#a0a0a0" metalness={0.4} roughness={0.5} />
589 |     </mesh>
590 |     {ioPortPositions.map((z, i) => (
591 |       <mesh key={`ioport-${i}`} position={[1.55, 0.025, z]}>
592 |         <boxGeometry args={[0.01, 0.01, 0.06]} />
593 |         <meshStandardMaterial color="#555" />
594 |       </mesh>
595 |     ))}
596 |     {sataPositions.map((z, i) => (
597 |       <mesh key={`sata-${i}`} position={[1.3, 0.025, z]}>
598 |         <boxGeometry args={[0.04, 0.015, 0.02]} />
599 |         <meshStandardMaterial color="#444" />
600 |       </mesh>
601 |     ))}

```

```

602 |     <mesh position={[[-0.5, 0.025, -0.1]]}>
603 |     <boxGeometry args={[0.5, 0.008, 0.03]} />
604 |     <meshStandardMaterial color="#555" />
605 |   </mesh>
606 |   {capClusterPositions.map((x, i) => (
607 |     <mesh key={`capclust-${i}`} position={[x, 0.025, 0.3]}>
608 |     <cylinderGeometry args={[0.025, 0.025, 0.015, 8]} />
609 |     <meshStandardMaterial color="#666" />
610 |   </mesh>
611 | ))}
612 |   <mesh position={[1.4, 0.025, -0.1]}>
613 |     <boxGeometry args={[0.02, 0.015, 0.2]} />
614 |     <meshStandardMaterial color="#333" />
615 |   </mesh>
616 |   {mountPositions.map(([x, z], i) => (
617 |     <mesh key={`mount-${i}`} position={[x, 0, z]}>
618 |     <cylinderGeometry args={[0.03, 0.03, 0.01, 8]} />
619 |     <meshStandardMaterial color="#888" metalness={0.4} />
620 |   </mesh>
621 | ))}
622 |   <mesh position={[[-0.5, 0.035, 0.95]]>
623 |     <boxGeometry args={[0.3, 0.025, 0.1]} />
624 |     <meshStandardMaterial color="#999" metalness={0.5} roughness={0.4} />
625 |   </mesh>
626 | </group>
627 | );
628 | }
629 |
630 | /* ---- Model Selector ---- */
631 | function ModelSelector({ componentType, brand, socket, isHovered }) {
632 |   switch (componentType) {
633 |     case 'CPU':
634 |       return <CPUModel brand={brand} socket={socket} isHovered={isHovered} />;
635 |     case 'GPU':
636 |       return <GPUModel isHovered={isHovered} />;
637 |     case 'RAM':
638 |       return <RAMModel brand={brand} isHovered={isHovered} />;
639 |     case 'SSD':
640 |       return <SSDModel isHovered={isHovered} />;
641 |     case 'PSU':
642 |       return <PSUModel isHovered={isHovered} />;
643 |     case 'MAINBOARD':
644 |       return <MainboardModel isHovered={isHovered} />;
645 |     default:
646 |       return null;
647 |   }
648 | }
649 |
650 | /* ---- Scene ---- */
651 | function Scene({ componentType, brand, socket, onReady, isHovered, setHovered }) {
652 |   const controlsRef = useRef();
653 |   const [interacted, setInteracted] = useState(false);
654 |
655 |   const handlePointerDown = useCallback(() => {
656 |     setInteracted(true);
657 |   }, []);
658 |
659 |   useEffect(() => {
660 |     if (onReady) onReady();
661 |   }, [onReady]);
662 |
663 |   return (
664 |     <>
665 |       <ambientLight intensity={0.5} />
666 |       <directionalLight position={[5, 8, 5]} intensity={1.2} castShadow shadow-mapSize-width={512} shadow-mapS...
667 |       <directionalLight position={[-3, 4, -3]} intensity={0.4} />
668 |       <pointLight position={[0, 3, 0]} intensity={0.3} />
669 |       <pointLight position={[0, -2, 2]} intensity={0.15} color="#4488ff" />
670 |       <OrbitControls
671 |         ref={controlsRef}
672 |         enablePan={true}
673 |         enableZoom={true}
674 |         enableRotate={true}
675 |         autoRotate={!interacted}
676 |         autoRotateSpeed={2.0}
677 |         minDistance={0.5}
678 |         maxDistance={5.0}
679 |         onPointerDown={handlePointerDown}
680 |       />
681 |       <gridHelper args={[6, 12, '#333', '#222']} position={[0, -0.5, 0]} opacity={0.3} transparent />
682 |       <group
683 |         position={[0, 0, 0]}
684 |         onPointerEnter={() => setHovered(true)}
685 |         onPointerLeave={() => setHovered(false)}
686 |       >
687 |         <Suspense fallback={null}>
688 |           <ModelSelector componentType={componentType} brand={brand} socket={socket} isHovered={isHovered} />
689 |         </Suspense>
690 |       </group>
691 |       {isHovered && (
692 |         <mesh position={[0, 0, 0]}>
693 |           <sphereGeometry args={[0.5, 16, 16]} />
694 |           <meshStandardMaterial
695 |             color={PART_COLORS[componentType] || '#00d4aa'}
696 |             transparent
697 |             opacity={0.08}
698 |           />
699 |         </mesh>
700 |       )}
701 |     </>
702 |   );
703 | }
704 |
705 | /* ---- Main 3D Canvas ---- */
706 | function ComponentPreview3DCanvas({ componentType, brand, socket, onReady }) {
707 |   const [isHovered, setHovered] = useState(false);
708 |   const [clicked, setClicked] = useState(false);
709 |   const containerRef = useRef(null);
710 |   const [dimensions, setDimensions] = useState({ width: 300, height: 300 });
711 |
712 |   useEffect(() => {
713 |     const el = containerRef.current;
714 |     if (!el) return;
715 |     const observer = new ResizeObserver((entries) => {
716 |       for (const entry of entries) {
717 |         const { width, height } = entry.contentRect;
718 |         if (width > 0 && height > 0) {
719 |           setDimensions({ width, height });
720 |         }
721 |       }
722 |     });
723 |     observer.observe(el);
724 |     return () => observer.disconnect();
725 |   }, []);

```

```

726 | const handleClick = useCallback(() => {
727 |   setClicked((prev) => !prev);
728 | }, []);
729 |
730 |
731 | return (
732 |   <div
733 |     ref={containerRef}
734 |     style={{
735 |       width: '100%',
736 |       height: '100%',
737 |       position: 'relative',
738 |       borderRadius: 'inherit',
739 |       overflow: 'hidden'
740 |     }}
741 |   >
742 |     <Canvas
743 |       camera={{ position: [2, 1.5, 2.5], fov: 40 }}
744 |       dpr={[1, 2]}
745 |       gl={{ antialias: true, alpha: true }}
746 |       style={{ width: '100%', height: '100%' }}
747 |       onCreated={({ gl }) => { gl.setClearColor(0x000000, 0); }}
748 |     >
749 |       <Scene
750 |         componentType={componentType}
751 |         brand={brand}
752 |         socket={socket}
753 |         onReady={onReady}
754 |         isHovered={isHovered}
755 |         setHovered={setHovered}
756 |       />
757 |     </Canvas>
758 |     <div
759 |       onClick={handleClick}
760 |       style={{
761 |         position: 'absolute',
762 |         top: 0,
763 |         left: 0,
764 |         width: '100%',
765 |         height: '100%',
766 |         cursor: 'pointer'
767 |       }}
768 |     />
769 |     {clicked && (
770 |       <div style={{
771 |         position: 'absolute',
772 |         bottom: 12,
773 |         left: '50%',
774 |         transform: 'translateX(-50%)',
775 |         background: 'rgba(0,0,0,0.7)',
776 |         color: PART_COLORS[componentType] || '#fff',
777 |         padding: '4px 12px',
778 |         borderRadius: 6,
779 |         fontSize: 13,
780 |         fontWeight: 600,
781 |         fontFamily: 'monospace',
782 |         pointerEvents: 'none',
783 |         backdropFilter: 'blur(4px)'
784 |       }}
785 |     > {componentType} {brand ? '\u00b7 ${brand}` : ''}
786 |     </div>
787 |   )}
788 | </div>
789 | );
790 | }
791 |
792 | /* ---- Dynamic import wrapper for SSR ---- */
793 | const DynamicComponentPreview3D = dynamic(
794 |   () => Promise.resolve({ default: ComponentPreview3DCanvas }),
795 |   { ssr: false }
796 | );
797 |
798 | /* ---- Exported default wrapper ---- */
799 | export default function ComponentPreview3D({
800 |   componentType = 'CPU',
801 |   componentId,
802 |   brand,
803 |   model,
804 |   style,
805 |   onReady
806 | }) {
807 |   const color = PART_COLORS[componentType] || '#666';
808 |
809 |   return (
810 |     <div style={{
811 |       background: 'var(--bg-surface, #111)',
812 |       border: '1px solid ${color}30',
813 |       borderRadius: 12,
814 |       overflow: 'hidden',
815 |       width: style?.width || 300,
816 |       height: style?.height || 300,
817 |       position: 'relative',
818 |       display: 'flex',
819 |       alignItems: 'center',
820 |       justifyContent: 'center',
821 |       ...style
822 |     }}>
823 |       <Suspense fallback={<LoadingSpinner />}>
824 |         <DynamicComponentPreview3D
825 |           componentType={componentType}
826 |           brand={brand || model?.brand}
827 |           socket={model?.socket}
828 |           onReady={onReady}
829 |         />
830 |       </Suspense>
831 |     </div>
832 |   );
833 | }
834 |
835 | export { PART_COLORS, ErrorFallback, LoadingSpinner, DynamicComponentPreview3D, CPUModel, GPUModel, RAMModel, ...
836 |

```

File: components/Components.tsx (640 lines, 22.0 KB)

```

1 | 'use client';
2 |
3 | import { useRef, useState, useCallback, useMemo, useEffect } from 'react';
4 | import { useFrame, useThree } from '@react-three/fiber';
5 | import { Html, Text, Float, Sparkles, RoundedBox } from '@react-three/drei';
6 | import * as THREE from 'three';
7 | import { useAssemblyStore, type ComponentType } from '@lib/useStore';
8 |
9 | const COMPONENT_COLORS: Record<ComponentType, { base: string; emissive: string }> = {

```



```

10  cpu: { base: '#c0c0c0', emissive: '#00d4aa' },
11  cooler: { base: '#2a2a3a', emissive: '#00aaff' },
12  ram: { base: '#1a1a2e', emissive: '#6366f1' },
13  gpu: { base: '#2d2d2d', emissive: '#ef4444' },
14  psu: { base: '#1a1a1a', emissive: '#f59e0b' },
15  ssd: { base: '#1a1a2e', emissive: '#22c55e' },
16  motherboard: { base: '#1a3a1a', emissive: '#8b5cf6' },
17  };
18
19  const SNAP_DISTANCE = 0.15;
20
21  const WORKBENCH_POSITIONS: Record<ComponentType, [number, number, number]> = {
22    cpu: [-1.4, 0.12, 1.8],
23    cooler: [-0.5, 0.12, 1.8],
24    ram: [0.5, 0.12, 1.8],
25    gpu: [1.4, 0.12, 1.8],
26    psu: [-1.0, 0.12, -1.8],
27    ssd: [1.0, 0.12, -1.8],
28    motherboard: [0, -0.15, 0],
29  };
30
31  function GoldPins({ count = 20, width = 0.35 }) {
32    const pins = useMemo(() => {
33      const arr = [];
34      for (let i = 0; i < count; i++) {
35        arr.push((i / count - 0.5) * width);
36      }
37      return arr;
38    }, [count, width]);
39    return (
40      <>
41        {pins.map((x, i) => (
42          <mesh key={i} position={[x, -0.02, 0]}>
43            <boxGeometry args={[0.008, 0.015, 0.01]} />
44            <meshStandardMaterial color="#d4a017" metalness={0.9} roughness={0.15} />
45          </mesh>
46        ))}
47      </>
48    );
49  }
50
51  function CpuModel({ hovered, position, installed }: { hovered: boolean; position: [number, number, number]; in...
52    return (
53      <group position={position}>
54        <RoundedBox args={[0.55, 0.04, 0.55]} radius={0.01} smoothness={2}>
55          <meshStandardMaterial
56            color="#c0c0c0"
57            metalness={0.75}
58            roughness={0.25}
59            emissive={hovered ? '#00d4aa' : '#000'}
60            emissiveIntensity={hovered ? 0.4 : 0}
61          />
62        </RoundedBox>
63        <mesh position={[0, 0.03, 0]}>
64          <RoundedBox args={[0.48, 0.005, 0.48]} radius={0.008}>
65            <meshStandardMaterial color="#e8e8e8" metalness={0.6} roughness={0.3} />
66          </RoundedBox>
67        </mesh>
68        <mesh position={[0.24, 0.035, 0.24]} rotation={[0, 0, Math.PI / 4]}>
69          <RoundedBox args={[0.025, 0.004, 0.025]} radius={0.003}>
70            <meshStandardMaterial color="#ef4444" metalness={0.3} roughness={0.5} />
71          </RoundedBox>
72        </mesh>
73        <mesh position={[0, -0.02, 0]}>
74          <RoundedBox args={[0.56, 0.012, 0.56]} radius={0.005}>
75            <meshStandardMaterial color="#2a1a0a" roughness={0.9} />
76          </RoundedBox>
77        </mesh>
78        <mesh position={[0, -0.035, 0]}>
79          <boxGeometry args={[0.4, 0.008, 0.4]} />
80          <meshStandardMaterial color="#d4a017" metalness={0.7} roughness={0.3} />
81        </mesh>
82        <GoldPins count={16} width={0.28} />
83        {installed && (
84          <pointLight position={[0, 0.06, 0]} intensity={0.2} color="#00d4aa" distance={0.3} />
85        )}
86      </group>
87    );
88  }
89
90  function CoolerModel({ hovered, position, installed, fanSpeed }: {
91    hovered: boolean; position: [number, number, number];
92    installed: boolean; fanSpeed: number;
93  }) {
94    const fanRef = useRef<THREE.Group>(null);
95    useFrame((_, delta) => {
96      if (fanRef.current && fanSpeed > 0) {
97        fanRef.current.rotation.y += delta * fanSpeed * 12;
98      }
99    });
100
101    return (
102      <group position={position}>
103        <RoundedBox args={[0.7, 0.03, 0.7]} radius={0.02}>
104          <meshStandardMaterial color="#1a1a1a" metalness={0.3} roughness={0.6} />
105        </RoundedBox>
106        <mesh position={[0, 0.03, 0]}>
107          <cylinderGeometry args={[0.3, 0.3, 0.04, 32]} />
108          <meshStandardMaterial
109            color="#2a2a3a"
110            metalness={0.4}
111            roughness={0.5}
112            emissive={hovered ? '#00aaff' : '#000'}
113            emissiveIntensity={hovered ? 0.3 : 0}
114          />
115        </mesh>
116        <group ref={fanRef} position={[0, 0.05, 0]}>
117          {Array.from({ length: 9 }).map((_, i) => {
118            const angle = (i / 9) * Math.PI * 2;
119            return (
120              <mesh
121                key={i}
122                position={[Math.sin(angle) * 0.2, 0, Math.cos(angle) * 0.2]}
123                rotation={[0, -angle, 0.3]}
124              >
125                <boxGeometry args={[0.04, 0.004, 0.16]} />
126                <meshStandardMaterial
127                  color={installed ? '#6b7280' : '#444'}
128                  metalness={0.3}
129                  roughness={0.5}
130                  transparent
131                  opacity={0.85}
132                />
133              </mesh>

```

```

134 |     });
135 |   });
136 | </group>
137 | <mesh position={[0, 0.045, 0]}>
138 |   <cylinderGeometry args={[0.06, 0.06, 0.008, 16]} />
139 |   <meshStandardMaterial color="#333" metalness={0.5} roughness={0.3}>
140 |     <meshStandardMaterial color="#333" metalness={0.5} roughness={0.3} />
141 |   </meshStandardMaterial>
142 | </mesh>
143 | {installed && (
144 |   <mesh position={[0, 0.07, 0]}>
145 |     <torusGeometry args={[0.28, 0.006, 8, 32]} />
146 |     <meshStandardMaterial
147 |       color="#00aaff"
148 |       emissive="#00aaff"
149 |       emissiveIntensity={0.5}
150 |       transparent
151 |       opacity={0.7}
152 |     </>
153 |   </mesh>
154 | )}
155 | </group>
156 | );
157 | }
158 |
159 | function RamModel({ hovered, position, installed }: {
160 |   hovered: boolean; position: [number, number, number]; installed: boolean;
161 | }) {
162 |   return (
163 |     <group position={position}>
164 |       <RoundedBox args={[0.07, 0.4, 0.018]} radius={0.005}>
165 |         <meshStandardMaterial color="#0a0a12" roughness={0.85} />
166 |       </RoundedBox>
167 |       <mesh position={[-0.045, 0, 0]}>
168 |         <RoundedBox args={[0.02, 0.32, 0.028]} radius={0.004}>
169 |           <meshStandardMaterial
170 |             color={hovered ? '#6366f1' : '#1a1a2e'}
171 |             metalness={0.5}
172 |             roughness={0.4}
173 |             emissive={hovered ? '#6366f1' : '#000'}
174 |             emissiveIntensity={hovered ? 0.5 : 0}
175 |           </>
176 |         </RoundedBox>
177 |       </mesh>
178 |       <mesh position={[0.045, 0, 0]}>
179 |         <RoundedBox args={[0.02, 0.32, 0.028]} radius={0.004}>
180 |           <meshStandardMaterial
181 |             color={hovered ? '#6366f1' : '#1a1a2e'}
182 |             metalness={0.5}
183 |             roughness={0.4}
184 |             emissive={hovered ? '#6366f1' : '#000'}
185 |             emissiveIntensity={hovered ? 0.5 : 0}
186 |           </>
187 |         </RoundedBox>
188 |       </mesh>
189 |       <mesh position={[0, -0.19, 0]}>
190 |         <boxGeometry args={[0.065, 0.05, 0.014]} />
191 |         <meshStandardMaterial color="#d4a017" metalness={0.85} roughness={0.15} />
192 |       </mesh>
193 |       {installed && (
194 |         <mesh position={[0, 0, 0.012]}>
195 |           <planeGeometry args={[0.04, 0.2]} />
196 |           <meshStandardMaterial
197 |             color="#6366f1"
198 |             emissive="#6366f1"
199 |             emissiveIntensity={0.3}
200 |             transparent
201 |             opacity={0.4}
202 |           </>
203 |         </mesh>
204 |       )}
205 |     </group>
206 |   );
207 | }
208 |
209 | function GpuModel({ hovered, position, installed, fanSpeed }: {
210 |   hovered: boolean; position: [number, number, number];
211 |   installed: boolean; fanSpeed: number;
212 | }) {
213 |   const fanRefs = useRef<(THREE.Group | null)[]>([null, null, null]);
214 |   useFrame((_, delta) => {
215 |     fanRefs.current.forEach((ref) => {
216 |       if (ref && fanSpeed > 0) ref.rotation.y += delta * fanSpeed * 15;
217 |     });
218 |   });
219 |
220 |   return (
221 |     <group position={position}>
222 |       <RoundedBox args={[1.5, 0.06, 0.6]} radius={0.01}>
223 |         <meshStandardMaterial
224 |           color="#2d2d2d"
225 |           roughness={0.85}
226 |           emissive={hovered ? '#ef4444' : '#000'}
227 |           emissiveIntensity={hovered ? 0.2 : 0}
228 |         </>
229 |       </RoundedBox>
230 |       <mesh position={[0, 0.04, 0]}>
231 |         <RoundedBox args={[1.4, 0.008, 0.5]} radius={0.005}>
232 |           <meshStandardMaterial color="#1a1a1a" roughness={0.9} />
233 |         </RoundedBox>
234 |       </mesh>
235 |       {[0.5, 0, 0.5].map((x, i) => (
236 |         <group key={i} position={[x, 0.075, 0]}>
237 |           <mesh>
238 |             <torusGeometry args={[0.15, 0.012, 8, 24]} />
239 |             <meshStandardMaterial
240 |               color={installed ? '#ef4444' : hovered ? '#ef4444' : '#444'}
241 |               emissive={installed ? '#ef4444' : '#000'}
242 |               emissiveIntensity={installed ? 0.4 : 0}
243 |               transparent
244 |               opacity={0.7}
245 |             </>
246 |           </mesh>
247 |           <group ref={el} => { fanRefs.current[i] = el; }>
248 |             {Array.from({ length: 9 }).map((_, j) => {
249 |               const angle = (j / 9) * Math.PI * 2;
250 |               return (
251 |                 <mesh>
252 |                   key={j}
253 |                   position={[Math.sin(angle) * 0.1, 0, Math.cos(angle) * 0.1]}
254 |                   rotation={[0, -angle, 0.3]}
255 |                 >
256 |                   <boxGeometry args={[0.025, 0.003, 0.08]} />
257 |                   <meshStandardMaterial color="#555" transparent opacity={0.7} />

```

```

258 |         </mesh>
259 |     });
260 | }}
261 | </group>
262 | </group>
263 | ))}
264 | <mesh position=[0, -0.035, 0]>
265 |   <RoundedBox args=[0.6, 0.012, 0.04] radius={0.005}>
266 |     <meshStandardMaterial color="#d4a017" metalness={0.8} roughness={0.2} />
267 |   </RoundedBox>
268 | </mesh>
269 | <mesh position=[0.75, 0, 0]>
270 |   <RoundedBox args=[0.02, 0.1, 0.08] radius={0.004}>
271 |     <meshStandardMaterial color="#333" metalness={0.3} roughness={0.5} />
272 |   </RoundedBox>
273 | </mesh>
274 | </group>
275 | );
276 | }
277 |
278 | function PsuModel({ hovered, position, installed }: {
279 |   hovered: boolean; position: [number, number, number]; installed: boolean;
280 | }) {
281 |   return (
282 |     <group position={position}>
283 |       <RoundedBox args=[1.0, 0.3, 0.6] radius={0.01}>
284 |         <meshStandardMaterial
285 |           color="#1a1a1a"
286 |           metalness={0.4}
287 |           roughness={0.5}
288 |           emissive={hovered ? '#f59e0b' : '#000'}
289 |           emissiveIntensity={hovered ? 0.25 : 0}
290 |         />
291 |       </RoundedBox>
292 |       <mesh position=[0, 0.14, 0]>
293 |         <torusGeometry args=[0.25, 0.008, 8, 24] />
294 |         <meshStandardMaterial color={hovered ? '#f59e0b' : '#333'} />
295 |       </mesh>
296 |       <mesh position=[0.52, 0.06, 0]>
297 |         <RoundedBox args=[0.02, 0.12, 0.2] radius={0.005}>
298 |           <meshStandardMaterial color="#111" />
299 |         </RoundedBox>
300 |       </mesh>
301 |       <mesh position=[0.54, 0.06, -0.08]>
302 |         <boxGeometry args=[0.015, 0.025, 0.02] />
303 |         <meshStandardMaterial color="#444" />
304 |       </mesh>
305 |       <mesh position=[0.54, 0.06, 0.08]>
306 |         <boxGeometry args=[0.015, 0.025, 0.02] />
307 |         <meshStandardMaterial color="#444" />
308 |       </mesh>
309 |       <mesh position=[-0.52, 0, 0]>
310 |         <RoundedBox args=[0.025, 0.08, 0.08] radius={0.004}>
311 |           <meshStandardMaterial color="#333" />
312 |         </RoundedBox>
313 |       </mesh>
314 |       {installed && (
315 |         <mesh position=[0, 0.16, 0]>
316 |           <pointLight intensity={0.15} color="#f59e0b" distance={0.4} />
317 |         </mesh>
318 |       )}
319 |     </group>
320 |   );
321 | }
322 |
323 | function SsdModel({ hovered, position, installed }: {
324 |   hovered: boolean; position: [number, number, number]; installed: boolean;
325 | }) {
326 |   return (
327 |     <group position={position}>
328 |       <RoundedBox args=[0.7, 0.012, 0.2] radius={0.005}>
329 |         <meshStandardMaterial
330 |           color="#0a0a12"
331 |           roughness={0.85}
332 |           emissive={hovered ? '#22c55e' : '#000'}
333 |           emissiveIntensity={hovered ? 0.4 : 0}
334 |         />
335 |       </RoundedBox>
336 |       <mesh position=[-0.1, 0.008, 0]>
337 |         <RoundedBox args=[0.15, 0.006, 0.1] radius={0.003}>
338 |           <meshStandardMaterial color="#333" roughness={0.7} />
339 |         </RoundedBox>
340 |       </mesh>
341 |       <mesh position=[0.37, 0, 0]>
342 |         <RoundedBox args=[0.03, 0.006, 0.16] radius={0.003}>
343 |           <meshStandardMaterial color="#d4a017" metalness={0.85} roughness={0.15} />
344 |         </RoundedBox>
345 |       </mesh>
346 |       <mesh position=[0.37, 0.004, 0]>
347 |         <boxGeometry args=[0.04, 0.003, 0.18] />
348 |         <meshStandardMaterial color="#b8860b" metalness={0.9} roughness={0.1} />
349 |       </mesh>
350 |     </group>
351 |   );
352 | }
353 |
354 | function MotherboardModel({ hovered, position, installed }: {
355 |   hovered: boolean; position: [number, number, number]; installed: boolean;
356 | }) {
357 |   return (
358 |     <group position={position}>
359 |       <RoundedBox args=[2.2, 0.015, 1.8] radius={0.01}>
360 |         <meshStandardMaterial
361 |           color="#0d2818"
362 |           roughness={0.9}
363 |           emissive={hovered ? '#8b5cf6' : '#000'}
364 |           emissiveIntensity={hovered ? 0.15 : 0}
365 |         />
366 |       </RoundedBox>
367 |       <mesh position=[0.5, 0.01, 0.35]>
368 |         <RoundedBox args=[0.5, 0.006, 0.5] radius={0.005}>
369 |           <meshStandardMaterial color="#b0b0b0" metalness={0.4} roughness={0.5} />
370 |         </RoundedBox>
371 |       </mesh>
372 |       <mesh position=[0.5, 0.014, 0.35]>
373 |         <RoundedBox args=[0.35, 0.003, 0.35] radius={0.003}>
374 |           <meshStandardMaterial color="#888" metalness={0.3} roughness={0.5} />
375 |         </RoundedBox>
376 |       </mesh>
377 |       <mesh position=[-0.65, 0.01, 0.4]>
378 |         <RoundedBox args=[0.2, 0.006, 1.0] radius={0.004}>
379 |           <meshStandardMaterial color="#222" roughness={0.8} />
380 |         </RoundedBox>
381 |       </mesh>

```

```

382 |     <mesh position={[0, 0.01, -0.4]}>
383 |     <RoundedBox args={[1.2, 0.006, 0.06]} radius={0.003}>
384 |     <meshStandardMaterial color="#444" roughness={0.6} />
385 |     </RoundedBox>
386 | </mesh>
387 | <mesh position={[0.7, 0.01, -0.5]}>
388 | <RoundedBox args={[0.06, 0.006, 0.06]} radius={0.003}>
389 | <meshStandardMaterial color="#c0c0c0" metalness={0.5} roughness={0.4} />
390 | </RoundedBox>
391 | </mesh>
392 | <mesh position={[-0.2, 0.012, -0.6]}>
393 | <cylinderGeometry args={[0.1, 0.1, 0.006, 16]} />
394 | <meshStandardMaterial color="#c0c0c0" metalness={0.5} roughness={0.4} />
395 | </mesh>
396 | <mesh position={[0.3, 0.012, -0.6]}>
397 | <cylinderGeometry args={[0.1, 0.1, 0.006, 16]} />
398 | <meshStandardMaterial color="#c0c0c0" metalness={0.5} roughness={0.4} />
399 | </mesh>
400 | </group>
401 | );
402 | }
403 |
404 | function ModelRenderer({ type, hovered, installed, fanSpeed }: {
405 |   type: ComponentType; hovered: boolean;
406 |   installed: boolean; fanSpeed: number;
407 | }) {
408 |   const pos: [number, number, number] = [0, 0, 0];
409 |   switch (type) {
410 |     case 'cpu': return <CpuModel hovered={hovered} position={pos} installed={installed} />;
411 |     case 'cooler': return <CoolerModel hovered={hovered} position={pos} installed={installed} fanSpeed={fanSpe...
412 |     case 'ram': return <RamModel hovered={hovered} position={pos} installed={installed} />;
413 |     case 'gpu': return <GpuModel hovered={hovered} position={pos} installed={installed} fanSpeed={fanSpeed} />;
414 |     case 'psu': return <PsmModel hovered={hovered} position={pos} installed={installed} />;
415 |     case 'ssd': return <SsdModel hovered={hovered} position={pos} installed={installed} />;
416 |     case 'motherboard': return <MotherboardModel hovered={hovered} position={pos} installed={installed} />;
417 |     default: return null;
418 |   }
419 | }
420 |
421 | function DraggableComponent({
422 |   type, slotId, initialPosition,
423 | }: {
424 |   type: ComponentType; slotId: string; initialPosition: [number, number, number];
425 | }) {
426 |   const groupRef = useRef<THREE.Group>(null);
427 |   const [hovered, setHovered] = useState(false);
428 |   const [dragging, setDragging] = useState(false);
429 |   const dragPos = useRef(new THREE.Vector3(...initialPosition));
430 |   const snapTarget = useRef<THREE.Vector3 | null>(null);
431 |   const snapProgress = useRef(0);
432 |   const isAnimating = useRef(false);
433 |
434 |   const { camera, pointer, scene } = useThree();
435 |   const store = useAssemblyStore();
436 |   const installed = store.components.find((c) => c.slotId === slotId)?.installed ?? false;
437 |   const slots = store.slots;
438 |   const fanSpeed = store.fanSpeed;
439 |   const slot = slots.find((s) => s.id === slotId);
440 |
441 |   const colors = COMPONENT_COLORS[type];
442 |
443 |   const trySnap = useCallback((pos: THREE.Vector3) => {
444 |     let closest = null;
445 |     let closestDist = SNAP_DISTANCE;
446 |
447 |     for (const s of slots) {
448 |       if (s.type !== type || s.locked) continue;
449 |       const alreadyOccupied = store.components.some(
450 |         (c) => c.slotId === s.id && c.installed
451 |       );
452 |       if (alreadyOccupied) continue;
453 |       const slotPos = new THREE.Vector3(...s.position);
454 |       const dist = pos.distanceTo(slotPos);
455 |       if (dist < closestDist) {
456 |         closestDist = dist;
457 |         closest = s;
458 |       }
459 |     }
460 |     return closest;
461 |   }, [type, slots, store.components]);
462 |
463 |   const playSnapSound = useCallback(() => {
464 |     try {
465 |       const audioCtx = new (window.AudioContext || (window as any).webkitAudioContext)();
466 |       const osc = audioCtx.createOscillator();
467 |       const gain = audioCtx.createGain();
468 |       osc.connect(gain);
469 |       gain.connect(audioCtx.destination);
470 |       osc.type = 'sine';
471 |       osc.frequency.setValueAtTime(1800, audioCtx.currentTime);
472 |       osc.frequency.exponentialRampToValueAtTime(400, audioCtx.currentTime + 0.15);
473 |       gain.gain.setValueAtTime(0.12, audioCtx.currentTime);
474 |       gain.gain.exponentialRampToValueAtTime(0.001, audioCtx.currentTime + 0.18);
475 |       osc.start(audioCtx.currentTime);
476 |       osc.stop(audioCtx.currentTime + 0.18);
477 |       setTimeout(() => audioCtx.close(), 500);
478 |     } catch {}
479 |   }, []);
480 |
481 |   const onPointerDown = useCallback((e: any) => {
482 |     e.stopPropagation();
483 |     if (installed || isAnimating.current) return;
484 |     setDragging(true);
485 |     store.setDraggedComponent(slotId);
486 |     (e.target as HTMLElement).setPointerCapture(e.pointerId);
487 |   }, [installed, slotId, store]);
488 |
489 |   const onPointerUp = useCallback(() => {
490 |     if (!dragging || isAnimating.current) return;
491 |     setDragging(false);
492 |     store.setDraggedComponent(null);
493 |
494 |     const pos = dragPos.current;
495 |     const target = trySnap(pos);
496 |     if (target) {
497 |       snapTarget.current = new THREE.Vector3(...target.position);
498 |       snapProgress.current = 0;
499 |       isAnimating.current = true;
500 |       playSnapSound();
501 |     }
502 |   }, [dragging, store, slotId, trySnap, playSnapSound]);
503 |
504 |   useFrame((_, delta) => {
505 |     if (isAnimating.current && snapTarget.current) {

```

```

506|     snapProgress.current = Math.min(snapProgress.current + delta * 4, 1);
507|     const eased = 1 - Math.pow(1 - snapProgress.current, 3);
508|     dragPos.current.lerp(snapTarget.current, eased);
509|     if (snapProgress.current >= 1) {
510|         isAnimating.current = false;
511|         store.installComponent(slotId, `comp_${slotId}`);
512|     }
513|     if (groupRef.current) {
514|         groupRef.current.position.copy(dragPos.current);
515|     }
516|     return;
517| }
518|
519| if (dragging && !installed) {
520|     const worldPos = new THREE.Vector3(pointer.x, pointer.y, 0.5).unproject(camera);
521|     const dir = worldPos.sub(camera.position).normalize();
522|     const dist = -camera.position.y / dir.y;
523|     const p = camera.position.clone().add(dir.multiplyScalar(dist));
524|     p.y = Math.max(0.02, p.y);
525|     p.x = Math.max(-2, Math.min(2, p.x));
526|     p.z = Math.max(-2, Math.min(2, p.z));
527|     dragPos.current.copy(p);
528| }
529|
530| if (groupRef.current) {
531|     if (!isAnimating.current) {
532|         groupRef.current.position.lerp(dragPos.current, 0.25);
533|     }
534| }
535| });
536|
537| if (installed && slot) {
538|     dragPos.current.set(slot.position[0], slot.position[1], slot.position[2]);
539|     if (groupRef.current) {
540|         groupRef.current.position.copy(dragPos.current);
541|     }
542| }
543|
544| return (
545|     <group>
546|         ref={groupRef}
547|         position={dragPos.current.toArray()}
548|         scale={dragging ? 1.15 : hovered && !installed ? 1.08 : 1}
549|         >
550|             <mesh>
551|                 visible={!installed}
552|                 position={[0, 0.1, 0]}
553|                 onPointerOver={(e) => { e.stopPropagation(); if (!installed) { setHovered(true); document.body.style.c...
554|                 onPointerOut={() => { if (!installed) { setHovered(false); document.body.style.cursor = 'default'; }}}
555|                 onPointerDown={onPointerDown}
556|                 onPointerUp={onPointerUp}
557|             >
558|                 <boxGeometry args={[0.8, 0.2, 0.8]} />
559|                 <meshBasicMaterial transparent opacity={0} />
560|             </mesh>
561|
562|             <ModelRenderer type={type} hovered={hovered && !installed} installed={installed} fanSpeed={fanSpeed} />
563|
564|             {hovered && !installed && !dragging && (
565|                 <Html position={[0, 0.35, 0]} center style={{ pointerEvents: 'none' }}>
566|                     <div style={{
567|                         background: 'rgba(0,0,0,0.85)', color: colors.emissive,
568|                         padding: '4px 12px', borderRadius: 6, fontSize: 11, fontWeight: 700,
569|                         border: '1px solid ${colors.emissive}55',
570|                         backdropFilter: 'blur(8px)',
571|                         whiteSpace: 'nowrap',
572|                         fontFamily: '"JetBrains Mono", monospace",
573|                     }}>
574|                         {slot?.label || type.toUpperCase()}
575|                     </div>
576|                 </Html>
577|             )}
578|
579|             {hovered && !installed && (
580|                 <mesh position={[0, 0, 0]} rotation={[-Math.PI / 2, 0, 0]}>
581|                     <ringGeometry args={[0.3, 0.38, 32]} />
582|                     <meshBasicMaterial color={colors.emissive} transparent opacity={0.15} side={THREE.DoubleSide} depthW...
583|                 </mesh>
584|             )}
585|
586|             {dragging && (
587|                 <Sparkles count={10} scale={0.6} size={3} speed={0.3} color={colors.emissive} opacity={0.5} />
588|             )}
589|         </group>
590|     );
591| }
592|
593| function SlotIndicator({ position, type, occupied }: {
594|     position: [number, number, number]; type: ComponentType; occupied: boolean;
595| }) {
596|     const meshRef = useRef<THREE.Mesh>(null);
597|     const colors = useAssemblyStore(COLORS[type]);
598|
599|     useFrame(({ clock }) => {
600|         if (meshRef.current && !occupied) {
601|             const p = (Math.sin(clock.getElapsedTime() * 2.5) + 1) / 2;
602|             (meshRef.current.material as THREE.MeshStandardMaterial).opacity = 0.1 + p * 0.25;
603|         }
604|     });
605|
606|     if (occupied) return null;
607|     return (
608|         <mesh ref={meshRef} position={[position[0], position[1] + 0.003, position[2]]} rotation={[-Math.PI / 2, 0, ...
609|         <ringGeometry args={[0.22, 0.3, 48]} />
610|         <meshStandardMaterial color={colors.emissive} transparent opacity={0.2} side={THREE.DoubleSide} depthWri...
611|         </mesh>
612|     );
613| }
614|
615| export default function Components() {
616|     const slots = useAssemblyStore((s) => s.slots);
617|     const components = useAssemblyStore((s) => s.components);
618|
619|     const slotList = useMemo(() =>
620|         slots.map((slot) => ({
621|             ...slot,
622|             occupied: components.some((c) => c.slotId === slot.id && c.installed),
623|         })),
624|         [slots, components]
625|     );
626|
627|     return (
628|         <group>
629|             {slotList.map((slot) => (

```

```

630 |         <group key={slot.id}>
631 |             <SlotIndicator position={slot.position} type={slot.type} occupied={slot.occupied} />
632 |             <DraggableComponent type={slot.type} slotId={slot.id} initialPosition={
633 |                 WORKBENCH_POSITIONS[slot.type] || [0, 0.1, 0]
634 |             } />
635 |         </group>
636 |     ))}
637 | </group>
638 | );
639 | }
640 |

```

File: components/dashboard/CertificateCard.tsx (66 lines, 3.2 KB)

```

1 | 'use client';
2 |
3 | import React from 'react';
4 | import { Award, Download, Share2, ShieldCheck, Calendar } from 'lucide-react';
5 | import { motion } from 'framer-motion';
6 |
7 | interface CertificateProps {
8 |   courseName: string;
9 |   issueDate: string;
10 |   verificationCode: string;
11 |   studentName: string;
12 | }
13 |
14 | export default function CertificateCard({ courseName, issueDate, verificationCode, studentName }: CertificateP...
15 |   return (
16 |     <motion.div
17 |       initial={{ opacity: 0, scale: 0.9 }}
18 |       animate={{ opacity: 1, scale: 1 }}
19 |       className="glass-panel p-8 border-brand-primary/30 relative overflow-hidden group hover:border-brand-pri...
20 |     >
21 |       {/* Decorative elements */}
22 |       <div className="absolute top-0 right-0 w-32 h-32 bg-brand-primary/5 rounded-full -mr-16 -mt-16 blur-2xl ...
23 |       <div className="absolute bottom-0 left-0 w-24 h-24 bg-brand-primary/5 rounded-full -ml-12 -mb-12 blur-2x...
24 |
25 |       <div className="relative z-10 flex flex-col gap-6">
26 |         <div className="flex justify-between items-start">
27 |           <div className="w-16 h-16 rounded-2xl bg-brand-primary/10 flex items-center justify-center text-bran...
28 |             <Award size={32} />
29 |           </div>
30 |           <div className="text-right">
31 |             <p className="text-xs font-mono text-text-muted">{verificationCode}</p>
32 |           </div>
33 |         </div>
34 |
35 |         <div>
36 |           <h3 className="text-2xl font-black text-white mb-2">{courseName}</h3>
37 |           <p className="text-sm text-text-secondary leading-relaxed">
38 |             Chéng nh - n <span className="text-white font-bold">{studentName}
39 |           </p>
40 |         </div>
41 |
42 |         <div className="flex items-center gap-6 py-4 border-y border-white/5">
43 |           <div className="flex items-center gap-2">
44 |             <Calendar size={14} className="text-brand-primary" />
45 |             <span className="text-xs text-text-muted">Ngày c ÷ p: <strong>
46 |           </div>
47 |           <div className="flex items-center gap-2">
48 |             <ShieldCheck size={14} className="text-green" />
49 |             <span className="text-xs text-text-muted">Tríng tháí: <strong>
50 |           </div>
51 |         </div>
52 |
53 |         <div className="flex gap-4">
54 |           <button className="flex-1 py-3 bg-brand-primary text-black rounded-xl font-black text-xs uppercase t...
55 |           <button className="px-4 py-3 bg-white/5 text-white rounded-xl hover:bg-white/10 transition-all flex ...
56 |           <button className="px-4 py-3 bg-white/5 text-white rounded-xl hover:bg-white/10 transition-all flex ...
57 |           <button className="px-4 py-3 bg-white/5 text-white rounded-xl hover:bg-white/10 transition-all flex ...
58 |           <button className="px-4 py-3 bg-white/5 text-white rounded-xl hover:bg-white/10 transition-all flex ...
59 |           <button className="px-4 py-3 bg-white/5 text-white rounded-xl hover:bg-white/10 transition-all flex ...
60 |           <button className="px-4 py-3 bg-white/5 text-white rounded-xl hover:bg-white/10 transition-all flex ...
61 |         </div>
62 |       </div>
63 |     </motion.div>
64 |   );
65 | }
66 |

```

File: components/dashboard/ExportReportButton.tsx (80 lines, 3.4 KB)

```

1 | 'use client';
2 |
3 | import React from 'react';
4 | import { PDFDownloadLink, Document, Page, Text, View, StyleSheet } from '@react-pdf/renderer';
5 | import { Download, FileText, Table as TableIcon } from 'lucide-react';
6 |
7 | // PDF Styles
8 | const styles = StyleSheet.create({
9 |   page: { padding: 40, backgroundColor: '#ffffff' },
10 |   header: { fontSize: 20, marginBottom: 20, textAlign: 'center' },
11 |   table: { display: 'flex', width: 'auto', borderStyle: 'solid', borderWidth: 1, borderRightWidth: 0, borderBo...
12 |   tableRow: { margin: 'flex', flexDirection: 'row' },
13 |   tableCol: { width: '25%', borderStyle: 'solid', borderWidth: 1, borderLeftWidth: 0, borderTopWidth: 0 },
14 |   tableCell: { margin: 'auto', marginTop: 5, fontSize: 10 }
15 | });
16 |
17 | // PDF Content
18 | const MyDocument = ({ data }: { data: any[] }) => (
19 |   <Document>
20 |     <Page size="A4" style={styles.page}>
21 |       <Text style={styles.header}>STUDENT PROGRESS REPORT - PC MASTER LMS</Text>
22 |       <View style={styles.table}>
23 |         <View style={styles.tableRow}>
24 |           <View style={styles.tableCol}><Text style={styles.tableCell}>Student</Text></View>
25 |           <View style={styles.tableCol}><Text style={styles.tableCell}>Progress (%)</Text></View>
26 |           <View style={styles.tableCol}><Text style={styles.tableCell}>Avg Score</Text></View>
27 |           <View style={styles.tableCol}><Text style={styles.tableCell}>Status</Text></View>
28 |         </View>
29 |         {data.map((row, i) => (
30 |           <View key={i} style={styles.tableRow}>
31 |             <View style={styles.tableCol}><Text style={styles.tableCell}>{row.name}</Text></View>
32 |             <View style={styles.tableCol}><Text style={styles.tableCell}>{row.progress}%</Text></View>
33 |             <View style={styles.tableCol}><Text style={styles.tableCell}>{row.avgScore}</Text></View>
34 |             <View style={styles.tableCol}><Text style={styles.tableCell}>{row.status}</Text></View>
35 |           </View>
36 |         ))}
37 |       </View>
38 |     </Page>

```

```

39 | </Document>
40 | );
41 |
42 | export default function ExportReportButton({ data, filename = 'report' }: { data: any[], filename?: string }) {
43 |   const [isMounted, setIsMounted] = React.useState(false);
44 |
45 |   React.useEffect(() => {
46 |     setIsMounted(true);
47 |   }, []);
48 |
49 |   const exportCSV = () => {
50 |     const headers = ['Student,Progress(%,AvgScore,Status\n'];
51 |     const rows = data.map(r => `${r.name},${r.progress},${r.avgScore},${r.status}\n`);
52 |     const blob = new Blob([headers.concat(rows).join('\n')], { type: 'text/csv;charset=utf-8;' });
53 |     const link = document.createElement('a');
54 |     link.href = URL.createObjectURL(blob);
55 |     link.download = `${filename}-${new Date().toISOString().split('T')[0]}.csv`;
56 |     link.click();
57 |   };
58 |
59 |   return (
60 |     <div className="flex gap-2">
61 |       <button
62 |         onClick={exportCSV}
63 |         className="flex items-center gap-2 px-4 py-2 bg-bg-elevated text-text-primary rounded-lg border border...
64 |       >
65 |         <TableIcon size={16} /> CSV
66 |       </button>
67 |
68 |       {isMounted && (
69 |         <PDFDownloadLink document={<MyDocument data={data} />} fileName={`_${filename}.pdf`}>
70 |           {({ loading }) => (
71 |             <button className="flex items-center gap-2 px-4 py-2 bg-brand-primary text-black rounded-lg font-b...
72 |               <FileText size={16} /> {loading ? '...' : 'PDF'}
73 |             </button>
74 |           )}
75 |         </PDFDownloadLink>
76 |       )}
77 |     </div>
78 |   );
79 | }
80 |

```

File: components/DiagnosisMode_part1.js (365 lines, 18.8 KB)

```

1 | py 'use client';
2 |
3 | import React, { useState, useEffect, useRef, useCallback } from 'react';
4 |
5 | const TOOLS = [
6 |   { id: 'multimeter', icon: 'i', label: 'Multimeter', label_vn: 'Đồng hồ',
7 |     { id: 'screwdriver', icon: '0=Y', label: 'Screwdriver', label_vn: 'Tua
8 |     { id: 'compressed_air', icon: '0=U', label: 'Compressed Air', label_vn:
9 |     { id: 'spare_psu', icon: '0=Y', label: 'Spare PSU', label_vn: 'Ngu
10 |    { id: 'bios_speaker', icon: '0=Y', label: 'BIOS Speaker', label_vn: 'Lo
11 | ];
12 |
13 | const SCENARIOS = [
14 |   {
15 |     id: 'no_power',
16 |     title: 'No Power',
17 |     title_vn: 'Không Lên Nguồn',
18 |     description: 'PC doesn\'t turn on when pressing the power button.',
19 |     description_vn: 'PC không lên khi bấm nút nguồn.',
20 |     symptoms: ['No fan spin', 'No LEDs', 'Power button does nothing'],
21 |     symptoms_vn: ['Quít không quay', 'èn không sáng', 'Nút nguồn không
22 |     correct_answer: 'psu',
23 |     hints: ['Check PSU switch', 'Check power cable', 'Try different outlet', 'PSU may be dead'],
24 |     hints_vn: ['Kiểm tra công tắc PSU', 'Kiểm tra dây nguồn', 'Thí ổ c-
25 |     difficulty: 'easy',
26 |     icon: '0=Y',
27 |     options: [
28 |       { id: 'psu', label: 'Dead PSU', label_vn: 'Nguồn chết', correct: t
29 |       { id: 'cpu', label: 'CPU fried', label_vn: 'CPU cháy' },
30 |       { id: 'ram', label: 'Loose RAM', label_vn: 'RAM lỏng' },
31 |       { id: 'case', label: 'Case short circuit', label_vn: 'Case bẽ ch-
32 |     ],
33 |     tool_hints: {
34 |       multimeter: 'PSU voltage rails read 0V on all lines.',
35 |       multimeter_vn: 'Các ắc ắc ắc PSU Ắt 0V.',
36 |       screwdriver: 'All components seated. Power header fine.',
37 |       screwdriver_vn: 'Linh kiện g- n ch- t. Sủ nãi nguồn bình th ắc ắc.',
38 |       compressed_air: 'No significant dust buildup.',
39 |       compressed_air_vn: 'Không có bãi b@n Ắng k Ắ.',
40 |       spare_psu: 'PC boots normally with known-good PSU.',
41 |       spare_psu_vn: 'PC ch- y bình th ắc ắc vủi nguồn khác.',
42 |       bios_speaker: 'No beep at all - no power to board.',
43 |       bios_speaker_vn: 'Không beep - board không có ắc n.',
44 |     },
45 |     beep_pattern: '... (silence)',
46 |     diagram: 'psu',
47 |     diagnosis_step: 1,
48 |   },
49 |   {
50 |     id: 'no_display',
51 |     title: 'No Display / Black Screen',
52 |     title_vn: 'Không Có Hình Ắnh',
53 |     description: 'PC turns on but monitor shows no signal.',
54 |     description_vn: 'PC lên nguồn nh Ắng màn hình không tín hi Ắu.',
55 |     symptoms: ['Fans spinning', 'LEDs on', 'No display output', 'No BIOS beep'],
56 |     symptoms_vn: ['Quít quay', 'èn sáng', 'Không có hình', 'Không tiếng
57 |     correct_answer: 'ram_reseat',
58 |     hints: ['Reset RAM modules', 'Try one RAM stick', 'Check GPU seating', 'Clear CMOS'],
59 |     hints_vn: ['G- n lii RAM', 'Thí tẮng thanh RAM', 'Ki Ắm tra GPU', 'Cle
60 |     difficulty: 'easy',
61 |     icon: '0=Yp',
62 |     options: [
63 |       { id: 'ram_reseat', label: 'Reset RAM', label_vn: 'G- n lii RAM',
64 |       { id: 'replace_cpu', label: 'Replace CPU', label_vn: 'Thay CPU' },
65 |       { id: 'replace_monitor', label: 'Replace monitor', label_vn: 'Thay màn hình' },
66 |       { id: 'reinstall_os', label: 'Reinstall Windows', label_vn: 'Cài t
67 |     ],
68 |     tool_hints: {
69 |       multimeter: 'Voltages normal. PSU working correctly.',
70 |       multimeter_vn: 'Ắc Ắp, bình th Ắng. Nguồn t Ắt.',
71 |       screwdriver: 'Reseating RAM fixed it - one stick was loose.',
72 |       screwdriver_vn: 'G- n lii RAM hẮt lẮi - mùt thanh bỂ lẮng.',
73 |       compressed_air: 'RAM slots cleaned. Minor dust.',
74 |       compressed_air_vn: 'Khe RAM Ắ vẮ sinh. Bãi nh Ắ.',
75 |       spare_psu: 'Same issue with spare PSU.',
76 |       spare_psu_vn: 'V Ắn lẮi vủi nguồn khác.',
77 |       bios_speaker: 'Continuous long beep - RAM error.',
78 |       bios_speaker_vn: 'Beep dài liên tẮc - lẮi RAM.',

```

```
79 | },
80 | beep_pattern: 'BEEP... BEEP... BEEP... (long repeating)',
81 | diagram: 'ram',
82 | diagnosis_step: 1,
83 | },
84 | {
85 | id: 'blue_screen',
86 | title: 'Blue Screen (CRITICAL_PROCESS_DIED)',
87 | title_vn: 'Màn Hình Xanh',
88 | description: 'PC blue-screens shortly after booting Windows.',
89 | description_vn: 'PC bẻ màn hình xanh sau khi vào Windows.',
90 | symptoms: ['BSOD on boot', 'CRITICAL_PROCESS_DIED', 'Boots to recovery', 'Safe mode crashes'],
91 | symptoms_vn: ['BSOD khi boot', 'Lỗi PROCESS_DIED', 'Vào recovery', 'Chạy Safe mode crash'],
92 | correct_answer: 'repair_os',
93 | hints: ['Try Startup Repair', 'Check recent drivers', 'Run SFC /scannow', 'Reinstall Windows'],
94 | hints_vn: ['Thử Startup Repair', 'Kiểm tra driver', 'Chạy SFC /scannow', 'Cài lại Windows'],
95 | difficulty: 'medium',
96 | icon: '00€',
97 | options: [
98 | { id: 'repair_os', label: 'Repair OS', label_vn: 'Sửa Windows' },
99 | { id: 'replace_ram', label: 'Replace RAM', label_vn: 'Thay RAM' },
100 | { id: 'replace_gpu', label: 'Replace GPU', label_vn: 'Thay GPU' },
101 | { id: 'replace_cpu', label: 'Replace CPU', label_vn: 'Thay CPU' },
102 | ],
103 | tool_hints: {
104 | multimeter: 'All voltages within spec.',
105 | multimeter_vn: 'Điện áp trong ngưỡng.',
106 | screwdriver: 'Hardware seated correctly.',
107 | screwdriver_vn: 'Phần cứng gắn chặt.',
108 | compressed_air: 'System clean, no overheating.',
109 | compressed_air_vn: 'Hệ thống sạch, không quá nhiệt.',
110 | spare_psu: 'BSOD still occurs with spare PSU.',
111 | spare_psu_vn: 'BSOD vẫn xảy ra với nguồn khác.',
112 | bios_speaker: 'Single short beep - POST OK, software issue.',
113 | bios_speaker_vn: 'Một beep ngắn - POST OK, lỗi phần mềm.',
114 | },
115 | beep_pattern: 'BEEP (single short - POST OK)',
116 | diagram: 'motherboard',
117 | diagnosis_step: 1,
118 | },
119 | {
120 | id: 'overheating',
121 | title: 'Overheating / Random Shutdowns',
122 | title_vn: 'Quá Nhiệt / Tắt Ngắt',
123 | description: 'PC randomly shuts down under load. Case feels hot.',
124 | description_vn: 'PC tắt ngắt đột ngột khi chạy nặng. Vỏ máy nóng.',
125 | symptoms: ['Random shutdowns', 'Loud fan then silence', 'Hot case', 'Crashes under load'],
126 | symptoms_vn: ['Tắt ngắt đột ngột', 'Quạt ồn rối rít', 'Vỏ nóng', 'Treo khi tải'],
127 | correct_answer: 'cooler_issue',
128 | hints: ['Check CPU cooler', 'Reapply thermal paste', 'Check fan curves', 'Clean dust'],
129 | hints_vn: ['Kiểm tra tản nhiệt', 'Bôi keo tản nhiệt', 'Kiểm tra quạt'],
130 | difficulty: 'medium',
131 | icon: '0<B!p',
132 | options: [
133 | { id: 'cooler_issue', label: 'Cooler failure / dust clogged', label_vn: 'Lỗi tản nhiệt / bụi tắc' },
134 | { id: 'psu_overload', label: 'PSU overload', label_vn: 'PSU quá tải' },
135 | { id: 'virus', label: 'Virus / malware', label_vn: 'Virus' },
136 | { id: 'ram_error', label: 'RAM error', label_vn: 'RAM lỗi' },
137 | ],
138 | tool_hints: {
139 | multimeter: 'PSU voltages fluctuate but in range.',
140 | multimeter_vn: 'Điện áp PSU dao động nhưng trong ngưỡng.',
141 | screwdriver: 'Cooler mounting screws loose. Heatsink not contacting.',
142 | screwdriver_vn: 'Đ ốc tản nhiệt lỏng. Không tiếp xúc.',
143 | compressed_air: 'Heatsink fins clogged with dust. Blocked airflow.',
144 | compressed_air_vn: 'Khe tản nhiệt bị bẩn. Không lưu thông khí.',
145 | spare_psu: 'Same shutdowns with spare PSU.',
146 | spare_psu_vn: 'Vẫn tắt ngắt với nguồn khác.',
147 | bios_speaker: 'Single short beep. BIOS overheat warning.',
148 | bios_speaker_vn: 'Một beep ngắn. BIOS báo quá nhiệt.',
149 | },
150 | beep_pattern: 'BEEP (short) + overheat LED',
151 | diagram: 'cpu_cooler',
152 | diagnosis_step: 2,
153 | },
154 | {
155 | id: 'gpu_artifacts',
156 | title: 'GPU Artifacts / Glitched Display',
157 | title_vn: 'Lỗi Hiển Thị GPU',
158 | description: 'Screen shows colored squares, lines, flickering.',
159 | description_vn: 'Màn hình có ô màu lị, nhấp nháy.',
160 | symptoms: ['Colored artifacts', 'Flickering', 'Lines across display', 'Driver crashes'],
161 | symptoms_vn: ['Ô màu lị', 'Nhấp nháy', 'Vạch ngang', 'Driver crash'],
162 | correct_answer: 'gpu_dying',
163 | hints: ['Check GPU temps', 'Try underclocking', 'Test another monitor', 'Check GPU seating'],
164 | hints_vn: ['Kiểm tra nhiệt GPU', 'Thử underclock', 'Dùng màn khác', 'Kiểm tra lắp GPU'],
165 | difficulty: 'medium',
166 | icon: '0<B@',
167 | options: [
168 | { id: 'gpu_dying', label: 'GPU failing / VRAM damaged', label_vn: 'GPU hỏng / VRAM hỏng' },
169 | { id: 'cable_loose', label: 'Loose display cable', label_vn: 'Dây cáp lỏng' },
170 | { id: 'monitor_broken', label: 'Monitor broken', label_vn: 'Màn hỏng' },
171 | { id: 'driver_issue', label: 'Driver issue only', label_vn: 'Lỗi driver' },
172 | ],
173 | tool_hints: {
174 | multimeter: 'GPU power lines stable.',
175 | multimeter_vn: 'Điện áp GPU ổn định.',
176 | screwdriver: 'GPU seated firmly. Reseating did not help.',
177 | screwdriver_vn: 'GPU gắn chặt. Gắn lại không hết.',
178 | compressed_air: 'GPU fans clean. No overheating.',
179 | compressed_air_vn: 'Quạt GPU sạch. Không quá nhiệt.',
180 | spare_psu: 'Artifacts persist with spare PSU.',
181 | spare_psu_vn: 'Lỗi vẫn còn với nguồn khác.',
182 | bios_speaker: 'One short beep. Artifacts in BIOS - hardware fail.',
183 | bios_speaker_vn: 'Một beep. Lỗi ngay trong BIOS - phần cứng.',
184 | },
185 | beep_pattern: 'BEEP (short)',
186 | diagram: 'gpu',
187 | diagnosis_step: 2,
188 | },
189 | {
190 | id: 'slow_pc',
191 | title: 'Slow PC / 100% Disk Usage',
192 | title_vn: 'Máy Chậm / Disk 100%',
193 | description: 'PC extremely slow. Task Manager shows 100% disk usage.',
194 | description_vn: 'Máy rất chậm. Task Manager hiển thị disk 100%.',
195 | symptoms: ['100% disk usage', 'Slow boot', 'Apps freeze', 'Explorer hangs'],
196 | symptoms_vn: ['Disk 100%', 'Chậm boot', 'Ứng dụng đơ', 'Explorer treo'],
197 | correct_answer: 'hdd_failing',
198 | hints: ['Check Task Manager', 'Run CHKDSK', 'Check S.M.A.R.T.', 'Replace drive'],
199 | hints_vn: ['Kiểm tra Task Manager', 'Chạy CHKDSK', 'Kiểm tra S.M.A.R.T.', 'Thay ổ cứng'],
200 | difficulty: 'medium',
201 | icon: '0=0',
202 | options: [
```



```
2 0 3 | { id: 'hdd_failing', label: 'HDD failing / SSD full', label_vn: 'HDD hỏng / SSD đầy' },
2 0 4 | { id: 'virus', label: 'Virus infection', label_vn: 'Nhiễm virus' },
2 0 5 | { id: 'ram_low', label: 'Not enough RAM', label_vn: 'Thiếu RAM' },
2 0 6 | { id: 'cpu_old', label: 'CPU too old', label_vn: 'CPU cũ' },
207 | },
208 | tool_hints: {
209 |   multimeter: 'Drive power normal.',
2 1 0 |     multimeter_vn: 'Nguồn ổ cứng bình thường.',
211 |   screwdriver: 'Cables secure. Drive mounted properly.',
2 1 2 |     screwdriver_vn: 'Cáp ch. t. ổ g-n úng.',
213 |   compressed_air: 'Drive clean and cool.',
2 1 4 |     compressed_air_vn: 'Ổ sạch và mát.',
215 |   spare_psu: 'Still slow with spare PSU.',
2 1 6 |     spare_psu_vn: 'V«n ch-m vñi nguồn khác.',
217 |   bios_speaker: 'Single beep. BIOS detects drive with high latency.',
2 1 8 |     bios_speaker_vn: 'Mút beep. BIOS thấy ổ nh«ng ù trA cao.',
219 | },
220 | beep_pattern: 'BEEP (short)',
221 | diagram: 'storage',
222 | diagnosis_step: 2,
223 | },
224 | {
225 |   id: 'no_boot_device',
226 |   title: 'No Boot Device Found',
2 2 7 |     title_vn: 'Không tìm thấy ổ khi Ừng',
228 |   description: 'Screen says "No boot device found" or "Reboot and Select proper Boot device".',
229 |   description_vn: 'Màn hình báo "No boot device found".',
230 |   symptoms: ['No boot device error', 'BIOS shows no drive', 'Was working yesterday'],
2 3 1 |     symptoms_vn: ['L«i không boot device', 'BIOS không thấy ổ', 'Hôm qua
232 |   correct_answer: 'ssd_dead',
233 |   hints: ['Check SATA/NVMe connections', 'Boot into BIOS', 'Try different SATA port', 'Check if drive appear...'],
2 3 4 |     hints_vn: ['KiA m tra k«t nñi SATA', 'Vào BIOS', 'Thí cõng SATA khác
235 |   difficulty: 'easy',
2 3 6 |     icon: '0=0',
237 |   options: [
2 3 8 |     { id: 'ssd_dead', label: 'SSD disconnected or dead', label_vn: 'SSD
2 3 9 |       { id: 'os_corrupt', label: 'Windows corrupted', label_vn: 'Windows
2 4 0 |       { id: 'bios_bug', label: 'BIOS settings wrong', label_vn: 'Cài
2 4 1 |       { id: 'cable_loose', label: 'Loose SATA cable', label_vn: 'Cáp SAT
242 |   ],
243 |   tool_hints: {
244 |     multimeter: 'SATA power correct voltage.',
2 4 5 |       multimeter_vn: 'Cáp nguồn SATA úng iChn áp.',
246 |     screwdriver: 'SSD not detected after reseating. Try known-working SSD.',
2 4 7 |       screwdriver_vn: 'SSD không thấy dù ã g-n lii.',
248 |     compressed_air: 'SATA ports cleaned.',
2 4 9 |       compressed_air_vn: 'Cõng SATA ã vÇ sinh.',
250 |     spare_psu: 'Same with spare PSU.',
2 5 1 |       spare_psu_vn: 'V«n l«i vñi nguồn khác.',
252 |     bios_speaker: 'Beep codes - missing storage device.',
2 5 3 |       bios_speaker_vn: 'Mã beep báo thi«u thi«t b« l«u trı.',
254 |   },
255 |   beep_pattern: 'BEEP (short) + "No boot device"',
256 |   diagram: 'storage',
257 |   diagnosis_step: 1,
258 | },
259 | {
260 |   id: 'boot_loop',
261 |   title: 'Boot Loop / Continuous Restart',
2 6 2 |     title_vn: 'Boot Loop / Khi Ừng Li«n Tắc',
263 |   description: 'PC turns on, shows BIOS splash, then restarts repeatedly.',
2 6 4 |     description_vn: 'PC b-t, thấy BIOS, r«i restart li«n tắc.',
265 |   symptoms: ['Continuous restart loop', 'Never reaches desktop', 'Started after BIOS change'],
2 6 6 |     symptoms_vn: ['Restart li«n tắc', 'Không vào desktop', 'Sau khi thay
267 |   correct_answer: 'bios_ram_instability',
268 |   hints: ['Clear CMOS', 'Boot with one RAM stick', 'Check XMP stability', 'Update BIOS'],
2 6 9 |     hints_vn: ['C-clear CMOS', 'Boot 1 thanh RAM', 'KiA m tra XMP', 'C-p nh
270 |   difficulty: 'hard',
2 7 1 |     icon: '0=Y',
272 |   options: [
2 7 3 |     { id: 'bios_ram_instability', label: 'Bad BIOS / RAM instability',
2 7 4 |       { id: 'psu_failing', label: 'PSU failing under load', label_vn: 'M
2 7 5 |       { id: 'gpu_driver', label: 'GPU driver crash loop', label_vn: 'Drı
2 7 6 |       { id: 'windows_update', label: 'Corrupted Windows update', label_v
277 |   ],
278 |   tool_hints: {
279 |     multimeter: 'PSU voltages fluctuate during boot.',
2 8 0 |       multimeter_vn: 'iChn áp PSU dao Ừng khi boot.',
281 |     screwdriver: 'Clearing CMOS resolved the boot loop.',
2 8 2 |       screwdriver_vn: 'C-clear CMOS ã kh-c phắc boot loop.',
283 |     compressed_air: 'System clean.',
2 8 4 |       compressed_air_vn: 'HÇ thñng sạch.',
285 |     spare_psu: 'Boot loop persists with spare PSU.',
2 8 6 |       spare_psu_vn: 'Boot loop v«n còn vñi nguồn khác.',
287 |     bios_speaker: 'Repeating short beeps - memory/BIOS error.',
2 8 8 |       bios_speaker_vn: 'Beep ng-n l-p lii - l«i RAM/BIOS.',
289 |   },
290 |   beep_pattern: 'BEEP BEEP BEEP BEEP (continuous short)',
291 |   diagram: 'motherboard',
292 |   diagnosis_step: 3,
293 | },
294 | {
295 |   id: 'strange_noise',
296 |   title: 'Strange Noise from PC',
2 9 7 |     title_vn: 'A m Thanh Li T« PC',
298 |   description: 'PC makes grinding or clicking noise. Performance normal.',
2 9 9 |     description_vn: 'PC phát tiếng li. HiÇu n ng bình thường.',
300 |   symptoms: ['Grinding noise', 'Buzzing from PSU area', 'Noise changes with load'],
3 0 1 |     symptoms_vn: ['Ti«ng lách cách', 'Ù t« nguồn', 'Ồn thay òi theo t«i
302 |   correct_answer: 'fan_bearing',
303 |   hints: ['Locate noise source', 'Stop fans one by one', 'Check HDD for clicks', 'Inspect fan blades'],
3 0 4 |     hints_vn: ['Xác ãnh v« trı ón', 'Đ«ng t«ng quit', 'KiA m tra HDD',
305 |   difficulty: 'easy',
3 0 6 |     icon: '0=Y',
307 |   options: [
3 0 8 |     { id: 'fan_bearing', label: 'Fan bearing worn out', label_vn: 'Bi
3 0 9 |     { id: 'hdd_damage', label: 'HDD head crash', label_vn: 'HDD s«u
3 1 0 |     { id: 'coil_whine', label: 'GPU coil whine', label_vn: 'Cu ùn c«m
3 1 1 |     { id: 'loose_screw', label: 'Loose screw vibrating', label_vn: 'Đo
312 |   ],
313 |   tool_hints: {
314 |     multimeter: 'No electrical issues.',
3 1 5 |       multimeter_vn: 'Không v«n Á iChn.',
316 |     screwdriver: 'Stopping CPU fan stops noise - bearing worn.',
3 1 7 |       screwdriver_vn: 'Đ«ng quit CPU h«t ón - b«c mòn.',
318 |     compressed_air: 'Dust blow did not help.',
3 1 9 |       compressed_air_vn: 'X«t bái không h«t.',
320 |     spare_psu: 'Noise persists.',
3 2 1 |       spare_psu_vn: 'Ồn v«n còn.',
322 |     bios_speaker: 'No beep errors. POST fine.',
3 2 3 |       bios_speaker_vn: 'Không l«i beep. POST bình thường.',
324 |   },
325 |   beep_pattern: 'BEEP (short - normal)',
326 |   diagram: 'fan',
```

```

327|     diagnosis_step: 1,
328|   },
329|   {
330|     id: 'no_wifi',
331|     title: 'Network Not Working / WiFi Missing',
332|     title_vn: 'Mạng không có WiFi',
333|     description: 'WiFi shows no networks. Ethernet works fine.',
334|     description_vn: 'WiFi không thấy mạng. Ethernet vẫn dùng được.',
335|     symptoms: ['No WiFi networks', 'Adapter missing in Device Manager', 'Driver error'],
336|     symptoms_vn: ['Không thấy WiFi', 'Thiếu adapter trong Device Manager', 'Lỗi driver'],
337|     correct_answer: 'wifi_driver',
338|     hints: ['Check WiFi enabled', 'Toggle Airplane mode', 'Check Device Manager', 'Reinstall driver'],
339|     hints_vn: ['Kiểm tra WiFi bật', 'Bật/tắt máy bay', 'Kiểm tra Device Manager', 'Cài lại driver'],
340|     difficulty: 'medium',
341|     icon: '🔌',
342|     options: [
343|       { id: 'wifi_driver', label: 'WiFi driver / antenna issue', label_vn: 'Driver WiFi / antenna', correct: true },
344|       { id: 'router_down', label: 'Router down', label_vn: 'Router hỏng', correct: false },
345|       { id: 'isp_outage', label: 'ISP outage', label_vn: 'Mất internet', correct: false },
346|       { id: 'windows_bug', label: 'Windows network bug', label_vn: 'Lỗi Windows', correct: false }
347|     ],
348|     tool_hints: {
349|       multimeter: 'WiFi card power correct.',
350|       multimeter_vn: 'Nguồn card WiFi đúng.',
351|       screwdriver: 'WiFi card seated. Antenna cable loose.',
352|       screwdriver_vn: 'Card gắn chặt. Cáp antenna lỏng.',
353|       compressed_air: 'Card cleaned. No damage.',
354|       compressed_air_vn: 'Card vệ sinh. Không hỏng.',
355|       spare_psu: 'WiFi still missing.',
356|       spare_psu_vn: 'WiFi vẫn không thấy.',
357|       bios_speaker: 'POST OK. No hardware errors.',
358|       bios_speaker_vn: 'POST OK. Không lỗi phần cứng.',
359|     },
360|     beep_pattern: 'BEEP (short - normal)',
361|     diagram: 'wifi',
362|     diagnosis_step: 2,
363|   },
364| ];
365|

```

File: components/DiagnosisMode.js (1456 lines, 70.5 KB)

```

1| py 'use client';
2|
3| import React, { useState, useEffect, useRef, useCallback } from 'react';
4|
5| const TOOLS = [
6|   { id: 'multimeter', icon: '🔌', label: 'Multimeter', label_vn: 'Đồng hồ đo điện', correct: true },
7|   { id: 'screwdriver', icon: '🔧', label: 'Screwdriver', label_vn: 'Tuốc nơ vít', correct: false },
8|   { id: 'compressed_air', icon: '💨', label: 'Compressed Air', label_vn: 'Khí nén', correct: false },
9|   { id: 'spare_psu', icon: '🔌', label: 'Spare PSU', label_vn: 'Nguồn dự phòng', correct: false },
10|   { id: 'bios_speaker', icon: '🔊', label: 'BIOS Speaker', label_vn: 'Loa BIOS', correct: false }
11| ];
12|
13| const SCENARIOS = [
14|   {
15|     id: 'no_power',
16|     title: 'No Power',
17|     title_vn: 'Không Lên Nguồn',
18|     description: 'PC does not turn on when pressing the power button. Fans do not spin.',
19|     description_vn: 'PC không lên khi bấm nút nguồn. Quạt không quay.',
20|     symptoms: ['No fan spin', 'No LEDs', 'Power button does nothing'],
21|     symptoms_vn: ['Quạt không quay', 'Không có đèn sáng', 'Nút nguồn không phản ứng'],
22|     correct_answer: 'psu',
23|     hints: ['Check if the PSU switch is on', 'Check power cable connection', 'Try a different power outlet', '...'],
24|     hints_vn: ['Kiểm tra công tắc nguồn PSU', 'Kiểm tra dây nguồn', 'Thử ổ cắm điện khác'],
25|     difficulty: 'easy',
26|     icon: '🔌',
27|     options: [
28|       { id: 'psu', label: 'Dead PSU (Nguồn chết)', label_vn: 'Nguồn chết', correct: true },
29|       { id: 'cpu', label: 'CPU fried', label_vn: 'CPU cháy', correct: false },
30|       { id: 'ram', label: 'Loose RAM', label_vn: 'RAM lỏng', correct: false },
31|       { id: 'case', label: 'Short circuit in case', label_vn: 'Case bị chập', correct: false }
32|     ],
33|     tool_hints: {
34|       multimeter: 'PSU voltage rails read 0V on all lines.',
35|       multimeter_vn: 'Các dây nguồn PSU đều 0V.',
36|       screwdriver: 'All components appear seated. Power button header looks fine.',
37|       screwdriver_vn: 'Các linh kiện gắn chặt. Sụn nút nguồn bình thường.',
38|       compressed_air: 'No significant dust buildup inside.',
39|       compressed_air_vn: 'Không có bụi bẩn đáng kể bên trong.',
40|       spare_psu: 'PC boots normally with a known-good PSU.',
41|       spare_psu_vn: 'PC hoạt động bình thường với nguồn khác.',
42|       bios_speaker: 'No beep at all - no power reaching the board.',
43|       bios_speaker_vn: 'Không có tiếng beep - bo mạch không có điện.',
44|     },
45|     beep_pattern: '... (silence)',
46|     diagram: 'psu',
47|     diagnosis_step: 1,
48|   },
49|   {
50|     id: 'no_display',
51|     title: 'No Display / Black Screen',
52|     title_vn: 'Không Có Hình Ảnh',
53|     description: 'PC turns on (fans spin, LEDs on) but monitor shows no signal.',
54|     description_vn: 'PC lên nguồn (quạt quay, đèn sáng) nhưng màn hình không có tín hiệu.',
55|     symptoms: ['Fans spinning', 'LEDs on', 'No display output', 'No BIOS beep'],
56|     symptoms_vn: ['Quạt quay', 'Đèn sáng', 'Không có hình ảnh', 'Không có tiếng beep'],
57|     correct_answer: 'ram_reseat',
58|     hints: ['Try reseating RAM modules', 'Try one RAM stick at a time', 'Check GPU is seated properly', 'Clear...'],
59|     hints_vn: ['Gắn lại RAM', 'Thử từng thanh RAM một', 'Kiểm tra GPU gắn đúng', 'Xóa bộ nhớ cache'],
60|     difficulty: 'easy',
61|     icon: '🖥️',
62|     options: [
63|       { id: 'ram_reseat', label: 'Reseat RAM modules', label_vn: 'Gắn lại RAM', correct: true },
64|       { id: 'replace_cpu', label: 'Replace CPU', label_vn: 'Thay CPU', correct: false },
65|       { id: 'replace_monitor', label: 'Replace monitor', label_vn: 'Thay màn hình', correct: false },
66|       { id: 'reinstall_os', label: 'Reinstall Windows', label_vn: 'Cài lại Windows', correct: false }
67|     ],
68|     tool_hints: {
69|       multimeter: 'Voltages are normal. PSU functioning correctly.',
70|       multimeter_vn: 'Điện áp bình thường. Nguồn hoạt động tốt.',
71|       screwdriver: 'Reseating RAM modules fixed the issue - one stick was loose.',
72|       screwdriver_vn: 'Gắn lại RAM đã khắc phục - một thanh bị lỏng.',
73|       compressed_air: 'RAM slots cleaned. Minor dust present.',
74|       compressed_air_vn: 'Khе RAM đã vệ sinh. Có bụi nhẹ.',
75|       spare_psu: 'Same issue with spare PSU - problem is not the PSU.',
76|       spare_psu_vn: 'Vẫn lỗi với nguồn khác - không phải do nguồn.',
77|       bios_speaker: 'Continuous long beep pattern - RAM error.',
78|       bios_speaker_vn: 'Tiếng beep dài liên tục - lỗi RAM.',
79|     },
80|     beep_pattern: 'BEEP... BEEP... BEEP... (long repeating)',
81|     diagram: 'ram',

```

```
82 |     diagnosis_step: 1,
83 | },
84 | {
85 |     id: 'blue_screen',
86 |     title: 'Blue Screen (CRITICAL_PROCESS_DIED)',
87 |     title_vn: 'Màn Hình Xanh (CRITICAL_PROCESS_DIED)',
88 |     description: 'PC blue-screens with CRITICAL_PROCESS_DIED error shortly after booting Windows.',
89 |     symptoms: ['BSOD on boot', 'CRITICAL_PROCESS_DIED error', 'Boots to recovery', 'Safe mode also crashes'],
90 |     correct_answer: 'repair_os',
91 |     hints: ['Try Startup Repair', 'Check for recent driver updates', 'Run SFC /scannow', 'Reinstall Windows if...'],
92 |     difficulty: 'medium',
93 |     icon: '0=0E',
94 |     options: [
95 |         { id: 'repair_os', label: 'Repair corrupted system files', label_vn: 'Sửa chữa tệp hệ thống bị hỏng' },
96 |         { id: 'replace_ram', label: 'Replace RAM', label_vn: 'Thay RAM' },
97 |         { id: 'replace_gpu', label: 'Replace GPU', label_vn: 'Thay GPU' },
98 |         { id: 'replace_cpu', label: 'Replace CPU', label_vn: 'Thay CPU' }
99 |     ],
100 |     tool_hints: {
101 |         multimeter: 'All voltage readings are within spec.',
102 |         screwdriver: 'Hardware appears correctly seated. Nothing loose.',
103 |         compressed_air: 'System is clean, no overheating signs.',
104 |         spare_psu: 'BSOD still occurs with spare PSU.',
105 |         bios_speaker: 'Single short beep - POST OK, problem is software.',
106 |         beep_pattern: 'BEEP (single short - POST OK)',
107 |         diagram: 'motherboard',
108 |         diagnosis_step: 1,
109 |     },
110 | },
111 | {
112 |     id: 'overheating',
113 |     title: 'Overheating / Random Shutdowns',
114 |     description: 'PC randomly shuts down under load. Case feels hot. Fans run loud then stop.',
115 |     symptoms: ['Random shutdowns', 'Loud fan noise then silence', 'Hot case', 'Crashes under load'],
116 |     correct_answer: 'cooler_issue',
117 |     hints: ['Check CPU cooler is spinning', 'Reapply thermal paste', 'Check fan curves in BIOS', 'Clean dust f...'],
118 |     difficulty: 'medium',
119 |     icon: '0<B!p',
120 |     options: [
121 |         { id: 'cooler_issue', label: 'CPU cooler failure / dust clogged', label_vn: 'Bộ tản nhiệt hỏng / bụi bẩn' },
122 |         { id: 'psu_overload', label: 'PSU overload', label_vn: 'PSU quá tải' },
123 |         { id: 'virus', label: 'Virus / malware', label_vn: 'Virus' },
124 |         { id: 'ram_error', label: 'RAM error', label_vn: 'RAM lỗi' }
125 |     ],
126 |     tool_hints: {
127 |         multimeter: 'PSU voltages fluctuate but stay in range.',
128 |         screwdriver: 'CPU cooler mounting screws are loose. Heatsink not making full contact.',
129 |         compressed_air: 'Heatsink fins completely clogged with dust. Airflow blocked.',
130 |         spare_psu: 'Same shutdowns with spare PSU.',
131 |         bios_speaker: 'Single short beep. Overheat warning in BIOS.',
132 |         beep_pattern: 'BEEP (short) + overheat warning LED',
133 |         diagram: 'cpu_cooler',
134 |         diagnosis_step: 2,
135 |     },
136 | },
137 | {
138 |     id: 'gpu_artifacts',
139 |     title: 'GPU Artifacts / Glitched Display',
140 |     description: 'Screen shows strange colored squares, lines, or flickering. Happens in games and desktop.',
141 |     symptoms: ['Colored artifacts on screen', 'Flickering', 'Lines across display', 'Driver crashes'],
142 |     correct_answer: 'gpu_dying',
143 |     hints: ['Check GPU temperatures', 'Try underclocking GPU', 'Test with another monitor', 'Check GPU seating'],
144 |     difficulty: 'medium',
145 |     icon: '0<B@',
146 |     options: [
147 |         { id: 'gpu_dying', label: 'GPU is failing / VRAM damaged', label_vn: 'GPU hỏng / VRAM hỏng' },
148 |         { id: 'cable_loose', label: 'Loose display cable', label_vn: 'Dây cáp lỏng' },
149 |         { id: 'monitor_broken', label: 'Monitor broken', label_vn: 'Màn hình hỏng' },
150 |         { id: 'driver_issue', label: 'Driver issue only', label_vn: 'Chỉ driver' }
151 |     ],
152 |     tool_hints: {
153 |         multimeter: 'GPU power delivery lines are stable.',
154 |         screwdriver: 'GPU is seated firmly. Reseating did not help.',
155 |         compressed_air: 'GPU fans are clean. Some dust but not overheating.',
156 |         spare_psu: 'Artifacts persist with spare PSU.',
157 |         bios_speaker: 'One short beep. Artifacts visible even in BIOS - hardware failure.',
158 |         beep_pattern: 'BEEP (short)',
159 |         diagram: 'gpu',
160 |         diagnosis_step: 2,
161 |     },
162 | },
163 | {
164 |     id: 'slow_pc',
165 |     title: 'Slow PC / 100% Disk Usage',
166 |     description: 'PC is extremely slow. Task Manager shows 100% disk usage constantly.',
167 |     symptoms: ['100% disk usage', 'Slow boot times', 'Applications freeze', 'File explorer hangs'],
168 |     correct_answer: 'hdd_failing',
169 |     hints: ['Check Task Manager disk usage', 'Run CHKDSK', 'Check S.M.A.R.T. status', 'Consider replacing the ...'],
170 |     difficulty: 'medium',
171 |     icon: '0=0',
172 |     options: [
173 |         { id: 'hdd_failing', label: 'HDD failing / SSD full', label_vn: 'Ổ cứng hỏng / SSD đầy' },
174 |         { id: 'virus', label: 'Virus infection', label_vn: 'Nhiễm virus' },
175 |         { id: 'ram_low', label: 'Not enough RAM', label_vn: 'Thiếu RAM' }
176 |     ]
177 | }
```

```
206 | { id: 'cpu_old', label: 'CPU too old', label_vn: 'CPU quá cũ' },
207 | },
208 | tool_hints: {
209 |   multimeter: 'Drive power is normal.',
210 |   multimeter_vn: 'Nguồn ổ cứng bình thường.',
211 |   screwdriver: 'Cable connections are secure. Drive mounted properly.',
212 |   screwdriver_vn: 'Kết nối cáp chính xác. Ổ cứng lắp đúng cách.',
213 |   compressed_air: 'Drive enclosure is clean and cool.',
214 |   compressed_air_vn: 'Vỏ ổ cứng sạch và mát.',
215 |   spare_psu: 'Still slow with spare PSU.',
216 |   spare_psu_vn: 'Vẫn chậm với nguồn khác.',
217 |   bios_speaker: 'Single short beep. BIOS detects the drive but with high latency.',
218 |   bios_speaker_vn: 'Một beep ngắn. BIOS phát hiện ổ cứng nhưng trễ.',
219 | },
220 | beep_pattern: 'BEEP (short)',
221 | diagram: 'storage',
222 | diagnosis_step: 2,
223 | },
224 | {
225 |   id: 'no_boot_device',
226 |   title: 'No Boot Device Found',
227 |   title_vn: 'Không tìm thấy ổ khi khởi động',
228 |   description: 'Screen says "No boot device found" or "Reboot and Select proper Boot device".',
229 |   description_vn: 'Màn hình báo "No boot device found" hoặc "Reboot and Select proper Boot device".',
230 |   symptoms: ['No boot device error', 'BIOS shows no drive', 'Clicking sound from HDD', 'Was working yesterday...'],
231 |   symptoms_vn: ['Lỗi không có boot device', 'BIOS không thấy ổ', 'Âm thanh nhấp nháy từ ổ cứng', 'Vừa hoạt động bình thường hôm trước'],
232 |   correct_answer: 'ssd_dead',
233 |   hints: ['Check SATA/NVMe connections', 'Boot into BIOS setup', 'Try a different SATA port', 'Check if drive is working'],
234 |   hints_vn: ['Kiểm tra kết nối SATA/NVMe', 'Vào BIOS setup', 'Thử cổng SATA khác', 'Kiểm tra ổ cứng có hoạt động không'],
235 |   difficulty: 'easy',
236 |   icon: '0=0',
237 |   options: [
238 |     { id: 'ssd_dead', label: 'SSD disconnected or dead', label_vn: 'SSD ngắt kết nối hoặc hỏng' },
239 |     { id: 'os_corrupt', label: 'Windows corrupted', label_vn: 'Windows bị hỏng' },
240 |     { id: 'bios_bug', label: 'BIOS settings wrong', label_vn: 'Cài đặt BIOS sai' },
241 |     { id: 'cable_loose', label: 'Loose SATA cable', label_vn: 'Cáp SATA lỏng' },
242 |   ],
243 |   tool_hints: {
244 |     multimeter: 'SATA power cable has correct voltage.',
245 |     multimeter_vn: 'Cáp nguồn SATA có điện áp đúng.',
246 |     screwdriver: 'SSD is not detected even after reseating. Try a known-working SSD.',
247 |     screwdriver_vn: 'SSD không được phát hiện dù đã lắp lại. Thử SSD khác.',
248 |     compressed_air: 'SATA ports cleaned. No visible damage.',
249 |     compressed_air_vn: 'Cổng SATA đã vệ sinh. Không hề hỏng thấy gì.',
250 |     spare_psu: 'Same issue with spare PSU.',
251 |     spare_psu_vn: 'Vẫn lỗi với nguồn khác.',
252 |     bios_speaker: 'Beep codes indicate missing storage device.',
253 |     bios_speaker_vn: 'Mã beep báo thiếu thiết bị lưu trữ.',
254 |   },
255 |   beep_pattern: 'BEEP (short) + "No boot device"',
256 |   diagram: 'storage',
257 |   diagnosis_step: 1,
258 | },
259 | },
260 | {
261 |   id: 'boot_loop',
262 |   title: 'Boot Loop / Continuous Restart',
263 |   title_vn: 'Boot Loop / Khởi động liên tục',
264 |   description: 'PC turns on, shows BIOS splash, then restarts repeatedly before reaching Windows.',
265 |   description_vn: 'PC bật lên, thấy BIOS splash, rồi khởi động lại liên tục trước khi vào Windows.',
266 |   symptoms: ['Continuous restart loop', 'Never reaches desktop', 'Sometimes shows windows repair', 'Started working fine yesterday'],
267 |   symptoms_vn: ['Vòng lặp khởi động liên tục', 'Không vào được desktop', 'Đôi khi hiển thị cửa sổ sửa chữa Windows', 'Vừa hoạt động bình thường hôm trước'],
268 |   correct_answer: 'bios_ram_instability',
269 |   hints: ['Clear CMOS by removing the battery', 'Try booting with one RAM stick', 'Check if XMP/DOCP profile is supported'],
270 |   hints_vn: ['Xóa CMOS bằng cách tháo pin', 'Thử boot với 1 thanh RAM', 'Kiểm tra xem XMP/DOCP có được hỗ trợ không'],
271 |   difficulty: 'hard',
272 |   icon: '0=Y',
273 |   options: [
274 |     { id: 'bios_ram_instability', label: 'Bad BIOS settings / RAM instability', label_vn: 'BIOS sai / RAM không ổn định' },
275 |     { id: 'psu_failing', label: 'PSU failing under load', label_vn: 'Nguồn điện hỏng khi tải' },
276 |     { id: 'gpu_driver', label: 'GPU driver crash loop', label_vn: 'Driver đồ họa bị lỗi' },
277 |     { id: 'windows_update', label: 'Corrupted Windows update', label_vn: 'Cập nhật Windows bị hỏng' },
278 |   ],
279 |   tool_hints: {
280 |     multimeter: 'PSU voltages fluctuate during boot attempt.',
281 |     multimeter_vn: 'Điện áp PSU dao động khi khởi động.',
282 |     screwdriver: 'Clearing CMOS by removing battery for 5 minutes resolved the boot loop.',
283 |     screwdriver_vn: 'Xóa CMOS bằng cách tháo pin 5 phút đã khắc phục được vòng lặp khởi động.',
284 |     compressed_air: 'System is clean. No dust issues.',
285 |     compressed_air_vn: 'Hệ thống sạch. Không có vấn đề bụi.',
286 |     spare_psu: 'Boot loop persists with spare PSU.',
287 |     spare_psu_vn: 'Vòng lặp khởi động vẫn còn với nguồn khác.',
288 |     bios_speaker: 'Repeating short beeps - memory or BIOS error.',
289 |     bios_speaker_vn: 'Beep ngắn lặp lại - lỗi RAM hoặc BIOS.',
290 |   },
291 |   beep_pattern: 'BEEP BEEP BEEP BEEP (continuous short)',
292 |   diagram: 'motherboard',
293 |   diagnosis_step: 3,
294 | },
295 | {
296 |   id: 'strange_noise',
297 |   title: 'Strange Noise from PC',
298 |   title_vn: 'Âm thanh lạ từ PC',
299 |   description: 'PC makes grinding, clicking, or buzzing noise. Performance seems normal otherwise.',
300 |   description_vn: 'PC phát ra tiếng lách cách, ù hoặc rít. Hiệu suất vẫn bình thường.',
301 |   symptoms: ['Grinding noise', 'Clicking sound', 'Buzzing from PSU area', 'Noise changes with load'],
302 |   symptoms_vn: ['Tiếng lách cách', 'Tiếng lộp cộp', 'Ồn từ vùng nguồn', 'Tiếng ồn thay đổi khi tải'],
303 |   correct_answer: 'fan_bearing',
304 |   hints: ['Locate where the noise is coming from', 'Stop fans one by one to identify', 'Check HDD for clicking'],
305 |   hints_vn: ['Xác định vị trí tiếng ồn', 'Dừng từng quạt để xác định', 'Kiểm tra ổ cứng có tiếng rít không'],
306 |   difficulty: 'easy',
307 |   icon: '0=Y',
308 |   options: [
309 |     { id: 'fan_bearing', label: 'Fan bearing worn out', label_vn: 'Bearing quạt hỏng' },
310 |     { id: 'hdd_damage', label: 'HDD head crash', label_vn: 'Ổ cứng bị hỏng đầu' },
311 |     { id: 'coil_whine', label: 'GPU coil whine (normal)', label_vn: 'Ồn cuộn dây đồ họa (bình thường)' },
312 |     { id: 'loose_screw', label: 'Loose screw vibrating', label_vn: 'Đ ốc lỏng rung' },
313 |   ],
314 |   tool_hints: {
315 |     multimeter: 'No electrical issues detected.',
316 |     multimeter_vn: 'Không phát hiện vấn đề điện.',
317 |     screwdriver: 'Stopping the CPU fan temporarily stops the noise. Bearing is worn.',
318 |     screwdriver_vn: 'Dừng quạt CPU tạm thời hết tiếng ồn. Béc quạt hỏng.',
319 |     compressed_air: 'Blowing out dust from the fan did not help.',
320 |     compressed_air_vn: 'Xịt bụi quạt không hết tiếng ồn.',
321 |     spare_psu: 'Noise still present with spare PSU.',
322 |     spare_psu_vn: 'Tiếng ồn vẫn còn với nguồn khác.',
323 |     bios_speaker: 'No beep errors. System POSTs fine.',
324 |     bios_speaker_vn: 'Không lỗi beep. System POST bình thường.',
325 |   },
326 |   beep_pattern: 'BEEP (short - normal)',
327 |   diagram: 'fan',
328 |   diagnosis_step: 1,
329 | },
330 | }
```

```

330 | id: 'no_wifi',
331 | title: 'Network Not Working / WiFi Missing',
332 | title_vn: 'Mạng không có WiFi',
333 | description: 'WiFi icon shows no networks available. Ethernet works fine.',
334 | description_vn: 'Biểu tượng WiFi không thấy mạng nào. Ethernet vẫn hoạt động.',
335 | symptoms: ['WiFi networks not showing', 'WiFi adapter missing in Device Manager', 'Physical WiFi switch off...'],
336 | symptoms_vn: ['Không thấy mạng WiFi', 'Thiếu WiFi adapter trong Device Manager', 'Công tắc WiFi vật lý tắt'],
337 | correct_answer: 'wifi_driver',
338 | hints: ['Check WiFi is enabled in Windows', 'Toggle Airplane mode', 'Check Device Manager for errors', 'Restart the router', 'Check the router\'s power button', 'Check the router\'s power button', 'Check the router\'s power button'],
339 | hints_vn: ['Kiểm tra WiFi đã bật chưa', 'Bật/tắt chế độ máy bay', 'Kiểm tra bộ điều khiển của bộ định tuyến', 'Kiểm tra nút nguồn của bộ định tuyến', 'Kiểm tra nút nguồn của bộ định tuyến', 'Kiểm tra nút nguồn của bộ định tuyến'],
340 | difficulty: 'medium',
341 | icon: '000',
342 | options: [
343 |   { id: 'wifi_driver', label: 'WiFi driver corrupted / antenna disconnected', label_vn: 'Bộ điều khiển WiFi bị hỏng / ăng-ten bị ngắt kết nối' },
344 |   { id: 'router_down', label: 'Router is down', label_vn: 'Bộ định tuyến đã tắt' },
345 |   { id: 'isp_outage', label: 'ISP internet outage', label_vn: 'Sự cố của nhà cung cấp dịch vụ Internet' },
346 |   { id: 'windows_bug', label: 'Windows network bug', label_vn: 'Lỗi mạng Windows' }
347 | ],
348 | tool_hints: {
349 |   multimeter: 'WiFi card power pins are getting correct voltage.',
350 |   multimeter_vn: 'Chân nguồn card WiFi có tiếp xúc đúng.',
351 |   screwdriver: 'WiFi card is seated. Antenna cables are connected but one seems loose.',
352 |   screwdriver_vn: 'Card WiFi gắn chặt. Cáp ăng-ten có vẻ lỏng lẻo.',
353 |   compressed_air: 'WiFi card cleaned. No visible damage.',
354 |   compressed_air_vn: 'Card WiFi đã vệ sinh. Không thấy hỏng hóc.',
355 |   spare_psu: 'WiFi still missing with spare PSU.',
356 |   spare_psu_vn: 'WiFi vẫn không thấy với nguồn khác.',
357 |   bios_speaker: 'POST OK. No hardware beep errors.',
358 |   bios_speaker_vn: 'POST OK. Không lỗi beep phần cứng.',
359 | },
360 | beep_pattern: 'BEEP (short - normal)',
361 | diagram: 'wifi',
362 | diagnosis_step: 2,
363 | },
364 | ],
365 | },
366 | const DIFFICULTY_MAP = {
367 |   easy: { label: 'Easy', label_vn: 'Dễ', color: '#28c76f' },
368 |   medium: { label: 'Medium', label_vn: 'Trung bình', color: '#ffa300' },
369 |   hard: { label: 'Hard', label_vn: 'Khó', color: '#e84855' },
370 | };
371 |
372 | function BeepAnimation({ pattern }) {
373 |   const count = (pattern.match(/BEEP/g) || []).length;
374 |   const isSilence = pattern.includes('silence');
375 |   return (
376 |     <div style={{ display: 'flex', alignItems: 'center', gap: '6px', height: '32px' }}>
377 |       {isSilence ? (
378 |         <span style={{ color: 'var(--text-muted)', fontSize: '12px', fontStyle: 'italic' }}>... (silence)</span>
379 |       ) : (
380 |         Array.from({ length: Math.min(count, 6) }).map((_, i) => (
381 |           <div
382 |             key={i}
383 |             style={{
384 |               width: '10px', height: '10px', borderRadius: '50%',
385 |               background: 'var(--success)',
386 |               animation: 'diagnosisBeepPulse 0.8s ease-in-out ' + (i * 0.15) + 's infinite',
387 |             }}
388 |           />
389 |         ))
390 |       )}
391 |     </div>
392 |   );
393 | }
394 |
395 | function ToolCard({ tool, selected, disabled, onClick, lang }) {
396 |   return (
397 |     <button
398 |       onClick={onClick}
399 |       disabled={disabled}
400 |       style={{
401 |         display: 'flex', flexDirection: 'column', alignItems: 'center', gap: '4px',
402 |         padding: '12px 8px', borderRadius: '10px', cursor: disabled ? 'default' : 'pointer',
403 |         background: selected ? 'rgb(0,212,170,0.15)' : 'var(--bg-elevated)',
404 |         border: selected ? '2px solid var(--success)' : '2px solid transparent',
405 |         opacity: disabled ? 0.4 : 1,
406 |         transition: 'all 0.2s', minWidth: '80px', fontFamily: 'inherit',
407 |         color: 'var(--text-primary)',
408 |       }}
409 |     >
410 |       <span style={{ fontSize: '24px' }}>{tool.icon}</span>
411 |       <span style={{ fontSize: '11px', fontWeight: 600, textAlign: 'center' }}>
412 |         {lang === 'vn' ? tool.label_vn : tool.label}
413 |       </span>
414 |     </button>
415 |   );
416 | }
417 |
418 | function SymptomPanel({ symptoms, symptomsVn, beepPattern, icon, title, titleVn, difficulty, description, desc...
419 |   const diff = DIFFICULTY_MAP[difficulty] || DIFFICULTY_MAP.easy;
420 |   return (
421 |     <div
422 |       style={{
423 |         background: 'var(--bg-surface)', borderRadius: '12px',
424 |         border: '1px solid var(--border-default)', overflow: 'hidden',
425 |         animation: 'diagnosisFadeIn 0.3s ease-out',
426 |       }}
427 |     >
428 |       <div
429 |         style={{
430 |           padding: '16px 20px', borderBottom: '1px solid var(--border-default)',
431 |           display: 'flex', alignItems: 'center', gap: '12px',
432 |         }}
433 |       >
434 |         <span style={{ fontSize: '32px' }}>{icon}</span>
435 |         <div style={{ flex: 1 }}>
436 |           <div style={{ fontSize: '16px', fontWeight: 700, color: 'var(--text-primary)' }}>
437 |             {lang === 'vn' ? titleVn : title}
438 |           </div>
439 |           <div style={{ fontSize: '12px', color: 'var(--text-muted)', marginTop: '2px' }}>
440 |             {lang === 'vn' ? descriptionVn : description}
441 |           </div>
442 |         </div>
443 |         <div
444 |           style={{
445 |             padding: '4px 10px', borderRadius: '6px', fontSize: '11px',
446 |             fontWeight: 700, background: diff.color + '22', color: diff.color,
447 |           }}
448 |       >
449 |         {lang === 'vn' ? diff.label_vn : diff.label}
450 |       </div>
451 |     </div>
452 |   );
453 | }

```

```

454 | <div style={{ padding: '16px 20px', display: 'flex', gap: '24px', flexWrap: 'wrap' }}>
455 | <div>
456 | <div style={{ fontSize: '11px', fontWeight: 700, color: 'var(--text-muted)', marginBottom: '8px', te...
457 |   { (lang === 'vn') ? 'Triệu chứng' : 'Symptoms' }
458 | </div>
459 | <div style={{ display: 'flex', flexDirection: 'column', gap: '6px' }}>
460 |   {{(lang === 'vn' ? symptomsVn : symptoms).map((s, i) => {
461 |     <div key={i} style={{ display: 'flex', alignItems: 'center', gap: '8px' }}>
462 |       <div
463 |         style={{
464 |           width: '6px', height: '6px', borderRadius: '50%',
465 |           background: 'var(--danger)', flexShrink: 0,
466 |           animation: 'diagnosisPulseWarning 1.5s ease-in-out infinite',
467 |         }}
468 |       />
469 |       <span style={{ fontSize: '13px', color: 'var(--text-secondary)' }}>{s}</span>
470 |     </div>
471 |   }}
472 | </div>
473 | </div>
474 | <div>
475 | <div style={{ fontSize: '11px', fontWeight: 700, color: 'var(--text-muted)', marginBottom: '8px', te...
476 |   {(lang === 'vn') ? 'Mã Beep' : 'Beep Pattern'}
477 | </div>
478 | <BeepAnimation pattern={beepPattern} />
479 | <div style={{ fontSize: '11px', color: 'var(--text-muted)', marginTop: '4px' }}>{beepPattern}</div>
480 | </div>
481 | </div>
482 | </div>
483 | );
484 | }
485 |
486 | function ComponentDiagram({ type }) {
487 |   const diagrams = {
488 |     psu: { label: 'PSU', label_vn: 'Nguồn', color: '#e84855', parts: [ 'P
489 |     ram: { label: 'RAM Slots', label_vn: 'Khe RAM', color: '#ffa300', parts: [ 'RAM Slot 1', 'RAM Slot 2', 'CPU...
490 |     motherboard: { label: 'Motherboard', label_vn: 'Bo mạch chủ', color:
491 |     cpu_cooler: { label: 'CPU Cooler', label_vn: 'Tản nhiệt CPU', color:
492 |     gpu: { label: 'GPU', label_vn: 'Card đồ họa', color: '#28c76f', part
493 |     storage: { label: 'Storage Drive', label_vn: 'Ổ cứng', color: '#ffa3
494 |     fan: { label: 'Case Fan', label_vn: 'Quạt case', color: '#4a90e2', p
495 |     wifi: { label: 'WiFi Card', label_vn: 'Card WiFi', color: '#28c76f', parts: [ 'Antenna', 'Chip', 'PCIe Slot...
496 |   };
497 |   const d = diagrams[type] || diagrams.psu;
498 |   return (
499 |     <div
500 |       style={{
501 |         background: 'var(--bg-elevated)', borderRadius: '10px', padding: '16px',
502 |         border: '1px solid var(--border-default)',
503 |       }}
504 |     >
505 |       <div style={{ fontSize: '11px', fontWeight: 700, color: 'var(--text-muted)', marginBottom: '10px', textT...
506 |         {d.label}
507 |       </div>
508 |       <div
509 |         style={{
510 |           display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '8px',
511 |           padding: '20px 12px', background: 'var(--bg-base)', borderRadius: '8px',
512 |           border: '2px solid ' + d.color + '44',
513 |         }}
514 |       >
515 |         <div style={{ fontSize: '36px', textAlign: 'center', color: d.color, opacity: 0.6 }}>
516 |           +!
517 |         </div>
518 |       </div>
519 |       <div style={{ display: 'flex', flexDirection: 'column', gap: '4px', marginTop: '10px' }}>
520 |         {{d.parts.map((part, i) => {
521 |           <div
522 |             key={i}
523 |             style={{
524 |               display: 'flex', alignItems: 'center', gap: '8px', padding: '4px 8px',
525 |               borderRadius: '6px', fontSize: '12px', color: 'var(--text-secondary)',
526 |               background: i === 0 ? d.color + '15' : 'transparent',
527 |             }}
528 |           >
529 |             <div style={{ width: '6px', height: '6px', borderRadius: '50%', background: i === 0 ? d.color : 'v...
530 |             <span style={{ fontWeight: i === 0 ? 700 : 400 }}>{part}</span>
531 |             {i === 0 && (
532 |               <span style={{ fontSize: '10px', color: d.color, fontWeight: 700, marginLeft: 'auto' }}>
533 |                 { '!•' + d.label }
534 |               </span>
535 |             )}
536 |           </div>
537 |         )}}
538 |       </div>
539 |     </div>
540 |   );
541 | }
542 |
543 | function Lives({ count }) {
544 |   return (
545 |     <div style={{ display: 'flex', gap: '4px', alignItems: 'center' }}>
546 |       {{Array.from({ length: 3 }).map((_, i) => {
547 |         <span
548 |           key={i}
549 |           style={{
550 |             fontSize: '20px', transition: 'all 0.3s',
551 |             opacity: i < count ? 1 : 0.2,
552 |             filter: i < count ? 'none' : 'grayscale(1)',
553 |           }}
554 |         >
555 |           'dp
556 |         </span>
557 |       )}}
558 |     </div>
559 |   );
560 | }
561 |
562 | function ProgressBar({ current, total, lang }) {
563 |   const pct = Math.round(((current) / total) * 100);
564 |   return (
565 |     <div style={{ display: 'flex', alignItems: 'center', gap: '10px', flex: 1 }}>
566 |       <div
567 |         style={{
568 |           flex: 1, height: '6px', background: 'var(--bg-elevated)',
569 |           borderRadius: '3px', overflow: 'hidden',
570 |         }}
571 |       >
572 |         <div
573 |           style={{
574 |             height: '100%', width: pct + '%',
575 |             background: 'linear-gradient(90deg, var(--success), var(--accent-blue))',
576 |             borderRadius: '3px', transition: 'width 0.4s ease-out',
577 |           }}

```

```

578 |     />
579 |   </div>
580 |   <span style={{ fontSize: '11px', color: 'var(--text-muted)', fontWeight: 600, whiteSpace: 'nowrap' }}>
581 |     {lang === 'vn' ? (current + 1) + '/' + total : (current + 1) + '/' + total}
582 |   </span>
583 | </div>
584 | );
585 | }
586 |
587 | function getGrade(score) {
588 |   if (score >= 450) return { grade: 'S', label: 'Excellent', label_vn: 'Giỏi' };
589 |   if (score >= 350) return { grade: 'A', label: 'Great', label_vn: 'Khá' };
590 |   if (score >= 250) return { grade: 'B', label: 'Good', label_vn: 'Khá', color: 'var(--accent-blue)' };
591 |   if (score >= 150) return { grade: 'C', label: 'Average', label_vn: 'Trung bình', color: 'var(--warning)' };
592 |   return { grade: 'F', label: 'Fail', label_vn: 'Trở lại', color: 'var(--danger)' };
593 | }
594 |
595 | function formatTime(seconds) {
596 |   const m = Math.floor(seconds / 60);
597 |   const s = seconds % 60;
598 |   return m.toString().padStart(2, '0') + ':' + s.toString().padStart(2, '0');
599 | }
600 |
601 | export default function DiagnosisMode({ lang = 'en', onComplete, onExit }) {
602 |   const [phase, setPhase] = useState('intro');
603 |   const [currentScenario, setCurrentScenario] = useState(0);
604 |   const [lives, setLives] = useState(3);
605 |   const [score, setScore] = useState(0);
606 |   const [selectedTool, setSelectedTool] = useState(null);
607 |   const [toolUsedThisStep, setToolUsedThisStep] = useState(false);
608 |   const [showResult, setShowResult] = useState(null);
609 |   const [feedback, setFeedback] = useState('');
610 |   const [completedScenarios, setCompletedScenarios] = useState([]);
611 |   const [elapsed, setElapsed] = useState(0);
612 |   const [finalResults, setFinalResults] = useState(null);
613 |   const [showExplanation, setShowExplanation] = useState(false);
614 |   const [diagnosisTab, setDiagnosisTab] = useState('game');
615 |   const [aiSymptoms, setAiSymptoms] = useState('');
616 |   const [aiDiagnosing, setAiDiagnosing] = useState(false);
617 |   const [aiResult, setAiResult] = useState(null);
618 |   const [aiError, setAiError] = useState('');
619 |   const timerRef = useRef(null);
620 |
621 |   const tr = useCallback((en, vn) => (lang === 'vn' ? vn : en), [lang]);
622 |
623 |   useEffect(() => {
624 |     if (!document.getElementById('diagnosis-mode-styles')) {
625 |       const style = document.createElement('style');
626 |       style.id = 'diagnosis-mode-styles';
627 |       style.textContent = [
628 |         '@keyframes diagnosisFadeIn { from { opacity: 0; transform: translateY(10px); } to { opacity: 1; trans...',
629 |         '@keyframes diagnosisPulseWarning { 0%, 100% { opacity: 1; transform: scale(1); } 50% { opacity: 0.5; ...',
630 |         '@keyframes diagnosisBeepPulse { 0%, 100% { transform: scale(1); opacity: 0.4; } 50% { transform: scal...',
631 |         '@keyframes diagnosisGlow { 0%, 100% { box-shadow: 0 0 5px var(--success); } 50% { box-shadow: 0 0 20p...',
632 |         '@keyframes diagnosisSlideIn { from { opacity: 0; transform: translateX(20px); } to { opacity: 1; tran...',
633 |         '@keyframes diagnosisShake { 0%, 100% { transform: translateX(0); } 25% { transform: translateX(-8px); ...',
634 |         '.diagnosis-terminal { background: #0a0e17; color: #00ff88; font-family: "Courier New", monospace; tex...',
635 |       ].join(' ');
636 |       document.head.appendChild(style);
637 |     }
638 |   }, []);
639 |
640 |   useEffect(() => {
641 |     if (phase === 'playing') {
642 |       timerRef.current = setInterval(() => {
643 |         setElapsed(function(e) { return e + 1; });
644 |       }, 1000);
645 |       return function() { clearInterval(timerRef.current); };
646 |     }
647 |   }, [phase]);
648 |
649 |   const handleStart = useCallback(function() {
650 |     setPhase('playing');
651 |     setCurrentScenario(0);
652 |     setLives(3);
653 |     setScore(0);
654 |     setCompletedScenarios([]);
655 |     setElapsed(0);
656 |     setSelectedTool(null);
657 |     setToolUsedThisStep(false);
658 |     setShowResult(null);
659 |     setShowExplanation(false);
660 |   }, []);
661 |
662 |   const handleToolSelect = useCallback(function(toolId) {
663 |     if (toolUsedThisStep) return;
664 |     setSelectedTool(toolId);
665 |     setToolUsedThisStep(true);
666 |   }, [toolUsedThisStep]);
667 |
668 |   const finishGame = useCallback(function() {
669 |     clearInterval(timerRef.current);
670 |     var g = getGrade(score);
671 |     setFinalResults({ score: score, scenariosCompleted: completedScenarios.length + 1, livesRemaining: lives, ...
672 |     setPhase('results');
673 |     if (onComplete) {
674 |       onComplete({ score: score, scenariosCompleted: SCENARIOS.length, livesRemaining: lives, grade: g.grade });
675 |     }
676 |   }, [score, lives, completedScenarios, onComplete]);
677 |
678 |   const handleScenarioFail = useCallback(function() {
679 |     setCompletedScenarios(function(c) { return c.concat([currentScenario]); });
680 |     if (currentScenario >= SCENARIOS.length - 1) {
681 |       finishGame();
682 |     } else {
683 |       setCurrentScenario(function(c) { return c + 1; });
684 |       setSelectedTool(null);
685 |       setToolUsedThisStep(false);
686 |       setShowResult(null);
687 |       setShowExplanation(false);
688 |     }
689 |   }, [currentScenario, finishGame]);
690 |
691 |   const handleNextScenario = useCallback(function() {
692 |     setCompletedScenarios(function(c) { return c.concat([currentScenario]); });
693 |     setShowResult(null);
694 |     setShowExplanation(false);
695 |     setSelectedTool(null);
696 |     setToolUsedThisStep(false);
697 |     if (currentScenario >= SCENARIOS.length - 1) {
698 |       finishGame();
699 |     } else {
700 |       setCurrentScenario(function(c) { return c + 1; });
701 |     }

```

```

702 | }, [currentScenario, finishGame]);
703 |
704 | const handleAnswer = useCallback(function(optionId) {
705 |   var scenario = SCENARIOS[currentScenario];
706 |   var isCorrect = optionId === scenario.correct_answer;
707 |   var selectedOption = scenario.options.find(function(o) { return o.id === optionId; });
708 |
709 |   if (isCorrect) {
710 |     var timeBonus = Math.max(0, 60 - (elapsed % 60)) * 2;
711 |     var livesBonus = lives * 25;
712 |     var stepScore = 100 + timeBonus + livesBonus;
713 |     setScore(function(s) { return s + stepScore; });
714 |     setShowResult('correct');
715 |     setFeedback(tr(
716 |       'Correct! The issue was: ' + selectedOption.label + '. ' + scenario.hints[scenario.hints.length - 1],
717 |       'Đúng! Vấn đề là: ' + selectedOption.label_vn + '. ' + scenario
718 |     ));
719 |     setShowExplanation(true);
720 |   } else {
721 |     var newLives = lives - 1;
722 |     setLives(newLives);
723 |     setShowResult('wrong');
724 |     setFeedback(tr(
725 |       'Wrong! ' + selectedOption.label + ' is not the issue.',
726 |       'Sai! ' + selectedOption.label_vn + ' không phải vấn đề.'
727 |     ));
728 |     if (newLives <= 0) {
729 |       setTimeout(function() { handleScenarioFail(); }, 1500);
730 |       return;
731 |     }
732 |   }
733 | }, [currentScenario, lives, elapsed, handleScenarioFail, tr]);
734 |
735 | var scenario = SCENARIOS[currentScenario];
736 |
737 | if (phase === 'intro') {
738 |   return (
739 |     <div
740 |       style={{
741 |         position: 'fixed', inset: 0, zIndex: 9999,
742 |         background: 'var(--bg-base)', display: 'flex', flexDirection: 'column',
743 |         overflow: 'auto',
744 |       }}
745 |     >
746 |       <div
747 |         style={{
748 |           flex: 1, display: 'flex', flexDirection: 'column', alignItems: 'center',
749 |           justifyContent: 'center', padding: '32px', maxWidth: '640px',
750 |           margin: '0 auto', textAlign: 'center', gap: '24px',
751 |         }}
752 |       >
753 |         <div style={{ fontSize: '72px' }}>Ý, </div>
754 |         <h1 style={{ fontSize: '32px', fontWeight: 900, color: 'var(--text-primary)', margin: 0, letterSpacing: ...
755 |           { tr('PC Diagnosis Mode', 'Chẩn đoán PC')} >
756 |         </h1>
757 |         <p style={{ fontSize: '14px', color: 'var(--text-secondary)', lineHeight: '1.6', margin: 0, maxWidth: ...
758 |           { tr(
759 |             'Become a PC technician! Diagnose 10 common PC problems by reading symptoms, using virtual tools...
760 |             'Trở thành kỹ thuật viên PC! Chẩn đoán 10 lỗi PC phổ biến
761 |           )}}
762 |         </p>
763 |
764 |         <div
765 |           style={{
766 |             display: 'grid', gridTemplateColumns: 'repeat(3, 1fr)', gap: '12px',
767 |             width: '100%', maxWidth: '420px',
768 |           }}
769 |         >
770 |           <div style={{ background: 'var(--bg-elevated)', borderRadius: '10px', padding: '12px', textAlign: ...
771 |             <div style={{ fontSize: '24px' }}>0=Ý </div>
772 |             <div style={{ fontSize: '12px', fontWeight: 700, color: 'var(--text-primary)', marginTop: '4px' ...
773 |             <div style={{ fontSize: '10px', color: 'var(--text-muted)' }}>
774 |               { tr('Scenarios', 'Kịch bản')}
775 |             </div>
776 |           </div>
777 |           <div style={{ background: 'var(--bg-elevated)', borderRadius: '10px', padding: '12px', textAlign: ...
778 |             <div style={{ fontSize: '24px' }}>0=Ý </div>
779 |             <div style={{ fontSize: '12px', fontWeight: 700, color: 'var(--text-primary)', marginTop: '4px' ...
780 |             <div style={{ fontSize: '10px', color: 'var(--text-muted)' }}>
781 |               { tr('Lives', 'Mình')}
782 |             </div>
783 |           </div>
784 |           <div style={{ background: 'var(--bg-elevated)', borderRadius: '10px', padding: '12px', textAlign: ...
785 |             <div style={{ fontSize: '24px' }}>0=Ý </div>
786 |             <div style={{ fontSize: '12px', fontWeight: 700, color: 'var(--text-primary)', marginTop: '4px' ...
787 |             <div style={{ fontSize: '10px', color: 'var(--text-muted)' }}>
788 |               { tr('Tools', 'Công cụ')}
789 |             </div>
790 |           </div>
791 |         </div>
792 |
793 |         <div style={{ display: 'flex', gap: '12px', marginTop: '8px' }}>
794 |           <button
795 |             onClick={handleStart}
796 |             style={{
797 |               padding: '14px 40px', background: 'var(--brand-primary)', color: '#fff',
798 |               border: 'none', borderRadius: '10px', fontSize: '15px', fontWeight: 700,
799 |               cursor: 'pointer', fontFamily: 'inherit',
800 |               transition: 'all 0.2s',
801 |             }}
802 |             type="button"
803 |           >
804 |             { tr('Start Diagnosis', 'Bắt đầu Chẩn đoán')}
805 |           </button>
806 |           <button
807 |             onClick={onExit}
808 |             style={{
809 |               padding: '14px 24px', background: 'transparent', color: 'var(--text-secondary)',
810 |               border: '1px solid var(--border-default)', borderRadius: '10px', fontSize: '15px',
811 |               fontWeight: 600, cursor: 'pointer', fontFamily: 'inherit',
812 |               transition: 'all 0.2s',
813 |             }}
814 |             type="button"
815 |           >
816 |             { tr('Exit', 'Thoát')}
817 |           </button>
818 |         </div>
819 |       </div>
820 |     </div>
821 |   );
822 | }
823 | );
824 |
825 |

```



```

826 | if (phase === 'results') {
827 |   var g = getGrade(finalResults ? finalResults.score : score);
828 |   return (
829 |     <div
830 |       style={{
831 |         position: 'fixed', inset: 0, zIndex: 9999,
832 |         background: 'var(--bg-base)', display: 'flex', flexDirection: 'column',
833 |         overflow: 'auto',
834 |       }}
835 |     >
836 |       <div
837 |         style={{
838 |           flex: 1, display: 'flex', flexDirection: 'column', alignItems: 'center',
839 |           justifyContent: 'center', padding: '32px', gap: '24px', maxWidth: '520px',
840 |           margin: '0 auto', textAlign: 'center',
841 |         }}
842 |       >
843 |         <div style={{ fontSize: '72px' }}><h1>
844 |           <h1 style={{ fontSize: '28px', fontWeight: 900, color: 'var(--text-primary)', margin: 0 }}>
845 |             { tr ( ' D i a g n o s i s   C o m p l e t e ! ' , ' C h ò n   o á n   H o à n   T ỷ t ! ' ) }
846 |           </h1>
847 |
848 |           <div
849 |             style={{
850 |               display: 'flex', alignItems: 'center', justifyContent: 'center',
851 |               width: '120px', height: '120px', borderRadius: '50%',
852 |               background: g.color + '22', border: '4px solid ' + g.color,
853 |             }}
854 |           >
855 |             <span style={{ fontSize: '48px', fontWeight: 900, color: g.color }}>
856 |               {g.grade}
857 |             </span>
858 |           </div>
859 |
860 |           <div
861 |             style={{
862 |               background: 'var(--bg-surface)', borderRadius: '12px', padding: '20px',
863 |               border: '1px solid var(--border-default)', width: '100%',
864 |               display: 'flex', flexDirection: 'column', gap: '12px',
865 |             }}
866 |           >
867 |             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
868 |               <span style={{ color: 'var(--text-muted)', fontSize: '13px' }}>
869 |                 <span style={{ color: 'var(--text-primary)', fontSize: '20px', fontWeight: 800 }}>
870 |                   {(finalResults ? finalResults.score : score).toLocaleString()}
871 |                 </span>
872 |               </div>
873 |             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
874 |               <span style={{ color: 'var(--text-muted)', fontSize: '13px' }}>
875 |                 <span style={{ color: 'var(--text-primary)', fontSize: '16px', fontWeight: 700 }}>
876 |                   {SCENARIOS.length}/{SCENARIOS.length}
877 |                 </span>
878 |               </div>
879 |             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
880 |               <span style={{ color: 'var(--text-muted)', fontSize: '13px' }}>
881 |                 <span style={{ display: 'flex', gap: '4px' }}>
882 |                   {Array.from({ length: 3 }).map(function(_, i) {
883 |                     return (
884 |                       <span key={i} style={{ opacity: i < (finalResults ? finalResults.livesRemaining : lives) ?...
885 |                         'dp
886 |                       </span>
887 |                     );
888 |                   })}
889 |                 </span>
890 |               </div>
891 |             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
892 |               <span style={{ color: 'var(--text-muted)', fontSize: '13px' }}>
893 |                 <span style={{ color: 'var(--text-primary)', fontSize: '16px', fontWeight: 700, fontFamily: 'mon...
894 |                   {formatTime(elapsed)}
895 |                 </span>
896 |               </div>
897 |             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
898 |               <span style={{ color: g.color, fontSize: '16px', fontWeight: 800 }}>{g.grade} - {lang === 'vn' ?...
899 |             </div>
900 |           </div>
901 |         </div>
902 |
903 |         <div style={{ display: 'flex', gap: '12px' }}>
904 |           <button
905 |             onClick={handleStart}
906 |             style={{
907 |               padding: '14px 32px', background: 'var(--brand-primary)', color: '#fff',
908 |               border: 'none', borderRadius: '10px', fontSize: '14px', fontWeight: 700,
909 |               cursor: 'pointer', fontFamily: 'inherit',
910 |             }}
911 |             type="button"
912 |           >
913 |             { tr ( ' P l a y   A g a i n ' , ' C h i i   L i i ' ) }
914 |           </button>
915 |           {onExit && (
916 |             <button
917 |               onClick={onExit}
918 |               style={{
919 |                 padding: '14px 24px', background: 'transparent', color: 'var(--text-secondary)',
920 |                 border: '1px solid var(--border-default)', borderRadius: '10px', fontSize: '14px',
921 |                 fontWeight: 600, cursor: 'pointer', fontFamily: 'inherit',
922 |               }}
923 |               type="button"
924 |             >
925 |               {tr('Exit', 'Thoát')}
926 |             </button>
927 |           )}
928 |         </div>
929 |       </div>
930 |     </div>
931 |   );
932 | }
933 |
934 | var toolHint = null;
935 | if (selectedTool && scenario.tool_hints) {
936 |   toolHint = scenario.tool_hints[selectedTool + (lang === 'vn' ? '_vn' : '')] || scenario.tool_hints[selecte...
937 | }
938 |
939 | var headerStyle = {
940 |   display: 'flex', alignItems: 'center', gap: '12px',
941 |   padding: '12px 20px', borderBottom: '1px solid var(--border-default)',
942 |   background: 'var(--bg-surface)',
943 | };
944 |
945 | return (
946 |   <div
947 |     style={{
948 |       position: 'fixed', inset: 0, zIndex: 9999,
949 |       background: 'var(--bg-base)', display: 'flex', flexDirection: 'column',

```

```
950 |         overflow: 'hidden',
951 |     }}
952 | }
953 | <div style={headerStyle}>
954 |     <button
955 |       onClick={onExit}
956 |       style={{
957 |         background: 'transparent', border: 'none', color: 'var(--text-muted)',
958 |         cursor: 'pointer', fontSize: '18px', padding: '4px', fontFamily: 'inherit',
959 |       }}
960 |       type="button"
961 |     >
962 |       9 6 2 |
963 |     </button>
964 |     <div style={{ display: 'flex', gap: '4px', background: 'var(--bg-elevated)', borderRadius: '8px', padd...
965 |     <button
966 |       onClick={() => setDiagnosisTab('game')}
967 |       style={{
968 |         padding: '6px 14px', borderRadius: '6px', border: 'none',
969 |         background: diagnosisTab === 'game' ? 'var(--brand-primary)' : 'transparent',
970 |         color: diagnosisTab === 'game' ? '#fff' : 'var(--text-muted)',
971 |         fontSize: '12px', fontWeight: 600, cursor: 'pointer', fontFamily: 'inherit',
972 |         transition: 'all 0.15s',
973 |       }}
974 |       type="button"
975 |     >
976 |       9 7 6 |
977 |     </button>
978 |     <button
979 |       onClick={() => setDiagnosisTab('ai')}
980 |       style={{
981 |         padding: '6px 14px', borderRadius: '6px', border: 'none',
982 |         background: diagnosisTab === 'ai' ? 'var(--brand-primary)' : 'transparent',
983 |         color: diagnosisTab === 'ai' ? '#fff' : 'var(--text-muted)',
984 |         fontSize: '12px', fontWeight: 600, cursor: 'pointer', fontFamily: 'inherit',
985 |         transition: 'all 0.15s',
986 |       }}
987 |       type="button"
988 |     >
989 |       9 8 9 |
990 |     </button>
991 |   </div>
992 |   {diagnosisTab === 'game' && (
993 |     <div>
994 |       <ProgressBar current={currentScenario} total={SCENARIOS.length} lang={lang} />
995 |       <Lives count={lives} />
996 |       <div style={{ fontSize: '14px', fontWeight: 700, color: 'var(--brand-primary)', fontFamily: 'monos...
997 |       {score.toLocaleString()}
998 |     </div>
999 |     <div style={{ fontSize: '11px', color: 'var(--text-muted)', fontFamily: 'monospace' }}>
1000 |       {formatTime(elapsed)}
1001 |     </div>
1002 |   </div>
1003 | )}
1004 | </div>
1005 |
1006 | <div
1007 |   style={{
1008 |     flex: 1, overflow: 'auto', padding: '20px',
1009 |     maxWidth: '900px', margin: '0 auto', width: '100%',
1010 |     display: 'flex', flexDirection: 'column', gap: '16px',
1011 |   }}
1012 | >
1013 |   {diagnosisTab === 'ai' ? (
1014 |     <div style={{ animation: 'diagnosisFadeIn 0.3s ease-out' }}>
1015 |       <div style={{
1016 |         background: 'var(--bg-surface)', borderRadius: '12px',
1017 |         border: '1px solid var(--border-default)', overflow: 'hidden', marginBottom: '16px',
1018 |       }}>
1019 |         <div className="diagnosis-terminal" style={{
1020 |           padding: '16px 20px', borderBottom: '1px solid rgba(0,255,136,0.15)',
1021 |         }}>
1022 |           <div style={{ fontSize: '11px', color: '#00ff88', opacity: 0.6, marginBottom: '4px' }}>
1023 |             {'>} AI_DIAGNOSIS_TERMINAL v2.0
1024 |           </div>
1025 |           <div style={{ fontSize: '11px', color: '#00ff88', opacity: 0.4, marginBottom: '12px' }}>
1026 |             {'>} M ô t ê l x i P C b ã n g t i c n g V i c t ă n h - n c h o n o á n .
1027 |           </div>
1028 |           <div style={{ position: 'relative' }}>
1029 |             <textarea
1030 |               value={aiSymptoms}
1031 |               onChange={(e) => setAiSymptoms(e.target.value)}
1032 |               placeholder={tr(
1033 |                 'Describe your PC issue... (e.g., "PC keeps beeping and wont turn on")',
1034 |                 'M ô t ê l x i P C c ç a b i n . . . ( V D : " M á y t í n h k ê u b í p b
1035 |               )}
1036 |               style={{
1037 |                 width: '100%', minHeight: '100px', padding: '14px',
1038 |                 background: '#0a0e17', color: '#00ff88', border: '1px solid rgba(0,255,136,0.2)',
1039 |                 borderRadius: '8px', fontSize: '14px', fontFamily: '"Courier New", monospace',
1040 |                 resize: 'vertical', outline: 'none', lineHeight: '1.6',
1041 |               }}
1042 |             </div>
1043 |             <div style={{
1044 |               position: 'absolute', bottom: '8px', right: '12px',
1045 |               fontSize: '10px', color: 'rgba(0,255,136,0.3)',
1046 |               fontFamily: '"Courier New", monospace',
1047 |             }}>
1048 |               { a i S y m p t o m s . l e n g t h } k ý t ñ
1049 |             </div>
1050 |           </div>
1051 |         </div>
1052 |         <div style={{ padding: '12px 20px', display: 'flex', gap: '8px', justifyContent: 'flex-end' }}>
1053 |           <button
1054 |             onClick={async () => {
1055 |               if (!aiSymptoms.trim()) return;
1056 |               setAiDiagnosing(true);
1057 |               setAiError('');
1058 |               setAiResult(null);
1059 |               try {
1060 |                 const res = await fetch('/api/ai/diagnose-symptoms', {
1061 |                   method: 'POST',
1062 |                   headers: { 'Content-Type': 'application/json' },
1063 |                   body: JSON.stringify({ symptoms: aiSymptoms }),
1064 |                 });
1065 |                 const data = await res.json();
1066 |                 if (data.error) throw new Error(data.error);
1067 |                 setAiResult(data);
1068 |               } catch (err) {
1069 |                 1 0 6 9 |
1070 |                 setAiError(err.message || 'L x i k ẽ t n ñ i A I ');
1071 |               } finally {
1072 |                 setAiDiagnosing(false);
1073 |               }
1074 |             }
1075 |           </button>
1076 |         </div>
1077 |       </div>
1078 |     )}
1079 |   )}
1080 | </div>
```



```

1198 |         </div>
1199 |         <p style={{ fontSize: '13px', lineHeight: '1.7', color: 'var(--text-secondary)', margin: 0...
1200 |             {aiResult.reason}
1201 |         </p>
1202 |     </div>
1203 |
1204 |     <div style={{
1205 |         padding: '16px', borderRadius: '10px',
1206 |         background: 'rgba(0,212,170,0.05)', border: '1px solid rgba(0,212,170,0.15)',
1207 |     }}>
1208 |         <div style={{ fontSize: '11px', fontWeight: 700, color: '#00d4aa', textTransform: 'upperca...
1209 |             { t r ( ' F i x   S t e p s ' , ' C á c   b ° Û c   k h ° c   p h á c ' ) }
1210 |         </div>
1211 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
1212 |             {aiResult.steps && aiResult.steps.map((step, i) => (
1213 |                 <div key={i} style={{
1214 |                     display: 'flex', alignItems: 'flex-start', gap: '10px',
1215 |                     padding: '10px 14px', borderRadius: '8px',
1216 |                     background: 'var(--bg-elevated)', border: '1px solid var(--border-default)',
1217 |                 }}>
1218 |                     <div style={{
1219 |                         width: '22px', height: '22px', borderRadius: '50%',
1220 |                         background: '#00d4aa', color: '#fff', fontSize: '11px',
1221 |                         fontWeight: 800, display: 'flex', alignItems: 'center',
1222 |                         justifyContent: 'center', flexShrink: 0,
1223 |                     }}>
1224 |                         {i + 1}
1225 |                     </div>
1226 |                     <span style={{ fontSize: '13px', color: 'var(--text-secondary)', lineHeight: '1.5' }}>
1227 |                         {step}
1228 |                     </span>
1229 |                 </div>
1230 |             ))
1231 |         </div>
1232 |     </div>
1233 |
1234 |     {aiResult.toolsNeeded && aiResult.toolsNeeded.length > 0 && (
1235 |         <div>
1236 |             <div style={{ fontSize: '11px', fontWeight: 700, color: 'var(--text-muted)', textTransfo...
1237 |                 { t r ( ' T o o l s   N e e d e d ' , ' D ã n g   c á c   s n   d ù n g ' ) }
1238 |             </div>
1239 |             <div style={{ display: 'flex', gap: '8px', flexWrap: 'wrap' }}>
1240 |                 {aiResult.toolsNeeded.map((tool, i) => (
1241 |                     <div key={i} style={{
1242 |                         padding: '6px 14px', borderRadius: '10px',
1243 |                         background: 'var(--bg-elevated)', border: '1px solid var(--border-default)',
1244 |                         fontSize: '12px', color: 'var(--text-secondary)', fontWeight: 500,
1245 |                     }}>
1246 |                         0=Y'   { t o o l }
1247 |                     </div>
1248 |                 ))
1249 |             </div>
1250 |         </div>
1251 |     )}
1252 |
1253 |     <div style={{ display: 'flex', gap: '8px', justifyContent: 'flex-end', marginTop: '8px' }}>
1254 |         <button
1255 |             onClick={() => { setAiSymptoms(''); setAiResult(null); }}
1256 |             style={{
1257 |                 padding: '10px 20px', borderRadius: '8px', border: '1px solid var(--border-default)',
1258 |                 background: 'transparent', color: 'var(--text-secondary)',
1259 |                 fontSize: '13px', fontWeight: 600, cursor: 'pointer', fontFamily: 'inherit',
1260 |             }}
1261 |             type="button"
1262 |         >
1263 |             { t r ( ' N e w   D i a g n o s i s ' , ' C h © n   o á n   m ù i ' ) }
1264 |         </button>
1265 |     </div>
1266 | </div>
1267 | </div>
1268 | ))
1269 | </div>
1270 | ) : (
1271 | <=
1272 | {scenario && (
1273 |     <SymptomPanel
1274 |         symptoms={scenario.symptoms}
1275 |         symptomsVn={scenario.symptoms_vn}
1276 |         beepPattern={scenario.beep_pattern}
1277 |         icon={scenario.icon}
1278 |         title={scenario.title}
1279 |         titleVn={scenario.title_vn}
1280 |         difficulty={scenario.difficulty}
1281 |         description={scenario.description}
1282 |         descriptionVn={scenario.description_vn}
1283 |         lang={lang}
1284 |     />
1285 | )}
1286 |
1287 | <div
1288 |     style={{
1289 |         background: 'var(--bg-surface)', borderRadius: '12px',
1290 |         border: '1px solid var(--border-default)', padding: '16px 20px',
1291 |     }}
1292 | >
1293 |     <div style={{ fontSize: '11px', fontWeight: 700, color: 'var(--text-muted)', marginBottom: '10px', t...
1294 |         { t r ( ' T o o l   b o x ' , ' H ù p   d ã n g   c á ' ) }
1295 |     </div>
1296 |     <div style={{ display: 'flex', gap: '8px', flexWrap: 'wrap' }}>
1297 |         {TOOLS.map(function(tool) {
1298 |             return (
1299 |                 <ToolCard
1300 |                     key={tool.id}
1301 |                     tool={tool}
1302 |                     selected={selectedTool === tool.id}
1303 |                     disabled={toolUsedThisStep}
1304 |                     onClick={function() { handleToolSelect(tool.id); }}
1305 |                     lang={lang}
1306 |                 />
1307 |             );
1308 |         })}
1309 |     </div>
1310 |
1311 |     {toolHint && (
1312 |         <div
1313 |             className="diagnosis-terminal"
1314 |             style={{
1315 |                 marginTop: '12px', padding: '12px 16px', borderRadius: '8px',
1316 |                 fontSize: '13px', lineHeight: '1.5',
1317 |                 animation: 'diagnosisFadeIn 0.2s ease-out',
1318 |             }}
1319 |         >
1320 |             <div style={{ color: '#00ff88', opacity: 0.5, fontSize: '11px', marginBottom: '4px' }}>
1321 |                 { ' > ' }   { t r ( ' T o o l   R e s u l t : ' , ' K ẽ t   q u ẽ   c ô n g   c á : ' ) }

```

```

1322|     </div>
1323|     <div>{toolHint}</div>
1324|   </div>
1325| )}
1326|
1327| {!toolUsedThisStep && selectedTool === null && (
1328|   <div style={{ color: 'var(--text-muted)', fontSize: '12px', marginTop: '10px', fontStyle: 'italic'...
1329|     { t r ( ' S e l e c t   a   t o o l   a b o v e   t o   i n s p e c t   t h e   s y s t e m . . . ' ,   ' C h í n
1330|   </div>
1331| )}
1332| </div>
1333|
1334| <div
1335|   style={{
1336|     background: 'var(--bg-surface)', borderRadius: '12px',
1337|     border: '1px solid var(--border-default)', padding: '16px 20px',
1338|   }}
1339| >
1340|   <div style={{ fontSize: '11px', fontWeight: 700, color: 'var(--text-muted)', marginBottom: '10px', t...
1341|     { t r ( ' W h a t   i s   t h e   p r o b l e m   ? ' ,   ' V ã n   Á l à   g ì   ? ' ) }
1342|   </div>
1343|
1344|   {showExplanation ? (
1345|     <div
1346|       style={{
1347|         animation: 'diagnosisFadeIn 0.3s ease-out',
1348|         display: 'flex', flexDirection: 'column', gap: '16px',
1349|       }}
1350|     >
1351|       {showResult === 'correct' ? (
1352|         <div
1353|           style={{
1354|             padding: '16px', borderRadius: '10px',
1355|             background: 'rgba(0,212,170,0.1)',
1356|             border: '1px solid var(--success)',
1357|             animation: 'diagnosisGlow 1.5s ease-in-out',
1358|           }}
1359|         >
1360|           <div style={{ display: 'flex', alignItems: 'center', gap: '8px', marginBottom: '8px' }}>
1361|             <span style={{ display: 'flex', alignSelf: 'center', gap: '8px', margin: 0 10px }}>
1362|               <span style={{ fontSize: '16px', fontWeight: 800, color: 'var(--success)' }}>
1363|                 { t r ( ' C o r r e c t   D i a g n o s i s ! ' ,   ' C h @ n   o á n   C h í n h   X á c ! ' ) }
1364|               </span>
1365|             </div>
1366|             <p style={{ color: 'var(--text-secondary)', fontSize: '13px', lineHeight: '1.5', margin: 0 }}>
1367|               {feedback}
1368|             </p>
1369|           </div>
1370|         ) : (
1371|           <div
1372|             style={{
1373|               padding: '16px', borderRadius: '10px',
1374|               background: 'rgba(232,72,85,0.1)',
1375|               border: '1px solid var(--danger)',
1376|               animation: 'diagnosisShake 0.4s ease-in-out',
1377|             }}
1378|           >
1379|             <div style={{ display: 'flex', alignItems: 'center', gap: '8px', marginBottom: '8px' }}>
1380|               <span style={{ display: 'flex', alignSelf: 'center', gap: '8px', margin: 0 10px }}>
1381|                 <span style={{ fontSize: '16px', fontWeight: 800, color: 'var(--danger)' }}>
1382|                   { t r ( ' W r o n g ! ' ,   ' S a i ! ' ) }
1383|                 </span>
1384|               </div>
1385|               <p style={{ color: 'var(--text-secondary)', fontSize: '13px', lineHeight: '1.5', margin: 0 }}>
1386|                 {feedback}
1387|               </p>
1388|             </div>
1389|           )}
1390|
1391|       {showResult === 'correct' && (
1392|         <div style={{ display: 'flex', gap: '8px', justifyContent: 'flex-end' }}>
1393|           <button
1394|             onClick={handleNextScenario}
1395|             style={{
1396|               padding: '12px 28px', background: 'var(--brand-primary)', color: '#fff',
1397|               border: 'none', borderRadius: '8px', fontSize: '14px', fontWeight: 700,
1398|               cursor: 'pointer', fontFamily: 'inherit',
1399|             }}
1400|           >
1401|             {currentScenario >= SCENARIOS.length - 1
1402|               ? t r ( ' V i e w   R e s u l t s ' ,   ' X e m   K ế t   Q u ế ' )
1403|               : t r ( ' N e x t   S c e n a r i o ' ,   ' K ế c h   b ế n   t i ếp ' ) }
1404|           </button>
1405|         </div>
1406|       )}
1407|     </div>
1408|   ) : (
1409|     <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
1410|       {scenario && scenario.options.map(function(option) {
1411|         var isSelected = false;
1412|         return (
1413|           <button
1414|             key={option.id}
1415|             onClick={function() { handleAnswer(option.id); }}
1416|             style={{
1417|               display: 'flex', alignItems: 'center', gap: '10px',
1418|               padding: '12px 16px', borderRadius: '10px',
1419|               background: 'var(--bg-elevated)', border: '1px solid var(--border-default)',
1420|               cursor: 'pointer', textAlign: 'left', fontFamily: 'inherit',
1421|               color: 'var(--text-primary)', fontSize: '13px',
1422|               transition: 'all 0.15s', width: '100%',
1423|             }}
1424|           >
1425|             type="button"
1426|           </div>
1427|           <div
1428|             style={{
1429|               width: '18px', height: '18px', borderRadius: '50%',
1430|               border: '2px solid var(--text-muted)', flexShrink: 0,
1431|               display: 'flex', alignItems: 'center', justifyContent: 'center',
1432|             }}
1433|           >
1434|             {isSelected && <div style={{ width: '10px', height: '10px', borderRadius: '50%', backgro...
1435|           </div>
1436|           <span>{lang === 'vn' ? option.label_vn : option.label}</span>
1437|         </button>
1438|       )};
1439|     </div>
1440|     <div style={{ color: 'var(--text-muted)', fontSize: '11px', textAlign: 'center', marginTop: '4px...
1441|       { t r ( ' U s e   a   t o o l   f r o m   t h e   t o o l b o x   t o   g e t   h i n t s   b e f o r e   a n s
1442|     </div>
1443|   </div>
1444| )}
1445| </div>

```

```

1446 |         {scenario && !showExplanation && (
1447 |             <ComponentDiagram type={scenario.diagram} />
1448 |         )}
1449 |     </>
1450 | }
1451 | </div>
1452 | </div>
1453 | </div>
1454 | );
1455 | }
1456 |

```

File: components/discussion/discussion/DiscussionPanel.tsx (115 lines, 4.6 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { useDiscussion, Thread } from '@hooks/useDiscussion'
5 | import { ThreadCard } from './ThreadCard'
6 | import { ThreadDetail } from './ThreadDetail'
7 | import { NewThreadModal } from './NewThreadModal'
8 | import { MessageSquare, Plus, RefreshCw } from 'lucide-react'
9 |
10 | interface DiscussionPanelProps {
11 |   lessonId: string
12 |   currentUserId: string
13 |   userRole: 'student' | 'teacher'
14 | }
15 |
16 | export default function DiscussionPanel({ lessonId, currentUserId, userRole }: DiscussionPanelProps) {
17 |   const [activeTab, setActiveTab] = useState<'all' | 'unanswered' | 'pinned'>('all')
18 |   const [selectedThread, setSelectedThread] = useState<Thread | null>(null)
19 |   const [showNewThread, setShowNewThread] = useState(false)
20 |   const { threads, isLoading, createThread } = useDiscussion(lessonId, activeTab)
21 |
22 |   const handleCreateThread = async (title: string, body: string, type: 'question' | 'discussion' | 'announceme...
23 |     await createThread(title, body, type, currentUserId)
24 |   }
25 |
26 |   if (selectedThread) {
27 |     return (
28 |       <div className="h-full">
29 |         <ThreadDetail
30 |           thread={selectedThread}
31 |           currentUserId={currentUserId}
32 |           onBack={() => setSelectedThread(null)}
33 |         />
34 |       </div>
35 |     )
36 |   }
37 |
38 |   return (
39 |     <div className="flex flex-col h-full rounded-2xl overflow-hidden" style={{ background: 'var(--bg-surface)' }}...
40 |     <div className="p-4 flex items-center justify-between flex-shrink-0" style={{ borderBottom: '1px solid v...
41 |     <h3 className="font-bold text-sm flex items-center gap-2" style={{ color: 'var(--text-primary)' }}>
42 |       <MessageSquare size={16} style={{ color: 'var(--brand-primary)' }} />
43 |       <span>Thẻo lun ({ threads.length }) </span>
44 |     </h3>
45 |     <button
46 |       onClick={() => setShowNewThread(true)}
47 |       className="flex items-center gap-1 text-xs px-2.5 py-1.5 rounded-lg transition-all font-semibold"
48 |       style={{ background: 'var(--brand-subtle)', border: '1px solid color-mix(in srgb, var(--brand-primar...
49 |     >
50 |       Plus size={14} />
51 |     </button>
52 |   </div>
53 |
54 |   <div className="flex px-4 py-2 gap-2 flex-shrink-0" style={{ borderBottom: '1px solid var(--border-subtl...
55 |   <button
56 |     onClick={() => setActiveTab('all')}
57 |     className="px-3 py-1 text-xs rounded-full font-medium transition-colors"
58 |     style={activeTab === 'all' ? { background: 'var(--brand-primary)', color: '#000' } : { color: 'var(--...
59 |   >
60 |     T  t  c  f
61 |   </button>
62 |
63 |   <button
64 |     {userRole === 'teacher' && (
65 |       <button
66 |         onClick={() => setActiveTab('unanswered')}
67 |         className="px-3 py-1 text-xs rounded-full font-medium transition-colors"
68 |         style={activeTab === 'unanswered' ? { background: 'var(--brand-primary)', color: '#000' } : { colo...
69 |       >
70 |         Ch  a  tr  l  y  i
71 |     </button>
72 |   )}
73 |   <button
74 |     onClick={() => setActiveTab('pinned')}
75 |     className="px-3 py-1 text-xs rounded-full font-medium transition-colors"
76 |     style={activeTab === 'pinned' ? { background: 'var(--brand-primary)', color: '#000' } : { color: 'va...
77 |   >
78 |     Ghim
79 |   </button>
80 | </div>
81 |
82 | <div className="flex-1 overflow-y-auto p-4 space-y-3">
83 |   {isLoading ? (
84 |     <div className="flex flex-col items-center justify-center py-12 gap-2" style={{ color: 'var(--text-m...
85 |     <RefreshCw size={24} className="animate-spin" style={{ color: 'var(--brand-primary)' }} />
86 |     <span className="text-xs">ang t  i  t  h  o  l  u  n . . . </span>
87 |   </div>
88 |   ) : threads.length === 0 ? (
89 |     <div className="text-center py-12 text-xs" style={{ color: 'var(--text-muted)' }}>
90 |       Kh  n  g  c  ó  c  á  u  h  i  i  n  à  o  .
91 |     </div>
92 |   ) : (
93 |     threads.map(thread => (
94 |       <ThreadCard
95 |         key={thread.id}
96 |         thread={thread}
97 |         onSelect={setSelectedThread}
98 |       />
99 |     ))
100 |   )}
101 | </div>
102 |
103 | <div>
104 |   <NewThreadModal
105 |     isOpen={showNewThread}
106 |     onClose={() => setShowNewThread(false)}
107 |     onSubmit={handleCreateThread}
108 |     userRole={userRole}
109 |   />

```

```

110 |     />
111 |   </div>
112 | )
113 | }
114 | export { DiscussionPanel }
115 |

```

File: components/discussion/discussion/NewThreadModal.tsx (127 lines, 5.2 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { X } from 'lucide-react'
5 |
6 | interface Props {
7 |   isOpen: boolean
8 |   onClose: () => void
9 |   onSubmit: (title: string, body: string, type: 'question' | 'discussion' | 'announcement') => Promise<void>
10 |   userRole: string
11 | }
12 |
13 | export function NewThreadModal({ isOpen, onClose, onSubmit, userRole }: Props) {
14 |   const [title, setTitle] = useState('')
15 |   const [body, setBody] = useState('')
16 |   const [type, setType] = useState<'question' | 'discussion' | 'announcement'>('question')
17 |   const [loading, setLoading] = useState(false)
18 |   const [error, setError] = useState('')
19 |
20 |   if (!isOpen) return null
21 |
22 |   const handleSubmit = async (e: React.FormEvent) => {
23 |     e.preventDefault()
24 |     if (!title.trim() || !body.trim()) return
25 |
26 |     setLoading(true)
27 |     setError('')
28 |     try {
29 |       await onSubmit(title, body, type)
30 |       setTitle('')
31 |       setBody('')
32 |       onClose()
33 |     } catch (err: any) {
34 |       setError(err.message || 'Lỗi khi gửi bài viết.')
35 |     } finally {
36 |       setLoading(false)
37 |     }
38 |   }
39 |
40 |   return (
41 |     <div className="fixed inset-0 z-50 flex items-center justify-center p-4" style={{ background: 'rgba(0,0,0,...
42 |       <div className="rounded-2xl w-full max-w-lg overflow-hidden shadow-2xl animate-in fade-in zoom-in-95 dur...
43 |
44 |       <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
45 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
46 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
47 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
48 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
49 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
50 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
51 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
52 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
53 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
54 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
55 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
56 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
57 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
58 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
59 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
60 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
61 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
62 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
63 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
64 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
65 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
66 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
67 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
68 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
69 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
70 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
71 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
72 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
73 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
74 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
75 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
76 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
77 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
78 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
79 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
80 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
81 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
82 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
83 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
84 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
85 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
86 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
87 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
88 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
89 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
90 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
91 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
92 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
93 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
94 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
95 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
96 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
97 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
98 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
99 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
100 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
101 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
102 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
103 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
104 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
105 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
106 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
107 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
108 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
109 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
110 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
111 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
112 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
113 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...
114 |         <div className="flex items-center justify-between px-6 py-4" style={{ border: '1px solid var(--border...

```

```

115 |         type="submit"
116 |         disabled={loading}
117 |         className="w-full font-bold py-3 rounded-xl transition-opacity disabled:opacity-50"
118 |         style={{ background: 'var(--brand-primary)', color: '#000' }}
119 |     >
120 |         { loading ? ' đang gửi...' : ' gửi' }
121 |     </button>
122 | </form>
123 | </div>
124 | </div>
125 | )
126 | }
127 |

```

File: components/discussion/discussion/ThreadCard.tsx (109 lines, 4.3 KB)

```

1 | 'use client'
2 |
3 | import React from 'react'
4 | import { Thread } from '@hooks/useDiscussion'
5 | import { MessageSquare, ArrowUp, User, Clock } from 'lucide-react'
6 |
7 | interface Props {
8 |   thread: Thread
9 |   onSelect: (thread: Thread) => void
10 | }
11 |
12 | export function ThreadCard({ thread, onSelect }: Props) {
13 |   const getBadgeStyle = () => {
14 |     switch (thread.type) {
15 |       case 'announcement':
16 |         return {
17 |           background: 'color-mix(in srgb, var(--accent-amber) 15%, transparent)',
18 |           color: 'var(--accent-amber)',
19 |           border: '1px solid color-mix(in srgb, var(--accent-amber) 25%, transparent)'
20 |         }
21 |       case 'discussion':
22 |         return {
23 |           background: 'color-mix(in srgb, var(--brand-primary) 15%, transparent)',
24 |           color: 'var(--brand-primary)',
25 |           border: '1px solid color-mix(in srgb, var(--brand-primary) 25%, transparent)'
26 |         }
27 |       default:
28 |         return {
29 |           background: 'color-mix(in srgb, var(--accent-blue) 15%, transparent)',
30 |           color: 'var(--accent-blue)',
31 |           border: '1px solid color-mix(in srgb, var(--accent-blue) 25%, transparent)'
32 |         }
33 |     }
34 |   }
35 |
36 |   const getBadgeText = () => {
37 |     switch (thread.type) {
38 |       case 'announcement': return 'Thông báo'
39 |       case 'discussion': return 'Thảo luận'
40 |       default: return 'Câu hỏi'
41 |     }
42 |   }
43 |
44 |   const timeAgo = (date: string) => {
45 |     const now = new Date()
46 |     const then = new Date(date)
47 |     const diffMs = now.getTime() - then.getTime()
48 |     const diffMins = Math.floor(diffMs / 60000)
49 |     if (diffMins < 1) return 'Vừa xong'
50 |     if (diffMins < 60) return `${diffMins} phút`
51 |     const diffHours = Math.floor(diffMins / 60)
52 |     if (diffHours < 24) return `${diffHours} giờ`
53 |     const diffDays = Math.floor(diffHours / 24)
54 |     if (diffDays < 7) return `${diffDays} ngày`
55 |     return then.toLocaleDateString('vi-VN')
56 |   }
57 |
58 |   return (
59 |     <div>
60 |       <button
61 |         onClick={() => onSelect(thread)}
62 |         className="p-4 rounded-xl transition-all cursor-pointer"
63 |         style={{ background: 'var(--bg-elevated)', border: '1px solid var(--border-default)' }}
64 |         onMouseOver={e => { e.currentTarget.style.borderColor = 'var(--border-strong)'; e.currentTarget.style.ba...
65 |         onMouseOut={e => { e.currentTarget.style.borderColor = 'var(--border-default)'; e.currentTarget.style.ba...
66 |       >
67 |         <div className="flex items-start justify-between gap-3 mb-2">
68 |           <h4 className="font-semibold line-clamp-1 transition-colors" style={{ color: 'var(--text-primary)' }}>
69 |             {thread.title}
70 |           </h4>
71 |           <span className="px-2 py-0.5 text-[10px] font-bold uppercase tracking-wider rounded" style={getBadgeSt...
72 |             {getBadgeText()}
73 |           </span>
74 |         </div>
75 |         <p className="text-xs line-clamp-2 mb-3" style={{ color: 'var(--text-muted)' }}>
76 |           {thread.body}
77 |         </p>
78 |         <div className="flex items-center justify-between text-xs" style={{ color: 'var(--text-muted)' }}>
79 |           <div className="flex items-center gap-2">
80 |             {thread.author?.avatar_url ? (
81 |               <img src={thread.author.avatar_url} alt="Avatar" className="w-5 h-5 rounded-full object-cover" />
82 |             ) : (
83 |               <div className="w-5 h-5 rounded-full flex items-center justify-center" style={{ background: 'var(-...
84 |                 <User size={10} style={{ color: 'var(--text-muted)' }} />
85 |               </div>
86 |             )}
87 |             <span className="truncate max-w-[100px]">{thread.author?.full...
88 |             {thread.author?.role === 'teacher' && (
89 |               <span className="text-[9px] px-1 rounded uppercase font-bold" style={{ background: 'var(--brand-su...
90 |             )}
91 |           </div>
92 |         </div>
93 |         <div className="flex items-center gap-3">
94 |           <span className="flex items-center gap-1">
95 |             <Clock size={12} /> {timeAgo(thread.created_at)}
96 |           </span>
97 |           <span className="flex items-center gap-1">
98 |             <ArrowUp size={14} /> {thread.upvote_count}
99 |           </span>
100 |          <span className="flex items-center gap-1">
101 |            <MessageSquare size={14} /> {thread.reply_count}
102 |          </span>
103 |        </div>
104 |      </div>
105 |    </div>
106 |  )
107 |

```



```
108 | }
109 |
```

File: components/discussion/discussion/ThreadDetail.tsx (164 lines, 7.7 KB)

```
1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { Thread, useThreadReplies } from '@/hooks/useDiscussion'
5 | import { ArrowLeft, Send, User } from 'lucide-react'
6 | import ReactMarkdown from 'react-markdown'
7 | import rehypeSanitize from 'rehype-sanitize'
8 | import remarkGfm from 'remark-gfm'
9 |
10 | interface Props {
11 |   thread: Thread
12 |   currentUserId: string
13 |   onBack: () => void
14 | }
15 |
16 | export function ThreadDetail({ thread, currentUserId, onBack }: Props) {
17 |   const { replies, addReply } = useThreadReplies(thread.id)
18 |   const [commentBody, setCommentBody] = useState('')
19 |   const [submitting, setSubmitting] = useState(false)
20 |   const [error, setError] = useState('')
21 |
22 |   const handleSend = async (e: React.FormEvent) => {
23 |     e.preventDefault()
24 |     if (!commentBody.trim()) return
25 |
26 |     setSubmitting(true)
27 |     setError('')
28 |     try {
29 |       await addReply(commentBody, currentUserId)
30 |       setCommentBody('')
31 |     } catch (err: any) {
32 |       setError(err.message || 'Lỗi khi gửi bình luận.')
33 |     } finally {
34 |       setSubmitting(false)
35 |     }
36 |   }
37 |
38 |   return (
39 |     <div className="flex flex-col h-full rounded-2xl overflow-hidden" style={{ background: 'var(--bg-surface)' }}...
40 |     <div className="flex items-center gap-3 px-4 py-3 flex-shrink-0" style={{ borderBottom: '1px solid var(--border-default)' }}...
41 |       <button
42 |         onClick={onBack}
43 |         className="p-1 rounded-lg transition-all"
44 |         style={{ color: 'var(--text-muted)' }}
45 |         onMouseOver={e => { e.currentTarget.style.color = 'var(--text-primary)'; e.currentTarget.style.backgrou...
46 |         onMouseOut={e => { e.currentTarget.style.color = 'var(--text-muted)'; e.currentTarget.style.backgrou...
47 |       >
48 |       <ArrowLeft size={18} />
49 |     </button>
50 |     <span className="text-xs font-medium" style={{ color: 'var(--text-muted)' }}>
51 |       </div>
52 |
53 |     <div className="flex-1 overflow-y-auto p-4 space-y-6">
54 |       <div className="flex items-center gap-3 text-xs" style={{ color: 'var(--text-muted)' }}>
55 |         <div className="space-y-3 pb-6" style={{ borderBottom: '1px solid var(--border-default)' }}>
56 |           <div className="flex items-center gap-3" style={{ color: 'var(--text-muted)' }}>
57 |             <img src={thread.author.avatar_url} alt="Avatar" className="w-6 h-6 rounded-full object-cover" />
58 |             <div className="w-6 h-6 rounded-full flex items-center justify-center" style={{ background: 'var(--text-muted)' }}>
59 |               <div className="flex items-center justify-center" style={{ color: 'var(--text-muted)' }}>
60 |                 <div>
61 |                   <div>
62 |                     <p className="font-semibold flex items-center gap-1" style={{ color: 'var(--text-primary)' }}>
63 |                       <div>
64 |                         <div>
65 |                           <div>
66 |                             <div>
67 |                               <p className="font-semibold flex items-center gap-1" style={{ color: 'var(--text-primary)' }}>
68 |                                 <div>
69 |                                   <div>
70 |                                     <div>
71 |                                       <div>
72 |                                         <div>
73 |                                           <div>
74 |                                             <div>
75 |                                             </div>
76 |                                           </div>
77 |                                         </div>
78 |                                       </div>
79 |                                     </div>
80 |                                   </div>
81 |                                 </div>
82 |                               </div>
83 |                             </div>
84 |                           </div>
85 |                         </div>
86 |                       </div>
87 |                     </div>
88 |                   </div>
89 |                 </div>
90 |               </div>
91 |             </div>
92 |           </div>
93 |         </div>
94 |       </div>
95 |       <div>
96 |         <div>
97 |           <div>
98 |             <div>
99 |               <div>
100 |                 <div>
101 |                   <div>
102 |                     <div>
103 |                       <div>
104 |                         <div>
105 |                           <div>
106 |                             <div>
107 |                               <div>
108 |                                 <div>
109 |                                   <div>
110 |                                     <div>
111 |                                       <div>
112 |                                         <div>
113 |                                           <div>
114 |                                             <div>
115 |                                             </div>
116 |                                           </div>
117 |                                         </div>
118 |                                       </div>
119 |                                     </div>
120 |                                   </div>
121 |                                 </div>
122 |                               </div>
123 |                             </div>
124 |                           </div>
125 |                         </div>
126 |                       </div>
127 |                     </div>
128 |                   </div>
129 |                 </div>
130 |               </div>
131 |             </div>
132 |           </div>
133 |         </div>
134 |       </div>
135 |     </div>
136 |   )
137 | }
138 |
```

```

119 |         <ReactMarkdown
120 |           remarkPlugins={[remarkGfm]}
121 |           rehypePlugins={[rehypeSanitize]}
122 |         >
123 |           {reply.body}
124 |         </ReactMarkdown>
125 |       </div>
126 |     </div>
127 |   )}}
128 | </div>
129 | }}
130 | </div>
131 | </div>
132 |
133 | {/* Comment Form */}
134 | <form onSubmit={handleSend} className="p-4 flex-shrink-0" style={{ borderTop: '1px solid var(--border-de...
135 |   {error && (
136 |     <div className="p-2 mb-2 text-xs rounded-lg" style={{ background: 'color-mix(in srgb, var(--danger) ...
137 |       {error}
138 |     </div>
139 |   )}
140 |   <div className="flex gap-2">
141 |     <input
142 |       type="text"
143 |       required
144 |       disabled={submitting}
145 |       value={commentBody}
146 |       onChange={(e) => setCommentBody(e.target.value)}
147 |       placeholder="N h - p n ù i d u n g c â u t r ê l ý i . . . "
148 |       className="flex-1 rounded-xl px-4 py-2.5 text-xs outline-none transition-colors"
149 |       style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)', color: 'var(-...
150 |     </div>
151 |     <button
152 |       type="submit"
153 |       disabled={submitting}
154 |       className="p-2.5 rounded-xl transition-opacity disabled:opacity-50 flex items-center justify-center"
155 |       style={{ background: 'var(--brand-primary)', color: '#000' }}
156 |     >
157 |       <Send size={16} />
158 |     </button>
159 |   </div>
160 | </form>
161 | </div>
162 | )
163 | }
164 |

```

File: components/discussion/DiscussionPanel.tsx (115 lines, 4.6 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { useDiscussion, Thread } from '@/hooks/useDiscussion'
5 | import { ThreadCard } from './ThreadCard'
6 | import { ThreadDetail } from './ThreadDetail'
7 | import { NewThreadModal } from './NewThreadModal'
8 | import { MessageSquare, Plus, RefreshCw } from 'lucide-react'
9 |
10 | interface DiscussionPanelProps {
11 |   lessonId: string
12 |   currentUserId: string
13 |   userRole: 'student' | 'teacher'
14 | }
15 |
16 | export default function DiscussionPanel({ lessonId, currentUserId, userRole }: DiscussionPanelProps) {
17 |   const [activeTab, setActiveTab] = useState<'all' | 'unanswered' | 'pinned'>('all')
18 |   const [selectedThread, setSelectedThread] = useState<Thread | null>(null)
19 |   const [showNewThread, setShowNewThread] = useState(false)
20 |   const { threads, isLoading, createThread } = useDiscussion(lessonId, activeTab)
21 |
22 |   const handleCreateThread = async (title: string, body: string, type: 'question' | 'discussion' | 'announceme...
23 |     await createThread(title, body, type, currentUserId)
24 |   }
25 |
26 |   if (selectedThread) {
27 |     return (
28 |       <div className="h-full">
29 |         <ThreadDetail
30 |           thread={selectedThread}
31 |           currentUserId={currentUserId}
32 |           onBack={() => setSelectedThread(null)}
33 |         />
34 |       </div>
35 |     )
36 |   }
37 |
38 |   return (
39 |     <div className="flex flex-col h-full rounded-2xl overflow-hidden" style={{ background: 'var(--bg-surface)'...
40 |       </* Header */>
41 |       <div className="p-4 flex items-center justify-between flex-shrink-0" style={{ borderBottom: '1px solid v...
42 |         <h3 className="font-bold text-sm flex items-center gap-2" style={{ color: 'var(--text-primary)' }}>
43 |           <MessageSquare size={16} style={{ color: 'var(--brand-primary)' }} />
44 |           <span> > T h r e a d s ( { t h r e a d s . l e n g t h } ) < / s p a n >
45 |         </h3>
46 |         <button
47 |           onClick={() => setShowNewThread(true)}
48 |           className="flex items-center gap-1 text-xs px-2.5 py-1.5 rounded-lg transition-all font-semibold"
49 |           style={{ background: 'var(--brand-subtle)', border: '1px solid color-mix(in srgb, var(--brand-primar...
50 |         >
51 |           <Plus size={14} /> H ã i á p
52 |         </button>
53 |       </div>
54 |
55 |       </* Tabs */>
56 |       <div className="flex px-4 py-2 gap-2 flex-shrink-0" style={{ borderBottom: '1px solid var(--border-subtl...
57 |         <button
58 |           onClick={() => setActiveTab('all')}
59 |           className="px-3 py-1 text-xs rounded-full font-medium transition-colors"
60 |           style={activeTab === 'all' ? { background: 'var(--brand-primary)', color: '#000' } : { color: 'var(-...
61 |         >
62 |           T ð t c Ë
63 |         </button>
64 |         {userRole === 'teacher' && (
65 |           <button
66 |             onClick={() => setActiveTab('unanswered')}
67 |             className="px-3 py-1 text-xs rounded-full font-medium transition-colors"
68 |             style={activeTab === 'unanswered' ? { background: 'var(--brand-primary)', color: '#000' } : { colo...
69 |           >
70 |             C h ° a t r ê l ý i
71 |           </button>
72 |         )}
73 |         <button
74 |           onClick={() => setActiveTab('pinned')}

```

```

75 |         className="px-3 py-1 text-xs rounded-full font-medium transition-colors"
76 |         style={activeTab === 'pinned' ? { background: 'var(--brand-primary)', color: '#000' } : { color: 'va...
77 |     >
78 |     Ghim
79 | </button>
80 | </div>
81 |
82 | { /* List Area */ }
83 | <div className="flex-1 overflow-y-auto p-4 space-y-3">
84 |     {isLoading ? (
85 |         <div className="flex flex-col items-center justify-center py-12 gap-2" style={{ color: 'var(--text-m...
86 |         <RefreshCw size={24} className="animate-spin" style={{ color: 'var(--brand-primary)' }} /> </span>
87 |         <span className="text-xs"> ang t Ē i t h Ē o l u - n . . . </span>
88 |     ) : threads.length === 0 ? (
89 |         <div className="text-center py-12 text-xs" style={{ color: 'var(--text-muted)' }}>
90 |         K h ô n g c ó c â u h ĩ i n à o .
91 |     ) : (
92 |         threads.map(thread => (
93 |             <ThreadCard
94 |                 key={thread.id}
95 |                 thread={thread}
96 |                 onSelect={setSelectedThread}
97 |             />
98 |         ))
99 |     )
100 | }
101 | </div>
102 |
103 | { /* Modal */ }
104 | <NewThreadModal
105 |     isOpen={showNewThread}
106 |     onClose={() => setShowNewThread(false)}
107 |     onSubmit={handleCreateThread}
108 |     userRole={userRole}
109 | />
110 | </div>
111 | )
112 | }
113 | }
114 | export { DiscussionPanel }
115 |

```

File: components/discussion/NewThreadModal.tsx (127 lines, 5.2 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { X } from 'lucide-react'
5 |
6 | interface Props {
7 |   isOpen: boolean
8 |   onClose: () => void
9 |   onSubmit: (title: string, body: string, type: 'question' | 'discussion' | 'announcement') => Promise<void>
10 |   userRole: string
11 | }
12 |
13 | export function NewThreadModal({ isOpen, onClose, onSubmit, userRole }: Props) {
14 |   const [title, setTitle] = useState('')
15 |   const [body, setBody] = useState('')
16 |   const [type, setType] = useState<'question' | 'discussion' | 'announcement'>('question')
17 |   const [loading, setLoading] = useState(false)
18 |   const [error, setError] = useState('')
19 |
20 |   if (!isOpen) return null
21 |
22 |   const handleSubmit = async (e: React.FormEvent) => {
23 |     e.preventDefault()
24 |     if (!title.trim() || !body.trim()) return
25 |
26 |     setLoading(true)
27 |     setError('')
28 |     try {
29 |       await onSubmit(title, body, type)
30 |       setTitle('')
31 |       setBody('')
32 |       onClose()
33 |     } catch (err: any) {
34 |       setError(err.message || 'L x i k h i n g c h c Á . ' )
35 |     } finally {
36 |       setLoading(false)
37 |     }
38 |   }
39 |
40 |   return (
41 |     <div className="fixed inset-0 z-50 flex items-center justify-center p-4" style={{ background: 'rgba(0,0,0,...
42 |       <div className="rounded-2xl w-full max-w-lg overflow-hidden shadow-2xl animate-in fade-in zoom-in-95 dur...
43 |         { /* Header */ }
44 |         <div className="flex items-center justify-between px-6 py-4" style={{ borderBottom: '1px solid var(--b...
45 |         <h3 className="font-bold text-lg" style={{ color: 'var(--text-...
46 |         <button
47 |           onClick={onClose}
48 |           className="p-1 rounded-lg transition-all"
49 |           style={{ color: 'var(--text-muted)' }}
50 |           onMouseOver={e => { e.currentTarget.style.color = 'var(--text-primary)'; e.currentTarget.style.bac...
51 |           onMouseOut={e => { e.currentTarget.style.color = 'var(--text-muted)'; e.currentTarget.style.backgr...
52 |         >
53 |         <X size={20} />
54 |       </button>
55 |     </div>
56 |
57 |     { /* Form */ }
58 |     <form onSubmit={handleSubmit} className="p-6 space-y-4">
59 |       {error && (
60 |         <div className="p-3 text-sm rounded-lg" style={{ background: 'color-mix(in srgb, var(--danger) 10%...
61 |         {error}
62 |       </div>
63 |     )}
64 |
65 |     <div>
66 |       <label className="block text-xs font-bold uppercase tracking-wider mb-2" style={{ color: 'var(--te...
67 |       T i Ē u Á t h Ē o l u - n
68 |     </label>
69 |     <input
70 |       type="text"
71 |       required
72 |       value={title}
73 |       onChange={(e) => setTitle(e.target.value)}
74 |     />
75 |     <div className="w-full rounded-xl px-4 py-3 text-sm outline-none transition-colors"
76 |       style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)', color: 'var...
77 |     />
78 |   </div>
79 |

```

```

80 |
81 |         <div className="grid grid-cols-2 gap-4">
82 |             <div>
83 |                 <label className="block text-xs font-bold uppercase tracking-wider mb-2" style={{ color: 'var(--...
84 |                     L o i i   c h  ç       Á
85 |                 </label>
86 |                 <select
87 |                     value={type}
88 |                     onChange={(e) => setType(e.target.value as any)}
89 |                     className="w-full rounded-xl px-4 py-3 text-sm outline-none transition-colors"
90 |                     style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)', color: 'v...
91 |                 >
92 |                     <option value="question">'S   C ầ u   h ỉ i</option>
93 |                     <option value="discussion">Th ẻ o   l u - n</option>
94 |                     { userRole === 'teacher' && <option value="announcement">
95 |
96 |                 </select>
97 |             </div>
98 |         </div>
99 |
100 |         <div>
101 |             <label className="block text-xs font-bold uppercase tracking-wider mb-2" style={{ color: 'var(--te...
102 |                 N ử i   d ứ n g   c h i   t i ẻ t   ( h  ×   t r ầ   M a r k d o w n )
103 |             </label>
104 |             <textarea
105 |                 required
106 |                 rows={5}
107 |                 value={body}
108 |                 onChange={(e) => setBody(e.target.value)}
109 |                 className="w-full rounded-xl px-4 py-3 text-sm outline-none transition-colors resize-none"
110 |                 style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)', color: 'var...
111 |             </div>
112 |
113 |             <button
114 |                 type="submit"
115 |                 disabled={loading}
116 |                 className="w-full font-bold py-3 rounded-xl transition-opacity disabled:opacity-50"
117 |                 style={{ background: 'var(--brand-primary)', color: '#000' }}
118 |             >
119 |                 { loading ? '   a n g   g í i . . . ' : '       n g   c h  ç       Á ' }
120 |             </button>
121 |         </form>
122 |     </div>
123 | </div>
124 | )
125 | }
126 |
127 |

```

File: components/discussion/ThreadCard.tsx (109 lines, 4.3 KB)

```

1 | 'use client'
2 |
3 | import React from 'react'
4 | import { Thread } from '@hooks/useDiscussion'
5 | import { MessageSquare, ArrowUp, User, Clock } from 'lucide-react'
6 |
7 | interface Props {
8 |   thread: Thread
9 |   onSelect: (thread: Thread) => void
10 | }
11 |
12 | export function ThreadCard({ thread, onSelect }: Props) {
13 |   const getBadgeStyle = () => {
14 |     switch (thread.type) {
15 |       case 'announcement':
16 |         return {
17 |           background: 'color-mix(in srgb, var(--accent-amber) 15%, transparent)',
18 |           color: 'var(--accent-amber)',
19 |           border: '1px solid color-mix(in srgb, var(--accent-amber) 25%, transparent)'
20 |         }
21 |       case 'discussion':
22 |         return {
23 |           background: 'color-mix(in srgb, var(--brand-primary) 15%, transparent)',
24 |           color: 'var(--brand-primary)',
25 |           border: '1px solid color-mix(in srgb, var(--brand-primary) 25%, transparent)'
26 |         }
27 |       default:
28 |         return {
29 |           background: 'color-mix(in srgb, var(--accent-blue) 15%, transparent)',
30 |           color: 'var(--accent-blue)',
31 |           border: '1px solid color-mix(in srgb, var(--accent-blue) 25%, transparent)'
32 |         }
33 |     }
34 |   }
35 |
36 |   const getBadgeText = () => {
37 |     switch (thread.type) {
38 |       case 'announcement': return 'Thông báo'
39 |       case 'discussion': return 'Th ẻ o   l u - n '
40 |       default: return 'C ầ u   h ỉ i '
41 |     }
42 |   }
43 |
44 |   const timeAgo = (date: string) => {
45 |     const now = new Date()
46 |     const then = new Date(date)
47 |     const diffMs = now.getTime() - then.getTime()
48 |     const diffMins = Math.floor(diffMs / 60000)
49 |     if (diffMins < 1) return 'V ẻ a   x o n g '
50 |     if (diffMins < 60) return ` ${diffMins} p t r ° ứ c `
51 |     const diffHours = Math.floor(diffMins / 60)
52 |     if (diffHours < 24) return ` ${diffHours} h t r ° ứ c `
53 |     const diffDays = Math.floor(diffHours / 24)
54 |     if (diffDays < 7) return ` ${diffDays} n g à y   t r ° ứ c `
55 |     return then.toLocaleDateString('vi-VN')
56 |   }
57 |
58 |   return (
59 |     <div>
60 |       <onClick={() => onSelect(thread)}
61 |       className="p-4 rounded-xl transition-all cursor-pointer"
62 |       style={{ background: 'var(--bg-elevated)', border: '1px solid var(--border-default)' }}
63 |       <onmouseover={e => { e.currentTarget.style.borderColor = 'var(--border-strong)'; e.currentTarget.style.ba...
64 |       <onmouseout={e => { e.currentTarget.style.borderColor = 'var(--border-default)'; e.currentTarget.style.ba...
65 |     >
66 |     <div className="flex items-start justify-between gap-3 mb-2">
67 |       <h4 className="font-semibold line-clamp-1 transition-colors" style={{ color: 'var(--text-primary)' }}>
68 |         {thread.title}
69 |       </h4>
70 |       <span className="px-2 py-0.5 text-[10px] font-bold uppercase tracking-wider rounded" style={getBadgeSt...
71 |         {getBadgeText()}
72 |       </span>

```

```

73 |     </div>
74 |
75 |     <p className="text-xs line-clamp-2 mb-3" style={{ color: 'var(--text-muted)' }}>
76 |       {thread.body}
77 |     </p>
78 |
79 |     <div className="flex items-center justify-between text-xs" style={{ color: 'var(--text-muted)' }}>
80 |       <div className="flex items-center gap-2">
81 |         {thread.author?.avatar_url ? (
82 |           <img src={thread.author.avatar_url} alt="Avatar" className="w-5 h-5 rounded-full object-cover" />
83 |         ) : (
84 |           <div className="w-5 h-5 rounded-full flex items-center justify-center" style={{ background: 'var(--text-muted)' }}>
85 |             <User size={10} style={{ color: 'var(--text-muted)' }} />
86 |           </div>
87 |         )}
88 |       <span className="truncate max-w-[100px]">{thread.author?.full_name}</span>
89 |       {thread.author?.role === 'teacher' && (
90 |         <span className="text-[9px] px-1 rounded uppercase font-bold" style={{ background: 'var(--brand-secondary)' }}>
91 |           {thread.author?.role}</span>
92 |       )}
93 |     </div>
94 |
95 |     <div className="flex items-center gap-3">
96 |       <span className="flex items-center gap-1">
97 |         <Clock size={12} /> {timeAgo(thread.created_at)}
98 |       </span>
99 |       <span className="flex items-center gap-1">
100 |         <ArrowUp size={14} /> {thread.upvote_count}
101 |       </span>
102 |       <span className="flex items-center gap-1">
103 |         <MessageSquare size={14} /> {thread.reply_count}
104 |       </span>
105 |     </div>
106 |   </div>
107 | )
108 | }
109 |

```

File: components/discussion/ThreadDetail.tsx (164 lines, 7.7 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { Thread, useThreadReplies } from '@/hooks/useDiscussion'
5 | import { ArrowLeft, Send, User } from 'lucide-react'
6 | import ReactMarkdown from 'react-markdown'
7 | import rehypeSanitize from 'rehype-sanitize'
8 | import remarkGfm from 'remark-gfm'
9 |
10 | interface Props {
11 |   thread: Thread
12 |   currentUserId: string
13 |   onBack: () => void
14 | }
15 |
16 | export function ThreadDetail({ thread, currentUserId, onBack }: Props) {
17 |   const { replies, addReply } = useThreadReplies(thread.id)
18 |   const [commentBody, setCommentBody] = useState('')
19 |   const [submitting, setSubmitting] = useState(false)
20 |   const [error, setError] = useState('')
21 |
22 |   const handleSend = async (e: React.FormEvent) => {
23 |     e.preventDefault()
24 |     if (!commentBody.trim()) return
25 |
26 |     setSubmitting(true)
27 |     setError('')
28 |     try {
29 |       await addReply(commentBody, currentUserId)
30 |       setCommentBody('')
31 |     } catch (err: any) {
32 |       setError(err.message || 'Lỗi khi gửi bình luận.')
33 |     } finally {
34 |       setSubmitting(false)
35 |     }
36 |   }
37 |
38 |   return (
39 |     <div className="flex flex-col h-full rounded-2xl overflow-hidden" style={{ background: 'var(--bg-surface)' }}>
40 |       <div className="flex items-center gap-3 px-4 py-3 flex-shrink-0" style={{ border: '1px solid var(--border-default)' }}>
41 |         <button
42 |           onClick={onBack}
43 |           className="p-1 rounded-lg transition-all"
44 |           style={{ color: 'var(--text-muted)' }}
45 |           onMouseOver={e => { e.currentTarget.style.color = 'var(--text-primary)' }}
46 |           onMouseOut={e => { e.currentTarget.style.color = 'var(--text-muted)' }}
47 |           style={{ background: 'var(--text-muted)' }}
48 |         >
49 |           <ArrowLeft size={18} />
50 |         </button>
51 |       </div>
52 |
53 |       <div>
54 |         <div className="flex-1 overflow-y-auto p-4 space-y-6">
55 |           <div className="space-y-3 pb-6" style={{ border: '1px solid var(--border-default)' }}>
56 |             <div className="flex items-center gap-3 text-xs" style={{ color: 'var(--text-muted)' }}>
57 |               {thread.author?.avatar_url ? (
58 |                 <img src={thread.author.avatar_url} alt="Avatar" className="w-6 h-6 rounded-full object-cover" />
59 |               ) : (
60 |                 <div className="w-6 h-6 rounded-full flex items-center justify-center" style={{ background: 'var(--text-muted)' }}>
61 |                   <User size={12} style={{ color: 'var(--text-muted)' }} />
62 |                 </div>
63 |               )}
64 |             <div>
65 |               <p className="font-semibold flex items-center gap-1" style={{ color: 'var(--text-primary)' }}>
66 |                 {thread.author?.role === 'teacher' && (
67 |                   <span className="text-[9px] px-1 rounded uppercase font-bold" style={{ background: 'var(--brand-secondary)' }}>
68 |                     {thread.author?.role}</span>
69 |                 )}
70 |                 <span className="text-[10px]" style={{ color: 'var(--text-muted)' }}>{new Date(thread.created_at).toLocaleDateString()}</span>
71 |               </p>
72 |             </div>
73 |             <h3 className="text-lg font-bold" style={{ color: 'var(--text-primary)' }}>{thread.title}</h3>
74 |
75 |             <div className="prose prose-invert max-w-none text-sm" style={{ color: 'var(--text-secondary)' }}>
76 |               <ReactMarkdown
77 |                 remarkPlugins={[remarkGfm]}
78 |                 rehypePlugins={[rehypeSanitize]}
79 |               />
80 |             </div>
81 |           </div>
82 |         </div>
83 |       </div>

```

```

84 |         {thread.body}
85 |       </ReactMarkdown>
86 |     </div>
87 |   </div>
88 |
89 |   {/* Replies List */}
90 |   <div className="space-y-4">
91 |     <h4 className="text-xs font-bold uppercase tracking-wider" style={{ color: 'var(--text-muted)' }}>Ph...
92 |
93 |     {replies.length === 0 ? (
94 |       <div className="text-center py-6 text-sm" style={{ color: 'var(--text-muted)' }}>
95 |         9 5 |         Ch ỏ a c ỏ ph ẻ n h ỏ i n ỏ o . H ỏ y l ỏ n g ỏ y i s u t i ẻ n t r ẻ l y i !
96 |       </div>
97 |     ) : (
98 |       <div className="space-y-4 pl-2" style={{ borderLeft: '1px solid var(--border-default)' }}>
99 |         {replies.map(reply => (
100 |           <div key={reply.id} className="space-y-2 p-3 rounded-xl" style={{ background: 'var(--bg-elevat...
101 |             <div className="flex items-center justify-between text-xs">
102 |               <div className="flex items-center gap-2" style={{ color: 'var(--text-muted)' }}>
103 |                 {reply.author?.avatar_url ? (
104 |                   <img src={reply.author.avatar_url} alt="Avatar" className="w-4 h-4 rounded-full object...
105 |                 ) : (
106 |                   <div className="w-4 h-4 rounded-full flex items-center justify-center" style={{ backgr...
107 |                     <User size={8} style={{ color: 'var(--text-muted)' }} />
108 |                   </div>
109 |                 )}
110 |                 <span className="font-semibold" style={{ color: 'var(--text-secondary)' }}>{reply.author...
111 |                 {reply.is_teacher_answer && (
112 |                   <span className="text-[8px] px-1 rounded uppercase font-bold" style={{ background: 'va...
113 |                 )}
114 |               </div>
115 |               <span className="text-[10px]" style={{ color: 'var(--text-muted)' }}>{new Date(reply.creat...
116 |             </div>
117 |
118 |             <div className="prose prose-invert max-w-none text-xs" style={{ color: 'var(--text-secondary...
119 |               <ReactMarkdown
120 |                 remarkPlugins={[remarkGfm]}
121 |                 rehypePlugins={[rehypeSanitize]}
122 |               >
123 |                 {reply.body}
124 |               </ReactMarkdown>
125 |             </div>
126 |           </div>
127 |         ))}
128 |       </div>
129 |     )}
130 |   </div>
131 | </div>
132 |
133 |   {/* Comment Form */}
134 |   <form onSubmit={handleSend} className="p-4 flex-shrink-0" style={{ borderTop: '1px solid var(--border-de...
135 |     {error && (
136 |       <div className="p-2 mb-2 text-xs rounded-lg" style={{ background: 'color-mix(in srgb, var(--danger) ...
137 |         {error}
138 |       </div>
139 |     )}
140 |     <div className="flex gap-2">
141 |       <input
142 |         type="text"
143 |         required
144 |         disabled={submitting}
145 |         value={commentBody}
146 |         onChange={(e) => setCommentBody(e.target.value)}
147 |         placeholder="N h - p n ỏ i d ử n g c ỏ u t r ẻ l y i . . ."
148 |         className="flex-1 rounded-xl px-4 py-2.5 text-xs outline-none transition-colors"
149 |         style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)', color: 'var(-...
150 |       </div>
151 |       <button
152 |         type="submit"
153 |         disabled={submitting}
154 |         className="p-2.5 rounded-xl transition-opacity disabled:opacity-50 flex items-center justify-center"
155 |         style={{ background: 'var(--brand-primary)', color: '#000' }}
156 |       >
157 |         <Send size={16} />
158 |       </button>
159 |     </div>
160 |   </form>
161 | </div>
162 | )
163 | }
164 |

```

File: components/ExamsList.js (140 lines, 8.0 KB)

```

1 | 'use client';
2 |
3 | import React, { useState, useEffect } from 'react';
4 | import { supabase } from '@lib/supabase';
5 | import { motion } from 'framer-motion';
6 | import { FileText, Clock, Trophy, ArrowRight, Loader2, Search, Calendar, ChevronRight } from 'lucide-react';
7 |
8 | const ExamsList = ({ lang, onBack }) => {
9 |   const [exams, setExams] = useState([]);
10 |   const [loading, setLoading] = useState(true);
11 |   const [searchTerm, setSearchTerm] = useState('');
12 |
13 |   useEffect(() => {
14 |     fetchExams();
15 |   }, []);
16 |
17 |   async function fetchExams() {
18 |     try {
19 |       setLoading(true);
20 |       const { data } = await supabase
21 |         .from('assignments')
22 |         .select(`
23 |           *
24 |           classes (name)
25 |         `)
26 |         .eq('is_published', true)
27 |         .order('created_at', { ascending: false });
28 |
29 |       setExams(data || []);
30 |     } catch (err) {
31 |       console.error('Error fetching exams:', err);
32 |     } finally {
33 |       setLoading(false);
34 |     }
35 |   }
36 |
37 |   const filteredExams = exams.filter(exam =>
38 |     exam.title.toLowerCase().includes(searchTerm.toLowerCase())
39 |   );

```

```

40 |
41 |     return (
42 |       <div style={{ width: '100%', height: '100%', display: 'flex', flexDirection: 'column', color: 'var(--t...
43 |         <div style={{ marginBottom: '32px' }}>
44 |           <h2 style={{ fontSize: '28px', fontWeight: 800, margin: '0 0 8px 0', color: '#00f3ff' }}>
45 |             { lang === 'en' ? 'Examination Center' : 'Trung tâm K
46 |           </h2>
47 |           <p style={{ color: 'var(--text-secondary)', margin: 0 }}>
48 |             { lang === 'en' ? 'Access your official tests and ass
49 |         </p>
50 |       </div>
51 |
52 |       <div style={{ position: 'relative', marginBottom: '24px' }}>
53 |         <Search style={{ position: 'absolute', left: '16px', top: '50%', transform: 'translateY(-50%)'...
54 |         <input
55 |           type="text"
56 |           placeholder = { lang === 'en' ? "Search exams..." : "T
57 |           value={searchTerm}
58 |           onChange={(e) => setSearchTerm(e.target.value)}
59 |           style={{
60 |             width: '100%', padding: '14px 14px 14px 48px', borderRadius: '14px',
61 |             background: 'var(--bg-elevated)', border: '1px solid var(--border-default)',
62 |             color: 'white', outline: 'none', transition: 'all 0.2s'
63 |           }}
64 |           onFocus={(e) => e.target.style.borderColor = 'var(--brand-primary)'}
65 |           onBlur={(e) => e.target.style.borderColor = 'var(--border-default)'}
66 |         </input>
67 |       </div>
68 |
69 |       {loading ? (
70 |         <div style={{ flex: 1, display: 'flex', justifyContent: 'center', alignItems: 'center' }}>
71 |           <Loader2 className="animate-spin" color="var(--brand-primary)" size={40} />
72 |         </div>
73 |       ) : filteredExams.length > 0 ? (
74 |         <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fill, minmax(320px, 1fr))', g...
75 |           {filteredExams.map((exam, idx) => (
76 |             <motion.div
77 |               key={exam.id}
78 |               initial={{ opacity: 0, y: 10 }}
79 |               animate={{ opacity: 1, y: 0 }}
80 |               transition={{ delay: idx * 0.05 }}
81 |               style={{
82 |                 background: 'var(--bg-surface)', border: '1px solid var(--border-subtle)',
83 |                 borderRadius: '20px', padding: '24px', cursor: 'pointer'
84 |                 transition: 'all 0.3s', position: 'relative', overflow: 'hidden'
85 |               }}
86 |               onMouseEnter={(e) => { e.currentTarget.style.borderColor = 'var(--brand-primary)';...
87 |               onMouseLeave={(e) => { e.currentTarget.style.borderColor = 'var(--border-subtle)';...
88 |             >
89 |               <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'flex-...
90 |                 <div style={{ padding: '10px', background: 'rgba(0, 243, 255, 0.1)', color: '#...
91 |                   <FileText size={20} />
92 |                 </div>
93 |                 <div style={{ textAlign: 'right' }}>
94 |                   <div style={{ fontSize: '10px', fontWeight: 800, color: 'var(--text-muted)...
95 |                     { lang === 'en' ? 'Duration' : 'T
96 |                   </div>
97 |                   <div style={{ fontSize: '12px', fontWeight: 700, color: 'var(--text-primar...
98 |                     <Clock size={12} /> 45'
99 |                   </div>
100 |                 </div>
101 |               </div>
102 |
103 |               <h3 style={{ fontSize: '18px', fontWeight: 700, margin: '0 0 6px 0', lineHeight: 1...
104 |                 <p style={{ fontSize: '13px', color: 'var(--text-secondary)', marginBottom: '20px'...
105 |                   <Calendar size={14} /> {exam.classes?.name || (lang === 'en' ? 'General' : 'To...
106 |                 </p>
107 |
108 |                 <div style={{ height: '1px', background: 'var(--border-subtle)', margin: '0 -24px ...
109 |
110 |                 <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'cente...
111 |                   <div style={{ display: 'flex', alignItems: 'center', gap: '6px' }}>
112 |                     <Trophy size={14} color="var(--warning)" />
113 |                     <span style={{ fontSize: '12px', fontWeight: 700, color: 'var(--text-muted)...
114 |                   </div>
115 |                   <button style={{
116 |                     padding: '8px 16px', background: '#00f3ff', color: '#000',
117 |                     fontSize: '11px', fontWeight: 800, borderRadius: '8px', border: 'none',
118 |                     cursor: 'pointer', display: 'flex', alignItems: 'center', gap: '6px',
119 |                     transition: 'all 0.2s'
120 |                   }}>
121 |                     { lang === 'en' ? 'START EXAM' : 'B
122 |                   </button>
123 |                 </div>
124 |               </motion.div>
125 |             )}}
126 |           </div>
127 |         ) : (
128 |           <div style={{ flex: 1, display: 'flex', flexDirection: 'column', alignItems: 'center', justify...
129 |             <FileText size={48} style={{ color: 'var(--text-muted)', marginBottom: '16px', opacity: 0...
130 |             <p style={{ color: 'var(--text-muted)', fontSize: '14px', fontWeight: 500 }}>
131 |               { lang === 'en' ? 'No exams found.' : 'Không tìm
132 |             </p>
133 |           </div>
134 |         )}
135 |       </div>
136 |     );
137 |   };
138 |
139 |   export default ExamsList;
140 |

```

File: components/FloatingChatbot.js (49 lines, 1.6 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { usePathname } from 'next/navigation'
5 | import { motion, AnimatePresence } from 'framer-motion'
6 | import { MessageCircle, X } from 'lucide-react'
7 | import VirtualAssistant from './VirtualAssistant'
8 |
9 | export default function FloatingChatbot() {
10 |   const [isOpen, setIsOpen] = useState(false)
11 |   const pathname = usePathname()
12 |   const hiddenPaths = ['/student/discussion']
13 |   if (hiddenPaths.some(p => pathname?.startsWith(p))) return null
14 |
15 |   return (
16 |     <
17 |       <AnimatePresence>
18 |       {!isOpen && (
19 |         <motion.button

```

```

20 |         initial={{ scale: 0, opacity: 0 }}
21 |         animate={{ scale: 1, opacity: 1 }}
22 |         exit={{ scale: 0, opacity: 0 }}
23 |         transition={{ type: 'spring', stiffness: 300, damping: 20 }}
24 |         onClick={() => setIsOpen(true)}
25 |         className="fixed bottom-20 right-4 z-[9999] w-14 h-14 rounded-full flex items-center justify-cente...
26 |         style={{
27 |             background: 'var(--brand-primary)',
28 |             borderColor: 'color-mix(in srgb, var(--brand-primary) 50%, white)',
29 |             boxShadow: '0 4px 20px color-mix(in srgb, var(--brand-primary) 40%, transparent)'
30 |         }}
31 |     >
32 |     <MessageCircle size={26} className="text-black" />
33 | </motion.button>
34 | })
35 | </AnimatePresence>
36 |
37 | <VirtualAssistant
38 |   lang="vi"
39 |   appMode={null}
40 |   cartItems={[]}
41 |   remainingBudget={0}
42 |   missionTitle={null}
43 |   isOpen={isOpen}
44 |   setIsOpen={setIsOpen}
45 | />
46 | </>
47 | )
48 | }
49 |

```

File: components/GameEngine.js (1175 lines, 54.9 KB)

```

1 | 'use client';
2 |
3 | import { useEffect, useRef, useState, useImperativeHandle, forwardRef, useCallback } from 'react';
4 | import confetti from 'canvas-confetti';
5 | import { IMAGE_MAP } from './imageAssets';
6 |
7 | const GRAB_THRESHOLD = 0.06;
8 | const RELEASE_THRESHOLD = 0.09;
9 |
10 | const imgCache = new Map();
11 |
12 | function getCacheImg(src) {
13 |   if (imgCache.has(src)) return imgCache.get(src);
14 |   const img = new Image();
15 |   img.crossOrigin = 'anonymous';
16 |   img.onload = () => imgCache.set(src, img);
17 |   img.onerror = () => imgCache.set(src, img);
18 |   img.src = src;
19 |   imgCache.set(src, img);
20 |   return null;
21 | }
22 |
23 | const IMAGE_MAP_TYPES = ['CPU', 'RAM', 'GPU', 'PSU', 'SSD', 'COOLER'];
24 |
25 | function randomImage(type) {
26 |   const images = IMAGE_MAP[type];
27 |   if (!images || images.length === 0) return '';
28 |   return images[Math.floor(Math.random() * images.length)];
29 | }
30 |
31 | function assignRandomImages(components) {
32 |   components.forEach(c => {
33 |     if (!c.imageUrl && IMAGE_MAP_TYPES.includes(c.type)) {
34 |       c.imageUrl = randomImage(c.type);
35 |     }
36 |   });
37 | }
38 |
39 | const GameEngine = forwardRef(({ landmarks, onGameEvent, purchasedItems, defaultPlaced, imageMode }, ref) => {
40 |   const canvasRef = useRef(null);
41 |
42 |   // Offscreen Canvas for caching static heavy elements
43 |   const bgCacheRef = useRef(null);
44 |
45 |   // Game State Ref - Init with or without purchasedItems
46 |   const stateRef = useRef(null);
47 |   if (stateRef.current === null) {
48 |     let initialComponents = [];
49 |     if (purchasedItems) {
50 |       let leftY = 50;
51 |       let rightY = 50;
52 |
53 |     purchasedItems.forEach((item) => {
54 |       if (item.type === 'Mainboard') return; // Don't spawn mainboard as draggable part
55 |
56 |       const spawnProps = (type) => {
57 |         let w, h;
58 |         if (type === 'CPU') { w = 80; h = 80; }
59 |         else if (type === 'RAM') { w = 25; h = 260; }
60 |         else if (type === 'GPU') { w = 320; h = 130; }
61 |         else if (type === 'SSD' || type === 'Storage') { w = 140; h = 35; }
62 |         else if (type === 'COOLER' || type === 'Cooler') { w = 90; h = 90; }
63 |         else if (type === 'PSU') { w = 180; h = 160; }
64 |         else { w = 100; h = 100; }
65 |         return { w, h };
66 |       };
67 |
68 |       const pushComp = (ctype) => {
69 |         const { w, h } = spawnProps(ctype);
70 |         let x, y;
71 |         if (initialComponents.length % 2 === 0) {
72 |           x = 20; y = leftY; leftY += h + 20;
73 |         } else {
74 |           x = 1050; y = rightY; rightY += h + 20;
75 |         }
76 |         initialComponents.push({
77 |           id: Math.random(),
78 |           type: ctype === 'Storage' ? 'SSD' : ctype === 'Cooler' ? 'COOLER' : ctype,
79 |           x, y, width: w, height: h, isGrabbed: false, isPlaced: false, itemData: item
80 |         });
81 |       };
82 |
83 |       pushComp(item.type);
84 |       if (item.type === 'RAM') { // Kits have 2 pieces
85 |         pushComp('RAM');
86 |       }
87 |     });
88 |   } else if (defaultPlaced) {
89 |     initialComponents = [
90 |       { id: 1, type: 'CPU', x: 615, y: 230, width: 80, height: 80, isGrabbed: false, isPlaced: true },

```



```

91 |         { id: 2, type: 'RAM', x: 800, y: 160, width: 25, height: 260, isGrabbed: false, isPlaced: true },
92 |         { id: 4, type: 'RAM', x: 870, y: 160, width: 25, height: 260, isGrabbed: false, isPlaced: true },
93 |         { id: 5, type: 'PSU', x: 135, y: 620, width: 180, height: 160, isGrabbed: false, isPlaced: true },
94 |         { id: 3, type: 'GPU', x: 455, y: 600, width: 320, height: 130, isGrabbed: false, isPlaced: true },
95 |         { id: 6, type: 'SSD', x: 515, y: 520, width: 140, height: 35, isGrabbed: false, isPlaced: true },
96 |         { id: 7, type: 'COOLER', x: 610, y: 225, width: 90, height: 90, isGrabbed: false, isPlaced: true },
97 |     ];
98 | } else {
99 |     initialComponents = [
100 |         // Left Side staging:
101 |         { id: 1, type: 'CPU', x: 20, y: 50, width: 80, height: 80, isGrabbed: false, isPlaced: false },
102 |         { id: 2, type: 'RAM', x: 20, y: 200, width: 25, height: 260, isGrabbed: false, isPlaced: false },
103 |         { id: 4, type: 'RAM', x: 100, y: 200, width: 25, height: 260, isGrabbed: false, isPlaced: false },
104 |         { id: 5, type: 'PSU', x: 20, y: 500, width: 180, height: 160, isGrabbed: false, isPlaced: false },
105 |         // Right Side staging:
106 |         { id: 3, type: 'GPU', x: 1000, y: 650, width: 320, height: 130, isGrabbed: false, isPlaced: false },
107 |         { id: 6, type: 'SSD', x: 1050, y: 50, width: 140, height: 35, isGrabbed: false, isPlaced: false },
108 |         { id: 7, type: 'COOLER', x: 1050, y: 150, width: 90, height: 90, isGrabbed: false, isPlaced: false },
109 |     ];
110 | }
111 |
112 | stateRef.current = {
113 |     components: initialComponents,
114 |     sockets: [
115 |         { id: 'socket-cpu', type: 'CPU', x: 615, y: 230, width: 80, height: 80 },
116 |         { id: 'socket-cooler', type: 'COOLER', x: 610, y: 225, width: 90, height: 90 },
117 |         { id: 'socket-ram-1', type: 'RAM', x: 800, y: 160, width: 25, height: 260, vertical: true },
118 |         { id: 'socket-ram-2', type: 'RAM', x: 870, y: 160, width: 25, height: 260, vertical: true },
119 |         { id: 'socket-gpu', type: 'GPU', x: 455, y: 600, width: 320, height: 130 },
120 |         { id: 'socket-psu', type: 'PSU', x: 135, y: 620, width: 200, height: 160 },
121 |         { id: 'socket-ssd', type: 'SSD', x: 515, y: 520, width: 140, height: 35 }
122 |     ]
123 | };
124 |
125 | // Expose methods to parent
126 | useImperativeHandle(ref, () => ({
127 |     spawnComponent: (type, count = 1) => {
128 |         const spawnOne = () => {
129 |             let w, h, spawnY = 100;
130 |             if (type === 'CPU') { w = 80; h = 80; }
131 |             else if (type === 'RAM') { w = 35; h = 160; spawnY = 250; }
132 |             else if (type === 'GPU') { w = 320; h = 130; spawnY = 450; }
133 |             else if (type === 'SSD') { w = 140; h = 35; spawnY = 320; }
134 |             else if (type === 'COOLER') { w = 90; h = 90; spawnY = 350; }
135 |             else if (type === 'PSU') { w = 180; h = 160; spawnY = 550; }
136 |
137 |             const offset = stateRef.current.components.filter(c => c.type === type && !c.isPlaced).length;
138 |             stateRef.current.components.push({
139 |                 id: Date.now() + Math.random(),
140 |                 type,
141 |                 x: 120 + offset * 30,
142 |                 y: spawnY + offset * 20,
143 |                 width: w, height: h,
144 |                 isGrabbed: false, isPlaced: false
145 |             });
146 |         };
147 |
148 |         for (let i = 0; i < count; i++) {
149 |             spawnOne();
150 |         }
151 |     },
152 |     placeComponent: (type) => {
153 |         const dims = { CPU: {w:80,h:80}, RAM: {w:25,h:260}, GPU: {w:320,h:130}, SSD: {w:140,h:35}, COOLER: {w:90,h:90} };
154 |         let comp = stateRef.current.components.find(c => c.type === type && !c.isPlaced);
155 |         if (!comp) {
156 |             const d = dims[type] || {w:80,h:80};
157 |             const offset = stateRef.current.components.filter(c => c.type === type && !c.isPlaced).length;
158 |             comp = { id: Date.now() + Math.random(), type, x: 120 + offset * 30, y: 100 + offset * 20, width: d.w, height: d.h, isGrabbed: false, isPlaced: false };
159 |             stateRef.current.components.push(comp);
160 |         }
161 |
162 |         const socket = stateRef.current.sockets.find(s => s.type === type && !stateRef.current.components.some(c2 => c2.isPlaced && c2 !== comp && Math.abs(c2.x - s.x) < 5 && !c2.isGrabbed));
163 |         if (!socket) return false;
164 |         comp.x = socket.x;
165 |         comp.y = socket.y;
166 |         comp.width = socket.width;
167 |         comp.height = socket.height;
168 |         comp.isPlaced = true;
169 |         comp.isGrabbed = false;
170 |         if (onGameEvent) onGameEvent('placed', type);
171 |         return true;
172 |     }
173 | });
174 |
175 | // Input States
176 | const mouseRef = useRef({ x: 0, y: 0, isDown: false });
177 | const landmarksRef = useRef(null);
178 | const cursorFilterRef = useRef({ x: null, y: null, alpha: 0.3 }); // Smooth but responsive cursor filter
179 | const grabLockRef = useRef({ x: null, y: null, active: false }); // Freeze cursor when pinching
180 | const pinchStateRef = useRef(false); // Hysteresis: track actual pinch state
181 |
182 | useEffect(() => {
183 |     landmarksRef.current = landmarks;
184 | }, [landmarks]);
185 |
186 | useEffect(() => {
187 |     if (imageMode && stateRef.current) {
188 |         assignRandomImages(stateRef.current.components);
189 |     }
190 | }, [imageMode]);
191 |
192 | const handleMouseMove = (e) => {
193 |     const canvas = canvasRef.current;
194 |     if (!canvas) return;
195 |     const rect = canvas.getBoundingClientRect();
196 |     const scale = canvas.width / rect.width;
197 |     mouseRef.current.x = (e.clientX - rect.left) * scale;
198 |     mouseRef.current.y = (e.clientY - rect.top) * scale;
199 | };
200 |
201 | const handleMouseDown = useCallback((e) => {
202 |     if (!canvasRef.current) return;
203 |     const rect = canvasRef.current.getBoundingClientRect();
204 |     const scale = canvasRef.current.width / rect.width;
205 |     const x = (e.clientX - rect.left) * scale;
206 |     const y = (e.clientY - rect.top) * scale;
207 |
208 |     mouseRef.current = { x, y, isDown: true };
209 |     canvasRef.current.style.cursor = 'grabbing';
210 | }, []);
211 |
212 |
213 |
214 |

```

```

215 | const handleMouseUp = useCallback(() => {
216 |     mouseRef.current.isDown = false;
217 |     if (canvasRef.current) {
218 |         canvasRef.current.style.cursor = 'grab';
219 |     }
220 | }, []);
221 |
222 | useEffect(() => {
223 |     const canvas = canvasRef.current;
224 |     if (!canvas) return;
225 |     const ctx = canvas.getContext('2d', { alpha: false }); // Optimize for no alpha bg
226 |     let animId;
227 |
228 |     // --- HELPER FUNCTIONS FOR 3D RENDERING ---
229 |     const draw3DBox = (context, x, y, width, height, depth, colorTop, colorFront, colorSide, edgeColor = '...
230 |     // Shadow (only if drawing to main context)
231 |     if (context === ctx) {
232 |         context.fillStyle = 'rgba(0,0,0,0.5)';
233 |         context.beginPath();
234 |         context.moveTo(x + depth, y - depth); context.lineTo(x + width + depth, y - depth); context.li...
235 |         context.fill();
236 |     }
237 |
238 |     context.lineWidth = 1;
239 |     context.strokeStyle = edgeColor;
240 |     context.lineJoin = 'round';
241 |
242 |     // Top Face
243 |     context.fillStyle = colorTop;
244 |     context.beginPath();
245 |     context.moveTo(x, y); context.lineTo(x + width, y);
246 |     context.lineTo(x + width + depth, y - depth); context.lineTo(x + depth, y - depth);
247 |     context.closePath();
248 |     context.fill(); context.stroke();
249 |
250 |     // Side Face
251 |     context.fillStyle = colorSide;
252 |     context.beginPath();
253 |     context.moveTo(x + width, y); context.lineTo(x + width + depth, y - depth);
254 |     context.lineTo(x + width + depth, y + height - depth); context.lineTo(x + width, y + height);
255 |     context.closePath();
256 |     context.fill(); context.stroke();
257 |
258 |     // Front Face
259 |     context.fillStyle = colorFront;
260 |     context.fillRect(x, y, width, height);
261 |     context.strokeRect(x, y, width, height);
262 | };
263 |
264 | const drawCylinder = (context, x, y, radius, height, colorDark, colorLight) => {
265 |     const grad = context.createLinearGradient(x - radius, y, x + radius, y);
266 |     grad.addColorStop(0, colorDark);
267 |     grad.addColorStop(0.5, colorLight);
268 |     grad.addColorStop(1, colorDark);
269 |
270 |     context.fillStyle = grad;
271 |     context.beginPath();
272 |     context.ellipse(x, y, radius, radius * 0.4, 0, 0, Math.PI * 2);
273 |     context.fill();
274 |
275 |     context.fillRect(x - radius, y, radius * 2, height);
276 |
277 |     context.beginPath();
278 |     context.ellipse(x, y + height, radius, radius * 0.4, 0, 0, Math.PI * 2);
279 |     context.fill();
280 | };
281 |
282 | // Static heavy elements cached to offscreen canvas
283 | const initBgCache = () => {
284 |     const offCanvas = document.createElement('canvas');
285 |     offCanvas.width = 1400;
286 |     offCanvas.height = 800;
287 |     const oCtx = offCanvas.getContext('2d');
288 |
289 |     // --- 0. Draw Background PC Case Chassis & Case Fans ---
290 |     const caseGrad = oCtx.createLinearGradient(0, 0, 1400, 800);
291 |     caseGrad.addColorStop(0, '#111827');
292 |     caseGrad.addColorStop(1, '#030712');
293 |     oCtx.fillStyle = caseGrad;
294 |     oCtx.fillRect(0, 0, 1400, 800);
295 |
296 |     oCtx.save();
297 |     oCtx.translate(-125, 0); // Center translation for whole Case + Motherboard
298 |
299 |     // Outer Case Boundary
300 |     draw3DBox(oCtx, 240, 20, 940, 780, 20, '#1f2937', '#111827', '#030712');
301 |
302 |     // Front panel inner bracket
303 |     oCtx.fillStyle = '#0f172a';
304 |     oCtx.fillRect(1130, 40, 40, 740);
305 |
306 |     // Cable Routing Grommets
307 |     oCtx.fillStyle = '#0a0a0a';
308 |     for (let i = 100; i < 600; i += 150) {
309 |         oCtx.beginPath(); oCtx.roundRect(1080, i, 40, 100, 20); oCtx.fill();
310 |         oCtx.beginPath(); oCtx.roundRect(650 + (i / 3), 730, 80, 30, 15); oCtx.fill();
311 |     }
312 |
313 |     // PSU Shroud
314 |     draw3DBox(oCtx, 240, 600, 940, 200, 10, '#0f172a', '#020617', '#000000');
315 |     oCtx.fillStyle = '#1e293b';
316 |     oCtx.fillRect(250, 610, 220, 180);
317 |     oCtx.strokeStyle = '#334155'; oCtx.lineWidth = 4;
318 |     oCtx.strokeRect(250, 610, 220, 180);
319 |
320 |
321 |     // Shift mainboard down to center CPU socket in view
322 |     oCtx.save();
323 |     oCtx.translate(0, 40);
324 |
325 |     // --- 1. Draw Elite Dark Motherboard Base ---
326 |     const mainboardGrad = oCtx.createLinearGradient(480, 50, 1150, 770);
327 |     mainboardGrad.addColorStop(0, '#1e1e24'); // Deep carbon slate
328 |     mainboardGrad.addColorStop(0.5, '#121214'); // Dark
329 |     mainboardGrad.addColorStop(1, '#09090b'); // Almost black shadow
330 |     oCtx.fillStyle = mainboardGrad;
331 |     oCtx.fillRect(500, 50, 650, 720);
332 |
333 |     // Motherboard metallic rim/border
334 |     oCtx.strokeStyle = '#3f3f46';
335 |     oCtx.lineWidth = 3;
336 |     oCtx.strokeRect(500, 50, 650, 720);
337 |
338 |     // Cyberpunk PCB Angled Decals

```

```

339| oCtx.fillStyle = 'rgba(255, 255, 255, 0.02)';
340| oCtx.beginPath(); oCtx.moveTo(500, 600); oCtx.lineTo(900, 50); oCtx.lineTo(500, 50); oCtx.fill();
341| oCtx.beginPath(); oCtx.moveTo(1150, 200); oCtx.lineTo(600, 770); oCtx.lineTo(1150, 770); oCtx.fill();
342|
343| // Motherboard metallic mounting standoffs (9 holes)
344| oCtx.fillStyle = '#9ca3af'; // silver screw
345| oCtx.strokeStyle = '#374151'; oCtx.lineWidth = 2;
346| [
347|   [520, 70], [825, 70], [1130, 70],
348|   [520, 380], [825, 380], [1130, 380],
349|   [520, 750], [825, 750], [1130, 750]
350| ].forEach(([sx, sy]) => {
351|   oCtx.beginPath(); oCtx.arc(sx, sy, 5, 0, Math.PI * 2); oCtx.fill(); oCtx.stroke();
352|   // Inner screw hole
353|   oCtx.fillStyle = '#030712';
354|   oCtx.beginPath(); oCtx.arc(sx, sy, 2, 0, Math.PI * 2); oCtx.fill(); oCtx.fillStyle = '#9ca3af';
355| });
356|
357| // Intense High-Density PCB Background Traces (Copper / Gold styling)
358| oCtx.strokeStyle = 'rgba(180, 83, 9, 0.3)'; // Copper traces
359| oCtx.lineWidth = 1;
360| oCtx.beginPath();
361|
362| // Generate heavily dense 90-degree/45-degree circuit traces
363| for (let i = 0; i < 250; i++) {
364|   let startX = 510 + Math.random() * 630;
365|   let startY = 60 + Math.random() * 700;
366|   let len1 = 15 + Math.random() * 50;
367|   let len2 = 10 + Math.random() * 30;
368|   oCtx.moveTo(startX, startY);
369|
370|   let direction = Math.random();
371|   if (direction > 0.75) {
372|     oCtx.lineTo(startX + len1, startY);
373|     oCtx.lineTo(startX + len1 + len2, startY + len2);
374|   } else if (direction > 0.5) {
375|     oCtx.lineTo(startX, startY + len1);
376|     oCtx.lineTo(startX - len2, startY + len1 + len2);
377|   } else if (direction > 0.25) {
378|     oCtx.lineTo(startX - len1, startY);
379|     oCtx.lineTo(startX - len1, startY - len2);
380|   } else {
381|     oCtx.lineTo(startX, startY - len1);
382|     oCtx.lineTo(startX + len2, startY - len1 - len2);
383|   }
384| }
385| oCtx.stroke();
386|
387| // Micro-vias and solder pads
388| oCtx.fillStyle = 'rgba(217, 119, 6, 0.4)'; // Dim Copper for pads
389| for (let i = 0; i < 300; i++) {
390|   let vx = 520 + Math.random() * 600;
391|   let vy = 70 + Math.random() * 680;
392|   oCtx.fillRect(vx, vy, 2, 2);
393| }
394| oCtx.fillStyle = '#fde047'; // Shiny Gold solder points
395| for (let i = 0; i < 60; i++) {
396|   let px = 550 + Math.random() * 550;
397|   let py = 100 + Math.random() * 600;
398|   oCtx.beginPath();
399|   oCtx.arc(px, py, 1.5, 0, Math.PI * 2);
400|   oCtx.fill();
401| }
402|
403| // 3D VRM Heatsinks (Aggressive Stealth Aluminum)
404| draw3DBox(oCtx, 580, 50, 240, 60, 15, '#3f3f46', '#27272a', '#18181b', '#000');
405| draw3DBox(oCtx, 580, 120, 50, 180, 15, '#3f3f46', '#27272a', '#18181b', '#000');
406|
407| // VRM Heatsink Fins Detail with Cyber-slits
408| oCtx.fillStyle = '#09090b';
409| for (let i = 590; i < 810; i += 12) { oCtx.fillRect(i, 58, 4, 44); oCtx.fillRect(i + 1, 65, 2, 30); }
410| for (let i = 130; i < 290; i += 12) { oCtx.fillRect(588, i, 34, 4); oCtx.fillRect(595, i + 1, 20, ...
411|
412| // CPU Socket Area Bracket (Silver Metal LGA1700/AM5 Style)
413| // Outer reinforcement plate
414| draw3DBox(oCtx, 715, 165, 130, 130, 8, '#a1a1aa', '#d4d4d8', '#71717a', '#52525b');
415|
416| // Inner hollow for pins
417| oCtx.fillStyle = '#111827';
418| oCtx.fillRect(735, 185, 90, 90);
419|
420| // Socket pins bed (LGA style - dense gold contact array)
421| oCtx.fillStyle = '#030712';
422| oCtx.fillRect(740, 190, 80, 80);
423| oCtx.fillStyle = '#fef08a'; // Bright gold pins
424| for (let x = 742; x < 818; x += 3) {
425|   for (let y = 192; y < 268; y += 3) {
426|     if ((x + y) % 7 !== 0) oCtx.fillRect(x, y, 1, 1);
427|   }
428| }
429|
430| // CPU Socket alignment triangle marker (gold corner)
431| oCtx.fillStyle = '#fbbf24';
432| oCtx.beginPath();
433| oCtx.moveTo(785, 182);
434| oCtx.lineTo(795, 182);
435| oCtx.lineTo(790, 190);
436| oCtx.closePath();
437| oCtx.fill();
438| oCtx.fillStyle = '#fff';
439| oCtx.font = 'bold 7px monospace';
440| oCtx.textAlign = 'center';
441| o C t x . f i l l T e x t ( '% ' , 7 9 0 , 1 9 3 ) ;
442|
443| // "LGA1700/AM5" text on socket
444| oCtx.fillStyle = '#94a3b8';
445| oCtx.font = 'bold 7px monospace';
446| oCtx.textAlign = 'left';
447| oCtx.fillText('LGA', 742, 205);
448| oCtx.fillText('1700', 742, 214);
449|
450| // Top Hinge Bracket & Bottom Locking Clasp (Silver)
451| draw3DBox(oCtx, 730, 160, 100, 15, 6, '#cbd5e1', '#f1f5f9', '#94a3b8'); // Top hinge
452| draw3DBox(oCtx, 750, 285, 60, 12, 6, '#cbd5e1', '#f1f5f9', '#94a3b8'); // Bottom lip
453|
454| // Locking Metal Lever (Right side) - improved detail
455| oCtx.strokeStyle = '#e2e8f0'; oCtx.lineWidth = 5;
456| oCtx.shadowBlur = 4; oCtx.shadowColor = 'rgba(0,0,0,0.5)';
457| oCtx.beginPath();
458| oCtx.moveTo(850, 280); // Bottom latch
459| oCtx.lineTo(865, 280);
460| oCtx.lineTo(865, 150); // Up arm
461| oCtx.lineTo(845, 150); // Top hook
462| oCtx.stroke();

```

```

463 | oCtx.shadowBlur = 0;
464 |
465 | // Load lever highlight
466 | oCtx.strokeStyle = '#fef08a';
467 | oCtx.lineWidth = 2;
468 | oCtx.beginPath();
469 | oCtx.moveTo(850, 280);
470 | oCtx.lineTo(865, 280);
471 | oCtx.stroke();
472 |
473 | // Strict Proportion: Exactly 4 RAM Slots (Dual Channel Configuration)
474 | // Alternating colors: Elite dark gray and jet black
475 | for (let i = 0; i < 4; i++) {
476 |     const startX = 890 + (i * 35); // 35px gap between DIMMs precisely scaled
477 |
478 |     // Active slots 2 and 4
479 |     const isOptimalSlot = (i === 1 || i === 3);
480 |     const baseColor = isOptimalSlot ? '#52525b' : '#18181b';
481 |     const shadowColor = isOptimalSlot ? '#3f3f46' : '#09090b';
482 |
483 |     draw3DBox(oCtx, startX, 120, 18, 260, 6, baseColor, shadowColor, '#000');
484 |
485 |     // Slot groove
486 |     oCtx.fillStyle = '#000'; oCtx.fillRect(startX + 4, 125, 10, 250);
487 |
488 |     // Top/Bottom clips (grey vs dark grey based on slot)
489 |     oCtx.fillStyle = isOptimalSlot ? '#9ca3af' : '#4b5563';
490 |     oCtx.fillRect(startX - 1, 108, 20, 12); oCtx.fillRect(startX - 1, 375, 20, 12);
491 |
492 |     // PCB Gold trace pins bordering slot
493 |     oCtx.fillStyle = '#f59e0b';
494 |     for (let py = 130; py < 370; py += 5) {
495 |         if (py !== 260) { // realistic notch gap
496 |             oCtx.fillRect(startX - 2, py, 3, 2);
497 |             oCtx.fillRect(startX + 17, py, 3, 2);
498 |         }
499 |     }
500 | }
501 |
502 | // GPU Armor Reinforced PCIe Slots (Steel/Metal)
503 | draw3DBox(oCtx, 560, 550, 400, 30, 8, '#e4e4e7', '#a1a1aa', '#71717a');
504 | oCtx.fillStyle = '#09090b'; oCtx.fillRect(570, 560, 380, 10);
505 | oCtx.fillStyle = '#fbbf24'; // PCIe gold pins
506 | for (let i = 575; i < 945; i += 4) oCtx.fillRect(i, 563, 2, 4);
507 | // Secondary simple slot
508 | draw3DBox(oCtx, 560, 650, 320, 20, 5, '#27272a', '#18181b', '#09090b');
509 |
510 | // M.2 NVMe Slot with Armor Heatsink Shield
511 | draw3DBox(oCtx, 640, 480, 150, 35, 6, '#3f3f46', '#27272a', '#18181b');
512 | oCtx.fillStyle = '#a1a1aa'; // Shield metallic logo accent
513 | oCtx.fillRect(660, 495, 60, 2); oCtx.fillRect(660, 500, 40, 2);
514 |
515 | // 3D CMOS Battery (Moved)
516 | drawCylinder(oCtx, 1000, 580, 22, 6, '#6b7280', '#e5e7eb');
517 | oCtx.beginPath(); oCtx.arc(1000, 580, 18, 0, Math.PI * 2); oCtx.strokeStyle = '#9ca3af'; oCtx.stro...
518 |
519 | // Scattered Metallic Chokes and Black Capacitors
520 | // Silver blocks around CPU
521 | for (let i = 0; i < 5; i++) draw3DBox(oCtx, 670, 180 + i * 20, 15, 15, 8, '#d4d4d8', '#a1a1aa', '#...
522 | for (let i = 0; i < 3; i++) draw3DBox(oCtx, 720 + i * 25, 120, 15, 15, 8, '#d4d4d8', '#a1a1aa', '#...
523 | // Premium Gold/Black Capacitors
524 | for (let i = 0; i < 8; i++) drawCylinder(oCtx, 610, 135 + i * 12, 5, 10, '#18181b', '#3f3f46');
525 | oCtx.fillStyle = '#f59e0b'; // Gold top caps
526 | for (let i = 0; i < 8; i++) oCtx.beginPath(), oCtx.arc(610, 135 + i * 12, 3, 0, Math.PI * 2), oCtx...
527 |
528 | // Southbridge Base (Chipset Heatsink)
529 | draw3DBox(oCtx, 960, 630, 110, 10, '#27272a', '#18181b', '#09090b', '#000');
530 | oCtx.fillStyle = '#18181b'; oCtx.fillRect(970, 640, 90, 90);
531 |
532 | // Real-life Steel/Silver Motherboard Screws (replaced green)
533 | const drawScrew = (x, y) => {
534 |     oCtx.fillStyle = '#f4f4f5'; // Bright silver
535 |     oCtx.beginPath(); oCtx.arc(x, y, 6, 0, Math.PI * 2); oCtx.fill();
536 |     oCtx.fillStyle = '#a1a1aa'; // Dark inner socket
537 |
538 |     oCtx.beginPath(); oCtx.arc(x, y, 4, 0, Math.PI * 2); oCtx.fill();
539 |     oCtx.strokeStyle = '#064e3b'; // Screw cross slots
540 |     oCtx.lineWidth = 1.5;
541 |     oCtx.beginPath(); oCtx.moveTo(x - 3, y - 3); oCtx.lineTo(x + 3, y + 3); oCtx.moveTo(x + 3, y -...
542 | };
543 | drawScrew(520, 70);
544 | drawScrew(1130, 70);
545 | drawScrew(520, 750);
546 | drawScrew(1130, 750);
547 | drawScrew(520, 410);
548 | drawScrew(1130, 410);
549 |
550 | oCtx.restore(); // end mainboard translate shift
551 | oCtx.restore(); // outer save (case translate)
552 | bgCacheRef.current = offCanvas;
553 | };
554 |
555 | if (!bgCacheRef.current) initBgCache();
556 |
557 | const drawAnimatedMotherboard = (ctx, time) => {
558 |     // Draw static baked layer
559 |     if (bgCacheRef.current) {
560 |         ctx.drawImage(bgCacheRef.current, 0, 0);
561 |     }
562 |
563 |
564 |     // Glowing Circuit Traces (Animated)
565 |     ctx.save();
566 |     ctx.translate(-125, 0);
567 |     ctx.translate(0, 40);
568 |
569 |     ctx.strokeStyle = '#d97706';
570 |     ctx.lineWidth = 1.5;
571 |     ctx.shadowBlur = 4;
572 |     ctx.shadowColor = '#f59e0b';
573 |     ctx.setLineDash([15, 20]);
574 |     ctx.lineDashOffset = -time * 10;
575 |     ctx.beginPath();
576 |     ctx.moveTo(600, 300); ctx.lineTo(550, 300); ctx.lineTo(550, 150); ctx.lineTo(520, 150);
577 |     ctx.moveTo(800, 200); ctx.lineTo(850, 200); ctx.lineTo(850, 100); ctx.lineTo(1100, 100);
578 |     ctx.moveTo(600, 350); ctx.lineTo(570, 380); ctx.lineTo(570, 480); ctx.lineTo(520, 480);
579 |     ctx.moveTo(800, 350); ctx.lineTo(850, 390); ctx.lineTo(1050, 390); ctx.lineTo(1050, 650);
580 |     ctx.stroke();
581 |     ctx.shadowBlur = 0;
582 |     ctx.setLineDash([]);
583 |
584 |     // Southbridge Pulsating Glow
585 |     const pulse = (Math.sin(time * 2) + 1) / 2;
586 |     ctx.shadowBlur = 10; ctx.shadowColor = `rgba(245, 158, 11, ${pulse})`;

```

```

587|         ctx.strokeStyle = '#f59e0b'; ctx.lineWidth = 2; ctx.strokeRect(970, 640, 90, 90);
588|         ctx.shadowBlur = 0;
589|
590|         // --- Component Slot Labels ---
591|         ctx.save();
592|         ctx.font = 'bold 11px monospace';
593|         ctx.textAlign = 'center';
594|         ctx.textBaseline = 'bottom';
595|         ctx.shadowBlur = 4;
596|         ctx.shadowColor = 'rgba(0,0,0,0.9)';
597|
598|         // CPU Socket Label
599|         ctx.fillStyle = '#fbbf24';
600|         ctx.fillText('CPU SOCKET', 780, 165);
601|         // Cooler Label
602|         ctx.fillStyle = '#60a5fa';
603|         ctx.fillText('COOLER', 780, 310);
604|         // RAM Slots Label
605|         ctx.fillStyle = '#34d399';
606|         ctx.fillText('DDR5 RAM (Kênh ôi !' Khe 2 & 4)', 960, 110);
607|
608|         // GPU Slot Label (PCIe)
609|         ctx.fillStyle = '#f472b6';
610|         ctx.fillText('PCIe x16 (GPU)', 760, 545);
611|         // M.2 SSD Label
612|         ctx.fillStyle = '#a78bfa';
613|         ctx.fillText('M.2 NVMe', 715, 475);
614|         // CMOS Battery Label
615|         ctx.fillStyle = '#94a3b8';
616|         ctx.font = '9px monospace';
617|         ctx.fillText('CMOS', 1000, 575);
618|         // Standoff screw labels
619|         ctx.font = '8px monospace';
620|         ctx.fillText('Screw 1', 520, 68);
621|         ctx.fillText('Screw 2', 1130, 68);
622|         ctx.fillText('Screw 3', 520, 748);
623|         ctx.fillText('Screw 4', 1130, 748);
624|         ctx.fillText('Screw 5', 520, 408);
625|         ctx.fillText('Screw 6', 1130, 408);
626|
627|         ctx.restore();
628|
629|         ctx.restore();
630|     };
631|
632|     const drawComponent = (ctx, comp, time) => {
633|         ctx.save();
634|
635|         // Image mode: draw actual 2D image instead of canvas art
636|         if (imageMode && comp.imageUrl) {
637|             ctx.translate(comp.x, comp.y);
638|             ctx.translate(comp.width / 2, comp.height / 2);
639|             if (comp.isGrabbed) {
640|                 ctx.scale(1.05, 1.05);
641|             }
642|             ctx.translate(-comp.width / 2, -comp.height / 2);
643|             ctx.shadowBlur = 6;
644|             ctx.shadowColor = 'rgba(0, 0, 0, 0.5)';
645|             ctx.shadowOffsetY = 4;
646|             let img = comp.imageUrl ? imgCache.get(comp.imageUrl) : null;
647|             if (img && img.complete && img.naturalWidth > 0) {
648|                 ctx.drawImage(img, 0, 0, comp.width, comp.height);
649|             } else {
650|                 if (!img) getCacheImg(comp.imageUrl);
651|                 ctx.fillStyle = '#333';
652|                 ctx.fillRect(0, 0, comp.width, comp.height);
653|                 ctx.fillStyle = '#666';
654|                 ctx.font = '12px monospace';
655|                 ctx.textAlign = 'center';
656|                 ctx.fillText(comp.type, comp.width / 2, comp.height / 2 + 4);
657|             }
658|             ctx.restore();
659|             return;
660|         }
661|
662|         ctx.translate(comp.x, comp.y);
663|
664|         // Realistic Drop Shadow - Native shadowBlur is EXTREMELY expensive on complex paths like GPU, cau...
665|         // When grabbed, we remove shadowBlur and rely entirely on scale + a fast native offset fill to en...
666|         ctx.translate(comp.width / 2, comp.height / 2);
667|         if (comp.isGrabbed) {
668|             ctx.scale(1.05, 1.05); // Simple scale is GPU accelerated and instant
669|             ctx.shadowBlur = 0;
670|             ctx.shadowOffsetY = 0;
671|         } else {
672|             ctx.shadowBlur = 6;
673|             ctx.shadowColor = 'rgba(0, 0, 0, 0.5)';
674|             ctx.shadowOffsetY = 4;
675|         }
676|         ctx.translate(-comp.width / 2, -comp.height / 2);
677|
678|         if (comp.type === 'CPU') {
679|             // 3D CPU IHS Base (Premium Matte Silver with metallic gradient)
680|             const ihsGrad = ctx.createLinearGradient(0, 0, comp.width, comp.height);
681|             ihsGrad.addColorStop(0, '#f4f4f5');
682|             ihsGrad.addColorStop(0.5, '#d4d4d8');
683|             ihsGrad.addColorStop(1, '#a1a1aa');
684|
685|             // High-End PCB Substrate (Deep Bronze/Brown like Ryzen/Intel bases)
686|             draw3DBox(ctx, 0, 0, comp.width, comp.height, 4, '#451a03', '#2e1503', '#1c0d02', '#000');
687|
688|             // Dense Gold Contacts array on bottom edge of substrate
689|             ctx.fillStyle = '#fef08a';
690|             for (let i = 4; i < comp.width - 4; i += 4) {
691|                 ctx.fillRect(i, 2, 2, 2);
692|                 ctx.fillRect(i, comp.height - 4, 2, 2);
693|                 ctx.fillRect(2, i, 2, 2);
694|                 ctx.fillRect(comp.width - 4, i, 2, 2);
695|             }
696|
697|             // Heat Spreader Surface (IHS)
698|             ctx.fillStyle = ihsGrad;
699|             ctx.shadowBlur = 5; ctx.shadowColor = 'rgba(0,0,0,0.6)';
700|             // Unique cut-out shape for IHS
701|             ctx.beginPath();
702|             ctx.moveTo(15, 10); ctx.lineTo(comp.width - 15, 10);
703|             ctx.lineTo(comp.width - 10, 15); ctx.lineTo(comp.width - 10, comp.height - 15);
704|             ctx.lineTo(comp.width - 15, comp.height - 10); ctx.lineTo(15, comp.height - 10);
705|             ctx.lineTo(10, comp.height - 15); ctx.lineTo(10, 15); ctx.closePath();
706|             ctx.fill();
707|             ctx.shadowBlur = 0;
708|
709|             // Intricate Heat Spreader Engravings (QR Code like square + alignments)
710|             ctx.strokeStyle = '#6b7280'; ctx.lineWidth = 1.5;

```

```

711|         ctx.beginPath();
712|         ctx.moveTo(15, 15); ctx.lineTo(35, 15); ctx.lineTo(35, 35); ctx.lineTo(15, 35); ctx.closePath(...
713|
714|         // Corner Alignment Triangle (Gold)
715|         ctx.fillStyle = '#eab308';
716|         ctx.beginPath();
717|         ctx.moveTo(5, 5); ctx.lineTo(12, 5); ctx.lineTo(5, 12); ctx.fill();
718|
719|     } else if (comp.type === 'RAM') {
720|         // Tall 3D RAM Stick (Black Aluminum)
721|         draw3DBox(ctx, 0, 0, comp.width, comp.height, 10, '#1f2937', '#111827', '#030712', '#374151');
722|
723|         // Advanced Aluminum Heatsink Grooves (Angular Cyber Cuts)
724|         ctx.strokeStyle = '#09090b'; ctx.lineWidth = 2;
725|         ctx.beginPath();
726|         for (let i = 15; i < comp.height - 15; i += 12) {
727|             ctx.moveTo(0, i); ctx.lineTo(10, i + 5); ctx.lineTo(comp.width - 10, i + 5); ctx.lineTo(co...
728|         }
729|         ctx.stroke();
730|
731|         // Bright ARGB LED Edge-Lit Diffuser
732|         const hue1 = (time * 120) % 360;
733|         const hue2 = (time * 120 + 80) % 360;
734|         const grad = ctx.createLinearGradient(0, 0, 0, comp.height);
735|         grad.addColorStop(0, `hsl(${hue1}, 100%, 65%)`);
736|         grad.addColorStop(0.5, 'ffffff'); // Crystal burst
737|         grad.addColorStop(1, `hsl(${hue2}, 100%, 65%)`);
738|
739|         // RGB Core Light
740|         ctx.shadowBlur = 20; ctx.shadowColor = `hsl(${hue1}, 100%, 60%)`;
741|         ctx.fillStyle = grad;
742|         ctx.beginPath(); ctx.roundRect(comp.width / 2 - 4, 2, 8, comp.height - 4, 4); ctx.fill();
743|         ctx.shadowBlur = 0;
744|
745|         // Gold PCB Connection Pins (bottom edge)
746|         ctx.fillStyle = '#eab308';
747|         for (let i = 10; i < comp.height - 10; i += 4) {
748|             if (i !== comp.height / 2 && i !== comp.height / 2 + 4) { // Notch gap
749|                 ctx.fillRect(-4, i, 4, 2); // Left edge pins
750|             }
751|         }
752|
753|     } else if (comp.type === 'GPU') {
754|         // Massive 3D GPU Block (Elite Stealth Armor)
755|         const gpuGrad = ctx.createLinearGradient(0, 0, comp.width, 0);
756|         gpuGrad.addColorStop(0, '#27272a'); gpuGrad.addColorStop(0.5, '#3f3f46'); gpuGrad.addColorStop...
757|         draw3DBox(ctx, 0, 0, comp.width, comp.height, 25, gpuGrad, '#18181b', '#09090b', '#000');
758|
759|         // Advanced Armor Plates and Indentations
760|         ctx.fillStyle = '#09090b';
761|         ctx.beginPath(); ctx.moveTo(10, 10); ctx.lineTo(comp.width - 10, 10); ctx.lineTo(comp.width - ...
762|
763|         // Cyberpunk Neon/Tech line accents
764|         ctx.strokeStyle = '#0ea5e9'; ctx.lineWidth = 2;
765|         ctx.shadowBlur = 5; ctx.shadowColor = '#0ea5e9';
766|         ctx.beginPath(); ctx.moveTo(25, 25); ctx.lineTo(120, 25); ctx.lineTo(140, 45); ctx.stroke();
767|         ctx.beginPath(); ctx.moveTo(comp.width - 25, comp.height - 25); ctx.lineTo(comp.width - 120, c...
768|         ctx.shadowBlur = 0;
769|
770|         // Exposed Copper Heatpipes (Multiple layers)
771|         ctx.strokeStyle = '#b45309'; ctx.lineWidth = 5;
772|         ctx.lineCap = 'round';
773|         [35, 45, comp.width - 45, comp.width - 35].forEach(x => {
774|             ctx.beginPath(); ctx.moveTo(x, 20); ctx.lineTo(x, comp.height - 20); ctx.stroke();
775|             // Highlights for pipes
776|             ctx.strokeStyle = '#f59e0b'; ctx.lineWidth = 2;
777|             ctx.beginPath(); ctx.moveTo(x - 1, 20); ctx.lineTo(x - 1, comp.height - 20); ctx.stroke();
778|             ctx.strokeStyle = '#b45309'; ctx.lineWidth = 5; // Reset
779|         });
780|
781|         // Advanced Tri-Fan Setup with 3D Depth & Reflection
782|         const fanRadii = 42;
783|         const fanCenters = [90, 160, 230];
784|         fanCenters.forEach(cx => {
785|             // Fan Well Depth
786|             ctx.fillStyle = '#030712';
787|             ctx.beginPath(); ctx.arc(cx, comp.height / 2, fanRadii + 5, 0, Math.PI * 2); ctx.fill();
788|
789|             // RGB Ring Around Fan - only pulse when GPU is placed in PCIe slot
790|             if (comp.isPlaced) {
791|                 const pulse = (Math.sin(time * 3 + cx) + 1) / 2;
792|                 ctx.strokeStyle = `rgba(16, 185, 129, ${0.4 + pulse * 0.6})`;
793|                 ctx.shadowBlur = 8; ctx.shadowColor = '#10b981';
794|             } else {
795|                 ctx.strokeStyle = 'rgba(71, 85, 105, 0.5)';
796|                 ctx.shadowBlur = 0;
797|             }
798|             ctx.lineWidth = 3;
799|             ctx.beginPath(); ctx.arc(cx, comp.height / 2, fanRadii + 1, 0, Math.PI * 2); ctx.stroke();
800|             ctx.shadowBlur = 0;
801|
802|             // Blade Spin - only spin when GPU is placed
803|             const gpuFanAngle = comp.isPlaced ? time * 30 : 0;
804|             ctx.save();
805|             ctx.translate(cx, comp.height / 2);
806|             ctx.rotate(gpuFanAngle);
807|
808|             for (let i = 0; i < 9; i++) {
809|                 ctx.rotate(Math.PI * 2 / 9);
810|                 ctx.beginPath();
811|                 ctx.moveTo(0, 0);
812|                 ctx.bezierCurveTo(fanRadii * 0.3, -18, fanRadii * 0.8, -12, fanRadii - 2, 0);
813|                 ctx.bezierCurveTo(fanRadii * 0.8, 10, fanRadii * 0.3, 10, 0, 0);
814|                 ctx.fillStyle = comp.isPlaced ? 'rgba(71, 85, 105, 0.85)' : 'rgba(51, 65, 85, 0.5)';
815|                 ctx.fill();
816|
817|                 const bladeGrad = ctx.createLinearGradient(0, -10, fanRadii, 10);
818|                 bladeGrad.addColorStop(0, comp.isPlaced ? 'rgba(156, 163, 175, 0.9)' : 'rgba(100, 116, ...
819|                 bladeGrad.addColorStop(1, comp.isPlaced ? 'rgba(75, 85, 99, 0.9)' : 'rgba(51, 65, 85, ...
820|                 ctx.fillStyle = bladeGrad;
821|                 ctx.fill();
822|
823|                 ctx.strokeStyle = comp.isPlaced ? 'rgba(255, 255, 255, 0.3)' : 'rgba(255,255,255,0.1)';
824|                 ctx.lineWidth = 1;
825|                 ctx.stroke();
826|             }
827|             ctx.restore();
828|
829|             // Fan Hub Center (Brushed metal)
830|             drawCylinder(ctx, cx, comp.height / 2, 16, 3, '#27272a', '#a1a1aa');
831|             // Hub Logo Ring
832|             ctx.beginPath(); ctx.arc(cx, comp.height / 2, 10, 0, Math.PI * 2); ctx.strokeStyle = '#090...
833|         });
834|

```

```

835 // Top PCIe Connector Edge (Hidden slightly behind body, drops exactly into slot)
836 ctx.fillStyle = '#f59e0b'; // Gold PCIe pins base
837 ctx.fillRect(comp.width / 2 - 70, -8, 140, 8);
838 // PCIe Pins detail
839 ctx.fillStyle = '#b45309';
840 for (let i = comp.width / 2 - 68; i < comp.width / 2 + 68; i += 3) {
841   if (i !== comp.width / 2 - 20) { // Small gap notch
842     ctx.fillRect(i, -8, 2, 6);
843   }
844 }
845 } else if (comp.type === 'SSD') {
846   // Elite NVMe M.2 SSD
847   const ssdGrad = ctx.createLinearGradient(0, 0, comp.width, comp.height);
848   ssdGrad.addColorStop(0, '#18181b'); ssdGrad.addColorStop(1, '#09090b');
849   draw3DBox(ctx, 0, 0, comp.width, comp.height, 4, ssdGrad, '#09090b', '#000', '#27272a');
850
851   // Label
852   ctx.fillStyle = '#0ea5e9'; // Cyan accent
853   ctx.font = 'bold 10px monospace';
854   ctx.fillText('NVMe M.2 1TB', 10, 22);
855
856   // Gold Pins
857   ctx.fillStyle = '#eab308';
858   for (let i = 2; i < 20; i += 3) ctx.fillRect(-3, i, 3, 2);
859   for (let i = 26; i < comp.height - 2; i += 3) ctx.fillRect(-3, i, 3, 2); // Gap for notch
860 } else if (comp.type === 'PSU') {
861   // Massive 3D PSU Power Box (Dark industrial aesthetic with gold trim)
862   const psuGrad = ctx.createLinearGradient(0, 0, comp.width, comp.height);
863   psuGrad.addColorStop(0, '#27272a'); psuGrad.addColorStop(1, '#09090b');
864   draw3DBox(ctx, 0, 0, comp.width, comp.height, 18, psuGrad, '#18181b', '#09090b', '#000');
865
866   // Gold Warning Trim
867   ctx.strokeStyle = '#eab308'; ctx.lineWidth = 2;
868   ctx.strokeRect(5, 5, comp.width - 10, comp.height - 10);
869
870   // Advanced Modular Port Grid Interface
871   ctx.fillStyle = '#09090b';
872   ctx.fillRect(10, 10, comp.width - 20, 30);
873   ctx.fillStyle = '#020617';
874   // 12 Modular cable ports
875   for (let row = 0; row < 2; row++) {
876     for (let col = 0; col < 6; col++) {
877       ctx.fillRect(15 + col * 26, 13 + row * 12, 20, 8);
878       // Gold pin inside port
879       ctx.fillStyle = '#fbbf24'; ctx.fillRect(18 + col * 26, 15 + row * 12, 14, 4); ctx.fill...
880     }
881   }
882
883   // Circular fan grille on top of PSU
884   const cx = comp.width / 2;
885   const cy = comp.height / 2 + 10;
886   ctx.fillStyle = '#0f172a';
887   ctx.beginPath(); ctx.arc(cx, cy, 55, 0, Math.PI * 2); ctx.fill();
888
889   // Concentric circles for grille
890   ctx.strokeStyle = '#334155'; ctx.lineWidth = 3;
891   for (let r = 15; r <= 50; r += 10) {
892     ctx.beginPath(); ctx.arc(cx, cy, r, 0, Math.PI * 2); ctx.stroke();
893   }
894
895   // 24-pin Braided Cable snake protruding from side (Cable Managed!)
896   ctx.strokeStyle = '#1e293b'; ctx.lineWidth = 14;
897   ctx.lineCap = 'round';
898   ctx.beginPath();
899   ctx.moveTo(comp.width, comp.height - 40);
900
901   if (comp.isPlaced) {
902     // Tuck cable immediately downwards and behind the back panel (Cable Management!)
903     ctx.bezierCurveTo(comp.width + 40, comp.height - 40, comp.width + 50, comp.height + 20, co...
904   } else {
905     // Flailing cable when grabbed
906     ctx.bezierCurveTo(comp.width + 60, comp.height - 50, comp.width + 80, -20, comp.width + 15...
907   }
908   ctx.stroke();
909
910   // Colorful wires inside the sleeve (only visible near the connector)
911   ctx.strokeStyle = '#ef4444'; ctx.lineWidth = 2; // Red wire
912   ctx.beginPath(); ctx.moveTo(comp.width, comp.height - 42);
913   if (comp.isPlaced) {
914     ctx.bezierCurveTo(comp.width + 40, comp.height - 40, comp.width + 50, comp.height + 20, co...
915   } else {
916     ctx.bezierCurveTo(comp.width + 60, comp.height - 50, comp.width + 80, -20, comp.width + 15...
917   }
918   ctx.stroke();
919
920 } else if (comp.type === 'COOLER') {
921   // Circular CPU Cooler with RGB and spinning fan
922   ctx.fillStyle = '#0f172a';
923   ctx.beginPath(); ctx.arc(comp.width / 2, comp.height / 2, 45, 0, Math.PI * 2); ctx.fill();
924   ctx.strokeStyle = '#334155'; ctx.lineWidth = 4;
925   ctx.stroke();
926
927   // RGB Ring - only glow when placed
928   ctx.strokeStyle = comp.isPlaced ? `hsl(${(time * 50) % 360}, 100%, 65%)` : '#334155';
929   ctx.shadowBlur = comp.isPlaced ? 10 : 0;
930   ctx.shadowColor = comp.isPlaced ? ctx.strokeStyle : 'transparent';
931   ctx.lineWidth = 3;
932   ctx.beginPath(); ctx.arc(comp.width / 2, comp.height / 2, 40, 0, Math.PI * 2); ctx.stroke();
933   ctx.shadowBlur = 0;
934
935   // Fan Blades - only spin when placed on CPU socket
936   const fanAngle = comp.isPlaced ? time * 40 : 0; // stationary if not placed
937   ctx.save();
938   ctx.translate(comp.width / 2, comp.height / 2);
939   ctx.rotate(fanAngle);
940   for (let i = 0; i < 9; i++) {
941     ctx.rotate(Math.PI * 2 / 9);
942     ctx.beginPath();
943     ctx.moveTo(0, 0);
944     ctx.bezierCurveTo(15, -15, 30, -10, 38, 0);
945     ctx.bezierCurveTo(30, 8, 15, 8, 0, 0);
946     ctx.fillStyle = comp.isPlaced ? 'rgba(148, 163, 184, 0.9)' : 'rgba(100, 116, 139, 0.7)';
947     ctx.fill();
948   }
949   ctx.restore();
950
951   // Center logo
952   ctx.fillStyle = '#0f172a';
953   ctx.beginPath(); ctx.arc(comp.width / 2, comp.height / 2, 12, 0, Math.PI * 2); ctx.fill();
954   ctx.strokeStyle = '#1e293b'; ctx.lineWidth = 2; ctx.stroke();
955   ctx.fillStyle = comp.isPlaced ? '#0ea5e9' : '#475569'; ctx.font = 'bold 8px Arial'; ctx.textAl...
956
957   // "Not installed" indicator when not placed
958   if (!comp.isPlaced) {

```

```

959 |         ctx.fillStyle = 'rgba(0,0,0,0.0)'; // transparent – just logic indicator
960 |     }
961 | }
962 |
963 | ctx.restore();
964 | };
965 |
966 | const updateAndDraw = () => {
967 |     const time = performance.now() / 1000;
968 |     const state = stateRef.current;
969 |     const w = canvas.width;
970 |     const h = canvas.height;
971 |
972 |     // Determine active cursor (Hand Tracking OR Mouse)
973 |     let activeCursorX = mouseRef.current.x;
974 |     let activeCursorY = mouseRef.current.y;
975 |     let isGrabbing = mouseRef.current.isDown;
976 |     let showHandCursor = false;
977 |
978 |     if (landmarksRef.current && landmarksRef.current.length > 0) {
979 |         const hand = landmarksRef.current[0];
980 |         const indexTip = hand[8];
981 |         const thumbTip = hand[4];
982 |         const distance = Math.hypot(indexTip.x - thumbTip.x, indexTip.y - thumbTip.y);
983 |
984 |         // Add sensitivity multiplier to make the cursor feel extremely fast and responsive
985 |         const SENSITIVITY = 1.0;
986 |         const offsetX = (indexTip.x - 0.5) * SENSITIVITY + 0.5;
987 |         const offsetY = (indexTip.y - 0.5) * SENSITIVITY + 0.5;
988 |
989 |         const rawX = Math.max(0, Math.min(1, offsetX)) * w;
990 |         const rawY = Math.max(0, Math.min(1, offsetY)) * h;
991 |
992 |         const filter = cursorFilterRef.current;
993 |         if (filter.x === null) {
994 |             filter.x = rawX;
995 |             filter.y = rawY;
996 |         } else {
997 |             filter.x += filter.alpha * (rawX - filter.x);
998 |             filter.y += filter.alpha * (rawY - filter.y);
999 |         }
1000 |
1001 |         activeCursorX = filter.x;
1002 |         activeCursorY = filter.y;
1003 |         // Hysteresis: easier to grab, harder to release (anti-jitter)
1004 |         if (pinchStateRef.current) {
1005 |             isGrabbing = distance < RELEASE_THRESHOLD;
1006 |             if (!isGrabbing) pinchStateRef.current = false;
1007 |         } else {
1008 |             isGrabbing = distance < GRAB_THRESHOLD;
1009 |             if (isGrabbing) pinchStateRef.current = true;
1010 |         }
1011 |         showHandCursor = true;
1012 |
1013 |         // Freeze cursor lúc b-t su pinch ã n-m chính xác, khi
1014 |         if (isGrabbing) {
1015 |             const hasGrabbed = state.components.some(c => c.isGrabbed);
1016 |             if (!hasGrabbed) {
1017 |                 if (!grabLockRef.current.active) {
1018 |                     grabLockRef.current = { x: activeCursorX, y: activeCursorY, active: true };
1019 |                 }
1020 |                 activeCursorX = grabLockRef.current.x;
1021 |                 activeCursorY = grabLockRef.current.y;
1022 |             }
1023 |         } else {
1024 |             grabLockRef.current = { x: null, y: null, active: false };
1025 |         }
1026 |     } else {
1027 |         cursorFilterRef.current.x = null;
1028 |     }
1029 |
1030 |     let hoveredCompType = null;
1031 |
1032 |     // Logic Update
1033 |     state.components.forEach(comp => {
1034 |         if (comp.isPlaced) return;
1035 |
1036 |         if (activeCursorX > comp.x && activeCursorX < comp.x + comp.width &&
1037 |             activeCursorY > comp.y && activeCursorY < comp.y + comp.height) {
1038 |             hoveredCompType = comp.type;
1039 |         }
1040 |
1041 |         if (comp.isGrabbed) {
1042 |             if (isGrabbing) {
1043 |                 comp.x = activeCursorX - comp.width / 2;
1044 |                 comp.y = activeCursorY - comp.height / 2;
1045 |             } else {
1046 |                 // Released
1047 |                 comp.isGrabbed = false;
1048 |
1049 |                 // Check Snap to closest Unoccupied Socket
1050 |                 let closestSocket = null;
1051 |                 let minDist = 120; // Very forgiving snap radius
1052 |
1053 |                 state.sockets.filter(s => s.type === comp.type).forEach(socket => {
1054 |                     if (comp.type === 'COOLER') {
1055 |                         const cpuPlaced = state.components.some(c => c.type === 'CPU' && c.isPlaced);
1056 |                         if (!cpuPlaced) return; // Prevent placing cooler before CPU
1057 |                     }
1058 |
1059 |                     // Check if this specific socket is already occupied by another placed component
1060 |                     const isOccupied = state.components.some(c => c.isPlaced && c.id !== comp.id && c....
1061 |                     if (!isOccupied) {
1062 |                         const dist = Math.hypot(comp.x - socket.x, comp.y - socket.y);
1063 |                         if (dist < minDist) {
1064 |                             minDist = dist;
1065 |                             closestSocket = socket;
1066 |                         }
1067 |                     }
1068 |                 });
1069 |
1070 |                 if (closestSocket) {
1071 |                     comp.x = closestSocket.x;
1072 |                     comp.y = closestSocket.y;
1073 |                     comp.isPlaced = true;
1074 |                     if (onGameEvent) onGameEvent('placed', comp.type);
1075 |
1076 |                     // Trigger Confetti
1077 |                     const allPlaced = state.components.every(c => c.isPlaced);
1078 |                     if (allPlaced && confetti) {
1079 |                         confetti({
1080 |                             particleCount: 200,
1081 |                             spread: 100,
1082 |                             origin: { y: 0.6 },

```



```

1083 |         colors: ['#38bdf8', '#ff0055', '#10b981', '#f59e0b']
1084 |     });
1085 |     if (onGameEvent) onGameEvent('COMPLETE', 'ALL');
1086 | }
1087 | }
1088 | }
1089 | } else {
1090 |     if (isGrabbing) {
1091 |         const holding = state.components.find(c => c.isGrabbed);
1092 |         if (!holding && hoveredCompType === comp.type) {
1093 |             comp.isGrabbed = true;
1094 |             if (onGameEvent) onGameEvent('grabbed', comp.type);
1095 |         }
1096 |     }
1097 | }
1098 | });
1099 |
1100 | // --- DRAWING PHASE ---
1101 | ctx.fillStyle = '#0f172a'; // Deep base color
1102 | ctx.fillRect(0, 0, w, h);
1103 |
1104 | // Render Motherboard (cached + dynamic)
1105 | drawAnimatedMotherboard(ctx, time);
1106 |
1107 | // Draw Glowing Sockets (Drop Zones) inside Motherboard - Only if empty
1108 | state.sockets.forEach(socket => {
1109 |     const componentPlaced = state.components.find(c => c.isPlaced && Math.abs(c.x - socket.x) < 5 ...
1110 |         if (componentPlaced) return; // Do not draw glowing socket if filled perfectly
1111 |
1112 |     ctx.save();
1113 |     ctx.translate(socket.x, socket.y);
1114 |     const pulse = (Math.sin(time * 4) + 1) / 2;
1115 |     ctx.strokeStyle = `rgba(16, 185, 129, ${0.5 + pulse * 0.5})`; // Emerald glowing pulsating border
1116 |     ctx.shadowBlur = 10; ctx.shadowColor = '#10b981';
1117 |     ctx.lineWidth = 3;
1118 |     ctx.setLineDash([15, 10]);
1119 |     ctx.lineDashOffset = -time * 25;
1120 |     ctx.strokeRect(0, 0, socket.width, socket.height);
1121 |     ctx.fillStyle = 'rgba(16, 185, 129, 0.08)';
1122 |     ctx.fillRect(0, 0, socket.width, socket.height);
1123 |     ctx.restore();
1124 | });
1125 |
1126 | // Draw Components (Z-index sorting: placed -> unplaced non-grabbed -> grabbed goes ON TOP)
1127 | state.components.filter(c => c.isPlaced).forEach(comp => drawComponent(ctx, comp, time));
1128 | state.components.filter(c => !c.isPlaced && !c.isGrabbed).forEach(comp => drawComponent(ctx, comp, ...
1129 | state.components.filter(c => c.isGrabbed).forEach(comp => drawComponent(ctx, comp, time));
1130 |
1131 | // Draw Active Cursor Indicator
1132 | if (showHandCursor) {
1133 |     ctx.beginPath();
1134 |     ctx.arc(activeCursorX, activeCursorY, 15, 0, 2 * Math.PI);
1135 |     ctx.fillStyle = isGrabbing ? 'rgba(234, 179, 8, 0.9)' : 'rgba(56, 189, 248, 0.9)';
1136 |     ctx.fill();
1137 |     ctx.strokeStyle = 'ffff';
1138 |     ctx.lineWidth = 3;
1139 |     ctx.stroke();
1140 | }
1141 |
1142 |     animId = requestAnimationFrame(updateAndDraw);
1143 | };
1144 |
1145 | // Start Loop
1146 | animId = requestAnimationFrame(updateAndDraw);
1147 |
1148 |     return () => cancelAnimationFrame(animId);
1149 | }, [onGameEvent]);
1150 |
1151 | return (
1152 |     <canvas
1153 |         ref={canvasRef}
1154 |         width={1400}
1155 |         height={800}
1156 |         onMouseMove={handleMouseMove}
1157 |         onMouseDown={handleMouseDown}
1158 |         onMouseUp={handleMouseUp}
1159 |         onMouseLeave={handleMouseUp}
1160 |         style={{
1161 |             width: '100%',
1162 |             maxHeight: '100%',
1163 |             aspectRatio: '7/4',
1164 |             background: '#0f172a',
1165 |             touchAction: 'none',
1166 |             cursor: 'crosshair'
1167 |         }}
1168 |     />
1169 | );
1170 | });
1171 |
1172 | GameEngine.displayName = 'GameEngine';
1173 |
1174 | export default GameEngine;
1175 |

```

File: components/GameScene.tsx (864 lines, 35.7 KB)

```

=====
1 | 'use client';
2 |
3 | import React, { useRef, useState, useEffect } from 'react';
4 | import { Canvas, useFrame, useThree, useLoader } from '@react-three/fiber';
5 | import * as THREE from 'three';
6 | import dynamic from 'next/dynamic';
7 | import { useAssemblyStore } from '@lib/useStore';
8 |
9 | const HeadTracker = dynamic(() => import('./HeadTracker'), { ssr: false });
10 |
11 | function CameraRig() {
12 |     const { camera } = useThree();
13 |     const cameraCoords = useAssemblyStore((s) => s.cameraCoords);
14 |     const keys = useRef({ w: false, a: false, s: false, d: false });
15 |     const euler = useRef(new THREE.Euler(0, 0, 0, 'YXZ'));
16 |     const moveSpeed = 4;
17 |     const faceSensitivity = 3.0;
18 |
19 |     const accumulatedYaw = useRef(0);
20 |     const prevFaceYaw = useRef(0);
21 |     const stillnessTimer = useRef(0);
22 |     const viewSaved = useRef(false);
23 |     const mainView2 = useRef({ position: new THREE.Vector3(), quaternion: new THREE.Quaternion() });
24 |     const initialized = useRef(false);
25 |
26 |     useEffect(() => {
27 |         const down = (e: KeyboardEvent) => { const k = e.key.toLowerCase(); if (k in keys.current) (keys.current a...

```

```

18 const up = (e: KeyboardEvent) => { const k = e.key.toLowerCaseCase(); if (k in keys.current) (keys.current as ...
19 window.addEventListener('keydown', down);
20 window.addEventListener('keyup', up);
21
22 camera.position.set(0, 1.6, 4.5);
23 euler.current.set(0, 0, 0, 'XYZ');
24 camera.quaternion.setFromEuler(euler.current);
25 initialized.current = true;
26
27 return () => { window.removeEventListener('keydown', down); window.removeEventListener('keyup', up); };
28 }, []);
29
30 useFrame((), dt) => {
31   const delta = Math.min(dt, 0.05);
32
33   const rawPitch = -cameraCoords.pitch * 2.2;
34   const rawYaw = cameraCoords.yaw * faceSensitivity;
35   const yawMovement = Math.abs(rawYaw - prevFaceYaw.current);
36
37   if (yawMovement > 0.008) {
38     const diff = (rawYaw - prevFaceYaw.current) * 0.6;
39     accumulatedYaw.current += diff;
40     prevFaceYaw.current = rawYaw;
41     stillnessTimer.current = 0;
42     viewSaved.current = false;
43   } else {
44     stillnessTimer.current += delta;
45     if (stillnessTimer.current >= 1.0 && !viewSaved.current && initialized.current) {
46       mainView2.current.position.copy(camera.position);
47       mainView2.current.quaternion.copy(camera.quaternion);
48       viewSaved.current = true;
49       stillnessTimer.current = 0;
50     }
51   }
52
53   euler.current.set(rawPitch, accumulatedYaw.current, 0, 'XYZ');
54   camera.quaternion.setFromEuler(euler.current);
55
56   const forward = new THREE.Vector3(0, 0, -1).applyQuaternion(camera.quaternion);
57   forward.y = 0; forward.normalize();
58   const right = new THREE.Vector3(1, 0, 0).applyQuaternion(camera.quaternion);
59   right.y = 0; right.normalize();
60   const move = new THREE.Vector3();
61   if (keys.current.w) move.add(forward);
62   if (keys.current.s) move.sub(forward);
63   if (keys.current.a) move.sub(right);
64   if (keys.current.d) move.add(right);
65   if (move.length() > 0) move.normalize().multiplyScalar(moveSpeed * delta);
66   camera.position.add(move);
67   camera.position.x = THREE.MathUtils.clamp(camera.position.x, -4.5, 4.5);
68   camera.position.z = THREE.MathUtils.clamp(camera.position.z, -4.5, 4.0);
69   camera.position.y = 1.6;
70 });
71
72 return null;
73 }
74
75 function LabBench({ position, size = [3.0, 0.9, 1.2], color = '#cbd5e1' }: { position: [number, number, number]...
76   const [w, h, d] = size;
77   return (
78     <group position={position}>
79       <mesh position={[0, h, 0]} castShadow>
80         <boxGeometry args={[w, 0.06, d]} />
81         <meshPhysicalMaterial color={color} roughness={0.2} metalness={0.4} />
82       </mesh>
83       <mesh position={[0, h + 0.03, 0]}>
84         <boxGeometry args={[w - 0.1, 0.008, d - 0.1]} />
85         <meshPhysicalMaterial color="#f1f5f9" roughness={0.1} metalness={0.3} />
86       </mesh>
87       {[[-w / 2 + 0.1, h / 2, -d / 2 + 0.1], [-w / 2 + 0.1, h / 2, d / 2 - 0.1], [w / 2 - 0.1, h / 2, -d / 2 ...
88         <mesh key={i} position={p}>
89           <cylinderGeometry args={[0.04, 0.05, h, 8]} />
90           <meshPhysicalMaterial color="#94a3b8" roughness={0.3} metalness={0.5} />
91         </mesh>
92       )]}
93     </group>
94   );
95 }
96
97 function Beaker({ position, color = '#60a5fa', height = 0.25, radius = 0.08 }: { position: [number, number, nu...
98   return (
99     <group position={position}>
100       <mesh position={[0, height / 2, 0]} castShadow>
101         <cylinderGeometry args={[radius * 0.85, radius, height, 16]} />
102         <meshPhysicalMaterial color="#e2e8f0" roughness={0.05} metalness={0.1} transparent opacity={0.4} />
103       </mesh>
104       <mesh position={[0, height * 0.35, 0]}>
105         <cylinderGeometry args={[radius * 0.75, radius * 0.75, height * 0.35, 16]} />
106         <meshPhysicalMaterial color={color} roughness={0.1} transparent opacity={0.7} />
107       </mesh>
108       <mesh position={[0, height + 0.02, 0]}>
109         <torusGeometry args={[radius * 0.8, 0.008, 8, 16]} />
110         <meshPhysicalMaterial color="#94a3b8" roughness={0.3} metalness={0.2} />
111       </mesh>
112     </group>
113   );
114 }
115
116 function Flask({ position, color = '#f97316' }: { position: [number, number, number]; color?: string }) {
117   return (
118     <group position={position}>
119       <mesh position={[0, 0.2, 0]}>
120         <cylinderGeometry args={[0.06, 0.06, 0.25, 12]} />
121         <meshPhysicalMaterial color="#e2e8f0" roughness={0.05} transparent opacity={0.35} />
122       </mesh>
123       <mesh position={[0, 0.06, 0]}>
124         <sphereGeometry args={[0.1, 12, 12]} />
125         <meshPhysicalMaterial color="#e2e8f0" roughness={0.05} transparent opacity={0.35} />
126       </mesh>
127       <mesh position={[0, 0.06, 0]}>
128         <sphereGeometry args={[0.08, 12, 12]} />
129         <meshPhysicalMaterial color={color} roughness={0.1} transparent opacity={0.6} />
130       </mesh>
131     </group>
132   );
133 }
134
135 function TestTubes({ position }: { position: [number, number, number] }) {
136   const colors = ['#ef4444', '#22c55e', '#3b82f6', '#eab308', '#a855f7'];
137   return (
138     <group position={position}>
139       {colors.map((c, i) => (
140         <group key={i} position={[ -0.1 + i * 0.05, 0, 0 ]}>
141           <mesh position={[0, 0.12, 0]}>

```

```

152         <cylinderGeometry args={[0.02, 0.025, 0.2, 8]} />
153         <meshPhysicalMaterial color="#e2e8f0" roughness={0.05} transparent opacity={0.3} />
154     </mesh>
155     <mesh position={[0, 0.06, 0]}>
156         <cylinderGeometry args={[0.018, 0.022, 0.08, 8]} />
157         <meshPhysicalMaterial color={c} roughness={0.1} transparent opacity={0.6} />
158     </mesh>
159 </group>
160 </>
161 </group>
162 </>
163 }
164
165 function Centrifuge({ position }: { position: [number, number, number] }) {
166     return (
167         <group position={position}>
168             <mesh position={[0, 0.2, 0]}>
169                 <boxGeometry args={[0.4, 0.4, 0.4]} />
170                 <meshPhysicalMaterial color="#94a3b8" roughness={0.2} metalness={0.6} />
171             </mesh>
172             <mesh position={[0, 0.42, 0]}>
173                 <cylinderGeometry args={[0.22, 0.22, 0.04, 20]} />
174                 <meshPhysicalMaterial color="#e2e8f0" roughness={0.3} />
175             </mesh>
176             <mesh position={[0, 0.44, 0]}>
177                 <cylinderGeometry args={[0.1, 0.15, 0.06, 20]} />
178                 <meshPhysicalMaterial color="#64748b" roughness={0.3} transparent opacity={0.5} />
179             </mesh>
180         </group>
181     );
182 }
183
184 function Microscope({ position }: { position: [number, number, number] }) {
185     return (
186         <group position={position}>
187             <mesh position={[0, 0.1, 0]}>
188                 <boxGeometry args={[0.2, 0.2, 0.25]} />
189                 <meshPhysicalMaterial color="#475569" roughness={0.2} metalness={0.5} />
190             </mesh>
191             <mesh position={[0, 0.25, 0.08]}>
192                 <cylinderGeometry args={[0.03, 0.03, 0.15, 8]} />
193                 <meshPhysicalMaterial color="#94a3b8" roughness={0.3} metalness={0.4} />
194             </mesh>
195             <mesh position={[0, 0.35, 0.12]}>
196                 <cylinderGeometry args={[0.06, 0.04, 0.04, 12]} />
197                 <meshPhysicalMaterial color="#1e293b" roughness={0.3} />
198             </mesh>
199             <mesh position={[0.12, 0.18, 0]}>
200                 <boxGeometry args={[0.04, 0.04, 0.15]} />
201                 <meshPhysicalMaterial color="#1e293b" roughness={0.3} />
202             </mesh>
203         </group>
204     );
205 }
206
207 function DistillationApparatus({ position }: { position: [number, number, number] }) {
208     return (
209         <group position={position}>
210             <!-- Base platform -->
211             <mesh position={[0, 0.03, 0]}>
212                 <boxGeometry args={[0.6, 0.06, 0.5]} />
213                 <meshPhysicalMaterial color="#64748b" roughness={0.2} metalness={0.6} />
214             </mesh>
215             <!-- Main boiling flask -->
216             <mesh position={[-0.15, 0.2, 0]}>
217                 <sphereGeometry args={[0.15, 16, 16]} />
218                 <meshPhysicalMaterial color="#e2e8f0" roughness={0.05} transparent opacity={0.35} />
219             </mesh>
220             <mesh position={[-0.15, 0.2, 0]}>
221                 <sphereGeometry args={[0.12, 16, 16]} />
222                 <meshPhysicalMaterial color="#f97316" roughness={0.1} transparent opacity={0.5} />
223             </mesh>
224             <!-- Neck / column -->
225             <mesh position={[-0.15, 0.4, 0]}>
226                 <cylinderGeometry args={[0.04, 0.05, 0.25, 12]} />
227                 <meshPhysicalMaterial color="#e2e8f0" roughness={0.05} transparent opacity={0.35} />
228             </mesh>
229             <!-- Condenser coil -->
230             <mesh position={[0.1, 0.5, 0]}>
231                 <cylinderGeometry args={[0.06, 0.06, 0.3, 12]} />
232                 <meshPhysicalMaterial color="#e2e8f0" roughness={0.05} transparent opacity={0.25} />
233             </mesh>
234             <mesh position={[0.1, 0.5, 0]}>
235                 <cylinderGeometry args={[0.03, 0.03, 0.3, 6]} />
236                 <meshPhysicalMaterial color="#94a3b8" roughness={0.2} metalness={0.4} />
237             </mesh>
238             <!-- Collection flask -->
239             <mesh position={[0.35, 0.18, 0]}>
240                 <sphereGeometry args={[0.12, 16, 16]} />
241                 <meshPhysicalMaterial color="#e2e8f0" roughness={0.05} transparent opacity={0.35} />
242             </mesh>
243             <mesh position={[0.35, 0.18, 0]}>
244                 <sphereGeometry args={[0.09, 16, 16]} />
245                 <meshPhysicalMaterial color="#22c55e" roughness={0.1} transparent opacity={0.5} />
246             </mesh>
247             <!-- Connector tube -->
248             <mesh position={[0.0, 0.35, 0]}>
249                 <boxGeometry args={[0.25, 0.02, 0.02]} />
250                 <meshPhysicalMaterial color="#94a3b8" roughness={0.2} metalness={0.3} />
251             </mesh>
252             <!-- Digital console -->
253             <mesh position={[0, 0.08, 0.28]}>
254                 <boxGeometry args={[0.3, 0.1, 0.04]} />
255                 <meshPhysicalMaterial color="#1e293b" roughness={0.2} />
256             </mesh>
257             <mesh position={[0, 0.08, 0.3]}>
258                 <planeGeometry args={[0.25, 0.06]} />
259                 <meshPhysicalMaterial color="#22d3ee" roughness={0.1} emissive="#22d3ee" emissiveIntensity={0.3} />
260             </mesh>
261         </group>
262     );
263 }
264
265 function RoboticArmSection() {
266     return (
267         <group position={[3.5, 0, 0]} rotation={[0, -Math.PI / 6, 0]}>
268             <!-- Base -->
269             <mesh position={[0, 0.08, 0]}>
270                 <cylinderGeometry args={[0.2, 0.25, 0.16, 16]} />
271                 <meshPhysicalMaterial color="#475569" roughness={0.2} metalness={0.7} />
272             </mesh>
273             <!-- Arm segment 1 -->
274             <mesh position={[0, 0.35, 0.08]} castShadow>
275                 <boxGeometry args={[0.12, 0.3, 0.12]} />

```

```

276 |     <meshPhysicalMaterial color="#94a3b8" roughness={0.2} metalness={0.6} />
277 |   </mesh>
278 |   { /* Joint */
279 |     <mesh position={[0, 0.52, 0.08]}>
280 |       <sphereGeometry args={[0.07, 12, 12]} />
281 |       <meshPhysicalMaterial color="#64748b" roughness={0.2} metalness={0.5} />
282 |     </mesh>
283 |     { /* Arm segment 2 */
284 |       <mesh position={[0.15, 0.58, 0.08]} rotation={[0, 0, -0.4]}>
285 |         <boxGeometry args={[0.25, 0.08, 0.08]} />
286 |         <meshPhysicalMaterial color="#94a3b8" roughness={0.2} metalness={0.6} />
287 |       </mesh>
288 |       { /* Gripper */
289 |         <mesh position={[0.3, 0.6, 0.08]}>
290 |           <boxGeometry args={[0.06, 0.1, 0.06]} />
291 |           <meshPhysicalMaterial color="#64748b" roughness={0.3} metalness={0.5} />
292 |         </mesh>
293 |         <mesh position={[0.33, 0.6, 0.04]}>
294 |           <boxGeometry args={[0.04, 0.06, 0.02]} />
295 |           <meshPhysicalMaterial color="#475569" roughness={0.3} metalness={0.5} />
296 |         </mesh>
297 |         <mesh position={[0.33, 0.6, 0.12]}>
298 |           <boxGeometry args={[0.04, 0.06, 0.02]} />
299 |           <meshPhysicalMaterial color="#475569" roughness={0.3} metalness={0.5} />
300 |         </mesh>
301 |         { /* Flask being held */
302 |           <mesh position={[0.38, 0.55, 0.08]}>
303 |             <sphereGeometry args={[0.06, 10, 10]} />
304 |             <meshPhysicalMaterial color="#e2e8f0" roughness={0.05} transparent opacity={0.35} />
305 |           </mesh>
306 |           <mesh position={[0.38, 0.55, 0.08]}>
307 |             <sphereGeometry args={[0.04, 10, 10]} />
308 |             <meshPhysicalMaterial color="#a855f7" roughness={0.1} transparent opacity={0.5} />
309 |           </mesh>
310 |           { /* Dispenser unit next to it */
311 |             <mesh position={[0.5, 0.2, -0.2]}>
312 |               <boxGeometry args={[0.2, 0.4, 0.2]} />
313 |               <meshPhysicalMaterial color="#64748b" roughness={0.2} metalness={0.6} />
314 |             </mesh>
315 |             <mesh position={[0.5, 0.42, -0.2]}>
316 |               <cylinderGeometry args={[0.04, 0.04, 0.05, 8]} />
317 |               <meshPhysicalMaterial color="#1e293b" roughness={0.3} />
318 |             </mesh>
319 |           </group>
320 |         </group>
321 |       </group>
322 |     }
323 |   }
324 |   function ComputerStation() {
325 |     const screenData = useRef(new THREE.CanvasTexture(document.createElement('canvas')));
326 |     const [screenReady, setScreenReady] = useState(false);
327 |
328 |     useEffect(() => {
329 |       const c = document.createElement('canvas');
330 |       c.width = 256; c.height = 160;
331 |       const ctx = c.getContext('2d'); if (!ctx) return;
332 |       ctx.fillStyle = '#0c1929'; ctx.fillRect(0, 0, 256, 160);
333 |       for (let i = 0; i < 30; i++) {
334 |         ctx.fillStyle = `hsl(190, 80%, ${40 + Math.random() * 40}%)`;
335 |         ctx.fillRect(10 + Math.random() * 200, 10 + Math.random() * 120, 3 + Math.random() * 20, 2 + Math.random() * 2);
336 |       }
337 |       for (let i = 0; i < 20; i++) {
338 |         ctx.beginPath();
339 |         ctx.arc(40 + Math.random() * 180, 30 + Math.random() * 100, 3 + Math.random() * 8, 0, Math.PI * 2);
340 |         ctx.strokeStyle = `hsl(140, 70%, 50%, ${0.3 + Math.random() * 0.4}%)`;
341 |         ctx.lineWidth = 1; ctx.stroke();
342 |         ctx.fillStyle = `hsl(140, 70%, 50%, ${0.1 + Math.random() * 0.2}%)`;
343 |         ctx.fill();
344 |       }
345 |       ctx.font = '10px monospace'; ctx.fillStyle = '#22d3ee'; ctx.textAlign = 'center';
346 |       ctx.fillText('MOLECULAR ANALYSIS v3.2', 128, 148);
347 |       const t = new THREE.CanvasTexture(c);
348 |       t.needsUpdate = true;
349 |       screenData.current = t;
350 |       setScreenReady(true);
351 |     }, []);
352 |
353 |     return (
354 |       <group position={[-3.5, 0, -0.5]} rotation={[0, 0.3, 0]}>
355 |         { /* Desk */
356 |           <mesh position={[0, 0.55, 0]} castShadow>
357 |             <boxGeometry args={[1.6, 0.04, 0.9]} />
358 |             <meshPhysicalMaterial color="#64748b" roughness={0.2} metalness={0.5} />
359 |           </mesh>
360 |           { ([[-0.7, 0.27, -0.38], [-0.7, 0.27, 0.38], [0.7, 0.27, -0.38], [0.7, 0.27, 0.38]] as const).map((p, i) => {
361 |             <mesh key={i} position={p}>
362 |               <cylinderGeometry args={[0.03, 0.04, 0.55, 8]} />
363 |               <meshPhysicalMaterial color="#94a3b8" roughness={0.3} metalness={0.4} />
364 |             </mesh>
365 |           })
366 |         }
367 |         { /* Main screen */
368 |           <mesh position={[-0.25, 0.95, -0.08]}>
369 |             <boxGeometry args={[0.5, 0.35, 0.03]} />
370 |             <meshPhysicalMaterial color="#0f172a" roughness={0.1} />
371 |           </mesh>
372 |           <mesh position={[-0.25, 0.95, -0.05]}>
373 |             <planeGeometry args={[0.45, 0.3]} />
374 |             {screenReady && <meshPhysicalMaterial map={screenData.current} roughness={0.05} />}
375 |           </mesh>
376 |           { /* Second screen */
377 |             <mesh position={[0.35, 0.9, -0.08]}>
378 |               <boxGeometry args={[0.4, 0.28, 0.03]} />
379 |               <meshPhysicalMaterial color="#0f172a" roughness={0.1} />
380 |             </mesh>
381 |             <mesh position={[0.35, 0.9, -0.05]}>
382 |               <planeGeometry args={[0.35, 0.23]} />
383 |               {screenReady && <meshPhysicalMaterial map={screenData.current} roughness={0.05} />}
384 |             </mesh>
385 |           }
386 |           { /* Keyboard */
387 |             <mesh position={[0, 0.58, 0.15]}>
388 |               <boxGeometry args={[0.35, 0.02, 0.12]} />
389 |               <meshPhysicalMaterial color="#1e293b" roughness={0.5} />
390 |             </mesh>
391 |           }
392 |           { /* Chair */
393 |             <mesh position={[-0.6, 0.25, 0.45]}>
394 |               <cylinderGeometry args={[0.15, 0.18, 0.05, 12]} />
395 |               <meshPhysicalMaterial color="#334155" roughness={0.7} />
396 |             </mesh>
397 |             <mesh position={[-0.6, 0.48, 0.45]}>
398 |               <cylinderGeometry args={[0.12, 0.12, 0.42, 12]} />
399 |               <meshPhysicalMaterial color="#1e293b" roughness={0.6} />
400 |             </mesh>
401 |             <mesh position={[-0.6, 0.7, 0.4]}>
402 |               <boxGeometry args={[0.3, 0.04, 0.25]} />
403 |               <meshPhysicalMaterial color="#334155" roughness={0.7} />
404 |             </mesh>
405 |           }
406 |         </group>
407 |       </group>
408 |     );
409 |   }
410 | }

```

```

400 |     </mesh>
401 |   </group>
402 | );
403 | }
404 |
405 | function Hologram() {
406 |   const [frame] = useState(() => Math.floor(Math.random() * 100));
407 |
408 |   return (
409 |     <group position={[-1.5, 1.0, -1.0]}>
410 |       <mesh position={[0, 0.02, 0]}>
411 |         <cylinderGeometry args={[0.12, 0.2, 0.04, 20]} />
412 |         <meshPhysicalMaterial color="#64748b" roughness={0.2} metalness={0.5} />
413 |       </mesh>
414 |       <mesh position={[0, 0.4, 0]}>
415 |         <cylinderGeometry args={[0.02, 0.02, 0.7, 8]} />
416 |         <meshPhysicalMaterial color="#22d3ee" roughness={0.1} emissive="#22d3ee" emissiveIntensity={0.3} trans...
417 |       </mesh>
418 |       /* Molecule hologram */
419 |       {[[[0, 0.65, 0], [-0.08, 0.75, 0.05], [0.1, 0.8, -0.03], [-0.05, 0.9, -0.04], [0.12, 0.85, 0.06], [0.0, ...
420 |         <mesh key={i} position={pos}>
421 |           <sphereGeometry args={[0.03, 8, 8]} />
422 |           <meshPhysicalMaterial color="#22d3ee" roughness={0.1} emissive="#22d3ee" emissiveIntensity={0.5} tra...
423 |         </mesh>
424 |       ]])
425 |       {[[[-0.04, 0.7, 0.03], [0.01, 0.72, -0.02], [-0.07, 0.72, -0.01], [0.05, 0.72, 0.01]] as const).map((pos...
426 |         <mesh key={i + 10} position={pos}>
427 |           <boxGeometry args={[0.06, 0.005, 0.005]} />
428 |           <meshPhysicalMaterial color="#67e8f9" roughness={0.1} emissive="#67e8f9" emissiveIntensity={0.3} tra...
429 |         </mesh>
430 |       ]])
431 |       /* Glow ring */
432 |       <mesh position={[0, 0.65, 0]} rotation={[Math.PI / 2, 0, 0]}>
433 |         <ringGeometry args={[0.08, 0.12, 24]} />
434 |         <meshPhysicalMaterial color="#22d3ee" roughness={0.1} emissive="#22d3ee" emissiveIntensity={0.4} trans...
435 |       </mesh>
436 |     </group>
437 |   );
438 | }
439 |
440 | function StorageCabinet({ position }: { position: [number, number, number] }) {
441 |   return (
442 |     <group position={position}>
443 |       <mesh position={[0, 0.6, 0]}>
444 |         <boxGeometry args={[1.0, 1.2, 0.5]} />
445 |         <meshPhysicalMaterial color="#64748b" roughness={0.2} metalness={0.6} />
446 |       </mesh>
447 |       <mesh position={[0, 0.6, 0.27]}>
448 |         <boxGeometry args={[0.9, 1.1, 0.02]} />
449 |         <meshPhysicalMaterial color="#94a3b8" roughness={0.3} metalness={0.3} />
450 |       </mesh>
451 |       /* Shelves visible through glass */
452 |       <mesh position={[0, 0.3, 0.27]}>
453 |         <boxGeometry args={[0.85, 0.02, 0.02]} />
454 |         <meshPhysicalMaterial color="#475569" roughness={0.4} />
455 |       </mesh>
456 |       <mesh position={[0, 0.7, 0.27]}>
457 |         <boxGeometry args={[0.85, 0.02, 0.02]} />
458 |         <meshPhysicalMaterial color="#475569" roughness={0.4} />
459 |       </mesh>
460 |       /* Hazmat symbol */
461 |       <mesh position={[0, 0.9, 0.3]}>
462 |         <planeGeometry args={[0.15, 0.15]} />
463 |         <meshPhysicalMaterial color="#ef4444" roughness={0.5} />
464 |       </mesh>
465 |       <mesh position={[0, 0.9, 0.3]}>
466 |         <planeGeometry args={[0.08, 0.08]} />
467 |         <meshPhysicalMaterial color="#fef2f2" roughness={0.5} />
468 |       </mesh>
469 |     </group>
470 |   );
471 | }
472 |
473 | function LabWindow() {
474 |   const [campusTex, setCampusTex] = useState<THREE.CanvasTexture | null>(null);
475 |
476 |   useEffect(() => {
477 |     const c = document.createElement('canvas');
478 |     c.width = 512; c.height = 384;
479 |     const ctx = c.getContext('2d'); if (!ctx) return;
480 |     // Sky
481 |     const sky = ctx.createLinearGradient(0, 0, 0, 384);
482 |     sky.addColorStop(0, '#94a3b8'); sky.addColorStop(0.5, '#cbd5e1'); sky.addColorStop(1, '#e2e8f0');
483 |     ctx.fillStyle = sky; ctx.fillRect(0, 0, 512, 384);
484 |     // Buildings
485 |     ctx.fillStyle = '#64748b'; ctx.fillRect(80, 160, 60, 140);
486 |     ctx.fillStyle = '#475569'; ctx.fillRect(100, 140, 50, 160);
487 |     ctx.fillStyle = '#64748b'; ctx.fillRect(170, 130, 80, 170);
488 |     ctx.fillStyle = '#334155'; ctx.fillRect(200, 110, 60, 190);
489 |     ctx.fillStyle = '#64748b'; ctx.fillRect(280, 150, 70, 150);
490 |     ctx.fillStyle = '#475569'; ctx.fillRect(320, 120, 50, 180);
491 |     ctx.fillStyle = '#64748b'; ctx.fillRect(380, 160, 60, 140);
492 |     // Windows on buildings
493 |     ctx.fillStyle = '#94a3b8';
494 |     for (let bx = 0; bx < 5; bx++) {
495 |       for (let by = 0; by < 4; by++) {
496 |         ctx.fillRect(290 + bx * 15, 160 + by * 25, 8, 14);
497 |       }
498 |     }
499 |     // Trees
500 |     ctx.fillStyle = '#4a7c59'; ctx.fillRect(30, 220, 15, 80);
501 |     ctx.beginPath(); ctx.arc(37, 210, 25, 0, Math.PI * 2); ctx.fillStyle = '#3b6b47'; ctx.fill();
502 |     ctx.fillStyle = '#4a7c59'; ctx.fillRect(440, 200, 18, 100);
503 |     ctx.beginPath(); ctx.arc(449, 190, 30, 0, Math.PI * 2); ctx.fillStyle = '#3b6b47'; ctx.fill();
504 |     // Greenery
505 |     ctx.fillStyle = '#5a8f6a'; ctx.fillRect(0, 300, 512, 84);
506 |     ctx.fillStyle = '#4a7c59'; ctx.fillRect(0, 310, 512, 20);
507 |     // Road
508 |     ctx.fillStyle = '#475569'; ctx.fillRect(0, 340, 512, 20);
509 |     ctx.fillStyle = '#eab308'; ctx.fillRect(200, 348, 8, 4); ctx.fillRect(300, 348, 8, 4);
510 |     const t = new THREE.CanvasTexture(c);
511 |     t.needsUpdate = true;
512 |     setCampusTex(t);
513 |   }, []);
514 |
515 |   return (
516 |     <group position={[4.85, 1.6, 1.5]}>
517 |       <mesh position={[0, 0, 0]}>
518 |         <boxGeometry args={[0.06, 1.8, 2.0]} />
519 |         <meshPhysicalMaterial color="#334155" roughness={0.8} side={THREE.DoubleSide} />
520 |       </mesh>
521 |       <mesh position={[0.02, 0, 0]} rotation={[0, -Math.PI / 2, 0]}>
522 |         <planeGeometry args={[2.0, 1.8]} />
523 |         <meshPhysicalMaterial color="#87ceeb" roughness={0.1} transparent opacity={0.2} />

```

```

524 |     </mesh>
525 |     <mesh position=[[0.03, 0, 0]] rotation=[[0, -Math.PI / 2, 0]]>
526 |       <planeGeometry args=[[1.8, 1.6]] />
527 |       {campusTex && <meshPhysicalMaterial map={campusTex} roughness={0.2} />}
528 |     </mesh>
529 |     { /* Frame */
530 |     <mesh position=[[0, 0.93, 0]]>
531 |       <boxGeometry args=[[0.08, 0.06, 2.04]] />
532 |       <meshPhysicalMaterial color="#475569" roughness={0.5} />
533 |     </mesh>
534 |     <mesh position=[[0, -0.93, 0]]>
535 |       <boxGeometry args=[[0.08, 0.06, 2.04]] />
536 |       <meshPhysicalMaterial color="#475569" roughness={0.5} />
537 |     </mesh>
538 |     <mesh position=[[0, 0, 1.03]]>
539 |       <boxGeometry args=[[0.08, 1.86, 0.06]] />
540 |       <meshPhysicalMaterial color="#475569" roughness={0.5} />
541 |     </mesh>
542 |     <mesh position=[[0, 0, -1.03]]>
543 |       <boxGeometry args=[[0.08, 1.86, 0.06]] />
544 |       <meshPhysicalMaterial color="#475569" roughness={0.5} />
545 |     </mesh>
546 |   </group>
547 | );
548 | }
549 |
550 | function Window() {
551 |   return (
552 |     <group position=[[4.85, 1.5, -1.5]]>
553 |       <mesh position=[[0, 0, 0]]>
554 |         <boxGeometry args=[[0.04, 1.6, 1.4]] />
555 |         <meshPhysicalMaterial color="#f8fafc" roughness={0.5} transparent opacity={0.15} />
556 |       </mesh>
557 |       <mesh position=[[0.01, 0, 0]]>
558 |         <planeGeometry args=[[1.5, 1.7]] />
559 |         <meshPhysicalMaterial color="#87ceeb" roughness={0.1} metalness={0.05} transparent opacity={0.35} />
560 |       </mesh>
561 |       <mesh position=[[0, 0.86, 0]]>
562 |         <boxGeometry args=[[0.06, 0.04, 1.44]] />
563 |         <meshPhysicalMaterial color="#e2e8f0" roughness={0.5} />
564 |       </mesh>
565 |       <mesh position=[[0, -0.86, 0]]>
566 |         <boxGeometry args=[[0.06, 0.04, 1.44]] />
567 |         <meshPhysicalMaterial color="#e2e8f0" roughness={0.5} />
568 |       </mesh>
569 |       <mesh position=[[0, 0, 0.73]]>
570 |         <boxGeometry args=[[0.06, 1.64, 0.04]] />
571 |         <meshPhysicalMaterial color="#e2e8f0" roughness={0.5} />
572 |       </mesh>
573 |       <mesh position=[[0, 0, -0.73]]>
574 |         <boxGeometry args=[[0.06, 1.64, 0.04]] />
575 |         <meshPhysicalMaterial color="#e2e8f0" roughness={0.5} />
576 |       </mesh>
577 |       <mesh position=[[0, 0, 0]]>
578 |         <boxGeometry args=[[0.06, 1.64, 1.44]] />
579 |         <meshPhysicalMaterial color="#475569" roughness={0.6} transparent opacity={0.08} />
580 |       </mesh>
581 |       <mesh position=[[0.02, 0.3, 0.15]]>
582 |         <planeGeometry args=[[0.5, 0.4]] />
583 |         <meshPhysicalMaterial color="#fbbf24" roughness={0.3} emissive="#fbbf24" emissiveIntensity={0.6} trans...
584 |       </mesh>
585 |       <lineSegments>
586 |         <edgesGeometry args=[[new THREE.BoxGeometry(0.08, 1.64, 1.44)]] />
587 |         <lineBasicMaterial color="#94a3b8" />
588 |       </lineSegments>
589 |     </group>
590 |   );
591 | }
592 |
593 | function LabPoster() {
594 |   const tex = useLoader(THREE.TextureLoader, '/lab-poster.png');
595 |   tex.colorSpace = THREE.SRGBColorSpace;
596 |   tex.anisotropy = 16;
597 |   tex.minFilter = THREE.LinearMipmapLinearFilter;
598 |   tex.magFilter = THREE.LinearFilter;
599 |
600 |   return (
601 |     <group position=[[4.85, 1.5, 0]]>
602 |       <mesh position=[[0, 0, 0]]>
603 |         <boxGeometry args=[[0.06, 1.45, 2.78]] />
604 |         <meshPhysicalMaterial color="#1e293b" roughness={0.6} metalness={0.3} />
605 |       </mesh>
606 |       <mesh position=[[0.031, 0, 0]] rotation=[[0, Math.PI / 2, 0]]>
607 |         <planeGeometry args=[[2.6, 1.3]] />
608 |         <meshPhysicalMaterial map={tex} roughness={0.15} metalness={0.05} side={THREE.DoubleSide} />
609 |       </mesh>
610 |       <mesh position=[[0, 0.67, 0]]>
611 |         <boxGeometry args=[[0.08, 0.05, 2.75]] />
612 |         <meshPhysicalMaterial color="#475569" roughness={0.4} metalness={0.5} />
613 |       </mesh>
614 |       <mesh position=[[0, -0.67, 0]]>
615 |         <boxGeometry args=[[0.08, 0.05, 2.75]] />
616 |         <meshPhysicalMaterial color="#475569" roughness={0.4} metalness={0.5} />
617 |       </mesh>
618 |       <mesh position=[[1.37, 0, 0]]>
619 |         <boxGeometry args=[[0.08, 1.4, 0.05]] />
620 |         <meshPhysicalMaterial color="#475569" roughness={0.4} metalness={0.5} />
621 |       </mesh>
622 |       <mesh position=[[1.37, 0, 0]]>
623 |         <boxGeometry args=[[0.08, 1.4, 0.05]] />
624 |         <meshPhysicalMaterial color="#475569" roughness={0.4} metalness={0.5} />
625 |       </mesh>
626 |     </group>
627 |   );
628 | }
629 |
630 | function LabPoster2() {
631 |   const tex = useLoader(THREE.TextureLoader, '/lab-poster2.jpg');
632 |   tex.colorSpace = THREE.SRGBColorSpace;
633 |   tex.anisotropy = 16;
634 |   tex.minFilter = THREE.LinearMipmapLinearFilter;
635 |   tex.magFilter = THREE.LinearFilter;
636 |
637 |   return (
638 |     <group position=[[4.85, 1.5, 0]]>
639 |       <mesh position=[[0, 0, 0]]>
640 |         <boxGeometry args=[[0.06, 1.5, 1.5]] />
641 |         <meshPhysicalMaterial color="#0f172a" roughness={0.7} metalness={0.2} />
642 |       </mesh>
643 |       <mesh position=[[0.031, 0, 0]] rotation=[[0, -Math.PI / 2, 0]]>
644 |         <planeGeometry args=[[1.4, 1.4]] />
645 |         <meshPhysicalMaterial map={tex} roughness={0.15} metalness={0.05} side={THREE.DoubleSide} />
646 |       </mesh>
647 |       <mesh position=[[0, 0.72, 0]]>

```

```

648     <boxGeometry args={[0.06, 0.04, 1.46]} />
649     <meshPhysicalMaterial color="#1e293b" roughness={0.4} metalness={0.4} />
650   </mesh>
651   <mesh position={[0, -0.72, 0]}>
652     <boxGeometry args={[0.06, 0.04, 1.46]} />
653     <meshPhysicalMaterial color="#1e293b" roughness={0.4} metalness={0.4} />
654   </mesh>
655   <mesh position={[0, 0, -0.72]}>
656     <boxGeometry args={[0.06, 1.46, 0.04]} />
657     <meshPhysicalMaterial color="#1e293b" roughness={0.4} metalness={0.4} />
658   </mesh>
659   <mesh position={[0, 0, 0.72]}>
660     <boxGeometry args={[0.06, 1.46, 0.04]} />
661     <meshPhysicalMaterial color="#1e293b" roughness={0.4} metalness={0.4} />
662   </mesh>
663 </group>
664 );
665 }
666
667 function Door() {
668   return (
669     <group position={[0, 0, 4.85]}>
670       <mesh position={[0, 1.1, 0]}>
671         <boxGeometry args={[1.2, 2.2, 0.06]} />
672         <meshPhysicalMaterial color="#94a3b8" roughness={0.3} metalness={0.4} />
673       </mesh>
674       <mesh position={[0, 1.1, 0.04]}>
675         <boxGeometry args={[1.15, 2.15, 0.01]} />
676         <meshPhysicalMaterial color="#e2e8f0" roughness={0.2} metalness={0.3} />
677       </mesh>
678       <mesh position={[-0.6, 1.1, 0]}>
679         <boxGeometry args={[0.04, 2.22, 0.1]} />
680         <meshPhysicalMaterial color="#475569" roughness={0.3} metalness={0.5} />
681       </mesh>
682       <mesh position={[0.6, 1.1, 0]}>
683         <boxGeometry args={[0.04, 2.22, 0.1]} />
684         <meshPhysicalMaterial color="#475569" roughness={0.3} metalness={0.5} />
685       </mesh>
686       <mesh position={[0, 2.21, 0]}>
687         <boxGeometry args={[1.24, 0.04, 0.1]} />
688         <meshPhysicalMaterial color="#475569" roughness={0.3} metalness={0.5} />
689       </mesh>
690       <mesh position={[0.45, 1.0, 0.08]}>
691         <sphereGeometry args={[0.04, 8, 8]} />
692         <meshPhysicalMaterial color="#94a3b8" roughness={0.2} metalness={0.8} />
693       </mesh>
694       <mesh position={[0, 1.1, 0.06]}>
695         <planeGeometry args={[0.1, 0.1]} />
696         <meshPhysicalMaterial color="#ef4444" roughness={0.5} transparent opacity={0.3} />
697       </mesh>
698     </group>
699   );
700 }
701
702 export default function GameScene() {
703   return (
704     <div className="w-full h-screen bg-[#f8f9fc] relative overflow-hidden">
705       <HeadTracker />
706       <Canvas shadows camera={{ position: [0, 1.6, 4.5], fov: 60, near: 0.1, far: 25 }}>
707         <color attach="background" args={['#ffffff']} />
708
709         <CameraRig />
710
711         <ambientLight intensity={1.4} color="#e8f4ff" />
712         <hemisphereLight args={['#ffffff', '#c8dfff', 0.8]} />
713         <directionalLight position={[8, 12, 6]} intensity={1.4} color="ffffff" castShadow shadow-mapSize={[10...
714         <directionalLight position={[-5, 8, 8]} intensity={1.2} color="ffffff" />
715         <pointLight position={[0, 3.5, 0]} intensity={1.6} color="ffffff" />
716         <pointLight position={[2, 3.5, -2]} intensity={0.8} color="ffffff" />
717         <pointLight position={[-2, 3.5, 2]} intensity={0.8} color="ffffff" />
718
719         <mesh position={[0, -0.01, 0]} rotation={[-Math.PI / 2, 0, 0]} receiveShadow>
720           <planeGeometry args={[10, 10]} />
721           <meshPhysicalMaterial color="#8ba0b8" roughness={0.3} />
722         </mesh>
723
724         <mesh position={[0, 0.002, 0]} rotation={[-Math.PI / 2, 0, 0]}>
725           <planeGeometry args={[10, 10]} />
726           <meshPhysicalMaterial color="#9ab0c8" roughness={0.4} transparent opacity={0.25} />
727         </mesh>
728
729         {[
730           { pos: [0, 0.01, -4.98], size: [10, 0.02, 0.015], c: '#6b8199' },
731           { pos: [0, 0.01, -3.32], size: [10, 0.02, 0.015], c: '#6b8199' },
732           { pos: [0, 0.01, -1.66], size: [10, 0.02, 0.015], c: '#6b8199' },
733           { pos: [0, 0.01, 0], size: [10, 0.02, 0.015], c: '#6b8199' },
734           { pos: [0, 0.01, 1.66], size: [10, 0.02, 0.015], c: '#6b8199' },
735           { pos: [0, 0.01, 3.32], size: [10, 0.02, 0.015], c: '#6b8199' },
736           { pos: [-4.98, 0.01, 0], size: [0.015, 0.02, 10], c: '#6b8199' },
737           { pos: [-3.32, 0.01, 0], size: [0.015, 0.02, 10], c: '#6b8199' },
738           { pos: [-1.66, 0.01, 0], size: [0.015, 0.02, 10], c: '#6b8199' },
739           { pos: [1.66, 0.01, 0], size: [0.015, 0.02, 10], c: '#6b8199' },
740           { pos: [3.32, 0.01, 0], size: [0.015, 0.02, 10], c: '#6b8199' },
741           { pos: [4.98, 0.01, 0], size: [0.015, 0.02, 10], c: '#6b8199' },
742         ] as const).map((item, i) => (
743           <mesh key={i} position={item.pos}>
744             <boxGeometry args={item.size} />
745             <meshPhysicalMaterial color={item.c} roughness={0.5} transparent opacity={0.6} />
746           </mesh>
747         )))
748
749         <mesh position={[0, 1.5, -5]} receiveShadow>
750           <boxGeometry args={[10, 3, 0.18]} />
751           <meshPhysicalMaterial color="#52525b" roughness={0.9} side={THREE.DoubleSide} />
752         </mesh>
753         <mesh position={[-2.8, 1.0, -4.9]}>
754           <boxGeometry args={[1.2, 0.8, 0.04]} />
755           <meshPhysicalMaterial color="#94a3b8" roughness={0.1} transparent opacity={0.35} metalness={0.2} />
756         </mesh>
757         <mesh position={[2.8, 1.0, -4.9]}>
758           <boxGeometry args={[1.2, 0.8, 0.04]} />
759           <meshPhysicalMaterial color="#94a3b8" roughness={0.1} transparent opacity={0.35} metalness={0.2} />
760         </mesh>
761         <mesh position={[0, 2.3, -4.9]}>
762           <boxGeometry args={[3, 0.4, 0.04]} />
763           <meshPhysicalMaterial color="#94a3b8" roughness={0.1} transparent opacity={0.25} metalness={0.2} />
764         </mesh>
765
766         <mesh position={[0, 1.5, 5]}>
767           <boxGeometry args={[10, 3, 0.18]} />
768           <meshPhysicalMaterial color="#52525b" roughness={0.9} side={THREE.DoubleSide} />
769         </mesh>
770         <mesh position={[3.5, 1.5, 4.9]}>
771           <boxGeometry args={[2, 1.2, 0.04]} />

```

```

772 |         <meshPhysicalMaterial color="#94a3b8" roughness={0.1} transparent opacity={0.35} metalness={0.2} />
773 |     </mesh>
774 |     <mesh position={[-3.5, 1.5, 4.9]}>
775 |         <boxGeometry args={[2, 1.2, 0.04]} />
776 |         <meshPhysicalMaterial color="#94a3b8" roughness={0.1} transparent opacity={0.35} metalness={0.2} />
777 |     </mesh>
778 |
779 |     <mesh position={[-5, 1.5, 0]}>
780 |         <boxGeometry args={[0.18, 3, 10]} />
781 |         <meshPhysicalMaterial color="#52525b" roughness={0.9} side={THREE.DoubleSide} />
782 |     </mesh>
783 |     <mesh position={[-4.9, 1.8, -2.8]}>
784 |         <boxGeometry args={[0.04, 1.0, 1.0]} />
785 |         <meshPhysicalMaterial color="#94a3b8" roughness={0.1} transparent opacity={0.35} metalness={0.2} />
786 |     </mesh>
787 |     <mesh position={[-4.9, 1.8, 2.8]}>
788 |         <boxGeometry args={[0.04, 1.0, 1.0]} />
789 |         <meshPhysicalMaterial color="#94a3b8" roughness={0.1} transparent opacity={0.35} metalness={0.2} />
790 |     </mesh>
791 |
792 |     <mesh position={[5, 1.5, 0]}>
793 |         <boxGeometry args={[0.18, 3, 10]} />
794 |         <meshPhysicalMaterial color="#52525b" roughness={0.9} side={THREE.DoubleSide} />
795 |     </mesh>
796 |     <mesh position={[4.9, 1.8, -3.8]}>
797 |         <boxGeometry args={[0.04, 1.0, 0.8]} />
798 |         <meshPhysicalMaterial color="#94a3b8" roughness={0.1} transparent opacity={0.35} metalness={0.2} />
799 |     </mesh>
800 |     <mesh position={[4.9, 1.8, 3.8]}>
801 |         <boxGeometry args={[0.04, 1.0, 0.8]} />
802 |         <meshPhysicalMaterial color="#94a3b8" roughness={0.1} transparent opacity={0.35} metalness={0.2} />
803 |     </mesh>
804 |
805 |     <mesh position={[0, 3, 0]} rotation={[Math.PI / 2, 0, 0]}>
806 |         <planeGeometry args={[10, 10]} />
807 |         <meshPhysicalMaterial color="#f1f5f9" roughness={0.8} />
808 |     </mesh>
809 |     <mesh position={[0, 2.98, -4]}>
810 |         <boxGeometry args={[0.15, 0.04, 3]} />
811 |         <meshPhysicalMaterial color="#f8faff" roughness={0.3} emissive="#dce8ff" emissiveIntensity={0.6} />
812 |     </mesh>
813 |     <mesh position={[0, 2.98, 0]}>
814 |         <boxGeometry args={[0.15, 0.04, 3]} />
815 |         <meshPhysicalMaterial color="#f8faff" roughness={0.3} emissive="#dce8ff" emissiveIntensity={0.6} />
816 |     </mesh>
817 |     <mesh position={[0, 2.98, 4]}>
818 |         <boxGeometry args={[0.15, 0.04, 3]} />
819 |         <meshPhysicalMaterial color="#f8faff" roughness={0.3} emissive="#dce8ff" emissiveIntensity={0.6} />
820 |     </mesh>
821 |
822 |     {([0, 0, -4.9], [0, 0, 4.9], [-4.9, 0, 0], [4.9, 0, 0]) as const}.map((pos, i) => (
823 |         <mesh key={i} position={pos}>
824 |             <boxGeometry args={i < 2 ? [10, 0.08, 0.02] : [0.02, 0.08, 10]} />
825 |             <meshPhysicalMaterial color="#6b8199" roughness={0.4} />
826 |         </mesh>
827 |     ))
828 |
829 |     /* Back wall: storage cabinets */
830 |     <StorageCabinet position={[-3.5, 0, -4.85]} />
831 |     <StorageCabinet position={[-2.3, 0, -4.85]} />
832 |     <StorageCabinet position={[2.5, 0, -4.85]} />
833 |     <StorageCabinet position={[3.7, 0, -4.85]} />
834 |
835 |     /* Center bench - main work table */
836 |     <LabBench position={[0, 0, -1.0]} size={[3.5, 0.9, 2.0]} />
837 |
838 |     /* Left bench with computer station */
839 |     <LabBench position={[-2.5, 0, 1.5]} size={[2.0, 0.9, 1.0]} />
840 |     <ComputerStation />
841 |
842 |     /* Right bench with equipment */
843 |     <LabBench position={[3.0, 0, 1.0]} size={[2.0, 0.9, 1.0]} />
844 |     <Centrifuge position={[3.0, 0.9, 1.0]} />
845 |     <Microscope position={[2.5, 0.9, 1.3]} />
846 |     <Beaker position={[3.5, 0.9, 1.2]} color="#22c55e" />
847 |     <Flask position={[3.0, 0.9, 0.6]} color="#ef4444" />
848 |
849 |     /* Robotic arm section */
850 |     <RoboticArmSection />
851 |     <LabBench position={[4.2, 0, -1.8]} size={[1.2, 0.9, 0.8]} color="#94a3b8" />
852 |     <Beaker position={[4.2, 0.9, -1.6]} color="#eab308" height={0.3} radius={0.1} />
853 |     <Beaker position={[4.6, 0.9, -1.8]} color="#ef4444" height={0.2} radius={0.07} />
854 |
855 |     /* Window (right wall, front corner) */
856 |     <LabWindow />
857 |     <LabPoster />
858 |     <LabPoster2 />
859 |     <Door />
860 | </Canvas>
861 | </div>
862 | );
863 | }
864 |

```

File: components/HandTracker.js (242 lines, 9.1 KB)

```

1 | 'use client';
2 |
3 | import { useEffect, useRef, useState } from 'react';
4 |
5 | const MODEL_URLS = [
6 |   'https://cdn.jsdelivr.net/npm/@mediapipe/tasks-vision@0.10.18/wasm',
7 |   'https://unpkg.com/@mediapipe/tasks-vision@0.10.18/wasm',
8 |   'https://cdn.jsdelivr.net/npm/@mediapipe/tasks-vision@0.10.32/wasm',
9 | ];
10 |
11 | const LANDMARKER_URLS = [
12 |   'https://storage.googleapis.com/mediapipe-models/hand_landmarker/float16/1/hand_landmarker.t...',
13 |   'https://cdn.jsdelivr.net/gh/simonsown/pc-master-lms@main/public/models/hand_landmarker.task',
14 | ];
15 |
16 | function checkWebGLSupport() {
17 |   try {
18 |     const canvas = document.createElement('canvas');
19 |     const gl = canvas.getContext('webgl2') || canvas.getContext('webgl');
20 |     if (!gl) return false;
21 |     return true;
22 |   } catch { return false; }
23 | }
24 |
25 | const HandTracker = ({ onLandmarks, numHands = 1 }) => {
26 |   const videoRef = useRef(null);
27 |   const canvasRef = useRef(null);

```



```

28 | const [handLandmarker, setHandLandmarker] = useState(null);
29 | const [webcamRunning, setWebcamRunning] = useState(false);
30 | const [error, setError] = useState(null);
31 | const [loading, setLoading] = useState(true);
32 | 3 2 | c o n s t [ l o a d i n g M e s s a g e , s e t L o a d i n g M e s s a g e ] = u s e S t a t e ( ' a n g k h ã i t i o A I . . . ' ) ;
33 | const retryCountRef = useRef(0);
34 | const streamRef = useRef(null);
35 | const animRef = useRef(null);
36 | const mountedRef = useRef(true);
37 | const frameSkipRef = useRef(0);
38 | const lastLMCallbackRef = useRef(null);
39 |
40 | useEffect(() => {
41 |   lastLMCallbackRef.current = onLandmarks;
42 | }, [onLandmarks]);
43 |
44 | useEffect(() => {
45 |   mountedRef.current = true;
46 |   let cancelled = false;
47 |
48 |   const loadModel = async () => {
49 |     setLoading(true);
50 |     5 0 | s e t L o a d i n g M e s s a g e ( ' a n g t ã i m ô h ì n h A I . . . ' ) ;
51 |     const hasWebGL = checkWebGLSupport();
52 |
53 |     let lastError = null;
54 |     for (let attempt = 0; attempt < MODEL_URLS.length * 2; attempt++) {
55 |       if (cancelled) return;
56 |       if (attempt > 0) {
57 |         await new Promise(r => setTimeout(r, Math.min(1000 * Math.pow(2, attempt), 8000)));
58 |       }
59 |
60 |       const visionUrlIdx = attempt % MODEL_URLS.length;
61 |       const modelUrlIdx = Math.floor(attempt / MODEL_URLS.length) % LANDMARKER_URLS.length;
62 |
63 |       try {
64 |         const { FilesetResolver, HandLandmarker } = await import('@mediapipe/tasks-vision');
65 |
66 |         6 7 | i f ( c a n c e l l e d ) r e t u r n ; s e t L o a d i n g M e s s a g e ( ` a n g t ã i A I . . . ( t h í $ { a t t e m p t + 1 } ) ` ) ;
67 |
68 |         const vision = await FilesetResolver.forVisionTasks(MODEL_URLS[visionUrlIdx]);
69 |         if (cancelled) return;
70 |
71 |         const isLowEnd = navigator.hardwareConcurrency && navigator.hardwareConcurrency <= 4;
72 |         const landmarker = await HandLandmarker.createFromOptions(vision, {
73 |           baseOptions: {
74 |             modelAssetPath: LANDMARKER_URLS[modelUrlIdx],
75 |             delegate: isLowEnd ? 'CPU' : (hasWebGL ? 'GPU' : 'CPU')
76 |           },
77 |           runningMode: 'VIDEO',
78 |           numHands: numHands
79 |         });
80 |
81 |         if (cancelled) return;
82 |         setHandLandmarker(landmarker);
83 |         setLoading(false);
84 |         setTimeout(() => setWebcamRunning(true), 100);
85 |         return;
86 |       } catch (err) {
87 |         lastError = err;
88 |         console.warn('HandTracker attempt ${attempt + 1} failed:', err.message);
89 |       }
90 |     }
91 |
92 |     if (!cancelled) {
93 |       9 5 | s e t E r r o r ( l a s t E r r o r . m e s s a g e . i n c l u d e s ( ' N e t w o r k E r r o r ' ) || l a s t E r r o r . m e s s a g e . i n c l u d e s ( ' F a i l e d t o f e t c h ' )
94 |       9 6 | : ' K h ô n g t h ã t ã i m ô h ì n h A I đ o m ì n g y ẽ u . V u i l ò n g t h í l i i h o . o
95 |       : ' K h ô n g t h ã t ã i m ô h ì n h A I . V u i l ò n g t h í l i i . ' ) ;
96 |
97 |       setLoading(false);
98 |     }
99 |   };
100 |
101 |   loadModel();
102 |   return () => { cancelled = true; mountedRef.current = false; };
103 | }, []);
104 |
105 | useEffect(() => {
106 |   if (!handLandmarker || !webcamRunning) return;
107 |   let cancelled = false;
108 |
109 |   const startCamera = async () => {
110 |     try {
111 |       1 2 | i f ( ! n a v i g a t o r . m e d i a D e v i c e s . g e t U s e r M e d i a ) {
112 |         s e t E r r o r ( ' T r ì n h d u y ẽ t k h ô n g h x t r ã c a m e r a . ' ) ;
113 |       }
114 |     }
115 |
116 |     const isLowEnd = navigator.hardwareConcurrency && navigator.hardwareConcurrency <= 4;
117 |     const stream = await navigator.mediaDevices.getUserMedia({
118 |       video: { width: { ideal: 320 }, height: { ideal: 240 }, facingMode: 'user', frameRate: { ideal: isLo...
119 |     });
120 |     if (cancelled) { stream.getTracks().forEach(t => t.stop()); return; }
121 |     streamRef.current = stream;
122 |
123 |     const video = videoRef.current;
124 |     if (!video) return;
125 |     video.srcObject = stream;
126 |     video.setAttribute('playsinline', '');
127 |     video.setAttribute('autoplay', '');
128 |
129 |     await video.play();
130 |
131 |     const canvas = canvasRef.current;
132 |     if (!canvas) return;
133 |
134 |     const predictLoop = async () => {
135 |       if (!mountedRef.current) return;
136 |
137 |       const isLowEnd = navigator.hardwareConcurrency && navigator.hardwareConcurrency <= 4;
138 |       const skipFrames = isLowEnd ? 3 : 2;
139 |       frameSkipRef.current++;
140 |       if (frameSkipRef.current % skipFrames !== 0) {
141 |         animRef.current = requestAnimationFrame(predictLoop);
142 |         return;
143 |       }
144 |
145 |       if (video.videoWidth > 0 && video.videoHeight > 0) {
146 |         canvas.width = video.videoWidth;
147 |         canvas.height = video.videoHeight;
148 |       }
149 |
150 |       try {
151 |         const results = await handLandmarker.detectForVideo(video, performance.now());

```

```

152 |
153 |         if (results.landmarks && results.landmarks.length > 0) {
154 |             const mirrored = results.landmarks.map((hand) =>
155 |                 hand.map(p => ({ ...p, x: 1 - p.x })))
156 |         );
157 |         if (lastLMCallbackRef.current) {
158 |             lastLMCallbackRef.current(mirrored);
159 |         }
160 |     } else {
161 |         if (lastLMCallbackRef.current) {
162 |             lastLMCallbackRef.current([]);
163 |         }
164 |     }
165 | } catch (e) {
166 |     console.warn('detect error:', e);
167 | }
168 | animRef.current = requestAnimationFrame(predictLoop);
169 | };
170 |
171 | predictLoop();
172 | } catch (err) {
173 |     if (cancelled) return;
174 |     const msg = err.name === 'NotAllowedError'
175 |       ? 'Quy Án truy c-p camera bẽ tẽ ch Ñĩ. Vui lòng c ỷp quy Án trong
176 |       : 'Không th ấ truy c-p camera: ' + err.message;
177 |     setError(msg);
178 |   };
179 | };
180 |
181 | startCamera();
182 | return () => { cancelled = true; if (animRef.current) cancelAnimationFrame(animRef.current); };
183 | }, [handLandmarker, webcamRunning]);
184 |
185 | useEffect(() => {
186 |   return () => {
187 |     if (streamRef.current) {
188 |       streamRef.current.getTracks().forEach(t => t.stop());
189 |       streamRef.current = null;
190 |     }
191 |   };
192 | }, []);
193 |
194 | const handleRetry = () => {
195 |   setError(null);
196 |   setHandLandmarker(null);
197 |   setWebcamRunning(false);
198 |   setLoading(true);
199 |   set Loading Message ( ' ang thí li . . . ' );
200 |   retryCountRef.current = 0;
201 | };
202 |
203 | if (error) {
204 |   return (
205 |     <div style={{ width: '100%', height: '100%', display: 'flex', flexDirection: 'column', alignItems: 'cent...
206 |       <svg width="48" height="48" viewBox="0 0 24 24" fill="none" stroke="#ef4444" strokeWidth="1.5">
207 |         <circle cx="12" cy="12" r="10" /><line x1="12" y1="8" x2="12" y2="12" /><line x1="12" y1="16" x2="12...
208 |       </svg>
209 |       <p style={{ fontSize: '13px', textAlign: 'center', margin: 0, maxWidth: '280px', lineHeight: 1.5 }}>{e...
210 |       <button onClick={handleRetry} style={{ padding: '8px 20px', background: '#00d4aa', color: '#000', bord...
211 |     </button>
212 |   </div>
213 | );
214 | };
215 | }
216 |
217 | return (
218 |   <div style={{ position: 'relative', width: '100%', height: '100%', backgroundColor: 'black', overflow: 'hi...
219 |     <video
220 |       ref={videoRef}
221 |       style={{ position: 'absolute', top: 0, left: 0, width: '100%', height: '100%', objectFit: 'cover', tra...
222 |       playsInline
223 |       muted
224 |     />
225 |     <canvas
226 |       ref={canvasRef}
227 |       style={{ position: 'absolute', top: 0, left: 0, width: '100%', height: '100%', transform: 'scaleX(-1)'...
228 |     />
229 |     {loading && (
230 |       <div style={{ position: 'absolute', inset: 0, display: 'flex', flexDirection: 'column', alignItems: 'c...
231 |       <div style={{ width: '32px', height: '32px', border: '3px solid #00d4aa', borderTopColor: 'transpare...
232 |       <span style={{ color: '#10b981', fontWeight: 600, fontSize: '13px' }}>{loadingMessage}</span>
233 |       <span style={{ color: '#666', fontSize: '11px' }}>S í d ăng m ô
234 |     </div>
235 |   )}
236 |   <style>{@keyframes spin { to { transform: rotate(360deg) } }}</style>
237 | </div>
238 | );
239 | };
240 |
241 | export default HandTracker;
242 |

```

File: components/HeadTracker.tsx (563 lines, 14.1 KB)

```

1 | 'use client';
2 |
3 | import { useEffect, useRef, useState, useCallback } from 'react';
4 | import { useAssemblyStore } from '@lib/useStore';
5 |
6 | const FACE_LANDMARKER_URL =
7 |   'https://storage.googleapis.com/mediapipe-models/face_landmarker/face_landmarker/float16/1/face_landmarker.t...
8 | const WASM_BASE = 'https://cdn.jsdelivr.net/npm/@mediapipe/tasks-vision@0.10.32/wasm';
9 |
10 | const LERP_FACTOR = 0.6;
11 | const PITCH_MIN = -0.6;
12 | const PITCH_MAX = 0.6;
13 | const YAW_MIN = -0.8;
14 | const YAW_MAX = 0.8;
15 |
16 | export default function HeadTracker() {
17 |   const setCameraCoords = useAssemblyStore((s) => s.setCameraCoords);
18 |   const cameraCoords = useAssemblyStore((s) => s.cameraCoords);
19 |
20 |   const videoRef = useRef<HTMLVideoElement>(null);
21 |   const canvasRef = useRef<HTMLCanvasElement>(null);
22 |   const streamRef = useRef<MediaStream | null>(null);
23 |   const landmarkerRef = useRef<any>(null);
24 |   const animRef = useRef<number>(0);
25 |   const mountedRef = useRef(true);
26 |
27 |   const [status, setStatus] = useState<'loading' | 'active' | 'error'>('loading');
28 |   const [errorMsg, setErrorMsg] = useState('');
29 |   const [hasWebcam, setHasWebcam] = useState(false);

```

```

30 | const [trackingFps, setTrackingFps] = useState(0);
31 |
32 | const smoothPitch = useRef(0);
33 | const smoothYaw = useRef(0);
34 | const fpsCount = useRef(0);
35 | const fpsTime = useRef(0);
36 |
37 | const drawFaceMesh = useCallback(
38 |   (landmarks: { x: number; y: number; z: number }[]) => {
39 |     const canvas = canvasRef.current;
40 |     if (!canvas) return;
41 |     const ctx = canvas.getContext('2d');
42 |     if (!ctx) return;
43 |
44 |     const w = canvas.width;
45 |     const h = canvas.height;
46 |     ctx.clearRect(0, 0, w, h);
47 |
48 |     ctx.strokeStyle = 'rgba(0, 212, 170, 0.6)';
49 |     ctx.lineWidth = 1;
50 |
51 |     const FACE_CONNECTIONS: [number, number][] = [
52 |       [10, 338],
53 |       [338, 297],
54 |       [297, 332],
55 |       [332, 284],
56 |       [284, 251],
57 |       [251, 389],
58 |       [389, 356],
59 |       [356, 454],
60 |       [454, 323],
61 |       [323, 361],
62 |       [361, 288],
63 |       [288, 397],
64 |       [397, 365],
65 |       [365, 379],
66 |       [379, 378],
67 |       [378, 400],
68 |       [400, 377],
69 |       [377, 152],
70 |       [152, 148],
71 |       [148, 176],
72 |       [176, 149],
73 |       [149, 150],
74 |       [150, 136],
75 |       [136, 172],
76 |       [172, 58],
77 |       [58, 132],
78 |       [132, 93],
79 |       [93, 234],
80 |       [234, 127],
81 |       [127, 162],
82 |       [162, 21],
83 |       [21, 54],
84 |       [54, 103],
85 |       [103, 67],
86 |       [67, 109],
87 |       [109, 10],
88 |       [10, 338],
89 |       [46, 53],
90 |       [53, 52],
91 |       [52, 65],
92 |       [65, 55],
93 |       [55, 285],
94 |       [285, 295],
95 |       [295, 282],
96 |       [282, 283],
97 |       [283, 276],
98 |       [276, 352],
99 |       [352, 346],
100 |      [346, 340],
101 |      [340, 330],
102 |      [330, 329],
103 |      [329, 450],
104 |      [450, 449],
105 |      [449, 448],
106 |      [448, 261],
107 |      [261, 446],
108 |      [446, 342],
109 |      [342, 445],
110 |      [445, 444],
111 |      [444, 443],
112 |      [443, 442],
113 |      [442, 441],
114 |      [441, 413],
115 |      [413, 414],
116 |      [414, 372],
117 |      [372, 383],
118 |      [383, 300],
119 |      [300, 301],
120 |      [301, 263],
121 |      [263, 466],
122 |      [466, 388],
123 |      [388, 387],
124 |      [387, 386],
125 |      [386, 385],
126 |      [385, 384],
127 |      [384, 398],
128 |      [398, 362],
129 |      [362, 382],
130 |      [382, 381],
131 |      [381, 380],
132 |      [380, 374],
133 |      [374, 373],
134 |      [373, 390],
135 |      [390, 249],
136 |      [249, 64],
137 |      [64, 98],
138 |      [98, 97],
139 |      [97, 96],
140 |      [96, 95],
141 |      [95, 94],
142 |      [94, 93],
143 |      [93, 132],
144 |      [116, 163],
145 |      [163, 123],
146 |      [123, 143],
147 |      [143, 155],
148 |      [155, 154],
149 |      [154, 157],
150 |      [157, 158],
151 |      [158, 159],
152 |      [159, 160],
153 |      [160, 161],

```

```

154|         [161, 243],
155|         [243, 173],
156|         [173, 133],
157|         [133, 245],
158|         [245, 244],
159|         [244, 246],
160|         [246, 247],
161|         [247, 248],
162|         [248, 249],
163|         [33, 246],
164|         [246, 161],
165|         [161, 160],
166|         [160, 159],
167|         [159, 158],
168|         [158, 157],
169|         [157, 173],
170|         [173, 133],
171|         [133, 155],
172|         [155, 154],
173|         [154, 153],
174|         [153, 145],
175|         [145, 144],
176|         [144, 163],
177|         [163, 116],
178|         [168, 9],
179|         [9, 8],
180|         [8, 168],
181|         [168, 6],
182|         [6, 197],
183|         [197, 195],
184|         [195, 5],
185|         [5, 4],
186|         [4, 1],
187|         [1, 19],
188|         [19, 20],
189|         [20, 24],
190|         [24, 4],
191|         [4, 5],
192|         [5, 195],
193|         [197, 6],
194|         [6, 168],
195|         [168, 8],
196|         [8, 9],
197|     ];
198|
199|     for (const [i, j] of FACE_CONNECTIONS) {
200|         const p1 = landmarks[i];
201|         const p2 = landmarks[j];
202|         if (p1 && p2) {
203|             ctx.beginPath();
204|             ctx.moveTo(p1.x * w, p1.y * h);
205|             ctx.lineTo(p2.x * w, p2.y * h);
206|             ctx.stroke();
207|         }
208|     }
209|
210|     ctx.fillStyle = 'rgba(0, 212, 170, 0.8)';
211|     const keyPoints = [1, 4, 5, 6, 8, 9, 10, 33, 133, 152, 159, 249, 263, 362, 386, 454, 466, 468, 473];
212|     for (const idx of keyPoints) {
213|         const p = landmarks[idx];
214|         if (p) {
215|             ctx.beginPath();
216|             ctx.arc(p.x * w, p.y * h, 2.5, 0, Math.PI * 2);
217|             ctx.fill();
218|         }
219|     }
220| },
221| []
222| );
223|
224| useEffect(() => {
225|     mountedRef.current = true;
226|     let cancelled = false;
227|
228|     const init = async () => {
229|         try {
230|             setStatus('loading');
231|             const { FilesetResolver, FaceLandmarker } = await import(
232|                 '@mediapipe/tasks-vision'
233|             );
234|
235|             if (cancelled) return;
236|
237|             const vision = await FilesetResolver.forVisionTasks(WASM_BASE);
238|             if (cancelled) return;
239|
240|             const landmarker = await FaceLandmarker.createFromOptions(vision, {
241|                 baseOptions: {
242|                     modelAssetPath: FACE_LANDMARKER_URL,
243|                     delegate: 'GPU',
244|                 },
245|                 runningMode: 'VIDEO',
246|                 numFaces: 1,
247|                 outputFaceBlendshapes: false,
248|             });
249|
250|             if (cancelled) return;
251|             landmarkerRef.current = landmarker;
252|
253|             const stream = await navigator.mediaDevices.getUserMedia({
254|                 video: {
255|                     width: { ideal: 320 },
256|                     height: { ideal: 240 },
257|                     facingMode: 'user',
258|                 },
259|             });
260|
261|             if (cancelled) {
262|                 stream.getTracks().forEach((t) => t.stop());
263|                 return;
264|             }
265|
266|             streamRef.current = stream;
267|             const video = videoRef.current;
268|             if (!video) return;
269|             video.srcObject = stream;
270|             video.setAttribute('playsinline', '');
271|             await video.play();
272|             setHasWebcam(true);
273|             setStatus('active');
274|             fpsTime.current = performance.now();
275|         } catch (err: any) {
276|             if (cancelled) return;
277|             const msg =

```

```

278 |         err.name === 'NotAllowedError'
279 |         ? 'Webcam access denied. Please allow camera permissions.'
280 |         : err.name === 'NotFoundError'
281 |         ? 'No webcam found.'
282 |         : `Face tracking init failed: ${err.message}`;
283 |     setErrMsg(msg);
284 |     setStatus('error');
285 | }
286 | };
287 |
288 | init();
289 | return () => {
290 |     cancelled = true;
291 |     mountedRef.current = false;
292 |     cancelAnimationFrame(animRef.current);
293 |     if (streamRef.current) {
294 |         streamRef.current.getTracks().forEach((t) => t.stop());
295 |         streamRef.current = null;
296 |     }
297 | };
298 | }, []);
299 |
300 | useEffect(() => {
301 |     if (status !== 'active' || !landmarkerRef.current) return;
302 |
303 |     let running = true;
304 |     const video = videoRef.current;
305 |
306 |     const predictLoop = async () => {
307 |         if (!running || !landmarkerRef.current || !video || video.readyState < 2)
308 |             return;
309 |
310 |         try {
311 |             const results = await landmarkerRef.current.detectForVideo(
312 |                 video,
313 |                 performance.now()
314 |             );
315 |
316 |             if (results.faceLandmarks && results.faceLandmarks.length > 0) {
317 |                 const landmarks = results.faceLandmarks[0];
318 |                 const nose = landmarks[1];
319 |                 const leftEye = landmarks[33];
320 |                 const rightEye = landmarks[263];
321 |                 const leftEar = landmarks[234];
322 |                 const rightEar = landmarks[454];
323 |                 const mouth = landmarks[13];
324 |
325 |                 if (nose && leftEye && rightEye && mouth) {
326 |                     const eyeMidX = (leftEye.x + rightEye.x) / 2;
327 |                     const eyeMidY = (leftEye.y + rightEye.y) / 2;
328 |
329 |                     const rawYaw = (nose.x - 0.5) * 2;
330 |                     const rawPitch = (nose.y - 0.5) * 2;
331 |
332 |                     smoothPitch.current +=
333 |                         LERP_FACTOR *
334 |                         (Math.max(PITCH_MIN, Math.min(PITCH_MAX, rawPitch)) -
335 |                          smoothPitch.current);
336 |                     smoothYaw.current +=
337 |                         LERP_FACTOR *
338 |                         (Math.max(YAW_MIN, Math.min(YAW_MAX, rawYaw)) -
339 |                          smoothYaw.current);
340 |
341 |                     setCameraCoords({
342 |                         pitch: smoothPitch.current,
343 |                         yaw: smoothYaw.current,
344 |                         roll: 0,
345 |                     });
346 |
347 |                     fpsCount.current++;
348 |                     const now = performance.now();
349 |                     if (now - fpsTime.current > 1000) {
350 |                         setTrackingFps(fpsCount.current);
351 |                         fpsCount.current = 0;
352 |                         fpsTime.current = now;
353 |                     }
354 |                 }
355 |
356 |                 drawFaceMesh(results.faceLandmarks[0]);
357 |             }
358 |         } catch (e) {
359 |         }
360 |
361 |         animRef.current = requestAnimationFrame(predictLoop);
362 |     };
363 |
364 |     predictLoop();
365 |     return () => {
366 |         running = false;
367 |         cancelAnimationFrame(animRef.current);
368 |     };
369 | }, [status, setCameraCoords, drawFaceMesh]);
370 |
371 | const handleRetry = () => {
372 |     setStatus('loading');
373 |     setErrMsg('');
374 |     setHasWebcam(false);
375 |     landmarkerRef.current = null;
376 |     window.location.reload();
377 | };
378 |
379 | return (
380 |     <div
381 |         style={{
382 |             position: 'fixed',
383 |             top: 16,
384 |             left: 16,
385 |             width: 200,
386 |             height: 150,
387 |             borderRadius: 12,
388 |             overflow: 'hidden',
389 |             zIndex: 9999,
390 |             border: '1px solid rgba(0, 212, 170, 0.3)',
391 |             boxShadow: '0 4px 24px rgba(0,0,0,0.5)',
392 |             background: '#000',
393 |         }}
394 |     >
395 |         <video
396 |             ref={videoRef}
397 |             muted
398 |             playsInline
399 |             style={{
400 |                 position: 'absolute',
401 |                 inset: 0,

```

```

402 |         width: '100%',
403 |         height: '100%',
404 |         objectFit: 'cover',
405 |         transform: 'scaleX(-1)',
406 |     }}
407 | />
408 | <canvas
409 |   ref={canvasRef}
410 |   width={320}
411 |   height={240}
412 |   style={{
413 |     position: 'absolute',
414 |     inset: 0,
415 |     width: '100%',
416 |     height: '100%',
417 |     transform: 'scaleX(-1)',
418 |     pointerEvents: 'none',
419 |   }}
420 | />
421 | {status === 'loading' && (
422 |   <div
423 |     style={{
424 |       position: 'absolute',
425 |       inset: 0,
426 |       display: 'flex',
427 |       flexDirection: 'column',
428 |       alignItems: 'center',
429 |       justifyContent: 'center',
430 |       gap: 8,
431 |       background: 'rgba(0,0,0,0.85)',
432 |     }}
433 |   >
434 |     <div
435 |       style={{
436 |         width: 24,
437 |         height: 24,
438 |         border: '2px solid rgba(0,212,170,0.3)',
439 |         borderTopColor: '#00d4aa',
440 |         borderRadius: '50%',
441 |         animation: 'ht-spin 0.8s linear infinite',
442 |       }}
443 |     />
444 |     <span style={{ color: '#00d4aa', fontSize: 10, fontWeight: 600 }}>
445 |       INITIALIZING AI TRACKING
446 |     </span>
447 |   </div>
448 | )}
449 | {status === 'error' && (
450 |   <div
451 |     style={{
452 |       position: 'absolute',
453 |       inset: 0,
454 |       display: 'flex',
455 |       flexDirection: 'column',
456 |       alignItems: 'center',
457 |       justifyContent: 'center',
458 |       gap: 8,
459 |       padding: 12,
460 |       background: 'rgba(0,0,0,0.9)',
461 |     }}
462 |   >
463 |     <span style={{ color: '#ef4444', fontSize: 9, textAlign: 'center' }}>
464 |       {errorMsg}
465 |     </span>
466 |     <button
467 |       onClick={handleRetry}
468 |       style={{
469 |         padding: '4px 12px',
470 |         background: '#00d4aa',
471 |         color: '#000',
472 |         border: 'none',
473 |         borderRadius: 6,
474 |         fontWeight: 700,
475 |         fontSize: 10,
476 |         cursor: 'pointer',
477 |         fontFamily: 'inherit',
478 |       }}
479 |     >
480 |       RETRY
481 |     </button>
482 |   </div>
483 | )}
484 | {status === 'active' && (
485 |   <div
486 |     <div
487 |       style={{
488 |         position: 'absolute',
489 |         top: 6,
490 |         left: 6,
491 |         display: 'flex',
492 |         alignItems: 'center',
493 |         gap: 6,
494 |         background: 'rgba(0,0,0,0.7)',
495 |         padding: '3px 8px',
496 |         borderRadius: 6,
497 |         backdropFilter: 'blur(4px)',
498 |       }}
499 |     >
500 |       <span
501 |         style={{
502 |           width: 6,
503 |           height: 6,
504 |           borderRadius: '50%',
505 |           background: '#10b981',
506 |           boxShadow: '0 0 6px #10b981',
507 |         }}
508 |       />
509 |       <span
510 |         style={{
511 |           color: '#10b981',
512 |           fontSize: 8,
513 |           fontWeight: 700,
514 |           letterSpacing: '0.5px',
515 |         }}
516 |       >
517 |         AI TRACKING ACTIVE
518 |       </span>
519 |     </div>
520 |     <div
521 |       style={{
522 |         position: 'absolute',
523 |         bottom: 6,
524 |         right: 6,
525 |         background: 'rgba(0,0,0,0.7)',

```

```

526 |         padding: '2px 6px',
527 |         borderRadius: 4,
528 |         fontSize: 8,
529 |         color: '#94a3b8',
530 |         fontFamily: 'monospace',
531 |     }}
532 | >
533 |     {trackingFps} FPS
534 | </div>
535 | <div>
536 |     style={{
537 |       position: 'absolute',
538 |       bottom: 6,
539 |       left: 6,
540 |       display: 'flex',
541 |       gap: 8,
542 |       background: 'rgba(0,0,0,0.7)',
543 |       padding: '2px 6px',
544 |       borderRadius: 4,
545 |       fontSize: 7,
546 |       fontFamily: 'monospace',
547 |       color: '#64748b',
548 |     }}
549 | >
550 |     <span>
551 |       P: {(cameraCoords.pitch * 100).toFixed(0)}
552 |     </span>
553 |     <span>
554 |       Y: {(cameraCoords.yaw * 100).toFixed(0)}
555 |     </span>
556 |   </div>
557 | </>
558 | })
559 | <style>{@keyframes ht-spin { to { transform: rotate(360deg); } }}</style>
560 | </div>
561 | });
562 | }
563 |

```

File: components/imageAssets.ts (126 lines, 4.9 KB)

```

1 | export type ComponentType = 'CPU' | 'RAM' | 'GPU' | 'PSU' | 'SSD' | 'COOLER' | 'Mainboard' | 'HDD';
2 |
3 | export const IMAGE_MAP: Record<string, string[]> = {
4 |   CPU: [
5 |     '/images/components/cpu_amd_front.png',
6 |     '/images/components/cpu_amd_iso.png',
7 |     '/images/components/cpu_intel_front.png',
8 |     '/images/components/cpu_intel_iso.png',
9 |     '/images/components/cpu_ryzen5_old_front.png',
10 |    '/images/components/cpu_xeon_old_front.png',
11 |    '/images/components/amd_ryzen_7_9800x3d.png',
12 |    '/images/components/amd_ryzen_9_9950x.png',
13 |    '/images/components/intel_core_i3_12100.png',
14 |    '/images/components/intel_core_i7_14700f.png',
15 |    '/images/components/intel_core_ultra_9_285k.png',
16 |    '/images/components/intel_xeon_e5_2667_v4.png',
17 |  ],
18 |   RAM: [
19 |     '/images/components/cudimm_ram.png',
20 |     '/images/components/ddr4_ecc_ram.png',
21 |     '/images/components/ddr4_ram_front.png',
22 |     '/images/components/ddr4_ram_non_rgb.png',
23 |     '/images/components/ddr5_ram_front.png',
24 |     '/images/components/ddr5_ram_rgb.png',
25 |     '/images/components/ram_ecc_front.png',
26 |     '/images/components/ram_rgb_front.png',
27 |     '/images/components/ram_rgb_iso.png',
28 |   ],
29 |   GPU: [
30 |     '/images/components/gpu_gtx_1650_front.png',
31 |     '/images/components/gpu_gtx_1660_front.png',
32 |     '/images/components/gpu_rtx_5090_front.png',
33 |     '/images/components/gpu_rtx_5090_iso.png',
34 |     '/images/components/gpu_rx580_front.png',
35 |     '/images/components/gpu_rx580_iso.png',
36 |     '/images/components/gpu_rx_7600_front.png',
37 |     '/images/components/gpu_rx_7600_iso.png',
38 |     '/images/components/gtx_1660_super.png',
39 |     '/images/components/nvidia_rtx_5090_front_view.png',
40 |     '/images/components/nvidia_rtx_5090_white_bg.png',
41 |     '/images/components/rtx_4060_itx.png',
42 |     '/images/components/rtx_5070_ti.png',
43 |     '/images/components/rtx_5080_16gb.png',
44 |     '/images/components/rtx_5090_32gb.png',
45 |     '/images/components/rtx_5090_pure_white_bg.png',
46 |     '/images/components/rtx_5090_rainbow_fans.png',
47 |     '/images/components/rtx_5090_rainbow_white_bg.png',
48 |     '/images/components/rx_7600_8gb.png',
49 |   ],
50 |   PSU: [
51 |     '/images/components/corsair_rm750e_psu.png',
52 |     '/images/components/msi_mpg_ai1600ts_psu.png',
53 |     '/images/components/psu_400w_iso.png',
54 |     '/images/components/psu_back_front.png',
55 |     '/images/components/psu_front.png',
56 |     '/images/components/psu_iso_new.png',
57 |     '/images/components/psu_side_front.png',
58 |     '/images/components/sfx_gold_psu.png',
59 |   ],
60 |   SSD: [
61 |     '/images/components/nvme_ssd_front.png',
62 |     '/images/components/nvme_ssd_front_new.png',
63 |     '/images/components/nvme_ssd_iso_new.png',
64 |     '/images/components/samsung_990_pro_ssd.png',
65 |     '/images/components/ssd_120gb_front.png',
66 |     '/images/components/ssd_2_5_front.png',
67 |     '/images/components/ssd_2_5_inch.png',
68 |   ],
69 |   COOLER: [
70 |     '/images/components/aio_cooler_front.png',
71 |     '/images/components/aio_pump_front.png',
72 |     '/images/components/air_cooler_front.png',
73 |     '/images/components/air_cooler_front_new.png',
74 |     '/images/components/air_cooler_iso_new.png',
75 |     '/images/components/arctic_liquid_freezer_iii_360.png',
76 |     '/images/components/intel_stock_cooler.png',
77 |     '/images/components/msi_mag_coreliquid_360.png',
78 |     '/images/components/stock_cooler_iso.png',
79 |     '/images/components/thermalright_peerless_assassin_120.png',
80 |   ],
81 |   Mainboard: [
82 |     '/images/components/high_end_mainboard.png',

```

```

83 | '/images/components/mainboard_a520_front.png',
84 | '/images/components/mainboard_a520_iso.png',
85 | '/images/components/mainboard_b550_front.png',
86 | '/images/components/mainboard_b550_iso.png',
87 | '/images/components/mainboard_b760_front.png',
88 | '/images/components/mainboard_b760_iso.png',
89 | '/images/components/mainboard_classic_green_front.png',
90 | '/images/components/mainboard_classic_green_iso.png',
91 | '/images/components/mainboard_front.png',
92 | '/images/components/mainboard_h610_front.png',
93 | '/images/components/mainboard_h610_iso.png',
94 | '/images/components/mainboard_iso.png',
95 | '/images/components/mainboard_military_green_front.png',
96 | '/images/components/mainboard_military_green_iso.png',
97 | ],
98 | hdd: [
99 |   '/images/components/hdd_3_5_inch.png',
100 |   '/images/components/hdd_iso.png',
101 |   '/images/components/hdd_0ld_front.png',
102 | ],
103 | };
104 |
105 | export function getRandomSrc(): string {
106 |   const all = Object.values(IMAGE_MAP).flat();
107 |   if (all.length === 0) return '';
108 |   return all[Math.floor(Math.random() * all.length)];
109 | }
110 |
111 | export function getRandomImage(type: string): string {
112 |   const images = IMAGE_MAP[type];
113 |   if (!images || images.length === 0) return '';
114 |   return images[Math.floor(Math.random() * images.length)];
115 | }
116 |
117 | export function preloadImages(type: string): Promise<HTMLImageElement[]> {
118 |   const urls = IMAGE_MAP[type] || [];
119 |   return Promise.all(urls.map(src => new Promise<HTMLImageElement>((resolve) => {
120 |     const img = new Image();
121 |     img.onload = () => resolve(img);
122 |     img.onerror = () => resolve(img);
123 |     img.src = src;
124 |   })));
125 | }
126 |

```

File: components/JoinClassModal.js (154 lines, 6.9 KB)

```

1 | 'use client';
2 |
3 | import React, { useState } from 'react';
4 | import { X, Users, Loader2, CheckCircle2 } from 'lucide-react';
5 | import { supabase } from '@lib/supabase';
6 |
7 | const JoinClassModal = ({ isOpen, onClose, lang }) => {
8 |   const [classCode, setClassCode] = useState('');
9 |   const [loading, setLoading] = useState(false);
10 |   const [error, setError] = useState(null);
11 |   const [success, setSuccess] = useState(false);
12 |
13 |   if (!isOpen) return null;
14 |
15 |   const handleJoin = async (e) => {
16 |     e.preventDefault();
17 |     setLoading(true);
18 |     setError(null);
19 |
20 |     try {
21 |       const { data: { user } } = await supabase.auth.getUser();
22 |       if (!user) throw new Error(lang === 'en' ? 'Please login first' : 'Vui lòng đăng nhập trước');
23 |
24 |       // 1. Tìm lớp học theo mã (dùng SECURITY DEFINER function)
25 |       const { data: classData, error: classErr } = await supabase
26 |         .rpc('fn_lookup_class_by_code', { p_code: classCode.toUpperCase() })
27 |         .single();
28 |
29 |       if (classErr || !classData) {
30 |         throw new Error(lang === 'en' ? 'Invalid class code' : 'Mã lớp học không hợp lệ');
31 |       }
32 |
33 |       // 2. Kiểm tra xem đã tham gia chưa
34 |       const { data: existingMember } = await supabase
35 |         .from('class_members')
36 |         .select('id')
37 |         .eq('class_id', classData.id)
38 |         .eq('student_id', user.id)
39 |         .single();
40 |
41 |       if (existingMember) {
42 |         throw new Error(lang === 'en' ? 'You are already in this class' : 'Bạn đã tham gia lớp học này rồi');
43 |       }
44 |
45 |       // 3. Thêm vào lớp
46 |       const { error: joinErr } = await supabase
47 |         .from('class_members')
48 |         .insert({
49 |           class_id: classData.id,
50 |           student_id: user.id,
51 |           status: 'active'
52 |         });
53 |
54 |       if (joinErr) throw joinErr;
55 |
56 |       setSuccess(true);
57 |       setTimeout(() => {
58 |         onClose();
59 |         setSuccess(false);
60 |         setClassCode('');
61 |       }, 2000);
62 |
63 |     } catch (err) {
64 |       setError(err.message);
65 |     } finally {
66 |       setLoading(false);
67 |     }
68 |   };
69 |
70 |   return (
71 |     <div style={{
72 |       position: 'fixed', top: 0, left: 0, width: '100vw', height: '100vh',
73 |       background: 'rgba(12, 15, 20, 0.85)', backdropFilter: 'blur(8px)',
74 |       zIndex: 9999, display: 'flex', justifyContent: 'center', alignItems: 'center', padding: '2rem'
75 |     }}>
76 |       <div style={{

```



```

77 |         background: 'var(--bg-surface)', border: '1px solid var(--border-strong)', borderRadius: '24px',
78 |         padding: '40px', maxWidth: '400px', width: '100%', color: 'var(--text-primary)',
79 |         position: 'relative', boxShadow: '0 20px 50px rgba(0,0,0,0.5)'
80 |     }}>
81 |     <button
82 |       onClick={onClose}
83 |       style={{ position: 'absolute', top: '24px', right: '24px', background: 'transparent', bord...
84 |     >X size={24} /></button>
85 |
86 |     <div style={{ textAlign: 'center', marginBottom: '32px' }}>
87 |       <div style={{
88 |         width: '64px', height: '64px', borderRadius: '20px', background: 'rgba(0, 243, 255, 0....
89 |         display: 'flex', alignItems: 'center', justifyContent: 'center', margin: '0 auto 20px'
90 |       }}>
91 |         <Users size={32} color="var(--brand-primary)" />
92 |       </div>
93 |       <h2 style={{ fontSize: '24px', fontWeight: 800, margin: 0 }}>
94 |         { lang === 'en' ? 'Join Class' : 'Tham gia lớp học' }
95 |       </h2>
96 |       <p style={{ color: 'var(--text-secondary)', fontSize: '14px', marginTop: '8px' }}>
97 |         { lang === 'en' ? 'Enter the code provided by you' : 'Nhập mã do bạn cung cấp' }
98 |     </p>
99 |   </div>
100 |
101 |   {success ? (
102 |     <div style={{ textAlign: 'center', padding: '20px' }}>
103 |       <CheckCircle2 size={48} color="var(--success)" style={{ margin: '0 auto 16px' }} />
104 |       <p style={{ color: 'var(--success)', fontWeight: 600 }}>
105 |         { lang === 'en' ? 'Successfully joined!' : 'Tham gia thành công!' }
106 |       </p>
107 |     </div>
108 |   ) : (
109 |     <form onSubmit={handleJoin} style={{ display: 'flex', flexDirection: 'column', gap: '20px'...
110 |       <div>
111 |         <input
112 |           type="text"
113 |           value={classCode}
114 |           onChange={(e) => setClassCode(e.target.value.toUpperCase())}
115 |           placeholder="VD: TIN001"
116 |           required
117 |           style={{
118 |             width: '100%', padding: '16px', borderRadius: '12px',
119 |             background: 'var(--bg-base)', border: '1px solid var(--border-strong)',
120 |             color: 'white', fontSize: '18px', fontWeight: 700, textAlign: 'center',
121 |             textTransform: 'uppercase', letterSpacing: '2px'
122 |           }}
123 |         />
124 |       </div>
125 |
126 |       {error && (
127 |         <p style={{ color: '#ef4444', fontSize: '13px', margin: 0, textAlign: 'center', fo...
128 |           {error}
129 |         </p>
130 |       )}
131 |
132 |       <button
133 |         type="submit"
134 |         disabled={loading}
135 |         style={{
136 |           width: '100%', padding: '16px', borderRadius: '12px',
137 |           background: 'var(--brand-primary)', color: '#000',
138 |           fontSize: '16px', fontWeight: 700, border: 'none',
139 |           cursor: loading ? 'not-allowed' : 'pointer',
140 |           display: 'flex', alignItems: 'center', justifyContent: 'center', gap: '10px',
141 |           transition: 'all 0.2s'
142 |         }}
143 |       >
144 |         {loading ? <Loader2 className="animate-spin" size={20} /> : (lang === 'en' ? 'Join...
145 |       </button>
146 |     </form>
147 |   )}
148 | </div>
149 | </div>
150 | );
151 | };
152 |
153 | export default JoinClassModal;
154 |

```

File: components/LearningMode.js (421 lines, 22.7 KB)

```

1 | 'use client';
2 |
3 | import { useState, useEffect, useRef } from 'react';
4 |
5 | import confetti from 'canvas-confetti';
6 | import GameEngine from '../GameEngine';
7 | import LearningRoadmap from '../LearningRoadmap';
8 | import HandTracker from '../HandTracker';
9 |
10 |
11 | const HARDWARE_DATA = {
12 |   // ... existing data ...
13 |   'CPU': {
14 |     en: "The Central Processing Unit (CPU) is the brain of the computer. It performs basic arithmetic, log...
15 |     vn: "Central Processing Unit (CPU) là bộ não của máy tính. Nó thực hiện các phép tính cơ bản và xử lý dữ liệu.
16 |   },
17 |   'RAM': {
18 |     en: "Random Access Memory (RAM) is a form of computer memory that can be read and changed in any order...
19 |     vn: "Random Access Memory (RAM) là bộ nhớ truy cập ngẫu nhiên, cho phép đọc và ghi dữ liệu theo bất kỳ thứ tự nào.
20 |   },
21 |   'GPU': {
22 |     en: "A Graphics Processing Unit (GPU) is a specialized electronic circuit designed to rapidly manipula...
23 |     vn: "Graphics Processing Unit (GPU) là một mạch chuyên dụng để xử lý đồ họa, tăng tốc độ hiển thị hình ảnh.
24 |   },
25 |   'Mainboard': {
26 |     en: "The Motherboard (Mainboard) is the main printed circuit board (PCB) in general-purpose computers ...
27 |     vn: "Bo mạch chủ (Mainboard) là bảng mạch in chính trong máy tính, kết nối tất cả các linh kiện khác.
28 |   },
29 |   'Screws': {
30 |     en: "Screws secure the components inside the case.",
31 |     vn: "Đ ốc vít giúp cố định các linh kiện bên trong vỏ case."
32 |   },
33 |   'PSU': {
34 |     en: "A Power Supply Unit (PSU) converts mains AC to low-voltage regulated DC power for the internal co...
35 |     vn: "Nguồn máy tính (PSU) chuyển đổi điện áp xoay chiều AC thành điện áp một chiều DC ổn định để cấp cho các linh kiện.
36 |   }
37 | };
38 |
39 | const getTargetForLevel = (levelId) => {
40 |   switch (levelId) {
41 |     case 1: return 'CPU';
42 |     case 2: return 'RAM';

```

```

43 |         case 3: return 'GPU';
44 |         case 4: return 'PSU';
45 |         case 5: return 'Screws'; // Final check
46 |         default: return 'CPU';
47 |     }
48 | };
49 |
50 | const getQuizTopicForLevel = (levelId) => {
51 |     switch (levelId) {
52 |         case 1: return 'CPU';
53 |         case 2: return 'RAM';
54 |         case 3: return 'GPU';
55 |         case 4: return 'PSU';
56 |         case 5: return 'Computer Case';
57 |         default: return 'CPU';
58 |     }
59 | };
60 |
61 | const LearningMode = ({
62 |     lang,
63 |     externalSelection,
64 |     appMode,
65 |     landmarks,
66 |     cameraEnabled,
67 |     onLandmarks,
68 |     onHover,
69 |     onGameEvent,
70 |     onTakeQuiz,
71 |     onBackToMenu,
72 | }) => {
73 |     // subMode: 'select' | 'roadmap' | 'assembly' | 'free'
74 |     const [subMode, setSubMode] = useState('select');
75 |     // view is 'roadmap' or 'assembly' (used in roadmap/structured learning)
76 |     const [view, setView] = useState('roadmap');
77 |     const [currentLevelId, setCurrentLevelId] = useState(null);
78 |     const [unlockedLevels, setUnlockedLevels] = useState([1]);
79 |     const [completedLevels, setCompletedLevels] = useState([]);
80 |     const [viewFull, setViewFull] = useState(false);
81 |
82 |     const processedPlacementRef = useRef(null);
83 |
84 |     // local block tracking
85 |     const [progress, setProgress] = useState([]);
86 |
87 |     useEffect(() => {
88 |         try {
89 |             const savedUnlocked = localStorage.getItem('unlockedLevels');
90 |             const savedCompleted = localStorage.getItem('completedLevels');
91 |             if (savedUnlocked) setUnlockedLevels(JSON.parse(savedUnlocked));
92 |             if (savedCompleted) setCompletedLevels(JSON.parse(savedCompleted));
93 |         } catch (e) {
94 |             console.error("Error loading progress from localStorage", e);
95 |             // Default values already set in useState
96 |         }
97 |     }, []);
98 |
99 |     const saveProgress = (newUnlocked, newCompleted) => {
100 |         setUnlockedLevels(newUnlocked);
101 |         setCompletedLevels(newCompleted);
102 |         localStorage.setItem('unlockedLevels', JSON.stringify(newUnlocked));
103 |         localStorage.setItem('completedLevels', JSON.stringify(newCompleted));
104 |     };
105 |
106 |     const handleNodeClick = (levelId) => {
107 |         setCurrentLevelId(levelId);
108 |         setView('assembly');
109 |         setProgress([]);
110 |     };
111 |
112 |     function handleQuizSuccess(levelId) {
113 |         const newCompleted = [...new Set([...completedLevels, levelId])];
114 |         if (levelId < 5) {
115 |             const nextLevel = levelId + 1;
116 |             const newUnlocked = [...new Set([...unlockedLevels, levelId, nextLevel])];
117 |             saveProgress(newUnlocked, newCompleted);
118 |         } else {
119 |             saveProgress(unlockedLevels, newCompleted); // level 5 complete
120 |         }
121 |
122 |         setView('roadmap');
123 |
124 |         // Firework effect
125 |         confetti({
126 |             particleCount: 150,
127 |             spread: 80,
128 |             origin: { y: 0.6 }
129 |         });
130 |     }
131 |
132 |     function triggerQuizForLevel(levelId, hardwareTarget) {
133 |         // We prompt the user with the quiz modal.
134 |         // If they succeed (score >= 8), handleQuizSuccess runs.
135 |         const quizTopic = getQuizTopicForLevel(levelId);
136 |         onTakeQuiz(quizTopic, levelId, (score) => {
137 |             if (score >= 8) {
138 |                 handleQuizSuccess(levelId);
139 |             }
140 |         });
141 |     }
142 |
143 |     useEffect(() => {
144 |         if (!externalSelection || !externalSelection.type || view !== 'assembly' || !currentLevelId) return;
145 |
146 |         // Prevent infinite loop by checking if we already processed this exact placement event via its ID
147 |         if (processedPlacementRef.current === externalSelection.id) return;
148 |         processedPlacementRef.current = externalSelection.id;
149 |
150 |         const target = getTargetForLevel(currentLevelId);
151 |         const type = externalSelection.type;
152 |
153 |         if (type === target || (target === 'Screws' && type === 'Mainboard')) {
154 |             if (target === 'RAM') {
155 |                 setProgress(prev => {
156 |                     const newProgress = [...prev, type];
157 |                     const ramPlaced = newProgress.filter(t => t === 'RAM').length;
158 |                     if (ramPlaced >= 2) {
159 |                         // Use timeout to let state settle before opening quiz modal
160 |                         setTimeout(() => triggerQuizForLevel(currentLevelId, target), 50);
161 |                     }
162 |                     return newProgress;
163 |                 });
164 |             } else {
165 |                 triggerQuizForLevel(currentLevelId, target);
166 |             }

```

```

167 |     }
168 | }, [externalSelection, view, currentLevelId]);
169 |
170 | // --- Sub-mode selector screen ---
171 | if (subMode === 'select') {
172 |     return (
173 |         <div style={{ width: '100%', maxWidth: '900px', margin: '0 auto', display: 'flex', flexDirection: ...
174 |             <h2 className="neon-text-blue" style={{ margin: 0, fontSize: '2.5rem', letterSpacing: '2px', t...
175 |                 { l a n g = = = ' e n ' ? ' P r a c t i c e   M o d e ' : ' C h ử   ù   L u y c n   T - p
176 |             </h2>
177 |             <p style={{ color: 'var(--text-muted)', textAlign: 'center', fontSize: '1.1rem' }}>
178 |                 { l a n g = = = ' e n ' ? ' C h o o s e   h o w   y o u   w a n t   t o   p r a c t i c e : '
179 |             </p>
180 |             <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '2rem', width: '100%' }}>
181 |                 /* Roadmap */
182 |                 <button
183 |                     onClick={() => setSubMode('roadmap')}
184 |                     className="glass-panel"
185 |                     style={{
186 |                         display: 'flex', flexDirection: 'column', alignItems: 'center', padding: '3rem 2re...
187 |                         cursor: 'pointer', border: '1px solid var(--accent-blue)', background: 'var(--bg-s...
188 |                         borderRadius: '16px', color: 'var(--text-primary)'
189 |                     }}
190 |                     onMouseOver={e => { e.currentTarget.style.borderColor = 'rgba(0,243,255,0.6)'; }}
191 |                     onMouseOut={e => { e.currentTarget.style.borderColor = 'var(--accent-blue)'; }}
192 |                 >
193 |                     <span style={{ fontSize: '3.5rem' }}>Yúp </span>
194 |                     <div style={{ textAlign: 'center' }}>
195 |                         <h3 style={{ color: 'var(--accent-blue)', marginBottom: '0.5rem', fontSize: '1.5re...
196 |                             { l a n g = = = ' e n ' ? ' L e a r n i n g   R o a d m a p ' : ' L
197 |                         </h3>
198 |                         <p style={{ color: 'var(--text-muted)', fontSize: '0.95rem' }}>
199 |                             { l a n g = = = ' e n ' ? ' S t e p - b y - s t e p   g u i d e d   m i
200 |                         </p>
201 |                     </div>
202 |                     <div style={{ height: '4px', width: '100%', background: 'var(--accent-blue)', boxShado...
203 |                 </button>
204 |
205 |                 /* Free Practice */
206 |                 <button
207 |                     onClick={() => setSubMode('free')}
208 |                     className="glass-panel"
209 |                     style={{
210 |                         display: 'flex', flexDirection: 'column', alignItems: 'center', padding: '3rem 2re...
211 |                         cursor: 'pointer', border: '1px solid var(--brand-primary)', background: 'var(--bg...
212 |                         borderRadius: '16px', color: 'var(--text-primary)'
213 |                     }}
214 |                     onMouseOver={e => { e.currentTarget.style.borderColor = 'rgba(52,211,153,0.6)'; }}
215 |                     onMouseOut={e => { e.currentTarget.style.borderColor = 'var(--brand-primary)'; }}
216 |                 >
217 |                     <span style={{ fontSize: '3.5rem' }}>Bàp </span>
218 |                     <div style={{ textAlign: 'center' }}>
219 |                         <h3 style={{ color: 'var(--brand-primary)', marginBottom: '0.5rem', fontSize: '1.5...
220 |                             { l a n g = = = ' e n ' ? ' F r e e   P r a c t i c e ' : ' L u y
221 |                         </h3>
222 |                         <p style={{ color: 'var(--text-muted)', fontSize: '0.95rem' }}>
223 |                             { l a n g = = = ' e n ' ? ' F r e e l y   a s s e m b l e   c o m p o r
224 |                         </p>
225 |                     </div>
226 |                     <div style={{ height: '4px', width: '100%', background: 'var(--brand-primary)', boxSha...
227 |                 </button>
228 |             </div>
229 |         );
230 |     }
231 | }
232 |
233 | // --- Free Practice sub-mode ---
234 | if (subMode === 'free') {
235 |     return (
236 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '1rem', width: '100%' }}>
237 |             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
238 |                 <h2 className="neon-text-green" style={{ margin: 0, fontSize: '18px' }}>
239 |                     { l a n g = = = ' e n ' ? ' B à p   F r e e   P r a c t i c e ' : ' B à p   L u y
240 |                 </h2>
241 |                 <button onClick={() => setSubMode('select')} style={{
242 |                     padding: '6px 14px', background: 'transparent', color: 'var(--text-primary)',
243 |                     border: '1px solid var(--border-default)', borderRadius: '8px', cursor: 'pointer', fon...
244 |                 }}>
245 |                     { l a n g = = = ' e n ' ? ' ! •   B a c k ' : ' ! •   Q u a y   l i i ' }
246 |                 </button>
247 |             </div>
248 |             <div style={{ display: 'flex', gap: '16px', alignItems: 'flex-start' }}>
249 |                 <div style={{ flex: 1, maxWidth: '800px' }}>
250 |                     <GameEngine
251 |                         onHover={onHover}
252 |                         onGameEvent={onGameEvent}
253 |                         landmarks={landmarks}
254 |                     />
255 |                 </div>
256 |                 <div style={{ width: '220px', flexShrink: 0 }}>
257 |                     <div style={{
258 |                         borderRadius: '12px', overflow: 'hidden',
259 |                         border: '1px solid rgba(0,212,170,0.25)',
260 |                         boxShadow: '0 0 16px rgba(0,212,170,0.15)',
261 |                         background: '#0f172a',
262 |                     }}>
263 |                         <div style={{ padding: '6px 10px', background: 'rgba(0,212,170,0.08)', borderBotto...
264 |                             <span style={{ fontSize: '10px', color: cameraEnabled ? '#10b981' : '#ef4444' }}>
265 |                                 <span>>0÷ Camera</span>
266 |                                 { cameraEnabled ? ' B - t ' : ' T - t ' }
267 |                             </span>
268 |                         </div>
269 |                         <div style={{ width: '100%', aspectRatio: '4/3', minHeight: '140px', position: 're...
270 |                             {cameraEnabled && <HandTracker onLandmarks={onLandmarks} numHands={1} />}
271 |                             {!cameraEnabled && {
272 |                                 <div style={{ width: '100%', height: '100%', display: 'flex', alignItems: ...
273 |                                     Camera T @ T
274 |                                 </div>
275 |                             )}
276 |                         </div>
277 |                     </div>
278 |                 </div>
279 |             </div>
280 |         );
281 |     }
282 | }
283 |
284 | // --- Full PC View (pre-assembled, full screen) ---
285 | if (viewFull) {
286 |     return (
287 |         <div style={{ position: 'fixed', inset: 0, zIndex: 9999, background: '#030712' }}>
288 |             <div style={{ position: 'absolute', top: 10, left: 10, zIndex: 10, display: 'flex', gap: 8 }}>
289 |                 <button onClick={() => setViewFull(false)} style={{
290 |                     padding: '6px 14px', background: 'rgba(255,255,255,0.08)', color: 'ffff',

```

```

291|         border: '1px solid rgba(255,255,255,0.15)', borderRadius: 8, cursor: 'pointer',
292|         fontSize: 12, fontWeight: 600, fontFamily: 'inherit', backdropFilter: 'blur(8px)',
293|     })>
294|         !• { lang === 'en' ? 'Back' : 'Quay l i i' }
295|     </button>
296|     <div style={{
297|         padding: '6px 14px', background: 'rgba(0,212,170,0.1)',
298|         border: '1px solid rgba(0,212,170,0.2)', borderRadius: 8,
299|         color: '#00d4aa', fontSize: 11, fontWeight: 600, fontFamily: 'inherit',
300|         backdropFilter: 'blur(8px)',
301|     }}>
302|         Ø=Ý¥p { lang === 'en' ? 'Full PC View' : 'Xem Full
303|     </div>
304| </div>
305| <div style={{ width: '100vw', height: '100vh' }}>
306|     <GameEngine defaultPlaced={true} landmarks={landmarks} />
307| </div>
308| </div>
309| );
310| }
311| }
312| // --- Roadmap sub-mode: show roadmap or level assembly ---
313| if (subMode === 'roadmap' && view === 'roadmap') {
314|     return (
315|         <div style={{ width: '100%', display: 'flex', flexDirection: 'column', gap: '1rem' }}>
316|         <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
317|             <h2 className="neon-text-green" style={{ margin: 0 }}>
318|                 { lang === 'en' ? 'Interactive Learning Roadmap'
319|             </h2>
320|             <div style={{ display: 'flex', gap: 8 }}>
321|                 <button onClick={() => setViewFull(true)} style={{
322|                     padding: '8px 18px', background: 'rgba(0,212,170,0.1)', color: '#00d4aa',
323|                     border: '1px solid rgba(0,212,170,0.3)', borderRadius: 8, cursor: 'pointer',
324|                     fontWeight: 600, fontSize: 12, fontFamily: 'inherit',
325|                 }}>
326|                     Ø=Ý¥p { lang === 'en' ? 'Full PC' : 'Xem Full
327|                 </button>
328|                 <button onClick={() => setSubMode('select')} style={{
329|                     padding: '8px 18px', background: 'transparent', color: 'var(--text-primary)',
330|                     border: '1px solid var(--border-default)', borderRadius: 8, cursor: 'pointer', fon...
331|                 }}>
332|                     { lang === 'en' ? '!• Back' : '!• Quay l i i' }
333|                 </button>
334|             </div>
335|             <div>
336|                 <LearningRoadmap
337|                     lang={lang}
338|                     unlockedLevels={unlockedLevels}
339|                     completedLevels={completedLevels}
340|                     onNodeClick={handleNodeClick}
341|                     landmarks={landmarks}
342|                 />
343|             </div>
344|         </div>
345|     );
346| }
347| // Assembly View
348| const targetHardware = getTargetForLevel(currentLevelId);
349|
350| return (
351|     <div className="practice-layout" style={{ display: 'flex', flexDirection: 'column', gap: '1rem' }}>
352|     <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
353|         <h2 className="neon-text-green" style={{ margin: 0, fontSize: '18px' }}>
354|             { lang === 'en' ? `Level ${currentLevelId}: Action`
355|         </h2>
356|         <button onClick={() => setView('roadmap')} style={{ padding: '6px 14px', background: 'transpar...
357|             { lang === 'en' ? '+ Back' : '+ V Á' }
358|         </button>
359|     </div>
360|
361|     <div style={{ display: 'flex', gap: '16px', alignItems: 'flex-start' }}>
362|         /* Left: Mainboard game area - compact */
363|         <div style={{ flex: 1, maxWidth: '800px', position: 'relative' }}>
364|             <GameEngine
365|                 onHover={onHover}
366|                 onGameEvent={onGameEvent}
367|                 landmarks={landmarks}
368|             />
369|         </div>
370|
371|         /* Right: Camera on top + Info below */
372|         <div style={{ width: '220px', flexShrink: 0, display: 'flex', flexDirection: 'column', gap: '1...
373|             /* Camera */
374|             <div style={{
375|                 borderRadius: '12px', overflow: 'hidden',
376|                 border: '1px solid rgba(0,212,170,0.25)',
377|                 boxShadow: '0 0 16px rgba(0,212,170,0.15)',
378|                 background: '#0f172a',
379|             }}>
380|                 <div style={{ padding: '6px 10px', background: 'rgba(0,212,170,0.08)', borderBottom: '...
381|                     <span style={{ fontSize: '10px', color: cameraEnabled ? '#10b981' : '#ef4444' }}>
382|                         { cameraEnabled ? 'B-t' : 'T-t' }
383|                     </span>
384|                 </div>
385|                 <div style={{ width: '100%', aspectRatio: '4/3', minHeight: '140px', position: 'relati...
386|                     {cameraEnabled && {
387|                         <HandTracker onLandmarks={onLandmarks} numHands={1} />
388|                     }}
389|                     {!cameraEnabled && {
390|                         <div style={{ width: '100%', height: '100%', display: 'flex', alignItems: 'cen...
391|                             Camera T @ T
392|                         </div>
393|                     }}
394|                 </div>
395|             </div>
396|
397|             /* Hardware Info Panel */
398|             <div className="glass-panel" style={{ padding: '14px', borderRadius: '12px', display: 'fle...
399|                 <h3 style={{ color: '#60a5fa', margin: 0, fontSize: '14px', fontWeight: 800 }}>{target...
400|                 <p style={{ color: 'var(--text-secondary)', lineHeight: 1.5, fontSize: '13px', margin: ...
401|                     {HARDWARE_DATA[targetHardware === 'Screws' ? 'Screws' : targetHardware]?.[lang]}
402|                 </p>
403|                 <p style={{ color: '#00d4aa', fontWeight: 700, fontSize: '12px', margin: 0 }}>
404|                     {lang === 'en'
405|                         ? (targetHardware === 'Screws' ? "Assemble to trigger the quiz!" : "Place the ...
406|                           : (targetHardware === 'Screws' ? "L - p
407|                     </p>
408|                 <button onClick={() => triggerQuizForLevel(currentLevelId, targetHardware)}
409|                     style={{ padding: '10px', background: 'var(--accent-blue)', color: '#fff', border:...
410|                         { lang === 'en' ? 'Start Quiz 'e' : 'L à m B à i
411|                 </button>
412|             </div>
413|         </div>
414|     </div>

```

```
415 |         </div>
416 |     </div>
417 | );
418 | };
419 |
420 | export default LearningMode;
421 |
```

File: components/LearningRoadmap.js (252 lines, 13.2 KB)

```
1 | import React, { useState, useEffect, useRef } from 'react';
2 | import { motion } from 'framer-motion';
3 |
4 | const LEVELS = [
5 |   { id: 1, titleEn: 'Level 1: The Heart', titleVn: 'Level 1: Trái tim', target: 'CPU', descEn: 'Install the ...', titleVn: 'Level 1: Trái tim', target: 'CPU', descEn: 'Install the ...' },
6 |   { id: 2, titleEn: 'Level 2: The Memory', titleVn: 'Level 2: Bộ nhớ', target: 'GPU', descEn: 'Upgrade the ...', titleVn: 'Level 2: Bộ nhớ', target: 'GPU', descEn: 'Upgrade the ...' },
7 |   { id: 3, titleEn: 'Level 3: The Graphics', titleVn: 'Level 3: Đồ họa', target: 'SSD', descEn: 'Upgrade the ...', titleVn: 'Level 3: Đồ họa', target: 'SSD', descEn: 'Upgrade the ...' },
8 |   { id: 4, titleEn: 'Level 4: The Power', titleVn: 'Level 4: Nguồn', target: 'PSU', descEn: 'Upgrade the ...', titleVn: 'Level 4: Nguồn', target: 'PSU', descEn: 'Upgrade the ...' },
9 |   { id: 5, titleEn: 'Level 5: Final Check', titleVn: 'Level 5: Hoàn tất', target: 'System', descEn: 'Final check and optimization', titleVn: 'Level 5: Hoàn tất', target: 'System', descEn: 'Final check and optimization' }
10 | ];
11 |
12 | // Simple 1D exponential moving average as a basic low-pass filter (simplified One Euro)
13 | class LowPassFilter {
14 |   constructor(alpha = 0.5) {
15 |     this.alpha = alpha;
16 |     this.lastVal = null;
17 |   }
18 |   filter(val) {
19 |     if (this.lastVal === null) {
20 |       this.lastVal = val;
21 |       return val;
22 |     }
23 |     this.lastVal = this.alpha * val + (1 - this.alpha) * this.lastVal;
24 |     return this.lastVal;
25 |   }
26 | }
27 |
28 | const LearningRoadmap = ({ lang, unlockedLevels = [1], completedLevels = [], onNodeClick, landmarks, trackingSensitivity }) => {
29 |   const containerRef = useRef(null);
30 |   const filterX = useRef(new LowPassFilter(0.15));
31 |   const filterY = useRef(new LowPassFilter(0.15));
32 |
33 |   const [cursorPos, setCursorPos] = useState({ x: -100, y: -100, isPinching: false });
34 |   const [hoveredNode, setHoveredNode] = useState(null);
35 |   const pinchActiveRef = useRef(false);
36 |
37 |   useEffect(() => {
38 |     if (!containerRef.current || !landmarks || landmarks.length === 0) return;
39 |
40 |     const hand = landmarks[0];
41 |     const indexTip = hand[8];
42 |     const thumbTip = hand[4];
43 |
44 |     const rect = containerRef.current.getBoundingClientRect();
45 |     const scaleX = (indexTip.x - 0.5) * trackingSensitivity + 0.5;
46 |     const scaleY = (indexTip.y - 0.5) * trackingSensitivity + 0.5;
47 |     const rawX = scaleX * rect.width;
48 |     const rawY = scaleY * rect.height;
49 |
50 |     const smoothX = filterX.current.filter(rawX);
51 |     const smoothY = filterY.current.filter(rawY);
52 |
53 |     const distance = Math.hypot(indexTip.x - thumbTip.x, indexTip.y - thumbTip.y);
54 |     // Hysteresis: easier to pinch, harder to release
55 |     const isPinching = pinchActiveRef.current
56 |       ? distance < 0.10
57 |       : distance < 0.07;
58 |     pinchActiveRef.current = isPinching;
59 |
60 |     setCursorPos({ x: smoothX, y: smoothY, isPinching });
61 |
62 |     // Node hit detection with padding
63 |     const PADDING = 30; // Vùng đệm cho phép chính xác hơn
64 |     let found = null;
65 |
66 |     // Nodes positions map based on index for simplicity structure below
67 |     const nodeCoords = [
68 |       { id: 1, x: 100, y: 100 },
69 |       { id: 2, x: 300, y: 100 },
70 |       { id: 3, x: 300, y: 350 },
71 |       { id: 4, x: 500, y: 350 },
72 |       { id: 5, x: 500, y: 600 },
73 |     ];
74 |
75 |     for (let node of nodeCoords) {
76 |       // approximate bounding box based on UI layout
77 |       const dx = Math.abs(smoothX - node.x - 50); // node center X offset
78 |       const dy = Math.abs(smoothY - node.y - 50); // node center Y offset
79 |       if (dx < 50 + PADDING && dy < 50 + PADDING) {
80 |         found = node.id;
81 |         break;
82 |       }
83 |     }
84 |
85 |     setHoveredNode(found);
86 |
87 |     if (isPinching && found && unlockedLevels.includes(found)) {
88 |       onNodeClick(found);
89 |     }
90 |   }, [landmarks, unlockedLevels, onNodeClick, trackingSensitivity]);
91 |
92 |   const getNodeState = (levelId) => {
93 |     if (completedLevels.includes(levelId)) return 'completed';
94 |     if (unlockedLevels.includes(levelId)) return 'active';
95 |     return 'locked';
96 |   };
97 |
98 |   const getNodeStyle = (state) => {
99 |     if (state === 'locked') return { bg: '#444', border: '#666', i: 0.2 };
100 |     if (state === 'active') return { bg: 'rgb(0, 123, 255, 0.2)', i: 0.2 };
101 |     if (state === 'completed') return { bg: 'rgb(0, 255, 100, 0.2)', i: 0.2 };
102 |   };
103 |
104 |   const [layoutScale, setLayoutScale] = useState(1);
105 |   useEffect(() => {
106 |     const updateScale = () => {
107 |       const hScale = window.innerWidth < 800 ? Math.min(1, window.innerWidth / 800) : 1;
108 |       setLayoutScale(hScale);
109 |     };
110 |     updateScale();
111 |     window.addEventListener('resize', updateScale);
112 |     return () => window.removeEventListener('resize', updateScale);
113 |   }, []);
```

```

114 |
115 | return (
116 |   <div ref={containerRef} style={{
117 |     width: '100%',
118 |     height: '100%', // Take full height available
119 |     maxHeight: '80vh', // Prevent overpowering the screen
120 |     background: '#0f172a',
121 |     backgroundImage: 'linear-gradient(#1e293b 1px, transparent 1px), linear-gradient(90deg, #1e293b 1p...',
122 |     backgroundSize: '40px 40px',
123 |     position: 'relative',
124 |     borderRadius: '16px',
125 |     border: '2px solid #334155',
126 |     overflowY: 'auto', // Allow scrolling down the roadmap
127 |     overflowX: 'hidden',
128 |     display: 'block', // Changed from flex strictly centering
129 |     padding: '20px'
130 |   }}>
131 |     /* Extended height map */
132 |     <div style={{ width: '800px', height: '800px', position: 'relative', transform: `scale(${layoutSca...
133 |       /* SVGs for Circuit Lines (Elongated for scroll) */
134 |       <svg style={{ position: 'absolute', top: 0, left: 0, width: '100%', height: '100%', pointerEve...
135 |         <path d="M 150 150 L 350 150" stroke={unlockedLevels.includes(2) ? '#00ccff' : '#444'} str...
136 |         style={{ filter: unlockedLevels.includes(2) ? 'drop-shadow(0 0 5px #00ccff)' : 'none',...
137 |         <path d="M 350 150 L 350 400" stroke={unlockedLevels.includes(3) ? '#00ccff' : '#444'} str...
138 |         style={{ filter: unlockedLevels.includes(3) ? 'drop-shadow(0 0 5px #00ccff)' : 'none',...
139 |         <path d="M 350 400 L 550 400" stroke={unlockedLevels.includes(4) ? '#00ccff' : '#444'} str...
140 |         style={{ filter: unlockedLevels.includes(4) ? 'drop-shadow(0 0 5px #00ccff)' : 'none',...
141 |         <path d="M 550 400 L 550 650" stroke={unlockedLevels.includes(5) ? '#00ccff' : '#444'} str...
142 |         style={{ filter: unlockedLevels.includes(5) ? 'drop-shadow(0 0 5px #00ccff)' : 'none',...
143 |       </svg>
144 |
145 |       /* Nodes */
146 |       {LEVELS.map((lvl, index) => {
147 |         const state = getNodeState(lvl.id);
148 |         const style = getNodeStyle(state);
149 |         const coords = [
150 |           { top: '100px', left: '100px' },
151 |           { top: '100px', left: '300px' },
152 |           { top: '350px', left: '300px' }, // Down
153 |           { top: '350px', left: '500px' },
154 |           { top: '600px', left: '500px' }, // Down further
155 |         ];
156 |
157 |         const isHovered = hoveredNode === lvl.id;
158 |
159 |         return (
160 |           <div key={lvl.id} style={{ position: 'absolute', ...coords }}>
161 |             <motion.div
162 |               onClick={() => { if (state !== 'locked') onNodeClick(lvl.id); }}
163 |               animate={state === 'active' ? { scale: [1, 1.1, 1], boxShadow: [style.glow, '0...
164 |               transition={{ repeat: Infinity, duration: 2 }}
165 |               style={{
166 |                 width: '100px', height: '100px',
167 |                 borderRadius: '50%',
168 |                 background: style.bg,
169 |                 border: `4px solid ${style.border}`,
170 |                 display: 'flex', alignItems: 'center', justifyContent: 'center',
171 |                 flexDirection: 'column',
172 |                 cursor: state === 'locked' ? 'not-allowed' : 'pointer',
173 |                 boxShadow: style.glow,
174 |                 position: 'relative',
175 |                 zIndex: 10
176 |               }}
177 |               whileHover={{ scale: state !== 'locked' ? 1.1 : 1 }}
178 |             >
179 |               <span style={{ fontSize: '2rem' }}>{style.icon}</span>
180 |               {state === 'completed' && (
181 |                 <div style={{ display: 'flex', gap: '2px', marginTop: '4px' }}>
182 |                   <span style={{ fontSize: '0.8rem' }}>1 8 2 |
183 |                   <span style={{ fontSize: '0.8rem' }}>1 8 3 |
184 |                   <span style={{ fontSize: '0.8rem' }}>1 8 4 |
185 |                 </div>
186 |               )
187 |             </motion.div>
188 |
189 |             /* Title Under Node */
190 |             <div style={{
191 |               position: 'absolute', top: '105px', left: '50%', transform: 'translateX(-50%)',
192 |               whiteSpace: 'nowrap', textAlign: 'center', fontSize: '0.85rem', fontWeight: 'b...
193 |               color: state === 'locked' ? '#64748b' : '#38bdf8',
194 |               textShadow: state === 'locked' ? 'none' : '0 2px 4px rgba(0,0,0,0.8)',
195 |               pointerEvents: 'none'
196 |             }}>
197 |               {lang === 'en' ? lvl.titleEn : lvl.titleVn}
198 |             </div>
199 |
200 |             /* Tooltip on hover */
201 |             {(isHovered || true) && (
202 |               <motion.div
203 |                 initial={{ opacity: 0, y: 10 }}
204 |                 animate={{ opacity: isHovered ? 1 : 0, y: isHovered ? 0 : 10 }}
205 |                 style={{
206 |                   position: 'absolute',
207 |                   top: '-90px', left: '50%', transform: 'translateX(-50%)',
208 |                   background: 'rgba(15, 23, 42, 0.95)',
209 |                   border: '1px solid #334155',
210 |                   padding: '10px', borderRadius: '8px',
211 |                   width: '200px', pointerEvents: 'none',
212 |                   color: 'white', zIndex: 20,
213 |                   boxShadow: '0 4px 6px rgba(0,0,0,0.5)',
214 |                   textAlign: 'center'
215 |                 }}
216 |               >
217 |                 <div style={{ fontWeight: 'bold', color: '#00ccff', margin: '4px 0' }}>
218 |                   {lang === 'en' ? lvl.titleEn : lvl.titleVn}
219 |                 </div>
220 |                 <div style={{ fontSize: '0.85rem', color: '#cbd5e1' }}>
221 |                   {lang === 'en' ? lvl.descEn : lvl.descVn}
222 |                 </div>
223 |               </motion.div>
224 |             )
225 |           </div>
226 |         );
227 |       })
228 |
229 |       /* Webcam Cursor overlay */
230 |       {landmarks && landmarks.length > 0 && (
231 |         <div style={{
232 |           position: 'absolute',
233 |           left: cursorPos.x - 10,
234 |           top: cursorPos.y - 10,
235 |           width: 20,
236 |           height: 20,
237 |           borderRadius: '50%',

```

```

238|         background: cursorPos.isPinching ? 'yellow' : 'cyan',
239|         border: '2px solid black',
240|         pointerEvents: 'none',
241|         zIndex: 200,
242|         boxShadow: cursorPos.isPinching ? '0 0 15px yellow' : '0 0 10px cyan',
243|         transition: 'background 0.2s'
244|       }} />
245|     </div>
246|   </div>
247| );
248| };
249|
250|
251| export default LearningRoadmap;
252|

```

[illegible]

```

106 |         animate={{ x: 0 }}
107 |         transition={{ type: 'spring', stiffness: 120, damping: 20 }}
108 |         className="w-full md:w-[240px] flex-shrink-0 bg-white border-b md:border-b-0 md:border-r borde...
109 |
110 |     <motion.button
111 |       whileHover={{ x: -4 }}
112 |       onClick={onBack}
113 |       className="flex items-center gap-2 text-slate-500 hover:text-slate-700 mb-6 text-sm font-m...
114 |     >
115 |       < ArrowLeft size={16} /> Quay lại
116 |     </motion.button>
117 |     < h4 className="text-xs font-bold uppercase tracking-wide"
118 |       {loading ? (
119 |         <div className="flex justify-center py-8">
120 |           <Loader2 className="animate-spin text-blue-600" size={20} />
121 |         </div>
122 |       ) : (
123 |         <nav className="flex flex-col gap-1">
124 |           {sections.map((s, i) => (
125 |             <a
126 |               key={s.id}
127 |               href={`#sec-${s.id}`}
128 |               className="flex items-center gap-3 px-3 py-2.5 rounded-xl text-slate-700 hover...
129 |             >
130 |               <span className="w-6 h-6 rounded-lg bg-blue-100 text-blue-700 flex items-cen...
131 |                 {i + 1}
132 |               </span>
133 |               <span className="truncate">{s.title}</span>
134 |             </a>
135 |           )}}
136 |         </nav>
137 |       )}
138 |     </div>
139 |     <div className="mt-auto pt-6">
140 |       <motion.button
141 |         whileHover={{ scale: 1.02 }}
142 |         whileTap={{ scale: 0.97 }}
143 |         onClick={handleComplete}
144 |         disabled={completing}
145 |         className={w-full py-3 px-4 rounded-xl font-bold text-sm flex items-center justify-ce...
146 |           isCompleted
147 |             ? 'bg-emerald-50 text-emerald-600 border-emerald-200'
148 |             : 'bg-blue-600 text-white border-blue-600 hover:bg-blue-700 shadow-lg shadow-b...
149 |       >
150 |         {completing ? (
151 |           <Loader2 className="animate-spin" size={16} />
152 |         ) : isCompleted ? (
153 |           < > < CheckCircle size={16} /> đã hoàn thành < />
154 |         ) : (
155 |           < > < Circle size={16} /> ảnh d... hoàn thành < />
156 |         )}
157 |       </motion.button>
158 |     </div>
159 |   </motion.aside>
160 |
161 |   /* Main content */
162 |   <div className="flex-1 overflow-y-auto">
163 |     <div className="max-w-4xl mx-auto px-4 sm:px-6 md:px-8 py-4 sm:py-6 md:py-8">
164 |       {lesson.thumbnail_url && (
165 |         <motion.div
166 |           initial={{ opacity: 0, y: 20 }}
167 |           animate={{ opacity: 1, y: 0 }}
168 |           className="w-full h-[260px] rounded-2xl overflow-hidden mb-8 shadow-lg"
169 |         >
170 |           <img src={lesson.thumbnail_url} className="w-full h-full object-cover" />
171 |         </motion.div>
172 |       )}
173 |       <motion.div
174 |         initial={{ opacity: 0, y: 16 }}
175 |         animate={{ opacity: 1, y: 0 }}
176 |         transition={{ delay: 0.1 }}
177 |         className="flex justify-between items-start mb-6"
178 |       >
179 |         <div>
180 |           <h1 className="text-3xl font-extrabold text-slate-900 mb-2">{lesson.title}</h1>
181 |           {lesson.description && (
182 |             <p className="text-slate-500 text-base leading-relaxed">{lesson.description}</p>
183 |           )}
184 |         </div>
185 |         {isCompleted && (
186 |           <div className="flex items-center gap-2 bg-emerald-50 text-emerald-600 px-4 py-2 r...
187 |             < CheckCircle size={16} /> đã hoàn thành
188 |           </div>
189 |         )}
190 |       </motion.div>
191 |       <div className="h-1 w-16 bg-gradient-to-r from-blue-500 to-emerald-400 rounded-full mb-10" />
192 |
193 |       {loading ? (
194 |         <div className="flex justify-center py-20">
195 |           <Loader2 className="animate-spin text-blue-600" size={40} />
196 |         </div>
197 |       ) : (
198 |         <motion.div
199 |           variants={containerVariants}
200 |           initial="hidden"
201 |           animate="visible"
202 |           className="flex flex-col gap-12"
203 |         >
204 |           {sections.map(s => (
205 |             <motion.section key={s.id} id={`sec-${s.id}`} variants={itemVariants}>
206 |               <div className="flex items-center gap-3 mb-5">
207 |                 <div className="p-2.5 rounded-xl bg-blue-100 text-blue-700">
208 |                   {s.type === 'video' && <Video size={18} />}
209 |                   {s.type === 'text' && <FileText size={18} />}
210 |                   {s.type === 'image' && <ImageIcon size={18} />}
211 |                   {s.type === 'pdf' && <FileSearch size={18} />}
212 |                   {s.type === 'embed' && <Code size={18} />}
213 |                 </div>
214 |                 <h2 className="text-xl font-bold text-slate-900">{s.title}</h2>
215 |               </div>
216 |               {s.type === 'video' && getYouTubeEmbed(s.content) && (
217 |                 <div className="w-full aspect-video rounded-2xl overflow-hidden bg-sla...
218 |                   <iframe src={getYouTubeEmbed(s.content)} className="w-full h-full ...
219 |                 </div>
220 |               )}
221 |               {s.type === 'text' && (
222 |                 <div className="bg-white border border-slate-200 rounded-2xl p-8 shado...
223 |                   {parseRichContent(s.content).map((part, i) => {
224 |                     if (part.type === 'video') {
225 |                       const embedUrl = getYouTubeEmbed(part.url);
226 |                       return embedUrl ? (
227 |                         <div key={i} className="w-full aspect-video rounded-xl...
228 |                           <iframe src={embedUrl} className="w-full h-full bo...
229 |

```



```

230 |         </div>
231 |     ) : null;
232 |
233 |     if (part.type === 'pdf') {
234 |         const embedUrl = getPdfEmbedUrl(part.url);
235 |         return embedUrl ? (
236 |             <div key={i} className="rounded-xl overflow-hidden bor...
237 |                 <iframe src={embedUrl} className="w-full h-[540px]...
238 |             </div>
239 |         ) : null;
240 |     }
241 |     return <SimpleMarkdown key={i} text={part.content} />;
242 |   }
243 | }
244 | </div>
245 |
246 | {s.type === 'image' && s.content && (
247 |   <div className="rounded-2xl overflow-hidden shadow-lg">
248 |     <img src={s.content} className="w-full h-auto block" />
249 |   </div>
250 | )}
251 | {s.type === 'pdf' && getPdfEmbedUrl(s.content) && (
252 |   <div className="bg-white rounded-2xl overflow-hidden border border-sla...
253 |     <iframe src={getPdfEmbedUrl(s.content)} className="w-full h-[540px]...
254 |     <div className="px-6 py-3 flex justify-end border-t border-slate-1...
255 |       <a href={s.content} target="_blank" className="text-blue-600 h...
256 |         <Maximize2 size={14} /> Toàn màn hình
257 |       </a>
258 |     </div>
259 |   </div>
260 | )}
261 | {s.type === 'embed' && s.content && (
262 |   <div className="rounded-2xl overflow-hidden border border-slate-200 bg...
263 |     <div dangerouslySetInnerHTML={{ __html: s.content }} />
264 |   </div>
265 | )}
266 | </motion.section>
267 | </motion.div>
268 | )}
269 |
270 | { / *   V i d e o   m ß   r ù n g   * / }
271 |
272 | (() => {
273 |   const EXTRA_VIDEOS = {
274 |     'cpu': [ { title: 'CPU là gì? - Giỏi thích ch...
275 |       'mainboard': [ { title: 'Bo mạch chủ là gì?' },
276 |       'ram': [ { title: 'RAM là gì? DDR4 vs DDR5', id: 'qTQ7vg1JFDA' } ],
277 |       'gpu': [ { title: 'Card đồ họa hoạt ùng th...
278 |       'ổ cứng': [ { title: 'SSD vs HDD - Nên ch...
279 |       'nguồn': [ { title: 'Nguồn máy tính - H...
280 |       'case': [ { title: 'Case máy tính - Các lo...
281 |   }
282 |   const matchKey = Object.keys(EXTRA_VIDEOS).find(k => lesson.title?.toLowerCase().inclu...
283 |   const videos = matchKey ? EXTRA_VIDEOS[matchKey] : null
284 |   if (!videos) return null
285 |   return (
286 |     <motion.div
287 |       initial={{ opacity: 0 }}
288 |       animate={{ opacity: 1 }}
289 |       transition={{ delay: 0.3 }}
290 |       className="mt-16 pt-10 border-t border-slate-200"
291 |     >
292 |       <h3 className="text-xl font-bold text-slate-900 mb-6 flex items-center gap-3">
293 |         <Sparkles size={22} className="text-...
294 |       </h3>
295 |       <div className="grid grid-cols-1 sm:grid-cols-2 gap-5">
296 |         {videos.map((v, i) => (
297 |           <motion.div
298 |             key={i}
299 |             whileHover={{ y: -4 }}
300 |             className="rounded-xl overflow-hidden bg-white border border-slate...
301 |           >
302 |             <div className="aspect-video">
303 |               <iframe src={`https://www.youtube-nocookie.com/embed/${v.id}`}...
304 |             </div>
305 |             <div className="p-4">
306 |               <p className="text-slate-800 font-semibold text-sm">{v.title}</p>
307 |             </div>
308 |           </motion.div>
309 |         ))}
310 |       </div>
311 |     </motion.div>
312 |   )}
313 | )}
314 |
315 | { / *   B o o k s   * / }
316 | {books.length > 0 && (
317 |   <motion.div
318 |     initial={{ opacity: 0 }}
319 |     animate={{ opacity: 1 }}
320 |     transition={{ delay: 0.4 }}
321 |     className="mt-16 pt-10 border-t border-slate-200"
322 |   >
323 |     <h3 className="text-xl font-bold text-slate-900 mb-6 flex items-center gap-3">
324 |       <Book size={22} className="text-blue-600...
325 |     </h3>
326 |     <div className="grid grid-cols-2 sm:grid-cols-3 md:grid-cols-4 lg:grid-cols-5 gap-4">
327 |       {books.map(book => (
328 |         <motion.div
329 |           key={book.id}
330 |           whileHover={{ y: -6, scale: 1.02 }}
331 |           onClick={() => setSelectedBook(book)}
332 |           className="bg-white rounded-xl p-4 border border-slate-200 cursor-poin...
333 |         >
334 |           {book.cover_image_url && (
335 |             <img src={book.cover_image_url} className="w-full aspect-[3/4] obj...
336 |           )}
337 |           <p className="text-sm font-bold text-slate-800 text-center line-clamp-...
338 |         </motion.div>
339 |       ))}
340 |     </div>
341 |   </motion.div>
342 | )}
343 |
344 | { / *   B o t t o m   c o m p l e t e   b u t t o n   * / }
345 | <motion.div
346 |   initial={{ opacity: 0 }}
347 |   animate={{ opacity: 1 }}
348 |   transition={{ delay: 0.5 }}
349 |   className="mt-16 pt-10 border-t border-slate-200 flex justify-center"
350 | >
351 |   <motion.button
352 |     whileHover={{ scale: 1.03 }}
353 |     whileTap={{ scale: 0.97 }}
354 |     onClick={handleComplete}

```

[illegible]

[illegible]

```

602|         {/* Badges */}
603|         {userId && (
604|             <div className="flex flex-wrap items-center gap-2">
605|                 <span className="inline-flex items-center gap-1.5 text-[11px] sm:text-xs font-...
606|                     0=00 {totalLessons} bài híc
607|                 </span>
608|                 <span className="inline-flex items-center gap-1.5 text-[11px] sm:text-xs font-...
609|                     {totalCompleted} hoàn thành
610|                 </span>
611|                 <span className="inline-flex items-center gap-1.5 text-[11px] sm:text-xs font-...
612|                     +P {totalLessons} > 0 ? Math.round((t
613|                 </span>
614|             </div>
615|         )}
616|     </div>
617| </div>
618|
619|
620|     {/* Tabs */}
621|     <div className="flex-shrink-0 bg-white border-b border-slate-200">
622|         <div className="max-w-7xl mx-auto px-4 sm:px-6 md:px-10 py-3 flex gap-2 overflow-x-auto scroll...
623|             {tabs.map(tab => {
624|                 const isActive = activeTab === tab.key;
625|                 return (
626|                     <motion.button
627|                         key={tab.key}
628|                         layout
629|                         onClick={() => setActiveTab(tab.key)}
630|                         className={`relative flex items-center gap-2.5 px-5 py-2.5 rounded-full text-s...
631|                             isActive
632|                             ? 'bg-gradient-to-r from-blue-600 to-emerald-500 text-white shadow-md'
633|                             : 'bg-slate-100 text-slate-600 hover:bg-slate-200'
634|                         `}
635|                     >
636|                         {tab.icon}
637|                         {tab.label}
638|                         <span className={`px-2 py-0.5 rounded-full text-[11px] font-extrabold ${
639|                             isActive
640|                             ? 'bg-white/20 text-white'
641|                             : 'bg-slate-200 text-slate-500'
642|                         `}>
643|                             {loading ? '...' : lessons[tab.key].length}
644|                         </span>
645|                         {!loading && lessons[tab.key].some(l => completedIds.has(l.id)) && (
646|                             <span className={`px-2 py-0.5 rounded-full text-[11px] font-extrabold ${
647|                                 isActive ? 'bg-white/20 text-white' : 'bg-emerald-100 text-emerald-600'
648|                             `}>
649|                                 {lessons[tab.key].filter(l => completedIds.has(l.id)).length}
650|                             </span>
651|                         )}
652|                     </motion.button>
653|                 );
654|             })}
655|         </div>
656|     </div>
657|
658|     {/* Content */}
659|     <div className="flex-1 overflow-y-auto px-4 sm:px-6 md:px-10 py-4 sm:py-6">
660|         <div className="max-w-7xl mx-auto">
661|             {loading ? (
662|                 <div className="flex justify-center py-24">
663|                     <div className="flex flex-col items-center gap-4">
664|                         <Loader2 className="animate-spin text-blue-600" size={44} />
665|                         <p className="text-slate-500 text-sm font-
666|                     </div>
667|                 </div>
668|             ) : currentLessons.length === 0 ? (
669|                 <EmptyState
670|                     tabKey={activeTab}
671|                     onExplore={() => {
672|                         const otherKey = activeTab === 'textbook' ? 'extended' : 'textbook';
673|                         if (lessons[otherKey].length > 0) setActiveTab(otherKey);
674|                     }}
675|                 />
676|             ) : (
677|                 <motion.div
678|                     variants={containerVariants}
679|                     initial="hidden"
680|                     animate="visible"
681|                 >
682|                     <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4 gap-5">
683|                         {currentLessons.map(lesson => (
684|                             <LessonCard
685|                                 key={lesson.id}
686|                                 lesson={lesson}
687|                                 isCompleted={completedIds.has(lesson.id)}
688|                                 onClick={() => setSelectedLesson(lesson)}
689|                             />
690|                         ))}
691|                     </div>
692|                 </motion.div>
693|             )}
694|         </div>
695|     </div>
696| </div>
697| );
698| }
699|

```

File: components/LessonComments.js (107 lines, 5.9 KB)

```

1| 'use client';
2| import React, { useState, useEffect } from 'react';
3| import { supabase } from '@lib/supabase';
4| import { Loader2, Send, Trash2, MessageCircle } from 'lucide-react';
5|
6| export default function LessonComments({ lessonId }) {
7|     const [comments, setComments] = useState([]);
8|     const [text, setText] = useState('');
9|     const [loading, setLoading] = useState(true);
10|    const [sending, setSending] = useState(false);
11|    const [userId, setUserId] = useState(null);
12|    const [userEmail, setUserEmail] = useState('');
13|
14|    useEffect(() => {
15|        const init = async () => {
16|            const { data: { user } } = await supabase.auth.getUser();
17|            if (user) { setUserId(user.id); setUserEmail(user.email); }
18|            fetchComments();
19|        };
20|        init();
21|    }, [lessonId]);
22|

```

```

23| const fetchComments = async () => {
24|   const { data } = await supabase
25|     .from('lesson_comments')
26|     .select('*')
27|     .eq('lesson_id', lessonId)
28|     .order('created_at', { ascending: true });
29|   setComments(data || []);
30|   setLoading(false);
31| };
32|
33| const handleSend = async () => {
34|   if (!text.trim() || !userId) return;
35|   setSending(true);
36|   const { data } = await supabase.from('lesson_comments').insert({
37|     lesson_id: lessonId, user_id: userId, content: text.trim(), user_email: userEmail
38|   }).select().single();
39|   if (data) setComments(prev => [...prev, data]);
40|   setText('');
41|   setSending(false);
42| };
43|
44| const handleDelete = async (id) => {
45|   await supabase.from('lesson_comments').delete().eq('id', id);
46|   setComments(prev => prev.filter(c => c.id !== id));
47| };
48|
49| if (loading) return <div style={{ display: 'flex', justifyContent: 'center', padding: '40px' }}><Loader2 c...
50|
51| return (
52|   <div style={{ display: 'flex', flexDirection: 'column', gap: '16px' }}>
53|     <div style={{ display: 'flex', alignItems: 'center', gap: '10px', marginBottom: '8px' }}>
54|       <MessageCircle size={20} color="#00f3ff" />
55|       <h3 style={{ margin: 0, fontSize: '1.1rem', fontWeight: 'bold' }}>
56|         Comments List
57|       </h3>
58|       <div style={{ display: 'flex', flexDirection: 'column', gap: '12px', maxHeight: '400px', overflowY: 'auto' }}>
59|         {comments.length === 0 && (
60|           <p style={{ color: '#4b5563', textAlign: 'center' }}>No comments yet.</p>
61|         )}
62|         {comments.map(c => (
63|           <div key={c.id} style={{ background: 'rgba(255,255,255,0.03)', border: '1px solid rgba(255,255,255,0.03)', padding: '10px', borderRadius: '10px' }}>
64|             <div style={{ width: '36px', height: '36px', borderRadius: '50%', background: 'rgb(0,0,0)' }}>
65|               {c.user_email || 'U'}[0].toUpperCase()
66|             </div>
67|             <div style={{ flex: 1 }}>
68|               <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
69|                 <span style={{ fontSize: '12px', font-weight: 'bold', color: '#00f3ff' }}>{c.user_...}</span>
70|                 <span style={{ display: 'flex', gap: '8px' }}>
71|                   <span style={{ display: 'flex', gap: '8px' }}>
72|                     <span style={{ fontSize: '11px', color: '#4b5563' }}>{new Date(c.created_a...)}</span>
73|                     <button onClick={() => handleDelete(c.id)} style={{ background: 'none', border: none, cursor: pointer }}>X</button>
74|                   </span>
75|                 </div>
76|               <div style={{ flex: 1 }}>{c.content}</div>
77|             </div>
78|             <p style={{ margin: 0, color: '#e0e6ed', fontSize: '14px', lineHeight: 1.6 }}>{c.c...}</p>
79|           </div>
80|         ))}
81|       </div>
82|       <div style={{ display: 'flex', gap: '10px', marginTop: '8px' }}>
83|         <input type="text" value={text} onChange={e => setText(e.target.value)} onKeyDown={e => e.key === 'Enter' && !e.shiftKey && handleSend()} placeholder="Nhập bình luận... (Enter để gửi)" style={{ flex: 1, background: 'rgba(255,255,255,0.04)', border: '1px solid rgba(255,255,255,0.03)', padding: '10px' }} />
84|         <button onClick={handleSend} disabled={!sending || !text.trim()} style={{ padding: '12px 20px', background: '#00f3ff', color: '#050a14', border: 'none' }}>Gửi</button>
85|         <button type="button" className="animate-spin" size={16} /> : <Send size={16} />
86|       </div>
87|     ) : (
88|       <p style={{ color: '#4b5563', textAlign: 'center', padding: '16px', background: 'rgba(255,255,255,0.03)', border: '1px solid rgba(255,255,255,0.03)' }}>Vui lòng <a href="/login" style={{ color: '#00f3ff' }}>đăng nhập</a> để bình luận.</p>
89|     )
90|   );
91| }

```

[illegible]

[illegible]

[illegible]

```

1 'use client';
2
3 import React, { useState, useEffect } from 'react';
4
5 const LoadingScreen = ({ onComplete }) => {
6   const [progress, setProgress] = useState(0);
7   const [step, setStep] = useState(0);
8   const [visible, setVisible] = useState(true);
9   const [rotation, setRotation] = useState(0);
10  const [glowPulse, setGlowPulse] = useState(0);
11
12  const loadingMessages = [
13    "      a n g       k h ß i       ù n g       h Ç       t h Ñ n g       g i £       l - p       P C       M a s t e r . . . " " í ý . . . " ,
14    |         "      a n g       t £ i       c j i       s ß       d i       l i Ç u       h i n       5 0 +       l i n h       k i Ç n       v - t       " " ,
15    |         "      a n g       k ç t       n Ñ i       T r ä       g i £ n g       A I       G u r u       h x       t r ä       2 4 / 7 . . . " ,
16    |         " H Ç       t h Ñ n g       p h ò n g       t h i       n g h i Ç m       3 D       ä       s µ n       s à n g ! "
17  ];
18
19  useEffect(() => {
20    let mounted = true;
21    let animFrame;
22    let glowFrame;
23
24    const rotate = () => {
25      if (!mounted) return;
26      setRotation(prev => (prev + 2) % 360);
27      animFrame = requestAnimationFrame(rotate);
28    };
29    animFrame = requestAnimationFrame(rotate);
30
31    const pulseGlow = () => {
32      if (!mounted) return;
33      setGlowPulse(prev => (prev + 0.02) % 1);
34      glowFrame = requestAnimationFrame(pulseGlow);
35    };
36    glowFrame = requestAnimationFrame(pulseGlow);
37
38    const runSequence = async () => {
39      await new Promise(r => setTimeout(r, 400));
40      if (!mounted) return;
41      setStep(1);
42
43      for (let i = 0; i <= 100; i += 5) {
44        await new Promise(r => setTimeout(r, 12));
45        if (!mounted) return;
46        setProgress(i);
47      }
48      await new Promise(r => setTimeout(r, 200));
49      if (!mounted) return;
50      setStep(2);
51      setProgress(0);
52
53      for (let i = 0; i <= 100; i += 10) {
54        await new Promise(r => setTimeout(r, 15));
55        if (!mounted) return;
56        setProgress(i);
57      }
58      await new Promise(r => setTimeout(r, 300));
59      if (!mounted) return;
60      setStep(3);
61
62      await new Promise(r => setTimeout(r, 600));
63      if (!mounted) return;
64
65      setVisible(false);
66      setTimeout(onComplete, 300);
67    };
68
69    runSequence();
70
71    return () => {
72      mounted = false;
73      if (animFrame) cancelAnimationFrame(animFrame);

```

```

74 |         if (glowFrame) cancelAnimationFrame(glowFrame);
75 |     };
76 | }, [onComplete]);
77 |
78 | if (!visible) return null;
79 |
80 | const ringCount = 4;
81 | const rings = [];
82 | const ringColors = ['#00d4aa', '#6366f1', '#a855f7', '#ffb900'];
83 | for (let i = 0; i < ringCount; i++) {
84 |     const baseSize = 110 + i * 40;
85 |     const delay = i * 0.6;
86 |     const direction = i % 2 === 0 ? 1 : -1;
87 |     const opacity = 0.6 - i * 0.1;
88 |     rings.push(
89 |         <div key={i} style={{
90 |             position: 'absolute',
91 |             width: `${baseSize}px`,
92 |             height: `${baseSize}px`,
93 |             borderRadius: '50%',
94 |             border: `2px solid transparent`,
95 |             borderTopColor: ringColors[i],
96 |             borderRightColor: i % 2 === 1 ? ringColors[i] : 'transparent',
97 |             borderBottomColor: i % 3 === 0 ? ringColors[i] : 'transparent',
98 |             transform: `rotate(${rotation * direction + delay * 30}deg)`,
99 |             transition: 'transform 0.05s linear',
100 |             opacity: opacity,
101 |             boxShadow: `0 0 ${15 + glowPulse * 20}px ${ringColors[i]}40, inset 0 0 ${10 + glowPulse * 15}p...
102 |             filter: `brightness(${1 + glowPulse * 0.3})`,
103 |         }} />
104 |     );
105 | }
106 |
107 | const particles = [];
108 | for (let i = 0; i < 16; i++) {
109 |     const angle = (i / 16) * 360;
110 |     const radius = 100 + glowPulse * 20;
111 |     const x = Math.cos((angle * Math.PI) / 180) * radius;
112 |     const y = Math.sin((angle * Math.PI) / 180) * radius;
113 |     const colors = ['#00d4aa', '#6366f1', '#a855f7', '#ffb900'];
114 |     particles.push(
115 |         <div key={`dot-${i}`} style={{
116 |             position: 'absolute',
117 |             width: '4px',
118 |             height: '4px',
119 |             borderRadius: '50%',
120 |             background: colors[i % 4],
121 |             transform: `translate(${x}px, ${y}px)`,
122 |             opacity: 0.2 + Math.sin((rotation * 2 + angle * 3) * Math.PI / 180) * 0.4,
123 |             boxShadow: `0 0 8px ${colors[i % 4]}`,
124 |             transition: 'opacity 0.1s ease',
125 |         }} />
126 |     );
127 | }
128 |
129 | const glowIntensity = 0.5 + glowPulse * 0.5;
130 |
131 | return (
132 |     <div
133 |         onClick={() => {
134 |             setVisible(false);
135 |             onComplete();
136 |         }}
137 |         style={{
138 |             position: 'fixed',
139 |             top: 0,
140 |             left: 0,
141 |             width: '100vw',
142 |             height: '100vh',
143 |             background: 'var(--bg-base)',
144 |             zIndex: 99999,
145 |             display: 'flex',
146 |             flexDirection: 'column',
147 |             justifyContent: 'center',
148 |             alignItems: 'center',
149 |             cursor: 'pointer',
150 |             opacity: visible ? 1 : 0,
151 |             transition: 'opacity 0.4s ease',
152 |             overflow: 'hidden'
153 |         }}
154 |     >
155 |         <div style={{
156 |             position: 'absolute', inset: 0, pointerEvents: 'none',
157 |             background:
158 |                 radial-gradient(circle at 30% 40%, rgba(0,212,170,${0.03 * glowIntensity}) 0%, transparent...
159 |                 radial-gradient(circle at 70% 60%, rgba(99,102,241,${0.03 * glowIntensity}) 0%, transparen...
160 |                 radial-gradient(circle at 50% 80%, rgba(168,85,247,${0.03 * glowIntensity}) 0%, transparen...
161 |                 radial-gradient(circle at 20% 70%, rgba(255,185,0,${0.02 * glowIntensity}) 0%, transparent...
162 |             ` />
163 |
164 |         <div style={{
165 |             display: 'flex',
166 |             flexDirection: 'column',
167 |             alignItems: 'center',
168 |             gap: '28px',
169 |             position: 'relative',
170 |             zIndex: 10
171 |         }}>
172 |             <div style={{
173 |                 position: 'relative',
174 |                 width: '180px',
175 |                 height: '180px',
176 |                 display: 'flex',
177 |                 alignItems: 'center',
178 |                 justifyContent: 'center',
179 |             }}>
180 |                 {rings}
181 |                 <div style={{
182 |                     position: 'absolute',
183 |                     display: 'flex',
184 |                     alignItems: 'center',
185 |                     justifyContent: 'center',
186 |                 }}>
187 |                     {particles}
188 |                 </div>
189 |                 <div style={{
190 |                     position: 'absolute',
191 |                     width: '130px',
192 |                     height: '130px',
193 |                     borderRadius: '50%',
194 |                     background: radial-gradient(circle, rgba(0,212,170,${0.12 + glowPulse * 0.08}) 0%, tr...
195 |                     display: 'flex',
196 |                     alignItems: 'center',
197 |

```



```

198 |         justifyContent: 'center',
199 |     }}>
200 |     
212 |   </div>
213 | </div>
214 |
215 | <div style={{ textAlign: 'center' }}>
216 |   <h1 style={{
217 |     color: '#ffffff',
218 |     fontSize: '24px',
219 |     fontWeight: 900,
220 |     letterSpacing: '3px',
221 |     margin: 0,
222 |     textShadow: `0 0 ${10 + glowPulse * 20}px rgba(0, 212, 170, ${0.2 + glowPulse * 0.3})`,
223 |   }}>
224 |     PC MASTER{' '}
225 |     <span style={{
226 |       background: 'linear-gradient(90deg, #00d4aa, #6366f1, #a855f7, #ffb900, #00d4aa)',
227 |       backgroundSize: '300% auto',
228 |       WebkitBackgroundClip: 'text',
229 |       WebkitTextFillColor: 'transparent',
230 |       backgroundClip: 'text',
231 |       animation: 'loadingGradient 2s linear infinite',
232 |     }}>
233 |       BUILDER
234 |     </span>
235 |   </h1>
236 |   <p style={{
237 |     fontSize: '10px',
238 |     color: 'rgba(255,255,255,0.4)',
239 |     textTransform: 'uppercase',
240 |     letterSpacing: '3px',
241 |     margin: 0,
242 |     fontWeight: 700,
243 |     textShadow: `0 0 10px rgba(0,212,170,0.2)`,
244 |   }}>
245 |     2 4 5 |
246 |
247 |   </p>
248 | </div>
249 |
250 | <div style={{
251 |   padding: '10px 24px',
252 |   background: 'rgba(255,255,255,0.04)',
253 |   border: '1px solid rgba(255,255,255,0.06)',
254 |   borderRadius: '14px',
255 |   minWidth: '280px',
256 |   textAlign: 'center',
257 |   backdropFilter: 'blur(4px)',
258 | }}>
259 |   <p style={{
260 |     background: 'linear-gradient(90deg, #00d4aa, #6366f1, #a855f7)',
261 |     backgroundSize: '200% auto',
262 |     WebkitBackgroundClip: 'text',
263 |     WebkitTextFillColor: 'transparent',
264 |     backgroundClip: 'text',
265 |     animation: 'loadingGradient 2s linear infinite',
266 |     fontSize: '13px',
267 |     fontWeight: 700,
268 |     margin: 0,
269 |   }} key={step}>
270 |     {loadingMessages[step]}
271 |   </p>
272 | </div>
273 |
274 | <div style={{
275 |   width: '300px',
276 |   height: '4px',
277 |   background: 'rgba(255,255,255,0.05)',
278 |   borderRadius: '10px',
279 |   overflow: 'hidden',
280 |   boxShadow: 'inset 0 0 6px rgba(0,0,0,0.3)',
281 | }}>
282 |   <div style={{
283 |     width: `${progress}%`,
284 |     height: '100%',
285 |     background: 'linear-gradient(90deg, #00d4aa, #6366f1, #a855f7, #ffb900)',
286 |     backgroundSize: '300% auto',
287 |     animation: 'loadingGradient 1.5s linear infinite',
288 |     transition: 'width 0.15s ease',
289 |     boxShadow: '0 0 16px rgba(0, 212, 170, 0.6), 0 0 30px rgba(99, 102, 241, 0.3)',
290 |   }} />
291 | </div>
292 |
293 | <div style={{
294 |   position: 'absolute',
295 |   bottom: '40px',
296 |   color: 'rgba(255,255,255,0.25)',
297 |   fontSize: '11px',
298 |   letterSpacing: '2px',
299 |   fontWeight: 600,
300 |   textTransform: 'uppercase',
301 |   animation: 'loadingPulse 2s ease-in-out infinite',
302 | }}>
303 |   3 0 3 |
304 |
305 | </div>
306 |
307 | <style>{`
308 |   @keyframes loadingGradient {
309 |     0% { background-position: 0% center; }
310 |     100% { background-position: 300% center; }
311 |   }
312 |   @keyframes loadingPulse {
313 |     0%, 100% { opacity: 0.3; }
314 |     50% { opacity: 0.8; }
315 |   }
316 | `}</style>
317 |
318 |   );
319 |
320 | export default LoadingScreen;
321 |

```

File: components/MacCompatibilityChecker.js (301 lines, 20.3 KB)

```
1 | 'use client';
2 |
3 | import { useState, useMemo } from 'react';
4 |
5 | const MAC_ORIG = [
6 |   { platform: 'Mac', id: 'macbook-air-m1', name: 'MacBook Air M1 (2020)', cpu: 'Apple M1', ram: '8GB Unified', ...
7 |   { platform: 'Mac', id: 'macbook-pro-14-m4', name: 'MacBook Pro 14" M4 Pro', cpu: 'Apple M4 Pro', ram: '24GB ...
8 |   { platform: 'Mac', id: 'mac-mini-m4', name: 'Mac Mini M4 (2024)', cpu: 'Apple M4', ram: '16GB Unified', gpu:...
9 | ];
10 |
11 | const PC_SYSTEMS = [
12 |   { platform: 'Win', id: 'win-budget', name: 'PC Siêu R», cpu: 'Xeon E5
13 |   { platform: 'Win', id: 'win-office', name: 'PC Văn Phòng', cpu: 'i3-14
14 |   { platform: 'Win', id: 'win-work', name: 'PC Office Tsm Trung', cpu: 'Ry
15 |   { platform: 'Win', id: 'win-gaming', name: 'PC Gaming Thun', cpu: 'Ry
16 |   { platform: 'Win', id: 'win-mini', name: 'PC Mini ITX', cpu: 'Ryzen 5 7600', ram: '16-32GB DDR5', gpu: 'RTX ...
17 |   { platform: 'Win', id: 'win-pro', name: 'PC Workstation Pro', cpu: 'i7-14700K / Ryzen 9 7950X', ram: '64GB D...
18 |   { platform: 'Win', id: 'win-server', name: 'PC NAS / Server', cpu: 'i3-N305 / Ryzen 3', ram: '8-16GB', gpu: ...
19 |   ...MAC_ORIG,
20 | ];
21 |
22 |
23 | const ALL_WORK_TYPES = [ 'Văn phòng', 'Gaming', 'Render', 'L-p trình', '
24 |
25 | function formatPrice(tr) {
26 |   return tr + 'tr';
27 | }
28 |
29 | function analyzePurpose(text) {
30 |   const w = { office: 0, render: 0, coding: 0, edit: 0 };
31 |   const k = text.toLowerCase();
32 |   if (k.match(/v n phòng|word|excel|email|web|hfc|sinh viên|híp|thuy ế t t
33 |   if (k.match(/game|ch i|valorant|lol|pubg|fortnite|aaa|competitive/i)) )
34 |   if (k.match(/render|blender|after effect|3d|maya|cad|vray|keyshot/i)) w.render = 10;
35 |   if (k.match(/l-p trình|code|dev|web|react|python|java|android|ios|dock
36 |   if (k.match(/đ ã ã g ph i m |edit|design|th i ế t k ế |photoshop|illustrator|figm
37 |   if (k.match(/ai|deep learning|machine learning|train|model/i)) { w.render = 10; w.coding = 10; }
38 |   if (Object.values(w).every(v => v === 0)) w.office = 5;
39 |   return w;
40 | }
41 |
42 | export default function MacCompatibilityChecker({ lang = 'vn' }) {
43 |   const [purpose, setPurpose] = useState('');
44 |   const [purposeDone, setPurposeDone] = useState(false);
45 |   const [selected, setSelected] = useState(null);
46 |   const [filterPlatform, setFilterPlatform] = useState('all');
47 |   const [showAll, setShowAll] = useState(false);
48 |
49 |   const analysis = useMemo(() => {
50 |     if (!purposeDone || !purpose.trim()) return null;
51 |     return analyzePurpose(purpose);
52 |   }, [purpose, purposeDone]);
53 |
54 |   const scored = useMemo(() => {
55 |     let list = [...PC_SYSTEMS];
56 |     if (filterPlatform !== 'all') list = list.filter(s => s.platform === filterPlatform);
57 |     return list.map(s => {
58 |       let totalScore = 0;
59 |       if (analysis) {
60 |         for (const k of Object.keys(analysis)) totalScore += (s.score[k] || 0) * (analysis[k] || 1);
61 |       } else {
62 |         totalScore = Object.values(s.score).reduce((a, b) => a + b, 0);
63 |       }
64 |       return { ...s, totalScore };
65 |     }).sort((a, b) => b.totalScore - a.totalScore);
66 |   }, [analysis, filterPlatform]);
67 |
68 |   const topMatch = scored[0];
69 |
70 |   return (
71 |     <div style={{
72 |       position: 'fixed', inset: 0, zIndex: 9999,
73 |       background: 'var(--bg-base)', display: 'flex', flexDirection: 'column',
74 |       fontFamily: 'system-ui, sans-serif', overflow: 'hidden',
75 |     }}>
76 |       <div style={{
77 |         display: 'flex', alignItems: 'center', justifyContent: 'space-between',
78 |         padding: '12px 24px', borderBottom: '1px solid var(--border-default)',
79 |         background: 'var(--bg-surface)', flexShrink: 0,
80 |       }}>
81 |         <div style={{ display: 'flex', alignItems: 'center', gap: 10 }}>
82 |           <span style={{ fontS i z e : 2 2 }}>0 ù i </span>
83 |           <div>
84 |             <div style={{ fontSize: 16, fontWeight: 800, color: 'var(--text-primary)' }}>
85 |               G ấ i Ỗ M ấ y T i n h
86 |             </div>
87 |             <div style={{ fontSize: 11, color: 'var(--text-muted)' }}>
88 |               C h i n c ư u h i n h p h ầ v ồ i n h u c s u & n g ầ n s á c h
89 |             </div>
90 |           </div>
91 |           <div>
92 |             <button onClick={() => window.history.back()}
93 |               style={{
94 |                 padding: '6px 14px', borderRadius: 8,
95 |                 border: '1px solid var(--border-default)',
96 |                 background: 'transparent', color: 'var(--text-muted)',
97 |                 fontSize: 12, cursor: 'pointer', fontFamily: 'inherit',
98 |               }}>
99 |             </div>
100 |
101 |           <div style={{ flex: 1, overflow: 'auto', padding: '16px 24px' }}>
102 |             <div style={{ maxWidth: 900, margin: '0 auto', display: 'flex', flexDirection: 'column', gap: 12 }}>
103 |               <div>
104 |                 <div style={{ flex: 1, overflow: 'auto', padding: '16px 24px' }}>
105 |                   <div style={{
106 |                     background: 'var(--bg-surface)', border: '1px solid var(--border-default)',
107 |                     borderRadius: 12, padding: '18px 22px',
108 |                   }}>
109 |                     <div style={{ fontSize: 13, fontWeight: 700, color: 'var(--text-primary)', marginBottom: 8, displa...
110 |                       <span style={{ fontS i z e : 1 8 }}>0 ù i </span> M ấ c í c h s í d ầ n g
111 |                     </div>
112 |                     <p style={{ fontSize: 11, color: 'var(--text-muted)', margin: '0 0 10px' }}>
113 |                       N h - p c ồ n g v i ế c b i n c s n l ầ m - t ồ i s ầ g ấ i Ỗ c ư u h i n h p h ầ h ầ p
114 |                     </p>
115 |                     <div style={{ display: 'flex', gap: 8 }}>
116 |                       <input value={purpose} onChange={e => setPurpose(e.target.value)}
117 |                         p l a c e h o l d e r = " V D : L - p t r i n h w e b , c h i i g a m e , đ ã ã g p h i m . . . "
118 |                       </div>
119 |                     <div style={{
120 |                       flex: 1, padding: '10px 14px', borderRadius: 8,
```

```

121 |         background: 'var(--bg-elevated)', border: '1px solid var(--border-default)',
122 |         color: 'var(--text-primary)', fontSize: 13, fontFamily: 'inherit', outline: 'none',
123 |     })
124 |     onKeyDown={e => e.key === 'Enter' && purpose.trim() && setPurposeDone(true)}
125 | />
126 | <button onClick={() => purpose.trim() && setPurposeDone(true)}
127 |   style={{
128 |     padding: '10px 20px', borderRadius: 8, fontSize: 12, fontWeight: 700,
129 |     background: 'var(--brand-primary)', border: 'none',
130 |     color: '#fff', cursor: 'pointer', fontFamily: 'inherit',
131 |   }}>
132 |     3 2 | G ā i ý
133 |   </button>
134 | </div>
135 | </div>
136 |
137 | /* Top match */
138 | {topMatch && analysis && (
139 |   <div style={{
140 |     background: 'var(--bg-surface)',
141 |     border: '1px solid ${topMatch.color}33', borderRadius: 12, padding: '14px 18px',
142 |     display: 'flex', alignItems: 'center', gap: 12,
143 |     animation: 'fadeIn 0.3s',
144 |   }}>
145 |     <div style={{
146 |       width: 40, height: 40, borderRadius: 10,
147 |       background: `${topMatch.color}15`, display: 'flex',
148 |       alignItems: 'center', justifyContent: 'center', fontSize: 20,
149 |     }}>{topMatch.icon}</div>
150 |     <div style={{ flex: 1 }}>
151 |       <div style={{ fontSize: 11, color: 'var(--text-muted)', marginBottom: 2 }}>
152 |         5 2 | & G ā i ý t Ñ t n h ¤ t c h o b i n
153 |       </div>
154 |       <div style={{ fontSize: 14, fontWeight: 700, color: 'var(--text-primary)' }}>{topMatch.name}</...
155 |       <div style={{ fontSize: 11, color: 'var(--text-secondary)', marginTop: 2 }}>
156 |         {topMatch.cpu} · {topMatch.gpu} · {topMatch.ram}
157 |       </div>
158 |     </div>
159 |     <div style={{ textAlign: 'right', flexShrink: 0 }}>
160 |       <div style={{ fontSize: 16, fontWeight: 800, color: topMatch.color }}>
161 |         {formatPrice(Math.round((topMatch.priceRange.min + topMatch.priceRange.max) / 2))}
162 |       </div>
163 |       <div style={{ fontSize: 10, color: 'var(--text-muted)' }}>
164 |         {formatPrice(topMatch.priceRange.min)}–{formatPrice(topMatch.priceRange.max)}
165 |       </div>
166 |     </div>
167 |   </div>
168 | )}
169 |
170 | /* Filters */
171 | <div style={{ display: 'flex', gap: 6, alignItems: 'center', flexWrap: 'wrap' }}>
172 |   <span style={{ fontSize: 10, fontWeight: 700, color: 'var(--text-muted)', textTransform: 'uppercas...
173 |     {[ 'all', 'windows', 'Mac' ].map(p => (
174 |       <button key={p} onClick={() => setFilterPlatform(p)}
175 |         style={{
176 |           padding: '4px 12px', borderRadius: 6, fontSize: 10, fontWeight: 600,
177 |           background: filterPlatform === p ? 'var(--brand-subtle)' : 'var(--bg-elevated)',
178 |           border: '1px solid ${filterPlatform === p ? 'var(--brand-primary)' : 'var(--border-default)'}',
179 |           color: filterPlatform === p ? 'var(--brand-primary)' : 'var(--text-muted)',
180 |           cursor: 'pointer', fontFamily: 'inherit',
181 |         }}>{ p === 'all' ? 'T ¥ t c £' : p }</button>
182 |       </div>
183 |     </div>
184 |
185 | /* System cards */
186 | <div style={{ display: 'flex', flexDirection: 'column', gap: 6 }}>
187 |   {(showAll ? scored : scored.slice(0, 6)).map((sys, i) => {
188 |     const avgPrice = Math.round((sys.priceRange.min + sys.priceRange.max) / 2);
189 |     const isTop = topMatch?.id === sys.id;
190 |     return (
191 |       <div key={sys.id}
192 |         onClick={() => setSelected(selected?.id === sys.id ? null : sys)}
193 |         style={{
194 |           background: 'var(--bg-surface)',
195 |           border: '1px solid ${isTop ? `${sys.color}33` : 'var(--border-default)'}',
196 |           borderRadius: 12, cursor: 'pointer', overflow: 'hidden',
197 |           animation: 'fadeInUp 0.3s ease-out ${i * 0.05}s both',
198 |         }}>
199 |         <div style={{ padding: '12px 16px', display: 'flex', gap: 12, alignItems: 'flex-start' }}>
200 |           <div style={{
201 |             width: 36, height: 36, borderRadius: 8,
202 |             background: `${sys.color}15`, display: 'flex',
203 |             alignItems: 'center', justifyContent: 'center', fontSize: 18, flexShrink: 0,
204 |           }}>{sys.icon}</div>
205 |           <div style={{ flex: 1, minWidth: 0 }}>
206 |             <div style={{ display: 'flex', alignItems: 'center', gap: 6, marginBottom: 2 }}>
207 |               <span style={{ fontSize: 14, fontWeight: 700, color: 'var(--text-primary)' }}>{sys.nam...
208 |               {isTop && (
209 |                 <span style={{ fontSize: 8, fontWeight: 700, padding: '1px 6px', borderRadius: 99, b...
210 |               )}
211 |               <span style={{
212 |                 fontSize: 8, fontWeight: 600, padding: '1px 6px', borderRadius: 99,
213 |                 background: sys.platform === 'Mac' ? 'rgba(85,85,85,0.15)' : 'rgba(59,130,246,0.15)',
214 |                 color: sys.platform === 'Mac' ? '#888' : '#3b82f6',
215 |               }}>{sys.platform}</span>
216 |             </div>
217 |             <div style={{ fontSize: 10, color: 'var(--text-secondary)', lineHeight: 1.5, marginBotto...
218 |               {sys.cpu} · {sys.gpu} · {sys.ram}
219 |             </div>
220 |             <div style={{ display: 'flex', gap: 4, flexWrap: 'wrap' }}>
221 |               {sys.suitable.slice(0, 4).map(sw => (
222 |                 <span key={sw} style={{ fontSize: 8, padding: '1px 6px', borderRadius: 99, backgroun...
223 |               )}}
224 |             </div>
225 |           </div>
226 |           /* Score bars */
227 |           <div style={{ display: 'flex', gap: 4, marginTop: 6 }}>
228 |             {Object.entries(sys.score).map(([k, v]) => (
229 |               <div key={k} style={{ flex: 1 }}>
230 |                 <div style={{ height: 3, borderRadius: 2, background: 'var(--border-subtle)', over...
231 |                 <div style={{ height: '100%', width: `${(v / 10) * 100}%`, background: v >= 8 ? ...
232 |                 <div style={{ fontSize: 7, color: 'var(--text-muted)', textTransform: 'uppercase', ...
233 |                 </div>
234 |               </div>
235 |             )}}
236 |           </div>
237 |           <div style={{ textAlign: 'right', flexShrink: 0 }}>
238 |             <div style={{ fontSize: 15, fontWeight: 800, color: sys.color }}>{formatPrice(avgPrice)}...
239 |             <div style={{ fontSize: 9, color: 'var(--text-muted)' }}>{formatPrice(sys.priceRange.min...
240 |             {analysis && (
241 |               <div style={{ marginTop: 4, fontSize: 10, fontWeight: 700, color: sys.totalScore > 20 ...
242 |                 { Math.round ( s y s . t o t a l s c o r e ) } i Ä m
243 |               </div>
244 |             )}

```

```

245 |         </div>
246 |     </div>
247 |
248 |     {selected?.id === sys.id && (
249 |       <div style={{ padding: '0 16px 14px', borderTop: '1px solid var(--border-subtle)' }}>
250 |         <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: 8, marginTop: 12 }}>
251 |           <div style={{ background: 'var(--bg-elevated)', borderRadius: 8, padding: '10px 12px' }}>
252 |             <div style={{ fontSize: 9, fontWeight: 700, color: 'var(--text-muted)', textTransform: 'uppercase' }}>
253 |               <div style={{ display: 'flex', flexDirection: 'column', gap: 3 }}>
254 |                 {[{ l: 'CPU', v: sys.cpu }, { l: 'GPU', v: sys.gpu }, { l: 'RAM', v: sys.ram }, { l: 'Storage', v: sys.storage }].map((item) => (
255 |                   <div key={spec.l} style={{ display: 'flex', gap: 6, fontSize: 10 }}>
256 |                     <span style={{ color: 'var(--text-muted)', minWidth: 40 }}>{spec.l}</span>
257 |                     <span style={{ color: 'var(--text-secondary)' }}>{spec.v}</span>
258 |                   </div>
259 |                 ))}
260 |               </div>
261 |             </div>
262 |             <div style={{ background: 'var(--bg-elevated)', borderRadius: 8, padding: '10px 12px' }}>
263 |               <div style={{ fontSize: 9, fontWeight: 700, color: 'var(--text-muted)', textTransform: 'uppercase' }}>
264 |                 <ul style={{ margin: 0, padding: '0 0 0 14px', display: 'flex', flexDirection: 'column', gap: 4 }}>
265 |                   {sys.features.map((f, i) => (
266 |                     <li key={i} style={{ fontSize: 10, color: 'var(--text-secondary)' }}>{f}</li>
267 |                   ))}
268 |                 </ul>
269 |               </div>
270 |             </div>
271 |             <div style={{ background: 'var(--bg-elevated)', borderRadius: 8, padding: '10px 12px', margin: 4 }}>
272 |               <div style={{ fontSize: 9, fontWeight: 700, color: 'var(--text-muted)', textTransform: 'uppercase' }}>
273 |                 <div style={{ display: 'flex', gap: 4, flexWrap: 'wrap' }}>
274 |                   {sys.suitable.map((sw) => (
275 |                     <span key={sw} style={{ fontSize: 9, padding: '2px 8px', borderRadius: 99, background: 'var(--bg-elevated)', color: 'var(--text-muted)', cursor: 'pointer', fontFamily: 'inherit' }}>{sw}</span>
276 |                   ))}
277 |                 </div>
278 |               </div>
279 |             </div>
280 |           </div>
281 |         </div>
282 |       </div>
283 |     ))}
284 |   </div>
285 |
286 |   {!showAll && scored.length > 6 && (
287 |     <button onClick={() => setShowAll(true)}
288 |       style={{
289 |         padding: '8px', borderRadius: 8, fontSize: 11, fontWeight: 600,
290 |         background: 'var(--bg-surface)', border: '1px solid var(--border-default)',
291 |         color: 'var(--text-muted)', cursor: 'pointer', fontFamily: 'inherit',
292 |       }}
293 |     >
294 |       </button>
295 |     </div>
296 |   </div>
297 | </div>
298 | </div>
299 | );
300 | }
301 |

```

File: components/MainMenu.js (348 lines, 19.8 KB)

```

1 | 'use client';
2 |
3 | import React, { useState, useEffect } from 'react';
4 | import { BookOpen, Cpu, ShoppingCart, Users, ArrowRight, Info, LogIn, LogOut, User, Award, Swords, Star, Troph...
5 | import { supabase } from '@lib/supabase';
6 | import Link from 'next/link';
7 | import { useRouter } from 'next/navigation';
8 | import { motion } from 'framer-motion';
9 | import { useIsMobile } from '@hooks/useIsMobile';
10 | import NotificationBell from './NotificationBell';
11 |
12 | const s = {
13 |   red: '#D32F2F',
14 |   navy: '#1A2F4A',
15 |   teal: '#0097A7',
16 |   orange: '#F5A623'
17 | };
18 |
19 | const fadeUp = (delay = 0) => ({
20 |   initial: { opacity: 0, y: 20 },
21 |   animate: { opacity: 1, y: 0 },
22 |   transition: { duration: 0.4, delay, ease: [0.25, 0.46, 0.45, 0.94] }
23 | });
24 |
25 | const MainMenu = ({ onStart, lang, onOpenLogin }) => {
26 |   const isMobile = useIsMobile();
27 |   const [user, setUser] = useState(null);
28 |   const [profile, setProfile] = useState(null);
29 |   const [loading, setLoading] = useState(true);
30 |   const router = useRouter();
31 |
32 |   useEffect(() => {
33 |     const checkUser = async () => {
34 |       const { data: { user: currentUser } } = await supabase.auth.getUser();
35 |       setUser(currentUser);
36 |       if (currentUser) {
37 |         const { data: userProfile } = await supabase
38 |           .from('profiles')
39 |           .select('role, full_name')
40 |           .eq('id', currentUser.id)
41 |           .single();
42 |         setProfile(userProfile);
43 |       }
44 |       setLoading(false);
45 |     };
46 |     checkUser();
47 |   }, []);
48 |
49 |   const handleSignOut = async () => {
50 |     await supabase.auth.signOut();
51 |     window.location.href = '/';
52 |   };
53 |
54 |   const modes = [
55 |     { id: 'course', title: 'B à i G i E n g', desc: 'L ý t h u y ̣ t p h ̣ n c é n g t' },
56 |     { id: 'learning', title: 'L u y ̣ n T - p', desc: 'L p r ̣ p t h ̣ n d o t h e o' },
57 |     { id: 'market', title: 'C h ̣ a M ̄ y T í n h', desc: 'M u a l ̣ i n h k i ̣ c n, t ̄ n' },
58 |     { id: 'multiplayer', title: '2 N g ̄ Y i C h i i', desc: 'N i k h ̄ a n g - a' }
59 |   ];
60 |
61 |   return (
62 |     <div style={{
63 |       display: 'flex', flexDirection: 'column', width: '100%', maxWidth: '1100px',

```

```

64 | margin: '0 auto', padding: isMobile ? '16px' : '32px 48px', minHeight: '100vh'
65 | }}>
66 | <motion.div {...fadeUp(0)} style={{ marginBottom: '16px' }}>
67 | <div style={{ position: 'relative', height: '32px', overflow: 'hidden', borderRadius: '8px', m...
68 | <div style={{
69 |   position: 'absolute', inset: 0,
70 |   background: 'linear-gradient(90deg, transparent 0%, rgba(0,212,170,0.04) 20%, rgba(99,...
71 | }} />
72 | {[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11].map(i => (
73 |   <motion.div key={i}
74 |     animate={{
75 |       left: ['-100px', 'calc(100% + 100px)'],
76 |       opacity: [0, 1, 1, 0],
77 |     }}
78 |     transition={{
79 |       duration: 3.0 - (i % 3) * 0.3,
80 |       repeat: Infinity,
81 |       delay: i * 0.25,
82 |       ease: 'linear',
83 |     }}
84 |     style={{
85 |       position: 'absolute', top: `${4 + (i % 4) * 7}px`,
86 |       width: `${6 + (i % 5) * 3}px`, height: '2px',
87 |       borderRadius: '2px',
88 |       background: i % 3 === 0
89 |         ? 'linear-gradient(90deg, var(--brand-primary), #6366f1)'
90 |         : i % 3 === 1
91 |         ? 'linear-gradient(90deg, #a855f7, var(--brand-primary))'
92 |         : 'linear-gradient(90deg, #ffb900, #f472b6)',
93 |       boxShadow: `0 0 ${8 + i * 2}px ${i % 3 === 0 ? 'rgba(0,212,170,0.6)' : i % 3 =...
94 |     }}
95 |   />
96 | ))}
97 | {[0, 1, 2, 3, 4, 5].map(i => (
98 |   <motion.div key={`dot-${i}`}
99 |     animate={{
100 |       left: ['-60px', 'calc(100% + 60px)'],
101 |       y: [0, -8, 0, 8, 0],
102 |     }}
103 |     transition={{
104 |       duration: 4.0 - i * 0.2,
105 |       repeat: Infinity,
106 |       delay: i * 0.5,
107 |       ease: 'easeInOut',
108 |     }}
109 |     style={{
110 |       position: 'absolute', top: '12px',
111 |       width: '5px', height: '5px', borderRadius: '50%',
112 |       background: i % 3 === 0 ? '#00d4aa' : i % 3 === 1 ? '#a855f7' : '#ffb900',
113 |       boxShadow: `0 0 14px ${i % 3 === 0 ? '#00d4aa' : i % 3 === 1 ? '#a855f7' : '#f...
114 |     }}
115 |   />
116 | ))}
117 | </div>
118 | <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'flex-start' }}>
119 |   <div>
120 |     <h1 style={{ fontSize: '24px', fontWeight: 800, color: 'var(--text-primary)', margin: ...
121 |       PC Master Builder
122 |     </h1>
123 |     <p style={{ fontSize: '13px', color: 'var(--text-muted)', margin: '2px 0 0' }}>
124 |       1 2 4 | M ò p h i n g l ò p r á p P C 2 D t í c h h ä p A I
125 |     </p>
126 |   </div>
127 |   <div>
128 |     {loading ? (
129 |       <div style={{ width: '40px' }} />
130 |     ) : user ? (
131 |       <div style={{ display: 'flex', alignItems: 'center', gap: '10px' }}>
132 |         <NotificationBell />
133 |         <Link href={profile?.role === 'teacher' ? '/teacher' : profile?.role === 'parent' ? '/...
134 |           style={{
135 |             display: 'flex', alignItems: 'center', gap: '8px', textDecoration: 'none',
136 |             padding: '6px 14px', borderRadius: '8px',
137 |             background: 'var(--bg-elevated)', border: '1px solid var(--border-default)'
138 |           }}>
139 |             <div style={{
140 |               width: '28px', height: '28px', borderRadius: '50%',
141 |               background: 'linear-gradient(135deg, var(--brand-primary), #6366f1)',
142 |               display: 'flex', alignItems: 'center',
143 |               justifyContent: 'center', color: 'fff'
144 |             }}>
145 |               <User size={14} />
146 |             </div>
147 |             <span style={{ fontSize: '13px', fontWeight: 600, color: 'var(--text-primary)' }}>
148 |               {profile?.full_name || user.email?.split('@')[0]}
149 |             </span>
150 |           </Link>
151 |           <button onClick={handleSignOut}
152 |             style={{
153 |               background: 'linear-gradient(135deg, var(--brand-primary), #6366f1)',
154 |               color: 'fff', border: 'none',
155 |               padding: '6px 12px', borderRadius: '6px', cursor: 'pointer',
156 |               fontSize: '12px', display: 'flex', alignItems: 'center',
157 |               gap: '4px', fontWeight: 600, fontFamily: 'inherit'
158 |             }}>
159 |             <LogOut size={14} /> Thoát
160 |           </button>
161 |         </div>
162 |       ) : (
163 |         <button onClick={onOpenLogin}
164 |           style={{
165 |             background: 'linear-gradient(90deg, var(--accent-amber), var(--accent-orange))',
166 |             color: 'fff', border: 'none', padding: '8px 18px', borderRadius: '8px',
167 |             fontWeight: 700, cursor: 'pointer', display: 'flex', alignItems: 'center',
168 |             gap: '6px', fontSize: '13px', fontFamily: 'inherit'
169 |           }}>
170 |           1 7 0 | < L o g I n s i z e = { 1 6 } / > n g n h - p
171 |         </button>
172 |       )
173 |     </div>
174 |   </div>
175 | </motion.div>
176 |
177 | <motion.div {...fadeUp(0.05)} style={{
178 |   borderBottom: '1px solid var(--border-default)',
179 |   margin: '20px 0'
180 | }} />
181 |
182 |
183 |
184 | {/ * 4 MODES GRID */}
185 | <motion.div {...fadeUp(0.15)} style={{ marginBottom: '24px' }}>
186 |   <div style={{
187 |     fontSize: '11px', fontWeight: 700, color: 'var(--text-muted)',

```

```

188 |         textTransform: 'uppercase', letterSpacing: '1.5px', marginBottom: '12px'
189 |     }}>
190 |         C h ử      ù
191 |     </div>
192 |     <div style={{
193 |         display: 'grid',
194 |         gridTemplateColumns: isMobile ? 'repeat(2, 1fr)' : 'repeat(4, 1fr)',
195 |         gap: isMobile ? '10px' : '12px'
196 |     }}>
197 |         {modes.map((mode, idx) => (
198 |             <motion.button
199 |                 key={mode.id}
200 |                 onClick={() => onStart(mode.id)}
201 |                 whileHover={{ y: -2, boxShadow: '0 4px 12px var(--shadow-color)' }}
202 |                 style={{
203 |                     display: 'flex', flexDirection: 'column',
204 |                     padding: '20px', borderRadius: '12px',
205 |                     border: '1px solid var(--border-default)',
206 |                     background: 'var(--bg-surface)',
207 |                     cursor: 'pointer', textAlign: 'left', width: '100%',
208 |                     fontFamily: 'inherit',
209 |                     transition: 'border-color 0.2s, box-shadow 0.2s'
210 |                 }}
211 |                 onMouseOver={e => { e.currentTarget.style.borderColor = mode.color; e.currentTarge...
212 |                 onMouseOut={e => { e.currentTarget.style.borderColor = 'var(--border-default)'; e...
213 |             >
214 |                 <div style={{
215 |                     width: '44px', height: '44px', borderRadius: '10px',
216 |                     background: `${mode.color}15`, color: mode.color,
217 |                     display: 'flex', alignItems: 'center', justifyContent: 'center',
218 |                     marginBottom: '14px', flexShrink: 0
219 |                 }}>
220 |                     <mode.Icon size={22} />
221 |                 </div>
222 |                 <div style={{ fontSize: '15px', fontWeight: 700, color: 'var(--text-primary)', mar...
223 |                     {mode.title}
224 |                 </div>
225 |                 <div style={{ fontSize: '12px', color: 'var(--text-muted)', lineHeight: 1.5, flex:...
226 |                     {mode.desc}
227 |                 </div>
228 |                 <div style={{ marginTop: '10px', display: 'flex', justifyContent: 'flex-end' }}>
229 |                     <ArrowRight size={14} style={{ color: 'var(--text-muted)' }} />
230 |                 </div>
231 |             </motion.button>
232 |         ))}
233 |     </div>
234 | </motion.div>
235 |
236 | /* EXTRA MODES */
237 | <motion.div {...fadeUp(0.25)} style={{ marginBottom: '24px' }}>
238 |     <div style={{
239 |         fontSize: '11px', fontWeight: 700, color: 'var(--text-muted)',
240 |         textTransform: 'uppercase', letterSpacing: '1.5px', marginBottom: '12px'
241 |     }}>
242 |         {lang === 'en' ? 'More' : 'Thêm'}
243 |     </div>
244 |     <div style={{ display: 'grid', gridTemplateColumns: isMobile ? '1fr' : 'repeat(2, 1fr)', gap: ...
245 |         {
246 |             { id: 'exams', title: 'Kó Thi', desc: 'Ki Ām tra
247 |             { id: 'challenge', title: 'Nhi Ćm Vă', desc: 'Nh
248 |         ].map((mode) => (
249 |             <motion.button
250 |                 key={mode.id}
251 |                 onClick={() => mode.id === 'practice' ? router.push('/practice') : onStart(mode.id)}
252 |                 whileHover={{ y: -1 }}
253 |                 style={{
254 |                     display: 'flex', alignItems: 'center', gap: '12px',
255 |                     padding: '12px 16px', borderRadius: '10px',
256 |                     border: '1px solid var(--border-default)',
257 |                     background: 'var(--bg-elevated)',
258 |                     cursor: 'pointer', textAlign: 'left', width: '100%',
259 |                     fontFamily: 'inherit',
260 |                     transition: 'border-color 0.2s'
261 |                 }}
262 |                 onMouseOver={e => { e.currentTarget.style.borderColor = mode.color; e.currentTarge...
263 |                 onMouseOut={e => { e.currentTarget.style.borderColor = 'var(--border-default)'; e...
264 |             >
265 |                 <div style={{
266 |                     width: '34px', height: '34px', borderRadius: '8px',
267 |                     background: `${mode.color}12`, color: mode.color,
268 |                     display: 'flex', alignItems: 'center', justifyContent: 'center', flexShrink: 0
269 |                 }}>
270 |                     <mode.Icon size={16} />
271 |                 </div>
272 |                 <div>
273 |                     <div style={{ fontSize: '13px', fontWeight: 600, color: 'var(--text-primary)' ...
274 |                     <div style={{ fontSize: '11px', color: 'var(--text-muted)' }}>{mode.desc}</div>
275 |                 </div>
276 |             </motion.button>
277 |         ))}
278 |     </div>
279 | </motion.div>
280 |
281 | /* PRO LEARNING SYSTEM */
282 | <motion.div {...fadeUp(0.2)} style={{ marginBottom: '24px' }}>
283 |     <div style={{
284 |         fontSize: '11px', fontWeight: 700, color: 'var(--accent-amber)',
285 |         textTransform: 'uppercase', letterSpacing: '1.5px', marginBottom: '12px',
286 |         display: 'flex', alignItems: 'center', gap: '6px'
287 |     }}>
288 |         <Zap size={13} /> Pro Learning System
289 |     </div>
290 |     <div style={{ display: 'grid', gridTemplateColumns: isMobile ? 'repeat(2, 1fr)' : 'repeat(3, 1...
291 |         {
292 |             { id: '/builder/os-install', title: 'Cài Windows
293 |             { id: '/builder/diagnosis', title: 'Chú n oán',
294 |             { id: '/builder/invoice-check', title: 'Quét Hóa
295 |             { id: '/builder/3d-viewer', title: 'PC Builder 3
296 |             { id: '/builder/common-mistakes', title: 'Lxí TH
297 |             { id: '/builder/mac-check', title: 'Găi - Ý Máy T
298 |             { id: '/builder/career-build', title: 'Ũc Mì &
299 |             { id: '/builder/3d-components', title: 'Kho Linh
300 |         }
301 |     ].map((mode) => (
302 |         <motion.a
303 |             key={mode.id}
304 |             href={mode.id}
305 |             whileHover={{ y: -2, boxShadow: '0 4px 16px rgba(0,0,0,0.2)' }}
306 |             style={{
307 |                 display: 'flex', alignItems: 'center', gap: '10px',
308 |                 padding: '14px', borderRadius: '10px',
309 |                 border: '1px solid var(--border-default)',
310 |                 background: 'var(--bg-surface)',
311 |                 cursor: 'pointer', textDecoration: 'none',

```

```

312 |         transition: 'border-color 0.2s, box-shadow 0.2s'
313 |     }}
314 |     onMouseOver={e => { e.currentTarget.style.borderColor = mode.color; e.currentTarge...
315 |     onMouseOut={e => { e.currentTarget.style.borderColor = 'var(--border-default)'; e....
316 |     >
317 |         <div style={{
318 |             width: {36px}', height: '36px', borderRadius: '8px',
319 |             background: `${mode.color}15`, color: mode.color,
320 |             display: 'flex', alignItems: 'center', justifyContent: 'center', flexShrink: 0
321 |         }}>
322 |             <mode.Icon size={18} />
323 |         </div>
324 |         <div>
325 |             <div style={{ fontSize: '12px', fontWeight: 600, color: 'var(--text-primary)' ...
326 |             <div style={{ fontSize: '10px', color: 'var(--text-muted)' }}>{mode.desc}</div>
327 |         </div>
328 |     </motion.a>
329 |     ))}
330 | </div>
331 | </motion.div>
332 |
333 |     <motion.div {...fadeUp(0.35)} style={{
334 |         marginTop: '12px', padding: '12px 16px', borderRadius: '10px',
335 |         background: `${s.orange}08`, border: '1px solid ${s.orange}20',
336 |         display: 'flex', gap: '10px', alignItems: 'flex-start'
337 |     }}>
338 |         <Info size={15} color={s.orange} style={{ flexShrink: 0, marginTop: '1px' }} />
339 |         <span style={{ fontSize: '12px', color: 'var(--text-muted)', lineHeight: 1.5 }}>
340 |             B i n c ó t h Ā d ũ n g W e b c a m h o . c   C h u ù t   Ā l - p   r á p .   i Ā u   c
341 |         </span>
342 |     </motion.div>
343 | </div>
344 | );
345 | };
346 |
347 | export default MainMenu;
348 |

```

File: components/Marketplace.js (335 lines, 23.4 KB)

```

1 | 'use client';
2 |
3 | import React, { useState, useMemo } from 'react';
4 | import { motion, AnimatePresence } from 'framer-motion';
5 | import componentsData from '../data/componentsData.json';
6 | import scenariosData from '../data/build-scenarios.json';
7 |
8 | const categories = ['CPU', 'Mainboard', 'RAM', 'GPU', 'PSU', 'Storage', 'Cooler'];
9 | const categoryLabels = { 'CPU': 'B ũ v i x í l ý', 'Mainboard': 'B o m i c h c h o
10 | const categoryIcons = { 'CPU': '&i', 'Mainboard': '0>Y', 'RAM': '0>Y', 'GPU': '0>Y' };
11 |
12 | const DIFFICULTY_COLORS = { easy: '#10b981', medium: '#f59e0b', hard: '#ef4444', expert: '#8b5cf6' };
13 | const DIFFICULTY_LABELS = { easy: 'D Ā', medium: 'T r u n g b i n h', hard: 'K h o
14 | const USE_CASE_LABELS = { office: '0=0% V n p h ò n g', gaming: '0<0% G a m i n g', v
15 |
16 | const Marketplace = ({ lang = 'en', onCheckout, onCancel }) => {
17 |     const [phase, setPhase] = useState('select'); // select | store | done
18 |     const [selectedScenario, setSelectedScenario] = useState(null);
19 |     const [cart, setCart] = useState([]);
20 |     const [budget, setBudget] = useState(0);
21 |     const [initialBudget, setInitialBudget] = useState(0);
22 |     const [activeTab, setActiveTab] = useState('CPU');
23 |     const [errorMsg, setErrorMsg] = useState('');
24 |
25 |     const s = {
26 |         bgBase: 'var(--bg-base)', bgSurface: 'var(--bg-surface)', bgElevated: 'var(--bg-elevated)',
27 |         textPrimary: 'var(--text-primary)', textMuted: 'var(--text-muted)', textSecondary: 'var(--text-seconda...
28 |         border: 'var(--border-default)', brand: 'var(--brand-primary)', success: 'var(--success)', danger: 'va...
29 |     };
30 |
31 |     const selectScenario = (scenario) => {
32 |         setSelectedScenario(scenario);
33 |         const b = scenario.budget_max;
34 |         setBudget(b);
35 |         setInitialBudget(b);
36 |         setCart([]);
37 |         setErrorMsg('');
38 |         setPhase('store');
39 |     };
40 |
41 |     const filteredProducts = useMemo(() => componentsData.filter(item => item.type === activeTab), [activeTab]);
42 |     const groupedProducts = useMemo(() => {
43 |         const groups = {};
44 |         for (const prod of filteredProducts) {
45 |             const key = prod.socket || prod.ramType || (prod.type === 'C
46 |             if (!groups[key]) groups[key] = [];
47 |             groups[key].push(prod);
48 |         }
49 |         return groups;
50 |     }, [filteredProducts]);
51 |
52 |     const validateCompat = (item) => {
53 |         if (item.type === 'Mainboard') {
54 |             const cpu = cart.find(c => c.type === 'CPU');
55 |             if (cpu && cpu.socket !== item.socket) return `Không t°ing t
56 |         }
57 |         if (item.type === 'CPU') {
58 |             const mb = cart.find(c => c.type === 'Mainboard');
59 |             if (mb && mb.socket !== item.socket) return `Không t°ing th
60 |         }
61 |         if (item.type === 'RAM') {
62 |             const mb = cart.find(c => c.type === 'Mainboard');
63 |             if (mb && mb.ramType && mb.ramType !== item.ramType) return
64 |         }
65 |         return null;
66 |     };
67 |
68 |     const handleAddToCart = (product) => {
69 |         if (cart.find(item => item.type === product.type)) { setErrorMsg
70 |         const compatError = validateCompat(product);
71 |         if (compatError) { setErrorMsg(compatError); return; }
72 |         if (budget - product.price < 0) { setErrorMsg('Không c ng n s á c
73 |         setErrorMsg('');
74 |         setCart(prev => [...prev, product]);
75 |         setBudget(prev => prev - product.price);
76 |     };
77 |
78 |     const handleRemoveFromCart = (product) => {
79 |         setCart(prev => prev.filter(c => c.id !== product.id));
80 |         setBudget(prev => prev + product.price);
81 |         setErrorMsg('');
82 |     };
83 |

```

```

84 | const handleCheckout = () => {
85 |   const required = ['CPU', 'Mainboard', 'RAM', 'PSU'];
86 |   const missing = required.filter(req => !cart.some(c => c.type === req));
87 |   if (missing.length > 0) { setErrorMsg(`Thiếu linh kiện thiết yếu`);
88 |     if (onCheckout) onCheckout({
89 |       purchasedItems: cart,
90 |       remainingBudget: budget,
91 |       scenario: selectedScenario,
92 |       scenarioName: selectedScenario?.name_vn || 'Tủ đồ'
93 |     });
94 |     setPhase('done');
95 |   };
96 |
97 |   return (
98 |     <div style={{ display: 'flex', flexDirection: 'column', width: '100%', maxWidth: '1300px', margin: '0 ...
99 |       {phase === 'select' && (
100 |         <motion.div initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }} style={{ flex: 1, o...
101 |           <div className="flex items-center gap-3 mb-2">
102 |             <span className="text-2xl"><ð&#x2013;/span>
103 |               <div>
104 |                 <h2 className="text-xl font-bold" style={{ color: s.textPrimary }}>{selectedScenario.name_vn}</h2>
105 |                 <p className="text-sm" style={{ color: s.textMuted }}>{selectedScenario.description_vn}</p>
106 |               </div>
107 |               <div className="grid gap-4 mt-4" style={{ gridTemplateColumns: 'repeat(auto-fill, minmax(3...
108 |                 {scenariosData.map((sc, i) => (
109 |                   <motion.div
110 |                     key={sc.id}
111 |                     initial={{ opacity: 0, y: 20 }}
112 |                     animate={{ opacity: 1, y: 0 }}
113 |                     transition={{ delay: i * 0.07 }}
114 |                     whileHover={{ y: -4, boxShadow: '0 12px 32px var(--shadow-hover)' }}
115 |                     onClick={() => selectScenario(sc)}
116 |                     className="rounded-2xl p-5 cursor-pointer"
117 |                     style={{ background: s.bgSurface, border: `1px solid ${s.border}` }}
118 |                   >
119 |                     <div className="flex items-start gap-4">
120 |                       <div className="flex-1">
121 |                         <h3 className="text-lg font-bold" style={{ color: s.textPrimary }}>{sc.name_vn}</h3>
122 |                         <p className="text-sm mt-1" style={{ color: s.textMuted, lineHeight: 1.1 }}>{sc.description_vn}</p>
123 |                         <div className="flex flex-wrap gap-2 mt-3">
124 |                           <span className="px-3 py-1 rounded-full text-xs font-bold" style={{ background: `${DIFFICULTY_COLORS[sc.difficulty]}20`, ...
125 |                             {DIFFICULTY_LABELS[sc.difficulty]}>{sc.difficulty}</span>
126 |                           <span className="px-3 py-1 rounded-full text-xs font-bold" style={{ background: 'transparent', color: s.textMuted, border: `1px solid ${s.border}` }}>{sc.requirements.use_case}</span>
127 |                         </div>
128 |                       <div>
129 |                         <div className="flex flex-wrap gap-1.5 mt-2">
130 |                           {sc.requirements.min_ram ? <span className="text-xs font-medium">{sc.requirements.min_ram}</span> : null}
131 |                           {sc.requirements.min_storage ? <span className="text-xs font-medium">{sc.requirements.min_storage}</span> : null}
132 |                           {sc.requirements.min_gpu ? <span className="text-xs font-medium">{sc.requirements.min_gpu}</span> : null}
133 |                           {sc.requirements.min_cores ? <span className="text-xs font-medium">{sc.requirements.min_cores}</span> : null}
134 |                         </div>
135 |                       </div>
136 |                     </div>
137 |                   </motion.div>
138 |                 )})
139 |               </div>
140 |             </div>
141 |             <button onClick={onCancel} className="mt-4 px-6 py-2 rounded-xl font-bold text-sm cursor-p...
142 |               style={{ background: 'transparent', color: s.textMuted, border: `1px solid ${s.border}` }}>Quay lại</button>
143 |           </motion.div>
144 |         ))
145 |       </div>
146 |     );
147 |   }
148 | }
149 |
150 | // Error handling
151 |
152 | // Animate presence
153 |
154 | // Store state
155 |
156 | // Header
157 |
158 | // Main Content
159 |
160 | // Footer
161 |
162 | // Tabs
163 |
164 | // Difficulty Labels
165 |
166 | // Requirements
167 |
168 | // Scenarios
169 |
170 | // Selected Scenario
171 |
172 | // Budget
173 |
174 | // Error Message
175 |
176 | // Animation
177 |
178 | // Store
179 |
180 | // Difficulty Labels
181 |
182 | // Requirements
183 |
184 | // Scenarios
185 |
186 | // Selected Scenario
187 |
188 | // Budget
189 |
190 | // Error Message
191 |
192 | // Animation
193 |
194 | // Store
195 |
196 | // Difficulty Labels
197 |
198 | // Requirements
199 |
200 | // Scenarios
201 |
202 | // Selected Scenario
203 |
204 | // Budget
205 |
206 | // Error Message
207 |

```



```

208 | <div className="flex gap-2 overflow-x-auto pb-1 shrink-0">
209 |   {categories.map(cat => (
210 |     <motion.button key={cat} onClick={() => setActiveTab(cat)} whileTap={{ sca...
211 |       style={{
212 |         padding: '8px 16px', display: 'flex', alignItems: 'center', gap: '...
213 |         background: activeTab === cat ? s.brand : s.bgElevated,
214 |         color: activeTab === cat ? '#fff' : s.textSecondary,
215 |         border: '1px solid ${activeTab === cat ? s.brand : s.border}',
216 |         borderRadius: '10px', cursor: 'pointer', fontWeight: 700, fontSize...
217 |         whiteSpace: 'nowrap', fontFamily: 'inherit'
218 |       }}
219 |       <span>{categoryIcons[cat]}</span> {categoryLabels[cat]}
220 |     </motion.button>
221 |   ))}
222 | </div>
223 |
224 | { /* Products */ }
225 | <div className="flex-1 overflow-y-auto">
226 |   {Object.entries(groupedProducts).map(([groupKey, products]) => (
227 |     <div key={groupKey} className="mb-4">
228 |       <div className="text-xs font-bold uppercase tracking-wider mb-2" style...
229 |       <div className="grid gap-3" style={{ gridTemplateColumns: 'repeat(auto...
230 |         {products.map(prod => {
231 |           const inCart = cart.some(c => c.id === prod.id);
232 |           return (
233 |             <motion.div key={prod.id} whileHover={{ y: -2 }} style={{
234 |               background: s.bgElevated, border: '1px solid ${inCart ...
235 |               borderRadius: '12px', padding: '14px', opacity: inCart...
236 |             }}>
237 |               {inCart && <div className="absolute top-2 right-2 text...
238 |               <h4 className="text-sm font-bold pr-10" style={{ color...
239 |               <div className="text-base font-black mt-1" style={{ co...
240 |               <p className="text-xs mt-1" style={{ color: s.textMuted...
241 |               <div className="flex flex-wrap gap-1 mt-1.5">
242 |                 {prod.socket && <Tag label={ Socket: ${prod.socket...
243 |                 {prod.ramType && <Tag label={ RAM: ${prod.ramType}...
244 |                 {prod.power && <Tag label={ TDP: ${prod.power}W } />}
245 |                 {prod.wattage && <Tag label={ ${prod.wattage}W } />}
246 |               </div>
247 |               <motion.button whileTap={{ scale: 0.95 }} disabled={in...
248 |                 style={{
249 |                   width: '100%', padding: '8px', marginTop: '8px...
250 |                   background: inCart ? s.bgElevated : s.brand, c...
251 |                   border: '1px solid ${inCart ? s.border : s.bra...
252 |                   cursor: inCart ? 'not-allowed' : 'pointer', fo...
253 |                 }}
254 |               </motion.button>
255 |             </motion.div>
256 |           );
257 |         }}
258 |       </div>
259 |     </div>
260 |   ))}
261 | </div>
262 |
263 | { /* Cart */ }
264 | <div className="w-72 shrink-0 flex flex-col gap-3 p-5 rounded-2xl overflow-y-auto"
265 |   style={{ background: s.bgSurface, border: '1px solid ${s.border}' }}>
266 |   <div className="flex justify-between items-center pb-3" style={{ borderBottom: '1p...
267 |     <h3 class s s N a m e = " t e x t - s m f o n t - b o l d " s t y l e =
268 |     <span className="text-xs font-bold" style={{ color: s.textMuted }}>{cart.length...
269 |   </div>
270 |   <div className="flex-1 overflow-y-auto space-y-2">
271 |     <AnimatePresence>
272 |       {cart.length === 0 ? (
273 |         <div className="text-center py-8" style={{ color: s.textMuted, fontSiz...
274 |         <div class s s N a m e = " t e x t - 3 x l m b .
275 |         G i l h à n g t r Ñ n g
276 |       ) : cart.map(item => (
277 |         <motion.div key={item.id} initial={{ opacity: 0, x: 20 }} animate={{ o...
278 |         className="flex justify-between items-center gap-2 p-3 rounded-xl"...
279 |         <div className="flex-1 min-w-0">
280 |           <div className="text-xs font-bold uppercase" style={{ color: s...
281 |           <div className="text-xs font-bold truncate" style={{ color: s...
282 |         </div>
283 |         <div className="flex items-center gap-2 shrink-0">
284 |           <span className="text-xs font-bold" style={{ color: s.success ...
285 |           <button onClick={() => handleRemoveFromCart(item)}
286 |             className="text-lg leading-none cursor-pointer" style={{ b...
287 |         </div>
288 |       </motion.div>
289 |     ))}
290 |   </AnimatePresence>
291 | </div>
292 | <div className="pt-3 space-y-2" style={{ borderTop: '1px solid ${s.border}' }}>
293 |   <motion.button onClick={handleCheckout} whileTap={{ scale: 0.97 }}
294 |     className="w-full py-3 rounded-xl font-bold text-sm text-white cursor-poin...
295 |     style={{ background: 'linear-gradient(135deg, ${s.brand}, #067a4d)', borde...
296 |     0 = b € T Ú Í P H Ò N G L A B
297 |   </motion.button>
298 |   <button onClick={() => setPhase('select')}
299 |     className="w-full py-2 rounded-xl font-bold text-xs cursor-pointer"
300 |     style={{ background: 'transparent', color: s.textMuted, border: '1px solid...
301 |     ! . Ò i n g h Á n g h i Ç p
302 |   </button>
303 | </div>
304 | </div>
305 |
306 | {
307 |   </div>
308 | }
309 |
310 | { phase === 'done' && (
311 |   <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }} className="flex items-center jus...
312 |   <div className="text-center p-8 rounded-2xl" style={{ background: s.bgSurface, border: '1p...
313 |     <div class s s N a m e = " t e x t - 6 x l m b - 4 " > ' < / d i v >
314 |     <h2 class s s N a m e = " t e x t - 2 x l f o n t - b o l d m b - 2 " s t y l e = {
315 |     <p class s s N a m e = " m b - 6 " s t y l e = { { color: s . t e x t M u t e d
316 |   </div>
317 | </motion.div>
318 |
319 | )};
320 | </div>
321 |
322 | );
323 | };
324 |
325 | function Tag({ label }) {
326 |   const s2 = { border: 'var(--border-default)', bg: 'var(--bg-elevated)', text: 'var(--text-muted)' };
327 |   return <span className="text-xs px-2 py-0.5 rounded font-semibold" style={{ background: s2.bg, color: s2.t...
328 | }
329 |
330 | function ReqTag({ label }) {
331 |   return <span className="text-[10px] px-2 py-0.5 rounded-full" style={{ background: 'rgba(245,158,11,0.1)',...

```

```

332 }
333
334 export default Marketplace;
335

```

File: components/MultiplayerEngine.js (350 lines, 18.5 KB)

```

1  'use client';
2
3  import React, { useState, useEffect, useMemo, useRef, useCallback } from 'react';
4  import GameEngine from '../GameEngine';
5
6  const P1_COLOR = '#6366f1';
7  const P2_COLOR = '#f59e0b';
8  const SUCCESS_COLOR = '#10b981';
9
10 const T = {
11   vi: {
12     title: 'CH % 0 2 NG - Ü I CH I',
13     subtitle: 'M x i ng ° Ý i é ng 1 bên - Gi i tay lên Ā s un s à ng !',
14     p1Label: 'NG - Ü I CH I 1',
15     p2Label: 'NG - Ü I CH I 2',
16     waiting: 'N a ng ch Ý ng ° Ý i ch i i . . .',
17     ready: 'S un s à ng !',
18     p1Ready: 'P 1 ā s un s à ng',
19     p2Ready: 'P 2 ā s un s à ng',
20     bothReady: 'C Ē 2 ā s un s à ng - B - t s u !',
21     startMatch: 'B @ T U',
22     starting: 'A NG B @ T U . . .',
23     p1Wins: 'P 1 T H @ NG !',
24     p2Wins: 'P 2 T H @ NG !',
25
26     vs: 'VS',
27     handOk: 'ā t h x y tay',
28     handRaised: 'ā g i i l é n',
29     handMissing: '° a tay v à o k h ung',
30     handsCount: 'C Ē 2 tay',
31     p1Desc: 'Gi i 1 tay lên c a o !',
32     p2Desc: 'Gi i 1 tay lên c a o !',
33   },
34   en: {
35     title: '2-PLAYER MODE',
36     subtitle: 'Stand on opposite sides - Raise your hand to ready up!',
37     p1Label: 'PLAYER 1',
38     p2Label: 'PLAYER 2',
39     waiting: 'Waiting for players...',
40     ready: 'Ready!',
41     p1Ready: 'P1 ready',
42     p2Ready: 'P2 ready',
43     bothReady: 'Both ready - Start!',
44     startMatch: 'START',
45     starting: 'STARTING...',
46     p1Wins: 'P1 WINS!',
47     p2Wins: 'P2 WINS!',
48     vs: 'VS',
49     handOk: 'Hand detected',
50     handRaised: 'Hand raised',
51     handMissing: 'Show your hand',
52     handsCount: 'Both hands',
53     p1Desc: 'Raise your hand!',
54     p2Desc: 'Raise your hand!',
55   },
56 };
57
58 const L = (key, lang) => (T[lang === 'en' ? 'en' : 'vi']?.[key] ?? T.vi[key]);
59
60 const LM_SMOOTH = 0.3;
61
62 function smoothLandmarks(raw, cache) {
63   if (!raw || raw.length === 0) return raw;
64   if (!cache || cache.length !== raw.length || cache[0]?.length !== raw[0]?.length) return raw;
65   return raw.map((hand, hi) => {
66     hand.map((lm, li) => {
67       const p = cache[hi]?.[li];
68       if (!p) return { ...lm };
69       return {
70         x: p.x + LM_SMOOTH * (lm.x - p.x),
71         y: p.y + LM_SMOOTH * (lm.y - p.y),
72         z: (p.z ?? 0) + LM_SMOOTH * ((lm.z ?? 0) - (p.z ?? 0)),
73       };
74     });
75   });
76 }
77
78 const MultiplayerEngine = ({ landmarks, onGameEvent, lang, trackingSensitivity }) => {
79   const smoothCacheRef = useRef(null);
80
81   const smoothedLandmarks = useMemo(() => {
82     if (!landmarks || landmarks.length === 0) {
83       smoothCacheRef.current = null;
84       return landmarks;
85     }
86     const cache = smoothCacheRef.current;
87     const smoothed = smoothLandmarks(landmarks, cache);
88     smoothCacheRef.current = smoothed;
89     return smoothed;
90   }, [landmarks]);
91
92   const [p1State, setP1State] = useState({ detected: false, ready: false, finished: false, time: 0 });
93   const [p2State, setP2State] = useState({ detected: false, ready: false, finished: false, time: 0 });
94   const [gameStarted, setGameStarted] = useState(false);
95   const [winner, setWinner] = useState(null);
96   const [countdown, setCountdown] = useState(0);
97
98   const timerRef = useRef(null);
99   const mountedRef = useRef(true);
100
101   // Spatial detection: P1 = left side (x<0.5), P2 = right side (x>=0.5)
102   const p1Landmarks = useMemo(() => {
103     if (!smoothedLandmarks || smoothedLandmarks.length === 0) return null;
104     const leftSide = smoothedLandmarks.filter(hand =>
105       hand[0] && hand[0].x !== undefined && hand[0].x < 0.5
106     );
107     return leftSide.length >= 1 ? leftSide : null;
108   }, [smoothedLandmarks]);
109
110   const p2Landmarks = useMemo(() => {
111     if (!smoothedLandmarks || smoothedLandmarks.length === 0) return null;
112     const rightSide = smoothedLandmarks.filter(hand =>
113       hand[0] && hand[0].x !== undefined && hand[0].x >= 0.5
114     );
115     return rightSide.length >= 1 ? rightSide : null;
116   }, [smoothedLandmarks]);
117
118   function isHandRaised(hand) {

```

```

117 |     return hand && hand[8] && hand[8].y !== undefined && hand[8].y < 0.35;
118 | }
119 |
120 | // Player detection
121 | useEffect(() => {
122 |   if (gameStarted) return;
123 |   const p1h = p1Landmarks;
124 |   const p1d = p1h && p1h.length >= 1;
125 |   const p1r = p1d && p1h.every(h => isHandRaised(h));
126 |   setP1State(prev => ({ ...prev, detected: !!p1d, ready: !!p1r }));
127 |
128 |   const p2h = p2Landmarks;
129 |   const p2d = p2h && p2h.length >= 1;
130 |   const p2r = p2d && p2h.every(h => isHandRaised(h));
131 |   setP2State(prev => ({ ...prev, detected: !!p2d, ready: !!p2r }));
132 | }, [p1Landmarks, p2Landmarks, gameStarted]);
133 |
134 | const canStart = p1State.ready && p2State.ready;
135 |
136 | const startGame = useCallback(() => {
137 |   setCountdown(3);
138 |   const cdInterval = setInterval(() => {
139 |     setCountdown(prev => {
140 |       if (prev <= 1) {
141 |         clearInterval(cdInterval);
142 |         setGameStarted(true);
143 |         setP1State(prev => ({ ...prev, time: 0, finished: false }));
144 |         setP2State(prev => ({ ...prev, time: 0, finished: false }));
145 |         setWinner(null);
146 |         const startTime = Date.now();
147 |         timerRef.current = setInterval(() => {
148 |           const elapsed = (Date.now() - startTime) / 1000).toFixed(1);
149 |           setP1State(prev => prev.finished ? prev : { ...prev, time: elapsed });
150 |           setP2State(prev => prev.finished ? prev : { ...prev, time: elapsed });
151 |         }, 100);
152 |         return 0;
153 |       }
154 |       return prev - 1;
155 |     });
156 |   }, 1000);
157 | }, []);
158 |
159 | const handlePlayerFinish = useCallback((playerId) => {
160 |   if (!gameStarted || winner) return;
161 |   const id = playerId === 1 ? 'P1' : 'P2';
162 |   if (playerId === 1) setP1State(prev => ({ ...prev, finished: true }));
163 |   else setP2State(prev => ({ ...prev, finished: true }));
164 |   setWinner(id);
165 |   clearInterval(timerRef.current);
166 |   if (onGameEvent) onGameEvent(`win_${playerId}`, 'all');
167 | }, [gameStarted, winner, onGameEvent]);
168 |
169 | const dotTrans = 'top 0.1s, left 0.1s';
170 |
171 | return (
172 |   <div style={{ width: '100%', display: 'flex', flexDirection: 'column', gap: '1rem' }}>
173 |     <div style={{
174 |       padding: '40px 32px', borderRadius: '24px',
175 |       background: 'linear-gradient(135deg, rgba(99,102,241,0.08), rgba(245,158,11,0.08))',
176 |       border: '1px solid rgba(255,255,255,0.08)',
177 |       display: 'flex', flexDirection: 'column', alignItems: 'center', gap: '24px',
178 |     }}>
179 |       <div style={{ textAlign: 'center' }}>
180 |         <h2 style={{ color: '#fff', fontSize: '28px', fontWeight: 800, margin: '0 0 6px 0', letterSpacing: '0.1em' }}>L ( ' t i t l e ' , l a n g )
181 |       </h2>
182 |       <p style={{ color: 'var(--text-secondary)', fontSize: '14px', margin: 0 }}>
183 |         L('subtitle', lang)
184 |       </p>
185 |     </div>
186 |
187 |     <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
188 |       <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
189 |         <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
190 |           <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
191 |             <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
192 |               <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
193 |                 <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
194 |                   <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
195 |                     <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
196 |                       <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
197 |                         <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
198 |                           <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
199 |                             <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
200 |                               <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
201 |                                 <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
202 |                                   <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
203 |                                     <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
204 |                                       <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
205 |                                         <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
206 |                                           <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
207 |                                             <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
208 |                                               <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
209 |                                                 <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
210 |                                                   <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
211 |                                                     <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
212 |                                                       <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
213 |                                                         <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
214 |                                                           <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
215 |                                                             <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
216 |                                                               <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
217 |                                                                 <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
218 |                                                                  <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
219 |                                                                    <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
220 |                                                                      <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
221 |                                                                        <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
222 |                                                                          <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
223 |                                                                            <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
224 |                                                                              <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
225 |                                                                                <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
226 |                                                                                  <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
227 |                                                                                    <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
228 |                                                                                      <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
229 |                                                                                        <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
230 |                                                                                          <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
231 |                                                                                           <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
232 |                                                                                             <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
233 |                                                                                              <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
234 |                                                                                               <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
235 |                                                                                                <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
236 |                                                                                                 <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
237 |                                                                                                  <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
238 |                                                                                                   <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
239 |                                                                                                    <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...
240 |                                                                                                     <div style={{ display: 'flex', gap: '24px', width: '100%', maxWidth: '800px', alignItems: ...

```

```

241 |         <div style={{ fontSize: '72px', fontWeight: 900, color: '#fff', textShadow: '0 0 40px ...
242 |             {countdown}
243 |         }>
244 |     ) : (
245 |         <button onClick={startGame} disabled={!canStart}
246 |             style={{
247 |                 padding: '16px 56px', fontSize: '20px', fontWeight: 800, letterSpacing: '2px',
248 |                 background: canStart ? 'linear-gradient(135deg, #00d4aa, #06b6d4)' : 'rgba(255...
249 |                 color: canStart ? '#000' : 'var(--text-muted)',
250 |                 border: 'none', borderRadius: '16px', cursor: canStart ? 'pointer' : 'not-allo...
251 |                 boxShadow: canStart ? '0 8px 32px rgba(0,212,170,0.4)' : 'none',
252 |             }}>
253 |                 { canStart ? 'ðŸŽ¥' + L('startMatch', lang) :
254 |             }>
255 |         )}
256 |     {!canStart && !countdown && (
257 |         <p style={{ color: 'var(--text-secondary)', fontSize: '13px', margin: 0 }}>
258 |             { p1State.ready && !p2State.ready ? 'ðŸŽŠ' : L('
259 |             { p1State.ready && p2State.ready ? 'ðŸŽŠ' : L('
260 |             L('waiting', lang)}
261 |         </p>
262 |     )}
263 | }
264 | </div>
265 | ) : (
266 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '16px', width: '100%' }}>
267 |         <div style={{
268 |             display: 'flex', justifyContent: 'space-between', alignItems: 'center',
269 |             padding: '16px 28px', borderRadius: '16px',
270 |             background: 'rgba(0,0,0,0.3)', border: '1px solid rgba(255,255,255,0.08)',
271 |         }}>
272 |             <div style={{ textAlign: 'center', flex: 1 }}>
273 |                 <div style={{ fontSize: '12px', color: P1_COLOR, fontWeight: 700, letterSpacing: '...
274 |                 <div style={{ fontSize: '32px', fontWeight: 800, color: '#fff', fontFamily: 'monos...
275 |                     { winner === 'P2' ? '#u' : ` ${p1State.time
276 |                 </div>
277 |             </div>
278 |             <div style={{
279 |                 padding: '6px 24px', borderRadius: '28px',
280 |                 background: winner ? 'rgba(16,185,129,0.2)' : 'rgba(255,255,255,0.04)',
281 |                 border: '1px solid ${winner ? SUCCESS_COLOR : 'rgba(255,255,255,0.08)'}\`,
282 |             }}>
283 |                 <span style={{ fontSize: '18px', fontWeight: 900, color: winner ? SUCCESS_COLOR : ...
284 |                     { winner ? 'ðŸŒˆ' : 'ðŸŒˆ' : 'P1' ? 'P1' :
285 |                 </span>
286 |             </div>
287 |             <div style={{ textAlign: 'center', flex: 1 }}>
288 |                 <div style={{ fontSize: '12px', color: P2_COLOR, fontWeight: 700, letterSpacing: '...
289 |                 <div style={{ fontSize: '32px', fontWeight: 800, color: '#fff', fontFamily: 'monos...
290 |                     { winner === 'P1' ? '#u' : ` ${p2State.time
291 |                 </div>
292 |             </div>
293 |         </div>
294 |
295 |         <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '16px', width: '100%' }}>
296 |             <div style={{
297 |                 position: 'relative', borderRadius: '16px', overflow: 'hidden',
298 |                 border: winner === 'P1' ? `3px solid ${SUCCESS_COLOR}` : `1px solid ${P1_COLOR}40`,
299 |             }}>
300 |                 <div style={{ position: 'absolute', top: '8px', left: '8px', zIndex: 100, padding:...
301 |                 <GameEngine landmarks={p1Landmarks}
302 |                     onGameEvent={(ev) => { if (ev === 'COMPLETE') handlePlayerFinish(1); }}
303 |                     trackingSensitivity={trackingSensitivity} />
304 |                 {winner === 'P1' && (
305 |                     <div style={{ position: 'absolute', inset: 0, background: 'rgba(16,185,129,0.1...
306 |                     <h2 style={{ color: '#fff', fontSize: '2.5rem', fontWeight: 900, textShado...
307 |                         ðŸŒˆ { L('p1Wins', lang) }
308 |                     </h2>
309 |                 </div>
310 |             </div>
311 |             <div style={{
312 |                 position: 'relative', borderRadius: '16px', overflow: 'hidden',
313 |                 border: winner === 'P2' ? `3px solid ${SUCCESS_COLOR}` : `1px solid ${P2_COLOR}40`,
314 |             }}>
315 |                 <div style={{ position: 'absolute', top: '8px', right: '8px', zIndex: 100, padding:...
316 |                 <GameEngine landmarks={p2Landmarks}
317 |                     onGameEvent={(ev) => { if (ev === 'COMPLETE') handlePlayerFinish(2); }}
318 |                     trackingSensitivity={trackingSensitivity} />
319 |                 {winner === 'P2' && (
320 |                     <div style={{ position: 'absolute', inset: 0, background: 'rgba(16,185,129,0.1...
321 |                     <h2 style={{ color: '#fff', fontSize: '2.5rem', fontWeight: 900, textShado...
322 |                         ðŸŒˆ { L('p2Wins', lang) }
323 |                     </h2>
324 |                 </div>
325 |             </div>
326 |         </div>
327 |
328 |         <div style={{
329 |             position: 'relative', width: '100%', height: '100%',
330 |             border: '1px solid #000',
331 |         }}>
332 |             <div style={{ position: 'absolute', top: '8px', left: '8px', zIndex: 100, padding:...
333 |             <div style={{ position: 'absolute', top: '8px', right: '8px', zIndex: 100, padding:...
334 |             <div style={{ position: 'absolute', bottom: '8px', left: '8px', zIndex: 100, padding:...
335 |             <div style={{ position: 'absolute', bottom: '8px', right: '8px', zIndex: 100, padding:...
336 |             <div style={{ position: 'absolute', top: '50%', left: '50%', transform: 'translate(-50%, -50%)',
337 |                 padding: '12px 36px', fontSize: '16px', fontWeight: 700,
338 |                 background: 'linear-gradient(135deg, #00d4aa, #06b6d4)',
339 |                 color: '#000', border: 'none', borderRadius: '12px', cursor: 'pointer',
340 |                 boxShadow: '0 4px 16px rgba(0,212,170,0.3)',
341 |             }}>
342 |                 Ch i i l i
343 |             </div>
344 |         </div>
345 |     )}
346 | }
347 | </div>
348 | );
349 | export default MultiplayerEngine;
350 |

```

File: components/Navbar.tsx (315 lines, 14.6 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect } from 'react'
4 | import Link from 'next/link'
5 | import { usePathname } from 'next/navigation'
6 | import { motion, AnimatePresence } from 'framer-motion'
7 | import { Home, BookOpen, Trophy, FileText, LayoutDashboard, Users, BarChart3, Bell, Menu, X, Settings, Logout,...
8 | import { getProfile, logout } from '@lib/auth-actions'
9 | import { NotificationBell } from '@components/notifications/NotificationBell'
10 |

```

```

11 export default function Navbar() {
12   const pathname = usePathname()
13
14   const [isScrolled, setIsScrolled] = useState(false)
15   const [mobileMenuOpen, setMobileMenuOpen] = useState(false)
16   const [userProfile, setUserProfile] = useState<any>(null)
17   const [loading, setLoading] = useState(true)
18   const [avatarDropdownOpen, setAvatarDropdownOpen] = useState(false)
19
20   // Fetch user profile on mount
21   useEffect(() => {
22     const fetchProfile = async () => {
23       try {
24         const profile = await getProfile()
25         setUserProfile(profile)
26       } catch (error) {
27         console.error("Failed to fetch profile", error)
28       } finally {
29         setLoading(false)
30       }
31     }
32     fetchProfile()
33   }, [])
34
35   // Handle scroll behavior
36   useEffect(() => {
37     const handleScroll = () => {
38       if (window.scrollY > 50) {
39         setIsScrolled(true)
40       } else {
41         setIsScrolled(false)
42       }
43     }
44     window.addEventListener('scroll', handleScroll)
45     return () => window.removeEventListener('scroll', handleScroll)
46   }, [])
47
48   // Menu items based on role
49   const role = userProfile?.role || 'student'
50
51   const studentMenu = [
52     { name: 'Trang chủ', path: '/student', icon: <Home size={18} /> },
53     { name: 'Bài học', path: '/courses', icon: <BookOpen size={18} /> },
54     { name: 'Kế hoạch', path: '/exams', icon: <FileText size={18} /> },
55     { name: 'Trắc nghiệm', path: '/quizz', icon: <HelpCircle size={18} /> },
56     { name: 'Xếp hạng', path: '/leaderboard', icon: <Trophy size={18} /> }
57   ]
58
59   const teacherMenu = [
60     { name: 'Dashboard', path: '/teacher', icon: <LayoutDashboard size={18} /> },
61     { name: 'Bài học', path: '/teacher/lessons', icon: <BookOpen size={18} /> },
62     { name: 'Ngân hàng', path: '/teacher/questions', icon: <FileText size={18} /> },
63     { name: 'Lớp học', path: '/teacher/classes', icon: <Users size={18} /> },
64     { name: 'Báo cáo', path: '/teacher/reports', icon: <BarChart3 size={18} /> },
65   ]
66
67   const activeMenu = role === 'teacher' ? teacherMenu : studentMenu
68
69   const handleLogout = async () => {
70     await logout()
71   }
72
73   // Don't show complex navbar on landing page or auth pages if not needed,
74   // but usually this navbar is placed inside protected layouts (e.g. app/(dashboard)/layout.tsx)
75   // We assume this component is used correctly by the parent.
76
77   return (
78     <
79       <nav
80         className={`fixed top-0 left-0 right-0 z-50 w-full border-b border-[#1e293b] transition-all duration-3...
81         isScrolled
82           ? 'bg-[#0f0f1a]/90 backdrop-blur-md py-2 shadow-lg'
83           : 'bg-[#0f0f1a] py-4'
84       }`
85     >
86       <div className="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">
87         <div className="flex flex-row items-center justify-between">
88
89           {/* LOGO – bên trái */}
90           <Link href="/" className="flex flex-row items-center gap-2 flex-shrink-0 group">
91             PC MASTER <span className="...
93           </Link>
94
95           { /* NAV LINKS – giữa, CHÈ HIỆN DESKTOP */ }
96           <div className="hidden md:flex flex-row items-center gap-1">
97             {activeMenu.map((item) => {
98               const isActive = pathname === item.path || pathname?.startsWith(item.path + '/')
99               return (
100                 <Link
101                   key={item.path}
102                   href={item.path}
103                   className={`relative px-4 py-2 rounded-full flex flex-row items-center gap-2 text-sm font-...
104                   isActive ? 'text-white' : 'text-slate-400 hover:text-white hover:bg-[#16213e]'
105                 >
106                   >
107                   {isActive && (
108                     <motion.div
109                       layoutId="navbar-indicator"
110                       className="absolute inset-0 bg-[#1e293b] rounded-full -z-10 border border-[#2a3655]"
111                       transition={{ type: "spring", stiffness: 400, damping: 30 }}
112                     />
113                   )}
114                   {item.icon}
115                   <span>{item.name}</span>
116                 </Link>
117               )
118             )}
119           </div>
120
121           { /* ACTIONS – bên phải */ }
122           <div className="flex flex-row items-center gap-2 sm:gap-4">
123
124             {/* Notifications */}
125             {userProfile && <NotificationBell userId={userProfile.id} />}
126
127             {/* User Avatar & Dropdown (Desktop) */}
128             <div className="hidden md:block relative">
129               {loading ? (
130                 <div className="w-9 h-9 rounded-full bg-[#1e293b] animate-pulse"></div>
131               ) : userProfile ? (
132                 <button
133                   onClick={() => setAvatarDropdownOpen(!avatarDropdownOpen)}
134
```

```

135 |         className="flex items-center gap-2 focus:outline-none"
136 |     >
137 |         <div className="w-9 h-9 rounded-full bg-gradient-to-tr from-[#00d2a0] to-[#00b4d8] p-[2p...
138 |         <div className="w-full h-full rounded-full bg-[#16213e] flex items-center justify-cent...
139 |             {userProfile.avatar_url ? (
140 |                 <img src={userProfile.avatar_url} alt="Avatar" className="w-full h-full object-cov...
141 |             ) : (
142 |                 <UserIcon size={16} className="text-slate-300" />
143 |             )}
144 |         </div>
145 |     </div>
146 | </button>
147 |
148 | <AnimatePresence>
149 |     {avatarDropdownOpen && (
150 |         <>
151 |             <div className="fixed inset-0 z-40" onClick={() => setAvatarDropdownOpen(false)}></div>
152 |             <motion.div
153 |                 initial={{ opacity: 0, y: 10, scale: 0.95 }}
154 |                 animate={{ opacity: 1, y: 0, scale: 1 }}
155 |                 exit={{ opacity: 0, y: 10, scale: 0.95 }}
156 |                 transition={{ duration: 0.15 }}
157 |                 className="absolute right-0 mt-2 w-56 bg-[#16213e] border border-[#1e293b] rounded...
158 |             >
159 |                 <div className="px-4 py-2 border-b border-[#1e293b] mb-2">
160 |                     <p className="text-sm font-bold text-white truncate">{userProfile.full_name}</p>
161 |                     <p className="text-xs text-slate-400 truncate">{userProfile.email}</p>
162 |                     <div className="mt-1 inline-block px-2 py-0.5 rounded text-[10px] font-bold uppe...
163 |                         {role}
164 |                     </div>
165 |                 </div>
166 |
167 |                 <Link href="/profile" onClick={() => setAvatarDropdownOpen(false)} className="flex...
168 |                     <UserIcon size={16} /> H Ó s i c á n h â n
169 |                 </Link>
170 |                 <Link href="/settings" onClick={() => setAvatarDropdownOpen(false)} className="fle...
171 |                     <Settings size={16} /> C à i . t
172 |                 </Link>
173 |
174 |                 <div className="h-[1px] bg-[#1e293b] my-2"></div>
175 |
176 |                 <button onClick={() => { setAvatarDropdownOpen(false); handleLogout(); }} classNam...
177 |                     <Logout size={16} /> n g x u ẏ t
178 |                 </button>
179 |             </motion.div>
180 |         </>
181 |     )}
182 | </AnimatePresence>
183 | </>
184 | ) : (
185 |     <Link href="/login" className="px-4 py-2 bg-[#00d2a0] text-black text-sm font-bold rounded-l...
186 |         n g n h - p
187 |     </Link>
188 | )}
189 | </div>
190 |
191 | { / *   H a m b u r g e r   -   C H Ê   M O B I L E   * / }
192 |
193 | <button
194 |     className="md:hidden p-2 text-slate-400 hover:text-white bg-[#16213e] rounded-lg border border...
195 |     onClick={() => setMobileMenuOpen(true)}
196 | >
197 |     <Menu size={20} />
198 | </button>
199 | </div>
200 | </div>
201 | </div>
202 | </nav>
203 |
204 | {/ * MOBILE MENU DRAWER */}
205 | <AnimatePresence>
206 |     {mobileMenuOpen && (
207 |         <>
208 |             <motion.div
209 |                 initial={{ opacity: 0 }}
210 |                 animate={{ opacity: 1 }}
211 |                 exit={{ opacity: 0 }}
212 |                 onClick={() => setMobileMenuOpen(false)}
213 |                 className="fixed inset-0 bg-black/60 backdrop-blur-sm z-[60] md:hidden"
214 |             />
215 |
216 |             <motion.div
217 |                 initial={{ x: '-100%' }}
218 |                 animate={{ x: 0 }}
219 |                 exit={{ x: '-100%' }}
220 |                 transition={{ type: 'spring', damping: 25, stiffness: 300 }}
221 |                 className="fixed top-0 left-0 bottom-0 w-[280px] bg-[#0f0f1a] border-r border-[#1e293b] z-[70] f...
222 |             >
223 |                 {/ * Drawer Header */}
224 |                 <div className="p-4 border-b border-[#1e293b] flex items-center justify-between bg-[#16213e]/50">
225 |                     <div className="flex items-center gap-3">
226 |                         {userProfile ? (
227 |                             <div className="w-10 h-10 rounded-full bg-[#1e293b] flex items-center justify-center borde...
228 |                             <UserIcon size={20} className="text-[#00d2a0]" />
229 |                         </div>
230 |                         ) : (
231 |                             
232 |                         )}
233 |                     <div>
234 |                         <div className="font-bold text-sm text-white">
235 |                             {userProfile ? userProfile.full_name : 'PC MASTER LMS'}
236 |                         </div>
237 |                         {userProfile && (
238 |                             <div className="text-xs text-[#00d2a0] uppercase font-bold">{role}</div>
239 |                         )}
240 |                     </div>
241 |                 </div>
242 |                 <button onClick={() => setMobileMenuOpen(false)} className="p-2 text-slate-400 hover:text-whit...
243 |                     <X size={20} />
244 |                 </button>
245 |             </div>
246 |
247 |             {/ * Drawer Links */}
248 |             <div className="flex-1 overflow-y-auto py-4 px-3 flex flex-col gap-1">
249 |                 {activeMenu.map(item => {
250 |                     const isActive = pathname === item.path || pathname?.startsWith(item.path + '/')
251 |                     return (
252 |                         <Link
253 |                             key={item.path}
254 |                             href={item.path}
255 |                             onClick={() => setMobileMenuOpen(false)}
256 |                             className={`flex items-center gap-3 px-4 py-3 rounded-xl font-medium transition-colors $${
257 |                                 isActive
258 |                                     ? 'bg-[#00d2a0]/10 text-[#00d2a0] border border-[#00d2a0]/20'

```

```

259 |         : 'text-slate-300 hover:bg-[#16213e] hover:text-white'
260 |       }}
261 |     >
262 |       {item.icon}
263 |       {item.name}
264 |     </Link>
265 |   )
266 | }}
267 |
268 | {userProfile && (
269 |   <div className="h-[1px] bg-[#1e293b] my-3 mx-2"></div>
270 |   <Link
271 |     href="/profile"
272 |     onClick={() => setMobileMenuOpen(false)}
273 |     className="flex items-center gap-3 px-4 py-3 rounded-xl font-medium text-slate-300 hover...
274 |   >
275 |     <UserIcon size={18} /> Hồ s i cá n h â n
276 |
277 |     <Link
278 |       href="/settings"
279 |       onClick={() => setMobileMenuOpen(false)}
280 |       className="flex items-center gap-3 px-4 py-3 rounded-xl font-medium text-slate-300 hover...
281 |     >
282 |       <Settings size={18} /> Cài . t
283 |
284 |     </Link>
285 |   </>
286 | )}
287 | </div>
288 |
289 | {/ * Drawer Footer */}
290 | <div className="p-4 border-t border-[#1e293b] bg-[#16213e]/30">
291 |   {userProfile ? (
292 |     <button
293 |       onClick={() => { setMobileMenuOpen(false); handleLogout(); }}
294 |       className="w-full flex items-center justify-center gap-2 py-3 rounded-xl font-bold bg-red-...
295 |     >
296 |       <Logout size={18} /> n g x u ̣ t
297 |     </button>
298 |   ) : (
299 |     <Link
300 |       href="/login"
301 |       onClick={() => setMobileMenuOpen(false)}
302 |       className="w-full flex items-center justify-center py-3 rounded-xl font-bold bg-[#00d2a0] ...
303 |     >
304 |       n g n h - p / n g k ý
305 |     </Link>
306 |   )}
307 | </div>
308 | </motion.div>
309 | </>
310 | )}
311 | </AnimatePresence>
312 | </>
313 | )
314 | }
315 |

```

File: components/NotificationBar.js (49 lines, 2.6 KB)

```

1 | 'use client';
2 | import React, { useState, useEffect } from 'react';
3 | import { supabase } from '@lib/supabase';
4 | import { Bell, X } from 'lucide-react';
5 |
6 | export default function NotificationBar() {
7 |   const [notifications, setNotifications] = useState([]);
8 |   const [dismissed, setDismissed] = useState(new Set());
9 |
10 |   useEffect(() => {
11 |     const fetch = async () => {
12 |       const { data } = await supabase
13 |         .from('notifications')
14 |         .select('*')
15 |         .eq('is_global', true)
16 |         .order('created_at', { ascending: false })
17 |         .limit(3);
18 |       setNotifications(data || []);
19 |     };
20 |     fetch();
21 |   }, []);
22 |
23 |   const visible = notifications.filter(n => !dismissed.has(n.id));
24 |   if (visible.length === 0) return null;
25 |
26 |   return (
27 |     <div style={{ position: 'fixed', top: '16px', right: '16px', zIndex: 500, display: 'flex', flexDirecti...
28 |       {visible.map(n => (
29 |         <div key={n.id} style={{ background: 'rgba(10,20,40,0.97)', border: '1px solid rgba(0,243,255,...
30 |         <style>{@keyframes slideInRight { from { transform: translateX(100px); opacity: 0; } to {...
31 |         <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'flex-start', ...
32 |           <div style={{ display: 'flex', gap: '10px', alignItems: 'flex-start' }}>
33 |             <Bell size={16} color="#00f3ff" style={{ flexShrink: 0, marginTop: '2px' }} />
34 |             <div>
35 |               <p style={{ margin: '0 0 4px 0', fontSize: '13px', fontWeight: 700, color: 'wh...
36 |                 {n.content && <p style={{ margin: 0, fontSize: '12px', color: '#8899aa', lineH...
37 |                 <p style={{ margin: '6px 0 0 0', fontSize: '11px', color: '#4b5563' }}>{new Da...
38 |             </div>
39 |           </div>
40 |           <button onClick={() => setDismissed(prev => new Set([...prev, n.id])} style={{ backgr...
41 |             <X size={16} />
42 |           </button>
43 |         </div>
44 |       </div>
45 |     )}
46 |   </div>
47 | );
48 | }
49 |

```

File: components/NotificationBell.tsx (122 lines, 5.5 KB)

```

1 | 'use client'
2 |
3 | import React, { useEffect, useState, useRef } from 'react'
4 | import { Bell, CheckCheck, X } from 'lucide-react'
5 | import { createClientComponentClient } from '@supabase/auth-helpers-nextjs'
6 | import { useNotifications } from '@hooks/useNotifications'
7 |
8 | export default function NotificationBell() {
9 |   const supabase = createClientComponentClient()

```

```

10 | const [userId, setUserId] = useState<string | null>(null)
11 | const [isOpen, setIsOpen] = useState(false)
12 | const dropdownRef = useRef<HTMLDivElement>(null)
13 |
14 | useEffect(() => {
15 |   let mounted = true
16 |   supabase.auth.getUser().then(({ data: { user } }) => {
17 |     if (mounted) setUserId(user?.id ?? null)
18 |   })
19 |   return () => { mounted = false }
20 | }, [supabase])
21 |
22 | useEffect(() => {
23 |   const handleClickOutside = (e: MouseEvent) => {
24 |     if (dropdownRef.current && !dropdownRef.current.contains(e.target as Node)) {
25 |       setIsOpen(false)
26 |     }
27 |   }
28 |   if (isOpen) {
29 |     document.addEventListener('mousedown', handleClickOutside)
30 |   }
31 |   return () => document.removeEventListener('mousedown', handleClickOutside)
32 | }, [isOpen])
33 |
34 | const { unreadCount, notifications, markAsRead, markAllAsRead } = useNotifications(userId ?? undefined)
35 |
36 | return (
37 |   <div ref={dropdownRef} style={{ position: 'relative', display: 'inline-block' }}>
38 |     <button
39 |       aria-label="Notifications"
40 |       className="p-2 rounded-md hover:bg-[rgba(255,255,255,0.02)]"
41 |       onClick={() => setIsOpen(prev => !prev)}
42 |       style={{ cursor: 'pointer' }}
43 |     >
44 |       <Bell size={18} />
45 |     </button>
46 |     {unreadCount > 0 && (
47 |       <span style={{ position: 'absolute', top: 0, right: 0, transform: 'translate(30%,-30%)' }}
48 |         className="text-xs bg-[#f03060] text-white rounded-full w-5 h-5 flex items-center justify-center">
49 |           {unreadCount > 99 ? '99+' : unreadCount}
50 |         </span>
51 |       )}
52 |     {isOpen && (
53 |       <div style={{
54 |         position: 'absolute', top: '100%', right: 0, marginTop: '8px',
55 |         width: '360px', maxHeight: '480px', overflow: 'hidden',
56 |         background: 'var(--bg-surface)', borderRadius: '12px',
57 |         border: '1px solid var(--border-strong)',
58 |         boxShadow: '0 12px 40px rgba(0,0,0,0.25)',
59 |         zIndex: 9999, display: 'flex', flexDirection: 'column',
60 |       }}>
61 |         <div style={{
62 |           display: 'flex', alignItems: 'center', justifyContent: 'space-between',
63 |           padding: '14px 16px', borderBottom: '1px solid var(--border-default)',
64 |         }}>
65 |           <span style={{ fontWeight: 700, fontSize: '14px' }}>Thông báo</span>
66 |           <div style={{ display: 'flex', gap: '8px', alignItems: 'center' }}>
67 |             {unreadCount > 0 && (
68 |               <button onClick={markAllAsRead}
69 |                 style={{ display: 'flex', alignItems: 'center', gap: '4px', padding: '4px 10px', borderRadiu...
70 |                 <span style={{ display: 'flex', gap: '4px', padding: '4px 10px', borderRadiu...
71 |               </button>
72 |             )}
73 |             <button onClick={() => setIsOpen(false)}
74 |               style={{ background: 'transparent', border: 'none', color: 'var(--text-muted)', cursor: 'point...
75 |             <span style={{ padding: '0 16px' }}>
76 |               <X size={16} />
77 |             </button>
78 |           </div>
79 |         </div>
80 |         <div style={{ overflowY: 'auto', flex: 1 }}>
81 |           {notifications.length === 0 ? (
82 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
83 |             <Bell size={24} style={{ margin: '0 auto 8px', opacity: 0.3 }} />
84 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
85 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
86 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
87 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
88 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
89 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
90 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
91 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
92 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
93 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
94 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
95 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
96 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
97 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
98 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
99 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
100 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
101 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
102 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
103 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
104 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
105 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
106 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
107 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
108 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
109 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
110 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
111 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
112 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
113 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
114 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
115 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
116 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
117 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
118 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
119 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
120 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
121 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...
122 |             <div style={{ padding: '40px 16px', textAlign: 'center', color: 'var(--text-muted)', fontSize: '...

```

File: components/notifications/NotificationBell.tsx (85 lines, 3.1 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useRef, useEffect } from 'react'
4 | import { Bell } from 'lucide-react'
5 | import { useNotifications } from '@hooks/useNotifications'
6 | import { NotificationItem } from './NotificationItem'
7 | import Link from 'next/link'

```



```

8 |
9 | export function NotificationBell({ userId }: { userId?: string }) {
10 |   const [isOpen, setIsOpen] = useState(false)
11 |   const { notifications, unreadCount, markAsRead, markAllAsRead } = useNotifications(userId)
12 |   const dropdownRef = useRef<HTMLDivElement>(null)
13 |
14 |   // Close dropdown when clicking outside
15 |   useEffect(() => {
16 |     function handleClickOutside(event: MouseEvent) {
17 |       if (dropdownRef.current && !dropdownRef.current.contains(event.target as Node)) {
18 |         setIsOpen(false)
19 |       }
20 |     }
21 |     document.addEventListener("mousedown", handleClickOutside)
22 |     return () => document.removeEventListener("mousedown", handleClickOutside)
23 |   }, [])
24 |
25 |   return (
26 |     <div className="relative" ref={dropdownRef}>
27 |       </* Bell button */>
28 |       <button
29 |         onClick={() => setIsOpen(!isOpen)}
30 |         className="relative p-2 text-slate-400 hover:text-white hover:bg-[#16213e] rounded-full transition-col...
31 |         aria-label={ `Thông báo ${unreadCount > 0 ? ` (${unreadCount} ch`
32 |       >
33 |         <Bell size={20} />
34 |         {unreadCount > 0 && (
35 |           <span className="
36 |             absolute top-1.5 right-1.5 w-2.5 h-2.5
37 |             bg-[#e84855] border-2 border-[#0f0f1a] rounded-full
38 |             "></span>
39 |         )}
40 |       </button>
41 |
42 |       { /* Dropdown - 5 notifications mõi nhĩt */ }
43 |       {isOpen && (
44 |         <div className="
45 |           absolute right-0 top-full mt-2 w-80
46 |           bg-[#1a1c25] border border-gray-800
47 |           rounded-xl shadow-2xl z-50 overflow-hidden
48 |         ">
49 |           <div className="flex items-center justify-between px-4 py-3 border-b border-gray-800 bg-[#16213e]/50">
50 |             <span className="text-sm font-bold text-[#dde0ed]">Thông báo</span>
51 |             {unreadCount > 0 && (
52 |               <button onClick={markAllAsRead} className="text-xs text-[#00d4aa] font-medium hover:underline">
53 |                 i c t ỹ t c ẽ
54 |               </button>
55 |             )}
56 |           </div>
57 |
58 |           <div className="max-h-[350px] overflow-y-auto">
59 |             {notifications.length === 0 ? (
60 |               <div className="p-6 text-center text-sm text-gray-400">
61 |                 C h ° a c ó t h ò n g b á o n à o
62 |               </div>
63 |             ) : (
64 |               notifications.slice(0, 5).map(n => (
65 |                 <NotificationItem key={n.id} notification={n} onRead={markAsRead} />
66 |               ))
67 |             )}
68 |           </div>
69 |
70 |           <Link
71 |             href="/notifications"
72 |             onClick={() => setIsOpen(false)}
73 |             className="
74 |               block text-center py-3 text-xs font-medium text-[#00d4aa]
75 |               border-t border-gray-800 bg-[#16213e]/30 hover:bg-[#16213e] transition-colors
76 |             "
77 |           >
78 |             X e m t ỹ t c ẽ t h ò n g b á o !'
79 |           </Link>
80 |         </div>
81 |       )}
82 |     </div>
83 |   )
84 | }
85 |

```

File: components/notifications/NotificationItem.tsx (64 lines, 2.5 KB)

```

1 | 'use client'
2 |
3 | import React from 'react'
4 | import Link from 'next/link'
5 | import { Notification } from '@/hooks/useNotifications'
6 | import { CheckCircle, BookOpen, Clock, MessageSquare, Trophy, FileText, Bell } from 'lucide-react'
7 |
8 | interface Props {
9 |   notification: Notification
10 |   onRead: (id: string) => void
11 | }
12 |
13 | export function NotificationItem({ notification, onRead }: Props) {
14 |   const getIcon = () => {
15 |     switch (notification.type) {
16 |       case 'quiz_graded': return <CheckCircle size={16} className="text-[#00d4aa]" />
17 |       case 'lesson_unlocked': return <BookOpen size={16} className="text-blue-400" />
18 |       case 'deadline_reminder': return <Clock size={16} className="text-orange-400" />
19 |       case 'teacher_message': return <MessageSquare size={16} className="text-purple-400" />
20 |       case 'achievement_earned': return <Trophy size={16} className="text-yellow-400" />
21 |       case 'certificate_ready': return <FileText size={16} className="text-pink-400" />
22 |       default: return <Bell size={16} className="text-gray-400" />
23 |     }
24 |   }
25 |
26 |   const timeAgo = (dateStr: string) => {
27 |     const diff = Math.floor((new Date().getTime() - new Date(dateStr).getTime()) / 1000)
28 |     if (diff < 60) return `V ẽ a x o n g`
29 |     if (diff < 3600) return `${Math.floor(diff / 60)} phút tr ° 0 c`
30 |     if (diff < 86400) return `${Math.floor(diff / 3600)} gi Ỗ tr ° 0 c`
31 |     return `${Math.floor(diff / 86400)} ng ả y tr ° 0 c`
32 |   }
33 |
34 |   return (
35 |     <Link
36 |       href={notification.action_url || '#'}
37 |       onClick={() => {
38 |         if (!notification.is_read) onRead(notification.id)
39 |       }}
40 |       className={`block px-4 py-3 border-b border-white/5 hover:bg-white/5 transition-colors ${!notification.i...
41 |     >
42 |     <div className="flex gap-3">

```

```

43 |         <div className="mt-1 flex-shrink-0">
44 |             {getIcon()}
45 |         </div>
46 |         <div className="flex-1 min-w-0">
47 |             <p className={`text-sm font-medium truncate ${!notification.is_read ? 'text-white font-bold' : 'text...`
48 |                 {notification.title}
49 |             </p>
50 |             <p className="text-xs text-gray-400 mt-0.5 line-clamp-2">
51 |                 {notification.body}
52 |             </p>
53 |             <p className="text-[10px] text-gray-500 mt-1">
54 |                 {timeAgo(notification.created_at)}
55 |             </p>
56 |         </div>
57 |         {!notification.is_read && (
58 |             <div className="w-2 h-2 bg-[#e84855] rounded-full mt-2 flex-shrink-0"></div>
59 |         )}
60 |     </div>
61 | </Link>
62 | )
63 | }
64 |

```

File: components/notifications/notifications/NotificationBell.tsx (85 lines, 3.1 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useRef, useEffect } from 'react'
4 | import { Bell } from 'lucide-react'
5 | import { useNotifications } from '@hooks/useNotifications'
6 | import { NotificationItem } from './NotificationItem'
7 | import Link from 'next/link'
8 |
9 | export function NotificationBell({ userId, filterTypes }: { userId?: string, filterTypes?: string[] }) {
10 |     const [isOpen, setIsOpen] = useState(false)
11 |     const { notifications, unreadCount, markAsRead, markAllAsRead } = useNotifications(userId, { filterTypes })
12 |     const dropdownRef = useRef<HTMLDivElement>(null)
13 |
14 |     // Close dropdown when clicking outside
15 |     useEffect(() => {
16 |         function handleClickOutside(event: MouseEvent) {
17 |             if (dropdownRef.current && !dropdownRef.current.contains(event.target as Node)) {
18 |                 setIsOpen(false)
19 |             }
20 |         }
21 |         document.addEventListener("mousedown", handleClickOutside)
22 |         return () => document.removeEventListener("mousedown", handleClickOutside)
23 |     }, [])
24 |
25 |     return (
26 |         <div className="relative" ref={dropdownRef}>
27 |             /* Bell button */
28 |             <button
29 |                 onClick={() => setIsOpen(!isOpen)}
30 |                 className="relative p-2 text-slate-400 hover:text-white hover:bg-[#16213e] rounded-full transition-col...
31 |                 aria-label={ `Thông báo ${unreadCount > 0 ? ` (${unreadCount} ch...`
32 |             >
33 |                 <Bell size={20} />
34 |                 {unreadCount > 0 && (
35 |                     <span className="
36 |                         absolute top-1.5 right-1.5 w-2.5 h-2.5
37 |                         bg-[#e84855] border-2 border-[#0f0f1a] rounded-full
38 |                     "></span>
39 |                 )}
40 |             </button>
41 |
42 |             { /* Dropdown - 5 notifications mõi nhét */
43 |             {isOpen && (
44 |                 <div className="
45 |                     absolute right-0 top-full mt-2 w-80
46 |                     bg-[#1a1c25] border border-gray-800
47 |                     rounded-xl shadow-2xl z-50 overflow-hidden
48 |                 ">
49 |                     <div className="flex items-center justify-between px-4 py-3 border-b border-gray-800 bg-[#16213e]/50">
50 |                         <span className="text-sm font-bold text-[#dde0ed]">Thông báo</span>
51 |                         {unreadCount > 0 && (
52 |                             <button onClick={markAllAsRead} className="text-xs text-[#00d4aa] font-medium hover:underline">
53 |                                 Íc t t c f
54 |                             </button>
55 |                         )}
56 |                     </div>
57 |
58 |                     <div className="max-h-[350px] overflow-y-auto">
59 |                         {notifications.length === 0 ? (
60 |                             <div className="p-6 text-center text-sm text-gray-400">
61 |                                 C h ° a c ó t h ò n g b á o n à o
62 |                             </div>
63 |                         ) : (
64 |                             notifications.slice(0, 5).map(n => (
65 |                                 <NotificationItem key={n.id} notification={n} onRead={markAsRead} />
66 |                             ))
67 |                         )}
68 |                     </div>
69 |
70 |                     <Link
71 |                         href="/notifications"
72 |                         onClick={() => setIsOpen(false)}
73 |                         className="
74 |                             block text-center py-3 text-xs font-medium text-[#00d4aa]
75 |                             border-t border-gray-800 bg-[#16213e]/30 hover:bg-[#16213e] transition-colors
76 |                         "
77 |                     >
78 |                         X e m t t c f t h ò n g b á o !'
79 |                     </Link>
80 |                 </div>
81 |             )}
82 |         </div>
83 |     )
84 | }
85 |

```

File: components/notifications/notifications/NotificationItem.tsx (64 lines, 2.5 KB)

```

1 | 'use client'
2 |
3 | import React from 'react'
4 | import Link from 'next/link'
5 | import { Notification } from '@hooks/useNotifications'
6 | import { CheckCircle, BookOpen, Clock, MessageSquare, Trophy, FileText, Bell } from 'lucide-react'
7 |
8 | interface Props {

```

```

9 | notification: Notification
10 | onRead: (id: string) => void
11 | }
12 |
13 | export function NotificationItem({ notification, onRead }: Props) {
14 |   const getIcon = () => {
15 |     switch (notification.type) {
16 |       case 'quiz_graded': return <CheckCircle size={16} className="text-[#00d4aa]" />
17 |       case 'lesson_unlocked': return <BookOpen size={16} className="text-blue-400" />
18 |       case 'deadline_reminder': return <Clock size={16} className="text-orange-400" />
19 |       case 'teacher_message': return <MessageSquare size={16} className="text-purple-400" />
20 |       case 'achievement_earned': return <Trophy size={16} className="text-yellow-400" />
21 |       case 'certificate_ready': return <FileText size={16} className="text-pink-400" />
22 |       default: return <Bell size={16} className="text-gray-400" />
23 |     }
24 |   }
25 |
26 |   const timeAgo = (dateStr: string) => {
27 |     const diff = Math.floor((new Date().getTime() - new Date(dateStr).getTime()) / 1000)
28 |     if (diff < 60) return 'Vừa xong'
29 |     if (diff < 3600) return `${Math.floor(diff / 60)} phút tr° 0 c`
30 |     if (diff < 86400) return `${Math.floor(diff / 3600)} g i ý tr° 0 c`
31 |     return `${Math.floor(diff / 86400)} ngày tr° 0 c`
32 |   }
33 |
34 |   return (
35 |     <Link
36 |       href={notification.action_url || '#'}
37 |       onClick={() => {
38 |         if (!notification.is_read) onRead(notification.id)
39 |       }}
40 |       className={`block px-4 py-3 border-b border-white/5 hover:bg-white/5 transition-colors ${!notification.i...
41 |     >
42 |       <div className="flex gap-3">
43 |         <div className="mt-1 flex-shrink-0">
44 |           {getIcon()}
45 |         </div>
46 |         <div className="flex-1 min-w-0">
47 |           <p className={`text-sm font-medium truncate ${!notification.is_read ? 'text-white font-bold' : 'text...
48 |             {notification.title}
49 |           </p>
50 |           <p className="text-xs text-gray-400 mt-0.5 line-clamp-2">
51 |             {notification.body}
52 |           </p>
53 |           <p className="text-[10px] text-gray-500 mt-1">
54 |             {timeAgo(notification.created_at)}
55 |           </p>
56 |         </div>
57 |       <div className="flex items-center justify-between">
58 |         {!notification.is_read && (
59 |           <div className="w-2 h-2 bg-[#e84855] rounded-full mt-2 flex-shrink-0"></div>
60 |         )}
61 |       </div>
62 |     </Link>
63 |   )
64 | }

```

File: components/notifications/notifications/ParentNotificationBell.tsx (21 lines, 0.4 KB)

```

1 | 'use client'
2 |
3 | import React from 'react'
4 | import { NotificationBell } from './NotificationBell'
5 |
6 | export function ParentNotificationBell({ userId }: { userId?: string }) {
7 |   return (
8 |     <NotificationBell
9 |       userId={userId}
10 |       filterTypes={[
11 |         'child_absent',
12 |         'child_low_score',
13 |         'child_completed_path',
14 |         'lesson_completed',
15 |         'quiz_graded',
16 |         'achievement_earned'
17 |       ]}
18 |     />
19 |   )
20 | }
21 |

```

File: components/Onboarding.tsx (145 lines, 4.9 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useEffect } from 'react'
4 | import { motion, AnimatePresence } from 'framer-motion'
5 | import { Sparkles, Cpu, BookOpen, Trophy, ArrowRight, X } from 'lucide-react'
6 | import confetti from 'canvas-confetti'
7 |
8 | export default function Onboarding() {
9 |   const [show, setShow] = useState(false)
10 |   const [step, setStep] = useState(1)
11 |
12 |   useEffect(() => {
13 |     const isDone = localStorage.getItem('onboarding_done')
14 |     if (!isDone) {
15 |       const timer = setTimeout(() => setShow(true), 1500)
16 |       return () => clearTimeout(timer)
17 |     }
18 |   }, [])
19 |
20 |   const handleNext = () => {
21 |     if (step === 4) {
22 |       handleClose()
23 |     } else {
24 |       setStep(step + 1)
25 |       if (step === 3) {
26 |         confetti({
27 |           particleCount: 100,
28 |           spread: 70,
29 |           origin: { y: 0.6 }
30 |         })
31 |       }
32 |     }
33 |   }
34 |
35 |   const handleClose = () => {
36 |     localStorage.setItem('onboarding_done', 'true')
37 |     setShow(false)
38 |   }

```

```

39 |   if (!show) return null
40 |
41 |
42 |   const steps = [
43 |     {
44 |       id: 1,
45 |       title: 'Chào mừng bạn đến với PC Master!',
46 |       desc: 'Nắm tắt những tính năng mới nhất dành cho bạn',
47 |       icon: <Sparkles className="text-yellow-400" size={48} />,
48 |       color: 'bg-yellow-400/10'
49 |     },
50 |     {
51 |       id: 2,
52 |       title: 'Phòng Lab 3D thực tế',
53 |       desc: 'Lắp ráp hơn 50+ linh kiện, thiết lập thông số v-t lý chính xác',
54 |       icon: <Cpu className="text-[#00d2a0]" size={48} />,
55 |       color: 'bg-[#00d2a0]/10'
56 |     },
57 |     {
58 |       id: 3,
59 |       title: 'Híct chuyên gia',
60 |       desc: 'Hết những bài giảng video đang tác và sẽ hỗ trợ cặ AI Guru',
61 |       icon: <BookOpen className="text-blue-400" size={48} />,
62 |       color: 'bg-blue-400/10'
63 |     },
64 |     {
65 |       id: 4,
66 |       title: 'Sẵn sàng chưa?',
67 |       desc: 'Bắt đầu hành trình chinh phục công nghệ của bạn ngay bây giờ',
68 |       icon: <Trophy className="text-purple-400" size={48} />,
69 |       color: 'bg-purple-400/10'
70 |     }
71 |   ]
72 |
73 |   const currentStep = steps[step - 1]
74 |
75 |   return (
76 |     <AnimatePresence>
77 |       <div className="fixed inset-0 z-[9999] flex items-center justify-center p-6">
78 |         <motion.div
79 |           initial={{ opacity: 0 }}
80 |           animate={{ opacity: 1 }}
81 |           exit={{ opacity: 0 }}
82 |           className="absolute inset-0 bg-black/80 backdrop-blur-md"
83 |           onClick={handleClose}
84 |         />
85 |
86 |         <motion.div
87 |           initial={{ opacity: 0, scale: 0.9, y: 20 }}
88 |           animate={{ opacity: 1, scale: 1, y: 0 }}
89 |           exit={{ opacity: 0, scale: 0.9, y: 20 }}
90 |           className="relative w-full max-w-lg bg-[#16213e] border border-[#1e293b] rounded-[32px] overflow-hid...
91 |         >
92 |           <button
93 |             onClick={handleClose}
94 |             className="absolute top-6 right-6 p-2 text-slate-500 hover:text-white transition-colors z-10"
95 |           >
96 |             <X size={20} />
97 |           </button>
98 |
99 |           <div className="p-8 md:p-12 text-center">
100 |             <motion.div
101 |               key={step}
102 |               initial={{ opacity: 0, scale: 0.5 }}
103 |               animate={{ opacity: 1, scale: 1 }}
104 |               className={`w-24 h-24 ${currentStep.color} rounded-3xl flex items-center justify-center mx-auto ...
105 |             >
106 |               {currentStep.icon}
107 |             </motion.div>
108 |
109 |             <motion.div
110 |               key={`text-${step}`}
111 |               initial={{ opacity: 0, x: 20 }}
112 |               animate={{ opacity: 1, x: 0 }}
113 |               className="space-y-4 mb-10"
114 |             >
115 |               <h2 className="text-3xl font-black text-white leading-tight">{currentStep.title}</h2>
116 |               <p className="text-slate-400 text-lg leading-relaxed">{currentStep.desc}</p>
117 |             </motion.div>
118 |
119 |             <div className="flex flex-col gap-4">
120 |               <button
121 |                 onClick={handleNext}
122 |                 className="w-full py-4 bg-[#00d2a0] hover:bg-[#00e6af] text-black font-black rounded-2xl trans...
123 |               >
124 |                 {step === 4 ? 'B@T ! U N G A Y' : 'T I ¼ P T ä C '}
125 |               <ArrowRight size={20} className="group-hover:translate-x-1 transition-transform" />
126 |             </button>
127 |
128 |             <div className="flex justify-center gap-2">
129 |               {steps.map((_, i) => (
130 |                 <div
131 |                   key={i}
132 |                   className={`h-1.5 rounded-full transition-all duration-300 ${
133 |                     i + 1 === step ? 'w-8 bg-[#00d2a0]' : 'w-2 bg-[#1e293b]'
134 |                   }`}
135 |                 />
136 |               ))}
137 |             </div>
138 |           </div>
139 |         </div>
140 |       </motion.div>
141 |     </AnimatePresence>
142 |   )
143 | }
144 |
145 |

```

File: components/OSInstallationSimulator.js (1129 lines, 43.4 KB)

```

1 | 'use client';
2 |
3 | import React, { useState, useEffect, useRef } from 'react';
4 |
5 | /**
6 |  * @typedef {import('@types/pc-components').Build} Build
7 |  * @typedef {import('@types/pc-components').CPUProduct} CPUProduct
8 |  * @typedef {import('@types/pc-components').GPUProduct} GPUProduct
9 |  * @typedef {import('@types/pc-components').RAMProduct} RAMProduct
10 |  * @typedef {import('@types/pc-components').SSDProduct} SSDProduct
11 |  */
12 |
13 | const WINDOWS_VERSION = 'Windows 11 Pro (24H2)';

```

```

14 |
15 | const WIFI_NETWORKS = [
16 |   { id: 1, name: 'Nhà riêng', strength: 4, secured: true },
17 |   { id: 2, name: 'Văn phòng PCM', strength: 5, secured: true },
18 |   { id: 3, name: 'WiFi Quán Cà Phê', strength: 2, secured: false },
19 |   { id: 4, name: 'Mình Lãng Giãng', strength: 3, secured: true },
20 |   { id: 5, name: '5G Hotspot', strength: 5, secured: true },
21 |   { id: 6, name: 'WiFi Trờng Híc', strength: 4, secured: true },
22 | ];
23 |
24 | const SETUP_STEPS = [
25 |   { label: 'Giới thiệu', icon: '0=00' },
26 |   { label: 'Ngôn ngữ', icon: '0<B' },
27 |   { label: 'Cài đặt', icon: '&"p' },
28 |   { label: 'Tài khoản', icon: '0=0d' },
29 |   { label: 'Quyền riêng tư', icon: '0=Y' },
30 |   { label: 'Mình', icon: '0=00' },
31 |   { label: 'Hoàn tất', icon: '0<B' },
32 | ];
33 |
34 | const INSTALL_MESSAGES = [
35 |   { min: 0, max: 12, text: 'ang sao chép t-p tin cài .t...'},
36 |   { min: 12, max: 28, text: 'ang cài .t tinh n ng Windows...'},
37 |   { min: 28, max: 45, text: 'ang áp đ ang cài .t h c th n ng...'},
38 |   { min: 45, max: 65, text: 'ang chu n b ã th i t b ã c c a b i n...'},
39 |   { min: 65, max: 82, text: 'ang th i t l - p t ù y c h ã n h...'},
40 |   { min: 82, max: 95, text: 'S _ p x o n g r ó i...'},
41 |   { min: 95, max: 100, text: 'ang hoàn t t...'},
42 | ];
43 |
44 | function LoadingSpinner({ size, color }) {
45 |   return (
46 |     <div
47 |       style={{
48 |         width: size || 40,
49 |         height: size || 40,
50 |         border: '3px solid rgba(255,255,255,0.15)',
51 |         borderTopColor: color || 'var(--accent-blue, #0078d4)',
52 |         borderRadius: '50%',
53 |         animation: 'ospin 0.8s linear infinite',
54 |       }}
55 |     />
56 |   );
57 | }
58 |
59 | function ToggleSwitch({ checked, onChange, disabled }) {
60 |   return (
61 |     <button
62 |       onClick={() => { if (!disabled) onChange(!checked); }}
63 |       disabled={disabled}
64 |       style={{
65 |         width: '44px', height: '24px',
66 |         background: checked ? '#0078d4' : 'rgba(255,255,255,0.15)',
67 |         borderRadius: '12px', border: 'none',
68 |         cursor: disabled ? 'default' : 'pointer',
69 |         position: 'relative', transition: 'background 0.2s',
70 |         padding: 0, flexShrink: 0,
71 |       }}
72 |       type="button"
73 |     >
74 |       <div
75 |         style={{
76 |           width: '18px', height: '18px',
77 |           background: 'white', borderRadius: '50%',
78 |           position: 'absolute', top: '3px',
79 |           left: checked ? '23px' : '3px',
80 |           transition: 'left 0.2s',
81 |           boxShadow: '0 1px 3px rgba(0,0,0,0.3)',
82 |         }}
83 |       />
84 |     </button>
85 |   );
86 | }
87 |
88 | function WiFiIcon({ strength }) {
89 |   return (
90 |     <span style={{ display: 'inline-flex', alignItems: 'flex-end', gap: '2px', height: '16px' }}>
91 |       {[1, 2, 3, 4, 5].map(i => (
92 |         <span
93 |           key={i}
94 |           style={{
95 |             width: '3px',
96 |             height: `${4 + i * 2.5}px`,
97 |             background: i <= strength ? '#fff' : 'rgba(255,255,255,0.2)',
98 |             borderRadius: '1px 1px 0 0',
99 |             transition: 'background 0.3s',
100 |           }}
101 |         />
102 |       ))}
103 |     </span>
104 |   );
105 | }
106 |
107 | const glassPanel = {
108 |   background: 'rgba(0, 103, 192, 0.15)',
109 |   backdropFilter: 'blur(12px)',
110 |   WebkitBackdropFilter: 'blur(12px)',
111 |   border: '1px solid rgba(255,255,255,0.12)',
112 |   borderRadius: '12px',
113 | };
114 |
115 | const win11Btn = (active) => ({
116 |   padding: '8px 32px',
117 |   background: active ? '#0078d4' : 'rgba(255,255,255,0.08)',
118 |   color: active ? 'white' : 'rgba(255,255,255,0.4)',
119 |   border: 'none',
120 |   borderRadius: '6px',
121 |   fontSize: '14px',
122 |   fontWeight: 600,
123 |   cursor: active ? 'pointer' : 'default',
124 |   fontFamily: 'inherit',
125 |   transition: 'all 0.15s',
126 |   opacity: active ? 1 : 0.5,
127 | });
128 |
129 | function StepIndicator({ currentStep, steps }) {
130 |   return (
131 |     <div
132 |       style={{
133 |         display: 'flex', justifyContent: 'center', alignItems: 'center',
134 |         gap: '8px', padding: '16px 20px',
135 |       }}
136 |     >
137 |       {steps.map((step, i) => (

```

```

138 |     <React.Fragment key={i}>
139 |       <div style={{ display: 'flex', alignItems: 'center', gap: '8px' }}>
140 |         <div
141 |           style={{
142 |             width: '32px', height: '32px', borderRadius: '50%',
143 |             display: 'flex', alignItems: 'center', justifyContent: 'center',
144 |             fontSize: '12px', fontWeight: 700,
145 |             background: i < currentStep
146 |               ? '#0078d4'
147 |               : i === currentStep
148 |                 ? 'rgba(0,120,212,0.3)'
149 |                 : 'rgba(255,255,255,0.08)',
150 |             color: i <= currentStep ? 'white' : 'rgba(255,255,255,0.3)',
151 |             border: i === currentStep ? '2px solid #0078d4' : '2px solid transparent',
152 |             transition: 'all 0.3s',
153 |           }}
154 |         >
155 |           { i < currentStep ? '' : step.icon }
156 |         </div>
157 |         <span
158 |           style={{
159 |             color: i <= currentStep ? 'white' : 'rgba(255,255,255,0.3)',
160 |             fontSize: '11px', fontWeight: i === currentStep ? 600 : 400,
161 |             transition: 'color 0.3s',
162 |           }}
163 |         >
164 |           {step.label}
165 |         </span>
166 |       </div>
167 |       {i < steps.length - 1 && (
168 |         <div
169 |           style={{
170 |             width: '32px', height: '2px',
171 |             background: i < currentStep ? '#0078d4' : 'rgba(255,255,255,0.1)',
172 |             transition: 'background 0.3s',
173 |           }}
174 |         >
175 |       </div>
176 |     </React.Fragment>
177 |   )}
178 | </div>
179 | );
180 | }
181 |
182 | function LanguageRegionStep({ lang, region, keyboard, onLangChange, onRegionChange, onKeyboardChange, onNext }) {
183 |   const languages = [ 'Tiếng Việt', 'English (United States)', 'English (
184 |   const regions = [ 'Việt Nam', 'United States', 'United Kingdom', 'Japan
185 |   const keyboards = [ 'Tiếng Việt (Telex)', 'Tiếng Việt (VNI)', 'US QWERTY
186 |
187 |   const selectStyle = (val, current) => ({
188 |     padding: '10px 16px',
189 |     width: '100%',
190 |     background: val === current ? 'rgba(0,120,212,0.2)' : 'rgba(255,255,255,0.06)',
191 |     border: val === current ? '1px solid #0078d4' : '1px solid rgba(255,255,255,0.1)',
192 |     borderRadius: '8px',
193 |     color: 'white',
194 |     cursor: 'pointer',
195 |     fontSize: '13px',
196 |     textAlign: 'left',
197 |     fontFamily: 'inherit',
198 |     transition: 'all 0.15s',
199 |   });
200 |
201 |   return (
202 |     <div
203 |       style={{
204 |         display: 'flex', flexDirection: 'column', alignItems: 'center',
205 |         gap: '24px', padding: '24px 20px 40px',
206 |         maxWidth: '480px', margin: '0 auto',
207 |         width: '100%',
208 |       }}
209 |     >
210 |       <div style={{ fontSize: '48px' }}></div>
211 |       <h2 style={{ color: 'white', fontSize: '24px', fontWeight: 300, ma
212 |     </div>
213 |     <div style={{ width: '100%' }}>
214 |       <div style={{ display: 'flex', flexDirection: 'column', gap: '4px', maxHeight: '140px', overflow: 'au...
215 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '4px', maxHeight: '140px', overflow: 'au...
216 |           {languages.map(l => (
217 |             <button key={l} onClick={() => onLangChange(l)} style={selectStyle(l, lang)} type="button">{l}</bu...
218 |           ))}
219 |         </div>
220 |       </div>
221 |
222 |       <div style={{ width: '100%' }}>
223 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '4px' }}>
224 |           {regions.map(r => (
225 |             <button key={r} onClick={() => onRegionChange(r)} style={selectStyle(r, region)} type="button">{r}...
226 |           ))}
227 |         </div>
228 |       </div>
229 |
230 |       <div style={{ width: '100%' }}>
231 |         <div style={{ display: 'flex', flexDirection: 'column', gap: '4px' }}>
232 |           <label style={{ color: 'white', fontSize: '12px', display: 'block', marginBottom: '6px' }}>Bàn phím</la...
233 |           <div style={{ display: 'flex', flexDirection: 'column', gap: '4px' }}>
234 |             {keyboards.map(k => (
235 |               <button key={k} onClick={() => onKeyboardChange(k)} style={selectStyle(k, keyboard)} type="button"...
236 |             ))}
237 |           </div>
238 |         </div>
239 |
240 |         <button onClick={onNext} style={{ ...win11Btn(true), padding: '12px 48px', fontSize: '15px', marginTop: ...
241 |           Tiếng Việt
242 |         </button>
243 |       </div>
244 |     );
245 |   }
246 |
247 |   function InstallingStep({ progress }) {
248 |     const msg = INSTALL_MESSAGES.find(m => progress >= m.min && progress < m.max) || INSTALL_MESSAGES[INSTALL_ME...
249 |
250 |     return (
251 |       <div
252 |         style={{
253 |           display: 'flex', flexDirection: 'column', alignItems: 'center',
254 |           justifyContent: 'center', gap: '32px', padding: '60px 20px',
255 |           minHeight: '400px',
256 |         }}
257 |       >
258 |         <LoadingSpinner size={60} color="#0078d4" />
259 |         <h2 style={{ color: 'white', fontSize: '22px', fontWeight: 300, margin: 0 }}>
260 |           Cài đặt Windows 11
261 |         </h2>

```

```
262 | <div style={{ width: '100%', maxWidth: '400px' }}>
263 |   <div style={{ height: '6px', background: 'rgba(255,255,255,0.1)', borderRadius: '3px', overflow: 'hidd...
264 |     <div
265 |       style={{
266 |         height: '100%', width: `${progress}%`,
267 |         background: 'linear-gradient(90deg, #0078d4, #00b4d8)',
268 |         borderRadius: '3px',
269 |         transition: 'width 0.15s ease-out',
270 |         boxShadow: '0 0 8px rgba(0,120,212,0.5)',
271 |       }}
272 |     />
273 |   </div>
274 |   <div style={{ display: 'flex', justifyContent: 'space-between', marginTop: '6px' }}>
275 |     <span style={{ color: '#888', fontSize: '12px' }}>{Math.round(progress)}%</span>
276 |     <span style={{ color: '#ccc', fontSize: '12px' }}>{msg.text}</span>
277 |   </div>
278 |   <div style={{ color: '#555', fontSize: '12px', textAlign: 'center', maxWidth: '360px', marginTop: '20px'...
279 |     PC Master Build &copy; {new Date().getFullYear()}
280 |   </div>
281 | </div>
282 | </div>
283 | );
284 | }
285 |
286 | function AccountSetupStep({ username, password, confirmPassword, onUsernameChange, onPasswordChange, onConfirm...
287 |   const [error, setError] = useState('');
288 |
289 |   const inputStyle = {
290 |     width: '100%', padding: '12px 16px', background: 'rgba(255,255,255,0.06)',
291 |     border: '1px solid rgba(255,255,255,0.12)', borderRadius: '8px',
292 |     color: 'white', fontSize: '15px', outline: 'none',
293 |     fontFamily: 'inherit', boxSizing: 'border-box',
294 |     transition: 'border-color 0.15s',
295 |   };
296 |
297 |   const handleNext = () => {
298 |     if (!username.trim()) { setError('Vui lòng nh p tên ng Ỗi dùng'); re
299 |     if (password && password !== confirmPassword) { setError('M-t kh Ỗu
300 |     setError('');
301 |     onNext();
302 |   };
303 |
304 |   return (
305 |     <div
306 |       style={{
307 |         display: 'flex', flexDirection: 'column', alignItems: 'center',
308 |         gap: '24px', padding: '40px 20px',
309 |         maxWidth: '420px', margin: '0 auto',
310 |       }}
311 |     >
312 |       <div style={{ fontSize: '48px' }}>0</div>
313 |       <h2 style={{ color: 'white', fontSize: '24px', fontWeight: 300, ma
314 |       <p style={{ color: 'aaa', fontSize: '13px', textAlign: 'center', margin: 0 }}>
315 |         Th i Ỗt l-p t Ỗn v Ỗ m-t kh Ỗu cho th i Ỗt b Ỗ n Ỗy
316 |       </p>
317 |
318 |       {error && (
319 |         <div style={{ color: 'var(--danger, #f46a6a)', fontSize: '13px', background: 'rgba(244,106,106,0.12)',...
320 |         {error}
321 |       </div>
322 |     )}
323 |
324 |     <div style={{ width: '100%', display: 'flex', flexDirection: 'column', gap: '16px' }}>
325 |       <div>
326 |         <label style={{ color: 'aaa', fontSize: '12px', display: 'blo
327 |         <input
328 |           value={username}
329 |           onChange={e => onUsernameChange(e.target.value)}
330 |           placeholder="Nh p t Ỗn c Ỗa b i n"
331 |         style={inputStyle}
332 |       />
333 |     </div>
334 |     <div>
335 |       <label style={{ color: 'aaa', fontSize: '12px', display: 'blo
336 |       <input
337 |         value={password}
338 |         onChange={e => onPasswordChange(e.target.value)}
339 |         type="password"
340 |         placeholder="Nh p m-t kh Ỗu"
341 |       style={inputStyle}
342 |     />
343 |   </div>
344 |   <div>
345 |     <label style={{ color: 'aaa', fontSize: '12px', display: 'blo
346 |     <input
347 |       value={confirmPassword}
348 |       onChange={e => onConfirmChange(e.target.value)}
349 |       type="password"
350 |       placeholder="Nh p l i i m-t kh Ỗu"
351 |     style={inputStyle}
352 |   />
353 | </div>
354 | </div>
355 |
356 |   <div style={{ display: 'flex', gap: '12px', marginTop: '8px' }}>
357 |     <button onClick={onBack} style={{ win11Btn() }} type="button">! Quay
358 |     <button onClick={handleNext} style={{ win11Btn(true) }} type="button"
359 |   </div>
360 | </div>
361 | );
362 | }
363 |
364 | function PrivacySettingsStep({ settings, onToggle, onNext, onBack }) {
365 |   const items = [
366 |     { key: 'location', label: 'V Ỗ tr i', desc: 'Cho ph Ỗp Windows v Ỗ Ỗng c
367 |     { key: 'diagnostics', label: 'Đ i l i Ỗu ch Ỗn o Ỗn', desc: 'Gi i đ i l i Ỗu
368 |     { key: 'ads', label: 'ID qu Ỗng c Ỗo', desc: 'Cho ph Ỗp Ỗng đ Ỗng s i đ Ỗn
369 |   ];
370 |
371 |   return (
372 |     <div
373 |       style={{
374 |         display: 'flex', flexDirection: 'column', alignItems: 'center',
375 |         gap: '24px', padding: '40px 20px',
376 |         maxWidth: '480px', margin: '0 auto',
377 |       }}
378 |     >
379 |       <div style={{ fontSize: '48px' }}>0</div>
380 |       <h2 style={{ color: 'white', fontSize: '24px', fontWeight: 300, ma
381 |       <p style={{ color: 'aaa', fontSize: '13px', textAlign: 'center', margin: 0 }}>
382 |         Ch i n c Ỗi t qu y Ỗn ri Ỗng t Ỗ cho th i Ỗt b Ỗ c Ỗa b i n. B i n c Ỗ th Ỗ th Ỗ
383 |       </p>
384 |
385 |       <div style={{ width: '100%', display: 'flex', flexDirection: 'column', gap: '8px', ...glassPanel, paddin...
```

```

386 |         {items.map(item => (
387 |             <div
388 |                 key={item.key}
389 |                 style={{
390 |                     display: 'flex', justifyContent: 'space-between', alignItems: 'center',
391 |                     padding: '12px 16px', borderRadius: '8px',
392 |                     background: 'rgba(0,0,0,0.15)',
393 |                 }}
394 |             >
395 |                 <div>
396 |                     <div style={{ color: 'white', fontSize: '14px', fontWeight: 600 }}>{item.label}</div>
397 |                     <div style={{ color: '#aaa', fontSize: '11px', marginTop: '2px' }}>{item.desc}</div>
398 |                 </div>
399 |                 <ToggleSwitch checked={settings[item.key]} onChange={() => onToggle(item.key)} />
400 |             </div>
401 |         ))}
402 |     </div>
403 |
404 |     <div style={{ display: 'flex', gap: '12px', marginTop: '8px' }}>
405 |         <button onClick={onBack} style={win11Btn()} type="button">Quay
406 |         <button onClick={onNext} style={win11Btn(true)} type="button">T
407 |     </div>
408 | </div>
409 | );
410 | }
411 |
412 | function NetworkSetupStep({ selectedNetwork, isConnected, onSelect, onConnect, onSkip, onBack }) {
413 |     return (
414 |         <div
415 |             style={{
416 |                 display: 'flex', flexDirection: 'column', alignItems: 'center',
417 |                 gap: '24px', padding: '40px 20px',
418 |                 maxWidth: '480px', margin: '0 auto',
419 |             }}
420 |         >
421 |             <div style={{ fontSize: '48px' }}>0</div>
422 |             <h2 style={{ color: 'white', fontSize: '24px', fontWeight: 300, margin: 0 }}>
423 |                 <p style={{ color: '#aaa', fontSize: '13px', textAlign: 'center', margin: 0 }}>
424 |                     Chín mǐng Wǐ Fǐ À k ò t n Ě i Ĭ n t Ě r n ě t
425 |                 </p>
426 |
427 |                 <div style={{ width: '100%', display: 'flex', flexDirection: 'column', gap: '6px', ...glassPanel, paddin...
428 |                     {WIFI_NETWORKS.map(net => (
429 |                         <button
430 |                             key={net.id}
431 |                             onClick={() => onSelect(net)}
432 |                             style={{
433 |                                 display: 'flex', justifyContent: 'space-between', alignItems: 'center',
434 |                                 padding: '12px 16px', borderRadius: '8px', cursor: 'pointer',
435 |                                 background: selectedNetwork?.id === net.id ? 'rgba(0,120,212,0.2)' : 'rgba(0,0,0,0.15)',
436 |                                 border: selectedNetwork?.id === net.id ? '1px solid #0078d4' : '1px solid transparent',
437 |                                 color: 'white', fontSize: '14px', fontFamily: 'inherit',
438 |                                 textAlign: 'left', transition: 'all 0.15s', width: '100%',
439 |                             }}
440 |                             type="button"
441 |                         >
442 |                             <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
443 |                                 <WiFIcon strength={net.strength} />
444 |                                 <span>{net.name}</span>
445 |                             </div>
446 |                             {net.secured
447 |                                 ? <span style={{ color: '#888', fontSize: '11px' }}>0=Ý B Ē ō
448 |                                 : <span style={{ color: 'var(--warning, #f59e0b)', fontSize:
449 |                             </button>
450 |                         </div>
451 |                     ))}
452 |                 </div>
453 |
454 |                 {isConnected && (
455 |                     <div style={{ color: 'var(--success, #28c76f)', fontSize: '13px', display: 'flex', alignItems: 'center...
456 |                     <span></span> </span> ā k ò t n Ě i v Ū i <strong>{selectedNetwork?.name}</strong>
457 |                 </div>
458 |             )}
459 |
460 |             <div style={{ display: 'flex', gap: '12px', marginTop: '4px' }}>
461 |                 <button onClick={onBack} style={win11Btn()} type="button">Quay
462 |                 {selectedNetwork && isConnected && (
463 |                     <button onClick={onConnect} style={win11Btn(true)} type="button">
464 |                 )}
465 |                 <button onClick={onSkip} style={{ ...win11Btn(true), background: 'transparent', border: '1px solid rgb...
466 |                 B Ĭ qu a !'
467 |             </button>
468 |         </div>
469 |     </div>
470 | );
471 | }
472 |
473 | function SpecRow({ icon, label, value }) {
474 |     return (
475 |         <div
476 |             style={{
477 |                 display: 'flex', alignItems: 'flex-start', gap: '12px',
478 |                 padding: '10px 12px', borderRadius: '8px',
479 |                 background: 'rgba(0,0,0,0.15)',
480 |             }}
481 |         >
482 |             <span style={{ fontSize: '16px', flexShrink: 0, marginTop: '1px' }}>{icon}</span>
483 |             <div>
484 |                 <div style={{ color: '#aaa', fontSize: '11px', fontWeight: 700, textTransform: 'uppercase', marginBott...
485 |                 <div style={{ color: 'white', fontSize: '13px', lineHeight: '1.4' }}>{value}</div>
486 |             </div>
487 |         </div>
488 |     );
489 | }
490 |
491 | function SpecsApp({ build, onClose }) {
492 |     const cpu = build?.cpu;
493 |     const gpu = build?.gpu;
494 |     const ram = build?.ram || [];
495 |     const ssd = build?.ssd || [];
496 |     const hdd = build?.hdd || [];
497 |
498 |     const totalRamGb = ram.reduce((sum, r) => sum + (r.ram_capacity_gb || 0), 0);
499 |     const totalStorageGb = [...ssd, ...hdd].reduce((sum, s) => sum + (s.ssd_capacity_gb || 0), 0);
500 |
501 |     const cpuScore = (cpu?.cpu_cores || 4) * (cpu?.cpu_threads || 8) * ((cpu?.cpu_boost_ghz || 3) * 100)
502 |         + (cpu?.cpu_cinebench_r23_multi || 10000) / 10;
503 |     const gpuScore = (gpu?.gpu_vram_gb || 4) * 2000 + (gpu?.gpu_benchmark_1080p || 5000);
504 |     const ramScore = totalRamGb * 50 + (ram[0]?.ram_speed_mhz || 3200) * 2;
505 |     const storageScore = totalStorageGb * 2 + (ssd[0]?.ssd_read_mbps || 3000);
506 |     const performanceScore = Math.round((cpuScore + gpuScore + ramScore + storageScore) / 5);
507 |
508 |     const getPerformanceLevel = (score) => {
509 |         if (score >= 15000) return { label: 'X u * t s ^ c ^', color: 'var(--success'

```



```

511|         if (score >= 10000) return { label: 'Tốt', color: '#0078d4' };
512|         if (score >= 6000) return { label: 'Trung bình', color: 'var(--warning, #f59e0b)' };
513|         return { label: 'Cải thiện', color: '#888' };
514|     };
515|     const perf = getPerformanceLevel(performanceScore);
516|
517|     return (
518|         <div
519|             style={{
520|                 position: 'fixed', top: '50%', left: '50%', transform: 'translate(-50%, -50%)',
521|                 zIndex: 100, width: '520px', maxWidth: '90vw', maxHeight: '80vh', overflowY: 'auto',
522|                 ...glassPanel, padding: '24px',
523|                 boxShadow: '0 20px 60px rgba(0,0,0,0.6)',
524|                 animation: 'osfadein 0.3s ease-out',
525|             }}
526|         >
527|             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom: '20p...
528|                 <h3 style={{ color: 'white', margin: 0, fontSize: '16px', fontWeight: 700, display: 'flex', alignItems...
529|                     0=0» Th ông t in h ệ t h ã n g
530|                 </h3>
531|                 {onClose && (
532|                     <button
533|                         onClick={onClose}
534|                         style={{
535|                             background: 'rgba(255,255,255,0.1)', border: 'none',
536|                             color: 'aaa', width: '28px', height: '28px', borderRadius: '6px',
537|                             cursor: 'pointer', fontSize: '16px',
538|                             display: 'flex', alignItems: 'center', justifyContent: 'center',
539|                         }}
540|                         type="button"
541|                     >
542|                         '
543|                     </button>
544|                 )}
545|             </div>
546|
547|             <div
548|                 style={{
549|                     ...glassPanel, padding: '16px', marginBottom: '16px', textAlign: 'center',
550|                     border: '1px solid ${perf.color}33',
551|                 }}
552|             >
553|                 <div style={{ color: perf.color, fontSize: '28px', fontWeight: 800, letterSpacing: '-1px' }}>
554|                     {performanceScore.toLocaleString()}
555|                 </div>
556|                 <div style={{ color: perf.color, fontSize: '12px', fontWeight: 600, marginTop: '4px' }}>
557|                     {perf.label}
558|                 </div>
559|                 <div style={{ color: '#888', fontSize: '11px', marginTop: '4px'
560|             </div>
561|
562|             <div style={{ display: 'flex', flexDirection: 'column', gap: '8px' }}>
563|                 <SpecRow icon="0>pY" label="H ệ i Á u hành" value={WINDOWS_VERSION}
564|                 {cpu && (
565|                     <SpecRow
566|                         icon="&i"
567|                         label="CPU"
568|                         value={` ${cpu.brand} ${cpu.model} (${cpu.cpu_cores} nhân, $
569|                     )}
570|                 )}
571|                 {gpu && (
572|                     <SpecRow
573|                         icon="0<B@ "
574|                         label="GPU"
575|                         value={` ${gpu.brand} ${gpu.model} (${gpu.gpu_vram_gb}GB VRAM)`}
576|                     )}
577|                 {ram.length > 0 && (
578|                     <SpecRow
579|                         icon="0>Yà "
580|                         label="RAM"
581|                         value={` ${totalRamGb}GB DDR${ram[0].ram_ddr_type || ''} ${ram[0].ram_speed_mhz || ''}MHz`}
582|                     )}
583|                 )}
584|                 <SpecRow
585|                     icon="0=U% "
586|                     label="L ộ u t r i "
587|                     value={` ${totalStorageGb}GB (${ssd.length} SSD${hdd.length > 0 ? ', ${hdd.length} HDD' : ''})`}
588|                 )}
589|                 {ssd[0] && (
590|                     <SpecRow
591|                         icon="0>p€ "
592|                         label="SSD"
593|                         value={` ${ssd[0].brand} ${ssd[0].model} - ${ssd[0].ssd_capacity}
594|                     )}
595|                 )}
596|             </div>
597|         </div>
598|     );
599| }
600|
601|
602| function DesktopPhase({ build, username, onExit }) {
603|     const [openApp, setOpenApp] = useState(null);
604|     const [startOpen, setStartOpen] = useState(false);
605|     const [time, setTime] = useState('');
606|     const [showWelcome, setShowWelcome] = useState(true);
607|
608|     useEffect(() => {
609|         const t = setInterval(
610|             () => setTime(new Date().toLocaleTimeString('vi-VN', { hour: '2-digit', minute: '2-digit' })),
611|             1000,
612|         );
613|         setTime(new Date().toLocaleTimeString('vi-VN', { hour: '2-digit', minute: '2-digit' }));
614|         return () => clearInterval(t);
615|     }, []);
616|
617|     useEffect(() => {
618|         if (showWelcome) {
619|             const timer = setTimeout(() => setShowWelcome(false), 4000);
620|             return () => clearTimeout(timer);
621|         }
622|     }, [showWelcome]);
623|
624|     const wallpaperColors = 'linear-gradient(135deg, #0f1b2d 0%, #0d2137 40%, #1a0533 100%)';
625|
626|     const desktopApps = [
627|         { id: 'specs', icon: '0=U»', name: 'PC Specs' },
628|         { id: 'browser', icon: '0<B', name: 'Edge' },
629|         { id: 'files', icon: '0=UÀ', name: 'File Explorer' },
630|         { id: 'store', icon: '0>p0', name: 'Store' },
631|     ];
632|
633|     const appWindows = {

```

```
634 | specs: (  
635 |   <SpecsApp build={build} onClose={() => setOpenApp(null)} />  
636 | ),  
637 | browser: (  
638 |   <div  
639 |     style={{  
640 |       position: 'fixed', top: '50%', left: '50%', transform: 'translate(-50%, -50%)',  
641 |       ...glassPanel, padding: '24px', zIndex: 100,  
642 |       boxShadow: '0 20px 60px rgba(0,0,0,0.6)',  
643 |       minWidth: '400px', textAlign: 'center',  
644 |     }}  
645 |   >  
646 |     <div style={{ fontSize: '48px', marginBottom: '12px' }}>B</div>  
647 |     <h3 style={{ color: 'white', margin: '0 8px 0', fontWeight: 600 }}>Microsoft Edge</h3>  
648 |     <p style={{ color: 'white', fontSize: '13px' }}>Trinh duy t web m  
649 |     <button onClick={() => setOpenApp(null)} style={{ ...win11Btn(true), marginTop: '12px' }} type="button...  
650 |   </div>  
651 | ),  
652 | files: (  
653 |   <div  
654 |     style={{  
655 |       position: 'fixed', top: '50%', left: '50%', transform: 'translate(-50%, -50%)',  
656 |       ...glassPanel, padding: '24px', zIndex: 100,  
657 |       boxShadow: '0 20px 60px rgba(0,0,0,0.6)',  
658 |       minWidth: '380px',  
659 |     }}  
660 |   >  
661 |     <h3 style={{ color: 'white', margin: '0 0 12px 0', fontSize: '15  
662 |     <div style={{ display: 'flex', flexDirection: 'column', gap: '6px', marginBottom: '16px' }}>  
663 |       {'Desktop', 'Downloads', 'Documents', 'Pictures', 'Music'}.map(f => (  
664 |         <div key={f} style={{ color: '#ccc', padding: '8px 12px', borderRadius: '6px', background: 'rgba(2...  
665 |         <div style={{ color: 'white', padding: '0 4px' }}>f</div>  
666 |       </div>  
667 |     </div>  
668 |     <button onClick={() => setOpenApp(null)} style={{ win11Btn(true) }}  
669 |   </div>  
670 | ),  
671 | store: (  
672 |   <div  
673 |     style={{  
674 |       position: 'fixed', top: '50%', left: '50%', transform: 'translate(-50%, -50%)',  
675 |       ...glassPanel, padding: '24px', zIndex: 100,  
676 |       boxShadow: '0 20px 60px rgba(0,0,0,0.6)',  
677 |       minWidth: '400px', textAlign: 'center',  
678 |     }}  
679 |   >  
680 |     <div style={{ fontSize: '48px', marginBottom: '12px' }}>B</div>  
681 |     <h3 style={{ color: 'white', margin: '0 8px 0', fontWeight: 600 }}>Microsoft Store</h3>  
682 |     <p style={{ color: 'white', fontSize: '13px' }}>Hàng nghìn ứng d  
683 |     <button onClick={() => setOpenApp(null)} style={{ ...win11Btn(true), marginTop: '12px' }} type="button...  
684 |   </div>  
685 | ),  
686 | );  
687 |  
688 |  
689 | return (  
690 |   <div  
691 |     style={{  
692 |       position: 'fixed', inset: 0, zIndex: 9999,  
693 |       background: wallpaperColors,  
694 |       fontFamily: '"Segoe UI", "Segoe UI Variable", sans-serif",  
695 |       userSelect: 'none', overflow: 'hidden',  
696 |     }}  
697 |     onClick={() => { if (startOpen) setStartOpen(false); }}  
698 |   >  
699 |     {showWelcome && (  
700 |       <div  
701 |         style={{  
702 |           position: 'absolute', inset: 0, zIndex: 50,  
703 |           display: 'flex', flexDirection: 'column', alignItems: 'center', justifyContent: 'center',  
704 |           background: 'rgba(0,0,0,0.7)', backdropFilter: 'blur(8px)',  
705 |           animation: 'osfadein 0.5s',  
706 |         }}  
707 |       >  
708 |         <div style={{ fontSize: '64px', marginBottom: '16px' }}>B</div>  
709 |         <h1 style={{ color: 'white', fontSize: '32px', fontWeight: 300, margin: 0 }}>  
710 |           Chào mừng đến với Windows 11  
711 |         </h1>  
712 |         <p style={{ color: 'white', fontSize: '14px', marginTop: '12px' }}>  
713 |           {username || 'PC Master User'}  
714 |         </p>  
715 |       </div>  
716 |     </div>  
717 |     <div  
718 |       style={{  
719 |         padding: '20px', display: 'flex', flexDirection: 'column', gap: '16px',  
720 |         width: '80px', position: 'relative', zIndex: 10,  
721 |       }}  
722 |     >  
723 |     {desktopApps.map(app => (  
724 |       <div  
725 |         key={app.id}  
726 |         onDoubleClick={() => setOpenApp(app)}  
727 |         style={{  
728 |           display: 'flex', flexDirection: 'column', alignItems: 'center',  
729 |           gap: '4px', cursor: 'pointer', padding: '8px', borderRadius: '6px',  
730 |           color: 'white',  
731 |         }}  
732 |         onMouseOver={e => { e.currentTarget.style.background = 'rgba(255,255,255,0.1)'; }}  
733 |         onMouseOut={e => { e.currentTarget.style.background = 'transparent'; }}  
734 |       >  
735 |         <span style={{ fontSize: '32px' }}>{app.icon}</span>  
736 |         <span style={{ fontSize: '11px', textAlign: 'center', textShadow: '1px 1px 2px black' }}>  
737 |           {app.name}  
738 |         </span>  
739 |       </div>  
740 |     </div>  
741 |   </div>  
742 |   </div>  
743 |   {openApp && appWindows[openApp.id]}  
744 |  
745 |   <div  
746 |     style={{  
747 |       position: 'fixed', bottom: 0, left: 0, right: 0, height: '48px',  
748 |       background: 'rgba(32,32,32,0.95)', backdropFilter: 'blur(20px)',  
749 |       webkitBackdropFilter: 'blur(20px)',  
750 |       borderTop: '1px solid rgba(255,255,255,0.1)',  
751 |       display: 'flex', alignItems: 'center', justifyContent: 'space-between',  
752 |       padding: '0 16px', zIndex: 200,  
753 |     }}  
754 |     onClick={e => e.stopPropagation()}  
755 |   >  
756 |   <div style={{ display: 'flex', alignItems: 'center', gap: '4px' }}>  
757 |
```

```

758 |         <button
759 |             onClick={() => setStartOpen(v => !v)}
760 |             style={{
761 |                 background: 'transparent', border: 'none', fontSize: '22px',
762 |                 cursor: 'pointer', padding: '6px 10px', borderRadius: '6px',
763 |                 transition: 'background 0.15s',
764 |             }}
765 |             type="button"
766 |         >
7 6 7 |             0>PỖ
768 |     </button>
769 |     {desktopApps.map(app => (
770 |         <button
771 |             key={app.id}
772 |             onClick={() => setOpenApp(app)}
773 |             title={app.name}
774 |             style={{
775 |                 background: openApp?.id === app.id ? 'rgba(255,255,255,0.15)' : 'transparent',
776 |                 border: 'none', fontSize: '20px', padding: '6px 10px', borderRadius: '6px',
777 |                 cursor: 'pointer', fontFamily: 'inherit',
778 |                 borderBottom: openApp?.id === app.id ? '2px solid #0078d4' : '2px solid transparent',
779 |             }}
780 |             type="button"
781 |         >
782 |             {app.icon}
783 |         </button>
784 |     ))}
785 | </div>
786 |
787 | <div style={{ display: 'flex', alignItems: 'center', gap: '12px' }}>
788 |     <button
789 |         onClick={onExit}
790 |         style={{
791 |             background: 'rgba(239,68,68,0.8)', color: 'white', border: 'none',
792 |             padding: '4px 14px', borderRadius: '6px', cursor: 'pointer',
793 |             fontSize: '12px', fontWeight: 700, fontFamily: 'inherit',
794 |         }}
795 |         type="button"
796 |     >
7 9 7 |         #0 T - t m á y
798 |     </button>
799 |     <div style={{ color: 'white', fontSize: '12px', textAlign: 'right' }}>
800 |         <div style={{ fontWeight: 600 }}>{time}</div>
801 |         <div style={{ color: 'aaa', fontSize: '10px' }}>
802 |             {new Date().toLocaleDateString('vi-VN')}
803 |         </div>
804 |     </div>
805 | </div>
806 | </div>
807 |
808 | {startOpen && (
809 |     <div
810 |         onClick={e => e.stopPropagation()}
811 |         style={{
812 |             position: 'fixed', bottom: '56px', left: '50%', transform: 'translateX(-50%)',
813 |             width: '520px', background: 'rgba(36,36,36,0.98)', backdropFilter: 'blur(30px)',
814 |             WebkitBackdropFilter: 'blur(30px)',
815 |             borderRadius: '12px', padding: '24px',
816 |             border: '1px solid rgba(255,255,255,0.1)',
817 |             boxShadow: '0 24px 80px rgba(0,0,0,0.6)',
818 |             zIndex: 300, animation: 'osfadein 0.15s ease-out',
819 |         }}
820 |     >
821 |         <div style={{ display: 'flex', alignItems: 'center', gap: '12px', marginBottom: '20px' }}>
822 |             <input
8 2 3 |                 p l a c e h o l d e r = " T i m k i z m é n g d a n g , c à i . t . . . "
824 |                 style={{
825 |                     flex: 1, padding: '10px 16px', borderRadius: '8px', border: 'none',
826 |                     background: 'rgba(255,255,255,0.08)', color: 'white', fontSize: '13px',
827 |                     outline: 'none', fontFamily: 'inherit',
828 |                 }}
829 |             />
830 |         </div>
831 |         <div
832 |             style={{
833 |                 color: 'rgba(255,255,255,0.6)', fontWeight: 600, marginBottom: '12px',
834 |                 fontSize: '12px', textTransform: 'uppercase', letterSpacing: '1px',
835 |             }}
836 |         >
8 3 7 |             è n g d a n g
838 |     </div>
839 |     <div style={{ display: 'grid', gridTemplateColumns: 'repeat(4,1fr)', gap: '8px', marginBottom: '16px...' }}>
840 |         {desktopApps.map(app => (
841 |             <button
842 |                 key={app.id}
843 |                 onClick={() => { setOpenApp(app); setStartOpen(false); }}
844 |                 style={{
845 |                     background: 'transparent', border: 'none', cursor: 'pointer',
846 |                     display: 'flex', flexDirection: 'column', alignItems: 'center',
847 |                     gap: '4px', padding: '12px 8px', borderRadius: '8px',
848 |                     color: 'white', fontFamily: 'inherit',
849 |                     transition: 'background 0.15s',
850 |                 }}
851 |                 onMouseOver={e => { e.currentTarget.style.background = 'rgba(255,255,255,0.08)'; }}
852 |                 onMouseOut={e => { e.currentTarget.style.background = 'transparent'; }}
853 |                 type="button"
854 |             >
855 |                 <span style={{ fontSize: '28px' }}>{app.icon}</span>
856 |                 <span style={{ fontSize: '11px' }}>{app.name}</span>
857 |             </button>
858 |         ))}
859 |     </div>
860 |     <div
861 |         style={{
862 |             borderTop: '1px solid rgba(255,255,255,0.06)', paddingTop: '12px',
863 |             display: 'flex', justifyContent: 'space-between', alignItems: 'center',
864 |         }}
865 |     >
866 |         <span style={{ color: 'aaa', fontSize: '13px' }}>
8 6 7 |             0=Úd { u s e r n a m e | | ' P C M a s t e r U s e r ' }
868 |         </span>
869 |         <button
870 |             onClick={onExit}
871 |             style={{
872 |                 background: 'transparent', border: 'none', color: 'aaa',
873 |                 cursor: 'pointer', fontSize: '13px', fontFamily: 'inherit',
874 |             }}
875 |             type="button"
876 |         >
8 7 7 |             #0 T - t m á y
878 |     </button>
879 | </div>
880 | </div>
881 | })

```

```

882 |     </div>
883 |   );
884 | }
885 |
886 | function IntroStep({ onNext }) {
887 |   const [phase, setPhase] = useState(0);
888 |   const phases = [
889 |     { icon: '0=0æ', title: 'Bìn vèa mua mùt bù PC mŭi!', desc: 'Linh kiçn',
890 |     { icon: '0=0ç', title: 'Nh'ng máy tĩnh ch' a có hç i Ấu hành', desc: 'N
891 |     { icon: '0>by', title: 'Cài Windows 11 bèn quy Ắn - an toàn, bèo m-t',
892 |   ];
893 |
894 |   useEffect(() => {
895 |     if (phase < phases.length - 1) {
896 |       const t = setTimeout(() => setPhase(p => p + 1), 2500);
897 |       return () => clearTimeout(t);
898 |     }
899 |   }, [phase]);
900 |
901 |   return (
902 |     <div style={{
903 |       display: 'flex', flexDirection: 'column', alignItems: 'center',
904 |       justifyContent: 'center', gap: 20, padding: '40px 20px',
905 |       minHeight: 400, maxWidth: 600, margin: '0 auto',
906 |     }}>
907 |       <div style={{
908 |         fontSize: 72, animation: 'ospulse 2s ease-in-out infinite',
909 |         transition: 'all 0.4s',
910 |       }}>{phases[phase].icon}</div>
911 |       <h2 style={{ color: 'white', fontSize: 22, fontWeight: 300, margin: 0, textAlign: 'center' }}>
912 |         {phases[phase].title}
913 |       </h2>
914 |       <p style={{ color: '#aaa', fontSize: 14, textAlign: 'center', lineHeight: 1.6, margin: 0 }}>
915 |         {phases[phase].desc}
916 |       </p>
917 |
918 |       {phase === phases.length - 1 && (
919 |         <div style={{
920 |           animation: 'osfadein 0.5s', display: 'flex', flexDirection: 'column',
921 |           gap: 14, width: '100%', marginTop: 8,
922 |         }}>
923 |           <div style={{
924 |             background: 'rgba(0,120,212,0.1)', border: '1px solid rgba(0,120,212,0.2)',
925 |             borderRadius: 10, padding: '14px 18px',
926 |           }}>
927 |             <div style={{ fontSize: 12, fontWeight: 700, color: '#0078d4', marginBottom: 6 }}>
928 |               0=0 Ắ T Ế i Windows 11 bèn quy Ắn (chính th Ờng)
929 |             </div>
930 |             <ul style={{ margin: 0, padding: '0 0 0 16px', display: 'flex', flexDirection: 'column', gap: 4 }}>
931 |               <li style={{ fontSize: 12, color: '#aaa' }}>
932 |                 <a href="https://www.microsoft.com/software-download/windows11" target="_blank" rel="noopener ...
933 |                   style={{ color: '#3b82f6' }}>
934 |                   Microsoft.com - T ịo USB cài Win chính th ế c
935 |                 </a>
936 |               </li>
937 |               <li style={{ fontSize: 12, color: '#aaa' }}>
938 |                 <a href="https://www.microsoft.com/en-us/evalcenter/evaluate-windows-11-enterprise" target="_b...
939 |                   style={{ color: '#3b82f6' }}>
940 |                   Trung tâm ănh giá - Windows 11 Enterprise Evaluation
941 |                 </a>
942 |               </li>
943 |             </ul>
944 |           </div>
945 |           <button onClick={onNext}
946 |             style={{
947 |               padding: '12px 48px', borderRadius: 8, fontSize: 15, fontWeight: 600,
948 |               background: '#0078d4', border: 'none', color: 'white', cursor: 'pointer',
949 |               fontFamily: 'inherit', alignSelf: 'center',
950 |             }}>
951 |             B - t ậ u c à i . t !'
952 |           </button>
953 |         </div>
954 |       )}
955 |     </div>
956 |   );
957 | }
958 |
959 | export default function OSInstallationSimulator({ build, onComplete, onExit }) {
960 |   const [step, setStep] = useState(0);
961 |   const [lang, setLang] = useState('T ịng Viçt');
962 |   const [region, setRegion] = useState('Viçt Nam');
963 |   const [keyboard, setKeyboard] = useState('T ịng Viçt (Telex)');
964 |   const [installProgress, setInstallProgress] = useState(0);
965 |   const [username, setUsername] = useState('');
966 |   const [password, setPassword] = useState('');
967 |   const [confirmPassword, setConfirmPassword] = useState('');
968 |   const [privacySettings, setPrivacySettings] = useState({ location: true, diagnostics: false, ads: false });
969 |   const [selectedNetwork, setSelectedNetwork] = useState(null);
970 |   const [isConnected, setIsConnected] = useState(false);
971 |   const [desktopReady, setDesktopReady] = useState(false);
972 |
973 |   useEffect(() => {
974 |     if (step === 2) {
975 |       setInstallProgress(0);
976 |       const timer = setInterval(() => {
977 |         setInstallProgress(p => {
978 |           if (p >= 100) {
979 |             clearInterval(timer);
980 |             setTimeout(() => setStep(3), 500);
981 |             return 100;
982 |           }
983 |           const increment = p < 80 ? 1.5 : p < 95 ? 0.8 : 0.3;
984 |           return Math.min(100, p + increment);
985 |         });
986 |       }, 80);
987 |       return () => clearInterval(timer);
988 |     }
989 |   }, [step]);
990 |
991 |   useEffect(() => {
992 |     if (!document.getElementById('os-sim-styles')) {
993 |       const style = document.createElement('style');
994 |       style.id = 'os-sim-styles';
995 |       style.textContent = `
996 |         @keyframes ospin { to { transform: rotate(360deg); } }
997 |         @keyframes osfadein { from { opacity: 0; transform: scale(0.95); } to { opacity: 1; transform: scale(1...
998 |         @keyframes ospulse { 0%, 100% { opacity: 1; } 50% { opacity: 0.5; } }
999 |         @keyframes osslideup { from { opacity: 0; transform: translateY(16px); } to { opacity: 1; transform: t...
1000 |         input::placeholder { color: #666; }
1001 |         ::-webkit-scrollbar { width: 6px; }
1002 |         ::-webkit-scrollbar-track { background: transparent; }
1003 |         ::-webkit-scrollbar-thumb { background: rgba(255,255,255,0.15); border-radius: 3px; }
1004 |         ::-webkit-scrollbar-thumb:hover { background: rgba(255,255,255,0.25); }
1005 |       `;
1006 |     }

```

```

1006 |     document.head.appendChild(style);
1007 | }
1008 | }, []);
1009 |
1010 | const handleNext = () => {
1011 |     setStep(s => Math.min(s + 1, 6));
1012 | };
1013 |
1014 | const handleBack = () => {
1015 |     setStep(s => Math.max(s - 1, 0));
1016 | };
1017 |
1018 | const handleNetworkConnect = () => {
1019 |     if (selectedNetwork) setIsConnected(true);
1020 | };
1021 |
1022 | const handleNetworkSkip = () => {
1023 |     setIsConnected(false);
1024 |     setSelectedNetwork(null);
1025 |     setStep(6);
1026 | };
1027 |
1028 | useEffect(() => {
1029 |     if (step === 6 && !desktopReady) {
1030 |         setDesktopReady(true);
1031 |         if (onComplete) {
1032 |             setTimeout(() => onComplete(), 1000);
1033 |         }
1034 |     }
1035 | }, [step, desktopReady, onComplete]);
1036 |
1037 | const renderStep = () => {
1038 |     switch (step) {
1039 |         case 0:
1040 |             return <IntroStep onNext={handleNext} />;
1041 |         case 1:
1042 |             return (
1043 |                 <LanguageRegionStep
1044 |                     lang={lang}
1045 |                     region={region}
1046 |                     keyboard={keyboard}
1047 |                     onLangChange={setLang}
1048 |                     onRegionChange={setRegion}
1049 |                     onKeyboardChange={setKeyboard}
1050 |                     onNext={handleNext}
1051 |                 />
1052 |             );
1053 |         case 2:
1054 |             return <InstallingStep progress={installProgress} />;
1055 |         case 3:
1056 |             return (
1057 |                 <AccountSetupStep
1058 |                     username={username}
1059 |                     password={password}
1060 |                     confirmPassword={confirmPassword}
1061 |                     onUsernameChange={setUsername}
1062 |                     onPasswordChange={setPassword}
1063 |                     onConfirmChange={setConfirmPassword}
1064 |                     onNext={handleNext}
1065 |                     onBack={handleBack}
1066 |                 />
1067 |             );
1068 |         case 4:
1069 |             return (
1070 |                 <PrivacySettingsStep
1071 |                     settings={privacySettings}
1072 |                     onToggle={(key) => setPrivacySettings(s => ({ ...s, [key]: !s[key] })})
1073 |                     onNext={handleNext}
1074 |                     onBack={handleBack}
1075 |                 />
1076 |             );
1077 |         case 5:
1078 |             return (
1079 |                 <NetworkSetupStep
1080 |                     selectedNetwork={selectedNetwork}
1081 |                     isConnected={isConnected}
1082 |                     onSelect={setSelectedNetwork}
1083 |                     onConnect={handleNetworkConnect}
1084 |                     onSkip={handleNetworkSkip}
1085 |                     onBack={handleBack}
1086 |                 />
1087 |             );
1088 |         case 6:
1089 |             return <DesktopPhase build={build} username={username} onExit={onExit} />;
1090 |         default:
1091 |             return null;
1092 |     }
1093 | };
1094 |
1095 | return (
1096 |     <div
1097 |         style={{
1098 |             position: 'fixed', inset: 0, zIndex: 9999,
1099 |             background: 'linear-gradient(135deg, #0f1b2d 0%, #0d2137 40%, #1a0533 100%)',
1100 |             fontFamily: '"Segoe UI", "Segoe UI Variable", -apple-system, sans-serif',
1101 |             display: 'flex', flexDirection: 'column',
1102 |             overflow: 'auto',
1103 |         }}
1104 |     >
1105 |         {step < 6 && (
1106 |             <div
1107 |                 style={{
1108 |                     display: 'flex', alignItems: 'center', gap: '12px',
1109 |                     padding: '12px 24px',
1110 |                     borderBottom: '1px solid rgba(255,255,255,0.06)',
1111 |                     background: 'rgba(0,0,0,0.2)',
1112 |                 }}
1113 |             >
1114 |                 1 1 4 | <span style={{ color: 'white', fontSize: '13px', fontWeight: 600 }}>
1115 |                     <span style={{ color: 'white', fontSize: '13px', fontWeight: 600 }}>
1116 |                         1 1 6 | T r i n h c à i t W i n d o w s 1 1
1117 |                     </span>
1118 |                 </div>
1119 |             )}
1120 |
1121 |             {step < 6 && <StepIndicator currentStep={step} steps={SETUP_STEPS} />}
1122 |
1123 |             <div style={{ flex: 1, display: 'flex', flexDirection: 'column', justifyContent: 'center', animation: 'o...
1124 |                 {renderStep()}
1125 |             </div>
1126 |         </div>
1127 |     );
1128 | }
1129 |

```

File: components/PageTransition.tsx (30 lines, 0.8 KB)

```
1 'use client'
2
3 import { motion, AnimatePresence } from 'framer-motion'
4 import { usePathname } from 'next/navigation'
5
6 const variants = {
7   initial: { opacity: 0, y: 24, scale: 0.98 },
8   animate: { opacity: 1, y: 0, scale: 1 },
9   exit: { opacity: 0, y: -16, scale: 0.98 },
10 }
11
12 export default function PageTransition({ children }: { children: React.ReactNode }) {
13   const pathname = usePathname()
14   return (
15     <AnimatePresence mode="wait">
16       <motion.div
17         key={pathname}
18         variants={variants}
19         initial="initial"
20         animate="animate"
21         exit="exit"
22         transition={{ duration: 0.3, ease: [0.16, 1, 0.3, 1] }}
23         style={{ width: '100%' }}
24       >
25         {children}
26       </motion.div>
27     </AnimatePresence>
28   )
29 }
30
```

File: components/parent/ChildCard.tsx (110 lines, 5.0 KB)

```
1 "use client"
2
3 import React from 'react'
4 import Link from 'next/link'
5
6 export interface ChildSummary {
7   student_id: string
8   student_name: string
9   student_code: string
10  school_name: string | null
11  xp: number
12  level: number
13  avatar_url: string | null
14  lessons_completed: number
15  lessons_started: number
16  total_time_seconds: number | null
17  last_active: string | null
18 }
19
20 function formatTimeAgo(dateString: string | null) {
21   if (!dateString) return 'Chưa bắt đầu học'
22   const date = new Date(dateString)
23   const now = new Date()
24   const diffMs = now.getTime() - date.getTime()
25   if (isNaN(diffMs)) return 'Chưa bắt đầu học'
26   const diffSecs = Math.floor(diffMs / 1000)
27   const diffMins = Math.floor(diffSecs / 60)
28   const diffHours = Math.floor(diffMins / 60)
29   const diffDays = Math.floor(diffHours / 24)
30
31   if (diffSecs < 60) return 'Vừa xong'
32   if (diffMins < 60) return `${diffMins} phút trước`
33   if (diffHours < 24) return `${diffHours} giờ trước`
34   return `${diffDays} ngày trước`
35 }
36
37 export function ChildCard({ child }: { child: ChildSummary }) {
38   const progressPercent = child.lessons_started > 0
39   ? Math.round((child.lessons_completed / child.lessons_started) * 100) : 0
40   const lastActiveText = formatTimeAgo(child.last_active)
41
42   // Kiểm tra hoạt động trong vòng 24 giờ qua
43   const isRecentlyActive = child.last_active
44   ? new Date(child.last_active) > new Date(Date.now() - 86400000)
45   : false
46
47   return (
48     <Link
49       href={`/${parent}/children/${child.student_id}`}
50       className="block rounded-xl p-5 transition-all duration-200 group"
51       style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-subtle)' }}
52     >
53       { /* Avatar + Tên */ }
54       <div className="flex items-center gap-3 mb-4">
55         {child.avatar_url ? (
56           <img src={child.avatar_url} className="w-12 h-12 rounded-full object-cover" style={{ border: '1px so...
57         ) : (
58           <div className="w-12 h-12 rounded-full flex items-center justify-center text-xl font-bold uppercase"...
59             {child.student_name[0] || 'S'}
60           </div>
61         )}
62       <div className="min-w-0 flex-1">
63         <div className="font-bold truncate transition-colors" style={{ color: 'var(--text-primary)' }}>{chil...
64         <div className="text-xs truncate" style={{ color: 'var(--text-...
65       </div>
66       <div className="text-[10px] font-bold px-2 py-0.5 rounded-full uppercase tracking-wider" style={{ back...
67       C # p { child.level || 1 }
68     </div>
69   </div>
70
71   { /* Stats Grid */ }
72   <div className="grid grid-cols-2 gap-3 mb-4">
73     <div className="rounded-xl p-3 text-center" style={{ background: 'var(--bg-elevated)', border: '1px so...
74       <div className="text-xl font-black" style={{ color: 'var(--brand-primary)' }}>{child.lessons_complet...
75       <div className="text-[10px] uppercase font-bold tracking-wider" style={{ color: 'var(--text-muted)' }} ...
76     </div>
77     <div className="rounded-xl p-3 text-center" style={{ background: 'var(--bg-elevated)', border: '1px so...
78       <div className="text-xl font-black" style={{ color: 'var(--text-primary)' }}>
79         {Math.round((child.total_time_seconds ?? 0) / 3600)}h
80       </div>
81       <div className="text-[10px] uppercase font-bold tracking-wider" style={{ color: 'var(--text-muted)' }} ...
82     </div>
83   </div>
84
85   { /* Progress Bar */ }
```

```
86 | <div className="mb-4">
87 |   <div className="flex justify-between text-[11px] font-bold mb-1" style={{ color: 'var(--text-muted)' }}>
88 |     8 8 | <span>Tiền ù hoàn thành bài</span>
89 |     <span style={{ color: 'var(--text-primary)' }}>{progressPercent}%</span>
90 |   </div>
91 |   <div className="h-2 rounded-full overflow-hidden" style={{ background: 'var(--bg-elevated)', border: '...
92 |     <div
93 |       className="h-full bg-gradient-to-r from-[#00d4aa] to-[#00b4d8] rounded-full transition-all duratio...
94 |       style={{ width: `${progressPercent}%` }}
95 |     </div>
96 |   </div>
97 | </div>
98 |
99 | /* Last Active Footer */
100 | <div className="text-[11px] flex items-center gap-2 pt-3" style={{ color: 'var(--text-muted)', borderTop...
101 |   <span className="w-2 h-2 rounded-full" style={{
102 |     background: isRecentlyActive ? 'var(--brand-primary)' : 'var(--text-muted)',
103 |     boxShadow: isRecentlyActive ? '0 0 8px var(--brand-primary)' : 'none'
104 |   }} />
105 |   <span>Hoit ùng : {lastActiveText}</span>
106 | </div>
107 | </Link>
108 | )
109 | }
110 |
```

File: components/parent/ChildProgressChart.tsx (135 lines, 5.3 KB)

```
1 | 'use client'
2 |
3 | import React, { useMemo } from 'react'
4 |
5 | interface DailyProgress {
6 |   date: string
7 |   count: number
8 | }
9 |
10 | interface ChildProgressChartProps {
11 |   data: DailyProgress[]
12 | }
13 |
14 | export default function ChildProgressChart({ data }: ChildProgressChartProps) {
15 |   const chartHeight = 200
16 |   const chartWidth = 500
17 |
18 |   const processedData = useMemo(() => {
19 |     if (!data || data.length === 0) return []
20 |     // Giỏi hì n h i Ầ n t h Ế 1 5 i Ầ m g Ầ n n h ỷ t Ầ b i Ầ u ó t h o ầ n g ầ n g
21 |     return data.slice(-15)
22 |   }, [data])
23 |
24 |   const { pointsPath, areaPath, dots } = useMemo(() => {
25 |     if (processedData.length === 0) {
26 |       return { pointsPath: '', areaPath: '', dots: [] }
27 |     }
28 |
29 |     const maxVal = Math.max(...processedData.map((d) => d.count), 1)
30 |     const xStep = chartWidth / (processedData.length - 1 || 1)
31 |
32 |     const coords = processedData.map((d, index) => {
33 |       const x = index * xStep
34 |       // S V G ồ ồ t r ề n c ù n g , n ề n l - t n g ầ c t í a ù y
35 |       const y = chartHeight - (d.count / maxVal) * (chartHeight - 40) - 20
36 |       return { x, y, ...d }
37 |     })
38 |
39 |     // V ỷ ồ Ỗ n g L i n e q u a c ầ c i Ầ m
40 |     const lineParts = coords.map((c, i) => `${i === 0 ? 'M' : 'L'} ${c.x.toFixed(1)} ${c.y.toFixed(1)} `)
41 |     const linePath = lineParts.join('')
42 |
43 |     // V ỷ v ù n g A r e a d ồ i ồ Ỗ n g L i n e
44 |     const firstX = coords[0].x.toFixed(1)
45 |     const lastX = coords[coords.length - 1].x.toFixed(1)
46 |     const areaPath = `${linePath} L ${lastX} ${chartHeight} L ${firstX} ${chartHeight} Z`
47 |
48 |     return { pointsPath: linePath, areaPath, dots: coords }
49 |   }, [processedData])
50 |
51 |   if (processedData.length === 0) {
52 |     return (
53 |       <div className="h-48 flex items-center justify-center text-xs rounded-2xl" style={{ color: 'var(--text-m...
54 |         C h ầ c ó d i ầ l i ầ u h i ầ t - p t r o n g 3 ồ n g ầ y q u a
55 |       </div>
56 |     )
57 |   }
58 |
59 |   return (
60 |     <div className="rounded-2xl p-5" style={{ background: 'var(--bg-surface)', border: '1px solid var(--border...
61 |       <div className="flex justify-between items-center mb-6">
62 |         <div>
63 |           <div className="text-[10px] uppercase font-bold tracking-wider mt-0.5" style={{ color: 'var(--text-m...
64 |             <div className="text-[10px] font-bold tracking-wider mt-0.5" style={{ color: 'var(--text-m...
65 |           </div>
66 |           <div className="flex items-center gap-1.5 text-xs font-semibold px-2.5 py-1 rounded-full" style={{ col...
67 |             <span className="w-1.5 h-1.5 rounded-full animate-ping" style={{ background: 'var(--brand-primary)' }} ...
68 |             3 ồ n g ầ y g Ầ n ầ y
69 |           </div>
70 |         </div>
71 |
72 |         <div className="relative w-full overflow-hidden">
73 |           <svg
74 |             viewBox={`0 0 ${chartWidth} ${chartHeight}`}
75 |             className="w-full h-auto overflow-visible"
76 |           >
77 |             <defs>
78 |               { /* G r a d i e n t c h o ồ Ỗ n g L i n e */ }
79 |               <linearGradient id="line-gradient" x1="0" y1="0" x2="1" y2="0">
80 |                 <stop offset="0%" stopColor="#00d4aa" />
81 |                 <stop offset="100%" stopColor="#00b4d8" />
82 |               </linearGradient>
83 |               { /* G r a d i e n t c h o v ù n g b ề n d ồ i */ }
84 |               <linearGradient id="area-gradient" x1="0" y1="0" x2="0" y2="1">
85 |                 <stop offset="0%" stopColor="#00d4aa" stopOpacity="0.25" />
86 |                 <stop offset="100%" stopColor="#00d4aa" stopOpacity="0.00" />
87 |               </linearGradient>
88 |             </defs>
89 |
90 |             { /* ồ Ỗ n g l ồ i G r i d n g a n g */ }
91 |             <line x1="0" y1={chartHeight - 20} x2={chartWidth} y2={chartHeight - 20} stroke="rgba(255,255,255,0.00...
92 |             <line x1="0" y1={chartHeight / 2} x2={chartWidth} y2={chartHeight / 2} stroke="rgba(255,255,255,0.03...
93 |             <line x1="0" y1="20" x2={chartWidth} y2="20" stroke="rgba(255,255,255,0.03)" strokeWidth="1" />
94 |
95 |             { /* V ù n g A r e a p h i ầ d ồ i */ }
```

```

96 |         <path d={areaPath} fill="url(#area-gradient)" />
97 |
98 |         { / *      ° Ý n g   v ½   L i n e   c h í n h   * / }
99 |         <path
100 |           d={pointsPath}
101 |           fill="none"
102 |           stroke="url(#line-gradient)"
103 |           strokeWidth="3"
104 |           strokeLinecap="round"
105 |           strokeLinejoin="round"
106 |         />
107 |
108 |         { / *      C á c   i Ầ m   D o t s   t r ê n      ° Ý n g   L i n e   * / }
109 |         {dots.map((dot, index) => (
110 |           <g key={index} className="group/dot cursor-pointer">
111 |             <circle
112 |               cx={dot.x}
113 |               cy={dot.y}
114 |               r="5"
115 |               fill="#111318"
116 |               stroke="#00d4aa"
117 |               strokeWidth="2"
118 |               className="transition-all duration-150 hover:r-7"
119 |             />
120 |             <title>`${dot.date}: ${dot.count} bài`</title>
121 |           </g>
122 |         ))}
123 |       </svg>
124 |     </div>
125 |
126 |     { / *      N h ẫ n   t r ầ c   X   * / }
127 |     <div className="flex justify-between mt-3 px-1 text-[10px] text-[#636678] font-bold">
128 |       <span>{processedData[0].date}</span>
129 |       <span>{processedData[Math.floor(processedData.length / 2)].date}</span>
130 |       <span>{processedData[processedData.length - 1].date}</span>
131 |     </div>
132 |   </div>
133 | )
134 | }
135 |

```

File: components/parent/ParentNotificationsView.tsx (292 lines, 12.1 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useTransition } from 'react'
4 | import { Bell, Check, Trash2, Calendar, BookOpen, Award, Trophy, UserMinus } from 'lucide-react'
5 |
6 | interface StudentProfile {
7 |   id: string
8 |   full_name: string
9 |   avatar_url: string | null
10 | }
11 |
12 | interface Notification {
13 |   id: string
14 |   user_id: string
15 |   type: string
16 |   title: string
17 |   body: string
18 |   data: any
19 |   action_url: string
20 |   is_read: boolean
21 |   created_at: string
22 | }
23 |
24 | interface ParentNotificationsViewProps {
25 |   initialNotifications: Notification[]
26 |   students: StudentProfile[]
27 |   markAllAsRead: () => Promise<void>
28 |   deleteAll: () => Promise<void>
29 |   markAsRead: (id: string) => Promise<void>
30 |   deleteSingle: (id: string) => Promise<void>
31 | }
32 |
33 | type TabType = 'all' | 'lesson' | 'quiz' | 'absent'
34 |
35 | export default function ParentNotificationsView({
36 |   initialNotifications,
37 |   students,
38 |   markAllAsRead,
39 |   deleteAll,
40 |   markAsRead,
41 |   deleteSingle
42 | }: ParentNotificationsViewProps) {
43 |   const [activeTab, setActiveTab] = useState<TabType>('all')
44 |   const [notifications, setNotifications] = useState<Notification[]>(initialNotifications)
45 |   const [isPending, startTransition] = useTransition()
46 |
47 |   // T i o   m a p   t r a   c é u   h í c   s i n h   t h e o   i d
48 |   const studentMap = React.useMemo(() => {
49 |     const map = new Map<string, StudentProfile>()
50 |     students.forEach((s) => map.set(s.id, s))
51 |     return map
52 |   }, [students])
53 |
54 |   // L í c   t h ò n g   b á o   đ ã   t r ê n   t a b   a n g   c h í n
55 |   const filteredNotifications = React.useMemo(() => {
56 |     return notifications.filter((notif) => {
57 |       if (activeTab === 'all') return true
58 |       if (activeTab === 'lesson') return notif.type === 'lesson_completed'
59 |       if (activeTab === 'quiz') return notif.type === 'quiz_graded' || notif.type === 'child_low_score'
60 |       if (activeTab === 'absent') return notif.type === 'child_absent'
61 |       return false
62 |     })
63 |   }, [notifications, activeTab])
64 |
65 |   // U I   i c o n   c h o   t ề n g   l o i i   t h ò n g   b á o
66 |   const getIcon = (type: string) => {
67 |     switch (type) {
68 |       case 'lesson_completed':
69 |         return <BookOpen size={16} className="text-blue-400" />
70 |       case 'quiz_graded':
71 |         return <Award size={16} className="text-[#00d4aa]" />
72 |       case 'child_low_score':
73 |         return <Award size={16} className="text-red-400" />
74 |       case 'child_absent':
75 |         return <UserMinus size={16} className="text-orange-400" />
76 |       default:
77 |         return <Bell size={16} className="text-[#636678]" />
78 |     }
79 |   }
80 |

```



```

81 | // á n h d ð u t ð t c ð l à ã í c
82 | const handleMarkAllRead = () => {
83 |   startTransition(async () => {
84 |     await markAllAsRead()
85 |     setNotifications((prev) => prev.map((n) => ({ ...n, is_read: true })))
86 |   })
87 | }
88 |
89 | // X ó a t ð t c ð t h ò n g b á o
90 | const handleDeleteAll = () => {
91 |   if (!confirm('Bin muÑn x ó a t ð t c ð t h ò n g b á o?')) return
92 |   startTransition(async () => {
93 |     await deleteAll()
94 |     setNotifications([])
95 |   })
96 | }
97 |
98 | // á n h d ð u í c m ù t t h ò n g b á o
99 | const handleMarkSingleRead = async (id: string) => {
100 |   await markAsRead(id)
101 |   setNotifications((prev) =>
102 |     prev.map((n) => (n.id === id ? { ...n, is_read: true } : n))
103 |   )
104 | }
105 |
106 | // X ó a m ù t t h ò n g b á o
107 | const handleDeleteSingle = async (id: string) => {
108 |   await deleteSingle(id)
109 |   setNotifications((prev) => prev.filter((n) => n.id !== id))
110 | }
111 |
112 | const unreadCount = notifications.filter((n) => !n.is_read).length
113 |
114 | return (
115 |   <div className="space-y-6">
116 |     <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-4 pb-6" style={{ borderBot...
117 |       <div>
118 |         <h1 className="text-2xl md:text-3xl font-bold flex items-center gap-2" style={{ color: 'var(--text-p...
119 |         <Bell size={24} style={{ color: 'var(--brand-primary)' }} />
120 |         T h ò n g b á o P h ã h u y n h
121 |       </h1>
122 |       <p className="mt-1 text-sm" style={{ color: 'var(--text-muted)' }}>
123 |         Q u ê n l ý t h ò n g b á o t i ð n ù , i ã m q u i z , s t r e a k s v à t h à n h t í c h
124 |       </p>
125 |     </div>
126 |     {notifications.length > 0 && (
127 |       <div className="flex gap-2 self-start sm:self-auto">
128 |         <button
129 |           onClick={handleMarkAllRead}
130 |           disabled={isPending}
131 |           className="inline-flex items-center gap-1.5 px-3 py-2 rounded-xl text-xs font-bold transition-al...
132 |           style={{ background: 'rgba(0, 212, 170, 0.1)', border: '1px solid rgba(0, 212, 170, 0.2)', color...
133 |         >
134 |           <Check size={14} />
135 |           ã í c t ð t c ð
136 |         </button>
137 |         <button
138 |           onClick={handleDeleteAll}
139 |           disabled={isPending}
140 |           className="inline-flex items-center gap-1.5 px-3 py-2 rounded-xl text-xs font-bold transition-al...
141 |           style={{ background: 'rgba(239, 68, 68, 0.1)', border: '1px solid rgba(239, 68, 68, 0.2)', color...
142 |         >
143 |           <Trash2 size={14} />
144 |           X ó a t ð t c ð
145 |         </button>
146 |       </div>
147 |     )}
148 |   </div>
149 |   <div className="flex items-center gap-1 overflow-x-auto pb-2 no-scrollbar scroll-smooth" style={{ border...
150 |     <div>
151 |       {
152 |         { key: 'all', label: 'T ð t c ð ' },
153 |         { key: 'lesson', label: 'T i ð n ù h í c ' },
154 |         { key: 'quiz', label: 'i ã m s Ñ ' },
155 |         { key: 'absent', label: 'N g h é h í c ' }
156 |       }
157 |     </div>
158 |     .map((tab) => (
159 |       <button
160 |         key={tab.key}
161 |         onClick={() => setActiveTab(tab.key as TabType)}
162 |         className="px-4 py-2 rounded-xl text-xs font-bold transition-all duration-150 whitespace-nowrap bo...
163 |         style={{
164 |           background: activeTab === tab.key ? 'rgba(0, 212, 170, 0.1)' : 'transparent',
165 |           borderColor: activeTab === tab.key ? 'rgba(0, 212, 170, 0.3)' : 'transparent',
166 |           color: activeTab === tab.key ? 'var(--brand-primary)' : 'var(--text-muted)'
167 |         }}
168 |       >
169 |         {tab.label}
170 |       </button>
171 |     ))
172 |   </div>
173 |   <div>
174 |     <div>
175 |       <div>
176 |         <div className="rounded-2xl p-10 md:p-14 text-center my-6" style={{ background: 'var(--bg-surface)', b...
177 |         <div className="w-14 h-14 rounded-full flex items-center justify-center mx-auto mb-4" style={{ backg...
178 |         <Bell size={24} />
179 |       </div>
180 |       <h3 className="text-base font-bold mb-1" style={{ color: 'var(--text-primary)' }}>
181 |         K h ò n g t i m t h ð y t h ò n g b á o
182 |       </h3>
183 |       <p className="text-xs max-w-xs mx-auto" style={{ color: 'var(--text-muted)' }}>
184 |         K h ò n g c ó t h ò n g b á o n à o k h ú p v ù i d a n h m á c b ù l í c h i ç n t i i c q a
185 |     </p>
186 |   </div>
187 | ) : (
188 |   <div className="space-y-3">
189 |     {filteredNotifications.map((notif) => {
190 |       const studentId = notif.data?.student_id
191 |       const student = studentId ? studentMap.get(studentId) : null
192 |
193 |       return (
194 |         <div
195 |           key={notif.id}
196 |           className="rounded-xl p-4 transition-all flex items-start gap-4"
197 |           style={{
198 |             background: 'var(--bg-surface)',
199 |             border: notif.is_read ? '1px solid var(--border-subtle)' : '1px solid rgba(0, 212, 170, 0.2)',
200 |             opacity: notif.is_read ? 0.6 : 1
201 |           }}
202 |         >
203 |
204 |

```

```

206 | 2 0 5 | { / * L o i i i c o n * / }
207 | <div className="w-8 h-8 rounded-lg flex items-center justify-center flex-shrink-0 mt-0.5" styl...
208 | {getIcon(notif.type)}
209 | </div>
210 | 2 1 0 | { / * A v a t a r & T ê n h í c s í n h * / }
211 | {student && (
212 |   <div className="flex-shrink-0 relative hidden sm:block">
213 |     {student.avatar_url ? (
214 |       <img
215 |         src={student.avatar_url}
216 |         className="w-10 h-10 rounded-full object-cover"
217 |         style={{ border: '1px solid var(--border-default)' }}
218 |         alt={student.full_name}
219 |       />
220 |     ) : (
221 |       <div className="w-10 h-10 rounded-full flex items-center justify-center text-sm font-bol...
222 |         {student.full_name[0] || 'S'}
223 |       </div>
224 |     )}
225 |     <span className="absolute -bottom-1 -right-1 text-[10px] px-1 rounded-md font-bold upperca...
226 |       Con
227 |     </span>
228 |   </div>
229 | )}
230 | 2 3 1 | { / * N ù i d u n g t h ò n g b á o * / }
231 | <div className="flex-1 min-w-0">
232 |   <div className="flex items-center gap-2">
233 |     <span className="font-bold text-sm" style={{ color: 'var(--text-primary)' }}>{notif.title}...
234 |     {student && (
235 |       <span className="text-[10px] font-bold sm:hidden" style={{ color: 'var(--brand-primary)'}}...
236 |         ({student.full_name})
237 |       </span>
238 |     )}
239 |   </div>
240 |   {student && (
241 |     <div className="text-[10px] font-bold mt-0.5 hidden sm:block" style={{ color: 'var(--text-...
242 |       H í c s í n h : <span style={{ color: 'var(--text-primary)' }}>{student.full_name}
243 |     </div>
244 |   )}
245 |   <div className="text-xs mt-1.5 leading-relaxed" style={{ color: 'var(--text-secondary)' }}>{...
246 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
247 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
248 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
249 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
250 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
251 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
252 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
253 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
254 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
255 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
256 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
257 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
258 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
259 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
260 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
261 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
262 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
263 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
264 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
265 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
266 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
267 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
268 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
269 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
270 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
271 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
272 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
273 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
274 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
275 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
276 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
277 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
278 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
279 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
280 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
281 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
282 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
283 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
284 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
285 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
286 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
287 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
288 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
289 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
290 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
291 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...
292 |     <div className="flex items-center gap-1.5 text-[9px] font-bold mt-3" style={{ color: 'var(--...

```

File: components/parent/ParentSidebar.tsx (131 lines, 5.2 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import Link from 'next/link'
5 | import { usePathname } from 'next/navigation'
6 | import {
7 |   LayoutDashboard,
8 |   Users,
9 |   Link2,
10 |  Bell,
11 |  User,
12 |  Logout,
13 |  Menu,
14 |  X,
15 |  Sparkles
16 | } from 'lucide-react'
17 | import { logout } from '@lib/auth-actions'
18 |
19 | const parentNavItems = [
20 |   { icon: LayoutDashboard, label: 'Tổng quan', href: '/parent/dashboard' },
21 |   { icon: Users, label: 'Con c ả t 0 i', href: '/parent/children' },
22 |   { icon: Link2, label: 'Liên k ết c 0 n', href: '/parent/link-child' },
23 |   { icon: Bell, label: 'Thông báo', href: '/parent/notifications' },
24 |   { icon: User, label: 'H 0 s i', href: '/parent/profile' },
25 | ]
26 |
27 | export default function ParentSidebar() {
28 |   const pathname = usePathname()
29 |   const [isMobileOpen, setIsMobileOpen] = useState(false)
30 |
31 |   return (
32 |     <

```

```

33 |      /* Nút Hamburger trên Mobile */
34 |      <button
35 |        onClick={() => setIsMobileOpen(true)}
36 |        className="md:hidden fixed top-4 left-4 z-50 p-2 rounded-lg transition-colors"
37 |        style={{ background: 'var(--bg-elevated)', border: '1px solid var(--border-default)', color: 'var(--te...
38 |      >
39 |      <Menu size={20} />
40 |    </button>
41 |
42 |    /* Backdrop cho Mobile */
43 |    {isMobileOpen && (
44 |      <div
45 |        onClick={() => setIsMobileOpen(false)}
46 |        className="md:hidden fixed inset-0 z-40 bg-black/60 backdrop-blur-sm"
47 |      />
48 |    )}
49 |
50 |    /* Sidebar Container */
51 |    <aside
52 |      className={`fixed md:sticky top-0 left-0 z-50 flex flex-col w-64 h-screen transition-transform duratio...
53 |      isMobileOpen ? 'translate-x-0' : '-translate-x-full'
54 |    }>
55 |      style={{ background: 'var(--bg-surface)', borderRight: '1px solid var(--border-subtle)' }}
56 |    >
57 |      /* Header */
58 |      <div className="h-16 px-6 flex items-center justify-between" style={{ borderBottom: '1px solid var(--b...
59 |        <Link href="/" className="flex items-center gap-3">
60 |          
61 |          <span className="font-bold text-sm tracking-wide" style={{ color: 'var(--text-primary)' }}>
62 |            PC Master <span style={{ color: 'var(--brand-primary)' }}>Parent</span>
63 |          </span>
64 |        </Link>
65 |        <button
66 |          onClick={() => setIsMobileOpen(false)}
67 |          className="md:hidden" style={{ color: 'var(--text-muted)' }}
68 |        >
69 |          <X size={18} />
70 |        </button>
71 |      </div>
72 |
73 |      /* Navigation Items */
74 |      <nav className="flex-1 px-4 py-6 space-y-1.5 overflow-y-auto">
75 |        <div className="px-3 mb-2 text-[10px] font-bold tracking-wider uppercase" style={{ color: 'var(--tex...
76 |          M e n u   P h ã   H u y n h
77 |        </div>
78 |        {parentNavItems.map((item) => {
79 |          const icon = item.icon
80 |          const isActive = pathname === item.href || pathname.startsWith(item.href + '/')
81 |          return (
82 |            <Link
83 |              key={item.href}
84 |              href={item.href}
85 |              onClick={() => setIsMobileOpen(false)}
86 |              className={`flex items-center gap-3 px-3 py-2.5 rounded-lg text-sm font-medium transition-all ...
87 |                isActive
88 |                ? 'border-[#00d4aa]'
89 |                : 'border-transparent'
90 |              `}
91 |              style={{
92 |                background: isActive ? 'rgba(0, 212, 170, 0.1)' : 'transparent',
93 |                color: isActive ? 'var(--brand-primary)' : 'var(--text-muted)'
94 |              }}
95 |            >
96 |              <Icon size={18} style={{ color: isActive ? 'var(--brand-primary)' : 'var(--text-muted)' }} />
97 |              <span>{item.label}</span>
98 |            </Link>
99 |          )
100 |        })}
101 |      </nav>
102 |
103 |      { / *   F o o t e r   S i d e b a r   v ù i   N ú t   L o g   O u t   * / }
104 |      <div className="p-4" style={{ borderTop: '1px solid var(--border-subtle)', background: 'var(--bg-eleva...
105 |        <div className="flex items-center gap-3 px-2 mb-4">
106 |          <div className="w-9 h-9 rounded-full flex items-center justify-center text-sm font-bold" style={{ ...
107 |            P
108 |          </div>
109 |          <div className="overflow-hidden">
110 |            <div className="text-xs font-bold truncate" style={{ color: 'var(--text-primary)' }}>
111 |              <div className="text-[10px] truncate" style={{ color: 'var(--text-primary)' }}>
112 |            </div>
113 |          </div>
114 |          <button
115 |            onClick={async () => {
116 |              if (confirm('Bạn muốn đăng xuất không?')) {
117 |                await logout()
118 |              }
119 |            }}
120 |            className="flex items-center justify-center gap-2 w-full px-4 py-2.5 text-xs font-bold transition-...
121 |            style={{ background: 'rgba(239, 68, 68, 0.1)', border: '1px solid rgba(239, 68, 68, 0.2)', color: ...
122 |          >
123 |            <Logout size={14} />
124 |            n g   x u ý t
125 |          </button>
126 |        </div>
127 |      </aside>
128 |    </>
129 |  )
130 | }
131 |

```

File: components/PartPickerSidebar.js (164 lines, 7.5 KB)

```

1 | 'use client';
2 |
3 | import { useState, useMemo } from 'react';
4 | import { motion, AnimatePresence } from 'framer-motion';
5 | import componentsData from '../data/componentsData.json';
6 | import { COMPONENT_DATA } from '@utils/i18nData';
7 | import ComponentPreview from './ComponentPreview';
8 |
9 | const PART_ICONS = {
10 |   CPU: 'CPU', RAM: 'RAM', GPU: 'GPU', SSD: 'SSD', PSU: 'PSU', COOLER: 'Cooler'
11 | };
12 |
13 | const PART_COLORS = {
14 |   CPU: '#00d4aa', RAM: '#6366f1', GPU: '#ef4444',
15 |   SSD: '#22c55e', PSU: '#f59e0b', COOLER: '#06b6d4'
16 | };
17 |
18 | export default function PartPickerSidebar({ lang, onSelect, placedCounts }) {
19 |   const [expandedType, setExpandedType] = useState(null);
20 |   const [selectedModel, setSelectedModel] = useState(null);
21 |   const data = COMPONENT_DATA[lang] || COMPONENT_DATA.en;

```

```

22 | const parts = ['CPU', 'GPU', 'RAM', 'SSD', 'PSU', 'COOLER'];
23 |
24 | const modelsByType = useMemo(() => {
25 |   const map = {};
26 |   parts.forEach(type => {
27 |     map[type] = componentsData.filter(c => c.type === type);
28 |   });
29 |   return map;
30 | }, []);
31 |
32 | const handleSelect = (type, modelId = null) => {
33 |   if (modelId) {
34 |     onSelect(type, modelId);
35 |     const model = modelsByType[type]?.find(m => m.id === modelId);
36 |     setSelectedModel(model);
37 |     setExpandedType(null);
38 |   } else {
39 |     setExpandedType(expandedType === type ? null : type);
40 |     setSelectedModel(null);
41 |   }
42 | };
43 |
44 | return (
45 |   <div style={{
46 |     width: '240px', minWidth: '240px',
47 |     background: 'var(--bg-surface)',
48 |     borderRight: '1px solid var(--border-subtle)',
49 |     borderRadius: '12px', padding: '16px 12px',
50 |     display: 'flex', flexDirection: 'column',
51 |     maxHeight: 'calc(100vh - 120px)',
52 |     overflowY: 'auto', gap: '4px'
53 |   }}>
54 |     <h3 style={{ margin: '0 0 4px 0', fontSize: '13px', color: 'var(--text-muted)', textTransform: 'uppercase...
55 |       { lang === 'en' ? 'Component Tray' : 'Khay linh kiç n' }}
56 |   </h3>
57 |   <p style={{ margin: '0 0 12px 0', fontSize: '11px', color: 'var(--text-muted)', lineHeight: 1.4 }}>
58 |     {lang === 'en'
59 |       ? 'Click to expand and pick a specific model.'
60 |       : 'Nhấn vào chọn model cần thã.'}}
61 |   </p>
62 |
63 |   {parts.map(type => {
64 |     const info = data[type];
65 |     const count = placedCounts[type] || 0;
66 |     const models = modelsByType[type] || [];
67 |     const isExpanded = expandedType === type;
68 |
69 |     return (
70 |       <div key={type} style={{ marginBottom: isExpanded ? '4px' : '0' }}>
71 |         <button
72 |           onClick={() => handleSelect(type)}
73 |           style={{
74 |             display: 'flex', alignItems: 'center', gap: '10px',
75 |             padding: '10px 12px',
76 |             background: isExpanded ? 'var(--bg-hover)' : 'var(--bg-elevated)',
77 |             border: '1px solid ${isExpanded ? PART_COLORS[type] : 'var(--border-default)'}',
78 |             borderRadius: '8px', cursor: 'pointer',
79 |             transition: 'all 0.2s', textAlign: 'left',
80 |             width: '100%', color: 'var(--text-primary)',
81 |             fontSize: '13px', fontFamily: 'inherit',
82 |             position: 'relative'
83 |           }}
84 |           onMouseOver={e => { if (!isExpanded) { e.currentTarget.style.borderColor = PART_COLORS[type]; e...
85 |           onMouseOut={e => { if (!isExpanded) { e.currentTarget.style.borderColor = 'var(--border-default)...
86 |         >
87 |         <span style={{ fontSize: '20px' }}>{PART_ICONS[type]}</span>
88 |         <div style={{ flex: 1, minWidth: 0 }}>
89 |           <div style={{ fontWeight: 600, fontSize: '12px' }}>{type}</div>
90 |           <div style={{ fontSize: '10px', color: 'var(--text-muted)', whiteSpace: 'nowrap', overflow: 'h...
91 |             {info?.title || type} · {models.length} model
92 |           </div>
93 |         </div>
94 |         {count > 0 && (
95 |           <span style={{ background: PART_COLORS[type] + '20', color: PART_COLORS[type], padding: '1px 7...
96 |             {count}
97 |           </span>
98 |         )}
99 |         <span style={{ color: 'var(--text-muted)', fontSize: '10px', transition: 'transform 0.2s', trans...
100 |           %4
101 |       </span>
102 |     </button>
103 |
104 |     { /* Component Preview */ }
105 |     {isExpanded && (
106 |       <div style={{ margin: '8px 0', padding: '0 4px' }}>
107 |         <ComponentPreview type={type} model={null} />
108 |       </div>
109 |     )}
110 |     { /* Nested scrollable model list */ }
111 |     <AnimatePresence>
112 |       {isExpanded && (
113 |         <motion.div
114 |           initial={{ height: 0, opacity: 0 }}
115 |           animate={{ height: 'auto', opacity: 1 }}
116 |           exit={{ height: 0, opacity: 0 }}
117 |           transition={{ duration: 0.2 }}
118 |           style={{ overflow: 'hidden' }}
119 |         >
120 |           <div style={{
121 |             maxHeight: '220px', overflowY: 'auto',
122 |             marginTop: '4px', marginLeft: '8px',
123 |             display: 'flex', flexDirection: 'column', gap: '3px',
124 |             paddingRight: '4px'
125 |           }}>
126 |             {models.map(model => (
127 |               <button
128 |                 key={model.id}
129 |                 onClick={() => handleSelect(type, model.id)}
130 |                 style={{
131 |                   display: 'flex', justifyContent: 'space-between', alignItems: 'center',
132 |                   padding: '8px 10px', background: 'var(--bg-elevated)',
133 |                   border: '1px solid var(--border-subtle)', borderRadius: '6px',
134 |                   cursor: 'pointer', textAlign: 'left', width: '100%',
135 |                   color: 'var(--text-primary)', fontSize: '11px', fontFamily: 'inherit',
136 |                   transition: 'all 0.15s'
137 |                 }}
138 |                 onMouseOver={e => { e.currentTarget.style.borderColor = PART_COLORS[type]; e.currentTa...
139 |                 onMouseOut={e => { e.currentTarget.style.borderColor = 'var(--border-subtle)'; e.curre...
140 |               >
141 |                 <div style={{ flex: 1, minWidth: 0 }}>
142 |                   <div style={{ fontWeight: 600, fontSize: '11px', overflow: 'hidden', textOverflow: '...
143 |                     <div style={{ fontSize: '9px', color: 'var(--text-muted)' }}>
144 |                       {model.socket && 'Socket: ${model.socket}'}
145 |                       {model.ramType && 'RAM: ${model.ramType}'}

```

```

146 |         {model.power && `${model.power}W`}
147 |     </div>
148 | </div>
149 |     <span style={{ fontSize: '10px', color: PART_COLORS[type], fontWeight: 700, flexShrink...
150 |         {(model.price || 0).toLocaleString()}
151 |     </span>
152 | </button>
153 |     ))}
154 | </div>
155 | </motion.div>
156 | ))}
157 | </AnimatePresence>
158 | </div>
159 | );
160 | }}}
161 | </div>
162 | };
163 | }
164 |

```

File: components/PartPickerSidebarV2.js (366 lines, 17.2 KB)

```

1 | 'use client';
2 |
3 | import { useState, useMemo } from 'react';
4 | import { useBuildStore } from '@lib/build-store';
5 | import { getPriorityIssues } from '@lib/compatibility-engine';
6 |
7 | const PART_CONFIG = {
8 |   CPU: { icon: 'i', color: '#00d4aa', label: 'CPU', labelVn: 'CPU' },
9 |   GPU: { icon: '0<B@', color: '#ef4444', label: 'GPU', labelVn: 'Card',
10 |   MAINBOARD: { icon: '0=Y', color: '#6366f1', label: 'Mainboard', labelVn: 'Mainboard' },
11 |   RAM: { icon: '0>Yà', color: '#22c55e', label: 'RAM', labelVn: 'RAM' },
12 |   SSD: { icon: '0=U%', color: '#06b6d4', label: 'SSD', labelVn: 'Ổ cứng' },
13 |   PSU: { icon: '0=Y', color: '#f59e0b', label: 'PSU', labelVn: 'Nguồn' },
14 |   COOLER: { icon: '0=Uæ', color: '#8b5cf6', label: 'Cooler', labelVn: 'Tản' },
15 |   CASE: { icon: '0=Uæ', color: '#ec4899', label: 'Case', labelVn: 'Vỏ máy' },
16 | };
17 |
18 | function SpecRow({ label, value, color }) {
19 |   if (!value && value !== 0) return null;
20 |   return (
21 |     <div style={{ display: 'flex', justifyContent: 'space-between', padding: '2px 0', fontSize: '11px' }}>
22 |       <span style={{ color: 'var(--text-muted)' }}>{label}</span>
23 |       <span style={{ color: color || 'var(--text-primary)', fontWeight: 600 }}>{value}</span>
24 |     </div>
25 |   );
26 | }
27 |
28 | export default function PartPickerSidebarV2({ lang = 'vn' }) {
29 |   const [expandedType, setExpandedType] = useState(null);
30 |   const [products, setProducts] = useState({});
31 |   const [loading, setLoading] = useState({});
32 |   const [searchQuery, setSearchQuery] = useState('');
33 |
34 |   const store = useBuildStore();
35 |
36 |   const selectedParts = useMemo(() => ({
37 |     CPU: store.cpu,
38 |     GPU: store.gpu,
39 |     MAINBOARD: store.mainboard,
40 |     RAM: store.ram,
41 |     SSD: store.ssd,
42 |     PSU: store.psu,
43 |     COOLER: store.cooler,
44 |     CASE: store.case,
45 |   })), [store.cpu, store.gpu, store.mainboard, store.ram, store.ssd, store.psu, store.cooler, store.case]);
46 |
47 |   const { errors, warnings } = useMemo(
48 |     () => getPriorityIssues(store.issues),
49 |     [store.issues]
50 |   );
51 |
52 |   const fetchProducts = async (type) => {
53 |     if (products[type]) return;
54 |     setLoading(prev => ({ ...prev, [type]: true }));
55 |     try {
56 |       const params = new URLSearchParams({ category: type, limit: '50' });
57 |       if (searchQuery) params.set('q', searchQuery);
58 |       const res = await fetch(`/api/products?${params}`);
59 |       const data = await res.json();
60 |       if (data.products) setProducts(prev => ({ ...prev, [type]: data.products }));
61 |     } catch (err) {
62 |       console.error('Failed to fetch products:', err);
63 |     } finally {
64 |       setLoading(prev => ({ ...prev, [type]: false }));
65 |     }
66 |   };
67 |
68 |   const handleSelect = (type, product) => {
69 |     if (!product) return;
70 |     switch (type) {
71 |       case 'CPU': store.setCPU(product); break;
72 |       case 'GPU': store.setGPU(product); break;
73 |       case 'MAINBOARD': store.setMainboard(product); break;
74 |       case 'RAM': store.addRAM(product); break;
75 |       case 'SSD': store.addSSD(product); break;
76 |       case 'PSU': store.setPSU(product); break;
77 |       case 'COOLER': store.setCooler(product); break;
78 |       case 'CASE': store.setCase(product); break;
79 |     }
80 |     setExpandedType(null);
81 |   };
82 |
83 |   const handleRemove = (type, index) => {
84 |     switch (type) {
85 |       case 'CPU': store.setCPU(null); break;
86 |       case 'GPU': store.setGPU(null); break;
87 |       case 'MAINBOARD': store.setMainboard(null); break;
88 |       case 'RAM': store.removeRAM(index); break;
89 |       case 'SSD': store.removeSSD(index); break;
90 |       case 'PSU': store.setPSU(null); break;
91 |       case 'COOLER': store.setCooler(null); break;
92 |       case 'CASE': store.setCase(null); break;
93 |     }
94 |   };
95 |
96 |   const toggleExpand = (type) => {
97 |     if (expandedType === type) {
98 |       setExpandedType(null);
99 |     } else {
100 |       setExpandedType(type);
101 |       fetchProducts(type);

```

```

102 |   }
103 | };
104 |
105 | const getSpecs = (type, product) => {
106 |   switch (type) {
107 |     case 'CPU': return [
108 |       { label: 'Socket', value: product.cpu_socket },
109 |       { label: 'Nhân / Lõi', value: `${product.cpu_cores}C / ${product.cpu_base_ghz} / ${product.cpu_boost_ghz} GHz`,
110 |         { label: 'Clock', value: `${product.cpu_base_ghz} / ${product.cpu_boost_ghz} GHz` },
111 |         { label: 'TDP', value: `${product.cpu_tdp_watts}W`, color: product.cpu_tdp_watts > 125 ? 'var(--warnin...' },
112 |         { label: 'RAM', value: `${product.cpu_ram_type} ${product.cpu_max_ram_mhz}MHz` },
113 |         { label: 'iGPU', value: product.cpu_has_igpu ? product.cpu_igpu_model || 'Có' : 'Không' },
114 |       ];
115 |     case 'GPU': return [
116 |       { label: 'VRAM', value: `${product.gpu_vram_gb}GB` },
117 |       { label: 'TDP', value: `${product.gpu_tdp_watts}W`,
118 |         { label: 'Chiều dài', value: `${product.gpu_length_mm}mm`, color: 'var(--danger)' },
119 |         { label: 'PSU yêu cầu', value: `${product.gpu_recommended_psu}W` },
120 |       ];
121 |     case 'MAINBOARD': return [
122 |       { label: 'Socket', value: product.mb_socket },
123 |       { label: 'Chipset', value: product.mb_chipset },
124 |       { label: 'Form Factor', value: product.mb_form_factor },
125 |       { label: 'RAM', value: `${product.mb_ram_type} ${product.mb_max_ram_mhz}MHz`,
126 |         { label: 'Khe RAM', value: `${product.mb_ram_slots} khe / tối đa ${product.mb_max_ram_slots} khe` },
127 |       ];
128 |     case 'M.2 / SATA': return [
129 |       { label: 'M.2 / SATA', value: `${product.mb_m2_slots} M.2 · ${product.mb_sata_ports} SATA` },
130 |       { label: 'WiFi', value: product.mb_has_wifi ? product.mb_wifi_version || 'Có' : 'Không' },
131 |     ];
132 |     case 'RAM': return [
133 |       { label: 'Loại', value: product.ram_ddr_type },
134 |       { label: 'Dung lượng', value: `${product.ram_capacity_gb}GB × ${product.ram_speed_mhz}MHz` },
135 |       { label: 'CL', value: product.ram_cl_latency },
136 |       { label: 'XMP/EXPO', value: product.ram_has_xmp ? 'XMP' : product.ram_has_expo ? 'EXPO' : 'Không' },
137 |     ];
138 |     case 'SSD': return [
139 |       { label: 'Loại', value: product.ssd_interface === 'm2_nvme' ? 'M.2 NVMe' : 'SATA' },
140 |       { label: 'Dung lượng', value: `${product.ssd_capacity_gb}GB` },
141 |       { label: 'Tốc độ đọc', value: `${product.ssd_read_mbps} / ${product.ssd_write_mbps}MB/s` },
142 |       { label: 'DRAM', value: product.ssd_has_dram_cache ? 'Có' : 'Không' },
143 |     ];
144 |     case 'PSU': return [
145 |       { label: 'Công suất', value: `${product.psu_wattage}W`, color: 'var(--danger)' },
146 |       { label: 'Chuẩn', value: product.psu_efficiency },
147 |       { label: 'Modular', value: product.psu_modular === 'full_modular' ? 'Full' : product.psu_modular === 'semi_modular' ? 'Semi' : 'Non-Modular' },
148 |       { label: 'Dài', value: `${product.psu_length_mm}mm` },
149 |     ];
150 |     case 'COOLER': return [
151 |       { label: 'Loại', value: product.cooler_type === 'aio' ? 'AIO' : 'Tower' },
152 |       { label: 'TDP tối đa', value: `${product.cooler_max_tdp_support}W` },
153 |       { label: 'Cao', value: product.cooler_height_mm ? `${product.cooler_height_mm}mm` : '-' },
154 |       { label: 'RGB', value: product.cooler_has_rgb ? 'Có' : 'Không' },
155 |     ];
156 |     case 'CASE': return [
157 |       { label: 'Form h × t × r', value: product.case_supported_forms?.join(' / ') || 'Standard' },
158 |       { label: 'GPU tối đa', value: `${product.case_max_gpu_length_mm}mm` },
159 |       { label: 'Cooler tối đa', value: `${product.case_max_cooler_height_mm}mm` },
160 |       { label: 'AIO h × t × r', value: product.case_supports_aio?.map(s => `${s.h} × ${s.t} × ${s.r}mm`)?.join(' / ') || 'Không' },
161 |       { label: 'Kính', value: product.case_side_panel === 'glass' ? 'Kính' : 'Nhựa' },
162 |     ];
163 |     default: return [];
164 |   }
165 | };
166 |
167 | const renderSelectedPart = (type) => {
168 |   const items = selectedParts[type];
169 |   if (!items || (Array.isArray(items) && items.length === 0)) return null;
170 |
171 |   const arr = Array.isArray(items) ? items : [items];
172 |   return arr.map((item, idx) => {
173 |     <div key={idx} style={{
174 |       display: 'flex', alignItems: 'center', gap: '8px',
175 |       padding: '6px 8px', marginTop: '4px',
176 |       background: PART_CONFIG[type]?.color + '12',
177 |       borderRadius: '6px', border: '1px solid ${PART_CONFIG[type]?.color}30',
178 |       fontSize: '11px',
179 |     }}>
180 |       <div style={{ flex: 1, minWidth: 0 }}>
181 |         <div style={{ fontWeight: 600, color: 'var(--text-primary)', overflow: 'hidden', textOverflow: 'elli...' }}>
182 |           {item.full_name || item.name}
183 |         </div>
184 |         <div style={{ color: 'var(--text-muted)' }}>
185 |           {((item.price_vnd || 0).toLocaleString())}
186 |         </div>
187 |       </div>
188 |       <button
189 |         onClick={() => handleRemove(type, idx)}
190 |         style={{
191 |           background: 'rgb(239,68,68,0.1)', border: 'none', color: '#ef4444',
192 |           borderRadius: '4px', padding: '2px 6px', cursor: 'pointer', fontSize: '11px', fontFamily: 'inherit',
193 |         }}>
194 |         ✕
195 |       </button>
196 |     </div>
197 |   ));
198 | };
199 |
200 |
201 | return (
202 |   <div style={{
203 |     width: '280px', minWidth: '280px',
204 |     background: 'var(--bg-surface)',
205 |     borderRight: '1px solid var(--border-subtle)',
206 |     borderRadius: '12px',
207 |     display: 'flex', flexDirection: 'column',
208 |     maxHeight: 'calc(100vh - 100px)',
209 |     overflow: 'hidden',
210 |   }}>
211 |     <div style={{ flex: 1, flexDirection: 'column' }}>
212 |       <div style={{ padding: '16px', borderBottom: '1px solid var(--border-subtle)', background: 'var(--bg-elevated)' }}>
213 |         <div style={{ display: 'flex', justify-content: space-between, align-items: center }}>
214 |           <div style={{ flex: 1 }}>
215 |             <div style={{ display: 'flex', justify-content: space-between }}>
216 |               <div style={{ flex: 1 }}>
217 |                 <div style={{ display: 'flex', justify-content: space-between }}>
218 |                   <div style={{ flex: 1 }}>
219 |                     <div style={{ display: 'flex', justify-content: space-between }}>
220 |                       <div style={{ flex: 1 }}>
221 |                         <div style={{ display: 'flex', justify-content: space-between }}>
222 |                           <div style={{ flex: 1 }}>
223 |                             <div style={{ display: 'flex', justify-content: space-between }}>
224 |                               <div style={{ flex: 1 }}>
225 |                                 <div style={{ display: 'flex', justify-content: space-between }}>

```

```

226 |     /* Search */
227 |     <div style={{ padding: '8px 12px' }}>
228 |         <input
229 |             type="text"
230 |             value={searchQuery}
231 |             onChange={e => setSearchQuery(e.target.value)}
232 |             placeholder={lang === 'en' ? 'Search components...' : 'Tìm linh kiện'}
233 |             style={{
234 |                 width: '100%', padding: '6px 10px', borderRadius: '6px',
235 |                 border: '1px solid var(--border-default)',
236 |                 background: 'var(--bg-elevated)', color: 'var(--text-primary)',
237 |                 fontSize: '12px', fontFamily: 'inherit', outline: 'none',
238 |             }}
239 |         />
240 |     </div>
241 |
242 |     /* Component list */
243 |     <div style={{ flex: 1, overflowY: 'auto', padding: '0 12px 12px' }}>
244 |         {Object.entries(PART_CONFIG).map(([type, cfg]) => {
245 |             const isExpanded = expandedType === type;
246 |             const selected = selectedParts[type];
247 |             const hasSelected = selected && (Array.isArray(selected) ? selected.length > 0 : true);
248 |             const list = products[type];
249 |             const isLoading = loading[type];
250 |
251 |             return (
252 |                 <div key={type} style={{ marginBottom: '4px' }}>
253 |                     <button
254 |                         onClick={() => toggleExpand(type)}
255 |                         style={{
256 |                             display: 'flex', alignItems: 'center', gap: '8px',
257 |                             width: '100%', padding: '8px 10px',
258 |                             background: isExpanded ? 'var(--bg-hover)' : 'hasSelected ? cfg.color + '10' : 'var(--bg-elevated)',
259 |                             border: '1px solid ' + (isExpanded ? cfg.color : hasSelected ? cfg.color + '40' : 'var(--border-default)'),
260 |                             borderRadius: '8px', cursor: 'pointer',
261 |                             textAlign: 'left', fontFamily: 'inherit',
262 |                             color: 'var(--text-primary)', fontSize: '12px',
263 |                             transition: 'all 0.15s',
264 |                         }}
265 |                     >
266 |                         <span style={{ fontSize: '16px' }}>{cfg.icon}</span>
267 |                         <div style={{ flex: 1, minWidth: 0 }}>
268 |                             <div style={{ fontWeight: 600, fontSize: '12px' }}>
269 |                                 {lang === 'en' ? cfg.label : cfg.labelVn}
270 |                             </div>
271 |                             {hasSelected && (
272 |                                 <div style={{ fontSize: '10px', color: cfg.color, overflow: 'hidden', textOverflow: 'ellipsis' }}
273 |                                     {Array.isArray(selected) ? ` ${selected.length} ${lang === 'vi' ? 'đã chọn' : 'items selected'}` : ''}
274 |                                 )}
275 |                             </div>
276 |                             <span style={{
277 |                                 color: 'var(--text-muted)', fontSize: '9px',
278 |                                 transition: 'transform 0.2s', transform: isExpanded ? 'rotate(180deg)' : 'none',
279 |                             }}>
280 |                                 %%%
281 |                             </span>
282 |                         </div>
283 |                     </button>
284 |
285 |                     {renderSelectedPart(type)}
286 |
287 |                     {isExpanded && (
288 |                         <div style={{
289 |                             marginTop: '4px', maxHeight: '300px', overflowY: 'auto',
290 |                             display: 'flex', flexDirection: 'column', gap: '3px',
291 |                             padding: '4px', background: 'var(--bg-elevated)', borderRadius: '8px',
292 |                         }}>
293 |                             {isLoading ? (
294 |                                 <div style={{ textAlign: 'center', padding: '16px', color: 'var(--text-muted)', fontSize: ... }}
295 |                                     {lang === 'en' ? 'Loading...' : 'Đang tải...'}
296 |                                 </div>
297 |                             ) : !list || list.length === 0 ? (
298 |                                 <div style={{ textAlign: 'center', padding: '16px', color: 'var(--text-muted)', fontSize: ... }}
299 |                                     {lang === 'en' ? 'No components found' : 'Không tìm thấy linh kiện'}
300 |                                 </div>
301 |                             ) : (
302 |                                 list.map(product => (
303 |                                     <button
304 |                                         key={product.id}
305 |                                         onClick={() => handleSelect(type, product)}
306 |                                         style={{
307 |                                             display: 'flex', flexDirection: 'column', gap: '2px',
308 |                                             padding: '8px 10px', background: 'var(--bg-surface)',
309 |                                             border: '1px solid var(--border-subtle)', borderRadius: '6px',
310 |                                             cursor: 'pointer', textAlign: 'left', fontFamily: 'inherit',
311 |                                             color: 'var(--text-primary)', fontSize: '11px',
312 |                                             transition: 'all 0.1s',
313 |                                         }}
314 |                                         onMouseOver={e => { e.currentTarget.style.borderColor = cfg.color; e.currentTarget.style.backgroundColor = 'var(--bg-elevated)'; e.currentTarget.style.borderColor = 'var(--border-subtle)'; e.currentTarget.style.backgroundColor = 'var(--bg-surface)'; }}
315 |                                         >
316 |                                             <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>
317 |                                                 <span style={{ fontWeight: 600, fontSize: '11px', overflow: 'hidden', textOverflow: 'ellipsis' }}>
318 |                                                     {product.full_name || product.name}
319 |                                                 </span>
320 |                                                 <span style={{ color: cfg.color, fontWeight: 700, fontSize: '10px', flexShrink: 0, margin: 0 10px }}>
321 |                                                     {product.price_vnd || 0}.toLocaleString()
322 |                                                 </span>
323 |                                             </div>
324 |                                             <div style={{ display: 'flex', gap: '6px', flexWrap: 'wrap' }}>
325 |                                                 {getSpecs(type, product).slice(0, 3).map((spec, si) => (
326 |                                                     <span key={si} style={{
327 |                                                         background: 'var(--bg-elevated)', padding: '1px 5px',
328 |                                                         borderRadius: '3px', fontSize: '9px', color: 'var(--text-muted)',
329 |                                                     }}>
330 |                                                         {spec.label}: {spec.value}
331 |                                                     </span>
332 |                                                 ))}
333 |                                             </div>
334 |                                         </button>
335 |                                     )}
336 |                                 </div>
337 |                             )}
338 |                         </div>
339 |                     )}
340 |                 </div>
341 |             );
342 |         })}
343 |     </div>
344 |
345 |     /* Budget Summary */
346 |     <div style={{
347 |         padding: '12px 16px', borderTop: '1px solid var(--border-subtle)',
348 |         background: 'var(--bg-elevated)',
349 |     }}>

```

```

350|         <div style={{ display: 'flex', justifyContent: 'space-between', marginBottom: '4px', fontSize: '12px' }}>
351|             <span style={{ color: 'var(--text-muted)' }}>{ lang === 'en' ?
352|                 <span style={{ fontWeight: 700, color: store.totalPrice > store.budget ? 'var(--danger)' : 'var(--te...
353|                     { (store.totalPrice || 0) .toLocaleString() }
354|                 </span>
355|             </div>
356|             <div style={{ display: 'flex', justifyContent: 'space-between', fontSize: '12px' }}>
357|                 <span style={{ color: 'var(--text-muted)' }}>TDP</span>
358|                 <span style={{ fontWeight: 600, color: 'var(--text-primary)' }}>
359|                     {store.totalTdp}W
360|                 </span>
361|             </div>
362|         </div>
363|     </div>
364| );
365| }
366|

```

File: components/PcCase.tsx (272 lines, 8.7 KB)

```

1| 'use client';
2|
3| import { useRef, useMemo } from 'react';
4| import { useFrame } from '@react-three/fiber';
5| import { RoundedBox } from '@react-three/drei';
6| import * as THREE from 'three';
7| import { useAssemblyStore } from '@lib/useStore';
8|
9| function GlassPanel({ position, size }: { position: [number, number, number]; size: [number, number] }) {
10|     return (
11|         <mesh position={position}>
12|             <planeGeometry args={size} />
13|             <meshPhysicalMaterial
14|                 color="#88ccff"
15|                 metalness={0.3}
16|                 roughness={0.05}
17|                 transparent
18|                 opacity={0.18}
19|                 envMapIntensity={1.5}
20|                 side={THREE.DoubleSide}
21|             />
22|         </mesh>
23|     );
24| }
25|
26| function CaseFan({ position, fanSpeed, rgbActive }: {
27|     position: [number, number, number];
28|     fanSpeed: number;
29|     rgbActive: boolean;
30| }) {
31|     const bladeRef = useRef<THREE.Group>(null);
32|     const hue = useMemo(() => Math.floor(Math.random() * 360), []);
33|     const color1 = `hsl(${hue}, 100%, 60%)`;
34|     const color2 = `hsl(${(hue + 60) % 360}, 100%, 60%)`;
35|
36|     useFrame((_, delta) => {
37|         if (bladeRef.current && fanSpeed > 0) {
38|             bladeRef.current.rotation.z += delta * fanSpeed * 25;
39|         }
40|     });
41|
42|     const ringColor = rgbActive ? color1 : '#445588';
43|     const ringIntensity = rgbActive ? 1 : 0;
44|
45|     return (
46|         <group position={position}>
47|             <mesh>
48|                 <torusGeometry args={[0.14, 0.015, 8, 28]} />
49|                 <meshPhysicalMaterial
50|                     color={ringColor}
51|                     emissive={ringColor}
52|                     emissiveIntensity={ringIntensity}
53|                     metalness={0.5}
54|                     roughness={0.3}
55|                     transparent
56|                     opacity={rgbActive ? 1 : 0.4}
57|                 />
58|             </mesh>
59|             <group ref={bladeRef} position={[0, 0, 0]}>
60|                 {Array.from({ length: 7 }).map((_, i) => {
61|                     const angle = (i / 7) * Math.PI * 2;
62|                     return (
63|                         <mesh>
64|                             key={i}
65|                             position={[Math.sin(angle) * 0.09, Math.cos(angle) * 0.09, 0]}
66|                             rotation={[0, 0, angle + Math.PI / 2]}
67|                         >
68|                             <planeGeometry args={[0.04, 0.18]} />
69|                             <meshPhysicalMaterial
70|                                 color={rgbActive ? '#667799' : '#445566'}
71|                                 metalness={0.6}
72|                                 roughness={0.3}
73|                                 transparent
74|                                 opacity={0.85}
75|                                 side={THREE.DoubleSide}
76|                             />
77|                         </mesh>
78|                     );
79|                 })}
80|             <mesh>
81|                 <cylinderGeometry args={[0.025, 0.025, 0.008, 12]} />
82|                 <meshStandardMaterial color="#667799" metalness={0.5} roughness={0.3} />
83|             </mesh>
84|         </group>
85|         {rgbActive && (
86|             <
87|                 <pointLight position={[0, 0, 0]} intensity={0.3} color={color1} distance={0.8} />
88|                 <pointLight position={[0, 0, 0]} intensity={0.15} color={color2} distance={0.5} />
89|             </>
90|         )}
91|     </group>
92| );
93| }
94|
95| function MotherboardTray() {
96|     return (
97|         <group position={[0, 0.015, 0]}>
98|             <RoundedBox args={[2.2, 0.008, 1.8]} radius={0.005}>
99|                 <meshPhysicalMaterial
100|                     color="#1a3a2a"
101|                     roughness={0.8}
102|                     metalness={0.2}
103|                     envMapIntensity={0.4}

```



```

104 |     />
105 |   </RoundedBox>
106 |   <mesh position=[[0.5, 0.002, 0.35]]>
107 |     <RoundedBox args=[[0.5, 0.004, 0.5]] radius={0.005}>
108 |       <meshStandardMaterial color="#c0c0c0" metalness={0.6} roughness={0.3} />
109 |     </RoundedBox>
110 |   </mesh>
111 |   <mesh position=[[-0.6, 0.002, 0.35]]>
112 |     <RoundedBox args=[[0.18, 0.004, 0.9]] radius={0.004}>
113 |       <meshStandardMaterial color="#334455" roughness={0.7} />
114 |     </RoundedBox>
115 |   </mesh>
116 |   <mesh position=[[0, 0.002, -0.4]]>
117 |     <RoundedBox args=[[1.2, 0.004, 0.04]] radius={0.003}>
118 |       <meshStandardMaterial color="#556677" roughness={0.5} metalness={0.3} />
119 |     </RoundedBox>
120 |   </mesh>
121 |   <mesh position=[[0.7, 0.002, -0.5]]>
122 |     <RoundedBox args=[[0.04, 0.004, 0.04]] radius={0.003}>
123 |       <meshStandardMaterial color="#d0d0d0" metalness={0.7} roughness={0.2} />
124 |     </RoundedBox>
125 |   </mesh>
126 |   <mesh position=[[-0.2, 0.003, -0.6]]>
127 |     <cylinderGeometry args=[[0.06, 0.06, 0.004, 16]] />
128 |     <meshStandardMaterial color="#d0d0d0" metalness={0.7} roughness={0.2} />
129 |   </mesh>
130 |   <mesh position=[[0.3, 0.003, -0.6]]>
131 |     <cylinderGeometry args=[[0.06, 0.06, 0.004, 16]] />
132 |     <meshStandardMaterial color="#d0d0d0" metalness={0.7} roughness={0.2} />
133 |   </mesh>
134 |   {Array.from({ length: 6 }).map((_, i) => {
135 |     const x = -0.9 + (i % 3) * 0.9;
136 |     const z = -0.7 + Math.floor(i / 3) * 1.4;
137 |     return (
138 |       <mesh key={i} position=[[x, -0.002, z]]>
139 |         <cylinderGeometry args=[[0.012, 0.015, 0.006, 8]] />
140 |         <meshStandardMaterial color="aaaaac" metalness={0.7} roughness={0.2} />
141 |       </mesh>
142 |     );
143 |   })}
144 |   <mesh position=[[0.5, 0.003, 0.35]]>
145 |     <RoundedBox args=[[0.35, 0.002, 0.35]] radius={0.003}>
146 |       <meshStandardMaterial color="#999" metalness={0.4} roughness={0.4} />
147 |     </RoundedBox>
148 |   </mesh>
149 | </group>
150 | );
151 | }
152 |
153 | export default function PcCase() {
154 |   const bootStatus = useAssemblyStore((s) => s.bootStatus);
155 |   const fanSpeed = useAssemblyStore((s) => s.fanSpeed);
156 |   const rgbActive = useAssemblyStore((s) => s.rgbActive);
157 |   const isDefaultRgb = bootStatus === 'idle';
158 |
159 |   const glowColor = useMemo(() => {
160 |     if (bootStatus === 'success') return '#00ffcc';
161 |     if (bootStatus === 'failed') return '#ff4466';
162 |     return '#4488ff';
163 |   }, [bootStatus]);
164 |
165 |   const glowIntensity = useMemo(() => {
166 |     if (bootStatus === 'success') return 1;
167 |     if (bootStatus === 'failed') return 0.6;
168 |     return 0.2;
169 |   }, [bootStatus]);
170 |
171 |   const caseColor = '#2a3a5c';
172 |   const caseMetalness = 0.7;
173 |   const caseRoughness = 0.25;
174 |
175 |   return (
176 |     <group position=[[0, 0.35, 0]]>
177 |       <RoundedBox args=[[2.6, 1.3, 2.0]] radius={0.02}>
178 |         <meshPhysicalMaterial
179 |           color={caseColor}
180 |           metalness={caseMetalness}
181 |           roughness={caseRoughness}
182 |           envMapIntensity={0.8}
183 |         />
184 |       </RoundedBox>
185 |
186 |       <mesh position=[[0, 0.66, 0]]>
187 |         <RoundedBox args=[[2.62, 0.015, 2.02]] radius={0.005}>
188 |           <meshPhysicalMaterial color="#3a4a6c" metalness={0.7} roughness={0.2} />
189 |         </RoundedBox>
190 |       </mesh>
191 |
192 |       <mesh position=[[0, -0.66, 0]]>
193 |         <RoundedBox args=[[2.62, 0.015, 2.02]] radius={0.005}>
194 |           <meshPhysicalMaterial color="#1a2a4a" metalness={0.5} roughness={0.3} />
195 |         </RoundedBox>
196 |       </mesh>
197 |
198 |       <mesh position=[[1.31, 0, 0]]>
199 |         <planeGeometry args=[[2.02, 1.3]] />
200 |         <meshPhysicalMaterial
201 |           color="#2a3a5c"
202 |           metalness={0.6}
203 |           roughness={0.3}
204 |           side={THREE.DoubleSide}
205 |         />
206 |       </mesh>
207 |
208 |       <GlassPanel position=[[-1.305, 0, 0]] size=[[1.98, 1.26]] as [number, number] />
209 |
210 |       <mesh position=[[0, 0, 1.01]]>
211 |         <planeGeometry args=[[2.5, 1.15]] />
212 |         <meshPhysicalMaterial
213 |           color="#3a4a6c"
214 |           metalness={0.7}
215 |           roughness={0.2}
216 |           side={THREE.DoubleSide}
217 |         />
218 |       </mesh>
219 |
220 |       <mesh position=[[0, 0, -1.01]]>
221 |         <planeGeometry args=[[2.5, 1.15]] />
222 |         <meshPhysicalMaterial
223 |           color="#1a2a4a"
224 |           metalness={0.4}
225 |           roughness={0.5}
226 |           side={THREE.DoubleSide}
227 |         />

```

```

228 |     </mesh>
229 |
230 |     <mesh position=[[0, 0.04, 1.015]]>
231 |       <planeGeometry args=[[0.35, 0.025]] />
232 |       <meshPhysicalMaterial
233 |         color={glowColor}
234 |         transparent
235 |         opacity={glowIntensity}
236 |         emissive={glowColor}
237 |         emissiveIntensity={glowIntensity * 3}
238 |       />
239 |     </mesh>
240 |
241 |     <mesh position=[[0, 0, 1.015]]>
242 |       <planeGeometry args=[[0.12, 0.012]] />
243 |       <meshPhysicalMaterial
244 |         color={glowColor}
245 |         transparent
246 |         opacity={glowIntensity * 0.5}
247 |         emissive={glowColor}
248 |         emissiveIntensity={glowIntensity}
249 |       />
250 |     </mesh>
251 |
252 |     <pointLight position=[[0, 0.3, 0.8]] intensity={glowIntensity} color={glowColor} distance={2.5} decay={0...
253 |
254 |     <CaseFan position=[[-0.7, 0.4, 1.01]] fanSpeed={fanSpeed || (isDefaultRgb ? 0.3 : 0)} rgbActive={rgbActi...
255 |     <CaseFan position=[[0.7, 0.4, 1.01]] fanSpeed={fanSpeed || (isDefaultRgb ? 0.3 : 0)} rgbActive={rgbActiv...
256 |     <CaseFan position=[[-0.7, -0.3, 1.01]] fanSpeed={fanSpeed || (isDefaultRgb ? 0.3 : 0)} rgbActive={rgbAct...
257 |     <CaseFan position=[[0.7, -0.3, 1.01]] fanSpeed={fanSpeed || (isDefaultRgb ? 0.3 : 0)} rgbActive={rgbActi...
258 |
259 |     <MotherboardTray />
260 |
261 |     <mesh position=[[0, 0.2, 1.015]] rotation=[[0, 0, 0]]>
262 |       <boxGeometry args=[[0.08, 0.01, 0.01]] />
263 |       <meshPhysicalMaterial
264 |         color="#ffffff"
265 |         emissive={glowColor}
266 |         emissiveIntensity={glowIntensity * 2}
267 |       />
268 |     </mesh>
269 |   </group>
270 | );
271 | }
272 |

```

File: components/PdfViewer.tsx (125 lines, 4.9 KB)

```

1 | 'use client'
2 |
3 | import React, { useState, useCallback } from 'react'
4 | import { ChevronLeft, ChevronRight, ZoomIn, ZoomOut, Loader2 } from 'lucide-react'
5 |
6 | export default function PdfViewer({ url, title }: { url: string; title?: string }) {
7 |   const [page, setPage] = useState(1)
8 |   const [totalPages, setTotalPages] = useState(0)
9 |   const [loading, setLoading] = useState(true)
10 |   const [scale, setScale] = useState(1)
11 |
12 |   if (!url) return null
13 |
14 |   const isGoogleDrive = url.includes('drive.google.com')
15 |   const isPdf = url.toLowerCase().endsWith('.pdf')
16 |
17 |   let embedUrl = url
18 |   if (isGoogleDrive) {
19 |     const id = url.match(/\/d\/(.+)\(\/?(?:view|preview)\/\)?\.[1] || url.match(/id=(.+)(&|$)/)?\.[1]
20 |     if (id) {
21 |       embedUrl = `https://docs.google.com/gview?url=https://drive.google.com/uc?export=download&id=${id}&embed...
22 |     }
23 |   }
24 |
25 |   if (isGoogleDrive || !isPdf) {
26 |     return (
27 |       <div style={{ borderRadius: '12px', overflow: 'hidden', border: '1px solid var(--border-subtle)' }}>
28 |         <iframe
29 |           src={embedUrl}
30 |           style={{ width: '100%', height: '80vh', border: 'none' }}
31 |           title={title || 'PDF Viewer'}
32 |           loading="lazy"
33 |           allowFullScreen
34 |         />
35 |       </div>
36 |     )
37 |   }
38 |
39 |   return (
40 |     <div className="pdf-viewer" style={{
41 |       borderRadius: '12px', overflow: 'hidden',
42 |       border: '1px solid var(--border-subtle)',
43 |       background: 'var(--bg-elevated)'
44 |     }}>
45 |       <div style={{
46 |         display: 'flex', alignItems: 'center', justifyContent: 'space-between',
47 |         padding: '8px 16px', background: 'var(--bg-surface)',
48 |         borderBottom: '1px solid var(--border-subtle)'
49 |       }}>
50 |         <div style={{ display: 'flex', alignItems: 'center', gap: '8px' }}>
51 |           <button
52 |             onClick={() => setPage(p => Math.max(1, p - 1))}
53 |             disabled={page <= 1}
54 |             style={{
55 |               padding: '6px 10px', borderRadius: '6px', border: '1px solid var(--border-default)',
56 |               background: 'var(--bg-elevated)', cursor: page <= 1 ? 'not-allowed' : 'pointer',
57 |               color: page <= 1 ? 'var(--text-muted)' : 'var(--text-primary)', opacity: page <= 1 ? 0.4 : 1
58 |             }}
59 |           >
60 |             <ChevronLeft size={16} />
61 |           </button>
62 |           <span style={{ fontSize: '13px', fontWeight: 600, color: 'var(--text-primary)' }}>
63 |             Trang {page} / {totalPages} || '--'
64 |           </span>
65 |           <button
66 |             onClick={() => setPage(p => Math.min(totalPages, p + 1))}
67 |             disabled={page >= totalPages}
68 |             style={{
69 |               padding: '6px 10px', borderRadius: '6px', border: '1px solid var(--border-default)',
70 |               background: 'var(--bg-elevated)', cursor: page >= totalPages ? 'not-allowed' : 'pointer',
71 |               color: page >= totalPages ? 'var(--text-muted)' : 'var(--text-primary)', opacity: page >= totalP...
72 |             }}
73 |           >
74 |             <ChevronRight size={16} />
75 |           </button>

```

```

76 | </div>
77 | <div style={{ display: 'flex', alignItems: 'center', gap: '8px' }}>
78 |   <button
79 |     onClick={() => setScale(s => Math.max(0.5, s - 0.25))}
80 |     style={{ padding: '6px 8px', borderRadius: '6px', border: 'none', background: 'transparent', curso...
81 |   >
82 |     <ZoomOut size={16} />
83 |   </button>
84 |   <span style={{ fontSize: '12px', color: 'var(--text-muted)', fontWeight: 600 }}>{Math.round(scale * ...
85 |   <button
86 |     onClick={() => setScale(s => Math.min(2, s + 0.25))}
87 |     style={{ padding: '6px 8px', borderRadius: '6px', border: 'none', background: 'transparent', curso...
88 |   >
89 |     <ZoomIn size={16} />
90 |   </button>
91 | </div>
92 | </div>
93 | <div style={{
94 |   position: 'relative', width: '100%', minHeight: '300px',
95 |   display: 'flex', justifyContent: 'center', alignItems: 'flex-start',
96 |   padding: '16px', overflow: 'auto',
97 |   background: 'var(--bg-base)'
98 | }}>
99 |   {loading && (
100 |     <div style={{
101 |       position: 'absolute', top: '50%', left: '50%', transform: 'translate(-50%, -50%)',
102 |       display: 'flex', flexDirection: 'column', alignItems: 'center', gap: '8px',
103 |       color: 'var(--text-muted)'
104 |     }}>
105 |       <loader2 size={24} style={{ animation: 'spin 1s linear infinite' }} />
106 |     <span style={{ fontSize: '12px' }}> ang t ê i P D F . . . </span>
107 |   )}
108 |   <iframe
109 |     src={` ${embedUrl}#page=${page}`}
110 |     style={{
111 |       width: '100%', maxWidth: '800px', height: '70vh', border: 'none',
112 |       borderRadius: '8px', opacity: loading ? 0 : 1,
113 |       transition: 'opacity 0.3s'
114 |     }}
115 |     onLoad={() => setLoading(false)}
116 |     title={title || 'PDF'}
117 |     loading="lazy"
118 |   />
119 | </div>
120 | <style>{@keyframes spin { to { transform: rotate(360deg) } }}</style>
121 | </div>
122 | )
123 | }
124 | }
125 |

```

File: components/price-compare/ChatInput.tsx (48 lines, 1.7 KB)

```

1 | 'use client'
2 |
3 | import { useState } from 'react'
4 | import { Send, Loader2 } from 'lucide-react'
5 |
6 | interface ChatInputProps {
7 |   onSend: (query: string) => void
8 |   loading: boolean
9 | }
10 |
11 | export default function ChatInput({ onSend, loading }: ChatInputProps) {
12 |   const [value, setValue] = useState('')
13 |
14 |   function handleSubmit() {
15 |     if (!value.trim() || loading) return
16 |     onSend(value.trim())
17 |     setValue('')
18 |   }
19 |
20 |   return (
21 |     <div style={{ display: 'flex', gap: '8px', padding: '12px', borderRadius: '16px', background: 'var(--bg-el...
22 |     <input
23 |       value={value}
24 |       onChange={e => setValue(e.target.value)}
25 |       onKeyDown={e => e.key === 'Enter' && handleSubmit()}
26 |       placeholder="Nhập tên linh kiện (ví dụ: So sánh giá CPU i5 12400)"
27 |       disabled={loading}
28 |       style={{
29 |         flex: 1, padding: '12px 16px', borderRadius: '12px', border: 'none',
30 |         background: 'var(--bg-surface)', color: 'var(--text-primary)',
31 |         fontSize: '14px', outline: 'none', fontFamily: 'inherit'
32 |       }}
33 |     />
34 |     <button onClick={handleSubmit} disabled={loading || !value.trim()}
35 |       style={{
36 |         padding: '12px 20px', borderRadius: '12px', border: 'none',
37 |         background: loading ? 'var(--text-muted)' : 'var(--brand-primary)',
38 |         color: '#000', cursor: loading ? 'not-allowed' : 'pointer',
39 |         fontWeight: 700, fontSize: '14px', display: 'flex', alignItems: 'center', gap: '6px',
40 |         fontFamily: 'inherit'
41 |       }}
42 |     >{loading ? <Loader2 size={16} style={{ animation: 'spin 1s linear infinite' }} /> : <Send size={16} />}
43 |   </button>
44 | </div>
45 | )
46 | }
47 | }
48 |

```

File: components/price-compare/ChatMessage.tsx (88 lines, 3.0 KB)

```

1 | 'use client'
2 |
3 | import { AlertTriangle } from 'lucide-react'
4 | import ComponentCard from './ComponentCard'
5 | import PriceTable from './PriceTable'
6 |
7 | interface PriceInfo {
8 |   site: string
9 |   price: string
10 |   link: string
11 |   rating: number
12 | }
13 |
14 | interface Recommended {
15 |   site: string
16 |   price: string
17 |   link: string

```

```

18 }
19
20 interface Component {
21   id: string
22   exact_name: string
23   estimated_price: string
24   image_url: string
25 }
26
27 interface ChatMessageProps {
28   role: 'user' | 'assistant'
29   content: string
30   components?: Component[]
31   prices?: PriceInfo[]
32   recommended?: Recommended
33   onComponentClick?: (component: Component) => void
34   error?: boolean
35 }
36
37 export default function ChatMessage({ role, content, components, prices, recommended, onComponentClick, error ...
38   const isUser = role === 'user'
39
40   if (error) {
41     return (
42       <div style={{ display: 'flex', gap: '12px', padding: '16px', borderRadius: '14px', background: 'rgba(239...
43       <AlertTriangle size={20} style={{ color: '#ef4444', flexShrink: 0, marginTop: '2px' }} />
44       <div>
45         <p style={{ fontSize: '13px', color: 'var(--text-muted)', margin: 0, lineHeight: '1.5' }}>{content}</p>
46         <div style={{ fontSize: '14px', fontWeight: 600, color: '#ef4444' }}>{error}</div>
47       </div>
48     )
49   }
50
51   return (
52     <div style={{
53       display: 'flex', flexDirection: 'column', gap: '12px',
54       padding: isUser ? '12px 16px' : '16px',
55       borderRadius: '14px',
56       background: isUser ? 'var(--bg-surface)' : 'transparent',
57       border: isUser ? '1px solid var(--border-default)' : 'none',
58       maxWidth: isUser ? '80%' : '100%',
59       alignSelf: isUser ? 'flex-end' : 'flex-start'
60     }}>
61     <div style={{
62       display: 'flex', gap: '12px', padding: '16px',
63       border: isUser ? '1px solid var(--border-default)' : 'none',
64       background: isUser ? 'var(--bg-surface)' : 'transparent',
65       borderRadius: '14px',
66       align-items: 'center', justify-content: 'center'
67     }}>
68     <div style={{ width: '20px', height: '20px', borderRadius: '6px', background: 'var(--brand-primary)' }}>
69       Chuyền gia PC
70     </div>
71     <p style={{ fontSize: '14px', color: 'var(--text-primary)', margin: 0, lineHeight: '1.6', whiteSpace: 'p...
72     {components && components.length > 0 && (
73       <div style={{ display: 'grid', gridTemplateColumns: `repeat($Math.min(components.length, 3)), 1fr`, ...
74       {components.map((comp) => (
75         <ComponentCard
76           key={comp.id}
77           exact_name={comp.exact_name}
78           estimated_price={comp.estimated_price}
79           image_url={comp.image_url}
80           onClick={() => onComponentClick?.(comp)}
81         />
82       ))}
83     </div>
84     <p style={{ fontSize: '14px', color: 'var(--text-primary)', margin: 0, lineHeight: '1.6', whiteSpace: 'p...
85     {prices && prices.length > 0 && recommended && (
86       <PriceTable prices={prices} recommended={recommended} />
87     )}
88   )

```

File: components/price-compare/ComponentCard.tsx (49 lines, 2.1 KB)

```

1 'use client'
2
3 import { ImageIcon } from 'lucide-react'
4
5 interface ComponentCardProps {
6   exact_name: string
7   estimated_price: string
8   image_url: string
9   onClick: () => void
10 }
11
12 export default function ComponentCard({ exact_name, estimated_price, image_url, onClick }: ComponentCardProps) {
13   return (
14     <div onClick={onClick}
15       style={{
16         padding: '16px', borderRadius: '14px', cursor: 'pointer',
17         background: 'var(--bg-elevated)', border: '1px solid var(--border-default)',
18         color: 'var(--text-primary)', transition: 'all 0.2s'
19       }}
20     <div style={{
21       width: '100%', height: '100px', borderRadius: '10px', marginBottom: '10px',
22       background: 'var(--bg-base)', display: 'flex', align-items: 'center', justify-content: 'center',
23       overflow: 'hidden'
24     }}>
25     <img
26       src={image_url}
27       alt={exact_name}
28       style={{ width: '100%', height: '100%', objectFit: 'cover' }}
29     />
30     <div style={{
31       width: '100%', height: '100px', borderRadius: '10px', marginBottom: '10px',
32       background: 'var(--bg-base)', display: 'flex', align-items: 'center', justify-content: 'center',
33       overflow: 'hidden'
34     }}>
35     <div style={{
36       width: '100%', height: '100px', borderRadius: '10px', marginBottom: '10px',
37       background: 'var(--bg-base)', display: 'flex', align-items: 'center', justify-content: 'center',
38       overflow: 'hidden'
39     }}>
40     <div style={{
41       width: '100%', height: '100px', borderRadius: '10px', marginBottom: '10px',
42       background: 'var(--bg-base)', display: 'flex', align-items: 'center', justify-content: 'center',
43       overflow: 'hidden'
44     }}>
45     <div style={{
46       width: '100%', height: '100px', borderRadius: '10px', marginBottom: '10px',
47       background: 'var(--bg-base)', display: 'flex', align-items: 'center', justify-content: 'center',
48       overflow: 'hidden'
49     }}>

```

File: components/price-compare/ImageModal.tsx (41 lines, 1.5 KB)

```
1 | 'use client'
2 |
3 | import { X } from 'lucide-react'
4 |
5 | interface ImageModalProps {
6 |   component: { exact_name: string; image_url: string } | null
7 |   onClose: () => void
8 | }
9 |
10 | export default function ImageModal({ component, onClose }: ImageModalProps) {
11 |   if (!component) return null
12 |
13 |   return (
14 |     <div className="fixed inset-0 z-[9999] flex items-center justify-center p-4"
15 |       onClick={onClose}
16 |       style={{ background: 'rgba(0,0,0,0.75)' }}>
17 |     <div className="relative max-w-2xl w-full rounded-3xl overflow-hidden shadow-2xl"
18 |       onClick={e => e.stopPropagation()}
19 |       style={{ background: 'var(--bg-surface)', border: '1px solid var(--border-default)' }}>
20 |       <div className="flex items-center justify-between p-4 border-b" style={{ borderColor: 'var(--border-de...' }}>
21 |         <h3 className="font-bold text-sm truncate" style={{ color: 'var(--text-primary)' }}>{component.exact...
22 |         <button onClick={onClose}
23 |           className="p-2 rounded-xl transition-colors" style={{ color: 'var(--text-muted)' }}>
24 |           <X size={20} />
25 |         </button>
26 |       </div>
27 |       <div className="relative w-full" style={{ height: '400px' }}>
28 |         <img
29 |           src={component.image_url}
30 |           alt={component.exact_name}
31 |           style={{ width: '100%', height: '100%', objectFit: 'contain' }}
32 |           onError={(e: any) => {
33 |             e.target.src = 'https://images.unsplash.com/photo-1591799264318-7e6ef8ddb7ea?w=500&q=80'
34 |           }}
35 |         />
36 |       </div>
37 |     </div>
38 |   )
39 | }
40 |
41 |
```

File: components/price-compare/PriceTable.tsx (304 lines, 9.7 KB)

```
1 | 'use client'
2 |
3 | import { useEffect } from 'react'
4 | import { motion } from 'framer-motion'
5 | import { CheckCircle, ExternalLink, Star, RefreshCw } from 'lucide-react'
6 |
7 | interface PriceInfo {
8 |   site: string
9 |   price: string
10 |   link: string
11 |   rating: number
12 | }
13 |
14 | interface Recommended {
15 |   site: string
16 |   price: string
17 |   link: string
18 | }
19 |
20 | interface PriceTableProps {
21 |   prices: PriceInfo[]
22 |   recommended: Recommended
23 |   onRefresh?: () => void
24 |   isUpdating?: boolean
25 | }
26 |
27 | const SITE_COLORS: Record<string, { bg: string; text: string; label: string }> = {
28 |   'GearVN': { bg: '#fff3e0', text: '#e65100', label: 'GearVN' },
29 |   'Phong Vi': { bg: '#e3f2fd', text: '#0d47a1', label: 'Phong Vi' },
30 |   'An Phát': { bg: '#ffebee', text: '#b71c1c', label: 'An Phát' },
31 |   'Hoàng Hà': { bg: '#e8f5e9', text: '#1b5e20', label: 'Hoàng Hà' },
32 |   'FPT Shop': { bg: '#fff8e1', text: '#f57f17', label: 'FPT Shop' },
33 | }
34 |
35 | export default function PriceTable({ prices, recommended, onRefresh, isUpdating }: PriceTableProps) {
36 |   const getNumericPrice = (priceStr: string): number => {
37 |     const num = priceStr.replace(/^[^0-9]/g, '')
38 |     return parseInt(num, 10) || 1
39 |   }
40 |
41 |   const numericPrices = prices.map(p => getNumericPrice(p.price))
42 |   const maxPrice = Math.max(...numericPrices)
43 |   const minPrice = Math.min(...numericPrices)
44 |   const totalPrice = numericPrices.reduce((a, b) => a + b, 0)
45 |   const avgPrice = Math.round(totalPrice / numericPrices.length)
46 |
47 |   useEffect(() => {
48 |     if (!onRefresh) return
49 |     const interval = setInterval(() => {
50 |       onRefresh()
51 |     }, 30000)
52 |     return () => clearInterval(interval)
53 |   }, [onRefresh])
54 |
55 |   return (
56 |     <motion.div
57 |       initial={{ opacity: 0, y: 10 }}
58 |       animate={{ opacity: 1, y: 0 }}
59 |       style={{
60 |         marginTop: '16px',
61 |         borderRadius: '16px',
62 |         overflow: 'hidden',
63 |         border: '1px solid var(--border-default)',
64 |         background: 'var(--bg-elevated)',
65 |       }}
66 |     >
67 |     <div style={{
68 |       padding: '12px 16px',
69 |       background: 'var(--bg-surface)',
70 |       borderBottom: '1px solid var(--border-default)',
71 |       fontSize: '13px',
72 |       fontWeight: 700,
73 |       color: 'var(--text-primary)',
74 |       display: 'flex',
```

```

75 |         alignItems: 'center',
76 |         gap: '8px',
77 |     }}>
78 | < ExternalLink size={16} /> So sánh giá tế các trang th°ing mi i
79 | </div>
80 |
81 | <div style={{ padding: '12px' }}>
82 |   {prices.map((item, i) => {
83 |     const numericPrice = getNumericPrice(item.price)
84 |     const barWidth = (numericPrice / maxPrice) * 100
85 |     const isBest = numericPrice === minPrice
86 |     const siteStyle = SITE_COLORS[item.site] || { bg: '#f5f5f5', text: '#333', label: item.site }
87 |
88 |     return (
89 |       <motion.div
90 |         key={item.site}
91 |         initial={{ opacity: 0, x: -10 }}
92 |         animate={{ opacity: 1, x: 0 }}
93 |         transition={{ delay: i * 0.1 }}
94 |         style={{
95 |           padding: '10px 12px',
96 |           borderRadius: '12px',
97 |           marginBottom: '8px',
98 |           background: isBest ? 'rgba(34,197,94,0.08)' : 'transparent',
99 |           border: isBest ? '1px solid rgba(34,197,94,0.3)' : '1px solid transparent',
100 |           position: 'relative',
101 |         }}
102 |       >
103 |         {isBest && (
104 |           <div style={{
105 |             position: 'absolute',
106 |             top: '-8px',
107 |             right: '12px',
108 |             background: '#22c55e',
109 |             color: 'fff',
110 |             fontSize: '10px',
111 |             fontWeight: 800,
112 |             padding: '2px 8px',
113 |             borderRadius: '8px',
114 |             display: 'flex',
115 |             alignItems: 'center',
116 |             gap: '4px',
117 |           }}>
118 |             < CheckCircle size={10} /> Giá t Ñ t nh ¥ t
119 |           </div>
120 |         )}
121 |
122 |         <div style={{ display: 'flex', alignItems: 'center', gap: '12px', marginBottom: '8px' }}>
123 |           <a
124 |             href={item.link}
125 |             target="_blank"
126 |             rel="noopener noreferrer"
127 |             style={{
128 |               textDecoration: 'none',
129 |               padding: '4px 12px',
130 |               borderRadius: '8px',
131 |               fontSize: '12px',
132 |               fontWeight: 700,
133 |               background: siteStyle.bg,
134 |               color: siteStyle.text,
135 |               whiteSpace: 'nowrap',
136 |               flexShrink: 0,
137 |             }}
138 |           >
139 |             {siteStyle.label}
140 |           </a>
141 |
142 |           <div style={{
143 |             fontSize: '16px',
144 |             fontWeight: 800,
145 |             color: '#22c55e',
146 |             flexShrink: 0,
147 |           }}>
148 |             {item.price}
149 |           </div>
150 |
151 |           <div style={{
152 |             display: 'flex',
153 |             alignItems: 'center',
154 |             gap: '4px',
155 |             fontSize: '11px',
156 |             color: 'var(--text-muted)',
157 |             marginLeft: 'auto',
158 |           }}>
159 |             <Star size={12} style={{ color: '#f59e0b', fill: '#f59e0b' }} />
160 |             {item.rating}
161 |           </div>
162 |         </div>
163 |
164 |         <div style={{
165 |           height: '6px',
166 |           borderRadius: '3px',
167 |           background: 'var(--bg-base)',
168 |           overflow: 'hidden',
169 |         }}>
170 |           <motion.div
171 |             initial={{ width: 0 }}
172 |             animate={{ width: `${barWidth}%` }}
173 |             transition={{ duration: 0.5, delay: i * 0.1 }}
174 |             style={{
175 |               height: '100%',
176 |               borderRadius: '3px',
177 |               background: isBest
178 |                 ? 'linear-gradient(90deg, #22c55e, #16a34a)'
179 |                 : 'linear-gradient(90deg, var(--brand-primary), var(--brand-primary))',
180 |               opacity: isBest ? 1 : 0.4,
181 |             }}
182 |           />
183 |         </div>
184 |
185 |         <a
186 |           href={item.link}
187 |           target="_blank"
188 |           rel="noopener noreferrer"
189 |           style={{
190 |             display: 'inline-flex',
191 |             alignItems: 'center',
192 |             gap: '4px',
193 |             marginTop: '6px',
194 |             fontSize: '11px',
195 |             color: 'var(--brand-primary)',
196 |             textDecoration: 'none',
197 |           }}
198 |         >

```

```

1 9 9 | <ExternalLink size={10} /> M8 trang
200| </a>
201| </motion.div>
202| )
203| }}}
204| </div>
205|
206| { /* Average Price Bar */
207| <div style={{
208|   padding: '10px 16px',
209|   borderTop: '1px solid var(--border-default)',
210|   borderBottom: '1px solid var(--border-default)',
211|   background: 'rgba(34,197,94,0.04)',
212|   display: 'flex',
213|   alignItems: 'center',
214|   justifyContent: 'space-between',
215|   gap: '8px',
216|   fontSize: '13px',
217| }}>
218|   <span style={{ color: 'var(--text-muted)' }}>
219|     Giá trung bình: { ' ' }
220|     <strong style={{ color: '#22c55e', fontSize: '15px' }}>
221|       { avgPrice.toLocaleString( 'vi-VN' ) } «
222|     </strong>
223|   </span>
224|   <span style={{ color: 'var(--text-muted)' }}>
225|     Tổng giá: { ' ' }
226|     <strong style={{ color: '#22c55e', fontSize: '15px' }}>
227|       { totalPrice.toLocaleString( 'vi-VN' ) } «
228|     </strong>
229|   </span>
230|   {onRefresh && (
231|     <motion.button
232|       whileHover={{ scale: 1.05 }}
233|       whileTap={{ scale: 0.95 }}
234|       onClick={onRefresh}
235|       disabled={isUpdating}
236|       style={{
237|         display: 'inline-flex',
238|         alignItems: 'center',
239|         gap: '6px',
240|         padding: '6px 16px',
241|         borderRadius: '8px',
242|         border: 'none',
243|         background: isUpdating ? 'rgba(34,197,94,0.4)' : '#22c55e',
244|         color: '#fff',
245|         fontSize: '12px',
246|         fontWeight: 700,
247|         cursor: isUpdating ? 'not-allowed' : 'pointer',
248|         fontFamily: 'inherit',
249|         transition: 'all 0.2s',
250|       }}
251|     >
252|       <RefreshCw size={14} style={{ animation: isUpdating ? 'spin 1s linear infinite' : 'none' }} />
253|       { isUpdating ? ' đang cập nhật...' : 'Thay thế' }
254|     </motion.button>
255|   )}
256| </div>
257|
258| {recommended && (
259|   <div style={{
260|     padding: '12px 16px',
261|     borderTop: '1px solid var(--border-default)',
262|     background: 'rgba(34,197,94,0.05)',
263|     display: 'flex',
264|     alignItems: 'center',
265|     gap: '8px',
266|     fontSize: '13px',
267|   }}>
268|     <CheckCircle size={16} style={{ color: '#22c55e', flexShrink: 0 }} />
269|     <span style={{ color: 'var(--text-muted)' }}>
270|       <strong style={{ color: 'var(--text-primary)' }}>Khuyến dùng:</strong>{ ' ' }
271|       Mua t i i <strong style={{ color: '#22c55e' }}>{ recommended.s
272|       <strong style={{ color: '#22c55e' }}>{recommended.price}</strong>
273|     </span>
274|     <a
275|       href={recommended.link}
276|       target="_blank"
277|       rel="noopener noreferrer"
278|       style={{
279|         marginLeft: 'auto',
280|         padding: '4px 12px',
281|         borderRadius: '8px',
282|         background: 'var(--brand-primary)',
283|         color: '#000',
284|         fontSize: '11px',
285|         fontWeight: 700,
286|         textDecoration: 'none',
287|         whiteSpace: 'nowrap',
288|       }}
289|     >
290|       ın trang
291|     </a>
292|   </div>
293| )}
294|
295| <style jsx>{`
296|   @keyframes spin {
297|     from { transform: rotate(0deg); }
298|     to { transform: rotate(360deg); }
299|   }
300| </style>
301| </motion.div>
302| )
303| }
304|

```

File: components/profile/AvatarUpload.tsx (95 lines, 3.2 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { createBrowserClient } from '@supabase/ssr'
5 | import { User, Camera, Loader } from 'lucide-react'
6 | import { toast } from 'react-hot-toast'
7 |
8 | interface Props {
9 |   userId: string
10 |   currentAvatarUrl?: string
11 |   onUploadSuccess: (url: string) => void
12 | }
13 |
14 | export function AvatarUpload({ userId, currentAvatarUrl, onUploadSuccess }: Props) {

```

```

15 | const [uploading, setUploading] = useState(false)
16 | const supabase = createBrowserClient(
17 |   process.env.NEXT_PUBLIC_SUPABASE_URL!,
18 |   process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
19 | )
20 |
21 | const handleFileChange = async (e: React.ChangeEvent<HTMLInputElement>) => {
22 |   const file = e.target.files?.[0]
23 |   if (!file) return
24 |
25 |   if (!file.type.startsWith('image/')) {
26 |     toast.error('Chếch nh - n Enh i i di Çn Enh ding Enh (J P E G / P N G / G I F)')
27 |     return
28 |   }
29 |   if (file.size > 2 * 1024 * 1024) {
30 |     toast.error('K ích th ò c Enh t Ñ i a 2 M B . ')
31 |     return
32 |   }
33 |
34 |   setUploading(true)
35 |   try {
36 |     const ext = file.name.split('.').pop()
37 |     const path = `${userId}/avatar-${Date.now()}.${ext}`
38 |
39 |     const { error: uploadError } = await supabase.storage
40 |       .from('avatars')
41 |       .upload(path, file, { upsert: true })
42 |
43 |     if (uploadError) throw uploadError
44 |
45 |     const { data } = supabase.storage.from('avatars').getPublicUrl(path)
46 |     const publicUrl = data.publicUrl
47 |
48 |     const { error: dbError } = await supabase
49 |       .from('profiles')
50 |       .update({ avatar_url: publicUrl })
51 |       .eq('id', userId)
52 |
53 |     if (dbError) throw dbError
54 |
55 |     onUploadSuccess(publicUrl)
56 |     toast.success('C - p nh - t Enh i i di Çn thành công!')
57 |   } catch (err: any) {
58 |     toast.error(err.message || 'L ã i khi t ã i Enh l ê n . ')
59 |   } finally {
60 |     setUploading(false)
61 |   }
62 | }
63 |
64 | return (
65 |   <div className="flex flex-col items-center gap-4">
66 |     <div className="relative group">
67 |       <div className="w-24 h-24 rounded-full overflow-hidden flex items-center justify-center" style={{ border: 1px solid #ccc, background-color: #f0f0f0 }}>
68 |         {currentAvatarUrl ? (
69 |           <img
70 |             src={currentAvatarUrl}
71 |             alt="Avatar"
72 |             className="w-full h-full object-cover"
73 |             referrerPolicy="no-referrer"
74 |           />
75 |         ) : (
76 |           <User size={40} style={{ color: 'var(--text-muted)' }} />
77 |         )}
78 |       </div>
79 |
80 |       <label className="absolute bottom-0 right-0 p-2 rounded-full cursor-pointer hover:opacity-90 transition">
81 |         {uploading ? <Loader size={14} className="animate-spin" /> : <Camera size={14} />}
82 |         <input
83 |           type="file"
84 |           accept="image/*"
85 |           onChange={handleFileChange}
86 |           disabled={uploading}
87 |           className="hidden"
88 |         />
89 |       </label>
90 |     </div>
91 |     <span className="text-[10px] font-bold" style={{ color: 'var(--text-muted)' }}>
92 |       </div>
93 |   )
94 | }
95 |

```

File: components/profile/PreferencesForm.tsx (129 lines, 5.1 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { updatePreferences } from '@actions/profile'
5 | import { toast } from 'react-hot-toast'
6 |
7 | interface Props {
8 |   preferences: {
9 |     email_notifications: boolean
10 |    push_notifications: boolean
11 |    weekly_digest: boolean
12 |    theme: 'dark' | 'light' | 'system'
13 |    language: 'vi' | 'en'
14 |  }
15 | }
16 |
17 | export function PreferencesForm({ preferences }: Props) {
18 |   const [emailNotif, setEmailNotif] = useState(preferences.email_notifications)
19 |   const [pushNotif, setPushNotif] = useState(preferences.push_notifications)
20 |   const [weeklyDigest, setWeeklyDigest] = useState(preferences.weekly_digest)
21 |   const [theme, setTheme] = useState<'dark' | 'light' | 'system'>(preferences.theme || 'dark')
22 |   const [language, setLanguage] = useState<'vi' | 'en'>(preferences.language || 'vi')
23 |   const [saving, setSaving] = useState(false)
24 |
25 |   const handleSubmit = async (e: React.FormEvent) => {
26 |     e.preventDefault()
27 |     setSaving(true)
28 |     try {
29 |       await updatePreferences({
30 |         email_notifications: emailNotif,
31 |         push_notifications: pushNotif,
32 |         weekly_digest: weeklyDigest,
33 |         theme,
34 |         language
35 |       })
36 |     } catch (err: any) {
37 |       toast.error('L ã i khi l ò u c ã i . t . ')
38 |     } finally {
39 |       setSaving(false)

```



```

40 |     setSaving(false)
41 |   }
42 | }
43 |
44 | return (
45 |   <form onSubmit={handleSubmit} className="space-y-6">
46 |     {/* Notifications toggles */}
47 |     <div className="space-y-4">
48 |       4 9 |       <h4 classN ame = "text - xs font - bold text - gray - 400 uppercase tracking-wider">Thông báo qua Email</h4>
49 |       <div classN ame = "flex items-center justify-between p-3 bg-[#1e202f]/30 border border-gray-850 rounded...">
50 |         <div>
51 |           <p classN ame = "text-xs font-semibold text-white">Thông báo qua Email</p>
52 |           <input type="checkbox" checked={emailNotif} onChange={(e) => setEmailNotif(e.target.checked)} className="w-4 h-4 rounded accent-[#00d4aa]" />
53 |         </div>
54 |         <div>
55 |           <p classN ame = "text - xs font - semibold text - white">Thông báo qua Email</p>
56 |           <input type="checkbox" checked={pushNotif} onChange={(e) => setPushNotif(e.target.checked)} className="w-4 h-4 rounded accent-[#00d4aa]" />
57 |         </div>
58 |       </div>
59 |       <div classN ame = "flex items-center justify-between p-3 bg-[#1e202f]/30 border border-gray-850 rounded...">
60 |         <div>
61 |           <p classN ame = "text - xs font - semibold text - white">Thông báo qua Email</p>
62 |           <input type="checkbox" checked={weeklyDigest} onChange={(e) => setWeeklyDigest(e.target.checked)} className="w-4 h-4 rounded accent-[#00d4aa]" />
63 |         </div>
64 |         <div>
65 |           <p classN ame = "text - xs font - semibold text - white">Thông báo qua Email</p>
66 |           <input type="checkbox" checked={weeklyDigest} onChange={(e) => setWeeklyDigest(e.target.checked)} className="w-4 h-4 rounded accent-[#00d4aa]" />
67 |         </div>
68 |       </div>
69 |       <div classN ame = "flex items-center justify-between p-3 bg-[#1e202f]/30 border border-gray-850 rounded...">
70 |         <div>
71 |           <p classN ame = "text - xs font - semibold text - white">Thông báo qua Email</p>
72 |           <input type="checkbox" checked={weeklyDigest} onChange={(e) => setWeeklyDigest(e.target.checked)} className="w-4 h-4 rounded accent-[#00d4aa]" />
73 |         </div>
74 |         <div>
75 |           <p classN ame = "text - xs font - semibold text - white">Thông báo qua Email</p>
76 |           <input type="checkbox" checked={weeklyDigest} onChange={(e) => setWeeklyDigest(e.target.checked)} className="w-4 h-4 rounded accent-[#00d4aa]" />
77 |         </div>
78 |       </div>
79 |       <div classN ame = "flex items-center justify-between p-3 bg-[#1e202f]/30 border border-gray-850 rounded...">
80 |         <div>
81 |           <p classN ame = "text - xs font - semibold text - white">Thông báo qua Email</p>
82 |           <input type="checkbox" checked={weeklyDigest} onChange={(e) => setWeeklyDigest(e.target.checked)} className="w-4 h-4 rounded accent-[#00d4aa]" />
83 |         </div>
84 |         <div>
85 |           <p classN ame = "text - xs font - semibold text - white">Thông báo qua Email</p>
86 |           <input type="checkbox" checked={weeklyDigest} onChange={(e) => setWeeklyDigest(e.target.checked)} className="w-4 h-4 rounded accent-[#00d4aa]" />
87 |         </div>
88 |       </div>
89 |     </div>
90 |     {/* Theme and Language grid */}
91 |     <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">
92 |       <div>
93 |         9 4 |         <label classN ame = "block text - xs font - bold text - gray - 400 uppercase">Chọn Theme</label>
94 |         <select value={theme} onChange={(e) => setTheme(e.target.value as any)} className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white focus:outline-none">
95 |           <option value="dark">Dark Mode</option>
96 |           <option value="light">Light Mode</option>
97 |           <option value="system">System Default</option>
98 |         </select>
99 |         <div>
100 |           <label classN ame = "block text - xs font - bold text - gray - 400 uppercase">Chọn Ngôn ngữ</label>
101 |           <select value={language} onChange={(e) => setLanguage(e.target.value as any)} className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white focus:outline-none">
102 |             <option value="vi">Tiếng Việt</option>
103 |             <option value="en">English</option>
104 |           </select>
105 |         </div>
106 |         <button type="submit" disabled={!saving} className="w-full py-3 bg-[#00d4aa] text-[#0d0e13] font-bold text-xs rounded-xl hover:opacity-90 transition">
107 |           {saving ? 'Đang lưu...' : 'Lưu cài đặt'}
108 |         </button>
109 |       </div>
110 |     </div>
111 |   </form>
112 | )
113 | }
114 |
115 |
116 |
117 |
118 |
119 |
120 |
121 |
122 |
123 |
124 |
125 |
126 |
127 |
128 |
129 |

```

File: components/profile/profile/AvatarUpload.tsx (100 lines, 3.2 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { createClientComponentClient } from '@supabase/auth-helpers-nextjs'
5 | import { User, Camera, Loader } from 'lucide-react'
6 | import { toast } from 'react-hot-toast'
7 |
8 | interface Props {
9 |   userId: string
10 |   currentAvatarUrl?: string
11 |   onUploadSuccess: (url: string) => void
12 | }
13 |
14 | export function AvatarUpload({ userId, currentAvatarUrl, onUploadSuccess }: Props) {
15 |   const [uploading, setUploading] = useState(false)
16 |   const supabase = createClientComponentClient()
17 |
18 |   const handleFileChange = async (e: React.ChangeEvent<HTMLInputElement>) => {
19 |     const file = e.target.files?.[0]
20 |     if (!file) return
21 |
22 |     // 1. Client-side Validation (Rule 3)
23 |     if (!file.type.startsWith('image/')) {
24 |       toast.error('Chỉ chấp nhận hình ảnh (JPEG/PNG/GIF)')
25 |       return
26 |     }
27 |     if (file.size > 2 * 1024 * 1024) {
28 |       toast.error('Kích thước ảnh tối đa 2MB.')
29 |       return
30 |     }

```

```

31 |     setUploading(true)
32 |   try {
33 |     const ext = file.name.split('.').pop()
34 |     const path = `${userId}/avatar-${Date.now()}.${ext}`
35 |
36 |     // 2. Upload to storage bucket 'avatars'
37 |     const { error: uploadError } = await supabase.storage
38 |       .from('avatars')
39 |       .upload(path, file, { upsert: true })
40 |
41 |     if (uploadError) throw uploadError
42 |
43 |     // 3. Get Public URL
44 |     const { data } = supabase.storage.from('avatars').getPublicUrl(path)
45 |     const publicUrl = data.publicUrl
46 |
47 |     // 4. Update Profile Table
48 |     const { error: dbError } = await supabase
49 |       .from('profiles')
50 |       .update({ avatar_url: publicUrl })
51 |       .eq('id', userId)
52 |
53 |     if (dbError) throw dbError
54 |
55 |     onUploadSuccess(publicUrl)
56 |   } catch (err: any) {
57 |     toast.success('C - p n h - t ĩ nh i i đĩ Ç n t h à n h c õ n g ! ' )
58 |   } catch (err: any) {
59 |     toast.error(err.message || 'L x i k h i t ĩ i ĩ nh l ĩ n . ' )
60 |   } finally {
61 |     setUploading(false)
62 |   }
63 | }
64 |
65 | return (
66 |   <div className="flex flex-col items-center gap-4">
67 |     <div className="relative group">
68 |       <div className="w-24 h-24 rounded-full overflow-hidden border-2 border-gray-800 bg-[#1e202f]/50 flex i...
69 |         {currentAvatarUrl ? (
70 |           <img
71 |             src={currentAvatarUrl}
72 |             alt="Avatar"
73 |             className="w-full h-full object-cover"
74 |             referrerPolicy="no-referrer" // Rule 3
75 |           />
76 |         ) : (
77 |           <User size={40} className="text-gray-600" />
78 |         )}
79 |       </div>
80 |
81 |       {/ * Floating Upload Trigger */}
82 |       <label className="
83 |         absolute bottom-0 right-0 p-2 bg-[#00d4aa] text-[#0d0e13] rounded-full cursor-pointer
84 |         hover:opacity-90 transition-opacity shadow-lg flex items-center justify-center
85 |       ">
86 |         {uploading ? <Loader size={14} className="animate-spin" /> : <Camera size={14} />}
87 |         <input
88 |           type="file"
89 |           accept="image/*"
90 |           onChange={handleFileChange}
91 |           disabled={uploading}
92 |           className="hidden"
93 |         />
94 |       </label>
95 |     </div>
96 |     <span className="text-[10px] text-gray-500 font-bold">Ch ĩ p n h - n ĩ p
97 |   </div>
98 | )
99 | }
100 |

```

File: components/profile/profile/PreferencesForm.tsx (129 lines, 5.1 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { updatePreferences } from '@actions/profile'
5 | import { toast } from 'react-hot-toast'
6 |
7 | interface Props {
8 |   preferences: {
9 |     email_notifications: boolean
10 |    push_notifications: boolean
11 |    weekly_digest: boolean
12 |    theme: 'dark' | 'light' | 'system'
13 |    language: 'vi' | 'en'
14 |  }
15 | }
16 |
17 | export function PreferencesForm({ preferences }: Props) {
18 |   const [emailNotif, setEmailNotif] = useState(preferences.email_notifications)
19 |   const [pushNotif, setPushNotif] = useState(preferences.push_notifications)
20 |   const [weeklyDigest, setWeeklyDigest] = useState(preferences.weekly_digest)
21 |   const [theme, setTheme] = useState<'dark' | 'light' | 'system'>(preferences.theme || 'dark')
22 |   const [language, setLanguage] = useState<'vi' | 'en'>(preferences.language || 'vi')
23 |   const [saving, setSaving] = useState(false)
24 |
25 |   const handleSubmit = async (e: React.FormEvent) => {
26 |     e.preventDefault()
27 |     setSaving(true)
28 |     try {
29 |       await updatePreferences({
30 |         email_notifications: emailNotif,
31 |         push_notifications: pushNotif,
32 |         weekly_digest: weeklyDigest,
33 |         theme,
34 |         language
35 |       })
36 |     } catch (err: any) {
37 |       toast.success(' ĩ l ° u c ĩ u h ĩ nh c à i . t ! ' )
38 |     } catch (err: any) {
39 |       toast.error(' L x i k h i l ° u c à i . t . ' )
40 |     } finally {
41 |       setSaving(false)
42 |     }
43 |   }
44 |
45 |   return (
46 |     <form onSubmit={handleSubmit} className="space-y-6">
47 |       {/ * Notifications toggles */}
48 |       <div className="space-y-4">
49 |         <h4 className="text-xs font-bold text-gray-400 uppercase tracking
50 |

```

```

51 | <label className="flex items-center justify-between p-3 bg-[#1e202f]/30 border border-gray-850 rounded...
52 | <div>
53 |   <p className="text-xs font-semibold text-white">Thông báo qua Email</p>
54 |   <p className="text-[10px] text-gray-500">Nh-n thông báo ch ư
55 | </div>
56 | <input
57 |   type="checkbox"
58 |   checked={emailNotif}
59 |   onChange={(e) => setEmailNotif(e.target.checked)}
60 |   className="w-4 h-4 rounded accent-[#00d4aa]"
61 | />
62 | </label>
63 |
64 | <label className="flex items-center justify-between p-3 bg-[#1e202f]/30 border border-gray-850 rounded...
65 | <div>
66 |   <p className="text-xs font-semibold text-white">Thông báo
67 |   <p className="text-[10px] text-gray-500">Hi  ần th ể thông báo
68 | </div>
69 | <input
70 |   type="checkbox"
71 |   checked={pushNotif}
72 |   onChange={(e) => setPushNotif(e.target.checked)}
73 |   className="w-4 h-4 rounded accent-[#00d4aa]"
74 | />
75 | </label>
76 |
77 | <label className="flex items-center justify-between p-3 bg-[#1e202f]/30 border border-gray-850 rounded...
78 | <div>
79 |   <p className="text-xs font-semibold text-white">Báo cáo h  c
80 |   <p className="text-[10px] text-gray-500">Nh-n báo cáo tóm t
81 | </div>
82 | <input
83 |   type="checkbox"
84 |   checked={weeklyDigest}
85 |   onChange={(e) => setWeeklyDigest(e.target.checked)}
86 |   className="w-4 h-4 rounded accent-[#00d4aa]"
87 | />
88 | </label>
89 | </div>
90 |
91 | {/ * Theme and Language grid */}
92 | <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">
93 | <div>
94 |   <label className="block text-xs font-bold text-gray-400 uppercase">
95 |     <select
96 |       value={theme}
97 |       onChange={(e) => setTheme(e.target.value as any)}
98 |       className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white f...
99 |     >
100 |       <option value="dark">Đ   Dark Mode </option>
101 |       <option value="light">Đ   Light Mode </option>
102 |       <option value="system">Đ   System Default </option>
103 |     </select>
104 |   </div>
105 |
106 | <div>
107 |   <label className="block text-xs font-bold text-gray-400 uppercase">
108 |     <select
109 |       value={language}
110 |       onChange={(e) => setLanguage(e.target.value as any)}
111 |       className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white f...
112 |     >
113 |       <option value="vi">Đ   Tiếng Vi  ệ t </option>
114 |       <option value="en">Đ   English </option>
115 |     </select>
116 |   </div>
117 | </div>
118 |
119 | <button
120 |   type="submit"
121 |   disabled={saving}
122 |   className="w-full py-3 bg-[#00d4aa] text-[#0d0e13] font-bold text-xs rounded-xl hover:opacity-90 trans...
123 | >
124 |   { saving ? '  ăng l  u... ' : ' L  u c  u h  nh ' }
125 | </button>
126 | </form>
127 | )
128 | }
129 |

```

File: components/profile/profile/ProfileForm.tsx (115 lines, 4.2 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { updateProfile } from '@actions/profile'
5 | import { toast } from 'react-hot-toast'
6 |
7 | interface Props {
8 |   profile: {
9 |     full_name: string
10 |    bio?: string
11 |    school?: string
12 |    grade?: string
13 |    email: string
14 |  }
15 | }
16 |
17 | export function ProfileForm({ profile }: Props) {
18 |   const [fullName, setFullName] = useState(profile.full_name || '')
19 |   const [bio, setBio] = useState(profile.bio || '')
20 |   const [school, setSchool] = useState(profile.school || '')
21 |   const [grade, setGrade] = useState(profile.grade || '')
22 |   const [saving, setSaving] = useState(false)
23 |
24 |   const handleSubmit = async (e: React.FormEvent) => {
25 |     e.preventDefault()
26 |     setSaving(true)
27 |     try {
28 |       await updateProfile({
29 |         full_name: fullName,
30 |         bio: bio.slice(0, 200),
31 |         school,
32 |         grade
33 |       })
34 |     } catch (err: any) {
35 |       toast.success('  ă l  u thông tin cá nh  n ! ' )
36 |       toast.error(' L  x i khi l  u thông tin . ' )
37 |     } finally {
38 |       setSaving(false)
39 |     }
40 |   }
41 | }

```

```

42 |   return (
43 |     <form onSubmit={handleSubmit} className="space-y-4">
44 |       <div>
45 |         <label className="block text-xs font-bold text-gray-400 uppercase">
46 |           <input
47 |             type="email"
48 |             disabled
49 |             value={profile.email}
50 |             className="w-full bg-[#1e202f]/30 border border-gray-850 text-gray-500 rounded-xl px-4 py-3 text-xs ...
51 |           />
52 |         </div>
53 |
54 |         <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">
55 |           <div>
56 |             <label className="block text-xs font-bold text-gray-400 uppercase">
57 |               <input
58 |                 type="text"
59 |                 required
60 |                 value={fullName}
61 |                 onChange={(e) => setFullName(e.target.value)}
62 |                 placeholder="VD: Nguyễn Văn A"
63 |                 className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white p...
64 |               />
65 |             </div>
66 |
67 |             <div>
68 |               <label className="block text-xs font-bold text-gray-400 uppercase">
69 |                 <input
70 |                   type="text"
71 |                   value={grade}
72 |                   onChange={(e) => setGrade(e.target.value)}
73 |                   placeholder="VD: 10A1, 11B2..."
74 |                   className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white p...
75 |                 />
76 |               </div>
77 |             </div>
78 |
79 |             <div>
80 |               <label className="block text-xs font-bold text-gray-400 uppercase">
81 |                 <input
82 |                   type="text"
83 |                   value={school}
84 |                   onChange={(e) => setSchool(e.target.value)}
85 |                   placeholder="Tên trường THPT..."
86 |                   className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white pla...
87 |                 />
88 |               </div>
89 |
90 |             <div>
91 |               <div className="flex justify-between items-center mb-2">
92 |                 <label className="block text-xs font-bold text-gray-400 uppercase">
93 |                   <span className="text-[10px] text-gray-500 font-bold">{bio.length}/200</span>
94 |                 </div>
95 |                 <textarea
96 |                   rows={3}
97 |                   value={bio}
98 |                   maxLength={200}
99 |                   onChange={(e) => setBio(e.target.value)}
100 |                   placeholder="Mô tả vài dòng giới thiệu về bản thân..."
101 |                   className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white pla...
102 |                 />
103 |               </div>
104 |
105 |               <button
106 |                 type="submit"
107 |                 disabled={saving}
108 |                 className="w-full py-3 bg-[#00d4aa] text-[#0d0e13] font-bold text-xs rounded-xl hover:opacity-90 trans...
109 |               >
110 |                 {saving ? 'Đang lưu...' : 'Lưu thay đổi'}
111 |             </button>
112 |           </form>
113 |         )
114 |       }
115 |     )

```

File: components/profile/profile/SecurityForm.tsx (114 lines, 4.5 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { changePassword } from '@actions/profile'
5 | import { createClientComponentClient } from '@supabase/auth-helpers-nextjs'
6 | import { toast } from 'react-hot-toast'
7 | import { Logout } from 'lucide-react'
8 |
9 | export function SecurityForm() {
10 |   const [currentPass, setCurrentPass] = useState('')
11 |   const [newPass, setNewPass] = useState('')
12 |   const [confirmPass, setConfirmPass] = useState('')
13 |   const [loading, setLoading] = useState(false)
14 |   const supabase = createClientComponentClient()
15 |
16 |   const handlePasswordChange = async (e: React.FormEvent) => {
17 |     e.preventDefault()
18 |
19 |     if (newPass !== confirmPass) {
20 |       toast.error('Mật khẩu xác nhận không khớp!')
21 |       return
22 |     }
23 |
24 |     setLoading(true)
25 |     try {
26 |       await changePassword(currentPass, newPass)
27 |       toast.success('Đổi mật khẩu thành công!')
28 |       setCurrentPass('')
29 |       setNewPass('')
30 |       setConfirmPass('')
31 |     } catch (err: any) {
32 |       toast.error(err.message || 'Lỗi khi đổi mật khẩu.')
33 |     } finally {
34 |       setLoading(false)
35 |     }
36 |   }
37 |
38 |   const handleLogoutAllDevices = async () => {
39 |     const { error } = await supabase.auth.signOut({ scope: 'global' })
40 |     if (!error) {
41 |       toast.success('Bạn đã ngắt kết nối tất cả thiết bị!')
42 |       window.location.href = '/login'
43 |     } else {
44 |       toast.error('Lỗi khi ngắt kết nối.')
45 |     }
46 |   }

```

```

47 |
48 |     return (
49 |       <div className="space-y-6">
50 |
51 |         {/* Change Password Form */}
52 |         <form onSubmit={handlePasswordChange} className="space-y-4">
53 |           <h4 className="text-xs font-bold text-gray-400 uppercase tracking-wider">
54 |             <div>
55 |               <label className="block text-xs font-bold text-gray-400 uppercase">
56 |                 <input
57 |                   type="password"
58 |                   required
59 |                   value={currentPass}
60 |                   onChange={(e) => setCurrentPass(e.target.value)}
61 |                   className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white focus:outline-none cursor-not-allowed"
62 |                 />
63 |               </div>
64 |
65 |               <div>
66 |                 <label className="block text-xs font-bold text-gray-400 uppercase">
67 |                   <input
68 |                     type="password"
69 |                     required
70 |                     value={newPass}
71 |                     onChange={(e) => setNewPass(e.target.value)}
72 |                     className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white focus:outline-none cursor-not-allowed"
73 |                   />
74 |                 </div>
75 |
76 |                 <div>
77 |                   <label className="block text-xs font-bold text-gray-400 uppercase">
78 |                     <input
79 |                       type="password"
80 |                       required
81 |                       value={confirmPass}
82 |                       onChange={(e) => setConfirmPass(e.target.value)}
83 |                       className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white focus:outline-none cursor-not-allowed"
84 |                     />
85 |                   </div>
86 |
87 |                   <button
88 |                     type="submit"
89 |                     disabled={loading}
90 |                     className="w-full py-3 bg-[#00d4aa] text-[#0d0e13] font-bold text-xs rounded-xl hover:opacity-90 transition"
91 |                   />
92 |                 </div>
93 |               </form>
94 |             {loading ? 'Đang tải...' : 'Cấp nhật mật khẩu'}
95 |           </div>
96 |
97 |           {/* Global Signout */}
98 |           <div className="pt-6 border-t border-white/5 space-y-4">
99 |             <p className="text-xs font-bold text-red-400 uppercase tracking-wider">
100 |               <p className="text-[10px] text-gray-500">
101 |                 <button
102 |                   onClick={handleLogoutAllDevices}
103 |                   className="w-full py-3 bg-red-500/10 border border-red-500/20 text-red-400 font-bold text-xs rounded-xl"
104 |                 />
105 |                 <button
106 |                   type="button"
107 |                   size={14} />
108 |                 </div>
109 |               </div>
110 |             </div>
111 |           </div>
112 |         )
113 |       }
114 |     )

```

File: components/profile/ProfileForm.tsx (121 lines, 4.8 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { updateProfile } from '@actions/profile'
5 | import { toast } from 'react-hot-toast'
6 |
7 | interface Props {
8 |   profile: {
9 |     full_name: string
10 |    bio?: string
11 |    school?: string
12 |    grade?: string
13 |    email: string
14 |  }
15 | }
16 |
17 | export function ProfileForm({ profile }: Props) {
18 |   const [fullName, setFullName] = useState(profile.full_name || '')
19 |   const [bio, setBio] = useState(profile.bio || '')
20 |   const [school, setSchool] = useState(profile.school || '')
21 |   const [grade, setGrade] = useState(profile.grade || '')
22 |   const [saving, setSaving] = useState(false)
23 |
24 |   const handleSubmit = async (e: React.FormEvent) => {
25 |     e.preventDefault()
26 |     setSaving(true)
27 |     try {
28 |       await updateProfile({
29 |         full_name: fullName,
30 |         bio: bio.slice(0, 200),
31 |         school,
32 |         grade
33 |       })
34 |     } catch (err: any) {
35 |       toast.error('Lỗi khi lưu thông tin.')
36 |     } finally {
37 |       setSaving(false)
38 |     }
39 |   }
40 |
41 |   return (
42 |     <form onSubmit={handleSubmit} className="space-y-4">
43 |       <div>
44 |         <label className="block text-xs font-bold uppercase tracking-wider mb-2" style={{ color: 'var(--text-m...
45 |         <input
46 |           type="email"
47 |           disabled
48 |           value={profile.email}
49 |           className="w-full rounded-xl px-4 py-3 text-xs focus:outline-none cursor-not-allowed"
50 |           style={{ background: 'color-mix(in srgb, var(--bg-elevated) 30%, transparent)', border: '1px solid v...
51 |         />
52 |       </div>

```

```

53 |     </div>
54 |
55 |     <div className="grid grid-cols-1 sm:grid-cols-2 gap-4">
56 |         <div>
57 |             <label className="block text-xs font-bold uppercase tracking-wider mb-2" style={{ color: 'var(--text...
58 |             <input
59 |                 type="text"
60 |                 required
61 |                 value={fullName}
62 |                 onChange={(e) => setFullName(e.target.value)}
63 |                 placeholder="VD: Nguyễn Văn A"
64 |                 className="w-full rounded-xl px-4 py-3 text-xs focus:outline-none transition-colors"
65 |                 style={{ background: 'color-mix(in srgb, var(--bg-elevated) 50%, transparent)', border: '1px solid...
66 |             </div>
67 |         </div>
68 |
69 |         <div>
70 |             <label className="block text-xs font-bold uppercase tracking-wider mb-2" style={{ color: 'var(--text...
71 |             <input
72 |                 type="text"
73 |                 value={grade}
74 |                 onChange={(e) => setGrade(e.target.value)}
75 |                 placeholder="VD: 10A1, 11B2..."
76 |                 className="w-full rounded-xl px-4 py-3 text-xs focus:outline-none transition-colors"
77 |                 style={{ background: 'color-mix(in srgb, var(--bg-elevated) 50%, transparent)', border: '1px solid...
78 |             </div>
79 |         </div>
80 |     </div>
81 |
82 |     <div>
83 |         <label className="block text-xs font-bold uppercase tracking-wider mb-2" style={{ color: 'var(--text-m...
84 |         <input
85 |             type="text"
86 |             value={school}
87 |             onChange={(e) => setSchool(e.target.value)}
88 |             placeholder="Tên trường THPT..."
89 |             className="w-full rounded-xl px-4 py-3 text-xs focus:outline-none transition-colors"
90 |             style={{ background: 'color-mix(in srgb, var(--bg-elevated) 50%, transparent)', border: '1px solid v...
91 |         </div>
92 |
93 |         <div>
94 |             <div className="flex justify-between items-center mb-2">
95 |                 <label className="block text-xs font-bold uppercase tracking-wider" style={{ color: 'var(--text-mute...
96 |                 <span className="text-[10px] font-bold" style={{ color: 'var(--text-muted)' }}>{bio.length}/200</span>
97 |             </div>
98 |             <textarea
99 |                 rows={3}
100 |                 value={bio}
101 |                 maxLength={200}
102 |                 onChange={(e) => setBio(e.target.value)}
103 |                 placeholder="Mô tả vài dòng giới thiệu về bản thân..."
104 |                 className="w-full rounded-xl px-4 py-3 text-xs focus:outline-none transition-colors resize-none"
105 |                 style={{ background: 'color-mix(in srgb, var(--bg-elevated) 50%, transparent)', border: '1px solid v...
106 |             </div>
107 |
108 |             <button
109 |                 type="submit"
110 |                 disabled={saving}
111 |                 className="w-full py-3 font-bold text-xs rounded-xl hover:opacity-90 transition-opacity disabled:opaci...
112 |                 style={{ background: 'var(--brand-primary)', color: 'var(--bg-base)' }}
113 |             >
114 |                 {saving ? 'Đang lưu...' : 'Lưu thay đổi'}
115 |             </button>
116 |         </form>
117 |     )
118 | }
119 |
120 | }
121 |

```

File: components/profile/SecurityForm.tsx (114 lines, 4.5 KB)

```

1 | 'use client'
2 |
3 | import React, { useState } from 'react'
4 | import { changePassword } from '@actions/profile'
5 | import { createClientComponentClient } from '@supabase/auth-helpers-nextjs'
6 | import { toast } from 'react-hot-toast'
7 | import { Logout } from 'lucide-react'
8 |
9 | export function SecurityForm() {
10 |     const [currentPass, setCurrentPass] = useState('')
11 |     const [newPass, setNewPass] = useState('')
12 |     const [confirmPass, setConfirmPass] = useState('')
13 |     const [loading, setLoading] = useState(false)
14 |     const supabase = createClientComponentClient()
15 |
16 |     const handlePasswordChange = async (e: React.FormEvent) => {
17 |         e.preventDefault()
18 |
19 |         if (newPass !== confirmPass) {
20 |             toast.error('Mật khẩu xác nhận không khớp!')
21 |             return
22 |         }
23 |
24 |         setLoading(true)
25 |         try {
26 |             await changePassword(currentPass, newPass)
27 |             toast.success('Đổi mật khẩu thành công!')
28 |             setCurrentPass('')
29 |             setNewPass('')
30 |             setConfirmPass('')
31 |         } catch (err: any) {
32 |             toast.error(err.message || 'Lỗi khi đổi mật khẩu.')
33 |         } finally {
34 |             setLoading(false)
35 |         }
36 |     }
37 |
38 |     const handleLogoutAllDevices = async () => {
39 |         const { error } = await supabase.auth.signOut({ scope: 'global' })
40 |         if (!error) {
41 |             toast.success('Đã ngắt kết nối tất cả thiết bị!')
42 |             window.location.href = '/login'
43 |         } else {
44 |             toast.error('Lỗi khi ngắt kết nối.')
45 |         }
46 |     }
47 |
48 |     return (
49 |         <div className="space-y-6">
50 |
51 |             {/* Change Password Form */}

```

```

52|     <form onSubmit={handlePasswordChange} className="space-y-4">
53|         <h4 className="text-xs font-bold text-gray-400 uppercase tracking-wider">
54|             <div>
55|                 <label className="block text-xs font-bold text-gray-400 uppercase">
56|                     <input
57|                         type="password"
58|                         required
59|                         value={currentPass}
60|                         onChange={(e) => setCurrentPass(e.target.value)}
61|                         className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white focus:outline-none focus:ring-2 focus:ring-gray-500"
62|                     />
63|                 </div>
64|                 <div>
65|                     <label className="block text-xs font-bold text-gray-400 uppercase">
66|                         <input
67|                             type="password"
68|                             required
69|                             value={newPass}
70|                             onChange={(e) => setNewPass(e.target.value)}
71|                             placeholder="T Ñi thi Ấu 8 ký tñ, 1 chĩ hoa, 1 s Ñ"
72|                             className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white focus:outline-none focus:ring-2 focus:ring-gray-500"
73|                         />
74|                     </div>
75|                     <div>
76|                         <label className="block text-xs font-bold text-gray-400 uppercase">
77|                             <input
78|                                 type="password"
79|                                 required
80|                                 value={confirmPass}
81|                                 onChange={(e) => setConfirmPass(e.target.value)}
82|                                 className="w-full bg-[#1e202f]/50 border border-gray-800 rounded-xl px-4 py-3 text-xs text-white focus:outline-none focus:ring-2 focus:ring-gray-500"
83|                             />
84|                         </div>
85|                         <button
86|                             type="submit"
87|                             disabled={loading}
88|                             className="w-full py-3 bg-[#00d4aa] text-[#0d0e13] font-bold text-xs rounded-xl hover:opacity-90 transition-colors"
89|                         >
90|                             {loading ? 'ang thñc thi...' : 'C-p nh-t m-t kh@u'}
91|                         </button>
92|                     </form>
93|                 </div>
94|                 <div className="pt-6 border-t border-white/5 space-y-4">
95|                     <h4 className="text-xs font-bold text-red-400 uppercase tracking-wider">
96|                         <p className="text-[10px] text-gray-500">N ử nghi ng ý tài kho ển
97|                     </div>
98|                     <button
99|                         onClick={handleLogoutAllDevices}
100|                         className="w-full py-3 bg-red-500/10 border border-red-500/20 text-red-400 font-bold text-xs rounded-xl"
101|                     >
102|                         <Logout size={14} /> ng xu ýt khĩi t ýt c ế thi ỉt b ế
103|                     </button>
104|                 </div>
105|             </div>
106|         )
107|     }
108| }
109|
110|
111|
112|
113|
114|

```

File: components/ProLearningSystem.js (582 lines, 29.5 KB)

```

1| 'use client';
2|
3| import { useState, useEffect, useCallback } from 'react';
4| import Link from 'next/link';
5|
6| const SKILLS = [
7|     {
8|         id: 'scenarios',
9|         title: 'Build Scenarios',
10|         titleVn: 'K ế ch ế ãn Build',
11|         desc: 'X ấ y ã ãng c ầ u h ãnh P C theo y ề u c ầ u th ãc t ử ',
12|         descEn: 'Build PCs for real-world requirements',
13|         icon: '\u083D\uDEE0\uFE0F',
14|         route: '/builder/scenarios',
15|         category: 'planning',
16|         color: '#22c55e',
17|         xp: 150,
18|         prerequisites: [],
19|     },
20|     {
21|         id: 'diagnosis',
22|         title: 'Diagnosis Mode',
23|         titleVn: 'C ầ n o ả n l ầ i',
24|         desc: 'P ả t h ã c ã v ầ s ã l ầ i p ả n c ế ng m ầ y t ãnh ',
25|         descEn: 'Detect and fix hardware issues',
26|         icon: '\u083D\uDD0D',
27|         route: '/builder/diagnosis',
28|         category: 'troubleshooting',
29|         color: '#3742fa',
30|         xp: 200,
31|         prerequisites: ['scenarios'],
32|     },
33|     {
34|         id: 'os-install',
35|         title: 'OS Installation',
36|         titleVn: 'C ầ i . t H H ',
37|         desc: 'C ầ i . t Windows , v ầ driver cho P C m ầ i r ầp ',
38|         descEn: 'Install Windows and drivers for new PC',
39|         icon: '\u083D\uDCBB',
40|         route: '/builder/os-install',
41|         category: 'software',
42|         color: '#00cec9',
43|         xp: 120,
44|         prerequisites: ['diagnosis'],
45|     },
46|     {
47|         id: 'thermal',
48|         title: 'Thermal Heatmap',
49|         titleVn: 'B ế n ó N h ã t ',
50|         desc: 'M ồ p h ãnh n h ã t ử v ầ l ườ ng k h ã trong case ',
51|         descEn: 'Simulate temperature and airflow inside case',
52|         icon: '\u083D\uDD25',
53|         route: '/builder/thermal',
54|         category: 'optimization',
55|         color: '#ff6348',
56|         xp: 180,
57|         prerequisites: ['os-install'],

```

```

58 | },
59 | {
60 |   id: 'mac-check',
61 |   title: 'Mac Compatibility',
62 |   titleVn: 'Toing thich Mac',
63 |   desc: 'Kiám tra phhsn mám toing thich voi Mac',
64 |   descEn: 'Check software compatibility on Mac',
65 |   icon: '\uD83C\uDF4E',
66 |   route: '/builder/mac-check',
67 |   category: 'compatibility',
68 |   color: '#A29BFE',
69 |   xp: 80,
70 |   prerequisites: ['thermal'],
71 | },
72 | ];
73 |
74 | const CATEGORIES = [
75 |   { id: 'planning', label: 'L - p kz ho i ch', labelEn: 'Planning', color: '#FFD700' },
76 |   { id: 'assembly', label: 'L - p r á p', labelEn: 'Assembly', color: '#FFD700' },
77 |   { id: 'troubleshooting', label: 'Chhoooon oán', labelEn: 'Troubleshooting', color: '#000000' },
78 |   { id: 'software', label: 'Phhsn mám', labelEn: 'Software', color: '#000000' },
79 |   { id: 'optimization', label: 'Thihu', labelEn: 'Optimization', color: '#000000' },
80 |   { id: 'compatibility', label: 'Toing thich', labelEn: 'Compatibility', color: '#000000' },
81 | ];
82 |
83 | const STORAGE_KEY = 'pcm_pro_learning';
84 |
85 | function loadProgress() {
86 |   if (typeof window === 'undefined') return {};
87 |   try {
88 |     const raw = localStorage.getItem(STORAGE_KEY);
89 |     return raw ? JSON.parse(raw) : {};
90 |   } catch { return {}; }
91 | }
92 |
93 | function saveProgress(data) {
94 |   try {
95 |     localStorage.setItem(STORAGE_KEY, JSON.stringify(data));
96 |   } catch {}
97 | }
98 |
99 | function getLevel(xp) {
100 |   if (xp < 100) return { level: 1, nextAt: 100, progress: xp / 100 };
101 |   if (xp < 250) return { level: 2, nextAt: 250, progress: (xp - 100) / 150 };
102 |   if (xp < 500) return { level: 3, nextAt: 500, progress: (xp - 250) / 250 };
103 |   if (xp < 830) return { level: 4, nextAt: 830, progress: (xp - 500) / 330 };
104 |   return { level: 5, nextAt: null, progress: 1 };
105 | }
106 |
107 | function SkillNode({ skill, status, progress, lang, onSelect, selected }) {
108 |   const isSelected = selected === skill.id;
109 |   const isLocked = status === 'locked';
110 |   const isComplete = status === 'mastered';
111 |   const isActive = status === 'available' || status === 'in-progress';
112 |
113 |   const borderColor = isComplete ? skill.color : isSelected ? skill.color : isLocked ? 'rgba(255,255,255,0.06)...
114 |   const bgColor = isComplete ? `${skill.color}15` : isSelected ? `${skill.color}20` : isLocked ? 'rgba(255,255...
115 |
116 |   return (
117 |     <div onClick={() => !isLocked && onSelect(skill.id)}
118 |       onMouseOver={e => { if (!isLocked && !isSelected) { e.currentTarget.style.borderColor = skill.color + '9...
119 |       onMouseOut={e => { if (!isLocked && !isSelected) { e.currentTarget.style.borderColor = 'rgba(255,255,255...
120 |       style={{
121 |         display: 'flex', alignItems: 'center', gap: 12, padding: '14px 18px',
122 |         borderRadius: 12, border: '1.5px solid ${borderColor}',
123 |         background: bgColor, cursor: isLocked ? 'not-allowed' : 'pointer',
124 |         transition: 'all 0.25s', opacity: isLocked ? 0.35 : 1,
125 |         position: 'relative', overflow: 'hidden',
126 |       }}>
127 |         {isComplete && (
128 |           <div style={{ position: 'absolute', top: 0, right: 0, width: 40, height: 40, background: 'linear-gradi...
129 |         )}
130 |         <div style={{
131 |           fontSize: 28, width: 44, height: 44, display: 'flex', alignItems: 'center',
132 |           justifyContent: 'center', borderRadius: 10, background: isComplete ? `${skill.color}22` : 'rgba(255,25...
133 |           border: '1px solid ${isComplete ? skill.color + '44' : 'rgba(255,255,255,0.06)'}',
134 |           flexShrink: 0,
135 |         }}>
136 |           {isLocked ? '\uD83D\uDD12' : skill.icon}
137 |         </div>
138 |         <div style={{ flex: 1, minWidth: 0 }}>
139 |           <div style={{ display: 'flex', alignItems: 'center', gap: 8, marginBottom: 2 }}>
140 |             <span style={{ color: isLocked ? 'rgba(255,255,255,0.3)' : 'var(--text-primary, #dde0ed)', fontSize: ...
141 |               {lang === 'vn' ? skill.titleVn : skill.title}
142 |             </span>
143 |             {isComplete && <span style={{ fontSize: 10, color: skill.color, fontWeight: 700 }}>\u2713</span>}
144 |             {isActive && !isComplete && <span style={{ fontSize: 8, padding: '1px 6px', borderRadius: 99, backgr...
145 |           </div>
146 |           <div style={{ color: isLocked ? 'rgba(255,255,255,0.2)' : 'var(--text-muted, #636678)', fontSize: 11, ...
147 |             {lang === 'vn' ? skill.desc : skill.descEn}
148 |           </div>
149 |           {isComplete && (
150 |             <div style={{ marginTop: 4, display: 'flex', alignItems: 'center', gap: 6 }}>
151 |               <div style={{ flex: 1, height: 3, borderRadius: 2, background: 'rgba(255,255,255,0.06)' }}>
152 |                 <div style={{ width: '100%', height: '100%', borderRadius: 2, background: 'linear-gradient(90deg...
153 |               </div>
154 |               <span style={{ fontSize: 9, color: skill.color, fontWeight: 600, fontFamily: 'monospace' }}>100%</...
155 |             </div>
156 |           )}
157 |           {isActive && !isComplete && progress > 0 && (
158 |             <div style={{ marginTop: 4, display: 'flex', alignItems: 'center', gap: 6 }}>
159 |               <div style={{ flex: 1, height: 3, borderRadius: 2, background: 'rgba(255,255,255,0.06)' }}>
160 |                 <div style={{ width: `${progress}%`, height: '100%', borderRadius: 2, background: 'linear-gradi...
161 |               </div>
162 |               <span style={{ fontSize: 9, color: skill.color, fontWeight: 600, fontFamily: 'monospace' }}>{progr...
163 |             </div>
164 |           )}
165 |         </div>
166 |         <div style={{ fontSize: 10, color: 'rgba(255,255,255,0.25)', fontWeight: 700, fontFamily: 'monospace', w...
167 |           +{skill.xp}XP
168 |         </div>
169 |         {!isLocked && (
170 |           <div style={{ fontSize: 14, color: 'rgba(255,255,255,0.15)', marginLeft: 4, flexShrink: 0 }}>\u203A</div>
171 |         )}
172 |       </div>
173 |     );
174 | }
175 |
176 | export default function ProLearningSystem() {
177 |   const [lang, setLang] = useState('vn');
178 |   const [progress, setProgress] = useState({});
179 |   const [selected, setSelected] = useState(null);
180 |   const [tab, setTab] = useState('dashboard');
181 |   const [earnedXp, setEarnedXp] = useState(0);

```



```

182 |
183 | useEffect(() => {
184 |   const p = loadProgress();
185 |   setProgress(p);
186 |   let xp = 0;
187 |   Object.entries(p).forEach(([id, data]) => {
188 |     if (data.mastered) {
189 |       const skill = SKILLS.find(s => s.id === id);
190 |       if (skill) xp += skill.xp;
191 |     }
192 |   });
193 |   setEarnedXp(xp);
194 | }, []);
195 |
196 | const userLevel = getLevel(earnedXp);
197 | const masteredCount = Object.values(progress).filter(p => p.mastered).length;
198 | const inProgressCount = Object.values(progress).filter(p => p.inProgress).length;
199 | const totalSkills = SKILLS.length;
200 |
201 | const getSkillStatus = useCallback((skillId) => {
202 |   const skill = SKILLS.find(s => s.id === skillId);
203 |   if (!skill) return 'locked';
204 |   const p = progress[skillId];
205 |   if (p?.mastered) return 'mastered';
206 |   if (p?.inProgress) return 'in-progress';
207 |   const prereqsMet = skill.prerequisites.every(preId => progress[preId]?.mastered);
208 |   if (skill.prerequisites.length === 0 || prereqsMet) return 'available';
209 |   return 'locked';
210 | }, [progress]);
211 |
212 | const getSkillProgress = useCallback((skillId) => {
213 |   const p = progress[skillId];
214 |   return p?.progress || 0;
215 | }, [progress]);
216 |
217 | const handleSelect = useCallback((skillId) => {
218 |   setSelected(prev => prev === skillId ? null : skillId);
219 | }, []);
220 |
221 | const handleMasterSkill = useCallback((skillId) => {
222 |   const skill = SKILLS.find(s => s.id === skillId);
223 |   if (!skill) return;
224 |   const status = getSkillStatus(skillId);
225 |   if (status !== 'available' && status !== 'in-progress') return;
226 |   const newProgress = { ...progress };
227 |   newProgress[skillId] = { mastered: true, inProgress: false, progress: 100 };
228 |   setProgress(newProgress);
229 |   saveProgress(newProgress);
230 |   setEarnedXp(prev => prev + skill.xp);
231 |   setSelected(null);
232 | }, [progress, getSkillStatus]);
233 |
234 | const handleProgress = useCallback((skillId, pct) => {
235 |   const skill = SKILLS.find(s => s.id === skillId);
236 |   if (!skill) return;
237 |   const newProgress = { ...progress };
238 |   newProgress[skillId] = { mastered: false, inProgress: true, progress: pct };
239 |   setProgress(newProgress);
240 |   saveProgress(newProgress);
241 | }, [progress]);
242 |
243 | const resetAll = useCallback(() => {
244 |   setProgress({});
245 |   saveProgress({});
246 |   setEarnedXp(0);
247 |   setSelected(null);
248 | }, []);
249 |
250 | const selectedSkill = selected ? SKILLS.find(s => s.id === selected) : null;
251 | const selectedStatus = selected ? getSkillStatus(selected) : null;
252 |
253 | const t = (vn, en) => lang === 'vn' ? vn : en;
254 |
255 | return (
256 |   <div style={{
257 |     minHeight: '100vh', background: 'var(--bg-base, #080910)',
258 |     color: 'var(--text-primary, #d8dbe8)',
259 |     fontFamily: '"Be Vietnam Pro", "Inter", sans-serif',
260 |   }}>
261 |     <style>{
262 |       @keyframes slideUp { from { opacity: 0; transform: translateY(12px); } to { opacity: 1; transform: tra...
263 |       @keyframes fadeIn { from { opacity: 0; } to { opacity: 1; } }
264 |       @keyframes pulseGlow { 0%,100% { box-shadow: 0 0 8px rgba(0,212,170,0.2); } 50% { box-shadow: 0 0 20px...
265 |       .skill-detail { animation: slideUp 0.3s ease-out; }
266 |       * { scrollbar-width: thin; scrollbar-color: rgba(255,255,255,0.06) transparent; }
267 |     }</style>
268 |
269 |     /* Header */
270 |     <div style={{
271 |       display: 'flex', alignItems: 'center', justifyContent: 'space-between',
272 |       padding: '14px 28px', borderBottom: '1px solid rgba(255,255,255,0.06)',
273 |       background: 'var(--bg-surface, #0f1018)',
274 |     }}>
275 |       <div style={{ display: 'flex', alignItems: 'center', gap: 12 }}>
276 |         <div style={{
277 |           width: 36, height: 36, borderRadius: 10,
278 |           background: 'linear-gradient(135deg, #00d4aa, #00aaff)',
279 |           display: 'flex', alignItems: 'center', justifyContent: 'center',
280 |           fontSize: 18, fontWeight: 800, color: '#000',
281 |         }}>P</div>
282 |         <div>
283 |           <div style={{ fontSiz e: 16, fontW eigh t: 800, letterSpacing: 0.15 }}>
284 |             <div style={{ fontSiz e: 11, color: 'var(--text-muted, #5a5d72)' }}>
285 |               </div>
286 |             </div>
287 |             <div style={{ display: 'flex', alignItems: 'center', gap: 10 }}>
288 |               <div style={{
289 |                 display: 'flex', background: 'rgba(255,255,255,0.04)', borderRadius: 8,
290 |                 border: '1px solid rgba(255,255,255,0.06)', overflow: 'hidden',
291 |               }}>
292 |                 {[ 'dashboard', 'skills' ].map(tabId => (
293 |                   <button key={tabId} onClick={() => { setTab(tabId); setSelected(null); }}
294 |                     style={{
295 |                       padding: '6px 16px', fontSize: 12, fontWeight: 600, cursor: 'pointer',
296 |                       background: tab === tabId ? 'rgba(0,212,170,0.12)' : 'transparent',
297 |                       border: 'none', color: tab === tabId ? '#00d4aa' : 'var(--text-muted, #5a5d72)',
298 |                       fontFamily: 'inherit', transition: 'all 0.2s',
299 |                     }}
300 |                   { tabId === 'dashboard' ? t('T  o  n  g  q  u  a  n', 'Dashboard') : t
301 |                 )})
302 |               </div>
303 |             </div>
304 |             <button onClick={() => setLang(prev => prev === 'vn' ? 'en' : 'vn')}
305 |               style={{

```

```

306|         padding: '6px 12px', borderRadius: 8, fontSize: 12, fontWeight: 600,
307|         background: 'rgba(255,255,255,0.06)', border: '1px solid rgba(255,255,255,0.1)',
308|         color: 'var(--text-muted, #5a5d72)', cursor: 'pointer', fontFamily: 'inherit',
309|     }}>
310|     {lang === 'vn' ? 'EN' : 'VN'}
311| </button>
312| <Link href="/builder"
313|     style={{
314|         padding: '6px 16px', borderRadius: 8, fontSize: 12, fontWeight: 600,
315|         background: 'rgba(0,212,170,0.12)', border: '1px solid rgba(0,212,170,0.2)',
316|         color: '#00d4aa', cursor: 'pointer', textDecoration: 'none', fontFamily: 'inherit',
317|         transition: 'all 0.2s',
318|     }}
319|     onMouseOver={e => { e.currentTarget.style.background = 'rgba(0,212,170,0.2)'; }}
320|     onMouseOut={e => { e.currentTarget.style.background = 'rgba(0,212,170,0.12)'; }}
321| >{t('Vào Builder', 'Open Builder')}

```

```

430 |         <button onClick={() => setTab('skills')}
431 |             style={{
432 |                 padding: '10px 24px', borderRadius: 10, fontSize: 13, fontWeight: 700,
433 |                 background: 'rgba(0,212,170,0.12)', border: '1px solid rgba(0,212,170,0.2)',
434 |                 color: '#00d4aa', cursor: 'pointer', fontFamily: 'inherit',
435 |                 transition: 'all 0.2s',
436 |             }}
437 |             onMouseOver={e => { e.currentTarget.style.background = 'rgba(0,212,170,0.2)'; }}
438 |             onMouseOut={e => { e.currentTarget.style.background = 'rgba(0,212,170,0.12)'; }}
439 |         > { t ( ' X e m t h t c e k ù n n g ' , ' V i e w A l l S k i l l s ' ) } < / b u t t o n >
440 |
441 |         <button onClick={resetAll}
442 |             style={{
443 |                 padding: '10px 24px', borderRadius: 10, fontSize: 13, fontWeight: 600,
444 |                 background: 'rgba(255,255,255,0.04)', border: '1px solid rgba(255,255,255,0.08)',
445 |                 color: 'var(--text-muted, #5a5d72)', cursor: 'pointer', fontFamily: 'inherit',
446 |                 transition: 'all 0.2s',
447 |             }}
448 |             onMouseOver={e => { e.currentTarget.style.background = 'rgba(255,255,255,0.08)'; }}
449 |             onMouseOut={e => { e.currentTarget.style.background = 'rgba(255,255,255,0.04)'; }}
450 |         > { t ( ' T l i t i ò n U ' , ' R e s e t P r o g r e s s ' ) } < / b u t t o n >
451 |     </div>
452 | )}
453 |
454 | {tab === 'skills' && (
455 |     <div style={{ animation: 'fadeIn 0.3s' }}>
456 |         { /* Category filter row */ }
457 |         <div style={{ display: 'flex', gap: 8, marginBottom: 20, flexWrap: 'wrap' }}>
458 |             { C A T E G O R I E S . m a p ( c a t => (
459 |                 <div key={cat.id} style={{
460 |                     padding: '4px 12px', borderRadius: 99, fontSize: 10, fontWeight: 600,
461 |                     background: `${cat.color}12`, border: '1px solid ${cat.color}33',
462 |                     color: cat.color,
463 |                 }}>
464 |                     {lang === 'vn' ? cat.label : cat.labelEn}
465 |                 </div>
466 |             ) ) }
467 |         </div>
468 |
469 |         <div style={{ display: 'flex', flexDirection: 'column', gap: 10 }}>
470 |             {SKILLS.map(skill => {
471 |                 const status = getSkillStatus(skill.id);
472 |                 const pct = getSkillProgress(skill.id);
473 |                 return (
474 |                     <SkillNode
475 |                         key={skill.id}
476 |                         skill={skill}
477 |                         status={status}
478 |                         progress={pct}
479 |                         lang={lang}
480 |                         onSelect={handleSelect}
481 |                         selected={selected}
482 |                     />
483 |                 );
484 |             ) } }
485 |         </div>
486 |
487 |         {selectedSkill && (
488 |             <div className="skill-detail" style={{
489 |                 marginTop: 20, padding: '20px 24px', borderRadius: 14,
490 |                 background: 'var(--bg-surface, #0f1018)',
491 |                 border: '1px solid ${selectedSkill.color}33',
492 |                 boxShadow: '0 0 30px ${selectedSkill.color}08',
493 |             }}>
494 |                 <div style={{ display: 'flex', alignItems: 'flex-start', gap: 16 }}>
495 |                     <div style={{
496 |                         fontSize: 42, width: 60, height: 60, display: 'flex', alignItems: 'center',
497 |                         justifyContent: 'center', borderRadius: 12, background: `${selectedSkill.color}15`,
498 |                         border: '1px solid ${selectedSkill.color}33', flexShrink: 0,
499 |                     }}>
500 |                         {selectedSkill.icon}
501 |                     </div>
502 |                     <div style={{ flex: 1 }}>
503 |                         <div style={{ fontSize: 18, fontWeight: 800, marginBottom: 2 }}>
504 |                             {lang === 'vn' ? selectedSkill.titleVn : selectedSkill.title}
505 |                         </div>
506 |                         <div style={{ fontSize: 12, color: 'var(--text-muted, #5a5d72)', marginBottom: 10 }}>
507 |                             {lang === 'vn' ? selectedSkill.desc : selectedSkill.descEn}
508 |                         </div>
509 |                         <div style={{ display: 'flex', gap: 10, flexWrap: 'wrap', marginBottom: 14 }}>
510 |                             <div style={{ padding: '3px 10px', borderRadius: 6, fontSize: 10, fontWeight: 600, backg...
511 |                                 +{selectedSkill.xp} XP
512 |                             </div>
513 |                             <div style={{ padding: '3px 10px', borderRadius: 6, fontSize: 10, fontWeight: 600, backg...
514 |                                 { t ( ' D a n h m à c : ' , ' C a t e g o r y : ' ) } { l a n g = = = ' v n ' ? C
515 |                             </div>
516 |                             <div style={{ padding: '3px 10px', borderRadius: 6, fontSize: 10, fontWeight: 600, backg...
517 |                                 { t ( ' T r i n g t h à i : ' , ' S t a t u s : ' ) } { s e l e c t e d S t a t u s = =
518 |                             </div>
519 |                         </div>
520 |
521 |                         {selectedSkill.prerequisites.length > 0 && (
522 |                             <div style={{ marginBottom: 14 }}>
523 |                                 <div style={{ fontSize: 10, fontWeight: 600, color: 'var(--text-muted, #5a5d72)', marg...
524 |                                     { t ( ' Y ê u c s u t r ò c : ' , ' P r e r e q u i s i t e s : ' ) }
525 |                                 </div>
526 |                                 <div style={{ display: 'flex', gap: 6, flexWrap: 'wrap' }}>
527 |                                     {selectedSkill.prerequisites.map(preId => {
528 |                                         const pre = SKILLS.find(s => s.id === preId);
529 |                                         const preStatus = getSkillStatus(preId);
530 |                                         return (
531 |                                             <div key={preId} style={{
532 |                                                 padding: '2px 8px', borderRadius: 4, fontSize: 10, fontWeight: 600,
533 |                                                 background: preStatus === 'mastered' ? 'rgba(34,197,94,0.1)' : 'rgba(239,68,68...
534 |                                                 border: '1px solid ${preStatus === 'mastered' ? 'rgba(34,197,94,0.2)' : 'rgba(...
535 |                                                 color: preStatus === 'mastered' ? '#22c55e' : '#ef4444',
536 |                                             )}>
537 |                                                 {preStatus === 'mastered' ? '\u2713' : '\u2717'}
538 |                                                 {lang === 'vn' ? pre?.titleVn : pre?.title}
539 |                                             </div>
540 |                                         );
541 |                                     } ) }
542 |                                 </div>
543 |                             </div>
544 |                         ) }
545 |
546 |                     <div style={{ display: 'flex', gap: 10, alignItems: 'center' }}>
547 |                         <Link href={selectedSkill.route}
548 |                             style={{
549 |                                 padding: '10px 24px', borderRadius: 8, fontSize: 13, fontWeight: 700,
550 |                                 background: 'linear-gradient(135deg, ${selectedSkill.color}, ${selectedSkill.color}8...
551 |                                 border: 'none', color: '#fff', cursor: 'pointer', textDecoration: 'none',
552 |                                 fontFamily: 'inherit', transition: 'all 0.2s',
553 |                             }}

```

```

554 |         onMouseOver={e => { e.currentTarget.style.opacity = '0.85'; }}
555 |         onMouseOut={e => { e.currentTarget.style.opacity = '1'; }}
556 |     >
557 |         { selectedStatus === 'mastered' ? t('Ôn t - p', 'Re
558 |         </Link>
559 |         {selectedStatus !== 'mastered' && (
560 |         <button onClick={() => handleMasterSkill(selectedSkill.id)}
561 |             style={{
562 |                 padding: '10px 20px', borderRadius: 8, fontSize: 12, fontWeight: 600,
563 |                 background: 'rgba(34,197,94,0.1)', border: '1px solid rgba(34,197,94,0.2)',
564 |                 color: '#22c55e', cursor: 'pointer', fontFamily: 'inherit',
565 |                 transition: 'all 0.2s',
566 |             }}
567 |             onMouseOver={e => { e.currentTarget.style.background = 'rgba(34,197,94,0.2)'; }}
568 |             onMouseOut={e => { e.currentTarget.style.background = 'rgba(34,197,94,0.1)'; }}
569 |         > { t('ành đ ã u ho àn t h ành', 'Mark Complete') } </button>
570 |         </div>
571 |     </div>
572 | </div>
573 | </div>
574 | </div>
575 | </div>
576 | </div>
577 | </div>
578 | </div>
579 | </div>
580 | </div>
581 | </div>
582 | </div>

```

File: components/quiz/QuestionCard.tsx (86 lines, 3.2 KB)

```

1 | 'use client';
2 |
3 | import React from 'react';
4 | import { motion } from 'framer-motion';
5 | import { Question, QuestionOption } from '@types/quiz';
6 |
7 | interface QuestionCardProps {
8 |   question: Question;
9 |   selectedOptionId?: string;
10 |   textAnswer?: string;
11 |   onSelectOption: (optionId: string) => void;
12 |   onTextAnswerChange: (text: string) => void;
13 |   disabled?: boolean;
14 | }
15 |
16 | const QuestionCard: React.FC<QuestionCardProps> = ({
17 |   question,
18 |   selectedOptionId,
19 |   textAnswer,
20 |   onSelectOption,
21 |   onTextAnswerChange,
22 |   disabled = false
23 | }) => {
24 |   return (
25 |     <div className="glass-panel p-6 mb-4">
26 |       <div className="flex justify-between items-start mb-4">
27 |         <span className="text-sm font-semibold text-brand-light bg-brand-subtle px-2 py-1 rounded">
28 |           {question.points} {question.points > 1 ? 'points' : 'point'}
29 |         </span>
30 |         <span className="text-xs text-text-muted uppercase tracking-wider">
31 |           {question.type.replace('_', ' ')}
32 |         </span>
33 |       </div>
34 |
35 |       <h3 className="text-lg font-medium text-text-primary mb-6 leading-relaxed">
36 |         {question.content}
37 |       </h3>
38 |
39 |       {/* Multiple Choice or True/False */}
40 |       {(question.type === 'multiple_choice' || question.type === 'true_false') && (
41 |         <div className="grid gap-3">
42 |           {question.options?.map((option) => (
43 |             <button
44 |               key={option.id}
45 |               onClick={() => !disabled && onSelectOption(option.id)}
46 |               disabled={disabled}
47 |               className={`w-full text-left p-4 rounded-lg border transition-all duration-200 ${
48 |                 selectedOptionId === option.id
49 |                   ? 'bg-brand-subtle border-brand-primary text-brand-light shadow-[0_0_15px_rgba(0,198,174,0.2)]'
50 |                   : 'bg-bg-elevated border-border-default text-text-secondary hover:border-brand-light hover:b...
51 |                 } ${disabled ? 'cursor-not-allowed opacity-80' : 'cursor-pointer'}`}
52 |             >
53 |               <div className="flex items-center gap-3">
54 |                 <div className={`w-5 h-5 rounded-full border-2 flex items-center justify-center transition-col...
55 |                   selectedOptionId === option.id ? 'border-brand-primary' : 'border-text-muted'
56 |                 `}>
57 |                   {selectedOptionId === option.id && (
58 |                     <div className="w-2.5 h-2.5 rounded-full bg-brand-primary" />
59 |                   )}
60 |                 </div>
61 |                 <span>{option.content}</span>
62 |               </div>
63 |             </button>
64 |           )]}
65 |         </div>
66 |       )}
67 |
68 |       {/* Fill in the blank */}
69 |       {question.type === 'fill_blank' && (
70 |         <div className="mt-4">
71 |           <input
72 |             type="text"
73 |             value={textAnswer || ''}
74 |             onChange={(e) => onTextAnswerChange(e.target.value)}
75 |             disabled={disabled}
76 |             placeholder="Type your answer here..."
77 |             className="w-full p-4 bg-bg-elevated border border-border-default rounded-lg text-text-primary foc...
78 |           />
79 |         </div>
80 |       )}
81 |     </div>
82 |   );
83 | };
84 |
85 | export default QuestionCard;
86 |

```

File: components/quiz/ScoreReport.tsx (77 lines, 3.0 KB)

```

1 'use client';
2
3 import React from 'react';
4 import { motion } from 'framer-motion';
5 import { Trophy, XCircle, CheckCircle2, ChevronRight } from 'lucide-react';
6 import Link from 'next/link';
7
8 interface ScoreReportProps {
9   score: number;
10  totalPoints: number;
11  passingScore: number;
12  attemptId: string;
13 }
14
15 const ScoreReport: React.FC<ScoreReportProps> = ({ score, totalPoints, passingScore, attemptId }) => {
16   const isPassed = score >= passingScore;
17   const percentage = (score / totalPoints) * 100;
18
19   return (
20     <div className="glass-panel p-8 text-center max-w-2xl mx-auto mt-10">
21       <motion.div
22         initial={{ scale: 0.8, opacity: 0 }}
23         animate={{ scale: 1, opacity: 1 }}
24         transition={{ type: 'spring', damping: 12 }}
25         className="mb-8"
26       >
27         <div className={`mx-auto w-24 h-24 rounded-full flex items-center justify-center mb-6 ${
28           isPassed ? 'bg-success/10 text-success shadow-[0_0_30px_rgba(16,185,129,0.3)]' : 'bg-danger/10 text-...
29         }`} >
30           {isPassed ? <Trophy size={48} /> : <XCircle size={48} />}
31         </div>
32
33         <h2 className="text-3xl font-bold text-text-primary mb-2">
34           {isPassed ? 'Congratulations!' : 'Good Effort!'}
35         </h2>
36         <p className="text-text-secondary">
37           {isPassed
38             ? 'You have successfully passed the quiz.'
39             : 'You did not reach the passing score this time.'}
40         </p>
41       </motion.div>
42
43       <div className="grid grid-cols-2 gap-4 mb-10">
44         <div className="bg-bg-elevated p-6 rounded-2xl border border-border-subtle">
45           <p className="text-xs text-text-muted uppercase font-bold mb-1">Your Score</p>
46           <p className={`text-4xl font-bold ${isPassed ? 'text-success' : 'text-danger'} `}>
47             {score}<span className="text-lg text-text-muted">/<totalPoints></span>
48           </p>
49         </div>
50         <div className="bg-bg-elevated p-6 rounded-2xl border border-border-subtle">
51           <p className="text-xs text-text-muted uppercase font-bold mb-1">Percentage</p>
52           <p className="text-4xl font-bold text-brand-light">
53             {Math.round(percentage)}%
54           </p>
55         </div>
56       </div>
57
58       <div className="flex flex-col sm:flex-row gap-4 justify-center">
59         <Link
60           href={`/student/quiz/${attemptId}/results`}
61           className="flex items-center justify-center gap-2 px-8 py-4 bg-brand-primary text-white rounded-xl f...
62         >
63           Review Answers <ChevronRight size={18} />
64         </Link>
65         <Link
66           href="/student"
67           className="flex items-center justify-center gap-2 px-8 py-4 bg-bg-elevated text-text-primary rounded...
68         >
69           Back to Dashboard
70         </Link>
71       </div>
72     </div>
73   );
74 };
75
76 export default ScoreReport;
77

```

File: components/quiz/TimerDisplay.tsx (68 lines, 2.2 KB)

```

1 'use client';
2
3 import React, { useState, useEffect } from 'react';
4 import { Timer } from 'lucide-react';
5
6 interface TimerDisplayProps {
7   initialMinutes: number;
8   onTimeUp: () => void;
9   isPaused?: boolean;
10 }
11
12 const TimerDisplay: React.FC<TimerDisplayProps> = ({ initialMinutes, onTimeUp, isPaused = false }) => {
13   const [secondsLeft, setSecondsLeft] = useState(initialMinutes * 60);
14
15   useEffect(() => {
16     if (isPaused || secondsLeft <= 0) return;
17
18     const timer = setInterval(() => {
19       setSecondsLeft((prev) => {
20         if (prev <= 1) {
21           clearInterval(timer);
22           onTimeUp();
23           return 0;
24         }
25         return prev - 1;
26       });
27     }, 1000);
28
29     return () => clearInterval(timer);
30   }, [secondsLeft, isPaused, onTimeUp]);
31
32   const formatTime = (seconds: number) => {
33     const mins = Math.floor(seconds / 60);
34     const secs = seconds % 60;
35     return `${mins}:${secs < 10 ? '0' : ''}${secs}`;
36   };
37
38   const progress = (secondsLeft / (initialMinutes * 60)) * 100;
39   const isUrgent = secondsLeft < 60; // Less than 1 minute
40
41   return (
42     <div className="glass-panel p-4 flex items-center justify-between sticky top-4 z-50">
43       <div className="flex items-center gap-3">

```

```

44 |         <div className={`p-2 rounded-lg ${isUrgent ? 'bg-danger/10 text-danger' : 'bg-brand-subtle text-brand-...
45 |             <Timer size={20} className={isUrgent ? 'animate-pulse' : ''} />
46 |         </div>
47 |         <div>
48 |             <p className="text-xs text-text-muted uppercase font-bold tracking-widest">Time Remaining</p>
49 |             <p className={`text-xl font-mono font-bold ${isUrgent ? 'text-danger' : 'text-text-primary'}}>
50 |                 {formatTime(secondsLeft)}
51 |             </p>
52 |         </div>
53 |     </div>
54 |
55 |     <div className="flex-1 max-w-[200px] ml-6 hidden md:block">
56 |         <div className="h-1.5 w-full bg-bg-elevated rounded-full overflow-hidden">
57 |             <div
58 |                 className={`h-full transition-all duration-1000 ease-linear rounded-full ${isUrgent ? 'bg-danger' : ...
59 |                 style={{ width: `${progress}%` }}
60 |             </div>
61 |         </div>
62 |     </div>
63 | </div>
64 | );
65 | };
66 |
67 | export default TimerDisplay;
68 |

```

File: components/QuizModal.js (339 lines, 16.9 KB)

```

1 | 'use client';
2 |
3 | import { useState, useEffect, useMemo } from 'react';
4 | import { motion, AnimatePresence } from 'framer-motion';
5 | import { supabase } from '@lib/supabase';
6 | import { QUIZ_BANK } from '../data/quiz-bank';
7 |
8 | const TOPIC_MAP = {
9 |   'CPU': ['quiz-cpu-arch'],
10 |   'RAM': ['quiz-ram-deep'],
11 |   'GPU': ['quiz-gpu-advanced'],
12 |   'PSU': ['quiz-psu-cooling', 'quiz-cooling-advanced'],
13 |   'Mainboard': ['quiz-motherboard'],
14 |   'Storage': ['quiz-storage'],
15 |   'SSD': ['quiz-storage'],
16 |   'Computer Case': ['quiz-case'],
17 |   'Cooler': ['quiz-case', 'quiz-cooling-advanced'],
18 |   'general': ['quiz-cpu-arch', 'quiz-ram-deep', 'quiz-gpu-advanced', 'quiz-psu-cooling', 'quiz-motherboard', '...
19 | };
20 |
21 | function shuffleArray(arr) {
22 |   const a = [...arr];
23 |   for (let i = a.length - 1; i > 0; i--) {
24 |     const j = Math.floor(Math.random() * (i + 1));
25 |     [a[i], a[j]] = [a[j], a[i]];
26 |   }
27 |   return a;
28 | }
29 |
30 | function getQuestionsForTopic(topic, count = 10) {
31 |   const quizIds = TOPIC_MAP[topic] || TOPIC_MAP['general'];
32 |   let allQuestions = [];
33 |   for (const qid of quizIds) {
34 |     const quiz = QUIZ_BANK.find(q => q.id === qid);
35 |     if (quiz) {
36 |       allQuestions.push(...quiz.questions.map((qq, idx) => ({
37 |         id: `${qid}-${idx}`,
38 |         ...qq
39 |       })));
40 |     }
41 |   }
42 |   if (allQuestions.length === 0) {
43 |     allQuestions = QUIZ_BANK.flatMap(q =>
44 |       q.questions.map((qq, idx) => ({ id: `${q.id}-${idx}`, ...qq })))
45 |   );
46 | }
47 | const shuffled = shuffleArray(allQuestions);
48 | return shuffled.slice(0, Math.min(count, shuffled.length));
49 | }
50 |
51 | function formatQuestion(q) {
52 |   const letters = ['A', 'B', 'C', 'D'];
53 |   return {
54 |     question: q.q,
55 |     options: q.options.map((opt, i) => `${letters[i]}. ${opt}`),
56 |     correctAnswer: letters[q.answer],
57 |     explanation: `áp án úng là ${letters[q.answer]}: ${q.options[q.answer]}`;
58 |   };
59 | }
60 |
61 | const QuizModal = ({ onClose, lang, topic, level, onSuccess, onScore }) => {
62 |   const [questions, setQuestions] = useState([]);
63 |   const [currentIndex, setCurrentIndex] = useState(0);
64 |   const [score, setScore] = useState(0);
65 |   const [loading, setLoading] = useState(true);
66 |   const [selectedOption, setSelectedOption] = useState(null);
67 |   const [result, setResult] = useState(null);
68 |   const [view, setView] = useState('quiz');
69 |
70 |   useEffect(() => {
71 |     const raw = getQuestionsForTopic(topic || 'general', 10);
72 |     const formatted = raw.map(formatQuestion);
73 |     setQuestions(formatted);
74 |     setLoading(false);
75 |   }, [topic]);
76 |
77 |   useEffect(() => {
78 |     if (view === 'score') {
79 |       if (onScore) onScore(score);
80 |       const saveResult = async () => {
81 |         const { data: { user } } = await supabase.auth.getUser();
82 |         if (user) {
83 |           await supabase.from('quiz_results').insert({
84 |             user_id: user.id,
85 |             topic: topic,
86 |             score: score,
87 |             total: 10,
88 |             lang: lang
89 |           });
90 |         }
91 |       };
92 |       saveResult();
93 |     }
94 |   }, [view]);
95 | }

```

```

96 | const handleAnswer = (option) => {
97 |   if (selectedOption || view === 'score') return;
98 |   setSelectedOption(option);
99 |   const currentQ = questions[currentIndex];
100 |   if (option === currentQ.correctAnswer) {
101 |     setResult('correct');
102 |     setScore(prev => prev + 1);
103 |   } else {
104 |     setResult('incorrect');
105 |   }
106 | };
107 |
108 | const nextQuestion = () => {
109 |   if (currentIndex < 9) {
110 |     setCurrentIndex(prev => prev + 1);
111 |     setSelectedOption(null);
112 |     setResult(null);
113 |   } else {
114 |     setView('score');
115 |   }
116 | };
117 |
118 | const retryQuiz = () => {
119 |   setCurrentIndex(0);
120 |   setScore(0);
121 |   setSelectedOption(null);
122 |   setResult(null);
123 |   setView('quiz');
124 |   const raw = getQuestionsForTopic(topic || 'general', 10);
125 |   const formatted = raw.map(formatQuestion);
126 |   setQuestions(formatted);
127 | };
128 |
129 | const currentQuestion = questions[currentIndex];
130 |
131 | return (
132 |   <div style={{
133 |     position: 'fixed',
134 |     inset: 0,
135 |     background: 'rgba(0,0,0,0.85)',
136 |     backdropFilter: 'blur(8px)',
137 |     zIndex: 300,
138 |     display: 'flex',
139 |     alignItems: 'center',
140 |     justifyContent: 'center',
141 |     padding: '2rem'
142 |   }}>
143 |     <motion.div
144 |       initial={{ opacity: 0, y: 50, scale: 0.95 }}
145 |       animate={{ opacity: 1, y: 0, scale: 1 }}
146 |       exit={{ opacity: 0, y: 50, scale: 0.95 }}
147 |       className="glass-panel"
148 |       style={{
149 |         width: '100%',
150 |         maxWidth: '650px',
151 |         padding: '3rem',
152 |         border: '1px solid #555',
153 |         boxShadow: '0 0 50px rgba(139, 92, 246, 0.4)',
154 |         position: 'relative'
155 |       }}
156 |     >
157 |       <button
158 |         onClick={onClose}
159 |         style={{
160 |           position: 'absolute',
161 |           top: '1.5rem',
162 |           right: '1.5rem',
163 |           background: 'none',
164 |           border: 'none',
165 |           color: '#aaa',
166 |           fontSize: '1.5rem',
167 |           cursor: 'pointer'
168 |         }}
169 |       > < / b u t t o n >
170 |
171 |       {loading ? (
172 |         <div style={{ textAlign: 'center', padding: '3rem' }}>
173 |           <div className="neon-text-purple" style={{ animation: 'pulse 1.5s infinite', fontSize:...
174 |             { l a n g   = = =   ' e n '   ?   ' L o a d i n g   q u e s t i o n s . . . '   :   '
175 |           </div>
176 |         </div>
177 |       ) : questions.length === 0 ? (
178 |         <div style={{ color: 'red', textAlign: 'center', padding: '1rem' }}>
179 |           { l a n g   = = =   ' e n '   ?   ' N o   q u e s t i o n s   a v a i l a b l e   f o r   t h i
180 |         </div>
181 |       ) : view === 'score' ? (
182 |         <motion.div
183 |           initial={{ opacity: 0, scale: 0.95 }}
184 |           animate={{ opacity: 1, scale: 1 }}
185 |           style={{ textAlign: 'center', display: 'flex', flexDirection: 'column', alignItems: 'c...
186 |         >
187 |           <h2 style={{ fontSize: '24px', fontWeight: 700, color: 'var(--text-primary)', marginBo...
188 |             { l a n g   = = =   ' e n '   ?   ' Q U I Z   C O M P L E T E D '   :   ' H O À N   T H
189 |           </h2>
190 |
191 |           <div style={{ position: 'relative', width: '160px', height: '160px', marginBottom: '24...
192 |             <svg width="160" height="160" viewBox="0 0 160 160" style={{ transform: 'rotate(-9...
193 |               <circle cx="80" cy="80" r="70" fill="none" stroke="var(--border-strong)" strok...
194 |               <circle cx="80" cy="80" r="70" fill="none" stroke={score >= 8 ? 'var(--success...
195 |               strokeDasharray="439.8" strokeDashoffset={439.8 - (439.8 * score) / 10}
196 |               strokeLinecap="round" style={{ transition: 'stroke-dashoffset 1s ease-out'...
197 |             </svg>
198 |             <div style={{ position: 'absolute', top: 0, left: 0, width: '100%', height: '100%'...
199 |               <span style={{ fontSize: '40px', fontWeight: 800, color: score >= 8 ? 'var(--s...
200 |               <span style={{ fontSize: '14px', color: 'var(--text-muted)', fontWeight: 600 }...
201 |             </div>
202 |           </div>
203 |
204 |           <p style={{ color: 'var(--text-secondary)', marginBottom: '32px', fontSize: '15px', ma...
205 |             {score >= 8
206 |               ? ( l a n g   = = =   ' e n '   ?   ' E x c e l l e n t !   Y o u   d e m o n
207 |               : ( l a n g   = = =   ' e n '   ?   ' G o o d   e f f o r t !   K e e p   l e
208 |           </p>
209 |
210 |           <p style={{ color: 'var(--success)', fontSize: '14px', fontWeight: 700, marginBottom: ...
211 |             + { ( s c o r e   *   1 0 0 0 0 0 ) . t o L o c a l e S t r i n g ( ) }   «   { l a n g
212 |           </p>
213 |
214 |           <div style={{ display: 'flex', gap: '16px', width: '100%', justifyContent: 'center' }}>
215 |             <button
216 |               onClick={retryQuiz}
217 |               style={{
218 |                 padding: '12px 24px', background: 'var(--bg-elevated)', color: 'var(--text...
219 |                 border: '1px solid var(--border-default)', borderRadius: '8px', fontWeight...

```

```

220 |         cursor: 'pointer', transition: 'all 0.2s'
221 |     }}
222 |     onMouseOver={e => e.currentTarget.style.background = 'var(--bg-hover)'}
223 |     onMouseOut={e => e.currentTarget.style.background = 'var(--bg-elevated)'}
224 | >
225 |         { lang === 'en' ? 'Retry' : 'Thi lii' }
226 |     </button>
227 |     <button
228 |       onClick={() => {
229 |         if (score >= 8 && onSuccess) onSuccess(score);
230 |         onClose();
231 |       }}
232 |       style={{
233 |         padding: '12px 24px', background: 'var(--brand-primary)', color: 'white',
234 |         border: 'none', borderRadius: '8px', fontWeight: 600,
235 |         cursor: 'pointer', transition: 'all 0.2s', boxShadow: '0 4px 14px rgba(99,...
236 |       }}
237 |       onMouseOver={e => e.currentTarget.style.transform = 'translateY(-2px)'}
238 |       onMouseOut={e => e.currentTarget.style.transform = 'translateY(0)'}
239 |     >
240 |       {lang === 'en' ? 'Complete' : 'Hoàn thành'}
241 |     </button>
242 |   </div>
243 | </motion.div>
244 | ) : currentQuestion ? (
245 |   <AnimatePresence mode="wait">
246 |     <motion.div
247 |       key={currentIndex}
248 |       initial={{ opacity: 0, x: 20 }}
249 |       animate={{ opacity: 1, x: 0 }}
250 |       exit={{ opacity: 0, x: -20 }}
251 |     >
252 |       <div style={{ display: 'flex', justifyContent: 'space-between', marginBottom: '2re...
253 |         <span>{ lang === 'en' ? 'Question' : 'Câu hỏi' }
254 |         <span>{ lang === 'en' ? 'Score: ' + score : 'Điểm: ' + score }
255 |       </div>
256 |
257 |       <h2 className="neon-text-blue" style={{ marginBottom: '2rem', fontSize: '1.4rem', ...
258 |         {currentQuestion.question}
259 |       </h2>
260 |
261 |       <div style={{ display: 'grid', gap: '1rem' }}>
262 |         {currentQuestion.options.map((opt, idx) => {
263 |           let bgColor = 'var(--bg-elevated)';
264 |           let borderColor = 'var(--border-default)';
265 |
266 |           if (selectedOption) {
267 |             const optionLetter = opt.charAt(0);
268 |             if (optionLetter === currentQuestion.correctAnswer) {
269 |               bgColor = 'rgba(0, 255, 157, 0.2)';
270 |               borderColor = 'var(--accent-green)';
271 |             } else if (selectedOption === optionLetter && result === 'incorrect') {
272 |               bgColor = 'rgba(255, 0, 85, 0.2)';
273 |               borderColor = '#ff0055';
274 |             }
275 |           }
276 |
277 |           return (
278 |             <motion.button
279 |               whileHover={{ scale: selectedOption ? 1 : 1.02 }}
280 |               whileTap={{ scale: selectedOption ? 1 : 0.98 }}
281 |               key={idx}
282 |               onClick={() => handleAnswer(opt.charAt(0))}
283 |               style={{
284 |                 padding: '1.2rem',
285 |                 textAlign: 'left',
286 |                 background: bgColor,
287 |                 border: '1px solid ' + borderColor,
288 |                 color: 'var(--text-primary)',
289 |                 borderRadius: '8px',
290 |                 cursor: selectedOption ? 'default' : 'pointer',
291 |                 transition: 'background-color 0.2s, border-color 0.2s',
292 |                 fontSize: '1rem'
293 |               }}
294 |             >
295 |               {opt}
296 |             </motion.button>
297 |           );
298 |         })}
299 |     </div>
300 |
301 |     {result && (
302 |       <motion.div
303 |         initial={{ opacity: 0, y: 10 }}
304 |         animate={{ opacity: 1, y: 0 }}
305 |         style={{ marginTop: '2rem', padding: '1.5rem', background: 'rgba(255,255,2...
306 |       >
307 |         <strong style={{ color: result === 'correct' ? 'var(--accent-green)' : '#f...
308 |           {result === 'correct' ? (lang === 'en' ? 'CORRECT!' : 'CHÍNH XÁC!) : ...
309 |         </strong>
310 |         <p style={{ marginTop: '0.5rem', color: 'var(--text-secondary)', fontSize:...
311 |         <motion.button
312 |           whileHover={{ scale: 1.05 }}
313 |           whileTap={{ scale: 0.95 }}
314 |           onClick={nextQuestion}
315 |           style={{
316 |             marginTop: '1.5rem',
317 |             padding: '0.8rem 1.8rem',
318 |             background: 'var(--primary-neon)',
319 |             border: 'none',
320 |             borderRadius: '4px',
321 |             color: 'black',
322 |             fontWeight: 'bold',
323 |             cursor: 'pointer'
324 |           }}
325 |         >
326 |           {lang === 'en' ? (currentIndex === 9 ? 'Show Score' : 'Next Question')...
327 |         </motion.button>
328 |       </motion.div>
329 |     )}
330 |   </motion.div>
331 |   </AnimatePresence>
332 | ) : null}
333 | </motion.div>
334 | </div>
335 | );
336 | };
337 |
338 | export default QuizModal;
339 |

```



```

1 | 'use client';
2 |
3 | import React, { useEffect, useState } from 'react';
4 | import { RealtimeProvider } from '@lib/realtime-provider';
5 |
6 | export default function RealtimeWrapper({ children }) {
7 |   const [userId, setUserId] = useState(null);
8 |   useEffect(() => {
9 |     const url = process.env.NEXT_PUBLIC_SUPABASE_URL;
10 |    const key = process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY;
11 |    if (!url || !key) return;
12 |    import('@supabase/ssr').then(({ createBrowserClient }) => {
13 |      const client = createBrowserClient(url, key);
14 |      client.auth.getUser().then(({ data }) => {
15 |        if (data?.user) setUserId(data.user.id);
16 |      });
17 |    });
18 |   }, []);
19 |   return <RealtimeProvider userId={userId}>{children}</RealtimeProvider>;
20 | }
21 |

```

File: components/SessionManager.tsx (9 lines, 0.2 KB)

```

1 | 'use client'
2 |
3 | import { useSessionTimeout } from '@hooks/useSessionTimeout'
4 |
5 | export default function SessionManager() {
6 |   useSessionTimeout(30) // 30 minutes
7 |   return null
8 | }
9 |

```

File: components/SketchfabViewer.js (26 lines, 0.8 KB)

```

1 | 'use client';
2 |
3 | export default function SketchfabViewer() {
4 |   return (
5 |     <div style={{
6 |       width: '100%', height: '100%', position: 'relative',
7 |       background: '#0a0a0f', borderRadius: 12, overflow: 'hidden',
8 |       border: '1px solid var(--border-default)',
9 |     }}>
10 |     <iframe
11 |       title="Dream Computer Setup"
12 |       style={{ width: '100%', height: '100%', border: 'none' }}
13 |       src="https://sketchfab.com/models/82f78bbaf2d34f01af854a52151dbf49/embed"
14 |       allow="autoplay; fullscreen; xr-spatial-tracking"
15 |       allowFullScreen
16 |       mozallowfullscreen="true"
17 |       webkitallowfullscreen="true"
18 |       xr-spatial-tracking
19 |       web-share
20 |       execution-while-out-of-viewport
21 |       execution-while-not-rendered
22 |     />
23 |   </div>
24 | );
25 | }
26 |

```

File: components/StudentDashboardContent.tsx (105 lines, 5.5 KB)

```

1 | 'use client'
2 |
3 | import React, { useEffect, useState, useCallback } from 'react'
4 | import Link from 'next/link'
5 | import { usePathname } from 'next/navigation'
6 | import { GraduationCap, LayoutDashboard, Cpu, History, Logout, Users, BookOpen, ChevronDown, User, Sun, Moon, ...
7 | import { logout } from '@lib/auth-actions'
8 | import { createBrowserClient } from '@supabase/ssr'
9 | import dynamic from 'next/dynamic'
10 |
11 | const StudentDashboard = dynamic(() => import('../app/(dashboard)/student/page'), { ssr: false })
12 |
13 | export default function StudentDashboardContent({ onClose }: { onClose: () => void }) {
14 |   const [hasClass, setHasClass] = useState(false)
15 |   const [profile, setProfile] = useState<any>(null)
16 |   const [theme, setThemeState] = useState<'light' | 'dark'>('light')
17 |   const pathname = '/student/dashboard'
18 |   const supabase = createBrowserClient(
19 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
20 |     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
21 |   )
22 |
23 |   useEffect(() => {
24 |     const saved = (localStorage.getItem('theme') as 'light' | 'dark') || 'light'
25 |     setThemeState(saved)
26 |     document.documentElement.setAttribute('data-theme', saved)
27 |   }, [])
28 |
29 |   useEffect(() => {
30 |     const checkClass = async () => {
31 |       const { data: { user } } = await supabase.auth.getUser()
32 |       if (user) {
33 |         const { data: p } = await supabase.from('profiles').select('*').eq('id', user.id).single()
34 |         if (p) { setProfile(p); if (p.school_id || p.class_id) setHasClass(true) }
35 |       }
36 |     }
37 |     checkClass()
38 |   }, [])
39 |
40 |   const toggle = useCallback(() => {
41 |     const next = theme === 'light' ? 'dark' : 'light'
42 |     setThemeState(next)
43 |     document.documentElement.setAttribute('data-theme', next)
44 |     localStorage.setItem('theme', next)
45 |   }, [theme])
46 |
47 |   const navItems = [
48 |     { href: '/student/dashboard', label: 'Bảng điều khiển', icon: LayoutDashboard },
49 |     { href: '/student/classes', label: 'Lớp học', icon: Users },
50 |     { href: '/student/lessons', label: 'Bài giảng', icon: BookOpen },
51 |     { href: '/builder', label: 'Thiết kế', icon: Cpu },
52 |     { href: '/student/history', label: 'Lịch sử', icon: History }
53 |   ]
54 | }
55 |

```

```

56 |   return (
57 |     <div style={{ position: 'fixed', inset: 0, zIndex: 9999, display: 'flex', minHeight: '100vh', background: ...
58 |       <aside style={{ width: '280px', background: 'linear-gradient(180deg, #031f3b 0%, #1a2f53 100%)', display: ...
59 |         <div style={{ padding: '24px', display: 'flex', alignItems: 'center', gap: '12px', borderBottom: '1px ...
60 |           <div style={{ width: '40px', height: '40px', borderRadius: '10px', background: 'var(--brand-primary)...
61 |             <GraduationCap size={20} color="#fff" />
62 |           </div>
63 |         </div>
64 |         <div style={{ fontWeight: 800, fontSize: '18px', color: '#fff' }}>PC Master</div>
65 |       </div>
66 |     </div>
67 |   </div>
68 |
69 |     <nav style={{ padding: '16px 12px', display: 'flex', flexDirection: 'column', gap: '4px', flex: 1 }}>
70 |       {navItems.map(({ href, label, icon }) => {
71 |         const active = pathname === href
72 |         return (
73 |           <Link key={href} href={href} style={{
74 |             display: 'flex', alignItems: 'center', gap: '12px', padding: '12px 16px', borderRadius: '10px'...
75 |             background: active ? 'rgba(8,158,96,0.15)' : 'transparent',
76 |             color: active ? '#fff' : 'rgba(255,255,255,0.6)',
77 |             transition: 'all 0.2s'
78 |           }}>
79 |             <Icon size={20} /> {label}
80 |           </Link>
81 |         )
82 |       })}
83 |     </nav>
84 |
85 |     <div style={{ padding: '16px', borderTop: '1px solid rgba(255,255,255,0.08)', display: 'flex', flexDir...
86 |       <button onClick={toggle} style={{ display: 'flex', alignItems: 'center', gap: '12px', padding: '12px...
87 |         {theme === 'light' ? <Moon size={18} /> : <Sun size={18} />}
88 |       </button>
89 |       <button onClick={onClose} style={{ display: 'flex', alignItems: 'center', gap: '12px', padding: '12p...
90 |         <X size={18} /> Quay lại Builder
91 |       </button>
92 |       <button onClick={() => logout()} style={{ display: 'flex', alignItems: 'center', gap: '12px', paddin...
93 |         <Logout size={18} /> Đăng xuất
94 |       </button>
95 |     </div>
96 |   </aside>
97 |
98 |   <main style={{ flex: 1, padding: '32px', overflowY: 'auto', minWidth: 0 }}>
99 |     <StudentDashboard />
100 |   </main>
101 | </div>
102 | )
103 | }
104 | }
105 |

```

File: components/ui/BackButton.tsx (31 lines, 0.7 KB)

```

1 | 'use client'
2 |
3 | import React from 'react'
4 | import { ArrowLeft } from 'lucide-react'
5 | import { useRouter } from 'next/navigation'
6 |
7 | interface BackButtonProps {
8 |   href?: string
9 |   label?: string
10 |   className?: string
11 | }
12 |
13 | export function BackButton({ href, label = 'Quay lại', className = 'flex'
14 |   const router = useRouter()
15 |   return (
16 |     <button
17 |       onClick={() => {
18 |         if (href) {
19 |           router.push(href)
20 |         } else {
21 |           router.back()
22 |         }
23 |       }}
24 |       className={className}
25 |     >
26 |       <ArrowLeft size={16} />
27 |       {label}
28 |     </button>
29 |   )
30 | }
31 |

```

File: components/VirtualAssistant.js (270 lines, 12.4 KB)

```

1 | 'use client';
2 |
3 | import React, { useState, useEffect, useRef, useCallback } from 'react';
4 | import { motion, AnimatePresence } from 'framer-motion';
5 | import ReactMarkdown from 'react-markdown';
6 | import { Bot, Minimize2, Maximize2, X, Send, Copy, Sparkles, CheckCheck } from 'lucide-react';
7 | import { QUICK_ACTIONS, WELCOME_MESSAGE } from '../utils/aiConstants';
8 |
9 | // Typing indicator
10 | function TypingDots() {
11 |   return (
12 |     <div className="flex gap-1.5 items-center px-1 py-2">
13 |       <motion.span animate={{ y: [0, -5, 0] }} transition={{ repeat: Infinity, duration: 0.6, delay: 0 }} clas...
14 |       <motion.span animate={{ y: [0, -5, 0] }} transition={{ repeat: Infinity, duration: 0.6, delay: 0.2 }} cl...
15 |       <motion.span animate={{ y: [0, -5, 0] }} transition={{ repeat: Infinity, duration: 0.6, delay: 0.4 }} cl...
16 |     </div>
17 |   );
18 | }
19 |
20 | // Markdown Custom Components
21 | const mdComponents = {
22 |   p: ({ children }) => <p className="mb-2 leading-relaxed text-[15px]">{children}</p>,
23 |   strong: ({ children }) => <strong className="font-bold text-[#00d2a0]">{children}</strong>,
24 |   li: ({ children }) => <li className="ml-4 list-disc mb-1">{children}</li>,
25 |   ul: ({ children }) => <ul className="my-2">{children}</ul>,
26 |   code: ({ children }) => <code className="bg-black/30 px-1.5 py-0.5 rounded text-[#00b4d8] text-[13px] font-m...
27 |   pre: ({ children }) => <pre className="bg-[#0f0f1a] p-3 rounded-lg overflow-x-auto my-2 border border-[#2a36...
28 | };
29 |
30 | const VirtualAssistant = ({ lang = 'vi', appMode, cartItems = [], remainingBudget, missionTitle, isOpen, setIs...
31 |   const [isMinimized, setIsMinimized] = useState(false);
32 |   const [messages, setMessages] = useState([WELCOME_MESSAGE]);
33 |   const [input, setInput] = useState('');
34 |   const [isLoading, setIsLoading] = useState(false);

```

```

35 | const [copiedIndex, setCopiedIndex] = useState(null);
36 | const messagesEndRef = useRef(null);
37 | const inputRef = useRef(null);
38 | const sendingRef = useRef(false);
39 |
40 | // Auto-scroll
41 | useEffect(() => {
42 |   if (isOpen && !isMinimized) {
43 |     messagesEndRef.current?.scrollIntoView({ behavior: 'smooth' });
44 |   }
45 | }, [messages, isLoading, isOpen, isMinimized]);
46 |
47 | // Focus input
48 | useEffect(() => {
49 |   if (isOpen && !isMinimized) {
50 |     setTimeout(() => inputRef.current?.focus(), 300);
51 |   }
52 | }, [isOpen, isMinimized]);
53 |
54 | const copyToClipboard = (text, index) => {
55 |   navigator.clipboard.writeText(text);
56 |   setCopiedIndex(index);
57 |   setTimeout(() => setCopiedIndex(null), 2000);
58 | };
59 |
60 | const sendMessage = useCallback(async (text) => {
61 |   const trimmed = (text || input).trim();
62 |   if (!trimmed || sendingRef.current) return;
63 |   sendingRef.current = true;
64 |
65 |   const userMsg = { role: 'user', content: trimmed };
66 |
67 |   setMessages(prev => [...prev, userMsg]);
68 |   setInput('');
69 |   setIsLoading(true);
70 |
71 |   const prevMessages = messages.concat(userMsg);
72 |   const historyForAPI = prevMessages.filter(m => m.content !== WELCOME_MESSAGE.content).slice(-10);
73 |
74 |   try {
75 |     const res = await fetch('/api/ai/chat', {
76 |       method: 'POST',
77 |       headers: { 'Content-Type': 'application/json' },
78 |       body: JSON.stringify({
79 |         message: trimmed,
80 |         history: historyForAPI.slice(0, -1),
81 |         context: { appMode, cartItems, remainingBudget, missionTitle }
82 |       })
83 |     });
84 |     const data = await res.json();
85 |     setMessages(prev => [...prev, {
86 |       role: 'assistant',
87 |       content: data.reply || data.error || 'Xin lxi, tôi g.p sñ cñ k z t
88 |     }]);
89 |   } catch {
90 |     setMessages(prev => [...prev, {
91 |       role: 'assistant',
92 |       content: '& p L x i k z t n Ñ i máy chặ . Vui lòng thí l i i sau . '
93 |     }]);
94 |   } finally {
95 |     setIsLoading(false);
96 |     sendingRef.current = false;
97 |   }
98 | }, [input, appMode, cartItems, remainingBudget, missionTitle]);
99 |
100 | const handleKeyDown = (e) => {
101 |   if (e.key === 'Enter' && !e.shiftKey) {
102 |     e.preventDefault();
103 |     sendMessage();
104 |   }
105 | };
106 |
107 | return (
108 |   <AnimatePresence>
109 |     {isOpen && (
110 |       <motion.div
111 |         initial={{ y: 20, opacity: 0, scale: 0.95 }}
112 |         animate={{ y: 0, opacity: 1, scale: 1 }}
113 |         exit={{ y: 20, opacity: 0, scale: 0.95 }}
114 |         transition={{ type: 'spring', stiffness: 300, damping: 25 }}
115 |         className={`fixed bottom-4 right-4 z-[9999] bg-[#16213e]/95 backdrop-blur-xl border border-[#1e293b]...
116 |           isMinimized ? 'w-[320px] h-[60px] rounded-2xl' : 'w-[380px] h-[650px] max-h-[85vh] rounded-2xl sm:...
117 |         `}
118 |       >
119 |         { /* HEADER */ }
120 |         <div
121 |           onClick={() => setIsMinimized(m => !m)}
122 |           className={`flex items-center justify-between px-5 py-3.5 cursor-pointer shrink-0 transition-color...
123 |             isMinimized ? '' : 'border-b border-[#1e293b] bg-gradient-to-r from-[#00d2a0]/10 to-transparent ...
124 |           `}
125 |         >
126 |           <div className="flex items-center gap-3">
127 |             <div className="relative">
128 |               <div className="w-10 h-10 rounded-full bg-[#00d2a0]/20 flex items-center justify-center text-[...
129 |                 <Bot size={22} />
130 |               </div>
131 |               <div className="absolute bottom-0 right-0 w-3 h-3 bg-green-500 border-2 border-[#16213e] round...
132 |             </div>
133 |             <div>
134 |               <h3 className="font-bold text-white leading-tight">AI Guru <span className="text-xl leading-no...
135 |               <p className="text-[11px] text-green-400 font-medium uppercase tracking-wider">
136 |                 { is Loading ? ' ang so in tin . . ' : ' Tr ñ c tuy z n ' }
137 |             </p>
138 |             </div>
139 |           </div>
140 |           <div className="flex items-center gap-2">
141 |             <button
142 |               onClick={(e) => { e.stopPropagation(); setIsMinimized(!isMinimized); }}
143 |               className="p-1.5 text-slate-400 hover:text-white hover:bg-[#2a3655] rounded-md transition-colors"
144 |             >
145 |               {isMinimized ? <Maximize2 size={16} /> : <Minimize2 size={16} />}
146 |             </button>
147 |             <button
148 |               onClick={(e) => { e.stopPropagation(); setIsOpen(false); }}
149 |               className="p-1.5 text-slate-400 hover:text-red-400 hover:bg-red-400/10 rounded-md transition-c...
150 |             >
151 |               <X size={18} />
152 |             </button>
153 |           </div>
154 |         </div>
155 |       )
156 |     } /* BODY */
157 |     { !isMinimized && (
158 |       <>

```

```

159 |     /* Context Alert */
160 |     {appMode && (
161 |       <div className="bg-[#00d2a0]/10 border-b border-[#00d2a0]/20 px-4 py-2 flex items-center gap-2...
162 |       <Sparkles size={14} />
163 |     <span> AI      ang theo doi: {missionTitle} | 'Thức hành t
164 |   </div>
165 | )}
166 |
167 |   /* Chat Area */
168 |   <div className="flex-1 overflow-y-auto p-4 flex flex-col gap-4 scrollbar-hide">
169 |     {messages.map((msg, i) => (
170 |       <div key={i} className={`flex ${msg.role === 'user' ? 'justify-end' : 'justify-start'} group`}>
171 |
172 |         {msg.role === 'assistant' && (
173 |           <div className="w-8 h-8 rounded-full bg-[#1e293b] flex items-center justify-center mr-2 ...
174 |             <Bot size={16} className="text-[#00d2a0]" />
175 |           </div>
176 |         )}
177 |
178 |         <div className={`relative max-w-[85%] rounded-2xl px-4 py-3 ${
179 |           msg.role === 'user'
180 |             ? 'bg-[#00d2a0] text-black rounded-tr-sm shadow-[0_4px_15px_rgba(0,210,160,0.2)]'
181 |             : 'bg-[#1e293b] text-slate-200 rounded-tl-sm border border-[#2a3655]'
182 |           }`}>
183 |           {msg.role === 'assistant' ? (
184 |             <div className="markdown-body">
185 |               <ReactMarkdown components={mdComponents}>{msg.content}</ReactMarkdown>
186 |             </div>
187 |           ) : (
188 |             <p className="text-[15px] leading-relaxed">{msg.content}</p>
189 |           )}
190 |         )}
191 |
192 |         /* Copy Button (Only for assistant) */
193 |         {msg.role === 'assistant' && (
194 |           <button
195 |             onClick={() => copyToClipboard(msg.content, i)}
196 |             className="absolute -right-10 top-2 p-1.5 text-slate-500 hover:text-[#00d2a0] bg-[#1...
197 |             title="Copy"
198 |             >
199 |               {copiedIndex === i ? <CheckCheck size={14} className="text-[#00d2a0]" /> : <Copy siz...
200 |             </button>
201 |           )}
202 |         </div>
203 |       )}
204 |
205 |       {isLoading && (
206 |         <div className="flex justify-start">
207 |           <div className="w-8 h-8 rounded-full bg-[#1e293b] flex items-center justify-center mr-2 sh...
208 |             <Bot size={16} className="text-[#00d2a0]" />
209 |           </div>
210 |           <div className="bg-[#1e293b] rounded-2xl rounded-tl-sm px-4 py-2 border border-[#2a3655]">
211 |             <TypingDots />
212 |           </div>
213 |         </div>
214 |       )}
215 |     <div ref={messagesEndRef} />
216 |   </div>
217 |
218 |   /* Quick Actions */
219 |   <div className="px-3 pb-2 pt-1 flex gap-2 overflow-x-auto scrollbar-hide shrink-0">
220 |     {QUICK_ACTIONS?.map((qa, i) => (
221 |       <button
222 |         key={i}
223 |         onClick={() => sendMessage(qa.prompt)}
224 |         disabled={isLoading}
225 |         className="shrink-0 px-3 py-1.5 bg-[#1e293b] hover:bg-[#2a3655] text-[#00b4d8] text-[12px]...
226 |         >
227 |           {qa.label}
228 |         </button>
229 |       )}
230 |     </div>
231 |
232 |     /* Input Area */
233 |     <div className="p-3 border-t border-[#1e293b] bg-[#16213e] rounded-b-2xl shrink-0">
234 |       <div className="relative flex items-end gap-2 bg-[#0f0f1a] rounded-xl border border-[#2a3655] ...
235 |         <textarea
236 |           ref={inputRef}
237 |           value={input}
238 |           onChange={e => setInput(e.target.value)}
239 |           onKeyDown={handleKeyDown}
240 |           placeholder="H  i  AI  Guru  v  Á  linh  ki  Ç  n  .  .  ."
241 |           disabled={isLoading}
242 |           rows={1}
243 |           className="flex-1 max-h-[120px] bg-transparent border-none text-white text-[14px] placehol...
244 |           onInput={e => {
245 |             e.target.style.height = 'auto';
246 |             e.target.style.height = Math.min(e.target.scrollHeight, 120) + 'px';
247 |           }}
248 |         />
249 |         <button
250 |           onClick={() => sendMessage()}
251 |           disabled={isLoading || !input.trim()}
252 |           className="shrink-0 w-10 h-10 mb-0.5 mr-0.5 rounded-lg bg-[#00d2a0] text-black flex items-...
253 |           >
254 |             <Send size={18} className={input.trim() && !isLoading ? "ml-1" : ""} />
255 |           </button>
256 |       </div>
257 |       <div className="text-center mt-2">
258 |         <span className="text-[10px] text-slate-500"> AI có th
259 |       </div>
260 |     </div>
261 |   )}
262 | </motion.div>
263 | )}
264 | </AnimatePresence>
265 | );
266 | };
267 |
268 | export default VirtualAssistant;
269 |
270 |

```

File: components/VoiceController.tsx (290 lines, 11.5 KB)

```

1 | 'use client';
2 |
3 | import React, { useState, useEffect, useRef, useCallback } from 'react';
4 |
5 | interface VoiceControllerProps {
6 |   lang: string;
7 |   onCommand: (action: string, params?: any) => void;
8 | }

```

```

9 |
10 | interface CommandDef {
11 |     keywords: string[];
12 |     action: string;
13 |     params?: any;
14 |     response: string;
15 | }
16 |
17 | const COMMANDS: CommandDef[] = [
18 |     {
19 |         keywords: ['l-p', 'c p u', 'g-n', 'c p u', 't h ê m', 'c p u', 'b ù', 'v i', 'x í', 'l ý', 'i n s', 'a l',
20 |         keywords: ['l-p', 'r a m', 'g-n', 'r a m', 't h ê m', 'r a m', 'h í', 'n a', 'h d', 'm à', 'n h', 't a l', 'r a m',
21 |         keywords: ['l-p', 'n g u', 'g-n', 'n g u', 't h ê m', 'n g u', 't h ê m', 'n g u', 't h ê m', 'n g u', 't h ê m', 'n g u',
22 |         keywords: ['l-p', 't t', 'g-n', 't t', 't h ê m', 't t', 't h ê m', 't t', 't h ê m', 't t', 't h ê m', 't t', 't h ê m', 't t',
23 |         keywords: ['l-p', 's s d', 'g-n', 's s d', 't t', 't h ê m', 't t', 't h ê m', 't t', 't h ê m', 't t', 't h ê m', 't t',
24 |         keywords: ['t h a o', 'c p u', 'g-n', 'c p u', 't h ê m', 'c p u', 't h ê m', 'c p u', 't h ê m', 'c p u', 't h ê m', 'c p u',
25 |         keywords: ['t h a o', 'r a m', 'g-n', 'r a m', 't h ê m', 'r a m', 't h ê m', 'r a m', 't h ê m', 'r a m', 't h ê m', 'r a m',
26 |         keywords: ['t h a o', 'g p u', 'g-n', 'g p u', 't h ê m', 'g p u', 't h ê m', 'g p u', 't h ê m', 'g p u', 't h ê m', 'g p u',
27 |         keywords: ['t h a o', 'n g u', 'g-n', 'n g u', 't h ê m', 'n g u', 't h ê m', 'n g u', 't h ê m', 'n g u', 't h ê m', 'n g u',
28 |         keywords: ['t h a o', 't t', 'g-n', 't t', 't h ê m', 't t', 't h ê m', 't t', 't h ê m', 't t', 't h ê m', 't t',
29 |         keywords: ['t h a o', 's s d', 'g-n', 's s d', 't h ê m', 's s d', 't h ê m', 's s d', 't h ê m', 's s d', 't h ê m', 's s d',
30 |         keywords: ['x ó a', 'h t', 'g-n', 'h t', 't h ê m', 'h t', 't h ê m', 'h t', 't h ê m', 'h t', 't h ê m', 'h t',
31 |         keywords: ['k i Á m', 't r a', 't', 'g-n', 't r a', 't', 'g-n', 't r a', 't', 'g-n', 't r a', 't', 'g-n', 't r a', 't',
32 |         keywords: ['k h ß i', 'Ü n g', 'g-n', 'Ü n g', 't h ê m', 'Ü n g', 't h ê m', 'Ü n g', 't h ê m', 'Ü n g', 't h ê m', 'Ü n g',
33 | ];
34 |
35 | export default function VoiceController({ lang, onCommand }: VoiceControllerProps) {
36 |     const [isListening, setIsListening] = useState(false);
37 |     const [feedback, setFeedback] = useState('');
38 |     const [transcript, setTranscript] = useState('');
39 |     const recognitionRef = useRef<any>(null);
40 |     const listenRef = useRef(false);
41 |     const onCommandRef = useRef(onCommand);
42 |     const viVoiceRef = useRef<SpeechSynthesisVoice | null>(null);
43 |     onCommandRef.current = onCommand;
44 |
45 |     const findViVoice = useCallback(() => {
46 |         const voices = window.speechSynthesis.getVoices();
47 |         let voice = voices.find(v => v.lang.startsWith('vi') && (v.name.includes('Natural') || v.name.includes('On...
48 |         if (!voice) voice = voices.find(v => v.lang.startsWith('vi'));
49 |         return voice || null;
50 |     }, []);
51 |
52 |     useEffect(() => {
53 |         if (typeof window === 'undefined' || !window.speechSynthesis) return;
54 |         const loadVoices = () => {
55 |             viVoiceRef.current = findViVoice();
56 |         };
57 |         loadVoices();
58 |         window.speechSynthesis.onvoiceschanged = loadVoices;
59 |         return () => { window.speechSynthesis.onvoiceschanged = null; };
60 |     }, [findViVoice]);
61 |
62 |     const speakText = useCallback((text: string) => {
63 |         if (typeof window === 'undefined') return;
64 |         const playViaGoogle = () => {
65 |             const audio = new Audio(`/api/tts?text=${encodeURIComponent(text)}&lang=vi`);
66 |             audio.volume = 1.0;
67 |             audio.play().catch(() => {
68 |                 playViaSpeechSynth();
69 |             });
70 |         };
71 |         const playViaSpeechSynth = () => {
72 |             if (!window.speechSynthesis) return;
73 |             window.speechSynthesis.cancel();
74 |             const utterance = new SpeechSynthesisUtterance(text);
75 |             utterance.lang = 'vi-VN';
76 |             utterance.rate = 0.7;
77 |             utterance.pitch = 1.0;
78 |             utterance.volume = 1.0;
79 |             let v = viVoiceRef.current;
80 |             if (!v) {
81 |                 v = findViVoice();
82 |                 viVoiceRef.current = v;
83 |             }
84 |             if (v) utterance.voice = v;
85 |             window.speechSynthesis.speak(utterance);
86 |         };
87 |         playViaGoogle();
88 |     }, [findViVoice]);
89 |
90 |     const findBestMatch = useCallback((text: string): CommandDef | null => {
91 |         const normalized = text
92 |             .toLowerCase()
93 |             .normalize('NFD')
94 |             .replace(/[\u0300-\u036f]/g, '');
95 |         . r e p l a c e ( / / g , ' d ' )
96 |         .trim();
97 |
98 |         let bestMatch: CommandDef | null = null;
99 |         let bestScore = 0;
100 |
101 |         for (const cmd of COMMANDS) {
102 |             for (const keyword of cmd.keywords) {
103 |                 const kw = keyword
104 |                     .toLowerCase()
105 |                     .normalize('NFD')
106 |                     .replace(/[\u0300-\u036f]/g, '');
107 |                 . r e p l a c e ( / / g , ' d ' )
108 |                 .trim();
109 |
110 |                 if (normalized.includes(kw)) {
111 |                     const score = kw.length / normalized.length;
112 |                     if (score > bestScore) {
113 |                         bestScore = score;
114 |                         bestMatch = cmd;
115 |                     }
116 |                 }
117 |             }
118 |         }
119 |         return bestMatch;
120 |     }, []);
121 |
122 |     const startListening = useCallback(() => {
123 |         if (!recognitionRef.current) return;
124 |         listenRef.current = true;
125 |         setTranscript('');
126 |         try {
127 |             recognitionRef.current.start();
128 |         } catch {
129 |             setTimeout(() => {
130 |                 try { recognitionRef.current?.start(); } catch {}
131 |             }, 200);
132 |         }

```

```

133 | }, []);
134 |
135 | const stopListening = useCallback(() => {
136 |   listenRef.current = false;
137 |   try { recognitionRef.current?.stop(); } catch {}
138 | }, []);
139 |
140 | useEffect(() => {
141 |   if (typeof window === 'undefined') return;
142 |   const SpeechRecognition = (window as any).SpeechRecognition || (window as any).webkitSpeechRecognition;
143 |   if (!SpeechRecognition) {
144 |     1 4 4 | setFeedback(lang === 'vn' ? 'Trình duyệt không hỗ trợ' : 'Browser
145 |     return;
146 |   }
147 |
148 |   const rec = new SpeechRecognition();
149 |   rec.lang = 'vi-VN';
150 |   rec.continuous = false;
151 |   rec.interimResults = false;
152 |   rec.maxAlternatives = 5;
153 |
154 |   rec.onstart = () => {
155 |     1 5 6 | setIsListening(true);
156 |     setFeedback(lang === 'vn' ? 'Đang nghe...' : 'Listening...');
157 |   };
158 |
159 |   rec.onend = () => {
160 |     setIsListening(false);
161 |     if (listenRef.current) {
162 |       setTimeout(() => startListening(), 300);
163 |     } else {
164 |       1 6 4 | setFeedback(lang === 'vn' ? 'Đừng nghe' : 'Stopped');
165 |     }
166 |   };
167 |
168 |   rec.onerror = (event: any) => {
169 |     if (event.error === 'no-speech') {
170 |       if (listenRef.current) {
171 |         setTimeout(() => startListening(), 200);
172 |       }
173 |       return;
174 |     }
175 |     setIsListening(false);
176 |     if (event.error === 'not-allowed') {
177 |       1 7 7 | setFeedback(lang === 'vn' ? 'Vui lòng cấp quyền microphone' : 'P
178 |     } else {
179 |       1 7 9 | setFeedback(lang === 'vn' ? 'Lỗi: ' + event.error : 'Error: ' +
180 |     }
181 |   };
182 |
183 |   rec.onresult = (event: any) => {
184 |     let bestText = '';
185 |     let bestConfidence = 0;
186 |
187 |     for (let i = event.resultIndex; i < event.results.length; i++) {
188 |       if (event.results[i].isFinal) {
189 |         for (let j = 0; j < event.results[i].length; j++) {
190 |           const alt = event.results[i][j];
191 |           if (alt.confidence > bestConfidence) {
192 |             bestConfidence = alt.confidence;
193 |             bestText = alt.transcript;
194 |           }
195 |         }
196 |       }
197 |     }
198 |
199 |     if (bestText) {
200 |       setTranscript(bestText);
201 |       const matched = findBestMatch(bestText);
202 |       if (matched) {
203 |         2 0 3 | setFeedback(` ${matched.response} `);
204 |         speakText(matched.response);
205 |         onCommandRef.current(matched.action, matched.params);
206 |       } else {
207 |         2 0 7 | setFeedback(lang === 'vn' ? 'Không hiểu: "${bestText}"' : 'Un
208 |         2 0 8 | speakText(lang === 'vn' ? 'Xin hãy nói rõ lệnh' : 'Please spea
209 |       }
210 |     }
211 |   };
212 |
213 |   recognitionRef.current = rec;
214 | }, [lang, speakText, findBestMatch, startListening]);
215 |
216 | const toggleListening = () => {
217 |   if (!recognitionRef.current) {
218 |     2 1 8 | setFeedback(lang === 'vn' ? 'Trình duyệt không hỗ trợ' : 'Browser
219 |   }
220 |   return;
221 |   if (isListening || listenRef.current) {
222 |     stopListening();
223 |   } else {
224 |     startListening();
225 |   }
226 | };
227 |
228 | return (
229 |   <div style={{
230 |     position: 'fixed', bottom: '100px', left: '24px', zIndex: 999,
231 |     display: 'flex', flexDirection: 'column', gap: '8px', alignItems: 'flex-start',
232 |   }}>
233 |     <button
234 |       onClick={toggleListening}
235 |       2 3 5 | title={lang === 'vn' ? 'Ấu khi ấn bng gióng nói' : 'Voice cont
236 |       style={{
237 |         width: '56px', height: '56px', borderRadius: '50%',
238 |         display: 'flex', alignItems: 'center', justifyContent: 'center',
239 |         border: '2px solid ${isListening ? '#ef4444' : 'var(--border-default)}',
240 |         background: isListening
241 |           ? 'rgba(239,68,68,0.15)'
242 |           : 'var(--bg-surface)',
243 |         cursor: 'pointer', transition: 'all 0.2s',
244 |         boxShadow: isListening
245 |           ? '0 0 24px rgba(239,68,68,0.4), 0 0 60px rgba(239,68,68,0.15)'
246 |           : '0 4px 12px rgba(0,0,0,0.15)',
247 |         position: 'relative',
248 |       }}
249 |     >
250 |       <svg width="24" height="24" viewBox="0 0 24 24" fill="none" stroke={isListening ? '#ef4444' : 'var(--t...
251 |       <path d="M12 1a3 3 0 0 0 3 3v8a3 3 0 0 0 6 0V4a3 3 0 0 0 3 3z" />
252 |       <path d="M19 10v2a7 7 0 0 1 14 0v-2" />
253 |       <line x1="12" y1="19" x2="12" y2="23" />
254 |       <line x1="8" y1="23" x2="16" y2="23" />
255 |     </svg>
256 |     {isListening && (

```

```

257 |         <span style={{
258 |             position: 'absolute', inset: '-4px', borderRadius: '50%',
259 |             border: '2px solid #ef4444', animation: 'voicePulse 1.5s ease-in-out infinite',
260 |         }} />
261 |     })
262 | </button>
263 |
264 |     {feedback && (
265 |         <div style={{
266 |             padding: '8px 14px', borderRadius: '8px',
267 |             background: 'rgba(0,0,0,0.85)', backdropFilter: 'blur(8px)',
268 |             border: '1px solid rgba(0,212,170,0.3)',
269 |             fontSiz e: ' 1 2 p x ', color: feedback.startsWit h( ' ' ) ? '# 2 2 c 5 5 e '
270 |             maxWidth: '300px', fontFamily: 'monospace',
271 |             boxShadow: '0 4px 20px rgba(0,0,0,0.3)',
272 |         }}>
273 |             {isListening && (
274 |                 <div style={{ display: 'flex', alignItems: 'center', gap: '6px', marginBottom: transcript ? '4px' ...
275 |                     <span style={{ width: '6px', height: '6px', borderRadius: '50%', background: '#ef4444', animatio...
276 |                     < s p a n > { l a n g == ' v n ' ? ' ' a n g n g h e . . . ' : ' L i s t e n i n g . . . ' } < / s
277 |                 </div>
278 |             )}
279 |             {transcript && (
280 |                 <div style={{ marginTop: '4px', color: 'rgba(255,255,255,0.7)', fontSize: '11px' }}>
281 |                     &quot;{transcript}&quot;;
282 |                 </div>
283 |             )}
284 |             <div style={{ marginTop: '4px', fontSize: '11px' }}>{feedback}</div>
285 |         </div>
286 |     )}
287 | </div>
288 | );
289 | }
290 |

```

File: components/WebcamCursor.js (95 lines, 3.2 KB)

```

1 | 'use client';
2 |
3 | import { useEffect, useRef, useState } from 'react';
4 |
5 | const PINCH_THRESHOLD = 0.07; // Pinch to click
6 | const PINCH_RELEASE = 0.10; // Release threshold (hysteresis prevents jitter)
7 |
8 | const WebcamCursor = ({ landmarks, enabled, trackingSensitivity = 1.0 }) => {
9 |     const [cursorPos, setCursorPos] = useState({ x: -100, y: -100 });
10 |     const [isPinching, setIsPinching] = useState(false);
11 |     const wasPinchingRef = useRef(false);
12 |     const filterRef = useRef({ x: null, y: null, alpha: 0.12 }); // High smoothing coefficient
13 |
14 |     useEffect(() => {
15 |         if (!enabled || !landmarks || !landmarks.length) {
16 |             filterRef.current = { x: null, y: null, alpha: 0.12 };
17 |             setCursorPos({ x: -100, y: -100 });
18 |             return;
19 |         }
20 |
21 |         const hand = landmarks[0];
22 |         const tips = [4, 8, 12, 16, 20];
23 |         let sumX = 0;
24 |         let sumY = 0;
25 |
26 |         tips.forEach(i => {
27 |             sumX += hand[i].x;
28 |             sumY += hand[i].y;
29 |         });
30 |
31 |         const avgX = sumX / tips.length;
32 |         const avgY = sumY / tips.length;
33 |
34 |         const scaleX = (avgX - 0.5) * trackingSensitivity + 0.5;
35 |         const scaleY = (avgY - 0.5) * trackingSensitivity + 0.5;
36 |         const rawX = scaleX * window.innerWidth;
37 |         const rawY = scaleY * window.innerHeight;
38 |
39 |         const f = filterRef.current;
40 |         if (f.x === null) {
41 |             f.x = rawX;
42 |             f.y = rawY;
43 |         } else {
44 |             // Apply exponential moving average (EMA)
45 |             f.x += f.alpha * (rawX - f.x);
46 |             f.y += f.alpha * (rawY - f.y);
47 |         }
48 |
49 |         setCursorPos({ x: f.x, y: f.y });
50 |
51 |         const thumbTip = hand[4];
52 |         const indexTip = hand[8];
53 |         const distance = Math.hypot(indexTip.x - thumbTip.x, indexTip.y - thumbTip.y);
54 |         // Hysteresis: easier to pinch, harder to release
55 |         const currentPinch = wasPinchingRef.current
56 |             ? distance < PINCH_RELEASE
57 |             : distance < PINCH_THRESHOLD;
58 |         setIsPinching(currentPinch);
59 |
60 |         if (currentPinch && !wasPinchingRef.current) {
61 |             // Try to find the element under cursor
62 |             // We use standard document.elementFromPoint. Ensure the cursor itself has pointer-events: none
63 |             const el = document.elementFromPoint(f.x, f.y);
64 |             if (el && typeof el.click === 'function') {
65 |                 el.click();
66 |             }
67 |         }
68 |
69 |         wasPinchingRef.current = currentPinch;
70 |
71 |     }, [landmarks, enabled, trackingSensitivity]);
72 |
73 |     if (!enabled) return null;
74 |
75 |     return (
76 |         <div style={{
77 |             position: 'fixed',
78 |             left: 0,
79 |             top: 0,
80 |             width: '24px',
81 |             height: '24px',
82 |             borderRadius: '50%',
83 |             backgroundColor: isPinching ? 'rgba(255, 255, 0, 0.8)' : 'rgba(0, 255, 255, 0.5)',
84 |             border: '2px solid white',
85 |             transform: `translate(${cursorPos.x - 12}px, ${cursorPos.y - 12}px)`,
86 |             pointerEvents: 'none',

```

```

87|         zIndex: 9999,
88|         transition: 'transform 0.05s linear, background-color 0.1s',
89|         boxShadow: isPinching ? '0 0 20px yellow' : '0 0 15px cyan'
90|     }} />
91| );
92| };
93|
94| export default WebcamCursor;
95|

```

File: components/WindowsSimulator.js (484 lines, 32.8 KB)

```

1| 'use client';
2| import React, { useState, useEffect, useRef } from 'react';
3| import { GLOSSARY } from '../data/glossary';
4|
5| const APPS = [
6|     { id: 'browser', icon: '0<B', name: 'Edge', color: '#0078d4' },
7|     { id: 'youtube', icon: '%b', name: 'YouTube', color: '#ff0000' },
8|     { id: 'wiki', icon: '0=U0', name: 'Wiki PC', color: '#10b981' },
9|     { id: 'music', icon: '0<Bµ', name: 'Music', color: '#1db954' },
10|     { id: 'explorer', icon: '0=UÁ', name: 'Explorer', color: '#ffb900' },
11|     { id: 'calculator', icon: '0=Y", name: 'Calculator', color: '#0078d4' },
12|     { id: 'notepad', icon: '0=Üÿ', name: 'Notepad', color: '#333' },
13|     { id: 'specs', icon: '0=U»', name: 'PC Specs', color: '#10b981' },
14| ];
15|
16| function CalculatorApp() {
17|     const [display, setDisplay] = useState('0');
18|     const [prev, setPrev] = useState(null);
19|     const [op, setOp] = useState(null);
20|     const btn = (v, style = {}) => (
21|         <button onClick={() => {
22|             if (v === 'C') { setDisplay('0'); setPrev(null); setOp(null); return; }
23|             if ([ '+', '-', 'x', '/' ].includes(v)) { setPrev(parseFloat(display)); setOp(v); setDisplay('0'); return; }
24|             if (v === '=') {
25|                 if (prev === null || !op) return;
26|                 const a = prev, b = parseFloat(display);
27|                 const r = op === '+' ? a + b : op === '-' ? a - b : op === 'x' ? a * b : op === '/' ? a / b : 'Err';
28|                 setDisplay(String(r)); setPrev(null); setOp(null); return;
29|             }
30|             setDisplay(prev => prev === '0' ? v : prev + v);
31|         }} style={{ padding: '12px', background: style.bg || '#3a3a3a', color: style.color || 'white', border: 'none', ...
32|     });
33|     return (
34|         <div style={{ background: '#1e1e1e', borderRadius: '8px', padding: '16px', width: '260px' }}>
35|             <div style={{ background: '#000', color: 'white', padding: '12px 16px', borderRadius: '6px', fontSize: ...
36|                 <div style={{ display: 'grid', gridTemplateColumns: 'repeat(4, 1fr)', gap: '6px' }}>
37|                     {btn('C', {bg: '#d83b3b', color: 'white'})} {btn('%', {bg: '#555'})} {btn('±', {bg: '#555'})} {btn('÷'...
38|                     {btn('7')} {btn('8')} {btn('9')} {btn('x', {bg: '#0078d4'})}
39|                     {btn('4')} {btn('5')} {btn('6')} {btn('-', {bg: '#0078d4'})}
40|                     {btn('1')} {btn('2')} {btn('3')} {btn('+', {bg: '#0078d4'})}
41|                     <button onClick={() => setDisplay(d => d.length > 1 ? d.slice(0, -1) : '0')} style={{ padding: '12px', ...
42|                         {btn('0')} {btn('.')} {btn('=' , {bg: '#0078d4'})}
43|                     </div>
44|                 </div>
45|             </div>
46|         );
47|
48| function BrowserApp() {
49|     const [url, setUrl] = useState('https://google.com');
50|     const [showIframe, setShowIframe] = useState(true);
51|     const [key, setKey] = useState(0);
52|     const suggestions = [
53|         { label: '0=ÿ Google', url: 'https://google.com' },
54|         { label: '0=U0 Wikipedia PC', url: 'https://en.wikipedia.org/wiki/' },
55|         { label: '0>Ya Wiki CPU', url: 'https://vi.wikipedia.org/wiki/B%E' },
56|         { label: '0<Bµ Wiki GPU', url: 'https://en.wikipedia.org/wiki/Grap' },
57|         { label: '0=U» Wiki RAM', url: 'https://en.wikipedia.org/wiki/Ran' },
58|     ];
59|     return (
60|         <div style={{ width: '650px', maxWidth: '90vw', display: 'flex', flexDirection: 'column' }}>
61|             <div style={{ background: '#f3f3f3', padding: '8px 12px', display: 'flex', gap: '8px', alignItems: 'cen...
62|                 <input value={url} onChange={e => setUrl(e.target.value)} onKeyDown={e => { if (e.key === 'Enter'...
63|                     p l a c e h o l d e r = " N h - p U R L . . . "
64|                 style={{ flex: 1, borderRadius: '20px', padding: '6px 16px', fontSize: '13px', border: '1px sol...
65|                 <button onClick={() => { setShowIframe(true); setKey(k => k + 1); }} style={{ background: '#0078d...
66|             </div>
67|             <div style={{ display: 'flex', gap: '4px', padding: '4px 8px', background: '#fafafa', borderBottom: '1p...
68|                 {suggestions.map((s, i) => (
69|                     <button key={i} onClick={() => { setUrl(s.url); setShowIframe(true); setKey(k => k + 1); }}
70|                         style={{ background: '#e8e8e8', border: 'none', borderRadius: '12px', padding: '2px 10px', ...
71|                         {s.label}
72|                     </button>
73|                 ))}
74|             </div>
75|             {showIframe && (
76|                 <iframe key={key} src={url} style={{ width: '100%', height: '420px', border: 'none' }} title="Bro...
77|             )}
78|         </div>
79|     );
80| }
81|
82| function YouTubeApp() {
83|     const [embedUrl, setEmbedUrl] = useState('');
84|     const videos = [
85|         { label: 'H° Ò ng d « n l - p r á p PC', id: 'Zvz77q1aLBHG' },
86|         { label: 'C á ch ch í n CPU 2026', id: 'y1F0pPvYFnA' },
87|         { label: 'B u i l d PC G a m i n g t š m t r u n g', id: 'P0YCHg0N0P4' },
88|         { label: 'C á ch ép x u n g RAM', id: 'd_0e07D9FD8' },
89|         { label: 'T ê n n h i ệ t n ộ c A I O', id: 'k04Z4N4nqso' },
90|         { label: 'So sánh SSD NVME vs SATA', id: 'dj0BBWuqW5w' },
91|         { label: 'H° Ò ng d « n ch í n n g u ộ n PSU', id: 'EaZxtvB6_jE' },
92|     ];
93|     return (
94|         <div style={{ width: '640px', maxWidth: '90vw', display: 'flex', flexDirection: 'column', maxHeight: '70vh'...
95|             <div style={{ background: '#202020', padding: '12px', borderRadius: '6px 6px 0 0', color: 'white' }}>
96|                 <p style={{ margin: 0, fontSize: '13px' }}> %b Ch í n v i d e o
97|             </div>
98|             <div style={{ background: '#111', padding: '12px', overflow: 'auto', display: 'flex', flexDirection: '...
99|                 {videos.map((v, i) => (
100|                     <button key={i} onClick={() => setEmbedUrl('https://www.youtube.com/embed/${v.id}?autoplay...
101|                         style={{ background: embedUrl ? 'includes(v.id)?'#333':'#1a1a1a', color: '#ddd', border: '1...
102|                         <span style={{ color: '#ff0000' }}> %b </span> { v . t
103|                     </button>
104|                 ))}
105|             </div>
106|             {embedUrl && (
107|                 <iframe key={embedUrl} src={embedUrl} style={{ width: '100%', height: '350px', border: 'none', bo...
108|             )}
109|         </div>
110|     );
111| }

```



```

112 function WikiApp() {
113   const [search, setSearch] = useState('');
114   const [selected, setSelected] = useState(null);
115   const entries = Object.entries(GLOSSARY).sort(([a],[b]) => a.localeCompare(b));
116   const filtered = search ? entries.filter(([k,v]) => k.toLowerCase().includes(search.toLowerCase())) || v.to...
117
118   return (
119     <div style={{ width:'550px', maxWidth:'90vw', maxHeight:'70vh', display:'flex', flexDirection:'column'...
120       <div style={{ padding:'12px', background:'#16213e' }}>
121         <div style={{ display:'flex', alignItems:'center', gap:'8px', marginBottom:'8px' }}>
122           <span style={{ color: '#00d4aa', fontWeight: 700, font
123             <span style={{ color: '#00d4aa', fontW
124             </span>
125           </div>
126           <input value={search} onChange={e => { setSearch(e.target.value); setSelected(null); }}
127             style={{ width:'100%', padding:'8px 14px', borderRadius:'8px', border:'1px solid #2a2a4a',...
128           </div>
129           <div style={{ display:'flex', flex:1, overflow:'hidden' }}>
130             <div style={{ width:'180px', overflow:'auto', borderRight:'1px solid rgba(255,255,255,0.05)',...
131               {filtered.map(([key]) => (
132                 <div key={key} onClick={() => setSelected(key)}
133                   style={{ padding:'6px 10px', borderRadius:'6px', cursor:'pointer', fontSize:'12px'...
134                     background: selected===key ? 'rgba(0,212,170,0.15)' : 'transparent',
135                     color: selected===key ? '#00d4aa' : '#8a8aa0', fontWeight: selected===key ? 70...
136                   {key}
137                 </div>
138               ))}
139             </div>
140             <div style={{ flex:1, overflowY:'auto', padding:'16px', color:'#c0c0d0' }}>
141               {selected ? (
142                 <div>
143                   <h3 style={{ color:'#00d4aa', margin:'0 0 8px 0', fontSize:'18px' }}>{selected}</h3>
144                   <p style={{ fontSize:'14px', lineHeight:1.6, color:'#a0a0b0' }}>{GLOSSARY[selected...
145                 </div>
146               ) : (
147                 <div style={{ textAlign:'center', padding:'40px 0', color:'#6a6a80' }}>
148                   <div style={{ fontS
149                     <div style={{ fontS
150                     <p style={{ fontS
151                     </p>
152                   </div>
153                 </div>
154               </div>
155             </div>
156           );
157   }
158
159   function MusicApp() {
160     const audioRef = useRef(null);
161     const [currentIdx, setCurrentIdx] = useState(null);
162     const [playing, setPlaying] = useState(false);
163     const tracks = [
164       { title: 'Lofi Chill', url: 'https://www.soundhelix.com/examples/mp3/SoundHelix-Song-1.mp3' },
165       { title: 'Th° gi°n', url: 'https://www.soundhelix.com/examples/m
166       { title: 'Beats i°n t°i', url: 'https://www.soundhelix.com/exam
167       { title: 'Jazz', url: 'https://www.soundhelix.com/examples/mp3/SoundHelix-Song-4.mp3' },
168       { title: 'Piano', url: 'https://www.soundhelix.com/examples/mp3/SoundHelix-Song-5.mp3' },
169     ];
170     useEffect(() => {
171       if (audioRef.current) {
172         if (playing && currentIdx !== null) { audioRef.current.play().catch(() => setPlaying(false)); }
173         else { audioRef.current.pause(); }
174       }
175     }, [currentIdx, playing]);
176
177     return (
178       <div style={{ width:'360px', background:'#121212', borderRadius:'8px', padding:'20px', color:'white' }}>
179         <h3 style={{ margin:'0 0 16px 0', fontSize:'16px', display:'flex', alignItems:'center', gap:'8px' ...
180         <div style={{ display:'flex', flexDirection:'column', gap:'6px' }}>
181           {tracks.map((t,i) => (
182             <button key={i} onClick={() => { setCurrentIdx(i); setPlaying(true); }}
183               style={{ background: currentIdx===i && playing ? '#1db954' : '#282828', color: current...
184               border:'none', borderRadius:'8px', padding:'10px 14px', cursor:'pointer', textAlig...
185             <span>{currentIdx===i && playing ? '0<sup>Y</sup>' : '0<sup>µ</sup>'}</span>
186             {t.title}
187           </button>
188           ))}
189         </div>
190         {currentIdx !== null && (
191           <div style={{ marginTop:'12px', display:'flex', gap:'8px', alignItems:'center' }}>
192             <button onClick={() => setPlaying(p => !p)} style={{ background:'#1db954', color:'#000', b...
193               <span>{playing ? '0<sup>P</sup>' : '0<sup>A</sup>'}</span>
194             </button>
195             <button onClick={() => { setCurrentIdx(null); setPlaying(false); }} style={{ background:'#...
196             </div>
197           )}
198           <audio ref={audioRef} src={currentIdx !== null ? tracks[currentIdx].url : ''} controls style={{ wi...
199         </div>
200       );
201   }
202
203   function ExplorerApp({ cart }) {
204     const folders = ['Desktop', 'Downloads', 'Documents', 'Pictures', 'Music'];
205     return (
206       <div style={{ width:'460px', color:'#333' }}>
207         <div style={{ background:'#f3f3f3', padding:'6px 12px', fontSize:'12px', borderBottom:'1px solid #...
208           {['File', 'Home', 'Share', 'View'].map(m => <span key={m} style={{ cursor:'pointer' }}>{m}</span>)}
209         </div>
210         <div style={{ display:'flex', background:'white', borderRadius:'0 0 6px 6px', minHeight:'200px' }}>
211           <div style={{ width:'120px', background:'#fafafa', padding:'8px', borderRight:'1px solid #ddd'...
212             {folders.map(f => <div key={f} style={{ padding:'4px 8px', cursor:'pointer', borderRight:...
213             </div>
214           <div style={{ flex:1, padding:'10px' }}>
215             <p style={{ fontS
216               {(cart||[]).map((c,i) => <div key={i} style={{ fontS
217               (!cart||cart.length==0) && <p style={{ color:'#999
218             </div>
219           </div>
220         </div>
221       );
222   }
223
224   function NotepadApp() {
225     const [text, setText] = useState('0<sup>Ų</sup> Ghi chú\ n\ nMáy tính ã °ăc l- p
226     return (
227       <div style={{ width:'400px' }}>
228         <div style={{ background:'#f0f0f0', padding:'4px 12px', fontSize:'12px', borderRadius:'6px 6px 0 0...
229           {['File', 'Edit', 'Format', 'View', 'Help'].map(m => <span key={m} style={{ cursor:'pointer' }}>{m...
230         </div>
231         <textarea value={text} onChange={e => setText(e.target.value)} style={{ width:'100%', minHeight:'2...
232       </div>
233     );
234   }
235

```

```

236 function SpecsApp({ cart, scenarioName }) {
237   return (
238     <div style={{ width:'440px', background:'#1a1a2e', color:'white', borderRadius:'8px', padding:'20px' }}>
239       {scenarioName && (
240         <div style={{ background:'rgba(0,212,170,0.08)', border:'1px solid rgba(0,212,170,0.2)', border...
241         2 4 1 |
242           <span style={{ fontSize:'13px', color:'#00d4aa', fontWeight:600 }}>{scenarioName}</span>
243         </div>
244       )}
245       <h3 style={{ color:'#00d4aa', margin:'0 0 16px 0', display:'flex', alignItems:'center', gap:'8px' ...
246       <div style={{ display:'flex', flexDirection:'column', gap:'10px' }}>
247         <div style={{ background:'rgba(0,212,170,0.05)', border:'1px solid rgba(0,212,170,0.1)', border...
248         <div style={{ fontSize:'11px', color:'#6b7280', fontWeight:700, textTransform:'uppercase',...
249         <div style={{ color:'#e0e6ed' }}>Windows 11 Pro (64-bit)</div>
250       </div>
251       <div style={{ background:'rgba(0,212,170,0.05)', border:'1px solid rgba(0,212,170,0.1)', border...
252       <div style={{ fontSize:'11px', color:'#6b7280', fontWeight:700, textTransform:'uppercase',...
253         <div style={{ color:'#10b981' }}>All components work</div>
254         <div style={{ color:'#10b981' }}>Drivers installed</div>
255         <div style={{ color:'#10b981' }}>System stable</div>
256       </div>
257       {(cart||[]).map((c,i) => (
258         <div key={i} style={{ background:'rgba(255,255,255,0.03)', border:'1px solid rgba(255,255,...
259         <span style={{ fontSize:'12px', color:'#6b7280', textTransform:'uppercase', fontWeight...
260         <span style={{ color:'#e0e6ed', fontSize:'13px' }}>{c.name || 'Installed'}</span>
261       </div>
262     ))}
263   </div>
264 </div>
265 );
266 }
267
268 function WindowApp({ app, cart, scenarioName, onClose }) {
269   const components = {
270     calculator: CalculatorApp,
271     browser: BrowserApp,
272     youtube: YouTubeApp,
273     wiki: WikiApp,
274     music: MusicApp,
275     explorer: () => <ExplorerApp cart={cart} />,
276     notepad: NotepadApp,
277     specs: () => <SpecsApp cart={cart} scenarioName={scenarioName} />,
278   };
279   const AppComponent = components[app.id];
280   return (
281     <div style={{ position:'fixed', top:'50%', left:'50%', transform:'translate(-50%,-50%)', zIndex:100, b...
282     <div style={{ background:'#202020', display:'flex', justifyContent:'space-between', alignItems:'ce...
283     <span style={{ color:'white', fontSize:'13px', fontWeight:600 }}>{app.icon} {app.name}</span>
284     <button onClick={onClose} style={{ width:'14px', height:'14px', borderRadius:'50%', background...
285   </div>
286   <div style={{ background:'#1e1e1e', overflow:'auto', maxHeight:'calc(90vh - 36px)' }}>
287     {AppComponent && <AppComponent cart={cart} />
288   </div>
289 </div>
290 );
291 }
292
293 export default function WindowsSimulator({ cart, scenarioName, onExit }) {
294   const [phase, setPhase] = useState('post');
295   const [progress, setProgress] = useState(0);
296   const [openApp, setOpenApp] = useState(null);
297   const [time, setTime] = useState('');
298   const [startOpen, setStartOpen] = useState(false);
299   const [bootMsgs, setBootMsgs] = useState([]);
300
301   const counts = {};
302   (cart||[]).forEach(c => { const t = c.type || c; counts[t] = (counts[t]||0)+1; });
303   const hasCPU = counts['CPU'] >= 1;
304   const hasRAM = counts['RAM'] >= 1;
305   const hasGPU = counts['GPU'] >= 1;
306   const hasSSD = counts['SSD'] >= 1 || counts['Storage'] >= 1;
307   const hasPSU = counts['PSU'] >= 1;
308   const allOK = hasCPU && hasRAM && hasGPU && hasSSD && hasPSU;
309
310   useEffect(() => {
311     const t = setInterval(() => setTime(new Date().toLocaleTimeString('vi-VN', { hour:'2-digit', minute:'2...
312     setTime(new Date().toLocaleTimeString('vi-VN', { hour:'2-digit', minute:'2-digit' }));
313     return () => clearInterval(t);
314   }, []);
315
316   // POST sequence
317   useEffect(() => {
318     if (phase !== 'post') return;
319     const lines = [
320       { msg: 'PC Master BIOS v2.04 (UEFI)', delay: 100 },
321       { msg: 'CPU: $ { has CPU ? 'Intel / AMD Detected' : 'NOT FOUND' }', delay: 100 },
322       { msg: 'RAM: $ { has RAM ? 'Stick(s) detected' : 'NOT FOUND' }', delay: 100 },
323       { msg: 'GPU: $ { has GPU ? 'PCIe device detected' : 'NOT FOUND' }', delay: 100 },
324       { msg: 'Storage: $ { has SSD ? 'NVMe / SATA detected' : 'NOT FOUND' }', delay: 100 },
325       { msg: 'PSU: $ { has PSU ? 'Power OK' : 'POWER FAILURE' }', delay: 100 },
326     ];
327     const timers = lines.map(({msg, delay}) => setTimeout(() => setBootMsgs(prev => [...prev, msg]), delay));
328     const final = setTimeout(() => {
329       if (allOK) {
330         setBootMsgs(prev => [...prev, '', 'POST OK - BEEP (1 short beep)', 'Starting boot sequence...']);
331         setTimeout(() => setPhase('boot'), 1500);
332       } else {
333         setBootMsgs(prev => [...prev, '', 'FATAL: POST failed - check components']);
334         setTimeout(() => setPhase('error'), 1000);
335       }
336     }, 3000);
337     return () => { timers.forEach(clearTimeout); clearTimeout(final); };
338   }, [phase]);
339
340   useEffect(() => {
341     if (phase !== 'boot') return;
342     const t = setInterval(() => setProgress(p => { if (p>=100) { clearInterval(t); setTimeout(() => setPha...
343     return () => clearInterval(t);
344   }, [phase]);
345
346   if (phase === 'post') return (
347     <div style={{ position:'fixed', inset:0, zIndex:9999, background:'#0a0a0a', display:'flex', alignItems...
348     <div style={{ width:'600px', maxWidth:'90vw', background:'#000', border:'1px solid #1a1a1a', borde...
349     <pre style={{ color:'#00ff00', fontSize:'12px', lineHeight:1.3, marginBottom:'20px' }}>
350 {` PC Master BIOS v2.04 (UEFI) - 2026
351 =====
352 CPU : Intel/AMD [OK]
353 Memory: Testing...
354 </pre>
355 <div style={{ color:'#c0c0c0', fontSize:'13px', lineHeight:1.8, fontFamily:'inherit' }}>
356 {bootMsgs.map((msg, i) => (
357   <div key={i} style={{ color: msg.startsWith('FATAL')||msg.includes('FAILURE')||msg.inc...
358   ))}
359 {bootMsgs.length <= 8 && <span style={{ color:'#00ff00' }}></span>

```

```

360 |         </div>
361 |     </div>
362 | </div>
363 | );
364 |
365 | if (phase === 'error') return (
366 |     <div style={{ position:'fixed', inset:0, zIndex:9999, background:'#0a0000', display:'flex', flexDirect...
367 |         <div style={{ position:'fixed', inset:0, zIndex:9999, background:'#0a0000', display:'flex', flexDirect...
368 |             <h2 style={{ color:'#ef4444', margin:0, fontSize:'22px' }}>POST FAILED - SYSTEM HALTED</h2>
369 |             <div style={{ color:'#ff6666', lineHeight:2, fontSize:'14px' }}>
370 |                 { ! has CPU && <div> CPU : Not detected </div> }
371 |                 { ! has RAM && <div> RAM : Not detected </div> }
372 |                 { ! has GPU && <div> GPU : Not detected </div> }
373 |                 { ! has SSD && <div> Storage : Not detected </div> }
374 |                 { ! has PSU && <div> PSU : Power failure </div> }
375 |             </div>
376 |             <p style={{ color:'#ff8888', fontSize:'13px', maxWidth:'400px', textAlign:'center' }}>System canno...
377 |             <button onClick={onExit} style={{ padding:'10px 30px', background:'#ef4444', color:'white', border...
378 |         </div>
379 |     );
380 |
381 | if (phase === 'boot') return (
382 |     <div style={{ position:'fixed', inset:0, zIndex:9999, background:'#000', display:'flex', flexDirection...
383 |         <div style={{ display:'flex', alignItems:'center', gap:'16px' }}>
384 |             <div style={{ width:'48px', height:'48px', borderRadius:'8px', background:'linear-gradient(135...
385 |             <div style={{ color:'white', fontSize:'28px', fontWeight:300, letterSpacing:'2px' }}>Windows 1...
386 |         </div>
387 |         <div style={{ width:'200px', height:'4px', background:'#333', borderRadius:'2px', overflow:'hidden...
388 |         <div style={{ height:'100%', width:'${progress}%', background:'#0078d4', borderRadius:'2px', t...
389 |     </div>
390 |     <div style={{ color:'#666', fontSize:'13px' }}>Starting up...</div>
391 | </div>
392 | );
393 |
394 | if (phase === 'login') return (
395 |     <div style={{ position:'fixed', inset:0, zIndex:9999, background:'linear-gradient(135deg,#0078d4,#003c...
396 |         <div style={{ width:'72px', height:'72px', borderRadius:'50%', background:'rgba(255,255,255,0.15)'...
397 |         <div style={{ color:'white', fontSize:'20px', fontWeight:300 }}>PC Master User</div>
398 |         { scenarioName && <div style={{ color:'rgba(255,255,255,0.5)' }}>
399 |             <button onClick={() => setPhase('desktop')}
400 |                 style={{ background:'rgba(255,255,255,0.12)', color:'white', border:'1px solid rgba(255,255,25...
401 |                 Sign in !
402 |             </button>
403 |         </div>
404 |     );
405 |
406 | const wallpaperColors = 'linear-gradient(135deg,#0f1b2d 0%,#0d2137 40%,#1a0533 100%)';
407 | return (
408 |     <div style={{ position:'fixed', inset:0, zIndex:9999, background:wallpaperColors, fontFamily:'Segoe U...
409 |         /* Desktop Icons */
410 |         <div style={{ padding:'16px', display:'flex', flexDirection:'column', gap:'4px', width:'80px' }}>
411 |             {APPS.map(app => (
412 |                 <div key={app.id} onClick={() => { setOpenApp(app); setStartOpen(false); }}
413 |                 style={{ display:'flex', flexDirection:'column', alignItems:'center', gap:'2px', curso...
414 |                 onMouseOver={e => e.currentTarget.style.background = 'rgba(255,255,255,0.1)'}
415 |                 onMouseOut={e => e.currentTarget.style.background = 'transparent'}
416 |                 <span style={{ fontSize:'28px' }}>{app.icon}</span>
417 |                 <span style={{ fontSize:'11px', textAlign:'center', textShadow:'0 1px 3px rgba(0,0,0,0...
418 |             </div>
419 |             ))}
420 |         </div>
421 |
422 |         /* Open App Window */
423 |         {openApp && <WindowApp app={openApp} cart={cart} scenarioName={scenarioName} onClose={() => setOpe...
424 |
425 |         /* Taskbar */
426 |         <div onClick={e => e.stopPropagation()} style={{ position:'fixed', bottom:0, left:0, right:0, heig...
427 |         <div style={{ display:'flex', alignItems:'center', gap:'2px' }}>
428 |             /* Start Button */
429 |             <button onClick={() => setStartOpen(v => !v)} style={{ background:'transparent', border:'n...
430 |             onMouseOver={e => e.currentTarget.style.background = 'rgba(255,255,255,0.1)'}
431 |             onMouseOut={e => e.currentTarget.style.background = 'transparent'}
432 |             <span style={{ width:'24px', height:'24px', borderRadius:'4px', background:'linear-gra...
433 |         </button>
434 |         /* Search */
435 |         <button onClick={() => alert('0=Y M B Edge A search')}
436 |         <span>Search</span>
437 |     </div>
438 |     /* Taskbar Apps */
439 |     {APPS.map(app => (
440 |         <button key={app.id} onClick={() => setOpenApp(app)} title={app.name}
441 |         style={{ background: openApp?.id === app.id ? 'rgba(255,255,255,0.15)' : 'transparen...
442 |         onMouseOver={e => e.currentTarget.style.background = 'rgba(255,255,255,0.08)'}
443 |         onMouseOut={e => { if (openApp?.id !== app.id) e.currentTarget.style.background = ...
444 |         {app.icon}
445 |     </button>
446 |     ))}
447 | </div>
448 |     /* System Tray */
449 |     <div style={{ display:'flex', alignItems:'center', gap:'8px' }}>
450 |         <button onClick={onExit} style={{ background:'rgba(239,68,68,0.8)', color:'white', border:...
451 |         #0 Shut Down
452 |     </button>
453 |     <div style={{ color:'white', fontSize:'12px', textAlign:'right', lineHeight:1.2 }}>
454 |         <div style={{ fontWeight:600 }}>{time}</div>
455 |         <div style={{ color:'#888', fontSize:'10px' }}>{new Date().toLocaleDateString('vi-VN')}...
456 |     </div>
457 | </div>
458 |
459 |
460 |     /* Start Menu */
461 |     {startOpen && (
462 |         <div onClick={e => e.stopPropagation()} style={{ position:'fixed', bottom:'56px', left:'8px', ...
463 |         <div style={{ color:'white', fontWeight:600, marginBottom:'12px', fontSize:'13px' }}>Pinne...
464 |         <div style={{ display:'grid', gridTemplateColumns:'repeat(4,1fr)', gap:'8px', marginBottom...
465 |             {APPS.map(app => (
466 |                 <button key={app.id} onClick={() => { setOpenApp(app); setStartOpen(false); }}
467 |                 style={{ background:'transparent', border:'none', cursor:'pointer', display:'f...
468 |                 onMouseOver={e => e.currentTarget.style.background = 'rgba(255,255,255,0.08)'}
469 |                 onMouseOut={e => e.currentTarget.style.background = 'transparent'}
470 |                 <span style={{ fontSize:'28px' }}>{app.icon}</span>
471 |                 <span style={{ fontSize:'10px', color:'#ccc' }}>{app.name}</span>
472 |             </button>
473 |             ))}
474 |         </div>
475 |         <div style={{ borderTop:'1px solid rgba(255,255,255,0.08)', paddingTop:'12px', display:'fl...
476 |         <span style={{ color:'#aaa', fontSize:'12px' }}>
477 |             <button onClick={onExit} style={{ background:'transparent', border:'none', color:'#888...
478 |         </div>
479 |     </div>
480 |     ))}
481 | </div>
482 | );
483 | }

```

File: lib/auth-actions.ts (295 lines, 8.6 KB)

```

1 | 'use server'
2 |
3 | import { createClient } from '@lib/supabase-ssr-server'
4 | import { revalidatePath } from 'next/cache'
5 | import { redirect } from 'next/navigation'
6 |
7 | export async function login(formData: FormData) {
8 |   const supabase = await createClient()
9 |   const email = formData.get('email') as string
10 |   const password = formData.get('password') as string
11 |
12 |   if (!email || !password) {
13 |     return { error: 'Vui lòng nh - p s y ç email và m - t kh@u' }
14 |   }
15 |
16 |   // Admin login via Supabase Auth
17 |   if (email === 'admin') {
18 |     const adminEmail = process.env.ADMIN_EMAIL || 'admin@pcmaster.com'
19 |     const { error: adminError } = await supabase.auth.signInWithPassword({
20 |       email: adminEmail,
21 |       password: password,
22 |     })
23 |     if (adminError) {
24 |       return { error: 'Thông tin ng nh - p không chính xác' }
25 |     }
26 |     return { success: true, redirectUrl: '/admin' }
27 |   }
28 |
29 |   // 1. Th ã c hi Ç n ng nh - p th ã ng b ã ng Supabase Auth
30 |   const { data, error } = await supabase.auth.signInWithPassword({
31 |     email,
32 |     password,
33 |   })
34 |
35 |   if (error) {
36 |     let friendlyError = 'Email ho . c m - t kh@u không chính xác. Vui lòng k i Á n ç thông tin b - t bu ò c'
37 |     const errMsg = error.message.toLowerCase()
38 |
39 |     if (errMsg.includes('confirm') || errMsg.includes('email_not_confirmed')) {
40 |       friendlyError = 'T ã i kh o ã n c ç ã b i n ch ò a ò ã c x á c nh - n qua Email. V
41 |     } else if (errMsg.includes('invalid') || errMsg.includes('credentials')) {
42 |       friendlyError = 'Email ho . c m - t kh@u không chính xác. Vui lòng k i Á n ç thông tin b - t bu ò c'
43 |     } else {
44 |       friendlyError = error.message
45 |     }
46 |     return { error: friendlyError }
47 |   }
48 |
49 |   const user = data.user
50 |
51 |   // 2. T ã ù ng c ó u h ù P r o f i l e n ù u b Ë t h i ù u ( ví d ã : d o R L S ho . c T r i g g e r
52 |   if (user) {
53 |     const { data: profile } = await supabase
54 |       .from('profiles')
55 |       .select('role')
56 |       .eq('id', user.id)
57 |       .maybeSingle()
58 |
59 |     if (!profile) {
60 |       const role = user.user_metadata?.role || 'student'
61 |       const fullName = user.user_metadata?.full_name || email.split('@')[0]
62 |
63 |       const { error: rescueError } = await supabase.from('profiles').upsert({
64 |         id: user.id,
65 |         email: email,
66 |         full_name: fullName,
67 |         role: role,
68 |         updated_at: new Date().toISOString()
69 |       })
70 |
71 |       if (rescueError) {
72 |         console.error('Profile Rescue Failed:', rescueError)
73 |       }
74 |     }
75 |   }
76 |
77 |   // L ã y r o l e c u ã i c ù ng
78 |   const { data: finalProfile } = await supabase
79 |     .from('profiles')
80 |     .select('role')
81 |     .eq('id', user.id)
82 |     .single()
83 |
84 |   const userRole = finalProfile?.role || user!.user_metadata?.role || 'student'
85 |
86 |   if (userRole === 'student') {
87 |     return { success: true, redirectUrl: '/builder' }
88 |   } else if (userRole === 'parent') {
89 |     return { success: true, redirectUrl: '/parent' }
90 |   } else {
91 |     return { success: true, redirectUrl: `/${userRole}` }
92 |   }
93 | }
94 |
95 | export async function register(formData: FormData) {
96 |   const supabase = await createClient()
97 |   const email = formData.get('email') as string
98 |   const password = formData.get('password') as string
99 |   const fullName = formData.get('full_name') as string
100 |   const role = formData.get('role') as string || 'student'
101 |   const schoolCode = formData.get('school_code') as string
102 |   const schoolName = formData.get('school_name') as string
103 |   const classCode = formData.get('class_code') as string
104 |
105 |   if (!email || !password || !fullName) {
106 |     return { error: 'Vui lòng i Á n ç thông tin b - t bu ò c' }
107 |   }
108 |
109 |   let schoolId = null;
110 |   let classIdToJoin = null;
111 |
112 |   // 1. K i Á m t r a m ã t r ò Ý ng ( n ù u c ó )
113 |   if (schoolCode) {
114 |     const { data: school } = await supabase
115 |       .from('schools')
116 |       .select('id')
117 |       .eq('code', schoolCode)
118 |       .single()
119 |

```

```

120 |     if (school) schoolId = school.id;
121 | }
122 |
123 | // 3.      n g k ý t à i k h o ã n   A u t h
124 | const { data, error } = await supabase.auth.signUp({
125 |   email,
126 |   password,
127 |   options: {
128 |     data: {
129 |       full_name: fullName,
130 |       role: role,
131 |     }
132 |   }
133 | })
134 |
135 | if (error) return { error: error.message }
136 |
137 | if (data.user) {
138 |   let profileError = null;
139 |   const { error: updateError } = await supabase
140 |     .from('profiles')
141 |     .update({
142 |       full_name: fullName,
143 |       role: role,
144 |       school_id: schoolId,
145 |       school_name: schoolName,
146 |       updated_at: new Date().toISOString()
147 |     })
148 |     .eq('id', data.user.id);
149 |
150 |   if (updateError) {
151 |     console.warn('Profile UPDATE failed, falling back to UPSERT:', updateError.message);
152 |     const { error: upsertError } = await supabase.from('profiles').upsert({
153 |       id: data.user.id,
154 |       email: email,
155 |       full_name: fullName,
156 |       role: role,
157 |       school_id: schoolId,
158 |       school_name: schoolName,
159 |       updated_at: new Date().toISOString()
160 |     });
161 |     profileError = upsertError;
162 |   }
163 |
164 |   if (profileError) {
165 |     console.error('Profile Creation/Update Error:', profileError);
166 |   }
167 |
168 |   if (role === 'parent' && classCode) {
169 |     const { data: student } = await supabase
170 |       .from('profiles')
171 |       .select('id')
172 |       .eq('role', 'student')
173 |       .or(`email.eq.${classCode},id.eq.${classCode}`)
174 |       .maybeSingle();
175 |
176 |     if (student) {
177 |       await supabase.from('parent_student_links').insert({
178 |         parent_id: data.user.id,
179 |         student_id: student.id,
180 |         relationship: 'parent'
181 |       });
182 |     }
183 |   }
184 |
185 |   // Auto sign-in after registration
186 |   const { error: signInError } = await supabase.auth.signInWithPassword({ email, password });
187 |   if (signInError) {
188 |     console.error('Auto sign-in after registration failed:', signInError);
189 |   }
190 | }
191 |
192 | const userRole = role || 'student'
193 | if (userRole === 'student') {
194 |   return { success: true, redirectUrl: '/builder' }
195 | } else if (userRole === 'parent') {
196 |   return { success: true, redirectUrl: '/parent' }
197 | } else {
198 |   return { success: true, redirectUrl: `/${userRole}` }
199 | }
200 | }
201 |
202 | export async function logout() {
203 |   const supabase = await createClient()
204 |   await supabase.auth.signOut()
205 |   revalidatePath('/', 'layout')
206 |   redirect('/login')
207 | }
208 |
209 | export async function getProfile() {
210 |   const supabase = await createClient()
211 |   const { data: { user } } = await supabase.auth.getUser()
212 |
213 |   if (!user) return null;
214 |
215 |   const { data: profile } = await supabase
216 |     .from('profiles')
217 |     .select('*', schools('*'))
218 |     .eq('id', user.id)
219 |     .single()
220 |
221 |   return profile;
222 | }
223 |
224 | export async function signInWithGoogle() {
225 |   const supabase = await createClient()
226 |   const origin = process.env.NEXT_PUBLIC_SITE_URL
227 |   || process.env.VERCEL_URL
228 |   || 'http://localhost:3000'
229 |   const redirectUrl = `${origin}/auth/callback`
230 |
231 |   const { data, error } = await supabase.auth.signInWithOAuth({
232 |     provider: 'google',
233 |     options: {
234 |       redirectTo: redirectUrl,
235 |     },
236 |   })
237 |
238 |   if (error) {
239 |     console.error('Google OAuth error:', error)
240 |     return { error: error.message }
241 |   }
242 |
243 |   if (data.url) {

```

```

244 |     redirect(data.url)
245 | }
246 | }
247 |
248 | export async function completeOAuthRegistration(formData: FormData) {
249 |     const supabase = await createClient()
250 |     const { data: { user } } = await supabase.auth.getUser()
251 |
252 |     if (!user) {
253 |         return { error: 'Không tìm thấy phiên đăng nhập' }
254 |     }
255 |
256 |     const role = formData.get('role') as string || 'student'
257 |     const fullName = formData.get('full_name') as string || user.user_metadata?.full_name || user.email?.split('...
258 |     const schoolCode = formData.get('school_code') as string
259 |     const schoolName = formData.get('school_name') as string
260 |
261 |     let schoolId = null;
262 |     if (schoolCode) {
263 |         const { data: school } = await supabase
264 |             .from('schools')
265 |             .select('id')
266 |             .eq('code', schoolCode)
267 |             .single()
268 |
269 |         if (school) schoolId = school.id;
270 |     }
271 |
272 |     // Cập nhật profile cho tài khoản Google đã tạo
273 |     const { error: profileError } = await supabase.from('profiles').upsert({
274 |         id: user.id,
275 |         email: user.email,
276 |         full_name: fullName,
277 |         role: role,
278 |         school_id: schoolId,
279 |         school_name: schoolName,
280 |         updated_at: new Date().toISOString()
281 |     });
282 |
283 |     if (profileError) {
284 |         return { error: 'Lỗi khi cập nhật hồ sơ: ' + profileError.message }
285 |     }
286 |
287 |     revalidatePath('/', 'layout')
288 |
289 |     if (role === 'student') {
290 |         redirect('/builder')
291 |     } else {
292 |         redirect(`/${role}`)
293 |     }
294 | }
295 |

```

File: lib/auth/rbac.ts (84 lines, 2.4 KB)

```

=====
1 | import { createClient } from '@lib/supabase-ssr-server'
2 | import { redirect } from 'next/navigation'
3 | import React from 'react'
4 |
5 | export type UserRole = 'student' | 'teacher' | 'admin' | 'parent' | 'component_creator' | 'guest'
6 |
7 | export interface AuthUser {
8 |     id: string
9 |     email: string
10 |    role: UserRole
11 |    full_name: string
12 |    student_code?: string
13 | }
14 |
15 | // Dùng trong Server Components để lấy user hiện tại
16 | export async function getCurrentUser(): Promise<AuthUser | null> {
17 |     const supabase = await createClient()
18 |     const { data: { user } } = await supabase.auth.getUser()
19 |     if (!user) return null
20 |
21 |     const { data: profile } = await supabase
22 |         .from('profiles')
23 |         .select('id, full_name, role, student_code')
24 |         .eq('id', user.id)
25 |         .single()
26 |
27 |     if (!profile) return null
28 |
29 |     return {
30 |         id: user.id,
31 |         email: user.email,
32 |         role: profile.role as UserRole,
33 |         full_name: profile.full_name,
34 |         student_code: profile.student_code ?? undefined,
35 |     }
36 | }
37 |
38 | // Guard cho Server Components - redirect nếu không đúng role
39 | export async function requireRole(allowedRoles: UserRole[]): Promise<AuthUser> {
40 |     const user = await getCurrentUser()
41 |     if (!user) redirect('/login')
42 |     if (!allowedRoles.includes(user.role)) {
43 |         redirect(getDashboardByRole(user.role))
44 |     }
45 |     return user
46 | }
47 |
48 | // Dùng trong Server Action - throw error thay vì redirect
49 | export async function verifyRole(allowedRoles: UserRole[]): Promise<AuthUser> {
50 |     const user = await getCurrentUser()
51 |     if (!user) throw new Error('UNAUTHORIZED: Not authenticated')
52 |     if (!allowedRoles.includes(user.role)) {
53 |         throw new Error(`FORBIDDEN: Role ${user.role} cannot perform this action`)
54 |     }
55 |     return user
56 | }
57 |
58 | export function getDashboardByRole(role: UserRole): string {
59 |     const map: Record<UserRole, string> = {
60 |         admin: '/admin/dashboard',
61 |         teacher: '/teacher/dashboard',
62 |         parent: '/parent/dashboard',
63 |         student: '/student/dashboard',
64 |         component_creator: '/creator',
65 |         guest: '/login',
66 |     }
67 |     return map[role] ?? '/login'
68 | }

```

```

69 | 7 0 | // Component bẻo vậ - dùng trong JSX Server Components
71 | export async function RoleGuard({
72 |   allowedRoles,
73 |   children,
74 |   fallback = null,
75 | }: {
76 |   allowedRoles: UserRole[]
77 |   children: React.ReactNode
78 |   fallback?: React.ReactNode
79 | }) {
80 |   const user = await getCurrentUser()
81 |   if (!user || !allowedRoles.includes(user.role)) return fallback
82 |   return children
83 | }
84 |

```

File: lib/build-store.ts (204 lines, 6.8 KB)

```

1 | import { create } from 'zustand';
2 | import {
3 |   type CPUProduct,
4 |   type GPUProduct,
5 |   type MainboardProduct,
6 |   type RAMProduct,
7 |   type PSUProduct,
8 |   type CaseProduct,
9 |   type CoolerProduct,
10 | type SSDProduct,
11 | type AnyProduct,
12 | type CompatibilityIssue,
13 | type BuildScenario,
14 | } from '@types/pc-components';
15 | import {
16 |   checkCompatibility,
17 |   calculateCompatibilityScore,
18 |   calculateTotalTdp,
19 |   calculateTotalPrice,
20 |   estimatePerformance,
21 | } from './compatibility-engine';
22 |
23 | interface BuildState {
24 |   cpu: CPUProduct | null;
25 |   gpu: GPUProduct | null;
26 |   mainboard: MainboardProduct | null;
27 |   ram: RAMProduct[];
28 |   psu: PSUProduct | null;
29 |   case: CaseProduct | null;
30 |   cooler: CoolerProduct | null;
31 |   ssd: SSDProduct[];
32 |   buildName: string;
33 |   activeScenario: BuildScenario | null;
34 |   budget: number;
35 |
36 |   issues: CompatibilityIssue[];
37 |   score: number;
38 |   totalTdp: number;
39 |   totalPrice: number;
40 |   performance: { gaming: number; productivity: number; workstation: number; value: number };
41 |
42 |   setCPU: (cpu: CPUProduct | null) => void;
43 |   setGPU: (gpu: GPUProduct | null) => void;
44 |   setMainboard: (mb: MainboardProduct | null) => void;
45 |   addRAM: (ram: RAMProduct) => void;
46 |   removeRAM: (index: number) => void;
47 |   setPSU: (psu: PSUProduct | null) => void;
48 |   setCase: (c: CaseProduct | null) => void;
49 |   setCooler: (cooler: CoolerProduct | null) => void;
50 |   addSSD: (ssd: SSDProduct) => void;
51 |   removeSSD: (index: number) => void;
52 |   clearAll: () => void;
53 |   setBuildName: (name: string) => void;
54 |   setActiveScenario: (s: BuildScenario | null) => void;
55 |   setBudget: (b: number) => void;
56 |   recalculate: () => void;
57 |   loadBuild: (build: Partial<BuildState>) => void;
58 |   saveToSupabase: () => Promise<string | null>;
59 |   loadFromSupabase: (id: string) => Promise<void>;
60 | }
61 |
62 | export const useBuildStore = create<BuildState>((set, get) => ({
63 |   cpu: null,
64 |   gpu: null,
65 |   mainboard: null,
66 |   ram: [],
67 |   psu: null,
68 |   case: null,
69 |   cooler: null,
70 |   ssd: [],
71 |   buildName: 'Build m0i',
72 |   activeScenario: null,
73 |   budget: 15000000,
74 |
75 |   issues: [],
76 |   score: 0,
77 |   totalTdp: 0,
78 |   totalPrice: 0,
79 |   performance: { gaming: 0, productivity: 0, workstation: 0, value: 0 },
80 |
81 |   setCPU: (cpu) => { set({ cpu }); get().recalculate(); },
82 |   setGPU: (gpu) => { set({ gpu }); get().recalculate(); },
83 |   setMainboard: (mainboard) => { set({ mainboard }); get().recalculate(); },
84 |   addRAM: (ram) => { set((s) => ({ ram: [...s.ram, ram] })); get().recalculate(); },
85 |   removeRAM: (index) => { set((s) => ({ ram: s.ram.filter((_, i) => i !== index) })); get().recalculate(); },
86 |   setPSU: (psu) => { set({ psu }); get().recalculate(); },
87 |   setCase: (c) => { set({ case: c }); get().recalculate(); },
88 |   setCooler: (cooler) => { set({ cooler }); get().recalculate(); },
89 |   addSSD: (ssd) => { set((s) => ({ ssd: [...s.ssd, ssd] })); get().recalculate(); },
90 |   removeSSD: (index) => { set((s) => ({ ssd: s.ssd.filter((_, i) => i !== index) })); get().recalculate(); },
91 |   clearAll: () => set({
92 |     cpu: null, gpu: null, mainboard: null, ram: [], psu: null,
93 |     case: null, cooler: null, ssd: [], issues: [], score: 0,
94 |     totalTdp: 0, totalPrice: 0, performance: { gaming: 0, productivity: 0, workstation: 0, value: 0 }
95 |   }),
96 |   setBuildName: (buildName) => set({ buildName }),
97 |   setActiveScenario: (activeScenario) => set({ activeScenario }),
98 |   setBudget: (budget) => set({ budget }),
99 |
100 |   recalculate: () => {
101 |     const state = get();
102 |     const allComponents: AnyProduct[] = [
103 |       state.cpu, state.gpu, state.mainboard, state.psu, state.case, state.cooler,
104 |       ...state.ram, ...state.ssd

```

```

105 | ].filter(Boolean) as AnyProduct[];
106 |
107 | const issues = checkCompatibility({
108 |   cpu: state.cpu, gpu: state.gpu, mainboard: state.mainboard,
109 |   ram: state.ram, psu: state.psu, case: state.case,
110 |   cooler: state.cooler, ssd: state.ssd,
111 | });
112 |
113 | const score = calculateCompatibilityScore(issues);
114 | const totalTdp = calculateTotalTdp({ cpu: state.cpu, gpu: state.gpu });
115 | const totalPrice = calculateTotalPrice(allComponents);
116 | const performance = estimatePerformance({ cpu: state.cpu, gpu: state.gpu, ram: state.ram });
117 |
118 | set({ issues, score, totalTdp, totalPrice, performance });
119 | },
120 |
121 | loadBuild: (build) => {
122 |   set({
123 |     cpu: build.cpu ?? null,
124 |     gpu: build.gpu ?? null,
125 |     mainboard: build.mainboard ?? null,
126 |     ram: build.ram ?? [],
127 |     psu: build.psu ?? null,
128 |     case: build.case ?? null,
129 |     cooler: build.cooler ?? null,
130 |     ssd: build.ssd ?? [],
131 |     buildName: build.buildName ?? 'Build moi',
132 |   });
133 |   get().recalculate();
134 | },
135 |
136 | saveToSupabase: async () => {
137 |   const state = get();
138 |   try {
139 |     const { createClient } = await import('@lib/supabase-client');
140 |     const supabase = createClient();
141 |     const { data: { user } } = await supabase.auth.getUser();
142 |     if (!user) return null;
143 |
144 |     const buildData = {
145 |       user_id: user.id,
146 |       name: state.buildName,
147 |       cpu_id: state.cpu?.id || null,
148 |       gpu_id: state.gpu?.id || null,
149 |       mainboard_id: state.mainboard?.id || null,
150 |       ram_ids: state.ram.map(r => r.id),
151 |       psu_id: state.psu?.id || null,
152 |       case_id: state.case?.id || null,
153 |       cooler_id: state.cooler?.id || null,
154 |       ssd_ids: state.ssd.map(s => s.id),
155 |       compatibility_score: state.score,
156 |       total_tdp: state.totalTdp,
157 |       total_price: state.totalPrice,
158 |     };
159 |
160 |     const { data, error } = await supabase
161 |       .from('saved_builds')
162 |       .upsert(buildData)
163 |       .select('id')
164 |       .single();
165 |
166 |     if (error) throw error;
167 |     return data?.id || null;
168 |   } catch (err) {
169 |     console.error('Failed to save build:', err);
170 |     return null;
171 |   }
172 | },
173 |
174 | loadFromSupabase: async (id: string) => {
175 |   try {
176 |     const { createClient } = await import('@lib/supabase-client');
177 |     const supabase = createClient();
178 |     const { data, error } = await supabase
179 |       .from('saved_builds')
180 |       .select('*', cpu:products!cpu_id(*), gpu:products!gpu_id(*), mainboard:products!mainboard_id(*), ram:pr...
181 |       .eq('id', id)
182 |       .single();
183 |
184 |     if (error) throw error;
185 |     if (!data) return;
186 |
187 |     set({
188 |       buildName: data.name,
189 |       cpu: data.cpu as CPUProduct,
190 |       gpu: data.gpu as GPUProduct,
191 |       mainboard: data.mainboard as MainboardProduct,
192 |       ram: (data.ram || []) as RAMProduct[],
193 |       psu: data.psu as PSUPProduct,
194 |       case: data.case as CaseProduct,
195 |       cooler: data.cooler as CoolerProduct,
196 |       ssd: (data.ssd || []) as SSDProduct[],
197 |     });
198 |     get().recalculate();
199 |   } catch (err) {
200 |     console.error('Failed to load build:', err);
201 |   }
202 | },
203 | }));
204 |

```

File: lib/certificates.ts (235 lines, 7.2 KB)

```

1 | 'use server'
2 |
3 | import { createClient } from '@lib/supabase-ssr-server'
4 | import { PDFDocument, StandardFonts, rgb } from 'pdf-lib'
5 | import QRCode from 'qrcode'
6 |
7 | export async function issueCertificate(userId: string, pathId: string) {
8 |   const supabase = await createClient()
9 |
10 |   // 1. Check if certificate already exists
11 |   const { data: existing } = await supabase
12 |     .from('certificates')
13 |     .select('id, pdf_url')
14 |     .eq('student_id', userId)
15 |     .eq('path_id', pathId)
16 |     .maybeSingle()
17 |
18 |   if (existing) return existing
19 |
20 |   // 2. Validate all lessons completed

```



```

21 | const { data: pathItems } = await supabase
22 |   .from('path_items')
23 |   .select('id, item_id, item_type')
24 |   .eq('path_id', pathId)
25 |
26 | const lessons = (pathItems || []).filter(i => i.item_type === 'lesson')
27 | const { data: progress } = await supabase
28 |   .from('lesson_progress')
29 |   .select('lesson_id')
30 |   .eq('student_id', userId)
31 |   .eq('status', 'completed')
32 |
33 | const completedLessonIds = (progress || []).map(p => p.lesson_id)
34 | const allCompleted = lessons.every(l => completedLessonIds.includes(l.item_id))
35 |
36 | if (!allCompleted) {
37 |   throw new Error('Bạn chưa hoàn thành tất cả bài học cần thi<
38 | }
39 |
40 | // 3. Snapshot student and course information
41 | const { data: profile } = await supabase
42 |   .from('profiles')
43 |   .select('full_name')
44 |   .eq('id', userId)
45 |   .single()
46 |
47 | const { data: path } = await supabase
48 |   .from('learning_paths')
49 |   .select('title')
50 |   .eq('id', pathId)
51 |   .single()
52 |
53 | // Calculate final score based on student's quizzes in this path
54 | const quizzes = (pathItems || []).filter(i => i.item_type === 'quiz')
55 | let totalScore = 0
56 | if (quizzes.length > 0) {
57 |   const quizIds = quizzes.map(q => q.item_id)
58 |   const { data: attempts } = await supabase
59 |     .from('quiz_attempts')
60 |     .select('score')
61 |     .eq('student_id', userId)
62 |     .eq('status', 'submitted')
63 |     .in('quiz_id', quizIds)
64 |
65 |   const sum = (attempts || []).reduce((acc, a) => acc + (a.score || 0), 0)
66 |   totalScore = sum / (quizzes.length || 1)
67 | } else {
68 |   totalScore = 100
69 | }
70 |
71 | // Generate certificate number
72 | const chars = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789'
73 | let randomPart = ''
74 | for (let i = 0; i < 5; i++) {
75 |   randomPart += chars.charAt(Math.floor(Math.random() * chars.length))
76 | }
77 | const certNumber = `PCM-2026-${randomPart}`
78 | const verifyUrl = `${process.env.NEXT_PUBLIC_SITE_URL || 'https://localhost:3000'}/verify/${certNumber}`
79 |
80 | // Create certificate record
81 | const { data: cert, error: insertErr } = await supabase
82 |   .from('certificates')
83 |   .insert({
84 |     student_id: userId,
85 |     path_id: pathId,
86 |     certificate_number: certNumber,
87 |     student_name: profile?.full_name || 'H<
88 |     course_title: path?.title || 'L<
89 |     final_score: totalScore,
90 |     verify_url: verifyUrl,
91 |     completion_date: new Date().toISOString().split('T')[0]
92 |   })
93 |   .select()
94 |   .single()
95 |
96 | if (insertErr || !cert) throw insertErr
97 |
98 | // 4. Node.js Direct PDF Generation Fallback (Ensures it works immediately!)
99 | try {
100 |   const pdfDoc = await PDFDocument.create()
101 |   const page = pdfDoc.addPage([842, 595]) // A4 Landscape
102 |
103 |   const fontBold = await pdfDoc.embedFont(StandardFonts.HelveticaBold)
104 |   const fontRegular = await pdfDoc.embedFont(StandardFonts.Helvetica)
105 |
106 |   // Background
107 |   page.drawRectangle({
108 |     x: 0, y: 0, width: 842, height: 595,
109 |     color: rgb(0.05, 0.06, 0.075) // #0d0e13
110 |   })
111 |
112 |   // Glowing border
113 |   page.drawRectangle({
114 |     x: 20, y: 20, width: 802, height: 555,
115 |     borderColor: rgb(0, 0.83, 0.67), // #00d4aa
116 |     borderWidth: 2
117 |   })
118 |
119 |   // Content text drawing
120 |   page.drawText('CHUNG NHAN HOAN THANH', {
121 |     x: 250, y: 460, size: 28, font: fontBold,
122 |     color: rgb(0, 0.83, 0.67)
123 |   })
124 |
125 |   page.drawText('Chung chi nay duoc trao cho', {
126 |     x: 330, y: 400, size: 14, font: fontRegular,
127 |     color: rgb(0.6, 0.6, 0.6)
128 |   })
129 |
130 |   page.drawText(cert.student_name, {
131 |     x: 260, y: 340, size: 36, font: fontBold,
132 |     color: rgb(0.87, 0.88, 0.93)
133 |   })
134 |
135 |   page.drawText('Vi da hoan thanh xuat sac khoa hoc', {
136 |     x: 320, y: 290, size: 12, font: fontRegular,
137 |     color: rgb(0.6, 0.6, 0.6)
138 |   })
139 |
140 |   page.drawText(cert.course_title, {
141 |     x: 260, y: 240, size: 20, font: fontBold,
142 |     color: rgb(0, 0.83, 0.67)
143 |   })
144 |

```

```

145 | page.drawText(`Diem so: ${cert.final_score}% | Ngay cap: ${cert.completion_date}`, {
146 |   x: 280, y: 180, size: 11, font: fontRegular,
147 |   color: rgb(0.5, 0.5, 0.5)
148 | })
149 |
150 | page.drawText(`Ma so chung chi: ${cert.certificate_number}`, {
151 |   x: 300, y: 140, size: 11, font: fontRegular,
152 |   color: rgb(0.5, 0.5, 0.5)
153 | })
154 |
155 | // Generate QR Code verify url
156 | const qrDataUrl = await QRCode.toDataURL(verifyUrl)
157 | const qrImageBytes = Uint8Array.from(atob(qrDataUrl.split(',')[1]), c => c.charCodeAt(0))
158 | const qrImage = await pdfDoc.embedPng(qrImageBytes)
159 |
160 | page.drawImage(qrImage, {
161 |   x: 690, y: 40, width: 90, height: 90
162 | })
163 |
164 | const pdfBytes = await pdfDoc.save()
165 |
166 | // Upload to Supabase Storage Bucket 'certificates'
167 | const { error: uploadErr } = await supabase.storage
168 |   .from('certificates')
169 |   .upload(`${cert.id}.pdf`, pdfBytes, {
170 |     contentType: 'application/pdf',
171 |     upsert: true
172 |   })
173 |
174 | if (uploadErr) throw uploadErr
175 |
176 | const publicUrl = supabase.storage
177 |   .from('certificates').getPublicUrl(`${cert.id}.pdf`).data.publicUrl
178 |
179 | // Update pdf_url in DB
180 | await supabase
181 |   .from('certificates')
182 |   .update({ pdf_url: publicUrl })
183 |   .eq('id', cert.id)
184 |
185 | return { id: cert.id, pdf_url: publicUrl }
186 | } catch (err) {
187 |   console.error("PDF Generation error:", err)
188 |   return cert
189 | }
190 | }
191 |
192 | export async function issueCustomCertificate(data: {
193 |   studentId: string;
194 |   courseTitle: string;
195 |   finalScore: number;
196 |   stickerUrl: string;
197 |   completionDate: string;
198 | }) {
199 |   const supabase = await createClient()
200 |
201 |   // Generate certificate number
202 |   const chars = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789'
203 |   let randomPart = ''
204 |   for (let i = 0; i < 5; i++) {
205 |     randomPart += chars.charAt(Math.floor(Math.random() * chars.length))
206 |   }
207 |   const certNumber = `PCM-2026-${randomPart}`
208 |   const verifyUrl = `${process.env.NEXT_PUBLIC_SITE_URL || 'https://localhost:3000'}/verify/${certNumber}`
209 |
210 |   // Get student profile info
211 |   const { data: profile } = await supabase
212 |     .from('profiles')
213 |     .select('full_name')
214 |     .eq('id', data.studentId)
215 |     .single()
216 |
217 |   const { data: cert, error: insertErr } = await supabase
218 |     .from('certificates')
219 |     .insert({
220 |       student_id: data.studentId,
221 |       certificate_number: certNumber,
222 |       student_name: profile?.full_name || 'Híc viên PC Master',
223 |       course_title: data.courseTitle,
224 |       final_score: data.finalScore,
225 |       pdf_url: data.stickerUrl, // Lưu ảnh dán/sticker vào tr°Ýng pdf_url
226 |       verify_url: verifyUrl,
227 |       completion_date: data.completionDate || new Date().toISOString().split('T')[0]
228 |     })
229 |     .select()
230 |     .single()
231 |
232 |   if (insertErr || !cert) throw insertErr
233 |   return cert
234 | }
235 |

```

File: lib/compatibility-engine.ts (325 lines, 12.8 KB)

```

1 | import {
2 |   type AnyProduct,
3 |   type CPUProduct,
4 |   type GPUProduct,
5 |   type MainboardProduct,
6 |   type PSUProduct,
7 |   type CaseProduct,
8 |   type CoolerProduct,
9 |   type RAMProduct,
10 |   type SSDProduct,
11 |   type Build,
12 |   type CompatibilityIssue,
13 |   type CompatibilityIssueType,
14 |   type Severity,
15 | } from '@types/pc-components';
16 |
17 | function issue(
18 |   severity: Severity,
19 |   check_type: CompatibilityIssueType,
20 |   message: string,
21 |   fix: string
22 | ): CompatibilityIssue {
23 |   return { severity, check_type, message, fix };
24 | }
25 |
26 | export function checkCompatibility(build: {
27 |   cpu: CPUProduct | null;
28 |   gpu: GPUProduct | null;
29 |   mainboard: MainboardProduct | null;

```

```

30 | ram: RAMProduct[];
31 | psu: PSUProduct | null;
32 | case: CaseProduct | null;
33 | cooler: CoolerProduct | null;
34 | ssd: SSDProduct[];
35 | }): CompatibilityIssue[] {
36 | const issues: CompatibilityIssue[] = [];
37 | const { cpu, gpu, mainboard, ram, psu, case: case_, cooler, ssd } = build;
38 |
39 | if (!cpu) return issues;
40 | if (!mainboard) return issues;
41 |
42 | // 1. Socket check
43 | if (cpu.cpu_socket !== mainboard.mb_socket) {
44 |   issues.push(issue(
45 |     'error', 'socket_mismatch',
46 |     `CPU socket ${cpu.cpu_socket} không tương thích với Mainboard socket`,
47 |     `Chín Mainboard ${cpu.cpu_socket} hỗ trợ CPU ${mainboard.mb_socket}`
48 |   ));
49 | }
50 |
51 | // 2. RAM type CPU vs Mainboard
52 | if (cpu.cpu_ram_type !== mainboard.mb_ram_type) {
53 |   issues.push(issue(
54 |     'error', 'ram_type_mismatch',
55 |     `CPU hỗ trợ ${cpu.cpu_ram_type} nhưng Mainboard hỗ trợ ${mainboard.mb_ram_type}`,
56 |     `Chín CPU ${mainboard.mb_ram_type} hỗ trợ Mainboard ${cpu.cpu_ram_type}`
57 |   ));
58 | }
59 |
60 | // 3. RAM quantity vs slots
61 | if (ram.length > mainboard.mb_ram_slots) {
62 |   issues.push(issue(
63 |     'error', 'too_many_ram',
64 |     `Có ${ram.length} thanh RAM nhưng Mainboard chỉ có ${mainboard.mb_ram_slots} khe cắm`,
65 |     `Giới hạn ${mainboard.mb_ram_slots} thanh RAM hỗ trợ Mainboard`
66 |   ));
67 | }
68 |
69 | // 4. RAM type match
70 | if (ram.length > 0 && mainboard) {
71 |   const mismatchedRam = ram.some(r => r.ram_ddr_type !== mainboard.mb_ram_type);
72 |   if (mismatchedRam) {
73 |     issues.push(issue(
74 |       'error', 'ram_type_mismatch',
75 |       `Loại RAM không khớp với Mainboard (yêu cầu ${mainboard.mb_ram_type}`,
76 |       `Chín RAM ${mainboard.mb_ram_type}`
77 |     ));
78 |   }
79 | }
80 |
81 | // 5. GPU checks
82 | if (gpu) {
83 |   let totalTdp = (cpu.cpu_tdp_watts || 0) + (gpu.gpu_tdp_watts || 0);
84 |
85 |   // GPU length vs case
86 |   if (case_ && gpu.gpu_length_mm > case_.case_max_gpu_length_mm) {
87 |     issues.push(issue(
88 |       'error', 'gpu_too_long',
89 |       `GPU ${gpu.full_name} (dài ${gpu.gpu_length_mm}mm) quá dài cho case`,
90 |       `Chín GPU ngắn hơn ${case_.case_max_gpu_length_mm}mm hỗ trợ case`
91 |     ));
92 |   }
93 |
94 |   // PSU check
95 |   if (psu) {
96 |     const safeWattage = psu.psu_wattage * 0.8;
97 |     if (totalTdp > safeWattage) {
98 |       issues.push(issue(
99 |         'error', 'psu_insufficient',
100 |         `Tổng TDP ${totalTdp}W vượt quá 80% công suất PSU (${safeWattage}W`,
101 |         `Năng suất lên ít nhất ${Math.ceil(totalTdp * 1.25)}W`
102 |       ));
103 |     }
104 |
105 |     if (gpu.gpu_recommended_psu > psu.psu_wattage) {
106 |       issues.push(issue(
107 |         'warning', 'psu_insufficient',
108 |         `GPU ${gpu.full_name} khuyến nghị PSU tối thiểu ${gpu.gpu_recommended_psu}W`,
109 |         `Năng suất lên ${gpu.gpu_recommended_psu}W`
110 |       ));
111 |     }
112 |
113 |     // GPU power connectors vs PSU
114 |     for (const conn of gpu.gpu_power_connectors) {
115 |       const psuMatch = psu.psu_connectors.find(
116 |         c => c.type.toLowerCase().includes(conn.type.toLowerCase().replace('pin', '').trim())
117 |       );
118 |       if (!psuMatch || psuMatch.qty < conn.qty) {
119 |         issues.push(issue(
120 |           'warning', 'psu_insufficient',
121 |           `GPU cần ${conn.qty} x ${conn.type} nhưng PSU không có`,
122 |           `Chín PSU có cổng nguồn cho GPU`
123 |         ));
124 |       }
125 |     }
126 |   }
127 |
128 |   // No iGPU + No display
129 |   if (!cpu.cpu_has_igpu && !gpu) {
130 |     issues.push(issue(
131 |       'warning', 'no_display_output',
132 |       `CPU ${cpu.full_name} không có iGPU và chưa chèn GPU rời - sẽ không hoạt động`,
133 |       `Thêm GPU rời hỗ trợ chèn CPU có iGPU (non-F)`
134 |     ));
135 |   }
136 |
137 |   // Bottleneck estimation
138 |   if (gpu.gpu_benchmark_1080p && cpu.cpu_cinebench_r23_multi) {
139 |     const gpuPower = gpu.gpu_benchmark_1080p;
140 |     const cpuPower = cpu.cpu_cinebench_r23_multi;
141 |     if (cpuPower > 0 && gpuPower > 0) {
142 |       const ratio = gpuPower / cpuPower;
143 |       if (ratio > 2.5) {
144 |         issues.push(issue(
145 |           'warning', 'bottleneck',
146 |           `GPU minh họa CPU nặng hơn (tỷ lệ ${ratio.toFixed(1)}x) - CPU là yếu tố hạn chế`,
147 |           `Cân nhắc CPU minh họa GPU thấp hơn`
148 |         ));
149 |       } else if (ratio < 0.4) {
150 |         issues.push(issue(
151 |           'warning', 'bottleneck',
152 |           `CPU quá mạnh so với GPU (tỷ lệ ${ratio.toFixed(1)}x) - GPU là yếu tố hạn chế`,
153 |           `Cân nhắc GPU minh họa CPU thấp hơn`

```

```

154 |         ));
155 |     }
156 | }
157 | }
158 | } else if (!cpu.cpu_has_igpu) {
159 |     issues.push(issue(
160 |         'warning', 'no_display_output',
161 |         'CPU ${cpu.full_name} không có iGPU và ch°a ch°n GPU r°i',
162 |         'Th°m GPU r°i ho·c ch°n CPU có iGPU (non-F)'
163 |     ));
164 | }
165 |
166 | // 6. Cooler checks
167 | if (cooler) {
168 |     if (cooler.cooler_max_tdp_support < (cpu.cpu_tdp_watts || 0) * 0.9) {
169 |         issues.push(issue(
170 |             'warning', 'cooler_insufficient',
171 |             'CPU TDP ${cpu.cpu_tdp_watts}W g°n chim gi°i hin t°n nhi°t c°a C
172 |             'C°n nh°c n°ng c°p coo°ler (AIO ho·c khi hi°u n°ng cao)'
173 |         ));
174 |     }
175 |
176 |     if (case_ && cooler.cooler_height_mm > case_.case_max_cooler_height_mm) {
177 |         issues.push(issue(
178 |             'error', 'cooler_too_tall',
179 |             'Cooler cao ${cooler.cooler_height_mm}mm quá cao cho Case (ch° h
180 |             'Ch°n cooler th°p hin ${case_.case_max_cooler_height_mm}mm ho·c
181 |         ));
182 |     }
183 |
184 |     if (!cooler.cooler_supported_sockets.includes(cpu.cpu_socket)) {
185 |         issues.push(issue(
186 |             'error', 'cooler_socket_mismatch',
187 |             'Cooler không h× tr° socket ${cpu.cpu_socket}',
188 |             'Ch°n cooler h× tr° socket ${cpu.cpu_socket}'
189 |         ));
190 |     }
191 | }
192 |
193 | // 7. Form factor check
194 | if (case_ && mainboard) {
195 |     if (!case_.case_supported_forms.includes(mainboard.mb_form_factor)) {
196 |         issues.push(issue(
197 |             'error', 'form_factor_mismatch',
198 |             'Mainboard ${mainboard.mb_form_factor} không v°a Case (h× tr°: S
199 |             'Ch°n Mainboard ${case_.case_supported_forms[0]} || 'ATX' ho·c
200 |         ));
201 |     }
202 | }
203 |
204 | // 8. SSD M.2 slot count
205 | const m2Ssds = ssd.filter(s => s.ssd_interface === 'm2_nvme' || s.ssd_interface === 'm2_sata');
206 | if (m2Ssds.length > (mainboard.mb_m2_slots || 0)) {
207 |     issues.push(issue(
208 |         'warning', 'not_enough_m2_slots',
209 |         'C° ${m2Ssds.length} SSD M.2 nh°ng Mainboard ch° có ${mainboard.m
210 |         'Dùng SSD SATA cho các ổ còn l°i ho·c n°ng c°p Mainboard'
211 |     ));
212 | }
213 |
214 | // 9. PSU length vs case
215 | if (psu && case_ && psu.psu_length_mm > case_.case_max_psu_length_mm) {
216 |     issues.push(issue(
217 |         'error', 'psu_insufficient',
218 |         'PSU dài ${psu.psu_length_mm}mm quá dài cho Case (h× tr° t°i a $
219 |         'Ch°n PSU ng°n hin ${case_.case_max_psu_length_mm}mm'
220 |     ));
221 | }
222 |
223 | return issues;
224 | }
225 |
226 | export function calculateCompatibilityScore(issues: CompatibilityIssue[]): number {
227 |     if (issues.length === 0) return 100;
228 |
229 |     let score = 100;
230 |     for (const issue of issues) {
231 |         if (issue.severity === 'error') score -= 25;
232 |         else if (issue.severity === 'warning') score -= 10;
233 |         else score -= 5;
234 |     }
235 |     return Math.max(0, score);
236 | }
237 |
238 | export function calculateTotalTdp(build: {
239 |     cpu: CPUProduct | null;
240 |     gpu: GPUProduct | null;
241 | }): number {
242 |     let tdp = 0;
243 |     if (build.cpu) tdp += build.cpu.cpu_tdp_watts;
244 |     if (build.gpu) tdp += build.gpu.gpu_tdp_watts;
245 |     return tdp;
246 | }
247 |
248 | export function calculateTotalPrice(components: AnyProduct[]): number {
249 |     return components.reduce((sum, c) => sum + (c.price_vnd || 0), 0);
250 | }
251 |
252 | export function getPriorityIssues(issues: CompatibilityIssue[]): {
253 |     errors: CompatibilityIssue[];
254 |     warnings: CompatibilityIssue[];
255 |     info: CompatibilityIssue[];
256 | } {
257 |     return {
258 |         errors: issues.filter(i => i.severity === 'error'),
259 |         warnings: issues.filter(i => i.severity === 'warning'),
260 |         info: issues.filter(i => i.severity === 'info'),
261 |     };
262 | }
263 |
264 | export function estimatePerformance(build: {
265 |     cpu: CPUProduct | null;
266 |     gpu: GPUProduct | null;
267 |     ram: RAMProduct[];
268 | }): {
269 |     gaming: number;
270 |     productivity: number;
271 |     workstation: number;
272 |     value: number;
273 | } {
274 |     const cpuPower = build.cpu?.cpu_cinebench_r23_multi || 0;
275 |     const gpuPower = build.gpu?.gpu_benchmark_1080p || 0;
276 |     const ramSpeed = build.ram.reduce((s, r) => s + (r.ram_speed_mhz || 0), 0) / Math.max(build.ram.length, 1);
277 |     const hasGpu = !!build.gpu;

```

```

278 | const totalPrice = calculateTotalPrice([
279 |   build.cpu, build.gpu, ...build.ram
280 | ].filter(Boolean) as AnyProduct[]);
281 |
282 | const gaming = hasGpu
283 |   ? Math.round((gpuPower * 0.7 + cpuPower * 0.3) / 100)
284 |   : Math.round((cpuPower * 0.3) / 100);
285 |
286 | const productivity = Math.round((cpuPower * 0.6 + ramSpeed * 0.4) / 100);
287 | const workstation = Math.round((cpuPower * 0.5 + gpuPower * 0.3 + ramSpeed * 0.2) / 100);
288 | const value = totalPrice > 0
289 |   ? Math.round((gaming + productivity + workstation) / 3) * 10000000 / totalPrice * 100
290 |   : 0;
291 |
292 | return {
293 |   gaming: Math.min(100, Math.max(0, gaming)),
294 |   productivity: Math.min(100, Math.max(0, productivity)),
295 |   workstation: Math.min(100, Math.max(0, workstation)),
296 |   value: Math.min(100, Math.max(0, value)),
297 | };
298 | }
299 |
300 | export function getComponentRecommendations(
301 |   budget: number,
302 |   useCase: 'gaming' | 'productivity' | 'workstation' | 'budget'
303 | ): Partial<Build> {
304 |   const recommendations: Record<string, Partial<Build>> = {
305 |     budget: {
306 |       cpu: { full_name: 'Intel Core i3-12100F', cpu_socket: 'LGA1700', cpu_cores: 4, cpu_threads: 8, cpu_base...
307 |       gpu: { full_name: 'NVIDIA GTX 1650', gpu_vram_gb: 4, gpu_tdp_watts: 75, gpu_length_mm: 170, gpu_benchmar...
308 |     },
309 |     gaming: {
310 |       cpu: { full_name: 'AMD Ryzen 5 7600', cpu_socket: 'AM5', cpu_cores: 6, cpu_threads: 12, cpu_base_ghz: 3....
311 |       gpu: { full_name: 'NVIDIA RTX 4060', gpu_vram_gb: 8, gpu_tdp_watts: 115, gpu_length_mm: 240, gpu_benchma...
312 |     },
313 |     productivity: {
314 |       cpu: { full_name: 'Intel Core i7-13700K', cpu_socket: 'LGA1700', cpu_cores: 16, cpu_threads: 24, cpu_bas...
315 |       gpu: { full_name: 'NVIDIA RTX 4060 Ti', gpu_vram_gb: 16, gpu_tdp_watts: 160, gpu_length_mm: 268, gpu_ben...
316 |     },
317 |     workstation: {
318 |       cpu: { full_name: 'AMD Ryzen 9 7950X', cpu_socket: 'AM5', cpu_cores: 16, cpu_threads: 32, cpu_base_ghz: ...
319 |       gpu: { full_name: 'NVIDIA RTX 4080 Super', gpu_vram_gb: 16, gpu_tdp_watts: 320, gpu_length_mm: 310, gpu_...
320 |     },
321 |   };
322 |
323 |   return recommendations[useCase] || recommendations.budget;
324 | }
325 |

```

File: lib/exam-actions.ts (85 lines, 2.9 KB)

```

1 | 'use server'
2 |
3 | import { createClient } from '@lib/supabase-ssr-server'
4 | import { redirect } from 'next/navigation'
5 | import { revalidatePath } from 'next/cache'
6 |
7 | export async function startExamAttempt(examId: string) {
8 |   const supabase = await createClient()
9 |   const { data: { user } } = await supabase.auth.getUser()
10 |   if (!user) return { error: 'Unauthorized' }
11 |
12 |   const { data, error } = await supabase
13 |     .from('exam_attempts')
14 |     .insert({
15 |       user_id: user.id,
16 |       exam_id: examId,
17 |       status: 'in_progress',
18 |       started_at: new Date().toISOString()
19 |     })
20 |     .select()
21 |     .single()
22 |
23 |   if (error) return { error: error.message }
24 |   return { attemptId: data.id }
25 | }
26 |
27 | export async function submitExamAttempt(attemptId: string, answers: any, score: number, timeSpent: number) {
28 |   const supabase = await createClient()
29 |   const { data: { user } } = await supabase.auth.getUser()
30 |   if (!user) return { error: 'Unauthorized' }
31 |
32 |   // 1. Save answers and calculate final score
33 |   const { error } = await supabase
34 |     .from('exam_attempts')
35 |     .update({
36 |       score: score,
37 |       status: score >= 50 ? 'passed' : 'failed',
38 |       completed_at: new Date().toISOString(),
39 |       time_spent: timeSpent,
40 |       // answers: answers // Assuming there's a JSON column for this
41 |     })
42 |     .eq('id', attemptId)
43 |
44 |   if (error) return { error: error.message }
45 |
46 |   // 2. Add XP to profile
47 |   const { data: profile } = await supabase.from('profiles').select('xp, level').eq('id', user.id).single()
48 |   const currentXp = (profile?.xp || 0) + score
49 |   const now = new Date().toISOString()
50 |
51 |   const { data: levelDefs } = await supabase
52 |     .from('level_definitions')
53 |     .select('level')
54 |     .lte('xp_required', currentXp)
55 |     .order('xp_required', { ascending: false })
56 |     .limit(1)
57 |   const newLevel = (levelDefs?.[0] as any)?.level || 1
58 |
59 |   await supabase.from('profiles').update({ xp: currentXp, level: newLevel, updated_at: now }).eq('id', user.id)
60 |
61 |   await supabase.from('leaderboard_cache').upsert({
62 |     user_id: user.id, total_xp: currentXp, level: newLevel, updated_at: now,
63 |   }, { onConflict: 'user_id' })
64 |
65 |   await supabase.from('learning_stats').update({ total_xp: currentXp, updated_at: now }).eq('user_id', user.id)
66 |
67 |   await supabase.from('xp_transactions').insert({
68 |     user_id: user.id, amount: score, reason: 'Hoàn thành bài thi', reference_type: 'exam_completed', reference...
69 |   })
70 |
71 |   if (newLevel > (profile?.level || 1)) {
72 |     const { data: levelDef } = await supabase.from('level_definitions').select('title, icon').eq('level', newL...

```

```

73|     if (levelDef) {
74|       await supabase.from('notifications').insert({
75|         user_id: user.id, type: 'success', title: 'Lên CẤP!',
76|         body: 'Chúc mừng! Bạn đã đạt cấp độ mới!',
77|         data: { level: newLevel, icon: (levelDef as any).icon },
78|       })
79|     }
80|   }
81| }
82| revalidatePath('/student')
83| return { success: true }
84| }
85|

```

File: lib/grading/auto-grade.ts (107 lines, 3.4 KB)

```

1| export interface PCConfig {
2|   components: string[];
3|   total_price: number;
4|   total_tdp: number;
5|   ram_gb: number;
6|   cpu?: string;
7|   gpu?: string;
8| }
9|
10| export interface AssignmentRequirements {
11|   budget_max?: number;
12|   tdp_max?: number;
13|   min_ram_gb?: number;
14|   required_components?: string[];
15|   performance_min?: number;
16| }
17|
18| export interface GradingResult {
19|   score: number;
20|   breakdown: {
21|     criterion: string;
22|     passed: boolean;
23|     points: number;
24|     feedback: string;
25|   }[];
26| }
27|
28| export function autoGrade(submission: PCConfig, requirements: AssignmentRequirements, maxScore: number = 100):...
29|   const breakdown = [];
30|   let score = 0;
31|
32|   // 1. Budget check (30 points)
33|   if (requirements.budget_max) {
34|     const passed = submission.total_price <= requirements.budget_max;
35|     const points = passed ? 30 : 0;
36|     score += points;
37|     breakdown.push({
38|       criterion: 'Ngân sách',
39|       passed,
40|       points,
41|       feedback: passed
42|         ? 'Đạt yêu cầu: Tổng giá trị trong ngân sách (Cụ thể: ${submission.total_price.toLocaleString('vi-VN')} VNĐ <= ${requirements.budget_max.toLocaleString('vi-VN')} VNĐ)'
43|         : `Chưa đạt yêu cầu: Tổng giá trị vượt ngân sách ${Math.abs(submission.total_price - requirements.budget_max).toLocaleString('vi-VN')} VNĐ`;
44|     });
45|   }
46|
47|   // 2. TDP check (20 points)
48|   if (requirements.tdp_max) {
49|     const passed = submission.total_tdp <= requirements.tdp_max;
50|     const points = passed ? 20 : 0;
51|     score += points;
52|     breakdown.push({
53|       criterion: 'Tiêu thụ điện (TDP)',
54|       passed,
55|       points,
56|       feedback: passed
57|         ? `Đạt yêu cầu: TDP hấp thụ (${submission.total_tdp}W) <= ${requirements.tdp_max}W`
58|         : `Chưa đạt yêu cầu: TDP quá cao (${submission.total_tdp}W > ${requirements.tdp_max}W, yêu cầu <= ${requirements.tdp_max}W)`;
59|     });
60|   }
61|
62|   // 3. RAM check (15 points)
63|   if (requirements.min_ram_gb) {
64|     const passed = submission.ram_gb >= requirements.min_ram_gb;
65|     const points = passed ? 15 : 0;
66|     score += points;
67|     breakdown.push({
68|       criterion: 'Dung lượng RAM',
69|       passed,
70|       points,
71|       feedback: passed
72|         ? `Đạt yêu cầu: RAM lắp đặt (${submission.ram_gb}GB) >= ${requirements.min_ram_gb}GB`
73|         : `Chưa đạt yêu cầu: Thiếu RAM (${submission.ram_gb}GB < ${requirements.min_ram_gb}GB, yêu cầu >= ${requirements.min_ram_gb}GB)`;
74|     });
75|   }
76|
77|   // 4. Required components check (35 points)
78|   if (requirements.required_components && requirements.required_components.length > 0) {
79|     const missing = requirements.required_components.filter(c => !submission.components.some(sc => sc.includes(c)));
80|     const passed = missing.length === 0;
81|     const points = passed ? 35 : 0;
82|     score += points;
83|     breakdown.push({
84|       criterion: 'Linh kiện bắt buộc',
85|       passed,
86|       points,
87|       feedback: passed
88|         ? `Đạt yêu cầu: Có đủ các linh kiện yêu cầu`
89|         : `Chưa đạt yêu cầu: Thiếu linh kiện (${missing.join(', ')});`;
90|     });
91|   }
92|
93|   // Normalize score to maxScore (default 100) if not all criteria are present
94|   const totalPossible = breakdown.reduce((acc, curr) => acc + (curr.points > 0 || !curr.passed ? (curr.criterion === 'Linh kiện bắt buộc' ? curr.points : curr.points * 0.5) : 0), 0);
95|   const normalizedScore = totalPossible > 0 ? Math.round((score / totalPossible) * maxScore) : maxScore;
96|
97|   return {
98|     score: normalizedScore,
99|     breakdown
100|   };
101| };
102|
103| export function formatVND(amount: number) {
104|   return new Intl.NumberFormat('vi-VN', { style: 'currency', currency: 'VND' }).format(amount);
105| }
106|
107|

```

File: lib/guru-state.tsx (64 lines, 1.9 KB)

```
1 | 'use client';
2 |
3 | import { createContext, useContext, useState, useCallback, ReactNode } from 'react';
4 |
5 | interface GuruState {
6 |   message: string;
7 |   trigger: number;
8 |   silentMessage: string;
9 |   silentTrigger: number;
10 |   lang: string;
11 |   openTrigger: number;
12 |   setMessage: (msg: string) => void;
13 |   setSilentMessage: (msg: string) => void;
14 |   setLang: (l: string) => void;
15 |   incrementTrigger: () => void;
16 |   incrementSilentTrigger: () => void;
17 |   openChat: () => void;
18 | }
19 |
20 | const GuruContext = createContext<GuruState | null>(null);
21 |
22 | export function GuruProvider({ children }: { children: ReactNode }) {
23 |   const [message, setMessageState] = useState('');
24 |   const [trigger, setTrigger] = useState(0);
25 |   const [silentMessage, setSilentMessageState] = useState('');
26 |   const [silentTrigger, setSilentTrigger] = useState(0);
27 |   const [lang, setLangState] = useState('vn');
28 |   const [openTrigger, setOpenTrigger] = useState(0);
29 |
30 |   const setMessage = useCallback((msg: string) => {
31 |     setMessageState(msg);
32 |     setTrigger(t => t + 1);
33 |   }, []);
34 |
35 |   const setSilentMessage = useCallback((msg: string) => {
36 |     setSilentMessageState(msg);
37 |     setSilentTrigger(t => t + 1);
38 |   }, []);
39 |
40 |   const setLang = useCallback((l: string) => {
41 |     setLangState(l);
42 |   }, []);
43 |
44 |   const incrementTrigger = useCallback(() => setTrigger(t => t + 1), []);
45 |   const incrementSilentTrigger = useCallback(() => setSilentTrigger(t => t + 1), []);
46 |   const openChat = useCallback(() => setOpenTrigger(t => t + 1), []);
47 |
48 |   return (
49 |     <GuruContext.Provider value={{
50 |       message, trigger, silentMessage, silentTrigger, lang, openTrigger,
51 |       setMessage, setSilentMessage, setLang,
52 |       incrementTrigger, incrementSilentTrigger, openChat
53 |     }}>
54 |       {children}
55 |     </GuruContext.Provider>
56 |   );
57 | }
58 |
59 | export function useGuru(): GuruState {
60 |   const ctx = useContext(GuruContext);
61 |   if (!ctx) throw new Error('useGuru must be used within GuruProvider');
62 |   return ctx;
63 | }
64 |
```

File: lib/learning-actions.ts (100 lines, 2.5 KB)

```
1 | 'use server'
2 |
3 | import { createServerClient } from '@supabase/ssr'
4 | import { cookies } from 'next/headers'
5 | import { redirect } from 'next/navigation'
6 |
7 |
8 | export async function completeLessonAction(lessonId: string) {
9 |   const cookieStore = await cookies()
10 |   const supabase = createServerClient(
11 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
12 |     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!,
13 |     {
14 |       cookies: {
15 |         get(name: string) {
16 |           return cookieStore.get(name)?.value
17 |         },
18 |       },
19 |     }
20 |   )
21 |
22 |   const { data: { user } } = await supabase.auth.getUser()
23 |   if (!user) throw new Error('Unauthorized')
24 |
25 |   await supabase
26 |     .from('lesson_progress')
27 |     .upsert({
28 |       student_id: user.id,
29 |       lesson_id: lessonId,
30 |       status: 'completed',
31 |       completion_percentage: 100,
32 |       completed_at: new Date().toISOString()
33 |     }, { onConflict: 'student_id,lesson_id' })
34 |
35 |   // Redirect to dashboard or next lesson
36 |   redirect('/student')
37 | }
38 |
39 | export async function startBuilderSession(lessonId?: string) {
40 |   const cookieStore = await cookies()
41 |   const supabase = createServerClient(
42 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
43 |     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!,
44 |     {
45 |       cookies: {
46 |         get(name: string) {
47 |           return cookieStore.get(name)?.value
48 |         },
49 |       },
50 |     }
51 |   )
52 | }
```

```

52 |
53 | const { data: { user } } = await supabase.auth.getUser()
54 | if (!user) return null
55 |
56 | const { data } = await supabase
57 |   .from('builder_sessions')
58 |   .insert({
59 |     student_id: user.id,
60 |     lesson_id: lessonId || null,
61 |     started_at: new Date().toISOString()
62 |   })
63 |   .select('id')
64 |   .single()
65 |
66 | return data?.id
67 | }
68 |
69 | export async function endBuilderSession(sessionId: string, data: { components_used: any, tdp_calculated: numbe...
70 | if (!sessionId) return
71 | const cookieStore = await cookies()
72 | const supabase = createServerClient(
73 |   process.env.NEXT_PUBLIC_SUPABASE_URL!,
74 |   process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!,
75 |   {
76 |     cookies: {
77 |       get(name: string) {
78 |         return cookieStore.get(name)?.value
79 |       },
80 |     },
81 |   }
82 | )
83 |
84 | const { data: { user } } = await supabase.auth.getUser()
85 |
86 | await supabase
87 |   .from('builder_sessions')
88 |   .update({
89 |     ended_at: new Date().toISOString(),
90 |     components_used: data.components_used,
91 |     tdp_calculated: data.tdp_calculated,
92 |     compatibility_score: data.compatibility_score
93 |   })
94 |   .eq('id', sessionId)
95 |
96 | if (user) {
97 |   // Session ended
98 | }
99 | }
100 |

```

File: lib/lesson-actions.ts (123 lines, 4.3 KB)

```

1 | 'use server'
2 |
3 | import { createClient } from '@lib/supabase-ssr-server'
4 | import { revalidatePath } from 'next/cache'
5 |
6 | async function addXpAndLevel(userId: string, xp: number, type?: string, refId?: string, desc?: string) {
7 |   try {
8 |     const supabase = await createClient()
9 |     const now = new Date().toISOString()
10 |    const today = now.slice(0, 10)
11 |
12 |    const { data: profile } = await supabase.from('profiles').select('xp, level').eq('id', userId).single()
13 |    const currentXp = (profile?.xp || 0) + xp
14 |
15 |    const { data: levelDefs } = await supabase
16 |      .from('level_definitions')
17 |      .select('level')
18 |      .lte('xp_required', currentXp)
19 |      .order('xp_required', { ascending: false })
20 |      .limit(1)
21 |    const newLevel = (levelDefs?.[0] as any)?.level || 1
22 |
23 |    await supabase.from('profiles').update({ xp: currentXp, level: newLevel, updated_at: now }).eq('id', userId)
24 |
25 |    await supabase.from('leaderboard_cache').upsert({
26 |      user_id: userId, total_xp: currentXp, level: newLevel, updated_at: now,
27 |    }, { onConflict: 'user_id' })
28 |
29 |    await supabase.from('learning_stats').update({ total_xp: currentXp, updated_at: now }).eq('user_id', userId)
30 |
31 |    if (type) {
32 |      await supabase.from('xp_transactions').insert({
33 |        user_id: userId, amount: xp, reason: desc || type, reference_type: type, reference_id: refId,
34 |      })
35 |    }
36 |
37 |    if (newLevel > (profile?.level || 1)) {
38 |      const { data: levelDef } = await supabase.from('level_definitions').select('*').eq('level', newLevel).si...
39 |      if (levelDef) {
40 |        await supabase.from('notifications').insert({
41 |          user_id: userId, type: 'success', title: 'Lên CẤP!',
42 |          body: `Chúc mừng! Bạn đã đạt cấp ${newLevel}: ${(levelDef as any).icon}`,
43 |          data: { level: newLevel, icon: (levelDef as any).icon },
44 |        })
45 |      }
46 |    }
47 |
48 |    return { success: true, xp: currentXp, level: newLevel }
49 |  } catch (e) {
50 |    console.error('addXpAndLevel error:', e)
51 |    return { success: false, error: String(e) }
52 |  }
53 | }
54 |
55 | export async function saveLessonProgress(lessonId: string, score: number, isCompleted: boolean) {
56 |   const supabase = await createClient()
57 |   const { data: { user } } = await supabase.auth.getUser()
58 |
59 |   if (!user) return { error: 'Unauthorized' }
60 |
61 |   const { data, error } = await supabase
62 |     .from('lesson_progress')
63 |     .upsert({
64 |       student_id: user.id,
65 |       lesson_id: lessonId,
66 |       score: score,
67 |       is_completed: isCompleted,
68 |       status: isCompleted ? 'completed' : 'in_progress',
69 |       completion_percentage: isCompleted ? 100 : 0,
70 |       last_accessed: new Date().toISOString()
71 |     }, {

```



```

72 |         onConflict: 'student_id, lesson_id'
73 |     })
74 |
75 |     if (error) {
76 |         console.error('Error saving progress:', error)
77 |         return { error: error.message }
78 |     }
79 |
80 |     if (isCompleted) {
81 |         await addXpAndLevel(user.id, score || 50, 'lesson_completed', lessonId)
82 |
83 |         const now = new Date()
84 |         const today = now.toISOString().slice(0, 10)
85 |         const { data: stats } = await supabase.from('learning_stats').select('').eq('user_id', user.id).single()
86 |         if (stats) {
87 |             const dayOfWeek = now.getDay()
88 |             const activity = stats.weekly_activity || [0, 0, 0, 0, 0, 0, 0]
89 |             activity[dayOfWeek === 0 ? 6 : dayOfWeek - 1] = (activity[dayOfWeek === 0 ? 6 : dayOfWeek - 1] || 0) + 15
90 |             await supabase.from('learning_stats').update({
91 |                 current_streak: stats.last_active_date === today ? stats.current_streak : (stats.last_active_date === ...
92 |                     last_active_date: today,
93 |                     total_study_minutes: (stats.total_study_minutes || 0) + 15,
94 |                     weekly_activity: activity,
95 |                     updated_at: now.toISOString(),
96 |                 }).eq('user_id', user.id)
97 |             } else {
98 |                 const dayOfWeek = now.getDay()
99 |                 const activity = [0, 0, 0, 0, 0, 0, 0]
100 |                 activity[dayOfWeek === 0 ? 6 : dayOfWeek - 1] = 15
101 |                 await supabase.from('learning_stats').insert({
102 |                     user_id: user.id,
103 |                     current_streak: 1,
104 |                     last_active_date: today,
105 |                     total_study_minutes: 15,
106 |                     weekly_activity: activity,
107 |                     total_xp: score || 50,
108 |                 })
109 |             }
110 |         }
111 |
112 |         revalidatePath(`/lessons/${lessonId}`)
113 |         revalidatePath('/builder')
114 |
115 |         return { success: true }
116 |     }
117 |
118 |     function getYesterday(): string {
119 |         const d = new Date()
120 |         d.setDate(d.getDate() - 1)
121 |         return d.toISOString().slice(0, 10)
122 |     }
123 |

```

File: lib/parent-actions.ts (198 lines, 6.3 KB)

```

1 | // Path: lib/parent-actions.ts
2 | 'use server'
3 | import { createServerClient } from '@supabase/ssr'
4 | import { cookies } from 'next/headers'
5 | import { revalidatePath } from 'next/cache'
6 |
7 | async function getServerSupabase() {
8 |     const url = process.env.NEXT_PUBLIC_SUPABASE_URL
9 |     const key = process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY
10 |    if (!url || !key) return null
11 |    const cookieStore = await cookies()
12 |    return createServerClient(url, key, {
13 |        cookies: {
14 |            getAll() { return cookieStore.getAll() },
15 |            setAll(cookiesToSet) {
16 |                cookiesToSet.forEach(({ name, value, options }) =>
17 |                    cookieStore.set(name, value, options))
18 |            }
19 |        }
20 |    })
21 | }
22 |
23 |
24 | // Find student by code
25 | export async function findStudentByCode(code: string) {
26 |     try {
27 |         const supabase = await getServerSupabase()
28 |         const { data: { user }, error: authErr } = await supabase.auth.getUser()
29 |         if (!user || authErr) return { error: 'Chưa đăng nhập' }
30 |
31 |         // UPPERCASE and trim whitespace
32 |         const cleanCode = code.toUpperCase().trim()
33 |
34 |         const { data, error } = await supabase
35 |             .rpc('find_student_by_code', { input_code: cleanCode })
36 |
37 |         if (error) {
38 |             console.error('Error finding student:', error)
39 |             return { error: 'Lỗi hệ thống khi tìm kiếm' }
40 |         }
41 |
42 |         if (!data || data.length === 0) {
43 |             return { error: 'Không tìm thấy học sinh với mã này' }
44 |         }
45 |
46 |         return { student: data[0] }
47 |     } catch (err: any) {
48 |         return { error: err.message || 'Lỗi không xác định' }
49 |     }
50 | }
51 |
52 | // Request linking to student
53 | export async function requestLinkToStudent(
54 |     studentId: string,
55 |     relationship: string = 'parent'
56 | ) {
57 |     try {
58 |         const supabase = await getServerSupabase()
59 |         const { data: { user }, error: authErr } = await supabase.auth.getUser()
60 |         if (!user || authErr) return { error: 'Chưa đăng nhập' }
61 |
62 |         // Verify role is parent
63 |         const { data: profile, error: profErr } = await supabase
64 |             .from('profiles')
65 |             .select('role, full_name')
66 |             .eq('id', user.id)
67 |             .single()
68 |

```

```

69 |   if (profErr || profile?.role !== 'parent') {
70 |       return { error: 'Bên không phải tài khoản phải huynh' }
71 |   }
72 |
73 |   // Verify student exists and is student
74 |   const { data: student, error: studErr } = await supabase
75 |     .from('profiles')
76 |     .select('id, role, full_name')
77 |     .eq('id', studentId)
78 |     .single()
79 |
80 |   if (studErr || !student || student.role !== 'student') {
81 |       return { error: 'Hic sinh không tồn tại hoặc không hợp lệ' }
82 |   }
83 |
84 |   // Create link request (status = pending)
85 |   const { error: linkErr } = await supabase
86 |     .from('parent_student_links')
87 |     .upsert({
88 |       parent_id: user.id,
89 |       student_id: studentId,
90 |       relationship,
91 |       status: 'pending',
92 |       linked_at: new Date().toISOString()
93 |     }, { onConflict: 'parent_id,student_id' })
94 |
95 |   if (linkErr) {
96 |       console.error('Link creation error:', linkErr)
97 |       return { error: 'Không thể gửi yêu cầu liên kết' }
98 |   }
99 |
100 |   // Create notification for student
101 |   const { error: notifErr } = await supabase
102 |     .from('notifications')
103 |     .insert({
104 |       user_id: studentId,
105 |       type: 'parent_link_request',
106 |       title: 'Phải huynh muốn theo dõi bên',
107 |       body: `${profile?.full_name} muốn liên kết tài khoản của theo dõi
108 |       data: { parent_id: user.id, relationship },
109 |       action_url: '/student/profile/privacy'
110 |     })
111 |
112 |   if (notifErr) {
113 |       console.error('Notification error:', notifErr)
114 |   }
115 |
116 |   revalidatePath('/parent/dashboard')
117 |   return { success: true }
118 | } catch (err: any) {
119 |   return { error: err.message || 'Lỗi không xác định' }
120 | }
121 |
122 |
123 | // Student confirms parent link
124 | export async function confirmParentLink(parentId: string) {
125 |   try {
126 |     const supabase = await getServerSupabase()
127 |     const { data: { user }, error: authErr } = await supabase.auth.getUser()
128 |     if (!user || authErr) return { error: 'Chưa đăng nhập' }
129 |
130 |     // Update link status to active
131 |     const { error: linkErr } = await supabase
132 |       .from('parent_student_links')
133 |       .update({
134 |         status: 'active',
135 |         confirmed_at: new Date().toISOString()
136 |       })
137 |       .eq('parent_id', parentId)
138 |       .eq('student_id', user.id)
139 |       .eq('status', 'pending')
140 |
141 |     if (linkErr) {
142 |       console.error('Confirm link error:', linkErr)
143 |       return { error: 'Không thể xác nhận liên kết' }
144 |     }
145 |
146 |     // Fetch student's name for notification
147 |     const { data: profile } = await supabase
148 |       .from('profiles')
149 |       .select('full_name')
150 |       .eq('id', user.id)
151 |       .single()
152 |
153 |     // Send confirmation notification to parent
154 |     await supabase.from('notifications').insert({
155 |       user_id: parentId,
156 |       type: 'parent_link_confirmed',
157 |       title: 'Hic sinh đã xác nhận liên kết',
158 |       body: `${profile?.full_name} || 'Hic sinh'`,
159 |       action_url: '/parent/dashboard'
160 |     })
161 |
162 |     revalidatePath('/student/profile/privacy')
163 |     return { success: true }
164 | } catch (err: any) {
165 |   return { error: err.message || 'Lỗi không xác định' }
166 | }
167 |
168 |
169 | // Student or Parent revokes link
170 | export async function revokeParentLink(parentId: string) {
171 |   try {
172 |     const supabase = await getServerSupabase()
173 |     const { data: { user }, error: authErr } = await supabase.auth.getUser()
174 |     if (!user || authErr) return { error: 'Chưa đăng nhập' }
175 |
176 |     // Perform revoke on the record where student_id is current user (or parent_id is current user)
177 |     const { error: linkErr } = await supabase
178 |       .from('parent_student_links')
179 |       .update({
180 |         status: 'revoked',
181 |         revoked_at: new Date().toISOString(),
182 |         revoked_by: user.id
183 |       })
184 |       .eq('parent_id', parentId)
185 |       .eq('student_id', user.id)
186 |
187 |     if (linkErr) {
188 |       console.error('Revoke link error:', linkErr)
189 |       return { error: 'Không thể thu hồi quyền' }
190 |     }
191 |
192 |     revalidatePath('/student/profile/privacy')

```

```

193|     return { success: true }
194|   } catch (err: any) {
195|     return { error: err.message || 'Lỗi không xác định' }
196|   }
197| }
198|

```

File: lib/quiz-engine.ts (86 lines, 2.6 KB)

```

1| export function shuffleArray<T>(array: T[]): T[] {
2|   const shuffled = [...array];
3|   for (let i = shuffled.length - 1; i > 0; i--) {
4|     const j = Math.floor(Math.random() * (i + 1));
5|     [shuffled[i], shuffled[j]] = [shuffled[j], shuffled[i]];
6|   }
7|   return shuffled;
8| }
9|
10| export type QuestionType = 'single_choice' | 'multiple_choice' | 'true_false' | 'fill_blank' | 'ordering';
11|
12| export interface CorrectOption {
13|   id: string;
14|   is_correct: boolean;
15|   content: string;
16|   order?: number;
17| }
18|
19| export interface StudentAnswer {
20|   questionId: string;
21|   selectedOptionIds?: string[];
22|   textAnswer?: string;
23|   orderingAnswer?: string[];
24|   timeSpentSeconds?: number;
25| }
26|
27| export function scoreQuestion(
28|   type: QuestionType,
29|   studentAnswer: StudentAnswer,
30|   correctOptions: CorrectOption[],
31|   maxPoints: number
32| ): number {
33|   if (!studentAnswer) return 0;
34|
35|   switch (type) {
36|     case 'single_choice':
37|       case 'true_false': {
38|         const correctOpt = correctOptions.find(o => o.is_correct);
39|         if (!correctOpt) return 0;
40|         if (studentAnswer.selectedOptionIds && studentAnswer.selectedOptionIds[0] === correctOpt.id) {
41|           return maxPoints;
42|         }
43|         return 0;
44|       }
45|
46|     case 'multiple_choice': {
47|       const correctIds = correctOptions.filter(o => o.is_correct).map(o => o.id);
48|       const studentIds = studentAnswer.selectedOptionIds || [];
49|       if (correctIds.length === 0) return 0;
50|       // Must select all correct and NO incorrect
51|       const allCorrectSelected = correctIds.every(id => studentIds.includes(id));
52|       const noIncorrectSelected = studentIds.every(id => correctIds.includes(id));
53|
54|       if (allCorrectSelected && noIncorrectSelected) {
55|         return maxPoints;
56|       }
57|       // Optional: partial credit could go here
58|       return 0;
59|     }
60|
61|     case 'fill_blank': {
62|       const correctText = correctOptions.find(o => o.is_correct)?.content;
63|       if (!correctText || !studentAnswer.textAnswer) return 0;
64|       const cleanExpected = correctText.trim().toLowerCase();
65|       const cleanActual = studentAnswer.textAnswer.trim().toLowerCase();
66|       if (cleanExpected === cleanActual) {
67|         return maxPoints;
68|       }
69|       return 0;
70|     }
71|
72|     case 'ordering': {
73|       const correctOrder = correctOptions.sort((a, b) => (a.order || 0) - (b.order || 0)).map(o => o.id);
74|       const studentOrder = studentAnswer.orderingAnswer || [];
75|       if (correctOrder.length !== studentOrder.length) return 0;
76|       for (let i = 0; i < correctOrder.length; i++) {
77|         if (correctOrder[i] !== studentOrder[i]) return 0;
78|       }
79|       return maxPoints;
80|     }
81|
82|     default:
83|       return 0;
84|   }
85| }
86|

```

File: lib/realtime-provider.tsx (429 lines, 21.1 KB)

```

1| 'use client';
2|
3| import React, { createContext, useContext, useEffect, useState, useCallback, useRef } from 'react';
4|
5| interface LevelDef {
6|   level: number;
7|   title: string;
8|   xp_required: number;
9|   color_hex: string;
10|   icon: string;
11|   description: string;
12| }
13|
14| interface Quest {
15|   id: string;
16|   title: string;
17|   description: string;
18|   xp_reward: number;
19|   type: string;
20|   difficulty: string;
21|   icon: string;
22|   requirement_type: string;
23|   requirement_value: number;

```

```

24 |   is_active: boolean;
25 |   expires_at: string;
26 | }
27 |
28 | interface UserQuest {
29 |   id: string;
30 |   quest_id: string;
31 |   is_completed: boolean;
32 |   completed_at: string;
33 |   xp_earned: number;
34 |   progress: number;
35 |   daily_quests: Quest;
36 | }
37 |
38 | interface Title {
39 |   id: string;
40 |   title_id: string;
41 |   unlocked_at: string;
42 |   is_active: boolean;
43 |   player_titles: {
44 |     title: string;
45 |     description: string;
46 |     icon: string;
47 |     xp_bonus: number;
48 |   };
49 | }
50 |
51 | interface XpTransaction {
52 |   id: string;
53 |   amount: number;
54 |   reason: string;
55 |   reference_type: string;
56 |   created_at: string;
57 | }
58 |
59 | interface RealtimeState {
60 |   completedLessons: Set<string>;
61 |   xp: number;
62 |   streak: number;
63 |   studyMinutes: number;
64 |   weeklyActivity: number[];
65 |   averageScore: number | null;
66 |   completedLessonCount: number;
67 |   totalQuizAttempts: number;
68 |   level: number;
69 |   levelTitle: string;
70 |   levelIcon: string;
71 |   levelColor: string;
72 |   levelProgress: number;
73 |   xpToNext: number;
74 |   xpInLevel: number;
75 |   levelDefs: LevelDef[];
76 |   quests: UserQuest[];
77 |   titles: Title[];
78 |   xpHistory: XpTransaction[];
79 |   allQuests: Quest[];
80 | }
81 |
82 | interface RealtimeContextType {
83 |   state: RealtimeState;
84 |   refetch: () => Promise<void>;
85 | }
86 |
87 | const defaultLevelDefs: LevelDef[] = [
88 |   { level: 1, title: 'Tân Thạc', xp_required: 0, color_hex: '#94a3b8', icon: 'Tân Thạc' },
89 |   { level: 2, title: 'Thạc Sĩ', xp_required: 100, color_hex: '#647e94', icon: 'Thạc Sĩ' },
90 |   { level: 3, title: 'Tiến Sĩ', xp_required: 200, color_hex: '#4a6981', icon: 'Tiến Sĩ' },
91 |   { level: 4, title: 'Hội Đồng Giáo sư', xp_required: 300, color_hex: '#3a5a71', icon: 'Hội Đồng Giáo sư' },
92 |   { level: 5, title: 'Thầy Giáo', xp_required: 400, color_hex: '#2a4a61', icon: 'Thầy Giáo' },
93 |   { level: 6, title: 'Kỹ sư', xp_required: 500, color_hex: '#1a3a51', icon: 'Kỹ sư' },
94 |   { level: 7, title: 'Kỹ sư Chuyên gia', xp_required: 600, color_hex: '#0a2a41', icon: 'Kỹ sư Chuyên gia' },
95 |   { level: 8, title: 'Kỹ sư Senior', xp_required: 700, color_hex: '#0a2a41', icon: 'Kỹ sư Senior' },
96 |   { level: 9, title: 'Kỹ sư Chuyên gia', xp_required: 800, color_hex: '#0a2a41', icon: 'Kỹ sư Chuyên gia' },
97 |   { level: 10, title: 'Kỹ sư Senior', xp_required: 900, color_hex: '#0a2a41', icon: 'Kỹ sư Senior' },
98 |   { level: 11, title: 'Kỹ sư Chuyên gia', xp_required: 1000, color_hex: '#0a2a41', icon: 'Kỹ sư Chuyên gia' },
99 |   { level: 12, title: 'Kỹ sư Senior', xp_required: 1100, color_hex: '#0a2a41', icon: 'Kỹ sư Senior' },
100 |   { level: 13, title: 'Kỹ sư Chuyên gia', xp_required: 1200, color_hex: '#0a2a41', icon: 'Kỹ sư Chuyên gia' },
101 |   { level: 14, title: 'Kỹ sư Senior', xp_required: 1300, color_hex: '#0a2a41', icon: 'Kỹ sư Senior' },
102 |   { level: 15, title: 'Kỹ sư Chuyên gia', xp_required: 1400, color_hex: '#0a2a41', icon: 'Kỹ sư Chuyên gia' },
103 |   { level: 16, title: 'Kỹ sư Senior', xp_required: 1500, color_hex: '#0a2a41', icon: 'Kỹ sư Senior' },
104 |   { level: 17, title: 'Kỹ sư Chuyên gia', xp_required: 1600, color_hex: '#0a2a41', icon: 'Kỹ sư Chuyên gia' },
105 |   { level: 18, title: 'Kỹ sư Senior', xp_required: 1700, color_hex: '#0a2a41', icon: 'Kỹ sư Senior' },
106 |   { level: 19, title: 'Kỹ sư Chuyên gia', xp_required: 1800, color_hex: '#0a2a41', icon: 'Kỹ sư Chuyên gia' },
107 |   { level: 20, title: 'Kỹ sư Senior', xp_required: 1900, color_hex: '#0a2a41', icon: 'Kỹ sư Senior' },
108 | ];
109 |
110 | const FALLBACK_QUESTS: Quest[] = [
111 |   { id: 'fallback-lesson', title: 'Hoàn thành 1 bài học', description: 'Hoàn thành 1 bài học' },
112 |   { id: 'fallback-pc', title: 'Lập trình PC', description: 'Lập trình PC' },
113 |   { id: 'fallback-quiz', title: 'Làm quiz', description: 'Hoàn thành một bài quiz' },
114 |   { id: 'fallback-exam', title: 'Vượt qua kỳ thi', description: 'Vượt qua kỳ thi' },
115 |   { id: 'fallback-streak', title: 'Hít thở 3 ngày', description: 'Duy trì streak 3 ngày' },
116 |   { id: 'fallback-discussion', title: 'Tham gia thảo luận', description: 'Tham gia thảo luận' },
117 | ];
118 |
119 | function calcLevel(xp: number, levelDefs: LevelDef[]) {
120 |   let lvl = levelDefs[0];
121 |   for (const d of levelDefs) {
122 |     if (xp >= d.xp_required) lvl = d;
123 |   }
124 |   const nextLevel = levelDefs.find(d => d.level === lvl.level + 1);
125 |   const xpInLevel = xp - lvl.xp_required;
126 |   const xpToNext = nextLevel ? nextLevel.xp_required - lvl.xp_required : xpInLevel + 1;
127 |   const levelProgress = xpToNext > 0 ? Math.min((xpInLevel / xpToNext) * 100, 100) : 100;
128 |   return { level: lvl.level, levelTitle: lvl.title, levelIcon: lvl.icon, levelColor: lvl.color_hex, levelProgress: levelProgress };
129 | }
130 |
131 | const RealtimeContext = createContext<RealtimeContextType>({
132 |   state: {
133 |     completedLessons: new Set(), xp: 0, streak: 0, studyMinutes: 0, weeklyActivity: [],
134 |     averageScore: null, completedLessonCount: 0, totalQuizAttempts: 0,
135 |     level: 1, levelTitle: 'Tân Thạc', levelIcon: 'Tân Thạc', levelColor: '#94a3b8',
136 |     levelDefs: defaultLevelDefs, quests: [], titles: [], xpHistory: [], allQuests: [],
137 |   },
138 |   refetch: async () => {},
139 | });
140 |
141 | export function useRealtime() {
142 |   return useContext(RealtimeContext);
143 | }
144 |
145 | export function RealtimeProvider({ children, userId }: { children: React.ReactNode; userId?: string | null }) {
146 |   const [state, setState] = useState<RealtimeState>({
147 |     completedLessons: new Set(),

```

```

148 |     xp: 0,
149 |     streak: 0,
150 |     studyMinutes: 0,
151 |     weeklyActivity: [0, 0, 0, 0, 0, 0, 0, 0],
152 |     averageScore: null,
153 |     completedLessonCount: 0,
154 |     totalQuizAttempts: 0,
155 |     level: 1,
156 |     levelTitle: 'Tân Thç',
157 |     levelIcon: '0<ß1',
158 |     levelColor: '#94a3b8',
159 |     levelProgress: 0,
160 |     xpToNext: 100,
161 |     xpInLevel: 0,
162 |     levelDefs: defaultLevelDefs,
163 |     quests: [],
164 |     titles: [],
165 |     xpHistory: [],
166 |     allQuests: [],
167 |   });
168 |
169 |   const supabaseRef = useRef<any>(null);
170 |   const [clientReady, setClientReady] = useState(false);
171 |   useEffect(() => {
172 |     import('@supabase/ssr').then(({ createBrowserClient }) => {
173 |       supabaseRef.current = createBrowserClient(
174 |         process.env.NEXT_PUBLIC_SUPABASE_URL!,
175 |         process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
176 |       );
177 |       setClientReady(true);
178 |     });
179 |   }, []);
180 |
181 |   const trackDailyLogin = useCallback(async (supabase: any, userId: string) => {
182 |     try {
183 |       const today = new Date().toISOString().slice(0, 10);
184 |       const yesterday = new Date(Date.now() - 86400000).toISOString().slice(0, 10);
185 |       const { data: profile } = await supabase.from('profiles').select('last_active_date, streak').eq('i...
186 |       if (!profile) return
187 |
188 |       const lastActive = profile.last_active_date?.slice(0, 10);
189 |       if (lastActive === today) return null
190 |
191 |       let newStreak = 1;
192 |       if (lastActive === yesterday) {
193 |         newStreak = (profile.streak || 0) + 1;
194 |       }
195 |
196 |       await supabase.from('profiles').update({ last_active_date: today, streak: newStreak }).eq('id', us...
197 |       await supabase.from('learning_stats').upsert({
198 |         user_id: userId, current_streak: newStreak, last_active_date: today, updated_at: new Date().to...
199 |       }, { onConflict: 'user_id' });
200 |
201 |       if (newStreak > 1 && newStreak % 5 === 0) {
202 |         const bonusXp = newStreak * 10;
203 |         await supabase.from('xp_transactions').insert({
204 |           user_id: userId, amount: bonusXp, reason: 'Th° Bng s t
205 |         });
206 |       }
207 |
208 |       return newStreak;
209 |     } catch (e) {
210 |       console.error('Track daily login error:', e);
211 |       return null;
212 |     }
213 |   }, []);
214 |
215 |   const autoCompleteQuests = useCallback(async (supabase: any, userId: string, currentQuests: UserQuest[], a...
216 |     try {
217 |       const { data: lessonProgress } = await supabase.from('lesson_progress').select('lesson_id, status'...
218 |       const completedLessonCount = (lessonProgress || []).filter((p: any) => p.status === 'completed').l...
219 |
220 |       const { data: quizAttempts } = await supabase.from('quiz_attempts').select('id').eq('student_id', ...
221 |       const quizCount = (quizAttempts || []).length;
222 |
223 |       let buildCount = 0;
224 |       try { const { data: builderSessions } = await supabase.from('builder_sessions').select('id').eq('u...
225 |
226 |       let discussionCount = 0;
227 |       try { const { data: discussions } = await supabase.from('discussions').select('id').eq('user_id', ...
228 |
229 |       let passedExamCount = 0;
230 |       try { const { data: examAttempts } = await supabase.from('exam_attempts').select('id, status').eq(...
231 |
232 |       const { data: stats } = await supabase.from('learning_stats').select('current_streak').eq('user_id...
233 |       const currentStreak = stats?.current_streak || 0;
234 |
235 |       let newXp = currentXp;
236 |       let leveledUp = false;
237 |
238 |       for (const questDef of allQuestsDefs) {
239 |         const existingQuest = currentQuests.find((q: UserQuest) => q.quest_id === questDef.id);
240 |         if (existingQuest?.is_completed) continue;
241 |
242 |         let requirementMet = false;
243 |         switch (questDef.requirement_type) {
244 |           case 'lessons': requirementMet = completedLessonCount >= questDef.requirement_value; break;
245 |           case 'quizzes': requirementMet = quizCount >= questDef.requirement_value; break;
246 |           case 'builds': requirementMet = buildCount >= questDef.requirement_value; break;
247 |           case 'streak': requirementMet = currentStreak >= questDef.requirement_value; break;
248 |           case 'discussions': requirementMet = discussionCount >= questDef.requirement_value; break;
249 |           case 'exams': requirementMet = passedExamCount >= questDef.requirement_value; break;
250 |           default: requirementMet = false;
251 |         }
252 |
253 |         if (requirementMet) {
254 |           if (existingQuest) {
255 |             await supabase.from('user_quests').update({ is_completed: true, completed_at: new Date...
256 |           } else {
257 |             await supabase.from('user_quests').insert({ user_id: userId, quest_id: questDef.id, is...
258 |           }
259 |           newXp += questDef.xp_reward;
260 |           await supabase.from('xp_transactions').insert({ user_id: userId, amount: questDef.xp_rewar...
261 |         } else if (existingQuest) {
262 |           const progress = calculateQuestProgress(questDef, completedLessonCount, quizCount, buildCo...
263 |           if (progress !== existingQuest.progress) {
264 |             await supabase.from('user_quests').update({ progress }).eq('id', existingQuest.id);
265 |           }
266 |         }
267 |       }
268 |
269 |       if (newXp > currentXp) {
270 |         const { data: profile } = await supabase.from('profiles').select('xp, level').eq('id', userId)...
271 |         const totalXp = (profile?.xp || 0) + (newXp - currentXp);

```

```

272 |         const { data: levelDefs } = await supabase.from('level_definitions').select('level').lte('xp_r...
273 |         const newLevel = (levelDefs?.[0] as any)?.level || 1
274 |         await supabase.from('profiles').update({ xp: totalXp, level: newLevel, updated_at: new Date()...
275 |         await supabase.from('leaderboard_cache').upsert({ user_id: userId, total_xp: totalXp, level: n...
276 |         await supabase.from('learning_stats').update({ total_xp: totalXp, updated_at: new Date().toISO...
277 |         if (newLevel > currentLevel) {
278 |             leveledUp = true
279 |         }
280 |     }
281 |
282 |     return { xp: newXp, leveledUp }
283 | } catch (e) {
284 |     console.error('Auto-complete quests error:', e)
285 |     return null
286 | }
287 | }, [])
288 |
289 | function calculateQuestProgress(quest: Quest, completedLessons: number, quizzes: number, builds: number, d...
290 |     switch (quest.requirement_type) {
291 |         case 'lessons': return Math.min(completedLessons, quest.requirement_value)
292 |         case 'quizzes': return Math.min(quizzes, quest.requirement_value)
293 |         case 'builds': return Math.min(builds, quest.requirement_value)
294 |         case 'discussions': return Math.min(discussions, quest.requirement_value)
295 |         case 'streak': return Math.min(streak, quest.requirement_value)
296 |         case 'exams': return Math.min(exams || 0, quest.requirement_value)
297 |         default: return 0
298 |     }
299 | }
300 |
301 | const fetchData = useCallback(async () => {
302 |     if (!userId || !supabaseRef.current) return;
303 |     const supabase = supabaseRef.current;
304 |
305 |     // Track daily login & streak
306 |     await trackDailyLogin(supabase, userId);
307 |
308 |     const [
309 |         progress, statsRes, profile, levelDefs, userQuests, activeQuests, titles, xpHistory,
310 |     ] = await Promise.all([
311 |         supabase.from('lesson_progress').select('lesson_id').eq('student_id', userId),
312 |         supabase.from('learning_stats').select('*').eq('user_id', userId).single(),
313 |         supabase.from('profiles').select('xp, level').eq('id', userId).single(),
314 |         supabase.from('level_definitions').select('*').order('level', { ascending: true }),
315 |         supabase.from('user_quests').select('*').eq('user_id', userId).order('created_at'...
316 |         supabase.from('daily_quests').select('*').eq('is_active', true).order('xp_reward', { ascending: fa...
317 |         supabase.from('user_titles').select('*').eq('user_id', userId),
318 |         supabase.from('xp_transactions').select('*').eq('user_id', userId).order('created_at', { ascending...
319 |     ]);
320 |
321 |     let xp = profile?.data?.xp || statsRes?.data?.total_xp || 0;
322 |     const defs = (levelDefs.data as LevelDef[]) || defaultLevelDefs;
323 |     const levelInfo = calcLevel(xp, defs);
324 |     const currentQuests = (userQuests.data || []) as UserQuest[];
325 |     const allQuestsDefs = (() => {
326 |         const fetched = (activeQuests.data || []) as Quest[];
327 |         return fetched.length >= 5 ? fetched : [...fetched, ...FALLBACK_QUESTS].slice(0, 5);
328 |     })();
329 |
330 |     // Auto-complete quests if requirements are met
331 |     try {
332 |         const result = await autoCompleteQuests(supabase, userId, currentQuests, allQuestsDefs, xp, levelI...
333 |         if (result && result.xp > xp) {
334 |             xp = result.xp
335 |         }
336 |     } catch (e) {
337 |         console.error('Auto-complete error:', e)
338 |     }
339 |
340 |     // Fetch quiz attempts for average score
341 |     let avgScore: number | null = null
342 |     let completedCount = 0
343 |     let totalQuizAttempts = 0
344 |     try {
345 |         const progressData = progress.data || []
346 |         completedCount = progressData.filter((p: any) => p.status === 'completed' || p.status === 'in_prog...
347 |
348 |         const { data: quizData } = await supabase.from('quiz_attempts').select('score').eq('student_id', u...
349 |         if (quizData && quizData.length > 0) {
350 |             totalQuizAttempts = quizData.length
351 |             avgScore = Math.round(quizData.reduce((sum: number, q: any) => sum + (q.score || 0), 0) / quiz...
352 |         }
353 |     } catch (e) {}
354 |
355 |     const finalLevelInfo = calcLevel(xp, defs);
356 |
357 |     setState(prev => ({
358 |         ...prev,
359 |         completedLessons: new Set((progress.data || []).map((p: any) => p.lesson_id)),
360 |         xp,
361 |         streak: statsRes.data?.current_streak || 0,
362 |         studyMinutes: statsRes.data?.total_study_minutes || 0,
363 |         weeklyActivity: statsRes.data?.weekly_activity || [0, 0, 0, 0, 0, 0, 0, 0],
364 |         averageScore: avgScore,
365 |         completedLessonCount: completedCount,
366 |         totalQuizAttempts,
367 |         ...finalLevelInfo,
368 |         levelDefs: defs,
369 |         quests: currentQuests,
370 |         allQuests: allQuestsDefs,
371 |         titles: (titles.data || []) as Title[],
372 |         xpHistory: (xpHistory.data || []) as XpTransaction[],
373 |     }));
374 | }, [userId]);
375 |
376 | useEffect(() => {
377 |     if (!supabaseRef.current) return;
378 |     fetchData();
379 |     if (!userId) return;
380 |
381 |     const channel = supabaseRef.current
382 |         .channel('realtime-global')
383 |         .on(
384 |             'postgres_changes',
385 |             { event: '*', schema: 'public', table: 'lesson_progress', filter: `student_id=eq.${userId}` },
386 |             () => fetchData()
387 |         )
388 |         .on(
389 |             'postgres_changes',
390 |             { event: '*', schema: 'public', table: 'learning_stats', filter: `user_id=eq.${userId}` },
391 |             () => fetchData()
392 |         )
393 |         .on(
394 |             'postgres_changes',
395 |             { event: '*', schema: 'public', table: 'profiles', filter: `id=eq.${userId}` },

```

```

396 |         () => fetchData()
397 |     )
398 |     .on(
399 |       'postgres_changes',
400 |       { event: 'INSERT', schema: 'public', table: 'xp_transactions', filter: `user_id=eq.${userId}` },
401 |       () => fetchData()
402 |     )
403 |     .on(
404 |       'postgres_changes',
405 |       { event: '*', schema: 'public', table: 'user_requests', filter: `user_id=eq.${userId}` },
406 |       () => fetchData()
407 |     )
408 |     .on(
409 |       'postgres_changes',
410 |       { event: '*', schema: 'public', table: 'user_titles', filter: `user_id=eq.${userId}` },
411 |       () => fetchData()
412 |     )
413 |     .on(
414 |       'postgres_changes',
415 |       { event: 'INSERT', schema: 'public', table: 'quiz_attempts', filter: `student_id=eq.${userId}` },
416 |       () => fetchData()
417 |     )
418 |     .subscribe();
419 |
420 |     return () => { supabaseRef.current?.removeChannel(channel); };
421 |   }, [userId, clientReady]);
422 |
423 |   return (
424 |     <RealtimeContext.Provider value={{ state, refetch: fetchData }}>
425 |       {children}
426 |     </RealtimeContext.Provider>
427 |   );
428 | }
429 |

```

File: lib/session-time.tsx (58 lines, 1.5 KB)

```

1 | 'use client'
2 |
3 | import React, { createContext, useContext, useState, useEffect, useRef } from 'react'
4 |
5 | const SESSION_START_KEY = 'session_start_time'
6 |
7 | interface SessionTimeContextValue {
8 |   elapsed: number
9 |   formatted: string
10 | }
11 |
12 | const SessionTimeContext = createContext<SessionTimeContextValue>({
13 |   elapsed: 0,
14 |   formatted: '00:00:00',
15 | })
16 |
17 | export function useSessionTime() {
18 |   return useContext(SessionTimeContext)
19 | }
20 |
21 | function formatDuration(seconds: number): string {
22 |   const h = Math.floor(seconds / 3600)
23 |   const m = Math.floor((seconds % 3600) / 60)
24 |   const s = seconds % 60
25 |   return `${h.toString().padStart(2, '0')}:${m.toString().padStart(2, '0')}:${s.toString().padStart(2, '0')}`
26 | }
27 |
28 | export function SessionTimeProvider({ children }: { children: React.ReactNode }) {
29 |   const [elapsed, setElapsed] = useState(0)
30 |   const startTimeRef = useRef<number>(0)
31 |
32 |   useEffect(() => {
33 |     let storedStart = sessionStorage.getItem(SESSION_START_KEY)
34 |     if (!storedStart) {
35 |       storedStart = String(Date.now())
36 |       sessionStorage.setItem(SESSION_START_KEY, storedStart)
37 |     }
38 |     startTimeRef.current = Number(storedStart)
39 |
40 |     const timer = setInterval(() => {
41 |       setElapsed(Math.floor((Date.now() - startTimeRef.current) / 1000))
42 |     }, 1000)
43 |
44 |     return () => clearInterval(timer)
45 |   }, [])
46 |
47 |   const value: SessionTimeContextValue = {
48 |     elapsed,
49 |     formatted: formatDuration(elapsed),
50 |   }
51 |
52 |   return (
53 |     <SessionTimeContext.Provider value={value}>
54 |       {children}
55 |     </SessionTimeContext.Provider>
56 |   )
57 | }
58 |

```

File: lib/sketchfab-models.js (109 lines, 7.9 KB)

```

1 | 'use client';
2 |
3 | export const COLOR_THEMES = {
4 |   case: { label: 'Case', labelEn: 'Case', color: '#3b82f6', bg: '#0a1420', gradient: 'linear-gradient(135deg, ...
5 |   computerParts: { label: 'Computer Parts', labelEn: 'Computer Parts', color: '#f59e0b', bg: '#1a1008', gradie...
6 |   fullComponent: { label: 'Full Component', labelEn: 'Full Component', color: '#10b981', bg: '#0a1a12', gradie...
7 |   psu: { label: 'PSU', labelEn: 'Power Supply', color: '#f97316', bg: '#1c1000', gradient: 'linear-gradient(13...
8 |   mainboard: { label: 'Mainboard', labelEn: 'Mainboard', color: '#22d3ee', bg: '#0a1a1e', gradient: 'linear-gr...
9 |   cpu: { label: 'CPU', labelEn: 'CPU', color: '#ef4444', bg: '#1a0a0a', gradient: 'linear-gradient(135deg, #ef...
10 |   gpu: { label: 'GPU', labelEn: 'GPU', color: '#a855f7', bg: '#1a0a1e', gradient: 'linear-gradient(135deg, #a8...
11 |   ram: { label: 'RAM', labelEn: 'RAM', color: '#f59e0b', bg: '#1c1000', gradient: 'linear-gradient(135deg, #f5...
12 |   cooler: { label: 'Cooler', labelEn: 'Cooler', color: '#06b6d4', bg: '#080e14', gradient: 'linear-gradient(13...
13 | };
14 |
15 | export const CATEGORIES = [
16 |   {
17 |     id: 'case',
18 |     analyzedBy: 'Nguyễn Văn A - Kỹ thuật viên phần cứng',
19 |     knowledge: 'Phần biệt các loại vi case theo kích th° (ATX, Micro-A
20 |   },
21 |   { id: '1', title: 'Dream Computer Setup', embedId: '82f78bbaf2d34f01af854a52151dbf49', description: 'Cas...
22 |   { id: '2', title: 'Computer Components', embedId: 'ae35c58af1ae4e91aea4a81c7c225147', description: 'Case...
23 |   { id: '3', title: 'Tower Gaming Computer', embedId: '085d3328dc2e4618bfea9ebce7fd63a5', description: 'Ca...

```

```
24 |     { id: '4', title: 'Computer Case', embedId: 'cfb12fe17c64464e8fb69a60d20c1d8d', description: 'Case máy t...
25 |   },
26 | },
27 | {
28 |   id: 'computerParts',
29 |   analyzedBy: 'Trần Thế B - Giảng viên Kỹ thuật máy tính',
30 |   knowledge: 'Tổng quan về các bộ phận cấu thành máy tính',
31 |   items: [
32 |     { id: '1', title: 'Computer Parts (Built PC)', embedId: 'bd6fb0ed93f3475b890a099fedecf351', description:...
33 |   ],
34 | },
35 | {
36 |   id: 'fullComponent',
37 |   analyzedBy: 'Lê Văn C - Chuyên gia về cấu hình PC',
38 |   knowledge: 'Nhìn diện tích linh kiện trong mũt hệ thống máy tính:
39 |   items: [
40 |     { id: '1', title: 'Computer Components (Full)', embedId: '4daeb72925d140809bdfd51634a1908e', description...
41 |   ],
42 | },
43 | {
44 |   id: 'psu',
45 |   analyzedBy: 'Phạm Thế D - Kỹ sư điện tử',
46 |   knowledge: 'Tìm hiểu về bộ nguồn máy tính (PSU): công suất (wattage)
47 |   items: [
48 |     { id: '1', title: 'PSU', embedId: '26fead1e09ae41c08d8596aea24dca0',
49 |     { id: '2', title: 'PSU Power Supply Unit', embedId: '69cccd1be3a774
50 |   ],
51 | },
52 | {
53 |   id: 'mainboard',
54 |   analyzedBy: 'Hoàng Văn E - Kỹ thuật viên sửa chữa mainboard',
55 |   knowledge: 'Cấu tạo và chức năng của mainboard: chipset, socket CPU,
56 |   items: [
57 |     { id: '1', title: 'MotherBoard + Components', embedId: '3bc94057328243d4b341a55f59160f8a', description: ...
58 |     { id: '2', title: 'Asus Mainboard', embedId: '87723202fb8f4a66a1b6b69bd901f50c', description: 'Mainboard...
59 |     { id: '3', title: 'Mainboard Dreamer', embedId: '64939d99ac544ed79bba7ef1d88b283', description: 'Mainbo...
60 |     { id: '4', title: 'Mainboard', embedId: '3675514fef19434493df6e8cdfa916dd', description: 'Mainboard tiêu...
61 |     { id: '5', title: 'Mainboard Asus ROG Strix', embedId: '0f3029d57f8d470f99153c50f0285e53', description: ...
62 |     { id: '6', title: 'Mainboard (6)', embedId: 'd880ab7a3ffb49d29ef5f4fe3aa1da2b', description: 'Mainboard ...
63 |   ],
64 | },
65 | {
66 |   id: 'cpu',
67 |   analyzedBy: 'Ngô Văn F - Chuyên gia vi xử lý',
68 |   knowledge: 'So sánh kiến trúc CPU Intel và AMD: số nhân, luồng, xung
69 |   items: [
70 |     { id: '1', title: 'INTEL CPU', embedId: 'ec11f729b05848488522ba6f8c6f254f', description: 'CPU Intel - ki...
71 |     { id: '2', title: 'Intel Core i3-3220', embedId: '34a8a97940514abb90e29c6d6a2579cd', description: 'CPU I...
72 |     { id: '3', title: 'AMD 9600x CPU', embedId: 'b433826fb4b1432dbcb18a90b89dede71', description: 'CPU AMD Ry...
73 |   ],
74 | },
75 | {
76 |   id: 'gpu',
77 |   analyzedBy: 'Nguyễn Thế G - Chuyên gia đồ họa & render',
78 |   knowledge: 'Kiến thức về card đồ họa (GPU): kiến trúc, VRAM, xung nh
79 |   items: [
80 |     { id: '1', title: 'GPU', embedId: '0124232c57f047b98a0afeb7f0ac1a2
81 |     { id: '2', title: 'Nvidia RTX 3090', embedId: '22158616a1a4455917ee8e1e8fc4b09', description: 'GPU Nvid...
82 |     { id: '3', title: 'RX 480 GPU', embedId: '61e69f50bcd44c7284bef31a8f21c6a7', description: 'GPU AMD Radeo...
83 |     { id: '4', title: 'RTX 3060 TI Zotac', embedId: 'a5a78b9331b14c61b6bc0e99afd21b1c', description: 'GPU Zo...
84 |     { id: '5', title: 'GPU RX 580', embedId: '6b3db05319be48028321ac94c4834bcb', description: 'GPU AMD Radeo...
85 |   ],
86 | },
87 | {
88 |   id: 'ram',
89 |   analyzedBy: 'Bùi Văn H - Kỹ sư hệ thống',
90 |   knowledge: 'Phân biệt các loại RAM: DDR3, DDR4, DDR5 - dung lượng, b
91 |   items: [
92 |     { id: '1', title: 'Crucial 8GB DDR4', embedId: '6fa0d3b8c5684741be7d7e396e01bc9f', description: 'RAM Cru...
93 |     { id: '2', title: 'RAM', embedId: '50ca01ea09cf43318b84e176a127bd09', description: 'RAM máy tính'},
94 |     { id: '3', title: 'RAM DDR3', embedId: 'd2715d60f139421eb6028efd51
95 |     { id: '4', title: 'Old Computer RAM', embedId: '9d3016df1c44484096c589b75a53936e', description: 'RAM máy...
96 |   ],
97 | },
98 | {
99 |   id: 'cooler',
100 |   analyzedBy: 'Vũ Thế I - Chuyên gia tản nhiệt & overclock',
101 |   knowledge: 'Các loại tản nhiệt CPU: tản khí (air cooler) vs tản nhi
102 |   items: [
103 |     { id: '1', title: 'Cooler RGB', embedId: '5da990ba782943eb985723c3
104 |     { id: '2', title: 'Corsair H150i Elite', embedId: 'faa8f55407404f0
105 |     { id: '3', title: 'CPU Cooler', embedId: '672a0a74a98c452a862016be
106 |   ],
107 | },
108 | ];
109 |
```

File: lib/supabase-client.ts (9 lines, 0.2 KB)

```
1 | import { createBrowserClient } from '@supabase/ssr'
2 |
3 | export function createClient() {
4 |   return createBrowserClient(
5 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
6 |     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
7 |   )
8 | }
9 |
```

File: lib/supabase-helpers-mock.ts (6 lines, 0.1 KB)

```
1 | import { createClient } from './supabase-ssr-client'
2 |
3 | export function createClientComponentClient() {
4 |   return createClient()
5 | }
6 |
```

File: lib/supabase-service.ts (15 lines, 0.3 KB)

```
1 | import { createClient } from '@supabase/supabase-js'
2 |
3 | export function createServiceClient() {
4 |   return createClient(
5 |     process.env.NEXT_PUBLIC_SUPABASE_URL!,
6 |     process.env.SUPABASE_SERVICE_ROLE_KEY!,
7 |     {
8 |       auth: {
9 |         autoRefreshToken: false,
10 |        persistSession: false,
```



```

11|     },
12|   }
13| )
14| }
15|

```

File: lib/supabase-ssr-client.ts (9 lines, 0.2 KB)

```

1| import { createBrowserClient } from '@supabase/ssr'
2|
3| export function createClient() {
4|   return createBrowserClient(
5|     process.env.NEXT_PUBLIC_SUPABASE_URL!,
6|     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
7|   )
8| }
9|

```

File: lib/supabase-ssr-middleware.ts (119 lines, 3.5 KB)

```

1| import { createServerClient, type CookieOptions } from '@supabase/ssr'
2| import { NextResponse, type NextRequest } from 'next/server'
3|
4| const ROLE_ROUTES: Record<string, string[]> = {
5|   admin: ['/admin'],
6|   teacher: ['/teacher'],
7|   student: ['/student', '/builder'],
8|   parent: ['/parent'],
9| }
10|
11| export async function updateSession(request: NextRequest) {
12|   let supabaseResponse = NextResponse.next({
13|     request,
14|   })
15|
16|   if (!process.env.NEXT_PUBLIC_SUPABASE_URL || !process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY) {
17|     console.error("CRITICAL: Missing Supabase environment variables in Middleware!")
18|     return supabaseResponse
19|   }
20|
21|   const supabase = createServerClient(
22|     process.env.NEXT_PUBLIC_SUPABASE_URL,
23|     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY,
24|     {
25|       cookies: {
26|         get(name: string) {
27|           return request.cookies.get(name)?.value
28|         },
29|         set(name: string, value: string, options: CookieOptions) {
30|           request.cookies.set({
31|             name,
32|             value,
33|             ...options,
34|           })
35|           supabaseResponse = NextResponse.next({
36|             request: {
37|               headers: request.headers,
38|             },
39|           })
40|           supabaseResponse.cookies.set({
41|             name,
42|             value,
43|             ...options,
44|           })
45|         },
46|         remove(name: string, options: CookieOptions) {
47|           request.cookies.set({
48|             name,
49|             value: '',
50|             ...options,
51|           })
52|           supabaseResponse = NextResponse.next({
53|             request: {
54|               headers: request.headers,
55|             },
56|           })
57|           supabaseResponse.cookies.set({
58|             name,
59|             value: '',
60|             ...options,
61|           })
62|         },
63|       },
64|     })
65|   }
66|
67|   // Kiểm tra user session
68|   const { data: { user } } = await supabase.auth.getUser()
69|   const currentPath = request.nextUrl.pathname
70|
71|   // Bỏ qua các route public
72|   if (currentPath.startsWith('/_next') || currentPath.startsWith('/api') || currentPath === '/' || currentPath...)
73|     return supabaseResponse;
74| }
75|
76| // Nếu chưa đăng nhập mà vào route cần auth => Redirect login
77| if (!user) {
78|   const url = request.nextUrl.clone()
79|   url.pathname = '/login'
80|   return NextResponse.redirect(url)
81| }
82|
83| // Lấy role từ bảng profiles hoặc metadata của user
84| const { data: profile } = await supabase
85|   .from('profiles')
86|   .select('role')
87|   .eq('id', user.id)
88|   .single()
89|
90| // Nếu thiếu role từ profile, dùng role từ user_metadata
91| const userRole = profile?.role || user.user_metadata?.role || 'student';
92|
93| let isAllowed = false;
94| // Common routes allowed for ALL authenticated users
95| const commonRoutes = ['/profile', '/lessons']
96|
97| if (commonRoutes.some(route => currentPath.startsWith(route))) {
98|   isAllowed = true;
99| } else if (ROLE_ROUTES[userRole]) {
100|   for (const route of ROLE_ROUTES[userRole]) {
101|     if (currentPath.startsWith(route)) {

```

```

102 |         isAllowed = true;
103 |         break;
104 |     }
105 | }
106 | }
107 |
108 | // Nếu không ỏ ỏ c phép, redirect vỏ dashboard tỏ ing ỏ ng
109 | if (!isAllowed) {
110 |     const url = request.nextUrl.clone()
111 |     url.pathname = `/${userRole}`;
112 | }
113 | // Student cỏ route . c biỏ t lỏ /builder hoỏ c /student
114 | if (userRole === 'student') url.pathname = '/builder';
115 | return NextResponse.redirect(url)
116 | }
117 | return supabaseResponse
118 | }
119 |

```

File: lib/supabase-ssr-server.ts (30 lines, 0.8 KB)

```

1 | import { createServerClient } from '@supabase/ssr'
2 | import { cookies } from 'next/headers'
3 |
4 | export async function createClient() {
5 |     const cookieStore = await cookies()
6 |
7 |     return createServerClient(
8 |         process.env.NEXT_PUBLIC_SUPABASE_URL!,
9 |         process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!,
10 |         {
11 |             cookies: {
12 |                 getAll() {
13 |                     return cookieStore.getAll()
14 |                 },
15 |                 setAll(cookiesToSet) {
16 |                     cookiesToSet.forEach(({ name, value, options }) => {
17 |                         try {
18 |                             cookieStore.set(name, value, options)
19 |                         } catch {
20 |                             // The `set` method was called from a Server Component.
21 |                             // This can be ignored if you have middleware refreshing
22 |                             // user sessions.
23 |                         }
24 |                     })
25 |                 },
26 |             },
27 |         })
28 |     }
29 | }
30 |

```

File: lib/supabase.js (8 lines, 0.4 KB)

```

1 | import { createBrowserClient } from '@supabase/ssr'
2 |
3 | const supabaseUrl = process.env.NEXT_PUBLIC_SUPABASE_URL || 'https://placeholder.supabase.co'
4 | const supabaseAnonKey = process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY || 'placeholder'
5 |
6 | // Dùng Browser Client ỏ ỏ ng bỏ hoàn toàn vỏ i Middleware (Cookies)
7 | export const supabase = createBrowserClient(supabaseUrl, supabaseAnonKey)
8 |

```

File: lib/tracking.tsx (42 lines, 1.3 KB)

```

1 | 'use client'
2 |
3 | import React, { useEffect, useState } from 'react'
4 | import { startBuilderSession, endBuilderSession } from '@app/actions/trackingActions'
5 |
6 | // HOC for adding tracking to the builder component
7 | export function withTracking(WrappedComponent: React.ComponentType<any>) {
8 |     return function TrackingWrapper(props: any) {
9 |         const [sessionId, setSessionId] = useState<string | null>(null)
10 |
11 |         useEffect(() => {
12 |             let activeSession: string | null = null;
13 |
14 |             const initTracking = async () => {
15 |                 try {
16 |                     const sid = await startBuilderSession(props.lessonId)
17 |                     activeSession = sid
18 |                     setSessionId(sid)
19 |                 } catch (e) {
20 |                     console.error("Failed to start builder session tracking", e)
21 |                 }
22 |             }
23 |
24 |             initTracking()
25 |
26 |             return () => {
27 |                 if (activeSession) {
28 |                     // Send final stats when unmounting
29 |                     // In a real app, you would pass refs or state values here
30 |                     endBuilderSession(activeSession, {
31 |                         components_used: [], // Mock data
32 |                         tdp_calculated: 0,
33 |                         compatibility_score: 100
34 |                     }).catch(console.error)
35 |                 }
36 |             }
37 |         }, [props.lessonId])
38 |
39 |         return <WrappedComponent {...props} sessionId={sessionId} />
40 |     }
41 | }
42 |

```

File: lib/useStore.ts (350 lines, 8.7 KB)

```

1 | 'use client';
2 |
3 | import { create } from 'zustand';
4 |
5 | export type ComponentType = 'cpu' | 'cooler' | 'ram' | 'gpu' | 'psu' | 'ssd' | 'motherboard';
6 |
7 | export interface ComponentSlot {
8 |     id: string;

```

```

9 |   type: ComponentType;
10 |   label: string;
11 |   position: [number, number, number];
12 |   rotation: [number, number, number];
13 |   size: [number, number, number];
14 |   dependsOn?: ComponentType[];
15 |   locked: boolean;
16 | }
17 |
18 | export interface AssemblyComponent {
19 |   id: string;
20 |   slotId: string;
21 |   type: ComponentType;
22 |   name: string;
23 |   installed: boolean;
24 |   position: [number, number, number];
25 |   rotation: [number, number, number];
26 | }
27 |
28 | export interface CameraCoords {
29 |   pitch: number;
30 |   yaw: number;
31 |   roll: number;
32 | }
33 |
34 | export type BootStatus = 'idle' | 'booting' | 'success' | 'failed';
35 |
36 | export interface BootError {
37 |   code: string;
38 |   message: string;
39 |   component: ComponentType;
40 | }
41 |
42 | export interface AssemblyState {
43 |   components: AssemblyComponent[];
44 |   slots: ComponentSlot[];
45 |   cameraCoords: CameraCoords;
46 |   bootStatus: BootStatus;
47 |   bootErrors: BootError[];
48 |   hoveredComponent: string | null;
49 |   draggedComponent: string | null;
50 |   rgbActive: boolean;
51 |   fanSpeed: number;
52 | }
53 |
54 | interface StoreActions {
55 |   setCameraCoords: (coords: Partial<CameraCoords>) => void;
56 |   installComponent: (slotId: string, componentId: string) => boolean;
57 |   removeComponent: (slotId: string) => void;
58 |   setHoveredComponent: (id: string | null) => void;
59 |   setDraggedComponent: (id: string | null) => void;
60 |   triggerBootCheck: () => void;
61 |   resetBoot: () => void;
62 |   resetAssembly: () => void;
63 |   setRgbActive: (active: boolean) => void;
64 |   setFanSpeed: (speed: number) => void;
65 |   checkDependencies: (slotId: string) => { ok: boolean; missing: ComponentType[] };
66 |   getInstalledParts: () => AssemblyComponent[];
67 |   isSlotAvailable: (slotId: string) => boolean;
68 | }
69 |
70 | const DEFAULT_SLOTS: ComponentSlot[] = [
71 |   {
72 |     id: 'motherboard_1',
73 |     type: 'motherboard',
74 |     label: 'Mainboard',
75 |     position: [0, 0.02, 0],
76 |     rotation: [0, 0, 0],
77 |     size: [2.4, 0.02, 2.0],
78 |     dependsOn: [],
79 |     locked: false,
80 |   },
81 |   {
82 |     id: 'cpu_1',
83 |     type: 'cpu',
84 |     label: 'CPU',
85 |     position: [0, 0.12, 0.25],
86 |     rotation: [0, 0, 0],
87 |     size: [0.6, 0.06, 0.6],
88 |     dependsOn: ['motherboard'],
89 |     locked: true,
90 |   },
91 |   {
92 |     id: 'cooler_1',
93 |     type: 'cooler',
94 |     label: 'CPU Cooler',
95 |     position: [0, 0.25, 0.25],
96 |     rotation: [0, 0, 0],
97 |     size: [0.7, 0.15, 0.7],
98 |     dependsOn: ['cpu'],
99 |     locked: true,
100 |   },
101 |   {
102 |     id: 'ram_1',
103 |     type: 'ram',
104 |     label: 'RAM Slot A1',
105 |     position: [-0.55, 0.08, 0.45],
106 |     rotation: [0, 0, 0],
107 |     size: [0.08, 0.4, 0.02],
108 |     dependsOn: ['motherboard'],
109 |     locked: true,
110 |   },
111 |   {
112 |     id: 'ram_2',
113 |     type: 'ram',
114 |     label: 'RAM Slot A2',
115 |     position: [-0.42, 0.08, 0.45],
116 |     rotation: [0, 0, 0],
117 |     size: [0.08, 0.4, 0.02],
118 |     dependsOn: ['motherboard'],
119 |     locked: true,
120 |   },
121 |   {
122 |     id: 'ram_3',
123 |     type: 'ram',
124 |     label: 'RAM Slot B1',
125 |     position: [0.42, 0.08, 0.45],
126 |     rotation: [0, 0, 0],
127 |     size: [0.08, 0.4, 0.02],
128 |     dependsOn: ['motherboard'],
129 |     locked: true,
130 |   },
131 |   {
132 |     id: 'ram_4',

```

```

133 |     type: 'ram',
134 |     label: 'RAM Slot B2',
135 |     position: [0.55, 0.08, 0.45],
136 |     rotation: [0, 0, 0],
137 |     size: [0.08, 0.4, 0.02],
138 |     dependsOn: ['motherboard'],
139 |     locked: true,
140 |   },
141 |   {
142 |     id: 'gpu_1',
143 |     type: 'gpu',
144 |     label: 'GPU (PCIe x16)',
145 |     position: [0, 0.06, -0.5],
146 |     rotation: [0, 0, 0],
147 |     size: [1.6, 0.15, 0.6],
148 |     dependsOn: ['motherboard'],
149 |     locked: true,
150 |   },
151 |   {
152 |     id: 'psu_1',
153 |     type: 'psu',
154 |     label: 'Power Supply',
155 |     position: [0, -0.2, 1.2],
156 |     rotation: [0, 0, 0],
157 |     size: [1.2, 0.4, 0.7],
158 |     dependsOn: ['motherboard'],
159 |     locked: true,
160 |   },
161 |   {
162 |     id: 'ssd_1',
163 |     type: 'ssd',
164 |     label: 'M.2 NVMe SSD',
165 |     position: [0.35, 0.06, -0.15],
166 |     rotation: [0, 0, 0],
167 |     size: [0.8, 0.02, 0.2],
168 |     dependsOn: ['motherboard'],
169 |     locked: true,
170 |   },
171 | ];
172 |
173 | const INITIAL_COMPONENTS: AssemblyComponent[] = DEFAULT_SLOTS.map((slot) => ({
174 |   id: `comp_${slot.id}`,
175 |   slotId: slot.id,
176 |   type: slot.type,
177 |   name: slot.label,
178 |   installed: false,
179 |   position: [...slot.position] as [number, number, number],
180 |   rotation: [...slot.rotation] as [number, number, number],
181 | }));
182 |
183 | export const useAssemblyStore = create<AssemblyState & StoreActions>((set, get) => ({
184 |   components: INITIAL_COMPONENTS,
185 |   slots: DEFAULT_SLOTS,
186 |   cameraCoords: { pitch: 0, yaw: 0, roll: 0 },
187 |   bootStatus: 'idle',
188 |   bootErrors: [],
189 |   hoveredComponent: null,
190 |   draggedComponent: null,
191 |   rgbActive: false,
192 |   fanSpeed: 0,
193 |
194 |   setCameraCoords: (coords) =>
195 |     set((state) => ({
196 |       cameraCoords: { ...state.cameraCoords, ...coords },
197 |     })),
198 |
199 |   checkDependencies: (slotId) => {
200 |     const state = get();
201 |     const slot = state.slots.find((s) => s.id === slotId);
202 |     if (!slot || !slot.dependsOn || slot.dependsOn.length === 0) {
203 |       return { ok: true, missing: [] };
204 |     }
205 |     const installed = state.components.filter((c) => c.installed);
206 |     const missing = slot.dependsOn.filter(
207 |       (depType) => !installed.some((c) => c.type === depType)
208 |     );
209 |     return { ok: missing.length === 0, missing };
210 |   },
211 |
212 |   installComponent: (slotId, componentId) => {
213 |     const state = get();
214 |     const depCheck = state.checkDependencies(slotId);
215 |     if (!depCheck.ok) return false;
216 |
217 |     const slot = state.slots.find((s) => s.id === slotId);
218 |     if (!slot || slot.locked) return false;
219 |
220 |     const existing = state.components.find(
221 |       (c) => c.slotId === slotId && c.installed
222 |     );
223 |     if (existing) return false;
224 |
225 |     set((state) => ({
226 |       components: state.components.map((c) =>
227 |         c.id === componentId
228 |           ? { ...c, installed: true, position: [...slot.position] as [number, number, number] }
229 |           : c
230 |       ),
231 |     }));
232 |     return true;
233 |   },
234 |
235 |   removeComponent: (slotId) =>
236 |     set((state) => ({
237 |       components: state.components.map((c) =>
238 |         c.slotId === slotId && c.installed
239 |           ? { ...c, installed: false }
240 |           : c
241 |       ),
242 |     })),
243 |
244 |   setHoveredComponent: (id) => set({ hoveredComponent: id }),
245 |   setDraggedComponent: (id) => set({ draggedComponent: id }),
246 |
247 |   triggerBootCheck: () => {
248 |     const state = get();
249 |     const installed = state.components.filter((c) => c.installed);
250 |     const errors: BootError[] = [];
251 |
252 |     if (!installed.some((c) => c.type === 'cpu')) {
253 |       errors.push({
254 |         code: '0x01',
255 |         message: 'CPU not detected in socket',
256 |         component: 'cpu',

```

```

257 |     });
258 |   }
259 |   if (!installed.some((c) => c.type === 'cooler')) {
260 |     errors.push({
261 |       code: '0x02',
262 |       message: 'CPU Cooler missing - thermal shutdown risk',
263 |       component: 'cooler',
264 |     });
265 |   }
266 |   if (!installed.some((c) => c.type === 'ram')) {
267 |     errors.push({
268 |       code: '0x03',
269 |       message: 'No RAM modules detected in any DIMM slot',
270 |       component: 'ram',
271 |     });
272 |   }
273 |   const ramCount = installed.filter((c) => c.type === 'ram').length;
274 |   if (ramCount > 0 && ramCount < 2) {
275 |     errors.push({
276 |       code: '0x04',
277 |       message: `Only ${ramCount} RAM module(s) installed - single-channel mode`,
278 |       component: 'ram',
279 |     });
280 |   }
281 |   if (!installed.some((c) => c.type === 'gpu')) {
282 |     errors.push({
283 |       code: '0x05',
284 |       message: 'No GPU detected - no display output available',
285 |       component: 'gpu',
286 |     });
287 |   }
288 |   if (!installed.some((c) => c.type === 'psu')) {
289 |     errors.push({
290 |       code: '0x06',
291 |       message: 'Power Supply Unit not connected',
292 |       component: 'psu',
293 |     });
294 |   }
295 |   if (!installed.some((c) => c.type === 'ssd')) {
296 |     errors.push({
297 |       code: '0x07',
298 |       message: 'No boot drive detected (M.2 SSD missing)',
299 |       component: 'ssd',
300 |     });
301 |   }
302 | }
303 |
304 | if (errors.length > 0) {
305 |   set({ bootStatus: 'booting' });
306 |   setTimeout(() => {
307 |     set({ bootStatus: 'failed', bootErrors: errors, rgbActive: false, fanSpeed: 0 });
308 |   }, 1500);
309 | } else {
310 |   set({ bootStatus: 'booting' });
311 |   setTimeout(() => {
312 |     set({
313 |       bootStatus: 'success',
314 |       bootErrors: [],
315 |       rgbActive: true,
316 |       fanSpeed: 1,
317 |     });
318 |   }, 2000);
319 | }
320 |
321 | resetBoot: () =>
322 |   set({
323 |     bootStatus: 'idle',
324 |     bootErrors: [],
325 |     rgbActive: false,
326 |     fanSpeed: 0,
327 |   }),
328 |
329 | resetAssembly: () =>
330 |   set({
331 |     components: INITIAL_COMPONENTS.map((c) => ({ ...c })),
332 |     bootStatus: 'idle',
333 |     bootErrors: [],
334 |     rgbActive: false,
335 |     fanSpeed: 0,
336 |   }),
337 |
338 | setRgbActive: (active) => set({ rgbActive: active }),
339 | setFanSpeed: (speed) => set({ fanSpeed: speed }),
340 |
341 | getInstalledParts: () => get().components.filter((c) => c.installed),
342 |
343 | isSlotAvailable: (slotId) => {
344 |   const state = get();
345 |   const slot = state.slots.find((s) => s.id === slotId);
346 |   if (!slot || slot.locked) return false;
347 |   return !state.components.some((c) => c.slotId === slotId && c.installed);
348 | },
349 | }));
350 |

```

```

1 // =====
2 // PC Master AI — Constants, System Prompt & Quick Actions
3 // =====
4
5 | export const SYSTEM_PROMPT = `Bìn là **PC Master AI**, trã lý chuyên ng
6
7 | **Kiến thức:**
8 | - Nắm rõ toàn bộ bài giảng trong Chũ Híc T-p (CPU, Mainboard, RAM, GR
9 | - Biết danh sách linh kiện trong Chũ Máy Tính (Celeron G5905, i3-12100F,
10 | - Giá linh kiện tính bằng VNĐ (ví dụ: i3-12100F = 2.250.000 VNĐ, RTX 306
11
12 | **Nhicm vã:**
13 | 1. T° vãn c° hình-PC theo ngãn sách cá thã (°u tiên t°ing thích Socket,
14 | 2. Giĩ áp th-c°m-c kù thu-t vã linh kiện máy tính.
15 | 3. H° Õng đ«n dùng các b°c l°p ráp trong Chũ Õ Luyçn T-p.
16 | 4. Kiã m tra tnh° t°ing thích (Socket CPU-Mainboard, loii RAM DDR4/DDR5,
17
18 | **Phong cách:**
19 | - Trã lýi ng-n gĩn, súc tích, dùng thu-t ngĩ chuyên môn dã hiãu.
20 | - Kèm emoji công nghç (0=Ýp 0>Yã &i 0<B® 0=U% 0=Y' 0=U*).
21 | - Dùng B£ng Markdown khi so sánh nhĩu linh kiện.
22 | - Nçu ng°Yi dùng hĩi ngoài chç Á máy tính, khèo léo tẽ chñi: "Tõi ché c
23
24 | export const QUICK_ACTIONS = [
25 | {
26 |         label: '0=UÚ web có nhĩng gi?',

```

```

28| 2 7 |},      prompt: 'Website PC Master Builder có những tính năng gì? H° Òng
29| 3 0 |},      label: '0<3~ Tính n ng n òi b - t',
30| 3 1 |},      prompt: 'Các tính n ng n òi b - t c ça website PC Master Builder là
32| 3 4 |},      label: '0=00 H° Òng d«n sí d ang',
33| 3 5 |},      prompt: 'H° Òng d«n t òi cách sí d ang website PC Master Builder t
36| 3 8 |},      label: '0=0° T° v ¥n c ¥u hình 10 tri Çu',
37| 3 9 |},      prompt: 'T òi có ng ân sách 10 tri Çu V N , hãy t° v ¥n c ¥u hình PC p
40| 4 2 |},      label: '0=Y' Ki Ām tra t° ing thích',
41| 4 3 |},      prompt: 'Gi úp t òi ki Ām tra: CPU Intel Core i5 - 13600K (LGA1700) c
44| 4 6 |},      label: '0>Yà Gi Ēi thích v Ā CPU', S N nhān (Cores) và Xung nh Ēp (Clock
45| 4 7 |},      prompt: 'Gi Ēi thích i n gi Ēn: S N nhān (Cores) và Xung nh Ēp (Clock
48| 48 |},
49| 49 |},
50| 50 |},
51| 51 |}, export const WELCOME_MESSAGE = {
52| 52 |}, role: 'assistant',
53| 53 |}, content: `Xin chào! Tôi là **PC Master AI** 0>Y \n\nTôi có th Ā giúp b
54| 54 |},
55| 55 |},

```

File: utils/HandFilter.js (159 lines, 4.9 KB)

```

1| /**
2| * utils/HandFilter.js
3| */
4| * Module líc nhi Āu t ĩa Ò bān tay cho Web/MediaPipe.
5| * - Sí d ang thu - t toán One Euro Filter cho vi Çc làm m° āt ch Ñng Ò tr Ā.
6| * - Deadzone: Ng° āng lo i i b ĩ rung Òng vi m ò.
7| * - Hand Lost: Gi i t ĩa Ò khi tay b Ē r ýi kh ĩi kh ung hình trong th ýi gi ā
8| */
9|
10| class LowPassFilter {
11|   constructor(alpha) {
12|     this.setAlpha(alpha);
13|     this.y = null;
14|     this.s = null;
15|   }
16|
17|   setAlpha(alpha) {
18|     if (alpha <= 0.0 || alpha > 1.0) {
19|       throw new Error('Alpha must be in (0.0, 1.0]');
20|     }
21|     this.alpha = alpha;
22|   }
23|
24|   reset() {
25|     this.y = null;
26|     this.s = null;
27|   }
28|
29|   filter(value) {
30|     if (this.y === null) {
31|       this.s = value;
32|     } else {
33|       this.s = this.alpha * value + (1.0 - this.alpha) * this.s;
34|     }
35|     this.y = value;
36|     return this.s;
37|   }
38|
39|   lastValue() {
40|     return this.s;
41|   }
42| }
43|
44| class OneEuroFilter {
45|   constructor(minCutoff = 1.0, beta = 0.007, dCutoff = 1.0) {
46|     this.minCutoff = minCutoff;
47|     this.beta = beta;
48|     this.dCutoff = dCutoff;
49|     this.xFilter = new LowPassFilter(this.alpha(this.minCutoff));
50|     this.dxFilter = new LowPassFilter(this.alpha(this.dCutoff));
51|     this.lastTime = null;
52|   }
53|
54|   alphaWithDt(cutoff, dt) {
55|     const tau = 1.0 / (2 * Math.PI * cutoff);
56|     return 1.0 / (1.0 + tau / dt);
57|   }
58|
59|   alpha(cutoff) {
60|     return this.alphaWithDt(cutoff, 1.0 / 60.0); // fps m . c Ē nh
61|   }
62|
63|   reset() {
64|     this.xFilter.reset();
65|     this.dxFilter.reset();
66|     this.lastTime = null;
67|   }
68|
69|   filter(x) {
70|     let dx = 0.0;
71|     let dt = 1.0 / 60.0;
72|
73|     const now = Date.now();
74|     if (this.lastTime !== null) {
75|       dt = (now - this.lastTime) / 1000.0;
76|       if (dt <= 0) dt = 1e - 4; // Tr ānh l x i chia cho 0
77|       dx = (x - this.xFilter.lastValue()) / dt;
78|     }
79|     this.lastTime = now;
80|
81|     const edx = this.dxFilter.filter(dx);
82|     const cutoff = this.minCutoff + this.beta * Math.abs(edx);
83|
84|     this.xFilter.setAlpha(this.alphaWithDt(cutoff, dt));
85|     return this.xFilter.filter(x);
86|   }
87| }
88|
89| export default class HandFilter {
90|   constructor(config = {}) {
91|     const {

```

```

92 |         minCutoff = 1.0, // Tỉ số giảm tốc độ (tỉ lệ giữa tốc độ hiện tại và tốc độ tối đa)
93 |         beta = 0.007, // Hệ số giảm tốc độ (tỉ lệ giữa tốc độ hiện tại và tốc độ tối đa)
94 |         dCutoff = 1.0, // Damping coefficient (tỉ lệ giữa tốc độ hiện tại và tốc độ tối đa)
95 |         deadzone = 2.0, // (pixel) Ngưỡng ngừng cập nhật (điểm chết)
96 |         lossTimeout = 300 // (ms) Thời gian giới hạn nếu không
97 |     } = config;
98 |
99 |     this.filterX = new OneEuroFilter(minCutoff, beta, dCutoff);
100 |     this.filterY = new OneEuroFilter(minCutoff, beta, dCutoff);
101 |
102 |     this.deadzone = deadzone;
103 |     this.lossTimeout = lossTimeout;
104 |
105 |     this.lastValidPos = null;
106 |     this.lastSeenTime = 0;
107 |     this.isVisible = false;
108 | }
109 |
110 | /**
111 |  * @param {number|null} rawX - Tọa độ X (pixel). Nếu null vĩa mĩa
112 |  * @param {number|null} rawY - Tọa độ Y (pixel).
113 |  * @returns {{x: number, y: number}|null}
114 |  */
115 | update(rawX, rawY) {
116 |     const now = Date.now();
117 |
118 |     // 1. Mất track tay (Hand Lost logic)
119 |     if (rawX === null || rawY === null || rawX === undefined || rawY === undefined) {
120 |         if (this.isVisible && (now - this.lastSeenTime > this.lossTimeout)) {
121 |             this.isVisible = false;
122 |             this.filterX.reset();
123 |             this.filterY.reset();
124 |             this.lastValidPos = null;
125 |         }
126 |         // Trong thời gian 300ms, trả về vị trí cũ (nếu có)
127 |         return this.isVisible ? { x: this.lastValidPos.x, y: this.lastValidPos.y } : null;
128 |     }
129 |
130 |     this.isVisible = true;
131 |     this.lastSeenTime = now;
132 |
133 |     // 2. Vùng chết (Deadzone logic)
134 |     if (this.lastValidPos) {
135 |         const dx = rawX - this.lastValidPos.rawX;
136 |         const dy = rawY - this.lastValidPos.rawY;
137 |         const dist = Math.hypot(dx, dy);
138 |
139 |         // Bỏ qua khung hình nếu lệch chĩa vĩa qua deadzone
140 |         if (dist < this.deadzone) {
141 |             return { x: this.lastValidPos.x, y: this.lastValidPos.y };
142 |         }
143 |     }
144 |
145 |     // 3. One Euro Filter
146 |     const filteredX = this.filterX.filter(rawX);
147 |     const filteredY = this.filterY.filter(rawY);
148 |
149 |     this.lastValidPos = {
150 |         x: filteredX,
151 |         y: filteredY,
152 |         rawX, // Lưu lại rawX, rawY để tính khoảng cách cho lần update
153 |         rawY
154 |     };
155 |
156 |     return { x: filteredX, y: filteredY };
157 | }
158 | }
159 |

```

File: utils/i18nData.js (80 lines, 3.8 KB)

```

1 | export const COMPONENT_DATA = {
2 |   en: {
3 |     CPU: {
4 |       title: "Central Processing Unit (CPU)",
5 |       desc: "The 'brain' of the computer. It executes instructions and processes data.",
6 |       funFact: "Modern CPUs can execute billions of instructions per second!"
7 |     },
8 |     RAM: {
9 |       title: "Random Access Memory (RAM)",
10 |       desc: "Short-term memory. It stores data that the CPU needs to access quickly.",
11 |       funFact: "More RAM allows you to run more applications simultaneously."
12 |     },
13 |     GPU: {
14 |       title: "Graphics Processing Unit (GPU)",
15 |       desc: "Renders images, video, and games. Essential for gaming and creative work.",
16 |       funFact: "GPUs are also used for training Artificial Intelligence models!"
17 |     },
18 |     SSD: {
19 |       title: "Solid State Drive (SSD)",
20 |       desc: "Fast storage for your OS and applications. Much faster than old hard drives.",
21 |       funFact: "NVMe SSDs can reach speeds of 7000 MB/s!"
22 |     },
23 |     PSU: {
24 |       title: "Power Supply Unit (PSU)",
25 |       desc: "Converts power from the wall outlet into usable power for all components.",
26 |       funFact: "A good PSU can last through multiple computer builds!"
27 |     },
28 |     COOLER: {
29 |       title: "CPU Cooler",
30 |       desc: "Keeps your CPU from overheating by dissipating heat through fins and a fan.",
31 |       funFact: "Some high-end coolers can keep a CPU under 30°C at idle!"
32 |     },
33 |   },
34 |   vn: {
35 |     CPU: {
36 |       title: "Bộ Vi Xử Lý (CPU)",
37 |       desc: "Bộ não của máy tính. Nó thực hiện các lệnh và xử lý dữ liệu.",
38 |       funFact: "CPU hiện đại có thể thực hiện hàng tỷ lệnh mỗi giây!"
39 |     },
40 |     RAM: {
41 |       title: "Bộ Nhớ Truy Cập Nhanh (RAM)",
42 |       desc: "Bộ nhớ tạm thời. Lưu trữ dữ liệu mà CPU cần truy cập nhanh.",
43 |       funFact: "Nhiều RAM hơn giúp bạn chạy nhiều ứng dụng cùng lúc."
44 |     },
45 |     GPU: {
46 |       title: "Card Đồ Họa (GPU)",
47 |       desc: "Xử lý hình ảnh, video và game. Cần thiết cho chơi game và đồ họa.",
48 |       funFact: "GPU cũng được dùng để huấn luyện Trí Tuệ Nhân Tạo."
49 |     },
50 |     SSD: {
51 |       title: "Ổ Cứng Thể Rắn (SSD)",
52 |       desc: "Lưu trữ nhanh cho hệ điều hành và ứng dụng. Nhanh hơn ổ cứng."

```

```

54| 5 3 | }, funFact: "SSD NVMe có thể ít tốn 7000 MB/s!"
55| PSU: {
56|     title: "Bù Nguồn (PSU)",
57|     desc: "Chuyển đổi điện thành điện năng sử dụng",
58|     funFact: "Mút PSU tiết kiệm điện qua nhiều Ý tưởng"
59| },
60| COOLER: {
61|     title: "Tản Nhiệt CPU",
62|     desc: "Giữ cho CPU không bị quá nhiệt bằng cách tản nhiệt",
63|     funFact: "Tản nhiệt cao cấp có thể giảm CPU độ 30°C"
64| }
65| }
66| };
67|
68| export const GURU_MESSAGES = {
69|   en: {
70|     welcome: "Welcome, Builder! Use your hand to pick up components.",
71|     grabbed: (item) => `You've got the ${item}. Locate the glowing green socket on the motherboard.`,
72|     placed: (item) => `Excellent work! The ${item} is installed correctly.`
73|   },
74|   vn: {
75|     welcome: "Chào bạn! Hãy dùng tay lắp các linh kiện nhé.",
76|     grabbed: (item) => `Bạn đã lấy được ${item}. Hãy tìm khe cắm phát",
77|     placed: (item) => `Tuyệt vời! ${item} đã lắp đúng vị trí.`
78|   }
79| };
80|

```

File: utils/searchKnowledge.js (25 lines, 0.8 KB)

```

1| import { KNOWLEDGE_BASE } from '../data/knowledgeBase';
2|
3| /**
4|  * Tìm kiếm các oin kiến thức liên quan dựa trên topic.
5|  * Sử dụng Keyword Matching in google.
6|  */
7| export function searchKnowledge(topic) {
8|   if (!topic) return "";
9|
10|   const query = topic.toUpperCase();
11|
12|   // Tìm các chunk có chứa topic trong tên hoặc nội dung
13|   const relatedChunks = KNOWLEDGE_BASE.filter(item =>
14|     item.topic.toUpperCase().includes(query) ||
15|     query.includes(item.topic.toUpperCase())
16|   );
17|
18|   if (relatedChunks.length > 0) {
19|     return relatedChunks.map(chunk => chunk.content).join("\n\n");
20|   }
21|
22|   // Nếu không tìm thấy chính xác, trả về toàn bộ dữ liệu làm context (full)
23|   return KNOWLEDGE_BASE.map(chunk => chunk.content).join("\n\n");
24| }
25|

```

File: utils/translations.ts (162 lines, 11.0 KB)

```

1| type Lang = 'en' | 'vn'
2|
3| const translations: Record<string, { en: string; vn: string }> = {
4|   // Navigation
5|   'nav.home': { en: 'Home', vn: 'Trang chủ' },
6|   'nav.back': { en: 'Back', vn: 'Thoát' },
7|   'nav.backToBuilder': { en: 'Back to Builder', vn: 'Quay lại Builder' },
8|   'nav.backToHome': { en: 'Back to home', vn: 'Quay lại trang chủ' },
9|
10|   // Auth
11|   'auth.login': { en: 'Login', vn: 'Đăng nhập' },
12|   'auth.register': { en: 'Register', vn: 'Đăng ký' },
13|   'auth.logout': { en: 'Logout', vn: 'Đăng xuất' },
14|   'auth.email': { en: 'Email', vn: 'Email' },
15|   'auth.password': { en: 'Password', vn: 'Mật khẩu' },
16|   'auth.confirmPassword': { en: 'Confirm Password', vn: 'Xác nhận mật khẩu' },
17|   'auth.fullName': { en: 'Full Name', vn: 'Họ và tên' },
18|   'auth.googleLogin': { en: 'Continue with Google', vn: 'Tiếp tục với Google' },
19|   'auth.googleNote': { en: 'First login: Email auto-saved, just enter name + role. No password needed!', vn: '...'},
20|   'auth.welcomeBack': { en: 'Welcome back! Login to continue learning.', vn: 'Chào mừng trở lại! Đăng nhập để tiếp tục học tập.' },
21|   'auth.noAccount': { en: 'Don't have an account?', vn: 'Chưa có tài khoản?' },
22|   'auth.hasAccount': { en: 'Already have an account?', vn: 'Đã có tài khoản?' },
23|   'auth.registerNow': { en: 'Register now', vn: 'Đăng ký ngay' },
24|   'auth.forgotPassword': { en: 'Forgot password?', vn: 'Quên mật khẩu?' },
25|   'auth.orViaEmail': { en: 'Or via email', vn: 'Hoặc qua email' },
26|   'auth.adminLogin': { en: 'Login as Admin', vn: 'Đăng nhập với vai trò Admin' },
27|   'auth.adminPassword': { en: 'Admin Password', vn: 'Mật khẩu Admin' },
28|   'auth.enterAdminPass': { en: 'Enter admin password', vn: 'Nhập mật khẩu Admin' },
29|   'auth.loginSuccess': { en: 'Login successful!', vn: 'Đăng nhập thành công!' },
30|   'auth.loginFailed': { en: 'Email or password is incorrect', vn: 'Email hoặc mật khẩu không đúng' },
31|   'auth.processing': { en: 'Processing...', vn: 'Đang xử lý...' },
32|
33|   // Builder
34|   'builder.title': { en: 'PC Master Builder', vn: 'PC Master Builder' },
35|   'builder.freePractice': { en: 'Free Practice', vn: 'Luyện Tập Miễn Phí' },
36|   'builder.practiceMode': { en: 'Practice Mode', vn: 'Luyện Tập' },
37|   'builder.lecture': { en: 'Lecture Course', vn: 'Bài Giảng' },
38|   'builder.marketplace': { en: 'Marketplace', vn: 'Chợ Hàng' },
39|   'builder.multiplayer': { en: '2-Player Versus', vn: '2 Người Chơi' },
40|   'builder.exams': { en: 'Exams', vn: 'Kiểm Tra' },
41|   'builder.challenge': { en: 'Daily Challenge', vn: 'Thử Thách' },
42|   'builder.components': { en: 'Component Library', vn: 'Thư Viện Linh Kiện' },
43|   'builder.checklist': { en: 'Checklist', vn: 'Nhật Ký' },
44|
45|   // Career Build
46|   'career.title': { en: 'Career & PC Build', vn: 'Nghĩa Nghiệp & Cấu Hình' },
47|   'career.subtitle': { en: 'Select your dream career, AI will suggest the best PC build for you!', vn: 'Chọn nghề nghiệp bạn thích, AI sẽ đề xuất cấu hình PC phù hợp nhất!' },
48|   'career.orType': { en: 'Or type your dream:', vn: 'Hoặc nhập ước mơ của bạn:' },
49|   'career.placeholder': { en: 'e.g. I want to be an AI engineer, game developer', vn: 'Ví dụ: Tôi muốn trở thành kỹ sư AI, nhà phát triển game' },
50|   'career.suggest': { en: 'Suggest', vn: 'Gợi ý' },
51|   'career.analyzing': { en: 'AI is analyzing your career...', vn: 'AI đang phân tích sự nghiệp của bạn...' },
52|   'career.suggestFor': { en: 'Suggested Build For', vn: 'Cấu Hình Đề Xuất Cho' },
53|   'career.totalCost': { en: 'Estimated Total Cost', vn: 'Tổng chi phí ước tính' },
54|   'career.startBuild': { en: 'Start assembling this config!', vn: 'Bắt đầu lắp ráp cấu hình này!' },
55|   'career.chooseOther': { en: 'Choose another career', vn: 'Chọn nghề nghiệp khác' },
56|   'career.tips': { en: 'Tip:', vn: 'Mẹo:' },
57|
58|   // Invoice
59|   'invoice.title': { en: 'Invoice Scanner & Price Check', vn: 'Quét Hóa Đơn & Kiểm Tra Giá' },
60|   'invoice.subtitle': { en: 'Upload an invoice photo, AI will detect items and compare with market prices!', vn: 'Upload an invoice photo, AI will detect items and compare with market prices!' },
61|   'invoice.drop': { en: 'Drop invoice image here', vn: 'Kéo thả ảnh hóa đơn vào đây' },
62|   'invoice.orclick': { en: 'Or click to select file', vn: 'Hoặc nhấp để chọn tệp' }

```



```

63 |         'invoice.or': { en: 'or', vn: 'hoặc' },
64 |         'invoice.camerarai': { en: 'Capture Invoice with AI', vn: 'Chụp ảnh H...'},
65 |         'invoice.scannerAI': { en: 'Scan Invoice with AI', vn: 'Quét hóa B...'},
66 |         'invoice.scanning': { en: 'Scanning image', vn: 'Quét ảnh kha...'},
67 |         'invoice.scannerDesc': { en: 'AI is detecting components and comparing p...', vn: 'Trình phân tích và so sánh các thành phần' },
68 |         'invoice.chooseOther': { en: 'Choose other image', vn: 'Chọn ảnh khác' },
69 |         'invoice.results': { en: 'Price Comparison Results', vn: 'Kết Quả So S...'},
70 |         'invoice.totalInvoice': { en: 'Invoice Total', vn: 'Tổng hóa đơn' },
71 |         'invoice.marketPrice': { en: 'Market Price', vn: 'Giá thị trường' },
72 |         'invoice.diff': { en: 'Difference', vn: 'Chênh lệch' },
73 |         'invoice.diffPct': { en: 'Diff %', vn: '% chênh' },
74 |         'invoice.aiverdict': { en: 'AI Verdict', vn: 'Ánh giá AI' },
75 |         'invoice.reasonable': { en: 'Reasonable', vn: 'Hợp lý' },
76 |         'invoice.expensive': { en: 'Slightly expensive', vn: 'Hơi đắt' },
77 |         'invoice.overpriced': { en: 'Overpriced', vn: 'Chật chẽm' },
78 |         'invoice.manualEntry': { en: 'Manual Entry', vn: 'Nhập tay' },
79 |         'invoice.manualDesc': { en: 'AI could not detect the image. Please enter component details manually.', vn: '...' },
80 |         'invoice.addItem': { en: '+ Add component', vn: '+ Thêm linh kiện' },
81 |         'invoice.compare': { en: 'Compare prices', vn: 'So sánh giá' },
82 |         'invoice.cancel': { en: 'Cancel', vn: 'Hủy' },
83 |         'invoice.scanAnother': { en: 'Scan another invoice', vn: 'Quét hóa đơn khác' },
84 |         'invoice.top': { en: 'Back to top', vn: 'Lên đầu trang' },
85 |
86 | // Hand Tracking
87 |         'handtracking.off': { en: 'OFF', vn: 'TẮT' },
88 |         'handtracking.on': { en: 'ON', vn: 'BẬT' },
89 |         'handtracking.toggle': { en: 'Toggle', vn: 'B - t / T - t' },
90 |         'handtracking.label': { en: 'Hand Tracking', vn: 'Hand Tracking' },
91 |         'handtracking.waiting': { en: 'Waiting for players...', vn: 'Chờ chờ' },
92 |         'handtracking.raiseHand': { en: 'Raise your hand high!', vn: 'Giơ tay cao' },
93 |         'handtracking.ready': { en: 'Ready!', vn: 'Sẵn sàng!' },
94 |
95 | // Multiplayer
96 |         'multiplayer.title': { en: 'MULTIPLAYER MODE', vn: 'CHƠI 2 NGƯỜI CH...'},
97 |         'multiplayer.instruction': { en: 'Player 1 = Left side of screen. Player 2 = Right side of screen (Each pla...', vn: 'Hướng dẫn chơi' },
98 |         'multiplayer.startMatch': { en: 'START MATCH', vn: 'BẮT ĐẦU THI ĐUA' },
99 |         'multiplayer.readHint': { en: 'Both players raise BOTH hands to be re...', vn: 'Cả hai người chơi phải giơ cả hai tay để được nhận lại' },
100 |         'multiplayer.p1Win': { en: 'PLAYER 1 WINS!', vn: 'P1 THẮNG!' },
101 |         'multiplayer.p2Win': { en: 'PLAYER 2 WINS!', vn: 'P2 THẮNG!' },
102 |         'multiplayer.p1Label': { en: 'PLAYER 1', vn: 'NGƯỜI CHƠI 1' },
103 |         'multiplayer.p2Label': { en: 'PLAYER 2', vn: 'NGƯỜI CHƠI 2' },
104 |
105 | // VR 3D
106 |         'vr.title': { en: 'VR MODE', vn: 'CHƠI VR' },
107 |         'vr.explore': { en: 'Explore', vn: 'Khám phá' },
108 |         'vr.build': { en: 'Start Assembly', vn: 'Bắt đầu lắp ráp' },
109 |         'vr.building': { en: 'Building...', vn: 'Đang lắp ráp' },
110 |         'vr.wasdHint': { en: 'WASD to move, webcam to look around', vn: 'WASD để di chuyển, webc...'},
111 |         'vr.headActive': { en: 'Webcam head tracking active', vn: 'Webcam theo dõi đầu đang hoạt động' },
112 |         'vr.mouseHint': { en: 'Move mouse to control hand', vn: 'Di chuyển chuột để điều khiển tay' },
113 |         'vr.wasdMove': { en: 'Move', vn: 'Di chuyển' },
114 |         'vr.wasdLook': { en: 'Head tracking to look', vn: 'Quay đầu nhìn' },
115 |         'vr.parts': { en: 'Parts', vn: 'Linh kiện' },
116 |         'vr.missingParts': { en: 'Missing parts!', vn: 'Thiếu linh kiện!' },
117 |         'vr.systemOnline': { en: 'System Online!', vn: 'Hệ thống hoạt động!' },
118 |         'vr.powerOn': { en: 'POWER ON', vn: 'KHỞI ĐỘNG' },
119 |         'vr.enter': { en: 'ENTER VR WORKSHOP', vn: 'Vào Phòng VR' },
120 |         'vr.welcome': { en: 'Build your PC in a VR workshop. Use WASD to move, webcam to look around, drag parts int...', vn: 'Xây dựng PC của bạn trong phòng thí nghiệm VR. Dùng WASD để di chuyển, webc...'},
121 |         'vr.step1': { en: 'Toggle Explore to walk around the room (WASD + head)', vn: 'Chuyển đổi Khám phá để đi bộ quanh phòng (WASD + đầu)' },
122 |         'vr.step2': { en: 'Click Start Assembly to build your PC', vn: 'Nhấn Bắt đầu Lắp ráp để xây dựng PC của bạn' },
123 |         'vr.step3': { en: 'Drag parts into the case - press POWER!', vn: 'Kéo các thành phần vào vỏ máy - nhấn KHỞI ĐỘNG' },
124 |
125 | // Common
126 |         'common.loading': { en: 'Loading...', vn: 'Đang tải...' },
127 |         'common.error': { en: 'Error', vn: 'Lỗi' },
128 |         'common.success': { en: 'Success', vn: 'Thành công' },
129 |         'common.save': { en: 'Save', vn: 'Lưu' },
130 |         'common.cancel': { en: 'Cancel', vn: 'Hủy' },
131 |         'common.confirm': { en: 'Confirm', vn: 'Xác nhận' },
132 |         'common.search': { en: 'Search', vn: 'Tìm kiếm' },
133 |         'common.noData': { en: 'No data', vn: 'Không có dữ liệu' },
134 |
135 | // Profile
136 |         'profile.complete': { en: 'Complete Profile', vn: 'Hoàn thiện hồ sơ' },
137 |         'profile.emailAuto': { en: 'Email auto-saved from Google account', vn: 'Email tự động lưu từ tài khoản Google' },
138 |         'profile.role': { en: 'What is your role?', vn: 'Bạn tham gia với vai trò gì?' },
139 |         'profile.student': { en: 'Student', vn: 'Học viên' },
140 |         'profile.teacher': { en: 'Teacher', vn: 'Giáo viên' },
141 |         'profile.parent': { en: 'Parent', vn: 'Phụ huynh' },
142 |         'profile.finish': { en: 'Complete', vn: 'Hoàn tất' },
143 |         'profile.nextTimeAuto': { en: 'Next login with Google will go straight in without re-entering info.', vn: 'Lần đăng nhập tiếp theo với Google sẽ đi thẳng vào mà không cần nhập lại thông tin.' },
144 |         'profile.dob': { en: 'Date of Birth', vn: 'Ngày sinh' },
145 |         'profile.province': { en: 'Province / City', vn: 'Tỉnh / Thành phố' },
146 |         'profile.school': { en: 'School', vn: 'Trường học' },
147 |         'profile.schoolName': { en: 'School name (optional)', vn: 'Tên trường (tùy chọn)' },
148 |
149 | }
150 |
151 | export function t(key: string, lang: Lang): string {
152 |     const entry = translations[key]
153 |     if (!entry) return key
154 |     return entry[lang]
155 | }
156 |
157 | export function useT(lang: Lang) {
158 |     return (key: string) => t(key, lang)
159 | }
160 |
161 | export type { Lang }
162 |

```

File: `utils/word-parser.ts` (108 lines, 3.5 KB)

```

1 | export function parseQuestionsFromText(text: string) {
2 |     const lines = text.split('\n').filter(l => l.trim())
3 |     const questions: {
4 |         content: string
5 |         type: 'single_choice' | 'true_false' | 'fill_blank'
6 |         points: number
7 |         explanation: string
8 |         options: { content: string; isCorrect: boolean }[]
9 |     }[] = []
10 |
11 |     let currentQuestion: any = null
12 |     let currentOptions: { content: string; isCorrect: boolean }[] = []
13 |     let correctAnswerText = ''
14 |
15 |     for (const line of lines) {
16 |         const trimmed = line.trim()
17 |
18 |         const questionMatch = trimmed.match(/^(?:Câu)?(\d+)[.])\s*(.+)/i)
19 |         if (questionMatch) {
20 |             if (currentQuestion && currentOptions.length > 0) {

```

```

21 |         currentQuestion.options = currentOptions
22 |         questions.push(currentQuestion)
23 |     }
24 |     currentOptions = []
25 |     correctAnswerText = ''
26 |     currentQuestion = {
27 |         content: questionMatch[2].trim(),
28 |         type: 'single_choice',
29 |         points: 10,
30 |         explanation: '',
31 |         options: []
32 |     }
33 |     continue
34 | }
35 |
36 | const optionMatch = trimmed.match(/^( [A-E] ) [ . ] \ ] \ s * ( . * ) / )
37 | if (optionMatch && currentQuestion) {
38 |     currentOptions.push({
39 |         content: optionMatch[2].trim(),
40 |         isCorrect: false
41 |     })
42 |     if (optionMatch[1] === ' ') {
43 |         currentOptions[currentOptions.length - 1].isCorrect = true
44 |     }
45 |     continue
46 | }
47 |
48 | const isTrueFalse = trimmed.match(/^( úng | Sai | True | False ) \ b / i )
49 | if (isTrueFalse && currentQuestion && currentOptions.length === 0) {
50 |     currentQuestion.type = 'true_false'
51 |     currentOptions.push(
52 |         { content: ' úng ', isCorrect: trimmed.match(/^( úng | True ) \ b / i ) ?
53 |         { content: 'Sai', isCorrect: trimmed.match(/^(Sai|False)\b/i) ? true : false }
54 |     )
55 |     continue
56 | }
57 |
58 | const correctMatch = trimmed.match(/^( ? : á p \ s * án | Correct | Answer ) \ s * / )
59 | if (correctMatch && currentQuestion) {
60 |     correctAnswerText = correctMatch[1].toUpperCase()
61 |     const correctIndices: number[] = []
62 |     const letters = correctAnswerText.split('')
63 |     for (const letter of letters) {
64 |         const idx = letter.charCodeAt(0) - 65
65 |         if (idx >= 0 && idx < currentOptions.length) {
66 |             correctIndices.push(idx)
67 |         }
68 |     }
69 |     for (let i = 0; i < currentOptions.length; i++) {
70 |         currentOptions[i].isCorrect = correctIndices.includes(i)
71 |     }
72 |     if (correctIndices.length > 1) {
73 |         currentQuestion.type = 'multiple_choice'
74 |     }
75 |     continue
76 | }
77 |
78 | const fillBlankMatch = trimmed.match(/^( i Á n \ s * ( ? : v à o \ s * c h x \ s * t r Ñ n g |
79 | if (fillBlankMatch && currentQuestion) {
80 |     currentQuestion.type = 'fill_blank'
81 |     currentQuestion.options = [{ content: fillBlankMatch[1].trim(), isCorrect: true }]
82 |     continue
83 | }
84 |
85 | if (currentQuestion && !questionMatch && !optionMatch && trimmed) {
86 |     if (trimmed.match(/^( ? : G i Ē i \ s * t h i c h | E x p l a n a t i o n ) \ s * [ : = ] / i ) ) {
87 |         currentQuestion.explanation = trimmed.replace(/^( ? : G i Ē i \ s * t h i c h |
88 |     }
89 | }
90 | }
91 |
92 | if (currentQuestion && currentOptions.length > 0) {
93 |     currentQuestion.options = currentOptions
94 |     questions.push(currentQuestion)
95 | }
96 |
97 | return questions
98 | }
99 |
100 | export async function parseDocx(file: File): Promise<{ text: string; questions: Return<typeof parseQuestio...
101 |     const mammoth = await import('mammoth')
102 |     const arrayBuffer = await file.arrayBuffer()
103 |     const result = await mammoth.extractRawText({ arrayBuffer })
104 |     const text = result.value
105 |     const questions = parseQuestionsFromText(text)
106 |     return { text, questions }
107 | }
108 |

```

File: types/pc-components.ts (227 lines, 6.1 KB)

```

1 | export type ComponentCategory =
2 |     | 'CPU' | 'GPU' | 'MAINBOARD' | 'RAM' | 'PSU' | 'CASE' | 'COOLER' | 'SSD' | 'HDD';
3 |
4 | export type SocketType =
5 |     | 'LGA1700' | 'LGA1851' | 'LGA1200' | 'LGA1151'
6 |     | 'AM5' | 'AM4' | 'STR5' | 'STRX4';
7 |
8 | export type FormFactor = 'ATX' | 'mATX' | 'ITX' | 'eATX' | 'Mini-ITX' | 'XL-ATX';
9 | export type DDRType = 'DDR4' | 'DDR5';
10 | export type PCIeGen = '3.0' | '4.0' | '5.0';
11 | export type PSUEfficiency =
12 |     | '80plus_white' | '80plus_bronze' | '80plus_silver'
13 |     | '80plus_gold' | '80plus_platinum' | '80plus_titanium';
14 | export type PSUSubModular = 'non_modular' | 'semi_modular' | 'full_modular';
15 | export type CoolerType = 'air' | 'aio' | 'custom_loop';
16 | export type StorageInterface = 'm2_nvme' | 'm2_sata' | 'sata_ssd' | 'sata_hdd';
17 | export type Severity = 'error' | 'warning' | 'info' | 'success';
18 |
19 | interface BaseProduct {
20 |     id: string;
21 |     category: ComponentCategory;
22 |     brand: string;
23 |     model: string;
24 |     full_name: string;
25 |     price_vnd: number;
26 |     image_url?: string;
27 |     image_angles?: string[];
28 |     video_url?: string;
29 |     release_year?: number;
30 | }
31 |
32 | export interface CPUProduct extends BaseProduct {

```

```

33 | category: 'CPU';
34 | cpu_socket: SocketType;
35 | cpu_tdp_watts: number;
36 | cpu_cores: number;
37 | cpu_threads: number;
38 | cpu_base_ghz: number;
39 | cpu_boost_ghz: number;
40 | cpu_l3_cache_mb: number;
41 | cpu_has_igpu: boolean;
42 | cpu_igpu_model?: string;
43 | cpu_ram_type: DDRType;
44 | cpu_max_ram_mhz: number;
45 | cpu_generation: string;
46 | cpu_cinebench_r23_single?: number;
47 | cpu_cinebench_r23_multi?: number;
48 | }
49 |
50 | export interface GPUProduct extends BaseProduct {
51 |   category: 'GPU';
52 |   gpu_vram_gb: number;
53 |   gpu_tdp_watts: number;
54 |   gpu_pcie_gen: PCIEGen;
55 |   gpu_length_mm: number;
56 |   gpu_width_mm?: number;
57 |   gpu_height_mm?: number;
58 |   gpu_power_connectors: { type: string; qty: number }[];
59 |   gpu_benchmark_1080p?: number;
60 |   gpu_display_outputs: string[];
61 |   gpu_recommended_psu: number;
62 | }
63 |
64 | export interface MainboardProduct extends BaseProduct {
65 |   category: 'MAINBOARD';
66 |   mb_socket: SocketType;
67 |   mb_chipset: string;
68 |   mb_form_factor: FormFactor;
69 |   mb_ram_slots: number;
70 |   mb_max_ram_gb: number;
71 |   mb_ram_type: DDRType;
72 |   mb_max_ram_mhz: number;
73 |   mb_m2_slots: number;
74 |   mb_m2_pcie_gens: PCIEGen[];
75 |   mb_pcie_x16_slots: number;
76 |   mb_pcie_x16_gens: PCIEGen[];
77 |   mb_sata_ports: number;
78 |   mb_has_wifi: boolean;
79 |   mb_wifi_version?: string;
80 |   mb_usb_ports: { type: string; qty: number }[];
81 |   mb_front_panel_headers: { type: string; pin: number }[];
82 | }
83 |
84 | export interface PSUProduct extends BaseProduct {
85 |   category: 'PSU';
86 |   psu_wattage: number;
87 |   psu_efficiency: PSUEfficiency;
88 |   psu_modular: PSUSubModular;
89 |   psu_connectors: { type: string; qty: number }[];
90 |   psu_has_12vhpwr: boolean;
91 |   psu_length_mm: number;
92 |   psu_fan_size_mm?: number;
93 | }
94 |
95 | export interface CaseProduct extends BaseProduct {
96 |   category: 'CASE';
97 |   case_supported_forms: FormFactor[];
98 |   case_max_gpu_length_mm: number;
99 |   case_max_cooler_height_mm: number;
100 | case_max_psu_length_mm: number;
101 | case_drive_bays: { type: string; qty: number }[];
102 | case_fan_slots: { size: number; qty: number; type: string }[];
103 | case_supports_aio: number[];
104 | case_mesh_front: boolean;
105 | case_has_cable_management: boolean;
106 | case_side_panel: string;
107 | case_weight_kg?: number;
108 | }
109 |
110 | export interface CoolerProduct extends BaseProduct {
111 |   category: 'COOLER';
112 |   cooler_max_tdp_support: number;
113 |   cooler_height_mm: number;
114 |   cooler_supported_sockets: SocketType[];
115 |   cooler_type: CoolerType;
116 |   cooler_aio_size_mm?: number;
117 |   cooler_fan_rpm_max?: number;
118 |   cooler_fan_noise_dba?: number;
119 |   cooler_fan_qty?: number;
120 |   cooler_has_rgb: boolean;
121 | }
122 |
123 | export interface RAMProduct extends BaseProduct {
124 |   category: 'RAM';
125 |   ram_ddr_type: DDRType;
126 |   ram_speed_mhz: number;
127 |   ram_cl_latency: number;
128 |   ram_capacity_gb: number;
129 |   ram_kit_count: number;
130 |   ram_has_xmp: boolean;
131 |   ram_has_expo: boolean;
132 |   ram_voltage_v?: number;
133 |   ram_height_mm?: number;
134 | }
135 |
136 | export interface SSDProduct extends BaseProduct {
137 |   category: 'SSD';
138 |   ssd_interface: StorageInterface;
139 |   ssd_pcie_gen?: PCIEGen;
140 |   ssd_read_mbps: number;
141 |   ssd_write_mbps: number;
142 |   ssd_has_dram_cache: boolean;
143 |   ssd_capacity_gb: number;
144 |   ssd_form_factor: string;
145 |   ssd_nand_type?: string;
146 |   ssd_tbw?: number;
147 |   ssd_max_seq_read?: number;
148 |   ssd_max_seq_write?: number;
149 |   ssd_max_rand_read?: number;
150 |   ssd_max_rand_write?: number;
151 | }
152 |
153 | export type AnyProduct =
154 |   | CPUProduct | GPUProduct | MainboardProduct | PSUProduct
155 |   | CaseProduct | CoolerProduct | RAMProduct | SSDProduct;
156 |

```

```

157 export type CompatibilityIssueType =
158   | 'socket_mismatch'
159   | 'ram_type_mismatch'
160   | 'gpu_too_long'
161   | 'psu_insufficient'
162   | 'cooler_insufficient'
163   | 'cooler_too_tall'
164   | 'cooler_socket_mismatch'
165   | 'form_factor_mismatch'
166   | 'no_display_output'
167   | 'too_many_ram'
168   | 'not_enough_m2_slots'
169   | 'bottleneck'
170   | 'budget_exceeded'
171   | 'success';
172
173 export interface CompatibilityIssue {
174   severity: Severity;
175   check_type: CompatibilityIssueType;
176   message: string;
177   fix: string;
178 }
179
180 export interface Build {
181   id?: string;
182   name: string;
183   cpu: CPUProduct | null;
184   gpu: GPUProduct | null;
185   mainboard: MainboardProduct | null;
186   ram: RAMProduct[];
187   psu: PSUProduct | null;
188   case: CaseProduct | null;
189   cooler: CoolerProduct | null;
190   ssd: SSDProduct[];
191   hdd?: SSDProduct[];
192   total_price: number;
193   total_tdp: number;
194   compatibility_score: number;
195   compatibility_issues: CompatibilityIssue[];
196   created_at?: string;
197 }
198
199 export interface BuildScenario {
200   id: string;
201   name: string;
202   name_vn: string;
203   description: string;
204   description_vn: string;
205   icon: string;
206   difficulty: 'easy' | 'medium' | 'hard' | 'expert';
207   budget_min: number;
208   budget_max: number;
209   requirements: {
210     min_cores?: number;
211     min_gpu_power?: number;
212     min_ram_gb?: number;
213     min_storage_gb?: number;
214     require_igpu?: boolean;
215     use_case?: string;
216     min_cinebench_multi?: number;
217     max_tdp?: number;
218   };
219   weight_components: {
220     cpu: number;
221     gpu: number;
222     ram: number;
223     storage: number;
224     psu: number;
225   };
226 }
227

```

File: types/quiz.ts (63 lines, 1.3 KB)

```

=====
1 export type QuestionType = 'multiple_choice' | 'true_false' | 'fill_blank';
2 export type AttemptStatus = 'in_progress' | 'submitted' | 'timed_out';
3
4 export interface QuestionBank {
5   id: string;
6   teacher_id: string;
7   title: string;
8   subject?: string;
9   created_at: string;
10 }
11
12 export interface Question {
13   id: string;
14   bank_id: string;
15   content: string;
16   type: QuestionType;
17   points: number;
18   order: number;
19   created_at: string;
20   options?: QuestionOption[];
21 }
22
23 export interface QuestionOption {
24   id: string;
25   question_id: string;
26   content: string;
27   is_correct?: boolean; // Optional because we hide it from students
28   created_at: string;
29 }
30
31 export interface Quiz {
32   id: string;
33   lesson_id?: string;
34   assignment_id?: string;
35   title: string;
36   time_limit_minutes: number;
37   passing_score: number;
38   randomize: boolean;
39   max_attempts: number;
40   created_at: string;
41   questions?: Question[];
42 }
43
44 export interface QuizAttempt {
45   id: string;
46   quiz_id: string;
47   student_id: string;
48   started_at: string;
49   submitted_at?: string;

```

```

50 |   score: number;
51 |   status: AttemptStatus;
52 | }
53 |
54 | export interface QuizAnswer {
55 |   id: string;
56 |   attempt_id: string;
57 |   question_id: string;
58 |   selected_option_id?: string;
59 |   text_answer?: string;
60 |   is_correct: boolean;
61 |   points_earned: number;
62 | }
63 |

```

File: hooks/hooks/useClassroomUpdates.ts (49 lines, 1.2 KB)

```

1 | 'use client';
2 |
3 | import { useEffect, useState } from 'react';
4 | import { createClient } from '@lib/supabase-ssr-client';
5 |
6 | export function useClassroomUpdates(teacherId: string) {
7 |   const [lastUpdate, setLastUpdate] = useState<Date | null>(null);
8 |   const supabase = createClient();
9 |
10 |   useEffect(() => {
11 |     if (!teacherId) return;
12 |
13 |     // 1. Subscribe to quiz attempts
14 |     const quizChannel = supabase
15 |       .channel('classroom-updates')
16 |       .on(
17 |         'postgres_changes',
18 |         {
19 |           event: '*',
20 |           schema: 'public',
21 |           table: 'quiz_attempts',
22 |         },
23 |         (payload) => {
24 |           console.log('Real-time Quiz Update:', payload);
25 |           setLastUpdate(new Date());
26 |         }
27 |       )
28 |       .on(
29 |         'postgres_changes',
30 |         {
31 |           event: '*',
32 |           schema: 'public',
33 |           table: 'lesson_progress',
34 |         },
35 |         (payload) => {
36 |           console.log('Real-time Progress Update:', payload);
37 |           setLastUpdate(new Date());
38 |         }
39 |       )
40 |       .subscribe();
41 |
42 |     return () => {
43 |       supabase.removeChannel(quizChannel);
44 |     };
45 |   }, [teacherId, supabase]);
46 |
47 |   return { lastUpdate };
48 | }
49 |

```

File: hooks/hooks/useDiscussion.ts (219 lines, 5.9 KB)

```

1 | 'use client'
2 |
3 | import { useState, useEffect } from 'react'
4 | import { createClientComponentClient } from '@supabase/auth-helpers-nextjs'
5 |
6 | export type Thread = {
7 |   id: string
8 |   lesson_id: string
9 |   author_id: string
10 |   title: string
11 |   body: string
12 |   type: 'question' | 'discussion' | 'announcement'
13 |   status: 'open' | 'resolved' | 'pinned' | 'closed'
14 |   upvote_count: number
15 |   reply_count: number
16 |   is_deleted: boolean
17 |   created_at: string
18 |   updated_at: string
19 |   author?: {
20 |     full_name: string
21 |     avatar_url: string
22 |     role: string
23 |   }
24 | }
25 |
26 | export type Reply = {
27 |   id: string
28 |   thread_id: string
29 |   parent_reply_id: string | null
30 |   author_id: string
31 |   body: string
32 |   is_teacher_answer: boolean
33 |   upvote_count: number
34 |   is_deleted: boolean
35 |   created_at: string
36 |   author?: {
37 |     full_name: string
38 |     avatar_url: string
39 |     role: string
40 |   }
41 | }
42 |
43 | export function useDiscussion(lessonId: string, activeTab: 'all' | 'unanswered' | 'pinned') {
44 |   const [threads, setThreads] = useState<Thread[]>([])
45 |   const [isLoading, setIsLoading] = useState(true)
46 |   const supabase = createClientComponentClient()
47 |
48 |   useEffect(() => {
49 |     const fetchThreads = async () => {
50 |       setIsLoading(true)
51 |       let query = supabase
52 |         .from('discussion_threads')

```

```

53 |     .select(`
54 |         *,
55 |         author:profiles(full_name, avatar_url, role)
56 |     `)
57 |     .eq('lesson_id', lessonId)
58 |     .eq('is_deleted', false)
59 |
60 |     if (activeTab === 'pinned') {
61 |         query = query.eq('status', 'pinned')
62 |     } else if (activeTab === 'unanswered') {
63 |         query = query.eq('reply_count', 0)
64 |     }
65 |
66 |     const { data, error } = await query.order('created_at', { ascending: false })
67 |     if (!error && data) {
68 |         setThreads(data as any[])
69 |     }
70 |     setIsLoading(false)
71 | }
72 |
73 | fetchThreads()
74 |
75 | // Realtime channel for new threads
76 | const channel = supabase
77 | .channel(`lesson-discussion-${lessonId}`)
78 | .on('postgres_changes', {
79 |     event: '*',
80 |     schema: 'public',
81 |     table: 'discussion_threads',
82 |     filter: `lesson_id=eq.${lessonId}`
83 | }, () => {
84 |     fetchThreads()
85 | })
86 | .subscribe()
87 |
88 | return () => {
89 |     supabase.removeChannel(channel)
90 | }
91 | }, [lessonId, activeTab, supabase])
92 |
93 | const createThread = async (title: string, body: string, type: string, authorId: string) => {
94 |     // Basic Rate Limit Check (5 threads per hour)
95 |     const oneHourAgo = new Date(Date.now() - 3600000).toISOString()
96 |     const { count } = await supabase
97 |     .from('discussion_threads')
98 |     .select('id', { count: 'exact', head: true })
99 |     .eq('author_id', authorId)
100 |     .gt('created_at', oneHourAgo)
101 |
102 |     if (count !== null && count >= 5) {
103 |         throw new Error('Bạn đã gửi bình luận quá 5 bài viết mỗi giờ.')
104 |     }
105 |
106 |     const { data, error } = await supabase
107 |     .from('discussion_threads')
108 |     .insert({
109 |         lesson_id: lessonId,
110 |         author_id: authorId,
111 |         title,
112 |         body,
113 |         type
114 |     })
115 |     .select()
116 |     .single()
117 |
118 |     if (error) throw error
119 |     return data
120 | }
121 |
122 | return { threads, isLoading, createThread }
123 | }
124 |
125 | export function useThreadReplies(threadId: string) {
126 |     const [replies, setReplies] = useState<Reply[]>([])
127 |     const supabase = createClientComponentClient()
128 |
129 |     useEffect(() => {
130 |         const fetchReplies = async () => {
131 |             const { data, error } = await supabase
132 |             .from('discussion_replies')
133 |             .select(`
134 |                 *,
135 |                 author:profiles(full_name, avatar_url, role)
136 |             `)
137 |             .eq('thread_id', threadId)
138 |             .eq('is_deleted', false)
139 |             .order('created_at', { ascending: true })
140 |
141 |             if (!error && data) {
142 |                 setReplies(data as any[])
143 |             }
144 |         }
145 |
146 |         fetchReplies()
147 |
148 |         // Realtime replies subscription
149 |         const channel = supabase
150 |         .channel(`thread-replies-${threadId}`)
151 |         .on('postgres_changes', {
152 |             event: 'INSERT',
153 |             schema: 'public',
154 |             table: 'discussion_replies',
155 |             filter: `thread_id=eq.${threadId}`
156 |         }, (payload) => {
157 |             const fetchNewReplyAuthor = async () => {
158 |                 const { data } = await supabase
159 |                 .from('profiles')
160 |                 .select('full_name, avatar_url, role')
161 |                 .eq('id', payload.new.author_id)
162 |                 .single()
163 |
164 |                 const newReply: Reply = {
165 |                     ...payload.new as Reply,
166 |                     author: data || { full_name: 'Anonymous', avatar_url: '', role: 'user' }
167 |                 }
168 |                 setReplies(prev => [...prev, newReply])
169 |             }
170 |             fetchNewReplyAuthor()
171 |         })
172 |         .subscribe()
173 |
174 |         return () => {
175 |             supabase.removeChannel(channel)
176 |         }

```

```

177 | }, [threadId, supabase])
178 |
179 | const addReply = async (body: string, authorId: string, parentReplyId: string | null = null) => {
180 |   // Rate limit check (20 replies per hour)
181 |   const oneHourAgo = new Date(Date.now() - 3600000).toISOString()
182 |   const { count } = await supabase
183 |     .from('discussion_replies')
184 |     .select('id', { count: 'exact', head: true })
185 |     .eq('author_id', authorId)
186 |     .gt('created_at', oneHourAgo)
187 |
188 |   if (count !== null && count >= 20) {
189 |     throw new Error('Bạn đã gửi bình luận quá nhiều.')
190 |   }
191 |
192 |   // Check if author is teacher
193 |   const { data: profile } = await supabase
194 |     .from('profiles')
195 |     .select('role')
196 |     .eq('id', authorId)
197 |     .single()
198 |
199 |   const isTeacher = profile?.role === 'teacher'
200 |
201 |   const { data, error } = await supabase
202 |     .from('discussion_replies')
203 |     .insert({
204 |       thread_id: threadId,
205 |       parent_reply_id: parentReplyId,
206 |       author_id: authorId,
207 |       body,
208 |       is_teacher_answer: isTeacher
209 |     })
210 |     .select()
211 |     .single()
212 |
213 |   if (error) throw error
214 |   return data
215 | }
216 |
217 | return { replies, addReply }
218 | }
219 |

```

File: hooks/hooks/useNotifications.ts (92 lines, 2.7 KB)

```

1 | 'use client'
2 |
3 | import { useState, useEffect } from 'react'
4 | import { createClientComponentClient } from '@supabase/auth-helpers-nextjs'
5 | import { toast } from 'react-hot-toast'
6 |
7 | export type Notification = {
8 |   id: string
9 |   user_id: string
10 |   type: string
11 |   title: string
12 |   body: string
13 |   data: any
14 |   action_url: string
15 |   is_read: boolean
16 |   created_at: string
17 | }
18 |
19 | export function useNotifications(userId?: string, options?: { filterTypes?: string[] }) {
20 |   const [notifications, setNotifications] = useState<Notification[]>([])
21 |   const [unreadCount, setUnreadCount] = useState(0)
22 |   const supabase = createClientComponentClient()
23 |
24 |   useEffect(() => {
25 |     if (!userId) return;
26 |
27 |     const loadNotifications = async () => {
28 |       let query = supabase
29 |         .from('notifications')
30 |         .select('*')
31 |         .eq('user_id', userId)
32 |         .order('created_at', { ascending: false })
33 |         .limit(20)
34 |
35 |       if (options?.filterTypes && options.filterTypes.length > 0) {
36 |         query = query.in('type', options.filterTypes)
37 |       }
38 |
39 |       const { data } = await query
40 |       if (data) {
41 |         setNotifications(data as Notification[])
42 |         setUnreadCount(data.filter(n => !n.is_read).length)
43 |       }
44 |     }
45 |     loadNotifications()
46 |
47 |     const channel = supabase
48 |       .channel('notifications_channel')
49 |       .on('postgres_changes', {
50 |         event: 'INSERT',
51 |         schema: 'public',
52 |         table: 'notifications',
53 |         filter: `user_id=eq.${userId}`
54 |       }, (payload) => {
55 |         const newNotif = payload.new as Notification
56 |         if (options?.filterTypes && !options.filterTypes.includes(newNotif.type)) {
57 |           return
58 |         }
59 |         setNotifications(prev => [...prev, newNotif])
60 |         setUnreadCount(prev => prev + 1)
61 |         toast(newNotif.title, { icon: '🔔' })
62 |       })
63 |     .subscribe()
64 |
65 |     return () => {
66 |       supabase.removeChannel(channel)
67 |     }
68 |   }, [userId, supabase, options?.filterTypes?.join(',')])
69 |
70 |   const markAsRead = async (notificationId: string) => {
71 |     setNotifications(prev =>
72 |       prev.map(n => n.id === notificationId ? { ...n, is_read: true } : n)
73 |     )
74 |     setUnreadCount(prev => Math.max(0, prev - 1))
75 |     await supabase.from('notifications').update({ is_read: true }).eq('id', notificationId)
76 |   }
77 |

```

```

78 | const markAllAsRead = async () => {
79 |   const unreadIds = notifications.filter(n => !n.is_read).map(n => n.id)
80 |   if (unreadIds.length === 0) return
81 |
82 |   setNotifications(prev => prev.map(n => ({ ...n, is_read: true })))
83 |   setUnreadCount(0)
84 |
85 |   await supabase.from('notifications')
86 |     .update({ is_read: true })
87 |     .in('id', unreadIds)
88 | }
89 |
90 | return { notifications, unreadCount, markAsRead, markAllAsRead }
91 | }
92 |

```

File: hooks/hooks/useSessionTimeout.ts (41 lines, 1.1 KB)

```

1 | 'use client'
2 |
3 | import { useEffect, useCallback } from 'react'
4 | import { useRouter } from 'next/navigation'
5 | import { logout } from '@lib/auth-actions'
6 |
7 | export function useSessionTimeout(timeoutMinutes: number = 30) {
8 |   const router = useRouter()
9 |   const timeoutMs = timeoutMinutes * 60 * 1000
10 |
11 |   const handleLogout = useCallback(async () => {
12 |     await logout()
13 |     router.push('/login?session=expired')
14 |   }, [router])
15 |
16 |   useEffect(() => {
17 |     let timeoutId: NodeJS.Timeout
18 |
19 |     const resetTimer = () => {
20 |       if (timeoutId) clearTimeout(timeoutId)
21 |       timeoutId = setTimeout(handleLogout, timeoutMs)
22 |     }
23 |
24 |     // Events to reset the timer
25 |     const events = ['mousedown', 'keydown', 'scroll', 'touchstart']
26 |
27 |     events.forEach(event => {
28 |       window.addEventListener(event, resetTimer)
29 |     })
30 |
31 |     resetTimer() // Initialize timer
32 |
33 |     return () => {
34 |       if (timeoutId) clearTimeout(timeoutId)
35 |       events.forEach(event => {
36 |         window.removeEventListener(event, resetTimer)
37 |       })
38 |     }
39 |   }, [handleLogout, timeoutMs])
40 | }
41 |

```

File: hooks/useChildRealtime.ts (265 lines, 8.9 KB)

```

1 | // Path: hooks/useChildRealtime.ts
2 | 'use client'
3 | import { useEffect, useState, useCallback, useRef } from 'react'
4 | import { createClient } from '@lib/supabase-ssr-client'
5 | import { RealtimeChannel } from '@supabase/supabase-js'
6 |
7 | export interface RecentActivity {
8 |   id: string
9 |   type: 'lesson_completed' | 'lesson_started' | 'quiz_graded' | 'login'
10 |   title: string
11 |   timestamp: string
12 |   score?: number
13 |   lessonTitle?: string
14 | }
15 |
16 | export interface ChildRealtimeData {
17 |   isOnline: boolean
18 |   lastSeen: string | null
19 |   lessonsCompleted: number
20 |   totalLessonsStarted: number
21 |   totalTimeSeconds: number
22 |   averageQuizScore: number | null
23 |   recentActivities: RecentActivity[]
24 |   connectionStatus: 'connecting' | 'connected' | 'disconnected' | 'error'
25 | }
26 |
27 | export function useChildRealtime(studentId: string | null): ChildRealtimeData {
28 |   const [data, setData] = useState<ChildRealtimeData>({
29 |     isOnline: false,
30 |     lastSeen: null,
31 |     lessonsCompleted: 0,
32 |     totalLessonsStarted: 0,
33 |     totalTimeSeconds: 0,
34 |     averageQuizScore: null,
35 |     recentActivities: [],
36 |     connectionStatus: 'connecting',
37 |   })
38 |
39 |   const channelRef = useRef<RealtimeChannel | null>(null)
40 |   const supabase = createClient()
41 |
42 |   // Fetch initial data
43 |   const fetchInitialData = useCallback(async () => {
44 |     if (!studentId) return
45 |
46 |     setData(prev => ({ ...prev, connectionStatus: 'connecting' }))
47 |
48 |     try {
49 |       const [progressRes, quizRes, profileRes] = await Promise.all([
50 |         supabase
51 |           .from('lesson_progress')
52 |           .select('status, time_spent_seconds, completed_at, last_accessed')
53 |           .eq('student_id', studentId)
54 |           .order('last_accessed', { ascending: false }),
55 |         supabase
56 |           .from('quiz_attempts')
57 |           .select('score, max_score, submitted_at, quiz_title, status')
58 |           .eq('student_id', studentId)
59 |           .eq('status', 'graded')

```



```

60 |         .order('submitted_at', { ascending: false })
61 |         .limit(10),
62 |     supabase
63 |         .from('profiles')
64 |         .select('is_online, last_seen_at')
65 |         .eq('id', studentId)
66 |         .single(),
67 |     ])
68 |
69 |     const progress = progressRes.data ?? []
70 |     const quizzes = quizRes.data ?? []
71 |     const profile = profileRes.data
72 |
73 |     const completed = progress.filter(p => p.status === 'completed')
74 |     const totalTime = progress.reduce((sum, p) => sum + (p.time_spent_seconds ?? 0), 0)
75 |     const avgScore = quizzes.length > 0
76 |       ? quizzes.reduce((sum, q) => sum + (Number(q.score) ?? 0), 0) / quizzes.length
77 |       : null
78 |
79 |     // Build recent activities list
80 |     const activities: RecentActivity[] = []
81 |
82 |     // From lesson progress (last 5 completed/active)
83 |     progress.forEach(p => {
84 |       if (p.status === 'completed' && p.completed_at) {
85 |         activities.push({
86 |           id: `lesson-${p.completed_at}`,
87 |           type: 'lesson_completed',
88 |           title: 'Hoàn thành bài híc',
89 |           timestamp: p.completed_at,
90 |         })
91 |       } else if (p.status === 'in_progress' && p.last_accessed) {
92 |         activities.push({
93 |           id: `lesson-started-${p.last_accessed}`,
94 |           type: 'lesson_started',
95 |           title: 'ang híc bài',
96 |           timestamp: p.last_accessed,
97 |         })
98 |       }
99 |     })
100 |
101 |     // From quiz attempts (last 5 graded)
102 |     quizzes.forEach(q => {
103 |       if (q.submitted_at) {
104 |         activities.push({
105 |           id: `quiz-${q.submitted_at}`,
106 |           type: 'quiz_graded',
107 |           title: `K t qu : ${q.quiz_title}`,
108 |           timestamp: q.submitted_at,
109 |           score: Number(q.score) ?? 0,
110 |         })
111 |       }
112 |     })
113 |
114 |     // Sort activities descending by timestamp
115 |     activities.sort((a, b) =>
116 |       new Date(b.timestamp).getTime() - new Date(a.timestamp).getTime()
117 |     )
118 |
119 |     setData({
120 |       isOnline: profile?.is_online ?? false,
121 |       lastSeen: profile?.last_seen_at ?? null,
122 |       lessonsCompleted: completed.length,
123 |       totalLessonsStarted: progress.length,
124 |       totalTimeSeconds: totalTime,
125 |       averageQuizScore: avgScore,
126 |       recentActivities: activities.slice(0, 8),
127 |       connectionStatus: 'connecting', // will update after WebSocket connects
128 |     })
129 |   } catch (err) {
130 |     console.error('Error fetching initial data:', err)
131 |     setData(prev => ({ ...prev, connectionStatus: 'error' }))
132 |   }
133 | }, [studentId])
134 |
135 | useEffect(() => {
136 |   if (!studentId) {
137 |     setData({
138 |       isOnline: false,
139 |       lastSeen: null,
140 |       lessonsCompleted: 0,
141 |       totalLessonsStarted: 0,
142 |       totalTimeSeconds: 0,
143 |       averageQuizScore: null,
144 |       recentActivities: [],
145 |       connectionStatus: 'disconnected',
146 |     })
147 |     return
148 |   }
149 |
150 |   fetchInitialData()
151 |
152 |   // 1. Create a specific child monitoring realtime channel
153 |   const channelName = `parent-watch-${studentId}`
154 |   const channel = supabase.channel(channelName)
155 |
156 |   channelRef.current = channel
157 |
158 |   // 2. Subscribe to lesson_progress table changes
159 |   channel
160 |     .on('postgres_changes', {
161 |       event: '*',
162 |       schema: 'public',
163 |       table: 'lesson_progress',
164 |       filter: `student_id=eq.${studentId}`,
165 |     }, (payload) => {
166 |       if (payload.eventType === 'UPDATE' || payload.eventType === 'INSERT') {
167 |         const record = payload.new as any
168 |
169 |         setData(prev => {
170 |           const isComp = record.status === 'completed'
171 |           const newActivity: RecentActivity = {
172 |             id: `lesson-${record.lesson_id}-${Date.now()}`,
173 |             type: isComp ? 'lesson_completed' : 'lesson_started',
174 |             title: isComp ? 'Hoàn thành bài híc' : 'ang híc bài',
175 |             timestamp: new Date().toISOString(),
176 |           }
177 |
178 |           // Recalculate lessons completed & started
179 |           const completedCount = isComp ? prev.lessonsCompleted + 1 : prev.lessonsCompleted
180 |
181 |           return {
182 |             ...prev,
183 |             lessonsCompleted: completedCount,

```

```

184 |         totalTimeSeconds: prev.totalTimeSeconds + (record.time_spent_seconds ?? 0),
185 |         recentActivities: [newActivity, ...prev.recentActivities].slice(0, 8),
186 |     })
187 | })
188 | }
189 | })
190 |
191 | // 3. Subscribe to quiz_attempts table changes (when graded)
192 | .on('postgres_changes', {
193 |   event: '*',
194 |   schema: 'public',
195 |   table: 'quiz_attempts',
196 |   filter: `student_id=eq.${studentId}`,
197 | }, (payload) => {
198 |   if (payload.eventType === 'UPDATE' || payload.eventType === 'INSERT') {
199 |     const record = payload.new as any
200 |     if (record.status === 'graded') {
201 |       setData(prev => {
202 |         const newActivity: RecentActivity = {
203 |           id: `quiz-${record.id}-${Date.now()}`,
204 |           type: 'quiz_graded',
205 |           t i t l e : ` K ÷ t q u e : ${record.quiz_title}`,
206 |           timestamp: record.submitted_at ?? new Date().toISOString(),
207 |           score: Number(record.score) ?? 0,
208 |         }
209 |
210 |         // Recalculate average quiz score
211 |         const currentQuizActivities = prev.recentActivities.filter(a => a.type === 'quiz_graded')
212 |         const scores = currentQuizActivities.map(a => a.score ?? 0)
213 |         scores.unshift(Number(record.score) ?? 0)
214 |         const avgScore = scores.reduce((s, sc) => s + sc, 0) / scores.length
215 |
216 |         return {
217 |           ...prev,
218 |           averageQuizScore: avgScore,
219 |           recentActivities: [newActivity, ...prev.recentActivities].slice(0, 8),
220 |         }
221 |       })
222 |     }
223 |   }
224 | })
225 |
226 | // 4. Subscribe to profile table changes (online / active state)
227 | .on('postgres_changes', {
228 |   event: 'UPDATE',
229 |   schema: 'public',
230 |   table: 'profiles',
231 |   filter: `id=eq.${studentId}`,
232 | }, (payload) => {
233 |   const record = payload.new as any
234 |   setData(prev => ({
235 |     ...prev,
236 |     isOnline: record.is_online ?? false,
237 |     lastSeen: record.last_seen_at ?? prev.lastSeen,
238 |   })))
239 | })
240 |
241 | // 5. Connect and subscribe to channel updates
242 | .subscribe((status) => {
243 |   if (status === 'SUBSCRIBED') {
244 |     setData(prev => ({ ...prev, connectionStatus: 'connected' }))
245 |   } else if (status === 'CHANNEL_ERROR') {
246 |     setData(prev => ({ ...prev, connectionStatus: 'error' }))
247 |   } else if (status === 'CLOSED') {
248 |     setData(prev => ({ ...prev, connectionStatus: 'disconnected' }))
249 |   } else {
250 |     setData(prev => ({ ...prev, connectionStatus: 'connecting' }))
251 |   }
252 | })
253 |
254 | // 6. Cleanup channel on unmount or child selection change (Channel Cleanup Rule)
255 | return () => {
256 |   if (channelRef.current) {
257 |     supabase.removeChannel(channelRef.current)
258 |     channelRef.current = null
259 |   }
260 | }
261 | }, [studentId, fetchInitialData])
262 |
263 | return data
264 | }
265 |

```

File: hooks/useClassroomUpdates.ts (49 lines, 1.2 KB)

```

1 | 'use client';
2 |
3 | import { useEffect, useState } from 'react';
4 | import { createClient } from '@lib/supabase-ssr-client';
5 |
6 | export function useClassroomUpdates(teacherId: string) {
7 |   const [lastUpdate, setLastUpdate] = useState<Date | null>(null);
8 |   const supabase = createClient();
9 |
10 |   useEffect(() => {
11 |     if (!teacherId) return;
12 |
13 |     // 1. Subscribe to quiz attempts
14 |     const quizChannel = supabase
15 |       .channel('classroom-updates')
16 |       .on(
17 |         'postgres_changes',
18 |         {
19 |           event: '*',
20 |           schema: 'public',
21 |           table: 'quiz_attempts',
22 |         },
23 |         (payload) => {
24 |           console.log('Real-time Quiz Update:', payload);
25 |           setLastUpdate(new Date());
26 |         }
27 |       )
28 |       .on(
29 |         'postgres_changes',
30 |         {
31 |           event: '*',
32 |           schema: 'public',
33 |           table: 'lesson_progress',
34 |         },
35 |         (payload) => {
36 |           console.log('Real-time Progress Update:', payload);
37 |           setLastUpdate(new Date());
38 |         }

```

```

39 |     )
40 |     .subscribe();
41 |
42 |     return () => {
43 |       supabase.removeChannel(quizChannel);
44 |     };
45 |   }, [teacherId, supabase]);
46 |
47 |   return { lastUpdate };
48 | }
49 |

```

File: hooks/useComponentUpdates.ts (105 lines, 3.3 KB)

```

1 | 'use client';
2 |
3 | import { useEffect, useCallback } from 'react';
4 | import toast from 'react-hot-toast';
5 | import { createClient } from '@lib/supabase-client';
6 |
7 | interface UseComponentUpdatesOptions {
8 |   onComponentPublished?: (component: any) => void;
9 |   onComponentUpdated?: (component: any) => void;
10 |  onComponentDeleted?: (componentId: string) => void;
11 | }
12 |
13 | export function useComponentUpdates(options: UseComponentUpdatesOptions = {}) {
14 |   const subscribeToComponents = useCallback(() => {
15 |     const supabase = createClient();
16 |     const channel = supabase
17 |       .channel('components-updates')
18 |       .on(
19 |         'postgres_changes',
20 |         { event: 'INSERT', schema: 'public', table: 'components', filter: 'is_published=eq.true' },
21 |         (payload) => {
22 |           const { new: newComponent } = payload;
23 |           toast.success(`New component published: ${newComponent.name}`, {
24 |             duration: 5000,
25 |             icon: '🔍',
26 |           });
27 |           options.onComponentPublished?.(newComponent);
28 |         }
29 |       )
30 |       .on(
31 |         'postgres_changes',
32 |         { event: 'UPDATE', schema: 'public', table: 'components' },
33 |         (payload) => {
34 |           const { new: updatedComponent } = payload;
35 |           if (updatedComponent.is_published) {
36 |             toast(`Component specs updated: ${updatedComponent.name}`, {
37 |               duration: 5000,
38 |               icon: '🔄',
39 |             });
40 |             options.onComponentUpdated?.(updatedComponent);
41 |           }
42 |         }
43 |       )
44 |       .on(
45 |         'postgres_changes',
46 |         { event: 'DELETE', schema: 'public', table: 'components' },
47 |         (payload) => {
48 |           const { old: oldComponent } = payload;
49 |           toast.error(`Component removed: ${oldComponent.name}`, {
50 |             duration: 5000,
51 |             icon: '🗑️',
52 |           });
53 |           options.onComponentDeleted?.(oldComponent.id);
54 |         }
55 |       )
56 |       .subscribe();
57 |
58 |   return () => {
59 |     supabase.removeChannel(channel);
60 |   };
61 | }, [options]);
62 |
63 | useEffect(() => {
64 |   const unsubscribe = subscribeToComponents();
65 |   return unsubscribe;
66 | }, [subscribeToComponents]);
67 | }
68 |
69 | export function useComponentLock(componentId: string | null) {
70 |   const lockComponent = useCallback(async (userId: string) => {
71 |     if (!componentId) return false;
72 |     const res = await fetch('/api/components/lock', {
73 |       method: 'POST',
74 |       headers: { 'Content-Type': 'application/json' },
75 |       body: JSON.stringify({ componentId, userId, action: 'lock' }),
76 |     });
77 |     const data = await res.json();
78 |     if (!data.locked) {
79 |       toast.error(`${data.message}: ${data.lockedBy}`, { duration: 4000 });
80 |       return false;
81 |     }
82 |     return true;
83 |   }, [componentId]);
84 |
85 |   const releaseLock = useCallback(async (userId: string) => {
86 |     if (!componentId) return;
87 |     await fetch('/api/components/lock', {
88 |       method: 'POST',
89 |       headers: { 'Content-Type': 'application/json' },
90 |       body: JSON.stringify({ componentId, userId, action: 'release' }),
91 |     });
92 |   }, [componentId]);
93 |
94 |   const heartbeat = useCallback(async (userId: string) => {
95 |     if (!componentId) return;
96 |     await fetch('/api/components/lock', {
97 |       method: 'POST',
98 |       headers: { 'Content-Type': 'application/json' },
99 |       body: JSON.stringify({ componentId, userId, action: 'heartbeat' }),
100 |     });
101 |   }, [componentId]);
102 |
103 |   return { lockComponent, releaseLock, heartbeat };
104 | }
105 |

```

File: hooks/useDiscussion.ts (219 lines, 5.9 KB)

```

=====
1 | 'use client'
2 |
3 | import { useState, useEffect } from 'react'
4 | import { createClientComponentClient } from '@supabase/auth-helpers-nextjs'
5 |
6 | export type Thread = {
7 |   id: string
8 |   lesson_id: string
9 |   author_id: string
10 |  title: string
11 |  body: string
12 |  type: 'question' | 'discussion' | 'announcement'
13 |  status: 'open' | 'resolved' | 'pinned' | 'closed'
14 |  upvote_count: number
15 |  reply_count: number
16 |  is_deleted: boolean
17 |  created_at: string
18 |  updated_at: string
19 |  author?: {
20 |    full_name: string
21 |    avatar_url: string
22 |    role: string
23 |  }
24 | }
25 |
26 | export type Reply = {
27 |   id: string
28 |   thread_id: string
29 |   parent_reply_id: string | null
30 |   author_id: string
31 |   body: string
32 |   is_teacher_answer: boolean
33 |   upvote_count: number
34 |   is_deleted: boolean
35 |   created_at: string
36 |   author?: {
37 |     full_name: string
38 |     avatar_url: string
39 |     role: string
40 |   }
41 | }
42 |
43 | export function useDiscussion(lessonId: string, activeTab: 'all' | 'unanswered' | 'pinned') {
44 |   const [threads, setThreads] = useState<Thread[]>([])
45 |   const [isLoading, setIsLoading] = useState(true)
46 |   const supabase = createClientComponentClient()
47 |
48 |   useEffect(() => {
49 |     const fetchThreads = async () => {
50 |       setIsLoading(true)
51 |       let query = supabase
52 |         .from('discussion_threads')
53 |         .select(`
54 |           *,
55 |           author:profiles(full_name, avatar_url, role)
56 |         `)
57 |         .eq('lesson_id', lessonId)
58 |         .eq('is_deleted', false)
59 |
60 |       if (activeTab === 'pinned') {
61 |         query = query.eq('status', 'pinned')
62 |       } else if (activeTab === 'unanswered') {
63 |         query = query.eq('reply_count', 0)
64 |       }
65 |
66 |       const { data, error } = await query.order('created_at', { ascending: false })
67 |       if (!error && data) {
68 |         setThreads(data as any[])
69 |       }
70 |       setIsLoading(false)
71 |     }
72 |
73 |     fetchThreads()
74 |
75 |     // Realtime channel for new threads
76 |     const channel = supabase
77 |       .channel(`lesson-discussion-${lessonId}`)
78 |       .on('postgres_changes', {
79 |         event: '*',
80 |         schema: 'public',
81 |         table: 'discussion_threads',
82 |         filter: `lesson_id=eq.${lessonId}`
83 |       }, () => {
84 |         fetchThreads()
85 |       })
86 |       .subscribe()
87 |
88 |     return () => {
89 |       supabase.removeChannel(channel)
90 |     }
91 |   }, [lessonId, activeTab, supabase])
92 |
93 |   const createThread = async (title: string, body: string, type: string, authorId: string) => {
94 |     // Basic Rate Limit Check (5 threads per hour)
95 |     const oneHourAgo = new Date(Date.now() - 3600000).toISOString()
96 |     const { count } = await supabase
97 |       .from('discussion_threads')
98 |       .select('id', { count: 'exact', head: true })
99 |       .eq('author_id', authorId)
100 |      .gt('created_at', oneHourAgo)
101 |
102 |     if (count !== null && count >= 5) {
103 |       throw new Error('Bạn đã gửi bình luận quá 5 bài viết mỗi giờ.')
104 |     }
105 |
106 |     const { data, error } = await supabase
107 |       .from('discussion_threads')
108 |       .insert({
109 |         lesson_id: lessonId,
110 |         author_id: authorId,
111 |         title,
112 |         body,
113 |         type
114 |       })
115 |       .select()
116 |       .single()
117 |
118 |     if (error) throw error
119 |     return data
120 |   }
121 |
122 |   return { threads, isLoading, createThread }
123 | }

```

```

124 |
125 | export function useThreadReplies(threadId: string) {
126 |   const [replies, setReplies] = useState<Reply[]>([])
127 |   const supabase = createClientComponentClient()
128 |
129 |   useEffect(() => {
130 |     const fetchReplies = async () => {
131 |       const { data, error } = await supabase
132 |         .from('discussion_replies')
133 |         .select(`
134 |           *
135 |           author:profiles(full_name, avatar_url, role)
136 |         `)
137 |         .eq('thread_id', threadId)
138 |         .eq('is_deleted', false)
139 |         .order('created_at', { ascending: true })
140 |
141 |       if (!error && data) {
142 |         setReplies(data as any[])
143 |       }
144 |     }
145 |
146 |     fetchReplies()
147 |
148 |     // Realtime replies subscription
149 |     const channel = supabase
150 |       .channel(`thread-replies-${threadId}`)
151 |       .on('postgres_changes', {
152 |         event: 'INSERT',
153 |         schema: 'public',
154 |         table: 'discussion_replies',
155 |         filter: `thread_id=eq.${threadId}`
156 |       }, (payload) => {
157 |         const fetchNewReplyAuthor = async () => {
158 |           const { data } = await supabase
159 |             .from('profiles')
160 |             .select('full_name, avatar_url, role')
161 |             .eq('id', payload.new.author_id)
162 |             .single()
163 |
164 |           const newReply: Reply = {
165 |             ...((payload.new as Reply),
166 |               {
167 |                 author: data || { full_name: ' 'n danh', avatar_url: ' ', role: ' '
168 |               }
169 |             )
170 |             setReplies(prev => [...prev, newReply])
171 |             fetchNewReplyAuthor()
172 |           })
173 |           .subscribe()
174 |
175 |         return () => {
176 |           supabase.removeChannel(channel)
177 |         }, [threadId, supabase])
178 |
179 |         const addReply = async (body: string, authorId: string, parentReplyId: string | null = null) => {
180 |           // Rate limit check (20 replies per hour)
181 |           const oneHourAgo = new Date(Date.now() - 3600000).toISOString()
182 |           const { count } = await supabase
183 |             .from('discussion_replies')
184 |             .select('id', { count: 'exact', head: true })
185 |             .eq('author_id', authorId)
186 |             .gt('created_at', oneHourAgo)
187 |
188 |           if (count !== null && count >= 20) {
189 |             throw new Error('Bạn đã gửi bình luận quá nhiều.')
190 |           }
191 |
192 |           // Check if author is teacher
193 |           const { data: profile } = await supabase
194 |             .from('profiles')
195 |             .select('role')
196 |             .eq('id', authorId)
197 |             .single()
198 |
199 |           const isTeacher = profile?.role === 'teacher'
200 |
201 |           const { data, error } = await supabase
202 |             .from('discussion_replies')
203 |             .insert({
204 |               thread_id: threadId,
205 |               parent_reply_id: parentReplyId,
206 |               author_id: authorId,
207 |               body,
208 |               is_teacher_answer: isTeacher
209 |             })
210 |             .select()
211 |             .single()
212 |
213 |           if (error) throw error
214 |           return data
215 |         }
216 |
217 |         return { replies, addReply }
218 |       }
219 |

```

File: hooks/useIsMobile.ts (17 lines, 0.4 KB)

```

1 | 'use client'
2 |
3 | import { useState, useEffect } from 'react'
4 |
5 | export function useIsMobile(breakpoint = 768) {
6 |   const [isMobile, setIsMobile] = useState(false)
7 |
8 |   useEffect(() => {
9 |     const check = () => setIsMobile(window.innerWidth < breakpoint)
10 |    check()
11 |    window.addEventListener('resize', check)
12 |    return () => window.removeEventListener('resize', check)
13 |  }, [breakpoint])
14 |
15 |   return isMobile
16 | }
17 |

```

File: hooks/useNotifications.ts (92 lines, 2.6 KB)

```

1 | 'use client'
2 |

```

```

3 | import { useState, useEffect } from 'react'
4 | import { createClientComponentClient } from '@supabase/auth-helpers-nextjs'
5 | import { toast } from 'react-hot-toast'
6 |
7 | export type Notification = {
8 |   id: string
9 |   user_id: string
10 |   type: string
11 |   title: string
12 |   body: string
13 |   data: any
14 |   action_url: string
15 |   is_read: boolean
16 |   created_at: string
17 | }
18 |
19 | export function useNotifications(userId?: string, options?: { filterTypes?: string[] }) {
20 |   const [notifications, setNotifications] = useState<Notification[]>([])
21 |   const [unreadCount, setUnreadCount] = useState(0)
22 |   const supabase = createClientComponentClient()
23 |
24 |   const filterTypesStr = options?.filterTypes?.join(',') || ''
25 |
26 |   useEffect(() => {
27 |     if (!userId) return;
28 |
29 |     const loadNotifications = async () => {
30 |       let query = supabase
31 |         .from('notifications')
32 |         .select('*')
33 |         .eq('user_id', userId)
34 |
35 |       if (options?.filterTypes && options.filterTypes.length > 0) {
36 |         query = query.in('type', options.filterTypes)
37 |       }
38 |
39 |       const { data } = await query
40 |         .order('created_at', { ascending: false })
41 |         .limit(20)
42 |
43 |       if (data) {
44 |         setNotifications(data as Notification[])
45 |         setUnreadCount(data.filter(n => !n.is_read).length)
46 |       }
47 |     }
48 |     loadNotifications()
49 |
50 |     const channel = supabase
51 |       .channel('notifications_channel')
52 |       .on('postgres_changes', {
53 |         event: 'INSERT',
54 |         schema: 'public',
55 |         table: 'notifications',
56 |         filter: `user_id=eq.${userId}`
57 |       }, (payload) => {
58 |         const newNotif = payload.new as Notification
59 |         setNotifications(prev => [newNotif, ...prev])
60 |         setUnreadCount(prev => prev + 1)
61 |         toast(newNotif.title, { icon: '🔔' })
62 |       })
63 |     .subscribe()
64 |
65 |     return () => {
66 |       supabase.removeChannel(channel)
67 |     }
68 |   }, [userId, supabase, filterTypesStr])
69 |
70 |   const markAsRead = async (notificationId: string) => {
71 |     setNotifications(prev =>
72 |       prev.map(n => n.id === notificationId ? { ...n, is_read: true } : n)
73 |     )
74 |     setUnreadCount(prev => Math.max(0, prev - 1))
75 |     await supabase.from('notifications').update({ is_read: true }).eq('id', notificationId)
76 |   }
77 |
78 |   const markAllAsRead = async () => {
79 |     const unreadIds = notifications.filter(n => !n.is_read).map(n => n.id)
80 |     if (unreadIds.length === 0) return
81 |
82 |     setNotifications(prev => prev.map(n => ({ ...n, is_read: true })))
83 |     setUnreadCount(0)
84 |
85 |     await supabase.from('notifications')
86 |       .update({ is_read: true })
87 |       .in('id', unreadIds)
88 |   }
89 |
90 |   return { notifications, unreadCount, markAsRead, markAllAsRead }
91 | }
92 |

```

File: hooks/useParentNotifications.ts (161 lines, 5.0 KB)

```

1 | // Path: hooks/useParentNotifications.ts
2 | 'use client'
3 | import { useEffect, useState, useRef } from 'react'
4 | import { RealtimeChannel } from '@supabase/supabase-js'
5 |
6 | export interface NotificationData {
7 |   id: string
8 |   user_id: string
9 |   type: string
10 |   title: string
11 |   body: string
12 |   data: any
13 |   action_url: string | null
14 |   is_read: boolean
15 |   created_at: string
16 | }
17 |
18 | export function useParentNotifications(parentId: string | null) {
19 |   const [notifications, setNotifications] = useState<NotificationData[]>([])
20 |   const [unreadCount, setUnreadCount] = useState(0)
21 |   const [supabase, setSupabase] = useState<any>(null)
22 |   const channelRef = useRef<RealtimeChannel | null>(null)
23 |
24 |   useEffect(() => {
25 |     import('@lib/supabase-ssr-client').then(m => setSupabase(m.createClient()))
26 |   }, [])
27 |
28 |   // Fetch initial notifications
29 |   useEffect(() => {
30 |     if (!parentId || !supabase) return

```

```

31 |
32 | async function fetchNotifications() {
33 |   try {
34 |     const { data, error } = await supabase
35 |       .from('notifications')
36 |       .select('*')
37 |       .eq('user_id', parentId)
38 |       .order('created_at', { ascending: false })
39 |       .limit(20)
40 |
41 |     if (error) throw error
42 |
43 |     const list = data ?? []
44 |     setNotifications(list)
45 |     setUnreadCount(list.filter(n => !n.is_read).length)
46 |   } catch (err) {
47 |     console.error('Error fetching initial notifications:', err)
48 |   }
49 | }
50 |
51 | fetchNotifications()
52 | }, [parentId])
53 |
54 | // Subscribe to real-time notification changes
55 | useEffect(() => {
56 |   if (!parentId || !supabase) return
57 |
58 |   const channelName = `parent-notif-${parentId}`
59 |   const channel = supabase.channel(channelName)
60 |   channelRef.current = channel
61 |
62 |   channel
63 |     .on('postgres_changes', {
64 |       event: '*',
65 |       schema: 'public',
66 |       table: 'notifications',
67 |       filter: 'user_id=eq.${parentId}',
68 |     }, (payload) => {
69 |       if (payload.eventType === 'INSERT') {
70 |         const newNotif = payload.new as NotificationData
71 |
72 |         // Play warning sound or trigger vibration for alert events (like child low score)
73 |         try {
74 |           if (newNotif.type === 'child_low_score') {
75 |             // Trigger browser vibration API
76 |             if (typeof window !== 'undefined' && navigator.vibrate) {
77 |               navigator.vibrate([200, 100, 200])
78 |             }
79 |             // Play a quick warning alert sound
80 |             const audioCtx = new (window.AudioContext || (window as any).webkitAudioContext)()
81 |             const osc = audioCtx.createOscillator()
82 |             const gain = audioCtx.createGain()
83 |             osc.type = 'sine'
84 |             osc.frequency.setValueAtTime(300, audioCtx.currentTime) // Low warning beep
85 |             gain.gain.setValueAtTime(0.1, audioCtx.currentTime)
86 |             osc.connect(gain)
87 |             gain.connect(audioCtx.destination)
88 |             osc.start()
89 |             osc.stop(audioCtx.currentTime + 0.3)
90 |           }
91 |         } catch (soundErr) {
92 |           console.warn('AudioContext sound failed to play:', soundErr)
93 |         }
94 |
95 |         setNotifications(prev => [newNotif, ...prev])
96 |         setUnreadCount(prev => prev + 1)
97 |       } else if (payload.eventType === 'UPDATE') {
98 |         const updatedNotif = payload.new as NotificationData
99 |         setNotifications(prev =>
100 |           prev.map(n => n.id === updatedNotif.id ? updatedNotif : n)
101 |         )
102 |         // Recalculate unread count
103 |         setNotifications(prev => {
104 |           const list = prev.map(n => n.id === updatedNotif.id ? updatedNotif : n)
105 |           setUnreadCount(list.filter(n => !n.is_read).length)
106 |           return list
107 |         })
108 |       }
109 |     })
110 |     .subscribe()
111 |
112 |   return () => {
113 |     if (channelRef.current) {
114 |       supabase.removeChannel(channelRef.current)
115 |       channelRef.current = null
116 |     }
117 |   }
118 | }, [parentId])
119 |
120 | // Mark all notifications as read
121 | const markAllAsRead = async () => {
122 |   if (!parentId || !supabase) return
123 |   try {
124 |     const { error } = await supabase
125 |       .from('notifications')
126 |       .update({ is_read: true })
127 |       .eq('user_id', parentId)
128 |       .eq('is_read', false)
129 |
130 |     if (error) throw error
131 |
132 |     setNotifications(prev => prev.map(n => ({ ...n, is_read: true })))
133 |     setUnreadCount(0)
134 |   } catch (err) {
135 |     console.error('Error marking notifications as read:', err)
136 |   }
137 | }
138 |
139 | // Mark single notification as read
140 | const markAsRead = async (notificationId: string) => {
141 |   if (!supabase) return
142 |   try {
143 |     const { error } = await supabase
144 |       .from('notifications')
145 |       .update({ is_read: true })
146 |       .eq('id', notificationId)
147 |
148 |     if (error) throw error
149 |
150 |     setNotifications(prev =>
151 |       prev.map(n => n.id === notificationId ? { ...n, is_read: true } : n)
152 |     )
153 |     setUnreadCount(prev => Math.max(0, prev - 1))
154 |   } catch (err) {

```

```

155 |     console.error('Error marking notification as read:', err)
156 |   }
157 | }
158 |
159 | return { notifications, unreadCount, markAllAsRead, markAsRead }
160 | }
161 |

```

File: hooks/useProctorEngine.ts (435 lines, 14.5 KB)

```

1 | 'use client'
2 |
3 | import { useState, useEffect, useRef, useCallback } from 'react'
4 | import { supabase } from '@/lib/supabase'
5 |
6 | const SNAPSHOT_INTERVAL = 60000
7 | const FACE_CHECK_INTERVAL = 1500
8 | const AUDIO_CHECK_INTERVAL = 3000
9 | const IDLE_CHECK_INTERVAL = 5000
10 | const SCREEN_REC_CHECK_INTERVAL = 30000
11 | const MAX_VIOLATIONS = 5
12 | const IDLE_THRESHOLD = 30
13 | const AUDIO_THRESHOLD = 0.15
14 | const GAZE_YAW_THRESHOLD = 0.4
15 | const GAZE_PITCH_THRESHOLD = 0.35
16 |
17 | export type ViolationType = 'tab_switch' | 'fullscreen_exit' | 'face_not_visible' | 'multiple_faces' | 'lookin...
18 |
19 | export interface ProctorState {
20 |   status: 'idle' | 'loading' | 'active' | 'error'
21 |   faceDetected: boolean
22 |   faceMatch: boolean
23 |   multipleFaces: boolean
24 |   lookingAway: boolean
25 |   violations: number
26 |   maxViolations: number
27 |   cameraError: string | null
28 |   stream: MediaStream | null
29 |   identityCapture: string | null
30 |   gazeDirection: { yaw: number; pitch: number }
31 |   audioLevel: number
32 |   isIdle: boolean
33 |   idleTime: number
34 | }
35 |
36 | interface ProctorOptions {
37 |   examId: string
38 |   attemptId: string
39 |   identityPhoto?: string | null
40 |   onViolation?: (type: ViolationType) => void
41 |   onMaxViolations?: () => void
42 | }
43 |
44 | const FACE_LANDMARKER_URL = 'https://storage.googleapis.com/mediapipe-models/face_landmarker/face_landmarker/f...
45 | const WASM_BASE = 'https://cdn.jsdelivr.net/npm/@mediapipe/tasks-vision@0.10.32/wasm'
46 |
47 | export function useProctorEngine({ examId, attemptId, identityPhoto, onViolation, onMaxViolations }: ProctorOp...
48 |   const [state, setState] = useState<ProctorState>({
49 |     status: 'idle',
50 |     faceDetected: false,
51 |     faceMatch: false,
52 |     multipleFaces: false,
53 |     lookingAway: false,
54 |     violations: 0,
55 |     maxViolations: MAX_VIOLATIONS,
56 |     cameraError: null,
57 |     stream: null,
58 |     identityCapture: null,
59 |     gazeDirection: { yaw: 0, pitch: 0 },
60 |     audioLevel: 0,
61 |     isIdle: false,
62 |     idleTime: 0,
63 |   })
64 |
65 |   const videoRef = useRef<HTMLVideoElement>(null)
66 |   const canvasRef = useRef<HTMLCanvasElement>(null)
67 |   const streamRef = useRef<MediaStream | null>(null)
68 |   const landmarkerRef = useRef<any>(null)
69 |   const faceCheckTimerRef = useRef<any>(null)
70 |   const snapshotTimerRef = useRef<any>(null)
71 |   const mountedRef = useRef(true)
72 |   const violationCountRef = useRef(0)
73 |
74 |   const audioContextRef = useRef<AudioContext | null>(null)
75 |   const analyserRef = useRef<AnalyserNode | null>(null)
76 |   const audioDataRef = useRef<Uint8Array<ArrayBuffer> | null>(null)
77 |   const idleTimerRef = useRef<number>(0)
78 |   const audioCheckTimerRef = useRef<any>(null)
79 |   const idleCheckTimerRef = useRef<any>(null)
80 |   const lastActivityRef = useRef<number>(Date.now())
81 |   const activityHandlerRef = useRef<(() => void) | null>(null)
82 |   const screenRecCheckTimerRef = useRef<any>(null)
83 |   const vmCheckedRef = useRef<boolean>(false)
84 |
85 |   const capturePhoto = useCallback(() => {
86 |     if (videoRef.current && canvasRef.current) {
87 |       const ctx = canvasRef.current.getContext('2d')
88 |       if (ctx) {
89 |         ctx.drawImage(videoRef.current, 0, 0, 320, 240)
90 |         return canvasRef.current.toDataURL('image/png')
91 |       }
92 |     }
93 |     return null
94 |   }, [])
95 |
96 |   const recordViolation = useCallback(async (type: ViolationType, data?: string) => {
97 |     if (!mountedRef.current) return
98 |     violationCountRef.current += 1
99 |
100 |     setState(prev => ({ ...prev, violations: violationCountRef.current }))
101 |
102 |     const snapshot = capturePhoto()
103 |     const violationData = {
104 |       type,
105 |       message: data || `Vi phạm lan ${violationCountRef.current}`,
106 |       snapshot: snapshot || null,
107 |       violationCount: violationCountRef.current,
108 |     }
109 |
110 |     supabase.from('exam_logs').insert({
111 |       exam_id: examId,
112 |       attempt_id: attemptId,
113 |       event_type: type,

```



```

114     event_data: JSON.stringify(violationData),
115     created_at: new Date().toISOString(),
116   }).then()
117
118   onViolation?.(type)
119
120   if (violationCountRef.current >= MAX_VIOLATIONS) {
121     onMaxViolations?.()
122   }
123 }, [examId, attemptId, capturePhoto, onViolation, onMaxViolations])
124
125 const startCamera = useCallback(async () => {
126   try {
127     setState(prev => ({ ...prev, status: 'loading', cameraError: null }))
128     let stream: MediaStream
129     try {
130       stream = await navigator.mediaDevices.getUserMedia({
131         video: { width: { ideal: 240 }, height: { ideal: 180 }, facingMode: 'user', frameRate: { ideal: 15 } },
132         audio: true,
133       })
134     } catch {
135       stream = await navigator.mediaDevices.getUserMedia({
136         video: { width: { ideal: 240 }, height: { ideal: 180 }, facingMode: 'user', frameRate: { ideal: 15 } },
137       })
138     }
139     if (!mountedRef.current) { stream.getTracks().forEach(t => t.stop()); return }
140     streamRef.current = stream
141     if (videoRef.current) {
142       videoRef.current.srcObject = stream
143       await videoRef.current.play()
144     }
145     setState(prev => ({ ...prev, status: 'active', stream }))
146     return stream
147   } catch (err: any) {
148     const msg = err.name === 'NotAllowedError'
149       ? 'Bạn đã từ chối quyền Camera. Vui lòng cho phép Camera trong trình duyệt!'
150       : err.name === 'NotFoundError'
151       ? 'Không tìm thấy Webcam. Vui lòng kết nối Webcam và thử lại.'
152       : 'Lỗi camera: ' + (err.message || 'Lỗi không xác định')
153     setState(prev => ({ ...prev, status: 'error', cameraError: msg }))
154     return null
155   }
156 }, [])
157
158 const stopCamera = useCallback(() => {
159   if (streamRef.current) {
160     streamRef.current.getTracks().forEach(t => t.stop())
161     streamRef.current = null
162   }
163   setState(prev => ({ ...prev, stream: null, status: 'idle' }))
164 }, [])
165
166 const initMediaPipe = useCallback(async () => {
167   try {
168     const { FilesetResolver, FaceLandmarker } = await import('@mediapipe/tasks-vision')
169     const vision = await FilesetResolver.forVisionTasks(WASM_BASE)
170     const landmarker = await FaceLandmarker.createFromOptions(vision, {
171       baseOptions: { modelAssetPath: FACE_LANDMARKER_URL, delegate: 'GPU' },
172       runningMode: 'VIDEO',
173       numFaces: 2,
174       outputFaceBlendshapes: false,
175     })
176     if (mountedRef.current) landmarkerRef.current = landmarker
177   } catch (e) {
178     console.warn('MediaPipe init failed, falling back to canvas detection:', e)
179   }
180 }, [])
181
182 const detectFace = useCallback(() => {
183   if (!videoRef.current || !canvasRef.current) return { face: false, multiple: false, yaw: 0, pitch: 0 }
184   const video = videoRef.current
185   const canvas = canvasRef.current
186   const ctx = canvas.getContext('2d')
187   if (!ctx) return { face: false, multiple: false, yaw: 0, pitch: 0 }
188
189   const w = video.videoWidth || 240
190   const h = video.videoHeight || 180
191   canvas.width = w
192   canvas.height = h
193   ctx.drawImage(video, 0, 0, w, h)
194   const imageData = ctx.getImageData(0, 0, w, h)
195   const data = imageData.data
196   let skinPixels = 0
197   const total = data.length / 4
198   for (let i = 0; i < data.length; i += 16) {
199     const r = data[i], g = data[i + 1], b = data[i + 2]
200     if (r > 95 && g > 40 && b > 20 && r > g && r > b && Math.abs(r - g) > 15) skinPixels++
201   }
202   const ratio = skinPixels / (total / 4)
203   return { face: ratio > 0.05, multiple: false, yaw: 0, pitch: 0 }
204 }, [])
205
206 const setupAudioDetection = useCallback(() => {
207   const stream = streamRef.current
208   if (!stream) return
209   try {
210     const audioTrack = stream.getAudioTracks()[0]
211     if (!audioTrack) return
212
213     const audioContext = new AudioContext()
214     const source = audioContext.createMediaStreamSource(stream)
215     const analyser = audioContext.createAnalyser()
216     analyser.fftSize = 256
217     source.connect(analyser)
218
219     audioContextRef.current = audioContext
220     analyserRef.current = analyser
221     audioDataRef.current = new Uint8Array(analyser.frequencyBinCount) as Uint8Array<ArrayBuffer>
222
223     audioCheckTimerRef.current = setInterval(() => {
224       if (!analyserRef.current || !audioDataRef.current) return
225       const sum = audioDataRef.current.reduce((a, b) => a + b, 0)
226       const average = sum / audioDataRef.current.length
227       const level = average / 255
228
229       setState(prev => ({ ...prev, audioLevel: level }))
230
231       if (level > AUDIO_THRESHOLD) {
232         recordViolation('audio_anomaly', `Phát hiện âm thanh bất thường (mức độ: ${level.toFixed(2)})`)
233       }
234     }, AUDIO_CHECK_INTERVAL)
235   } catch {
236   }
237 }

```

```

238 | }, [recordViolation])
239 |
240 | const setupIdleDetection = useCallback(() => {
241 |     const updateActivity = () => {
242 |         lastActivityRef.current = Date.now()
243 |         idleTimeRef.current = 0
244 |     }
245 |     activityHandlerRef.current = updateActivity
246 |
247 |     window.addEventListener('mousemove', updateActivity)
248 |     window.addEventListener('mousedown', updateActivity)
249 |     window.addEventListener('keydown', updateActivity)
250 |     window.addEventListener('touchstart', updateActivity)
251 |
252 |     idleCheckTimerRef.current = setInterval(() => {
253 |         const elapsed = (Date.now() - lastActivityRef.current) / 1000
254 |         idleTimeRef.current = elapsed
255 |         setState(prev => ({ ...prev, isIdle: elapsed > IDLE_THRESHOLD, idleTime: Math.floor(elapsed) }))
256 |
257 |         if (Math.floor(elapsed) === IDLE_THRESHOLD + 1) {
258 |             recordViolation('prolonged_idle', `Không có hoạt động chuột/bàn phím trong ${Math.floor(elapsed)} giây`)
259 |         }
260 |     }, IDLE_CHECK_INTERVAL)
261 | }, [recordViolation])
262 |
263 | const detectScreenRecording = useCallback(() => {
264 |     try {
265 |         // @ts-ignore
266 |         if (navigator.mediaSession?.playbackState === 'recording') {
267 |             recordViolation('screen_recording', 'Phát hiện phân mềm ghi màn hình (mediaSession)')
268 |             return
269 |         }
270 |     } catch {}
271 |
272 |     try {
273 |         // @ts-ignore
274 |         if (window.screenRecording || window.__RECORDING__) {
275 |             recordViolation('screen_recording', 'Phát hiện phân mềm ghi màn hình (window flag)')
276 |             return
277 |         }
278 |     } catch {}
279 | }, [recordViolation])
280 |
281 | const detectVM = useCallback(() => {
282 |     if (vmCheckedRef.current) return
283 |     vmCheckedRef.current = true
284 |
285 |     const vmIndicators: string[] = []
286 |
287 |     if (navigator.webdriver) {
288 |         vmIndicators.push('webdriver')
289 |     }
290 |
291 |     const platform = navigator.platform?.toLowerCase() || ''
292 |     if (platform.includes('vm') || platform.includes('virtual')) {
293 |         vmIndicators.push('platform')
294 |     }
295 |
296 |     const { width, height } = window.screen
297 |     if ((width === 1024 && height === 768) || (width === 800 && height === 600)) {
298 |         vmIndicators.push('resolution')
299 |     }
300 |
301 |     if (navigator.hardwareConcurrency <= 2) {
302 |         vmIndicators.push('low_cores')
303 |     }
304 |
305 |     for (let i = 0; i < navigator.plugins.length; i++) {
306 |         const name = navigator.plugins[i].name.toLowerCase()
307 |         if (name.includes('vmware') || name.includes('virtualbox') || name.includes('hyper-v') || name.includes(...
308 |             vmIndicators.push('vmm_plugin')
309 |             break
310 |         }
311 |     }
312 |
313 |     if (vmIndicators.length >= 2) {
314 |         recordViolation('vm_detected', `Phát hiện máy ảo (dấu hiệu: ${vmIndicators.join(', ')}')`)
315 |     }
316 | }, [recordViolation])
317 |
318 | const startFaceDetection = useCallback(async () => {
319 |     await initMediaPipe()
320 |
321 |     const runDetection = async () => {
322 |         if (!mountedRef.current || !videoRef.current || videoRef.current.readyState < 2) return
323 |
324 |         try {
325 |             let face = false, multiple = false, yaw = 0, pitch = 0
326 |
327 |             if (landmarkerRef.current) {
328 |                 const results = await landmarkerRef.current.detectForVideo(videoRef.current, performance.now())
329 |                 if (results.faceLandmarks?.length > 0) {
330 |                     face = true
331 |                     multiple = results.faceLandmarks.length > 1
332 |
333 |                     const nose = results.faceLandmarks[0][1]
334 |                     if (nose) {
335 |                         yaw = (nose.x - 0.5) * 2
336 |                         pitch = (nose.y - 0.5) * 2
337 |                     }
338 |                 }
339 |             } else {
340 |                 const result = detectFace()
341 |                 face = result.face
342 |             }
343 |
344 |             if (!mountedRef.current) return
345 |
346 |             setState(prev => ({
347 |                 ...prev,
348 |                 faceDetected: face,
349 |                 multipleFaces: multiple,
350 |                 lookingAway: Math.abs(yaw) > GAZE_YAW_THRESHOLD || Math.abs(pitch) > GAZE_PITCH_THRESHOLD,
351 |                 gazeDirection: { yaw, pitch },
352 |             }))
353 |
354 |             if (multiple) {
355 |                 recordViolation('multiple_faces', 'Phát hiện nhiều hơn 1 khuôn mặt trong khung hình')
356 |             }
357 |
358 |             if (!face) {
359 |                 recordViolation('face_not_visible', 'Không phát hiện khuôn mặt trước camera')
360 |             } else if (Math.abs(yaw) > GAZE_YAW_THRESHOLD || Math.abs(pitch) > GAZE_PITCH_THRESHOLD) {
361 |                 recordViolation('looking_away', `Nhìn sang hướng khác (yaw=${yaw.toFixed(2)}, pitch=${pitch.toFixed(...

```

```

362 |     }
363 |   } catch (e) {
364 |   }
365 | }
366 |
367 | faceCheckTimerRef.current = setInterval(runDetection, FACE_CHECK_INTERVAL)
368 | snapshotTimerRef.current = setInterval(() => {
369 |   const photo = capturePhoto()
370 |   if (photo) {
371 |     supabase.from('exam_logs').insert({
372 |       exam_id: examId,
373 |       attempt_id: attemptId,
374 |       event_type: 'snapshot',
375 |       event_data: photo,
376 |       created_at: new Date().toISOString(),
377 |     }).then()
378 |   }
379 | }, SNAPSHOT_INTERVAL)
380 |
381 | setupAudioDetection()
382 | setupIdleDetection()
383 | screenRecCheckTimerRef.current = setInterval(detectScreenRecording, SCREEN_REC_CHECK_INTERVAL)
384 | detectVM()
385 | }, [initMediaPipe, detectFace, recordViolation, capturePhoto, examId, attemptId, setupAudioDetection, setupI...
386 |
387 | useEffect(() => {
388 |   mountedRef.current = true
389 |   return () => {
390 |     mountedRef.current = false
391 |     stopCamera()
392 |     clearInterval(faceCheckTimerRef.current)
393 |     clearInterval(snapshotTimerRef.current)
394 |     landmarkerRef.current = null
395 |
396 |     clearInterval(audioCheckTimerRef.current)
397 |     clearInterval(idleCheckTimerRef.current)
398 |     clearInterval(screenRecCheckTimerRef.current)
399 |
400 |     if (activityHandlerRef.current) {
401 |       window.removeEventListener('mousemove', activityHandlerRef.current)
402 |       window.removeEventListener('mousedown', activityHandlerRef.current)
403 |       window.removeEventListener('keydown', activityHandlerRef.current)
404 |       window.removeEventListener('touchstart', activityHandlerRef.current)
405 |       activityHandlerRef.current = null
406 |     }
407 |
408 |     if (audioContextRef.current) {
409 |       audioContextRef.current.close()
410 |       audioContextRef.current = null
411 |     }
412 |   }
413 | }, [stopCamera])
414 |
415 | const captureIdentity = useCallback(() => {
416 |   const photo = capturePhoto()
417 |   if (photo) {
418 |     setState(prev => ({ ...prev, identityCapture: photo }))
419 |   }
420 |   return photo
421 | }, [capturePhoto])
422 |
423 | return {
424 |   state,
425 |   videoRef,
426 |   canvasRef,
427 |   startCamera,
428 |   stopCamera,
429 |   startFaceDetection,
430 |   recordViolation,
431 |   captureIdentity,
432 |   capturePhoto,
433 | }
434 | }
435 |

```

File: hooks/useSessionTimeout.ts (41 lines, 1.1 KB)

```

1 | 'use client'
2 |
3 | import { useEffect, useCallback } from 'react'
4 | import { useRouter } from 'next/navigation'
5 | import { logout } from '@lib/auth-actions'
6 |
7 | export function useSessionTimeout(timeoutMinutes: number = 30) {
8 |   const router = useRouter()
9 |   const timeoutMs = timeoutMinutes * 60 * 1000
10 |
11 |   const handleLogout = useCallback(async () => {
12 |     await logout()
13 |     router.push('/login?session=expired')
14 |   }, [router])
15 |
16 |   useEffect(() => {
17 |     let timeoutId: NodeJS.Timeout
18 |
19 |     const resetTimer = () => {
20 |       if (timeoutId) clearTimeout(timeoutId)
21 |       timeoutId = setTimeout(handleLogout, timeoutMs)
22 |     }
23 |
24 |     // Events to reset the timer
25 |     const events = ['mousedown', 'keydown', 'scroll', 'touchstart']
26 |
27 |     events.forEach(event => {
28 |       window.addEventListener(event, resetTimer)
29 |     })
30 |
31 |     resetTimer() // Initialize timer
32 |
33 |     return () => {
34 |       if (timeoutId) clearTimeout(timeoutId)
35 |       events.forEach(event => {
36 |         window.removeEventListener(event, resetTimer)
37 |       })
38 |     }
39 |   }, [handleLogout, timeoutMs])
40 | }
41 |

```

File: providers/QueryProvider.tsx (21 lines, 0.5 KB)

```

1 | 'use client';
2 |
3 | import { QueryClient, QueryClientProvider } from '@tanstack/react-query';
4 | import { useState } from 'react';
5 |
6 | export default function QueryProvider({ children }: { children: React.ReactNode }) {
7 |   const [queryClient] = useState(() => new QueryClient({
8 |     defaultOptions: {
9 |       queries: {
10 |         staleTime: 60 * 1000,
11 |       },
12 |     },
13 |   }));
14 |
15 |   return (
16 |     <QueryClientProvider client={queryClient}>
17 |       {children}
18 |     </QueryClientProvider>
19 |   );
20 | }
21 |

```

File: data/assemblyErrors.ts (607 lines, 26.4 KB)

```

1 | export interface AssemblyVideo {
2 |   videoId: string;
3 |   title: string;
4 |   channelName: string;
5 |   startSeconds: number;
6 |   endSeconds?: number;
7 |   language: 'vi' | 'en';
8 |   description: string;
9 | }
10 |
11 | export interface AssemblyError {
12 |   id: string;
13 |   code: string;
14 |   severity: 'critical' | 'warning' | 'info';
15 |   title: string;
16 |   description: string;
17 |   howToFix: string;
18 |   youtubeVideo: AssemblyVideo[];
19 |   animation: {
20 |     type: 'shake' | 'red-flash' | 'arrow-wrong' | 'highlight-both' | 'cross-mark';
21 |     targetComponents: string[];
22 |     durationMs: number;
23 |   };
24 |   affectedParts: string[];
25 |   suggestedFix?: {
26 |     replaceComponent: string;
27 |     recommendation: string;
28 |   };
29 | };
30 |
31 | export const ASSEMBLY_ERRORS: AssemblyError[] = [
32 |   {
33 |     id: 'err-cpu-socket-001',
34 |     code: 'CPU_SOCKET_MISMATCH',
35 |     severity: 'critical',
36 |     title: 'CPU không khớp socket Mainboard',
37 |     description: 'CPU Intel LGA1700 không thể g_n vào Mainboard AMD AM5',
38 |     howToFix: '1. Kiểm tra socket của CPU (LGA1700, AM5, AM4...) \n 2. Ch
39 |     youtubeVideo: [
40 |       {
41 |         videoId: 'PXalc9AYIcg',
42 |         title: 'CPU Socket Explained - LGA vs AM4 vs AM5',
43 |         channelName: 'Linus Tech Tips',
44 |         startSeconds: 45,
45 |         endSeconds: 210,
46 |         language: 'en',
47 |         description: 'Giới thích rõ sñ khác biệt giữa các loại socket CP
48 |       },
49 |       {
50 |         videoId: 'dcuLV5SpzHE',
51 |         title: 'Lxi socket CPU th_Ýng g.p khi l_p máy tính',
52 |         channelName: 'PC Vi_çt',
53 |         startSeconds: 30,
54 |         language: 'vi',
55 |         description: 'H_Ưng d«n kiểm tra socket CPU b«ng tiếng Vi_çt'
56 |       },
57 |     ],
58 |     animation: { type: 'highlight-both', targetComponents: ['cpu', 'motherboard'], durationMs: 2000 },
59 |     affectedParts: ['cpu', 'motherboard']
60 |   },
61 |   {
62 |     id: 'err-ram-type-002',
63 |     code: 'RAM_TYPE_MISMATCH',
64 |     severity: 'critical',
65 |     title: 'RAM DDR4 c_m vào Mainboard DDR5 (ho_c ng_ăc l_i)',
66 |     description: 'DDR4 và DDR5 có khe l_çch nhau - v-t lý không thể c_m v
67 |     howToFix: '1. Kiểm tra Mainboard h_x tr_ă DDR4 hay DDR5 (không có boar
68 |     youtubeVideo: [
69 |       {
70 |         videoId: 'R7C09V9rp0k',
71 |         title: 'DDR4 vs DDR5 RAM - What You Need to Know',
72 |         channelName: 'GamersNexus',
73 |         startSeconds: 120,
74 |         endSeconds: 340,
75 |         language: 'en',
76 |         description: 'So sánh DDR4 và DDR5, cách phân biệt'
77 |       },
78 |     ],
79 |     animation: { type: 'red-flash', targetComponents: ['ram', 'motherboard'], durationMs: 1500 },
80 |     affectedParts: ['ram', 'motherboard']
81 |   },
82 |   {
83 |     id: 'err-psu-watt-003',
84 |     code: 'PSU_INSUFFICIENT_WATTAGE',
85 |     severity: 'critical',
86 |     title: 'PSU không c_ i_çn cho toàn b_ build',
87 |     description: 'Tổng TDP của CPU + GPU + các linh_ki_çn khác v_ăc quá s
88 |     howToFix: '1. Tính tổng TDP: CPU TDP + GPU TDP + 50W (RAM, SSD, fans
89 |     youtubeVideo: [
90 |       {
91 |         videoId: 'qyLCuYkZGA0',
92 |         title: 'How Much Power Does Your PC Need?',
93 |         channelName: 'Linus Tech Tips',
94 |         startSeconds: 60,
95 |         endSeconds: 280,
96 |         language: 'en',
97 |         description: 'Cách tính wattage PSU c_sñ thi_çt'
98 |       },
99 |     ]

```

```
100 |         videoId: 'HmVfm-UrEJM',
101 |         title: 'Cách ch ỉn ngu ỏn PSU ỏng watt cho PC',
102 |         channelName: 'Th ẻ Gi ỏi Di ỏng Tech',
103 |         startSeconds: 45,
104 |         language: 'vi',
105 |         description: 'H ỏng d ỏn ti ẻng Vi ẻt c ỏch t ỉnh v ỏ ch ỉn PSU ph ỏ h ỏp',
106 |     },
107 |     ],
108 |     animation: { type: 'red-flash', targetComponents: ['psu', 'cpu', 'gpu'], durationMs: 2500 },
109 |     affectedParts: ['psu', 'cpu', 'gpu']
110 | },
111 | {
112 |     id: 'err-gpu-size-004',
113 |     code: 'GPU_TOO_LONG_FOR_CASE',
114 |     severity: 'critical',
115 |     title: 'GPU d ỏi h ỉn Case cho ph ẻp',
116 |     description: 'GPU c ỏ ch ỏi ỏu d ỏi v -t, lý l ỏn h ỉn kho ẻng tr ẻng t ẻi a c ỏ',
117 |     howToFix: '1. Ki ỏm tra "Max GPU Length" trong spec c ỏa Case \n 2. Ki ỏm',
118 |     youtubeVideo: [
119 |         {
120 |             videoId: 'BblBMJFNmFY',
121 |             title: 'GPU Size vs Case Clearance - Common Mistake',
122 |             channelName: 'JayzTwoCents',
123 |             startSeconds: 30,
124 |             endSeconds: 200,
125 |             language: 'en',
126 |             description: 'L ỏ GPU d ỏi h ỉn case v ỏ c ỏch tr ỏnh'
127 |         },
128 |     ],
129 |     animation: { type: 'arrow-wrong', targetComponents: ['gpu', 'case'], durationMs: 2000 },
130 |     affectedParts: ['gpu', 'case']
131 | },
132 | {
133 |     id: 'err-cooler-height-005',
134 |     code: 'COOLER_TOO_TALL_FOR_CASE',
135 |     severity: 'critical',
136 |     title: 'Cooler CPU cao h ỉn Case cho ph ẻp',
137 |     description: 'Cooler tower air c ỏ ch ỏi ỏu c ỏo v ỏ ỏt qu ỏ CPU clearance c ỏ',
138 |     howToFix: '1. Ki ỏm tra "Max CPU Cooler Height" trong spec Case \n 2. C',
139 |     youtubeVideo: [
140 |         {
141 |             videoId: 'Hmb2QFE0MODE',
142 |             title: 'Air Cooler vs Case - Height Clearance Guide',
143 |             channelName: 'Hardware Canucks',
144 |             startSeconds: 90,
145 |             language: 'en',
146 |             description: 'C ỏch o v ỏ ki ỏm tra clearance cooler vs case'
147 |         },
148 |     ],
149 |     animation: { type: 'cross-mark', targetComponents: ['cooler', 'case'], durationMs: 1800 },
150 |     affectedParts: ['cooler', 'case']
151 | },
152 | {
153 |     id: 'err-no-display-006',
154 |     code: 'NO_DISPLAY_OUTPUT',
155 |     severity: 'critical',
156 |     title: 'Kh ỏng c ỏ ngu ỏn xu ỏt h ỉnh',
157 |     description: 'CPU kh ỏng c ỏ iGPU (integrated graphics) v ỏ c ỏng kh ỏng',
158 |     howToFix: '1. CPU Intel Core c ỏ ch ỏ "F" (i5-13600KF) = KH ỏNG c ỏ iGPU',
159 |     youtubeVideo: [
160 |         {
161 |             videoId: 'RN9sqc7mEWE',
162 |             title: 'Integrated vs Dedicated GPU Explained',
163 |             channelName: 'Tech Quickie',
164 |             startSeconds: 20,
165 |             endSeconds: 180,
166 |             language: 'en',
167 |             description: 'Gi ẻi th ỉch iGPU v ỏ khi n ỏo c ỏn GPU r ẻi'
168 |         },
169 |     ],
170 |     animation: { type: 'red-flash', targetComponents: ['cpu', 'motherboard'], durationMs: 2000 },
171 |     affectedParts: ['cpu', 'gpu']
172 | },
173 | {
174 |     id: 'err-ram-slot-007',
175 |     code: 'RAM_WRONG_DUAL_CHANNEL_SLOT',
176 |     severity: 'warning',
177 |     title: 'RAM c -m sai slot - m ỏt dual channel',
178 |     description: 'V ỏi 2 thanh RAM, ph ẻi c -m v ỏo slot A2+B2 (th ỏng l ỏ s',
179 |     howToFix: '1. Xem manual mainboard ỏ bi ẻt ỏng slot dual channel \n 2',
180 |     youtubeVideo: [
181 |         {
182 |             videoId: '0FKJLpZ5ByY',
183 |             title: 'RAM Dual Channel - Which Slots to Use?',
184 |             channelName: 'Linus Tech Tips',
185 |             startSeconds: 180,
186 |             endSeconds: 360,
187 |             language: 'en',
188 |             description: 'H ỏng d ỏn c -m RAM ỏng slot dual channel'
189 |         },
190 |     ],
191 |     animation: { type: 'arrow-wrong', targetComponents: ['ram'], durationMs: 2000 },
192 |     affectedParts: ['ram', 'motherboard']
193 | },
194 | {
195 |     id: 'err-psu-connector-008',
196 |     code: 'PSU_MISSING_CONNECTORS',
197 |     severity: 'critical',
198 |     title: 'PSU th ỏu s ỏ c ỏp ngu ỏn cho GPU',
199 |     description: 'GPU c ỏn c ỏc s ỏ c ỏp ngu ỏn PCIe power connector nh ỏng P',
200 |     howToFix: '1. Ki ỏm tra s ỏ l ỏ ỏng PCIe power connector c ỏa PSU \n 2. GPU',
201 |     youtubeVideo: [
202 |         {
203 |             videoId: '2CkU4-T1R9c',
204 |             title: 'PSU Cables Explained - What You Need',
205 |             channelName: 'Bitwit',
206 |             startSeconds: 120,
207 |             language: 'en',
208 |             description: 'Gi ẻi th ỉch c ỏc l ỏi c ỏp ngu ỏn PSU'
209 |         },
210 |     ],
211 |     animation: { type: 'shake', targetComponents: ['psu', 'gpu'], durationMs: 1500 },
212 |     affectedParts: ['psu', 'gpu']
213 | },
214 | {
215 |     id: 'err-form-factor-009',
216 |     code: 'FORM_FACTOR_MISMATCH',
217 |     severity: 'critical',
218 |     title: 'Mainboard kh ỏng v ỏa Case',
219 |     description: 'Mainboard c ỏ form factor (ATX, mATX, ITX) kh ỏng t ỏ ỏng',
220 |     howToFix: '1. Ki ỏm tra form factor c ỏa Case (h ỏ tr ỏ: ATX, mATX, ITX)',
221 |     youtubeVideo: [
222 |         {
223 |             videoId: 'SJXTjNy_b64',
```

```
224|         title: 'Motherboard Form Factors Explained',
225|         channelName: 'TechQuickie',
226|         startSeconds: 60,
227|         endSeconds: 240,
228|         language: 'en',
229|         description: 'Giới thích form factor ATX, MATX, ITX'
230|     },
231| ],
232|     animation: { type: 'cross-mark', targetComponents: ['motherboard', 'case'], durationMs: 2000 },
233|     affectedParts: ['motherboard', 'case']
234| },
235| {
236|     id: 'err-cooler-socket-010',
237|     code: 'COOLER_SOCKET_MISMATCH',
238|     severity: 'critical',
239|     title: 'Cooler không h× trā socket CPU',
240|     description: 'Cooler không bao gồm mounting bracket cho socket c×a CPU',
241|     howToFix: '1. Kiểm tra danh sách socket h× trā c×a Cooler\ n2. N×u t×i
242|     youtubeVideo: [
243|         {
244|             videoId: 'dk_3tE3E_DI',
245|             title: 'CPU Cooler Compatibility – Socket Guide',
246|             channelName: 'GamersNexus',
247|             startSeconds: 180,
248|             language: 'en',
249|             description: 'Kiểm tra t×ing thích socket c×a cooler'
250|         }
251|     ],
252|     animation: { type: 'highlight-both', targetComponents: ['cooler', 'cpu'], durationMs: 2000 },
253|     affectedParts: ['cooler', 'cpu']
254| },
255| {
256|     id: 'err-m2-slot-011',
257|     code: 'NOT_ENOUGH_M2_SLOTS',
258|     severity: 'warning',
259|     title: 'Thi×u slot M.2 cho SSD',
260|     description: 'S× ã l× ãng SSD M.2 v× ãt quá s×n slot M.2 có s×n trê× Mainboard',
261|     howToFix: '1. Kiểm tra s× slot M.2 trê× Mainboard\ n2. Dùng SSD SATA'
262|     youtubeVideo: [
263|         {
264|             videoId: 'RkSuk9IzT0s',
265|             title: 'M.2 vs SATA SSD – Which One to Choose?',
266|             channelName: 'Linus Tech Tips',
267|             startSeconds: 60,
268|             language: 'en',
269|             description: 'So sánh M.2 NVMe và SATA SSD'
270|         }
271|     ],
272|     animation: { type: 'red-flash', targetComponents: ['ssd', 'motherboard'], durationMs: 1500 },
273|     affectedParts: ['ssd', 'motherboard']
274| },
275| {
276|     id: 'err-bottleneck-012',
277|     code: 'CPU_GPU_BOTTLENECK',
278|     severity: 'warning',
279|     title: 'CPU và GPU m×t cân b×ng – ngh×n c× chāi',
280|     description: 'T× l× s×c m×nh giā CPU và GPU quá chê×h l×ch. M×t c×n',
281|     howToFix: '1. GPU quá m×nh (>2.5x CPU): nâng c×p CPU ho×c giē× GPU\ n'
282|     youtubeVideo: [
283|         {
284|             videoId: '4Vymf3DkI4Y',
285|             title: 'CPU Bottleneck Explained – How to Fix',
286|             channelName: 'JayzTwoCents',
287|             startSeconds: 60,
288|             endSeconds: 300,
289|             language: 'en',
290|             description: 'Giới thích bottleneck CPU/GPU và cách kh×c phāc'
291|         }
292|     ],
293|     animation: { type: 'arrow-wrong', targetComponents: ['cpu', 'gpu'], durationMs: 2000 },
294|     affectedParts: ['cpu', 'gpu']
295| },
296| {
297|     id: 'err-cooler-tdp-013',
298|     code: 'COOLER_INSUFFICIENT_TDP',
299|     severity: 'warning',
300|     title: 'Cooler không c× t×n nhi×t cho CPU',
301|     description: 'TDP t×i c×a Cooler th×p h×n TDP c×a CPU. CPU s× bē',
302|     howToFix: '1. Cooler c×n h× trā TDP "e CPU TDP × 0.9 (buffer)\ n2. V×i'
303|     youtubeVideo: [
304|         {
305|             videoId: 't6ono8R_nAM',
306|             title: 'CPU Cooler TDP Ratings Explained',
307|             channelName: 'Hardware Canucks',
308|             startSeconds: 200,
309|             language: 'en',
310|             description: 'Cách íc và hiāu TDP trê× CPU cooler'
311|         }
312|     ],
313|     animation: { type: 'red-flash', targetComponents: ['cooler', 'cpu'], durationMs: 2000 },
314|     affectedParts: ['cooler', 'cpu']
315| },
316| {
317|     id: 'err-psu-length-014',
318|     code: 'PSU_TOO_LONG_FOR_CASE',
319|     severity: 'critical',
320|     title: 'PSU dài h×n Case cho phép',
321|     description: 'Chiāu dài v-t lý c×a PSU v× ãt quá khoēng tr×ng t×i a',
322|     howToFix: '1. Kiểm tra max PSU length c×a Case\ n2. Ch×n PSU có chiāu'
323|     youtubeVideo: [
324|         {
325|             videoId: 'IEs0ILC_8jg',
326|             title: 'PSU Size Guide – ATX vs SFX',
327|             channelName: 'Optimum Tech',
328|             startSeconds: 60,
329|             language: 'en',
330|             description: 'H× ãng d×n ch×n kích th×c PSU phù hāp v×i case'
331|         }
332|     ],
333|     animation: { type: 'cross-mark', targetComponents: ['psu', 'case'], durationMs: 1800 },
334|     affectedParts: ['psu', 'case']
335| },
336| {
337|     id: 'err-ram-speed-015',
338|     code: 'RAM_SPEED_MISMATCH',
339|     severity: 'info',
340|     title: 'RAM ch×y không úng t×c ù t×i a',
341|     description: 'RAM có t×c ù cao h×n Mainboard h× trā. RAM s× t× ùng',
342|     howToFix: '1. Kiểm tra max RAM speed c×a Mainboard\ n2. N×u mu×n t×c'
343|     youtubeVideo: [
344|         {
345|             videoId: 'Jv5g0KRxh-E',
346|             title: 'What is XMP? RAM Speed Explained',
347|             channelName: 'Tech Quickie',
```

```
348|         startSeconds: 30,
349|         endSeconds: 180,
350|         language: 'en',
351|         description: 'Giới thiệu XMP/EXPO và tản nhiệt RAM'
352|     },
353| ],
354|     animation: { type: 'shake', targetComponents: ['ram', 'motherboard'], durationMs: 1000 },
355|     affectedParts: ['ram', 'motherboard']
356| },
357| {
358|     id: 'err-cpu-cooler-paste-016',
359|     code: 'NO_THERMAL_PASTE',
360|     severity: 'info',
361|     title: 'Chào có keo tản nhiệt cho CPU',
362|     description: 'Không có thermal paste giữa CPU IHS và cooler baseplate',
363|     howToFix: '1. Hết keo tản nhiệt. Áp dụng thermal paste (pre-applied)'.
364|     youtubeVideo: [
365|         {
366|             videoId: '5QxNwWNH7o',
367|             title: 'How to Apply Thermal Paste - Perfect Guide',
368|             channelName: 'GamersNexus',
369|             startSeconds: 60,
370|             endSeconds: 240,
371|             language: 'en',
372|             description: 'Cách bôi keo tản nhiệt đúng cách'
373|         }
374|     ],
375|     animation: { type: 'red-flash', targetComponents: ['cpu', 'cooler'], durationMs: 1500 },
376|     affectedParts: ['cpu', 'cooler']
377| },
378| {
379|     id: 'err-standoff-017',
380|     code: 'MISSING_STANDOFFS',
381|     severity: 'warning',
382|     title: 'Thiếu standoff cho Mainboard',
383|     description: 'Mainboard đã lắp sẵn standoff, lắp vào case mà không có standoff',
384|     howToFix: '1. Luôn gắn standoff trước khi lắp mainboard vào case\n2.
385|     youtubeVideo: [
386|         {
387|             videoId: 'N0J6Gjxl-CI',
388|             title: 'How to Install Motherboard - Standoffs Guide',
389|             channelName: "Paul's Hardware",
390|             startSeconds: 120,
391|             endSeconds: 300,
392|             language: 'en',
393|             description: 'Hướng dẫn lắp mainboard đúng cách'
394|         }
395|     ],
396|     animation: { type: 'shake', targetComponents: ['motherboard', 'case'], durationMs: 1500 },
397|     affectedParts: ['motherboard', 'case']
398| },
399| {
400|     id: 'err-pcie-riser-018',
401|     code: 'GPU_PCIE_RISER_ISSUE',
402|     severity: 'warning',
403|     title: 'Cáp PCIe riser có thể gây mất tín hiệu',
404|     description: 'Sử dụng cáp PCIe riser kém chất lượng có thể gây mất tín hiệu',
405|     howToFix: '1. Dùng PCIe riser chính hãng (Phanteks, Cooler Master, Thermaltake)\n2. Set BIOS sang PCIe Gen...
406|     youtubeVideo: [
407|         {
408|             videoId: 'J_frJZCU1TY',
409|             title: 'PCIe Riser Cable Issues Fixed!',
410|             channelName: 'GamersNexus',
411|             startSeconds: 90,
412|             language: 'en',
413|             description: 'Khắc phục lỗi PCIe riser cable'
414|         }
415|     ],
416|     animation: { type: 'arrow-wrong', targetComponents: ['gpu', 'motherboard'], durationMs: 2000 },
417|     affectedParts: ['gpu', 'motherboard']
418| },
419| {
420|     id: 'err-ram-voltage-019',
421|     code: 'RAM_VOLTAGE_HIGH',
422|     severity: 'info',
423|     title: 'RAM có điện áp cao hơn mức khuyến nghị',
424|     description: 'RAM XMP/EXPO có thể chạy ở điện áp 1.35V-1.5V, cao hơn mức khuyến nghị',
425|     howToFix: '1. Vào BIOS enable XMP/EXPO. Giảm điện áp RAM chỉ về mức mặc định'.
426|     youtubeVideo: [
427|         {
428|             videoId: '4c5P5F5D3Fs',
429|             title: 'RAM Voltage and XMP Explained',
430|             channelName: 'Tech Deals',
431|             startSeconds: 120,
432|             language: 'en',
433|             description: 'Giới thiệu về điện áp RAM và XMP'
434|         }
435|     ],
436|     animation: { type: 'shake', targetComponents: ['ram', 'motherboard'], durationMs: 1000 },
437|     affectedParts: ['ram', 'motherboard']
438| },
439| {
440|     id: 'err-case-fan-020',
441|     code: 'CASE_FAN_MISMATCH',
442|     severity: 'info',
443|     title: 'Quạt case không đồng bộ',
444|     description: 'Các quạt case không cùng loại (PWM vs DC) hoặc không có đầu nối PWM (4-pin)',
445|     howToFix: '1. Dùng toàn bộ quạt PWM (4-pin). Kiểm tra cài đặt BIOS về quạt case'.
446|     youtubeVideo: [
447|         {
448|             videoId: 'akR_dU4qGso',
449|             title: 'Case Fans Guide - PWM vs DC',
450|             channelName: 'Hardware Canucks',
451|             startSeconds: 90,
452|             language: 'en',
453|             description: 'So sánh quạt PWM và DC cho case'
454|         }
455|     ],
456|     animation: { type: 'red-flash', targetComponents: ['case'], durationMs: 1200 },
457|     affectedParts: ['case']
458| },
459| {
460|     id: 'err-aio-orientation-021',
461|     code: 'AIO_MOUNTING_ERROR',
462|     severity: 'warning',
463|     title: 'Gắn AIO cooler sai hướng',
464|     description: 'AIO liquid cooler nếu gắn sai hướng (pump về phía trái case)',
465|     howToFix: '1. Pump (block trên CPU) không được lắp phía sau case'.
466|     youtubeVideo: [
467|         {
468|             videoId: 'qgN88TQSTGQ',
469|             title: 'AIO Cooler Orientation Guide - Don't Do This!',
470|             channelName: 'JayzTwoCents',
471|             startSeconds: 60,
```

```

472|         endSeconds: 360,
473|         language: 'en',
474|         description: 'H° Ũng d«n g~n AIO cooler úng cách'
475|     },
476|     ],
477|     animation: { type: 'arrow-wrong', targetComponents: ['cooler', 'cpu'], durationMs: 2000 },
478|     affectedParts: ['cooler', 'cpu']
479| },
480| {
481|     id: 'err-front-panel-022',
482|     code: 'FRONT_PANEL_WRONG',
483|     severity: 'warning',
484|     title: 'C~m sai dây front panel',
485|     description: 'Các dây Power SW, Reset SW, HDD LED, Power LED tẽ case',
486|     howToFix: '1. Xem manual mainboard Á biẽt si Ồ chân front panel\`n2.
487|     youtubeVideo: [
488|         {
489|             videoId: '2zVPxGq_ar4',
490|             title: 'Front Panel Connectors Guide',
491|             channelName: 'Paul's Hardware',
492|             startSeconds: 180,
493|             endSeconds: 360,
494|             language: 'en',
495|             description: 'H° Ũng d«n c~m dây front panel case'
496|         }
497|     ],
498|     animation: { type: 'shake', targetComponents: ['motherboard', 'case'], durationMs: 1500 },
499|     affectedParts: ['motherboard', 'case']
500| },
501| {
502|     id: 'err-cable-management-023',
503|     code: 'POOR_CABLE_MANAGEMENT',
504|     severity: 'info',
505|     title: 'Quên lý dây cáp kém',
506|     description: 'Dây cáp trong case bẽa bũn, cẽn luống khí, gây t ng nh
507|     howToFix: '1. i dây sau mainboard tray (cable routing hole)\`n2. Dũn
508|     youtubeVideo: [
509|         {
510|             videoId: 'P6SYFWH3Y04',
511|             title: 'PC Cable Management for Beginners',
512|             channelName: 'Bitwit',
513|             startSeconds: 120,
514|             endSeconds: 480,
515|             language: 'en',
516|             description: 'H° Ũng d«n quên lý dây cáp trong case'
517|         }
518|     ],
519|     animation: { type: 'red-flash', targetComponents: ['case'], durationMs: 1800 },
520|     affectedParts: ['case']
521| },
522| {
523|     id: 'err-bios-settings-024',
524|     code: 'BIOS_SETTINGS_WRONG',
525|     severity: 'warning',
526|     title: 'Cài .t BIOS không t Ñi °u',
527|     description: 'BIOS ch°a c~u hình úng: ch°a b-t XMP, ch°a set b
528|     howToFix: '1. Vào BIOS (Del / F2 khi khẽi ùng)\`n2. B-t XMP / EXPO cho P
529|     youtubeVideo: [
530|         {
531|             videoId: 'b-pFHL4_F8',
532|             title: 'BIOS Settings You Should Change Right Now',
533|             channelName: 'Linus Tech Tips',
534|             startSeconds: 60,
535|             endSeconds: 360,
536|             language: 'en',
537|             description: 'Các cài .t BIOS quan trẽng csn thay Ồi'
538|         }
539|     ],
540|     animation: { type: 'shake', targetComponents: ['motherboard'], durationMs: 1200 },
541|     affectedParts: ['motherboard']
542| },
543| {
544|     id: 'err-gpu-sag-025',
545|     code: 'GPU_SAG_RISK',
546|     severity: 'info',
547|     title: 'GPU có nguy ci bẽ vũng (GPU sag)',
548|     description: 'GPU n~ng (RTX 4080 / 4090 trẽ, lãn) có thÃ bẽ vũng do trĩ
549|     howToFix: '1. Dũng GPU support bracket (chãn) á GPU) i kèm trong hũ
550|     youtubeVideo: [
551|         {
552|             videoId: 'qNcnlVBey84',
553|             title: 'GPU Sag Fix - Best Anti-Sag Solutions',
554|             channelName: 'JayzTwoCents',
555|             startSeconds: 60,
556|             endSeconds: 240,
557|             language: 'en',
558|             description: 'Các giẽi pháp ch Ñĩng vũng GPU'
559|         }
560|     ],
561|     animation: { type: 'cross-mark', targetComponents: ['gpu', 'motherboard'], durationMs: 1500 },
562|     affectedParts: ['gpu', 'motherboard']
563| },
564| ];
565|
566| export function findErrorByCode(code: string): AssemblyError | undefined {
567|     return ASSEMBLY_ERRORS.find((e) => e.code === code);
568| }
569|
570| export function findErrorsByPart(partType: string): AssemblyError[] {
571|     return ASSEMBLY_ERRORS.filter((e) => e.affectedParts.includes(partType));
572| }
573|
574| export function getErrorAnimationStyle(error: AssemblyError): Record<string, string | number> {
575|     const base: Record<string, string | number> = {
576|         position: 'absolute',
577|         inset: 0,
578|         pointerEvents: 'none',
579|         zIndex: 100,
580|     };
581|     switch (error.animation.type) {
582|         case 'shake':
583|             return {
584|                 ...base,
585|                 animation: `shake ${error.animation.durationMs}ms ease-in-out`,
586|             };
587|         case 'red-flash':
588|             return {
589|                 ...base,
590|                 background: 'rgba(239, 68, 68, 0.3)',
591|                 animation: `redFlash ${error.animation.durationMs}ms ease-in-out`,
592|             };
593|         case 'cross-mark':
594|             return {
595|                 ...base,

```



```

596|         display: 'flex',
597|         alignItems: 'center',
598|         justifyContent: 'center',
599|         fontSize: '4rem',
600|         color: '#ef4444',
601|         animation: `crossMark ${error.animation.durationMs}ms ease-out`,
602|     };
603|     default:
604|         return base;
605| }
606| }
607|

```

File: data/build-scenarios.json (183 lines, 5.3 KB)

```

1| [
2|   {
3|     "id": "office-budget",
4|     "name": "Office Budget PC",
5|     "description": "Build a reliable office PC for under 10 million VND. Perfect for Word, Excel, web browsing...",
6|     "icon": "🖨️",
7|     "difficulty": "easy",
8|     "budget_min": 3000000,
9|     "budget_max": 10000000,
10|     "requirements": {
11|       "min_ram_gb": 8,
12|       "min_storage_gb": 240,
13|       "require_igpu": true,
14|       "use_case": "office",
15|       "max_tdp": 150
16|     },
17|     "weight_components": {
18|       "cpu": 10,
19|       "gpu": 0,
20|       "ram": 20,
21|       "storage": 15,
22|       "psu": 10
23|     }
24|   },
25|   {
26|     "id": "budget-gaming",
27|     "name": "Budget Gaming Rig",
28|     "description": "Build a capable gaming PC for ~15 million VND. Should run Valorant, LOL, and CS2 at 1080p.",
29|     "icon": "🎮",
30|     "difficulty": "easy",
31|     "budget_min": 10000000,
32|     "budget_max": 18000000,
33|     "requirements": {
34|       "min_ram_gb": 16,
35|       "min_gpu_power": 6000,
36|       "min_storage_gb": 480,
37|       "use_case": "gaming",
38|       "min_cores": 6
39|     },
40|     "weight_components": {
41|       "cpu": 30,
42|       "gpu": 40,
43|       "ram": 10,
44|       "storage": 5,
45|       "psu": 15
46|     }
47|   },
48|   {
49|     "id": "mainstream-gaming",
50|     "name": "Mainstream Gaming",
51|     "description": "Build a balanced high-end gaming PC around 25 million. Handles AAA titles at 1440p ultra s...",
52|     "icon": "🎮",
53|     "difficulty": "medium",
54|     "budget_min": 18000000,
55|     "budget_max": 30000000,
56|     "requirements": {
57|       "min_ram_gb": 16,
58|       "min_gpu_power": 15000,
59|       "min_storage_gb": 1000,
60|       "use_case": "gaming",
61|       "min_cores": 8,
62|       "min_cinebench_multi": 18000
63|     },
64|     "weight_components": {
65|       "cpu": 35,
66|       "gpu": 40,
67|       "ram": 10,
68|       "storage": 5,
69|       "psu": 10
70|     }
71|   },
72|   {
73|     "id": "workstation",
74|     "name": "Content Creator Workstation",
75|     "description": "Build a powerful workstation for video editing, 3D rendering, and graphic design.",
76|     "icon": "💻",
77|     "difficulty": "hard",
78|     "budget_min": 25000000,
79|     "budget_max": 50000000,
80|     "requirements": {
81|       "min_ram_gb": 32,
82|       "min_gpu_power": 25000,
83|       "min_storage_gb": 2000,
84|       "use_case": "workstation",
85|       "min_cores": 12,
86|       "min_cinebench_multi": 30000,
87|       "max_tdp": 400
88|     },
89|     "weight_components": {
90|       "cpu": 35,
91|       "gpu": 30,
92|       "ram": 15,
93|       "storage": 10,
94|       "psu": 10
95|     }
96|   },
97|   {
98|     "id": "mini-itx",
99|     "name": "Mini ITX Build",
100|     "description": "Build a compact mini PC for office or light gaming.",
101|     "icon": "🖨️",
102|     "difficulty": "easy",
103|     "budget_min": 3000000,
104|     "budget_max": 10000000,
105|     "requirements": {
106|       "min_ram_gb": 8,
107|       "min_storage_gb": 240,
108|       "require_igpu": true,
109|       "use_case": "office",
110|       "max_tdp": 150
111|     },
112|     "weight_components": {
113|       "cpu": 10,
114|       "gpu": 0,
115|       "ram": 20,
116|       "storage": 15,
117|       "psu": 10
118|     }
119|   }
120| ]

```

```
109|         "description": "Build a compact Mini ITX system in a small form factor case under 20L volume.",
110|         "description_vn": "L_p PC Mini ITX trong case nhĩ gĩn d°0i 20L.",
111|         "icon": "0=Uæ",
112|         "difficulty": "expert",
113|         "budget_min": 15000000,
114|         "budget_max": 40000000,
115|         "requirements": {
116|             "min_ram_gb": 16,
117|             "min_gpu_power": 12000,
118|             "min_storage_gb": 1000,
119|             "use_case": "gaming",
120|             "max_tdp": 250
121|         },
122|         "weight_components": {
123|             "cpu": 25,
124|             "gpu": 30,
125|             "ram": 10,
126|             "storage": 10,
127|             "psu": 15
128|         },
129|     },
130|     {
131|         "id": "silent-pc",
132|         "name": "Silent Build",
133|         "name_vn": "PC Si ẽu Y ẽn T) nh",
134|         "description": "Build a near-silent PC with excellent cooling and low noise levels for home theater or off...
135|         "description_vn": "PC gĩn nh° im l.ng vũĩ tĩn nhiçt tĩt cho rip hát
136|         "icon": "0=Y",
137|         "difficulty": "medium",
138|         "budget_min": 12000000,
139|         "budget_max": 30000000,
140|         "requirements": {
141|             "min_ram_gb": 16,
142|             "min_storage_gb": 480,
143|             "require_igpu": true,
144|             "use_case": "productivity",
145|             "max_tdp": 100
146|         },
147|         "weight_components": {
148|             "cpu": 20,
149|             "gpu": 0,
150|             "ram": 15,
151|             "storage": 10,
152|             "psu": 15
153|         },
154|     },
155|     {
156|         "id": "ultimate-battlestation",
157|         "name": "Ultimate Battlestation",
158|         "name_vn": "D  ̃n PC  ̃nh Cao",
159|         "description": "No budget limit. Build the absolute best PC money can buy. 4K gaming, extreme VR, AI workl...
160|         "description_vn": "Khõng giũĩ hĩn ng  ̃n sách. L_p PC - t nhçt có th  ̃.
161|         "icon": "0=UQ",
162|         "difficulty": "hard",
163|         "budget_min": 40000000,
164|         "budget_max": 100000000,
165|         "requirements": {
166|             "min_ram_gb": 32,
167|             "min_gpu_power": 35000,
168|             "min_storage_gb": 2000,
169|             "use_case": "workstation",
170|             "min_cores": 16,
171|             "min_cinebench_multi": 35000,
172|             "max_tdp": 600
173|         },
174|         "weight_components": {
175|             "cpu": 30,
176|             "gpu": 35,
177|             "ram": 10,
178|             "storage": 10,
179|             "psu": 15
180|         },
181|     }
182| ]
183|
```

File: data/componentsData.json (5002 lines, 1039.3 KB)

```
1| [
2|   {
3|     "id": "cpu_intel_12100F_1",
4|     "name": "Intel Core i3-12100F",
5|     "type": "CPU",
6|     "price": 2730000,
7|     "socket": "LGA1700",
8|     "power": 125,
9|     "desc": "2 nh  ̃n / 4 luõng, 125W TDP, Socket LGA1700"
10|   },
11|   {
12|     "id": "cpu_intel_12100T_2",
13|     "name": "Intel Core i3-12100T",
14|     "type": "CPU",
15|     "price": 3550000,
16|     "socket": "LGA1700",
17|     "power": 75,
18|     "desc": "18 nh  ̃n / 36 luõng, 75W TDP, Socket LGA1700"
19|   },
20|   {
21|     "id": "cpu_intel_12100std_3",
22|     "name": "Intel Core i3-12100",
23|     "type": "CPU",
24|     "price": 4510000,
25|     "socket": "LGA1700",
26|     "power": 75,
27|     "desc": "14 nh  ̃n / 28 luõng, 75W TDP, Socket LGA1700"
28|   },
29|   {
30|     "id": "cpu_intel_12100F_4",
31|     "name": "Intel Core i3-12100F",
32|     "type": "CPU",
33|     "price": 1580000,
34|     "socket": "LGA1700",
35|     "power": 65,
36|     "desc": "8 nh  ̃n / 16 luõng, 65W TDP, Socket LGA1700"
37|   },
38|   {
39|     "id": "cpu_intel_12100T_5",
40|     "name": "Intel Core i3-12100T",
41|     "type": "CPU",
42|     "price": 2680000,
43|     "socket": "LGA1700",
44|     "power": 55,
45|     "desc": "14 nh  ̃n / 28 luõng, 55W TDP, Socket LGA1700"
46|   }
47| ]
```

```

46| },
47| {
48|     "id": "cpu_intel_12100std_6",
49|     "name": "Intel Core i3-12100",
50|     "type": "CPU",
51|     "price": 4770000,
52|     "socket": "LGA1700",
53|     "power": 95,
54|     "desc": "18 nhân / 36 luồng, 95W TDP, Socket LGA1700"
55| },
56| {
57|     "id": "cpu_intel_12100F_7",
58|     "name": "Intel Core i3-12100F",
59|     "type": "CPU",
60|     "price": 3260000,
61|     "socket": "LGA1700",
62|     "power": 65,
63|     "desc": "14 nhân / 28 luồng, 65W TDP, Socket LGA1700"
64| },
65| {
66|     "id": "cpu_intel_12100T_8",
67|     "name": "Intel Core i3-12100T",
68|     "type": "CPU",
69|     "price": 2670000,
70|     "socket": "LGA1700",
71|     "power": 120,
72|     "desc": "14 nhân / 28 luồng, 120W TDP, Socket LGA1700"
73| },
74| {
75|     "id": "cpu_intel_12100std_9",
76|     "name": "Intel Core i3-12100",
77|     "type": "CPU",
78|     "price": 2800000,
79|     "socket": "LGA1700",
80|     "power": 95,
81|     "desc": "4 nhân / 8 luồng, 95W TDP, Socket LGA1700"
82| },
83| {
84|     "id": "cpu_intel_12100F_10",
85|     "name": "Intel Core i3-12100F",
86|     "type": "CPU",
87|     "price": 930000,
88|     "socket": "LGA1700",
89|     "power": 65,
90|     "desc": "16 nhân / 32 luồng, 65W TDP, Socket LGA1700"
91| },
92| {
93|     "id": "cpu_intel_12100T_11",
94|     "name": "Intel Core i3-12100T",
95|     "type": "CPU",
96|     "price": 4090000,
97|     "socket": "LGA1700",
98|     "power": 125,
99|     "desc": "2 nhân / 4 luồng, 125W TDP, Socket LGA1700"
100| },
101| {
102|     "id": "cpu_intel_12100std_12",
103|     "name": "Intel Core i3-12100",
104|     "type": "CPU",
105|     "price": 4880000,
106|     "socket": "LGA1700",
107|     "power": 120,
108|     "desc": "18 nhân / 36 luồng, 120W TDP, Socket LGA1700"
109| },
110| {
111|     "id": "cpu_intel_13100F_13",
112|     "name": "Intel Core i3-13100F",
113|     "type": "CPU",
114|     "price": 1440000,
115|     "socket": "LGA1700",
116|     "power": 125,
117|     "desc": "6 nhân / 12 luồng, 125W TDP, Socket LGA1700"
118| },
119| {
120|     "id": "cpu_intel_13100T_14",
121|     "name": "Intel Core i3-13100T",
122|     "type": "CPU",
123|     "price": 2380000,
124|     "socket": "LGA1700",
125|     "power": 65,
126|     "desc": "16 nhân / 32 luồng, 65W TDP, Socket LGA1700"
127| },
128| {
129|     "id": "cpu_intel_13100std_15",
130|     "name": "Intel Core i3-13100",
131|     "type": "CPU",
132|     "price": 3280000,
133|     "socket": "LGA1700",
134|     "power": 65,
135|     "desc": "16 nhân / 32 luồng, 65W TDP, Socket LGA1700"
136| },
137| {
138|     "id": "cpu_intel_13100F_16",
139|     "name": "Intel Core i3-13100F",
140|     "type": "CPU",
141|     "price": 1210000,
142|     "socket": "LGA1700",
143|     "power": 95,
144|     "desc": "4 nhân / 8 luồng, 95W TDP, Socket LGA1700"
145| },
146| {
147|     "id": "cpu_intel_13100T_17",
148|     "name": "Intel Core i3-13100T",
149|     "type": "CPU",
150|     "price": 3180000,
151|     "socket": "LGA1700",
152|     "power": 85,
153|     "desc": "8 nhân / 16 luồng, 85W TDP, Socket LGA1700"
154| },
155| {
156|     "id": "cpu_intel_13100std_18",
157|     "name": "Intel Core i3-13100",
158|     "type": "CPU",
159|     "price": 4850000,
160|     "socket": "LGA1700",
161|     "power": 125,
162|     "desc": "6 nhân / 12 luồng, 125W TDP, Socket LGA1700"
163| },
164| {
165|     "id": "cpu_intel_13100F_19",
166|     "name": "Intel Core i3-13100F",
167|     "type": "CPU",
168|     "price": 1010000,
169|     "socket": "LGA1700",

```

```
170| "power": 95,
171| "desc": "14 nhân / 28 luồng, 95W TDP, Socket LGA1700"
172| },
173| {
174| "id": "cpu_intel_13100T_20",
175| "name": "Intel Core i3-13100T",
176| "type": "CPU",
177| "price": 3650000,
178| "socket": "LGA1700",
179| "power": 120,
180| "desc": "16 nhân / 32 luồng, 120W TDP, Socket LGA1700"
181| },
182| {
183| "id": "cpu_intel_13100std_21",
184| "name": "Intel Core i3-13100",
185| "type": "CPU",
186| "price": 1800000,
187| "socket": "LGA1700",
188| "power": 85,
189| "desc": "6 nhân / 12 luồng, 85W TDP, Socket LGA1700"
190| },
191| {
192| "id": "cpu_intel_13100F_22",
193| "name": "Intel Core i3-13100F",
194| "type": "CPU",
195| "price": 2600000,
196| "socket": "LGA1700",
197| "power": 95,
198| "desc": "16 nhân / 32 luồng, 95W TDP, Socket LGA1700"
199| },
200| {
201| "id": "cpu_intel_13100T_23",
202| "name": "Intel Core i3-13100T",
203| "type": "CPU",
204| "price": 1710000,
205| "socket": "LGA1700",
206| "power": 75,
207| "desc": "14 nhân / 28 luồng, 75W TDP, Socket LGA1700"
208| },
209| {
210| "id": "cpu_intel_13100std_24",
211| "name": "Intel Core i3-13100",
212| "type": "CPU",
213| "price": 4480000,
214| "socket": "LGA1700",
215| "power": 105,
216| "desc": "12 nhân / 24 luồng, 105W TDP, Socket LGA1700"
217| },
218| {
219| "id": "cpu_intel_14100F_25",
220| "name": "Intel Core i3-14100F",
221| "type": "CPU",
222| "price": 930000,
223| "socket": "LGA1851",
224| "power": 95,
225| "desc": "12 nhân / 24 luồng, 95W TDP, Socket LGA1851"
226| },
227| {
228| "id": "cpu_intel_14100T_26",
229| "name": "Intel Core i3-14100T",
230| "type": "CPU",
231| "price": 2800000,
232| "socket": "LGA1851",
233| "power": 95,
234| "desc": "16 nhân / 32 luồng, 95W TDP, Socket LGA1851"
235| },
236| {
237| "id": "cpu_intel_14100std_27",
238| "name": "Intel Core i3-14100",
239| "type": "CPU",
240| "price": 2950000,
241| "socket": "LGA1851",
242| "power": 95,
243| "desc": "18 nhân / 36 luồng, 95W TDP, Socket LGA1851"
244| },
245| {
246| "id": "cpu_intel_14100F_28",
247| "name": "Intel Core i3-14100F",
248| "type": "CPU",
249| "price": 4000000,
250| "socket": "LGA1851",
251| "power": 65,
252| "desc": "4 nhân / 8 luồng, 65W TDP, Socket LGA1851"
253| },
254| {
255| "id": "cpu_intel_14100T_29",
256| "name": "Intel Core i3-14100T",
257| "type": "CPU",
258| "price": 4450000,
259| "socket": "LGA1851",
260| "power": 125,
261| "desc": "10 nhân / 20 luồng, 125W TDP, Socket LGA1851"
262| },
263| {
264| "id": "cpu_intel_14100std_30",
265| "name": "Intel Core i3-14100",
266| "type": "CPU",
267| "price": 4430000,
268| "socket": "LGA1851",
269| "power": 75,
270| "desc": "16 nhân / 32 luồng, 75W TDP, Socket LGA1851"
271| },
272| {
273| "id": "cpu_intel_14100F_31",
274| "name": "Intel Core i3-14100F",
275| "type": "CPU",
276| "price": 2890000,
277| "socket": "LGA1851",
278| "power": 95,
279| "desc": "10 nhân / 20 luồng, 95W TDP, Socket LGA1851"
280| },
281| {
282| "id": "cpu_intel_14100T_32",
283| "name": "Intel Core i3-14100T",
284| "type": "CPU",
285| "price": 3660000,
286| "socket": "LGA1851",
287| "power": 120,
288| "desc": "8 nhân / 16 luồng, 120W TDP, Socket LGA1851"
289| },
290| {
291| "id": "cpu_intel_14100std_33",
292| "name": "Intel Core i3-14100",
293| "type": "CPU",
```

```

294|     "price": 4400000,
295|     "socket": "LGA1851",
296|     "power": 105,
297|     "desc": "2 nhân / 4 luồng, 105W TDP, Socket LGA1851"
298| },
299| {
300|     "id": "cpu_intel_14100F_34",
301|     "name": "Intel Core i3-14100F",
302|     "type": "CPU",
303|     "price": 4160000,
304|     "socket": "LGA1851",
305|     "power": 85,
306|     "desc": "8 nhân / 16 luồng, 85W TDP, Socket LGA1851"
307| },
308| {
309|     "id": "cpu_intel_14100T_35",
310|     "name": "Intel Core i3-14100T",
311|     "type": "CPU",
312|     "price": 9600000,
313|     "socket": "LGA1851",
314|     "power": 125,
315|     "desc": "16 nhân / 32 luồng, 125W TDP, Socket LGA1851"
316| },
317| {
318|     "id": "cpu_intel_14100std_36",
319|     "name": "Intel Core i3-14100",
320|     "type": "CPU",
321|     "price": 10400000,
322|     "socket": "LGA1851",
323|     "power": 105,
324|     "desc": "4 nhân / 8 luồng, 105W TDP, Socket LGA1851"
325| },
326| {
327|     "id": "cpu_intel_15100F_37",
328|     "name": "Intel Core i3-15100F",
329|     "type": "CPU",
330|     "price": 3070000,
331|     "socket": "LGA1851",
332|     "power": 85,
333|     "desc": "2 nhân / 4 luồng, 85W TDP, Socket LGA1851"
334| },
335| {
336|     "id": "cpu_intel_15100T_38",
337|     "name": "Intel Core i3-15100T",
338|     "type": "CPU",
339|     "price": 1710000,
340|     "socket": "LGA1851",
341|     "power": 75,
342|     "desc": "8 nhân / 16 luồng, 75W TDP, Socket LGA1851"
343| },
344| {
345|     "id": "cpu_intel_15100std_39",
346|     "name": "Intel Core i3-15100",
347|     "type": "CPU",
348|     "price": 1480000,
349|     "socket": "LGA1851",
350|     "power": 95,
351|     "desc": "10 nhân / 20 luồng, 95W TDP, Socket LGA1851"
352| },
353| {
354|     "id": "cpu_intel_15100F_40",
355|     "name": "Intel Core i3-15100F",
356|     "type": "CPU",
357|     "price": 1430000,
358|     "socket": "LGA1851",
359|     "power": 125,
360|     "desc": "10 nhân / 20 luồng, 125W TDP, Socket LGA1851"
361| },
362| {
363|     "id": "cpu_intel_15100T_41",
364|     "name": "Intel Core i3-15100T",
365|     "type": "CPU",
366|     "price": 4240000,
367|     "socket": "LGA1851",
368|     "power": 125,
369|     "desc": "18 nhân / 36 luồng, 125W TDP, Socket LGA1851"
370| },
371| {
372|     "id": "cpu_intel_15100std_42",
373|     "name": "Intel Core i3-15100",
374|     "type": "CPU",
375|     "price": 3030000,
376|     "socket": "LGA1851",
377|     "power": 95,
378|     "desc": "4 nhân / 8 luồng, 95W TDP, Socket LGA1851"
379| },
380| {
381|     "id": "cpu_intel_15100F_43",
382|     "name": "Intel Core i3-15100F",
383|     "type": "CPU",
384|     "price": 3870000,
385|     "socket": "LGA1851",
386|     "power": 75,
387|     "desc": "10 nhân / 20 luồng, 75W TDP, Socket LGA1851"
388| },
389| {
390|     "id": "cpu_intel_15100T_44",
391|     "name": "Intel Core i3-15100T",
392|     "type": "CPU",
393|     "price": 1740000,
394|     "socket": "LGA1851",
395|     "power": 120,
396|     "desc": "14 nhân / 28 luồng, 120W TDP, Socket LGA1851"
397| },
398| {
399|     "id": "cpu_intel_15100std_45",
400|     "name": "Intel Core i3-15100",
401|     "type": "CPU",
402|     "price": 2320000,
403|     "socket": "LGA1851",
404|     "power": 60,
405|     "desc": "4 nhân / 8 luồng, 60W TDP, Socket LGA1851"
406| },
407| {
408|     "id": "cpu_intel_15100F_46",
409|     "name": "Intel Core i3-15100F",
410|     "type": "CPU",
411|     "price": 2800000,
412|     "socket": "LGA1851",
413|     "power": 125,
414|     "desc": "12 nhân / 24 luồng, 125W TDP, Socket LGA1851"
415| },
416| {
417|     "id": "cpu_intel_15100T_47",

```

```

418| "name": "Intel Core i3-15100T",
419| "type": "CPU",
420| "price": 3900000,
421| "socket": "LGA1851",
422| "power": 95,
423| "desc": "16 nhân / 32 luồng, 95W TDP, Socket LGA1851"
424| },
425| {
426| "id": "cpu_intel_15100std_48",
427| "name": "Intel Core i3-15100",
428| "type": "CPU",
429| "price": 3260000,
430| "socket": "LGA1851",
431| "power": 95,
432| "desc": "12 nhân / 24 luồng, 95W TDP, Socket LGA1851"
433| },
434| {
435| "id": "cpu_intel_12400K_49",
436| "name": "Intel Core i5-12400K",
437| "type": "CPU",
438| "price": 7500000,
439| "socket": "LGA1700",
440| "power": 85,
441| "desc": "4 nhân / 8 luồng, 85W TDP, Socket LGA1700"
442| },
443| {
444| "id": "cpu_intel_12400KF_50",
445| "name": "Intel Core i5-12400KF",
446| "type": "CPU",
447| "price": 7480000,
448| "socket": "LGA1700",
449| "power": 95,
450| "desc": "16 nhân / 32 luồng, 95W TDP, Socket LGA1700"
451| },
452| {
453| "id": "cpu_intel_12400T_51",
454| "name": "Intel Core i5-12400T",
455| "type": "CPU",
456| "price": 3390000,
457| "socket": "LGA1700",
458| "power": 105,
459| "desc": "14 nhân / 28 luồng, 105W TDP, Socket LGA1700"
460| },
461| {
462| "id": "cpu_intel_12400F_52",
463| "name": "Intel Core i5-12400F",
464| "type": "CPU",
465| "price": 3840000,
466| "socket": "LGA1700",
467| "power": 55,
468| "desc": "16 nhân / 32 luồng, 55W TDP, Socket LGA1700"
469| },
470| {
471| "id": "cpu_intel_12400std_53",
472| "name": "Intel Core i5-12400",
473| "type": "CPU",
474| "price": 8580000,
475| "socket": "LGA1700",
476| "power": 75,
477| "desc": "16 nhân / 32 luồng, 75W TDP, Socket LGA1700"
478| },
479| {
480| "id": "cpu_intel_12600K_54",
481| "name": "Intel Core i5-12600K",
482| "type": "CPU",
483| "price": 4010000,
484| "socket": "LGA1700",
485| "power": 85,
486| "desc": "4 nhân / 8 luồng, 85W TDP, Socket LGA1700"
487| },
488| {
489| "id": "cpu_intel_12600KF_55",
490| "name": "Intel Core i5-12600KF",
491| "type": "CPU",
492| "price": 2190000,
493| "socket": "LGA1700",
494| "power": 60,
495| "desc": "6 nhân / 12 luồng, 60W TDP, Socket LGA1700"
496| },
497| {
498| "id": "cpu_intel_12600T_56",
499| "name": "Intel Core i5-12600T",
500| "type": "CPU",
501| "price": 9920000,
502| "socket": "LGA1700",
503| "power": 75,
504| "desc": "16 nhân / 32 luồng, 75W TDP, Socket LGA1700"
505| },
506| {
507| "id": "cpu_intel_12600F_57",
508| "name": "Intel Core i5-12600F",
509| "type": "CPU",
510| "price": 2290000,
511| "socket": "LGA1700",
512| "power": 105,
513| "desc": "14 nhân / 28 luồng, 105W TDP, Socket LGA1700"
514| },
515| {
516| "id": "cpu_intel_12600std_58",
517| "name": "Intel Core i5-12600",
518| "type": "CPU",
519| "price": 8470000,
520| "socket": "LGA1700",
521| "power": 120,
522| "desc": "2 nhân / 4 luồng, 120W TDP, Socket LGA1700"
523| },
524| {
525| "id": "cpu_intel_12400K_59",
526| "name": "Intel Core i5-12400K",
527| "type": "CPU",
528| "price": 6600000,
529| "socket": "LGA1700",
530| "power": 65,
531| "desc": "12 nhân / 24 luồng, 65W TDP, Socket LGA1700"
532| },
533| {
534| "id": "cpu_intel_12400KF_60",
535| "name": "Intel Core i5-12400KF",
536| "type": "CPU",
537| "price": 2350000,
538| "socket": "LGA1700",
539| "power": 105,
540| "desc": "12 nhân / 24 luồng, 105W TDP, Socket LGA1700"
541| },

```

```

542| {
543|   "id": "cpu_intel_12400T_61",
544|   "name": "Intel Core i5-12400T",
545|   "type": "CPU",
546|   "price": 3910000,
547|   "socket": "LGA1700",
548|   "power": 85,
549|   "desc": "8 nhân / 16 luồng, 85W TDP, Socket LGA1700"
550| },
551| {
552|   "id": "cpu_intel_12400F_62",
553|   "name": "Intel Core i5-12400F",
554|   "type": "CPU",
555|   "price": 1880000,
556|   "socket": "LGA1700",
557|   "power": 85,
558|   "desc": "4 nhân / 8 luồng, 85W TDP, Socket LGA1700"
559| },
560| {
561|   "id": "cpu_intel_12400std_63",
562|   "name": "Intel Core i5-12400",
563|   "type": "CPU",
564|   "price": 8780000,
565|   "socket": "LGA1700",
566|   "power": 85,
567|   "desc": "10 nhân / 20 luồng, 85W TDP, Socket LGA1700"
568| },
569| {
570|   "id": "cpu_intel_12600K_64",
571|   "name": "Intel Core i5-12600K",
572|   "type": "CPU",
573|   "price": 6420000,
574|   "socket": "LGA1700",
575|   "power": 75,
576|   "desc": "4 nhân / 8 luồng, 75W TDP, Socket LGA1700"
577| },
578| {
579|   "id": "cpu_intel_12600KF_65",
580|   "name": "Intel Core i5-12600KF",
581|   "type": "CPU",
582|   "price": 4540000,
583|   "socket": "LGA1700",
584|   "power": 65,
585|   "desc": "10 nhân / 20 luồng, 65W TDP, Socket LGA1700"
586| },
587| {
588|   "id": "cpu_intel_12600T_66",
589|   "name": "Intel Core i5-12600T",
590|   "type": "CPU",
591|   "price": 5160000,
592|   "socket": "LGA1700",
593|   "power": 105,
594|   "desc": "8 nhân / 16 luồng, 105W TDP, Socket LGA1700"
595| },
596| {
597|   "id": "cpu_intel_12600F_67",
598|   "name": "Intel Core i5-12600F",
599|   "type": "CPU",
600|   "price": 6240000,
601|   "socket": "LGA1700",
602|   "power": 65,
603|   "desc": "6 nhân / 12 luồng, 65W TDP, Socket LGA1700"
604| },
605| {
606|   "id": "cpu_intel_12600std_68",
607|   "name": "Intel Core i5-12600",
608|   "type": "CPU",
609|   "price": 6060000,
610|   "socket": "LGA1700",
611|   "power": 75,
612|   "desc": "2 nhân / 4 luồng, 75W TDP, Socket LGA1700"
613| },
614| {
615|   "id": "cpu_intel_12400K_69",
616|   "name": "Intel Core i5-12400K",
617|   "type": "CPU",
618|   "price": 6430000,
619|   "socket": "LGA1700",
620|   "power": 120,
621|   "desc": "6 nhân / 12 luồng, 120W TDP, Socket LGA1700"
622| },
623| {
624|   "id": "cpu_intel_12400KF_70",
625|   "name": "Intel Core i5-12400KF",
626|   "type": "CPU",
627|   "price": 7530000,
628|   "socket": "LGA1700",
629|   "power": 125,
630|   "desc": "4 nhân / 8 luồng, 125W TDP, Socket LGA1700"
631| },
632| {
633|   "id": "cpu_intel_12400T_71",
634|   "name": "Intel Core i5-12400T",
635|   "type": "CPU",
636|   "price": 7650000,
637|   "socket": "LGA1700",
638|   "power": 125,
639|   "desc": "12 nhân / 24 luồng, 125W TDP, Socket LGA1700"
640| },
641| {
642|   "id": "cpu_intel_12400F_72",
643|   "name": "Intel Core i5-12400F",
644|   "type": "CPU",
645|   "price": 1330000,
646|   "socket": "LGA1700",
647|   "power": 65,
648|   "desc": "14 nhân / 28 luồng, 65W TDP, Socket LGA1700"
649| },
650| {
651|   "id": "cpu_intel_12400std_73",
652|   "name": "Intel Core i5-12400",
653|   "type": "CPU",
654|   "price": 1500000,
655|   "socket": "LGA1700",
656|   "power": 85,
657|   "desc": "2 nhân / 4 luồng, 85W TDP, Socket LGA1700"
658| },
659| {
660|   "id": "cpu_intel_12600K_74",
661|   "name": "Intel Core i5-12600K",
662|   "type": "CPU",
663|   "price": 6960000,
664|   "socket": "LGA1700",
665|   "power": 65,

```

```

667| 6 6 6 | "desc": "4 nhân / 8 luồng, 65W TDP, Socket LGA1700"
668| },
669| {
670|   "id": "cpu_intel_12600KF_75",
671|   "name": "Intel Core i5-12600KF",
672|   "type": "CPU",
673|   "price": 8870000,
674|   "socket": "LGA1700",
675|   "power": 65,
676| 6 7 5 | "desc": "4 nhân / 8 luồng, 65W TDP, Socket LGA1700"
677| },
678| {
679|   "id": "cpu_intel_12600T_76",
680|   "name": "Intel Core i5-12600T",
681|   "type": "CPU",
682|   "price": 2880000,
683|   "socket": "LGA1700",
684|   "power": 55,
685| 6 8 4 | "desc": "4 nhân / 8 luồng, 55W TDP, Socket LGA1700"
686| },
687| {
688|   "id": "cpu_intel_12600F_77",
689|   "name": "Intel Core i5-12600F",
690|   "type": "CPU",
691|   "price": 8550000,
692|   "socket": "LGA1700",
693|   "power": 105,
694| 6 9 3 | "desc": "14 nhân / 28 luồng, 105W TDP, Socket LGA1700"
695| },
696| {
697|   "id": "cpu_intel_12600std_78",
698|   "name": "Intel Core i5-12600",
699|   "type": "CPU",
700|   "price": 8360000,
701|   "socket": "LGA1700",
702|   "power": 60,
703| 7 0 2 | "desc": "2 nhân / 4 luồng, 60W TDP, Socket LGA1700"
704| },
705| {
706|   "id": "cpu_intel_12400K_79",
707|   "name": "Intel Core i5-12400K",
708|   "type": "CPU",
709|   "price": 1920000,
710|   "socket": "LGA1700",
711|   "power": 75,
712| 7 1 1 | "desc": "14 nhân / 28 luồng, 75W TDP, Socket LGA1700"
713| },
714| {
715|   "id": "cpu_intel_12400KF_80",
716|   "name": "Intel Core i5-12400KF",
717|   "type": "CPU",
718|   "price": 4830000,
719|   "socket": "LGA1700",
720|   "power": 55,
721| 7 2 0 | "desc": "10 nhân / 20 luồng, 55W TDP, Socket LGA1700"
722| },
723| {
724|   "id": "cpu_intel_12400T_81",
725|   "name": "Intel Core i5-12400T",
726|   "type": "CPU",
727|   "price": 6000000,
728|   "socket": "LGA1700",
729|   "power": 125,
730| 7 2 9 | "desc": "2 nhân / 4 luồng, 125W TDP, Socket LGA1700"
731| },
732| {
733|   "id": "cpu_intel_12400F_82",
734|   "name": "Intel Core i5-12400F",
735|   "type": "CPU",
736|   "price": 6320000,
737|   "socket": "LGA1700",
738|   "power": 55,
739| 7 3 8 | "desc": "4 nhân / 8 luồng, 55W TDP, Socket LGA1700"
740| },
741| {
742|   "id": "cpu_intel_12400std_83",
743|   "name": "Intel Core i5-12400",
744|   "type": "CPU",
745|   "price": 6300000,
746|   "socket": "LGA1700",
747|   "power": 120,
748| 7 4 7 | "desc": "12 nhân / 24 luồng, 120W TDP, Socket LGA1700"
749| },
750| {
751|   "id": "cpu_intel_12600K_84",
752|   "name": "Intel Core i5-12600K",
753|   "type": "CPU",
754|   "price": 7860000,
755|   "socket": "LGA1700",
756|   "power": 95,
757| 7 5 6 | "desc": "8 nhân / 16 luồng, 95W TDP, Socket LGA1700"
758| },
759| {
760|   "id": "cpu_intel_12600KF_85",
761|   "name": "Intel Core i5-12600KF",
762|   "type": "CPU",
763|   "price": 7410000,
764|   "socket": "LGA1700",
765|   "power": 65,
766| 7 6 5 | "desc": "2 nhân / 4 luồng, 65W TDP, Socket LGA1700"
767| },
768| {
769|   "id": "cpu_intel_12600T_86",
770|   "name": "Intel Core i5-12600T",
771|   "type": "CPU",
772|   "price": 5630000,
773|   "socket": "LGA1700",
774|   "power": 65,
775| 7 7 4 | "desc": "14 nhân / 28 luồng, 65W TDP, Socket LGA1700"
776| },
777| {
778|   "id": "cpu_intel_12600F_87",
779|   "name": "Intel Core i5-12600F",
780|   "type": "CPU",
781|   "price": 7060000,
782|   "socket": "LGA1700",
783|   "power": 120,
784| 7 8 3 | "desc": "6 nhân / 12 luồng, 120W TDP, Socket LGA1700"
785| },
786| {
787|   "id": "cpu_intel_12600std_88",
788|   "name": "Intel Core i5-12600",
789|   "type": "CPU",
790|   "price": 7670000,

```



```
790| "socket": "LGA1700",
791| "power": 120,
792| "desc": "18 nhân / 36 luồng, 120W TDP, Socket LGA1700"
793| },
794| {
795| "id": "cpu_intel_13400K_89",
796| "name": "Intel Core i5-13400K",
797| "type": "CPU",
798| "price": 2390000,
799| "socket": "LGA1700",
800| "power": 95,
801| "desc": "14 nhân / 28 luồng, 95W TDP, Socket LGA1700"
802| },
803| {
804| "id": "cpu_intel_13400KF_90",
805| "name": "Intel Core i5-13400KF",
806| "type": "CPU",
807| "price": 3850000,
808| "socket": "LGA1700",
809| "power": 65,
810| "desc": "14 nhân / 28 luồng, 65W TDP, Socket LGA1700"
811| },
812| {
813| "id": "cpu_intel_13400T_91",
814| "name": "Intel Core i5-13400T",
815| "type": "CPU",
816| "price": 2840000,
817| "socket": "LGA1700",
818| "power": 75,
819| "desc": "12 nhân / 24 luồng, 75W TDP, Socket LGA1700"
820| },
821| {
822| "id": "cpu_intel_13400F_92",
823| "name": "Intel Core i5-13400F",
824| "type": "CPU",
825| "price": 1270000,
826| "socket": "LGA1700",
827| "power": 75,
828| "desc": "12 nhân / 24 luồng, 75W TDP, Socket LGA1700"
829| },
830| {
831| "id": "cpu_intel_13400std_93",
832| "name": "Intel Core i5-13400",
833| "type": "CPU",
834| "price": 3820000,
835| "socket": "LGA1700",
836| "power": 125,
837| "desc": "18 nhân / 36 luồng, 125W TDP, Socket LGA1700"
838| },
839| {
840| "id": "cpu_intel_13600K_94",
841| "name": "Intel Core i5-13600K",
842| "type": "CPU",
843| "price": 9270000,
844| "socket": "LGA1700",
845| "power": 85,
846| "desc": "2 nhân / 4 luồng, 85W TDP, Socket LGA1700"
847| },
848| {
849| "id": "cpu_intel_13600KF_95",
850| "name": "Intel Core i5-13600KF",
851| "type": "CPU",
852| "price": 2180000,
853| "socket": "LGA1700",
854| "power": 125,
855| "desc": "16 nhân / 32 luồng, 125W TDP, Socket LGA1700"
856| },
857| {
858| "id": "cpu_intel_13600T_96",
859| "name": "Intel Core i5-13600T",
860| "type": "CPU",
861| "price": 6410000,
862| "socket": "LGA1700",
863| "power": 75,
864| "desc": "14 nhân / 28 luồng, 75W TDP, Socket LGA1700"
865| },
866| {
867| "id": "cpu_intel_13600F_97",
868| "name": "Intel Core i5-13600F",
869| "type": "CPU",
870| "price": 9430000,
871| "socket": "LGA1700",
872| "power": 105,
873| "desc": "14 nhân / 28 luồng, 105W TDP, Socket LGA1700"
874| },
875| {
876| "id": "cpu_intel_13600std_98",
877| "name": "Intel Core i5-13600",
878| "type": "CPU",
879| "price": 3980000,
880| "socket": "LGA1700",
881| "power": 105,
882| "desc": "14 nhân / 28 luồng, 105W TDP, Socket LGA1700"
883| },
884| {
885| "id": "cpu_intel_13400K_99",
886| "name": "Intel Core i5-13400K",
887| "type": "CPU",
888| "price": 5590000,
889| "socket": "LGA1700",
890| "power": 75,
891| "desc": "12 nhân / 24 luồng, 75W TDP, Socket LGA1700"
892| },
893| {
894| "id": "cpu_intel_13400KF_100",
895| "name": "Intel Core i5-13400KF",
896| "type": "CPU",
897| "price": 3630000,
898| "socket": "LGA1700",
899| "power": 65,
900| "desc": "18 nhân / 36 luồng, 65W TDP, Socket LGA1700"
901| },
902| {
903| "id": "cpu_intel_13400T_101",
904| "name": "Intel Core i5-13400T",
905| "type": "CPU",
906| "price": 4860000,
907| "socket": "LGA1700",
908| "power": 65,
909| "desc": "18 nhân / 36 luồng, 65W TDP, Socket LGA1700"
910| },
911| {
912| "id": "cpu_intel_13400F_102",
913| "name": "Intel Core i5-13400F",
```

```

914|      "type": "CPU",
915|      "price": 9210000,
916|      "socket": "LGA1700",
917|      "power": 95,
918|      "desc": "2 nhân / 4 luồng, 95W TDP, Socket LGA1700"
919|    },
920|    {
921|      "id": "cpu_intel_13400std_103",
922|      "name": "Intel Core i5-13400",
923|      "type": "CPU",
924|      "price": 4920000,
925|      "socket": "LGA1700",
926|      "power": 75,
927|      "desc": "8 nhân / 16 luồng, 75W TDP, Socket LGA1700"
928|    },
929|    {
930|      "id": "cpu_intel_13600K_104",
931|      "name": "Intel Core i5-13600K",
932|      "type": "CPU",
933|      "price": 5190000,
934|      "socket": "LGA1700",
935|      "power": 125,
936|      "desc": "4 nhân / 8 luồng, 125W TDP, Socket LGA1700"
937|    },
938|    {
939|      "id": "cpu_intel_13600KF_105",
940|      "name": "Intel Core i5-13600KF",
941|      "type": "CPU",
942|      "price": 2710000,
943|      "socket": "LGA1700",
944|      "power": 75,
945|      "desc": "4 nhân / 8 luồng, 75W TDP, Socket LGA1700"
946|    },
947|    {
948|      "id": "cpu_intel_13600T_106",
949|      "name": "Intel Core i5-13600T",
950|      "type": "CPU",
951|      "price": 9910000,
952|      "socket": "LGA1700",
953|      "power": 60,
954|      "desc": "12 nhân / 24 luồng, 60W TDP, Socket LGA1700"
955|    },
956|    {
957|      "id": "cpu_intel_13600F_107",
958|      "name": "Intel Core i5-13600F",
959|      "type": "CPU",
960|      "price": 6360000,
961|      "socket": "LGA1700",
962|      "power": 60,
963|      "desc": "14 nhân / 28 luồng, 60W TDP, Socket LGA1700"
964|    },
965|    {
966|      "id": "cpu_intel_13600std_108",
967|      "name": "Intel Core i5-13600",
968|      "type": "CPU",
969|      "price": 4420000,
970|      "socket": "LGA1700",
971|      "power": 60,
972|      "desc": "16 nhân / 32 luồng, 60W TDP, Socket LGA1700"
973|    },
974|    {
975|      "id": "cpu_intel_13400K_109",
976|      "name": "Intel Core i5-13400K",
977|      "type": "CPU",
978|      "price": 8290000,
979|      "socket": "LGA1700",
980|      "power": 55,
981|      "desc": "18 nhân / 36 luồng, 55W TDP, Socket LGA1700"
982|    },
983|    {
984|      "id": "cpu_intel_13400KF_110",
985|      "name": "Intel Core i5-13400KF",
986|      "type": "CPU",
987|      "price": 2020000,
988|      "socket": "LGA1700",
989|      "power": 60,
990|      "desc": "18 nhân / 36 luồng, 60W TDP, Socket LGA1700"
991|    },
992|    {
993|      "id": "cpu_intel_13400T_111",
994|      "name": "Intel Core i5-13400T",
995|      "type": "CPU",
996|      "price": 7690000,
997|      "socket": "LGA1700",
998|      "power": 75,
999|      "desc": "18 nhân / 36 luồng, 75W TDP, Socket LGA1700"
1000|    },
1001|    {
1002|      "id": "cpu_intel_13400F_112",
1003|      "name": "Intel Core i5-13400F",
1004|      "type": "CPU",
1005|      "price": 5080000,
1006|      "socket": "LGA1700",
1007|      "power": 95,
1008|      "desc": "14 nhân / 28 luồng, 95W TDP, Socket LGA1700"
1009|    },
1010|    {
1011|      "id": "cpu_intel_13400std_113",
1012|      "name": "Intel Core i5-13400",
1013|      "type": "CPU",
1014|      "price": 8650000,
1015|      "socket": "LGA1700",
1016|      "power": 60,
1017|      "desc": "18 nhân / 36 luồng, 60W TDP, Socket LGA1700"
1018|    },
1019|    {
1020|      "id": "cpu_intel_13600K_114",
1021|      "name": "Intel Core i5-13600K",
1022|      "type": "CPU",
1023|      "price": 4670000,
1024|      "socket": "LGA1700",
1025|      "power": 95,
1026|      "desc": "10 nhân / 20 luồng, 95W TDP, Socket LGA1700"
1027|    },
1028|    {
1029|      "id": "cpu_intel_13600KF_115",
1030|      "name": "Intel Core i5-13600KF",
1031|      "type": "CPU",
1032|      "price": 9640000,
1033|      "socket": "LGA1700",
1034|      "power": 95,
1035|      "desc": "12 nhân / 24 luồng, 95W TDP, Socket LGA1700"
1036|    },
1037|    {

```

```
1038 | "id": "cpu_intel_13600T_116",
1039 | "name": "Intel Core i5-13600T",
1040 | "type": "CPU",
1041 | "price": 9970000,
1042 | "socket": "LGA1700",
1043 | "power": 85, "desc": "8 nhân / 16 luồng, 85W TDP, Socket LGA1700"
1044 | },
1045 | {
1046 | "id": "cpu_intel_13600F_117",
1047 | "name": "Intel Core i5-13600F",
1048 | "type": "CPU",
1049 | "price": 9800000,
1050 | "socket": "LGA1700",
1051 | "power": 55, "desc": "10 nhân / 20 luồng, 55W TDP, Socket LGA1700"
1052 | },
1053 | {
1054 | "id": "cpu_intel_13600std_118",
1055 | "name": "Intel Core i5-13600",
1056 | "type": "CPU",
1057 | "price": 4500000,
1058 | "socket": "LGA1700",
1059 | "power": 65, "desc": "12 nhân / 24 luồng, 65W TDP, Socket LGA1700"
1060 | },
1061 | {
1062 | "id": "cpu_intel_13400K_119",
1063 | "name": "Intel Core i5-13400K",
1064 | "type": "CPU",
1065 | "price": 9800000,
1066 | "socket": "LGA1700",
1067 | "power": 55, "desc": "2 nhân / 4 luồng, 55W TDP, Socket LGA1700"
1068 | },
1069 | {
1070 | "id": "cpu_intel_13400KF_120",
1071 | "name": "Intel Core i5-13400KF",
1072 | "type": "CPU",
1073 | "price": 2030000,
1074 | "socket": "LGA1700",
1075 | "power": 60, "desc": "6 nhân / 12 luồng, 60W TDP, Socket LGA1700"
1076 | },
1077 | {
1078 | "id": "cpu_intel_13400T_121",
1079 | "name": "Intel Core i5-13400T",
1080 | "type": "CPU",
1081 | "price": 1490000,
1082 | "socket": "LGA1700",
1083 | "power": 85, "desc": "16 nhân / 32 luồng, 85W TDP, Socket LGA1700"
1084 | },
1085 | {
1086 | "id": "cpu_intel_13400F_122",
1087 | "name": "Intel Core i5-13400F",
1088 | "type": "CPU",
1089 | "price": 8300000,
1090 | "socket": "LGA1700",
1091 | "power": 85, "desc": "16 nhân / 32 luồng, 85W TDP, Socket LGA1700"
1092 | },
1093 | {
1094 | "id": "cpu_intel_13400std_123",
1095 | "name": "Intel Core i5-13400",
1096 | "type": "CPU",
1097 | "price": 4300000,
1098 | "socket": "LGA1700",
1099 | "power": 55, "desc": "18 nhân / 36 luồng, 55W TDP, Socket LGA1700"
1100 | },
1101 | {
1102 | "id": "cpu_intel_13600K_124",
1103 | "name": "Intel Core i5-13600K",
1104 | "type": "CPU",
1105 | "price": 6510000,
1106 | "socket": "LGA1700",
1107 | "power": 125, "desc": "4 nhân / 8 luồng, 125W TDP, Socket LGA1700"
1108 | },
1109 | {
1110 | "id": "cpu_intel_13600KF_125",
1111 | "name": "Intel Core i5-13600KF",
1112 | "type": "CPU",
1113 | "price": 8100000,
1114 | "socket": "LGA1700",
1115 | "power": 65, "desc": "18 nhân / 36 luồng, 65W TDP, Socket LGA1700"
1116 | },
1117 | {
1118 | "id": "cpu_intel_13600T_126",
1119 | "name": "Intel Core i5-13600T",
1120 | "type": "CPU",
1121 | "price": 2300000,
1122 | "socket": "LGA1700",
1123 | "power": 95, "desc": "14 nhân / 28 luồng, 95W TDP, Socket LGA1700"
1124 | },
1125 | {
1126 | "id": "cpu_intel_13600F_127",
1127 | "name": "Intel Core i5-13600F",
1128 | "type": "CPU",
1129 | "price": 8210000,
1130 | "socket": "LGA1700",
1131 | "power": 55, "desc": "10 nhân / 20 luồng, 55W TDP, Socket LGA1700"
1132 | },
1133 | {
1134 | "id": "cpu_intel_13600std_128",
1135 | "name": "Intel Core i5-13600",
1136 | "type": "CPU",
1137 | "price": 8950000,
1138 | "socket": "LGA1700",
1139 | "power": 85, "desc": "14 nhân / 28 luồng, 85W TDP, Socket LGA1700"
1140 | },
1141 | {
1142 | "id": "cpu_intel_14400K_129",
1143 | "name": "Intel Core i5-14400K",
1144 | "type": "CPU",
1145 | "price": 8260000,
1146 | "socket": "LGA1851",
1147 | "power": 65, "desc": "12 nhân / 24 luồng, 65W TDP, Socket LGA1851"
1148 | }
```

```
1162| },
1163| {
1164|     "id": "cpu_intel_14400KF_130",
1165|     "name": "Intel Core i5-14400KF",
1166|     "type": "CPU",
1167|     "price": 6460000,
1168|     "socket": "LGA1851",
1169|     "power": 85,
1170|     "desc": "18 nhân / 36 luồng, 85W TDP, Socket LGA1851"
1171| },
1172| {
1173|     "id": "cpu_intel_14400T_131",
1174|     "name": "Intel Core i5-14400T",
1175|     "type": "CPU",
1176|     "price": 7240000,
1177|     "socket": "LGA1851",
1178|     "power": 60,
1179|     "desc": "10 nhân / 20 luồng, 60W TDP, Socket LGA1851"
1180| },
1181| {
1182|     "id": "cpu_intel_14400F_132",
1183|     "name": "Intel Core i5-14400F",
1184|     "type": "CPU",
1185|     "price": 8560000,
1186|     "socket": "LGA1851",
1187|     "power": 55,
1188|     "desc": "2 nhân / 4 luồng, 55W TDP, Socket LGA1851"
1189| },
1190| {
1191|     "id": "cpu_intel_14400std_133",
1192|     "name": "Intel Core i5-14400",
1193|     "type": "CPU",
1194|     "price": 3110000,
1195|     "socket": "LGA1851",
1196|     "power": 65,
1197|     "desc": "12 nhân / 24 luồng, 65W TDP, Socket LGA1851"
1198| },
1199| {
1200|     "id": "cpu_intel_14600K_134",
1201|     "name": "Intel Core i5-14600K",
1202|     "type": "CPU",
1203|     "price": 5150000,
1204|     "socket": "LGA1851",
1205|     "power": 60,
1206|     "desc": "6 nhân / 12 luồng, 60W TDP, Socket LGA1851"
1207| },
1208| {
1209|     "id": "cpu_intel_14600KF_135",
1210|     "name": "Intel Core i5-14600KF",
1211|     "type": "CPU",
1212|     "price": 4410000,
1213|     "socket": "LGA1851",
1214|     "power": 125,
1215|     "desc": "8 nhân / 16 luồng, 125W TDP, Socket LGA1851"
1216| },
1217| {
1218|     "id": "cpu_intel_14600T_136",
1219|     "name": "Intel Core i5-14600T",
1220|     "type": "CPU",
1221|     "price": 3260000,
1222|     "socket": "LGA1851",
1223|     "power": 105,
1224|     "desc": "8 nhân / 16 luồng, 105W TDP, Socket LGA1851"
1225| },
1226| {
1227|     "id": "cpu_intel_14600F_137",
1228|     "name": "Intel Core i5-14600F",
1229|     "type": "CPU",
1230|     "price": 2050000,
1231|     "socket": "LGA1851",
1232|     "power": 105,
1233|     "desc": "6 nhân / 12 luồng, 105W TDP, Socket LGA1851"
1234| },
1235| {
1236|     "id": "cpu_intel_14600std_138",
1237|     "name": "Intel Core i5-14600",
1238|     "type": "CPU",
1239|     "price": 6510000,
1240|     "socket": "LGA1851",
1241|     "power": 120,
1242|     "desc": "18 nhân / 36 luồng, 120W TDP, Socket LGA1851"
1243| },
1244| {
1245|     "id": "cpu_intel_14400K_139",
1246|     "name": "Intel Core i5-14400K",
1247|     "type": "CPU",
1248|     "price": 7680000,
1249|     "socket": "LGA1851",
1250|     "power": 105,
1251|     "desc": "14 nhân / 28 luồng, 105W TDP, Socket LGA1851"
1252| },
1253| {
1254|     "id": "cpu_intel_14400KF_140",
1255|     "name": "Intel Core i5-14400KF",
1256|     "type": "CPU",
1257|     "price": 9760000,
1258|     "socket": "LGA1851",
1259|     "power": 85,
1260|     "desc": "16 nhân / 32 luồng, 85W TDP, Socket LGA1851"
1261| },
1262| {
1263|     "id": "cpu_intel_14400T_141",
1264|     "name": "Intel Core i5-14400T",
1265|     "type": "CPU",
1266|     "price": 2860000,
1267|     "socket": "LGA1851",
1268|     "power": 75,
1269|     "desc": "2 nhân / 4 luồng, 75W TDP, Socket LGA1851"
1270| },
1271| {
1272|     "id": "cpu_intel_14400F_142",
1273|     "name": "Intel Core i5-14400F",
1274|     "type": "CPU",
1275|     "price": 7730000,
1276|     "socket": "LGA1851",
1277|     "power": 75,
1278|     "desc": "8 nhân / 16 luồng, 75W TDP, Socket LGA1851"
1279| },
1280| {
1281|     "id": "cpu_intel_14400std_143",
1282|     "name": "Intel Core i5-14400",
1283|     "type": "CPU",
1284|     "price": 2460000,
1285|     "socket": "LGA1851",
```

```
1286| "power": 85,
1287| "desc": "12 nhân / 24 luồng, 85W TDP, Socket LGA1851"
1288| },
1289| {
1290| "id": "cpu_intel_14600K_144",
1291| "name": "Intel Core i5-14600K",
1292| "type": "CPU",
1293| "price": 5560000,
1294| "socket": "LGA1851",
1295| "power": 60,
1296| "desc": "8 nhân / 16 luồng, 60W TDP, Socket LGA1851"
1297| },
1298| {
1299| "id": "cpu_intel_14600KF_145",
1300| "name": "Intel Core i5-14600KF",
1301| "type": "CPU",
1302| "price": 4470000,
1303| "socket": "LGA1851",
1304| "power": 55,
1305| "desc": "18 nhân / 36 luồng, 55W TDP, Socket LGA1851"
1306| },
1307| {
1308| "id": "cpu_intel_14600T_146",
1309| "name": "Intel Core i5-14600T",
1310| "type": "CPU",
1311| "price": 2810000,
1312| "socket": "LGA1851",
1313| "power": 65,
1314| "desc": "18 nhân / 36 luồng, 65W TDP, Socket LGA1851"
1315| },
1316| {
1317| "id": "cpu_intel_14600F_147",
1318| "name": "Intel Core i5-14600F",
1319| "type": "CPU",
1320| "price": 3680000,
1321| "socket": "LGA1851",
1322| "power": 75,
1323| "desc": "2 nhân / 4 luồng, 75W TDP, Socket LGA1851"
1324| },
1325| {
1326| "id": "cpu_intel_14600std_148",
1327| "name": "Intel Core i5-14600",
1328| "type": "CPU",
1329| "price": 1340000,
1330| "socket": "LGA1851",
1331| "power": 125,
1332| "desc": "6 nhân / 12 luồng, 125W TDP, Socket LGA1851"
1333| },
1334| {
1335| "id": "cpu_intel_14400K_149",
1336| "name": "Intel Core i5-14400K",
1337| "type": "CPU",
1338| "price": 1680000,
1339| "socket": "LGA1851",
1340| "power": 120,
1341| "desc": "8 nhân / 16 luồng, 120W TDP, Socket LGA1851"
1342| },
1343| {
1344| "id": "cpu_intel_14400KF_150",
1345| "name": "Intel Core i5-14400KF",
1346| "type": "CPU",
1347| "price": 7230000,
1348| "socket": "LGA1851",
1349| "power": 65,
1350| "desc": "12 nhân / 24 luồng, 65W TDP, Socket LGA1851"
1351| },
1352| {
1353| "id": "cpu_intel_14400T_151",
1354| "name": "Intel Core i5-14400T",
1355| "type": "CPU",
1356| "price": 6350000,
1357| "socket": "LGA1851",
1358| "power": 55,
1359| "desc": "18 nhân / 36 luồng, 55W TDP, Socket LGA1851"
1360| },
1361| {
1362| "id": "cpu_intel_14400F_152",
1363| "name": "Intel Core i5-14400F",
1364| "type": "CPU",
1365| "price": 6660000,
1366| "socket": "LGA1851",
1367| "power": 75,
1368| "desc": "6 nhân / 12 luồng, 75W TDP, Socket LGA1851"
1369| },
1370| {
1371| "id": "cpu_intel_14400std_153",
1372| "name": "Intel Core i5-14400",
1373| "type": "CPU",
1374| "price": 5360000,
1375| "socket": "LGA1851",
1376| "power": 75,
1377| "desc": "8 nhân / 16 luồng, 75W TDP, Socket LGA1851"
1378| },
1379| {
1380| "id": "cpu_intel_14600K_154",
1381| "name": "Intel Core i5-14600K",
1382| "type": "CPU",
1383| "price": 5180000,
1384| "socket": "LGA1851",
1385| "power": 60,
1386| "desc": "18 nhân / 36 luồng, 60W TDP, Socket LGA1851"
1387| },
1388| {
1389| "id": "cpu_intel_14600KF_155",
1390| "name": "Intel Core i5-14600KF",
1391| "type": "CPU",
1392| "price": 890000,
1393| "socket": "LGA1851",
1394| "power": 125,
1395| "desc": "14 nhân / 28 luồng, 125W TDP, Socket LGA1851"
1396| },
1397| {
1398| "id": "cpu_intel_14600T_156",
1399| "name": "Intel Core i5-14600T",
1400| "type": "CPU",
1401| "price": 9190000,
1402| "socket": "LGA1851",
1403| "power": 105,
1404| "desc": "16 nhân / 32 luồng, 105W TDP, Socket LGA1851"
1405| },
1406| {
1407| "id": "cpu_intel_14600F_157",
1408| "name": "Intel Core i5-14600F",
1409| "type": "CPU",
```

```
1410| "price": 5310000,
1411| "socket": "LGA1851",
1412| "power": 75, "desc": "4 nhân / 8 luồng, 75W TDP, Socket LGA1851"
1413| },
1414| {
1415| "id": "cpu_intel_14600std_158",
1416| "name": "Intel Core i5-14600",
1417| "type": "CPU",
1418| "price": 6000000,
1419| "socket": "LGA1851",
1420| "power": 125, "desc": "12 nhân / 24 luồng, 125W TDP, Socket LGA1851"
1421| },
1422| {
1423| "id": "cpu_intel_14400K_159",
1424| "name": "Intel Core i5-14400K",
1425| "type": "CPU",
1426| "price": 2570000,
1427| "socket": "LGA1851",
1428| "power": 60, "desc": "2 nhân / 4 luồng, 60W TDP, Socket LGA1851"
1429| },
1430| {
1431| "id": "cpu_intel_14400KF_160",
1432| "name": "Intel Core i5-14400KF",
1433| "type": "CPU",
1434| "price": 8720000,
1435| "socket": "LGA1851",
1436| "power": 60, "desc": "16 nhân / 32 luồng, 60W TDP, Socket LGA1851"
1437| },
1438| {
1439| "id": "cpu_intel_14400T_161",
1440| "name": "Intel Core i5-14400T",
1441| "type": "CPU",
1442| "price": 9790000,
1443| "socket": "LGA1851",
1444| "power": 85, "desc": "12 nhân / 24 luồng, 85W TDP, Socket LGA1851"
1445| },
1446| {
1447| "id": "cpu_intel_14400F_162",
1448| "name": "Intel Core i5-14400F",
1449| "type": "CPU",
1450| "price": 5710000,
1451| "socket": "LGA1851",
1452| "power": 95, "desc": "14 nhân / 28 luồng, 95W TDP, Socket LGA1851"
1453| },
1454| {
1455| "id": "cpu_intel_14400std_163",
1456| "name": "Intel Core i5-14400",
1457| "type": "CPU",
1458| "price": 6040000,
1459| "socket": "LGA1851",
1460| "power": 55, "desc": "10 nhân / 20 luồng, 55W TDP, Socket LGA1851"
1461| },
1462| {
1463| "id": "cpu_intel_14600K_164",
1464| "name": "Intel Core i5-14600K",
1465| "type": "CPU",
1466| "price": 3890000,
1467| "socket": "LGA1851",
1468| "power": 105, "desc": "18 nhân / 36 luồng, 105W TDP, Socket LGA1851"
1469| },
1470| {
1471| "id": "cpu_intel_14600KF_165",
1472| "name": "Intel Core i5-14600KF",
1473| "type": "CPU",
1474| "price": 6880000,
1475| "socket": "LGA1851",
1476| "power": 60, "desc": "2 nhân / 4 luồng, 60W TDP, Socket LGA1851"
1477| },
1478| {
1479| "id": "cpu_intel_14600T_166",
1480| "name": "Intel Core i5-14600T",
1481| "type": "CPU",
1482| "price": 7720000,
1483| "socket": "LGA1851",
1484| "power": 105, "desc": "16 nhân / 32 luồng, 105W TDP, Socket LGA1851"
1485| },
1486| {
1487| "id": "cpu_intel_14600F_167",
1488| "name": "Intel Core i5-14600F",
1489| "type": "CPU",
1490| "price": 6060000,
1491| "socket": "LGA1851",
1492| "power": 75, "desc": "6 nhân / 12 luồng, 75W TDP, Socket LGA1851"
1493| },
1494| {
1495| "id": "cpu_intel_14600std_168",
1496| "name": "Intel Core i5-14600",
1497| "type": "CPU",
1498| "price": 5420000,
1499| "socket": "LGA1851",
1500| "power": 120, "desc": "4 nhân / 8 luồng, 120W TDP, Socket LGA1851"
1501| },
1502| {
1503| "id": "cpu_intel_15400K_169",
1504| "name": "Intel Core i5-15400K",
1505| "type": "CPU",
1506| "price": 9800000,
1507| "socket": "LGA1851",
1508| "power": 75, "desc": "2 nhân / 4 luồng, 75W TDP, Socket LGA1851"
1509| },
1510| {
1511| "id": "cpu_intel_15400KF_170",
1512| "name": "Intel Core i5-15400KF",
1513| "type": "CPU",
1514| "price": 5340000,
1515| "socket": "LGA1851",
1516| "power": 105, "desc": "6 nhân / 12 luồng, 105W TDP, Socket LGA1851"
1517| },
1518| {
1519| "id": "cpu_intel_15400T_171",
```

```

1534|      "name": "Intel Core i5-15400T",
1535|      "type": "CPU",
1536|      "price": 3740000,
1537|      "socket": "LGA1851",
1538|      "power": 105,
1539|      "desc": "2 nhân / 4 luồng, 105W TDP, Socket LGA1851"
1540|    },
1541|    {
1542|      "id": "cpu_intel_15400F_172",
1543|      "name": "Intel Core i5-15400F",
1544|      "type": "CPU",
1545|      "price": 5300000,
1546|      "socket": "LGA1851",
1547|      "power": 85,
1548|      "desc": "8 nhân / 16 luồng, 85W TDP, Socket LGA1851"
1549|    },
1550|    {
1551|      "id": "cpu_intel_15400std_173",
1552|      "name": "Intel Core i5-15400",
1553|      "type": "CPU",
1554|      "price": 9150000,
1555|      "socket": "LGA1851",
1556|      "power": 120,
1557|      "desc": "6 nhân / 12 luồng, 120W TDP, Socket LGA1851"
1558|    },
1559|    {
1560|      "id": "cpu_intel_15600K_174",
1561|      "name": "Intel Core i5-15600K",
1562|      "type": "CPU",
1563|      "price": 3260000,
1564|      "socket": "LGA1851",
1565|      "power": 95,
1566|      "desc": "18 nhân / 36 luồng, 95W TDP, Socket LGA1851"
1567|    },
1568|    {
1569|      "id": "cpu_intel_15600KF_175",
1570|      "name": "Intel Core i5-15600KF",
1571|      "type": "CPU",
1572|      "price": 6970000,
1573|      "socket": "LGA1851",
1574|      "power": 95,
1575|      "desc": "6 nhân / 12 luồng, 95W TDP, Socket LGA1851"
1576|    },
1577|    {
1578|      "id": "cpu_intel_15600T_176",
1579|      "name": "Intel Core i5-15600T",
1580|      "type": "CPU",
1581|      "price": 9170000,
1582|      "socket": "LGA1851",
1583|      "power": 95,
1584|      "desc": "2 nhân / 4 luồng, 95W TDP, Socket LGA1851"
1585|    },
1586|    {
1587|      "id": "cpu_intel_15600F_177",
1588|      "name": "Intel Core i5-15600F",
1589|      "type": "CPU",
1590|      "price": 1420000,
1591|      "socket": "LGA1851",
1592|      "power": 95,
1593|      "desc": "2 nhân / 4 luồng, 95W TDP, Socket LGA1851"
1594|    },
1595|    {
1596|      "id": "cpu_intel_15600std_178",
1597|      "name": "Intel Core i5-15600",
1598|      "type": "CPU",
1599|      "price": 940000,
1600|      "socket": "LGA1851",
1601|      "power": 55,
1602|      "desc": "18 nhân / 36 luồng, 55W TDP, Socket LGA1851"
1603|    },
1604|    {
1605|      "id": "cpu_intel_15400K_179",
1606|      "name": "Intel Core i5-15400K",
1607|      "type": "CPU",
1608|      "price": 3570000,
1609|      "socket": "LGA1851",
1610|      "power": 125,
1611|      "desc": "4 nhân / 8 luồng, 125W TDP, Socket LGA1851"
1612|    },
1613|    {
1614|      "id": "cpu_intel_15400KF_180",
1615|      "name": "Intel Core i5-15400KF",
1616|      "type": "CPU",
1617|      "price": 2250000,
1618|      "socket": "LGA1851",
1619|      "power": 75,
1620|      "desc": "14 nhân / 28 luồng, 75W TDP, Socket LGA1851"
1621|    },
1622|    {
1623|      "id": "cpu_intel_15400T_181",
1624|      "name": "Intel Core i5-15400T",
1625|      "type": "CPU",
1626|      "price": 9180000,
1627|      "socket": "LGA1851",
1628|      "power": 65,
1629|      "desc": "8 nhân / 16 luồng, 65W TDP, Socket LGA1851"
1630|    },
1631|    {
1632|      "id": "cpu_intel_15400F_182",
1633|      "name": "Intel Core i5-15400F",
1634|      "type": "CPU",
1635|      "price": 5400000,
1636|      "socket": "LGA1851",
1637|      "power": 65,
1638|      "desc": "2 nhân / 4 luồng, 65W TDP, Socket LGA1851"
1639|    },
1640|    {
1641|      "id": "cpu_intel_15400std_183",
1642|      "name": "Intel Core i5-15400",
1643|      "type": "CPU",
1644|      "price": 2610000,
1645|      "socket": "LGA1851",
1646|      "power": 85,
1647|      "desc": "8 nhân / 16 luồng, 85W TDP, Socket LGA1851"
1648|    },
1649|    {
1650|      "id": "cpu_intel_15600K_184",
1651|      "name": "Intel Core i5-15600K",
1652|      "type": "CPU",
1653|      "price": 4910000,
1654|      "socket": "LGA1851",
1655|      "power": 120,
1656|      "desc": "18 nhân / 36 luồng, 120W TDP, Socket LGA1851"
1657|    },

```

```
1658 | {
1659 |   "id": "cpu_intel_15600KF_185",
1660 |   "name": "Intel Core i5-15600KF",
1661 |   "type": "CPU",
1662 |   "price": 5070000,
1663 |   "socket": "LGA1851",
1664 |   "power": 120,
1665 |   "desc": "8 nhân / 16 luồng, 120W TDP, Socket LGA1851"
1666 | },
1667 | {
1668 |   "id": "cpu_intel_15600T_186",
1669 |   "name": "Intel Core i5-15600T",
1670 |   "type": "CPU",
1671 |   "price": 4220000,
1672 |   "socket": "LGA1851",
1673 |   "power": 125,
1674 |   "desc": "6 nhân / 12 luồng, 125W TDP, Socket LGA1851"
1675 | },
1676 | {
1677 |   "id": "cpu_intel_15600F_187",
1678 |   "name": "Intel Core i5-15600F",
1679 |   "type": "CPU",
1680 |   "price": 3460000,
1681 |   "socket": "LGA1851",
1682 |   "power": 75,
1683 |   "desc": "6 nhân / 12 luồng, 75W TDP, Socket LGA1851"
1684 | },
1685 | {
1686 |   "id": "cpu_intel_15600std_188",
1687 |   "name": "Intel Core i5-15600",
1688 |   "type": "CPU",
1689 |   "price": 3290000,
1690 |   "socket": "LGA1851",
1691 |   "power": 95,
1692 |   "desc": "14 nhân / 28 luồng, 95W TDP, Socket LGA1851"
1693 | },
1694 | {
1695 |   "id": "cpu_intel_15400K_189",
1696 |   "name": "Intel Core i5-15400K",
1697 |   "type": "CPU",
1698 |   "price": 3050000,
1699 |   "socket": "LGA1851",
1700 |   "power": 120,
1701 |   "desc": "12 nhân / 24 luồng, 120W TDP, Socket LGA1851"
1702 | },
1703 | {
1704 |   "id": "cpu_intel_15400KF_190",
1705 |   "name": "Intel Core i5-15400KF",
1706 |   "type": "CPU",
1707 |   "price": 7180000,
1708 |   "socket": "LGA1851",
1709 |   "power": 55,
1710 |   "desc": "10 nhân / 20 luồng, 55W TDP, Socket LGA1851"
1711 | },
1712 | {
1713 |   "id": "cpu_intel_15400T_191",
1714 |   "name": "Intel Core i5-15400T",
1715 |   "type": "CPU",
1716 |   "price": 4320000,
1717 |   "socket": "LGA1851",
1718 |   "power": 95,
1719 |   "desc": "16 nhân / 32 luồng, 95W TDP, Socket LGA1851"
1720 | },
1721 | {
1722 |   "id": "cpu_intel_15400F_192",
1723 |   "name": "Intel Core i5-15400F",
1724 |   "type": "CPU",
1725 |   "price": 5570000,
1726 |   "socket": "LGA1851",
1727 |   "power": 95,
1728 |   "desc": "14 nhân / 28 luồng, 95W TDP, Socket LGA1851"
1729 | },
1730 | {
1731 |   "id": "cpu_intel_15400std_193",
1732 |   "name": "Intel Core i5-15400",
1733 |   "type": "CPU",
1734 |   "price": 5100000,
1735 |   "socket": "LGA1851",
1736 |   "power": 75,
1737 |   "desc": "12 nhân / 24 luồng, 75W TDP, Socket LGA1851"
1738 | },
1739 | {
1740 |   "id": "cpu_intel_15600K_194",
1741 |   "name": "Intel Core i5-15600K",
1742 |   "type": "CPU",
1743 |   "price": 6450000,
1744 |   "socket": "LGA1851",
1745 |   "power": 95,
1746 |   "desc": "8 nhân / 16 luồng, 95W TDP, Socket LGA1851"
1747 | },
1748 | {
1749 |   "id": "cpu_intel_15600KF_195",
1750 |   "name": "Intel Core i5-15600KF",
1751 |   "type": "CPU",
1752 |   "price": 7010000,
1753 |   "socket": "LGA1851",
1754 |   "power": 95,
1755 |   "desc": "10 nhân / 20 luồng, 95W TDP, Socket LGA1851"
1756 | },
1757 | {
1758 |   "id": "cpu_intel_15600T_196",
1759 |   "name": "Intel Core i5-15600T",
1760 |   "type": "CPU",
1761 |   "price": 4780000,
1762 |   "socket": "LGA1851",
1763 |   "power": 85,
1764 |   "desc": "6 nhân / 12 luồng, 85W TDP, Socket LGA1851"
1765 | },
1766 | {
1767 |   "id": "cpu_intel_15600F_197",
1768 |   "name": "Intel Core i5-15600F",
1769 |   "type": "CPU",
1770 |   "price": 7150000,
1771 |   "socket": "LGA1851",
1772 |   "power": 55,
1773 |   "desc": "18 nhân / 36 luồng, 55W TDP, Socket LGA1851"
1774 | },
1775 | {
1776 |   "id": "cpu_intel_15600std_198",
1777 |   "name": "Intel Core i5-15600",
1778 |   "type": "CPU",
1779 |   "price": 9090000,
1780 |   "socket": "LGA1851",
1781 |   "power": 120,
```



```
1 7 8 2 | "desc": "18 nhân / 36 luồng, 120W TDP, Socket LGA1851"
1783 | },
1784 | {
1785 |   "id": "cpu_intel_15400K_199",
1786 |   "name": "Intel Core i5-15400K",
1787 |   "type": "CPU",
1788 |   "price": 5360000,
1789 |   "socket": "LGA1851",
1790 |   "power": 60,
1 7 9 1 | "desc": "14 nhân / 28 luồng, 60W TDP, Socket LGA1851"
1792 | },
1793 | {
1794 |   "id": "cpu_intel_15400KF_200",
1795 |   "name": "Intel Core i5-15400KF",
1796 |   "type": "CPU",
1797 |   "price": 3740000,
1798 |   "socket": "LGA1851",
1799 |   "power": 55,
1 8 0 0 | "desc": "6 nhân / 12 luồng, 55W TDP, Socket LGA1851"
1801 | },
1802 | {
1803 |   "id": "cpu_intel_15400T_201",
1804 |   "name": "Intel Core i5-15400T",
1805 |   "type": "CPU",
1806 |   "price": 8150000,
1807 |   "socket": "LGA1851",
1808 |   "power": 105,
1 8 0 9 | "desc": "14 nhân / 28 luồng, 105W TDP, Socket LGA1851"
1810 | },
1811 | {
1812 |   "id": "cpu_intel_15400F_202",
1813 |   "name": "Intel Core i5-15400F",
1814 |   "type": "CPU",
1815 |   "price": 9320000,
1816 |   "socket": "LGA1851",
1817 |   "power": 75,
1 8 1 8 | "desc": "4 nhân / 8 luồng, 75W TDP, Socket LGA1851"
1819 | },
1820 | {
1821 |   "id": "cpu_intel_15400std_203",
1822 |   "name": "Intel Core i5-15400",
1823 |   "type": "CPU",
1824 |   "price": 8970000,
1825 |   "socket": "LGA1851",
1826 |   "power": 85,
1 8 2 7 | "desc": "16 nhân / 32 luồng, 85W TDP, Socket LGA1851"
1828 | },
1829 | {
1830 |   "id": "cpu_intel_15600K_204",
1831 |   "name": "Intel Core i5-15600K",
1832 |   "type": "CPU",
1833 |   "price": 1710000,
1834 |   "socket": "LGA1851",
1835 |   "power": 120,
1 8 3 6 | "desc": "14 nhân / 28 luồng, 120W TDP, Socket LGA1851"
1837 | },
1838 | {
1839 |   "id": "cpu_intel_15600KF_205",
1840 |   "name": "Intel Core i5-15600KF",
1841 |   "type": "CPU",
1842 |   "price": 7240000,
1843 |   "socket": "LGA1851",
1844 |   "power": 55,
1 8 4 5 | "desc": "8 nhân / 16 luồng, 55W TDP, Socket LGA1851"
1846 | },
1847 | {
1848 |   "id": "cpu_intel_15600T_206",
1849 |   "name": "Intel Core i5-15600T",
1850 |   "type": "CPU",
1851 |   "price": 9830000,
1852 |   "socket": "LGA1851",
1853 |   "power": 55,
1 8 5 4 | "desc": "4 nhân / 8 luồng, 55W TDP, Socket LGA1851"
1855 | },
1856 | {
1857 |   "id": "cpu_intel_15600F_207",
1858 |   "name": "Intel Core i5-15600F",
1859 |   "type": "CPU",
1860 |   "price": 940000,
1861 |   "socket": "LGA1851",
1862 |   "power": 105,
1 8 6 3 | "desc": "4 nhân / 8 luồng, 105W TDP, Socket LGA1851"
1864 | },
1865 | {
1866 |   "id": "cpu_intel_15600std_208",
1867 |   "name": "Intel Core i5-15600",
1868 |   "type": "CPU",
1869 |   "price": 6420000,
1870 |   "socket": "LGA1851",
1871 |   "power": 55,
1 8 7 2 | "desc": "10 nhân / 20 luồng, 55W TDP, Socket LGA1851"
1873 | },
1874 | {
1875 |   "id": "cpu_intel_12700K_209",
1876 |   "name": "Intel Core i7-12700K",
1877 |   "type": "CPU",
1878 |   "price": 13310000,
1879 |   "socket": "LGA1700",
1880 |   "power": 55,
1 8 8 1 | "desc": "8 nhân / 16 luồng, 55W TDP, Socket LGA1700"
1882 | },
1883 | {
1884 |   "id": "cpu_intel_12700KF_210",
1885 |   "name": "Intel Core i7-12700KF",
1886 |   "type": "CPU",
1887 |   "price": 12130000,
1888 |   "socket": "LGA1700",
1889 |   "power": 75,
1 8 9 0 | "desc": "14 nhân / 28 luồng, 75W TDP, Socket LGA1700"
1891 | },
1892 | {
1893 |   "id": "cpu_intel_12700T_211",
1894 |   "name": "Intel Core i7-12700T",
1895 |   "type": "CPU",
1896 |   "price": 6250000,
1897 |   "socket": "LGA1700",
1898 |   "power": 65,
1 8 9 9 | "desc": "4 nhân / 8 luồng, 65W TDP, Socket LGA1700"
1900 | },
1901 | {
1902 |   "id": "cpu_intel_12700F_212",
1903 |   "name": "Intel Core i7-12700F",
1904 |   "type": "CPU",
1905 |   "price": 6160000,
```

```
1906| "socket": "LGA1700",
1907| "power": 105,
1908| "desc": "12 nhân / 24 luồng, 105W TDP, Socket LGA1700"
1909| },
1910| {
1911| "id": "cpu_intel_12700std_213",
1912| "name": "Intel Core i7-12700",
1913| "type": "CPU",
1914| "price": 12610000,
1915| "socket": "LGA1700",
1916| "power": 60,
1917| "desc": "2 nhân / 4 luồng, 60W TDP, Socket LGA1700"
1918| },
1919| {
1920| "id": "cpu_intel_12700K_214",
1921| "name": "Intel Core i7-12700K",
1922| "type": "CPU",
1923| "price": 8110000,
1924| "socket": "LGA1700",
1925| "power": 120,
1926| "desc": "4 nhân / 8 luồng, 120W TDP, Socket LGA1700"
1927| },
1928| {
1929| "id": "cpu_intel_12700KF_215",
1930| "name": "Intel Core i7-12700KF",
1931| "type": "CPU",
1932| "price": 1840000,
1933| "socket": "LGA1700",
1934| "power": 120,
1935| "desc": "16 nhân / 32 luồng, 120W TDP, Socket LGA1700"
1936| },
1937| {
1938| "id": "cpu_intel_12700T_216",
1939| "name": "Intel Core i7-12700T",
1940| "type": "CPU",
1941| "price": 7870000,
1942| "socket": "LGA1700",
1943| "power": 65,
1944| "desc": "6 nhân / 12 luồng, 65W TDP, Socket LGA1700"
1945| },
1946| {
1947| "id": "cpu_intel_12700F_217",
1948| "name": "Intel Core i7-12700F",
1949| "type": "CPU",
1950| "price": 6980000,
1951| "socket": "LGA1700",
1952| "power": 55,
1953| "desc": "6 nhân / 12 luồng, 55W TDP, Socket LGA1700"
1954| },
1955| {
1956| "id": "cpu_intel_12700std_218",
1957| "name": "Intel Core i7-12700",
1958| "type": "CPU",
1959| "price": 11450000,
1960| "socket": "LGA1700",
1961| "power": 75,
1962| "desc": "10 nhân / 20 luồng, 75W TDP, Socket LGA1700"
1963| },
1964| {
1965| "id": "cpu_intel_12700K_219",
1966| "name": "Intel Core i7-12700K",
1967| "type": "CPU",
1968| "price": 2450000,
1969| "socket": "LGA1700",
1970| "power": 55,
1971| "desc": "8 nhân / 16 luồng, 55W TDP, Socket LGA1700"
1972| },
1973| {
1974| "id": "cpu_intel_12700KF_220",
1975| "name": "Intel Core i7-12700KF",
1976| "type": "CPU",
1977| "price": 12020000,
1978| "socket": "LGA1700",
1979| "power": 55,
1980| "desc": "12 nhân / 24 luồng, 55W TDP, Socket LGA1700"
1981| },
1982| {
1983| "id": "cpu_intel_12700T_221",
1984| "name": "Intel Core i7-12700T",
1985| "type": "CPU",
1986| "price": 10960000,
1987| "socket": "LGA1700",
1988| "power": 95,
1989| "desc": "10 nhân / 20 luồng, 95W TDP, Socket LGA1700"
1990| },
1991| {
1992| "id": "cpu_intel_12700F_222",
1993| "name": "Intel Core i7-12700F",
1994| "type": "CPU",
1995| "price": 14270000,
1996| "socket": "LGA1700",
1997| "power": 75,
1998| "desc": "10 nhân / 20 luồng, 75W TDP, Socket LGA1700"
1999| },
2000| {
2001| "id": "cpu_intel_12700std_223",
2002| "name": "Intel Core i7-12700",
2003| "type": "CPU",
2004| "price": 12820000,
2005| "socket": "LGA1700",
2006| "power": 55,
2007| "desc": "16 nhân / 32 luồng, 55W TDP, Socket LGA1700"
2008| },
2009| {
2010| "id": "cpu_intel_12700K_224",
2011| "name": "Intel Core i7-12700K",
2012| "type": "CPU",
2013| "price": 13150000,
2014| "socket": "LGA1700",
2015| "power": 120,
2016| "desc": "10 nhân / 20 luồng, 120W TDP, Socket LGA1700"
2017| },
2018| {
2019| "id": "cpu_intel_12700KF_225",
2020| "name": "Intel Core i7-12700KF",
2021| "type": "CPU",
2022| "price": 3060000,
2023| "socket": "LGA1700",
2024| "power": 65,
2025| "desc": "14 nhân / 28 luồng, 65W TDP, Socket LGA1700"
2026| },
2027| {
2028| "id": "cpu_intel_12700T_226",
2029| "name": "Intel Core i7-12700T",
```

```
2030| "type": "CPU",
2031| "price": 9960000,
2032| "socket": "LGA1700",
2033| "power": 60, "desc": "12 nhân / 24 luồng, 60W TDP, Socket LGA1700"
2034| },
2035| {
2036| "id": "cpu_intel_12700F_227",
2037| "name": "Intel Core i7-12700F",
2038| "type": "CPU",
2039| "price": 9050000,
2040| "socket": "LGA1700",
2041| "power": 95, "desc": "4 nhân / 8 luồng, 95W TDP, Socket LGA1700"
2042| },
2043| {
2044| "id": "cpu_intel_12700std_228",
2045| "name": "Intel Core i7-12700",
2046| "type": "CPU",
2047| "price": 6810000,
2048| "socket": "LGA1700",
2049| "power": 75, "desc": "4 nhân / 8 luồng, 75W TDP, Socket LGA1700"
2050| },
2051| {
2052| "id": "cpu_intel_13700K_229",
2053| "name": "Intel Core i7-13700K",
2054| "type": "CPU",
2055| "price": 7320000,
2056| "socket": "LGA1700",
2057| "power": 55, "desc": "14 nhân / 28 luồng, 55W TDP, Socket LGA1700"
2058| },
2059| {
2060| "id": "cpu_intel_13700KF_230",
2061| "name": "Intel Core i7-13700KF",
2062| "type": "CPU",
2063| "price": 8670000,
2064| "socket": "LGA1700",
2065| "power": 65, "desc": "16 nhân / 32 luồng, 65W TDP, Socket LGA1700"
2066| },
2067| {
2068| "id": "cpu_intel_13700T_231",
2069| "name": "Intel Core i7-13700T",
2070| "type": "CPU",
2071| "price": 9890000,
2072| "socket": "LGA1700",
2073| "power": 95, "desc": "10 nhân / 20 luồng, 95W TDP, Socket LGA1700"
2074| },
2075| {
2076| "id": "cpu_intel_13700F_232",
2077| "name": "Intel Core i7-13700F",
2078| "type": "CPU",
2079| "price": 13830000,
2080| "socket": "LGA1700",
2081| "power": 65, "desc": "14 nhân / 28 luồng, 65W TDP, Socket LGA1700"
2082| },
2083| {
2084| "id": "cpu_intel_13700std_233",
2085| "name": "Intel Core i7-13700",
2086| "type": "CPU",
2087| "price": 14310000,
2088| "socket": "LGA1700",
2089| "power": 60, "desc": "6 nhân / 12 luồng, 60W TDP, Socket LGA1700"
2090| },
2091| {
2092| "id": "cpu_intel_13700K_234",
2093| "name": "Intel Core i7-13700K",
2094| "type": "CPU",
2095| "price": 8040000,
2096| "socket": "LGA1700",
2097| "power": 55, "desc": "4 nhân / 8 luồng, 55W TDP, Socket LGA1700"
2098| },
2099| {
2100| "id": "cpu_intel_13700KF_235",
2101| "name": "Intel Core i7-13700KF",
2102| "type": "CPU",
2103| "price": 4470000,
2104| "socket": "LGA1700",
2105| "power": 120, "desc": "8 nhân / 16 luồng, 120W TDP, Socket LGA1700"
2106| },
2107| {
2108| "id": "cpu_intel_13700T_236",
2109| "name": "Intel Core i7-13700T",
2110| "type": "CPU",
2111| "price": 5480000,
2112| "socket": "LGA1700",
2113| "power": 85, "desc": "14 nhân / 28 luồng, 85W TDP, Socket LGA1700"
2114| },
2115| {
2116| "id": "cpu_intel_13700F_237",
2117| "name": "Intel Core i7-13700F",
2118| "type": "CPU",
2119| "price": 10540000,
2120| "socket": "LGA1700",
2121| "power": 105, "desc": "16 nhân / 32 luồng, 105W TDP, Socket LGA1700"
2122| },
2123| {
2124| "id": "cpu_intel_13700std_238",
2125| "name": "Intel Core i7-13700",
2126| "type": "CPU",
2127| "price": 14600000,
2128| "socket": "LGA1700",
2129| "power": 105, "desc": "16 nhân / 32 luồng, 105W TDP, Socket LGA1700"
2130| },
2131| {
2132| "id": "cpu_intel_13700K_239",
2133| "name": "Intel Core i7-13700K",
2134| "type": "CPU",
2135| "price": 10760000,
2136| "socket": "LGA1700",
2137| "power": 120, "desc": "4 nhân / 8 luồng, 120W TDP, Socket LGA1700"
2138| },
2139| {
2140| "id": "cpu_intel_13700K_239",
2141| "name": "Intel Core i7-13700K",
2142| "type": "CPU",
2143| "price": 10760000,
2144| "socket": "LGA1700",
2145| "power": 120, "desc": "4 nhân / 8 luồng, 120W TDP, Socket LGA1700"
2146| },
2147| {
2148| "id": "cpu_intel_13700K_239",
2149| "name": "Intel Core i7-13700K",
2150| "type": "CPU",
2151| "price": 10760000,
2152| "socket": "LGA1700",
2153| "power": 120, "desc": "4 nhân / 8 luồng, 120W TDP, Socket LGA1700"
2154| },
2155| {
```

```

2154 | "id": "cpu_intel_13700KF_240",
2155 | "name": "Intel Core i7-13700KF",
2156 | "type": "CPU",
2157 | "price": 2290000,
2158 | "socket": "LGA1700",
2159 | "power": 55,
2160 | "desc": "12 nhân / 24 luồng, 55W TDP, Socket LGA1700"
2161 | },
2162 | {
2163 | "id": "cpu_intel_13700T_241",
2164 | "name": "Intel Core i7-13700T",
2165 | "type": "CPU",
2166 | "price": 10330000,
2167 | "socket": "LGA1700",
2168 | "power": 120,
2169 | "desc": "6 nhân / 12 luồng, 120W TDP, Socket LGA1700"
2170 | },
2171 | {
2172 | "id": "cpu_intel_13700F_242",
2173 | "name": "Intel Core i7-13700F",
2174 | "type": "CPU",
2175 | "price": 13940000,
2176 | "socket": "LGA1700",
2177 | "power": 120,
2178 | "desc": "4 nhân / 8 luồng, 120W TDP, Socket LGA1700"
2179 | },
2180 | {
2181 | "id": "cpu_intel_13700std_243",
2182 | "name": "Intel Core i7-13700",
2183 | "type": "CPU",
2184 | "price": 3410000,
2185 | "socket": "LGA1700",
2186 | "power": 85,
2187 | "desc": "6 nhân / 12 luồng, 85W TDP, Socket LGA1700"
2188 | },
2189 | {
2190 | "id": "cpu_intel_13700K_244",
2191 | "name": "Intel Core i7-13700K",
2192 | "type": "CPU",
2193 | "price": 8880000,
2194 | "socket": "LGA1700",
2195 | "power": 85,
2196 | "desc": "14 nhân / 28 luồng, 85W TDP, Socket LGA1700"
2197 | },
2198 | {
2199 | "id": "cpu_intel_13700KF_245",
2200 | "name": "Intel Core i7-13700KF",
2201 | "type": "CPU",
2202 | "price": 2270000,
2203 | "socket": "LGA1700",
2204 | "power": 125,
2205 | "desc": "4 nhân / 8 luồng, 125W TDP, Socket LGA1700"
2206 | },
2207 | {
2208 | "id": "cpu_intel_13700T_246",
2209 | "name": "Intel Core i7-13700T",
2210 | "type": "CPU",
2211 | "price": 14370000,
2212 | "socket": "LGA1700",
2213 | "power": 55,
2214 | "desc": "12 nhân / 24 luồng, 55W TDP, Socket LGA1700"
2215 | },
2216 | {
2217 | "id": "cpu_intel_13700F_247",
2218 | "name": "Intel Core i7-13700F",
2219 | "type": "CPU",
2220 | "price": 10870000,
2221 | "socket": "LGA1700",
2222 | "power": 120,
2223 | "desc": "18 nhân / 36 luồng, 120W TDP, Socket LGA1700"
2224 | },
2225 | {
2226 | "id": "cpu_intel_13700std_248",
2227 | "name": "Intel Core i7-13700",
2228 | "type": "CPU",
2229 | "price": 12280000,
2230 | "socket": "LGA1700",
2231 | "power": 125,
2232 | "desc": "2 nhân / 4 luồng, 125W TDP, Socket LGA1700"
2233 | },
2234 | {
2235 | "id": "cpu_intel_14700K_249",
2236 | "name": "Intel Core i7-14700K",
2237 | "type": "CPU",
2238 | "price": 8270000,
2239 | "socket": "LGA1851",
2240 | "power": 125,
2241 | "desc": "16 nhân / 32 luồng, 125W TDP, Socket LGA1851"
2242 | },
2243 | {
2244 | "id": "cpu_intel_14700KF_250",
2245 | "name": "Intel Core i7-14700KF",
2246 | "type": "CPU",
2247 | "price": 8650000,
2248 | "socket": "LGA1851",
2249 | "power": 65,
2250 | "desc": "10 nhân / 20 luồng, 65W TDP, Socket LGA1851"
2251 | },
2252 | {
2253 | "id": "cpu_intel_14700T_251",
2254 | "name": "Intel Core i7-14700T",
2255 | "type": "CPU",
2256 | "price": 13880000,
2257 | "socket": "LGA1851",
2258 | "power": 105,
2259 | "desc": "4 nhân / 8 luồng, 105W TDP, Socket LGA1851"
2260 | },
2261 | {
2262 | "id": "cpu_intel_14700F_252",
2263 | "name": "Intel Core i7-14700F",
2264 | "type": "CPU",
2265 | "price": 6600000,
2266 | "socket": "LGA1851",
2267 | "power": 55,
2268 | "desc": "12 nhân / 24 luồng, 55W TDP, Socket LGA1851"
2269 | },
2270 | {
2271 | "id": "cpu_intel_14700std_253",
2272 | "name": "Intel Core i7-14700",
2273 | "type": "CPU",
2274 | "price": 1350000,
2275 | "socket": "LGA1851",
2276 | "power": 65,
2277 | "desc": "14 nhân / 28 luồng, 65W TDP, Socket LGA1851"

```

```
2278 | },
2279 | {
2280 |     "id": "cpu_intel_14700K_254",
2281 |     "name": "Intel Core i7-14700K",
2282 |     "type": "CPU",
2283 |     "price": 9110000,
2284 |     "socket": "LGA1851",
2285 |     "power": 105,
2286 |     "desc": "16 nhân / 32 luồng, 105W TDP, Socket LGA1851"
2287 | },
2288 | {
2289 |     "id": "cpu_intel_14700KF_255",
2290 |     "name": "Intel Core i7-14700KF",
2291 |     "type": "CPU",
2292 |     "price": 10000000,
2293 |     "socket": "LGA1851",
2294 |     "power": 105,
2295 |     "desc": "4 nhân / 8 luồng, 105W TDP, Socket LGA1851"
2296 | },
2297 | {
2298 |     "id": "cpu_intel_14700T_256",
2299 |     "name": "Intel Core i7-14700T",
2300 |     "type": "CPU",
2301 |     "price": 6880000,
2302 |     "socket": "LGA1851",
2303 |     "power": 75,
2304 |     "desc": "8 nhân / 16 luồng, 75W TDP, Socket LGA1851"
2305 | },
2306 | {
2307 |     "id": "cpu_intel_14700F_257",
2308 |     "name": "Intel Core i7-14700F",
2309 |     "type": "CPU",
2310 |     "price": 13800000,
2311 |     "socket": "LGA1851",
2312 |     "power": 65,
2313 |     "desc": "12 nhân / 24 luồng, 65W TDP, Socket LGA1851"
2314 | },
2315 | {
2316 |     "id": "cpu_intel_14700std_258",
2317 |     "name": "Intel Core i7-14700",
2318 |     "type": "CPU",
2319 |     "price": 9790000,
2320 |     "socket": "LGA1851",
2321 |     "power": 55,
2322 |     "desc": "10 nhân / 20 luồng, 55W TDP, Socket LGA1851"
2323 | },
2324 | {
2325 |     "id": "cpu_intel_14700K_259",
2326 |     "name": "Intel Core i7-14700K",
2327 |     "type": "CPU",
2328 |     "price": 14800000,
2329 |     "socket": "LGA1851",
2330 |     "power": 125,
2331 |     "desc": "18 nhân / 36 luồng, 125W TDP, Socket LGA1851"
2332 | },
2333 | {
2334 |     "id": "cpu_intel_14700KF_260",
2335 |     "name": "Intel Core i7-14700KF",
2336 |     "type": "CPU",
2337 |     "price": 1500000,
2338 |     "socket": "LGA1851",
2339 |     "power": 95,
2340 |     "desc": "10 nhân / 20 luồng, 95W TDP, Socket LGA1851"
2341 | },
2342 | {
2343 |     "id": "cpu_intel_14700T_261",
2344 |     "name": "Intel Core i7-14700T",
2345 |     "type": "CPU",
2346 |     "price": 11100000,
2347 |     "socket": "LGA1851",
2348 |     "power": 55,
2349 |     "desc": "18 nhân / 36 luồng, 55W TDP, Socket LGA1851"
2350 | },
2351 | {
2352 |     "id": "cpu_intel_14700F_262",
2353 |     "name": "Intel Core i7-14700F",
2354 |     "type": "CPU",
2355 |     "price": 2390000,
2356 |     "socket": "LGA1851",
2357 |     "power": 60,
2358 |     "desc": "14 nhân / 28 luồng, 60W TDP, Socket LGA1851"
2359 | },
2360 | {
2361 |     "id": "cpu_intel_14700std_263",
2362 |     "name": "Intel Core i7-14700",
2363 |     "type": "CPU",
2364 |     "price": 13040000,
2365 |     "socket": "LGA1851",
2366 |     "power": 95,
2367 |     "desc": "8 nhân / 16 luồng, 95W TDP, Socket LGA1851"
2368 | },
2369 | {
2370 |     "id": "cpu_intel_14700K_264",
2371 |     "name": "Intel Core i7-14700K",
2372 |     "type": "CPU",
2373 |     "price": 4550000,
2374 |     "socket": "LGA1851",
2375 |     "power": 95,
2376 |     "desc": "6 nhân / 12 luồng, 95W TDP, Socket LGA1851"
2377 | },
2378 | {
2379 |     "id": "cpu_intel_14700KF_265",
2380 |     "name": "Intel Core i7-14700KF",
2381 |     "type": "CPU",
2382 |     "price": 3090000,
2383 |     "socket": "LGA1851",
2384 |     "power": 85,
2385 |     "desc": "2 nhân / 4 luồng, 85W TDP, Socket LGA1851"
2386 | },
2387 | {
2388 |     "id": "cpu_intel_14700T_266",
2389 |     "name": "Intel Core i7-14700T",
2390 |     "type": "CPU",
2391 |     "price": 7470000,
2392 |     "socket": "LGA1851",
2393 |     "power": 120,
2394 |     "desc": "16 nhân / 32 luồng, 120W TDP, Socket LGA1851"
2395 | },
2396 | {
2397 |     "id": "cpu_intel_14700F_267",
2398 |     "name": "Intel Core i7-14700F",
2399 |     "type": "CPU",
2400 |     "price": 7020000,
2401 |     "socket": "LGA1851",
```

```
2402| "power": 125,
2403|   "desc": "10 nhân / 20 luồng, 125W TDP, Socket LGA1851"
2404| },
2405| {
2406|   "id": "cpu_intel_14700std_268",
2407|   "name": "Intel Core i7-14700",
2408|   "type": "CPU",
2409|   "price": 6650000,
2410|   "socket": "LGA1851",
2411|   "power": 75,
2412|   "desc": "16 nhân / 32 luồng, 75W TDP, Socket LGA1851"
2413| },
2414| {
2415|   "id": "cpu_intel_15700K_269",
2416|   "name": "Intel Core i7-15700K",
2417|   "type": "CPU",
2418|   "price": 13000000,
2419|   "socket": "LGA1851",
2420|   "power": 55,
2421|   "desc": "10 nhân / 20 luồng, 55W TDP, Socket LGA1851"
2422| },
2423| {
2424|   "id": "cpu_intel_15700KF_270",
2425|   "name": "Intel Core i7-15700KF",
2426|   "type": "CPU",
2427|   "price": 5170000,
2428|   "socket": "LGA1851",
2429|   "power": 125,
2430|   "desc": "12 nhân / 24 luồng, 125W TDP, Socket LGA1851"
2431| },
2432| {
2433|   "id": "cpu_intel_15700T_271",
2434|   "name": "Intel Core i7-15700T",
2435|   "type": "CPU",
2436|   "price": 2990000,
2437|   "socket": "LGA1851",
2438|   "power": 55,
2439|   "desc": "10 nhân / 20 luồng, 55W TDP, Socket LGA1851"
2440| },
2441| {
2442|   "id": "cpu_intel_15700F_272",
2443|   "name": "Intel Core i7-15700F",
2444|   "type": "CPU",
2445|   "price": 7510000,
2446|   "socket": "LGA1851",
2447|   "power": 105,
2448|   "desc": "4 nhân / 8 luồng, 105W TDP, Socket LGA1851"
2449| },
2450| {
2451|   "id": "cpu_intel_15700std_273",
2452|   "name": "Intel Core i7-15700",
2453|   "type": "CPU",
2454|   "price": 11640000,
2455|   "socket": "LGA1851",
2456|   "power": 75,
2457|   "desc": "10 nhân / 20 luồng, 75W TDP, Socket LGA1851"
2458| },
2459| {
2460|   "id": "cpu_intel_15700K_274",
2461|   "name": "Intel Core i7-15700K",
2462|   "type": "CPU",
2463|   "price": 11190000,
2464|   "socket": "LGA1851",
2465|   "power": 55,
2466|   "desc": "18 nhân / 36 luồng, 55W TDP, Socket LGA1851"
2467| },
2468| {
2469|   "id": "cpu_intel_15700KF_275",
2470|   "name": "Intel Core i7-15700KF",
2471|   "type": "CPU",
2472|   "price": 3610000,
2473|   "socket": "LGA1851",
2474|   "power": 60,
2475|   "desc": "16 nhân / 32 luồng, 60W TDP, Socket LGA1851"
2476| },
2477| {
2478|   "id": "cpu_intel_15700T_276",
2479|   "name": "Intel Core i7-15700T",
2480|   "type": "CPU",
2481|   "price": 9060000,
2482|   "socket": "LGA1851",
2483|   "power": 60,
2484|   "desc": "14 nhân / 28 luồng, 60W TDP, Socket LGA1851"
2485| },
2486| {
2487|   "id": "cpu_intel_15700F_277",
2488|   "name": "Intel Core i7-15700F",
2489|   "type": "CPU",
2490|   "price": 3550000,
2491|   "socket": "LGA1851",
2492|   "power": 85,
2493|   "desc": "2 nhân / 4 luồng, 85W TDP, Socket LGA1851"
2494| },
2495| {
2496|   "id": "cpu_intel_15700std_278",
2497|   "name": "Intel Core i7-15700",
2498|   "type": "CPU",
2499|   "price": 10430000,
2500|   "socket": "LGA1851",
2501|   "power": 75,
2502|   "desc": "4 nhân / 8 luồng, 75W TDP, Socket LGA1851"
2503| },
2504| {
2505|   "id": "cpu_intel_15700K_279",
2506|   "name": "Intel Core i7-15700K",
2507|   "type": "CPU",
2508|   "price": 1550000,
2509|   "socket": "LGA1851",
2510|   "power": 85,
2511|   "desc": "8 nhân / 16 luồng, 85W TDP, Socket LGA1851"
2512| },
2513| {
2514|   "id": "cpu_intel_15700KF_280",
2515|   "name": "Intel Core i7-15700KF",
2516|   "type": "CPU",
2517|   "price": 7070000,
2518|   "socket": "LGA1851",
2519|   "power": 65,
2520|   "desc": "6 nhân / 12 luồng, 65W TDP, Socket LGA1851"
2521| },
2522| {
2523|   "id": "cpu_intel_15700T_281",
2524|   "name": "Intel Core i7-15700T",
2525|   "type": "CPU",
```

```
2526 | "price": 5820000,
2527 | "socket": "LGA1851",
2528 | "power": 85, "desc": "18 nhân / 36 luồng, 85W TDP, Socket LGA1851"
2529 | },
2530 | {
2531 | "id": "cpu_intel_15700F_282",
2532 | "name": "Intel Core i7-15700F",
2533 | "type": "CPU",
2534 | "price": 6910000,
2535 | "socket": "LGA1851",
2536 | "power": 105, "desc": "4 nhân / 8 luồng, 105W TDP, Socket LGA1851"
2537 | },
2538 | {
2539 | "id": "cpu_intel_15700std_283",
2540 | "name": "Intel Core i7-15700",
2541 | "type": "CPU",
2542 | "price": 14520000,
2543 | "socket": "LGA1851",
2544 | "power": 65, "desc": "10 nhân / 20 luồng, 65W TDP, Socket LGA1851"
2545 | },
2546 | {
2547 | "id": "cpu_intel_15700K_284",
2548 | "name": "Intel Core i7-15700K",
2549 | "type": "CPU",
2550 | "price": 6220000,
2551 | "socket": "LGA1851",
2552 | "power": 55, "desc": "2 nhân / 4 luồng, 55W TDP, Socket LGA1851"
2553 | },
2554 | {
2555 | "id": "cpu_intel_15700KF_285",
2556 | "name": "Intel Core i7-15700KF",
2557 | "type": "CPU",
2558 | "price": 14090000,
2559 | "socket": "LGA1851",
2560 | "power": 105, "desc": "12 nhân / 24 luồng, 105W TDP, Socket LGA1851"
2561 | },
2562 | {
2563 | "id": "cpu_intel_15700T_286",
2564 | "name": "Intel Core i7-15700T",
2565 | "type": "CPU",
2566 | "price": 12470000,
2567 | "socket": "LGA1851",
2568 | "power": 125, "desc": "2 nhân / 4 luồng, 125W TDP, Socket LGA1851"
2569 | },
2570 | {
2571 | "id": "cpu_intel_15700F_287",
2572 | "name": "Intel Core i7-15700F",
2573 | "type": "CPU",
2574 | "price": 7240000,
2575 | "socket": "LGA1851",
2576 | "power": 120, "desc": "18 nhân / 36 luồng, 120W TDP, Socket LGA1851"
2577 | },
2578 | {
2579 | "id": "cpu_intel_15700std_288",
2580 | "name": "Intel Core i7-15700",
2581 | "type": "CPU",
2582 | "price": 5900000,
2583 | "socket": "LGA1851",
2584 | "power": 85, "desc": "10 nhân / 20 luồng, 85W TDP, Socket LGA1851"
2585 | },
2586 | {
2587 | "id": "cpu_intel_12900K_289",
2588 | "name": "Intel Core i9-12900K",
2589 | "type": "CPU",
2590 | "price": 21960000,
2591 | "socket": "LGA1700",
2592 | "power": 65, "desc": "6 nhân / 12 luồng, 65W TDP, Socket LGA1700"
2593 | },
2594 | {
2595 | "id": "cpu_intel_12900KF_290",
2596 | "name": "Intel Core i9-12900KF",
2597 | "type": "CPU",
2598 | "price": 10180000,
2599 | "socket": "LGA1700",
2600 | "power": 65, "desc": "2 nhân / 4 luồng, 65W TDP, Socket LGA1700"
2601 | },
2602 | {
2603 | "id": "cpu_intel_12900KS_291",
2604 | "name": "Intel Core i9-12900KS",
2605 | "type": "CPU",
2606 | "price": 24270000,
2607 | "socket": "LGA1700",
2608 | "power": 105, "desc": "2 nhân / 4 luồng, 105W TDP, Socket LGA1700"
2609 | },
2610 | {
2611 | "id": "cpu_intel_12900T_292",
2612 | "name": "Intel Core i9-12900T",
2613 | "type": "CPU",
2614 | "price": 17440000,
2615 | "socket": "LGA1700",
2616 | "power": 65, "desc": "16 nhân / 32 luồng, 65W TDP, Socket LGA1700"
2617 | },
2618 | {
2619 | "id": "cpu_intel_12900F_293",
2620 | "name": "Intel Core i9-12900F",
2621 | "type": "CPU",
2622 | "price": 10080000,
2623 | "socket": "LGA1700",
2624 | "power": 120, "desc": "12 nhân / 24 luồng, 120W TDP, Socket LGA1700"
2625 | },
2626 | {
2627 | "id": "cpu_intel_12900std_294",
2628 | "name": "Intel Core i9-12900",
2629 | "type": "CPU",
2630 | "price": 12870000,
2631 | "socket": "LGA1700",
2632 | "power": 75, "desc": "18 nhân / 36 luồng, 75W TDP, Socket LGA1700"
2633 | },
2634 | {
2635 | "id": "cpu_intel_12900K_295",
```

```
2650| "name": "Intel Core i9-12900K",
2651| "type": "CPU",
2652| "price": 5950000,
2653| "socket": "LGA1700",
2654| "power": 65,
2655| "desc": "12 nhân / 24 luồng, 65W TDP, Socket LGA1700"
2656| },
2657| {
2658| "id": "cpu_intel_12900KF_296",
2659| "name": "Intel Core i9-12900KF",
2660| "type": "CPU",
2661| "price": 8820000,
2662| "socket": "LGA1700",
2663| "power": 105,
2664| "desc": "12 nhân / 24 luồng, 105W TDP, Socket LGA1700"
2665| },
2666| {
2667| "id": "cpu_intel_12900KS_297",
2668| "name": "Intel Core i9-12900KS",
2669| "type": "CPU",
2670| "price": 14620000,
2671| "socket": "LGA1700",
2672| "power": 60,
2673| "desc": "14 nhân / 28 luồng, 60W TDP, Socket LGA1700"
2674| },
2675| {
2676| "id": "cpu_intel_12900T_298",
2677| "name": "Intel Core i9-12900T",
2678| "type": "CPU",
2679| "price": 17630000,
2680| "socket": "LGA1700",
2681| "power": 120,
2682| "desc": "2 nhân / 4 luồng, 120W TDP, Socket LGA1700"
2683| },
2684| {
2685| "id": "cpu_intel_12900F_299",
2686| "name": "Intel Core i9-12900F",
2687| "type": "CPU",
2688| "price": 9250000,
2689| "socket": "LGA1700",
2690| "power": 120,
2691| "desc": "4 nhân / 8 luồng, 120W TDP, Socket LGA1700"
2692| },
2693| {
2694| "id": "cpu_intel_12900std_300",
2695| "name": "Intel Core i9-12900",
2696| "type": "CPU",
2697| "price": 5940000,
2698| "socket": "LGA1700",
2699| "power": 125,
2700| "desc": "16 nhân / 32 luồng, 125W TDP, Socket LGA1700"
2701| },
2702| {
2703| "id": "cpu_intel_12900K_301",
2704| "name": "Intel Core i9-12900K",
2705| "type": "CPU",
2706| "price": 14780000,
2707| "socket": "LGA1700",
2708| "power": 105,
2709| "desc": "10 nhân / 20 luồng, 105W TDP, Socket LGA1700"
2710| },
2711| {
2712| "id": "cpu_intel_12900KF_302",
2713| "name": "Intel Core i9-12900KF",
2714| "type": "CPU",
2715| "price": 20740000,
2716| "socket": "LGA1700",
2717| "power": 125,
2718| "desc": "10 nhân / 20 luồng, 125W TDP, Socket LGA1700"
2719| },
2720| {
2721| "id": "cpu_intel_12900KS_303",
2722| "name": "Intel Core i9-12900KS",
2723| "type": "CPU",
2724| "price": 19940000,
2725| "socket": "LGA1700",
2726| "power": 125,
2727| "desc": "4 nhân / 8 luồng, 125W TDP, Socket LGA1700"
2728| },
2729| {
2730| "id": "cpu_intel_12900T_304",
2731| "name": "Intel Core i9-12900T",
2732| "type": "CPU",
2733| "price": 20770000,
2734| "socket": "LGA1700",
2735| "power": 105,
2736| "desc": "4 nhân / 8 luồng, 105W TDP, Socket LGA1700"
2737| },
2738| {
2739| "id": "cpu_intel_12900F_305",
2740| "name": "Intel Core i9-12900F",
2741| "type": "CPU",
2742| "price": 13440000,
2743| "socket": "LGA1700",
2744| "power": 85,
2745| "desc": "16 nhân / 32 luồng, 85W TDP, Socket LGA1700"
2746| },
2747| {
2748| "id": "cpu_intel_12900std_306",
2749| "name": "Intel Core i9-12900",
2750| "type": "CPU",
2751| "price": 10270000,
2752| "socket": "LGA1700",
2753| "power": 85,
2754| "desc": "18 nhân / 36 luồng, 85W TDP, Socket LGA1700"
2755| },
2756| {
2757| "id": "cpu_intel_12900K_307",
2758| "name": "Intel Core i9-12900K",
2759| "type": "CPU",
2760| "price": 6680000,
2761| "socket": "LGA1700",
2762| "power": 55,
2763| "desc": "2 nhân / 4 luồng, 55W TDP, Socket LGA1700"
2764| },
2765| {
2766| "id": "cpu_intel_12900KF_308",
2767| "name": "Intel Core i9-12900KF",
2768| "type": "CPU",
2769| "price": 5450000,
2770| "socket": "LGA1700",
2771| "power": 120,
2772| "desc": "12 nhân / 24 luồng, 120W TDP, Socket LGA1700"
2773| },
```



```
2774 | {
2775 |   "id": "cpu_intel_12900KS_309",
2776 |   "name": "Intel Core i9-12900KS",
2777 |   "type": "CPU",
2778 |   "price": 23480000,
2779 |   "socket": "LGA1700",
2780 |   "power": 65,
2781 |   "desc": "4 nhân / 8 luồng, 65W TDP, Socket LGA1700"
2782 | },
2783 | {
2784 |   "id": "cpu_intel_12900T_310",
2785 |   "name": "Intel Core i9-12900T",
2786 |   "type": "CPU",
2787 |   "price": 14250000,
2788 |   "socket": "LGA1700",
2789 |   "power": 105,
2790 |   "desc": "18 nhân / 36 luồng, 105W TDP, Socket LGA1700"
2791 | },
2792 | {
2793 |   "id": "cpu_intel_12900F_311",
2794 |   "name": "Intel Core i9-12900F",
2795 |   "type": "CPU",
2796 |   "price": 9640000,
2797 |   "socket": "LGA1700",
2798 |   "power": 75,
2799 |   "desc": "2 nhân / 4 luồng, 75W TDP, Socket LGA1700"
2800 | },
2801 | {
2802 |   "id": "cpu_intel_12900std_312",
2803 |   "name": "Intel Core i9-12900",
2804 |   "type": "CPU",
2805 |   "price": 24730000,
2806 |   "socket": "LGA1700",
2807 |   "power": 125,
2808 |   "desc": "14 nhân / 28 luồng, 125W TDP, Socket LGA1700"
2809 | },
2810 | {
2811 |   "id": "cpu_intel_13900K_313",
2812 |   "name": "Intel Core i9-13900K",
2813 |   "type": "CPU",
2814 |   "price": 13160000,
2815 |   "socket": "LGA1700",
2816 |   "power": 75,
2817 |   "desc": "16 nhân / 32 luồng, 75W TDP, Socket LGA1700"
2818 | },
2819 | {
2820 |   "id": "cpu_intel_13900KF_314",
2821 |   "name": "Intel Core i9-13900KF",
2822 |   "type": "CPU",
2823 |   "price": 1650000,
2824 |   "socket": "LGA1700",
2825 |   "power": 65,
2826 |   "desc": "12 nhân / 24 luồng, 65W TDP, Socket LGA1700"
2827 | },
2828 | {
2829 |   "id": "cpu_intel_13900KS_315",
2830 |   "name": "Intel Core i9-13900KS",
2831 |   "type": "CPU",
2832 |   "price": 10280000,
2833 |   "socket": "LGA1700",
2834 |   "power": 85,
2835 |   "desc": "2 nhân / 4 luồng, 85W TDP, Socket LGA1700"
2836 | },
2837 | {
2838 |   "id": "cpu_intel_13900T_316",
2839 |   "name": "Intel Core i9-13900T",
2840 |   "type": "CPU",
2841 |   "price": 23740000,
2842 |   "socket": "LGA1700",
2843 |   "power": 125,
2844 |   "desc": "10 nhân / 20 luồng, 125W TDP, Socket LGA1700"
2845 | },
2846 | {
2847 |   "id": "cpu_intel_13900F_317",
2848 |   "name": "Intel Core i9-13900F",
2849 |   "type": "CPU",
2850 |   "price": 22260000,
2851 |   "socket": "LGA1700",
2852 |   "power": 60,
2853 |   "desc": "2 nhân / 4 luồng, 60W TDP, Socket LGA1700"
2854 | },
2855 | {
2856 |   "id": "cpu_intel_13900std_318",
2857 |   "name": "Intel Core i9-13900",
2858 |   "type": "CPU",
2859 |   "price": 10770000,
2860 |   "socket": "LGA1700",
2861 |   "power": 60,
2862 |   "desc": "14 nhân / 28 luồng, 60W TDP, Socket LGA1700"
2863 | },
2864 | {
2865 |   "id": "cpu_intel_13900K_319",
2866 |   "name": "Intel Core i9-13900K",
2867 |   "type": "CPU",
2868 |   "price": 23100000,
2869 |   "socket": "LGA1700",
2870 |   "power": 85,
2871 |   "desc": "16 nhân / 32 luồng, 85W TDP, Socket LGA1700"
2872 | },
2873 | {
2874 |   "id": "cpu_intel_13900KF_320",
2875 |   "name": "Intel Core i9-13900KF",
2876 |   "type": "CPU",
2877 |   "price": 12200000,
2878 |   "socket": "LGA1700",
2879 |   "power": 85,
2880 |   "desc": "12 nhân / 24 luồng, 85W TDP, Socket LGA1700"
2881 | },
2882 | {
2883 |   "id": "cpu_intel_13900KS_321",
2884 |   "name": "Intel Core i9-13900KS",
2885 |   "type": "CPU",
2886 |   "price": 13660000,
2887 |   "socket": "LGA1700",
2888 |   "power": 55,
2889 |   "desc": "2 nhân / 4 luồng, 55W TDP, Socket LGA1700"
2890 | },
2891 | {
2892 |   "id": "cpu_intel_13900T_322",
2893 |   "name": "Intel Core i9-13900T",
2894 |   "type": "CPU",
2895 |   "price": 5200000,
2896 |   "socket": "LGA1700",
2897 |   "power": 55,
```

```
2898 | "desc": "18 nhân / 36 luồng, 55W TDP, Socket LGA1700"
2899 | },
2900 | {
2901 |   "id": "cpu_intel_13900F_323",
2902 |   "name": "Intel Core i9-13900F",
2903 |   "type": "CPU",
2904 |   "price": 7570000,
2905 |   "socket": "LGA1700",
2906 |   "power": 125,
2907 | "desc": "4 nhân / 8 luồng, 125W TDP, Socket LGA1700"
2908 | },
2909 | {
2910 |   "id": "cpu_intel_13900std_324",
2911 |   "name": "Intel Core i9-13900",
2912 |   "type": "CPU",
2913 |   "price": 23480000,
2914 |   "socket": "LGA1700",
2915 |   "power": 105,
2916 | "desc": "10 nhân / 20 luồng, 105W TDP, Socket LGA1700"
2917 | },
2918 | {
2919 |   "id": "cpu_intel_13900K_325",
2920 |   "name": "Intel Core i9-13900K",
2921 |   "type": "CPU",
2922 |   "price": 10120000,
2923 |   "socket": "LGA1700",
2924 |   "power": 65,
2925 | "desc": "4 nhân / 8 luồng, 65W TDP, Socket LGA1700"
2926 | },
2927 | {
2928 |   "id": "cpu_intel_13900KF_326",
2929 |   "name": "Intel Core i9-13900KF",
2930 |   "type": "CPU",
2931 |   "price": 4630000,
2932 |   "socket": "LGA1700",
2933 |   "power": 60,
2934 | "desc": "6 nhân / 12 luồng, 60W TDP, Socket LGA1700"
2935 | },
2936 | {
2937 |   "id": "cpu_intel_13900KS_327",
2938 |   "name": "Intel Core i9-13900KS",
2939 |   "type": "CPU",
2940 |   "price": 13330000,
2941 |   "socket": "LGA1700",
2942 |   "power": 55,
2943 | "desc": "14 nhân / 28 luồng, 55W TDP, Socket LGA1700"
2944 | },
2945 | {
2946 |   "id": "cpu_intel_13900T_328",
2947 |   "name": "Intel Core i9-13900T",
2948 |   "type": "CPU",
2949 |   "price": 5460000,
2950 |   "socket": "LGA1700",
2951 |   "power": 95,
2952 | "desc": "6 nhân / 12 luồng, 95W TDP, Socket LGA1700"
2953 | },
2954 | {
2955 |   "id": "cpu_intel_13900F_329",
2956 |   "name": "Intel Core i9-13900F",
2957 |   "type": "CPU",
2958 |   "price": 10320000,
2959 |   "socket": "LGA1700",
2960 |   "power": 60,
2961 | "desc": "2 nhân / 4 luồng, 60W TDP, Socket LGA1700"
2962 | },
2963 | {
2964 |   "id": "cpu_intel_13900std_330",
2965 |   "name": "Intel Core i9-13900",
2966 |   "type": "CPU",
2967 |   "price": 23970000,
2968 |   "socket": "LGA1700",
2969 |   "power": 65,
2970 | "desc": "4 nhân / 8 luồng, 65W TDP, Socket LGA1700"
2971 | },
2972 | {
2973 |   "id": "cpu_intel_13900K_331",
2974 |   "name": "Intel Core i9-13900K",
2975 |   "type": "CPU",
2976 |   "price": 17290000,
2977 |   "socket": "LGA1700",
2978 |   "power": 125,
2979 | "desc": "12 nhân / 24 luồng, 125W TDP, Socket LGA1700"
2980 | },
2981 | {
2982 |   "id": "cpu_intel_13900KF_332",
2983 |   "name": "Intel Core i9-13900KF",
2984 |   "type": "CPU",
2985 |   "price": 3710000,
2986 |   "socket": "LGA1700",
2987 |   "power": 55,
2988 | "desc": "6 nhân / 12 luồng, 55W TDP, Socket LGA1700"
2989 | },
2990 | {
2991 |   "id": "cpu_intel_13900KS_333",
2992 |   "name": "Intel Core i9-13900KS",
2993 |   "type": "CPU",
2994 |   "price": 21370000,
2995 |   "socket": "LGA1700",
2996 |   "power": 120,
2997 | "desc": "16 nhân / 32 luồng, 120W TDP, Socket LGA1700"
2998 | },
2999 | {
3000 |   "id": "cpu_intel_13900T_334",
3001 |   "name": "Intel Core i9-13900T",
3002 |   "type": "CPU",
3003 |   "price": 5490000,
3004 |   "socket": "LGA1700",
3005 |   "power": 60,
3006 | "desc": "2 nhân / 4 luồng, 60W TDP, Socket LGA1700"
3007 | },
3008 | {
3009 |   "id": "cpu_intel_13900F_335",
3010 |   "name": "Intel Core i9-13900F",
3011 |   "type": "CPU",
3012 |   "price": 9720000,
3013 |   "socket": "LGA1700",
3014 |   "power": 95,
3015 | "desc": "14 nhân / 28 luồng, 95W TDP, Socket LGA1700"
3016 | },
3017 | {
3018 |   "id": "cpu_intel_13900std_336",
3019 |   "name": "Intel Core i9-13900",
3020 |   "type": "CPU",
3021 |   "price": 18850000,
```

```
3022| "socket": "LGA1700",
3023| "power": 125,
3024| | " d e s c " : " 1 6   n h â n   /   3 2   l u ó n g ,   1 2 5 W   T D P ,   S o c k e t   L G A 1 7 0 0 "
3025| },
3026| {
3027|   "id": "cpu_intel_14900K_337",
3028|   "name": "Intel Core i9-14900K",
3029|   "type": "CPU",
3030|   "price": 5530000,
3031|   "socket": "LGA1851",
3032|   "power": 65,
3033| | " d e s c " : " 2   n h â n   /   4   l u ó n g ,   6 5 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3034| },
3035| {
3036|   "id": "cpu_intel_14900KF_338",
3037|   "name": "Intel Core i9-14900KF",
3038|   "type": "CPU",
3039|   "price": 21970000,
3040|   "socket": "LGA1851",
3041|   "power": 125,
3042| | " d e s c " : " 1 8   n h â n   /   3 6   l u ó n g ,   1 2 5 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3043| },
3044| {
3045|   "id": "cpu_intel_14900KS_339",
3046|   "name": "Intel Core i9-14900KS",
3047|   "type": "CPU",
3048|   "price": 18740000,
3049|   "socket": "LGA1851",
3050|   "power": 55,
3051| | " d e s c " : " 1 2   n h â n   /   2 4   l u ó n g ,   5 5 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3052| },
3053| {
3054|   "id": "cpu_intel_14900T_340",
3055|   "name": "Intel Core i9-14900T",
3056|   "type": "CPU",
3057|   "price": 10440000,
3058|   "socket": "LGA1851",
3059|   "power": 125,
3060| | " d e s c " : " 4   n h â n   /   8   l u ó n g ,   1 2 5 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3061| },
3062| {
3063|   "id": "cpu_intel_14900F_341",
3064|   "name": "Intel Core i9-14900F",
3065|   "type": "CPU",
3066|   "price": 2930000,
3067|   "socket": "LGA1851",
3068|   "power": 95,
3069| | " d e s c " : " 1 2   n h â n   /   2 4   l u ó n g ,   9 5 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3070| },
3071| {
3072|   "id": "cpu_intel_14900std_342",
3073|   "name": "Intel Core i9-14900",
3074|   "type": "CPU",
3075|   "price": 2070000,
3076|   "socket": "LGA1851",
3077|   "power": 75,
3078| | " d e s c " : " 1 8   n h â n   /   3 6   l u ó n g ,   7 5 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3079| },
3080| {
3081|   "id": "cpu_intel_14900K_343",
3082|   "name": "Intel Core i9-14900K",
3083|   "type": "CPU",
3084|   "price": 18450000,
3085|   "socket": "LGA1851",
3086|   "power": 60,
3087| | " d e s c " : " 1 4   n h â n   /   2 8   l u ó n g ,   6 0 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3088| },
3089| {
3090|   "id": "cpu_intel_14900KF_344",
3091|   "name": "Intel Core i9-14900KF",
3092|   "type": "CPU",
3093|   "price": 5910000,
3094|   "socket": "LGA1851",
3095|   "power": 85,
3096| | " d e s c " : " 1 6   n h â n   /   3 2   l u ó n g ,   8 5 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3097| },
3098| {
3099|   "id": "cpu_intel_14900KS_345",
3100|   "name": "Intel Core i9-14900KS",
3101|   "type": "CPU",
3102|   "price": 6020000,
3103|   "socket": "LGA1851",
3104|   "power": 95,
3105| | " d e s c " : " 8   n h â n   /   1 6   l u ó n g ,   9 5 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3106| },
3107| {
3108|   "id": "cpu_intel_14900T_346",
3109|   "name": "Intel Core i9-14900T",
3110|   "type": "CPU",
3111|   "price": 1920000,
3112|   "socket": "LGA1851",
3113|   "power": 120,
3114| | " d e s c " : " 1 8   n h â n   /   3 6   l u ó n g ,   1 2 0 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3115| },
3116| {
3117|   "id": "cpu_intel_14900F_347",
3118|   "name": "Intel Core i9-14900F",
3119|   "type": "CPU",
3120|   "price": 13680000,
3121|   "socket": "LGA1851",
3122|   "power": 60,
3123| | " d e s c " : " 1 0   n h â n   /   2 0   l u ó n g ,   6 0 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3124| },
3125| {
3126|   "id": "cpu_intel_14900std_348",
3127|   "name": "Intel Core i9-14900",
3128|   "type": "CPU",
3129|   "price": 20520000,
3130|   "socket": "LGA1851",
3131|   "power": 125,
3132| | " d e s c " : " 1 2   n h â n   /   2 4   l u ó n g ,   1 2 5 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3133| },
3134| {
3135|   "id": "cpu_intel_14900K_349",
3136|   "name": "Intel Core i9-14900K",
3137|   "type": "CPU",
3138|   "price": 15910000,
3139|   "socket": "LGA1851",
3140|   "power": 120,
3141| | " d e s c " : " 1 4   n h â n   /   2 8   l u ó n g ,   1 2 0 W   T D P ,   S o c k e t   L G A 1 8 5 1 "
3142| },
3143| {
3144|   "id": "cpu_intel_14900KF_350",
3145|   "name": "Intel Core i9-14900KF",
```



```
3270|      "id": "cpu_intel_15900T_364",
3271|      "name": "Intel Core i9-15900T",
3272|      "type": "CPU",
3273|      "price": 13460000,
3274|      "socket": "LGA1851",
3275|      "power": 75,
3276|      "desc": "4 nhân / 8 luồng, 75W TDP, Socket LGA1851"
3277|    },
3278|    {
3279|      "id": "cpu_intel_15900F_365",
3280|      "name": "Intel Core i9-15900F",
3281|      "type": "CPU",
3282|      "price": 19050000,
3283|      "socket": "LGA1851",
3284|      "power": 95,
3285|      "desc": "12 nhân / 24 luồng, 95W TDP, Socket LGA1851"
3286|    },
3287|    {
3288|      "id": "cpu_intel_15900std_366",
3289|      "name": "Intel Core i9-15900",
3290|      "type": "CPU",
3291|      "price": 23750000,
3292|      "socket": "LGA1851",
3293|      "power": 95,
3294|      "desc": "18 nhân / 36 luồng, 95W TDP, Socket LGA1851"
3295|    },
3296|    {
3297|      "id": "cpu_intel_15900K_367",
3298|      "name": "Intel Core i9-15900K",
3299|      "type": "CPU",
3300|      "price": 9240000,
3301|      "socket": "LGA1851",
3302|      "power": 55,
3303|      "desc": "14 nhân / 28 luồng, 55W TDP, Socket LGA1851"
3304|    },
3305|    {
3306|      "id": "cpu_intel_15900KF_368",
3307|      "name": "Intel Core i9-15900KF",
3308|      "type": "CPU",
3309|      "price": 12390000,
3310|      "socket": "LGA1851",
3311|      "power": 105,
3312|      "desc": "12 nhân / 24 luồng, 105W TDP, Socket LGA1851"
3313|    },
3314|    {
3315|      "id": "cpu_intel_15900KS_369",
3316|      "name": "Intel Core i9-15900KS",
3317|      "type": "CPU",
3318|      "price": 17260000,
3319|      "socket": "LGA1851",
3320|      "power": 60,
3321|      "desc": "12 nhân / 24 luồng, 60W TDP, Socket LGA1851"
3322|    },
3323|    {
3324|      "id": "cpu_intel_15900T_370",
3325|      "name": "Intel Core i9-15900T",
3326|      "type": "CPU",
3327|      "price": 4780000,
3328|      "socket": "LGA1851",
3329|      "power": 65,
3330|      "desc": "18 nhân / 36 luồng, 65W TDP, Socket LGA1851"
3331|    },
3332|    {
3333|      "id": "cpu_intel_15900F_371",
3334|      "name": "Intel Core i9-15900F",
3335|      "type": "CPU",
3336|      "price": 4160000,
3337|      "socket": "LGA1851",
3338|      "power": 95,
3339|      "desc": "10 nhân / 20 luồng, 95W TDP, Socket LGA1851"
3340|    },
3341|    {
3342|      "id": "cpu_intel_15900std_372",
3343|      "name": "Intel Core i9-15900",
3344|      "type": "CPU",
3345|      "price": 17480000,
3346|      "socket": "LGA1851",
3347|      "power": 60,
3348|      "desc": "12 nhân / 24 luồng, 60W TDP, Socket LGA1851"
3349|    },
3350|    {
3351|      "id": "cpu_intel_15900K_373",
3352|      "name": "Intel Core i9-15900K",
3353|      "type": "CPU",
3354|      "price": 6410000,
3355|      "socket": "LGA1851",
3356|      "power": 105,
3357|      "desc": "8 nhân / 16 luồng, 105W TDP, Socket LGA1851"
3358|    },
3359|    {
3360|      "id": "cpu_intel_15900KF_374",
3361|      "name": "Intel Core i9-15900KF",
3362|      "type": "CPU",
3363|      "price": 5640000,
3364|      "socket": "LGA1851",
3365|      "power": 75,
3366|      "desc": "16 nhân / 32 luồng, 75W TDP, Socket LGA1851"
3367|    },
3368|    {
3369|      "id": "cpu_intel_15900KS_375",
3370|      "name": "Intel Core i9-15900KS",
3371|      "type": "CPU",
3372|      "price": 10580000,
3373|      "socket": "LGA1851",
3374|      "power": 60,
3375|      "desc": "16 nhân / 32 luồng, 60W TDP, Socket LGA1851"
3376|    },
3377|    {
3378|      "id": "cpu_intel_15900T_376",
3379|      "name": "Intel Core i9-15900T",
3380|      "type": "CPU",
3381|      "price": 11520000,
3382|      "socket": "LGA1851",
3383|      "power": 120,
3384|      "desc": "2 nhân / 4 luồng, 120W TDP, Socket LGA1851"
3385|    },
3386|    {
3387|      "id": "cpu_intel_15900F_377",
3388|      "name": "Intel Core i9-15900F",
3389|      "type": "CPU",
3390|      "price": 18430000,
3391|      "socket": "LGA1851",
3392|      "power": 85,
3393|      "desc": "18 nhân / 36 luồng, 85W TDP, Socket LGA1851"
```

```
3394| },
3395| {
3396|     "id": "cpu_intel_15900std_378",
3397|     "name": "Intel Core i9-15900",
3398|     "type": "CPU",
3399|     "price": 17830000,
3400|     "socket": "LGA1851",
3401|     "power": 85,
3402|     "desc": "8 nhân / 16 luồng, 85W TDP, Socket LGA1851"
3403| },
3404| {
3405|     "id": "cpu_intel_15900K_379",
3406|     "name": "Intel Core i9-15900K",
3407|     "type": "CPU",
3408|     "price": 1150000,
3409|     "socket": "LGA1851",
3410|     "power": 85,
3411|     "desc": "10 nhân / 20 luồng, 85W TDP, Socket LGA1851"
3412| },
3413| {
3414|     "id": "cpu_intel_15900KF_380",
3415|     "name": "Intel Core i9-15900KF",
3416|     "type": "CPU",
3417|     "price": 6620000,
3418|     "socket": "LGA1851",
3419|     "power": 125,
3420|     "desc": "18 nhân / 36 luồng, 125W TDP, Socket LGA1851"
3421| },
3422| {
3423|     "id": "cpu_intel_15900KS_381",
3424|     "name": "Intel Core i9-15900KS",
3425|     "type": "CPU",
3426|     "price": 7890000,
3427|     "socket": "LGA1851",
3428|     "power": 85,
3429|     "desc": "16 nhân / 32 luồng, 85W TDP, Socket LGA1851"
3430| },
3431| {
3432|     "id": "cpu_intel_15900T_382",
3433|     "name": "Intel Core i9-15900T",
3434|     "type": "CPU",
3435|     "price": 15060000,
3436|     "socket": "LGA1851",
3437|     "power": 95,
3438|     "desc": "4 nhân / 8 luồng, 95W TDP, Socket LGA1851"
3439| },
3440| {
3441|     "id": "cpu_intel_15900F_383",
3442|     "name": "Intel Core i9-15900F",
3443|     "type": "CPU",
3444|     "price": 5550000,
3445|     "socket": "LGA1851",
3446|     "power": 85,
3447|     "desc": "12 nhân / 24 luồng, 85W TDP, Socket LGA1851"
3448| },
3449| {
3450|     "id": "cpu_intel_15900std_384",
3451|     "name": "Intel Core i9-15900",
3452|     "type": "CPU",
3453|     "price": 7750000,
3454|     "socket": "LGA1851",
3455|     "power": 55,
3456|     "desc": "6 nhân / 12 luồng, 55W TDP, Socket LGA1851"
3457| },
3458| {
3459|     "id": "cpu_amd_ryzen_3_5000_5300g_385",
3460|     "name": "AMD Ryzen 3 50005300G",
3461|     "type": "CPU",
3462|     "price": 1550000,
3463|     "socket": "AM4",
3464|     "power": 85,
3465|     "desc": "8 nhân / 16 luồng, 85W TDP, Socket AM4"
3466| },
3467| {
3468|     "id": "cpu_amd_ryzen_3_5000_5400g_386",
3469|     "name": "AMD Ryzen 3 50005400G",
3470|     "type": "CPU",
3471|     "price": 1630000,
3472|     "socket": "AM4",
3473|     "power": 120,
3474|     "desc": "16 nhân / 32 luồng, 120W TDP, Socket AM4"
3475| },
3476| {
3477|     "id": "cpu_amd_ryzen_3_5000_7300x_387",
3478|     "name": "AMD Ryzen 3 50007300X",
3479|     "type": "CPU",
3480|     "price": 3520000,
3481|     "socket": "AM4",
3482|     "power": 85,
3483|     "desc": "8 nhân / 16 luồng, 85W TDP, Socket AM4"
3484| },
3485| {
3486|     "id": "cpu_amd_ryzen_3_5000_8300g_388",
3487|     "name": "AMD Ryzen 3 50008300G",
3488|     "type": "CPU",
3489|     "price": 4730000,
3490|     "socket": "AM4",
3491|     "power": 105,
3492|     "desc": "6 nhân / 12 luồng, 105W TDP, Socket AM4"
3493| },
3494| {
3495|     "id": "cpu_amd_ryzen_3_5000_9300x_389",
3496|     "name": "AMD Ryzen 3 50009300X",
3497|     "type": "CPU",
3498|     "price": 4060000,
3499|     "socket": "AM4",
3500|     "power": 170,
3501|     "desc": "6 nhân / 12 luồng, 170W TDP, Socket AM4"
3502| },
3503| {
3504|     "id": "cpu_amd_ryzen_3_7000_5300g_390",
3505|     "name": "AMD Ryzen 3 70005300G",
3506|     "type": "CPU",
3507|     "price": 3710000,
3508|     "socket": "AM5",
3509|     "power": 170,
3510|     "desc": "12 nhân / 24 luồng, 170W TDP, Socket AM5"
3511| },
3512| {
3513|     "id": "cpu_amd_ryzen_3_7000_5400g_391",
3514|     "name": "AMD Ryzen 3 70005400G",
3515|     "type": "CPU",
3516|     "price": 1970000,
3517|     "socket": "AM5",
```

```
3518| "power": 120,
3519|   "desc": "8 nhân / 16 luồng, 120W TDP, Socket AM5"
3520| },
3521| {
3522|   "id": "cpu_amd_ryzen_3_7000_7300x_392",
3523|   "name": "AMD Ryzen 3 70007300X",
3524|   "type": "CPU",
3525|   "price": 2460000,
3526|   "socket": "AM5",
3527|   "power": 120,
3528|   "desc": "16 nhân / 32 luồng, 120W TDP, Socket AM5"
3529| },
3530| {
3531|   "id": "cpu_amd_ryzen_3_7000_8300g_393",
3532|   "name": "AMD Ryzen 3 70008300G",
3533|   "type": "CPU",
3534|   "price": 2450000,
3535|   "socket": "AM5",
3536|   "power": 65,
3537|   "desc": "4 nhân / 8 luồng, 65W TDP, Socket AM5"
3538| },
3539| {
3540|   "id": "cpu_amd_ryzen_3_7000_9300x_394",
3541|   "name": "AMD Ryzen 3 70009300X",
3542|   "type": "CPU",
3543|   "price": 2430000,
3544|   "socket": "AM5",
3545|   "power": 120,
3546|   "desc": "12 nhân / 24 luồng, 120W TDP, Socket AM5"
3547| },
3548| {
3549|   "id": "cpu_amd_ryzen_3_8000_5300g_395",
3550|   "name": "AMD Ryzen 3 80005300G",
3551|   "type": "CPU",
3552|   "price": 2530000,
3553|   "socket": "AM5",
3554|   "power": 105,
3555|   "desc": "4 nhân / 8 luồng, 105W TDP, Socket AM5"
3556| },
3557| {
3558|   "id": "cpu_amd_ryzen_3_8000_5400g_396",
3559|   "name": "AMD Ryzen 3 80005400G",
3560|   "type": "CPU",
3561|   "price": 3200000,
3562|   "socket": "AM5",
3563|   "power": 105,
3564|   "desc": "4 nhân / 8 luồng, 105W TDP, Socket AM5"
3565| },
3566| {
3567|   "id": "cpu_amd_ryzen_3_8000_7300x_397",
3568|   "name": "AMD Ryzen 3 80007300X",
3569|   "type": "CPU",
3570|   "price": 4320000,
3571|   "socket": "AM5",
3572|   "power": 170,
3573|   "desc": "6 nhân / 12 luồng, 170W TDP, Socket AM5"
3574| },
3575| {
3576|   "id": "cpu_amd_ryzen_3_8000_8300g_398",
3577|   "name": "AMD Ryzen 3 80008300G",
3578|   "type": "CPU",
3579|   "price": 3490000,
3580|   "socket": "AM5",
3581|   "power": 170,
3582|   "desc": "12 nhân / 24 luồng, 170W TDP, Socket AM5"
3583| },
3584| {
3585|   "id": "cpu_amd_ryzen_3_8000_9300x_399",
3586|   "name": "AMD Ryzen 3 80009300X",
3587|   "type": "CPU",
3588|   "price": 4590000,
3589|   "socket": "AM5",
3590|   "power": 170,
3591|   "desc": "4 nhân / 8 luồng, 170W TDP, Socket AM5"
3592| },
3593| {
3594|   "id": "cpu_amd_ryzen_3_9000_5300g_400",
3595|   "name": "AMD Ryzen 3 90005300G",
3596|   "type": "CPU",
3597|   "price": 4150000,
3598|   "socket": "AM5",
3599|   "power": 85,
3600|   "desc": "4 nhân / 8 luồng, 85W TDP, Socket AM5"
3601| },
3602| {
3603|   "id": "cpu_amd_ryzen_3_9000_5400g_401",
3604|   "name": "AMD Ryzen 3 90005400G",
3605|   "type": "CPU",
3606|   "price": 3520000,
3607|   "socket": "AM5",
3608|   "power": 95,
3609|   "desc": "6 nhân / 12 luồng, 95W TDP, Socket AM5"
3610| },
3611| {
3612|   "id": "cpu_amd_ryzen_3_9000_7300x_402",
3613|   "name": "AMD Ryzen 3 90007300X",
3614|   "type": "CPU",
3615|   "price": 2280000,
3616|   "socket": "AM5",
3617|   "power": 85,
3618|   "desc": "6 nhân / 12 luồng, 85W TDP, Socket AM5"
3619| },
3620| {
3621|   "id": "cpu_amd_ryzen_3_9000_8300g_403",
3622|   "name": "AMD Ryzen 3 90008300G",
3623|   "type": "CPU",
3624|   "price": 1600000,
3625|   "socket": "AM5",
3626|   "power": 120,
3627|   "desc": "12 nhân / 24 luồng, 120W TDP, Socket AM5"
3628| },
3629| {
3630|   "id": "cpu_amd_ryzen_3_9000_9300x_404",
3631|   "name": "AMD Ryzen 3 90009300X",
3632|   "type": "CPU",
3633|   "price": 1980000,
3634|   "socket": "AM5",
3635|   "power": 95,
3636|   "desc": "12 nhân / 24 luồng, 95W TDP, Socket AM5"
3637| },
3638| {
3639|   "id": "cpu_amd_ryzen_5_5000_5500_405",
3640|   "name": "AMD Ryzen 5 50005500",
3641|   "type": "CPU",
```

```
3642|    "price": 4320000,
3643|    "socket": "AM4",
3644|    "power": 105,
3645|    "desc": "6 nhân / 12 luồng, 105W TDP, Socket AM4"
3646|  },
3647|  {
3648|    "id": "cpu_amd_ryzen_5_5000_5600_406",
3649|    "name": "AMD Ryzen 5 50005600",
3650|    "type": "CPU",
3651|    "price": 5470000,
3652|    "socket": "AM4",
3653|    "power": 85,
3654|    "desc": "8 nhân / 16 luồng, 85W TDP, Socket AM4"
3655|  },
3656|  {
3657|    "id": "cpu_amd_ryzen_5_5000_5600x_407",
3658|    "name": "AMD Ryzen 5 50005600X",
3659|    "type": "CPU",
3660|    "price": 3680000,
3661|    "socket": "AM4",
3662|    "power": 95,
3663|    "desc": "4 nhân / 8 luồng, 95W TDP, Socket AM4"
3664|  },
3665|  {
3666|    "id": "cpu_amd_ryzen_5_5000_7500f_408",
3667|    "name": "AMD Ryzen 5 50007500F",
3668|    "type": "CPU",
3669|    "price": 7960000,
3670|    "socket": "AM4",
3671|    "power": 105,
3672|    "desc": "12 nhân / 24 luồng, 105W TDP, Socket AM4"
3673|  },
3674|  {
3675|    "id": "cpu_amd_ryzen_5_5000_7600_409",
3676|    "name": "AMD Ryzen 5 50007600",
3677|    "type": "CPU",
3678|    "price": 6390000,
3679|    "socket": "AM4",
3680|    "power": 95,
3681|    "desc": "4 nhân / 8 luồng, 95W TDP, Socket AM4"
3682|  },
3683|  {
3684|    "id": "cpu_amd_ryzen_5_5000_7600x_410",
3685|    "name": "AMD Ryzen 5 50007600X",
3686|    "type": "CPU",
3687|    "price": 2980000,
3688|    "socket": "AM4",
3689|    "power": 85,
3690|    "desc": "6 nhân / 12 luồng, 85W TDP, Socket AM4"
3691|  },
3692|  {
3693|    "id": "cpu_amd_ryzen_5_5000_8400f_411",
3694|    "name": "AMD Ryzen 5 50008400F",
3695|    "type": "CPU",
3696|    "price": 7620000,
3697|    "socket": "AM4",
3698|    "power": 95,
3699|    "desc": "4 nhân / 8 luồng, 95W TDP, Socket AM4"
3700|  },
3701|  {
3702|    "id": "cpu_amd_ryzen_5_5000_8500g_412",
3703|    "name": "AMD Ryzen 5 50008500G",
3704|    "type": "CPU",
3705|    "price": 6840000,
3706|    "socket": "AM4",
3707|    "power": 65,
3708|    "desc": "6 nhân / 12 luồng, 65W TDP, Socket AM4"
3709|  },
3710|  {
3711|    "id": "cpu_amd_ryzen_5_5000_8600g_413",
3712|    "name": "AMD Ryzen 5 50008600G",
3713|    "type": "CPU",
3714|    "price": 4250000,
3715|    "socket": "AM4",
3716|    "power": 65,
3717|    "desc": "8 nhân / 16 luồng, 65W TDP, Socket AM4"
3718|  },
3719|  {
3720|    "id": "cpu_amd_ryzen_5_5000_9600_414",
3721|    "name": "AMD Ryzen 5 50009600",
3722|    "type": "CPU",
3723|    "price": 6060000,
3724|    "socket": "AM4",
3725|    "power": 65,
3726|    "desc": "4 nhân / 8 luồng, 65W TDP, Socket AM4"
3727|  },
3728|  {
3729|    "id": "cpu_amd_ryzen_5_5000_9600x_415",
3730|    "name": "AMD Ryzen 5 50009600X",
3731|    "type": "CPU",
3732|    "price": 3870000,
3733|    "socket": "AM4",
3734|    "power": 95,
3735|    "desc": "8 nhân / 16 luồng, 95W TDP, Socket AM4"
3736|  },
3737|  {
3738|    "id": "cpu_amd_ryzen_5_7000_5500_416",
3739|    "name": "AMD Ryzen 5 70005500",
3740|    "type": "CPU",
3741|    "price": 4270000,
3742|    "socket": "AM5",
3743|    "power": 95,
3744|    "desc": "4 nhân / 8 luồng, 95W TDP, Socket AM5"
3745|  },
3746|  {
3747|    "id": "cpu_amd_ryzen_5_7000_5600_417",
3748|    "name": "AMD Ryzen 5 70005600",
3749|    "type": "CPU",
3750|    "price": 2940000,
3751|    "socket": "AM5",
3752|    "power": 105,
3753|    "desc": "6 nhân / 12 luồng, 105W TDP, Socket AM5"
3754|  },
3755|  {
3756|    "id": "cpu_amd_ryzen_5_7000_5600x_418",
3757|    "name": "AMD Ryzen 5 70005600X",
3758|    "type": "CPU",
3759|    "price": 3970000,
3760|    "socket": "AM5",
3761|    "power": 120,
3762|    "desc": "8 nhân / 16 luồng, 120W TDP, Socket AM5"
3763|  },
3764|  {
3765|    "id": "cpu_amd_ryzen_5_7000_7500f_419",
```



```
3766 | "name": "AMD Ryzen 5 70007500F",
3767 | "type": "CPU",
3768 | "price": 5800000,
3769 | "socket": "AM5",
3770 | "power": 85,
3771 | "desc": "4 nhân / 8 luồng, 85W TDP, Socket AM5"
3772 | },
3773 | {
3774 | "id": "cpu_amd_ryzen_5_7000_7600_420",
3775 | "name": "AMD Ryzen 5 70007600",
3776 | "type": "CPU",
3777 | "price": 7180000,
3778 | "socket": "AM5",
3779 | "power": 95,
3780 | "desc": "12 nhân / 24 luồng, 95W TDP, Socket AM5"
3781 | },
3782 | {
3783 | "id": "cpu_amd_ryzen_5_7000_7600x_421",
3784 | "name": "AMD Ryzen 5 70007600X",
3785 | "type": "CPU",
3786 | "price": 7280000,
3787 | "socket": "AM5",
3788 | "power": 120,
3789 | "desc": "4 nhân / 8 luồng, 120W TDP, Socket AM5"
3790 | },
3791 | {
3792 | "id": "cpu_amd_ryzen_5_7000_8400f_422",
3793 | "name": "AMD Ryzen 5 70008400F",
3794 | "type": "CPU",
3795 | "price": 5930000,
3796 | "socket": "AM5",
3797 | "power": 105,
3798 | "desc": "6 nhân / 12 luồng, 105W TDP, Socket AM5"
3799 | },
3800 | {
3801 | "id": "cpu_amd_ryzen_5_7000_8500g_423",
3802 | "name": "AMD Ryzen 5 70008500G",
3803 | "type": "CPU",
3804 | "price": 7860000,
3805 | "socket": "AM5",
3806 | "power": 170,
3807 | "desc": "12 nhân / 24 luồng, 170W TDP, Socket AM5"
3808 | },
3809 | {
3810 | "id": "cpu_amd_ryzen_5_7000_8600g_424",
3811 | "name": "AMD Ryzen 5 70008600G",
3812 | "type": "CPU",
3813 | "price": 6410000,
3814 | "socket": "AM5",
3815 | "power": 120,
3816 | "desc": "4 nhân / 8 luồng, 120W TDP, Socket AM5"
3817 | },
3818 | {
3819 | "id": "cpu_amd_ryzen_5_7000_9600_425",
3820 | "name": "AMD Ryzen 5 70009600",
3821 | "type": "CPU",
3822 | "price": 5570000,
3823 | "socket": "AM5",
3824 | "power": 120,
3825 | "desc": "8 nhân / 16 luồng, 120W TDP, Socket AM5"
3826 | },
3827 | {
3828 | "id": "cpu_amd_ryzen_5_7000_9600x_426",
3829 | "name": "AMD Ryzen 5 70009600X",
3830 | "type": "CPU",
3831 | "price": 7970000,
3832 | "socket": "AM5",
3833 | "power": 170,
3834 | "desc": "16 nhân / 32 luồng, 170W TDP, Socket AM5"
3835 | },
3836 | {
3837 | "id": "cpu_amd_ryzen_5_8000_5500_427",
3838 | "name": "AMD Ryzen 5 80005500",
3839 | "type": "CPU",
3840 | "price": 7550000,
3841 | "socket": "AM5",
3842 | "power": 170,
3843 | "desc": "8 nhân / 16 luồng, 170W TDP, Socket AM5"
3844 | },
3845 | {
3846 | "id": "cpu_amd_ryzen_5_8000_5600_428",
3847 | "name": "AMD Ryzen 5 80005600",
3848 | "type": "CPU",
3849 | "price": 3700000,
3850 | "socket": "AM5",
3851 | "power": 120,
3852 | "desc": "4 nhân / 8 luồng, 120W TDP, Socket AM5"
3853 | },
3854 | {
3855 | "id": "cpu_amd_ryzen_5_8000_5600x_429",
3856 | "name": "AMD Ryzen 5 80005600X",
3857 | "type": "CPU",
3858 | "price": 7520000,
3859 | "socket": "AM5",
3860 | "power": 95,
3861 | "desc": "16 nhân / 32 luồng, 95W TDP, Socket AM5"
3862 | },
3863 | {
3864 | "id": "cpu_amd_ryzen_5_8000_7500f_430",
3865 | "name": "AMD Ryzen 5 80007500F",
3866 | "type": "CPU",
3867 | "price": 5800000,
3868 | "socket": "AM5",
3869 | "power": 85,
3870 | "desc": "6 nhân / 12 luồng, 85W TDP, Socket AM5"
3871 | },
3872 | {
3873 | "id": "cpu_amd_ryzen_5_8000_7600_431",
3874 | "name": "AMD Ryzen 5 80007600",
3875 | "type": "CPU",
3876 | "price": 4790000,
3877 | "socket": "AM5",
3878 | "power": 120,
3879 | "desc": "8 nhân / 16 luồng, 120W TDP, Socket AM5"
3880 | },
3881 | {
3882 | "id": "cpu_amd_ryzen_5_8000_7600x_432",
3883 | "name": "AMD Ryzen 5 80007600X",
3884 | "type": "CPU",
3885 | "price": 6930000,
3886 | "socket": "AM5",
3887 | "power": 120,
3888 | "desc": "12 nhân / 24 luồng, 120W TDP, Socket AM5"
3889 | },
```

```
3890| {
3891|   "id": "cpu_amd_ryzen_5_8000_8400f_433",
3892|   "name": "AMD Ryzen 5 80008400F",
3893|   "type": "CPU",
3894|   "price": 6600000,
3895|   "socket": "AM5",
3896|   "power": 95,
3897|   "desc": "16 nhân / 32 luồng, 95W TDP, Socket AM5"
3898| },
3899| {
3900|   "id": "cpu_amd_ryzen_5_8000_8500g_434",
3901|   "name": "AMD Ryzen 5 80008500G",
3902|   "type": "CPU",
3903|   "price": 7300000,
3904|   "socket": "AM5",
3905|   "power": 170,
3906|   "desc": "4 nhân / 8 luồng, 170W TDP, Socket AM5"
3907| },
3908| {
3909|   "id": "cpu_amd_ryzen_5_8000_8600g_435",
3910|   "name": "AMD Ryzen 5 80008600G",
3911|   "type": "CPU",
3912|   "price": 7300000,
3913|   "socket": "AM5",
3914|   "power": 85,
3915|   "desc": "6 nhân / 12 luồng, 85W TDP, Socket AM5"
3916| },
3917| {
3918|   "id": "cpu_amd_ryzen_5_8000_9600_436",
3919|   "name": "AMD Ryzen 5 80009600",
3920|   "type": "CPU",
3921|   "price": 7780000,
3922|   "socket": "AM5",
3923|   "power": 120,
3924|   "desc": "8 nhân / 16 luồng, 120W TDP, Socket AM5"
3925| },
3926| {
3927|   "id": "cpu_amd_ryzen_5_8000_9600x_437",
3928|   "name": "AMD Ryzen 5 80009600X",
3929|   "type": "CPU",
3930|   "price": 7490000,
3931|   "socket": "AM5",
3932|   "power": 65,
3933|   "desc": "16 nhân / 32 luồng, 65W TDP, Socket AM5"
3934| },
3935| {
3936|   "id": "cpu_amd_ryzen_5_9000_5500_438",
3937|   "name": "AMD Ryzen 5 90005500",
3938|   "type": "CPU",
3939|   "price": 5970000,
3940|   "socket": "AM5",
3941|   "power": 120,
3942|   "desc": "6 nhân / 12 luồng, 120W TDP, Socket AM5"
3943| },
3944| {
3945|   "id": "cpu_amd_ryzen_5_9000_5600_439",
3946|   "name": "AMD Ryzen 5 90005600",
3947|   "type": "CPU",
3948|   "price": 6330000,
3949|   "socket": "AM5",
3950|   "power": 65,
3951|   "desc": "6 nhân / 12 luồng, 65W TDP, Socket AM5"
3952| },
3953| {
3954|   "id": "cpu_amd_ryzen_5_9000_5600x_440",
3955|   "name": "AMD Ryzen 5 90005600X",
3956|   "type": "CPU",
3957|   "price": 2660000,
3958|   "socket": "AM5",
3959|   "power": 65,
3960|   "desc": "8 nhân / 16 luồng, 65W TDP, Socket AM5"
3961| },
3962| {
3963|   "id": "cpu_amd_ryzen_5_9000_7500f_441",
3964|   "name": "AMD Ryzen 5 90007500F",
3965|   "type": "CPU",
3966|   "price": 2990000,
3967|   "socket": "AM5",
3968|   "power": 120,
3969|   "desc": "6 nhân / 12 luồng, 120W TDP, Socket AM5"
3970| },
3971| {
3972|   "id": "cpu_amd_ryzen_5_9000_7600_442",
3973|   "name": "AMD Ryzen 5 90007600",
3974|   "type": "CPU",
3975|   "price": 6850000,
3976|   "socket": "AM5",
3977|   "power": 95,
3978|   "desc": "12 nhân / 24 luồng, 95W TDP, Socket AM5"
3979| },
3980| {
3981|   "id": "cpu_amd_ryzen_5_9000_7600x_443",
3982|   "name": "AMD Ryzen 5 90007600X",
3983|   "type": "CPU",
3984|   "price": 5980000,
3985|   "socket": "AM5",
3986|   "power": 85,
3987|   "desc": "4 nhân / 8 luồng, 85W TDP, Socket AM5"
3988| },
3989| {
3990|   "id": "cpu_amd_ryzen_5_9000_8400f_444",
3991|   "name": "AMD Ryzen 5 90008400F",
3992|   "type": "CPU",
3993|   "price": 5710000,
3994|   "socket": "AM5",
3995|   "power": 105,
3996|   "desc": "12 nhân / 24 luồng, 105W TDP, Socket AM5"
3997| },
3998| {
3999|   "id": "cpu_amd_ryzen_5_9000_8500g_445",
4000|   "name": "AMD Ryzen 5 90008500G",
4001|   "type": "CPU",
4002|   "price": 6990000,
4003|   "socket": "AM5",
4004|   "power": 65,
4005|   "desc": "8 nhân / 16 luồng, 65W TDP, Socket AM5"
4006| },
4007| {
4008|   "id": "cpu_amd_ryzen_5_9000_8600g_446",
4009|   "name": "AMD Ryzen 5 90008600G",
4010|   "type": "CPU",
4011|   "price": 7720000,
4012|   "socket": "AM5",
4013|   "power": 105,
```

```
4 0 1 4 | "desc": "4 nhân / 8 luồng, 105W TDP, Socket AM5"
4015 | },
4016 | {
4017 |   "id": "cpu_amd_ryzen_5_9000_9600_447",
4018 |   "name": "AMD Ryzen 5 90009600",
4019 |   "type": "CPU",
4020 |   "price": 6650000,
4021 |   "socket": "AM5",
4022 |   "power": 170,
4 0 2 3 | "desc": "6 nhân / 12 luồng, 170W TDP, Socket AM5"
4024 | },
4025 | {
4026 |   "id": "cpu_amd_ryzen_5_9000_9600x_448",
4027 |   "name": "AMD Ryzen 5 90009600X",
4028 |   "type": "CPU",
4029 |   "price": 6560000,
4030 |   "socket": "AM5",
4031 |   "power": 120,
4 0 3 2 | "desc": "12 nhân / 24 luồng, 120W TDP, Socket AM5"
4033 | },
4034 | {
4035 |   "id": "cpu_amd_ryzen_7_5000_5700x_449",
4036 |   "name": "AMD Ryzen 7 50005700X",
4037 |   "type": "CPU",
4038 |   "price": 13030000,
4039 |   "socket": "AM4",
4040 |   "power": 105,
4 0 4 1 | "desc": "8 nhân / 16 luồng, 105W TDP, Socket AM4"
4042 | },
4043 | {
4044 |   "id": "cpu_amd_ryzen_7_5000_5700x3d_450",
4045 |   "name": "AMD Ryzen 7 50005700X3D",
4046 |   "type": "CPU",
4047 |   "price": 15300000,
4048 |   "socket": "AM4",
4049 |   "power": 65,
4 0 5 0 | "desc": "4 nhân / 8 luồng, 65W TDP, Socket AM4, 3D V-Cache"
4051 | },
4052 | {
4053 |   "id": "cpu_amd_ryzen_7_5000_5800x_451",
4054 |   "name": "AMD Ryzen 7 50005800X",
4055 |   "type": "CPU",
4056 |   "price": 10250000,
4057 |   "socket": "AM4",
4058 |   "power": 120,
4 0 5 9 | "desc": "16 nhân / 32 luồng, 120W TDP, Socket AM4"
4060 | },
4061 | {
4062 |   "id": "cpu_amd_ryzen_7_5000_5800x3d_452",
4063 |   "name": "AMD Ryzen 7 50005800X3D",
4064 |   "type": "CPU",
4065 |   "price": 17004000,
4066 |   "socket": "AM4",
4067 |   "power": 85,
4 0 6 8 | "desc": "8 nhân / 16 luồng, 85W TDP, Socket AM4, 3D V-Cache"
4069 | },
4070 | {
4071 |   "id": "cpu_amd_ryzen_7_5000_7700_453",
4072 |   "name": "AMD Ryzen 7 50007700",
4073 |   "type": "CPU",
4074 |   "price": 13580000,
4075 |   "socket": "AM4",
4076 |   "power": 85,
4 0 7 7 | "desc": "12 nhân / 24 luồng, 85W TDP, Socket AM4"
4078 | },
4079 | {
4080 |   "id": "cpu_amd_ryzen_7_5000_7700x_454",
4081 |   "name": "AMD Ryzen 7 50007700X",
4082 |   "type": "CPU",
4083 |   "price": 8720000,
4084 |   "socket": "AM4",
4085 |   "power": 85,
4 0 8 6 | "desc": "4 nhân / 8 luồng, 85W TDP, Socket AM4"
4087 | },
4088 | {
4089 |   "id": "cpu_amd_ryzen_7_5000_7800x3d_455",
4090 |   "name": "AMD Ryzen 7 50007800X3D",
4091 |   "type": "CPU",
4092 |   "price": 11448000,
4093 |   "socket": "AM4",
4094 |   "power": 105,
4 0 9 5 | "desc": "8 nhân / 16 luồng, 105W TDP, Socket AM4, 3D V-Cache"
4096 | },
4097 | {
4098 |   "id": "cpu_amd_ryzen_7_5000_8700g_456",
4099 |   "name": "AMD Ryzen 7 50008700G",
4100 |   "type": "CPU",
4101 |   "price": 8250000,
4102 |   "socket": "AM4",
4103 |   "power": 95,
4 1 0 4 | "desc": "6 nhân / 12 luồng, 95W TDP, Socket AM4"
4105 | },
4106 | {
4107 |   "id": "cpu_amd_ryzen_7_5000_9700x_457",
4108 |   "name": "AMD Ryzen 7 50009700X",
4109 |   "type": "CPU",
4110 |   "price": 9130000,
4111 |   "socket": "AM4",
4112 |   "power": 105,
4 1 1 3 | "desc": "16 nhân / 32 luồng, 105W TDP, Socket AM4"
4114 | },
4115 | {
4116 |   "id": "cpu_amd_ryzen_7_5000_9800x3d_458",
4117 |   "name": "AMD Ryzen 7 50009800X3D",
4118 |   "type": "CPU",
4119 |   "price": 9300000,
4120 |   "socket": "AM4",
4121 |   "power": 85,
4 1 2 2 | "desc": "6 nhân / 12 luồng, 85W TDP, Socket AM4, 3D V-Cache"
4123 | },
4124 | {
4125 |   "id": "cpu_amd_ryzen_7_7000_5700x_459",
4126 |   "name": "AMD Ryzen 7 70005700X",
4127 |   "type": "CPU",
4128 |   "price": 12370000,
4129 |   "socket": "AM5",
4130 |   "power": 95,
4 1 3 1 | "desc": "8 nhân / 16 luồng, 95W TDP, Socket AM5"
4132 | },
4133 | {
4134 |   "id": "cpu_amd_ryzen_7_7000_5700x3d_460",
4135 |   "name": "AMD Ryzen 7 70005700X3D",
4136 |   "type": "CPU",
4137 |   "price": 7392000,
```

```
4138 | "socket": "AM5",
4139 | "power": 120,
4140 | | " d e s c " : " 4 n h â n / 8 l u ó n g , 1 2 0 W T D P , S o c k e t A M 5 , 3 D V - C a c h e "
4141 | },
4142 | {
4143 | | "id": "cpu_amd_ryzen_7_7000_5800x_461",
4144 | | "name": "AMD Ryzen 7 70005800X",
4145 | | "type": "CPU",
4146 | | "price": 7440000,
4147 | | "socket": "AM5",
4148 | | "power": 120,
4149 | | " d e s c " : " 1 6 n h â n / 3 2 l u ó n g , 1 2 0 W T D P , S o c k e t A M 5 "
4150 | },
4151 | {
4152 | | "id": "cpu_amd_ryzen_7_7000_5800x3d_462",
4153 | | "name": "AMD Ryzen 7 70005800X3D",
4154 | | "type": "CPU",
4155 | | "price": 14688000,
4156 | | "socket": "AM5",
4157 | | "power": 170,
4158 | | " d e s c " : " 8 n h â n / 1 6 l u ó n g , 1 7 0 W T D P , S o c k e t A M 5 , 3 D V - C a c h e "
4159 | },
4160 | {
4161 | | "id": "cpu_amd_ryzen_7_7000_7700_463",
4162 | | "name": "AMD Ryzen 7 70007700",
4163 | | "type": "CPU",
4164 | | "price": 5890000,
4165 | | "socket": "AM5",
4166 | | "power": 105,
4167 | | " d e s c " : " 8 n h â n / 1 6 l u ó n g , 1 0 5 W T D P , S o c k e t A M 5 "
4168 | },
4169 | {
4170 | | "id": "cpu_amd_ryzen_7_7000_7700x_464",
4171 | | "name": "AMD Ryzen 7 70007700X",
4172 | | "type": "CPU",
4173 | | "price": 8280000,
4174 | | "socket": "AM5",
4175 | | "power": 120,
4176 | | " d e s c " : " 1 6 n h â n / 3 2 l u ó n g , 1 2 0 W T D P , S o c k e t A M 5 "
4177 | },
4178 | {
4179 | | "id": "cpu_amd_ryzen_7_7000_7800x3d_465",
4180 | | "name": "AMD Ryzen 7 70007800X3D",
4181 | | "type": "CPU",
4182 | | "price": 16152000,
4183 | | "socket": "AM5",
4184 | | "power": 65,
4185 | | " d e s c " : " 6 n h â n / 1 2 l u ó n g , 6 5 W T D P , S o c k e t A M 5 , 3 D V - C a c h e "
4186 | },
4187 | {
4188 | | "id": "cpu_amd_ryzen_7_7000_8700g_466",
4189 | | "name": "AMD Ryzen 7 70008700G",
4190 | | "type": "CPU",
4191 | | "price": 9810000,
4192 | | "socket": "AM5",
4193 | | "power": 120,
4194 | | " d e s c " : " 1 6 n h â n / 3 2 l u ó n g , 1 2 0 W T D P , S o c k e t A M 5 "
4195 | },
4196 | {
4197 | | "id": "cpu_amd_ryzen_7_7000_9700x_467",
4198 | | "name": "AMD Ryzen 7 70009700X",
4199 | | "type": "CPU",
4200 | | "price": 13320000,
4201 | | "socket": "AM5",
4202 | | "power": 105,
4203 | | " d e s c " : " 1 6 n h â n / 3 2 l u ó n g , 1 0 5 W T D P , S o c k e t A M 5 "
4204 | },
4205 | {
4206 | | "id": "cpu_amd_ryzen_7_7000_9800x3d_468",
4207 | | "name": "AMD Ryzen 7 70009800X3D",
4208 | | "type": "CPU",
4209 | | "price": 15156000,
4210 | | "socket": "AM5",
4211 | | "power": 95,
4212 | | " d e s c " : " 8 n h â n / 1 6 l u ó n g , 9 5 W T D P , S o c k e t A M 5 , 3 D V - C a c h e "
4213 | },
4214 | {
4215 | | "id": "cpu_amd_ryzen_7_8000_5700x_469",
4216 | | "name": "AMD Ryzen 7 80005700X",
4217 | | "type": "CPU",
4218 | | "price": 11830000,
4219 | | "socket": "AM5",
4220 | | "power": 95,
4221 | | " d e s c " : " 1 2 n h â n / 2 4 l u ó n g , 9 5 W T D P , S o c k e t A M 5 "
4222 | },
4223 | {
4224 | | "id": "cpu_amd_ryzen_7_8000_5700x3d_470",
4225 | | "name": "AMD Ryzen 7 80005700X3D",
4226 | | "type": "CPU",
4227 | | "price": 6876000,
4228 | | "socket": "AM5",
4229 | | "power": 65,
4230 | | " d e s c " : " 6 n h â n / 1 2 l u ó n g , 6 5 W T D P , S o c k e t A M 5 , 3 D V - C a c h e "
4231 | },
4232 | {
4233 | | "id": "cpu_amd_ryzen_7_8000_5800x_471",
4234 | | "name": "AMD Ryzen 7 80005800X",
4235 | | "type": "CPU",
4236 | | "price": 10260000,
4237 | | "socket": "AM5",
4238 | | "power": 105,
4239 | | " d e s c " : " 8 n h â n / 1 6 l u ó n g , 1 0 5 W T D P , S o c k e t A M 5 "
4240 | },
4241 | {
4242 | | "id": "cpu_amd_ryzen_7_8000_5800x3d_472",
4243 | | "name": "AMD Ryzen 7 80005800X3D",
4244 | | "type": "CPU",
4245 | | "price": 9900000,
4246 | | "socket": "AM5",
4247 | | "power": 85,
4248 | | " d e s c " : " 6 n h â n / 1 2 l u ó n g , 8 5 W T D P , S o c k e t A M 5 , 3 D V - C a c h e "
4249 | },
4250 | {
4251 | | "id": "cpu_amd_ryzen_7_8000_7700_473",
4252 | | "name": "AMD Ryzen 7 80007700",
4253 | | "type": "CPU",
4254 | | "price": 12570000,
4255 | | "socket": "AM5",
4256 | | "power": 170,
4257 | | " d e s c " : " 8 n h â n / 1 6 l u ó n g , 1 7 0 W T D P , S o c k e t A M 5 "
4258 | },
4259 | {
4260 | | "id": "cpu_amd_ryzen_7_8000_7700x_474",
4261 | | "name": "AMD Ryzen 7 80007700X",
```

```
4262| "type": "CPU",
4263| "price": 9060000,
4264| "socket": "AM5",
4265| "power": 95,
4266| "desc": "12 nhân / 24 luồng, 95W TDP, Socket AM5"
4267| },
4268| {
4269| "id": "cpu_amd_ryzen_7_8000_7800x3d_475",
4270| "name": "AMD Ryzen 7 80007800X3D",
4271| "type": "CPU",
4272| "price": 12384000,
4273| "socket": "AM5",
4274| "power": 105,
4275| "desc": "12 nhân / 24 luồng, 105W TDP, Socket AM5, 3D V-Cache"
4276| },
4277| {
4278| "id": "cpu_amd_ryzen_7_8000_8700g_476",
4279| "name": "AMD Ryzen 7 80008700G",
4280| "type": "CPU",
4281| "price": 10980000,
4282| "socket": "AM5",
4283| "power": 95,
4284| "desc": "6 nhân / 12 luồng, 95W TDP, Socket AM5"
4285| },
4286| {
4287| "id": "cpu_amd_ryzen_7_8000_9700x_477",
4288| "name": "AMD Ryzen 7 80009700X",
4289| "type": "CPU",
4290| "price": 11560000,
4291| "socket": "AM5",
4292| "power": 170,
4293| "desc": "6 nhân / 12 luồng, 170W TDP, Socket AM5"
4294| },
4295| {
4296| "id": "cpu_amd_ryzen_7_8000_9800x3d_478",
4297| "name": "AMD Ryzen 7 80009800X3D",
4298| "type": "CPU",
4299| "price": 12240000,
4300| "socket": "AM5",
4301| "power": 65,
4302| "desc": "8 nhân / 16 luồng, 65W TDP, Socket AM5, 3D V-Cache"
4303| },
4304| {
4305| "id": "cpu_amd_ryzen_7_9000_5700x_479",
4306| "name": "AMD Ryzen 7 90005700X",
4307| "type": "CPU",
4308| "price": 9560000,
4309| "socket": "AM5",
4310| "power": 85,
4311| "desc": "6 nhân / 12 luồng, 85W TDP, Socket AM5"
4312| },
4313| {
4314| "id": "cpu_amd_ryzen_7_9000_5700x3d_480",
4315| "name": "AMD Ryzen 7 90005700X3D",
4316| "type": "CPU",
4317| "price": 8076000,
4318| "socket": "AM5",
4319| "power": 65,
4320| "desc": "6 nhân / 12 luồng, 65W TDP, Socket AM5, 3D V-Cache"
4321| },
4322| {
4323| "id": "cpu_amd_ryzen_7_9000_5800x_481",
4324| "name": "AMD Ryzen 7 90005800X",
4325| "type": "CPU",
4326| "price": 5270000,
4327| "socket": "AM5",
4328| "power": 95,
4329| "desc": "12 nhân / 24 luồng, 95W TDP, Socket AM5"
4330| },
4331| {
4332| "id": "cpu_amd_ryzen_7_9000_5800x3d_482",
4333| "name": "AMD Ryzen 7 90005800X3D",
4334| "type": "CPU",
4335| "price": 6132000,
4336| "socket": "AM5",
4337| "power": 120,
4338| "desc": "6 nhân / 12 luồng, 120W TDP, Socket AM5, 3D V-Cache"
4339| },
4340| {
4341| "id": "cpu_amd_ryzen_7_9000_7700_483",
4342| "name": "AMD Ryzen 7 90007700",
4343| "type": "CPU",
4344| "price": 12900000,
4345| "socket": "AM5",
4346| "power": 105,
4347| "desc": "8 nhân / 16 luồng, 105W TDP, Socket AM5"
4348| },
4349| {
4350| "id": "cpu_amd_ryzen_7_9000_7700x_484",
4351| "name": "AMD Ryzen 7 90007700X",
4352| "type": "CPU",
4353| "price": 7650000,
4354| "socket": "AM5",
4355| "power": 170,
4356| "desc": "6 nhân / 12 luồng, 170W TDP, Socket AM5"
4357| },
4358| {
4359| "id": "cpu_amd_ryzen_7_9000_7800x3d_485",
4360| "name": "AMD Ryzen 7 90007800X3D",
4361| "type": "CPU",
4362| "price": 17772000,
4363| "socket": "AM5",
4364| "power": 95,
4365| "desc": "6 nhân / 12 luồng, 95W TDP, Socket AM5, 3D V-Cache"
4366| },
4367| {
4368| "id": "cpu_amd_ryzen_7_9000_8700g_486",
4369| "name": "AMD Ryzen 7 90008700G",
4370| "type": "CPU",
4371| "price": 14210000,
4372| "socket": "AM5",
4373| "power": 85,
4374| "desc": "12 nhân / 24 luồng, 85W TDP, Socket AM5"
4375| },
4376| {
4377| "id": "cpu_amd_ryzen_7_9000_9700x_487",
4378| "name": "AMD Ryzen 7 90009700X",
4379| "type": "CPU",
4380| "price": 8330000,
4381| "socket": "AM5",
4382| "power": 120,
4383| "desc": "4 nhân / 8 luồng, 120W TDP, Socket AM5"
4384| },
4385| {
```

```
4386 | "id": "cpu_amd_ryzen_7_9000_9800x3d_488",
4387 | "name": "AMD Ryzen 7 90009800X3D",
4388 | "type": "CPU",
4389 | "price": 17208000,
4390 | "socket": "AM5",
4391 | "power": 105,
4392 | "desc": "12 nhân / 24 luồng, 105W TDP, Socket AM5, 3D V-Cache"
4393 | },
4394 | {
4395 | "id": "cpu_amd_ryzen_9_5000_5900x_489",
4396 | "name": "AMD Ryzen 9 50005900X",
4397 | "type": "CPU",
4398 | "price": 13340000,
4399 | "socket": "AM4",
4400 | "power": 95,
4401 | "desc": "4 nhân / 8 luồng, 95W TDP, Socket AM4"
4402 | },
4403 | {
4404 | "id": "cpu_amd_ryzen_9_5000_5950x_490",
4405 | "name": "AMD Ryzen 9 50005950X",
4406 | "type": "CPU",
4407 | "price": 14100000,
4408 | "socket": "AM4",
4409 | "power": 170,
4410 | "desc": "12 nhân / 24 luồng, 170W TDP, Socket AM4"
4411 | },
4412 | {
4413 | "id": "cpu_amd_ryzen_9_5000_7900_491",
4414 | "name": "AMD Ryzen 9 50007900",
4415 | "type": "CPU",
4416 | "price": 12620000,
4417 | "socket": "AM4",
4418 | "power": 105,
4419 | "desc": "16 nhân / 32 luồng, 105W TDP, Socket AM4"
4420 | },
4421 | {
4422 | "id": "cpu_amd_ryzen_9_5000_7900x_492",
4423 | "name": "AMD Ryzen 9 50007900X",
4424 | "type": "CPU",
4425 | "price": 11640000,
4426 | "socket": "AM4",
4427 | "power": 120,
4428 | "desc": "16 nhân / 32 luồng, 120W TDP, Socket AM4"
4429 | },
4430 | {
4431 | "id": "cpu_amd_ryzen_9_5000_7950x_493",
4432 | "name": "AMD Ryzen 9 50007950X",
4433 | "type": "CPU",
4434 | "price": 11770000,
4435 | "socket": "AM4",
4436 | "power": 95,
4437 | "desc": "12 nhân / 24 luồng, 95W TDP, Socket AM4"
4438 | },
4439 | {
4440 | "id": "cpu_amd_ryzen_9_5000_7950x3d_494",
4441 | "name": "AMD Ryzen 9 50007950X3D",
4442 | "type": "CPU",
4443 | "price": 11472000,
4444 | "socket": "AM4",
4445 | "power": 120,
4446 | "desc": "16 nhân / 32 luồng, 120W TDP, Socket AM4, 3D V-Cache"
4447 | },
4448 | {
4449 | "id": "cpu_amd_ryzen_9_5000_9900x_495",
4450 | "name": "AMD Ryzen 9 50009900X",
4451 | "type": "CPU",
4452 | "price": 18980000,
4453 | "socket": "AM4",
4454 | "power": 65,
4455 | "desc": "12 nhân / 24 luồng, 65W TDP, Socket AM4"
4456 | },
4457 | {
4458 | "id": "cpu_amd_ryzen_9_5000_9950x_496",
4459 | "name": "AMD Ryzen 9 50009950X",
4460 | "type": "CPU",
4461 | "price": 18170000,
4462 | "socket": "AM4",
4463 | "power": 120,
4464 | "desc": "16 nhân / 32 luồng, 120W TDP, Socket AM4"
4465 | },
4466 | {
4467 | "id": "cpu_amd_ryzen_9_7000_5900x_497",
4468 | "name": "AMD Ryzen 9 70005900X",
4469 | "type": "CPU",
4470 | "price": 8490000,
4471 | "socket": "AM5",
4472 | "power": 105,
4473 | "desc": "16 nhân / 32 luồng, 105W TDP, Socket AM5"
4474 | },
4475 | {
4476 | "id": "cpu_amd_ryzen_9_7000_5950x_498",
4477 | "name": "AMD Ryzen 9 70005950X",
4478 | "type": "CPU",
4479 | "price": 15370000,
4480 | "socket": "AM5",
4481 | "power": 120,
4482 | "desc": "6 nhân / 12 luồng, 120W TDP, Socket AM5"
4483 | },
4484 | {
4485 | "id": "cpu_amd_ryzen_9_7000_7900_499",
4486 | "name": "AMD Ryzen 9 70007900",
4487 | "type": "CPU",
4488 | "price": 15380000,
4489 | "socket": "AM5",
4490 | "power": 105,
4491 | "desc": "4 nhân / 8 luồng, 105W TDP, Socket AM5"
4492 | },
4493 | {
4494 | "id": "cpu_amd_ryzen_9_7000_7900x_500",
4495 | "name": "AMD Ryzen 9 70007900X",
4496 | "type": "CPU",
4497 | "price": 20840000,
4498 | "socket": "AM5",
4499 | "power": 120,
4500 | "desc": "4 nhân / 8 luồng, 120W TDP, Socket AM5"
4501 | },
4502 | {
4503 | "id": "cpu_amd_ryzen_9_7000_7950x_501",
4504 | "name": "AMD Ryzen 9 70007950X",
4505 | "type": "CPU",
4506 | "price": 21680000,
4507 | "socket": "AM5",
4508 | "power": 105,
4509 | "desc": "12 nhân / 24 luồng, 105W TDP, Socket AM5"
```

```
4510 | },
4511 | {
4512 |   "id": "cpu_amd_ryzen_9_7000_7950x3d_502",
4513 |   "name": "AMD Ryzen 9 70007950X3D",
4514 |   "type": "CPU",
4515 |   "price": 12252000,
4516 |   "socket": "AM5",
4517 |   "power": 65,
4518 |   "desc": "8 nhân / 16 luồng, 65W TDP, Socket AM5, 3D V-Cache"
4519 | },
4520 | {
4521 |   "id": "cpu_amd_ryzen_9_7000_9900x_503",
4522 |   "name": "AMD Ryzen 9 70009900X",
4523 |   "type": "CPU",
4524 |   "price": 15470000,
4525 |   "socket": "AM5",
4526 |   "power": 95,
4527 |   "desc": "8 nhân / 16 luồng, 95W TDP, Socket AM5"
4528 | },
4529 | {
4530 |   "id": "cpu_amd_ryzen_9_7000_9950x_504",
4531 |   "name": "AMD Ryzen 9 70009950X",
4532 |   "type": "CPU",
4533 |   "price": 11270000,
4534 |   "socket": "AM5",
4535 |   "power": 85,
4536 |   "desc": "6 nhân / 12 luồng, 85W TDP, Socket AM5"
4537 | },
4538 | {
4539 |   "id": "cpu_amd_ryzen_9_9000_5900x_505",
4540 |   "name": "AMD Ryzen 9 90005900X",
4541 |   "type": "CPU",
4542 |   "price": 17720000,
4543 |   "socket": "AM5",
4544 |   "power": 170,
4545 |   "desc": "16 nhân / 32 luồng, 170W TDP, Socket AM5"
4546 | },
4547 | {
4548 |   "id": "cpu_amd_ryzen_9_9000_5950x_506",
4549 |   "name": "AMD Ryzen 9 90005950X",
4550 |   "type": "CPU",
4551 |   "price": 20980000,
4552 |   "socket": "AM5",
4553 |   "power": 105,
4554 |   "desc": "8 nhân / 16 luồng, 105W TDP, Socket AM5"
4555 | },
4556 | {
4557 |   "id": "cpu_amd_ryzen_9_9000_7900_507",
4558 |   "name": "AMD Ryzen 9 90007900",
4559 |   "type": "CPU",
4560 |   "price": 15920000,
4561 |   "socket": "AM5",
4562 |   "power": 85,
4563 |   "desc": "4 nhân / 8 luồng, 85W TDP, Socket AM5"
4564 | },
4565 | {
4566 |   "id": "cpu_amd_ryzen_9_9000_7900x_508",
4567 |   "name": "AMD Ryzen 9 90007900X",
4568 |   "type": "CPU",
4569 |   "price": 24900000,
4570 |   "socket": "AM5",
4571 |   "power": 85,
4572 |   "desc": "12 nhân / 24 luồng, 85W TDP, Socket AM5"
4573 | },
4574 | {
4575 |   "id": "cpu_amd_ryzen_9_9000_7950x_509",
4576 |   "name": "AMD Ryzen 9 90007950X",
4577 |   "type": "CPU",
4578 |   "price": 23450000,
4579 |   "socket": "AM5",
4580 |   "power": 65,
4581 |   "desc": "8 nhân / 16 luồng, 65W TDP, Socket AM5"
4582 | },
4583 | {
4584 |   "id": "cpu_amd_ryzen_9_9000_7950x3d_510",
4585 |   "name": "AMD Ryzen 9 90007950X3D",
4586 |   "type": "CPU",
4587 |   "price": 21580000,
4588 |   "socket": "AM5",
4589 |   "power": 95,
4590 |   "desc": "8 nhân / 16 luồng, 95W TDP, Socket AM5, 3D V-Cache"
4591 | },
4592 | {
4593 |   "id": "cpu_amd_ryzen_9_9000_9900x_511",
4594 |   "name": "AMD Ryzen 9 90009900X",
4595 |   "type": "CPU",
4596 |   "price": 23670000,
4597 |   "socket": "AM5",
4598 |   "power": 65,
4599 |   "desc": "16 nhân / 32 luồng, 65W TDP, Socket AM5"
4600 | },
4601 | {
4602 |   "id": "cpu_amd_ryzen_9_9000_9950x_512",
4603 |   "name": "AMD Ryzen 9 90009950X",
4604 |   "type": "CPU",
4605 |   "price": 16050000,
4606 |   "socket": "AM5",
4607 |   "power": 95,
4608 |   "desc": "4 nhân / 8 luồng, 95W TDP, Socket AM5"
4609 | },
4610 | {
4611 |   "id": "cpu_amd_ryzen_threadripper_7000_7960x_513",
4612 |   "name": "AMD Ryzen Threadripper 70007960X",
4613 |   "type": "CPU",
4614 |   "price": 75180000,
4615 |   "socket": "sTR5",
4616 |   "power": 350,
4617 |   "desc": "64 nhân / 128 luồng, 350W TDP, Socket sTR5"
4618 | },
4619 | {
4620 |   "id": "cpu_amd_ryzen_threadripper_7000_7970x_514",
4621 |   "name": "AMD Ryzen Threadripper 70007970X",
4622 |   "type": "CPU",
4623 |   "price": 42100000,
4624 |   "socket": "sTR5",
4625 |   "power": 250,
4626 |   "desc": "16 nhân / 32 luồng, 250W TDP, Socket sTR5"
4627 | },
4628 | {
4629 |   "id": "cpu_amd_ryzen_threadripper_7000_7980x_515",
4630 |   "name": "AMD Ryzen Threadripper 70007980X",
4631 |   "type": "CPU",
4632 |   "price": 31670000,
4633 |   "socket": "sTR5",
```

```
4634| "power": 350,
4635|   "desc": "16 nhân / 32 luồng, 350W TDP, Socket sTR5"
4636| },
4637| {
4638|   "id": "cpu_amd_ryzen_threadripper_7000_7990x_516",
4639|   "name": "AMD Ryzen Threadripper 70007990X",
4640|   "type": "CPU",
4641|   "price": 30450000,
4642|   "socket": "sTR5",
4643|   "power": 250,
4644|   "desc": "128 nhân / 256 luồng, 250W TDP, Socket sTR5"
4645| },
4646| {
4647|   "id": "cpu_amd_ryzen_threadripper_7000_7965wx_517",
4648|   "name": "AMD Ryzen Threadripper 70007965WX",
4649|   "type": "CPU",
4650|   "price": 18730000,
4651|   "socket": "sTR5",
4652|   "power": 280,
4653|   "desc": "32 nhân / 64 luồng, 280W TDP, Socket sTR5"
4654| },
4655| {
4656|   "id": "cpu_amd_ryzen_threadripper_7000_7975wx_518",
4657|   "name": "AMD Ryzen Threadripper 70007975WX",
4658|   "type": "CPU",
4659|   "price": 47360000,
4660|   "socket": "sTR5",
4661|   "power": 350,
4662|   "desc": "96 nhân / 192 luồng, 350W TDP, Socket sTR5"
4663| },
4664| {
4665|   "id": "cpu_amd_ryzen_threadripper_7000_7985wx_519",
4666|   "name": "AMD Ryzen Threadripper 70007985WX",
4667|   "type": "CPU",
4668|   "price": 44920000,
4669|   "socket": "sTR5",
4670|   "power": 350,
4671|   "desc": "128 nhân / 256 luồng, 350W TDP, Socket sTR5"
4672| },
4673| {
4674|   "id": "cpu_amd_ryzen_threadripper_7000_7995wx_520",
4675|   "name": "AMD Ryzen Threadripper 70007995WX",
4676|   "type": "CPU",
4677|   "price": 10770000,
4678|   "socket": "sTR5",
4679|   "power": 300,
4680|   "desc": "16 nhân / 32 luồng, 300W TDP, Socket sTR5"
4681| },
4682| {
4683|   "id": "cpu_amd_epyc_4004_4124p_521",
4684|   "name": "AMD EPYC 40044124P",
4685|   "type": "CPU",
4686|   "price": 68080000,
4687|   "socket": "SP5",
4688|   "power": 240,
4689|   "desc": "24 nhân / 48 luồng, 240W TDP, Socket SP5"
4690| },
4691| {
4692|   "id": "cpu_amd_epyc_4004_4344p_522",
4693|   "name": "AMD EPYC 40044344P",
4694|   "type": "CPU",
4695|   "price": 20920000,
4696|   "socket": "SP5",
4697|   "power": 240,
4698|   "desc": "16 nhân / 32 luồng, 240W TDP, Socket SP5"
4699| },
4700| {
4701|   "id": "cpu_amd_epyc_4004_4484px_523",
4702|   "name": "AMD EPYC 40044484PX",
4703|   "type": "CPU",
4704|   "price": 42470000,
4705|   "socket": "SP5",
4706|   "power": 240,
4707|   "desc": "64 nhân / 128 luồng, 240W TDP, Socket SP5"
4708| },
4709| {
4710|   "id": "cpu_amd_epyc_4004_9124_524",
4711|   "name": "AMD EPYC 40049124",
4712|   "type": "CPU",
4713|   "price": 65160000,
4714|   "socket": "SP5",
4715|   "power": 225,
4716|   "desc": "32 nhân / 64 luồng, 225W TDP, Socket SP5"
4717| },
4718| {
4719|   "id": "cpu_amd_epyc_4004_9254_525",
4720|   "name": "AMD EPYC 40049254",
4721|   "type": "CPU",
4722|   "price": 110230000,
4723|   "socket": "SP5",
4724|   "power": 200,
4725|   "desc": "24 nhân / 48 luồng, 200W TDP, Socket SP5"
4726| },
4727| {
4728|   "id": "cpu_amd_epyc_4004_9374f_526",
4729|   "name": "AMD EPYC 40049374F",
4730|   "type": "CPU",
4731|   "price": 45620000,
4732|   "socket": "SP5",
4733|   "power": 280,
4734|   "desc": "32 nhân / 64 luồng, 280W TDP, Socket SP5"
4735| },
4736| {
4737|   "id": "cpu_amd_epyc_4004_9554_527",
4738|   "name": "AMD EPYC 40049554",
4739|   "type": "CPU",
4740|   "price": 104730000,
4741|   "socket": "SP5",
4742|   "power": 225,
4743|   "desc": "96 nhân / 192 luồng, 225W TDP, Socket SP5"
4744| },
4745| {
4746|   "id": "cpu_amd_epyc_4004_9654_528",
4747|   "name": "AMD EPYC 40049654",
4748|   "type": "CPU",
4749|   "price": 39160000,
4750|   "socket": "SP5",
4751|   "power": 280,
4752|   "desc": "96 nhân / 192 luồng, 280W TDP, Socket SP5"
4753| },
4754| {
4755|   "id": "cpu_amd_epyc_4004_9684x_529",
4756|   "name": "AMD EPYC 40049684X",
4757|   "type": "CPU",
```



```
4758| "price": 95470000,  
4759| "socket": "SP5",  
4760| "power": 240,  
4761| " d e s c " : " 1 6   n h â n   /   3 2   l u ó n g ,   2 4 0 W   T D P ,   S o c k e t   S P 5   "  
4762| },  
4763| {  
4764| "id": "cpu_amd_epyc_8004_4124p_530",  
4765| "name": "AMD EPYC 80044124P",  
4766| "type": "CPU",  
4767| "price": 31080000,  
4768| "socket": "AM5",  
4769| "power": 240,  
4770| " d e s c " : " 2 4   n h â n   /   4 8   l u ó n g ,   2 4 0 W   T D P ,   S o c k e t   A M 5   "  
4771| },  
4772| {  
4773| "id": "cpu_amd_epyc_8004_4344p_531",  
4774| "name": "AMD EPYC 80044344P",  
4775| "type": "CPU",  
4776| "price": 33040000,  
4777| "socket": "AM5",  
4778| "power": 280,  
4779| " d e s c " : " 3 2   n h â n   /   6 4   l u ó n g ,   2 8 0 W   T D P ,   S o c k e t   A M 5   "  
4780| },  
4781| {  
4782| "id": "cpu_amd_epyc_8004_4484px_532",  
4783| "name": "AMD EPYC 80044484PX",  
4784| "type": "CPU",  
4785| "price": 90360000,  
4786| "socket": "AM5",  
4787| "power": 240,  
4788| " d e s c " : " 6 4   n h â n   /   1 2 8   l u ó n g ,   2 4 0 W   T D P ,   S o c k e t   A M 5   "  
4789| },  
4790| {  
4791| "id": "cpu_amd_epyc_8004_9124_533",  
4792| "name": "AMD EPYC 80049124",  
4793| "type": "CPU",  
4794| "price": 97300000,  
4795| "socket": "AM5",  
4796| "power": 280,  
4797| " d e s c " : " 3 2   n h â n   /   6 4   l u ó n g ,   2 8 0 W   T D P ,   S o c k e t   A M 5   "  
4798| },  
4799| {  
4800| "id": "cpu_amd_epyc_8004_9254_534",  
4801| "name": "AMD EPYC 80049254",  
4802| "type": "CPU",  
4803| "price": 75330000,  
4804| "socket": "AM5",  
4805| "power": 240,  
4806| " d e s c " : " 6 4   n h â n   /   1 2 8   l u ó n g ,   2 4 0 W   T D P ,   S o c k e t   A M 5   "  
4807| },  
4808| {  
4809| "id": "cpu_amd_epyc_8004_9374f_535",  
4810| "name": "AMD EPYC 80049374F",  
4811| "type": "CPU",  
4812| "price": 37550000,  
4813| "socket": "AM5",  
4814| "power": 200,  
4815| " d e s c " : " 3 2   n h â n   /   6 4   l u ó n g ,   2 0 0 W   T D P ,   S o c k e t   A M 5   "  
4816| },  
4817| {  
4818| "id": "cpu_amd_epyc_8004_9554_536",  
4819| "name": "AMD EPYC 80049554",  
4820| "type": "CPU",  
4821| "price": 16160000,  
4822| "socket": "AM5",  
4823| "power": 300,  
4824| " d e s c " : " 6 4   n h â n   /   1 2 8   l u ó n g ,   3 0 0 W   T D P ,   S o c k e t   A M 5   "  
4825| },  
4826| {  
4827| "id": "cpu_amd_epyc_8004_9654_537",  
4828| "name": "AMD EPYC 80049654",  
4829| "type": "CPU",  
4830| "price": 28590000,  
4831| "socket": "AM5",  
4832| "power": 300,  
4833| " d e s c " : " 1 2 8   n h â n   /   2 5 6   l u ó n g ,   3 0 0 W   T D P ,   S o c k e t   A M 5   "  
4834| },  
4835| {  
4836| "id": "cpu_amd_epyc_8004_9684x_538",  
4837| "name": "AMD EPYC 80049684X",  
4838| "type": "CPU",  
4839| "price": 26070000,  
4840| "socket": "AM5",  
4841| "power": 300,  
4842| " d e s c " : " 3 2   n h â n   /   6 4   l u ó n g ,   3 0 0 W   T D P ,   S o c k e t   A M 5   "  
4843| },  
4844| {  
4845| "id": "cpu_amd_epyc_9004_4124p_539",  
4846| "name": "AMD EPYC 90044124P",  
4847| "type": "CPU",  
4848| "price": 69510000,  
4849| "socket": "AM5",  
4850| "power": 280,  
4851| " d e s c " : " 1 6   n h â n   /   3 2   l u ó n g ,   2 8 0 W   T D P ,   S o c k e t   A M 5   "  
4852| },  
4853| {  
4854| "id": "cpu_amd_epyc_9004_4344p_540",  
4855| "name": "AMD EPYC 90044344P",  
4856| "type": "CPU",  
4857| "price": 82560000,  
4858| "socket": "AM5",  
4859| "power": 300,  
4860| " d e s c " : " 1 6   n h â n   /   3 2   l u ó n g ,   3 0 0 W   T D P ,   S o c k e t   A M 5   "  
4861| },  
4862| {  
4863| "id": "cpu_amd_epyc_9004_4484px_541",  
4864| "name": "AMD EPYC 90044484PX",  
4865| "type": "CPU",  
4866| "price": 72320000,  
4867| "socket": "AM5",  
4868| "power": 280,  
4869| " d e s c " : " 2 4   n h â n   /   4 8   l u ó n g ,   2 8 0 W   T D P ,   S o c k e t   A M 5   "  
4870| },  
4871| {  
4872| "id": "cpu_amd_epyc_9004_9124_542",  
4873| "name": "AMD EPYC 90049124",  
4874| "type": "CPU",  
4875| "price": 49980000,  
4876| "socket": "AM5",  
4877| "power": 280,  
4878| " d e s c " : " 2 4   n h â n   /   4 8   l u ó n g ,   2 8 0 W   T D P ,   S o c k e t   A M 5   "  
4879| },  
4880| {  
4881| "id": "cpu_amd_epyc_9004_9254_543",
```

```
4882 | "name": "AMD EPYC 90049254",
4883 | "type": "CPU",
4884 | "price": 49360000,
4885 | "socket": "AM5",
4886 | "power": 360,
4887 | "desc": "32 nhân / 64 luồng, 360W TDP, Socket AM5"
4888 | },
4889 | {
4890 | "id": "cpu_amd_epyc_9004_9374f_544",
4891 | "name": "AMD EPYC 90049374F",
4892 | "type": "CPU",
4893 | "price": 52800000,
4894 | "socket": "AM5",
4895 | "power": 300,
4896 | "desc": "64 nhân / 128 luồng, 300W TDP, Socket AM5"
4897 | },
4898 | {
4899 | "id": "cpu_amd_epyc_9004_9554_545",
4900 | "name": "AMD EPYC 90049554",
4901 | "type": "CPU",
4902 | "price": 65800000,
4903 | "socket": "AM5",
4904 | "power": 360,
4905 | "desc": "24 nhân / 48 luồng, 360W TDP, Socket AM5"
4906 | },
4907 | {
4908 | "id": "cpu_amd_epyc_9004_9654_546",
4909 | "name": "AMD EPYC 90049654",
4910 | "type": "CPU",
4911 | "price": 86720000,
4912 | "socket": "AM5",
4913 | "power": 280,
4914 | "desc": "24 nhân / 48 luồng, 280W TDP, Socket AM5"
4915 | },
4916 | {
4917 | "id": "cpu_amd_epyc_9004_9684x_547",
4918 | "name": "AMD EPYC 90049684X",
4919 | "type": "CPU",
4920 | "price": 32630000,
4921 | "socket": "AM5",
4922 | "power": 240,
4923 | "desc": "128 nhân / 256 luồng, 240W TDP, Socket AM5"
4924 | },
4925 | {
4926 | "id": "gpu_nvidia_3050_asus_1",
4927 | "name": "ASUS GeForce RTX 3050 8GB",
4928 | "type": "GPU",
4929 | "price": 3310000,
4930 | "power": 130,
4931 | "desc": "RTX 3050 8GB VRAM, 130W TDP"
4932 | },
4933 | {
4934 | "id": "gpu_nvidia_3050_msi_2",
4935 | "name": "MSI GeForce RTX 3050 8GB",
4936 | "type": "GPU",
4937 | "price": 7500000,
4938 | "power": 75,
4939 | "desc": "RTX 3050 8GB VRAM, 75W TDP"
4940 | },
4941 | {
4942 | "id": "gpu_nvidia_3050_gigabyte_3",
4943 | "name": "Gigabyte GeForce RTX 3050 8GB",
4944 | "type": "GPU",
4945 | "price": 4720000,
4946 | "power": 100,
4947 | "desc": "RTX 3050 8GB VRAM, 100W TDP"
4948 | },
4949 | {
4950 | "id": "gpu_nvidia_3050_colorful_4",
4951 | "name": "Colorful GeForce RTX 3050 8GB",
4952 | "type": "GPU",
4953 | "price": 4160000,
4954 | "power": 75,
4955 | "desc": "RTX 3050 8GB VRAM, 75W TDP"
4956 | },
4957 | {
4958 | "id": "gpu_nvidia_3050_zotac_5",
4959 | "name": "ZOTAC GeForce RTX 3050 8GB",
4960 | "type": "GPU",
4961 | "price": 4650000,
4962 | "power": 100,
4963 | "desc": "RTX 3050 8GB VRAM, 100W TDP"
4964 | },
4965 | {
4966 | "id": "gpu_nvidia_3050_pny_6",
4967 | "name": "PNY GeForce RTX 3050 8GB",
4968 | "type": "GPU",
4969 | "price": 7490000,
4970 | "power": 150,
4971 | "desc": "RTX 3050 8GB VRAM, 150W TDP"
4972 | },
4973 | {
4974 | "id": "gpu_nvidia_3050_palit_7",
4975 | "name": "Palit GeForce RTX 3050 8GB",
4976 | "type": "GPU",
4977 | "price": 4390000,
4978 | "power": 100,
4979 | "desc": "RTX 3050 8GB VRAM, 100W TDP"
4980 | },
4981 | {
4982 | "id": "gpu_nvidia_3050_galax_8",
4983 | "name": "Galax GeForce RTX 3050 8GB",
4984 | "type": "GPU",
4985 | "price": 6420000,
4986 | "power": 130,
4987 | "desc": "RTX 3050 8GB VRAM, 130W TDP"
4988 | },
4989 | {
4990 | "id": "gpu_nvidia_3050_inno3d_9",
4991 | "name": "Inno3D GeForce RTX 3050 8GB",
4992 | "type": "GPU",
4993 | "price": 4990000,
4994 | "power": 150,
4995 | "desc": "RTX 3050 8GB VRAM, 150W TDP"
4996 | },
4997 | {
4998 | "id": "gpu_nvidia_3050_powercolor_10",
4999 | "name": "PowerColor GeForce RTX 3050 8GB",
5000 | "type": "GPU",
5001 |
5002 | // ---- FILE TRUNCATED: 41084 lines total, showing first 5000 ----
```

```
1 export interface ComponentDimension {
2   width_mm: number;
3   height_mm: number;
4   depth_mm: number;
5 }
6
7 export interface ComponentColor {
8   pcb: string;
9   heatsink?: string;
10  shroud?: string;
11  led?: string;
12  connector: string;
13  accent?: string;
14 }
15
16 export interface ComponentRenderSpec {
17   id: string;
18   name: string;
19   type: 'cpu' | 'gpu' | 'ram' | 'motherboard' | 'psu' | 'case' | 'cooler' | 'ssd' | 'hdd';
20   brand: string;
21   model: string;
22   dimensions: ComponentDimension;
23   colors: ComponentColor;
24   connectors: {
25     id: string;
26     label: string;
27     type: string;
28     position: { x: number; y: number };
29     facing: 'top' | 'bottom' | 'left' | 'right';
30     color: string;
31   }[];
32   imageTop?: string;
33   imageSide?: string;
34   techSpecs: Record<string, string | number>;
35 }
36
37 export const COMPONENT_SPECS: ComponentRenderSpec[] = [
38   {
39     id: 'intel-i5-13600k',
40     name: 'Intel Core i5-13600K',
41     type: 'cpu',
42     brand: 'Intel',
43     model: 'Core i5-13600K',
44     dimensions: { width_mm: 37.5, height_mm: 37.5, depth_mm: 4.5 },
45     colors: {
46       pcb: '#1a472a',
47       heatsink: '#c0c0c0',
48       connector: '#d4af37',
49       accent: '#0071c5',
50     },
51     connectors: [
52       { id: 'lga1700', label: 'LGA1700 Socket', type: 'cpu_socket',
53         position: { x: 50, y: 50 }, facing: 'bottom', color: '#d4af37' }
54     ],
55     techSpecs: { socket: 'LGA1700', tdp: 125, cores: 14, threads: 20 }
56   },
57   {
58     id: 'amd-ryzen5-7600x',
59     name: 'AMD Ryzen 5 7600X',
60     type: 'cpu',
61     brand: 'AMD',
62     model: 'Ryzen 5 7600X',
63     dimensions: { width_mm: 40, height_mm: 40, depth_mm: 4.7 },
64     colors: {
65       pcb: '#1a2744',
66       heatsink: '#a8a8a8',
67       connector: '#b8860b',
68       accent: '#ed1c24',
69     },
70     connectors: [
71       { id: 'am5', label: 'AM5 Socket', type: 'cpu_socket',
72         position: { x: 50, y: 50 }, facing: 'bottom', color: '#b8860b' }
73     ],
74     techSpecs: { socket: 'AM5', tdp: 105, cores: 6, threads: 12 }
75   },
76   {
77     id: 'rtx-4060',
78     name: 'NVIDIA RTX 4060',
79     type: 'gpu',
80     brand: 'NVIDIA',
81     model: 'RTX 4060 (ASUS DUAL)',
82     dimensions: { width_mm: 200, height_mm: 120, depth_mm: 40 },
83     colors: {
84       pcb: '#1a1a2e',
85       heatsink: '#2d2d2d',
86       shroud: 'ffffff',
87       connector: '#000000',
88       accent: '#76b900',
89       led: '#76b900',
90     },
91     connectors: [
92       { id: 'pcie-x16', label: 'PCIe x16', type: 'pcie_x16',
93         position: { x: 90, y: 85 }, facing: 'bottom', color: '#ffcc00' },
94       { id: 'pwr-8pin', label: '8-pin Power', type: 'pcie_power',
95         position: { x: 15, y: 5 }, facing: 'top', color: '#ff6600' },
96       { id: 'dp1', label: 'DisplayPort 1', type: 'display',
97         position: { x: 8, y: 75 }, facing: 'left', color: '#00aaff' },
98       { id: 'dp2', label: 'DisplayPort 2', type: 'display',
99         position: { x: 18, y: 75 }, facing: 'left', color: '#00aaff' },
100      { id: 'hdmi', label: 'HDMI 2.1', type: 'display',
101        position: { x: 28, y: 75 }, facing: 'left', color: '#00aa44' },
102    ],
103    techSpecs: { vram_gb: 8, tdp: 115, pcie_gen: 4, length_mm: 200 }
104  },
105  {
106    id: 'rx-7600',
107    name: 'AMD Radeon RX 7600',
108    type: 'gpu',
109    brand: 'AMD',
110    model: 'RX 7600 (Sapphire PULSE)',
111    dimensions: { width_mm: 190, height_mm: 115, depth_mm: 38 },
112    colors: {
113      pcb: '#0d1117',
114      heatsink: '#3d3d3d',
115      shroud: '#1a1a1a',
116      connector: '#000000',
117      accent: '#ed1c24',
118      led: '#ed1c24',
119    },
120    connectors: [
121      { id: 'pcie-x16', label: 'PCIe x16', type: 'pcie_x16',
122        position: { x: 90, y: 85 }, facing: 'bottom', color: '#ffcc00' },
```

```
123 |     { id: 'pwr-8pin', label: '8-pin Power', type: 'pcie_power',
124 |       position: { x: 20, y: 5 }, facing: 'top', color: '#ff6600' },
125 |   ],
126 |   techSpecs: { vram_gb: 8, tdp: 165, pcie_gen: 4, length_mm: 190 }
127 | },
128 | {
129 |   id: 'asus-b760m-atx',
130 |   name: 'ASUS PRIME B760M-A',
131 |   type: 'motherboard',
132 |   brand: 'ASUS',
133 |   model: 'PRIME B760M-A',
134 |   dimensions: { width_mm: 244, height_mm: 244, depth_mm: 25 },
135 |   colors: {
136 |     pcb: '#1c2a1c',
137 |     heatsink: '#c0c0c0',
138 |     connector: '#ffcc00',
139 |     accent: '#ffd700',
140 |   },
141 |   connectors: [
142 |     { id: 'cpu-socket', label: 'LGA1700 CPU Socket', type: 'cpu_socket',
143 |       position: { x: 35, y: 25 }, facing: 'top', color: '#d4af37' },
144 |     { id: 'dimm-a1', label: 'DIMM A1 (DDR5)', type: 'ram_slot',
145 |       position: { x: 65, y: 20 }, facing: 'top', color: '#4a90d9' },
146 |     { id: 'dimm-b1', label: 'DIMM B1 (DDR5)', type: 'ram_slot',
147 |       position: { x: 72, y: 20 }, facing: 'top', color: '#4a90d9' },
148 |     { id: 'pcie-x16-1', label: 'PCIe x16 Slot 1', type: 'pcie_x16',
149 |       position: { x: 40, y: 58 }, facing: 'top', color: '#ff6b00' },
150 |     { id: 'm2-1', label: 'M.2_1 (PCIe 4.0)', type: 'm2',
151 |       position: { x: 50, y: 45 }, facing: 'top', color: '#00ccc4' },
152 |     { id: 'atx-24pin', label: '24-pin ATX Power', type: 'atx_power',
153 |       position: { x: 88, y: 35 }, facing: 'right', color: '#ff4400' },
154 |     { id: 'cpu-8pin', label: '8-pin CPU EPS', type: 'cpu_power',
155 |       position: { x: 20, y: 5 }, facing: 'top', color: '#ff8800' },
156 |   ],
157 |   techSpecs: { socket: 'LGA1700', chipset: 'B760', form_factor: 'mATX', ram_slots: 2, max_ram_gb: 64, ram_ty...
158 | },
159 | {
160 |   id: 'kingston-fury-16gb-ddr5',
161 |   name: 'Kingston FURY Beast 16GB DDR5',
162 |   type: 'ram',
163 |   brand: 'Kingston',
164 |   model: 'FURY Beast DDR5-5200',
165 |   dimensions: { width_mm: 133, height_mm: 35, depth_mm: 8 },
166 |   colors: {
167 |     pcb: '#1a1a1a',
168 |     heatsink: '#2c2c2c',
169 |     connector: '#d4af37',
170 |     accent: '#ff6600',
171 |   },
172 |   connectors: [
173 |     { id: 'dimm-edge', label: 'DIMM Edge Connector', type: 'dimm',
174 |       position: { x: 50, y: 98 }, facing: 'bottom', color: '#d4af37' }
175 |   ],
176 |   techSpecs: { type: 'DDR5', speed_mhz: 5200, capacity_gb: 16, cl: 40 }
177 | },
178 | {
179 |   id: 'corsair-rm750x',
180 |   name: 'Corsair RM750x 750W',
181 |   type: 'psu',
182 |   brand: 'Corsair',
183 |   model: 'RM750x 80+ Gold',
184 |   dimensions: { width_mm: 150, height_mm: 86, depth_mm: 160 },
185 |   colors: {
186 |     pcb: '#1a1a1a',
187 |     heatsink: '#000000',
188 |     connector: '#ff4400',
189 |     accent: '#ffff00',
190 |   },
191 |   connectors: [
192 |     { id: 'atx-24pin-out', label: '24-pin ATX Output', type: 'atx_power',
193 |       position: { x: 20, y: 50 }, facing: 'right', color: '#ff4400' },
194 |     { id: 'eps-8pin-out', label: '8-pin EPS CPU Output', type: 'cpu_power',
195 |       position: { x: 35, y: 50 }, facing: 'right', color: '#ff8800' },
196 |     { id: 'pcie-6p2-out', label: 'PCIe 6+2 Pin', type: 'pcie_power',
197 |       position: { x: 50, y: 50 }, facing: 'right', color: '#ff6600' },
198 |   ],
199 |   techSpecs: { wattage: 750, efficiency: '80+ Gold', modular: 'Full', pcie_connectors: 4, sata_connectors: 6 }
200 | },
201 | {
202 |   id: 'samsung-980-pro-1tb',
203 |   name: 'Samsung 980 PRO 1TB',
204 |   type: 'ssd',
205 |   brand: 'Samsung',
206 |   model: '980 PRO NVMe M.2',
207 |   dimensions: { width_mm: 80, height_mm: 22, depth_mm: 3.5 },
208 |   colors: {
209 |     pcb: '#1a1a2e',
210 |     heatsink: '#c0c0c0',
211 |     connector: '#d4af37',
212 |     accent: '#1428a0',
213 |   },
214 |   connectors: [
215 |     { id: 'm2-key-m', label: 'M.2 Key-M Connector', type: 'm2',
216 |       position: { x: 92, y: 50 }, facing: 'right', color: '#d4af37' }
217 |   ],
218 |   techSpecs: { interface: 'M.2 NVMe', pcie_gen: 4, read_mbps: 7000, write_mbps: 5100, capacity_gb: 1000 }
219 | },
220 | ];
221 |
222 | export const MM_TO_PX = 2.5;
223 | export const MM_TO_3D = 0.01;
224 |
```

File: data/glossary.ts (63 lines, 6.7 KB)

```
1 | export const GLOSSARY: Record<string, string> = {
2 |   'CPU': 'Bộ vi xử lý trung tâm (Central Processing Unit) - "bộ não" của máy tính, xử lý các lệnh và dữ liệu từ phần cứng và phần mềm.',
3 |   'GPU': 'Bộ xử lý đồ họa (Graphics Processing Unit) - chuyên xử lý các tác vụ liên quan đến đồ họa, bao gồm việc hiển thị hình ảnh trên màn hình.',
4 |   'RAM': 'Bộ nhớ truy cập ngẫu nhiên (Random Access Memory) - lưu trữ dữ liệu tạm thời để hệ thống có thể truy cập nhanh chóng khi cần thiết.',
5 |   'VRAM': 'Bộ nhớ video trên card đồ họa (Video Random Access Memory) - lưu trữ dữ liệu đồ họa để card đồ họa có thể xử lý nhanh chóng.',
6 |   'SSD': 'Ổ cứng thể rắn (Solid State Drive) - lưu trữ dữ liệu vĩnh viễn, không có bộ phận chuyển động cơ học như ổ cứng HDD, giúp tăng tốc độ truy cập dữ liệu.',
7 |   'NVMe': 'Giáo thức kết nối SSD qua PCIe, tận dụng băng thông cao của PCIe để đạt được tốc độ truyền dữ liệu nhanh hơn nhiều so với SATA và SAS.',
8 |   'HDD': 'Ổ cứng từ tính (Hard Disk Drive) - lưu trữ dữ liệu trên các đĩa từ quay, có bộ phận chuyển động cơ học để đọc và ghi dữ liệu, tốc độ chậm hơn SSD.',
9 |   'PSU': 'Bộ nguồn máy tính (Power Supply Unit) - chuyển đổi điện áp từ ổ cắm điện thành các mức điện áp phù hợp để cung cấp cho các thành phần của máy tính.',
10 |   'TDP': 'Công suất tiêu thụ tối đa (Thermal Design Power) - chỉ số cho biết mức công suất tối đa mà bộ tản nhiệt cần phải giải nhiệt cho bộ vi xử lý hoặc GPU để duy trì nhiệt độ hoạt động an toàn.',
11 |   'AIOS': 'Tế bào nhúng trong ổ cứng để quản lý nhiệt độ và hiệu suất hoạt động của ổ cứng, giúp tối ưu hóa hiệu suất và tuổi thọ của ổ cứng.',
12 |   'BIOS': 'Hệ thống nhúng trong bộ vi xử lý để khởi động và quản lý phần cứng của máy tính, bao gồm việc kiểm tra và khởi động các thành phần của máy tính.',
13 |   'UEFI': 'Giáo thức firmware mới hơn thay thế BIOS, giao diện đồ họa, hỗ trợ ổ cứng lớn hơn, khởi động nhanh hơn và hỗ trợ các tính năng tiên tiến hơn.',
14 |   'POST': 'Power-On Self-Test - quá trình kiểm tra phần cứng ngay sau khi khởi động máy tính để đảm bảo rằng tất cả các thành phần của máy tính đều hoạt động đúng cách.',
15 |   'VRM': 'Module điều chỉnh điện áp (Voltage Regulator Module) - trên mainboard, chuyển đổi điện áp từ ổ cắm điện thành các mức điện áp phù hợp để cung cấp cho các thành phần của máy tính.',
16 |   'PCIe': 'Peripheral Component Interconnect Express - chuẩn kết nối mở cho các thiết bị ngoại vi như ổ cứng, card đồ họa, card âm thanh, v.v., cho phép truyền dữ liệu nhanh chóng giữa các thiết bị và mainboard.',
17 |   'SATA': 'Serial ATA - chuẩn kết nối ổ cứng HDD/SSD và ổ quang, băng thông thấp hơn PCIe.',
18 |   'M.2': 'Chuẩn kết nối SSD NVMe nhúng trực tiếp lên mainboard, hỗ trợ tốc độ truyền dữ liệu nhanh chóng và tiết kiệm không gian lắp đặt.
```

```

1 9
2 0
3 1
4 2
5 3
6 4
7 5
8 6
9 7
10 8
11 9
12 0
13 1
14 2
15 3
16 4
17 5
18 6
19 7
20 8
21 9
22 0
23 1
24 2
25 3
26 4
27 5
28 6
29 7
30 8
31 9
32 0
33 1
34 2
35 3
36 4
37 5
38 6
39 7
40 8
41 9
42 0
43 1
44 2
45 3
46 4
47 5
48 6
49 7
50 }
51
52 export function findTerms(text: string): string[] {
53   const terms: string[] = []
54   const sorted = Object.keys(GLOSSARY).sort((a, b) => b.length - a.length)
55   const lower = text.toLowerCase()
56   for (const term of sorted) {
57     if (lower.includes(term.toLowerCase())) {
58       terms.push(term)
59     }
60   }
61   return terms
62 }
63

```

File: data/key-questions.ts (149 lines, 19.6 KB)

```

1 export interface KeyQuestion {
2   q: string
3   a: string
4 }
5
6 export const KEY_QUESTIONS: Record<string, KeyQuestion[]> = {
7   'BỘ vi xử lý (CPU)': [
8     { q: 'CPU Enhancement Intel hay AMD? ', a: 'CPU nâng cấp Intel hay AMD? ', },
9     { q: 'CPU Cooling System? ', a: 'CPU Cooling System? ', },
10    { q: 'Socket CPU? ', a: 'Socket CPU? ', },
11    { q: 'TDP? ', a: 'TDP? ', },
12  ],
13  'RAM & SSD': [
14    { q: 'Nên chèn DDR4 hay DDR5? ', a: 'DDR5 mới hơn, tiết kiệm năng lượng (48GB trở lên), ', },
15    { q: 'Bao nhiêu RAM? ', a: '8GB trở lên, ', },
16    { q: 'SSD NVMe hay SATA? ', a: 'NVMe nhanh hơn, ', },
17    { q: 'Nên mua SSD dung lượng bao nhiêu? ', a: '250GB trở lên, ', },
18  ],
19  'Card đồ họa (GPU)': [
20    { q: 'GPU quan trọng không? ', a: 'GPU là linh kiện quan trọng, ', },
21    { q: 'Nên mua card đồ họa rời hay tích hợp? ', a: 'Card đồ họa rời cho hiệu suất cao hơn, ', },
22    { q: 'VRAM bao nhiêu? ', a: '4GB trở lên, ', },
23    { q: 'Ray Tracing? ', a: 'Ray Tracing: mô phỏng ánh sáng thực tế, ', },
24  ],
25  'Nguồn & Tủ Case': [
26    { q: 'Chèn PSU công suất bao nhiêu? ', a: 'Tủ case thông thoáng, TDP linh kiện, ', },
27    { q: '80 Plus? ', a: 'Chèn PSU 80 Plus trở lên, ', },
28    { q: 'Nên dùng loại tủ case nào? ', a: 'Tủ case có quạt tản nhiệt, ', },
29    { q: 'Nguồn modular? ', a: 'Nguồn modular có dây cáp tháo rời, ', },
30  ],
31  'Bo mạch chủ (Motherboard)': [
32    { q: 'Chipset? ', a: 'Chipset theo socket CPU (Intel Z600/B760, ', },
33    { q: 'ATX, M-ATX, ITX? ', a: 'H610: chipset rẻ, B660/B760: chipset trung cấp, ', },
34    { q: 'VRM, tản nhiệt? ', a: 'VRM (Voltage Regulation Module) quan trọng, ', },
35    { q: 'M.2 slots? ', a: 'M.2 slots cho SSD, ', },
36  ],
37  'Lưu trữ (Storage)': [
38    { q: 'SSD hay HDD? ', a: 'SSD: cài Windows, ổ cứng, ', },
39    { q: 'Caching? ', a: 'Caching: tăng tốc độ truy cập, ', },
40    { q: 'Nên chèn SSD tra cứu? ', a: 'NVMe PCIe 4.0: CrystalDiskInfo (Windows), ', },
41  ],
42  'Thẻ mạng (Network)': [
43    { q: 'Cáp mạng? ', a: 'Cáp mạng Cat 5e trở lên, ', },
44    { q: 'Chuyển đổi? ', a: 'Chuyển đổi từ LAN sang Wi-Fi, ', },
45    { q: 'Sau khi lắp xong? ', a: 'Kiểm tra đèn LED mainboard, ', },
46    { q: 'Sử dụng? ', a: 'Sử dụng? ', },
47  ],
48  'Màn hình & Khung': [
49    { q: 'Nên dùng màn hình? ', a: 'Màn hình? ', },
50    { q: 'Router? ', a: 'Router? ', },
51    { q: 'Caching? ', a: 'Caching? ', },
52  ],
53  'Thị trường & Giá': [
54    { q: 'Caching? ', a: 'Caching? ', },
55    { q: 'Bàn phím? ', a: 'Bàn phím? ', },
56    { q: 'DPI? ', a: 'DPI? ', },
57  ],
58  'Hàng hóa & Phụ kiện': [
59    { q: 'UEFI BIOS? ', a: 'UEFI BIOS? ', },
60    { q: 'Caching? ', a: 'Caching? ', },
61    { q: 'Secure Boot? ', a: 'Secure Boot? ', },
62    { q: 'Nên? ', a: 'Nên? ', },
63  ],
64  'Cổng & Kết nối': [
65    { q: 'Hỗ trợ? ', a: 'Hỗ trợ? ', },
66    { q: 'Nên? ', a: 'Nên? ', },
67    { q: 'HDMI? ', a: 'HDMI? ', },
68  ],
69  'Hàng hóa & Phụ kiện': [
70    { q: 'Cẩn thận? ', a: 'Cẩn thận? ', },
71    { q: 'Tai nghe? ', a: 'Tai nghe? ', },
72    { q: 'Loa? ', a: 'Loa? ', },
73  ],
74  'Vỏ Case & Quạt': [
75    { q: 'Caching? ', a: 'Caching? ', },
76    { q: 'Vỏ Case? ', a: 'Vỏ Case? ', },
77    { q: 'Quạt? ', a: 'Quạt? ', },
78  ],
79  'Hàng hóa & Phụ kiện': [
80    { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
81    { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
82    { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
83  ],
84  'Hàng hóa & Phụ kiện': [
85    { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
86    { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
87    { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
88  ],
89  'Hàng hóa & Phụ kiện': [
90    { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
91    { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
92    { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
93  ],
94  'Hàng hóa & Phụ kiện': [
95    { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
96    { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
97    { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
98  ],
99  'Hàng hóa & Phụ kiện': [
100   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
101   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
102   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
103  ],
104  'Hàng hóa & Phụ kiện': [
105   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
106   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
107   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
108  ],
109  'Hàng hóa & Phụ kiện': [
110   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
111   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
112   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
113  ],
114  'Hàng hóa & Phụ kiện': [
115   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
116   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
117   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
118  ],
119  'Hàng hóa & Phụ kiện': [
120   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
121   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
122   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
123  ],
124  'Hàng hóa & Phụ kiện': [
125   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
126   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
127   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
128  ],
129  'Hàng hóa & Phụ kiện': [
130   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
131   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
132   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
133  ],
134  'Hàng hóa & Phụ kiện': [
135   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
136   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
137   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
138  ],
139  'Hàng hóa & Phụ kiện': [
140   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
141   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
142   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
143  ],
144  'Hàng hóa & Phụ kiện': [
145   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
146   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
147   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
148  ],
149  'Hàng hóa & Phụ kiện': [
150   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
151   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
152   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
153  ],
154  'Hàng hóa & Phụ kiện': [
155   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
156   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
157   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
158  ],
159  'Hàng hóa & Phụ kiện': [
160   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
161   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
162   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
163  ],
164  'Hàng hóa & Phụ kiện': [
165   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
166   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
167   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
168  ],
169  'Hàng hóa & Phụ kiện': [
170   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
171   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
172   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
173  ],
174  'Hàng hóa & Phụ kiện': [
175   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
176   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
177   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
178  ],
179  'Hàng hóa & Phụ kiện': [
180   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
181   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
182   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
183  ],
184  'Hàng hóa & Phụ kiện': [
185   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
186   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
187   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
188  ],
189  'Hàng hóa & Phụ kiện': [
190   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
191   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
192   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
193  ],
194  'Hàng hóa & Phụ kiện': [
195   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
196   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
197   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
198  ],
199  'Hàng hóa & Phụ kiện': [
200   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
201   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
202   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
203  ],
204  'Hàng hóa & Phụ kiện': [
205   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
206   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
207   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
208  ],
209  'Hàng hóa & Phụ kiện': [
210   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
211   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
212   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
213  ],
214  'Hàng hóa & Phụ kiện': [
215   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
216   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
217   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
218  ],
219  'Hàng hóa & Phụ kiện': [
220   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
221   { q: 'Phụ kiện? ', a: 'Phụ kiện? ', },
222   { q: 'Hàng hóa? ', a: 'Hàng hóa? ', },
223 
```

```
77 | { q: 'Cá ch bÑ tr í g i t', case: 'tÑ i t ° u?', tñ: 'Hút vào: m. t c t r ° U c + á y. t'
78 | ],
79 | { q: 'á n h g i á h i Ç u n n g ' : [ benchmark quan tr í n g? ', a: 'CPU: Cinebench (rende
80 | { q: 'Cá c ph ã n m Á m là b e n c h m a r k q u a n t r í n g? ', a: 'CPU: Cinebench (rende
81 | { q: 'FPS nh i ê u là h c h i i á c? ', a: '30 FPS: C h i i á c ( c o n s o l e ) .
82 | { q: 'Cá ch b a í c k ã t q u é b e n c h m a r k? ', a: 'So s á n h c i Á m s Ñ v ù i c á c c ÷ u
83 | ],
84 | 'Laptop': [
85 | { q: 'N ê n m u a l a p t o p h a y d e s k t o p? ', a: 'Laptop: di ù n g , g í n n h í , t j
86 | { q: 'Cá c ð o n g l a p t o p p h ô b i ã n? ', a: 'Ultrabook: m í n g n h í , p i n l á u (
87 | { q: 'Thunderbolt có g i c b i ã t? ', a: 'B o n g t h ô n g 4 0 G b p s , t r u y Á n
88 | ],
89 | 'É p x u n g & t ñ i ° u': [ ung CPU à k h ô n g? ', a: 'CPU, Ý i m ù i á c ó t u r b o b o o s t p
90 | { q: 'Cá c nh i ê u é p C P U à k h ô n g? ', a: 'CPU, Ý i m ù i á c ó t u r b o b o o s t p
91 | { q: 'N h i c t ù t i n g C P U à k h ô n g? ', a: 'CPU, Ý i m ù i á c ó t u r b o b o o s t p
92 | { q: 'U n d e r v o l t i n g l à g i? ', a: 'G i é m i c ñ , á p C P U / G P U 3 0 - 4 0 ° C . X e m p h
93 | ],
94 | 'T ã n n h i c t n â n g c a o ': [ t e r c o o l i n g l à g i? ', a: 'H Ç t h ñ n g t ã n n h i c t n ° U c
95 | { q: 'C ú t o m s l o o p w a t e r c o o l i n g l à g i? ', a: 'H Ç t h ñ n g t ã n n h i c t n ° U c
96 | { q: 'A I O v s C ú o n g t ã n n ° U c t h ã n ñ a o? ', a: 'A I O : t h ñ n g , c ã n b é o ð á n g
97 | { q: 'C ã n b é o ð á n g t ã n n ° U c t h ã n ñ a o? ', a: 'A I O : t h ñ n g , c ã n b é o ð á n g
98 | ],
99 | 'PC Gaming': [
100 | { q: 'C ÷ u h í n h P C g a m i n g 1 5 t r i Ç u? ', a: 'CPU: AMD Ryzen 5 7600. GPU:
101 | { q: 'C ÷ u h í n h P C g a m i n g 3 0 t r i Ç u? ', a: 'CPU: Intel i5-14600K / Ryzen
102 | { q: 'Y ç u t ñ n ñ a o q u a n t r í n g n h í t c h o g a m i n g? ', a: 'GPU là quan tr í n g
103 | ],
104 | 'M á y c h c & W o r k s t a t i o n': [ ã n g t h ã n ñ a o? ', a: 'S e r v e r ð ù n g C P U X e o n / E P Y C
105 | { q: 'E E C C R A M l à g i? ', a: 'R A M c ó k h é , n n g p h á t h i Ç n v à s i á l x i t ñ
106 | { q: 'R A I D n ê n ð ù n g l o i i ñ a o? ', a: 'R A I D 0: g ú p ô !' n h a n h n h ° n g k h ô n
107 | { q: 'R A I D n ê n ð ù n g l o i i ñ a o? ', a: 'R A I D 0: g ú p ô !' n h a n h n h ° n g k h ô n
108 | ],
109 | 'C á p & ÷ u c n ñ i': [ á p n g u ó n c ã n b i ã t? ', a: '24-pin ATX: c ÷ p n g u ó n m a i n b
110 | { q: 'Cá c l o i i c á g e m e n t l à g i? ', a: 'S p x ç p ð á y c á p g í n g á n g t r o n g c
111 | { q: 'Cá b l e l o i m a n c á g e m e n t l à g i? ', a: 'S p x ç p ð á y c á p g í n g á n g t r o n g c
112 | { q: 'N ê n ð ù n g c á p n ñ i ñ a o c h o , m à n h í n h? ', a: 'H D M I 2 . i : 4 K 144Hz h o
113 | ],
114 | 'X í l ý s ã n ñ i': [ b o o t , x í l ý t h ã n ñ a o? ', a: '1. K i Á m t r a n g u ó n ( q u i t n g
115 | { q: 'P C k h ô n g [ b o o t , x í l ý t h ã n ñ a o? ', a: '1. K i Á m t r a n g u ó n ( q u i t n g
116 | { q: 'M á y h í n h x a n h ( B S O D ) f i x t h ã n ñ a o? ', a: 'G h i l i m á l x i ( V D : M
117 | { q: 'M á y c h i y c h - m , k h - c p h á c r a s a o? ', a: 'K i Á m t r a s é c k h i e S S D / M
118 | { q: 'N h i c t U c a o k h i c h i i g a m e? ', a: 'K i Á m t r a q u i t t ã n n h i c t C P U
119 | ],
120 | 'VR/AR': [
121 | { q: 'P C c ã n c ÷ u h í n h g i c h o V R? ', a: 'T ñ i t h i Á u : R T X 3 0 6 0 + i 5 / R y z e
122 | { q: 'V R H e a d s e t ñ a o p h ô b i ã n? ', a: 'M e t a Q u e s t 3 : k h ô n g ð á y , g i á p h
123 | { q: 'A R k h á c V R t h ã n ñ a o? ', a: 'A R ( T h ñ c t ã t n g c ° Ý n g ) : c h ô n g h í n h
124 | ],
125 | 'L É c h s í P C': [ ñ h í n g c ú t m ñ c q u a n t r í n g t r o n g l É c h s í P C? ', a: '1945: E N I A C (
126 | { q: 'N h í n g c ú t m ñ c q u a n t r í n g t r o n g l É c h s í P C? ', a: '1945: E N I A C (
127 | { q: ' Ë n h l u - t M o o r e q u a n t r í n g t r o n g l É c h s í P C? ', a: 'S Ñ l ñ á n g t r a n s i s t o r t r ê n c h i p t r
128 | ],
129 | 'i o é c & a n t o à n': [
130 | { q: 'B é o v Ç ð i l i Ç u c á n h a n t h ã n ñ a o? ', a: ' ð ù n g m - t k h © u m í n h ( c h i
131 | { q: 'X í l ý r á c t t h ã i i c ñ n t i ú n g c á c h? ', a: 'K h ô n g v é t l í n h k i Ç n v à
132 | { q: 'B é n q u y Á n p h ã n m Á m? ', a: 'P h ã n m Á m , c ó b é n q u y Á n : W i n d o w s , O f f
133 | ],
134 | 'T r í t u Ç n h a n t i o': [ L e a r n i n g k h á c n h a u t h ã n ñ a o? ', a: 'A I : m á y m ô p h i
135 | { q: 'A I v à M a c h i n e L e a r n i n g k h á c n h a u t h ã n ñ a o? ', a: 'A I : m á y m ô p h i
136 | { q: 'G P U q u a n t r í n g v ù i A I t h ã n ñ a o? ', a: 'G P U c ó h à n g n g h i n l ò i x í
137 | { q: 'C á c é n g ð á n g A I p h ô b i ã n? ', a: 'C h a t G P T / G e m i n i ( c h a t b o t ) , M i d
138 | ],
139 | default: [
140 | { q: 'L à m s a o Á b - t s u h í c? ', a: '1. n g n h - p t à i k h o é n . 2. V à o s
141 | { q: 'K i é m n g à n s á c h t h ã n ñ a o? ', a: 'L à m q u i z t r c n g h i c m , m x i i Á m =
142 | { q: 'C á c h s í ð á n g g l o s s a r y? ', a: 'T r o n g k h i l à m q u i z , t h u - t n g i
143 | ],
144 | }
145 | }
146 | export function getKeyQuestions(lessonTitle: string): KeyQuestion[] {
147 | return KEY_QUESTIONS[lessonTitle] || KEY_QUESTIONS.default
148 | }
149 | }
```

File: data/knowledgeBase.js (70 lines, 4.3 KB)

```
1 | export const KNOWLEDGE_BASE = [
2 | {
3 |   topic: "CPU",
4 |   content: `CPU (Central Processing Unit) là bộ não của máy tính. AM
5 |   C á c l o i i s o c k e t p h ô b i ã n : I n t e l : ð ù n g L G A ( n h ° L G A 1 7 0 0 ) , A M D ð ù n g A M 4
6 |   C o r e s ( n h a n ) v à T h r e a d s ( l u ó n g ) : C à n g n h i Á u n h a n l u ó n g , t h i x í l ý
7 |   X u n g n h é p ( C l o c k s p e e d ) : T í n h b ± n g G H z , q u y ç t Ë n h t ñ c ù x í l ý c ç a
8 |   T D P ( T h e r m a l D e s i g n P o w e r ) : C h é s Ñ t h ó á t n h i c t , g í ú p c h í n t ã n n h i c t
9 |   C a c h e : B ù n h ù Ç m g í ú p C P U : t r u y c - p ð i l i Ç u n h a n h h í n . `
10 | },
11 | {
12 |   topic: "RAM",
13 |   content: `RAM (Random Access Memory) là bộ nhớ tạm thời của hç thñng
14 |   D D R 4 v s D D R 5 : D D R 5 c ó t ñ c ù ( b u s ) c a o h í n v à i c ñ á p t h ÷ p h í n D D R 4 .
15 |   ð ù n g l ñ á n g : P h ô b i ã n l à 8 G B , 1 6 G B , 3 2 G B . 1 6 G B l à m é c t i ê u c h u © n c h o
16 |   D u a l C h a n n e l : C - m 2 t h a n h R A M g í ú p b ñ g t h ô n g t ñ g g ÷ p ð i , c é i t h i c
17 |   B u s R A M : T ñ c ù t r u y Á n t ã i ð i l i Ç u , v í ð á 3 2 0 0 M H z ( D D R 4 ) h o c 5 6 0 0 M H
18 | },
19 | {
20 |   topic: "GPU",
21 |   content: `GPU (Graphics Processing Unit) x í l ý h í n h Ë n h v à ó h í a .
22 |   V R A M : B ù n h ù v í d e o , v í ð á 8 G B , 1 2 G B G D D R 6 . Q u a n t r í n g k h i c h i i g a m e
23 |   C U D A C o r e s ( N V I D I A ) , h o c S t r e a m P r o c e s s o r s ( A M D ) : S Ñ l ñ á n g n h a n x í
24 |   K ã t n ñ i : H s u h ç t ð ù n g k h e c - m P C I e x 1 6 t r ê n m a i n b o a r d .
25 |   T D P : G P U t h ° Ý n g l à l í n h k i Ç n n g ñ n i c ñ n h í t t r o n g m á y t í n h . `
26 | },
27 | {
28 |   topic: "PSU",
29 |   content: `PSU (Power Supply Unit) là bộ nguồn c÷p i c ñ c h o t o à n b ù n
30 |   W a t t a g e : C o n g s u ý t Ë n h m é c ( v í ð á 5 0 0 W , 6 5 0 W , 7 5 0 W ) . P h é i l ù n h í n t
31 |   H i Ç u s u ý t 8 0 P l u s : C á c c h u © n W h i t e , B r o n z e , S i l v e r , G o l d , P l a t i n u m ,
32 |   M o d u l a r : N g u ó n F u l l - M o d u l a r c h o p h é p t h á o r ý i t ÷ t c é ð á y c á p Á i c
33 | },
34 | {
35 |   topic: "SSD",
36 |   content: `SSD (Solid State Drive) là ổ c ñ g t h Á r - n , t ñ c ù c a o h í n
37 |   N V M e v s S A T A : S S D N V M e ( M . 2 ) c ó t ñ c ù c ñ c n h a n h ( l ã n c ñ n 7 0 0 0 M B / s ) ,
38 |   C o n g k ã t n ñ i : M . 2 ( g - n t r í c t i ) c ó p l ê n m a i n b o a r d ) h o c S A T A ( ð ù n g ð á y
39 |   T ñ c ù í c g h i : Y ç u t ñ q u y ç t t i Ë n h t h ý i g i a n k h ã i ð ù n g W i n v à m ð é n g
40 | },
41 | {
42 |   topic: "Mainboard",
43 |   content: `Mainboard (Bo mạch chç) k ã t n ñ i m í i l í n h k i Ç n l i i v ù i n h a u
44 |   C h i p s e t : V í ð á H 6 1 0 ( p h ô t h ô n g ) , B 7 6 0 ( t ñ m t r u n g ) , Z 7 9 0 ( c a o c ÷ p c h o
45 |   F o r m F a c t o r : A T X ( l ù n ) , M i c r o - A T X ( v é a ) , M i n i - I T X ( n h í ) .
46 |   S l o t : K h e c - m R A M ( 2 h o c 4 k h e ) , k h e P C I e c h o G P U , v à c á c k h e c - m s
```

```

47| },
48| {
49|   topic: "CASE",
50|   content: `Case (Vĩ máy tính) bèo vç linh kiện và quyế t enh tĩnh th
51|   Kĩch th° Ớc (Form Factor): Full Tower (lũn nhựt), Mid Tower (phổ biẻn
52|   Air flow: Luồng khõng khí trong máy, quan trĩng, Ặ giĩ linh kiện mấ t
53|   Kĩnh c° Ỗng lĩc (Tempered Glass): Giú p khoe linh kiện bên trong và
54| },
55| {
56|   topic: "COOLING",
57|   content: `Hç thĩng làm mát giú p CPU và GPU khõng bẻ quá nhiçt. bẻ và
58|   Tĩn nhiçt khĩ (Air Cooling): Dùng tháp tĩn nhiçt và quĩt, bẻ và
59|   Tĩn nhiçt n° Ớc (AIO/Liquid Cooling): Dùng dụng đẻch làm mát và radia
60|   Quit Case (Case Fans): Th° Ỗng cớ kĩch th° Ớc 120mm ho·c 140mm, l_p th
61| },
62| {
63|   topic: "MONITOR",
64|   content: `Màn hình hiẶn thẻ hình Ặnh tẻ máy tĩnh.
65|   Ớ phầ n giẻi: Full HD (1080p), 2K (1440p), 4K (2160p).
66|   Tĩn sñ quẻt (Refresh Rate): Tĩnh b±ng Hz (60Hz, 144Hz, 240Hz). Càng ph
67|   Tĩm nẶn (Panel Type): IPS (màu 1p, gớc nhĩn rửng), VA ( Ớ t° ĩng ph
68| },
69| ];
70|

```

File: data/quiz-bank-full.ts (1741 lines, 84.6 KB)

```

1| export const FULL_QUIZ_BANK = {
2|   'cpu-arch': {
3|     id: 'cpu-arch',
4|     title: 'Kĩ Ặn Trú c CPU & T-p LẶnh',
5|     icon: '0=y',
6|     color: '#6366f1',
7|     description: 'Kĩ Ặn thẻc vẶ bỪ vi xĩ lý, kĩ Ặn trú c, t-p lẶnh và cõng
8|   },
9|   questions: [
10|     {
11|       id: 'q1',
12|       text: 'CPU là viẻt t-t cặ tẻ gì?',
13|       options: ['A. Central Processing Unit', 'B. Computer Personal Unit', 'C. Central Program Utility', 'D....
14|       correct: 0,
15|       explanation: 'CPU là viẻt t-t cặ Central Processing Unit (BỪ xĩ lý)
16|     },
17|     {
18|       id: 'q2',
19|       text: 'ĩn vẻ o tĩn Ớ xung nhẻp CPU là gì?',
20|       options: ['A. GHz (Gigahertz)', 'B. MB (Megabyte)', 'C. GB/s (Gigabyte trên giẻ)', 'D. RPM (Võng trên...
21|       correct: 0,
22|       explanation: 'Tĩn Ớ CPU °Ặ o b±ng GHz - sñ t÷ chu kớ m xĩ giẻ
23|     },
24|     {
25|       id: 'q3',
26|       text: 'Cache L1, L2, L3 °Ặ t Ặ Ặu trong hç thĩng?',
27|       options: ['A. Trong CPU', 'B. Trên Mainboard', 'C. Trong RAM', 'D. Trong GPU'],
28|       correct: 0,
29|       explanation: 'BỪ nhỪ Ặm Cache °Ặ tích hẶp trĩc tiẻp trong CPU
30|     },
31|     {
32|       id: 'q4',
33|       text: 'CPU nào sau Ặy thuỚ dõng sẶn phõm cặ AMD?',
34|       options: ['A. Core i7', 'B. Ryzen 7', 'C. Xeon', 'D. Pentium'],
35|       correct: 1,
36|       explanation: 'Ryzen 7 là dõng CPU cặ AMD, Core i7 và Pentium là
37|     },
38|     {
39|       id: 'q5',
40|       text: 'Hyper-Threading là cõng nghç cho phẻp CPU làm gì?',
41|       options: ['A. Xĩ lý Ặ luớng hiẻu quẻ hĩn', 'B. Tĩng xung' nhẻp t
42|       correct: 0,
43|       explanation: 'Hyper-Threading cho phẻp m xĩ lờĩ v-t lý xĩ lý 2 lu
44|     },
45|     {
46|       id: 'q6',
47|       text: 'Socket LGA 1700 dùng cho CPU Intel thẻ hç nào?',
48|       options: ['A. Thẻ hç 12/13/14', 'B. Thẻ hç 10/11', 'C. Thẻ hç 8/
49|       correct: 0,
50|       explanation: 'LGA 1700 °Ặ Intel sí dẶng cho Core thẻ hç 12 (A
51|     },
52|     {
53|       id: 'q7',
54|       text: 'TDP là viẻt t-t cặ thu-t ngĩ gì?',
55|       options: ['A. Thermal Design Power', 'B. Total Data Processing', 'C. Technology Development Plan', 'D....
56|       correct: 0,
57|       explanation: 'TDP (Thermal Design Power) là cõng suỰt nhiçt thĩ
58|     },
59|     {
60|       id: 'q8',
61|       text: 'CPU AMD sí dẶng socket nào phổ biẻn nhựt hiẶn nay?',
62|       options: ['A. AM5', 'B. LGA 1700', 'C. AM4', 'D. TR4'],
63|       correct: 0,
64|       explanation: 'AMD Ặng sí dẶng socket AM5 cho dõng Ryzen 7000/90
65|     },
66|     {
67|       id: 'q9',
68|       text: 'Sñ nhầ n (cores) nhiỂu hĩn mang liẻ lẶĩ ích gì?',
69|       options: ['A. Ặ nhiẶm và xĩ lý song song tĩn hĩn', 'B. Tĩn Ớ
70|       correct: 0,
71|       explanation: 'NhiỂu nhầ n hĩn giú p CPU xĩ lý nhiỂu tác vẶ song s
72|     },
73|     {
74|       id: 'q10',
75|       text: 'CPU Intel Core i5 th° Ỗng phử hẶp vỚĩ Nĩ t° Ặng nào?',
76|       options: ['A. Ng° Ỗ dĩ dùng phổ thõng và gaming tĩm trung', 'B. Máy
77|       correct: 0,
78|       explanation: 'Core i5 là lẶĩ chĩn cân b±ng giẻ giẻ Ặ và hiẻu n
79|     },
80|   ],
81|   'ram-storage': {
82|     id: 'ram-storage',
83|     title: 'RAM & BỪ NhỪ',
84|     icon: '0>y',
85|     color: '#8b5cf6',
86|     description: 'Kĩ Ặn thẻc vẶ bỪ nhỪ trong, RAM, thõng sñ và cõng nghç
87|   },
88|   questions: [
89|     {
90|       id: 'q1',
91|       text: 'RAM là viẻt t-t cặ thu-t ngĩ nào?',
92|       options: ['A. Random Access Memory', 'B. Read Access Memory', 'C. Rapid Application Module', 'D. Real ...
93|       correct: 0,
94|       explanation: 'RAM là Random Access Memory - bỪ nhỪ truy c-p ng«
95|     },
96|     {
97|       id: 'q2',

```

```
97 |         text: 'DDR5 RAM có tần số bus bao nhiêu?',
98 |         options: ['A. 4800-8400 MT/s', 'B. 2400-3200 MT/s', 'C. 1600-2133 MT/s', 'D. 800-1066 MT/s'],
99 |         correct: 0,
100 |         explanation: 'DDR5 có tần số bus từ 4800 MT/s đến 8400 MT/s tùy phiên bản',
101 |     },
102 |     {
103 |         id: 'q3',
104 |         text: 'Dung lượng RAM phổ biến cho PC gaming hiện nay là bao nhiêu?',
105 |         options: ['A. 16-32GB', 'B. 4-8GB', 'C. 64-128GB', 'D. 2-4GB'],
106 |         correct: 0,
107 |         explanation: '16-32GB RAM là tiêu chuẩn cho gaming hiện tại',
108 |     },
109 |     {
110 |         id: 'q4',
111 |         text: 'CAS Latency (CL) thấp có ý nghĩa gì?',
112 |         options: ['A. Thời gian truy cập nhanh hơn', 'B. Dung lượng lớn hơn', 'C. Tốc độ xử lý nhanh hơn', 'D. Tiết kiệm năng lượng'],
113 |         correct: 0,
114 |         explanation: 'CL thấp đồng nghĩa với thời gian truy cập nhanh và hiệu năng cao',
115 |     },
116 |     {
117 |         id: 'q5',
118 |         text: 'Dual Channel là gì?',
119 |         options: ['A. Chạy 2 thanh RAM cùng lúc để tăng băng thông', 'B. Chạy 2 thanh RAM song song để tăng dung lượng', 'C. Chạy 2 thanh RAM để tiết kiệm năng lượng', 'D. Chạy 2 thanh RAM để giảm nhiệt độ'],
120 |         correct: 0,
121 |         explanation: 'Dual Channel sử dụng 2 thanh RAM song song để tăng băng thông',
122 |     },
123 |     {
124 |         id: 'q6',
125 |         text: 'RAM DDR4 có bao nhiêu chân?',
126 |         options: ['A. 288 chân', 'B. 240 chân', 'C. 260 chân', 'D. 204 chân'],
127 |         correct: 0,
128 |         explanation: 'DDR4 có 288 chân, trong khi DDR3 chỉ có 240 chân',
129 |     },
130 |     {
131 |         id: 'q7',
132 |         text: 'Loại RAM nào tiết kiệm điện nhất?',
133 |         options: ['A. LPDDR5 (Low Power DDR5)', 'B. DDR4', 'C. DDR5', 'D. DDR3'],
134 |         correct: 0,
135 |         explanation: 'LPDDR5 đặc biệt tiết kiệm điện cho các thiết bị di động',
136 |     },
137 |     {
138 |         id: 'q8',
139 |         text: 'ECC RAM thường được sử dụng trong môi trường nào?',
140 |         options: ['A. Máy chủ / Server yêu cầu độ tin cậy cao', 'B. PC Gaming', 'C. Điện thoại thông minh', 'D. Máy tính để bàn'],
141 |         correct: 0,
142 |         explanation: 'ECC RAM có khả năng phát hiện và sửa lỗi, rất quan trọng cho server',
143 |     },
144 |     {
145 |         id: 'q9',
146 |         text: 'XMP là công nghệ ép xung RAM từ hãng nào?',
147 |         options: ['A. Intel (Extreme Memory Profile)', 'B. AMD', 'C. Corsair', 'D. Kingston'],
148 |         correct: 0,
149 |         explanation: 'XMP là công nghệ của Intel, AMD có công nghệ tương tự gọi là D.O.C',
150 |     },
151 |     {
152 |         id: 'q10',
153 |         text: 'Dung lượng RAM tối thiểu Windows 11 Home là bao nhiêu?',
154 |         options: ['A. 128GB', 'B. 64GB', 'C. 256GB', 'D. 32GB'],
155 |         correct: 0,
156 |         explanation: 'Windows 11 Home yêu cầu tối thiểu 128GB RAM',
157 |     },
158 |     ],
159 |     },
160 |     'gpu-graphics': {
161 |         id: 'gpu-graphics',
162 |         title: 'Card đồ họa & xử lý đồ họa',
163 |         icon: '🖥️',
164 |         color: '#ef4444',
165 |         description: 'Kiến thức về GPU, VRAM, công nghệ đồ họa và render',
166 |         questions: [
167 |             {
168 |                 id: 'q1',
169 |                 text: 'GPU là viết tắt của thuật ngữ nào?',
170 |                 options: ['A. Graphics Processing Unit', 'B. General Processing Unit', 'C. Graphical Program Utility', 'D. Graphics Processing Utility'],
171 |                 correct: 0,
172 |                 explanation: 'GPU là Graphics Processing Unit - đơn vị xử lý đồ họa',
173 |             },
174 |             {
175 |                 id: 'q2',
176 |                 text: 'VRAM là bộ nhớ của thành phần nào?',
177 |                 options: ['A. Card đồ họa (GPU)', 'B. CPU', 'C. Mainboard', 'D. Nguồn'],
178 |                 correct: 0,
179 |                 explanation: 'VRAM (Video RAM) là bộ nhớ riêng của card đồ họa',
180 |             },
181 |             {
182 |                 id: 'q3',
183 |                 text: 'Ray Tracing là công nghệ gì?',
184 |                 options: ['A. Mô phỏng ánh sáng vật lý chân thực', 'B. Công nghệ đồ họa đơn giản', 'C. Công nghệ AI', 'D. Công nghệ render'],
185 |                 correct: 0,
186 |                 explanation: 'Ray Tracing mô phỏng ánh sáng để tạo ra hình ảnh chân thực',
187 |             },
188 |             {
189 |                 id: 'q4',
190 |                 text: 'DLSS là công nghệ AI của hãng nào?',
191 |                 options: ['A. NVIDIA', 'B. AMD', 'C. Intel', 'D. Apple'],
192 |                 correct: 0,
193 |                 explanation: 'DLSS (Deep Learning Super Sampling) là công nghệ của NVIDIA',
194 |             },
195 |             {
196 |                 id: 'q5',
197 |                 text: 'CUDA Core là kiến trúc tính toán của hãng nào?',
198 |                 options: ['A. NVIDIA', 'B. AMD', 'C. Intel', 'D. Qualcomm'],
199 |                 correct: 0,
200 |                 explanation: 'CUDA là nền tảng tính toán song song của NVIDIA',
201 |             },
202 |             {
203 |                 id: 'q6',
204 |                 text: 'PCIe 5.0 có băng thông gấp mấy lần PCIe 4.0?',
205 |                 options: ['A. Gấp đôi (2x)', 'B. Gấp 3 lần', 'C. Gấp 1.5 lần', 'D. Gấp 4 lần'],
206 |                 correct: 0,
207 |                 explanation: 'PCIe 5.0 có băng thông 32 GT/s, gấp đôi so với 16 GT/s của PCIe 4.0',
208 |             },
209 |             {
210 |                 id: 'q7',
211 |                 text: 'RTX 4090 sử dụng kiến trúc GPU nào?',
212 |                 options: ['A. Ada Lovelace', 'B. Ampere', 'C. Turing', 'D. Pascal'],
213 |                 correct: 0,
214 |                 explanation: 'RTX 4090 dùng kiến trúc Ada Lovelace, ra mắt năm 2024',
215 |             },
216 |             {
217 |                 id: 'q8',
218 |                 text: 'FidelityFX Super Resolution (FSR) là công nghệ của hãng nào?',
219 |                 options: ['A. AMD', 'B. NVIDIA', 'C. Intel', 'D. Microsoft'],
220 |                 correct: 0,
```



```
2 2 1 | explanation: 'FSR là công nghệ upscaling ùc quyan cça AMD, Ñi
222 | },
223 | {
224 | id: 'q9',
2 2 5 | text: 'TDP (công suất tiêu thã) cça RTX 4090 không bao nhiêu?',
226 | options: ['A. 450W', 'B. 250W', 'C. 350W', 'D. 550W'],
227 | correct: 0,
2 2 8 | explanation: 'RTX 4090 có TDP 450W, yêu cầu nguồn tñi thiãu 850W
229 | },
230 | {
231 | id: 'q10',
2 3 2 | text: 'Ù phân giãi 4K có kích th°Ùc bao nhiêu pixel?',
233 | options: ['A. 3840x2160', 'B. 2560x1440', 'C. 1920x1080', 'D. 5120x2880'],
234 | correct: 0,
2 3 5 | explanation: '4K UHD có ù phân giãi 3840x2160, g¥p 4 lãn Full H
236 | },
237 | ]
238 | },
239 | 'motherboard': {
240 | id: 'motherboard',
2 4 1 | title: 'Bø Mich Chç & Chipset',
2 4 2 | icon: '0=Y',
243 | color: '#10b981',
2 4 4 | description: 'Kiøn théc vÃ mainboard, chipset, khe c¬m và kút nñi',
245 | questions: [
246 | {
247 | id: 'q1',
2 4 8 | text: 'Chipset Z790 dành cho CPU Intel thç hç nào?',
2 4 9 | options: ['A. Thç hç 12/13/14', 'B. Thç hç 10/11', 'C. Thç hç 8/
250 | correct: 0,
2 5 1 | explanation: 'Z790 hç trã Intel Core thç hç 12, 13 và 14 vôi kh
252 | },
253 | {
254 | id: 'q2',
2 5 5 | text: 'Form factor mainboard phổ biãn nhçt hiçn nay là gì?',
256 | options: ['A. ATX', 'B. Mini-ITX', 'C. E-ATX', 'D. Micro-ATX'],
257 | correct: 0,
2 5 8 | explanation: 'ATX là chuøn phổ biãn nhçt vôi kích th°Ùc 305x244m
259 | },
260 | {
261 | id: 'q3',
2 6 2 | text: 'Khe M.2 trên mainboard dùng ã kút nñi thiçt bẽ gì?',
263 | options: ['A. SSD NVMe', 'B. RAM', 'C. GPU', 'D. CPU'],
264 | correct: 0,
2 6 5 | explanation: 'Khe M.2 chç ýçu dùng cho SSD NVMe tñc ù cao'
266 | },
267 | {
268 | id: 'q4',
2 6 9 | text: 'VRM trên mainboard có chçc n ng gì?',
2 7 0 | options: ['A. C¥p içn Õn ãnh cho CPU', 'B. Tãn nhiçt chipset',
271 | correct: 0,
2 7 2 | explanation: 'VRM (Voltage Regulator Module) chuyãn Õi và Õn
273 | },
274 | {
275 | id: 'q5',
2 7 6 | text: 'Khe PCIe x16 th°Ýng òãc dùng cho thiçt bẽ nào?',
2 7 7 | options: ['A. Card Õ hĩa (GPU)', 'B. SSD M.2', 'C. RAM', 'D. Ca
278 | correct: 0,
2 7 9 | explanation: 'Khe PCIe x16 có b ng thõng cao nhçt, th°Ýng dùng
280 | },
281 | {
282 | id: 'q6',
2 8 3 | text: 'USB 3.2 Gen 2 có tñc ù truyãn tñi a là bao nhiêu?',
284 | options: ['A. 10 Gbps', 'B. 5 Gbps', 'C. 20 Gbps', 'D. 40 Gbps'],
285 | correct: 0,
2 8 6 | explanation: 'USB 3.2 Gen 2 có tñc ù 10 Gbps, g¥p Õi Gen 1'
287 | },
288 | {
289 | id: 'q7',
2 9 0 | text: 'BIOS là viçt t¬t cça thu¬t ngĩ nào?',
291 | options: ['A. Basic Input Output System', 'B. Binary Integrated Operating System', 'C. Base Input Outp...
292 | correct: 0,
2 9 3 | explanation: 'BIOS là Basic Input/Output System - hç thñng xuçt
294 | },
295 | {
296 | id: 'q8',
2 9 7 | text: 'Pin CMOS trên mainboard có tác dãn gì?',
2 9 8 | options: ['A. Giãi cài t BIOS khi t¬t máy', 'B. C¥p nguồn cho
299 | correct: 0,
3 0 0 | explanation: 'Pin CMOS giúp l°u cài t BIOS và thÝi gian hç thñ
301 | },
302 | {
303 | id: 'q9',
3 0 4 | text: 'Mainboard ATX th°Ýng có tñi a bao nhiêu khe RAM?',
305 | options: ['A. 4 khe', 'B. 2 khe', 'C. 6 khe', 'D. 8 khe'],
306 | correct: 0,
3 0 7 | explanation: 'Mainboard ATX tiçu chuøn th°Ýng có 4 khe RAM DIMM
308 | },
309 | {
310 | id: 'q10',
3 1 1 | text: 'Chuøn WiFi mÕi nhçt th°Ýng tích hãp trên mainboard cao cã
312 | options: ['A. WiFi 6E / WiFi 7', 'B. WiFi 5 (802.11ac)', 'C. WiFi 4 (802.11n)', 'D. Bluetooth 5.0'],
313 | correct: 0,
3 1 4 | explanation: 'Mainboard cao c¥p hiçn nay tích hãp WiFi 6E (6GHz)
315 | },
316 | ],
317 | },
318 | 'psu-cooling': {
319 | id: 'psu-cooling',
3 2 0 | title: 'Nguồn & Tãn Nhiçt',
3 2 1 | icon: 'Dp',
322 | color: '#06b6d4',
3 2 3 | description: 'Kiøn théc vÃ nguồn içn, tãn nhiçt, quit và quãn lý nh
324 | questions: [
325 | {
326 | id: 'q1',
3 2 7 | text: 'PSU ãt chuøn 80 Plus Gold có hiçu suất tñi thiãu là bao
328 | options: ['A. 87%', 'B. 80%', 'C. 90%', 'D. 85%'],
329 | correct: 0,
3 3 0 | explanation: '80 Plus Gold yêu cầu hiçu suất tñi thiãu 87% ß tã
331 | },
332 | {
333 | id: 'q2',
3 3 4 | text: 'Loiji tãn nhiçt nào hiçu quẽ nhçt cho CPU hiçu n ng cao?',
3 3 5 | options: ['A. Tãn nhiçt n°Ùc AIO', 'B. Tãn nhiçt khí (air cooler
336 | correct: 0,
3 3 7 | explanation: 'AIO water cooling hiçu quẽ hìn air cooler vôi CPU
338 | },
339 | {
340 | id: 'q3',
3 4 1 | text: 'Nguồn içn modular là gì?',
3 4 2 | options: ['A. Nguồn có dây cấp tháo rÝi òãc', 'B. Nguồn içn d
343 | correct: 0,
3 4 4 | explanation: 'PSU modular cho phép tháo rÝi dây cấp không dùng
```

```
345| },
346| {
347|   id: 'q4',
348|   text: 'Thermal Paste (keo tén nhiệ t) có tác dăng gì?',
349|   options: ['A. Truy Án nhiệ t tề CPU lên ử tén nhiệ t', 'B. D«n iC
350|   correct: 0,
351|   explanation: 'Thermal paste l¥p s¥ khe hỖ giũa CPU và tén nhiệ t
352| },
353| {
354|   id: 'q5',
355|   text: 'PSU ATX 3.0 h¥ trầ chu©n cáp nguốn mụi nào?',
356|   options: ['A. 12VHPWR (12+4 pin)', 'B. ATX12VO', 'C. SFX', 'D. EPS12V'],
357|   correct: 0,
358|   explanation: 'ATX 3.0 giũ thiụ chu©n 12VHPWR cho GPU th¿ hç m
359| },
360| {
361|   id: 'q6',
362|   text: 'Kích th° Ộc quit tén nhiệ t khi phỗ biẻn nh¥t là bao nhiẻu?',
363|   options: ['A. 120mm', 'B. 80mm', 'C. 140mm', 'D. 92mm'],
364|   correct: 0,
365|   explanation: 'Quit 120mm là kích th° Ộc tiẻu chu©n phỗ biẻn nh¥t
366| },
367| {
368|   id: 'q7',
369|   text: 'Nhiệ t Ừ CPU an toản khi chii game là bao nhiẻu?',
370|   options: ['A. 60-80°C', 'B. 90-100°C', 'C. 40-50°C', 'D. Trên 100°C'],
371|   correct: 0,
372|   explanation: 'CPU hoit Ừng an toản ở 60-80°C khi chii game'
373| },
374| {
375|   id: 'q8',
376|   text: 'Liquid Metal (kim loii lĩng) là gì?',
377|   options: ['A. Kem tén nhiệ t tề kim loii lĩng d«n nhiệ t siẻu tñt
378|   correct: 0,
379|   explanation: 'Liquid Metal là hấp kim lĩng có Ừ d«n nhiệ t r¥t c
380| },
381| {
382|   id: 'q9',
383|   text: 'Fan Controller có chẻc n ng gì?',
384|   options: ['A. iẢu chẻnh tñc Ừ quit thç công ho.c tñ Ừng', 'B.
385|   correct: 0,
386|   explanation: 'Fan Controller cho phẻp kiẢm soát chính xác tñc Ừ
387| },
388| {
389|   id: 'q10',
390|   text: 'PSU 850W ç cho c¥u hình PC nào sau Ầy?',
391|   options: ['A. RTX 4070 + Intel Core i5', 'B. RTX 4090 + Intel Core i9', 'C. RTX 3060 + Intel Core i3',...
392|   correct: 0,
393|   explanation: '850W ç cho tsm trung-cao nh° RTX 4070 + i5, nh°nç
394| },
395| },
396| },
397| 'storage-devices': {
398|   id: 'storage-devices',
399|   title: 'Ồ Cẻng SSD/HDD',
400|   icon: '0=0%',
401|   color: '#f59e0b',
402|   description: 'Kiẻn thẻc vẢ Ồ cẻng, SSD, HDD, chu©n k¿t nñi và l°u tr
403| questions: [
404|   {
405|     id: 'q1',
406|     text: 'SSD NVMe PCIe 4.0 có tñc Ừ íc tuşn tñ tñi a khoẻng bao
407|     options: ['A. 7000 MB/s', 'B. 500 MB/s', 'C. 2000 MB/s', 'D. 15000 MB/s'],
408|     correct: 0,
409|     explanation: 'SSD NVMe PCIe 4.0 í tñc Ừ íc ~7000 MB/s'
410|   },
411|   {
412|     id: 'q2',
413|     text: 'HDD 7200RPM có tñc Ừ íc tuşn tñ khoẻng bao nhiẻu?',
414|     options: ['A. ~160 MB/s', 'B. ~500 MB/s', 'C. ~50 MB/s', 'D. ~1000 MB/s'],
415|     correct: 0,
416|     explanation: 'HDD 7200RPM í tñc Ừ íc tuşn tñ khoẻng 160 MB/s
417|   },
418|   {
419|     id: 'q3',
420|     text: 'Chu©n SATA III có b ng thông tñi a là bao nhiẻu?',
421|     options: ['A. 6 Gbps', 'B. 3 Gbps', 'C. 12 Gbps', 'D. 1.5 Gbps'],
422|     correct: 0,
423|     explanation: 'SATA III có b ng thông 6 Gbps (~600 MB/s thñc t¿)
424|   },
425|   {
426|     id: 'q4',
427|     text: 'SSD NVMe PCIe 3.0 có tñc Ừ íc tuşn tñ khoẻng bao nhiẻu?
428|     options: ['A. ~3500 MB/s', 'B. ~7000 MB/s', 'C. ~10000 MB/s', 'D. ~2000 MB/s'],
429|     correct: 0,
430|     explanation: 'NVMe PCIe 3.0 í tñc Ừ ~3500 MB/s, b ng mùt nĩa
431|   },
432|   {
433|     id: 'q5',
434|     text: 'TBW là chẻ sñ o gì trên Ồ SSD?',
435|     options: ['A. Tuỗi thĩ ghi cça SSD (Terabytes Written)', 'B. Tñc
436|     correct: 0,
437|     explanation: 'TBW cho biẻt tỗng l°ăng đi liẻu có thẢ ghi vào SSD
438|   },
439|   {
440|     id: 'q6',
441|     text: 'Các loii NAND Flash phỗ biẻn trong SSD là gì?',
442|     options: ['A. TLC, QLC, MLC, SLC', 'B. DDR, SDR, NAND', 'C. NVMe, SATA, SAS', 'D. AIO, Air, Water'],
443|     correct: 0,
444|     explanation: 'Các loii NAND Flash gổm SLC, MLC, TLC, QLC vủi m-t
445|   },
446|   {
447|     id: 'q7',
448|     text: 'Dung l°ăng Ồ cẻng phỗ biẻn nh¥t cho PC ng°Ýi dùng hiẻn na
449|     options: ['A. 1TB - 2TB', 'B. 128GB - 256GB', 'C. 4TB - 8TB', 'D. 500GB'],
450|     correct: 0,
451|     explanation: '1-2TB là dung l°ăng phỗ biẻn, cân b±ng giũa giá và
452|   },
453|   {
454|     id: 'q8',
455|     text: 'Công nghç S.M.A.R.T trên Ồ cẻng dùng Ả làm gì?',
456|     options: ['A. Theo dõi và dñ oản tình trĩng sẻc khĩe Ồ cẻng',
457|     correct: 0,
458|     explanation: 'S.M.A.R.T theo dõi các chẻ sñ sẻc khĩe Ả cẻnh bảc
459|   },
460|   {
461|     id: 'q9',
462|     text: 'SSD khác HDD ở iẢm ci bên nào?',
463|     options: ['A. Không có )a quay ci hĩc', 'B. Dung l°ăng l°u trĩ
464|     correct: 0,
465|     explanation: 'SSD dùng chip nhũ Flash, không có )a quay nh° HDD
466|   },
467|   {
468|     id: 'q10',
```

```
4 6 9 | text: 'Chuẩn ổ cứng 2.5 inch thông dụng nào?',
4 7 0 | options: ['A. Laptop và ổ cứng di động', 'B. PC Desktop', 'C. Máy
471 | correct: 0, explanation: 'Ổ 2.5 inch phổ biến trong laptop và ổ cứng di
473 | },
474 | ],
475 | },
476 | 'pc-assembly': {
477 | id: 'pc-assembly',
478 | title: 'Quy Trình Lắp Ráp PC',
479 | icon: '🔧',
480 | color: '#f97316',
481 | description: 'Kiến thức về quy trình, lưu ý khi lắp ráp
482 | questions: [
483 | {
484 | id: 'q1',
485 | text: 'Bộ phận nào cần lắp vào mainboard?',
486 | options: ['A. Lắp CPU lên mainboard', 'B. Lắp nguồn (PSU) vào case',
487 | correct: 0, explanation: 'Nên lắp CPU lên mainboard trước khi lắp mainboard
488 | },
489 | {
490 | id: 'q2',
491 | text: 'Nên bôi thermal paste (keo tản nhiệt) ở đâu?',
492 | options: ['A. Trên IHS (nắp kim loại) của CPU', 'B. Trên chân socket',
493 | correct: 0, explanation: 'Bôi thermal paste lên giữa IHS CPU và tản nhiệt
494 | },
495 | {
496 | id: 'q3',
497 | text: 'Dây 24-pin ATX cấp nguồn cho thành phần nào?',
498 | options: ['A. Mainboard', 'B. CPU', 'C. GPU', 'D. Ổ cứng SSD'],
499 | correct: 0, explanation: 'Sử dụng 24-pin ATX cấp nguồn chính cho mainboard'
500 | },
501 | {
502 | id: 'q4',
503 | text: 'Dây EPS 8-pin (4+4) cấp nguồn cho thành phần nào?',
504 | options: ['A. CPU', 'B. GPU', 'C. Mainboard', 'D. Quạt case'],
505 | correct: 0, explanation: 'Cáp EPS 8-pin cung cấp nguồn riêng cho CPU'
506 | },
507 | {
508 | id: 'q5',
509 | text: 'Khi lắp RAM cần chú ý điều gì quan trọng?',
510 | options: ['A. Cẩn thận khay trên thanh RAM với khe cắm', 'B.
511 | correct: 0, explanation: 'RAM có khay chốt về vị trí cắm để tránh cắm sai
512 | },
513 | {
514 | id: 'q6',
515 | text: 'Tinh iChíp (ESD) có thể gây hại gì cho linh kiện PC?',
516 | options: ['A. Hỏng linh kiện vĩnh viễn', 'B. Tăng nhiệt độ linh kiện',
517 | correct: 0, explanation: 'Tinh iChíp có thể phá hủy linh kiện nhạy cảm như CPU
518 | },
519 | {
520 | id: 'q7',
521 | text: 'Công cụ nào giúp chỉnh tinh iChíp khi lắp ráp PC?',
522 | options: ['A. Dây đeo tay ESD (chống tĩnh điện)', 'B. Găng tay da',
523 | correct: 0, explanation: 'Dây đeo tay ESD giúp xả tĩnh điện để tránh
524 | },
525 | {
526 | id: 'q8',
527 | text: 'Sau khi lắp ráp xong PC, bộ phận nào cần kiểm tra đầu tiên?',
528 | options: ['A. Kiểm tra nguồn và xem có POST không', 'B. Cài đặt
529 | correct: 0, explanation: 'Kiểm tra POST (Power-On Self-Test) để xác nhận
530 | },
531 | {
532 | id: 'q9',
533 | text: 'POST là viết tắt của thuật ngữ nào?',
534 | options: ['A. Power-On Self-Test', 'B. Program Operating System Test', 'C. Processor Optimization Stan...
535 | correct: 0, explanation: 'POST là quá trình tự kiểm tra khi khởi động máy
536 | },
537 | {
538 | id: 'q10',
539 | text: 'Đèn LED trên mainboard báo lỗi nào?',
540 | options: ['A. Debug LED', 'B. RGB LED', 'C. Status LED', 'D. Power LED'],
541 | correct: 0, explanation: 'Debug LED hiển thị mã lỗi giúp chẩn đoán
542 | },
543 | },
544 | },
545 | },
546 | },
547 | },
548 | },
549 | },
550 | },
551 | },
552 | },
553 | },
554 | },
555 | },
556 | },
557 | },
558 | },
559 | },
560 | },
561 | },
562 | },
563 | },
564 | },
565 | },
566 | },
567 | },
568 | },
569 | },
570 | },
571 | },
572 | },
573 | },
574 | },
575 | },
576 | },
577 | },
578 | },
579 | },
580 | },
581 | },
582 | },
583 | },
584 | },
585 | },
586 | },
587 | },
588 | },
589 | },
590 | },
591 | },
592 | },
```

```
594| 5 9 3 | correct: 0, options: ['A. S-p x ập thê tñ °u tiên khBi ùng tē thiêt bē', 'B.
595| 5 9 5 | }, explanation: 'Boot Order xác ěnh thiêt bē nào °ăc khBi ùng t
596| 597 | { id: 'q6', text: 'Legacy Boot khác UEFI Boot nh° thđ nào?',
598| 5 9 9 | options: ['A. Legacy dùng MBR (ci), UEFI dùng GPT (hiçn ii)',
600| 6 0 0 | correct: 0, explanation: 'Legacy sí dăng bēng phân vùng MBR, UEFI sí dăng GR
601| 6 0 2 | },
603| 604 | { id: 'q7', text: 'CMOS Clear (hay Clear RTC) có tác dăng gì?',
605| 6 0 6 | options: ['A. Reset cài .t BIOS vẢ m.c ěnh', 'B. Xóa toàn bđ c
607| 6 0 7 | correct: 0, explanation: 'Clear CMOS giúp khôi phắc cài .t BIOS gñc khi g.p
609| 6 0 9 | },
610| 611 | { id: 'q8', text: 'Tính n ng BIOS Flashback cho phép làm gì?',
612| 6 1 3 | options: ['A. C-p nh-t BIOS mà không cñn CPU/RAM', 'B. KhBi ùng
614| 6 1 4 | correct: 0, explanation: 'Flashback cho phép níp BIOS ngay cē khi không có C
615| 6 1 6 | },
617| 618 | { id: 'q9', text: 'VT-d (Intel Virtualization Technology for Directed I/O) l
619| 6 2 0 | options: ['A. Intel', 'B. AMD', 'C. NVIDIA', 'D. Microsoft'],
621| 622 | correct: 0, explanation: 'VT-d là công nghç ěo hóa I/O cça Intel, AMD có IOM
623| 6 2 3 | },
624| 625 | { id: 'q10', text: 'Resizable BAR (Re-Size BAR) giúp cēi thiçn iẤu gì?',
626| 6 2 7 | options: ['A. CPU có thẢ truy c-p toàn bđ VRAM GPU', 'B. T ng d
628| 6 2 8 | correct: 0, explanation: 'Resizable BAR cho phép CPU truy c-p toàn bđ VRAM,
629| 6 3 0 | },
631| 632 | ],
633| 634 | },
635| 636 | 'network': {
637| 6 3 6 | id: 'network', title: 'Ming Máy Tính',
638| 6 3 7 | color: '#3b82f6', icon: '0<β',
639| 6 3 9 | description: 'Kiçn thêc vẢ ming máy tính, giao thêc, thiêt bē ming v
640| 641 | questions: [
642| 642 | { id: 'q1', text: 'IP là viçt t-t cça thu-t ngi nào?',
644| 644 | options: ['A. Internet Protocol', 'B. Internal Processing', 'C. Input Protocol', 'D. Integrated Progra...
645| 645 | correct: 0, explanation: 'IP (Internet Protocol) là giao thêc Internet cñt l
647| 648 | },
649| 649 | { id: 'q2', text: 'Cáp Ethernet Cat6 có tñc ù tñi a là bao nhiêu?',
651| 651 | options: ['A. 10 Gbps', 'B. 1 Gbps', 'C. 100 Mbps', 'D. 40 Gbps'],
652| 652 | correct: 0, explanation: 'Cat6 h x trẢ 10 Gbps ß khoēng cách tñi a 55m'
654| 655 | },
656| 656 | { id: 'q3', text: 'WiFi 6 (802.11ax) hoit ùng tñt ß đēi tñn nào?',
658| 658 | options: ['A. 2.4GHz và 5GHz', 'B. ChẾ 2.4GHz', 'C. ChẾ 5GHz',
659| 660 | explanation: 'WiFi 6 h x trẢ cē 2.4GHz và 5GHz, WiFi 6E thēm b ng
661| 662 | },
663| 663 | { id: 'q4', text: 'Router (bđ ěnh tuyçn) có chēc n ng chính là gì?',
665| 665 | options: ['A. ěnh tuyçn và chuyẢn tiçp gói tin giĩa các ming',
666| 667 | explanation: 'Router chuyẢn tiçp gói tin giĩa các ming khác nhau
668| 669 | },
670| 670 | { id: 'q5', text: 'B ng tñn WiFi 5GHz có °u iẤm gì so vđi 2.4GHz?',
672| 672 | options: ['A. Tñc ù cao hìn, ít nhiẤu hìn', 'B. KhẾ n ng xuyên
673| 674 | explanation: '5GHz có tñc ù cao và ít nhiẤu hìn nh°ng phim vi h
675| 676 | },
677| 677 | { id: 'q6', text: 'Bđ giao thêc TCP/IP là gì?',
679| 679 | options: ['A. Bđ giao thêc nẢn tēng cho Internet', 'B. Phñn mẢm
680| 681 | explanation: 'TCP/IP là bđ giao thêc làm nẢn tēng cho toàn bđ Ir
682| 683 | },
684| 684 | { id: 'q7', text: 'LAN là viçt t-t cça thu-t ngi nào?',
686| 686 | options: ['A. Local Area Network', 'B. Large Area Network', 'C. Logical Access Node', 'D. Link Access ...
687| 687 | correct: 0, explanation: 'LAN là ming các bđ kçt nñi các thiêt bē trong phim
689| 690 | },
691| 691 | { id: 'q8', text: 'Switch khác Hub ß iẤm quan trĩng nào?',
693| 693 | options: ['A. Switch chuyẢn micth thông minh, Hub kém hiçu quē',
694| 695 | explanation: 'Switch chẾ gíi dĩ liçu çn úng thiêt bē ích, còn
696| 697 | },
698| 698 | { id: 'q9', text: 'Ping là chẾ sñ o gì trong ming?',
700| 700 | options: ['A. ù trẢ (latency) kçt nñi ming', 'B. Tñc ù downloa
701| 702 | explanation: 'Ping o thỖi gian phēn hđi cça gói tin, tính bēng
703| 704 | },
705| 705 | { id: 'q10', text: 'DNS (Domain Name System) dùng Ả làm gì?',
707| 707 | options: ['A. Phân giēi tên miẢn thành Ả chẾ IP', 'B. Mã hóa c
708| 709 | explanation: 'DNS chuyẢn đđi tên miẢn (google.com) thành Ả chẾ
710| 711 | },
712| 713 | },
714| 714 | 'peripherals': {
715| 715 | id: 'peripherals', title: 'Thiêt Bē Ngoi Vi',
716| 716 | icon: '0=Y+β',
```

```
717| color: '#ec4899',
718|   description: 'Kiến thức về chuột, bàn phím, màn hình, webcam và thíc
719| questions: [
720|   {
721|     id: 'q1',
722|     text: 'Monitor có Response Time (thời gian đáp ứng), thấp mang l
723|     options: ['A. Giảm hiện tượng bóng mờ (ghosting)', 'B. Màu sắc h
724|     correct: 0,
725|     explanation: 'Response Time thấp giúp giảm bóng mờ khi chuyển ứ
726|   },
727|   {
728|     id: 'q2',
729|     text: 'Tần số quét (Hz) của màn hình ảnh hưởng đến điều gì?',
730|     options: ['A. Độ mượt của chuyển động trên màn hình', 'B. Độ ph
731|     correct: 0,
732|     explanation: 'Tần số quét cao (144Hz+) cho hình ảnh chuyển độn
733|   },
734|   {
735|     id: 'q3',
736|     text: 'Bàn phím cơ (Mechanical Keyboard) sử dụng, cơ chế gì?',
737|     options: ['A. Switch lò xo v-t lý cho từng phím', 'B. Màng cao su
738|     correct: 0,
739|     explanation: 'Bàn phím cơ dùng switch v-t lý riêng biệt cho m
740|   },
741|   {
742|     id: 'q4',
743|     text: 'DPI của chuột máy tính là chế độ gì?',
744|     options: ['A. Độ nhạy và tốc độ di chuyển con trỏ', 'B. Tốc độ
745|     correct: 0,
746|     explanation: 'DPI càng cao, con trỏ di chuyển càng xa khi chu
747|   },
748|   {
749|     id: 'q5',
750|     text: 'Webcam 1080p có độ phân giải là bao nhiêu?',
751|     options: ['A. 1920x1080', 'B. 1280x720', 'C. 3840x2160', 'D. 640x480'],
752|     correct: 0,
753|     explanation: '1080p Full HD có độ phân giải 1920x1080 pixel'
754|   },
755|   {
756|     id: 'q6',
757|     text: 'Cổng USB Type-C có tính reversible (đảo ngược) không?',
758|     options: ['A. Có', 'B. Không'],
759|     correct: 0,
760|     explanation: 'USB-C có thiết kế 7 pin xén, có thể cắm theo b
761|   },
762|   {
763|     id: 'q7',
764|     text: 'Bluetooth 5.0 có phạm vi kết nối tối đa trong nhà là bao
765|     options: ['A. ~40m', 'B. ~10m', 'C. ~50m', 'D. ~100m'],
766|     correct: 0,
767|     explanation: 'Bluetooth 5.0 có phạm vi lên tới 40m trong nhà (g
768|   },
769|   {
770|     id: 'q8',
771|     text: 'Cổng HDMI 2.1 hỗ trợ độ phân giải tối đa là bao nhiêu?',
772|     options: ['A. 8K @60Hz', 'B. 4K @60Hz', 'C. 1080p @240Hz', 'D. 5K @30Hz'],
773|     correct: 0,
774|     explanation: 'HDMI 2.1 hỗ trợ 8K @60Hz hoặc 4K @120Hz'
775|   },
776|   {
777|     id: 'q9',
778|     text: 'Cổng DisplayPort khác HDMI ở điểm chính nào?',
779|     options: ['A. Băng thông cao hơn, không thu phí bản quyền', 'B.
780|     correct: 0,
781|     explanation: 'DisplayPort thường có băng thông cao hơn và không
782|   },
783|   {
784|     id: 'q10',
785|     text: 'Microphone condenser (tĩnh điện) phù hợp cho mục đích nào?',
786|     options: ['A. Thu âm phòng thu chất lượng cao', 'B. Chơi game th
787|     correct: 0,
788|     explanation: 'Micro condenser có độ nhạy cao, phù hợp thu âm tr
789|   },
790| ],
791| os-windows: {
792|   id: 'os-windows',
793|   title: 'Hệ điều hành Windows',
794|   icon: '🖥️',
795|   color: '#0078d4',
796|   description: 'Kiến thức về hệ điều hành Windows, cài đặt, cấu hình v
797| questions: [
798|   {
799|     id: 'q1',
800|     text: 'Hệ điều hành là gì?',
801|     options: ['A. Phần mềm quản lý tài nguyên phần cứng và phần mềm',
802|     correct: 0,
803|     explanation: 'Hệ điều hành quản lý tài nguyên và cung cấp giao đ
804|   },
805|   {
806|     id: 'q2',
807|     text: 'Windows 11 yêu cầu tối thiểu RAM bao nhiêu?',
808|     options: ['A. 4GB', 'B. 8GB', 'C. 2GB', 'D. 16GB'],
809|     correct: 0,
810|     explanation: 'Windows 11 yêu cầu tối thiểu 4GB RAM'
811|   },
812|   {
813|     id: 'q3',
814|     text: 'File hệ thống quản trị của Windows là gì?',
815|     options: ['A. NTOSKRNL.EXE (Windows kernel)', 'B. WIN.COM', 'C. SYSTEM.INI', 'D. BOOT.INI'],
816|     correct: 0,
817|     explanation: 'NTOSKRNL.EXE là nhân (kernel) chính của Windows'
818|   },
819|   {
820|     id: 'q4',
821|     text: 'Công cụ nào dùng để phân vùng ổ cứng trong Windows?',
822|     options: ['A. Disk Management (diskmgmt.msc)', 'B. Task Manager', 'C. Device Manager', 'D. Registry Ed...'],
823|     correct: 0,
824|     explanation: 'Disk Management cho phép tạo, xóa, thu nhỏ và mở r
825|   },
826|   {
827|     id: 'q5',
828|     text: 'BSOD (Blue Screen of Death) thường xuất hiện khi nào?',
829|     options: ['A. Khi Windows gặp lỗi hệ thống nghiêm trọng', 'B. Khi
830|     correct: 0,
831|     explanation: 'BSOD xuất hiện khi hệ thống gặp lỗi không thể ph
832|   },
833|   {
834|     id: 'q6',
835|     text: 'Safe Mode trong Windows có tác dụng gì?',
836|     options: ['A. Khởi động với tối thiểu driver và dịch vụ để khắc
837|     correct: 0,
838|     explanation: 'Safe Mode giúp chẩn đoán và khắc phục lỗi cũ
839|   },
840| ],
```

```
841| {
842|   id: 'q7',
843|   text: 'NTFS là viết tắt của hệ thống file nào?',
844|   options: ['A. New Technology File System', 'B. Network File System', 'C. Windows File System', 'D. Nor...'],
845|   correct: 0,
846|   explanation: 'NTFS là hệ thống tệp tin công nghệ mới của Windows',
847| },
848| {
849|   id: 'q8',
850|   text: 'Công cụ nào dùng để kiểm tra lỗi cứng trong Windows?',
851|   options: ['A. CHKDSK (Check Disk)', 'B. DEFRAG', 'C. SFC (System File Checker)', 'D. DISM'],
852|   correct: 0,
853|   explanation: 'CHKDSK kiểm tra và sửa lỗi hệ thống tệp tin trên ổ đĩa',
854| },
855| {
856|   id: 'q9',
857|   text: 'Registry Editor (regedit) dùng để làm gì?',
858|   options: ['A. Chỉnh sửa cơ sở dữ liệu cấu hình Windows', 'B. Quản lý tài nguyên hệ thống', 'C. Sửa lỗi phần cứng', 'D. Cài đặt phần mềm'],
859|   correct: 0,
860|   explanation: 'Registry lưu trữ cấu hình hệ thống, phần mềm và người dùng',
861| },
862| {
863|   id: 'q10',
864|   text: 'Windows Update có chức năng gì?',
865|   options: ['A. Cập nhật bên và bảo mật và tính năng mới', 'B. Diệt virus', 'C. Kiểm tra ổ cứng', 'D. Tối ưu hóa ổ cứng'],
866|   correct: 0,
867|   explanation: 'Windows Update cung cấp bên và bảo mật, sửa lỗi và tính năng mới',
868| },
869| },
870| },
871| },
872| },
873| },
874| },
875| },
876| },
877| },
878| },
879| },
880| },
881| },
882| },
883| },
884| },
885| },
886| },
887| },
888| },
889| },
890| },
891| },
892| },
893| },
894| },
895| },
896| },
897| },
898| },
899| },
900| },
901| },
902| },
903| },
904| },
905| },
906| },
907| },
908| },
909| },
910| },
911| },
912| },
913| },
914| },
915| },
916| },
917| },
918| },
919| },
920| },
921| },
922| },
923| },
924| },
925| },
926| },
927| },
928| },
929| },
930| },
931| },
932| },
933| },
934| },
935| },
936| },
937| },
938| },
939| },
940| },
941| },
942| },
943| },
944| },
945| },
946| },
947| },
948| },
949| },
950| },
951| },
952| },
953| },
954| },
955| },
956| },
957| },
958| },
959| },
960| },
961| },
962| },
963| },
964| },
```

```
965|         id: 'q2',
966|         text: 'Soldered RAM (RAM hàn) trên laptop có .c iẤm gì?',
967|         options: ['A. ắc hàn cN Enh trên mainboard, không thA năng c',
968|         correct: 0,
969|         explanation: 'Soldered RAM ắc hàn chăt, không thA thay thê ho
970|         },
971|         {
972|         id: 'q3',
973|         text: 'Ultrabook là dòng laptop nh° thê nào?',
974|         options: ['A. Mĩng nh°, hiệu n ng cân b±ng, thYi l°ăng pin tNt',
975|         correct: 0,
976|         explanation: 'Ultrabook là tiêu chu©n Intel cho laptop mĩng nh°,
977|         },
978|         {
979|         id: 'q4',
980|         text: 'CỔng Thunderbolt 4 có b ng thông tNĩ a là bao nhiêu?',
981|         options: ['A. 40 Gbps', 'B. 20 Gbps', 'C. 10 Gbps', 'D. 80 Gbps'],
982|         correct: 0,
983|         explanation: 'Thunderbolt 4 và 3 Áu có b ng thông 40 Gbps'
984|         },
985|         {
986|         id: 'q5',
987|         text: 'TGP (Total Graphics Power) là chẾ sNĩ o gì trên laptop?',
988|         options: ['A. CỔng suYt tiêu thA cçạ GPU trên laptop', 'B. TNc
989|         correct: 0,
990|         explanation: 'TGP cho biết méc içn n ng GPU laptop ắc phép t
991|         },
992|         {
993|         id: 'q6',
994|         text: 'eGPU (External GPU) là gì?',
995|         options: ['A. Card ó hĩa rYi kút nNĩ qua Thunderbolt/USB4', 'B.
996|         correct: 0,
997|         explanation: 'eGPU cho phép laptop sí dăng GPU desktop mĩnh qua
998|         },
999|         {
1000|         id: 'q7',
1001|         text: 'Laptop gaming th°Yng có trĩng l°ăng khoēng bao nhiêu?',
1002|         options: ['A. 2-3 kg', 'B. 1-1.5 kg', 'C. 3-4 kg', 'D. D°úi 1 kg
1003|         correct: 0,
1004|         explanation: 'Laptop gaming n ng 2-3kg do tēn nhiçt và linh kiç
1005|         },
1006|         {
1007|         id: 'q8',
1008|         text: 'Pin Lithium Polymer (Li-Po) khác Lithium Ion (Li-Ion) B
1009|         options: ['A. D»o hĩn, nh° hĩn, thiçt kç linh hoit hĩn', 'B. Dur
1010|         correct: 0,
1011|         explanation: 'Li-Po có thA chĩ tĩo dĩng mĩng, d»o, nh° hĩn Li-Io
1012|         },
1013|         {
1014|         id: 'q9',
1015|         text: 'VRM trên laptop th°Yng có .c iẤm gì?',
1016|         options: ['A. ít pha hĩn và tēn nhiçt kém hĩn desktop', 'B. NhiA
1017|         correct: 0,
1018|         explanation: 'VRM laptop in giēn hĩn do giõi hĩn không gian và
1019|         },
1020|         {
1021|         id: 'q10',
1022|         text: 'Laptop workstation (nh° Dell Precision, HP ZBook) dùng ch
1023|         options: ['A. Ó hĩa chuyên nghiçp, CAD, render, khoa hĩc', 'B.
1024|         correct: 0,
1025|         explanation: 'Workstation ắc chēng nh-n ISV cho phsĩn mÁm chuyē
1026|         },
1027|     ],
1028|     'monitor-display': {
1029|     id: 'monitor-display',
1030|     title: 'Màn Hình & Công Nghç HiẤn Thē',
1031|     icon: '0=Y#p',
1032|     color: '#f43f5e',
1033|     description: 'Kiçn thēc vÁ màn hình, tYm nẤn, ù phân giēi và công r
1034|     questions: [
1035|     {
1036|     id: 'q1',
1037|     text: 'TYm nẤn IPS có °u iẤm chính là gì?',
1038|     options: ['A. Góc nhĩn rỪng, màu s_c chính xác', 'B. Tĩn sNĩ quết
1039|     correct: 0,
1040|     explanation: 'IPS nõi tiēng vỦi góc nhĩn rỪng (178°) và màu s_c
1041|     },
1042|     {
1043|     id: 'q2',
1044|     text: 'CỔng nghç OLED khác LCD B iẤm cĩ bēn nào?',
1045|     options: ['A. Mxi pixel tĩn phăt sắng, không cĩn ãn nẤn', 'B. Cĩ
1046|     correct: 0,
1047|     explanation: 'OLED có iẤm Enh tĩn phăt sắng, cho màu en sâu và
1048|     },
1049|     {
1050|     id: 'q3',
1051|     text: 'HDR (High Dynamic Range) là công nghç gì?',
1052|     options: ['A. Đēi t°ing phēn Ung rỪng, màu s_c sNĩng ùng hĩn',
1053|     correct: 0,
1054|     explanation: 'HDR mB rỪng đēi t°ing phēn và màu s_c so vỦi SDR'
1055|     },
1056|     {
1057|     id: 'q4',
1058|     text: 'FreeSync là công nghç óng bù hóa cçạ hăng nào?',
1059|     options: ['A. AMD', 'B. NVIDIA', 'C. Intel', 'D. Samsung'],
1060|     correct: 0,
1061|     explanation: 'FreeSync là công nghç chNĩng xē hĩnh cçạ AMD, dùng
1062|     },
1063|     {
1064|     id: 'q5',
1065|     text: 'G-Sync là công nghç óng bù hóa cçạ hăng nào?',
1066|     options: ['A. NVIDIA', 'B. AMD', 'C. Intel', 'D. LG'],
1067|     correct: 0,
1068|     explanation: 'G-Sync là công nghç ùc quyÁn cçạ NVIDIA, yêu cĩu
1069|     },
1070|     {
1071|     id: 'q6',
1072|     text: ' ù phç màu sRGB 100% ngh)a là gì?',
1073|     options: ['A. Màn hình hiẤn thē ắc toàn' bù đēi màu chu©n sRGB
1074|     correct: 0,
1075|     explanation: 'sRGB 100% ēm bēo tái tĩo chính xác đēi màu tiêu c
1076|     },
1077|     {
1078|     id: 'q7',
1079|     text: 'Màn hình 144Hz phù hăp nhYt cho Nĩ t°ăng nào?',
1080|     options: ['A. Game thç FPS (b-n sũng góc nhĩn thē nhYt)', 'B. NH
1081|     correct: 0,
1082|     explanation: 'Tĩn sNĩ quết cao rYt quan trĩng cho game FPS cĩn ph
1083|     },
1084|     {
1085|     id: 'q8',
1086|     text: 'Màn hình cong (Curved Monitor) có °u iẤm gì?',
1087|     options: ['A. Trēi nghiçm xem nh-p vai, giēm mĩi m't', 'B. Tiçt
```

```
1089| correct: 0, explanation: 'Màn hình cong ôm sát tầm nhìn, tăng cảm giác immersion'
1090| },
1091| {
1092| id: 'q9',
1093| text: 'Tỷ lệ màn hình 21:9 đặc gì?',
1094| options: ['A. Ultrawide', 'B. Standard', 'C. CinemaScope', 'D. Widescreen'],
1095| correct: 0, explanation: '21:9 là tỷ lệ ultrawide, rộng hơn 16:9 tiêu chuẩn'
1096| },
1097| {
1098| id: 'q10',
1099| text: 'Màn hình có độ phân giải 2560x1440 đặc gì?',
1100| options: ['A. 1440p (QHD / 2K)', 'B. 1080p (Full HD)', 'C. 4K (UHD)', 'D. 720p (HD)'],
1101| correct: 0, explanation: '2560x1440 là QHD (Quad HD) hay còn gọi là 2K'
1102| },
1103| },
1104| },
1105| },
1106| },
1107| },
1108| },
1109| },
1110| },
1111| },
1112| },
1113| },
1114| },
1115| },
1116| },
1117| },
1118| },
1119| },
1120| },
1121| },
1122| },
1123| },
1124| },
1125| },
1126| },
1127| },
1128| },
1129| },
1130| },
1131| },
1132| },
1133| },
1134| },
1135| },
1136| },
1137| },
1138| },
1139| },
1140| },
1141| },
1142| },
1143| },
1144| },
1145| },
1146| },
1147| },
1148| },
1149| },
1150| },
1151| },
1152| },
1153| },
1154| },
1155| },
1156| },
1157| },
1158| },
1159| },
1160| },
1161| },
1162| },
1163| },
1164| },
1165| },
1166| },
1167| },
1168| },
1169| },
1170| },
1171| },
1172| },
1173| },
1174| },
1175| },
1176| },
1177| },
1178| },
1179| },
1180| },
1181| },
1182| },
1183| },
1184| },
1185| },
1186| },
1187| },
1188| },
1189| },
1190| },
1191| },
1192| },
1193| },
1194| },
1195| },
1196| },
1197| },
1198| },
1199| },
1200| },
1201| },
1202| },
1203| },
1204| },
1205| },
1206| },
1207| },
1208| },
1209| },
1210| },
1211| },
1212| },
```



```
1214| 1 2 1 3 | explanation: 'Vcore là içñ áp lõi CPU, t ng Vcore giúp ðn Ænh
1215| 1214| },
1216| 1215| {
1217| 1 2 1 7 | id: 'q4',
1218| 1 2 1 8 | text: 'T ng içñ áp (Vcore) quá cao có nguy ci gi?',
1219| 1218| options: ['A. Gây hĩng CPU do quá nhiçt ho.c quá áp', 'B. T ng h
1220| 1 2 2 0 | correct: 0,
1221| 1219| explanation: 'Vcore quá cao gây nhiçt ù cao và có thÃ phá hçy C
1222| 1220| },
1223| 1221| {
1224| 1 2 2 5 | id: 'q5',
1225| 1223| text: 'Thermal Throttling là gì?',
1226| 1224| options: ['A. CPU tñ giẽm xung nhẽp khi quá nhiçt Æ bẽo vç', 'B
1227| 1 2 2 7 | correct: 0,
1228| 1226| explanation: 'Thermal throttling bẽo vç CPU khi h° hĩng do nhiçt
1229| 1227| },
1230| 1228| {
1231| 1 2 3 1 | id: 'q6',
1232| 1229| text: 'Precision Boost Overdrive (PBO) là công nghç ép xung tñ
1233| 1230| options: ['A. AMD', 'B. Intel', 'C. NVIDIA', 'D. MSI'],
1234| 1 2 3 4 | correct: 0,
1235| 1231| explanation: 'PBO là công nghç ép xung tñ ùng cça AMD dành cho
1236| 1234| },
1237| 1235| {
1238| 1 2 3 8 | id: 'q7',
1239| 1236| text: 'Intel XTU (Extreme Tuning Utility) là phñn mÁm gi?',
1240| 1237| options: ['A. Ép xung và tính chÆnh CPU Intel', 'B. Test hiçu n
1241| 1238| correct: 0,
1242| 1239| explanation: 'Intel XTU cho phép ép xung và theo dõi CPU Intel trên Windows'
1243| 1240| },
1244| 1241| {
1245| 1 2 4 5 | id: 'q8',
1246| 1242| text: 'Undervolting là kù thu-t gi?',
1247| 1243| options: ['A. Giẽm içñ áp CPU Æ giẽm nhiçt, v«n giĩ hiçu n ng
1248| 1 2 4 8 | correct: 0,
1249| 1244| explanation: 'Undervolting giúp giẽm nhiçt và tiçt kiçm içñ mÀ
1250| 1247| },
1251| 1248| {
1252| 1 2 5 2 | id: 'q9',
1253| 1249| text: 'Memory Timing (thông sñ thÝi gian RAM) là gi?',
1254| 1250| options: ['A. Các thông sñ ù trÃ cça RAM nh° CL, tRCD, tRP...',
1255| 1 2 5 5 | correct: 0,
1256| 1251| explanation: 'Timings là các thông sñ ù trÃ, càng thÝp càng nha
1257| 1254| },
1258| 1255| {
1259| 1 2 5 9 | id: 'q10',
1260| 1256| text: '"Silicon Lottery" (xõ sñ silicon) nói vÁ iÁu gi?',
1261| 1257| options: ['A. Chçt l°ãng chip khác nhau giãa các CPU cùng', loii',
1262| 1 2 6 2 | correct: 0,
1263| 1258| explanation: 'Mxi chip có chçt l°ãng silicon khác nhau, Ænh h°B
1264| 1261| },
1265| 1262| },
1266| 1263| },
1267| 1264| },
1268| 1265| },
1269| 1266| },
1270| 1267| },
1271| 1268| },
1272| 1269| },
1273| 1270| },
1274| 1271| },
1275| 1272| },
1276| 1273| },
1277| 1274| },
1278| 1275| },
1279| 1276| },
1280| 1277| },
1281| 1278| },
1282| 1279| },
1283| 1280| },
1284| 1281| },
1285| 1282| },
1286| 1283| },
1287| 1284| },
1288| 1285| },
1289| 1286| },
1290| 1287| },
1291| 1288| },
1292| 1289| },
1293| 1290| },
1294| 1291| },
1295| 1292| },
1296| 1293| },
1297| 1294| },
1298| 1295| },
1299| 1296| },
1300| 1297| },
1301| 1298| },
1302| 1299| },
1303| 1300| },
1304| 1301| },
1305| 1302| },
1306| 1303| },
1307| 1304| },
1308| 1305| },
1309| 1306| },
1310| 1307| },
1311| 1308| },
1312| 1309| },
1313| 1310| },
1314| 1311| },
1315| 1312| },
1316| 1313| },
1317| 1314| },
1318| 1315| },
1319| 1316| },
1320| 1317| },
1321| 1318| },
1322| 1319| },
1323| 1320| },
1324| 1321| },
1325| 1322| },
1326| 1323| },
1327| 1324| },
1328| 1325| },
1329| 1326| },
1330| 1327| },
1331| 1328| },
1332| 1329| },
1333| 1330| },
1334| 1331| },
1335| 1332| },
1336| 1333| },
```

```
1337|         id: 'q10',
1338|         text: 'Geekbench ảnh giá hiệ n ng ß khía c nh nào?',
1339|         options: ['A. CPU in nhâ n và a nhâ n a nẤn tẾng', 'B. GPU gami
1340|         correct: 0,
1341|         explanation: 'Geekbench cho iẤm in nhâ n và a nhâ n, so sá nh
1342|     },
1343| },
1344| },
1345| 'pc-gaming': {
1346|     id: 'pc-gaming',
1347|     title: 'CỰu Hì nh PC Gaming',
1348|     icon: '0<ß',
1349|     color: '#dc2626',
1350|     description: 'Kiến théc vẤ xây đĩ ng cỰu hì nh PC chi i game cá c tẾm gi
1351|     questions: [
1352|         {
1353|             id: 'q1',
1354|             text: 'Card ó hĩa tẾm trung phù hấ p cho PC gaming 2024 là gì?',
1355|             options: ['A. RTX 4060 / RX 7600', 'B. RTX 4090', 'C. GT 1030', 'D. RTX 3060'],
1356|             correct: 0,
1357|             explanation: 'RTX 4060 và RX 7600 là lĩ a chín tẾm trung tỄt nhỰt
1358|         },
1359|         {
1360|             id: 'q2',
1361|             text: 'CPU gaming ó ấ c ảnh giá tỄt nhỰt hiệ n nay cho game thỰ?
1362|             options: ['A. AMD Ryzen 7 7800X3D', 'B. Intel Core i9-14900K', 'C. AMD Ryzen 5 7600', 'D. Intel Core i...
1363|             correct: 0,
1364|             explanation: '7800X3D vỦi 3D V-Cache cho hiệ n ng gaming v° ấ t t
1365|         },
1366|         {
1367|             id: 'q3',
1368|             text: 'Dung l° ẩ ng RAM tỄi thi Ấu cho PC gaming hiệ n nay là bao nh
1369|             options: ['A. 16GB', 'B. 8GB', 'C. 32GB', 'D. 64GB'],
1370|             correct: 0,
1371|             explanation: '16GB là tỄi thi Ấu cho gaming hiệ n i, 32GB khuyê n
1372|         },
1373|         {
1374|             id: 'q4',
1375|             text: 'Lo i i SSD nào cỄn thiế t cho PC gaming Ấ load game nhanh?
1376|             options: ['A. NVMe PCIe 4.0 / 5.0', 'B. SATA SSD', 'C. HDD 7200RPM', 'D. NVMe PCIe 3.0'],
1377|             correct: 0,
1378|             explanation: 'NVMe PCIe 4.0 vỦi tỄc ù cao giúp giế m thỰi gian t
1379|         },
1380|         {
1381|             id: 'q5',
1382|             text: 'Cồ ng sỰt nguồ n phù hấ p cho PC gaming tẾm trung là bao nh
1383|             options: ['A. 650W - 750W', 'B. 450W - 550W', 'C. 850W - 1000W', 'D. 300W - 400W'],
1384|             correct: 0,
1385|             explanation: '650 - 750W Ặ cho cỰu hì nh tẾm trung vỦi RTX 4060 / 40
1386|         },
1387|         {
1388|             id: 'q6',
1389|             text: 'TẾn nhiế t cho CPU gaming nên chín lo i nào?',
1390|             options: ['A. AIO 240mm ho c tẾn khiế tower kếp cao cỰp', 'B. TẾn
1391|             correct: 0,
1392|             explanation: 'AIO 240mm ho c tower kếp Ặ cho hỰu hỰt CPU gaming
1393|         },
1394|         {
1395|             id: 'q7',
1396|             text: 'Mainboard cho PC gaming nên chín chipset nào?',
1397|             options: ['A. B650 (AMD) ho c Z790 (Intel)', 'B. H610 (Intel) r
1398|             correct: 0,
1399|             explanation: 'B650 và Z790 cân bằ ng giũa tĩ nh n ng và giá cho ga
1400|         },
1401|         {
1402|             id: 'q8',
1403|             text: 'Khi chín case cho PC gaming cỄn l° u ý iẤu gì nhỰt?',
1404|             options: ['A. Luồ ng khiế tỄt và Ặ khồ ng gian cho GPU dài', 'B. M
1405|             correct: 0,
1406|             explanation: 'Case cỄn airflow tỄt và Ặ dài cho GPU gaming hiệ n
1407|         },
1408|         {
1409|             id: 'q9',
1410|             text: 'GPU nào Ặ mĩ nh Ấ chi i game ß ù phâ n giế i 4K 60FPS?',
1411|             options: ['A. RTX 4080 trẾ lên', 'B. RTX 4060', 'C. RX 7600', 'D
1412|             correct: 0,
1413|             explanation: '4K gaming yê u cَسُول GPU cao cỰp nh° RTX 4080 / 4090 ho
1414|         },
1415|         {
1416|             id: 'q10',
1417|             text: 'TẾ lẶ phâ n bỖ ngâ n sá ch CPU/GPU hấ p lý cho PC gaming là b
1418|             options: ['A. 30% CPU, 40% GPU', 'B. 50% CPU, 30% GPU', 'C. 20% CPU, 60% GPU', 'D. 40% CPU, 40% GPU'],
1419|             correct: 0,
1420|             explanation: 'GPU chiế m phَسُول ngâ n sá ch vì quan tỄt nhỰt ch
1421|         },
1422|     ],
1423| },
1424| 'server-workstation': {
1425|     id: 'server-workstation',
1426|     title: 'Má y ChẶ & Workstation',
1427|     icon: '0<BẤ',
1428|     color: '#7c3aed',
1429|     description: 'Kiến théc vẤ má y chẶ, workstation, server hardware và
1430|     questions: [
1431|         {
1432|             id: 'q1',
1433|             text: 'CPU th° Ỗ ng dũ ng cho má y chẶ (server) là dòng nào?',
1434|             options: ['A. Intel Xeon / AMD EPYC', 'B. Intel Core i9', 'C. AMD Ryzen 9', 'D. Intel Core i5'],
1435|             correct: 0,
1436|             explanation: 'Xeon và EPYC ó ấ c thiế t kẾ cho server vỦi nhiẤu lẶ
1437|         },
1438|         {
1439|             id: 'q2',
1440|             text: 'ECC RAM khác RAM th° Ỗ ng (non-ECC) ß iẤm gì?',
1441|             options: ['A. Có khẾ n ng phấ t hiệ n và sĩa l×i dĩ liệ u tỄ ù ng',
1442|             correct: 0,
1443|             explanation: 'ECC tỄ ù ng sĩa l×i bit, rỰt quan tỄt cho server
1444|         },
1445|         {
1446|             id: 'q3',
1447|             text: 'RAID 1 là gì?',
1448|             options: ['A. Nhâ n bỄn (mirror) dĩ liệ u giũa 2 ổ cẾng', 'B. Dĩ p
1449|             correct: 0,
1450|             explanation: 'RAID 1 ghi dĩ liệ u giỄ ng hỰt lên 2 ổ, nếu 1 ổ hĩ ng
1451|         },
1452|         {
1453|             id: 'q4',
1454|             text: 'Uptime trong ngĩ cỆnh server là gì?',
1455|             options: ['A. ThỰi gian hẶ thỄ ng ho i t ù ng liê n tắ c khồ ng giá n
1456|             correct: 0,
1457|             explanation: 'Uptime o thỰi gian server ho i t ù ng khồ ng restart
1458|         },
1459|         {
1460|             id: 'q5',
```

```
1461 | text: 'Kích thước rack mount server là bao nhiêu inch?',
1462 | options: ['A. 1U, 2U, 4U', 'B. ATX, Micro-ATX', 'C. Inch', 'D. Mét'],
1463 | correct: 0,
1464 | explanation: '1U = 1.75 inch chiều cao rack server'
1465 | },
1466 | {
1467 | id: 'q6',
1468 | text: 'Tính năng Hot Swap cho phép làm gì?',
1469 | options: ['A. Thay ổ cứng khi hệ thống đang chạy mà không tắt máy', 'B. Thay ổ cứng khi hệ thống tắt máy', 'C. Thay ổ cứng khi hệ thống đang chạy mà không cần tháo ra', 'D. Thay ổ cứng khi hệ thống đang chạy mà không cần khởi động lại'],
1470 | correct: 0,
1471 | explanation: 'Hot swap cho phép thay ổ cứng nóng mà không cần tắt máy'
1472 | },
1473 | {
1474 | id: 'q7',
1475 | text: 'Virtualization ( ảo hóa ) trong server cho phép làm gì?',
1476 | options: ['A. Chạy nhiều máy chủ ảo trên 1 máy chủ vật lý', 'B. Chạy nhiều máy chủ vật lý trên 1 máy chủ ảo', 'C. Chạy nhiều máy chủ ảo trên nhiều máy chủ vật lý', 'D. Chạy nhiều máy chủ vật lý trên nhiều máy chủ ảo'],
1477 | correct: 0,
1478 | explanation: ' ảo hóa chia 1 server vật lý thành nhiều VM ảo lập trình'
1479 | },
1480 | {
1481 | id: 'q8',
1482 | text: 'NAS (Network Attached Storage) dùng để làm gì?',
1483 | options: ['A. Lưu trữ tập trung qua mạng cho nhiều người dùng', 'B. Lưu trữ tập trung qua mạng cho một người dùng', 'C. Lưu trữ tập trung qua mạng cho nhiều người dùng và chia sẻ file qua LAN', 'D. Lưu trữ tập trung qua mạng cho một người dùng và chia sẻ file qua LAN'],
1484 | correct: 0,
1485 | explanation: 'NAS là thiết bị lưu trữ mạng, chia sẻ file qua LAN'
1486 | },
1487 | {
1488 | id: 'q9',
1489 | text: 'GPU trong workstation chuyên nghiệp dùng dòng nào?',
1490 | options: ['A. NVIDIA RTX/Quadro, AMD Radeon Pro', 'B. Card gaming', 'C. Card workstation', 'D. Card server'],
1491 | correct: 0,
1492 | explanation: 'Card workstation có driver ISV certified cho ứng dụng chuyên nghiệp'
1493 | },
1494 | {
1495 | id: 'q10',
1496 | text: 'UPS (Uninterruptible Power Supply) có tác dụng gì?',
1497 | options: ['A. Cung cấp điện tạm thời khi mất điện, bảo vệ thiết bị', 'B. Cung cấp điện tạm thời khi mất điện, bảo vệ thiết bị và tăng tuổi thọ', 'C. Cung cấp điện tạm thời khi mất điện, bảo vệ thiết bị và tăng tuổi thọ và giảm chi phí', 'D. Cung cấp điện tạm thời khi mất điện, bảo vệ thiết bị và tăng tuổi thọ và giảm chi phí và tăng an toàn'],
1498 | correct: 0,
1499 | explanation: 'UPS giải quyết hệ thống chạy tạm thời và tăng an toàn'
1500 | },
1501 | },
1502 | },
1503 | 'cable-connectivity': {
1504 | id: 'cable-connectivity',
1505 | title: 'Cáp Kết Nối & Chuẩn Giao Tiếp',
1506 | icon: '🔌',
1507 | color: '#64748b',
1508 | description: 'Kiến thức về cáp kết nối, chuẩn giao tiếp phần cứng và phần mềm'
1509 | questions: [
1510 | {
1511 | id: 'q1',
1512 | text: 'Cáp dữ liệu SATA có bao nhiêu chân?',
1513 | options: ['A. 7 chân', 'B. 15 chân', 'C. 4 chân', 'D. 24 chân'],
1514 | correct: 0,
1515 | explanation: 'SATA data cable có 7 chân, còn SATA power có 15 chân'
1516 | },
1517 | {
1518 | id: 'q2',
1519 | text: 'Cáp nguồn SATA (SATA power) có bao nhiêu chân?',
1520 | options: ['A. 15 chân', 'B. 7 chân', 'C. 4 chân', 'D. 24 chân'],
1521 | correct: 0,
1522 | explanation: 'SATA power 15 chân cung cấp 3.3V, 5V và 12V'
1523 | },
1524 | {
1525 | id: 'q3',
1526 | text: 'Cáp HDMI có hỗ trợ truyền tín hiệu âm thanh không?',
1527 | options: ['A. Có, HDMI tích hợp cả âm thanh và hình ảnh', 'B. Không, HDMI chỉ truyền hình ảnh', 'C. Có, HDMI tích hợp cả âm thanh và hình ảnh', 'D. Không, HDMI chỉ truyền âm thanh'],
1528 | correct: 0,
1529 | explanation: 'HDMI truyền cả video và audio qua một cáp duy nhất'
1530 | },
1531 | {
1532 | id: 'q4',
1533 | text: 'Cáp quang (Fiber Optic) truyền tín hiệu bằng gì?',
1534 | options: ['A. Xung ánh sáng qua sợi thủy tinh', 'B. Xung điện qua dây đồng', 'C. Xung âm thanh qua dây đồng', 'D. Xung từ trường qua dây đồng'],
1535 | correct: 0,
1536 | explanation: 'Fiber optic dùng ánh sáng, cho tốc độ cao và không nhiễu'
1537 | },
1538 | {
1539 | id: 'q5',
1540 | text: 'Cổng USB 3.0 có màu sắc đặc trưng là gì?',
1541 | options: ['A. Màu xanh dương (blue)', 'B. Màu đen', 'C. Màu đỏ', 'D. Màu trắng'],
1542 | correct: 0,
1543 | explanation: 'USB 3.0/3.1 thường có màu xanh dương để phân biệt'
1544 | },
1545 | {
1546 | id: 'q6',
1547 | text: 'Cáp Thunderbolt 3 sử dụng đầu nối dạng nào?',
1548 | options: ['A. USB Type-C', 'B. Mini DisplayPort', 'C. USB Type-A', 'D. Lightning'],
1549 | correct: 0,
1550 | explanation: 'Thunderbolt 3 dùng đầu nối USB Type-C, tích hợp PCI Express và DisplayPort'
1551 | },
1552 | {
1553 | id: 'q7',
1554 | text: 'RGB header trên mainboard có mấy loại chính?',
1555 | options: ['A. 4-pin 12V RGB và 3-pin 5V ARGB', 'B. 2-pin duy nhất', 'C. 4-pin 12V RGB và 3-pin 5V ARGB', 'D. 4-pin 12V RGB và 3-pin 5V ARGB'],
1556 | correct: 0,
1557 | explanation: 'RGB (4-pin 12V) và ARGB (3-pin 5V) là 2 chuẩn phổ biến'
1558 | },
1559 | {
1560 | id: 'q8',
1561 | text: 'Fan header (sạc quạt) trên mainboard thường là loại nào?',
1562 | options: ['A. 4-pin PWM (điều chỉnh tốc độ)', 'B. 3-pin DC cố định', 'C. 4-pin PWM cho phép mainboard điều chỉnh tốc độ quạt', 'D. 3-pin DC cố định'],
1563 | correct: 0,
1564 | explanation: '4-pin PWM cho phép mainboard điều chỉnh tốc độ quạt'
1565 | },
1566 | {
1567 | id: 'q9',
1568 | text: 'Cáp mạng RJ45 (Ethernet) có bao nhiêu chân?',
1569 | options: ['A. 8 chân (4 cặp xoắn)', 'B. 4 chân', 'C. 6 chân', 'D. 10 chân'],
1570 | correct: 0,
1571 | explanation: 'Cáp RJ45 có 8 chân tương ứng 4 cặp dây xoắn'
1572 | },
1573 | {
1574 | id: 'q10',
1575 | text: 'Cable management (quản lý dây) trong case PC là gì?',
1576 | options: ['A. Sắp xếp dây cáp gọn gàng để tiết kiệm không gian', 'B. Sắp xếp dây cáp gọn gàng để tiết kiệm không gian và tăng tuổi thọ', 'C. Sắp xếp dây cáp gọn gàng để tiết kiệm không gian và tăng tuổi thọ và giảm chi phí', 'D. Sắp xếp dây cáp gọn gàng để tiết kiệm không gian và tăng tuổi thọ và giảm chi phí và tăng an toàn'],
1577 | correct: 0,
1578 | explanation: 'Cable management giúp case gọn gàng và thoáng khí'
1579 | },
1580 | },
1581 | },
1582 | 'virtual-reality': {
1583 | id: 'virtual-reality',
1584 | title: 'Thực Tế Ảo (VR) & AR',
```

```
1585 | icon: '0>Y} ',
1586 | color: '#d946ef',
1587 | description: 'Kiến thức về thực tế ảo, thực tế tăng cường, thiết bị
1588 | questions: [
1589 | {
1590 |   id: 'q1',
1591 |   text: 'VR là viết tắt của thu-tĩnh gì?',
1592 |   options: ['A. Virtual Reality (Thực tế ảo)', 'B. Visual Reality'],
1593 |   correct: 0,
1594 |   explanation: 'VR tạo ra môi trường hoàn toàn ảo bằng cách sử dụng thiết bị',
1595 | },
1596 | {
1597 |   id: 'q2',
1598 |   text: 'Headset VR phổ biến nhất hiện nay là gì?',
1599 |   options: ['A. Meta Quest 3, Valve Index, PlayStation VR2', 'B. Samsung Galaxy Watch', 'C. Apple Vision...'],
1600 |   correct: 0,
1601 |   explanation: 'Meta Quest 3 là headset VR phổ biến nhất với giá p
1602 | },
1603 | {
1604 |   id: 'q3',
1605 |   text: 'Cấu hình PC tối thiểu cho VR gaming là gì?',
1606 |   options: ['A. RTX 3060 + Core i5 + 16GB RAM', 'B. GTX 1650 + Core i3 + 8GB RAM'],
1607 |   correct: 0,
1608 |   explanation: 'VR yêu cầu GPU mạnh mẽ ít nhất 90 FPS để phân giải độ
1609 | },
1610 | {
1611 |   id: 'q4',
1612 |   text: 'Inside-out tracking trong VR là gì?',
1613 |   options: ['A. Camera trên headset theo dõi vị trí, không cần beacon ngoài', 'B. Beacon ngoài theo dõi vị trí, không cần camera trên headset'],
1614 |   correct: 0,
1615 |   explanation: 'Inside-out dùng camera tích hợp trên headset để theo dõi vị trí',
1616 | },
1617 | {
1618 |   id: 'q5',
1619 |   text: 'FOV (Field of View) trong VR là gì?',
1620 |   options: ['A. Trông nhìn, góc quan sát trong headset VR', 'B. Tốc độ xử lý đồ họa'],
1621 |   correct: 0,
1622 |   explanation: 'FOV thường là 100-120 độ, càng rộng càng nhập vai',
1623 | },
1624 | {
1625 |   id: 'q6',
1626 |   text: 'AR (Augmented Reality) khác VR (Virtual Reality) ở điểm nào?',
1627 |   options: ['A. AR thêm nội dung ảo vào thế giới thực, VR thay thế thế giới thực bằng ảo', 'B. AR chỉ dùng camera, VR cần camera và beacon'],
1628 |   correct: 0,
1629 |   explanation: 'AR chồng lớp thông tin ảo lên thế giới thực (Pokemon GO), VR thay thế thế giới thực bằng ảo (Pokemon VR)',
1630 | },
1631 | {
1632 |   id: 'q7',
1633 |   text: 'Ù trã (lag) cao trong VR có thể gây ra vấn đề gì?',
1634 |   options: ['A. Motion sickness (say tàu xe ảo), chóng mặt', 'B. Hư hỏng thiết bị'],
1635 |   correct: 0,
1636 |   explanation: 'VR yêu cầu ù trã cực thấp để tránh gây khó chịu',
1637 | },
1638 | {
1639 |   id: 'q8',
1640 |   text: 'Tần số quét (refresh rate) tối thiểu của headset VR chấ
1641 |   options: ['A. 90Hz', 'B. 60Hz', 'C. 120Hz', 'D. 144Hz'],
1642 |   correct: 0,
1643 |   explanation: '90Hz là tối thiểu, 120Hz trở lên cho trải nghiệm v
1644 | },
1645 | {
1646 |   id: 'q9',
1647 |   text: 'SteamVR là nền tảng phần mềm của hãng nào?',
1648 |   options: ['A. Valve', 'B. Meta (Facebook)', 'C. Microsoft', 'D. Sony'],
1649 |   correct: 0,
1650 |   explanation: 'SteamVR của Valve hỗ trợ nhiều headset và game VR',
1651 | },
1652 | {
1653 |   id: 'q10',
1654 |   text: 'Tính năng Passthrough trên headset VR cho phép làm gì?',
1655 |   options: ['A. Nhìn thấy thế giới thực qua camera mà không cần kính VR', 'B. Nhìn thấy thế giới ảo qua camera mà không cần kính VR'],
1656 |   correct: 0,
1657 |   explanation: 'Passthrough giúp người dùng quan sát môi trường thực tế',
1658 | },
1659 | ],
1660 | },
1661 | cybersecurity: {
1662 |   id: 'cybersecurity',
1663 |   title: 'An Ninh Mạng Cơ Bản',
1664 |   icon: '0=Y',
1665 |   color: '#059669',
1666 |   description: 'Kiến thức cơ bản về an ninh mạng, bảo mật và phòng chống tấn công',
1667 |   questions: [
1668 |     {
1669 |       id: 'q1',
1670 |       text: 'Firewall (tường lửa) có chức năng gì?',
1671 |       options: ['A. Kiểm soát và lọc lưu lượng mạng vào/ra', 'B. Tăng tốc độ mạng'],
1672 |       correct: 0,
1673 |       explanation: 'Firewall ngăn chặn truy cập trái phép từ/đến mạng',
1674 |     },
1675 |     {
1676 |       id: 'q2',
1677 |       text: 'Malware là gì?',
1678 |       options: ['A. Phần mềm độc hại gây hại cho máy tính', 'B. Phần mềm diệt virus'],
1679 |       correct: 0,
1680 |       explanation: 'Malware bao gồm virus, trojan, ransomware, spyware',
1681 |     },
1682 |     {
1683 |       id: 'q3',
1684 |       text: 'Phishing là hình thức tấn công nào?',
1685 |       options: ['A. Lừa đảo lấy thông tin nhạy cảm qua email/website giả', 'B. Tấn công bằng mã độc'],
1686 |       correct: 0,
1687 |       explanation: 'Phishing gửi email/tổ chức uy tín để lừa nạn nhân cung cấp thông tin',
1688 |     },
1689 |     {
1690 |       id: 'q4',
1691 |       text: 'Mật khẩu mạnh nên có những yếu tố nào?',
1692 |       options: ['A. Kết hợp chữ hoa, chữ thường, số, ký tự đặc biệt, độ dài', 'B. Chỉ có chữ hoa và số'],
1693 |       correct: 0,
1694 |       explanation: 'Mật khẩu mạnh khó đoán và khó bẻ brute-force',
1695 |     },
1696 |     {
1697 |       id: 'q5',
1698 |       text: '2FA (Two-Factor Authentication) là gì?',
1699 |       options: ['A. Xác thực 2 yếu tố: mật khẩu + mã xác minh 2', 'B. Xác thực 2 yếu tố: mật khẩu + mã QR'],
1700 |       correct: 0,
1701 |       explanation: '2FA thêm lớp bảo vệ thứ 2 (SMS, app authenticator, khóa vật lý)',
1702 |     },
1703 |     {
1704 |       id: 'q6',
1705 |       text: 'Ransomware là loại mã độc gì?',
1706 |       options: ['A. Mã độc mã hóa dữ liệu và đòi tiền chuộc', 'B. Phần mềm diệt virus'],
1707 |       correct: 0,
1708 |       explanation: 'Ransomware mã hóa file nạn nhân, yêu cầu trả tiền để lấy lại dữ liệu'
1709 |     }
1710 |   ]
1711 | }
```

```
1709| },
1710| {
1711|   id: 'q7',
1712|   text: 'VPN (Virtual Private Network) có tác dụng gì?',
1713|   options: ['A. Mã hóa kết nối Internet và truyền tải dữ liệu', 'B. Tạo ra kết nối mạng', 'C. Cho phép truy cập từ xa', 'D. Bảo vệ dữ liệu'],
1714|   correct: 0,
1715|   explanation: 'VPN tạo kênh mã hóa an toàn giữa thiết bị và server để truyền tải dữ liệu.',
1716| },
1717| {
1718|   id: 'q8',
1719|   text: 'Xác thực sinh trắc học (biometric) trên PC là gì?',
1720|   options: ['A. Nhận dạng vân tay, khuôn mặt hoặc giọng nói', 'B. Mật khẩu', 'C. Thẻ thông minh', 'D. Mã QR'],
1721|   correct: 0,
1722|   explanation: 'Windows Hello hỗ trợ vân tay và nhận diện khuôn mặt để đăng nhập.',
1723| },
1724| {
1725|   id: 'q9',
1726|   text: 'Cách phát hiện email lừa đảo (phishing) là gì?',
1727|   options: ['A. Kiểm tra địa chỉ email, liên kết, và nội dung', 'B. Không mở email từ người lạ', 'C. Chỉ mở email từ người quen', 'D. Không nhấp vào liên kết lạ'],
1728|   correct: 0,
1729|   explanation: 'Email giả mạo có thể lừa đảo, liên kết lạ và nội dung đáng ngờ là dấu hiệu.',
1730| },
1731| {
1732|   id: 'q10',
1733|   text: 'C-p nh-t ph-s-n m-á-m và h-c i-á-u hành th-ó-yng xuy-ê-n có tác dụng gì?',
1734|   options: ['A. Vá các lỗ hổng bảo mật để ngăn chặn tấn công', 'B. Làm chậm hệ thống', 'C. Tăng cường bảo mật', 'D. Giảm tải hệ thống'],
1735|   correct: 0,
1736|   explanation: 'C-p nh-t vá l-x h-ó-ng zero-day và b-ê-o v-ậ-t tr-ô-c các m-á-m.',
1737| },
1738| }
1739| }
1740| };
1741|
```

File: data/quiz-bank.ts (453 lines, 41.4 KB)

```
1| export const QUIZ_BANK = [
2|   {
3|     id: 'quiz-cpu-arch',
4|     title: 'Kiến Trúc CPU & Typology',
5|     lessonTitle: 'Bộ vi xử lý (CPU)',
6|     estimated_minutes: 15, difficulty: 'Trung bình', xp: 120,
7|     questions: [
8|       {
9|         q: 'CPU là vi xử lý trung tâm của máy tính, đúng hay sai?',
10|        options: ['Central Processing Unit', 'Cache', 'Motherboard', 'RAM'],
11|        correct: 1,
12|        explanation: 'CPU là bộ vi xử lý trung tâm của máy tính.',
13|      },
14|      {
15|        q: 'Cache là bộ nhớ tạm để lưu trữ dữ liệu, đúng hay sai?',
16|        options: ['Cache', 'CPU', 'Motherboard', 'RAM'],
17|        correct: 1,
18|        explanation: 'Cache là bộ nhớ tạm để lưu trữ dữ liệu.',
19|      },
20|      {
21|        q: 'CPU nào sau đây có hiệu suất cao nhất?',
22|        options: ['Intel Core i7', 'AMD Ryzen 7', 'Apple M1', 'ARM Cortex-A78'],
23|        correct: 0,
24|        explanation: 'Intel Core i7 là CPU hiệu suất cao nhất trong các lựa chọn.',
25|      },
26|      {
27|        q: 'Hyper-Threading là gì?',
28|        options: ['Hyper-Threading', 'CPU', 'Motherboard', 'RAM'],
29|        correct: 1,
30|        explanation: 'Hyper-Threading là công nghệ cho phép CPU hoạt động như có nhiều lõi hơn.',
31|      },
32|      {
33|        q: 'Socket LGA là gì?',
34|        options: ['Socket LGA', 'CPU', 'Motherboard', 'RAM'],
35|        correct: 1,
36|        explanation: 'Socket LGA là loại chân cắm cho CPU.',
37|      },
38|      {
39|        q: 'CPU AMD sử dụng công nghệ nào?',
40|        options: ['AMD', 'Intel', 'ARM', 'MIPS'],
41|        correct: 1,
42|        explanation: 'CPU AMD sử dụng công nghệ Zen.',
43|      },
44|      {
45|        q: 'Sinh nhân (core) của CPU Intel Core i5 là bao nhiêu?',
46|        options: ['CPU Intel Core i5', '4', '6', '8'],
47|        correct: 0,
48|        explanation: 'Số nhân của CPU Intel Core i5 có thể khác nhau tùy phiên bản.',
49|      },
50|    ],
51|   },
52|   {
53|     id: 'quiz-ram-deep',
54|     title: 'RAM & Bộ Nhớ Trong',
55|     lessonTitle: 'RAM & SSD',
56|     estimated_minutes: 15, difficulty: 'Trung bình', xp: 120,
57|     questions: [
58|       {
59|         q: 'RAM là vi xử lý trung tâm của máy tính, đúng hay sai?',
60|        options: ['Random Access Memory', 'CPU', 'Motherboard', 'SSD'],
61|        correct: 0,
62|        explanation: 'RAM là bộ nhớ tạm để lưu trữ dữ liệu.',
63|      },
64|      {
65|        q: 'DDR5 có tốc độ nhanh hơn DDR4, đúng hay sai?',
66|        options: ['DDR5', 'DDR4', 'DDR3', 'DDR2'],
67|        correct: 1,
68|        explanation: 'DDR5 có tốc độ nhanh hơn DDR4.',
69|      },
70|      {
71|        q: 'Dùng ổ cứng RAM để lưu trữ dữ liệu, đúng hay sai?',
72|        options: ['RAM', 'CPU', 'Motherboard', 'SSD'],
73|        correct: 0,
74|        explanation: 'Ổ cứng là để lưu trữ dữ liệu lâu dài.',
75|      },
76|      {
77|        q: 'CAS Latency (CL) là gì?',
78|        options: ['CAS Latency', 'CPU', 'Motherboard', 'RAM'],
79|        correct: 1,
80|        explanation: 'CAS Latency là thời gian để truy cập dữ liệu.',
81|      },
82|      {
83|        q: 'Dual Channel là gì?',
84|        options: ['Dual Channel', 'CPU', 'Motherboard', 'RAM'],
85|        correct: 1,
86|        explanation: 'Dual Channel là công nghệ để tăng tốc độ truy cập dữ liệu.',
87|      },
88|      {
89|        q: 'RAM DDR4 có bao nhiêu chân?',
90|        options: ['288 chân', '240 chân', '260 chân', '204 chân'],
91|        correct: 0,
92|        explanation: 'Số chân của RAM DDR4 có thể khác nhau tùy phiên bản.',
93|      },
94|      {
95|        q: 'RAM nào tích hợp bộ nhớ cache?',
96|        options: ['LPDDR5', 'DDR4', 'DDR3', 'DDR2'],
97|        correct: 1,
98|        explanation: 'LPDDR5 là RAM tích hợp bộ nhớ cache.',
99|      },
100|      {
101|        q: 'ECC RAM là gì?',
102|        options: ['ECC RAM', 'CPU', 'Motherboard', 'RAM'],
103|        correct: 1,
104|        explanation: 'ECC RAM là RAM có tính năng sửa lỗi.',
105|      },
106|      {
107|        q: 'XMP là công nghệ gì?',
108|        options: ['XMP', 'CPU', 'Motherboard', 'RAM'],
109|        correct: 1,
110|        explanation: 'XMP là công nghệ để tăng tốc độ RAM.',
111|      },
112|      {
113|        q: 'Dùng ổ cứng RAM để lưu trữ dữ liệu, đúng hay sai?',
114|        options: ['RAM', 'CPU', 'Motherboard', 'SSD'],
115|        correct: 0,
116|        explanation: 'Ổ cứng là để lưu trữ dữ liệu lâu dài.',
117|      },
118|    ],
119|   },
120|   {
121|     id: 'quiz-gpu-advanced',
122|     title: 'Card Đồ Họa & Xử Lý Đồ Họa',
123|     lessonTitle: 'Card Đồ Họa (GPU)',
124|     estimated_minutes: 20, difficulty: 'Khó', xp: 200,
125|     questions: [
126|       {
127|         q: 'GPU là vi xử lý trung tâm của máy tính, đúng hay sai?',
128|        options: ['Graphics Processing Unit', 'CPU', 'Motherboard', 'RAM'],
129|        correct: 0,
130|        explanation: 'GPU là bộ vi xử lý chuyên xử lý đồ họa.',
131|      },
132|      {
133|        q: 'VRAM là bộ nhớ tạm để lưu trữ dữ liệu, đúng hay sai?',
134|        options: ['Card đồ họa', 'CPU', 'Motherboard', 'RAM'],
135|        correct: 1,
136|        explanation: 'VRAM là bộ nhớ tạm để lưu trữ dữ liệu đồ họa.',
137|      },
138|      {
139|        q: 'Ray Tracing là công nghệ gì?',
140|        options: ['Ray Tracing', 'CPU', 'Motherboard', 'RAM'],
141|        correct: 1,
142|        explanation: 'Ray Tracing là công nghệ để tạo ra hình ảnh thực tế.',
143|      },
144|      {
145|        q: 'DLSS là công nghệ gì?',
146|        options: ['DLSS', 'CPU', 'Motherboard', 'RAM'],
147|        correct: 1,
148|        explanation: 'DLSS là công nghệ để tăng tốc độ xử lý đồ họa.',
149|      },
150|      {
151|        q: 'CUDA Core là gì?',
152|        options: ['CUDA Core', 'CPU', 'Motherboard', 'RAM'],
153|        correct: 1,
154|        explanation: 'CUDA Core là đơn vị xử lý đồ họa của NVIDIA.',
155|      },
156|      {
157|        q: 'Giao thức kết nối GPU và CPU là gì?',
158|        options: ['PCIe', 'CPU', 'Motherboard', 'RAM'],
159|        correct: 1,
160|        explanation: 'PCIe là giao thức kết nối GPU và CPU.',
161|      },
162|      {
163|        q: 'RTX 4090 dùng kiến trúc nào?',
164|        options: ['Ada Lovelace', 'Ampere', 'Turing', 'Volta'],
165|        correct: 1,
166|        explanation: 'RTX 4090 dùng kiến trúc Ada Lovelace.',
167|      },
168|      {
169|        q: 'FidelityFX Super Resolution là gì?',
170|        options: ['FidelityFX Super Resolution', 'CPU', 'Motherboard', 'RAM'],
171|        correct: 1,
172|        explanation: 'FidelityFX Super Resolution là công nghệ để tăng tốc độ xử lý đồ họa.',
173|      },
174|      {
175|        q: 'TDP của RTX 4090 là bao nhiêu?',
176|        options: ['450W', '250W', '150W', '75W'],
177|        correct: 0,
178|        explanation: 'TDP của RTX 4090 là 450W.',
179|      },
180|      {
181|        q: 'Dùng ổ cứng RAM để lưu trữ dữ liệu, đúng hay sai?',
182|        options: ['RAM', 'CPU', 'Motherboard', 'SSD'],
183|        correct: 0,
184|        explanation: 'Ổ cứng là để lưu trữ dữ liệu lâu dài.',
185|      },
186|    ],
187|   },
188|   {
189|     id: 'quiz-psu-cooling',
190|     title: 'Nguồn Điện & Tản Nhiệt',
191|     lessonTitle: 'Nguồn Điện & Tản Nhiệt',
192|     estimated_minutes: 20, difficulty: 'Trung bình', xp: 150,
193|     questions: [
194|       {
195|         q: 'PSU 80 Plus Gold có hiệu suất chuyển đổi bao nhiêu?',
196|        options: ['87%', '85%', '83%', '81%'],
197|        correct: 0,
198|        explanation: 'PSU 80 Plus Gold có hiệu suất chuyển đổi 87%.',
199|      },
200|      {
201|        q: 'Loi điện trong nguồn điện là gì?',
202|        options: ['Loi điện', 'CPU', 'Motherboard', 'RAM'],
203|        correct: 1,
204|        explanation: 'Loi điện là thành phần quan trọng trong nguồn điện.',
205|      },
206|      {
207|        q: 'Nhiệt độ tối đa của CPU là bao nhiêu?',
208|        options: ['90°C', '85°C', '80°C', '75°C'],
209|        correct: 0,
210|        explanation: 'Nhiệt độ tối đa của CPU có thể khác nhau tùy phiên bản.',
211|      },
212|      {
213|        q: 'Thermal Paste có tác dụng gì?',
214|        options: ['Thermal Paste', 'CPU', 'Motherboard', 'RAM'],
215|        correct: 1,
216|        explanation: 'Thermal Paste có tác dụng tản nhiệt cho CPU.',
217|      },
218|      {
219|        q: 'PSU ATX3.0 hỗ trợ tải trọng bao nhiêu?',
220|        options: ['12VHPWR', 'CPU', 'Motherboard', 'RAM'],
221|        correct: 1,
222|        explanation: 'PSU ATX3.0 hỗ trợ tải trọng 12VHPWR.',
223|      },
219|      {
220|        q: 'Kích thước ổ cứng tối đa của CPU là bao nhiêu?',
221|        options: ['120mm', '110mm', '100mm', '90mm'],
222|        correct: 0,
223|        explanation: 'Kích thước ổ cứng tối đa của CPU có thể khác nhau tùy phiên bản.',
224|      },
225|      {
226|        q: 'Nhiệt độ tối đa của CPU là bao nhiêu?',
227|        options: ['90°C', '85°C', '80°C', '75°C'],
228|        correct: 0,
229|        explanation: 'Nhiệt độ tối đa của CPU có thể khác nhau tùy phiên bản.',
230|      },
231|      {
232|        q: 'Liquid Metal là gì?',
233|        options: ['Liquid Metal', 'CPU', 'Motherboard', 'RAM'],
234|        correct: 1,
235|        explanation: 'Liquid Metal là chất lỏng tản nhiệt.',
236|      },
237|      {
238|        q: 'Fan Controller dùng để làm gì?',
239|        options: ['Fan Controller', 'CPU', 'Motherboard', 'RAM'],
240|        correct: 1,
241|        explanation: 'Fan Controller dùng để điều khiển quạt tản nhiệt.',
242|      },
243|      {
244|        q: 'PSU 850W cho phép tải trọng bao nhiêu?',
245|        options: ['RTX 4070', 'CPU', 'Motherboard', 'RAM'],
246|        correct: 1,
247|        explanation: 'PSU 850W cho phép tải trọng RTX 4070.',
248|      },
249|    ],
250|   },
251|   {
252|     id: 'quiz-motherboard',
253|     title: 'Mainboard & Kết Nối',
254|     lessonTitle: 'Bộ mạch chủ',
255|     estimated_minutes: 15, difficulty: 'Trung bình', xp: 120,
256|     questions: [
257|       {
258|         q: 'Chipset Z790 dành cho CPU Intel nào?',
259|        options: ['Thế hệ 12', 'Thế hệ 11', 'Thế hệ 10', 'Thế hệ 9'],
260|        correct: 1,
261|        explanation: 'Chipset Z790 dành cho CPU Intel thế hệ 12.',
262|      },
263|       {
264|         q: 'Form factor mainboard ATX là gì?',
265|        options: ['ATX', 'Micro-ATX', 'Mini-ITX', 'Flex-ATX'],
266|        correct: 1,
267|        explanation: 'Form factor mainboard ATX là tiêu chuẩn phổ biến nhất.',
268|      },
269|       {
270|         q: 'M.2 Slot dùng để kết nối gì?',
271|        options: ['SSD NVMe', 'CPU', 'Motherboard', 'RAM'],
272|        correct: 1,
273|        explanation: 'M.2 Slot dùng để kết nối SSD NVMe.',
274|      },
275|       {
276|         q: 'VRM trên mainboard có chức năng gì?',
277|        options: ['VRM', 'CPU', 'Motherboard', 'RAM'],
278|        correct: 1,
279|        explanation: 'VRM trên mainboard có chức năng điều chỉnh điện áp cho CPU.',
280|      },
281|       {
282|         q: 'PCIe x16 slot dùng để kết nối gì?',
283|        options: ['Card đồ họa', 'CPU', 'Motherboard', 'RAM'],
284|        correct: 1,
285|        explanation: 'PCIe x16 slot dùng để kết nối card đồ họa.',
286|      },
287|       {
288|         q: 'USB 3.2 Gen 2 có tốc độ truyền dữ liệu bao nhiêu?',
289|        options: ['10 Gbps', '5 Gbps', '1 Gbps', '0.5 Gbps'],
290|        correct: 1,
291|        explanation: 'USB 3.2 Gen 2 có tốc độ truyền dữ liệu 10 Gbps.',
292|      },
293|       {
294|         q: 'BIOS là vi xử lý trung tâm của máy tính, đúng hay sai?',
295|        options: ['Basic Input Output System', 'CPU', 'Motherboard', 'RAM'],
296|        correct: 0,
297|        explanation: 'BIOS là firmware để khởi động máy tính.',
298|      },
299|       {
300|         q: 'CMOS battery dùng để làm gì?',
301|        options: ['Giữ cài đặt BIOS', 'CPU', 'Motherboard', 'RAM'],
302|        correct: 1,
303|        explanation: 'CMOS battery dùng để giữ cài đặt BIOS.',
304|      },
305|    ],
306|   },
307| ]
```

[illegible]

```
2 1 2 | { q : 'Codec âm thanh AL220 có hàng nào?', options: ['Realtek', 'Creative', 'Sonic', 'Harman'], correct: 'Realtek' }
2 1 3 | { q : '3.5mm jack audio codecs aptX có ở đâu?', options: ['Loa', 'Tai nghe', 'Microphone', 'Đầu nối'], correct: 'Loa' }
2 1 4 | { q : 'Bluetooth audio codecs aptX có ở đâu?', options: ['Loa', 'Tai nghe', 'Microphone', 'Đầu nối'], correct: 'Loa' }
2 1 5 | { q : 'Soundbar là?', options: ['Thiết bị âm thanh', 'Thiết bị âm thanh', 'Thiết bị âm thanh', 'Thiết bị âm thanh'], correct: 'Thiết bị âm thanh' }
216 | }
217 | {
218 | {
219 | id: 'quiz-case',
2 2 0 | title: 'Case & Quit Ten Nhịt',
2 2 1 | lessonTitle: 'Viết case',
2 2 2 | estimated_minutes: 10, difficulty: 'DA', xp: 80,
223 | questions: [
2 2 4 | { q : 'Case ATX có mấy loại?', options: ['ATX', 'Micro-ATX', 'Mini-ITX', 'Flex ATX'], correct: 'ATX' }
2 2 5 | { q : 'Quit case có mấy loại?', options: ['Quit', 'Micro-Quit', 'Mini-Quit', 'Flex Quit'], correct: 'Quit' }
2 2 6 | { q : 'Airflow có mấy loại?', options: ['Airflow', 'Micro-Airflow', 'Mini-Airflow', 'Flex Airflow'], correct: 'Airflow' }
2 2 7 | { q : 'Ten nhiet có mấy loại?', options: ['Ten', 'Micro-Ten', 'Mini-Ten', 'Flex Ten'], correct: 'Ten' }
2 2 8 | { q : 'AIO là gì?', options: ['AIO', 'Micro-AIO', 'Mini-AIO', 'Flex AIO'], correct: 'AIO' }
2 2 9 | { q : 'Kịch bản có mấy loại?', options: ['Kịch bản', 'Micro-Kịch bản', 'Mini-Kịch bản', 'Flex Kịch bản'], correct: 'Kịch bản' }
2 3 0 | { q : 'Dust filter có mấy loại?', options: ['Dust filter', 'Micro-Dust filter', 'Mini-Dust filter', 'Flex Dust filter'], correct: 'Dust filter' }
2 3 1 | { q : 'Temperatures có mấy loại?', options: ['Temperatures', 'Micro-Temperatures', 'Mini-Temperatures', 'Flex Temperatures'], correct: 'Temperatures' }
2 3 2 | { q : 'Quiet có mấy loại?', options: ['Quiet', 'Micro-Quiet', 'Mini-Quiet', 'Flex Quiet'], correct: 'Quiet' }
2 3 3 | { q : 'Static Pressure có mấy loại?', options: ['Static Pressure', 'Micro-Static Pressure', 'Mini-Static Pressure', 'Flex Static Pressure'], correct: 'Static Pressure' }
234 | }
235 | {
236 | {
237 | id: 'quiz-benchmark',
2 3 8 | title: 'Benchmark & Kiếm Tra Hiệu Năng',
2 3 9 | lessonTitle: 'Kiếm tra hiệu năng',
240 | estimated_minutes: 15, difficulty: 'Trung bình', xp: 120,
241 | questions: [
2 4 2 | { q : 'Cinebench là gì?', options: ['Cinebench', 'Micro-Cinebench', 'Mini-Cinebench', 'Flex Cinebench'], correct: 'Cinebench' }
2 4 3 | { q : '3DMark là gì?', options: ['3DMark', 'Micro-3DMark', 'Mini-3DMark', 'Flex 3DMark'], correct: '3DMark' }
2 4 4 | { q : 'CrystalDiskMark là gì?', options: ['CrystalDiskMark', 'Micro-CrystalDiskMark', 'Mini-CrystalDiskMark', 'Flex CrystalDiskMark'], correct: 'CrystalDiskMark' }
2 4 5 | { q : 'FPS là gì?', options: ['FPS', 'Micro-FPS', 'Mini-FPS', 'Flex FPS'], correct: 'FPS' }
2 4 6 | { q : '1% Low FPS là gì?', options: ['1% Low FPS', 'Micro-1% Low FPS', 'Mini-1% Low FPS', 'Flex 1% Low FPS'], correct: '1% Low FPS' }
2 4 7 | { q : 'Stress test có mấy loại?', options: ['Stress test', 'Micro-Stress test', 'Mini-Stress test', 'Flex Stress test'], correct: 'Stress test' }
2 4 8 | { q : 'HWMonitor có mấy loại?', options: ['HWMonitor', 'Micro-HWMonitor', 'Mini-HWMonitor', 'Flex HWMonitor'], correct: 'HWMonitor' }
2 4 9 | { q : 'OCCT là gì?', options: ['OCCT', 'Micro-OCCT', 'Mini-OCCT', 'Flex OCCT'], correct: 'OCCT' }
2 5 0 | { q : 'PassMark có mấy loại?', options: ['PassMark', 'Micro-PassMark', 'Mini-PassMark', 'Flex PassMark'], correct: 'PassMark' }
2 5 1 | { q : 'Geekbench là gì?', options: ['Geekbench', 'Micro-Geekbench', 'Mini-Geekbench', 'Flex Geekbench'], correct: 'Geekbench' }
252 | }
253 | {
254 | {
255 | id: 'quiz-laptop',
2 5 6 | title: 'Laptop & Linh Kiện Di Động',
257 | lessonTitle: 'Laptop',
258 | estimated_minutes: 15, difficulty: 'Trung bình', xp: 120,
259 | questions: [
2 6 0 | { q : 'MXM là gì?', options: ['MXM', 'Micro-MXM', 'Mini-MXM', 'Flex MXM'], correct: 'MXM' }
2 6 1 | { q : 'Soldered RAM là gì?', options: ['Soldered RAM', 'Micro-Soldered RAM', 'Mini-Soldered RAM', 'Flex Soldered RAM'], correct: 'Soldered RAM' }
2 6 2 | { q : 'Laptop Ultrabook là gì?', options: ['Laptop Ultrabook', 'Micro-Laptop Ultrabook', 'Mini-Laptop Ultrabook', 'Flex Laptop Ultrabook'], correct: 'Laptop Ultrabook' }
2 6 3 | { q : 'Thunderbolt 4 có mấy loại?', options: ['Thunderbolt 4', 'Micro-Thunderbolt 4', 'Mini-Thunderbolt 4', 'Flex Thunderbolt 4'], correct: 'Thunderbolt 4' }
2 6 4 | { q : 'TGP là gì?', options: ['TGP', 'Micro-TGP', 'Mini-TGP', 'Flex TGP'], correct: 'TGP' }
2 6 5 | { q : 'eGPU là gì?', options: ['eGPU', 'Micro-eGPU', 'Mini-eGPU', 'Flex eGPU'], correct: 'eGPU' }
2 6 6 | { q : 'Laptop gaming là gì?', options: ['Laptop gaming', 'Micro-Laptop gaming', 'Mini-Laptop gaming', 'Flex Laptop gaming'], correct: 'Laptop gaming' }
2 6 7 | { q : 'Pin laptop có mấy loại?', options: ['Pin laptop', 'Micro-Pin laptop', 'Mini-Pin laptop', 'Flex Pin laptop'], correct: 'Pin laptop' }
2 6 8 | { q : 'VRM trên laptop là gì?', options: ['VRM', 'Micro-VRM', 'Mini-VRM', 'Flex VRM'], correct: 'VRM' }
2 6 9 | { q : 'Laptop workstation là gì?', options: ['Laptop workstation', 'Micro-Laptop workstation', 'Mini-Laptop workstation', 'Flex Laptop workstation'], correct: 'Laptop workstation' }
270 | }
271 | {
272 | {
273 | id: 'quiz-overclocking',
274 | title: 'Ép Xung & Tuning',
275 | lessonTitle: 'Ép xung & Tuning',
276 | estimated_minutes: 20, difficulty: 'Khó', xp: 200,
277 | questions: [
2 7 8 | { q : 'Overclocking là gì?', options: ['Overclocking', 'Micro-Overclocking', 'Mini-Overclocking', 'Flex Overclocking'], correct: 'Overclocking' }
2 7 9 | { q : 'CPU multiplier là gì?', options: ['CPU multiplier', 'Micro-CPU multiplier', 'Mini-CPU multiplier', 'Flex CPU multiplier'], correct: 'CPU multiplier' }
2 8 0 | { q : 'Vcore là gì?', options: ['Vcore', 'Micro-Vcore', 'Mini-Vcore', 'Flex Vcore'], correct: 'Vcore' }
2 8 1 | { q : 'ICN áp cao quá có thể gây ra gì?', options: ['ICN áp cao quá', 'Micro-ICN áp cao quá', 'Mini-ICN áp cao quá', 'Flex ICN áp cao quá'], correct: 'ICN áp cao quá' }
2 8 2 | { q : 'Thermal Throttling là gì?', options: ['Thermal Throttling', 'Micro-Thermal Throttling', 'Mini-Thermal Throttling', 'Flex Thermal Throttling'], correct: 'Thermal Throttling' }
2 8 3 | { q : 'PBO (Precision Boost Overdrive) là gì?', options: ['PBO', 'Micro-PBO', 'Mini-PBO', 'Flex PBO'], correct: 'PBO' }
2 8 4 | { q : 'Intel XTU là gì?', options: ['Intel XTU', 'Micro-Intel XTU', 'Mini-Intel XTU', 'Flex Intel XTU'], correct: 'Intel XTU' }
2 8 5 | { q : 'Undervolting là gì?', options: ['Undervolting', 'Micro-Undervolting', 'Mini-Undervolting', 'Flex Undervolting'], correct: 'Undervolting' }
2 8 6 | { q : 'Memory Timing là gì?', options: ['Memory Timing', 'Micro-Memory Timing', 'Mini-Memory Timing', 'Flex Memory Timing'], correct: 'Memory Timing' }
2 8 7 | { q : 'Silicon Lottery là gì?', options: ['Silicon Lottery', 'Micro-Silicon Lottery', 'Mini-Silicon Lottery', 'Flex Silicon Lottery'], correct: 'Silicon Lottery' }
288 | }
289 | {
290 | {
291 | id: 'quiz-cooling-advanced',
2 9 2 | title: 'Hệ Thống Tản Nhiệt Nâng Cao',
2 9 3 | lessonTitle: 'Hệ thống tản nhiệt nâng cao',
294 | estimated_minutes: 20, difficulty: 'Khó', xp: 200,
295 | questions: [
2 9 6 | { q : 'Custom Loop water cooling là gì?', options: ['Custom Loop water cooling', 'Micro-Custom Loop water cooling', 'Mini-Custom Loop water cooling', 'Flex Custom Loop water cooling'], correct: 'Custom Loop water cooling' }
2 9 7 | { q : 'Pump là gì?', options: ['Pump', 'Micro-Pump', 'Mini-Pump', 'Flex Pump'], correct: 'Pump' }
2 9 8 | { q : 'Reservoir là gì?', options: ['Reservoir', 'Micro-Reservoir', 'Mini-Reservoir', 'Flex Reservoir'], correct: 'Reservoir' }
2 9 9 | { q : 'Coolant là gì?', options: ['Coolant', 'Micro-Coolant', 'Mini-Coolant', 'Flex Coolant'], correct: 'Coolant' }
3 0 0 | { q : 'Fitting là gì?', options: ['Fitting', 'Micro-Fitting', 'Mini-Fitting', 'Flex Fitting'], correct: 'Fitting' }
3 0 1 | { q : 'Radiators có mấy loại?', options: ['Radiators', 'Micro-Radiators', 'Mini-Radiators', 'Flex Radiators'], correct: 'Radiators' }
3 0 2 | { q : 'Phân bổ dòng nước trong custom loop là gì?', options: ['Phân bổ dòng nước', 'Micro-Phân bổ dòng nước', 'Mini-Phân bổ dòng nước', 'Flex Phân bổ dòng nước'], correct: 'Phân bổ dòng nước' }
3 0 3 | { q : 'Water block là gì?', options: ['Water block', 'Micro-Water block', 'Mini-Water block', 'Flex Water block'], correct: 'Water block' }
3 0 4 | { q : 'So sánh custom loop vs AIO?', options: ['Custom loop', 'Micro-Custom loop', 'Mini-Custom loop', 'Flex Custom loop'], correct: 'Custom loop' }
3 0 5 | { q : 'So sánh custom loop vs AIO?', options: ['Custom loop', 'Micro-Custom loop', 'Mini-Custom loop', 'Flex Custom loop'], correct: 'Custom loop' }
306 | }
307 | {
308 | {
309 | id: 'quiz-pc-gaming',
3 1 0 | title: 'Cấu Hình PC Gaming',
311 | lessonTitle: 'PC Gaming',
312 | estimated_minutes: 15, difficulty: 'Trung bình', xp: 150,
313 | questions: [
3 1 4 | { q : 'PC gaming có mấy loại?', options: ['PC gaming', 'Micro-PC gaming', 'Mini-PC gaming', 'Flex PC gaming'], correct: 'PC gaming' }
3 1 5 | { q : 'CPU gaming là gì?', options: ['CPU gaming', 'Micro-CPU gaming', 'Mini-CPU gaming', 'Flex CPU gaming'], correct: 'CPU gaming' }
3 1 6 | { q : 'Dùng RAM nào cho gaming?', options: ['Dùng RAM', 'Micro-Dùng RAM', 'Mini-Dùng RAM', 'Flex Dùng RAM'], correct: 'Dùng RAM' }
3 1 7 | { q : 'SSD có nên chơi game?', options: ['SSD', 'Micro-SSD', 'Mini-SSD', 'Flex SSD'], correct: 'SSD' }
3 1 8 | { q : 'PSU cho PC gaming là gì?', options: ['PSU', 'Micro-PSU', 'Mini-PSU', 'Flex PSU'], correct: 'PSU' }
3 1 9 | { q : 'Ten nhiet cho CPU gaming là gì?', options: ['Ten nhiet', 'Micro-Ten nhiet', 'Mini-Ten nhiet', 'Flex Ten nhiet'], correct: 'Ten nhiet' }
3 2 0 | { q : 'Mainboard cho CPU gaming là gì?', options: ['Mainboard', 'Micro-Mainboard', 'Mini-Mainboard', 'Flex Mainboard'], correct: 'Mainboard' }
3 2 1 | { q : 'Case gaming có mấy loại?', options: ['Case gaming', 'Micro-Case gaming', 'Mini-Case gaming', 'Flex Case gaming'], correct: 'Case gaming' }
3 2 2 | { q : 'GPU nào chơi 4K FPS tốt?', options: ['GPU', 'Micro-GPU', 'Mini-GPU', 'Flex GPU'], correct: 'GPU' }
3 2 3 | { q : 'Tết lễ CPU/GPU budget là gì?', options: ['Tết lễ CPU/GPU budget', 'Micro-Tết lễ CPU/GPU budget', 'Mini-Tết lễ CPU/GPU budget', 'Flex Tết lễ CPU/GPU budget'], correct: 'Tết lễ CPU/GPU budget' }
324 | }
325 | {
326 | {
327 | id: 'quiz-server',
3 2 8 | title: 'Máy Chủ & Workstation',
3 2 9 | lessonTitle: 'Máy chủ & Workstation',
330 | estimated_minutes: 15, difficulty: 'Khó', xp: 200,
331 | questions: [
3 3 2 | { q : 'CPU server là gì?', options: ['CPU server', 'Micro-CPU server', 'Mini-CPU server', 'Flex CPU server'], correct: 'CPU server' }
3 3 3 | { q : 'ECC RAM khác non-ECC ở đâu?', options: ['ECC RAM', 'Micro-ECC RAM', 'Mini-ECC RAM', 'Flex ECC RAM'], correct: 'ECC RAM' }
3 3 4 | { q : 'RAID 1 là gì?', options: ['RAID 1', 'Micro-RAID 1', 'Mini-RAID 1', 'Flex RAID 1'], correct: 'RAID 1' }
3 3 5 | { q : 'Uptime là gì?', options: ['Uptime', 'Micro-Uptime', 'Mini-Uptime', 'Flex Uptime'], correct: 'Uptime' }

```



```

3 | url: string
4 | logo: string
5 | type: 'online' | 'retail' | 'marketplace'
6 | rating: number
7 | }
8 |
9 | export const SHOPS: ShopInfo[] = [
10 | { name: 'GearVN', url: 'https://gearvn.com', logo: '/shops/gearvn.png', type: 'online', rating: 4.8 },
11 |   { name: 'Phong Vi', url: 'https://phongvu.vn', logo: '/shops/phongvu.png', type: 'online', rating: 4.6 },
12 | { name: 'An Phát', url: 'https://anphatpc.com.vn', logo: '/shops/anphat.png', type: 'online', rating: 4.6 },
13 | { name: 'Hoàng Hà', url: 'https://hoanghacpc.com.vn', logo: '/shops/hoangha.png', type: 'online', rating: 4.5 },
14 | { name: 'ThinkPro', url: 'https://thinkpro.vn', logo: '/shops/thinkpro.png', type: 'online', rating: 4.6 },
15 | { name: 'CellphoneS', url: 'https://cellphones.com.vn', logo: '/shops/cellphones.png', type: 'online', rating: 4.4 },
16 | { name: 'FPT Shop', url: 'https://fptshop.com.vn', logo: '/shops/fpt.png', type: 'online', rating: 4.3 },
17 | ]
18 |
19 | export interface ProductLink {
20 |   shop: string
21 |   url: string
22 |   price?: number
23 | }
24 |
25 | export interface ComponentShops {
26 |   [componentId: string]: ProductLink[]
27 | }
28 |
29 | const SEARCH_URLS: Record<string, (query: string) => string> = {
30 |   'GearVN': (q) => `https://gearvn.com/search?q=${encodeURIComponent(q)}`,
31 |   'Phong Vi': (q) => `https://phongvu.vn/search?q=${encodeURIComponent(q)}`,
32 |   'An Phát': (q) => `https://anphatpc.com.vn/tim-kiem?keyword=${encodeURIComponent(q)}`,
33 |   'Hoàng Hà': (q) => `https://hoanghacpc.com.vn/search?q=${encodeURIComponent(q)}`,
34 |   'ThinkPro': (q) => `https://thinkpro.vn/search?q=${encodeURIComponent(q)}`,
35 |   'CellphoneS': (q) => `https://cellphones.com.vn/tim-kiem?q=${encodeURIComponent(q)}`,
36 |   'FPT Shop': (q) => `https://fptshop.com.vn/tim-kiem?q=${encodeURIComponent(q)}`,
37 | }
38 |
39 | export function getComponentSearchLinks(name: string, type: string): ProductLink[] {
40 |   const query = `${name} ${type}`
41 |   const shops = ['GearVN', 'Phong Vi', 'An Phát', 'Hoàng Hà', 'ThinkPro', 'CellphoneS', 'FPT Shop']
42 |   return shops.map(shop => ({
43 |     shop,
44 |     url: SEARCH_URLS[shop](query),
45 |   }))
46 | }
47 |
48 | export const CATEGORY_LINKS: Record<string, ProductLink[]> = {
49 |   CPU: [
50 |     { shop: 'GearVN', url: 'https://gearvn.com/collections/cpu-bo-vi-xu-ly' },
51 |     { shop: 'Phong Vi', url: 'https://phongvu.vn/c/linh-kien-may-tinh/cpu-bo-vi-xu-ly' },
52 |     { shop: 'An Phát', url: 'https://anphatpc.com.vn/cpu-bo-vi-xu-ly' },
53 |     { shop: 'Hoàng Hà', url: 'https://hoanghacpc.com.vn/cpu-bo-vi-xu-ly' },
54 |     { shop: 'ThinkPro', url: 'https://thinkpro.vn/cpu-bo-vi-xu-ly' },
55 |     { shop: 'CellphoneS', url: 'https://cellphones.com.vn/linh-kien/cpu.html' },
56 |   ],
57 |   GPU: [
58 |     { shop: 'GearVN', url: 'https://gearvn.com/collections/vga-card-man-hinh' },
59 |     { shop: 'Phong Vi', url: 'https://phongvu.vn/c/linh-kien-may-tinh/vga-card-man-hinh' },
60 |     { shop: 'An Phát', url: 'https://anphatpc.com.vn/vga-card-man-hinh' },
61 |     { shop: 'Hoàng Hà', url: 'https://hoanghacpc.com.vn/vga-card-man-hinh' },
62 |     { shop: 'ThinkPro', url: 'https://thinkpro.vn/vga-card-man-hinh' },
63 |     { shop: 'CellphoneS', url: 'https://cellphones.com.vn/linh-kien/vga.html' },
64 |   ],
65 |   RAM: [
66 |     { shop: 'GearVN', url: 'https://gearvn.com/collections/ram-desktop' },
67 |     { shop: 'Phong Vi', url: 'https://phongvu.vn/c/linh-kien-may-tinh/ram-desktop' },
68 |     { shop: 'An Phát', url: 'https://anphatpc.com.vn/ram-may-tinh' },
69 |     { shop: 'Hoàng Hà', url: 'https://hoanghacpc.com.vn/ram' },
70 |     { shop: 'ThinkPro', url: 'https://thinkpro.vn/ram' },
71 |     { shop: 'CellphoneS', url: 'https://cellphones.com.vn/linh-kien/ram.html' },
72 |   ],
73 |   Mainboard: [
74 |     { shop: 'GearVN', url: 'https://gearvn.com/collections/mainboard-bo-mach-chu' },
75 |     { shop: 'Phong Vi', url: 'https://phongvu.vn/c/linh-kien-may-tinh/mainboard-bo-mach-chu' },
76 |     { shop: 'An Phát', url: 'https://anphatpc.com.vn/mainboard-bo-mach-chu' },
77 |     { shop: 'Hoàng Hà', url: 'https://hoanghacpc.com.vn/mainboard' },
78 |     { shop: 'ThinkPro', url: 'https://thinkpro.vn/mainboard' },
79 |   ],
80 |   Storage: [
81 |     { shop: 'GearVN', url: 'https://gearvn.com/collections/ssd-o-cung-ran' },
82 |     { shop: 'Phong Vi', url: 'https://phongvu.vn/c/linh-kien-may-tinh/ssd-o-cung-ran' },
83 |     { shop: 'An Phát', url: 'https://anphatpc.com.vn/o-cung-ssd' },
84 |     { shop: 'Hoàng Hà', url: 'https://hoanghacpc.com.vn/ssd' },
85 |     { shop: 'ThinkPro', url: 'https://thinkpro.vn/o-cung-ssd' },
86 |     { shop: 'CellphoneS', url: 'https://cellphones.com.vn/linh-kien/ssd.html' },
87 |   ],
88 |   PSU: [
89 |     { shop: 'GearVN', url: 'https://gearvn.com/collections/psu-nguon-may-tinh' },
90 |     { shop: 'Phong Vi', url: 'https://phongvu.vn/c/linh-kien-may-tinh/psu-nguon-may-tinh' },
91 |     { shop: 'An Phát', url: 'https://anphatpc.com.vn/nguon-may-tinh-psu' },
92 |     { shop: 'Hoàng Hà', url: 'https://hoanghacpc.com.vn/nguon-may-tinh' },
93 |     { shop: 'ThinkPro', url: 'https://thinkpro.vn/nguon-may-tinh' },
94 |   ],
95 |   Cooler: [
96 |     { shop: 'GearVN', url: 'https://gearvn.com/collections/tan-nhiet' },
97 |     { shop: 'Phong Vi', url: 'https://phongvu.vn/c/linh-kien-may-tinh/tan-nhiet' },
98 |     { shop: 'An Phát', url: 'https://anphatpc.com.vn/tan-nhiet' },
99 |     { shop: 'Hoàng Hà', url: 'https://hoanghacpc.com.vn/tan-nhiet' },
100 |     { shop: 'ThinkPro', url: 'https://thinkpro.vn/tan-nhiet' },
101 |   ],
102 |   Case: [
103 |     { shop: 'GearVN', url: 'https://gearvn.com/collections/case-vo-may-tinh' },
104 |     { shop: 'Phong Vi', url: 'https://phongvu.vn/c/linh-kien-may-tinh/case-vo-may-tinh' },
105 |     { shop: 'An Phát', url: 'https://anphatpc.com.vn/case-vo-may-tinh' },
106 |     { shop: 'Hoàng Hà', url: 'https://hoanghacpc.com.vn/case-vo-may-tinh' },
107 |     { shop: 'ThinkPro', url: 'https://thinkpro.vn/case-vo-may-tinh' },
108 |   ],
109 |   Monitor: [
110 |     { shop: 'GearVN', url: 'https://gearvn.com/collections/man-hinh' },
111 |     { shop: 'Phong Vi', url: 'https://phongvu.vn/c/man-hinh-may-tinh' },
112 |     { shop: 'An Phát', url: 'https://anphatpc.com.vn/man-hinh' },
113 |     { shop: 'Hoàng Hà', url: 'https://hoanghacpc.com.vn/man-hinh' },
114 |     { shop: 'ThinkPro', url: 'https://thinkpro.vn/man-hinh' },
115 |     { shop: 'CellphoneS', url: 'https://cellphones.com.vn/man-hinh-may-tinh.html' },
116 |     { shop: 'FPT Shop', url: 'https://fptshop.com.vn/man-hinh-may-tinh' },
117 |   ],
118 | }
119 |
120 | export function getCategoryLinks(type: string): ProductLink[] {
121 |   return CATEGORY_LINKS[type] || CATEGORY_LINKS['CPU']
122 | }
123 |

```

File: database/03_progress_sample.sql (55 lines, 2.0 KB)

```
-----
1 | -- Script to sample data cho trang Ti ın Ü H ıc T - p ( / student / progress )
2 | -- T H A Y T H % < student _ u u i d _ h e r e > ` b ą n g I D t h ń c t ı c ą a b ı n t ę b ę n g a u t h .
3 |
4 | DO $$
5 | DECLARE
6 |     student _ u i d U U I D := ' 0 0 0 0 0 0 0 0 - 0 0 0 0 - 0 0 0 0 - 0 0 0 0 - 0 0 0 0 0 0 0 0 0 0 0 0 ' ; -- i ń n I D
7 | BEGIN
8 |
9 |     -- 1 . X ỏ a d ỉ l ỉ ợ u m « u c ỉ ( n ỉ u c ỏ )
10 | DELETE FROM builder_sessions WHERE student_id = student_uid;
11 | DELETE FROM quiz_attempts WHERE student_id = student_uid;
12 | DELETE FROM lesson_progress WHERE student_id = student_uid;
13 |
14 | -- 2 . T h ę m d ỉ l ỉ ợ u l e s s o n _ p r o g r e s s ( 3 0 n g ą y g ỉ n ą y )
15 | FOR i IN 1..25 LOOP
16 |     INSERT INTO lesson_progress (student_id, lesson_id, status, completion_percentage, time_spent_seconds, las...
17 |     VALUES (
18 |         student_uid,
19 |         gen_random_uuid(), -- Mock lesson ID
20 |         CASE WHEN i <= 15 THEN 'completed' WHEN i <= 20 THEN 'in_progress' ELSE 'not_started' END,
21 |         CASE WHEN i <= 15 THEN 100 WHEN i <= 20 THEN (i * 5) ELSE 0 END,
22 |         CASE WHEN i <= 15 THEN 1200 + (i * 60) WHEN i <= 20 THEN 300 ELSE 0 END,
23 |         NOW() - (i || ' days')::INTERVAL,
24 |         CASE WHEN i <= 15 THEN NOW() - (i || ' days')::INTERVAL ELSE NULL END
25 |     );
26 | END LOOP;
27 |
28 | -- 3 . T h ę m d ỉ l ỉ ợ u q u i z _ a t t e m p t s
29 | FOR i IN 1..5 LOOP
30 |     INSERT INTO quiz_attempts (id, student_id, quiz_id, status, score, created_at)
31 |     VALUES (
32 |         gen_random_uuid(),
33 |         student_uid,
34 |         gen_random_uuid(),
35 |         'graded',
36 |         ( 6 0 + ( i * 8 ) ) , -- i ń m : 6 8 , 7 6 , 8 4 , 9 2 , 1 0 0
37 |         NOW() - (i * 2 || ' days')::INTERVAL
38 |     );
39 | END LOOP;
40 |
41 | -- 4 . T h ę m d ỉ l ỉ ợ u b u i l d e r _ s e s s i o n s ( H e a t m a p 9 0 n g ą y )
42 | FOR i IN 1..40 LOOP
43 |     INSERT INTO builder_sessions (student_id, started_at, ended_at, components_used, tdp_calculated, compati...
44 |     VALUES (
45 |         student_uid,
46 |         NOW() - (i * 2 || ' days')::INTERVAL,
47 |         (NOW() - (i * 2 || ' days')::INTERVAL) + ((10 + (i % 50)) || ' minutes')::INTERVAL,
48 |         '[]::jsonb',
49 |         350.5,
50 |         100
51 |     );
52 | END LOOP;
53 |
54 | END $$;
55 |
-----
```

File: database/20260513_dashboard_system.sql (1 lines, 0.0 KB)

```
-----
1 | last_accessed
-----
```

File: database/20260513_quiz_module.sql (132 lines, 5.2 KB)

```
-----
1 | -- =====
2 | -- QUIZ & QUESTION BANK MODULE SETUP
3 | -- =====
4 |
5 | -- 1 . Q u e s t i o n B a n k s ( N g ą n h ą n g ń t h ỉ )
6 | CREATE TABLE IF NOT EXISTS public.question_banks (
7 |     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
8 |     teacher_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE NOT NULL,
9 |     title TEXT NOT NULL,
10 |     subject TEXT,
11 |     created_at TIMESTAMPTZ DEFAULT NOW()
12 | );
13 |
14 | -- 2 . Q u e s t i o n s ( C ą u h ń i )
15 | CREATE TABLE IF NOT EXISTS public.questions (
16 |     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
17 |     bank_id UUID REFERENCES public.question_banks(id) ON DELETE CASCADE NOT NULL,
18 |     content TEXT NOT NULL,
19 |     type TEXT NOT NULL CHECK (type IN ('multiple_choice', 'true_false', 'fill_blank')),
20 |     points INTEGER DEFAULT 1,
21 |     "order" INTEGER DEFAULT 0,
22 |     created_at TIMESTAMPTZ DEFAULT NOW()
23 | );
24 |
25 | -- 3 . Q u e s t i o n O p t i o n s ( ń p ń n t r ń c n g h ỉ ợ m )
26 | CREATE TABLE IF NOT EXISTS public.question_options (
27 |     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
28 |     question_id UUID REFERENCES public.questions(id) ON DELETE CASCADE NOT NULL,
29 |     content TEXT NOT NULL,
30 |     is_correct BOOLEAN DEFAULT FALSE,
31 |     created_at TIMESTAMPTZ DEFAULT NOW()
32 | );
33 |
34 | -- 4 . Q u i z z e s ( B ą i k ỉ ń m t r ą )
35 | CREATE TABLE IF NOT EXISTS public.quizzes (
36 |     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
37 |     lesson_id UUID REFERENCES public.lessons(id) ON DELETE SET NULL, -- L ỉ
38 |     assignment_id UUID REFERENCES public.assignments(id) ON DELETE SET NULL
39 |     title TEXT NOT NULL,
40 |     time_limit_minutes INTEGER DEFAULT 15,
41 |     passing_score INTEGER DEFAULT 5,
42 |     randomize BOOLEAN DEFAULT TRUE,
43 |     max_attempts INTEGER DEFAULT 1,
44 |     created_at TIMESTAMPTZ DEFAULT NOW()
45 | );
46 |
47 | -- 5 . Q u i z A t t e m p t s ( L ỉ n l ą m b ą i )
48 | CREATE TABLE IF NOT EXISTS public.quiz_attempts (
49 |     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
50 |     quiz_id UUID REFERENCES public.quizzes(id) ON DELETE CASCADE NOT NULL,
51 |     student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE NOT NULL,
52 |     started_at TIMESTAMPTZ DEFAULT NOW(),
53 |     submitted_at TIMESTAMPTZ,
54 |     score INTEGER DEFAULT 0,
55 |     status TEXT DEFAULT 'in_progress' CHECK (status IN ('in_progress', 'submitted', 'timed_out'))
56 | );
57 |
-----
```

```

58 | -- 6. Quiz Answers (Chị t i ết bài làm)
59 | CREATE TABLE IF NOT EXISTS public.quiz_answers (
60 |   id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
61 |   attempt_id UUID REFERENCES public.quiz_attempts(id) ON DELETE CASCADE NOT NULL,
62 |   question_id UUID REFERENCES public.questions(id) ON DELETE CASCADE NOT NULL,
63 |   selected_option_id UUID REFERENCES public.question_options(id) ON DELETE SET NULL,
64 |   text_answer TEXT, -- Dùng cho fill_blank
65 |   is_correct BOOLEAN DEFAULT FALSE,
66 |   points_earned INTEGER DEFAULT 0
67 | );
68 |
69 | -- =====
70 | -- ROW LEVEL SECURITY (RLS)
71 | -- =====
72 |
73 | ALTER TABLE public.question_banks ENABLE ROW LEVEL SECURITY;
74 | ALTER TABLE public.questions ENABLE ROW LEVEL SECURITY;
75 | ALTER TABLE public.question_options ENABLE ROW LEVEL SECURITY;
76 | ALTER TABLE public.quizzes ENABLE ROW LEVEL SECURITY;
77 | ALTER TABLE public.quiz_attempts ENABLE ROW LEVEL SECURITY;
78 | ALTER TABLE public.quiz_answers ENABLE ROW LEVEL SECURITY;
79 |
80 | -- POLICIES FOR QUESTION BANKS (Teacher only)
81 | CREATE POLICY "Teachers can manage their own question banks" ON public.question_banks
82 |   FOR ALL USING (teacher_id = auth.uid());
83 |
84 | -- POLICIES FOR QUESTIONS (Inherit from Bank)
85 | CREATE POLICY "Teachers can manage questions in their banks" ON public.questions
86 |   FOR ALL USING (
87 |     EXISTS (SELECT 1 FROM public.question_banks WHERE id = bank_id AND teacher_id = auth.uid())
88 |   );
89 |
90 | -- POLICIES FOR OPTIONS (Inherit from Question)
91 | CREATE POLICY "Teachers can manage options for their questions" ON public.question_options
92 |   FOR ALL USING (
93 |     EXISTS (
94 |       SELECT 1 FROM public.questions q
95 |       JOIN public.question_banks b ON q.bank_id = b.id
96 |       WHERE q.id = question_id AND b.teacher_id = auth.uid()
97 |     )
98 |   );
99 |
100 | -- Students can only read options for quizzes they are part of (but NOT is_correct)
101 | CREATE POLICY "Students can view options" ON public.question_options
102 |   FOR SELECT USING (true); -- We will filter is_correct in the API/Server Action
103 |
104 | -- POLICIES FOR QUIZZES
105 | CREATE POLICY "Teachers can manage their quizzes" ON public.quizzes
106 |   FOR ALL USING (
107 |     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
108 |   );
109 |
110 | CREATE POLICY "Students can view quizzes" ON public.quizzes
111 |   FOR SELECT USING (true);
112 |
113 | -- POLICIES FOR ATTEMPTS
114 | CREATE POLICY "Students can manage own attempts" ON public.quiz_attempts
115 |   FOR ALL USING (student_id = auth.uid());
116 |
117 | CREATE POLICY "Teachers can view class attempts" ON public.quiz_attempts
118 |   FOR SELECT USING (
119 |     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
120 |   );
121 |
122 | -- POLICIES FOR ANSWERS
123 | CREATE POLICY "Students can manage own answers" ON public.quiz_answers
124 |   FOR ALL USING (
125 |     EXISTS (SELECT 1 FROM public.quiz_attempts WHERE id = attempt_id AND student_id = auth.uid())
126 |   );
127 |
128 | CREATE POLICY "Teachers can view answers" ON public.quiz_answers
129 |   FOR SELECT USING (
130 |     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
131 |   );
132 |

```

File: database/20260516_certificates_system.sql (18 lines, 0.8 KB)

```

1 | -- 1. Bảng Chứng chỉ (Certificates)
2 | CREATE TABLE IF NOT EXISTS public.certificates (
3 |   id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
4 |   student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE NOT NULL,
5 |   course_name TEXT NOT NULL,
6 |   issue_date TIMESTAMPTZ DEFAULT NOW(),
7 |   certificate_url TEXT, -- Link ến file PDF ho ặc trang web xác minh
8 |   verification_code TEXT UNIQUE NOT NULL
9 | );
10 |
11 | -- RLS cho certificates
12 | ALTER TABLE public.certificates ENABLE ROW LEVEL SECURITY;
13 | CREATE POLICY "Users can view their own certificates" ON public.certificates
14 |   FOR SELECT USING (auth.uid() = student_id);
15 |
16 | -- 2. C ập nh ập c ột Onboarded cho Profile (nếu command tr ớc ch ả a ch ị y)
17 | ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS is_onboarded BOOLEAN DEFAULT FALSE;
18 |

```

File: database/20260516_dashboard_views.sql (48 lines, 1.8 KB)

```

1 | -- 1. View tính toán tỉ lệ ừ h ịc t - p c ả a t ề ng h ịc sinh
2 | -- D ữ a tr ên s ố l ợ a ăng bài h ịc ả ho àn th ành và quiz ấ i t
3 | CREATE OR REPLACE VIEW public.student_course_progress AS
4 | SELECT
5 |   p.id as student_id,
6 |   p.full_name,
7 |   l.id as lesson_id,
8 |   l.title as lesson_title,
9 |   EXISTS (
10 |     SELECT 1 FROM quiz_attempts qa
11 |     JOIN quizzes q ON qa.quiz_id = q.id
12 |     WHERE qa.student_id = p.id
13 |     AND q.lesson_id = l.id
14 |     AND qa.score >= q.passing_score
15 |   ) as is_completed,
16 |   COALESCE(
17 |     (SELECT MAX(score) FROM quiz_attempts qa
18 |      JOIN quizzes q ON qa.quiz_id = q.id
19 |      WHERE qa.student_id = p.id AND q.lesson_id = l.id),
20 |     0
21 |   ) as best_score
22 | FROM profiles p

```

```

23| CROSS JOIN lessons l;
24|
25| -- 2. View tổng hợp hiệu suất lớp học cho Giáo viên
26| CREATE OR REPLACE VIEW public.teacher_class_stats AS
27| SELECT
28|     c.id as class_id,
29|     c.name as class_name,
30|     COUNT(DISTINCT cm.student_id) as total_students,
31|     AVG(qa.score) as avg_quiz_score,
32|     COUNT(DISTINCT qa.id) as total_submissions
33| FROM classes c
34| LEFT JOIN class_members cm ON c.id = cm.class_id
35| LEFT JOIN quizzes q ON q.lesson_id IS NOT NULL -- Gié ãnh quiz g-n v-oi
36| LEFT JOIN quiz_attempts qa ON qa.quiz_id = q.id AND qa.student_id = cm.student_id
37| GROUP BY c.id, c.name;
38|
39| -- 3. Bëng l-ou v-ot hoit ùng g-s-n -ây (Activity Log)
40| CREATE TABLE IF NOT EXISTS public.activity_logs (
41|     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
42|     user_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
43|     action_type TEXT NOT NULL, -- 'login', 'complete_lesson', 'submit_quiz', 'join_class'
44|     description TEXT,
45|     metadata JSONB,
46|     created_at TIMESTAMPTZ DEFAULT NOW()
47| );
48|

```

File: database/20260516_discussion_system.sql (19 lines, 1.0 KB)

```

=====
1| -- 1. ãm b-ou b-ëng bình l-u-n t-ôn t-i và có c-ýu trúc phân t-sng
2| CREATE TABLE IF NOT EXISTS public.lesson_comments (
3|     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
4|     lesson_id UUID REFERENCES public.lessons(id) ON DELETE CASCADE NOT NULL,
5|     user_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE NOT NULL,
6|     parent_id UUID REFERENCES public.lesson_comments(id) ON DELETE CASCADE
7|     content TEXT NOT NULL,
8|     created_at TIMESTAMPTZ DEFAULT NOW()
9| );
10|
11| -- B-t Realtime cho bình l-u-n
12| ALTER PUBLICATION supabase_realtime ADD TABLE lesson_comments;
13|
14| -- RLS cho bình l-u-n
15| ALTER TABLE public.lesson_comments ENABLE ROW LEVEL SECURITY;
16| CREATE POLICY "Anyone can view comments" ON public.lesson_comments FOR SELECT USING (true);
17| CREATE POLICY "Authenticated users can insert comments" ON public.lesson_comments FOR INSERT WITH CHECK (auth....
18| CREATE POLICY "Users can delete their own comments" ON public.lesson_comments FOR DELETE USING (auth.uid() = u...
19|

```

File: database/20260516_notifications_system.sql (32 lines, 1.4 KB)

```

=====
1| -- 1. B-ëng thông báo (Notifications)
2| CREATE TABLE IF NOT EXISTS public.notifications (
3|     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
4|     user_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE NOT NULL,
5|     title TEXT NOT NULL,
6|     message TEXT NOT NULL,
7|     type TEXT DEFAULT 'info' CHECK (type IN ('info', 'success', 'warning', 'error', 'assignment', 'quiz')),
8|     link TEXT, -- Link d-«-n -n bài h-íc / bài t-p li-ên quan
9|     is_read BOOLEAN DEFAULT FALSE,
10|     created_at TIMESTAMPTZ DEFAULT NOW()
11| );
12|
13| -- B-t realtime cho b-ëng notifications
14| ALTER PUBLICATION supabase_realtime ADD TABLE notifications;
15|
16| -- RLS cho notifications
17| ALTER TABLE public.notifications ENABLE ROW LEVEL SECURITY;
18| CREATE POLICY "Users can view their own notifications" ON public.notifications
19|     FOR SELECT USING (auth.uid() = user_id);
20| CREATE POLICY "System/Teachers can insert notifications" ON public.notifications
21|     FOR INSERT WITH CHECK (true);
22| CREATE POLICY "Users can update their own notifications (mark as read)" ON public.notifications
23|     FOR UPDATE USING (auth.uid() = user_id);
24|
25| -- 2. C-ýu trúc L-Ù trình h-íc t-p (Prerequisites)
26| -- Th-êm c-ùt bài h-íc ti-ên quy-çt vào b-ëng lessons
27| ALTER TABLE public.lessons
28| ADD COLUMN IF NOT EXISTS prerequisite_lesson_id UUID REFERENCES public.lessons(id) ON DELETE SET NULL;
29|
30| ALTER TABLE public.lessons
31| ADD COLUMN IF NOT EXISTS min_score_to_unlock INTEGER DEFAULT 5;
32|

```

File: database/20260517_achievements.sql (45 lines, 2.0 KB)

```

=====
1| -- =====
2| -- ACHIEVEMENTS & BADGES DATABASE SETUP
3| -- =====
4|
5| CREATE TABLE IF NOT EXISTS public.achievement_definitions (
6|     id TEXT PRIMARY KEY,
7|     title TEXT NOT NULL,
8|     description TEXT,
9|     icon TEXT,
10|     points INT DEFAULT 0,
11|     rarity TEXT CHECK (rarity IN ('common', 'uncommon', 'rare', 'legendary'))
12| );
13|
14| CREATE TABLE IF NOT EXISTS public.student_achievements (
15|     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
16|     student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
17|     achievement_id TEXT REFERENCES public.achievement_definitions(id) ON DELETE CASCADE,
18|     earned_at TIMESTAMPTZ DEFAULT NOW(),
19|     UNIQUE(student_id, achievement_id)
20| );
21|
22| -- Seed definitions
23| INSERT INTO public.achievement_definitions (id, title, description, icon, points, rarity)
24| VALUES
25| ( 'first_lesson', 'Kh-ĩi S-ù hành tr-ình', 'Hoàn thành bài gi-ëng S-ù ti-ên
26| ( 'quiz_ace', 'X-ùçt s-c', 'it i-ã m-tuyçt N-i (100%) trong m-ùt bài gi-iz
27| ( 'week_streak', 'Ki-ên tr-i 7 ngày', 'H-íc li-ên t-içp 7 ngày không ng-h-ĩ
28| ( 'pc_master', 'PC Master', 'Hoàn thành t-çt c-ë bài trong l-ù tr-ình', '0=Ý
29| ( 'lab_expert', 'Chuyên gia Lab', 'D-ãnh h-in 10 gi-Y trong Builder Lab',
30| ( 'helpful_member', 'Nh-ĩçt t-ình', '°-ãc upvote 10 l-s-n trong di-ã n-àn',
31| ON CONFLICT (id) DO UPDATE
32| SET
33|     title = EXCLUDED.title,
34|     description = EXCLUDED.description,

```

```

35 | icon = EXCLUDED.icon,
36 | points = EXCLUDED.points,
37 | rarity = EXCLUDED.rarity;
38 |
39 | -- Enable RLS
40 | ALTER TABLE public.achievement_definitions ENABLE ROW LEVEL SECURITY;
41 | ALTER TABLE public.student_achievements ENABLE ROW LEVEL SECURITY;
42 |
43 | CREATE POLICY "view_definitions" ON public.achievement_definitions FOR SELECT USING (true);
44 | CREATE POLICY "view_student_achievements" ON public.student_achievements FOR SELECT USING (true);
45 |

```

File: database/20260517_certificates.sql (42 lines, 1.4 KB)

```

1 | -- =====
2 | -- CERTIFICATES DATABASE SETUP
3 | -- =====
4 |
5 | CREATE TABLE IF NOT EXISTS public.certificates (
6 |   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
7 |   student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
8 |   path_id UUID REFERENCES public.learning_paths(id) ON DELETE CASCADE,
9 |   certificate_number TEXT UNIQUE NOT NULL,
10 |   issued_at TIMESTAMPTZ DEFAULT NOW(),
11 |   pdf_url TEXT,
12 |   verify_url TEXT,
13 |   student_name TEXT,
14 |   course_title TEXT,
15 |   completion_date DATE DEFAULT CURRENT_DATE,
16 |   final_score NUMERIC(5,2),
17 |   is_revoked BOOLEAN DEFAULT FALSE
18 | );
19 |
20 | -- Function to generate safe, random certificate number
21 | CREATE OR REPLACE FUNCTION generate_cert_number()
22 | RETURNS TEXT AS $$
23 | DECLARE
24 |   chars TEXT := 'ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789';
25 |   result TEXT := 'PCM-2026-';
26 |   i INT;
27 | BEGIN
28 |   FOR i IN 1..5 LOOP
29 |     result := result || substr(chars, floor(random() * length(chars) + 1)::INT, 1);
30 |   END LOOP;
31 |   RETURN result;
32 | END;
33 | $$ LANGUAGE plpgsql;
34 |
35 | -- Enable RLS
36 | ALTER TABLE public.certificates ENABLE ROW LEVEL SECURITY;
37 |
38 | CREATE POLICY "view_all_certificates" ON public.certificates FOR SELECT USING (true);
39 | CREATE POLICY "manage_certificates" ON public.certificates FOR ALL USING (
40 |   EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
41 | );
42 |

```

File: database/20260517_discussion_forum.sql (81 lines, 3.2 KB)

```

1 | -- =====
2 | -- DISCUSSION FORUM SETUP
3 | -- =====
4 |
5 | CREATE TABLE IF NOT EXISTS public.discussion_threads (
6 |   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
7 |   lesson_id UUID NOT NULL,
8 |   author_id UUID REFERENCES public.profiles(id) ON DELETE SET NULL,
9 |   title TEXT NOT NULL,
10 |   body TEXT NOT NULL,
11 |   type TEXT DEFAULT 'question' CHECK (type IN ('question', 'discussion', 'announcement')),
12 |   status TEXT DEFAULT 'open' CHECK (status IN ('open', 'resolved', 'pinned', 'closed')),
13 |   upvote_count INT DEFAULT 0,
14 |   reply_count INT DEFAULT 0,
15 |   is_deleted BOOLEAN DEFAULT FALSE,
16 |   created_at TIMESTAMPTZ DEFAULT NOW(),
17 |   updated_at TIMESTAMPTZ DEFAULT NOW()
18 | );
19 |
20 | CREATE TABLE IF NOT EXISTS public.discussion_replies (
21 |   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
22 |   thread_id UUID REFERENCES public.discussion_threads(id) ON DELETE CASCADE,
23 |   parent_reply_id UUID REFERENCES public.discussion_replies(id) ON DELETE CASCADE,
24 |   author_id UUID REFERENCES public.profiles(id) ON DELETE SET NULL,
25 |   body TEXT NOT NULL,
26 |   is_teacher_answer BOOLEAN DEFAULT FALSE,
27 |   upvote_count INT DEFAULT 0,
28 |   is_deleted BOOLEAN DEFAULT FALSE,
29 |   created_at TIMESTAMPTZ DEFAULT NOW()
30 | );
31 |
32 | CREATE TABLE IF NOT EXISTS public.discussion_upvotes (
33 |   user_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
34 |   thread_id UUID REFERENCES public.discussion_threads(id) ON DELETE CASCADE,
35 |   reply_id UUID REFERENCES public.discussion_replies(id) ON DELETE CASCADE,
36 |   created_at TIMESTAMPTZ DEFAULT NOW(),
37 |   CONSTRAINT one_target CHECK (
38 |     (thread_id IS NULL) != (reply_id IS NULL)
39 |   ),
40 |   UNIQUE(user_id, thread_id),
41 |   UNIQUE(user_id, reply_id)
42 | );
43 |
44 | -- Trigger to increment reply_count
45 | CREATE OR REPLACE FUNCTION update_reply_count()
46 | RETURNS TRIGGER AS $$
47 | BEGIN
48 |   UPDATE public.discussion_threads
49 |   SET reply_count = reply_count + 1
50 |   WHERE id = NEW.thread_id;
51 |   RETURN NEW;
52 | END;
53 | $$ LANGUAGE plpgsql;
54 |
55 | DROP TRIGGER IF EXISTS on_reply_insert ON public.discussion_replies;
56 | CREATE TRIGGER on_reply_insert
57 | AFTER INSERT ON public.discussion_replies
58 | FOR EACH ROW EXECUTE FUNCTION update_reply_count();
59 |
60 | -- RLS Enable
61 | ALTER TABLE public.discussion_threads ENABLE ROW LEVEL SECURITY;
62 | ALTER TABLE public.discussion_replies ENABLE ROW LEVEL SECURITY;

```

```

63 ALTER TABLE public.discussion_upvotes ENABLE ROW LEVEL SECURITY;
64
65 -- Policies
66 CREATE POLICY "view_threads" ON public.discussion_threads FOR SELECT USING (true);
67 CREATE POLICY "create_thread" ON public.discussion_threads FOR INSERT WITH CHECK (author_id = auth.uid());
68 CREATE POLICY "delete_own_or_teacher_thread" ON public.discussion_threads FOR UPDATE USING (
69     author_id = auth.uid() OR
70     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
71 );
72
73 CREATE POLICY "view_replies" ON public.discussion_replies FOR SELECT USING (true);
74 CREATE POLICY "create_reply" ON public.discussion_replies FOR INSERT WITH CHECK (author_id = auth.uid());
75 CREATE POLICY "delete_own_or_teacher_reply" ON public.discussion_replies FOR UPDATE USING (
76     author_id = auth.uid() OR
77     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
78 );
79
80 CREATE POLICY "manage_upvotes" ON public.discussion_upvotes FOR ALL USING (user_id = auth.uid());
81

```

File: database/20260517_learning_path.sql (115 lines, 3.5 KB)

```

1  -- =====
2  -- LEARNING PATH & PROGRESSIVE UNLOCKING SETUP
3  -- =====
4
5  CREATE TABLE IF NOT EXISTS public.learning_paths (
6      id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
7      class_id UUID,
8      title TEXT NOT NULL,
9      description TEXT,
10     is_active BOOLEAN DEFAULT TRUE,
11     created_by UUID REFERENCES public.profiles(id),
12     created_at TIMESTAMPTZ DEFAULT NOW()
13 );
14
15 CREATE TABLE IF NOT EXISTS public.path_items (
16     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
17     path_id UUID REFERENCES public.learning_paths(id) ON DELETE CASCADE,
18     item_type TEXT CHECK (item_type IN ('lesson','quiz','lab_session','milestone')),
19     item_id UUID, -- Nullable if milestone
20     title TEXT NOT NULL,
21     description TEXT,
22     "order" INT NOT NULL,
23     unlock_condition JSONB,
24     estimated_minutes INT,
25     is_optional BOOLEAN DEFAULT FALSE,
26     created_at TIMESTAMPTZ DEFAULT NOW(),
27     UNIQUE(path_id, "order")
28 );
29
30 -- PostgreSQL Function to calculate unlock states server-side
31 CREATE OR REPLACE FUNCTION public.get_unlocked_items(
32     p_student_id UUID,
33     p_path_id UUID
34 ) RETURNS TABLE (
35     id UUID,
36     item_id UUID,
37     is_unlocked BOOLEAN,
38     unlock_reason TEXT
39 ) AS $$
40 DECLARE
41     item RECORD;
42     cond JSONB;
43     cond_type TEXT;
44     cond_item_id UUID;
45     min_sc NUMERIC;
46     prev_completed BOOLEAN;
47     quiz_passed BOOLEAN;
48 BEGIN
49     FOR item IN
50         SELECT * FROM public.path_items WHERE path_id = p_path_id ORDER BY "order"
51     LOOP
52         cond := item.unlock_condition;
53         is_unlocked := TRUE;
54         unlock_reason := 'always_available';
55
56         IF cond IS NOT NULL THEN
57             cond_type := cond->>'type';
58
59             -- 1. Complete previous item
60             IF cond_type = 'complete_previous' THEN
61                 SELECT EXISTS (
62                     SELECT 1 FROM public.lesson_progress lp
63                     JOIN public.path_items pi ON pi.item_id = lp.lesson_id
64                     WHERE lp.student_id = p_student_id
65                         AND lp.status = 'completed'
66                         AND pi.path_id = p_path_id
67                         AND pi.order = item.order - 1
68                 ) INTO prev_completed;
69
70                 IF NOT prev_completed THEN
71                     is_unlocked := FALSE;
72                     unlock_reason := 'need_complete_previous';
73                 END IF;
74
75             -- 2. Score quiz above threshold
76             ELSIF cond_type = 'score_above' THEN
77                 cond_item_id := (cond->>'item_id')::UUID;
78                 min_sc := (cond->>'min_score')::NUMERIC;
79
80                 SELECT EXISTS (
81                     SELECT 1 FROM public.quiz_attempts
82                     WHERE student_id = p_student_id
83                         AND quiz_id = cond_item_id
84                         AND score >= min_sc
85                         AND status = 'submitted'
86                 ) INTO quiz_passed;
87
88                 IF NOT quiz_passed THEN
89                     is_unlocked := FALSE;
90                     unlock_reason := 'need_quiz_score_above_' || min_sc::TEXT;
91                 END IF;
92             END IF;
93         END IF;
94
95         id := item.id;
96         item_id := item.item_id;
97         RETURN NEXT;
98     END LOOP;
99 END;
100 $$ LANGUAGE plpgsql SECURITY DEFINER;
101

```

```

102 -- RLS Enable
103 ALTER TABLE public.learning_paths ENABLE ROW LEVEL SECURITY;
104 ALTER TABLE public.path_items ENABLE ROW LEVEL SECURITY;
105
106 CREATE POLICY "view_paths" ON public.learning_paths FOR SELECT USING (true);
107 CREATE POLICY "manage_paths" ON public.learning_paths FOR ALL USING (
108     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
109 );
110
111 CREATE POLICY "view_path_items" ON public.path_items FOR SELECT USING (true);
112 CREATE POLICY "manage_path_items" ON public.path_items FOR ALL USING (
113     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
114 );
115

```

File: database/20260517_profile_settings.sql (37 lines, 1.6 KB)

```

1 -- =====
2 -- PROFILE EXTENSION & USER PREFERENCES SETUP
3 -- =====
4
5 -- 1. Extend profiles table safely (RULE 1)
6 ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS bio TEXT;
7 ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS school TEXT;
8 ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS grade TEXT;
9 ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS updated_at TIMESTAMPTZ DEFAULT NOW();
10
11 -- 2. Create User Preferences Table
12 CREATE TABLE IF NOT EXISTS public.user_preferences (
13     user_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE PRIMARY KEY,
14     email_notifications BOOLEAN DEFAULT TRUE,
15     push_notifications BOOLEAN DEFAULT TRUE,
16     weekly_digest BOOLEAN DEFAULT TRUE,
17     show_profile_public BOOLEAN DEFAULT FALSE,
18     theme TEXT DEFAULT 'dark' CHECK (theme IN ('dark', 'light', 'system')),
19     language TEXT DEFAULT 'vi' CHECK (language IN ('vi', 'en')),
20     updated_at TIMESTAMPTZ DEFAULT NOW()
21 );
22
23 -- Enable RLS on preferences
24 ALTER TABLE public.user_preferences ENABLE ROW LEVEL SECURITY;
25
26 -- Preferences Policy
27 DROP POLICY IF EXISTS "own_preferences" ON public.user_preferences;
28 CREATE POLICY "own_preferences" ON public.user_preferences
29     FOR ALL USING (user_id = auth.uid());
30
31 -- 3. Storage Avatars policy setup (RULE 2)
32 -- Make sure the bucket 'avatars' is created under storage.buckets first.
33 -- This policy allows users to only upload inside their own folder.
34 -- Note: Create policy on storage.objects:
35 -- CREATE POLICY "own_avatar" ON storage.objects
36 --     FOR ALL USING (bucket_id = 'avatars' AND (storage.foldername(name))[1] = auth.uid()::text);
37

```

File: database/20260517_progress_seed_data.sql (115 lines, 5.2 KB)

```

1 -- =====
2 -- SAMPLE PROGRESS SEED DATA FOR TESTING PROGRESS VIEW
3 -- Run this in the Supabase SQL Editor
4 -- =====
5
6 -- 1. Insert sample lessons if they do not exist
7 INSERT INTO public.lessons (id, title, content, created_at)
8 VALUES
9     ( ' 1 1 1 1 1 1 1 1 - 1 1 1 1 - 1 1 1 1 - 1 1 1 1 - 1 1 1 1 1 1 1 1 1 1 ', ' Bài 1: Giới thiệu p h s n c é n g
10 | ( ' 2 2 2 2 2 2 2 2 - 2 2 2 2 - 2 2 2 2 - 2 2 2 2 - 2 2 2 2 2 2 2 2 2 2 ', ' Bài 2: Th ã c h à n h L p r á p C P U
11 | ( ' 3 3 3 3 3 3 3 3 - 3 3 3 3 - 3 3 3 3 - 3 3 3 3 - 3 3 3 3 3 3 3 3 3 3 ', ' Bài 3: B o m i c h c h c ( M a i n b o a
12 | ( ' 4 4 4 4 4 4 4 4 - 4 4 4 4 - 4 4 4 4 - 4 4 4 4 - 4 4 4 4 4 4 4 4 4 4 ', ' Quiz: P h s n c é n g c i b é n ', ' B
13 | ( ' 5 5 5 5 5 5 5 5 - 5 5 5 5 - 5 5 5 5 - 5 5 5 5 - 5 5 5 5 5 5 5 5 5 5 ', ' Bài 5: L p r á p R A M v à T ê n n h
14 ON CONFLICT (id) DO NOTHING;
15
16 -- 2. Insert sample quizzes if they do not exist
17 INSERT INTO public.quizzes (id, lesson_id, title, passing_score, created_at)
18 VALUES
19 | ( ' a 1 a 1 a 1 a 1 - a 1 a 1 - a 1 a 1 - a 1 a 1 - a 1 a 1 a 1 a 1 a 1 ', ' 4 4 4 4 4 4 4 4 - 4 4 4 4 - 4 4 4 4 - 4 4 4 4 - 4 4 4 4
20 ON CONFLICT (id) DO NOTHING;
21
22 -- 3. Insert mock progress data for the CURRENT LOGGED-IN USER
23 -- Note: Replace 'YOUR_USER_UUID' with the actual user id if running manually,
24 -- or use the query below which automatically targets all profiles as sample data.
25
26 DO $$
27 DECLARE
28     user_rec RECORD;
29 BEGIN
30     FOR user_rec IN SELECT id FROM public.profiles LOOP
31
32         -- Lesson 1: Completed 12 days ago
33         INSERT INTO public.lesson_progress (student_id, lesson_id, status, completion_percentage, time_spent_s...)
34         VALUES (
35             user_rec.id,
36             '11111111-1111-1111-111111111111',
37             'completed',
38             100,
39             3600, -- 1 hour
40             NOW() - INTERVAL '12 days',
41             NOW() - INTERVAL '12 days',
42         ) ON CONFLICT (student_id, lesson_id) DO UPDATE
43         SET status = 'completed', completion_percentage = 100, completed_at = NOW() - INTERVAL '12 days';
44
45         -- Lesson 2: Completed 10 days ago
46         INSERT INTO public.lesson_progress (student_id, lesson_id, status, completion_percentage, time_spent_s...)
47         VALUES (
48             user_rec.id,
49             '22222222-2222-2222-222222222222',
50             'completed',
51             100,
52             4200, -- 1.16 hours
53             NOW() - INTERVAL '10 days',
54             NOW() - INTERVAL '10 days',
55         ) ON CONFLICT (student_id, lesson_id) DO UPDATE
56         SET status = 'completed', completion_percentage = 100, completed_at = NOW() - INTERVAL '10 days';
57
58         -- Lesson 3: In progress
59         INSERT INTO public.lesson_progress (student_id, lesson_id, status, completion_percentage, time_spent_s...)
60         VALUES (
61             user_rec.id,
62             '33333333-3333-3333-3333-333333333333',
63             'in_progress',
64             45,

```

```

65         1800,
66         NOW() - INTERVAL '2 days',
67         NULL
68     ) ON CONFLICT (student_id, lesson_id) DO UPDATE
69     SET status = 'in_progress', completion_percentage = 45;
70
71     -- Quiz Attempt
72     INSERT INTO public.quiz_attempts (quiz_id, student_id, started_at, submitted_at, score, status)
73     VALUES (
74         'a1a1a1a1-a1a1-a1a1-a1a1-a1a1a1a1a1a1',
75         user_rec.id,
76         NOW() - INTERVAL '3 days 1 hour',
77         NOW() - INTERVAL '3 days',
78         85, -- 85 score
79         'submitted' -- matches CHECK constraints or standard submitted status
80     );
81
82     -- Builder Lab Sessions (Past 3 months heatmap simulation)
83     -- Session 1: 5 days ago (45 minutes)
84     INSERT INTO public.builder_sessions (student_id, lesson_id, started_at, ended_at, compatibility_score)
85     VALUES (
86         user_rec.id,
87         '22222222-2222-2222-2222-222222222222',
88         NOW() - INTERVAL '5 days 45 minutes',
89         NOW() - INTERVAL '5 days',
90         95
91     );
92
93     -- Session 2: 3 days ago (75 minutes)
94     INSERT INTO public.builder_sessions (student_id, lesson_id, started_at, ended_at, compatibility_score)
95     VALUES (
96         user_rec.id,
97         '22222222-2222-2222-2222-222222222222',
98         NOW() - INTERVAL '3 days 75 minutes',
99         NOW() - INTERVAL '3 days',
100        100
101    );
102
103    -- Session 3: Today (20 minutes)
104    INSERT INTO public.builder_sessions (student_id, lesson_id, started_at, ended_at, compatibility_score)
105    VALUES (
106        user_rec.id,
107        '55555555-5555-5555-5555-555555555555',
108        NOW() - INTERVAL '20 minutes',
109        NOW(),
110        90
111    );
112
113    END LOOP;
114 END $$;
115

```

File: database/20260517_quiz_engine_v2.sql (135 lines, 4.3 KB)

```

=====
1 | -- %%P%P%P  NG  AN  H  AN  G  C  AU  H  I  %P%P%P
2 | CREATE TABLE IF NOT EXISTS public.question_banks (
3 |   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
4 |   teacher_id UUID REFERENCES public.profiles(id) NOT NULL,
5 |   title TEXT NOT NULL,
6 |   subject TEXT DEFAULT 'Tin h c - L p r p PC',
7 |   description TEXT,
8 |   created_at TIMESTAMPTZ DEFAULT NOW(),
9 |   updated_at TIMESTAMPTZ DEFAULT NOW()
10 | );
11
12 | CREATE TABLE IF NOT EXISTS public.questions (
13 |   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
14 |   bank_id UUID REFERENCES public.question_banks(id) ON DELETE CASCADE,
15 |   content TEXT NOT NULL,
16 |   type TEXT CHECK (type IN (
17 |     'single_choice',
18 |     'multiple_choice',
19 |     'true_false',
20 |     'fill_blank',
21 |     'ordering'
22 |   )) NOT NULL,
23 |   difficulty TEXT CHECK (difficulty IN ('easy', 'medium', 'hard')) DEFAULT 'medium',
24 |   points INT DEFAULT 10,
25 |   explanation TEXT,
26 |   image_url TEXT,
27 |   tags TEXT[],
28 |   "order" INT DEFAULT 0,
29 |   created_at TIMESTAMPTZ DEFAULT NOW()
30 | );
31
32 | CREATE TABLE IF NOT EXISTS public.question_options (
33 |   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
34 |   question_id UUID REFERENCES public.questions(id) ON DELETE CASCADE,
35 |   content TEXT NOT NULL,
36 |   is_correct BOOLEAN DEFAULT FALSE,
37 |   "order" INT DEFAULT 0
38 | );
39
40 | -- %%P%P%P  A  TH  I  %P%P%P
41 | CREATE TABLE IF NOT EXISTS public.quizzes (
42 |   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
43 |   teacher_id UUID REFERENCES public.profiles(id) NOT NULL,
44 |   lesson_id UUID REFERENCES public.lessons(id),
45 |   class_id UUID,
46 |   title TEXT NOT NULL,
47 |   description TEXT,
48 |   time_limit_minutes INT,
49 |   passing_score INT DEFAULT 70,
50 |   randomize_questions BOOLEAN DEFAULT TRUE,
51 |   randomize_options BOOLEAN DEFAULT TRUE,
52 |   max_attempts INT DEFAULT 3,
53 |   show_answers_after TEXT DEFAULT 'after_submit'
54 |   CHECK (show_answers_after IN ('never', 'after_submit', 'after_deadline')),
55 |   available_from TIMESTAMPTZ,
56 |   available_until TIMESTAMPTZ,
57 |   is_published BOOLEAN DEFAULT FALSE,
58 |   created_at TIMESTAMPTZ DEFAULT NOW()
59 | );
60
61 | CREATE TABLE IF NOT EXISTS public.quiz_questions (
62 |   quiz_id UUID REFERENCES public.quizzes(id) ON DELETE CASCADE,
63 |   question_id UUID REFERENCES public.questions(id),
64 |   "order" INT,
65 |   PRIMARY KEY (quiz_id, question_id)
66 | );
67
68 | -- %%P%P%P  B  A  I  L  A  M  %P%P%P
69 | CREATE TABLE IF NOT EXISTS public.quiz_attempts (

```



```

70 | id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
71 | quiz_id UUID REFERENCES public.quizzes(id),
72 | student_id UUID REFERENCES public.profiles(id),
73 | started_at TIMESTAMPTZ DEFAULT NOW(),
74 | submitted_at TIMESTAMPTZ,
75 | time_taken_seconds INT,
76 | score NUMERIC(5,2),
77 | max_score INT,
78 | status TEXT DEFAULT 'in_progress'
79 | CHECK (status IN ('in_progress', 'submitted', 'graded', 'expired')),
80 | attempt_number INT DEFAULT 1
81 | );
82 |
83 | CREATE TABLE IF NOT EXISTS public.quiz_attempts (
84 | id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
85 | attempt_id UUID REFERENCES public.quiz_attempts(id) ON DELETE CASCADE,
86 | question_id UUID REFERENCES public.questions(id),
87 | selected_option_ids UUID[],
88 | text_answer TEXT,
89 | ordering_answer UUID[],
90 | is_correct BOOLEAN,
91 | points_earned NUMERIC(5,2) DEFAULT 0,
92 | time_spent_seconds INT
93 | );
94 |
95 | -- RLS Policies
96 | ALTER TABLE public.quiz_attempts ENABLE ROW LEVEL SECURITY;
97 | ALTER TABLE public.quiz_answers ENABLE ROW LEVEL SECURITY;
98 |
99 | CREATE POLICY "student_own_attempts" ON public.quiz_attempts
100 | FOR ALL USING (student_id = auth.uid());
101 |
102 | CREATE POLICY "student_own_answers" ON public.quiz_answers
103 | FOR ALL USING (
104 |     EXISTS (
105 |         SELECT 1 FROM public.quiz_attempts
106 |         WHERE id = attempt_id AND student_id = auth.uid()
107 |     )
108 | );
109 |
110 | CREATE POLICY "teacher_view_attempts" ON public.quiz_attempts
111 | FOR SELECT USING (
112 |     EXISTS (
113 |         SELECT 1 FROM public.profiles
114 |         WHERE id = auth.uid() AND role = 'teacher'
115 |     )
116 | );
117 |
118 | -- VIEW an toàn cho híc sinh
119 | CREATE OR REPLACE VIEW public.quiz_questions_for_student AS
120 | SELECT
121 |     q.id, q.content, q.type, q.points, q.image_url,
122 |     q.tags, qq.order,
123 |     json_agg(
124 |         json_build_object(
125 |             'id', qo.id,
126 |             'content', qo.content,
127 |             'order', qo.order
128 |         ) ORDER BY qo.order
129 |     ) AS options,
130 |     qq.quiz_id
131 | FROM public.questions q
132 | JOIN public.quiz_questions qq ON qq.question_id = q.id
133 | JOIN public.question_options qo ON qo.question_id = q.id
134 | GROUP BY q.id, q.content, q.type, q.points, q.image_url, q.tags, qq.order, qq.quiz_id;
135 |

```

File: database/20260517_user_flow_schema.sql (86 lines, 2.6 KB)

```

1 | -- Supabase SQL Migration: User Flow & Progress Tracking
2 |
3 | -- 1. Profiles Table (if not exists)
4 | CREATE TABLE IF NOT EXISTS profiles (
5 | id UUID REFERENCES auth.users PRIMARY KEY,
6 | full_name TEXT NOT NULL,
7 | role TEXT CHECK (role IN ('student', 'teacher')) NOT NULL,
8 | avatar_url TEXT,
9 | class_id UUID,
10 | created_at TIMESTAMPTZ DEFAULT NOW()
11 | );
12 |
13 | -- 2. Lessons Table (if not exists)
14 | CREATE TABLE IF NOT EXISTS lessons (
15 | id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
16 | title TEXT NOT NULL,
17 | content TEXT,
18 | course_id UUID,
19 | created_at TIMESTAMPTZ DEFAULT NOW()
20 | );
21 |
22 | -- DROP OLD TABLES TO AVOID COLUMN CONFLICTS
23 | DROP TABLE IF EXISTS lesson_progress CASCADE;
24 | DROP TABLE IF EXISTS builder_sessions CASCADE;
25 |
26 | -- 3. Lesson Progress
27 | CREATE TABLE IF NOT EXISTS lesson_progress (
28 | id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
29 | student_id UUID REFERENCES profiles(id),
30 | lesson_id UUID REFERENCES lessons(id),
31 | status TEXT DEFAULT 'not_started',
32 | completion_percentage INT DEFAULT 0,
33 | time_spent_seconds INT DEFAULT 0,
34 | last_accessed TIMESTAMPTZ,
35 | completed_at TIMESTAMPTZ,
36 | UNIQUE(student_id, lesson_id)
37 | );
38 |
39 | -- 4. Builder Lab Sessions
40 | CREATE TABLE IF NOT EXISTS builder_sessions (
41 | id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
42 | student_id UUID REFERENCES profiles(id),
43 | lesson_id UUID REFERENCES lessons(id),
44 | started_at TIMESTAMPTZ DEFAULT NOW(),
45 | ended_at TIMESTAMPTZ,
46 | components_used JSONB,
47 | tdp_calculated NUMERIC,
48 | compatibility_score INT
49 | );
50 |
51 | -- 5. Row Level Security (RLS) Policies
52 | ALTER TABLE profiles ENABLE ROW LEVEL SECURITY;
53 | ALTER TABLE lesson_progress ENABLE ROW LEVEL SECURITY;
54 | ALTER TABLE builder_sessions ENABLE ROW LEVEL SECURITY;

```

```

55 |
56 | -- Profile Policies
57 | CREATE POLICY "Public profiles are viewable by everyone."
58 |   ON profiles FOR SELECT
59 |   USING ( true );
60 |
61 | CREATE POLICY "Users can insert their own profile."
62 |   ON profiles FOR INSERT
63 |   WITH CHECK ( auth.uid() = id );
64 |
65 | CREATE POLICY "Users can update own profile."
66 |   ON profiles FOR UPDATE
67 |   USING ( auth.uid() = id );
68 |
69 | -- Lesson Progress Policies
70 | CREATE POLICY "student_own_progress" ON lesson_progress
71 |   FOR ALL USING (student_id = auth.uid());
72 |
73 | CREATE POLICY "teacher_view_class" ON lesson_progress
74 |   FOR SELECT USING (
75 |     EXISTS (SELECT 1 FROM profiles WHERE id = auth.uid() AND role = 'teacher')
76 |   );
77 |
78 | -- Builder Sessions Policies
79 | CREATE POLICY "student_own_sessions" ON builder_sessions
80 |   FOR ALL USING (student_id = auth.uid());
81 |
82 | CREATE POLICY "teacher_view_sessions" ON builder_sessions
83 |   FOR SELECT USING (
84 |     EXISTS (SELECT 1 FROM profiles WHERE id = auth.uid() AND role = 'teacher')
85 |   );
86 |

```

File: database/20260518_parent_notifications.sql (187 lines, 6.9 KB)

```

=====
1 | -- =====
2 | -- PROMPT 4 · PC MASTER LMS – PARENT NOTIFICATIONS & TRIGGERS
3 | -- Target Path: database/20260518_parent_notifications.sql
4 | -- =====
5 |
6 | -- %%% B-ÚC 1: C-P NH-T USER PREFERENCES   Ả HỒ TRẢ OPT-OUT %%%
7 | -- Thêm cột notification_settings (JSONB) cho bảng user_preferences
8 | ALTER TABLE public.user_preferences
9 | ADD COLUMN IF NOT EXISTS notification_settings JSONB
10 | DEFAULT '{"child_absent": true, "child_low_score": true, "child_completed_path": true, "lesson_completed": t...
11 |
12 | -- Tio bảng lờu trữ thông tin cooldown gửi email
13 | CREATE TABLE IF NOT EXISTS public.parent_email_cooldown (
14 |   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
15 |   student_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE CASCADE,
16 |   parent_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE CASCADE,
17 |   notification_type TEXT NOT NULL,
18 |   sent_at TIMESTAMPTZ DEFAULT NOW(),
19 |   UNIQUE(student_id, parent_id, notification_type)
20 | );
21 |
22 | -- B-t RLS cho bảng cooldown
23 | ALTER TABLE public.parent_email_cooldown ENABLE ROW LEVEL SECURITY;
24 |
25 | -- Cooldown Policy
26 | DROP POLICY IF EXISTS "system_all_cooldown" ON public.parent_email_cooldown;
27 | CREATE POLICY "system_all_cooldown" ON public.parent_email_cooldown
28 |   FOR ALL USING (true);
29 |
30 |
31 | -- %%% B-ÚC 2: TRIGGER KHI Híc SINH HOÀN THÀNH BÀI Híc %%%
32 | CREATE OR REPLACE FUNCTION notify_parent_on_lesson_complete()
33 | RETURNS TRIGGER AS $$
34 | DECLARE
35 |   parent_rec RECORD;
36 |   lesson_rec RECORD;
37 |   pref_val BOOLEAN;
38 | BEGIN
39 |   IF NEW.status = 'completed' AND OLD.status != 'completed' THEN
40 |     -- Láy tên bài híc
41 |     SELECT title INTO lesson_rec FROM lessons WHERE id = NEW.lesson_id;
42 |
43 |     -- Tim t¥t c£ phả huynh ang theo dõi híc sinh này
44 |     FOR parent_rec IN
45 |       SELECT psl.parent_id, up.notification_settings
46 |       FROM parent_student_links psl
47 |       LEFT JOIN user_preferences up ON up.user_id = psl.parent_id
48 |       WHERE psl.student_id = NEW.student_id AND psl.status = 'active'
49 |     LOOP
50 |       -- Check opt-out setting
51 |       pref_val := TRUE;
52 |       IF parent_rec.notification_settings IS NOT NULL AND parent_rec.notification_settings ? 'lesson_completed...'
53 |         pref_val := (parent_rec.notification_settings->'lesson_completed')::boolean;
54 |       END IF;
55 |
56 |       IF pref_val THEN
57 |         INSERT INTO notifications (user_id, type, title, body, data, action_url)
58 |         VALUES (
59 |           parent_rec.parent_id,
60 |           'lesson_completed',
61 |           'Con bin hoàn thành bài híc 0=00',
62 |           'Con bin vĩa hoàn thành bài "' || COALESCE(lesson_rec.title,
63 |             jsonb_build_object(
64 |               'student_id', NEW.student_id,
65 |               'lesson_id', NEW.lesson_id
66 |             ),
67 |             '/parent/children/' || NEW.student_id
68 |           );
69 |       END IF;
70 |     END LOOP;
71 |   END IF;
72 |   RETURN NEW;
73 | END;
74 | $$ LANGUAGE plpgsql SECURITY DEFINER;
75 |
76 | DROP TRIGGER IF EXISTS on_lesson_completed ON lesson_progress;
77 | CREATE TRIGGER on_lesson_completed
78 |   AFTER UPDATE ON lesson_progress
79 |   FOR EACH ROW EXECUTE FUNCTION notify_parent_on_lesson_complete();
80 |
81 |
82 | -- %%% B-ÚC 3: TRIGGER KHI CỎ K¥T QU£ QUIZ %%%
83 | CREATE OR REPLACE FUNCTION notify_parent_on_quiz_result()
84 | RETURNS TRIGGER AS $$
85 | DECLARE
86 |   parent_rec RECORD;
87 |   quiz_rec RECORD;
88 |   pref_val BOOLEAN;

```

```

89|   notif_type TEXT;
90|   notif_title TEXT;
91| BEGIN
92|   IF NEW.status = 'graded' THEN
93|     SELECT title INTO quiz_rec FROM quizzes WHERE id = NEW.quiz_id;
94|
95|     -- Quy ết  định  loại  thông  báo  dựa  trên  i ấ  m  s  ã
96|     IF NEW.score < 50 THEN
97|       notif_type := 'child_low_score';
98|       notif_title := '&p Con bìn có i ấ m th ấ p';
99|     ELSE
100|       notif_type := 'quiz_graded';
101|       notif_title := ' Con bìn có k ế t qu ế m ỗ i';
102|     END IF;
103|
104|     FOR parent_rec IN
105|       SELECT psl.parent_id, up.notification_settings
106|       FROM parent_student_links psl
107|       LEFT JOIN user_preferences up ON up.user_id = psl.parent_id
108|       WHERE psl.student_id = NEW.student_id AND psl.status = 'active'
109|     LOOP
110|       -- Check opt-out setting cho lo ị i  thông  báo  này
111|       pref_val := TRUE;
112|       IF parent_rec.notification_settings IS NOT NULL AND parent_rec.notification_settings ? notif_type THEN
113|         pref_val := (parent_rec.notification_settings->>notif_type)::boolean;
114|       END IF;
115|
116|       IF pref_val THEN
117|         INSERT INTO notifications (user_id, type, title, body, data, action_url)
118|         VALUES (
119|           parent_rec.parent_id,
120|           notif_type,
121|           notif_title,
122|           'K ế t qu ế ' || COALESCE(quiz_rec.title, 'Bài tr ậ c nghi ệ m') ||
123|             jsonb_build_object(
124|               'student_id', NEW.student_id,
125|               'attempt_id', NEW.id,
126|               'score', NEW.score
127|             ),
128|           '/parent/children/' || NEW.student_id
129|         );
130|       END IF;
131|     END LOOP;
132|   END IF;
133|   RETURN NEW;
134| END;
135| $$ LANGUAGE plpgsql SECURITY DEFINER;
136|
137| DROP TRIGGER IF EXISTS on_quiz_graded ON quiz_attempts;
138| CREATE TRIGGER on_quiz_graded
139| AFTER UPDATE ON quiz_attempts
140| FOR EACH ROW EXECUTE FUNCTION notify_parent_on_quiz_result();
141|
142|
143| -- %%% B Ứ C 4: TRIGGER KHI T TH Ầ NH T Ỉ CH M Ỗ I (CHAMPION) %%%
144| CREATE OR REPLACE FUNCTION notify_parent_on_achievement_earned()
145| RETURNS TRIGGER AS $$
146| DECLARE
147|   parent_rec RECORD;
148|   ach_rec RECORD;
149|   pref_val BOOLEAN;
150| BEGIN
151|   -- T ỉ m t ấ t c ế p ầ huynh ầ ng theo d ỗ i h ấ c sinh này
152|   FOR parent_rec IN
153|     SELECT psl.parent_id, up.notification_settings
154|     FROM parent_student_links psl
155|     LEFT JOIN user_preferences up ON up.user_id = psl.parent_id
156|     WHERE psl.student_id = NEW.student_id AND psl.status = 'active'
157|   LOOP
158|     -- Check opt-out setting
159|     pref_val := TRUE;
160|     IF parent_rec.notification_settings IS NOT NULL AND parent_rec.notification_settings ? 'achievement_earned...'
161|       pref_val := (parent_rec.notification_settings->>'achievement_earned')::boolean;
162|     END IF;
163|
164|     IF pref_val THEN
165|       INSERT INTO notifications (user_id, type, title, body, data, action_url)
166|       VALUES (
167|         parent_rec.parent_id,
168|         'achievement_earned',
169|         'Con bìn ấ t th ầ nh t ỉ ch m ỗ i! Ồ ấ',
170|         'Con bìn v ể a xu ấ t s ắ c m ầ kh ố a hu ỷ hi ệ u th ầ nh t ỉ ch m ỗ i trong h ấ c
171|           jsonb_build_object(
172|             'student_id', NEW.student_id,
173|             'achievement_id', NEW.achievement_id
174|           ),
175|         '/parent/children/' || NEW.student_id
176|       );
177|     END IF;
178|   END LOOP;
179|   RETURN NEW;
180| END;
181| $$ LANGUAGE plpgsql SECURITY DEFINER;
182|
183| DROP TRIGGER IF EXISTS on_achievement_earned ON student_achievements;
184| CREATE TRIGGER on_achievement_earned
185| AFTER INSERT ON student_achievements
186| FOR EACH ROW EXECUTE FUNCTION notify_parent_on_achievement_earned();
187|

```

File: database/20260518_profiles_insert_policy.sql (21 lines, 0.9 KB)

```

-----
1| -- 1. S ắ l ắ i ắ quy v ồ h ầ n (Infinite Recursion) c ắ b ằ ng profiles
2| DROP POLICY IF EXISTS "Admin read all profiles" ON public.profiles;
3| CREATE POLICY "Admin read all profiles" ON public.profiles
4| FOR SELECT USING (
5|   (auth.jwt() -> 'user_metadata' -> 'role') = 'admin'
6| );
7|
8| -- 2. Th ề m Policy cho p ầ p ầ INSERT v ầ o b ằ ng profiles
9| DROP POLICY IF EXISTS "Users insert own profile" ON public.profiles;
10| CREATE POLICY "Users insert own profile" ON public.profiles
11| FOR INSERT WITH CHECK (auth.uid() = id);
12|
13| -- 3. Ề m b ằ o RLS c ắ a profiles cho p ầ p ầ SELECT v ầ UPDATE ho ầ n ch ế nh
14| DROP POLICY IF EXISTS "Users read own profile" ON public.profiles;
15| CREATE POLICY "Users read own profile" ON public.profiles
16| FOR SELECT USING (auth.uid() = id);
17|
18| DROP POLICY IF EXISTS "Users update own profile" ON public.profiles;
19| CREATE POLICY "Users update own profile" ON public.profiles
20| FOR UPDATE USING (auth.uid() = id);
21|

```

File: database/20260525_daily_quiz.sql (30 lines, 1.0 KB)

```
1 | -- %P%P%P THÊM CỘT is_daily CHO BẢNG quizzes %P%P%P
2 | ALTER TABLE public.quizzes
3 | ADD COLUMN IF NOT EXISTS is_daily BOOLEAN DEFAULT FALSE;
4 |
5 | -- %P%P%P VIEW: daily_quiz_info (lấy bài quiz hàng ngày cho hôm nay) %P%P%P
6 | CREATE OR REPLACE VIEW public.daily_quiz_info AS
7 | SELECT
8 |   q.id,
9 |   q.title,
10 |  q.description,
11 |  q.time_limit_minutes,
12 |  q.passing_score,
13 |  q.created_at,
14 |  (SELECT COUNT(*)::int FROM public.quiz_attempts qa
15 |   WHERE qa.quiz_id = q.id AND qa.student_id = auth.uid()
16 |   AND qa.status = 'submitted')::int AS student_attempts
17 | FROM public.quizzes q
18 | WHERE q.is_daily = TRUE
19 |   AND q.is_published = TRUE
20 |   AND q.created_at::date = CURRENT_DATE
21 | ORDER BY q.created_at DESC
22 | LIMIT 1;
23 |
24 | -- %P%P%P RLS: cho phép tài khoản service role và admin tạo quiz daily %P%P%P
25 | CREATE POLICY IF NOT EXISTS "service_role_insert_daily" ON public.quizzes
26 |   FOR INSERT WITH CHECK (is_daily = TRUE);
27 |
28 | CREATE POLICY IF NOT EXISTS "select_daily_quiz" ON public.quizzes
29 |   FOR SELECT USING (is_daily = TRUE OR is_daily = FALSE);
30 |
```

File: database/20260525_enable_realtime.sql (14 lines, 0.6 KB)

```
1 | -- %P%P%P BẬT REALTIME CHO CÁC BẢNG %P%P%P
2 |
3 | -- BĂNG profiles (dùng cho leaderboard, online status)
4 | ALTER PUBLICATION supabase_realtime ADD TABLE IF NOT EXISTS profiles;
5 |
6 | -- BĂNG messages cho discussion (nếu dùng postgres_changes)
7 | ALTER PUBLICATION supabase_realtime ADD TABLE IF NOT EXISTS discussion_messages;
8 |
9 | -- BĂNG quiz_attempts (dùng cho dashboard realtime)
10 | ALTER PUBLICATION supabase_realtime ADD TABLE IF NOT EXISTS quiz_attempts;
11 |
12 | -- BĂNG lesson_progress (dùng cho dashboard realtime)
13 | ALTER PUBLICATION supabase_realtime ADD TABLE IF NOT EXISTS lesson_progress;
14 |
```

File: database/20260529_cpu_video_sources.sql (56 lines, 2.1 KB)

```
1 | -- =====
2 | -- THÊM CỘT NGUỒN (SOURCE) CHO LESSON_SECTIONS
3 | -- THÊM VIDEO CPU I ĐỘC VÀO BÀI 2
4 | -- Chạy trong Supabase SQL Editor
5 | -- =====
6 |
7 | -- 1. Thêm cột source (nguồn tham khảo) nếu chưa có
8 | ALTER TABLE public.lesson_sections
9 | ADD COLUMN IF NOT EXISTS source TEXT;
10 |
11 | -- 2. Cập nhật nguồn cho các section hiện có (nếu có dữ liệu)
12 | UPDATE public.lesson_sections
13 | SET source = 'Sách giáo khoa Tin h́c 10 - Kút năi tri th́c, Bài 2'
14 | WHERE lesson_id IN (SELECT id FROM public.lessons WHERE title LIKE '%CPU%');
15 | AND (source IS NULL OR source = '');
16 |
17 | UPDATE public.lesson_sections
18 | SET source = 'Intel ARK Database & Tài liệu kù thu-t'
19 | WHERE lesson_id IN (SELECT id FROM public.lessons WHERE title LIKE '%CPU%');
20 | AND (source IS NULL OR source = '');
21 |
22 | -- 3. Thêm section video CPU Intel Core i vào Bài 2 (Thực hành Lắp ráp CPU)
23 | -- Nếu bài 2 chưa có section video CPU
24 | DO $$
25 | DECLARE
26 |   lesson2_id UUID;
27 |   existing_count INT;
28 | BEGIN
29 |   SELECT id INTO lesson2_id FROM public.lessons
30 |   WHERE title LIKE '%Lắp ráp CPU%' OR title LIKE '%CPU%';
31 |   LIMIT 1;
32 |
33 |   IF lesson2_id IS NOT NULL THEN
34 |     SELECT COUNT(*) INTO existing_count
35 |     FROM public.lesson_sections
36 |     WHERE lesson_id = lesson2_id AND title LIKE '%Intel Core%';
37 |
38 |     IF existing_count = 0 THEN
39 |       INSERT INTO public.lesson_sections (lesson_id, title, type, content, order_index, source)
40 |       VALUES (
41 |         lesson2_id,
42 |         'video: CPU Intel Core i th́ h́c mŭi nh÷t',
43 |         'video',
44 |         'https://www.youtube.com/watch?v=6PAh4WYIqJY',
45 |         0,
46 |         'Intel Official - YouTube Channel'
47 |       );
48 |     END IF;
49 |   END IF;
50 | END $$;
51 |
52 | -- 4. Cập nhật nguồn cho t́t cđ các section khác
53 | UPDATE public.lesson_sections
54 | SET source = 'Sách giáo khoa Tin h́c 10 - Kút năi tri th́c'
55 | WHERE source IS NULL OR source = '';
56 |
```

File: database/20260610_products_full_specs.sql (378 lines, 14.8 KB)

```
1 | -- =====
2 | -- PC Master Builder - Products Table with Full Tech Specs
3 | -- Migration: 20260610_products_full_specs
4 | -- =====
5 |
6 | -- 1. ENUMS
```

```

7 CREATE TYPE component_category AS ENUM (
8 'CPU', 'GPU', 'MAINBOARD', 'RAM', 'PSU', 'CASE', 'COOLER', 'SSD', 'HDD'
9 );
10
11 CREATE TYPE socket_type AS ENUM (
12 'LGA1700', 'LGA1851', 'LGA1200', 'LGA1151', 'AM5', 'AM4', 'sTR5', 'sTRX4'
13 );
14
15 CREATE TYPE form_factor_type AS ENUM (
16 'ATX', 'mATX', 'ITX', 'eATX', 'Mini-ITX', 'XL-ATX'
17 );
18
19 CREATE TYPE ddr_type AS ENUM ('DDR4', 'DDR5');
20
21 CREATE TYPE psu_efficiency AS ENUM (
22 '80plus_white', '80plus_bronze', '80plus_silver',
23 '80plus_gold', '80plus_platinum', '80plus_titanium'
24 );
25
26 CREATE TYPE psu_modular AS ENUM ('non_modular', 'semi_modular', 'full_modular');
27
28 CREATE TYPE cooler_type AS ENUM ('air', 'aio', 'custom_loop');
29
30 CREATE TYPE storage_interface AS ENUM (
31 'm2_nvme', 'm2_sata', 'sata_ssd', 'sata_hdd'
32 );
33
34 CREATE TYPE pcie_gen AS ENUM ('3.0', '4.0', '5.0');
35
36 -- 2. DROP EXISTING products table if present
37 DROP TABLE IF EXISTS public.products CASCADE;
38
39 -- 3. MAIN PRODUCTS TABLE
40 CREATE TABLE public.products (
41 id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
42 category component_category NOT NULL,
43 brand TEXT NOT NULL,
44 model TEXT NOT NULL,
45 full_name TEXT NOT NULL, -- e.g. "Intel Core i5-13600K"
46 release_year SMALLINT,
47 image_url TEXT, -- real photo URL
48 image_angles JSONB DEFAULT '[]'::jsonb, -- ["top.jpg", "bottom.jpg", "side.jpg"]
49 video_url TEXT, -- 10s installation video
50 price_vnd BIGINT, -- base price in VND
51 created_at TIMESTAMPTZ DEFAULT now(),
52 updated_at TIMESTAMPTZ DEFAULT now(),
53
54 -- CPU-specific
55 cpu_socket socket_type,
56 cpu_tdp_watts SMALLINT,
57 cpu_cores SMALLINT,
58 cpu_threads SMALLINT,
59 cpu_base_ghz NUMERIC(4,1),
60 cpu_boost_ghz NUMERIC(4,1),
61 cpu_l3_cache_mb NUMERIC(4,1),
62 cpu_has_igpu BOOLEAN DEFAULT false,
63 cpu_igpu_model TEXT,
64 cpu_ram_type ddr_type, -- what RAM the memory controller supports
65 cpu_max_ram_mhz SMALLINT,
66 cpu_generation TEXT, -- "Raptor Lake", "Zen 4", etc.
67 cpu_cinebench_r23_single INT, -- benchmark score
68 cpu_cinebench_r23_multi INT,
69
70 -- GPU-specific
71 gpu_vram_gb NUMERIC(3,1),
72 gpu_tdp_watts SMALLINT,
73 gpu_pcie_gen pcie_gen,
74 gpu_length_mm SMALLINT, -- critical for case compatibility
75 gpu_width_mm SMALLINT,
76 gpu_height_mm SMALLINT,
77 gpu_power_connectors JSONB DEFAULT '[]'::jsonb, -- [{"type": "8pin", "qty": 2}]
78 gpu_benchmark_1080p INT, -- relative performance score
79 gpu_display_outputs JSONB DEFAULT '[]'::jsonb, -- [{"HDMI 2.1", "DP 1.4a"}]
80 gpu_recommended_psu SMALLINT, -- minimum PSU wattage recommended
81
82 -- Mainboard-specific
83 mb_socket socket_type,
84 mb_chipset TEXT,
85 mb_form_factor form_factor_type,
86 mb_ram_slots SMALLINT,
87 mb_max_ram_gb SMALLINT,
88 mb_ram_type ddr_type,
89 mb_max_ram_mhz SMALLINT,
90 mb_m2_slots SMALLINT,
91 mb_m2_pcie_gens JSONB DEFAULT '[]'::jsonb, -- ["4.0", "5.0"]
92 mb_pcie_x16_slots SMALLINT,
93 mb_pcie_x16_gens JSONB DEFAULT '[]'::jsonb,
94 mb_sata_ports SMALLINT,
95 mb_has_wifi BOOLEAN DEFAULT false,
96 mb_wifi_version TEXT,
97 mb_usb_ports JSONB DEFAULT '[]'::jsonb, -- [{"type": "USB 3.2 Gen2", "qty": 4}]
98 mb_front_panel_headers JSONB DEFAULT '[]'::jsonb, -- [{"type": "USB 3.0", "pin": 19}]
99
100 -- PSU-specific
101 psu_wattage SMALLINT,
102 psu_efficiency psu_efficiency,
103 psu_modular psu_modular,
104 psu_connectors JSONB DEFAULT '[]'::jsonb, -- [{"type": "PCIe 8-pin", "qty": 4}, {"type": "SATA", "qty": 8}]
105 psu_has_12vhpwr BOOLEAN DEFAULT false,
106 psu_length_mm SMALLINT,
107 psu_fan_size_mm SMALLINT,
108
109 -- Case-specific
110 case_supported_forms JSONB DEFAULT '[]'::jsonb, -- ["ATX", "mATX", "ITX"]
111 case_max_gpu_length_mm SMALLINT,
112 case_max_cooler_height_mm SMALLINT,
113 case_max_psu_length_mm SMALLINT,
114 case_drive_bays JSONB DEFAULT '[]'::jsonb, -- [{"type": "3.5", "qty": 2}, {"type": "2.5", "qty": 4}]
115 case_fan_slots JSONB DEFAULT '[]'::jsonb, -- [{"size": 120, "qty": 3, "type": "intake"}, {"size": 120, "qty": 1, "type": "exh..."}]
116 case_supports_aio JSONB DEFAULT '[]'::jsonb, -- [240, 280, 360]
117 case_mesh_front BOOLEAN DEFAULT false,
118 case_has_cable_management BOOLEAN DEFAULT true,
119 case_side_panel TEXT, -- "glass", "mesh", "solid", "acrylic"
120 case_weight_kg NUMERIC(4,1),
121
122 -- Cooler-specific
123 cooler_max_tdp_support SMALLINT,
124 cooler_height_mm SMALLINT,
125 cooler_supported_sockets JSONB DEFAULT '[]'::jsonb, -- ["LGA1700", "AM5"]
126 cooler_type cooler_type,
127 cooler_aio_size_mm SMALLINT, -- 120/240/280/360 if AIO
128 cooler_fan_rpm_max SMALLINT,
129 cooler_fan_noise_dba NUMERIC(3,1),
130 cooler_fan_qty SMALLINT,

```

```

131 cooler_has_rgb          BOOLEAN DEFAULT false,
132
133 -- RAM-specific
134 ram_ddr_type            ddr_type,
135 ram_speed_mhz           SMALLINT,
136 ram_cl_latency          SMALLINT,          -- CAS latency
137 ram_capacity_gb         SMALLINT,
138 ram_kit_count           SMALLINT DEFAULT 1, -- 1/2/4 sticks in package
139 ram_has_xmp             BOOLEAN DEFAULT false,
140 ram_has_expo            BOOLEAN DEFAULT false,
141 ram_voltage_v           NUMERIC(3,2),
142 ram_height_mm           SMALLINT,          -- for cooler clearance
143
144 -- SSD-specific
145 ssd_interface           storage_interface,
146 ssd_pcie_gen            pcie_gen,
147 ssd_read_mbps           INT,
148 ssd_write_mbps          INT,
149 ssd_has_dram_cache      BOOLEAN DEFAULT false,
150 ssd_capacity_gb         INT,
151 ssd_form_factor         TEXT,              -- "M.2 2280", "2.5 inch", "U.2"
152 ssd_nand_type           TEXT,              -- "TLC", "QLC", "MLC", "SLC"
153 ssd_tbw                 INT,              -- Terabytes Written endurance
154 ssd_max_seq_read        INT,
155 ssd_max_seq_write       INT,
156 ssd_max_rand_read       INT,
157 ssd_max_rand_write      INT,
158
159 -- HDD-specific
160 hdd_capacity_gb         INT,
161 hdd_rpm                 SMALLINT,          -- 5400/7200/10000
162 hdd_cache_mb            SMALLINT,
163 hdd_form_factor         TEXT,              -- "3.5 inch", "2.5 inch"
164 hdd_read_mbps           INT,
165 hdd_write_mbps          INT,
166
167 CONSTRAINT unique_product UNIQUE (category, brand, model)
168 );
169
170 -- 4. INDEXES
171 CREATE INDEX idx_products_category ON public.products(category);
172 CREATE INDEX idx_products_socket ON public.products(cpu_socket);
173 CREATE INDEX idx_products_mb_socket ON public.products(mb_socket);
174 CREATE INDEX idx_products_form_factor ON public.products(mb_form_factor);
175 CREATE INDEX idx_products_ram_type ON public.products(ram_ddr_type);
176 CREATE INDEX idx_products_case_forms ON public.products USING gin(case_supported_forms);
177
178 -- 5. RLS
179 ALTER TABLE public.products ENABLE ROW LEVEL SECURITY;
180
181 CREATE POLICY "Products are viewable by everyone"
182 ON public.products FOR SELECT
183 USING (true);
184
185 CREATE POLICY "Products are insertable by admin only"
186 ON public.products FOR INSERT
187 WITH CHECK (
188   auth.uid() IN (SELECT id FROM public.profiles WHERE role = 'admin')
189 );
190
191 CREATE POLICY "Products are updatable by admin only"
192 ON public.products FOR UPDATE
193 USING (auth.uid() IN (SELECT id FROM public.profiles WHERE role = 'admin'));
194
195 -- 6. TRIGGER: auto-update updated_at
196 CREATE OR REPLACE FUNCTION update_products_updated_at()
197 RETURNS TRIGGER AS $$
198 BEGIN
199   NEW.updated_at = now();
200   RETURN NEW;
201 END;
202 $$ LANGUAGE plpgsql;
203
204 DROP TRIGGER IF EXISTS trg_products_updated_at ON public.products;
205 CREATE TRIGGER trg_products_updated_at
206 BEFORE UPDATE ON public.products
207 FOR EACH ROW EXECUTE FUNCTION update_products_updated_at();
208
209 -- 7. COMPATIBILITY CHECK FUNCTION
210 -- Returns compatibility issues between components in a build
211 CREATE OR REPLACE FUNCTION check_build_compatibility(
212   p_cpu_id          UUID,
213   p_mb_id           UUID,
214   p_gpu_id          UUID DEFAULT NULL,
215   p_ram_ids         UUID[] DEFAULT '{}',
216   p_psu_id          UUID DEFAULT NULL,
217   p_case_id         UUID DEFAULT NULL,
218   p_cooler_id       UUID DEFAULT NULL,
219   p_ssd_ids         UUID[] DEFAULT '{}'
220 )
221 RETURNS TABLE(
222   severity TEXT,
223   check_type TEXT,
224   message TEXT,
225   fix TEXT
226 ) LANGUAGE plpgsql STABLE AS $$
227 DECLARE
228   cpu RECORD;
229   mb RECORD;
230   gpu RECORD;
231   psu RECORD;
232   case_rec RECORD;
233   cooler RECORD;
234   total_tdp INT := 0;
235   total_components INT;
236   gpu_length INT;
237   cooler_height INT;
238   wattage_available INT;
239 BEGIN
240   -- Fetch CPU
241   SELECT * INTO cpu FROM products WHERE id = p_cpu_id AND category = 'CPU';
242   IF NOT FOUND THEN RETURN; END IF;
243
244   -- Fetch Mainboard
245   SELECT * INTO mb FROM products WHERE id = p_mb_id AND category = 'MAINBOARD';
246   IF NOT FOUND THEN RETURN; END IF;
247
248   -- 1. Socket check
249   IF cpu.cpu_socket IS DISTINCT FROM mb.mb_socket THEN
250     RETURN QUERY SELECT
251       'error'::TEXT,
252       'socket_mismatch'::TEXT,
253       format('CPU %s (socket %s) không t°ing thich v°i Mainboard %s (socket %s)',
254         cpu.full_name, cpu.cpu_socket, mb.full_name, mb.mb_socket),

```

```

256|         format('Chín Mainboard socket %s ho·c CPU socket %s', cpu.cpu_socket);
257|     END IF;
258|
259|     -- 2. RAM type check
260|     IF cpu.cpu_ram_type IS DISTINCT FROM mb.mb_ram_type THEN
261|         RETURN QUERY SELECT
262|             'error'::TEXT,
263|             format('CPU h× trā %s nh°ng Mainboard ché h× trā %s',
264|                 cpu.cpu_ram_type, mb.mb_ram_type),
265|             format('Chín CPU %s ho·c Mainboard %s phù hăp', mb.mb_ram_type, cpu.cpu_socket);
266|     END IF;
267|
268|     -- 3. GPU checks
269|     IF p_gpu_id IS NOT NULL THEN
270|         SELECT * INTO gpu FROM products WHERE id = p_gpu_id AND category = 'GPU';
271|         IF FOUND THEN
272|             total_tdp := total_tdp + COALESCE(gpu.gpu_tdp_watts, 0);
273|             gpu_length := gpu.gpu_length_mm;
274|
275|             -- GPU length vs Case
276|             IF p_case_id IS NOT NULL THEN
277|                 SELECT * INTO case_rec FROM products WHERE id = p_case_id AND category = 'CASE';
278|                 IF FOUND AND gpu_length > case_rec.case_max_gpu_length_mm THEN
279|                     RETURN QUERY SELECT 'error'::TEXT, 'gpu_too_long'::TEXT,
280|                         format('GPU %s (dài %dmm) quá dài so v°i Case %s (h× trā tñ %dmm)',
281|                             gpu.full_name, gpu_length, case_rec.full_name, case_rec.case_max_gpu_length_mm),
282|                         format('Chín GPU ng·n hìn %dmm ho·c Case h× trā GPU dài hìn %dmm', gpu_length, case_rec.case_max_gpu_length_mm);
283|                     END IF;
284|                 END IF;
285|             END IF;
286|         END IF;
287|
288|     -- 4. PSU check
289|     total_tdp := total_tdp + COALESCE(cpu.cpu_tdp_watts, 0);
290|
291|     IF p_psu_id IS NOT NULL THEN
292|         SELECT * INTO psu FROM products WHERE id = p_psu_id AND category = 'PSU';
293|         IF FOUND THEN
294|             wattage_available := psu.psu_wattage * 0.8;
295|             IF total_tdp > wattage_available THEN
296|                 RETURN QUERY SELECT 'error'::TEXT, 'psu_insufficient'::TEXT,
297|                     format('Tổng TDP °üc tính %dw v°ất quá 80%% công suất PSU (%dw)', total_tdp, wattage_available),
298|                     format('Nâng PSU lên ít nh÷t %dw ho·c giēm TDP build', (total_tdp - wattage_available) / 0.8);
299|             END IF;
300|         END IF;
301|     END IF;
302|
303|     -- 5. Cooler check
304|     IF p_cooler_id IS NOT NULL THEN
305|         SELECT * INTO cooler FROM products WHERE id = p_cooler_id AND category = 'COOLER';
306|         IF FOUND THEN
307|             cooler_height := cooler.cooler_height_mm;
308|
309|             IF cpu.cpu_tdp_watts > cooler.cooler_max_tdp_support * 0.9 THEN
310|                 RETURN QUERY SELECT 'warning'::TEXT, 'cooler_insufficient'::TEXT,
311|                     format('CPU TDP %dw gñn ít gi÷i hìn tñ nh÷t cça Cooler %s (%dw)', cpu.cpu_tdp_watts, cooler.full_name, cooler.cooler_max_tdp_support),
312|                     format('Cần nh·c nâng cooler lên AIO ho·c cooler khi hi÷u n ng');
313|             END IF;
314|
315|             -- Cooler height vs Case
316|             IF p_case_id IS NOT NULL AND case_rec.id IS NOT NULL THEN
317|                 IF cooler_height > case_rec.case_max_cooler_height_mm THEN
318|                     RETURN QUERY SELECT 'error'::TEXT, 'cooler_too_tall'::TEXT,
319|                         format('Cooler %s (cao %dmm) quá cao cho Case %s (h× trā tñ %dmm)',
320|                             cooler.full_name, cooler_height, case_rec.full_name, case_rec.case_max_cooler_height_mm),
321|                         format('Chín cooler th÷p hìn %dmm ho·c Case rùng hìn', cooler_height, case_rec.case_max_cooler_height_mm);
322|                     END IF;
323|                 END IF;
324|
325|             -- Cooler socket vs CPU socket
326|             IF NOT cooler.cooler_supported_sockets @> to_jsonb(cpu.cpu_socket::TEXT) THEN
327|                 RETURN QUERY SELECT 'error'::TEXT, 'cooler_socket_mismatch'::TEXT,
328|                     format('Cooler %s không h× trā socket %s', cooler.full_name, cpu.cpu_socket),
329|                     format('Chín cooler h× trā socket %s', cpu.cpu_socket);
330|             END IF;
331|         END IF;
332|     END IF;
333|
334|     -- 6. Mainboard form factor vs Case
335|     IF p_case_id IS NOT NULL AND case_rec.id IS NOT NULL THEN
336|         IF NOT case_rec.case_supported_forms @> to_jsonb(mb.mb_form_factor::TEXT) THEN
337|             RETURN QUERY SELECT 'error'::TEXT, 'form_factor_mismatch'::TEXT,
338|                 format('Mainboard %s (form factor %s) không vëa Case %s (h× trā %s)',
339|                     mb.full_name, mb.mb_form_factor, case_rec.full_name, case_rec.case_supported_forms),
340|                 (SELECT string_agg(elem::TEXT, ', ') FROM jsonb_array_elements_text(case_rec.case_supported_for...));
341|             END IF;
342|         END IF;
343|     END IF;
344|
345|     -- 7. No iGPU + No GPU warning
346|     IF NOT cpu.cpu_has_igpu AND p_gpu_id IS NULL THEN
347|         RETURN QUERY SELECT 'warning'::TEXT, 'no_display_output'::TEXT,
348|             'CPU không có iGPU và ch°a chín GPU rýi - s÷ không có hình ãnh s÷',
349|             'Thêm GPU rýi ho·c chín CPU có iGPU (đòng Intel non-F)';
350|     END IF;
351|
352|     -- 8. RAM check
353|     IF array_length(p_ram_ids, 1) > 0 THEN
354|         IF array_length(p_ram_ids, 1) > mb.mb_ram_slots THEN
355|             RETURN QUERY SELECT 'error'::TEXT, 'too_many_ram'::TEXT,
356|                 format('Sñ thanh RAM (%d) v°ất quá sñ slot cça Mainboard (%d)',
357|                     array_length(p_ram_ids, 1), mb.mb_ram_slots);
358|             END IF;
359|         END IF;
360|     END IF;
361|
362|     -- 9. SSD M.2 check
363|     IF array_length(p_ssd_ids, 1) > 0 THEN
364|         total_components := (
365|             SELECT COUNT(*) FROM products WHERE id = ANY(p_ssd_ids)
366|         );
367|         IF total_components > mb.mb_m2_slots THEN
368|             RETURN QUERY SELECT 'warning'::TEXT, 'not_enough_m2_slots'::TEXT,
369|                 format('Sñ l°ãng SSD M.2 (%d) v°ất quá sñ slot M.2 cça Mainboard (%d)',
370|                     total_components, mb.mb_m2_slots),
371|                 format('Giēm SSD M.2 ho·c dùng SSD SATA cho các ò còn lii');
372|             END IF;
373|         END IF;
374|     END IF;
375|
376| END;
377| $$;
378|

```

File: database/20260610_saved_builds.sql (147 lines, 6.5 KB)

```
1  -- Saved Builds table (Idea #2 + #7)
2  CREATE TABLE IF NOT EXISTS public.saved_builds (
3      id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
4      user_id     UUID NOT NULL REFERENCES auth.users(id) ON DELETE CASCADE,
5      name        TEXT NOT NULL DEFAULT 'Build m0i',
6      cpu_id      UUID REFERENCES public.products(id) ON DELETE SET NULL,
7      gpu_id      UUID REFERENCES public.products(id) ON DELETE SET NULL,
8      mainboard_id UUID REFERENCES public.products(id) ON DELETE SET NULL,
9      ram_ids     UUID[] DEFAULT '{}',
10     psu_id       UUID REFERENCES public.products(id) ON DELETE SET NULL,
11     case_id      UUID REFERENCES public.products(id) ON DELETE SET NULL,
12     cooler_id    UUID REFERENCES public.products(id) ON DELETE SET NULL,
13     ssd_ids      UUID[] DEFAULT '{}',
14     compatibility_score INT DEFAULT 0,
15     total_tdp    INT DEFAULT 0,
16     total_price  BIGINT DEFAULT 0,
17     is_public    BOOLEAN DEFAULT false,
18     created_at   TIMESTAMPTZ DEFAULT now(),
19     updated_at   TIMESTAMPTZ DEFAULT now()
20 );
21
22 CREATE INDEX idx_saved_builds_user ON public.saved_builds(user_id);
23 CREATE INDEX idx_saved_builds_public ON public.saved_builds(is_public);
24
25 ALTER TABLE public.saved_builds ENABLE ROW LEVEL SECURITY;
26
27 CREATE POLICY "Users can view own builds"
28     ON public.saved_builds FOR SELECT USING (auth.uid() = user_id OR is_public = true);
29
30 CREATE POLICY "Users can insert own builds"
31     ON public.saved_builds FOR INSERT WITH CHECK (auth.uid() = user_id);
32
33 CREATE POLICY "Users can update own builds"
34     ON public.saved_builds FOR UPDATE USING (auth.uid() = user_id);
35
36 CREATE POLICY "Users can delete own builds"
37     ON public.saved_builds FOR DELETE USING (auth.uid() = user_id);
38
39 -- Build Scenarios table (Idea #7)
40 CREATE TABLE IF NOT EXISTS public.build_scenarios (
41     id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
42     name        TEXT NOT NULL,
43     name_vn     TEXT NOT NULL,
44     description  TEXT NOT NULL,
45     description_vn TEXT NOT NULL,
46     icon        TEXT NOT NULL DEFAULT '0=Ý',
47     difficulty   TEXT NOT NULL CHECK (difficulty IN ('easy', 'medium', 'hard', 'expert')),
48     budget_min  BIGINT DEFAULT 0,
49     budget_max  BIGINT DEFAULT 999999999,
50     requirements JSONB DEFAULT '{}':jsonb,
51     weight_components JSONB DEFAULT '{}':jsonb,
52     created_at  TIMESTAMPTZ DEFAULT now()
53 );
54
55 ALTER TABLE public.build_scenarios ENABLE ROW LEVEL SECURITY;
56 CREATE POLICY "Scenarios viewable by all" ON public.build_scenarios FOR SELECT USING (true);
57
58 -- Scenario scoring table
59 CREATE TABLE IF NOT EXISTS public.build_scores (
60     id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
61     build_id    UUID REFERENCES public.saved_builds(id) ON DELETE CASCADE,
62     scenario_id UUID REFERENCES public.build_scenarios(id) ON DELETE CASCADE,
63     user_id     UUID REFERENCES auth.users(id) ON DELETE CASCADE,
64     score       INT NOT NULL,
65     rank        INT,
66     created_at  TIMESTAMPTZ DEFAULT now(),
67     UNIQUE(build_id, scenario_id)
68 );
69
70 ALTER TABLE public.build_scores ENABLE ROW LEVEL SECURITY;
71 CREATE POLICY "Scores viewable by all" ON public.build_scores FOR SELECT USING (true);
72 CREATE POLICY "Users can insert own scores" ON public.build_scores FOR INSERT WITH CHECK (auth.uid() = user_id);
73
74 -- Cable Management scores (Idea #3)
75 CREATE TABLE IF NOT EXISTS public.cable_management_scores (
76     id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
77     user_id     UUID REFERENCES auth.users(id) ON DELETE CASCADE,
78     score       INT NOT NULL,
79     time_seconds INT,
80     cables_used INT,
81     difficulty   TEXT,
82     created_at  TIMESTAMPTZ DEFAULT now()
83 );
84
85 ALTER TABLE public.cable_management_scores ENABLE ROW LEVEL SECURITY;
86 CREATE POLICY "Cable scores viewable by all" ON public.cable_management_scores FOR SELECT USING (true);
87 CREATE POLICY "Users can insert own cable scores" ON public.cable_management_scores FOR INSERT WITH CHECK (auth.uid() = user_id);
88
89 -- OS Installation progress (Idea #4)
90 CREATE TABLE IF NOT EXISTS public.os_installation_progress (
91     id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
92     user_id     UUID REFERENCES auth.users(id) ON DELETE CASCADE,
93     step        INT DEFAULT 0,
94     completed   BOOLEAN DEFAULT false,
95     time_seconds INT,
96     errors_count INT DEFAULT 0,
97     created_at  TIMESTAMPTZ DEFAULT now(),
98     updated_at  TIMESTAMPTZ DEFAULT now()
99 );
100
101 ALTER TABLE public.os_installation_progress ENABLE ROW LEVEL SECURITY;
102 CREATE POLICY "OS progress viewable by owner" ON public.os_installation_progress FOR SELECT USING (auth.uid() = user_id);
103 CREATE POLICY "Users can insert own OS progress" ON public.os_installation_progress FOR INSERT WITH CHECK (auth.uid() = user_id);
104 CREATE POLICY "Users can update own OS progress" ON public.os_installation_progress FOR UPDATE USING (auth.uid() = user_id);
105
106 -- PC Diagnosis scenarios (Idea #6)
107 CREATE TABLE IF NOT EXISTS public.diagnosis_scenarios (
108     id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
109     title       TEXT NOT NULL,
110     title_vn    TEXT NOT NULL,
111     description  TEXT NOT NULL,
112     description_vn TEXT NOT NULL,
113     symptoms    TEXT[] DEFAULT '{}',
114     symptoms_vn TEXT[] DEFAULT '{}',
115     correct_answer TEXT NOT NULL,
116     hints       TEXT[] DEFAULT '{}',
117     hints_vn    TEXT[] DEFAULT '{}',
118     difficulty   TEXT NOT NULL CHECK (difficulty IN ('easy', 'medium', 'hard')),
119     icon        TEXT NOT NULL DEFAULT '0=Ý',
120     created_at  TIMESTAMPTZ DEFAULT now()
121 );
```



```

121 );
122
123 ALTER TABLE public.diagnosis_scenarios ENABLE ROW LEVEL SECURITY;
124 CREATE POLICY "Diagnosis viewable by all" ON public.diagnosis_scenarios FOR SELECT USING (true);
125
126 -- Thermal simulation data (Idea #5)
127 CREATE TABLE IF NOT EXISTS public.thermal_profiles (
128     id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
129     component_type    TEXT NOT NULL,
130     max_temp_c        INT NOT NULL,
131     idle_temp_c       INT NOT NULL,
132     load_temp_c       INT NOT NULL,
133     color_hex         TEXT DEFAULT '#ff4444',
134     created_at        TIMESTAMPTZ DEFAULT now()
135 );
136
137 ALTER TABLE public.thermal_profiles ENABLE ROW LEVEL SECURITY;
138 CREATE POLICY "Thermal profiles viewable by all" ON public.thermal_profiles FOR SELECT USING (true);
139
140 INSERT INTO public.thermal_profiles (component_type, max_temp_c, idle_temp_c, load_temp_c, color_hex) VALUES
141 ('CPU', 100, 40, 85, '#ff4444'),
142 ('GPU', 85, 35, 78, '#ff6600'),
143 ('RAM', 85, 30, 55, '#ffaa00'),
144 ('SSD', 70, 28, 55, '#ffcc00'),
145 ('PSU', 80, 30, 65, '#ff8800'),
146 ('CASE', 50, 25, 40, '#44aa00');
147

```

File: database/20260610_seed_products.sql (99 lines, 14.7 KB)

```

1  -- =====
2  -- PC Master Builder – Seed Data: Real Components (2026)
3  -- Migration: 20260610_seed_products
4  -- =====
5
6  INSERT INTO public.products (category, brand, model, full_name, release_year, price_vnd, cpu_socket, cpu_tdp_w...
7  -- Intel 14th Gen
8  ('CPU', 'Intel', 'i5-14600K', 'Intel Core i5-14600K', 2024, 84900000, 'LGA1700', 125, 14, 20, 3.5, 5.3, 24.0, t...
9  ('CPU', 'Intel', 'i5-14600KF', 'Intel Core i5-14600KF', 2024, 79900000, 'LGA1700', 125, 14, 20, 3.5, 5.3, 24.0, ...
10 ('CPU', 'Intel', 'i7-14700K', 'Intel Core i7-14700K', 2024, 129900000, 'LGA1700', 125, 20, 28, 3.4, 5.6, 33.0, ...
11 ('CPU', 'Intel', 'i9-14900K', 'Intel Core i9-14900K', 2024, 169900000, 'LGA1700', 125, 24, 32, 3.2, 6.0, 36.0, ...
12 -- Intel 13th Gen
13 ('CPU', 'Intel', 'i3-13100F', 'Intel Core i3-13100F', 2023, 27500000, 'LGA1700', 58, 4, 8, 3.4, 4.5, 12.0, fals...
14 ('CPU', 'Intel', 'i5-13600K', 'Intel Core i5-13600K', 2022, 74900000, 'LGA1700', 125, 14, 20, 3.5, 5.1, 24.0, t...
15 -- AMD Ryzen 7000
16 ('CPU', 'AMD', 'R5-7600', 'AMD Ryzen 5 7600', 2023, 65000000, 'AM5', 65, 6, 12, 3.8, 5.1, 32.0, true, 'DDR5', 5...
17 ('CPU', 'AMD', 'R5-7600X', 'AMD Ryzen 5 7600X', 2022, 71000000, 'AM5', 105, 6, 12, 4.7, 5.3, 32.0, true, 'DDR5'...
18 ('CPU', 'AMD', 'R7-7800X3D', 'AMD Ryzen 7 7800X3D', 2023, 118000000, 'AM5', 120, 8, 16, 4.2, 5.0, 96.0, true, '...
19 ('CPU', 'AMD', 'R9-7950X', 'AMD Ryzen 9 7950X', 2022, 176000000, 'AM5', 170, 16, 32, 4.5, 5.7, 64.0, true, 'DDR...
20 -- AMD Ryzen 5000
21 ('CPU', 'AMD', 'R5-5600', 'AMD Ryzen 5 5600', 2022, 33000000, 'AM4', 65, 6, 12, 3.5, 4.4, 32.0, false, 'DDR4', ...
22 ('CPU', 'AMD', 'R7-5800X3D', 'AMD Ryzen 7 5800X3D', 2022, 81000000, 'AM4', 105, 8, 16, 3.4, 4.5, 96.0, false, '...
23
24 -- GPUs
25 INSERT INTO public.products (category, brand, model, full_name, release_year, price_vnd, gpu_vram_gb, gpu_tdp...
26 ('GPU', 'NVIDIA', 'RTX 4060', 'NVIDIA GeForce RTX 4060', 2023, 82000000, 8, 115, '4.0', 200, 18000, 550, '['HDM...
27 ('GPU', 'NVIDIA', 'RTX 4060 Ti 8GB', 'NVIDIA GeForce RTX 4060 Ti 8GB', 2023, 110000000, 8, 160, '4.0', 240, 200...
28 ('GPU', 'NVIDIA', 'RTX 4060 Ti 16GB', 'NVIDIA GeForce RTX 4060 Ti 16GB', 2024, 135000000, 16, 160, '4.0', 240, ...
29 ('GPU', 'NVIDIA', 'RTX 4070 Super', 'NVIDIA GeForce RTX 4070 Super', 2024, 155000000, 12, 220, '4.0', 260, 2600...
30 ('GPU', 'NVIDIA', 'RTX 4070 Ti Super', 'NVIDIA GeForce RTX 4070 Ti Super', 2024, 190000000, 16, 285, '4.0', 285...
31 ('GPU', 'NVIDIA', 'RTX 4080 Super', 'NVIDIA GeForce RTX 4080 Super', 2024, 280000000, 16, 320, '4.0', 310, 3500...
32 ('GPU', 'NVIDIA', 'RTX 4090', 'NVIDIA GeForce RTX 4090', 2022, 420000000, 24, 450, '4.0', 357, 52000, 850, '['H...
33 ('GPU', 'AMD', 'RX 7600', 'AMD Radeon RX 7600', 2023, 65000000, 8, 165, '4.0', 204, 14000, 550, '['HDMI 2.1","D...
34 ('GPU', 'AMD', 'RX 7700 XT', 'AMD Radeon RX 7700 XT', 2023, 95000000, 12, 245, '4.0', 267, 18000, 650, '['HDMI...
35 ('GPU', 'AMD', 'RX 7800 XT', 'AMD Radeon RX 7800 XT', 2023, 125000000, 16, 263, '4.0', 267, 21500, 700, '['HDMI...
36 ('GPU', 'NVIDIA', 'GTX 1650', 'NVIDIA GeForce GTX 1650', 2019, 25000000, 4, 75, '3.0', 170, 6000, 300, '['HDMI'...
37
38 -- Mainboards / ATX / LGA1700
39 INSERT INTO public.products (category, brand, model, full_name, release_year, price_vnd, mb_socket, mb_chipset...
40 ('MAINBOARD', 'ASUS', 'TUF GAMING Z790-PLUS', 'ASUS TUF Gaming Z790-PLUS WiFi', 2023, 65000000, 'LGA1700', 'Z790...
41 ('MAINBOARD', 'MSI', 'PRO Z790-A MAX', 'MSI PRO Z790-A MAX WiFi', 2024, 58000000, 'LGA1700', 'Z790', 'ATX', 4, ...
42 ('MAINBOARD', 'Gigabyte', 'B760M DS3H', 'Gigabyte B760M DS3H WiFi', 2024, 32000000, 'LGA1700', 'B760', 'mATX', ...
43 ('MAINBOARD', 'ASUS', 'PRIME B760M-A', 'ASUS Prime B760M-A WiFi', 2024, 28000000, 'LGA1700', 'B760', 'mATX', 2...
44 -- AM5
45 ('MAINBOARD', 'ASUS', 'TUF GAMING B650-PLUS', 'ASUS TUF Gaming B650-PLUS WiFi', 2023, 52000000, 'AM5', 'B650', ...
46 ('MAINBOARD', 'MSI', 'MAG X670E TOMAHAWK', 'MSI MAG X670E Tomahawk WiFi', 2023, 75000000, 'AM5', 'X670E', 'ATX'...
47 -- AM4
48 ('MAINBOARD', 'ASUS', 'TUF GAMING B550M', 'ASUS TUF Gaming B550M WiFi', 2022, 28000000, 'AM4', 'B550', 'mATX', ...
49
50 -- RAM DDR5
51 INSERT INTO public.products (category, brand, model, full_name, release_year, price_vnd, ram_ddr_type, ram_spe...
52 ('RAM', 'Kingston', 'KF560C36-16', 'Kingston Fury Beast 16GB DDR5-6000', 2023, 12000000, 'DDR5', 6000, 36, 16, ...
53 ('RAM', 'Kingston', 'KF560C36-32', 'Kingston Fury Beast 32GB (2x16) DDR5-6000', 2023, 22000000, 'DDR5', 6000, 3...
54 ('RAM', 'G.Skill', 'F5-6000J3636F16G', 'G.Skill Trident Z5 32GB (2x16) DDR5-6000 CL36', 2023, 25000000, 'DDR5'...
55 ('RAM', 'Corsair', 'CMK32GX5M2B6000C36', 'Corsair Vengeance 32GB (2x16) DDR5-6000', 2023, 24000000, 'DDR5', 600...
56 ('RAM', 'G.Skill', 'F5-6000J3040G32G', 'G.Skill Trident Z5 Neo 64GB (2x32) DDR5-6000 CL30', 2024, 48000000, 'DD...
57 ('RAM', 'Kingston', 'KF560C30-64', 'Kingston Fury Beast 64GB (2x32) DDR5-6000', 2024, 45000000, 'DDR5', 6000, 3...
58
59 -- RAM DDR4
60 INSERT INTO public.products (category, brand, model, full_name, release_year, price_vnd, ram_ddr_type, ram_spe...
61 ('RAM', 'Kingston', 'KF432C16-8', 'Kingston Fury Beast 8GB DDR4-3200', 2022, 4000000, 'DDR4', 3200, 16, 8, 1, t...
62 ('RAM', 'Kingston', 'KF432C16-16', 'Kingston Fury Beast 16GB DDR4-3200', 2022, 7000000, 'DDR4', 3200, 16, 16, 1...
63 ('RAM', 'G.Skill', 'F4-3600C18D-32', 'G.Skill Ripjaws V 32GB (2x16) DDR4-3600', 2022, 13500000, 'DDR4', 3600, 1...
64
65 -- PSU
66 INSERT INTO public.products (category, brand, model, full_name, release_year, price_vnd, psu_wattage, psu_effi...
67 ('PSU', 'Corsair', 'CX550', 'Corsair CX550 550W 80+ Bronze', 2022, 12000000, 550, '80plus_bronze', 'semi_modula...
68 ('PSU', 'Corsair', 'RM650', 'Corsair RM650 650W 80+ Gold', 2023, 22000000, 650, '80plus_gold', 'full_modular', ...
69 ('PSU', 'Corsair', 'RM750', 'Corsair RM750 750W 80+ Gold', 2023, 26000000, 750, '80plus_gold', 'full_modular', ...
70 ('PSU', 'Corsair', 'RM850', 'Corsair RM850 850W 80+ Gold', 2023, 31000000, 850, '80plus_gold', 'full_modular', ...
71 ('PSU', 'Corsair', 'RM1000x', 'Corsair RM1000x 1000W 80+ Gold', 2024, 42000000, 1000, '80plus_gold', 'full_modu...
72 ('PSU', 'FSP', 'VITA GM 750', 'FSP VITA GM 750W 80+ Gold', 2024, 17500000, 750, '80plus_gold', 'full_modular', ...
73
74 -- Cases
75 INSERT INTO public.products (category, brand, model, full_name, release_year, price_vnd, case_supported_forms...
76 ('CASE', 'NZXT', 'H5 Flow', 'NZXT H5 Flow (2024)', 2024, 22000000, '['ATX","mATX","ITX"]':jsonb, 365, 165, 200...
77 ('CASE', 'Corsair', '4000D Airflow', 'Corsair 4000D Airflow', 2022, 24000000, '['ATX","mATX","ITX"]':jsonb, 36...
78 ('CASE', 'Lian Li', 'LANCOOL 216', 'Lian Li LANCOOL 216', 2023, 26000000, '['ATX","mATX","ITX"]':jsonb, 392, 1...
79 ('CASE', 'Fractal', 'Pop Air', 'Fractal Design Pop Air', 2023, 19000000, '['ATX","mATX","ITX"]':jsonb, 405, 17...
80 ('CASE', 'Cooler Master', 'MasterBox Q300L', 'Cooler Master MasterBox Q300L', 2020, 9000000, '['mATX","ITX"]':...
81
82 -- Coolers
83 INSERT INTO public.products (category, brand, model, full_name, release_year, price_vnd, cooler_max_tdp_suppor...
84 ('COOLER', 'Cooler Master', 'Hyper 212', 'Cooler Master Hyper 212 Black', 2022, 6500000, 150, 152, '['LGA1700"...
85 ('COOLER', 'Noctua', 'NH-D15', 'Noctua NH-D15', 2020, 18500000, 250, 165, '['LGA1700","LGA1200","LGA1151","AM5"...
86 ('COOLER', 'Deepcool', 'AK400', 'Deepcool AK400 Black', 2023, 5500000, 180, 155, '['LGA1700","LGA1200","AM5","A...
87 ('COOLER', 'Corsair', 'H100i RGB', 'Corsair H100i RGB Elite 240mm', 2023, 31000000, 250, NULL, '['LGA1700","LGA...
88 ('COOLER', 'Corsair', 'H150i RGB', 'Corsair H150i RGB Elite 360mm', 2023, 42000000, 300, NULL, '['LGA1700","LGA...
89 ('COOLER', 'NZXT', 'Kraken X63', 'NZXT Kraken X63 280mm', 2023, 38000000, 280, NULL, '['LGA1700","LGA1200","AM5...
90
91 -- SSDs
92 INSERT INTO public.products (category, brand, model, full_name, release_year, price_vnd, ssd_interface, ssd_pc...
93 ('SSD', 'Samsung', '990 Pro', 'Samsung 990 Pro 1TB NVMe', 2024, 32000000, 'm2_nvme', '4.0', 1000, 7450, 6900, t...

```

```
94 | ('SSD', 'Samsung', '990 Pro 2TB', 'Samsung 990 Pro 2TB NVMe', 2024, 5600000, 'm2_nvme', '4.0', 2000, 7450, 690...
95 | ('SSD', 'SK Hynix', 'P41 Plat', 'SK Hynix Platinum P41 1TB', 2023, 3000000, 'm2_nvme', '4.0', 1000, 7000, 6500...
96 | ('SSD', 'WD', 'SN850X', 'WD Black SN850X 1TB NVMe', 2023, 2900000, 'm2_nvme', '4.0', 1000, 7300, 6300, true, '...
97 | ('SSD', 'Kingston', 'NV3', 'Kingston NV3 1TB NVMe', 2024, 1400000, 'm2_nvme', '4.0', 1000, 6000, 4000, false, '...
98 | ('SSD', 'Crucial', 'MX500', 'Crucial MX500 1TB SATA', 2021, 1900000, 'sata_ssd', NULL, 1000, 560, 510, true, '...
99 |
```

File: database/20260621_gamification_system.sql (159 lines, 8.2 KB)

```
1  -- =====
2  -- GAMIFICATION SYSTEM: Levels, Titles, Ranks
3  -- Solo Leveling Inspired
4  -- =====
5
6  -- 1. Add columns to profiles table for level system
7  ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS xp INTEGER DEFAULT 0;
8  ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS level INTEGER DEFAULT 1;
9  ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS title_id TEXT;
10 ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS total quests_completed INTEGER DEFAULT 0;
11
12 -- 2. Level definitions table
13 CREATE TABLE IF NOT EXISTS public.level_definitions (
14     level INTEGER PRIMARY KEY,
15     title TEXT NOT NULL,
16     xp_required INTEGER NOT NULL,
17     color_hex TEXT DEFAULT '#00d4aa',
18     icon TEXT DEFAULT '&i',
19     description TEXT
20 );
21
22 INSERT INTO public.level_definitions (level, title, xp_required, color_hex, icon, description) VALUES
23 | (1, 'Tân Thập', 100, '#94a3b8', '0<B1', 'Ngôi múi b-t su hán h tr in h'),
24 | (2, 'Ngôi Hí C Vi Çt', 100, '#64748b', '0=U0', 'S m l à m q u e n t - p m x i k i r'),
25 | (3, 'Hí C Sinh Ch m Ch É', 300, '#22c55e', '0=U0', 'N - m v i t S u l à m q u e n t - p m x i k i r'),
26 | (4, 'Ngôi Hi Āu Bi Çt', 600, '#06b6d4', '0=Uj', 'C ó k ù n n g S u t h ã c h à n t N t'),
27 | (5, 'Thā L-p Ráp T-p', 1000, '#3b82f6', '0=Y', 'Am h i Ā u v Ā p h s n h ã n g P C t á y'),
28 | (6, 'Kù Thu-t Vi ěn', 1600, '#8b5cf6', '&mb', 'S é c m i n h v ò h i n - h i n g S o l o'),
29 | (7, 'Chuy ěn Gia PC', 2400, '#f59e0b', '0=Yp', 'H u y Ā n t h o i n h t r o n g c u n g Ó n g'),
30 | (8, 'Kù S. Xáy Dñng', 3400, '#f97316', '0<Bx', 'X á y, d ñ n g c ý t h ì n h P C t á y'),
31 | (9, 'Ki ěn Trú c S. H Ç', 4600, '#ef4444', '0<B0p', 'T h i ě t k ě h Ç t h ñ r'),
32 | (10, 'B-c Thsy PC', 6000, '#ec4899', '0=UQ', 'B - c t h s y, v Ā l p r á p m á y'),
33 | (11, 'Huy Ān Tho i i', 8000, '#00d4aa', '+P', 'H u y Ā n t h o i n h t r o n g c u n g Ó n g'),
34 | (12, 'Solo Leveling', 10000, '#00f3f', '&i', 'S é c m i n h v ò h i n - h i n g S o l o'),
35 | (13, 'R ěnh Thā S n H i n g S', 13000, '#a855f7', '0=Yáp', 'T h ā s i n - h i n g S u y'),
36 | (14, 'Chú a Thā B ò n g T N i', 17000, '#7c3aed', '0<B', 'C h i n g Ā b ó m f i l o à i'),
37 | (15, 'Qu ĩ c V. ing', 22000, '#fbbf24', '0=UQ', 'Q u ĩ c v ò h i n g Ā n g'),
38 | (16, 'Th S n', 28000, '#f6b6b', '0=Y%', 'T h S n L i n h t h Ā n g'),
39 | (17, 'V n h H ě n g', 35000, '#ff4500', '&b', 'S é c m i n h v ò h i n g S o l o'),
40 | (18, 'H ç y Di Çt', 43000, '#dc2626', '0=Ue', 'H ç y d i Ç t m i ě g i ě t h i'),
41 | (19, 'Si ěu Vi Çt', 52000, '#9333ea', '0<B', 'S i ěu v i Ç t t h ĩ c'),
42 | (20, 'B ět Ti', 65000, '#ffd700', '0<Bě', 'B ět t i t r o n g l ě c h s i'),
43 ON CONFLICT (level) DO UPDATE SET
44     title = EXCLUDED.title,
45     xp_required = EXCLUDED.xp_required,
46     color_hex = EXCLUDED.color_hex,
47     icon = EXCLUDED.icon,
48     description = EXCLUDED.description;
49
50 -- 3. Titles/Achievements table
51 CREATE TABLE IF NOT EXISTS public.player_titles (
52     id TEXT PRIMARY KEY,
53     title TEXT NOT NULL,
54     description TEXT,
55     icon TEXT DEFAULT '0<BA',
56     how_to_get TEXT,
57     xp_bonus INTEGER DEFAULT 0
58 );
59
60 INSERT INTO public.player_titles (id, title, description, icon, how_to_get, xp_bonus) VALUES
61 | ('first_lesson', 'Kh ĩi S u', 'Ho à n th à n h b à i h í c s u t i ě n', '0<B-', 'Ho à n t h à n h b à i h í c s u t i ě n', 100),
62 | ('diligent', 'C S n C ù', 'H í c', '7 n g à y l i ě n t i ě p', '0=Y%', 'D u y', 't r i s t r e a k'),
63 | ('perfect_score', 'H o à n H ě o', 'i t Ā m 10 t u y Ç t N i', '+P', 'i t 100%', 'Ho à n t h à n h 10 b à i h í c'),
64 | ('pc_builder', 'Th ā P C', 'L - p r á p t h à n h c ó n g P C s u t i ě n', '0=Yp', 'C ó 10 u p v o t e'),
65 | ('social', 'N ò i T i ě n g', 'N h - n 10 u p v o t e t ě d i Ā n Ā n', '0>Y', 'C ó 10 u p v o t e'),
66 | ('speed_demon', 'T N C Û', 'H o à n t h à n h q u i z t r o n g 2 p h ú t', '&i', 'L à m q u i z'),
67 | ('scholar', 'H í c G i ě', 'H o à n t h à n h 10 b à i h í c', '0=U0', 'H o à n t h à n h 10 b à i h í c'),
68 | ('collector', 'N h ā S. o u T - p', 'i t 5000 X P', '0=Už', 'T í c h l i y 5000 X P'),
69 | ('champion', 'N h ā V ò ě c h', 'é n g t o p 1 b ě n g x ě p h i n g', '0<Bě', 'i t h i'),
70 | ('quest_master', 'V u a N h i Ç m V ā', 'H o à n t h à n h 50 n h i Ç m v ā n g à y', '0=Ue', 'i t h i'),
71 ON CONFLICT (id) DO NOTHING;
72
73 -- 4. Player unlocked titles
74 CREATE TABLE IF NOT EXISTS public.user_titles (
75     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
76     user_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,
77     title_id TEXT REFERENCES public.player_titles(id) ON DELETE CASCADE,
78     unlocked_at TIMESTAMPTZ DEFAULT NOW(),
79     is_active BOOLEAN DEFAULT false,
80     UNIQUE(user_id, title_id)
81 );
82
83 -- 5. Enhanced daily quests table
84 ALTER TABLE public.daily_quests ADD COLUMN IF NOT EXISTS type TEXT DEFAULT 'daily';
85 ALTER TABLE public.daily_quests ADD COLUMN IF NOT EXISTS difficulty TEXT DEFAULT 'easy';
86 | A L T E R T A B L E p u b l i c . d a i l y _ q u e s t s A D D C O L U M N I F N O T E X I S T S i c o n T E X T D E F A U L T
87 ALTER TABLE public.daily_quests ADD COLUMN IF NOT EXISTS requirement_type TEXT;
88 ALTER TABLE public.daily_quests ADD COLUMN IF NOT EXISTS requirement_value INTEGER DEFAULT 1;
89 ALTER TABLE public.daily_quests ADD COLUMN IF NOT EXISTS is_active BOOLEAN DEFAULT true;
90 ALTER TABLE public.daily_quests ADD COLUMN IF NOT EXISTS expires_at TIMESTAMPTZ;
91
92 -- 6. Add XP tracking fields to user_quests
93 ALTER TABLE public.user_quests ADD COLUMN IF NOT EXISTS xp_earned INTEGER DEFAULT 0;
94 ALTER TABLE public.user_quests ADD COLUMN IF NOT EXISTS progress INTEGER DEFAULT 0;
95
96 -- 7. User quest streak tracking
97 ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS last_quest_date DATE;
98 ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS quest_streak INTEGER DEFAULT 0;
99
100 -- 8. Learning stats table (for realtime tracking)
101 CREATE TABLE IF NOT EXISTS public.learning_stats (
102     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
103     user_id UUID REFERENCES auth.users(id) ON DELETE CASCADE UNIQUE,
104     total_xp INTEGER DEFAULT 0,
105     current_streak INTEGER DEFAULT 0,
106     max_streak INTEGER DEFAULT 0,
107     total_study_minutes INTEGER DEFAULT 0,
108     weekly_activity INTEGER[] DEFAULT ARRAY[0,0,0,0,0,0,0],
109     last_active_date DATE,
110     created_at TIMESTAMPTZ DEFAULT NOW(),
111     updated_at TIMESTAMPTZ DEFAULT NOW()
112 );
113
114 -- 9. XP transaction log
```

```

115 CREATE TABLE IF NOT EXISTS public.xp_transactions (
116   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
117   user_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,
118   amount INTEGER NOT NULL,
119   reason TEXT NOT NULL,
120   reference_type TEXT,
121   reference_id TEXT,
122   created_at TIMESTAMPTZ DEFAULT NOW()
123 );
124
125 -- 10. Leaderboard cache table
126 CREATE TABLE IF NOT EXISTS public.leaderboard_cache (
127   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
128   user_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,
129   full_name TEXT,
130   avatar_url TEXT,
131   level INTEGER DEFAULT 1,
132   total_xp INTEGER DEFAULT 0,
133   title TEXT,
134   rank_position INTEGER,
135   updated_at TIMESTAMPTZ DEFAULT NOW()
136 );
137
138 -- Enable RLS
139 ALTER TABLE public.level_definitions ENABLE ROW LEVEL SECURITY;
140 ALTER TABLE public.player_titles ENABLE ROW LEVEL SECURITY;
141 ALTER TABLE public.user_titles ENABLE ROW LEVEL SECURITY;
142 ALTER TABLE public.learning_stats ENABLE ROW LEVEL SECURITY;
143 ALTER TABLE public.xp_transactions ENABLE ROW LEVEL SECURITY;
144 ALTER TABLE public.leaderboard_cache ENABLE ROW LEVEL SECURITY;
145
146 -- RLS Policies
147 CREATE POLICY "Everyone can view levels" ON public.level_definitions FOR SELECT USING (true);
148 CREATE POLICY "Everyone can view titles" ON public.player_titles FOR SELECT USING (true);
149 CREATE POLICY "Users can view their titles" ON public.user_titles FOR SELECT USING (auth.uid() = user_id);
150 CREATE POLICY "Users can manage their titles" ON public.user_titles FOR ALL USING (auth.uid() = user_id);
151 CREATE POLICY "Users can view their stats" ON public.learning_stats FOR SELECT USING (auth.uid() = user_id);
152 CREATE POLICY "Users can view own transactions" ON public.xp_transactions FOR SELECT USING (auth.uid() = user_id);
153 CREATE POLICY "Everyone can view leaderboard" ON public.leaderboard_cache FOR SELECT USING (true);
154
155 -- Enable realtime for relevant tables
156 ALTER PUBLICATION supabase_realtime ADD TABLE public.learning_stats;
157 ALTER PUBLICATION supabase_realtime ADD TABLE public.user_titles;
158 ALTER PUBLICATION supabase_realtime ADD TABLE public.xp_transactions;
159

```

File: database/20260622_n8n_integration.sql (451 lines, 13.4 KB)

```

1  -- =====
2  -- PC Master Builder - n8n Integration
3  -- Database Functions & Triggers
4  -- =====
5
6  -- -----
7  -- Fix: notifications table type constraint
8  -- The original constraint blocked all inserts.
9  -- Drop and recreate with proper allowed values.
10 -- -----
11 ALTER TABLE IF EXISTS public.notifications
12   DROP CONSTRAINT IF EXISTS notifications_type_check;
13
14 ALTER TABLE IF EXISTS public.notifications
15   ADD COLUMN IF NOT EXISTS body TEXT,
16   ADD COLUMN IF NOT EXISTS action_url TEXT,
17   ADD COLUMN IF NOT EXISTS data JSONB,
18   ALTER COLUMN type SET DEFAULT 'info';
19
20 ALTER TABLE IF EXISTS public.notifications
21   ADD CONSTRAINT notifications_type_check
22   CHECK (type IN ('info', 'success', 'warning', 'error', 'assignment', 'quiz', 'achievement_earned', 'lesson_c...
23
24 -- -----
25 -- Add Zalo fields to profiles for teacher Zalo notifications
26 -- -----
27 ALTER TABLE IF EXISTS public.profiles
28   ADD COLUMN IF NOT EXISTS zalo_phone TEXT,
29   ADD COLUMN IF NOT EXISTS zalo_notifications BOOLEAN DEFAULT false,
30   ADD COLUMN IF NOT EXISTS zalo_verified BOOLEAN DEFAULT false;
31
32 -- -----
33 -- Function: check_completed_paths()
34 -- n8n calls this to find students who completed
35 -- all lessons in a path but have no certificate
36 -- -----
37 CREATE OR REPLACE FUNCTION public.check_completed_paths()
38 RETURNS TABLE (
39   student_id UUID,
40   path_id UUID,
41   full_name TEXT,
42   path_title TEXT,
43   lesson_count INT,
44   completed_count INT,
45   avg_score NUMERIC
46 ) LANGUAGE plpgsql SECURITY DEFINER AS $$
47 BEGIN
48   RETURN QUERY
49     WITH path_stats AS (
50       SELECT
51         lp.student_id,
52         pi.path_id,
53         COUNT(DISTINCT pi.id)::INT AS total_lessons,
54         COUNT(DISTINCT lp.id) FILTER (WHERE lp.status = 'completed')::INT AS completed_lessons
55       FROM path_items pi
56       CROSS JOIN LATERAL (
57         SELECT lp_in.student_id, lp_in.id, lp_in.status
58         FROM lesson_progress lp_in
59         WHERE lp_in.lesson_id = pi.item_id
60         AND pi.item_type = 'lesson'
61       ) lp
62       GROUP BY lp.student_id, pi.path_id
63       HAVING COUNT(DISTINCT pi.id) = COUNT(DISTINCT lp.id) FILTER (WHERE lp.status = 'completed')
64     )
65     SELECT
66       ps.student_id,
67       ps.path_id,
68       p.full_name,
69       lp.title AS path_title,
70       ps.total_lessons,
71       ps.completed_lessons,
72       COALESCE(
73         (SELECT AVG(qa.score)::NUMERIC(5,2)
74          FROM quiz_attempts qa
75          JOIN quizzes q ON qa.quiz_id = q.id

```

```

76         WHERE qa.student_id = ps.student_id
77         AND qa.status = 'submitted'
78     ), 0
79 ) AS avg_score
80 FROM path_stats ps
81 JOIN profiles p ON p.id = ps.student_id
82 JOIN learning_paths lp ON lp.id = ps.path_id
83 WHERE NOT EXISTS (
84     SELECT 1 FROM certificates c
85     WHERE c.student_id = ps.student_id
86     AND c.path_id = ps.path_id
87     AND c.is_revoked = false
88 );
89 END;
90 $$;
91
92 -- -----
93 -- Function: get_students_low_performance()
94 -- n8n calls this for student alerts
95 -- -----
96 CREATE OR REPLACE FUNCTION public.get_students_low_performance()
97 RETURNS TABLE (
98     student_id UUID,
99     full_name TEXT,
100     email TEXT,
101     class_id UUID,
102     avg_score NUMERIC,
103     recent_quiz_count INT,
104     weakest_topic TEXT,
105     last_active_date DATE
106 ) LANGUAGE plpgsql SECURITY DEFINER AS $$
107 BEGIN
108     RETURN QUERY
109     WITH recent_scores AS (
110         SELECT
111             qa.student_id,
112             AVG(qa.score)::NUMERIC(5,2) AS avg_score,
113             COUNT(qa.id)::INT AS quiz_count,
114             MAX(qa.created_at)::DATE AS last_quiz_date
115         FROM quiz_attempts qa
116         WHERE qa.status = 'submitted'
117             AND qa.created_at >= NOW() - INTERVAL '7 days'
118         GROUP BY qa.student_id
119     ),
120     weakest AS (
121         SELECT DISTINCT ON (qa.student_id)
122             qa.student_id,
123             l.title AS weakest_lesson
124         FROM quiz_attempts qa
125         JOIN quizzes q ON qa.quiz_id = q.id
126         JOIN lessons l ON l.id = q.lesson_id
127         WHERE qa.status = 'submitted'
128             AND qa.score < 40
129             AND qa.created_at >= NOW() - INTERVAL '14 days'
130         ORDER BY qa.student_id, qa.score ASC
131     )
132     SELECT
133         p.id,
134         p.full_name,
135         p.email,
136         p.class_id,
137         COALESCE(rs.avg_score, 0),
138         COALESCE(rs.quiz_count, 0),
139         COALESCE(w.weakest_lesson, 'Chua co du lieu'),
140         p.last_quest_date
141     FROM profiles p
142     LEFT JOIN recent_scores rs ON rs.student_id = p.id
143     LEFT JOIN weakest w ON w.student_id = p.id
144     WHERE p.role = 'student'
145         AND (
146             rs.avg_score IS NOT NULL AND rs.avg_score < 50
147         )
148     ORDER BY rs.avg_score ASC NULLS LAST
149     LIMIT 100;
150 END;
151 $$;
152
153 -- -----
154 -- Function: get_weekly_stats_summary()
155 -- n8n calls this for weekly reports
156 -- -----
157 CREATE OR REPLACE FUNCTION public.get_weekly_stats_summary()
158 RETURNS TABLE (
159     metric TEXT,
160     value BIGINT
161 ) LANGUAGE plpgsql SECURITY DEFINER AS $$
162 DECLARE
163     week_start TIMESTAMPTZ := DATE_TRUNC('week', NOW()) - INTERVAL '1 week';
164     week_end TIMESTAMPTZ := DATE_TRUNC('week', NOW());
165 BEGIN
166     RETURN QUERY
167     SELECT 'active_students'::TEXT, COUNT(DISTINCT user_id)::BIGINT
168     FROM activity_logs
169     WHERE created_at >= week_start AND created_at < week_end
170
171     UNION ALL
172     SELECT 'new_lessons_completed'::TEXT, COUNT(*)::BIGINT
173     FROM lesson_progress
174     WHERE completed_at >= week_start AND completed_at < week_end
175         AND status = 'completed'
176
177     UNION ALL
178     SELECT 'quizzes_submitted'::TEXT, COUNT(*)::BIGINT
179     FROM quiz_attempts
180     WHERE submitted_at >= week_start AND submitted_at < week_end
181         AND status = 'submitted'
182
183     UNION ALL
184     SELECT 'certificates_issued'::TEXT, COUNT(*)::BIGINT
185     FROM certificates
186     WHERE issued_at >= week_start AND issued_at < week_end
187
188     UNION ALL
189     SELECT 'new_registrations'::TEXT, COUNT(*)::BIGINT
190     FROM profiles
191     WHERE created_at >= week_start AND created_at < week_end
192
193     UNION ALL
194     SELECT 'total_xp_earned'::TEXT, COALESCE(SUM(amount), 0)::BIGINT
195     FROM xp_transactions
196     WHERE created_at >= week_start AND created_at < week_end;
197 END;
198 $$;
199

```

```

200 -- -----
201 -- Function: get_student_weekly_summary()
202 -- For parent weekly reports - passes student_id
203 -- -----
204 CREATE OR REPLACE FUNCTION public.get_student_weekly_summary(
205     p_student_id UUID
206 ) RETURNS TABLE (
207     completed_lessons INT,
208     quiz_count INT,
209     avg_score NUMERIC,
210     total_xp_earned INT,
211     streak_days INT
212 ) LANGUAGE plpgsql SECURITY DEFINER AS $$
213 DECLARE
214     week_start TIMESTAMPTZ := DATE_TRUNC('week', NOW()) - INTERVAL '1 week';
215     week_end TIMESTAMPTZ := DATE_TRUNC('week', NOW());
216 BEGIN
217     RETURN QUERY
218     SELECT
219         (SELECT COUNT(*)::INT FROM lesson_progress
220          WHERE student_id = p_student_id
221           AND status = 'completed'
222           AND completed_at >= week_start AND completed_at < week_end) AS completed_lessons,
223         (SELECT COUNT(*)::INT FROM quiz_attempts
224          WHERE student_id = p_student_id
225           AND status = 'submitted'
226           AND submitted_at >= week_start AND submitted_at < week_end) AS quiz_count,
227         COALESCE(
228             (SELECT AVG(score)::NUMERIC(5,2) FROM quiz_attempts
229              WHERE student_id = p_student_id
230               AND status = 'submitted'
231               AND submitted_at >= week_start AND submitted_at < week_end)
232             , 0) AS avg_score,
233         COALESCE(
234             (SELECT SUM(amount)::INT FROM xp_transactions
235              WHERE user_id = p_student_id
236               AND created_at >= week_start AND created_at < week_end)
237             , 0) AS total_xp_earned,
238         COALESCE(
239             (SELECT quest_streak FROM profiles WHERE id = p_student_id)
240             , 0) AS streak_days;
241 END;
242 $$;
243
244 -- -----
245 -- Function: add_xp()
246 -- Safely add XP and update level/leaderboard
247 -- -----
248 CREATE OR REPLACE FUNCTION public.add_xp(
249     p_user_id UUID,
250     p_amount INT
251 ) RETURNS BOOLEAN LANGUAGE plpgsql SECURITY DEFINER AS $$
252 DECLARE
253     current_xp INT;
254     current_level INT;
255     new_level INT;
256 BEGIN
257     SELECT xp, level INTO current_xp, current_level
258     FROM profiles WHERE id = p_user_id;
259
260     current_xp := COALESCE(current_xp, 0) + p_amount;
261
262     SELECT MAX(l.level) INTO new_level
263     FROM level_definitions l
264     WHERE l.xp_required <= current_xp;
265
266     new_level := COALESCE(new_level, 1);
267
268     UPDATE profiles SET
269         xp = current_xp,
270         level = new_level,
271         updated_at = NOW()
272     WHERE id = p_user_id;
273
274     INSERT INTO leaderboard_cache (user_id, total_xp, level, updated_at)
275     VALUES (p_user_id, current_xp, new_level, NOW())
276     ON CONFLICT (user_id) DO UPDATE SET
277         total_xp = current_xp,
278         level = new_level,
279         updated_at = NOW();
280
281     UPDATE learning_stats SET
282         total_xp = current_xp,
283         updated_at = NOW()
284     WHERE user_id = p_user_id;
285
286     RETURN TRUE;
287 END;
288 $$;
289
290 -- -----
291 -- Function: get_n8n_pending_events()
292 -- Returns pending events that n8n should process
293 -- -----
294 CREATE OR REPLACE FUNCTION public.get_n8n_pending_events()
295 RETURNS TABLE (
296     event_id UUID,
297     event_type TEXT,
298     payload JSONB,
299     created_at TIMESTAMPTZ
300 ) LANGUAGE plpgsql SECURITY DEFINER AS $$
301 BEGIN
302     RETURN QUERY
303     SELECT
304         e.id,
305         e.event_type,
306         e.payload,
307         e.created_at
308     FROM n8n_events e
309     WHERE e.status = 'pending'
310     ORDER BY e.created_at ASC
311     LIMIT 50;
312 END;
313 $$;
314
315 -- -----
316 -- Table: n8n_events
317 -- Queue for events that need processing
318 -- -----
319 CREATE TABLE IF NOT EXISTS public.n8n_events (

```

```

324 id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
325 event_type TEXT NOT NULL,
326 payload JSONB NOT NULL DEFAULT '{}',
327 status TEXT DEFAULT 'pending' CHECK (status IN ('pending', 'processing', 'completed', 'failed')),
328 created_at TIMESTAMPTZ DEFAULT NOW(),
329 processed_at TIMESTAMPTZ
330 );
331
332 -- Enable RLS but allow service role only
333 ALTER TABLE public.n8n_events ENABLE ROW LEVEL SECURITY;
334 DO $$
335 BEGIN
336 IF NOT EXISTS (
337 SELECT 1 FROM pg_policies WHERE tablename = 'n8n_events' AND policyname = 'n8n service role only'
338 ) THEN
339 CREATE POLICY "n8n service role only" ON public.n8n_events
340 FOR ALL USING (true) WITH CHECK (true);
341 END IF;
342 END $$;
343
344 -- Index for n8n polling
345 CREATE INDEX IF NOT EXISTS idx_n8n_events_status ON public.n8n_events(status, created_at);
346
347 -- -----
348 -- Function: queue_n8n_event()
349 -- Called by triggers to queue events for n8n
350 -- -----
351 CREATE OR REPLACE FUNCTION public.queue_n8n_event()
352 RETURNS TRIGGER LANGUAGE plpgsql SECURITY DEFINER AS $$
353 DECLARE
354 event_type TEXT;
355 payload JSONB;
356 BEGIN
357 CASE TG_TABLE_NAME
358 WHEN 'lesson_progress' THEN
359 IF NEW.status = 'completed' THEN
360 event_type := 'lesson_completed';
361 payload := jsonb_build_object(
362 'student_id', NEW.student_id,
363 'lesson_id', NEW.lesson_id,
364 'completed_at', NEW.completed_at
365 );
366 ELSE
367 RETURN NULL;
368 END IF;
369
370 WHEN 'quiz_attempts' THEN
371 IF NEW.status = 'submitted' THEN
372 event_type := 'quiz_submitted';
373 payload := jsonb_build_object(
374 'student_id', NEW.student_id,
375 'quiz_id', NEW.quiz_id,
376 'attempt_id', NEW.id,
377 'score', NEW.score
378 );
379 ELSE
380 RETURN NULL;
381 END IF;
382
383 WHEN 'certificates' THEN
384 event_type := 'certificate_issued';
385 payload := jsonb_build_object(
386 'student_id', NEW.student_id,
387 'certificate_id', NEW.id,
388 'path_id', NEW.path_id
389 );
390 ELSE
391 RETURN NULL;
392 END CASE;
393
394 INSERT INTO public.n8n_events (event_type, payload)
395 VALUES (event_type, payload);
396
397 RETURN NULL;
398 END;
399 $$;
400
401 -- -----
402 -- Triggers: auto-queue events for n8n
403 -- -----
404 DROP TRIGGER IF EXISTS trg_n8n_lesson_complete ON public.lesson_progress;
405 CREATE TRIGGER trg_n8n_lesson_complete
406 AFTER INSERT OR UPDATE OF status ON public.lesson_progress
407 FOR EACH ROW
408 EXECUTE FUNCTION public.queue_n8n_event();
409
410 DROP TRIGGER IF EXISTS trg_n8n_quiz_submit ON public.quiz_attempts;
411 CREATE TRIGGER trg_n8n_quiz_submit
412 AFTER INSERT OR UPDATE OF status ON public.quiz_attempts
413 FOR EACH ROW
414 EXECUTE FUNCTION public.queue_n8n_event();
415
416 DROP TRIGGER IF EXISTS trg_n8n_certificate ON public.certificates;
417 CREATE TRIGGER trg_n8n_certificate
418 AFTER INSERT ON public.certificates
419 FOR EACH ROW
420 EXECUTE FUNCTION public.queue_n8n_event();
421
422 -- -----
423 -- Function: mark_event_processed()
424 -- n8n calls this after processing an event
425 -- -----
426 CREATE OR REPLACE FUNCTION public.mark_event_processed(
427 p_event_id UUID,
428 p_status TEXT DEFAULT 'completed'
429 ) RETURNS BOOLEAN LANGUAGE plpgsql SECURITY DEFINER AS $$
430 BEGIN
431 UPDATE n8n_events
432 SET status = p_status, processed_at = NOW()
433 WHERE id = p_event_id;
434 RETURN FOUND;
435 END;
436 $$;
437
438 -- -----
439 -- Grant permissions for n8n
440 -- -----
441 GRANT USAGE ON SCHEMA public TO anon, authenticated, service_role;
442 GRANT EXECUTE ON FUNCTION public.check_completed_paths() TO service_role;
443 GRANT EXECUTE ON FUNCTION public.get_students_low_performance() TO service_role;
444 GRANT EXECUTE ON FUNCTION public.get_weekly_stats_summary() TO service_role;
445 GRANT EXECUTE ON FUNCTION public.get_student_weekly_summary(UUID) TO service_role;
446 GRANT EXECUTE ON FUNCTION public.add_xp(UUID, INT) TO service_role;

```

```
448 GRANT EXECUTE ON FUNCTION public.get_n8n_pending_events() TO service_role;
449 GRANT EXECUTE ON FUNCTION public.mark_event_processed(UUID, TEXT) TO service_role;
450 GRANT SELECT, INSERT, UPDATE ON public.n8n_events TO service_role;
451
```

File: database/mega_prompt_02_classes.sql (117 lines, 4.6 KB)

```
=====
1 | -- 1. Bảng lớp học
2 | CREATE TABLE public.classes (
3 |     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
4 |     name TEXT NOT NULL,
5 |     code TEXT UNIQUE NOT NULL, -- "Lớp 10A1" - "Tin học" (6 kỳ thi, thi sinh)
6 |     teacher_id UUID REFERENCES public.profiles(id) NOT NULL,
7 |     school_id UUID REFERENCES public.schools(id),
8 |     grade TEXT NOT NULL, -- "10", "11", "12"
9 |     school_year TEXT NOT NULL, -- "2025-2026"
10 |    subject TEXT DEFAULT 'Tin học',
11 |    description TEXT,
12 |    is_active BOOLEAN DEFAULT TRUE,
13 |    max_students INTEGER DEFAULT 40,
14 |    created_at TIMESTAMPTZ DEFAULT NOW()
15 | );
16 |
17 | -- 2. Thành viên lớp học
18 | CREATE TABLE public.class_members (
19 |     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
20 |     class_id UUID REFERENCES public.classes(id) ON DELETE CASCADE,
21 |     student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
22 |     joined_at TIMESTAMPTZ DEFAULT NOW(),
23 |     status TEXT DEFAULT 'active' CHECK (status IN ('active', 'suspended', 'left')),
24 |     UNIQUE(class_id, student_id)
25 | );
26 |
27 | -- 3. Nhiệm vụ (assignments)
28 | CREATE TABLE public.assignments (
29 |     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
30 |     class_id UUID REFERENCES public.classes(id) ON DELETE CASCADE,
31 |     teacher_id UUID REFERENCES public.profiles(id),
32 |     title TEXT NOT NULL,
33 |     description TEXT,
34 |     type TEXT NOT NULL CHECK (type IN (
35 |         'build_config', -- L-p ráp PC theo yêu cầu
36 |         'optimize_budget', -- TÑi-ou trong ngân sách X
37 |         'minimize_tdp', -- Giэм TDP xuñng d°Ui Y watt
38 |         'maximize_perf', -- TÑi-a hiệun ng
39 |         'quiz', -- Tric nghĩch lý thuyt
40 |         'mixed' -- Kút hấp
41 |     )),
42 |     requirements JSONB NOT NULL DEFAULT '{}',
43 |     max_score INTEGER DEFAULT 100,
44 |     deadline TIMESTAMPTZ,
45 |     is_published BOOLEAN DEFAULT FALSE,
46 |     allow_late_submit BOOLEAN DEFAULT FALSE,
47 |     late_penalty_percent INTEGER DEFAULT 20,
48 |     created_at TIMESTAMPTZ DEFAULT NOW()
49 | );
50 |
51 | -- 4. Bài nộp
52 | CREATE TABLE public.submissions (
53 |     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
54 |     assignment_id UUID REFERENCES public.assignments(id) ON DELETE CASCADE,
55 |     student_id UUID REFERENCES public.profiles(id),
56 |     class_id UUID REFERENCES public.classes(id),
57 |     pc_config JSONB,
58 |     total_score INTEGER,
59 |     auto_score INTEGER, -- iẤm tính ã Ùng tē requirements
60 |     teacher_score INTEGER, -- iẤm giáo viên ch¥m thêm
61 |     feedback TEXT,
62 |     submitted_at TIMESTAMPTZ DEFAULT NOW(),
63 |     is_late BOOLEAN DEFAULT FALSE,
64 |     graded_at TIMESTAMPTZ,
65 |     UNIQUE(assignment_id, student_id)
66 | );
67 |
68 | -- 5. RLS Policies
69 | ALTER TABLE public.classes ENABLE ROW LEVEL SECURITY;
70 | ALTER TABLE public.class_members ENABLE ROW LEVEL SECURITY;
71 | ALTER TABLE public.assignments ENABLE ROW LEVEL SECURITY;
72 | ALTER TABLE public.submissions ENABLE ROW LEVEL SECURITY;
73 |
74 | -- Teacher ché th¥y lớp c¶a mình
75 | CREATE POLICY "Teacher sees own classes" ON public.classes
76 | FOR ALL USING (teacher_id = auth.uid());
77 |
78 | -- Student th¥y lớp mình tham gia
79 | CREATE POLICY "Student sees enrolled classes" ON public.classes
80 | FOR SELECT USING (
81 |     EXISTS (SELECT 1 FROM class_members WHERE class_id = id AND student_id = auth.uid())
82 | );
83 |
84 | -- Teacher th¥y t¥t c¶ class_members trong lớp c¶a mình
85 | CREATE POLICY "Teacher sees own class members" ON public.class_members
86 | FOR ALL USING (
87 |     EXISTS (SELECT 1 FROM classes WHERE id = class_id AND teacher_id = auth.uid())
88 | );
89 |
90 | -- Student ã tñ tñ member (join class) và xem member c¶a mình
91 | CREATE POLICY "Student sees own member status" ON public.class_members
92 | FOR SELECT USING (student_id = auth.uid());
93 |
94 | CREATE POLICY "Student can join class" ON public.class_members
95 | FOR INSERT WITH CHECK (student_id = auth.uid());
96 |
97 | -- Student ché th¥y bài nộp c¶a mình
98 | CREATE POLICY "Student sees own submissions" ON public.submissions
99 | FOR ALL USING (student_id = auth.uid());
100 |
101 | -- Teacher th¥y t¥t c¶ submission trong lớp c¶a mình
102 | CREATE POLICY "Teacher sees class submissions" ON public.submissions
103 | FOR ALL USING (
104 |     EXISTS (SELECT 1 FROM classes WHERE id = class_id AND teacher_id = auth.uid())
105 | );
106 |
107 | -- Teacher th¥y bài t-p c¶a lớp mình, c-p nh-t bài t-p
108 | CREATE POLICY "Teacher manages assignments" ON public.assignments
109 | FOR ALL USING (teacher_id = auth.uid());
110 |
111 | -- Student th¥y bài t-p c¶a lớp mình ang tham gia
112 | CREATE POLICY "Student sees published assignments" ON public.assignments
113 | FOR SELECT USING (
114 |     is_published = TRUE AND
115 |     EXISTS (SELECT 1 FROM class_members WHERE class_id = assignments.class_id AND student_id = auth.uid())
116 | );
```

File: database/megaprompt_fix.sql (117 lines, 4.7 KB)

```

1  -- Update profiles table to include school_name
2  ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS school_name TEXT;
3
4  -- Ensure lessons table exists with all necessary fields
5  CREATE TABLE IF NOT EXISTS public.lessons (
6      id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
7      teacher_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,
8      title TEXT NOT NULL,
9      description TEXT,
10     thumbnail_url TEXT,
11     is_published BOOLEAN DEFAULT false,
12     created_at TIMESTAMPTZ DEFAULT now(),
13     updated_at TIMESTAMPTZ DEFAULT now()
14 );
15
16 -- Ensure lesson_sections table exists (for video, pdf, quiz)
17 CREATE TABLE IF NOT EXISTS public.lesson_sections (
18     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
19     lesson_id UUID REFERENCES public.lessons(id) ON DELETE CASCADE,
20     title TEXT NOT NULL,
21     content_type TEXT NOT NULL, -- 'video', 'pdf', 'quiz', 'text'
22     content_url TEXT, -- for video/pdf
23     content_body TEXT, -- for text/quiz data
24     order_index INTEGER DEFAULT 0,
25     created_at TIMESTAMPTZ DEFAULT now()
26 );
27
28 -- Ensure assignments table exists
29 CREATE TABLE IF NOT EXISTS public.assignments (
30     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
31     teacher_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,
32     class_id UUID REFERENCES public.classes(id) ON DELETE CASCADE,
33     lesson_id UUID REFERENCES public.lessons(id) ON DELETE SET NULL,
34     title TEXT NOT NULL,
35     due_date TIMESTAMPTZ,
36     is_published BOOLEAN DEFAULT true,
37     created_at TIMESTAMPTZ DEFAULT now()
38 );
39
40 -- Enable RLS
41 ALTER TABLE public.lessons ENABLE ROW LEVEL SECURITY;
42 ALTER TABLE public.lesson_sections ENABLE ROW LEVEL SECURITY;
43 ALTER TABLE public.assignments ENABLE ROW LEVEL SECURITY;
44
45 -- Policies for lessons
46 CREATE POLICY "Teachers can manage their own lessons" ON public.lessons
47     FOR ALL USING (auth.uid() = teacher_id);
48 CREATE POLICY "Students can view published lessons" ON public.lessons
49     FOR SELECT USING (is_published = true OR auth.uid() = teacher_id);
50
51 -- Policies for lesson_sections
52 CREATE POLICY "Teachers can manage sections of their lessons" ON public.lesson_sections
53     FOR ALL USING (EXISTS (
54         SELECT 1 FROM public.lessons WHERE id = lesson_sections.lesson_id AND teacher_id = auth.uid()
55     ));
56 CREATE POLICY "Students can view sections of published lessons" ON public.lesson_sections
57     FOR SELECT USING (EXISTS (
58         SELECT 1 FROM public.lessons WHERE id = lesson_sections.lesson_id AND (is_published = true OR teacher_...
59     ));
60
61 -- Policies for assignments
62 CREATE POLICY "Teachers can manage their own assignments" ON public.assignments
63     FOR ALL USING (auth.uid() = teacher_id);
64 CREATE POLICY "Students can view assignments for their classes" ON public.assignments
65     FOR SELECT USING (EXISTS (
66         SELECT 1 FROM public.class_members WHERE class_id = assignments.class_id AND student_id = auth.uid()
67     ));
68
69 -- Daily Quests table
70 CREATE TABLE IF NOT EXISTS public.daily_questions (
71     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
72     title TEXT NOT NULL,
73     description TEXT,
74     xp_reward INTEGER DEFAULT 100,
75     created_at TIMESTAMPTZ DEFAULT now()
76 );
77
78 -- User Quest progress
79 CREATE TABLE IF NOT EXISTS public.user_questions (
80     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
81     user_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,
82     quest_id UUID REFERENCES public.daily_questions(id) ON DELETE CASCADE,
83     is_completed BOOLEAN DEFAULT false,
84     completed_at TIMESTAMPTZ,
85     created_at TIMESTAMPTZ DEFAULT now(),
86     UNIQUE(user_id, quest_id, created_at::date)
87 );
88
89 ALTER TABLE public.daily_questions ENABLE ROW LEVEL SECURITY;
90 ALTER TABLE public.user_questions ENABLE ROW LEVEL SECURITY;
91
92 CREATE POLICY "Everyone can view quests" ON public.daily_questions FOR SELECT USING (true);
93 CREATE POLICY "Users can view their own quest progress" ON public.user_questions FOR SELECT USING (auth.uid() = u...
94
95 -- Courses table
96 CREATE TABLE IF NOT EXISTS public.courses (
97     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
98     teacher_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,
99     title TEXT NOT NULL,
100    description TEXT,
101    thumbnail_url TEXT,
102    category TEXT,
103    level TEXT DEFAULT 'Co b?n',
104    is_published BOOLEAN DEFAULT false,
105    created_at TIMESTAMPTZ DEFAULT now()
106 );
107
108 -- Update lessons to belong to courses
109 ALTER TABLE public.lessons ADD COLUMN IF NOT EXISTS course_id UUID REFERENCES public.courses(id) ON DELETE CAS...
110
111 -- Update RLS for courses
112 ALTER TABLE public.courses ENABLE ROW LEVEL SECURITY;
113 CREATE POLICY "Everyone can view published courses" ON public.courses FOR SELECT USING (is_published = true 0...
114 CREATE POLICY "Teachers can manage their courses" ON public.courses FOR ALL USING (auth.uid() = teacher_id);
115
116
117

```


File: database/migrations/20260611_components_collaboration.sql (144 lines, 5.4 KB)

[illegible]

File: database/supabase_auth_roles.sql (84 lines, 2.9 KB)

File: database/supabase_final_fix.sql (29 lines, 1.2 KB)

```

1|      - - BỎ sung các bảng thi u h t cho Lesson Course
2| CREATE TABLE IF NOT EXISTS public.lesson_books (
3|     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
4|     lesson_id UUID REFERENCES public.lessons(id) ON DELETE CASCADE,
5|     title TEXT NOT NULL,
6|     description TEXT,
7|     cover_image_url TEXT,
8|     drive_embed_url TEXT,
9|     created_at TIMESTAMPTZ DEFAULT now()

```

```
10| );
11|
12| -- B - t R L S
13| ALTER TABLE public.lesson_books ENABLE ROW LEVEL SECURITY;
14|
15| CREATE POLICY "Cho phép mĩi ng°Ỡi íc sách bài giảng" ON public.lesson_b
16| FOR SELECT USING (true);
17|
18| CREATE POLICY "Giáo viên quản lý sách cặa mình" ON public.lesson_books
19| FOR ALL USING (EXISTS (
20| SELECT 1 FROM public.lessons WHERE id = lesson_books.lesson_id AND teacher_id = auth.uid()
21| ));
22|
23| -- Fix l_xi chính tẾ trong policy ci (n_zu có)
24| DROP POLICY IF EXISTS "Everyone can view quests" ON public.daily_quests;
25| CREATE POLICY "Everyone can view quests" ON public.daily_quests FOR SELECT USING (true);
26|
27| DROP POLICY IF EXISTS "Users can view their own quest progress" ON public.user_quests;
28| CREATE POLICY "Users can view their own quest progress" ON public.user_quests FOR SELECT USING (auth.uid() = u...
29|
```

File: database/supabase_new_features.sql (56 lines, 2.3 KB)

```
1| -- =====
2| -- 1. B C N G T I % N Ø H I C T - P ( ã c ó , c h i y l i c i n g k h ò n g s a o v ì d ù n g I F N O
3| =====
4| CREATE TABLE IF NOT EXISTS lesson_progress (
5| id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
6| user_id UUID REFERENCES auth.users ON DELETE CASCADE NOT NULL,
7| lesson_id UUID REFERENCES lessons ON DELETE CASCADE NOT NULL,
8| completed_at TIMESTAMP WITH TIME ZONE DEFAULT now(),
9| UNIQUE(user_id, lesson_id)
10| );
11|
12| -- =====
13| -- 2. B C N G B I N H L U - N B À I G I C N G ( M Ũ i )
14| =====
15| CREATE TABLE IF NOT EXISTS lesson_comments (
16| id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
17| lesson_id UUID REFERENCES lessons ON DELETE CASCADE NOT NULL,
18| user_id UUID REFERENCES auth.users ON DELETE CASCADE NOT NULL,
19| user_email TEXT,
20| content TEXT NOT NULL,
21| created_at TIMESTAMP WITH TIME ZONE DEFAULT now()
22| );
23|
24| -- B - t Row Level Security ( R L S ) c h o c o m m e n t s
25| ALTER TABLE lesson_comments ENABLE ROW LEVEL SECURITY;
26|
27| -- Cho phép ai cing °ăc íc bình lu - n
28| CREATE POLICY "Cho phép mĩi ng°Ỡi íc bình lu - n" ON lesson_comments
29| FOR SELECT USING (true);
30|
31| -- Cho phép ng°Ỡi dùng ng nh - p °ăc vi ết bình lu - n
32| CREATE POLICY "Cho phép user ng nh - p t i o bình lu - n" ON lesson_comments
33| FOR INSERT WITH CHECK (auth.uid() = user_id);
34|
35| -- Cho phép ng°Ỡi dùng x ó a bình lu - n cặa chính mình
36| CREATE POLICY "Cho phép user x ó a bình lu - n cặa mình" ON lesson_comments
37| FOR DELETE USING (auth.uid() = user_id);
38|
39| -- =====
40| -- 3. B C N G T H Ø N G B Á O T Ê G I Á O V I Ê N ( M Ũ i )
41| =====
42| CREATE TABLE IF NOT EXISTS notifications (
43| id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
44| title TEXT NOT NULL,
45| content TEXT,
46| is_global BOOLEAN DEFAULT true, -- N_zu true, t_ýt cẾ híc sinh Áu th_ýy
47| created_at TIMESTAMP WITH TIME ZONE DEFAULT now()
48| );
49|
50| -- B - t Row Level Security ( R L S ) c h o n o t i f i c a t i o n s
51| ALTER TABLE notifications ENABLE ROW LEVEL SECURITY;
52|
53| -- Cho phép mĩi ng°Ỡi íc thông báo
54| CREATE POLICY "Cho phép mĩi ng°Ỡi íc thông báo" ON notifications
55| FOR SELECT USING (true);
56|
```

File: database/supabase_seed.sql (32 lines, 2.1 KB)

```
1| -- T H Ê M D I L I Ę U M Ậ U ( S E E D D A T A ) C H O P C M A S T E R L M S
2|
3| -- 1. Th ê m t r Ộ y n g h í c m Ậ u
4| INSERT INTO public.schools (id, name, address, code)
5| VALUES
6| ( ' 1 1 1 1 1 1 1 1 - 1 1 1 1 - 1 1 1 1 - 1 1 1 1 - 1 1 1 1 1 1 1 1 1 1 ' , ' T H P T C h u y ề n L ê H ó n g P h o n g ' ,
7| ' 2 2 2 2 2 2 2 2 - 2 2 2 2 - 2 2 2 2 - 2 2 2 2 - 2 2 2 2 2 2 2 2 2 2 2 2 ' , ' T H P T C h u y ề n K h o a H í c T ấ N h i ế
8| ON CONFLICT (code) DO NOTHING;
9|
10| -- L Ộ u ý : S u p a b a s e A u t h q u ậ n l ý p a s s w o r d b ằ n g b c r y p t ß S c h e m a ' a u t
11| -- V i ệ c s e e d U s e r ( c Ế e m a i l / p a s s w o r d ) b ằ n g S Q L t r ấ c t i ậ p v ậ o a u t h . u s e r s
12| -- M I N H K H U Y Ề N B N N Ề N D ù N G G I Á O D I Ệ N T R Ầ N G W E B Ậ N G K Ỡ ( S I G N U P ) 1 Ø
13|
14| -- T u y n h i ề n , d Ộ Ừ i Ậ y l Ậ c á c h c h ề n t h ằ n g v ậ o A u t h c h o m Ộ i t r Ộ y n g d e v ( S
15| -- N_zu b i n ằ n g d ù n g đ ẫ Ậ n S u p a b a s e H o s t e d , b i n b Ậ t b u Ộ c p h Ế i d ù n g C l i ệ n
16| -- Ậ n g k Ỡ t ậ i k h o Ậ n , s Ậ u ó v ậ o b ằ n g p r o f i l e s Ậ Ộ i r o l e t h ằ n h ' a
17|
18| /*
19| INSERT INTO auth.users (id, email, encrypted_password, email_confirmed_at, raw_user_meta_data)
20| VALUES
21| ( '33333333-3333-3333-3333-333333333333', 'admin@lms.com', crypt('password123', gen_salt('bf')), now(), '{"fu...
22| ( '44444444-4444-4444-4444-444444444444', 'teacher1@lms.com', crypt('password123', gen_salt('bf')), now(), '{"...
23| ( '55555555-5555-5555-5555-555555555555', 'student1@lms.com', crypt('password123', gen_salt('bf')), now(), '{"...
24| ( '66666666-6666-6666-6666-666666666666', 'parent1@lms.com', crypt('password123', gen_salt('bf')), now(), '{"...
25| */
26|
27| -- 3. V í d ẫ l i ề n k ậ t p h ẫ h u y n h - h í c s i n h ( C h Ế c h i y ° ă c s Ậ u k h i t ậ i k h
28| /*
29| INSERT INTO public.parent_student_links (parent_id, student_id, relationship)
30| VALUES ( '66666666-6666-6666-6666-666666666666', '55555555-5555-5555-5555-555555555555', 'father' );
31| */
32|
```

File: supabase/functions/generate-certificate/index.ts (153 lines, 3.9 KB)

```
1| import { serve } from "https://deno.land/std@0.131.0/http/server.ts"
2| import { createClient } from "https://esm.sh/@supabase/supabase-js@2.39.8"
3| import { PDFDocument, StandardFonts, rgb } from "https://esm.sh/pdf-lib@1.17.1"
```

```

4 | import QRCode from 'https://esm.sh/qrcode@1.5.3'
5 |
6 | serve(async (req) => {
7 |   try {
8 |     const { certificateId } = await req.json()
9 |
10 |    // Initialize Supabase Admin client
11 |    const supabaseAdmin = createClient(
12 |      Deno.env.get('SUPABASE_URL') ?? '',
13 |      Deno.env.get('SUPABASE_SERVICE_ROLE_KEY') ?? ''
14 |    )
15 |
16 |    // 1. Fetch certificate details
17 |    const { data: cert, error: fetchErr } = await supabaseAdmin
18 |      .from('certificates')
19 |      .select('*')
20 |      .eq('id', certificateId)
21 |      .single()
22 |
23 |    if (fetchErr || !cert) {
24 |      return new Response(JSON.stringify({ error: "Certificate not found" }), { status: 404 })
25 |    }
26 |
27 |    // 2. Generate PDF
28 |    const pdfDoc = await PDFDocument.create()
29 |    const page = pdfDoc.addPage([842, 595]) // A4 Landscape
30 |
31 |    const fontBold = await pdfDoc.embedFont(StandardFonts.HelveticaBold)
32 |    const fontRegular = await pdfDoc.embedFont(StandardFonts.Helvetica)
33 |
34 |    // Dark Background
35 |    page.drawRectangle({
36 |      x: 0,
37 |      y: 0,
38 |      width: 842,
39 |      height: 595,
40 |      color: rgb(0.05, 0.06, 0.075) // #0d0e13
41 |    })
42 |
43 |    // Glowing Boarder
44 |    page.drawRectangle({
45 |      x: 20,
46 |      y: 20,
47 |      width: 802,
48 |      height: 555,
49 |      borderColor: rgb(0, 0.83, 0.67), // #00d4aa
50 |      borderWidth: 2
51 |    })
52 |
53 |    // Title
54 |    page.drawText('CHUNG CHI HOAN THANH', {
55 |      x: 260,
56 |      y: 470,
57 |      size: 28,
58 |      font: fontBold,
59 |      color: rgb(0, 0.83, 0.67)
60 |    })
61 |
62 |    page.drawText('Chung nhan hoc vien', {
63 |      x: 350,
64 |      y: 420,
65 |      size: 14,
66 |      font: fontRegular,
67 |      color: rgb(0.6, 0.6, 0.6)
68 |    })
69 |
70 |    // Student Name
71 |    page.drawText(cert.student_name, {
72 |      x: 300,
73 |      y: 360,
74 |      size: 32,
75 |      font: fontBold,
76 |      color: rgb(0.87, 0.88, 0.93)
77 |    })
78 |
79 |    // Course Title
80 |    page.drawText(`Da hoan thanh xuat sac khoa hoc`, {
81 |      x: 310,
82 |      y: 310,
83 |      size: 14,
84 |      font: fontRegular,
85 |      color: rgb(0.6, 0.6, 0.6)
86 |    })
87 |
88 |    page.drawText(cert.course_title, {
89 |      x: 270,
90 |      y: 260,
91 |      size: 20,
92 |      font: fontBold,
93 |      color: rgb(0, 0.83, 0.67)
94 |    })
95 |
96 |    // Info
97 |    page.drawText(`Diem: ${cert.final_score}% | Ngay cap: ${cert.completion_date}`, {
98 |      x: 280,
99 |      y: 200,
100 |      size: 12,
101 |      font: fontRegular,
102 |      color: rgb(0.6, 0.6, 0.6)
103 |    })
104 |
105 |    page.drawText(`Ma chung chi: ${cert.certificate_number}`, {
106 |      x: 310,
107 |      y: 160,
108 |      size: 12,
109 |      font: fontRegular,
110 |      color: rgb(0.6, 0.6, 0.6)
111 |    })
112 |
113 |    // 9. QR Code verify url
114 |    const qrDataUrl = await QRCode.toDataURL(cert.verify_url)
115 |    const qrImageBytes = Uint8Array.from(atob(qrDataUrl.split(',')[1]), c => c.charCodeAt(0))
116 |    const qrImage = await pdfDoc.embedPng(qrImageBytes)
117 |    page.drawImage(qrImage, {
118 |      x: 700,
119 |      y: 40,
120 |      width: 90,
121 |      height: 90
122 |    })
123 |
124 |    // Save PDF
125 |    const pdfBytes = await pdfDoc.save()
126 |
127 |    // 11. Upload to Supabase Storage

```

```

128|     const { error: uploadErr } = await supabaseAdmin.storage
129|       .from('certificates')
130|       .upload(`${certificateId}.pdf`, pdfBytes, {
131|         contentType: 'application/pdf',
132|         upsert: true
133|       })
134|
135|     if (uploadErr) throw uploadErr
136|
137|     const publicUrl = supabaseAdmin.storage
138|       .from('certificates').getPublicUrl(`${certificateId}.pdf`).data.publicUrl
139|
140|     // 12. Update Certificate PDF url
141|     await supabaseAdmin
142|       .from('certificates')
143|       .update({ pdf_url: publicUrl })
144|       .eq('id', certificateId)
145|
146|     return new Response(JSON.stringify({ pdfUrl: publicUrl }), {
147|       headers: { "Content-Type": "application/json" }
148|     })
149|   } catch (err: any) {
150|     return new Response(JSON.stringify({ error: err.message }), { status: 500 })
151|   }
152| })
153|

```

```

1| // Follow this setup guide to integrate the Deno language server with your editor:
2| // https://deno.land/manual/getting_started/setup_your_environment
3| // This enables autocomplete, go to definition, etc.
4|
5| // Setup type definitions for built-in Supabase Runtime APIs
6| import "@supabase/functions-js/edge-runtime.d.ts"
7|
8| console.log("Hello from Functions!")
9|
10| Deno.serve(async (req) => {
11|   const { name } = await req.json()
12|   const data = {
13|     message: `Hello ${name}!`,
14|   }
15|
16|   return new Response(
17|     JSON.stringify(data),
18|     { headers: { "Content-Type": "application/json" } },
19|   )
20| })
21|
22| /* To invoke locally:
23|
24| 1. Run `supabase start` (see: https://supabase.com/docs/reference/cli/supabase-start)
25| 2. Make an HTTP request:
26|
27| curl -i --location --request POST 'http://127.0.0.1:54321/functions/v1/send-notification' \
28|   --header 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJkdXBhYmFzZS1kZWZ1Iiwicm9sZS...' \
29|   --header 'Content-Type: application/json' \
30|   --data '{"name":"Functions"}'
31|
32| */
33|

```

```

1| -- =====
2| -- UNIFIED MIGRATION: LMS SCHEMA INITIALIZATION & PARENT ROLE SETUP
3| -- Target Path: supabase/migrations/20260518093525_create_parent_role.sql
4| -- =====
5|
6| -- 1. B ng schools
7| CREATE TABLE IF NOT EXISTS public.schools (
8|   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
9|   name TEXT NOT NULL,
10|  address TEXT,
11|  code TEXT UNIQUE NOT NULL, -- m  tr  ng 6 k  t 
12|  created_at TIMESTAMPTZ DEFAULT NOW()
13| );
14|
15| -- 2. B ng profiles (tham chi u auth.users v  schools)
16| CREATE TABLE IF NOT EXISTS public.profiles (
17|   id UUID REFERENCES auth.users(id) ON DELETE CASCADE PRIMARY KEY,
18|   email TEXT NOT NULL,
19|   full_name TEXT NOT NULL,
20|   role TEXT NOT NULL CHECK (role IN ('student', 'teacher', 'admin', 'parent')),
21|   created_at TIMESTAMPTZ DEFAULT NOW(),
22|   updated_at TIMESTAMPTZ DEFAULT NOW()
23| );
24|
25| -- B  sung c c tr  ng quan tr ng cho profiles n u b ng   t n t i tr  c
26| ALTER TABLE public.profiles
27| ADD COLUMN IF NOT EXISTS school_id UUID REFERENCES public.schools(id),
28| ADD COLUMN IF NOT EXISTS school_name TEXT,
29| ADD COLUMN IF NOT EXISTS student_code TEXT UNIQUE,
30| ADD COLUMN IF NOT EXISTS avatar_url TEXT,
31| ADD COLUMN IF NOT EXISTS bio TEXT,
32| ADD COLUMN IF NOT EXISTS date_of_birth DATE,
33| ADD COLUMN IF NOT EXISTS phone TEXT,
34| ADD COLUMN IF NOT EXISTS xp INTEGER DEFAULT 0,
35| ADD COLUMN IF NOT EXISTS level INTEGER DEFAULT 1;
36|
37| -- 3. B ng courses
38| CREATE TABLE IF NOT EXISTS public.courses (
39|   id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
40|   teacher_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,
41|   title TEXT NOT NULL,
42|   description TEXT,
43|   thumbnail_url TEXT,
44|   category TEXT,
45|   level TEXT DEFAULT 'C  b n',
46|   is_published BOOLEAN DEFAULT false,
47|   created_at TIMESTAMPTZ DEFAULT now()
48| );
49|
50| -- 4. B ng lessons
51| CREATE TABLE IF NOT EXISTS public.lessons (
52|   id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
53|   title TEXT NOT NULL,
54|   created_at TIMESTAMPTZ DEFAULT now(),
55|

```

```

57| updated_at TIMESTAMPTZ DEFAULT now()
58| );
59|
60| -- B ò s u n g c á c c ù t c h o l e s s o n s   Ā   ó n g   b ù
61| ALTER TABLE public.lessons
62| ADD COLUMN IF NOT EXISTS course_id UUID REFERENCES public.courses(id) ON DELETE CASCADE,
63| ADD COLUMN IF NOT EXISTS teacher_id UUID REFERENCES public.profiles(id) ON DELETE SET NULL,
64| ADD COLUMN IF NOT EXISTS description TEXT,
65| ADD COLUMN IF NOT EXISTS thumbnail_url TEXT,
66| ADD COLUMN IF NOT EXISTS is_published BOOLEAN DEFAULT false;
67|
68| -- 5 .   B ã n g   l e s s o n _ s e c t i o n s   ( c h é a   v i d e o ,   p d f ,   q u i z ,   t e x t )
69| CREATE TABLE IF NOT EXISTS public.lesson_sections (
70| id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
71| lesson_id UUID REFERENCES public.lessons(id) ON DELETE CASCADE,
72| title TEXT NOT NULL,
73| content_type TEXT NOT NULL, -- 'video', 'pdf', 'quiz', 'text'
74| content_url TEXT,
75| content_body TEXT,
76| order_index INTEGER DEFAULT 0,
77| created_at TIMESTAMPTZ DEFAULT now()
78| );
79|
80| -- 6 .   B ã n g   c l a s s e s   ( l ù p   h í c )
81| CREATE TABLE IF NOT EXISTS public.classes (
82| id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
83| name TEXT NOT NULL,
84| code TEXT UNIQUE NOT NULL, -- "TIN001" (t ã   s i n h )
85| teacher_id UUID REFERENCES public.profiles(id) NOT NULL,
86| created_at TIMESTAMPTZ DEFAULT NOW()
87| );
88|
89| ALTER TABLE public.classes
90| ADD COLUMN IF NOT EXISTS school_id UUID REFERENCES public.schools(id),
91| ADD COLUMN IF NOT EXISTS grade TEXT,
92| ADD COLUMN IF NOT EXISTS school_year TEXT,
93| ADD COLUMN IF NOT EXISTS subject TEXT DEFAULT 'T i n   h í c ',
94| ADD COLUMN IF NOT EXISTS description TEXT,
95| ADD COLUMN IF NOT EXISTS is_active BOOLEAN DEFAULT TRUE,
96| ADD COLUMN IF NOT EXISTS max_students INTEGER DEFAULT 40;
97|
98| -- 7 .   B ã n g   c l a s s _ m e m b e r s   ( t h à n h   v i ê n   l ù p )
99| CREATE TABLE IF NOT EXISTS public.class_members (
100| id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
101| class_id UUID REFERENCES public.classes(id) ON DELETE CASCADE,
102| student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
103| joined_at TIMESTAMPTZ DEFAULT NOW(),
104| status TEXT DEFAULT 'active' CHECK (status IN ('active', 'suspended', 'left')),
105| UNIQUE(class_id, student_id)
106| );
107|
108| -- 8 .   B ã n g   a s s i g n m e n t s   ( b à i   t - p )
109| CREATE TABLE IF NOT EXISTS public.assignments (
110| id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
111| title TEXT NOT NULL,
112| created_at TIMESTAMPTZ DEFAULT NOW()
113| );
114|
115| ALTER TABLE public.assignments
116| ADD COLUMN IF NOT EXISTS class_id UUID REFERENCES public.classes(id) ON DELETE CASCADE,
117| ADD COLUMN IF NOT EXISTS teacher_id UUID REFERENCES public.profiles(id) ON DELETE SET NULL,
118| ADD COLUMN IF NOT EXISTS lesson_id UUID REFERENCES public.lessons(id) ON DELETE SET NULL,
119| ADD COLUMN IF NOT EXISTS description TEXT,
120| ADD COLUMN IF NOT EXISTS type TEXT CHECK (type IN (
121| 'build_config', -- L - p   r á p   P C   t h e o   y ê u   c s u
122| 'optimize_budget', -- T Ñ i   ° u   t r o n g   n g â n   s á c h   X
123| 'minimize_tdp', -- G i   ¢ m   T D P   x u   N n g   đ °   Û i   Y   w a t t
124| 'maximize_perf', -- T Ñ i   ¢   a   h i   Ç u   n   n g
125| 'quiz', -- T r   ¢   n g h i   Ç m   l ý   t h u y   ¢ t
126| 'mixed', -- K ẻ t   h ấ p
127| )),
128| ADD COLUMN IF NOT EXISTS requirements JSONB DEFAULT '{}::jsonb',
129| ADD COLUMN IF NOT EXISTS max_score INTEGER DEFAULT 100,
130| ADD COLUMN IF NOT EXISTS deadline TIMESTAMPTZ,
131| ADD COLUMN IF NOT EXISTS due_date TIMESTAMPTZ,
132| ADD COLUMN IF NOT EXISTS is_published BOOLEAN DEFAULT FALSE,
133| ADD COLUMN IF NOT EXISTS allow_late_submit BOOLEAN DEFAULT FALSE,
134| ADD COLUMN IF NOT EXISTS late_penalty_percent INTEGER DEFAULT 20;
135|
136| -- 9 .   B ã n g   s u b m i s s i o n s   ( b à i   n ù p )
137| CREATE TABLE IF NOT EXISTS public.submissions (
138| id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
139| assignment_id UUID REFERENCES public.assignments(id) ON DELETE CASCADE,
140| student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
141| class_id UUID REFERENCES public.classes(id) ON DELETE CASCADE,
142| pc_config JSONB,
143| total_score INTEGER,
144| auto_score INTEGER,
145| teacher_score INTEGER,
146| feedback TEXT,
147| submitted_at TIMESTAMPTZ DEFAULT NOW(),
148| is_late BOOLEAN DEFAULT FALSE,
149| graded_at TIMESTAMPTZ,
150| UNIQUE(assignment_id, student_id)
151| );
152|
153| -- 10 .   B ã n g   p a r e n t _ s t u d e n t _ l i n k s   ( l i ê n   k ẻ t   p h ấ   h u y n h   -   h í c   s i n h )
154| CREATE TABLE IF NOT EXISTS public.parent_student_links (
155| id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
156| parent_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
157| student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
158| relationship TEXT DEFAULT 'parent' CHECK (relationship IN ('father', 'mother', 'guardian', 'sibling', 'other')),
159| linked_at TIMESTAMPTZ DEFAULT NOW(),
160| status TEXT DEFAULT 'active' CHECK (status IN ('active', 'revoked')),
161| revoked_at TIMESTAMPTZ,
162| revoked_by UUID REFERENCES public.profiles(id),
163| UNIQUE(parent_id, student_id)
164| );
165|
166| -- B ẻ o   ¢ m   c á c   c ù t   q u a n   t r ẻ   t ó n   t i i   t r o n g   p a r e n t _ s t u d e n t _ l i n k s   p h ò n g   t
167| ALTER TABLE public.parent_student_links
168| ADD COLUMN IF NOT EXISTS relationship TEXT DEFAULT 'parent' CHECK (relationship IN ('father', 'mother', 'guard...
169| ADD COLUMN IF NOT EXISTS status TEXT DEFAULT 'active' CHECK (status IN ('active', 'revoked')),
170| ADD COLUMN IF NOT EXISTS revoked_at TIMESTAMPTZ,
171| ADD COLUMN IF NOT EXISTS revoked_by UUID REFERENCES public.profiles(id),
172| ADD COLUMN IF NOT EXISTS linked_at TIMESTAMPTZ DEFAULT NOW();
173|
174| -- 11 .   B ã n g   l e s s o n _ p r o g r e s s   ( t i ẻ n   Û   h í c   b à i )
175| CREATE TABLE IF NOT EXISTS public.lesson_progress (
176| id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
177| student_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE CASCADE,
178| lesson_id UUID NOT NULL REFERENCES public.lessons(id) ON DELETE CASCADE,
179| status TEXT DEFAULT 'not_started' CHECK (status IN ('not_started', 'in_progress', 'completed')),
180| completion_percentage INT DEFAULT 0 CHECK (completion_percentage BETWEEN 0 AND 100),

```

```

181| time_spent_seconds INT DEFAULT 0,
182| last_accessed TIMESTAMPTZ,
183| completed_at TIMESTAMPTZ,
184| UNIQUE(student_id, lesson_id)
185| );
186|
187| -- 8. Em bèo các cột tỉ lệ học tập có sự
188| ALTER TABLE public.lesson_progress
189| ADD COLUMN IF NOT EXISTS completion_percentage INT DEFAULT 0 CHECK (completion_percentage BETWEEN 0 AND 100),
190| ADD COLUMN IF NOT EXISTS time_spent_seconds INT DEFAULT 0,
191| ADD COLUMN IF NOT EXISTS last_accessed TIMESTAMPTZ;
192|
193| -- 9. Bảng notifications (thông báo)
194| CREATE TABLE IF NOT EXISTS public.notifications (
195| id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
196| user_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE CASCADE,
197| type TEXT NOT NULL CHECK (type IN (
198| 'quiz_graded',
199| 'lesson_completed',
200| 'assignment_submitted',
201| 'achievement_earned',
202| 'child_absent',
203| 'child_low_score',
204| 'child_completed_path',
205| 'parent_link_request',
206| 'teacher_message',
207| 'system'
208| )),
209| title TEXT NOT NULL,
210| body TEXT NOT NULL,
211| data JSONB,
212| action_url TEXT,
213| is_read BOOLEAN DEFAULT FALSE,
214| created_at TIMESTAMPTZ DEFAULT NOW()
215| );
216|
217| -- 10. Bảng daily_quests (nhiệm vụ hàng ngày) và user_quests
218| CREATE TABLE IF NOT EXISTS public.daily_quests (
219| id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
220| title TEXT NOT NULL,
221| description TEXT,
222| xp_reward INTEGER DEFAULT 100,
223| created_at TIMESTAMPTZ DEFAULT now()
224| );
225|
226| CREATE TABLE IF NOT EXISTS public.user_quests (
227| id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
228| user_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,
229| quest_id UUID REFERENCES public.daily_quests(id) ON DELETE CASCADE,
230| is_completed BOOLEAN DEFAULT false,
231| completed_at TIMESTAMPTZ,
232| created_at TIMESTAMPTZ DEFAULT now()
233| );
234|
235| -- 11. Unique Index để tránh trùng lặp user_quests (Thay thế cho UNIQUE constraint)
236| CREATE UNIQUE INDEX IF NOT EXISTS user_quests_user_quest_date_idx
237| ON public.user_quests (user_id, quest_id, ((created_at AT TIME ZONE 'UTC')::date));
238|
239| -- 12. Hệ thống Quiz Engine
240| CREATE TABLE IF NOT EXISTS public.question_banks (
241| id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
242| teacher_id UUID REFERENCES public.profiles(id) NOT NULL,
243| title TEXT NOT NULL,
244| subject TEXT DEFAULT 'Tin học - Lập trình PC',
245| description TEXT,
246| created_at TIMESTAMPTZ DEFAULT NOW(),
247| updated_at TIMESTAMPTZ DEFAULT NOW()
248| );
249|
250| CREATE TABLE IF NOT EXISTS public.questions (
251| id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
252| bank_id UUID REFERENCES public.question_banks(id) ON DELETE CASCADE,
253| content TEXT NOT NULL,
254| type TEXT CHECK (type IN ('single_choice', 'multiple_choice', 'true_false', 'fill_blank', 'ordering')) NOT NULL,
255| difficulty TEXT CHECK (difficulty IN ('easy', 'medium', 'hard')) DEFAULT 'medium',
256| points INT DEFAULT 10,
257| explanation TEXT,
258| image_url TEXT,
259| tags TEXT[],
260| "order" INT DEFAULT 0,
261| created_at TIMESTAMPTZ DEFAULT NOW()
262| );
263|
264| CREATE TABLE IF NOT EXISTS public.question_options (
265| id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
266| question_id UUID REFERENCES public.questions(id) ON DELETE CASCADE,
267| content TEXT NOT NULL,
268| is_correct BOOLEAN DEFAULT FALSE,
269| "order" INT DEFAULT 0
270| );
271|
272| CREATE TABLE IF NOT EXISTS public.quizzes (
273| id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
274| teacher_id UUID REFERENCES public.profiles(id) NOT NULL,
275| lesson_id UUID REFERENCES public.lessons(id) ON DELETE SET NULL,
276| class_id UUID,
277| title TEXT NOT NULL,
278| description TEXT,
279| time_limit_minutes INT,
280| passing_score INT DEFAULT 70,
281| randomize_questions BOOLEAN DEFAULT TRUE,
282| randomize_options BOOLEAN DEFAULT TRUE,
283| max_attempts INT DEFAULT 3,
284| show_answers_after TEXT DEFAULT 'after_submit' CHECK (show_answers_after IN ('never', 'after_submit', 'after...
285| available_from TIMESTAMPTZ,
286| available_until TIMESTAMPTZ,
287| is_published BOOLEAN DEFAULT FALSE,
288| created_at TIMESTAMPTZ DEFAULT NOW()
289| );
290|
291| CREATE TABLE IF NOT EXISTS public.quiz_questions (
292| quiz_id UUID REFERENCES public.quizzes(id) ON DELETE CASCADE,
293| question_id UUID REFERENCES public.questions(id) ON DELETE CASCADE,
294| "order" INT,
295| PRIMARY KEY (quiz_id, question_id)
296| );
297|
298| CREATE TABLE IF NOT EXISTS public.quiz_attempts (
299| id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
300| quiz_id UUID REFERENCES public.quizzes(id) ON DELETE CASCADE,
301| student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
302| started_at TIMESTAMPTZ DEFAULT NOW(),
303| submitted_at TIMESTAMPTZ,
304| time_taken_seconds INT,

```

```

305 | score NUMERIC(5,2),
306 | max_score INT,
307 | status TEXT DEFAULT 'in_progress' CHECK (status IN ('in_progress','submitted','graded','expired')),
308 | attempt_number INT DEFAULT 1
309 | );
310 |
311 | CREATE TABLE IF NOT EXISTS public.quiz_answers (
312 | id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
313 | attempt_id UUID REFERENCES public.quiz_attempts(id) ON DELETE CASCADE,
314 | question_id UUID REFERENCES public.questions(id) ON DELETE CASCADE,
315 | selected_option_ids UUID[],
316 | text_answer TEXT,
317 | ordering_answer UUID[],
318 | is_correct BOOLEAN,
319 | points_earned NUMERIC(5,2) DEFAULT 0,
320 | time_spent_seconds INT
321 | );
322 |
3 2 3 | - - 1 5 . L ù tr ì nh h ì c ( L e a r n i n g P a t h )
324 | CREATE TABLE IF NOT EXISTS public.learning_paths (
325 | id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
326 | class_id UUID,
327 | title TEXT NOT NULL,
328 | description TEXT,
329 | is_active BOOLEAN DEFAULT TRUE,
330 | created_by UUID REFERENCES public.profiles(id) ON DELETE SET NULL,
331 | created_at TIMESTAMPTZ DEFAULT NOW()
332 | );
333 |
334 | CREATE TABLE IF NOT EXISTS public.path_items (
335 | id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
336 | path_id UUID REFERENCES public.learning_paths(id) ON DELETE CASCADE,
337 | item_type TEXT CHECK (item_type IN ('lesson','quiz','lab_session','milestone')),
338 | item_id UUID,
339 | title TEXT NOT NULL,
340 | description TEXT,
341 | "order" INT NOT NULL,
342 | unlock_condition JSONB,
343 | estimated_minutes INT,
344 | is_optional BOOLEAN DEFAULT FALSE,
345 | created_at TIMESTAMPTZ DEFAULT NOW(),
346 | UNIQUE(path_id, "order")
347 | );
348 |
3 4 9 | - - 1 6 . B ê ng d i   n   n / t h   o l u - n ( D i s c u s s i o n F o r u m )
350 | CREATE TABLE IF NOT EXISTS public.discussion_threads (
351 | id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
352 | lesson_id UUID NOT NULL,
353 | author_id UUID REFERENCES public.profiles(id) ON DELETE SET NULL,
354 | title TEXT NOT NULL,
355 | body TEXT NOT NULL,
356 | type TEXT DEFAULT 'question' CHECK (type IN ('question','discussion','announcement')),
357 | status TEXT DEFAULT 'open' CHECK (status IN ('open','resolved','pinned','closed')),
358 | upvote_count INT DEFAULT 0,
359 | reply_count INT DEFAULT 0,
360 | is_deleted BOOLEAN DEFAULT FALSE,
361 | created_at TIMESTAMPTZ DEFAULT NOW(),
362 | updated_at TIMESTAMPTZ DEFAULT NOW()
363 | );
364 |
365 | CREATE TABLE IF NOT EXISTS public.discussion_replies (
366 | id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
367 | thread_id UUID REFERENCES public.discussion_threads(id) ON DELETE CASCADE,
368 | parent_reply_id UUID REFERENCES public.discussion_replies(id) ON DELETE CASCADE,
369 | author_id UUID REFERENCES public.profiles(id) ON DELETE SET NULL,
370 | body TEXT NOT NULL,
371 | is_teacher_answer BOOLEAN DEFAULT FALSE,
372 | upvote_count INT DEFAULT 0,
373 | is_deleted BOOLEAN DEFAULT FALSE,
374 | created_at TIMESTAMPTZ DEFAULT NOW()
375 | );
376 |
377 | CREATE TABLE IF NOT EXISTS public.discussion_upvotes (
378 | user_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
379 | thread_id UUID REFERENCES public.discussion_threads(id) ON DELETE CASCADE,
380 | reply_id UUID REFERENCES public.discussion_replies(id) ON DELETE CASCADE,
381 | created_at TIMESTAMPTZ DEFAULT NOW(),
382 | CONSTRAINT one_target CHECK ((thread_id IS NULL) != (reply_id IS NULL)),
383 | UNIQUE(user_id, thread_id),
384 | UNIQUE(user_id, reply_id)
385 | );
386 |
387 |
3 8 8 | - - % % % B     C 2 : C   C F U N C T I O N & T R I G G E R Q U A N T R   I N G ( H   C S I N H & P H   H U Y N H )
389 |
3 9 0 | - - 1 .   m b   o v   i t r   p r o f i l e s _ r o l e _ c h e c k   y   c v   i t r  
391 | ALTER TABLE public.profiles DROP CONSTRAINT IF EXISTS profiles_role_check;
392 | ALTER TABLE public.profiles ADD CONSTRAINT profiles_role_check CHECK (role IN ('student', 'teacher', 'admin', ...
393 |
3 9 4 | - - 2 . S i n h m     n h d   n h P C M - X X X X X X X X t     n g
395 | CREATE OR REPLACE FUNCTION generate_student_code()
396 | RETURNS TEXT AS $$
397 | DECLARE
398 | chars TEXT := 'ABCDEFGHJKLMNPQRSTUVWXYZ23456789';
399 | result TEXT := 'PCM-';
400 | i INTEGER;
401 | BEGIN
402 | FOR i IN 1..8 LOOP
403 | result := result || substr(chars, floor(random() * length(chars) + 1)::int, 1);
404 | END LOOP;
405 | RETURN result;
406 | END;
407 | $$ LANGUAGE plpgsql;
408 |
4 0 9 | - - T r i g g e r s i n h s t u d e n t _ c o d e t r     c k h i t h   m m   i p r o f i l e
410 | CREATE OR REPLACE FUNCTION auto_generate_student_code()
411 | RETURNS TRIGGER AS $$
412 | BEGIN
413 | IF NEW.role = 'student' AND NEW.student_code IS NULL THEN
414 | LOOP
415 | NEW.student_code := generate_student_code();
416 | EXIT WHEN NOT EXISTS (
417 | SELECT 1 FROM profiles WHERE student_code = NEW.student_code
418 | );
419 | END LOOP;
420 | END IF;
421 | RETURN NEW;
422 | END;
423 | $$ LANGUAGE plpgsql;
424 |
425 | DROP TRIGGER IF EXISTS trigger_auto_student_code ON public.profiles;
426 | CREATE TRIGGER trigger_auto_student_code
427 | BEFORE INSERT ON public.profiles
428 | FOR EACH ROW EXECUTE FUNCTION auto_generate_student_code();

```



```

429| 3 0 | -- T ã ù ng i ả n s t u d e n t _ c o d e c h o h ấ c s i n h c i n ầ u b ể t h i ầ u
431| UPDATE public.profiles
432| SET student_code = generate_student_code()
433| WHERE role = 'student' AND student_code IS NULL;
434|
435| 3 5 | -- 3 . T r i g g e r t ã ù ng t i o P r o f i l e k h i n g ° Ỗ i d ầ ng n g k ỷ t ầ i k h o ầ n q u a s
436| CREATE OR REPLACE FUNCTION public.handle_new_user()
437| RETURNS TRIGGER AS $$
438| BEGIN
439| INSERT INTO public.profiles (id, email, full_name, role)
440| VALUES (
441| NEW.id,
442| NEW.email,
443| COALESCE(NEW.raw_user_meta_data->'full_name', split_part(NEW.email, '@', 1)),
444| COALESCE(NEW.raw_user_meta_data->'role', 'student')
445| );
446| RETURN NEW;
447| END;
448| $$ LANGUAGE plpgsql SECURITY DEFINER;
449|
450| DROP TRIGGER IF EXISTS on_auth_user_created ON auth.users;
451| CREATE TRIGGER on_auth_user_created
452| AFTER INSERT ON auth.users
453| FOR EACH ROW EXECUTE FUNCTION public.handle_new_user();
454|
455| 5 5 | -- 4 . T r i g g e r ồ ng b ầ s ầ l ầ ng c ầ u t r ể l Ỗ i t r o n g D i ầ n ầ n t h ầ o l u - n
456| CREATE OR REPLACE FUNCTION update_reply_count()
457| RETURNS TRIGGER AS $$
458| BEGIN
459| UPDATE public.discussion_threads
460| SET reply_count = reply_count + 1
461| WHERE id = NEW.thread_id;
462| RETURN NEW;
463| END;
464| $$ LANGUAGE plpgsql;
465|
466| DROP TRIGGER IF EXISTS on_reply_insert ON public.discussion_replies;
467| CREATE TRIGGER on_reply_insert
468| AFTER INSERT ON public.discussion_replies
469| FOR EACH ROW EXECUTE FUNCTION update_reply_count();
470|
471| 7 2 | -- % % % B ầ Ứ C 3 : H ầ T H ầ NG P H ầ N Q U Y ầ N B ầ O M ầ T R O W L E V E L S E C U R I T Y ( R L S ) % % %
473|
474| -- B - t R L S c h o c ầ c b ể ng
475| ALTER TABLE public.profiles ENABLE ROW LEVEL SECURITY;
476| ALTER TABLE public.schools ENABLE ROW LEVEL SECURITY;
477| ALTER TABLE public.parent_student_links ENABLE ROW LEVEL SECURITY;
478| ALTER TABLE public.lessons ENABLE ROW LEVEL SECURITY;
479| ALTER TABLE public.lesson_sections ENABLE ROW LEVEL SECURITY;
480| ALTER TABLE public.classes ENABLE ROW LEVEL SECURITY;
481| ALTER TABLE public.class_members ENABLE ROW LEVEL SECURITY;
482| ALTER TABLE public.assignments ENABLE ROW LEVEL SECURITY;
483| ALTER TABLE public.submissions ENABLE ROW LEVEL SECURITY;
484| ALTER TABLE public.lesson_progress ENABLE ROW LEVEL SECURITY;
485| ALTER TABLE public.notifications ENABLE ROW LEVEL SECURITY;
486| ALTER TABLE public.daily_quests ENABLE ROW LEVEL SECURITY;
487| ALTER TABLE public.user_quests ENABLE ROW LEVEL SECURITY;
488| ALTER TABLE public.quiz_attempts ENABLE ROW LEVEL SECURITY;
489| ALTER TABLE public.quiz_answers ENABLE ROW LEVEL SECURITY;
490| ALTER TABLE public.learning_paths ENABLE ROW LEVEL SECURITY;
491| ALTER TABLE public.path_items ENABLE ROW LEVEL SECURITY;
492| ALTER TABLE public.discussion_threads ENABLE ROW LEVEL SECURITY;
493| ALTER TABLE public.discussion_replies ENABLE ROW LEVEL SECURITY;
494| ALTER TABLE public.discussion_upvotes ENABLE ROW LEVEL SECURITY;
495|
496| 9 6 | -- C ầ I ầ T P O L I C I E S ( T ã ù ng x o ầ n ầ u t ồ n t i i ầ t r ầ nh l ầ i d u p l i c a t e )
497|
498| -- % % % P R O F I L E S & S C H O O L S % % %
499| DROP POLICY IF EXISTS "Users read own profile" ON public.profiles;
500| CREATE POLICY "Users read own profile" ON public.profiles FOR SELECT USING (auth.uid() = id);
501|
502| DROP POLICY IF EXISTS "Admin read all profiles" ON public.profiles;
503| CREATE POLICY "Admin read all profiles" ON public.profiles FOR SELECT USING (
504| EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'admin')
505| );
506|
507| DROP POLICY IF EXISTS "Users update own profile" ON public.profiles;
508| CREATE POLICY "Users update own profile" ON public.profiles FOR UPDATE USING (auth.uid() = id);
509|
510| DROP POLICY IF EXISTS "Schools are viewable by everyone" ON public.schools;
511| CREATE POLICY "Schools are viewable by everyone" ON public.schools FOR SELECT USING (true);
512|
513| -- % % % P A R E N T - S T U D E N T L I N K S % % %
514| DROP POLICY IF EXISTS "parent_own_links" ON public.parent_student_links;
515| CREATE POLICY "parent_own_links" ON public.parent_student_links FOR ALL USING (parent_id = auth.uid());
516|
517| DROP POLICY IF EXISTS "student_view_own_links" ON public.parent_student_links;
518| CREATE POLICY "student_view_own_links" ON public.parent_student_links FOR SELECT USING (student_id = auth.uid());
519|
520| DROP POLICY IF EXISTS "student_revoke_link" ON public.parent_student_links;
521| CREATE POLICY "student_revoke_link" ON public.parent_student_links FOR UPDATE USING (student_id = auth.uid())
522| WITH CHECK (status = 'revoked' AND revoked_by = auth.uid());
523|
524| -- % % % L E S S O N S & S E C T I O N S % % %
525| DROP POLICY IF EXISTS "teacher_own_lessons" ON public.lessons;
526| CREATE POLICY "teacher_own_lessons" ON public.lessons FOR ALL USING (teacher_id = auth.uid());
527|
528| DROP POLICY IF EXISTS "published_lessons_readable" ON public.lessons;
529| CREATE POLICY "published_lessons_readable" ON public.lessons FOR SELECT USING (is_published = TRUE);
530|
531| DROP POLICY IF EXISTS "lesson_sections_via_lesson" ON public.lesson_sections;
532| CREATE POLICY "lesson_sections_via_lesson" ON public.lesson_sections FOR SELECT USING (
533| EXISTS (SELECT 1 FROM public.lessons l WHERE l.id = lesson_id AND (l.is_published = TRUE OR l.teacher_id = a...
534| );
535|
536| DROP POLICY IF EXISTS "teacher_manage_sections" ON public.lesson_sections;
537| CREATE POLICY "teacher_manage_sections" ON public.lesson_sections FOR ALL USING (
538| EXISTS (SELECT 1 FROM public.lessons l WHERE l.id = lesson_id AND l.teacher_id = auth.uid())
539| );
540|
541| -- % % % C L A S S E S & M E M B E R S % % %
542| DROP POLICY IF EXISTS "teacher_own_classes" ON public.classes;
543| CREATE POLICY "teacher_own_classes" ON public.classes FOR ALL USING (teacher_id = auth.uid());
544|
545| DROP POLICY IF EXISTS "student_view_enrolled_class" ON public.classes;
546| CREATE POLICY "student_view_enrolled_class" ON public.classes FOR SELECT USING (
547| EXISTS (SELECT 1 FROM public.class_members cm WHERE cm.class_id = id AND cm.student_id = auth.uid())
548| );
549|
550| DROP POLICY IF EXISTS "teacher_manage_class_members" ON public.class_members;
551| CREATE POLICY "teacher_manage_class_members" ON public.class_members FOR ALL USING (
552| EXISTS (SELECT 1 FROM public.classes c WHERE c.id = class_id AND c.teacher_id = auth.uid())

```

```

553 );
554
555 DROP POLICY IF EXISTS "student_view_own_membership" ON public.class_members;
556 CREATE POLICY "student_view_own_membership" ON public.class_members FOR SELECT USING (student_id = auth.uid());
557
558 DROP POLICY IF EXISTS "student_can_join_class" ON public.class_members;
559 CREATE POLICY "student_can_join_class" ON public.class_members FOR INSERT WITH CHECK (student_id = auth.uid());
560
561 | - - % % % A S S I G N M E N T S & S U B M I S S I O N S % % %
562 DROP POLICY IF EXISTS "teacher_own_assignments" ON public.assignments;
563 CREATE POLICY "teacher_own_assignments" ON public.assignments FOR ALL USING (teacher_id = auth.uid());
564
565 DROP POLICY IF EXISTS "student_view_class_assignments" ON public.assignments;
566 CREATE POLICY "student_view_class_assignments" ON public.assignments FOR SELECT USING (
567     EXISTS (SELECT 1 FROM public.class_members cm WHERE cm.class_id = assignments.class_id AND cm.student_id = a...
568 );
569
570 DROP POLICY IF EXISTS "student_sees_own_submissions" ON public.submissions;
571 CREATE POLICY "student_sees_own_submissions" ON public.submissions FOR ALL USING (student_id = auth.uid());
572
573 DROP POLICY IF EXISTS "teacher_sees_class_submissions" ON public.submissions;
574 CREATE POLICY "teacher_sees_class_submissions" ON public.submissions FOR ALL USING (
575     EXISTS (SELECT 1 FROM public.classes c WHERE c.id = class_id AND c.teacher_id = auth.uid())
576 );
577
578 | - - % % % L E S S O N   P R O G R E S S   ( H í c   s í n h   &   P h â   h u y n h ) % % %
579 DROP POLICY IF EXISTS "student_own_progress" ON public.lesson_progress;
580 CREATE POLICY "student_own_progress" ON public.lesson_progress FOR ALL USING (student_id = auth.uid());
581
582 DROP POLICY IF EXISTS "teacher_view_class_progress" ON public.lesson_progress;
583 CREATE POLICY "teacher_view_class_progress" ON public.lesson_progress FOR SELECT USING (
584     EXISTS (
585         SELECT 1 FROM public.classes c
586         JOIN public.class_members cm ON cm.class_id = c.id
587         WHERE cm.student_id = lesson_progress.student_id AND c.teacher_id = auth.uid()
588     )
589 );
590
591 DROP POLICY IF EXISTS "parent_view_child_progress" ON public.lesson_progress;
592 CREATE POLICY "parent_view_child_progress" ON public.lesson_progress FOR SELECT USING (
593     EXISTS (
594         SELECT 1 FROM public.parent_student_links psl
595         WHERE psl.parent_id = auth.uid() AND psl.student_id = lesson_progress.student_id AND psl.status = 'active'
596     )
597 );
598
599 | - - % % % N O T I F I C A T I O N S % % %
600 DROP POLICY IF EXISTS "own_notifications" ON public.notifications;
601 CREATE POLICY "own_notifications" ON public.notifications FOR ALL USING (user_id = auth.uid());
602
603 | - - % % % Q U I Z   E N G I N E % % %
604 DROP POLICY IF EXISTS "student_own_attempts" ON public.quiz_attempts;
605 CREATE POLICY "student_own_attempts" ON public.quiz_attempts FOR ALL USING (student_id = auth.uid());
606
607 DROP POLICY IF EXISTS "student_own_answers" ON public.quiz_answers;
608 CREATE POLICY "student_own_answers" ON public.quiz_answers FOR ALL USING (
609     EXISTS (SELECT 1 FROM public.quiz_attempts WHERE id = attempt_id AND student_id = auth.uid())
610 );
611
612 DROP POLICY IF EXISTS "teacher_view_attempts" ON public.quiz_attempts;
613 CREATE POLICY "teacher_view_attempts" ON public.quiz_attempts FOR SELECT USING (
614     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
615 );
616
617 | - - % % % L E A R N I N G   P A T H % % %
618 DROP POLICY IF EXISTS "view_paths" ON public.learning_paths;
619 CREATE POLICY "view_paths" ON public.learning_paths FOR SELECT USING (true);
620
621 DROP POLICY IF EXISTS "manage_paths" ON public.learning_paths;
622 CREATE POLICY "manage_paths" ON public.learning_paths FOR ALL USING (
623     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
624 );
625
626 DROP POLICY IF EXISTS "view_path_items" ON public.path_items;
627 CREATE POLICY "view_path_items" ON public.path_items FOR SELECT USING (true);
628
629 DROP POLICY IF EXISTS "manage_path_items" ON public.path_items;
630 CREATE POLICY "manage_path_items" ON public.path_items FOR ALL USING (
631     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
632 );
633
634 | - - % % % D I S C U S S I O N   F O R U M S % % %
635 DROP POLICY IF EXISTS "view_threads" ON public.discussion_threads;
636 CREATE POLICY "view_threads" ON public.discussion_threads FOR SELECT USING (true);
637
638 DROP POLICY IF EXISTS "create_thread" ON public.discussion_threads;
639 CREATE POLICY "create_thread" ON public.discussion_threads FOR INSERT WITH CHECK (author_id = auth.uid());
640
641 DROP POLICY IF EXISTS "delete_own_or_teacher_thread" ON public.discussion_threads;
642 CREATE POLICY "delete_own_or_teacher_thread" ON public.discussion_threads FOR UPDATE USING (
643     author_id = auth.uid() OR EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
644 );
645
646 DROP POLICY IF EXISTS "view_replies" ON public.discussion_replies;
647 CREATE POLICY "view_replies" ON public.discussion_replies FOR SELECT USING (true);
648
649 DROP POLICY IF EXISTS "create_reply" ON public.discussion_replies;
650 CREATE POLICY "create_reply" ON public.discussion_replies FOR INSERT WITH CHECK (author_id = auth.uid());
651
652 DROP POLICY IF EXISTS "delete_own_or_teacher_reply" ON public.discussion_replies;
653 CREATE POLICY "delete_own_or_teacher_reply" ON public.discussion_replies FOR UPDATE USING (
654     author_id = auth.uid() OR EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
655 );
656
657 DROP POLICY IF EXISTS "manage_upvotes" ON public.discussion_upvotes;
658 CREATE POLICY "manage_upvotes" ON public.discussion_upvotes FOR ALL USING (user_id = auth.uid());
659
660 | - - % % % D A I L Y   Q U E S T S % % %
661 DROP POLICY IF EXISTS "Everyone can view quests" ON public.daily_quests;
662 CREATE POLICY "Everyone can view quests" ON public.daily_quests FOR SELECT USING (true);
663
664 DROP POLICY IF EXISTS "Users can view their own quest progress" ON public.user_quests;
665 CREATE POLICY "Users can view their own quest progress" ON public.user_quests FOR SELECT USING (auth.uid() = u...
666
667
668 | - - % % % B Ứ C   4 :   V I E W S   &   F U N C T I O N S   T I Ệ N   Í C H   D Ặ N H   C H O   C L I E N T S % % %
669
670 | - - 1.   V i e w   x e m   t ổng   q u a n   t i ệ n   ù   c ặ a   C o n   ( d ặ n h   c h o   P h ặ   h u y n h )
671 CREATE OR REPLACE VIEW public.child_summary_for_parent AS
672     SELECT
673         p.id AS student_id,
674         p.full_name AS student_name,
675         p.student_code,
676         p.school_name,

```

```

677 | p.xp,
678 | p.level,
679 | p.avatar_url,
680 | COUNT(DISTINCT lp.lesson_id) FILTER (WHERE lp.status = 'completed') AS lessons_completed,
681 | COUNT(DISTINCT lp.lesson_id) AS lessons_started,
682 | SUM(lp.time_spent_seconds) AS total_time_seconds,
683 | MAX(lp.last_accessed) AS last_active,
684 | psl.parent_id
685 | FROM public.profiles p
686 | JOIN public.parent_student_links psl ON psl.student_id = p.id
687 | LEFT JOIN public.lesson_progress lp ON lp.student_id = p.id
688 | WHERE psl.status = 'active' AND p.role = 'student'
689 | GROUP BY p.id, p.full_name, p.student_code, p.school_name,
690 | p.xp, p.level, p.avatar_url, psl.parent_id;
691 |
692 | -- 2. View lấy danh sách câu hỏi Quiz theo học sinh (cho học sinh làm bài)
693 | CREATE OR REPLACE VIEW public.quiz_questions_for_student AS
694 | SELECT
695 | q.id, q.content, q.type, q.points, q.image_url,
696 | q.tags, qq.order,
697 | json_agg(
698 | json_build_object(
699 | 'id', qo.id,
700 | 'content', qo.content,
701 | 'order', qo.order
702 | ) ORDER BY qo.order
703 | ) AS options,
704 | qq.quiz_id
705 | FROM public.questions q
706 | JOIN public.quiz_questions qq ON qq.question_id = q.id
707 | JOIN public.question_options qo ON qo.question_id = q.id
708 | GROUP BY q.id, q.content, q.type, q.points, q.image_url, q.tags, qq.order, qq.quiz_id;
709 |
710 | -- 3. Function cho phép học sinh tìm học sinh bằng mã PCM - XXXXXXXXX
711 | CREATE OR REPLACE FUNCTION public.find_student_by_code(input_code TEXT)
712 | RETURNS TABLE (
713 | student_id UUID,
714 | student_name TEXT,
715 | school_name TEXT,
716 | avatar_url TEXT
717 | )
718 | LANGUAGE plpgsql SECURITY DEFINER AS $$
719 | BEGIN
720 | RETURN QUERY
721 | SELECT p.id, p.full_name, p.school_name, p.avatar_url
722 | FROM public.profiles p
723 | WHERE p.student_code = UPPER(TRIM(input_code))
724 | AND p.role = 'student';
725 | END;
726 | $$;
727 |
728 | GRANT EXECUTE ON FUNCTION public.find_student_by_code TO authenticated;
729 |
730 | -- 4. Function tính trạng thái khóa lộ trình học (Learning Path)
731 | CREATE OR REPLACE FUNCTION public.get_unlocked_items(
732 | p_student_id UUID,
733 | p_path_id UUID
734 | ) RETURNS TABLE (
735 | id UUID,
736 | item_id UUID,
737 | is_unlocked BOOLEAN,
738 | unlock_reason TEXT
739 | ) AS $$
740 | DECLARE
741 | item RECORD;
742 | cond JSONB;
743 | cond_type TEXT;
744 | cond_item_id UUID;
745 | min_sc NUMERIC;
746 | prev_completed BOOLEAN;
747 | quiz_passed BOOLEAN;
748 | BEGIN
749 | FOR item IN
750 | SELECT * FROM public.path_items WHERE path_id = p_path_id ORDER BY "order"
751 | LOOP
752 | cond := item.unlock_condition;
753 | is_unlocked := TRUE;
754 | unlock_reason := 'always_available';
755 |
756 | IF cond IS NOT NULL THEN
757 | cond_type := cond->'type';
758 |
759 | -- Hoàn thành bài học trước đó
760 | IF cond_type = 'complete_previous' THEN
761 | SELECT EXISTS (
762 | SELECT 1 FROM public.lesson_progress lp
763 | JOIN public.path_items pi ON pi.item_id = lp.lesson_id
764 | WHERE lp.student_id = p_student_id
765 | AND lp.status = 'completed'
766 | AND pi.path_id = p_path_id
767 | AND pi.order = item.order - 1
768 | ) INTO prev_completed;
769 |
770 | IF NOT prev_completed THEN
771 | is_unlocked := FALSE;
772 | unlock_reason := 'need_complete_previous';
773 | END IF;
774 |
775 | -- item đã quiz thành công
776 | ELSIF cond_type = 'score_above' THEN
777 | cond_item_id := (cond->'item_id')::UUID;
778 | min_sc := (cond->'min_score')::NUMERIC;
779 |
780 | SELECT EXISTS (
781 | SELECT 1 FROM public.quiz_attempts
782 | WHERE student_id = p_student_id
783 | AND quiz_id = cond_item_id
784 | AND score >= min_sc
785 | AND status = 'submitted'
786 | ) INTO quiz_passed;
787 |
788 | IF NOT quiz_passed THEN
789 | is_unlocked := FALSE;
790 | unlock_reason := 'need_quiz_score_above_' || min_sc::TEXT;
791 | END IF;
792 | END IF;
793 | END LOOP;
794 |
795 | id := item.id;
796 | item_id := item.item_id;
797 | RETURN NEXT;
798 | END LOOP;
799 | END;
800 | $$ LANGUAGE plpgsql SECURITY DEFINER;

```

```

801|
802|
803| 8 0 3 | - - % % % B ^ Ú C 5 : H À N G â I C R O N J O B T ð Ø N G T H Ò N G B Á O C H O P H â H U Y N H % % %
804| DO $do$
805| BEGIN
806| IF EXISTS (SELECT 1 FROM pg_extension WHERE extname = 'pg_cron') THEN
807|     PERFORM cron.schedule(
808|         'notify-parent-absent',
809|         '0 20 * * *',
810|         $cron$
811|         INSERT INTO public.notifications (user_id, type, title, body, data, action_url)
812|         SELECT
813|             psl.parent_id,
814|             'child_absent',
815|             'Con bin ch^a h^ic trong 3 ngày',
816|             p.full_name || 'ch^a ng nh-p h^ic trong 3 ngày qua. H^y nh^c',
817|             jsonb_build_object('student_id', p.id, 'student_name', p.full_name),
818|             '/parent/children/' || p.id
819|         FROM public.profiles p
820|         JOIN public.parent_student_links psl ON psl.student_id = p.id
821|         LEFT JOIN public.lesson_progress lp ON lp.student_id = p.id
822|         AND lp.last_accessed > NOW() - INTERVAL '3 days'
823|         WHERE psl.status = 'active'
824|         AND p.role = 'student'
825|         AND lp.id IS NULL
826|         GROUP BY psl.parent_id, p.id, p.full_name;
827|         $cron$
828|     );
829| END IF;
830| END $do$;
831|

```

File: supabase/migrations/20260518094500_add_remaining_features.sql (134 lines, 5.9 KB)

```

1| -- =====
2| -- MIGRATION: ADD REMAINING FEATURES (PREFERENCES, BUILDER LAB, ACHIEVEMENTS & CERTIFICATES)
3| -- Target Path: supabase/migrations/20260518094500_add_remaining_features.sql
4| -- =====
5|
6| - - 1. B ê n g u s e r _ p r e f e r e n c e s ( T ù y c h ỉ n g i a o d i ệ n & t h ò n g b á o )
7| CREATE TABLE IF NOT EXISTS public.user_preferences (
8|     user_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE PRIMARY KEY,
9|     email_notifications BOOLEAN DEFAULT TRUE,
10|     push_notifications BOOLEAN DEFAULT TRUE,
11|     weekly_digest BOOLEAN DEFAULT TRUE,
12|     show_profile_public BOOLEAN DEFAULT FALSE,
13|     theme TEXT DEFAULT 'dark' CHECK (theme IN ('dark', 'light', 'system')),
14|     language TEXT DEFAULT 'vi' CHECK (language IN ('vi', 'en')),
15|     updated_at TIMESTAMPTZ DEFAULT NOW()
16| );
17|
18| - - 2. B ê n g b u i l d e r _ s e s s i o n s ( L ê c h s ỉ l ậ p r ấ p P C t h ỉ c h à n h t ỉ i B u i l d e r L a b )
19| CREATE TABLE IF NOT EXISTS public.builder_sessions (
20|     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
21|     student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
22|     lesson_id UUID REFERENCES public.lessons(id) ON DELETE SET NULL,
23|     started_at TIMESTAMPTZ DEFAULT NOW(),
24|     ended_at TIMESTAMPTZ,
25|     components_used JSONB DEFAULT '[]::jsonb',
26|     tdp_calculated NUMERIC,
27|     compatibility_score INT DEFAULT 100
28| );
29|
30| - - 3. B ê n g a c h i e v e m e n t _ d e f i n i t i o n s & s t u d e n t _ a c h i e v e m e n t s ( T h à n h t ỉ c h &
31| CREATE TABLE IF NOT EXISTS public.achievement_definitions (
32|     id TEXT PRIMARY KEY,
33|     title TEXT NOT NULL,
34|     description TEXT,
35|     icon TEXT,
36|     points INT DEFAULT 0,
37|     rarity TEXT CHECK (rarity IN ('common', 'uncommon', 'rare', 'legendary')),
38|     created_at TIMESTAMPTZ DEFAULT NOW()
39| );
40|
41| CREATE TABLE IF NOT EXISTS public.student_achievements (
42|     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
43|     student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
44|     achievement_id TEXT REFERENCES public.achievement_definitions(id) ON DELETE CASCADE,
45|     earned_at TIMESTAMPTZ DEFAULT NOW(),
46|     UNIQUE(student_id, achievement_id)
47| );
48|
49| - - N ỉ p h ỉ t d ỉ l ỉ ệ u ( S e e d ) c h o A c h i e v e m e n t s
50| INSERT INTO public.achievement_definitions (id, title, description, icon, points, rarity)
51| VALUES
52|     ('first_lesson', 'Kh ỉ i s u h à n h t r ỉ n h', 'Hoàn thành bài gi ệ n g s u t ỉ e n', '0=Yes'),
53|     ('quiz-ace', 'X u ỷ t s c', 'N ỉ i ( 1 0 0 % ) t r ỏ n g k h ỏ n g b à i q u i z', '0=Yes'),
54|     ('week-streak', 'K ỉ e n t r ỉ 7 n g à y', 'H ỉ c l ỉ e n t ỉ ậ p 7 n g à y k h ỏ n g n g h e', '0=Yes'),
55|     ('p_c_m_a_s_t_e_r', 'P C M a s t e r', 'H o à n t h à n h t ỷ t c ệ b à i t r ỏ n g l ậ t t r ỉ n h', '0=Yes'),
56|     ('lab-expert', 'C h u ỷ e n g ỉ a L a b', 'Đ à n h h ỉ n 1 0 g ỉ ỷ t r ỏ n g B u i l d e r L a b', '0=Yes'),
57|     ('helpful-member', 'N h ỉ ậ t t ỉ n h', ' ắ c u p v o t e 1 0 l ậ n t r ỏ n g d ỉ ả n à n', '0=Yes');
58| ON CONFLICT (id) DO UPDATE
59| SET
60|     title = EXCLUDED.title,
61|     description = EXCLUDED.description,
62|     icon = EXCLUDED.icon,
63|     points = EXCLUDED.points,
64|     rarity = EXCLUDED.rarity;
65|
66| - - 4. B ê n g c e r t i f i c a t e s ( C h é n g n h - n h o à n t h à n h l ậ t t r ỉ n h h ỉ c )
67| CREATE TABLE IF NOT EXISTS public.certificates (
68|     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
69|     student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
70|     path_id UUID REFERENCES public.learning_paths(id) ON DELETE CASCADE,
71|     certificate_number TEXT UNIQUE NOT NULL,
72|     issued_at TIMESTAMPTZ DEFAULT NOW(),
73|     pdf_url TEXT,
74|     verify_url TEXT,
75|     student_name TEXT,
76|     course_title TEXT,
77|     completion_date DATE DEFAULT CURRENT_DATE,
78|     final_score NUMERIC(5,2),
79|     is_revoked BOOLEAN DEFAULT FALSE
80| );
81|
82| - - H à m t ỉ ừ n g s ỉ n h s ỉ n c h é n g c h é n g « u n h ỉ e n
83| CREATE OR REPLACE FUNCTION public.generate_cert_number()
84| RETURNS TEXT AS $$
85| DECLARE
86|     chars TEXT := 'ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789';
87|     result TEXT := 'PCM-2026-';
88|     i INT;
89| BEGIN

```

```

90 | FOR i IN 1..5 LOOP
91 |     result := result || substr(chars, floor(random() * length(chars) + 1)::INT, 1);
92 | END LOOP;
93 | RETURN result;
94 | END;
95 | $$ LANGUAGE plpgsql;
96 |
97 |
98 | - - % % %   H Ệ  T H Ọ N G   P H Ầ N   Q U Y Ầ N   R O W   L E V E L   S E C U R I T Y   ( R L S )   &   P O L I C I E S   % % %
99 |
100 | ALTER TABLE public.user_preferences ENABLE ROW LEVEL SECURITY;
101 | ALTER TABLE public.builder_sessions ENABLE ROW LEVEL SECURITY;
102 | ALTER TABLE public.achievement_definitions ENABLE ROW LEVEL SECURITY;
103 | ALTER TABLE public.student_achievements ENABLE ROW LEVEL SECURITY;
104 | ALTER TABLE public.certificates ENABLE ROW LEVEL SECURITY;
105 |
106 | -- Preferences Policies
107 | DROP POLICY IF EXISTS "own_preferences" ON public.user_preferences;
108 | CREATE POLICY "own_preferences" ON public.user_preferences FOR ALL USING (user_id = auth.uid());
109 |
110 | -- Builder Sessions Policies
111 | DROP POLICY IF EXISTS "student_own_sessions" ON public.builder_sessions;
112 | CREATE POLICY "student_own_sessions" ON public.builder_sessions FOR ALL USING (student_id = auth.uid());
113 |
114 | DROP POLICY IF EXISTS "teacher_view_sessions" ON public.builder_sessions;
115 | CREATE POLICY "teacher_view_sessions" ON public.builder_sessions FOR SELECT USING (
116 |     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
117 | );
118 |
119 | -- Achievement Policies
120 | DROP POLICY IF EXISTS "view_definitions" ON public.achievement_definitions;
121 | CREATE POLICY "view_definitions" ON public.achievement_definitions FOR SELECT USING (true);
122 |
123 | DROP POLICY IF EXISTS "view_student_achievements" ON public.student_achievements;
124 | CREATE POLICY "view_student_achievements" ON public.student_achievements FOR SELECT USING (true);
125 |
126 | -- Certificate Policies
127 | DROP POLICY IF EXISTS "view_all_certificates" ON public.certificates;
128 | CREATE POLICY "view_all_certificates" ON public.certificates FOR SELECT USING (true);
129 |
130 | DROP POLICY IF EXISTS "manage_certificates" ON public.certificates;
131 | CREATE POLICY "manage_certificates" ON public.certificates FOR ALL USING (
132 |     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
133 | );
134 |

```

File: supabase/migrations/20260518095000_parent_notifications.sql (187 lines, 6.9 KB)

```

1 | -- =====
2 | -- PROMPT 4 - PC MASTER LMS - PARENT NOTIFICATIONS & TRIGGERS
3 | -- Target Path: supabase/migrations/20260518095000_parent_notifications.sql
4 | -- =====
5 |
6 | - - % % %   B Ứ C   1 :   C ẬP   N H ẬP   T   U S E R   P R E F E R E N C E S   Á   H Ồ   T R ả   O P T - O U T   % % %
7 | - -   T h êm   c Ụ T   n o t i f i c a t i o n _ s e t t i n g s   ( J S O N B )   c h o   b Ể N g   u s e r _ p r e f e r e n c e s
8 | ALTER TABLE public.user_preferences
9 | ADD COLUMN IF NOT EXISTS notification_settings JSONB
10 | DEFAULT '{"child_absent": true, "child_low_score": true, "child_completed_path": true, "lesson_completed": t...
11 |
12 | - -   T i ỏ   b Ể N g   l ờ u   t r ả   t h ỏ n g   t i n   c o o l d o w n   g i ả   e m a i l
13 | CREATE TABLE IF NOT EXISTS public.parent_email_cooldown (
14 |     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
15 |     student_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE CASCADE,
16 |     parent_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE CASCADE,
17 |     notification_type TEXT NOT NULL,
18 |     sent_at TIMESTAMPTZ DEFAULT NOW(),
19 |     UNIQUE(student_id, parent_id, notification_type)
20 | );
21 |
22 | - -   B - t   R L S   c h o   b Ể N g   c o o l d o w n
23 | ALTER TABLE public.parent_email_cooldown ENABLE ROW LEVEL SECURITY;
24 |
25 | -- Cooldown Policy
26 | DROP POLICY IF EXISTS "system_all_cooldown" ON public.parent_email_cooldown;
27 | CREATE POLICY "system_all_cooldown" ON public.parent_email_cooldown
28 | FOR ALL USING (true);
29 |
30 |
31 | - - % % %   B Ứ C   2 :   T R I G G E R   K H I   H Ỉ C   S I N H   H O Ầ N   T H Ầ N H   B ả I   H Ỉ C   % % %
32 | CREATE OR REPLACE FUNCTION notify_parent_on_lesson_complete()
33 | RETURNS TRIGGER AS $$
34 | DECLARE
35 |     parent_rec RECORD;
36 |     lesson_rec RECORD;
37 |     pref_val BOOLEAN;
38 | BEGIN
39 |     IF NEW.status = 'completed' AND OLD.status != 'completed' THEN
40 |         - -   L ấ y   t Ờ n   b ả i   h Ỉ c
41 |         SELECT title INTO lesson_rec FROM lessons WHERE id = NEW.lesson_id;
42 |
43 |         - -   T ỉ m   t ắ t   c Ể   p h ả   h u y ằ n h   ằ n g   t h ỏ   đ ỗ i   h Ỉ c   s i n h   n ằ y
44 |         FOR parent_rec IN
45 |             SELECT psl.parent_id, up.notification_settings
46 |             FROM parent_student_links psl
47 |             LEFT JOIN user_preferences up ON up.user_id = psl.parent_id
48 |             WHERE psl.student_id = NEW.student_id AND psl.status = 'active'
49 |         LOOP
50 |             -- Check opt-out setting
51 |             pref_val := TRUE;
52 |             IF parent_rec.notification_settings IS NOT NULL AND parent_rec.notification_settings ? 'lesson_completed...'
53 |                 pref_val := (parent_rec.notification_settings->'lesson_completed')::boolean;
54 |             END IF;
55 |
56 |             IF pref_val THEN
57 |                 INSERT INTO notifications (user_id, type, title, body, data, action_url)
58 |                 VALUES (
59 |                     parent_rec.parent_id,
60 |                     'lesson_completed',
61 |                     C ỏ n   b ỉ n   h o ằ n   t h Ầ n h   b ả i   h Ỉ c   Ờ Ờ ' ,
62 |                     C ỏ n   b ỉ n   v Ể ả   h o ằ n   t h Ầ n h   b ả i   " "   | |   C O A L E S C E ( l e s s o n _ r e c . t i t l e ,
63 |                     jsonb_build_object(
64 |                         'student_id', NEW.student_id,
65 |                         'lesson_id', NEW.lesson_id
66 |                     ),
67 |                     '/parent/children/' || NEW.student_id
68 |                 );
69 |             END IF;
70 |         END LOOP;
71 |     END IF;
72 |     RETURN NEW;
73 | END;
74 | $$ LANGUAGE plpgsql SECURITY DEFINER;
75 |

```

```

76| DROP TRIGGER IF EXISTS on_lesson_completed ON lesson_progress;
77| CREATE TRIGGER on_lesson_completed
78| AFTER UPDATE ON lesson_progress
79| FOR EACH ROW EXECUTE FUNCTION notify_parent_on_lesson_complete();
80|
81|
82| - - % % % B Û C 3 : T R I G G E R K H I C Ó K % T Q U Ç Q U I Z % % %
83| CREATE OR REPLACE FUNCTION notify_parent_on_quiz_result()
84| RETURNS TRIGGER AS $$
85| DECLARE
86| parent_rec RECORD;
87| quiz_rec RECORD;
88| pref_val BOOLEAN;
89| notif_type TEXT;
90| notif_title TEXT;
91| BEGIN
92| IF NEW.status = 'graded' THEN
93| SELECT title INTO quiz_rec FROM quizzes WHERE id = NEW.quiz_id;
94|
95| - - Q u y ÷ t Ê n h l o i i t h ò n g b á o d ñ a t r ê n i Ā m s Ñ
96| IF NEW.score < 50 THEN
97| notif_type := 'child_low_score';
98| n o t i f _ t i t l e := '&p Con bìn có i Ā m t h % p';
99| ELSE
100| notif_type := 'quiz_graded';
101| n o t i f _ t i t l e := ' ' Con bìn có k ÷ t q u Ç m ũ i';
102| END IF;
103|
104| FOR parent_rec IN
105| SELECT psl.parent_id, up.notification_settings
106| FROM parent_student_links psl
107| LEFT JOIN user_preferences up ON up.user_id = psl.parent_id
108| WHERE psl.student_id = NEW.student_id AND psl.status = 'active'
109| LOOP
110| - - C h e c k o p t - o u t s e t t i n g c h o l o i i t h ò n g b á o n à y
111| pref_val := TRUE;
112| IF parent_rec.notification_settings IS NOT NULL AND parent_rec.notification_settings ? notif_type THEN
113| pref_val := (parent_rec.notification_settings->>notif_type)::boolean;
114| END IF;
115|
116| IF pref_val THEN
117| INSERT INTO notifications (user_id, type, title, body, data, action_url)
118| VALUES (
119| parent_rec.parent_id,
120| notif_type,
121| notif_title,
122| ' K ÷ t q u Ç " ' || COALESCE(quiz_rec.title, 'Bài tr ã n g h i ÷ m') ||
123| jsonb_build_object(
124| 'student_id', NEW.student_id,
125| 'attempt_id', NEW.id,
126| 'score', NEW.score
127| ),
128| '/parent/children/' || NEW.student_id
129| );
130| END IF;
131| END LOOP;
132| END IF;
133| RETURN NEW;
134| END;
135| $$ LANGUAGE plpgsql SECURITY DEFINER;
136|
137| DROP TRIGGER IF EXISTS on_quiz_graded ON quiz_attempts;
138| CREATE TRIGGER on_quiz_graded
139| AFTER UPDATE ON quiz_attempts
140| FOR EACH ROW EXECUTE FUNCTION notify_parent_on_quiz_result();
141|
142|
143| - - % % % B Û C 4 : T R I G G E R K H I T T H Ā N H T Í C H M ũ I ( C H A M P I O N ) % % %
144| CREATE OR REPLACE FUNCTION notify_parent_on_achievement_earned()
145| RETURNS TRIGGER AS $$
146| DECLARE
147| parent_rec RECORD;
148| ach_rec RECORD;
149| pref_val BOOLEAN;
150| BEGIN
151| - - T ĩ m t % t c Ē p h Ā h u y n h ā n g t h e o đ o i h ĩ c s ĩ n h n à y
152| FOR parent_rec IN
153| SELECT psl.parent_id, up.notification_settings
154| FROM parent_student_links psl
155| LEFT JOIN user_preferences up ON up.user_id = psl.parent_id
156| WHERE psl.student_id = NEW.student_id AND psl.status = 'active'
157| LOOP
158| -- Check opt-out setting
159| pref_val := TRUE;
160| IF parent_rec.notification_settings IS NOT NULL AND parent_rec.notification_settings ? 'achievement_earned...'
161| pref_val := (parent_rec.notification_settings->>'achievement_earned')::boolean;
162| END IF;
163|
164| IF pref_val THEN
165| INSERT INTO notifications (user_id, type, title, body, data, action_url)
166| VALUES (
167| parent_rec.parent_id,
168| 'achievement_earned',
169| ' C ó n b ĩ n ĩ t t h ā n h t í c h m ũ i ! 0 < B Ē ',
170| ' C ó n b ĩ n v Ē a x u ģ t s ã c m ĩ k h ó a h u y h ĩ ç u t h ā n h t í c h m ũ i t r o n g h ĩ c
171| jsonb_build_object(
172| 'student_id', NEW.student_id,
173| 'achievement_id', NEW.achievement_id
174| ),
175| '/parent/children/' || NEW.student_id
176| );
177| END IF;
178| END LOOP;
179| RETURN NEW;
180| END;
181| $$ LANGUAGE plpgsql SECURITY DEFINER;
182|
183| DROP TRIGGER IF EXISTS on_achievement_earned ON student_achievements;
184| CREATE TRIGGER on_achievement_earned
185| AFTER INSERT ON student_achievements
186| FOR EACH ROW EXECUTE FUNCTION notify_parent_on_achievement_earned();
187|

```

File: supabase/migrations/20260524110500_parent_realtime_v2.sql (357 lines, 13.0 KB)

```

1| -- Path: supabase/migrations/20260524110500_parent_realtime_v2.sql
2| -- =====
3| -- SYSTEM UPGRADE: PARENT MONITORING REALTIME SYSTEM
4| -- =====
5|
6| - - % % % B Û C 1 : T H Ē M S T U D E N T _ C O D E V Ā P A R E N T R O L E V Ā O P R O F I L E S % % %
7| ALTER TABLE public.profiles
8| DROP CONSTRAINT IF EXISTS profiles_role_check;

```

```

9 |
10 | ALTER TABLE public.profiles
11 |     ADD CONSTRAINT profiles_role_check
12 |     CHECK (role IN ('student', 'teacher', 'admin', 'parent'));
13 |
14 | -- Thêm các cột bổ sung nếu cần
15 | ALTER TABLE public.profiles
16 |     ADD COLUMN IF NOT EXISTS student_code TEXT UNIQUE,
17 |     ADD COLUMN IF NOT EXISTS avatar_url TEXT,
18 |     ADD COLUMN IF NOT EXISTS phone TEXT,
19 |     ADD COLUMN IF NOT EXISTS is_online BOOLEAN DEFAULT FALSE,
20 |     ADD COLUMN IF NOT EXISTS last_seen_at TIMESTAMPTZ DEFAULT NOW();
21 |
22 | -- Function sinh student_code ngẫu nhiên 8 ký tự không nhism l«n
23 | CREATE OR REPLACE FUNCTION generate_student_code()
24 | RETURNS TEXT AS $$
25 | DECLARE
26 |     chars TEXT := 'ABCDEFGHIJKLMNOPQRSTUVWXYZ23456789'; -- Loii bĩ I, 0, 0,
27 |     result TEXT := 'PCM-';
28 |     i INTEGER;
29 | BEGIN
30 |     FOR i IN 1..8 LOOP
31 |         result := result || substr(chars, floor(random() * length(chars) + 1)::int, 1);
32 |     END LOOP;
33 |     RETURN result;
34 | END;
35 | $$ LANGUAGE plpgsql;
36 |
37 | -- Trigger: tñ sinh student_code khi ng ký role student
38 | CREATE OR REPLACE FUNCTION trigger_auto_student_code()
39 | RETURNS TRIGGER AS $$
40 | BEGIN
41 |     IF NEW.role = 'student' AND NEW.student_code IS NULL THEN
42 |         LOOP
43 |             NEW.student_code := generate_student_code();
44 |             EXIT WHEN NOT EXISTS (
45 |                 SELECT 1 FROM public.profiles WHERE student_code = NEW.student_code
46 |             );
47 |         END LOOP;
48 |     END IF;
49 |     RETURN NEW;
50 | END;
51 | $$ LANGUAGE plpgsql;
52 |
53 | DROP TRIGGER IF EXISTS auto_student_code ON public.profiles;
54 | CREATE TRIGGER auto_student_code
55 | BEFORE INSERT ON public.profiles
56 | FOR EACH ROW EXECUTE FUNCTION trigger_auto_student_code();
57 |
58 | -- Backfill mã cho các hĩc sinh ch°a có
59 | UPDATE public.profiles
60 | SET student_code = generate_student_code()
61 | WHERE role = 'student' AND student_code IS NULL;
62 |
63 |
64 | -- %%% B-ÚC 2: B¢NG LIÊN K%T PHẢ HUYNH - HỈC SINH %%%
65 | CREATE TABLE IF NOT EXISTS public.parent_student_links (
66 |     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
67 |     parent_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE CASCADE,
68 |     student_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE CASCADE,
69 |     relationship TEXT DEFAULT 'parent'
70 |     CHECK (relationship IN ('father', 'mother', 'guardian', 'sibling', 'other')),
71 |     status TEXT DEFAULT 'pending'
72 |     CHECK (status IN ('pending', 'active', 'revoked')),
73 |     linked_at TIMESTAMPTZ DEFAULT NOW(),
74 |     confirmed_at TIMESTAMPTZ,
75 |     revoked_at TIMESTAMPTZ,
76 |     revoked_by UUID REFERENCES public.profiles(id),
77 |     UNIQUE(parent_id, student_id)
78 | );
79 |
80 | -- B-t RLS
81 | ALTER TABLE public.parent_student_links ENABLE ROW LEVEL SECURITY;
82 |
83 | -- Cài t RLS Policies
84 | DROP POLICY IF EXISTS "parent_manage_own_links" ON public.parent_student_links;
85 | CREATE POLICY "parent_manage_own_links" ON public.parent_student_links
86 | FOR ALL USING (parent_id = auth.uid());
87 |
88 | DROP POLICY IF EXISTS "student_view_own_links" ON public.parent_student_links;
89 | CREATE POLICY "student_view_own_links" ON public.parent_student_links
90 | FOR SELECT USING (student_id = auth.uid());
91 |
92 | DROP POLICY IF EXISTS "student_update_link_status" ON public.parent_student_links;
93 | CREATE POLICY "student_update_link_status" ON public.parent_student_links
94 | FOR UPDATE USING (student_id = auth.uid())
95 | WITH CHECK (status IN ('active', 'revoked'));
96 |
97 |
98 | -- %%% B-ÚC 3: B¢NG THEO DÕI TI%N 0 HỈC T-P (LESSON PROGRESS) %%%
99 | CREATE TABLE IF NOT EXISTS public.lesson_progress (
100 |     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
101 |     student_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE CASCADE,
102 |     lesson_id UUID NOT NULL,
103 |     status TEXT DEFAULT 'not_started'
104 |     CHECK (status IN ('not_started', 'in_progress', 'completed')),
105 |     completion_percentage INT DEFAULT 0 CHECK (completion_percentage BETWEEN 0 AND 100),
106 |     time_spent_seconds INT DEFAULT 0,
107 |     last_accessed TIMESTAMPTZ DEFAULT NOW(),
108 |     completed_at TIMESTAMPTZ,
109 |     UNIQUE(student_id, lesson_id)
110 | );
111 |
112 | ALTER TABLE public.lesson_progress ENABLE ROW LEVEL SECURITY;
113 |
114 | DROP POLICY IF EXISTS "student_own_progress" ON public.lesson_progress;
115 | CREATE POLICY "student_own_progress" ON public.lesson_progress
116 | FOR ALL USING (student_id = auth.uid());
117 |
118 | DROP POLICY IF EXISTS "parent_view_child_progress" ON public.lesson_progress;
119 | CREATE POLICY "parent_view_child_progress" ON public.lesson_progress
120 | FOR SELECT USING (
121 |     EXISTS (
122 |         SELECT 1 FROM public.parent_student_links psl
123 |         WHERE psl.parent_id = auth.uid()
124 |         AND psl.student_id = lesson_progress.student_id
125 |         AND psl.status = 'active'
126 |     )
127 | );
128 |
129 | DROP POLICY IF EXISTS "teacher_view_class_progress" ON public.lesson_progress;
130 | CREATE POLICY "teacher_view_class_progress" ON public.lesson_progress
131 | FOR SELECT USING (
132 |     EXISTS (

```

```

133 | SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher'
134 | )
135 | );
136 |
137 |
138 | -- %%% B-ÚC 4: B C N G K % T Q U C T H I C æ A H I C S I N H ( Q U I Z A T T E M P T S ) % % %
139 | CREATE TABLE IF NOT EXISTS public.quiz_attempts (
140 |   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
141 |   student_id UUID NOT NULL REFERENCES public.profiles(id),
142 |   quiz_id UUID NOT NULL,
143 |   quiz_title TEXT NOT NULL,
144 |   started_at TIMESTAMPTZ DEFAULT NOW(),
145 |   submitted_at TIMESTAMPTZ,
146 |   score NUMERIC(5,2),
147 |   max_score INT DEFAULT 100,
148 |   status TEXT DEFAULT 'in_progress'
149 |   CHECK (status IN ('in_progress', 'submitted', 'graded')),
150 |   attempt_number INT DEFAULT 1
151 | );
152 |
153 | ALTER TABLE public.quiz_attempts ENABLE ROW LEVEL SECURITY;
154 |
155 | DROP POLICY IF EXISTS "student_own_attempts" ON public.quiz_attempts;
156 | CREATE POLICY "student_own_attempts" ON public.quiz_attempts
157 |   FOR ALL USING (student_id = auth.uid());
158 |
159 | DROP POLICY IF EXISTS "parent_view_child_quiz" ON public.quiz_attempts;
160 | CREATE POLICY "parent_view_child_quiz" ON public.quiz_attempts
161 |   FOR SELECT USING (
162 |     EXISTS (
163 |       SELECT 1 FROM public.parent_student_links psl
164 |       WHERE psl.parent_id = auth.uid()
165 |       AND psl.student_id = quiz_attempts.student_id
166 |       AND psl.status = 'active'
167 |     )
168 |   );
169 |
170 |
171 | -- %%% B-ÚC 5: B C N G T H Ô N G B Á O ( N O T I F I C A T I O N S ) % % %
172 | CREATE TABLE IF NOT EXISTS public.notifications (
173 |   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
174 |   user_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE CASCADE,
175 |   type TEXT NOT NULL CHECK (type IN (
176 |     'quiz_graded', 'lesson_completed', 'achievement_earned',
177 |     'child_absent_3days', 'child_low_score', 'child_completed_path',
178 |     'parent_link_request', 'parent_link_confirmed', 'teacher_message', 'system'
179 |   )),
180 |   title TEXT NOT NULL,
181 |   body TEXT NOT NULL,
182 |   data JSONB DEFAULT '{}',
183 |   action_url TEXT,
184 |   is_read BOOLEAN DEFAULT FALSE,
185 |   created_at TIMESTAMPTZ DEFAULT NOW()
186 | );
187 |
188 | ALTER TABLE public.notifications ENABLE ROW LEVEL SECURITY;
189 |
190 | DROP POLICY IF EXISTS "own_notifications" ON public.notifications;
191 | CREATE POLICY "own_notifications" ON public.notifications
192 |   FOR ALL USING (user_id = auth.uid());
193 |
194 |
195 | -- %%% B-ÚC 6: H À M T Ì M H Ì C S I N H Q U A M Æ A N T O À N % % %
196 | -- Drop tr 0 c n ư t 0 n t i i v 0 i s i g n a t u r e c i ( t r á n h l x i " c a n n o t c h a n g e r e t
197 | DROP FUNCTION IF EXISTS public.find_student_by_code(TEXT);
198 |
199 | CREATE OR REPLACE FUNCTION public.find_student_by_code(input_code TEXT)
200 | RETURNS TABLE (
201 |   student_id UUID,
202 |   student_name TEXT,
203 |   school_name TEXT,
204 |   avatar_url TEXT,
205 |   class_info TEXT
206 | )
207 | LANGUAGE plpgsql SECURITY DEFINER AS $$
208 | BEGIN
209 |   RETURN QUERY
210 |     SELECT
211 |       p.id,
212 |       p.full_name,
213 |       p.school_name,
214 |       p.avatar_url,
215 |       C O A L E S C E ( c . n a m e , ' C h ò a c ó l ộ p ' ) a s c l a s s _ i n f o
216 |     FROM public.profiles p
217 |     LEFT JOIN public.class_members cm ON cm.student_id = p.id
218 |     LEFT JOIN public.classes c ON c.id = cm.class_id
219 |     WHERE p.student_code = UPPER(TRIM(input_code))
220 |           AND p.role = 'student';
221 | END;
222 | $$;
223 |
224 | GRANT EXECUTE ON FUNCTION public.find_student_by_code TO authenticated;
225 |
226 |
227 | -- %%% B-ÚC 7: T R I G G E R S B @ N T H Ô N G B Á O T Ờ Ờ N G C H O P H ầ H U Y N H % % %
228 |
229 | -- Trigger 1: Thông báo khi con hoàn thành bài híc
230 | CREATE OR REPLACE FUNCTION notify_parents_on_lesson_complete()
231 | RETURNS TRIGGER AS $$
232 | DECLARE parent_rec RECORD;
233 | BEGIN
234 |   IF NEW.status = 'completed' AND (OLD.status IS NULL OR OLD.status != 'completed') THEN
235 |     FOR parent_rec IN
236 |       SELECT psl.parent_id, p.full_name as student_name
237 |       FROM public.parent_student_links psl
238 |       JOIN public.profiles p ON p.id = NEW.student_id
239 |       WHERE psl.student_id = NEW.student_id AND psl.status = 'active'
240 |     LOOP
241 |       INSERT INTO public.notifications
242 |         (user_id, type, title, body, data, action_url)
243 |       VALUES (
244 |         parent_rec.parent_id,
245 |         'lesson_completed',
246 |         ' Ờ Ờ C o n b i n h o à n t h à n h b à i h í c ',
247 |         p a r e n t _ r e c . s t u d e n t _ n a m e || ' v ề a h o à n t h à n h 1 b à i h í c m ô i ',
248 |         jsonb_build_object('student_id', NEW.student_id, 'lesson_id', NEW.lesson_id),
249 |         '/parent/children/' || NEW.student_id::text
250 |       );
251 |     END LOOP;
252 |   END IF;
253 |   RETURN NEW;
254 | END;
255 | $$ LANGUAGE plpgsql SECURITY DEFINER;
256 |

```



```

257 DROP TRIGGER IF EXISTS on_lesson_completed ON public.lesson_progress;
258 CREATE TRIGGER on_lesson_completed
259 AFTER INSERT OR UPDATE ON public.lesson_progress
260 FOR EACH ROW EXECUTE FUNCTION notify_parents_on_lesson_complete();
261
262 | -- Trigger 2: Thông báo i Âm thi
263 CREATE OR REPLACE FUNCTION notify_parents_on_quiz_graded()
264 RETURNS TRIGGER AS $$
265 DECLARE parent_rec RECORD;
266 BEGIN
267 IF NEW.status = 'graded' AND (OLD.status IS NULL OR OLD.status != 'graded') THEN
268 FOR parent_rec IN
269 SELECT psl.parent_id, p.full_name as student_name
270 FROM public.parent_student_links psl
271 JOIN public.profiles p ON p.id = NEW.student_id
272 WHERE psl.student_id = NEW.student_id AND psl.status = 'active'
273 LOOP
274 INSERT INTO public.notifications
275 (user_id, type, title, body, data, action_url)
276 VALUES (
277 parent_rec.parent_id,
278 CASE WHEN NEW.score < 50 THEN 'child_low_score' ELSE 'quiz_graded' END,
279 CASE WHEN NEW.score < 50
280 THEN '& p Con bin có i Âm th %p: ' || ROUND(NEW.score) || ' / ' ||
281 ROUND(NEW.score) || ' / ' || NEW.m
282 END,
283 parent_rec.student_name || ' ' ||
284 CASE WHEN NEW.score < 50 THEN 'chẾ it ' ELSE 'it ' END ||
285 ROUND(NEW.score) || ' i Âm trong bài ' || NEW.quiz_title || ' '
286 jsonb_build_object(
287 'student_id', NEW.student_id,
288 'attempt_id', NEW.id,
289 'score', NEW.score,
290 'quiz_title', NEW.quiz_title
291 ),
292 '/parent/children/' || NEW.student_id::text
293 );
294 END LOOP;
295 END IF;
296 RETURN NEW;
297 END;
298 $$ LANGUAGE plpgsql SECURITY DEFINER;
299
300 DROP TRIGGER IF EXISTS on_quiz_graded ON public.quiz_attempts;
301 CREATE TRIGGER on_quiz_graded
302 AFTER INSERT OR UPDATE ON public.quiz_attempts
303 FOR EACH ROW EXECUTE FUNCTION notify_parents_on_quiz_graded();
304
305 | -- % % % B - Ú C 8 : ENABLE REALTIME CHO CÁC B CNG LIÊN QUAN % % %
306 | -- B - t realtime cho lesson_progress, quiz_attempts, profiles, notifications
307 | -- An toàn: Ki Âm tra têng b Cng tr Ò Úc khi thêm, tránh l x i "already member
308
309 DO $$
310 DECLARE
311 pub_exists BOOLEAN;
312 BEGIN
313 SELECT EXISTS (SELECT 1 FROM pg_publication WHERE pubname = 'supabase_realtime') INTO pub_exists;
314
315 IF pub_exists THEN
316 -- lesson_progress
317 IF NOT EXISTS (
318 SELECT 1 FROM pg_publication_tables
319 WHERE pubname = 'supabase_realtime' AND schemaname = 'public' AND tablename = 'lesson_progress'
320 ) THEN
321 ALTER PUBLICATION supabase_realtime ADD TABLE public.lesson_progress;
322 END IF;
323
324 -- quiz_attempts
325 IF NOT EXISTS (
326 SELECT 1 FROM pg_publication_tables
327 WHERE pubname = 'supabase_realtime' AND schemaname = 'public' AND tablename = 'quiz_attempts'
328 ) THEN
329 ALTER PUBLICATION supabase_realtime ADD TABLE public.quiz_attempts;
330 END IF;
331
332 -- notifications
333 IF NOT EXISTS (
334 SELECT 1 FROM pg_publication_tables
335 WHERE pubname = 'supabase_realtime' AND schemaname = 'public' AND tablename = 'notifications'
336 ) THEN
337 ALTER PUBLICATION supabase_realtime ADD TABLE public.notifications;
338 END IF;
339
340 -- profiles
341 IF NOT EXISTS (
342 SELECT 1 FROM pg_publication_tables
343 WHERE pubname = 'supabase_realtime' AND schemaname = 'public' AND tablename = 'profiles'
344 ) THEN
345 ALTER PUBLICATION supabase_realtime ADD TABLE public.profiles;
346 END IF;
347
348 -- parent_student_links
349 IF NOT EXISTS (
350 SELECT 1 FROM pg_publication_tables
351 WHERE pubname = 'supabase_realtime' AND schemaname = 'public' AND tablename = 'parent_student_links'
352 ) THEN
353 ALTER PUBLICATION supabase_realtime ADD TABLE public.parent_student_links;
354 END IF;
355 END IF;
356 END $$;
357

```

File: supabase/migrations/20260524120000_fix_profiles_rls_and_recursion.sql (128 lines, 5.9 KB)

```

1 | -- Path: supabase/migrations/20260524120000_fix_profiles_rls_and_recursion.sql
2 | --
3 | -- SYSTEM UPGRADE: FIX PROFILES RLS INFINITE RECURSION & SIGNUP ERROR
4 | --
5 |
6 | -- % % % B - Ú C 1 : Si A Ô I RLS CHO B CNG PROFILES Ā TRÁNH INFINITE RECURSION
7 |
8 | -- G á b ĩ t % t c Ē p o l i c i e s c i t r ê n b Cng p r o f i l e s Ā t r á n h x u n g Û t
9 DROP POLICY IF EXISTS "Users read own profile" ON public.profiles;
10 DROP POLICY IF EXISTS "Admin read all profiles" ON public.profiles;
11 DROP POLICY IF EXISTS "Users update own profile" ON public.profiles;
12 DROP POLICY IF EXISTS "Allow public insert" ON public.profiles;
13 DROP POLICY IF EXISTS "profiles_select_own" ON public.profiles;
14 DROP POLICY IF EXISTS "profiles_select_admin" ON public.profiles;
15 DROP POLICY IF EXISTS "profiles_select_teacher" ON public.profiles;
16 DROP POLICY IF EXISTS "profiles_select_parent" ON public.profiles;
17 DROP POLICY IF EXISTS "profiles_insert_own" ON public.profiles;
18 DROP POLICY IF EXISTS "profiles_update_own" ON public.profiles;
19

```

```

20 | -- B-t Row Level Security cho profiles ( Em bèo ch-c ch-n ã b-t )
21 | ALTER TABLE public.profiles ENABLE ROW LEVEL SECURITY;
22 |
23 | -- 1.1. Policy SELECT: Cho phép ng°Yi dùng xem profile cça chính mình
24 | CREATE POLICY "profiles_select_own" ON public.profiles
25 | FOR SELECT USING (auth.uid() = id);
26 |
27 | -- 1.2. Policy SELECT: Cho phép Admin xem t¥t c£ profile (Sĩ dăng JWT me
28 | CREATE POLICY "profiles_select_admin" ON public.profiles
29 | FOR SELECT USING (
30 | COALESCE(auth.jwt() -> 'user_metadata' -> 'role', '') = 'admin'
31 | );
32 |
33 | -- 1.3. Policy SELECT: Cho phép Giáo viên xem profile cça Híc sinh trong
34 | CREATE POLICY "profiles_select_teacher" ON public.profiles
35 | FOR SELECT USING (
36 | COALESCE(auth.jwt() -> 'user_metadata' -> 'role', '') = 'teacher'
37 | AND role = 'student'
38 | );
39 |
40 | -- 1.4. Policy SELECT: Cho phép Phã huynh xem profile cça con ã liên k
41 | CREATE POLICY "profiles_select_parent" ON public.profiles
42 | FOR SELECT USING (
43 | COALESCE(auth.jwt() -> 'user_metadata' -> 'role', '') = 'parent'
44 | AND EXISTS (
45 | SELECT 1 FROM public.parent_student_links psl
46 | WHERE psl.parent_id = auth.uid()
47 | AND psl.student_id = public.profiles.id
48 | AND psl.status = 'active'
49 | )
50 | );
51 |
52 | -- 2. Policy INSERT: Cho phép chèn profile mõi ( Æ tránh lxi 'Database e
53 | -- Cho phép insert n÷u user ID trùng v÷i auth.uid() ho·c chèn tẽ hç thñ
54 | CREATE POLICY "profiles_insert_own" ON public.profiles
55 | FOR INSERT WITH CHECK (
56 | auth.uid() = id
57 | OR auth.uid() IS NULL
58 | OR (auth.jwt() -> 'role') = 'service_role'
59 | );
60 |
61 | -- 3. Policy UPDATE: Cho phép ng°Yi dùng tñ c-p nh-t thông tin cá nhâ
62 | CREATE POLICY "profiles_update_own" ON public.profiles
63 | FOR UPDATE USING (auth.uid() = id)
64 | WITH CHECK (auth.uid() = id);
65 |
66 |
67 | -- %%% B-ÚC 2: C-P NH-T RLS CHO CÁC B£NG LÚP HÍC & BÀI GI£NG Æ £M B£O
68 |
69 | -- Em bèo giáo viên toàn quyián qu£n lý lùp híc (INSERT, SELECT, UPDATE,
70 | DROP POLICY IF EXISTS "Teacher sees own classes" ON public.classes;
71 | DROP POLICY IF EXISTS "teacher_own_classes" ON public.classes;
72 | CREATE POLICY "teacher_own_classes" ON public.classes
73 | FOR ALL USING (teacher_id = auth.uid())
74 | WITH CHECK (teacher_id = auth.uid());
75 |
76 | -- Em bèo giáo viên toàn quyián qu£n lý bài gi£ng (INSERT, SELECT, UPDA
77 | DROP POLICY IF EXISTS "teacher_own_lessons" ON public.lessons;
78 | CREATE POLICY "teacher_own_lessons" ON public.lessons
79 | FOR ALL USING (teacher_id = auth.uid())
80 | WITH CHECK (teacher_id = auth.uid());
81 |
82 |
83 | -- %%% B-ÚC 3: CÀI ¶T RLS VÀ C-P TRÚC QUIZ Æ SÌA LỖI T O Æ THI & CÂU
84 |
85 | -- Em bèo giáo viên toàn quyián qu£n lý quizzes
86 | DROP POLICY IF EXISTS "Teachers can manage their quizzes" ON public.quizzes;
87 | DROP POLICY IF EXISTS "teachers_manage_quizzes" ON public.quizzes;
88 | CREATE POLICY "teachers_manage_quizzes" ON public.quizzes
89 | FOR ALL USING (teacher_id = auth.uid())
90 | WITH CHECK (teacher_id = auth.uid());
91 |
92 | -- Em bèo giáo viên toàn quyián qu£n lý câu hĩi trong Á thi và ngân hàn
93 | ALTER TABLE public.questions ENABLE ROW LEVEL security;
94 | ALTER TABLE public.question_options ENABLE ROW LEVEL security;
95 | ALTER TABLE public.quiz_questions ENABLE ROW LEVEL security;
96 |
97 | DROP POLICY IF EXISTS "teachers_manage_quizzes" ON public.quizzes;
98 |
99 | DROP POLICY IF EXISTS "teachers_manage_questions" ON public.questions;
100 | CREATE POLICY "teachers_manage_questions" ON public.questions
101 | FOR ALL USING (
102 | -- Giáo viên sß hĩu ngân hàng câu hĩi
103 | EXISTS (
104 | SELECT 1 FROM public.question_banks qb
105 | WHERE qb.id = bank_id AND qb.teacher_id = auth.uid()
106 | )
107 | OR bank_id IS NULL -- Cho phép tjo câu hĩi tñ do trong Á thi
108 | );
109 |
110 | DROP POLICY IF EXISTS "teachers_manage_question_options" ON public.question_options;
111 | CREATE POLICY "teachers_manage_question_options" ON public.question_options
112 | FOR ALL USING (
113 | EXISTS (
114 | SELECT 1 FROM public.questions q
115 | LEFT JOIN public.question_banks qb ON qb.id = q.bank_id
116 | WHERE q.id = question_id AND (qb.teacher_id = auth.uid() OR qb.id IS NULL)
117 | )
118 | );
119 |
120 | DROP POLICY IF EXISTS "teachers_manage_quiz_questions" ON public.quiz_questions;
121 | CREATE POLICY "teachers_manage_quiz_questions" ON public.quiz_questions
122 | FOR ALL USING (
123 | EXISTS (
124 | SELECT 1 FROM public.quizzes q
125 | WHERE q.id = quiz_id AND q.teacher_id = auth.uid()
126 | )
127 | );
128 |

```

File: supabase/migrations/20260524123000_fix_classes_qls_and_lesson_sections.sql (112 lines, 3.7 KB)

```

1 | -- =====
2 | -- EMERGENCY FIX: Infinite recursion in classes RLS
3 | -- Chiy file này trong Supabase SQL Editor
4 | -- =====
5 |
6 | -- B-ÚC 1: T-t RLS tĩm thỹi Æ xóa policy không b£ ch-n
7 | ALTER TABLE public.classes DISABLE ROW LEVEL SECURITY;
8 | ALTER TABLE public.class_members DISABLE ROW LEVEL SECURITY;
9 | ALTER TABLE public.submissions DISABLE ROW LEVEL SECURITY;
10 | ALTER TABLE public.assignments DISABLE ROW LEVEL SECURITY;
11 |

```

File: supabase/migrations/20260524130000_all_in_one_fix.sql (212 lines, 8.7 KB)

```

20| AS $$
21| BEGIN
22|     -- Thêm EXCEPTION ã trigger không bao giý crash dù profiles có l×i g
23| BEGIN
24|     INSERT INTO public.profiles (id, email, full_name, role, created_at, updated_at)
25|     VALUES (
26|         NEW.id,
27|         COALESCE(NEW.email, ''),
28|         COALESCE(NEW.raw_user_meta_data->'full_name', split_part(COALESCE(NEW.email, 'user@'), '@', 1)),
29|         COALESCE(NEW.raw_user_meta_data->'role', 'student'),
30|         NOW(),
31|         NOW()
32|     )
33|     -- Nếu profile ã tồn tii thì update , không crash
34| ON CONFLICT (id) DO UPDATE SET
35|     full_name = COALESCE(EXCLUDED.full_name, public.profiles.full_name),
36|     role = COALESCE(EXCLUDED.role, public.profiles.role),
37|     updated_at = NOW();
38| EXCEPTION WHEN OTHERS THEN
39|     -- Ghi log nh°ng KHÔNG fail trigger !' không bao giý block ng ký
40| RAISE WARNING 'handle_new_user failed for user %: ', NEW.id, SQLERRM;
41| END;
42|
43| RETURN NEW;
44| END;
45| $$;
46|
47| -- Recreate trigger
48| CREATE TRIGGER on_auth_user_created
49| AFTER INSERT ON auth.users
50| FOR EACH ROW EXECUTE FUNCTION public.handle_new_user();
51|
52| -- P H Ì N 2 : F I X p r o f i l e s R L S - m ß i n s e r t c h o t r i g g e r
53| -- P H Ì N 2 : F I X p r o f i l e s R L S - m ß i n s e r t c h o t r i g g e r
54| -- P H Ì N 2 : F I X p r o f i l e s R L S - m ß i n s e r t c h o t r i g g e r
55|
56| -- X ó a T O À N B Ø p o l i c i e s p r o f i l e s
57| DO $$
58| DECLARE pol RECORD;
59| BEGIN
60|     FOR pol IN
61|         SELECT policyname FROM pg_policies
62|         WHERE schemaname = 'public' AND tablename = 'profiles'
63|     LOOP
64|         EXECUTE format('DROP POLICY IF EXISTS %I ON public.profiles', pol.policyname);
65|     END LOOP;
66| END $$;
67|
68| ALTER TABLE public.profiles ENABLE ROW LEVEL SECURITY;
69|
70| -- I N S E R T : C h o p h é p t á y t c é ( t r i g g e r + c l i e n t Á u c s ñ )
71| CREATE POLICY "profiles_insert_open" ON public.profiles
72| FOR INSERT WITH CHECK (true);
73|
74| -- S E L E C T : C h é x e m c ç a c h í n h m í n h
75| CREATE POLICY "profiles_select_own" ON public.profiles
76| FOR SELECT USING (auth.uid() = id);
77|
78| -- SELECT: Admin dùng JWT metadata (tránh recursion)
79| CREATE POLICY "profiles_select_admin" ON public.profiles
80| FOR SELECT USING (
81|     COALESCE(auth.jwt() -> 'user_metadata' -> 'role', '') = 'admin'
82| );
83|
84| -- S E L E C T : T e a c h e r x e m h í c s í n h
85| CREATE POLICY "profiles_select_teacher" ON public.profiles
86| FOR SELECT USING (
87|     COALESCE(auth.jwt() -> 'user_metadata' -> 'role', '') = 'teacher'
88|     AND role = 'student'
89| );
90|
91| -- S E L E C T : P h â h u y n h x e m c o n
92| CREATE POLICY "profiles_select_parent" ON public.profiles
93| FOR SELECT USING (
94|     COALESCE(auth.jwt() -> 'user_metadata' -> 'role', '') = 'parent'
95|     AND EXISTS (
96|         SELECT 1 FROM public.parent_student_links psl
97|         WHERE psl.parent_id = auth.uid()
98|         AND psl.student_id = public.profiles.id
99|     )
100| );
101|
102| -- U P D A T E : C h é t ã c - p n h - t c ç a m í n h
103| CREATE POLICY "profiles_update_own" ON public.profiles
104| FOR UPDATE USING (auth.uid() = id)
105| WITH CHECK (auth.uid() = id);
106|
107| -- P H Ì N 3 : F I X c l a s s e s R L S - i n f i n i t e r e c u r s i o n
108| -- P H Ì N 3 : F I X c l a s s e s R L S - i n f i n i t e r e c u r s i o n
109| -- P H Ì N 3 : F I X c l a s s e s R L S - i n f i n i t e r e c u r s i o n
110|
111| ALTER TABLE public.classes DISABLE ROW LEVEL SECURITY;
112| ALTER TABLE public.class_members DISABLE ROW LEVEL SECURITY;
113| ALTER TABLE public.submissions DISABLE ROW LEVEL SECURITY;
114| ALTER TABLE public.assignments DISABLE ROW LEVEL SECURITY;
115|
116| -- X ó a T O À N B Ø p o l i c i e s
117| DO $$
118| DECLARE pol RECORD;
119| BEGIN
120|     FOR pol IN
121|         SELECT policyname, tablename FROM pg_policies
122|         WHERE schemaname = 'public'
123|         AND tablename IN ('classes', 'class_members', 'submissions', 'assignments')
124|     LOOP
125|         EXECUTE format('DROP POLICY IF EXISTS %I ON public.%I', pol.policyname, pol.tablename);
126|     END LOOP;
127| END $$;
128|
129| -- S E C U R I T Y D E F I N E R f u n c t i o n s ( b y p a s s R L S - p h á v ò n g l . p )
130| CREATE OR REPLACE FUNCTION public.fn_teacher_class_ids()
131| RETURNS UUID[] LANGUAGE sql SECURITY DEFINER STABLE
132| SET search_path = public AS $$
133| SELECT ARRAY(SELECT id FROM public.classes WHERE teacher_id = auth.uid());
134| $$;
135|
136| CREATE OR REPLACE FUNCTION public.fn_student_class_ids()
137| RETURNS UUID[] LANGUAGE sql SECURITY DEFINER STABLE
138| SET search_path = public AS $$
139| SELECT ARRAY(SELECT class_id FROM public.class_members WHERE student_id = auth.uid());
140| $$;
141|
142| -- B - t l i i R L S
143| ALTER TABLE public.classes ENABLE ROW LEVEL SECURITY;

```

[illegible]

```

1| -- =====
2| -- MIGRATION: ADD ACHIEVEMENT TABLES FOR STUDENT DASHBOARD
3| -- Target Path: supabase/migrations/20260524165000_add_student_dashboard_achievements.sql
4| -- =====
5|
6| -- 1. Bëng achievement definitions
7| CREATE TABLE IF NOT EXISTS public.achievement_definitions (
8|   id TEXT PRIMARY KEY,
9|   title TEXT NOT NULL,
10|  description TEXT,
11|  icon TEXT,
12|  points INT DEFAULT 0,
13|  rarity TEXT CHECK (rarity IN ('common', 'uncommon', 'rare', 'legendary')),
14|  created_at TIMESTAMPTZ DEFAULT NOW()
15| );
16|
17| -- 2. Bëng student achievements
18| CREATE TABLE IF NOT EXISTS public.student_achievements (
19|   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
20|   student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
21|   achievement_id TEXT REFERENCES public.achievement_definitions(id) ON DELETE CASCADE,
22|   earned_at TIMESTAMPTZ DEFAULT NOW(),
23|   UNIQUE(student_id, achievement_id)
24| );
25|
26| -- 3. Seed di liçu achievement definitions
27| INSERT INTO public.achievement_definitions (id, title, description, icon, points, rarity)
28| VALUES
29|   ('first_lesson', 'Khêi sù hành trính', 'Hòan thành bài giêng su tiêr', 10, 'common'),
30|   ('quiz_acce', 'Xúýt s c', 'it tụy Çt N i (100%) b ài trong m Ut b ài gi i', 20, 'common'),
31|   ('week_streak', 'Kiên trì 7 ngày', 'H i c li ên t i c p 7 ngày kh ông nh h E', 30, 'common'),
32|   ('pc_master', 'PC Master', 'H o à n h t h à n h t t c c b à i trong l ù trính', 50, 'uncommon'),
33|   ('lab_expert', 'Chuyên gia Lab', 'D à n h h i n 10 g i Y trong B u i l d e r L a b', 100, 'rare'),
34|   ('helpful_member', 'Nhiệt tình', 'ã c u p v o t e 10 l s n trong di An à n', 150, 'legendary');
35| ON CONFLICT (id) DO UPDATE
36| SET
37|   title = EXCLUDED.title,
38|   description = EXCLUDED.description,
39|   icon = EXCLUDED.icon,
40|   points = EXCLUDED.points,
41|   rarity = EXCLUDED.rarity;
42|
43| -- 4. Enable RLS for these tables
44| ALTER TABLE public.achievement_definitions ENABLE ROW LEVEL SECURITY;
45| ALTER TABLE public.student_achievements ENABLE ROW LEVEL SECURITY;
46|
47| -- 5. Policies
48| DROP POLICY IF EXISTS "view_definitions" ON public.achievement_definitions;
49| CREATE POLICY "view_definitions" ON public.achievement_definitions FOR SELECT USING (true);
50|
51| DROP POLICY IF EXISTS "view student achievements" ON public.student_achievements;

```



```

119 BEGIN
120 FOREACH tbl IN ARRAY ARRAY['public.lesson_progress','public.quiz_attempts','public.lessons','public.lesson_s...
121 LOOP
122 BEGIN
123 EXECUTE format('ALTER PUBLICATION supabase_realtime ADD TABLE %s', tbl);
124 EXCEPTION WHEN OTHERS THEN
125 RAISE NOTICE 'Table % already in publication, skipping', tbl;
126 END;
127 END LOOP;
128 END;
129 $$;
130

```

File: supabase/migrations/20260526100000_fix_join_class_lookup.sql (15 lines, 0.4 KB)

```

=====
1 | -- Fix: Cho phép híc sinh tim lúp theo mã code ã join
2 | -- T i o S E C U R I T Y D E F I N E R f u n c t i o n ( b y p a s s R L S ) ã l o o k u p c l a s s b y c o d e a n
3
4 CREATE OR REPLACE FUNCTION public.fn_lookup_class_by_code(p_code TEXT)
5 RETURNS TABLE(id UUID, name TEXT)
6 LANGUAGE sql
7 SECURITY DEFINER
8 STABLE
9 SET search_path = public
10 AS $$
11 SELECT id, name FROM public.classes WHERE code = p_code;
12 $$;
13
14 NOTIFY pgrst, 'reload schema';
15

```

File: supabase/migrations/20260527034907_enable-realtime-missing-tables.sql (22 lines, 0.6 KB)

```

=====
1 | -- Enable Realtime for admin dashboard tables
2 | DO $$
3 | BEGIN
4 | IF NOT EXISTS (
5 | SELECT 1 FROM pg_publication_tables
6 | WHERE pubname = 'supabase_realtime' AND schemaname = 'public' AND tablename = 'profiles'
7 | ) THEN
8 | ALTER PUBLICATION supabase_realtime ADD TABLE profiles;
9 | END IF;
10 | END $$;
11 | DO $$
12 | BEGIN
13 | IF EXISTS (SELECT 1 FROM pg_class WHERE relname = 'lessons' AND relkind = 'r') THEN
14 | IF NOT EXISTS (
15 | SELECT 1 FROM pg_publication_tables
16 | WHERE pubname = 'supabase_realtime' AND schemaname = 'public' AND tablename = 'lessons'
17 | ) THEN
18 | ALTER PUBLICATION supabase_realtime ADD TABLE lessons;
19 | END IF;
20 | END IF;
21 | END $$;
22 |

```

File: supabase/migrations/20260527035134_enable-realtime-final.sql (26 lines, 0.9 KB)

```

=====
1 | -- Enable Realtime for admin dashboard tables
2 | DO $$
3 | BEGIN
4 | IF EXISTS (SELECT 1 FROM pg_tables WHERE schemaname = 'public' AND tablename = 'profiles') THEN
5 | BEGIN
6 | EXECUTE 'ALTER PUBLICATION supabase_realtime ADD TABLE public.profiles';
7 | EXCEPTION WHEN OTHERS THEN
8 | RAISE NOTICE 'profiles: %', SQLERRM;
9 | END;
10 | END IF;
11 | IF EXISTS (SELECT 1 FROM pg_tables WHERE schemaname = 'public' AND tablename = 'lessons') THEN
12 | BEGIN
13 | EXECUTE 'ALTER PUBLICATION supabase_realtime ADD TABLE public.lessons';
14 | EXCEPTION WHEN OTHERS THEN
15 | RAISE NOTICE 'lessons: %', SQLERRM;
16 | END;
17 | END IF;
18 | IF EXISTS (SELECT 1 FROM pg_tables WHERE schemaname = 'public' AND tablename = 'exams') THEN
19 | BEGIN
20 | EXECUTE 'ALTER PUBLICATION supabase_realtime ADD TABLE public.exams';
21 | EXCEPTION WHEN OTHERS THEN
22 | RAISE NOTICE 'exams: %', SQLERRM;
23 | END;
24 | END IF;
25 | END $$;
26 |

```

File: supabase/migrations/20260527040000_add_require_camera.sql (2 lines, 0.1 KB)

```

=====
1 | ALTER TABLE public.quizzes ADD COLUMN IF NOT EXISTS require_camera BOOLEAN DEFAULT false;
2 |

```

File: supabase/migrations/20260527050000_add_certificates_and_discussion.sql (95 lines, 3.1 KB)

```

=====
1 | -- B ẽ n g d i s c u s s i o n _ m e s s a g e s c h o r e a l t i m e c h a t
2 | CREATE TABLE IF NOT EXISTS public.discussion_messages (
3 | id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
4 | channel TEXT NOT NULL DEFAULT 'global' CHECK (channel IN ('global', 'class')),
5 | class_id UUID REFERENCES public.classes(id) ON DELETE CASCADE,
6 | sender_id UUID REFERENCES public.profiles(id) ON DELETE SET NULL,
7 | sender_name TEXT,
8 | content TEXT NOT NULL,
9 | created_at TIMESTAMPTZ DEFAULT NOW()
10 | );
11 |
12 | ALTER TABLE public.discussion_messages ENABLE ROW LEVEL SECURITY;
13 |
14 | DO $$ BEGIN DROP POLICY IF EXISTS "view_messages" ON public.discussion_messages; EXCEPTION WHEN OTHERS THEN RA...
15 | CREATE POLICY "view_messages" ON public.discussion_messages
16 | FOR SELECT USING (true);
17 | DO $$ BEGIN DROP POLICY IF EXISTS "insert_messages" ON public.discussion_messages; EXCEPTION WHEN OTHERS THEN ...
18 | CREATE POLICY "insert_messages" ON public.discussion_messages
19 | FOR INSERT WITH CHECK (sender_id = auth.uid());
20 |
21 | -- B ẽ n g c e r t i f i c a t e s
22 | CREATE TABLE IF NOT EXISTS public.certificates (
23 | id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
24 | student_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
25 | path_id UUID REFERENCES public.learning_paths(id) ON DELETE CASCADE,

```

```

26 | certificate_number TEXT UNIQUE NOT NULL,
27 | issued_at TIMESTAMPTZ DEFAULT NOW(),
28 | pdf_url TEXT,
29 | verify_url TEXT,
30 | student_name TEXT,
31 | course_title TEXT,
32 | completion_date DATE DEFAULT CURRENT_DATE,
33 | final_score NUMERIC(5,2),
34 | is_revoked BOOLEAN DEFAULT FALSE
35 | );
36 |
37 | -- Function to generate safe, random certificate number
38 | CREATE OR REPLACE FUNCTION generate_cert_number()
39 | RETURNS TEXT AS $$
40 | DECLARE
41 |   chars TEXT := 'ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789';
42 |   result TEXT := 'PCM-2026-';
43 |   i INT;
44 | BEGIN
45 |   FOR i IN 1..5 LOOP
46 |     result := result || substr(chars, floor(random() * length(chars) + 1)::INT, 1);
47 |   END LOOP;
48 |   RETURN result;
49 | END;
50 | $$ LANGUAGE plpgsql;
51 |
52 | -- Enable RLS on certificates
53 | ALTER TABLE public.certificates ENABLE ROW LEVEL SECURITY;
54 |
55 | DO $$
56 | BEGIN
57 |   DROP POLICY IF EXISTS "view_all_certificates" ON public.certificates;
58 |   EXCEPTION WHEN OTHERS THEN
59 |     RAISE NOTICE 'view_all_certificates drop: %', SQLERRM;
60 | END;
61 | $$;
62 |
63 | CREATE POLICY "view_all_certificates" ON public.certificates
64 |   FOR SELECT USING (true);
65 |
66 | DO $$
67 | BEGIN
68 |   DROP POLICY IF EXISTS "manage_certificates" ON public.certificates;
69 |   EXCEPTION WHEN OTHERS THEN
70 |     RAISE NOTICE 'manage_certificates drop: %', SQLERRM;
71 | END;
72 | $$;
73 |
74 | CREATE POLICY "manage_certificates" ON public.certificates
75 |   FOR ALL USING (
76 |     EXISTS (SELECT 1 FROM public.profiles WHERE id = auth.uid() AND role = 'teacher')
77 |   );
78 |
79 | -- B - t r e a l t i m e   c h o t t c e b e n g m ũ i
80 | DO $$
81 | BEGIN
82 |   ALTER PUBLICATION supabase_realtime ADD TABLE discussion_messages;
83 |   EXCEPTION WHEN duplicate_object THEN
84 |     RAISE NOTICE 'discussion_messages already in publication';
85 | END;
86 | $$;
87 |
88 | DO $$
89 | BEGIN
90 |   ALTER PUBLICATION supabase_realtime ADD TABLE certificates;
91 |   EXCEPTION WHEN duplicate_object THEN
92 |     RAISE NOTICE 'certificates already in publication';
93 | END;
94 | $$;
95 |

```

File: supabase/migrations/20260528000000_add_lesson_class_assignment.sql (26 lines, 1.1 KB)

```

1 | -- Add lesson-class assignment junction table
2 | CREATE TABLE IF NOT EXISTS public.lesson_class_assignments (
3 |   id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
4 |   lesson_id UUID REFERENCES public.lessons(id) ON DELETE CASCADE NOT NULL,
5 |   class_id UUID REFERENCES public.classes(id) ON DELETE CASCADE NOT NULL,
6 |   assigned_at TIMESTAMPTZ DEFAULT NOW(),
7 |   UNIQUE(lesson_id, class_id)
8 | );
9 |
10 | ALTER TABLE public.lesson_class_assignments ENABLE ROW LEVEL SECURITY;
11 |
12 | -- Teachers can manage assignments for their lessons
13 | CREATE POLICY "Teacher manages lesson class assignments" ON public.lesson_class_assignments
14 |   FOR ALL USING (
15 |     EXISTS (SELECT 1 FROM public.lessons WHERE id = lesson_id AND teacher_id = auth.uid())
16 |   );
17 |
18 | -- Students see assignments for classes they belong to
19 | CREATE POLICY "Student sees own class lesson assignments" ON public.lesson_class_assignments
20 |   FOR SELECT USING (
21 |     EXISTS (SELECT 1 FROM public.class_members WHERE class_id = lesson_class_assignments.class_id AND stud...
22 |   );
23 |
24 | -- Add to realtime publication
25 | ALTER PUBLICATION supabase_realtime ADD TABLE public.lesson_class_assignments;
26 |

```

File: supabase/migrations/20260530000000_add_lessons_category_and_source.sql (5 lines, 0.3 KB)

```

1 | -- Add missing columns to lessons table for save functionality
2 | ALTER TABLE public.lessons ADD COLUMN IF NOT EXISTS category TEXT DEFAULT 'extended';
3 | ALTER TABLE public.lessons ADD COLUMN IF NOT EXISTS source_name TEXT;
4 | ALTER TABLE public.lessons ADD COLUMN IF NOT EXISTS source_url TEXT;
5 |

```

File: supabase/migrations/20260622_n8n_integration.sql (451 lines, 13.4 KB)

```

1 | -- =====
2 | -- PC Master Builder - n8n Integration
3 | -- Database Functions & Triggers
4 | -- =====
5 |
6 | -- -----
7 | -- Fix: notifications table type constraint
8 | -- The original constraint blocked all inserts.
9 | -- Drop and recreate with proper allowed values.
10 | -- -----

```



```

11 ALTER TABLE IF EXISTS public.notifications
12 DROP CONSTRAINT IF EXISTS notifications_type_check;
13
14 ALTER TABLE IF EXISTS public.notifications
15 ADD COLUMN IF NOT EXISTS body TEXT,
16 ADD COLUMN IF NOT EXISTS action_url TEXT,
17 ADD COLUMN IF NOT EXISTS data JSONB,
18 ALTER COLUMN type SET DEFAULT 'info';
19
20 ALTER TABLE IF EXISTS public.notifications
21 ADD CONSTRAINT notifications_type_check
22 CHECK (type IN ('info', 'success', 'warning', 'error', 'assignment', 'quiz', 'achievement_earned', 'lesson_c...
23
24 -----
25 -- Add Zalo fields to profiles for teacher Zalo notifications
26 -----
27 ALTER TABLE IF EXISTS public.profiles
28 ADD COLUMN IF NOT EXISTS zalo_phone TEXT,
29 ADD COLUMN IF NOT EXISTS zalo_notifications BOOLEAN DEFAULT false,
30 ADD COLUMN IF NOT EXISTS zalo_verified BOOLEAN DEFAULT false;
31
32 -----
33 -- Function: check_completed_paths()
34 -- n8n calls this to find students who completed
35 -- all lessons in a path but have no certificate
36 -----
37 CREATE OR REPLACE FUNCTION public.check_completed_paths()
38 RETURNS TABLE (
39     student_id UUID,
40     path_id UUID,
41     full_name TEXT,
42     path_title TEXT,
43     lesson_count INT,
44     completed_count INT,
45     avg_score NUMERIC
46 ) LANGUAGE plpgsql SECURITY DEFINER AS $$
47 BEGIN
48     RETURN QUERY
49     WITH path_stats AS (
50         SELECT
51             lp.student_id,
52             pi.path_id,
53             COUNT(DISTINCT pi.id)::INT AS total_lessons,
54             COUNT(DISTINCT lp.id) FILTER (WHERE lp.status = 'completed')::INT AS completed_lessons
55         FROM path_items pi
56         CROSS JOIN LATERAL (
57             SELECT lp_in.student_id, lp_in.id, lp_in.status
58             FROM lesson_progress lp_in
59             WHERE lp_in.lesson_id = pi.item_id
60             AND pi.item_type = 'lesson'
61         ) lp
62         GROUP BY lp.student_id, pi.path_id
63         HAVING COUNT(DISTINCT pi.id) = COUNT(DISTINCT lp.id) FILTER (WHERE lp.status = 'completed')
64     )
65     SELECT
66         ps.student_id,
67         ps.path_id,
68         p.full_name,
69         lp.title AS path_title,
70         ps.total_lessons,
71         ps.completed_lessons,
72         COALESCE(
73             (SELECT AVG(qa.score)::NUMERIC(5,2)
74              FROM quiz_attempts qa
75              JOIN quizzes q ON qa.quiz_id = q.id
76              WHERE qa.student_id = ps.student_id
77              AND qa.status = 'submitted'
78             ), 0
79         ) AS avg_score
80     FROM path_stats ps
81     JOIN profiles p ON p.id = ps.student_id
82     JOIN learning_paths lp ON lp.id = ps.path_id
83     WHERE NOT EXISTS (
84         SELECT 1 FROM certificates c
85         WHERE c.student_id = ps.student_id
86         AND c.path_id = ps.path_id
87         AND c.is_revoked = false
88     );
89 END;
90 $$;
91
92 -----
93 -- Function: get_students_low_performance()
94 -- n8n calls this for student alerts
95 -----
96 CREATE OR REPLACE FUNCTION public.get_students_low_performance()
97 RETURNS TABLE (
98     student_id UUID,
99     full_name TEXT,
100     email TEXT,
101     class_id UUID,
102     avg_score NUMERIC,
103     recent_quiz_count INT,
104     weakest_topic TEXT,
105     last_active_date DATE
106 ) LANGUAGE plpgsql SECURITY DEFINER AS $$
107 BEGIN
108     RETURN QUERY
109     WITH recent_scores AS (
110         SELECT
111             qa.student_id,
112             AVG(qa.score)::NUMERIC(5,2) AS avg_score,
113             COUNT(qa.id)::INT AS quiz_count,
114             MAX(qa.created_at)::DATE AS last_quiz_date
115         FROM quiz_attempts qa
116         WHERE qa.status = 'submitted'
117         AND qa.created_at >= NOW() - INTERVAL '7 days'
118         GROUP BY qa.student_id
119     ),
120     weakest AS (
121         SELECT DISTINCT ON (qa.student_id)
122             qa.student_id,
123             l.title AS weakest_lesson
124         FROM quiz_attempts qa
125         JOIN quizzes q ON qa.quiz_id = q.id
126         JOIN lessons l ON l.id = q.lesson_id
127         WHERE qa.status = 'submitted'
128         AND qa.score < 40
129         AND qa.created_at >= NOW() - INTERVAL '14 days'
130         ORDER BY qa.student_id, qa.score ASC
131     )
132     SELECT
133         p.id,
134         p.full_name,

```

```

135     p.email,
136     p.class_id,
137     COALESCE(rs.avg_score, 0),
138     COALESCE(rs.quiz_count, 0),
139     COALESCE(w.weakest_lesson, 'Chua co du lieu'),
140     p.last_quest_date
141 FROM profiles p
142 LEFT JOIN recent_scores rs ON rs.student_id = p.id
143 LEFT JOIN weakest w ON w.student_id = p.id
144 WHERE p.role = 'student'
145 AND (
146     rs.avg_score IS NOT NULL AND rs.avg_score < 50
147 )
148 ORDER BY rs.avg_score ASC NULLS LAST
149 LIMIT 100;
150 END;
151 $$;
152
153 -- -----
154 -- Function: get_weekly_stats_summary()
155 -- n8n calls this for weekly reports
156 -- -----
157 CREATE OR REPLACE FUNCTION public.get_weekly_stats_summary()
158 RETURNS TABLE (
159     metric TEXT,
160     value BIGINT
161 ) LANGUAGE plpgsql SECURITY DEFINER AS $$
162 DECLARE
163     week_start TIMESTAMPTZ := DATE_TRUNC('week', NOW()) - INTERVAL '1 week';
164     week_end TIMESTAMPTZ := DATE_TRUNC('week', NOW());
165 BEGIN
166     RETURN QUERY
167     SELECT 'active_students'::TEXT, COUNT(DISTINCT user_id)::BIGINT
168     FROM activity_logs
169     WHERE created_at >= week_start AND created_at < week_end
170
171     UNION ALL
172     SELECT 'new_lessons_completed'::TEXT, COUNT(*)::BIGINT
173     FROM lesson_progress
174     WHERE completed_at >= week_start AND completed_at < week_end
175     AND status = 'completed'
176
177     UNION ALL
178     SELECT 'quizzes_submitted'::TEXT, COUNT(*)::BIGINT
179     FROM quiz_attempts
180     WHERE submitted_at >= week_start AND submitted_at < week_end
181     AND status = 'submitted'
182
183     UNION ALL
184     SELECT 'certificates_issued'::TEXT, COUNT(*)::BIGINT
185     FROM certificates
186     WHERE issued_at >= week_start AND issued_at < week_end
187
188     UNION ALL
189     SELECT 'new_registrations'::TEXT, COUNT(*)::BIGINT
190     FROM profiles
191     WHERE created_at >= week_start AND created_at < week_end
192
193     UNION ALL
194     SELECT 'total_xp_earned'::TEXT, COALESCE(SUM(amount), 0)::BIGINT
195     FROM xp_transactions
196     WHERE created_at >= week_start AND created_at < week_end;
197 END;
198 $$;
199
200 -- -----
201 -- Function: get_student_weekly_summary()
202 -- For parent weekly reports - passes student_id
203 -- -----
204 CREATE OR REPLACE FUNCTION public.get_student_weekly_summary(
205     p_student_id UUID
206 ) RETURNS TABLE (
207     completed_lessons INT,
208     quiz_count INT,
209     avg_score NUMERIC,
210     total_xp_earned INT,
211     streak_days INT
212 ) LANGUAGE plpgsql SECURITY DEFINER AS $$
213 DECLARE
214     week_start TIMESTAMPTZ := DATE_TRUNC('week', NOW()) - INTERVAL '1 week';
215     week_end TIMESTAMPTZ := DATE_TRUNC('week', NOW());
216 BEGIN
217     RETURN QUERY
218     SELECT
219         (SELECT COUNT(*)::INT FROM lesson_progress
220          WHERE student_id = p_student_id
221          AND status = 'completed'
222          AND completed_at >= week_start AND completed_at < week_end) AS completed_lessons,
223
224         (SELECT COUNT(*)::INT FROM quiz_attempts
225          WHERE student_id = p_student_id
226          AND status = 'submitted'
227          AND submitted_at >= week_start AND submitted_at < week_end) AS quiz_count,
228
229         COALESCE(
230             (SELECT AVG(score)::NUMERIC(5,2) FROM quiz_attempts
231              WHERE student_id = p_student_id
232              AND status = 'submitted'
233              AND submitted_at >= week_start AND submitted_at < week_end)
234             , 0) AS avg_score,
235
236         COALESCE(
237             (SELECT SUM(amount)::INT FROM xp_transactions
238              WHERE user_id = p_student_id
239              AND created_at >= week_start AND created_at < week_end)
240             , 0) AS total_xp_earned,
241
242         COALESCE(
243             (SELECT quest_streak FROM profiles WHERE id = p_student_id)
244             , 0) AS streak_days;
245 END;
246 $$;
247
248 -- -----
249 -- Function: add_xp()
250 -- Safely add XP and update level/leaderboard
251 -- -----
252 CREATE OR REPLACE FUNCTION public.add_xp(
253     p_user_id UUID,
254     p_amount INT
255 ) RETURNS BOOLEAN LANGUAGE plpgsql SECURITY DEFINER AS $$
256 DECLARE
257     current_xp INT;
258     current_level INT;

```

```

259     new_level INT;
260 BEGIN
261     SELECT xp, level INTO current_xp, current_level
262     FROM profiles WHERE id = p_user_id;
263
264     current_xp := COALESCE(current_xp, 0) + p_amount;
265
266     SELECT MAX(l.level) INTO new_level
267     FROM level_definitions l
268     WHERE l.xp_required <= current_xp;
269
270     new_level := COALESCE(new_level, 1);
271
272     UPDATE profiles SET
273     xp = current_xp,
274     level = new_level,
275     updated_at = NOW()
276     WHERE id = p_user_id;
277
278     INSERT INTO leaderboard_cache (user_id, total_xp, level, updated_at)
279     VALUES (p_user_id, current_xp, new_level, NOW())
280     ON CONFLICT (user_id) DO UPDATE SET
281     total_xp = current_xp,
282     level = new_level,
283     updated_at = NOW();
284
285     UPDATE learning_stats SET
286     total_xp = current_xp,
287     updated_at = NOW()
288     WHERE user_id = p_user_id;
289
290     RETURN TRUE;
291 END;
292 $$;
293
294 -- -----
295 -- Function: get_n8n_pending_events()
296 -- Returns pending events that n8n should process
297 -- -----
298 CREATE OR REPLACE FUNCTION public.get_n8n_pending_events()
299 RETURNS TABLE (
300     event_id UUID,
301     event_type TEXT,
302     payload JSONB,
303     created_at TIMESTAMPTZ
304 ) LANGUAGE plpgsql SECURITY DEFINER AS $$
305 BEGIN
306     RETURN QUERY
307     SELECT
308     e.id,
309     e.event_type,
310     e.payload,
311     e.created_at
312     FROM n8n_events e
313     WHERE e.status = 'pending'
314     ORDER BY e.created_at ASC
315     LIMIT 50;
316 END;
317 $$;
318
319 -- -----
320 -- Table: n8n_events
321 -- Queue for events that need processing
322 -- -----
323 CREATE TABLE IF NOT EXISTS public.n8n_events (
324     id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
325     event_type TEXT NOT NULL,
326     payload JSONB NOT NULL DEFAULT '{}',
327     status TEXT DEFAULT 'pending' CHECK (status IN ('pending', 'processing', 'completed', 'failed')),
328     created_at TIMESTAMPTZ DEFAULT NOW(),
329     processed_at TIMESTAMPTZ
330 );
331
332 -- Enable RLS but allow service role only
333 ALTER TABLE public.n8n_events ENABLE ROW LEVEL SECURITY;
334 DO $$
335 BEGIN
336     IF NOT EXISTS (
337     SELECT 1 FROM pg_policies WHERE tablename = 'n8n_events' AND policyname = 'n8n service role only'
338     ) THEN
339     CREATE POLICY "n8n service role only" ON public.n8n_events
340     FOR ALL USING (true) WITH CHECK (true);
341     END IF;
342 END $$;
343
344 -- Index for n8n polling
345 CREATE INDEX IF NOT EXISTS idx_n8n_events_status ON public.n8n_events(status, created_at);
346
347 -- -----
348 -- Function: queue_n8n_event()
349 -- Called by triggers to queue events for n8n
350 -- -----
351 CREATE OR REPLACE FUNCTION public.queue_n8n_event()
352 RETURNS TRIGGER LANGUAGE plpgsql SECURITY DEFINER AS $$
353 DECLARE
354     event_type TEXT;
355     payload JSONB;
356 BEGIN
357     CASE TG_TABLE_NAME
358     WHEN 'lesson_progress' THEN
359     IF NEW.status = 'completed' THEN
360     event_type := 'lesson_completed';
361     payload := jsonb_build_object(
362     'student_id', NEW.student_id,
363     'lesson_id', NEW.lesson_id,
364     'completed_at', NEW.completed_at
365     );
366     ELSE
367     RETURN NULL;
368     END IF;
369
370     WHEN 'quiz_attempts' THEN
371     IF NEW.status = 'submitted' THEN
372     event_type := 'quiz_submitted';
373     payload := jsonb_build_object(
374     'student_id', NEW.student_id,
375     'quiz_id', NEW.quiz_id,
376     'attempt_id', NEW.id,
377     'score', NEW.score
378     );
379     ELSE
380     RETURN NULL;
381     END IF;
382

```

```

383| WHEN 'certificates' THEN
384|     event_type := 'certificate_issued';
385|     payload := jsonb_build_object(
386|         'student_id', NEW.student_id,
387|         'certificate_id', NEW.id,
388|         'path_id', NEW.path_id
389|     );
390|
391| ELSE
392|     RETURN NULL;
393| END CASE;
394|
395| INSERT INTO public.n8n_events (event_type, payload)
396| VALUES (event_type, payload);
397|
398| RETURN NULL;
399| END;
400| $$;
401|
402| -- -----
403| -- Triggers: auto-queue events for n8n
404| -- -----
405| DROP TRIGGER IF EXISTS trg_n8n_lesson_complete ON public.lesson_progress;
406| CREATE TRIGGER trg_n8n_lesson_complete
407| AFTER INSERT OR UPDATE OF status ON public.lesson_progress
408| FOR EACH ROW
409| EXECUTE FUNCTION public.queue_n8n_event();
410|
411| DROP TRIGGER IF EXISTS trg_n8n_quiz_submit ON public.quiz_attempts;
412| CREATE TRIGGER trg_n8n_quiz_submit
413| AFTER INSERT OR UPDATE OF status ON public.quiz_attempts
414| FOR EACH ROW
415| EXECUTE FUNCTION public.queue_n8n_event();
416|
417| DROP TRIGGER IF EXISTS trg_n8n_certificate ON public.certificates;
418| CREATE TRIGGER trg_n8n_certificate
419| AFTER INSERT ON public.certificates
420| FOR EACH ROW
421| EXECUTE FUNCTION public.queue_n8n_event();
422|
423| -- -----
424| -- Function: mark_event_processed()
425| -- n8n calls this after processing an event
426| -- -----
427| CREATE OR REPLACE FUNCTION public.mark_event_processed(
428|     p_event_id UUID,
429|     p_status TEXT DEFAULT 'completed'
430| ) RETURNS BOOLEAN LANGUAGE plpgsql SECURITY DEFINER AS $$
431| BEGIN
432|     UPDATE n8n_events
433|     SET status = p_status, processed_at = NOW()
434|     WHERE id = p_event_id;
435|     RETURN FOUND;
436| END;
437| $$;
438|
439| -- -----
440| -- Grant permissions for n8n
441| -- -----
442| GRANT USAGE ON SCHEMA public TO anon, authenticated, service_role;
443| GRANT EXECUTE ON FUNCTION public.check_completed_paths() TO service_role;
444| GRANT EXECUTE ON FUNCTION public.get_students_low_performance() TO service_role;
445| GRANT EXECUTE ON FUNCTION public.get_weekly_stats_summary() TO service_role;
446| GRANT EXECUTE ON FUNCTION public.get_student_weekly_summary(UUID) TO service_role;
447| GRANT EXECUTE ON FUNCTION public.add_xp(UUID, INT) TO service_role;
448| GRANT EXECUTE ON FUNCTION public.get_n8n_pending_events() TO service_role;
449| GRANT EXECUTE ON FUNCTION public.mark_event_processed(UUID, TEXT) TO service_role;
450| GRANT SELECT, INSERT, UPDATE ON public.n8n_events TO service_role;
451|

```

File: supabase/migrations/20260623_cleanup_new_exam_system.sql (74 lines, 2.8 KB)

```

=====
1 | -- %P%P%P CLEANUP: Xóa Zalo columns %P%P%P
2 | ALTER TABLE public.profiles DROP COLUMN IF EXISTS zalo_phone;
3 | ALTER TABLE public.profiles DROP COLUMN IF EXISTS zalo_notifications;
4 | ALTER TABLE public.profiles DROP COLUMN IF EXISTS zalo_verified;
5 |
6 | -- %P%P%P ADD: exam_published_notification_type %P%P%P
7 | ALTER TABLE public.notifications DROP CONSTRAINT IF EXISTS notifications_type_check;
8 | ALTER TABLE public.notifications ADD CONSTRAINT notifications_type_check CHECK (
9 |     type IN (
10 |         'info', 'success', 'warning', 'error',
11 |         'assignment', 'quiz', 'quiz_published',
12 |         'achievement_earned', 'lesson_completed',
13 |         'child_low_score', 'quiz_graded',
14 |         'exam_published', 'class_announcement'
15 |     )
16 | );
17 |
18 | -- %P%P%P ADD: streak_tracking_columns %P%P%P
19 | ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS streak_daily INT DEFAULT 0;
20 | ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS streak_last_date DATE;
21 | ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS total_exams_done INT DEFAULT 0;
22 | ALTER TABLE public.profiles ADD COLUMN IF NOT EXISTS total_exams_passed INT DEFAULT 0;
23 |
24 | -- %P%P%P FUNCTION: Update streak on quiz completion %P%P%P
25 | CREATE OR REPLACE FUNCTION public.update_streak_on_attempt()
26 | RETURNS TRIGGER AS $$
27 | BEGIN
28 |     IF NEW.status = 'submitted' OR NEW.status = 'graded' THEN
29 |         UPDATE public.profiles
30 |         SET
31 |             streak_daily = CASE
32 |                 WHEN streak_last_date = CURRENT_DATE - INTERVAL '1 day' THEN streak_daily + 1
33 |                 WHEN streak_last_date = CURRENT_DATE THEN streak_daily
34 |                 ELSE 1
35 |             END,
36 |             streak_last_date = CURRENT_DATE,
37 |             total_exams_done = total_exams_done + 1,
38 |             total_exams_passed = total_exams_passed + CASE WHEN NEW.score >= (SELECT passing_score FROM quizzes WHERE...)
39 |             WHERE id = NEW.student_id;
40 |     END IF;
41 |     RETURN NEW;
42 | END;
43 | $$ LANGUAGE plpgsql SECURITY DEFINER;
44 |
45 | DROP TRIGGER IF EXISTS on_quiz_attempt_submit ON public.quiz_attempts;
46 | CREATE TRIGGER on_quiz_attempt_submit
47 | AFTER UPDATE OF status ON public.quiz_attempts
48 | FOR EACH ROW
49 | WHEN (NEW.status = 'submitted' OR NEW.status = 'graded')
50 | EXECUTE FUNCTION public.update_streak_on_attempt();
51 |

```

```

5 2 | - - %P%P%P VIEW: quiz_questions_for_student (safe, hides is_correct) %P%P%P
53 CREATE OR REPLACE VIEW public.quiz_questions_for_student AS
54 SELECT
55   qq.quiz_id,
56   q.id AS question_id,
57   q.content,
58   q.type,
59   q.points,
60   q.difficulty,
61   q.explanation,
62   qq."order",
63   COALESCE(
64     JSONB_AGG(
65       JSONB_BUILD_OBJECT('id', qo.id, 'content', qo.content, 'order', qo."order")
66       ORDER BY qo."order"
67     ) FILTER (WHERE qo.id IS NOT NULL),
68     '[]::jsonb'
69   ) AS options
70 FROM public.quiz_questions qq
71 JOIN public.questions q ON q.id = qq.question_id
72 LEFT JOIN public.question_options qo ON qo.question_id = q.id
73 GROUP BY qq.quiz_id, q.id, q.content, q.type, q.points, q.difficulty, q.explanation, qq."order";
74

```

File: scripts/generate-components.mjs (547 lines, 26.6 KB)

```

1 import fs from 'fs';
2 import path from 'path';
3
4 import { fileURLToPath } from 'url';
5 const __dirname = path.dirname(fileURLToPath(import.meta.url));
6
7 const CPU_CORES = [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 24, 28, 32, 64];
8 const CPU_THREADS = [4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 48, 56, 64, 128];
9 const BASE_TDP = [35, 45, 55, 60, 65, 75, 85, 95, 105, 120, 125, 140, 150, 170, 200, 230, 250, 280, 350];
10
11 function rand(min, max) { return Math.floor(Math.random() * (max - min + 1)) + min; }
12 function pick(arr) { return arr[rand(0, arr.length - 1)]; }
13 function roundPrice(n) { return Math.round(n / 10000) * 10000; }
14
15 | / / % % % C P U S % % %
16 const CPU_BRANDS = ['Intel', 'AMD'];
17 const INTEL_LINES = [
18   { name: 'Core i3', genStart: 12, genEnd: 15, suffix: ['F', 'T', ''], base: [12100, 13100, 14100, 15100] },
19   { name: 'Core i5', genStart: 12, genEnd: 15, suffix: ['K', 'KF', 'T', 'F', ''], base: [12400, 12600, 13400, ...
20   { name: 'Core i7', genStart: 12, genEnd: 15, suffix: ['K', 'KF', 'T', 'F', ''], base: [12700, 13700, 14700, ...
21   { name: 'Core i9', genStart: 12, genEnd: 15, suffix: ['K', 'KF', 'KS', 'T', 'F', ''], base: [12900, 13900, 1...
22   { name: 'Core Ultra 5', genStart: 1, genEnd: 2, suffix: ['K', 'F', ''], base: [125, 225, 235] },
23   { name: 'Core Ultra 7', genStart: 1, genEnd: 2, suffix: ['K', 'F', ''], base: [165, 265, 275] },
24   { name: 'Core Ultra 9', genStart: 1, genEnd: 2, suffix: ['K', 'F', ''], base: [185, 285] },
25   { name: 'Pentium Gold', genStart: 7, genEnd: 9, suffix: ['', 'T'], base: [7400, 7500, 8500] },
26   { name: 'Celeron', genStart: 6, genEnd: 9, suffix: ['', 'T'], base: [6900, 7300, 7305] },
27 ];
28
29 const AMD_LINES = [
30   { name: 'Ryzen 3', gen: ['5000', '7000', '8000', '9000'], models: ['5300G', '5400G', '7300X', '8300G', '9300...
31   { name: 'Ryzen 5', gen: ['5000', '7000', '8000', '9000'], models: ['5500', '5600', '5600X', '7500F', '7600', ...
32   { name: 'Ryzen 7', gen: ['5000', '7000', '8000', '9000'], models: ['5700X', '5700X3D', '5800X', '5800X3D', ...
33   { name: 'Ryzen 9', gen: ['5000', '7000', '9000'], models: ['5900X', '5950X', '7900', '7900X', '7950X', '7950...
34   { name: 'Ryzen Threadripper', gen: ['7000'], models: ['7960X', '7970X', '7980X', '7990X', '7965WX', '7975WX'...
35   { name: 'EPYC', gen: ['4004', '8004', '9004'], models: ['4124P', '4344P', '4484PX', '9124', '9254', '9374F',...
36 ];
37
38 function generateCPUs() {
39   const cpus = [];
40   let id = 1;
41
42   // Intel
43   for (const line of INTEL_LINES) {
44     for (const gen of [12, 13, 14, 15]) {
45       if (gen < line.genStart || gen > line.genEnd) continue;
46       for (const base of line.base) {
47         const genPrefix = String(gen) + String(base).slice(-3);
48         for (const sfx of line.suffix) {
49           const name = `${line.name}-${genPrefix}${sfx}`;
50           const cores = pick(CPU_CORES.slice(0, 9));
51           const threads = cores * 2;
52           const tdp = pick(BASE_TDP.slice(2, 11));
53           const price = roundPrice(rand(800000, line.name.includes('i9') ? 25000000 : line.name.includes('i7')...
54           const socket = gen >= 14 ? 'LGA1851' : 'LGA1700';
55           cpus.push({
56             id: `cpu_intel_${genPrefix}${sfx} || 'std'`_${id++}`,
57             name: `Intel ${name}`,
58             type: 'CPU',
59             price,
60             socket,
61             power: tdp,
62 | desc: `${cores} nhân / ${threads} luồng, ${tdp}W TDP, Socket
63 | });
64 |   }
65 | }
66 | }
67 | }
68 |
69 | // AMD
70 | for (const line of AMD_LINES) {
71 |   for (const gen of line.gen) {
72 |     for (const model of line.models) {
73 |       const name = `AMD ${line.name} ${gen}${model}`;
74 |       const cores = line.name.includes('Threadripper') || line.name.includes('EPYC') ? pick([16, 24, 32, 64, ...
75 |       const threads = line.name.includes('Threadripper') || line.name.includes('EPYC') ? cores * 2 : cores * 2;
76 |       const tdp = line.name.includes('Threadripper') ? pick([250, 280, 300, 350]) : line.name.includes('EPYC...
77 |       const isX3D = model.includes('X3D');
78 |       const price = roundPrice(
79 |         line.name.includes('Threadripper') ? rand(10000000, 80000000) :
80 |         line.name.includes('EPYC') ? rand(15000000, 120000000) :
81 |         line.name.includes('Ryzen 9') ? rand(8000000, 25000000) :
82 |         line.name.includes('Ryzen 7') ? rand(5000000, 15000000) :
83 |         line.name.includes('Ryzen 5') ? rand(2500000, 8000000) :
84 |         rand(1500000, 5000000)
85 |       );
86 |       const socket = gen.startsWith('9') ? 'AM5' : gen.startsWith('8') ? 'AM5' : gen.startsWith('7') ? (line...
87 |       cpus.push({
88 |         id: `cpu_amd_${line.name.toLowerCase().replace(/ /g, '_')}_${gen}_${model.toLowerCase()}_${id++}`,
89 |         name,
90 |         type: 'CPU',
91 |         price: isX3D ? price * 1.2 : price,
92 |         socket,
93 |         power: tdp,
94 | | desc: `${cores} nhân / ${threads} luồng, ${tdp}W TDP, Socket
95 | | });
96 | | }
97 | | }

```

```

98 | }
99 |
100 | return cpus;
101 | }
102 |
103 | // % % % GPU s % % %
104 | function generateGPUs() {
105 |     const gpus = [];
106 |     const lines = [
107 |         { brand: 'NVIDIA', prefix: 'RTX', models: ['3050', '3060', '3060 Ti', '3070', '3070 Ti', '3080', '3080 Ti'...
108 |         { brand: 'NVIDIA', prefix: 'GTX', models: ['1630', '1650', '1650 Super', '1660', '1660 Super', '1660 Ti'...
109 |         { brand: 'AMD', prefix: 'RX', models: ['6400', '6500 XT', '6600', '6600 XT', '6650 XT', '6700 XT', '6750 X...
110 |         { brand: 'Intel', prefix: 'Arc', models: ['A310', 'A380', 'A580', 'A750', 'A770', 'B580', 'B770', 'B780'...]
111 |     ];
112 |
113 |     const brands = ['ASUS', 'MSI', 'Gigabyte', 'Colorful', 'ZOTAC', 'PNY', 'Palit', 'Galax', 'Inno3D', 'PowerCol...
114 |     let id = 1;
115 |
116 |     for (const line of lines) {
117 |         for (let i = 0; i < line.models.length; i++) {
118 |             const model = line.models[i];
119 |             const vram = line.vram[i];
120 |             for (const brand of brands) {
121 |                 const name = `${brand} NVIDIA ${line.prefix} ${model} ${vram}GB`; // default, will fix for AMD/Intel
122 |                 const isHighEnd = model.includes('90') || model.includes('80') || model.includes('70 Ti Super') || mod...
123 |                 const isMid = model.includes('70') || model.includes('60 Ti') || model.includes('7800') || model.inclu...
124 |                 const price = roundPrice(
125 |                     line.brand === 'NVIDIA' ?
126 |                     (isHighEnd ? rand(15000000, 75000000) : isMid ? rand(7000000, 16000000) : rand(3000000, 8000000)) :
127 |                     line.brand === 'AMD' ?
128 |                     (isHighEnd ? rand(12000000, 60000000) : isMid ? rand(6000000, 14000000) : rand(2500000, 7000000)) :
129 |                     rand(2000000, 10000000)
130 |                 );
131 |                 const tdp = isHighEnd ? pick([250, 300, 350, 400, 450]) : isMid ? pick([150, 180, 200, 220, 250]) : pi...
132 |
133 |                 let fullName;
134 |                 if (line.brand === 'Intel') {
135 |                     fullName = `${brand} Intel Arc ${model}`;
136 |                 } else {
137 |                     fullName = `${brand} ${line.brand === 'NVIDIA' ? `GeForce ${line.prefix}` : `Radeon ${line.prefix}`}`...
138 |                 }
139 |
140 |                 gpus.push({
141 |                     id: `gpu_${line.brand.toLowerCase()}_${model.toLowerCase().replace(/ /g, '_')}_${brand.toLowerCase()...
142 |                     name: fullName,
143 |                     type: 'GPU',
144 |                     price,
145 |                     power: tdp,
146 |                     desc: `${
147 |                         line.brand === 'NVIDIA' ? `${line.prefix} ${model}` :
148 |                         line.brand === 'AMD' ? `Radeon ${line.prefix} ${model}` :
149 |                         `Intel Arc ${model}`
150 |                     } ${vram}GB VRAM, ${tdp}W TDP`,
151 |                 });
152 |             }
153 |         }
154 |     }
155 |     return gpus;
156 | }
157 |
158 | // % % % RAM % % %
159 | function generateRAM() {
160 |     const rams = [];
161 |     const brands = ['Corsair', 'G.Skill', 'Kingston', 'Crucial', 'TeamGroup', 'ADATA', 'Patriot', 'Samsung', 'SK...
162 |     const ddr4Speeds = [2133, 2400, 2666, 2933, 3000, 3200, 3333, 3400, 3466, 3600, 3733, 3800, 4000, 4133, 4266...
163 |     const ddr5Speeds = [4800, 5200, 5400, 5600, 6000, 6200, 6400, 6600, 6800, 7000, 7200, 7400, 7600, 7800, 8000...
164 |     const sizes = [4, 8, 12, 16, 24, 32, 48, 64, 96, 128];
165 |     const sticks = [1, 2, 4, 8];
166 |     const lines = ['Value', 'Vengeance', 'Trident Z', 'Ripjaws', 'Fury', 'Ballistix', 'T-Force', 'XPG', 'Viper',...
167 |     let id = 1;
168 |
169 |     for (const brand of brands) {
170 |         for (const ddrType of ['DDR4', 'DDR5']) {
171 |             const speeds = ddrType === 'DDR4' ? ddr4Speeds : ddr5Speeds;
172 |             for (let si = 0; si < Math.min(sizes.length, 8); si++) {
173 |                 const sizeGB = sizes[si];
174 |                 for (const stick of [1, 2]) {
175 |                     const totalGB = sizeGB * stick;
176 |                     const speed = pick(speeds);
177 |                     const line = pick(lines);
178 |                     const price = roundPrice(ddrType === 'DDR5' ? rand(totalGB * 50000, totalGB * 120000) : rand(totalGB...
179 |                     const name = `${brand} ${line} ${totalGB}GB (${stick}x${sizeGB}GB) ${ddrType}-${speed}`;
180 |                     rams.push({
181 |                         id: `ram_${ddrType.toLowerCase()}_${brand.toLowerCase()}_${totalGB}gb_${id++}`,
182 |                         name,
183 |                         type: 'RAM',
184 |                         price,
185 |                         socket: ddrType === 'DDR4' ? 'DDR4' : 'DDR5',
186 |                         ramType: ddrType,
187 |                         size: `${totalGB}GB (${stick}x${sizeGB}GB)`,
188 |                         desc: `${ddrType}-${speed}MHz, CAS ${pick([14, 15, 16, 17, 18, 19, 20, 22, 24, 26, 28, 30, 32, 34,...
189 |                     });
190 |                 }
191 |             }
192 |         }
193 |     }
194 |     return rams;
195 | }
196 |
197 | // % % % Mainboards % % %
198 | function generateMainboards() {
199 |     const boards = [];
200 |     const brands = ['ASUS', 'MSI', 'Gigabyte', 'ASRock', 'Biostar', 'Maxsun', 'Colorful'];
201 |     const chipsets = [
202 |         { name: 'H610', socket: 'LGA1700', ddr: ['DDR4', 'DDR5'] },
203 |         { name: 'B660', socket: 'LGA1700', ddr: ['DDR4', 'DDR5'] },
204 |         { name: 'B760', socket: 'LGA1700', ddr: ['DDR4', 'DDR5'] },
205 |         { name: 'Z690', socket: 'LGA1700', ddr: ['DDR4', 'DDR5'] },
206 |         { name: 'Z790', socket: 'LGA1700', ddr: ['DDR4', 'DDR5'] },
207 |         { name: 'W680', socket: 'LGA1700', ddr: ['DDR5'] },
208 |         { name: 'B860', socket: 'LGA1851', ddr: ['DDR5'] },
209 |         { name: 'Z890', socket: 'LGA1851', ddr: ['DDR5'] },
210 |         { name: 'A520', socket: 'AM4', ddr: ['DDR4'] },
211 |         { name: 'B550', socket: 'AM4', ddr: ['DDR4'] },
212 |         { name: 'X570', socket: 'AM4', ddr: ['DDR4'] },
213 |         { name: 'A620', socket: 'AM5', ddr: ['DDR5'] },
214 |         { name: 'B650', socket: 'AM5', ddr: ['DDR5'] },
215 |         { name: 'B650E', socket: 'AM5', ddr: ['DDR5'] },
216 |         { name: 'X670', socket: 'AM5', ddr: ['DDR5'] },
217 |         { name: 'X670E', socket: 'AM5', ddr: ['DDR5'] },
218 |         { name: 'X870', socket: 'AM5', ddr: ['DDR5'] },
219 |         { name: 'X870E', socket: 'AM5', ddr: ['DDR5'] },
220 |         { name: 'TRX50', socket: 'sTR5', ddr: ['DDR5'] },
221 |         { name: 'WRX90', socket: 'sTR5', ddr: ['DDR5'] },

```

```

222 ];
223 const formFactors = ['Mini-ITX', 'Micro-ATX', 'ATX', 'E-ATX', 'Mini-DTX'];
224 let id = 1;
225
226 for (const brand of brands) {
227   for (const chip of chipsets) {
228     for (const ddr of chip.ddr) {
229       const ff = pick(formFactors);
230       const series = pick(['Prime', 'TUF Gaming', 'ROG Strix', 'Pro', 'MAG', 'MPG', 'MEG', 'Aorus', 'Master'...
231       const name = `${brand} ${series} ${chip.name} ${ddr} ${ff}`;
232       const price = roundPrice(
233         chip.name.startsWith('Z') || chip.name.startsWith('X') ? rand(3000000, 25000000) :
234         chip.name.startsWith('B') ? rand(1500000, 8000000) :
235         chip.name.startsWith('A') || chip.name.startsWith('H') ? rand(800000, 3500000) :
236         rand(5000000, 50000000)
237       );
238       boards.push({
239         id: `mb_${brand.toLowerCase()}_${chip.name.toLowerCase()}_${ddr.toLowerCase()}_${id++}`,
240         name,
241         type: 'Mainboard',
242         price,
243         socket: chip.socket,
244         ramType: ddr,
245         desc: `Chipset ${chip.name}, Socket ${chip.socket}, ${ff}, H x
246       });
247     }
248   }
249 }
250 return boards;
251 }
252
253 | / / % % % S t o r a g e % % %
254 function generateStorage() {
255   const storage = [];
256   const brands = ['Samsung', 'Western Digital', 'Seagate', 'Crucial', 'Kingston', 'SK Hynix', 'Solidigm', 'Tea...
257   let id = 1;
258
259   // NVMe SSDs
260   const nvmeLines = ['980', '990 EVO', '990 PRO', 'SN580', 'SN770', 'SN850X', 'T500', 'T700', 'KC3000', 'NV3',...
261   for (const brand of brands) {
262     for (const line of nvmeLines) {
263       for (const capacity of [256, 512, 1024, 2048, 4096, 8192]) {
264         if (Math.random() > 0.3) continue; // thin out
265         const gen = capacity >= 2048 ? pick([4, 5]) : pick([3, 4, 5]);
266         const name = `${brand} ${line} ${capacity} >= 1024 ? capacity / 1024 + 'TB' : capacity + 'GB' NVMe Gen...
267         const price = roundPrice(capacity * [60, 80, 100, 120, 150, 200][Math.min(gen, 5)]);
268         storage.push({
269           id: `ssd_nvme_${brand.toLowerCase()}_${line.toLowerCase()}_${capacity}gb_${id++}`,
270           name,
271           type: 'Storage',
272           price,
273           desc: `NVMe M.2 Gen${gen}, ${capacity} >= 1024 ? capacity / 1024 + 'TB' : capacity + 'GB'`,
274         });
275       }
276     }
277   }
278
279   // SATA SSDs
280   const sataLines = ['870 EVO', '870 QVO', 'BX500', 'MX500', 'SA400', 'A400', 'SU630', 'SU650', 'SU750', 'UV50...
281   for (const brand of brands.slice(0, 10)) {
282     for (const line of sataLines) {
283       for (const capacity of [240, 480, 960, 1920, 3840]) {
284         if (Math.random() > 0.3) continue;
285         const name = `${brand} ${line} ${capacity} >= 1000 ? capacity / 1000 + 'TB' : capacity + 'GB' SATA SSD`;
286         storage.push({
287           id: `ssd_sata_${brand.toLowerCase()}_${line.toLowerCase()}_${capacity}gb_${id++}`,
288           name,
289           type: 'Storage',
290           price: roundPrice(capacity * 40),
291           desc: `SATA III 2.5", ${capacity} >= 1000 ? capacity / 1000 + 'TB' : capacity + 'GB'`,
292         });
293       }
294     }
295   }
296
297   // HDDs
298   const hddBrands = ['Seagate', 'Western Digital', 'Toshiba'];
299   const hddLines = ['Barracuda', 'IronWolf', 'SkyHawk', 'WD Blue', 'WD Black', 'WD Red', 'WD Purple', 'Gold', ...
300   const hddCapacities = [500, 1000, 2000, 4000, 6000, 8000, 10000, 12000, 14000, 16000, 18000, 20000, 22000, 2...
301   for (const brand of hddBrands) {
302     for (const line of hddLines) {
303       for (const cap of hddCapacities) {
304         if (Math.random() > 0.25) continue;
305         const rpm = cap >= 4000 ? pick([5400, 7200]) : 7200;
306         const name = `${brand} ${line} ${cap} >= 1000 ? cap / 1000 + 'TB' : cap + 'GB' ${rpm}rpm`;
307         storage.push({
308           id: `hdd_${brand.toLowerCase()}_${line.toLowerCase()}_${cap}gb_${id++}`,
309           name,
310           type: 'Storage',
311           price: roundPrice(cap * [10, 15, 20, 25, 30, 35, 40][Math.min(Math.floor(cap / 2000), 6)]),
312           desc: `HDD ${rpm}rpm, ${cap} >= 1000 ? cap / 1000 + 'TB' : cap + 'GB', SATA III`,
313         });
314       }
315     }
316   }
317
318   return storage;
319 }
320
321 | / / % % % P S U % % %
322 function generatePSUs() {
323   const psus = [];
324   const brands = ['Corsair', 'EVGA', 'Seasonic', 'Cooler Master', 'be quiet!', 'Thermaltake', 'FSP', 'SilverSt...
325   const wattages = [300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 1000, 1050, 1100, 1200, 1...
326   const ratings = ['80+ White', '80+ Bronze', '80+ Gold', '80+ Platinum', '80+ Titanium'];
327   const series_map = {
328     'Corsair': ['CV', 'CX', 'RM', 'RMx', 'RMx Shift', 'HX', 'AX', 'SF'],
329     'EVGA': ['W1', 'GD', 'G5', 'G6', 'G7', 'P2', 'T2', 'SuperNOVA'],
330     'Seasonic': ['S12', 'Focus', 'Core', 'Prime', 'Vertex', 'Prime TX'],
331     'Cooler Master': ['Elite', 'MWE', 'V', 'XG'],
332     'be quiet!': ['System Power', 'Pure Power', 'Straight Power', 'Dark Power'],
333     'Thermaltake': ['Smart', 'Toughpower', 'Toughpower GF', 'Toughpower iRGB'],
334     'FSP': ['Hydro', 'Hydro G', 'Hydro PTM'],
335     'SilverStone': ['Strider', 'Strider Gold', 'Nightjar', 'SX'],
336     'NZXT': ['C', 'E'],
337     'Antec': ['Earthwatts', 'NeoECO', 'High Current', 'Signature'],
338     'Super Flower': ['Leadex', 'Leadex Platinum', 'Leadex Titanium'],
339     'Lian Li': ['SP'],
340     'MSI': ['MAG', 'MPG', 'A-G'],
341     'ASUS': ['TUF', 'ROG', 'ROG Thor'],
342     'Gigabyte': ['GP', 'UD', 'Aorus'],
343     'DeepCool': ['PF', 'DQ', 'PQ'],
344     'XPG': ['Core Reactor', 'CyberCore', 'Fusion'],
345   };

```

```

346 | let id = 1;
347 |
348 | for (const brand of brands) {
349 |   const series = series_map[brand] || ['Standard'];
350 |   for (const s of series) {
351 |     for (const rating of ratings.slice(0, brand.includes('Corsair') || brand.includes('Seasonic') || brand.i...
352 |       for (const watt of wattages) {
353 |         if (Math.random() > 0.2) continue;
354 |         if (rating === '80+ Titanium' && watt < 600) continue;
355 |         if (rating === '80+ Platinum' && watt < 400) continue;
356 |         const name = `${brand} ${s} ${watt}W ${rating}`;
357 |         const price = roundPrice(
358 |           rating.includes('Titanium') ? watt * 500 :
359 |           rating.includes('Platinum') ? watt * 400 :
360 |           rating.includes('Gold') ? watt * 300 :
361 |           rating.includes('Bronze') ? watt * 200 :
362 |           watt * 120
363 |         );
364 |         psus.push({
365 |           id: `psu_${brand.toLowerCase()}_${s.toLowerCase().replace(/ /g, '_')}_${watt}w_${id++}`,
366 |           name,
367 |           type: 'PSU',
368 |           price,
369 |           wattage: watt,
370 |           desc: `${watt}W ${rating}, Fully Modular`,
371 |         });
372 |       }
373 |     }
374 |   }
375 | }
376 | return psus;
377 | }
378 |
379 | // % % % C o o l e r s % % %
380 | function generateCoolers() {
381 |   const coolers = [];
382 |   const brands = ['Noctua', 'Cooler Master', 'NZXT', 'Corsair', 'be quiet!', 'DeepCool', 'Thermalright', 'ID-C...
383 |   let id = 1;
384 |
385 |   // Air coolers
386 |   const airlines = ['NH-D15', 'NH-U12S', 'NH-L9i', 'Hyper 212', 'Hyper 620S', 'MA620P', 'Pure Rock 2', 'Dark R...
387 |   for (const brand of brands.slice(0, 10)) {
388 |     for (const line of airlines) {
389 |       if (Math.random() > 0.5) continue;
390 |       const price = roundPrice(rand(150000, 250000));
391 |       const name = `${brand} ${line} (Air Cooler)`;
392 |       coolers.push({
393 |         id: `cooler_air_${brand.toLowerCase()}_${line.toLowerCase().replace(/ /g, '_')}_${id++}`,
394 |         name,
395 |         type: 'Cooler',
396 |         price,
397 |         desc: `T E n n h i ç t k h i , ${pick(['120mm', '140mm', '92mm', '80mm'])}`
398 |       });
399 |     }
400 |   }
401 |
402 |   // AIO liquid coolers
403 |   const aiolines = ['Kraken', 'Kraken Elite', 'Kraken X', 'iCUE H100i', 'iCUE H150i', 'iCUE H170i', 'Liquid Fr...
404 |   const radiatorSizes = [120, 240, 280, 360, 420, 480];
405 |   for (const brand of brands) {
406 |     for (const line of aiolines) {
407 |       if (Math.random() > 0.4) continue;
408 |       for (const rad of radiatorSizes) {
409 |         if (Math.random() > 0.5) continue;
410 |         const price = roundPrice(rand(800000, 1200000));
411 |         const name = `${brand} ${line} ${rad}mm AIO`;
412 |         coolers.push({
413 |           id: `cooler_aio_${brand.toLowerCase()}_${line.toLowerCase().replace(/ /g, '_')}_${rad}mm_${id++}`,
414 |           name,
415 |           type: 'Cooler',
416 |           price,
417 |           desc: `T E n n h i ç t n ° Ü c A I O ${rad}mm , ${pick(['RGB', 'ARGB', 'K...
418 |         });
419 |       }
420 |     }
421 |   }
422 |   return coolers;
423 | }
424 |
425 | // % % % C a s e s % % %
426 | function generateCases() {
427 |   const cases = [];
428 |   const brands = ['Corsair', 'NZXT', 'Lian Li', 'Fractal Design', 'Cooler Master', 'Phanteks', 'be quiet!', 'T...
429 |   const formFactors = ['Mini-ITX', 'Micro-ATX', 'Mid Tower', 'Full Tower', 'Super Tower', 'Mini Tower', 'SFF',...
430 |   let id = 1;
431 |
432 |   const caseSeries = {
433 |     'Corsair': ['4000D', '5000D', '7000D', 'iCUE 220T', 'iCUE 465X', 'Crystal 280X', 'Crystal 570X', 'Obsidian...
434 |     'NZXT': ['H5 Flow', 'H7 Flow', 'H9 Flow', 'H9 Elite', 'H510', 'H710', 'H700i', 'H1'],
435 |     'Lian Li': ['O11 Dynamic', 'O11 Dynamic EVO', 'O11 Dynamic XL', 'Lancool 205', 'Lancool 216', 'Lancool III...
436 |     'Fractal Design': ['Meshify 2', 'Define 7', 'North', 'Pop Air', 'Focus 2', 'Torrent', 'Era ITX', 'Terra'],
437 |     'Cooler Master': ['MasterBox Q300L', 'MasterBox MB520', 'MasterCase H500', 'COSMOS C700M', 'HAF 700', 'NR2...
438 |     'Phanteks': ['Eclipse G360A', 'Eclipse P400A', 'Eclipse P500A', 'Enthoo Pro 2', 'Evolv X', 'NV5', 'NV7', '...
439 |     'be quiet!': ['Pure Base 500', 'Pure Base 500DX', 'Shadow Base 800', 'Dark Base 701', 'Dark Base Pro 901'],
440 |     'Thermaltake': ['Core P3', 'Core P8', 'Level 20', 'Tower 100', 'Tower 200', 'Tower 300', 'Tower 500', 'Tow...
441 |     'SilverStone': ['Seta A1', 'Seta Q1', 'Seta H1', 'ALTA G1M', 'ALTA F2', 'Sugo 16', 'GD11', 'RM42'],
442 |     'DeepCool': ['CH370', 'CH510', 'CH560', 'CH780', 'MATREXX 40', 'MATREXX 50', 'MATREXX 70', 'LS520'],
443 |     'Jonsbo': ['D31', 'D41', 'D300', 'TK-1', 'TK-2', 'UMX6', 'VR4', 'N2', 'N3', 'N5'],
444 |   };
445 |
446 |   for (const brand of brands) {
447 |     const series = caseSeries[brand] || ['Standard', 'Pro', 'Elite'];
448 |     for (const s of series) {
449 |       const ff = pick(formFactors);
450 |       const name = `${brand} ${s} ${ff}`;
451 |       const price = roundPrice(
452 |         ff.includes('Full Tower') || ff.includes('Super Tower') ? rand(2000000, 2500000) :
453 |         ff.includes('Mid Tower') ? rand(500000, 5000000) :
454 |         rand(300000, 3000000)
455 |       );
456 |       cases.push({
457 |         id: `case_${brand.toLowerCase()}_${s.toLowerCase().replace(/ /g, '_')}_${id++}`,
458 |         name,
459 |         type: 'Case',
460 |         price,
461 |         desc: `${ff}, ${pick(['Tempered Glass', 'Mesh', 'Solid Panel', 'Dual Chamber'])}, ${pick(['ATX', 'E-AT...
462 |       });
463 |     }
464 |   }
465 |   return cases;
466 | }
467 |
468 | // % % % M o n i t o r s % % %
469 | function generateMonitors() {

```



```

470 const monitors = [];
471 const brands = ['Dell', 'LG', 'Samsung', 'ASUS', 'AOC', 'BenQ', 'ViewSonic', 'Acer', 'MSI', 'Gigabyte', 'HP'...
472 let id = 1;
473
474 const monitorSeries = {
475   'Dell': ['UltraSharp', 'P Series', 'S Series', 'Alienware', 'Gaming'],
476   'LG': ['UltraGear', 'UltraFine', 'UltraWide', 'Gram +view', 'MyView'],
477   'Samsung': ['Odyssey', 'Smart Monitor', 'ViewFinity', 'CRG9', 'G Series'],
478   'ASUS': ['ROG Swift', 'TUF Gaming', 'ProArt', 'Designo', 'VA'],
479   'AOC': ['Gaming', 'U Series', 'C Series', 'Agon', 'E Series', '24G2', '27G2'],
480 };
481
482 const resolutions = ['1920x1080', '2560x1080', '2560x1440', '3440x1440', '3840x1600', '3840x2160', '5120x216...
483 const sizes_inch = [21.5, 23.8, 24, 24.5, 25, 27, 28, 31.5, 32, 34, 38, 40, 42, 43, 48, 49, 54, 55, 57];
484 const refreshRates = [60, 75, 100, 120, 144, 165, 175, 200, 240, 360, 480, 540];
485 const panelTypes = ['IPS', 'VA', 'TN', 'OLED', 'QLED', 'Mini-LED', 'PLS', 'AHVA'];
486
487 for (const brand of brands) {
488   const series = monitorSeries[brand] || ['Standard', 'Pro', 'Ultra'];
489   for (const s of series) {
490     for (let i = 0; i < 3; i++) {
491       const size = pick(sizes_inch);
492       const res = pick(resolutions);
493       const hz = res.includes('2160') ? pick(refreshRates.slice(0, 5)) : res.includes('1440') ? pick(refresh...
494       const panel = pick(panelTypes);
495       const isUltrawide = res.includes('3440') || res.includes('3840x1600') || res.includes('5120');
496       const name = `${brand} ${s} ${size} ${res} ${hz}Hz ${panel}`;
497       const price = roundPrice(
498         res.includes('4320') ? rand(30000000, 150000000) :
499         res.includes('2160') ? rand(6000000, 50000000) :
500         res.includes('1440') ? rand(3000000, 25000000) :
501         rand(1500000, 12000000)
502       );
503       monitors.push({
504         id: `monitor_${brand.toLowerCase()}_${s.toLowerCase().replace(/ /g, '_')}_${size}_${id++}`,
505         name,
506         type: 'Monitor',
507         price,
508         desc: `${size}" ${panel}, ${res}, ${hz}Hz`,
509       });
510     }
511   }
512 }
513 return monitors;
514 }
515
5 1 6 | / / % % %   M a i n % % %
517 function main() {
518   const cpus = generateCPUs();
519   const gpus = generateGPUs();
520   const rams = generateRAM();
521   const boards = generateMainboards();
522   const storage = generateStorage();
523   const psus = generatePSUs();
524   const coolers = generateCoolers();
525   const cases = generateCases();
526   const monitors = generateMonitors();
527
528   const all = [...cpus, ...gpus, ...rams, ...boards, ...storage, ...psus, ...coolers, ...cases, ...monitors];
529
530   console.log(`Generated ${all.length} components:`);
531   console.log(`   CPUs: ${cpus.length}`);
532   console.log(`   GPUs: ${gpus.length}`);
533   console.log(`   RAM: ${rams.length}`);
534   console.log(`   Mainboards: ${boards.length}`);
535   console.log(`   Storage: ${storage.length}`);
536   console.log(`   PSUs: ${psus.length}`);
537   console.log(`   Coolers: ${coolers.length}`);
538   console.log(`   Cases: ${cases.length}`);
539   console.log(`   Monitors: ${monitors.length}`);
540
541   const outputPath = path.join(__dirname, '..', 'data', 'componentsData.json');
542   fs.writeFileSync(outputPath, JSON.stringify(all, null, 2), 'utf-8');
543   console.log(`\nWritten to: ${outputPath}`);
544 }
545
546 main();
547

```

File: scripts/seed.mjs (36 lines, 2.5 KB)

```

1 | import { createClient } from '@supabase/supabase-js'
2 |
3 | const supabaseUrl = 'https://ojjmdhrvivwvfqomonzd.supabase.co'
4 | const supabaseKey = 'eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJkdXBhYmFzZSIsInJlZiI6Im9qam1kaHJ2aXZ3dmZn...
5 |
6 | const supabase = createClient(supabaseUrl, supabaseKey)
7 |
8 | const QUESTS = [
9 |   { title: 'Hoan thanh 1 bai hoc', description: 'Xem het 1 bai giang bat ky', xp_reward: 50, type: 'daily', di...
10 |   { title: 'Lam bai trac nghiem', description: 'Hoan thanh 1 bai quiz bat ky', xp_reward: 30, type: 'daily', d...
11 |   { title: 'Tham gia thao luan', description: 'Dat cau hoi hoac tra loi trong dien dan', xp_reward: 20, type: ...
12 |   { title: 'Dat streak 3 ngay', description: 'Hoc lien tiep 3 ngay khong bo lo', xp_reward: 150, type: 'daily'...
13 |   { title: 'Kiem tra tuong thich', description: 'Chay kiem tra tuong thich cho 1 cau hinh PC', xp_reward: 40, ...
14 |   { title: 'Luyen tap 15 phut', description: 'Danh 15 phut trong PC Builder', xp_reward: 60, type: 'daily', di...
15 |   { title: 'Hoan thanh 3 bai hoc', description: 'Hoc lien tuc 3 bai giang trong ngay', xp_reward: 120, type: '...
16 | ]
17 |
18 | async function seed() {
19 |   console.log(`\`0=Y Bat dau seed du lieu...`)
20 |
21 |   const { data: existing } = await supabase.from('daily_quests').select('id').limit(1)
22 |   if (existing && existing.length > 0) {
23 |     console.log(`\` daily_quests da co du lieu, bo qua`)
24 |     return
25 |   }
26 |
27 |   const { error } = await supabase.from('daily_quests').insert(QUESTS)
28 |   if (error) {
29 |     console.log(`\`L Loi: ', error.message)
30 |   } else {
31 |     console.log(`\` Da them ' + QUESTS.length + ' nhiem vu hang ngay`)
32 |   }
33 | }
34 |
35 | seed().catch(console.error)
36 |

```

File: scripts/seed.sql (46 lines, 3.6 KB)

```

1 | -- =====

```

```

2 | - - SEED: C-p nh-t Level Definitions
3 | -----
4 | UPDATE level_definitions SET title = 'Tân Thới Hiệp, icon = '0<β1' WHERE level = 4;
5 | UPDATE level_definitions SET title = 'Ngã 5, icon = '0=Y', WHERE level = 5;
6 | UPDATE level_definitions SET title = 'Ngã 5, icon = '0=U', WHERE level = 6;
7 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=Y', WHERE level = 7;
8 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 8;
9 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 9;
10 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 10;
11 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 11;
12 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 12;
13 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 13;
14 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 14;
15 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 15;
16 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 16;
17 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 17;
18 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 18;
19 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 19;
20 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 20;
21 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 21;
22 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 22;
23 | UPDATE level_definitions SET title = 'Thị trấn, icon = '0=U', WHERE level = 23;

24 | -----
25 | -- SEED: Daily Quests
26 | -----
27 | -----
28 | INSERT INTO daily_questions (title, description, xp_reward, type, difficulty, icon, requirement_type, requirement_value, is_active)
29 | SELECT * FROM (VALUES
30 | ('Hàn Thới Hiệp, 1 bài híc', 'Xem híc 1 bài gíng bít kó', 50, 'daily', 'e
31 | ('Lập rap PC hoàn chỉnh', 'Lập rap c linh kichn tr ong', PC, Buil der', 80,
32 | ('Lập rap PC hoàn chỉnh', 'Lập rap c linh kichn tr ong', PC, Buil der', 80,
33 | ('Tham gia thđo lu-n', 't câu híc ho.c tré lýi trong diAn, àn', 20,
34 | ('it streak 3 ngày', 'Híc liên tiếp 3 ngày không bĩ lá', 150, 'daily',
35 | ('Kiếm tra t.ing thíc h', 'Chiy kiếm tra t.ing thíc h cho ng, 1 cýu', hinh PC
36 | ('Hoàn thành 3 bài híc', 'Híc liên tấc 3 bài gíng tr ong', 60, 'daily',
37 | ('Luy c n t - p 15 phút', 'Dành 15 phút trong PC Builder', 60, 'daily',

38 | ) AS q(title, description, xp_reward, type, difficulty, icon, requirement_type, requirement_value, is_active)
39 | WHERE NOT EXISTS (SELECT 1 FROM daily_questions LIMIT 1);
40 |
41 | -- =====
42 | -- Verify
43 | -----
44 | SELECT COUNT(*) AS quest_count FROM daily_questions;
45 | SELECT level, title, icon FROM level_definitions ORDER BY level;
46 |

```

File: eslint.config.mjs (17 lines, 0.4 KB)

```

1 | import { defineConfig, globalIgnores } from "eslint/config";
2 | import nextVitals from "eslint-config-next/core-web-vitals";
3 |
4 | const eslintConfig = defineConfig([
5 |   ...nextVitals,
6 |   // Override default ignores of eslint-config-next.
7 |   globalIgnores([
8 |     // Default ignores of eslint-config-next:
9 |     ".next/**",
10 |    "out/**",
11 |    "build/**",
12 |    "next-env.d.ts",
13 |  ]),
14 | ]);
15 |
16 | export default eslintConfig;
17 |

```

File: generate-pdf.mjs (179 lines, 4.3 KB)

```

1 | import { readFileSync, existsSync } from 'fs';
2 | import { glob } from 'tinyglobby';
3 | import { jsPDF } from 'jspdf';
4 | import path from 'path';
5 | import { fileURLToPath } from 'url';
6 |
7 | const __dirname = path.dirname(fileURLToPath(import.meta.url));
8 | const ROOT = __dirname;
9 |
10 | const includePatterns = [
11 |   'app/**/*.ts,tsx,js,jsx,css',
12 |   'components/**/*.ts,tsx,js,jsx',
13 |   'lib/**/*.ts,tsx,js',
14 |   'utils/**/*.ts,js',
15 |   'types/**/*.ts',
16 |   'hooks/**/*.ts',
17 |   'hooks/**/*.tsx',
18 |   'providers/**/*.tsx',
19 |   'data/**/*.ts,js,json',
20 |   'database/**/*.sql',
21 |   'supabase/migrations/**/*.sql',
22 |   'supabase/functions/**/*.ts',
23 |   'scripts/**/*.mjs,js,sql',
24 |   '*.ts,js,mjs',
25 |   'next.config.mjs',
26 |   'tailwind.config.js',
27 |   'postcss.config.js',
28 |   'tsconfig.json',
29 |   'eslint.config.mjs',
30 |   'vercel.json',
31 |   'package.json',
32 | ];
33 |
34 | const excludePatterns = [
35 |   'node_modules/**',
36 |   '.git/**',
37 |   '.next/**',
38 |   '.vercel/**',
39 |   '.vercel.bak/**',
40 |   'public/**',
41 |   'superpowers/**',
42 |   'n8n/**',
43 |   'GitNexus/**',
44 |   'next-env.d.ts',
45 |   'tsconfig.tsbuildinfo',
46 |   '*.tsbuildinfo',
47 | ];
48 |
49 | console.log('Scanning source files...');
50 |
51 | const allFiles = await glob(includePatterns, {
52 |   cwd: ROOT,
53 |   ignore: excludePatterns,

```

```

54 |     dot: true,
55 | });
56 |
57 | allFiles.sort();
58 |
59 | const srcDirs = ['app', 'components', 'lib', 'utils', 'types', 'hooks', 'providers', 'data', 'database', 'supabase', 'sc...
60 | allFiles.sort((a, b) => {
61 |     const aIdx = srcDirs.findIndex(d => a.startsWith(d));
62 |     const bIdx = srcDirs.findIndex(d => b.startsWith(d));
63 |     if (aIdx !== bIdx) {
64 |         if (aIdx === -1) return 1;
65 |         if (bIdx === -1) return -1;
66 |         return aIdx - bIdx;
67 |     }
68 |     return a.localeCompare(b);
69 | });
70 |
71 | console.log(`Found ${allFiles.length} source files.`);
72 |
73 | const doc = new jsPDF({
74 |     orientation: 'portrait',
75 |     unit: 'mm',
76 |     format: 'a4',
77 |     compress: true,
78 | });
79 |
80 | const PAGE_H = 297;
81 | const MARGIN = 10;
82 | const FONT_SIZE = 6;
83 | const LINE_HEIGHT = FONT_SIZE * 0.37;
84 | const MAX_CHARS = Math.floor((210 - 2 * MARGIN) / (FONT_SIZE * 0.28));
85 |
86 | doc.setFont('Courier', 'normal');
87 | doc.setFontSize(FONT_SIZE);
88 |
89 | let pageNum = 1;
90 | let cursorY = MARGIN;
91 |
92 | function addPage() {
93 |     doc.addPage();
94 |     pageNum++;
95 |     cursorY = MARGIN;
96 | }
97 |
98 | function ensureLine(n = 1) {
99 |     if (pageNum === 0) addPage();
100 |     if (cursorY + LINE_HEIGHT * n > PAGE_H - MARGIN) {
101 |         addPage();
102 |     }
103 | }
104 |
105 | function writeln(text = '') {
106 |     ensureLine();
107 |     doc.text(text, MARGIN, cursorY);
108 |     cursorY += LINE_HEIGHT;
109 | }
110 |
111 | function writeTitle(text) {
112 |     ensureLine(2);
113 |     const prevSize = doc.getFontSize();
114 |     doc.setFontSize(FONT_SIZE + 2);
115 |     doc.setFont('Courier', 'Bold');
116 |     doc.text(text, MARGIN, cursorY);
117 |     cursorY += LINE_HEIGHT * 1.5;
118 |     doc.setFont('Courier', 'normal');
119 |     doc.setFontSize(prevSize);
120 | }
121 |
122 | function writeSeparator() {
123 |     ensureLine();
124 |     doc.text('='.repeat(100), MARGIN, cursorY);
125 |     cursorY += LINE_HEIGHT;
126 | }
127 |
128 | function writeThinSep() {
129 |     ensureLine();
130 |     doc.text('-', repeat(100), MARGIN, cursorY);
131 |     cursorY += LINE_HEIGHT;
132 | }
133 |
134 | let totalSize = 0;
135 | const maxLinesPerFile = 5000;
136 |
137 | for (let idx = 0; idx < allFiles.length; idx++) {
138 |     const fp = allFiles[idx];
139 |     const fullPath = path.join(ROOT, fp);
140 |     try {
141 |         const content = readFileSync(fullPath, 'utf-8');
142 |         const size = Buffer.byteLength(content, 'utf-8');
143 |         totalSize += size;
144 |         const allLines = content.split(/\r?\n/);
145 |
146 |         if (allLines.length > maxLinesPerFile) {
147 |             const truncated = allLines.slice(0, maxLinesPerFile);
148 |             truncated.push('');
149 |             truncated.push('//---- FILE TRUNCATED: ${allLines.length} lines total, showing first ${maxLinesPerFile}...
150 |             allLines.length = 0;
151 |             allLines.push(...truncated);
152 |         }
153 |
154 |         writeTitle(`File: ${fp} (${allLines.length} lines, ${(size / 1024).toFixed(1)} KB`);
155 |         writeSeparator();
156 |
157 |         for (let li = 0; li < allLines.length; li++) {
158 |             const lineNum = String(li + 1).padStart(5, ' ');
159 |             let line = allLines[li];
160 |             if (line.length > MAX_CHARS) {
161 |                 line = line.substring(0, MAX_CHARS - 3) + '...';
162 |             }
163 |             writeln(`${lineNum} | ${line}`);
164 |         }
165 |
166 |         writeThinSep();
167 |         writeln('');
168 |     } catch (err) {
169 |         console.error(`Error reading ${fp}: ${err.message}`);
170 |     }
171 | }
172 |
173 | console.log(`\nTotal source size: ${(totalSize / 1024 / 1024).toFixed(2)} MB across ${allFiles.length} files`);
174 | console.log(`Total pages: ${pageNum}`);
175 |
176 | const outputPath = path.join(ROOT, 'PC-Master-LMS-Source-Code.pdf');
177 | doc.save(outputPath);

```

```
178 | console.log(`PDF saved to: ${outputPath}`);
179 |
```

File: inspect_schema.js (28 lines, 0.9 KB)

```
1 | const { createClient } = require('@supabase/supabase-js');
2 | const url = 'https://ojjmdhrvivwvfgomonzd.supabase.co';
3 | const anonKey = 'eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6Im9qam1kaHJ2aXZ3dmZnb21v...
4 |
5 | const supabase = createClient(url, anonKey);
6 |
7 | async function checkQuizzesTable() {
8 |   const { data, error } = await supabase
9 |     .from('quizzes')
10 |    .insert({
11 |      title: 'Test Quiz',
12 |      time_limit_minutes: 10,
13 |      max_attempts: 1,
14 |      is_published: false
15 |    })
16 |    .select();
17 |
18 |   if (error) {
19 |     console.log('Error code:', error.code);
20 |     console.log('Error message:', error.message);
21 |     console.log('Error details:', error.details);
22 |   } else {
23 |     console.log('Insert succeeded! Row:', data);
24 |   }
25 | }
26 |
27 | checkQuizzesTable();
28 |
```

File: next.config.mjs (49 lines, 2.1 KB)

```
1 | // Path: next.config.mjs
2 | /** @type {import('next').NextConfig} */
3 | const nextConfig = {
4 |   images: {
5 |     remotePatterns: [
6 |       {
7 |         protocol: 'https',
8 |         hostname: '*.supabase.co',
9 |         pathname: '/storage/v1/object/public/**',
10 |      },
11 |       {
12 |         protocol: 'https',
13 |         hostname: '*.supabase.in',
14 |         pathname: '/storage/v1/object/public/**',
15 |      },
16 |     ],
17 |   },
18 |   async headers() {
19 |     return [
20 |       {
21 |         source: '/(.*)',
22 |         headers: [
23 |           { key: 'X-Frame-Options', value: 'DENY' },
24 |           { key: 'X-Content-Type-Options', value: 'nosniff' },
25 |           { key: 'Referrer-Policy', value: 'strict-origin-when-cross-origin' },
26 |           { key: 'Strict-Transport-Security', value: 'max-age=63072000; includeSubDomains; preload' },
27 |           { key: 'Permissions-Policy', value: 'camera=(self), microphone=(self), geolocation=()' },
28 |           {
29 |             key: 'Content-Security-Policy',
30 |             value: [
31 |               "default-src 'self'",
32 |               "script-src 'self' 'unsafe-eval' 'unsafe-inline' https://cdn.jsdelivr.net https://accounts.google...",
33 |               "style-src 'self' 'unsafe-inline' https://fonts.googleapis.com",
34 |               "font-src 'self' https://fonts.gstatic.com https://*.sketchfab.com",
35 |               "img-src 'self' data: blob: https://*.supabase.co https://*.supabase.in https://*.googleusercontent...",
36 |               "connect-src 'self' https://*.supabase.co https://*.supabase.in wss://*.supabase.co https://acco...",
37 |               "frame-src 'self' https://accounts.google.com https://www.youtube.com https://sketchfab.com",
38 |               "worker-src 'self' blob: https://cdn.jsdelivr.net https://static.sketchfab.com",
39 |               "media-src 'self' blob: mediastream: https://*.sketchfab.com",
40 |             ].join('; '),
41 |           },
42 |         ],
43 |       },
44 |     ],
45 |   },
46 | }
47 | export default nextConfig
48 |
49 |
```

File: package.json (66 lines, 1.9 KB)

```
1 | {
2 |   "name": "pc-master-builder",
3 |   "description": "Educational PC building platform with Computer Vision",
4 |   "version": "0.1.0",
5 |   "private": true,
6 |   "scripts": {
7 |     "dev": "next dev",
8 |     "build": "next build",
9 |     "vercel-build": "next build",
10 |    "start": "next start",
11 |    "lint": "eslint"
12 |  },
13 |  "dependencies": {
14 |    "@dnd-kit/core": "^6.3.1",
15 |    "@dnd-kit/sortable": "^10.0.0",
16 |    "@dnd-kit/utilities": "^3.2.2",
17 |    "@google/generative-ai": "^0.24.1",
18 |    "@mediapipe/tasks-vision": "^0.10.32",
19 |    "@react-pdf/renderer": "^4.5.1",
20 |    "@react-three/drei": "^10.0.7",
21 |    "@react-three/fiber": "^9.2.0",
22 |    "@supabase/auth-helpers-nextjs": "^0.15.0",
23 |    "@supabase/ssr": "^0.10.2",
24 |    "@supabase/supabase-js": "^2.104.1",
25 |    "@tanstack/react-query": "^5.100.10",
26 |    "@tanstack/react-table": "^8.21.3",
27 |    "@vercel/analytics": "^2.0.1",
28 |    "canvas-confetti": "^1.9.4",
29 |    "date-fns": "^4.1.0",
30 |    "framer-motion": "^12.34.3",
31 |    "html2canvas": "^1.4.1",
32 |    "jspdf": "^4.2.1",

```

```

33|   "lucide-react": "^0.575.0",
34|   "mammoth": "^1.12.0",
35|   "nanoid": "^5.1.9",
36|   "next": "16.1.6",
37|   "pdf-lib": "^1.17.1",
38|   "pdfjs-dist": "^4.0.379",
39|   "pg": "^8.21.0",
40|   "qrcode": "^1.5.4",
41|   "qrcode.react": "^4.2.0",
42|   "react": "19.2.3",
43|   "react-dom": "19.2.3",
44|   "react-hot-toast": "^2.6.0",
45|   "react-markdown": "^10.1.0",
46|   "react-pdf": "^10.4.1",
47|   "recharts": "^3.8.1",
48|   "rehype-raw": "^7.0.0",
49|   "rehype-sanitize": "^6.0.0",
50|   "remark-gfm": "^4.0.1",
51|   "tesseract.js": "^7.0.0",
52|   "three": "^0.170.0",
53|   "zustand": "^5.0.14"
54| },
55| "devDependencies": {
56|   "@types/node": "25.6.0",
57|   "@types/qrcode": "^1.5.6",
58|   "autoprefixer": "^10.5.0",
59|   "eslint": "^9",
60|   "eslint-config-next": "16.1.6",
61|   "playwright": "^1.60.0",
62|   "postcss": "^8.5.14",
63|   "tailwindcss": "^3.4.19"
64| }
65| }
66|

```

File: postcss.config.js (7 lines, 0.1 KB)

```

1| module.exports = {
2|   plugins: {
3|     tailwindcss: {},
4|     autoprefixer: {},
5|   },
6| }
7|

```

File: proxy.ts (112 lines, 3.7 KB)

```

1| // Path: proxy.ts
2| import { createServerClient } from '@supabase/ssr'
3| import { NextResponse, type NextRequest } from 'next/server'
4|
5| // Role-based access control permissions map
6| const ROUTE_PERMISSIONS: Record<string, string[]> = {
7|   '/admin': ['admin'],
8|   '/teacher': ['teacher', 'admin'],
9|   '/student/discussion': ['student', 'teacher', 'admin'],
10|  '/student': ['student', 'admin'],
11|  '/parent': ['parent', 'admin'],
12|  '/builder': ['student', 'admin'],
13|  '/profile': ['student', 'teacher', 'admin', 'parent'],
14|  '/notifications': ['student', 'teacher', 'admin', 'parent'],
15|  '/leaderboard': ['student', 'teacher', 'admin', 'parent'],
16| }
17|
18| // Public routes that don't require user authentication
19| const PUBLIC_ROUTES = [
20|   '/', '/login', '/register', '/about',
21|   '/forgot-password', '/reset-password', '/verify-email',
22|   '/verify', '/check-email',
23|   '/admin', '/admin/settings', '/admin/users', '/admin/schools',
24|   '/auth/callback', '/api/auth/callback',
25|   '/onboarding',
26|   '/api',
27| ]
28|
29| export async function proxy(request: NextRequest) {
30|   const { pathname } = request.nextUrl
31|   let response = NextResponse.next({ request })
32|
33|   const supabase = createServerClient(
34|     process.env.NEXT_PUBLIC_SUPABASE_URL!,
35|     process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!,
36|     {
37|       cookies: {
38|         getAll() { return request.cookies.getAll() },
39|         setAll(cookiesToSet) {
40|           cookiesToSet.forEach(({ name, value }) =>
41|             request.cookies.set(name, value))
42|           response = NextResponse.next({ request })
43|           cookiesToSet.forEach(({ name, value, options }) =>
44|             response.cookies.set(name, value, options))
45|         },
46|       },
47|     },
48|   )
49|
50|   // Use getUser() instead of getSession() to securely verify JWT on the server
51|   const { data: { user } } = await supabase.auth.getUser()
52|
53|   const isPublicRoute = PUBLIC_ROUTES.some(r =>
54|     pathname === r || pathname.startsWith(r + '/')
55|   )
56|
57|   // Map roles to their respective active dashboard routes
58|   const dashMap: Record<string, string> = {
59|     admin: '/admin',
60|     teacher: '/teacher',
61|     parent: '/parent/dashboard',
62|     student: '/student',
63|   }
64|
65|   if (isPublicRoute) {
66|     if (user) {
67|       const { data: profile } = await supabase
68|         .from('profiles').select('role').eq('id', user.id).single()
69|       const role = profile?.role ?? 'student'
70|
71|       if (pathname === '/login' || pathname === '/register') {
72|         return NextResponse.redirect(
73|           new URL(dashMap[role] ?? '/student', request.url)
74|         )
75|       }
76|     }
77|   }
78|
79|   return response
80| }
81|

```

```

75 |     }
76 |
77 | // Ng n gi á o vi ê n v à p h ả h u y n h t r u y c - p v à o t r a n g b u i l d e r ( / b u i l d e
78 | if (pathname.startsWith('/builder') && (role === 'teacher' || role === 'parent')) {
79 |     return NextResponse.redirect(
80 |         new URL(dashMap[role] ?? '/student', request.url)
81 |     )
82 | }
83 | }
84 | return response
85 | }
86 |
87 | if (!user) {
88 |     const loginUrl = new URL('/login', request.url)
89 |     loginUrl.searchParams.set('redirect', pathname)
90 |     return NextResponse.redirect(loginUrl)
91 | }
92 |
93 | const { data: profile } = await supabase
94 |     .from('profiles').select('role').eq('id', user.id).single()
95 | const role = profile?.role ?? 'student'
96 |
97 | const matchedRoute = Object.keys(ROUTE_PERMISSIONS)
98 |     .find(r => pathname.startsWith(r))
99 |
100 | if (matchedRoute && !ROUTE_PERMISSIONS[matchedRoute].includes(role)) {
101 |     return NextResponse.redirect(new URL(dashMap[role] ?? '/student', request.url))
102 | }
103 |
104 | response.headers.set('x-user-role', role)
105 | response.headers.set('x-user-id', user.id)
106 | return response
107 | }
108 |
109 | export const config = {
110 |     matcher: ['/(?!_next/static|_next/image|favicon.ico|.*\\.(?:svg|png|jpg|jpeg|gif|webp)$).*'],
111 | }
112 |

```

File: scratch_profiles.ts (24 lines, 0.8 KB)

```

1 | import { createClient } from '@supabase/supabase-js'
2 |
3 | const supabaseUrl = 'https://ojjmdhrvivwvfgomonzd.supabase.co'
4 | const supabaseAnonKey = 'eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6Im9qam1kaHJ2aXZ3...'
5 |
6 | const supabase = createClient(supabaseUrl, supabaseAnonKey)
7 |
8 | async function run() {
9 |     console.log('Fetching profiles from Supabase...')
10 |     const { data, error } = await supabase
11 |         .from('profiles')
12 |         .select('id, email, full_name, role, created_at')
13 |         .order('created_at', { ascending: false })
14 |
15 |     if (error) {
16 |         console.error('Error:', error)
17 |     } else {
18 |         console.log('Profiles list:')
19 |         console.table(data)
20 |     }
21 | }
22 |
23 | run()
24 |

```

File: tailwind.config.js (24 lines, 0.4 KB)

```

1 | /** @type {import('tailwindcss').Config} */
2 | module.exports = {
3 |     content: [
4 |         './app/**/*.{js,ts,jsx,tsx,mdx}',
5 |         './components/**/*.{js,ts,jsx,tsx,mdx}',
6 |     ],
7 |     theme: {
8 |         extend: {
9 |             colors: {
10 |                 brand: {
11 |                     primary: '#00d2a0',
12 |                     secondary: '#00b4d8',
13 |                 },
14 |                 bg: {
15 |                     base: '#0f0f1a',
16 |                     surface: '#16213e',
17 |                 }
18 |             },
19 |         },
20 |     },
21 |     plugins: [],
22 | }
23 |
24 |

```

File: tsconfig.json (47 lines, 0.9 KB)

```

1 | {
2 |     "compilerOptions": {
3 |         "target": "ES2017",
4 |         "lib": [
5 |             "dom",
6 |             "dom.iterable",
7 |             "esnext"
8 |         ],
9 |         "allowJs": true,
10 |        "skipLibCheck": true,
11 |        "strict": false,
12 |        "noEmit": true,
13 |        "incremental": true,
14 |        "module": "esnext",
15 |        "esModuleInterop": true,
16 |        "moduleResolution": "node",
17 |        "resolveJsonModule": true,
18 |        "isolatedModules": true,
19 |        "jsx": "react-jsx",
20 |        "plugins": [
21 |            {
22 |                "name": "next"
23 |            }
24 |        ],
25 |        "paths": {

```

```

26|         "@/*": ["/.*"],
27|         "@/actions/*": ["/app/actions/*"],
28|         "@supabase/auth-helpers-nextjs": ["/lib/supabase-helpers-mock.ts"]
29|     },
30|     "baseUrl": ".",
31| },
32| "include": [
33|     "next-env.d.ts",
34|     ".next/types/**/*.ts",
35|     ".next/dev/types/**/*.ts",
36|     "**/*.mts",
37|     "**/*.ts",
38|     "**/*.tsx"
39| ],
40| "exclude": [
41|     "node_modules",
42|     "GitNexus",
43|     "superpowers",
44|     "supabase"
45| ]
46| }
47|

```

File: vercel.json (20 lines, 0.5 KB)

```

1| {
2|   "crons": [
3|     {
4|       "path": "/api/cron/daily-quiz",
5|       "schedule": "0 6 * * *"
6|     }
7|   ],
8|   "headers": [
9|     {
10|      "source": "/*",
11|      "headers": [
12|        { "key": "X-Frame-Options", "value": "DENY" },
13|        { "key": "X-Content-Type-Options", "value": "nosniff" },
14|        { "key": "Strict-Transport-Security", "value": "max-age=63072000; includeSubDomains; preload" },
15|        { "key": "Referrer-Policy", "value": "strict-origin-when-cross-origin" }
16|      ]
17|     }
18|   ]
19| }
20|

```